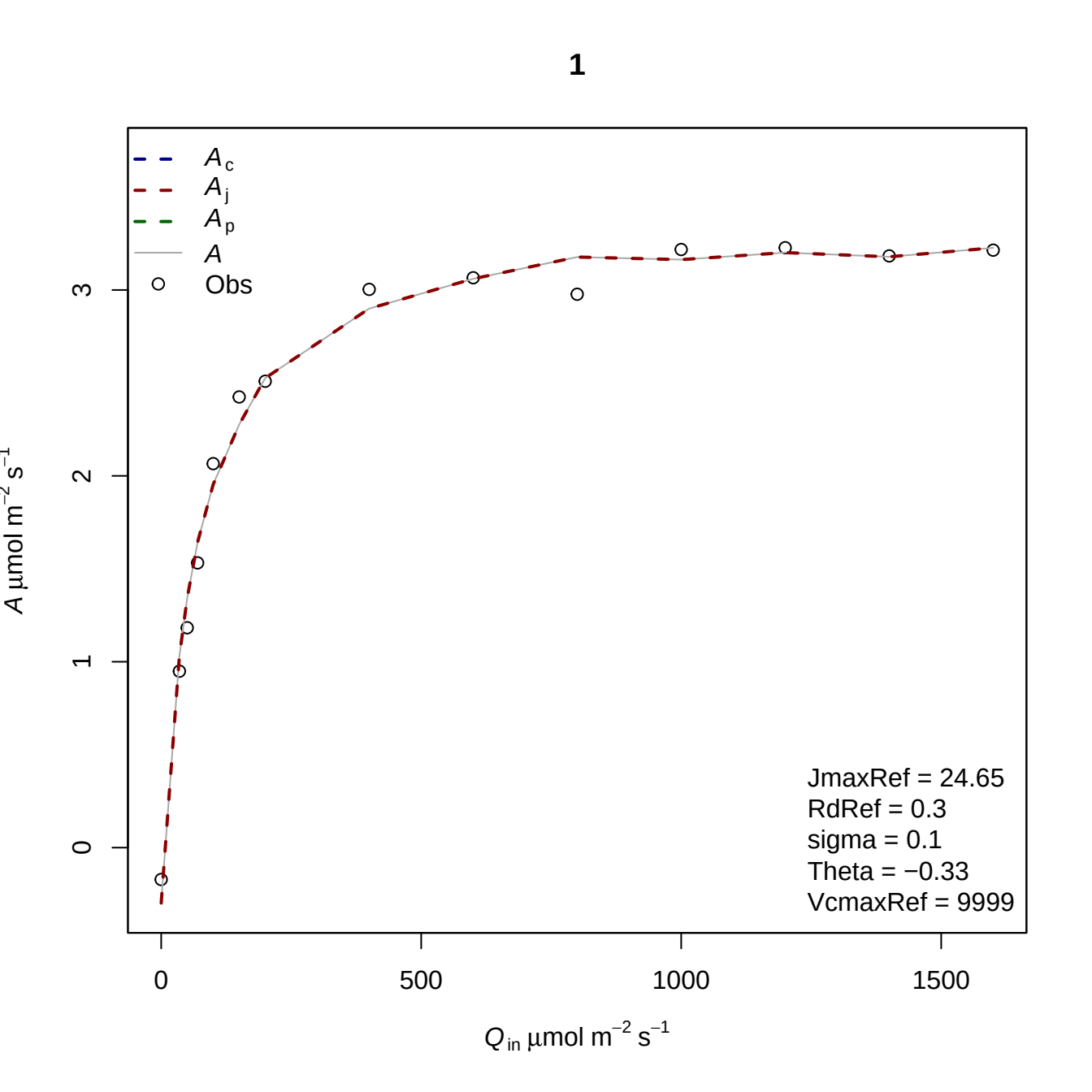
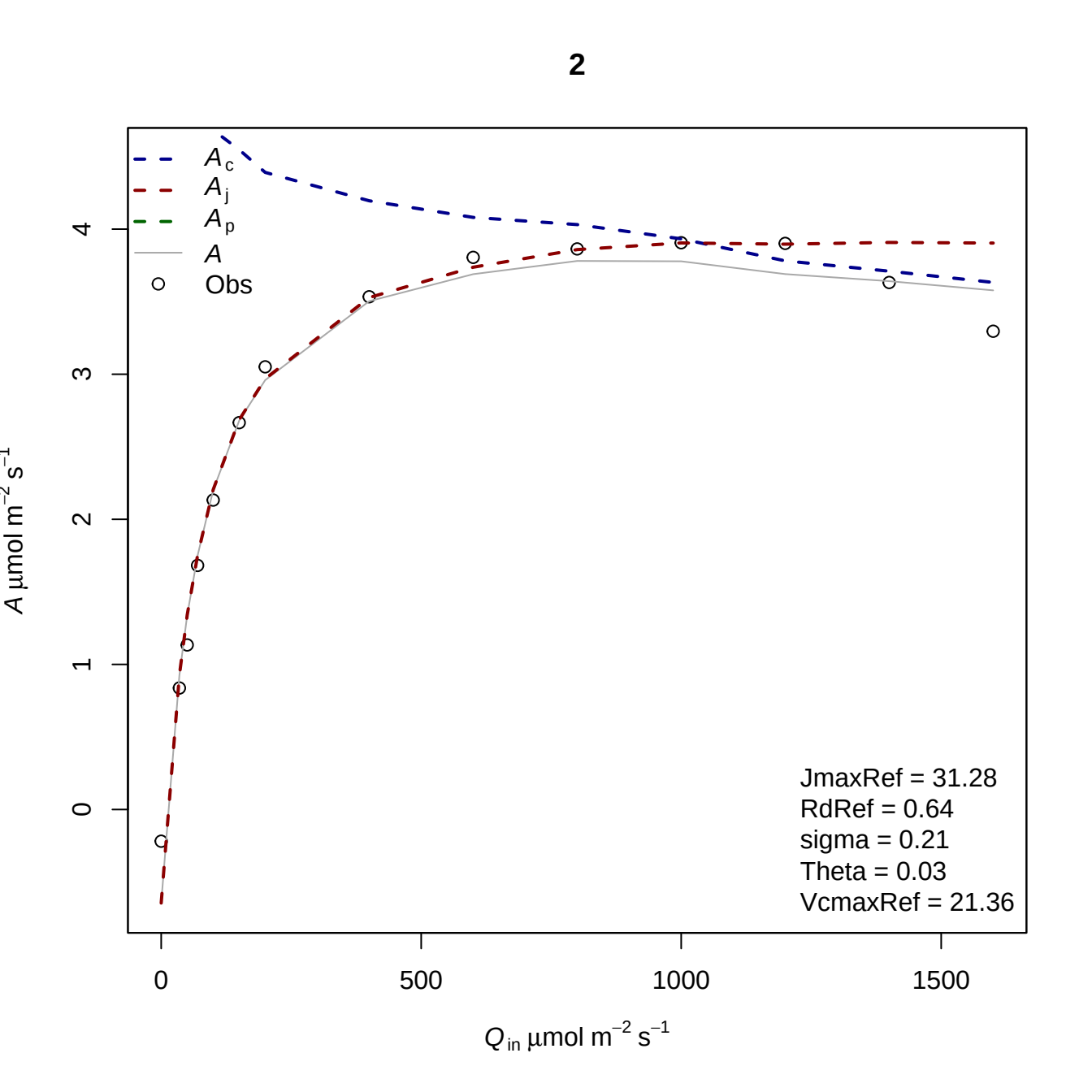
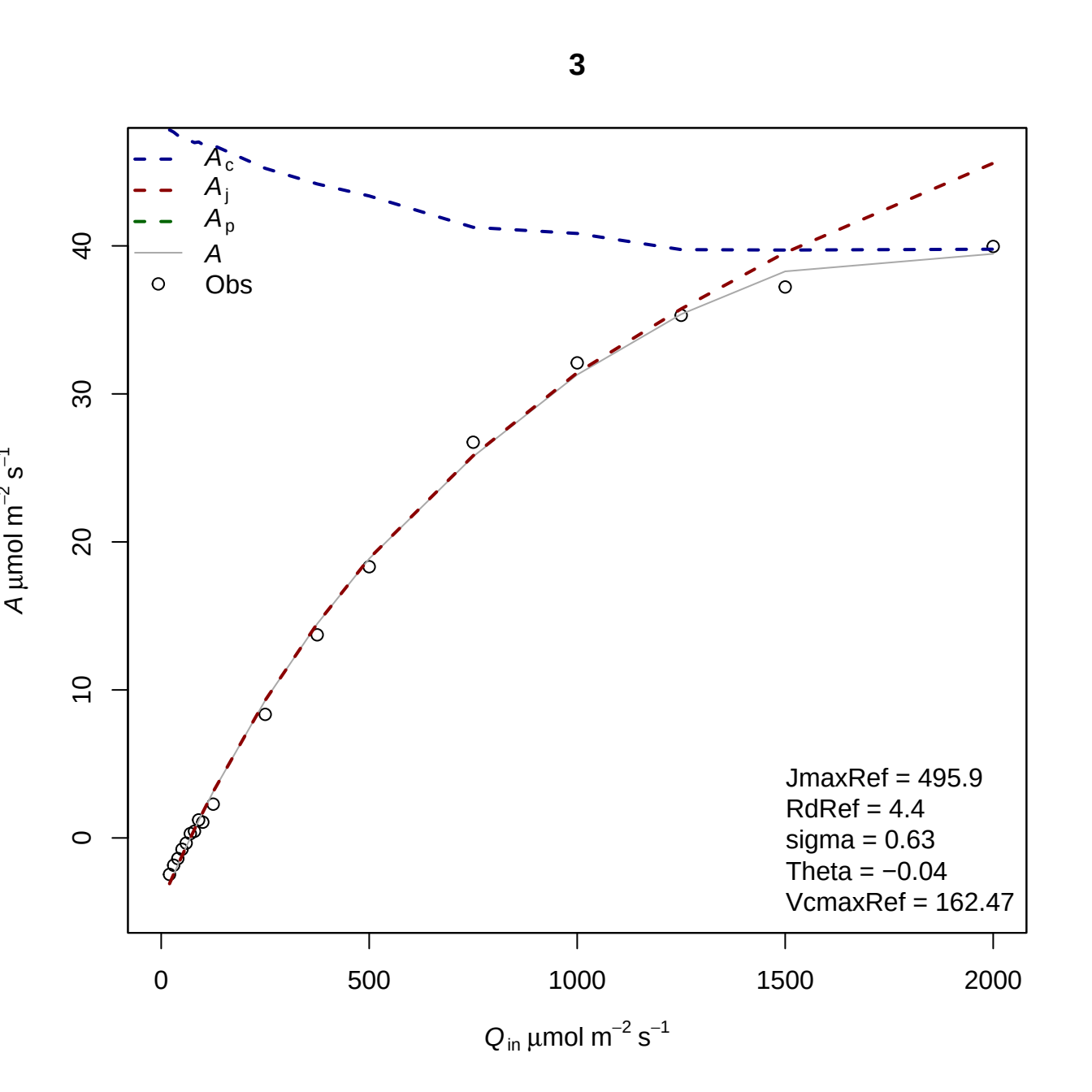
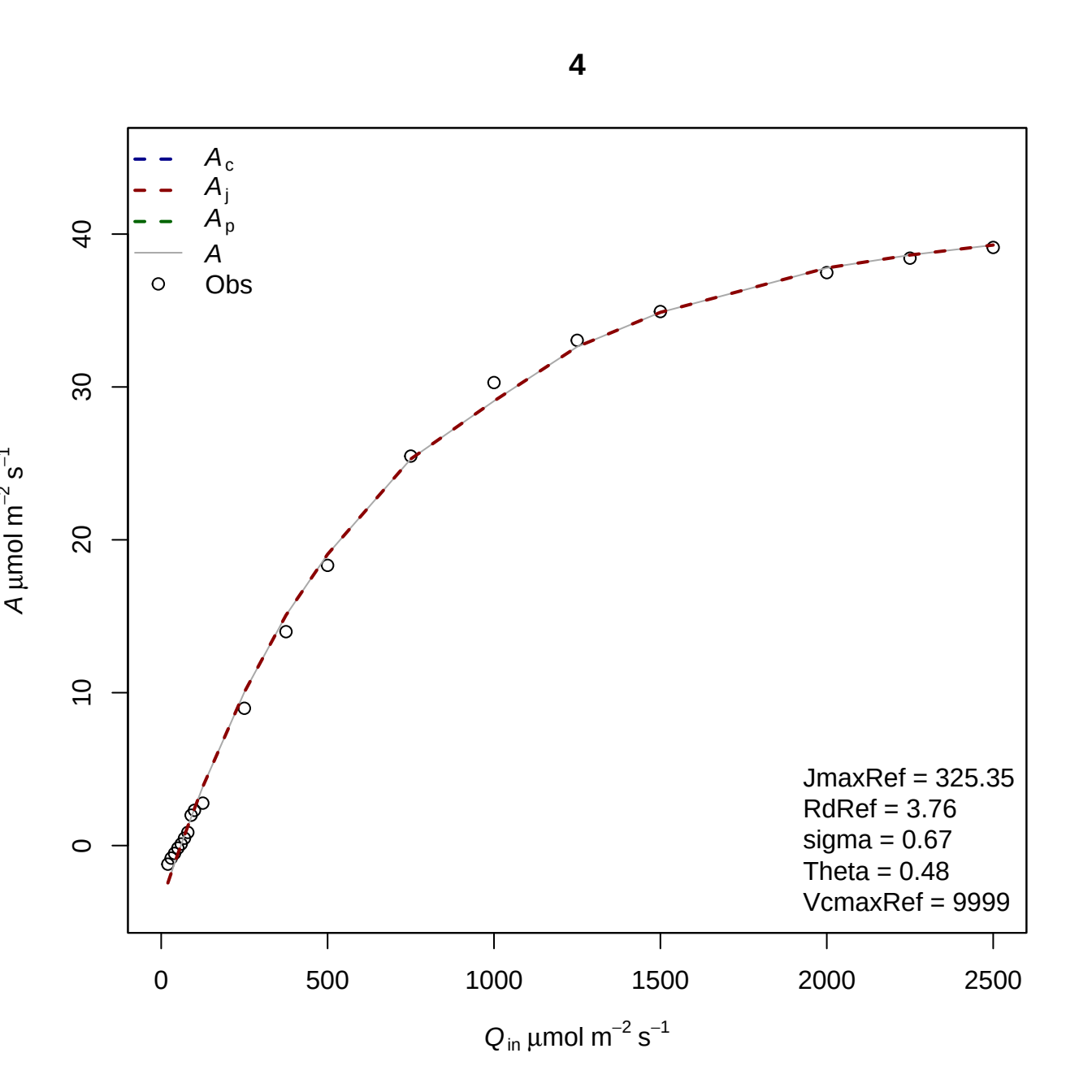
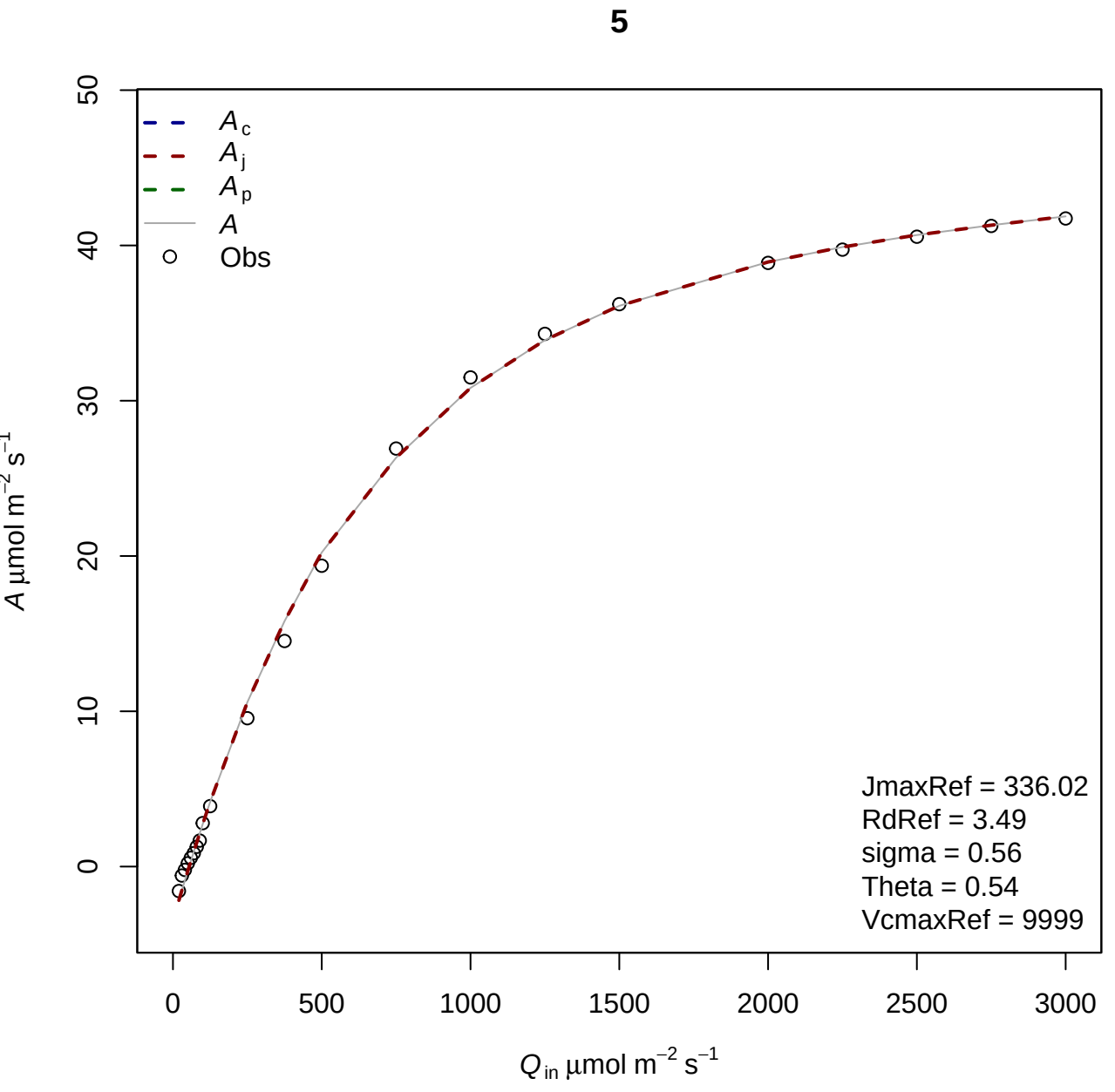
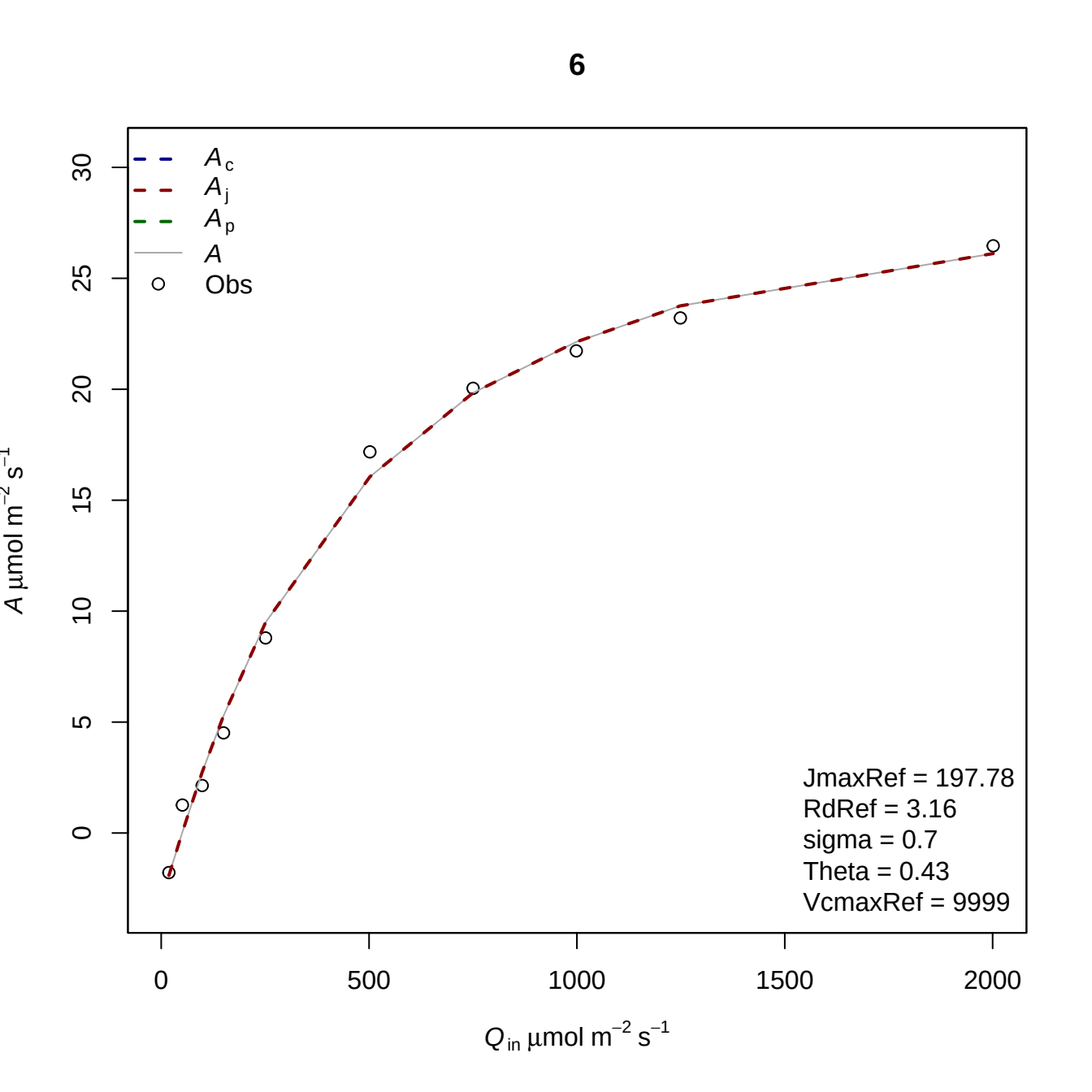

1
A_c
A_j
A_p
A
Obs
JmaxRef = 24.65
RdRef = 0.3
sigma = 0.1
Theta = −0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

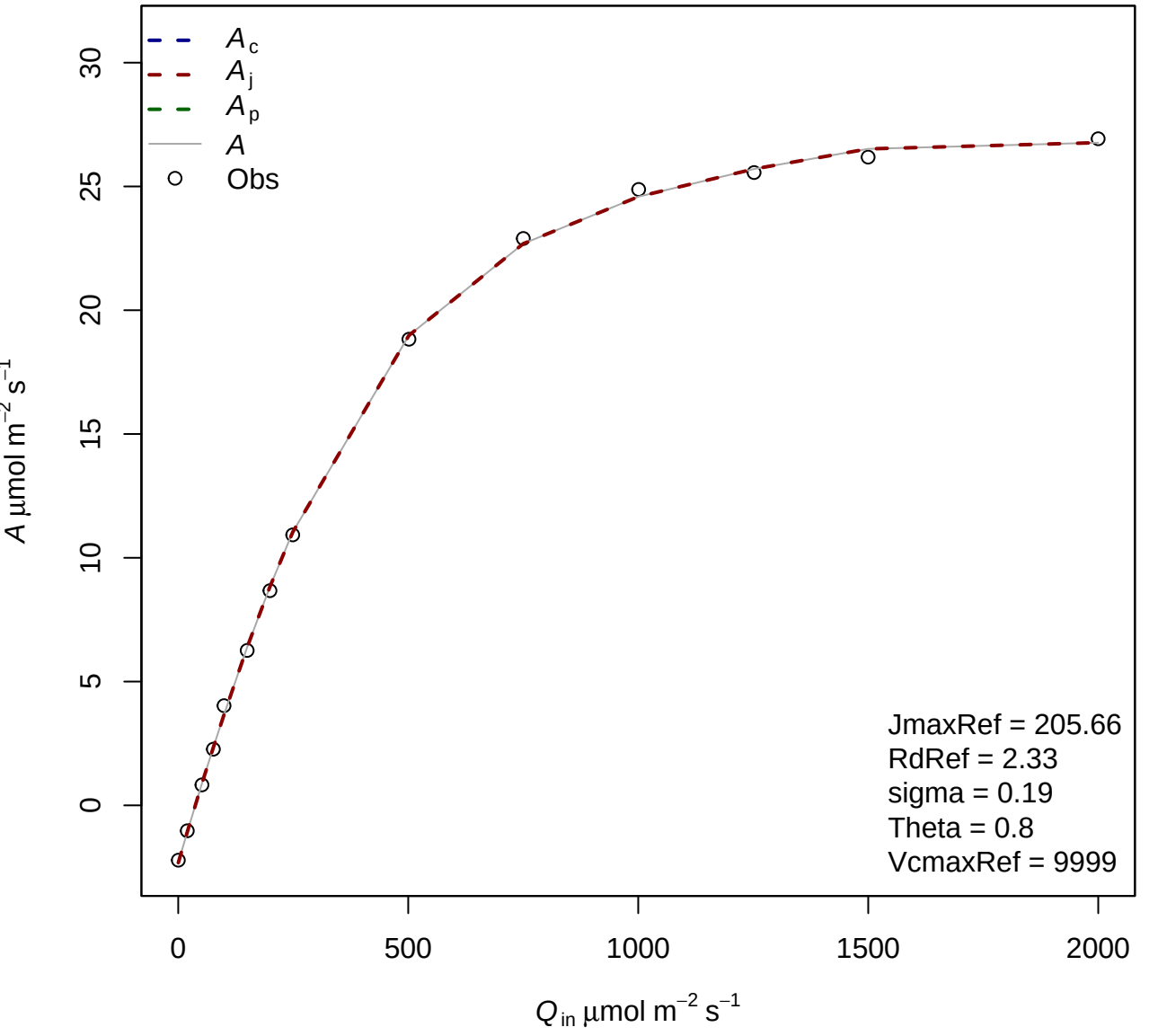

2
A_c
A_j
A_p
A
Obs
JmaxRef = 31.28
RdRef = 0.64
sigma = 0.21
Theta = 0.03
VcmaxRef = 21.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

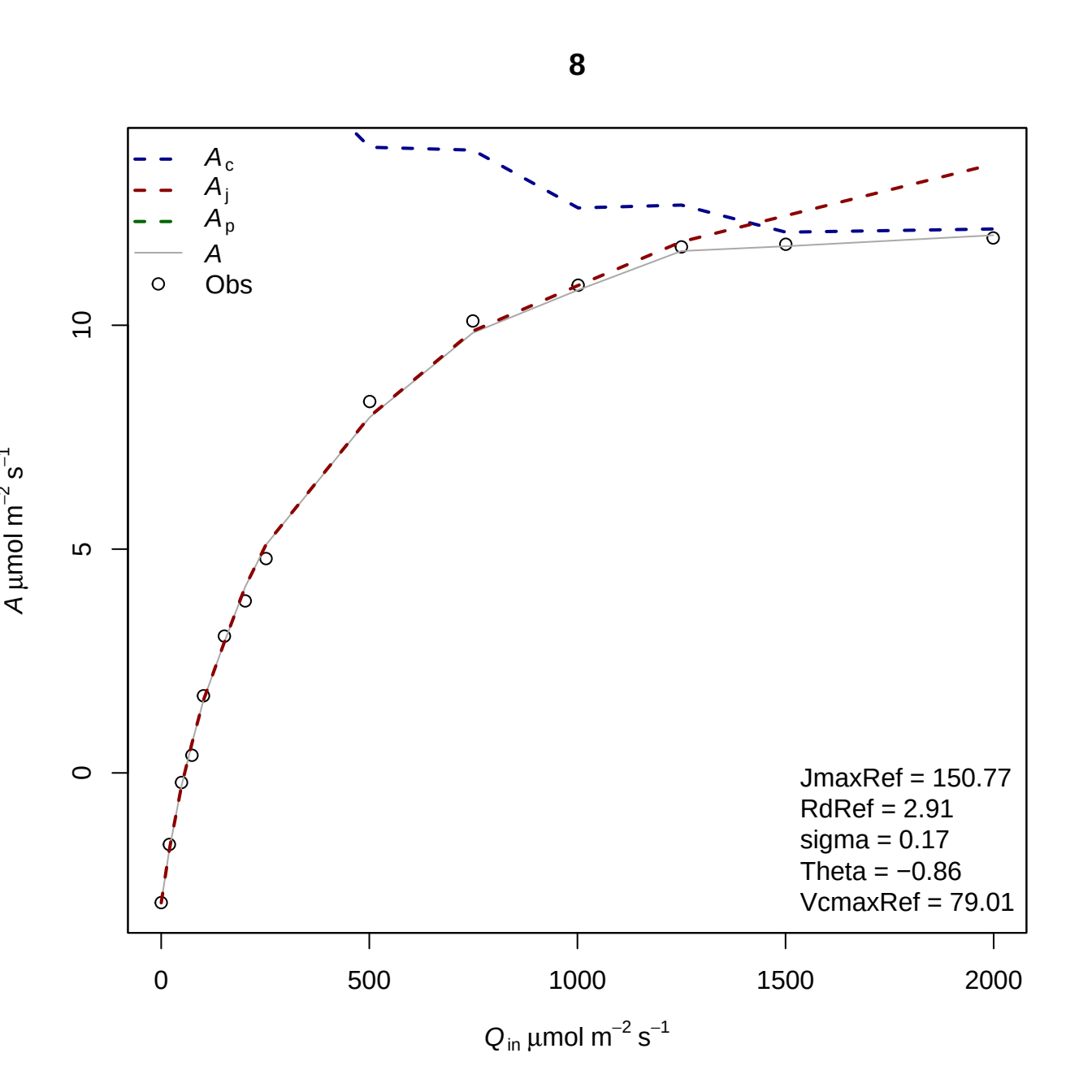

3
A_c
A_j
A_p
A
Obs
JmaxRef = 495.9
RdRef = 4.4
sigma = 0.63
Theta = −0.04
VcmaxRef = 162.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

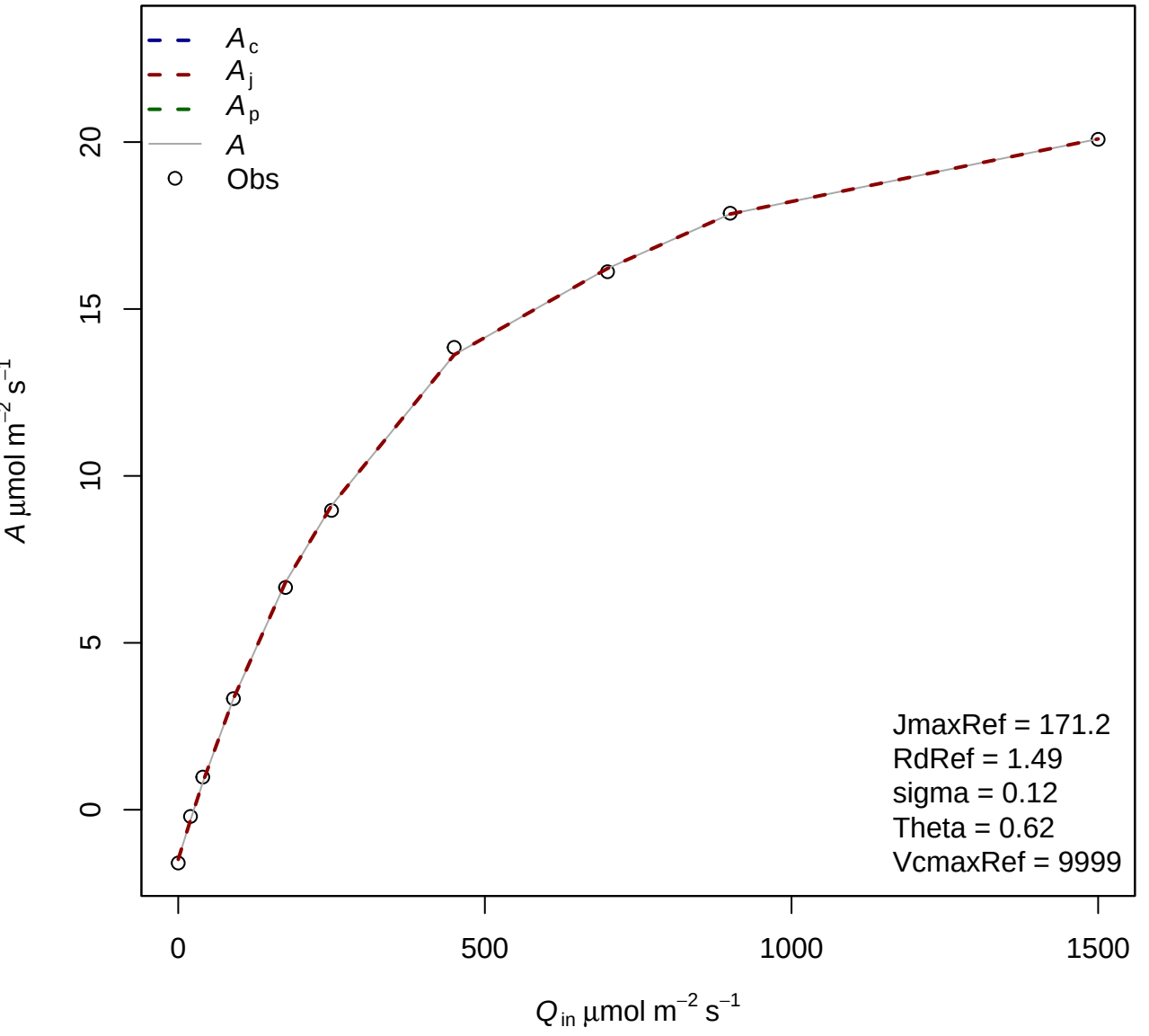

4
A_c
A_j
A_p
A
Obs
JmaxRef = 325.35
RdRef = 3.76
sigma = 0.67
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000
2500

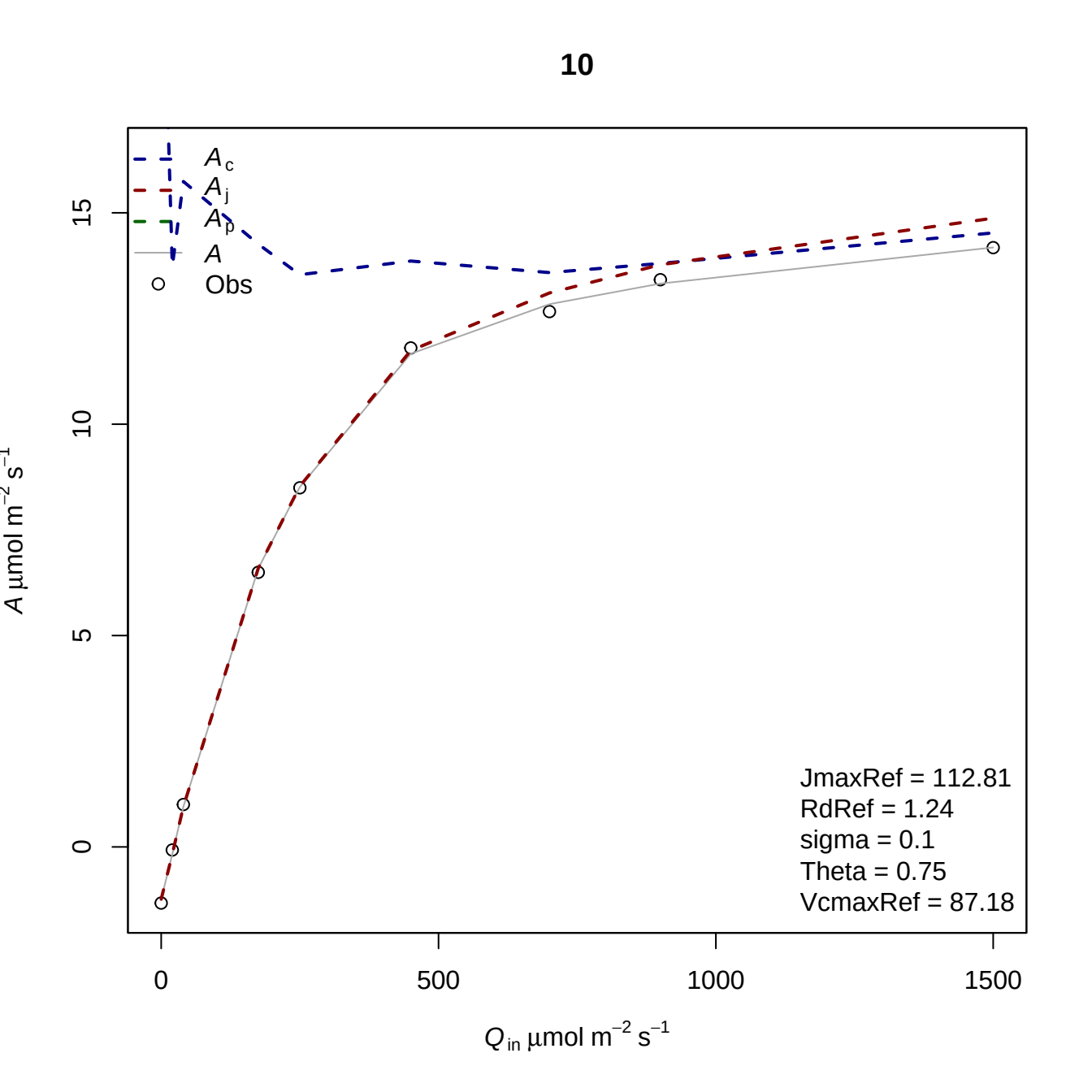

5
A_c
A_j
A_p
A
Obs
JmaxRef = 336.02
RdRef = 3.49
sigma = 0.56
Theta = 0.54
VcmaxRef = 9999
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

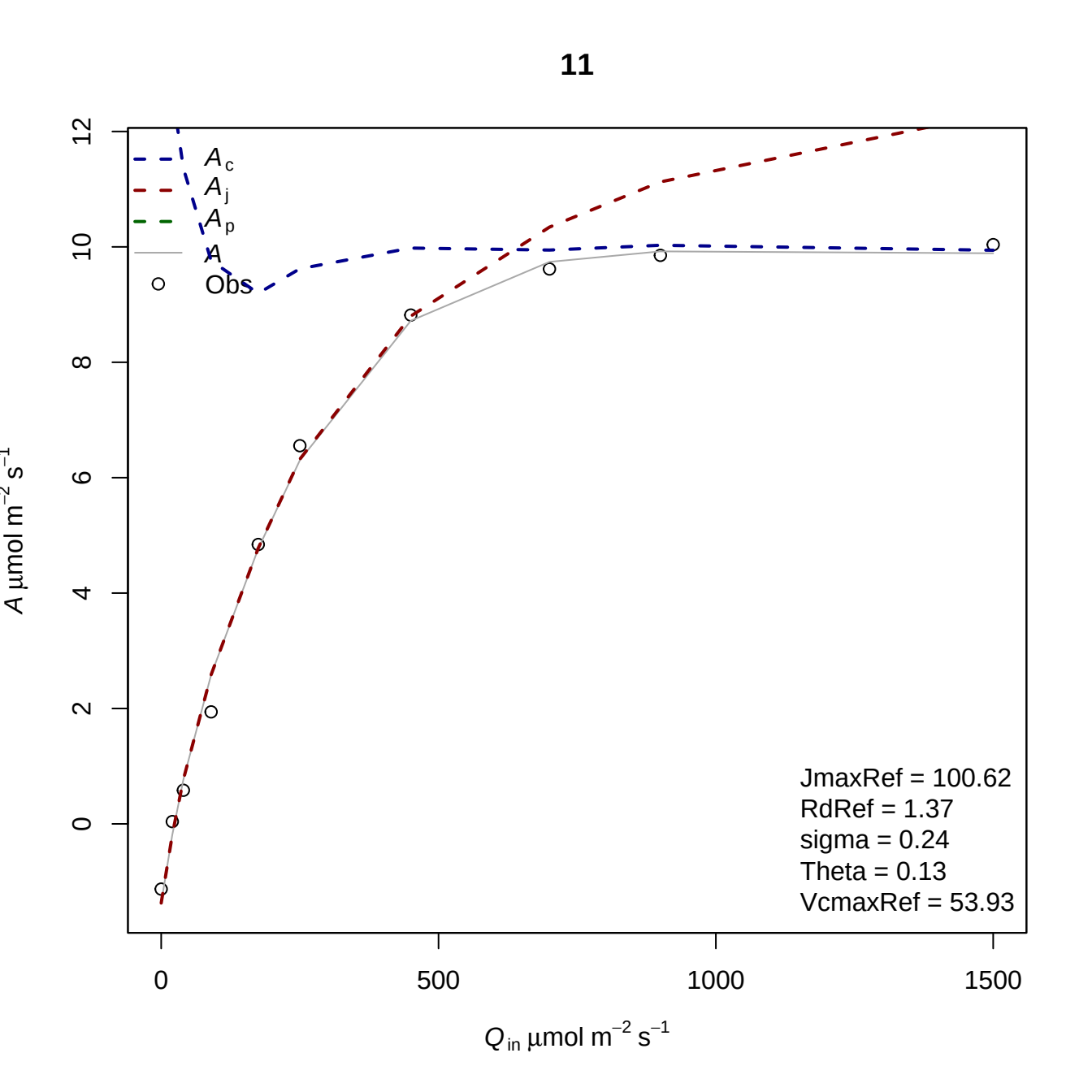

6
A_c
A_j
A_p
A
Obs
JmaxRef = 197.78
RdRef = 3.16
sigma = 0.7
Theta = 0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

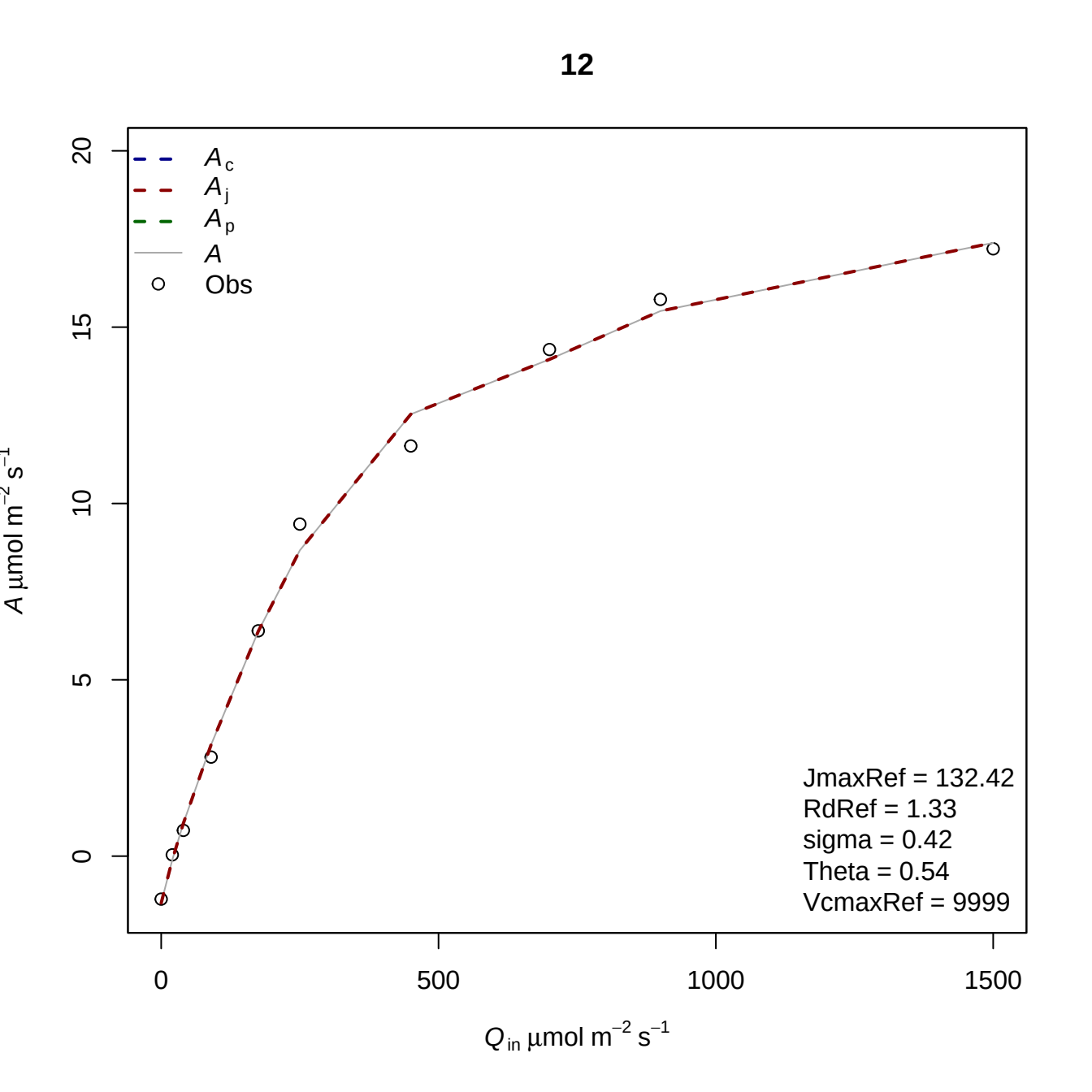

A_c
A_j
A_p
A
Obs
JmaxRef = 205.66
RdRef = 2.33
sigma = 0.19
Theta = 0.8
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

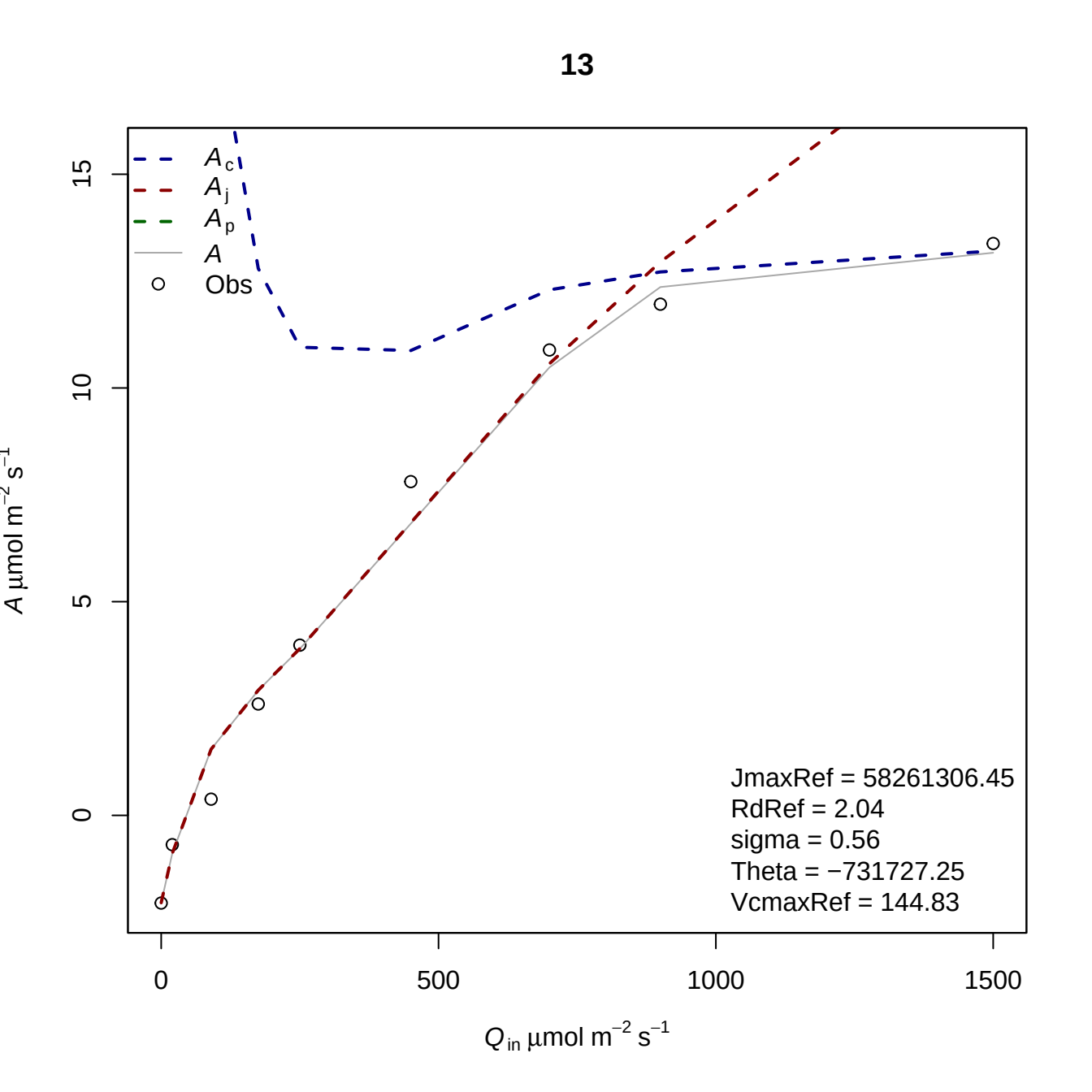

8
A_c
A_j
A_p
A
Obs
JmaxRef = 150.77
RdRef = 2.91
sigma = 0.17
Theta = −0.86
VcmaxRef = 79.01
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

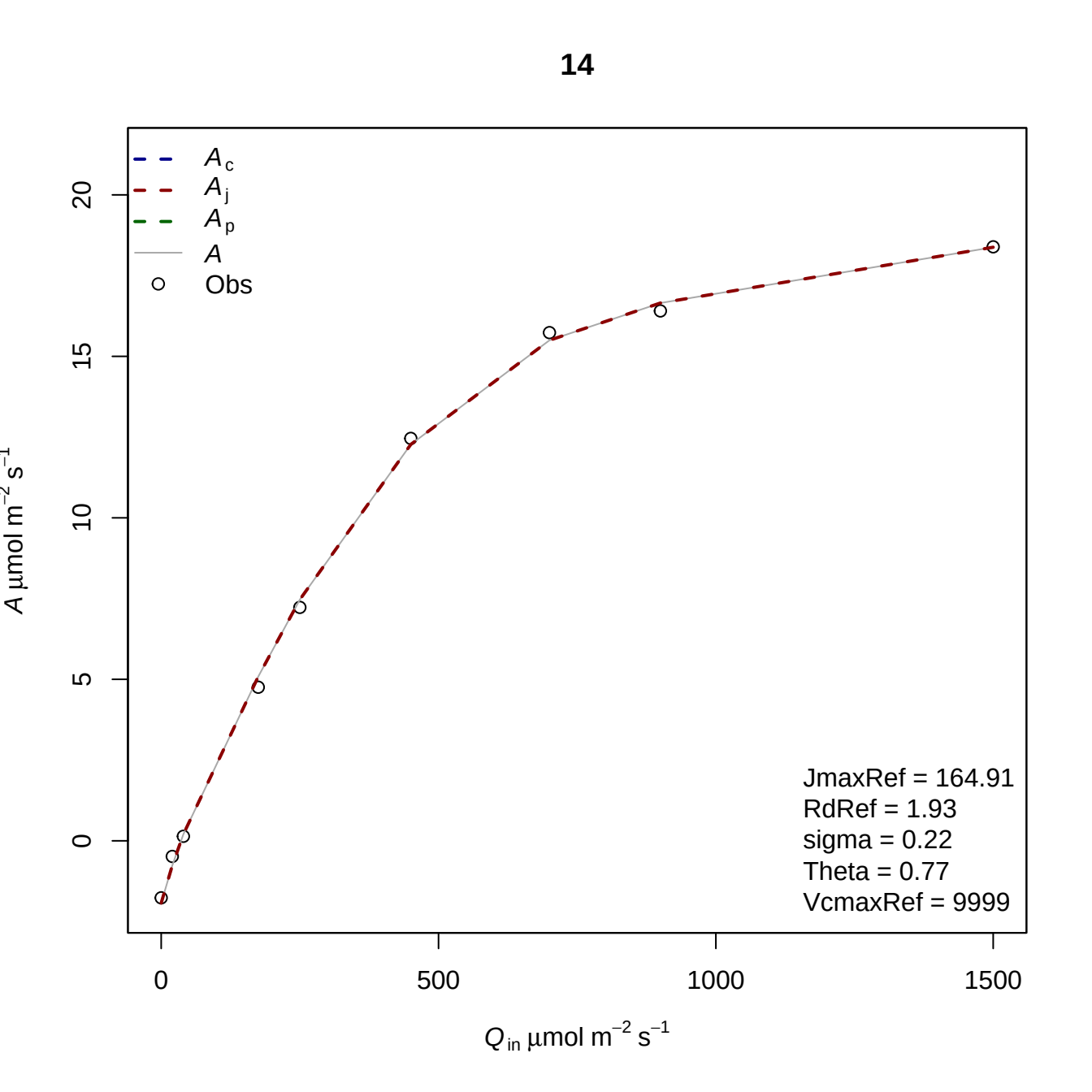

A_c
A_j
A_p
A
Obs
JmaxRef = 171.2
RdRef = 1.49
sigma = 0.12
Theta = 0.62
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

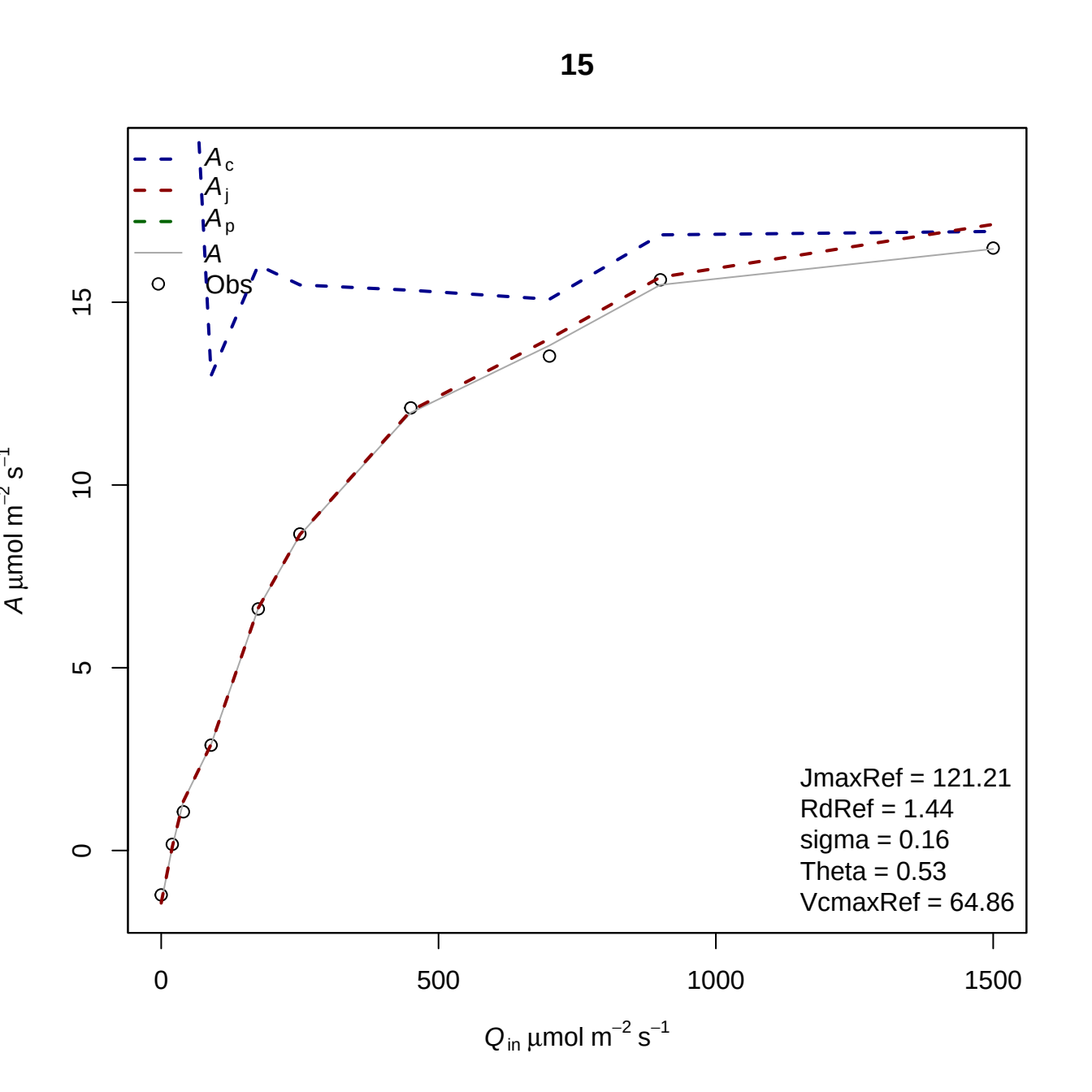

10
A_c
A_j
A_p
A
Obs
JmaxRef = 112.81
RdRef = 1.24
sigma = 0.1
Theta = 0.75
VcmaxRef = 87.18
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

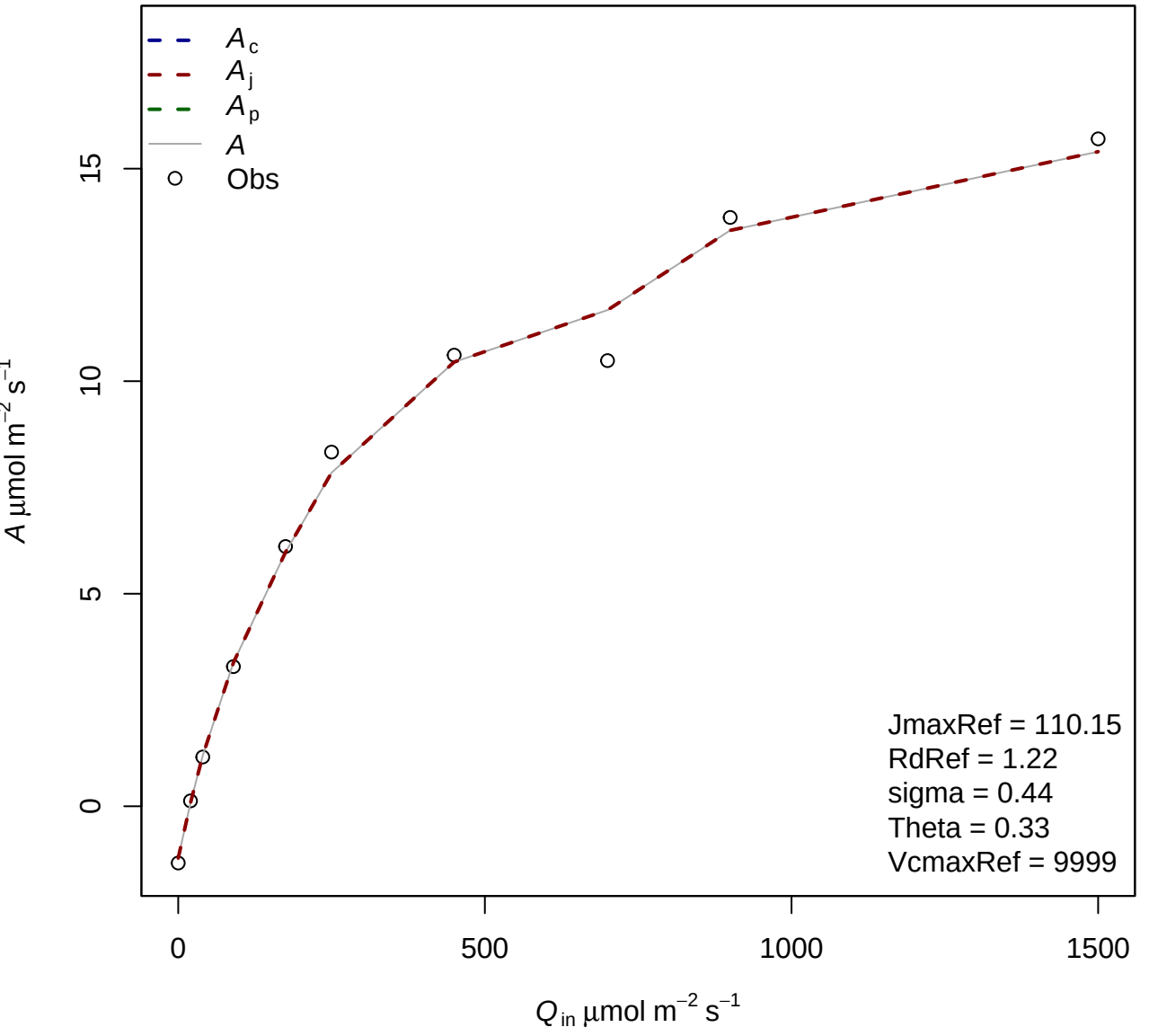

11
A_c
A_j
A_p
A
Obs
JmaxRef = 100.62
RdRef = 1.37
sigma = 0.24
Theta = 0.13
VcmaxRef = 53.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

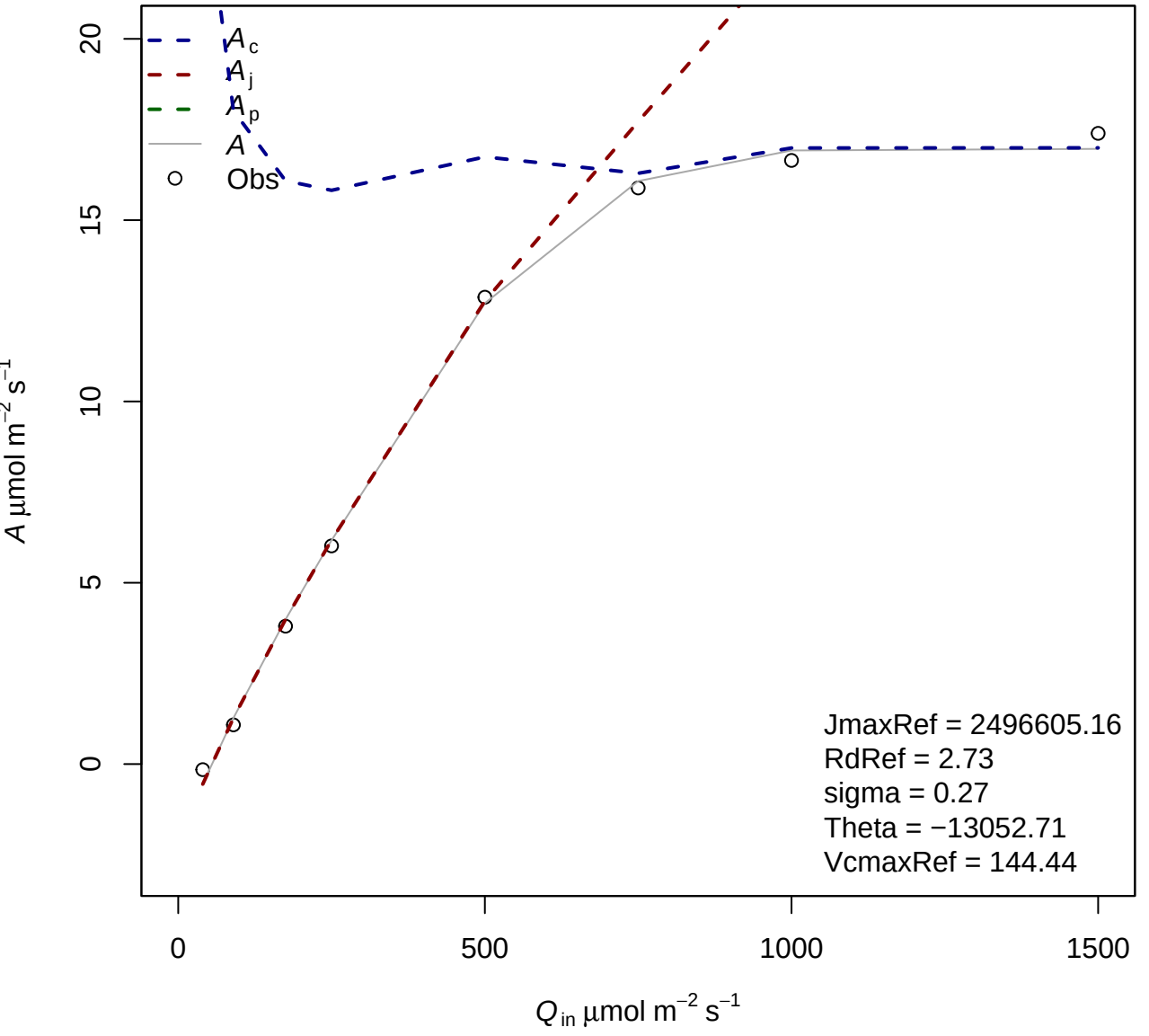

12
A_c
A_j
A_p
A
Obs
JmaxRef = 132.42
RdRef = 1.33
sigma = 0.42
Theta = 0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

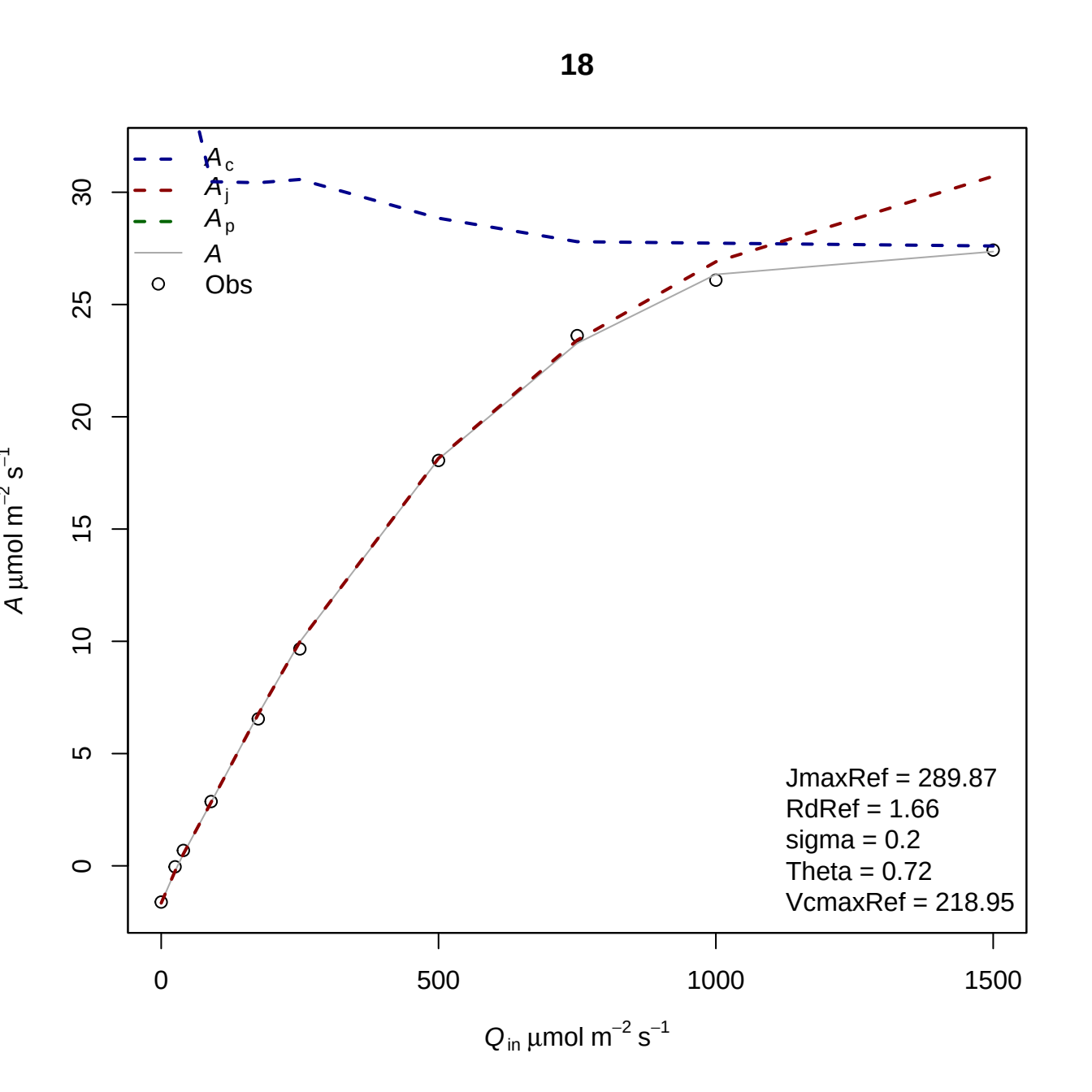

13
A_c
A_j
A_p
A
Obs
JmaxRef = 58261306.45
RdRef = 2.04
sigma = 0.56
Theta = −731727.25
VcmaxRef = 144.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

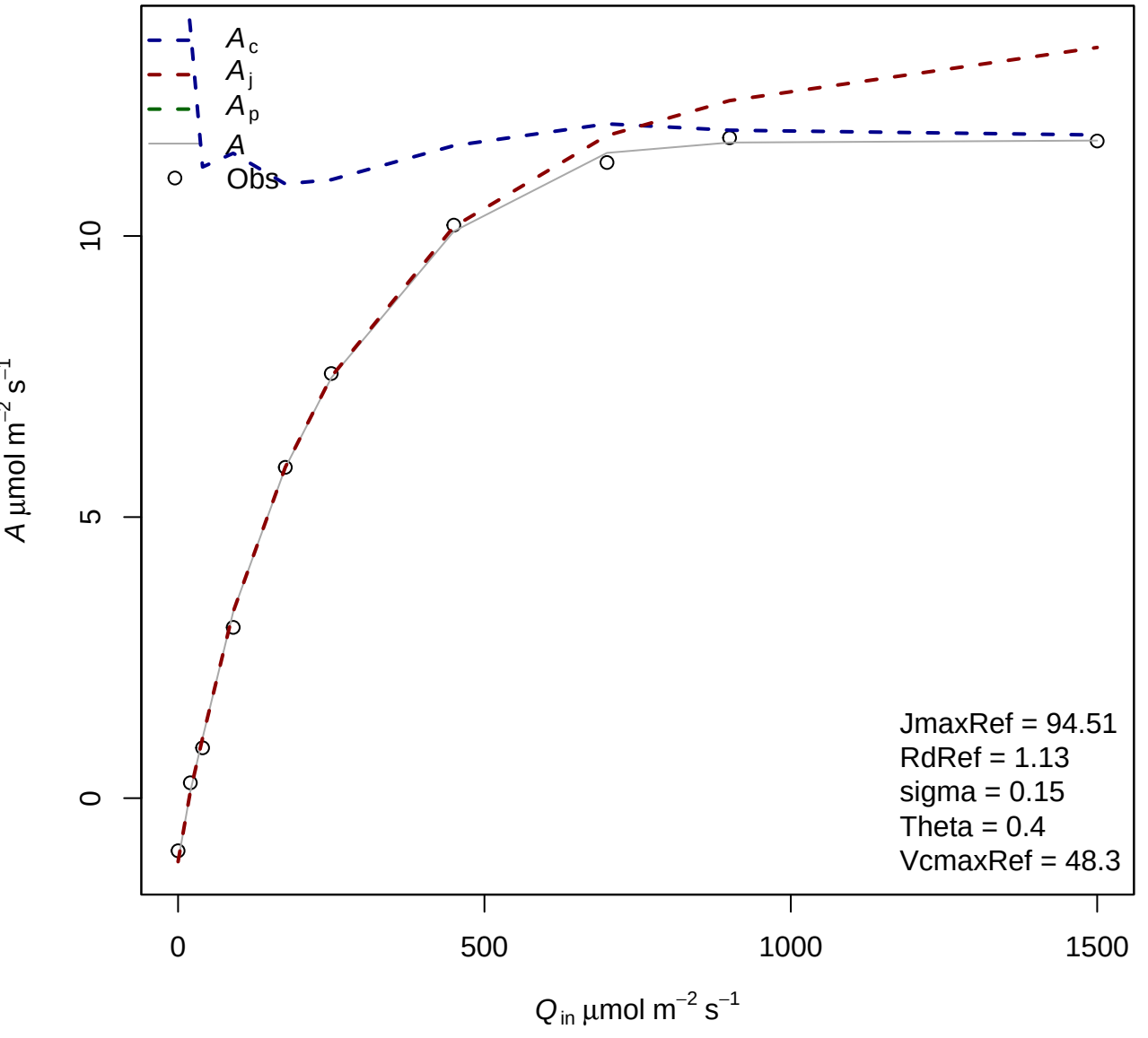

14
A_c
A_j
A_p
A
Obs
JmaxRef = 164.91
RdRef = 1.93
sigma = 0.22
Theta = 0.77
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

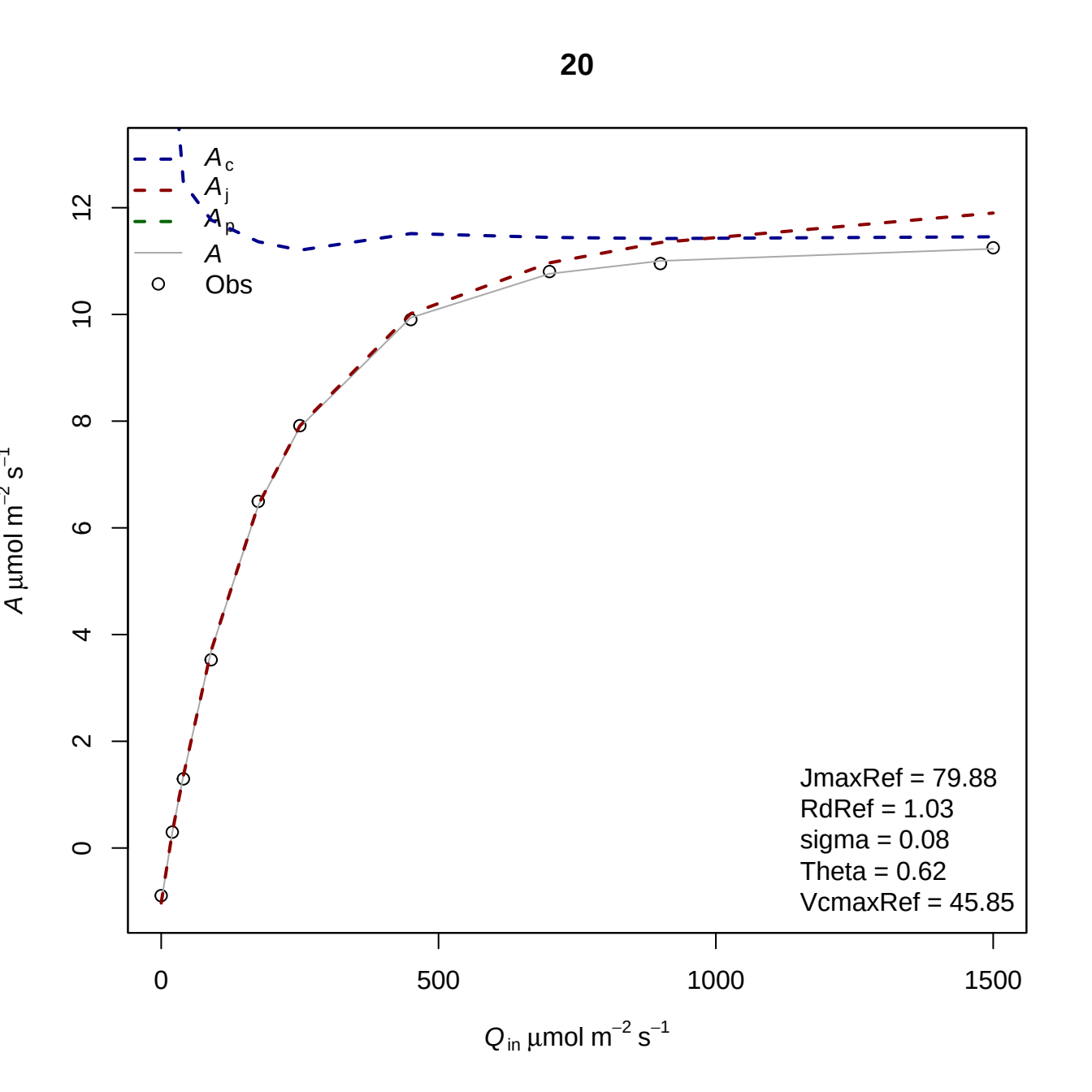

15
A_c
A_j
A_p
A
Obs
JmaxRef = 121.21
RdRef = 1.44
sigma = 0.16
Theta = 0.53
VcmaxRef = 64.86
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

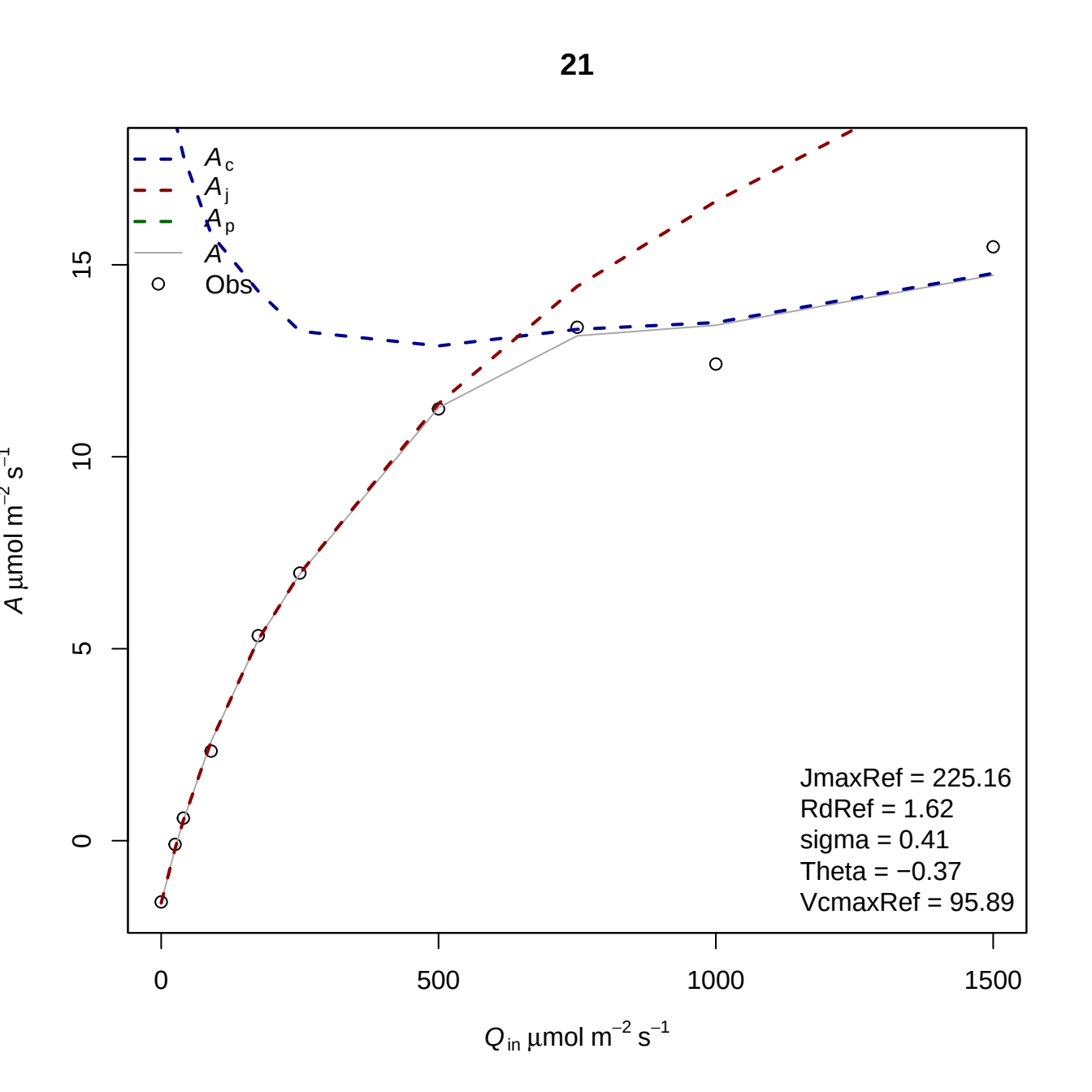

A_c
A_j
A_p
A
Obs
JmaxRef = 110.15
RdRef = 1.22
sigma = 0.44
Theta = 0.33
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

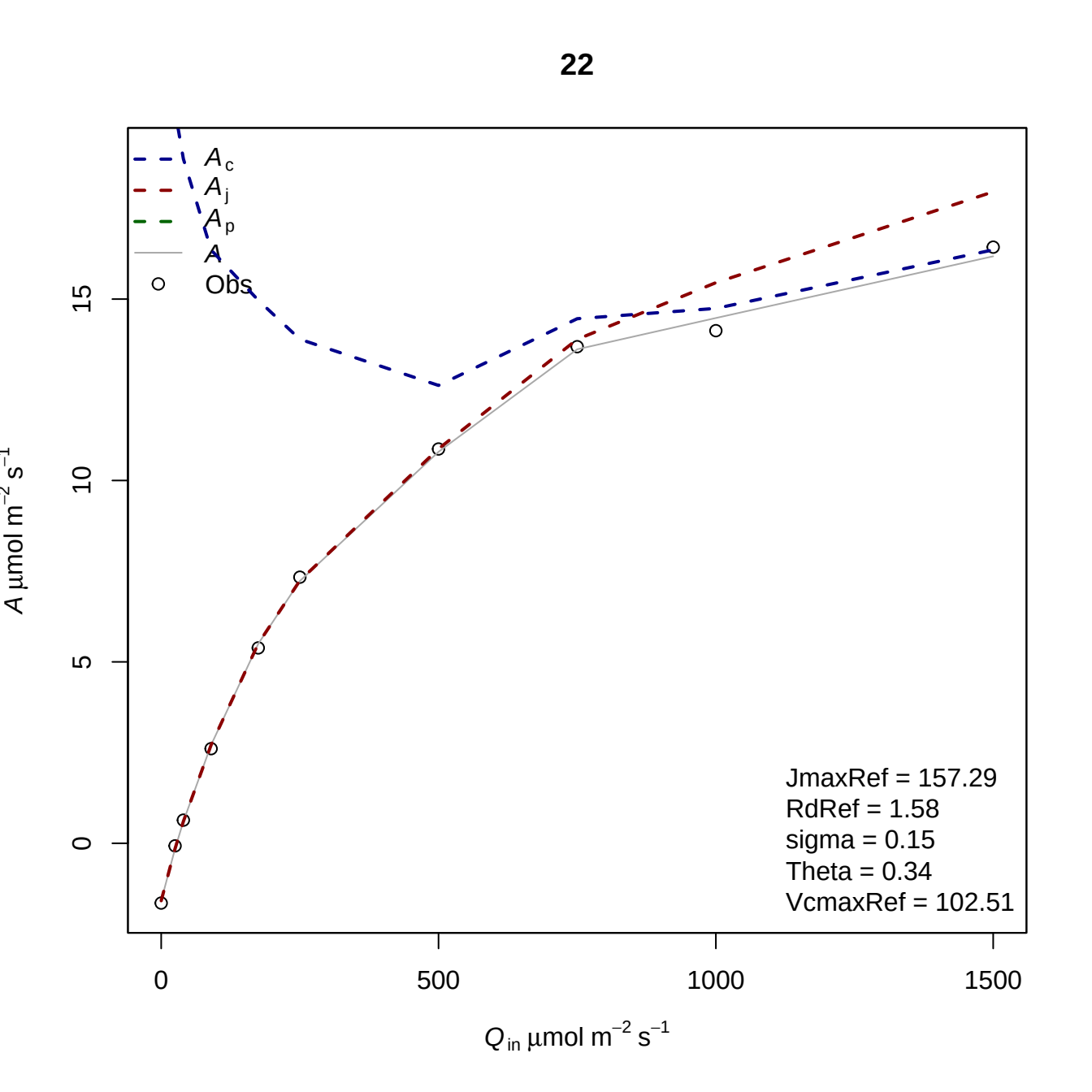

A_c
A_j
A_p
A
Obs
JmaxRef = 2496605.16
RdRef = 2.73
sigma = 0.27
Theta = −13052.71
VcmaxRef = 144.44
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

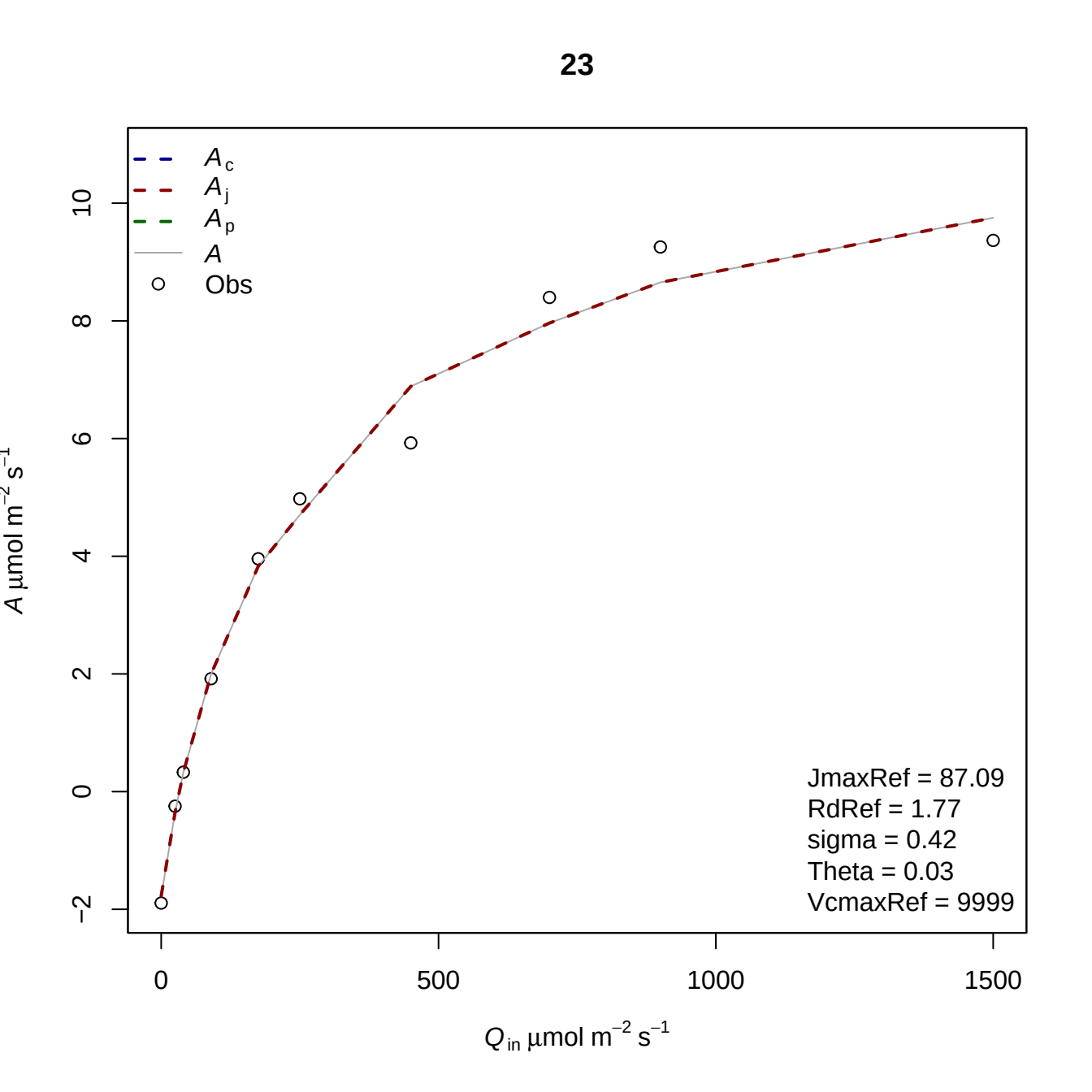

18
A_c
A_j
A_p
A
Obs
JmaxRef = 289.87
RdRef = 1.66
sigma = 0.2
Theta = 0.72
VcmaxRef = 218.95
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

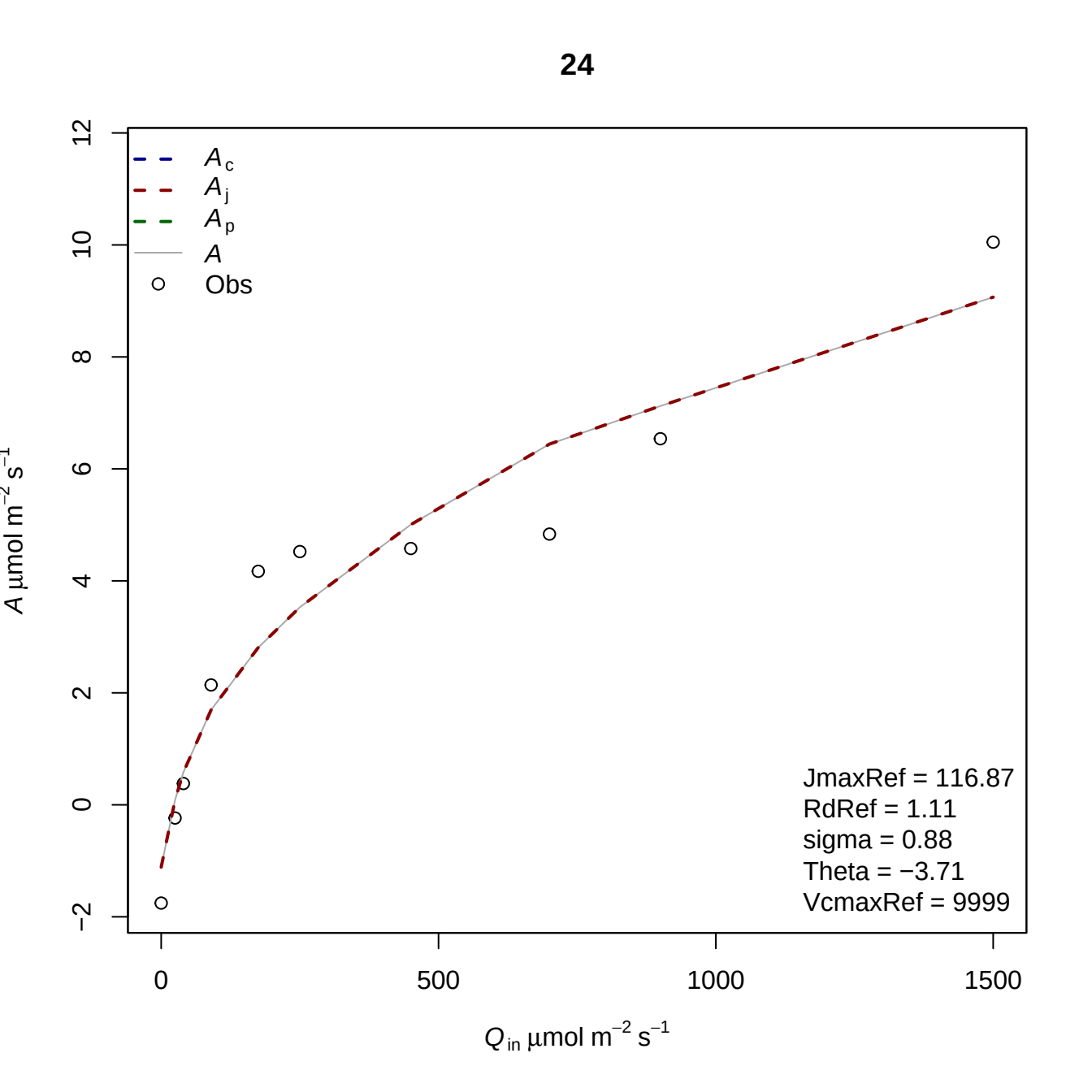

A_c
A_j
A_p
A
Obs
JmaxRef = 94.51
RdRef = 1.13
sigma = 0.15
Theta = 0.4
VcmaxRef = 48.3
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

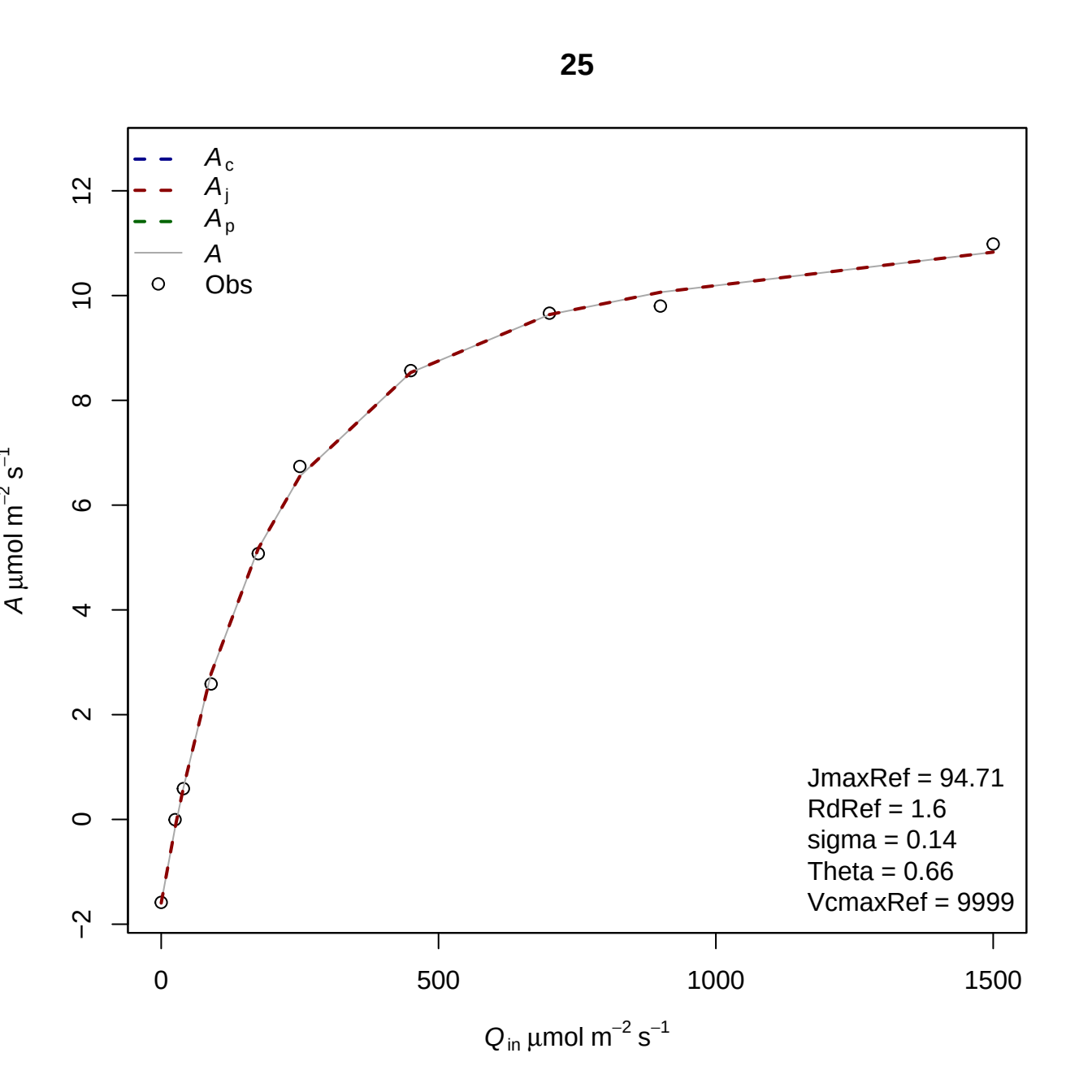

20
A_c
A_j
A_p
A
Obs
JmaxRef = 79.88
RdRef = 1.03
sigma = 0.08
Theta = 0.62
VcmaxRef = 45.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

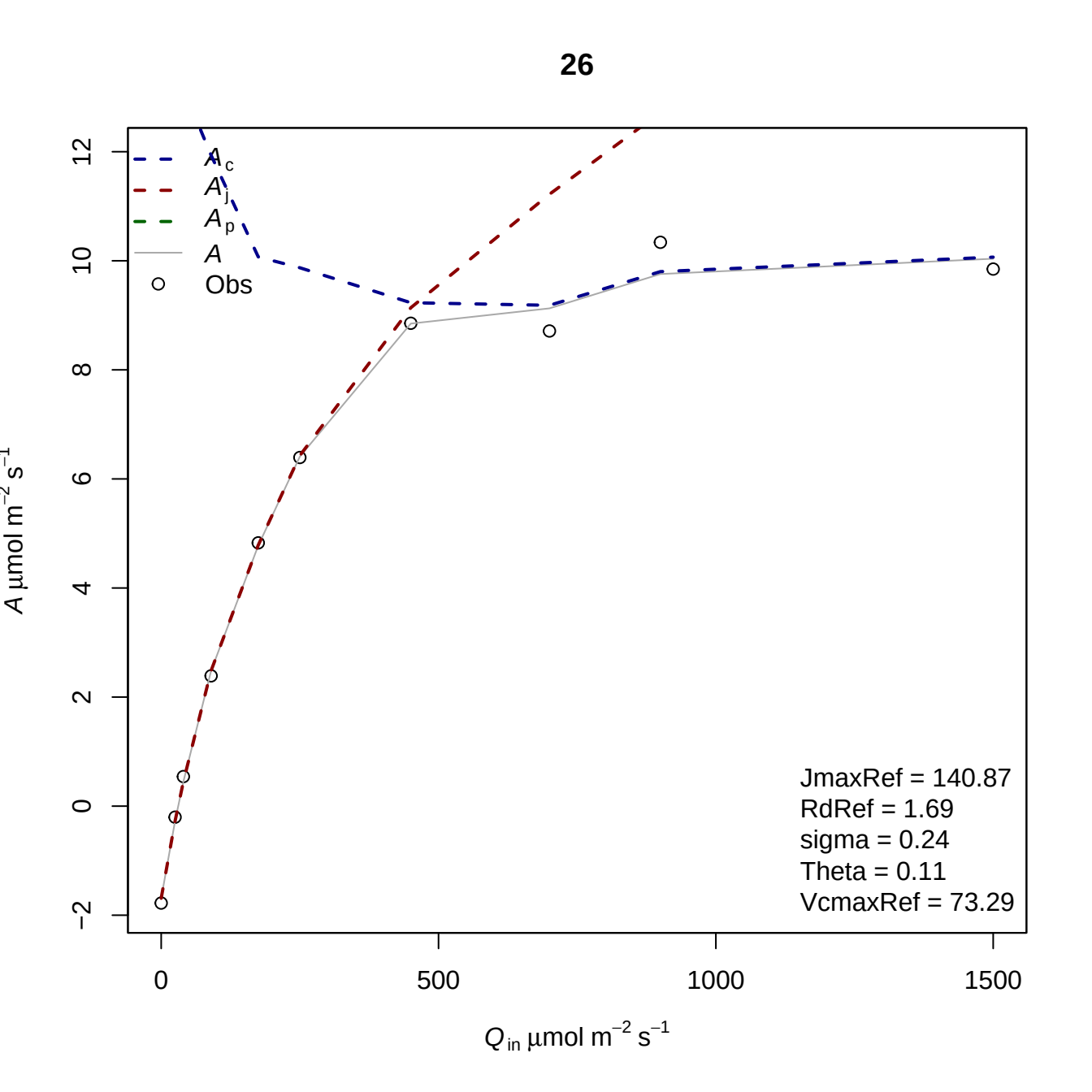

21
A_c
A_j
A_p
A
Obs
JmaxRef = 225.16
RdRef = 1.62
sigma = 0.41
Theta = −0.37
VcmaxRef = 95.89
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

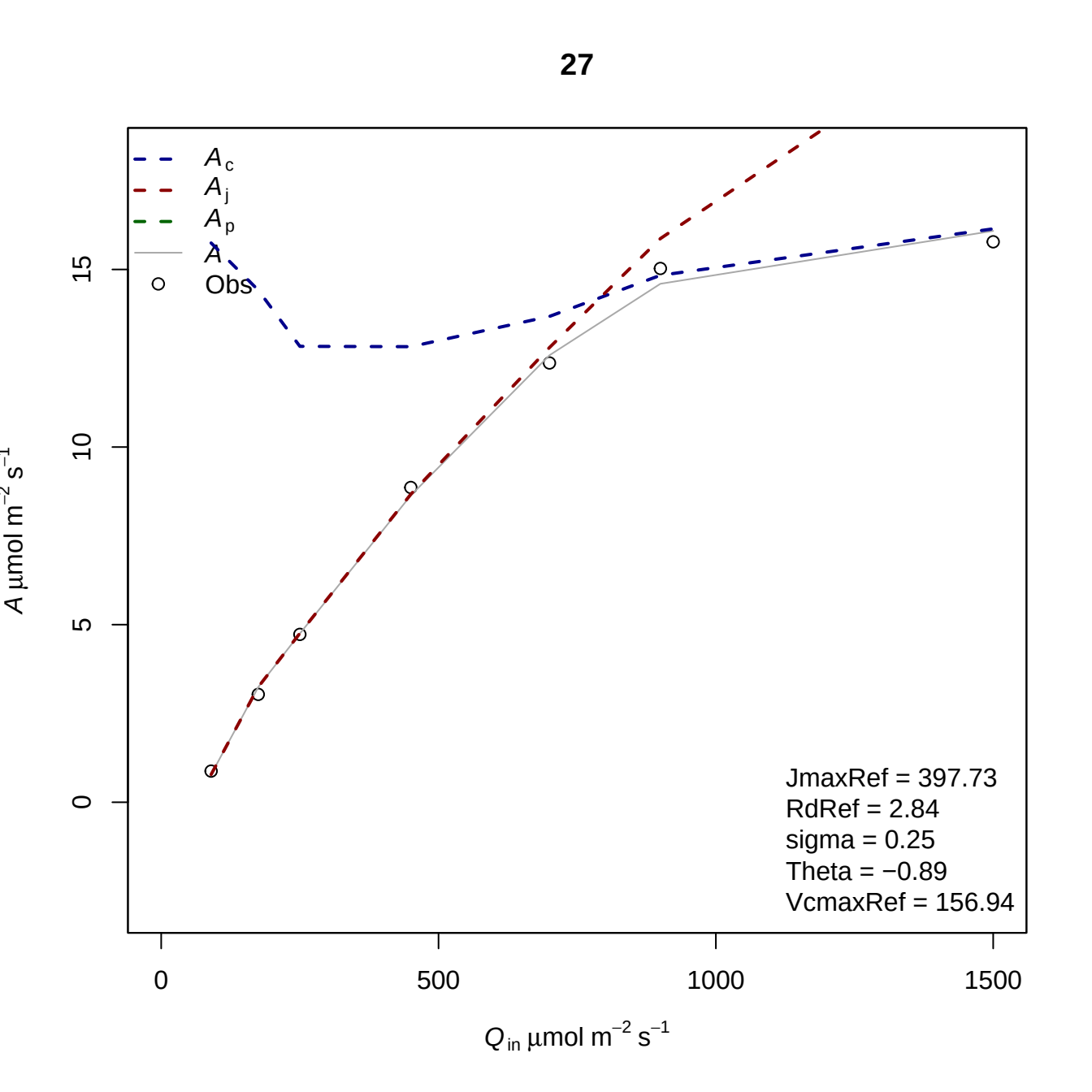

22
A_c
A_j
A_p
A
Obs
JmaxRef = 157.29
RdRef = 1.58
sigma = 0.15
Theta = 0.34
VcmaxRef = 102.51
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

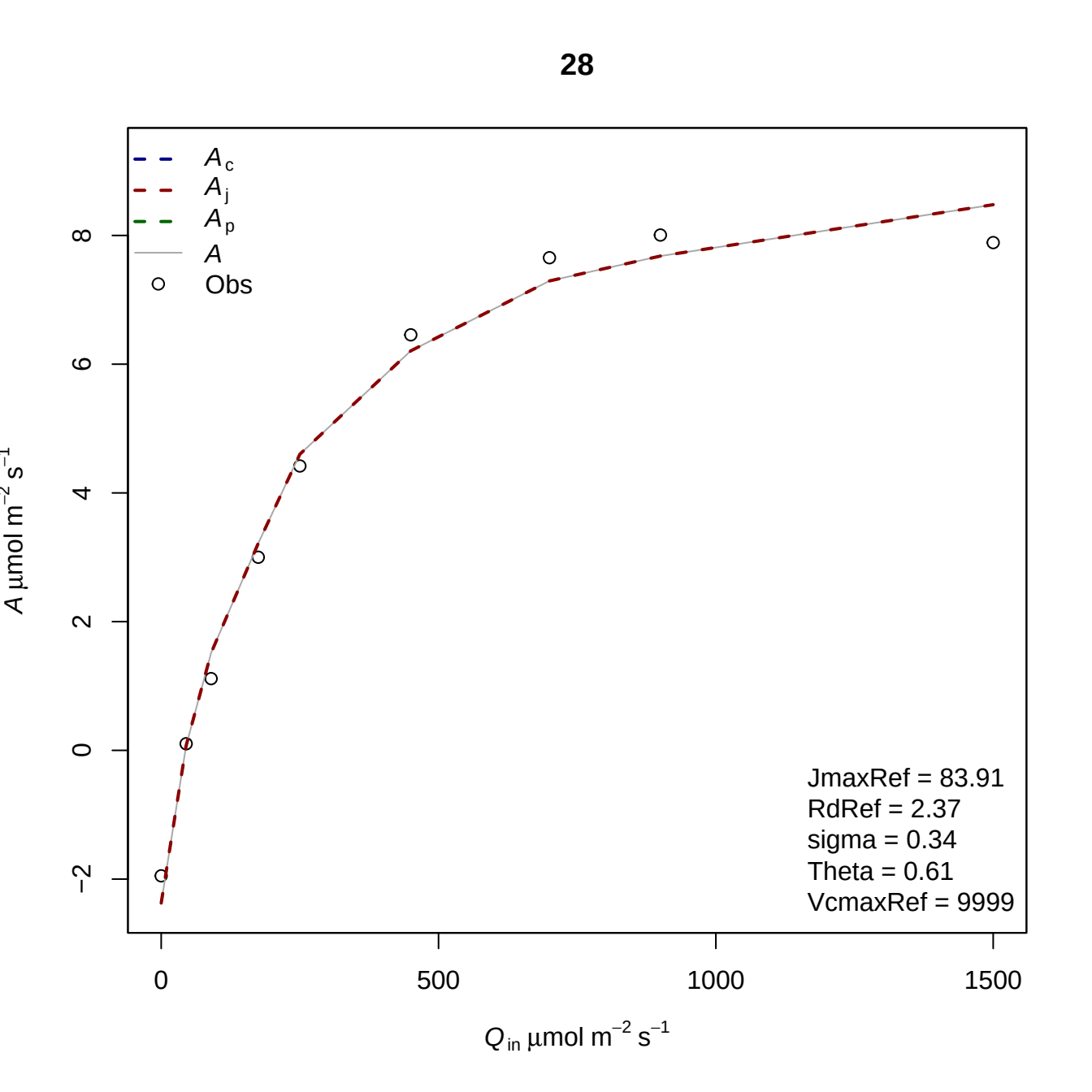

23
A_c
A_j
A_p
A
Obs
JmaxRef = 87.09
RdRef = 1.77
sigma = 0.42
Theta = 0.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

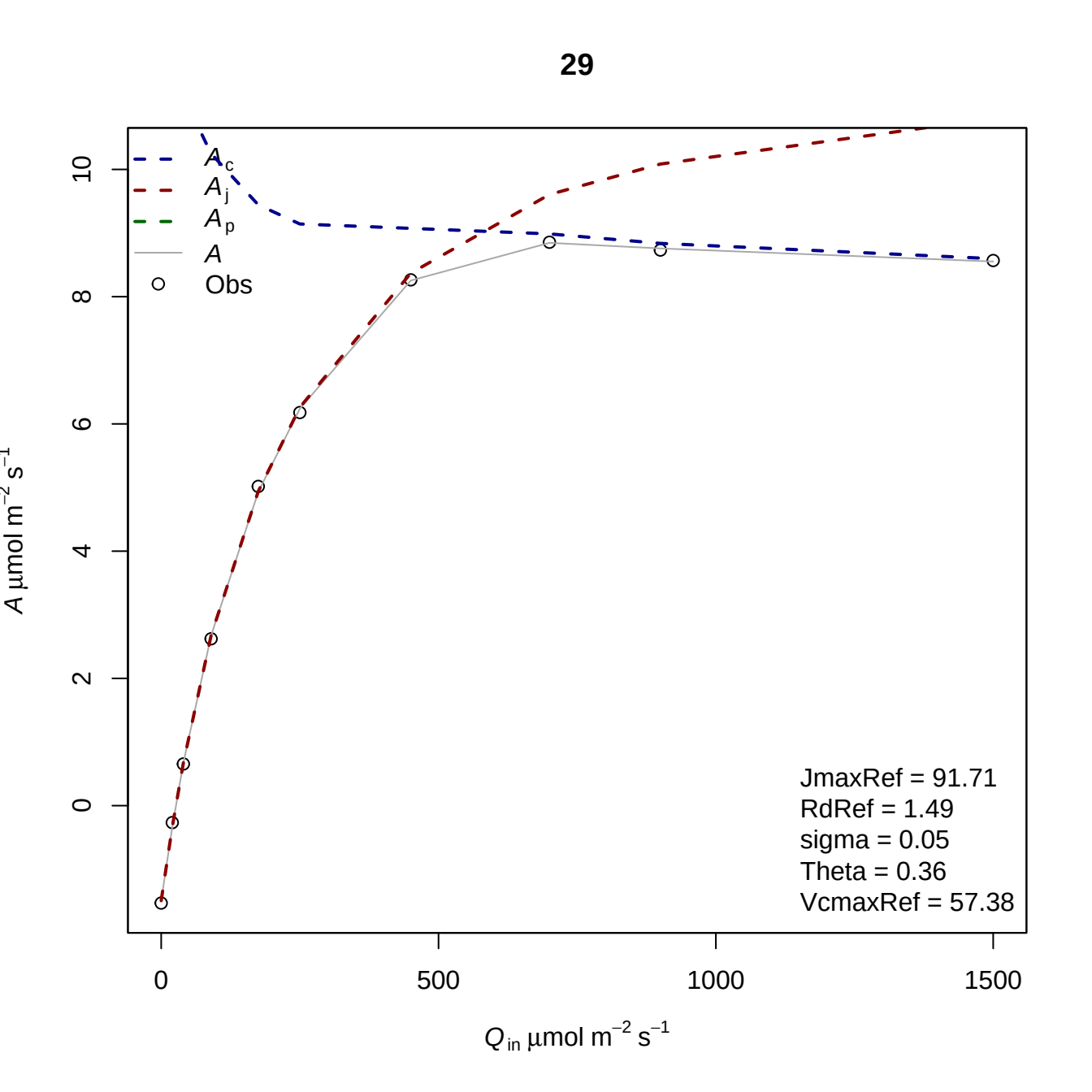

24
A_c
A_j
A_p
A
Obs
JmaxRef = 116.87
RdRef = 1.11
sigma = 0.88
Theta = −3.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

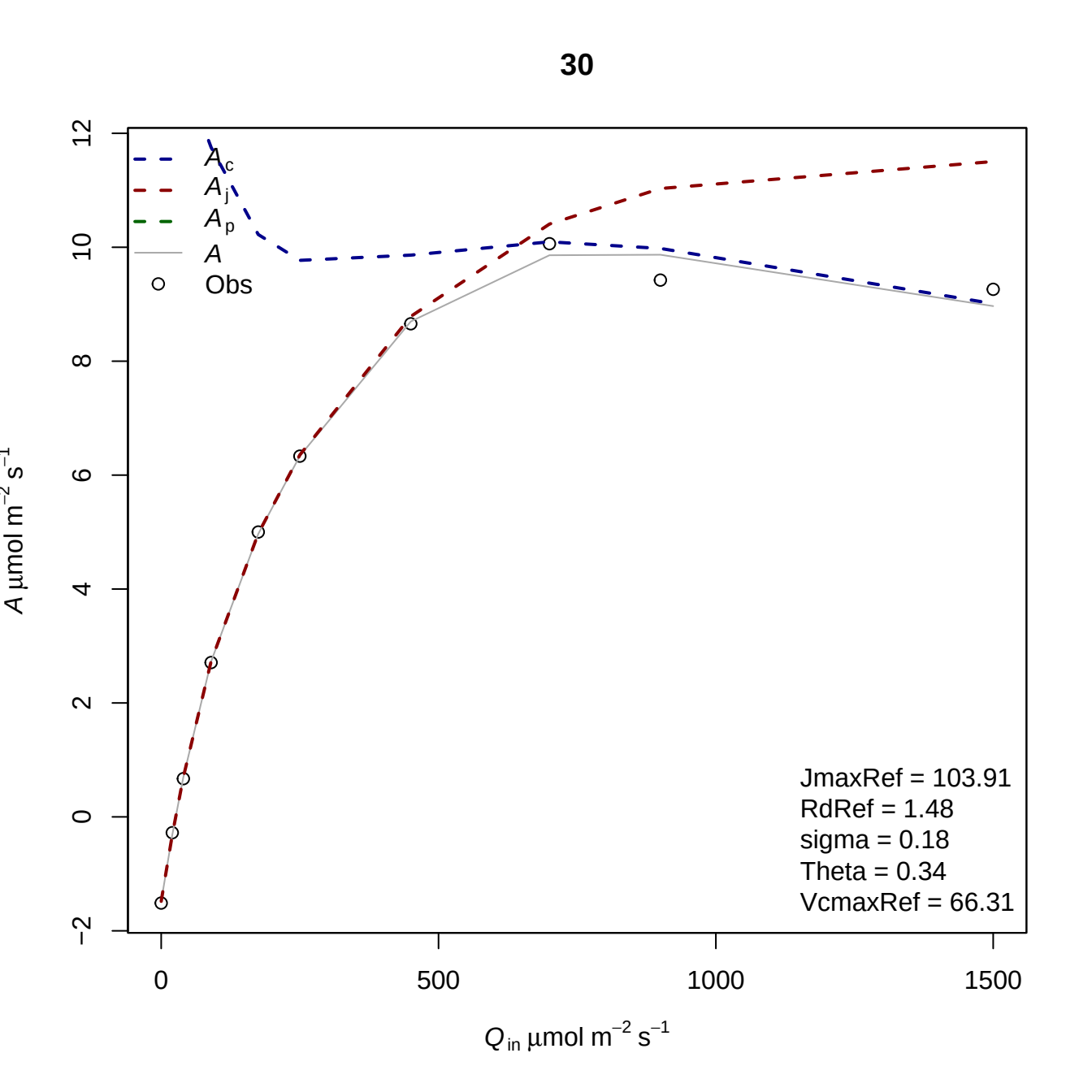

25
A_c
A_j
A_p
A
Obs
JmaxRef = 94.71
RdRef = 1.6
sigma = 0.14
Theta = 0.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

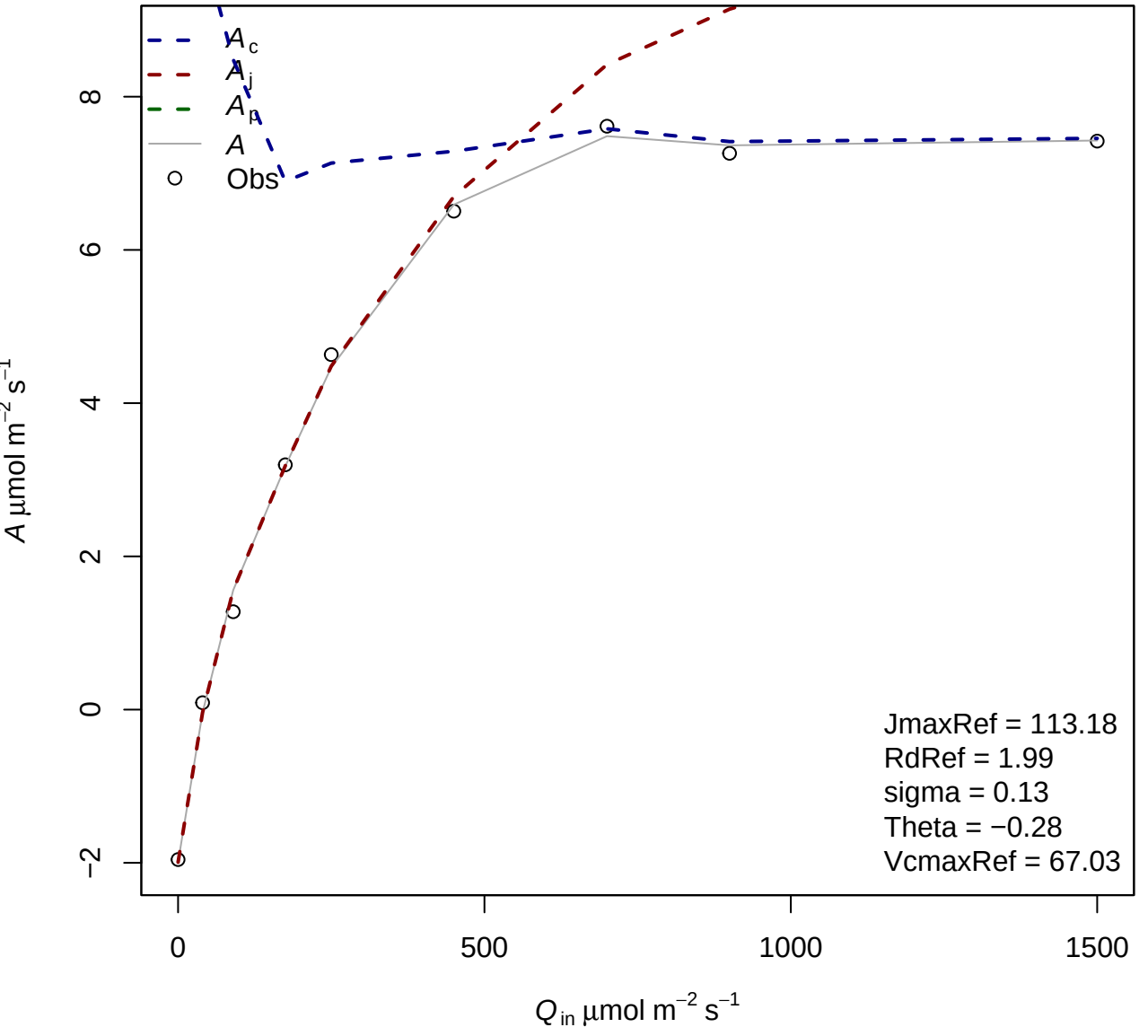

26
A_c
A_j
A_p
A
Obs
JmaxRef = 140.87
RdRef = 1.69
sigma = 0.24
Theta = 0.11
VcmaxRef = 73.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

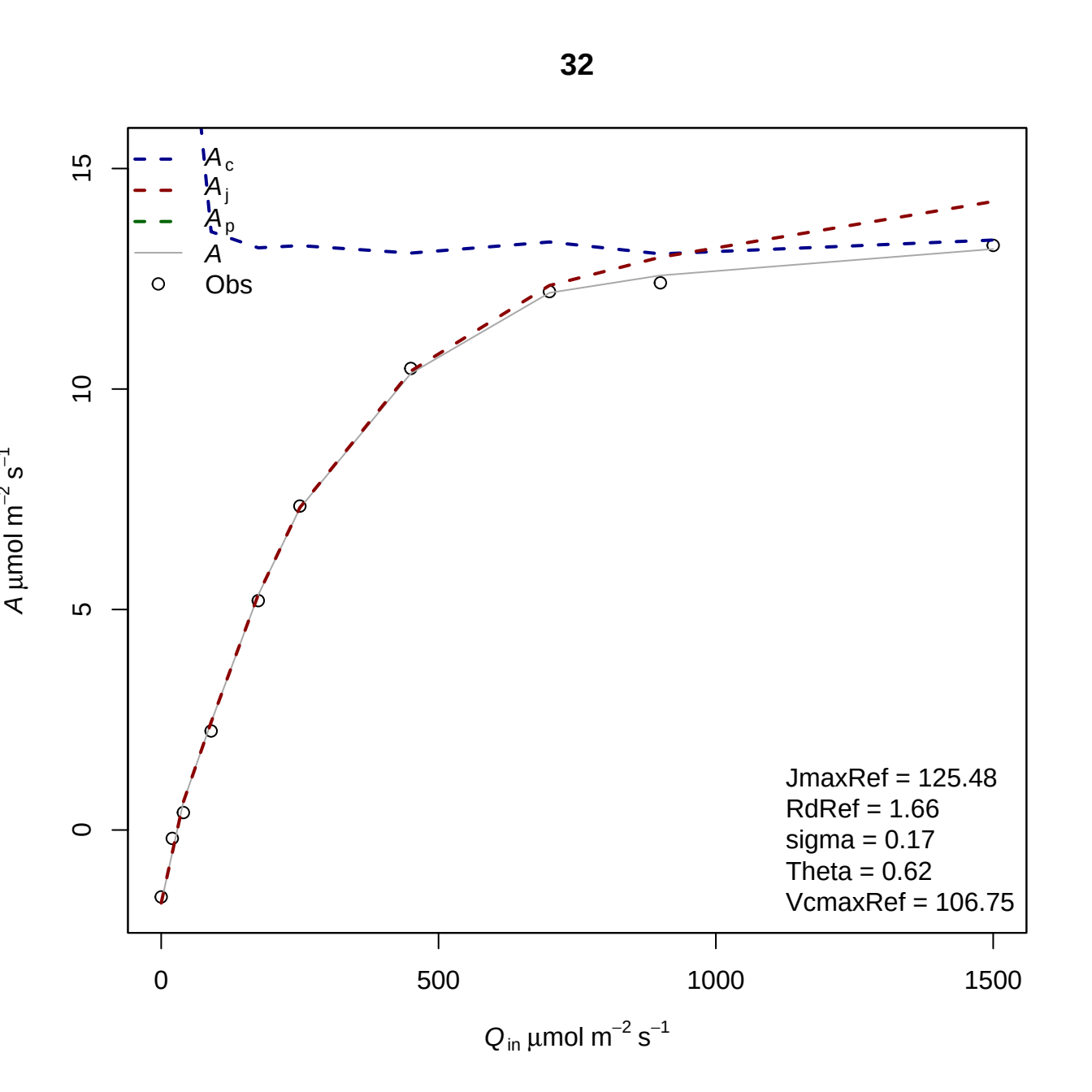

27
A_c
A_j
A_p
A
Obs
JmaxRef = 397.73
RdRef = 2.84
sigma = 0.25
Theta = −0.89
VcmaxRef = 156.94
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

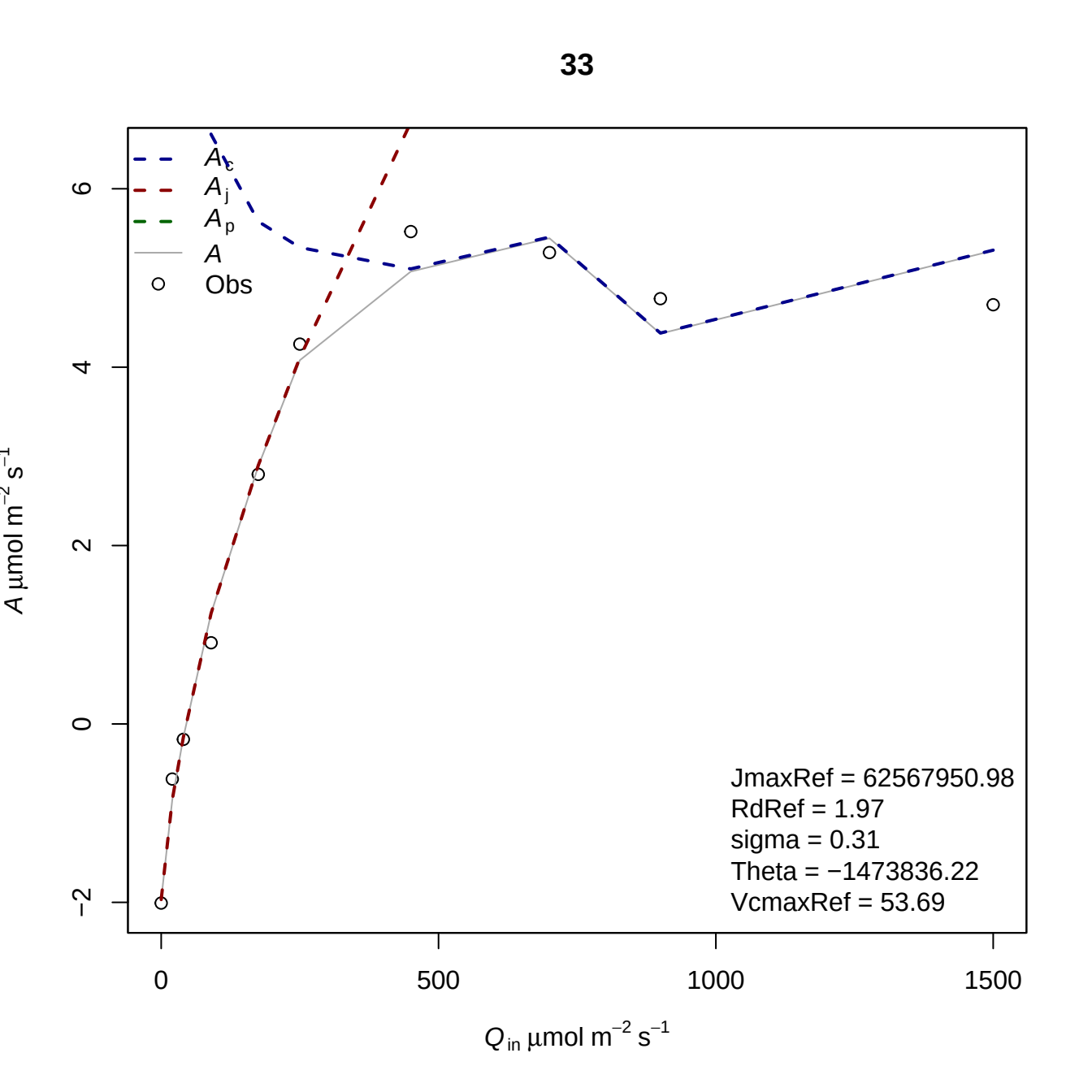

28
A_c
A_j
A_p
A
Obs
JmaxRef = 83.91
RdRef = 2.37
sigma = 0.34
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

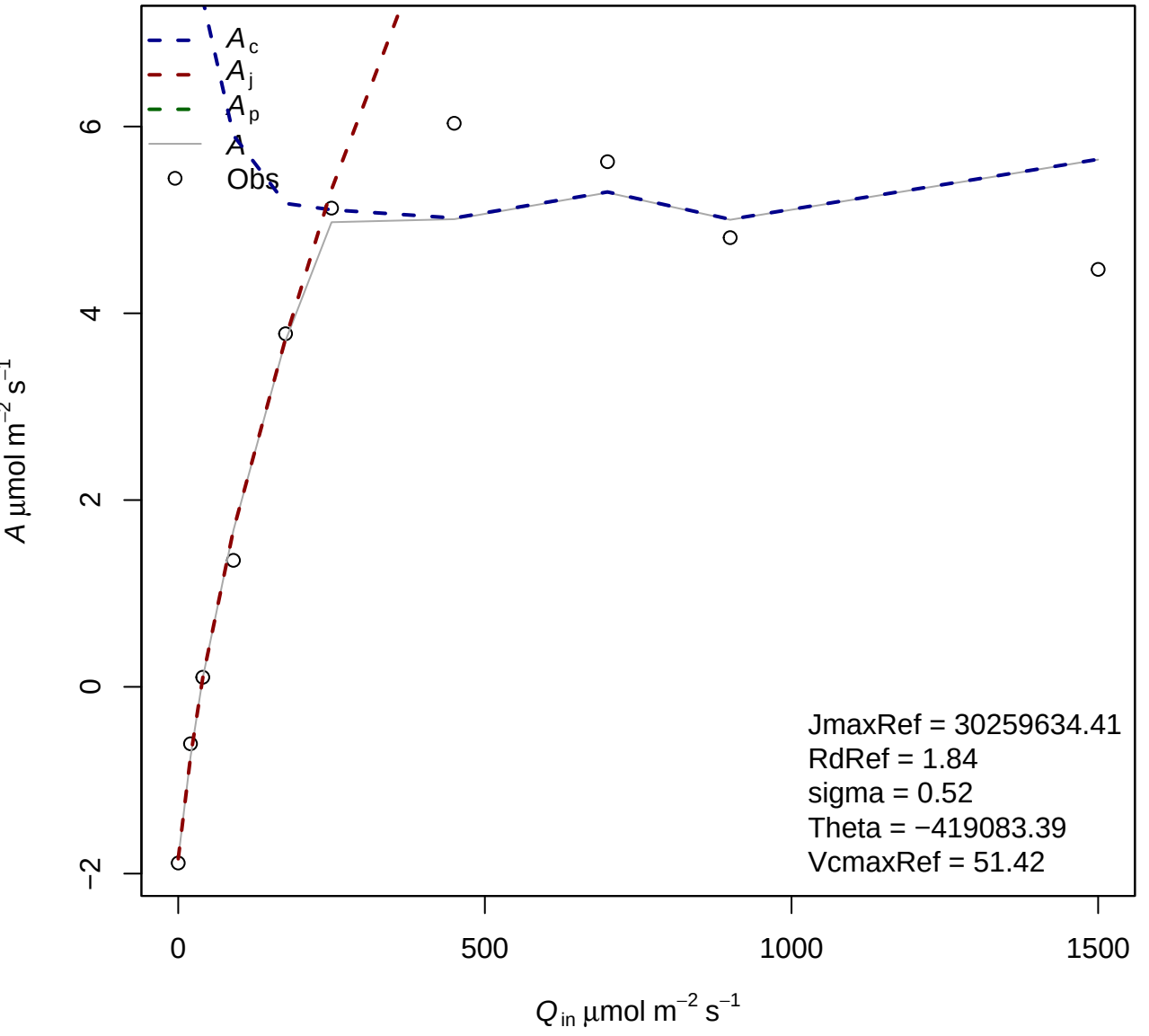

29
A_c
A_j
A_p
A
Obs
JmaxRef = 91.71
RdRef = 1.49
sigma = 0.05
Theta = 0.36
VcmaxRef = 57.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

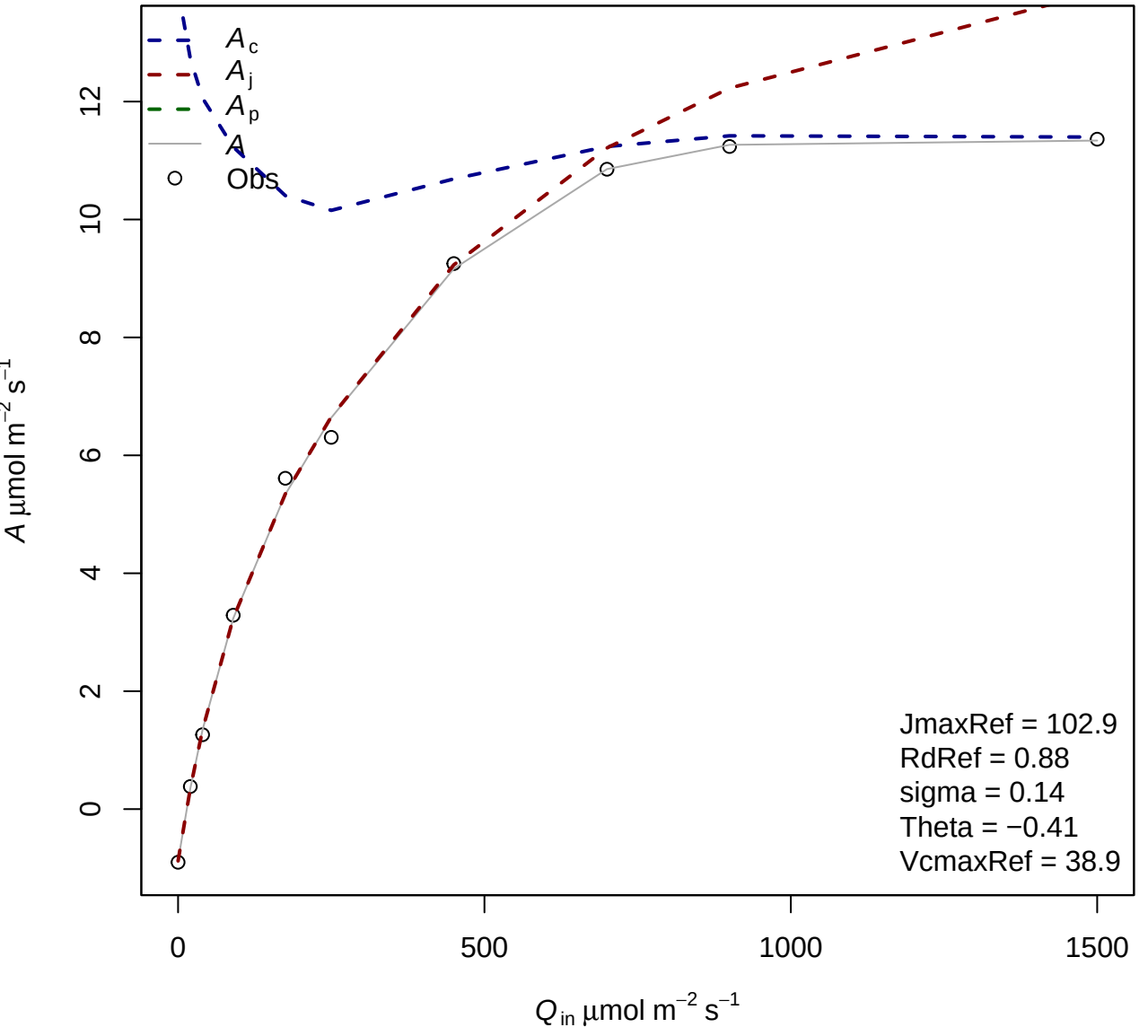

30
A_c
A_j
A_p
A
Obs
JmaxRef = 103.91
RdRef = 1.48
sigma = 0.18
Theta = 0.34
VcmaxRef = 66.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

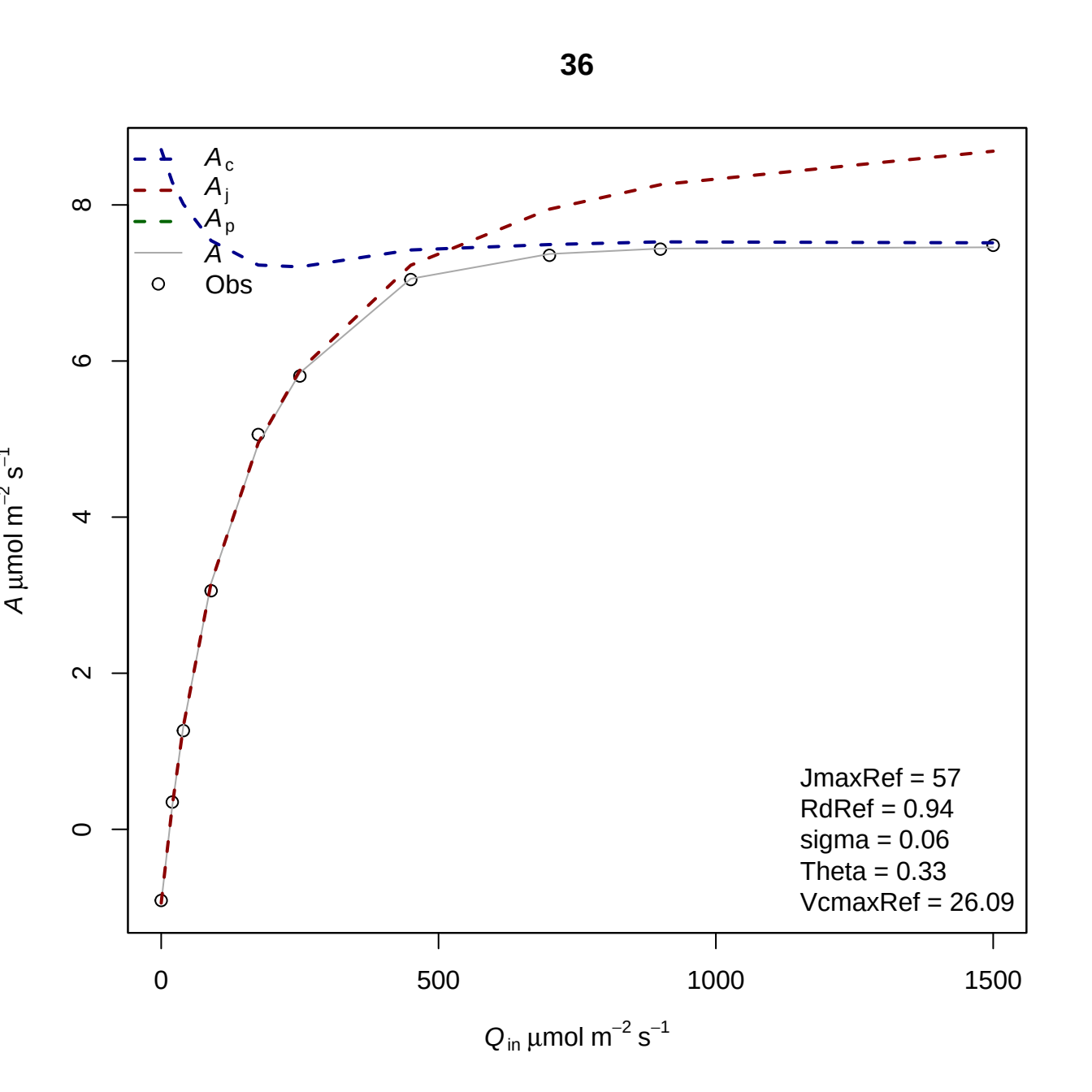

A_c
A_j
A_p
A
Obs
JmaxRef = 113.18
RdRef = 1.99
sigma = 0.13
Theta = −0.28
VcmaxRef = 67.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

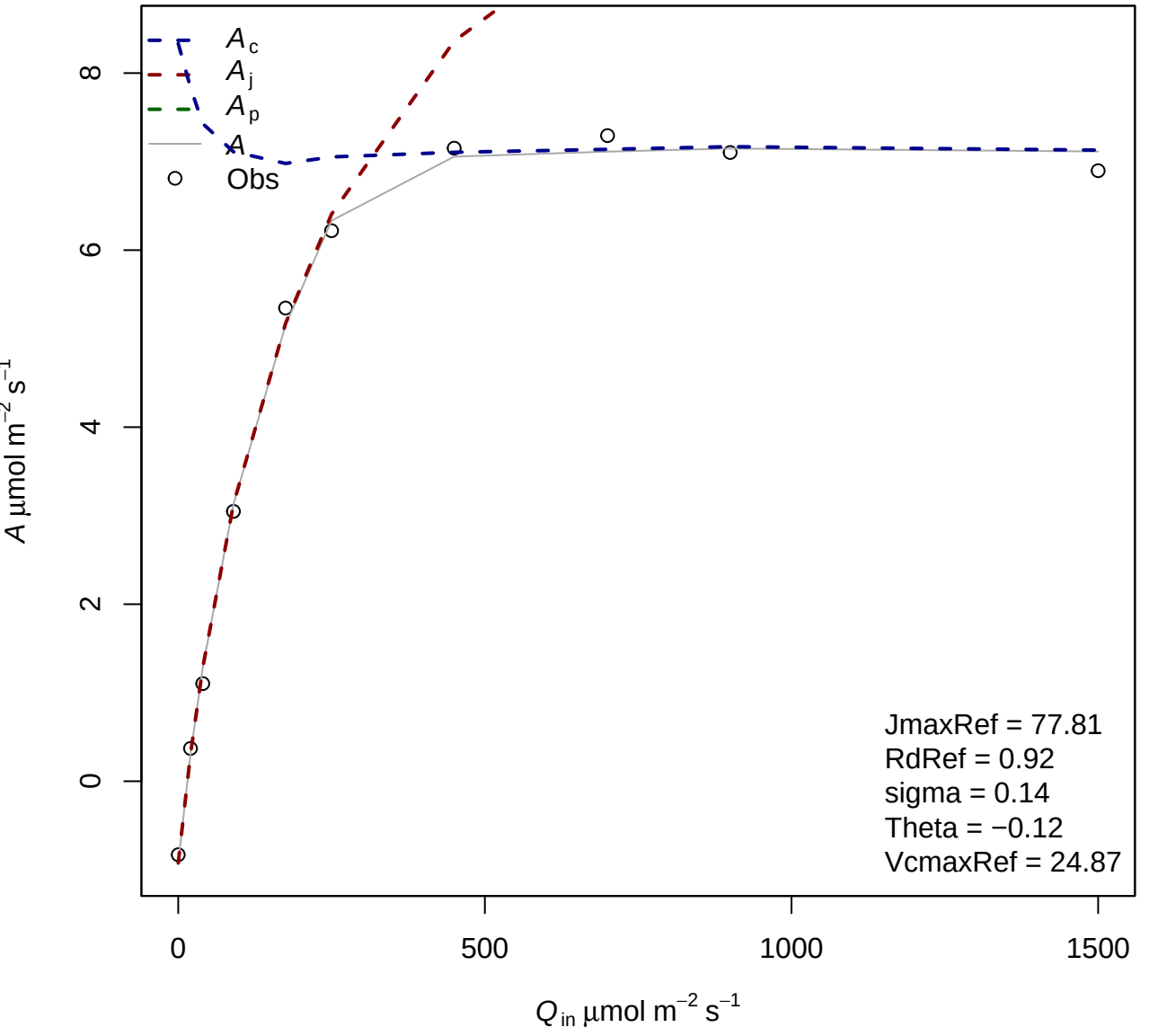

32
A_c
A_j
A_p
A
Obs
JmaxRef = 125.48
RdRef = 1.66
sigma = 0.17
Theta = 0.62
VcmaxRef = 106.75
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

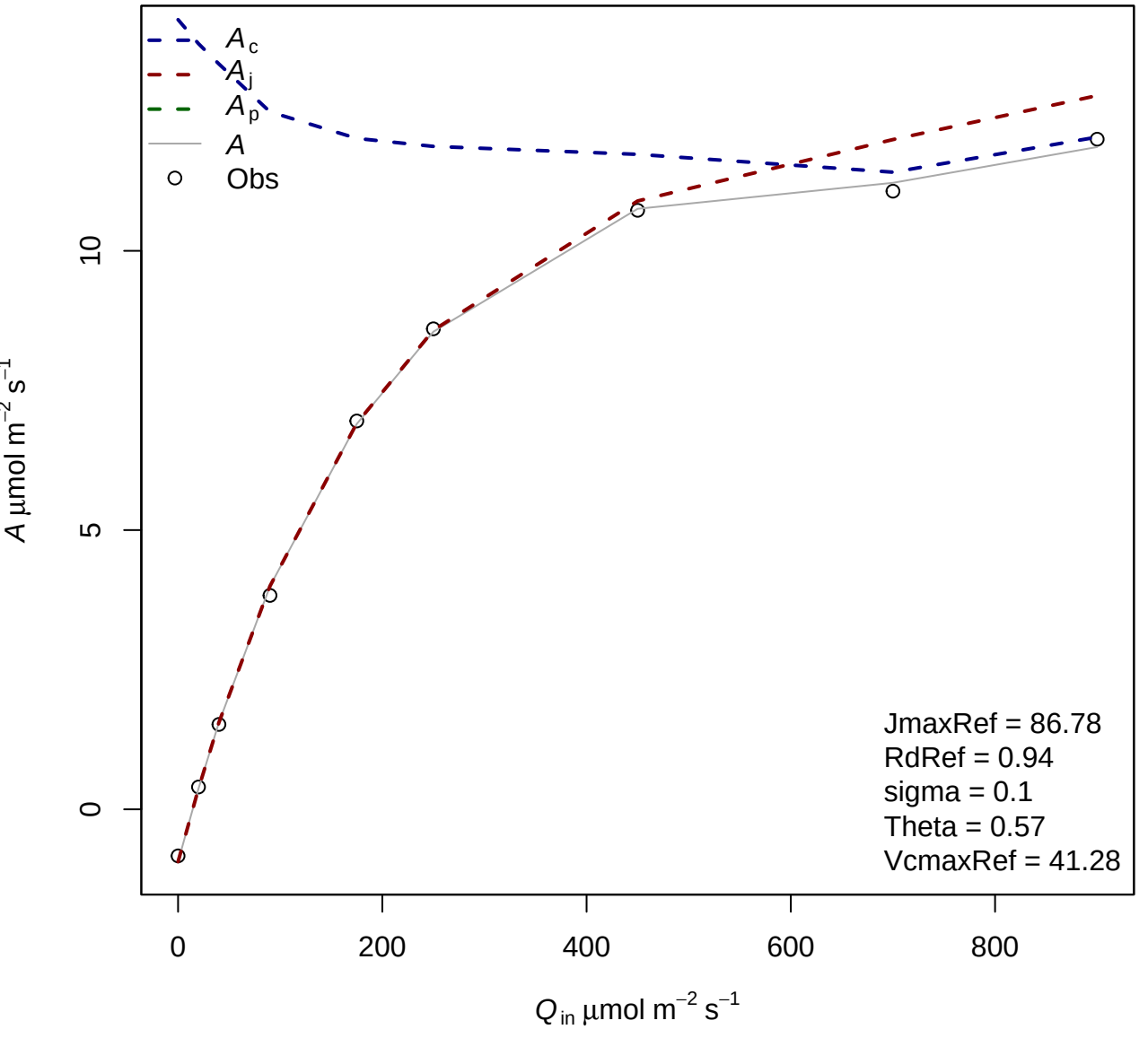

33
A_c
A_j
A_p
A
Obs
JmaxRef = 62567950.98
RdRef = 1.97
sigma = 0.31
Theta = −1473836.22
VcmaxRef = 53.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

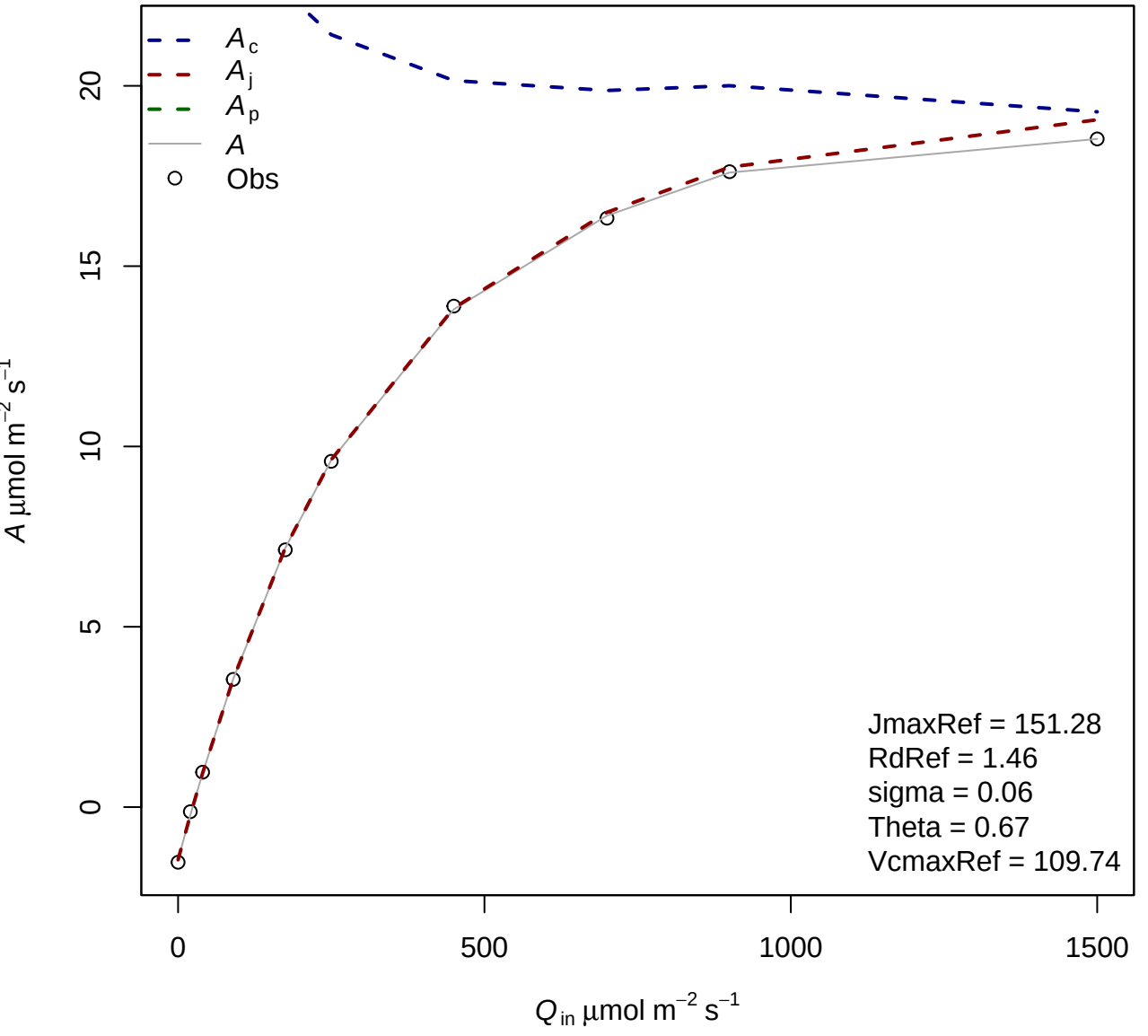

A_c
A_j
A_p
A
Obs
JmaxRef = 30259634.41
RdRef = 1.84
sigma = 0.52
Theta = −419083.39
VcmaxRef = 51.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

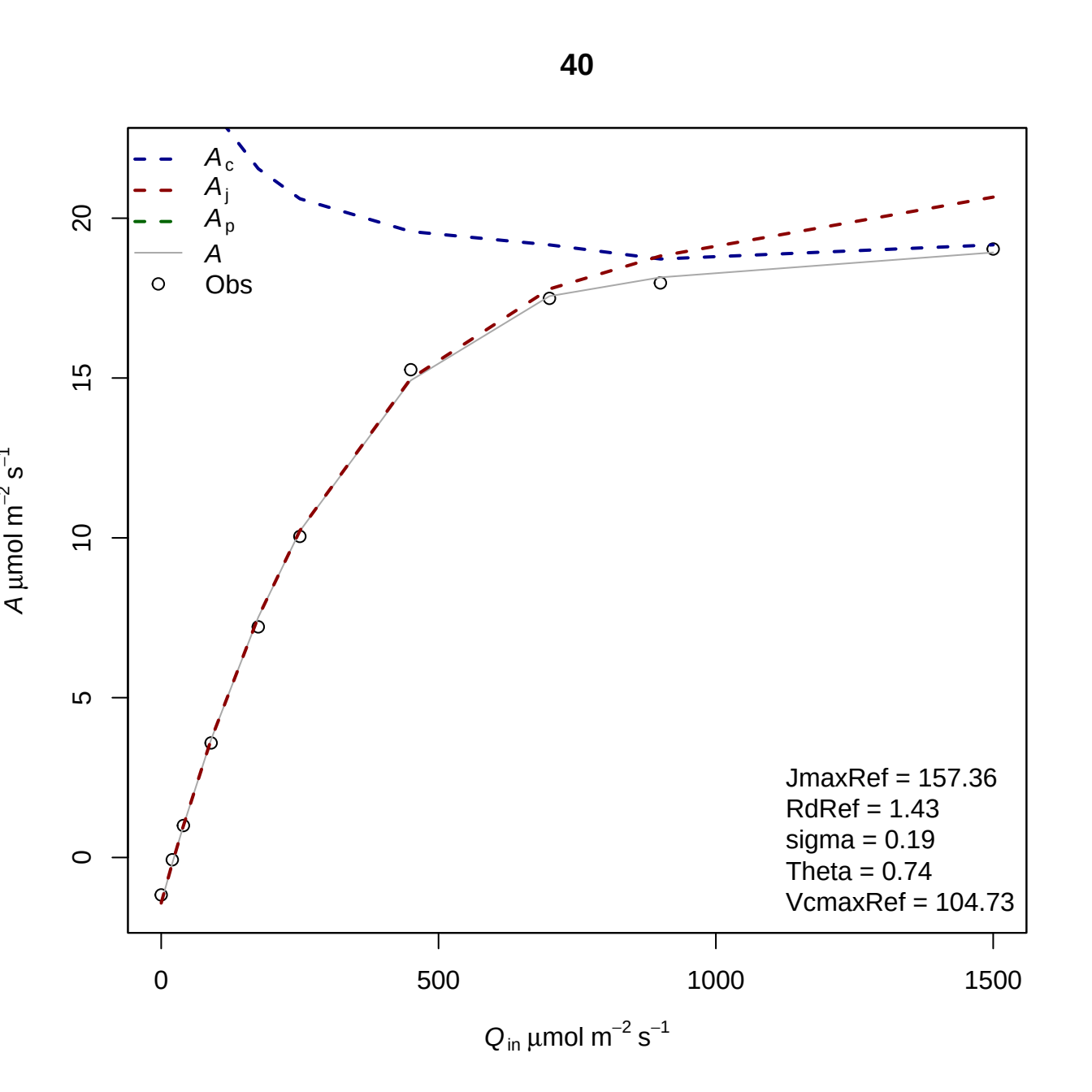

A_c
A_j
A_p
A
Obs
JmaxRef = 102.9
RdRef = 0.88
sigma = 0.14
Theta = −0.41
VcmaxRef = 38.9
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

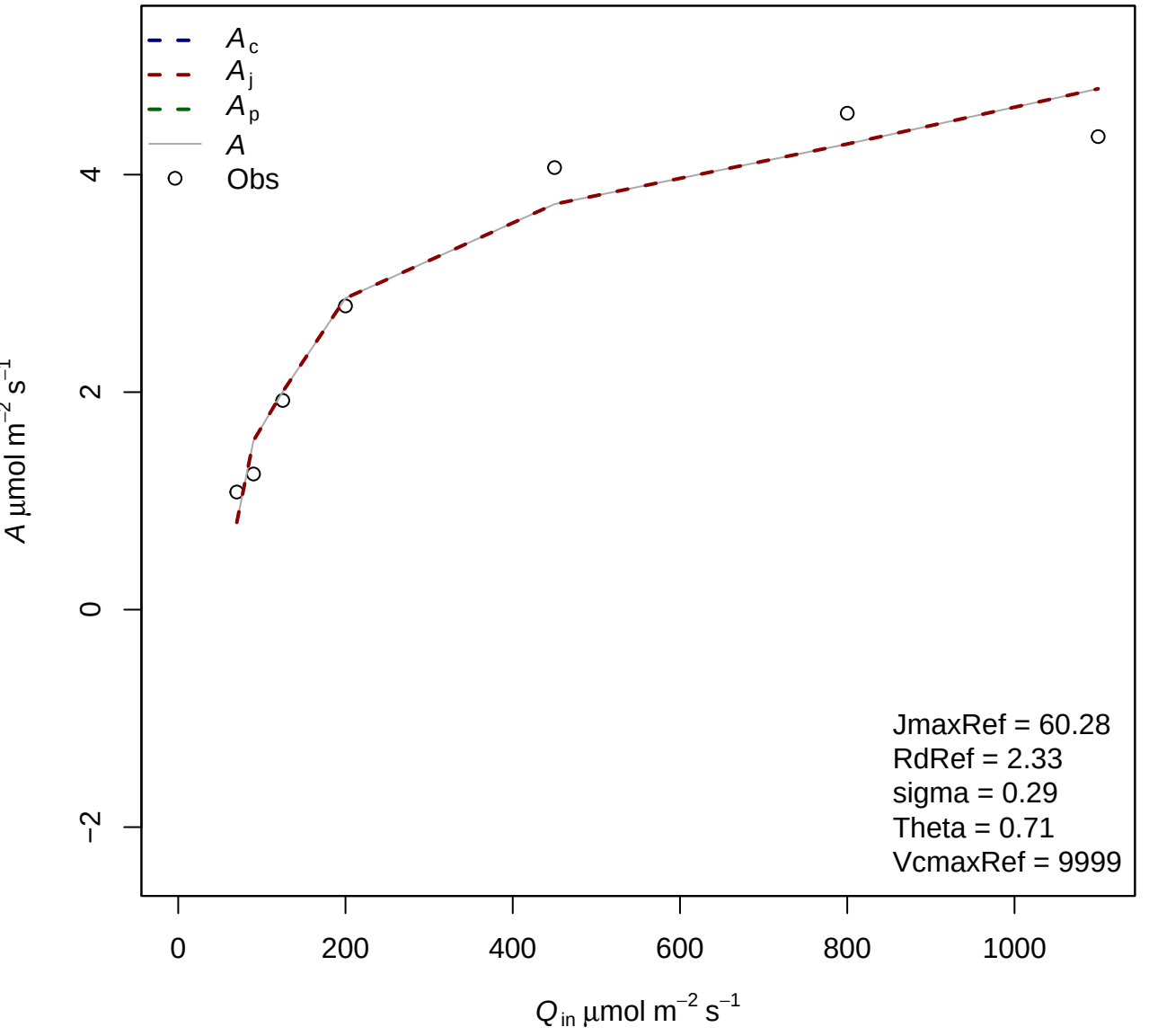

36
A_c
A_j
A_p
A
Obs
JmaxRef = 57
RdRef = 0.94
sigma = 0.06
Theta = 0.33
VcmaxRef = 26.09
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

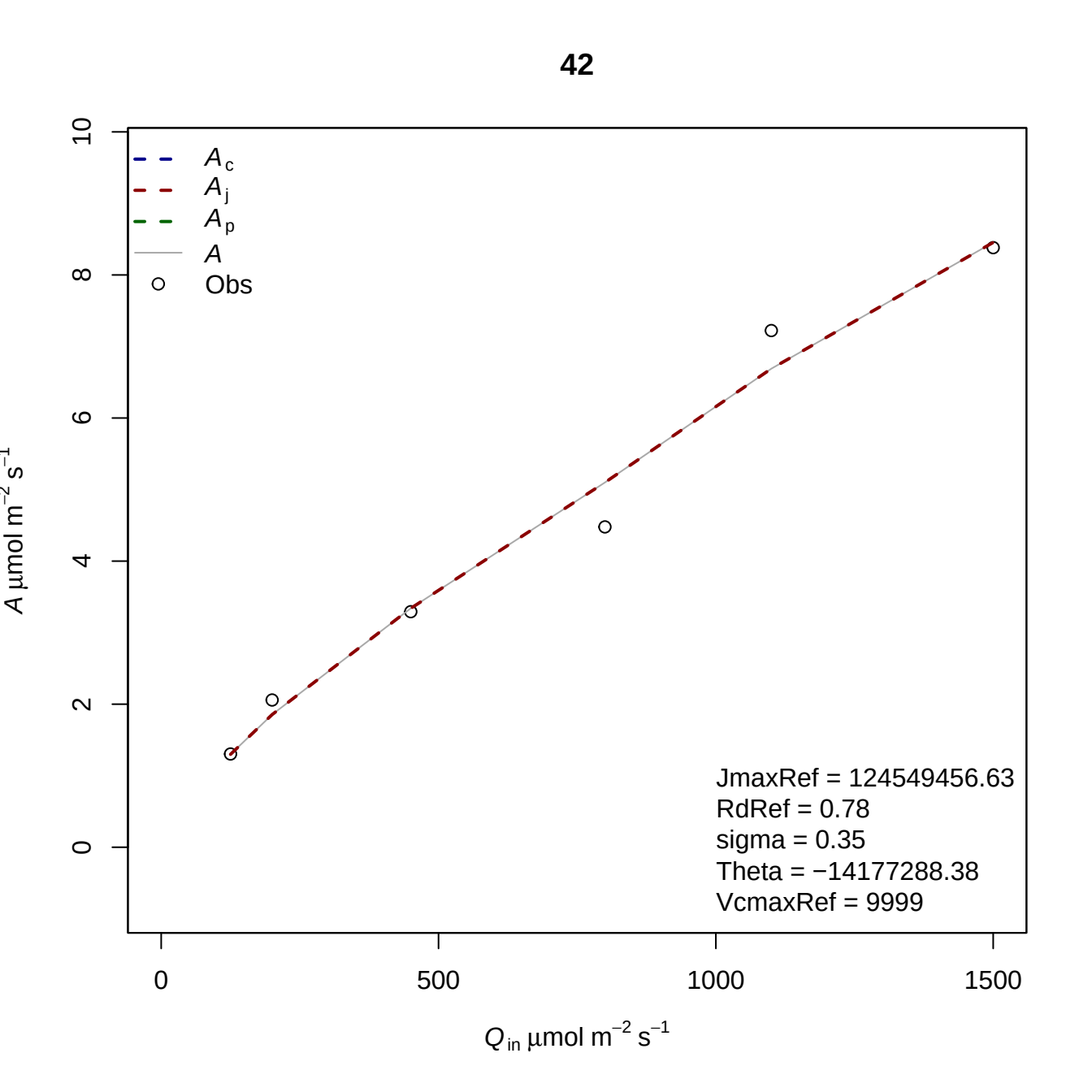

A_c
A_j
A_p
A
Obs
JmaxRef = 77.81
RdRef = 0.92
sigma = 0.14
Theta = −0.12
VcmaxRef = 24.87
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

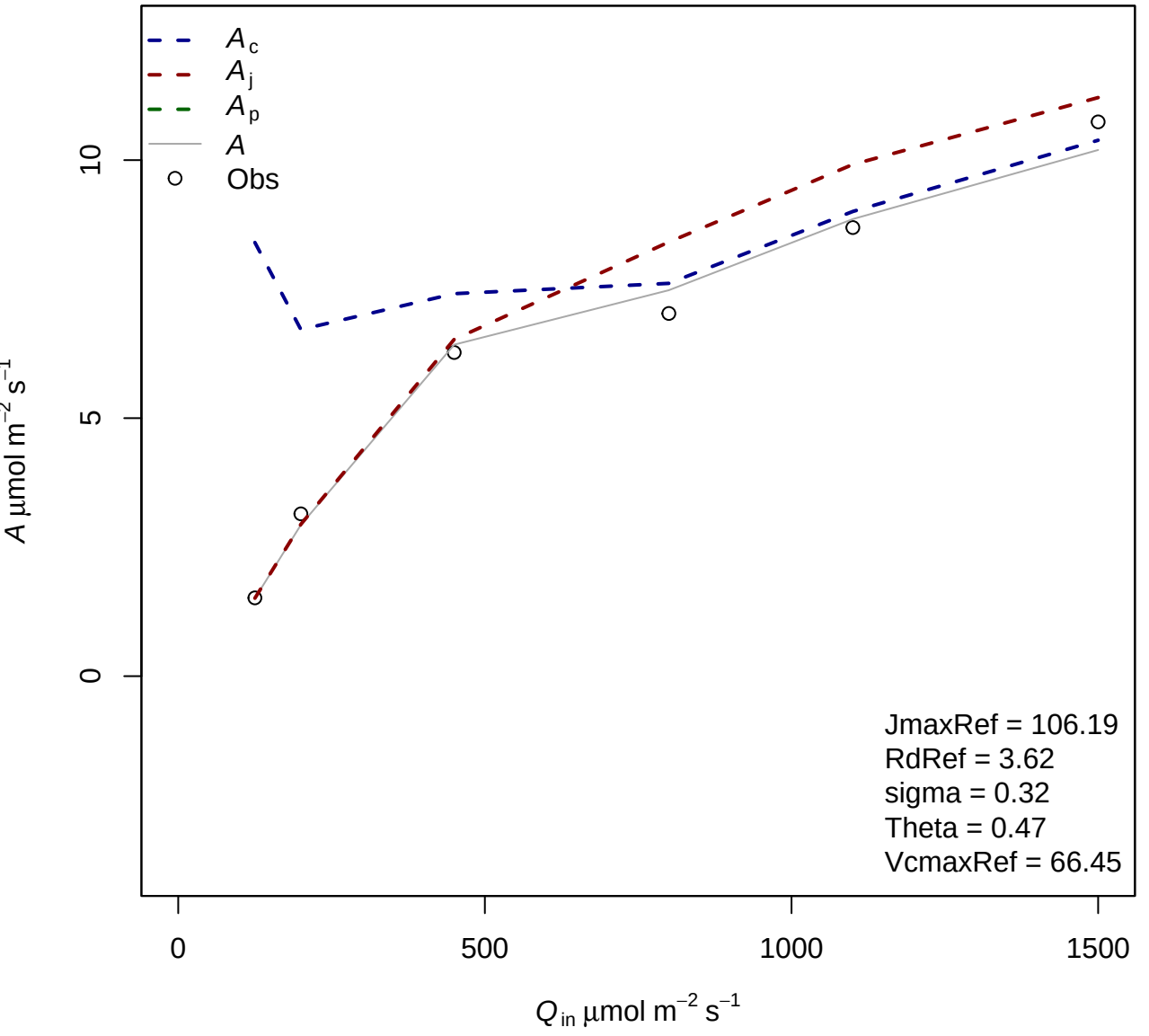

A_c
A_j
A_p
A
Obs
JmaxRef = 86.78
RdRef = 0.94
sigma = 0.1
Theta = 0.57
VcmaxRef = 41.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800

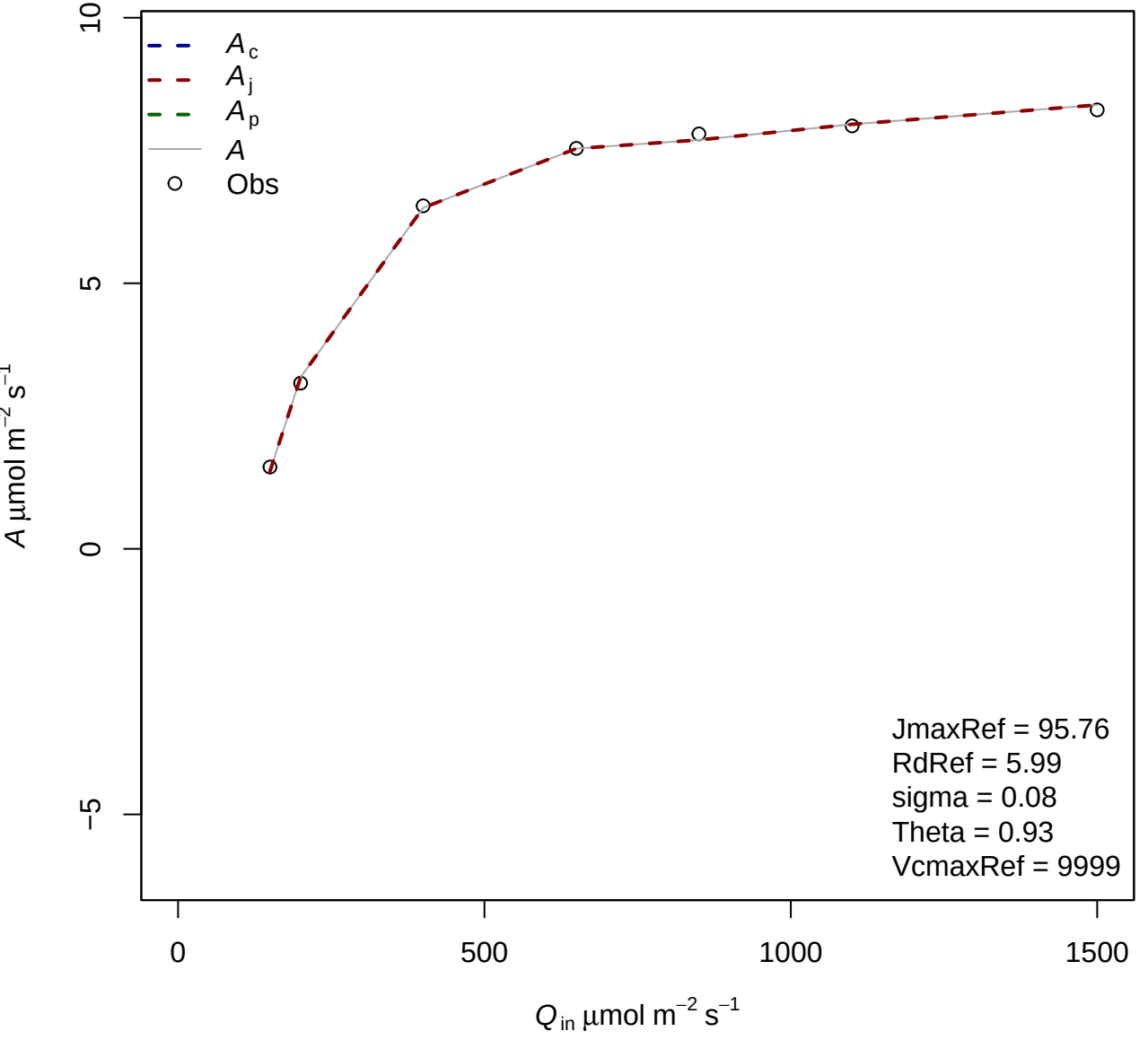

A_c
A_j
A_p
A
Obs
JmaxRef = 151.28
RdRef = 1.46
sigma = 0.06
Theta = 0.67
VcmaxRef = 109.74
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

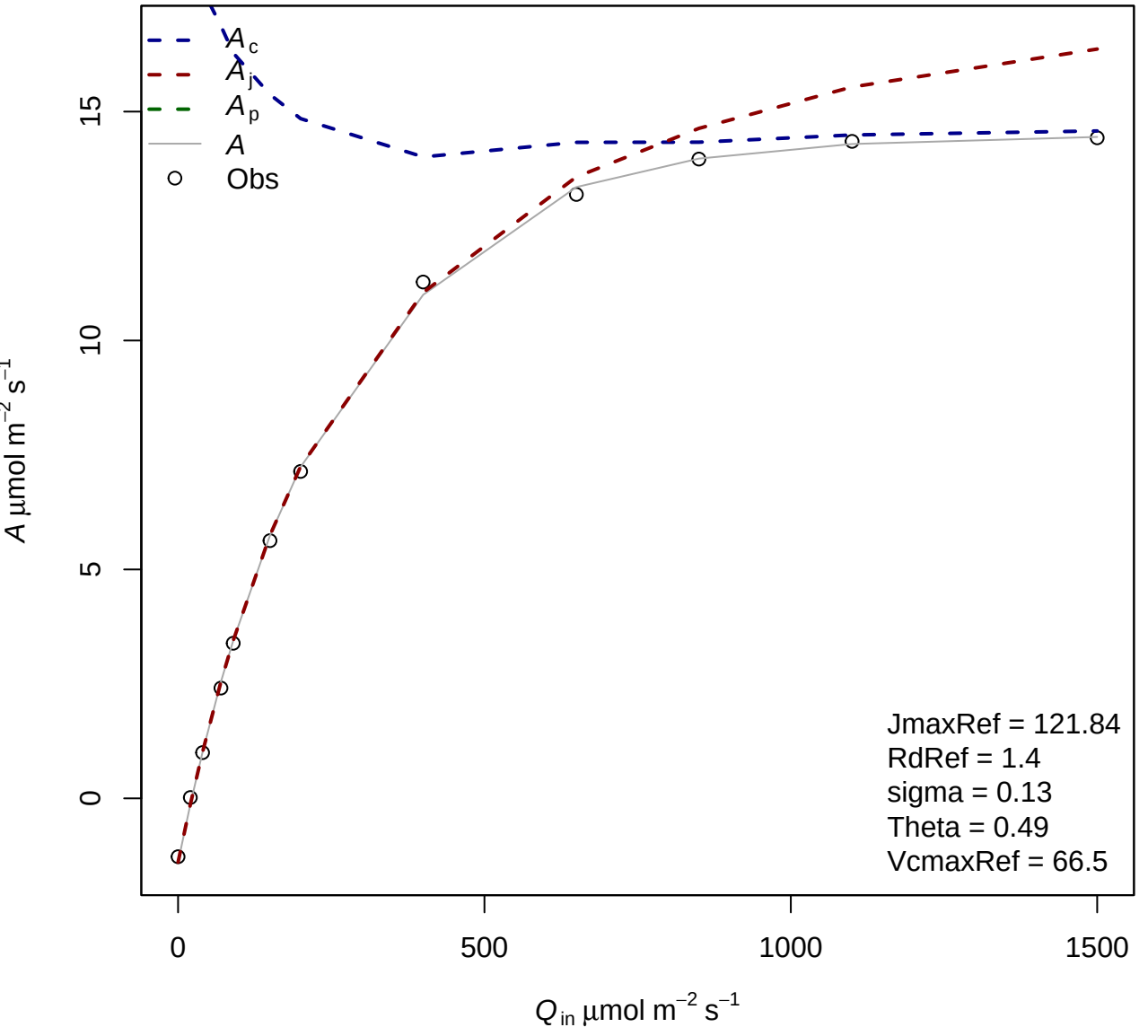

40
A_c
A_j
A_p
A
Obs
JmaxRef = 157.36
RdRef = 1.43
sigma = 0.19
Theta = 0.74
VcmaxRef = 104.73
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

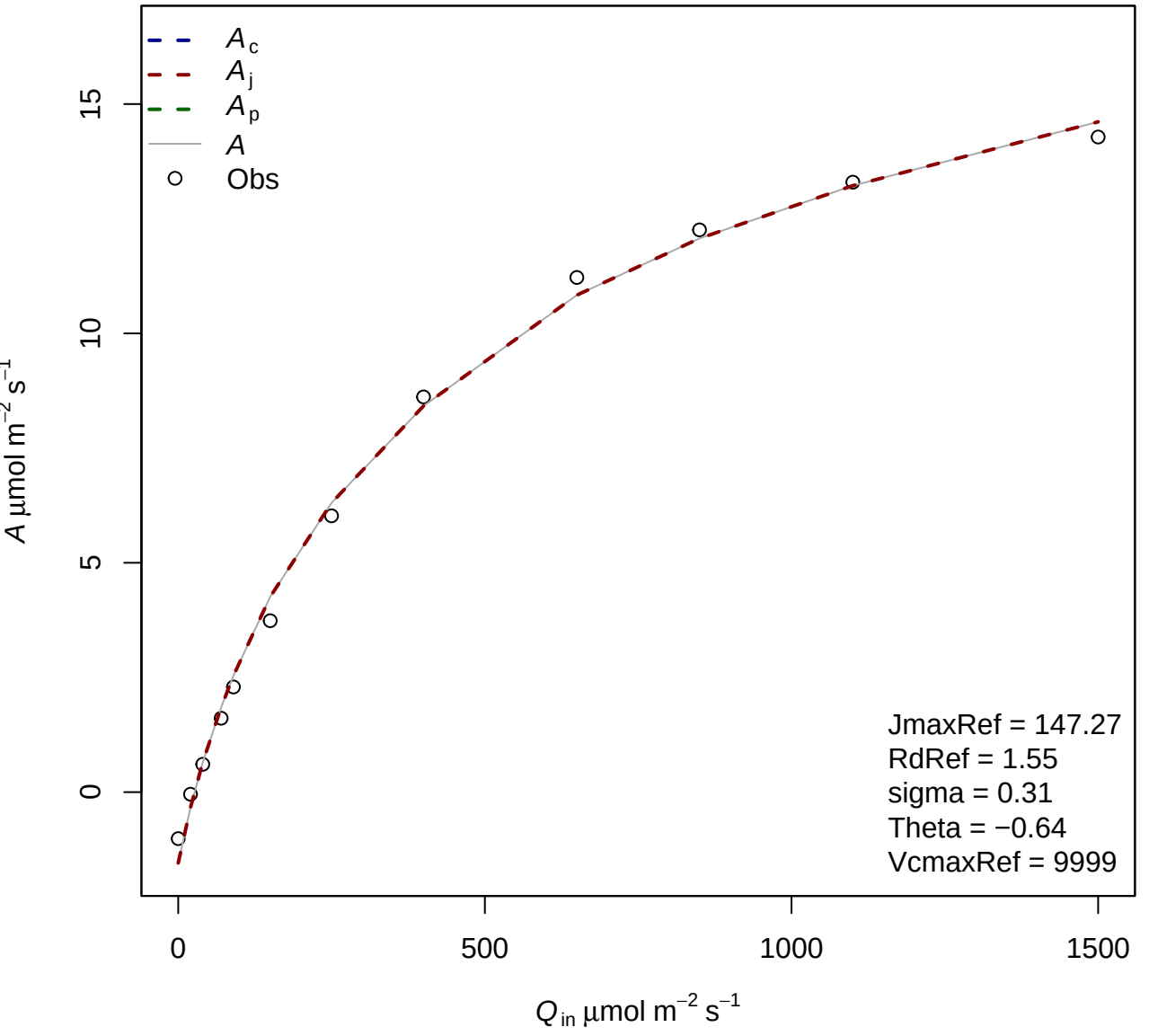

A_c
A_j
A_p
A
Obs
JmaxRef = 60.28
RdRef = 2.33
sigma = 0.29
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

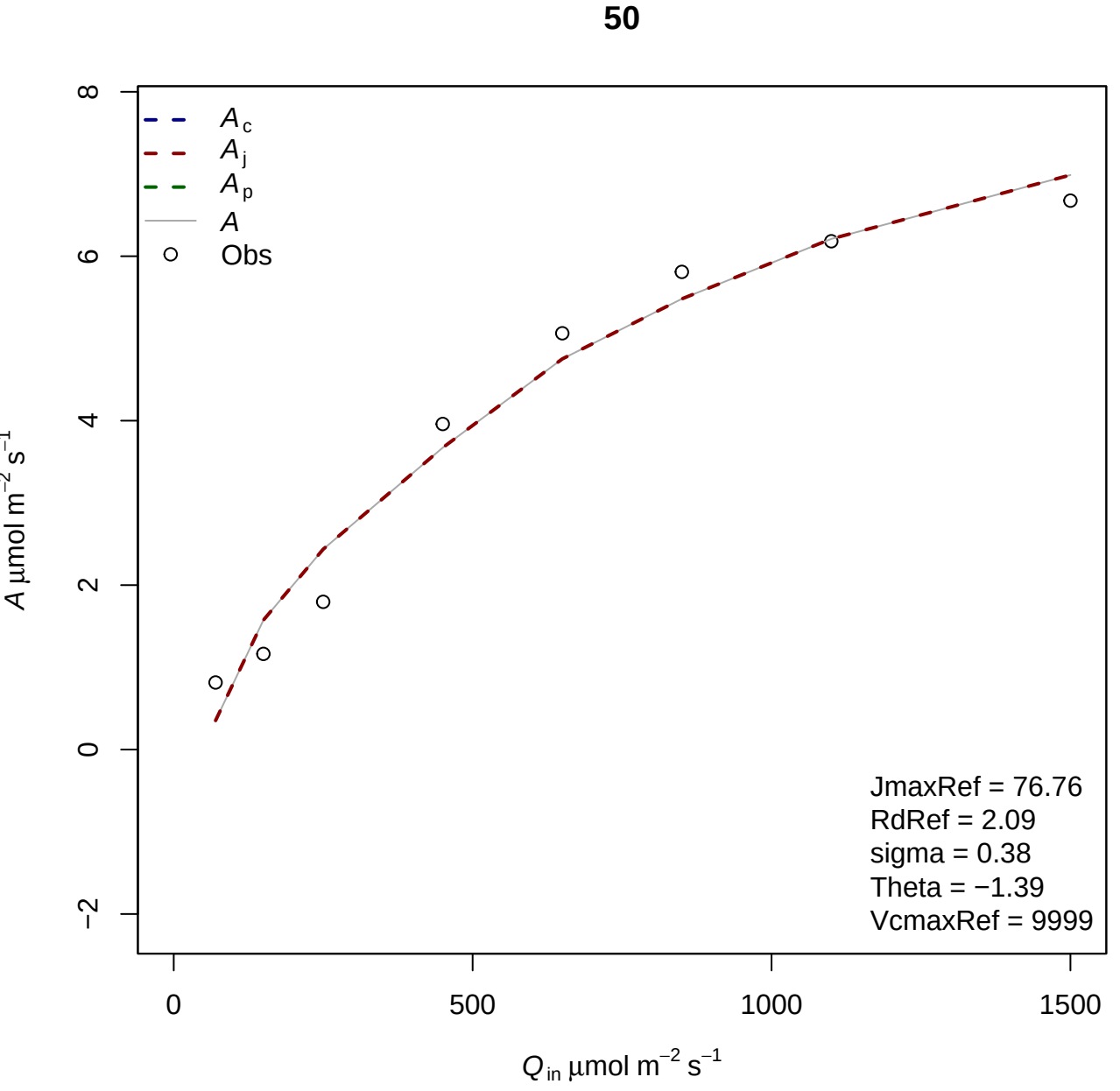

42
A_c
A_j
A_p
A
Obs
JmaxRef = 124549456.63
RdRef = 0.78
sigma = 0.35
Theta = −14177288.38
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

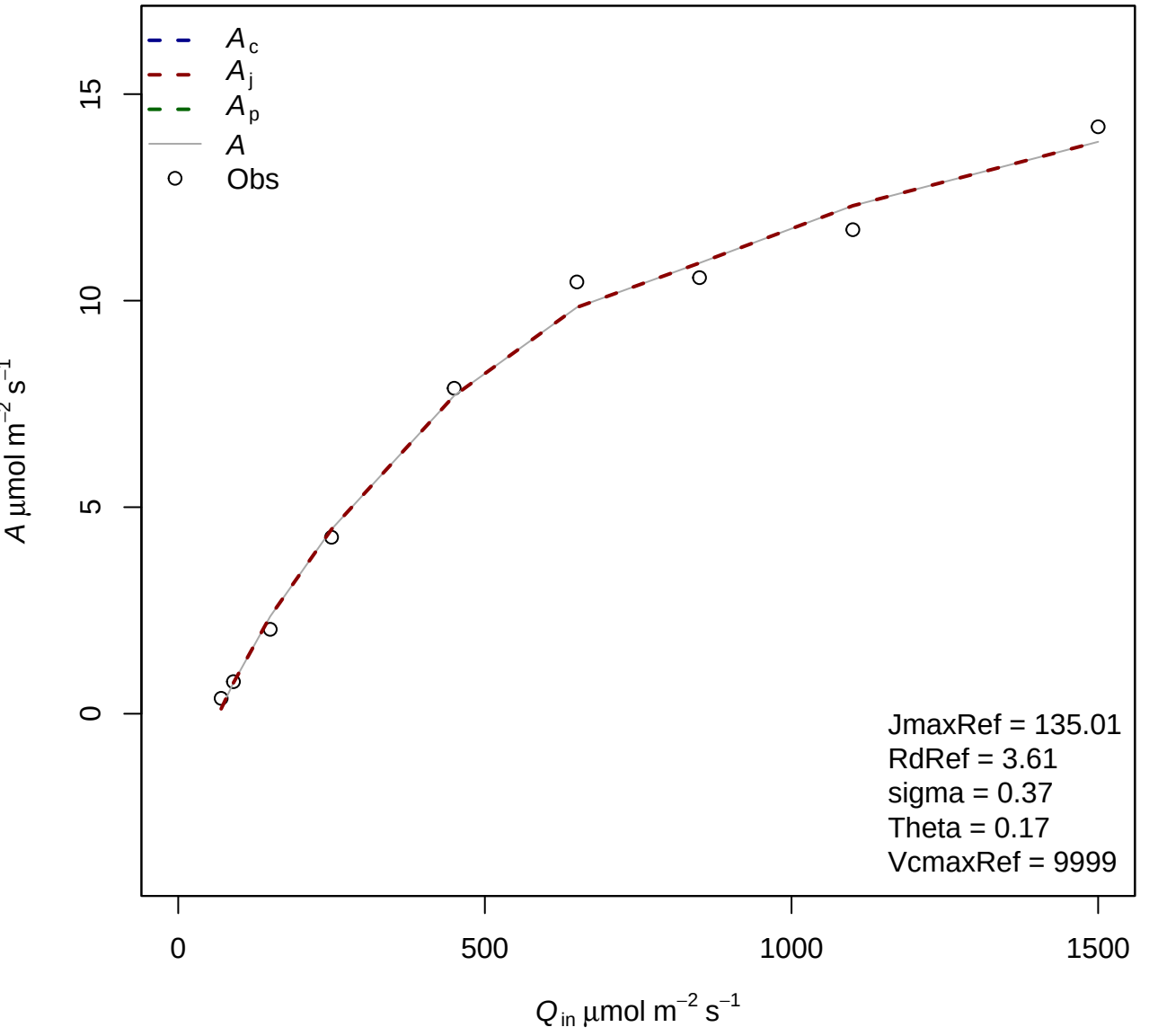

A_c
A_j
A_p
A
Obs
JmaxRef = 106.19
RdRef = 3.62
sigma = 0.32
Theta = 0.47
VcmaxRef = 66.45
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

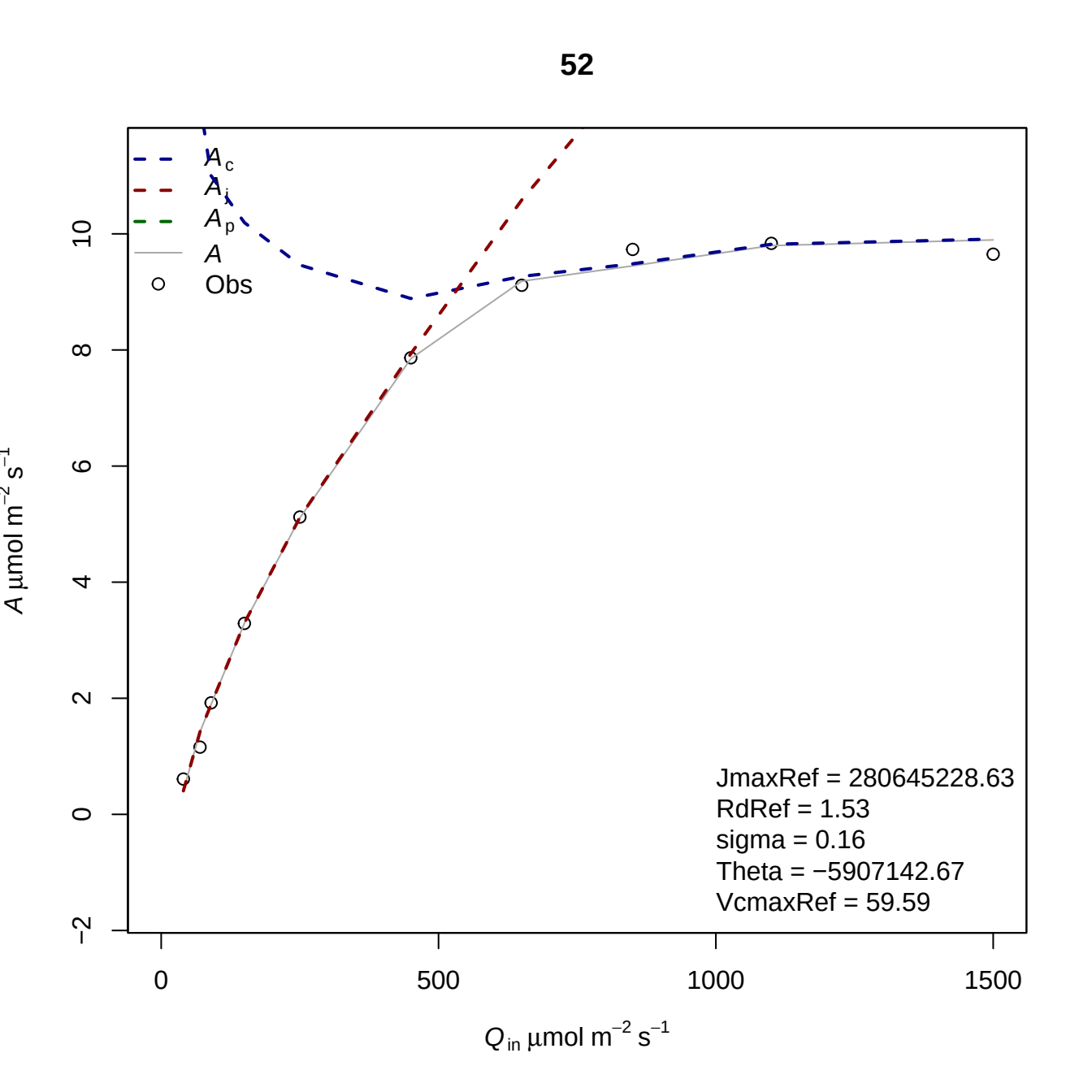

A_c
A_j
A_p
A
Obs
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 95.76
RdRef = 5.99
sigma = 0.08
Theta = 0.93
VcmaxRef = 9999

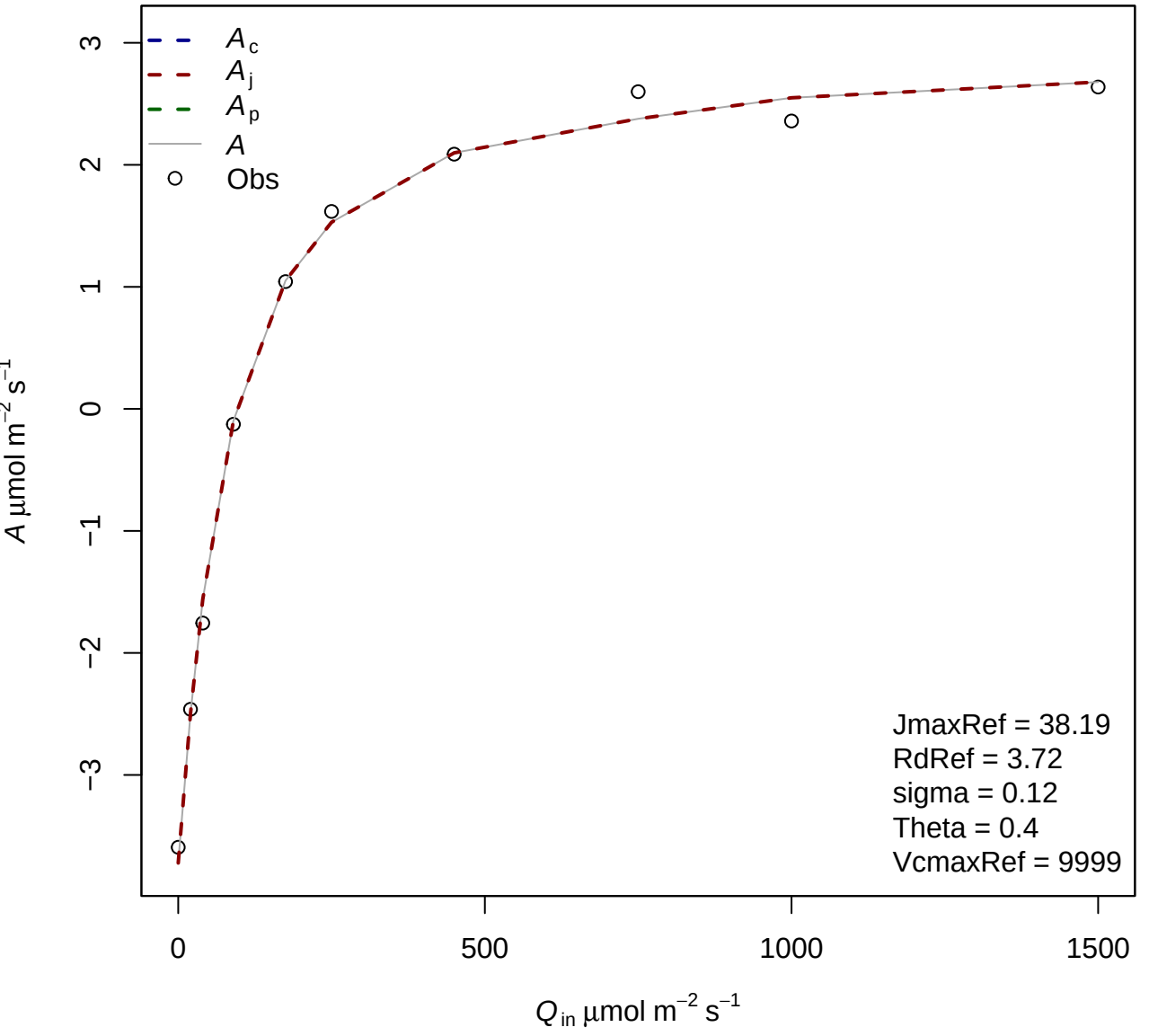

A_c
A_j
A_p
A
Obs
JmaxRef = 121.84
RdRef = 1.4
sigma = 0.13
Theta = 0.49
VcmaxRef = 66.5
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

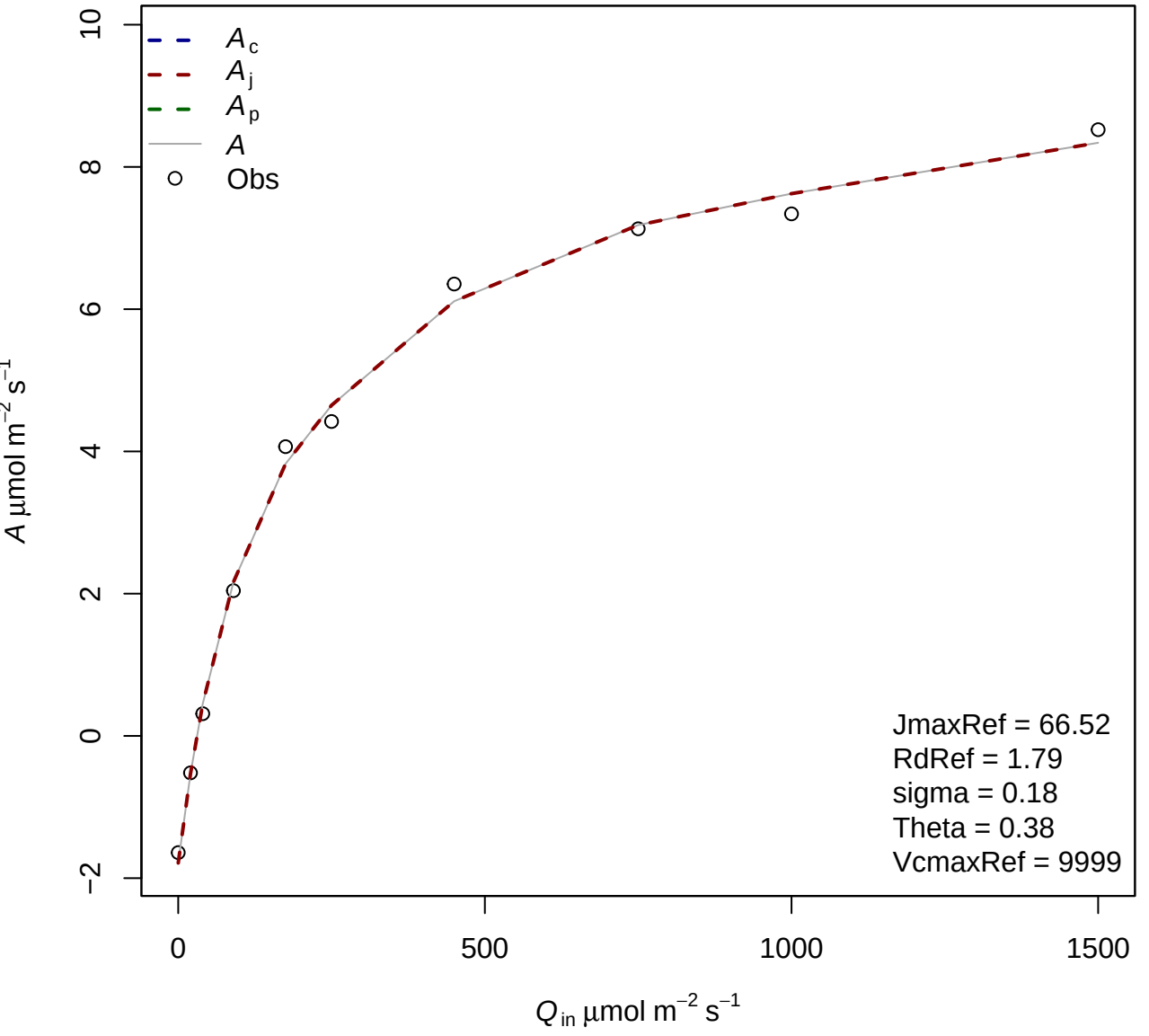

A_c
A_j
A_p
A
Obs
JmaxRef = 147.27
RdRef = 1.55
sigma = 0.31
Theta = −0.64
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

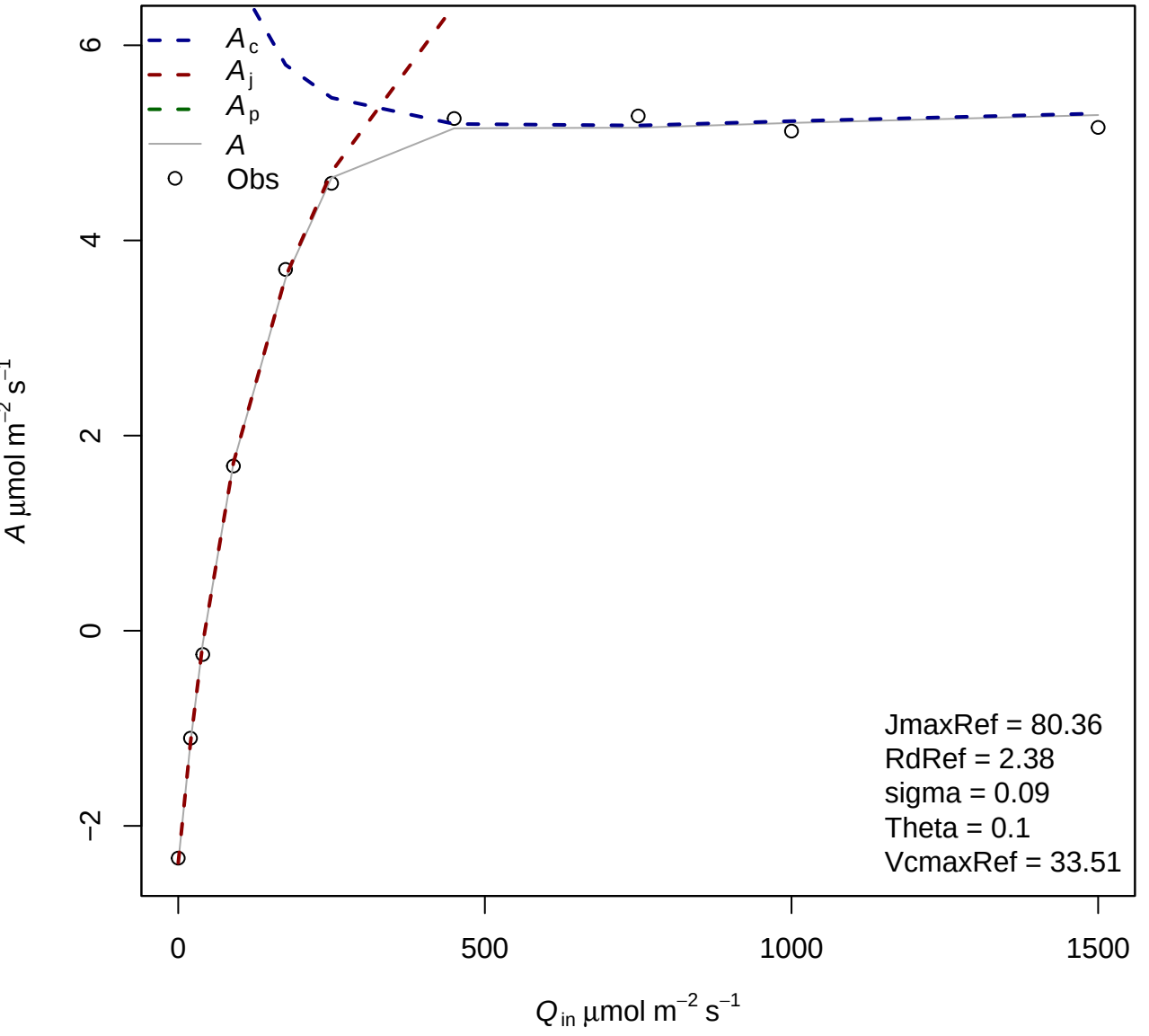

50
A_c
A_j
A_p
A
Obs
JmaxRef = 76.76
RdRef = 2.09
sigma = 0.38
Theta = −1.39
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

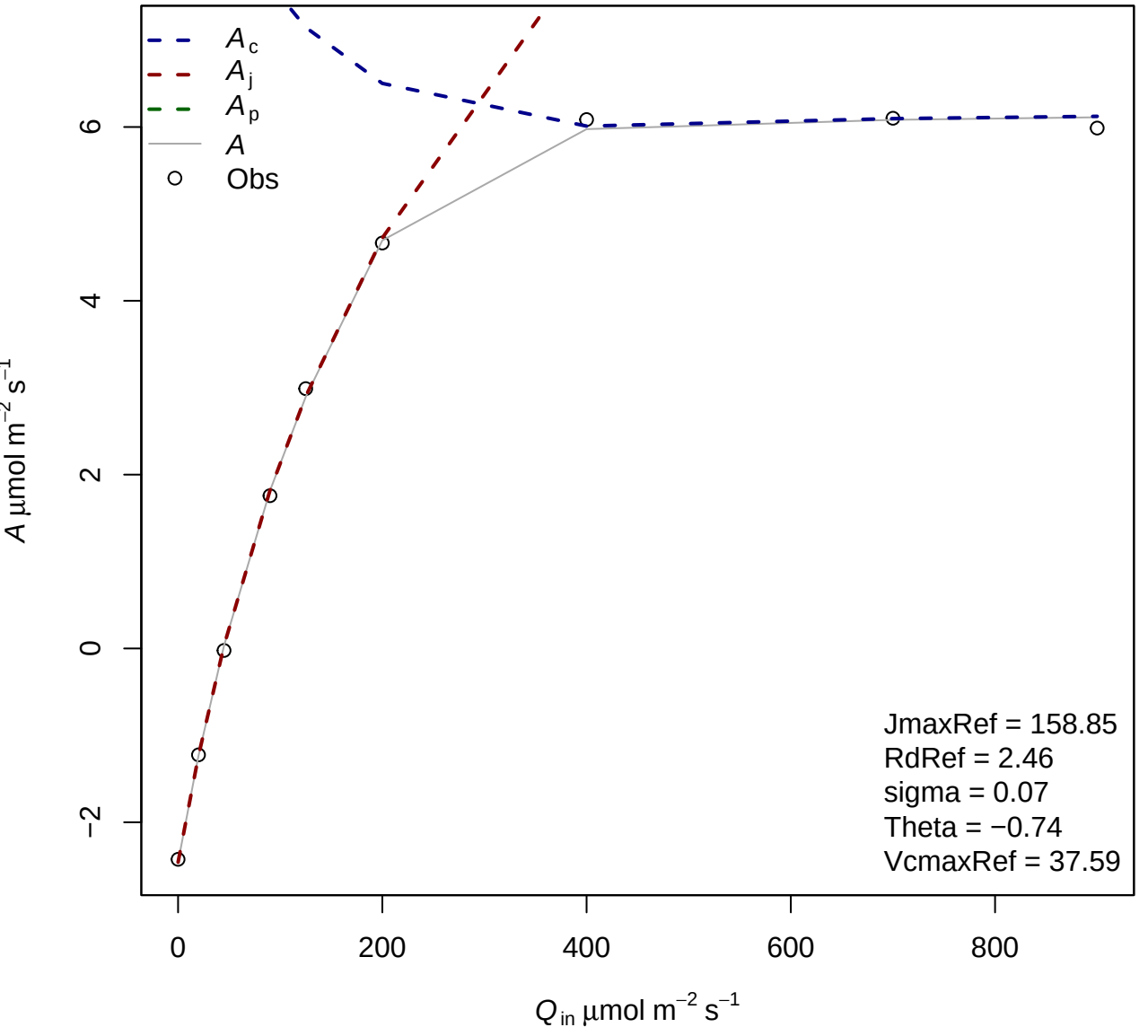

A_c
A_j
A_p
A
Obs
JmaxRef = 135.01
RdRef = 3.61
sigma = 0.37
Theta = 0.17
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

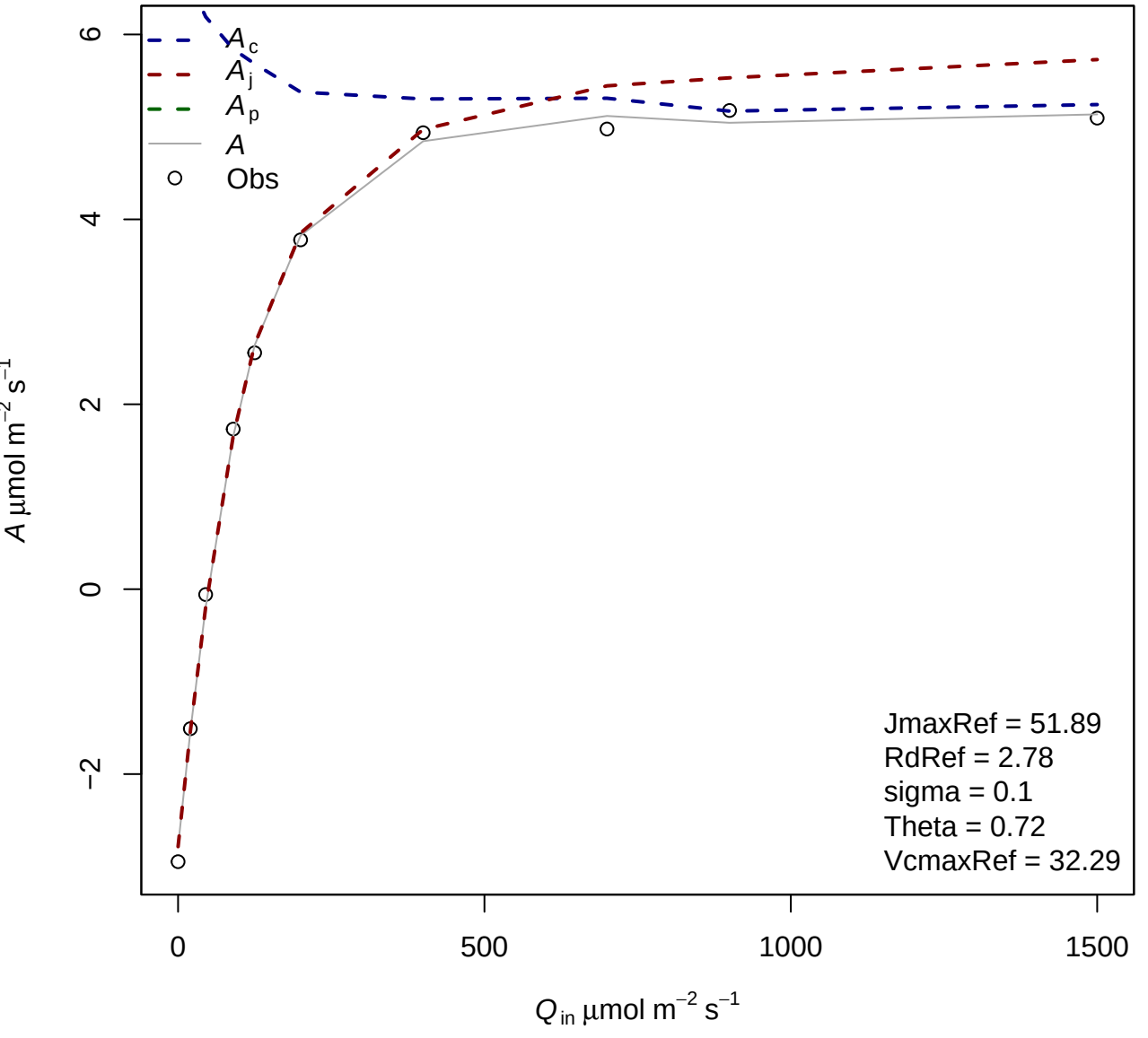

52
A_c
A_j
A_p
A
Obs
JmaxRef = 280645228.63
RdRef = 1.53
sigma = 0.16
Theta = −5907142.67
VcmaxRef = 59.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

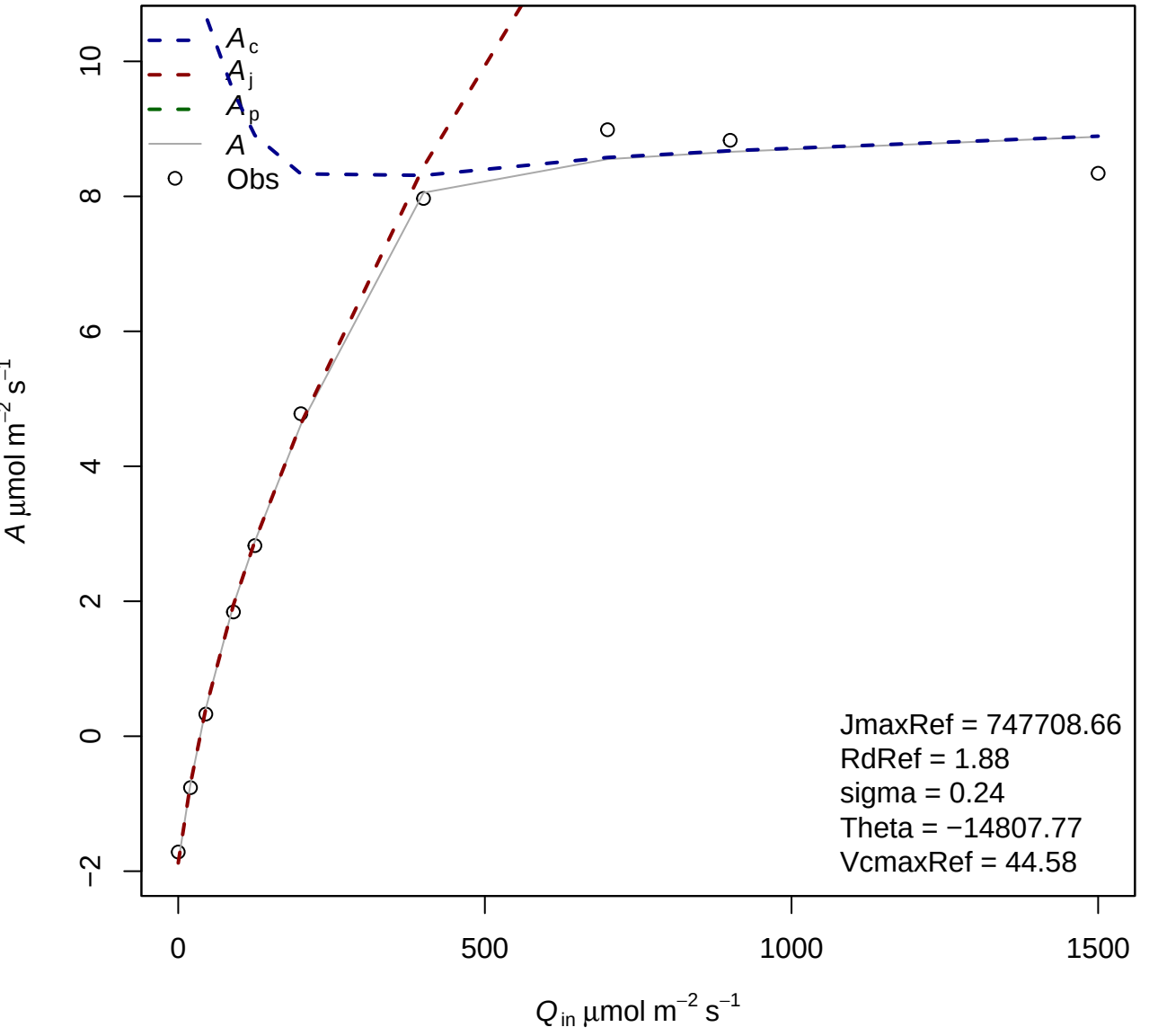

A_c
A_j
A_p
A
Obs
JmaxRef = 38.19
RdRef = 3.72
sigma = 0.12
Theta = 0.4
VcmaxRef = 9999
3
2
1
0
−1
−2
−3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

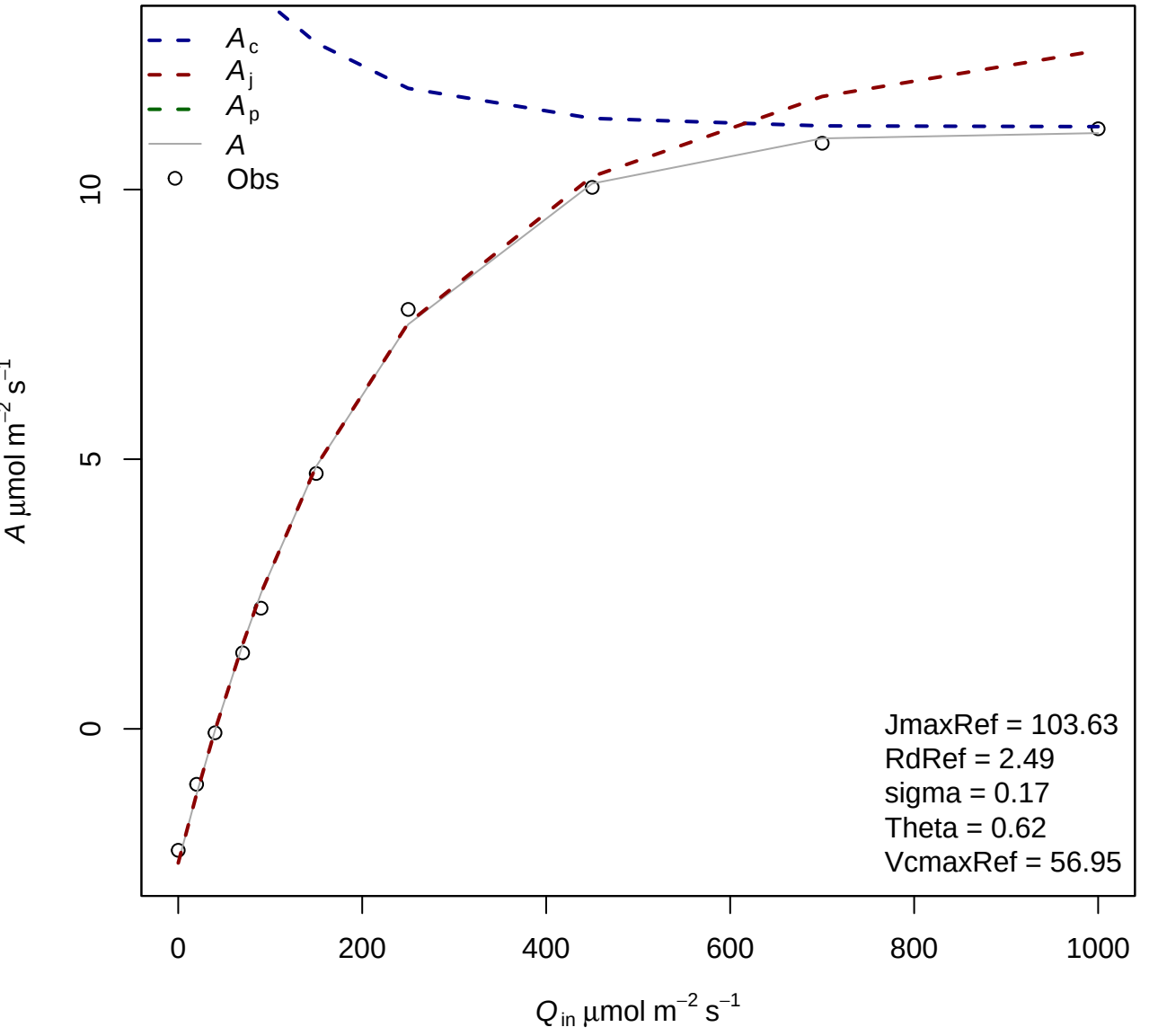

A_c
A_j
A_p
A
Obs
JmaxRef = 66.52
RdRef = 1.79
sigma = 0.18
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

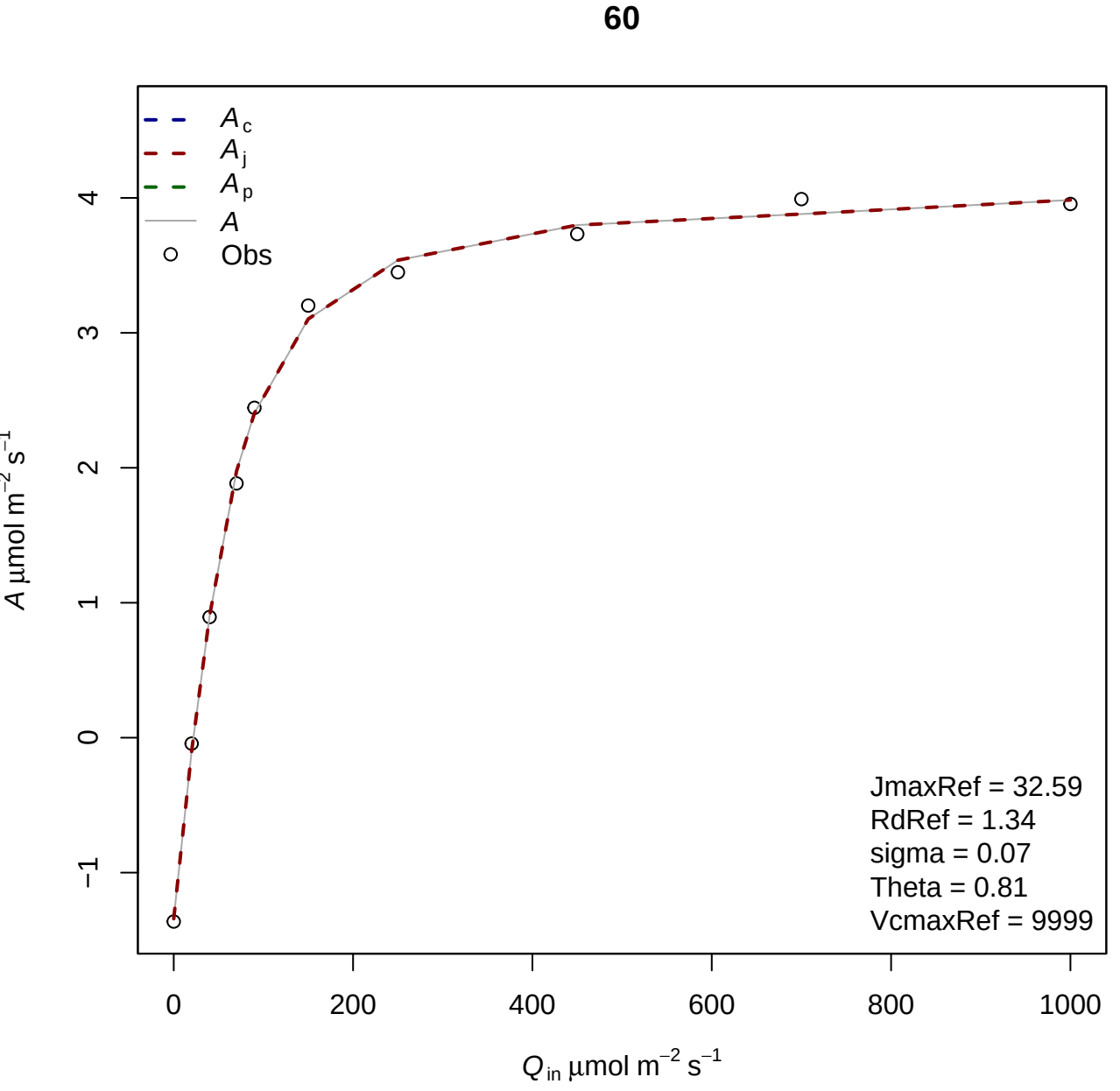

A_c
A_j
A_p
A
Obs
JmaxRef = 80.36
RdRef = 2.38
sigma = 0.09
Theta = 0.1
VcmaxRef = 33.51
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

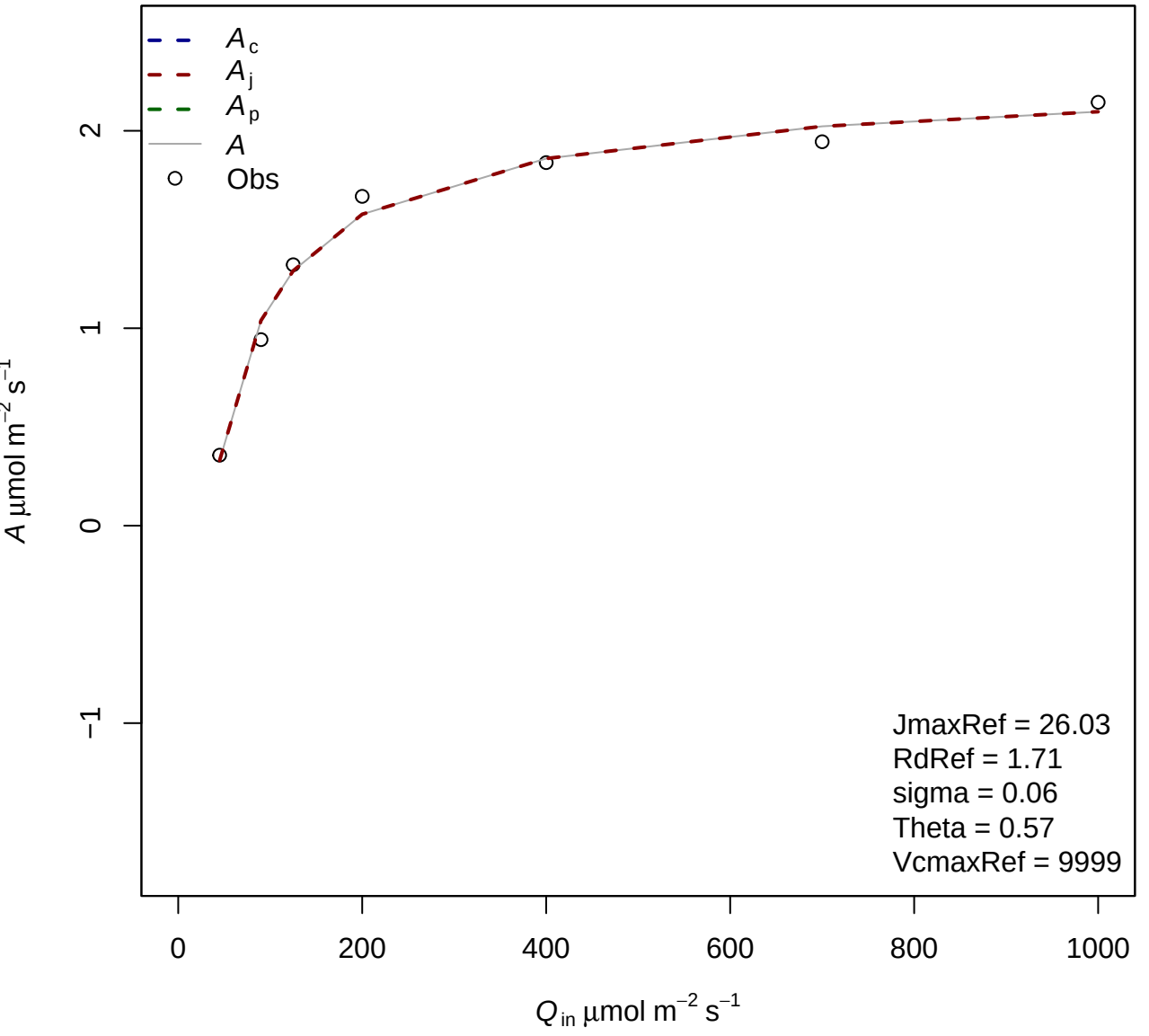

A_c
A_j
A_p
A
Obs
JmaxRef = 158.85
RdRef = 2.46
sigma = 0.07
Theta = −0.74
VcmaxRef = 37.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

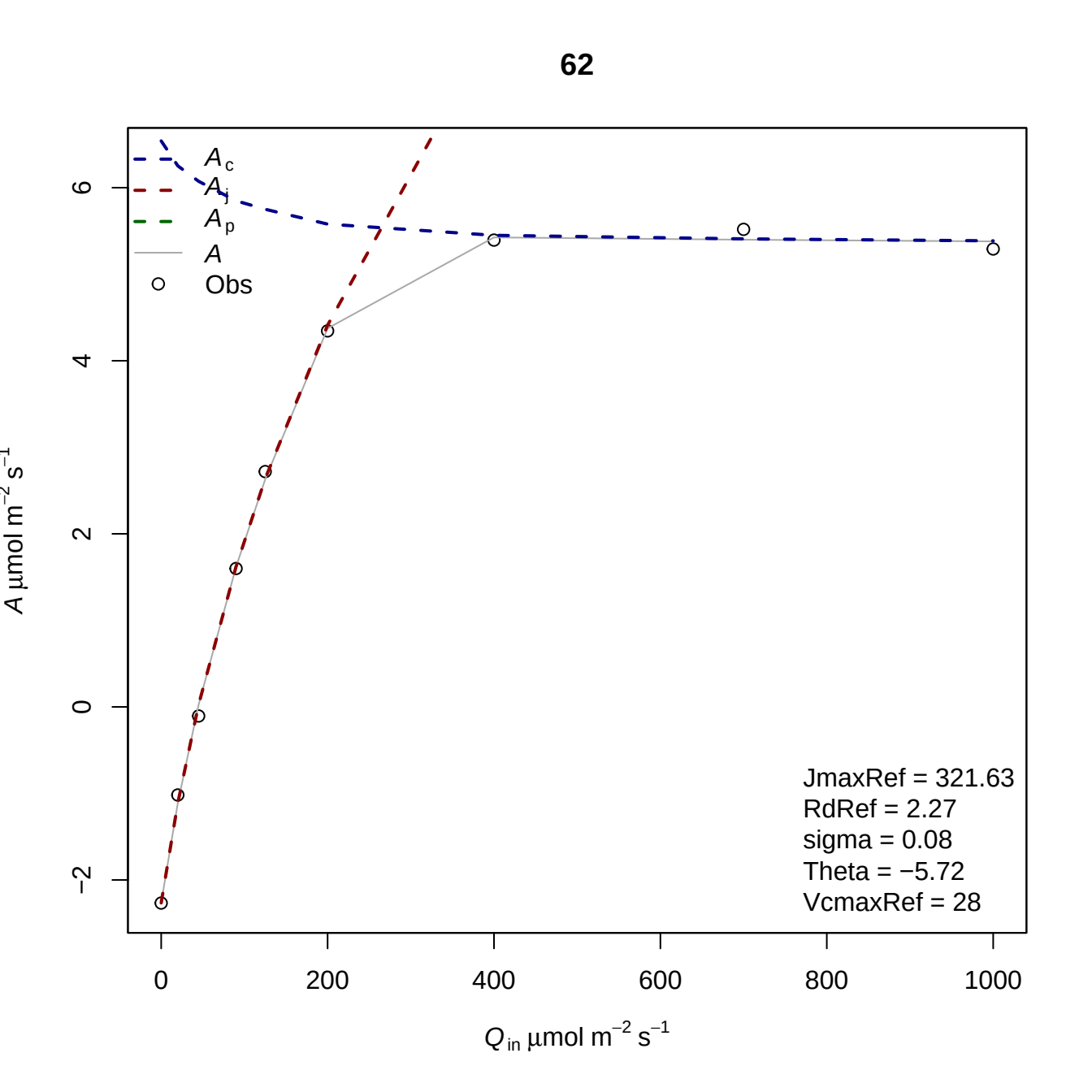

A_c
A_j
A_p
A
Obs
JmaxRef = 51.89
RdRef = 2.78
sigma = 0.1
Theta = 0.72
VcmaxRef = 32.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

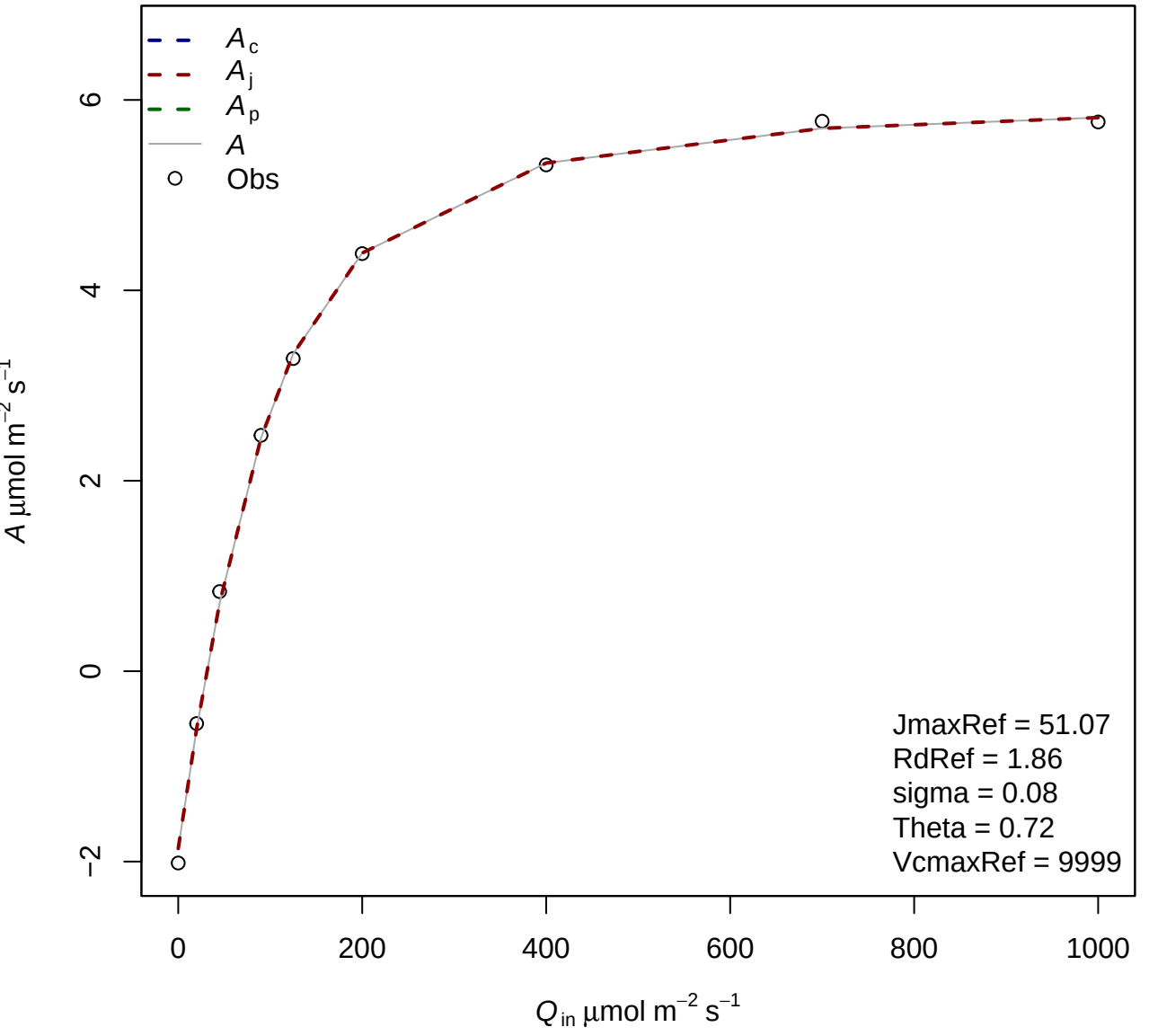

A_c
A_j
A_p
A
Obs
JmaxRef = 747708.66
RdRef = 1.88
sigma = 0.24
Theta = −14807.77
VcmaxRef = 44.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

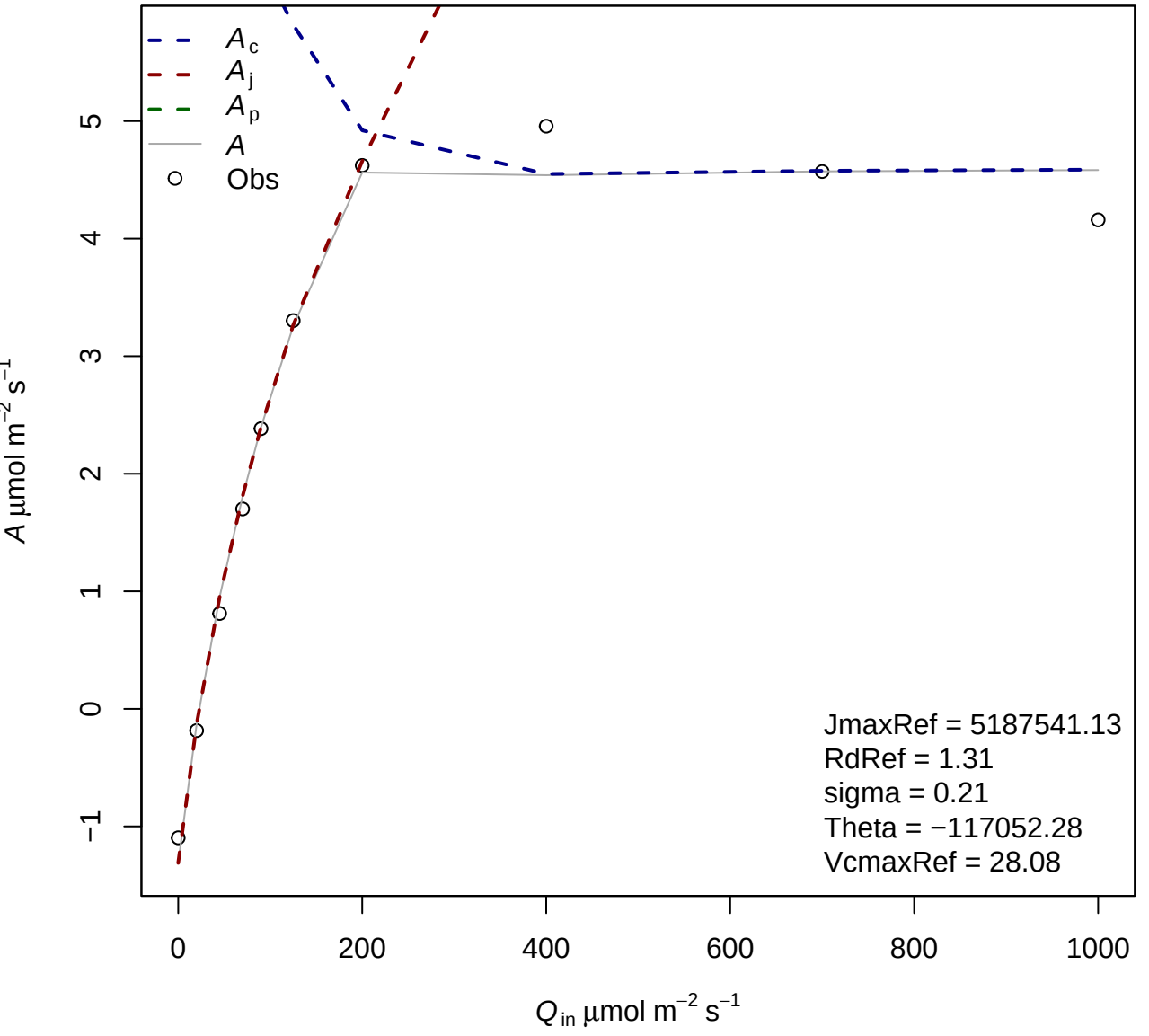

A_c
A_j
A_p
A
Obs
JmaxRef = 103.63
RdRef = 2.49
sigma = 0.17
Theta = 0.62
VcmaxRef = 56.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

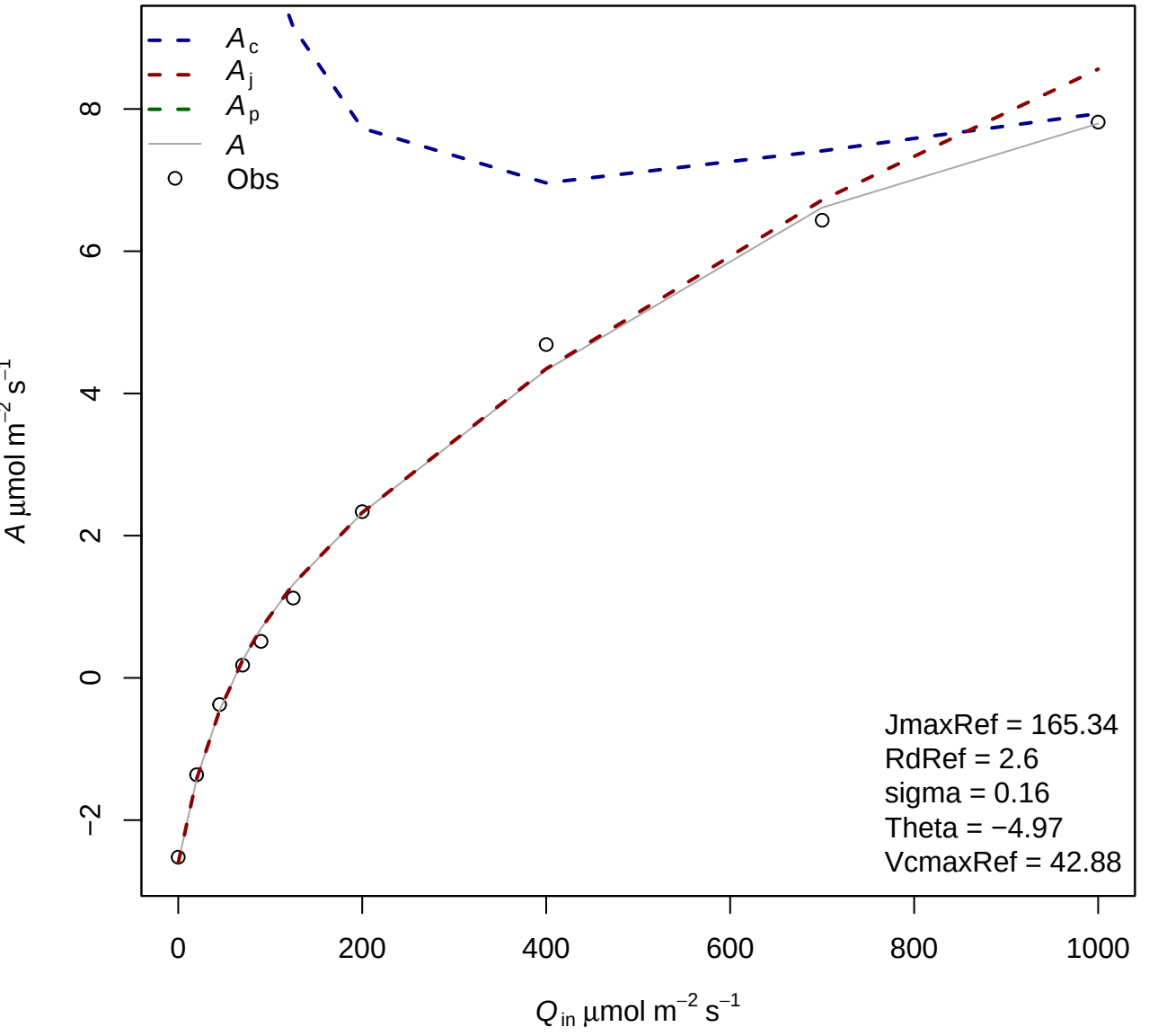

60
A_c
A_j
A_p
A
Obs
JmaxRef = 32.59
RdRef = 1.34
sigma = 0.07
Theta = 0.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

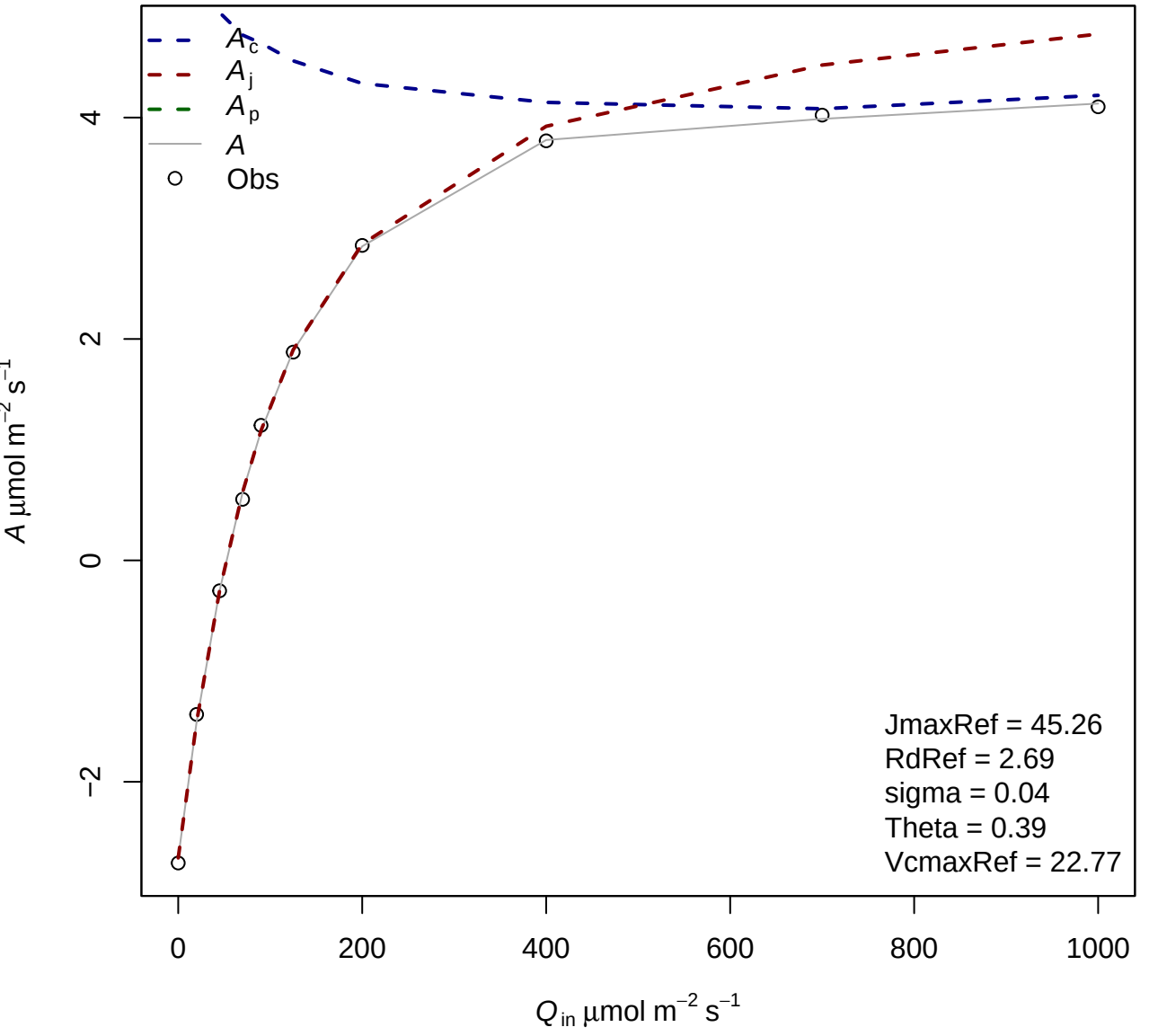

A_c
A_j
A_p
A
Obs
JmaxRef = 26.03
RdRef = 1.71
sigma = 0.06
Theta = 0.57
VcmaxRef = 9999
2
1
0
−1
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

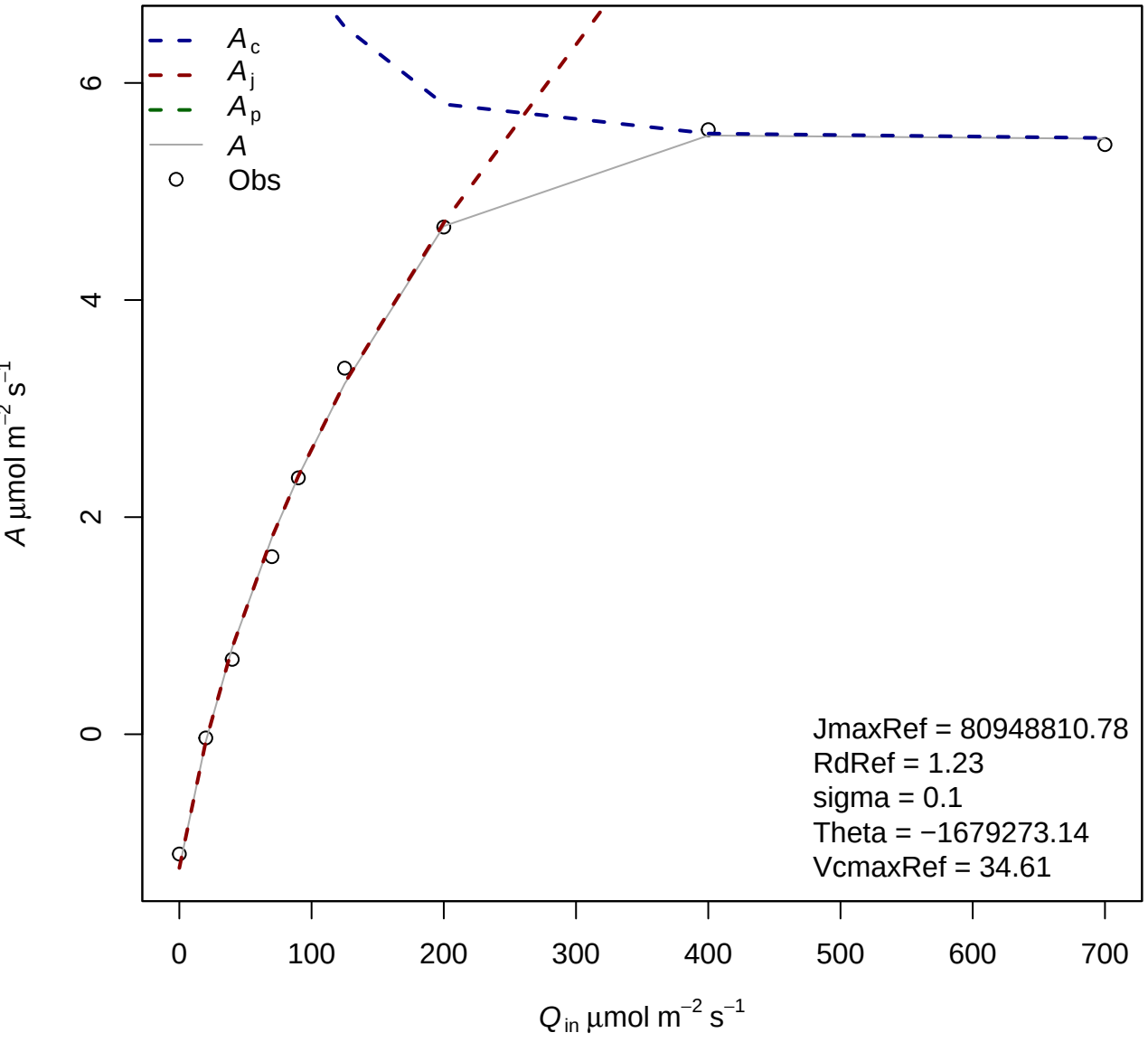

62
A_c
A_j
A_p
A
Obs
JmaxRef = 321.63
RdRef = 2.27
sigma = 0.08
Theta = −5.72
VcmaxRef = 28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

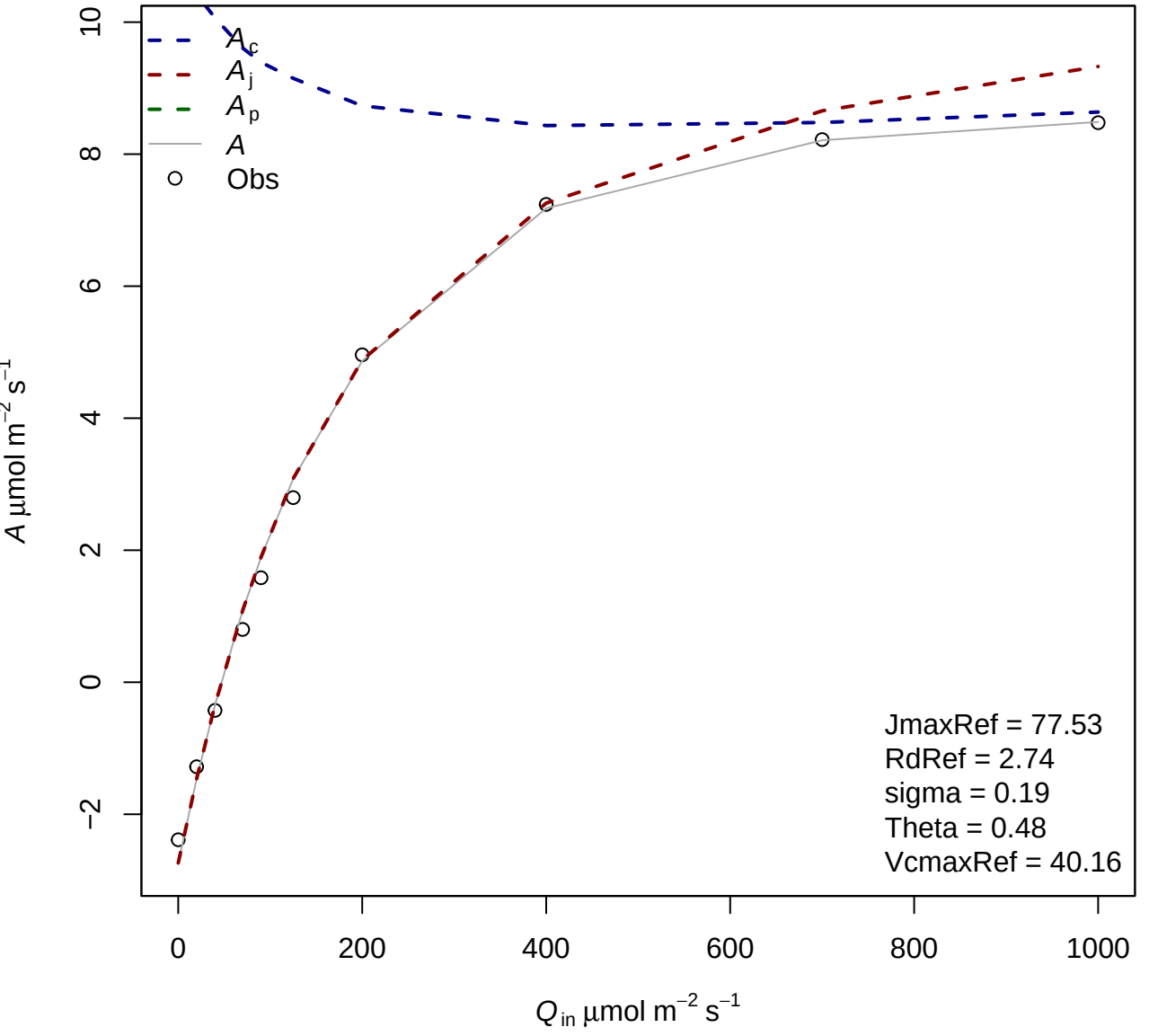

A_c
A_j
A_p
A
Obs
JmaxRef = 51.07
RdRef = 1.86
sigma = 0.08
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

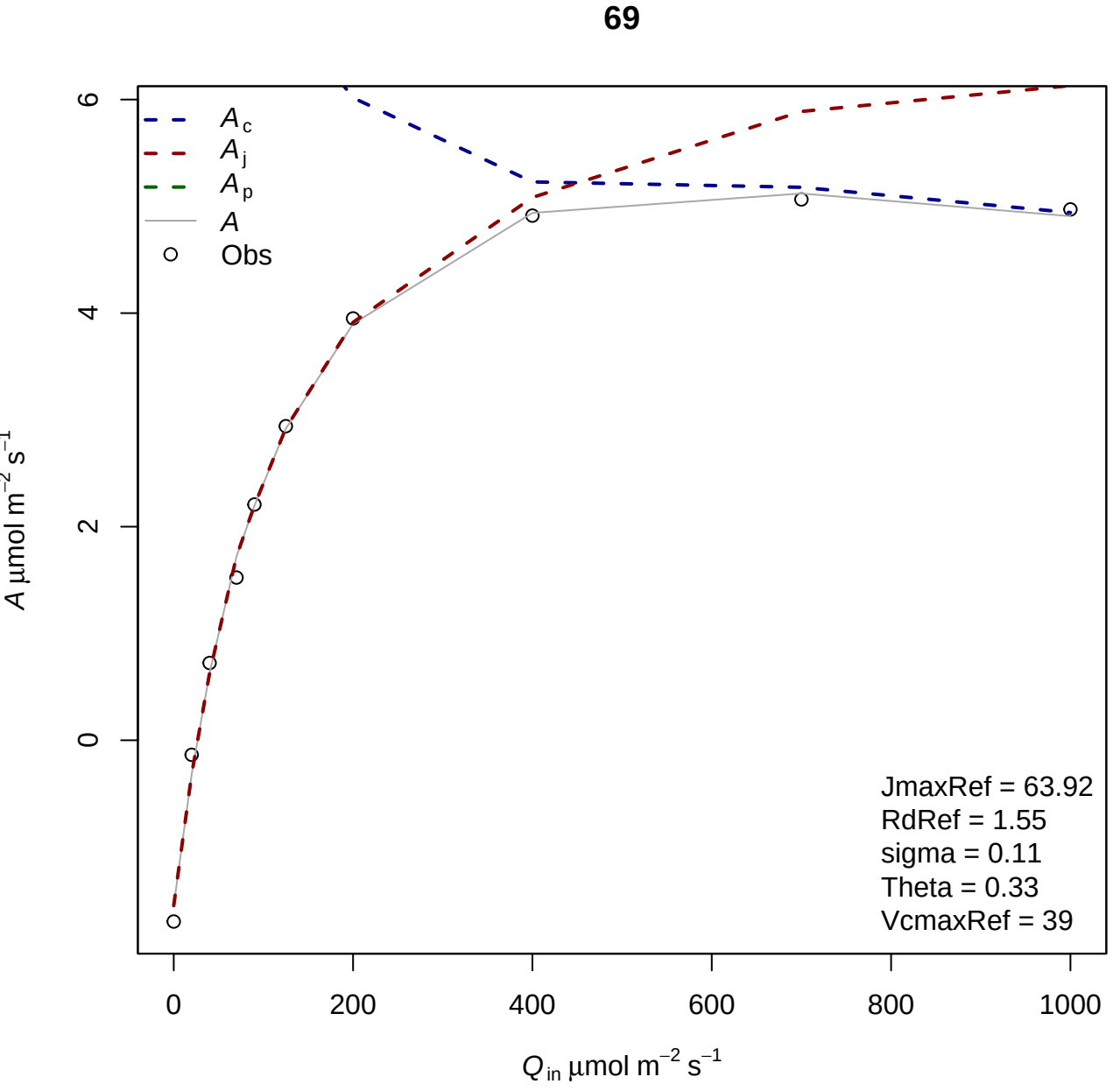

A_c
A_j
A_p
A
Obs
JmaxRef = 5187541.13
RdRef = 1.31
sigma = 0.21
Theta = −117052.28
VcmaxRef = 28.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000

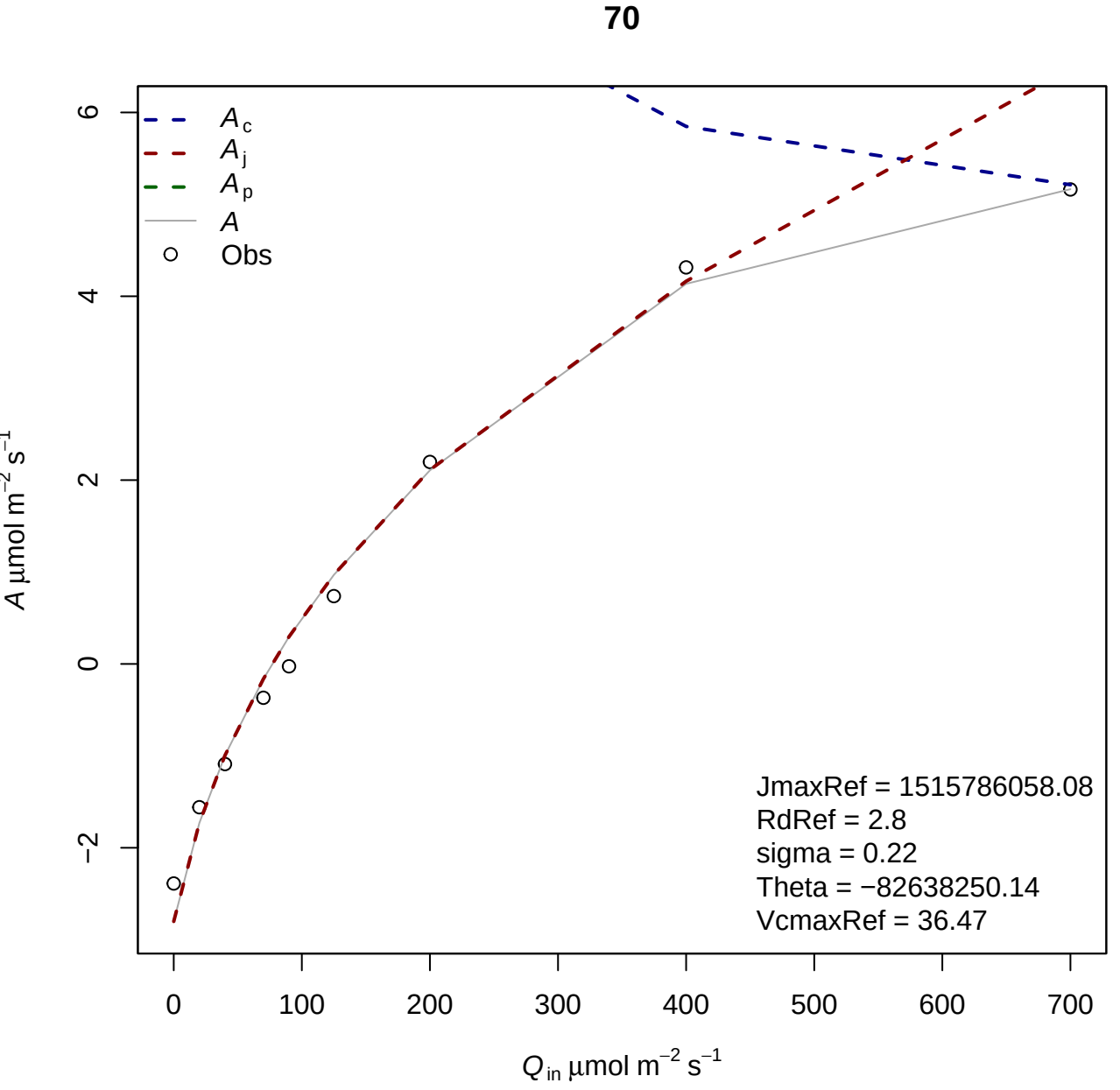

A_c
A_j
A_p
A
Obs
JmaxRef = 165.34
RdRef = 2.6
sigma = 0.16
Theta = −4.97
VcmaxRef = 42.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

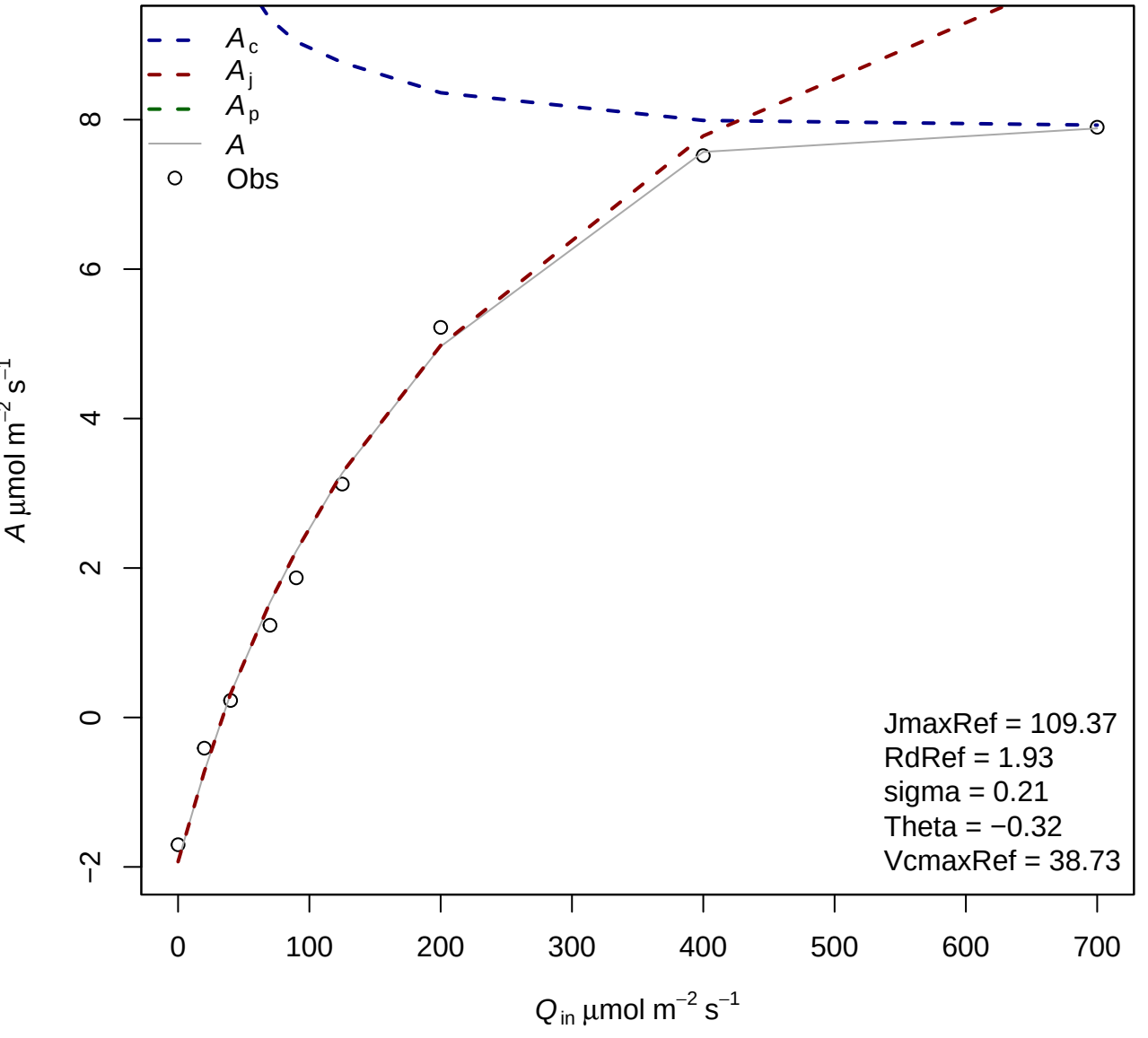

A_c
A_j
A_p
A
Obs
JmaxRef = 45.26
RdRef = 2.69
sigma = 0.04
Theta = 0.39
VcmaxRef = 22.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

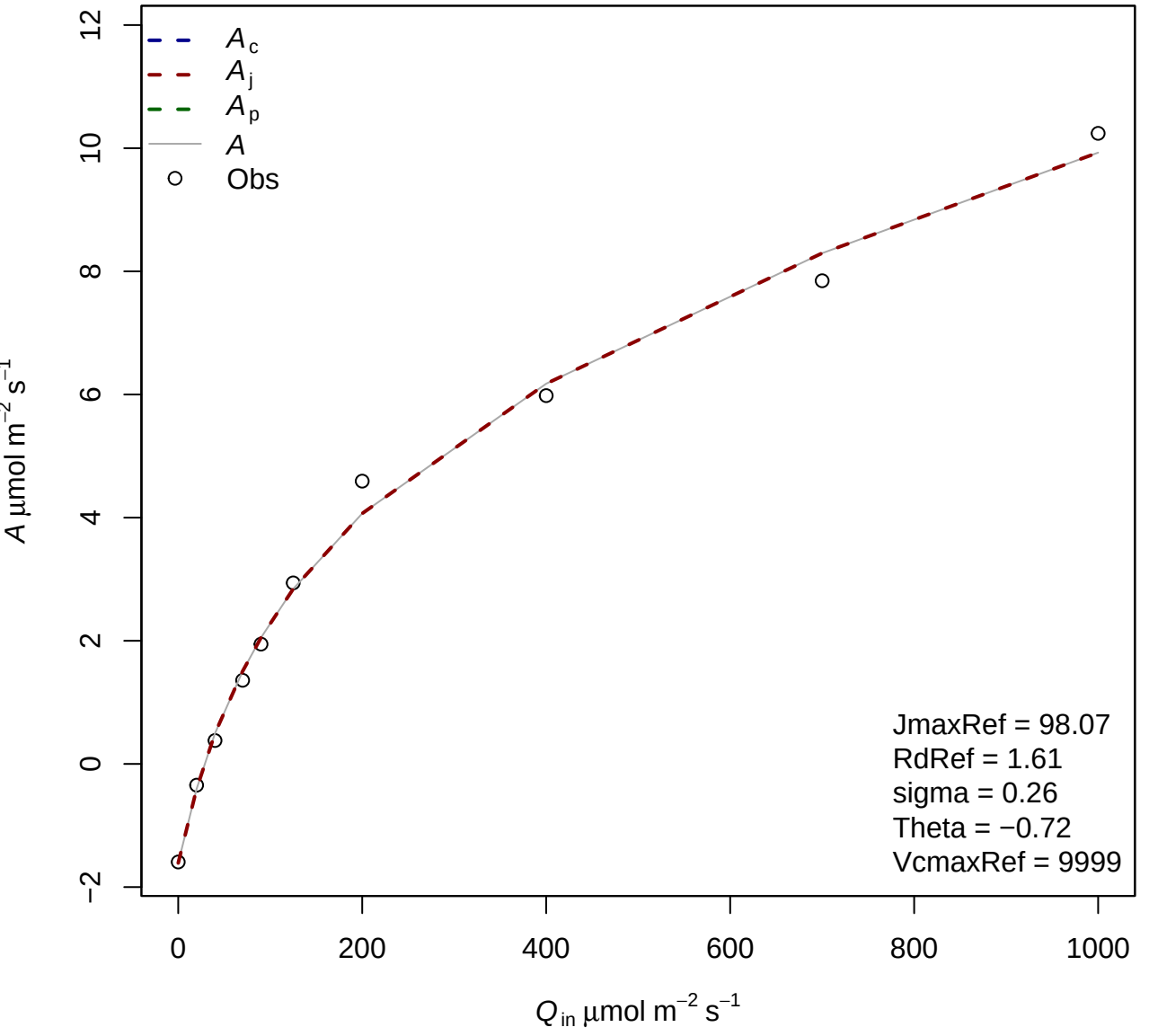

A_c
A_j
A_p
A
Obs
JmaxRef = 80948810.78
RdRef = 1.23
sigma = 0.1
Theta = −1679273.14
VcmaxRef = 34.61
0
2
4
6
0
100
200
300
400
500
600
700
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

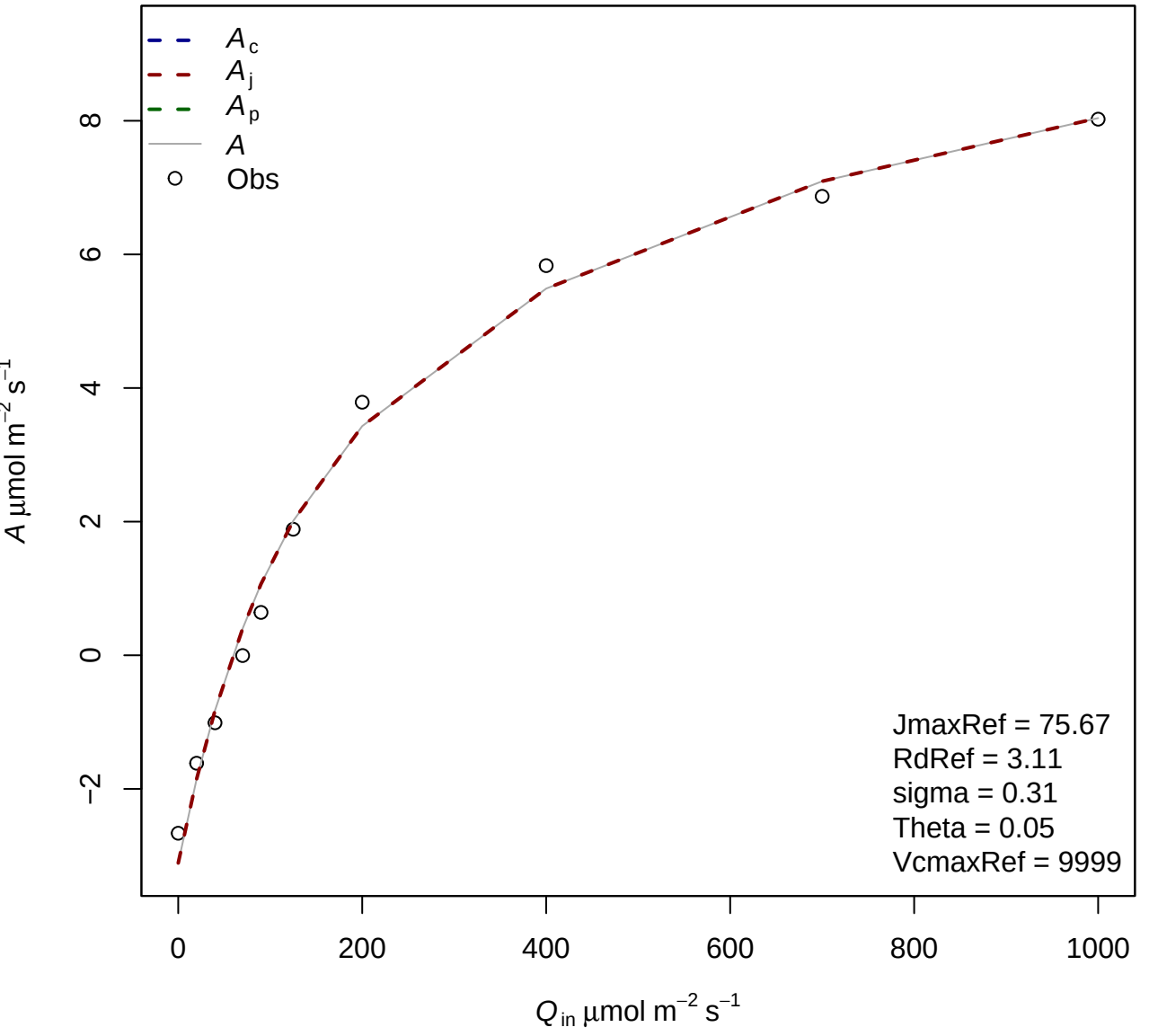

A_c
A_j
A_p
A
Obs
JmaxRef = 77.53
RdRef = 2.74
sigma = 0.19
Theta = 0.48
VcmaxRef = 40.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

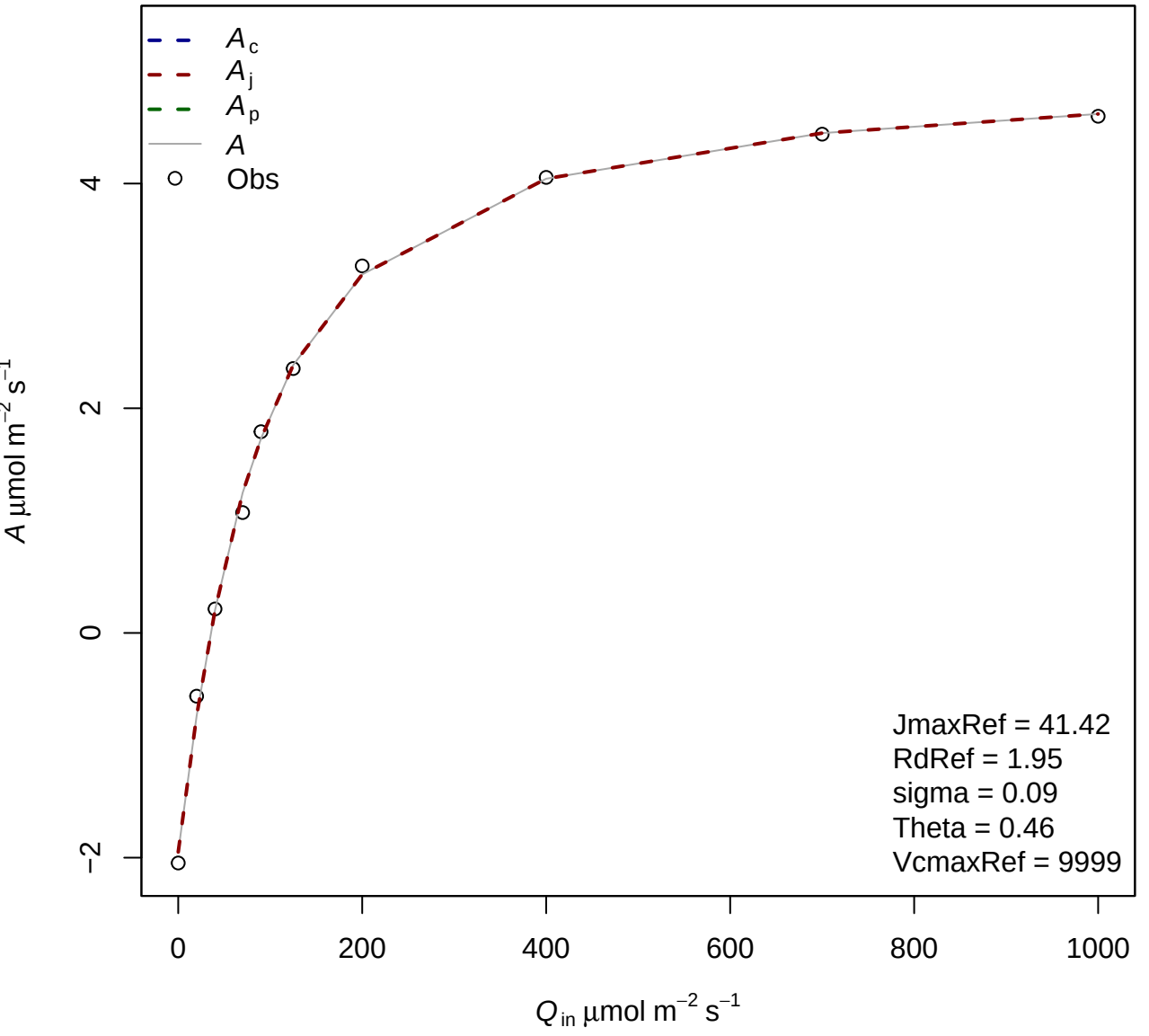

69
A_c
A_j
A_p
A
Obs
JmaxRef = 63.92
RdRef = 1.55
sigma = 0.11
Theta = 0.33
VcmaxRef = 39
0
2
4
6
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

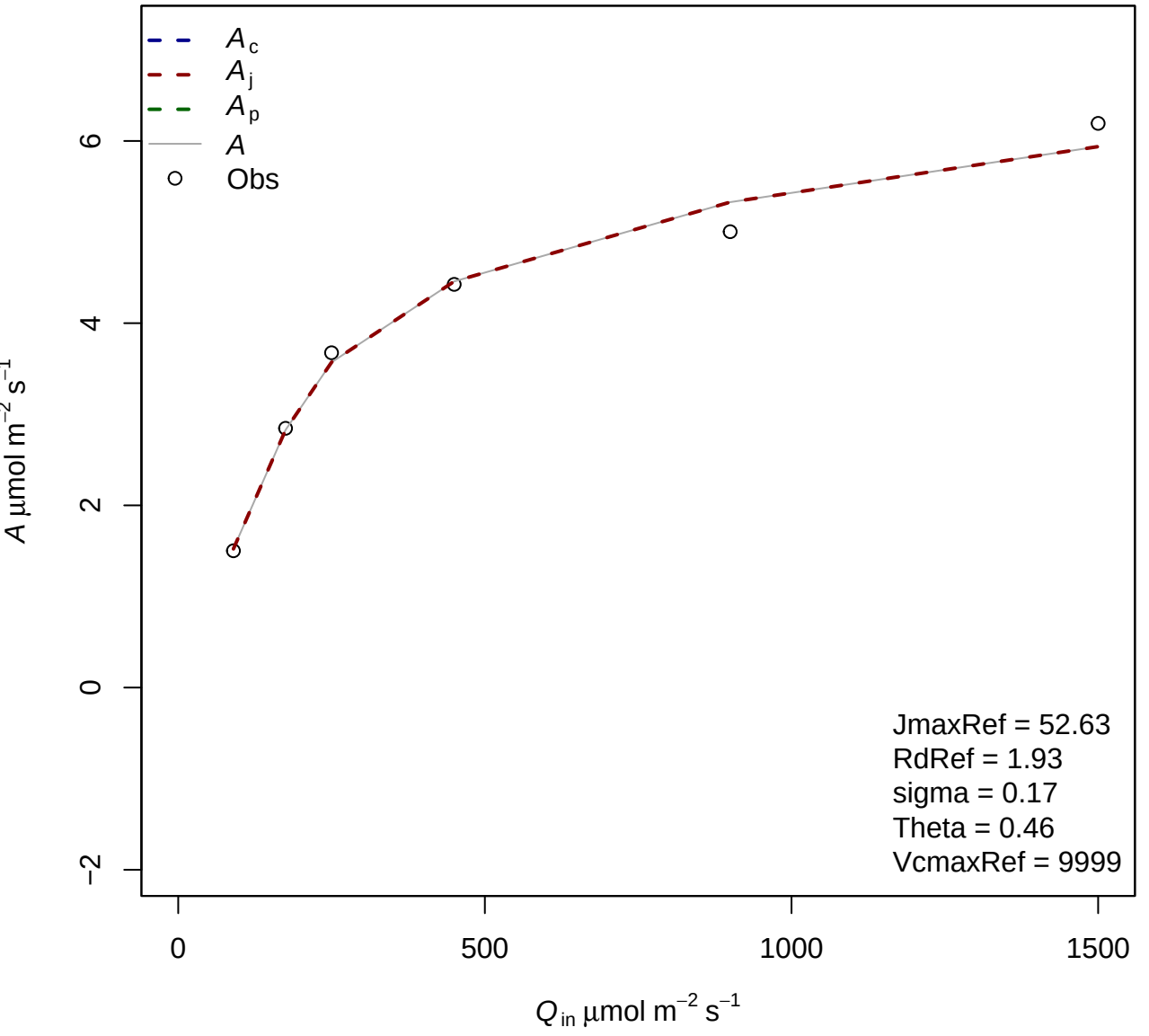

70
A_c
A_j
A_p
A
Obs
JmaxRef = 1515786058.08
RdRef = 2.8
sigma = 0.22
Theta = −82638250.14
VcmaxRef = 36.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
100
200
300
400
500
600
700

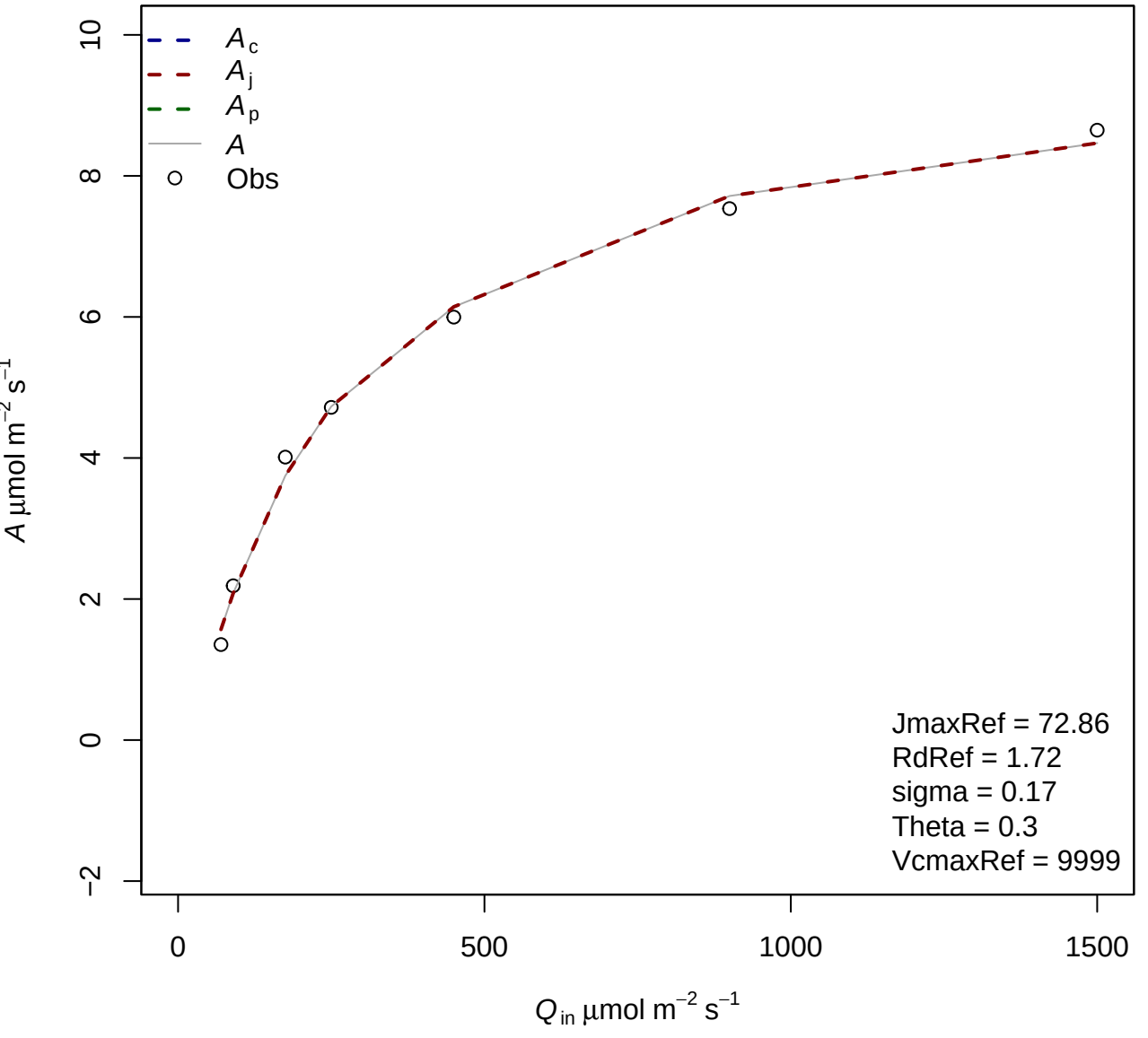

A_c
A_j
A_p
A
Obs
JmaxRef = 109.37
RdRef = 1.93
sigma = 0.21
Theta = −0.32
VcmaxRef = 38.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

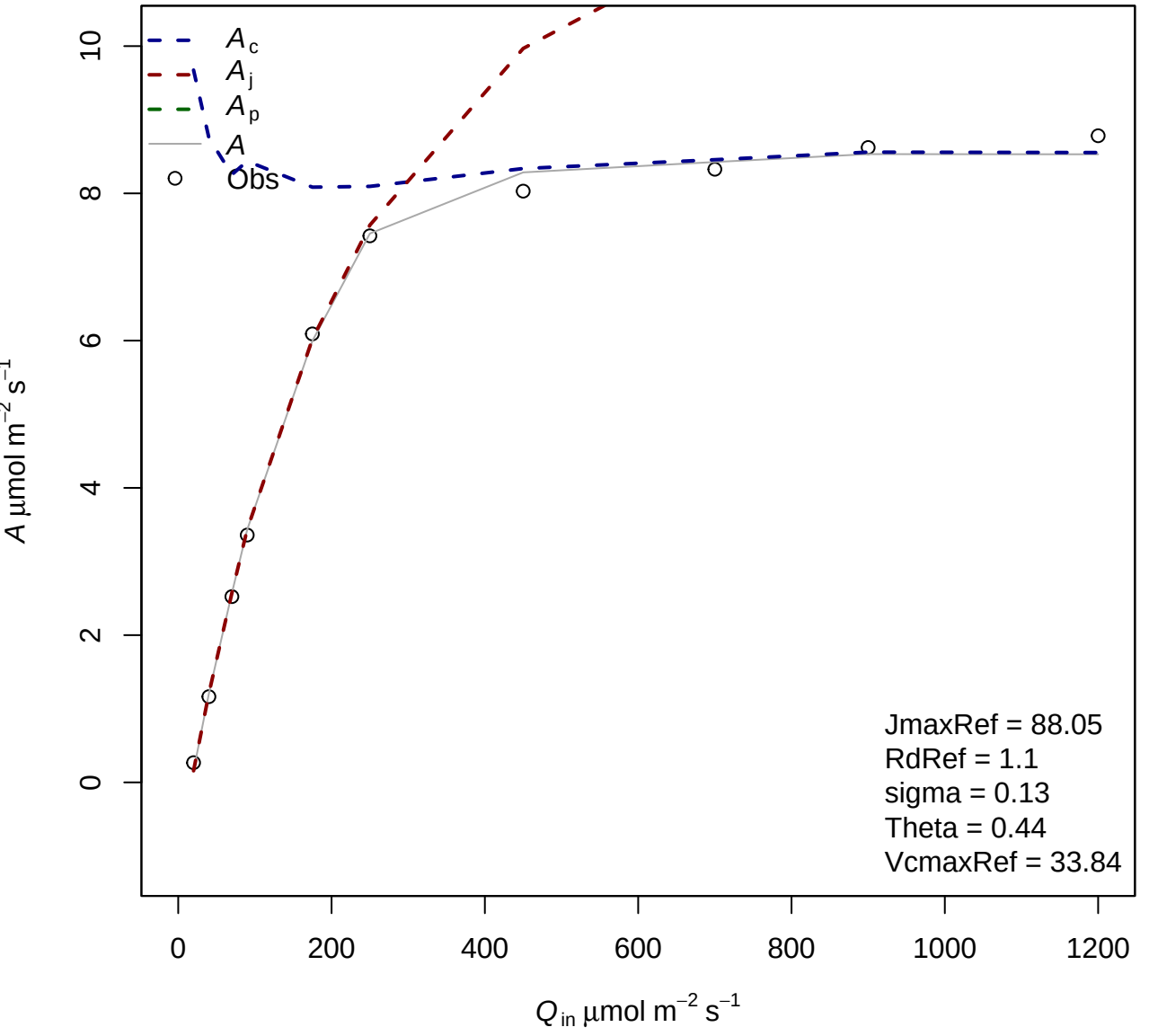

A_c
A_j
A_p
A
Obs
JmaxRef = 98.07
RdRef = 1.61
sigma = 0.26
Theta = −0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000

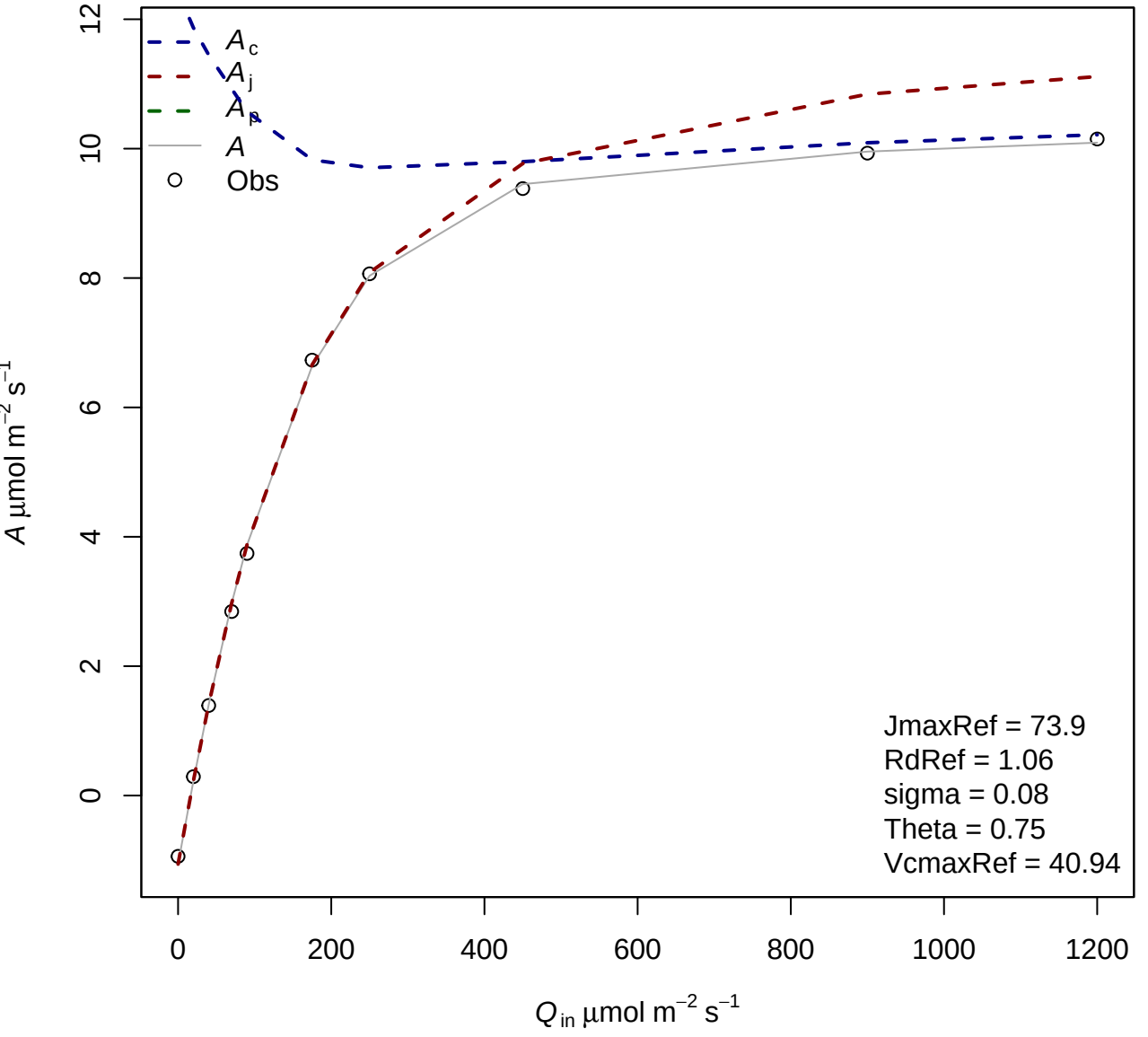

A_c
A_j
A_p
A
Obs
JmaxRef = 75.67
RdRef = 3.11
sigma = 0.31
Theta = 0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

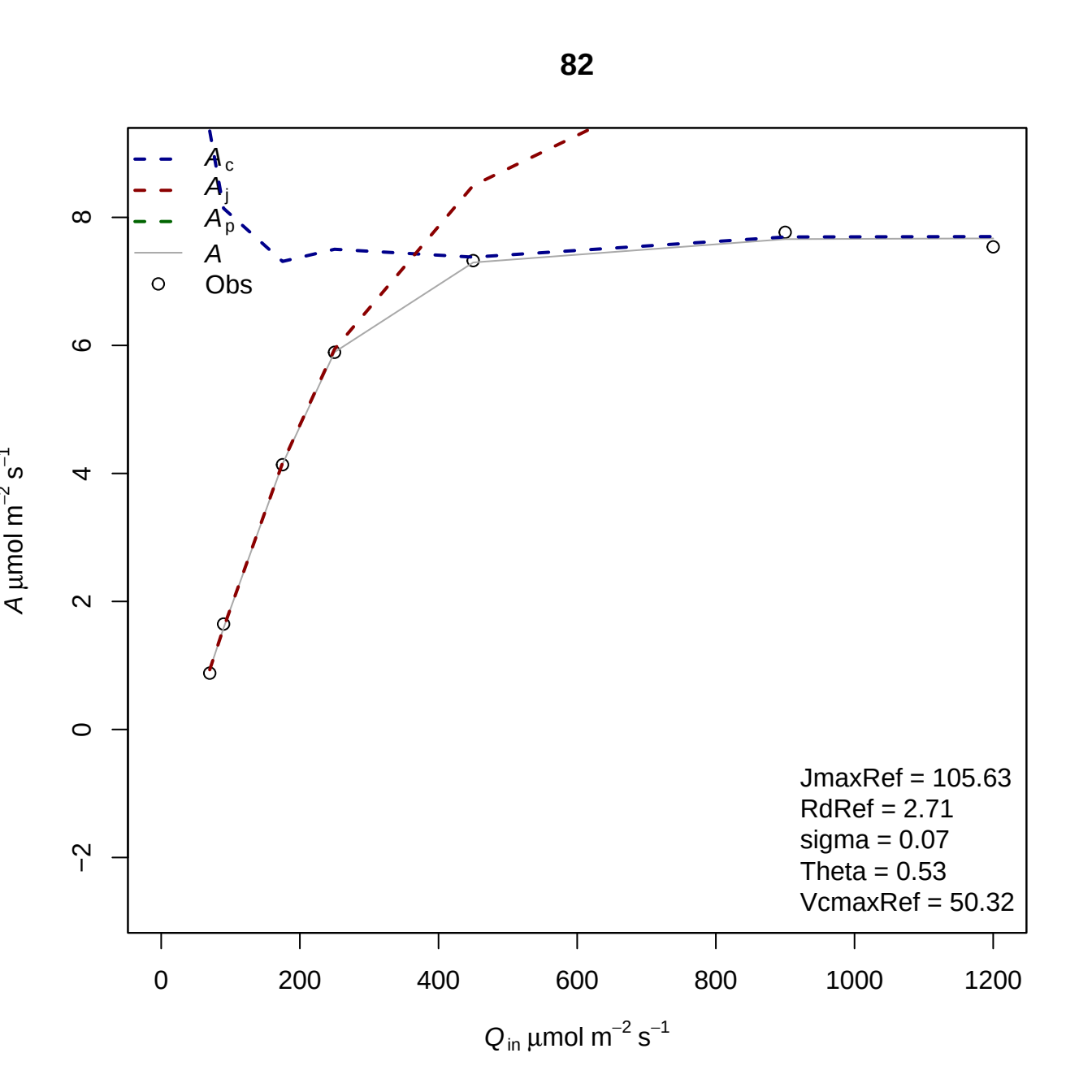

A_c
A_j
A_p
A
Obs
JmaxRef = 41.42
RdRef = 1.95
sigma = 0.09
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

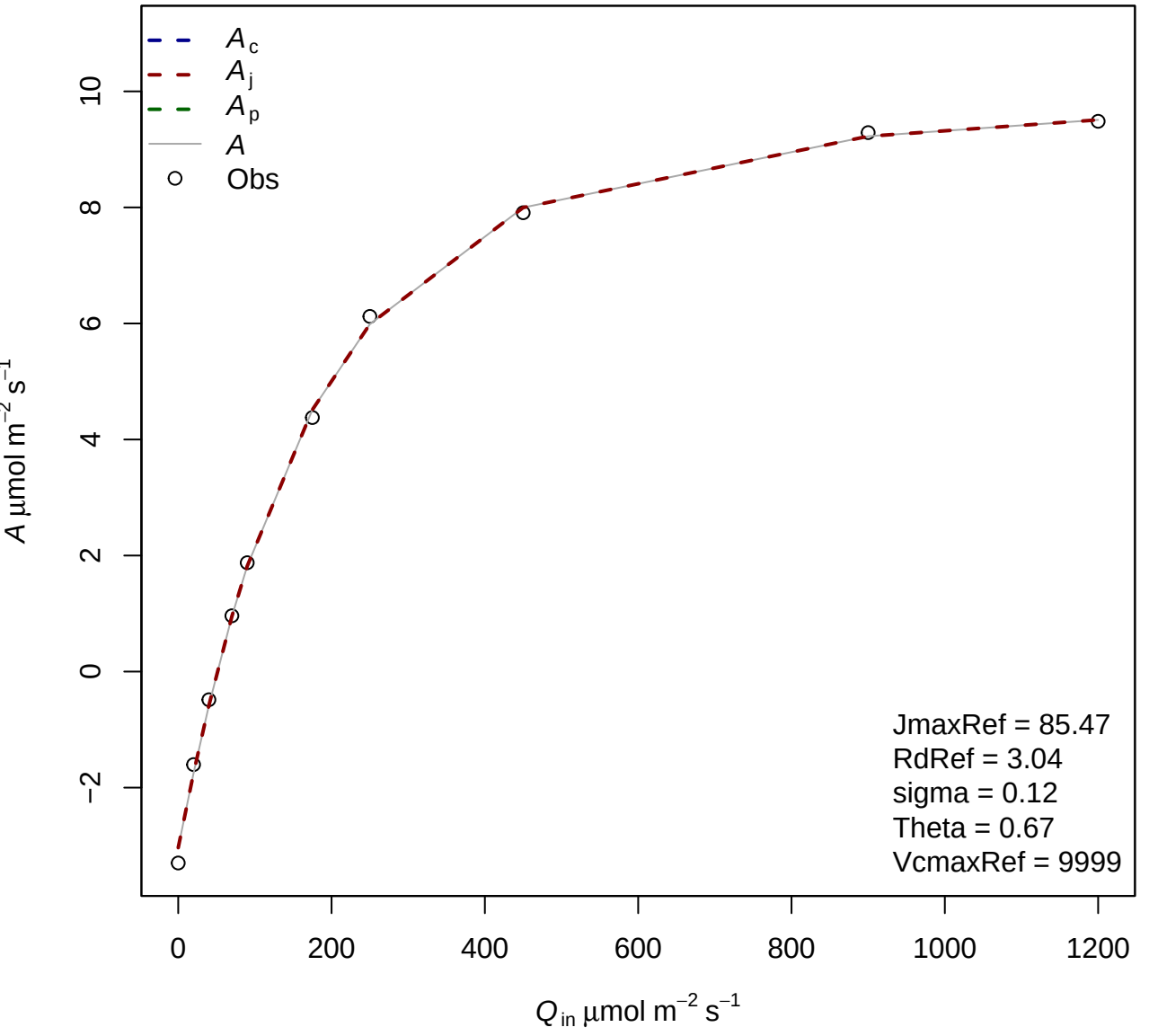

A_c
A_j
A_p
A
Obs
JmaxRef = 52.63
RdRef = 1.93
sigma = 0.17
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

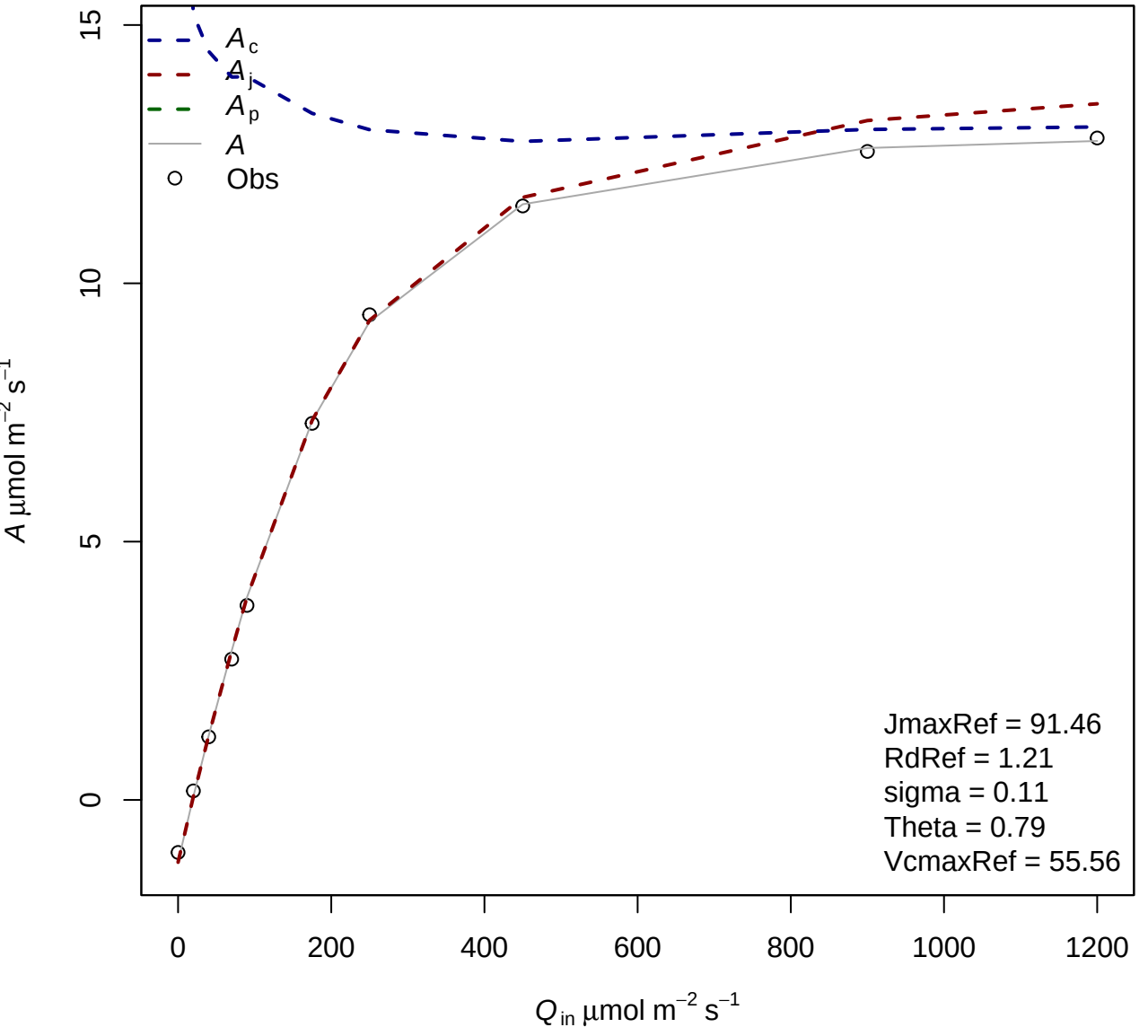

A_c
A_j
A_p
A
Obs
JmaxRef = 72.86
RdRef = 1.72
sigma = 0.17
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

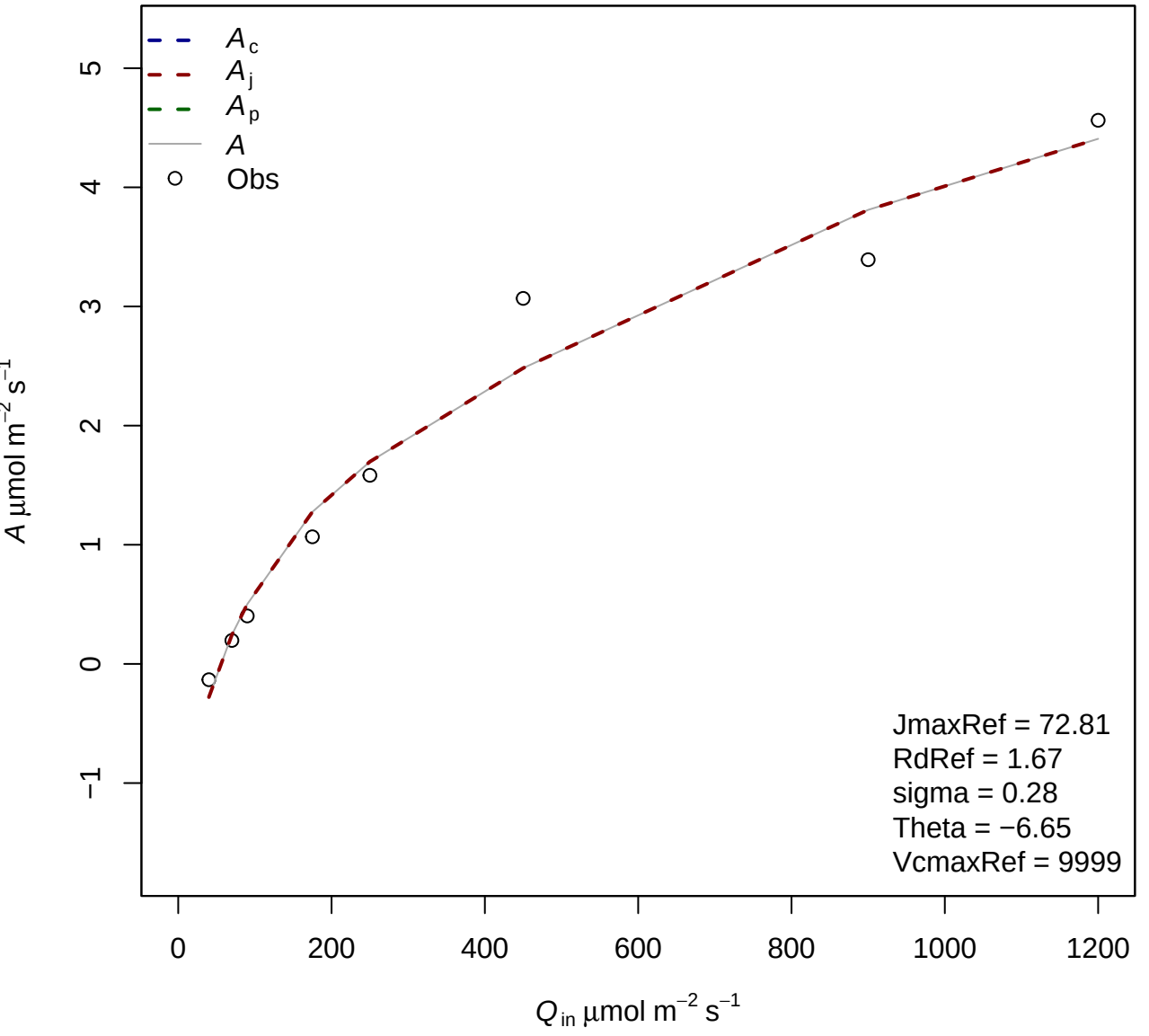

A_c
A_j
A_p
A
Obs
JmaxRef = 88.05
RdRef = 1.1
sigma = 0.13
Theta = 0.44
VcmaxRef = 33.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

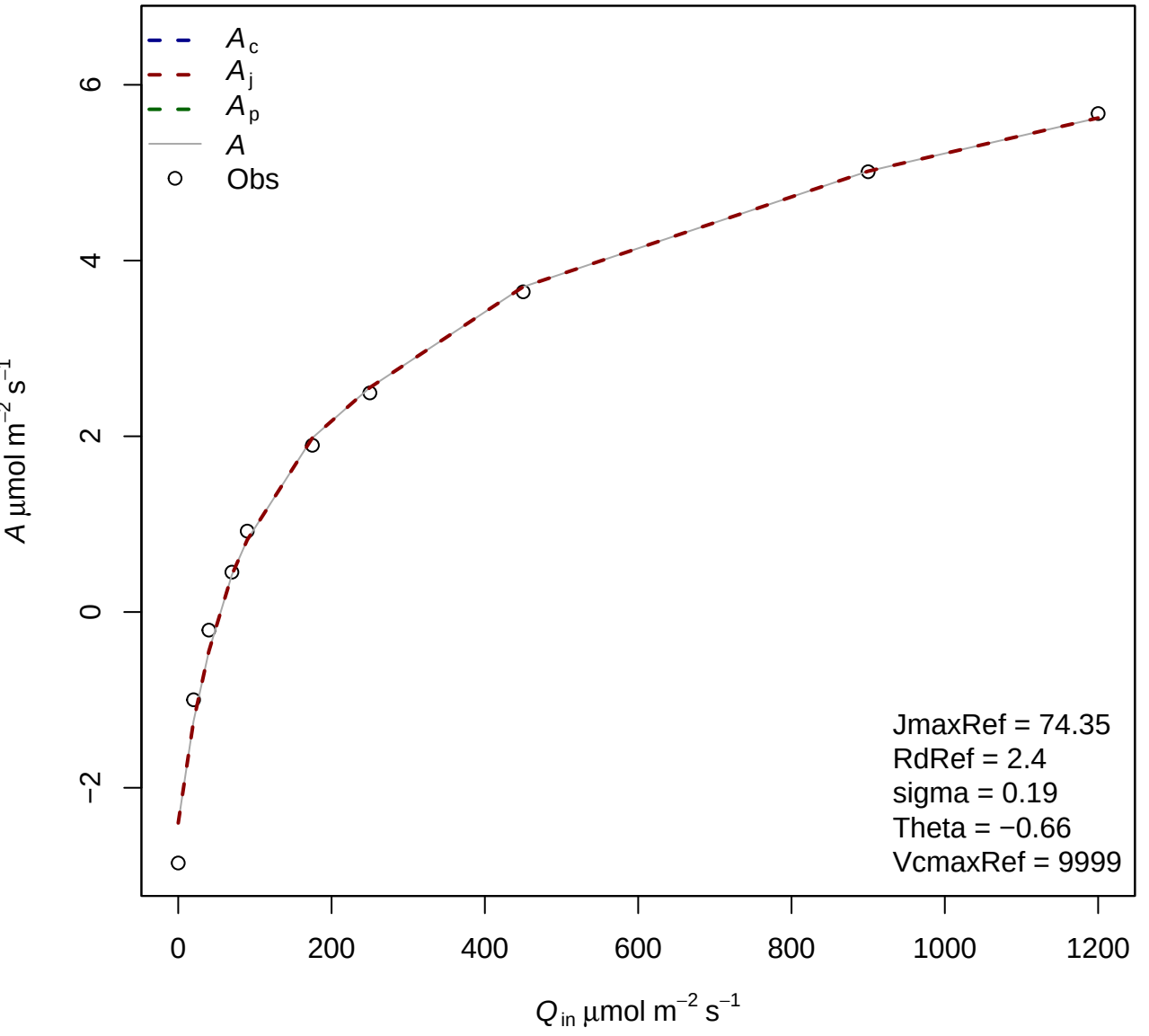

A_c
A_j
A_p
A
Obs
JmaxRef = 73.9
RdRef = 1.06
sigma = 0.08
Theta = 0.75
VcmaxRef = 40.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

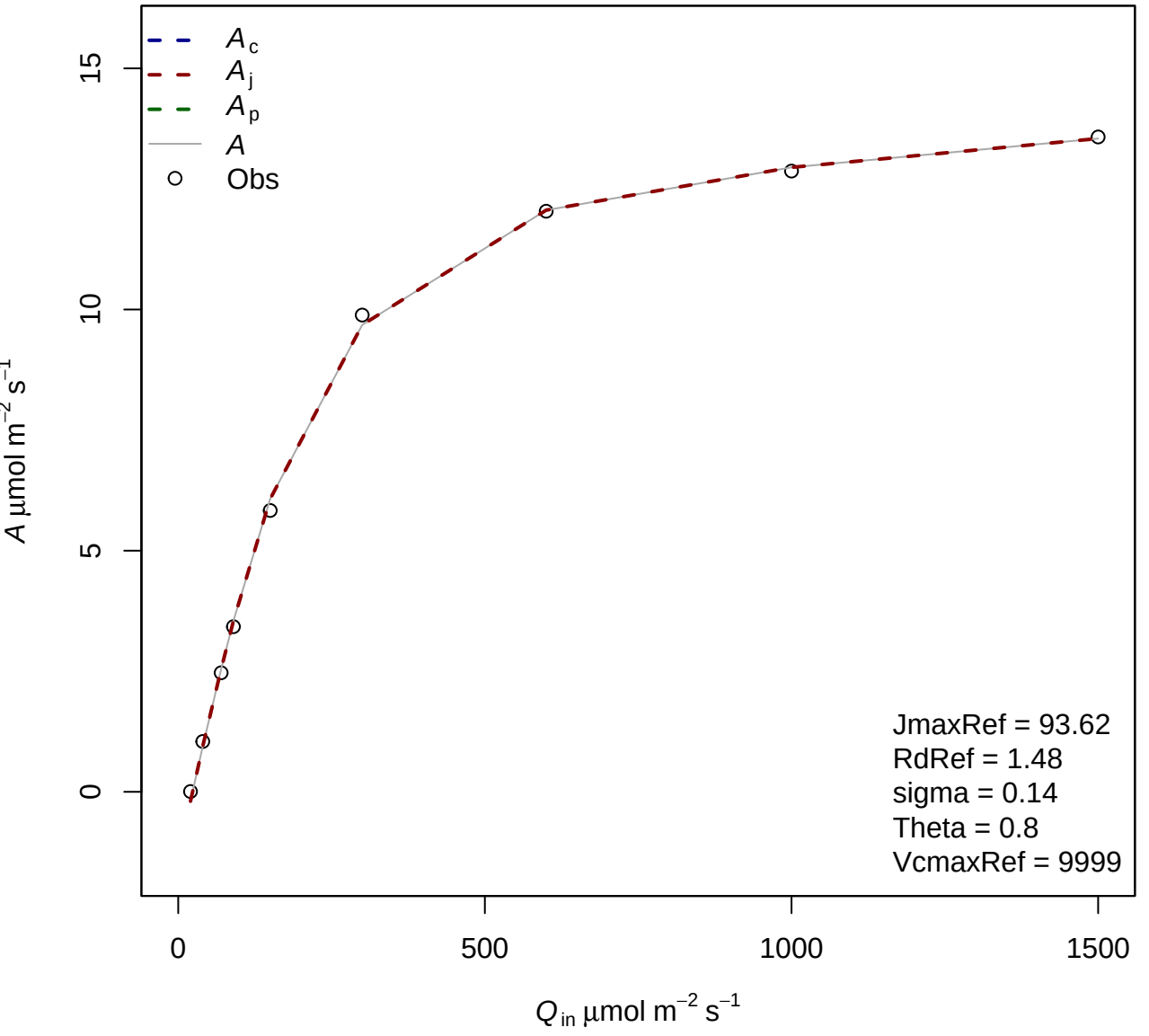

82
A_c
A_j
A_p
A
Obs
JmaxRef = 105.63
RdRef = 2.71
sigma = 0.07
Theta = 0.53
VcmaxRef = 50.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

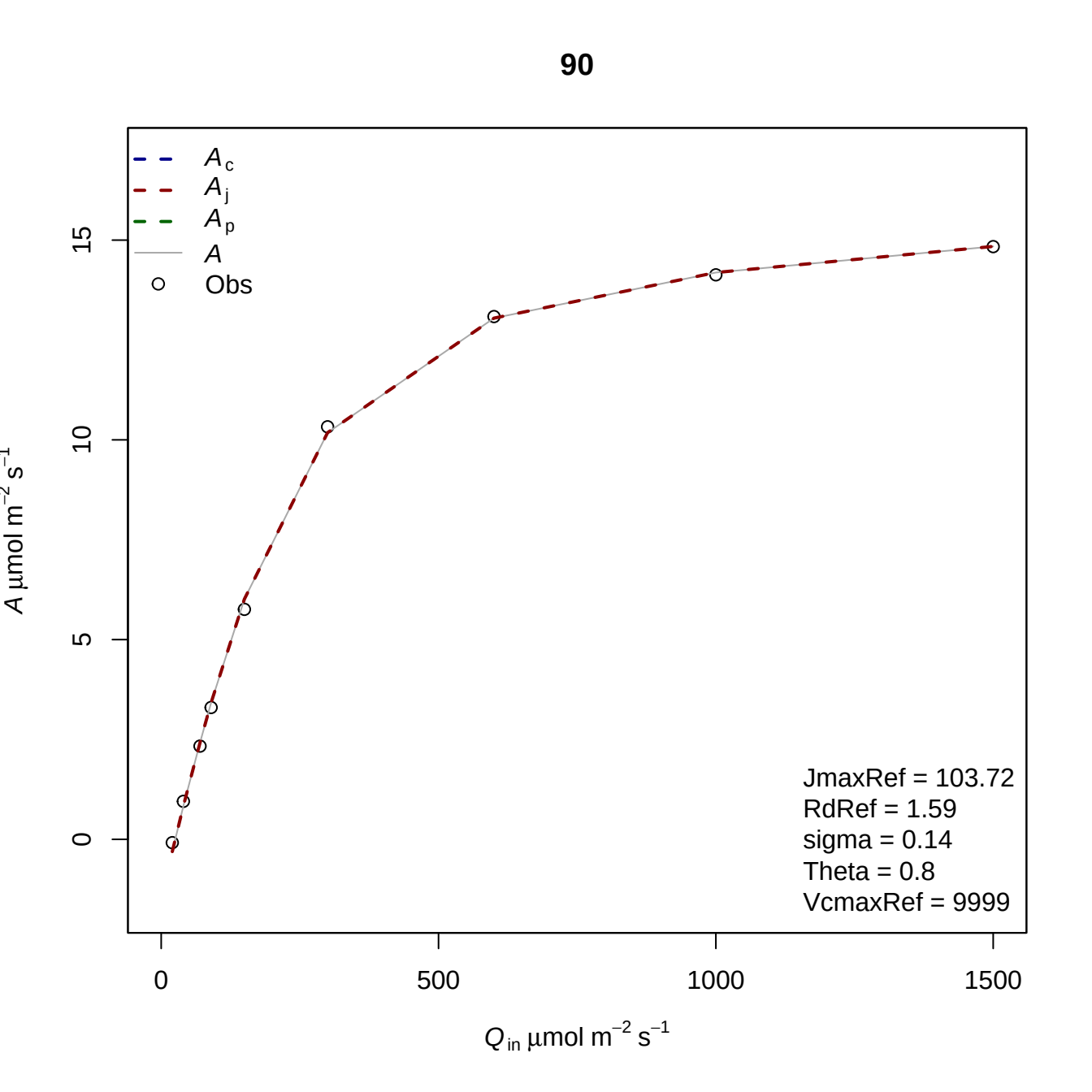

A_c
A_j
A_p
A
Obs
JmaxRef = 85.47
RdRef = 3.04
sigma = 0.12
Theta = 0.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

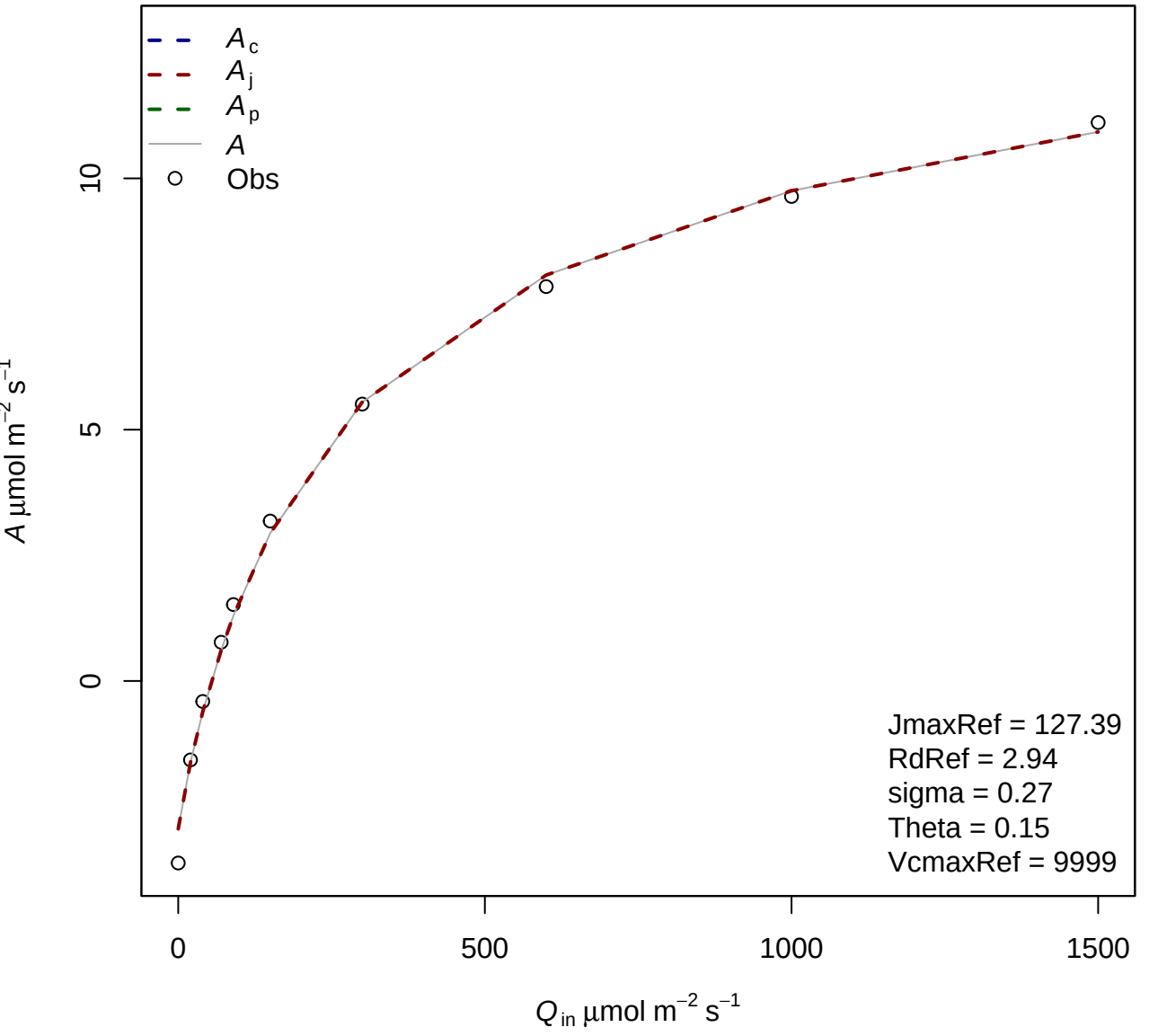

A_c
A_j
A_p
A
Obs
JmaxRef = 91.46
RdRef = 1.21
sigma = 0.11
Theta = 0.79
VcmaxRef = 55.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

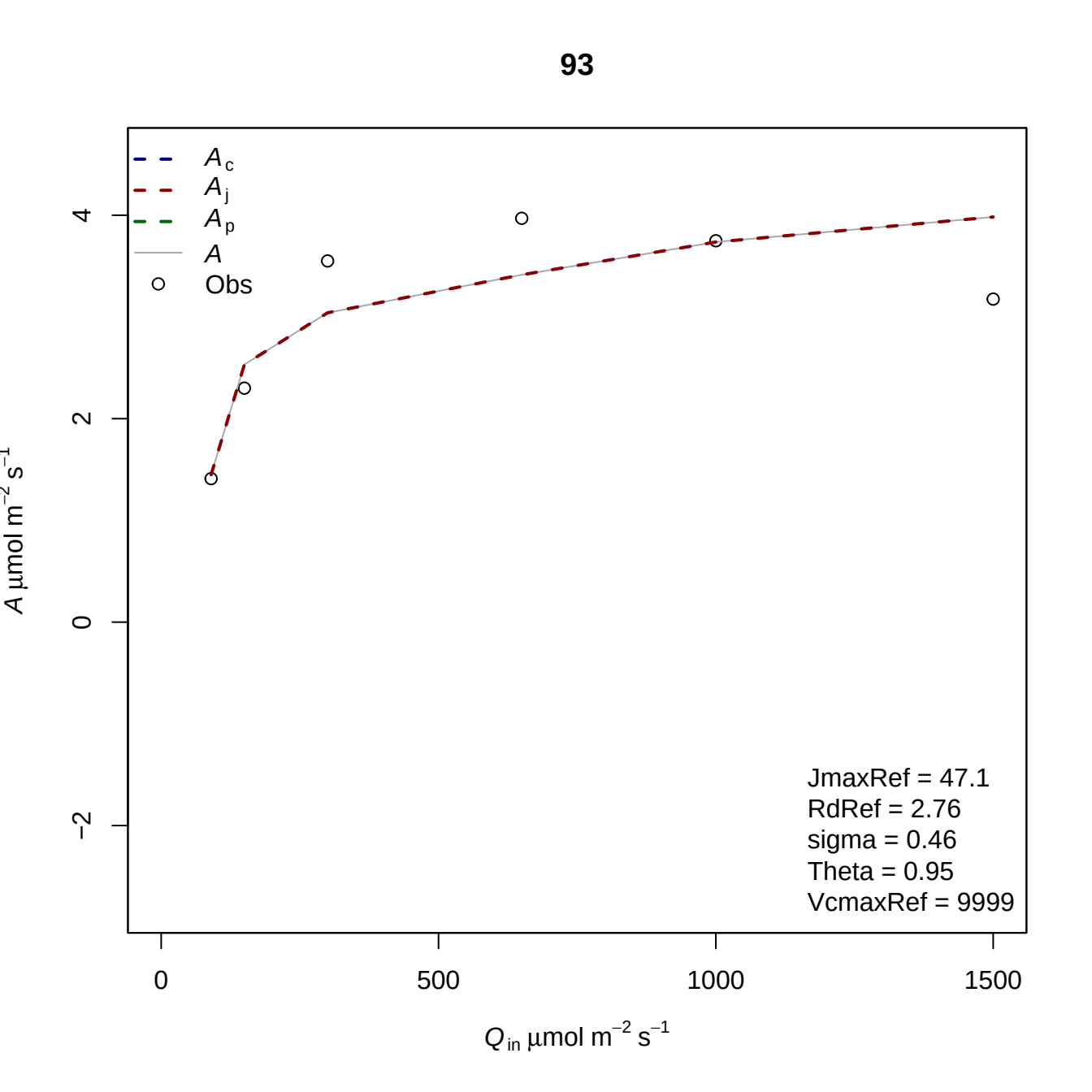

A_c
A_j
A_p
A
Obs
JmaxRef = 72.81
RdRef = 1.67
sigma = 0.28
Theta = −6.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

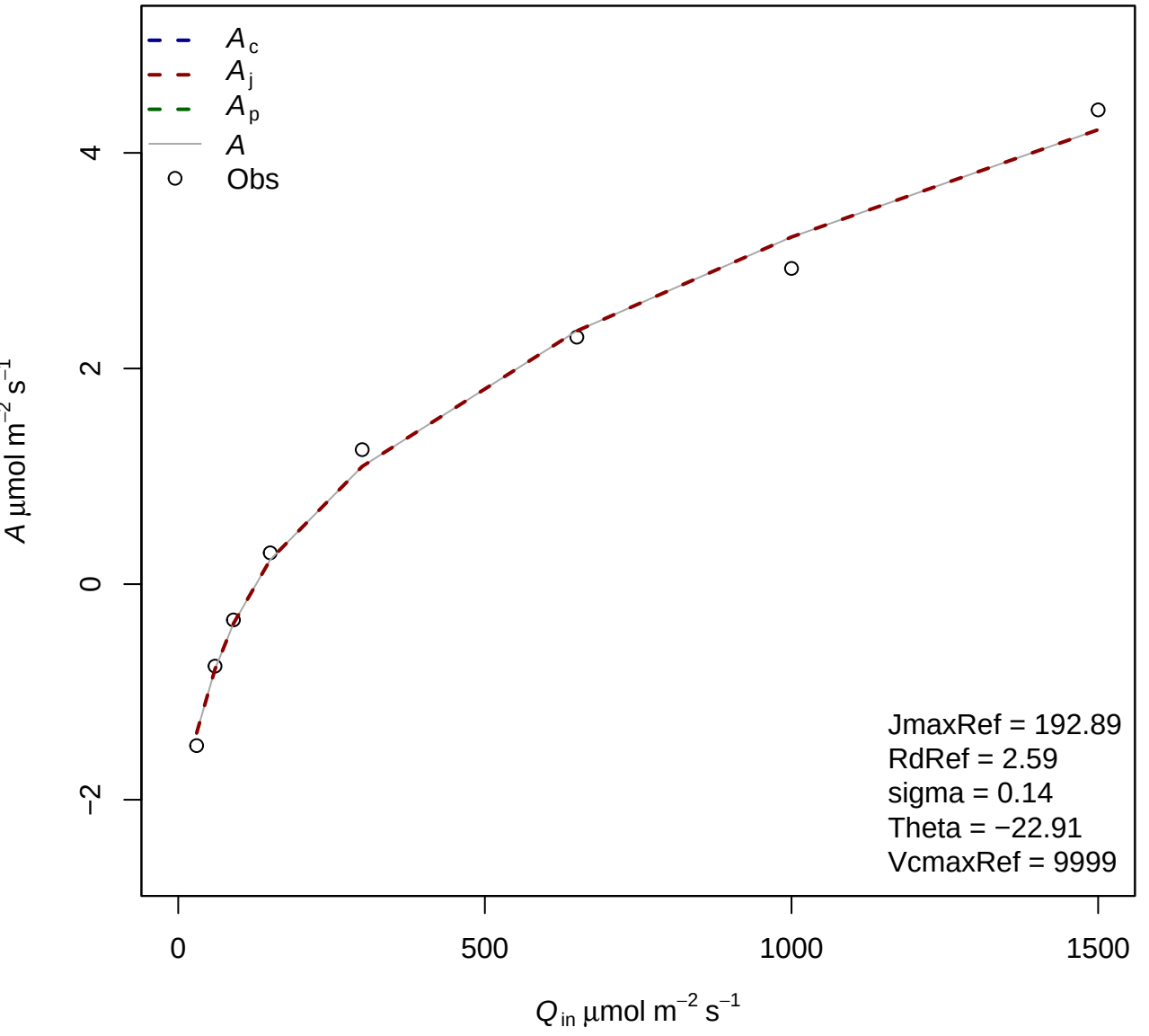

A_c
A_j
A_p
A
Obs
JmaxRef = 74.35
RdRef = 2.4
sigma = 0.19
Theta = −0.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

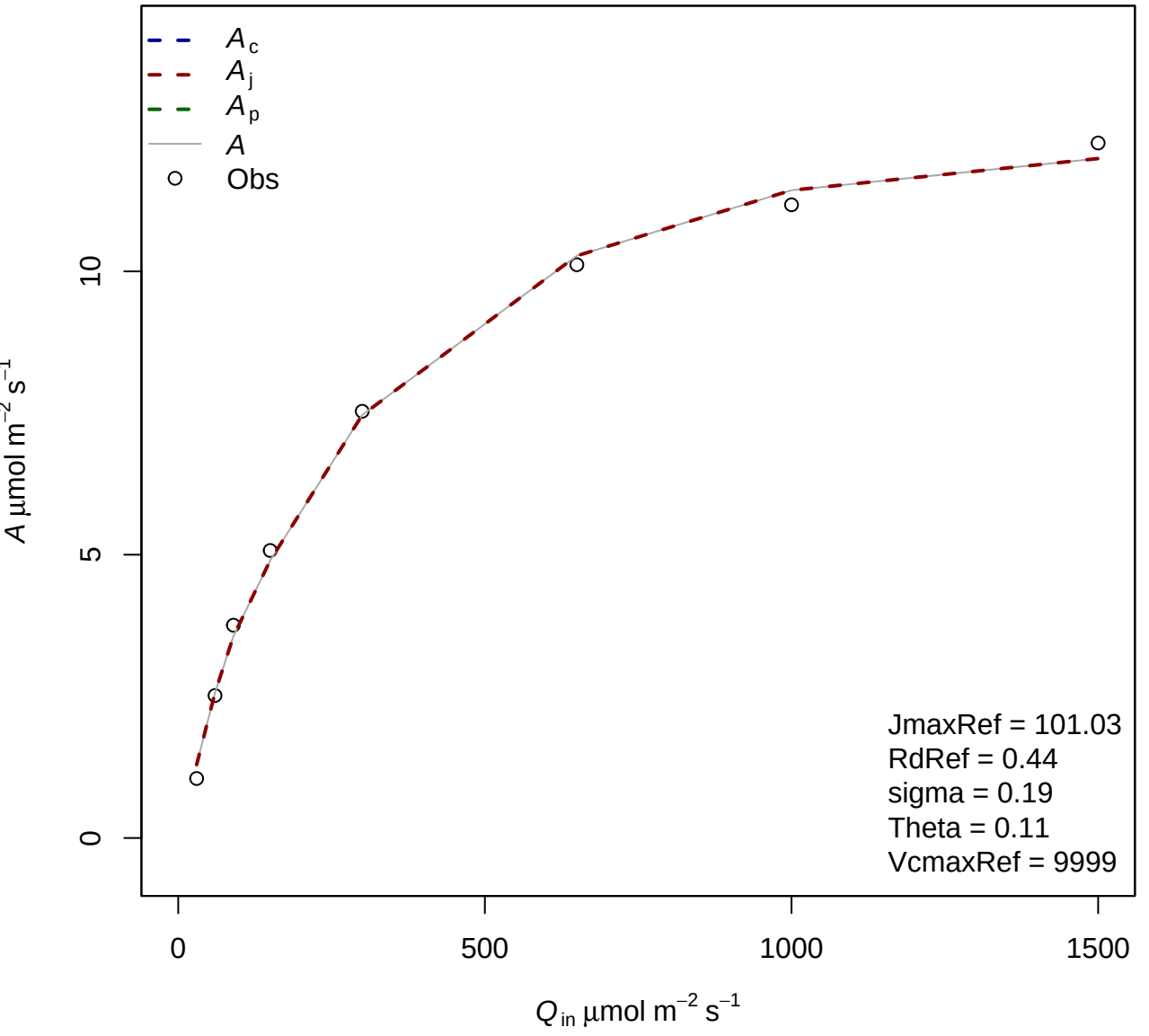

A_c
A_j
A_p
A
Obs
JmaxRef = 93.62
RdRef = 1.48
sigma = 0.14
Theta = 0.8
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

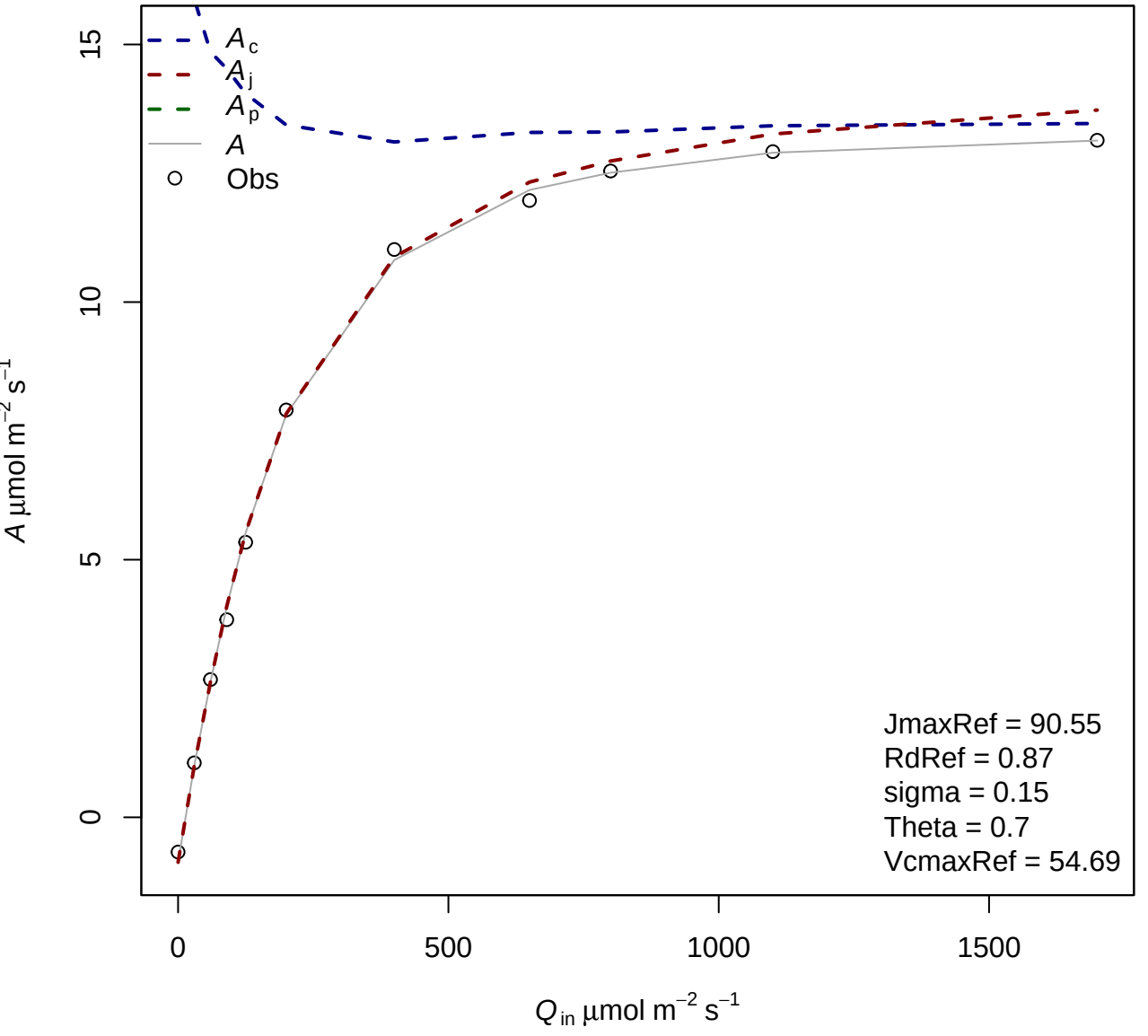

90
A_c
A_j
A_p
A
Obs
JmaxRef = 103.72
RdRef = 1.59
sigma = 0.14
Theta = 0.8
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

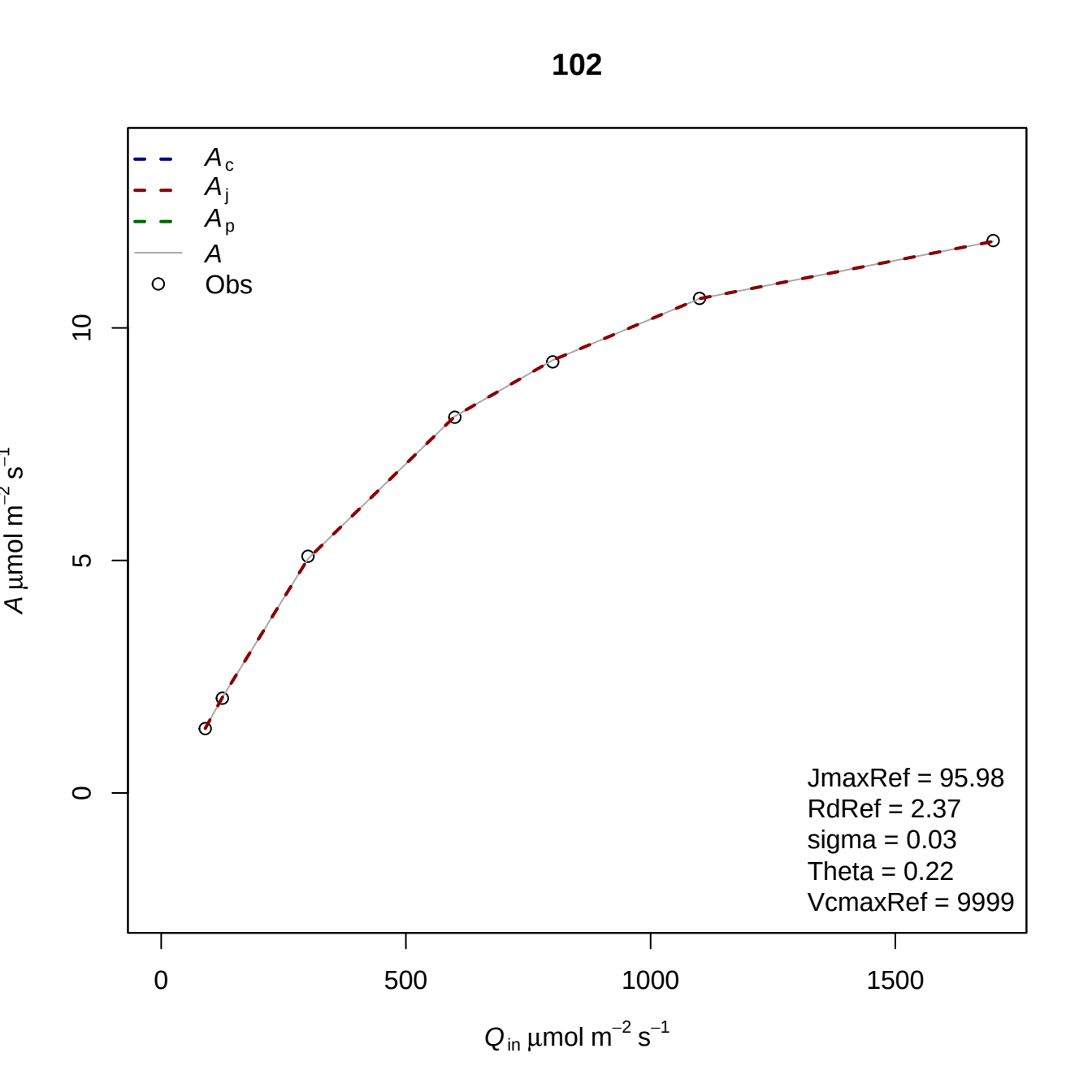

A_c
A_j
A_p
A
Obs
JmaxRef = 127.39
RdRef = 2.94
sigma = 0.27
Theta = 0.15
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

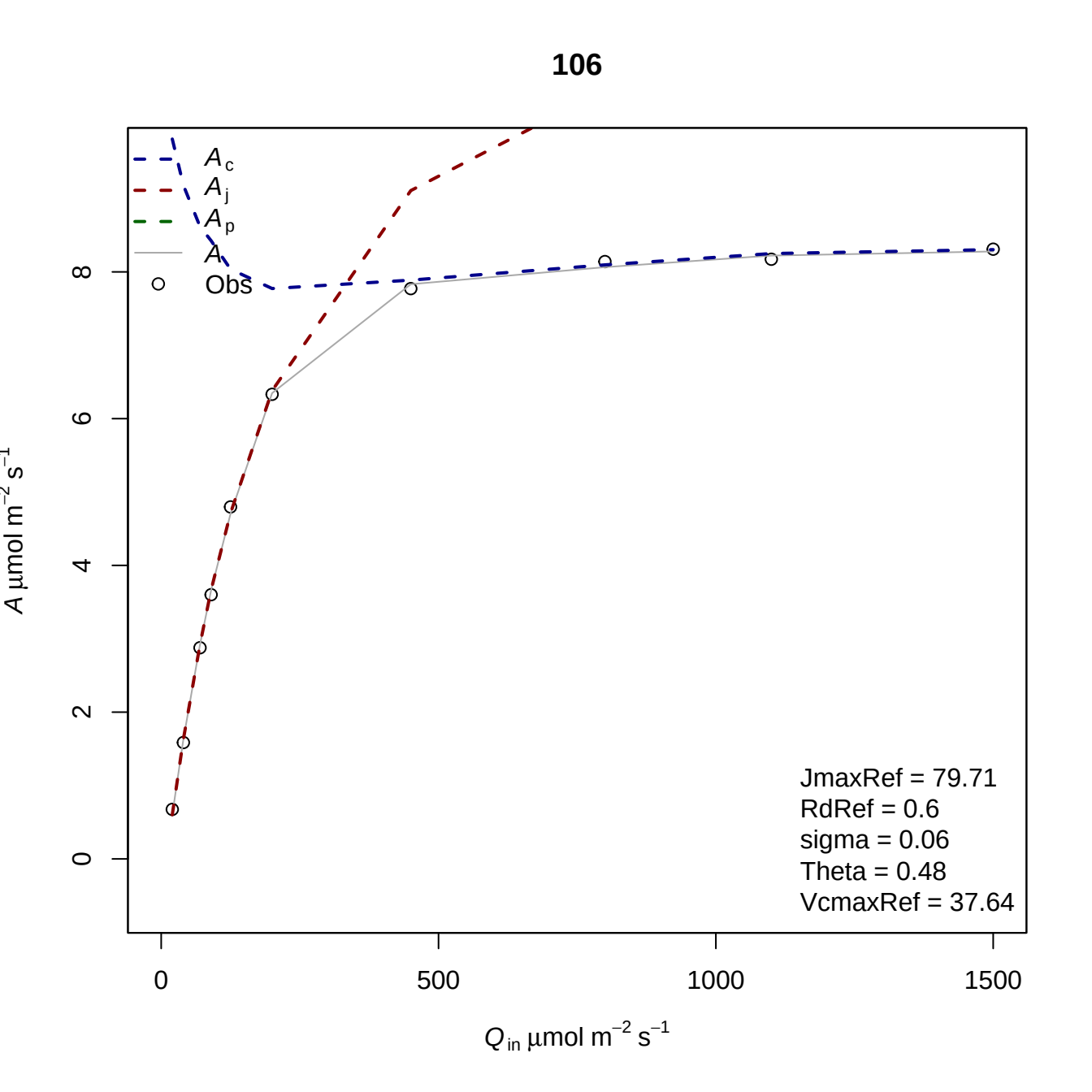

93
A_c
A_j
A_p
A
Obs
JmaxRef = 47.1
RdRef = 2.76
sigma = 0.46
Theta = 0.95
VcmaxRef = 9999
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

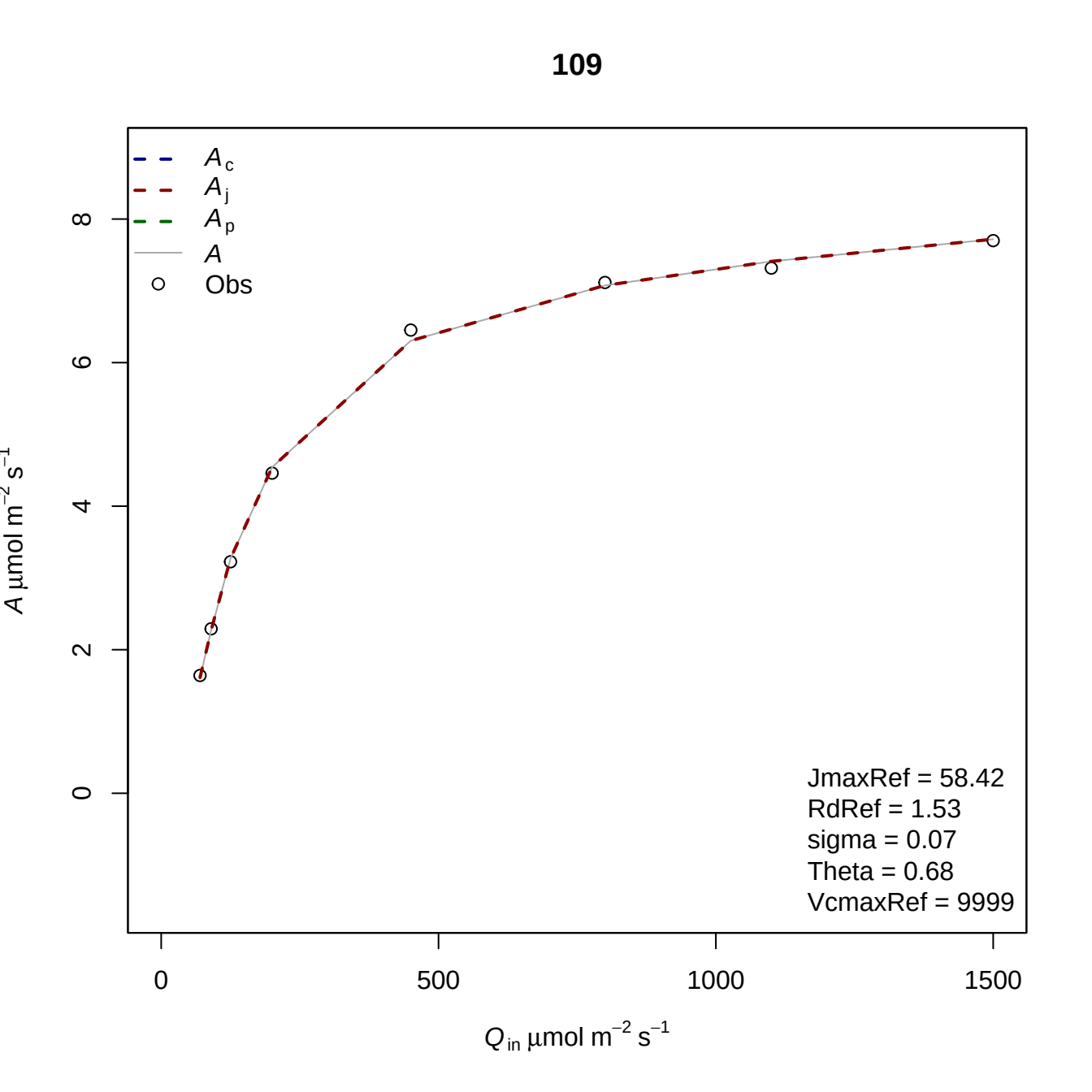

A_c
A_j
A_p
A
Obs
JmaxRef = 192.89
RdRef = 2.59
sigma = 0.14
Theta = −22.91
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

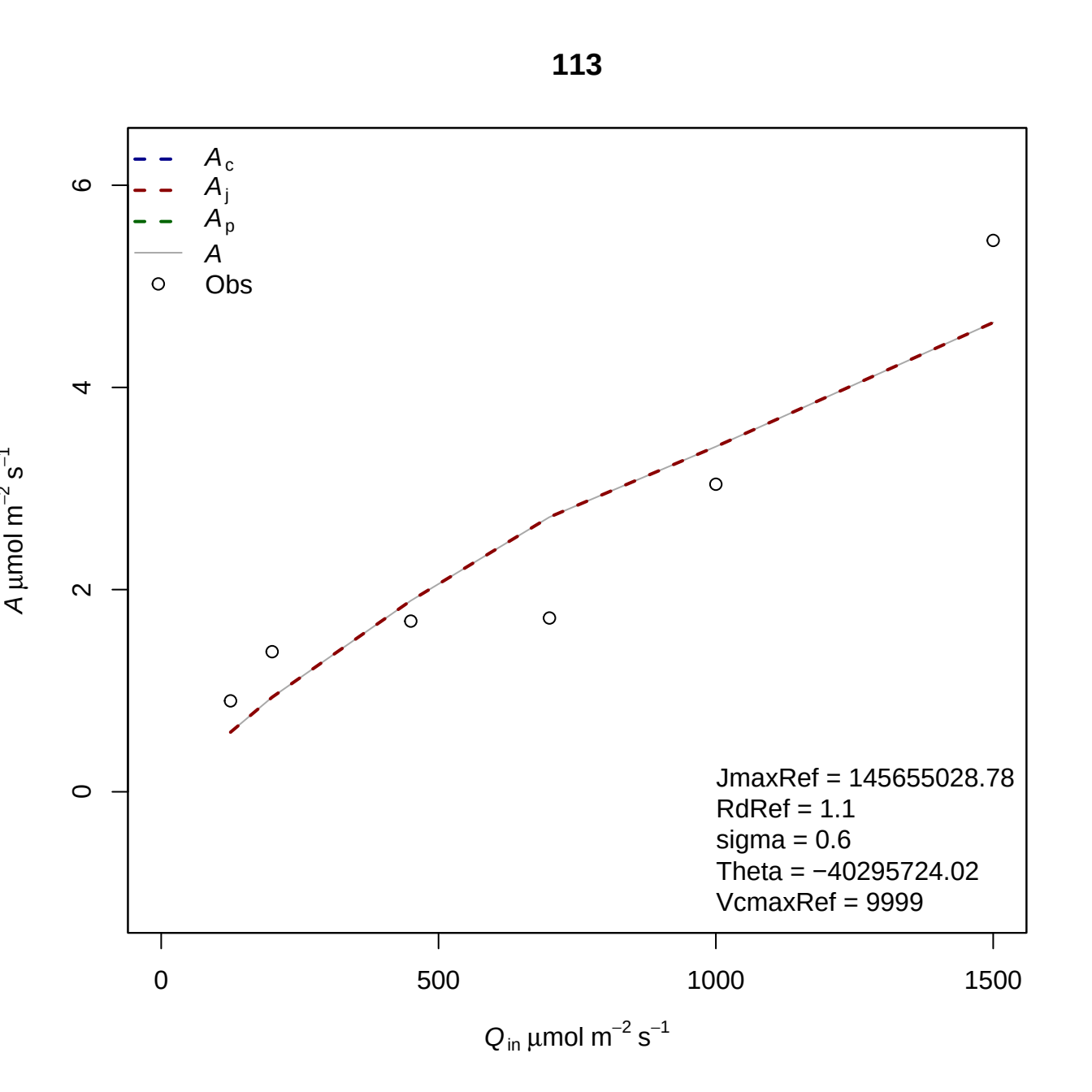

A_c
A_j
A_p
A
Obs
JmaxRef = 101.03
RdRef = 0.44
sigma = 0.19
Theta = 0.11
VcmaxRef = 9999
0
5
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

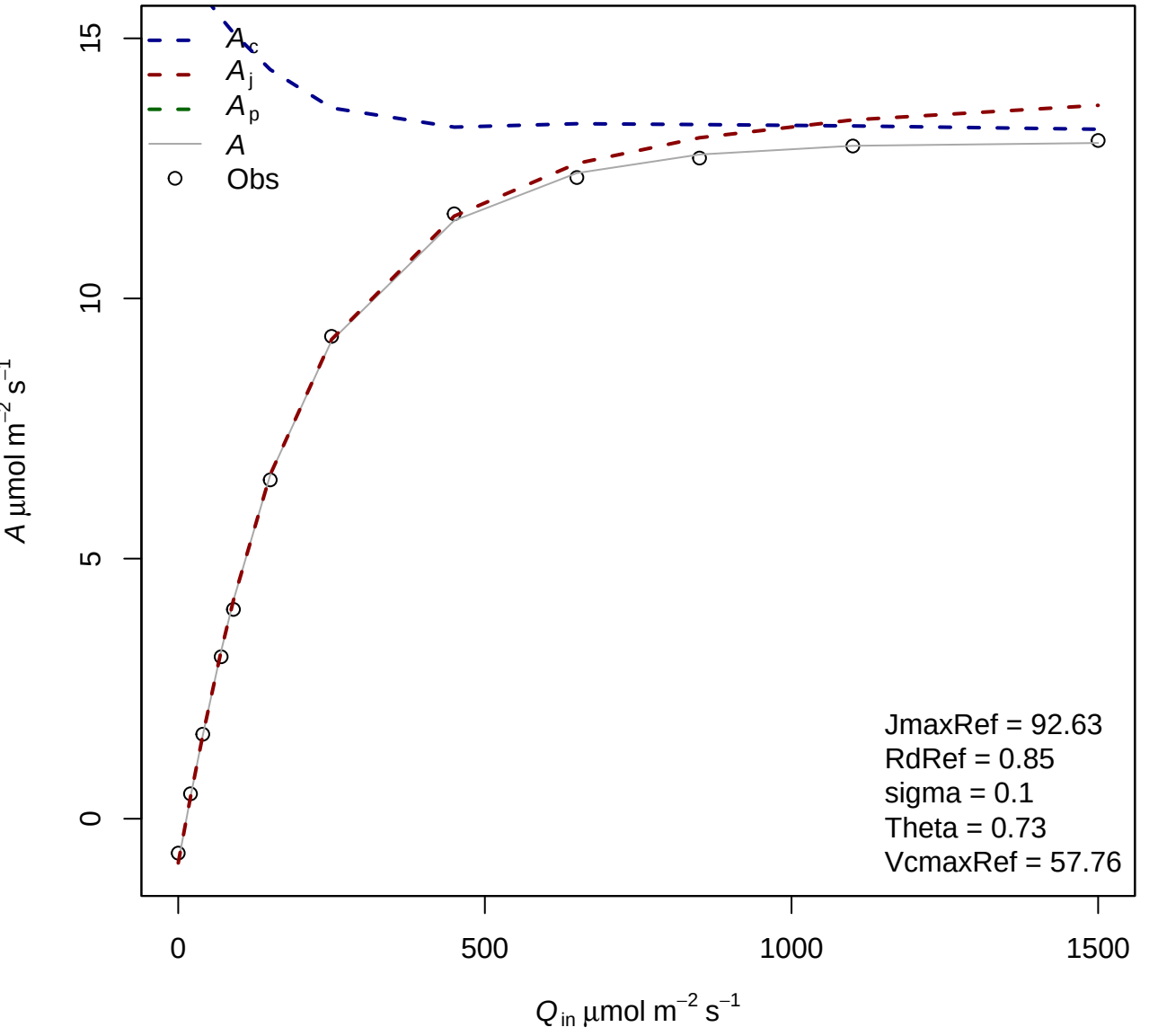

A_c
A_j
A_p
A
Obs
JmaxRef = 90.55
RdRef = 0.87
sigma = 0.15
Theta = 0.7
VcmaxRef = 54.69
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

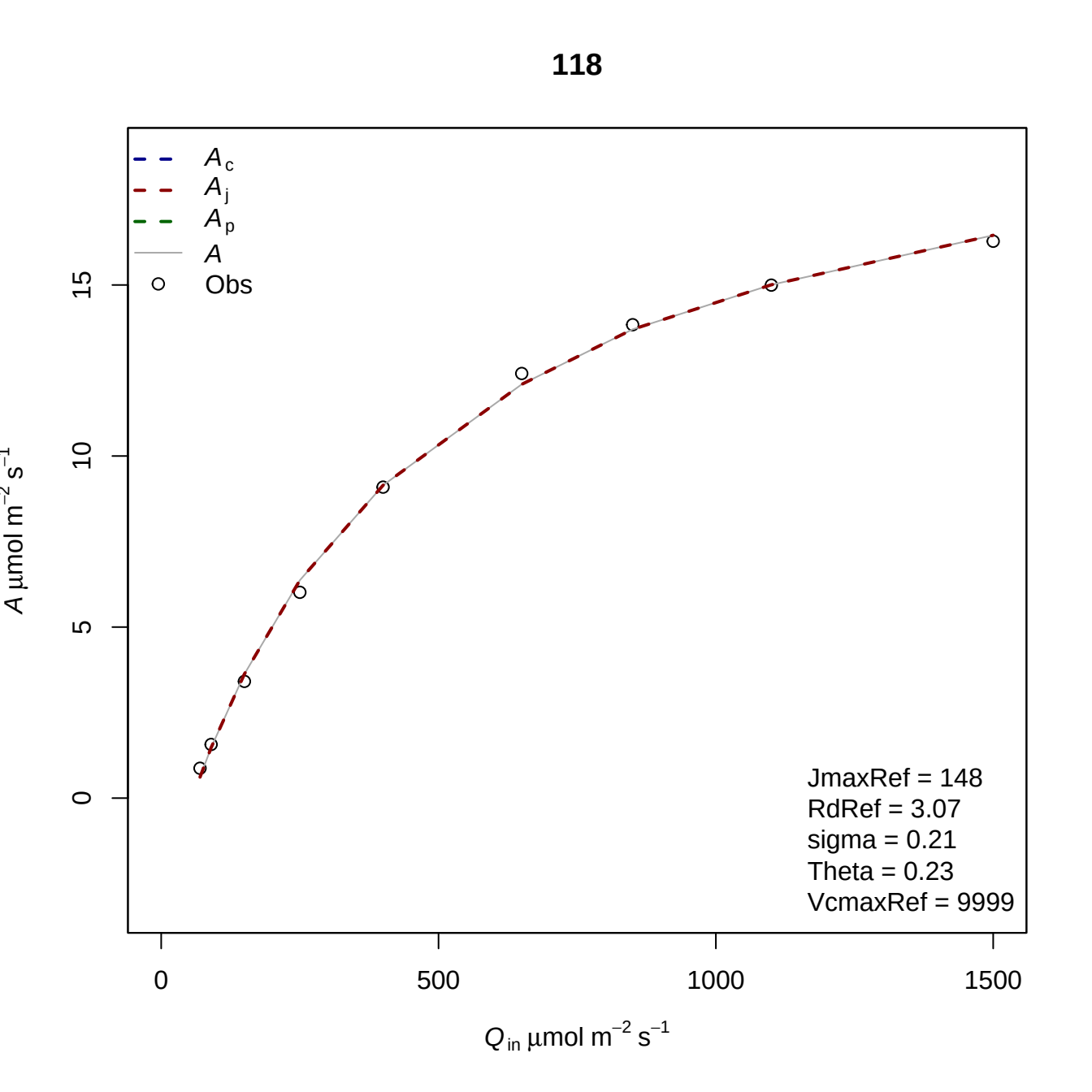

102
A_c
A_j
A_p
A
Obs
JmaxRef = 95.98
RdRef = 2.37
sigma = 0.03
Theta = 0.22
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

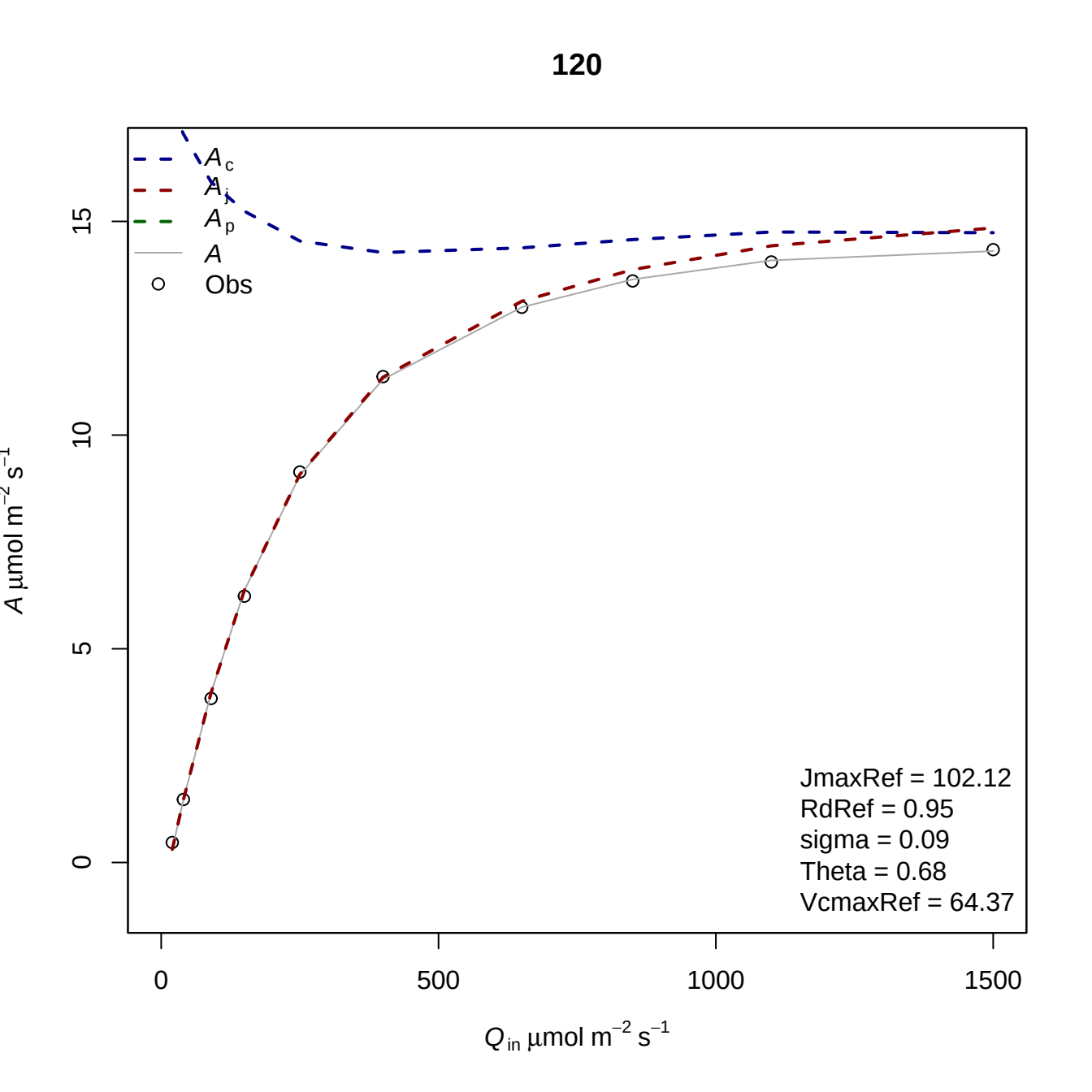

106
A_c
A_j
A_p
A
Obs
JmaxRef = 79.71
RdRef = 0.6
sigma = 0.06
Theta = 0.48
VcmaxRef = 37.64
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

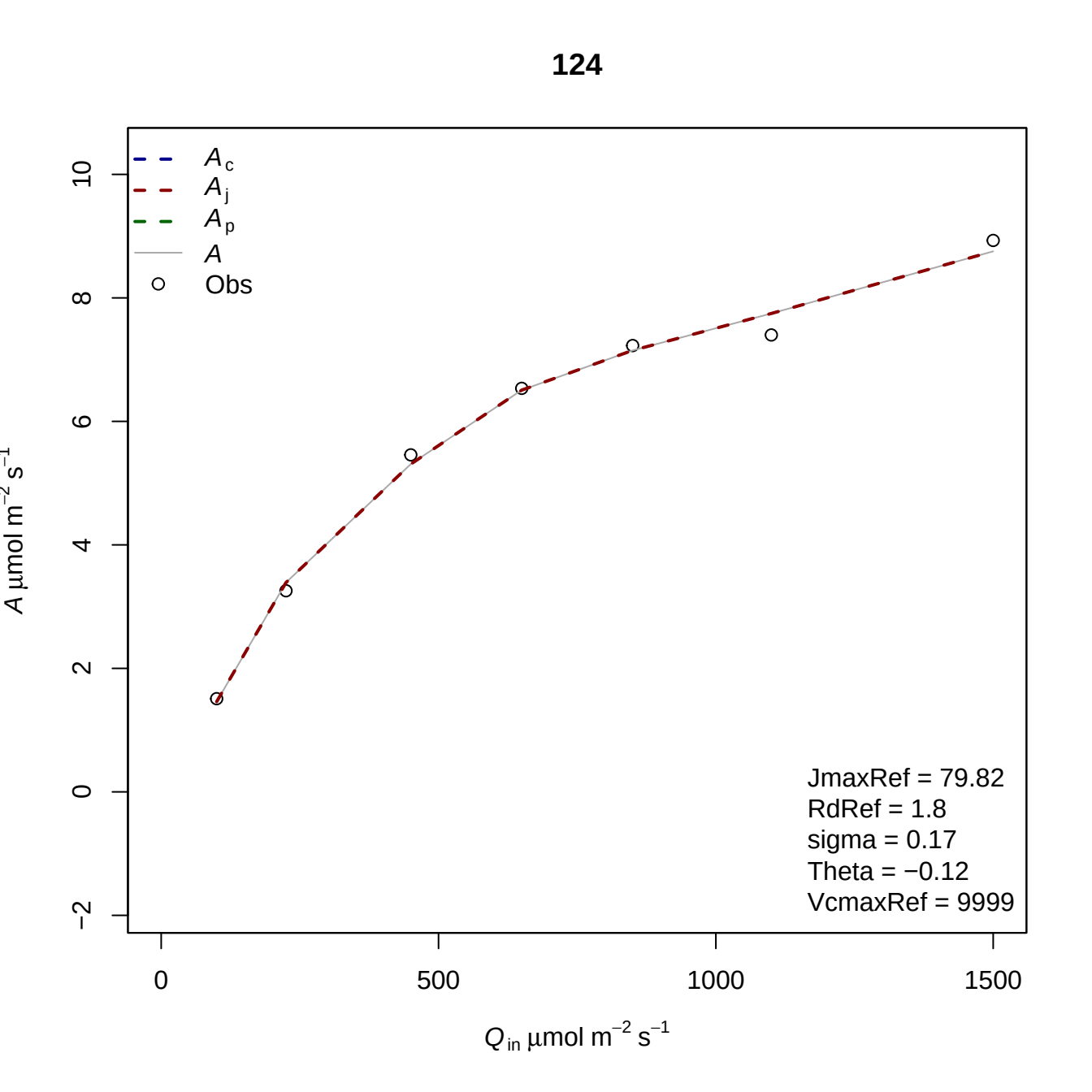

109
A_c
A_j
A_p
A
Obs
JmaxRef = 58.42
RdRef = 1.53
sigma = 0.07
Theta = 0.68
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

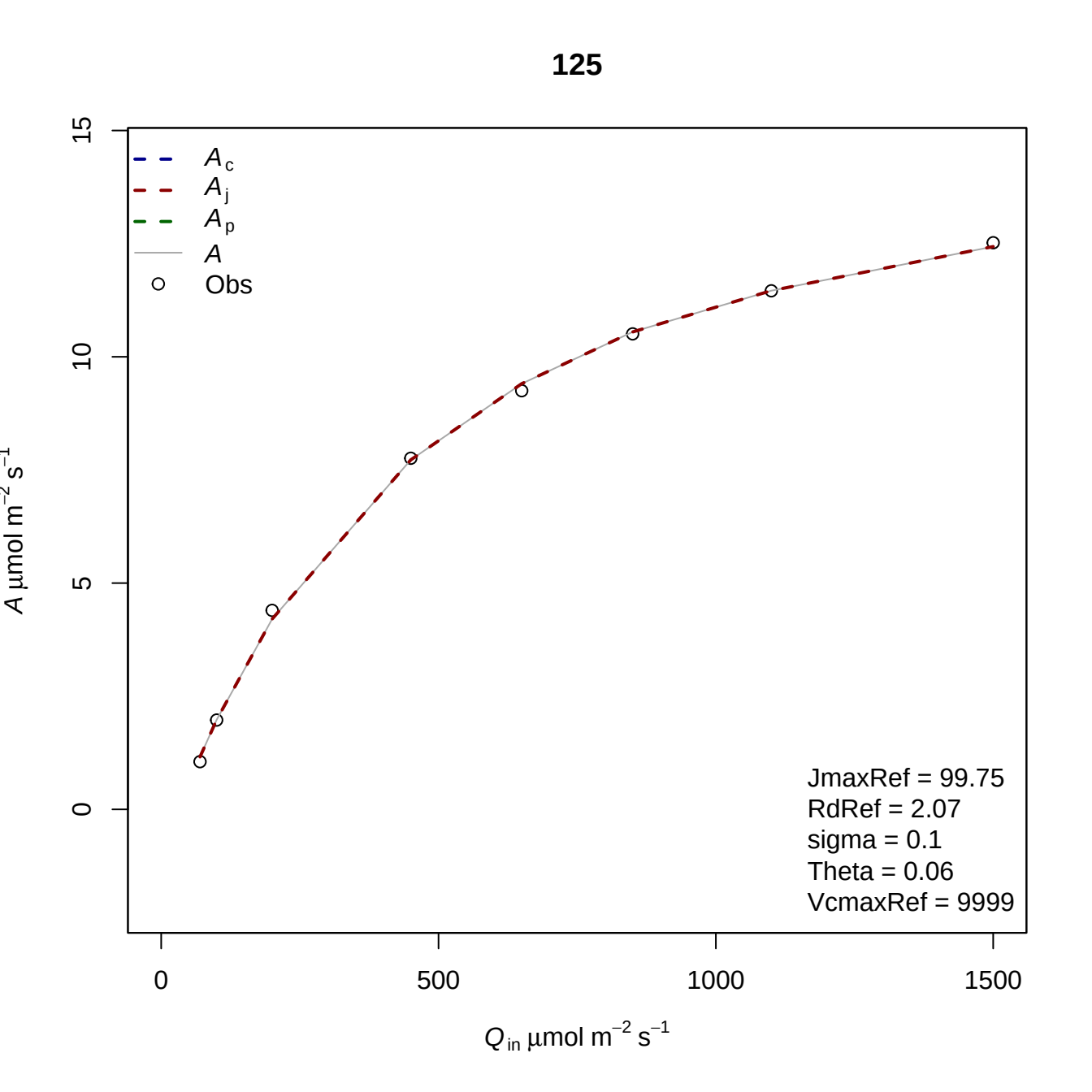

113
A_c
A_j
A_p
A
Obs
JmaxRef = 145655028.78
RdRef = 1.1
sigma = 0.6
Theta = −40295724.02
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

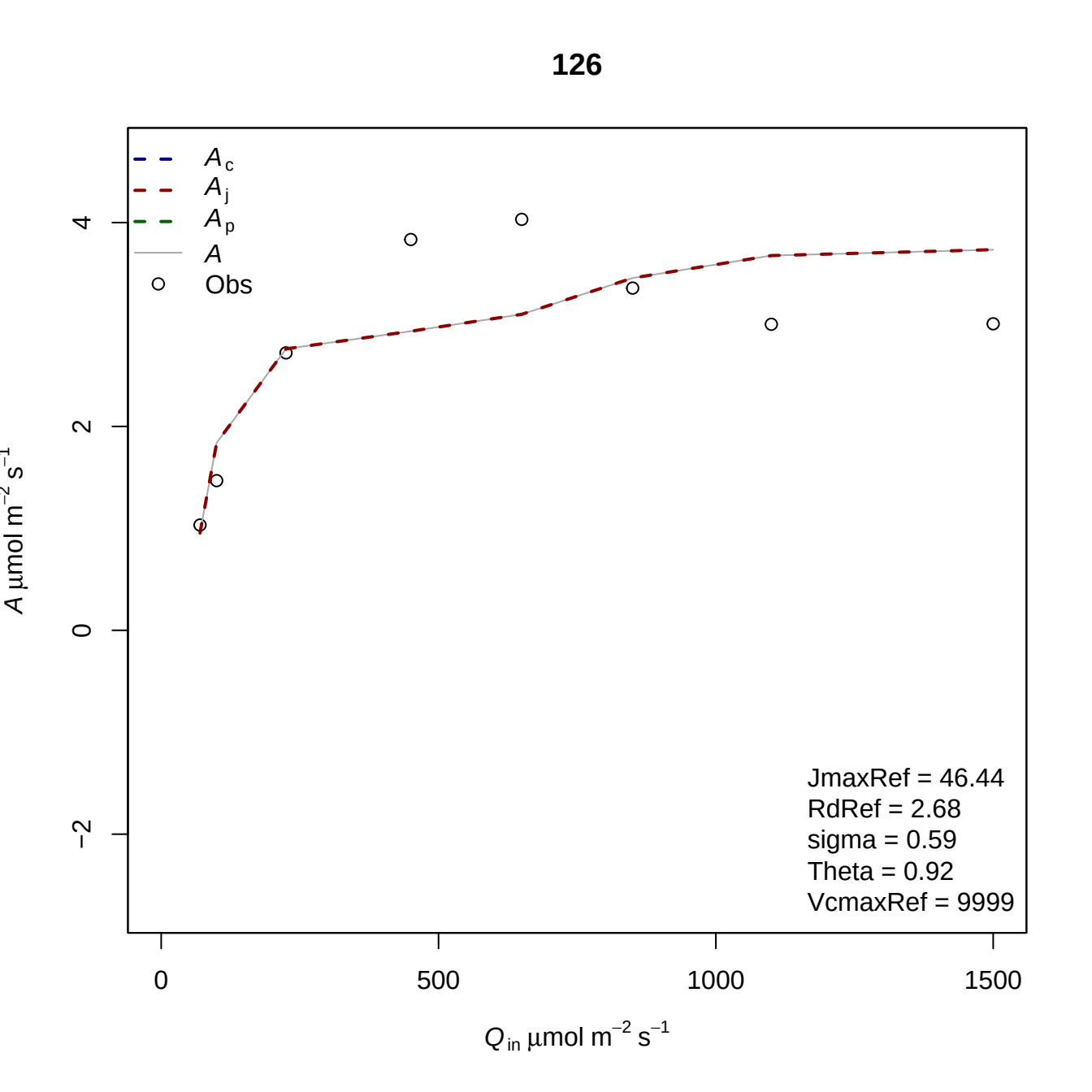

A_c
A_j
A_p
A
Obs
JmaxRef = 92.63
RdRef = 0.85
sigma = 0.1
Theta = 0.73
VcmaxRef = 57.76
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

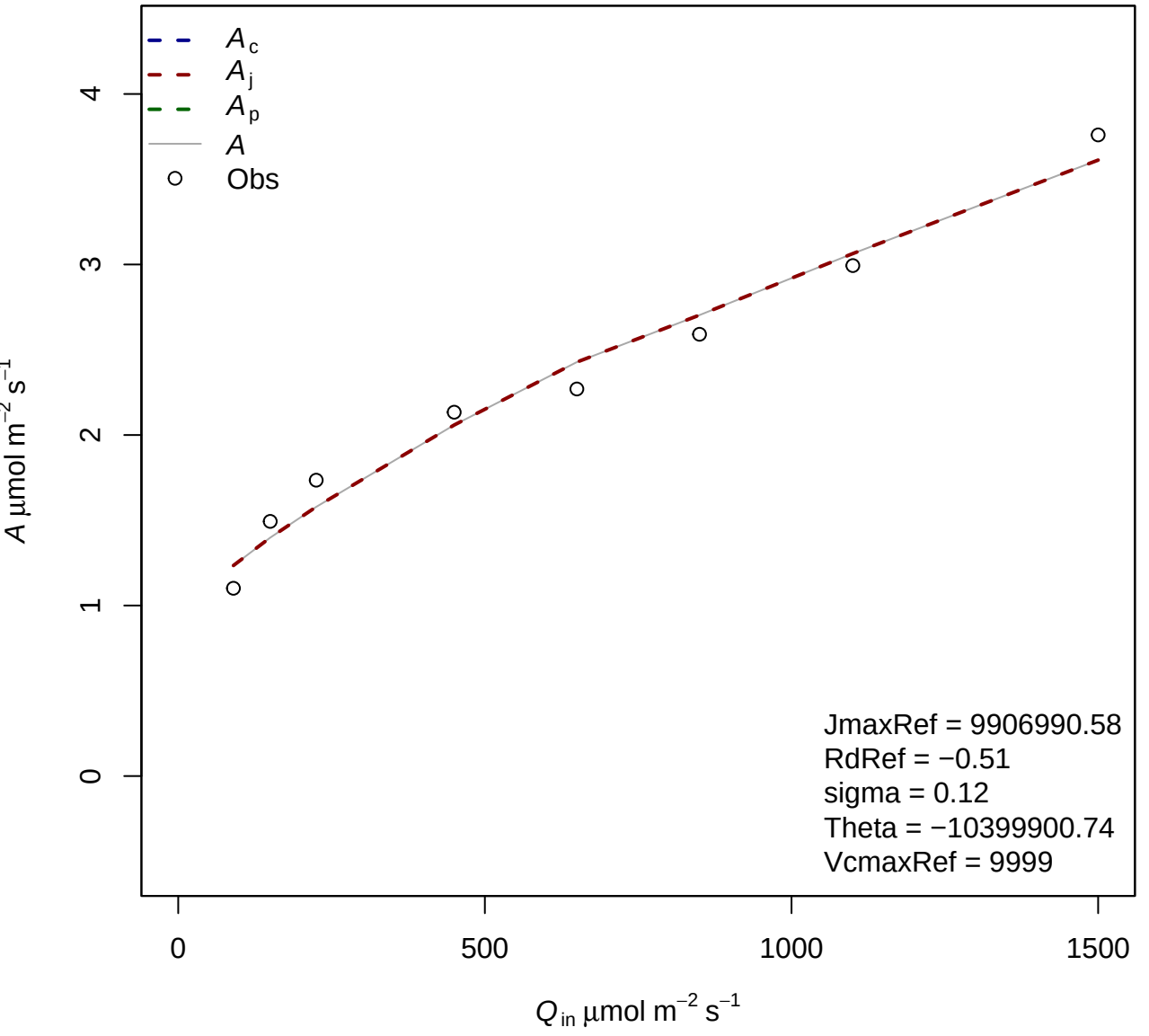

118
A_c
A_j
A_p
A
Obs
JmaxRef = 148
RdRef = 3.07
sigma = 0.21
Theta = 0.23
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

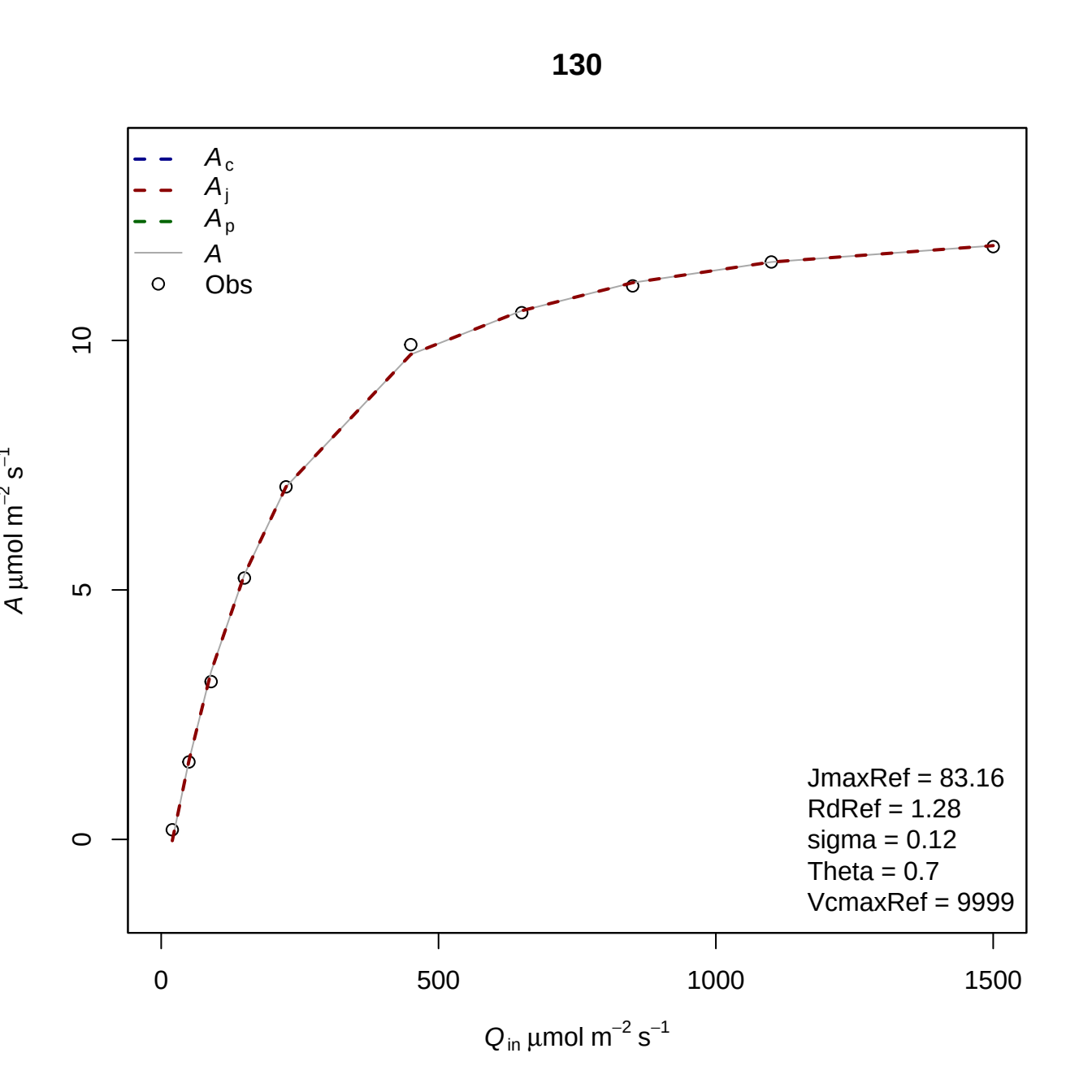

120
A_c
A_j
A_p
A
Obs
JmaxRef = 102.12
RdRef = 0.95
sigma = 0.09
Theta = 0.68
VcmaxRef = 64.37
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

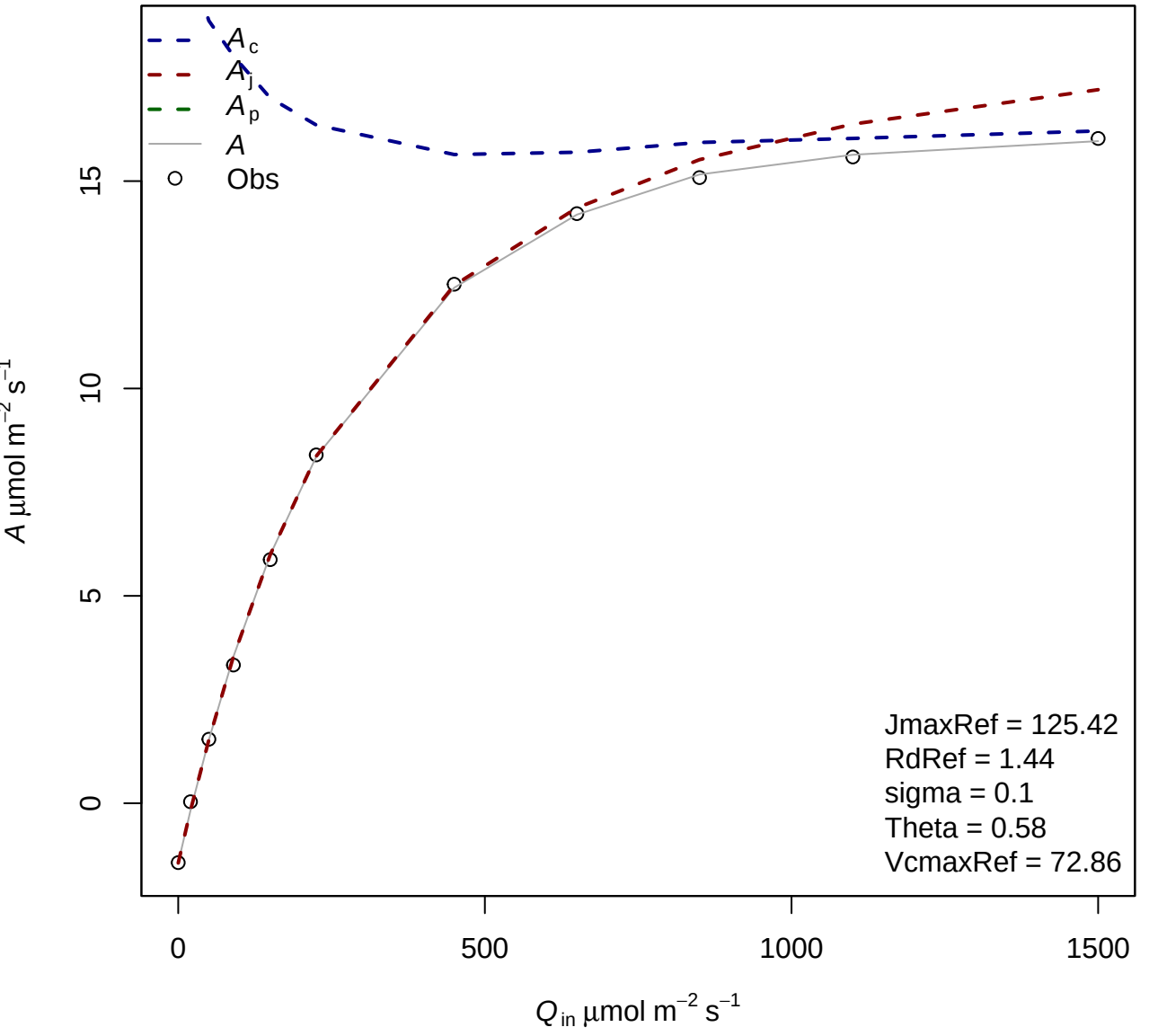

124
A_c
A_j
A_p
A
Obs
JmaxRef = 79.82
RdRef = 1.8
sigma = 0.17
Theta = −0.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

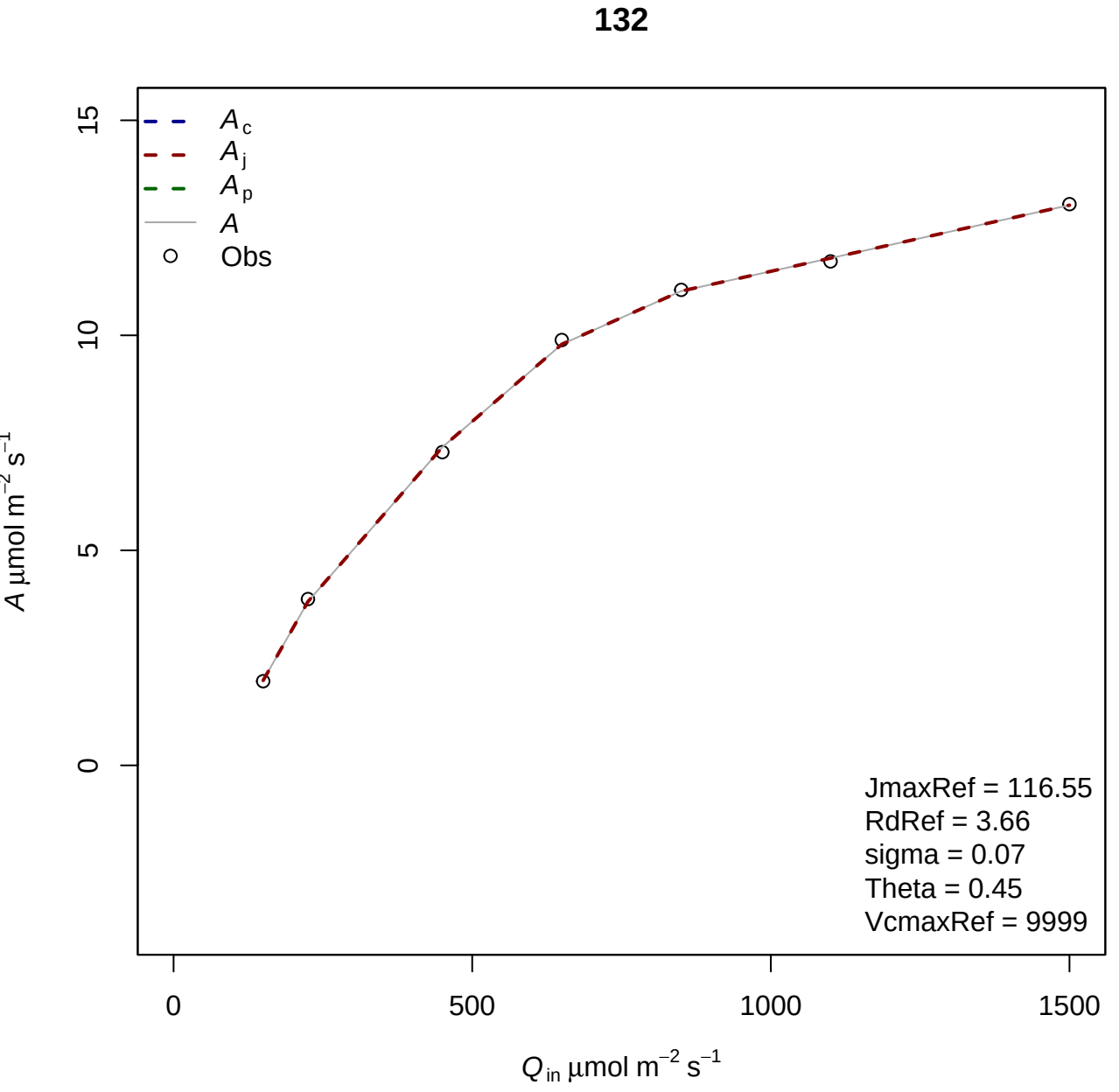

125
A_c
A_j
A_p
A
Obs
JmaxRef = 99.75
RdRef = 2.07
sigma = 0.1
Theta = 0.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

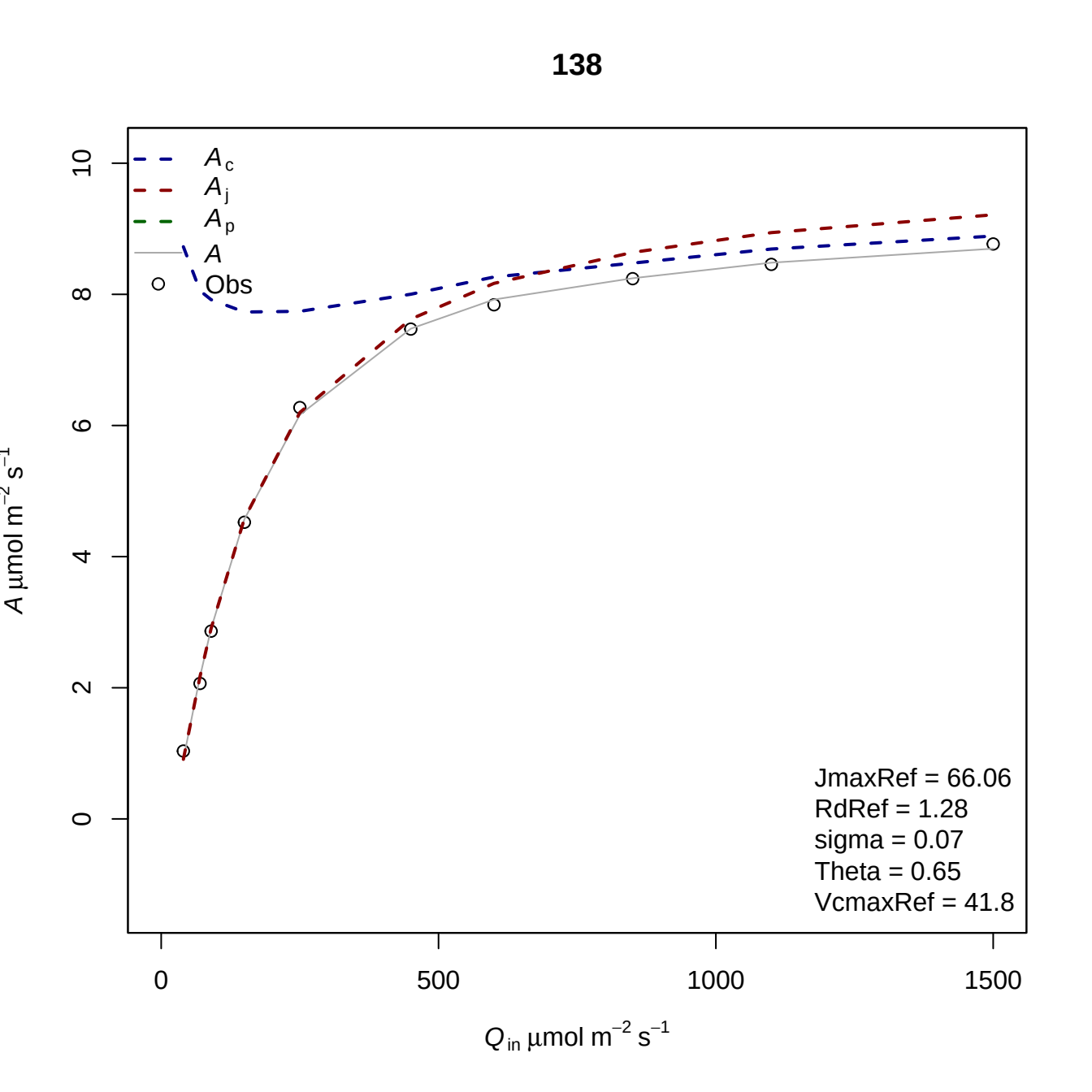

126
A_c
A_j
A_p
A
Obs
JmaxRef = 46.44
RdRef = 2.68
sigma = 0.59
Theta = 0.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
500
1000
1500

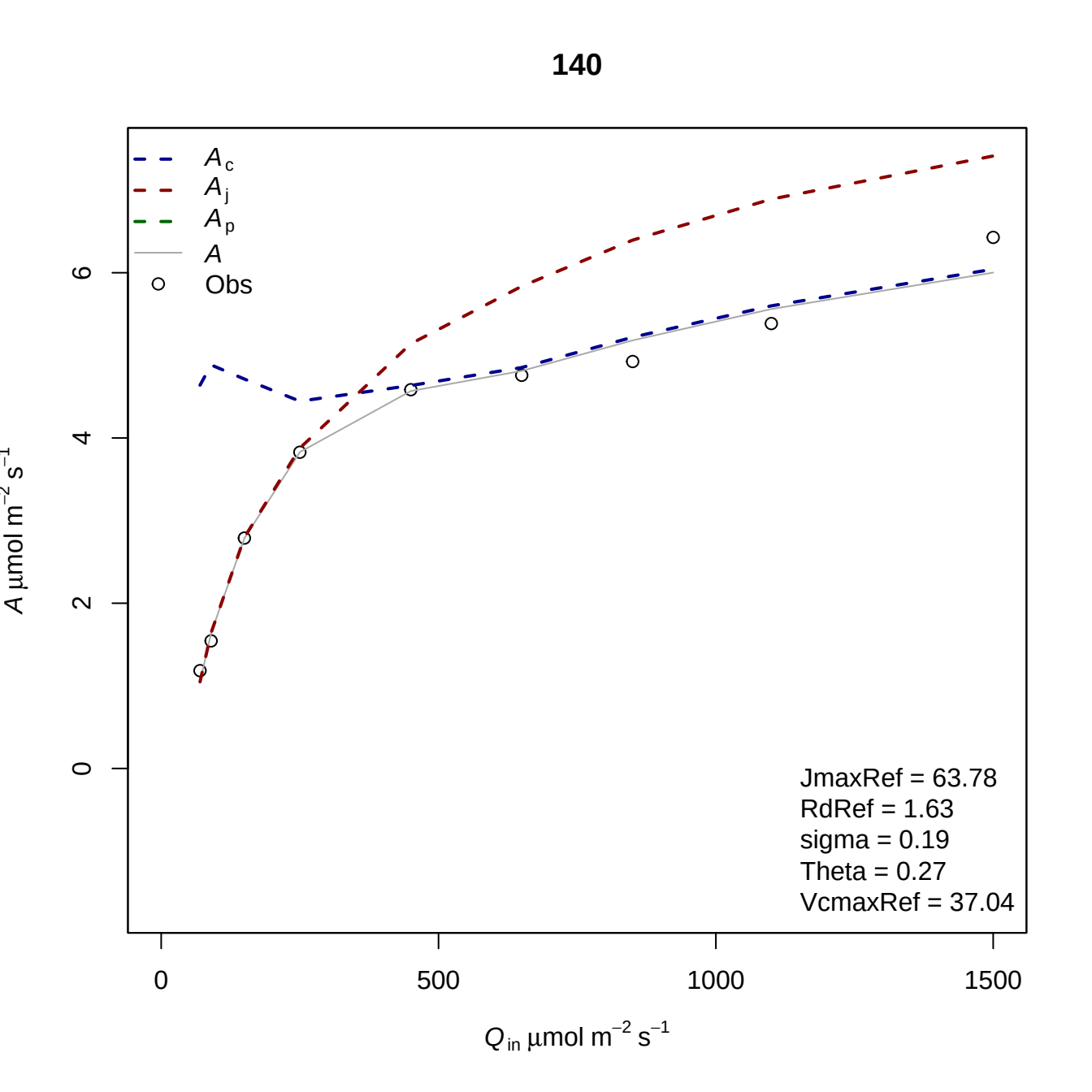

A_c
A_j
A_p
A
Obs
JmaxRef = 9906990.58
RdRef = −0.51
sigma = 0.12
Theta = −10399900.74
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

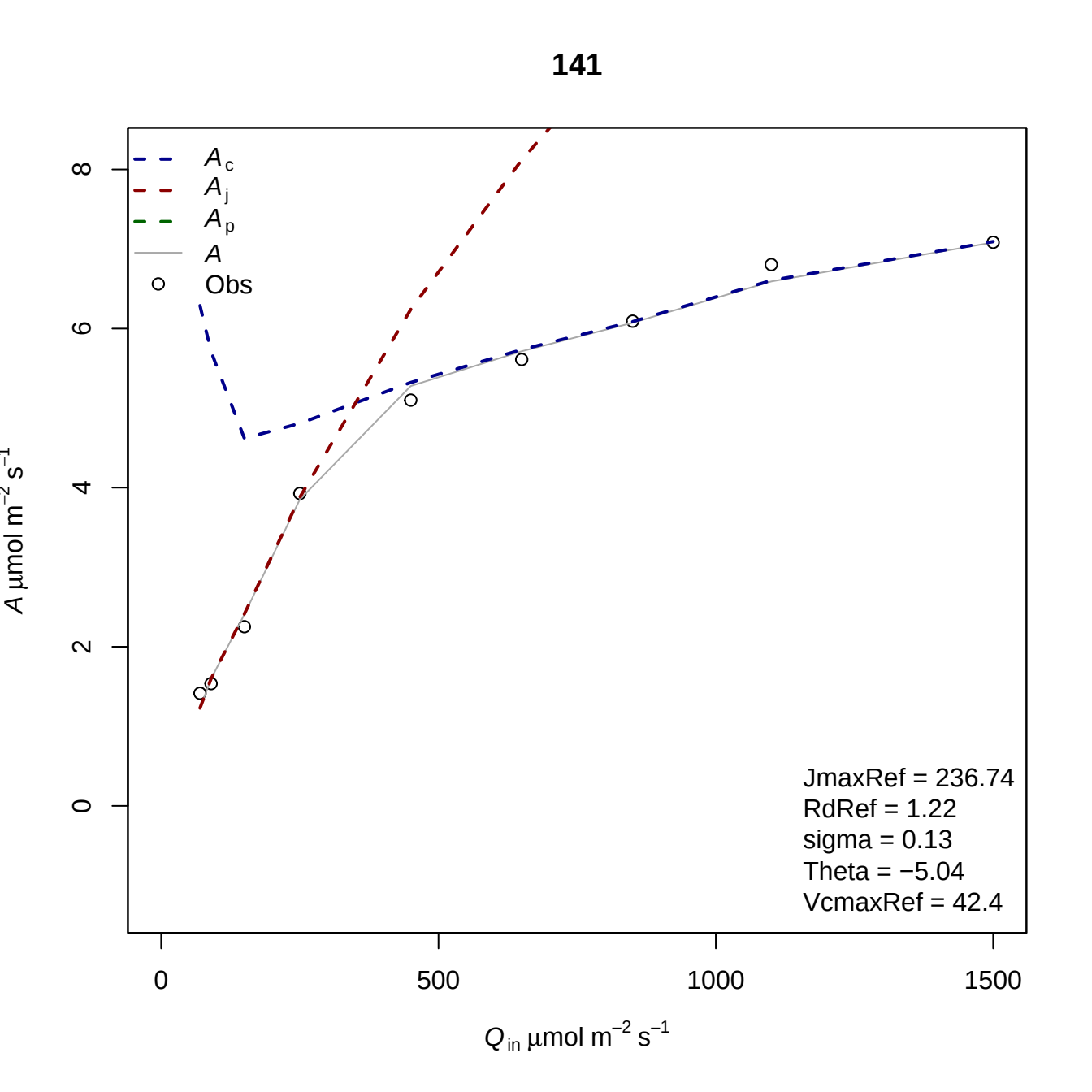

130
A_c
A_j
A_p
A
Obs
JmaxRef = 83.16
RdRef = 1.28
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

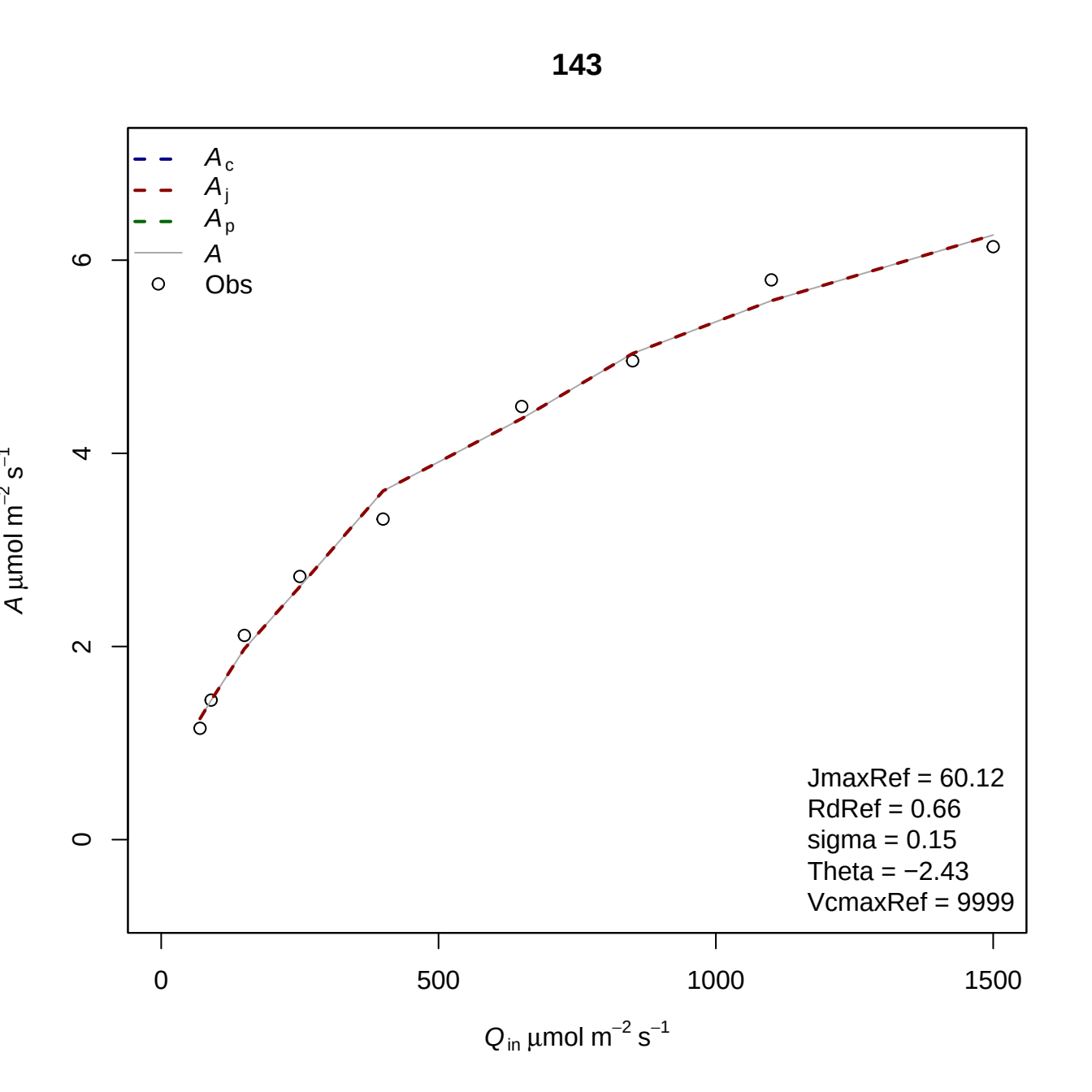

A_c
A_j
A_p
A
Obs
JmaxRef = 125.42
RdRef = 1.44
sigma = 0.1
Theta = 0.58
VcmaxRef = 72.86
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

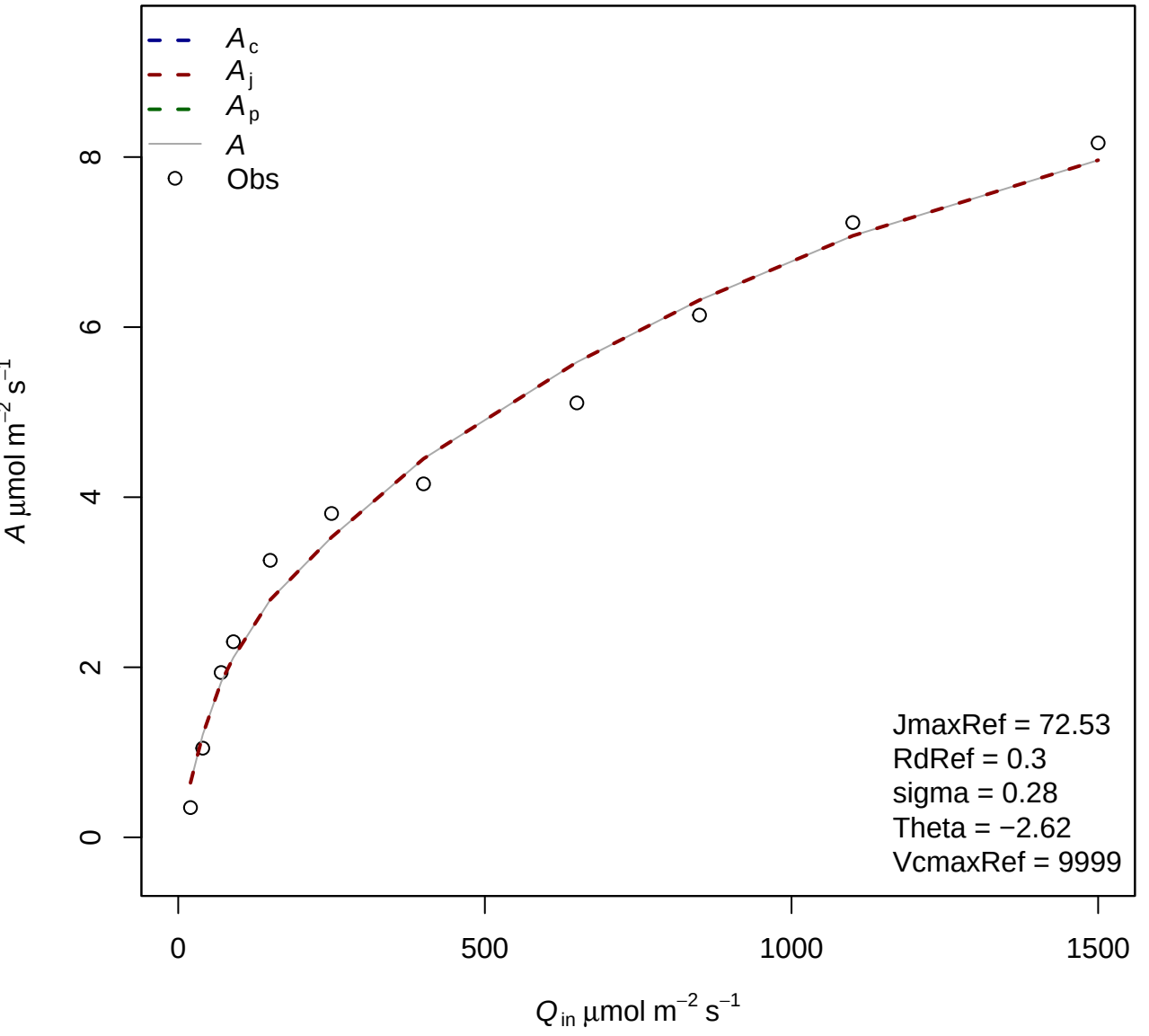

132
A_c
A_j
A_p
A
Obs
15
10
5
0
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 116.55
RdRef = 3.66
sigma = 0.07
Theta = 0.45
VcmaxRef = 9999

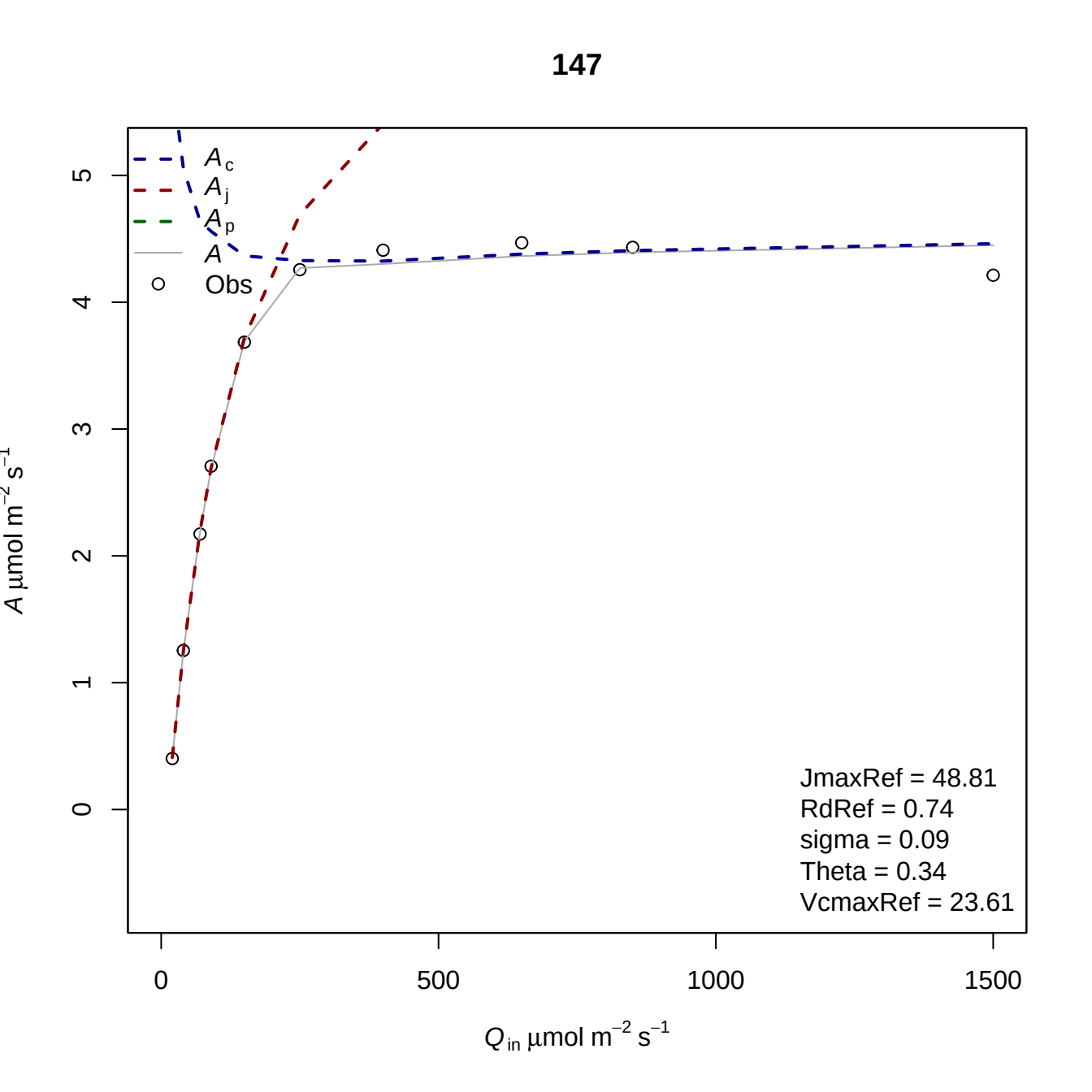

138
A_c
A_j
A_p
A
Obs
JmaxRef = 66.06
RdRef = 1.28
sigma = 0.07
Theta = 0.65
VcmaxRef = 41.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

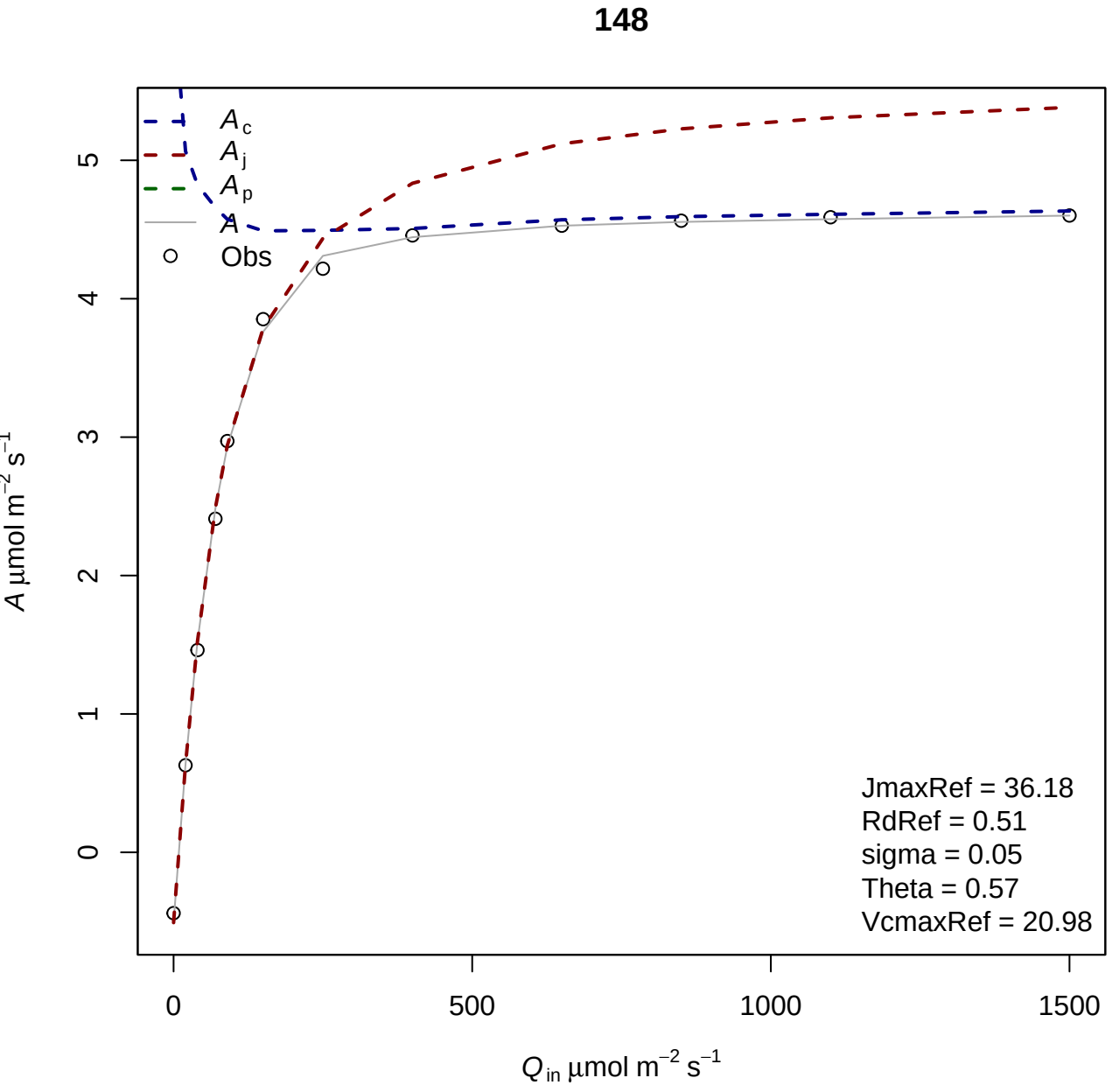

140
A_c
A_j
A_p
A
Obs
JmaxRef = 63.78
RdRef = 1.63
sigma = 0.19
Theta = 0.27
VcmaxRef = 37.04
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

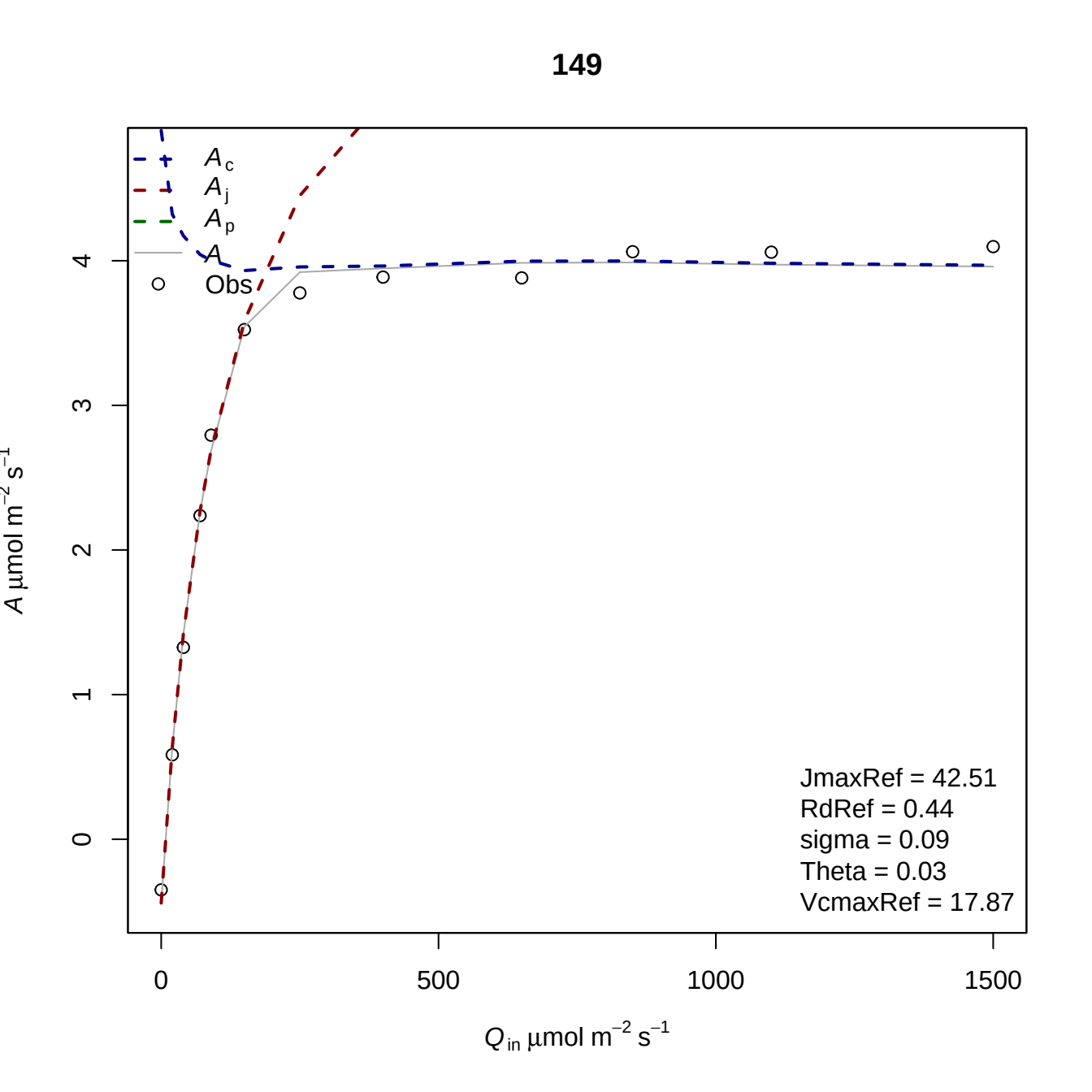

141
A_c
A_j
A_p
A
Obs
JmaxRef = 236.74
RdRef = 1.22
sigma = 0.13
Theta = −5.04
VcmaxRef = 42.4
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

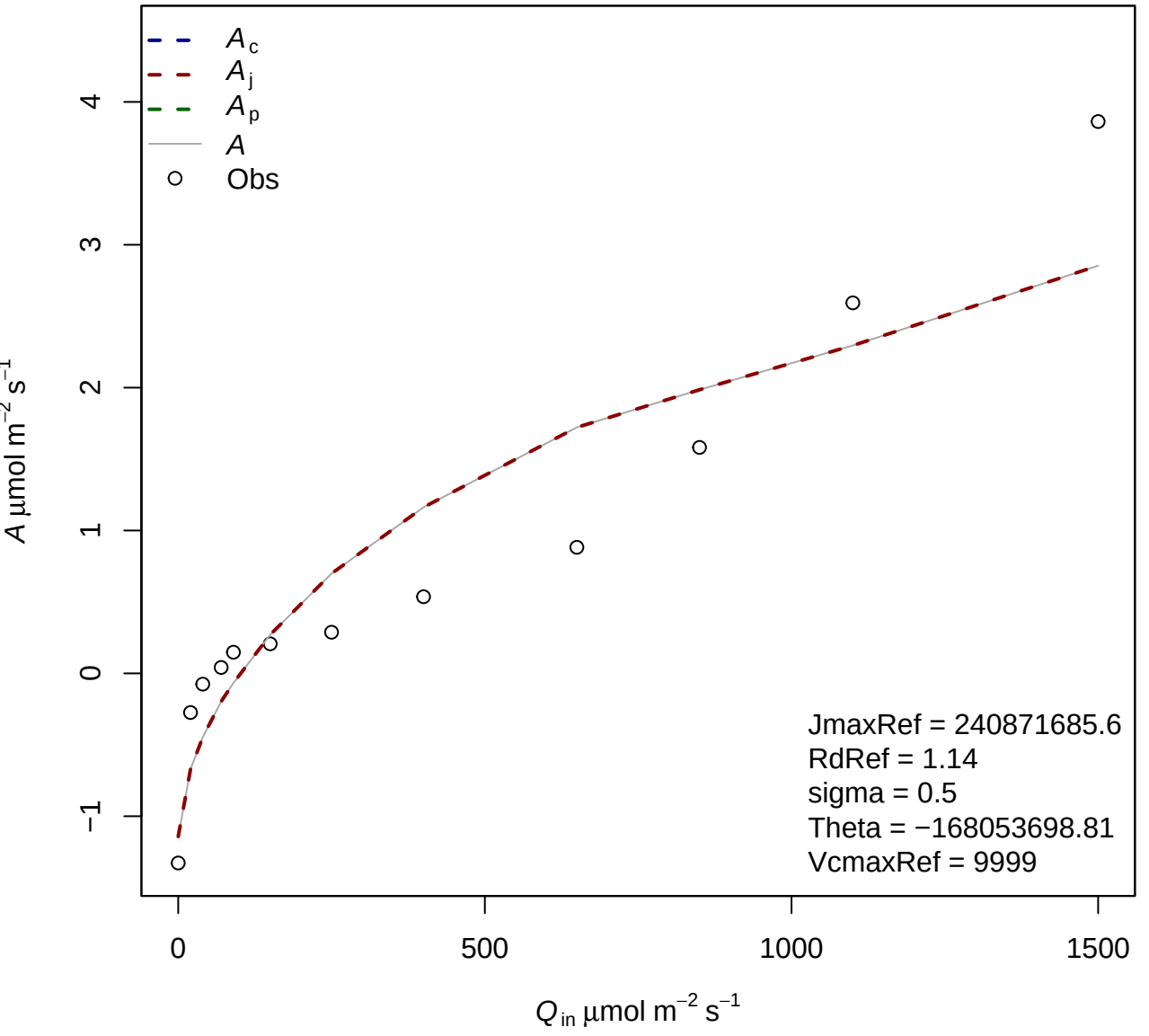

143
A_c
A_j
A_p
A
Obs
JmaxRef = 60.12
RdRef = 0.66
sigma = 0.15
Theta = −2.43
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

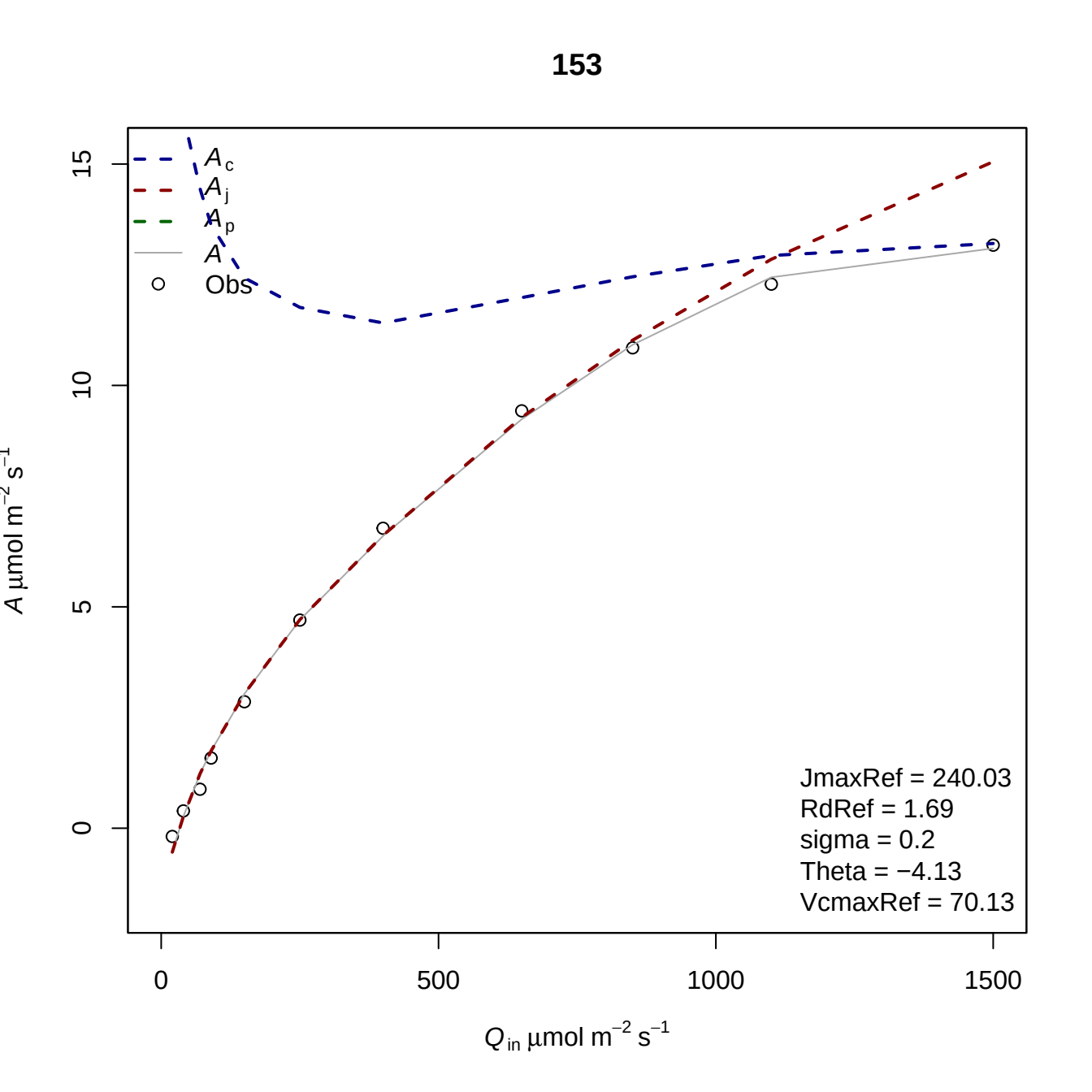

A_c
A_j
A_p
A
Obs
JmaxRef = 72.53
RdRef = 0.3
sigma = 0.28
Theta = −2.62
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

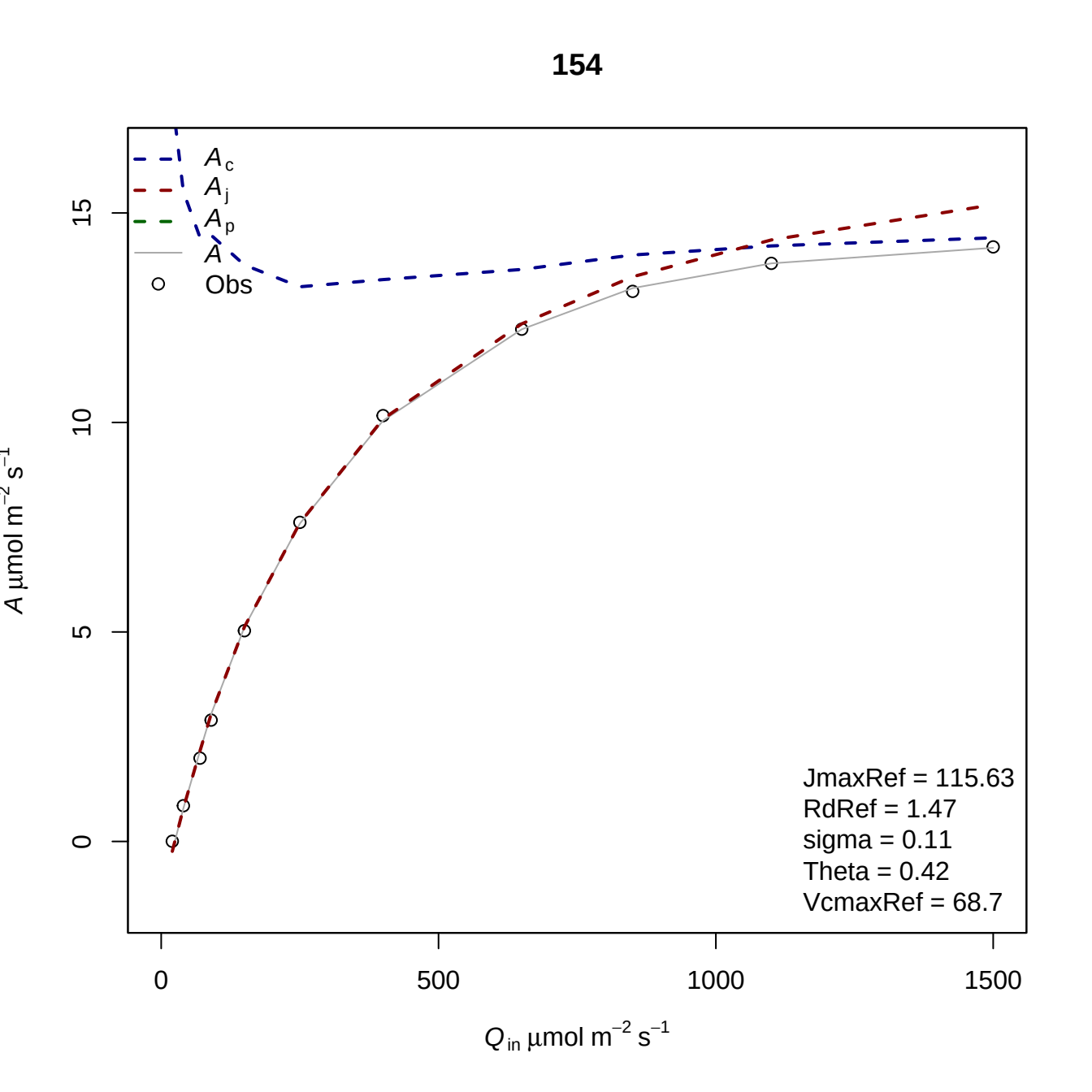

147
A_c
A_j
A_p
A
Obs
JmaxRef = 48.81
RdRef = 0.74
sigma = 0.09
Theta = 0.34
VcmaxRef = 23.61
0
1
2
3
4
5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

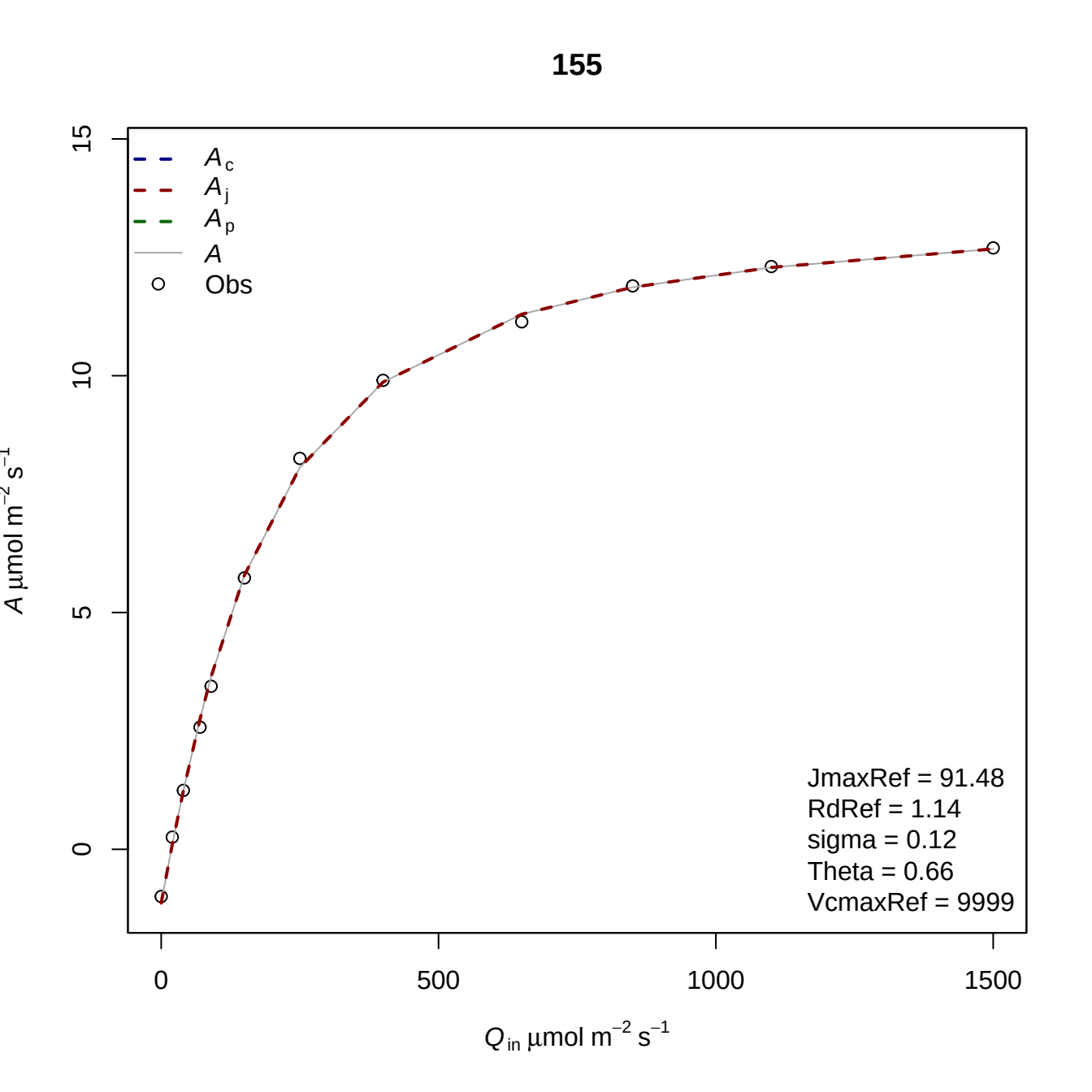

148
A_c
A_j
A_p
A
Obs
JmaxRef = 36.18
RdRef = 0.51
sigma = 0.05
Theta = 0.57
VcmaxRef = 20.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

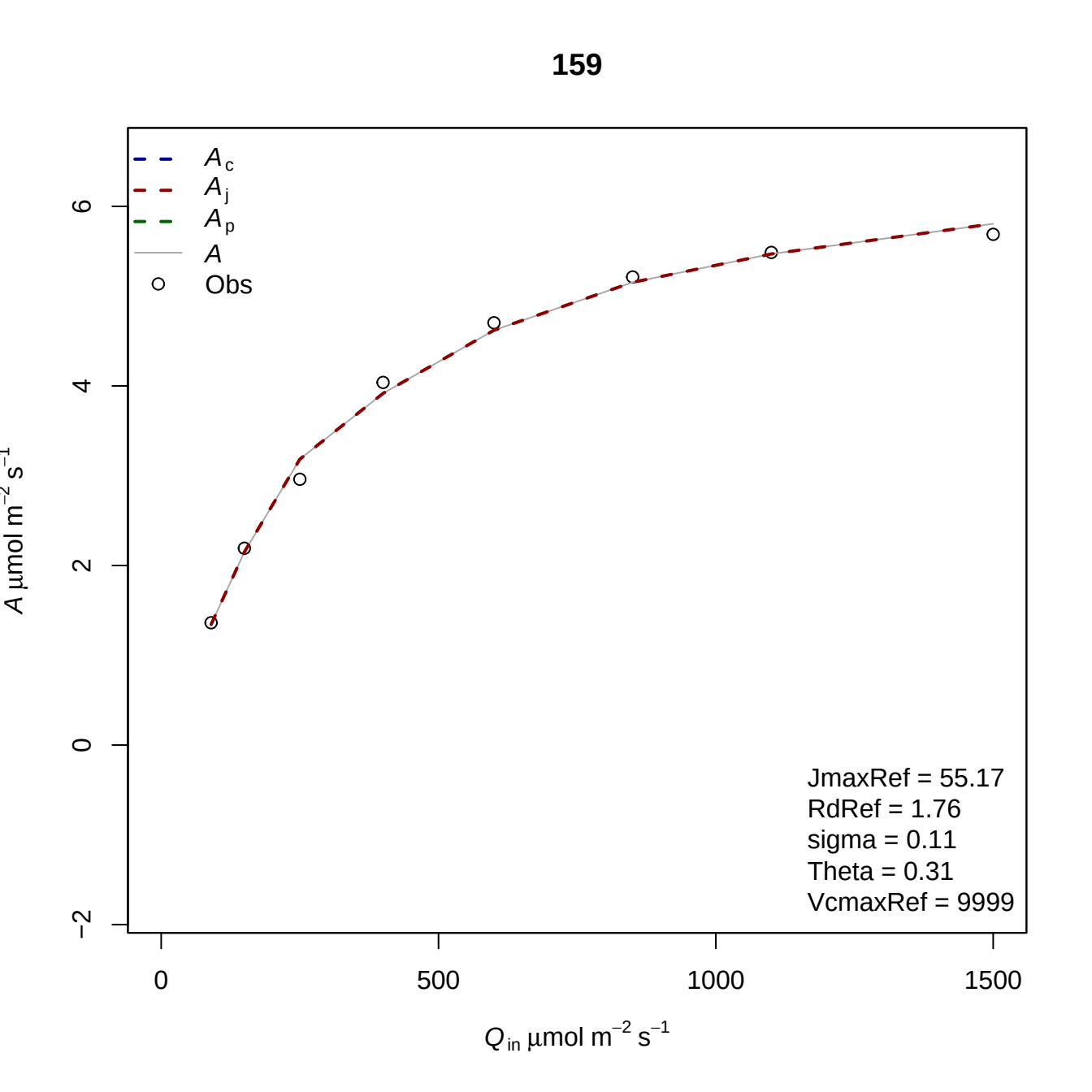

149
A_c
A_j
A_p
A
Obs
JmaxRef = 42.51
RdRef = 0.44
sigma = 0.09
Theta = 0.03
VcmaxRef = 17.87
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

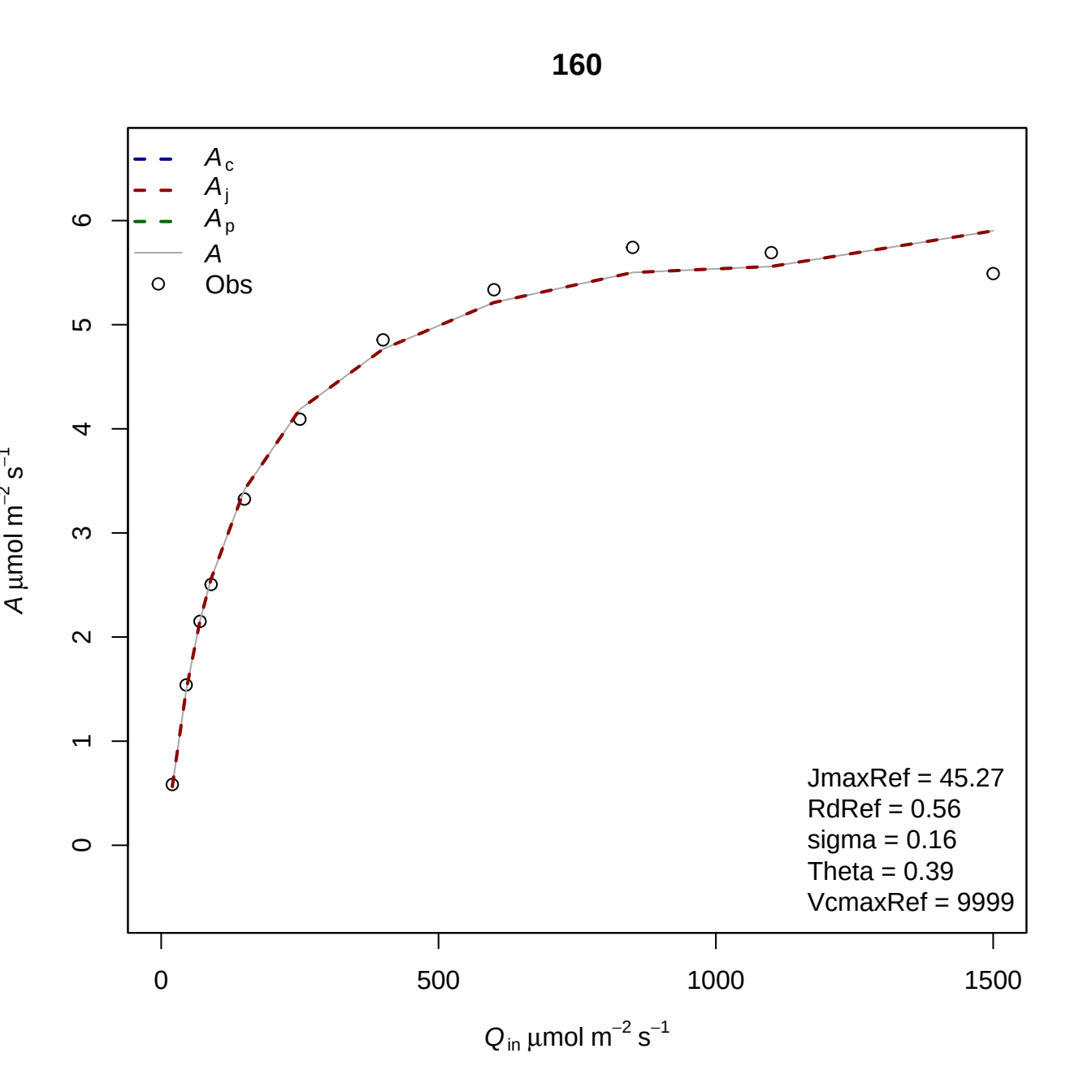

A_c
A_j
A_p
A
Obs
JmaxRef = 240871685.6
RdRef = 1.14
sigma = 0.5
Theta = −168053698.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

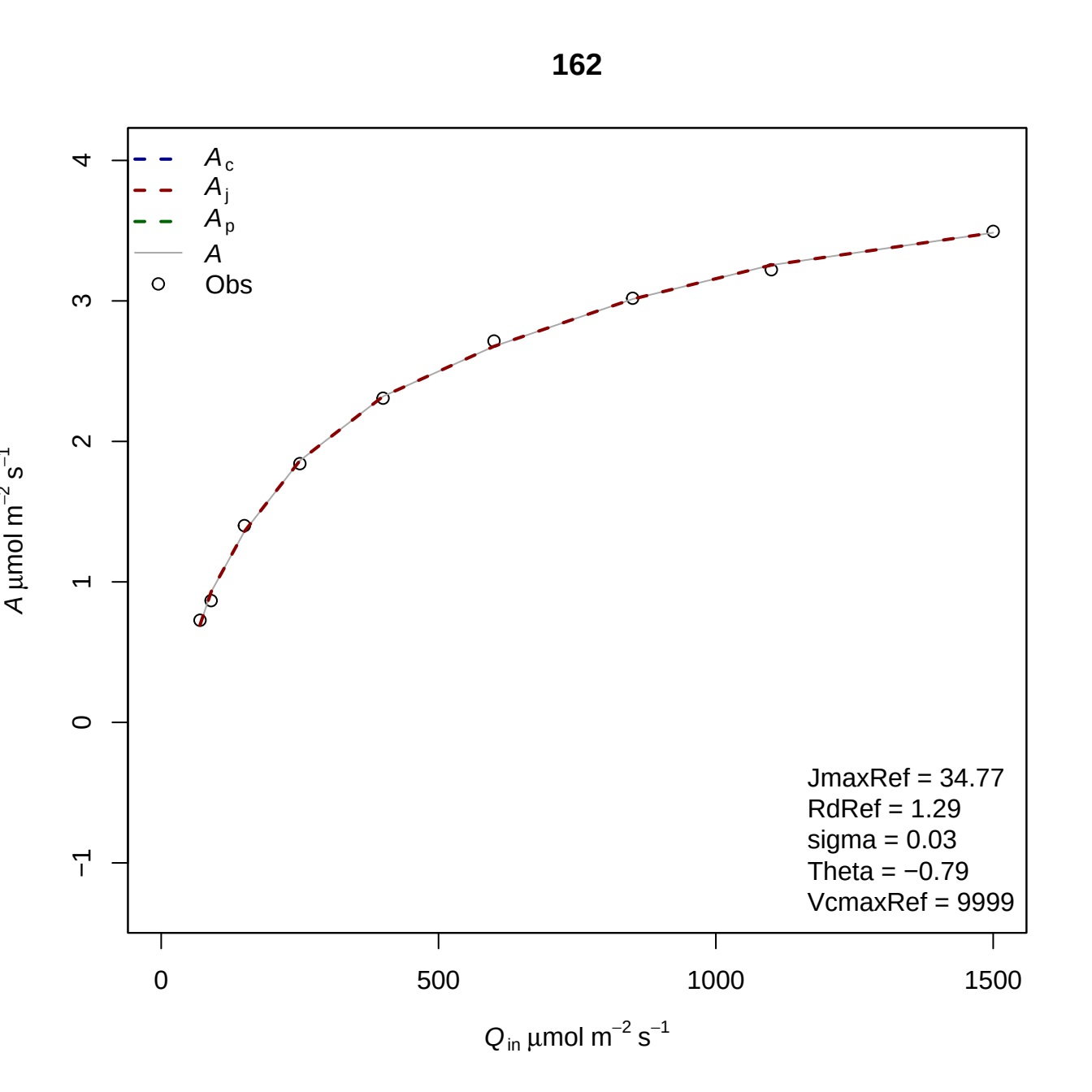

153
A_c
A_j
A_p
A
Obs
JmaxRef = 240.03
RdRef = 1.69
sigma = 0.2
Theta = −4.13
VcmaxRef = 70.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

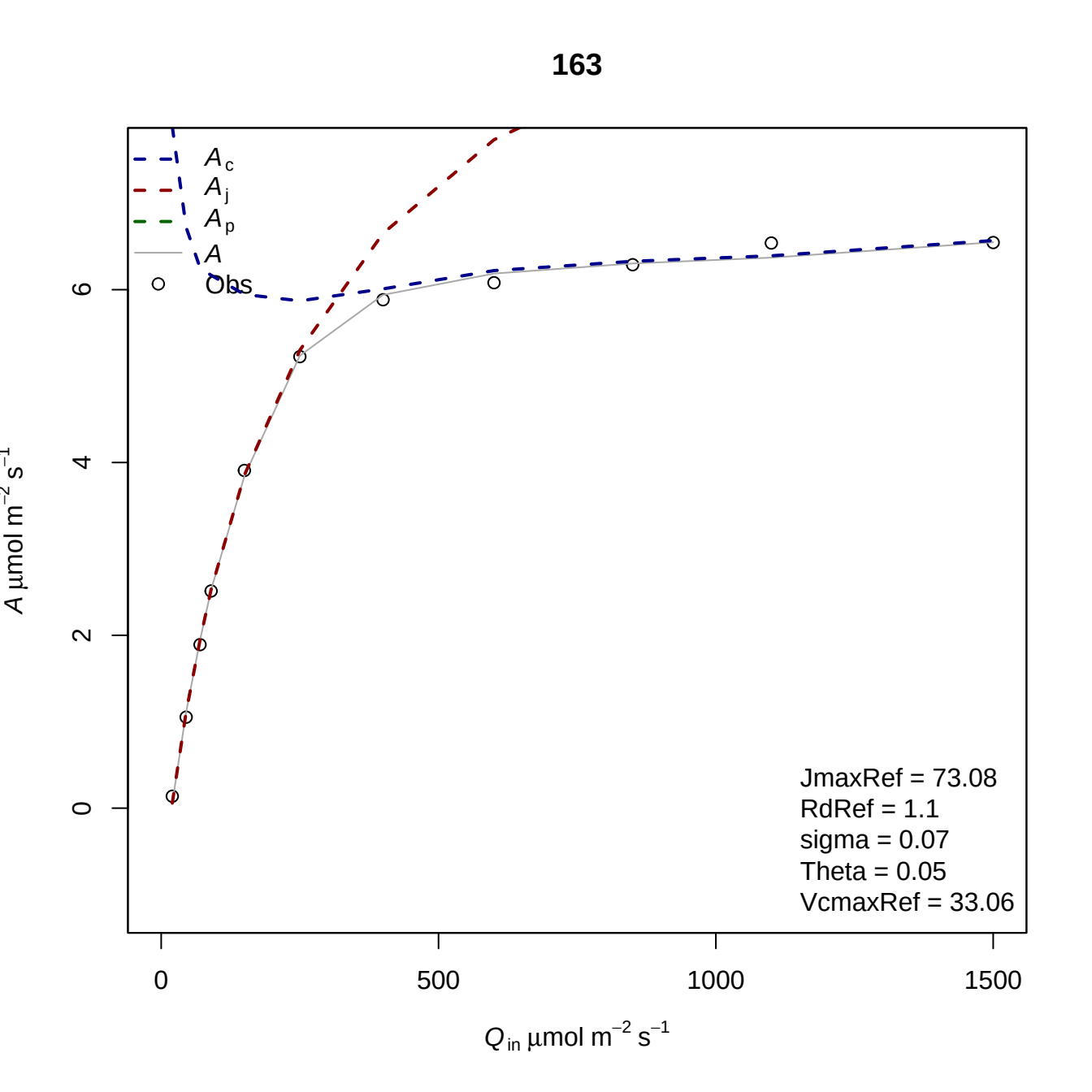

154
A_c
A_j
A_p
A
Obs
JmaxRef = 115.63
RdRef = 1.47
sigma = 0.11
Theta = 0.42
VcmaxRef = 68.7
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

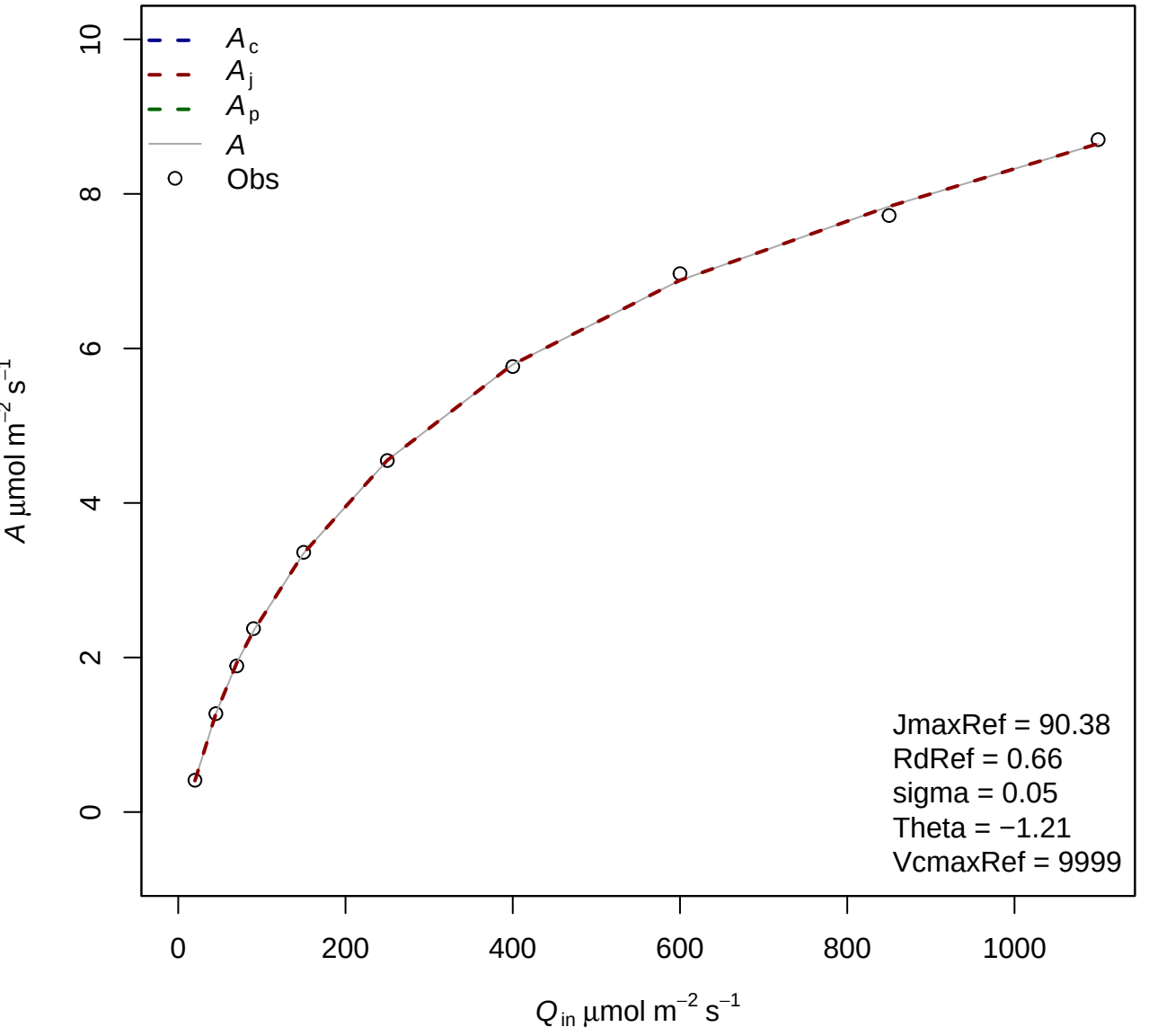

155
A_c
A_j
A_p
A
Obs
JmaxRef = 91.48
RdRef = 1.14
sigma = 0.12
Theta = 0.66
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

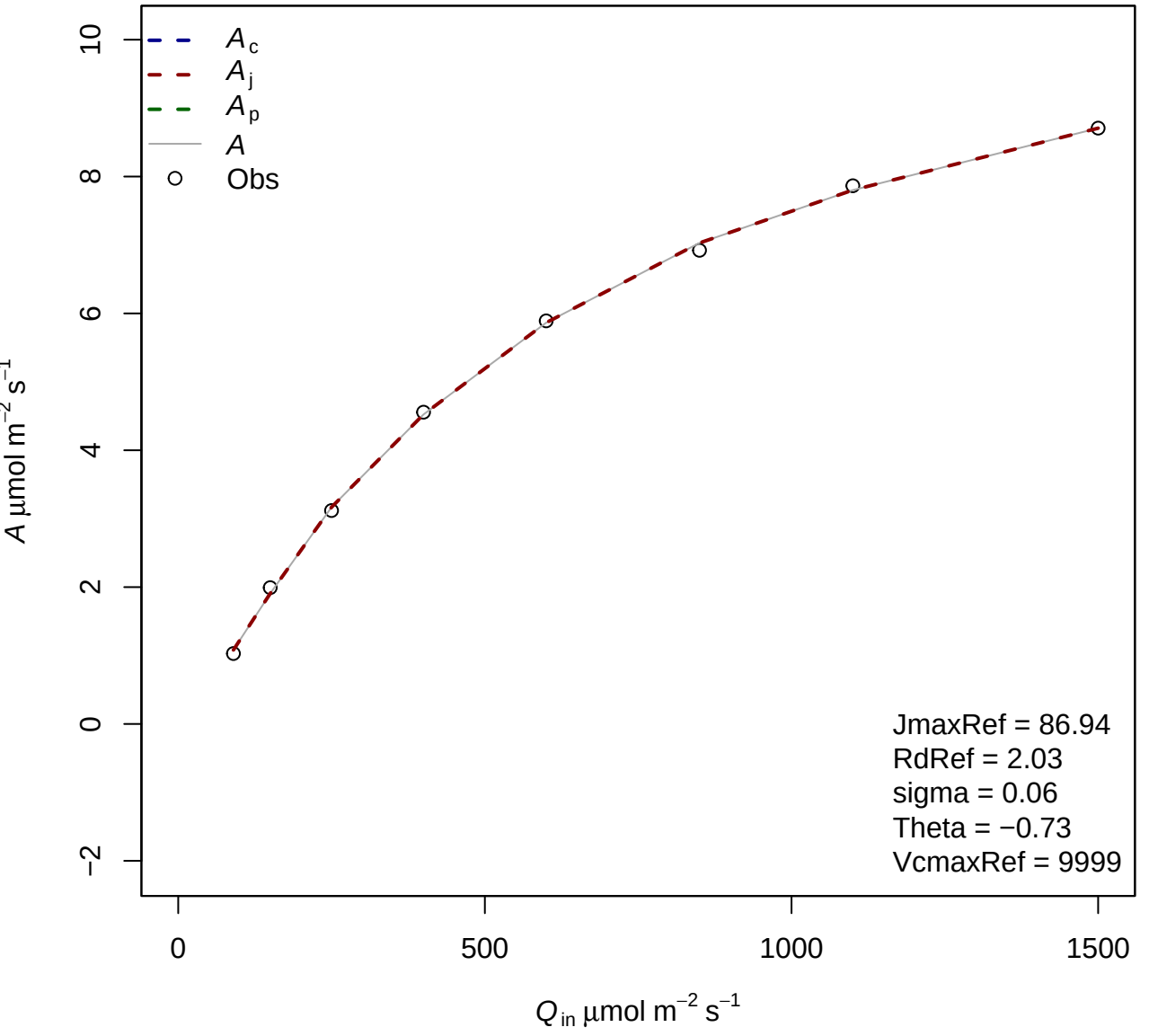

159
A_c
A_j
A_p
A
Obs
JmaxRef = 55.17
RdRef = 1.76
sigma = 0.11
Theta = 0.31
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

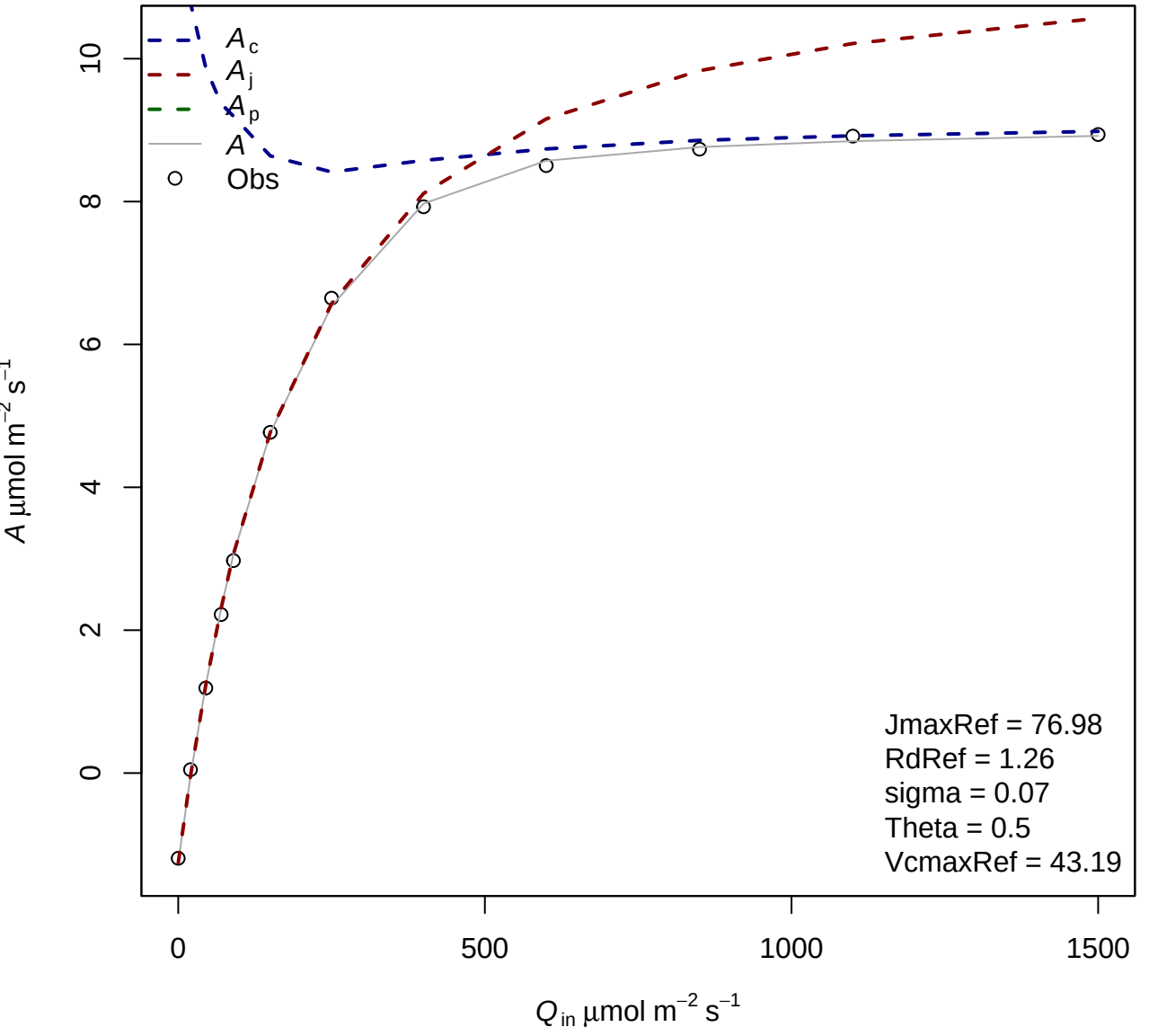

160
A_c
A_j
A_p
A
Obs
JmaxRef = 45.27
RdRef = 0.56
sigma = 0.16
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

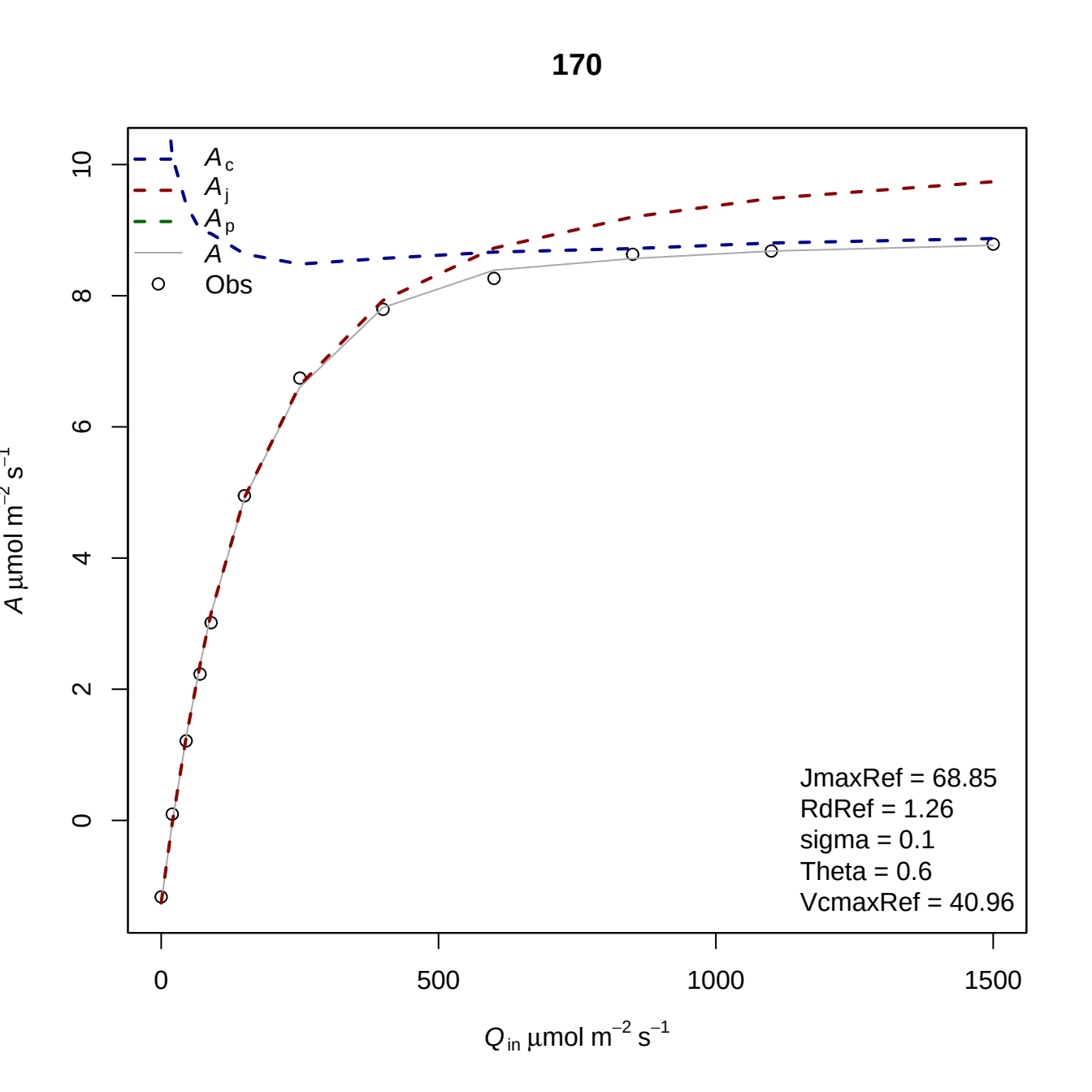

162
A_c
A_j
A_p
A
Obs
JmaxRef = 34.77
RdRef = 1.29
sigma = 0.03
Theta = −0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

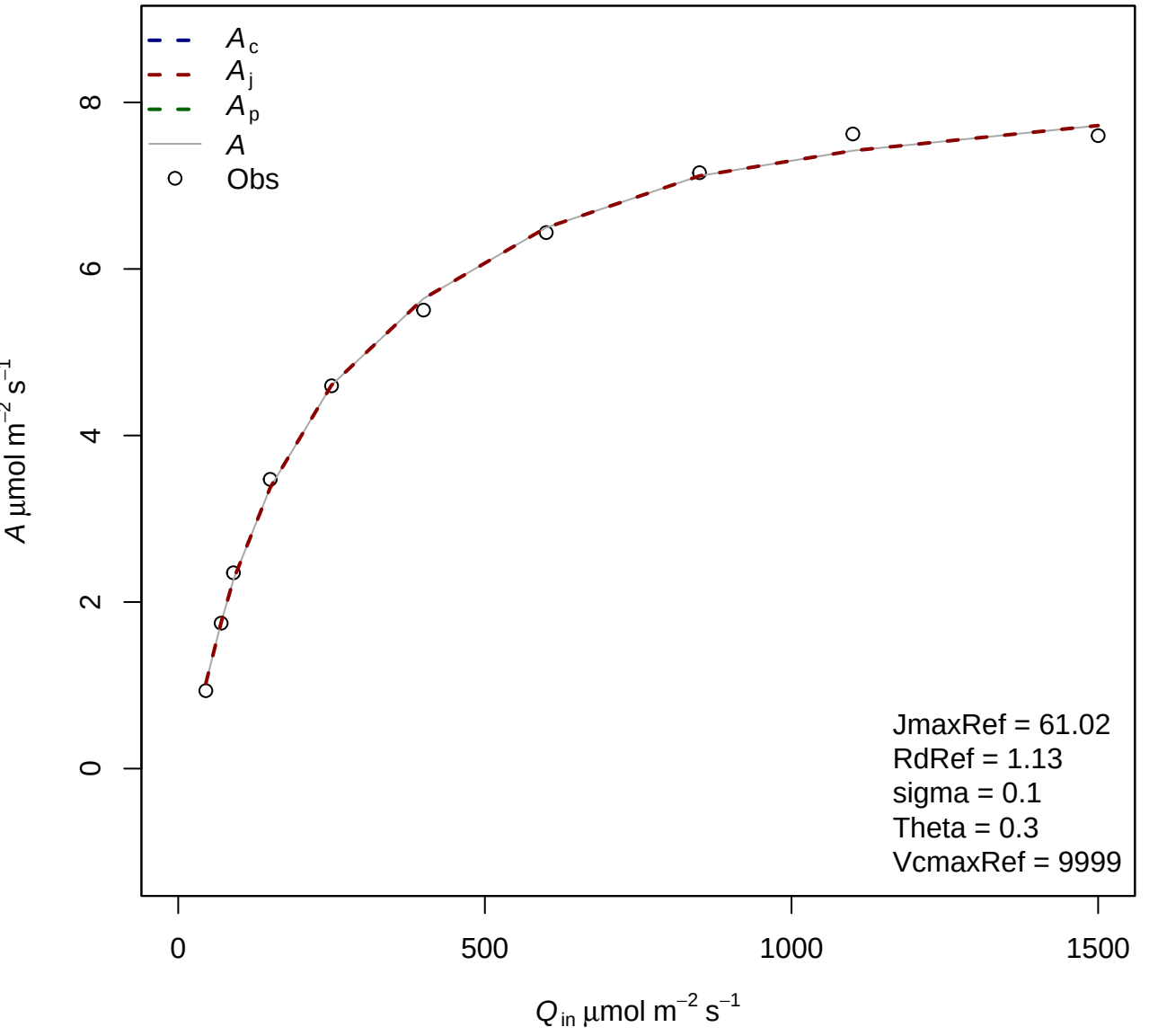

163
A_c
A_j
A_p
A
Obs
JmaxRef = 73.08
RdRef = 1.1
sigma = 0.07
Theta = 0.05
VcmaxRef = 33.06
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

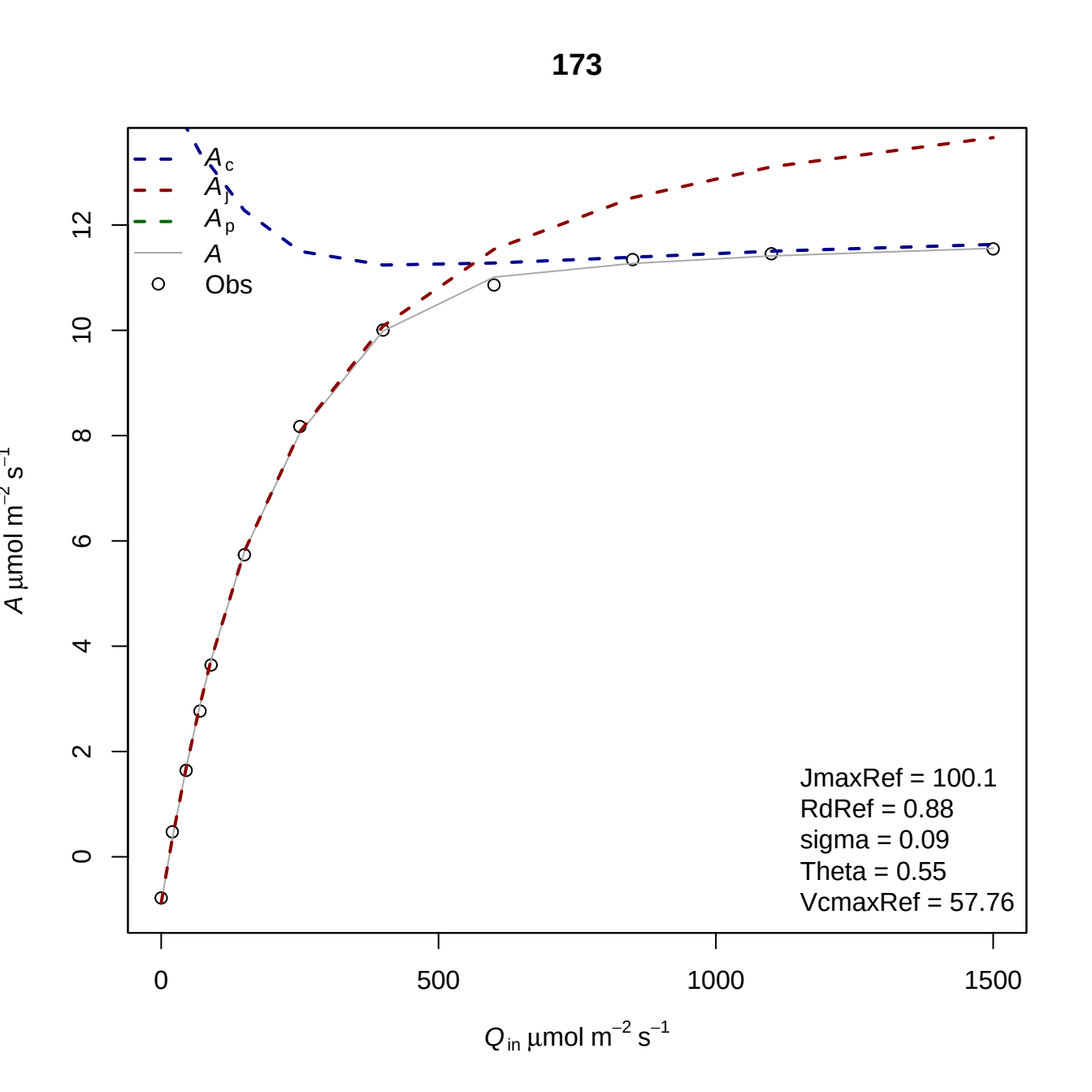

A_c
A_j
A_p
A
Obs
JmaxRef = 90.38
RdRef = 0.66
sigma = 0.05
Theta = −1.21
VcmaxRef = 9999
0
2
4
6
8
10
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

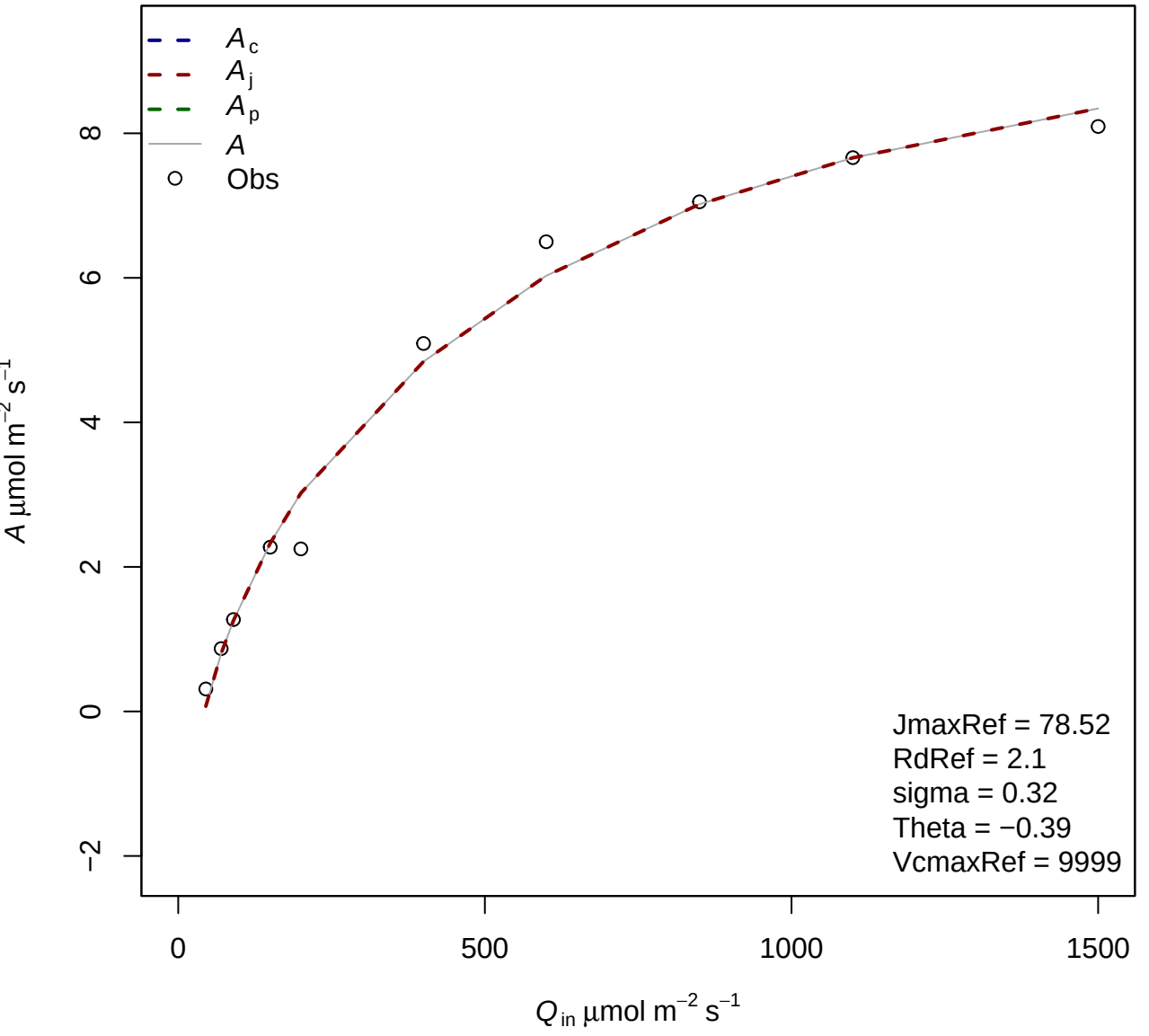

A_c
A_j
A_p
A
Obs
JmaxRef = 86.94
RdRef = 2.03
sigma = 0.06
Theta = −0.73
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

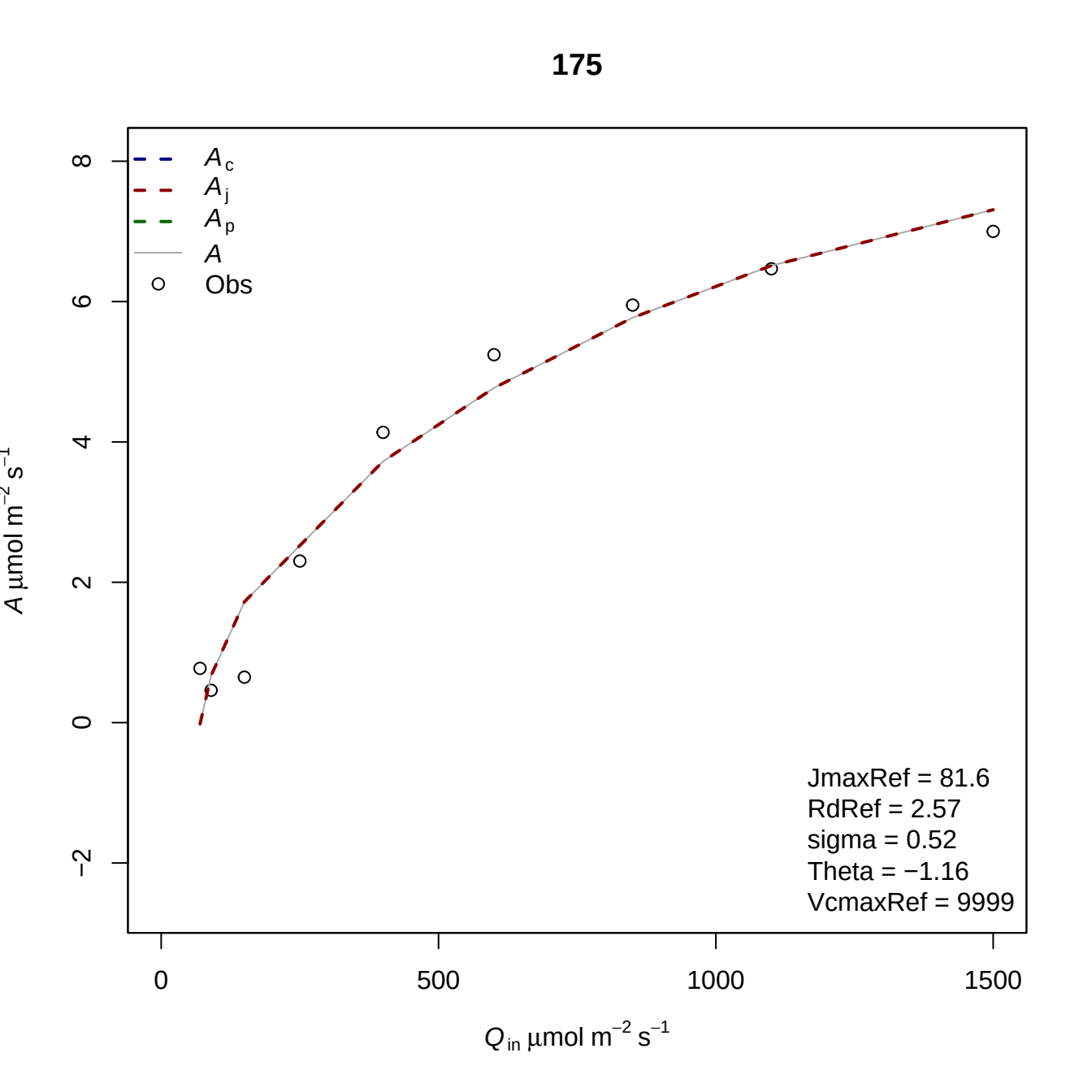

A_c
A_j
A_p
A
Obs
JmaxRef = 76.98
RdRef = 1.26
sigma = 0.07
Theta = 0.5
VcmaxRef = 43.19
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

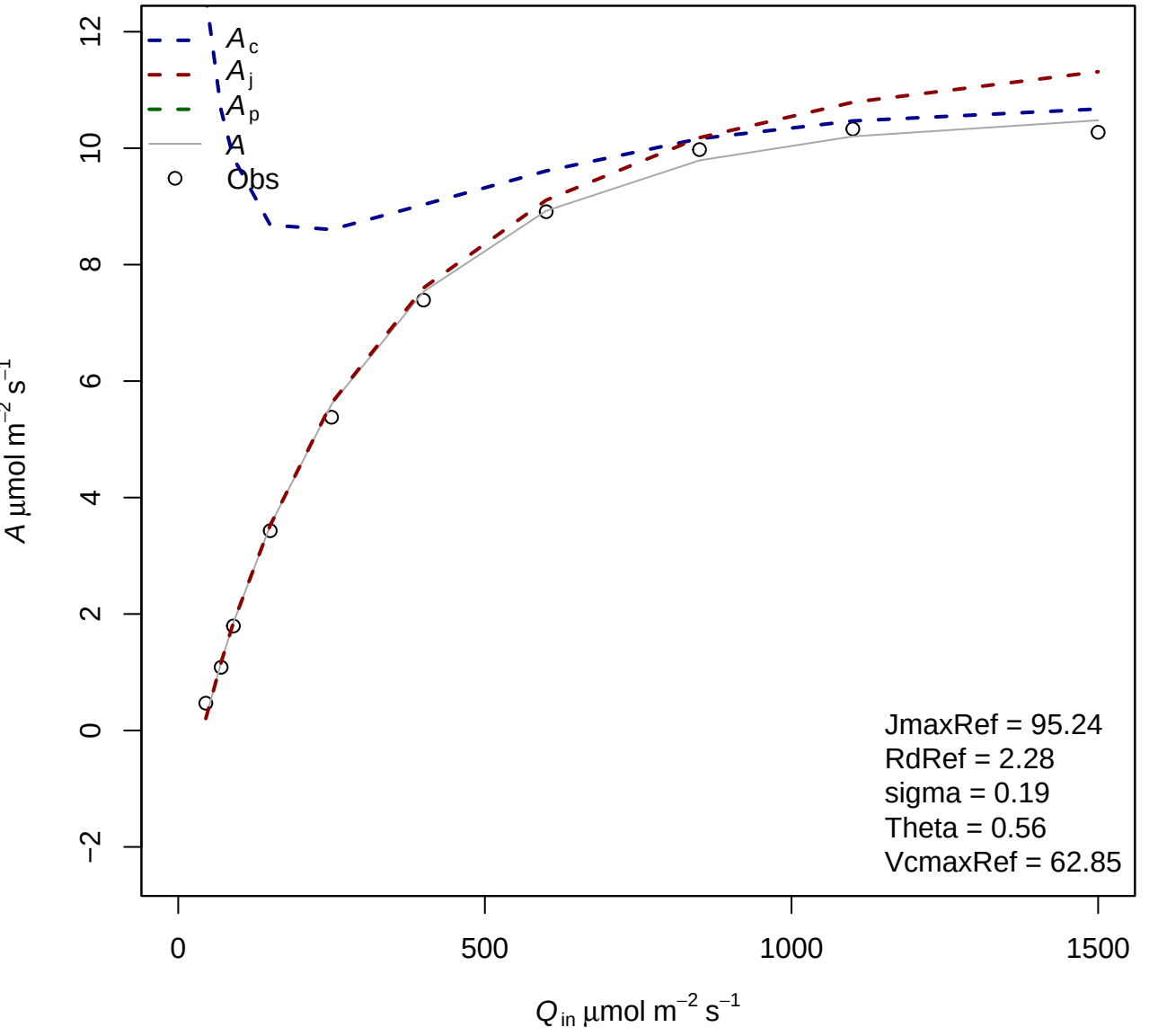

170
A_c
A_j
A_p
A
Obs
JmaxRef = 68.85
RdRef = 1.26
sigma = 0.1
Theta = 0.6
VcmaxRef = 40.96
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

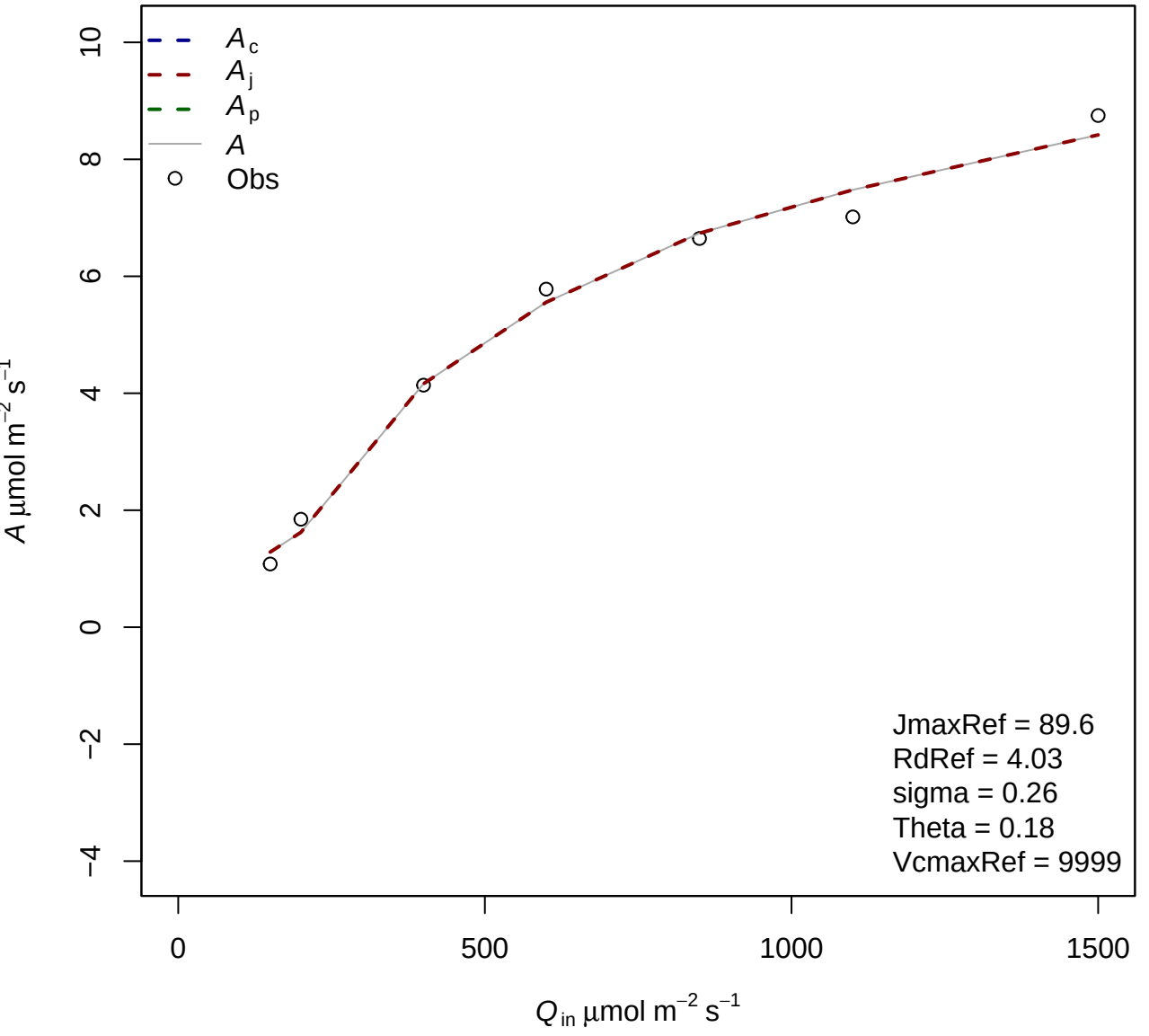

A_c
A_j
A_p
A
Obs
JmaxRef = 61.02
RdRef = 1.13
sigma = 0.1
Theta = 0.3
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

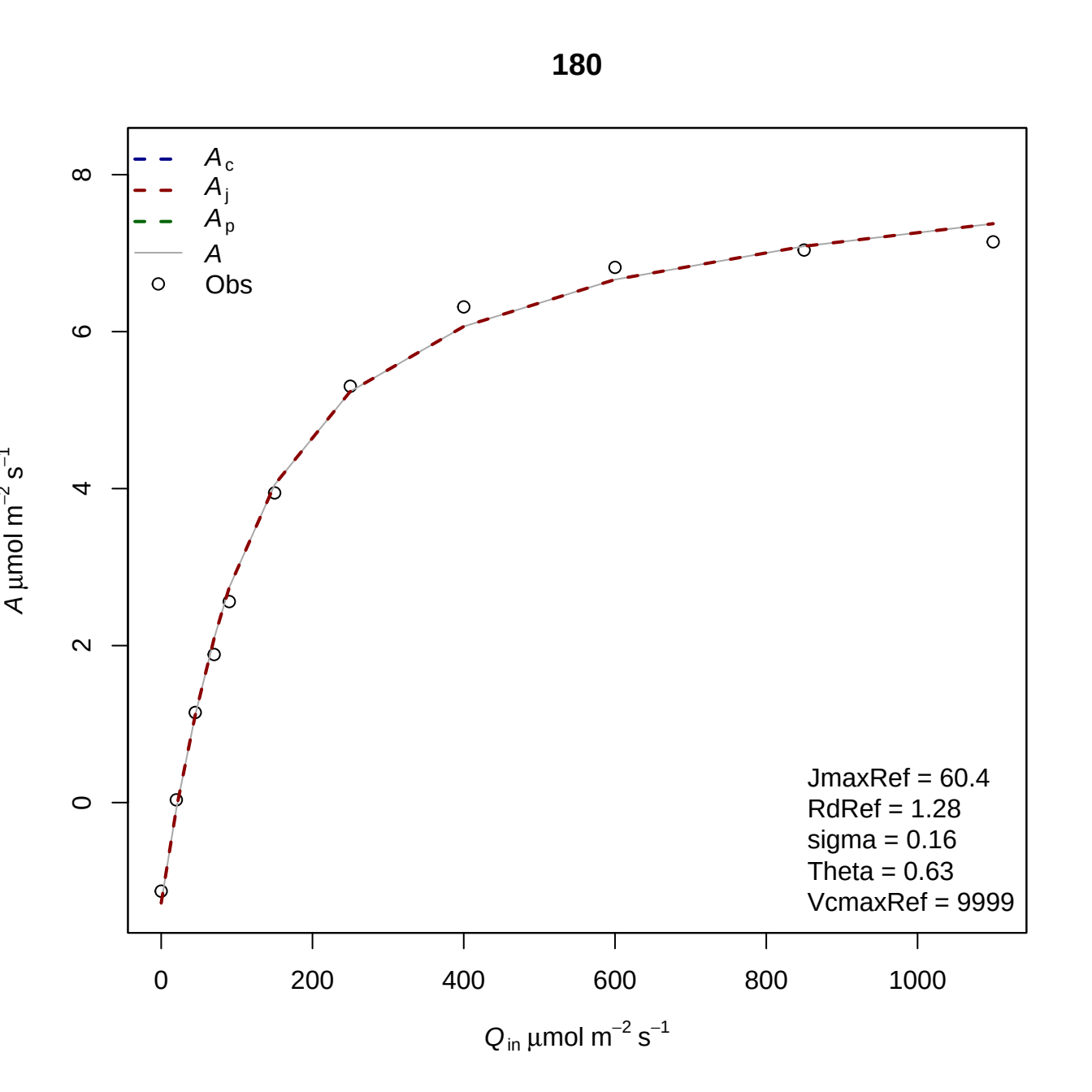

173
A_c
A_j
A_p
A
Obs
JmaxRef = 100.1
RdRef = 0.88
sigma = 0.09
Theta = 0.55
VcmaxRef = 57.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

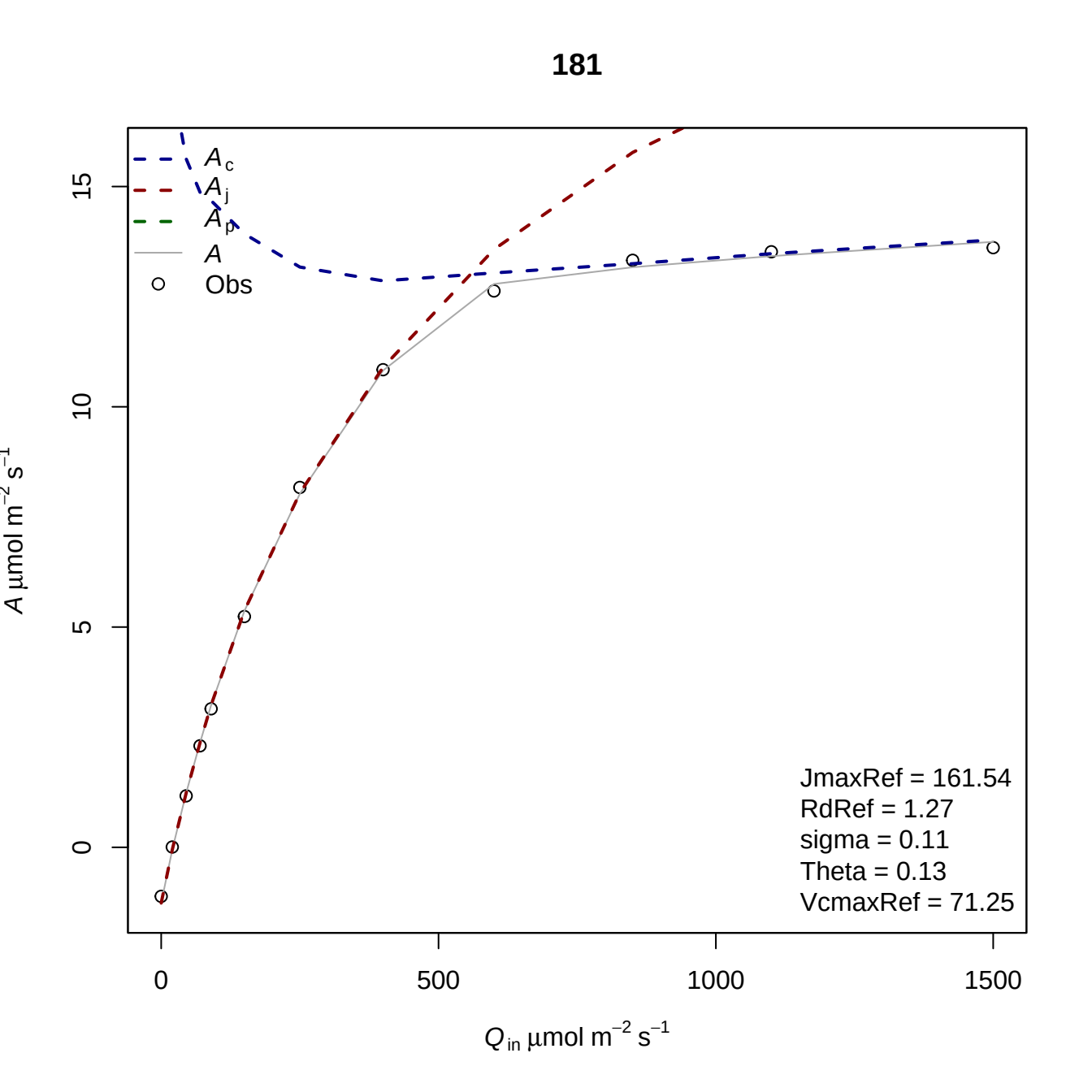

A_c
A_j
A_p
A
Obs
JmaxRef = 78.52
RdRef = 2.1
sigma = 0.32
Theta = −0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

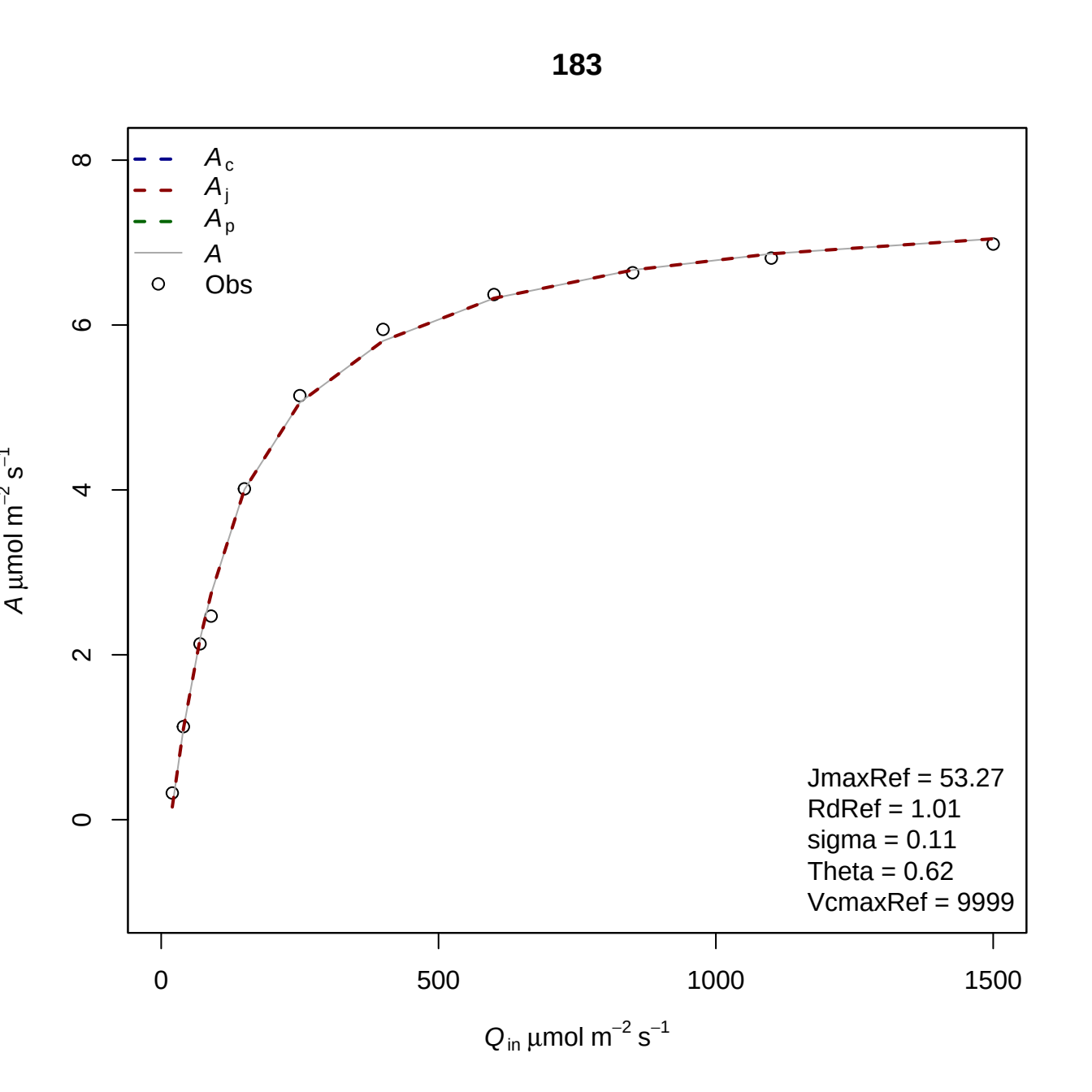

175
A_c
A_j
A_p
A
Obs
JmaxRef = 81.6
RdRef = 2.57
sigma = 0.52
Theta = −1.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

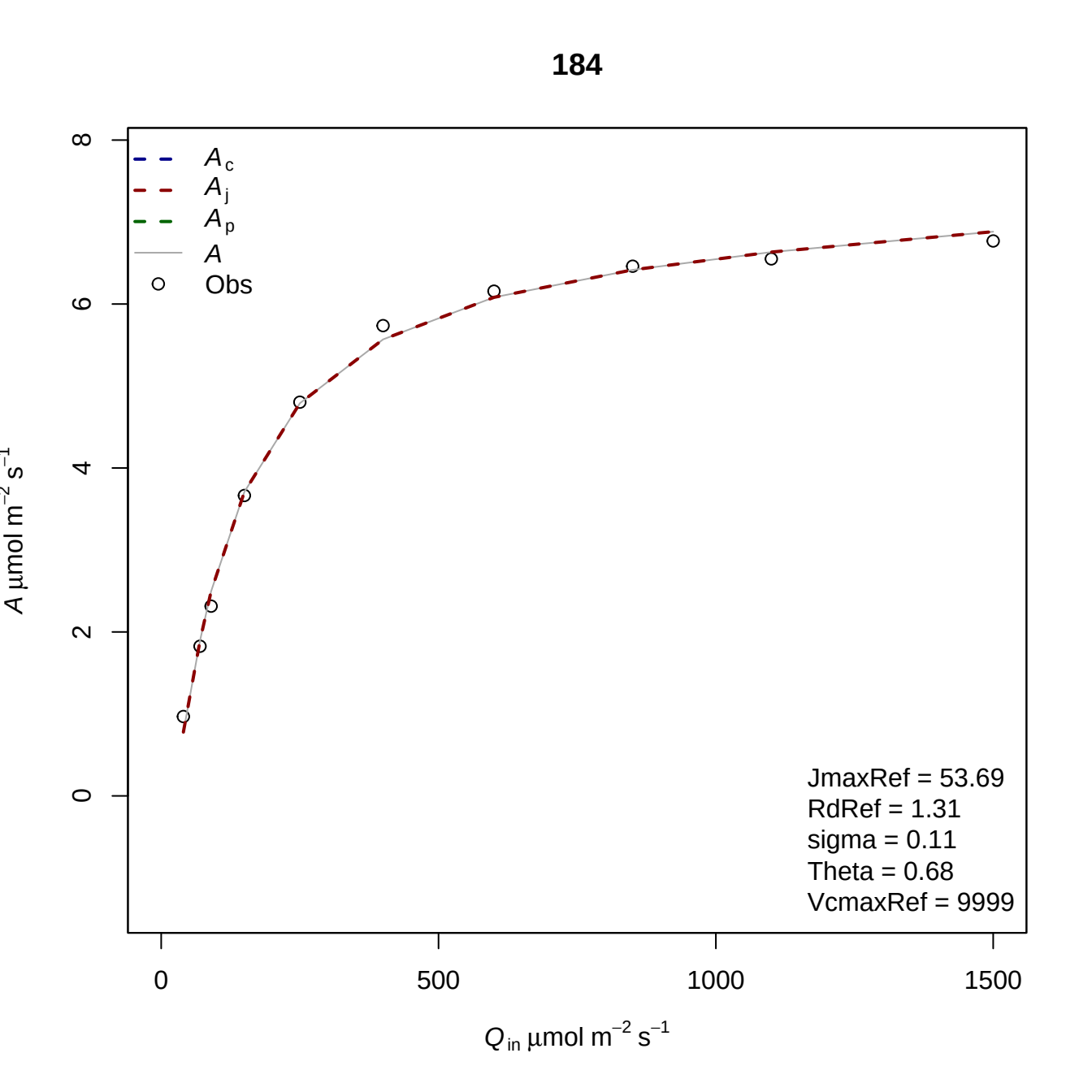

A_c
A_j
A_p
A
Obs
JmaxRef = 95.24
RdRef = 2.28
sigma = 0.19
Theta = 0.56
VcmaxRef = 62.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

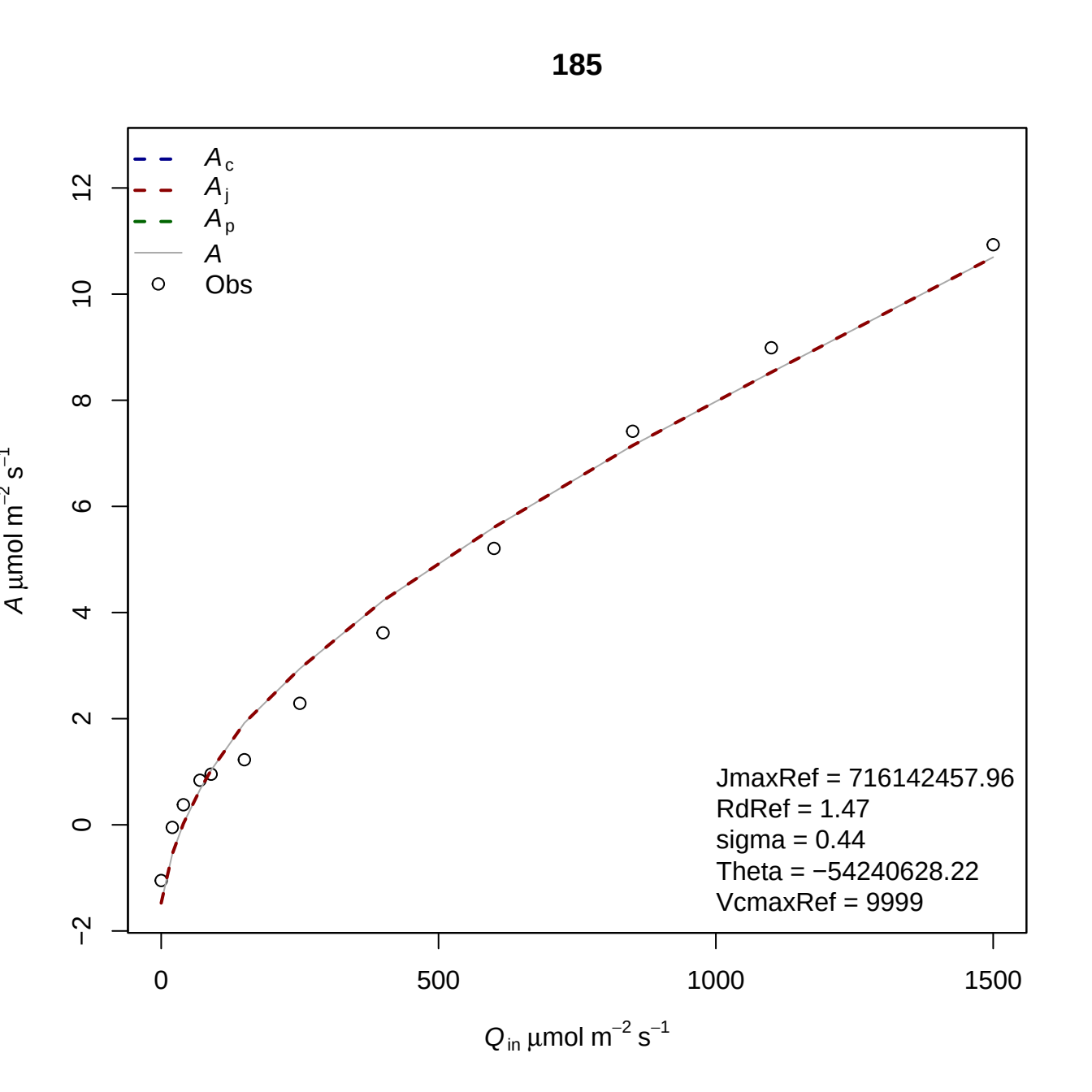

A_c
A_j
A_p
A
Obs
JmaxRef = 89.6
RdRef = 4.03
sigma = 0.26
Theta = 0.18
VcmaxRef = 9999
10
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

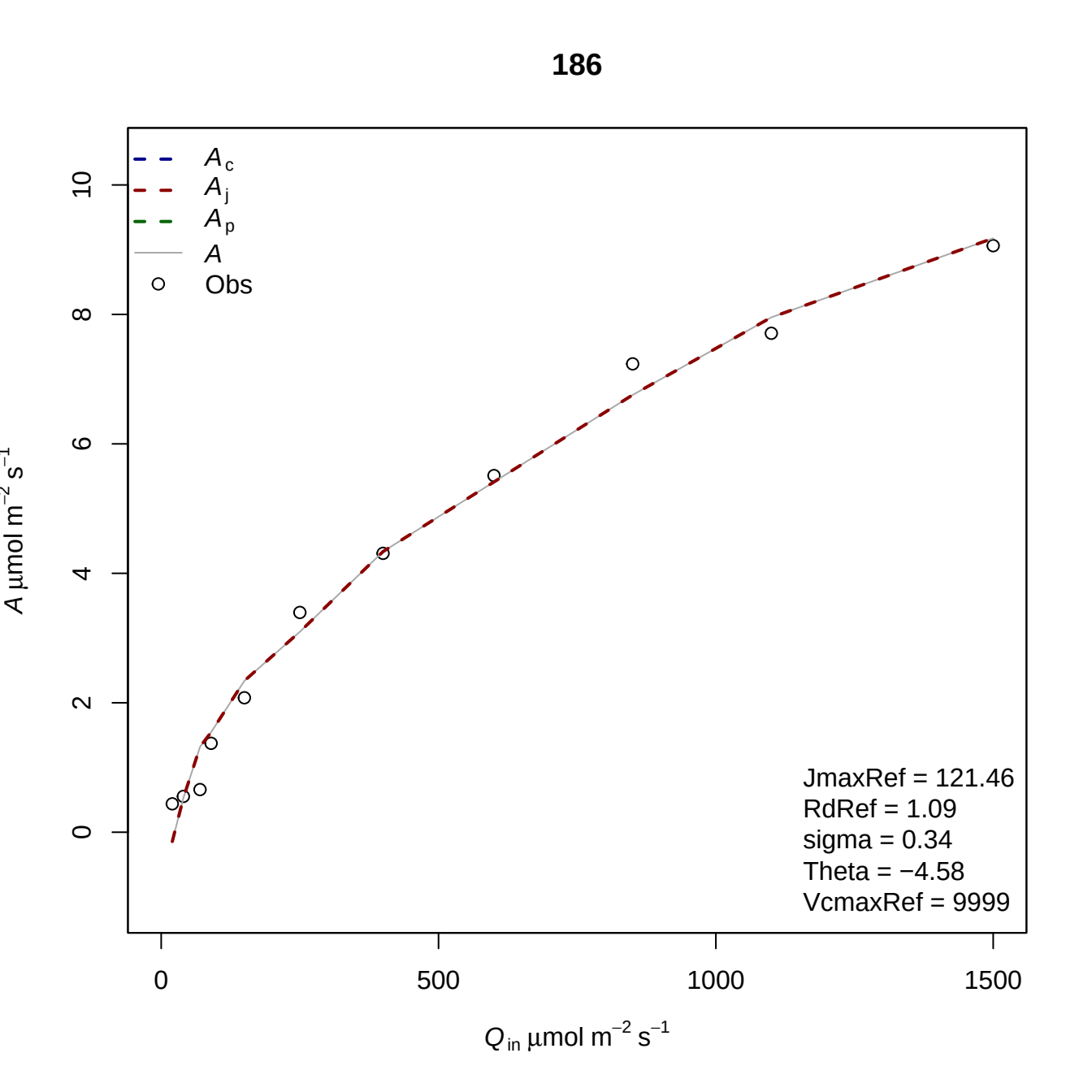

180
A_c
A_j
A_p
A
Obs
JmaxRef = 60.4
RdRef = 1.28
sigma = 0.16
Theta = 0.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000

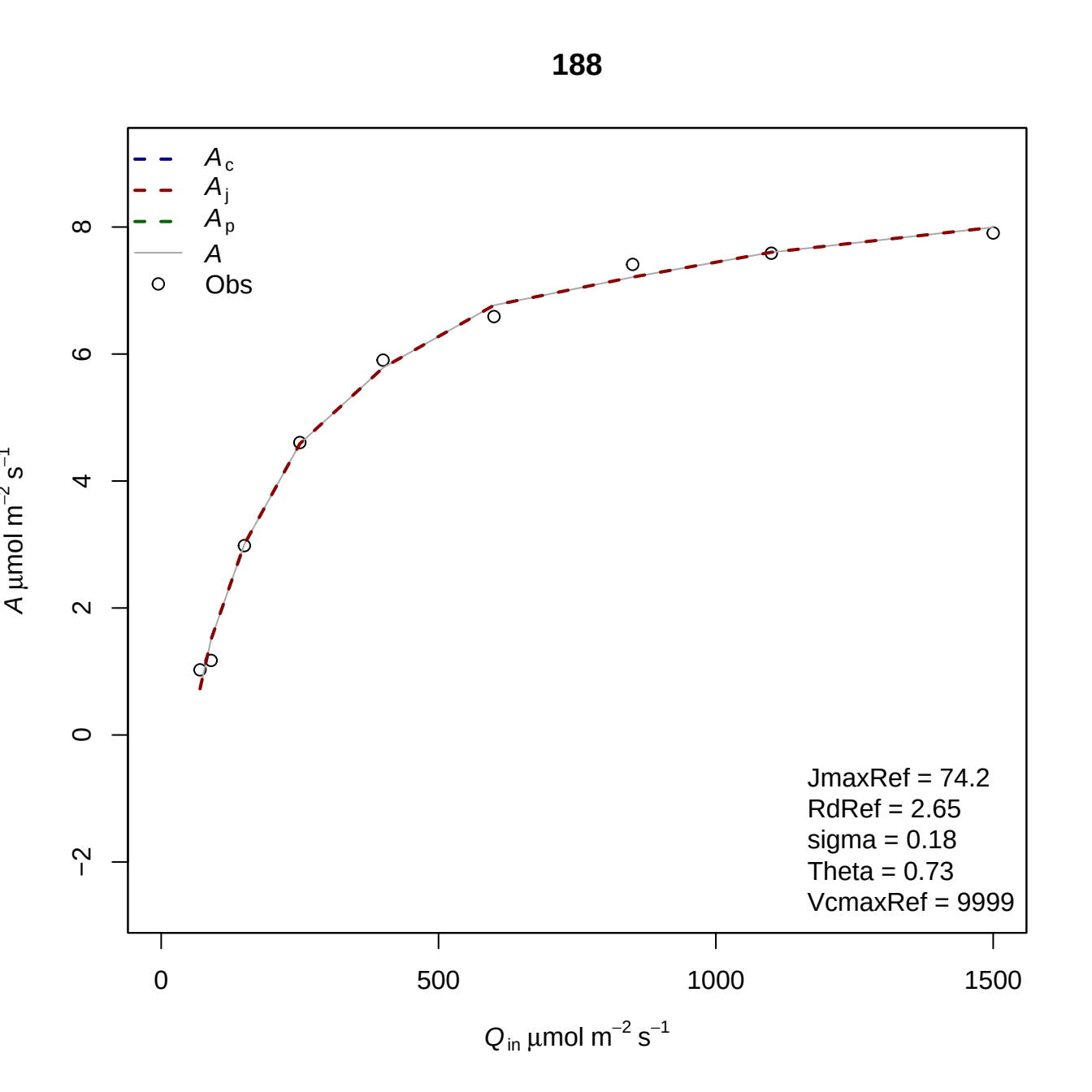

181
A_c
A_j
A_p
A
Obs
JmaxRef = 161.54
RdRef = 1.27
sigma = 0.11
Theta = 0.13
VcmaxRef = 71.25
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

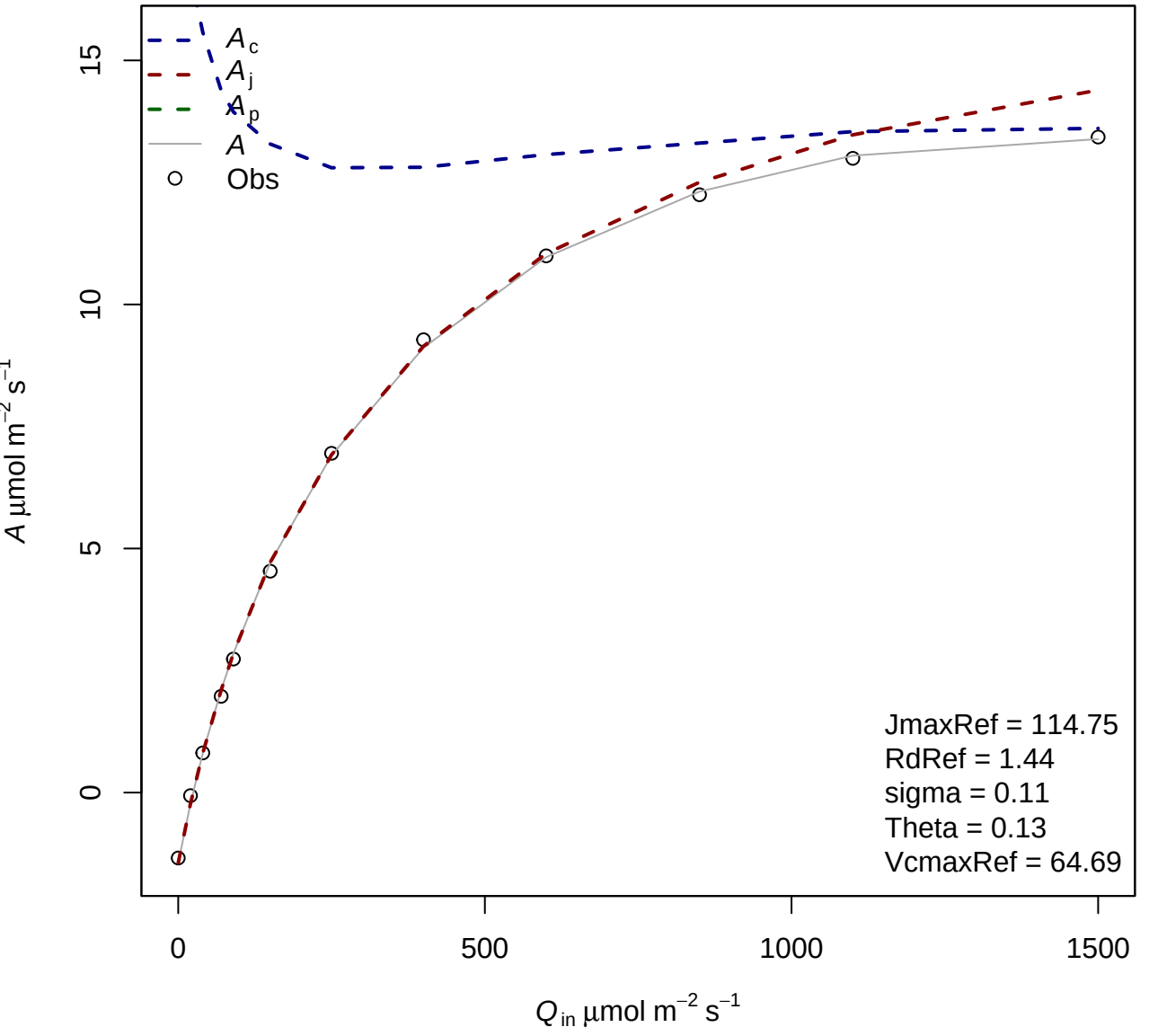

183
A_c
A_j
A_p
A
Obs
JmaxRef = 53.27
RdRef = 1.01
sigma = 0.11
Theta = 0.62
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

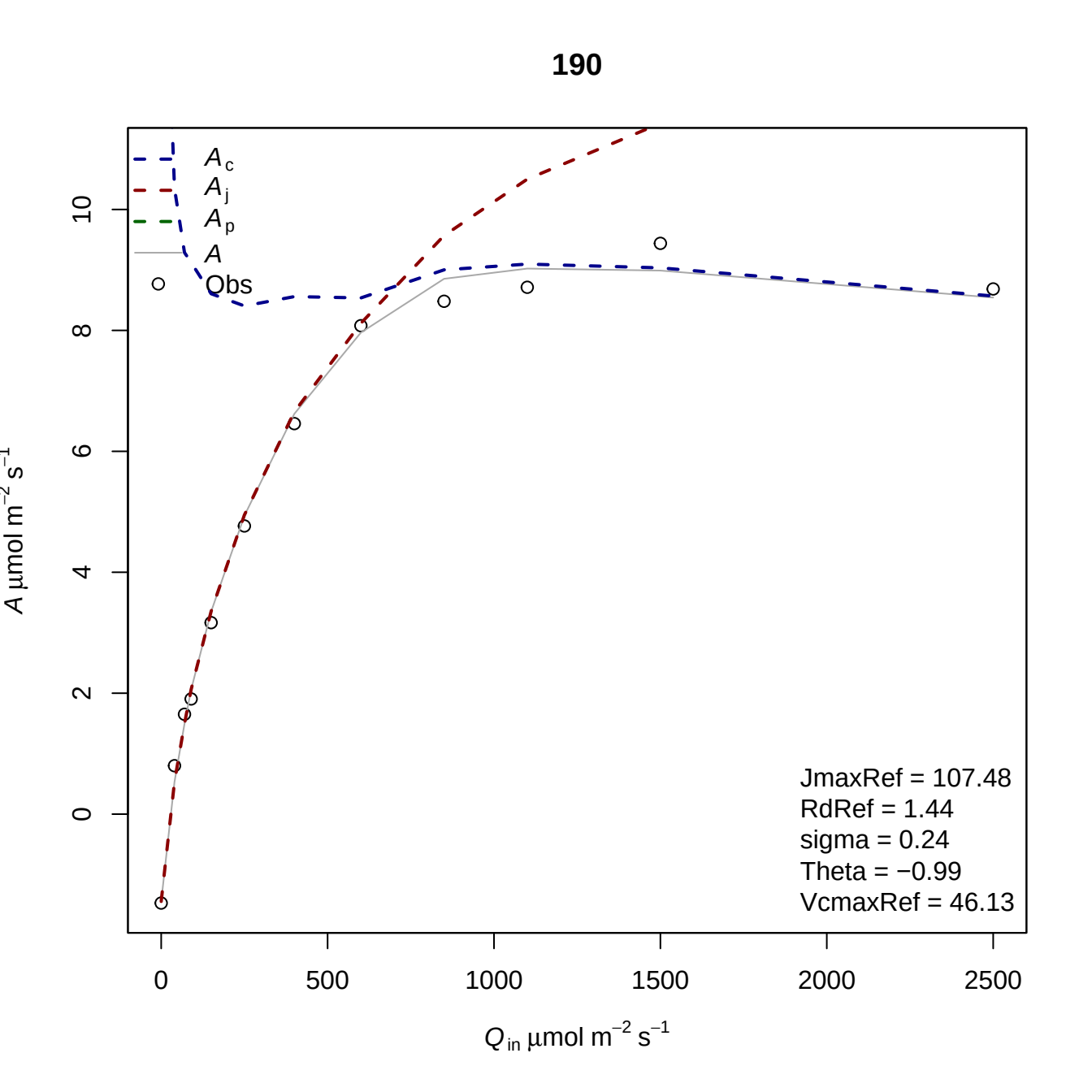

184
A_c
A_j
A_p
A
Obs
JmaxRef = 53.69
RdRef = 1.31
sigma = 0.11
Theta = 0.68
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

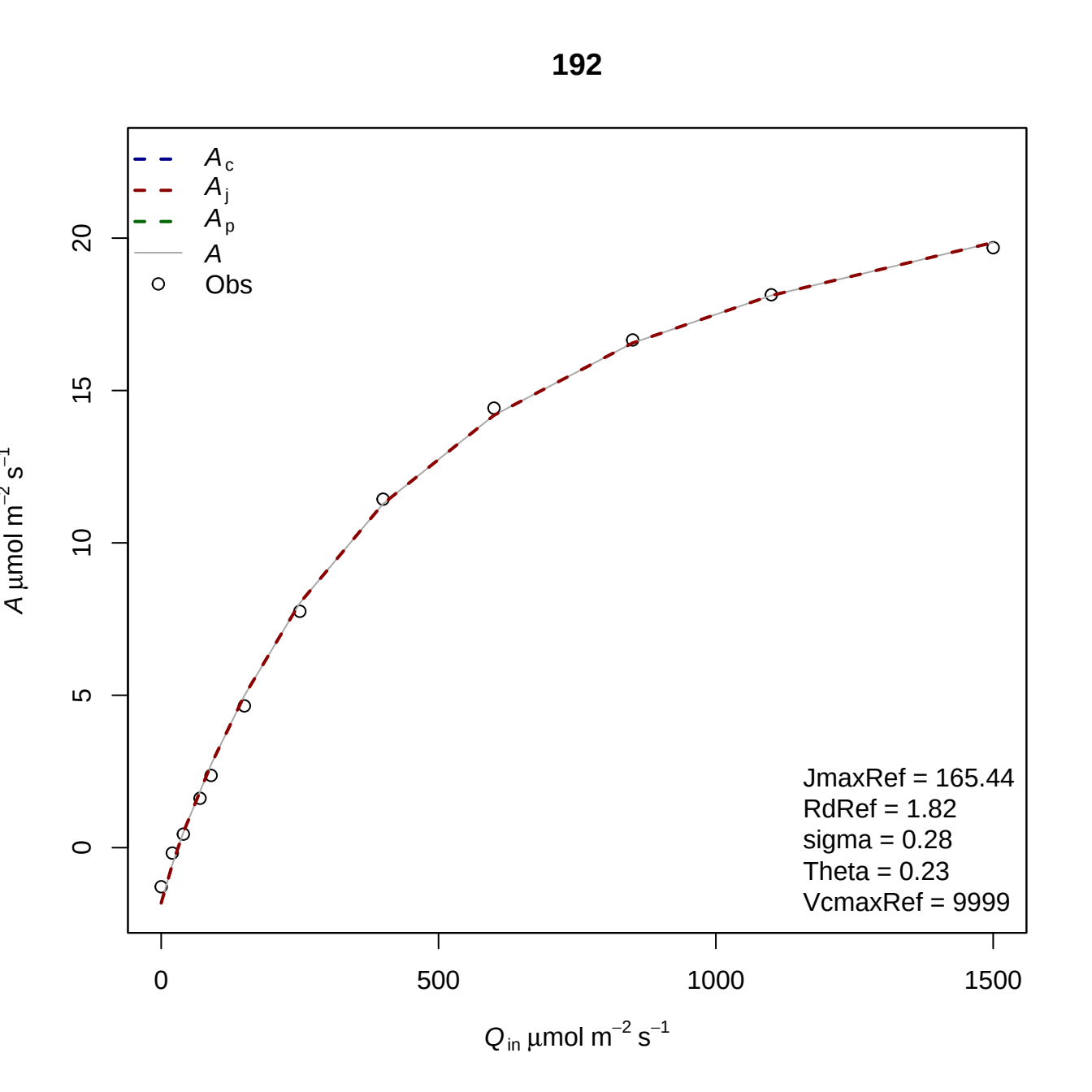

185
A_c
A_j
A_p
A
Obs
JmaxRef = 716142457.96
RdRef = 1.47
sigma = 0.44
Theta = −54240628.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

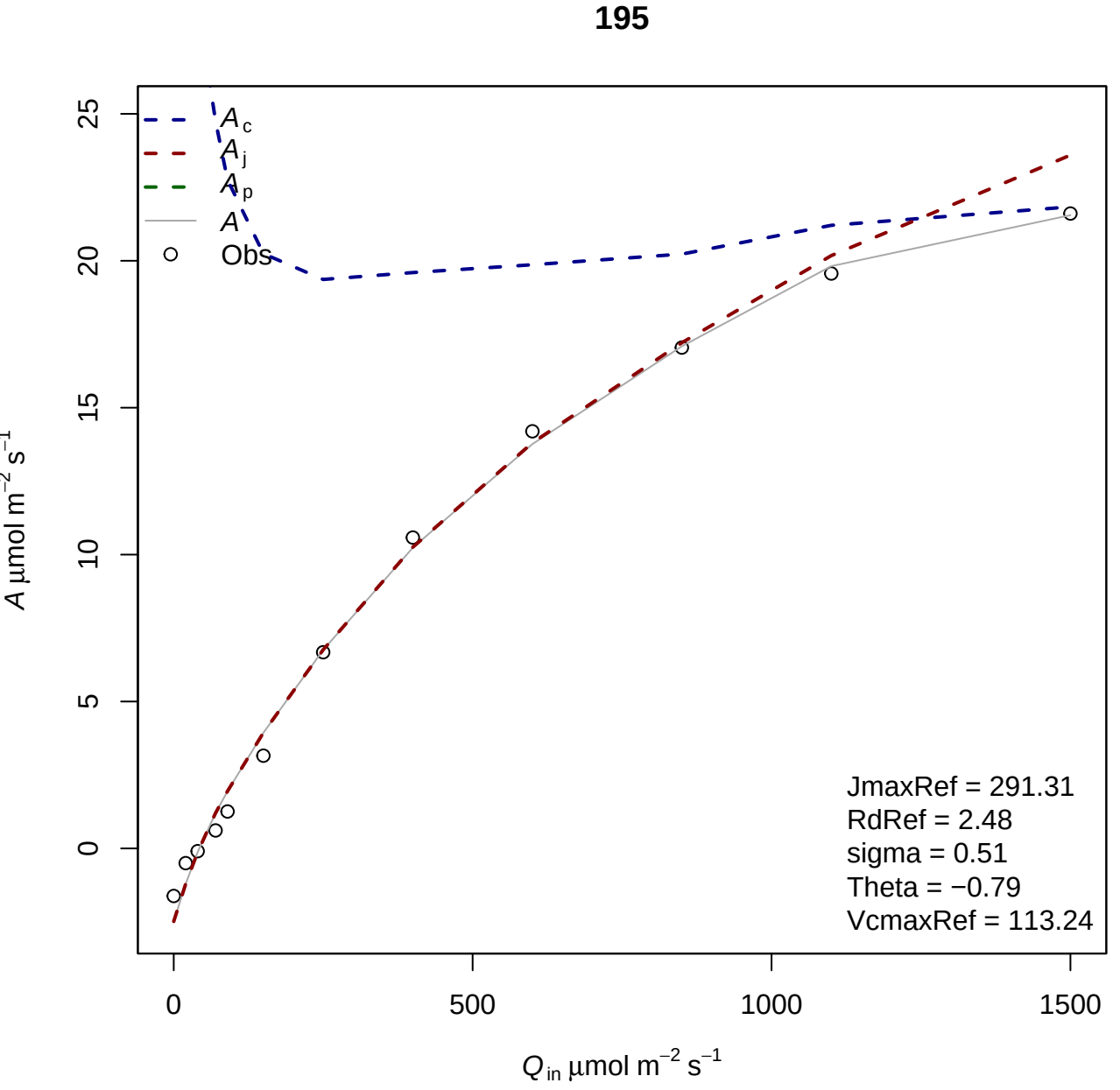

186
A_c
A_j
A_p
A
Obs
JmaxRef = 121.46
RdRef = 1.09
sigma = 0.34
Theta = −4.58
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

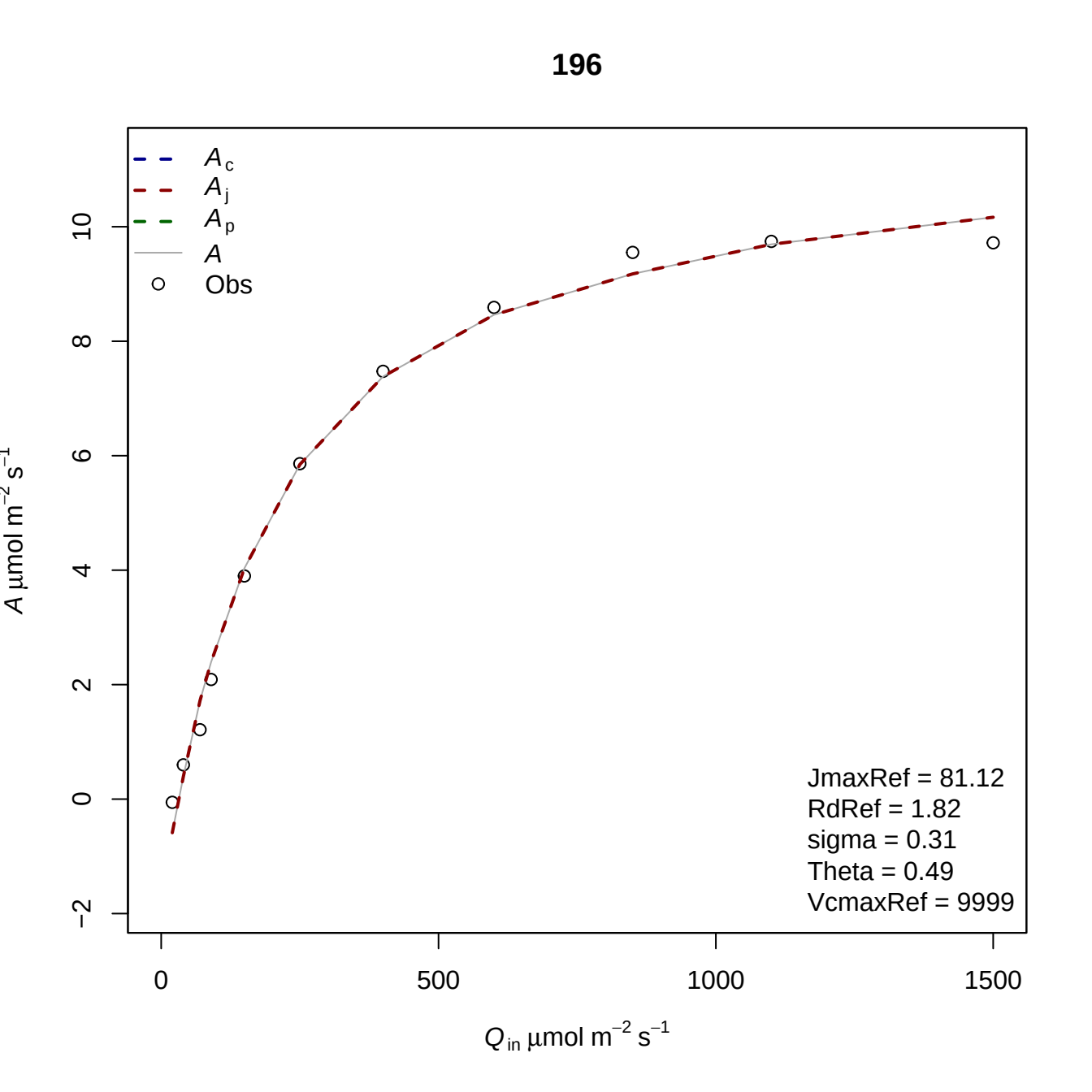

188
A_c
A_j
A_p
A
Obs
JmaxRef = 74.2
RdRef = 2.65
sigma = 0.18
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

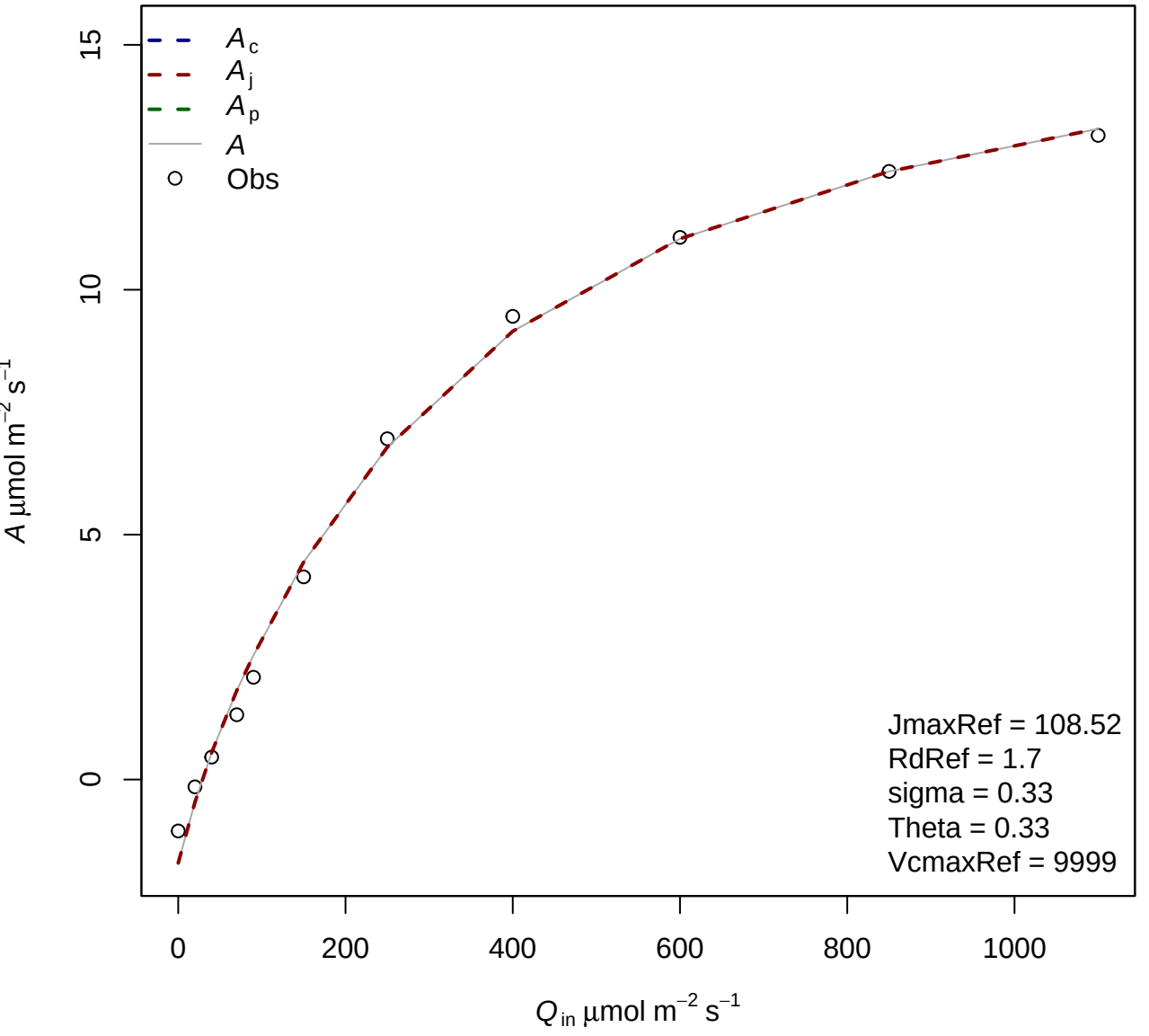

A_c
A_j
A_p
A
Obs
JmaxRef = 114.75
RdRef = 1.44
sigma = 0.11
Theta = 0.13
VcmaxRef = 64.69
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

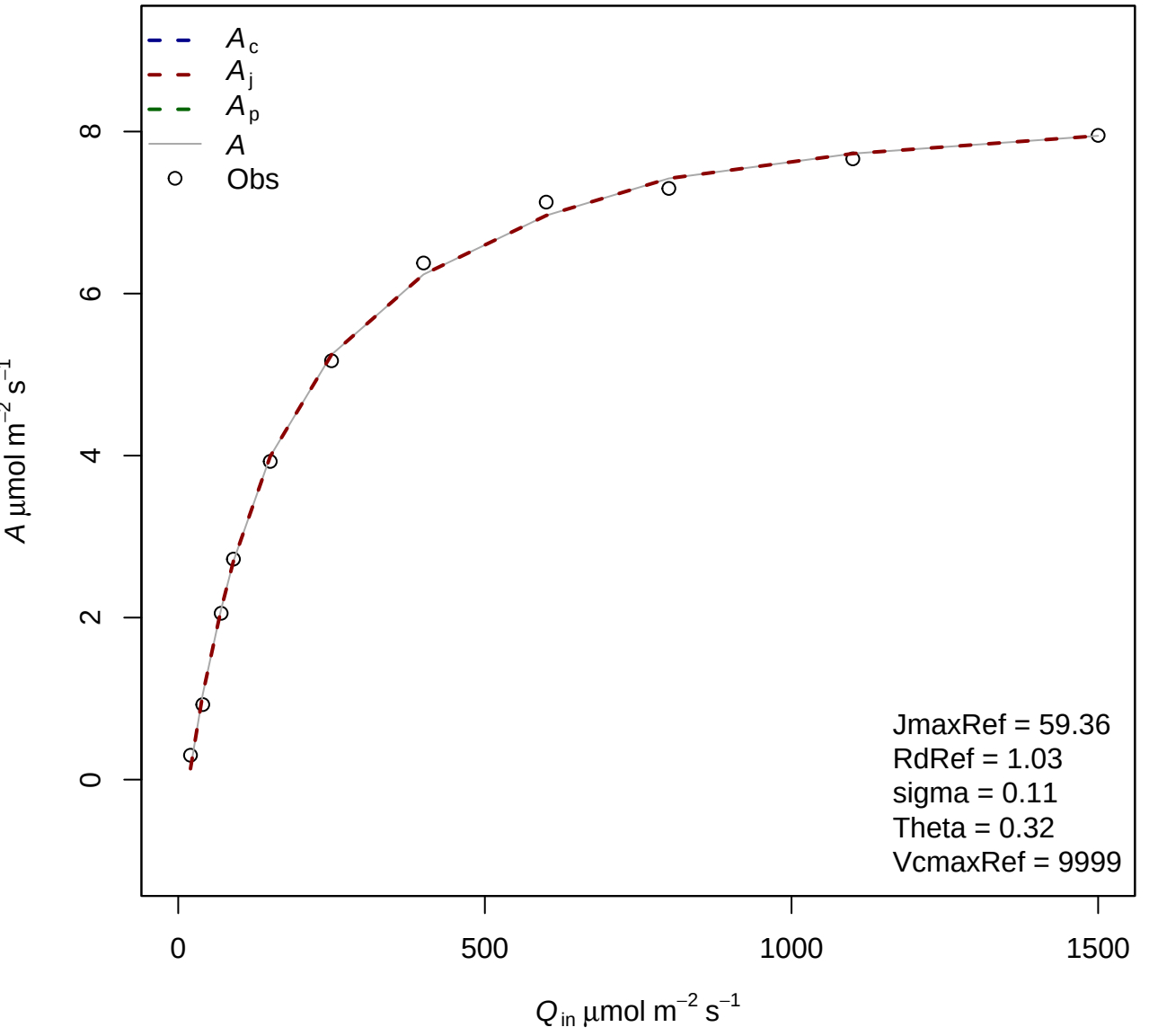

190
A_c
A_j
A_p
A
Obs
JmaxRef = 107.48
RdRef = 1.44
sigma = 0.24
Theta = −0.99
VcmaxRef = 46.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000
2500

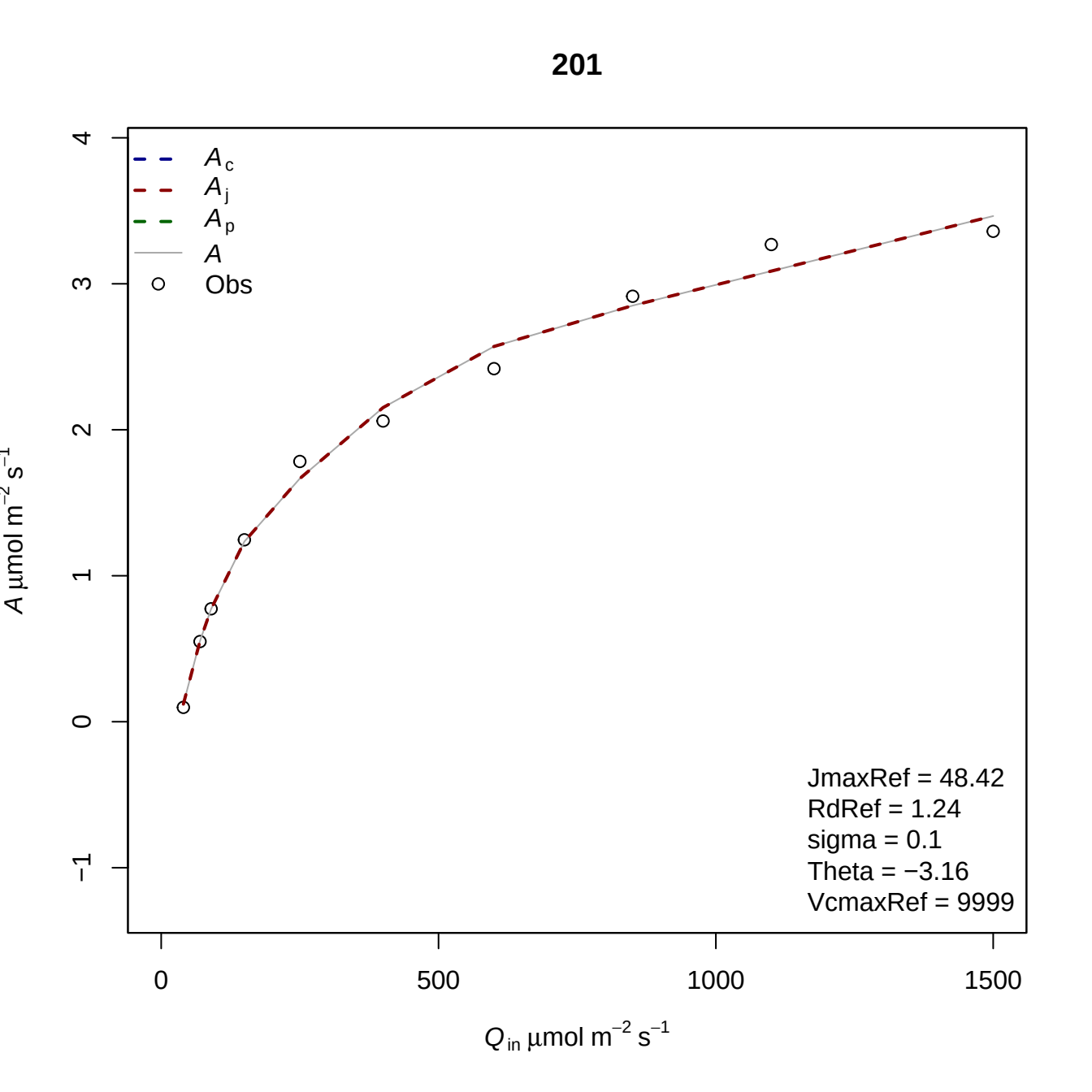

192
A_c
A_j
A_p
A
Obs
JmaxRef = 165.44
RdRef = 1.82
sigma = 0.28
Theta = 0.23
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

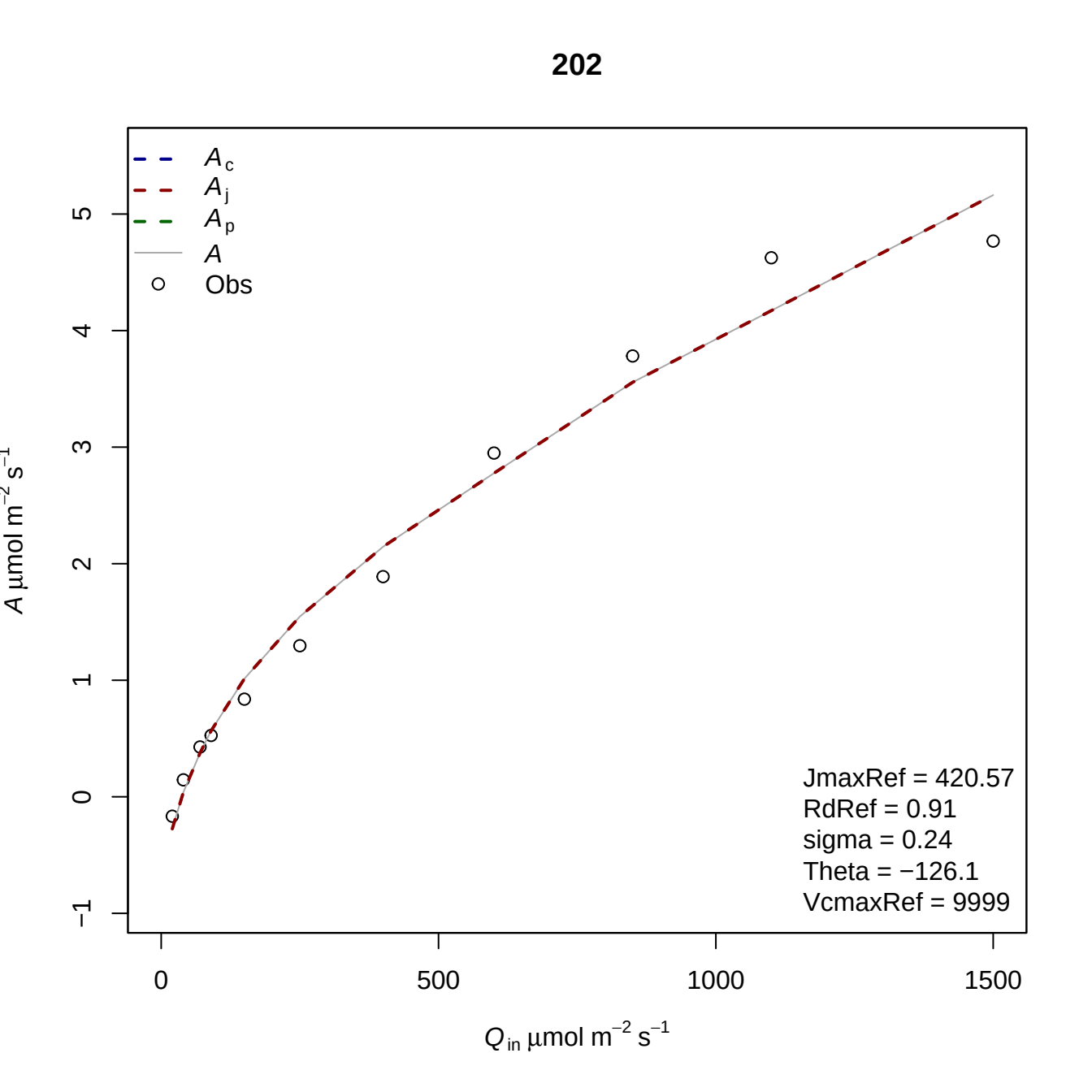

195
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
JmaxRef = 291.31
RdRef = 2.48
sigma = 0.51
Theta = −0.79
VcmaxRef = 113.24

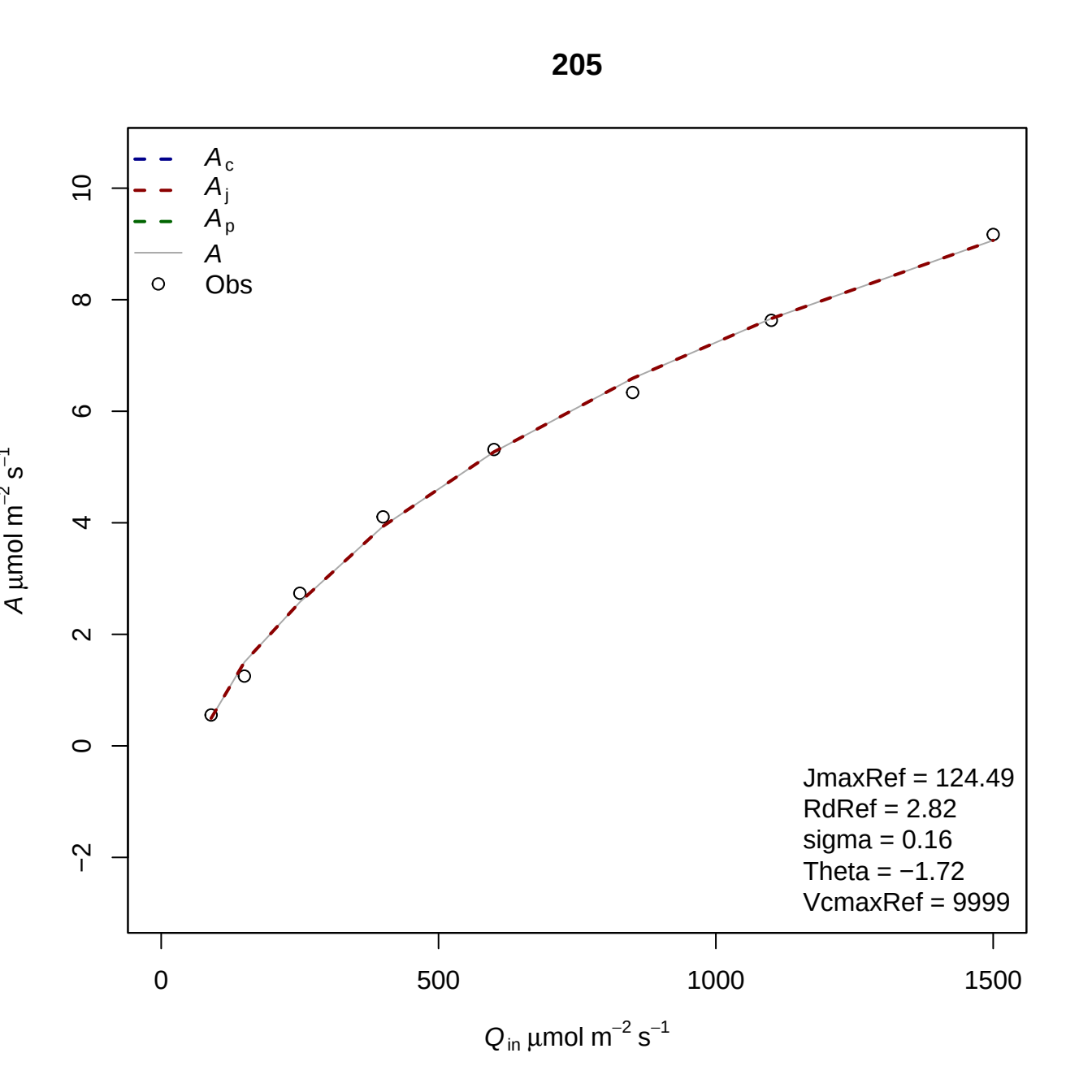

196
A_c
A_j
A_p
A
Obs
JmaxRef = 81.12
RdRef = 1.82
sigma = 0.31
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

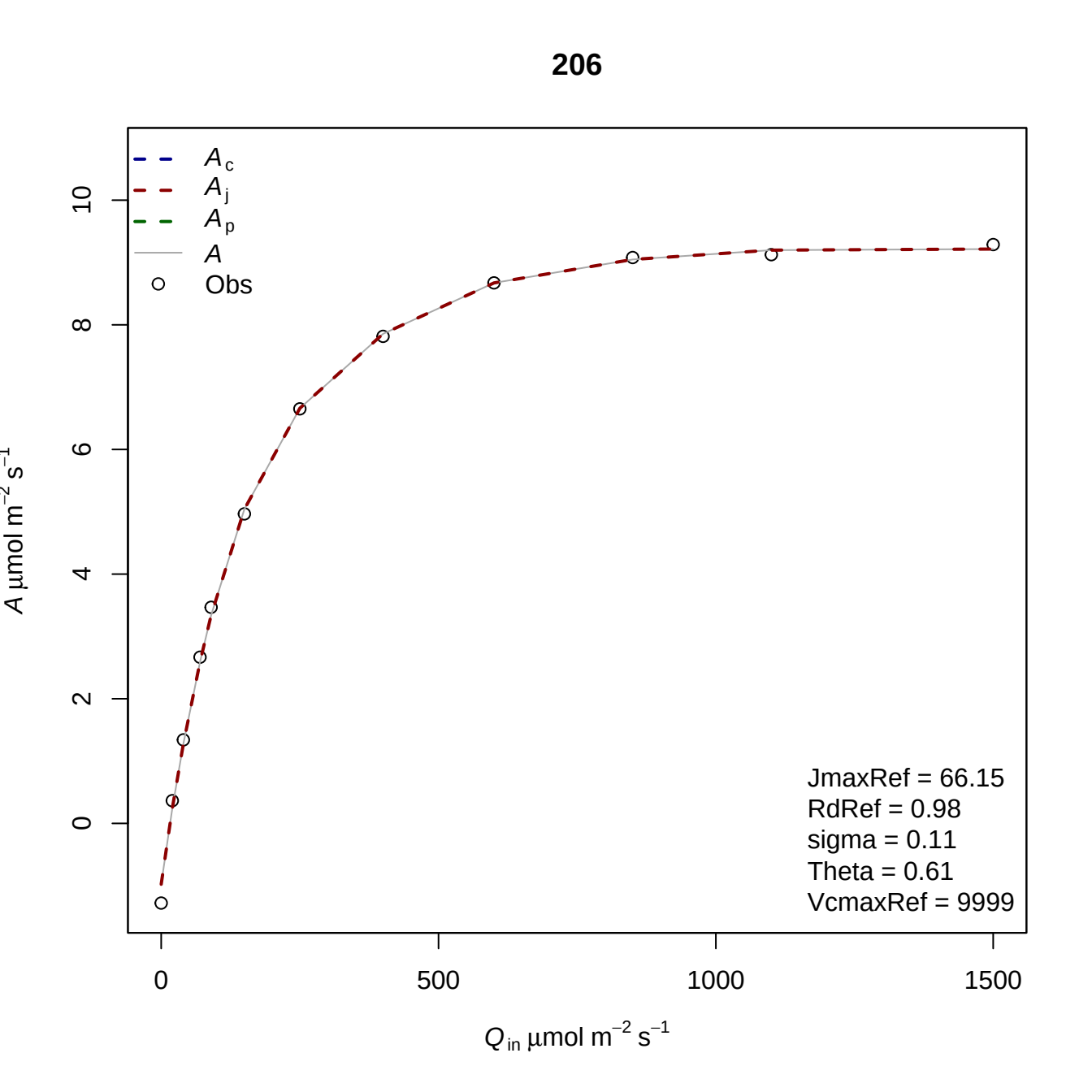

A_c
A_j
A_p
A
Obs
JmaxRef = 108.52
RdRef = 1.7
sigma = 0.33
Theta = 0.33
VcmaxRef = 9999
0
5
10
15
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

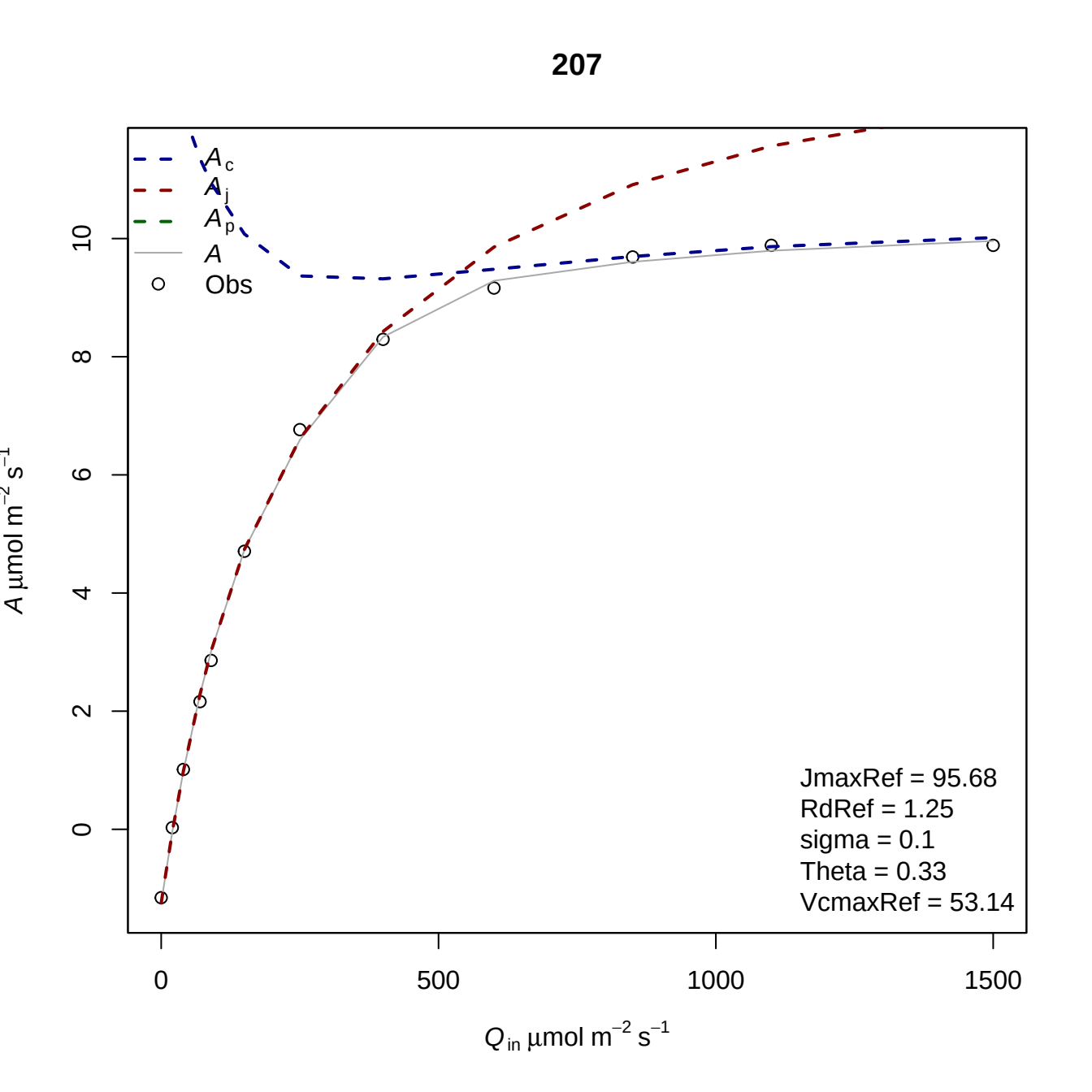

A_c
A_j
A_p
A
Obs
JmaxRef = 59.36
RdRef = 1.03
sigma = 0.11
Theta = 0.32
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

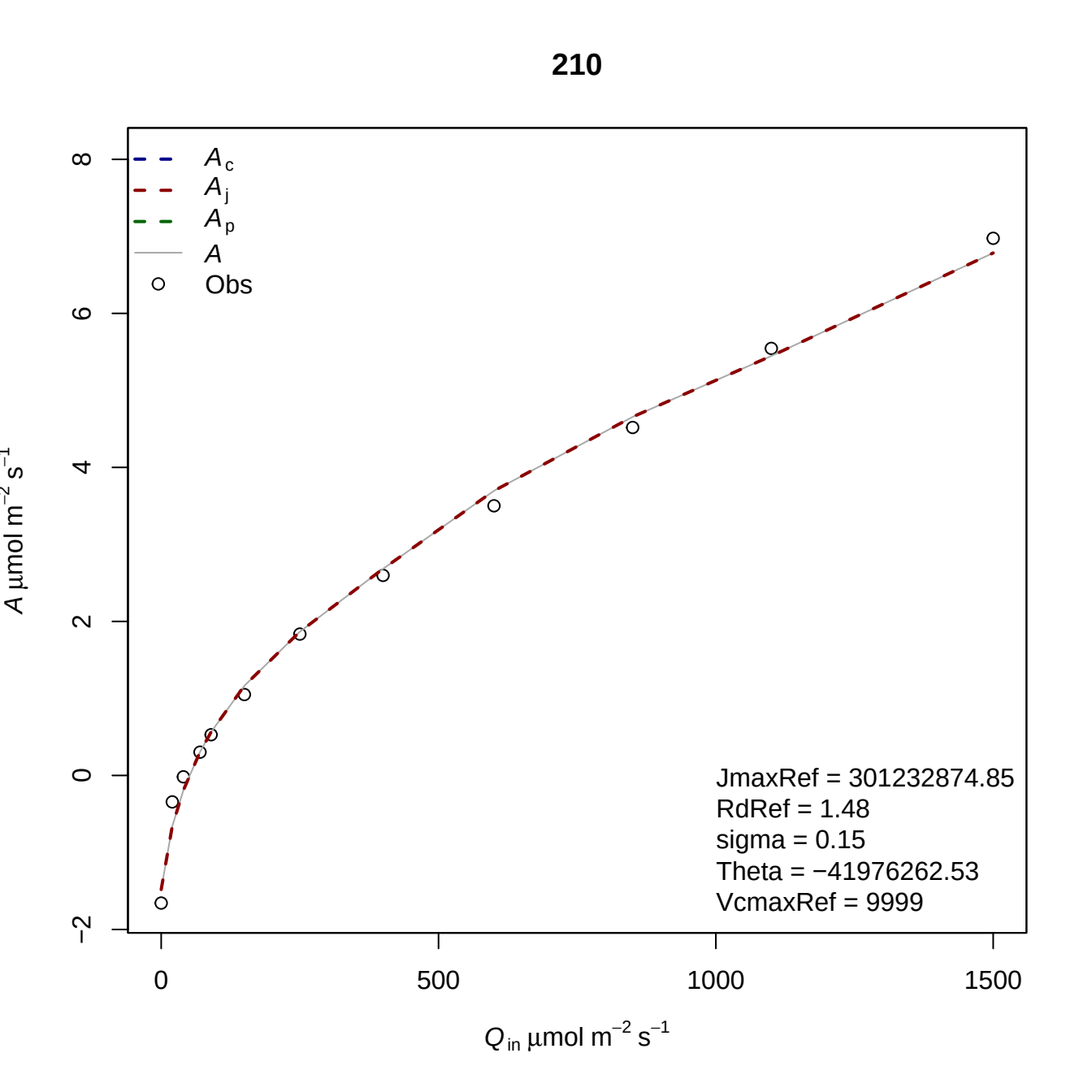

201
A_c
A_j
A_p
A
Obs
JmaxRef = 48.42
RdRef = 1.24
sigma = 0.1
Theta = −3.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

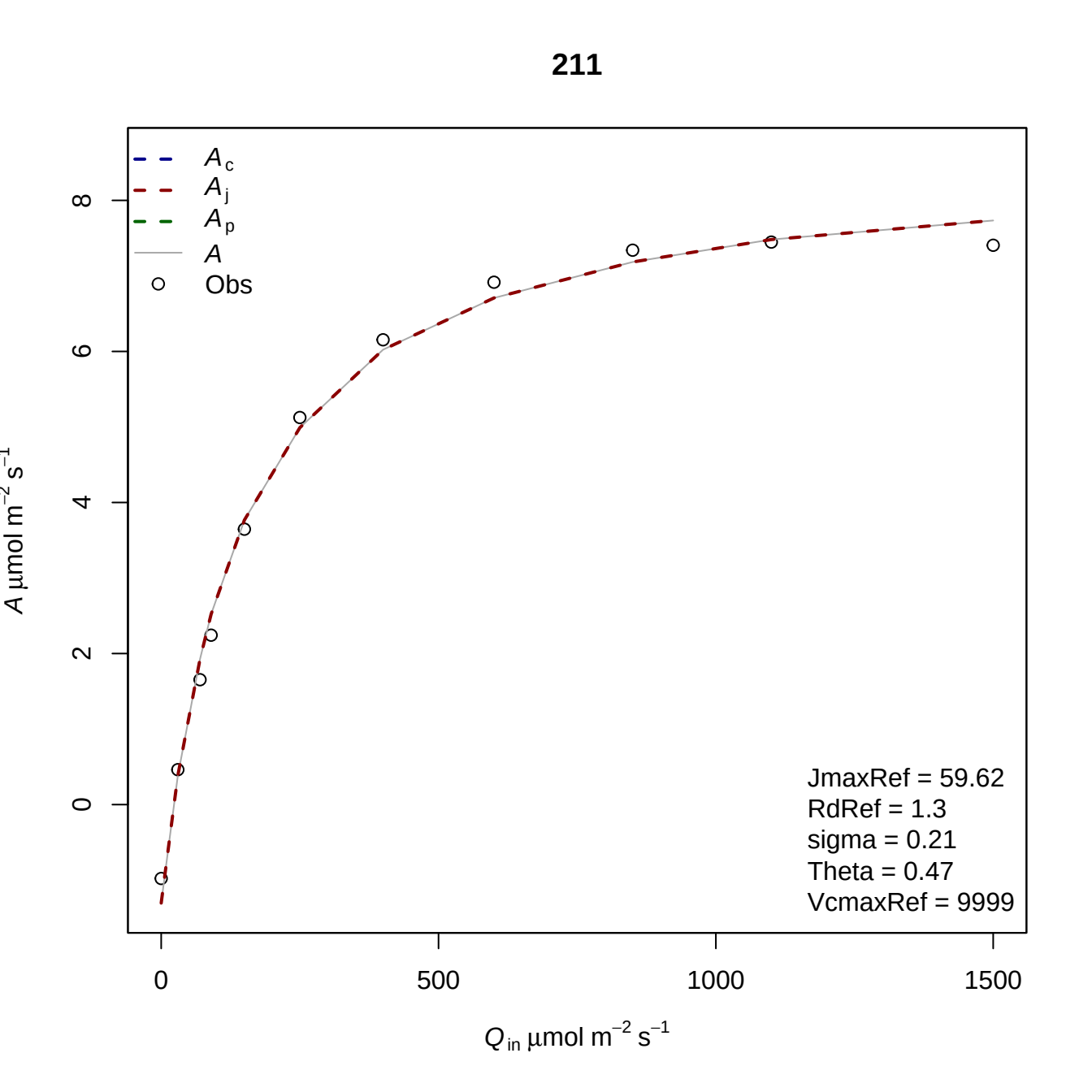

202
A_c
A_j
A_p
A
Obs
JmaxRef = 420.57
RdRef = 0.91
sigma = 0.24
Theta = −126.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

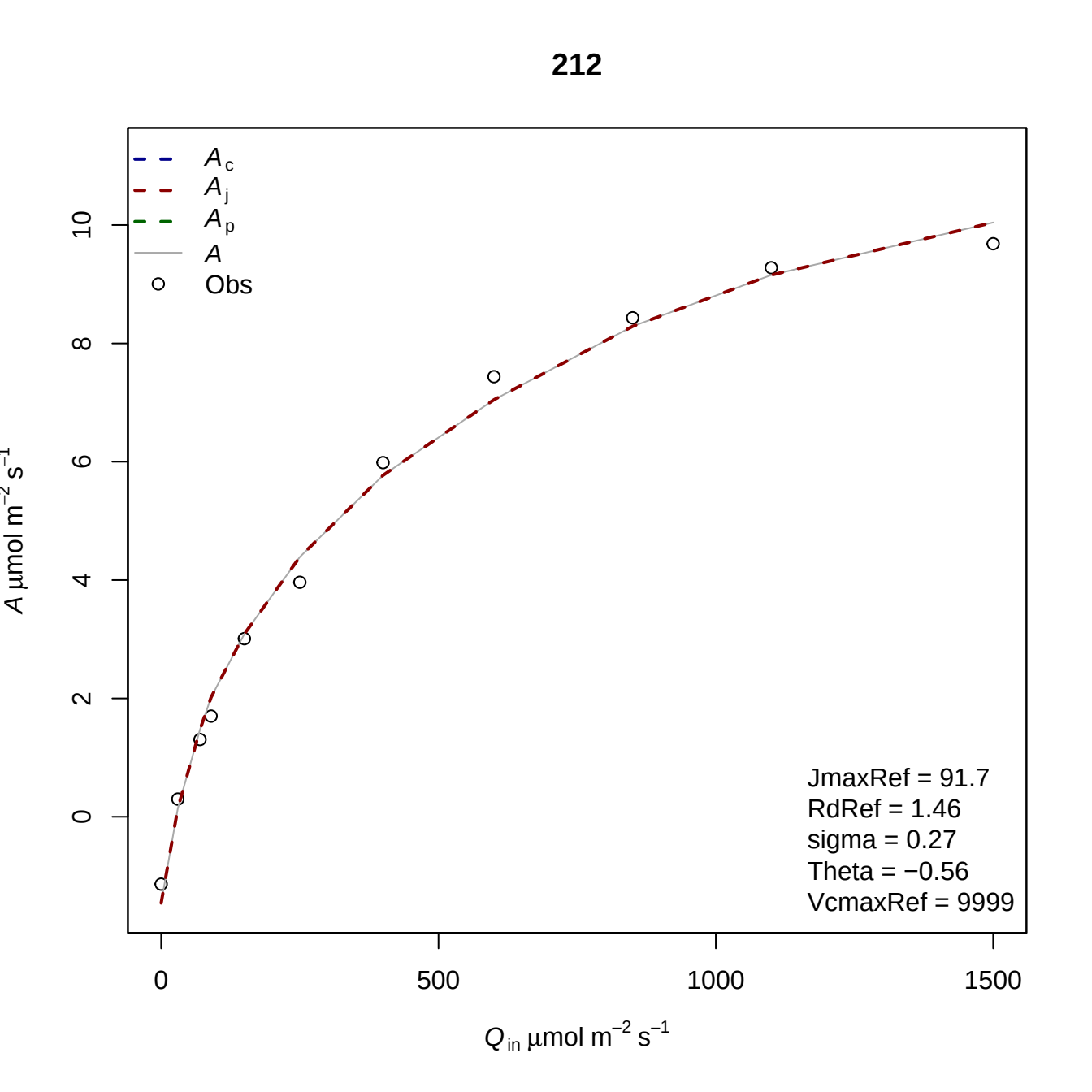

205
A_c
A_j
A_p
A
Obs
JmaxRef = 124.49
RdRef = 2.82
sigma = 0.16
Theta = −1.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

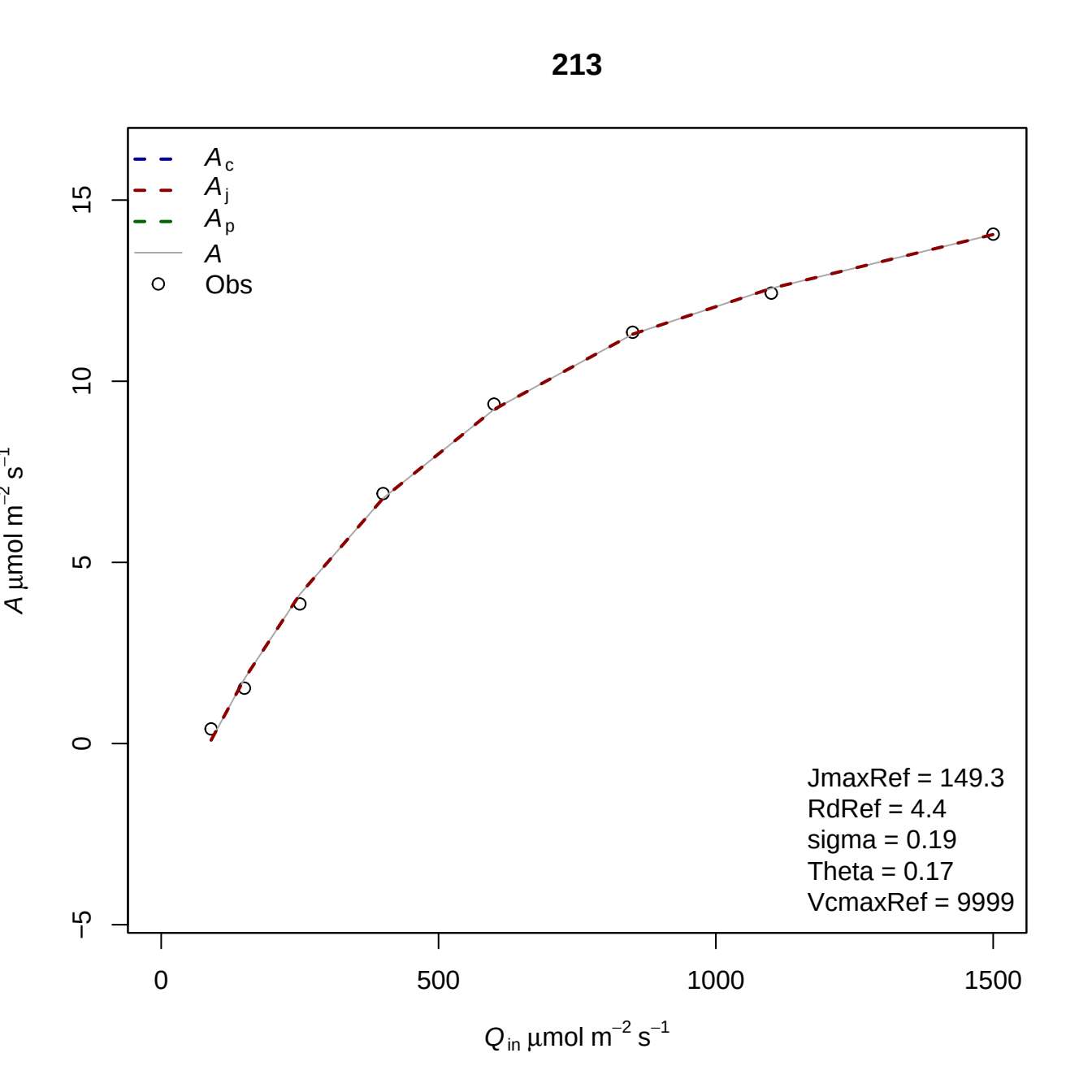

206
A_c
A_j
A_p
A
Obs
JmaxRef = 66.15
RdRef = 0.98
sigma = 0.11
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

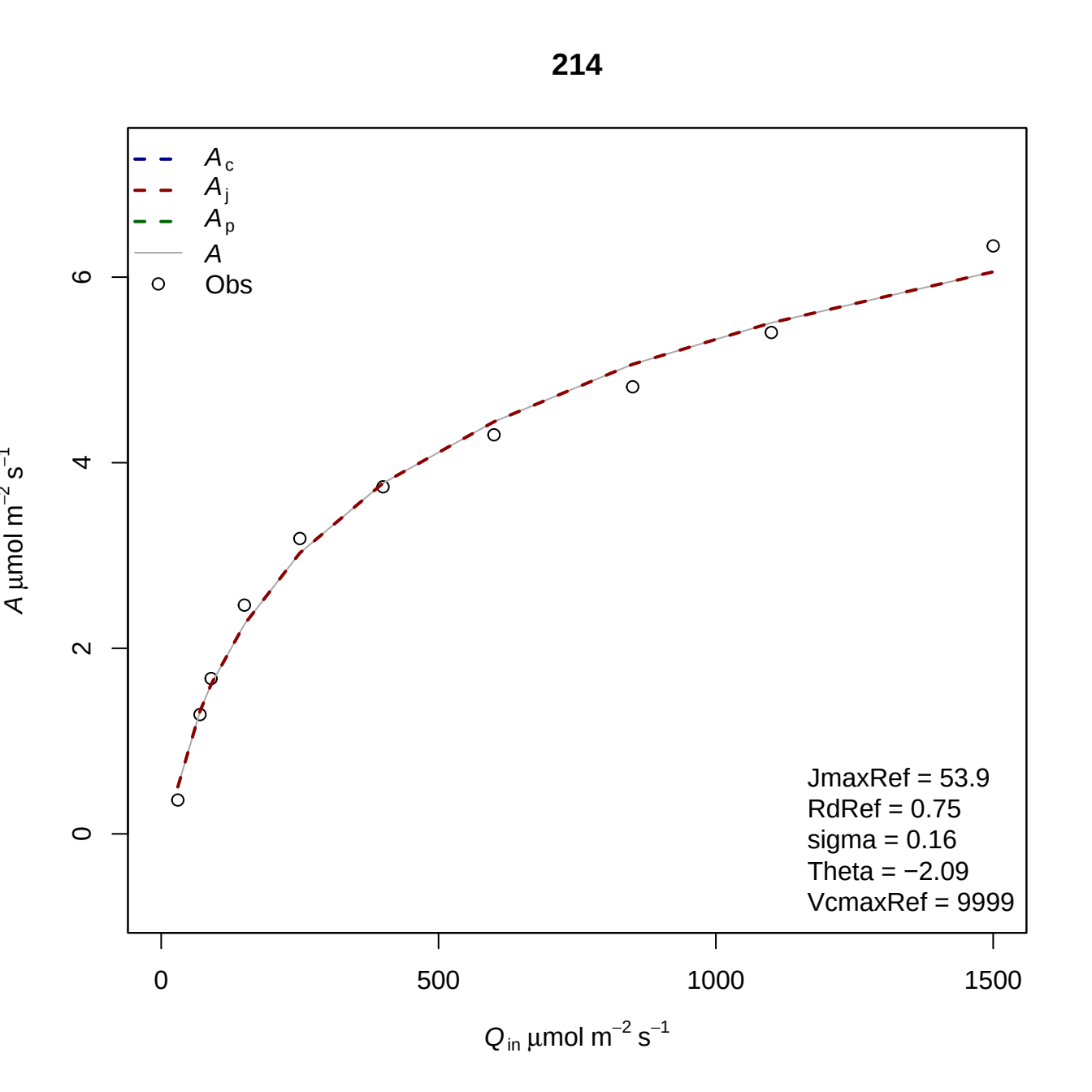

207
A_c
A_j
A_p
A
Obs
JmaxRef = 95.68
RdRef = 1.25
sigma = 0.1
Theta = 0.33
VcmaxRef = 53.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

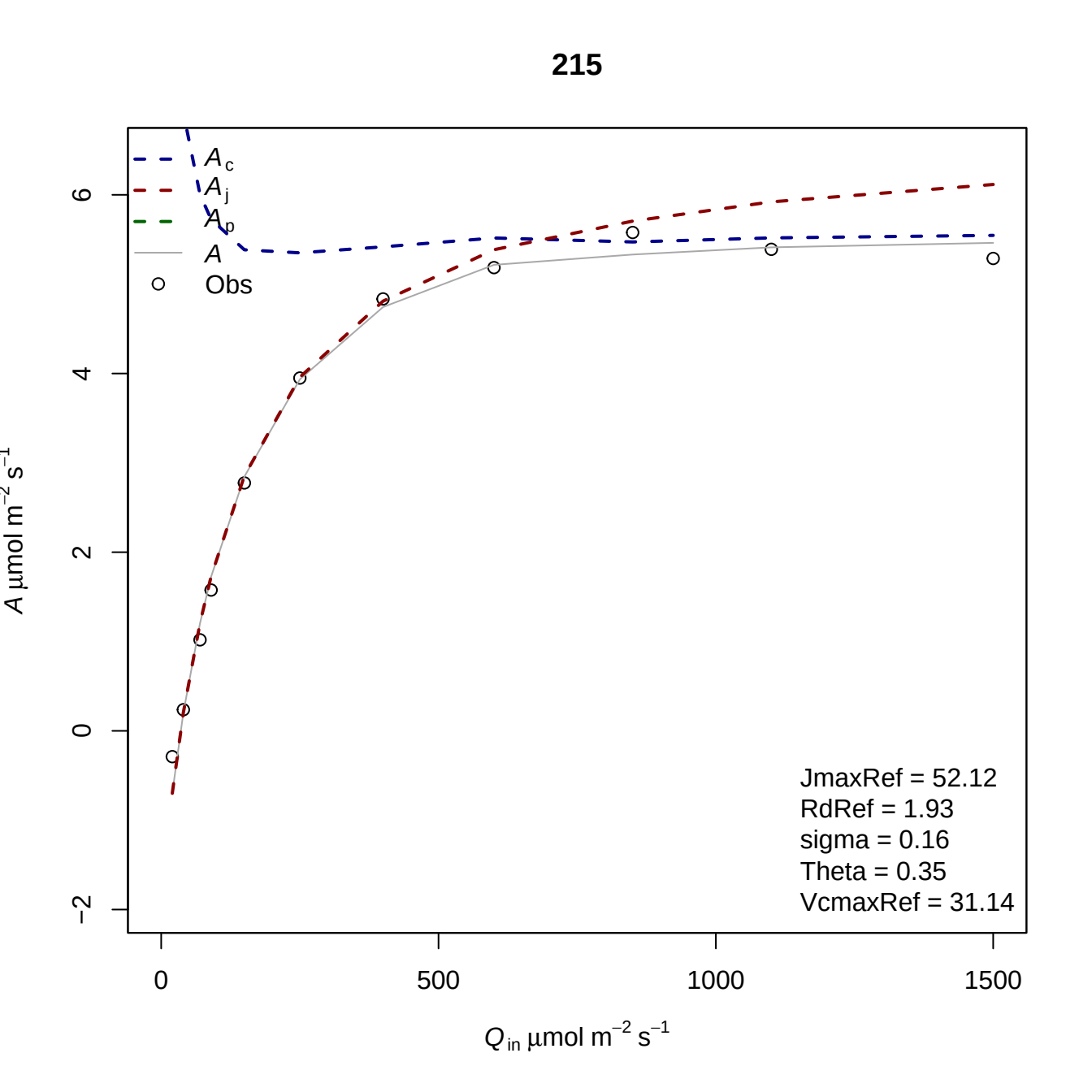

210
A_c
A_j
A_p
A
Obs
JmaxRef = 301232874.85
RdRef = 1.48
sigma = 0.15
Theta = −41976262.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

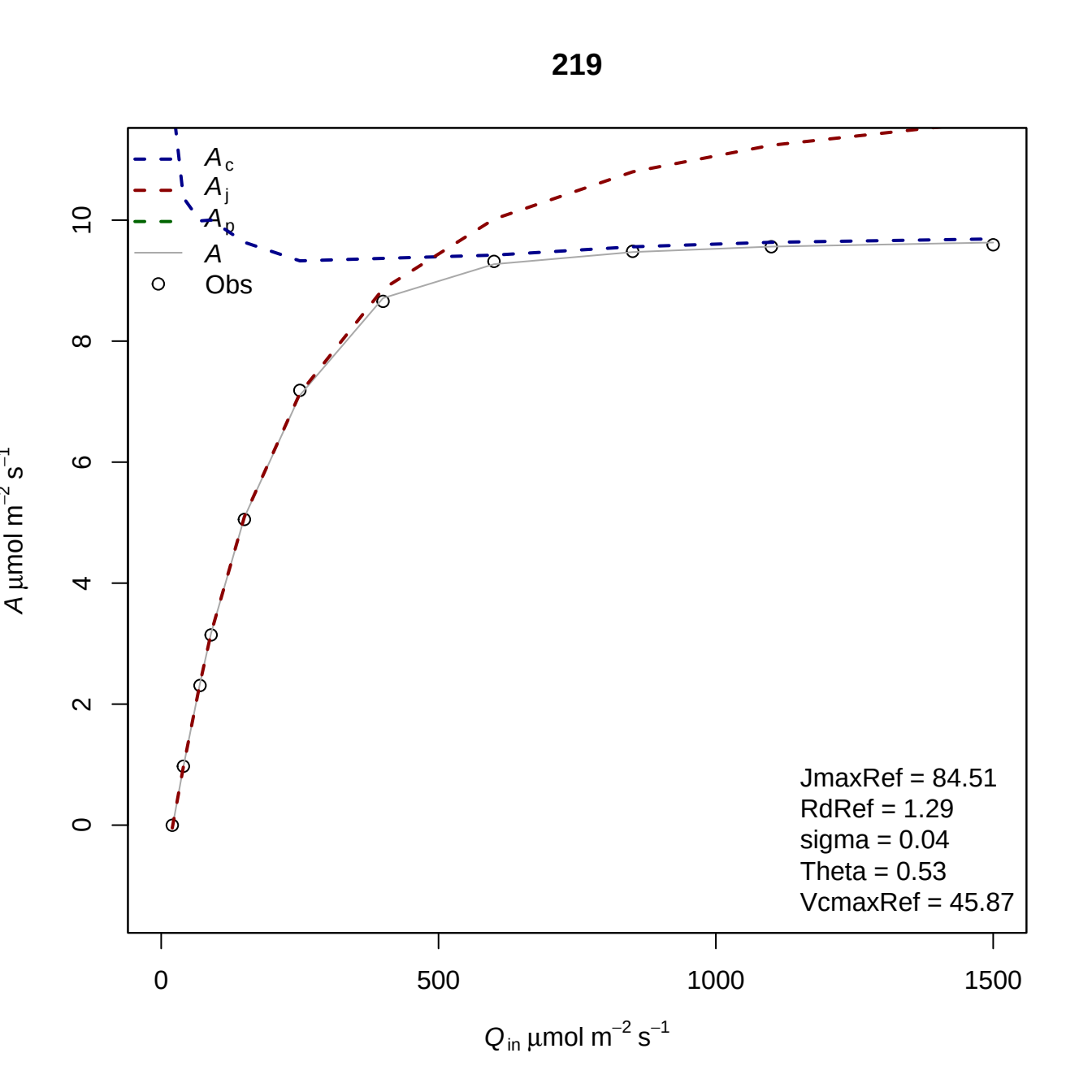

211
A_c
A_j
A_p
A
Obs
JmaxRef = 59.62
RdRef = 1.3
sigma = 0.21
Theta = 0.47
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

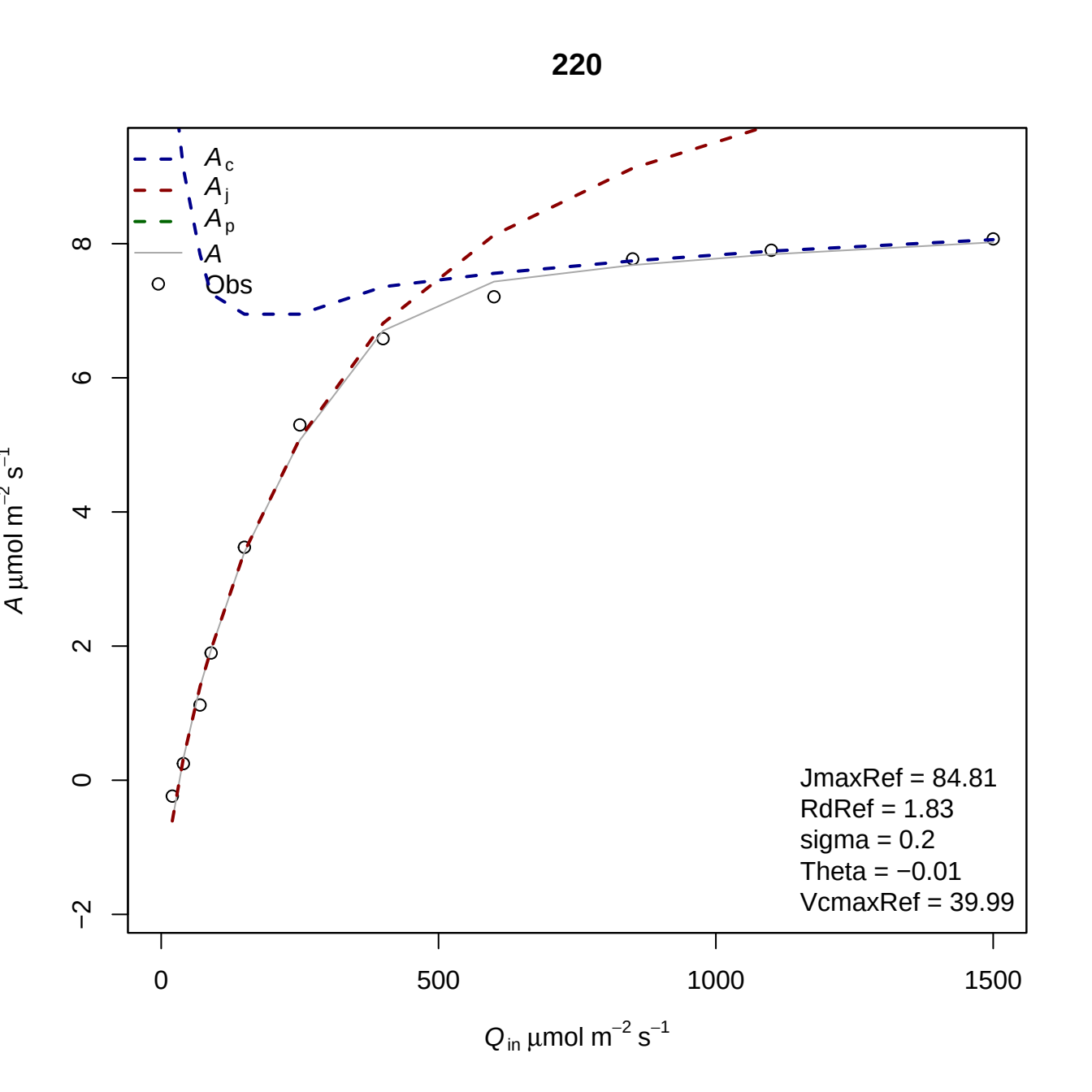

212
A_c
A_j
A_p
A
Obs
JmaxRef = 91.7
RdRef = 1.46
sigma = 0.27
Theta = −0.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

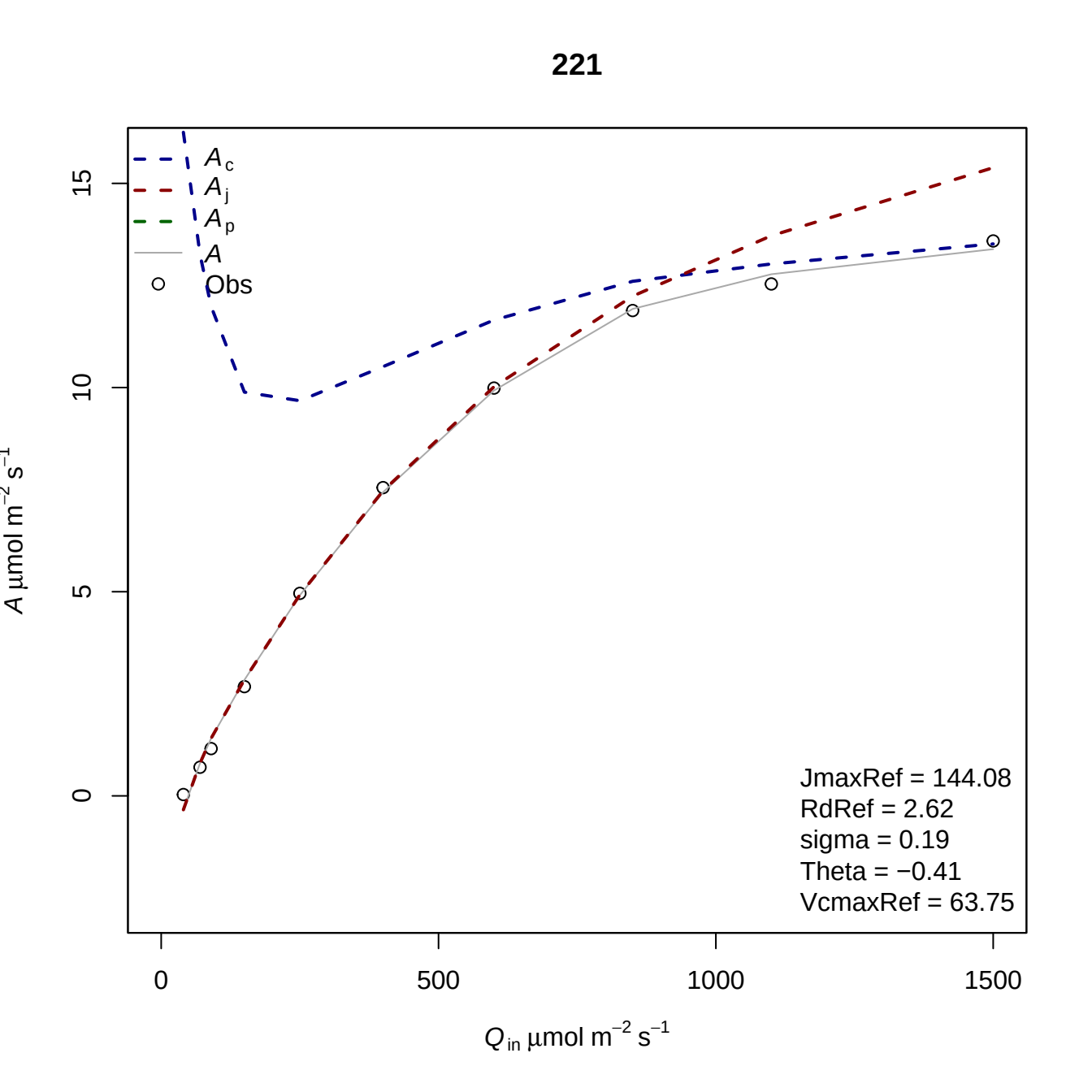

213
A_c
A_j
A_p
A
Obs
JmaxRef = 149.3
RdRef = 4.4
sigma = 0.19
Theta = 0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500

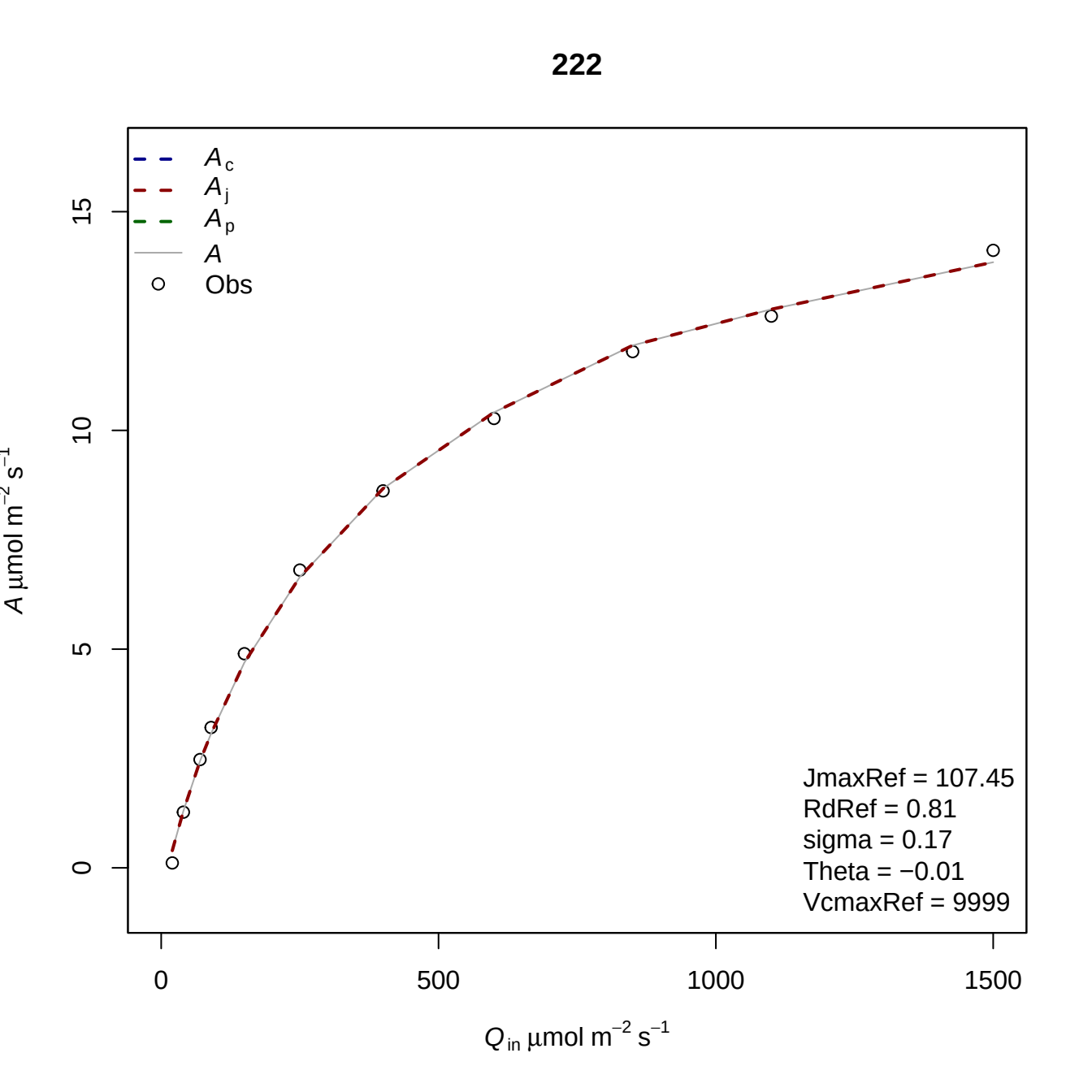

214
A_c
A_j
A_p
A
Obs
JmaxRef = 53.9
RdRef = 0.75
sigma = 0.16
Theta = −2.09
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

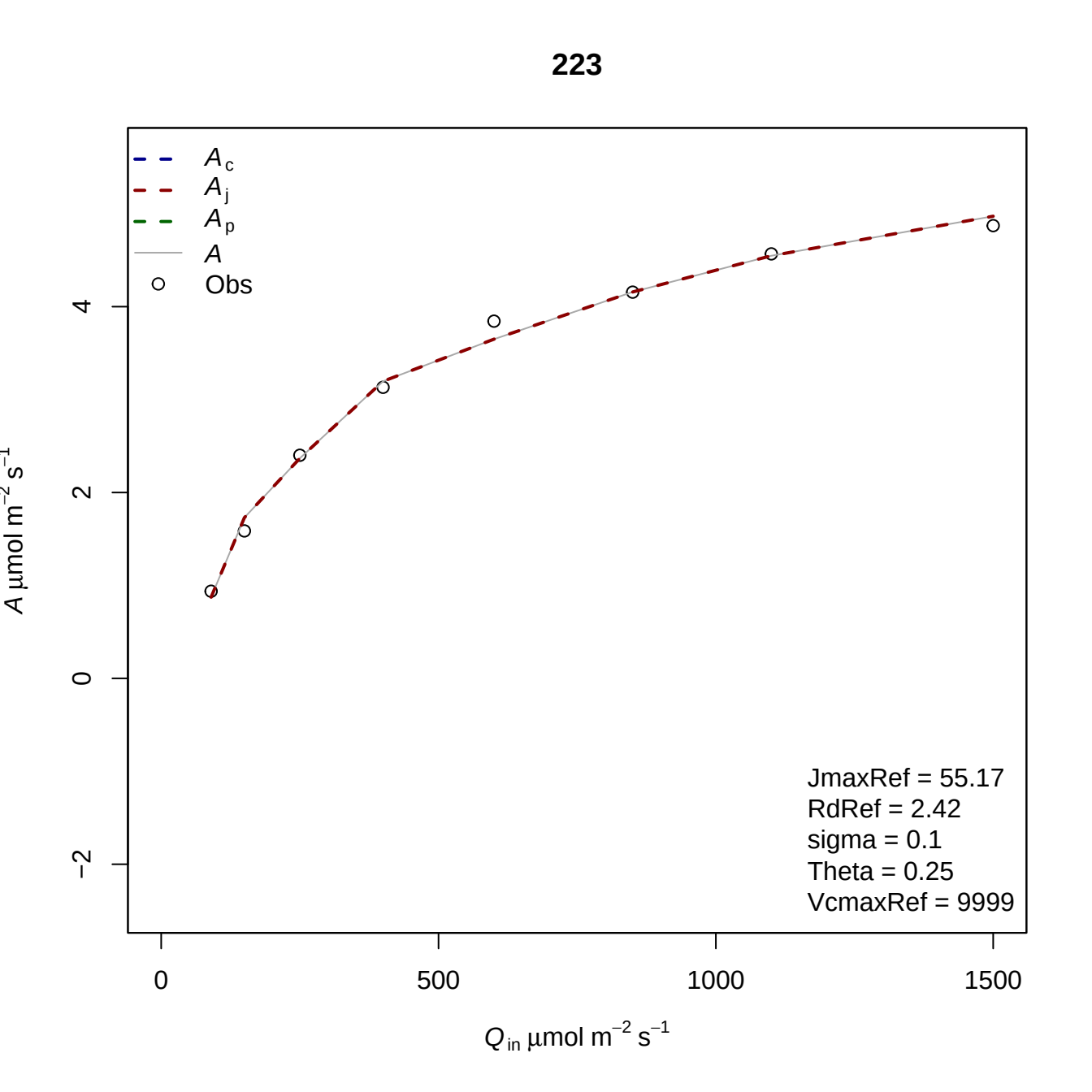

215
A_c
A_j
A_p
A
Obs
JmaxRef = 52.12
RdRef = 1.93
sigma = 0.16
Theta = 0.35
VcmaxRef = 31.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

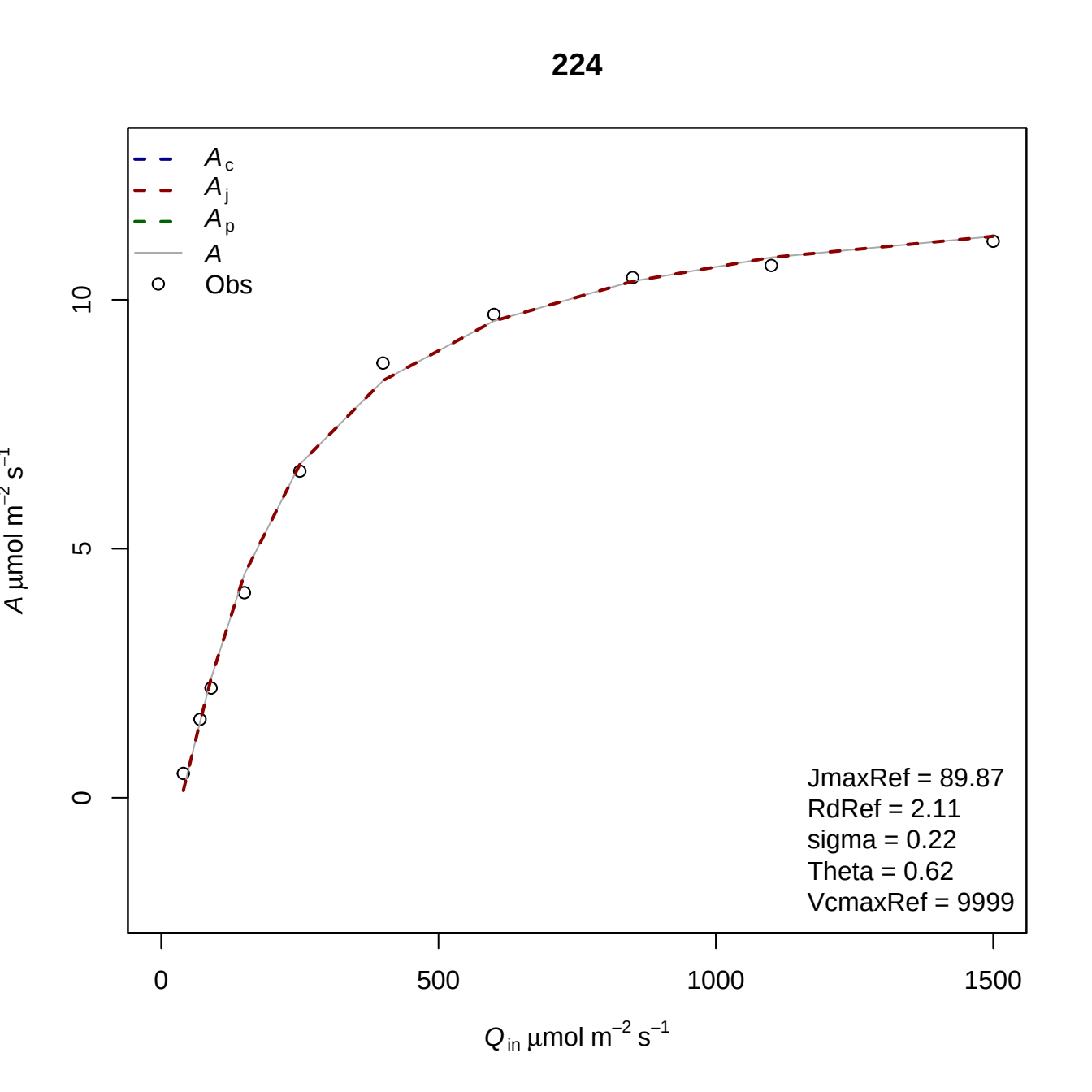

219
A_c
A_j
A_p
A
Obs
JmaxRef = 84.51
RdRef = 1.29
sigma = 0.04
Theta = 0.53
VcmaxRef = 45.87
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

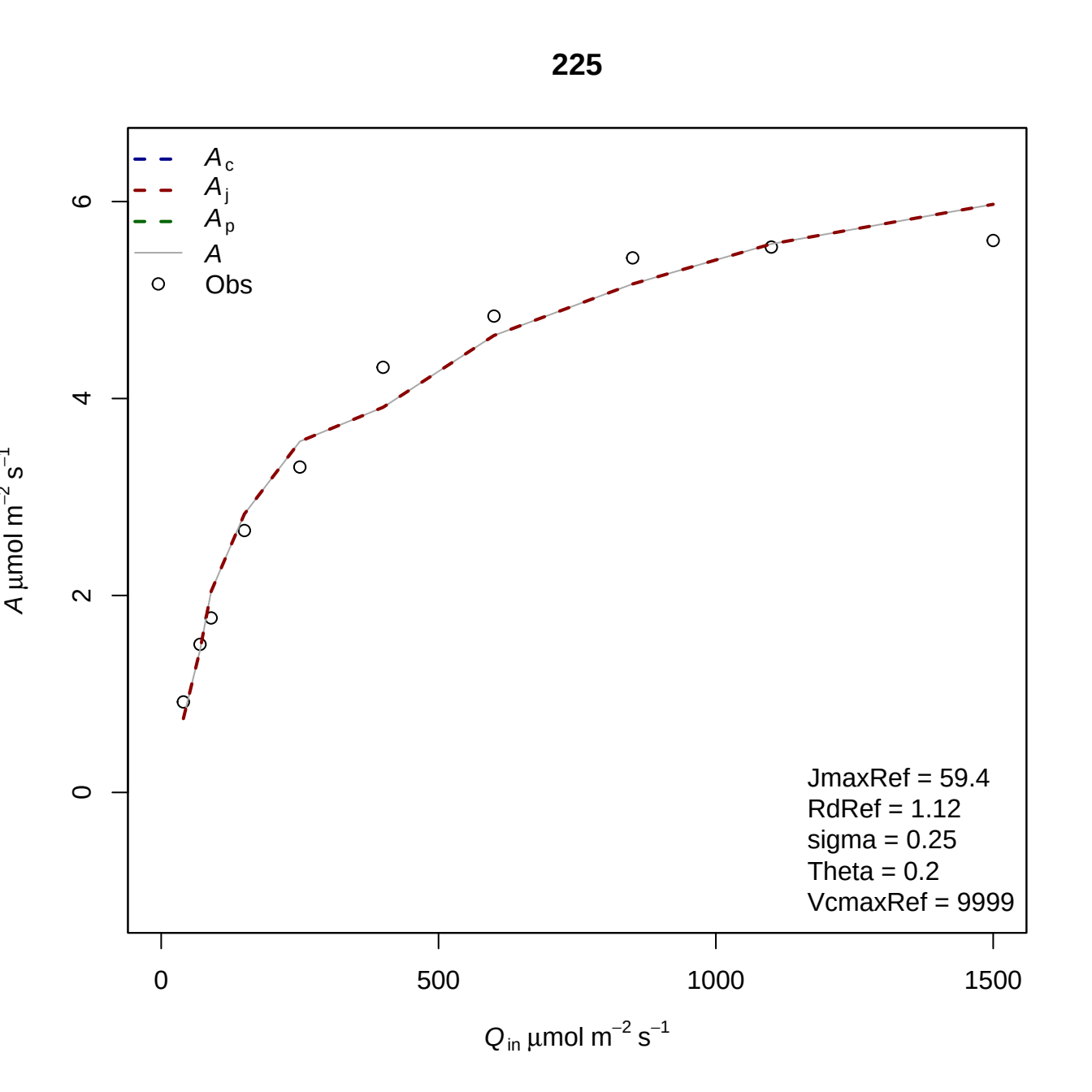

220
A_c
A_j
A_p
A
Obs
JmaxRef = 84.81
RdRef = 1.83
sigma = 0.2
Theta = −0.01
VcmaxRef = 39.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

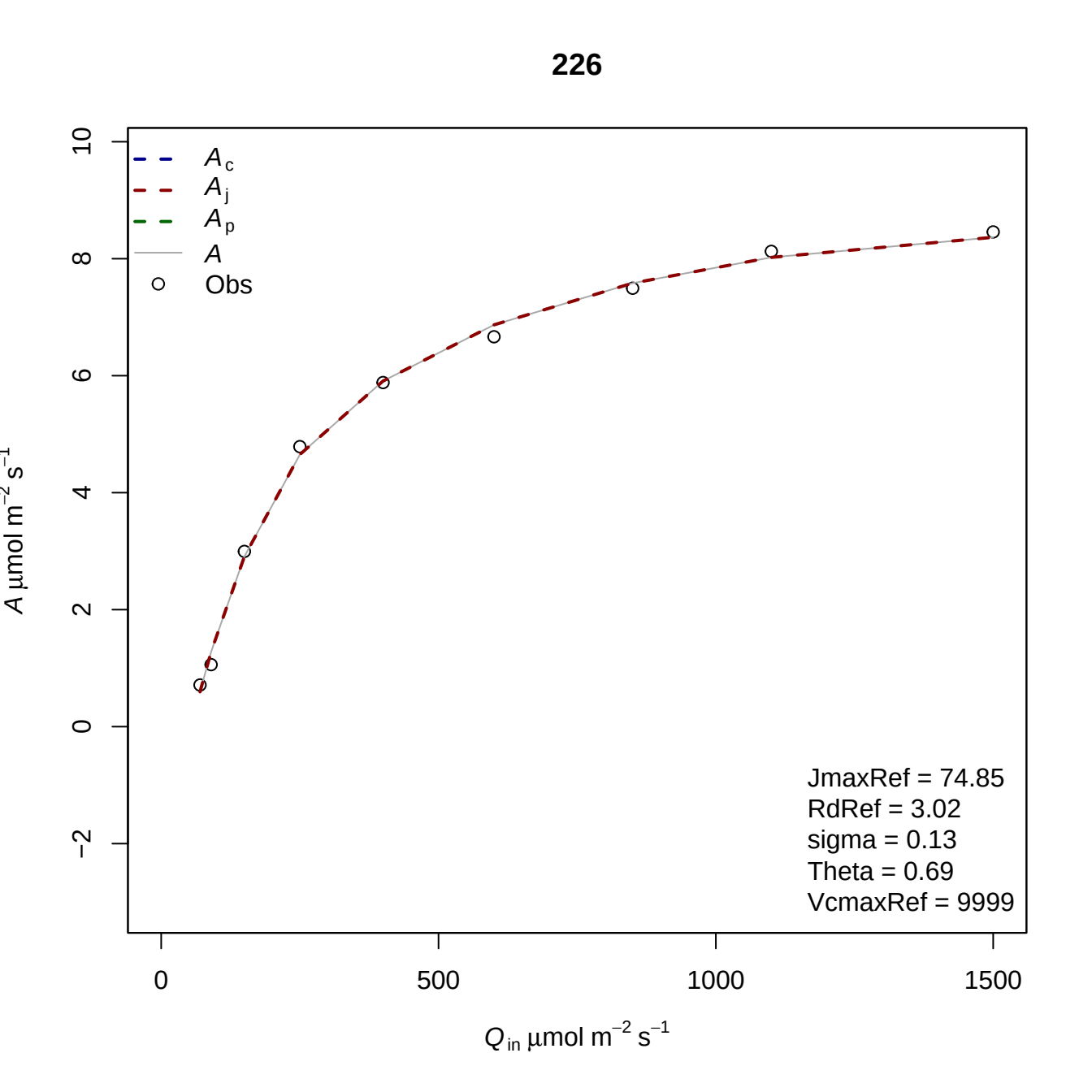

221
A_c
A_j
A_p
A
Obs
JmaxRef = 144.08
RdRef = 2.62
sigma = 0.19
Theta = −0.41
VcmaxRef = 63.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

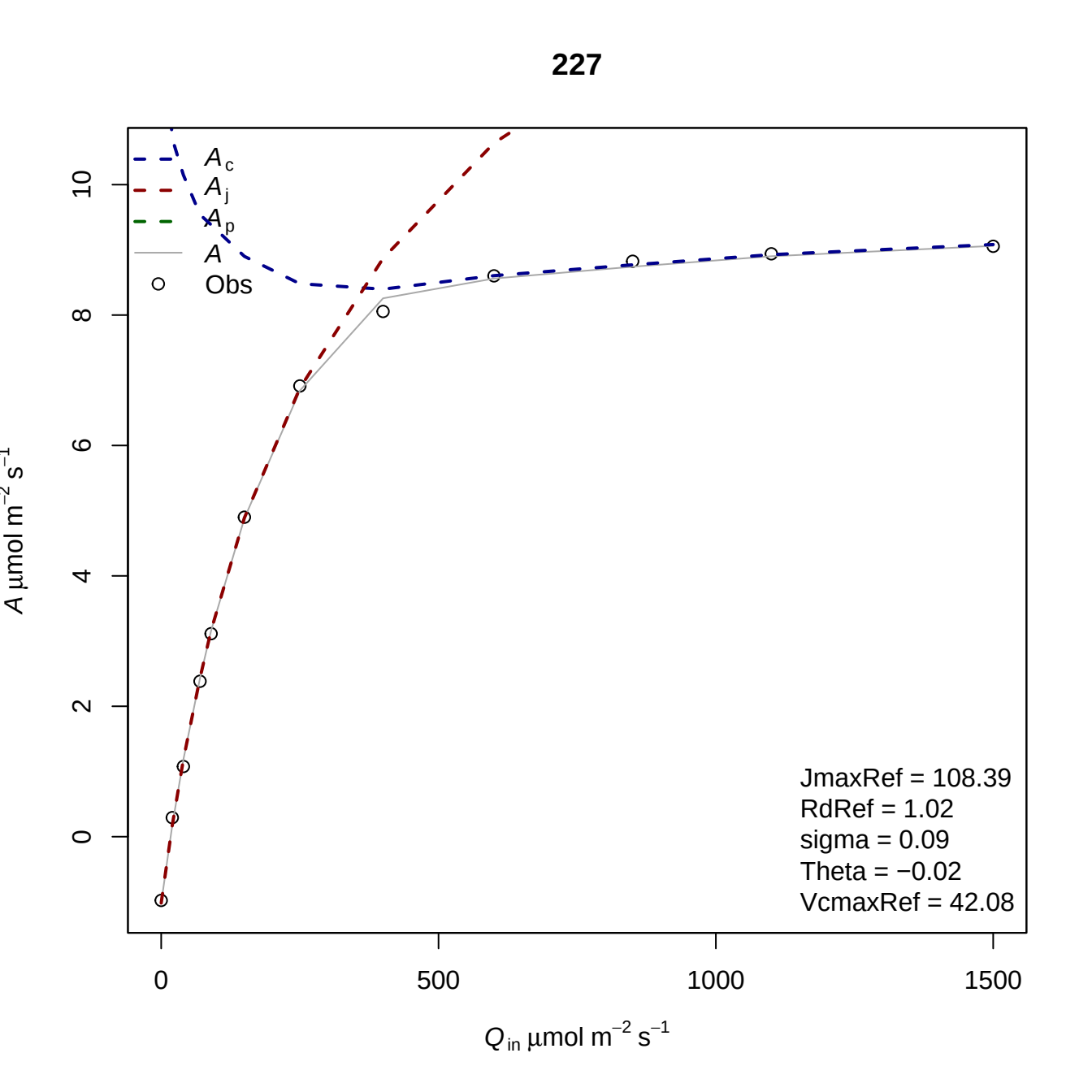

222
A_c
A_j
A_p
A
Obs
JmaxRef = 107.45
RdRef = 0.81
sigma = 0.17
Theta = −0.01
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

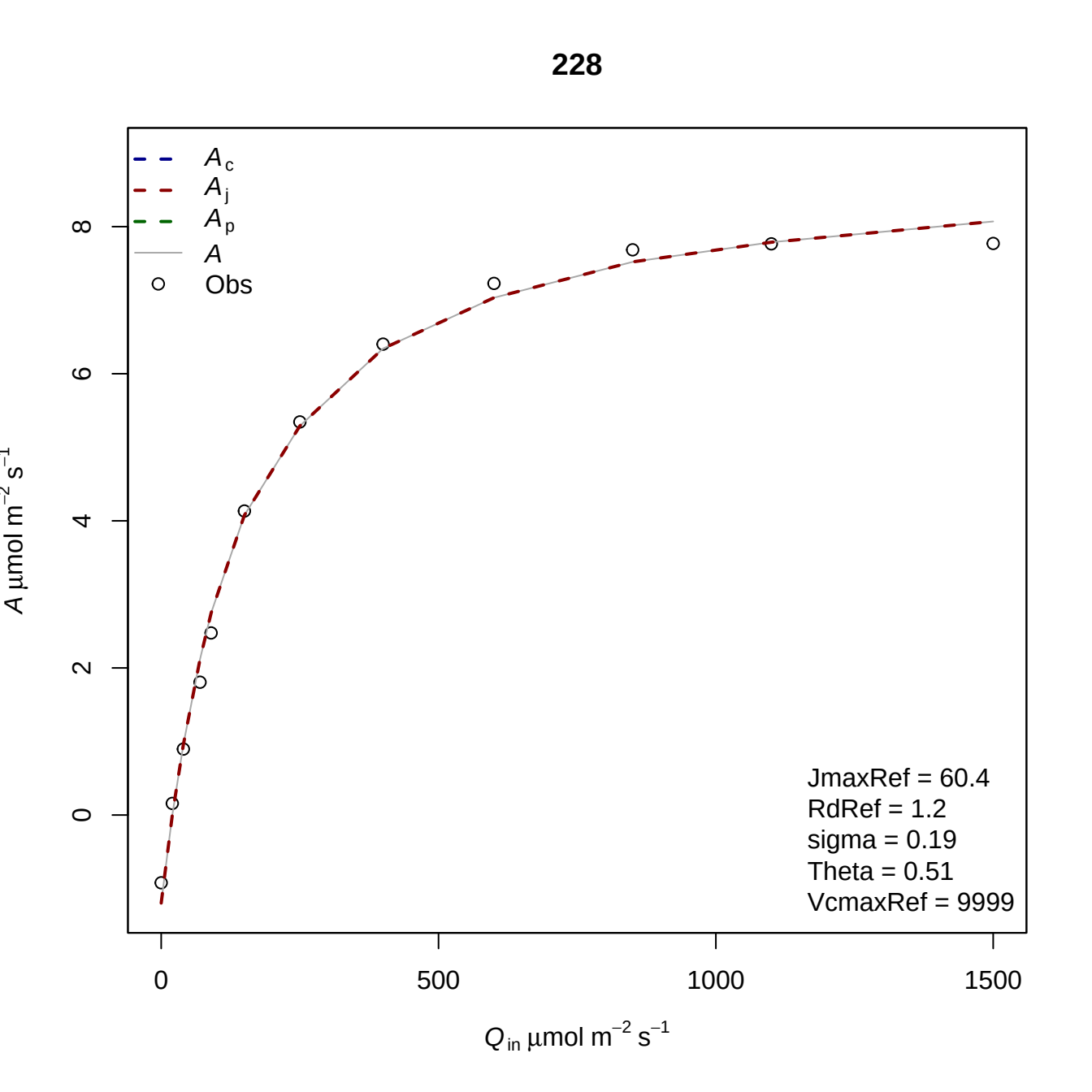

223
A_c
A_j
A_p
A
Obs
JmaxRef = 55.17
RdRef = 2.42
sigma = 0.1
Theta = 0.25
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

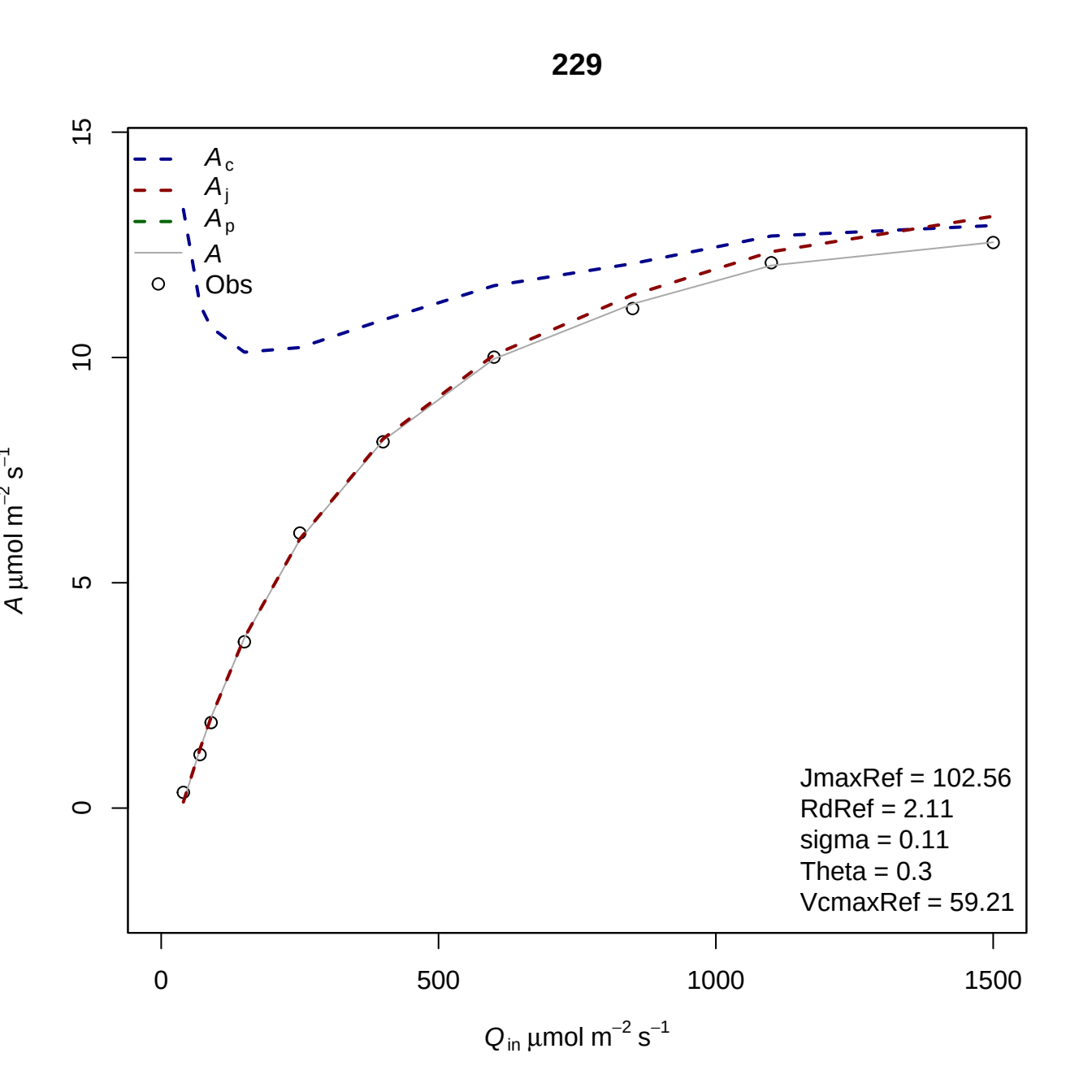

224
A_c
A_j
A_p
A
Obs
JmaxRef = 89.87
RdRef = 2.11
sigma = 0.22
Theta = 0.62
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

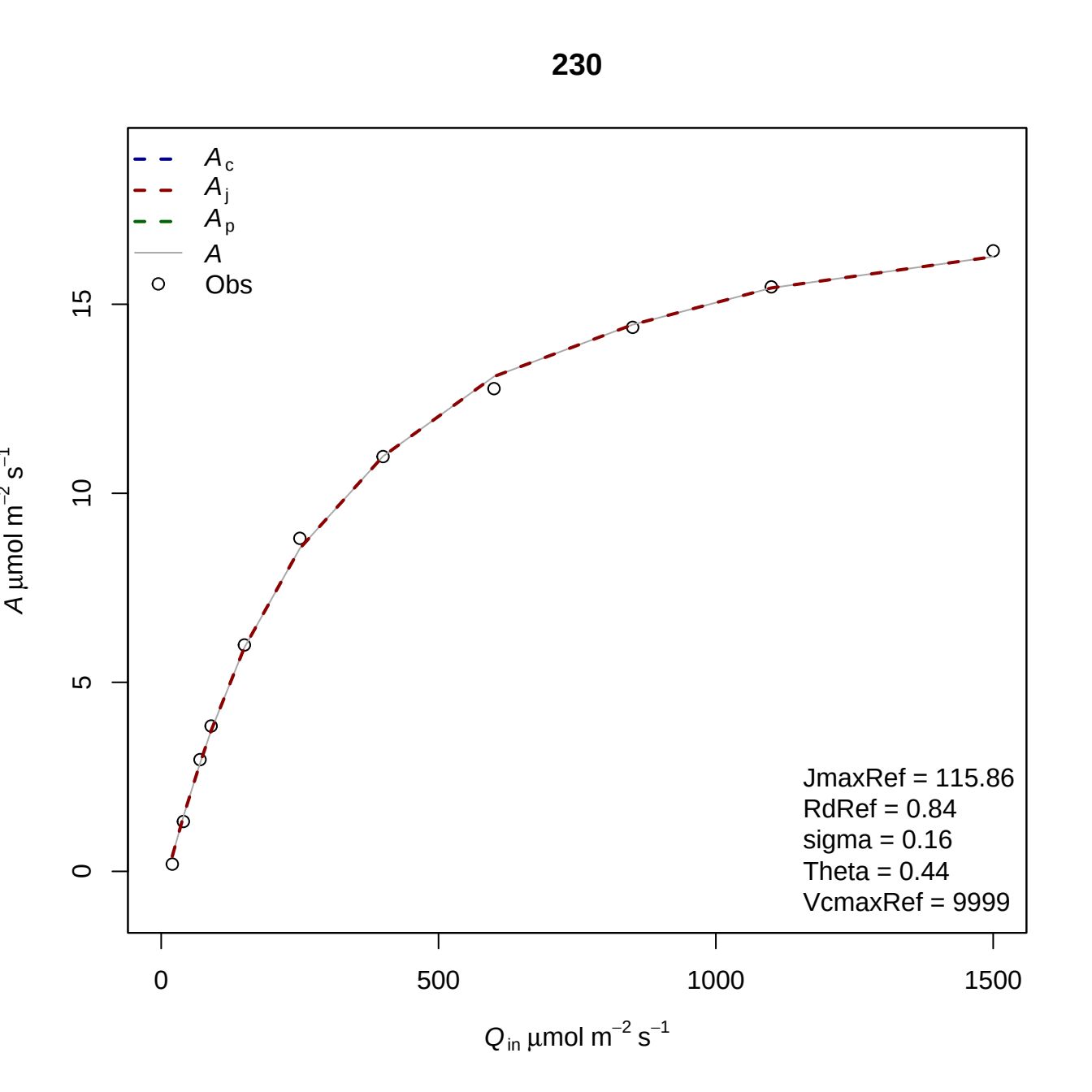

225
A_c
A_j
A_p
A
Obs
JmaxRef = 59.4
RdRef = 1.12
sigma = 0.25
Theta = 0.2
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

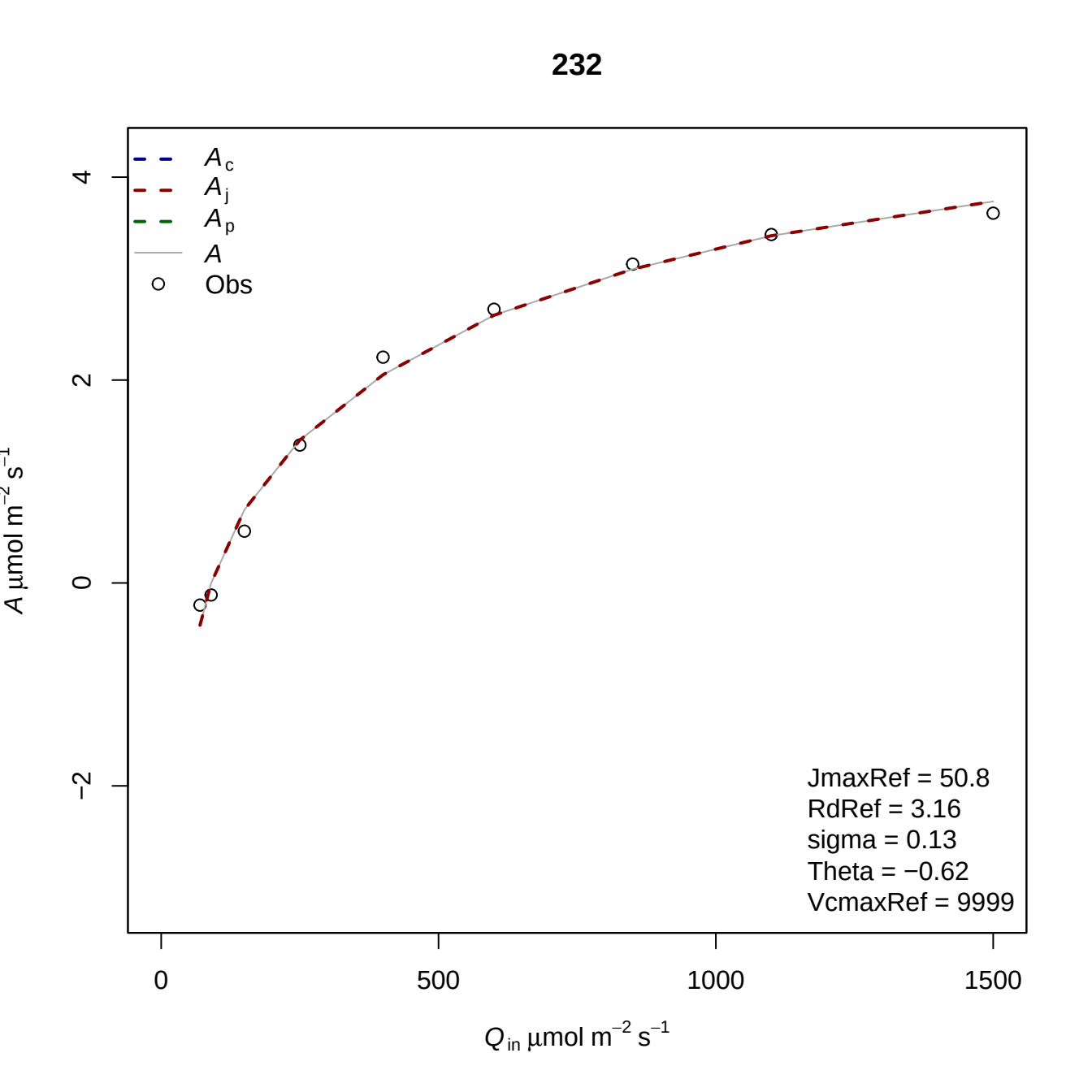

226
A_c
A_j
A_p
A
Obs
JmaxRef = 74.85
RdRef = 3.02
sigma = 0.13
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

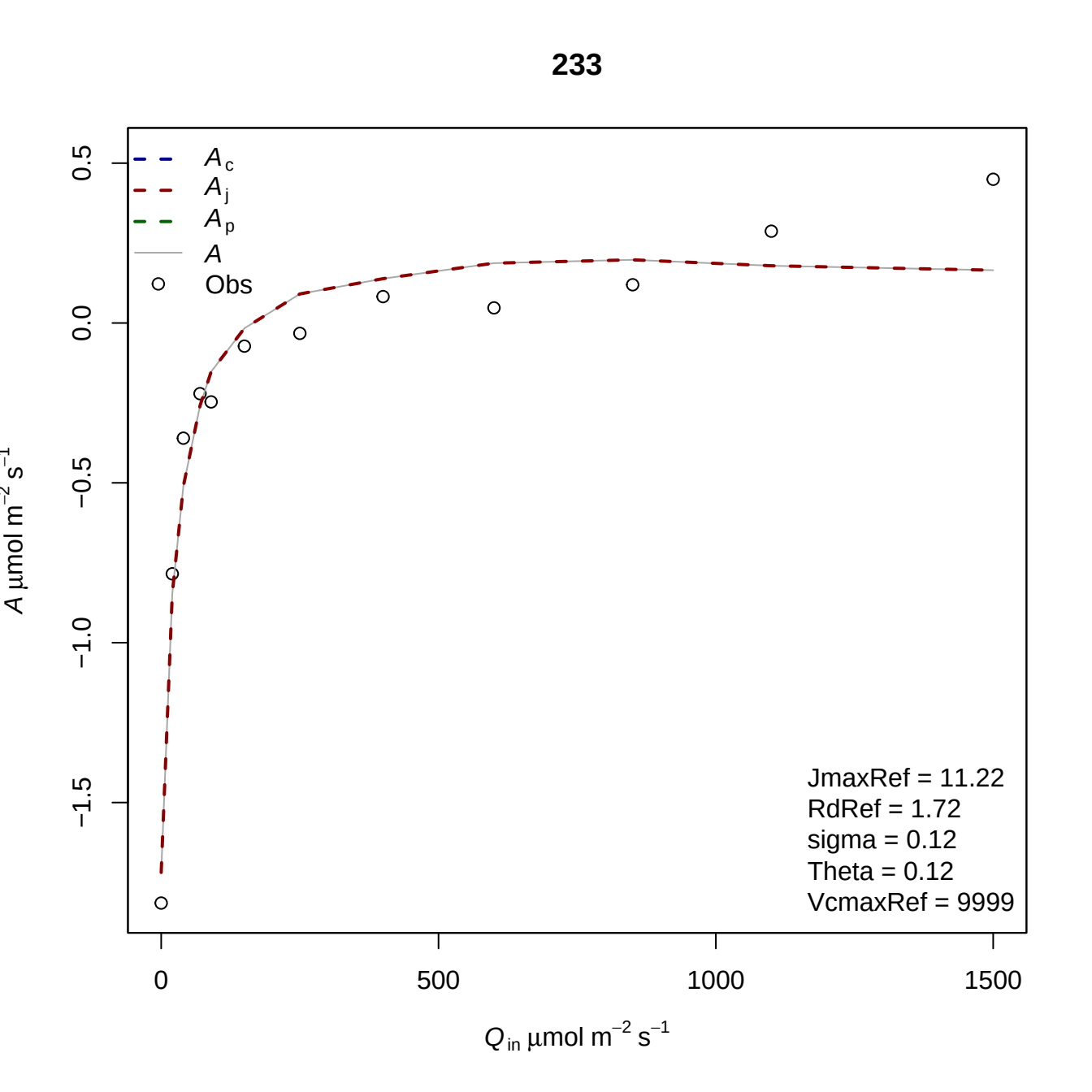

227
A_c
A_j
A_p
A
Obs
JmaxRef = 108.39
RdRef = 1.02
sigma = 0.09
Theta = −0.02
VcmaxRef = 42.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

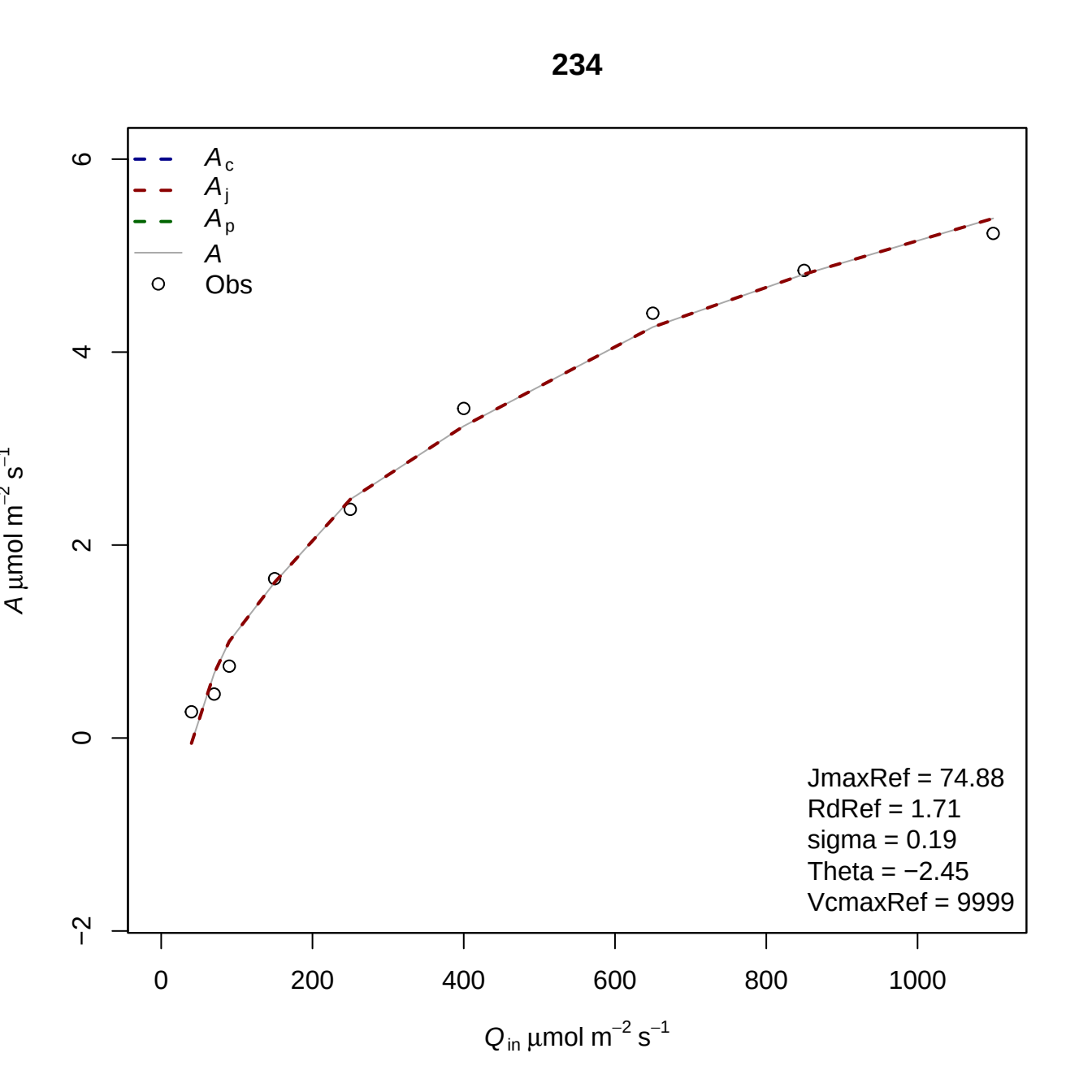

228
A_c
A_j
A_p
A
Obs
JmaxRef = 60.4
RdRef = 1.2
sigma = 0.19
Theta = 0.51
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

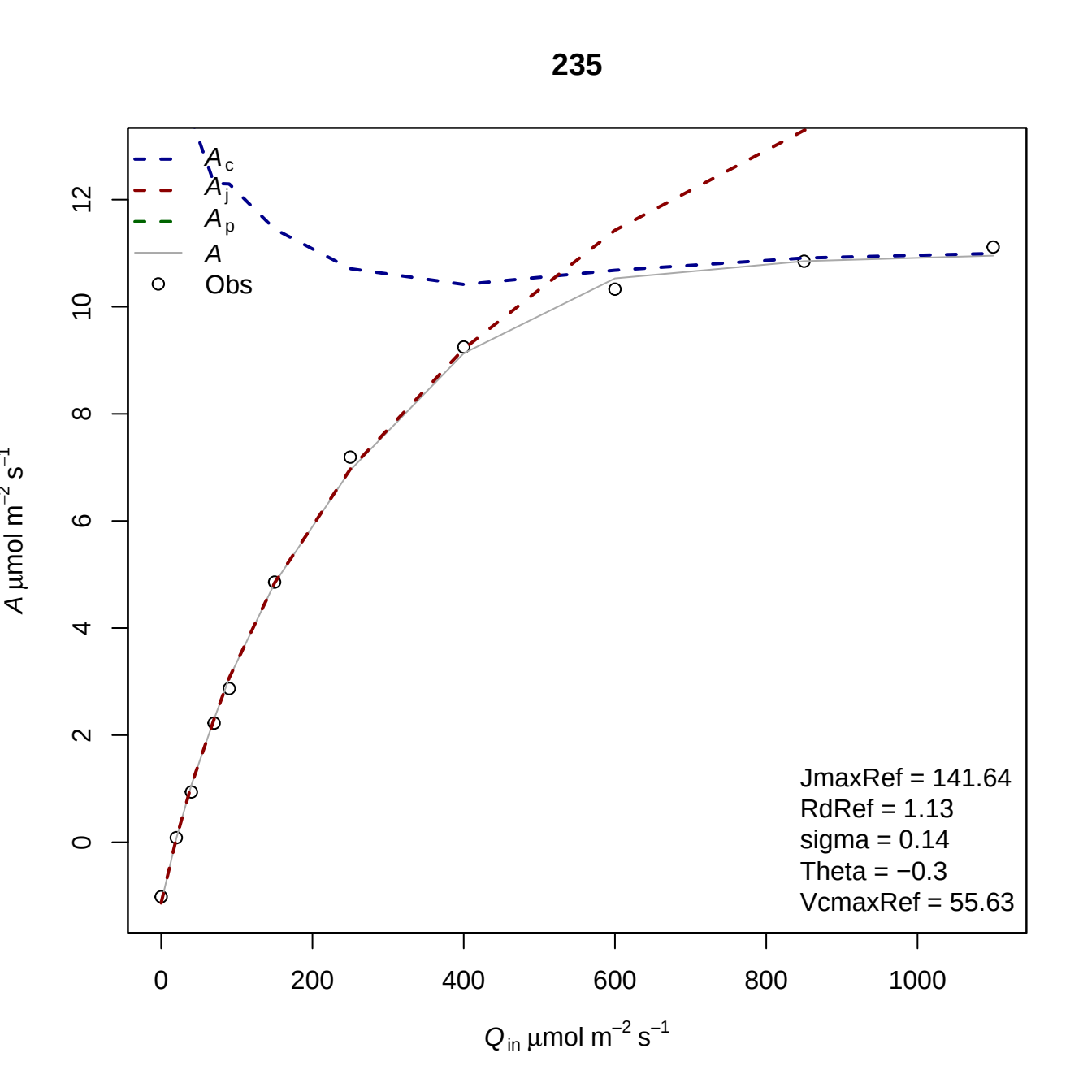

229
A_c
A_j
A_p
A
Obs
JmaxRef = 102.56
RdRef = 2.11
sigma = 0.11
Theta = 0.3
VcmaxRef = 59.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

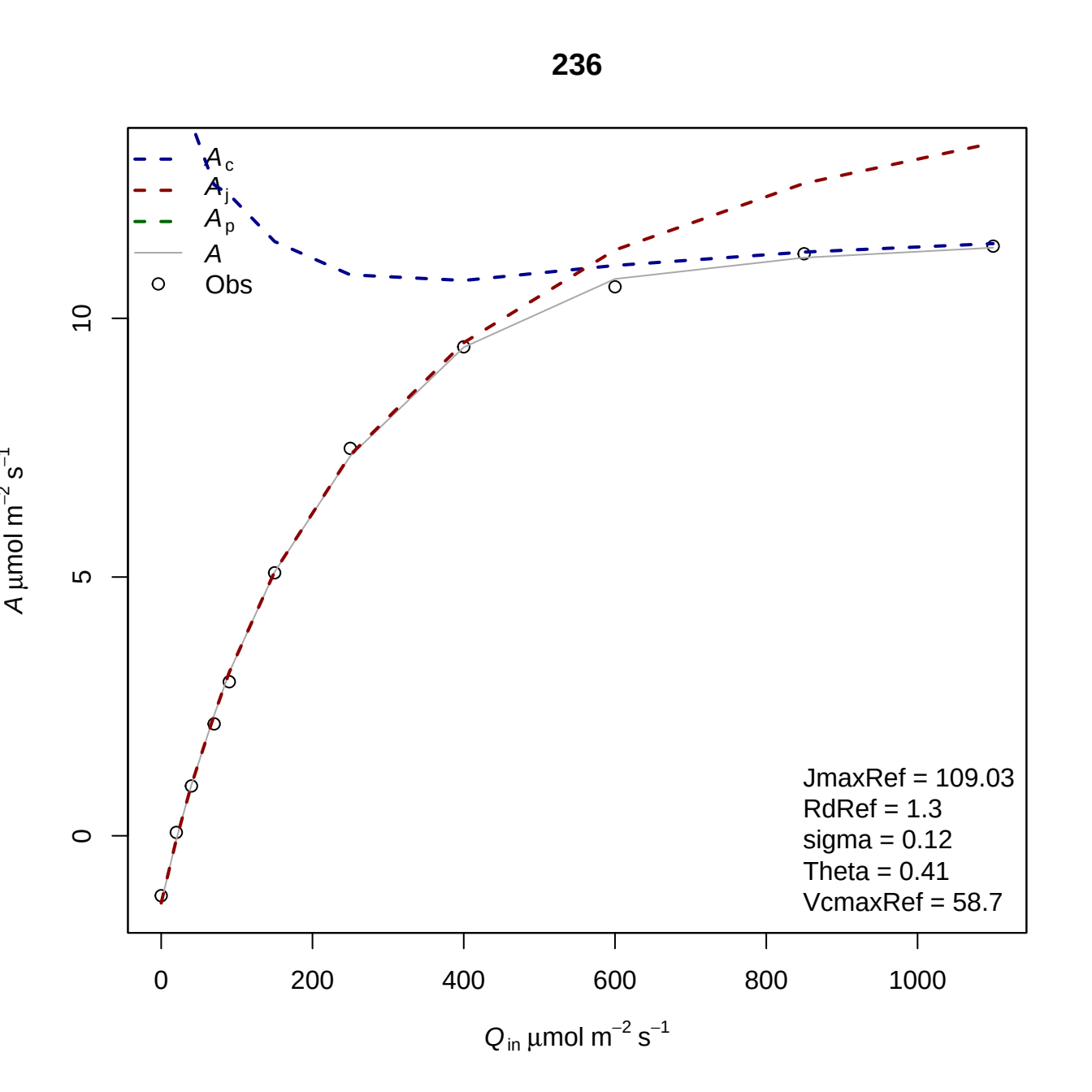

230
A_c
A_j
A_p
A
Obs
JmaxRef = 115.86
RdRef = 0.84
sigma = 0.16
Theta = 0.44
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

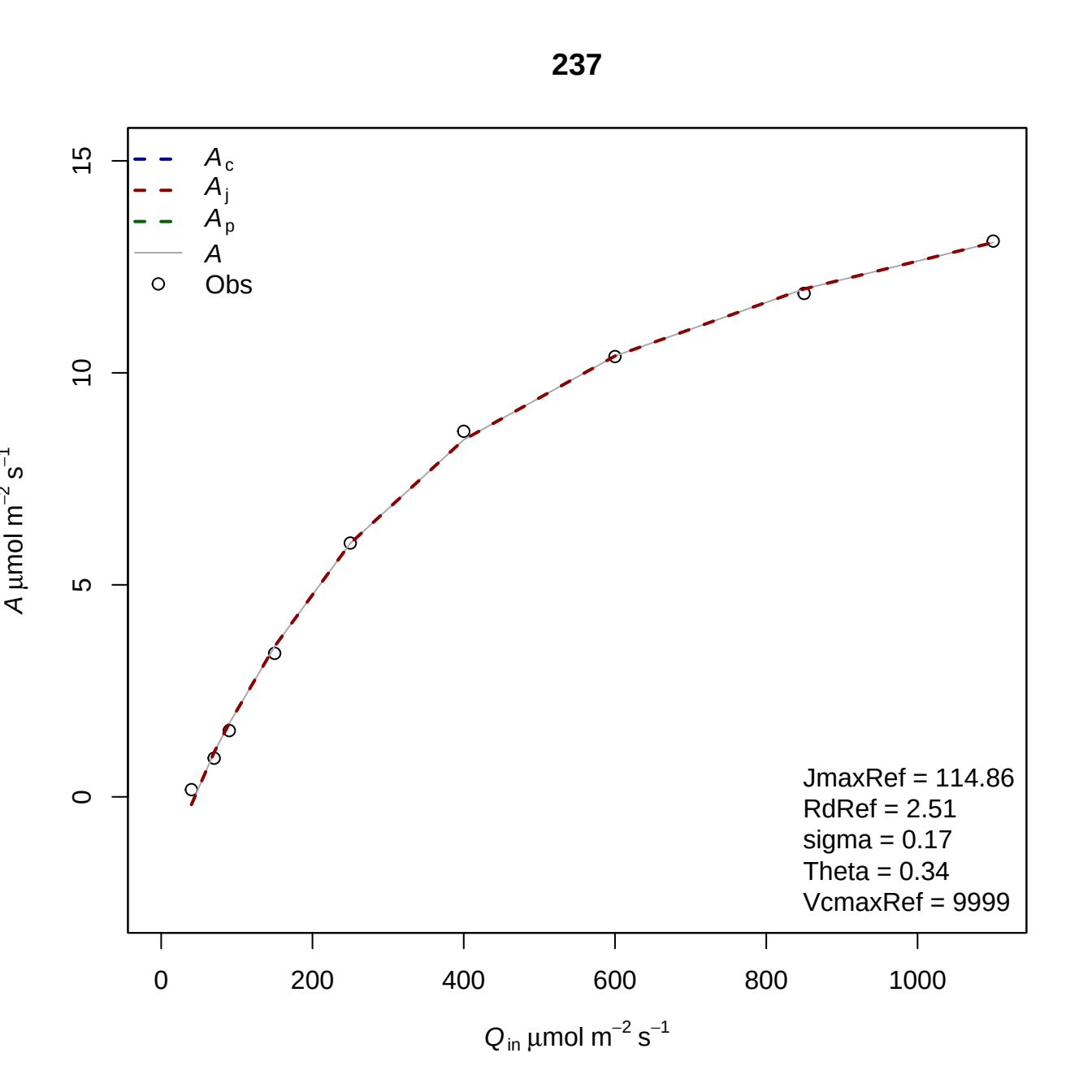

232
A_c
A_j
A_p
A
Obs
JmaxRef = 50.8
RdRef = 3.16
sigma = 0.13
Theta = −0.62
VcmaxRef = 9999
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

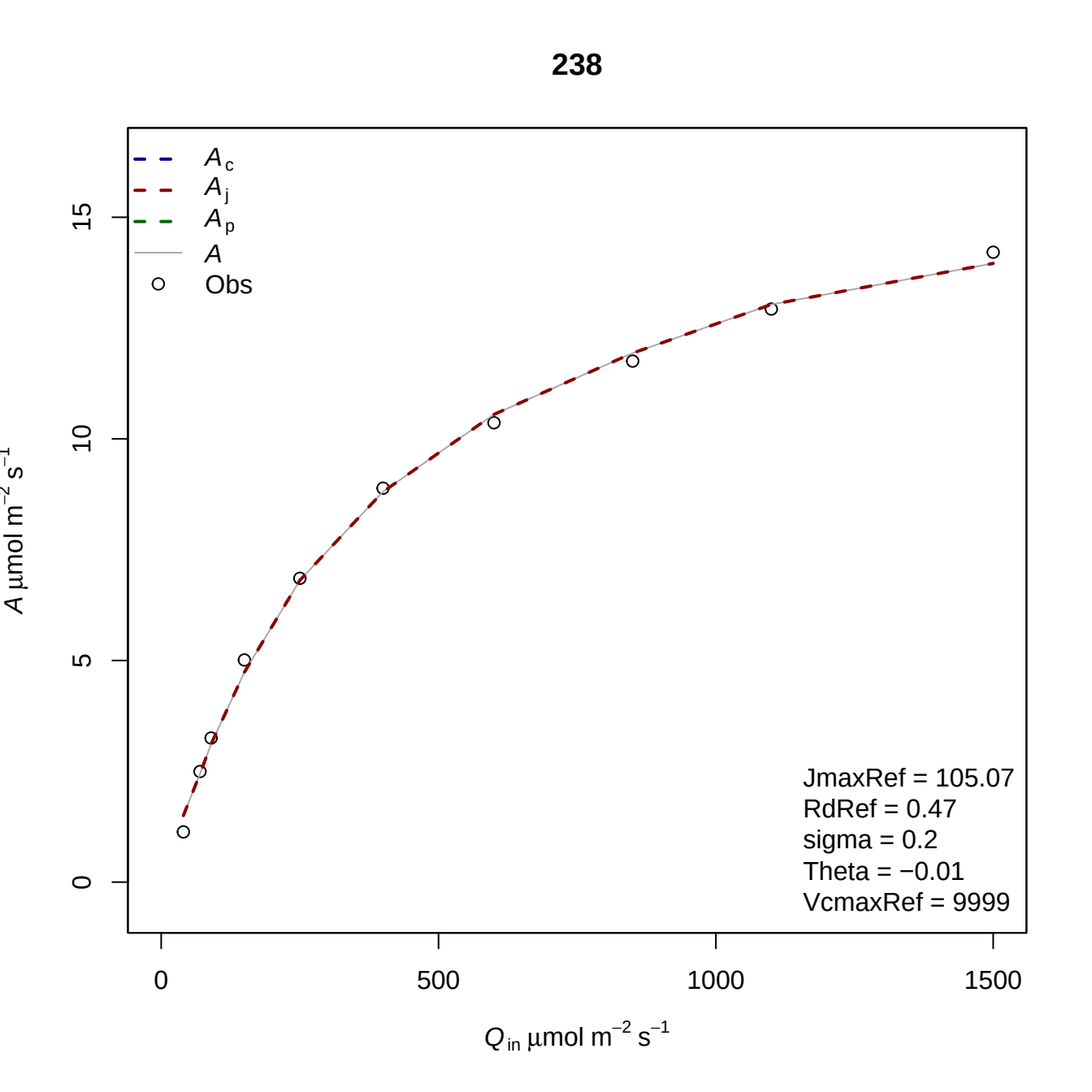

233
A_c
A_j
A_p
A
Obs
JmaxRef = 11.22
RdRef = 1.72
sigma = 0.12
Theta = 0.12
VcmaxRef = 9999
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

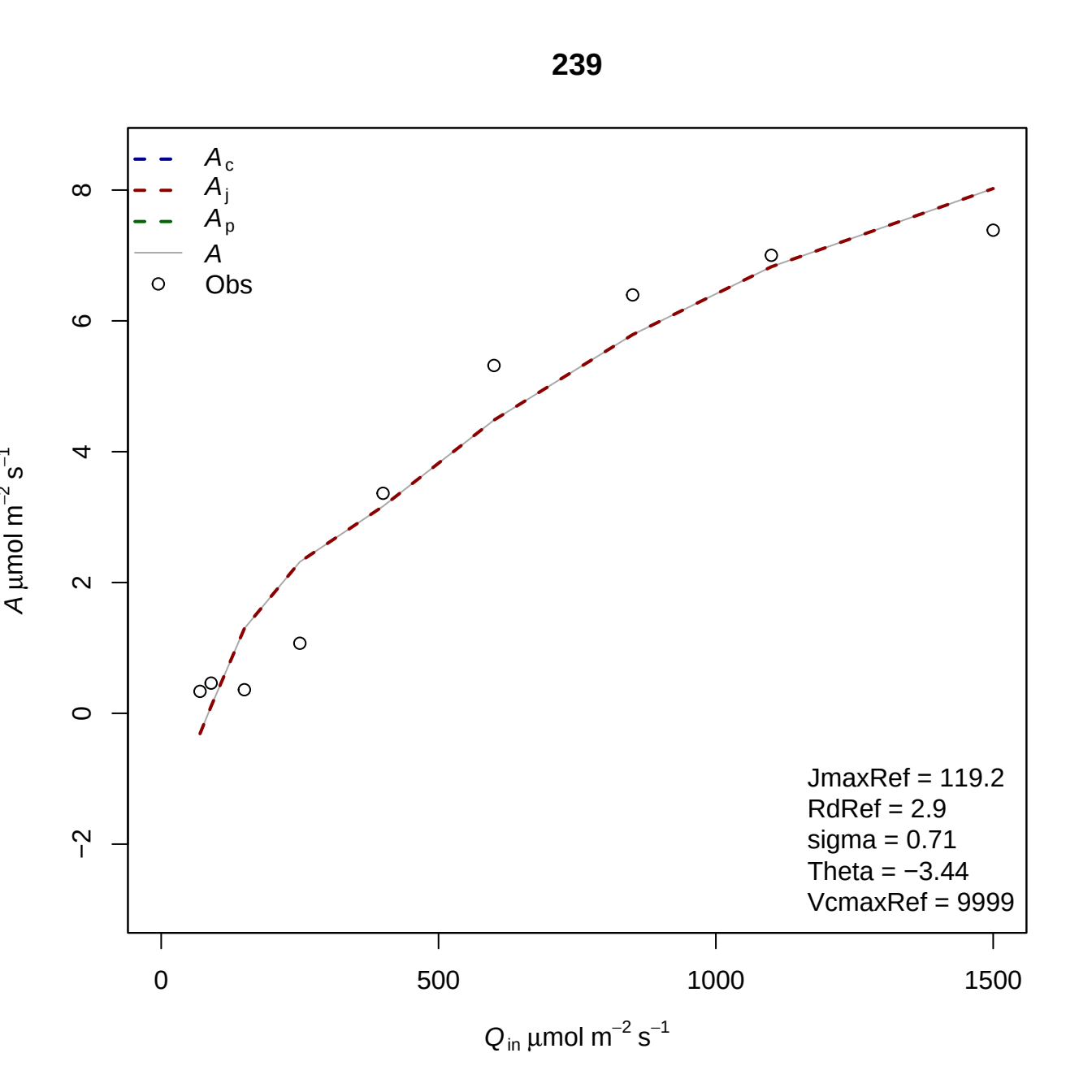

234
A_c
A_j
A_p
A
Obs
JmaxRef = 74.88
RdRef = 1.71
sigma = 0.19
Theta = −2.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

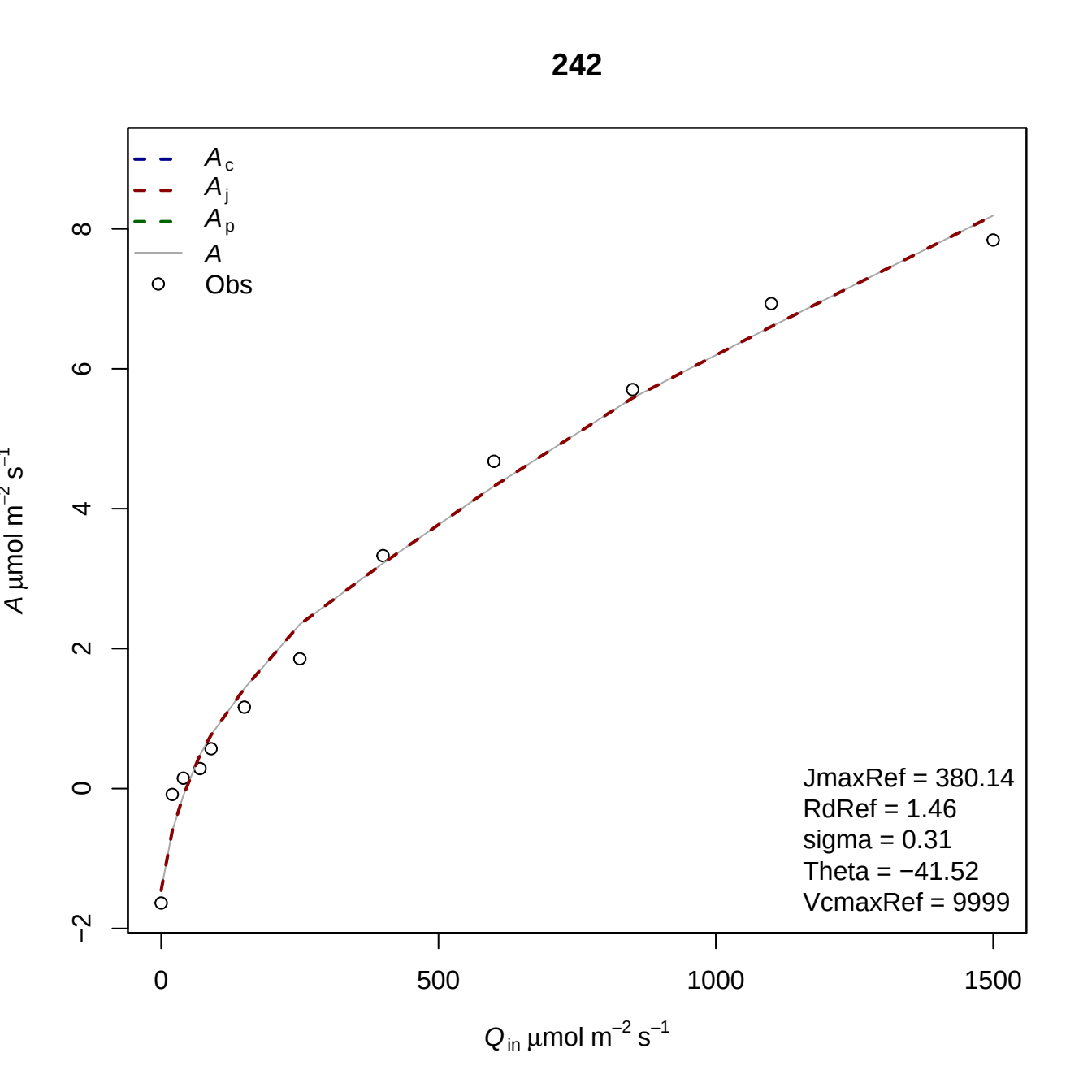

235
A_c
A_j
A_p
A
Obs
JmaxRef = 141.64
RdRef = 1.13
sigma = 0.14
Theta = −0.3
VcmaxRef = 55.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000

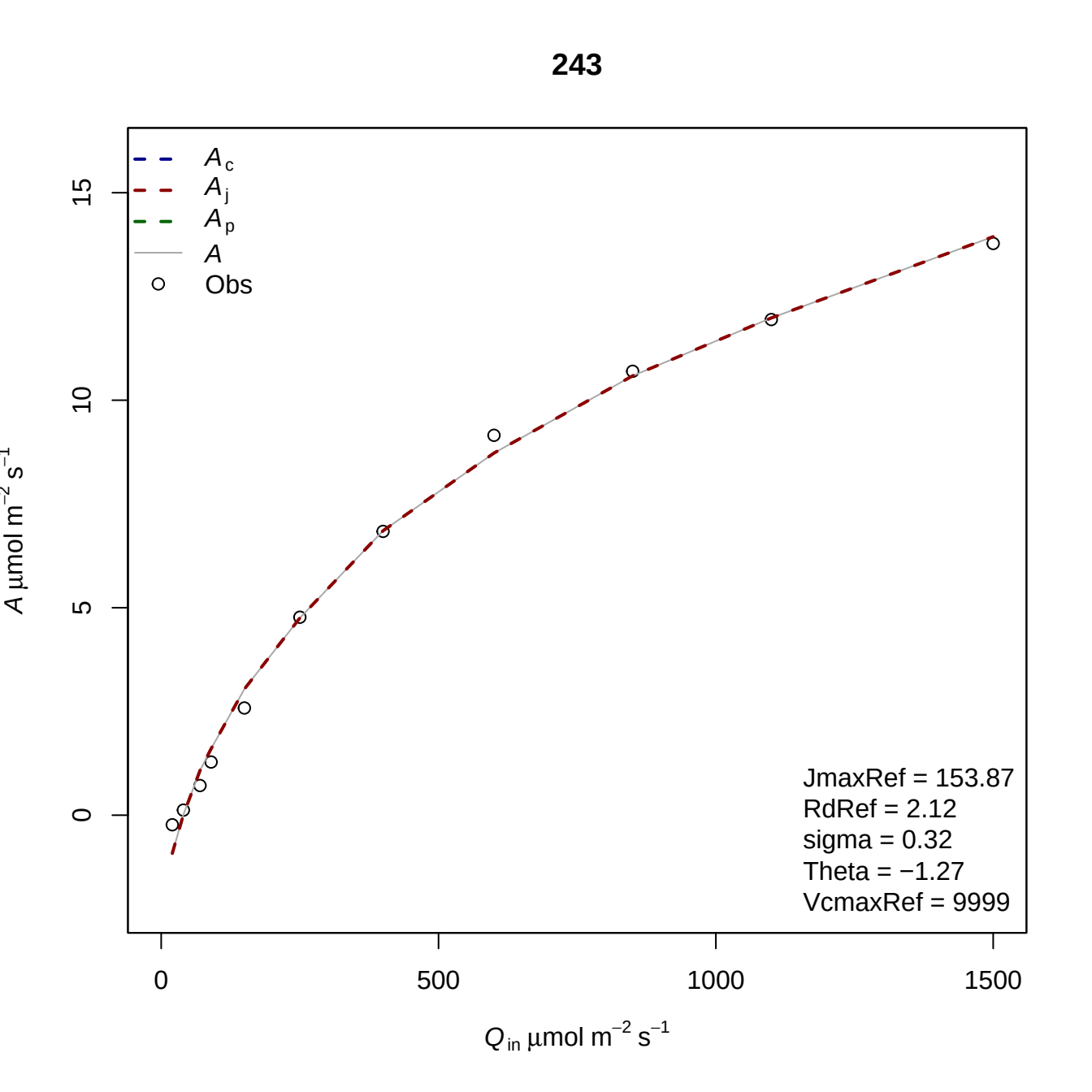

236
A_c
A_j
A_p
A
Obs
JmaxRef = 109.03
RdRef = 1.3
sigma = 0.12
Theta = 0.41
VcmaxRef = 58.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

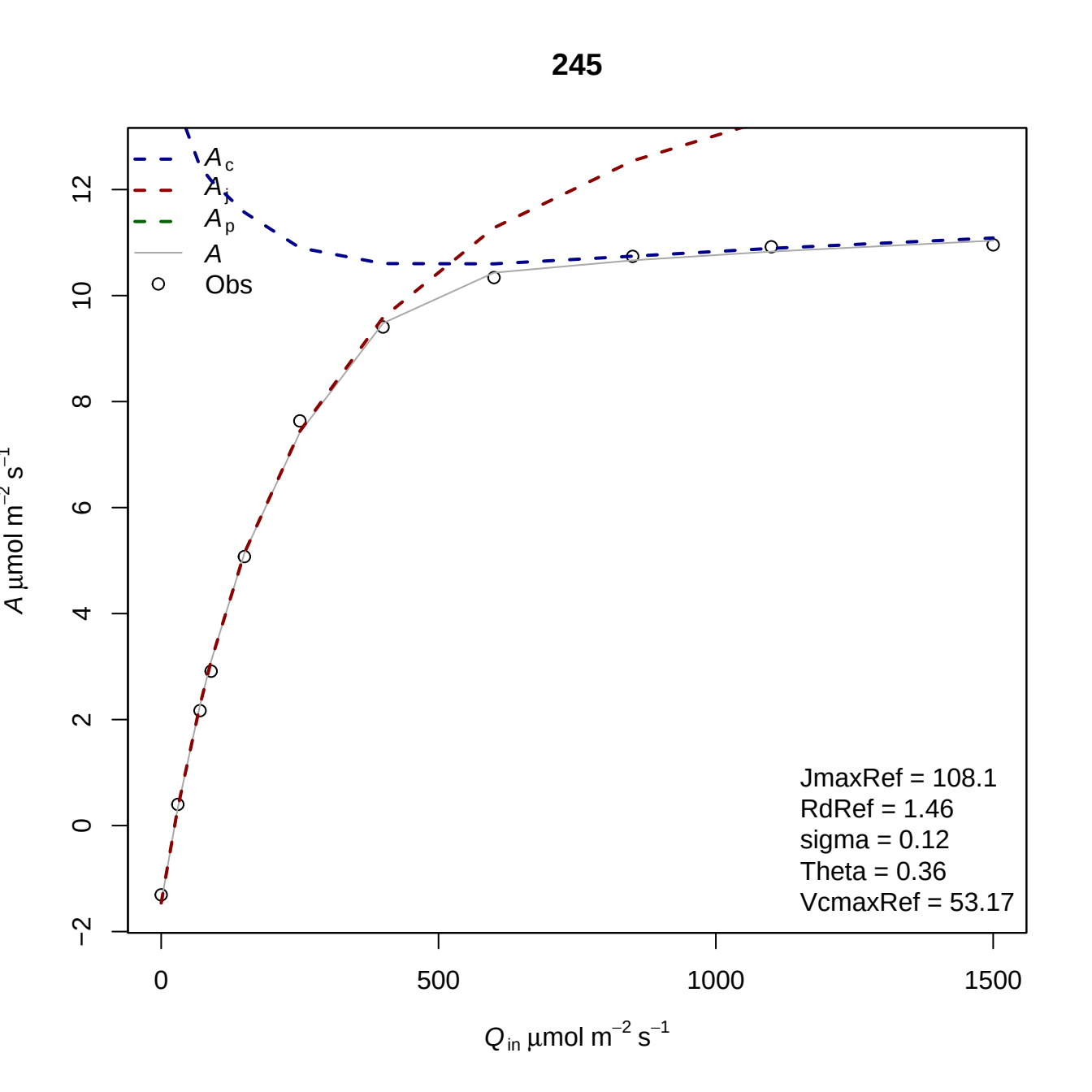

237
A_c
A_j
A_p
A
Obs
JmaxRef = 114.86
RdRef = 2.51
sigma = 0.17
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

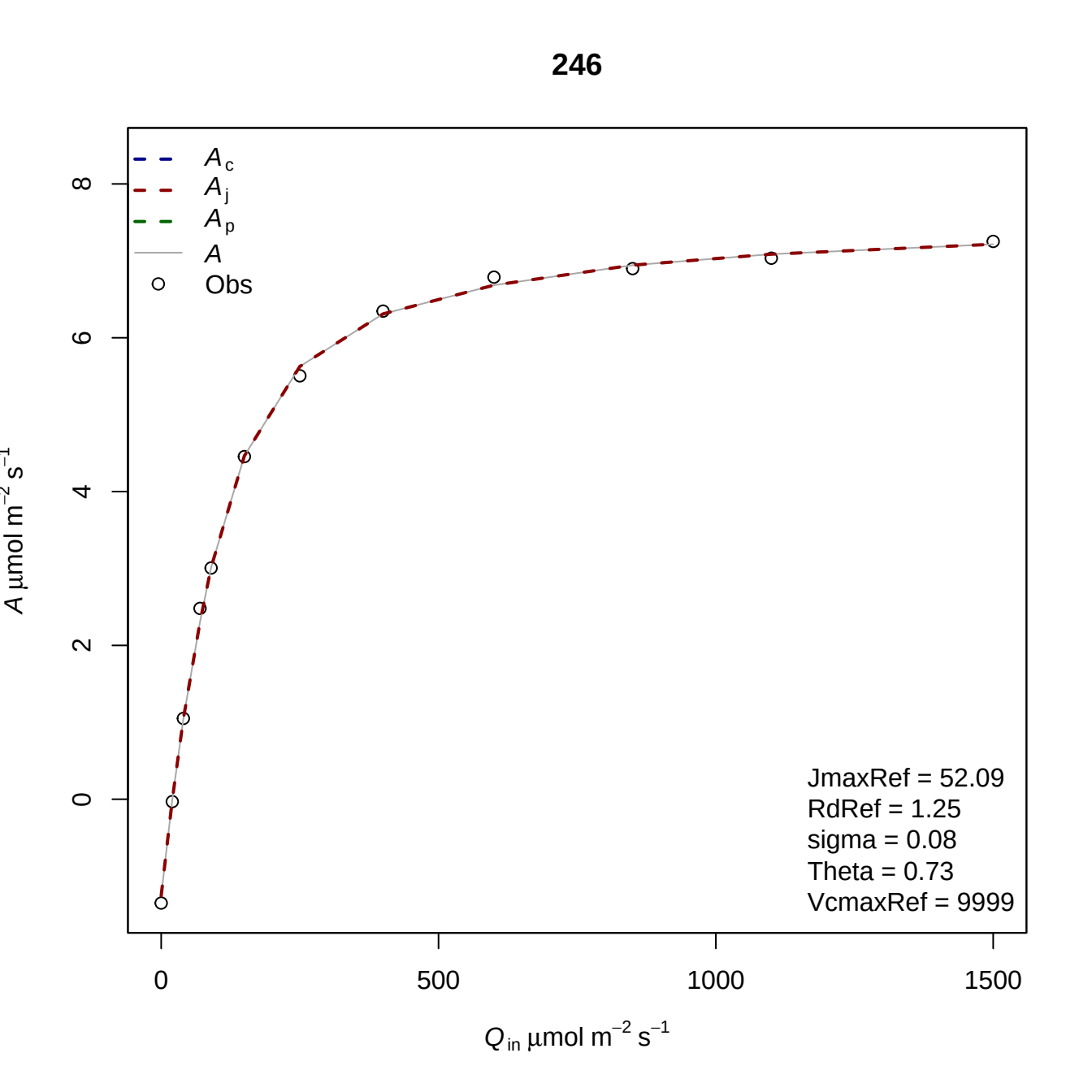

238
A_c
A_j
A_p
A
Obs
JmaxRef = 105.07
RdRef = 0.47
sigma = 0.2
Theta = −0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

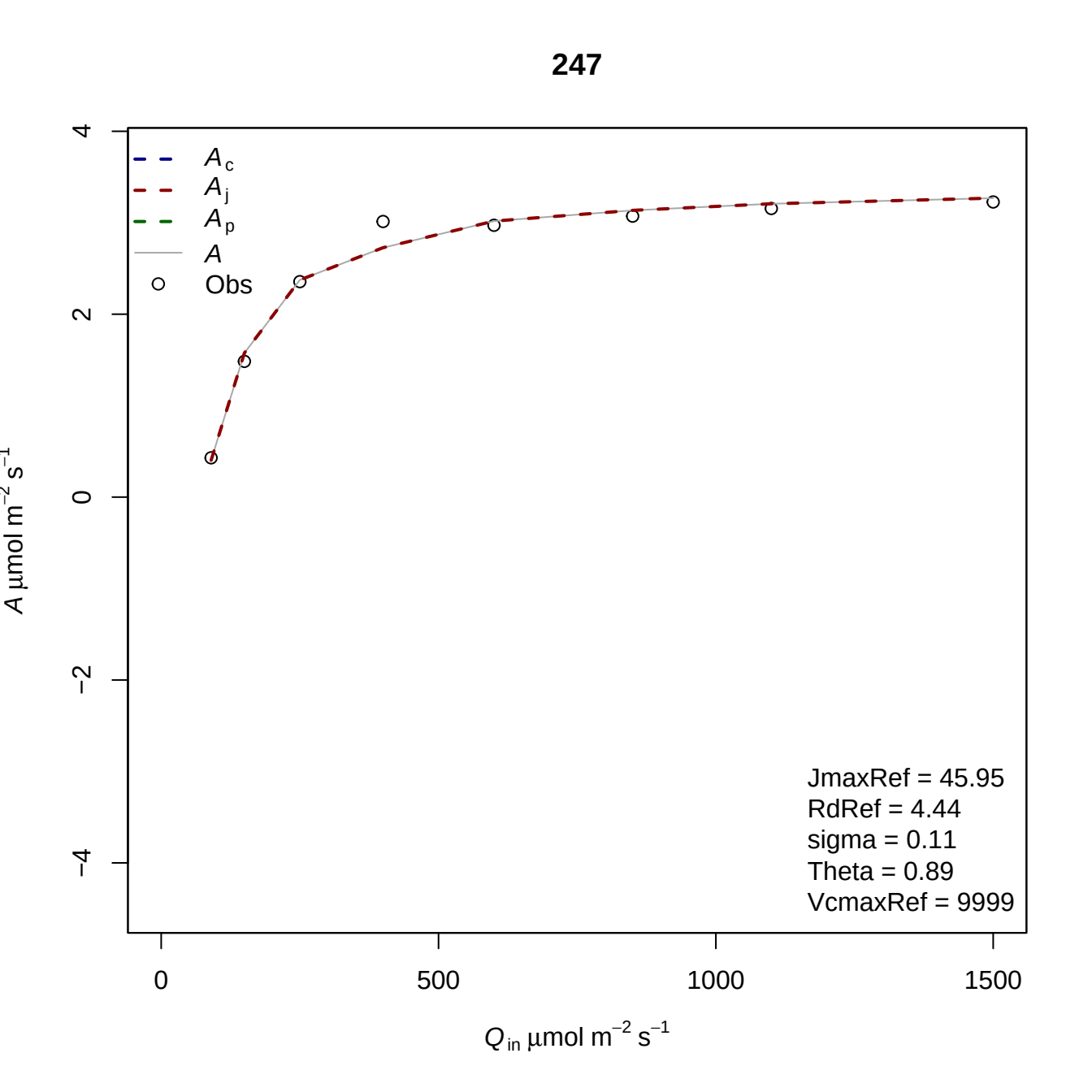

239
A_c
A_j
A_p
A
Obs
JmaxRef = 119.2
RdRef = 2.9
sigma = 0.71
Theta = −3.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

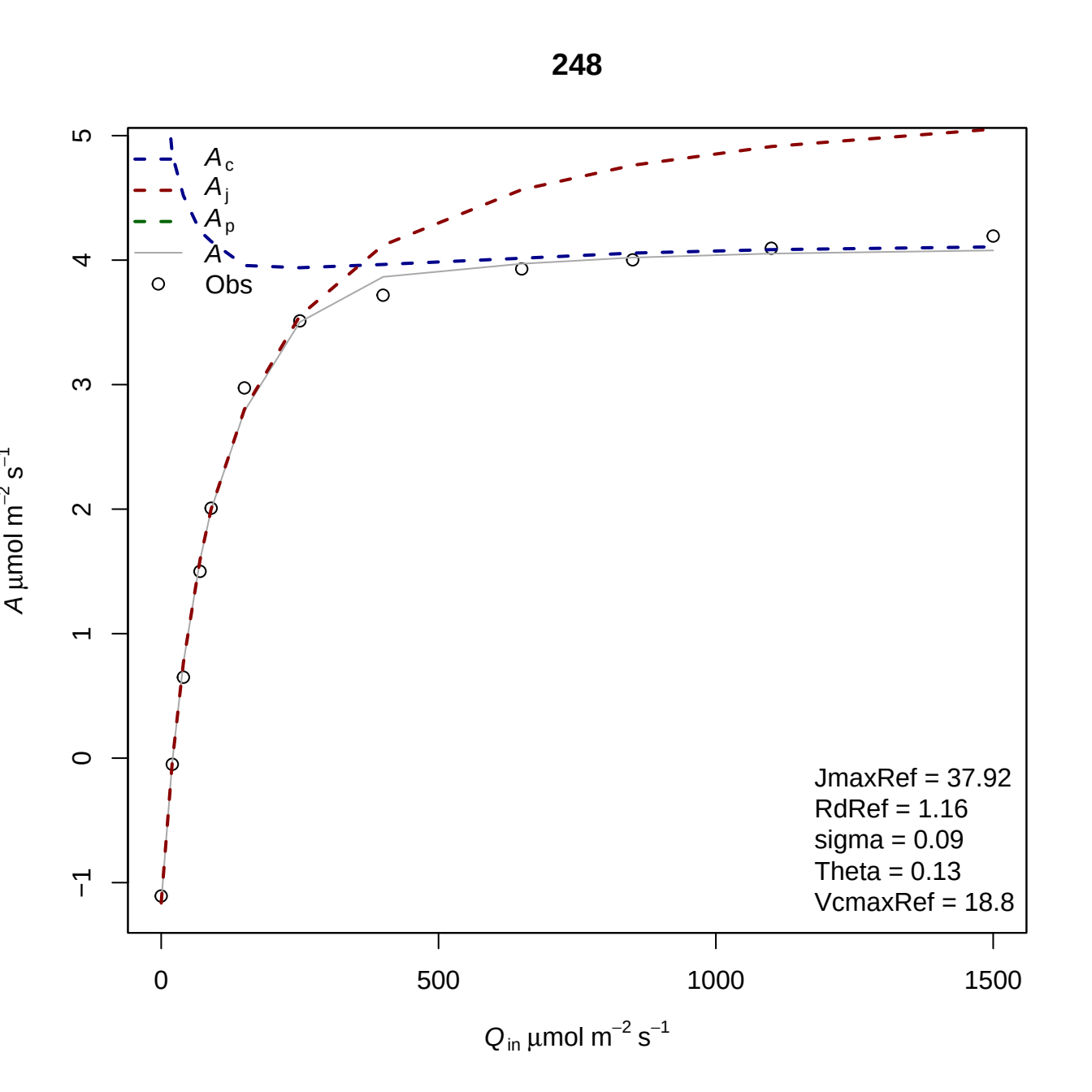

242
A_c
A_j
A_p
A
Obs
JmaxRef = 380.14
RdRef = 1.46
sigma = 0.31
Theta = −41.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

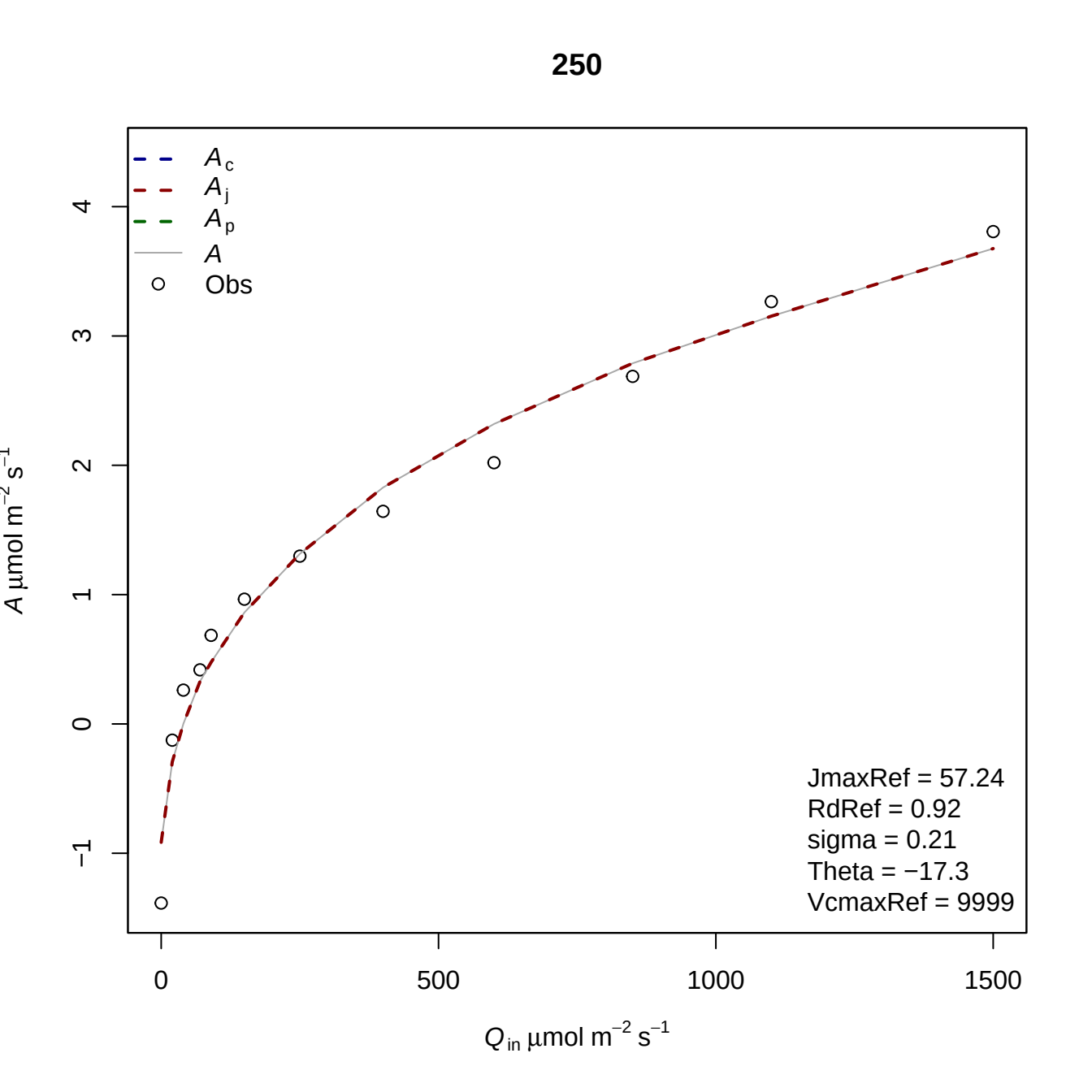

243
A_c
A_j
A_p
A
Obs
JmaxRef = 153.87
RdRef = 2.12
sigma = 0.32
Theta = −1.27
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

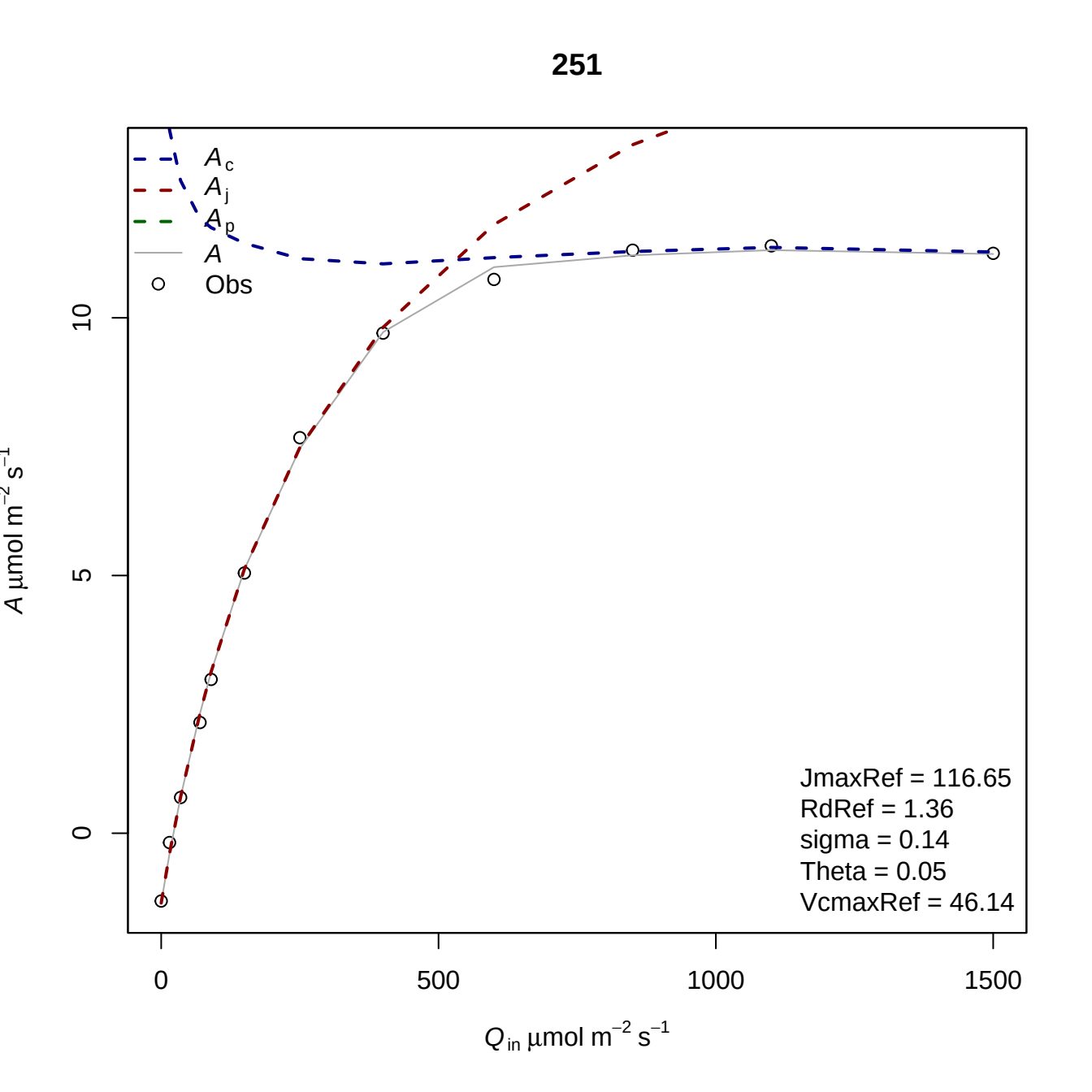

245
A_c
A_j
A_p
A
Obs
JmaxRef = 108.1
RdRef = 1.46
sigma = 0.12
Theta = 0.36
VcmaxRef = 53.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

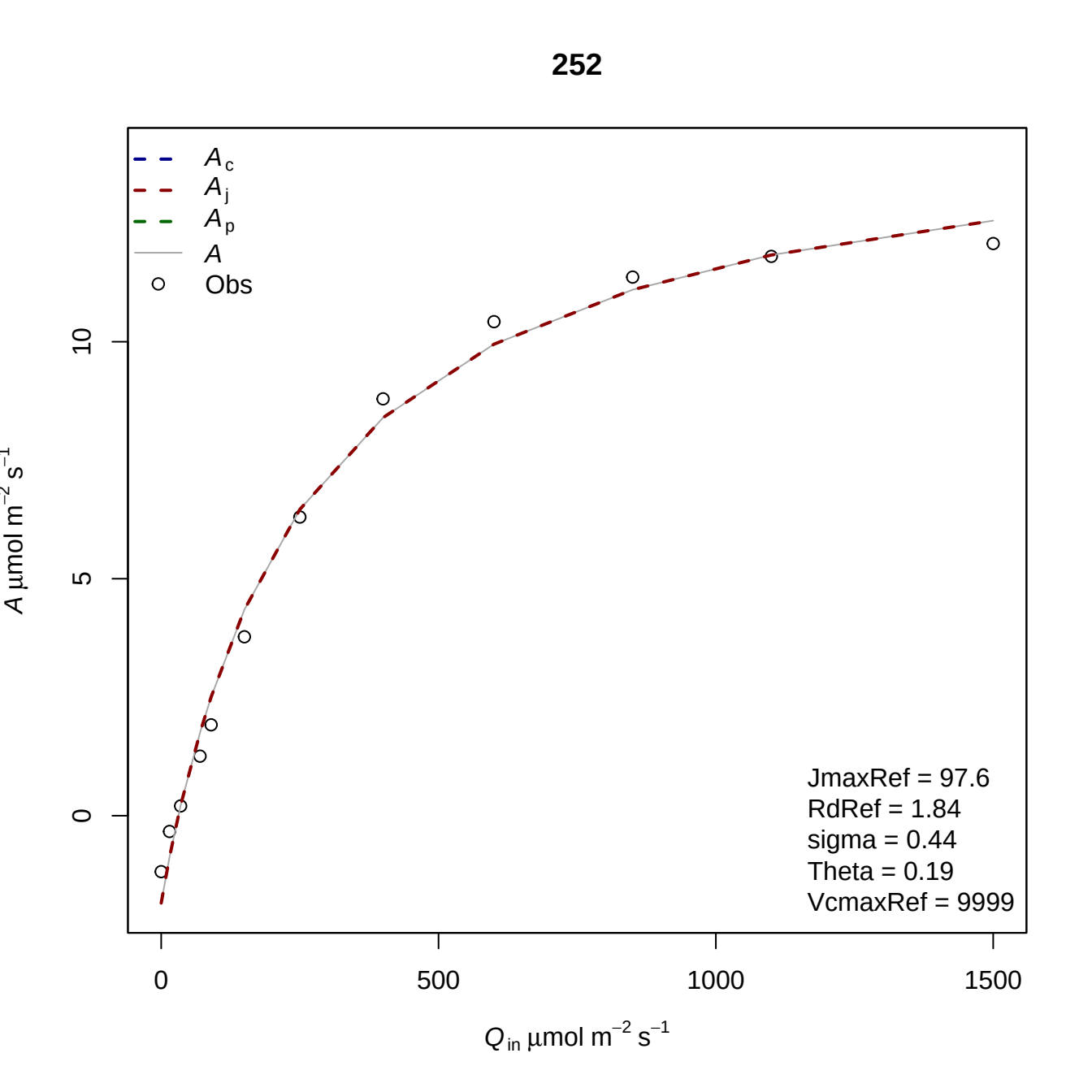

246
A_c
A_j
A_p
A
Obs
JmaxRef = 52.09
RdRef = 1.25
sigma = 0.08
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

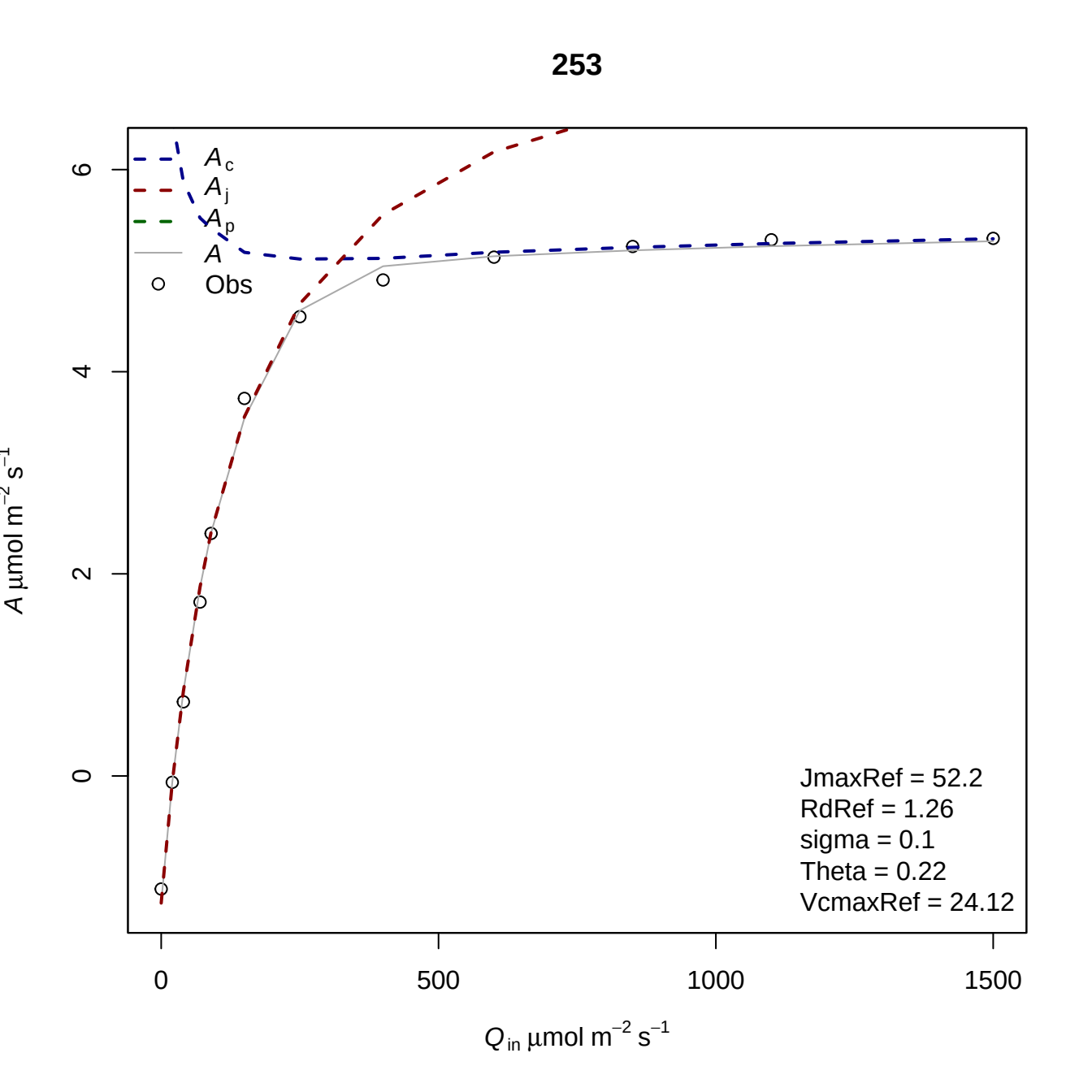

247
A_c
A_j
A_p
A
Obs
JmaxRef = 45.95
RdRef = 4.44
sigma = 0.11
Theta = 0.89
VcmaxRef = 9999
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

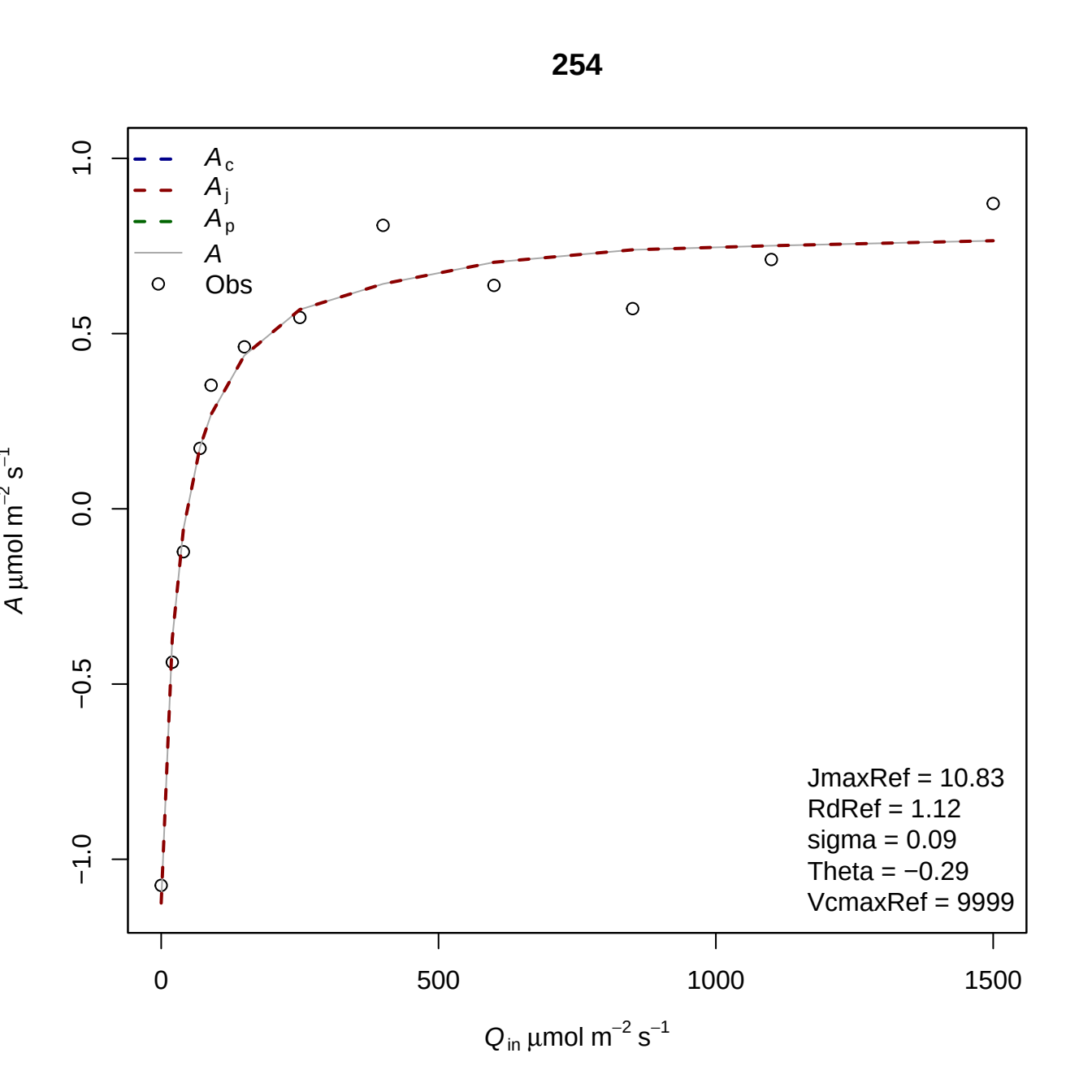

248
A_c
A_j
A_p
A
Obs
JmaxRef = 37.92
RdRef = 1.16
sigma = 0.09
Theta = 0.13
VcmaxRef = 18.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

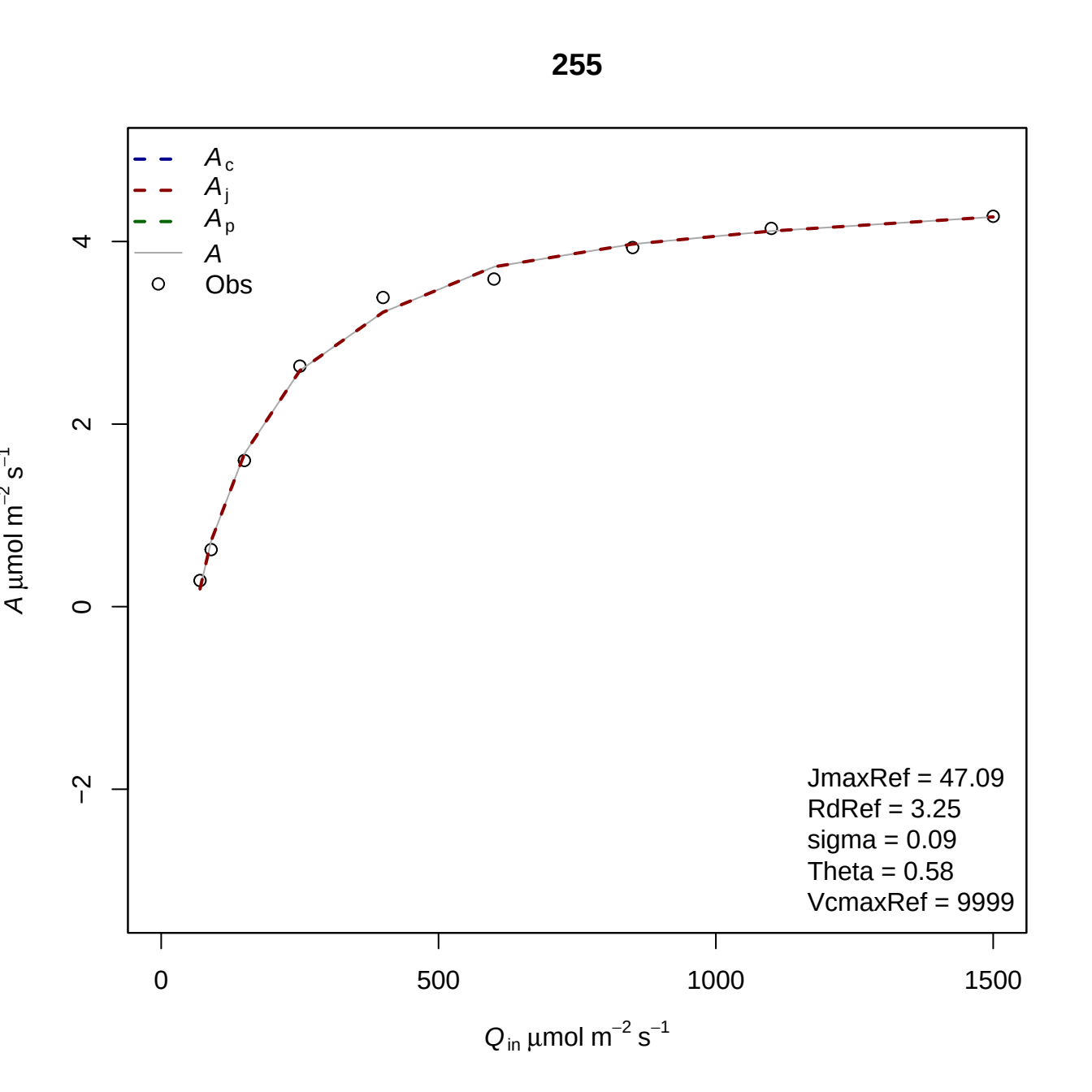

250
A_c
A_j
A_p
A
Obs
JmaxRef = 57.24
RdRef = 0.92
sigma = 0.21
Theta = −17.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

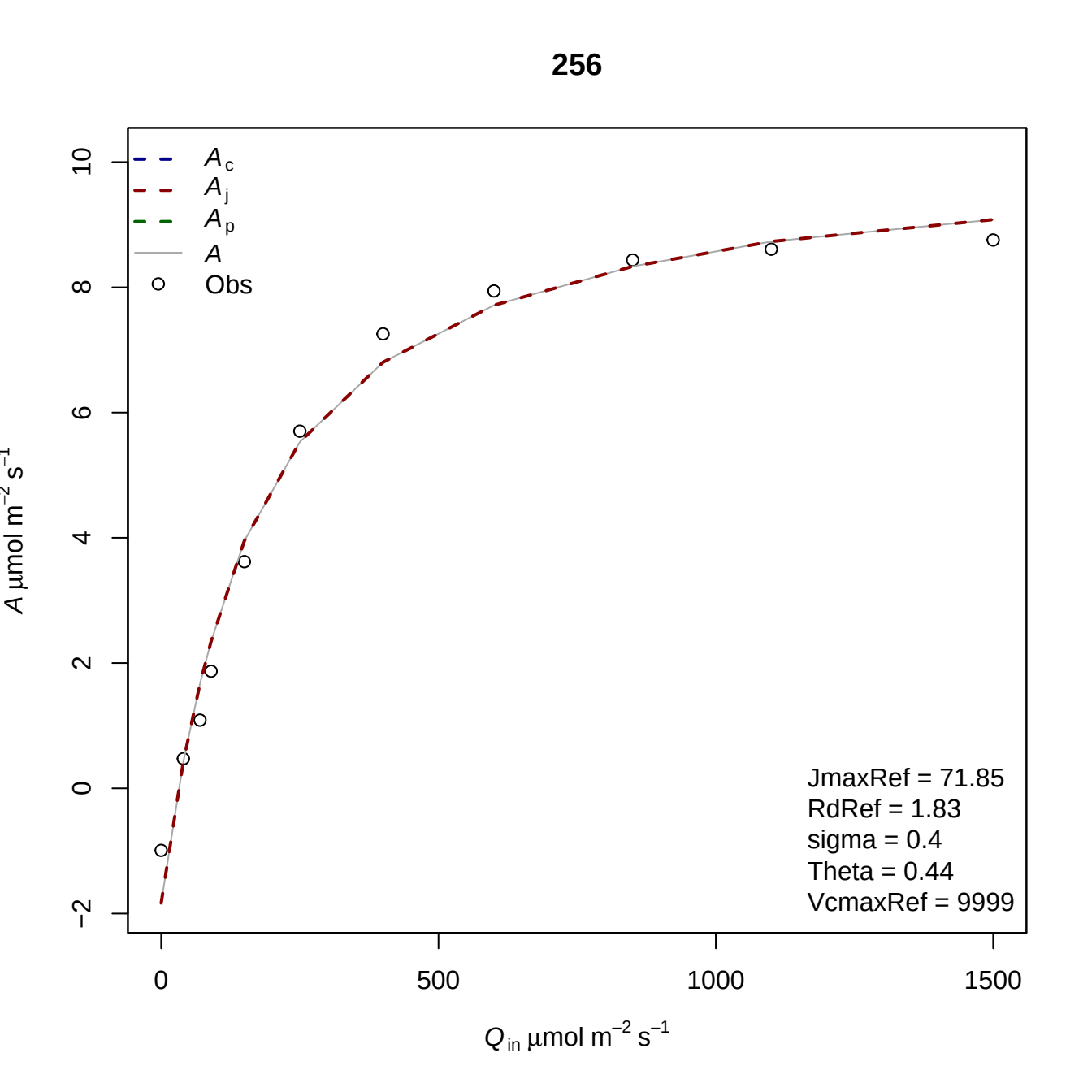

251
A_c
A_j
A_p
A
Obs
JmaxRef = 116.65
RdRef = 1.36
sigma = 0.14
Theta = 0.05
VcmaxRef = 46.14
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

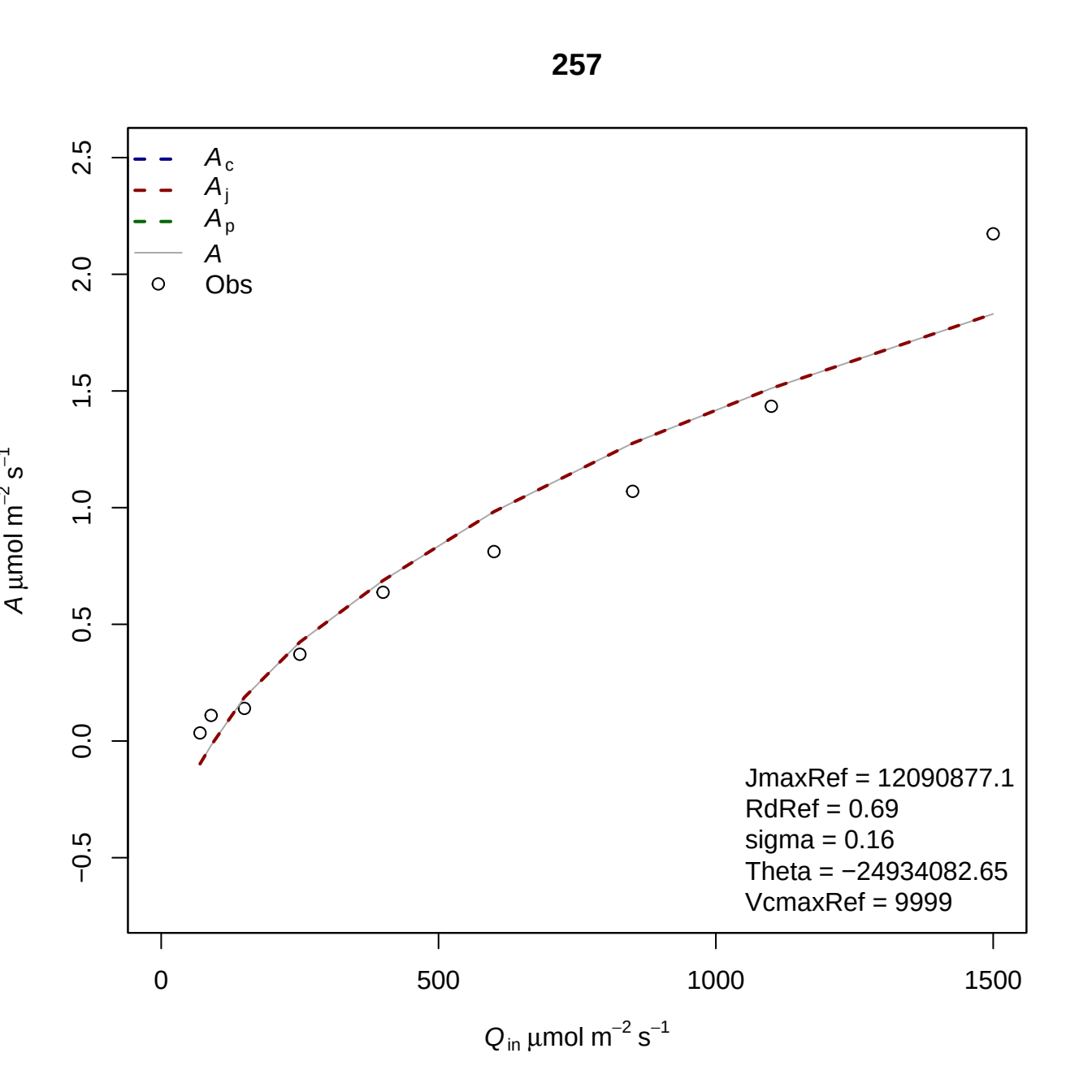

252
A_c
A_j
A_p
A
Obs
JmaxRef = 97.6
RdRef = 1.84
sigma = 0.44
Theta = 0.19
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

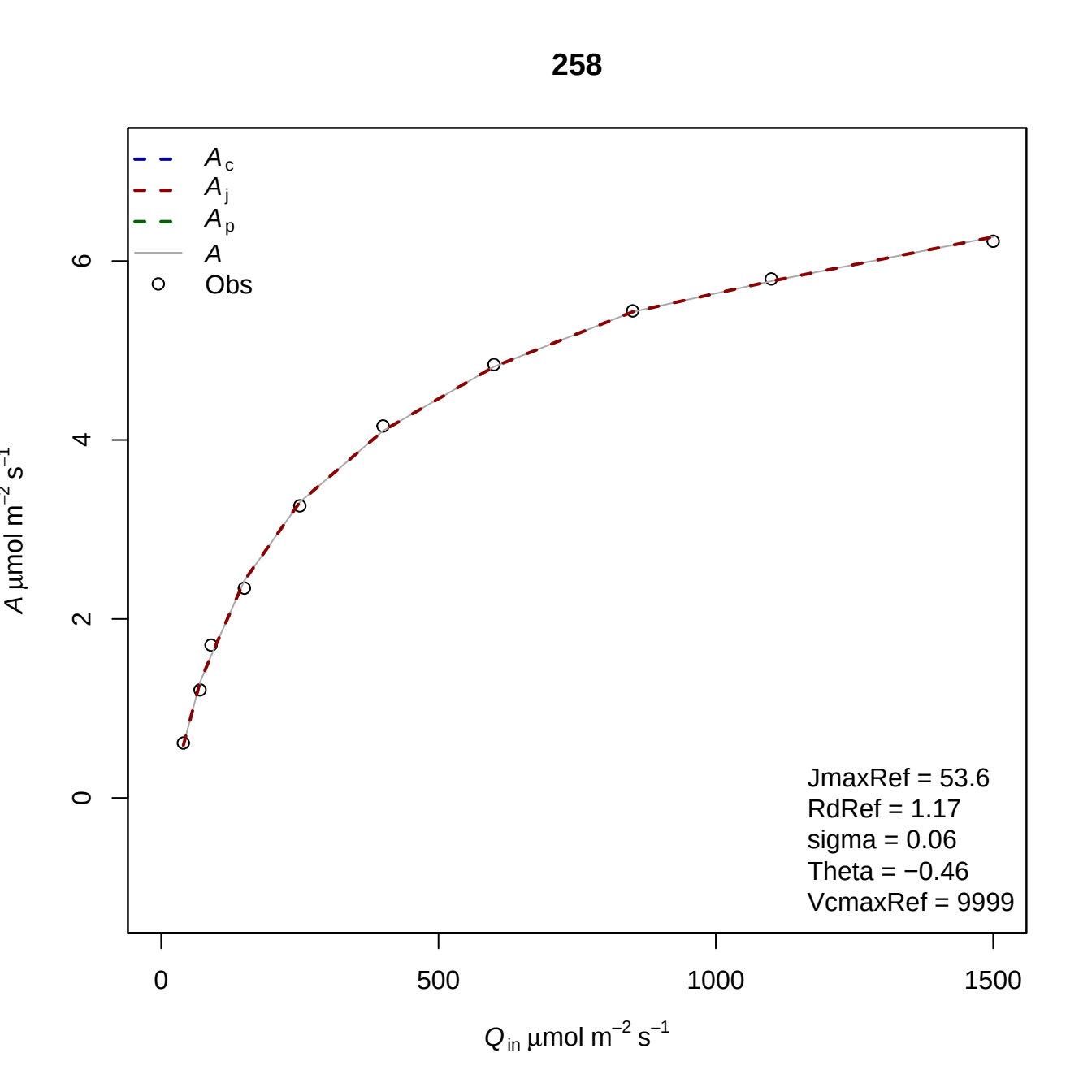

253
A_c
A_j
A_p
A
Obs
JmaxRef = 52.2
RdRef = 1.26
sigma = 0.1
Theta = 0.22
VcmaxRef = 24.12
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

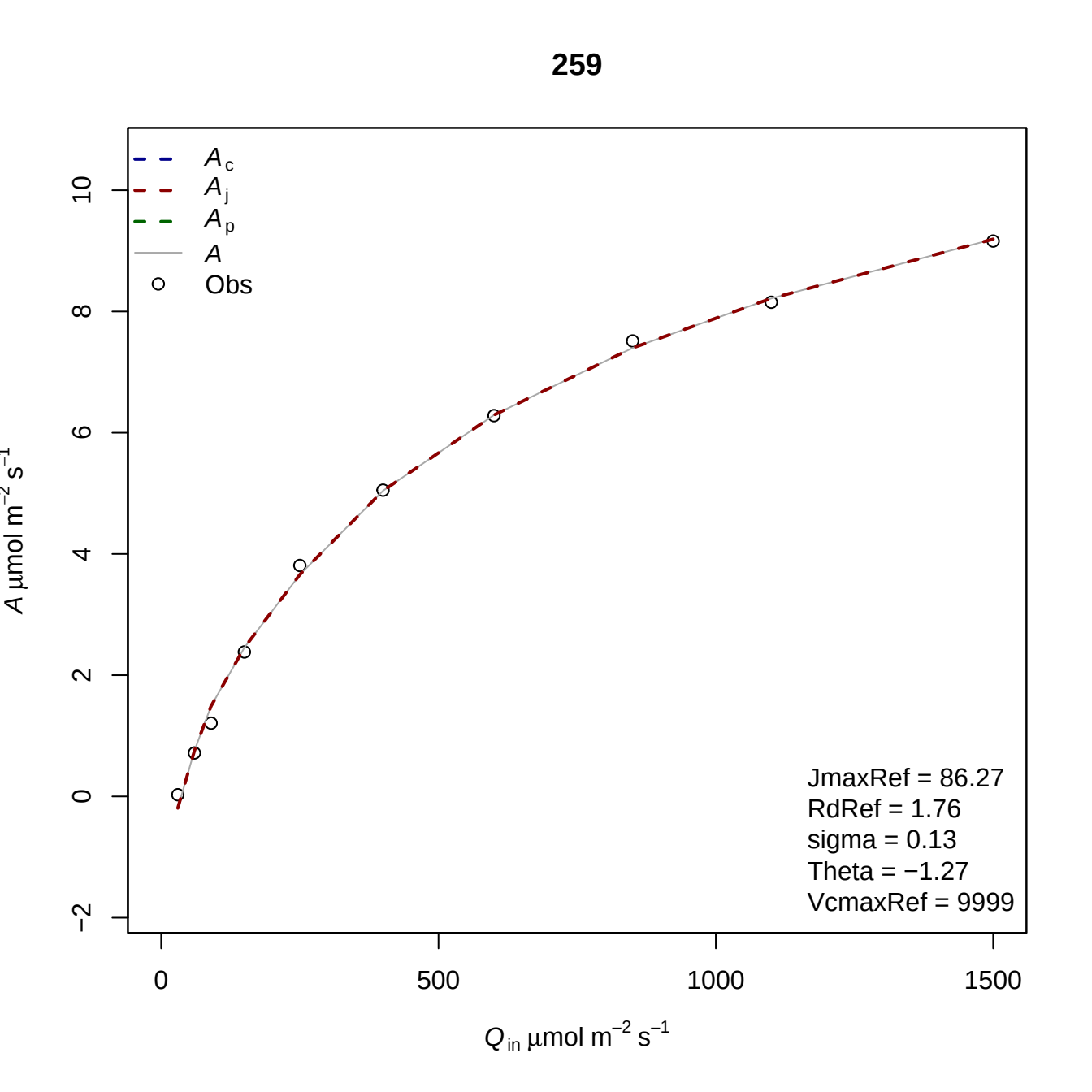

254
A_c
A_j
A_p
A
Obs
JmaxRef = 10.83
RdRef = 1.12
sigma = 0.09
Theta = −0.29
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

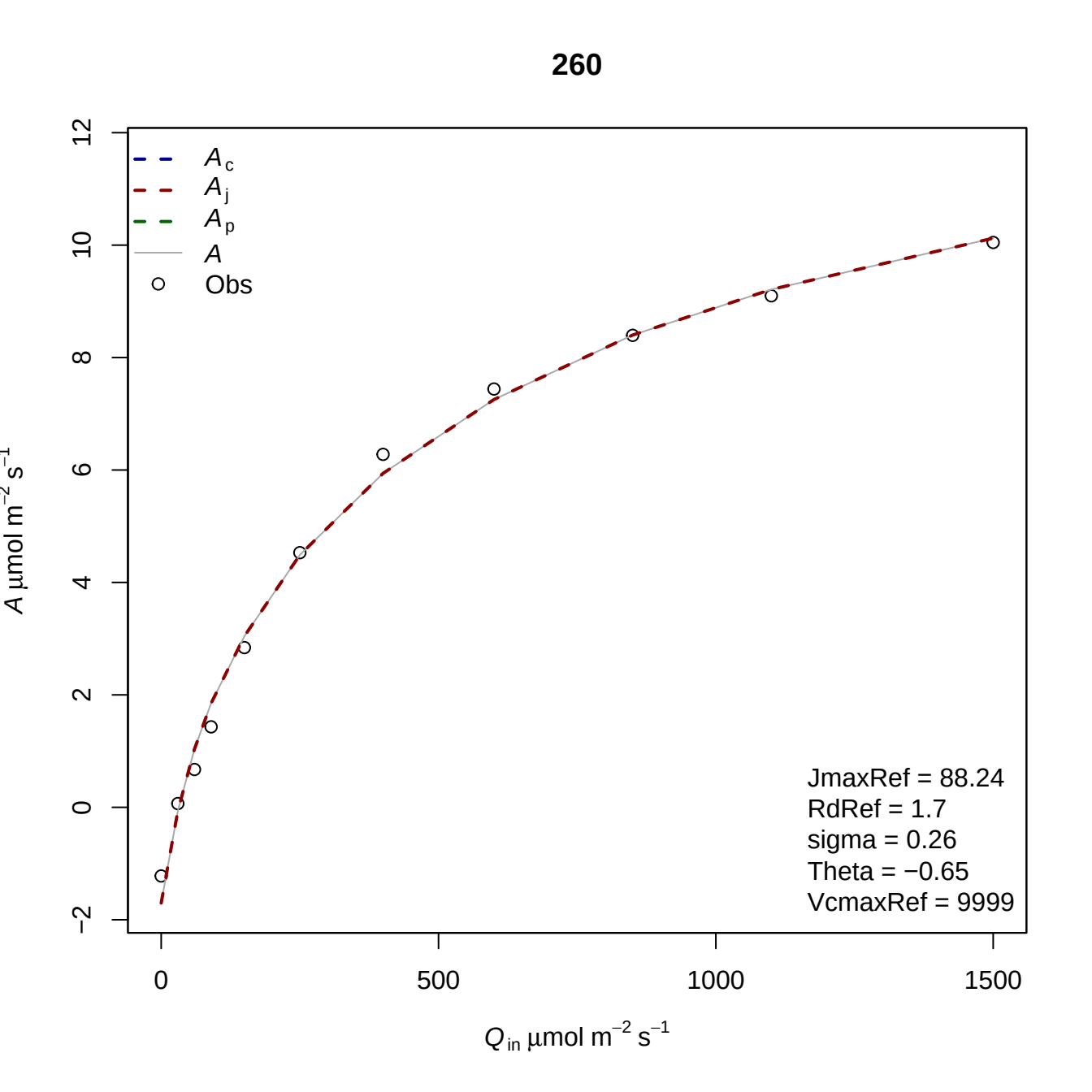

255
A_c
A_j
A_p
A
Obs
JmaxRef = 47.09
RdRef = 3.25
sigma = 0.09
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
500
1000
1500

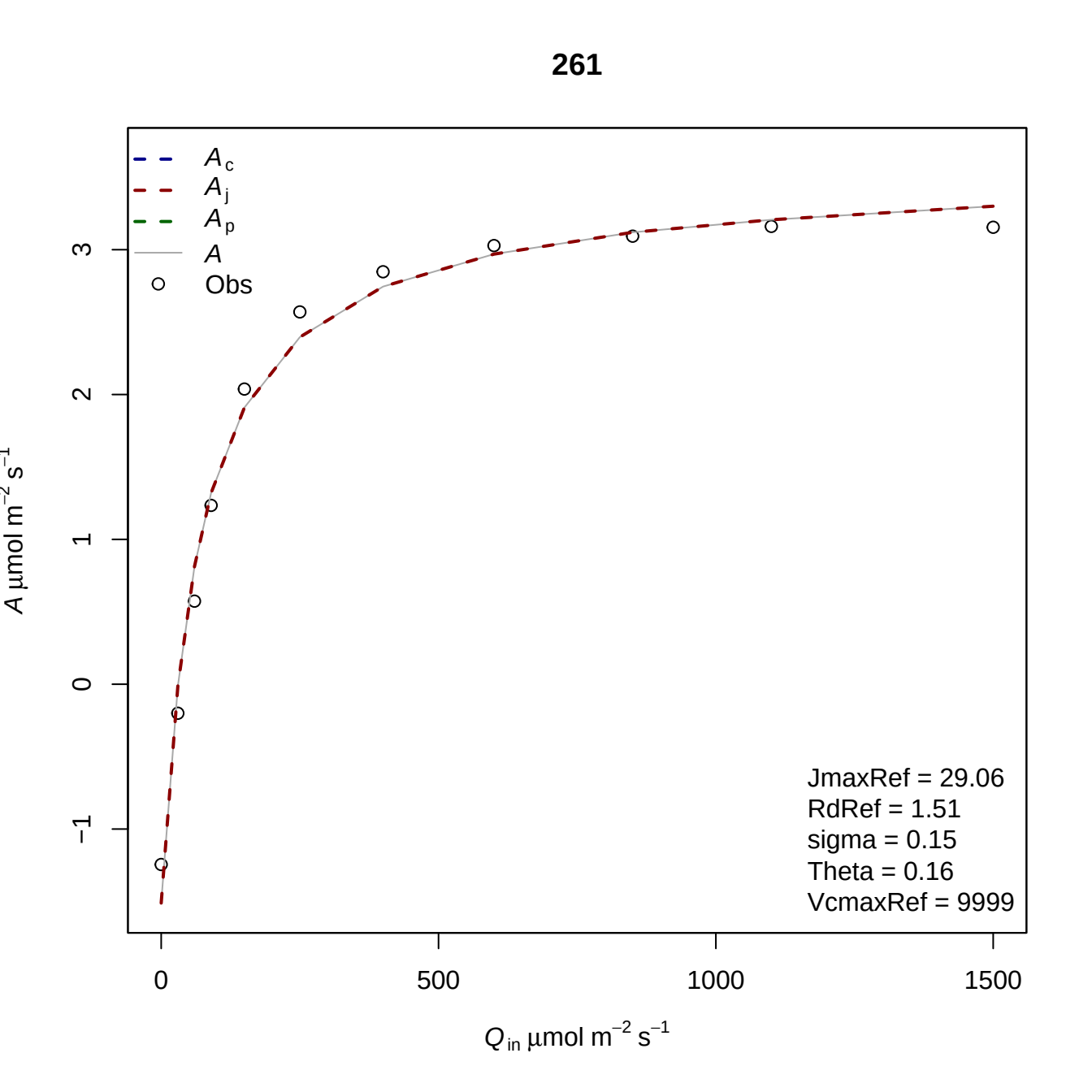

256
A_c
A_j
A_p
A
Obs
JmaxRef = 71.85
RdRef = 1.83
sigma = 0.4
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

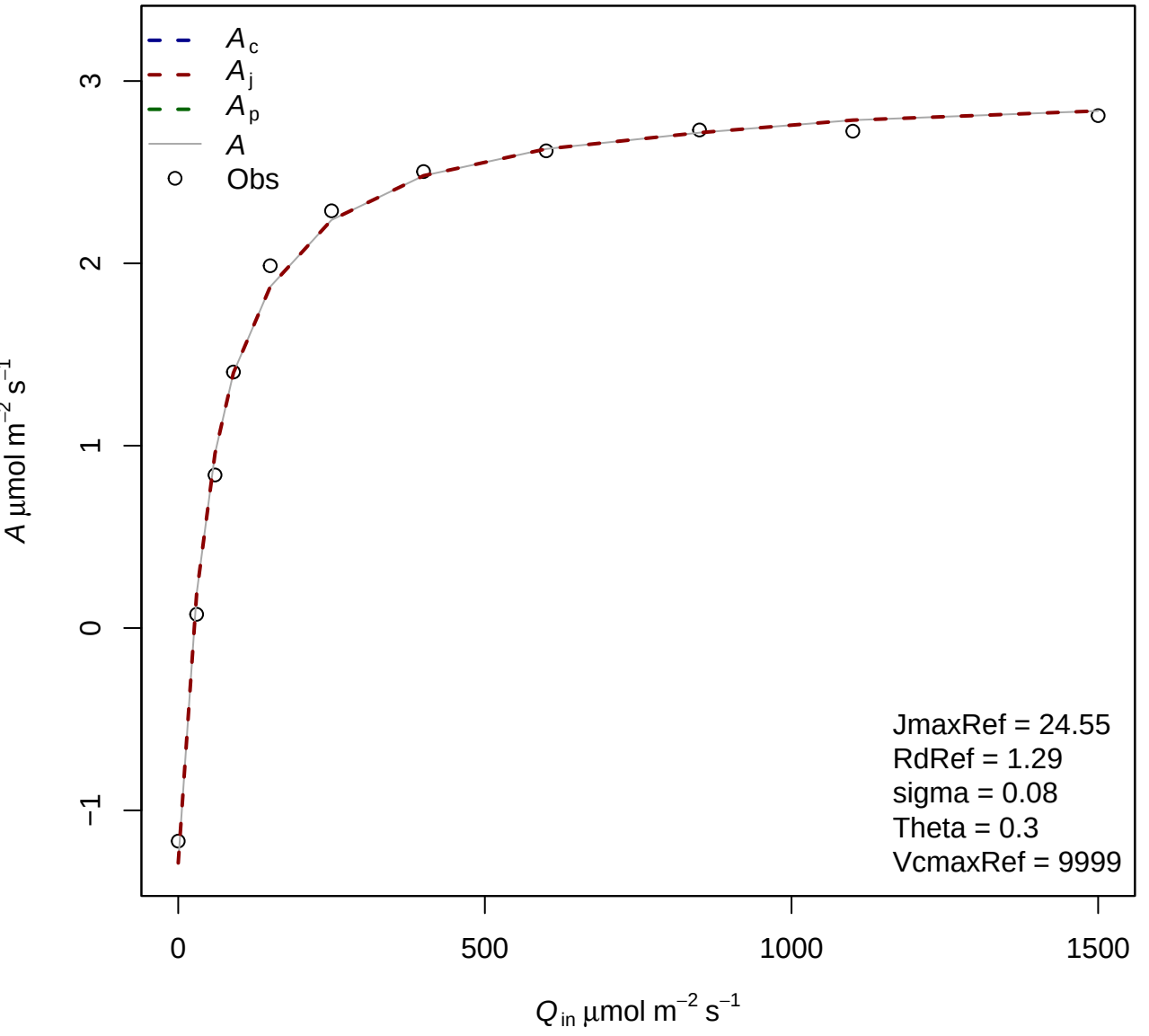

257
A_c
A_j
A_p
A
Obs
JmaxRef = 12090877.1
RdRef = 0.69
sigma = 0.16
Theta = −24934082.65
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

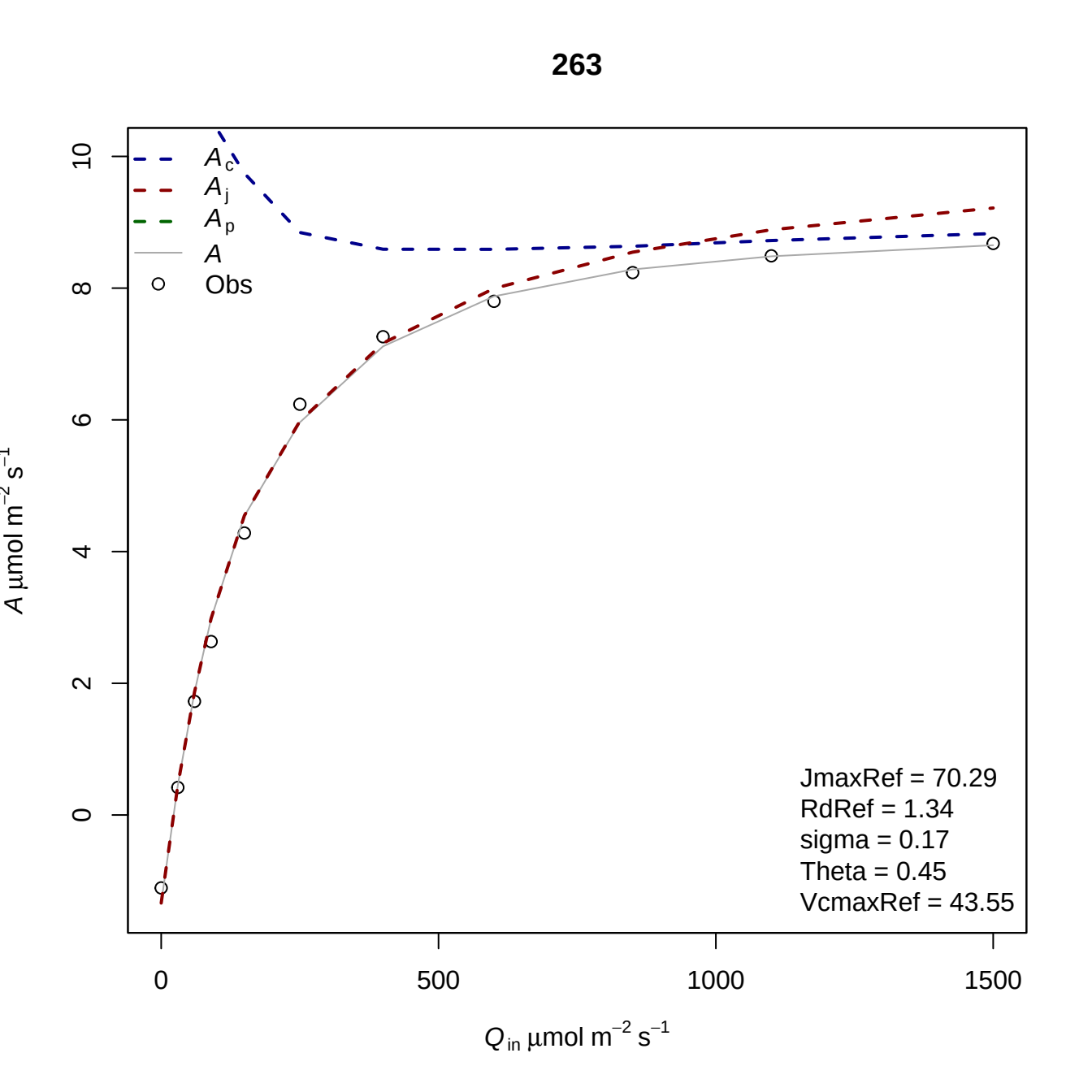

258
A_c
A_j
A_p
A
Obs
JmaxRef = 53.6
RdRef = 1.17
sigma = 0.06
Theta = −0.46
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

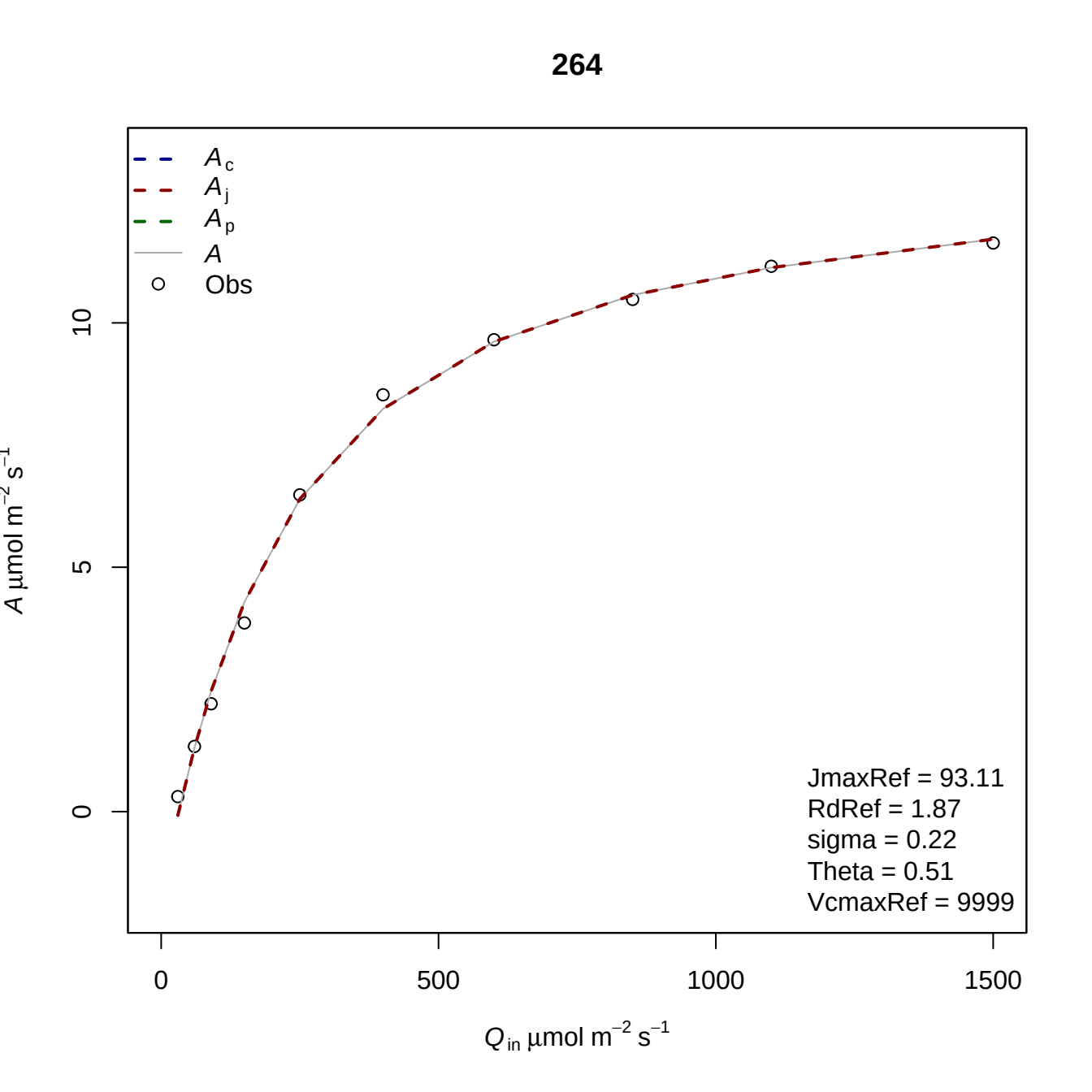

259
A_c
A_j
A_p
A
Obs
JmaxRef = 86.27
RdRef = 1.76
sigma = 0.13
Theta = −1.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

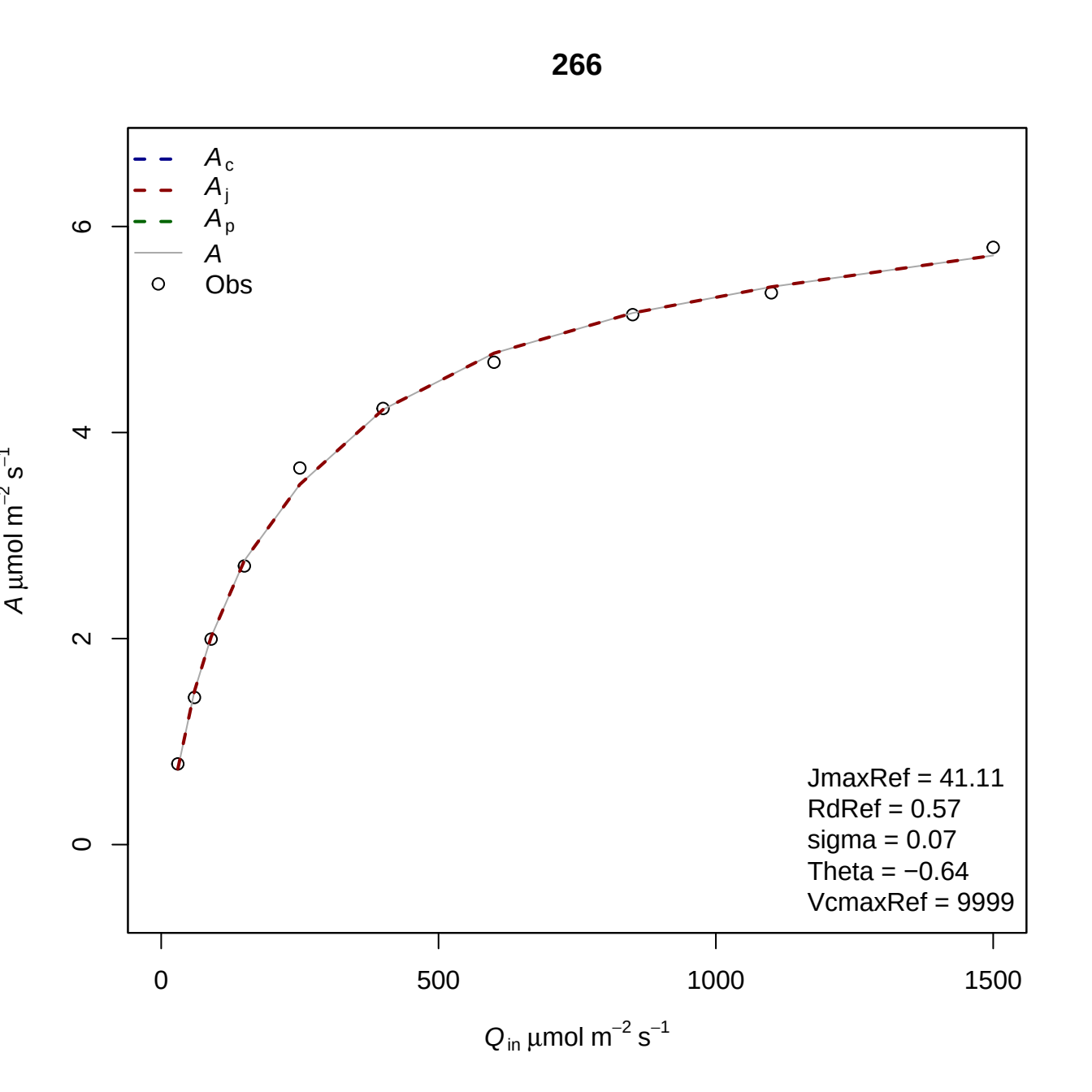

260
A_c
A_j
A_p
A
Obs
JmaxRef = 88.24
RdRef = 1.7
sigma = 0.26
Theta = −0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

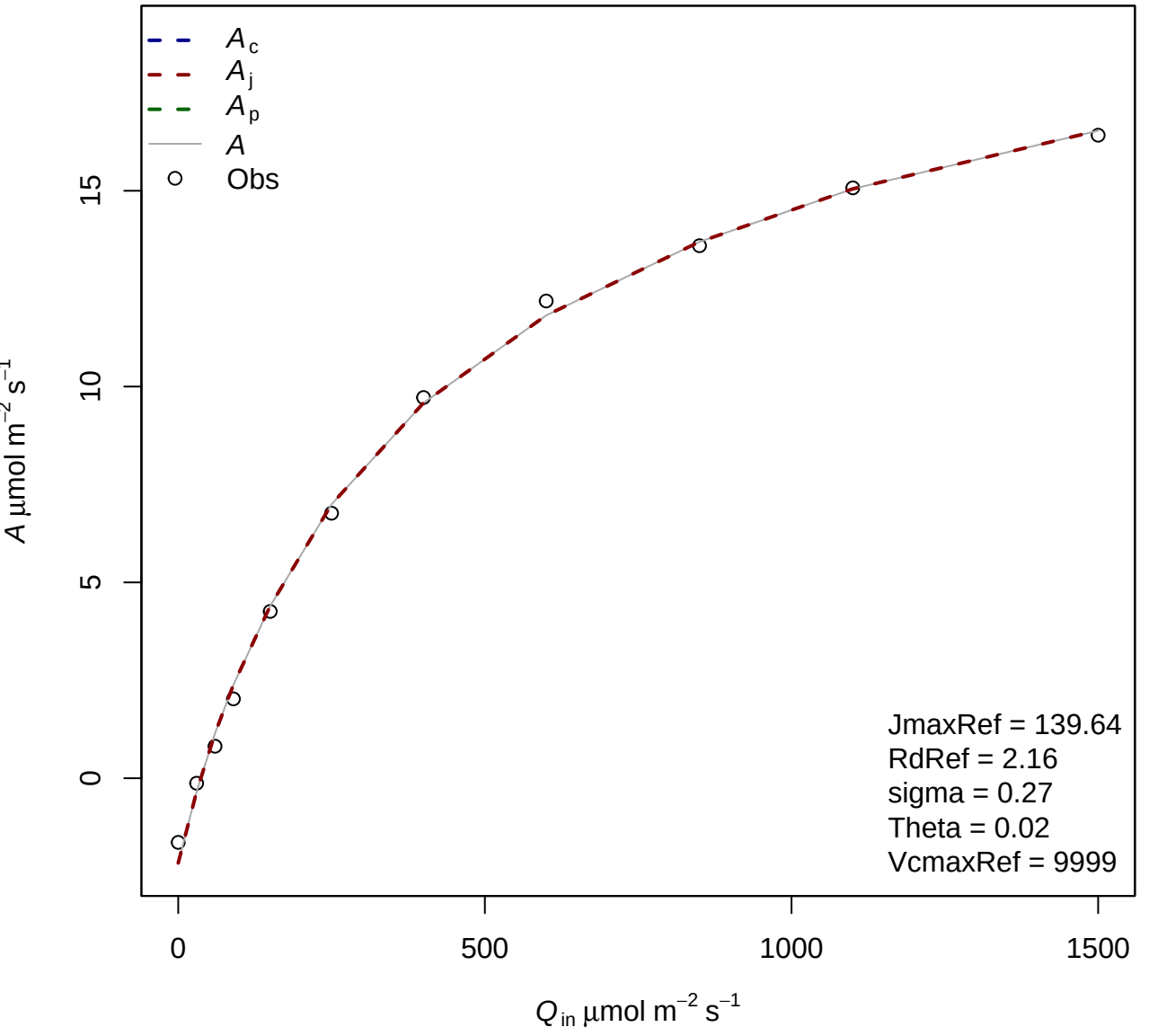

261
A_c
A_j
A_p
A
Obs
JmaxRef = 29.06
RdRef = 1.51
sigma = 0.15
Theta = 0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

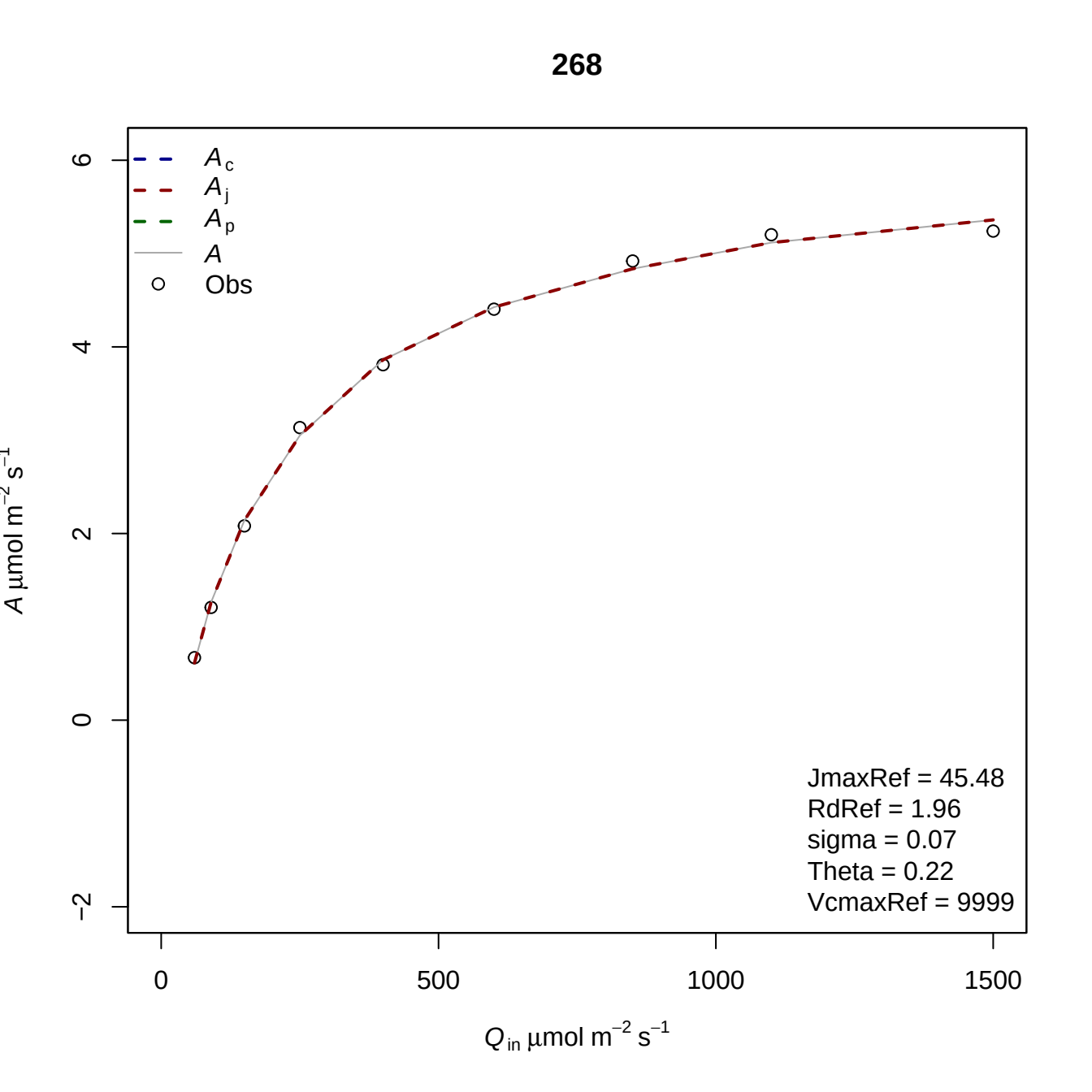

A_c
A_j
A_p
A
Obs
JmaxRef = 24.55
RdRef = 1.29
sigma = 0.08
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

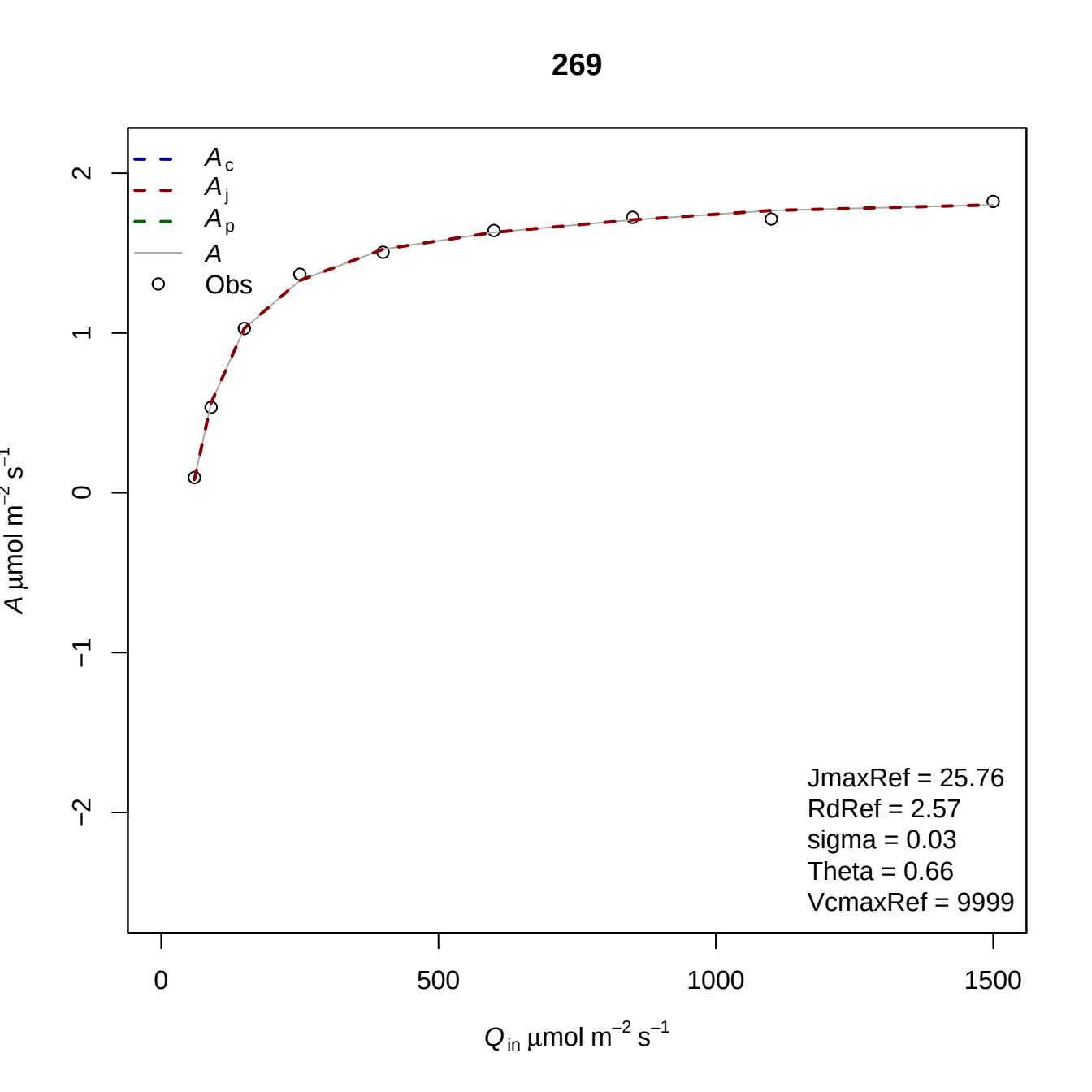

263
A_c
A_j
A_p
A
Obs
JmaxRef = 70.29
RdRef = 1.34
sigma = 0.17
Theta = 0.45
VcmaxRef = 43.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

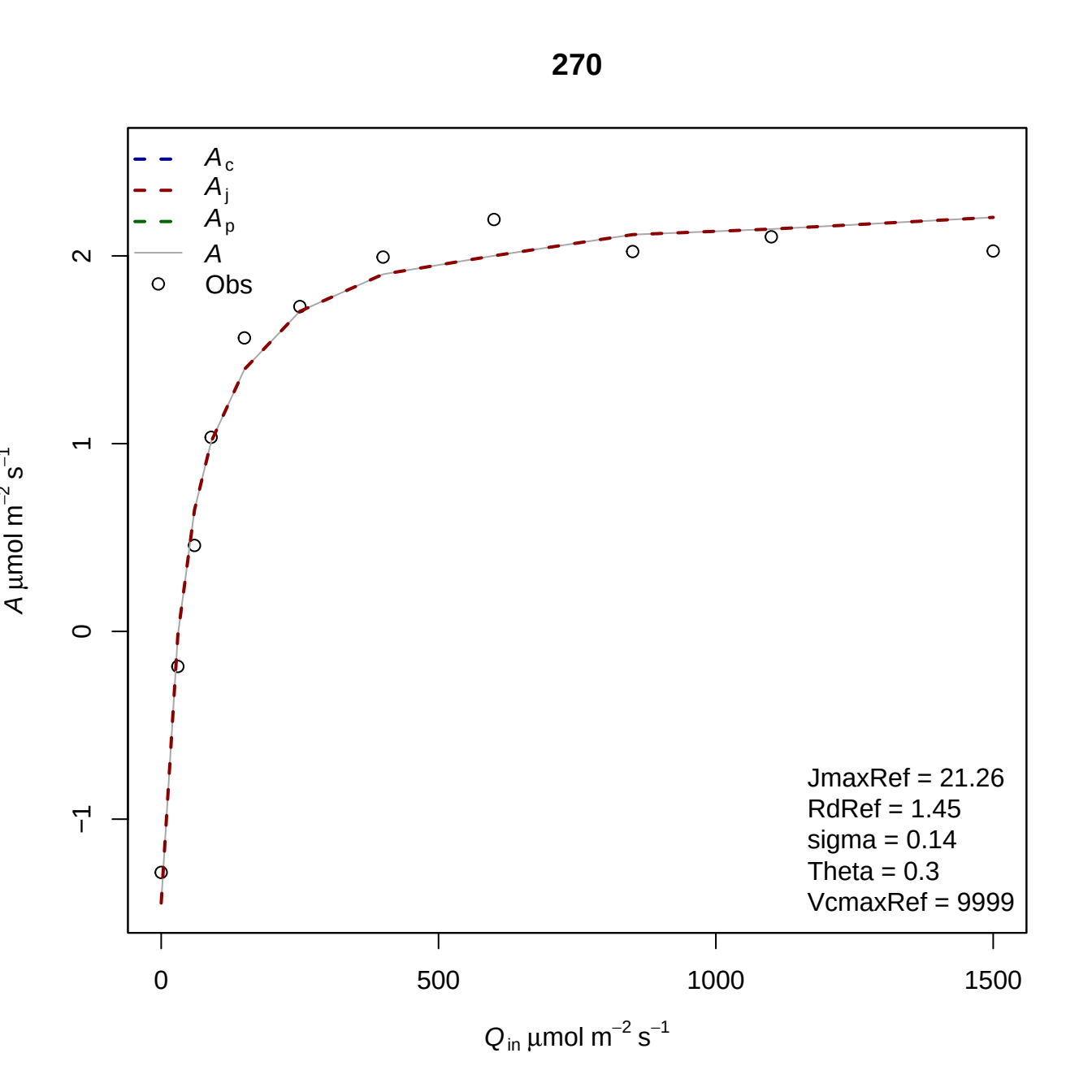

264
A_c
A_j
A_p
A
Obs
JmaxRef = 93.11
RdRef = 1.87
sigma = 0.22
Theta = 0.51
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

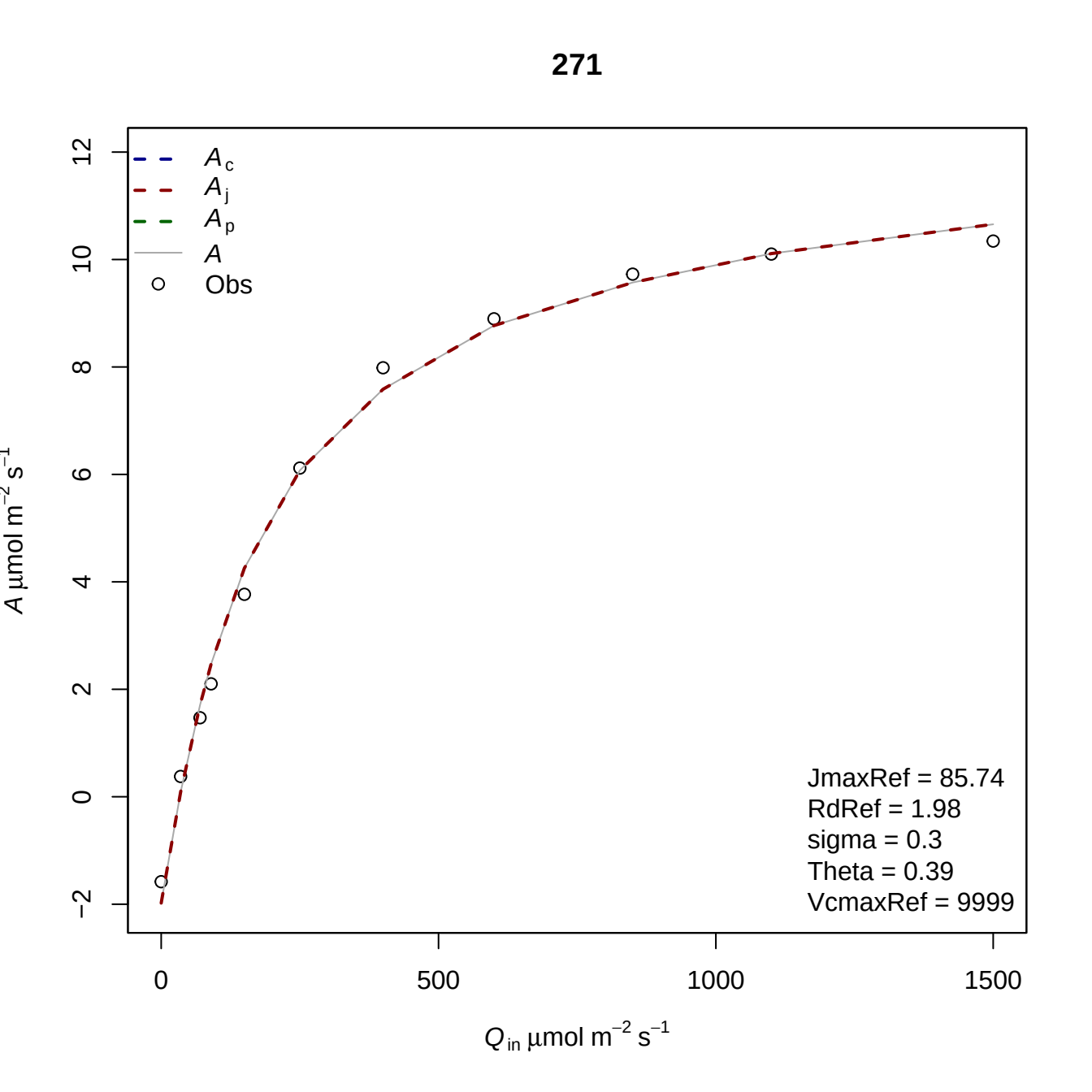

266
A_c
A_j
A_p
A
Obs
JmaxRef = 41.11
RdRef = 0.57
sigma = 0.07
Theta = −0.64
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

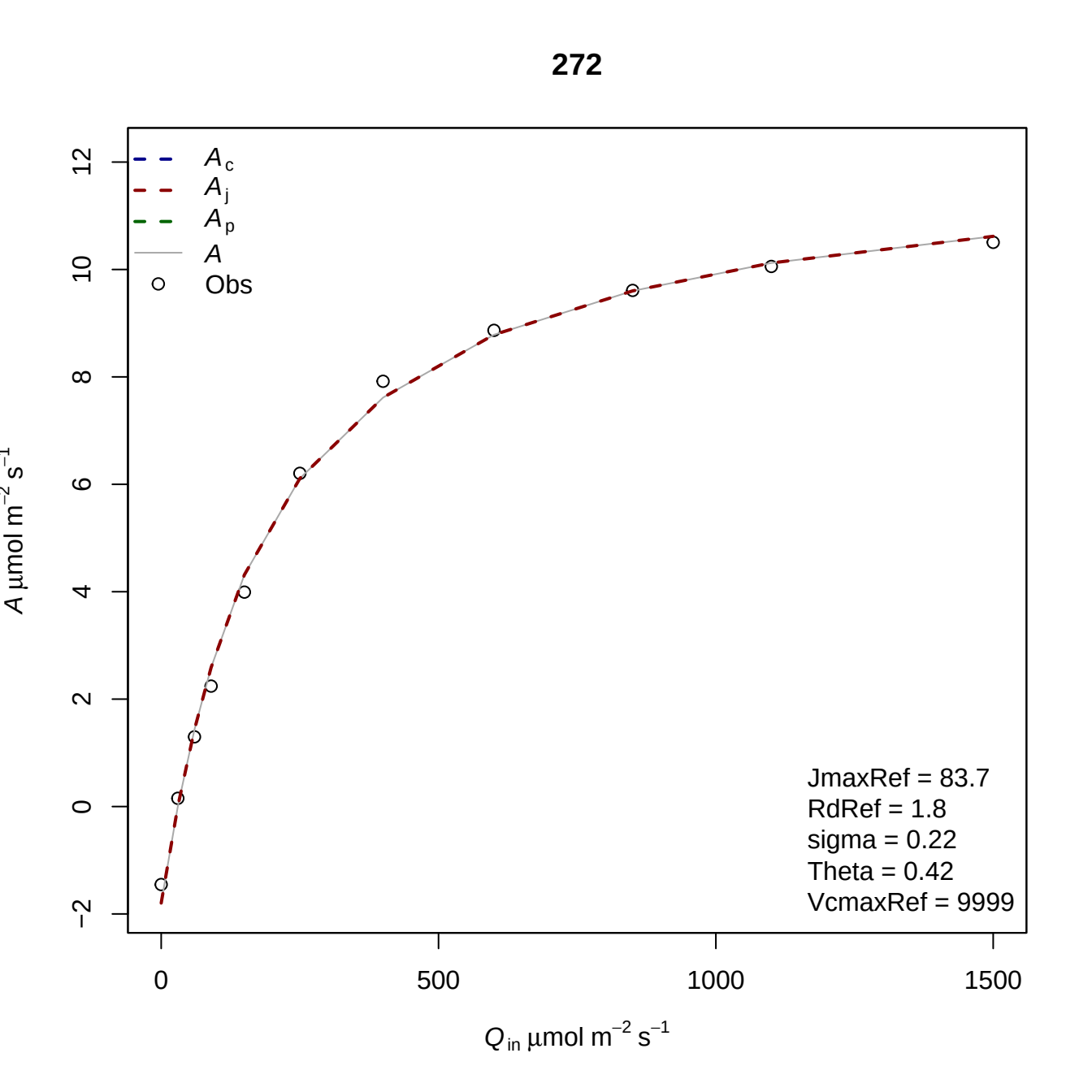

A_c
A_j
A_p
A
Obs
JmaxRef = 139.64
RdRef = 2.16
sigma = 0.27
Theta = 0.02
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

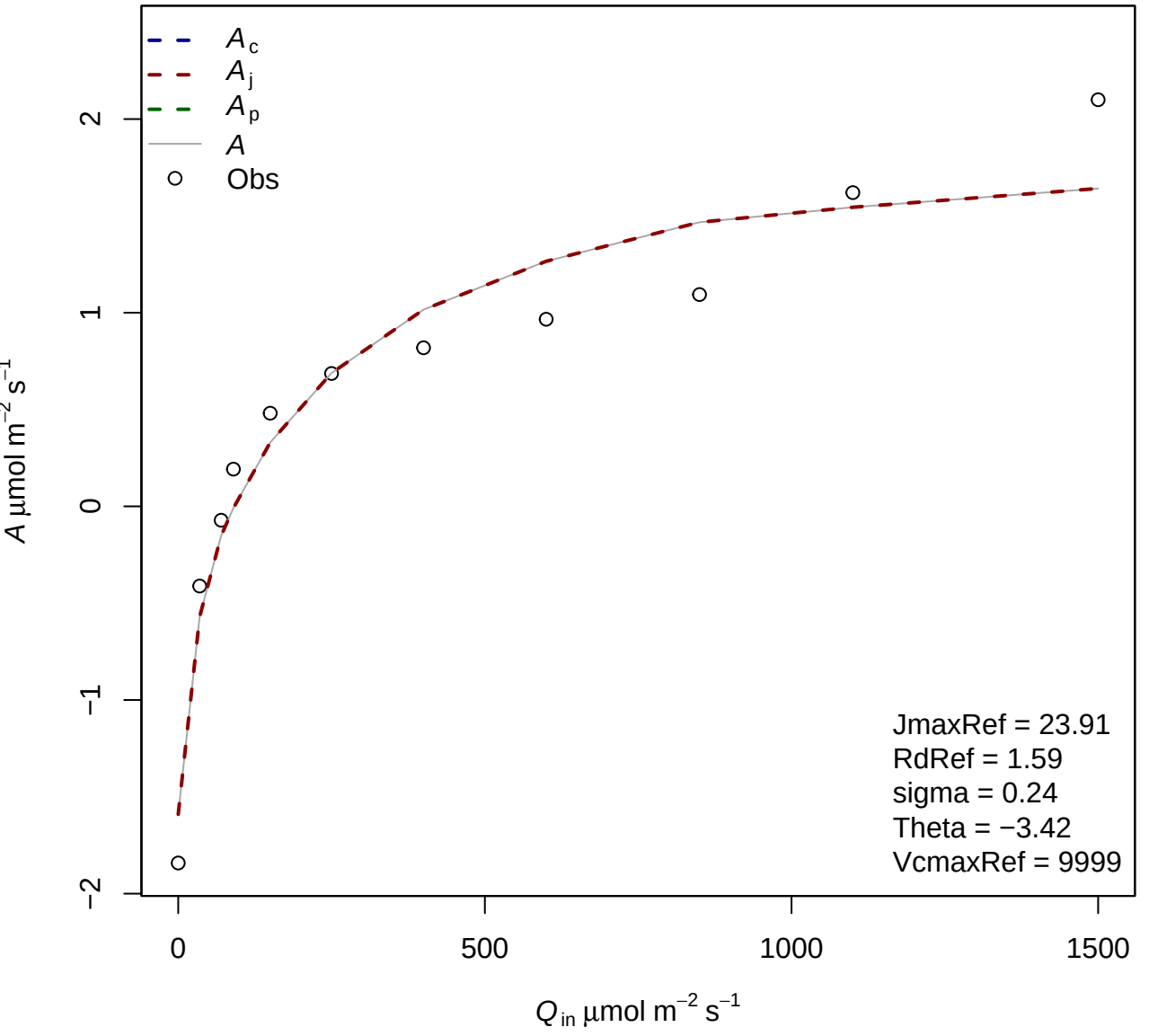

268
A_c
A_j
A_p
A
Obs
JmaxRef = 45.48
RdRef = 1.96
sigma = 0.07
Theta = 0.22
VcmaxRef = 9999
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

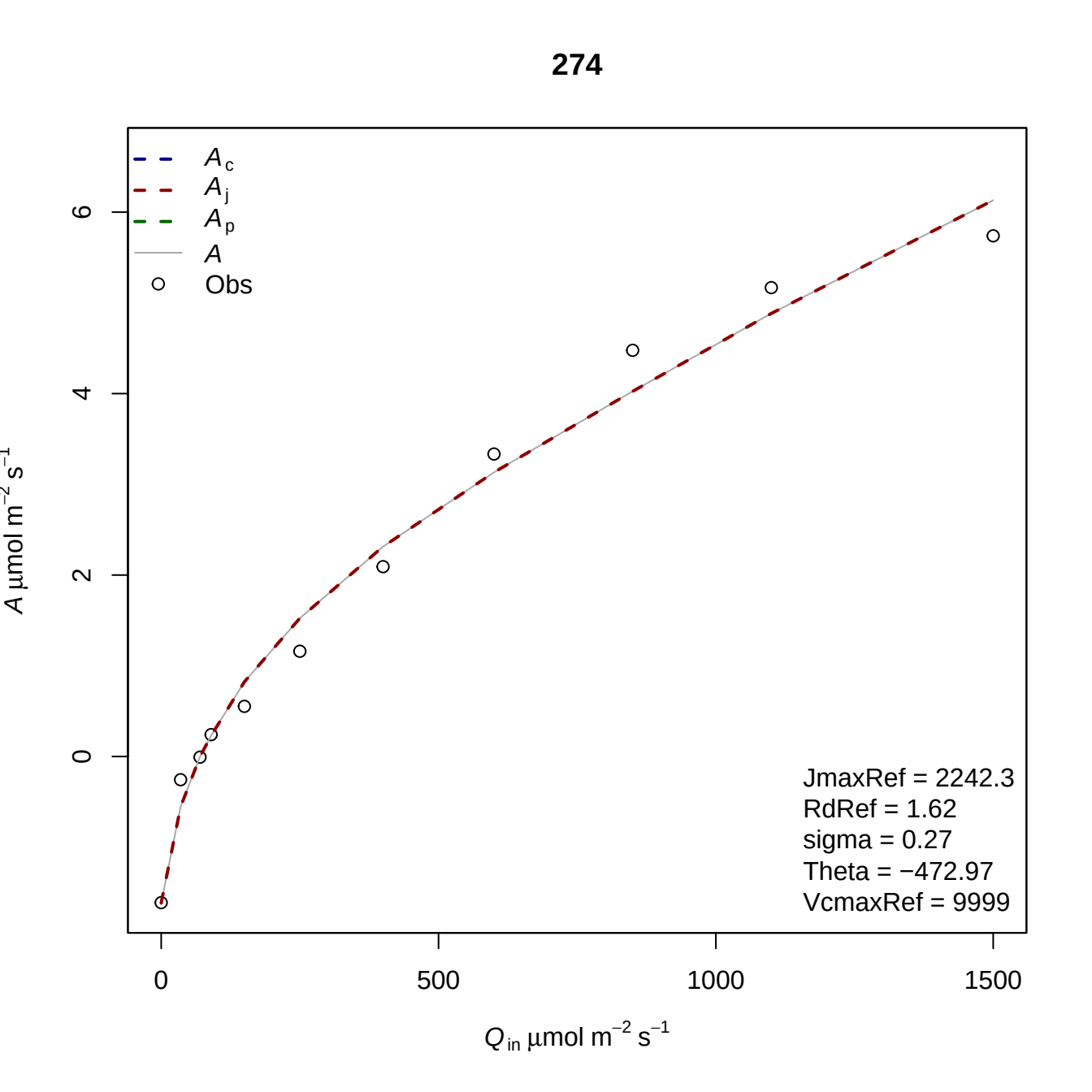

269
A_c
A_j
A_p
A
Obs
JmaxRef = 25.76
RdRef = 2.57
sigma = 0.03
Theta = 0.66
VcmaxRef = 9999
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

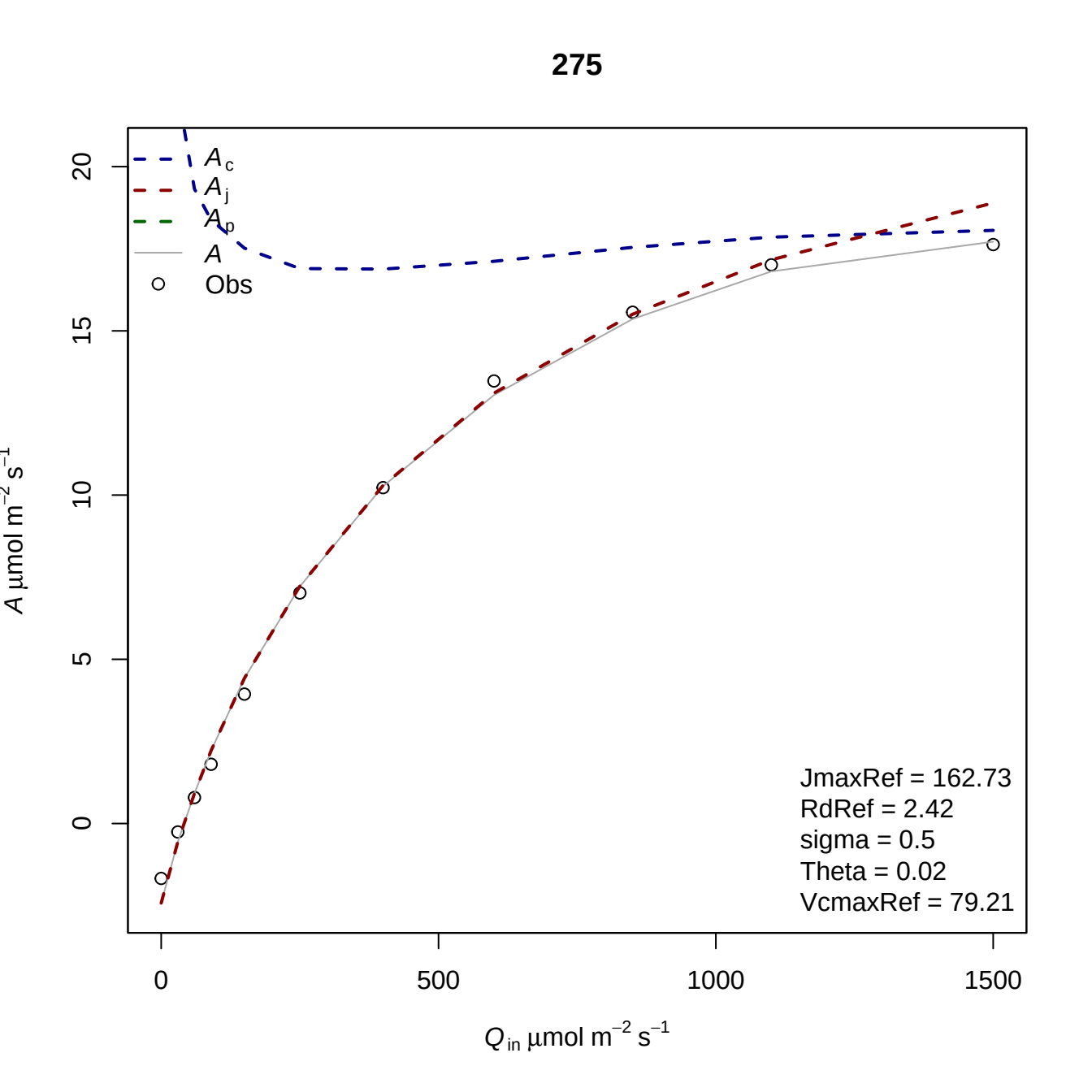

270
A_c
A_j
A_p
A
Obs
JmaxRef = 21.26
RdRef = 1.45
sigma = 0.14
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

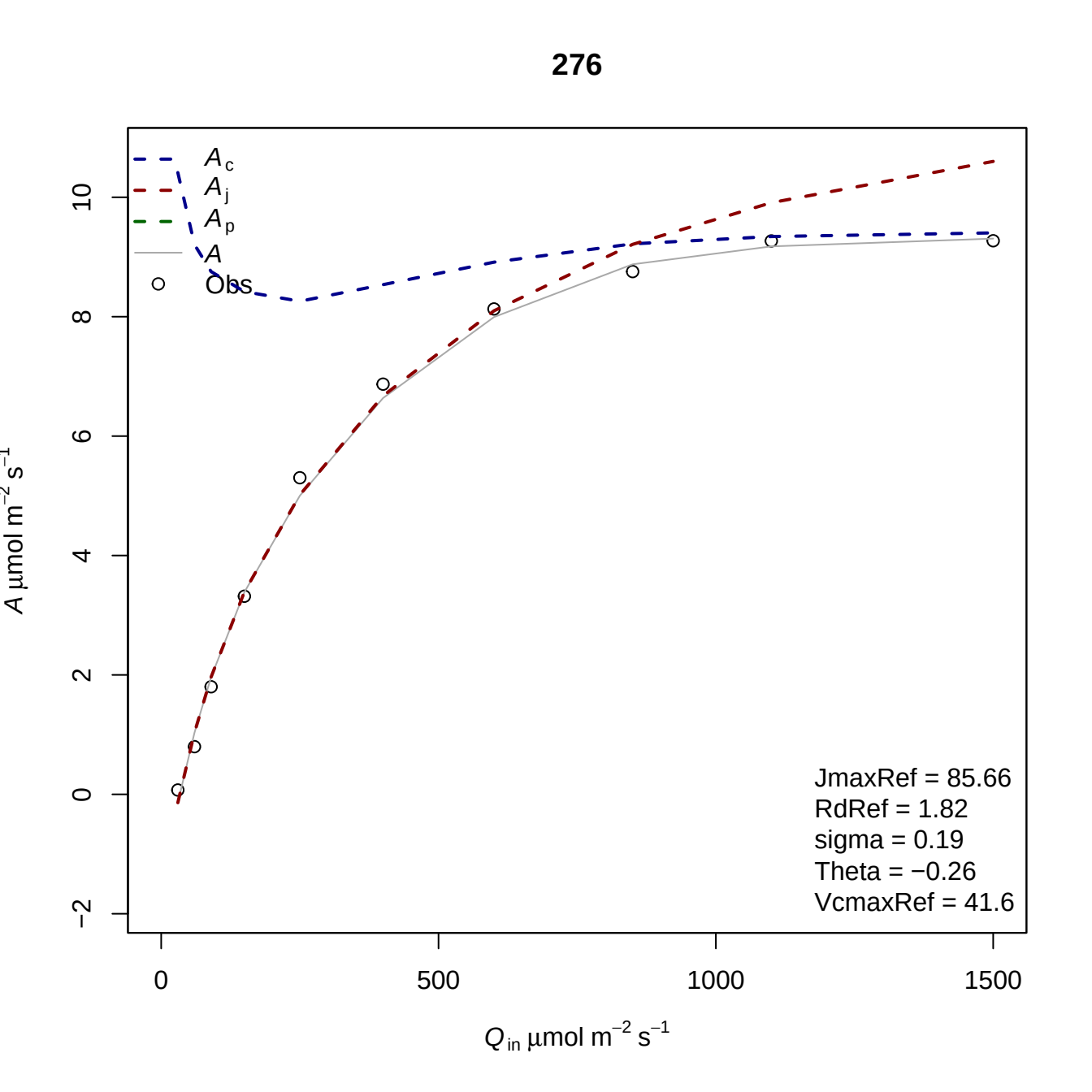

271
A_c
A_j
A_p
A
Obs
JmaxRef = 85.74
RdRef = 1.98
sigma = 0.3
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

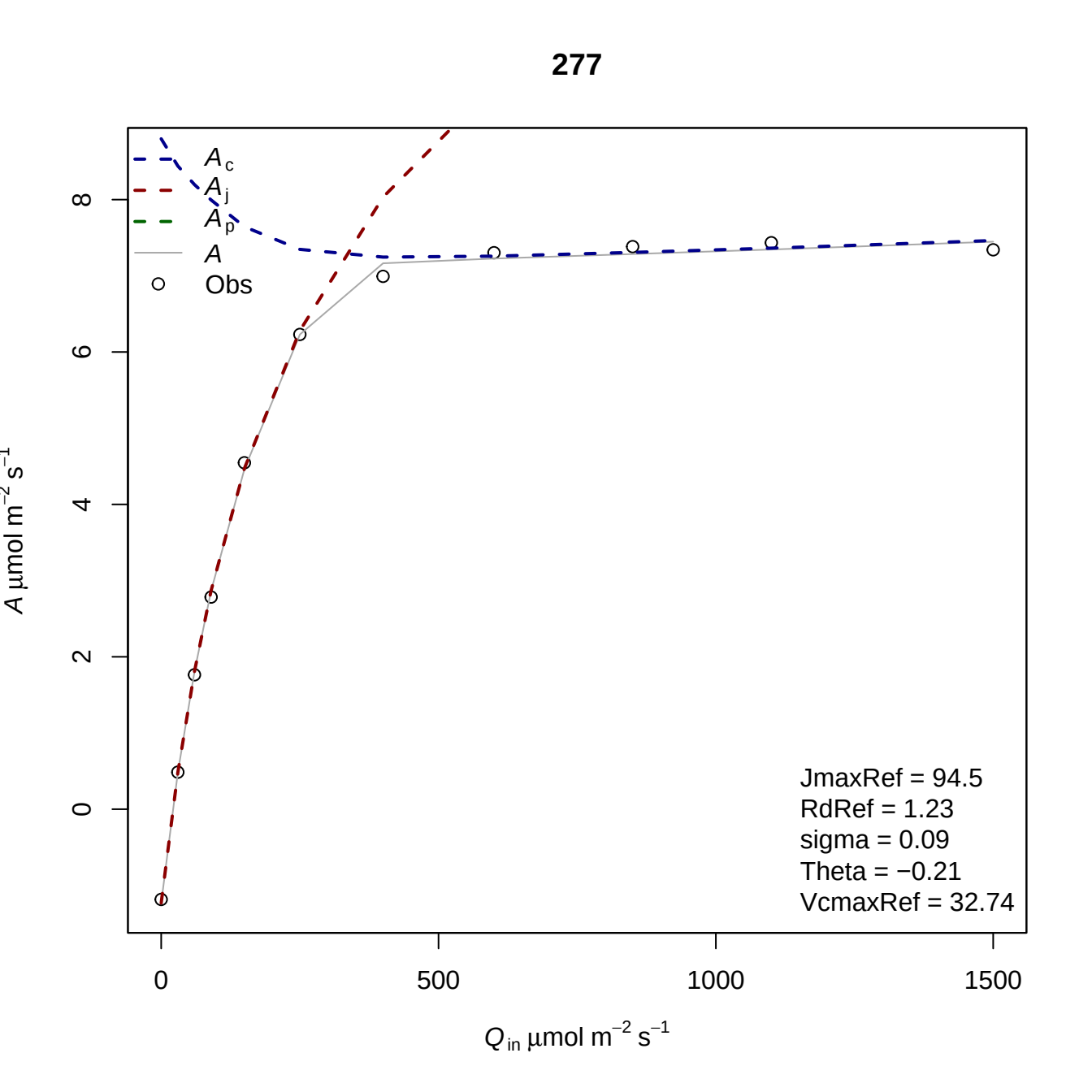

272
A_c
A_j
A_p
A
Obs
JmaxRef = 83.7
RdRef = 1.8
sigma = 0.22
Theta = 0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

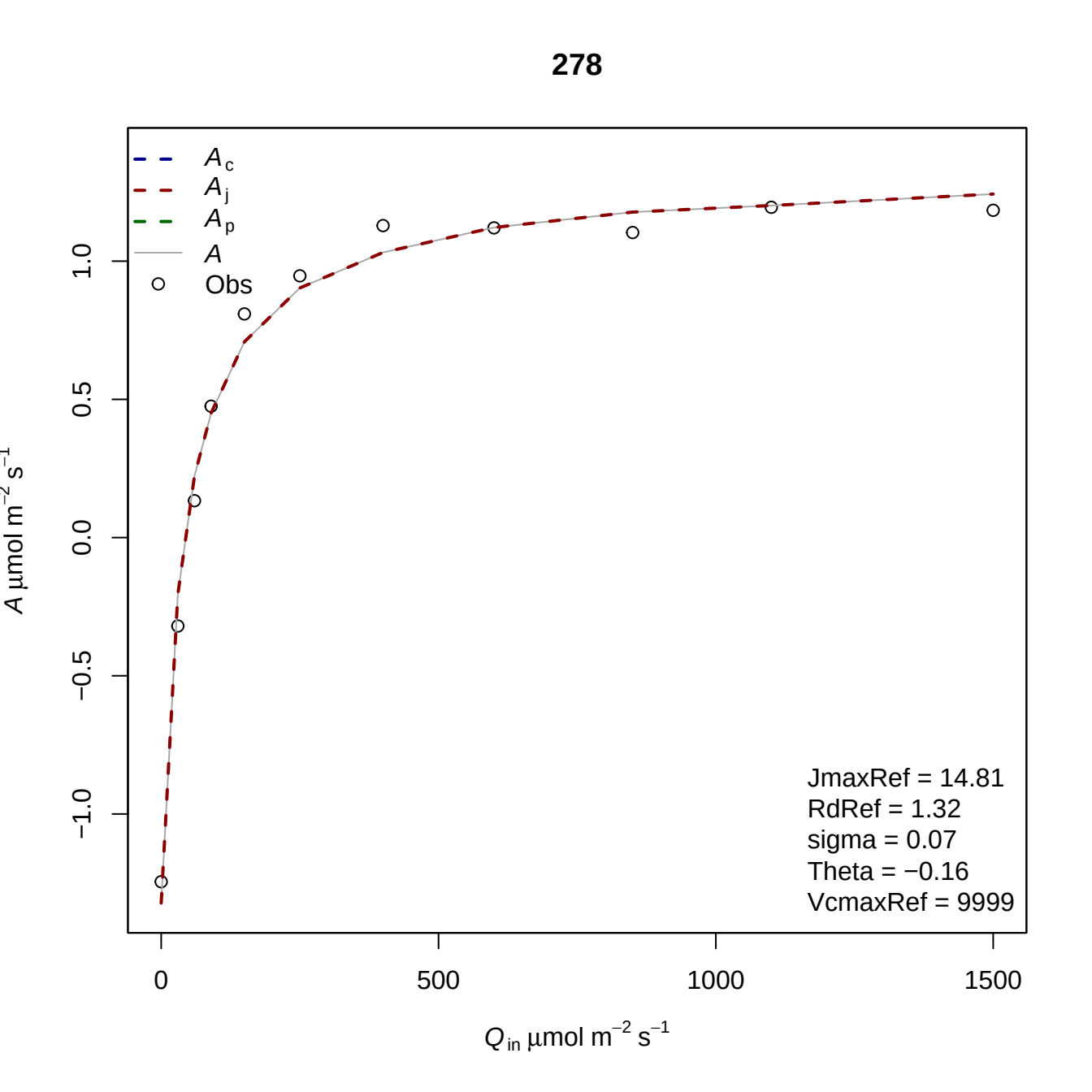

A_c
A_j
A_p
A
Obs
JmaxRef = 23.91
RdRef = 1.59
sigma = 0.24
Theta = −3.42
VcmaxRef = 9999
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

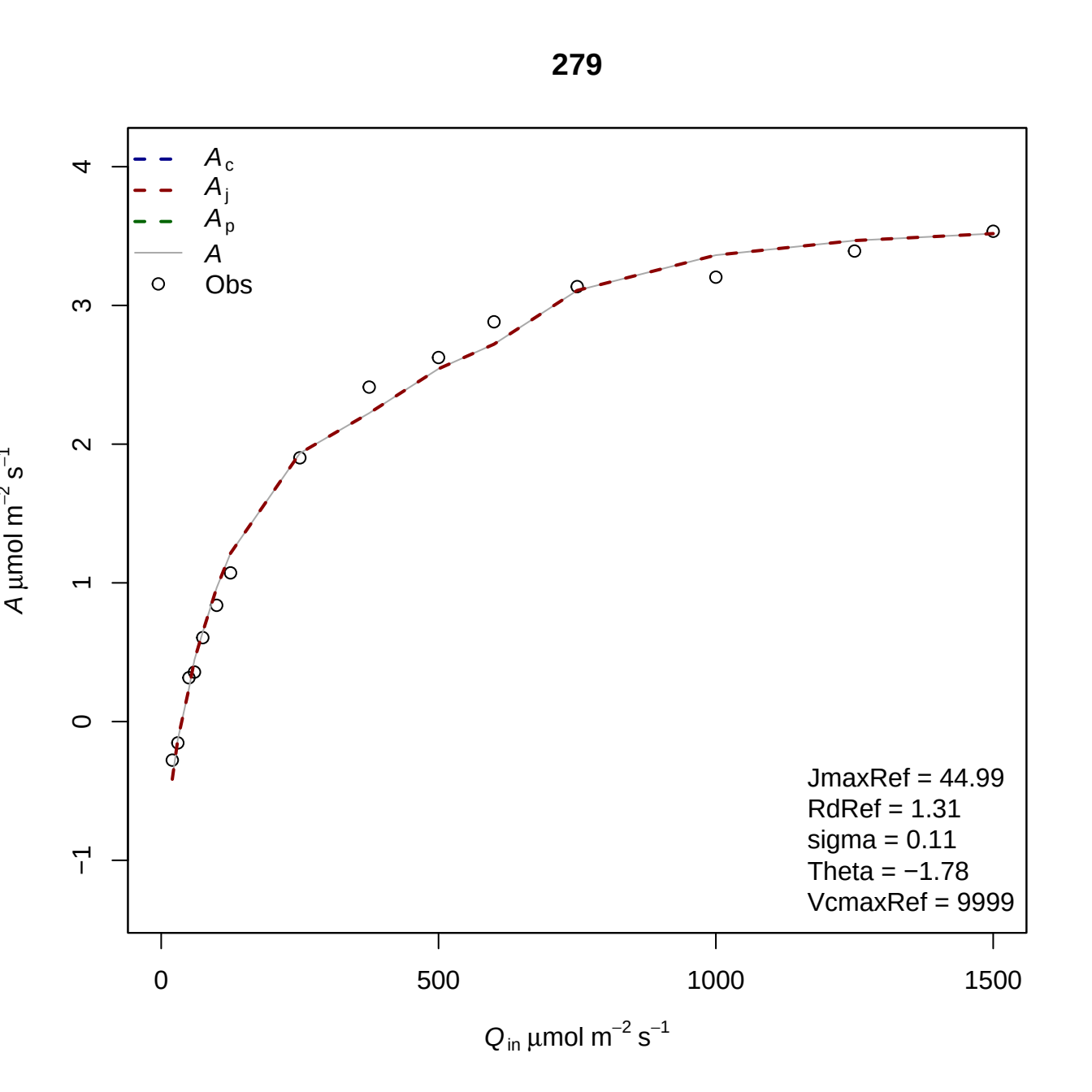

274
A_c
A_j
A_p
A
Obs
JmaxRef = 2242.3
RdRef = 1.62
sigma = 0.27
Theta = −472.97
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

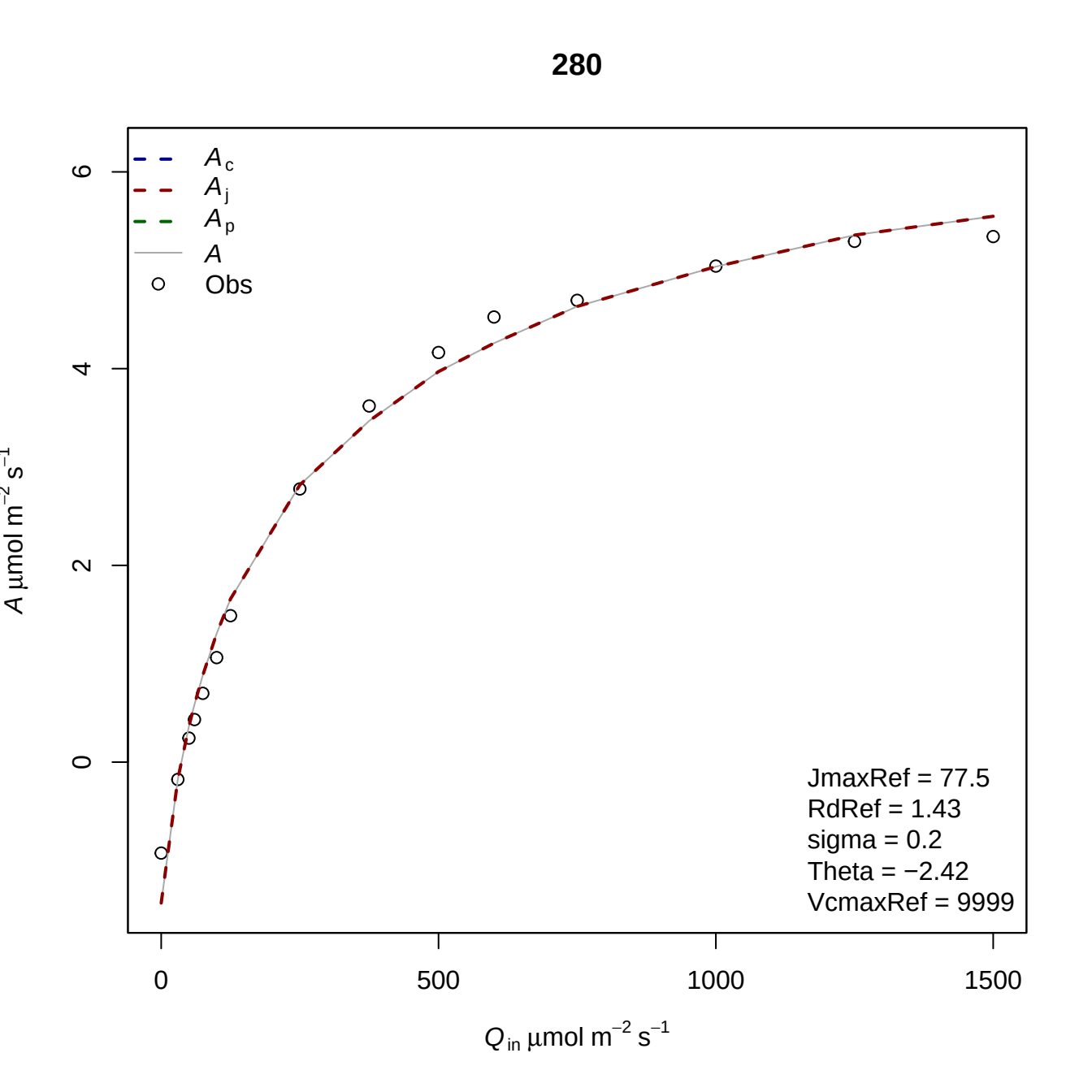

275
A_c
A_j
A_p
A
Obs
JmaxRef = 162.73
RdRef = 2.42
sigma = 0.5
Theta = 0.02
VcmaxRef = 79.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

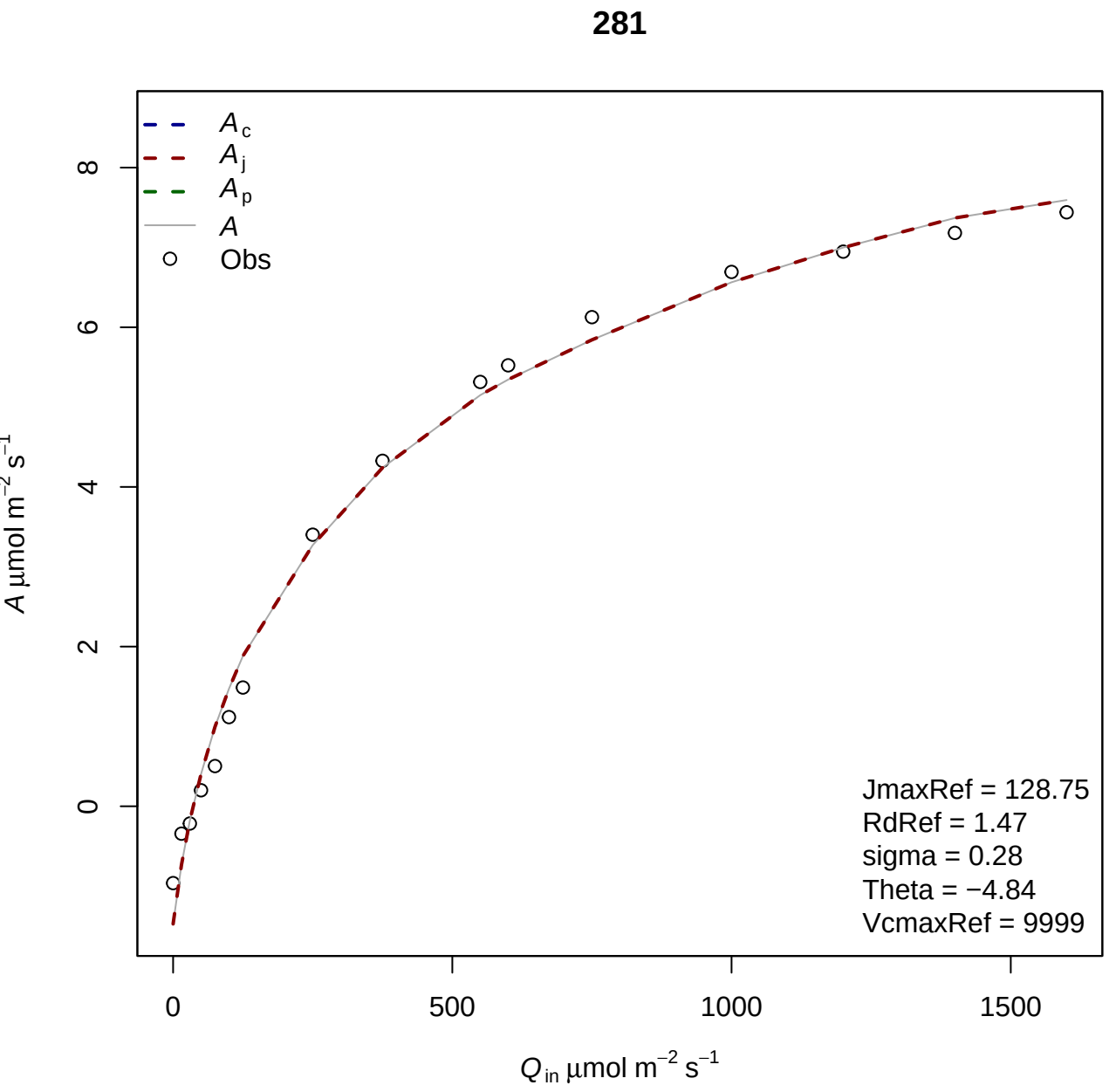

276
A_c
A_j
A_p
A
Obs
JmaxRef = 85.66
RdRef = 1.82
sigma = 0.19
Theta = −0.26
VcmaxRef = 41.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

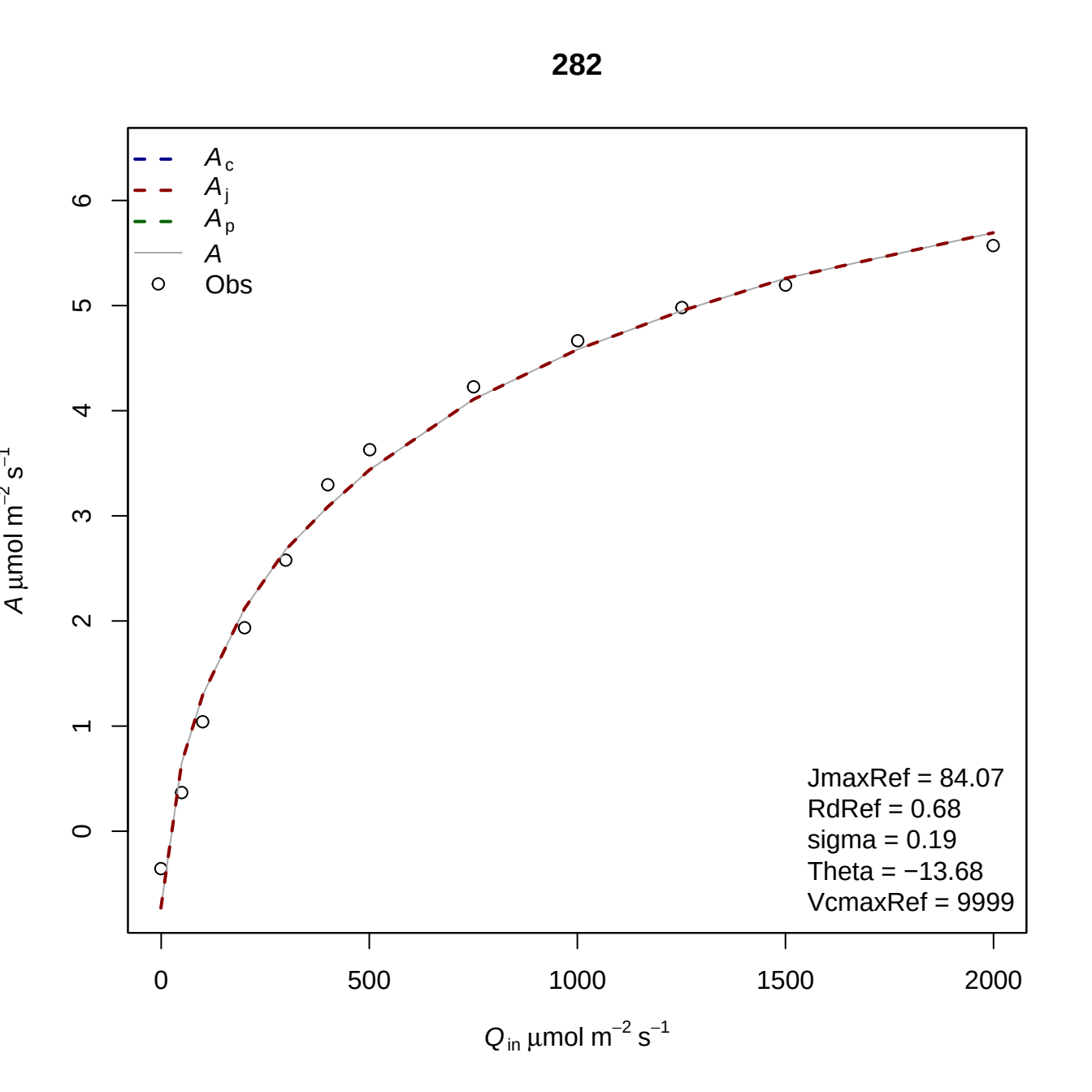

277
A_c
A_j
A_p
A
Obs
JmaxRef = 94.5
RdRef = 1.23
sigma = 0.09
Theta = −0.21
VcmaxRef = 32.74
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

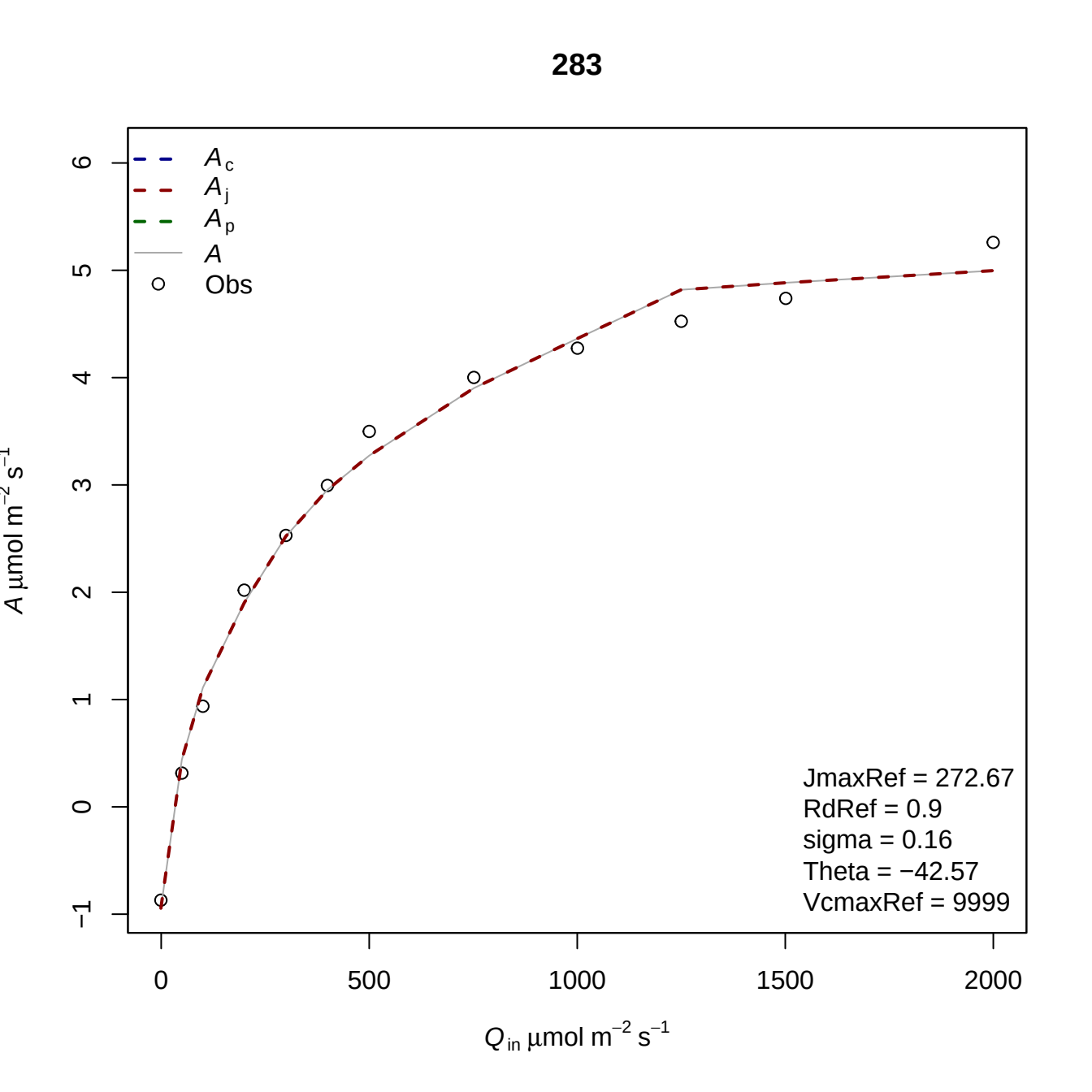

278
A_c
A_j
A_p
A
Obs
JmaxRef = 14.81
RdRef = 1.32
sigma = 0.07
Theta = −0.16
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

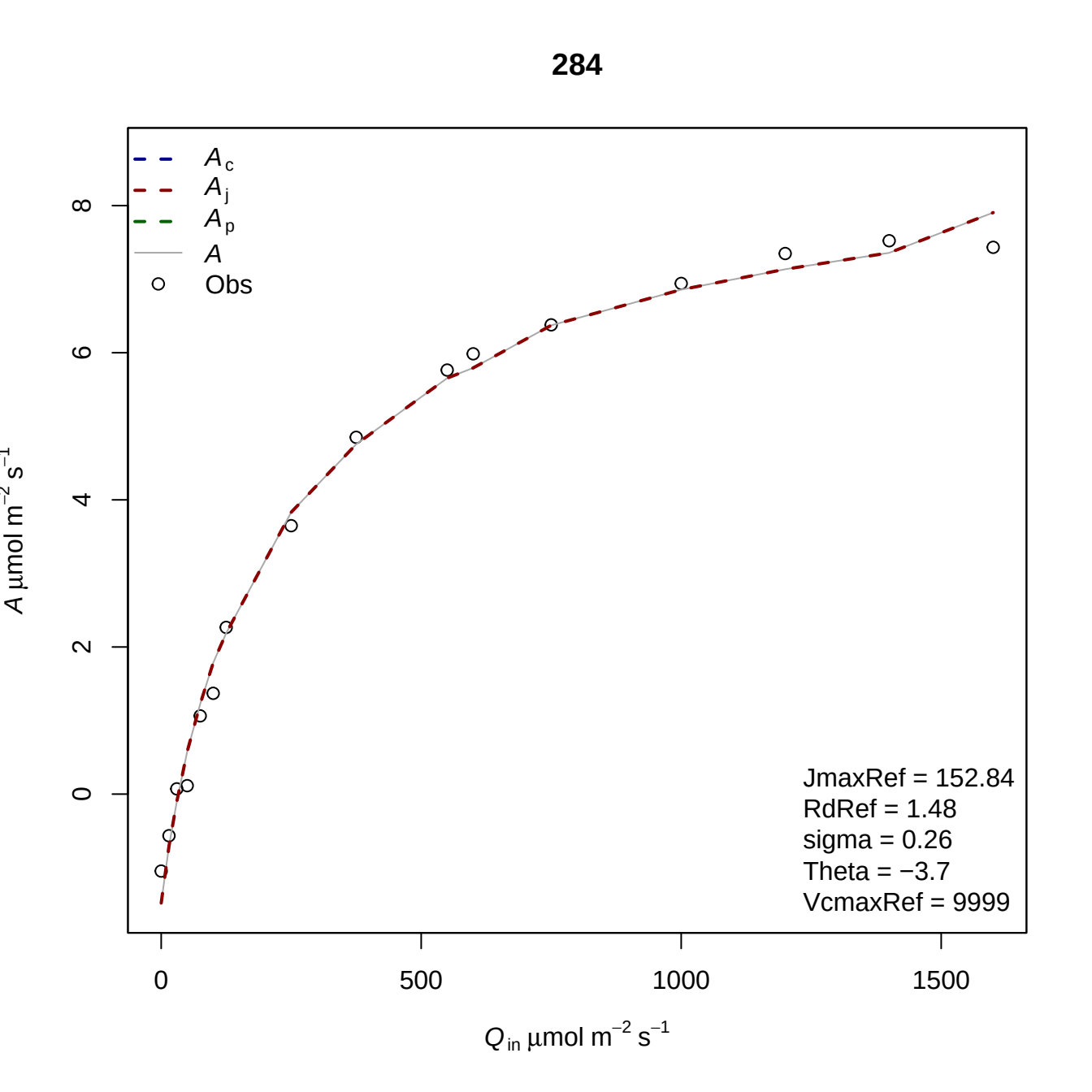

279
A_c
A_j
A_p
A
Obs
JmaxRef = 44.99
RdRef = 1.31
sigma = 0.11
Theta = −1.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

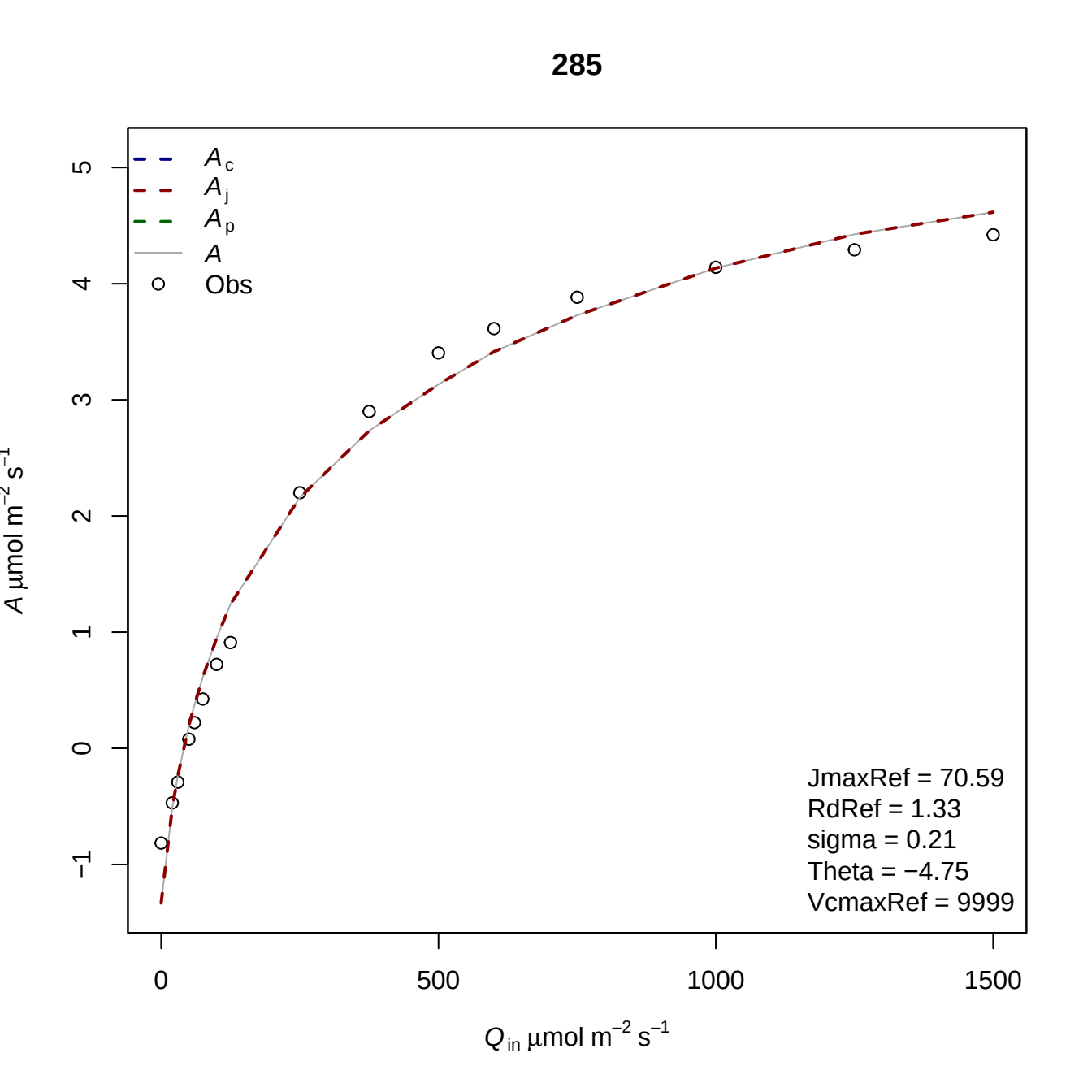

280
A_c
A_j
A_p
A
Obs
JmaxRef = 77.5
RdRef = 1.43
sigma = 0.2
Theta = −2.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

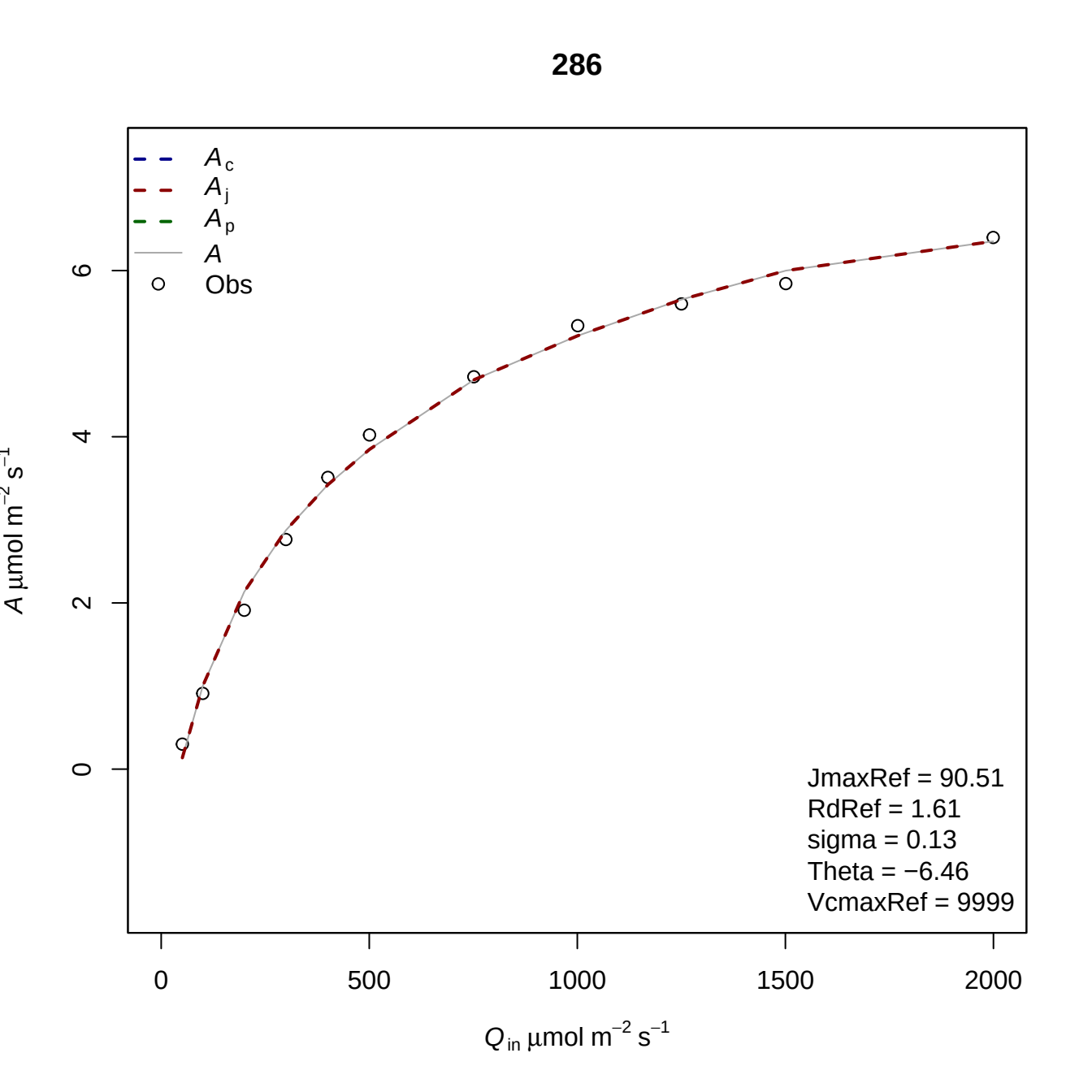

281
A_c
A_j
A_p
A
Obs
JmaxRef = 128.75
RdRef = 1.47
sigma = 0.28
Theta = −4.84
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

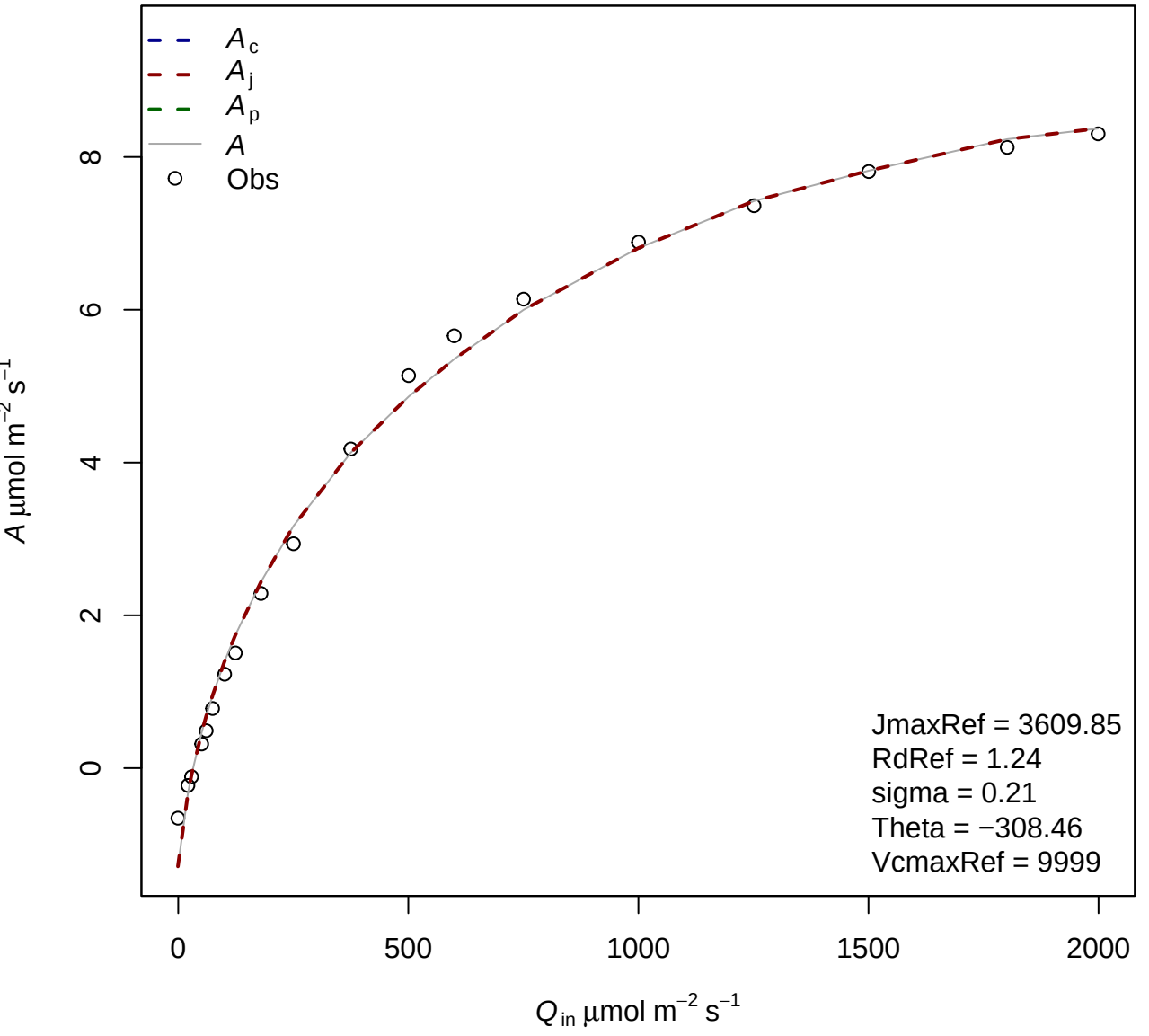

282
A_c
A_j
A_p
A
Obs
JmaxRef = 84.07
RdRef = 0.68
sigma = 0.19
Theta = −13.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

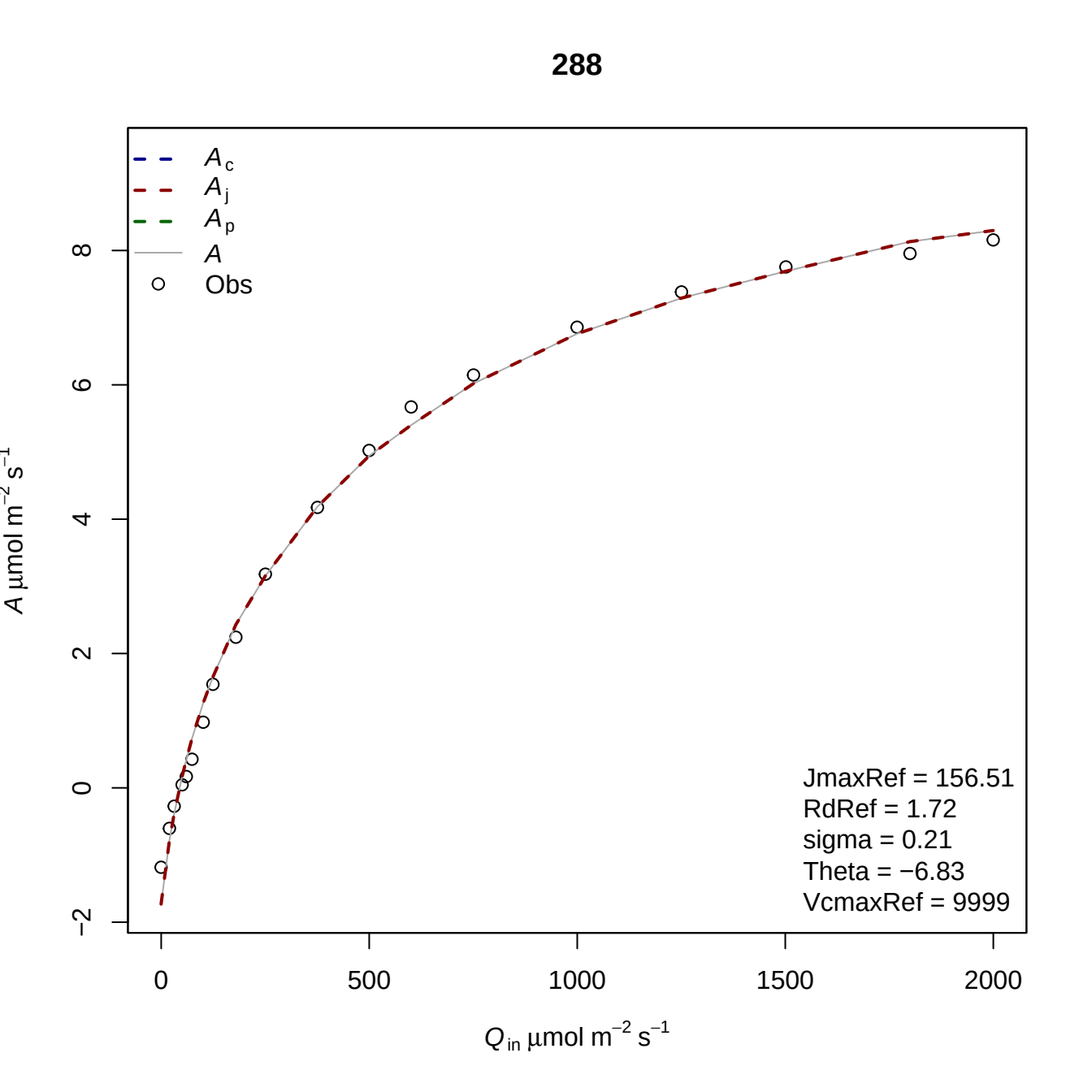

283
A_c
A_j
A_p
A
Obs
JmaxRef = 272.67
RdRef = 0.9
sigma = 0.16
Theta = −42.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

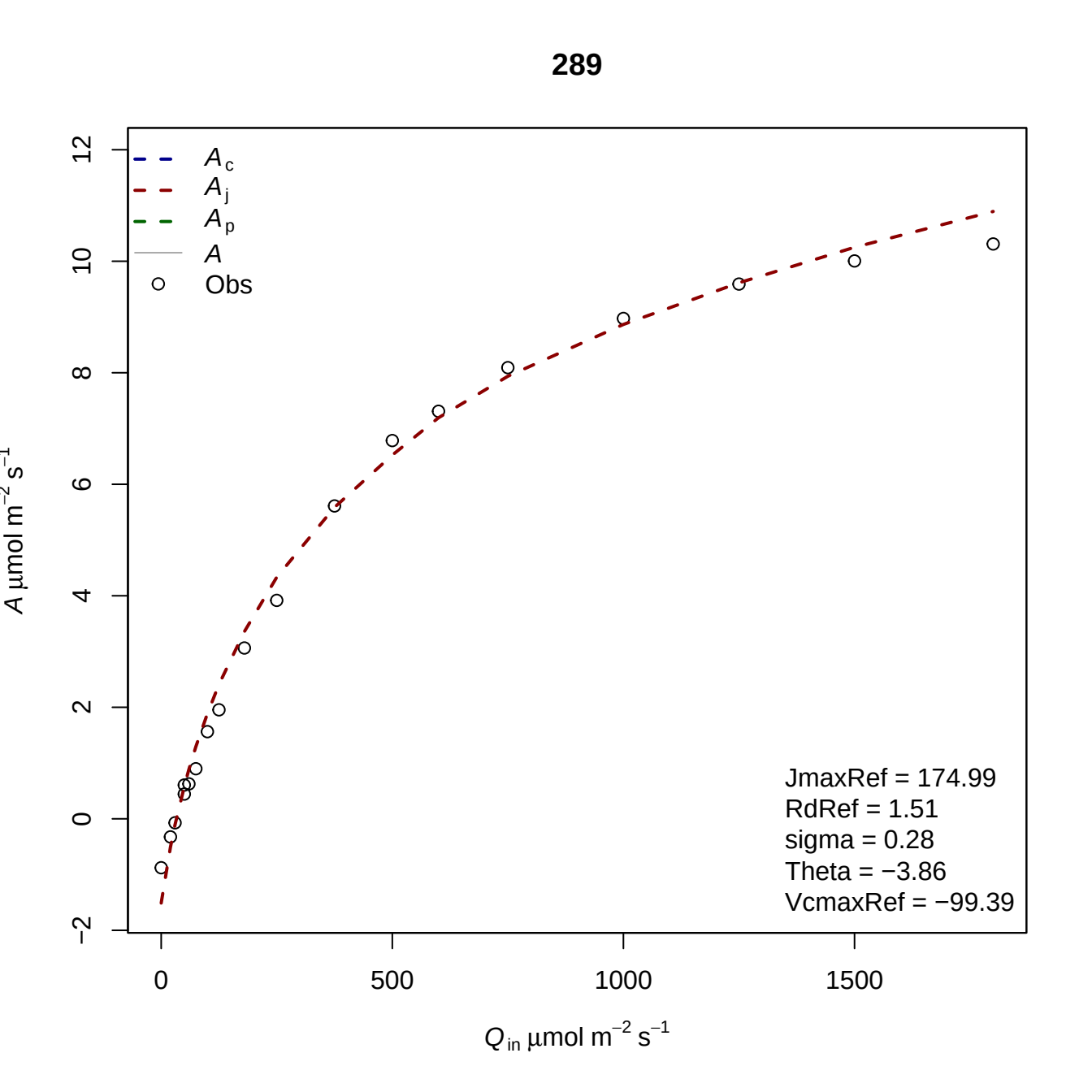

284
A_c
A_j
A_p
A
Obs
JmaxRef = 152.84
RdRef = 1.48
sigma = 0.26
Theta = −3.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

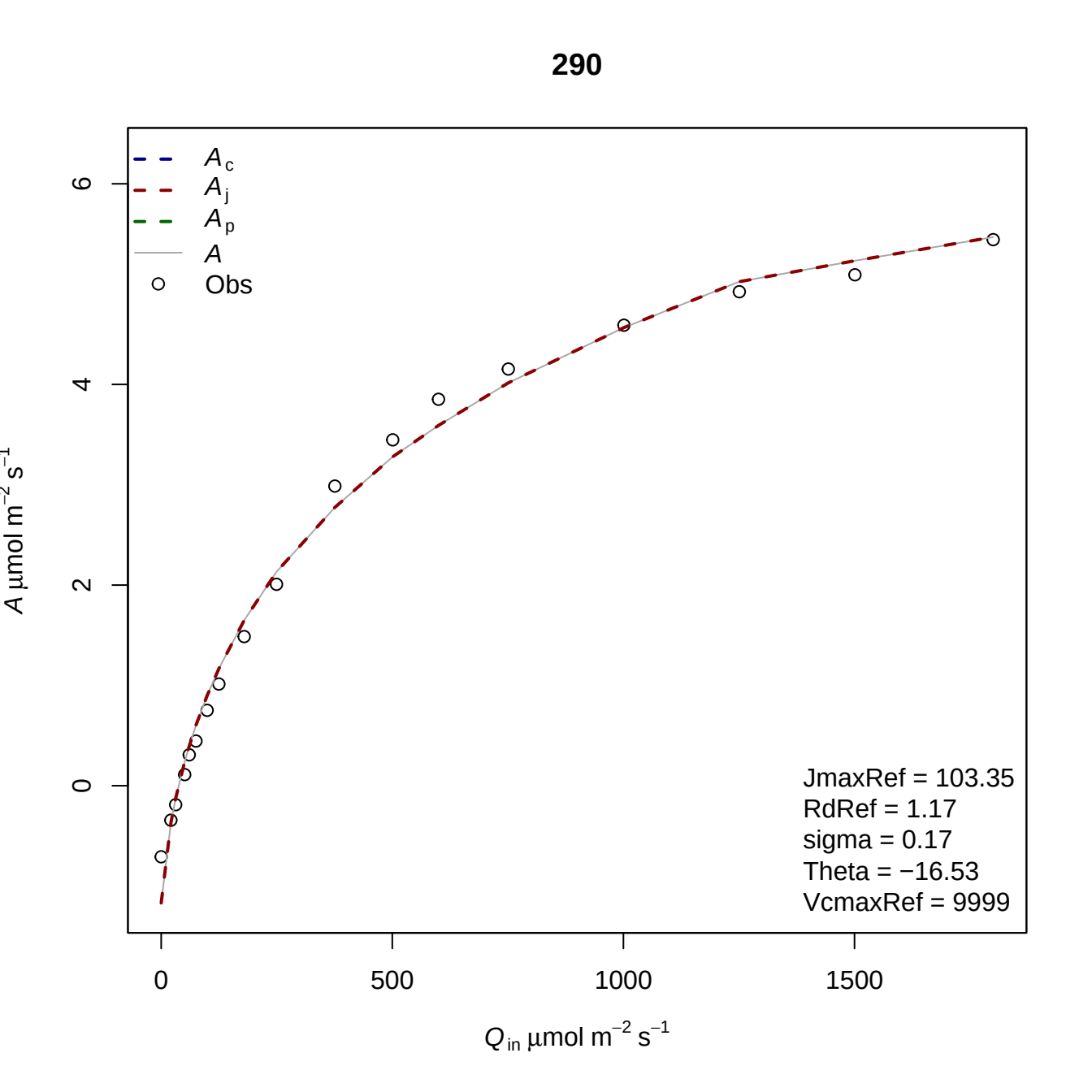

285
A_c
A_j
A_p
A
Obs
JmaxRef = 70.59
RdRef = 1.33
sigma = 0.21
Theta = −4.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

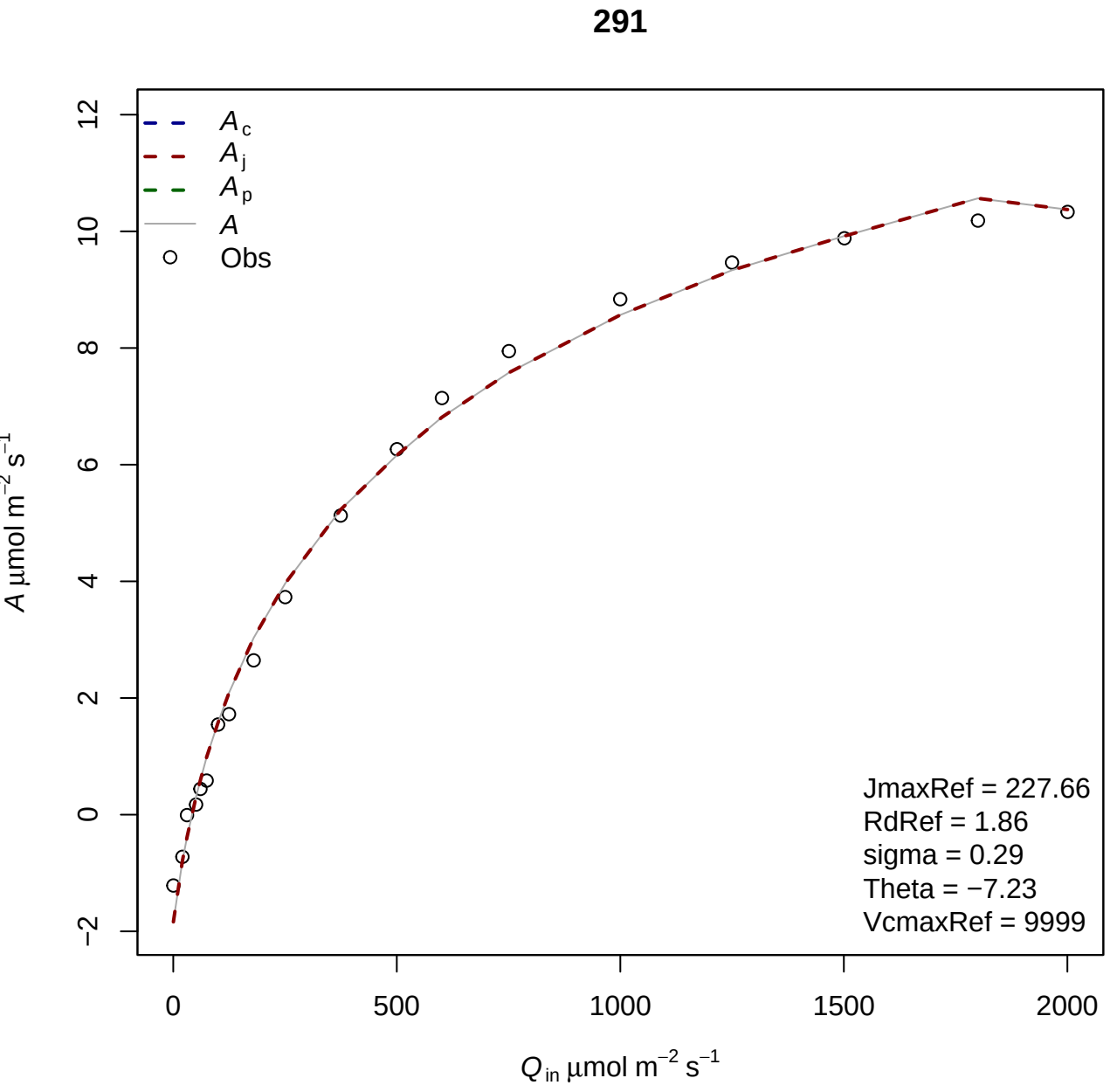

286
A_c
A_j
A_p
A
Obs
JmaxRef = 90.51
RdRef = 1.61
sigma = 0.13
Theta = −6.46
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

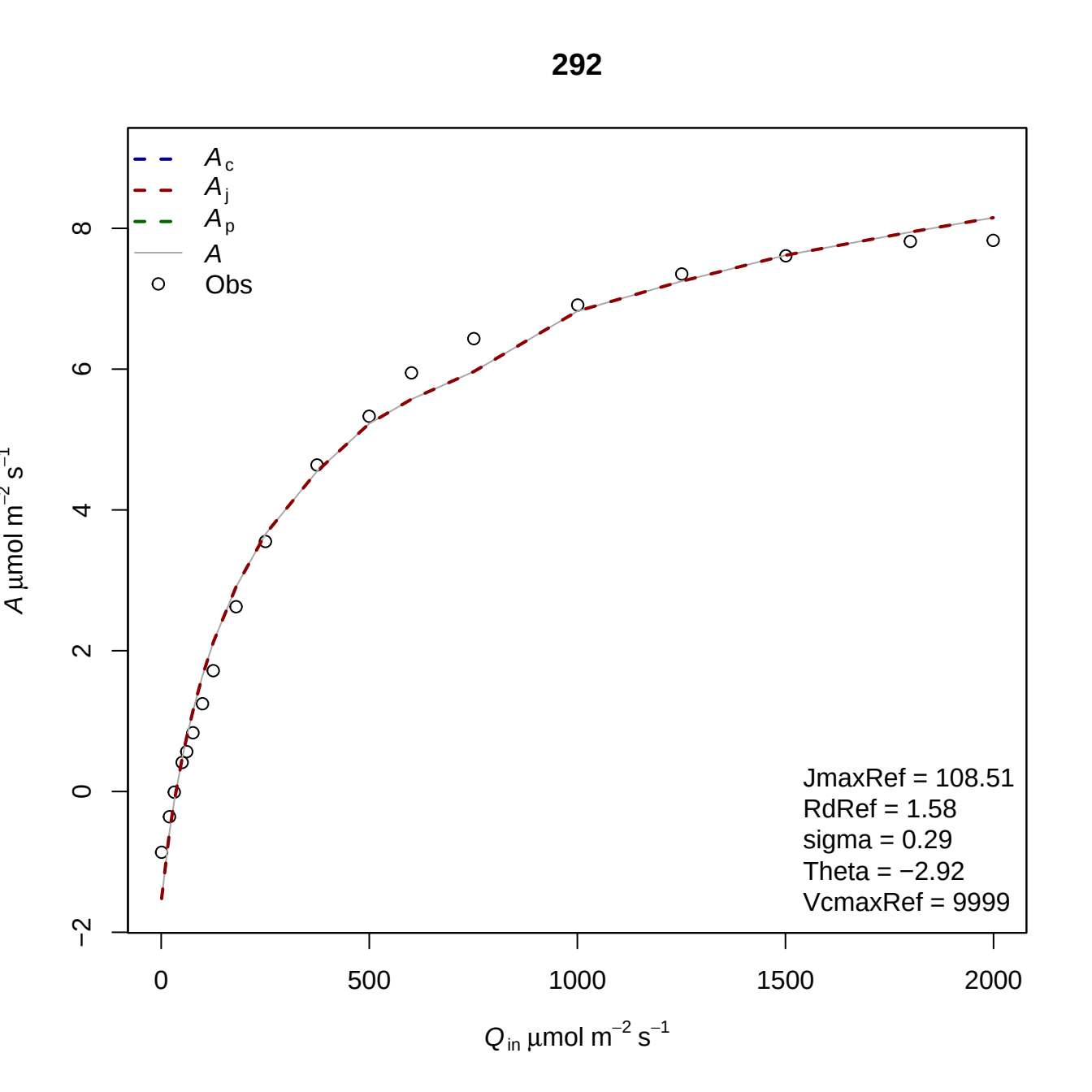

A_c
A_j
A_p
A
Obs
JmaxRef = 3609.85
RdRef = 1.24
sigma = 0.21
Theta = −308.46
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

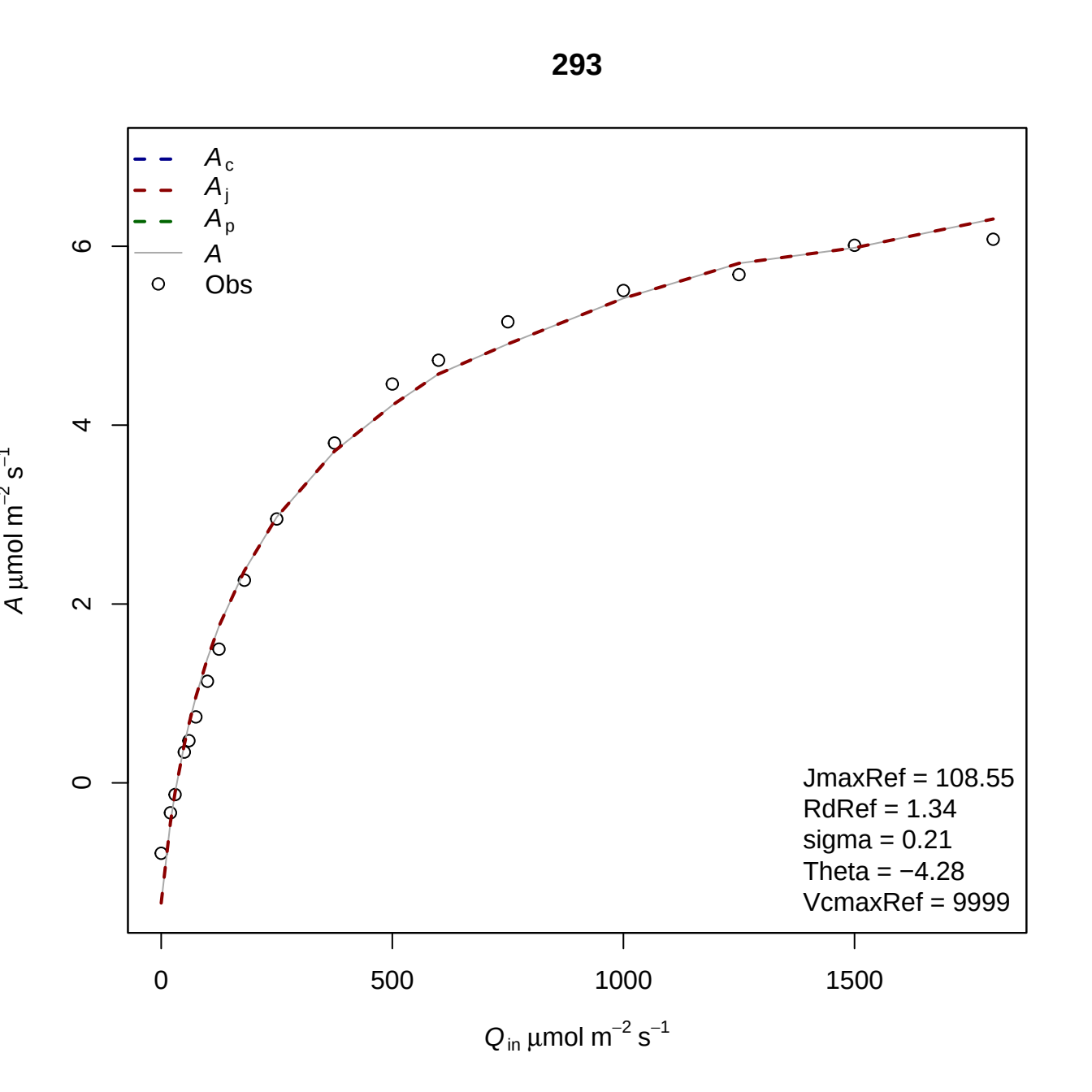

288
A_c
A_j
A_p
A
Obs
JmaxRef = 156.51
RdRef = 1.72
sigma = 0.21
Theta = −6.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

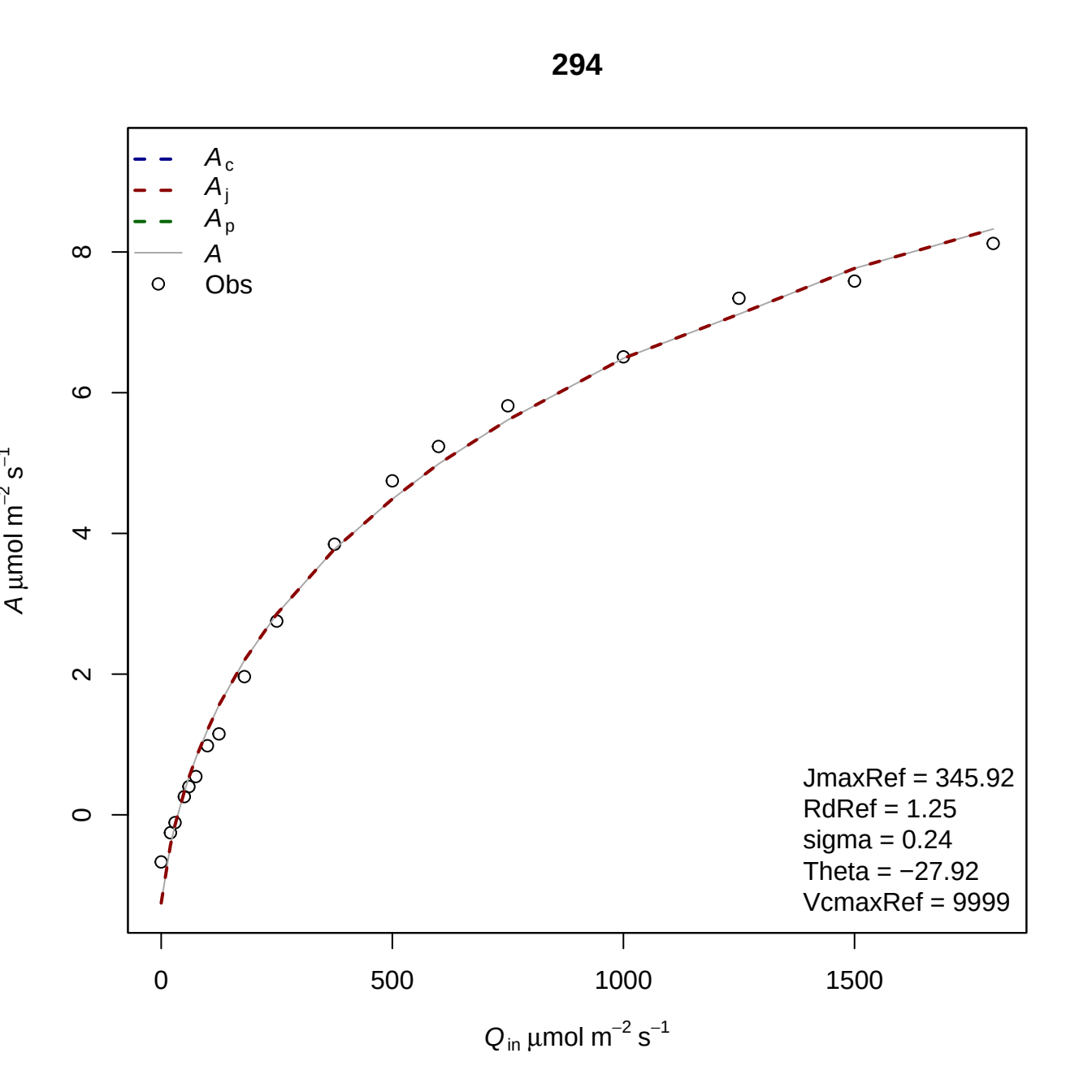

289
A_c
A_j
A_p
A
Obs
JmaxRef = 174.99
RdRef = 1.51
sigma = 0.28
Theta = −3.86
VcmaxRef = −99.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

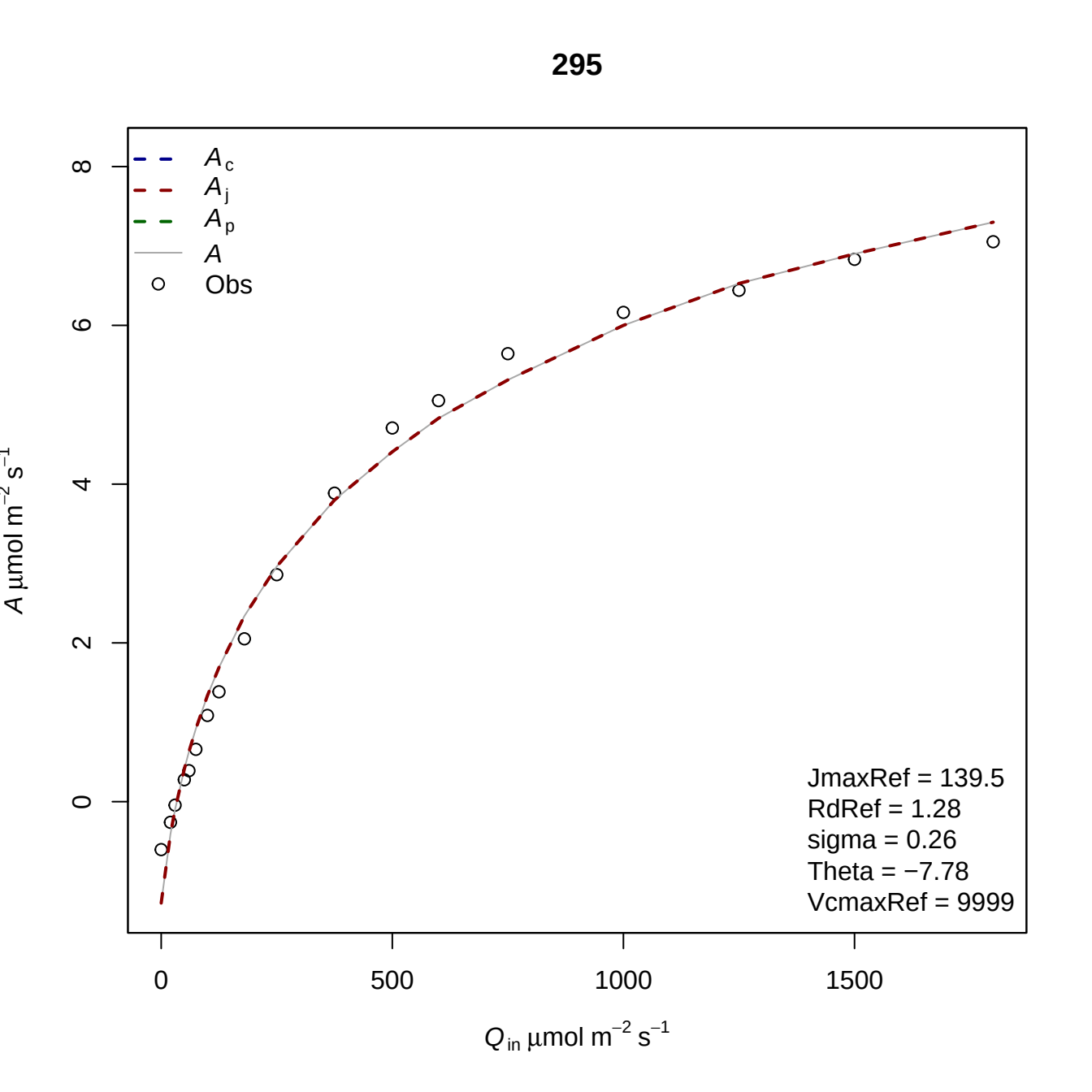

290
A_c
A_j
A_p
A
Obs
JmaxRef = 103.35
RdRef = 1.17
sigma = 0.17
Theta = −16.53
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

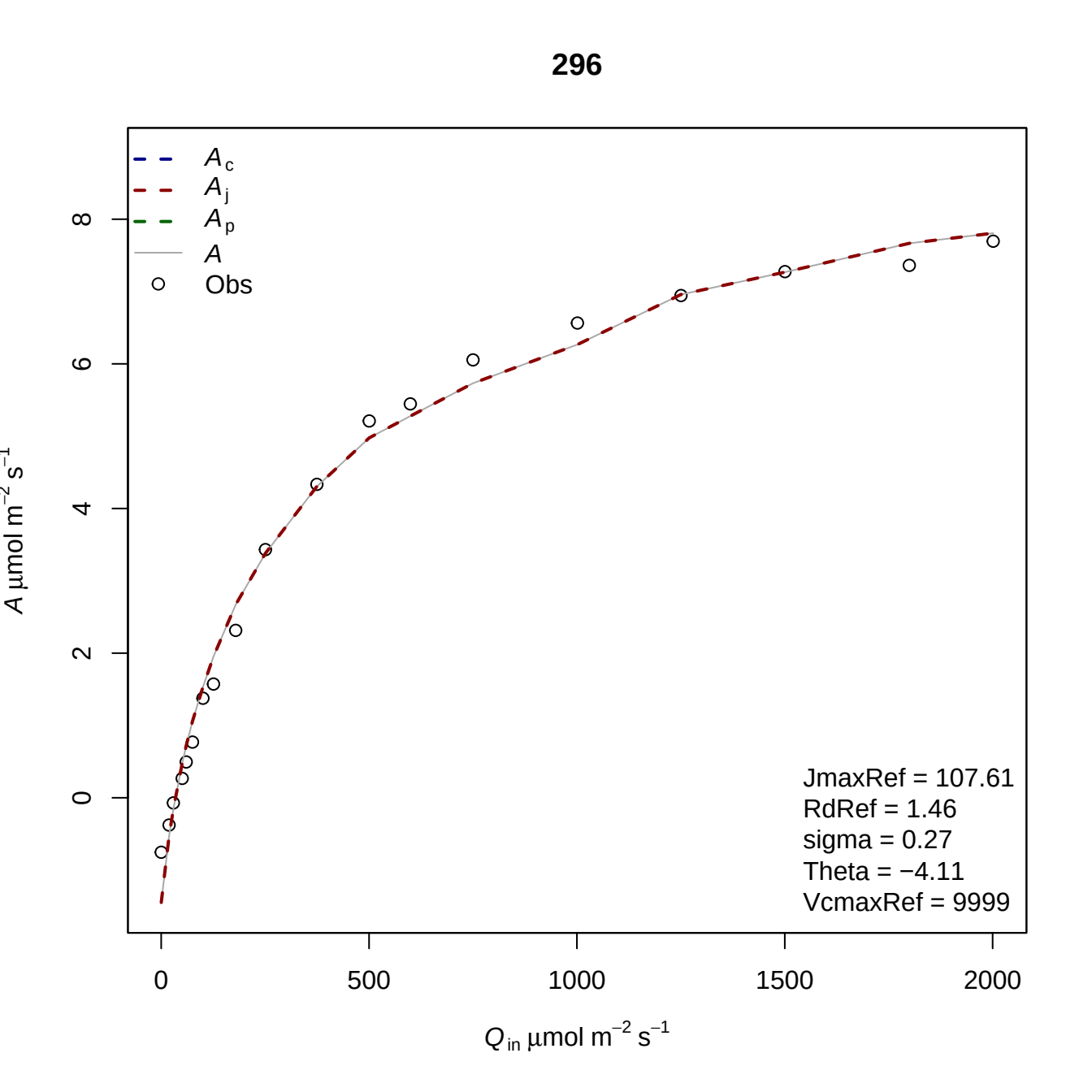

291
A_c
A_j
A_p
A
Obs
JmaxRef = 227.66
RdRef = 1.86
sigma = 0.29
Theta = −7.23
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

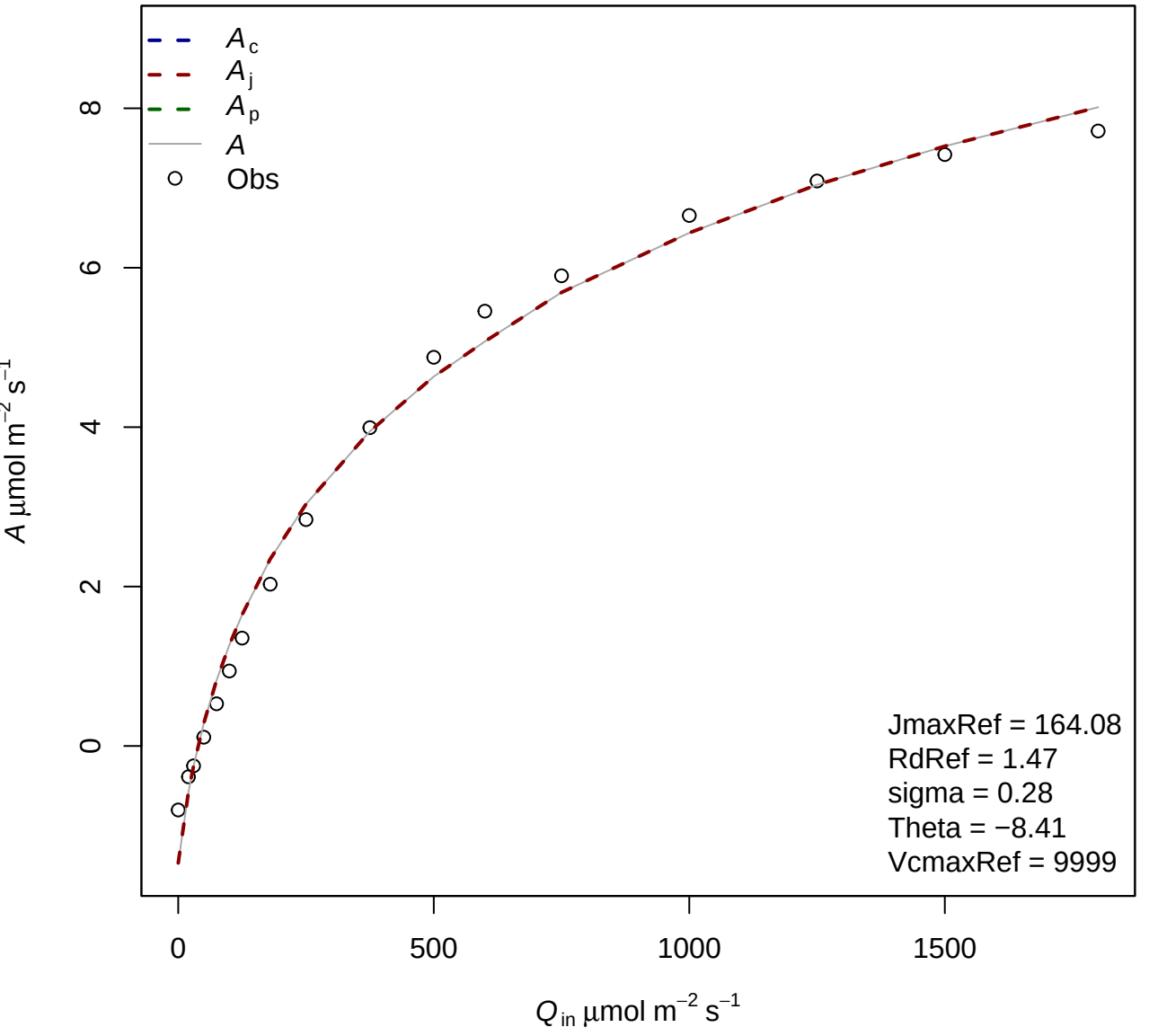

292
A_c
A_j
A_p
A
Obs
JmaxRef = 108.51
RdRef = 1.58
sigma = 0.29
Theta = −2.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

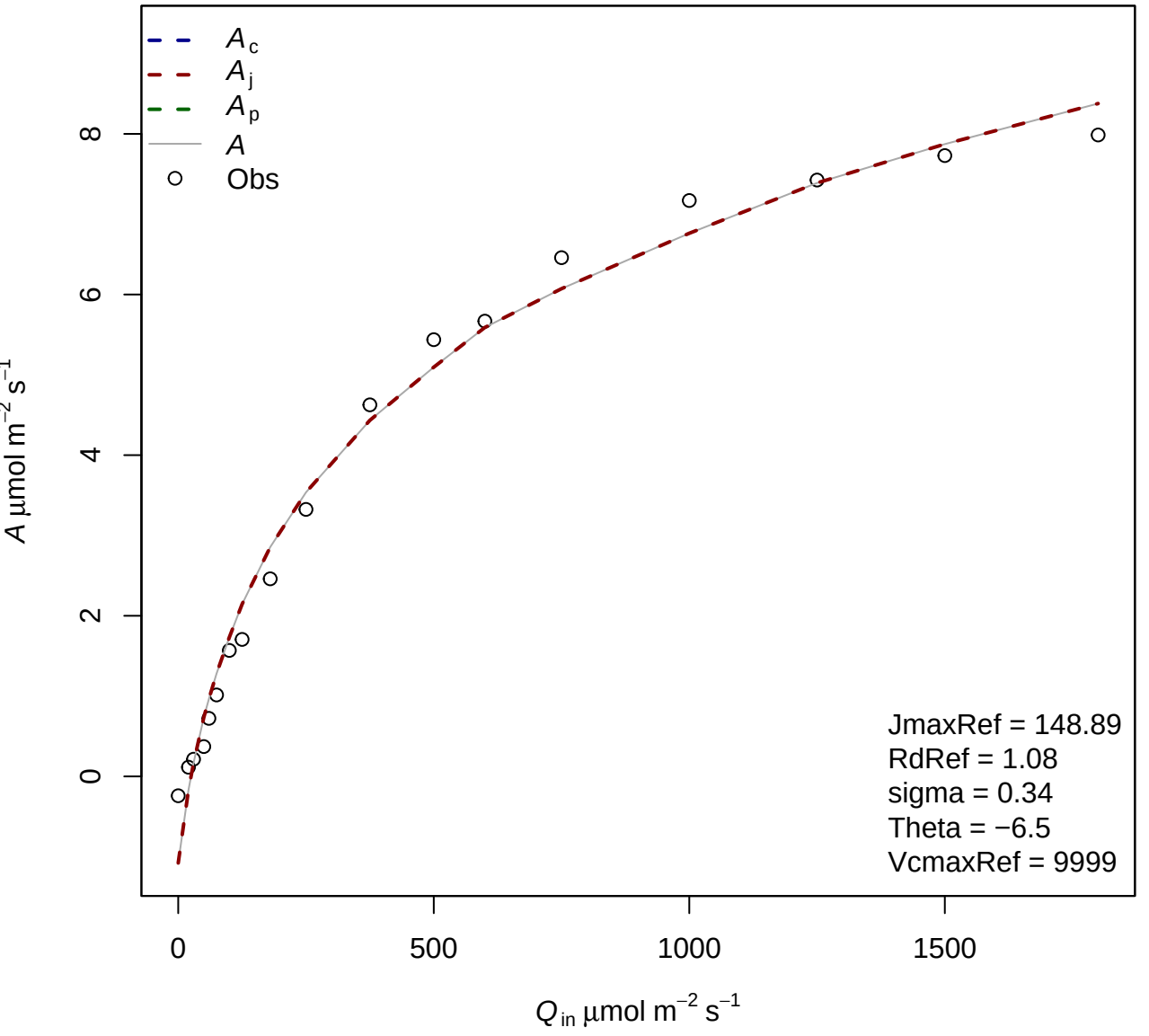

293
A_c
A_j
A_p
A
Obs
JmaxRef = 108.55
RdRef = 1.34
sigma = 0.21
Theta = −4.28
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

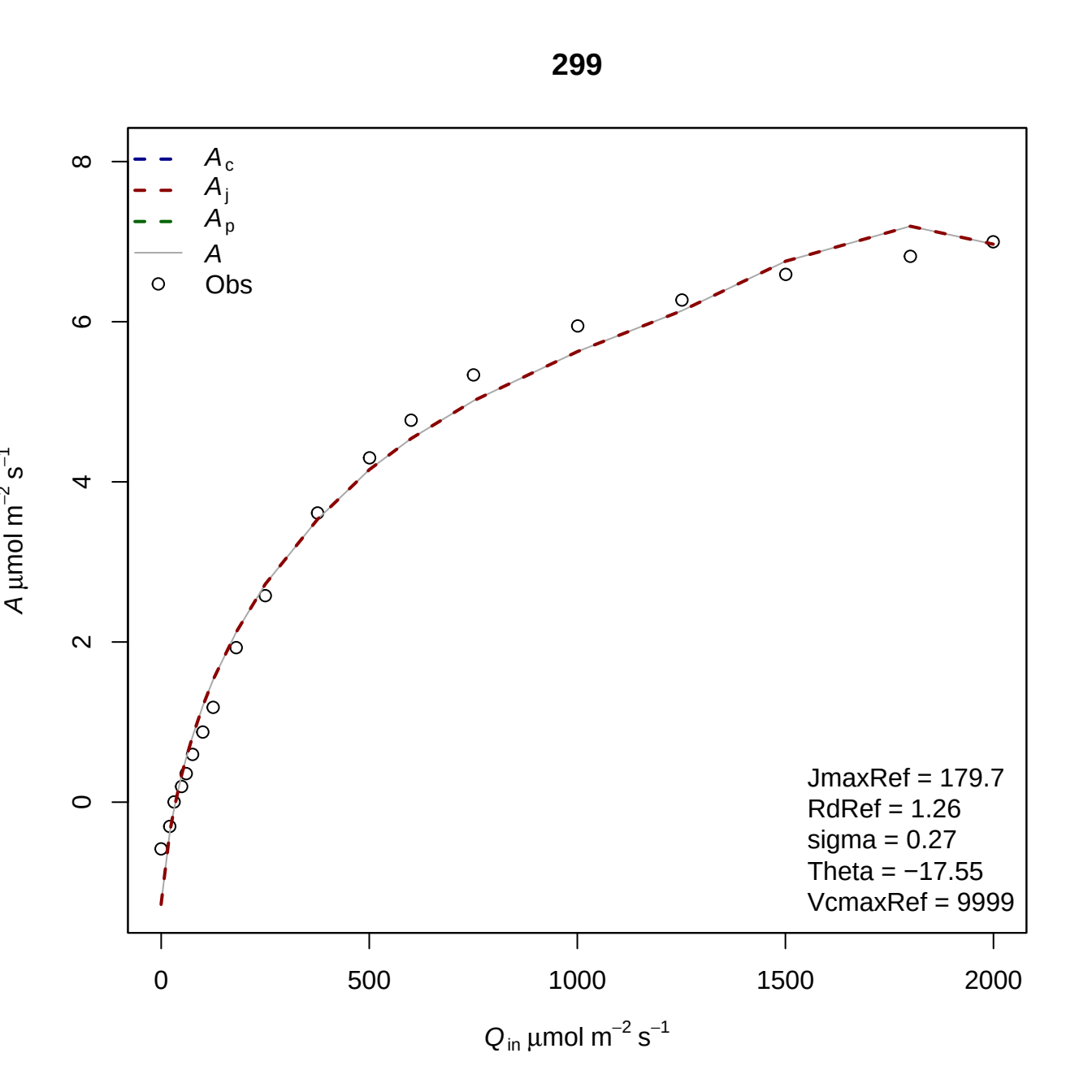

294
A_c
A_j
A_p
A
Obs
JmaxRef = 345.92
RdRef = 1.25
sigma = 0.24
Theta = −27.92
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

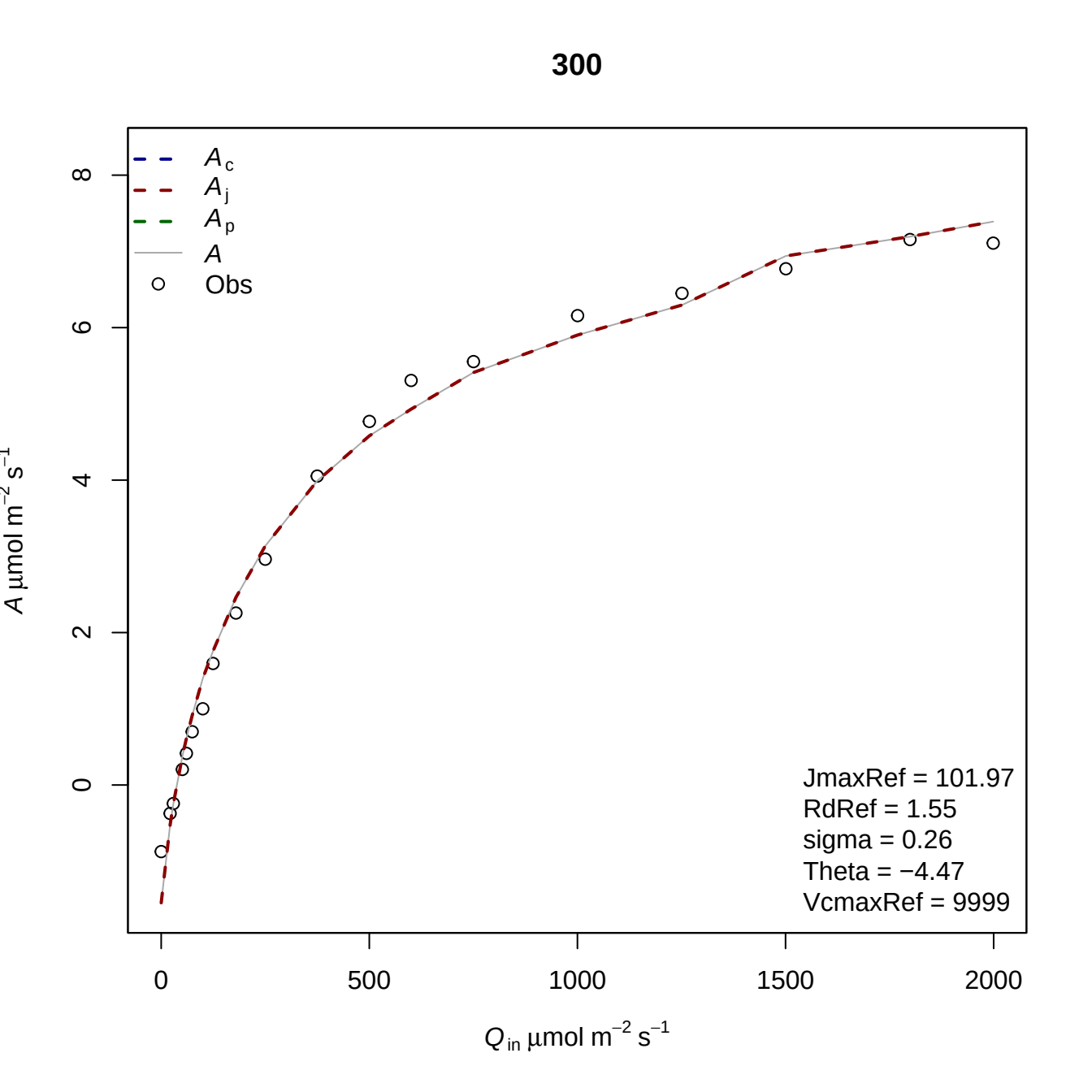

295
A_c
A_j
A_p
A
Obs
JmaxRef = 139.5
RdRef = 1.28
sigma = 0.26
Theta = −7.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

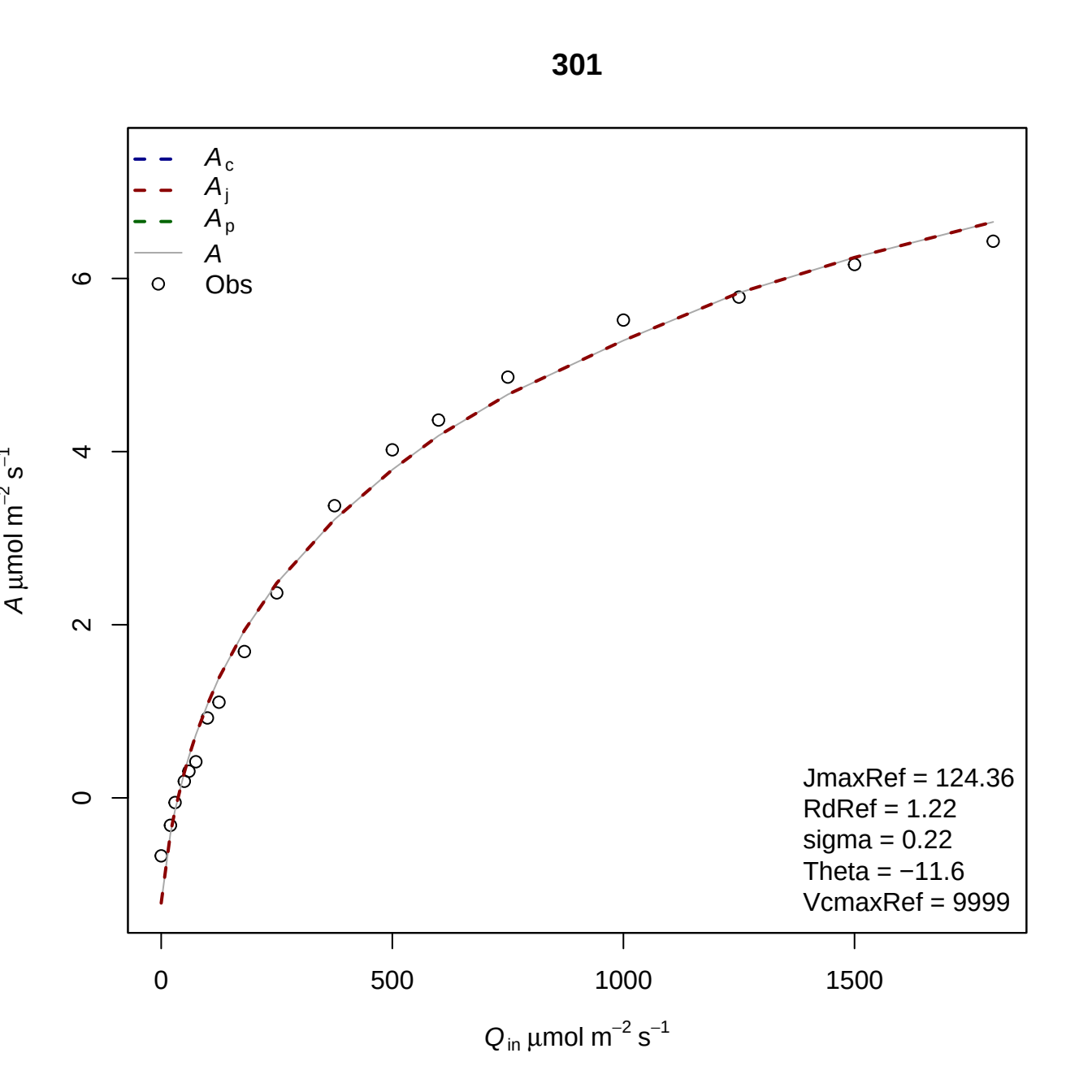

296
A_c
A_j
A_p
A
Obs
JmaxRef = 107.61
RdRef = 1.46
sigma = 0.27
Theta = −4.11
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

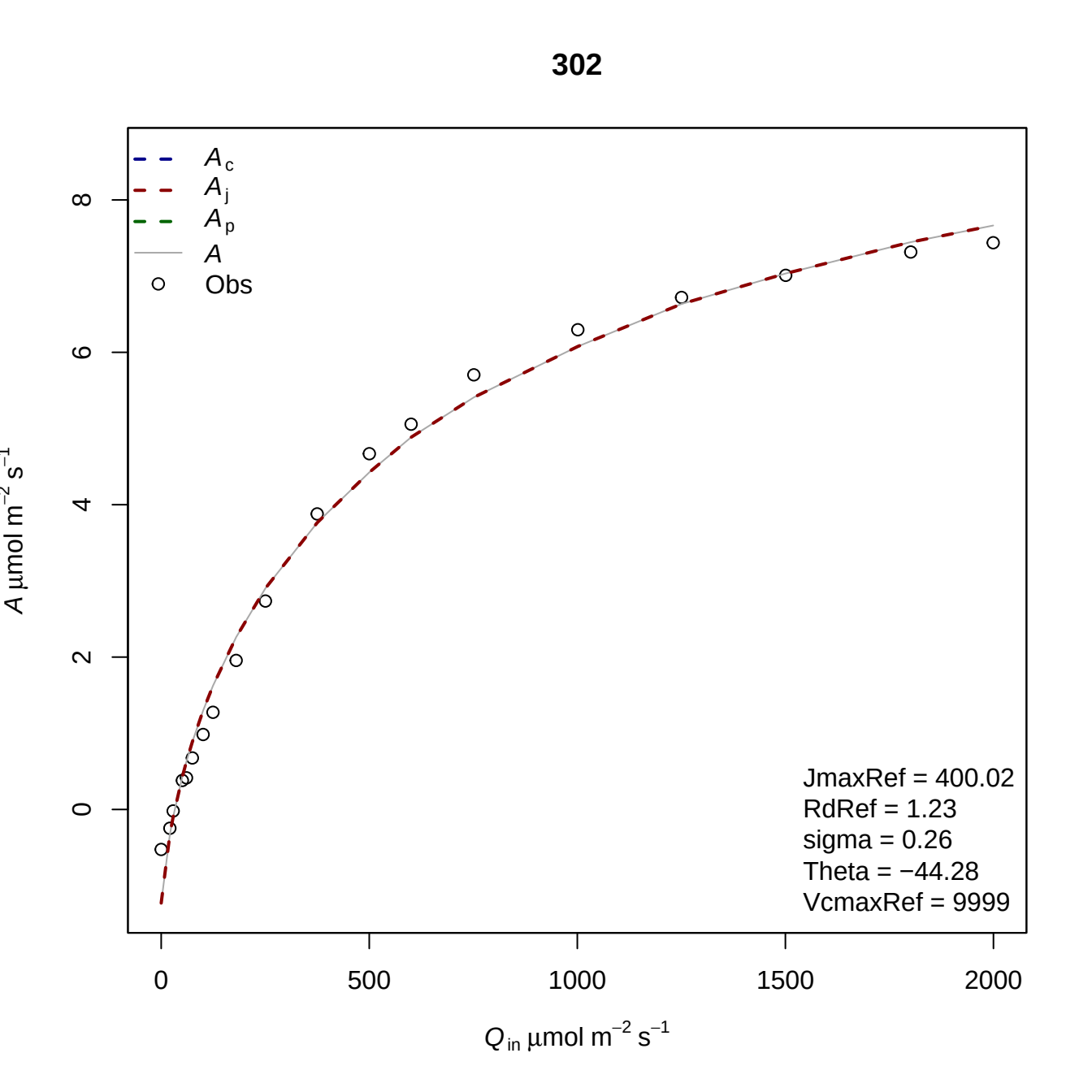

A_c
A_j
A_p
A
Obs
JmaxRef = 164.08
RdRef = 1.47
sigma = 0.28
Theta = −8.41
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

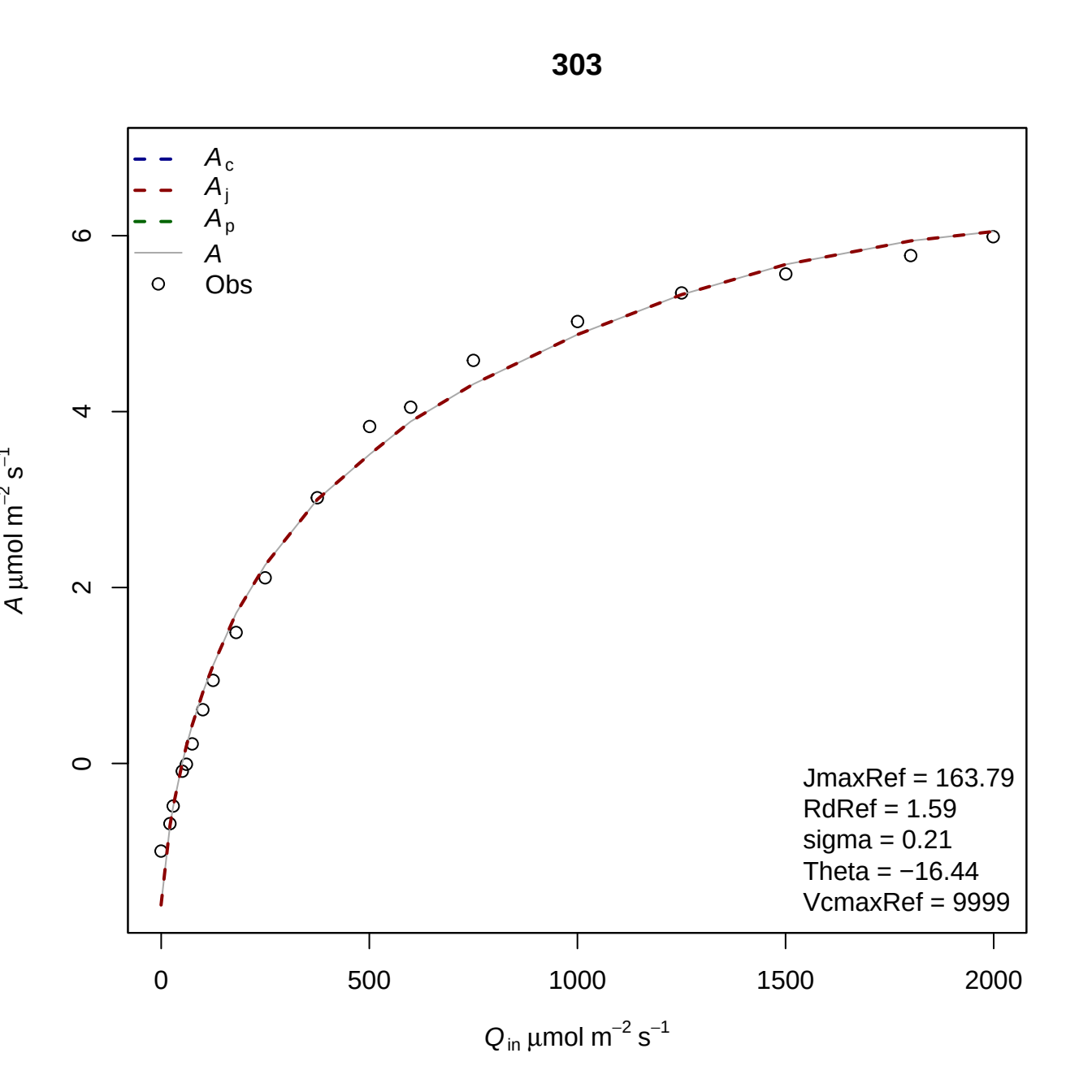

A_c
A_j
A_p
A
Obs
JmaxRef = 148.89
RdRef = 1.08
sigma = 0.34
Theta = −6.5
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

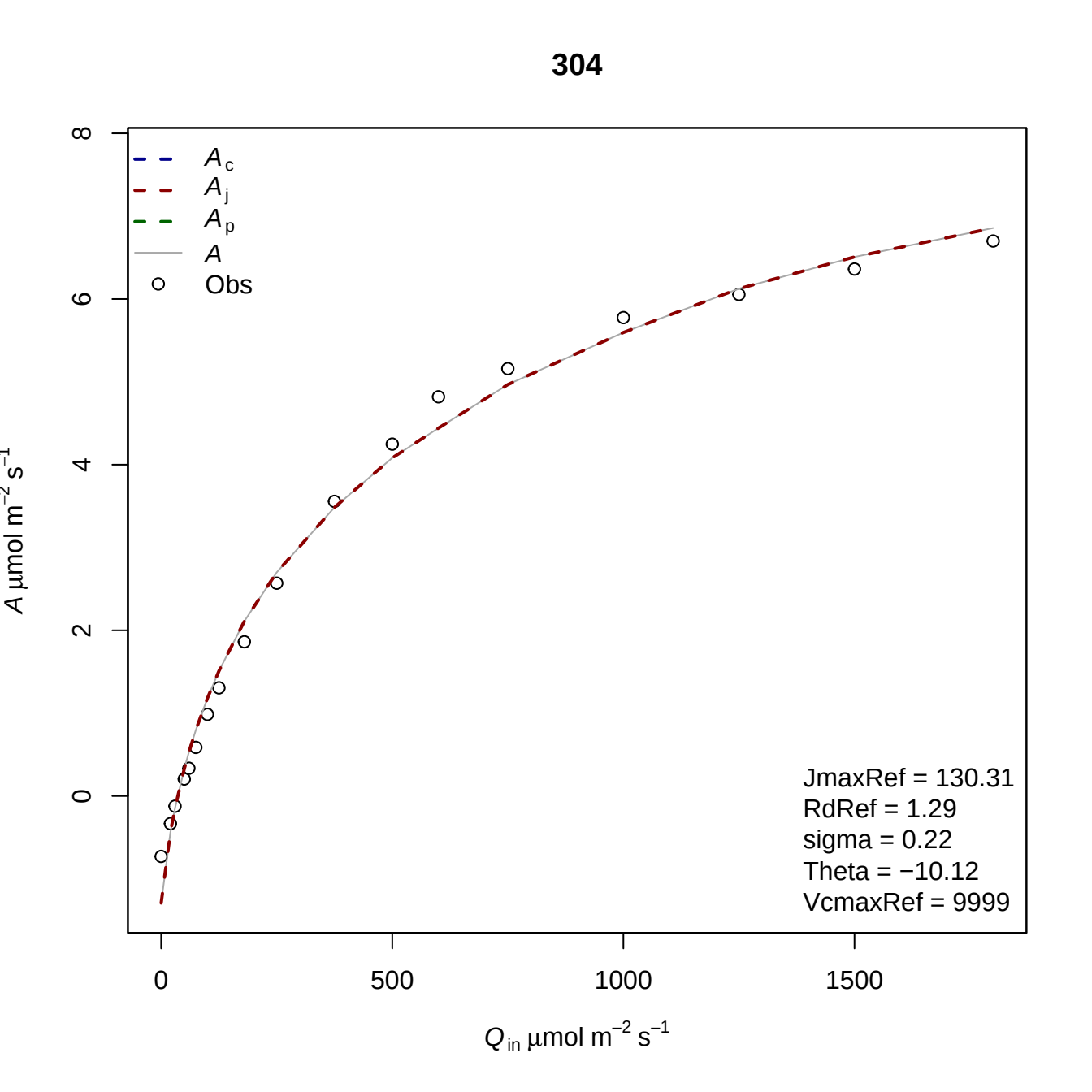

299
A_c
A_j
A_p
A
Obs
JmaxRef = 179.7
RdRef = 1.26
sigma = 0.27
Theta = −17.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

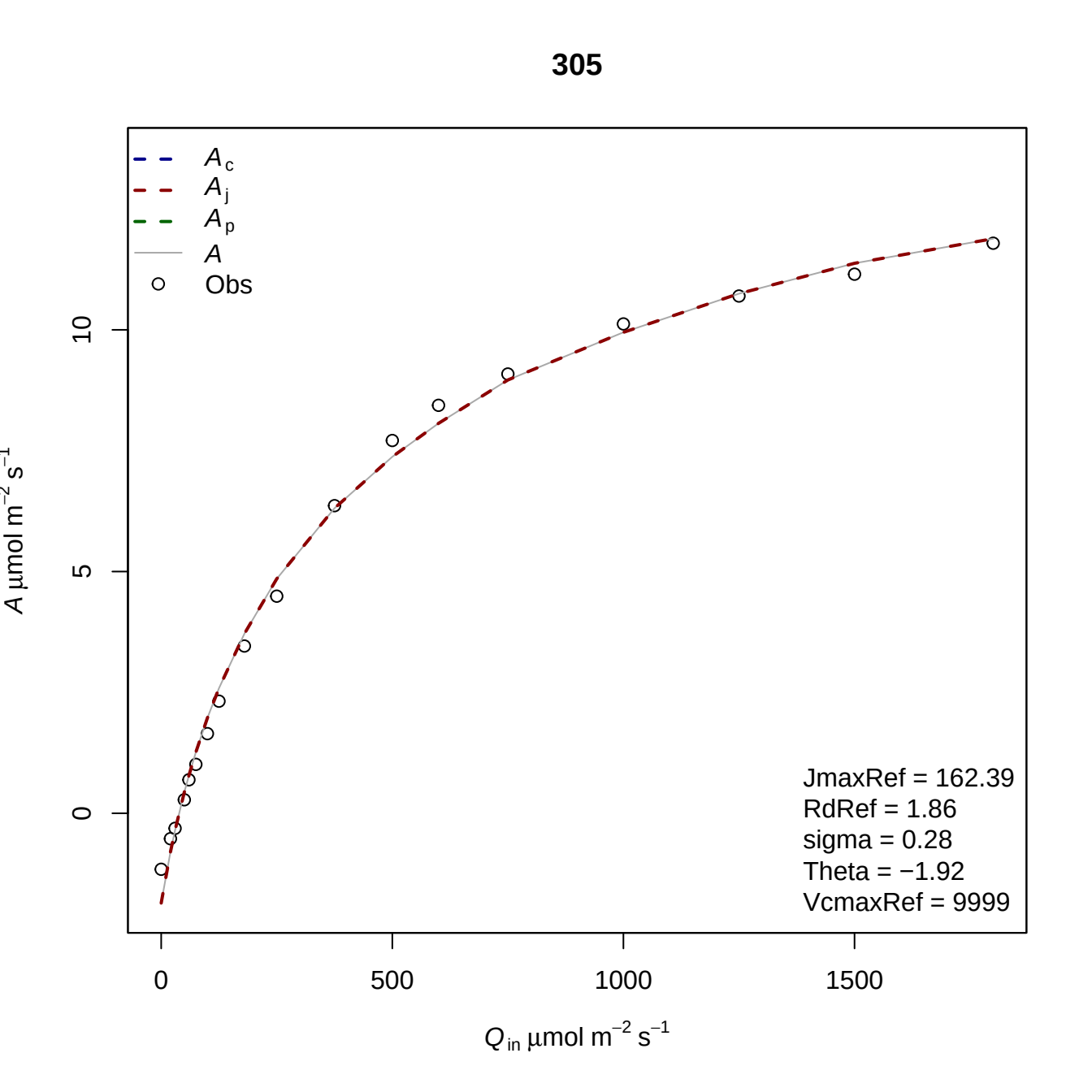

300
A_c
A_j
A_p
A
Obs
JmaxRef = 101.97
RdRef = 1.55
sigma = 0.26
Theta = −4.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

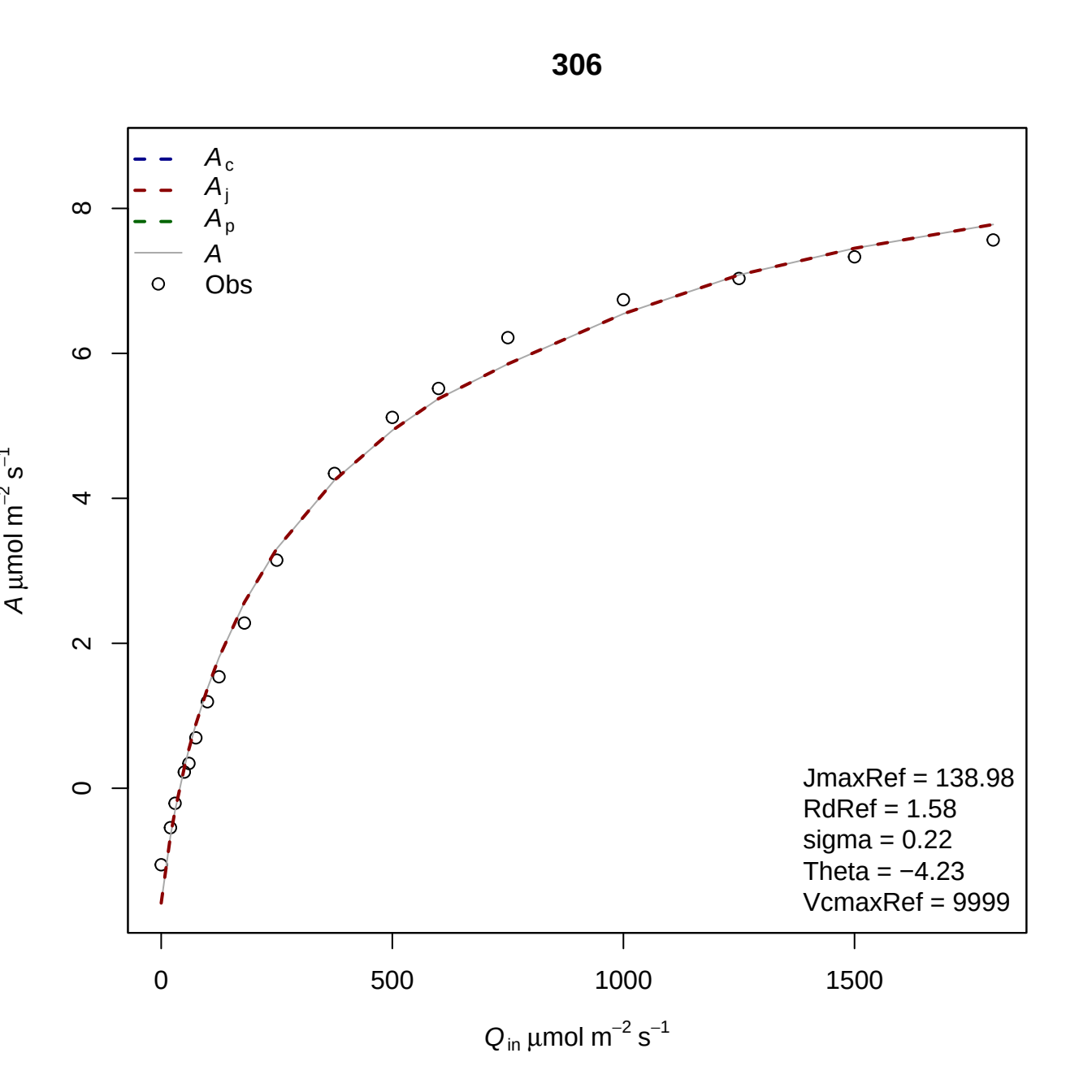

301
A_c
A_j
A_p
A
Obs
JmaxRef = 124.36
RdRef = 1.22
sigma = 0.22
Theta = −11.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

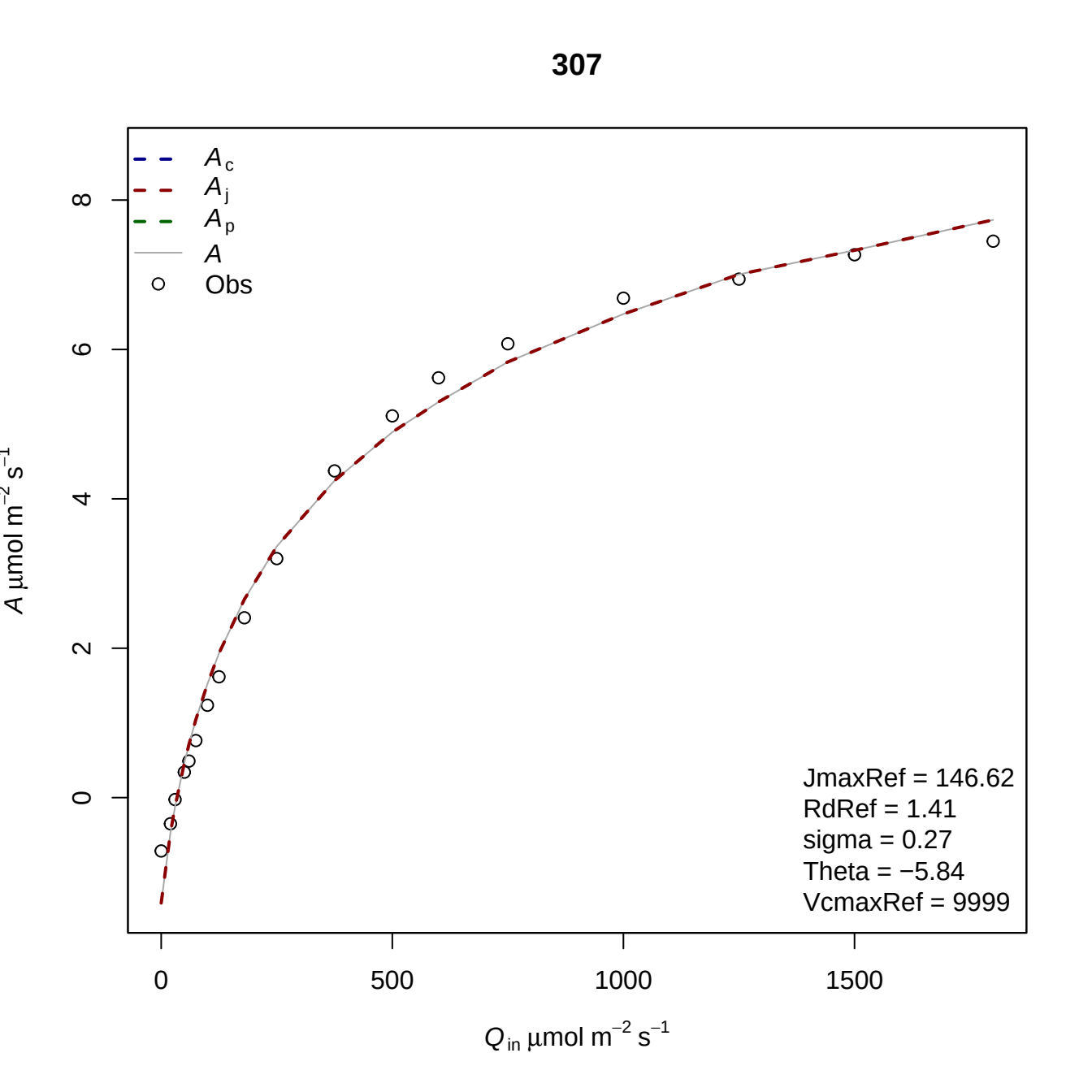

302
A_c
A_j
A_p
A
Obs
JmaxRef = 400.02
RdRef = 1.23
sigma = 0.26
Theta = −44.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

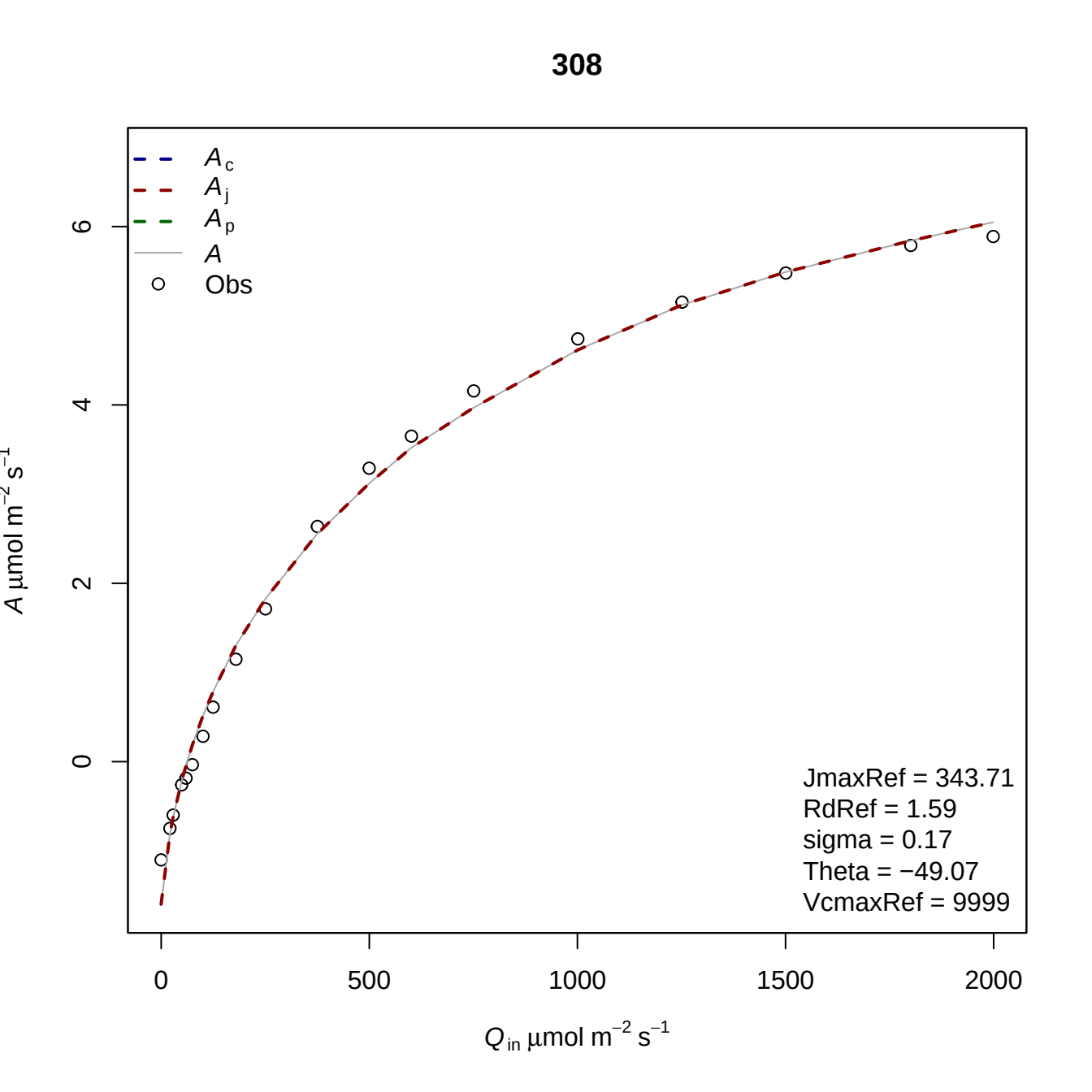

303
A_c
A_j
A_p
A
Obs
JmaxRef = 163.79
RdRef = 1.59
sigma = 0.21
Theta = −16.44
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

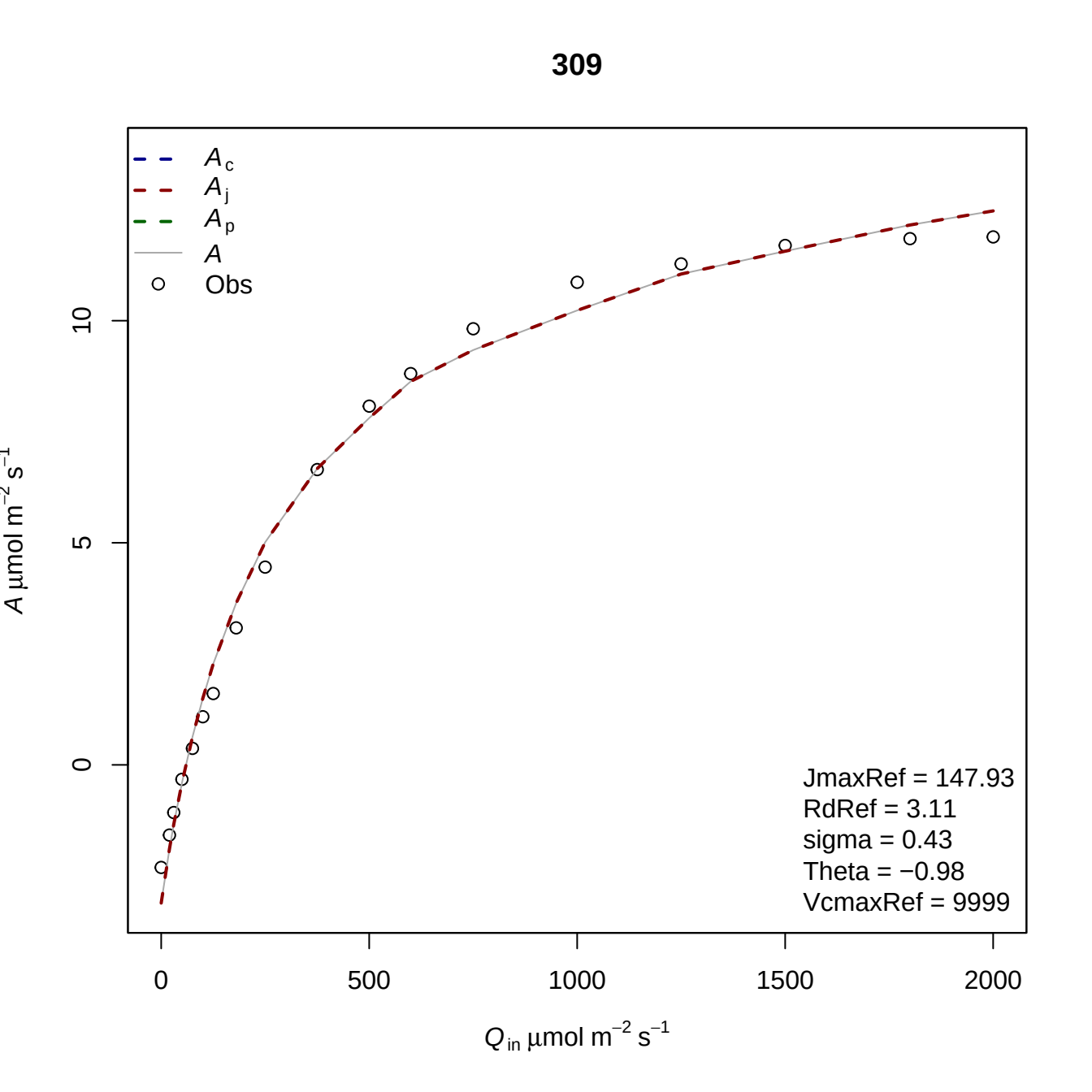

304
A_c
A_j
A_p
A
Obs
JmaxRef = 130.31
RdRef = 1.29
sigma = 0.22
Theta = −10.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

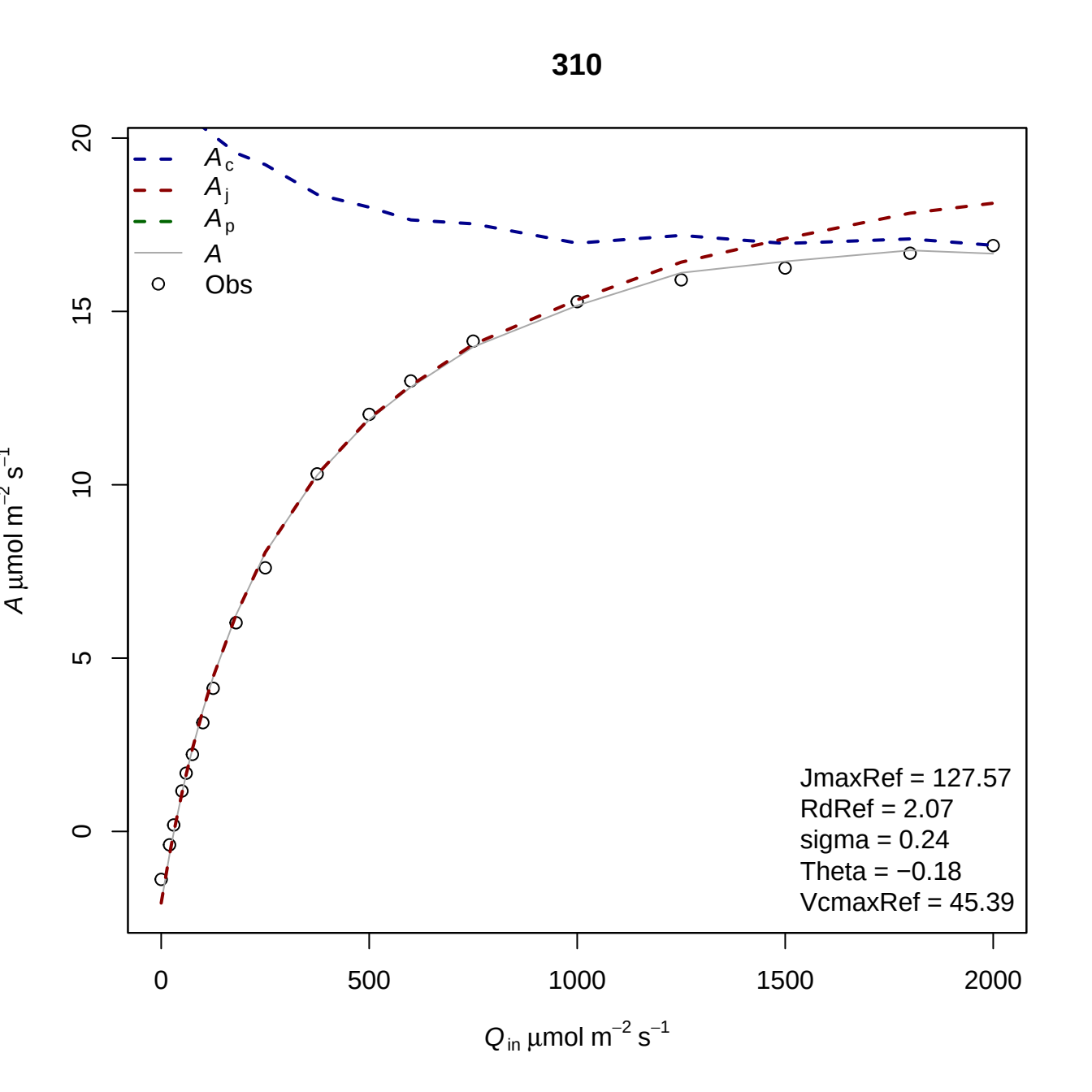

305
A_c
A_j
A_p
A
Obs
JmaxRef = 162.39
RdRef = 1.86
sigma = 0.28
Theta = −1.92
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

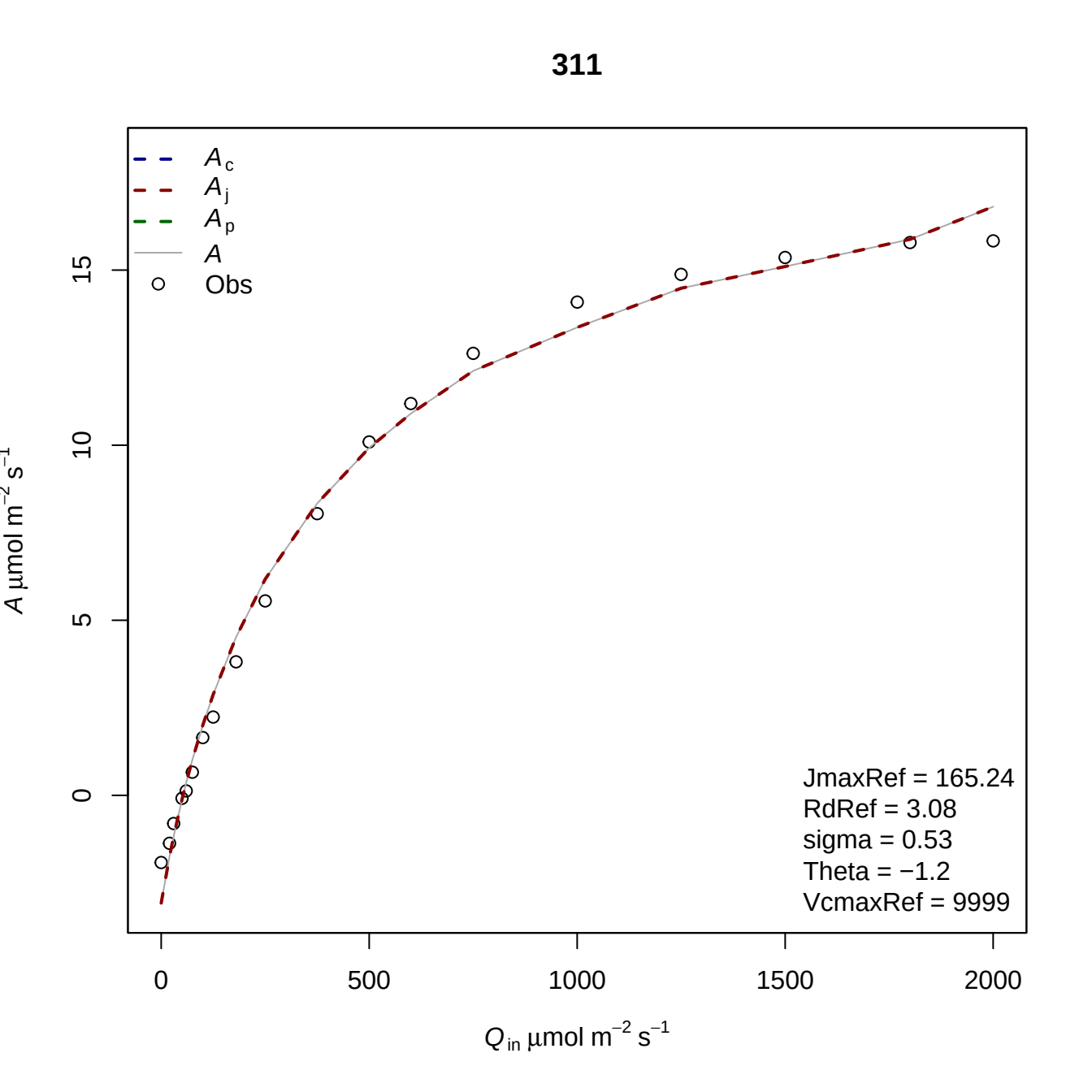

306
A_c
A_j
A_p
A
Obs
JmaxRef = 138.98
RdRef = 1.58
sigma = 0.22
Theta = −4.23
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

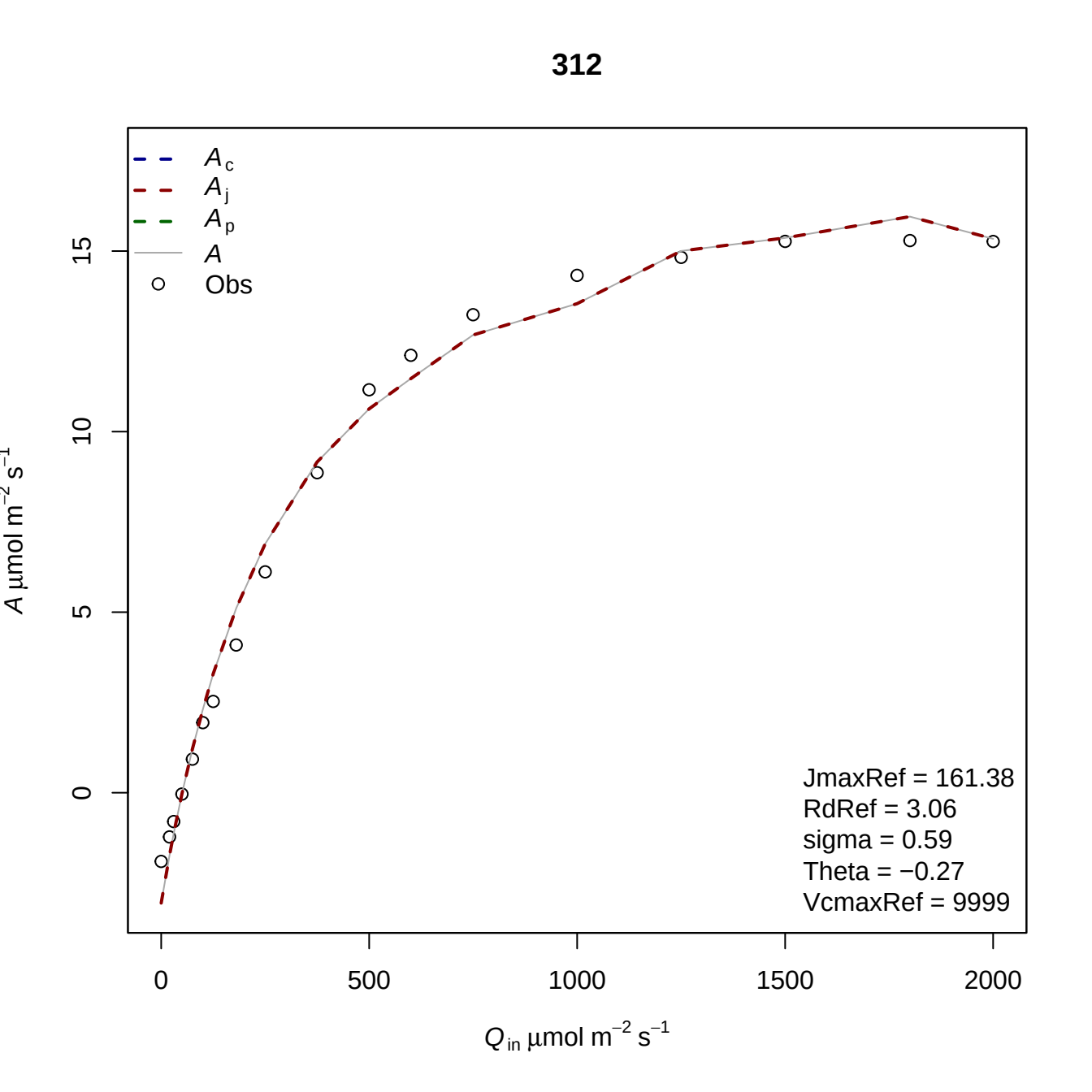

307
A_c
A_j
A_p
A
Obs
JmaxRef = 146.62
RdRef = 1.41
sigma = 0.27
Theta = −5.84
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

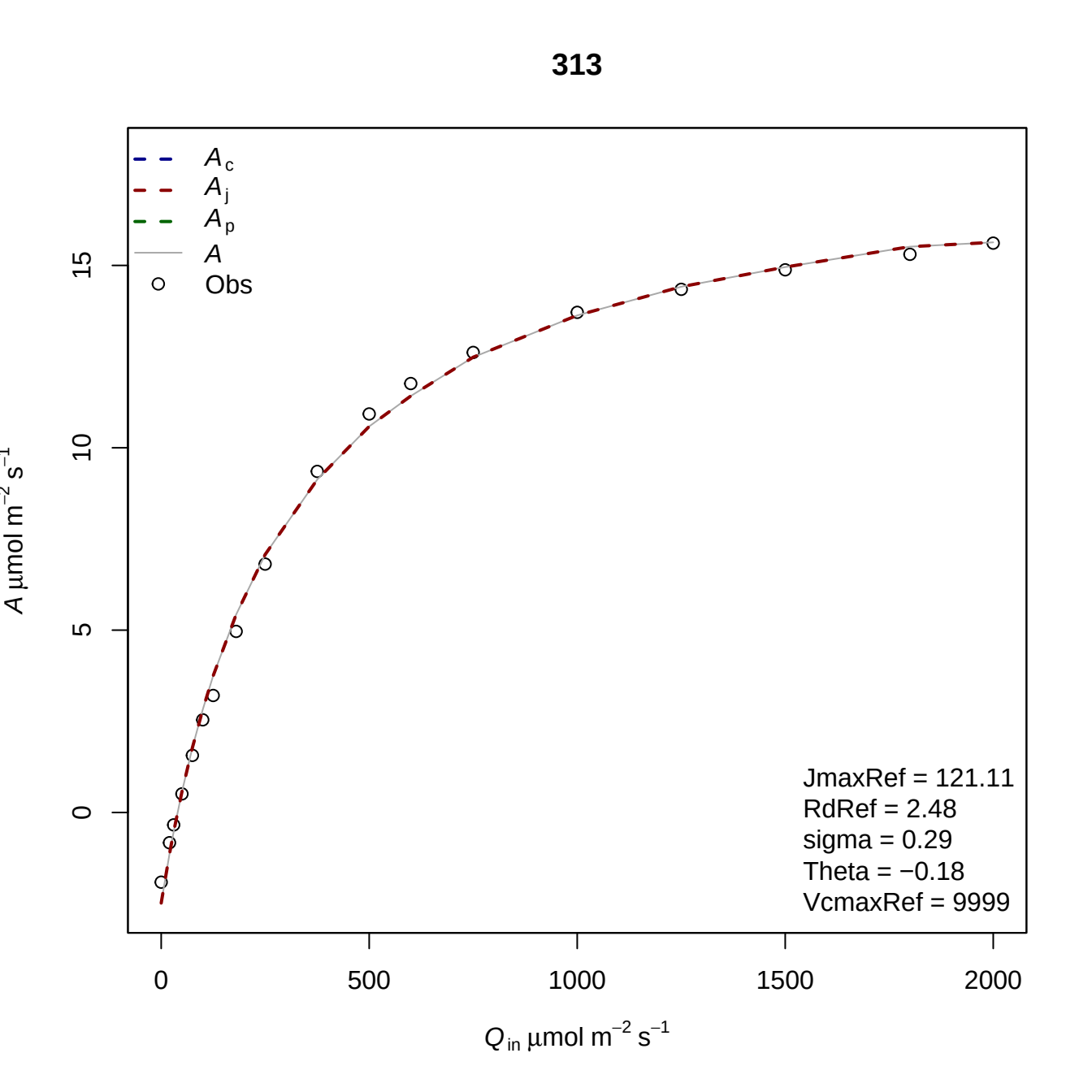

308
A_c
A_j
A_p
A
Obs
JmaxRef = 343.71
RdRef = 1.59
sigma = 0.17
Theta = −49.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

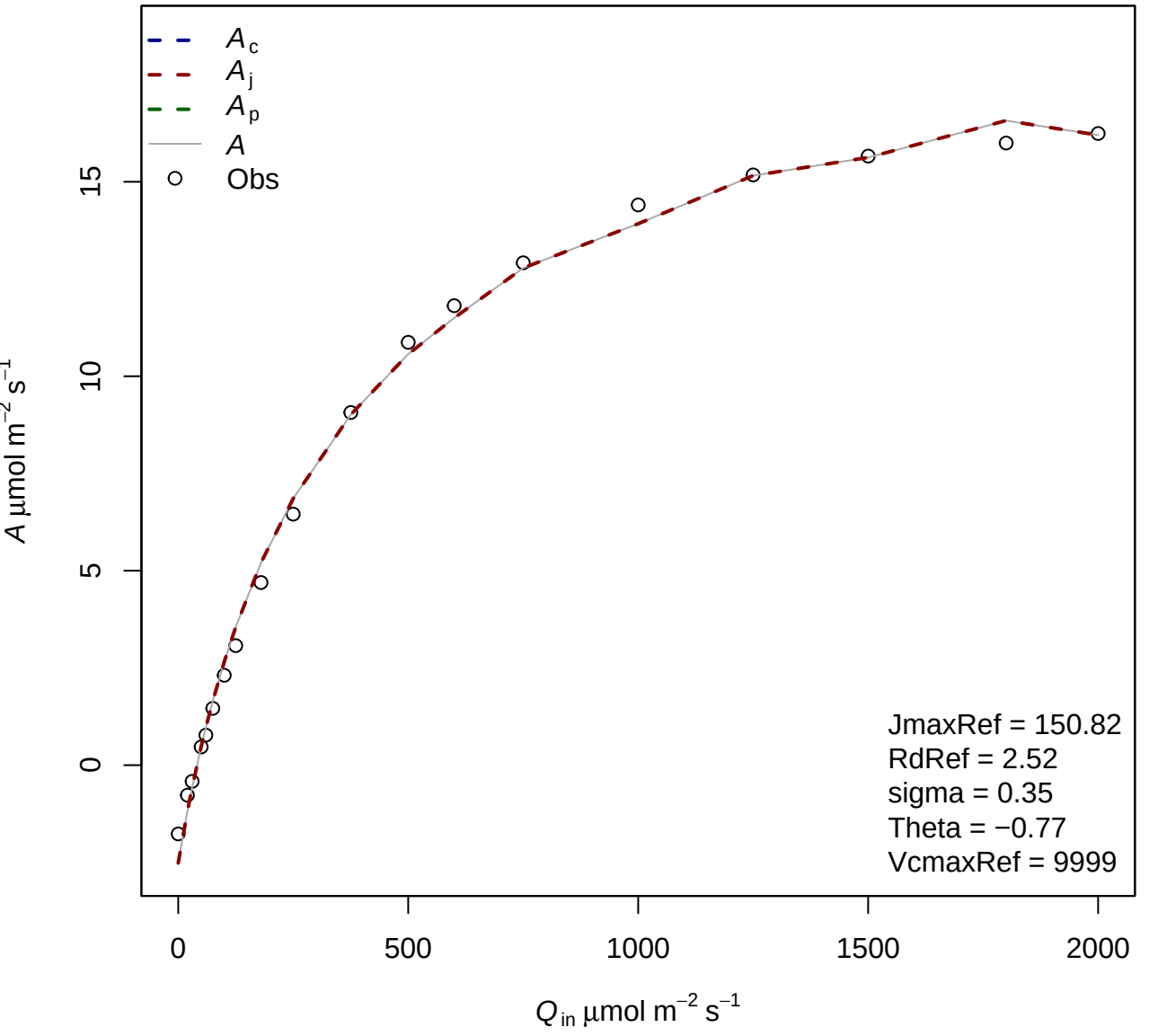

309
A_c
A_j
A_p
A
Obs
JmaxRef = 147.93
RdRef = 3.11
sigma = 0.43
Theta = −0.98
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

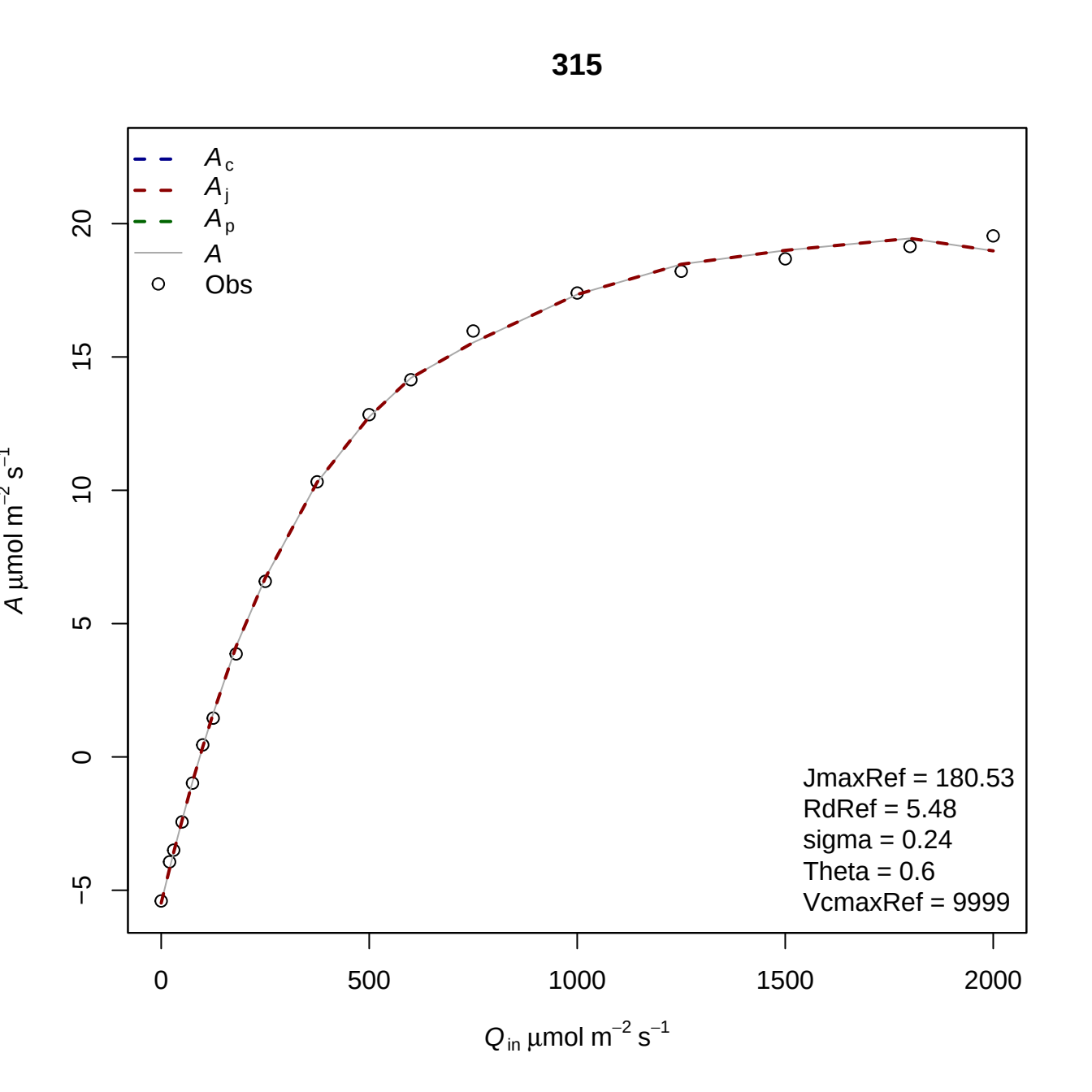

310
A_c
A_j
A_p
A
Obs
JmaxRef = 127.57
RdRef = 2.07
sigma = 0.24
Theta = −0.18
VcmaxRef = 45.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

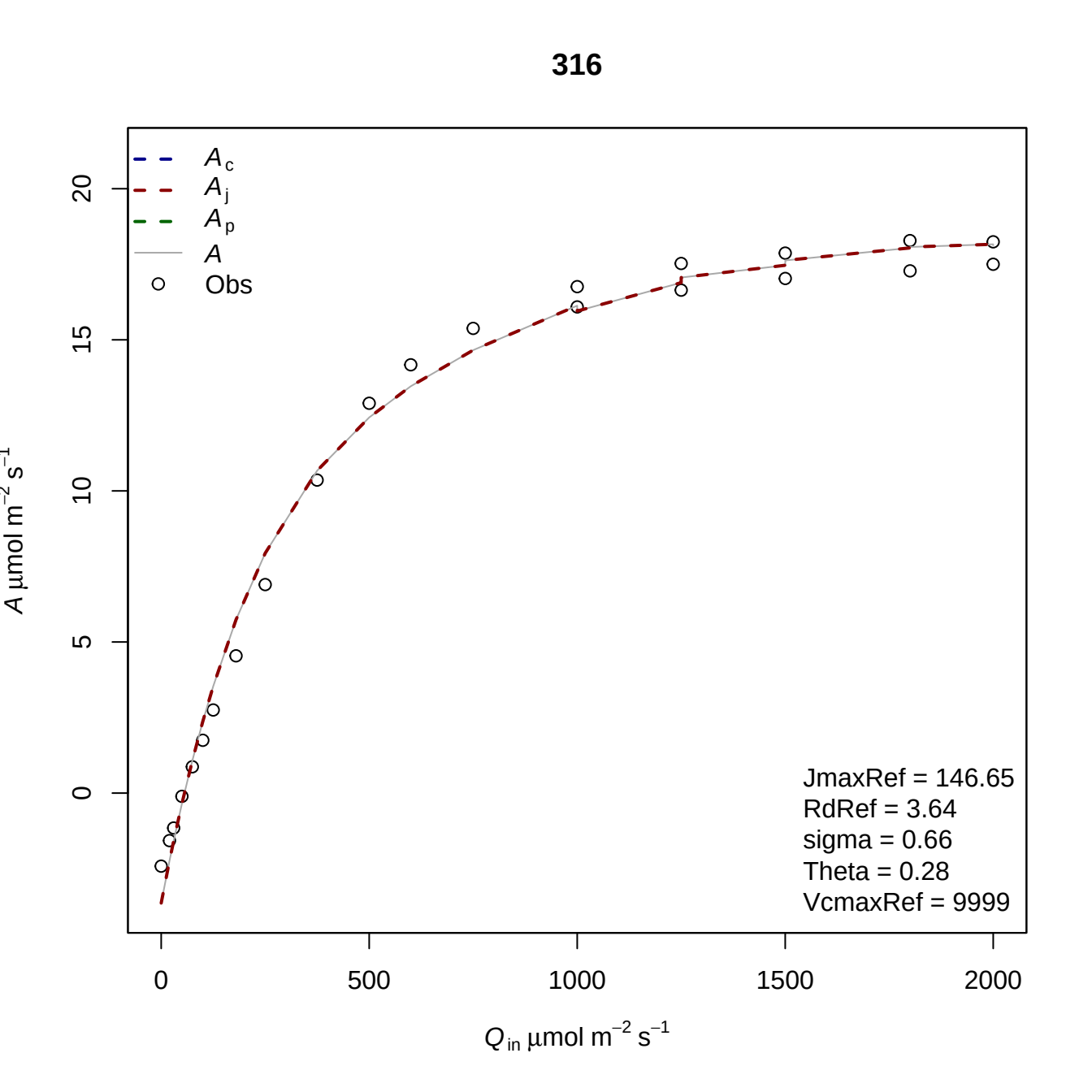

311
A_c
A_j
A_p
A
Obs
JmaxRef = 165.24
RdRef = 3.08
sigma = 0.53
Theta = −1.2
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

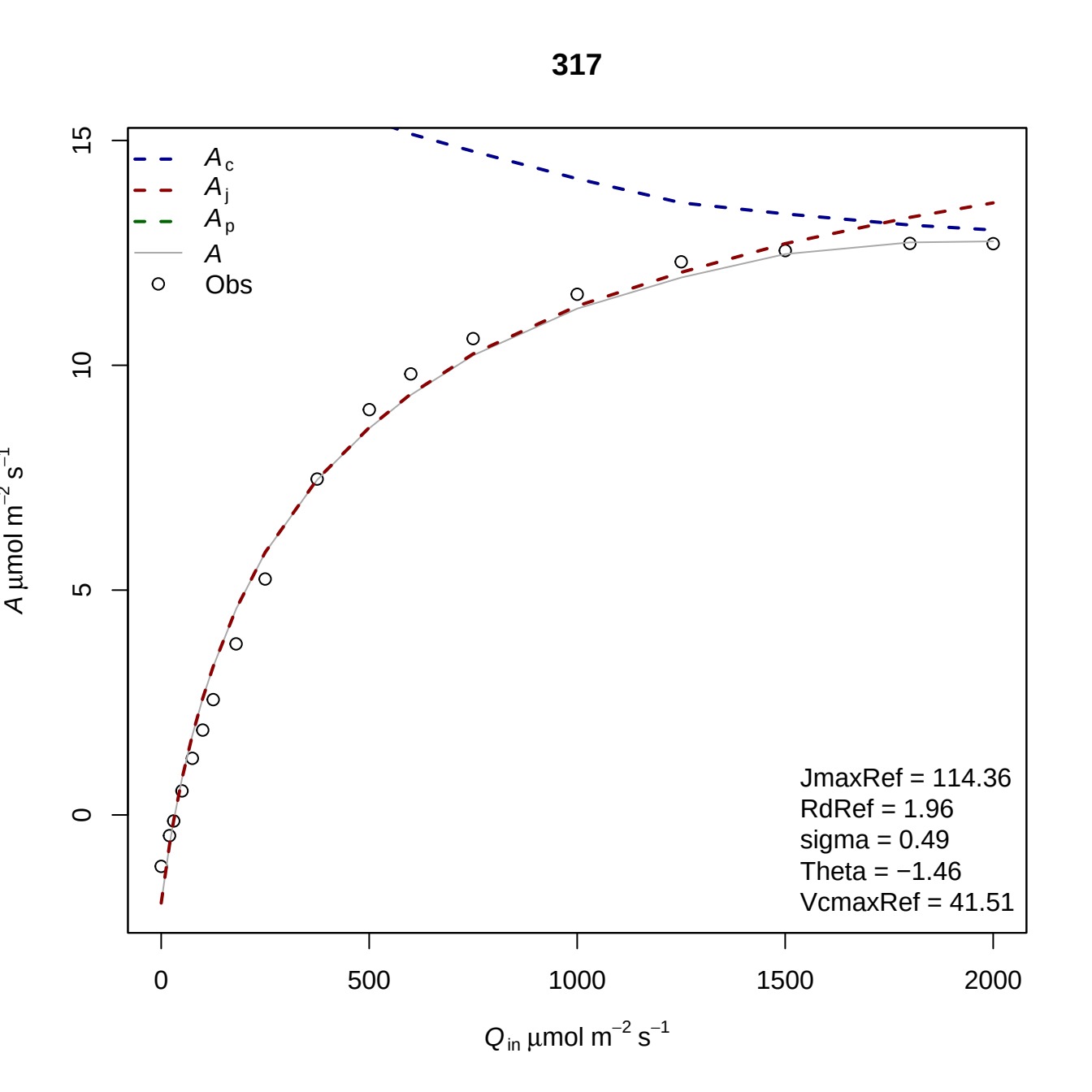

312
A_c
A_j
A_p
A
Obs
JmaxRef = 161.38
RdRef = 3.06
sigma = 0.59
Theta = −0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

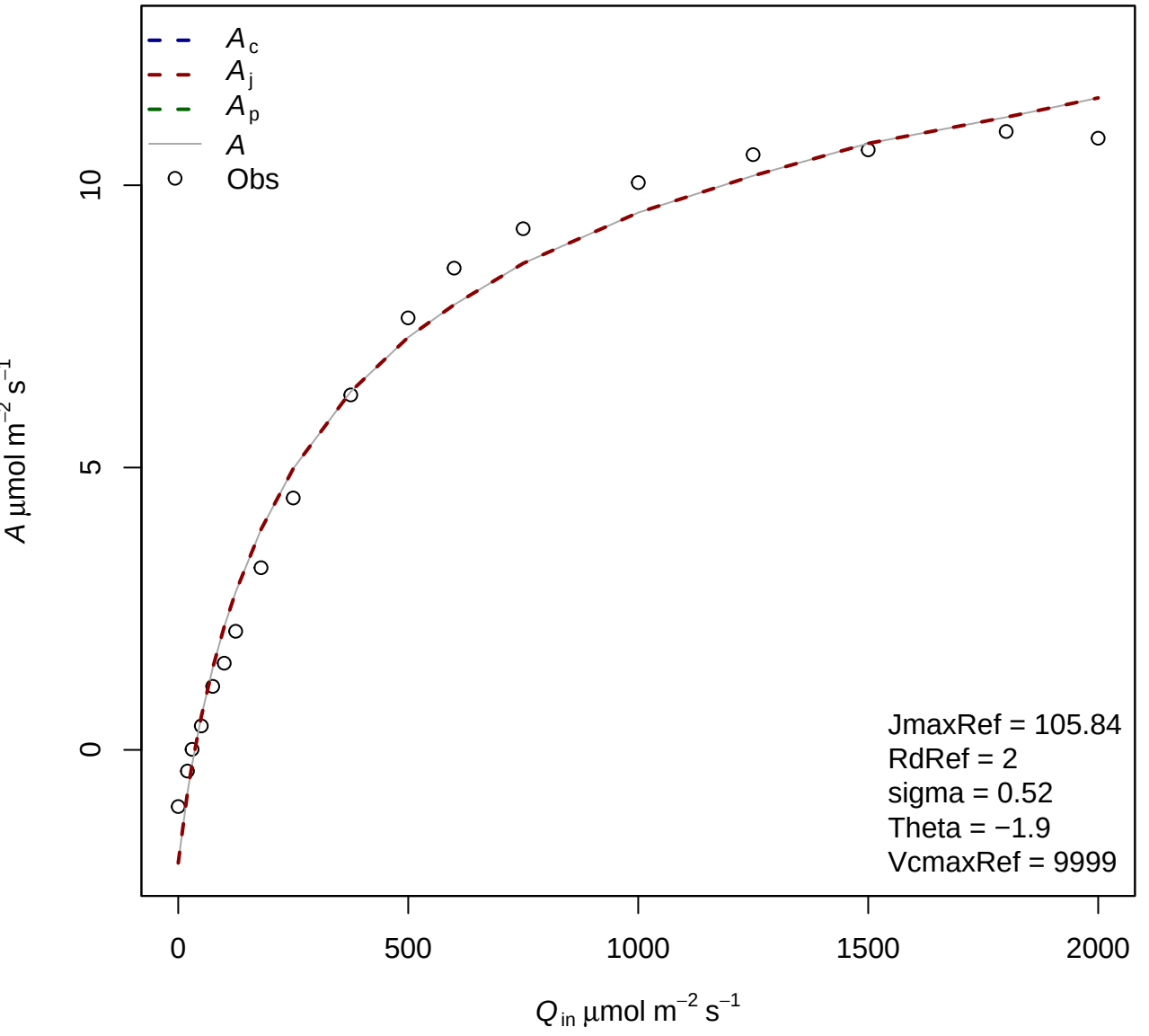

313
A_c
A_j
A_p
A
Obs
JmaxRef = 121.11
RdRef = 2.48
sigma = 0.29
Theta = −0.18
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

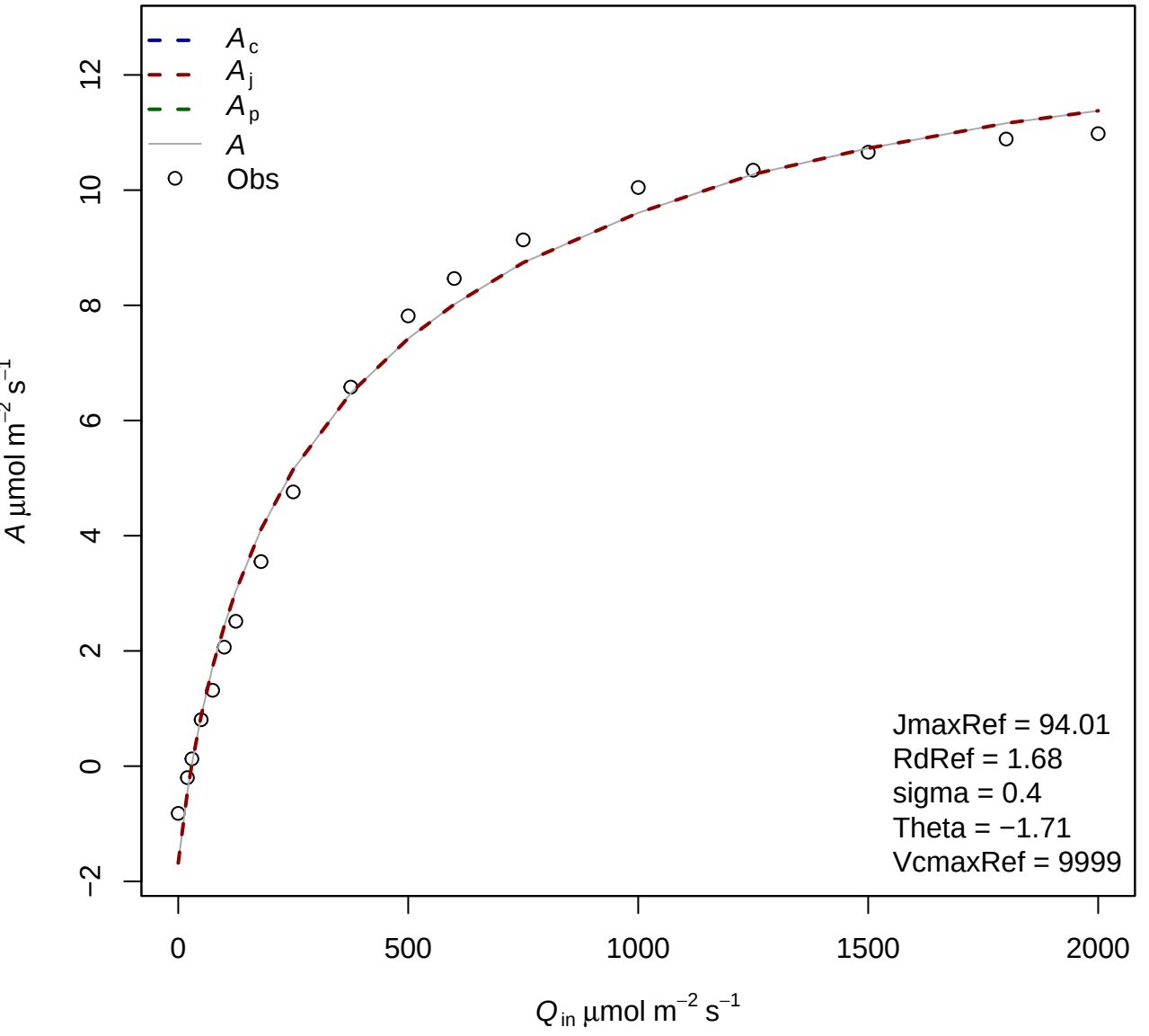

A_c
A_j
A_p
A
Obs
JmaxRef = 150.82
RdRef = 2.52
sigma = 0.35
Theta = −0.77
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

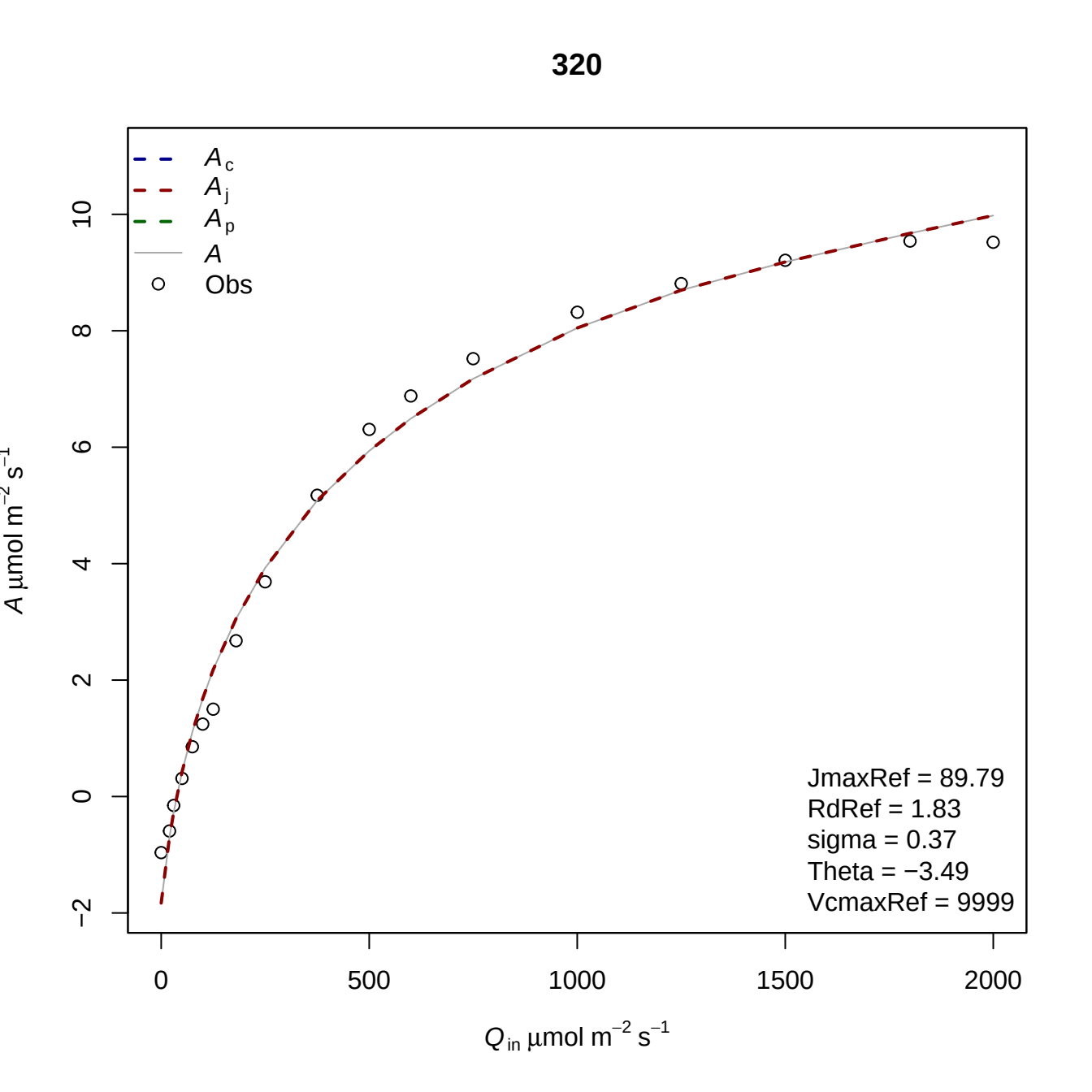

315
A_c
A_j
A_p
A
Obs
JmaxRef = 180.53
RdRef = 5.48
sigma = 0.24
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

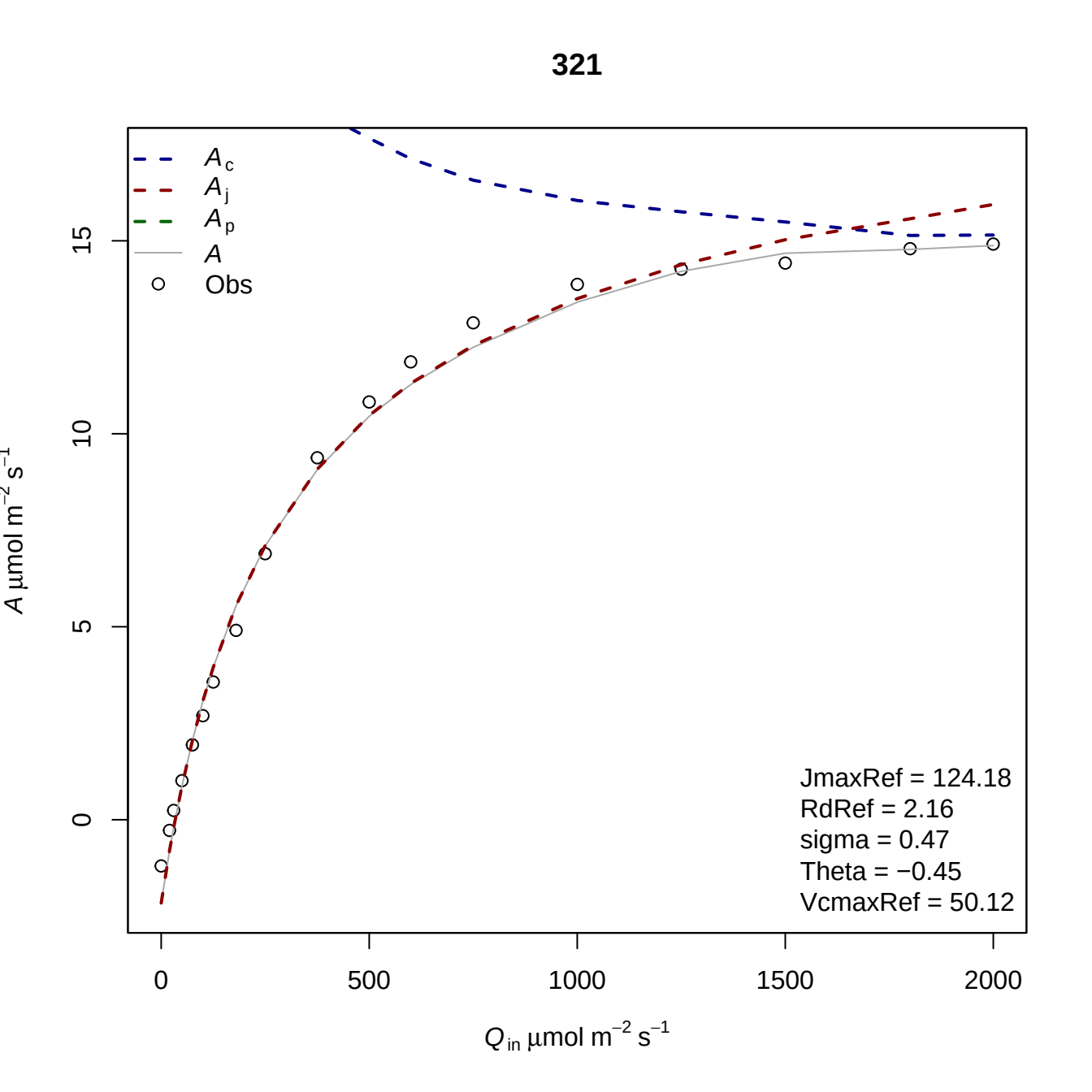

316
A_c
A_j
A_p
A
Obs
JmaxRef = 146.65
RdRef = 3.64
sigma = 0.66
Theta = 0.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

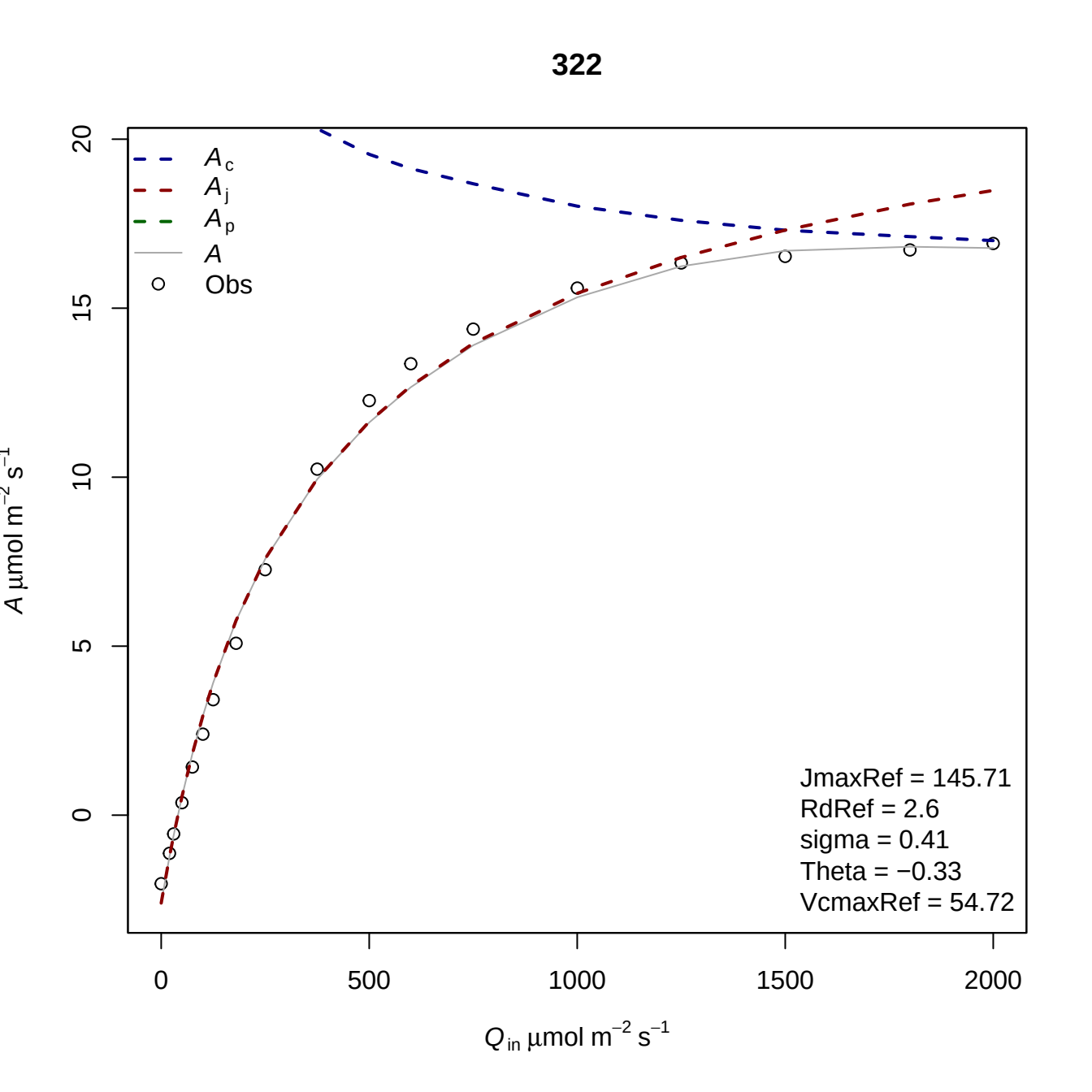

317
A_c
A_j
A_p
A
Obs
JmaxRef = 114.36
RdRef = 1.96
sigma = 0.49
Theta = −1.46
VcmaxRef = 41.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

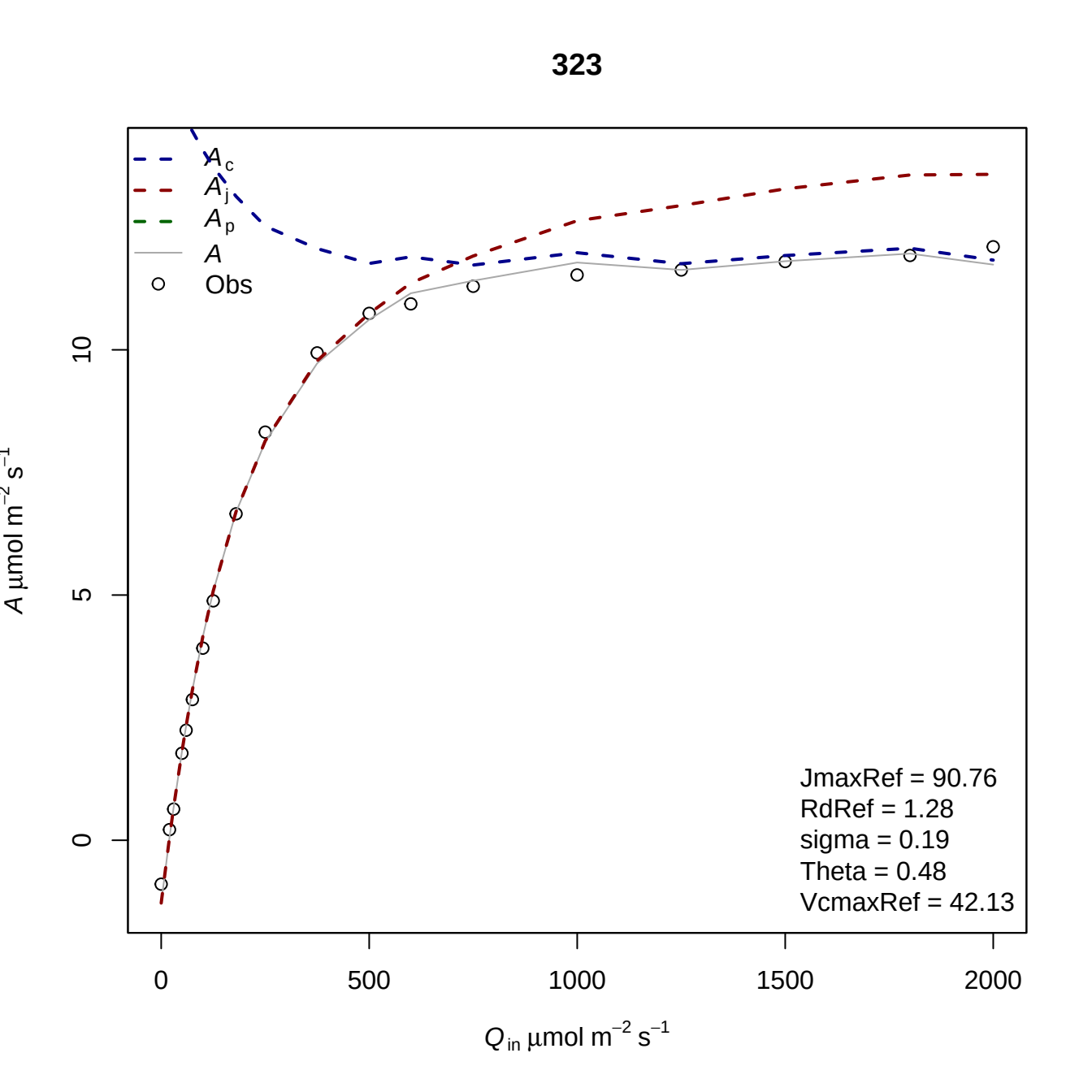

A_c
A_j
A_p
A
Obs
JmaxRef = 105.84
RdRef = 2
sigma = 0.52
Theta = −1.9
VcmaxRef = 9999
0
5
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

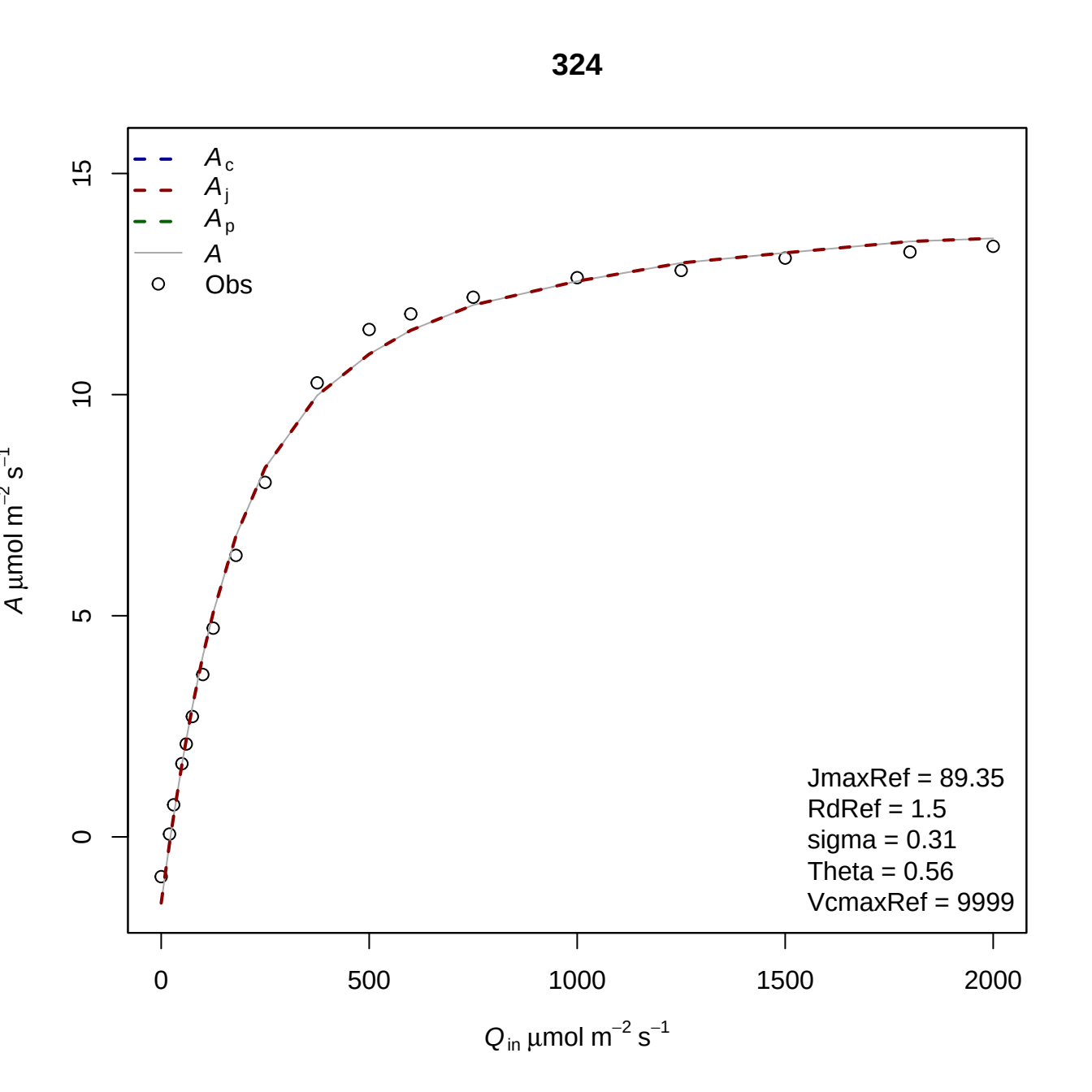

A_c
A_j
A_p
A
Obs
JmaxRef = 94.01
RdRef = 1.68
sigma = 0.4
Theta = −1.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

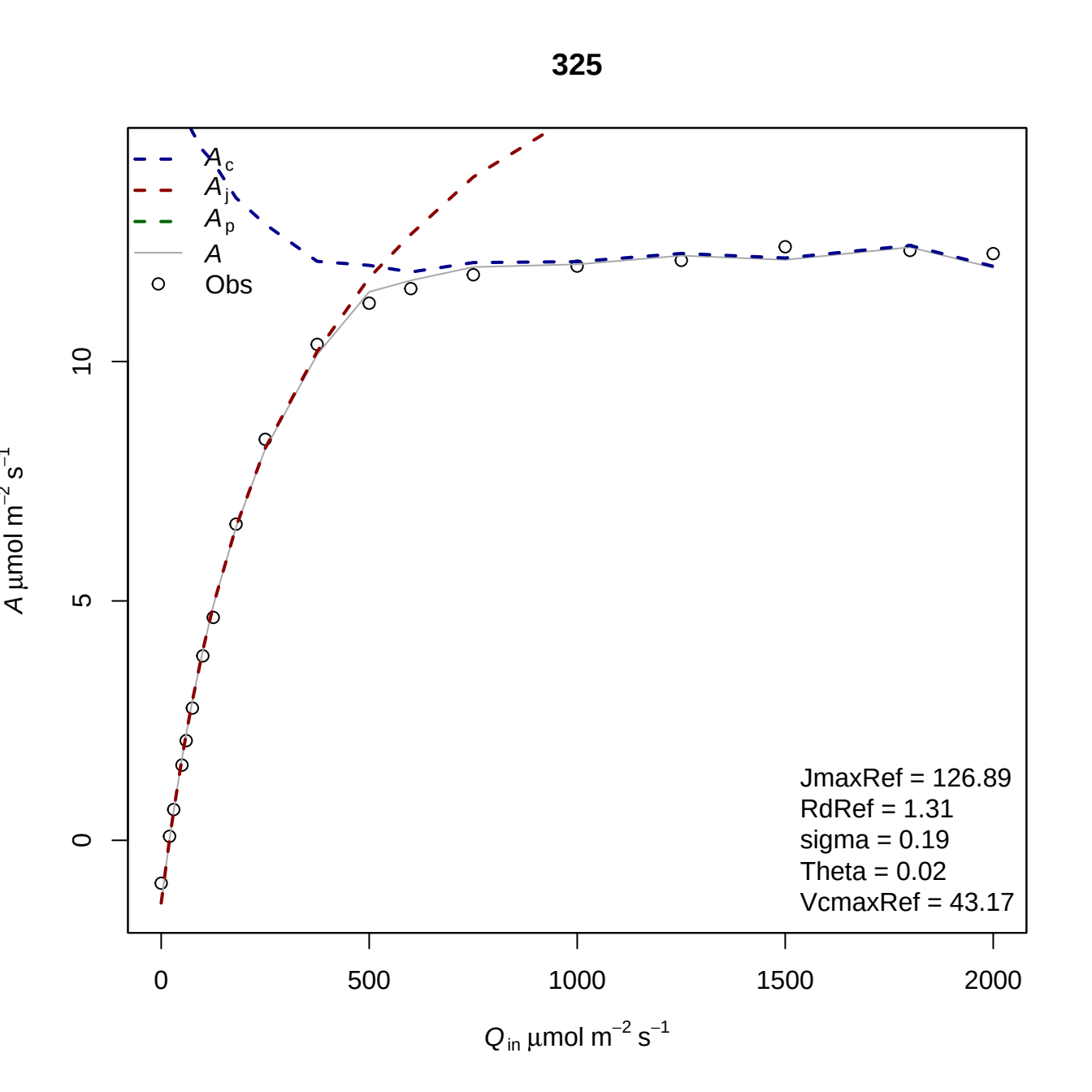

320
A_c
A_j
A_p
A
Obs
JmaxRef = 89.79
RdRef = 1.83
sigma = 0.37
Theta = −3.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

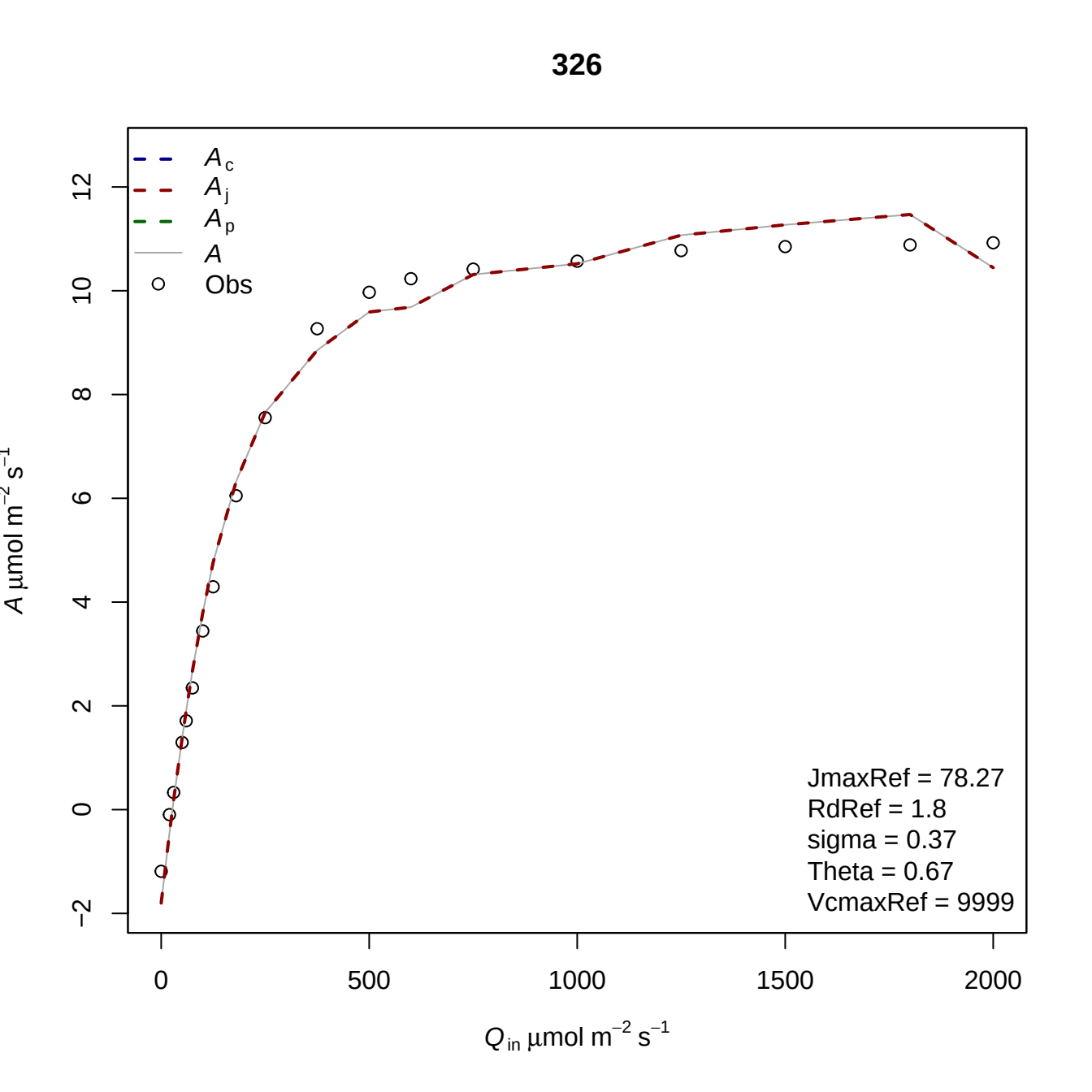

321
A_c
A_j
A_p
A
Obs
JmaxRef = 124.18
RdRef = 2.16
sigma = 0.47
Theta = −0.45
VcmaxRef = 50.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

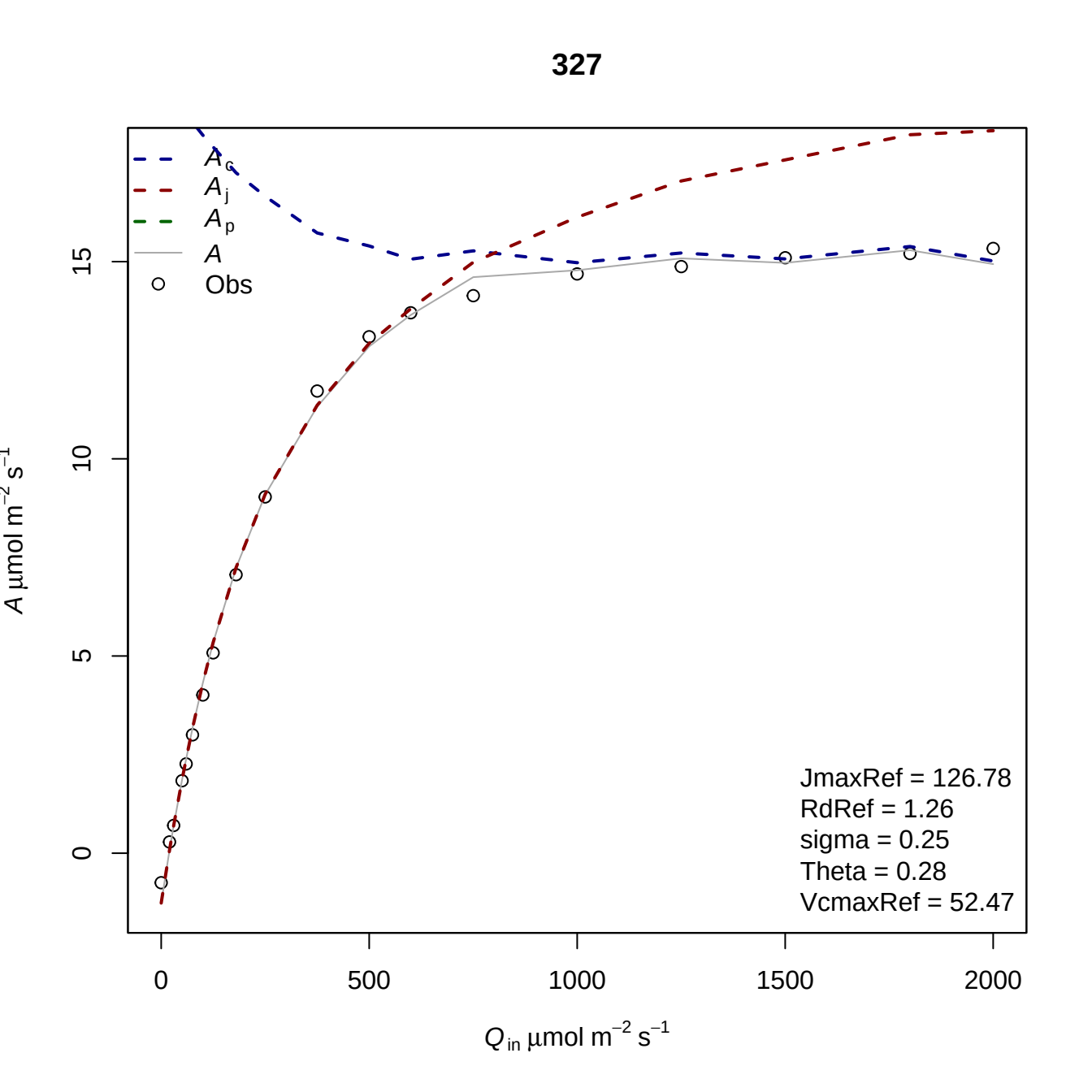

322
A_c
A_j
A_p
A
Obs
JmaxRef = 145.71
RdRef = 2.6
sigma = 0.41
Theta = −0.33
VcmaxRef = 54.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

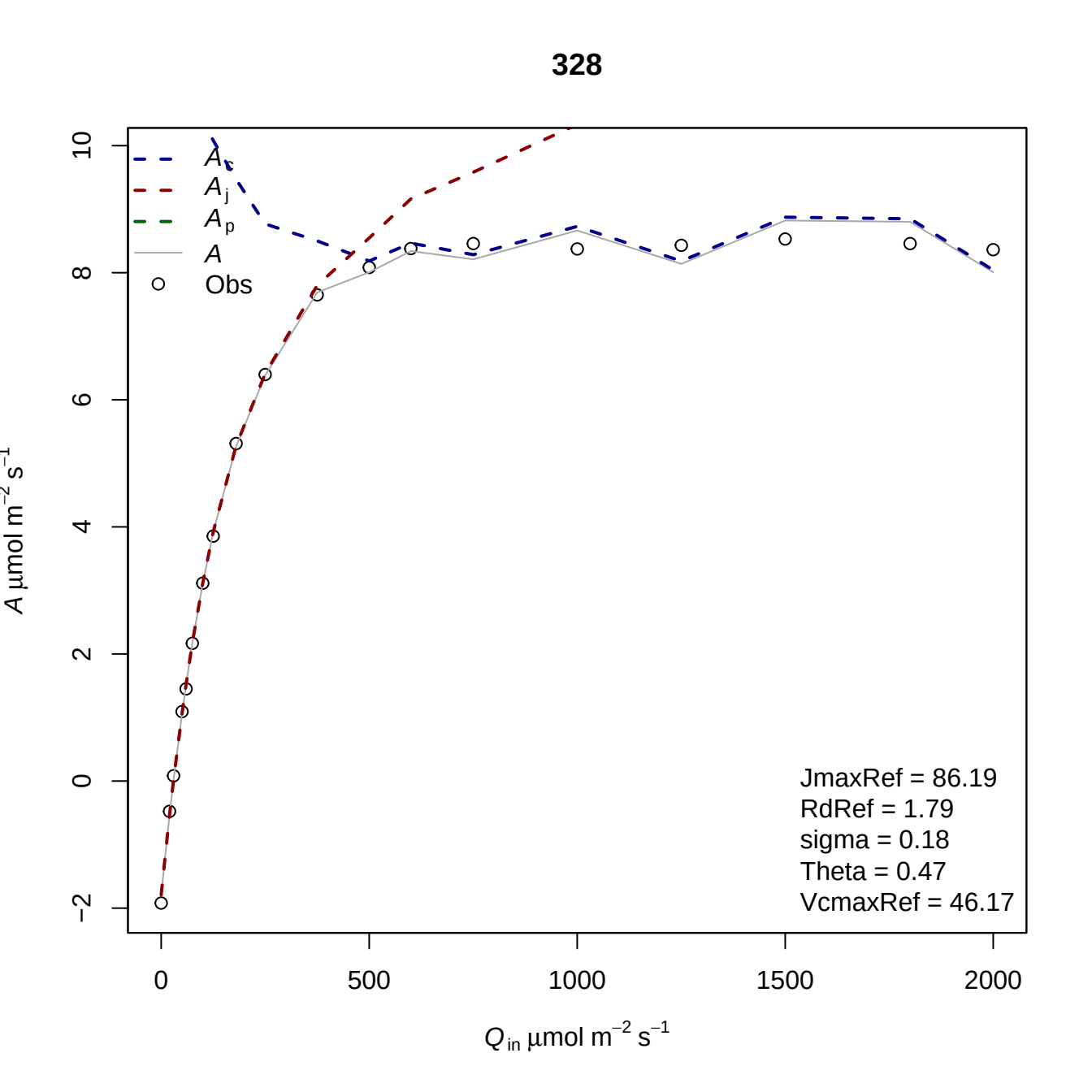

323
A_c
A_j
A_p
A
Obs
JmaxRef = 90.76
RdRef = 1.28
sigma = 0.19
Theta = 0.48
VcmaxRef = 42.13
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

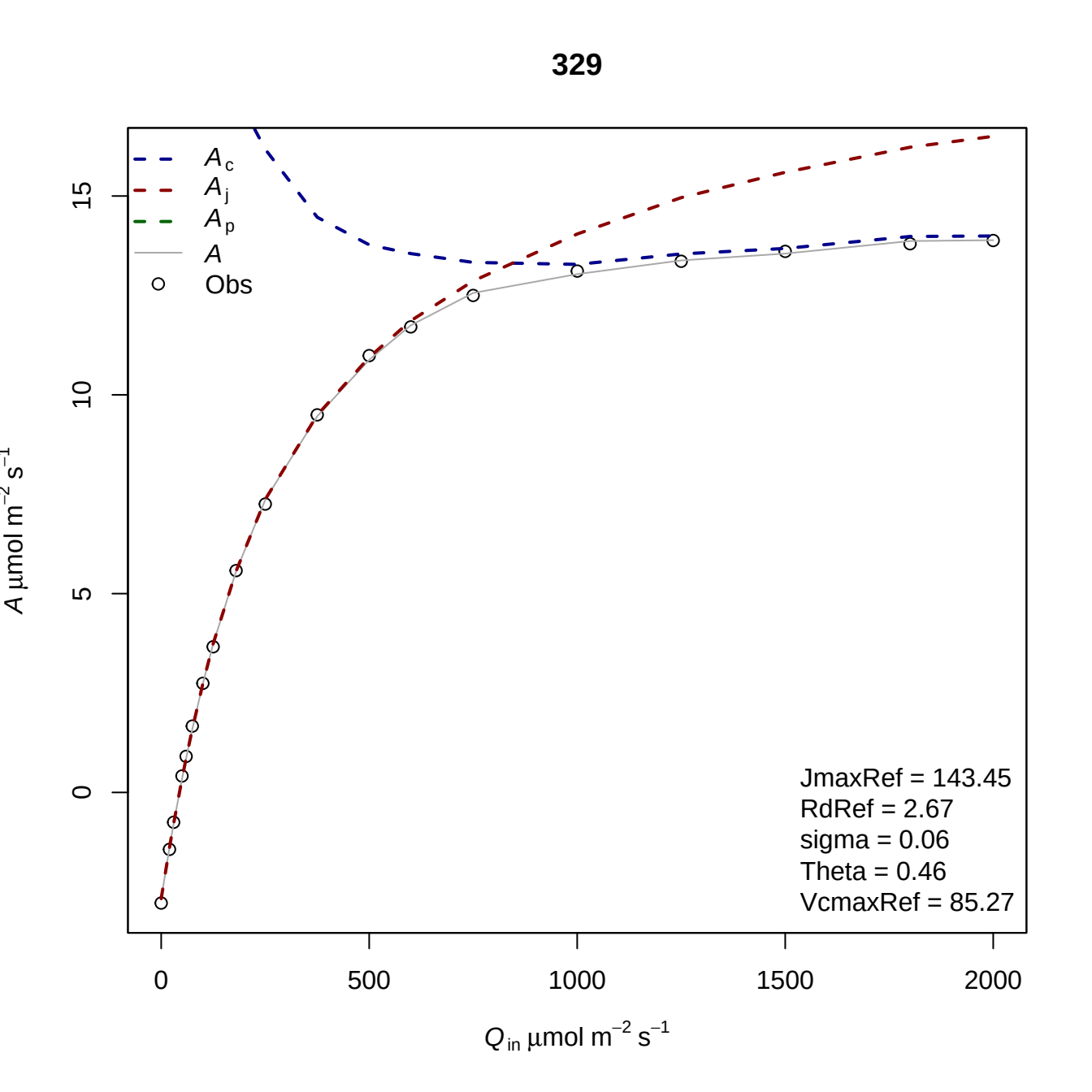

324
A_c
A_j
A_p
A
Obs
JmaxRef = 89.35
RdRef = 1.5
sigma = 0.31
Theta = 0.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

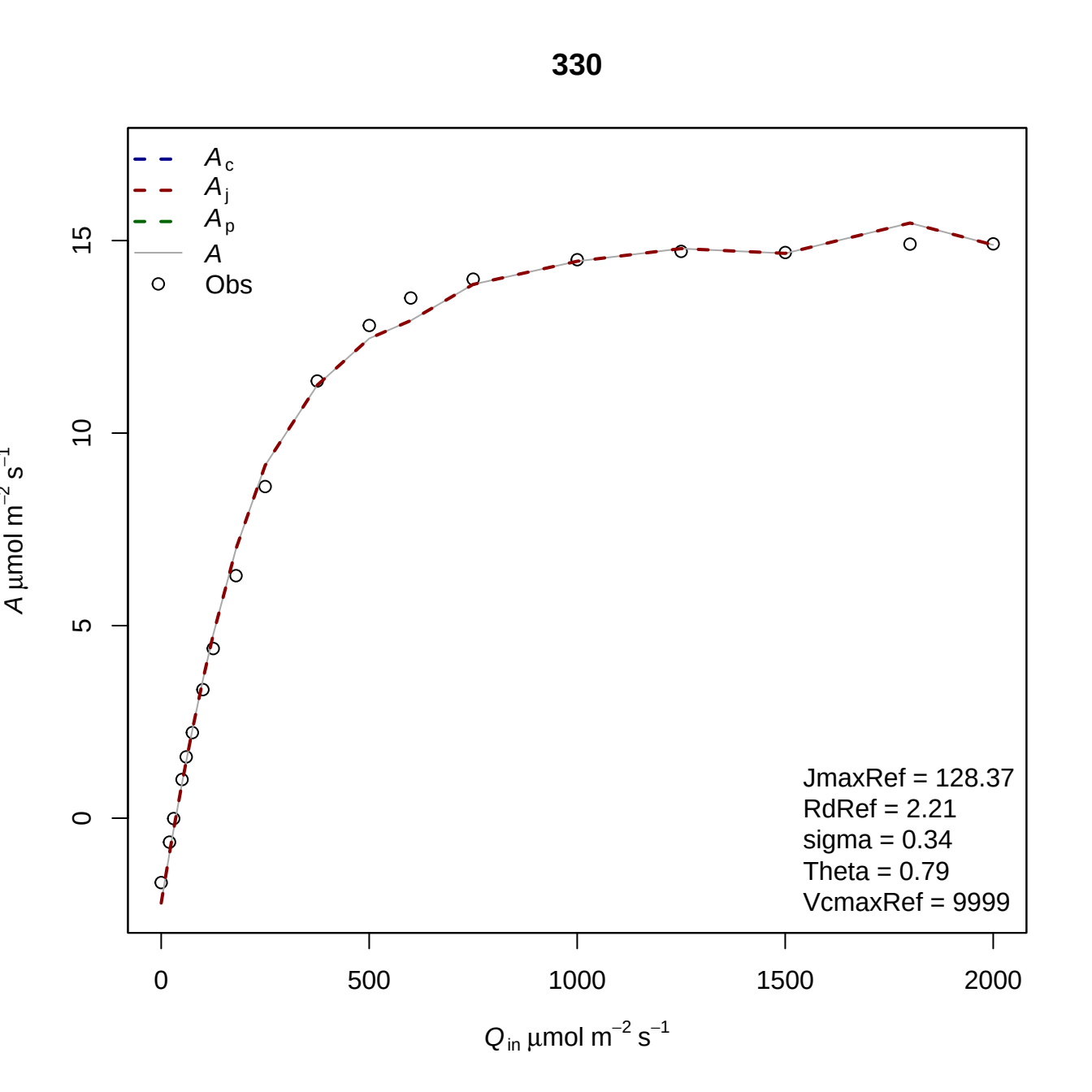

325
A_c
A_j
A_p
A
Obs
JmaxRef = 126.89
RdRef = 1.31
sigma = 0.19
Theta = 0.02
VcmaxRef = 43.17
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

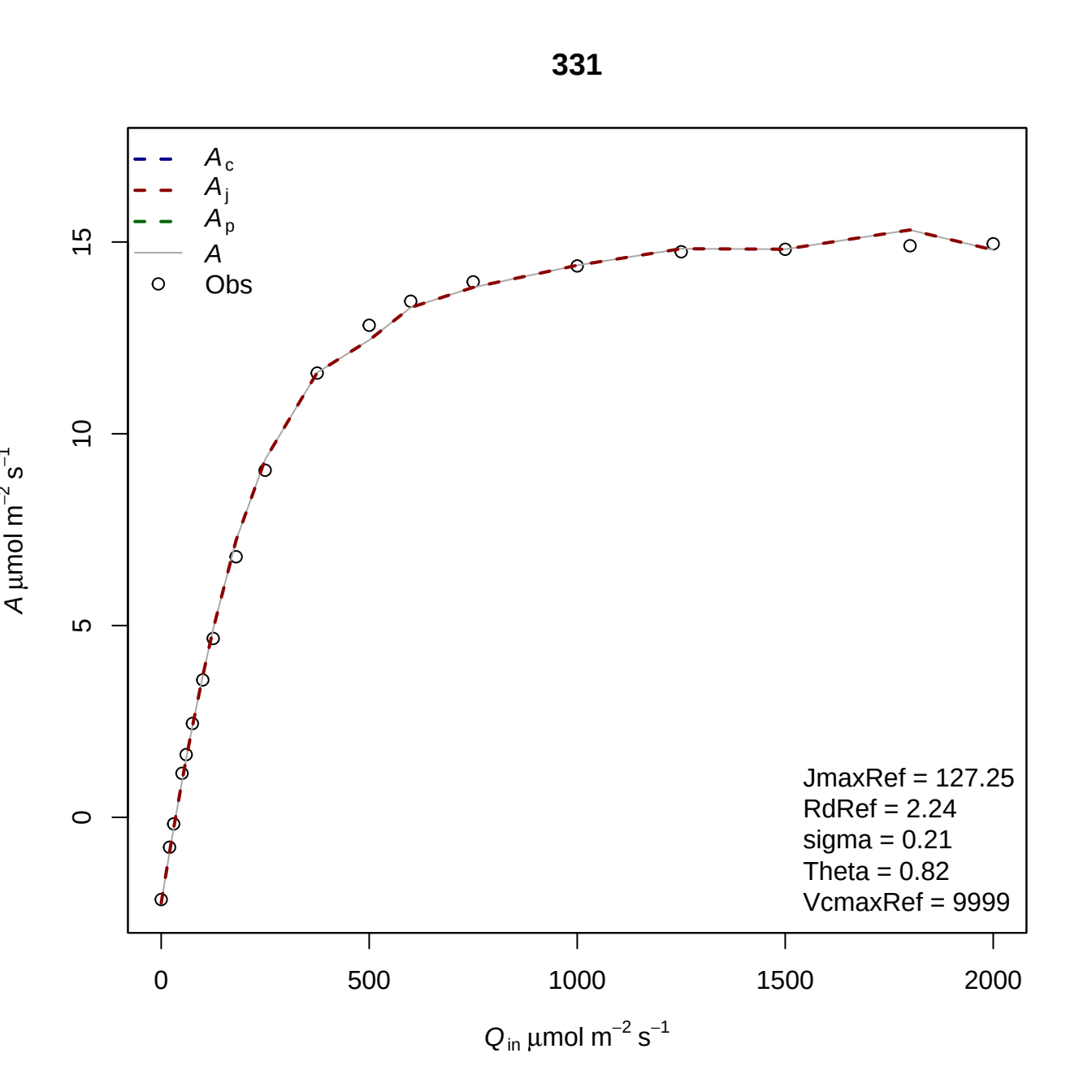

326
A_c
A_j
A_p
A
Obs
JmaxRef = 78.27
RdRef = 1.8
sigma = 0.37
Theta = 0.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

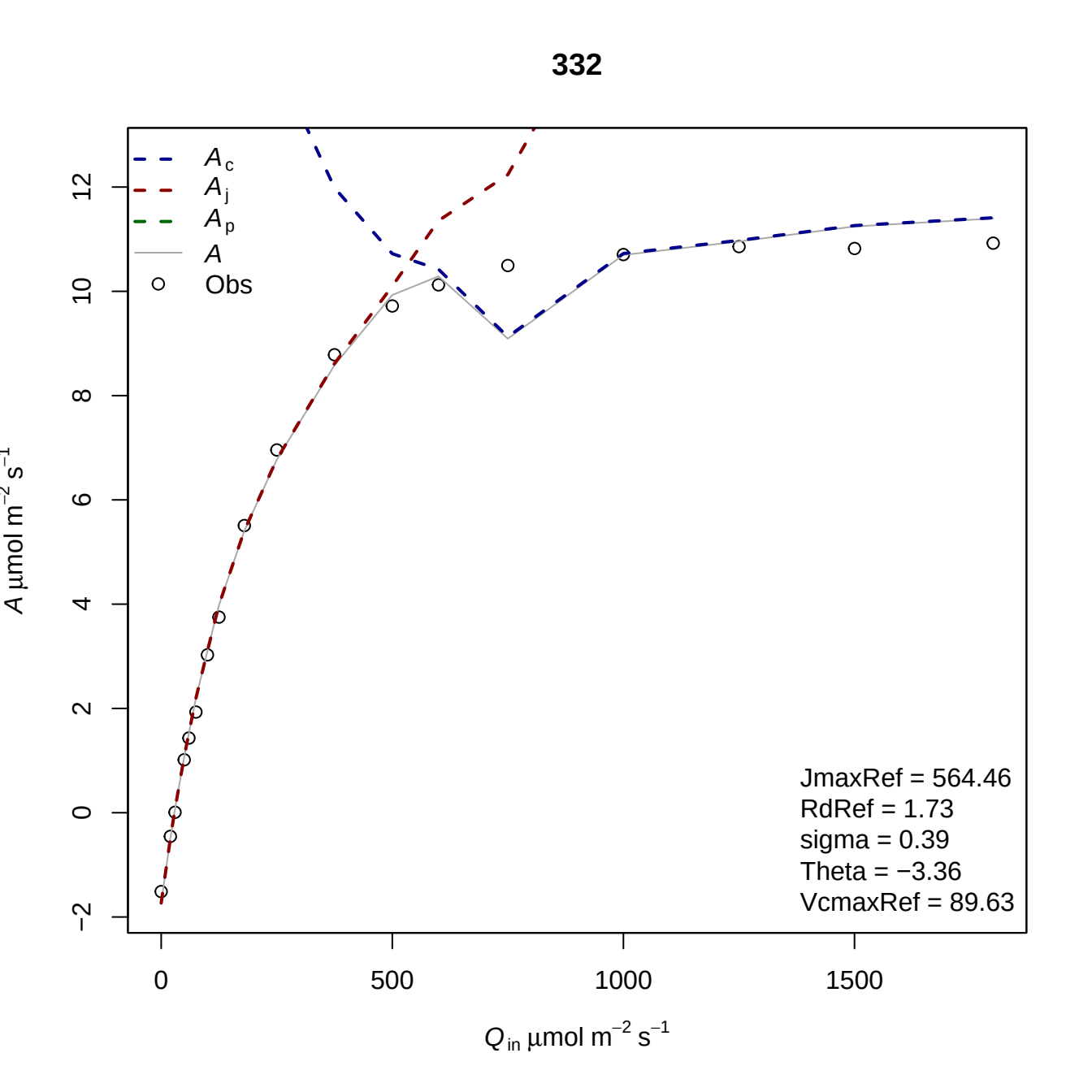

327
A_c
A_j
A_p
A
Obs
JmaxRef = 126.78
RdRef = 1.26
sigma = 0.25
Theta = 0.28
VcmaxRef = 52.47
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

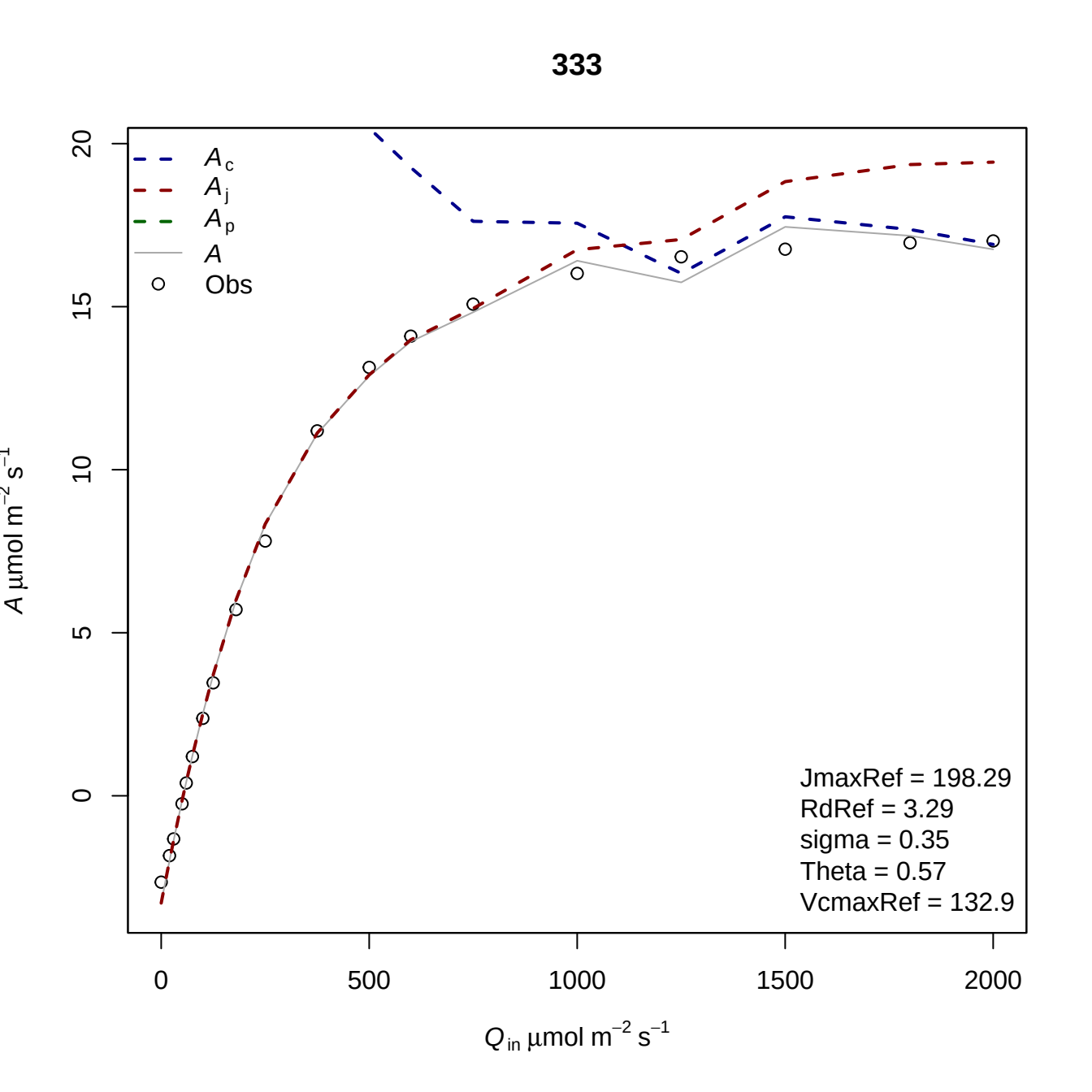

328
A_c
A_j
A_p
A
Obs
JmaxRef = 86.19
RdRef = 1.79
sigma = 0.18
Theta = 0.47
VcmaxRef = 46.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

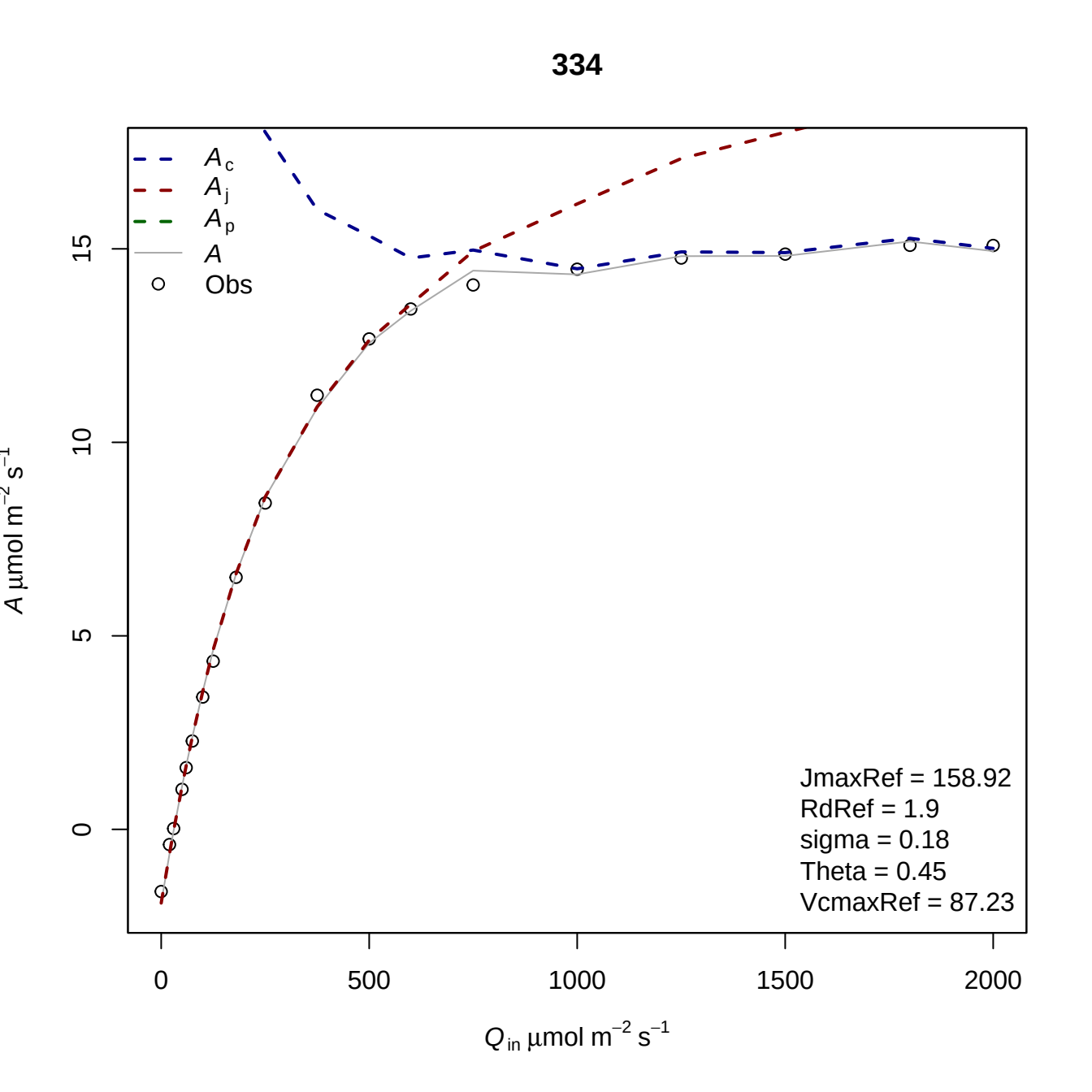

329
A_c
A_j
A_p
A
Obs
JmaxRef = 143.45
RdRef = 2.67
sigma = 0.06
Theta = 0.46
VcmaxRef = 85.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

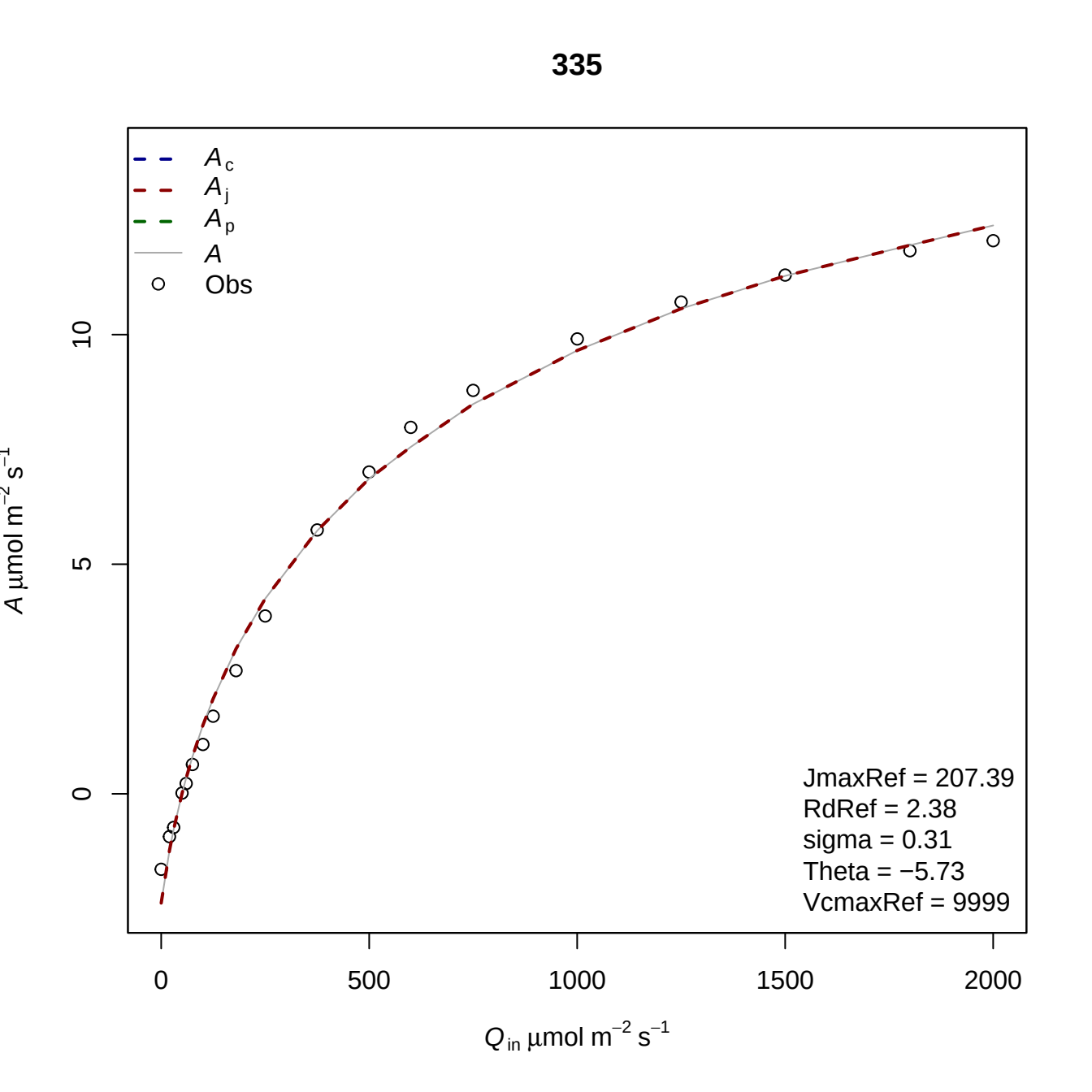

330
A_c
A_j
A_p
A
Obs
JmaxRef = 128.37
RdRef = 2.21
sigma = 0.34
Theta = 0.79
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

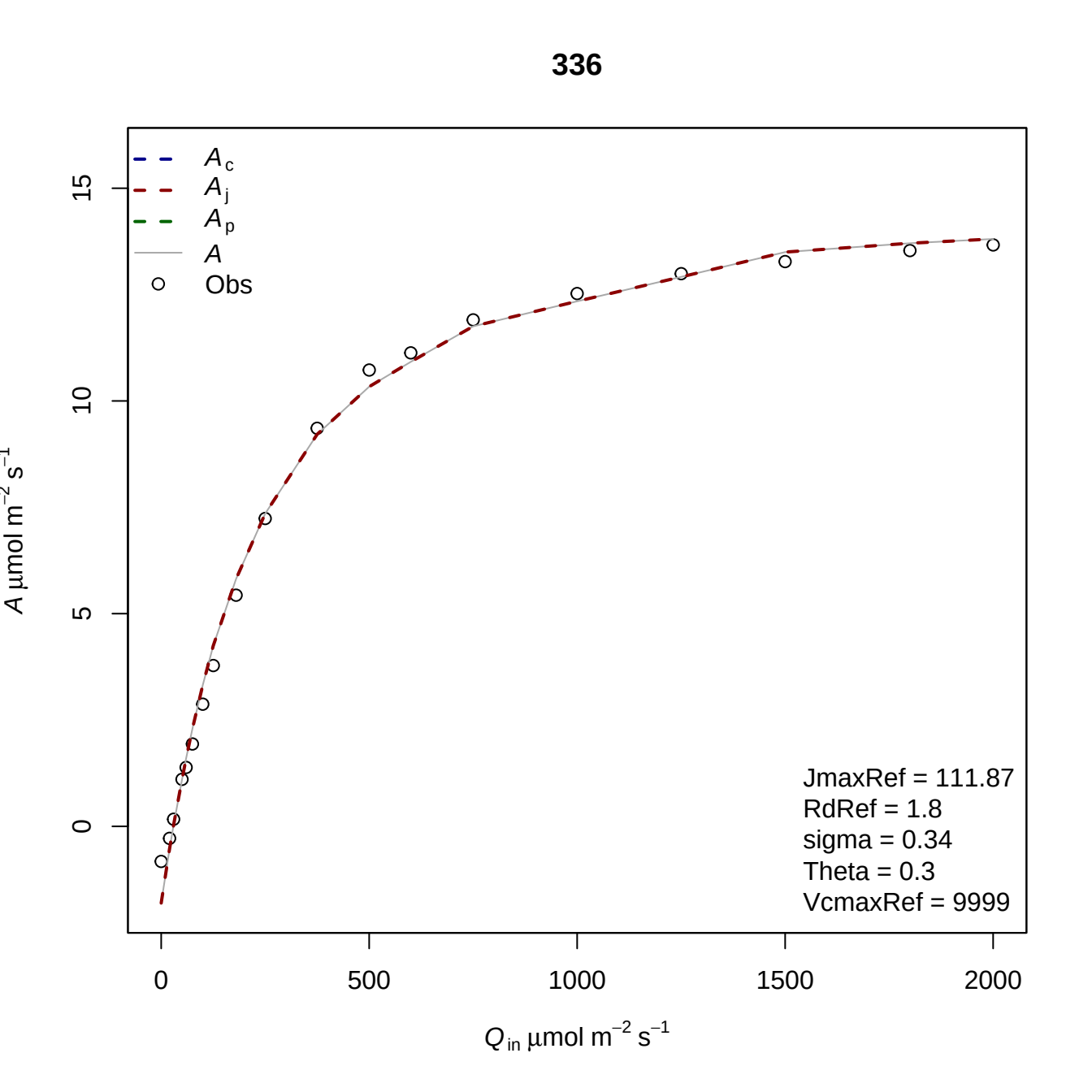

331
A_c
A_j
A_p
A
Obs
JmaxRef = 127.25
RdRef = 2.24
sigma = 0.21
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

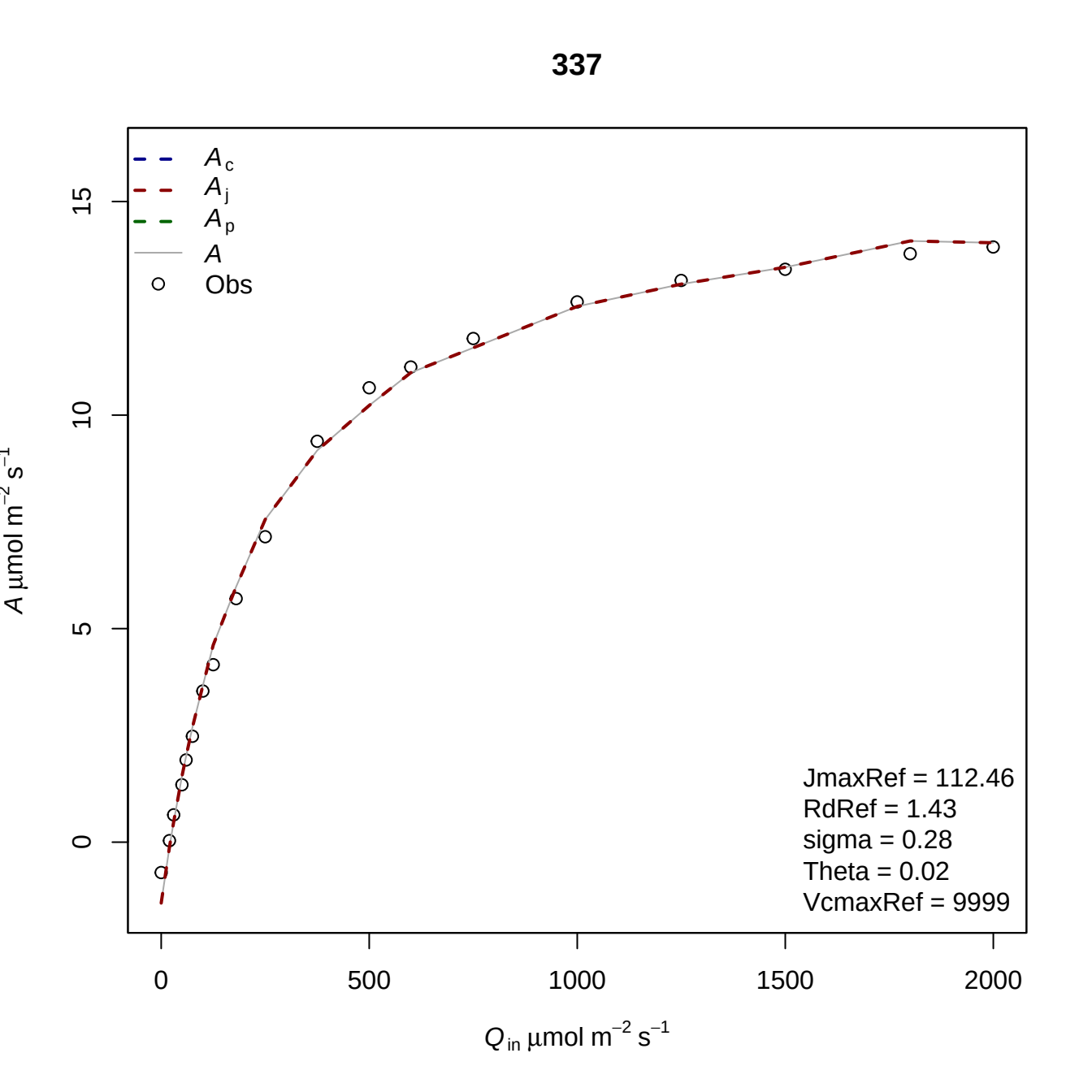

332
A_c
A_j
A_p
A
Obs
JmaxRef = 564.46
RdRef = 1.73
sigma = 0.39
Theta = −3.36
VcmaxRef = 89.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

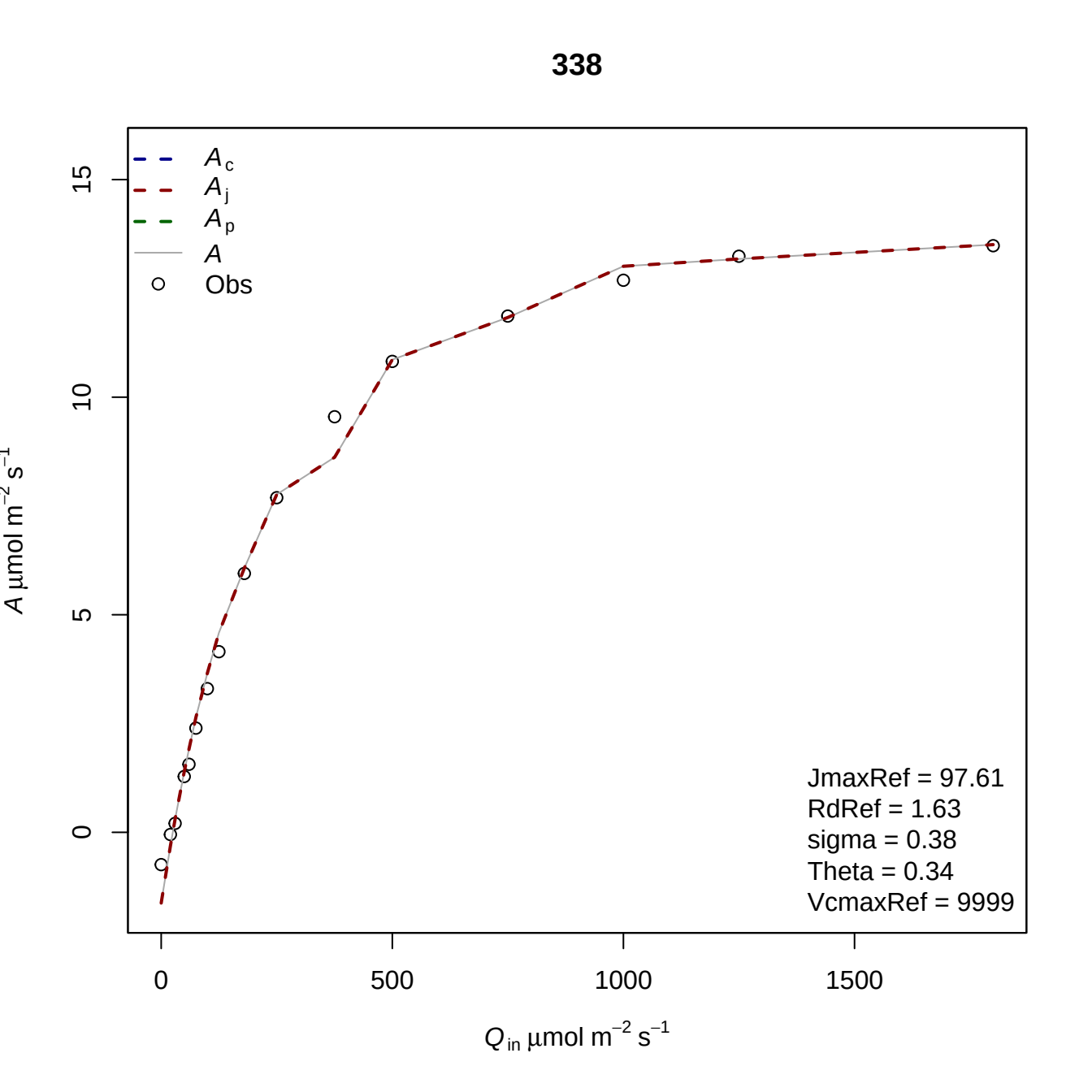

333
A_c
A_j
A_p
A
Obs
JmaxRef = 198.29
RdRef = 3.29
sigma = 0.35
Theta = 0.57
VcmaxRef = 132.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

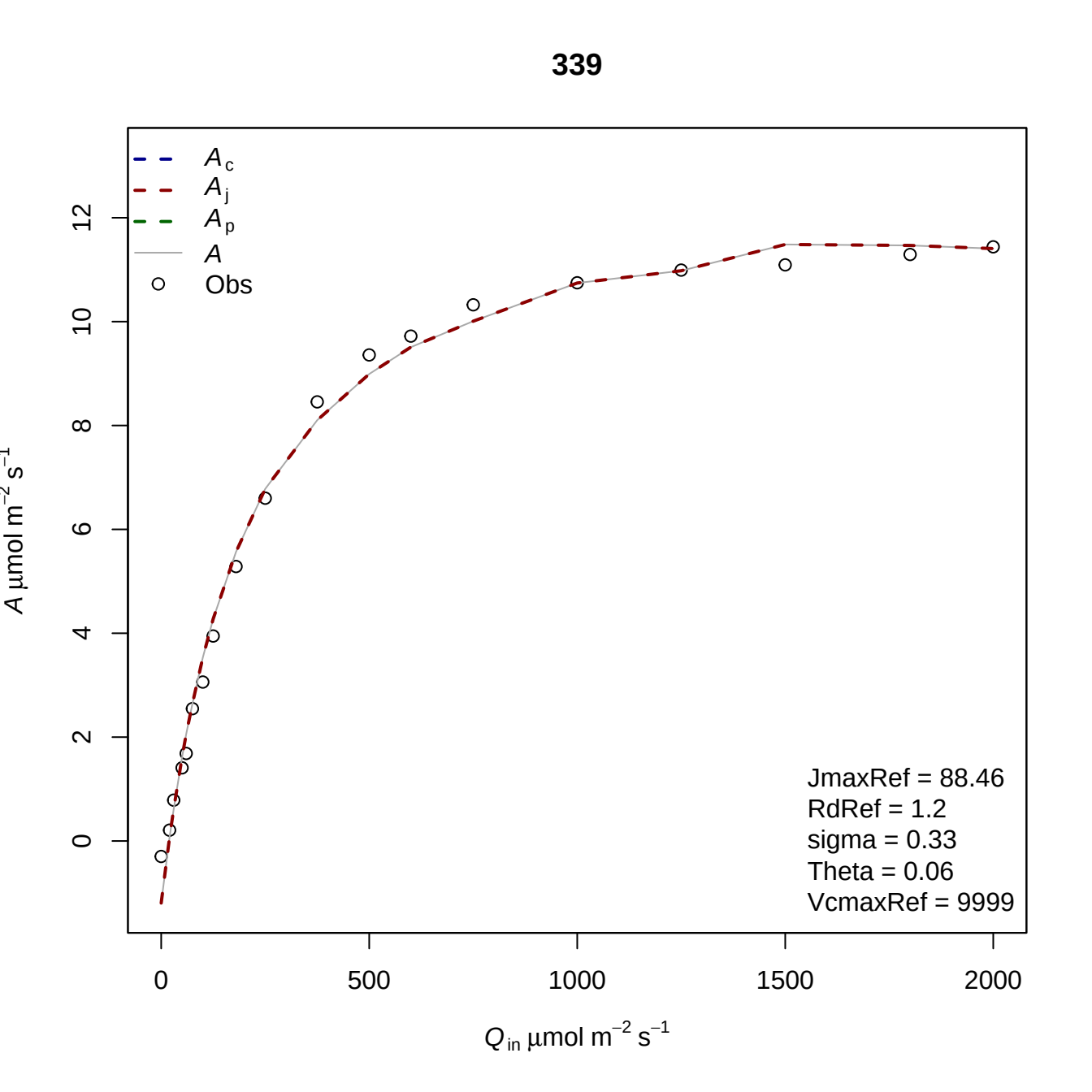

334
A_c
A_j
A_p
A
Obs
JmaxRef = 158.92
RdRef = 1.9
sigma = 0.18
Theta = 0.45
VcmaxRef = 87.23
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

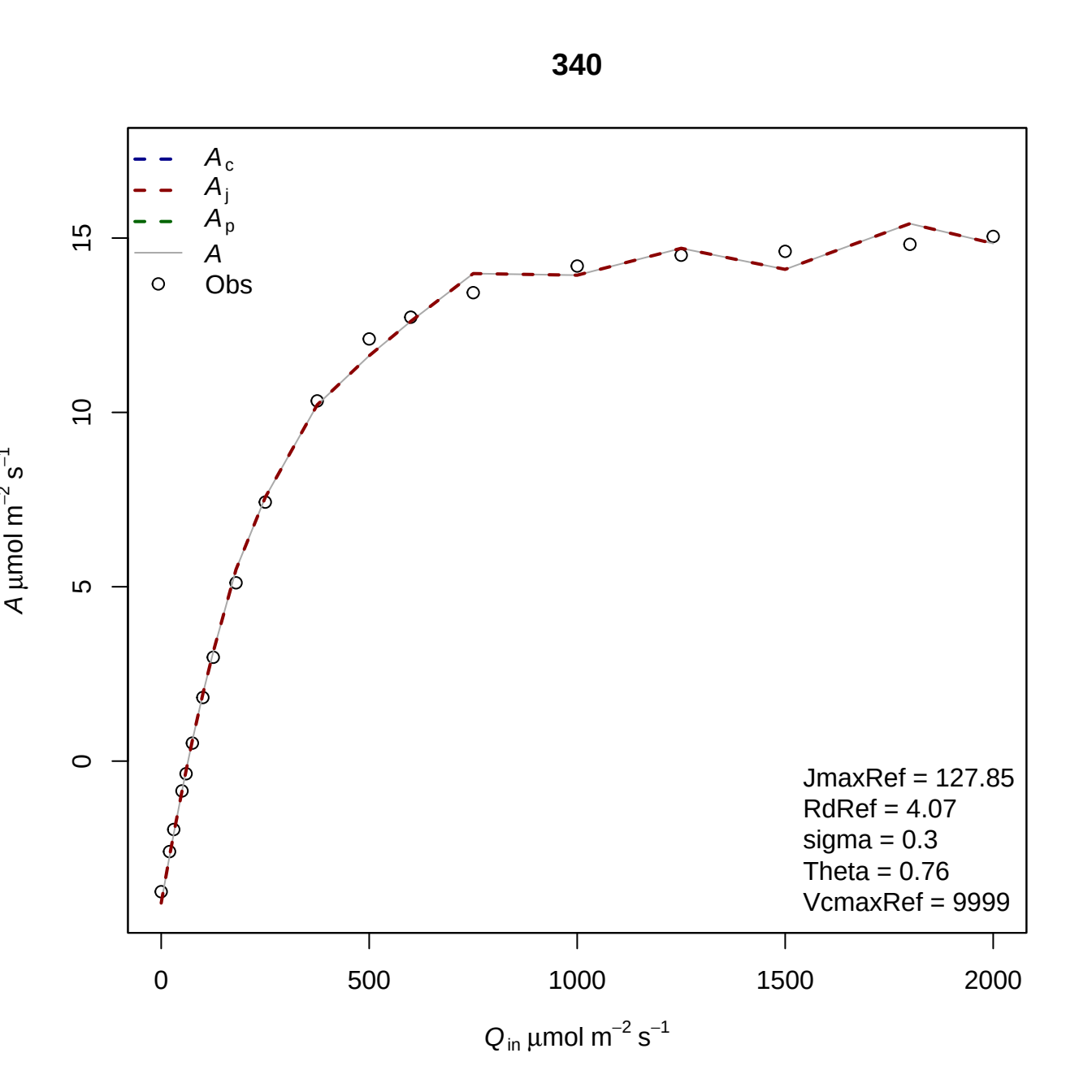

335
A_c
A_j
A_p
A
Obs
JmaxRef = 207.39
RdRef = 2.38
sigma = 0.31
Theta = −5.73
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

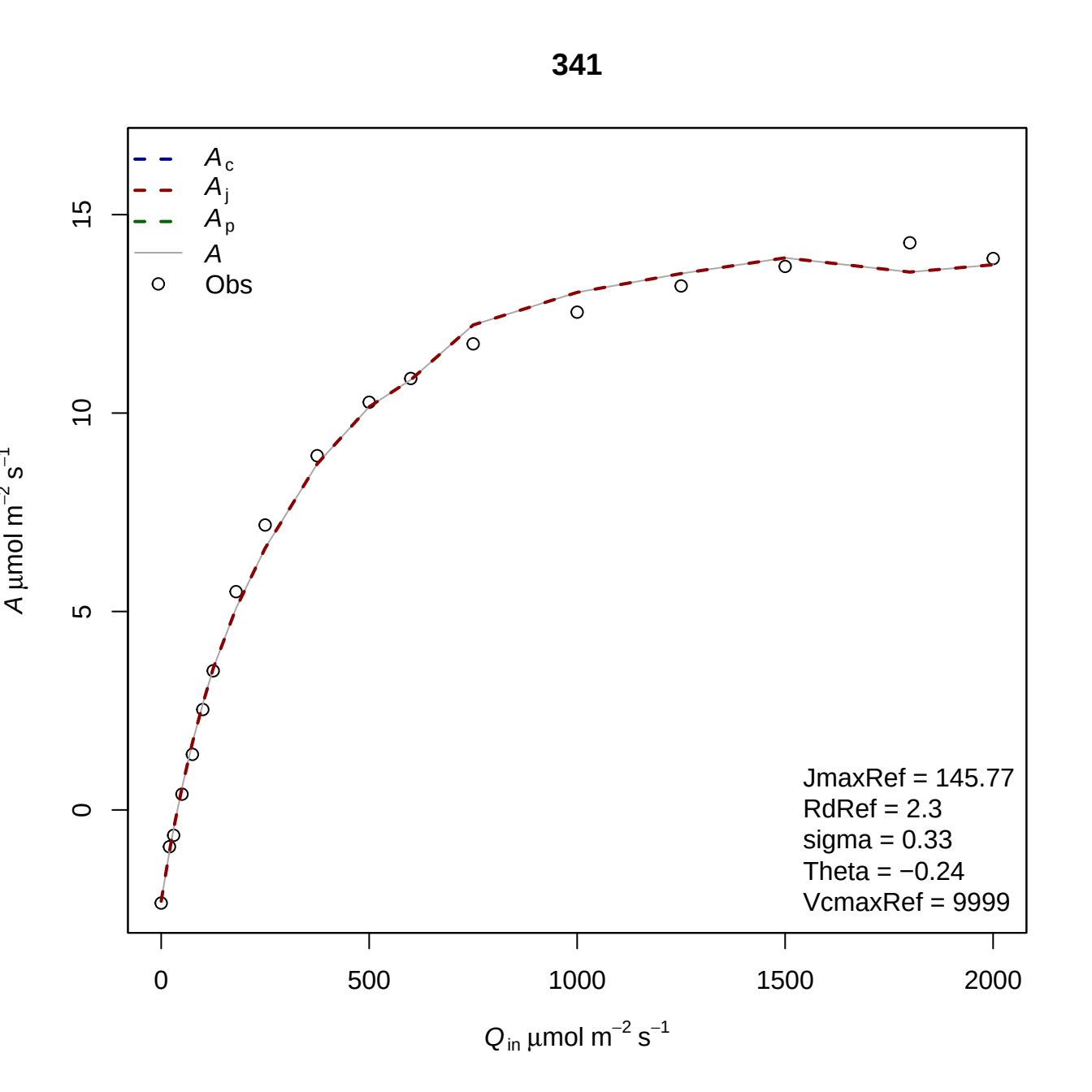

336
A_c
A_j
A_p
A
Obs
JmaxRef = 111.87
RdRef = 1.8
sigma = 0.34
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

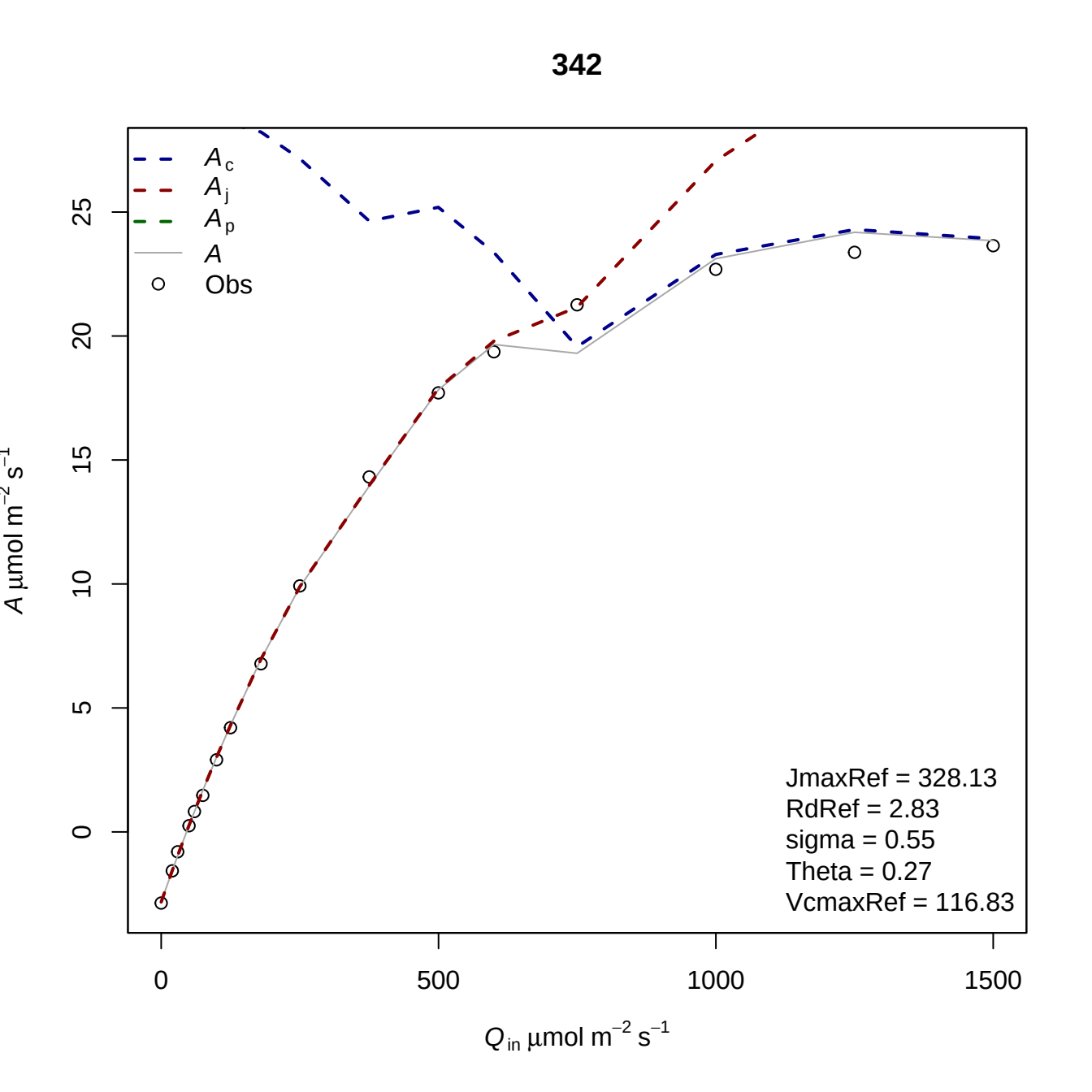

337
A_c
A_j
A_p
A
Obs
JmaxRef = 112.46
RdRef = 1.43
sigma = 0.28
Theta = 0.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

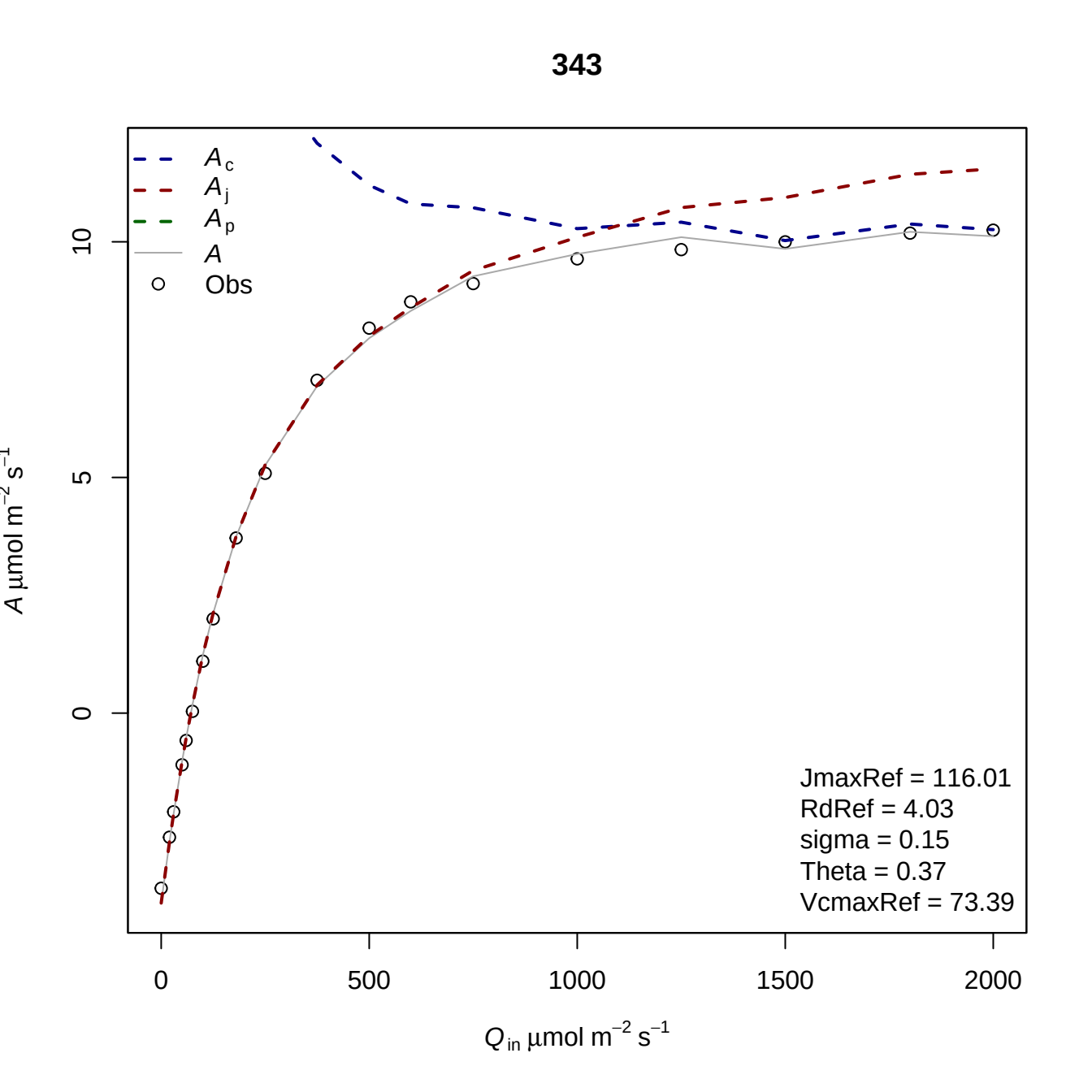

338
A_c
A_j
A_p
A
Obs
JmaxRef = 97.61
RdRef = 1.63
sigma = 0.38
Theta = 0.34
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

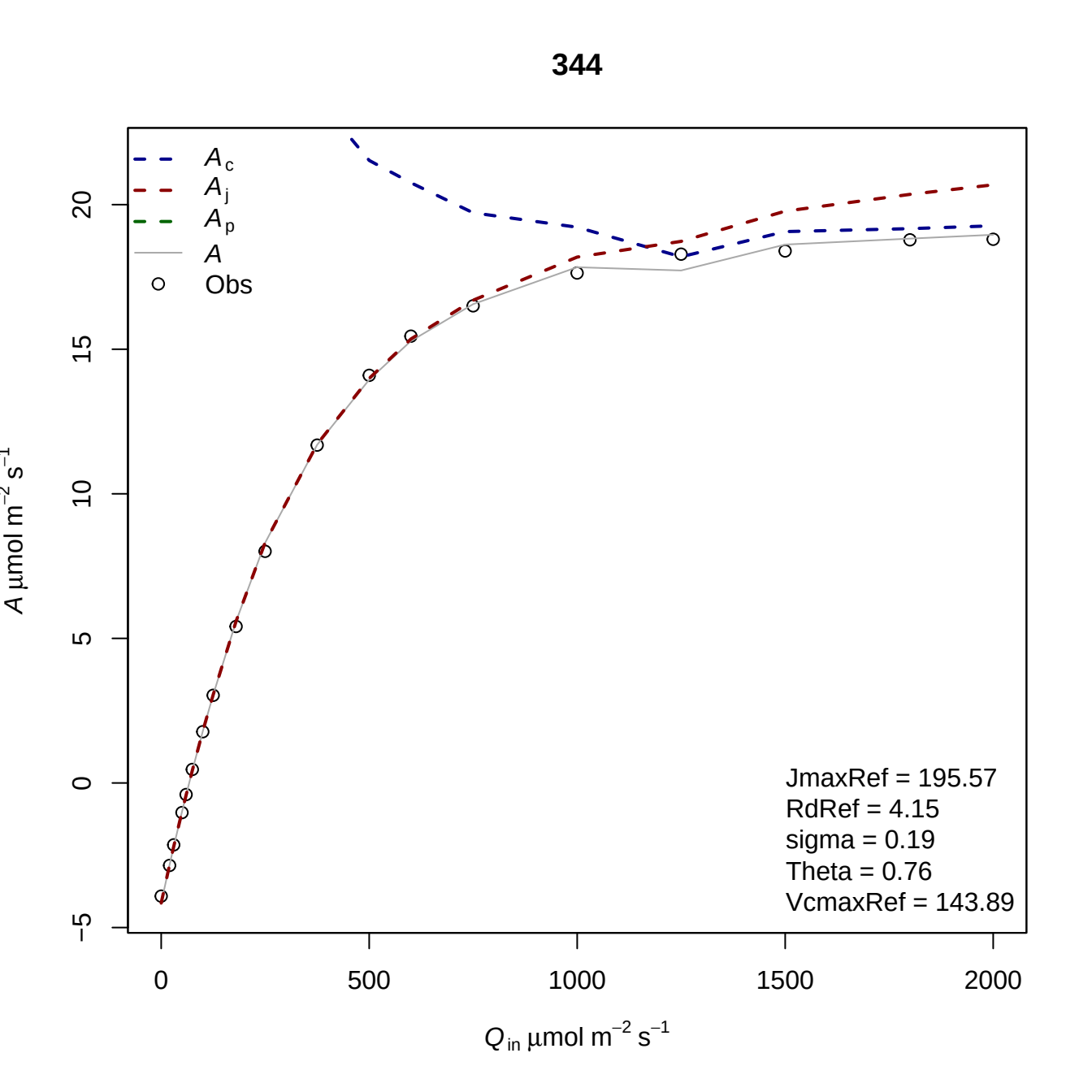

339
A_c
A_j
A_p
A
Obs
JmaxRef = 88.46
RdRef = 1.2
sigma = 0.33
Theta = 0.06
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

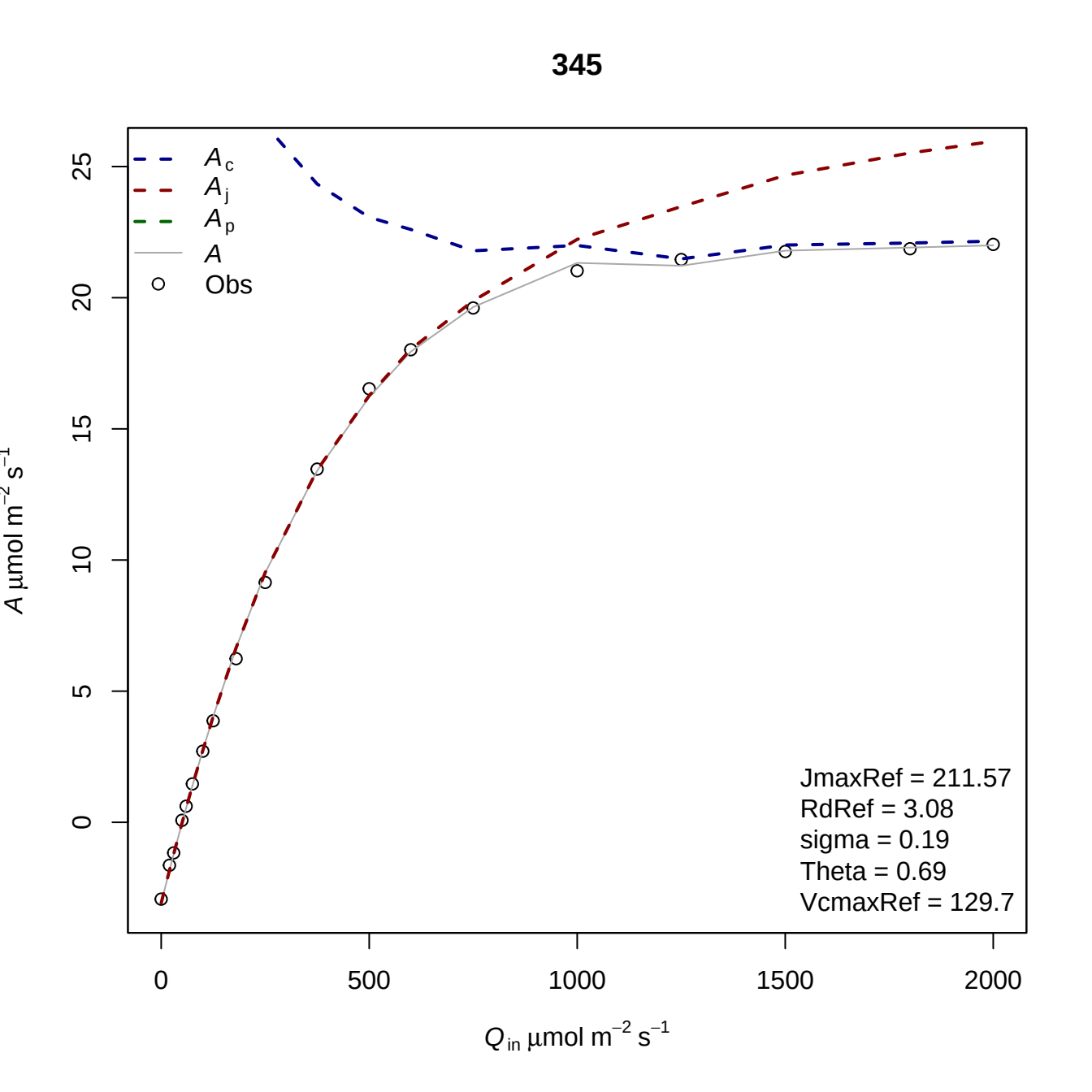

340
A_c
A_j
A_p
A
Obs
JmaxRef = 127.85
RdRef = 4.07
sigma = 0.3
Theta = 0.76
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

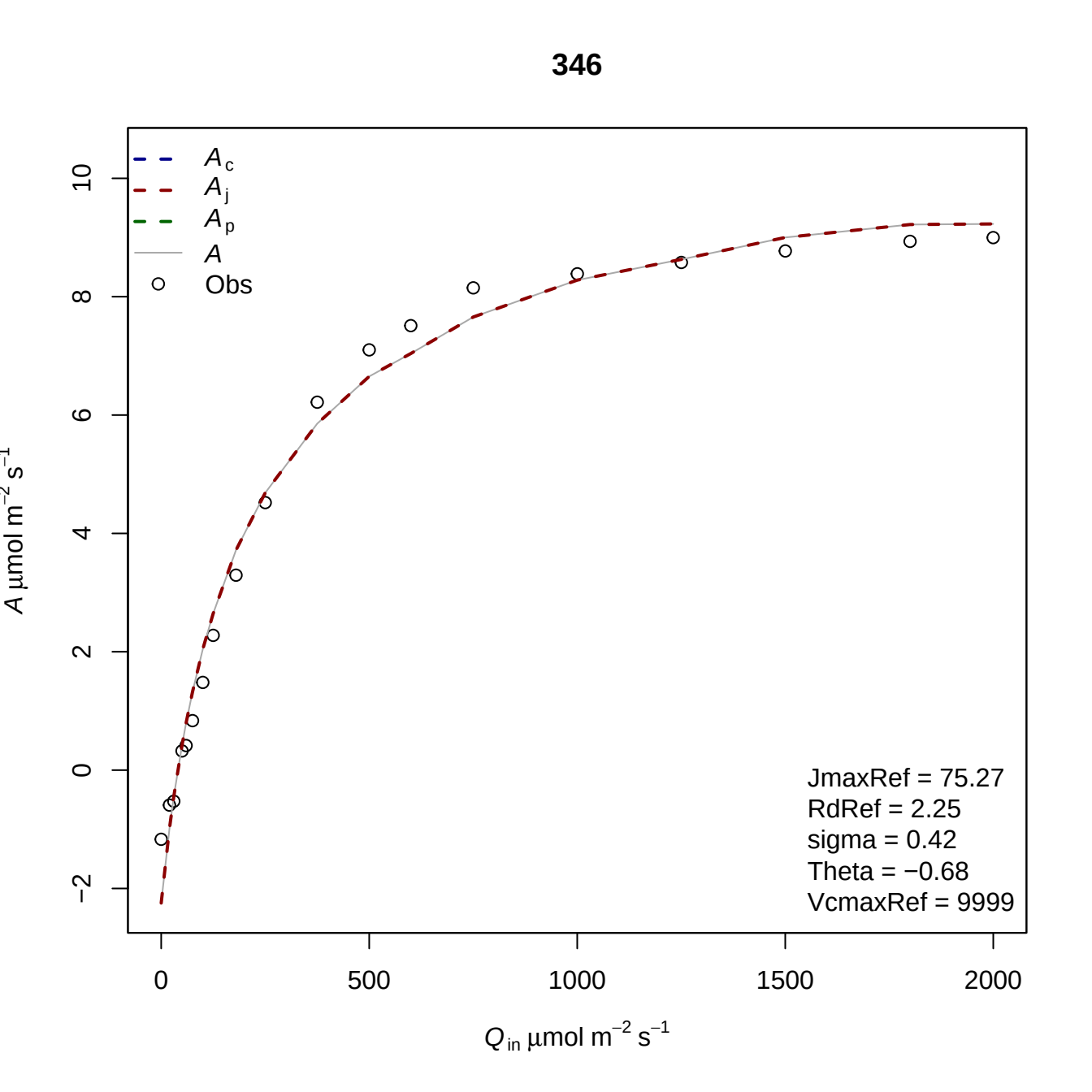

341
A_c
A_j
A_p
A
Obs
JmaxRef = 145.77
RdRef = 2.3
sigma = 0.33
Theta = −0.24
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

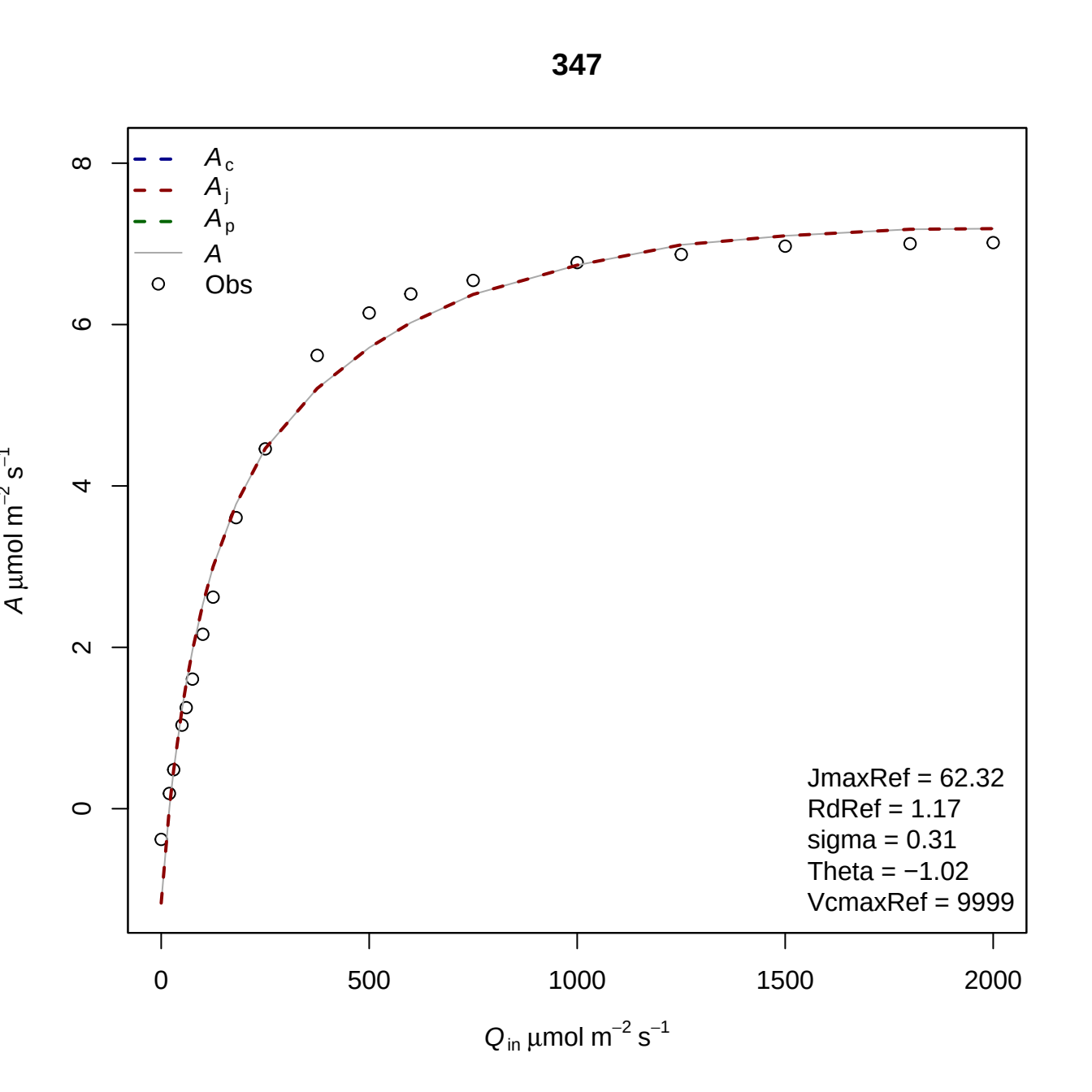

342
A_c
A_j
A_p
A
Obs
JmaxRef = 328.13
RdRef = 2.83
sigma = 0.55
Theta = 0.27
VcmaxRef = 116.83
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

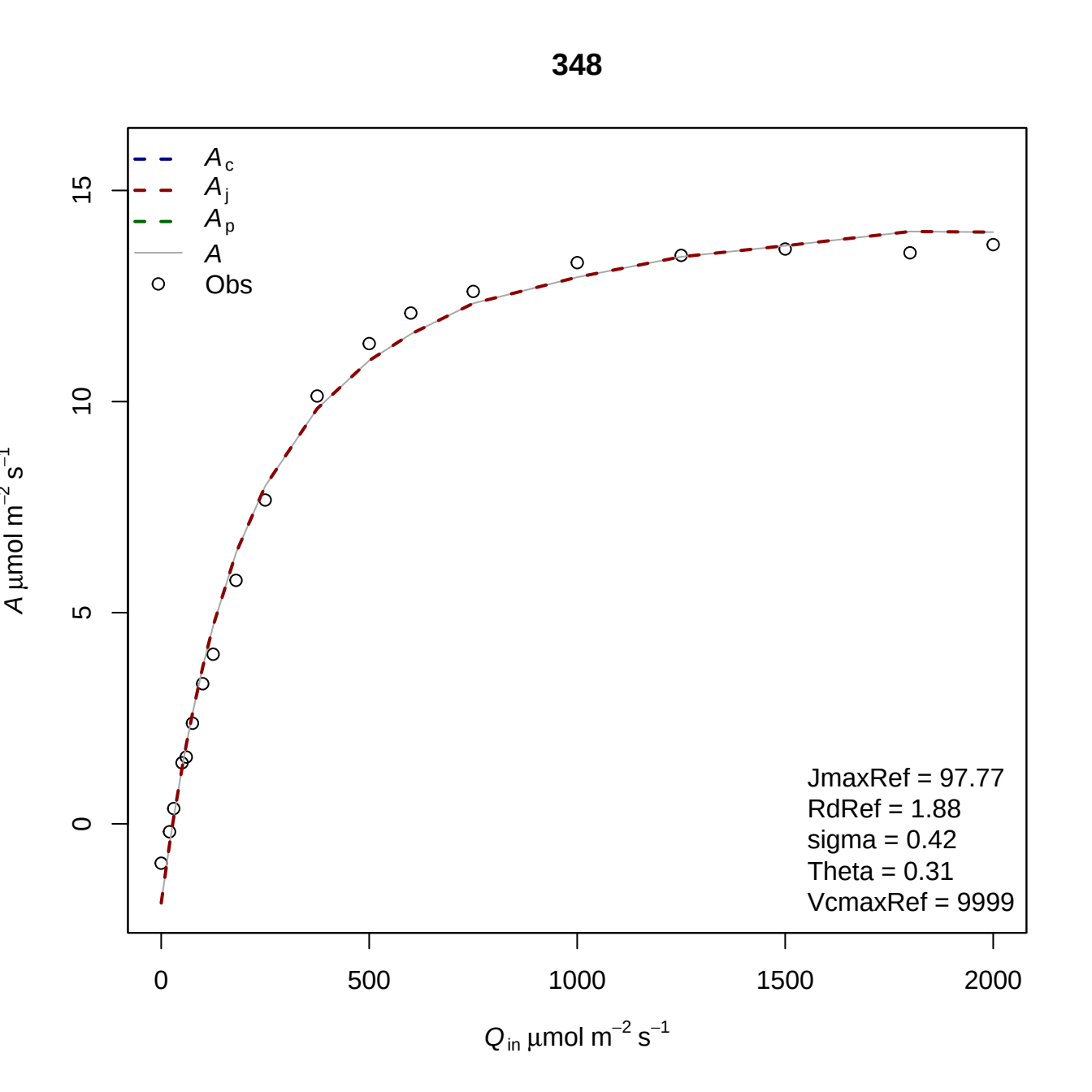

343
A_c
A_j
A_p
A
Obs
JmaxRef = 116.01
RdRef = 4.03
sigma = 0.15
Theta = 0.37
VcmaxRef = 73.39
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

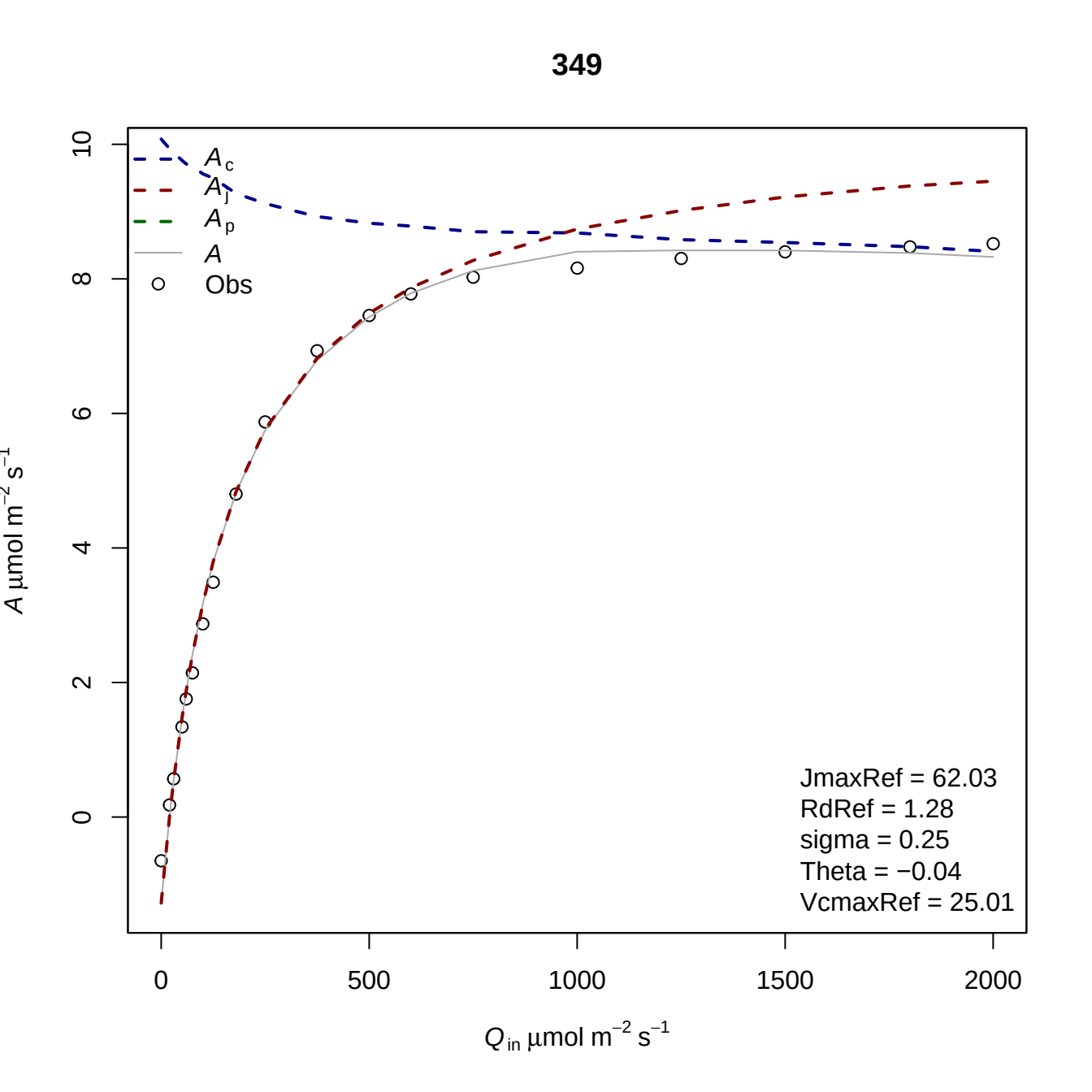

344
A_c
A_j
A_p
A
Obs
JmaxRef = 195.57
RdRef = 4.15
sigma = 0.19
Theta = 0.76
VcmaxRef = 143.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

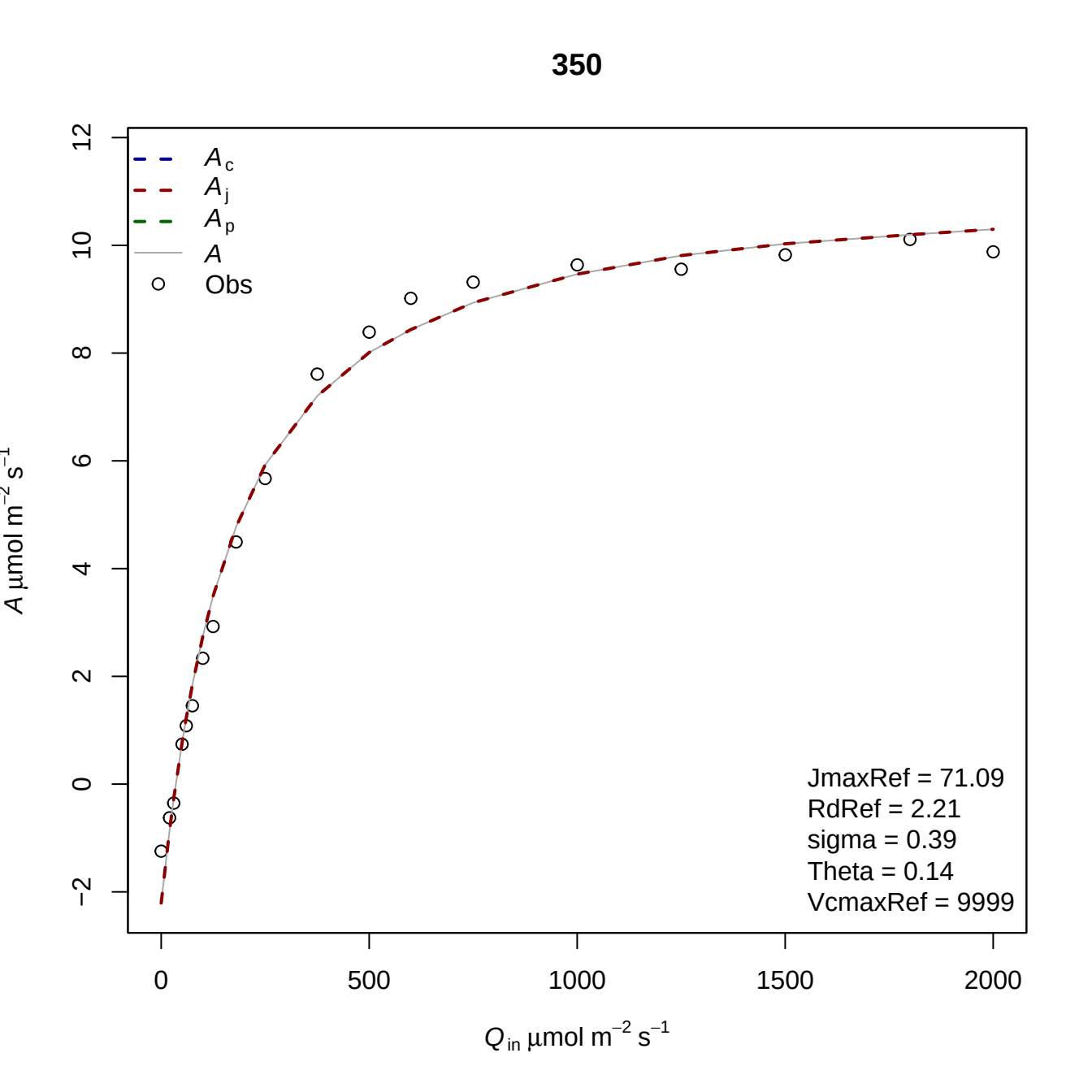

345
A_c
A_j
A_p
A
Obs
JmaxRef = 211.57
RdRef = 3.08
sigma = 0.19
Theta = 0.69
VcmaxRef = 129.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

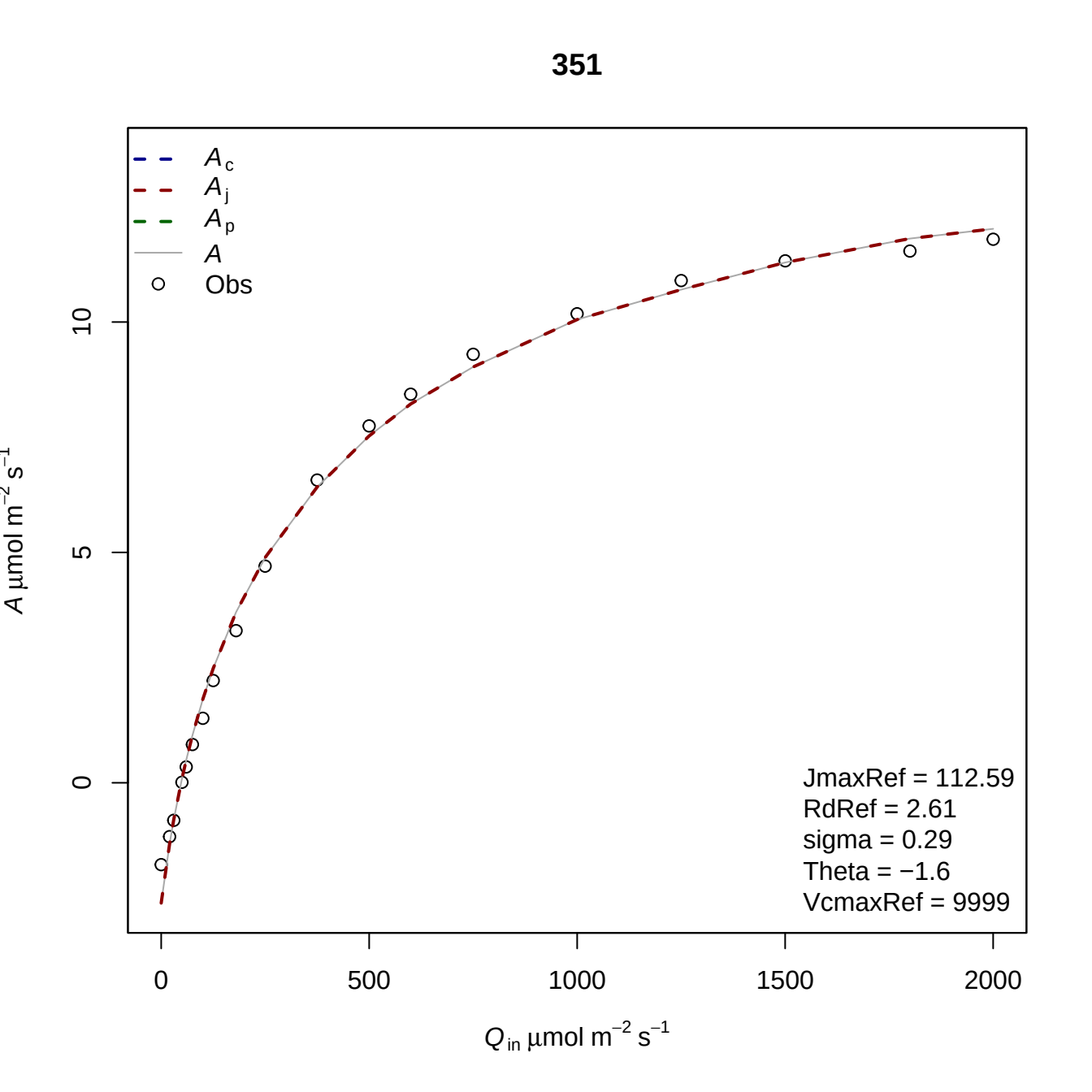

346
A_c
A_j
A_p
A
Obs
JmaxRef = 75.27
RdRef = 2.25
sigma = 0.42
Theta = −0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

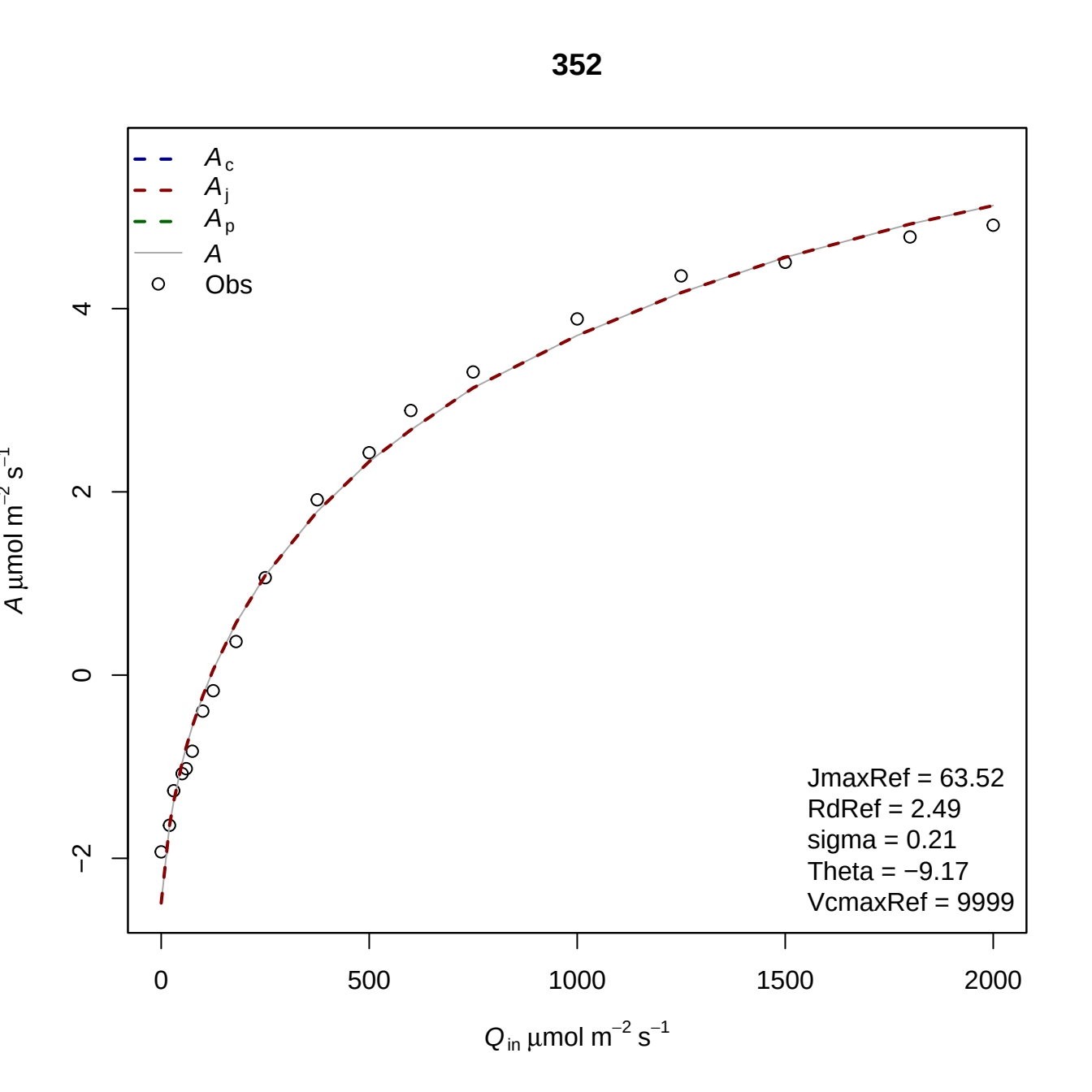

347
A_c
A_j
A_p
A
Obs
JmaxRef = 62.32
RdRef = 1.17
sigma = 0.31
Theta = −1.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

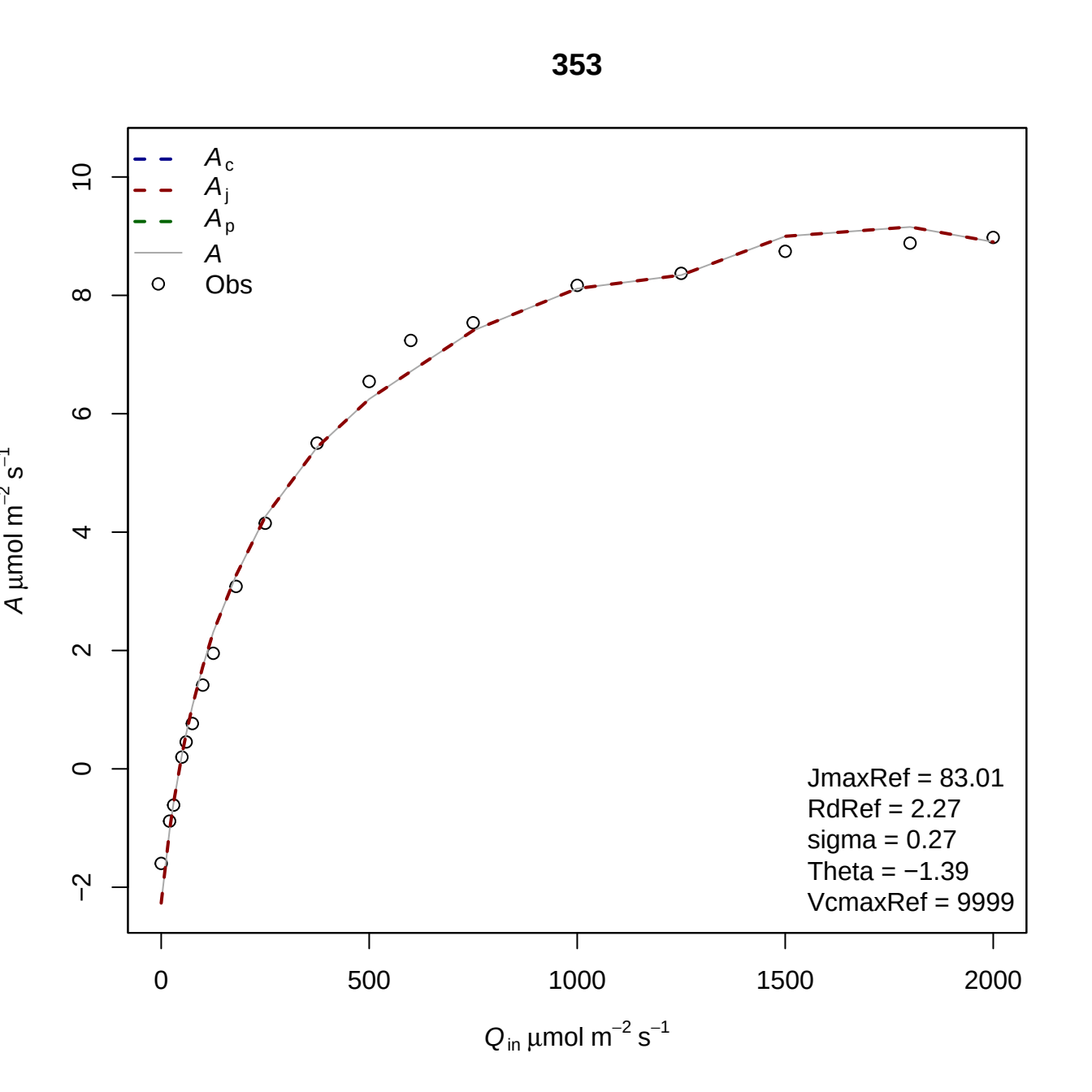

348
A_c
A_j
A_p
A
Obs
JmaxRef = 97.77
RdRef = 1.88
sigma = 0.42
Theta = 0.31
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

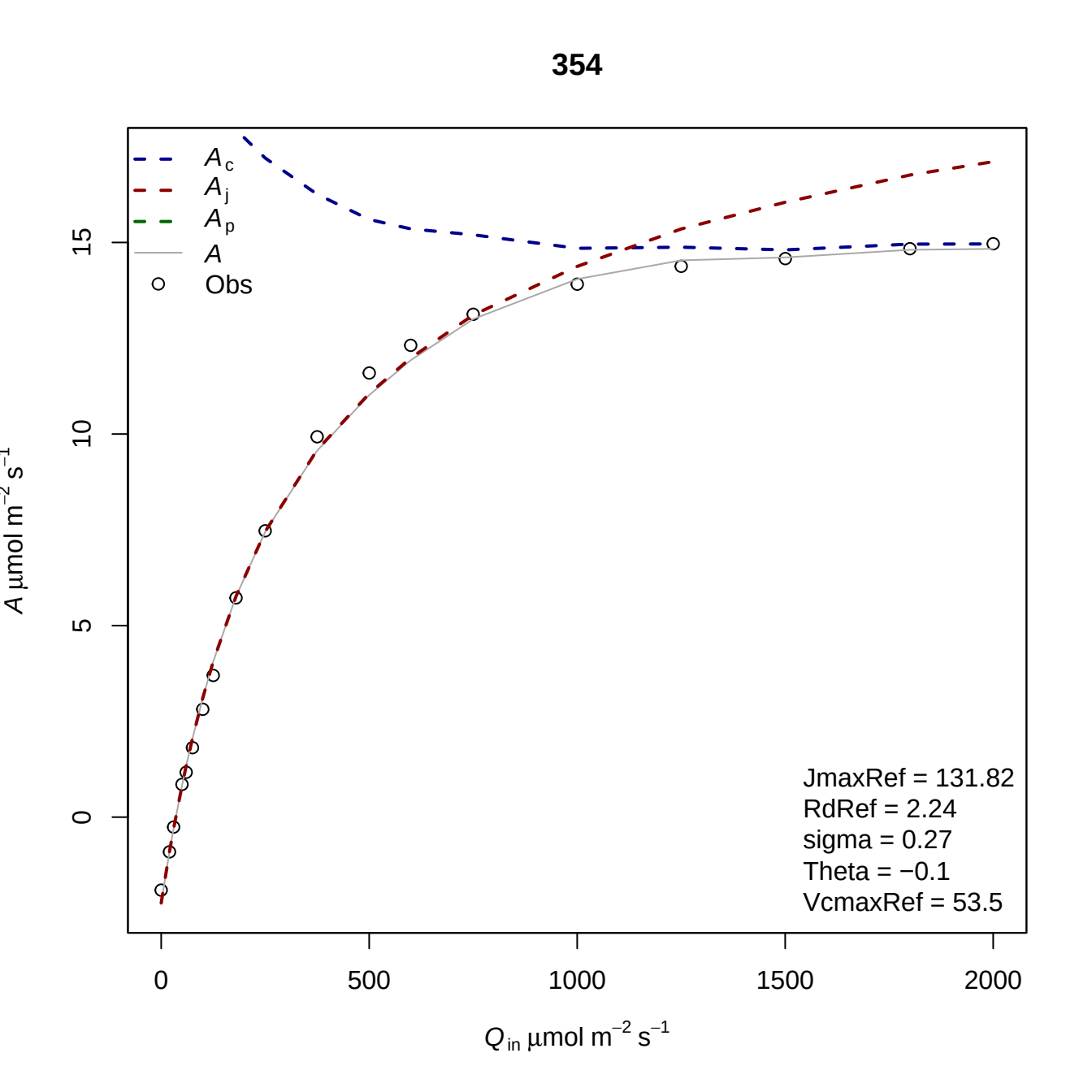

349
A_c
A_j
A_p
A
Obs
JmaxRef = 62.03
RdRef = 1.28
sigma = 0.25
Theta = −0.04
VcmaxRef = 25.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

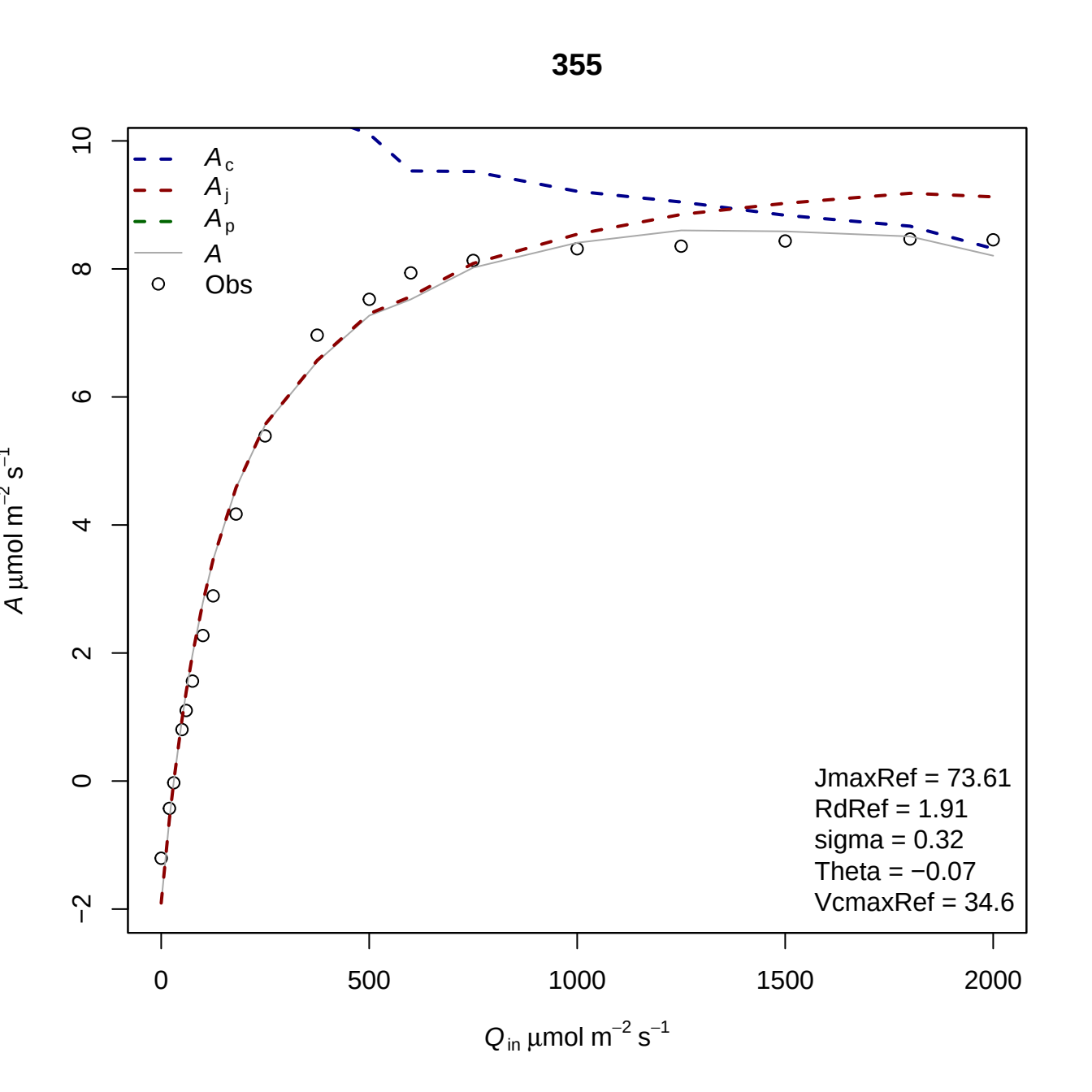

350
A_c
A_j
A_p
A
Obs
JmaxRef = 71.09
RdRef = 2.21
sigma = 0.39
Theta = 0.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

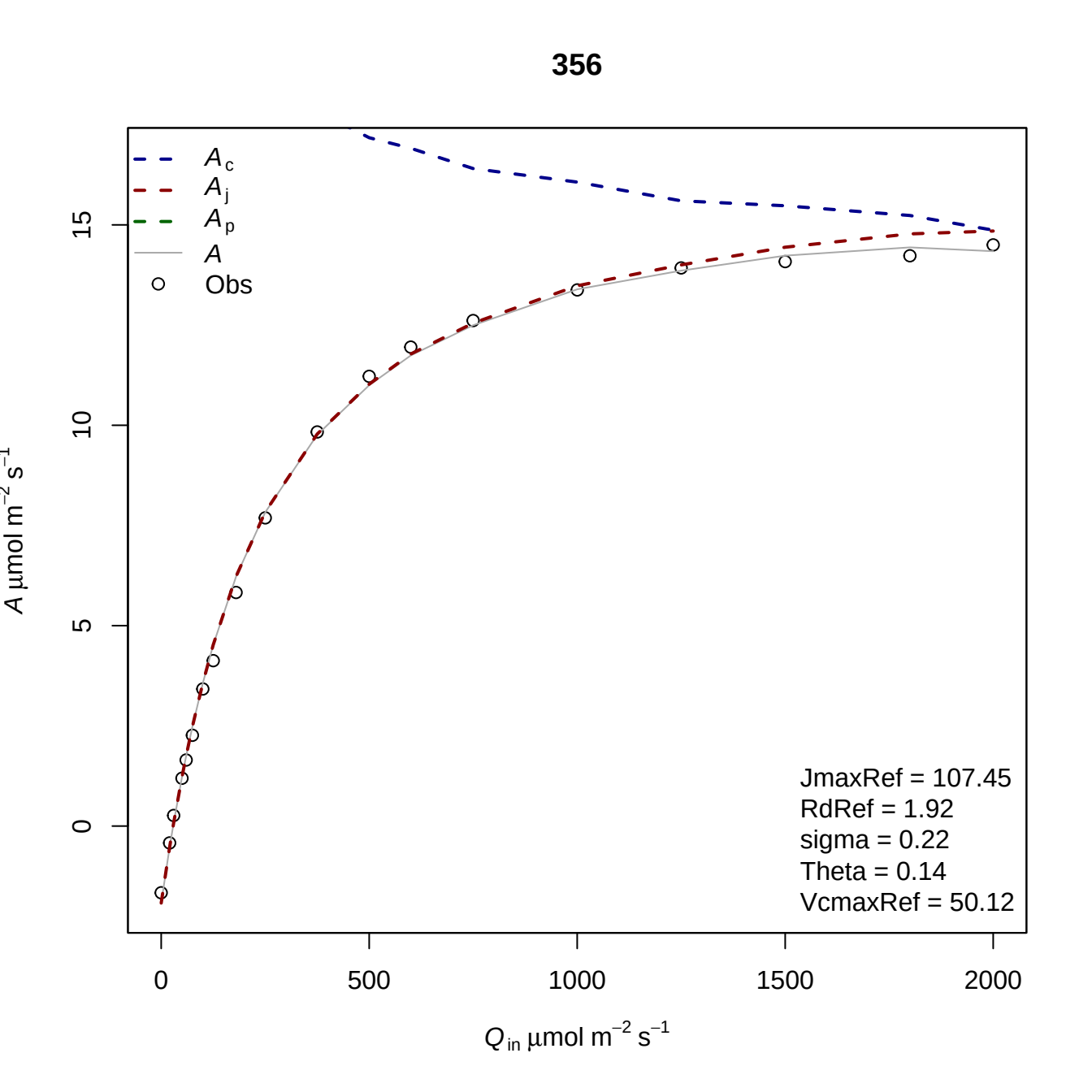

351
A_c
A_j
A_p
A
Obs
JmaxRef = 112.59
RdRef = 2.61
sigma = 0.29
Theta = −1.6
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

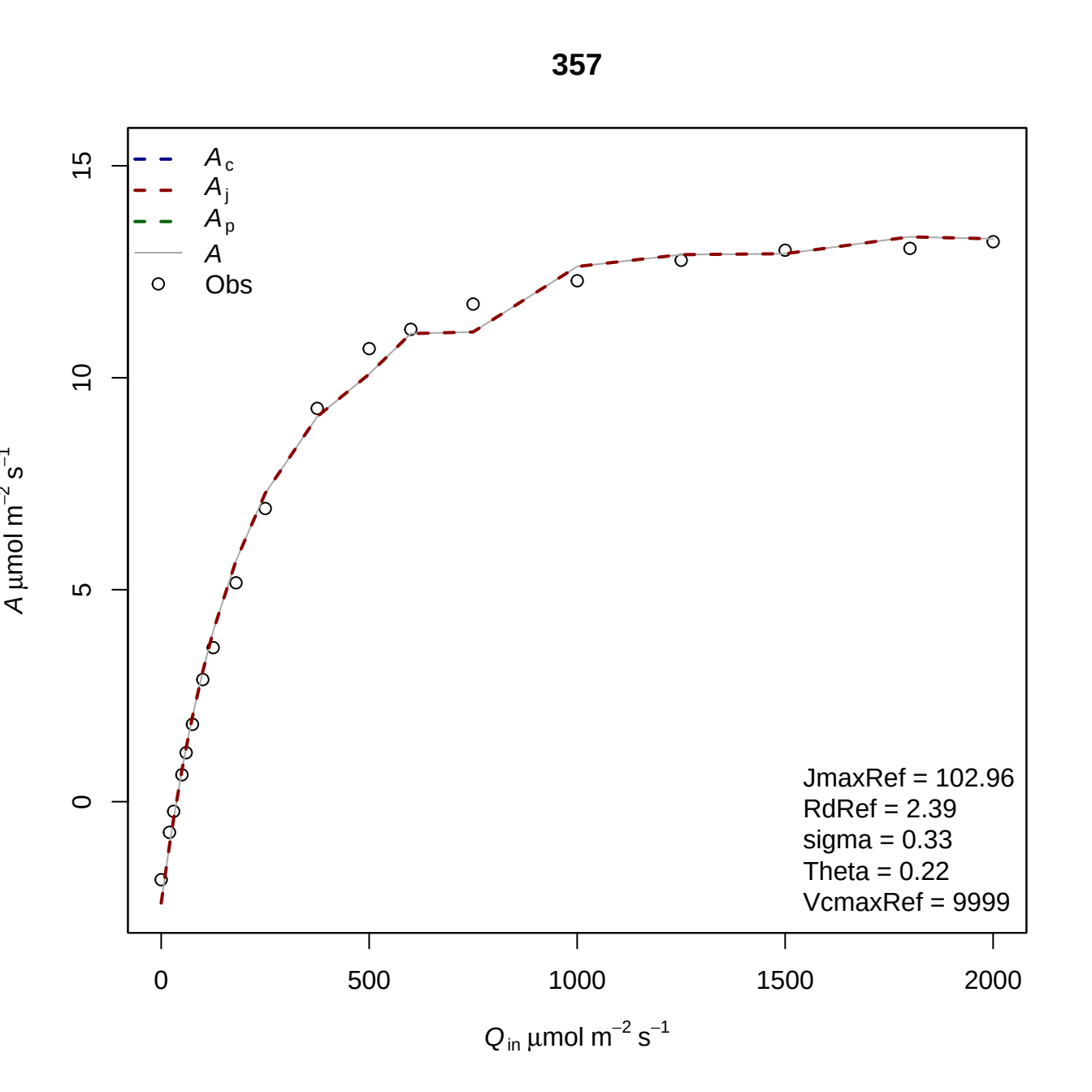

352
A_c
A_j
A_p
A
Obs
JmaxRef = 63.52
RdRef = 2.49
sigma = 0.21
Theta = −9.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

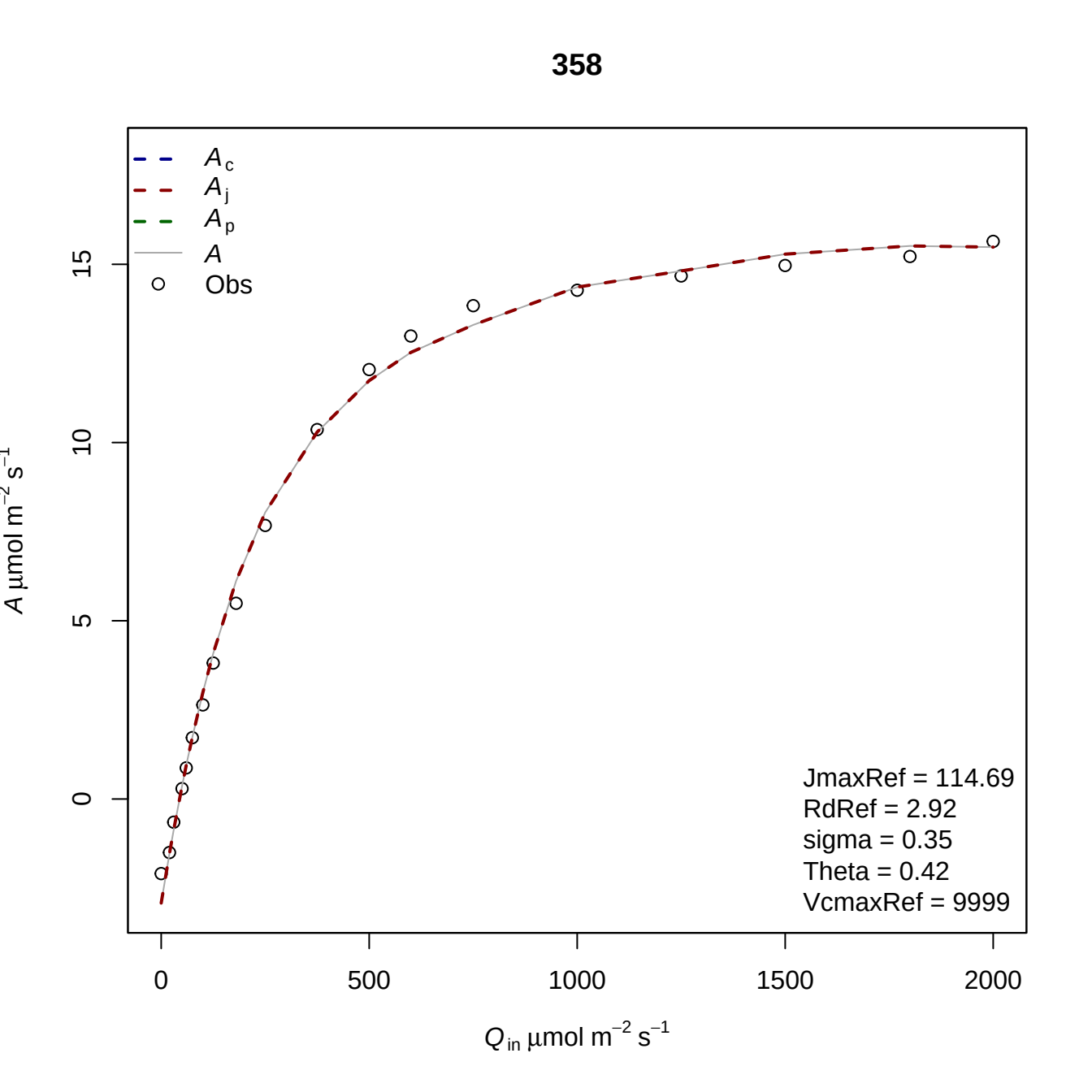

353
A_c
A_j
A_p
A
Obs
JmaxRef = 83.01
RdRef = 2.27
sigma = 0.27
Theta = −1.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

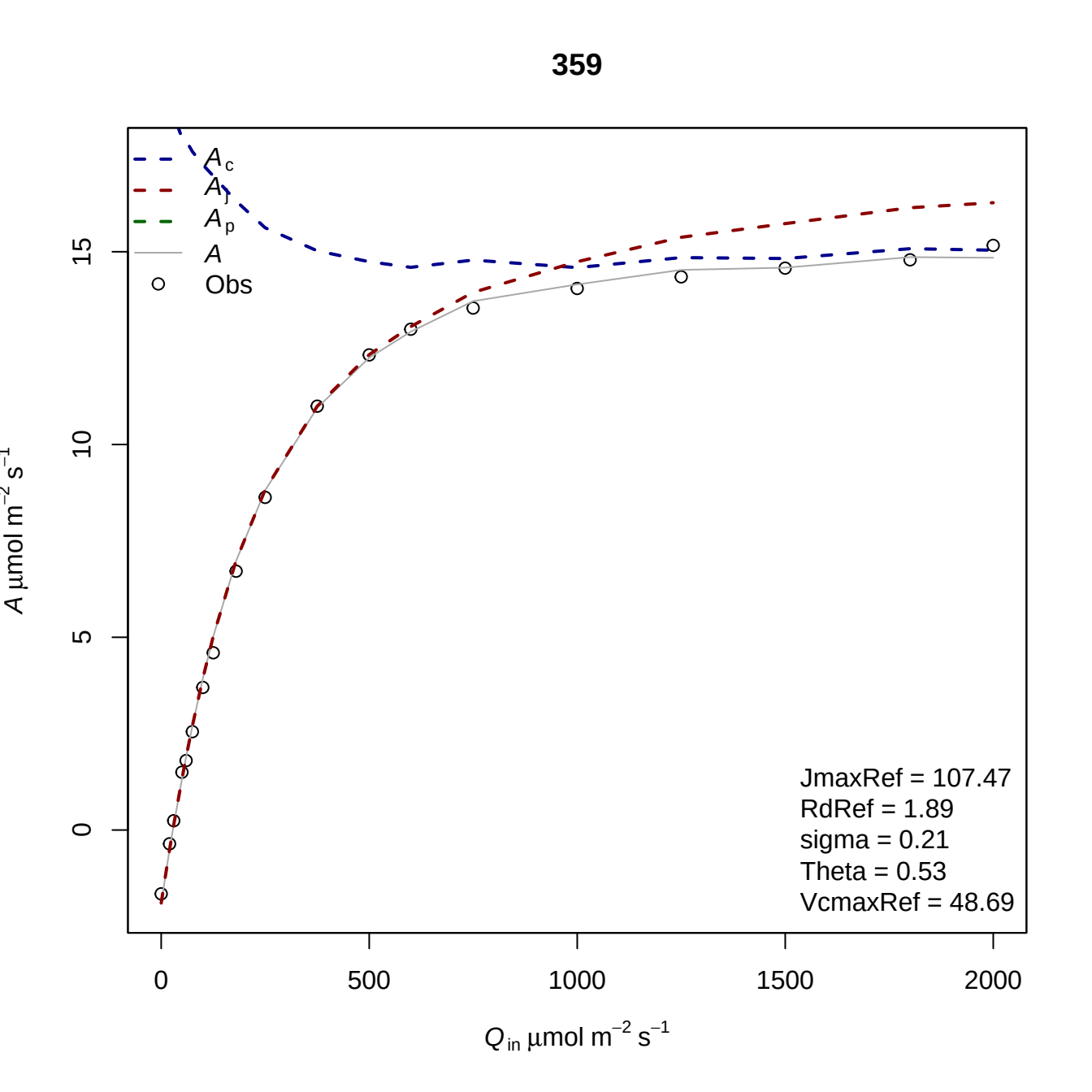

354
A_c
A_j
A_p
A
Obs
JmaxRef = 131.82
RdRef = 2.24
sigma = 0.27
Theta = −0.1
VcmaxRef = 53.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

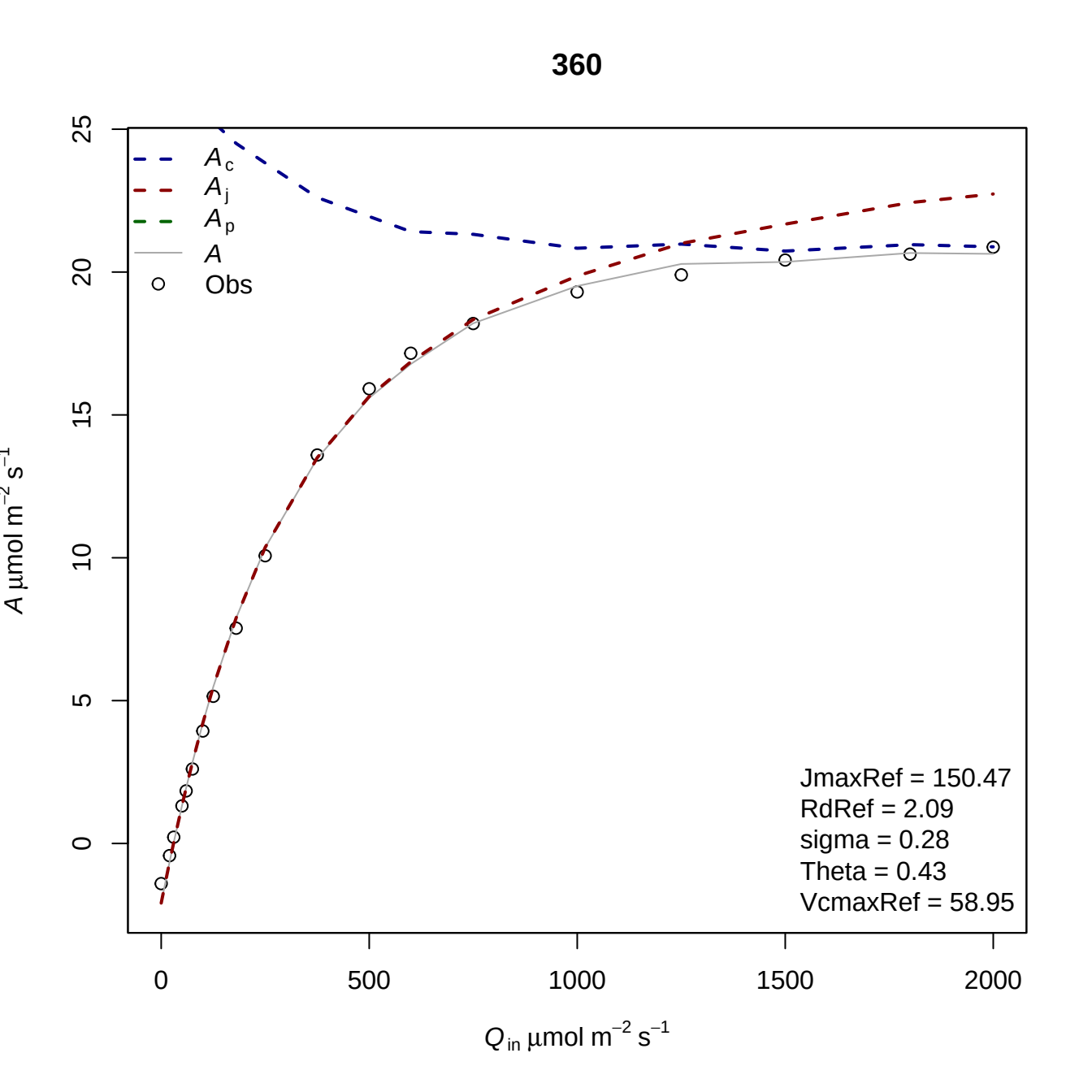

355
A_c
A_j
A_p
A
Obs
JmaxRef = 73.61
RdRef = 1.91
sigma = 0.32
Theta = −0.07
VcmaxRef = 34.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

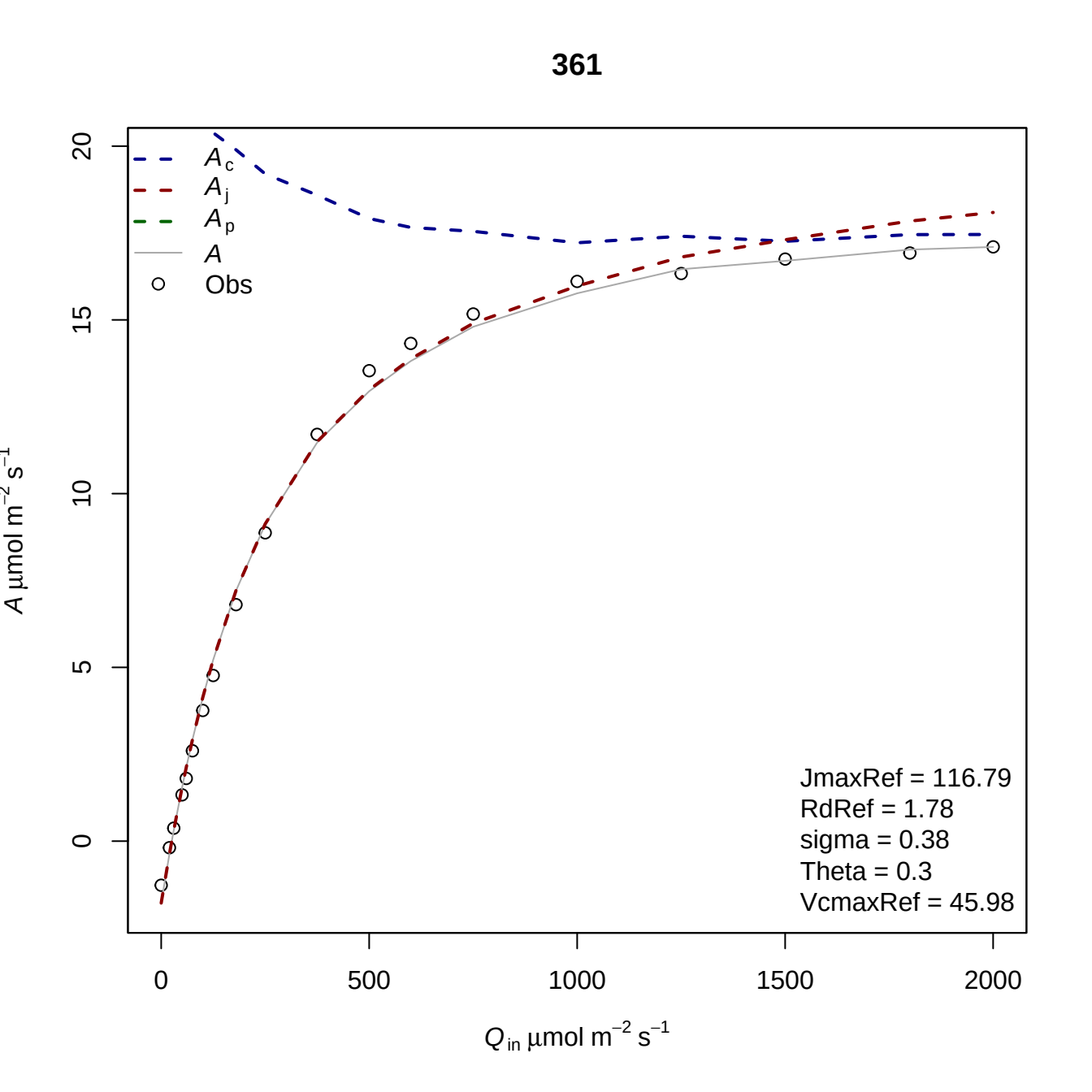

356
A_c
A_j
A_p
A
Obs
JmaxRef = 107.45
RdRef = 1.92
sigma = 0.22
Theta = 0.14
VcmaxRef = 50.12
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

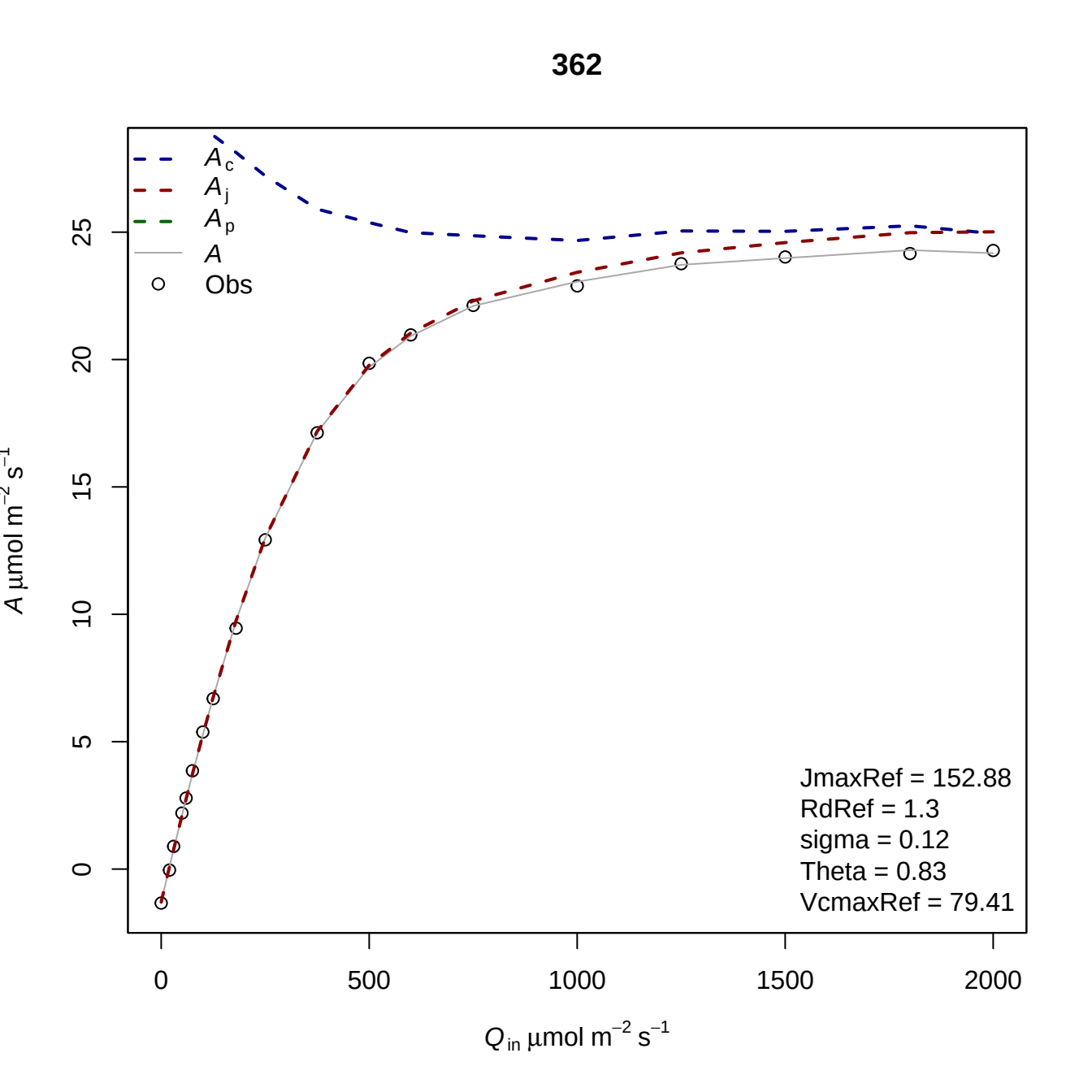

357
A_c
A_j
A_p
A
Obs
JmaxRef = 102.96
RdRef = 2.39
sigma = 0.33
Theta = 0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

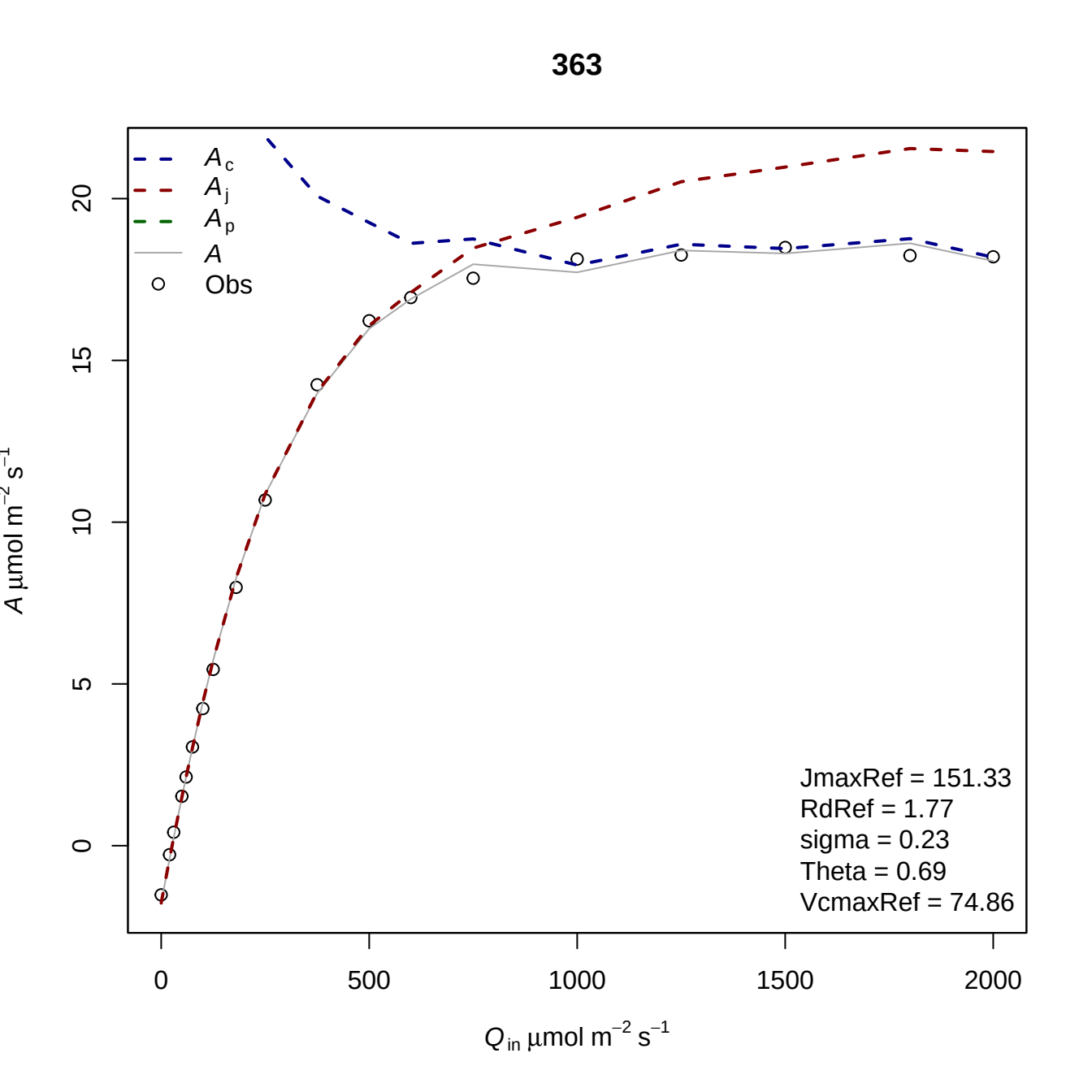

358
A_c
A_j
A_p
A
Obs
JmaxRef = 114.69
RdRef = 2.92
sigma = 0.35
Theta = 0.42
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

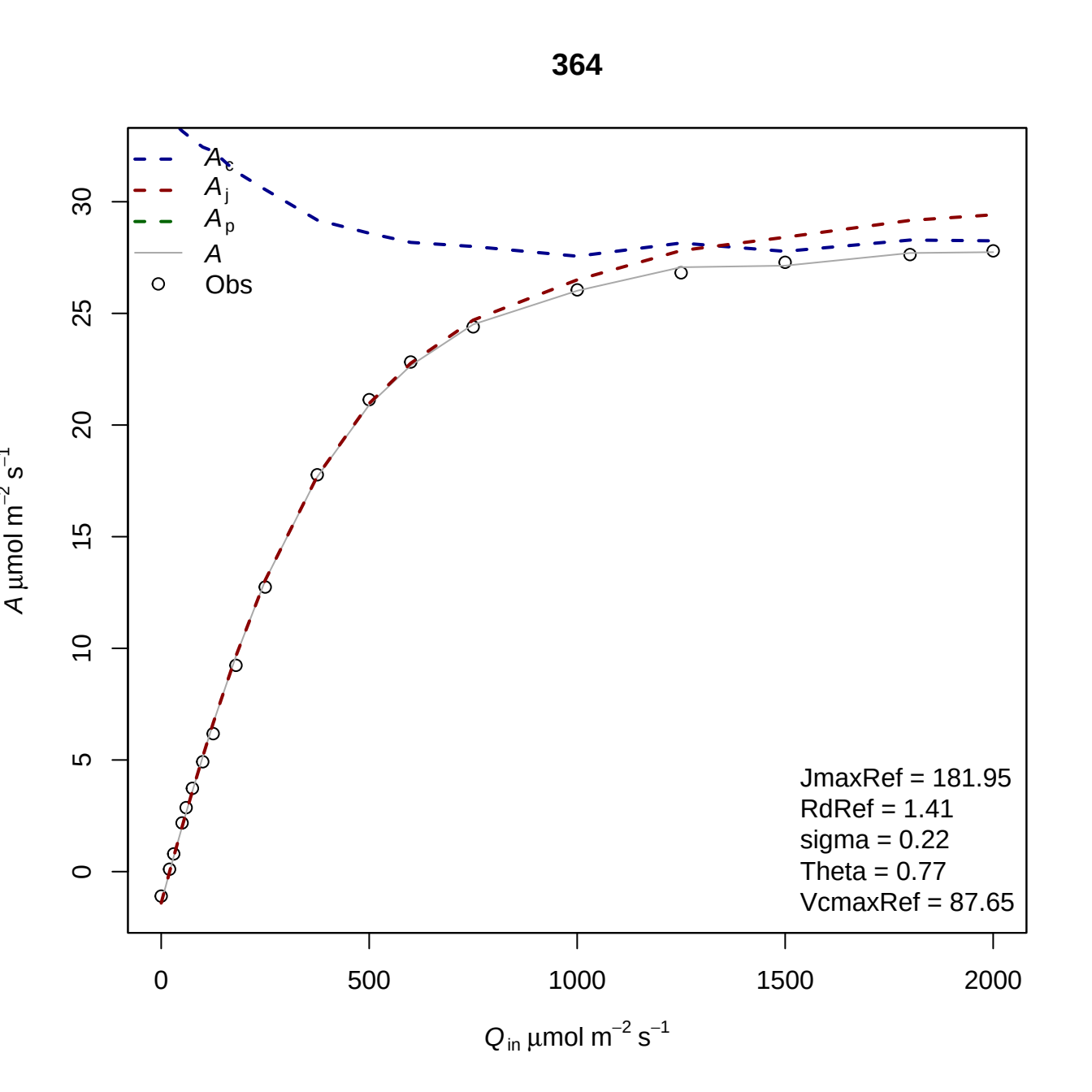

359
A_c
A_j
A_p
A
Obs
JmaxRef = 107.47
RdRef = 1.89
sigma = 0.21
Theta = 0.53
VcmaxRef = 48.69
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

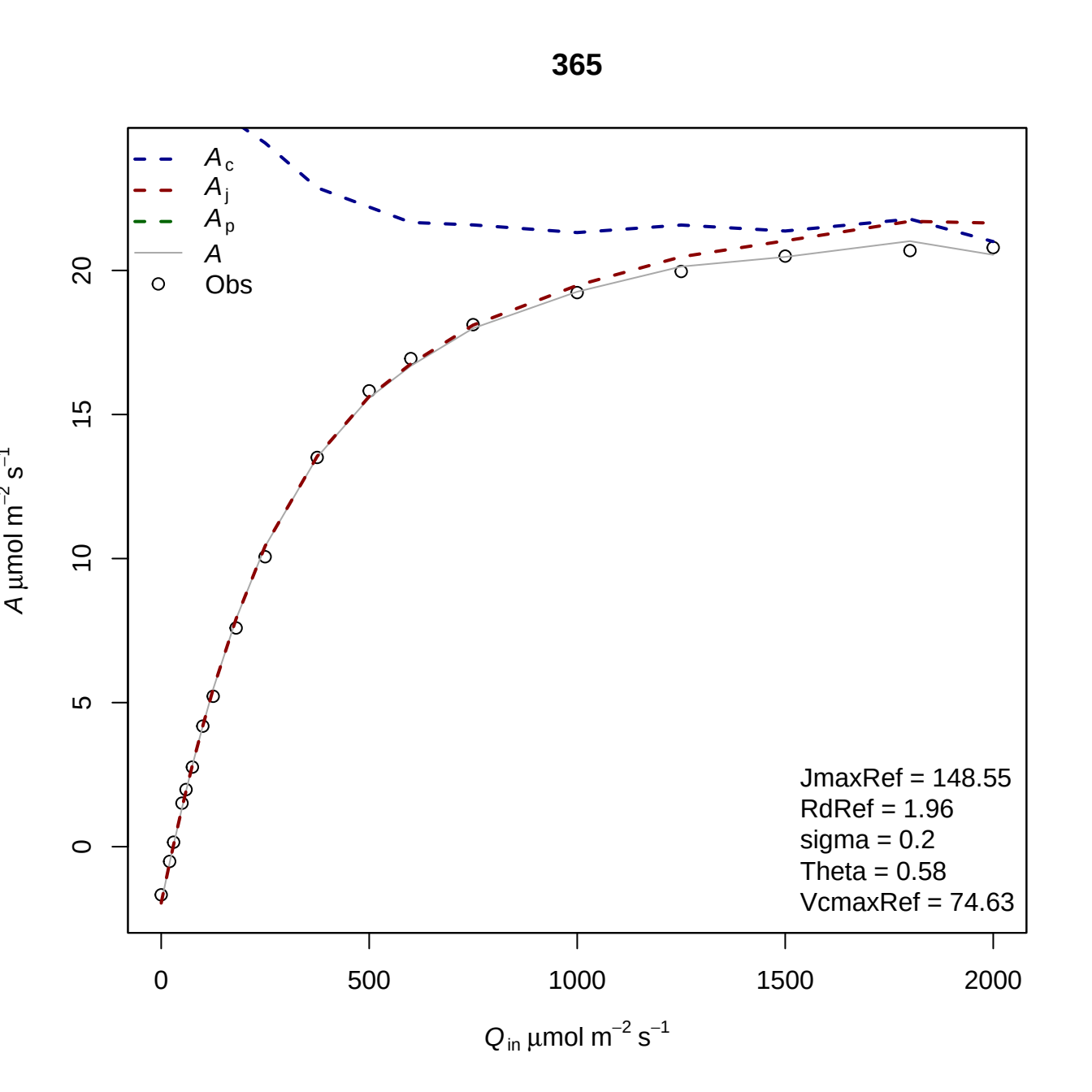

360
A_c
A_j
A_p
A
Obs
JmaxRef = 150.47
RdRef = 2.09
sigma = 0.28
Theta = 0.43
VcmaxRef = 58.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

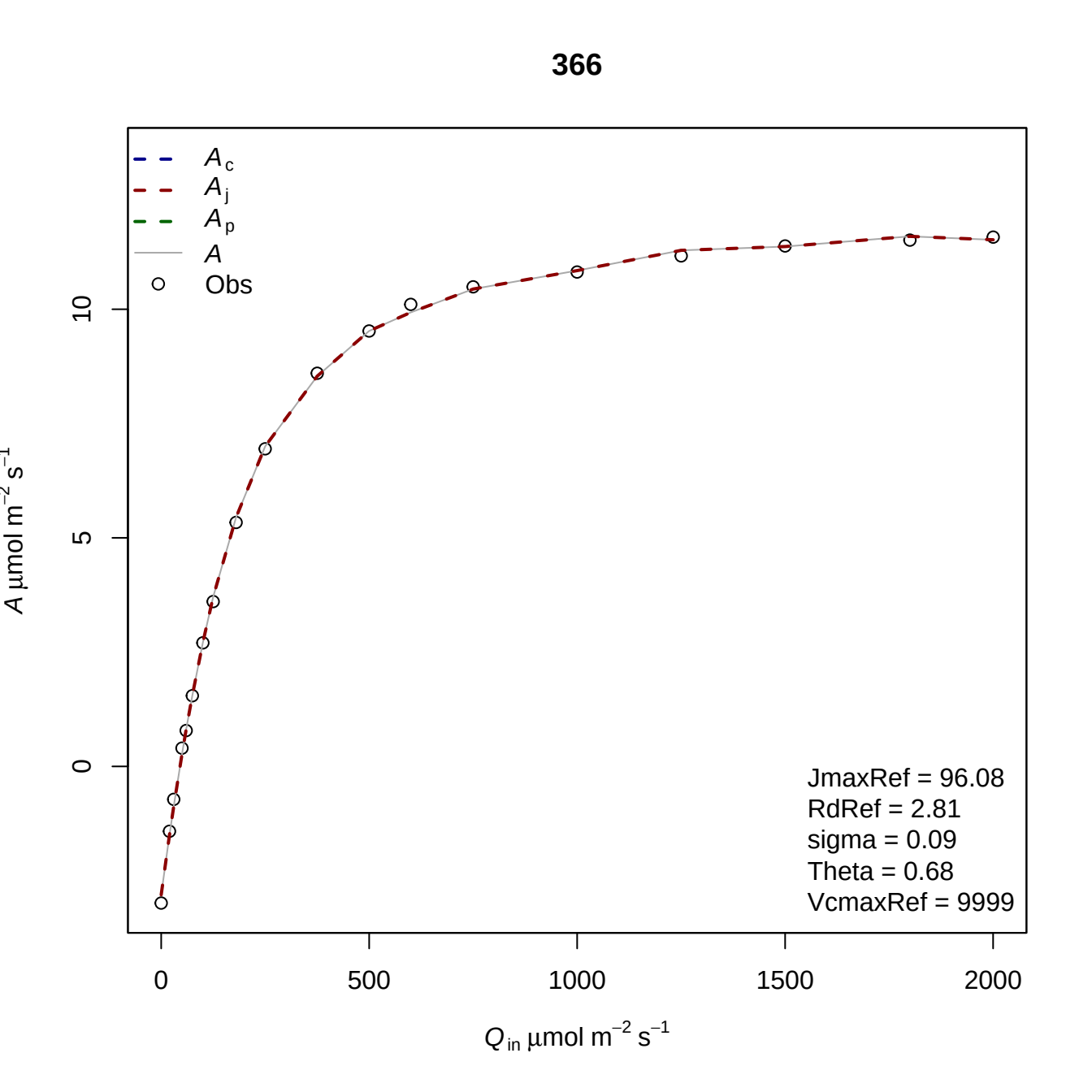

361
A_c
A_j
A_p
A
Obs
JmaxRef = 116.79
RdRef = 1.78
sigma = 0.38
Theta = 0.3
VcmaxRef = 45.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

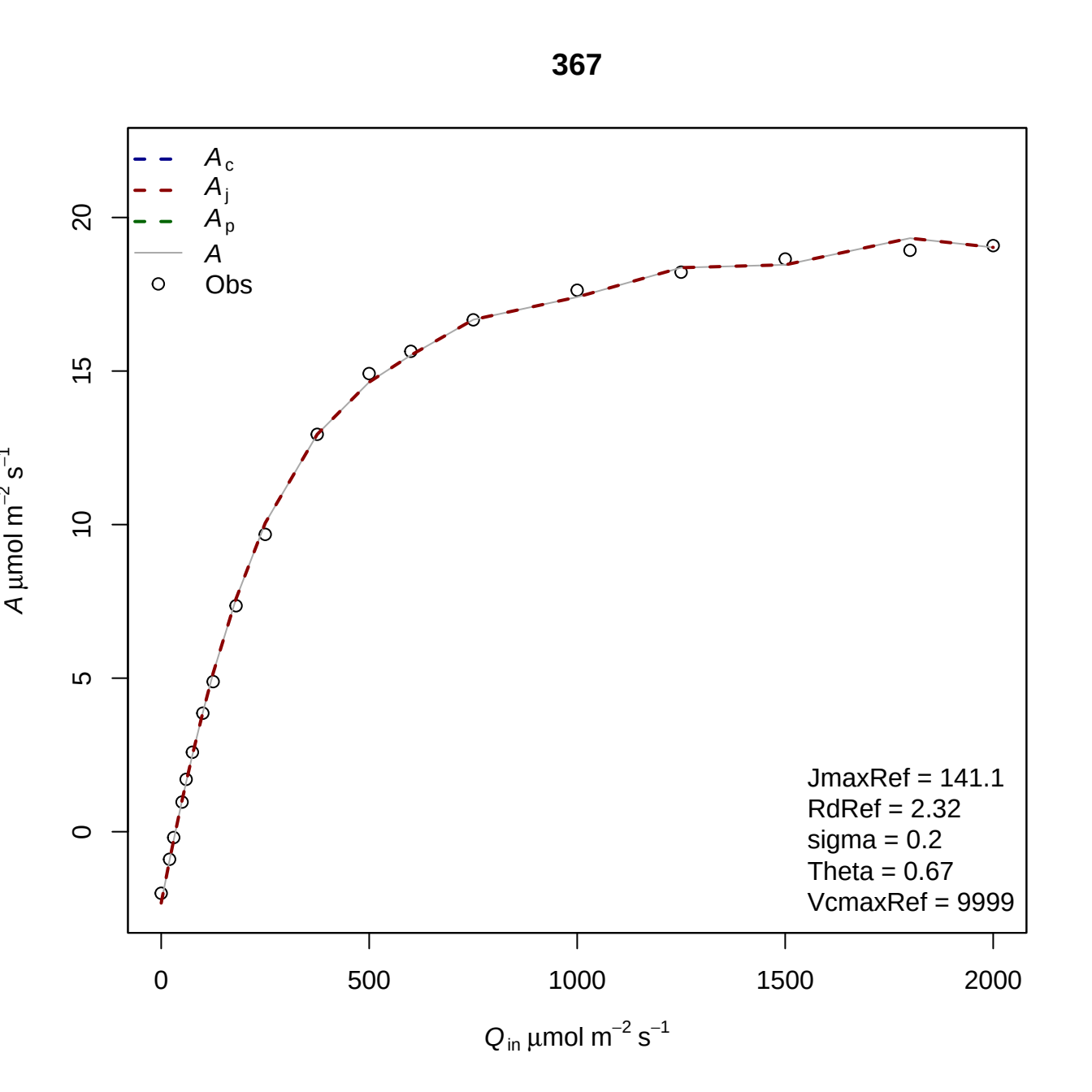

362
A_c
A_j
A_p
A
Obs
JmaxRef = 152.88
RdRef = 1.3
sigma = 0.12
Theta = 0.83
VcmaxRef = 79.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

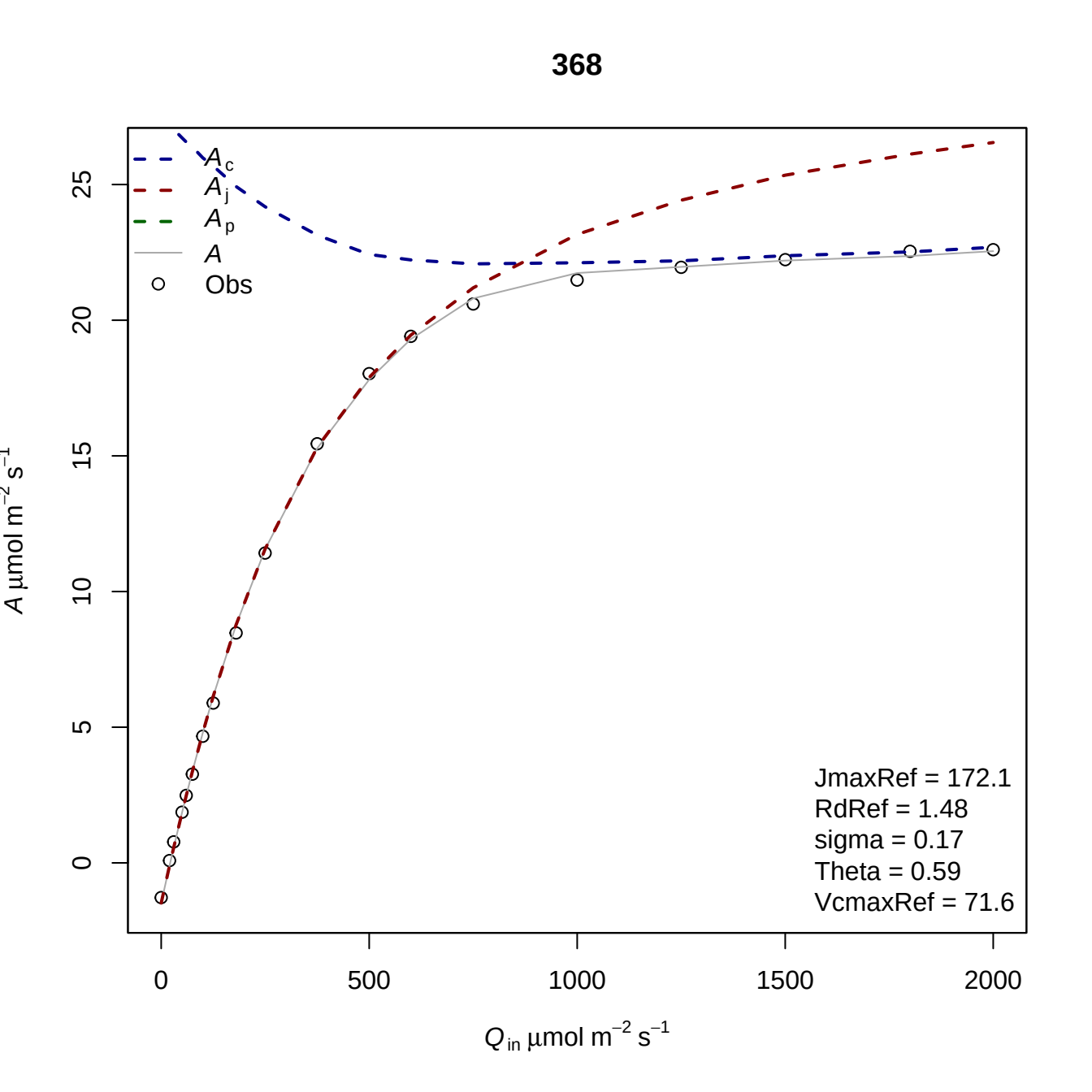

363
A_c
A_j
A_p
A
Obs
JmaxRef = 151.33
RdRef = 1.77
sigma = 0.23
Theta = 0.69
VcmaxRef = 74.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

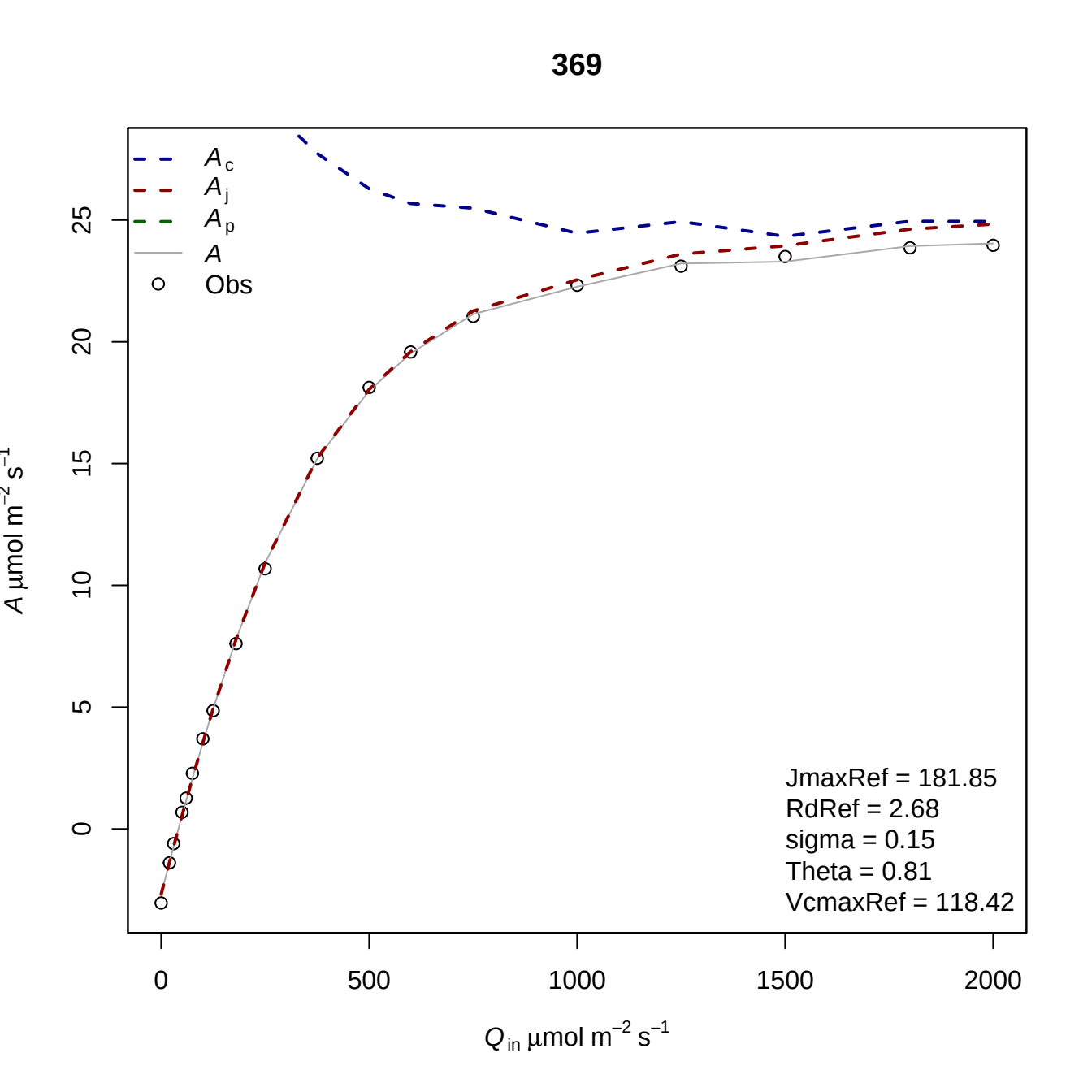

364
A_c
A_j
A_p
A
Obs
JmaxRef = 181.95
RdRef = 1.41
sigma = 0.22
Theta = 0.77
VcmaxRef = 87.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

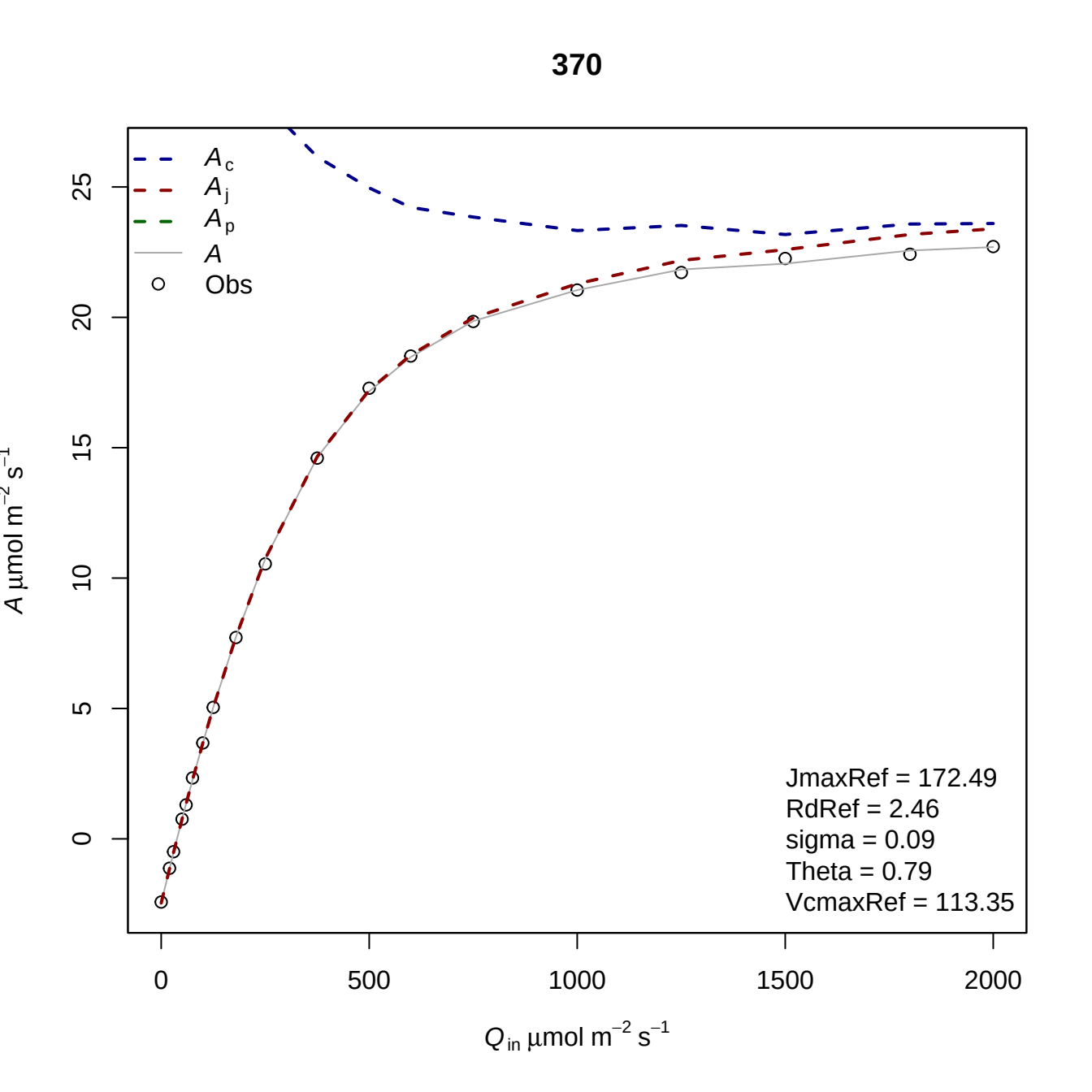

365
A_c
A_j
A_p
A
Obs
JmaxRef = 148.55
RdRef = 1.96
sigma = 0.2
Theta = 0.58
VcmaxRef = 74.63
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

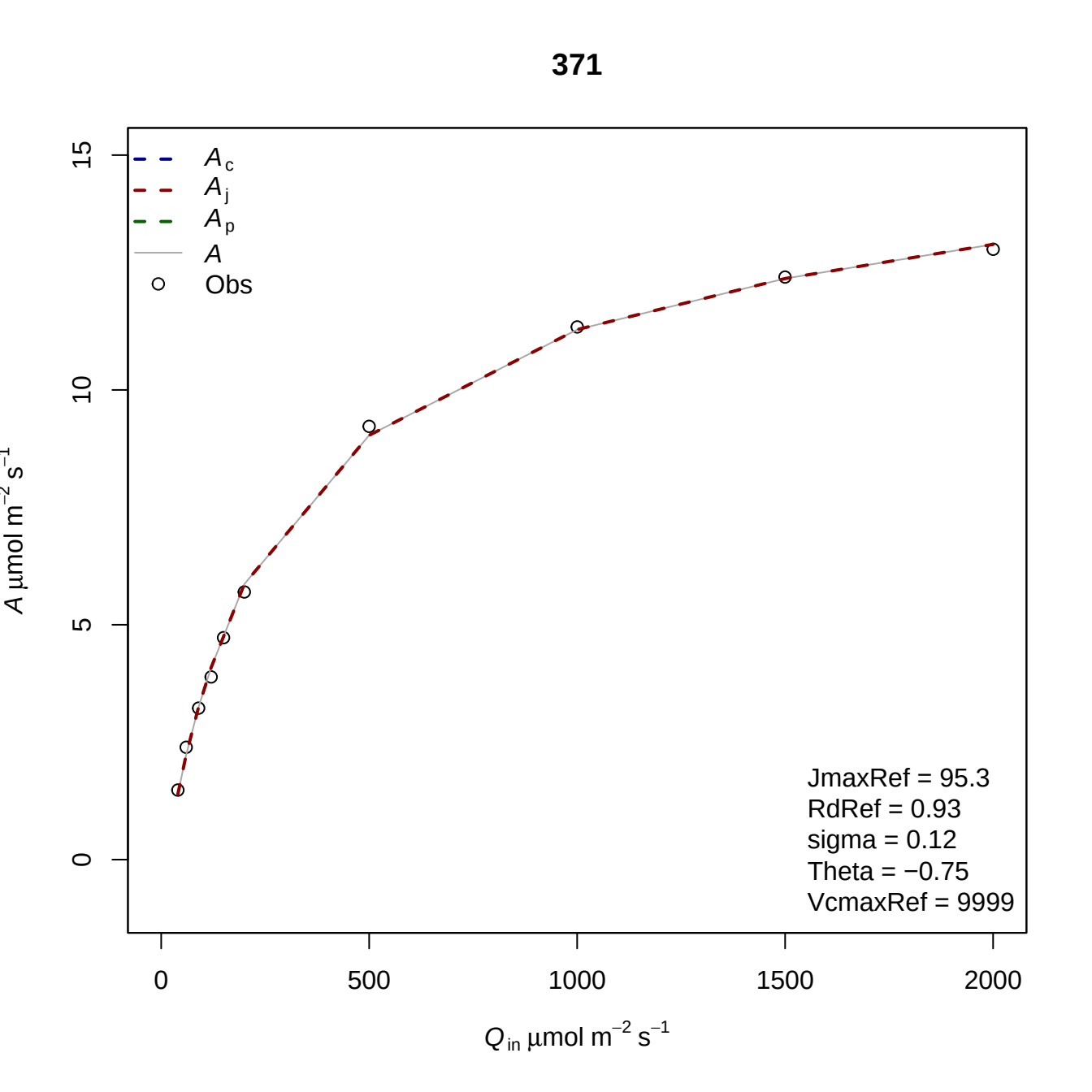

366
A_c
A_j
A_p
A
Obs
JmaxRef = 96.08
RdRef = 2.81
sigma = 0.09
Theta = 0.68
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

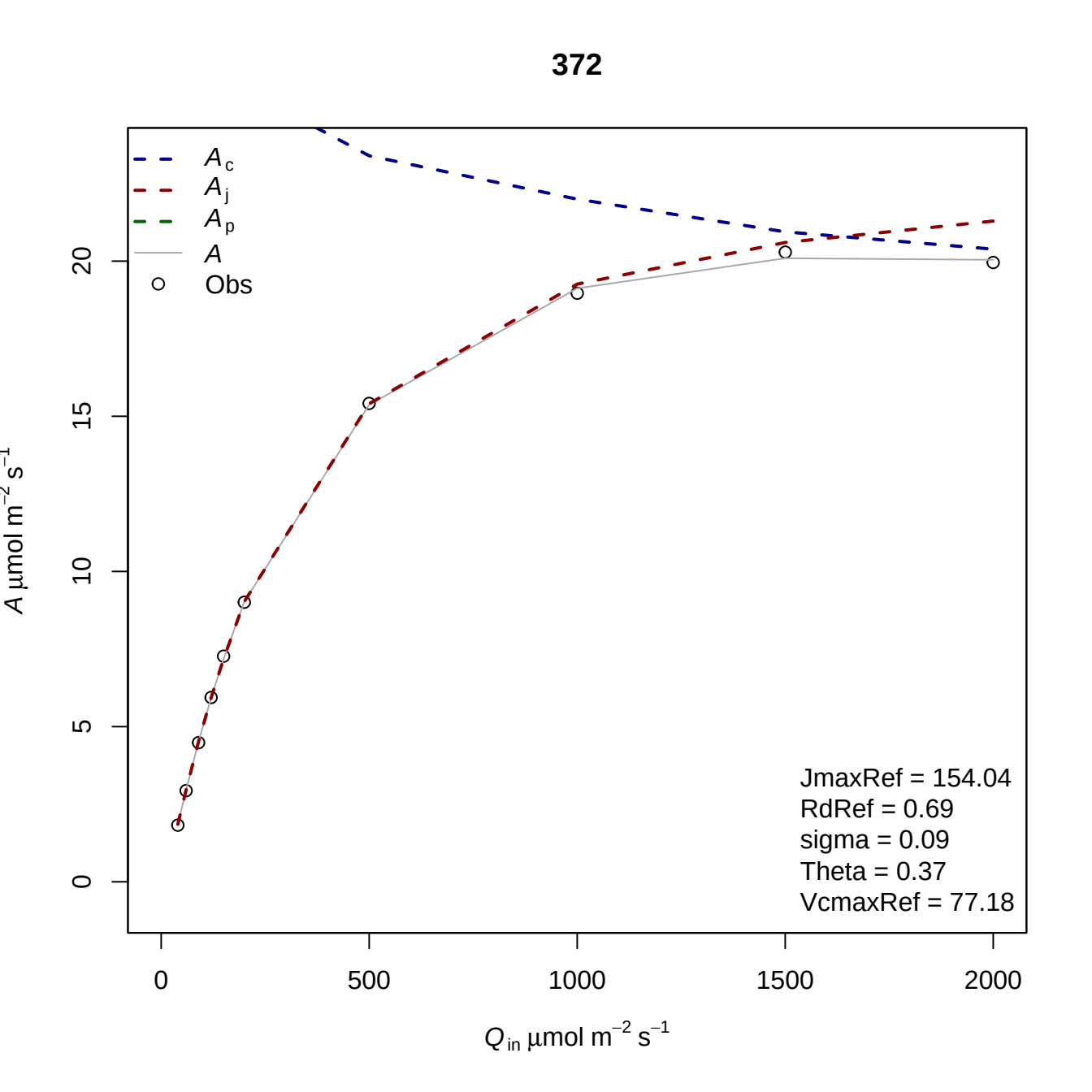

367
A_c
A_j
A_p
A
Obs
JmaxRef = 141.1
RdRef = 2.32
sigma = 0.2
Theta = 0.67
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

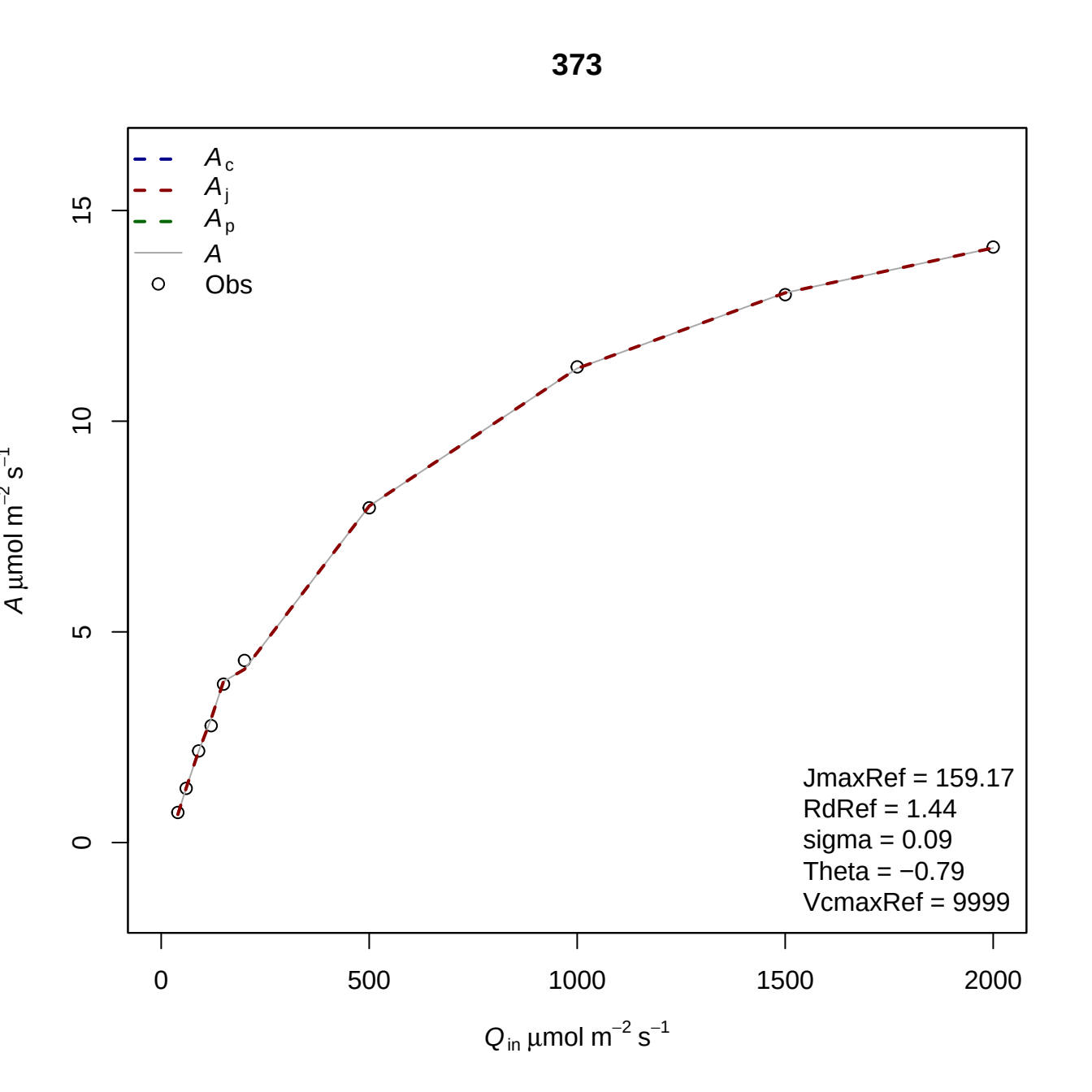

368
A_c
A_j
A_p
A
Obs
JmaxRef = 172.1
RdRef = 1.48
sigma = 0.17
Theta = 0.59
VcmaxRef = 71.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

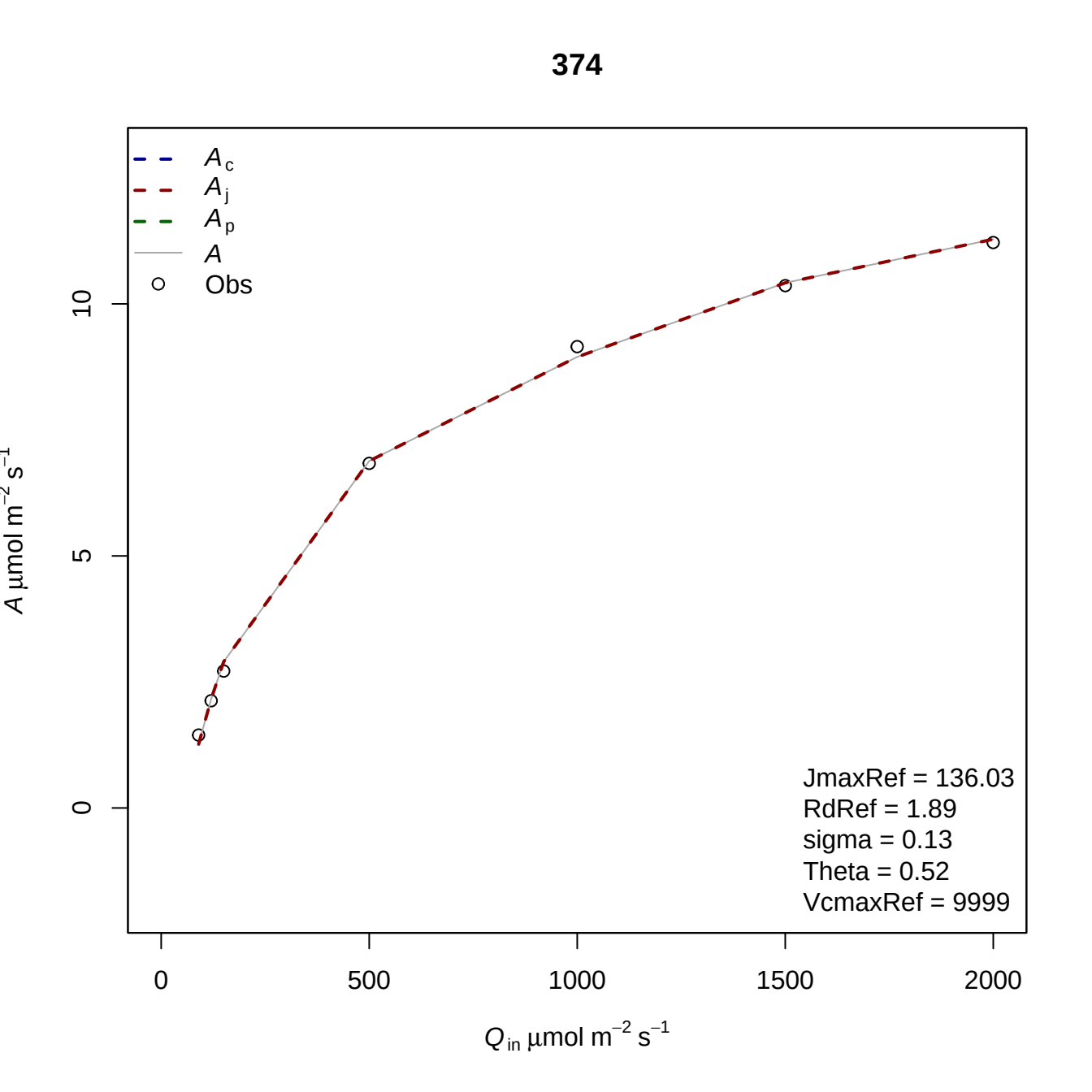

369
A_c
A_j
A_p
A
Obs
JmaxRef = 181.85
RdRef = 2.68
sigma = 0.15
Theta = 0.81
VcmaxRef = 118.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

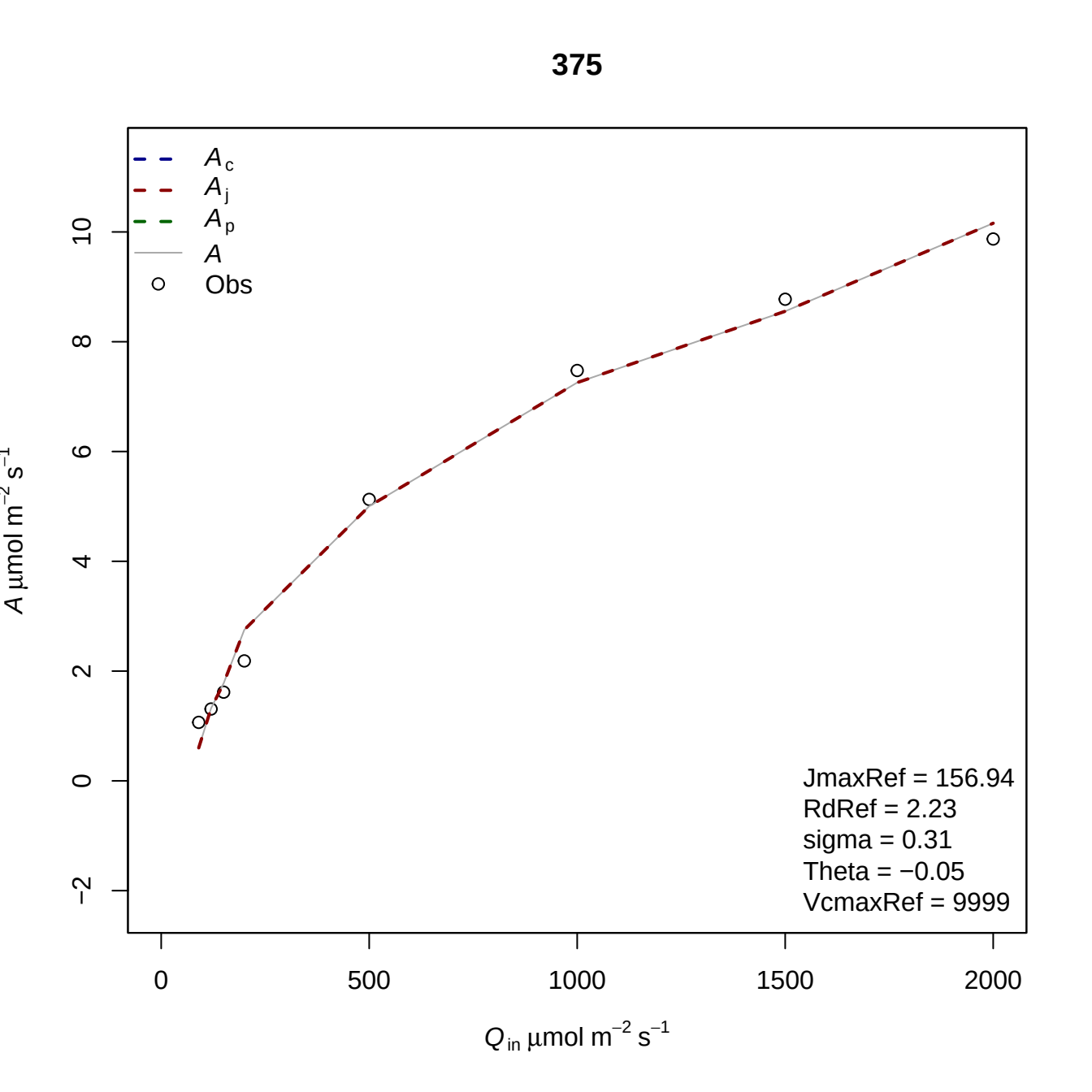

370
A_c
A_j
A_p
A
Obs
JmaxRef = 172.49
RdRef = 2.46
sigma = 0.09
Theta = 0.79
VcmaxRef = 113.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

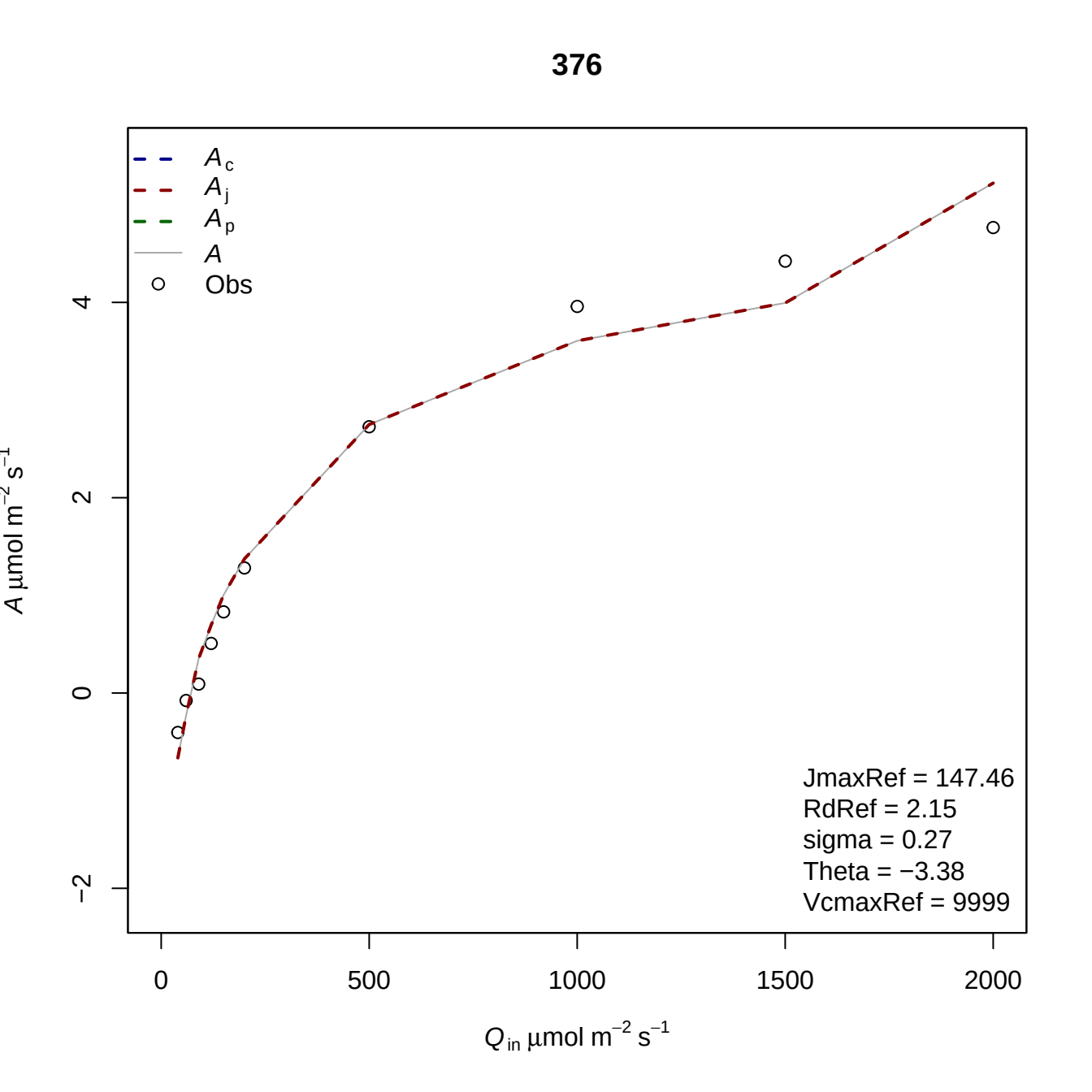

371
A_c
A_j
A_p
A
Obs
JmaxRef = 95.3
RdRef = 0.93
sigma = 0.12
Theta = −0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

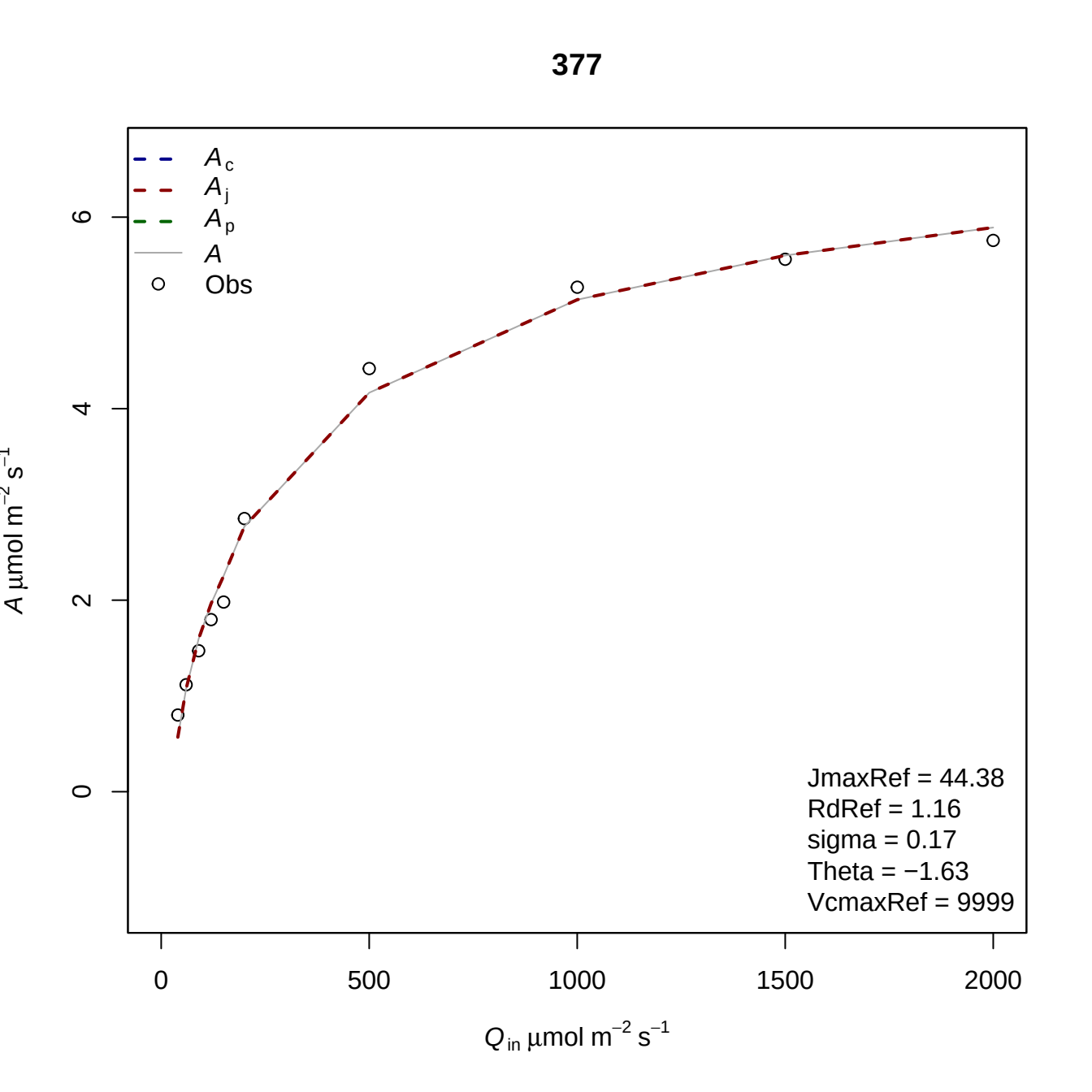

372
A_c
A_j
A_p
A
Obs
JmaxRef = 154.04
RdRef = 0.69
sigma = 0.09
Theta = 0.37
VcmaxRef = 77.18
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

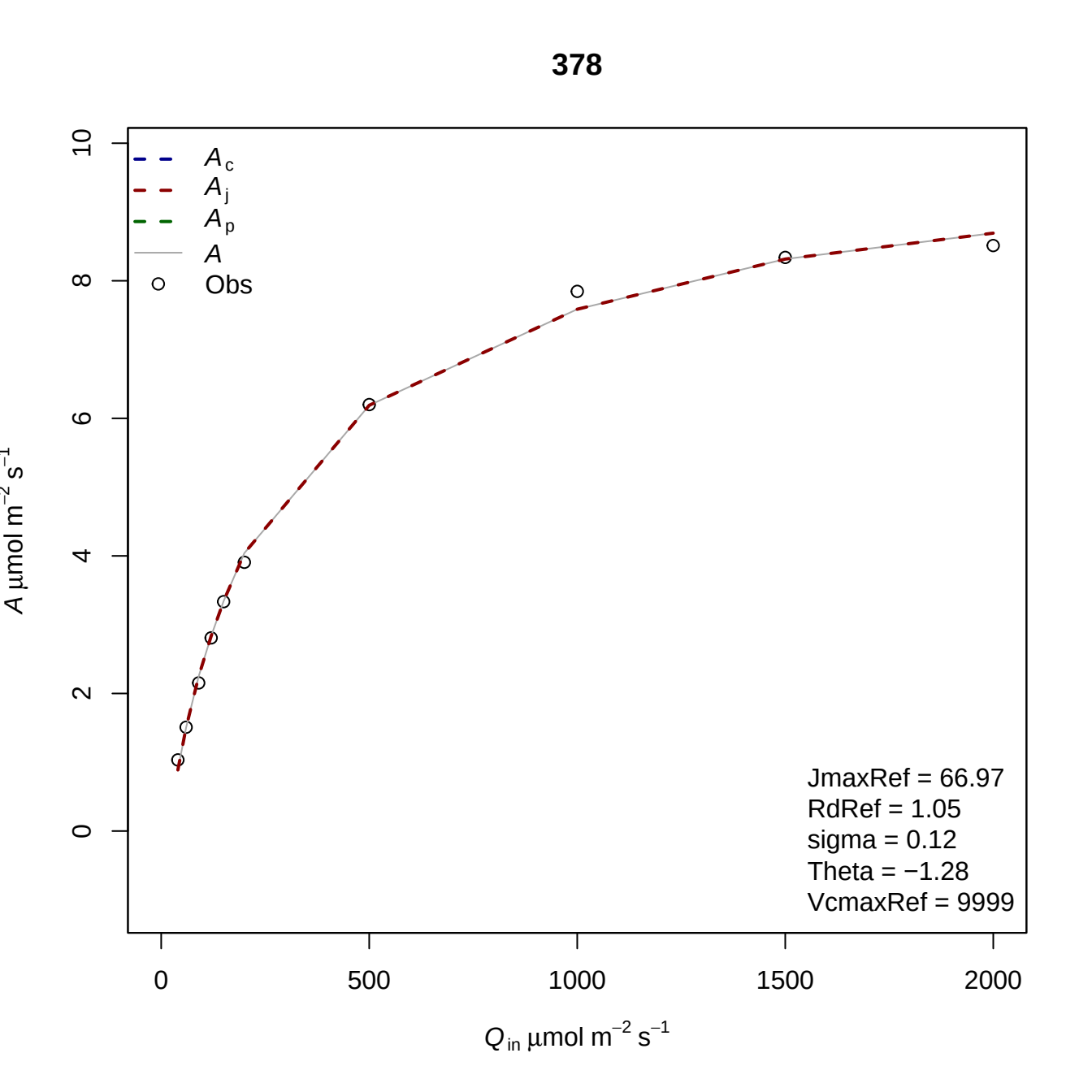

373
A_c
A_j
A_p
A
Obs
JmaxRef = 159.17
RdRef = 1.44
sigma = 0.09
Theta = −0.79
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

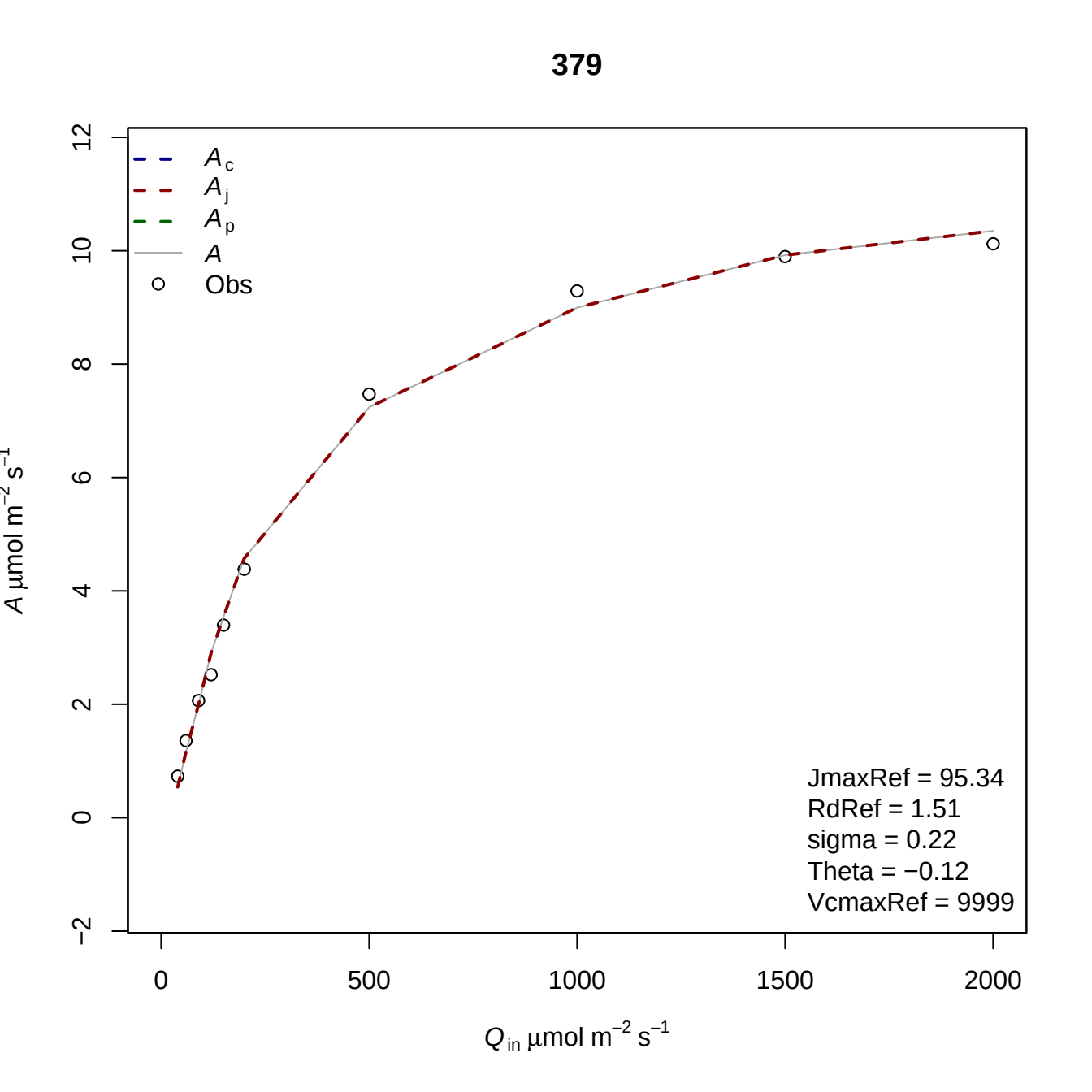

374
A_c
A_j
A_p
A
Obs
JmaxRef = 136.03
RdRef = 1.89
sigma = 0.13
Theta = 0.52
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

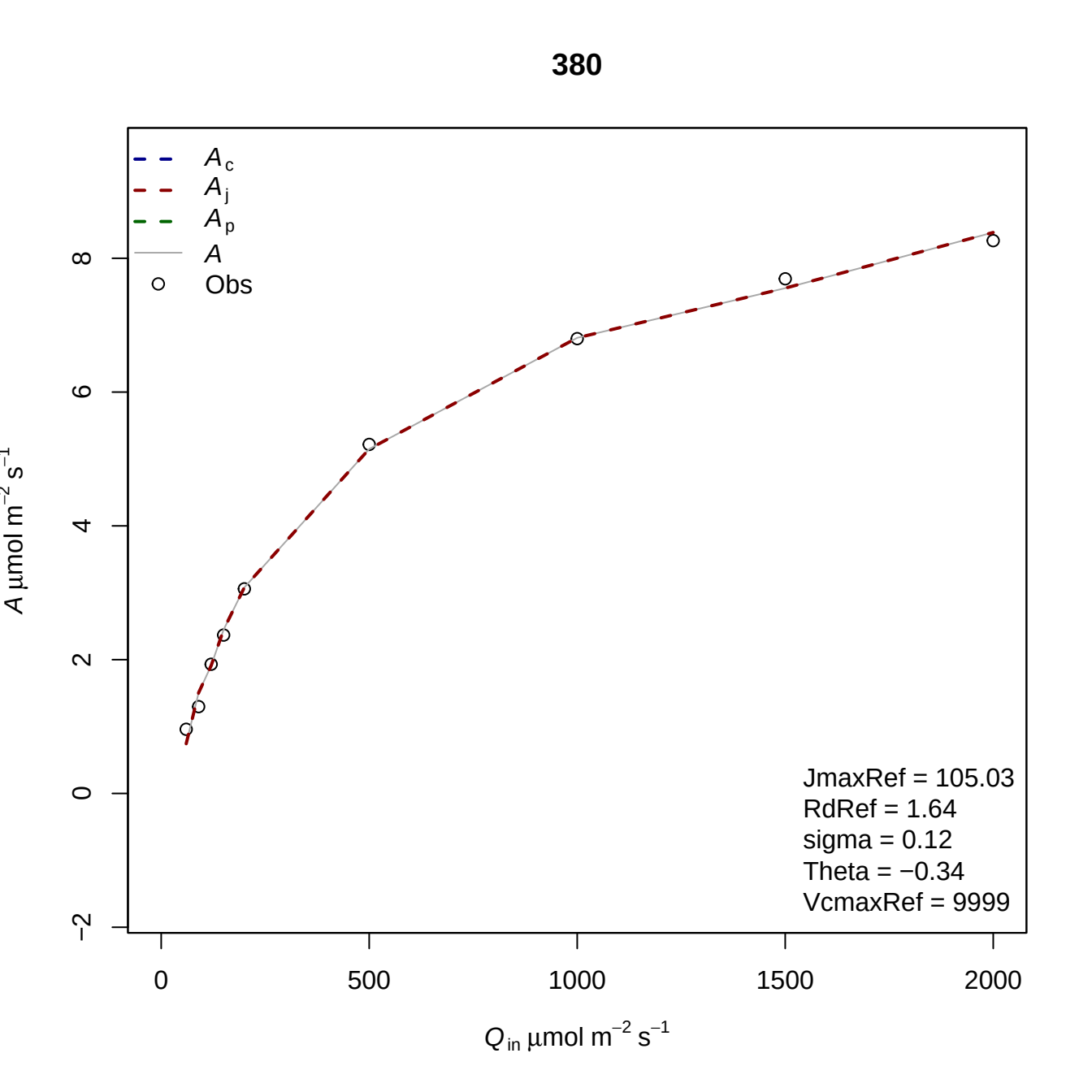

375
A_c
A_j
A_p
A
Obs
JmaxRef = 156.94
RdRef = 2.23
sigma = 0.31
Theta = −0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

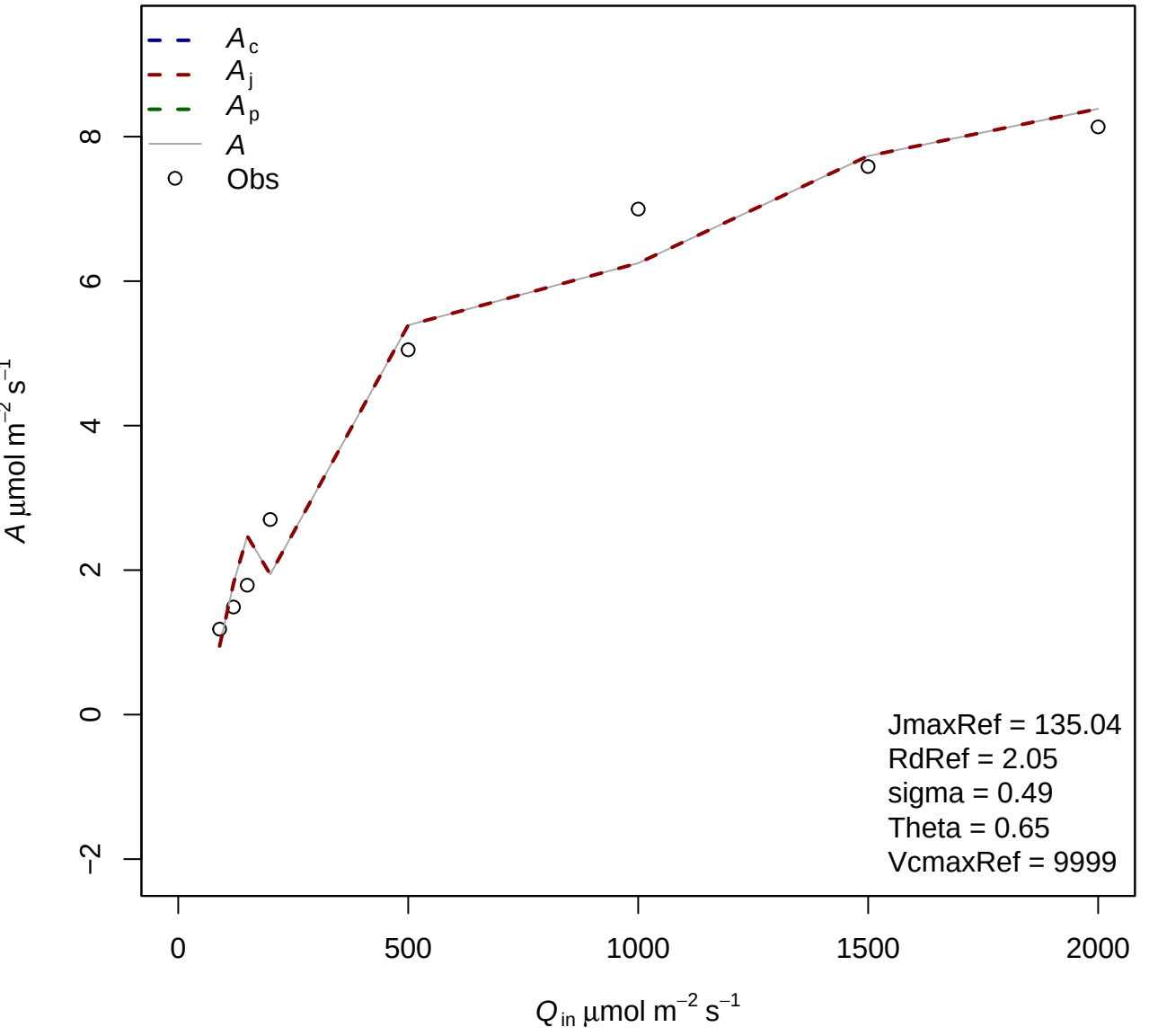

376
A_c
A_j
A_p
A
Obs
JmaxRef = 147.46
RdRef = 2.15
sigma = 0.27
Theta = −3.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

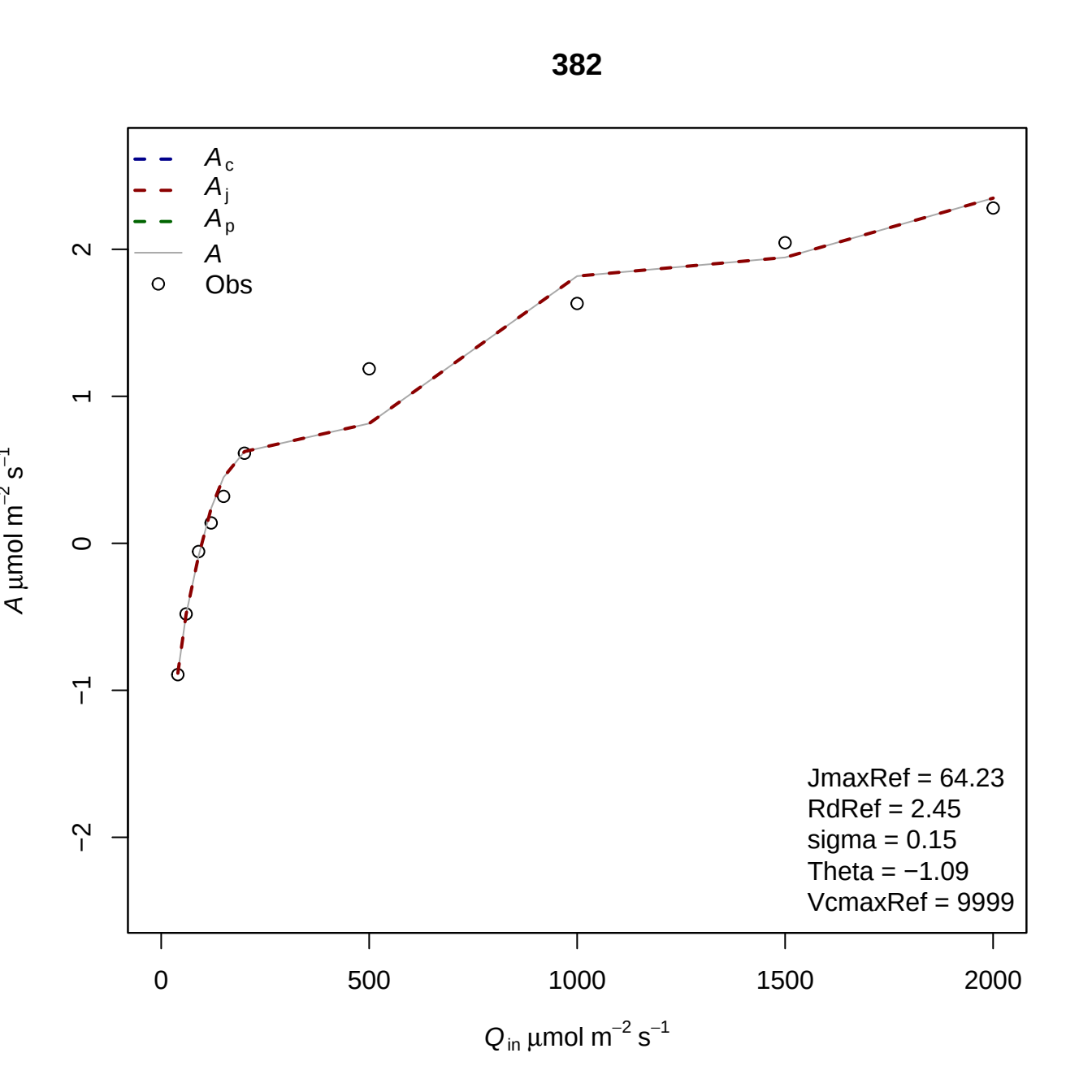

377
A_c
A_j
A_p
A
Obs
JmaxRef = 44.38
RdRef = 1.16
sigma = 0.17
Theta = −1.63
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

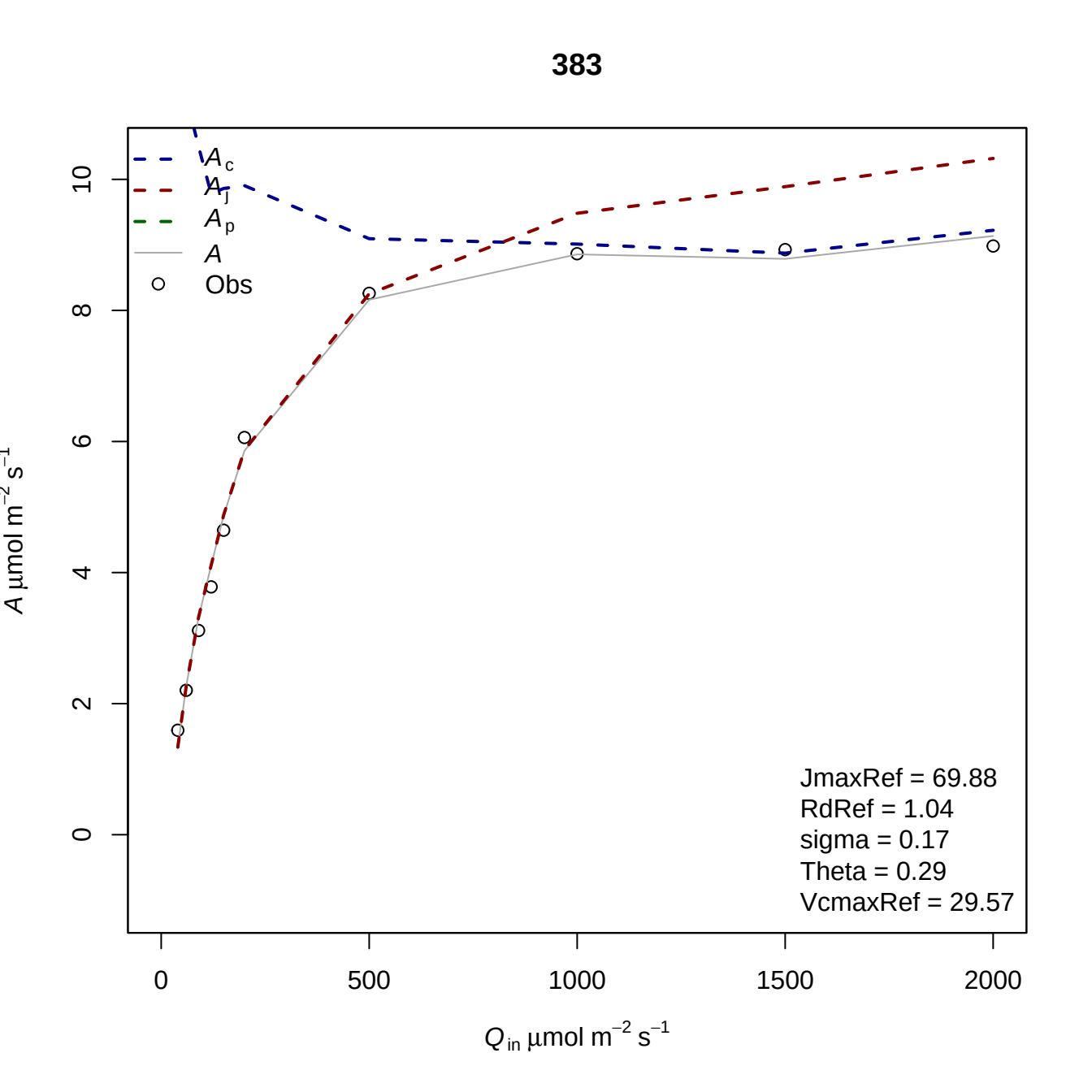

378
A_c
A_j
A_p
A
Obs
JmaxRef = 66.97
RdRef = 1.05
sigma = 0.12
Theta = −1.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

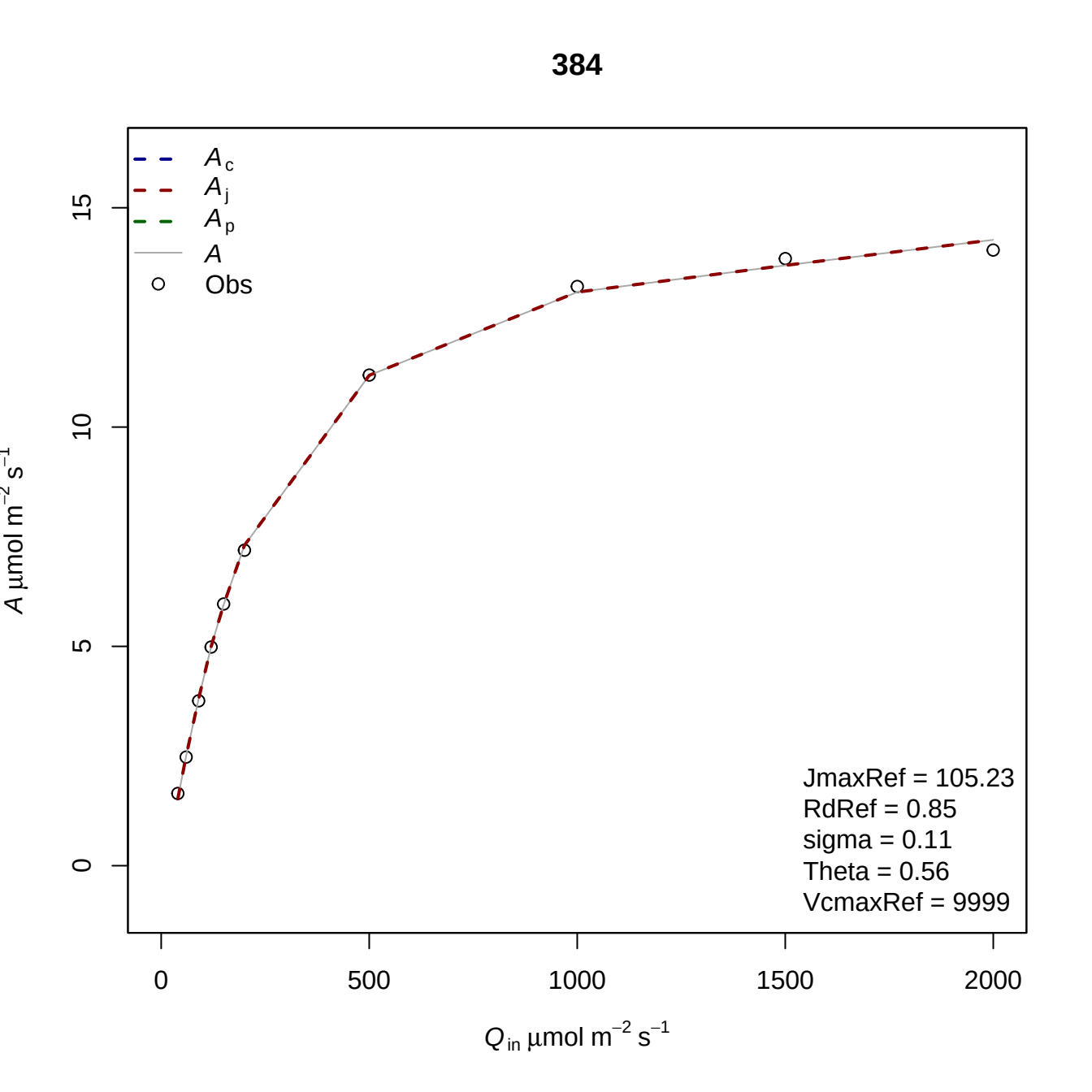

379
A_c
A_j
A_p
A
Obs
JmaxRef = 95.34
RdRef = 1.51
sigma = 0.22
Theta = −0.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

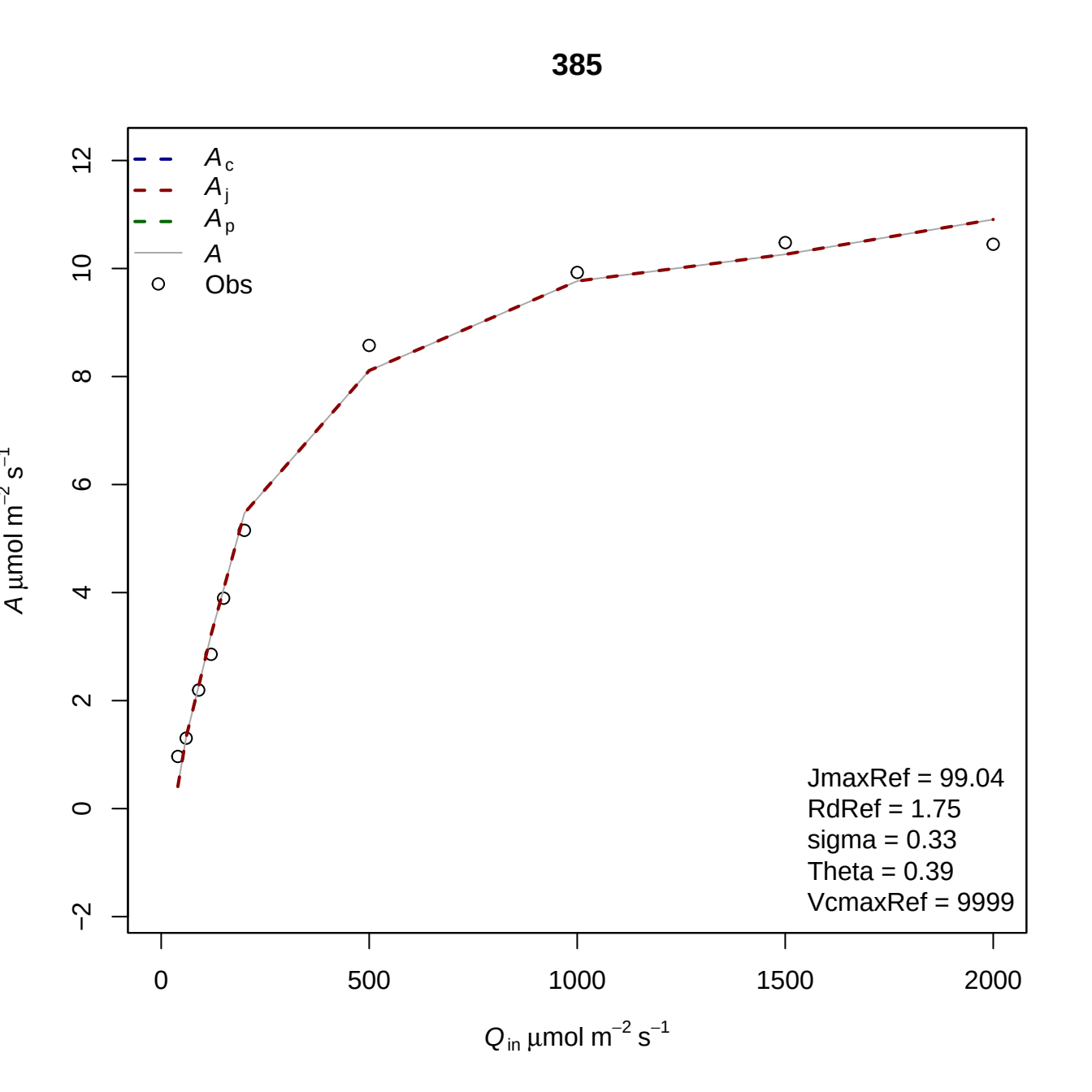

380
A_c
A_j
A_p
A
Obs
JmaxRef = 105.03
RdRef = 1.64
sigma = 0.12
Theta = −0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

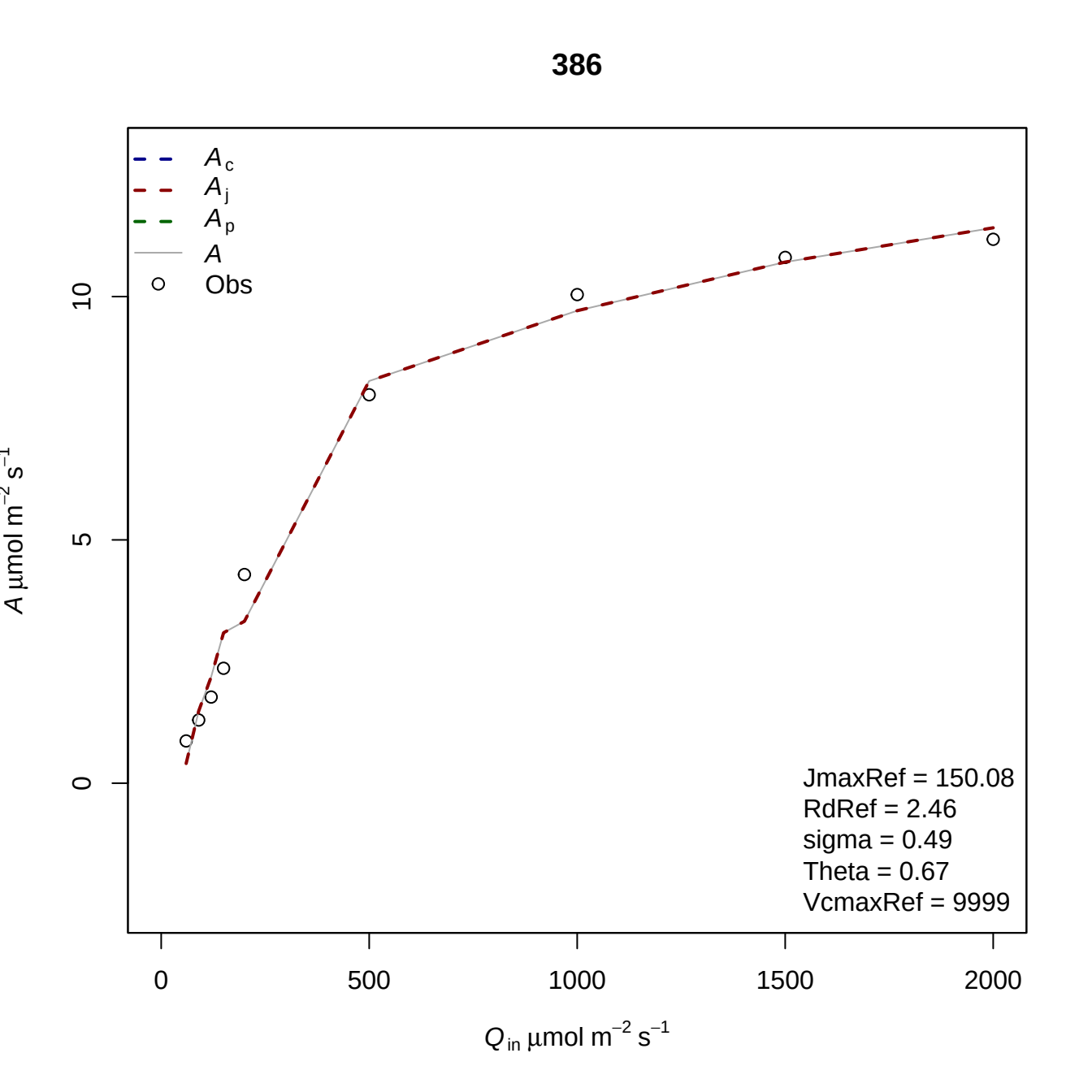

A_c
A_j
A_p
A
Obs
JmaxRef = 135.04
RdRef = 2.05
sigma = 0.49
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

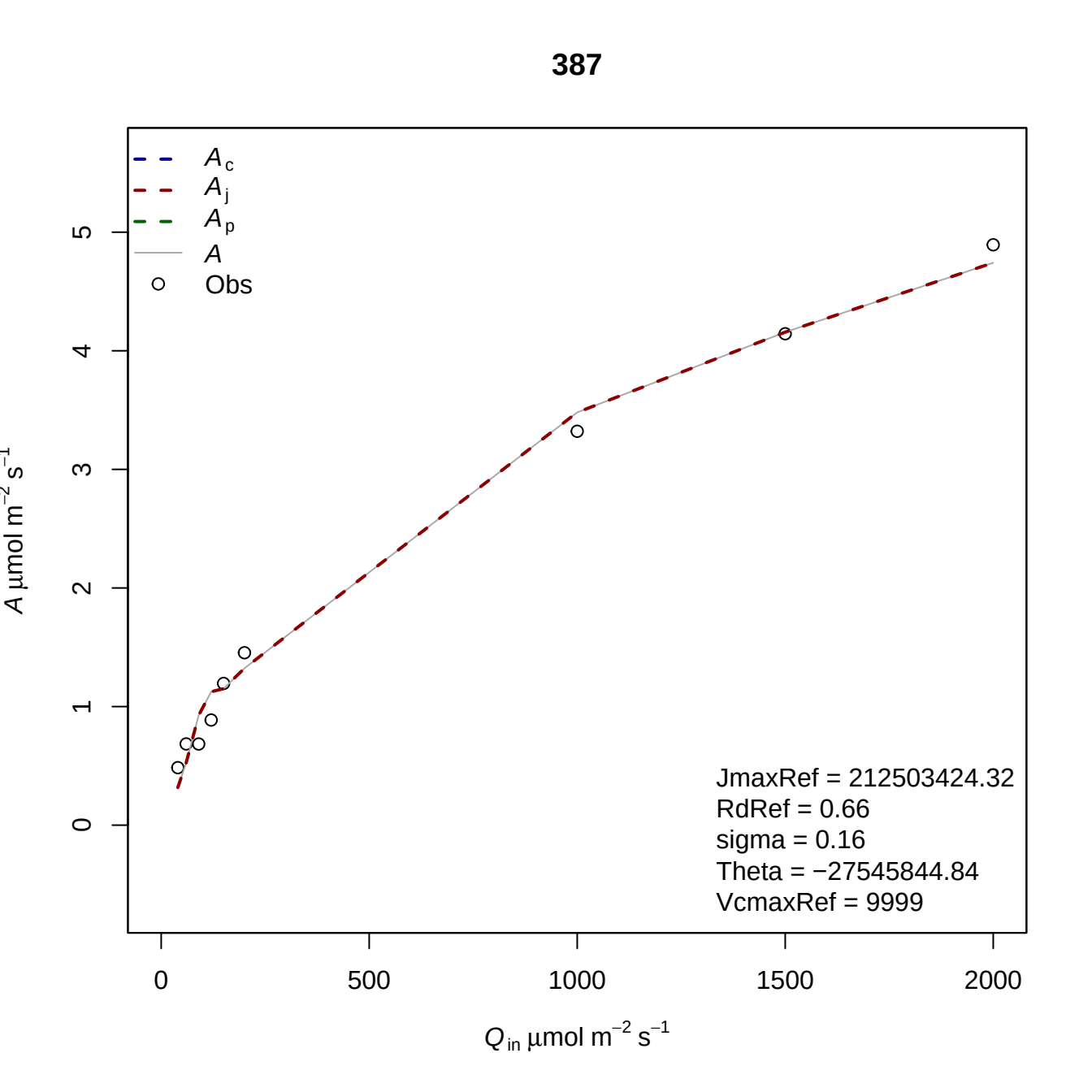

382
A_c
A_j
A_p
A
Obs
JmaxRef = 64.23
RdRef = 2.45
sigma = 0.15
Theta = −1.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500
2000

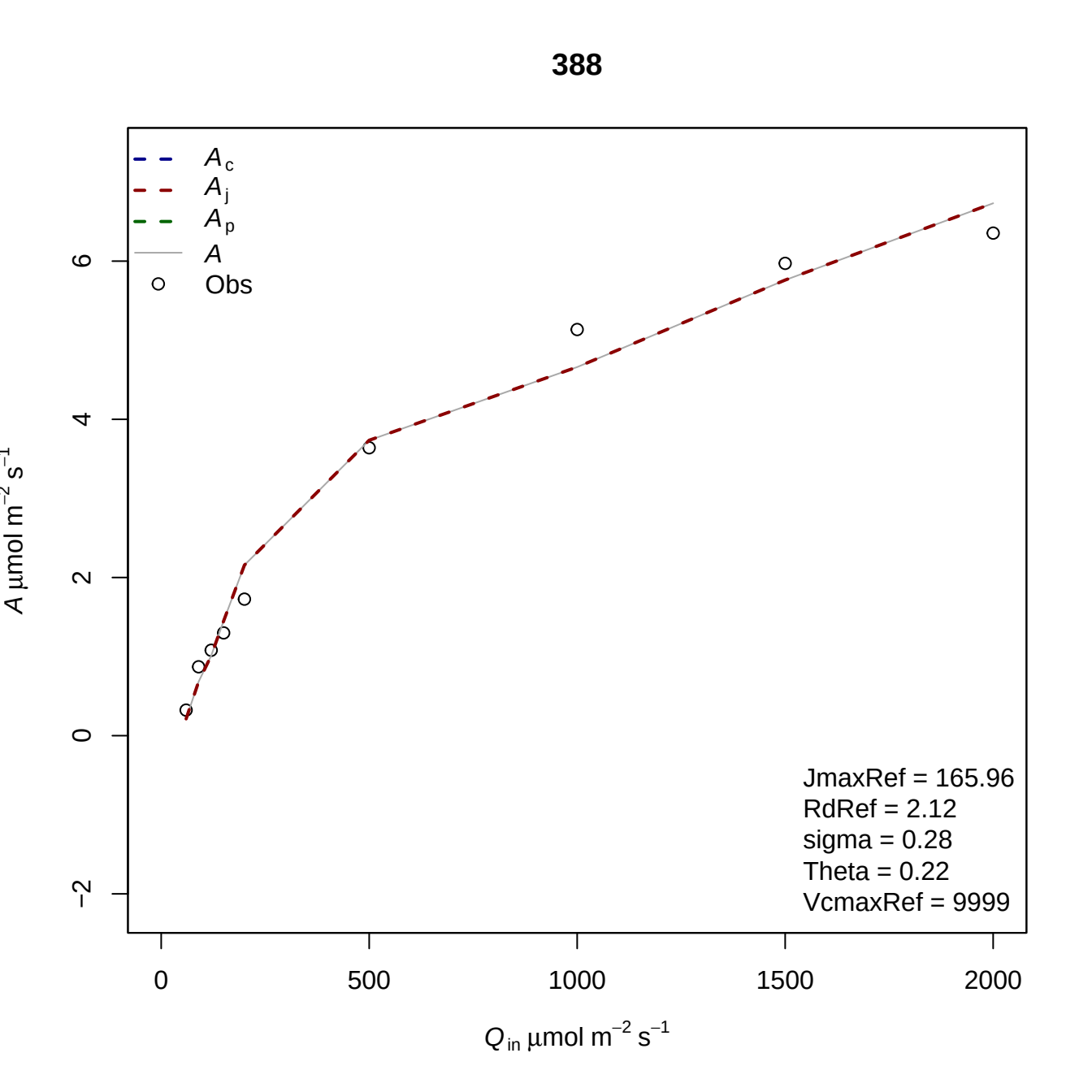

383
A_c
A_j
A_p
A
Obs
JmaxRef = 69.88
RdRef = 1.04
sigma = 0.17
Theta = 0.29
VcmaxRef = 29.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

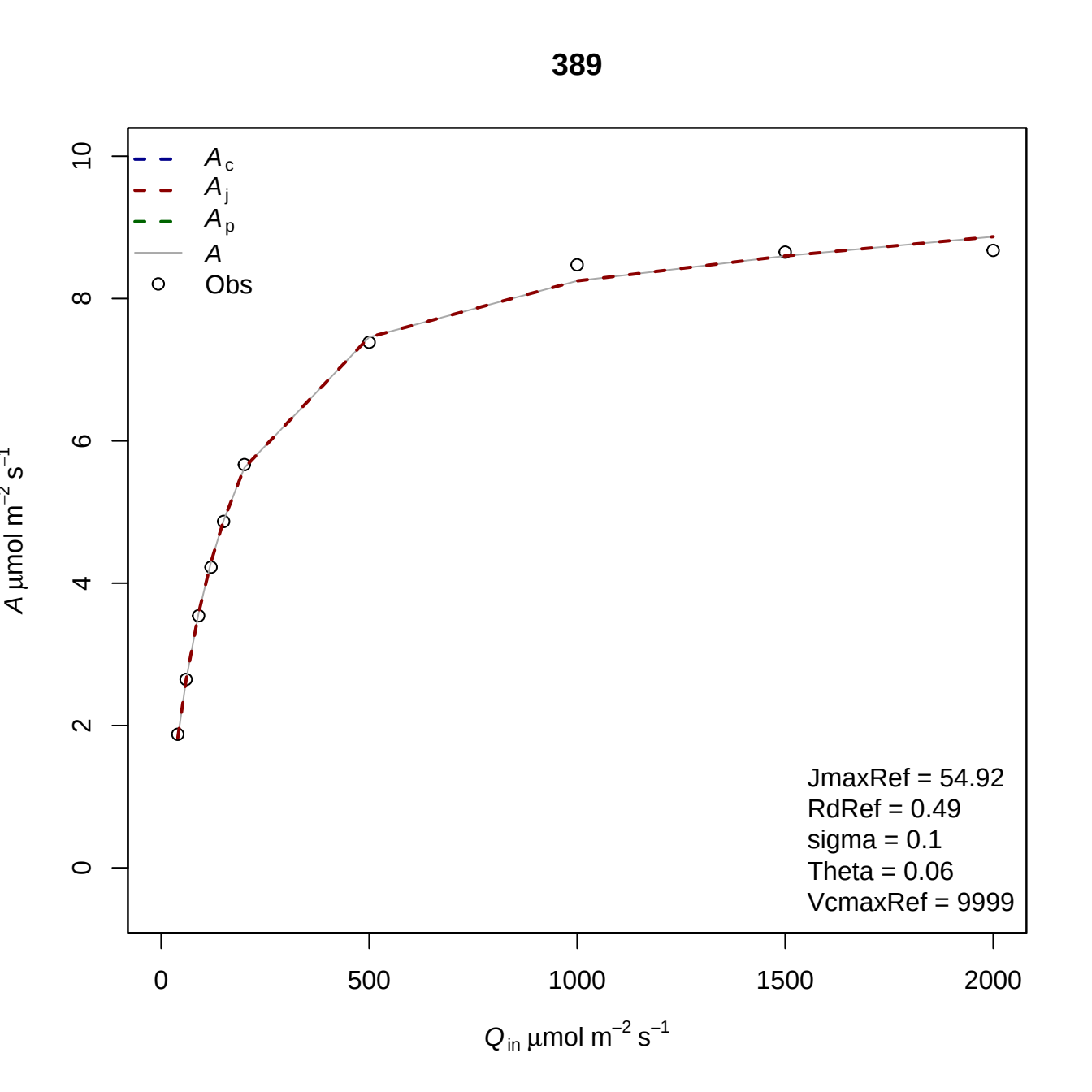

384
A_c
A_j
A_p
A
Obs
JmaxRef = 105.23
RdRef = 0.85
sigma = 0.11
Theta = 0.56
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

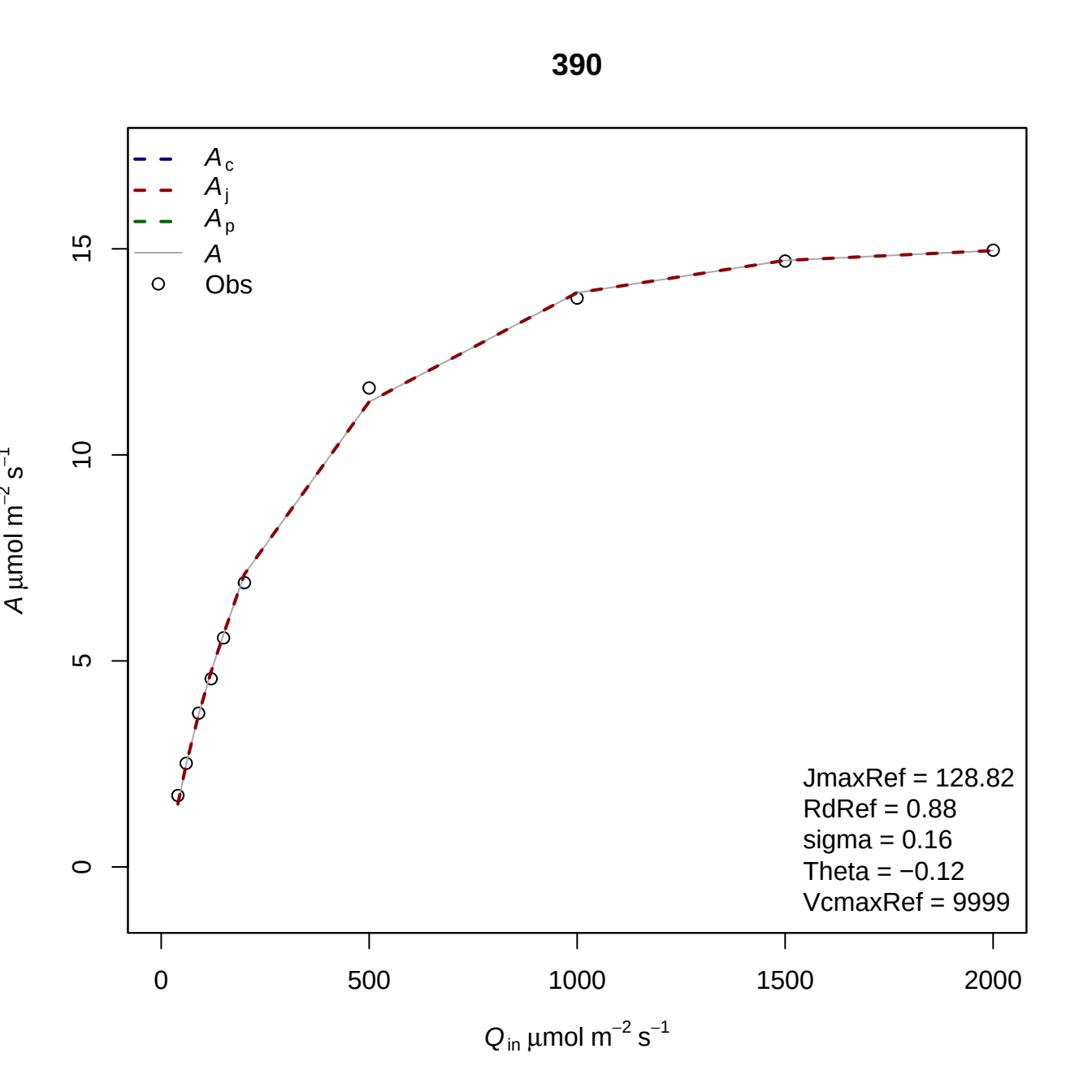

385
A_c
A_j
A_p
A
Obs
JmaxRef = 99.04
RdRef = 1.75
sigma = 0.33
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

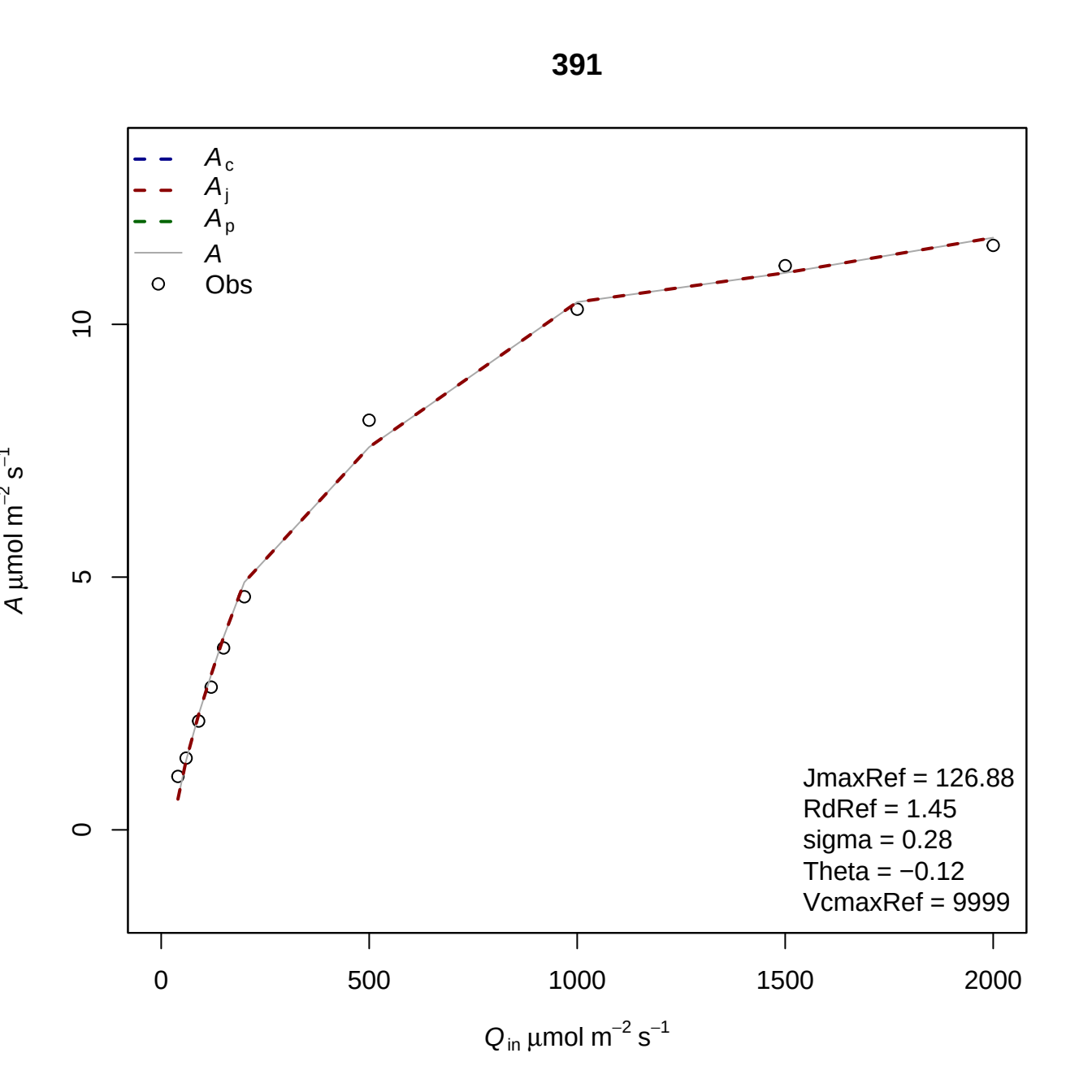

386
A_c
A_j
A_p
A
Obs
JmaxRef = 150.08
RdRef = 2.46
sigma = 0.49
Theta = 0.67
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

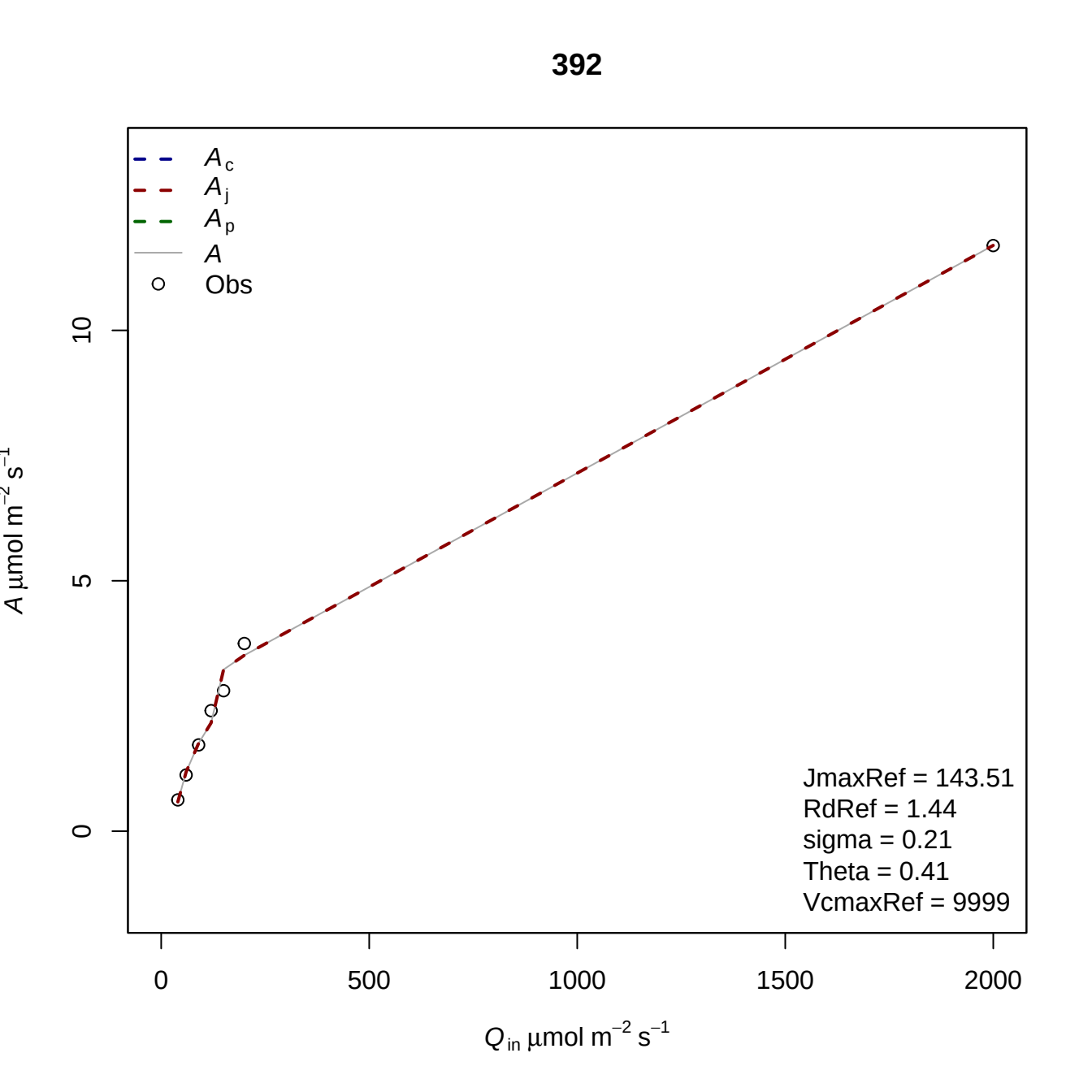

387
A_c
A_j
A_p
A
Obs
JmaxRef = 212503424.32
RdRef = 0.66
sigma = 0.16
Theta = −27545844.84
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

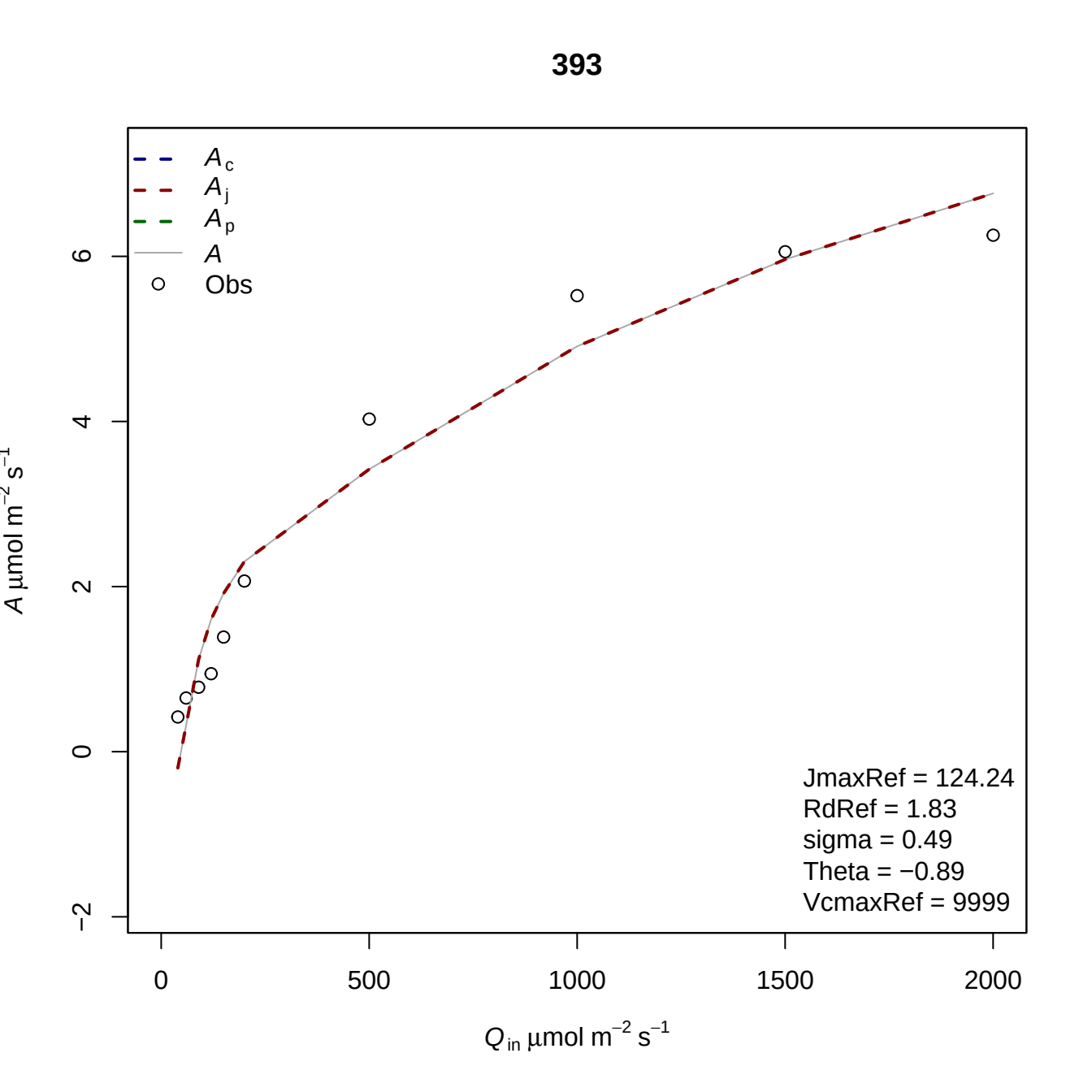

388
A_c
A_j
A_p
A
Obs
JmaxRef = 165.96
RdRef = 2.12
sigma = 0.28
Theta = 0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

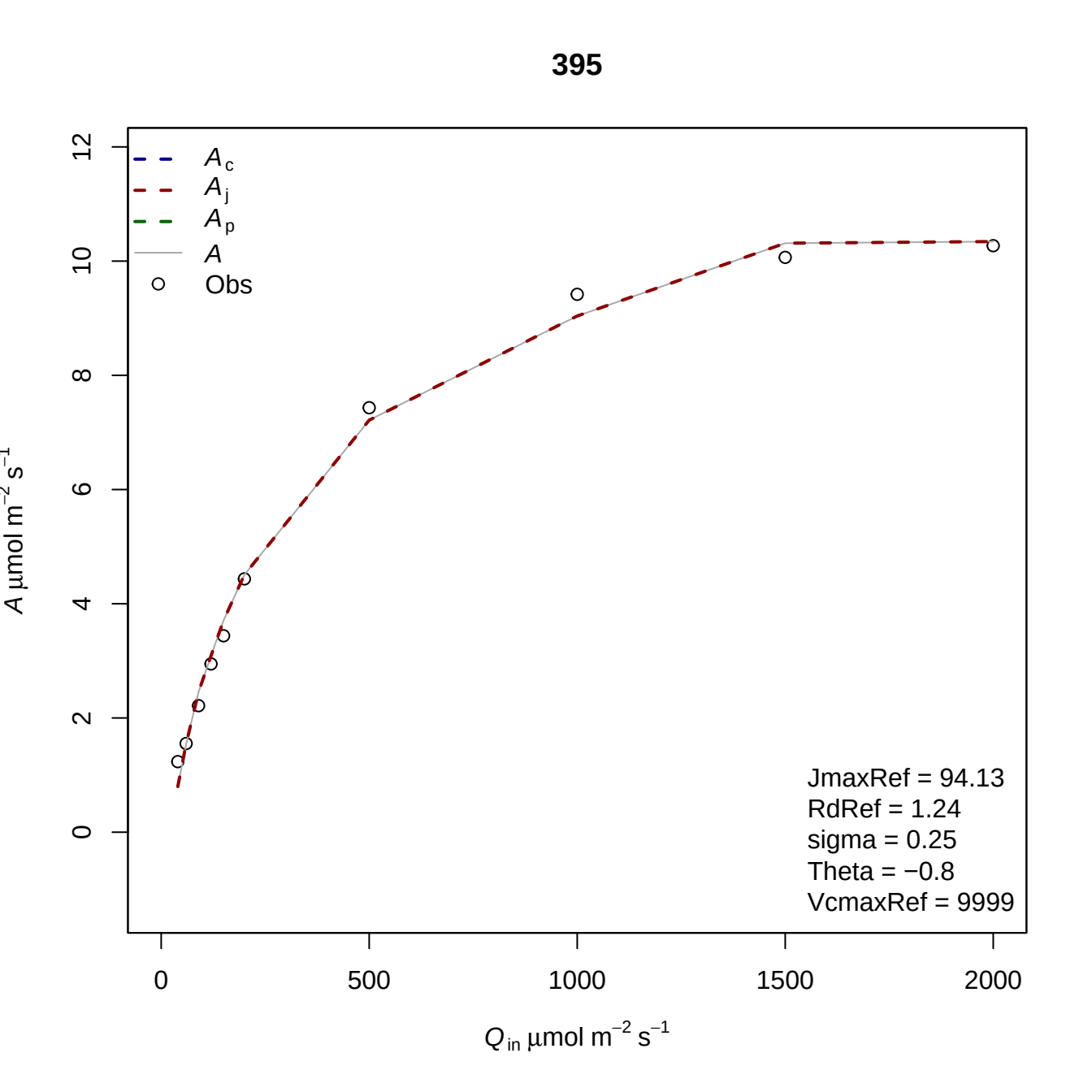

389
A_c
A_j
A_p
A
Obs
JmaxRef = 54.92
RdRef = 0.49
sigma = 0.1
Theta = 0.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

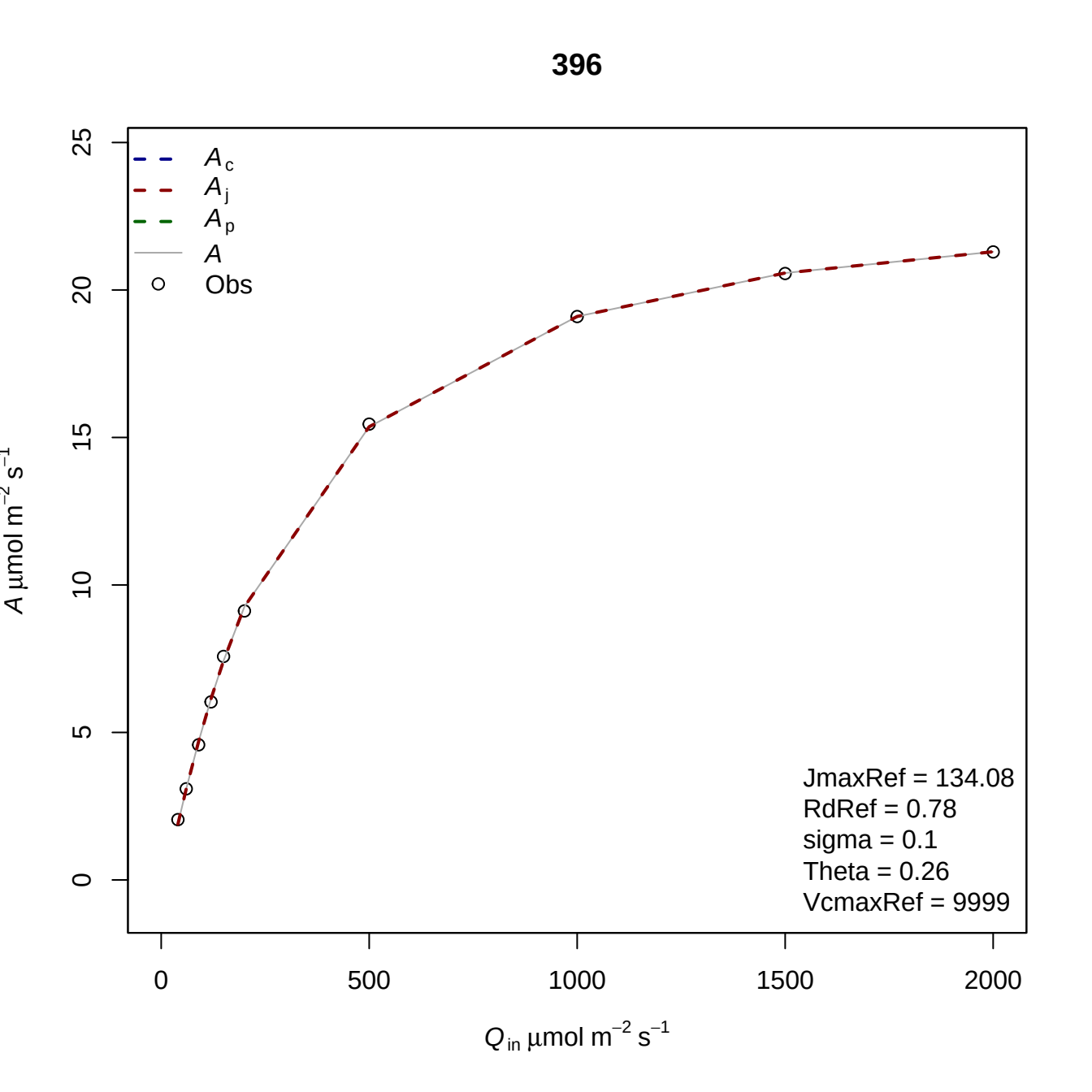

390
A_c
A_j
A_p
A
Obs
JmaxRef = 128.82
RdRef = 0.88
sigma = 0.16
Theta = −0.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

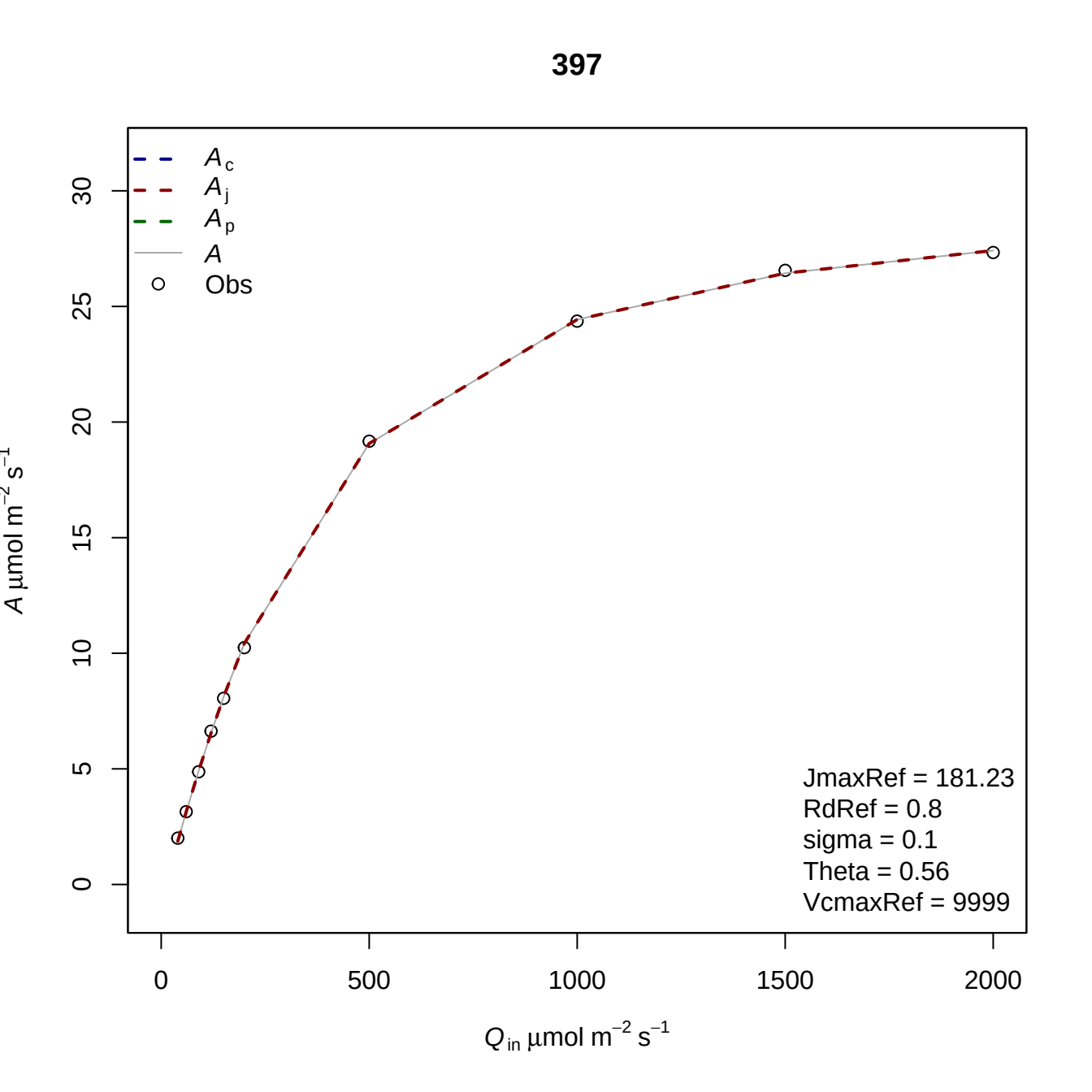

391
A_c
A_j
A_p
A
Obs
JmaxRef = 126.88
RdRef = 1.45
sigma = 0.28
Theta = −0.12
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

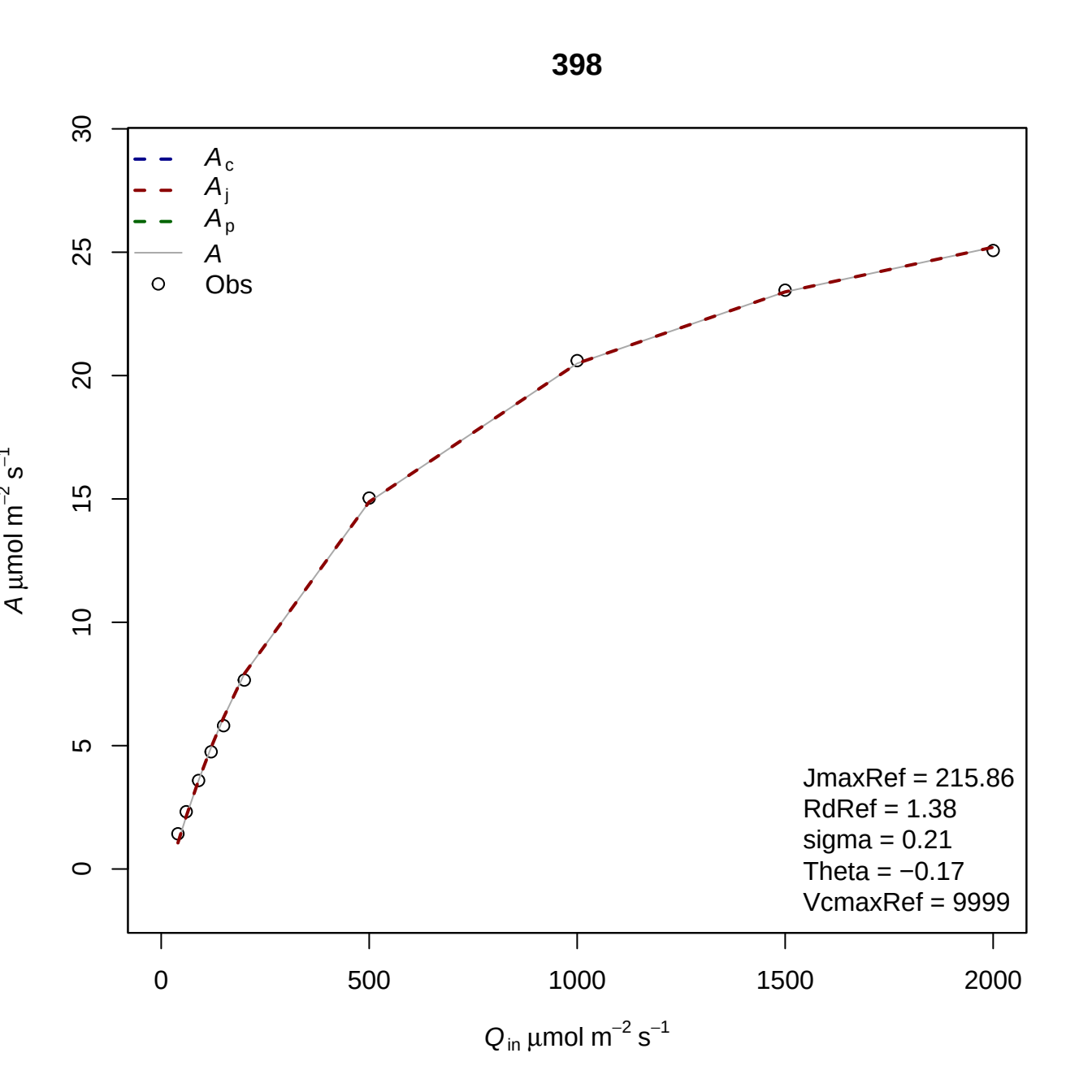

392
A_c
A_j
A_p
A
Obs
JmaxRef = 143.51
RdRef = 1.44
sigma = 0.21
Theta = 0.41
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

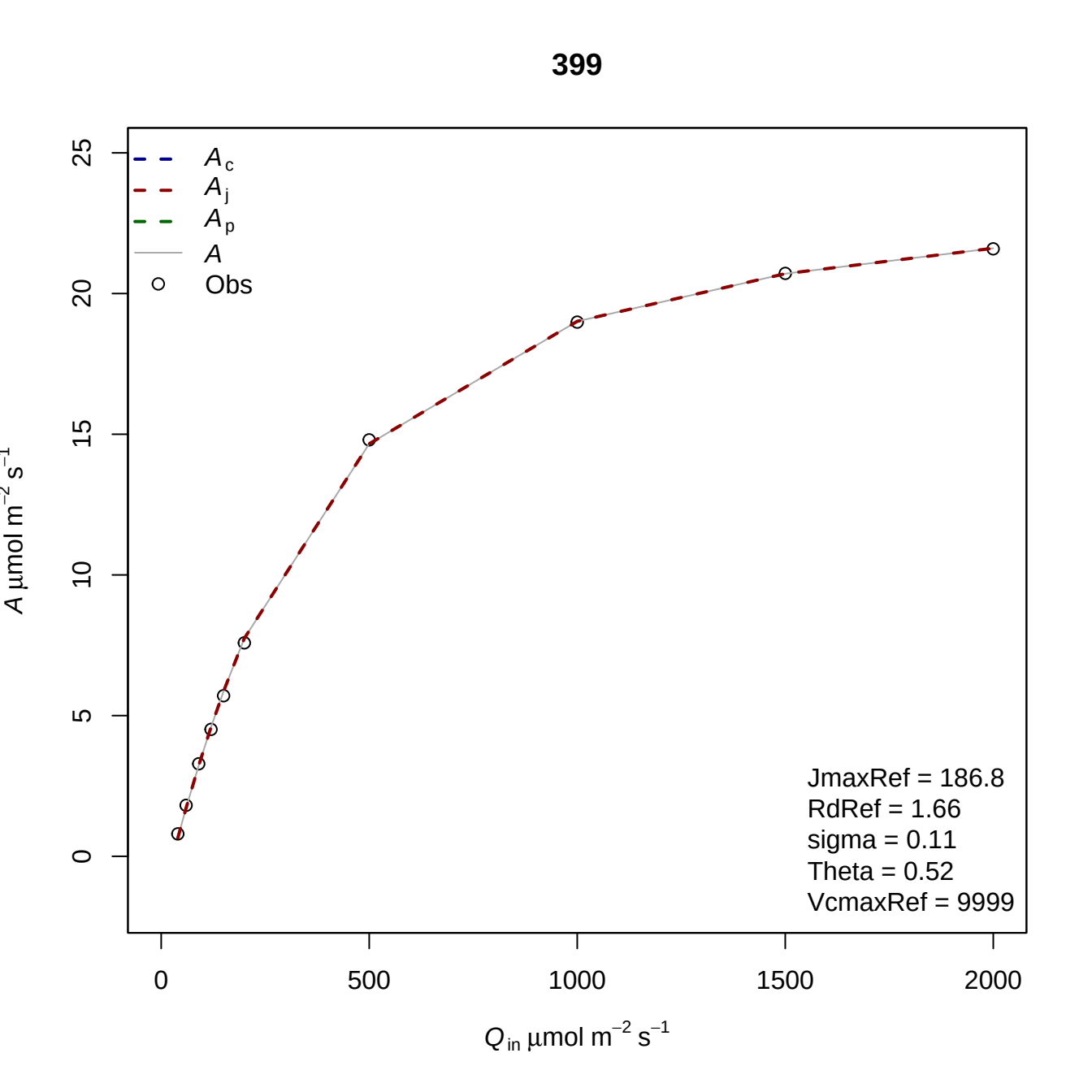

393
A_c
A_j
A_p
A
Obs
JmaxRef = 124.24
RdRef = 1.83
sigma = 0.49
Theta = −0.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

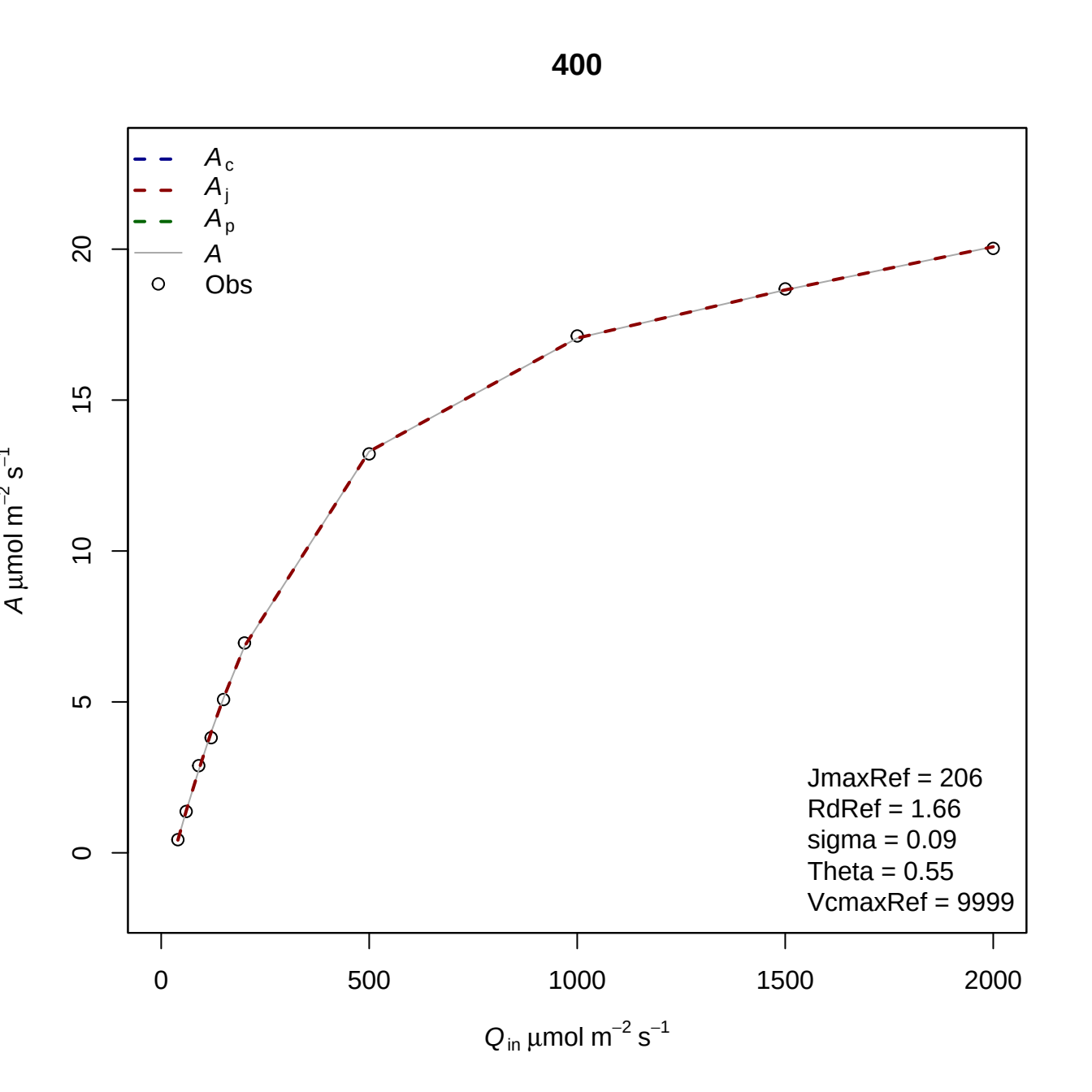

395
A_c
A_j
A_p
A
Obs
JmaxRef = 94.13
RdRef = 1.24
sigma = 0.25
Theta = −0.8
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

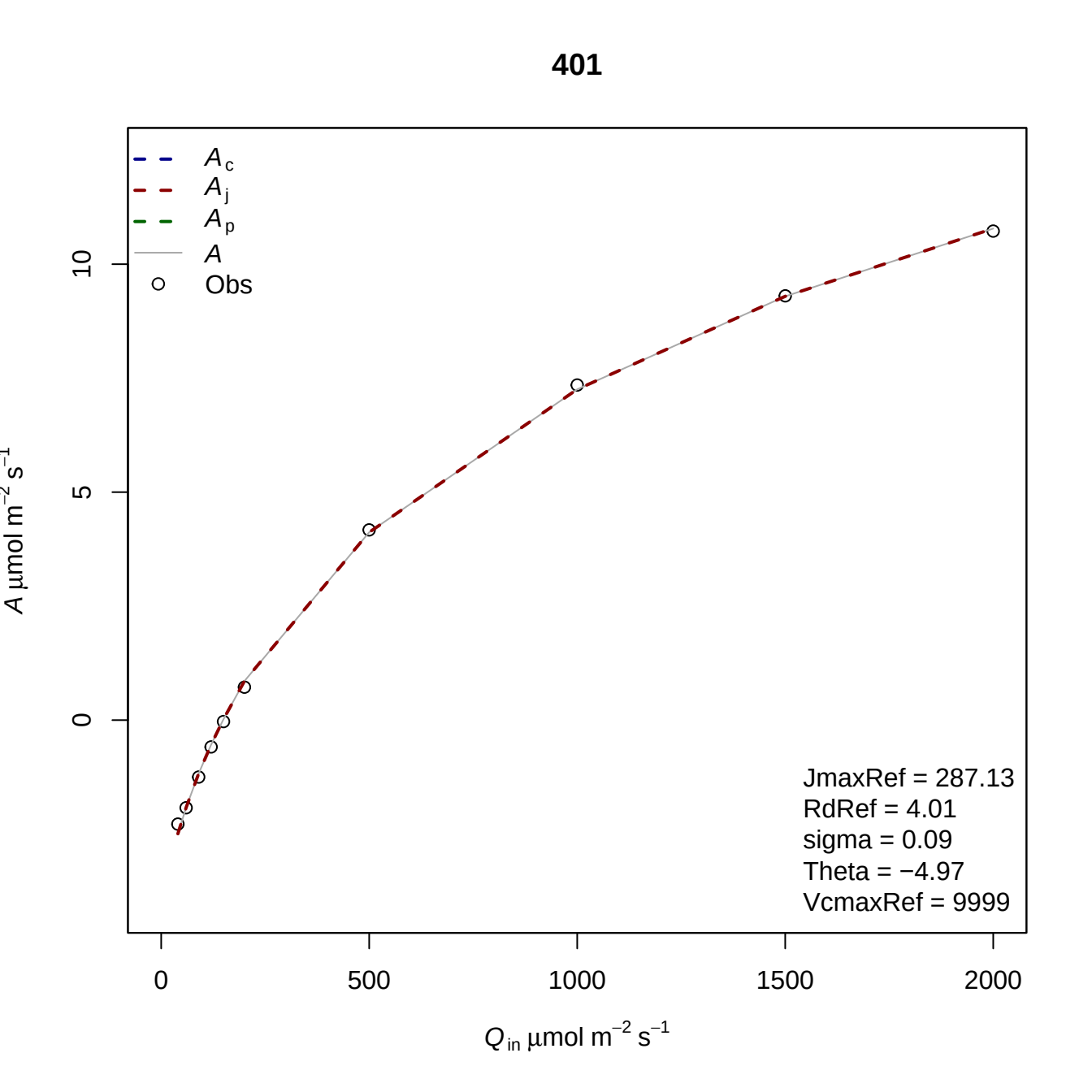

396
A_c
A_j
A_p
A
Obs
JmaxRef = 134.08
RdRef = 0.78
sigma = 0.1
Theta = 0.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

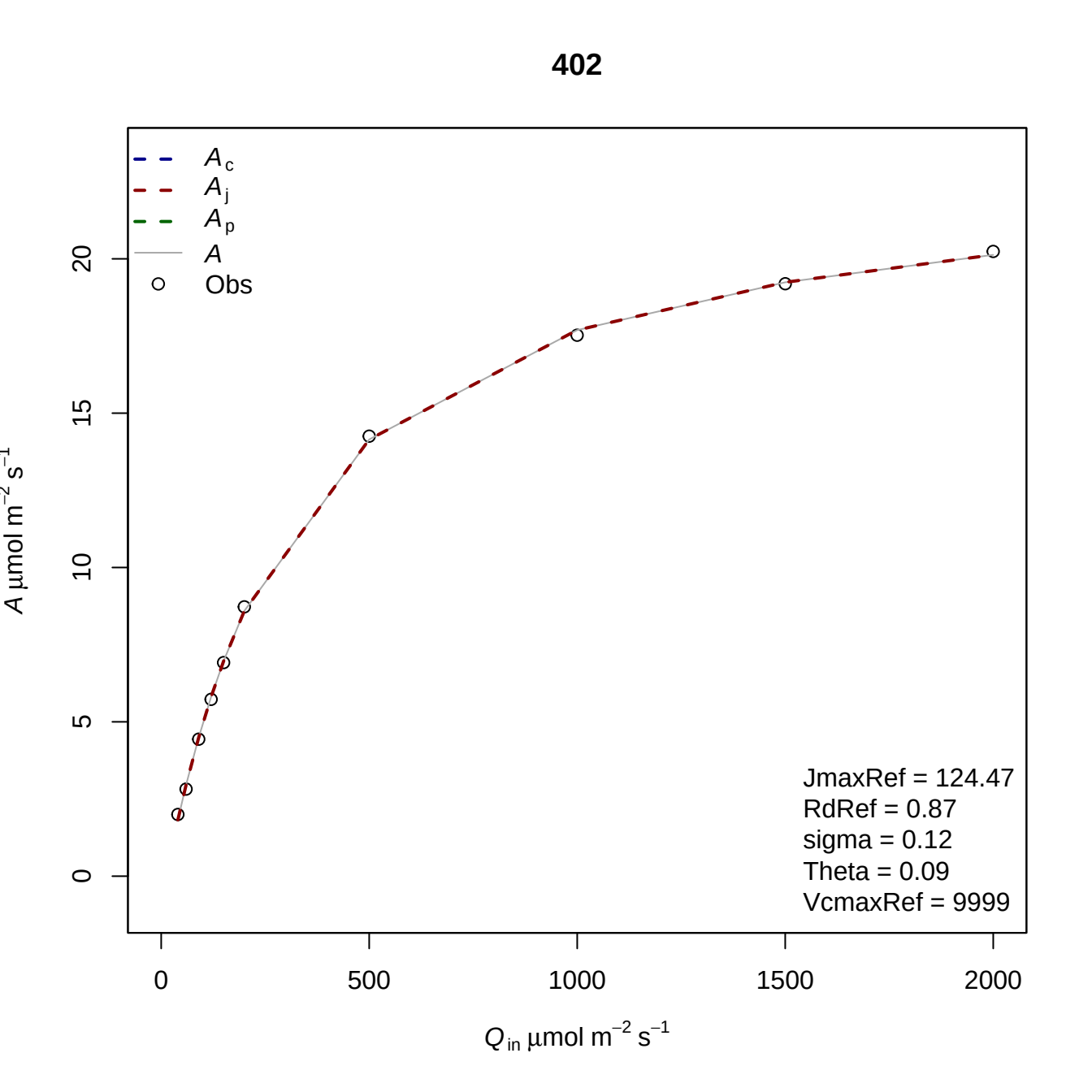

397
A_c
A_j
A_p
A
Obs
JmaxRef = 181.23
RdRef = 0.8
sigma = 0.1
Theta = 0.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

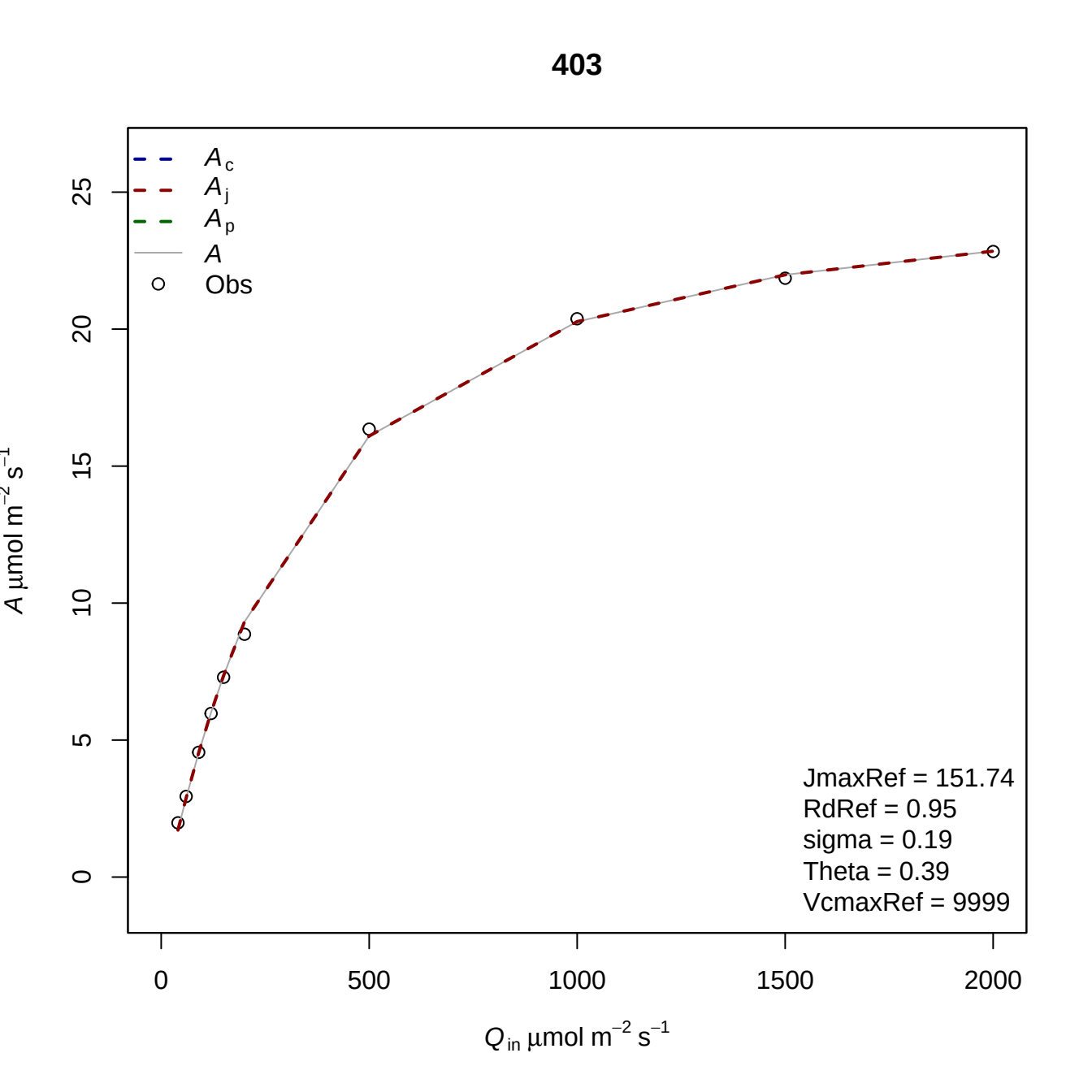

398
A_c
A_j
A_p
A
Obs
JmaxRef = 215.86
RdRef = 1.38
sigma = 0.21
Theta = −0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

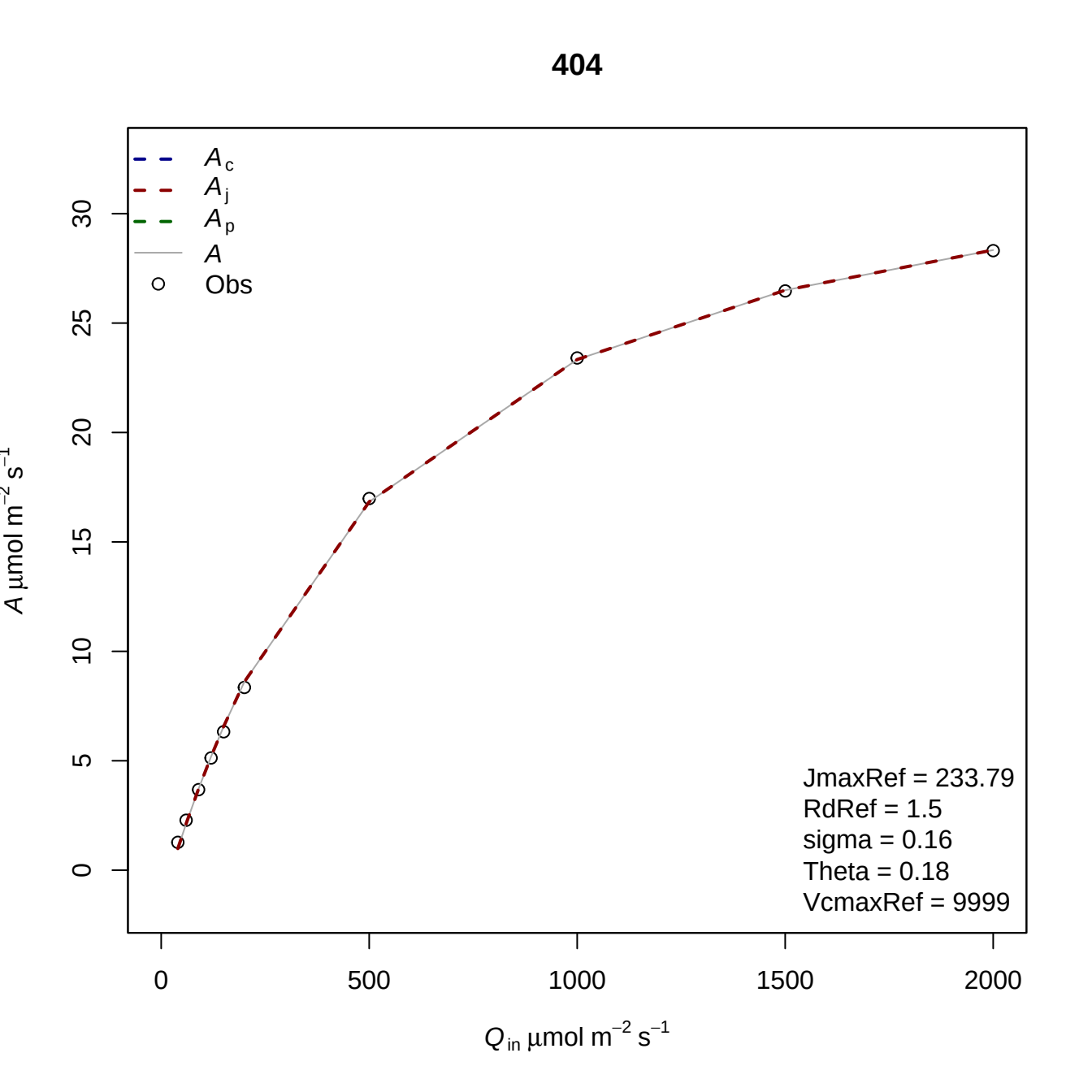

399
A_c
A_j
A_p
A
Obs
JmaxRef = 186.8
RdRef = 1.66
sigma = 0.11
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

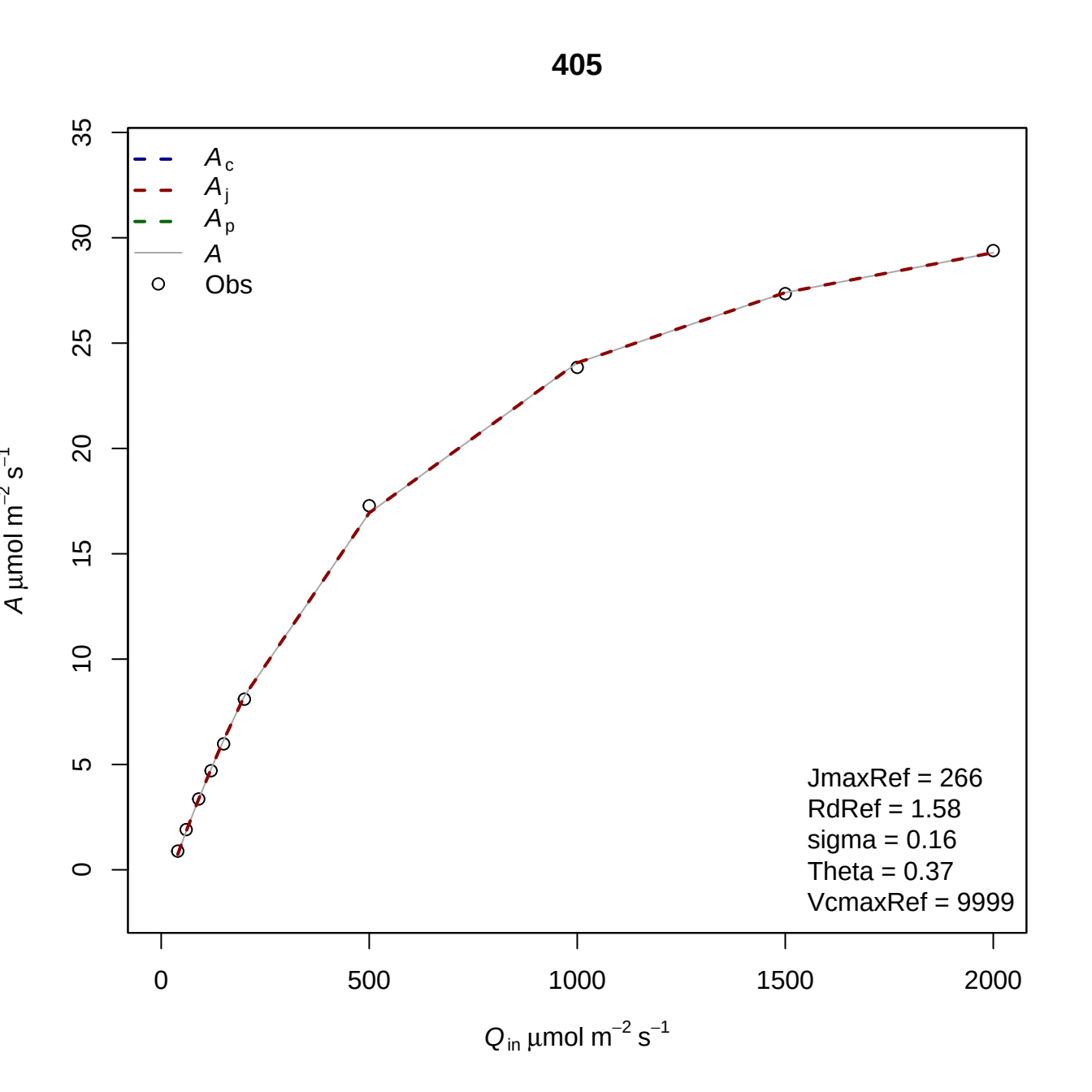

400
A_c
A_j
A_p
A
Obs
JmaxRef = 206
RdRef = 1.66
sigma = 0.09
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

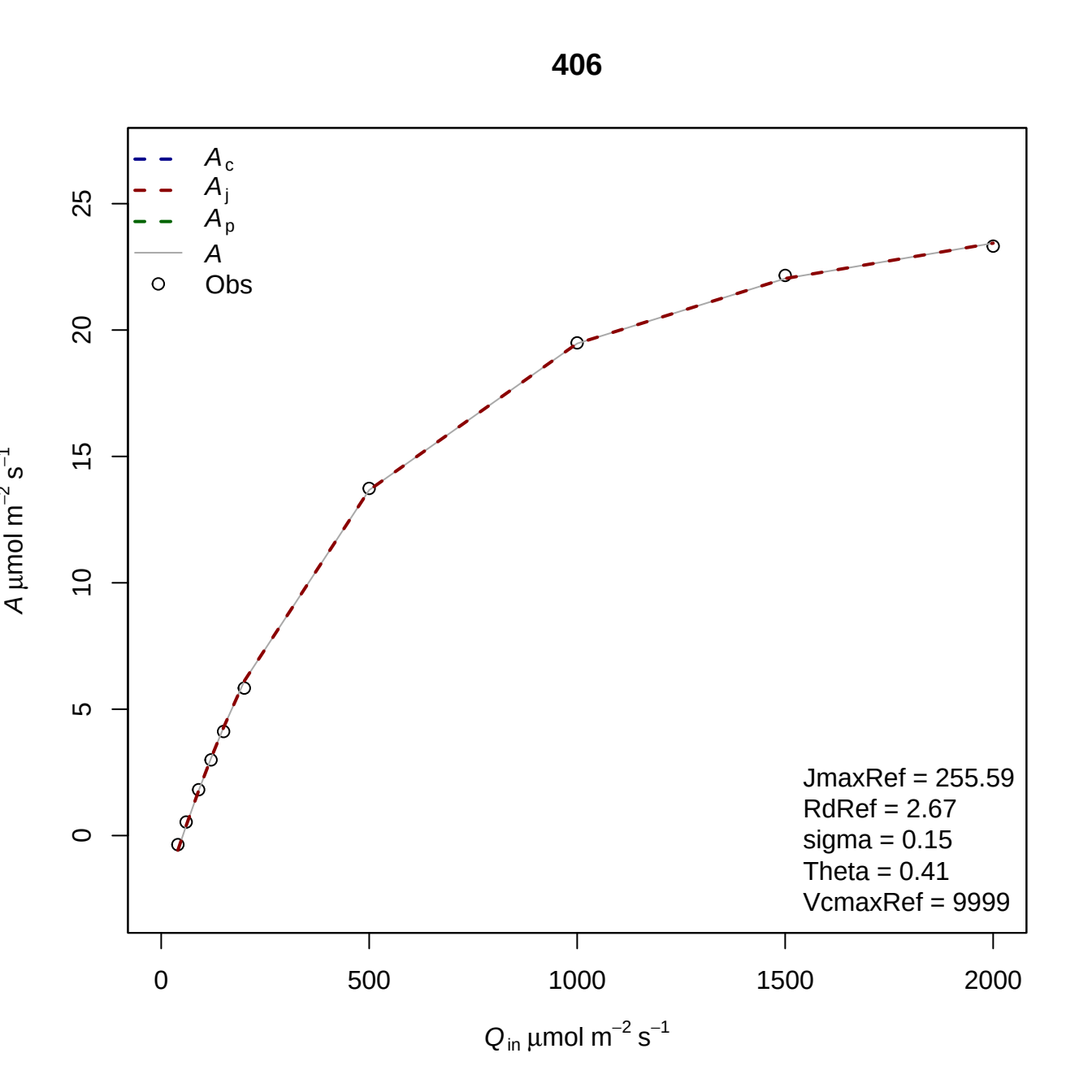

401
A_c
A_j
A_p
A
Obs
JmaxRef = 287.13
RdRef = 4.01
sigma = 0.09
Theta = −4.97
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

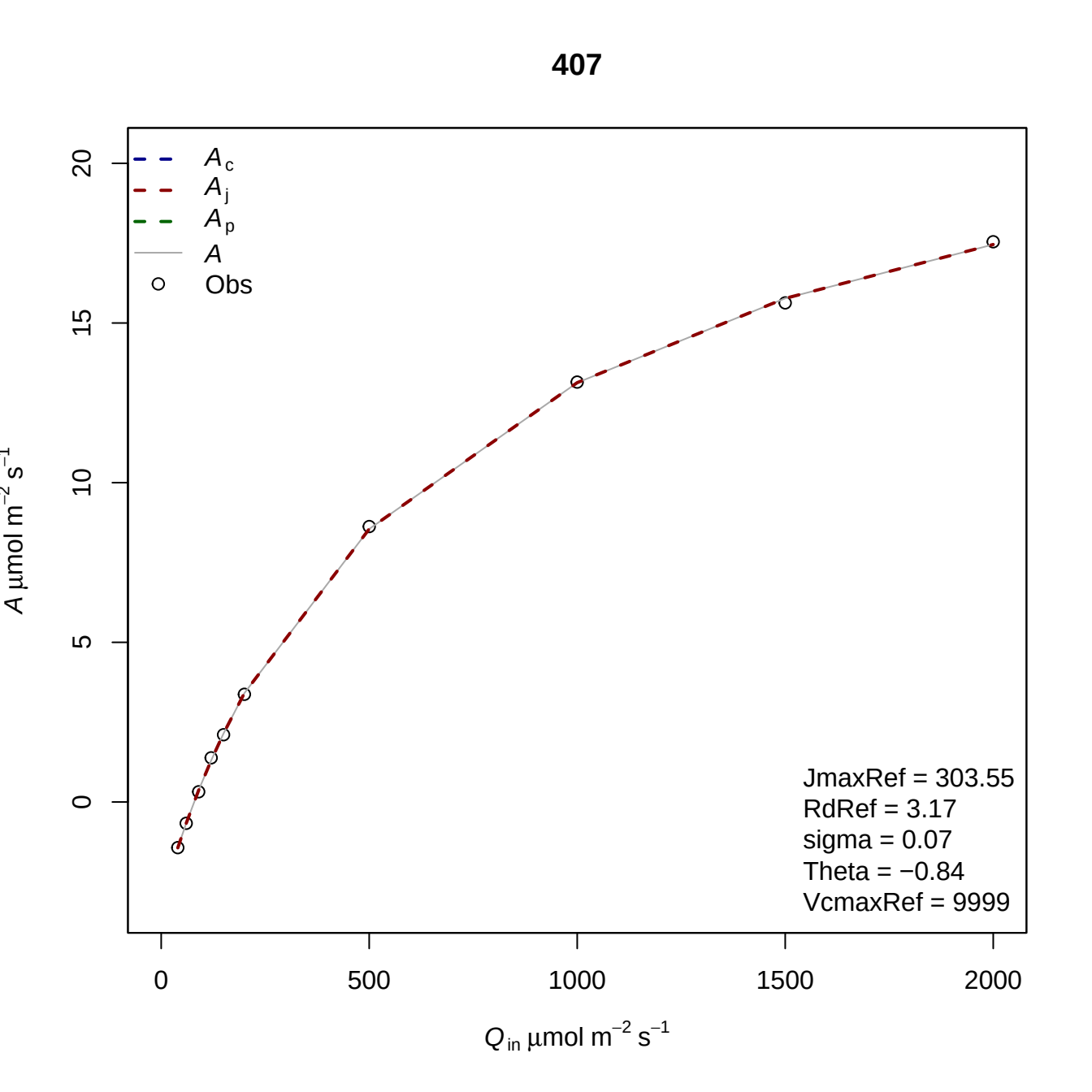

402
A_c
A_j
A_p
A
Obs
JmaxRef = 124.47
RdRef = 0.87
sigma = 0.12
Theta = 0.09
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

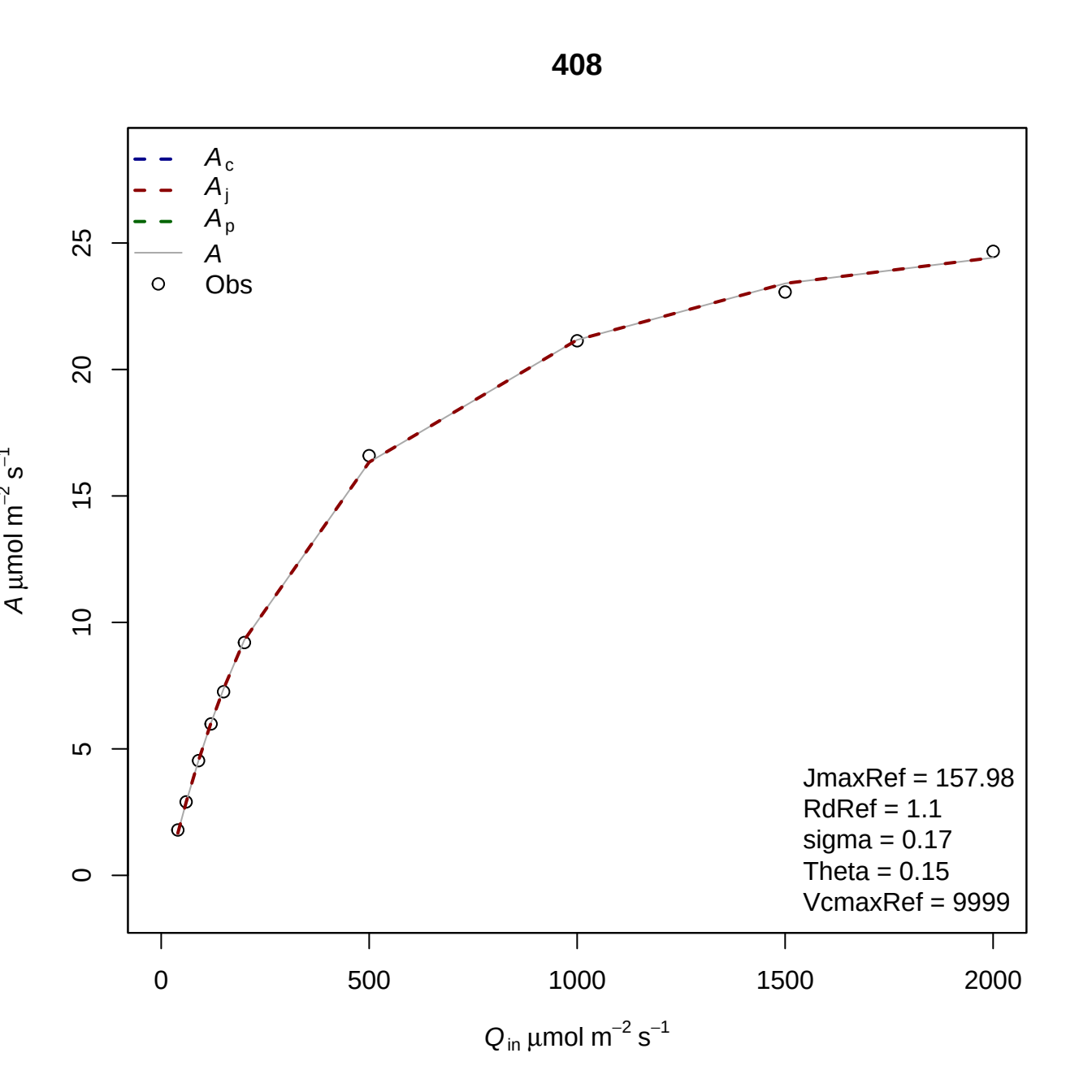

403
A_c
A_j
A_p
A
Obs
JmaxRef = 151.74
RdRef = 0.95
sigma = 0.19
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

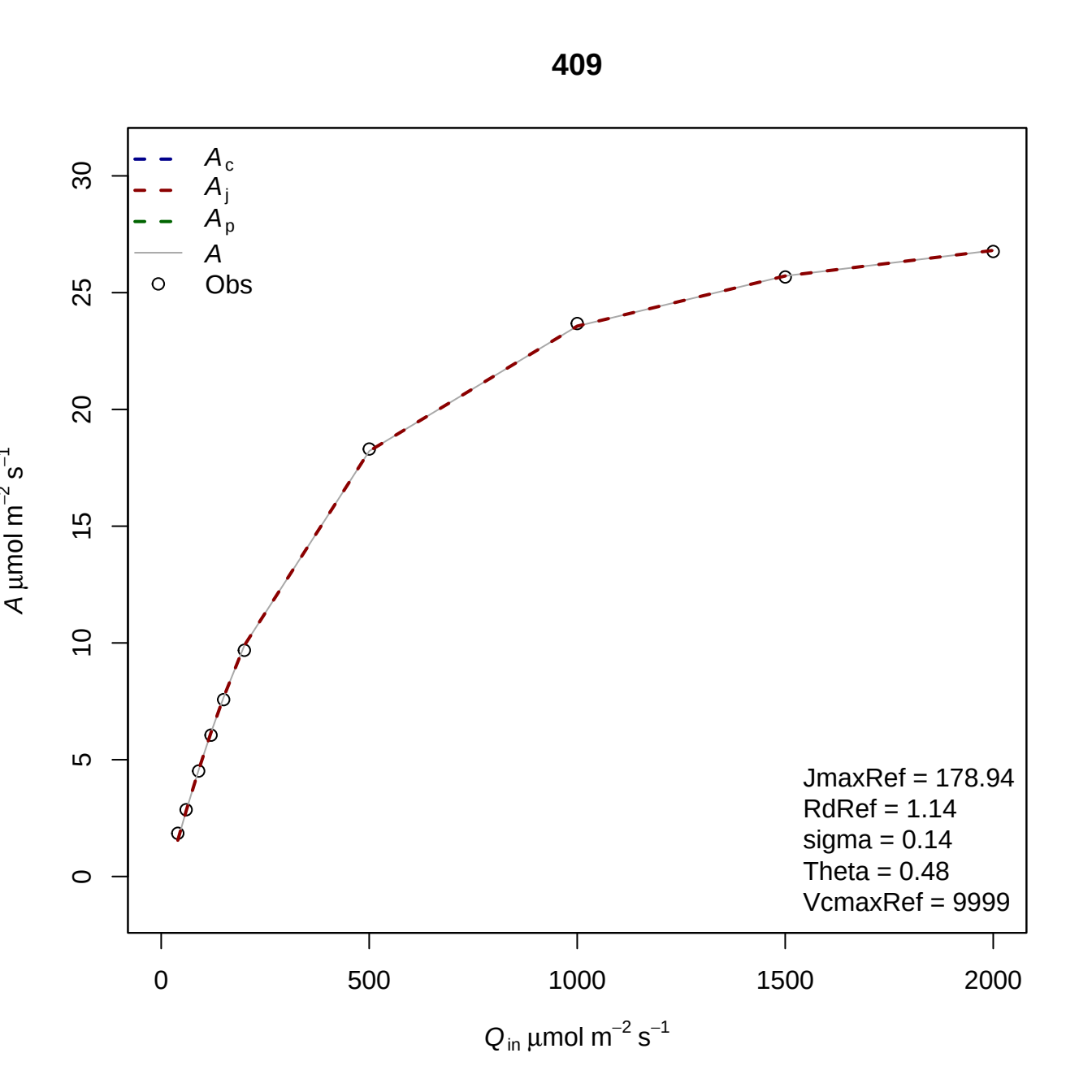

404
A_c
A_j
A_p
A
Obs
JmaxRef = 233.79
RdRef = 1.5
sigma = 0.16
Theta = 0.18
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

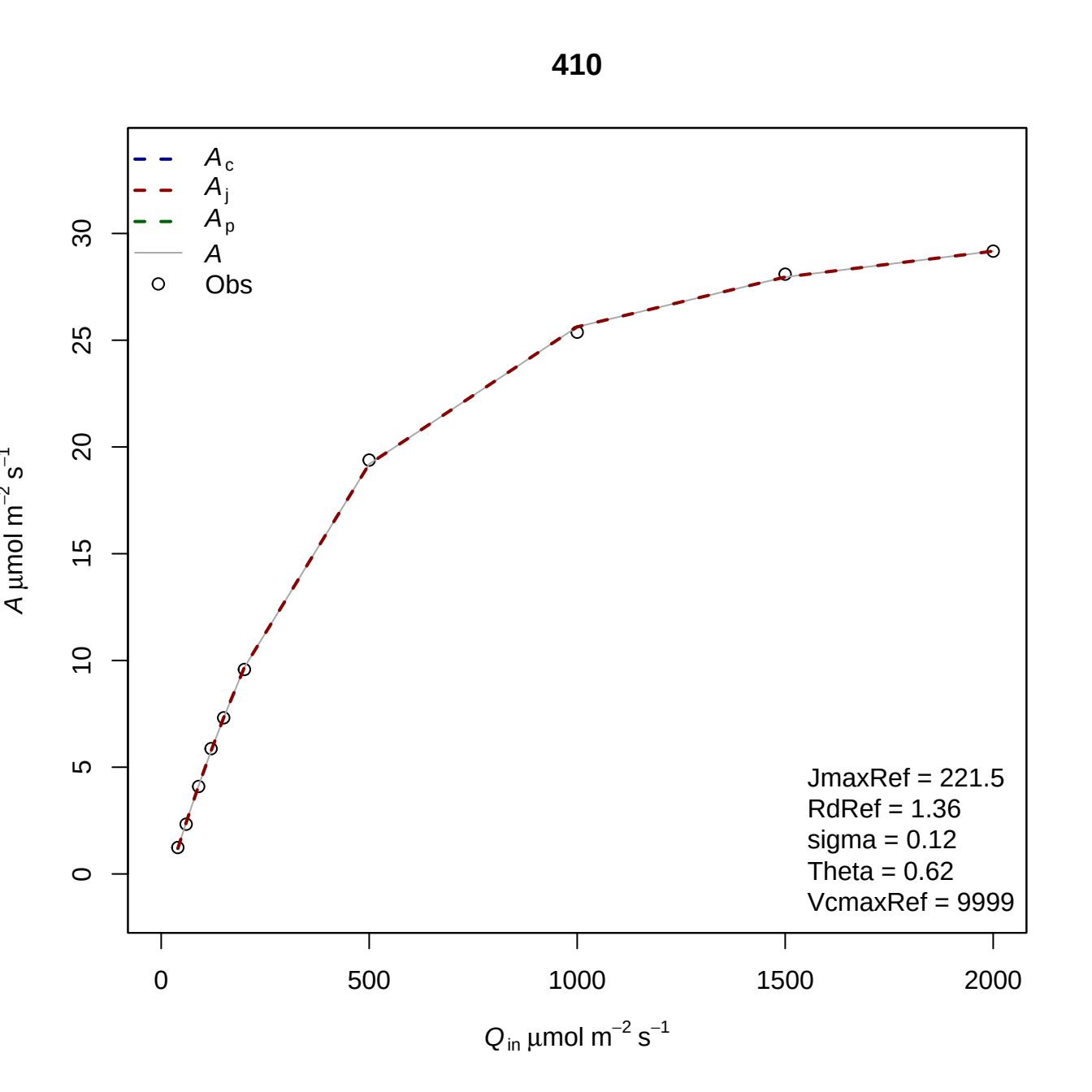

405
A_c
A_j
A_p
A
Obs
JmaxRef = 266
RdRef = 1.58
sigma = 0.16
Theta = 0.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

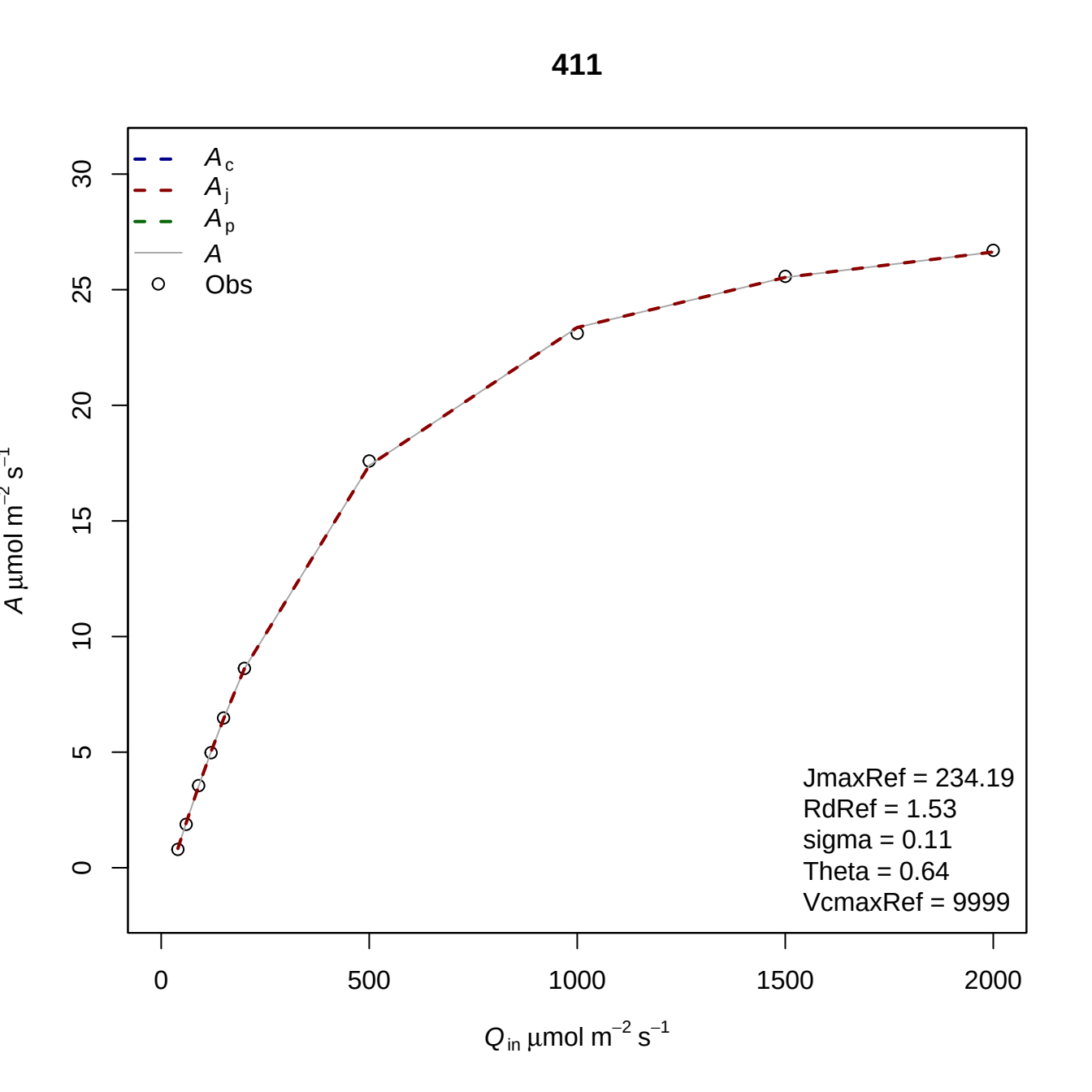

406
A_c
A_j
A_p
A
Obs
JmaxRef = 255.59
RdRef = 2.67
sigma = 0.15
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

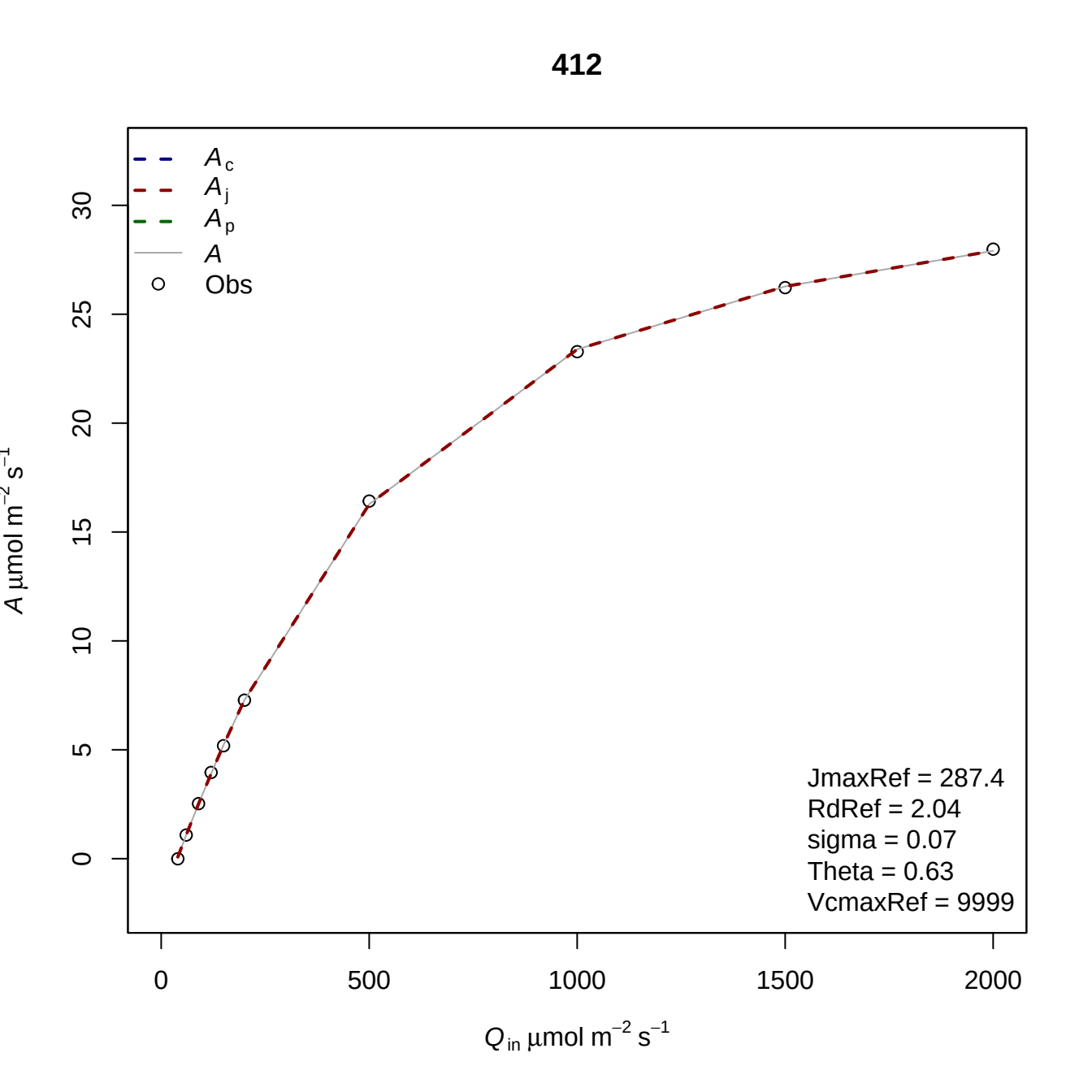

407
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 303.55
RdRef = 3.17
sigma = 0.07
Theta = −0.84
VcmaxRef = 9999

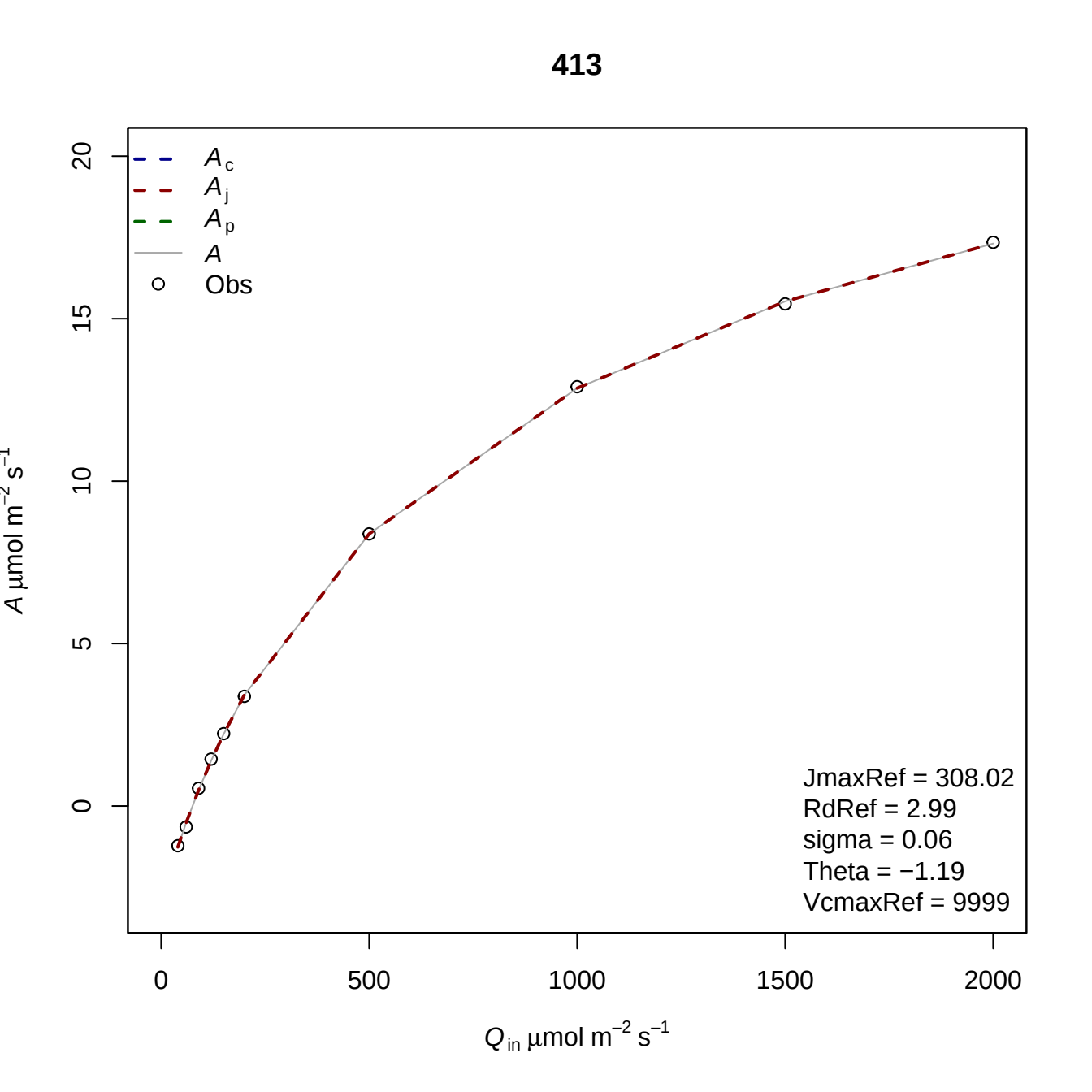

408
A_c
A_j
A_p
A
Obs
JmaxRef = 157.98
RdRef = 1.1
sigma = 0.17
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

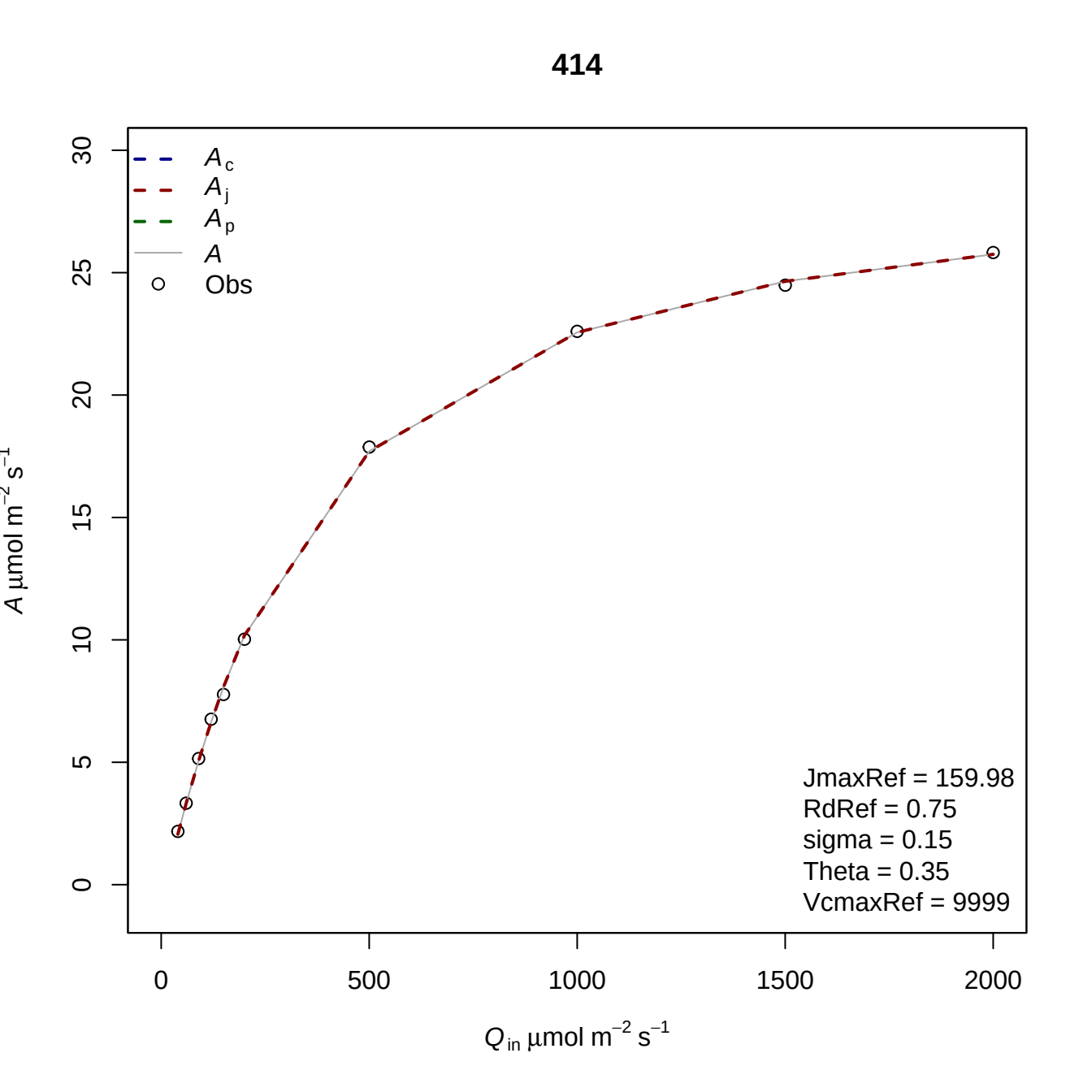

409
A_c
A_j
A_p
A
Obs
JmaxRef = 178.94
RdRef = 1.14
sigma = 0.14
Theta = 0.48
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

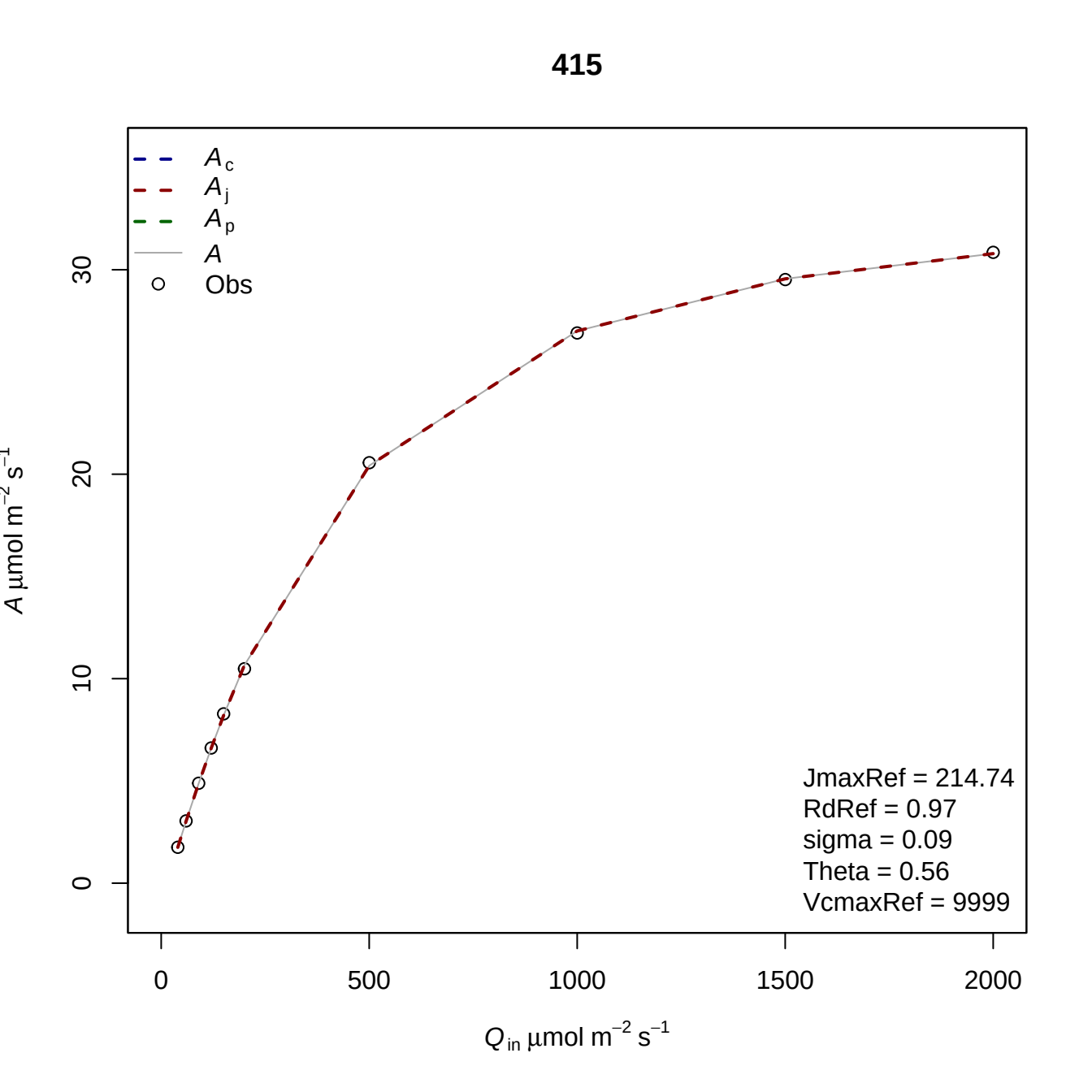

410
A_c
A_j
A_p
A
Obs
JmaxRef = 221.5
RdRef = 1.36
sigma = 0.12
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

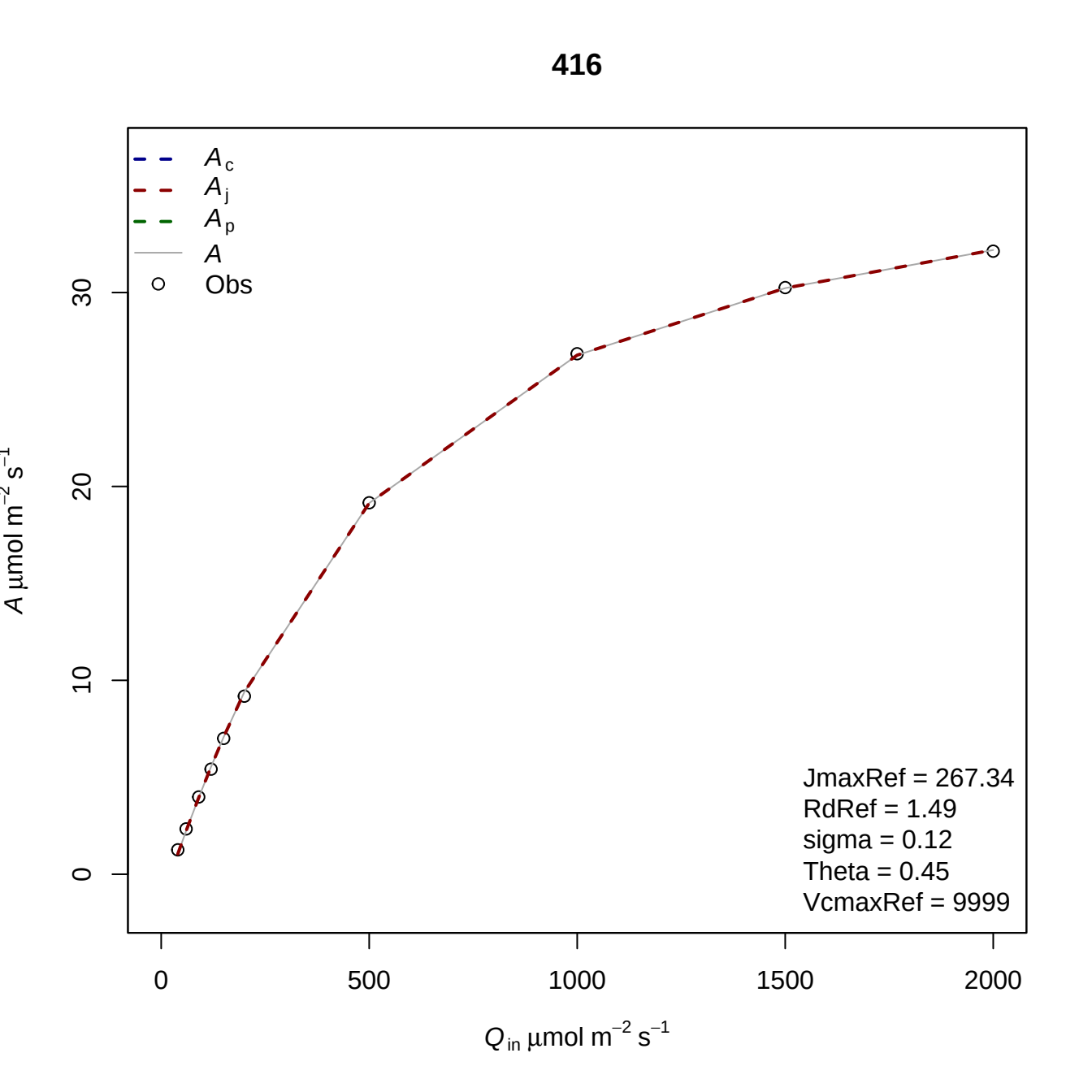

411
A_c
A_j
A_p
A
Obs
JmaxRef = 234.19
RdRef = 1.53
sigma = 0.11
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

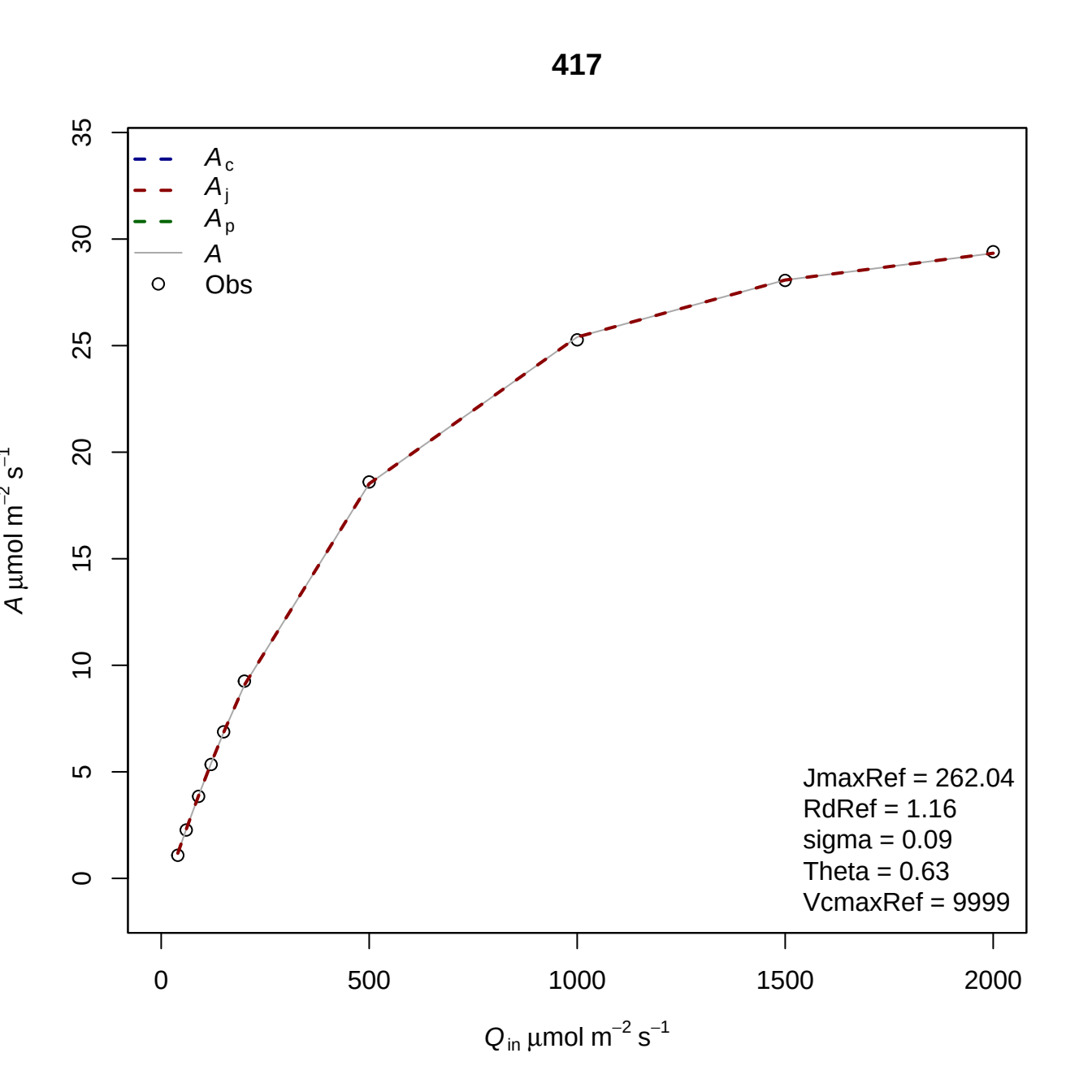

412
A_c
A_j
A_p
A
Obs
JmaxRef = 287.4
RdRef = 2.04
sigma = 0.07
Theta = 0.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

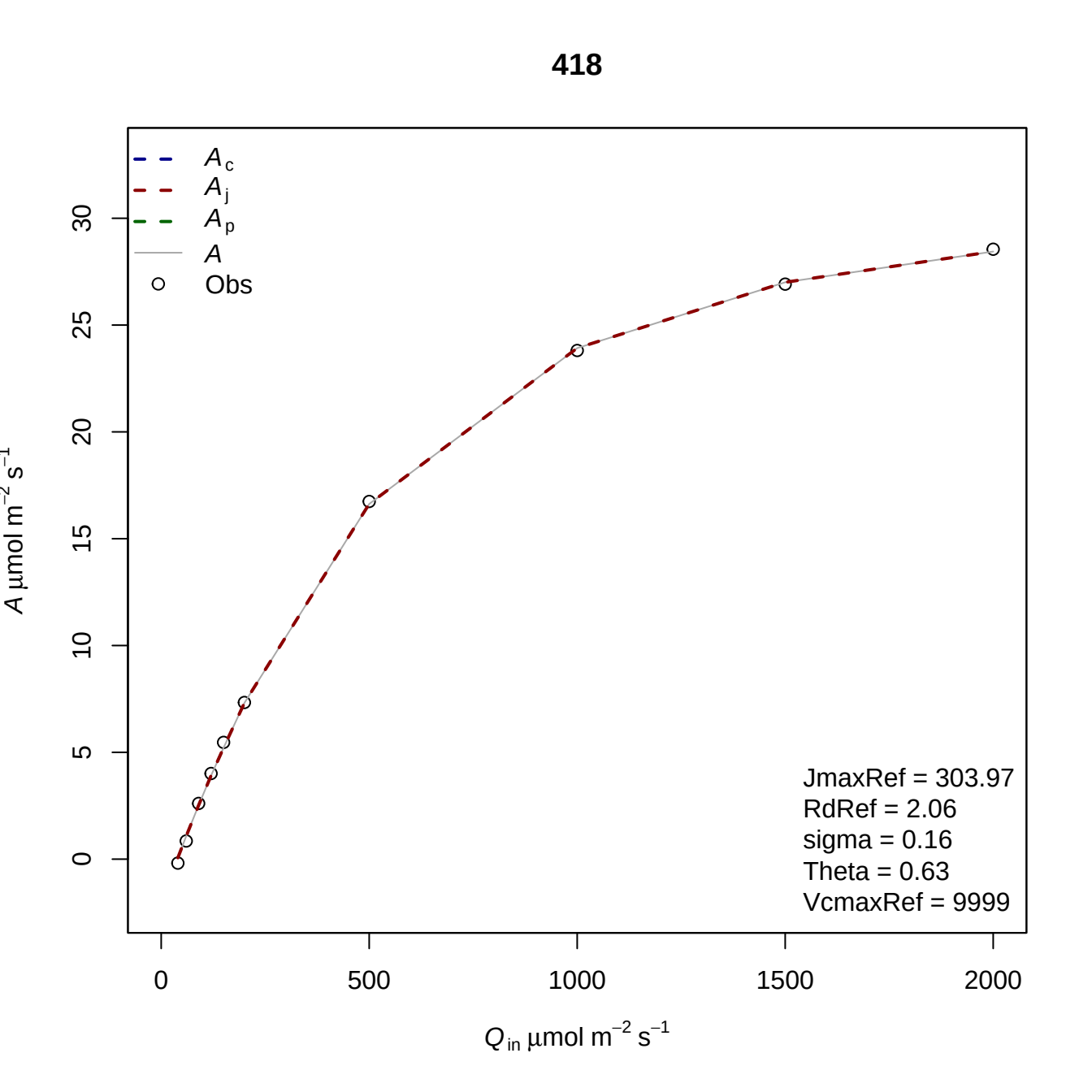

413
A_c
A_j
A_p
A
Obs
JmaxRef = 308.02
RdRef = 2.99
sigma = 0.06
Theta = −1.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

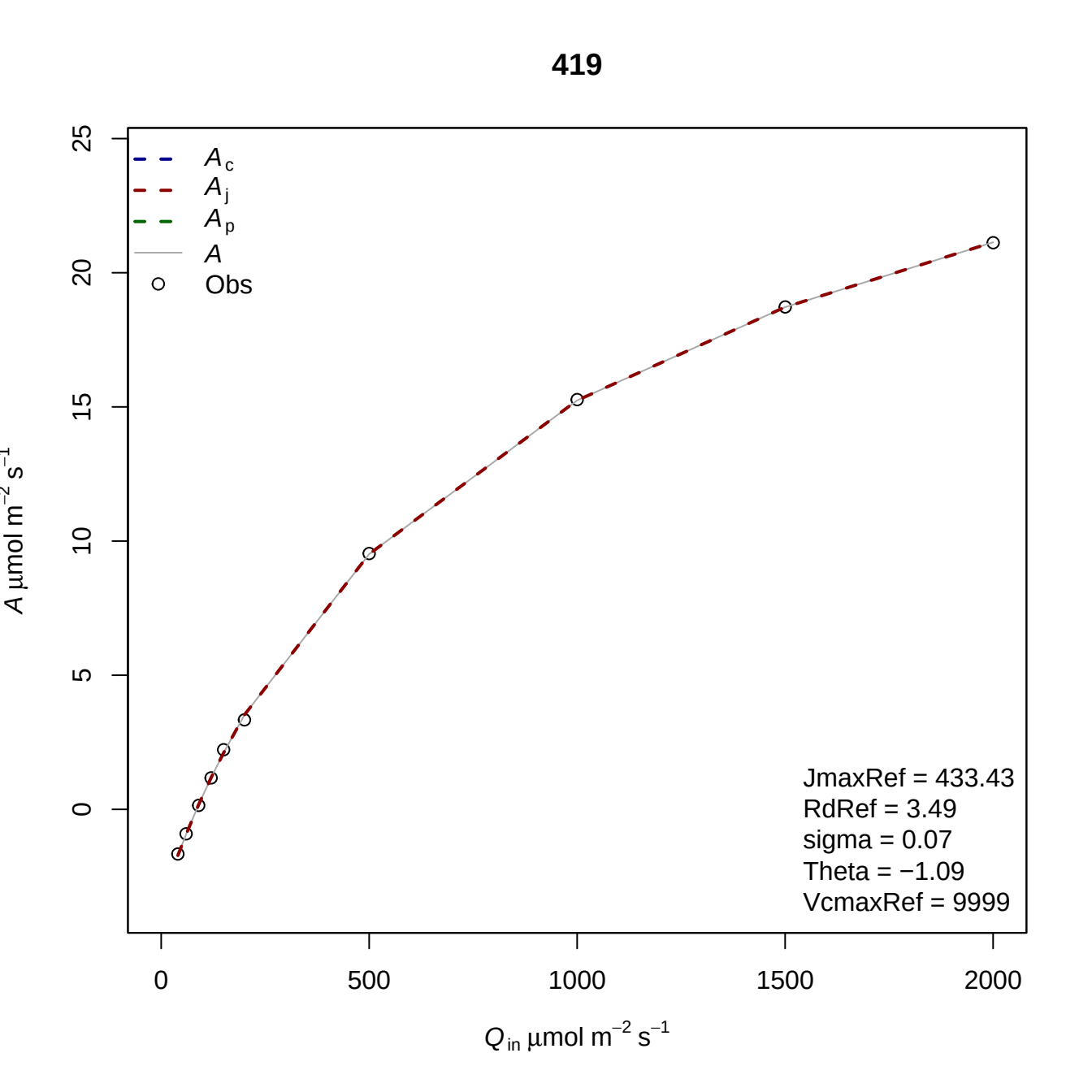

414
A_c
A_j
A_p
A
Obs
JmaxRef = 159.98
RdRef = 0.75
sigma = 0.15
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

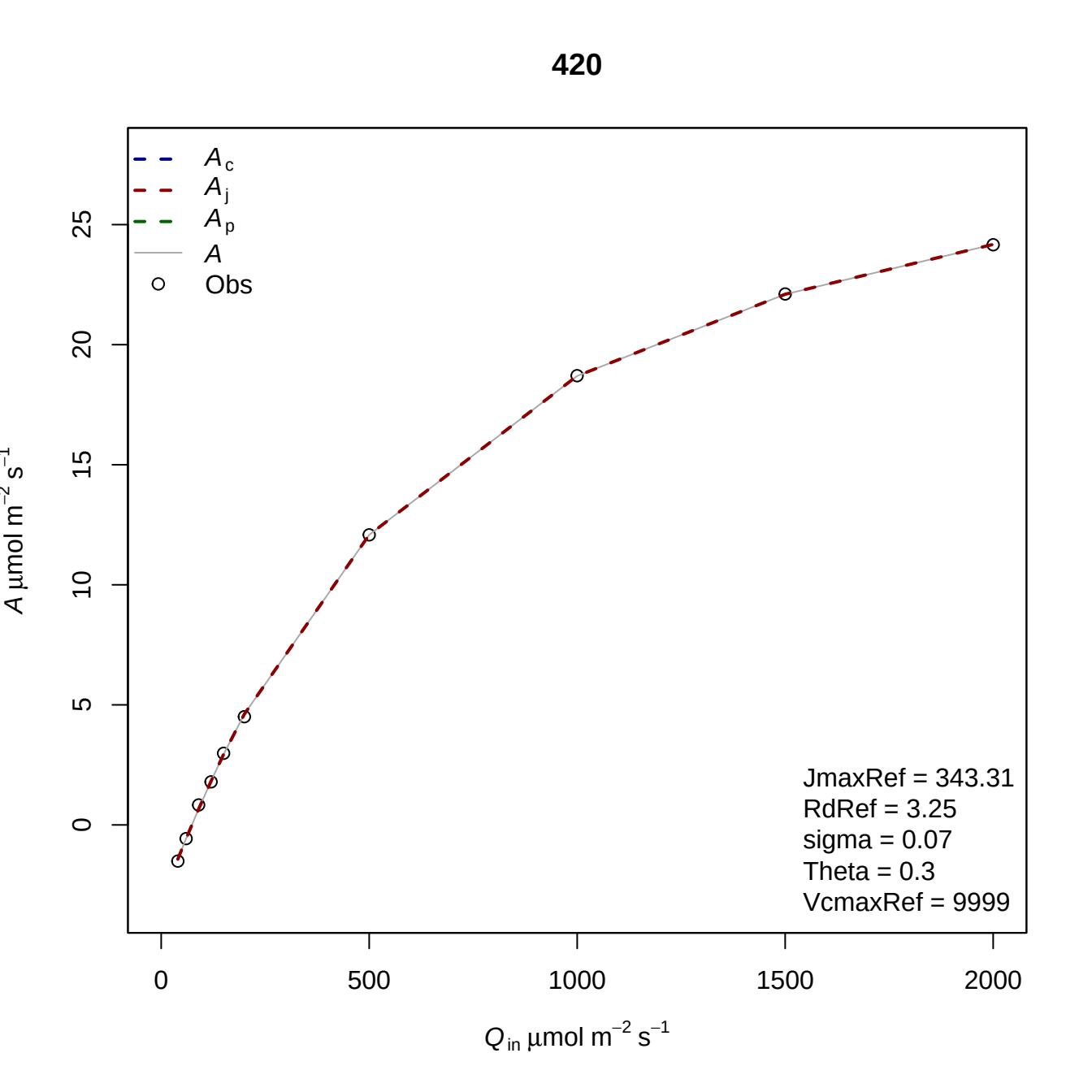

415
A_c
A_j
A_p
A
Obs
JmaxRef = 214.74
RdRef = 0.97
sigma = 0.09
Theta = 0.56
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

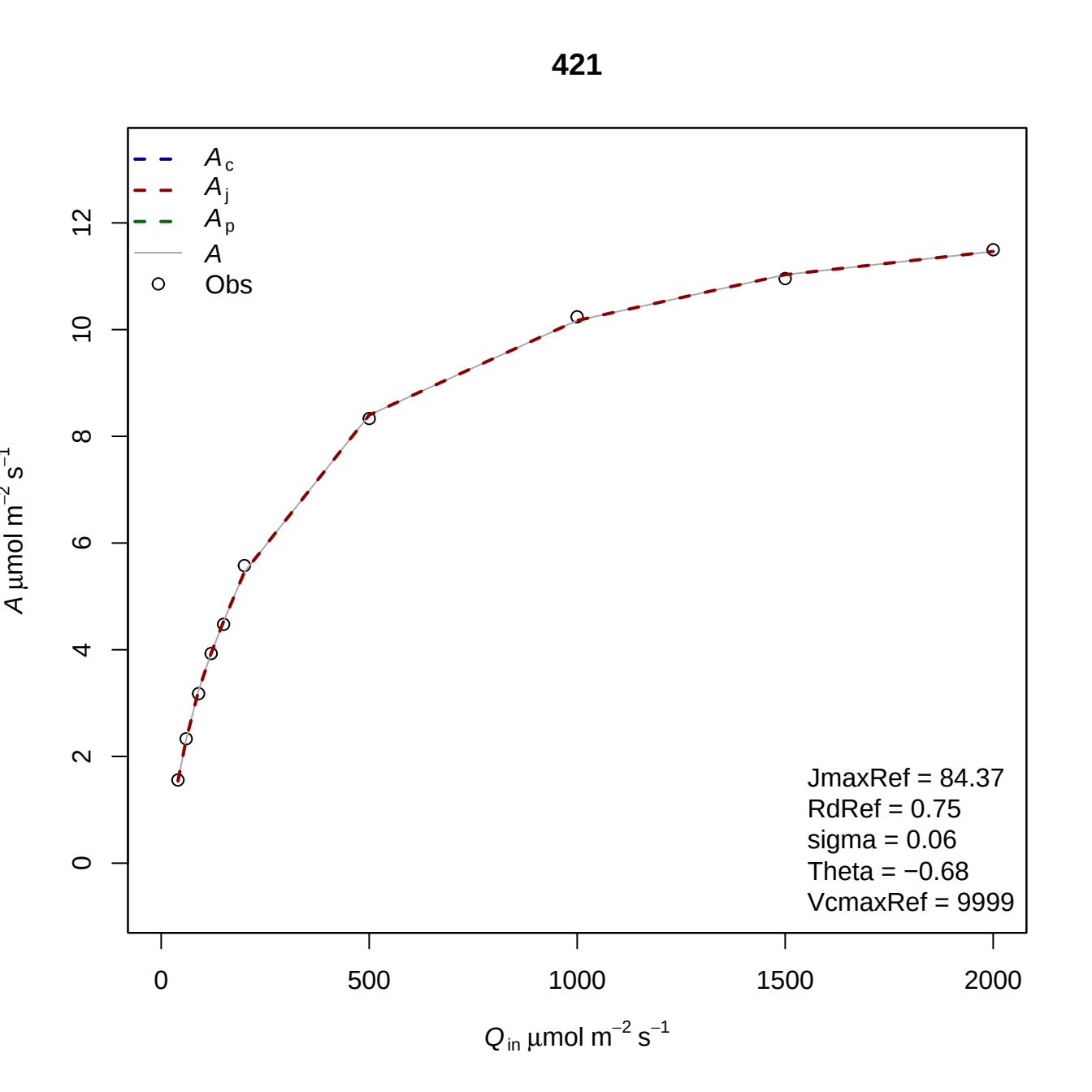

416
A_c
A_j
A_p
A
Obs
JmaxRef = 267.34
RdRef = 1.49
sigma = 0.12
Theta = 0.45
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

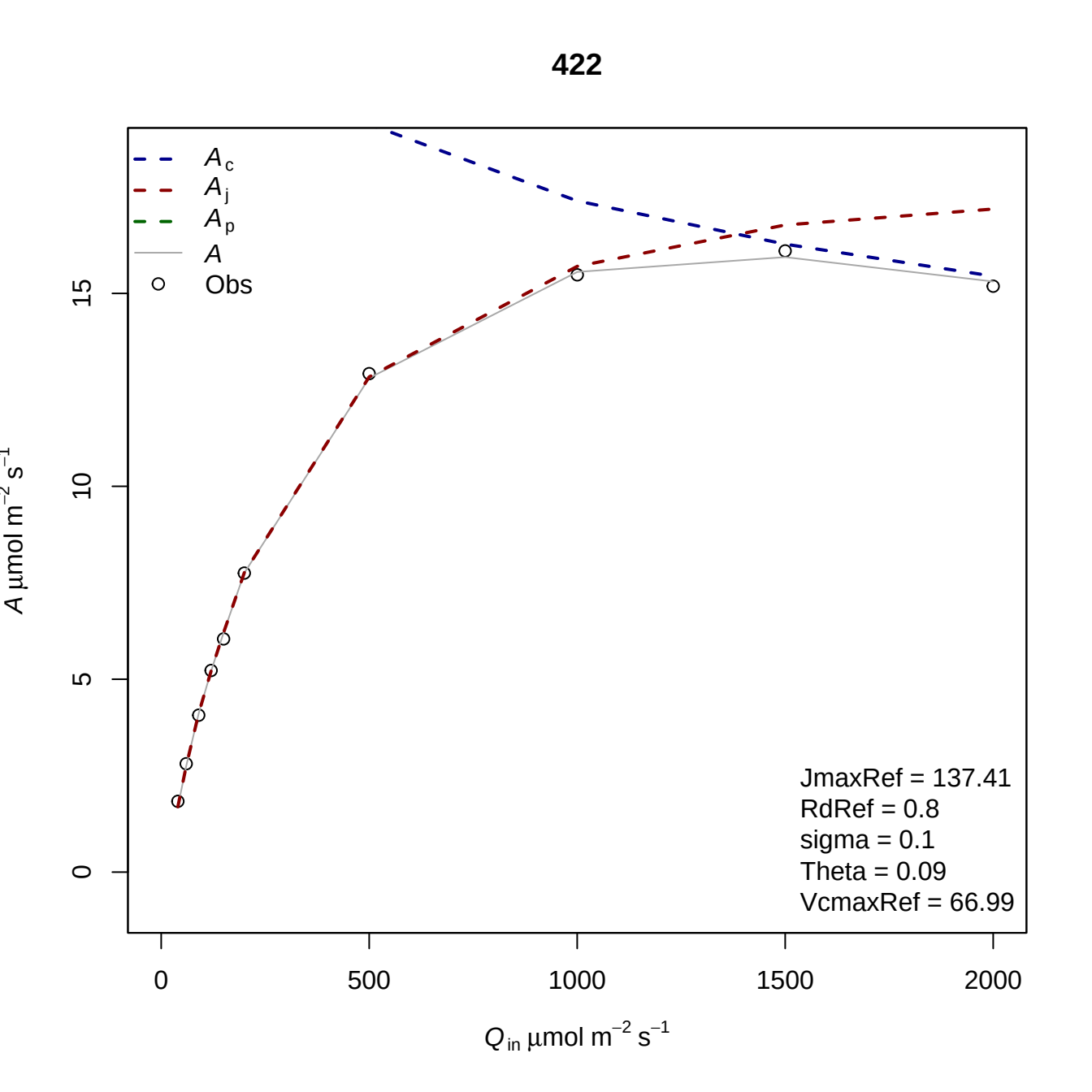

417
A_c
A_j
A_p
A
Obs
JmaxRef = 262.04
RdRef = 1.16
sigma = 0.09
Theta = 0.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

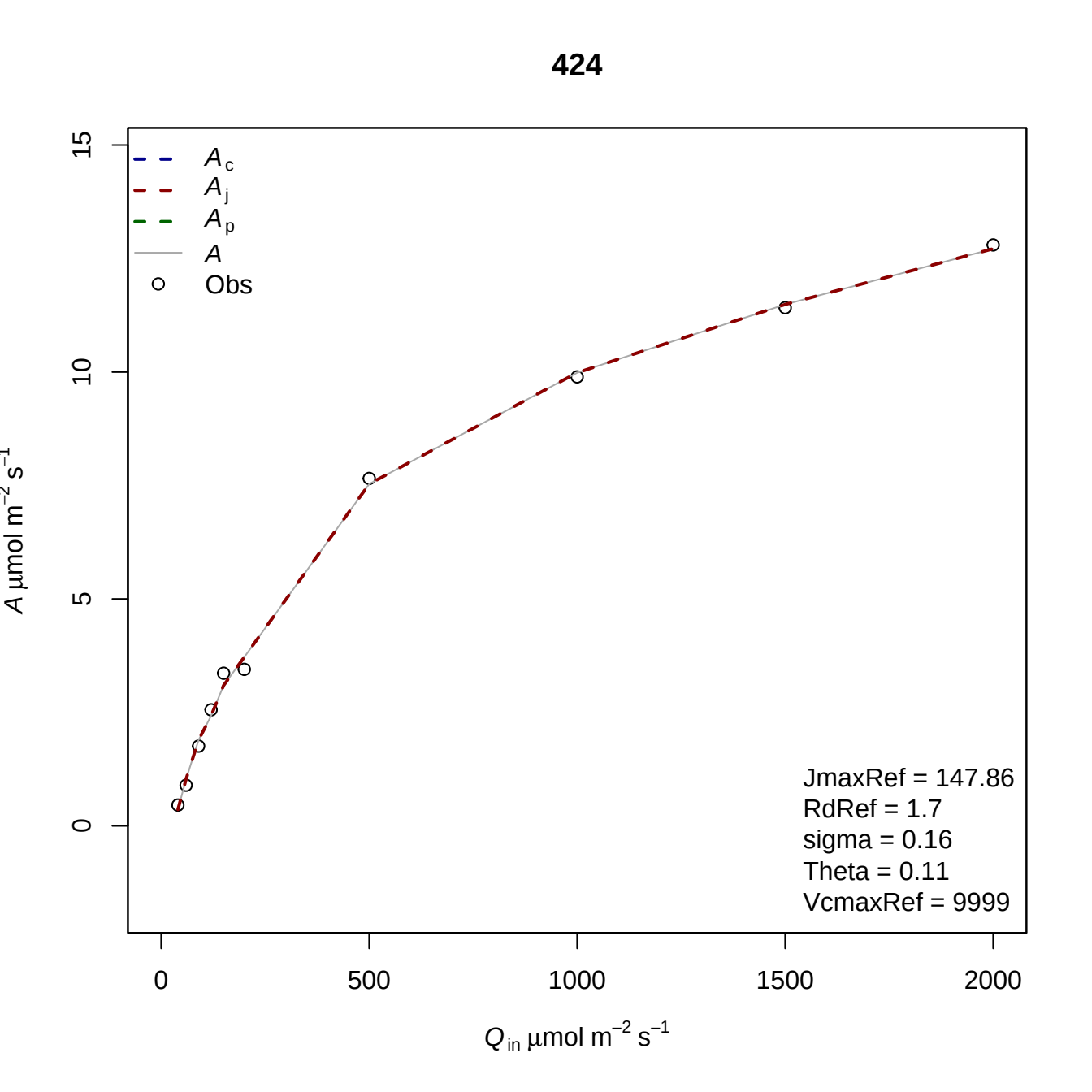

418
A_c
A_j
A_p
A
Obs
JmaxRef = 303.97
RdRef = 2.06
sigma = 0.16
Theta = 0.63
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

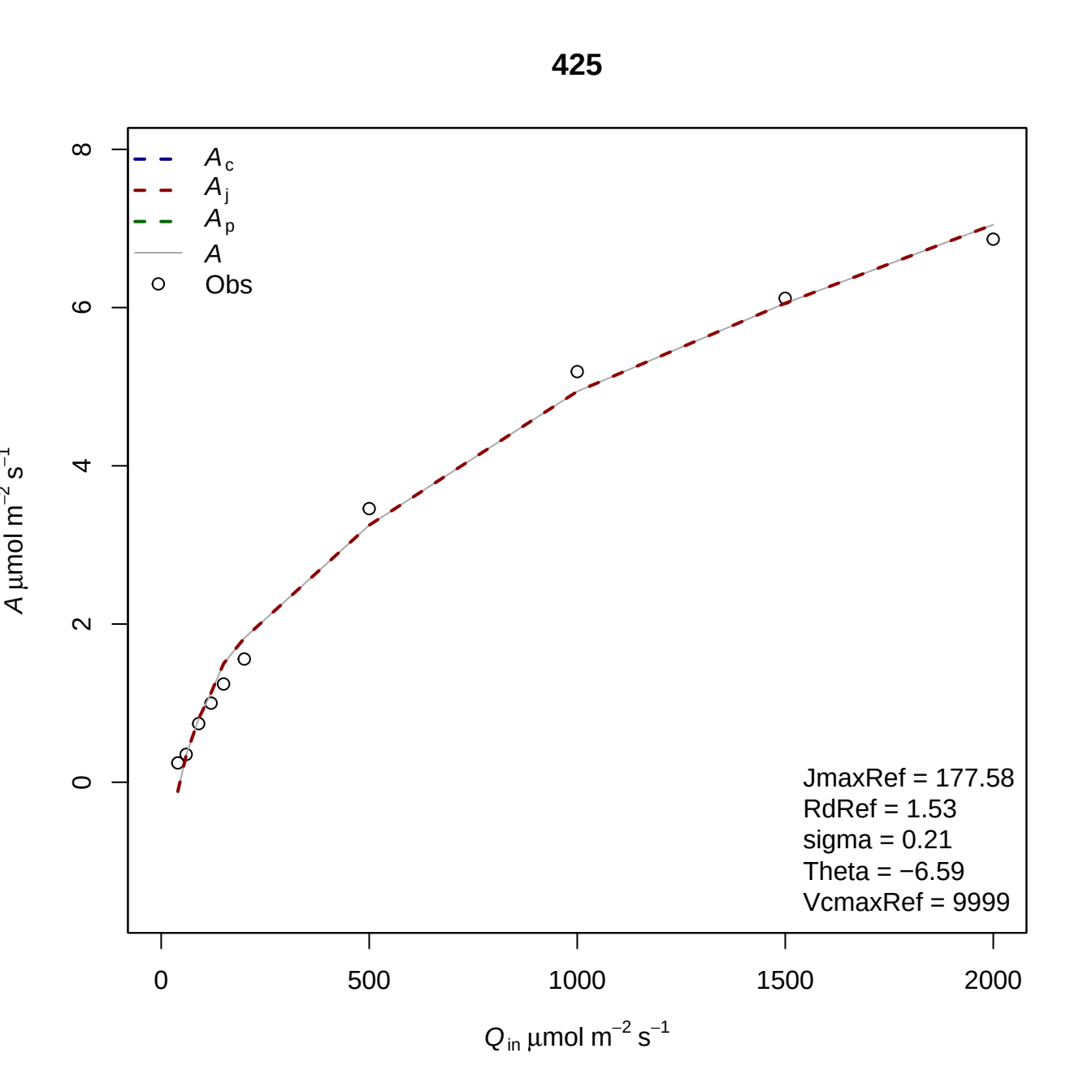

419
A_c
A_j
A_p
A
Obs
JmaxRef = 433.43
RdRef = 3.49
sigma = 0.07
Theta = −1.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

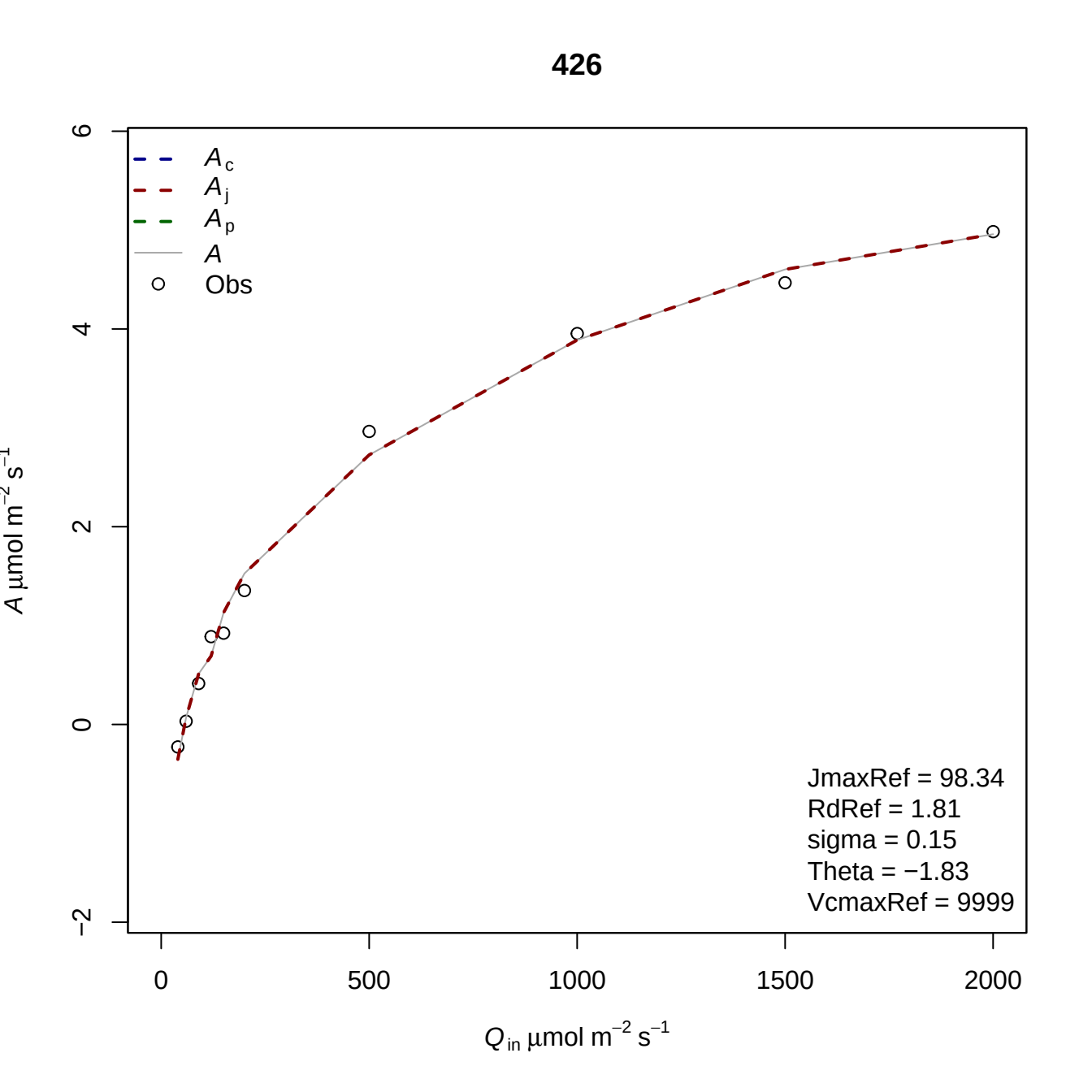

420
A_c
A_j
A_p
A
Obs
JmaxRef = 343.31
RdRef = 3.25
sigma = 0.07
Theta = 0.3
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

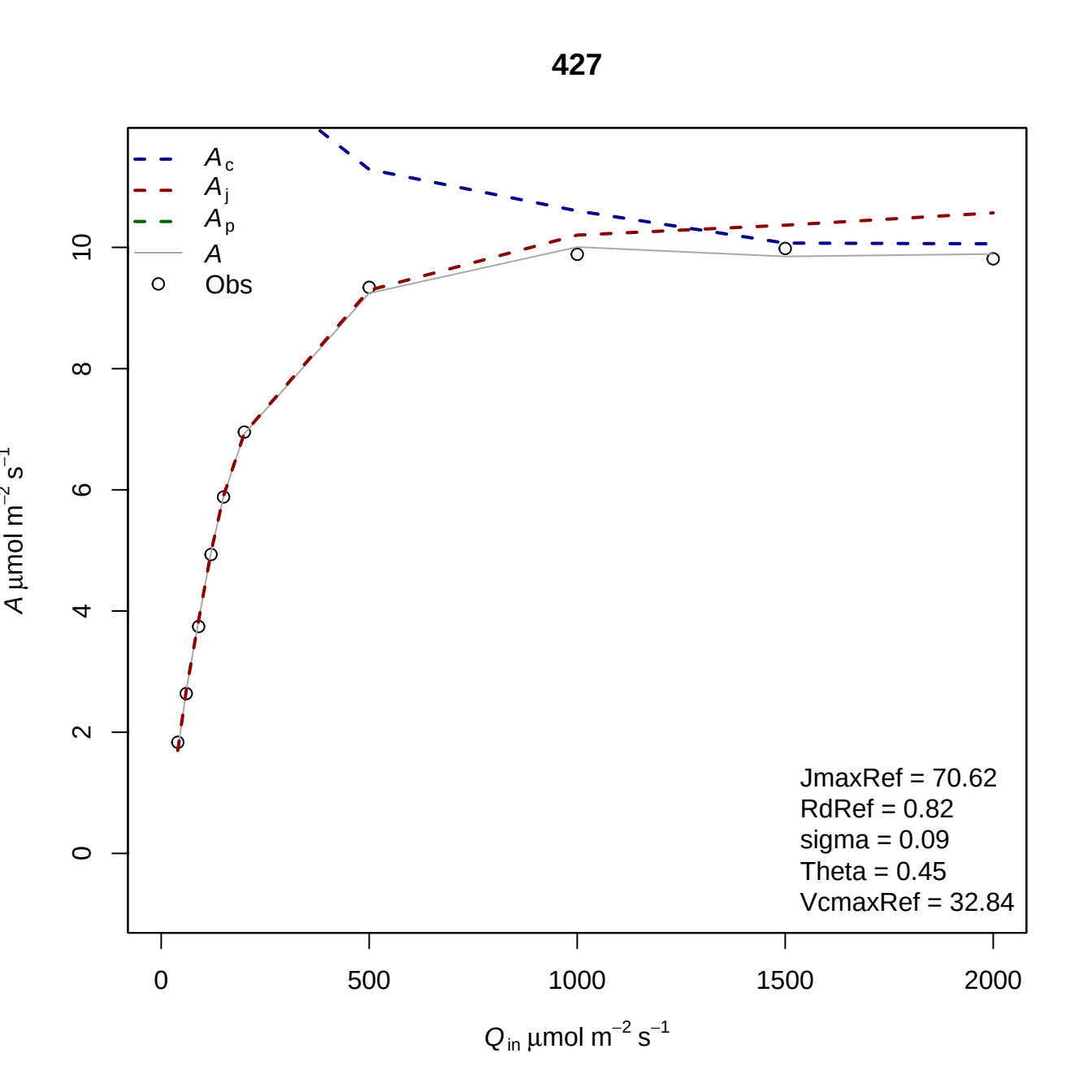

421
A_c
A_j
A_p
A
Obs
JmaxRef = 84.37
RdRef = 0.75
sigma = 0.06
Theta = −0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

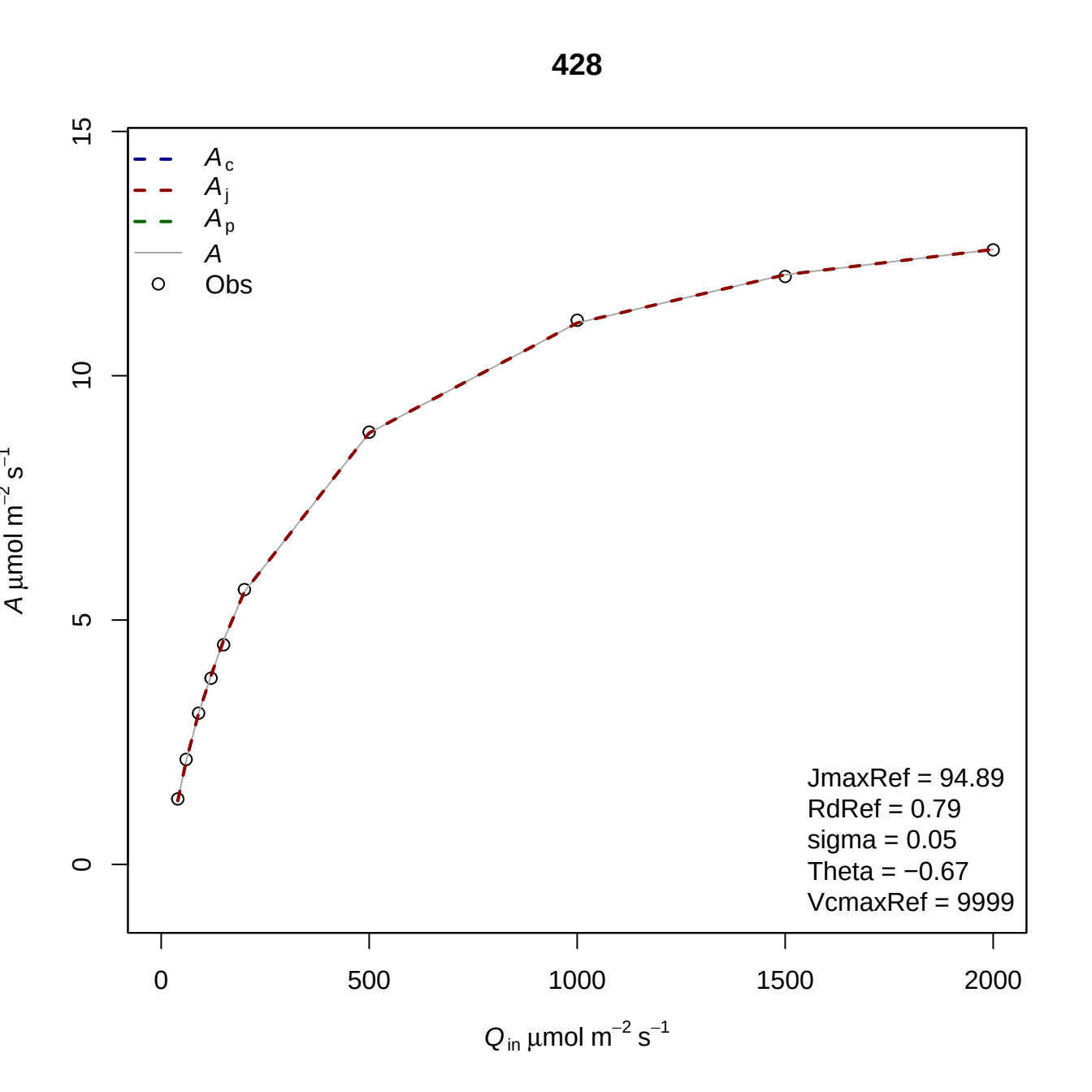

422
A_c
A_j
A_p
A
Obs
JmaxRef = 137.41
RdRef = 0.8
sigma = 0.1
Theta = 0.09
VcmaxRef = 66.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

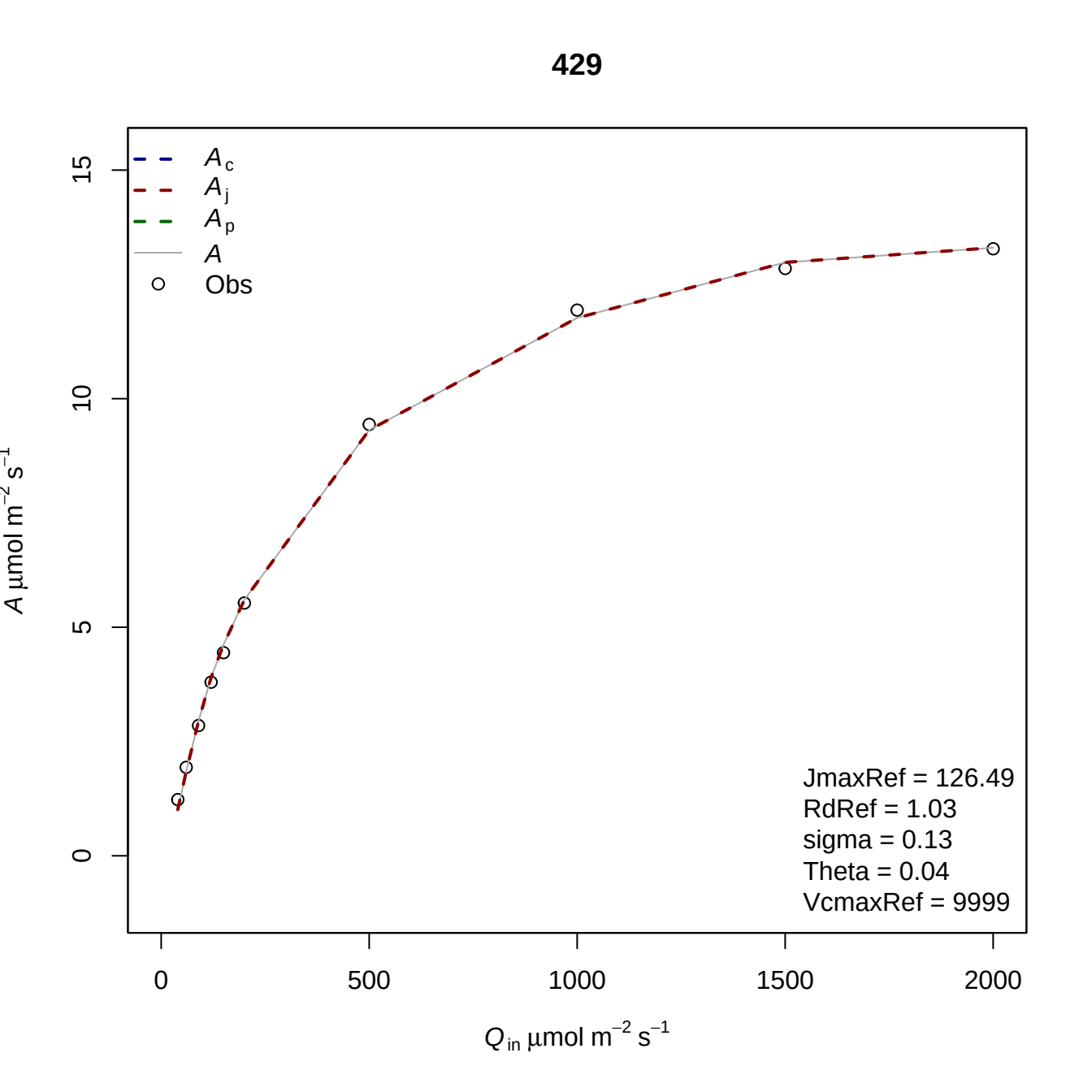

424
A_c
A_j
A_p
A
Obs
JmaxRef = 147.86
RdRef = 1.7
sigma = 0.16
Theta = 0.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

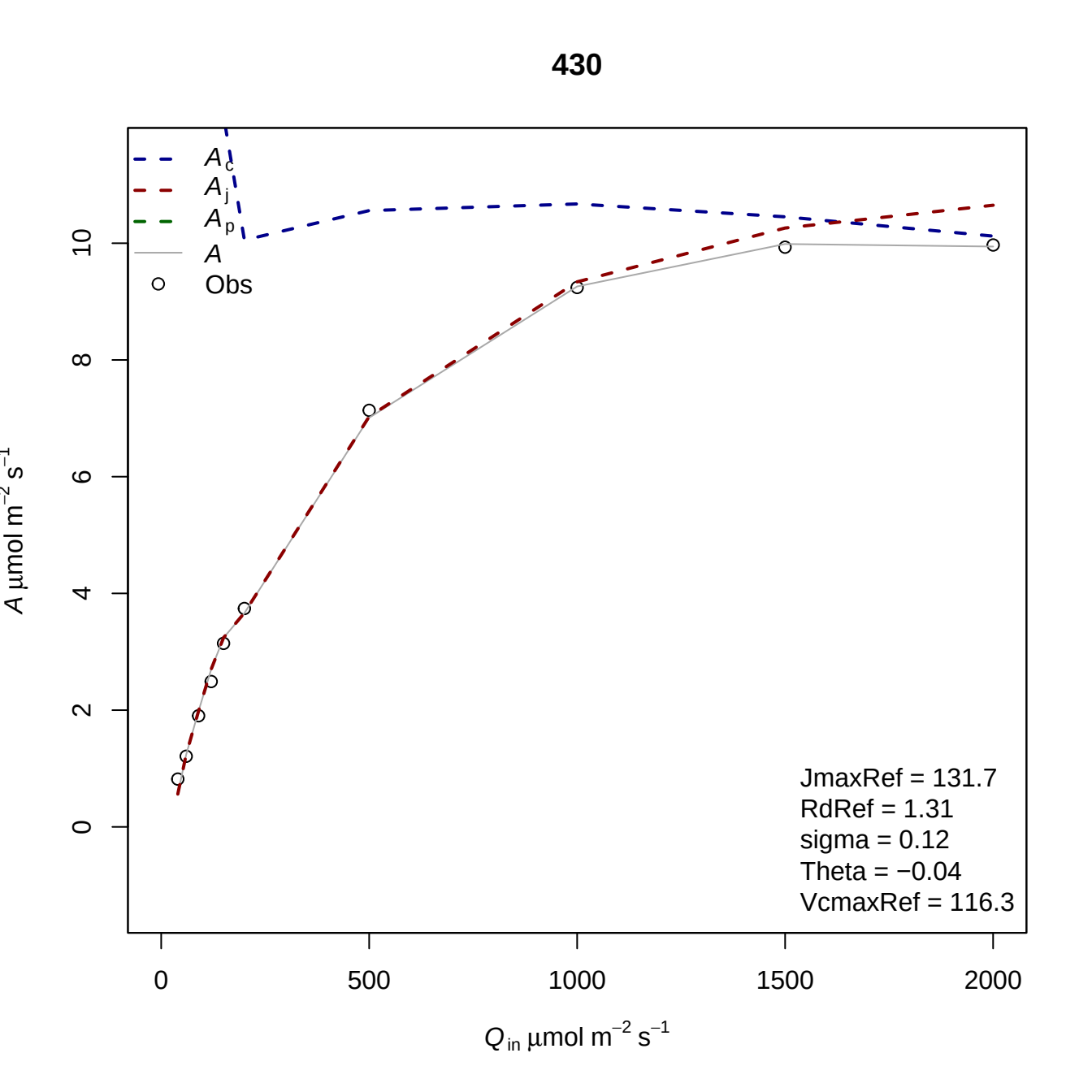

425
A_c
A_j
A_p
A
Obs
JmaxRef = 177.58
RdRef = 1.53
sigma = 0.21
Theta = −6.59
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

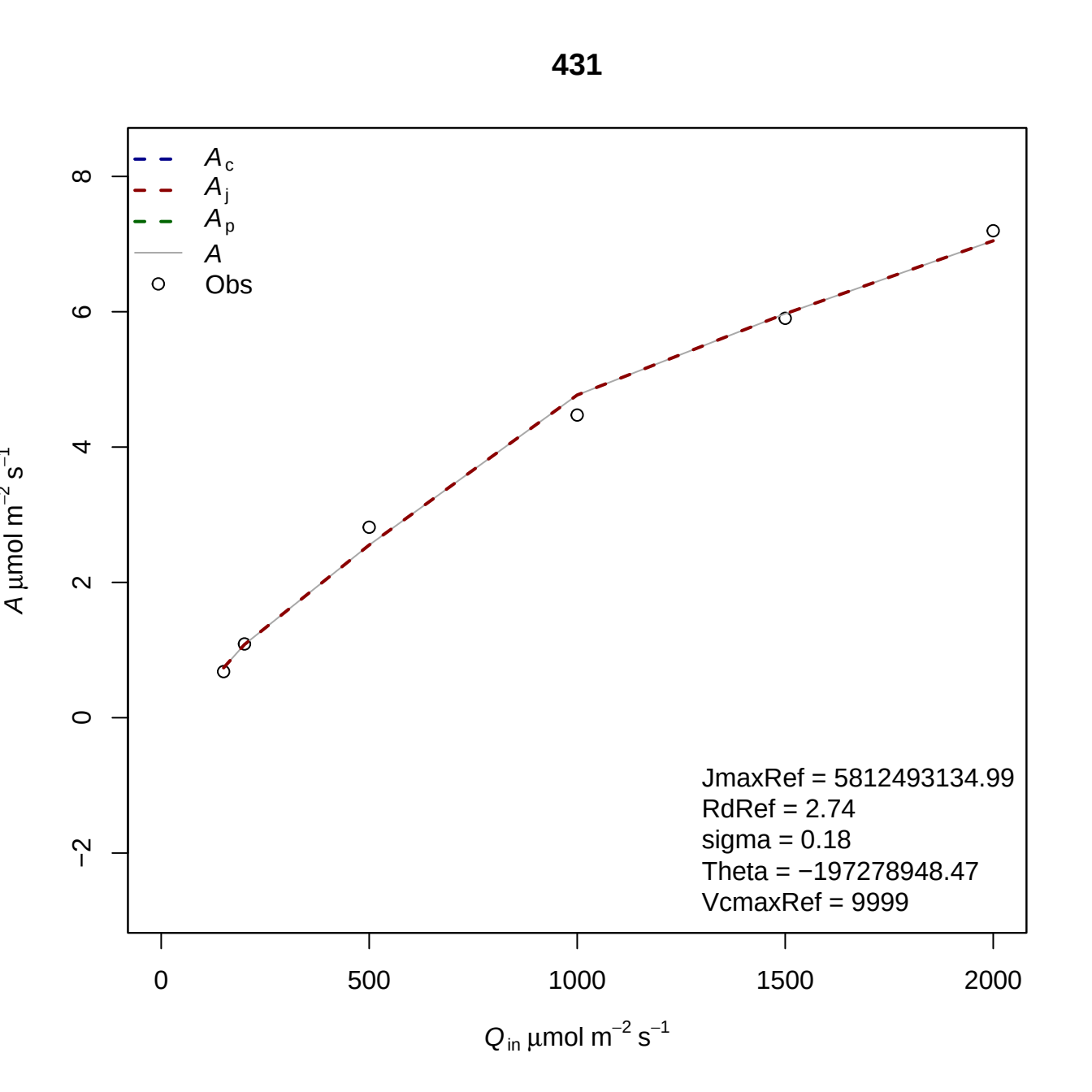

426
A_c
A_j
A_p
A
Obs
JmaxRef = 98.34
RdRef = 1.81
sigma = 0.15
Theta = −1.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

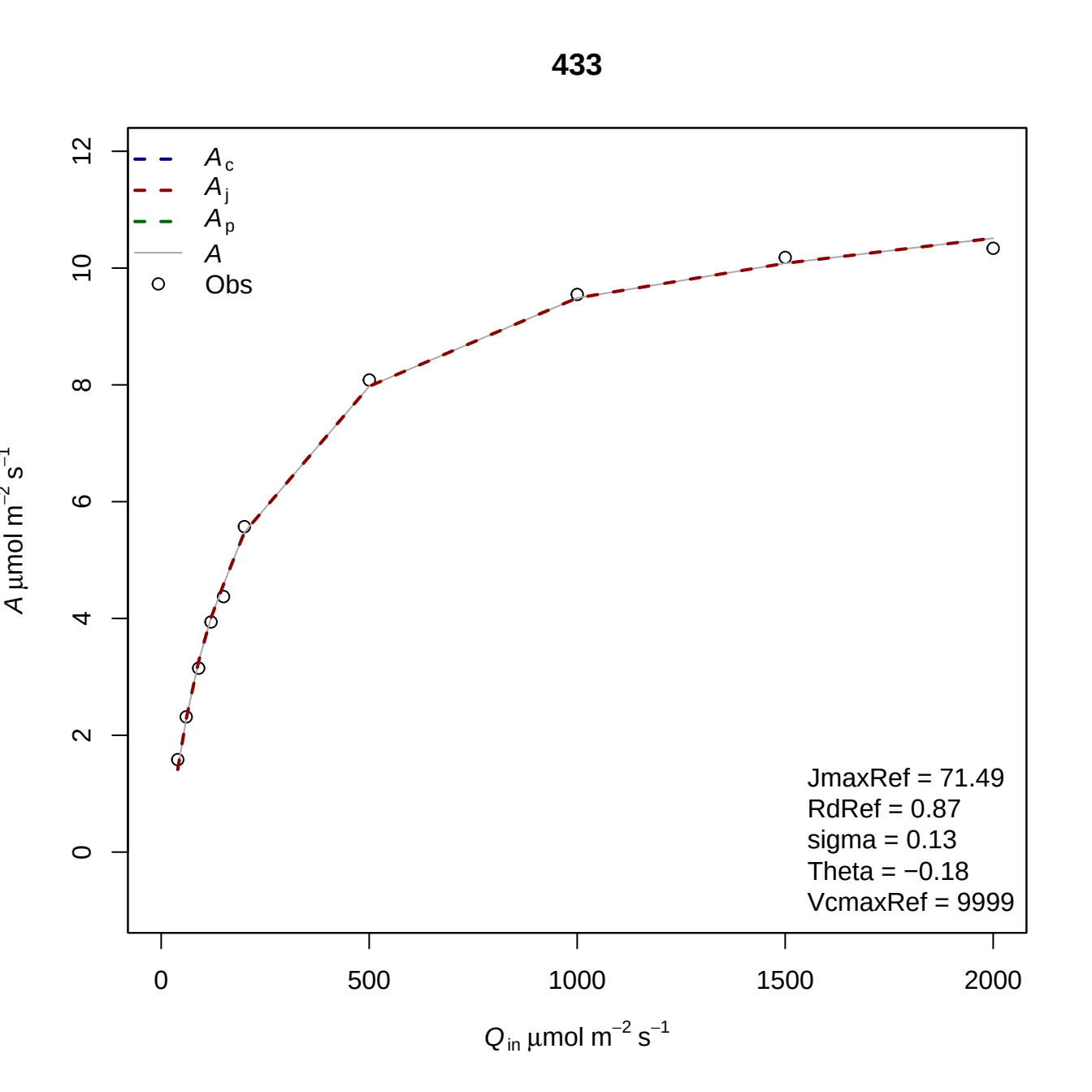

427
A_c
A_j
A_p
A
Obs
JmaxRef = 70.62
RdRef = 0.82
sigma = 0.09
Theta = 0.45
VcmaxRef = 32.84
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

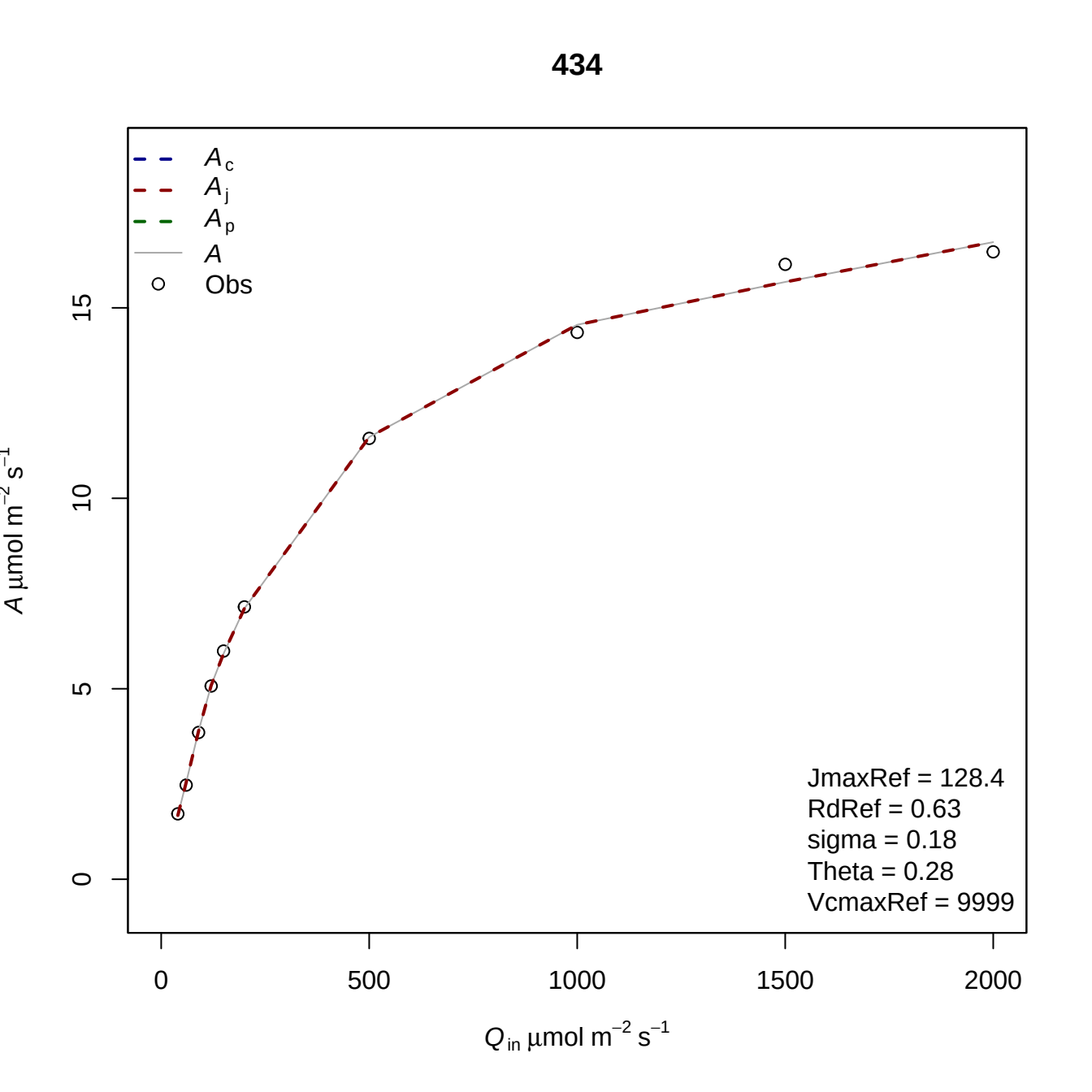

428
A_c
A_j
A_p
A
Obs
JmaxRef = 94.89
RdRef = 0.79
sigma = 0.05
Theta = −0.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

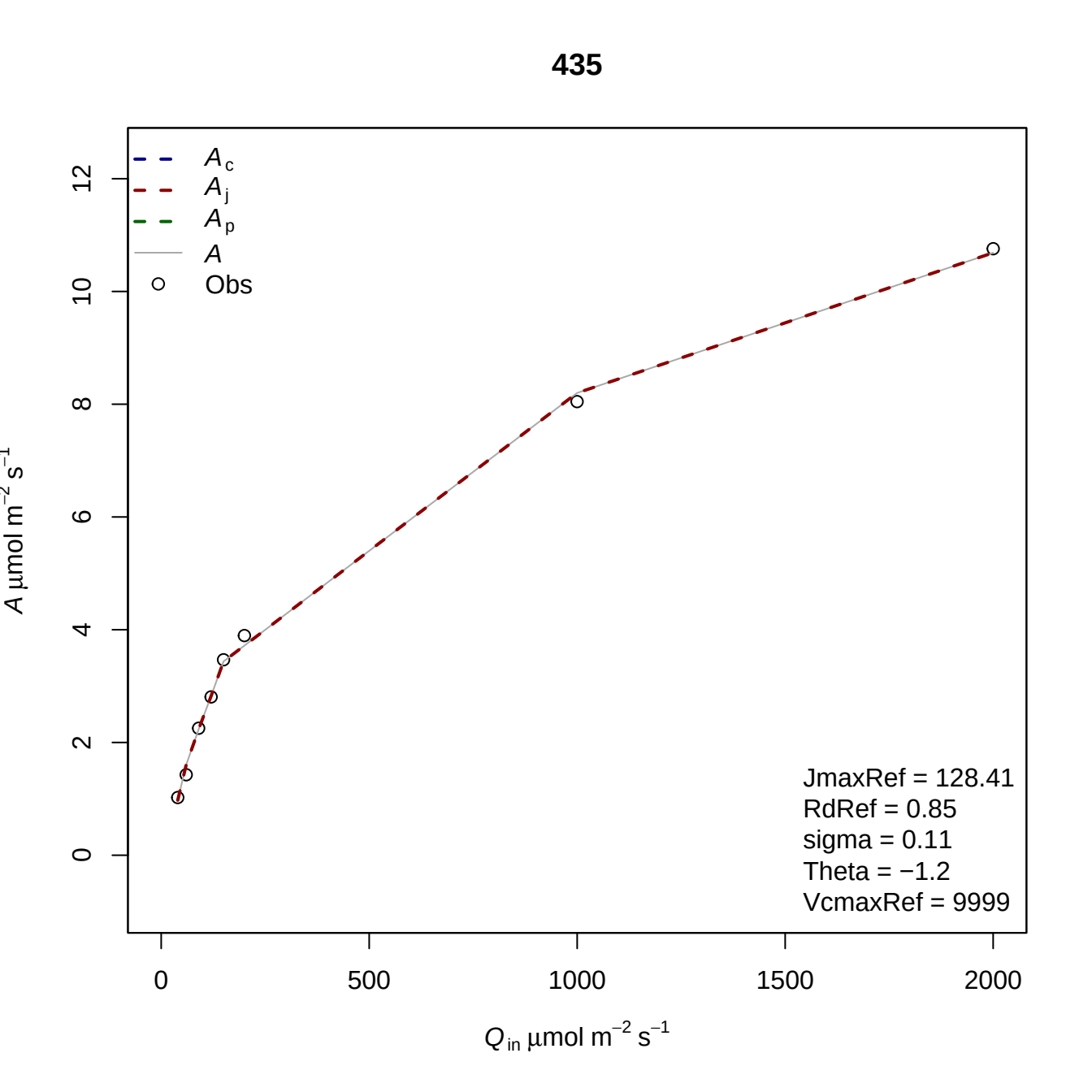

429
A_c
A_j
A_p
A
Obs
JmaxRef = 126.49
RdRef = 1.03
sigma = 0.13
Theta = 0.04
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

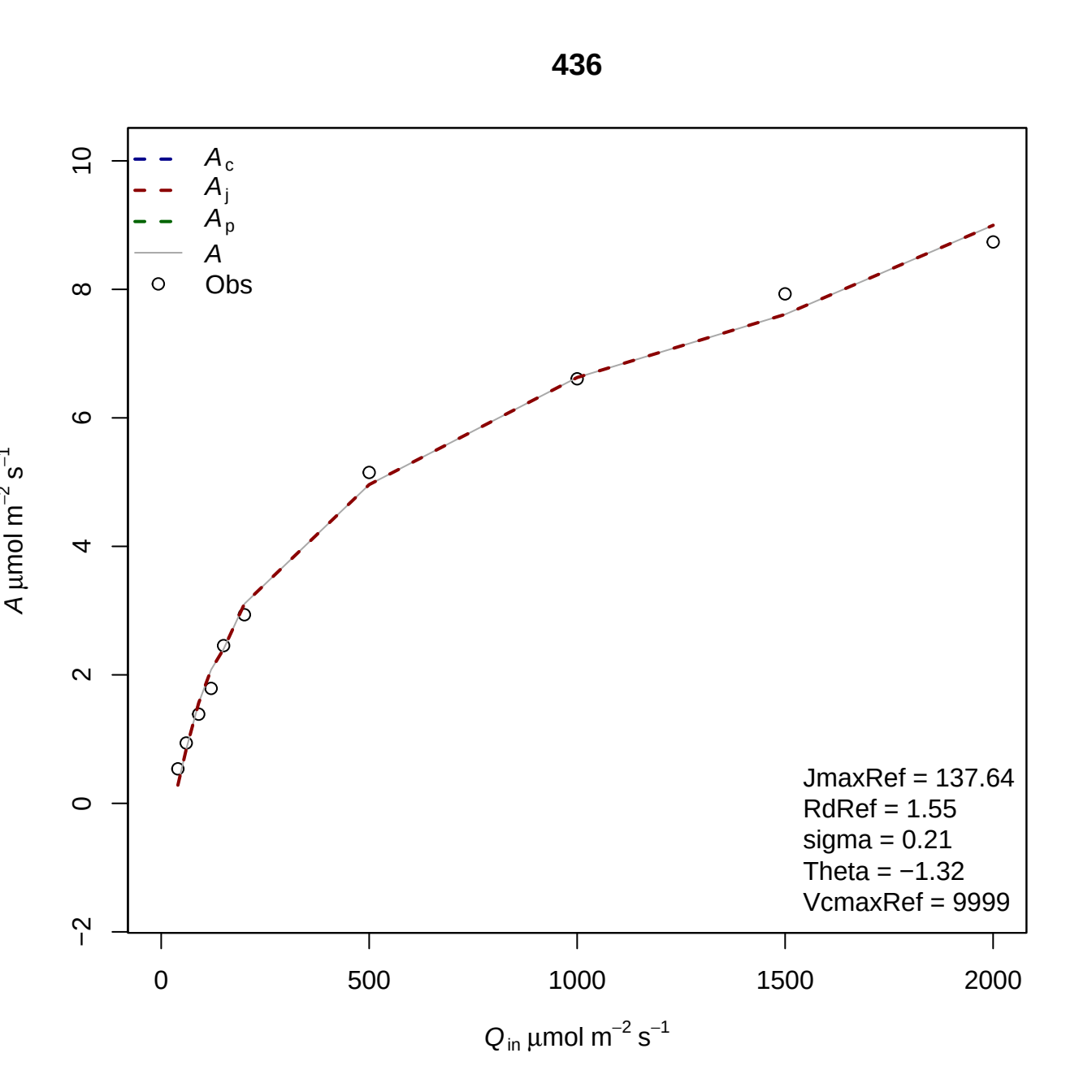

430
A_c
A_j
A_p
A
Obs
JmaxRef = 131.7
RdRef = 1.31
sigma = 0.12
Theta = −0.04
VcmaxRef = 116.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

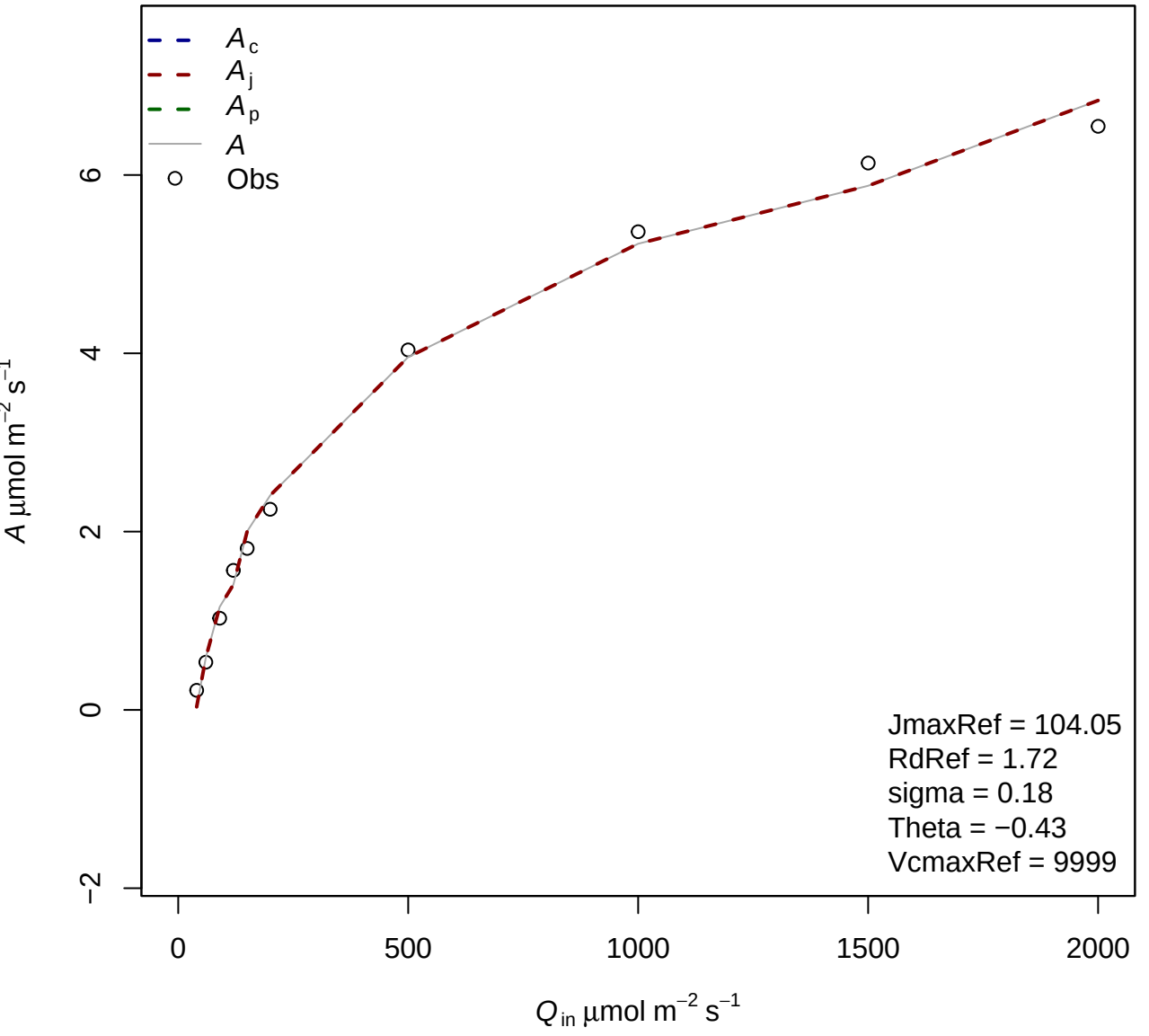

431
A_c
A_j
A_p
A
Obs
JmaxRef = 5812493134.99
RdRef = 2.74
sigma = 0.18
Theta = −197278948.47
VcmaxRef = 9999
8
6
4
2
0
−2
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

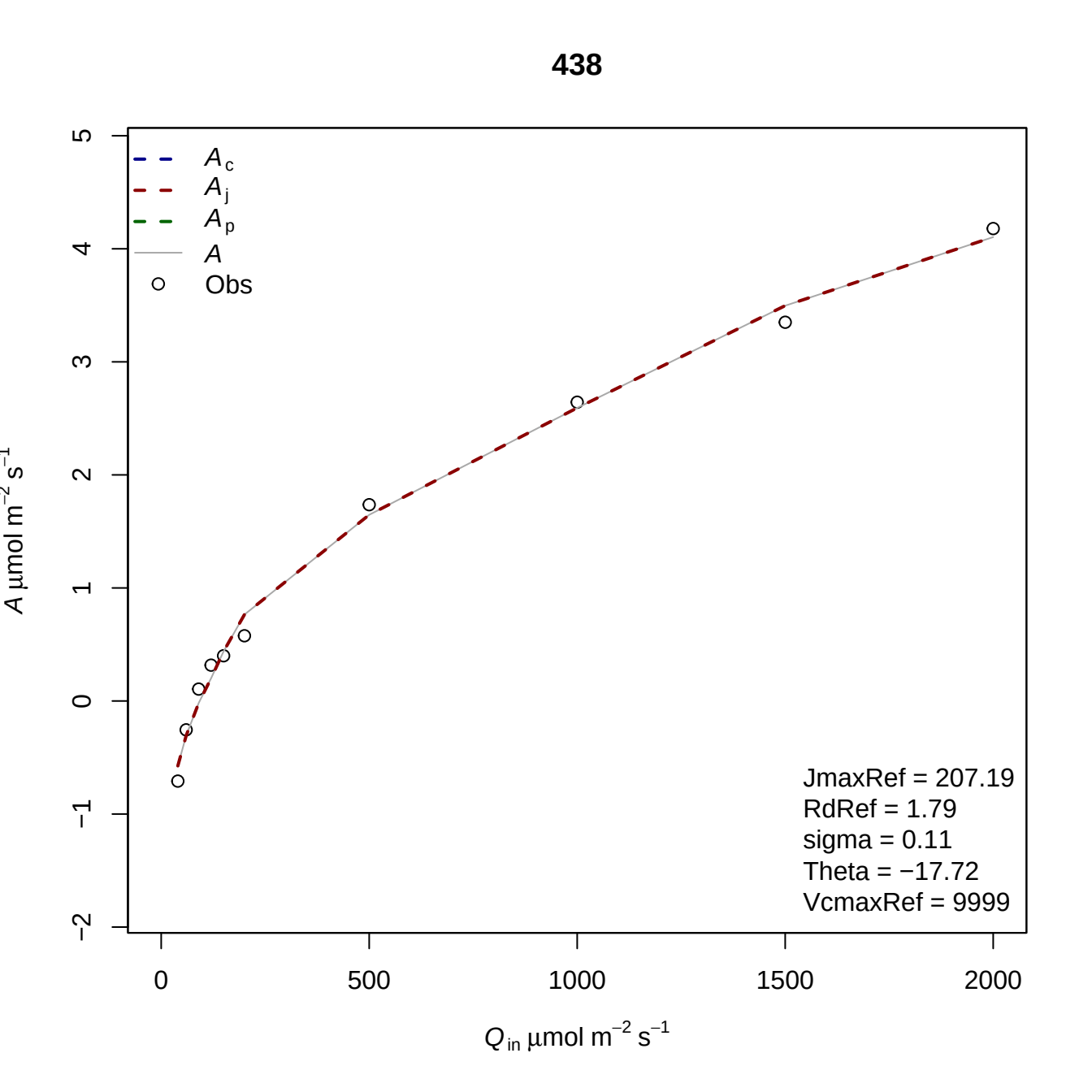

433
A_c
A_j
A_p
A
Obs
JmaxRef = 71.49
RdRef = 0.87
sigma = 0.13
Theta = −0.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

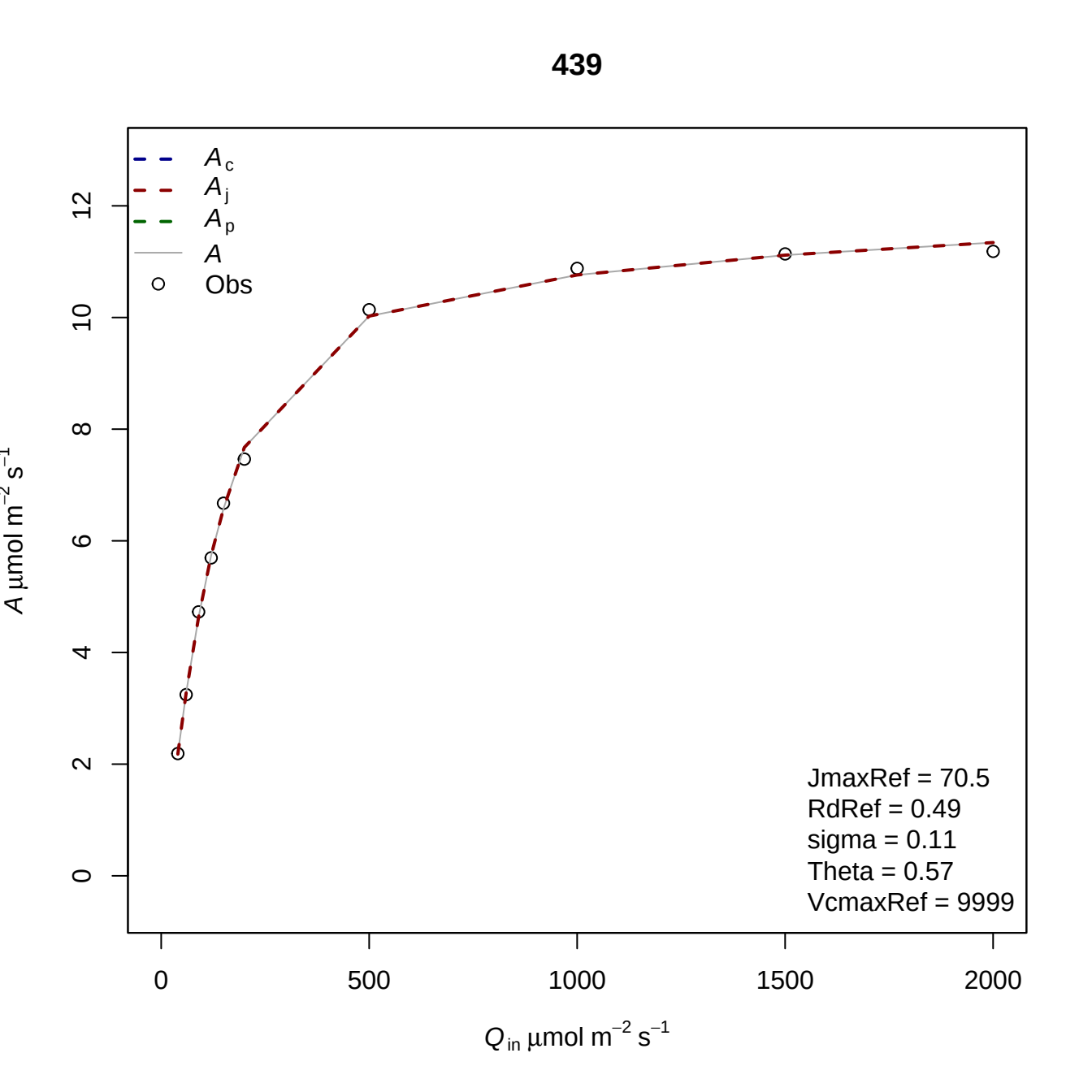

434
A_c
A_j
A_p
A
Obs
JmaxRef = 128.4
RdRef = 0.63
sigma = 0.18
Theta = 0.28
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

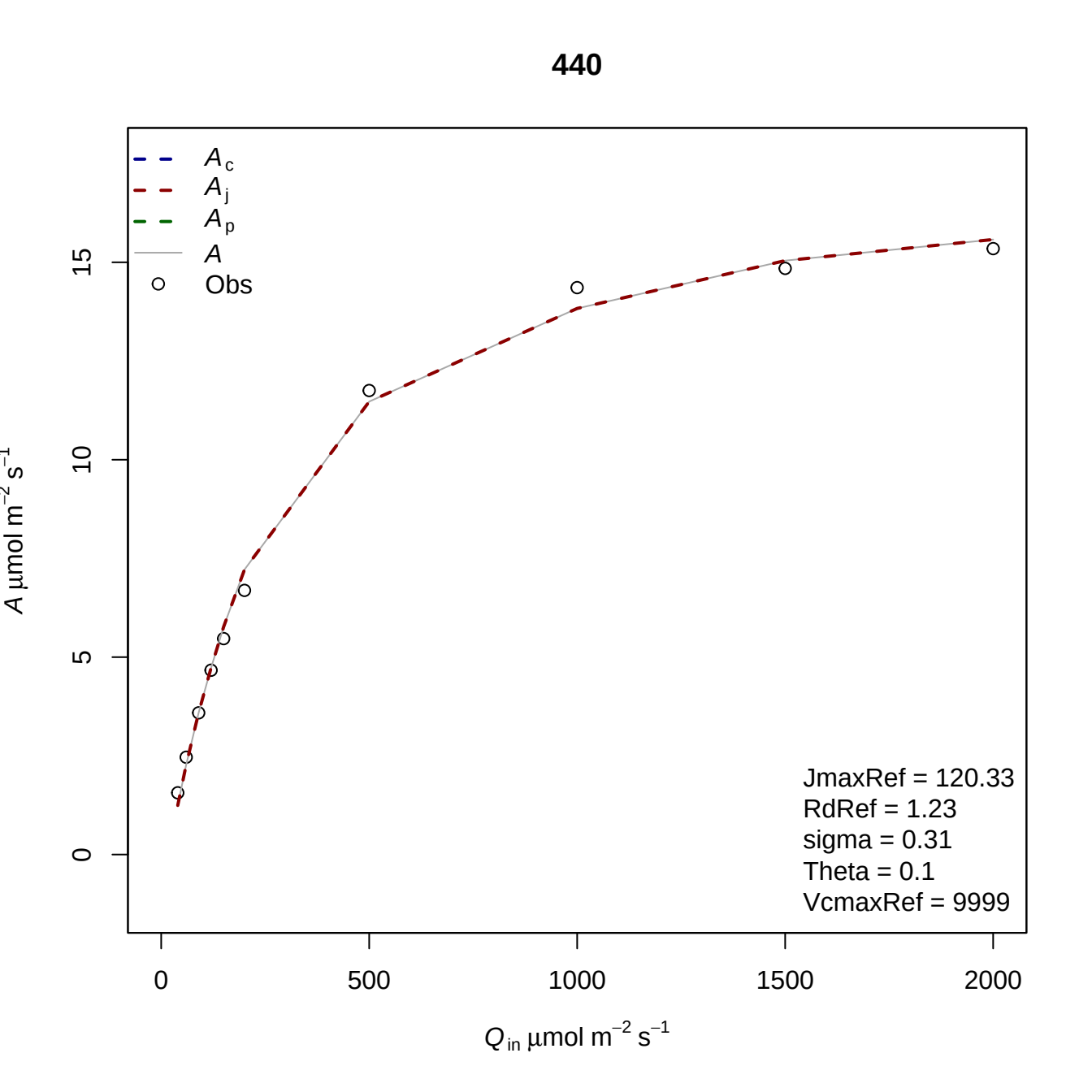

435
A_c
A_j
A_p
A
Obs
JmaxRef = 128.41
RdRef = 0.85
sigma = 0.11
Theta = −1.2
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

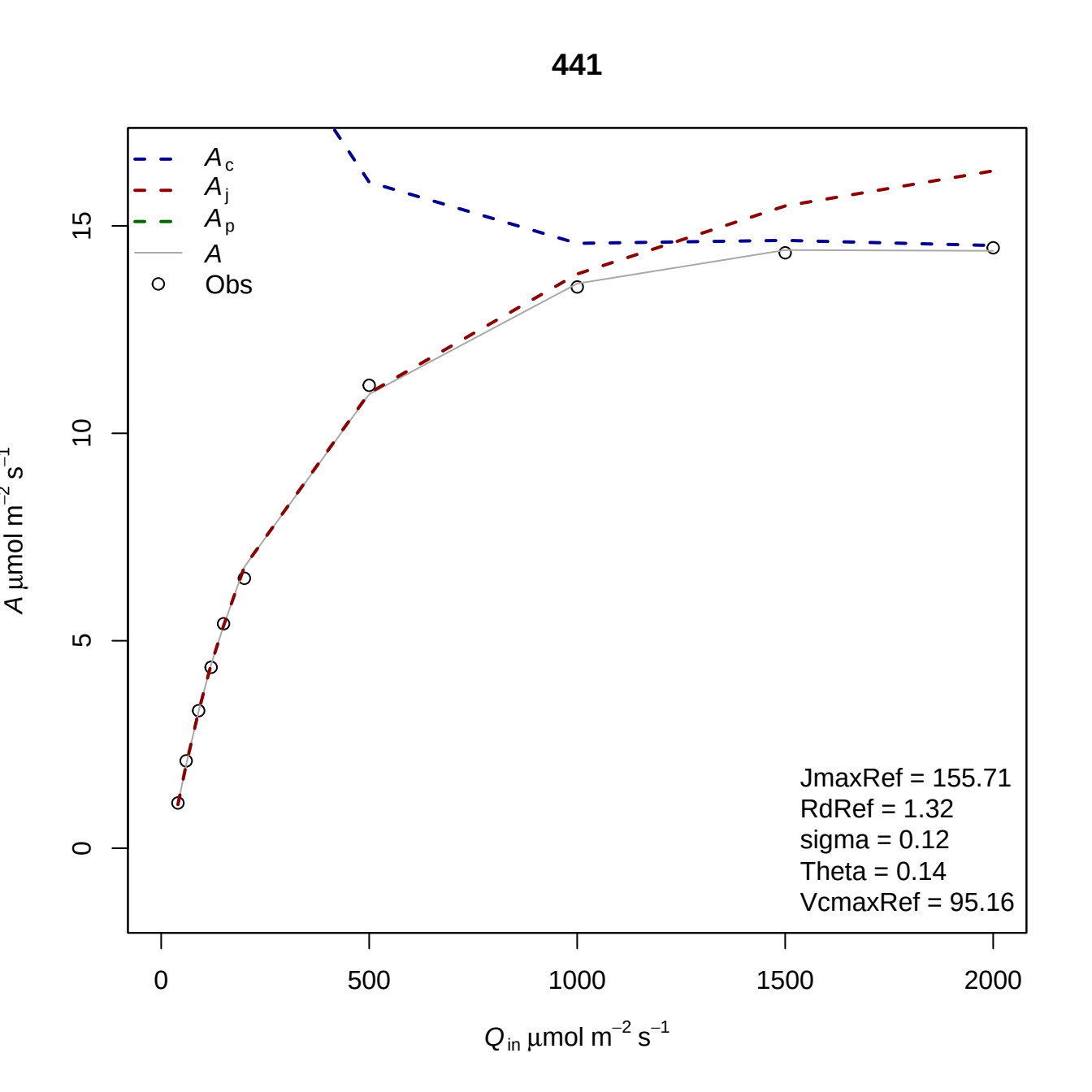

436
A_c
A_j
A_p
A
Obs
JmaxRef = 137.64
RdRef = 1.55
sigma = 0.21
Theta = −1.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

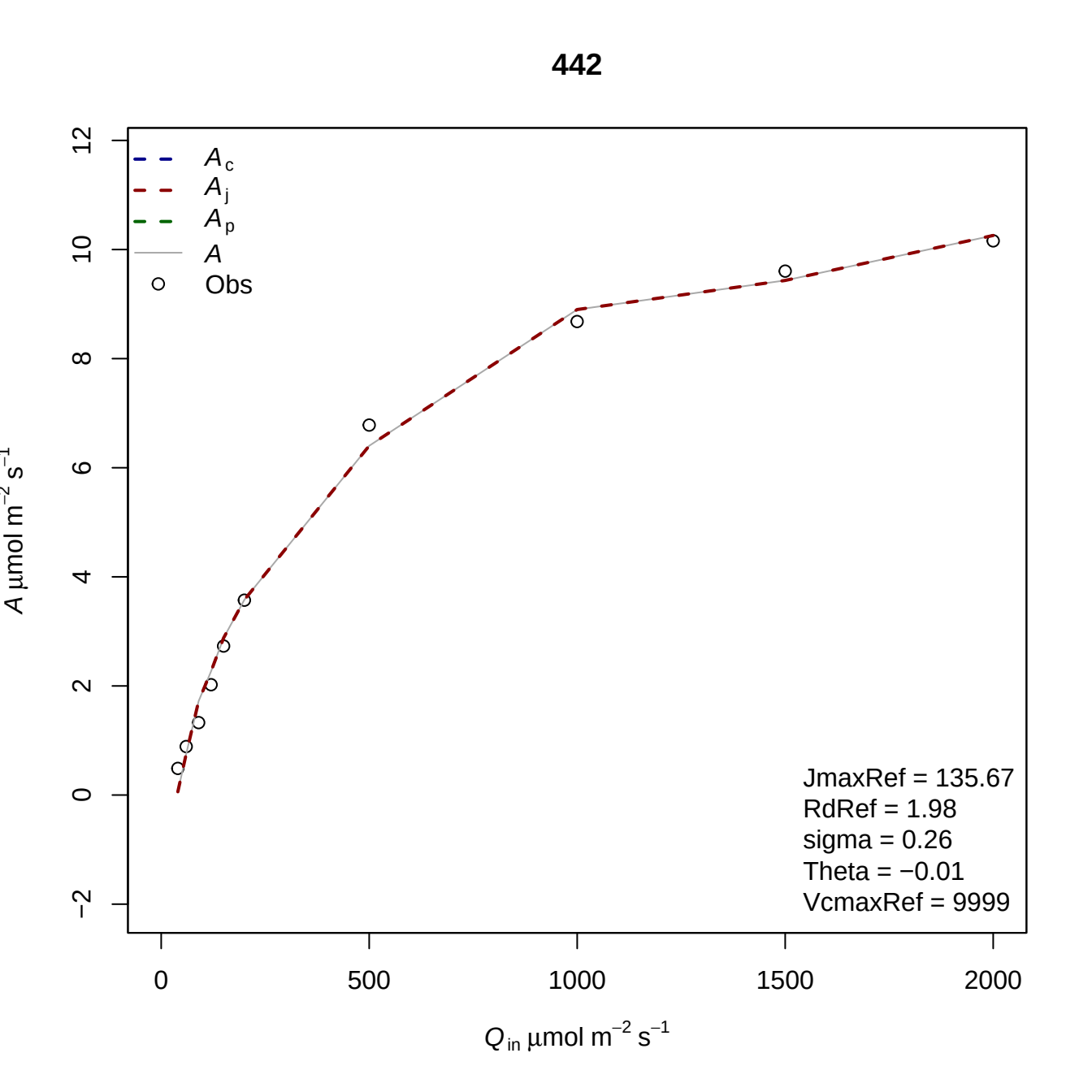

A_c
A_j
A_p
A
Obs
JmaxRef = 104.05
RdRef = 1.72
sigma = 0.18
Theta = −0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

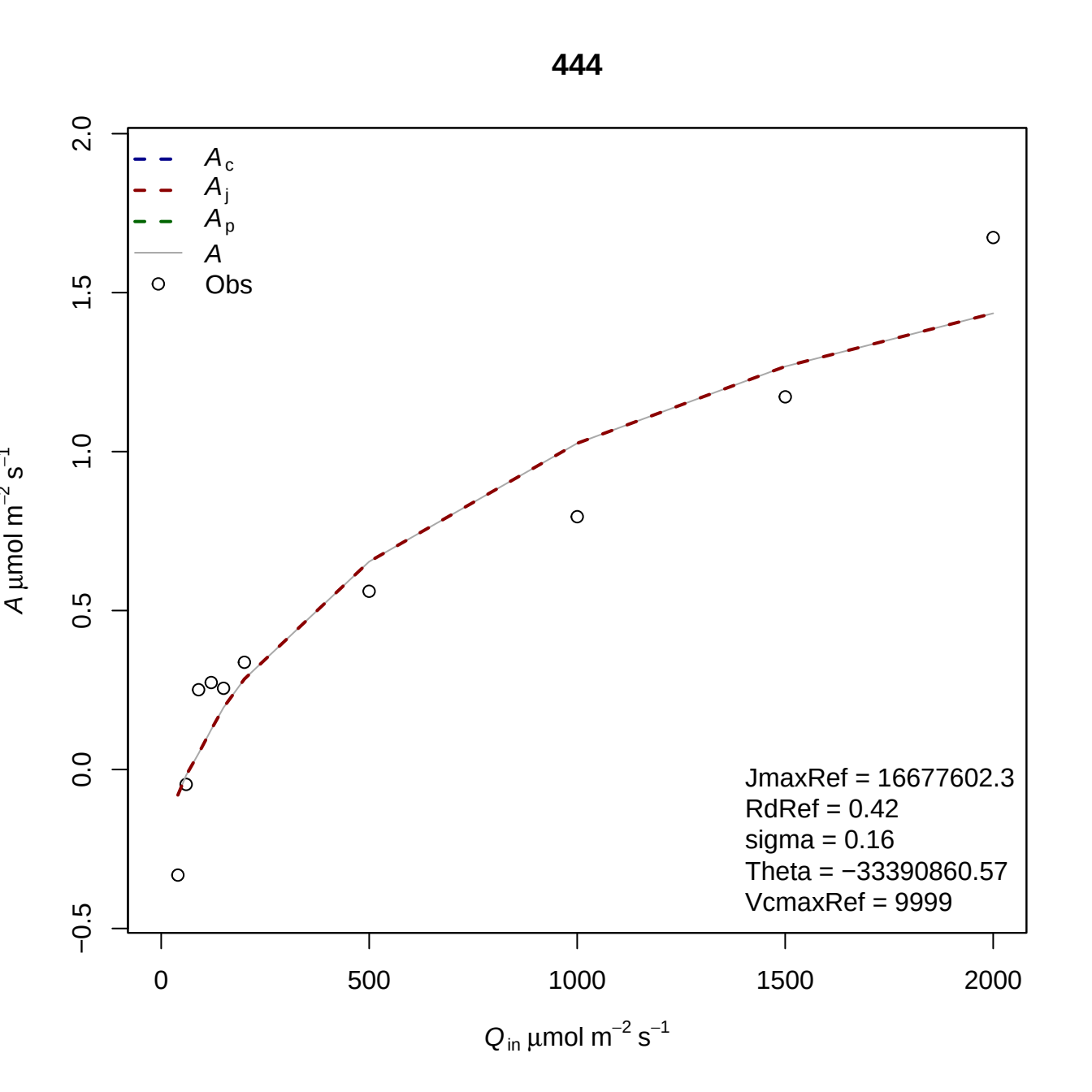

438
A_c
A_j
A_p
A
Obs
JmaxRef = 207.19
RdRef = 1.79
sigma = 0.11
Theta = −17.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

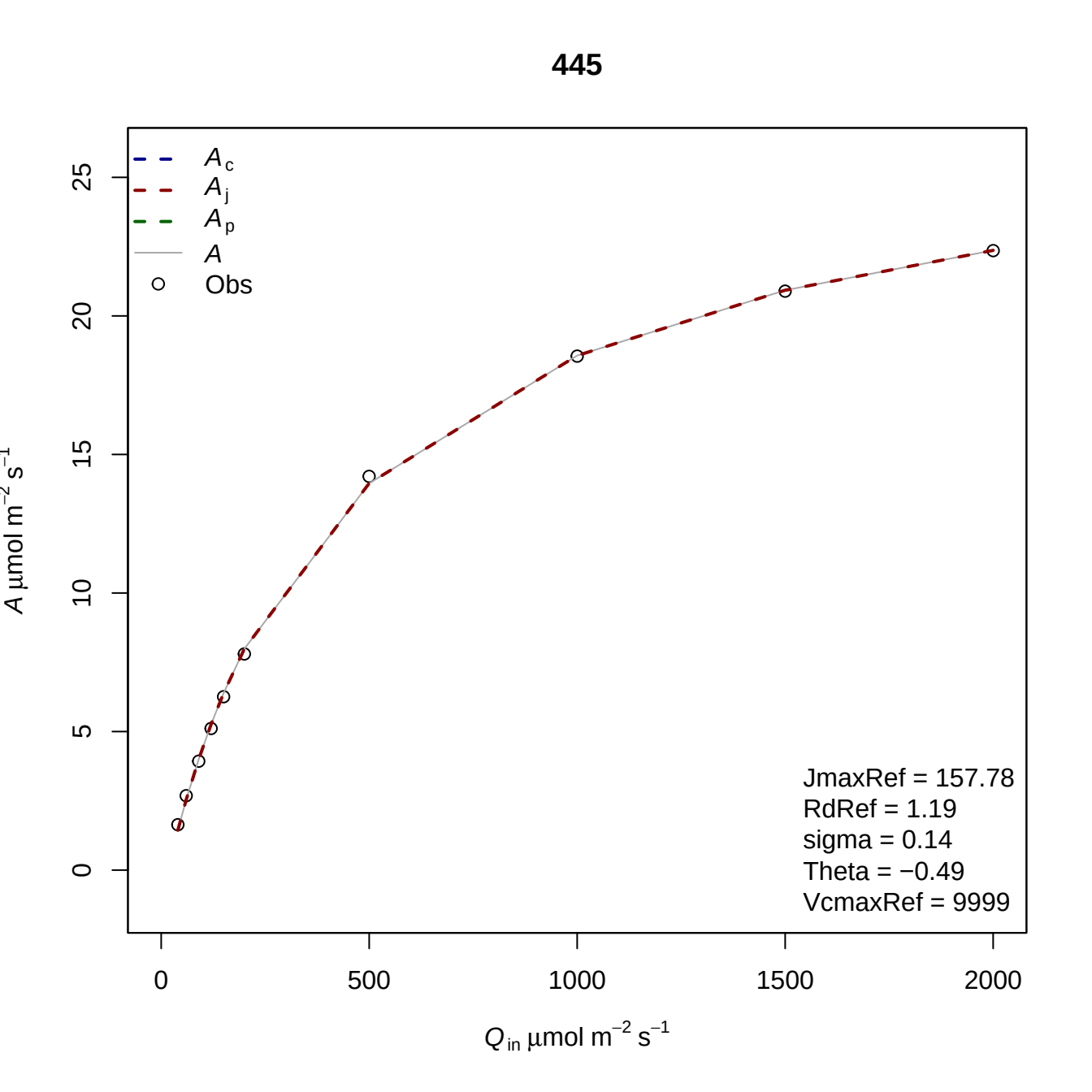

439
A_c
A_j
A_p
A
Obs
JmaxRef = 70.5
RdRef = 0.49
sigma = 0.11
Theta = 0.57
VcmaxRef = 9999
0
2
4
6
8
10
12
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

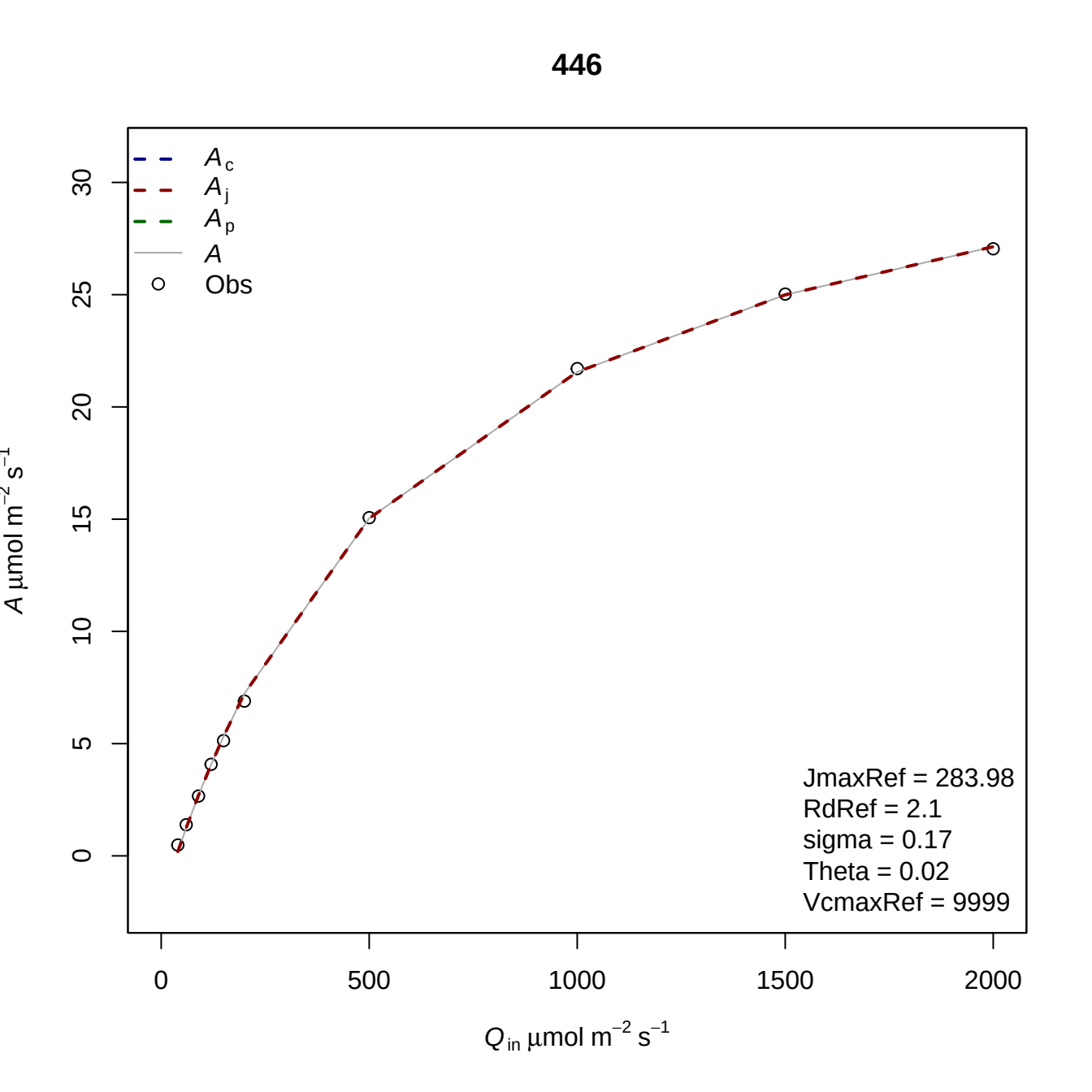

440
A_c
A_j
A_p
A
Obs
JmaxRef = 120.33
RdRef = 1.23
sigma = 0.31
Theta = 0.1
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

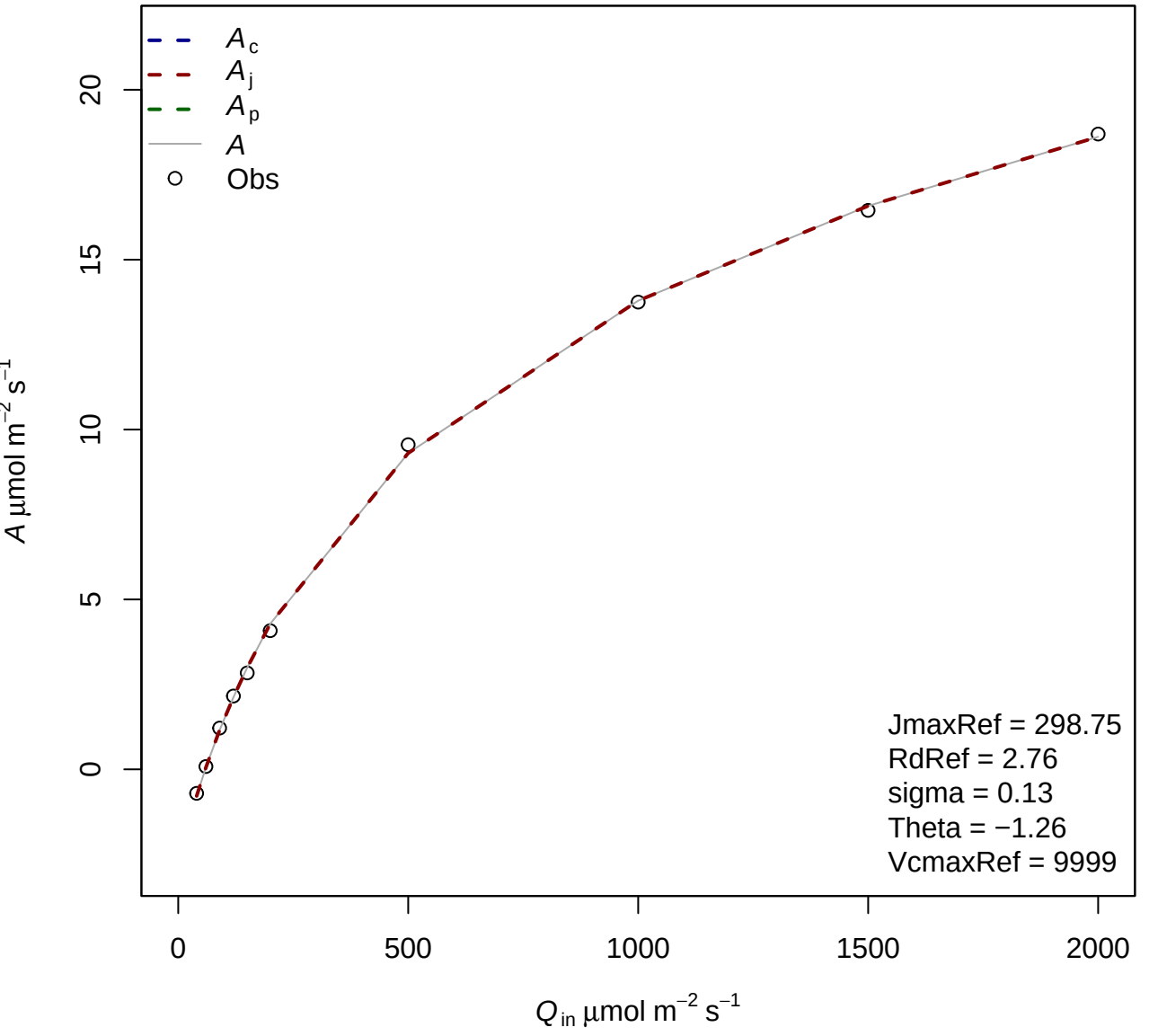

441
A_c
A_j
A_p
A
Obs
JmaxRef = 155.71
RdRef = 1.32
sigma = 0.12
Theta = 0.14
VcmaxRef = 95.16
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

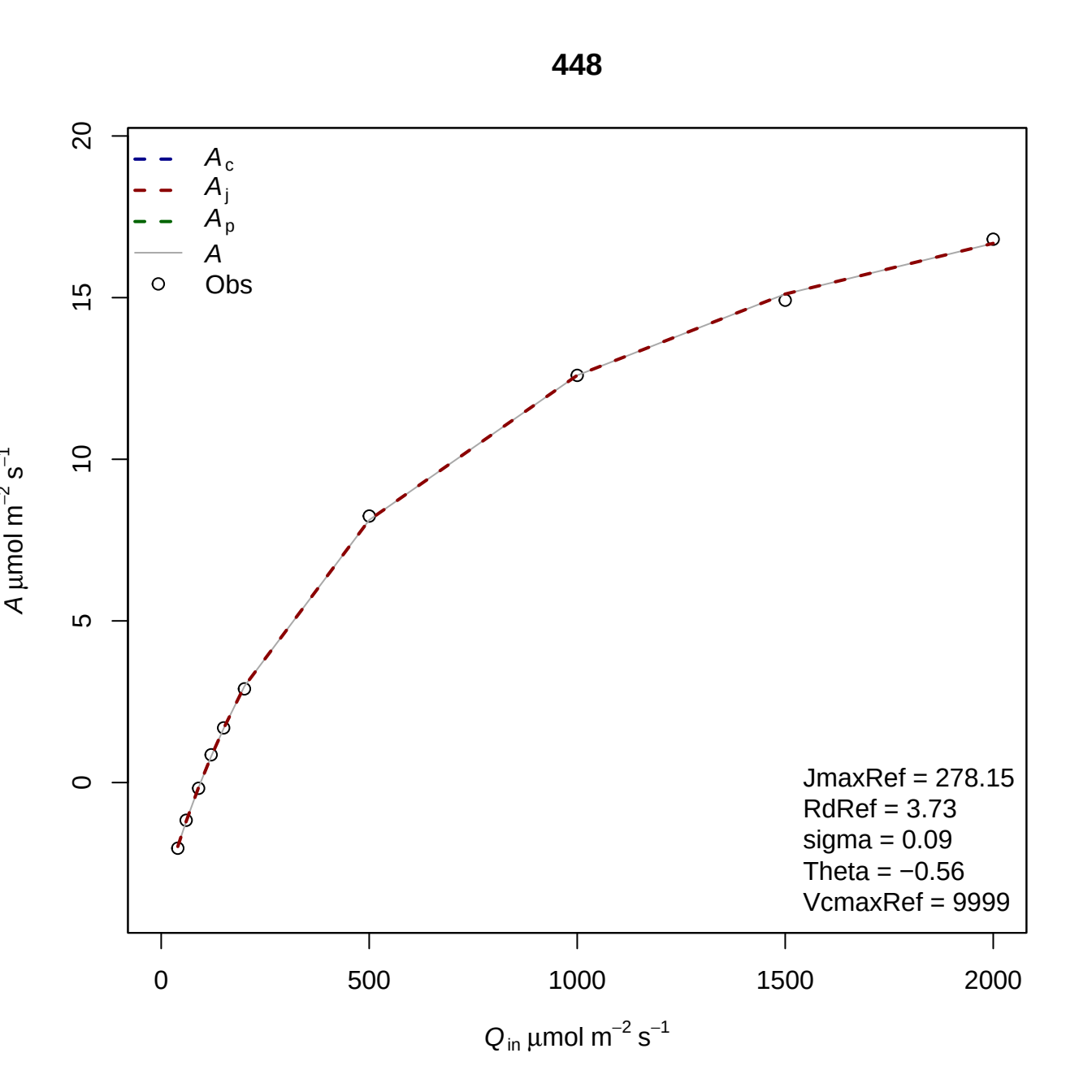

442
A_c
A_j
A_p
A
Obs
JmaxRef = 135.67
RdRef = 1.98
sigma = 0.26
Theta = −0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

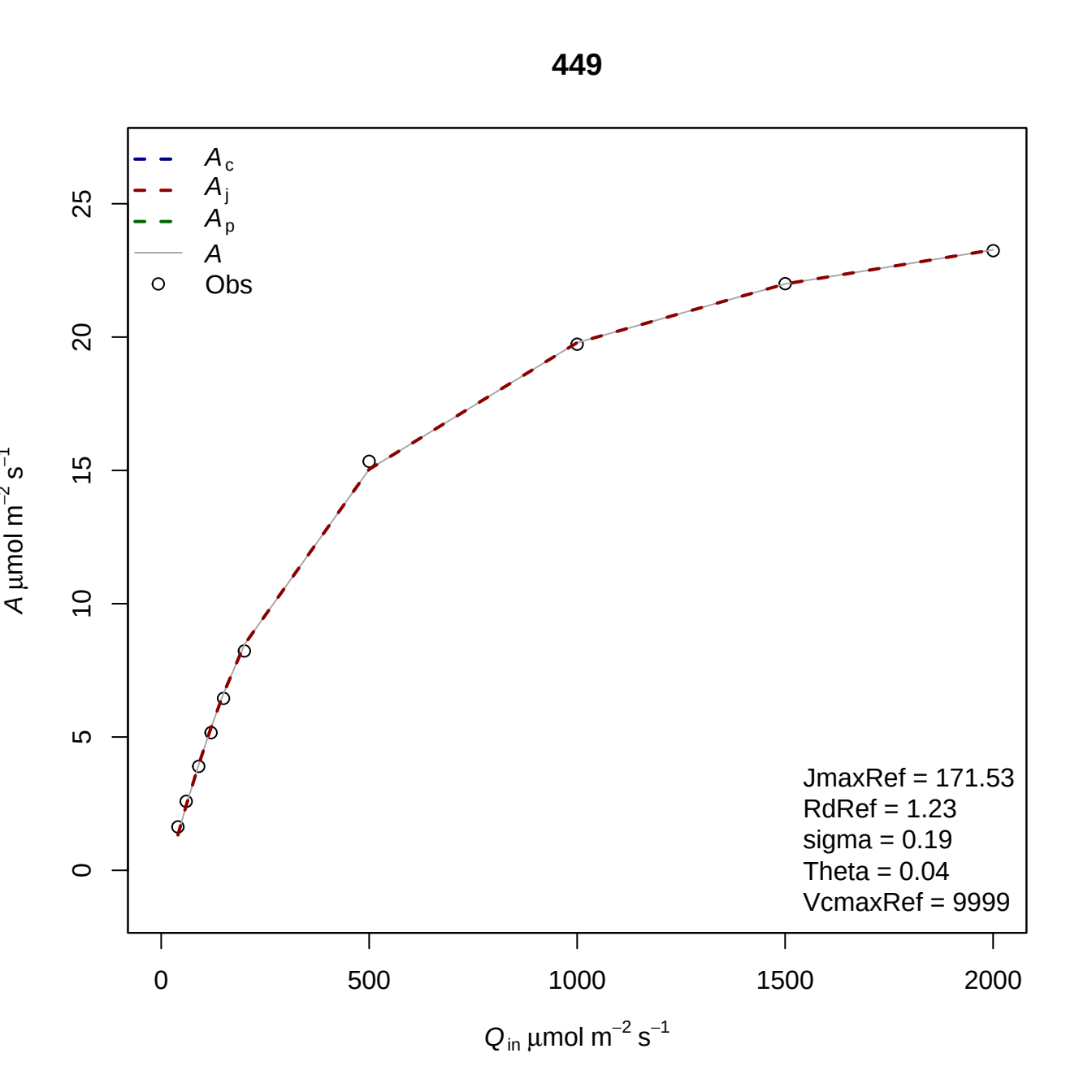

444
A_c
A_j
A_p
A
Obs
JmaxRef = 16677602.3
RdRef = 0.42
sigma = 0.16
Theta = −33390860.57
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

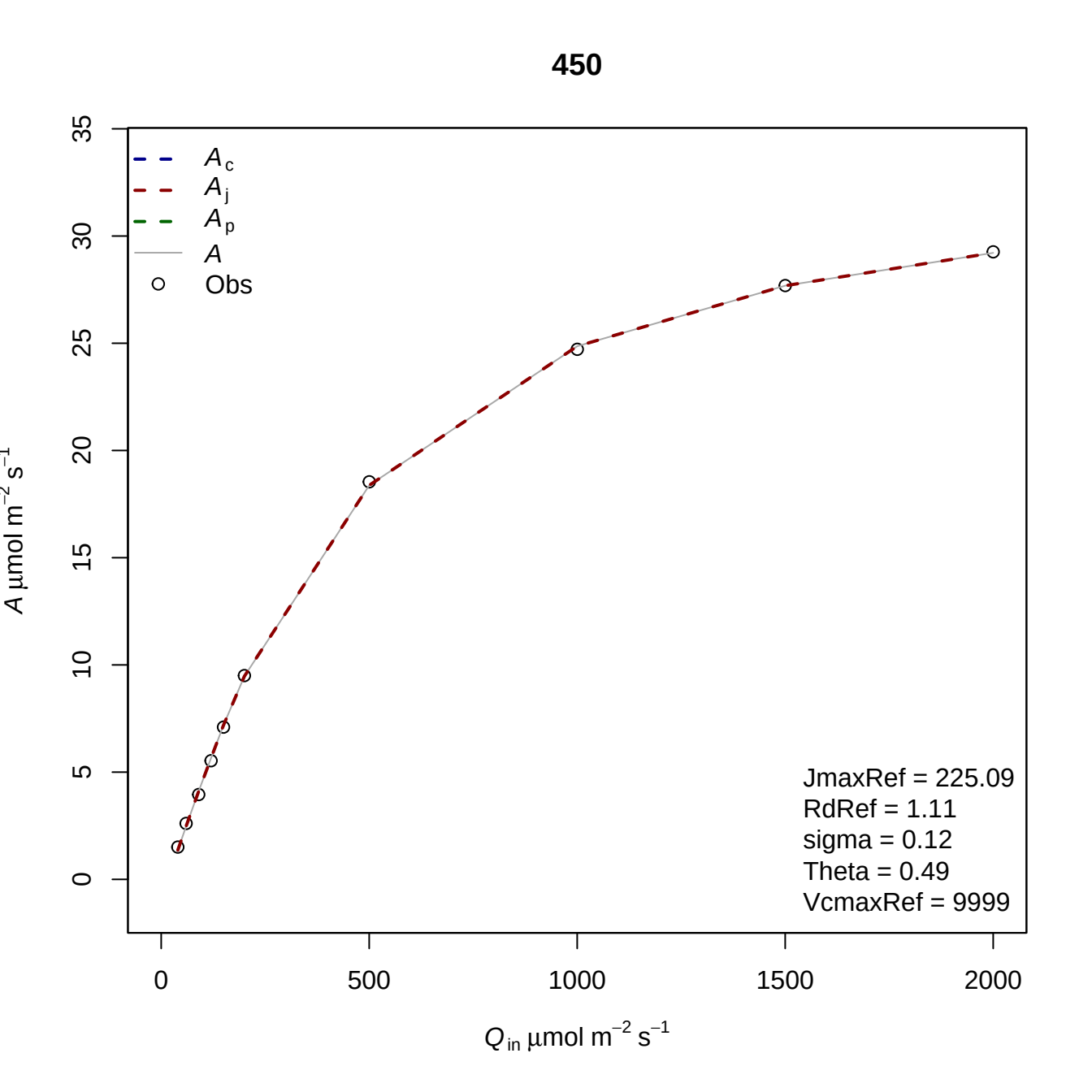

445
A_c
A_j
A_p
A
Obs
JmaxRef = 157.78
RdRef = 1.19
sigma = 0.14
Theta = −0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

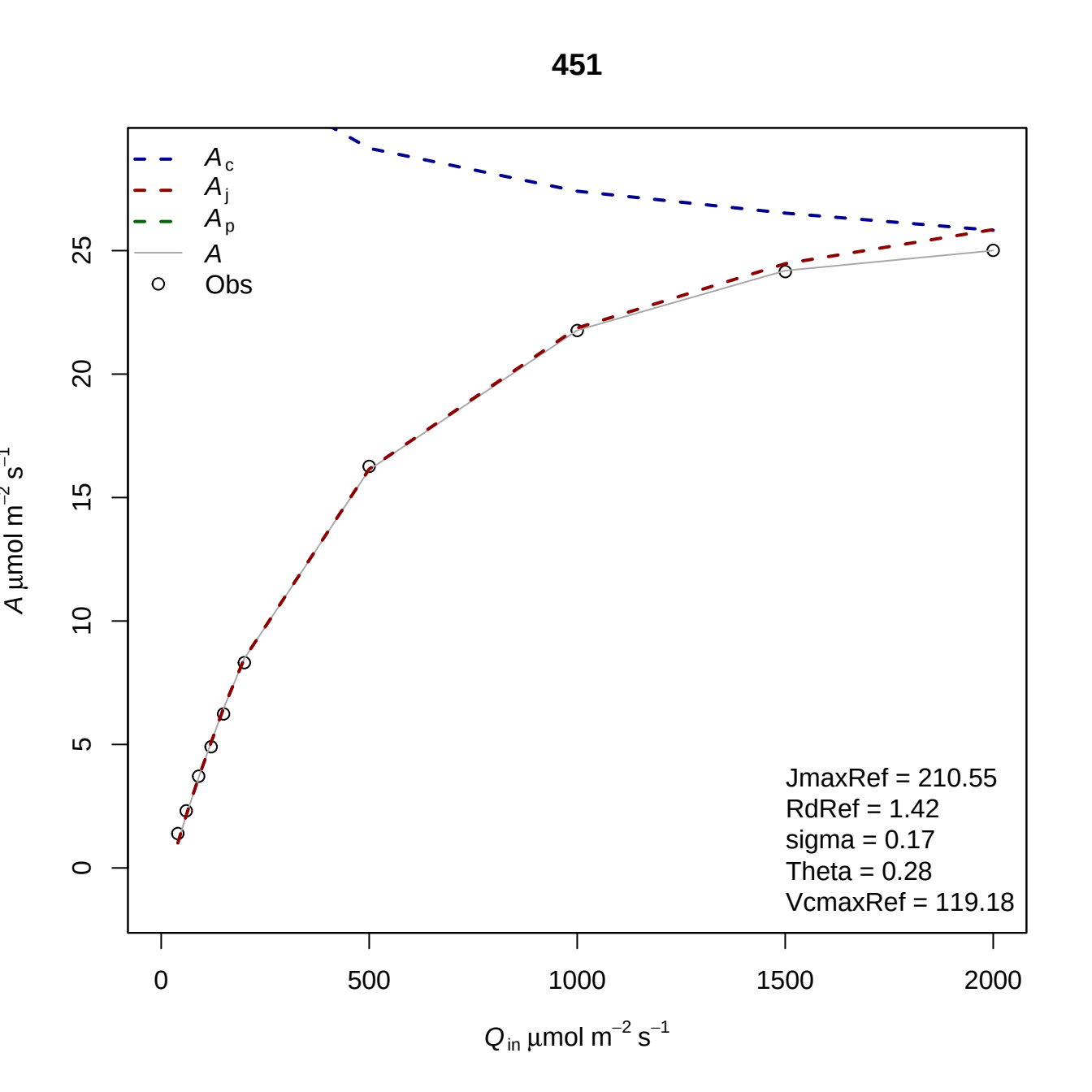

446
A_c
A_j
A_p
A
Obs
JmaxRef = 283.98
RdRef = 2.1
sigma = 0.17
Theta = 0.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

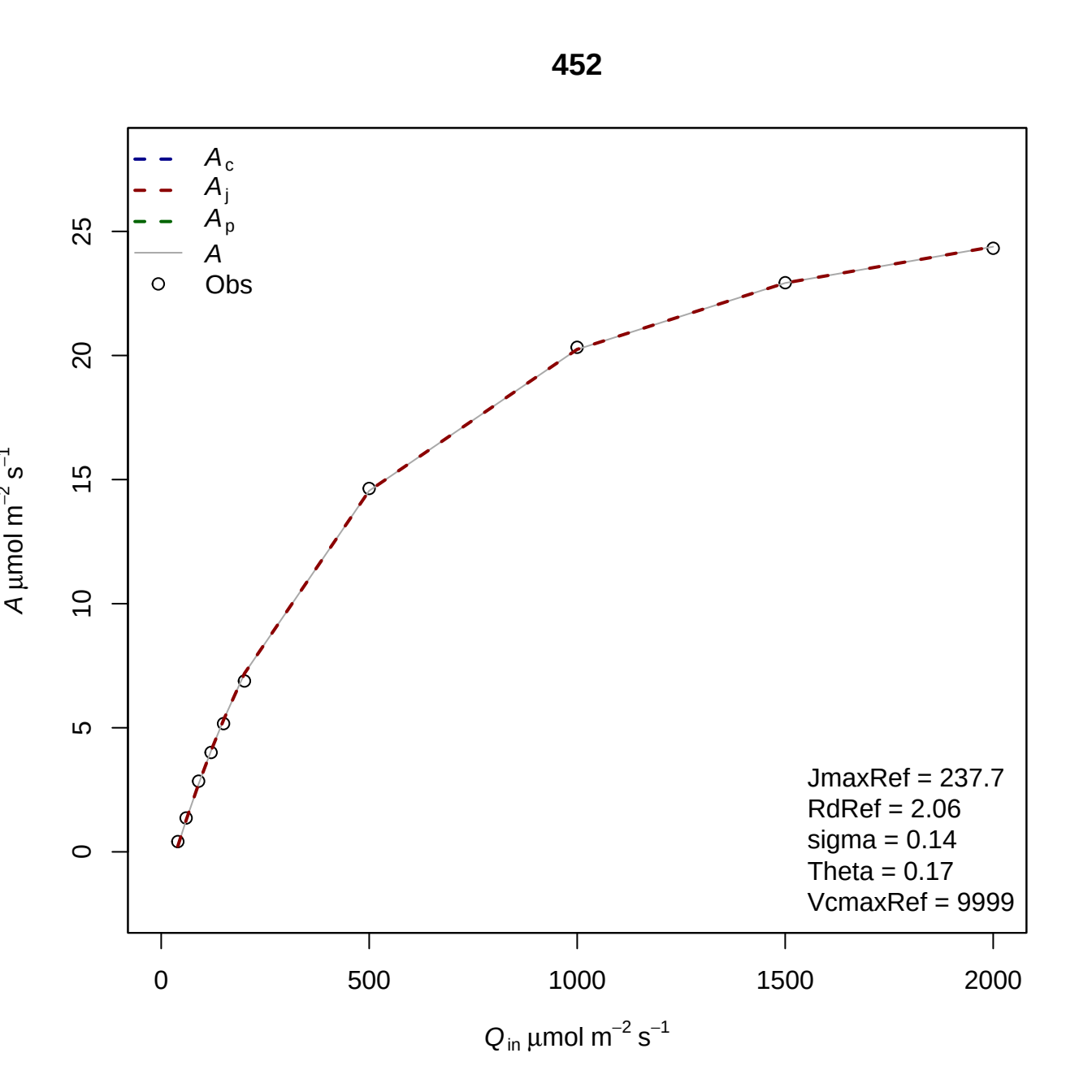

A_c
A_j
A_p
A
Obs
JmaxRef = 298.75
RdRef = 2.76
sigma = 0.13
Theta = −1.26
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

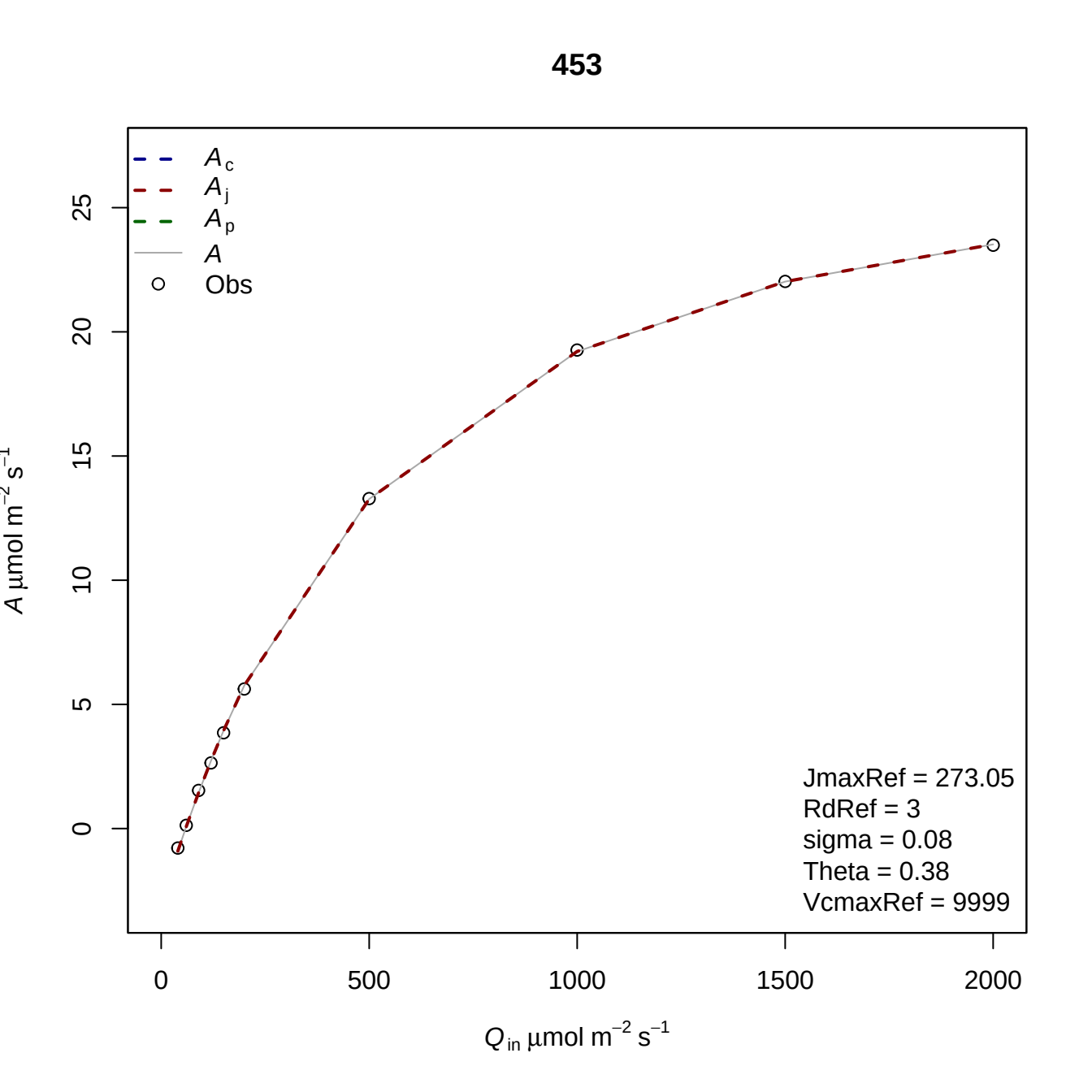

448
A_c
A_j
A_p
A
Obs
JmaxRef = 278.15
RdRef = 3.73
sigma = 0.09
Theta = −0.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

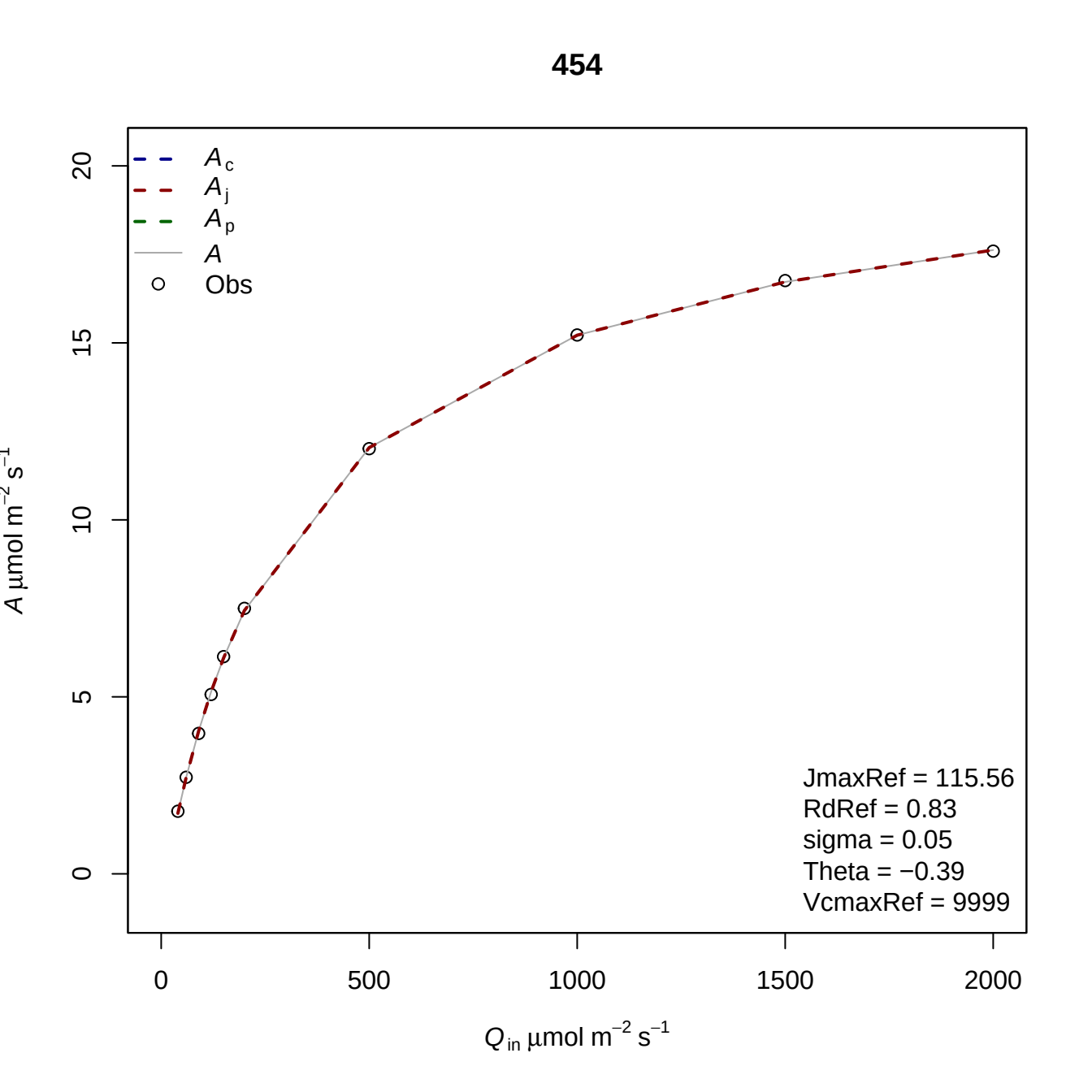

449
A_c
A_j
A_p
A
Obs
JmaxRef = 171.53
RdRef = 1.23
sigma = 0.19
Theta = 0.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

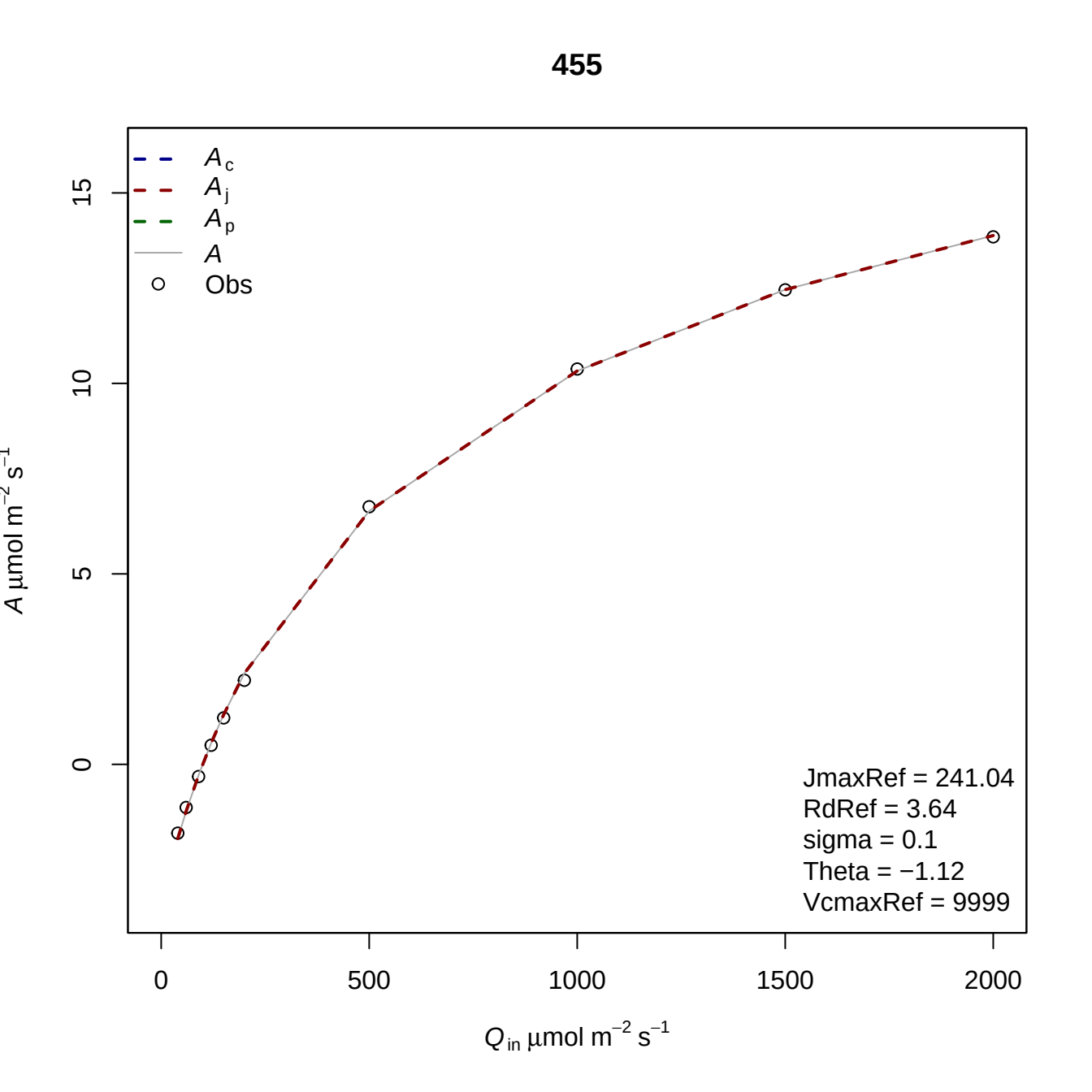

450
A_c
A_j
A_p
A
Obs
JmaxRef = 225.09
RdRef = 1.11
sigma = 0.12
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

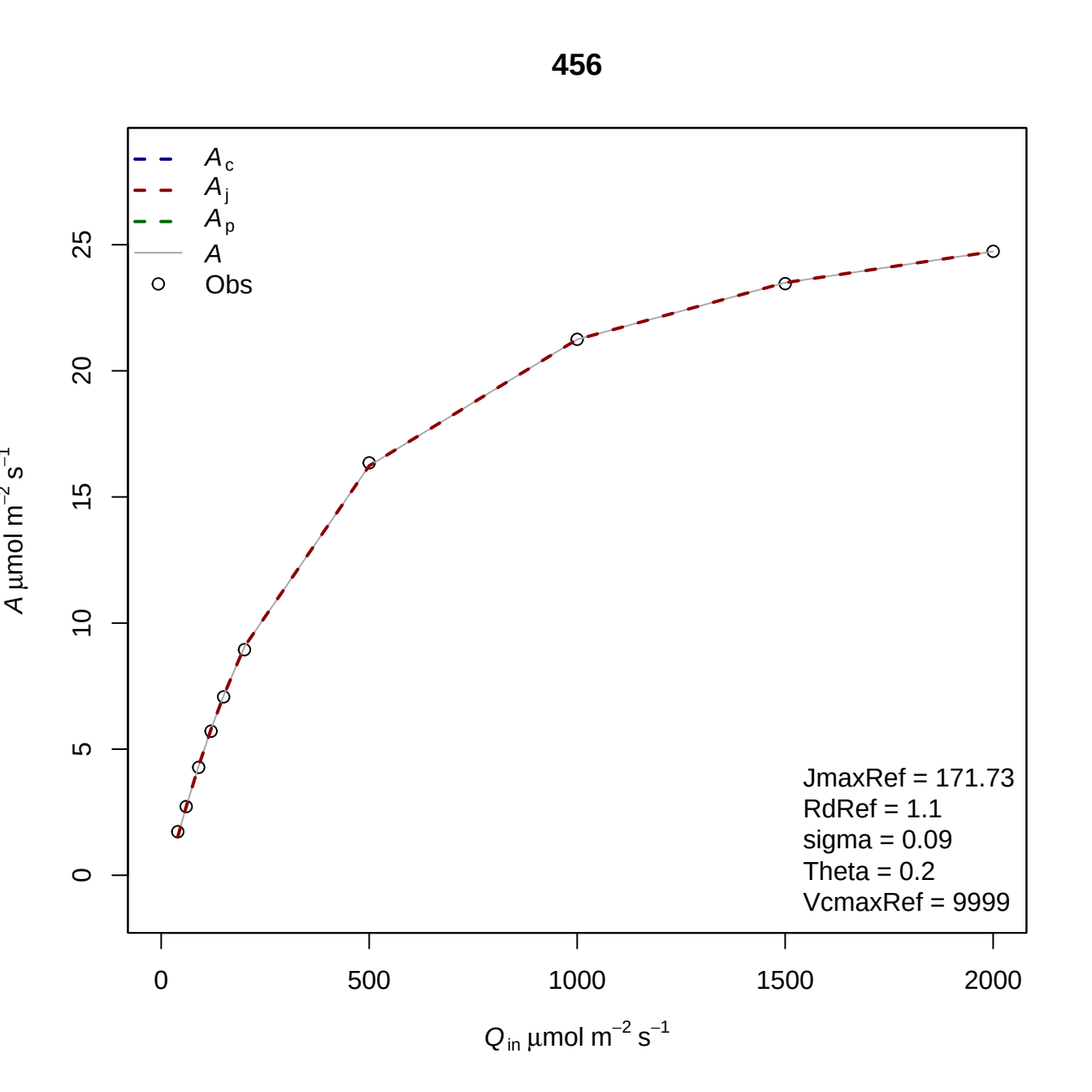

451
A_c
A_j
A_p
A
Obs
JmaxRef = 210.55
RdRef = 1.42
sigma = 0.17
Theta = 0.28
VcmaxRef = 119.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

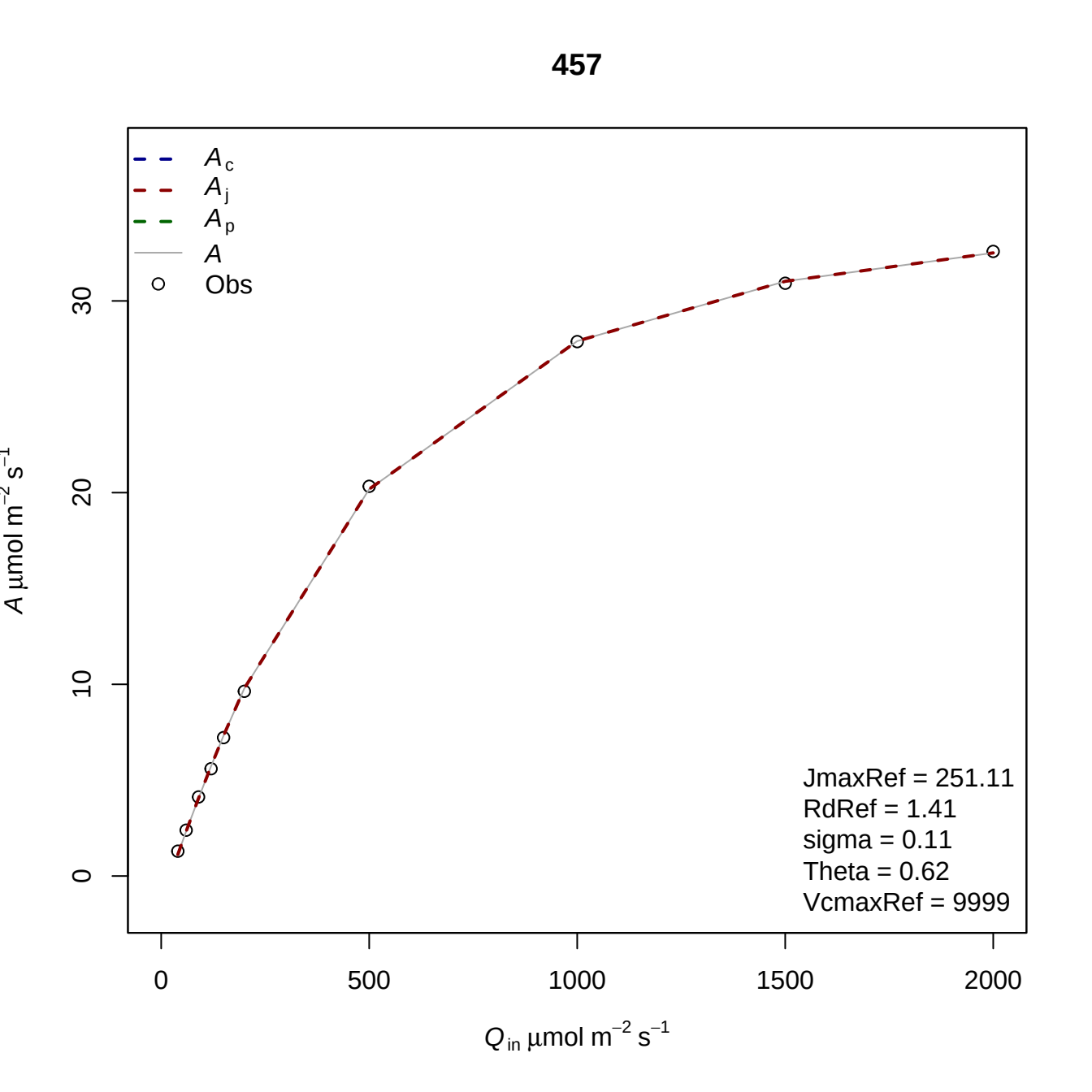

452
A_c
A_j
A_p
A
Obs
JmaxRef = 237.7
RdRef = 2.06
sigma = 0.14
Theta = 0.17
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

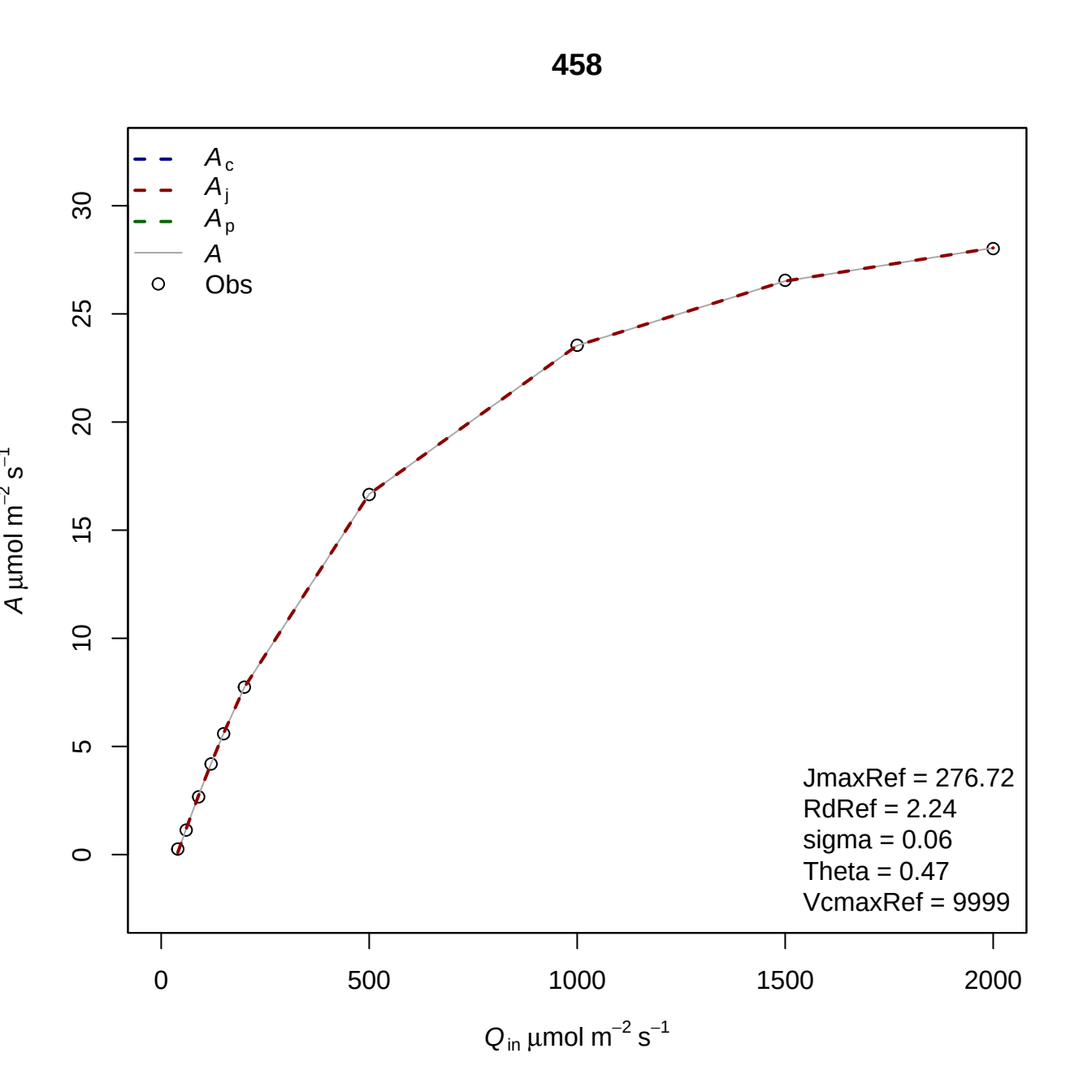

453
A_c
A_j
A_p
A
Obs
JmaxRef = 273.05
RdRef = 3
sigma = 0.08
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

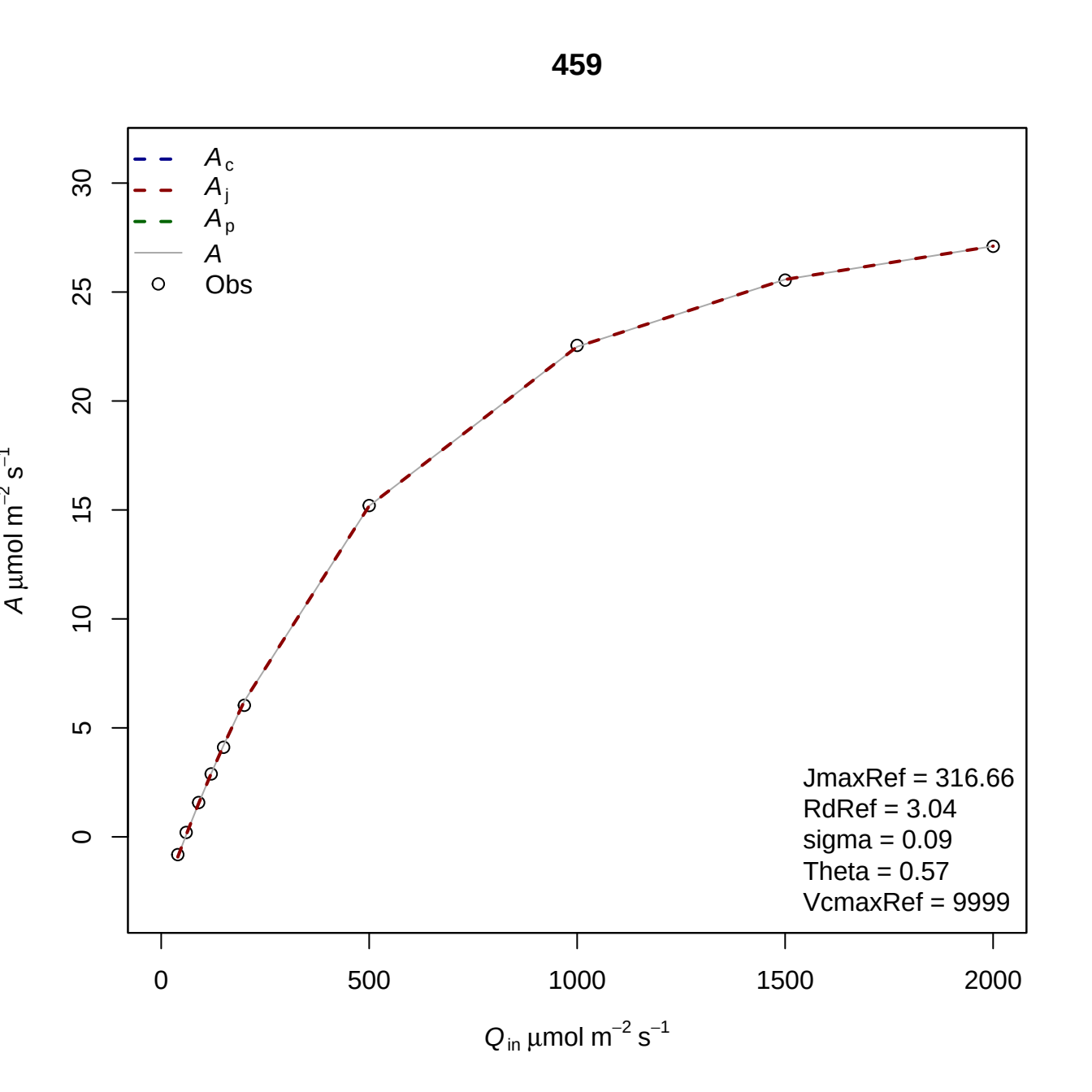

454
A_c
A_j
A_p
A
Obs
JmaxRef = 115.56
RdRef = 0.83
sigma = 0.05
Theta = −0.39
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

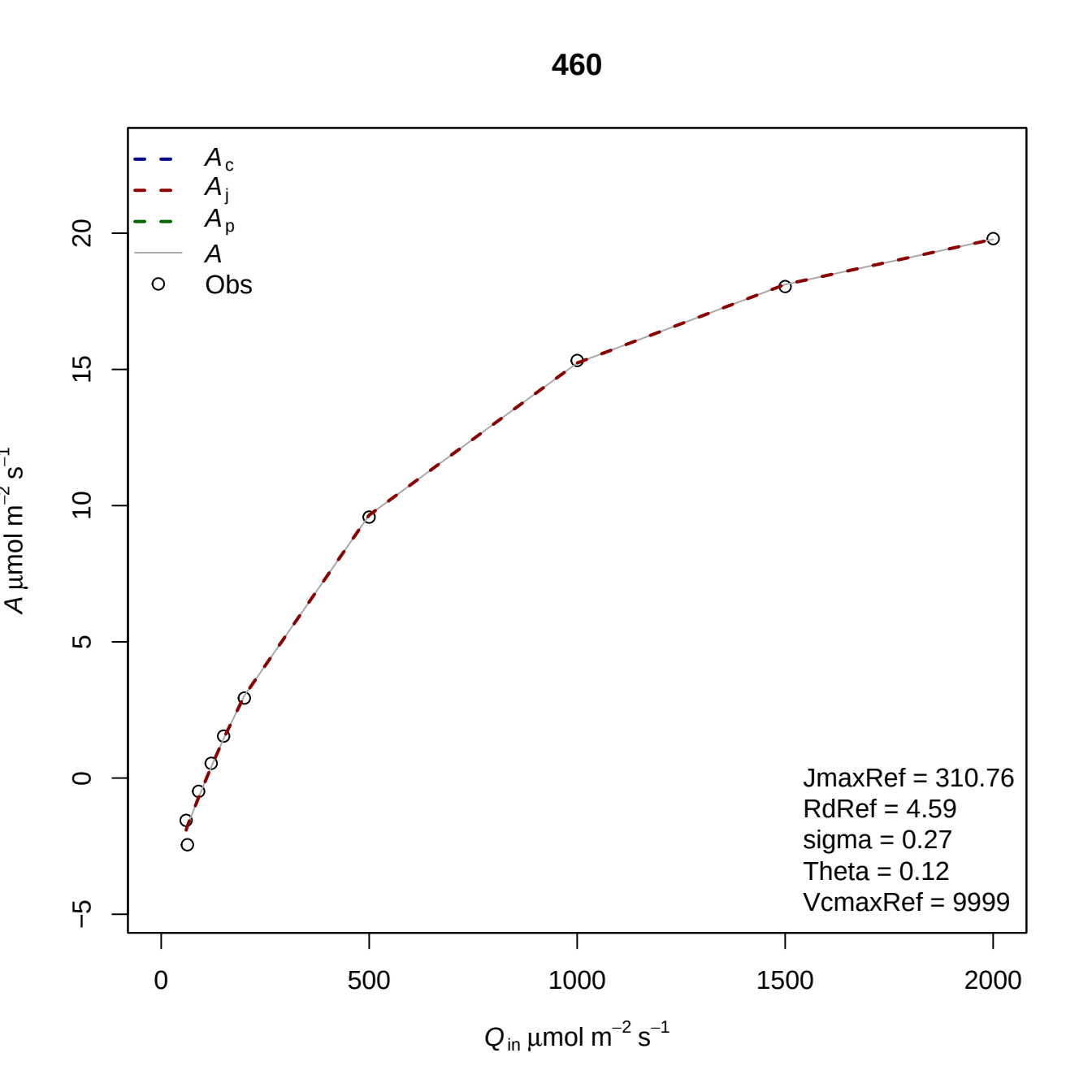

455
A_c
A_j
A_p
A
Obs
JmaxRef = 241.04
RdRef = 3.64
sigma = 0.1
Theta = −1.12
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

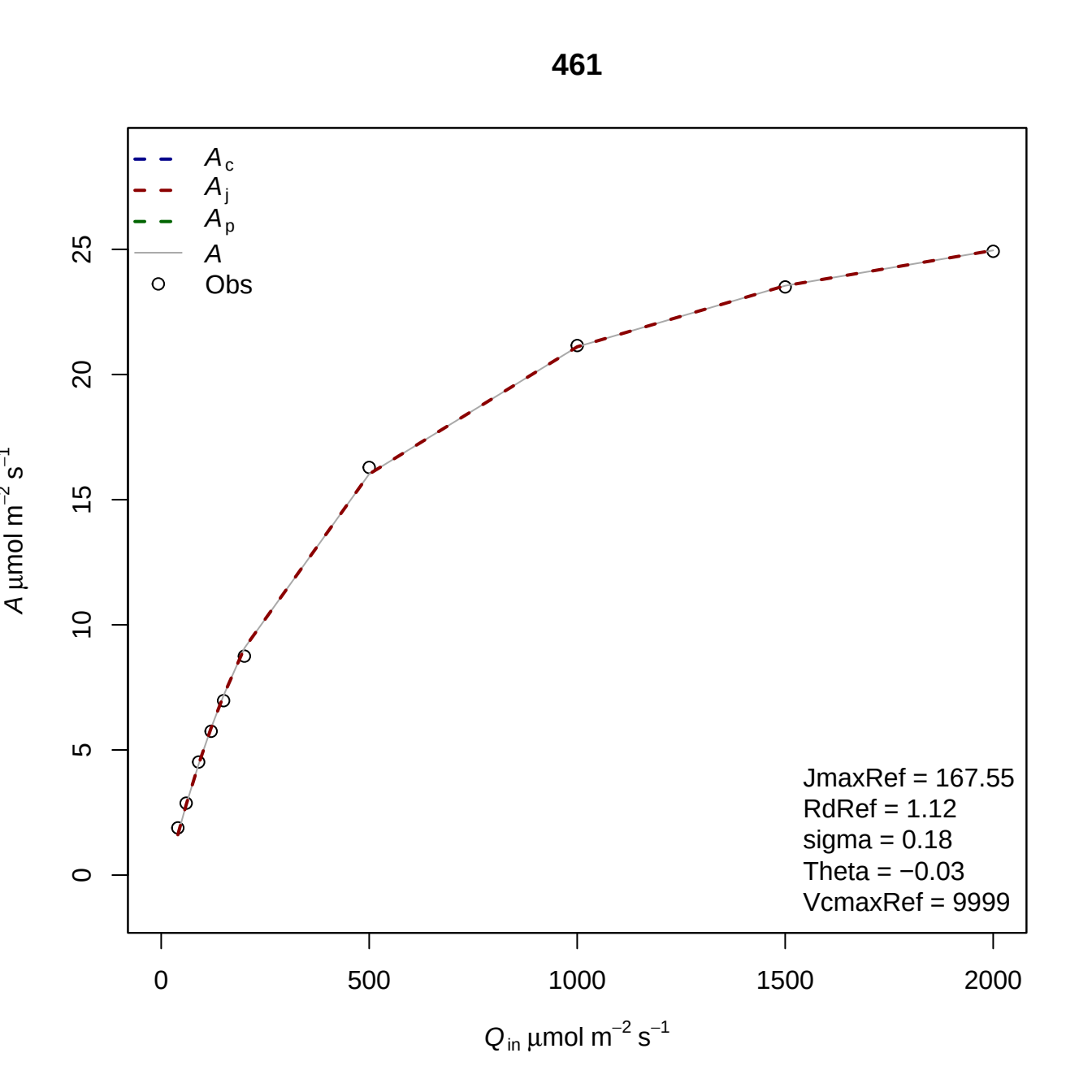

456
A_c
A_j
A_p
A
Obs
JmaxRef = 171.73
RdRef = 1.1
sigma = 0.09
Theta = 0.2
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

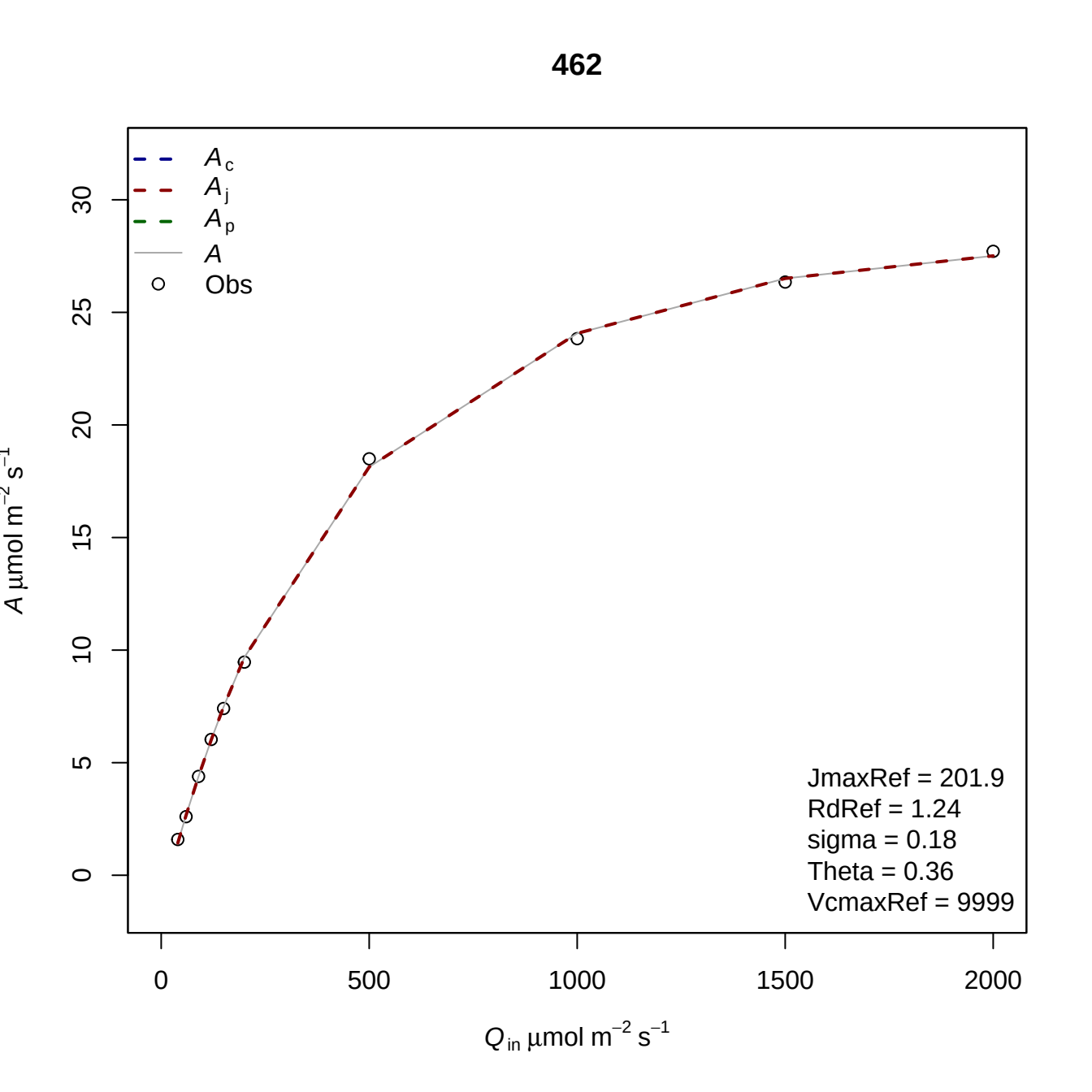

457
A_c
A_j
A_p
A
Obs
JmaxRef = 251.11
RdRef = 1.41
sigma = 0.11
Theta = 0.62
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

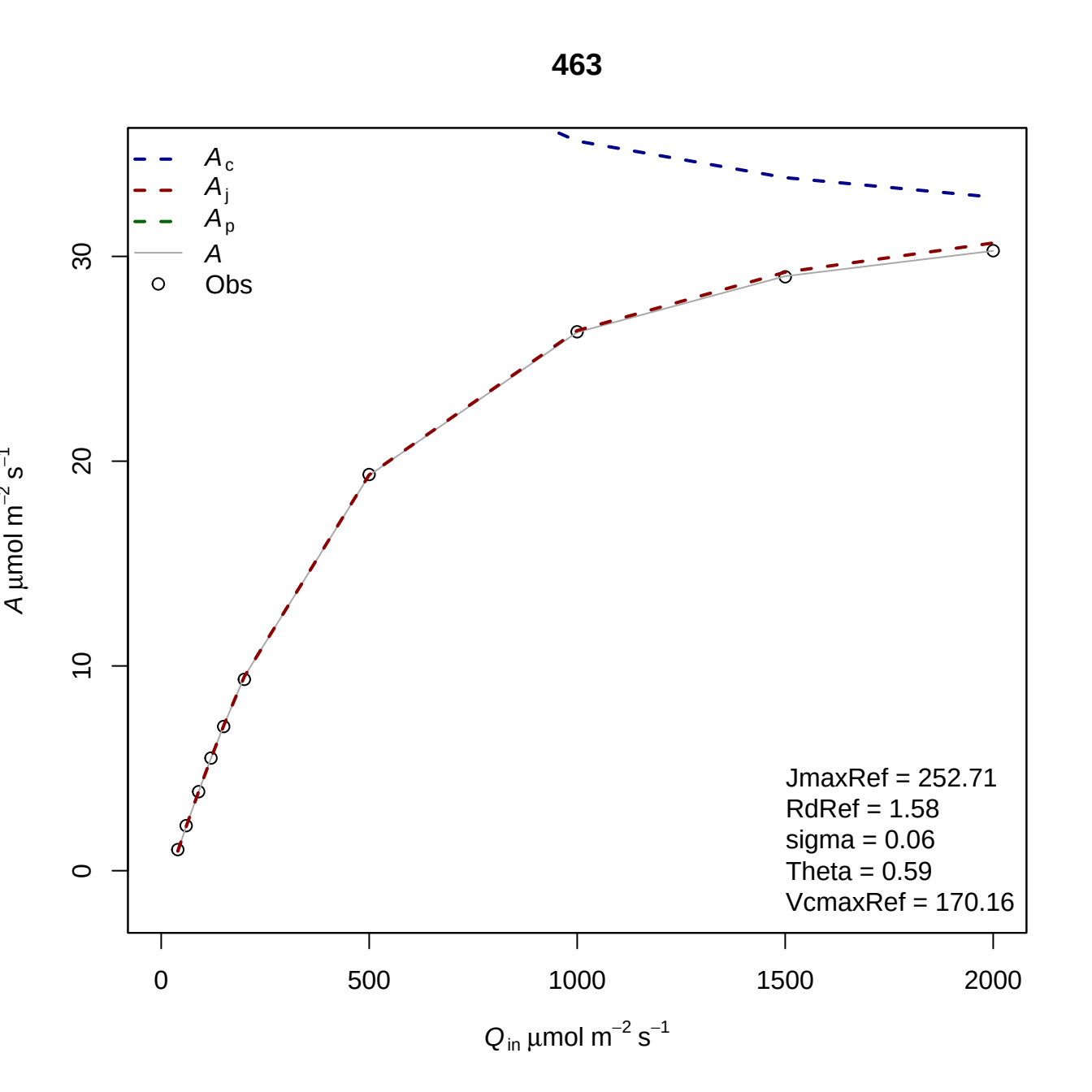

458
A_c
A_j
A_p
A
Obs
JmaxRef = 276.72
RdRef = 2.24
sigma = 0.06
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

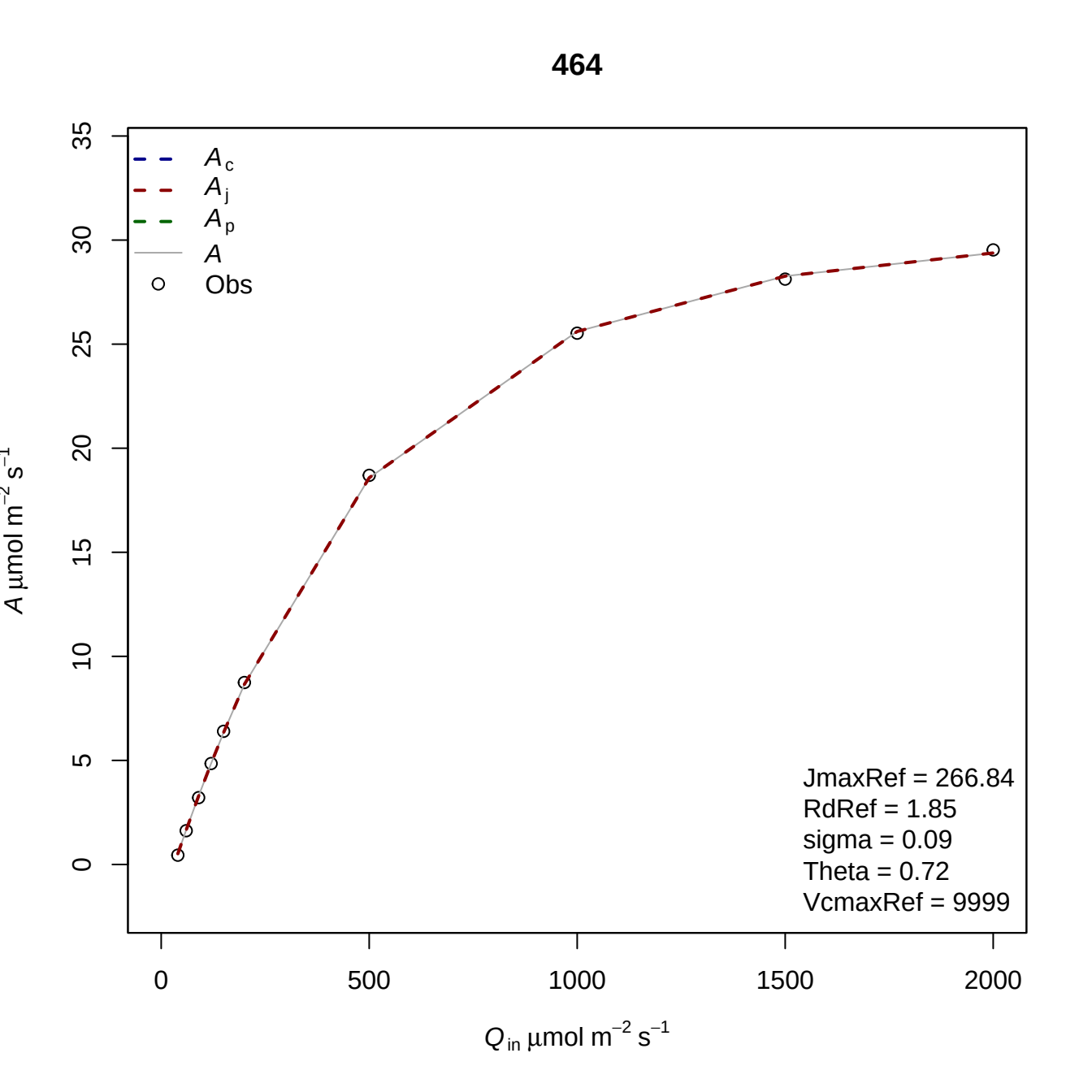

459
A_c
A_j
A_p
A
Obs
JmaxRef = 316.66
RdRef = 3.04
sigma = 0.09
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

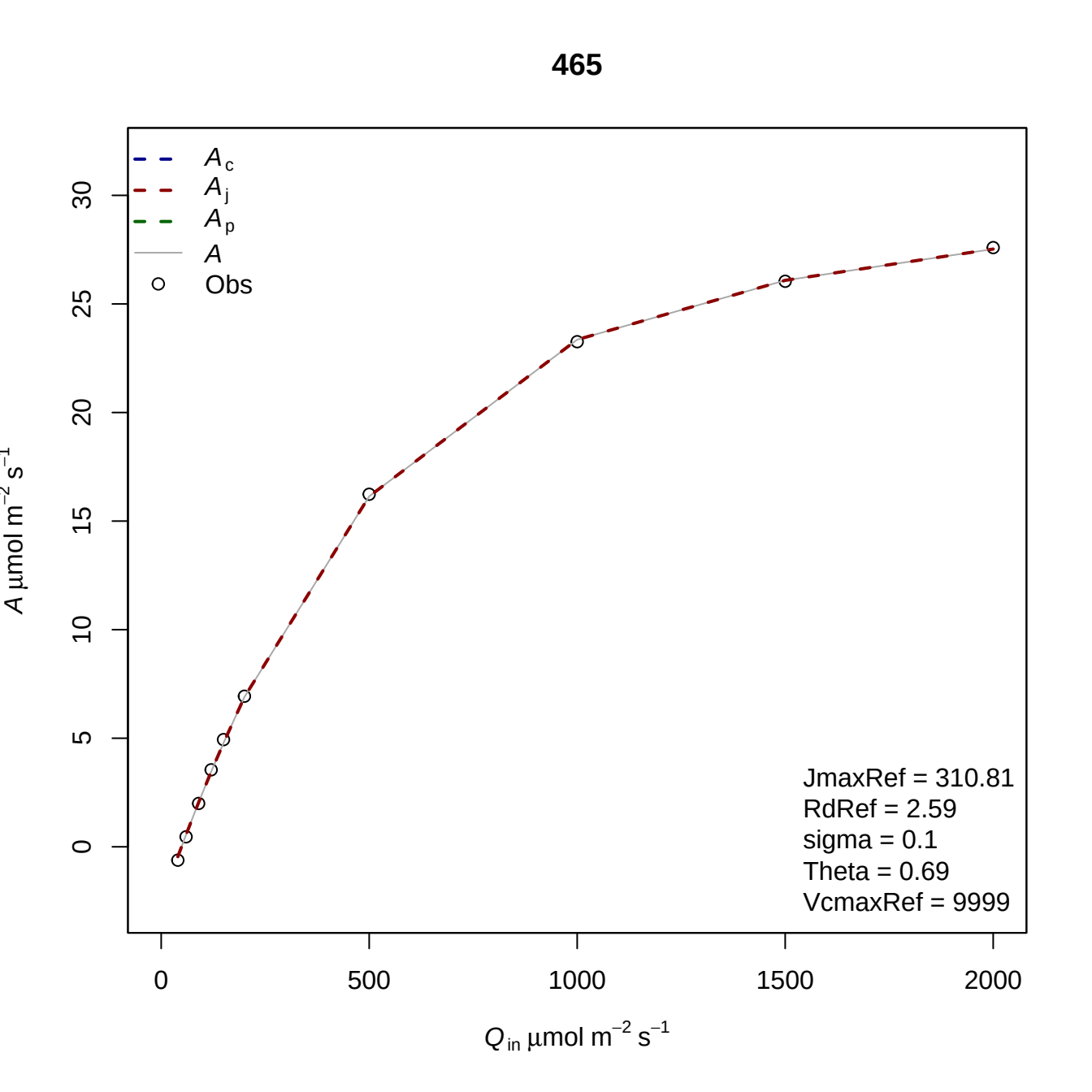

460
A_c
A_j
A_p
A
Obs
JmaxRef = 310.76
RdRef = 4.59
sigma = 0.27
Theta = 0.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

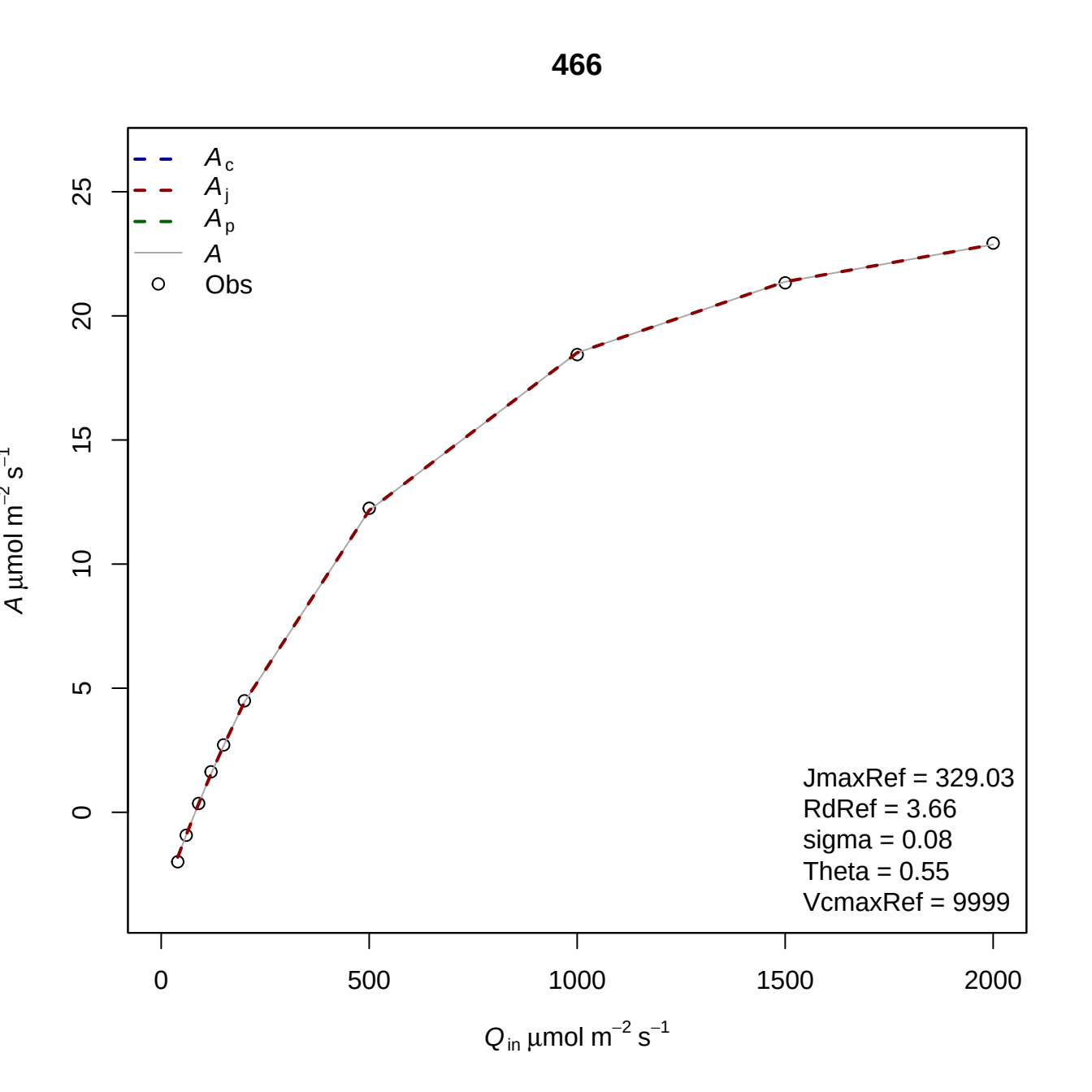

461
A_c
A_j
A_p
A
Obs
JmaxRef = 167.55
RdRef = 1.12
sigma = 0.18
Theta = −0.03
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

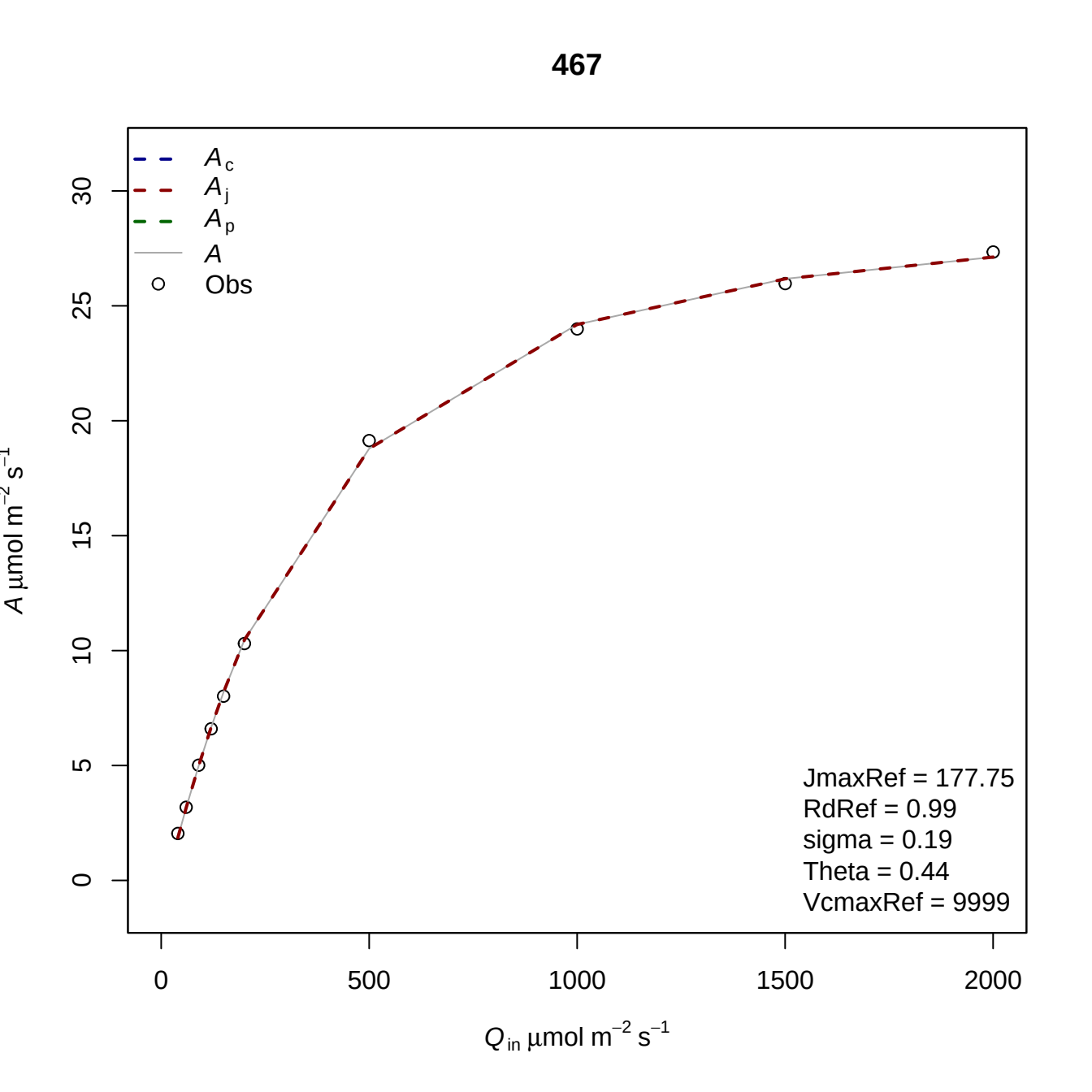

462
A_c
A_j
A_p
A
Obs
JmaxRef = 201.9
RdRef = 1.24
sigma = 0.18
Theta = 0.36
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

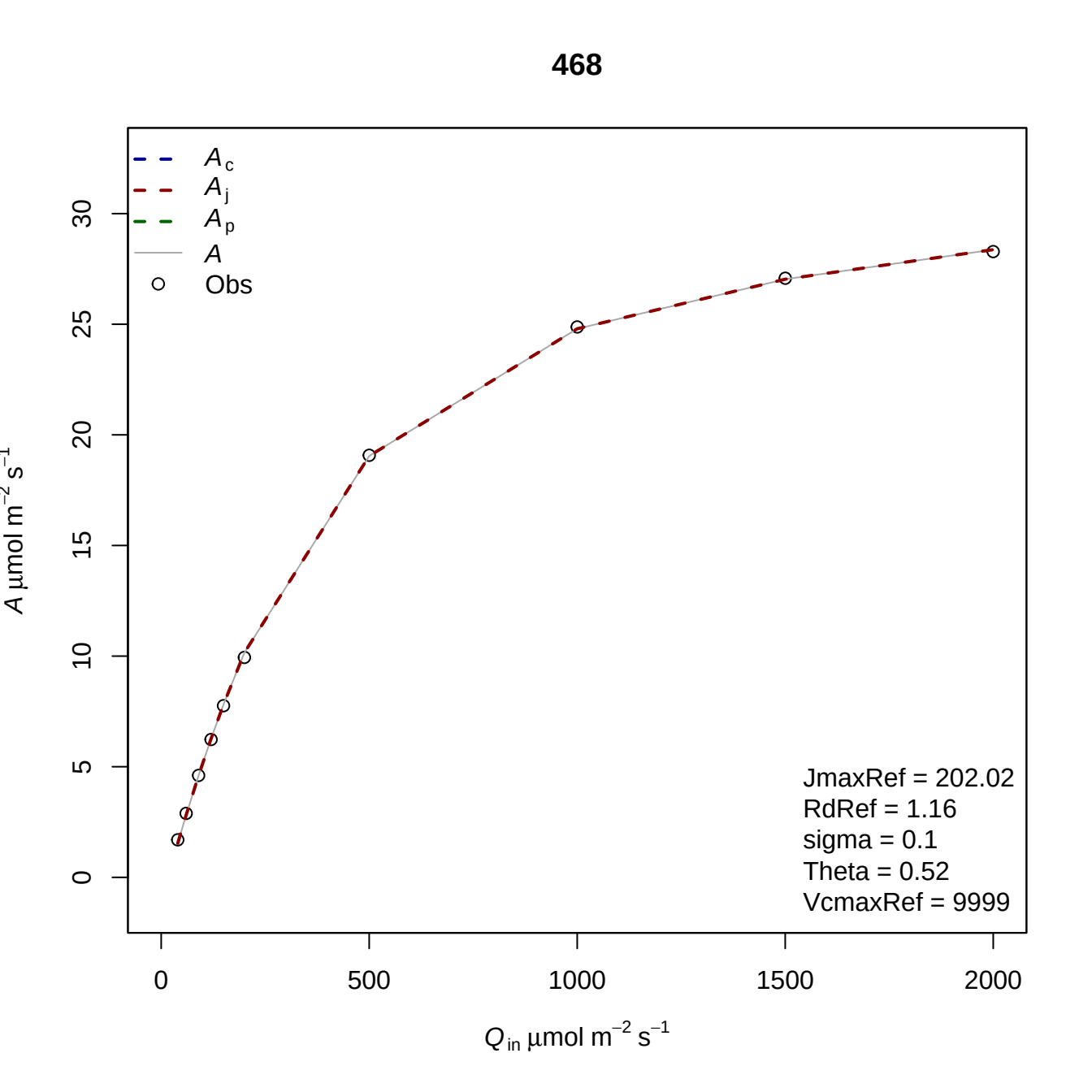

463
A_c
A_j
A_p
A
Obs
JmaxRef = 252.71
RdRef = 1.58
sigma = 0.06
Theta = 0.59
VcmaxRef = 170.16
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

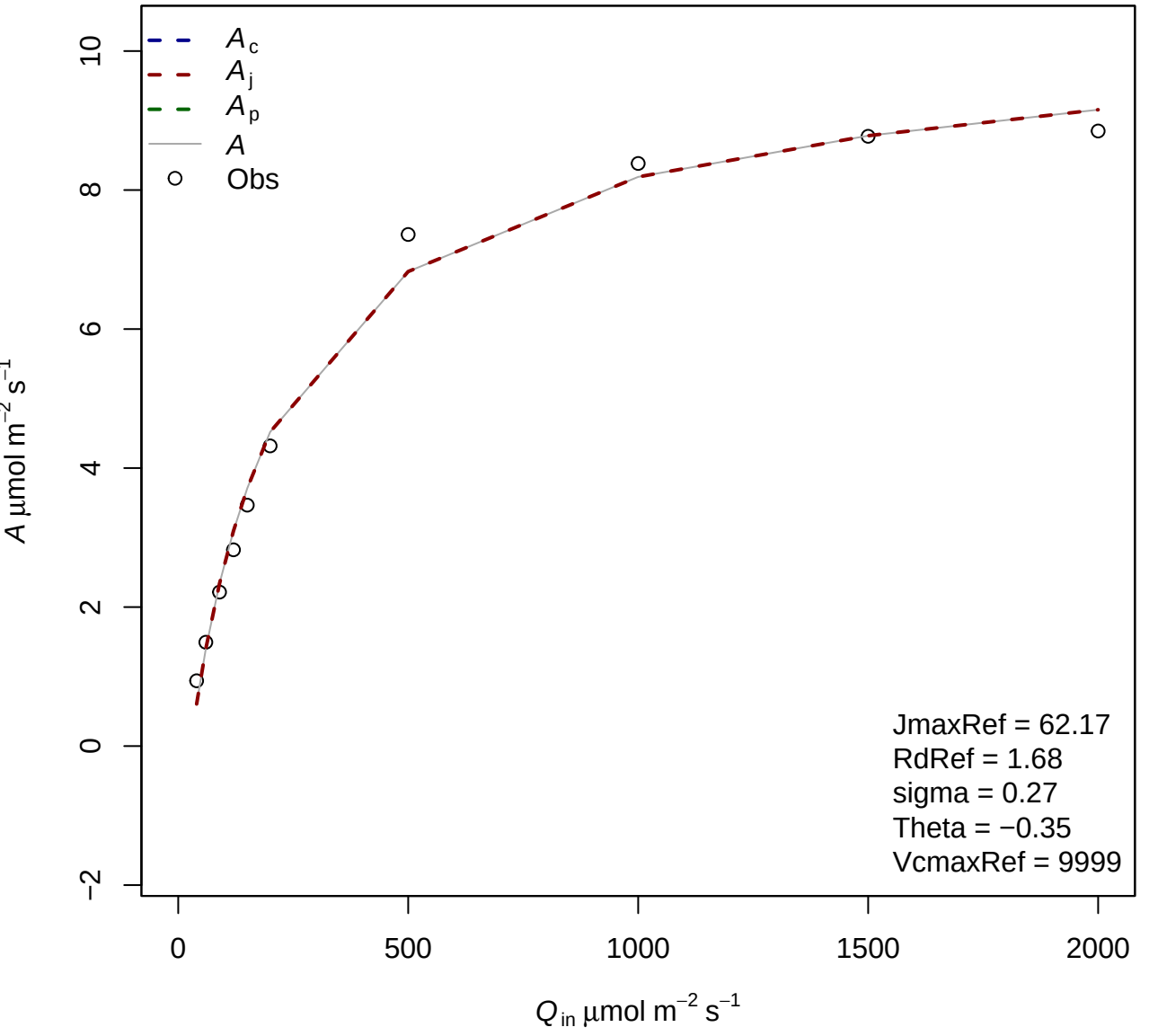

464
A_c
A_j
A_p
A
Obs
JmaxRef = 266.84
RdRef = 1.85
sigma = 0.09
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

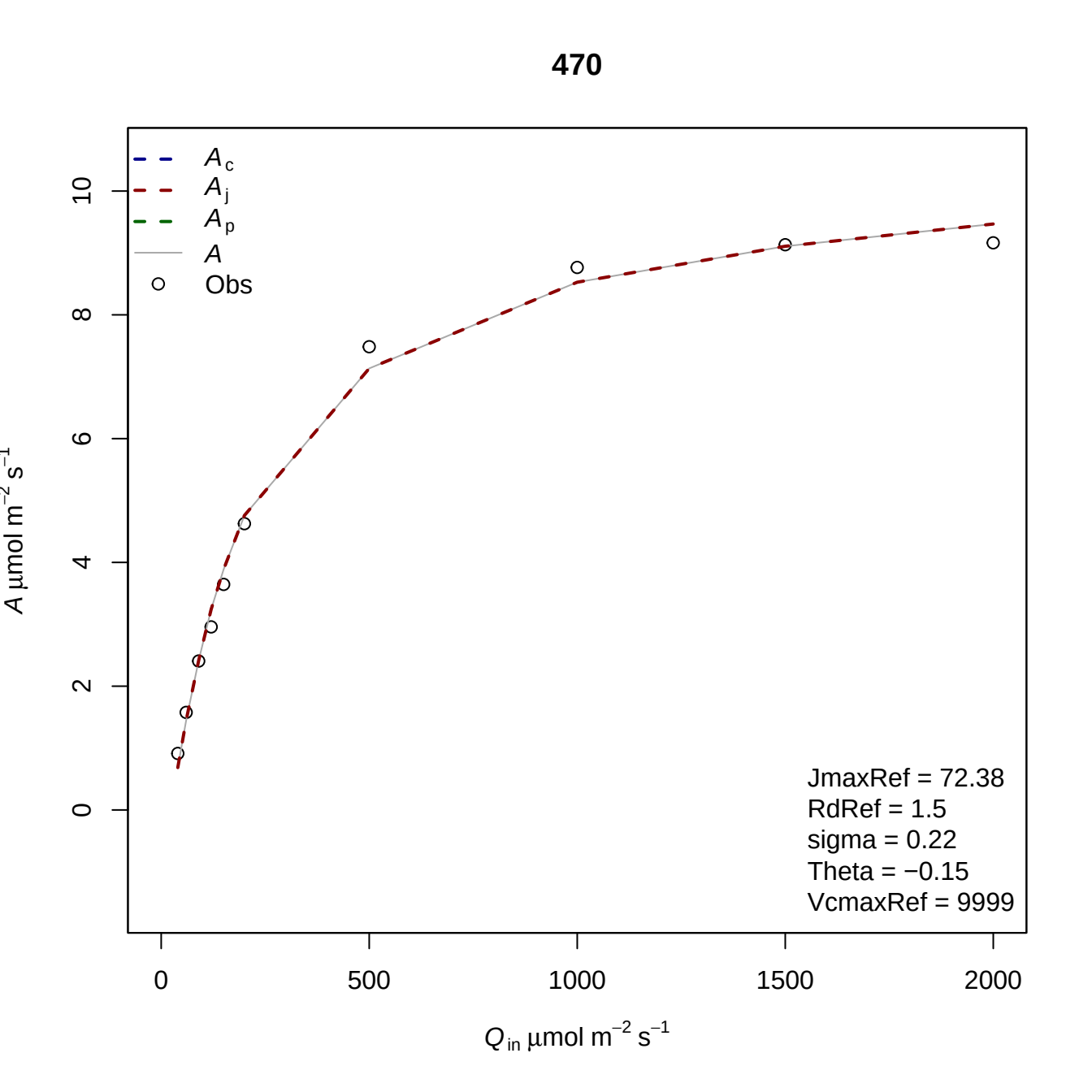

465
A_c
A_j
A_p
A
Obs
JmaxRef = 310.81
RdRef = 2.59
sigma = 0.1
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

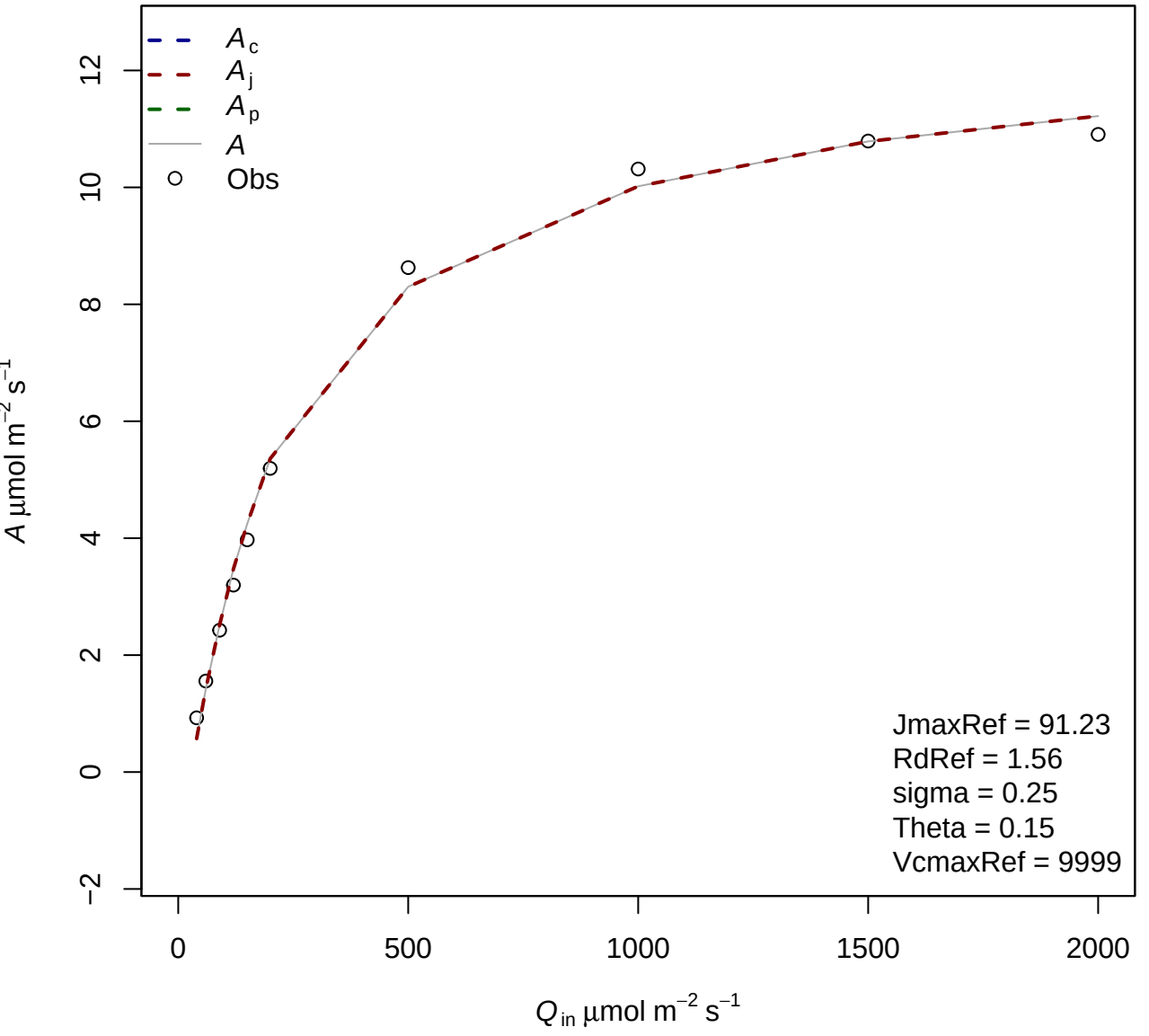

466
A_c
A_j
A_p
A
Obs
JmaxRef = 329.03
RdRef = 3.66
sigma = 0.08
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

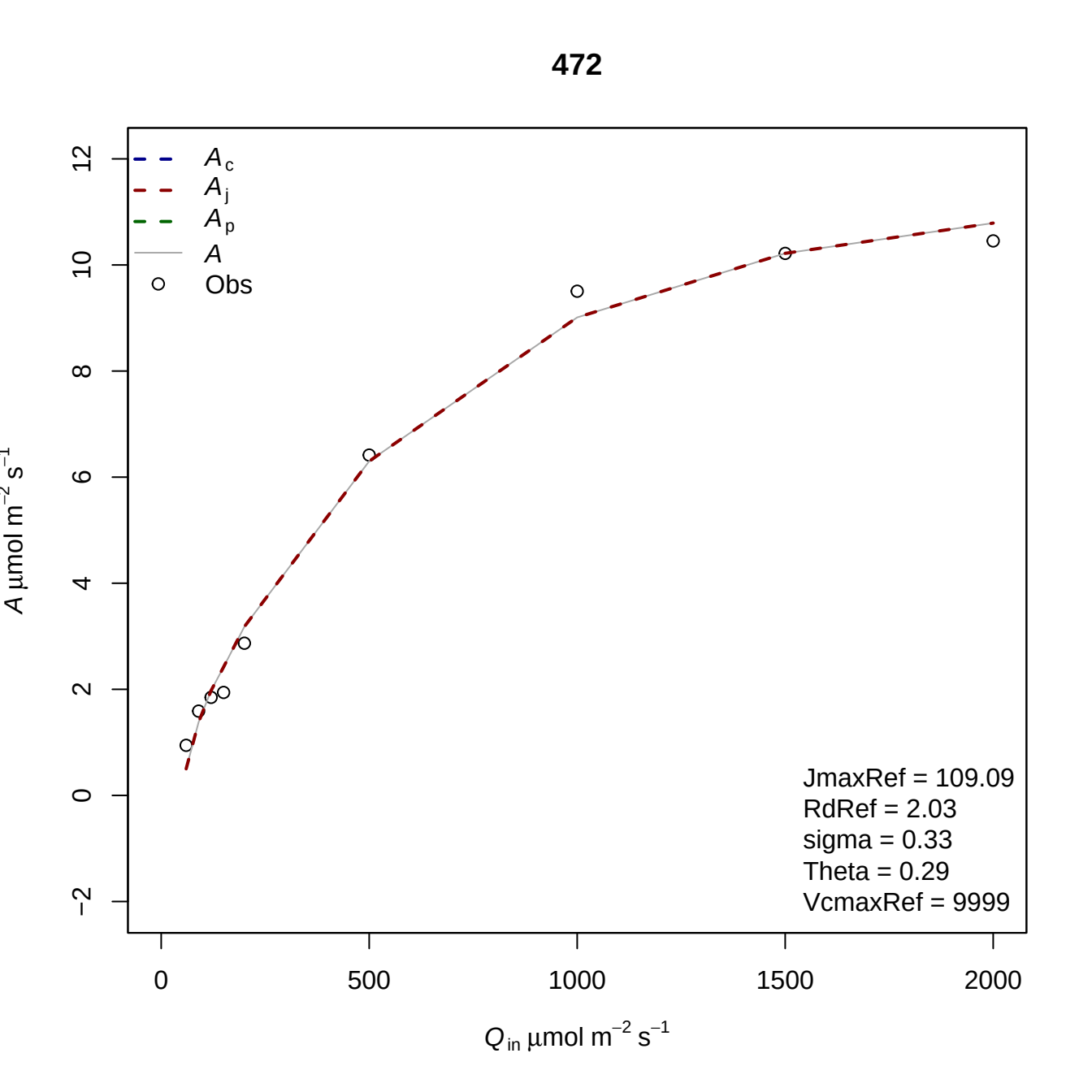

467
A_c
A_j
A_p
A
Obs
JmaxRef = 177.75
RdRef = 0.99
sigma = 0.19
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

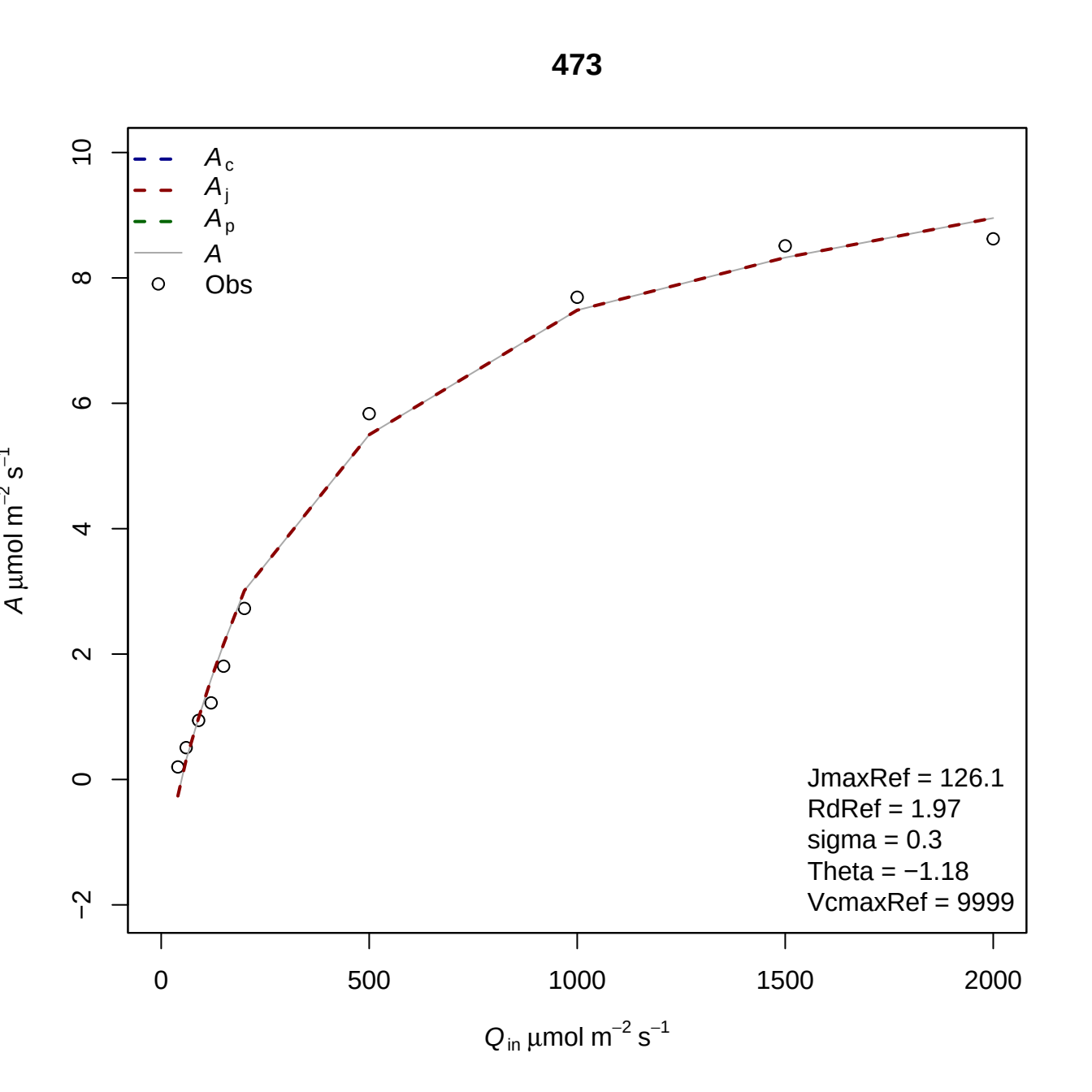

468
A_c
A_j
A_p
A
Obs
JmaxRef = 202.02
RdRef = 1.16
sigma = 0.1
Theta = 0.52
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

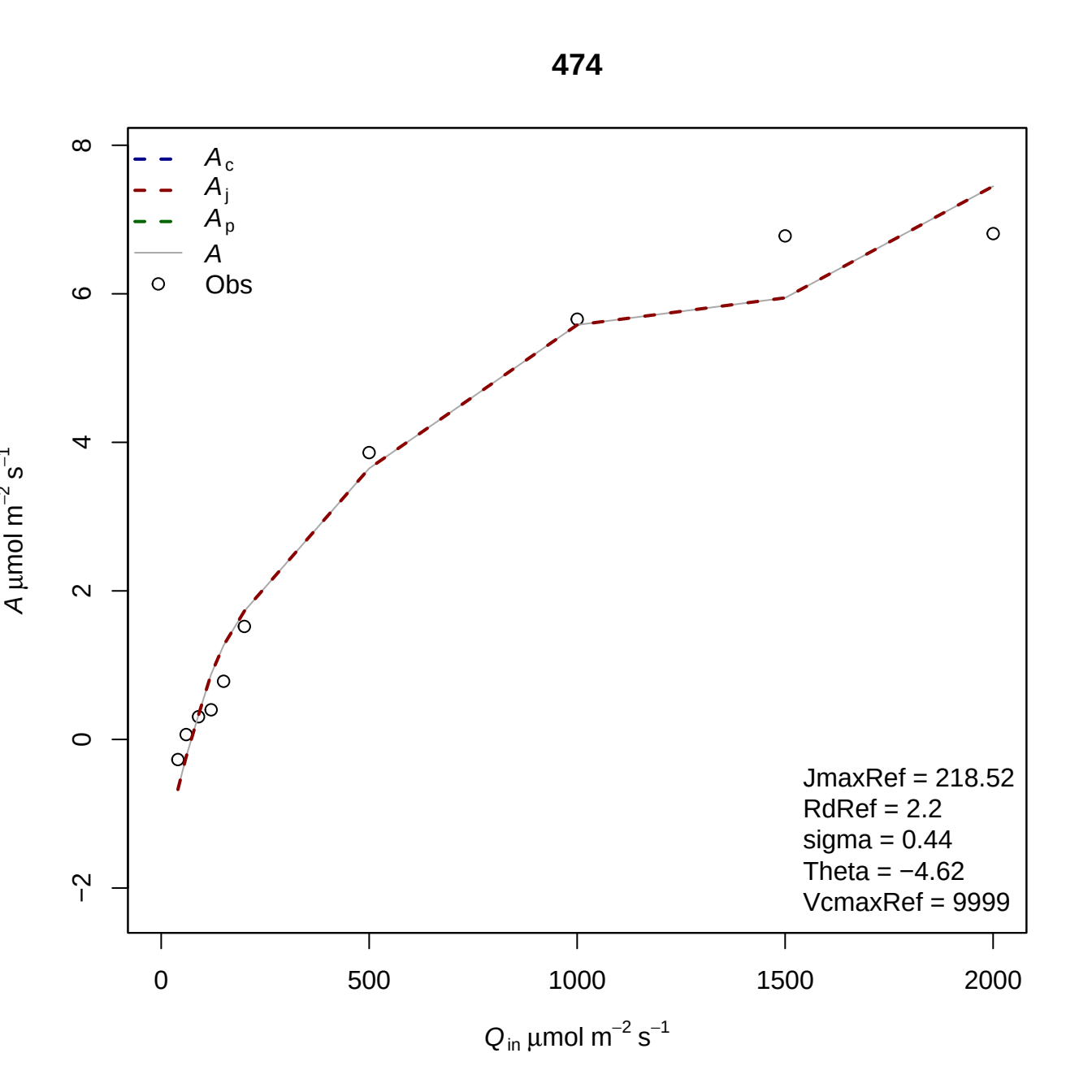

A_c
A_j
A_p
A
Obs
JmaxRef = 62.17
RdRef = 1.68
sigma = 0.27
Theta = −0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

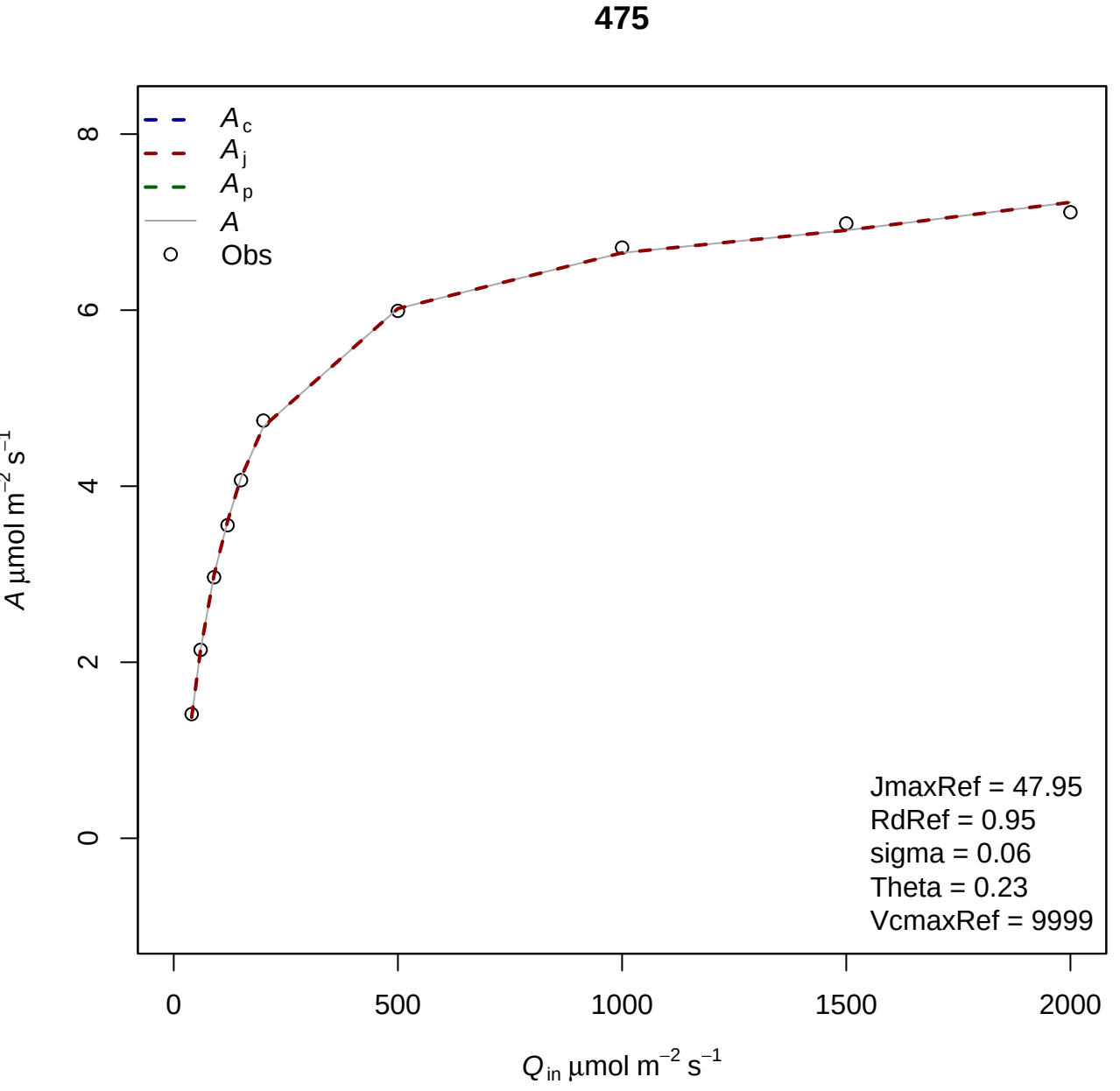

470
A_c
A_j
A_p
A
Obs
JmaxRef = 72.38
RdRef = 1.5
sigma = 0.22
Theta = −0.15
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

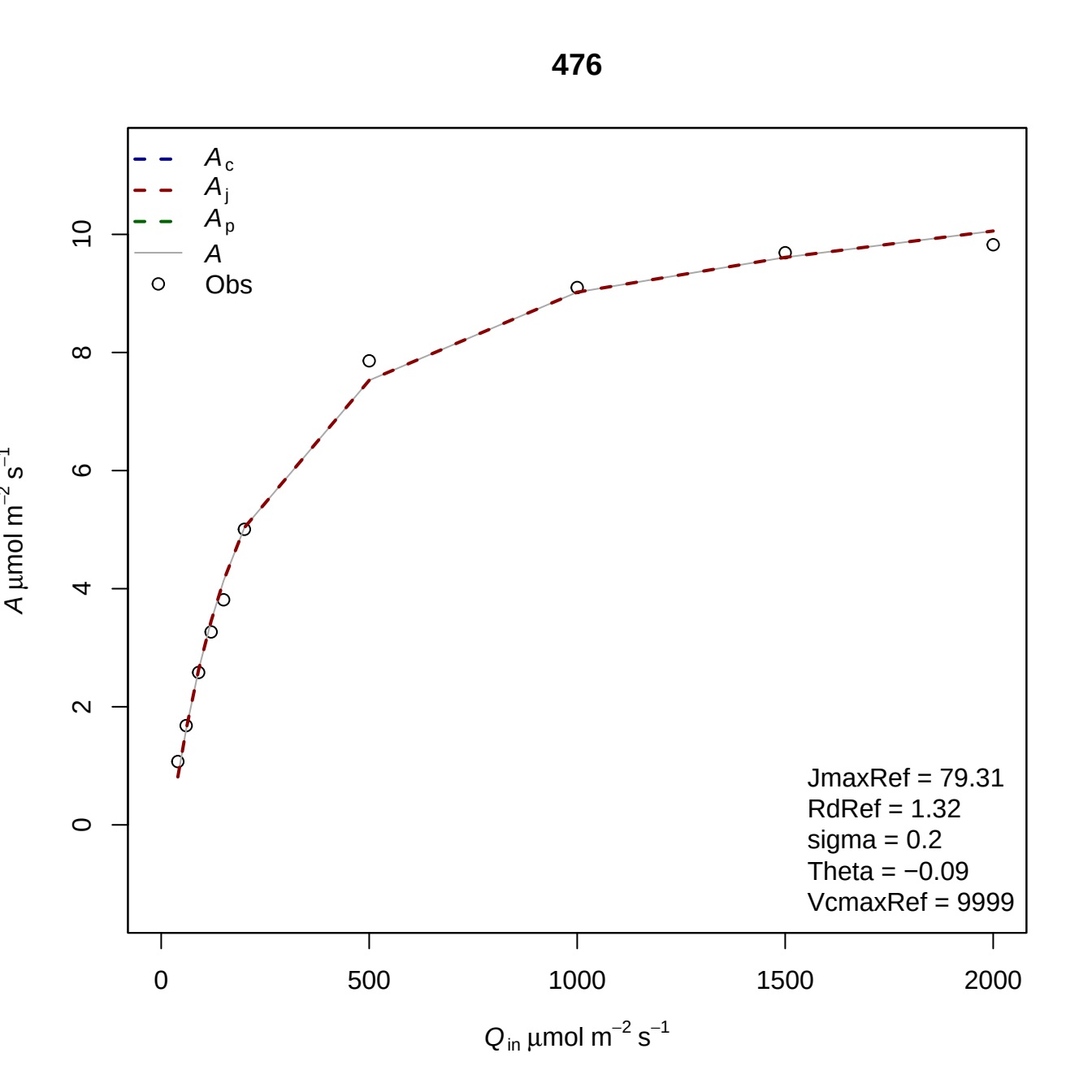

A_c
A_j
A_p
A
Obs
JmaxRef = 91.23
RdRef = 1.56
sigma = 0.25
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

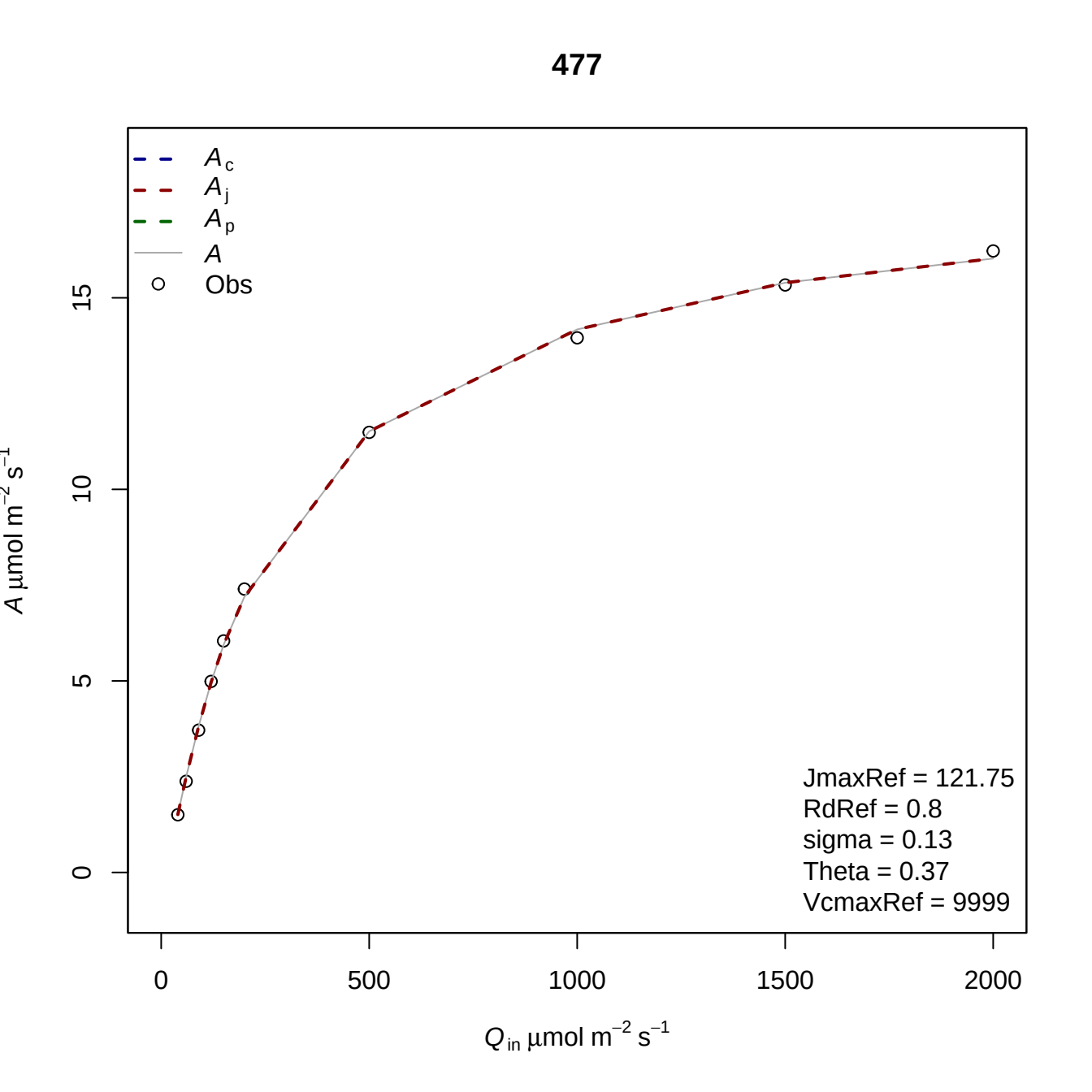

472
A_c
A_j
A_p
A
Obs
JmaxRef = 109.09
RdRef = 2.03
sigma = 0.33
Theta = 0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

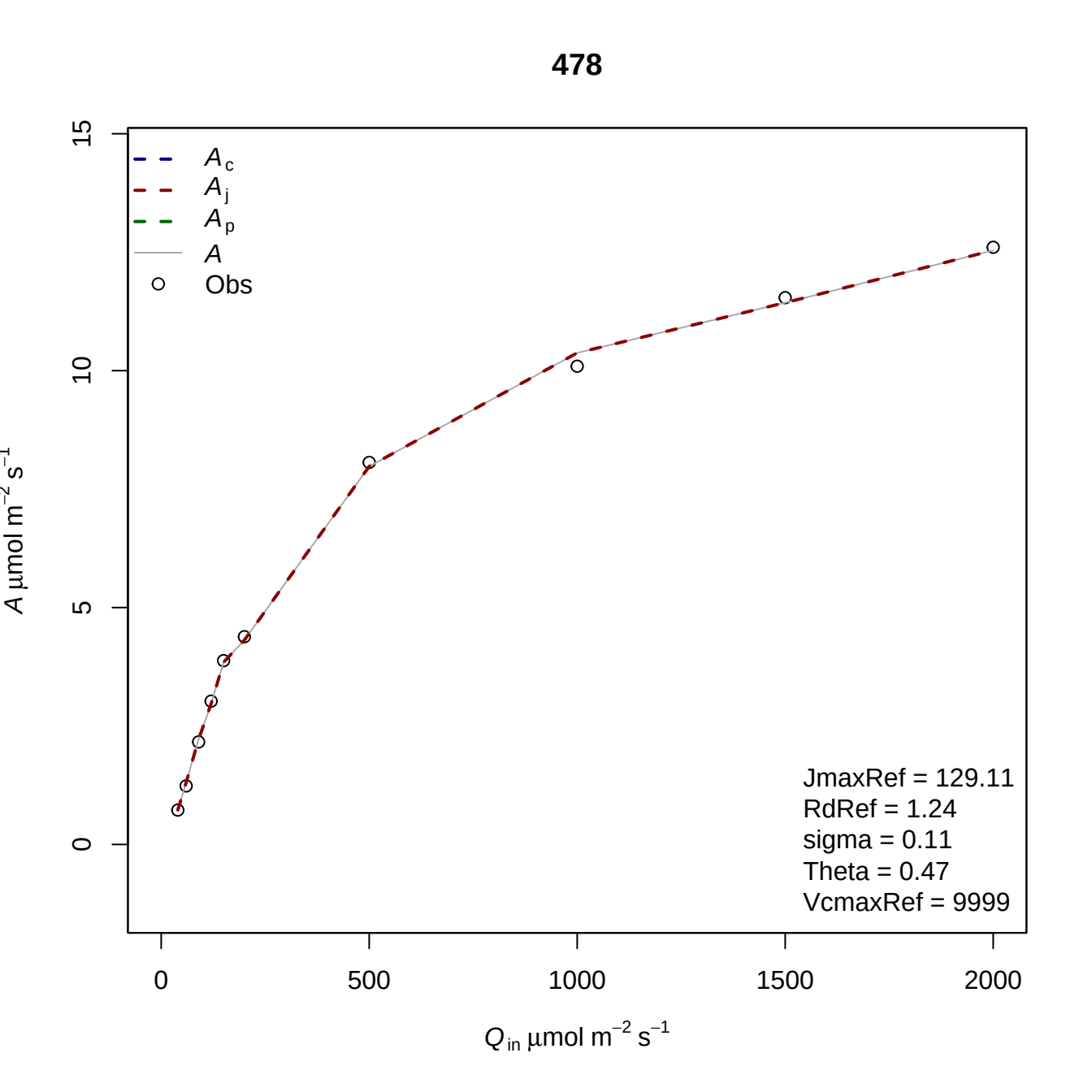

473
A_c
A_j
A_p
A
Obs
JmaxRef = 126.1
RdRef = 1.97
sigma = 0.3
Theta = −1.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

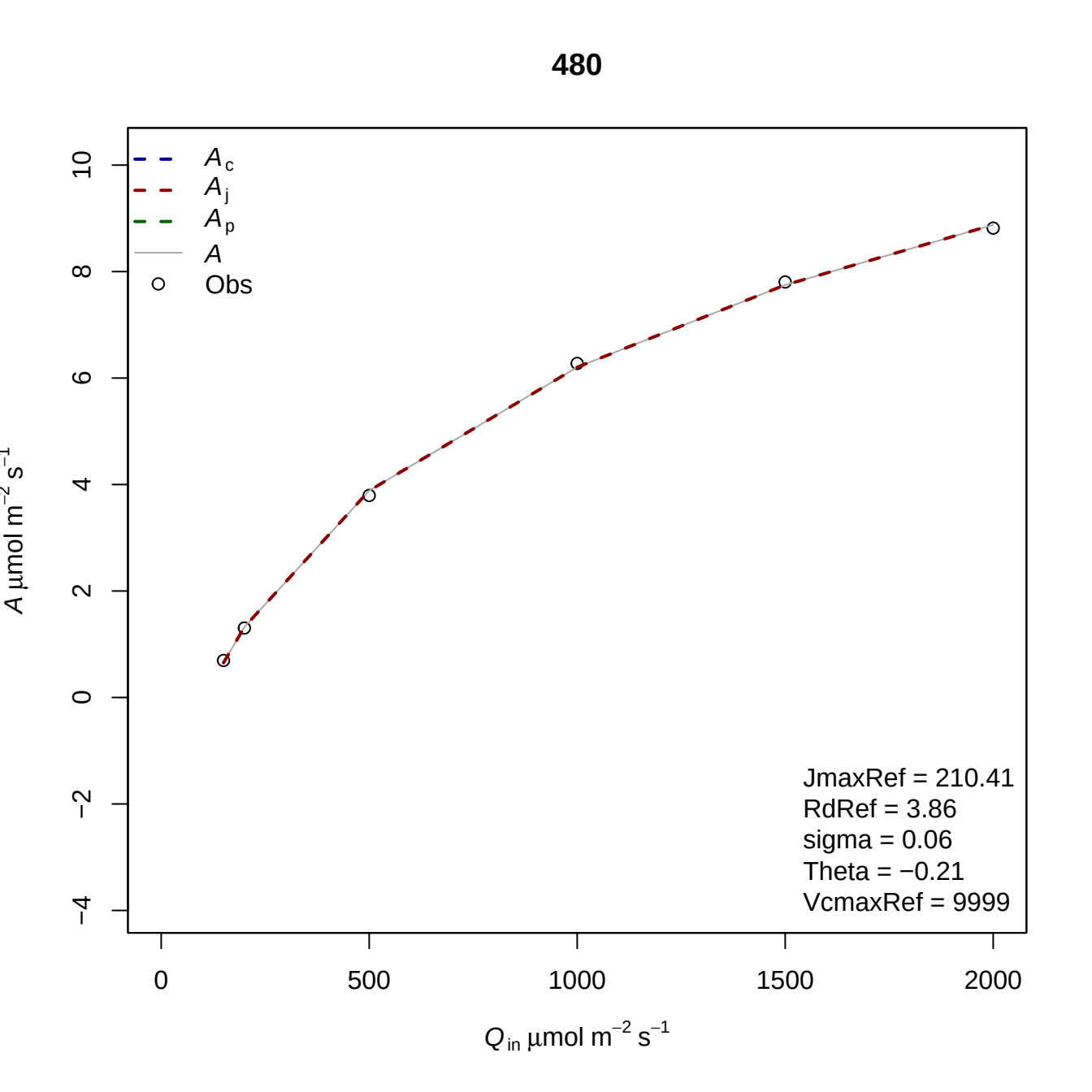

474
A_c
A_j
A_p
A
Obs
JmaxRef = 218.52
RdRef = 2.2
sigma = 0.44
Theta = −4.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

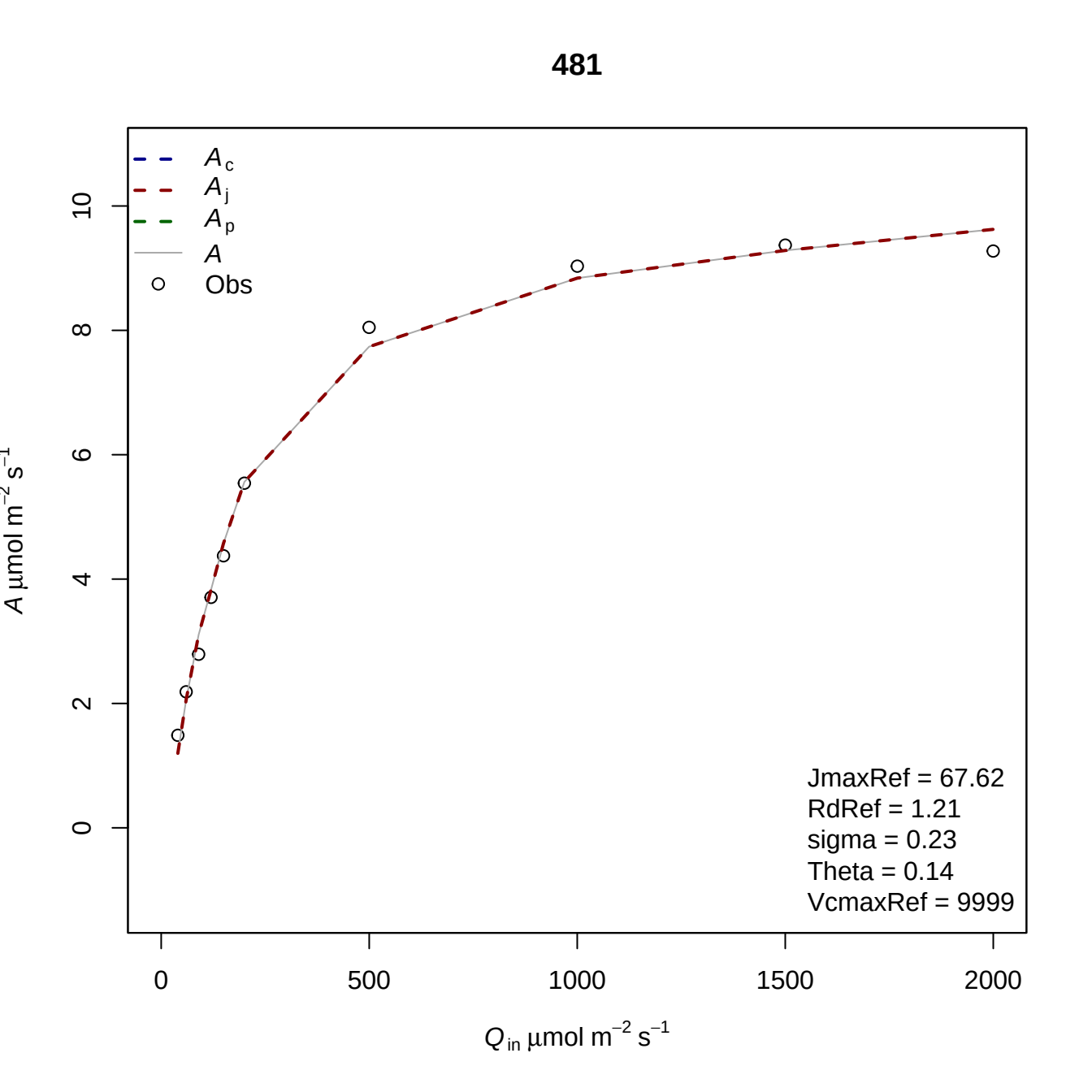

475
A_c
A_j
A_p
A
Obs
JmaxRef = 47.95
RdRef = 0.95
sigma = 0.06
Theta = 0.23
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

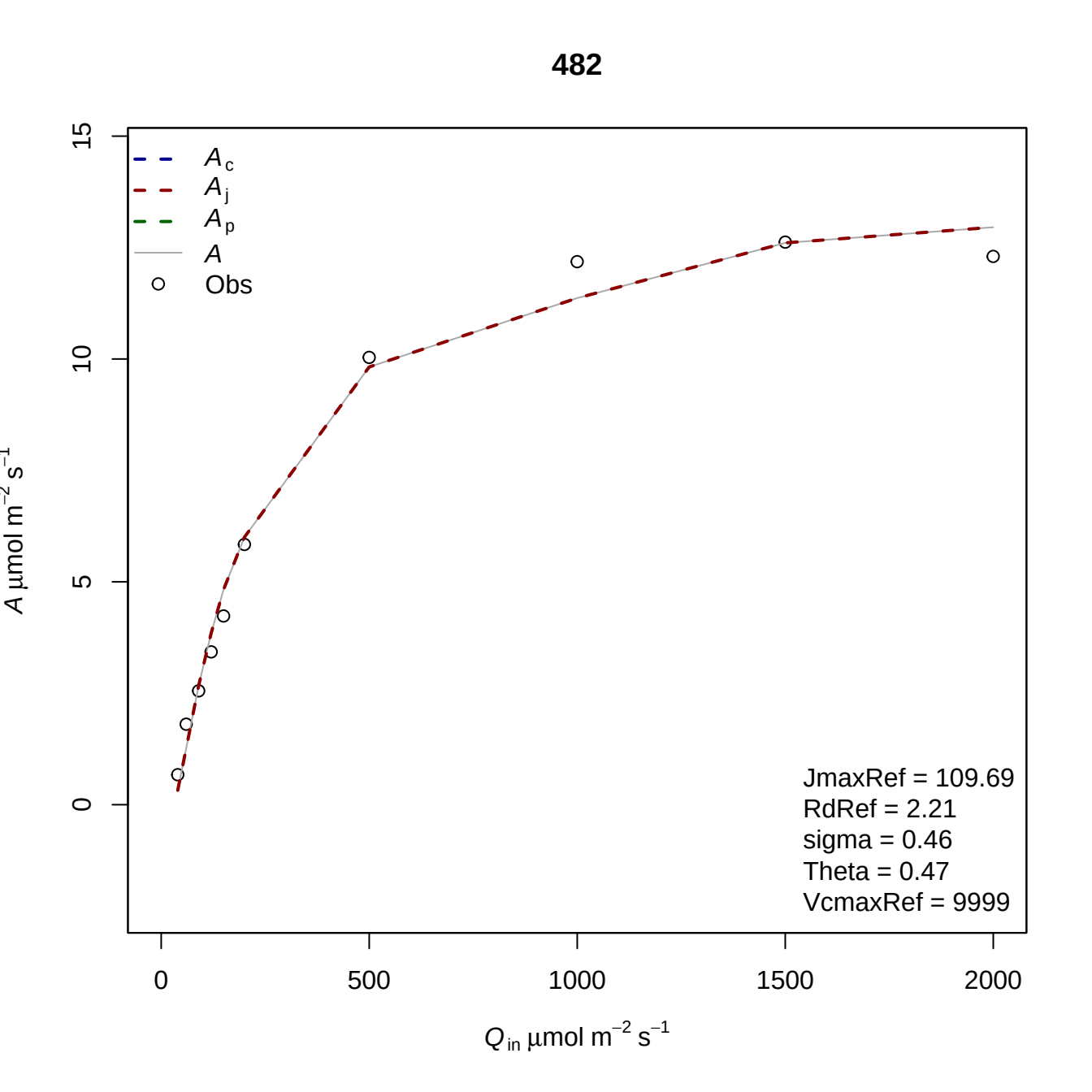

476
A_c
A_j
A_p
A
Obs
JmaxRef = 79.31
RdRef = 1.32
sigma = 0.2
Theta = −0.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

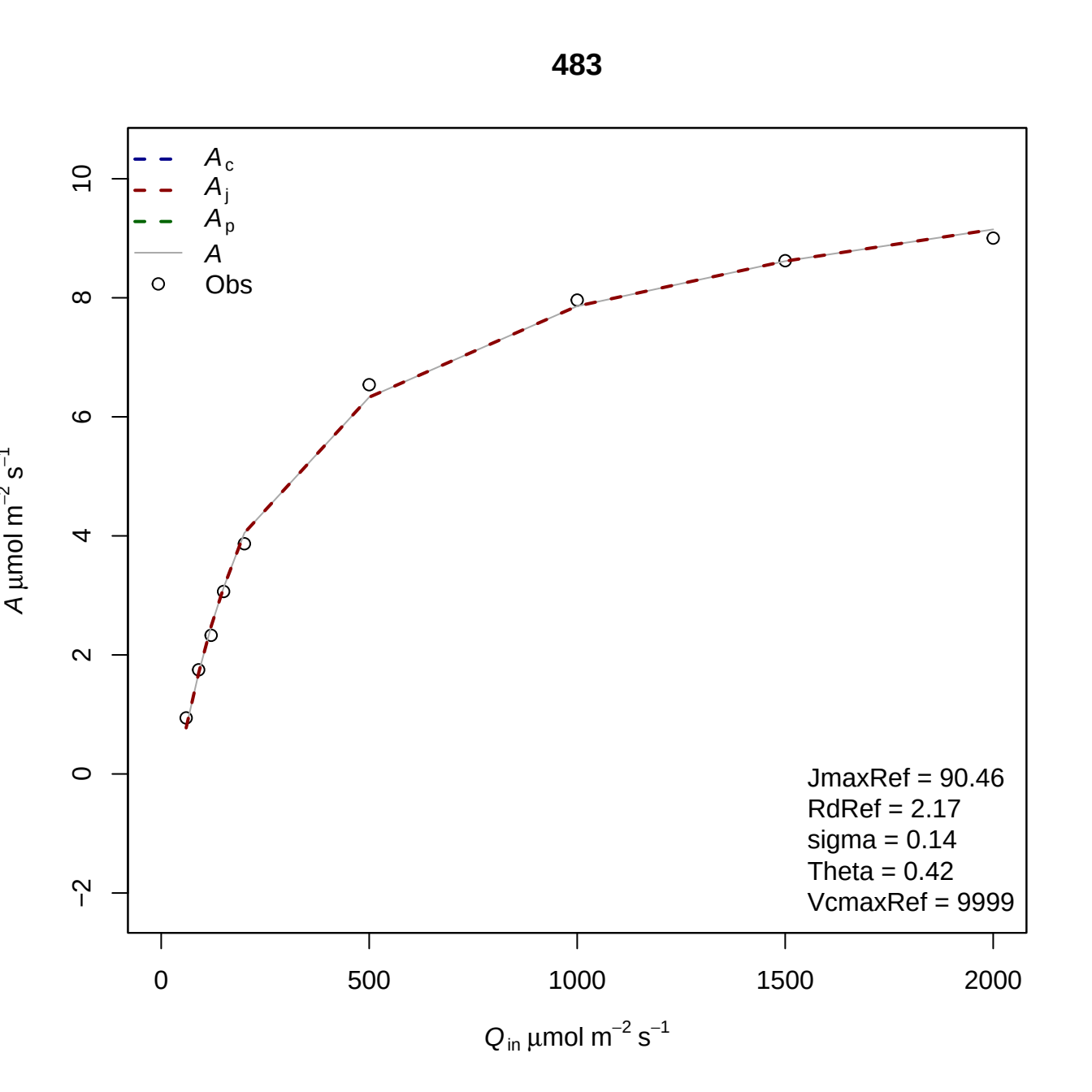

477
A_c
A_j
A_p
A
Obs
JmaxRef = 121.75
RdRef = 0.8
sigma = 0.13
Theta = 0.37
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

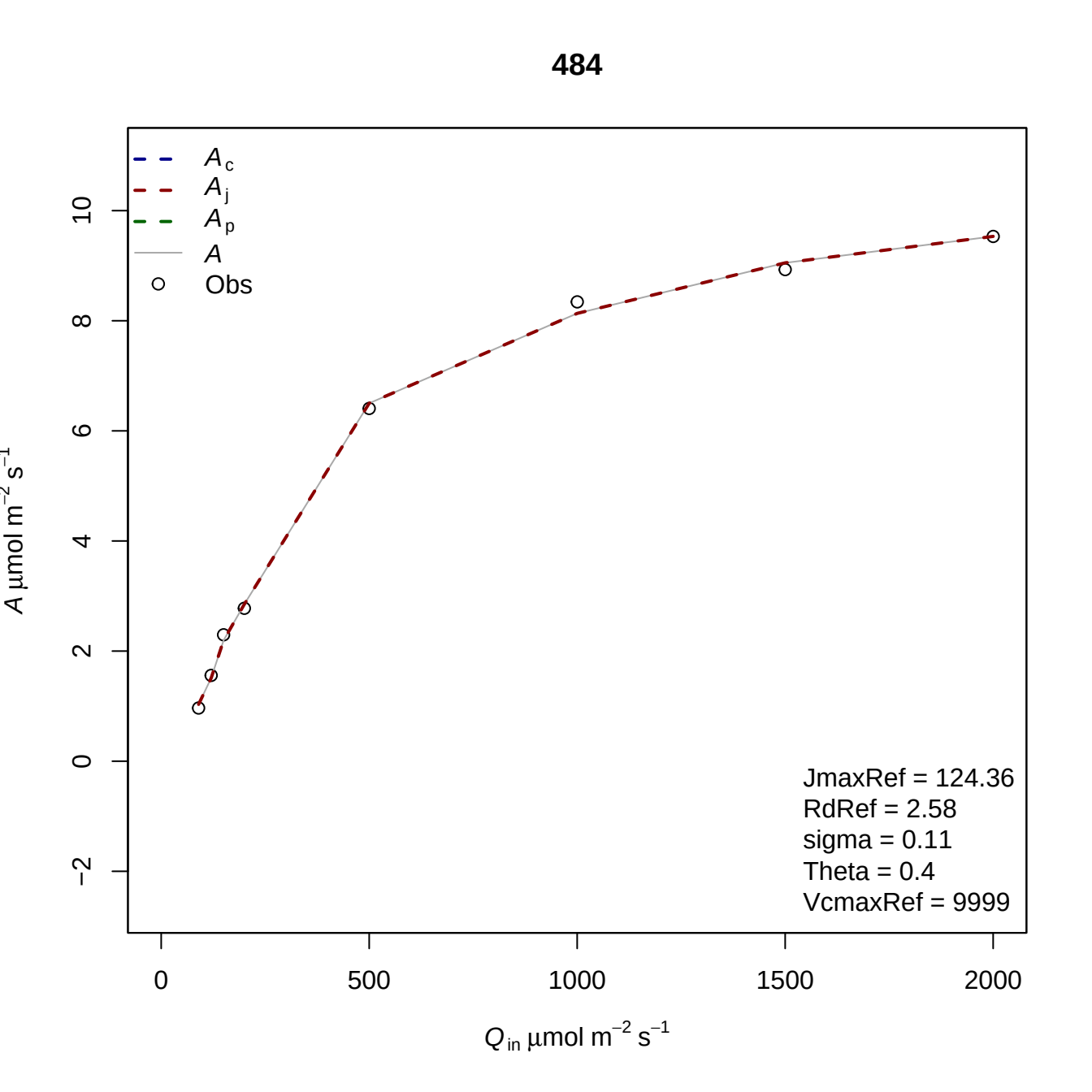

478
A_c
A_j
A_p
A
Obs
JmaxRef = 129.11
RdRef = 1.24
sigma = 0.11
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

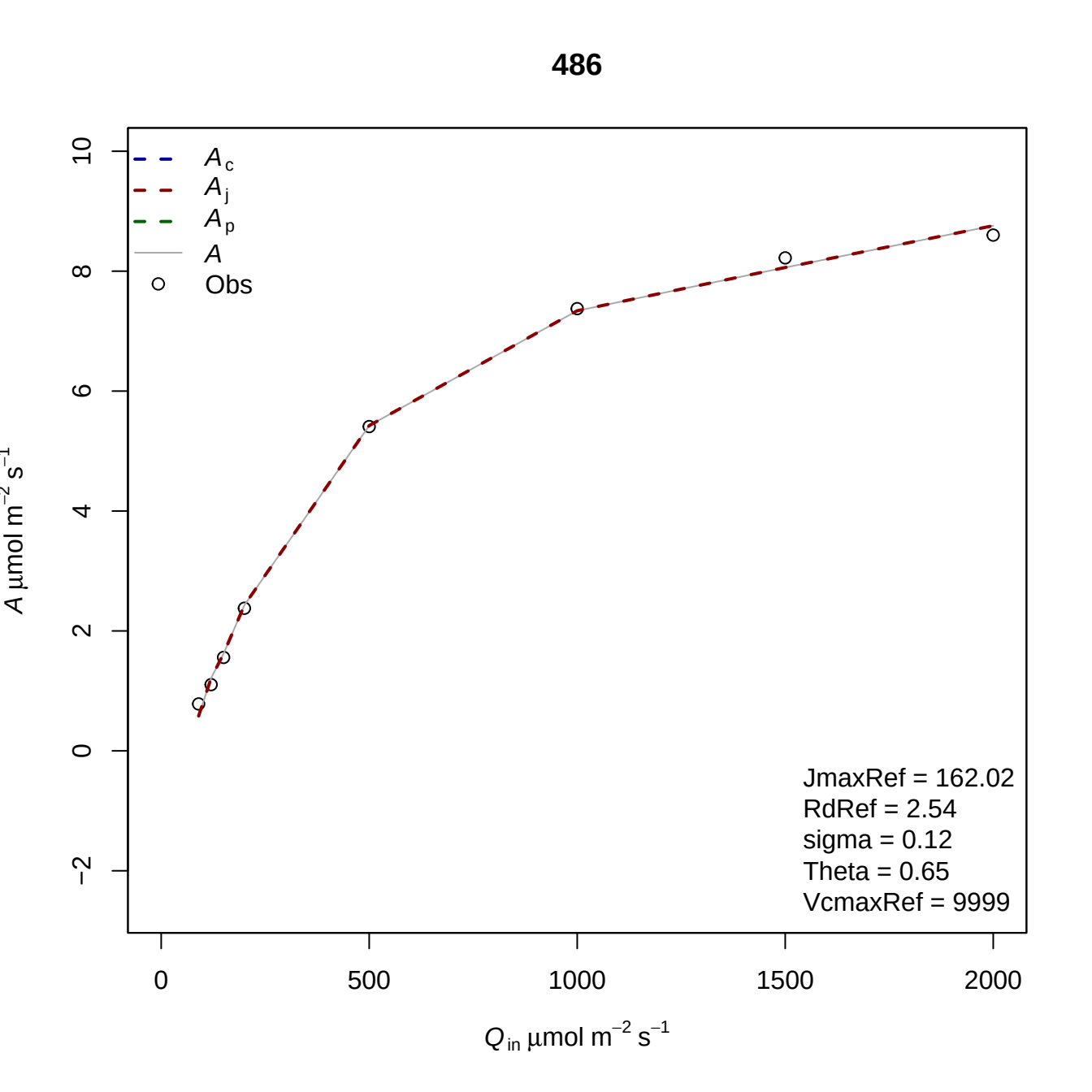

480
A_c
A_j
A_p
A
Obs
JmaxRef = 210.41
RdRef = 3.86
sigma = 0.06
Theta = −0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

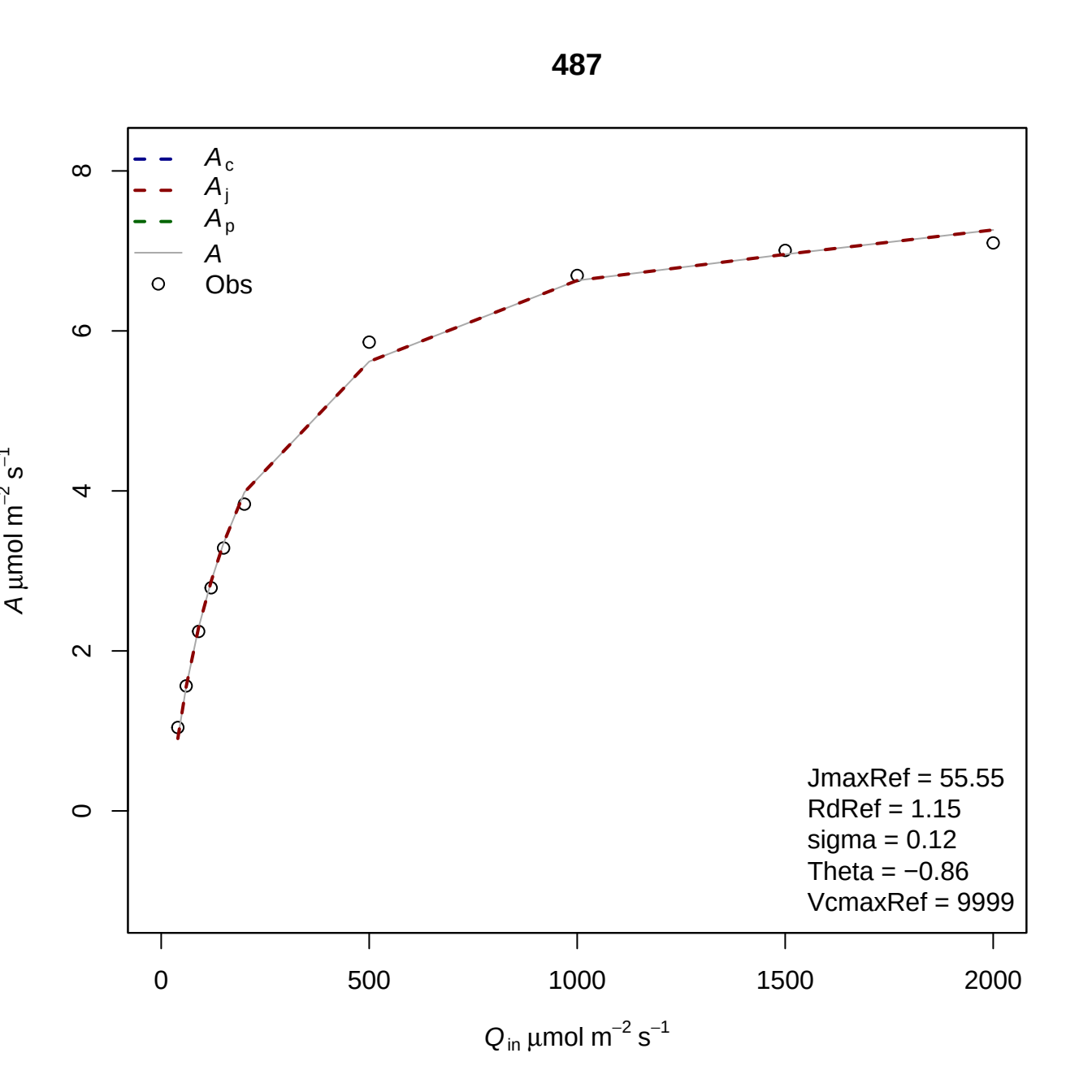

481
A_c
A_j
A_p
A
Obs
JmaxRef = 67.62
RdRef = 1.21
sigma = 0.23
Theta = 0.14
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

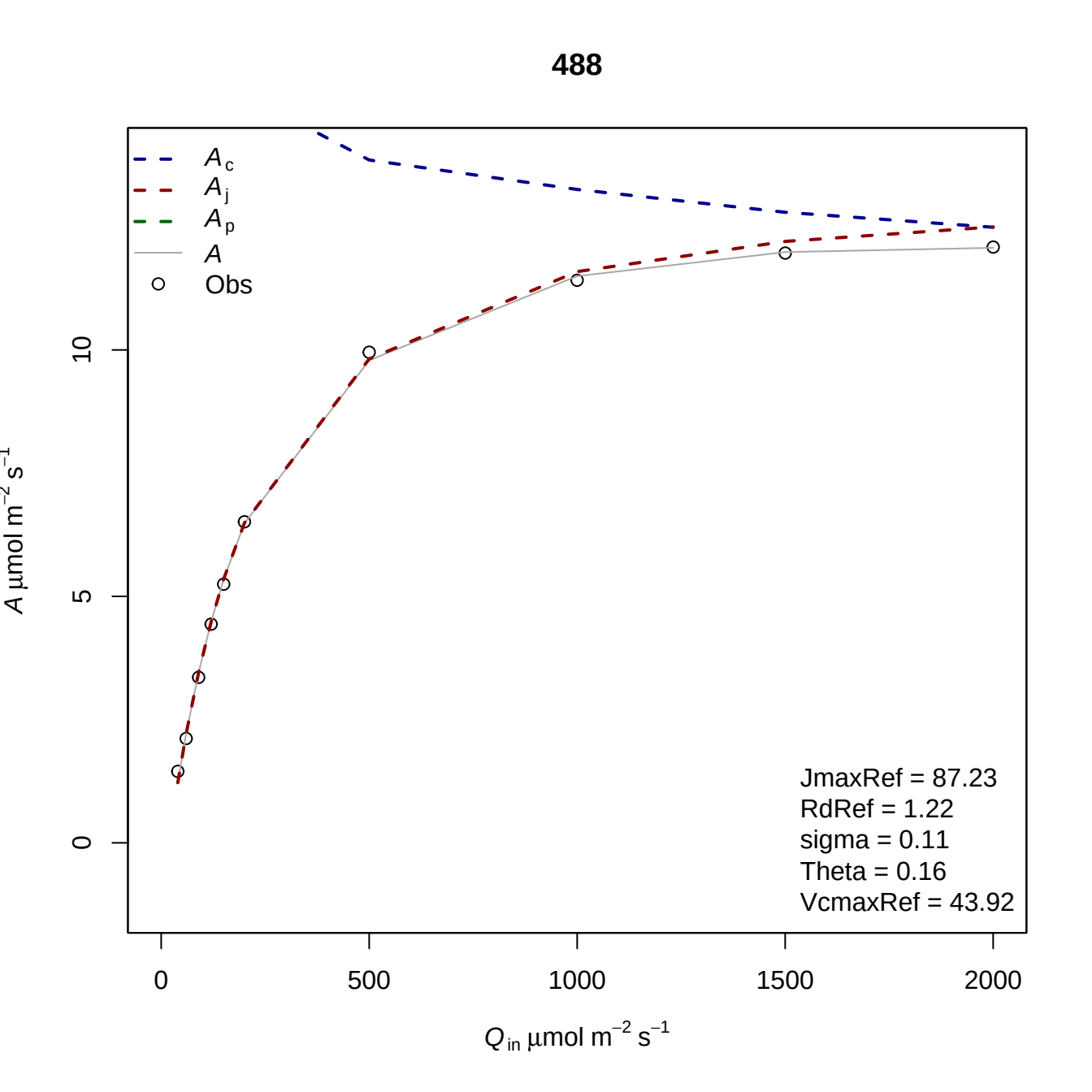

482
A_c
A_j
A_p
A
Obs
JmaxRef = 109.69
RdRef = 2.21
sigma = 0.46
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

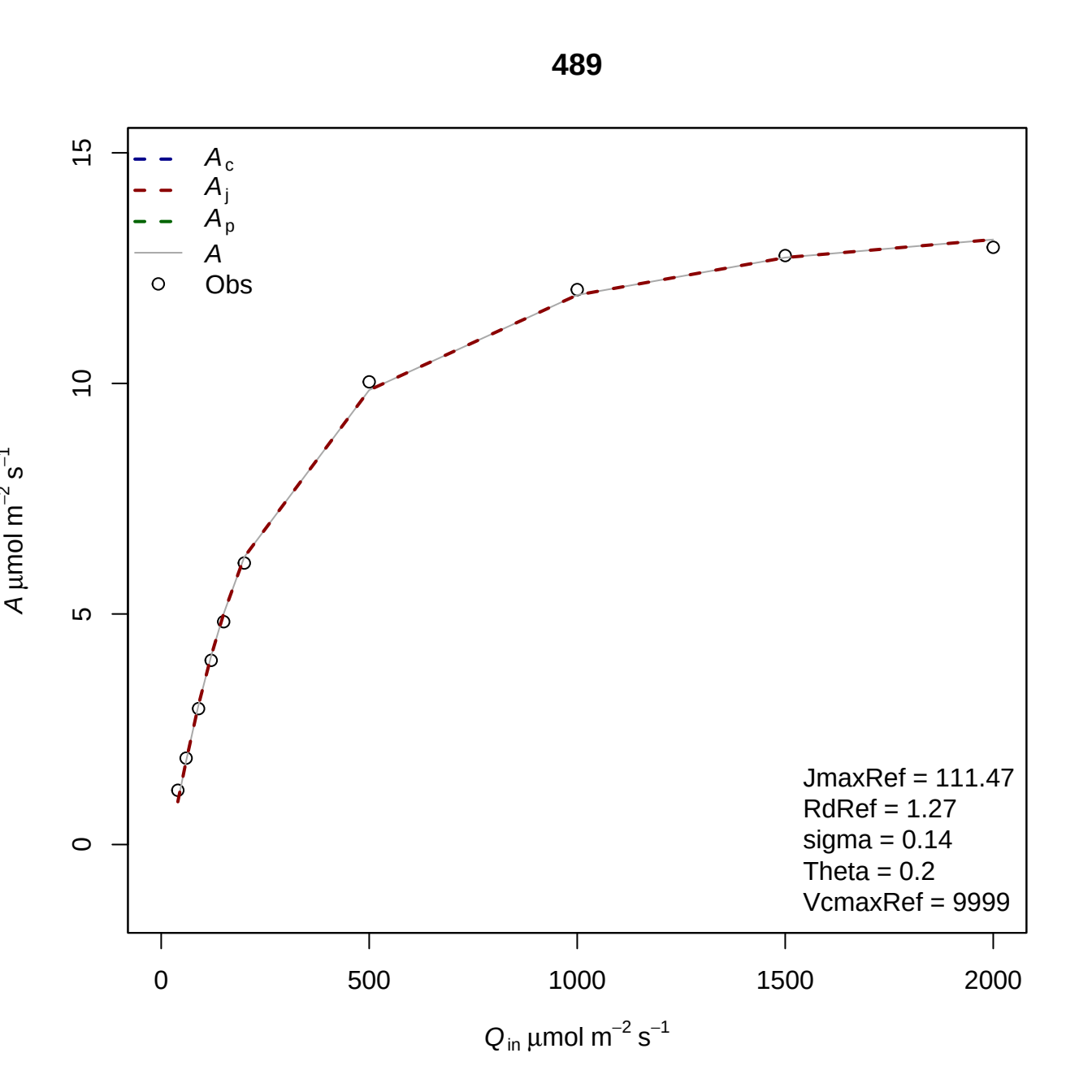

483
A_c
A_j
A_p
A
Obs
JmaxRef = 90.46
RdRef = 2.17
sigma = 0.14
Theta = 0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

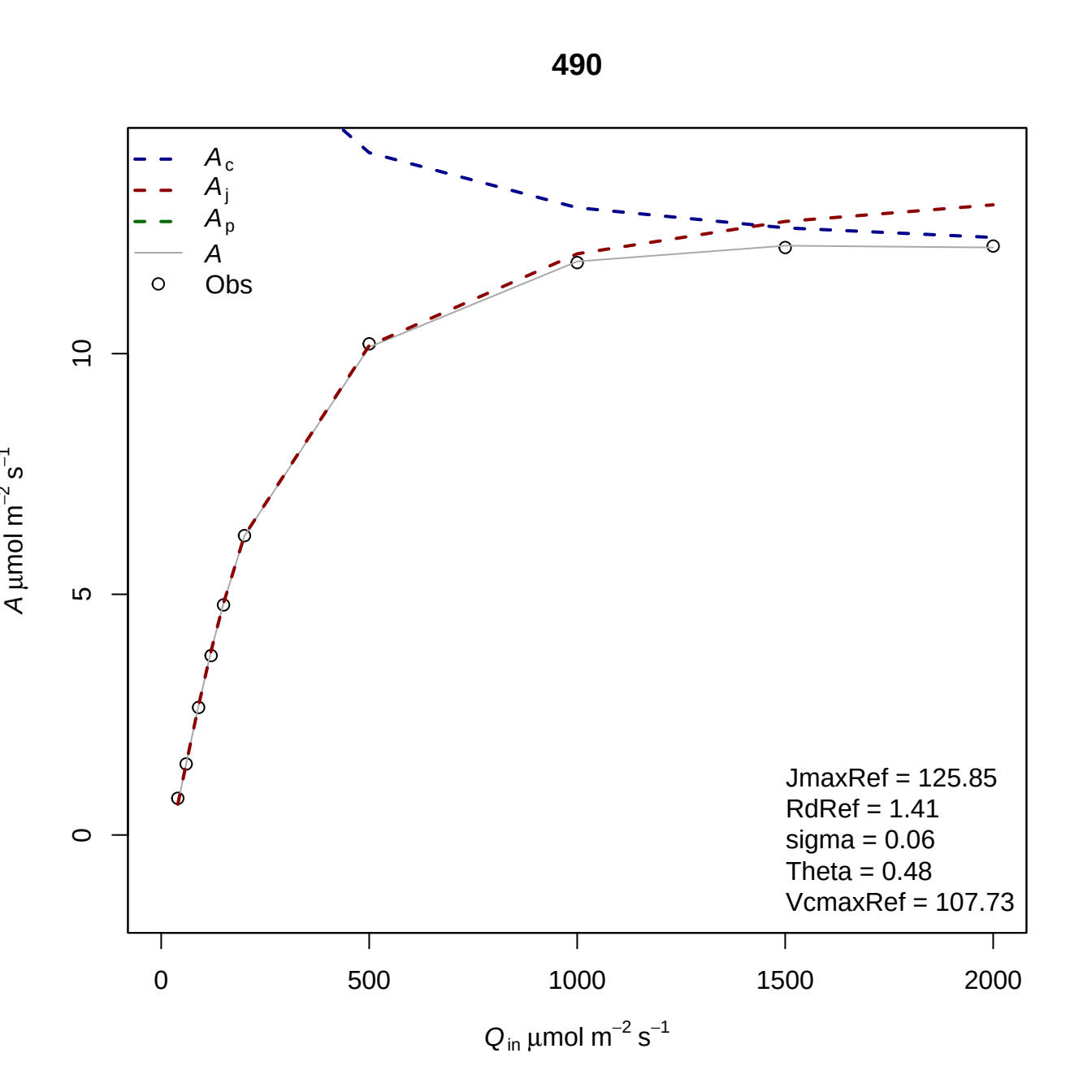

484
A_c
A_j
A_p
A
Obs
JmaxRef = 124.36
RdRef = 2.58
sigma = 0.11
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

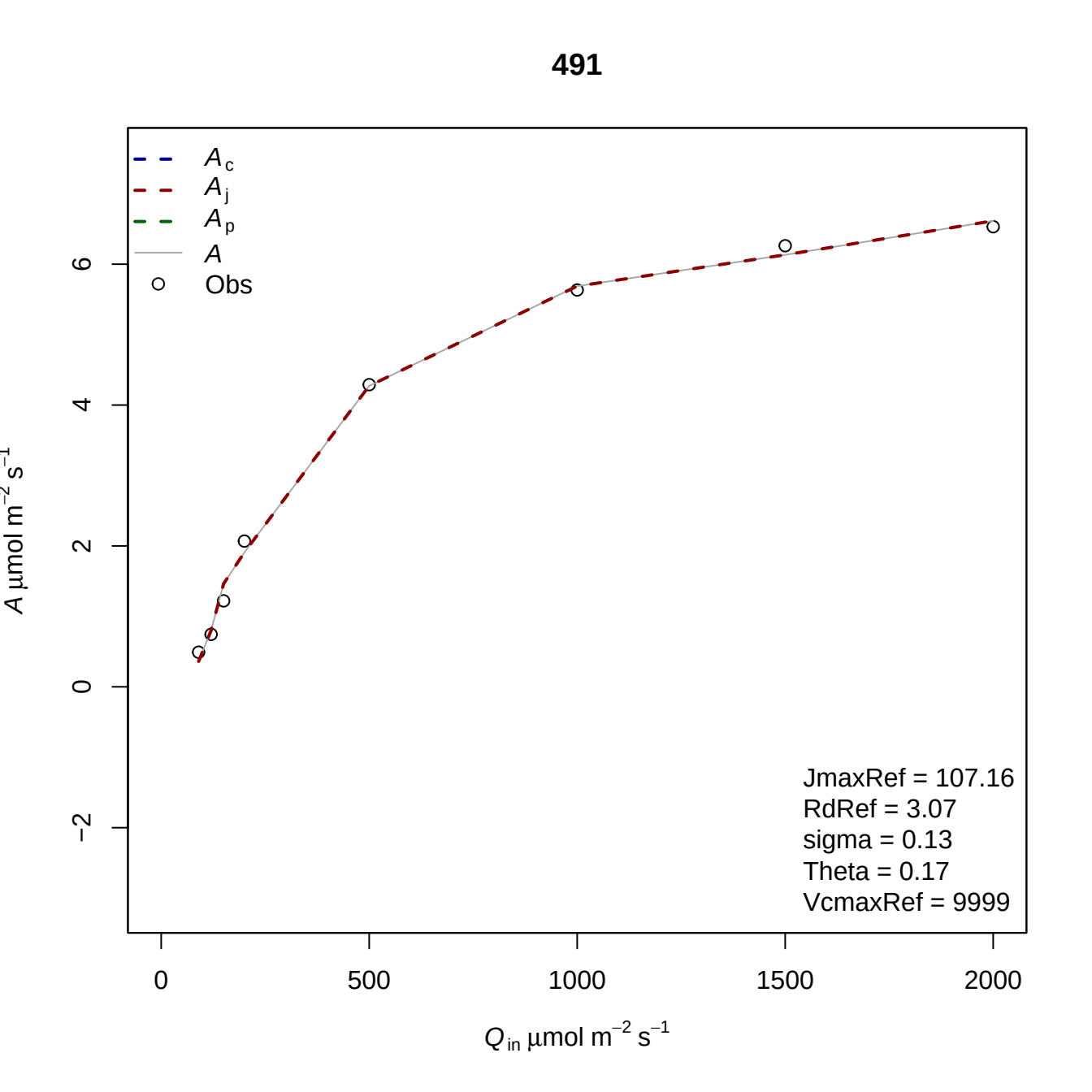

486
A_c
A_j
A_p
A
Obs
JmaxRef = 162.02
RdRef = 2.54
sigma = 0.12
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

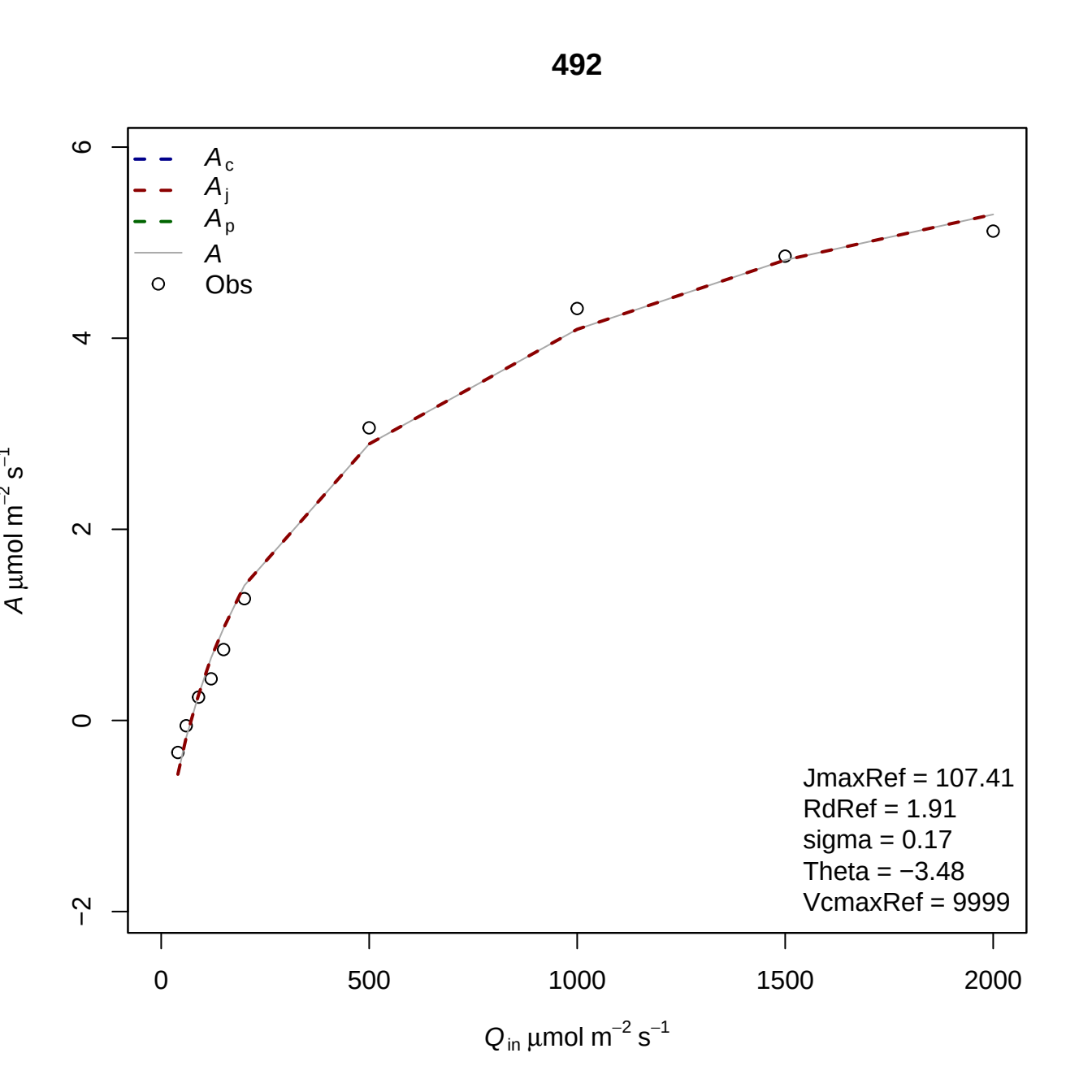

487
A_c
A_j
A_p
A
Obs
JmaxRef = 55.55
RdRef = 1.15
sigma = 0.12
Theta = −0.86
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

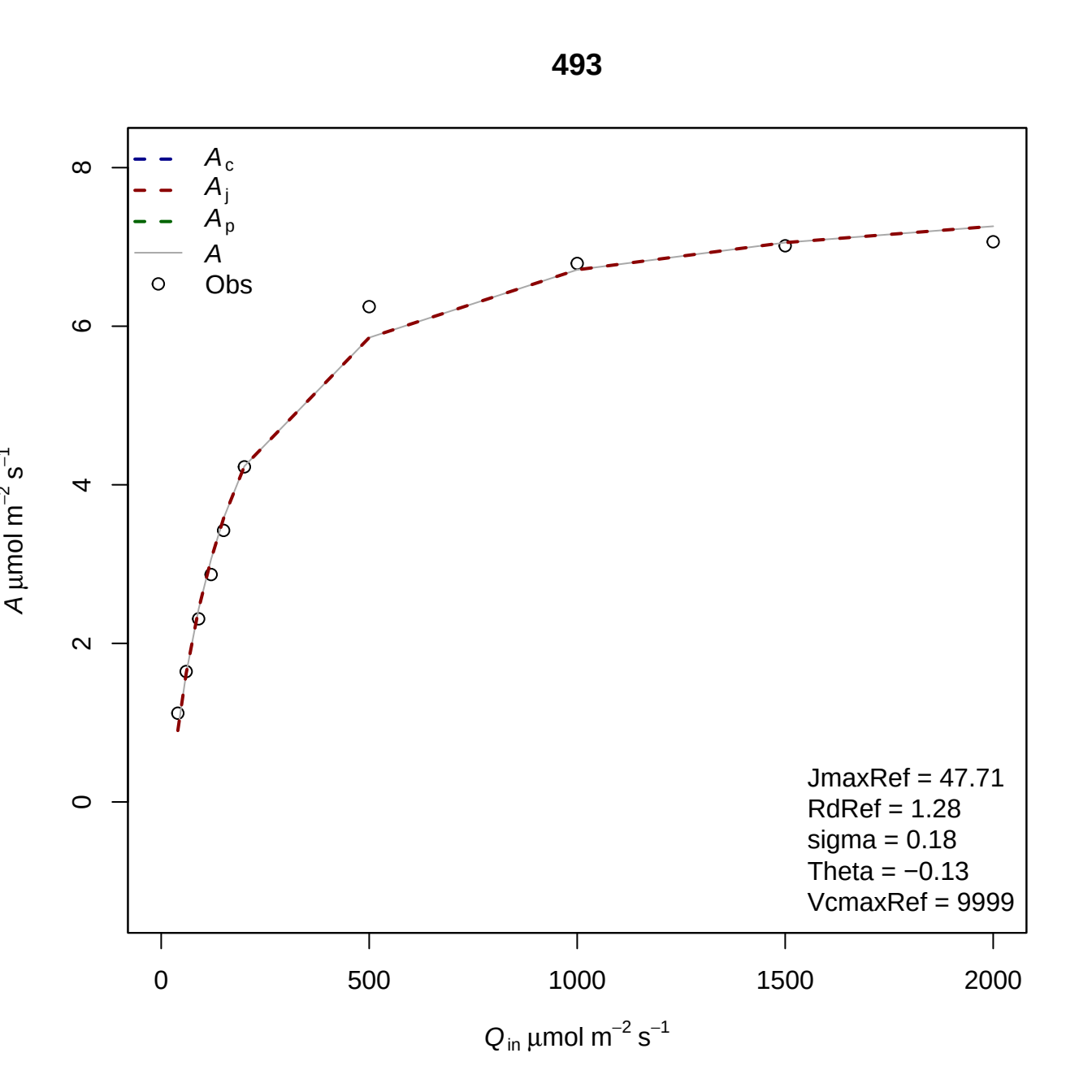

488
A_c
A_j
A_p
A
Obs
JmaxRef = 87.23
RdRef = 1.22
sigma = 0.11
Theta = 0.16
VcmaxRef = 43.92
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

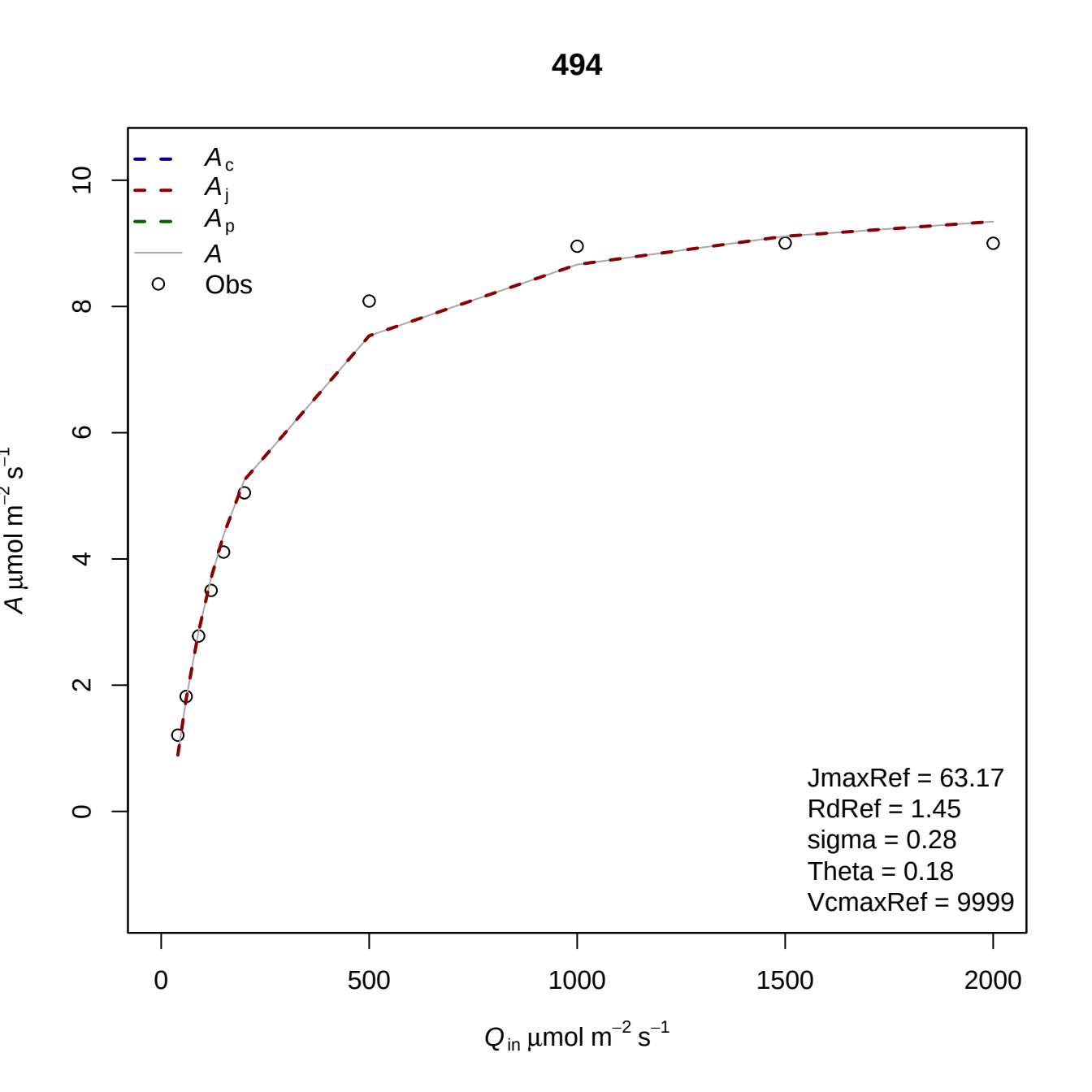

489
A_c
A_j
A_p
A
Obs
JmaxRef = 111.47
RdRef = 1.27
sigma = 0.14
Theta = 0.2
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

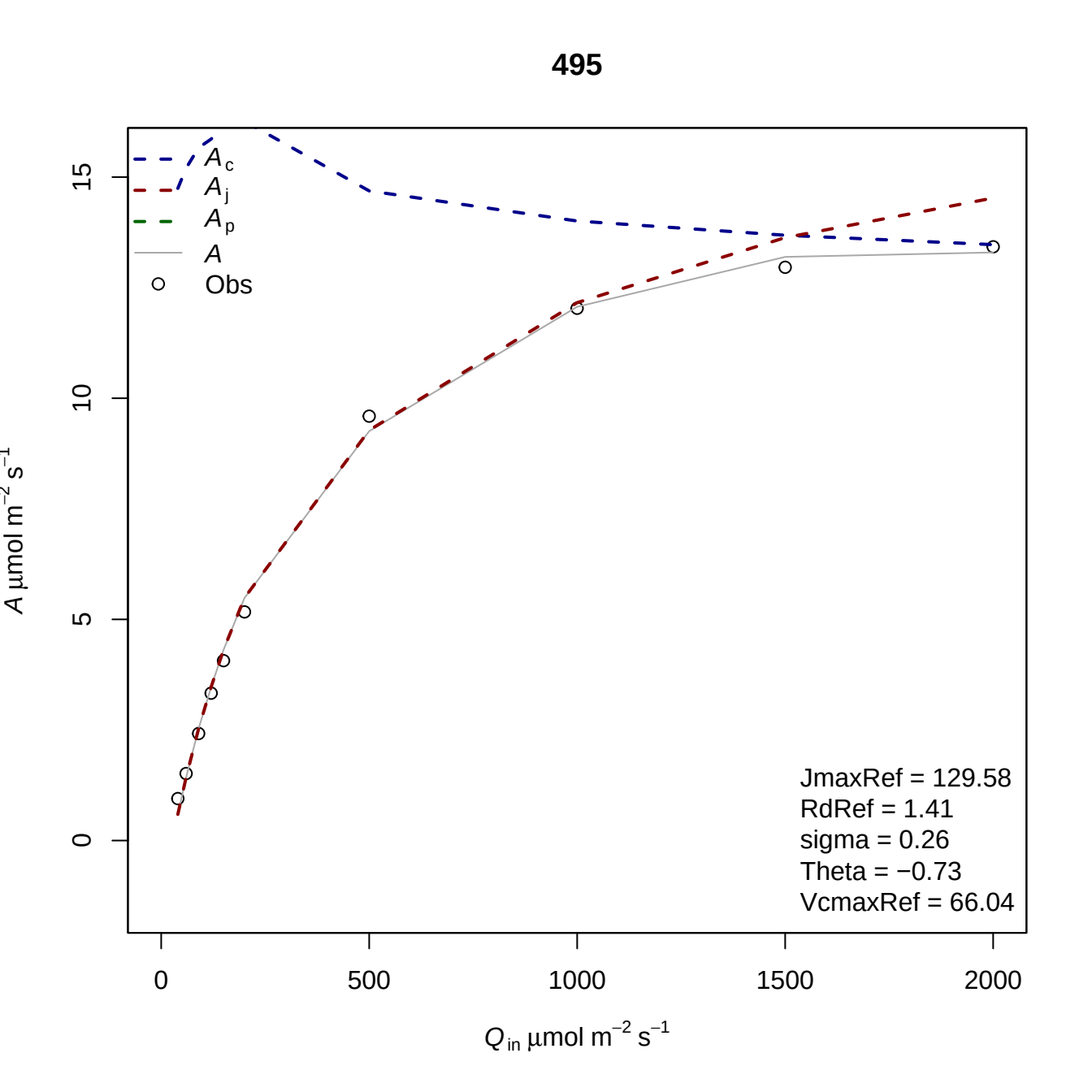

490
A_c
A_j
A_p
A
Obs
JmaxRef = 125.85
RdRef = 1.41
sigma = 0.06
Theta = 0.48
VcmaxRef = 107.73
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

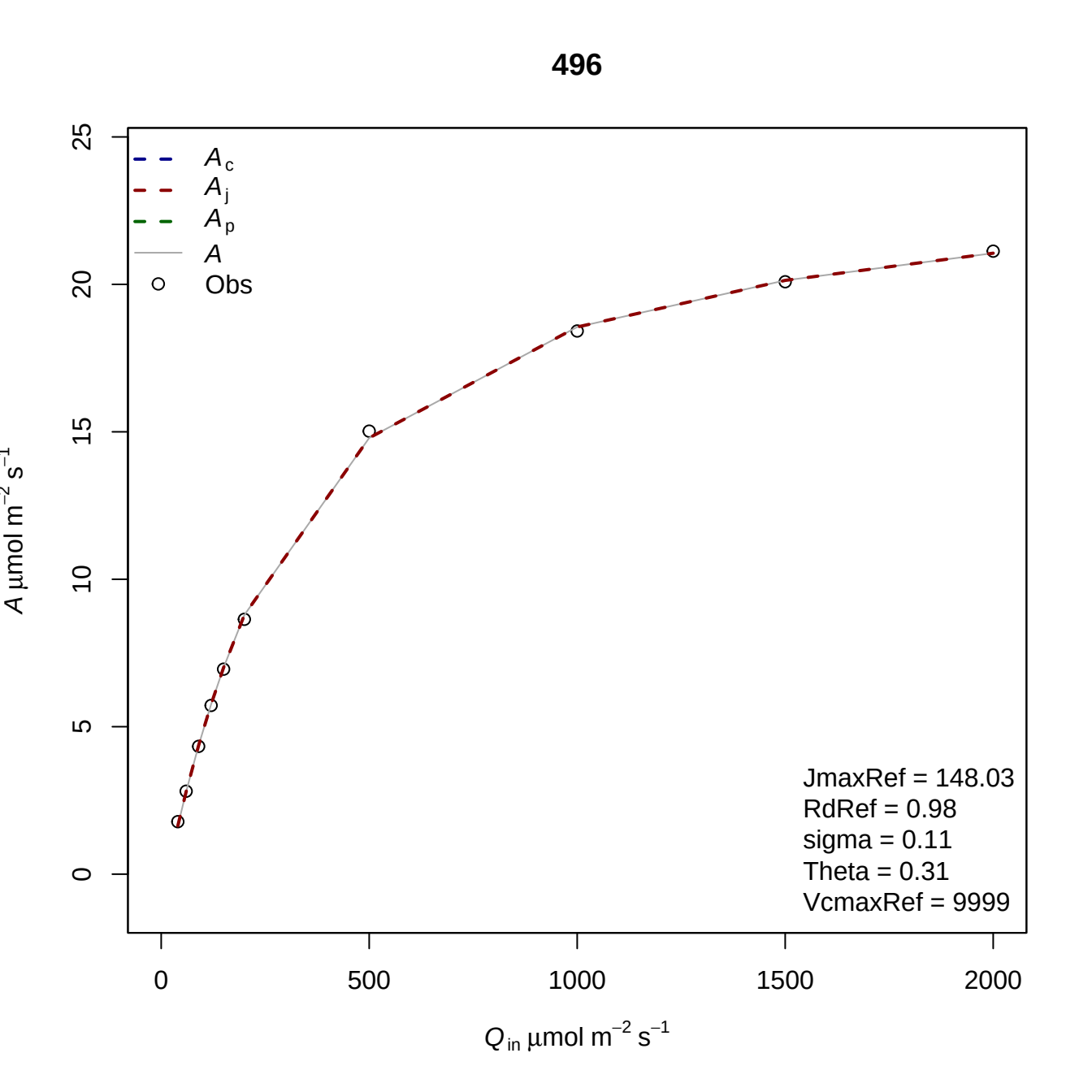

491
A_c
A_j
A_p
A
Obs
JmaxRef = 107.16
RdRef = 3.07
sigma = 0.13
Theta = 0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

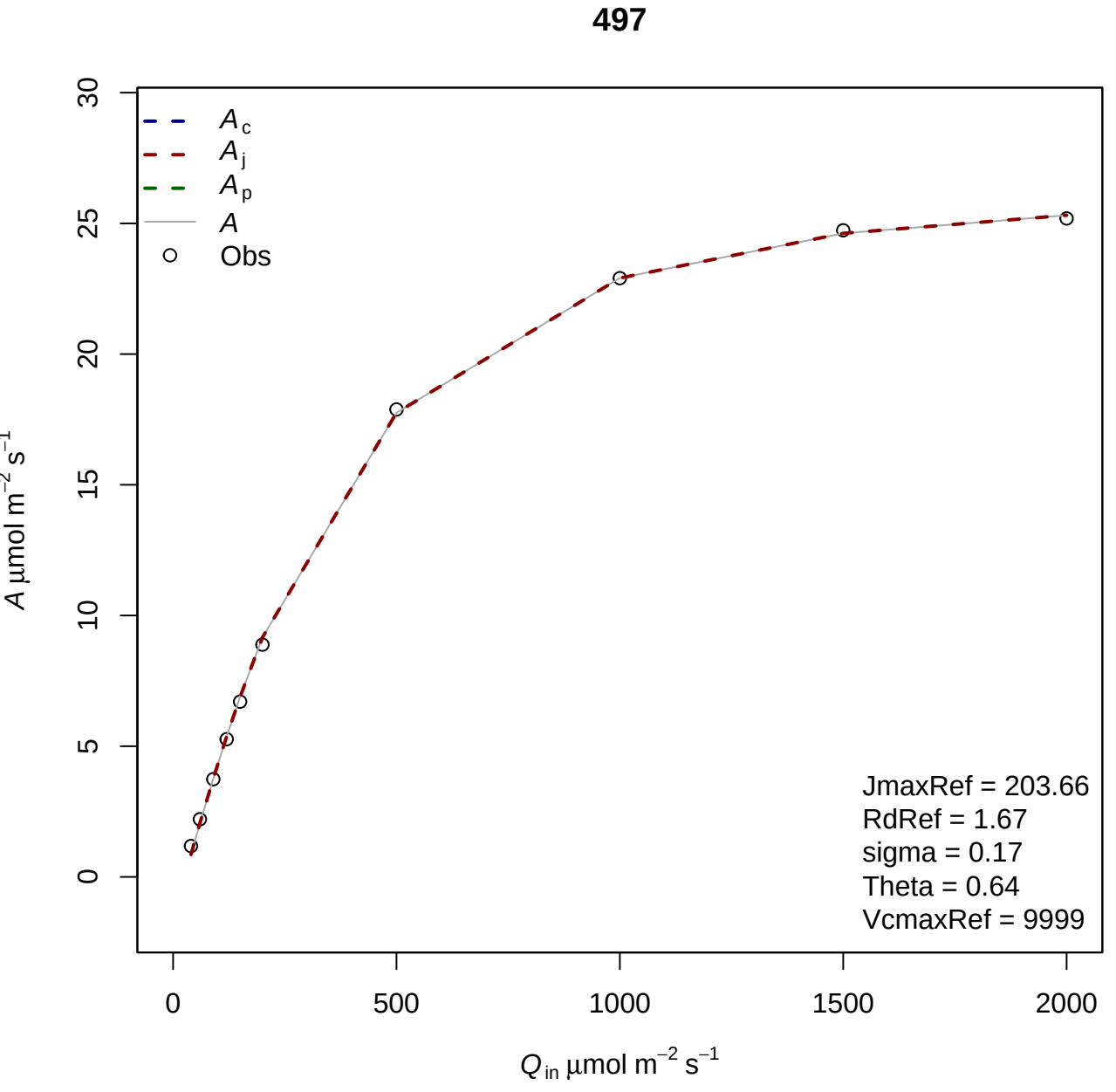

492
A_c
A_j
A_p
A
Obs
JmaxRef = 107.41
RdRef = 1.91
sigma = 0.17
Theta = −3.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

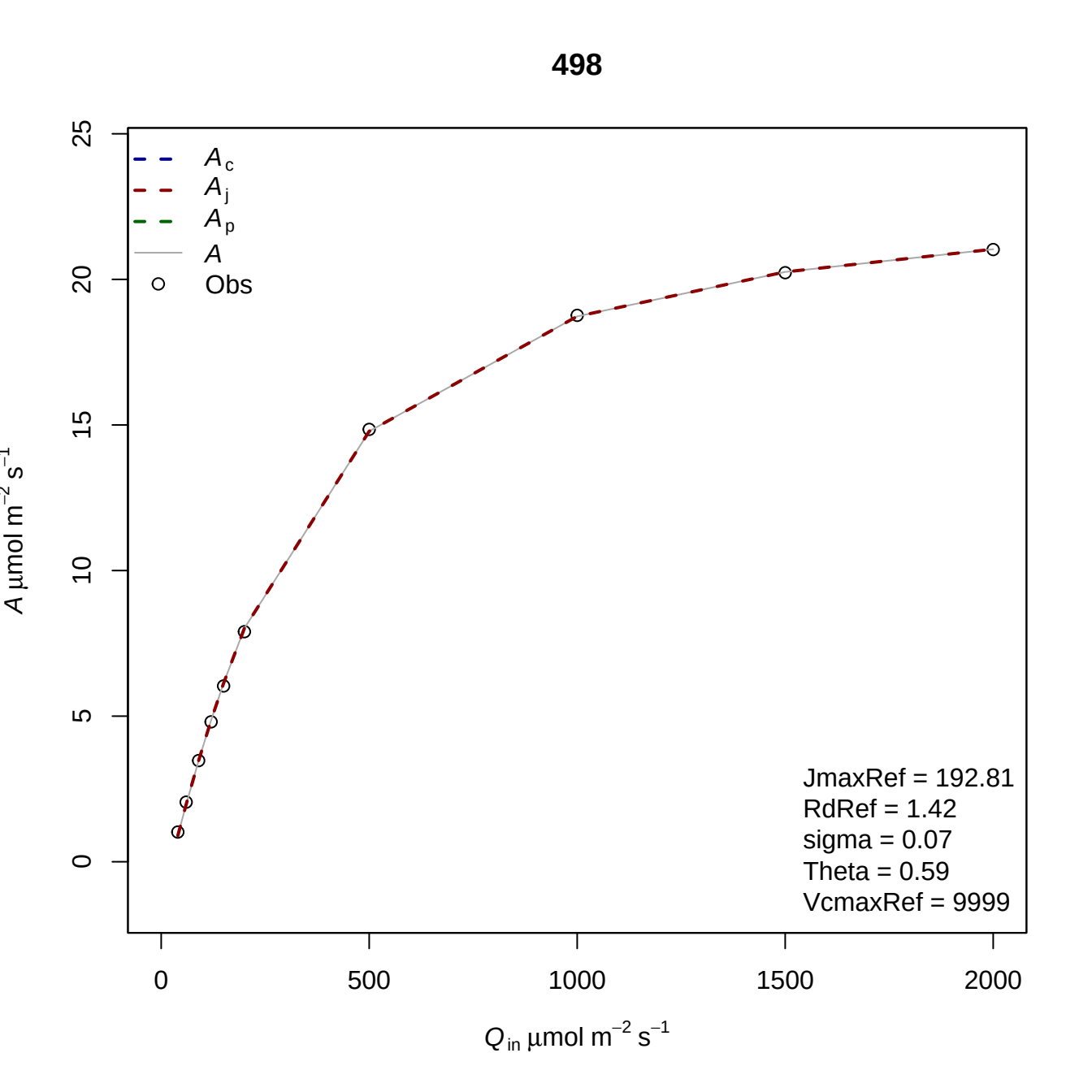

493
A_c
A_j
A_p
A
Obs
JmaxRef = 47.71
RdRef = 1.28
sigma = 0.18
Theta = −0.13
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

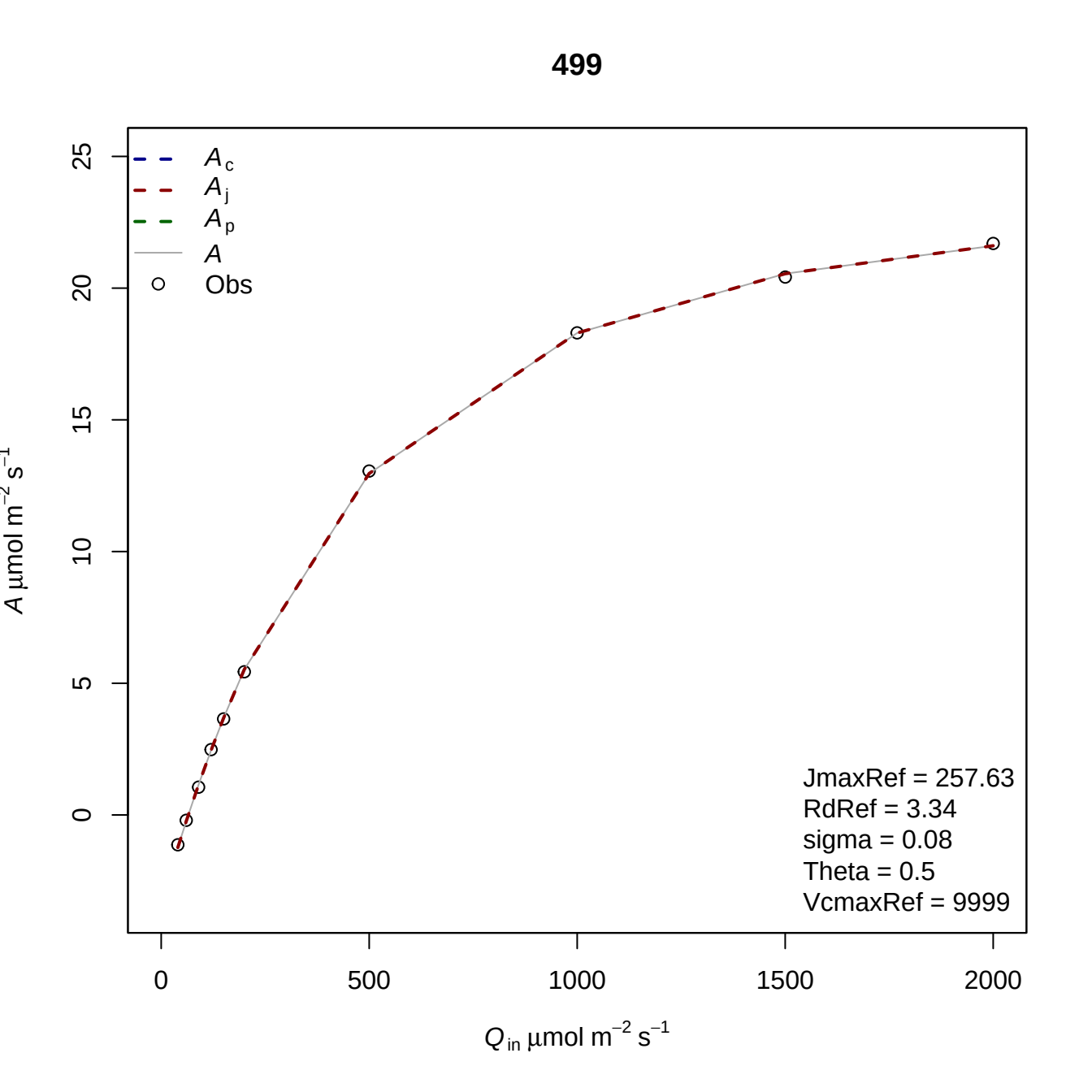

494
A_c
A_j
A_p
A
Obs
JmaxRef = 63.17
RdRef = 1.45
sigma = 0.28
Theta = 0.18
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

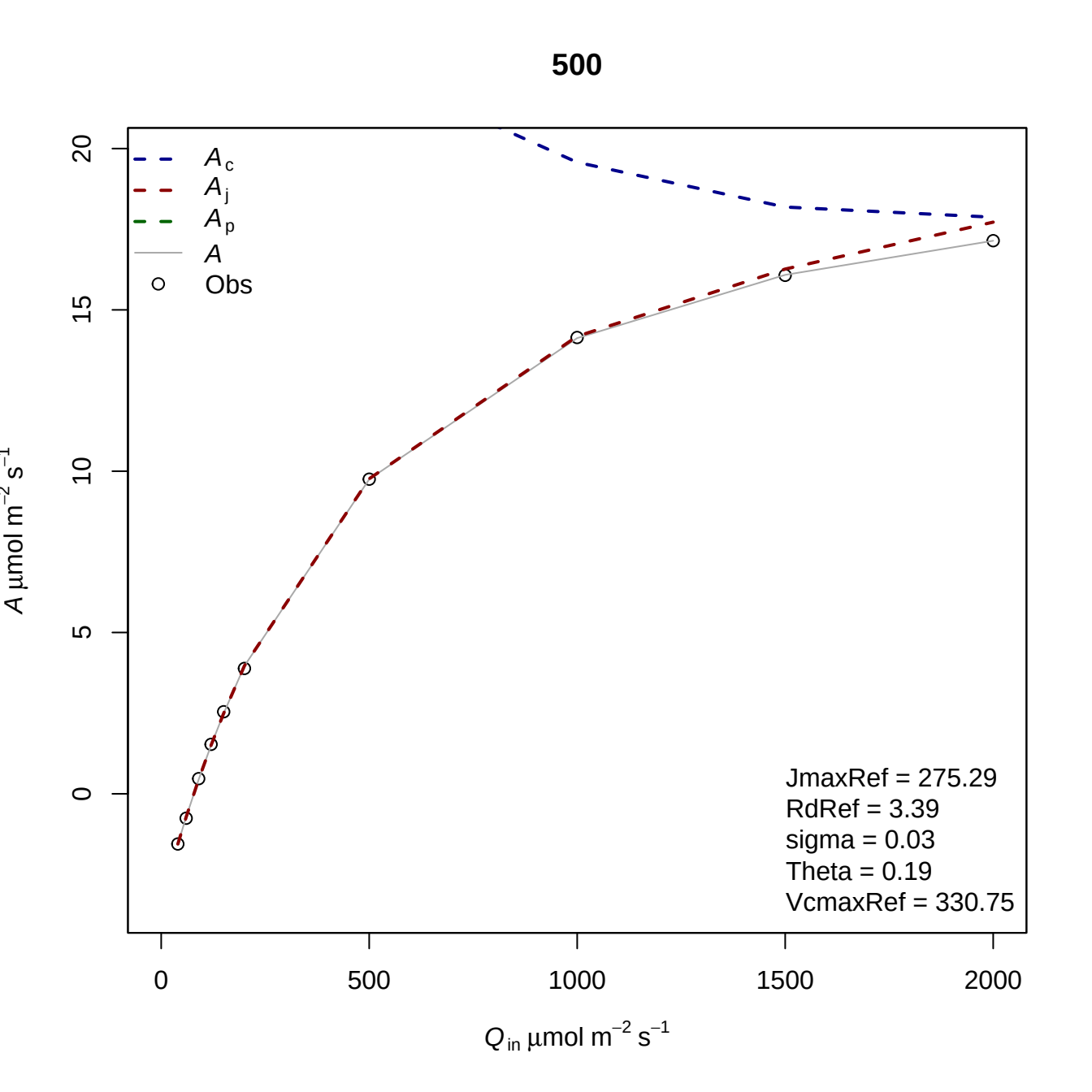

495
A_c
A_j
A_p
A
Obs
JmaxRef = 129.58
RdRef = 1.41
sigma = 0.26
Theta = −0.73
VcmaxRef = 66.04
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

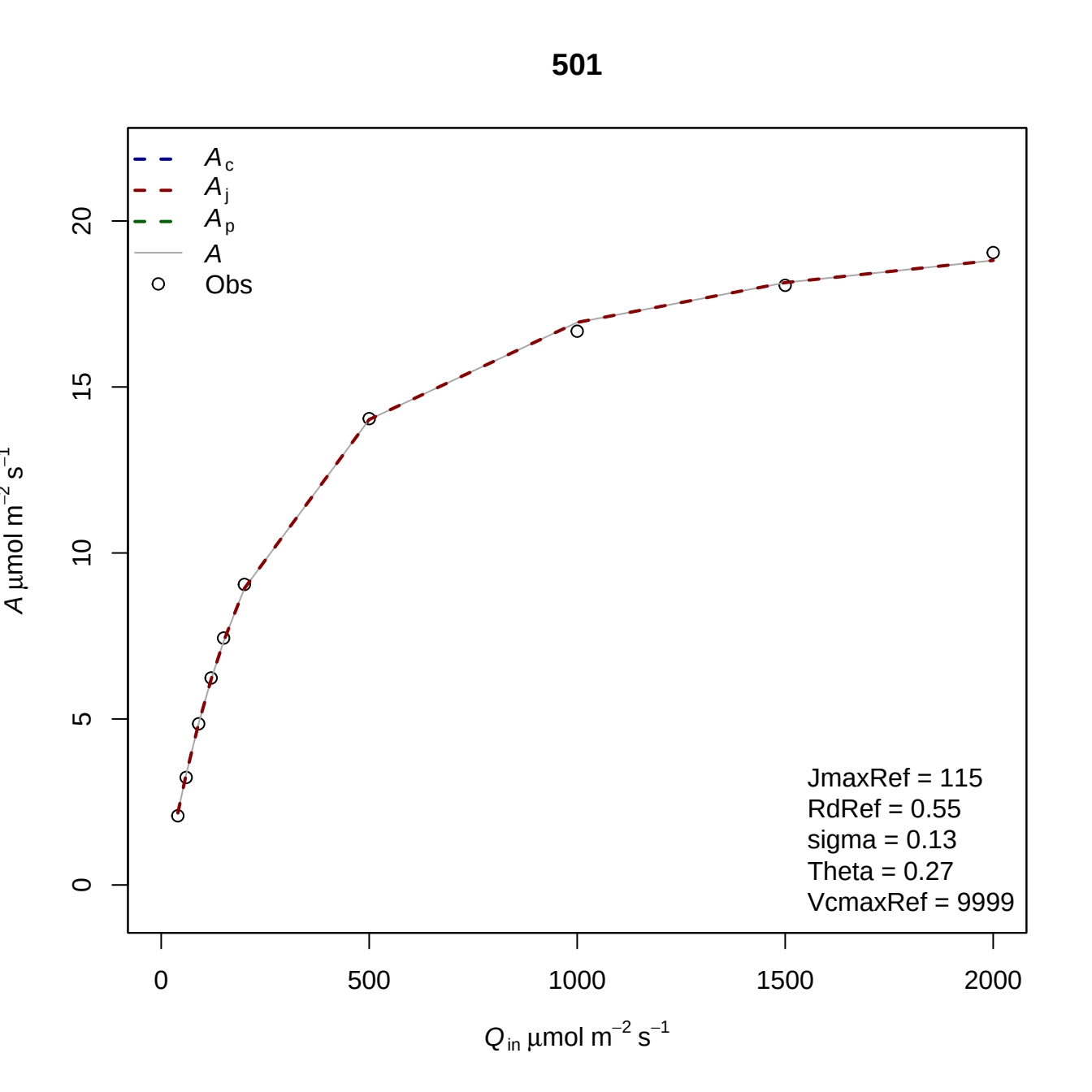

496
A_c
A_j
A_p
A
Obs
JmaxRef = 148.03
RdRef = 0.98
sigma = 0.11
Theta = 0.31
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

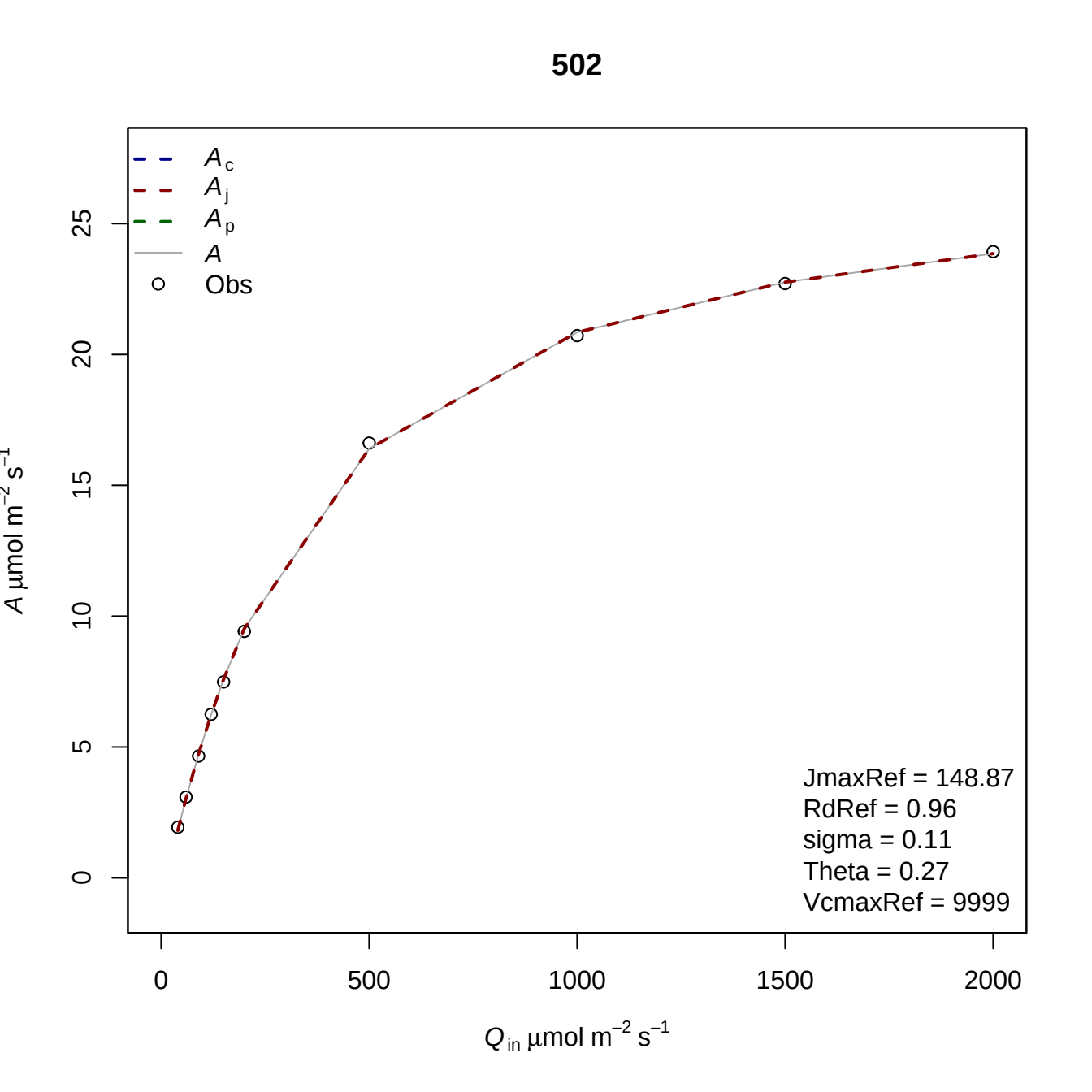

497
A_c
A_j
A_p
A
Obs
JmaxRef = 203.66
RdRef = 1.67
sigma = 0.17
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

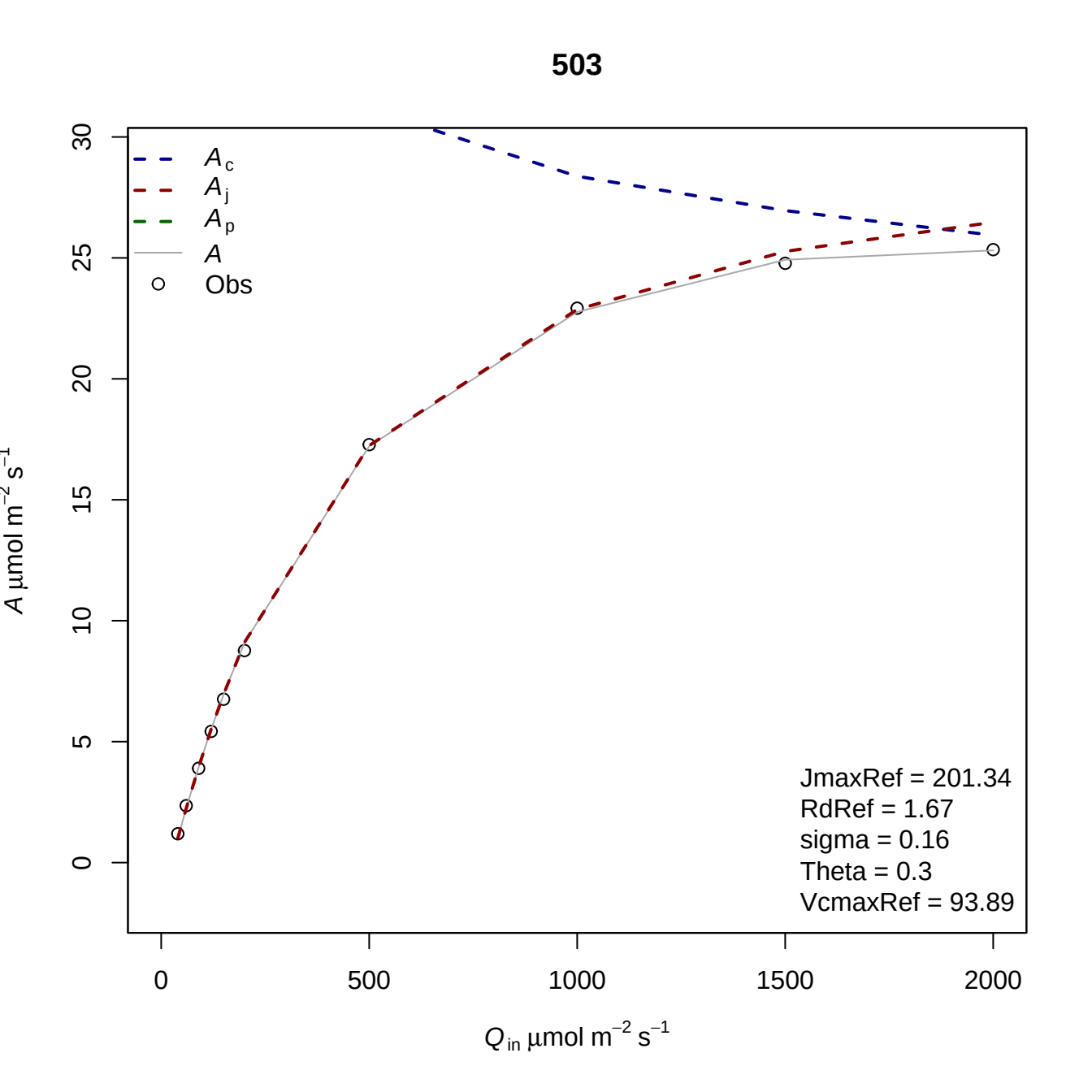

498
A_c
A_j
A_p
A
Obs
JmaxRef = 192.81
RdRef = 1.42
sigma = 0.07
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

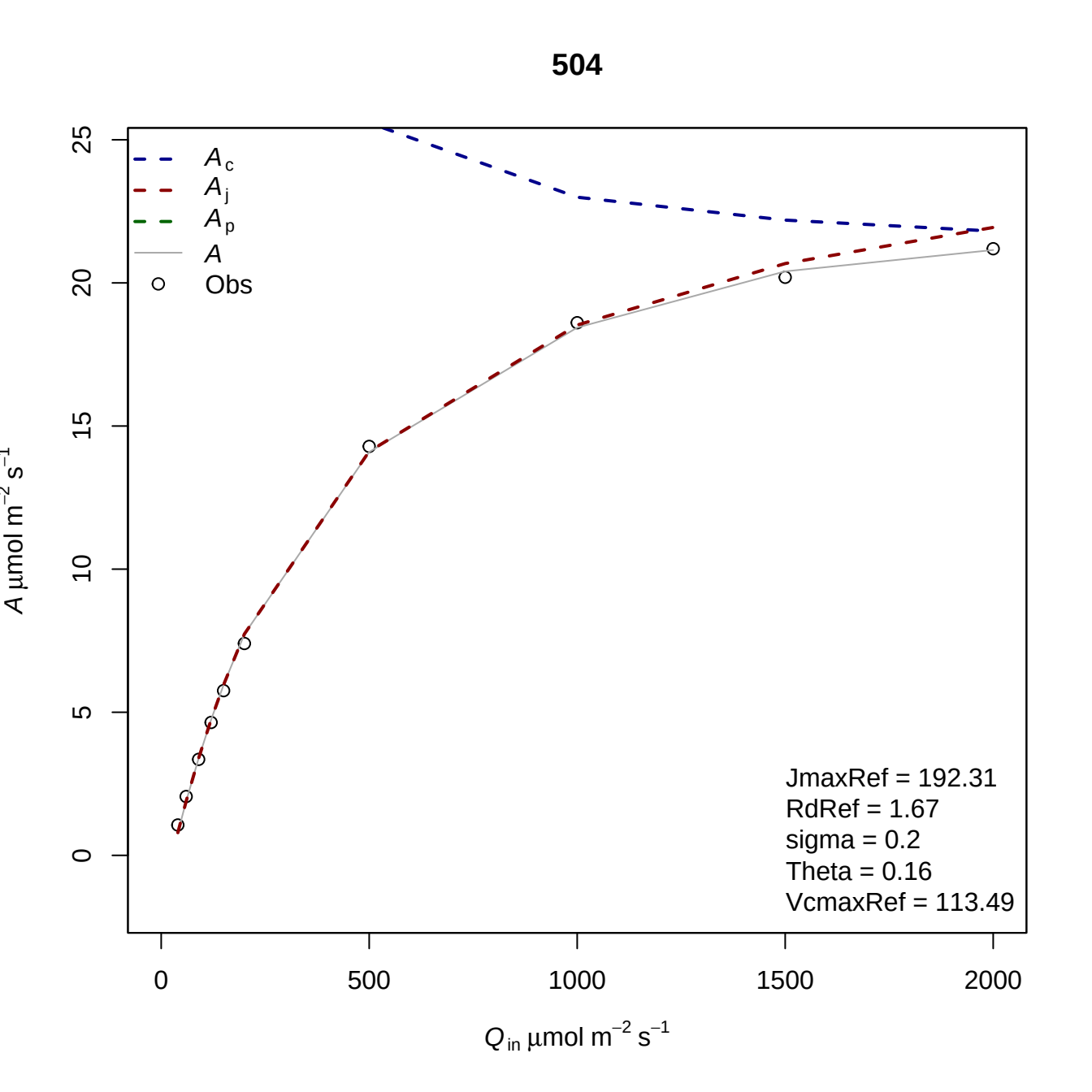

499
A_c
A_j
A_p
A
Obs
JmaxRef = 257.63
RdRef = 3.34
sigma = 0.08
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

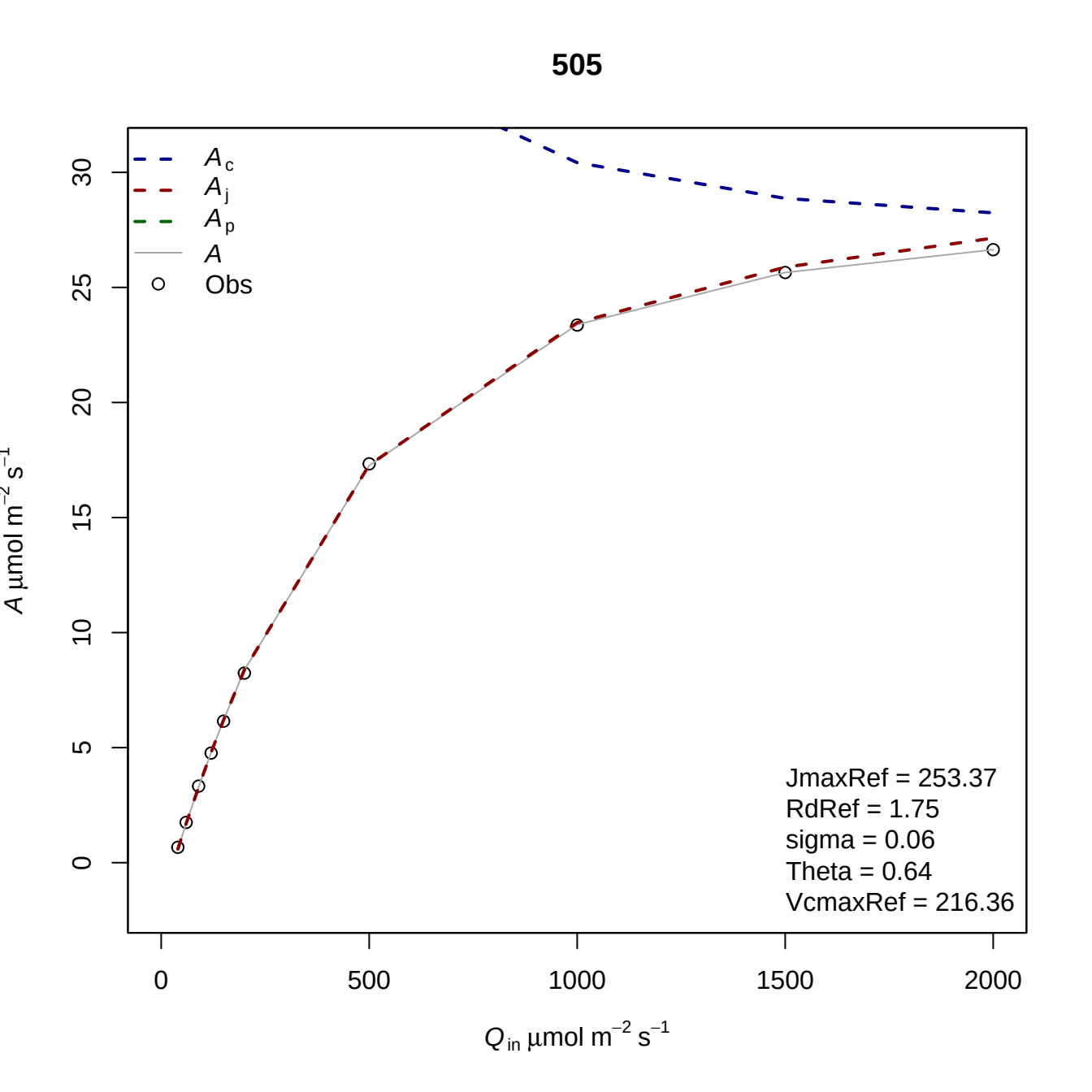

500
A_c
A_j
A_p
A
Obs
JmaxRef = 275.29
RdRef = 3.39
sigma = 0.03
Theta = 0.19
VcmaxRef = 330.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

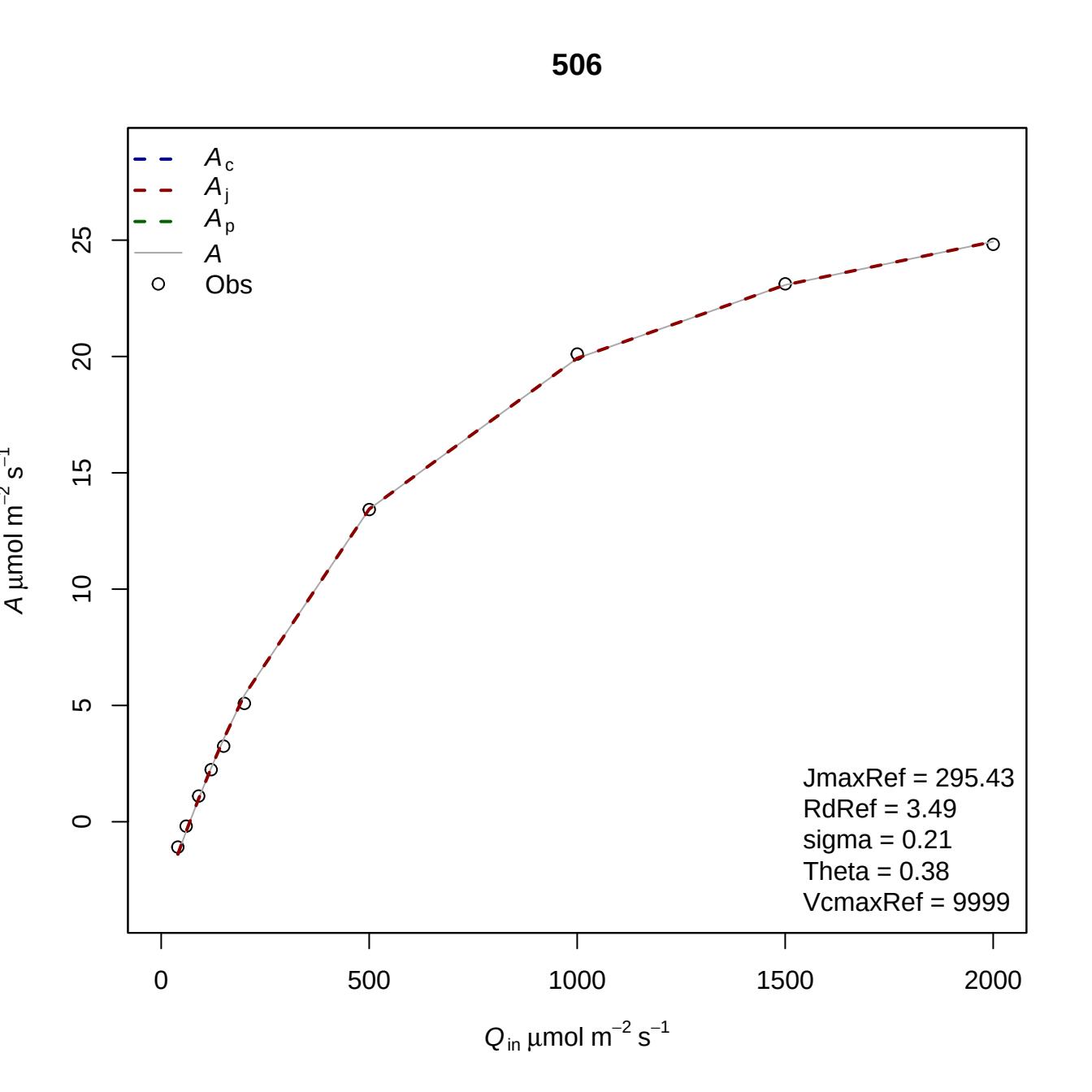

501
A_c
A_j
A_p
A
Obs
JmaxRef = 115
RdRef = 0.55
sigma = 0.13
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

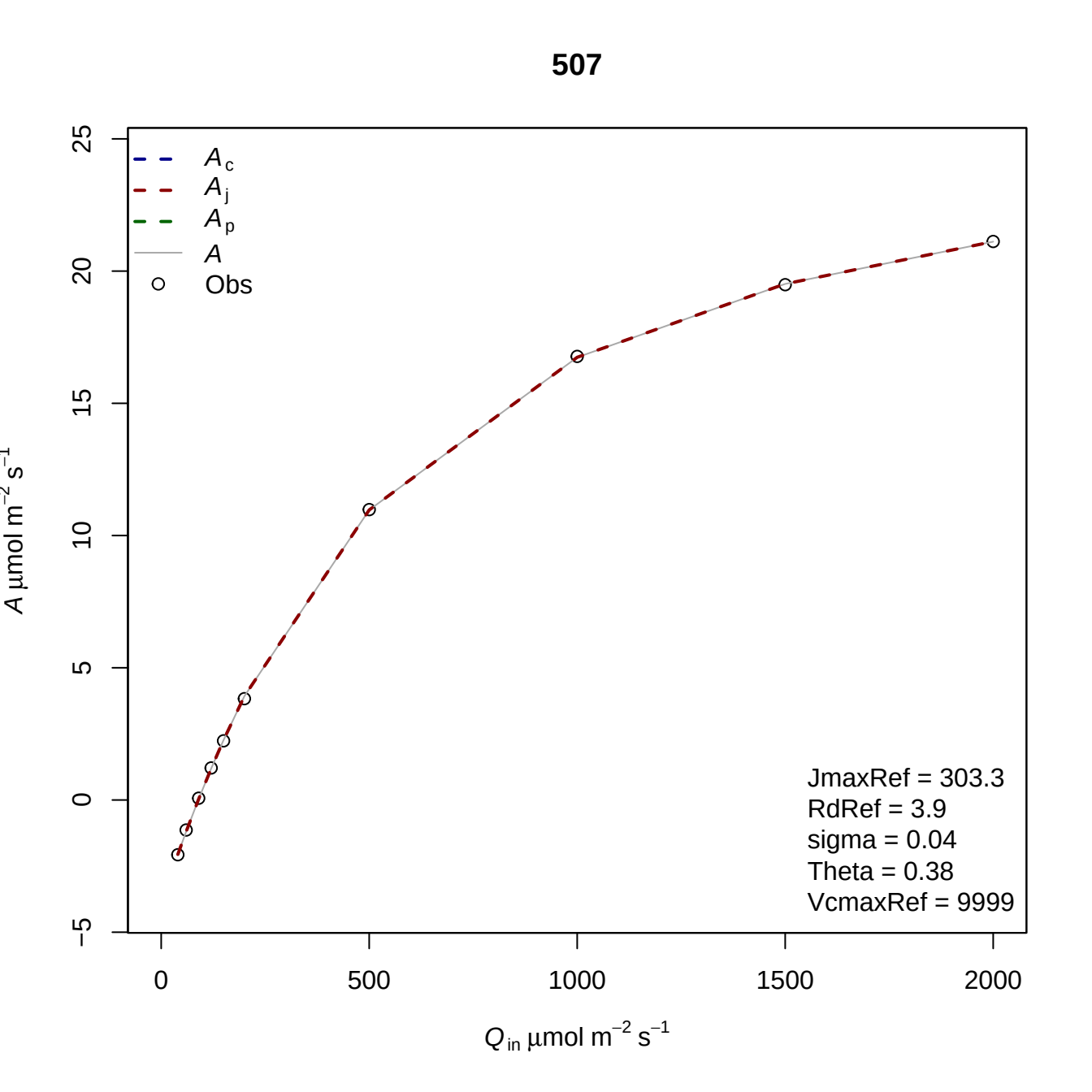

502
A_c
A_j
A_p
A
Obs
JmaxRef = 148.87
RdRef = 0.96
sigma = 0.11
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

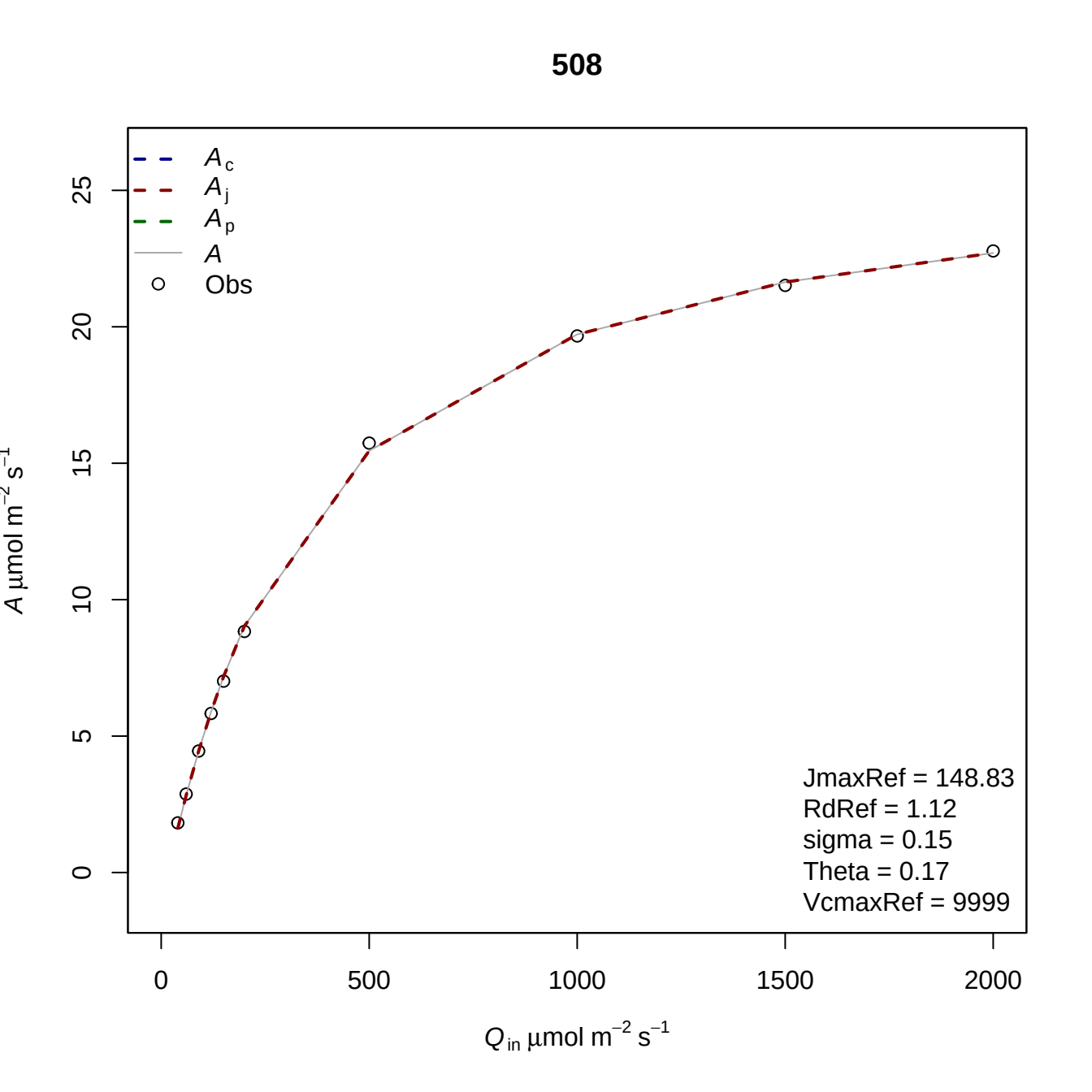

503
A_c
A_j
A_p
A
Obs
JmaxRef = 201.34
RdRef = 1.67
sigma = 0.16
Theta = 0.3
VcmaxRef = 93.89
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

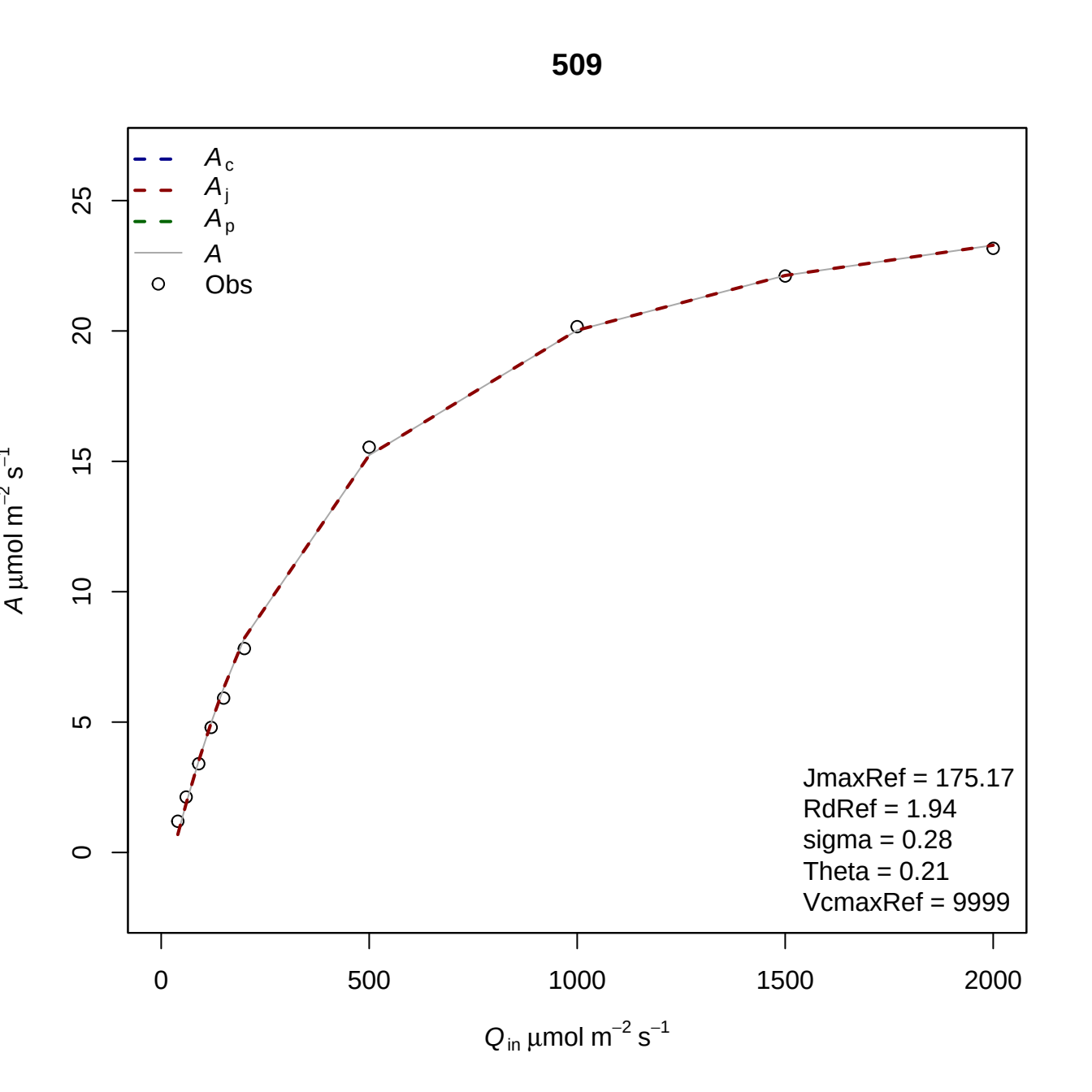

504
A_c
A_j
A_p
A
Obs
JmaxRef = 192.31
RdRef = 1.67
sigma = 0.2
Theta = 0.16
VcmaxRef = 113.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

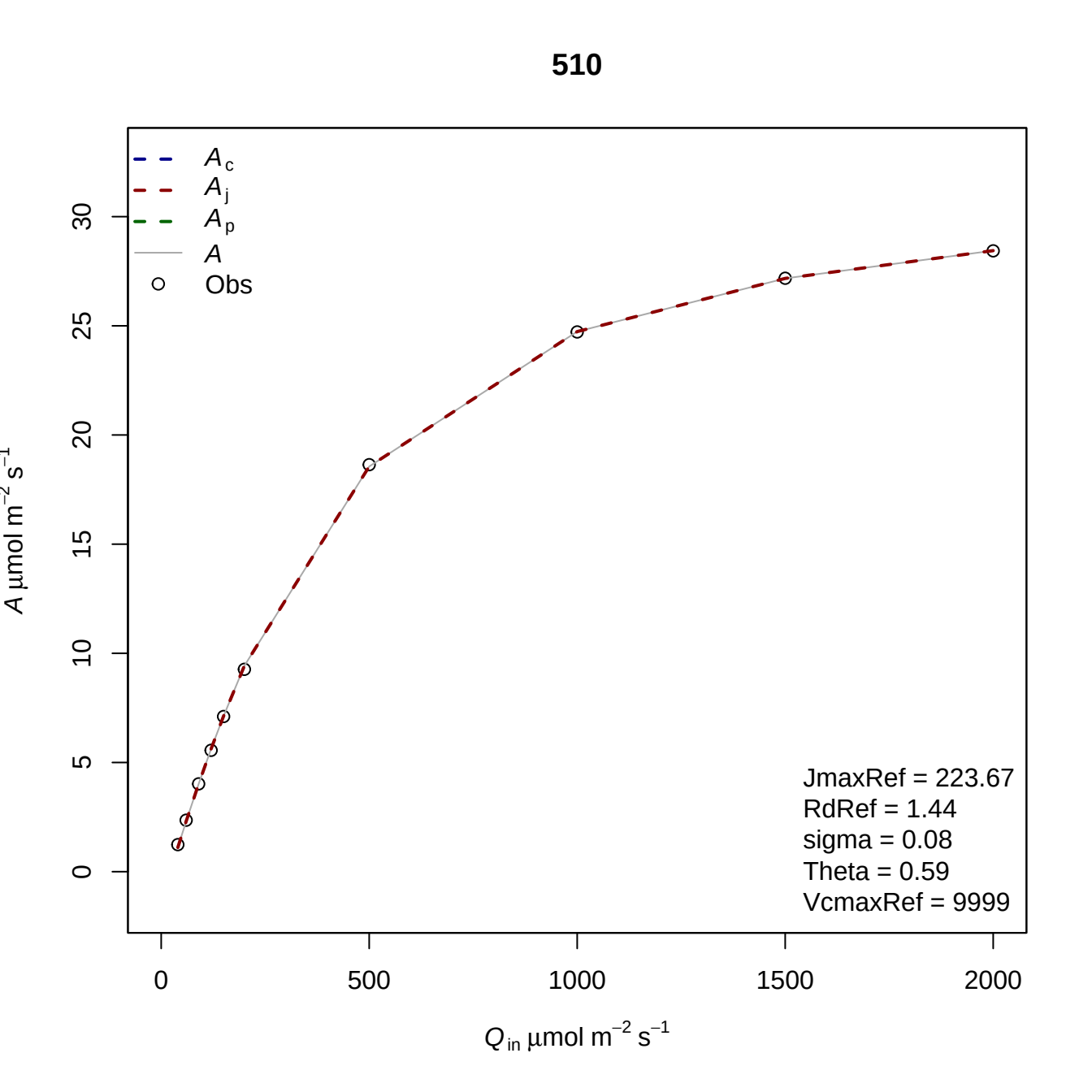

505
A_c
A_j
A_p
A
Obs
JmaxRef = 253.37
RdRef = 1.75
sigma = 0.06
Theta = 0.64
VcmaxRef = 216.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

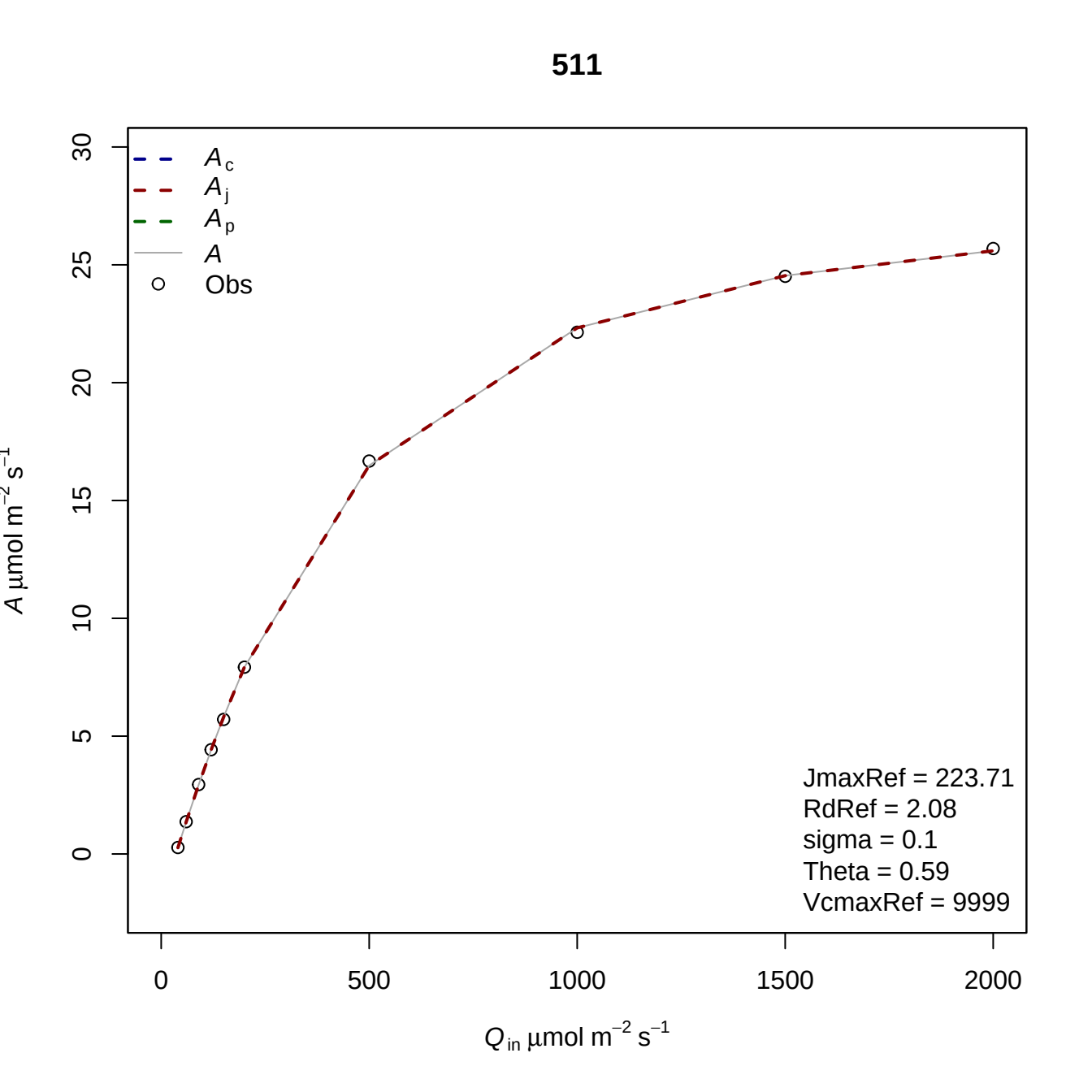

506
A_c
A_j
A_p
A
Obs
JmaxRef = 295.43
RdRef = 3.49
sigma = 0.21
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

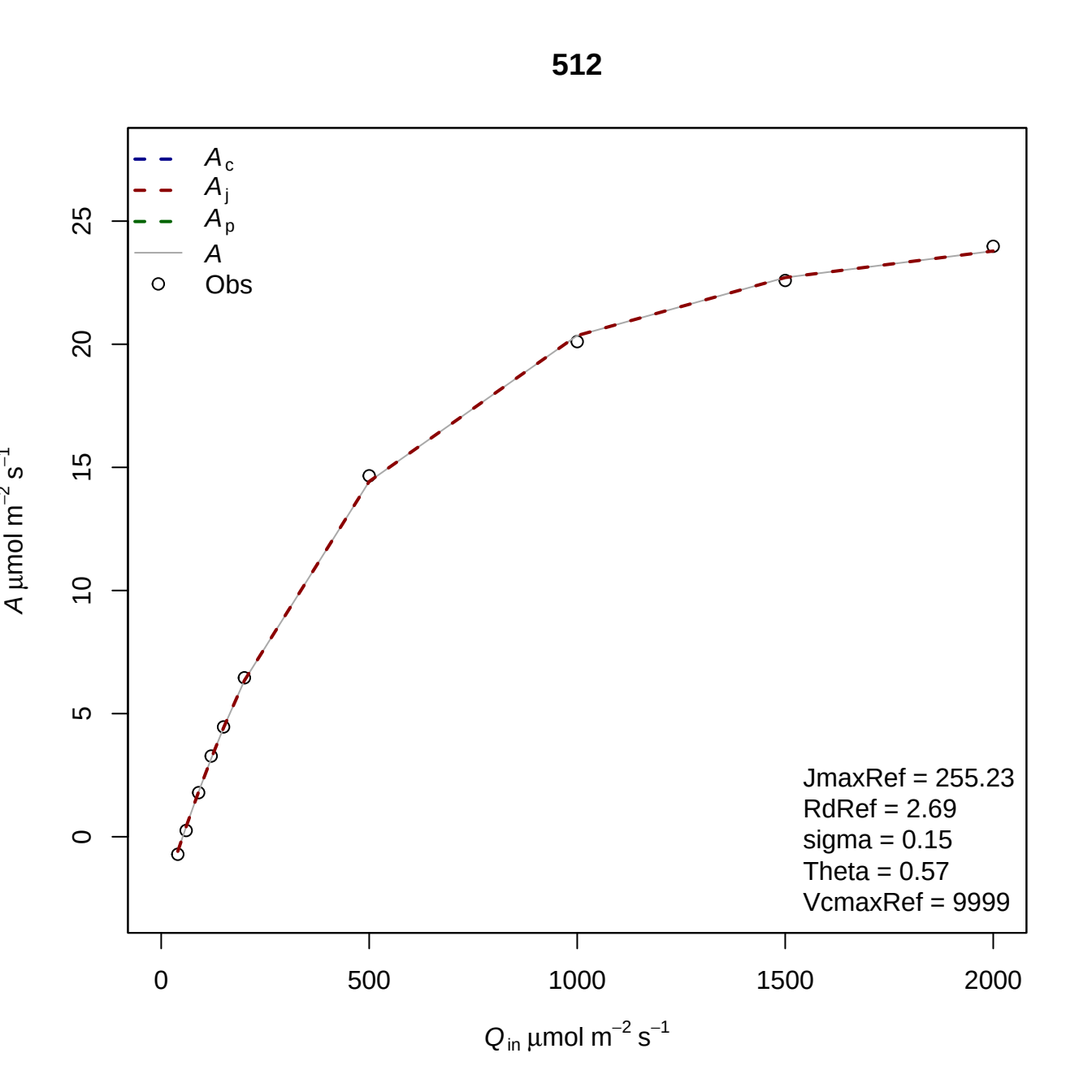

507
A_c
A_j
A_p
A
Obs
JmaxRef = 303.3
RdRef = 3.9
sigma = 0.04
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

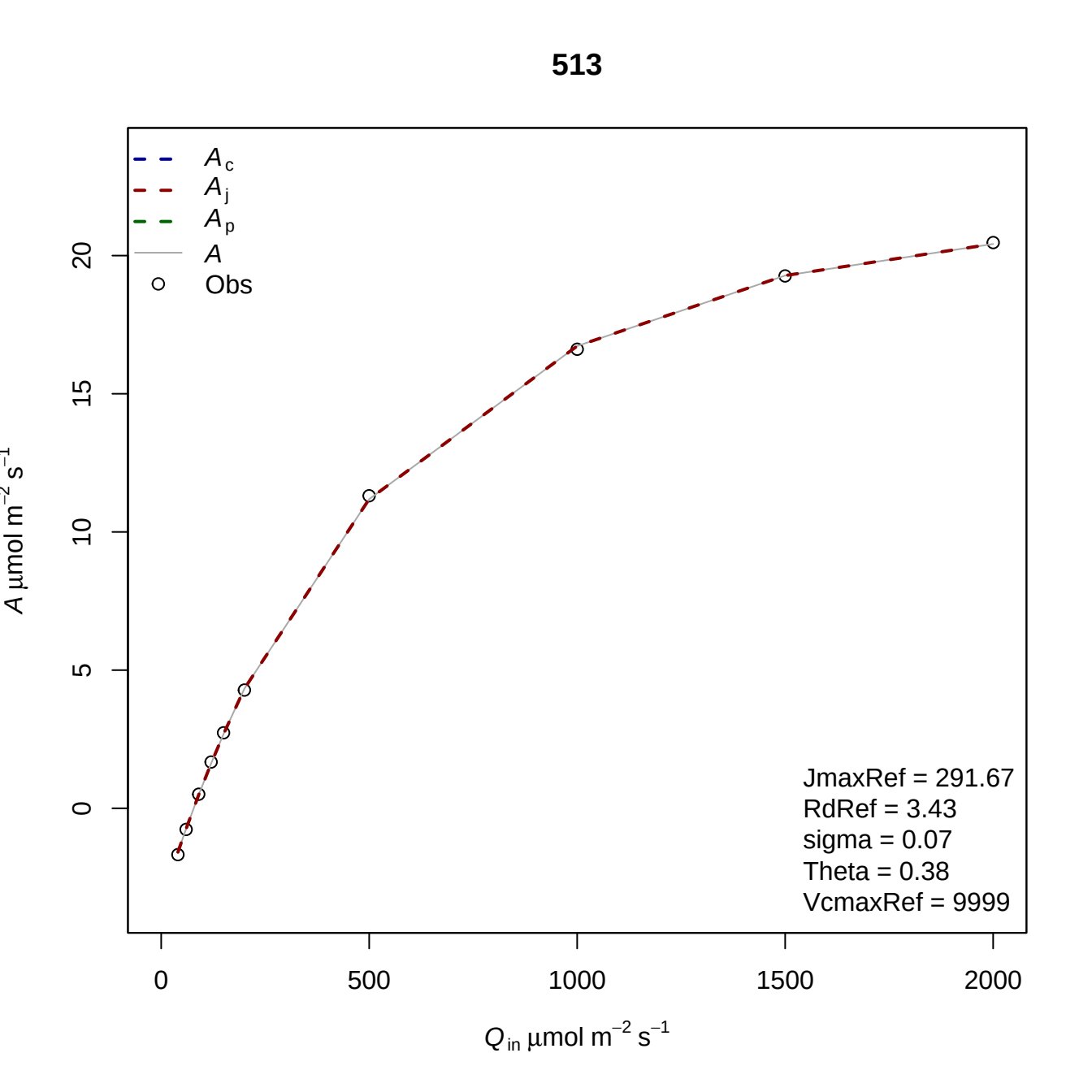

508
A_c
A_j
A_p
A
Obs
JmaxRef = 148.83
RdRef = 1.12
sigma = 0.15
Theta = 0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

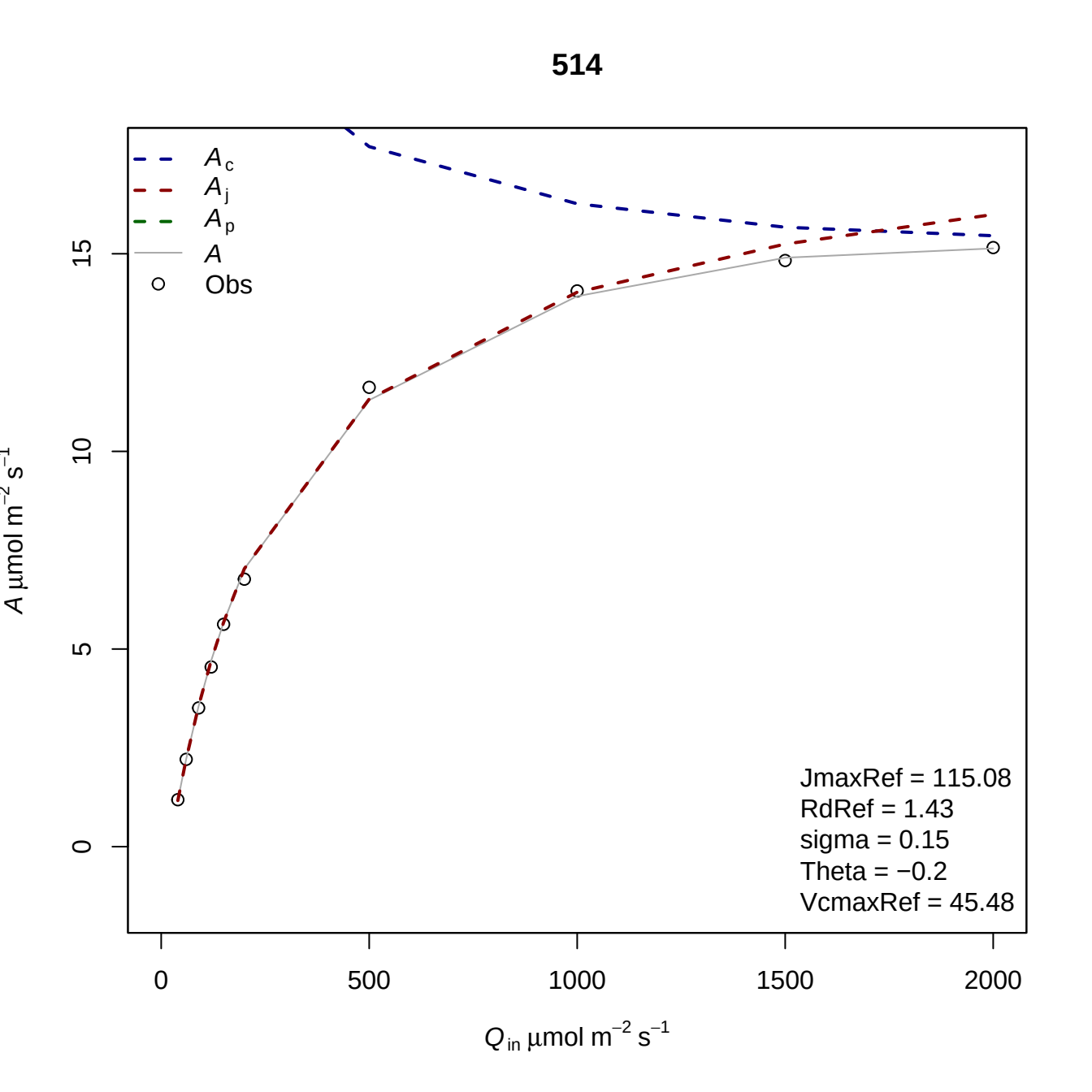

509
A_c
A_j
A_p
A
Obs
JmaxRef = 175.17
RdRef = 1.94
sigma = 0.28
Theta = 0.21
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

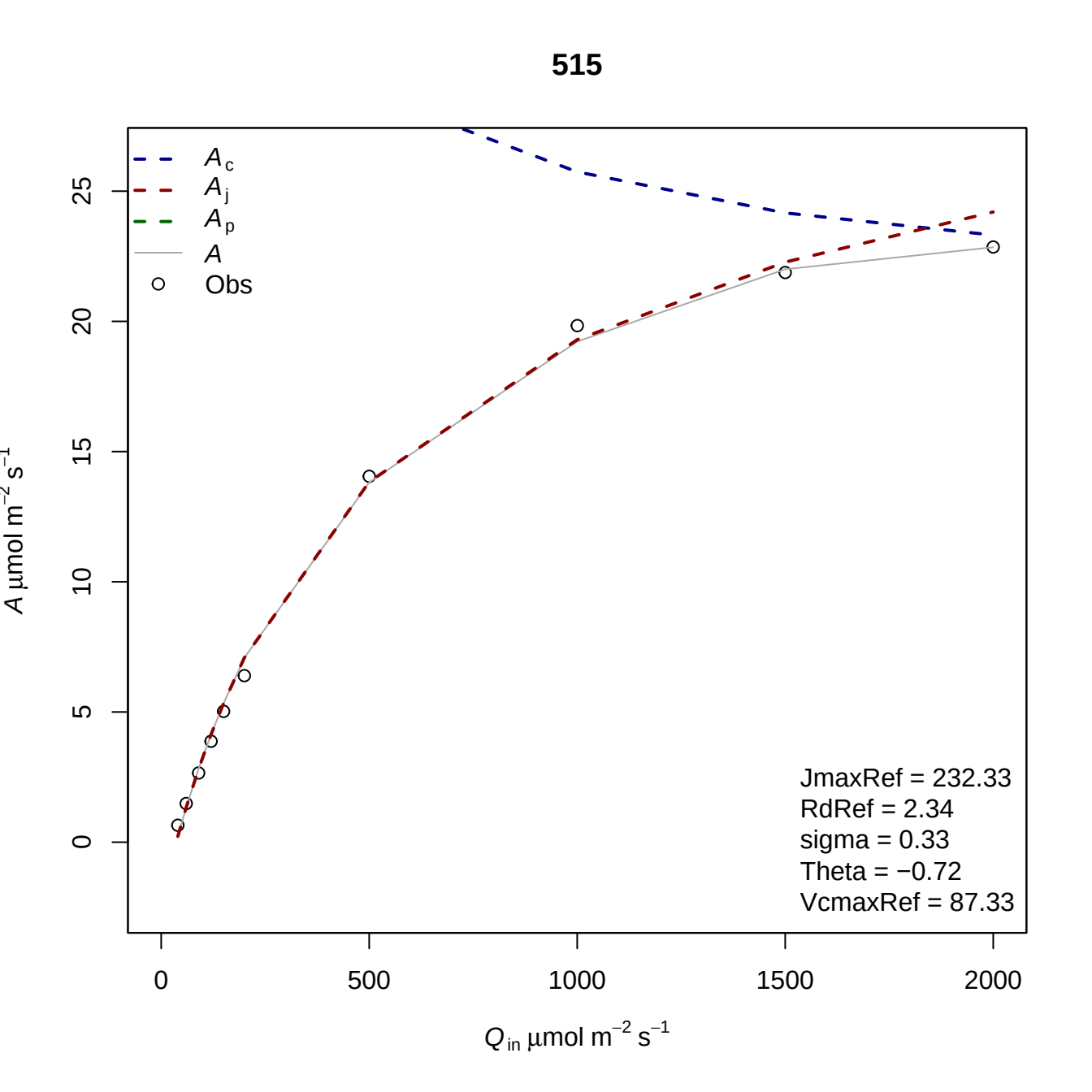

510
A_c
A_j
A_p
A
Obs
JmaxRef = 223.67
RdRef = 1.44
sigma = 0.08
Theta = 0.59
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

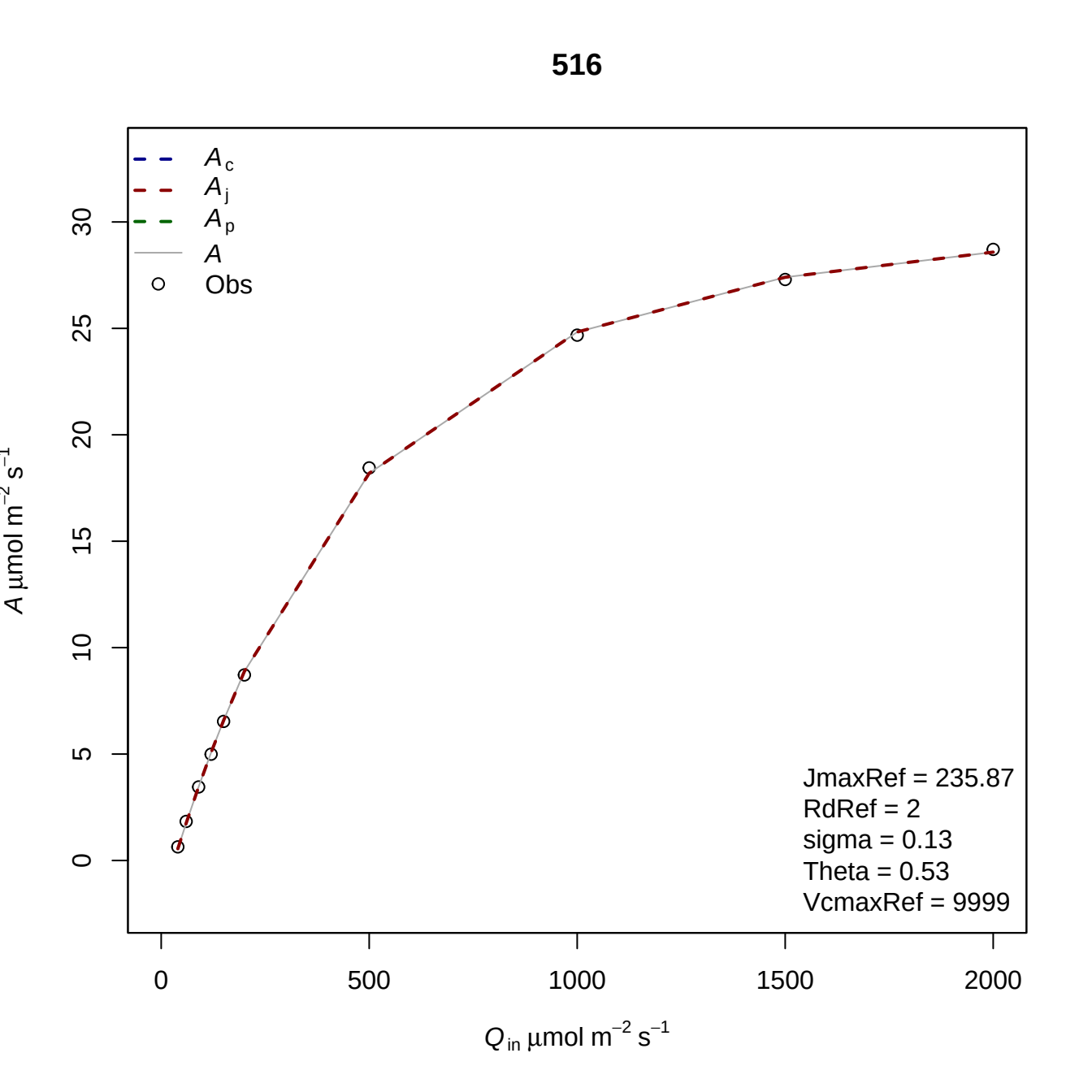

511
A_c
A_j
A_p
A
Obs
JmaxRef = 223.71
RdRef = 2.08
sigma = 0.1
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

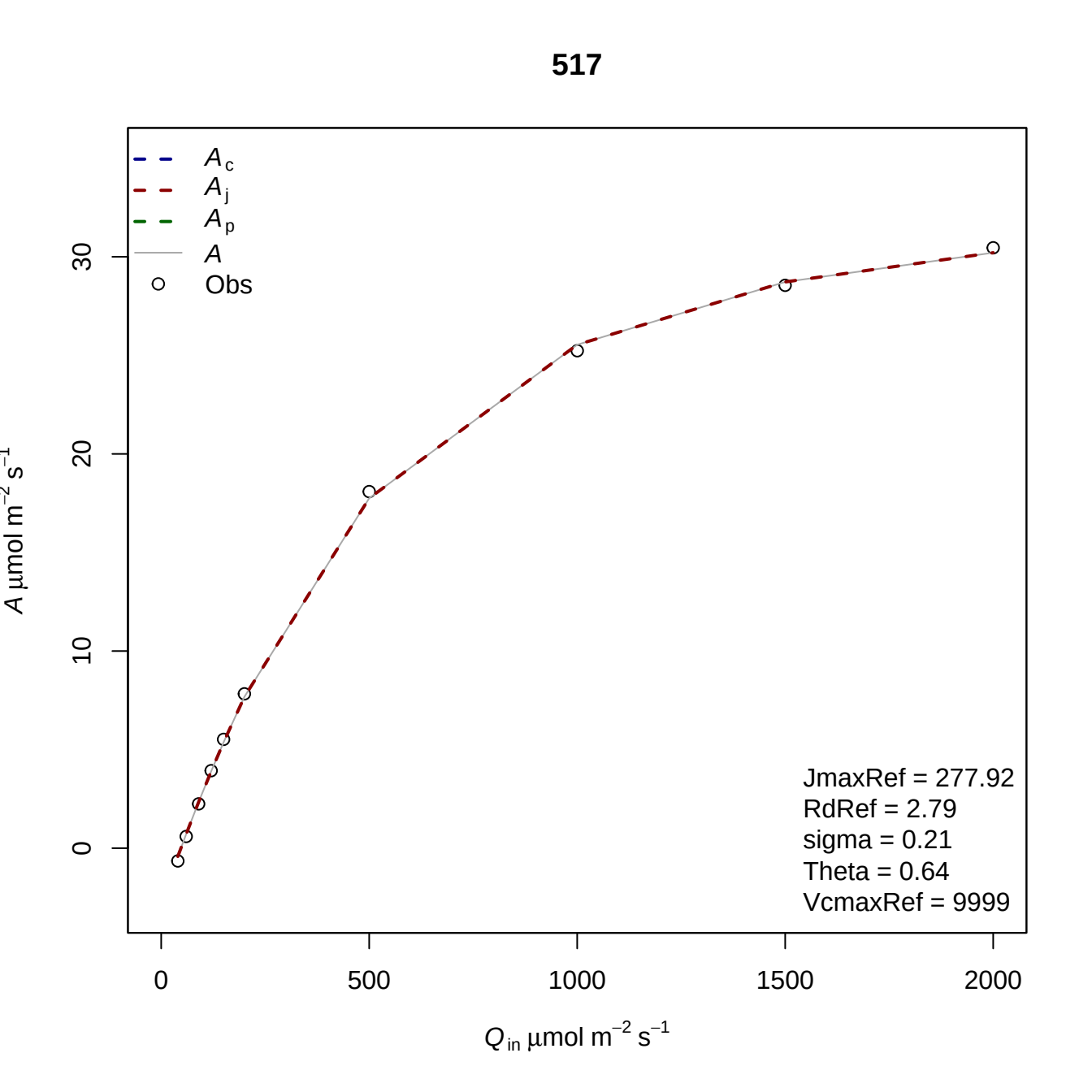

512
A_c
A_j
A_p
A
Obs
JmaxRef = 255.23
RdRef = 2.69
sigma = 0.15
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

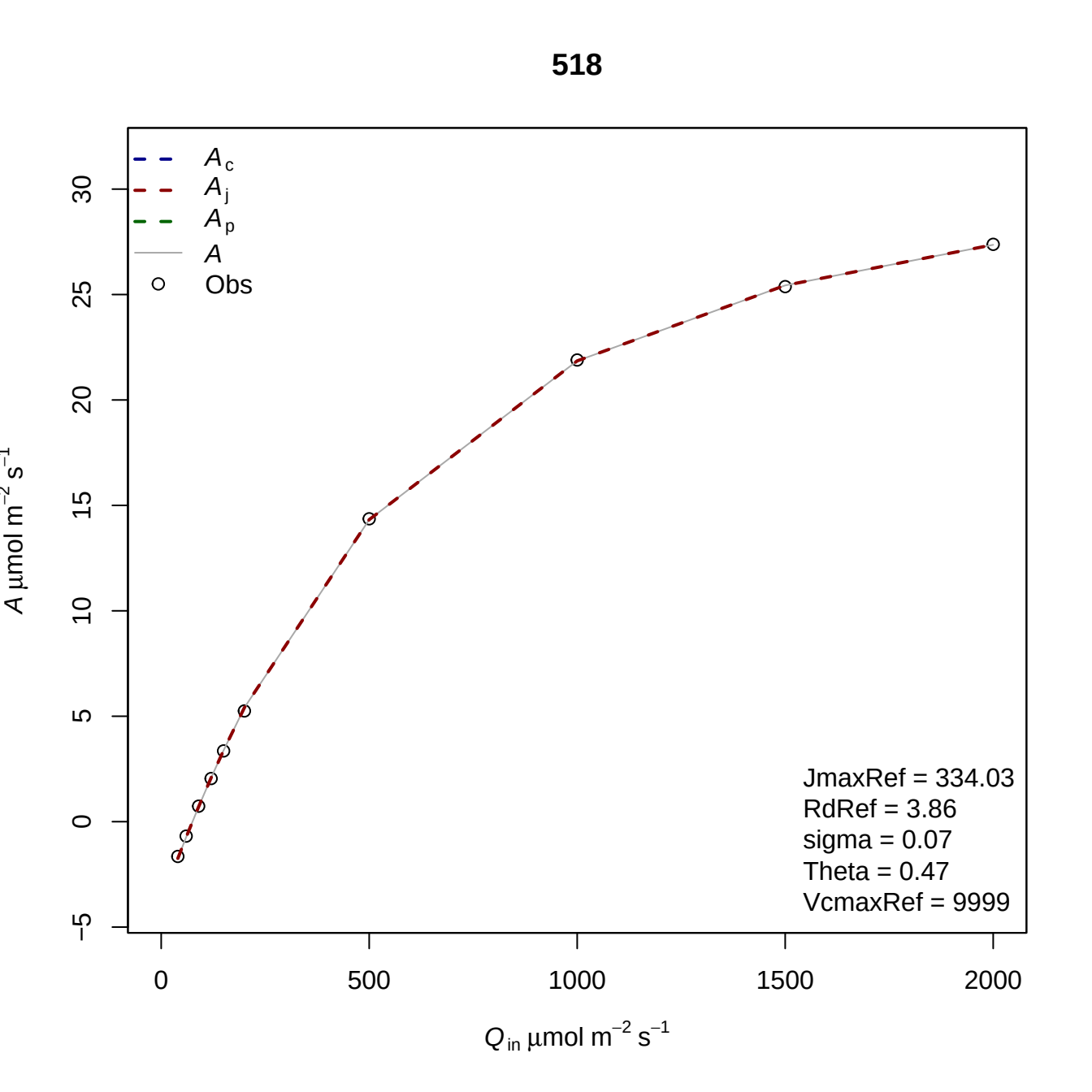

513
A_c
A_j
A_p
A
Obs
JmaxRef = 291.67
RdRef = 3.43
sigma = 0.07
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

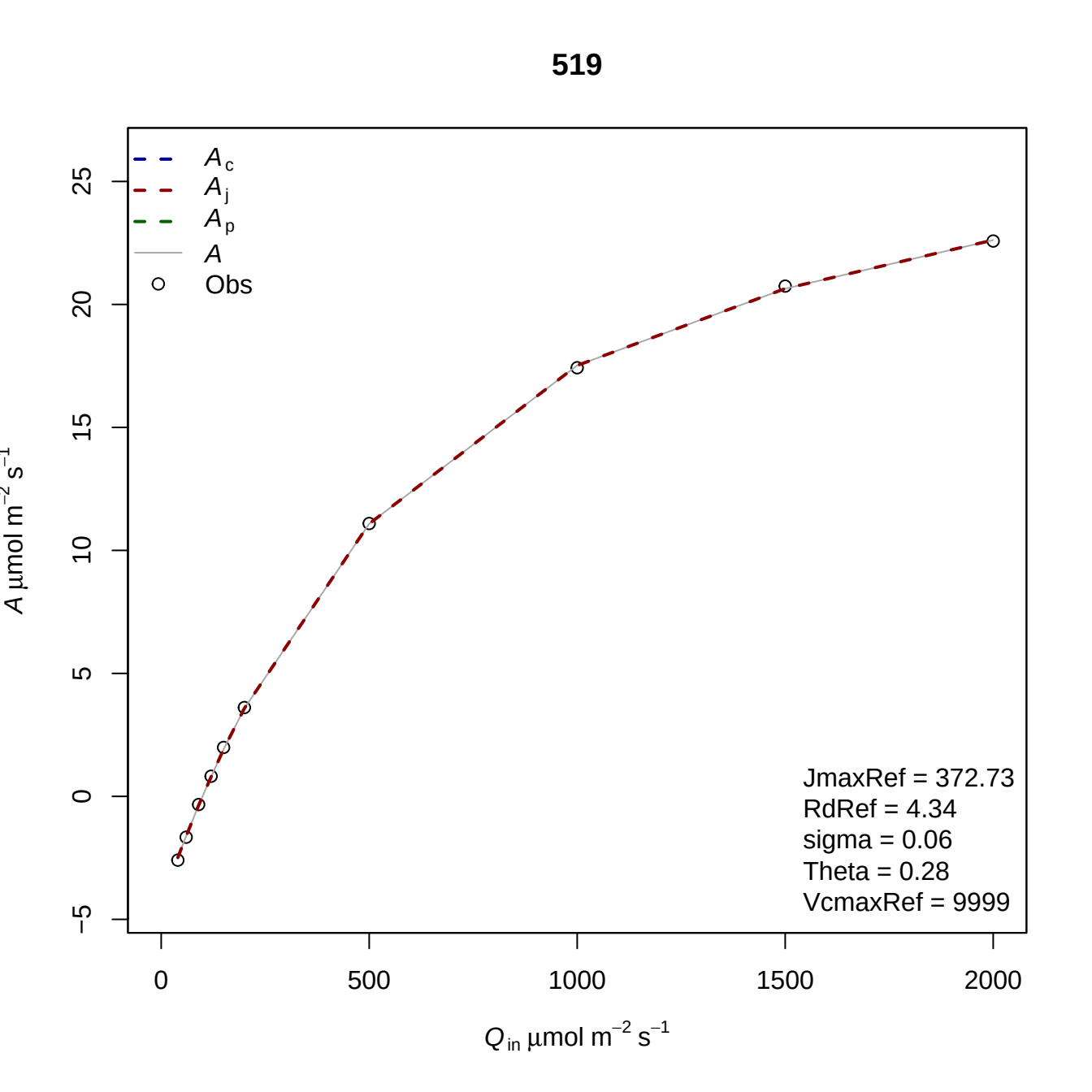

514
A_c
A_j
A_p
A
Obs
JmaxRef = 115.08
RdRef = 1.43
sigma = 0.15
Theta = −0.2
VcmaxRef = 45.48
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

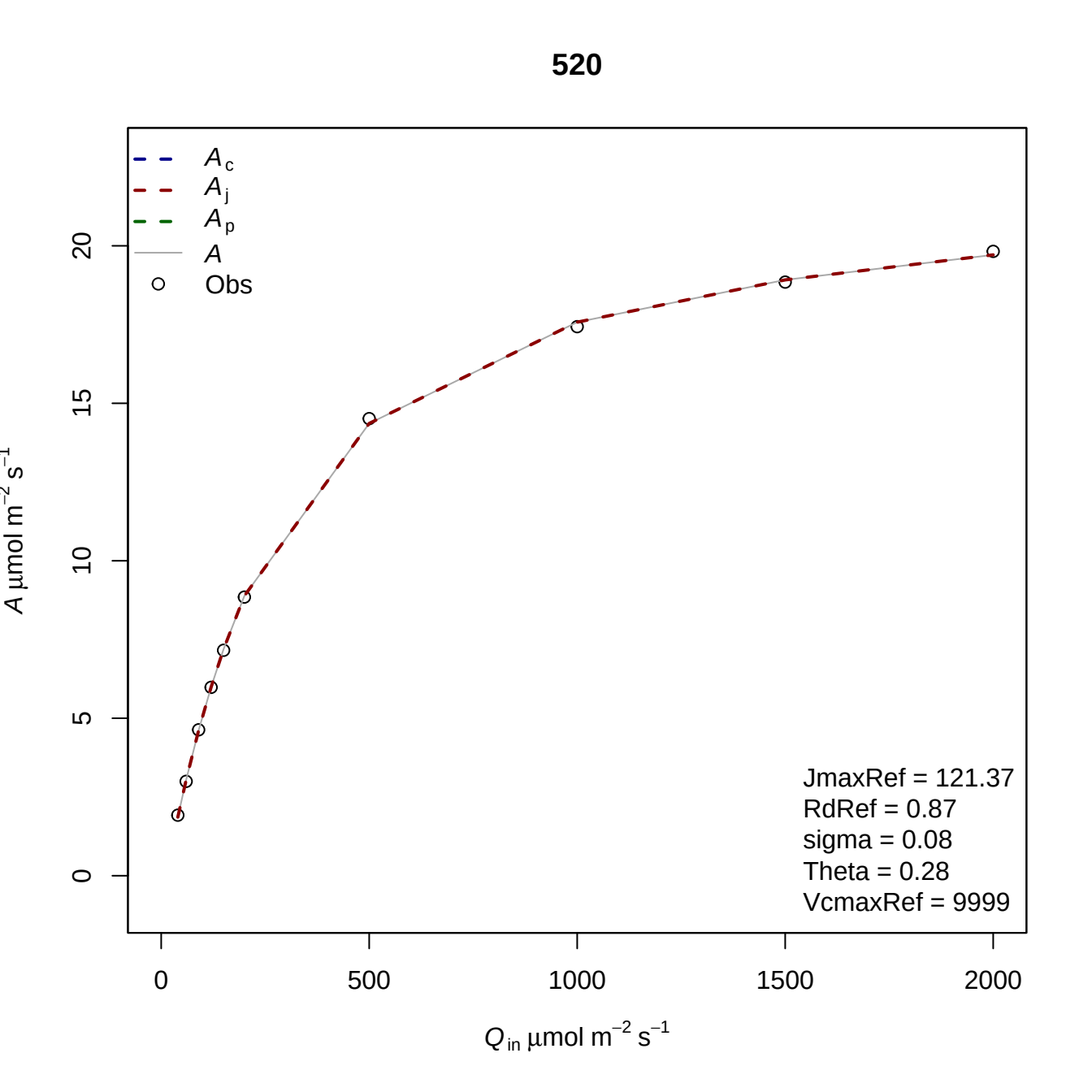

515
A_c
A_j
A_p
A
Obs
JmaxRef = 232.33
RdRef = 2.34
sigma = 0.33
Theta = −0.72
VcmaxRef = 87.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

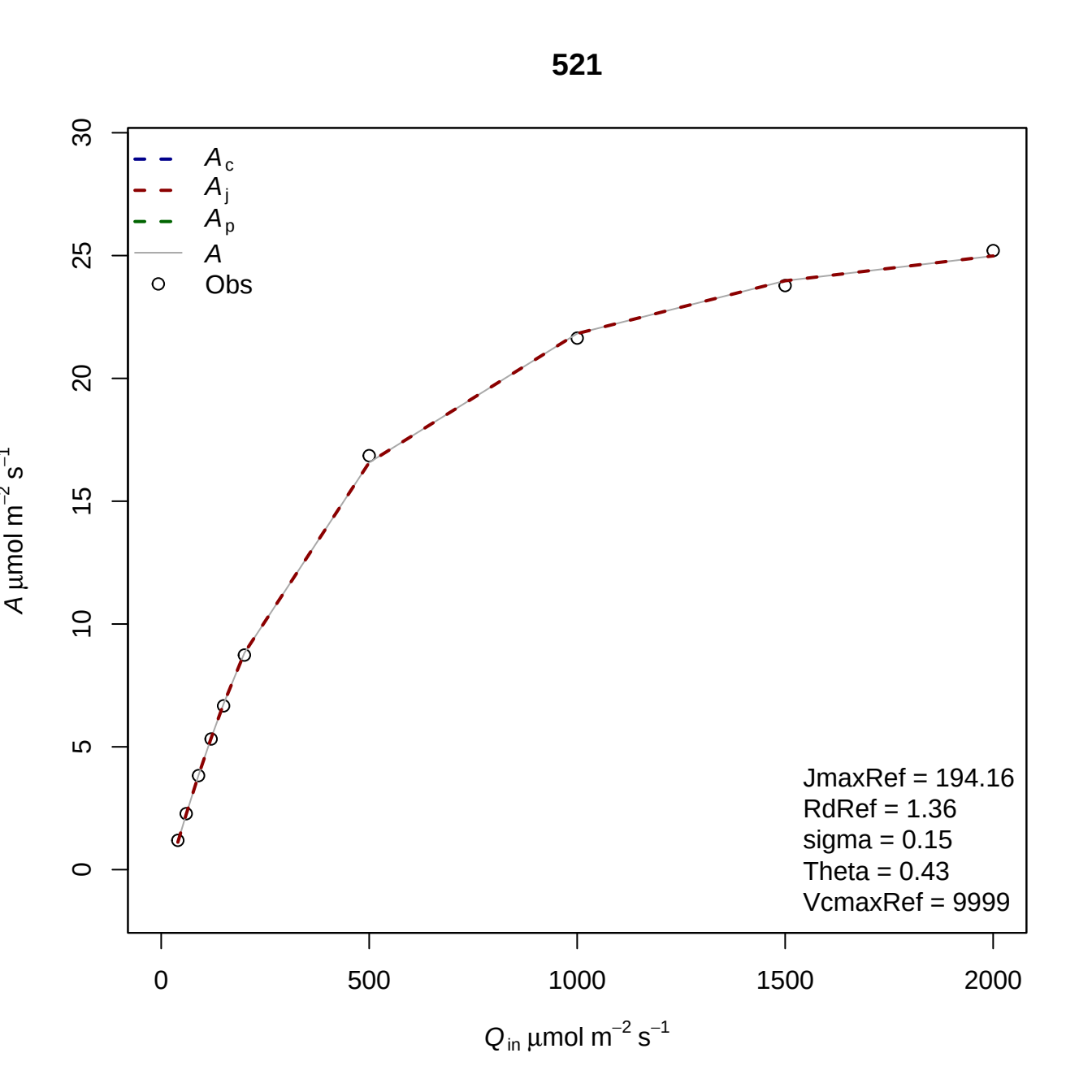

516
A_c
A_j
A_p
A
Obs
JmaxRef = 235.87
RdRef = 2
sigma = 0.13
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

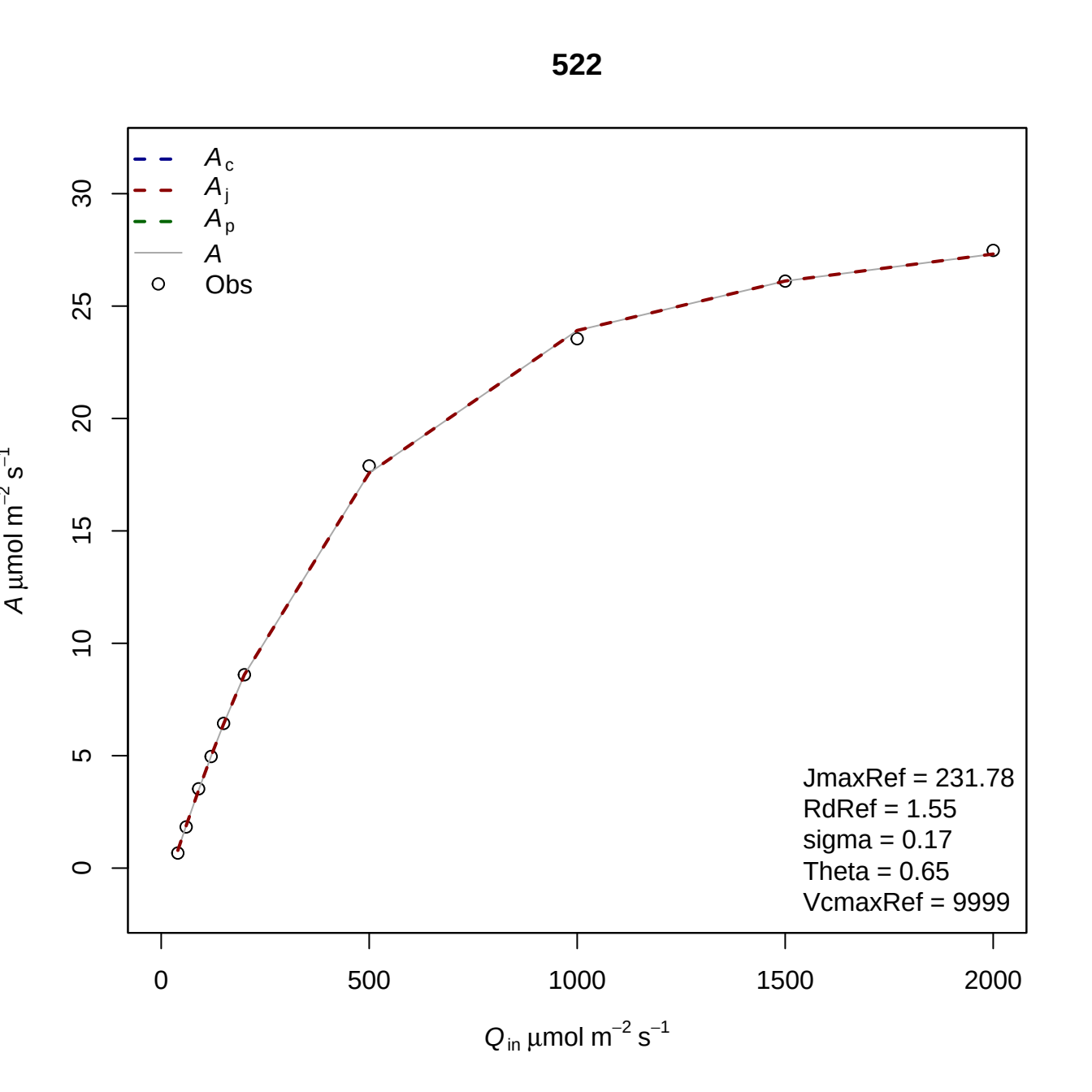

517
A_c
A_j
A_p
A
Obs
JmaxRef = 277.92
RdRef = 2.79
sigma = 0.21
Theta = 0.64
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

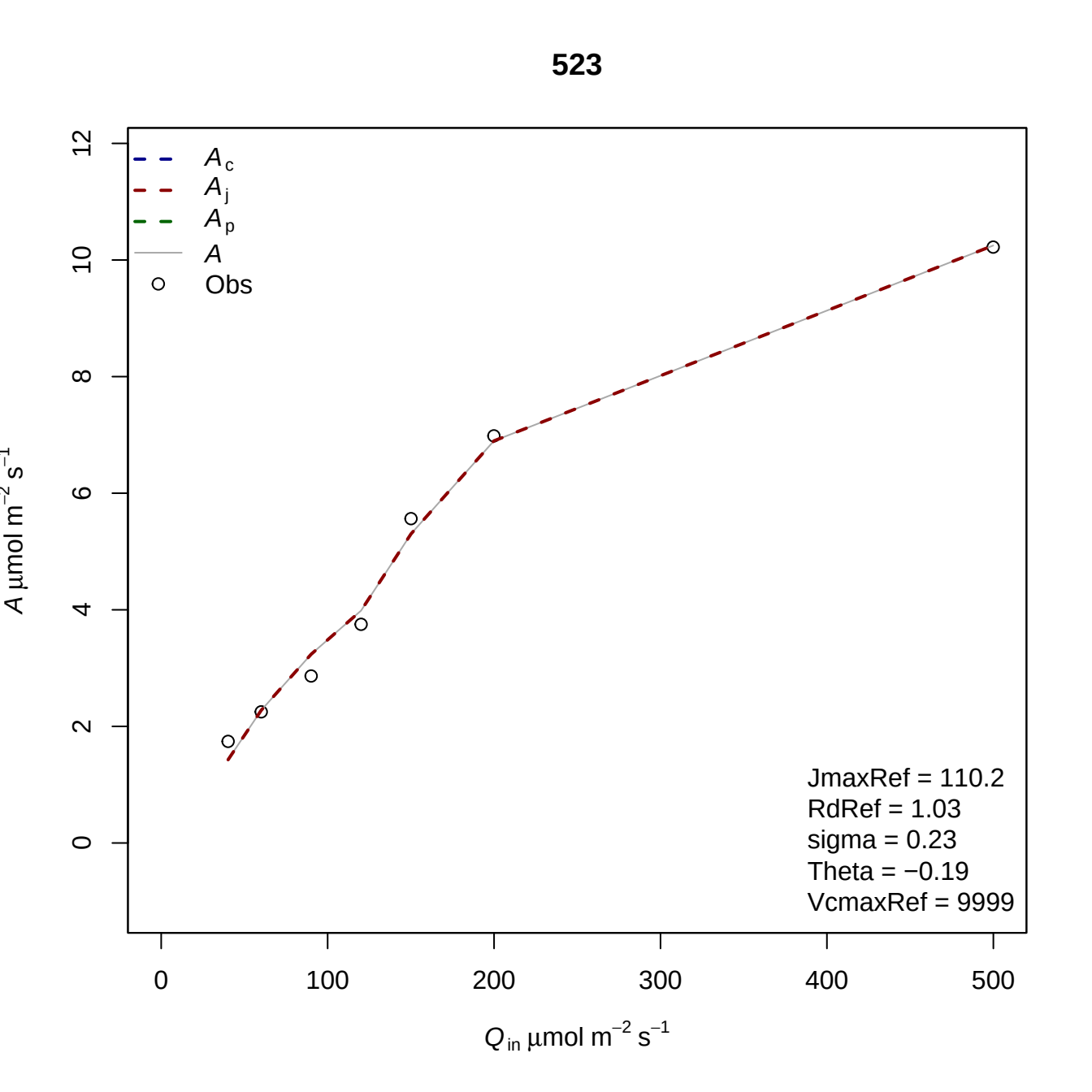

518
A_c
A_j
A_p
A
Obs
JmaxRef = 334.03
RdRef = 3.86
sigma = 0.07
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

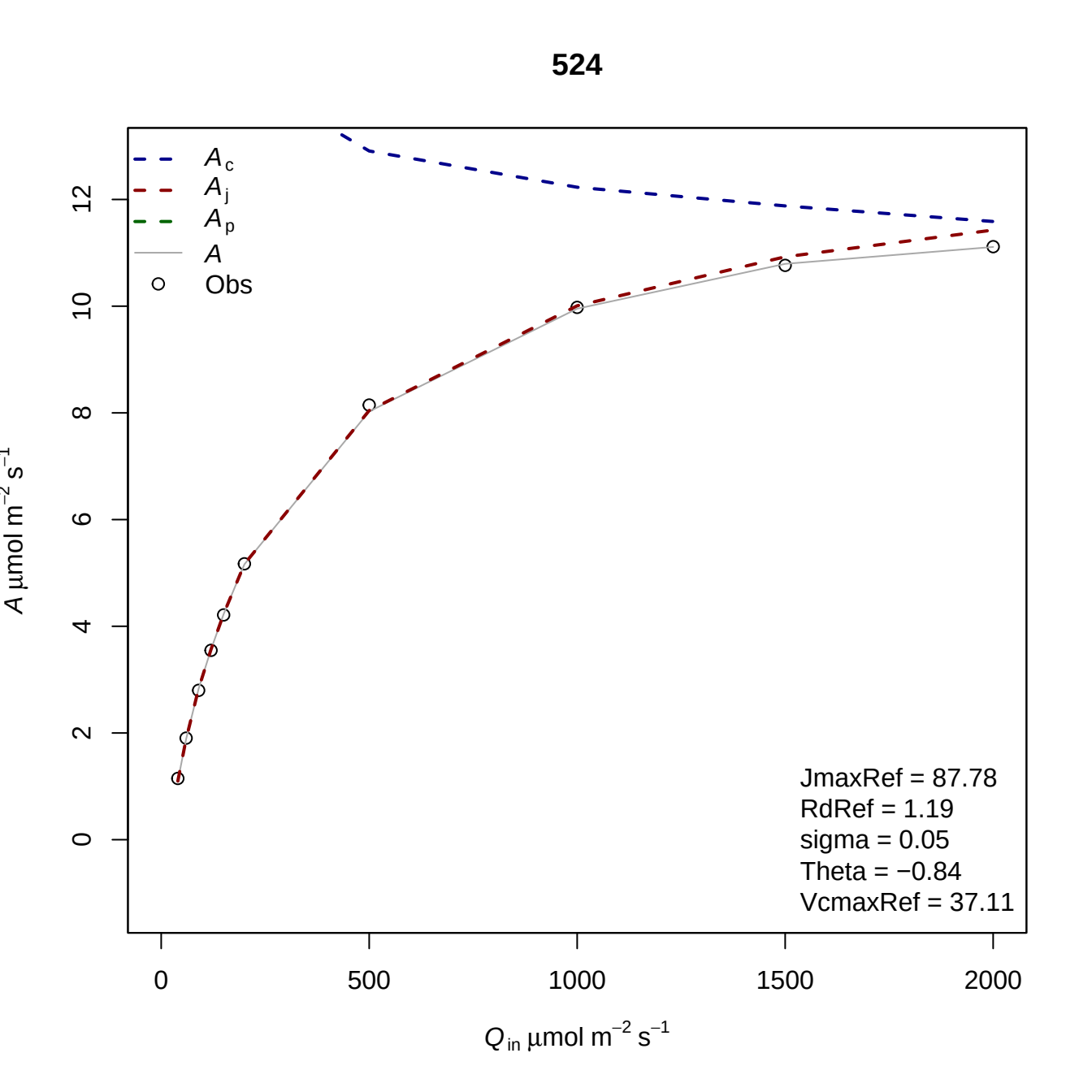

519
A_c
A_j
A_p
A
Obs
JmaxRef = 372.73
RdRef = 4.34
sigma = 0.06
Theta = 0.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

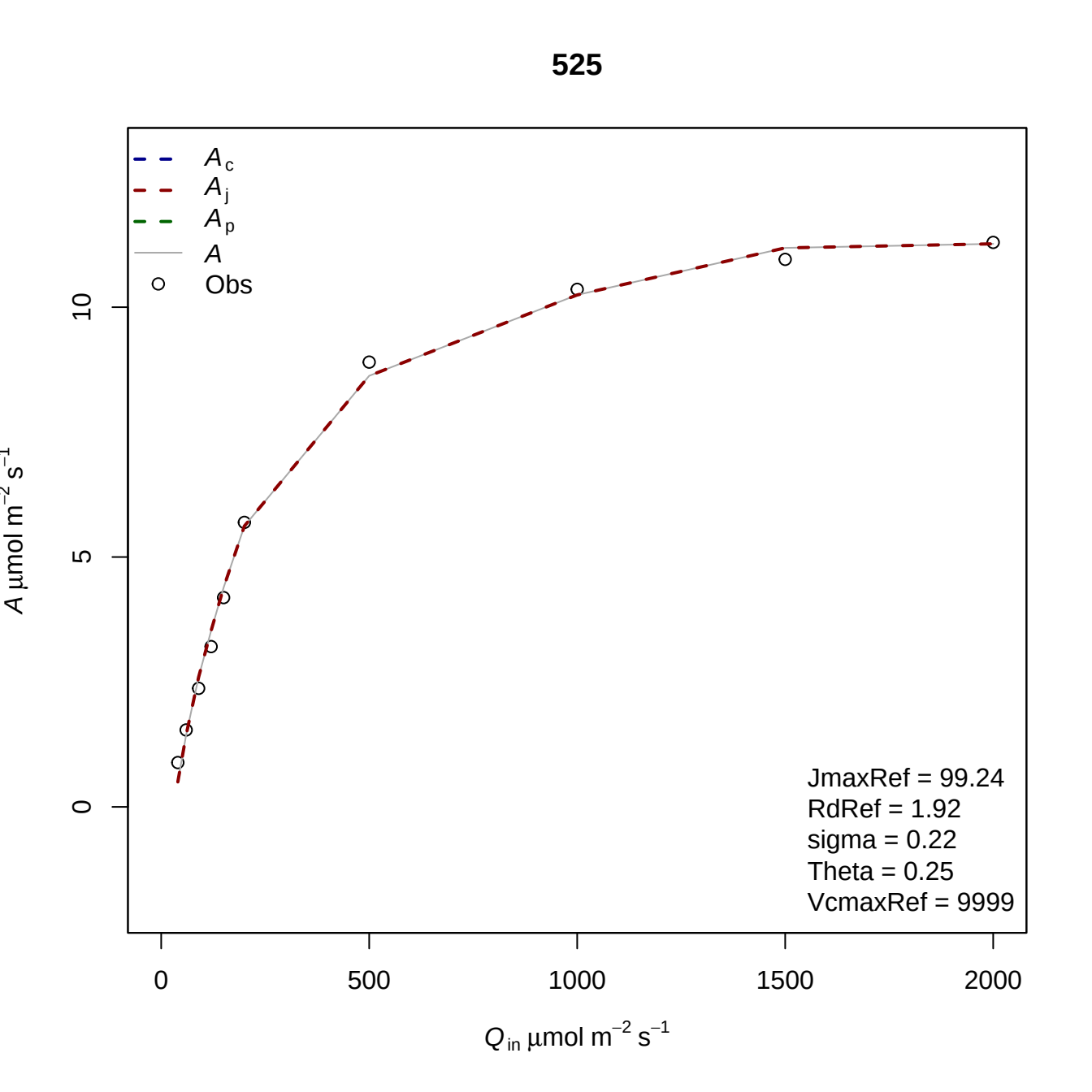

520
A_c
A_j
A_p
A
Obs
JmaxRef = 121.37
RdRef = 0.87
sigma = 0.08
Theta = 0.28
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

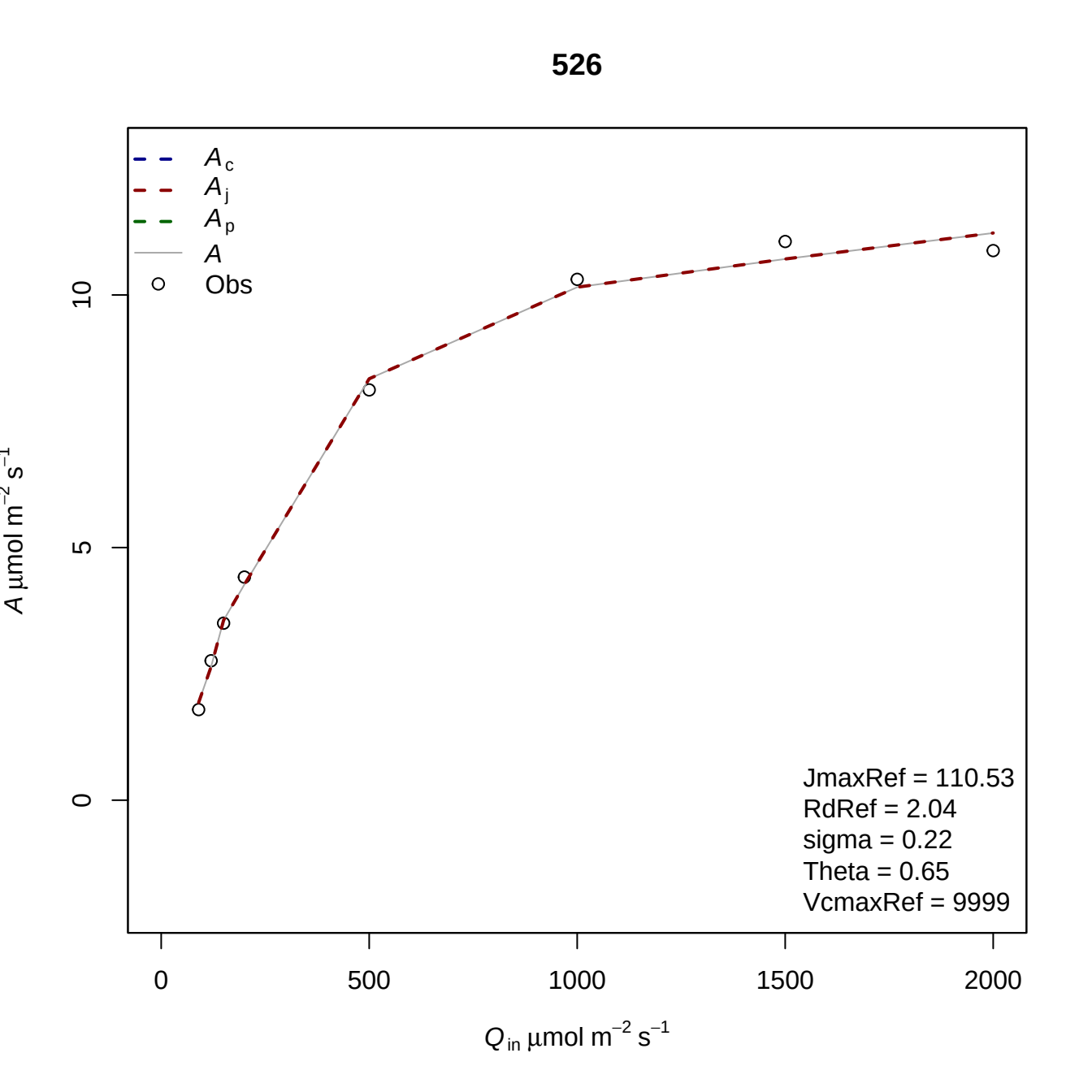

521
A_c
A_j
A_p
A
Obs
JmaxRef = 194.16
RdRef = 1.36
sigma = 0.15
Theta = 0.43
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

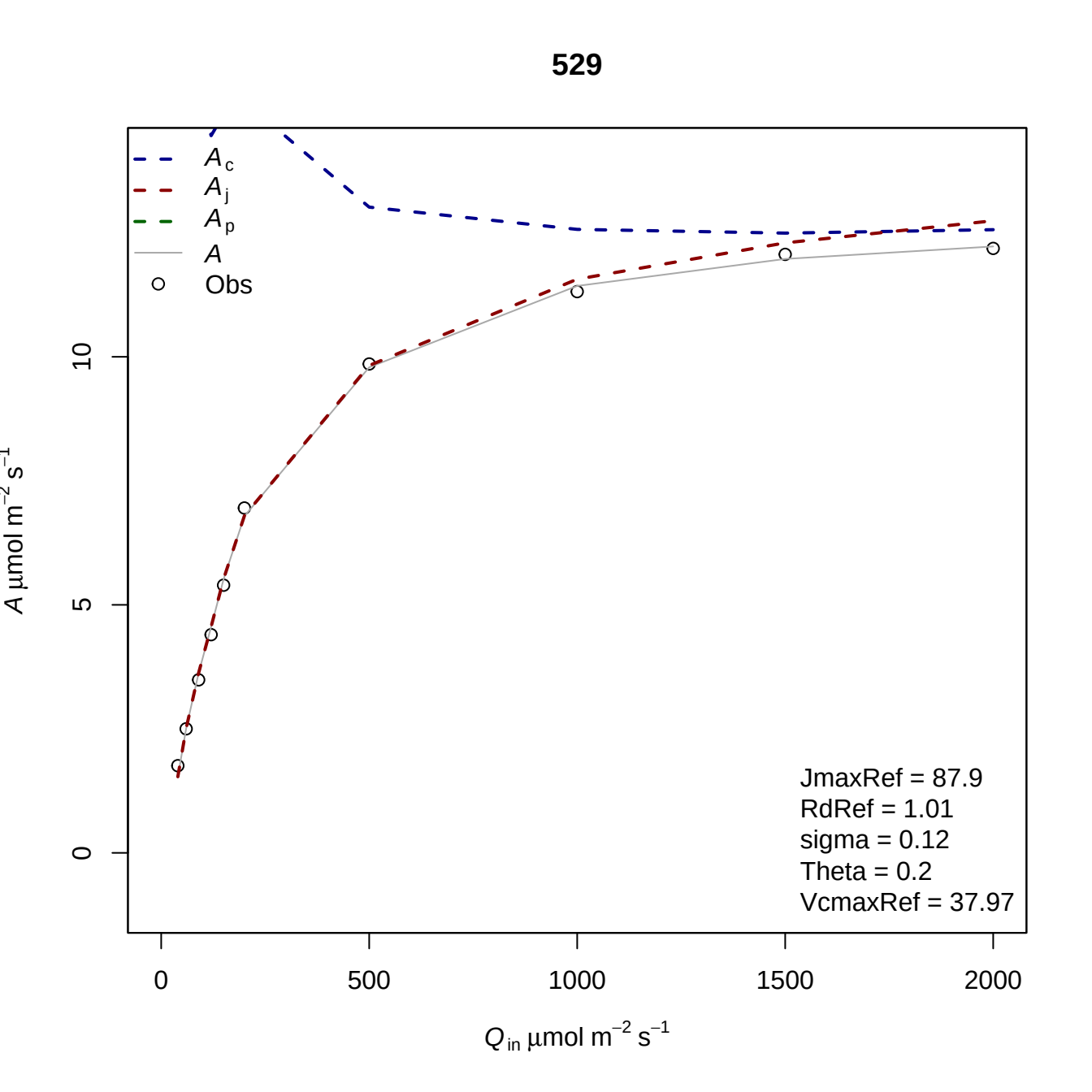

522
A_c
A_j
A_p
A
Obs
JmaxRef = 231.78
RdRef = 1.55
sigma = 0.17
Theta = 0.65
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

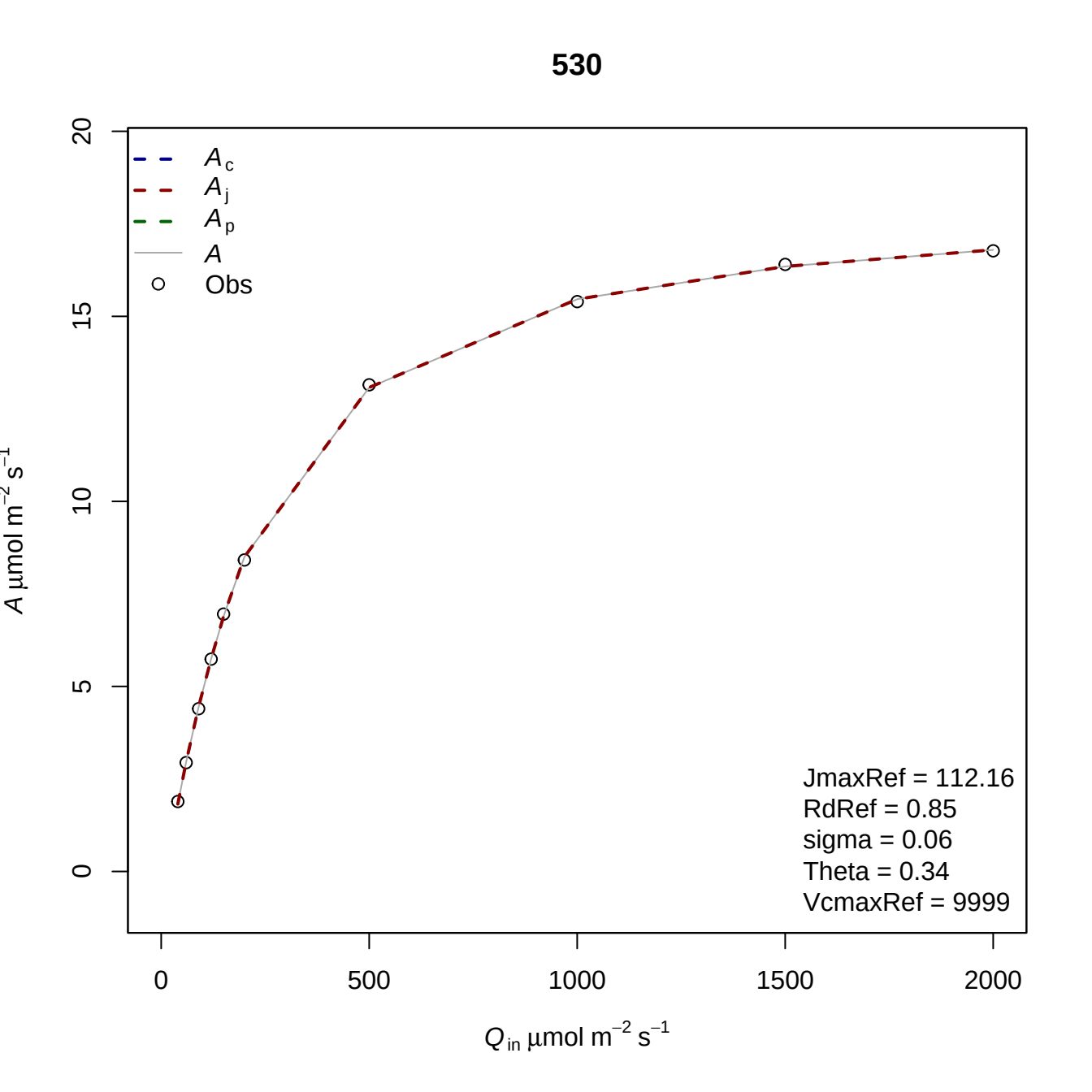

523
A_c
A_j
A_p
A
Obs
JmaxRef = 110.2
RdRef = 1.03
sigma = 0.23
Theta = −0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
100
200
300
400
500

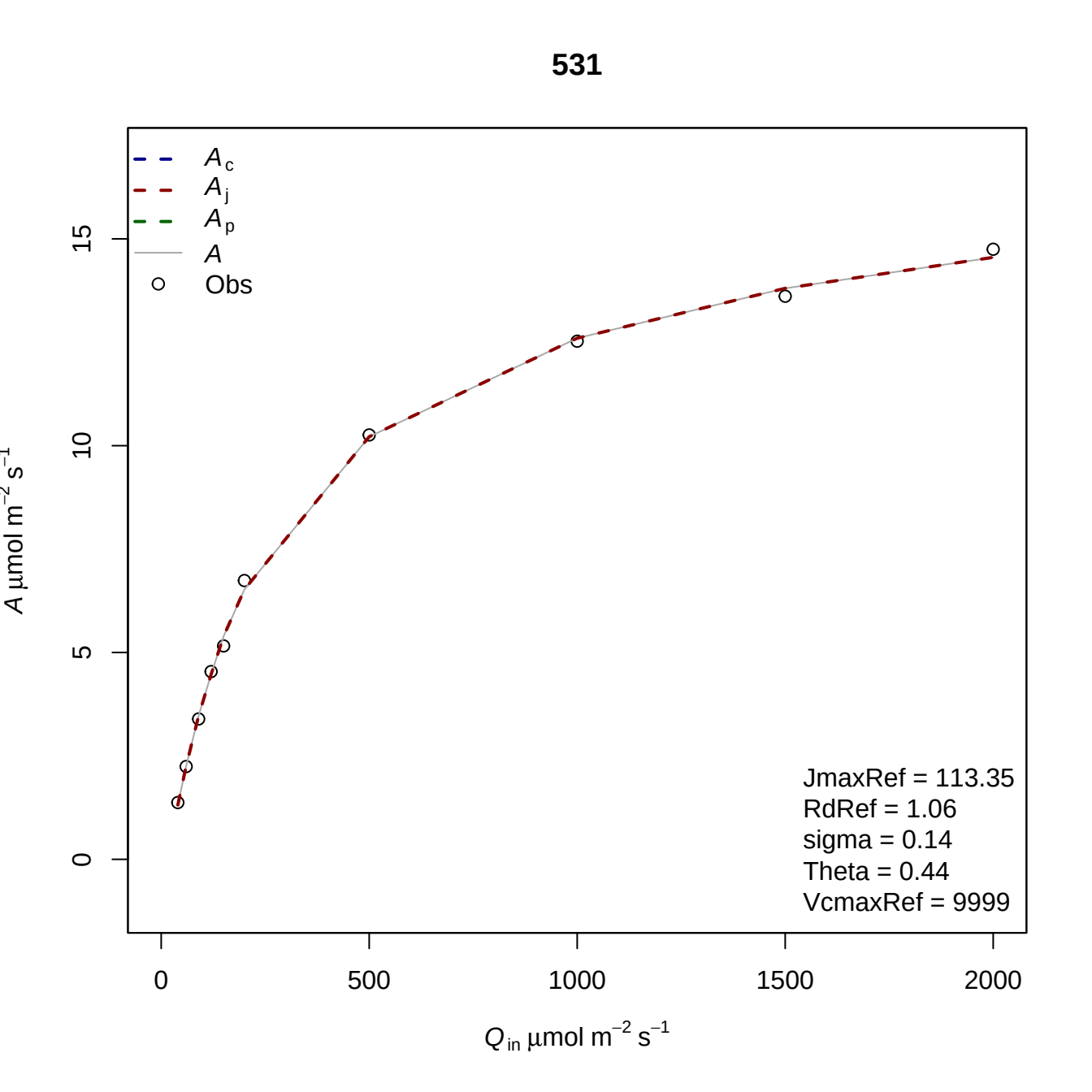

524
A_c
A_j
A_p
A
Obs
JmaxRef = 87.78
RdRef = 1.19
sigma = 0.05
Theta = −0.84
VcmaxRef = 37.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

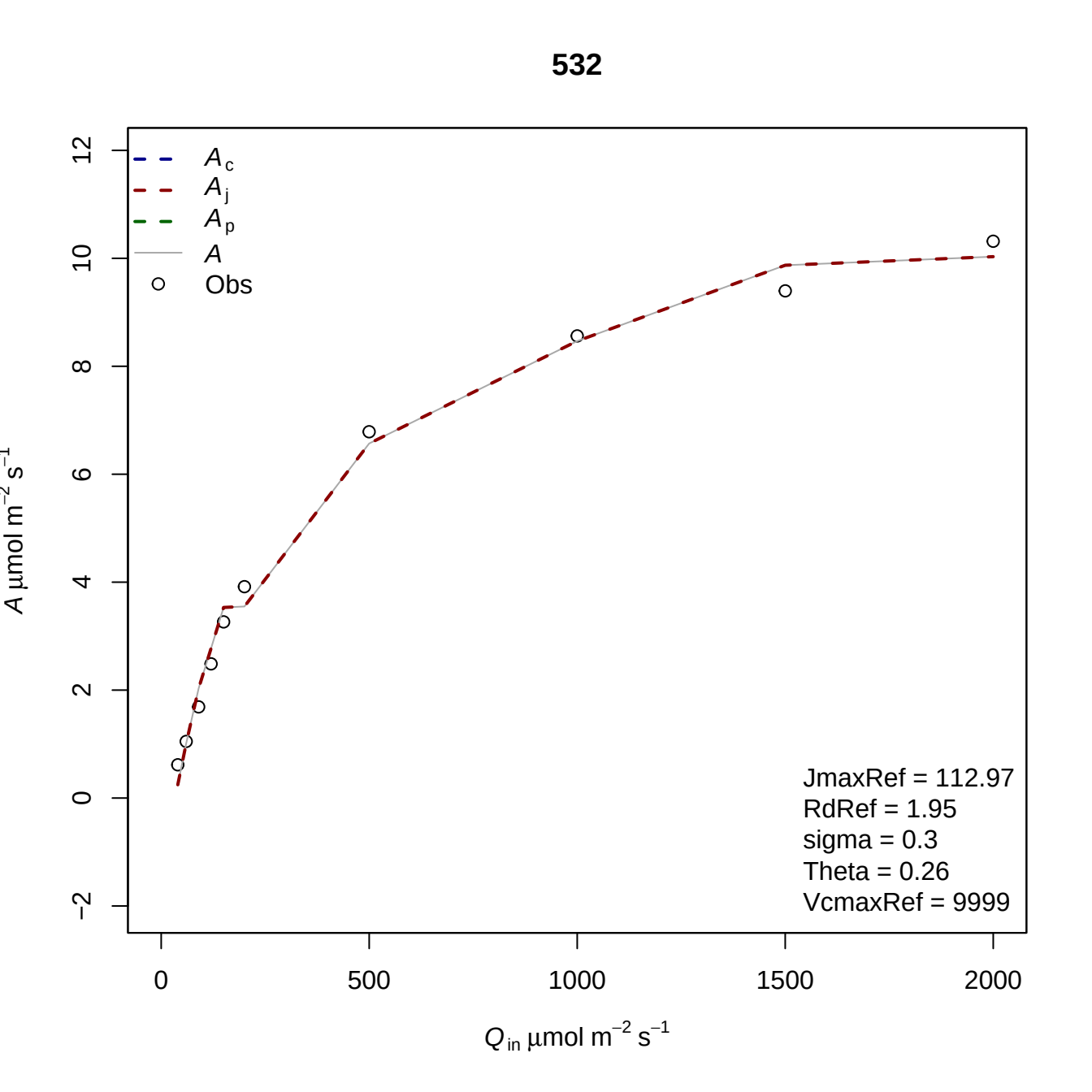

525
A_c
A_j
A_p
A
Obs
JmaxRef = 99.24
RdRef = 1.92
sigma = 0.22
Theta = 0.25
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

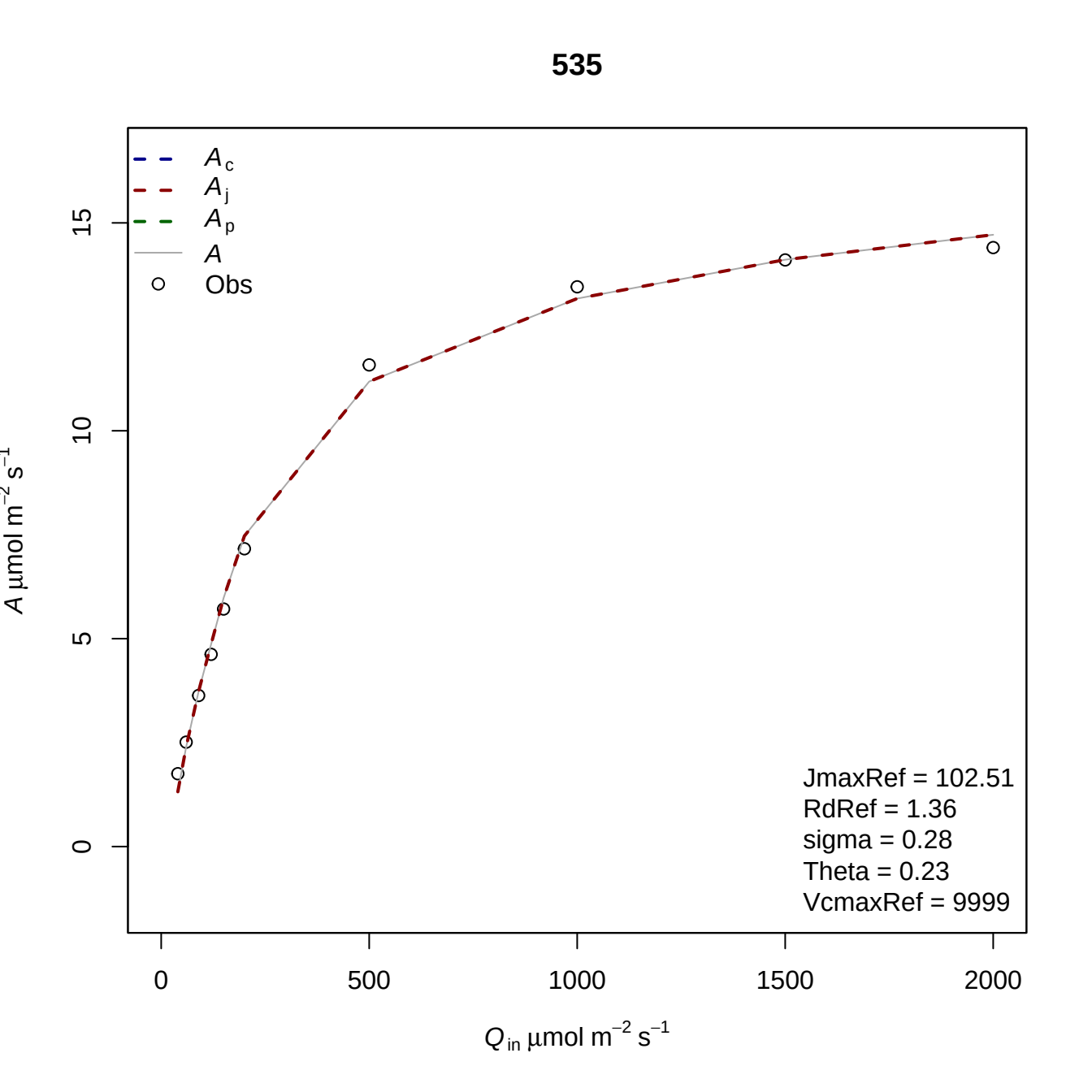

526
A_c
A_j
A_p
A
Obs
JmaxRef = 110.53
RdRef = 2.04
sigma = 0.22
Theta = 0.65
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

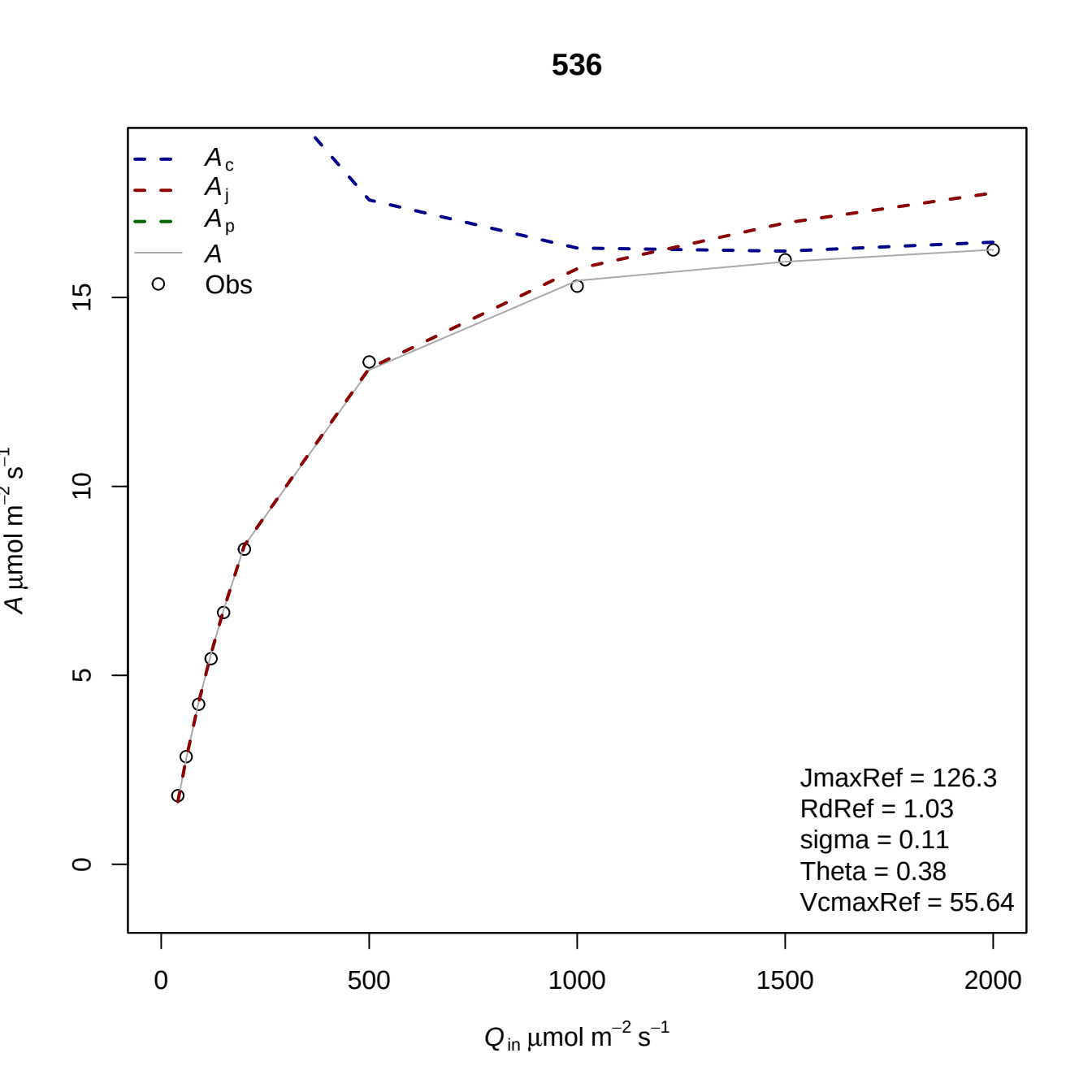

529
A_c
A_j
A_p
A
Obs
JmaxRef = 87.9
RdRef = 1.01
sigma = 0.12
Theta = 0.2
VcmaxRef = 37.97
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

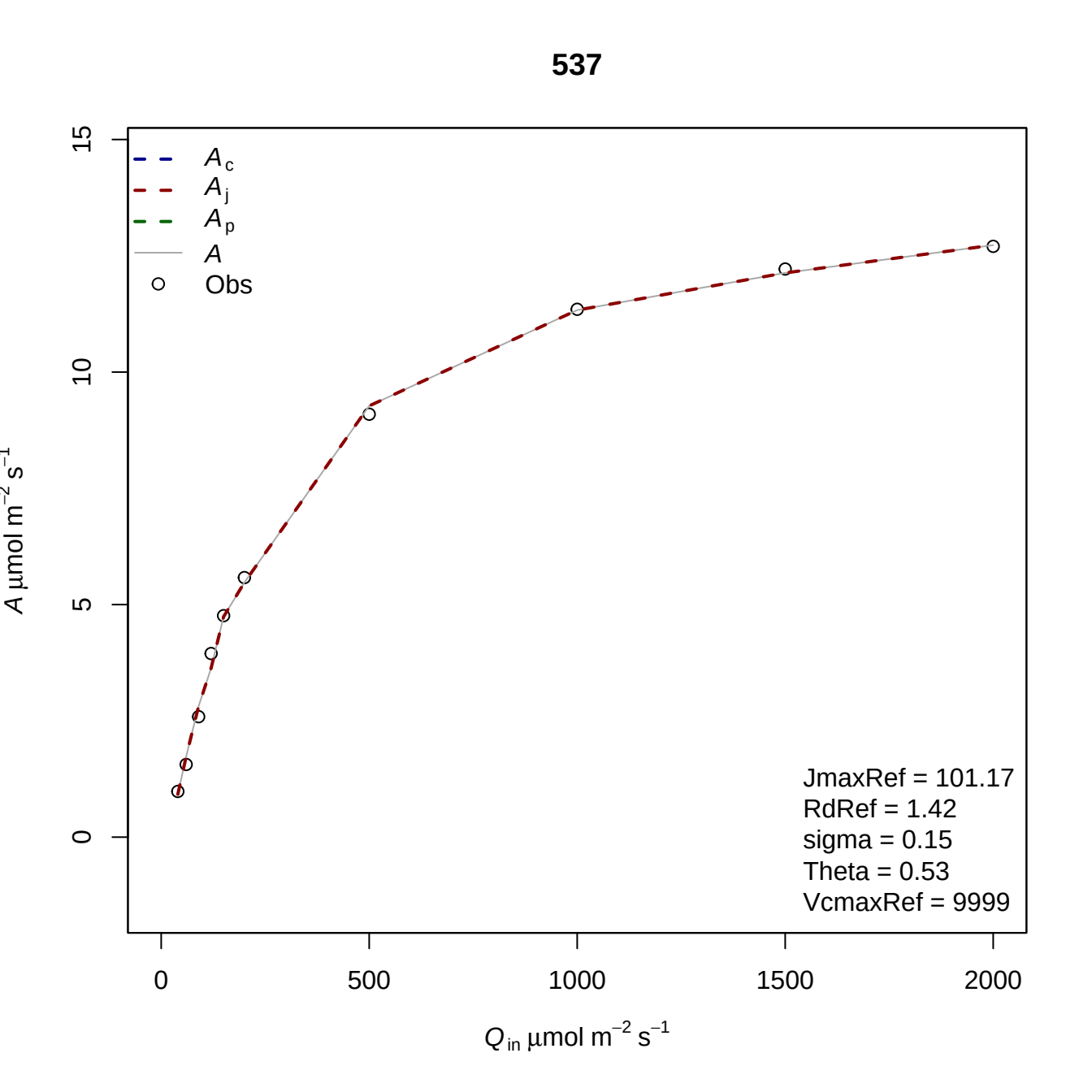

530
A_c
A_j
A_p
A
Obs
JmaxRef = 112.16
RdRef = 0.85
sigma = 0.06
Theta = 0.34
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

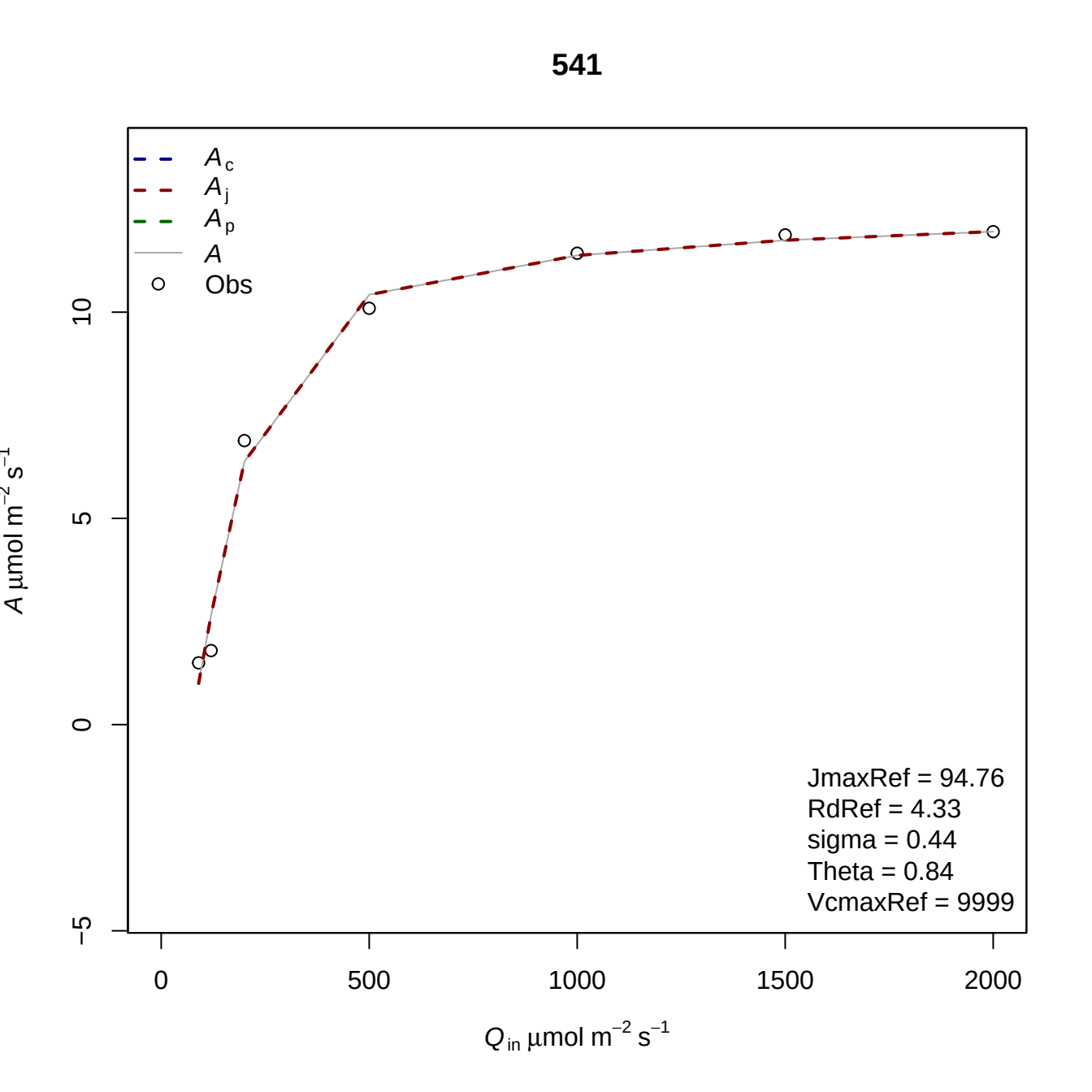

531
A_c
A_j
A_p
A
Obs
JmaxRef = 113.35
RdRef = 1.06
sigma = 0.14
Theta = 0.44
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

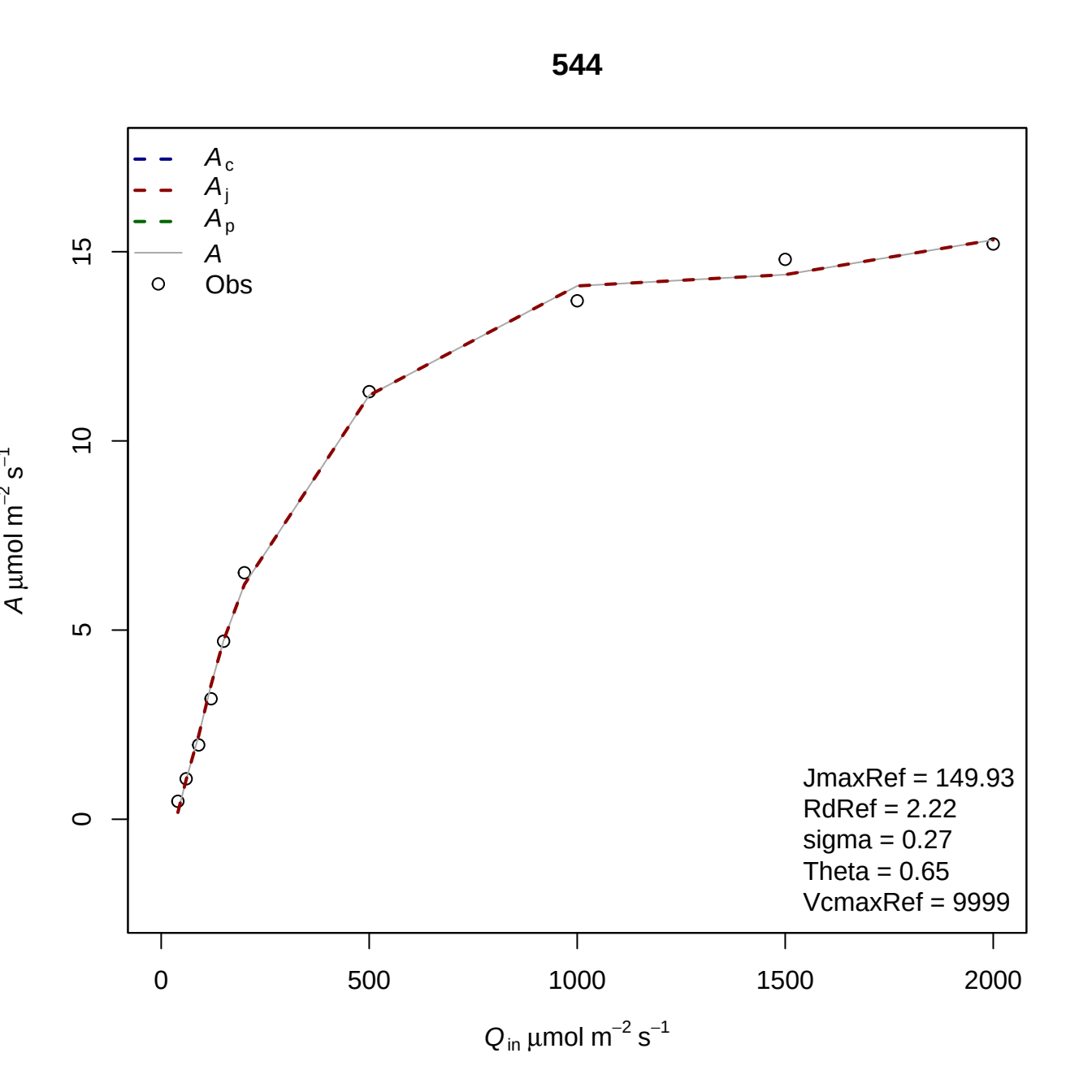

532
A_c
A_j
A_p
A
Obs
JmaxRef = 112.97
RdRef = 1.95
sigma = 0.3
Theta = 0.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

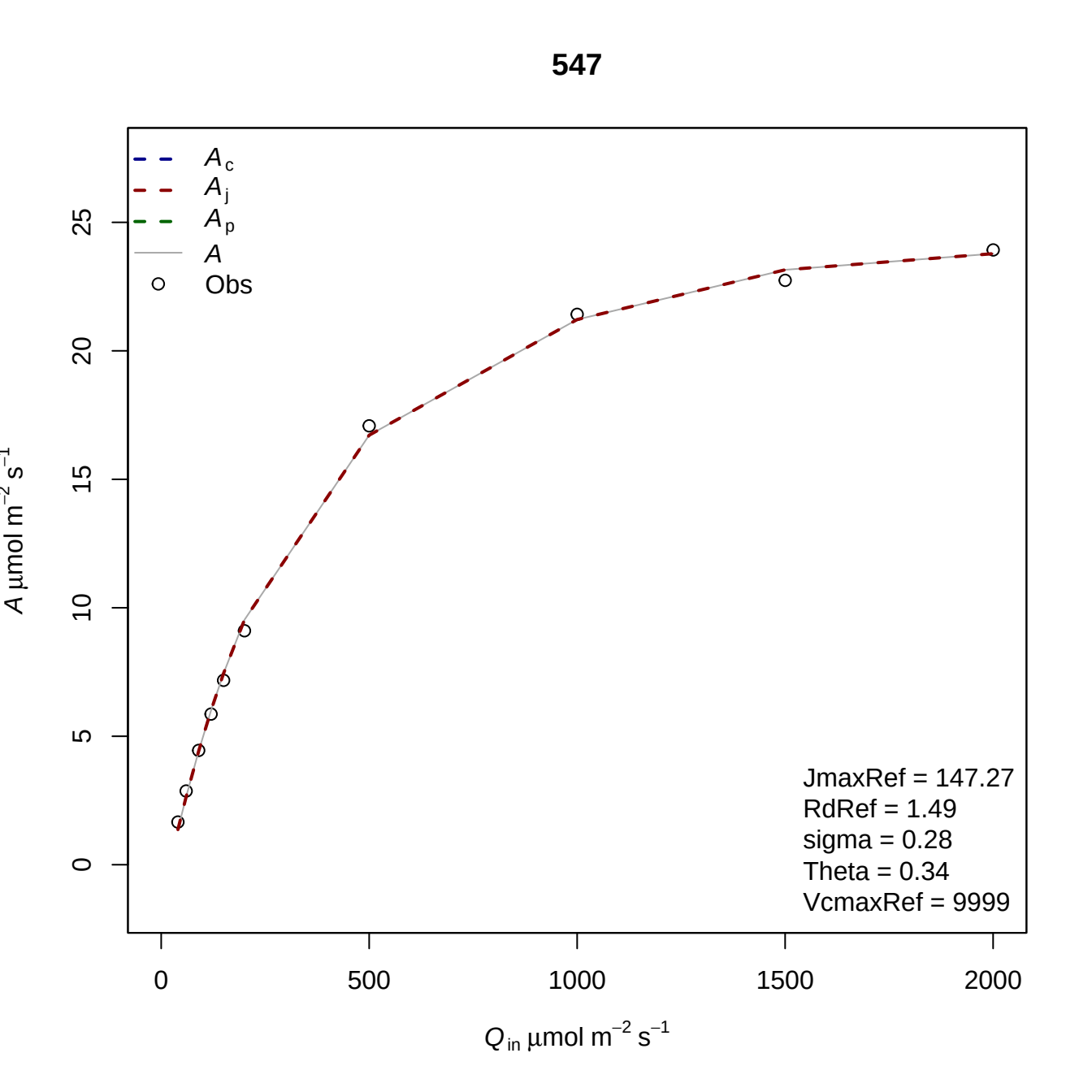

535
A_c
A_j
A_p
A
Obs
JmaxRef = 102.51
RdRef = 1.36
sigma = 0.28
Theta = 0.23
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

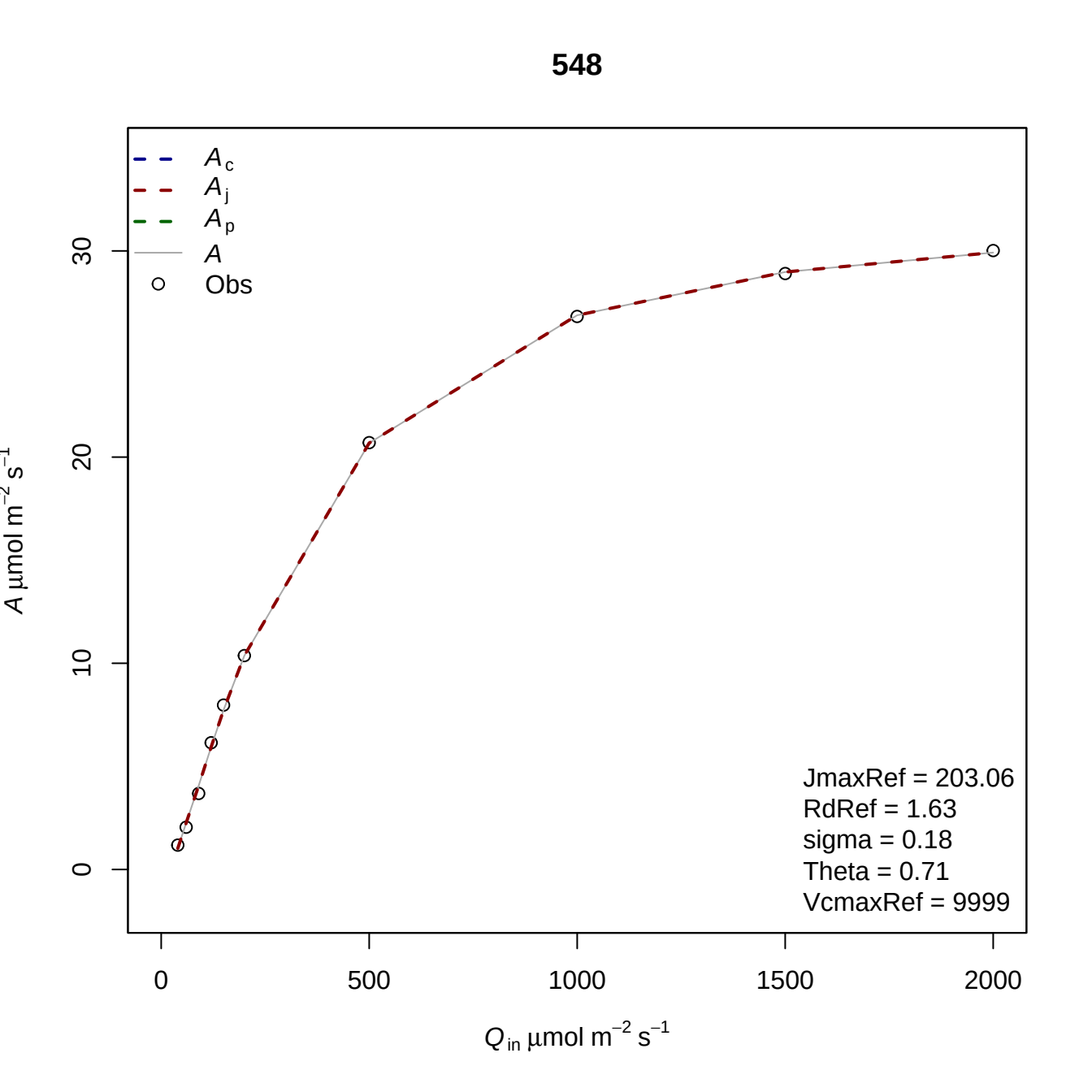

536
A_c
A_j
A_p
A
Obs
JmaxRef = 126.3
RdRef = 1.03
sigma = 0.11
Theta = 0.38
VcmaxRef = 55.64
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

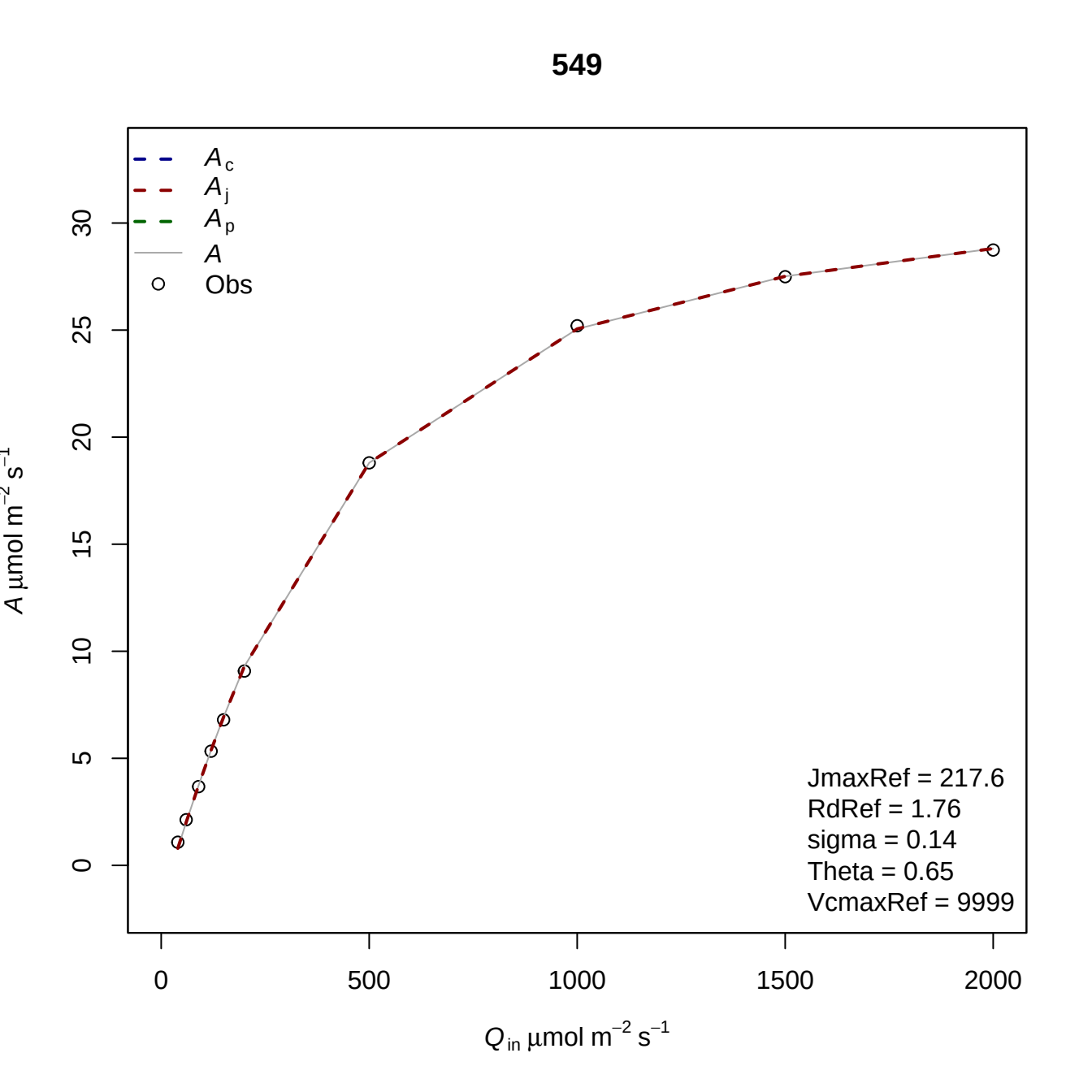

537
A_c
A_j
A_p
A
Obs
JmaxRef = 101.17
RdRef = 1.42
sigma = 0.15
Theta = 0.53
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

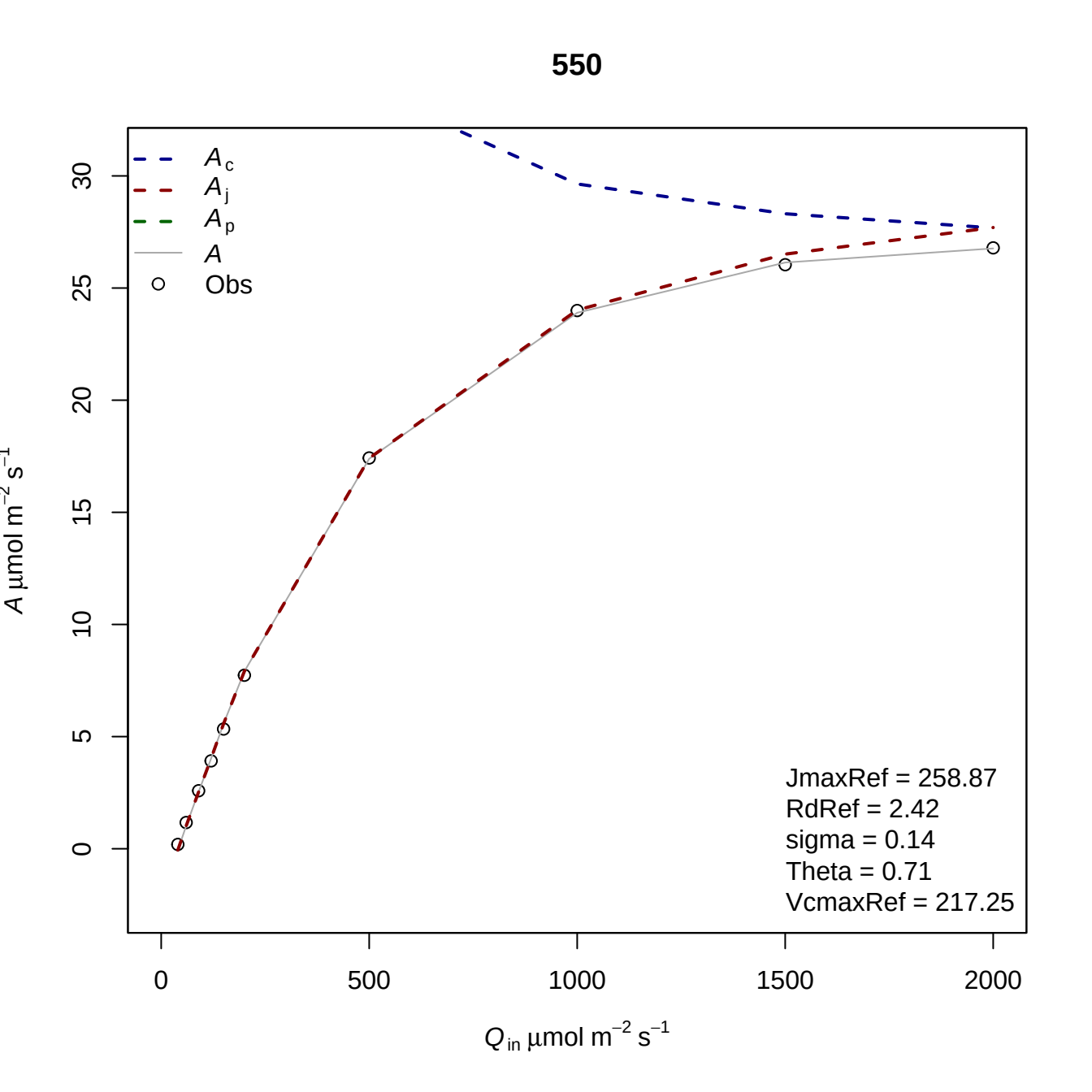

541
A_c
A_j
A_p
A
Obs
JmaxRef = 94.76
RdRef = 4.33
sigma = 0.44
Theta = 0.84
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

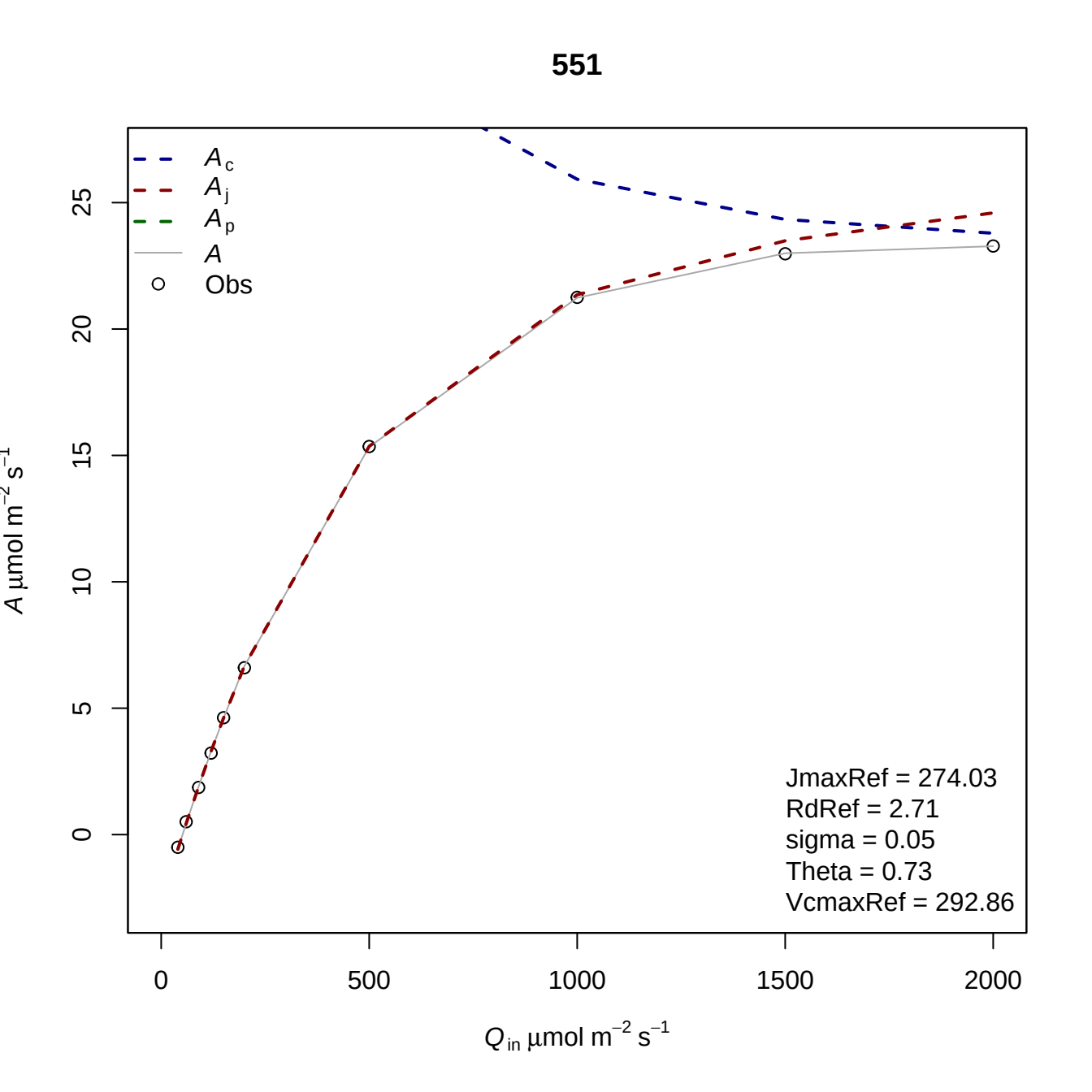

544
A_c
A_j
A_p
A
Obs
JmaxRef = 149.93
RdRef = 2.22
sigma = 0.27
Theta = 0.65
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

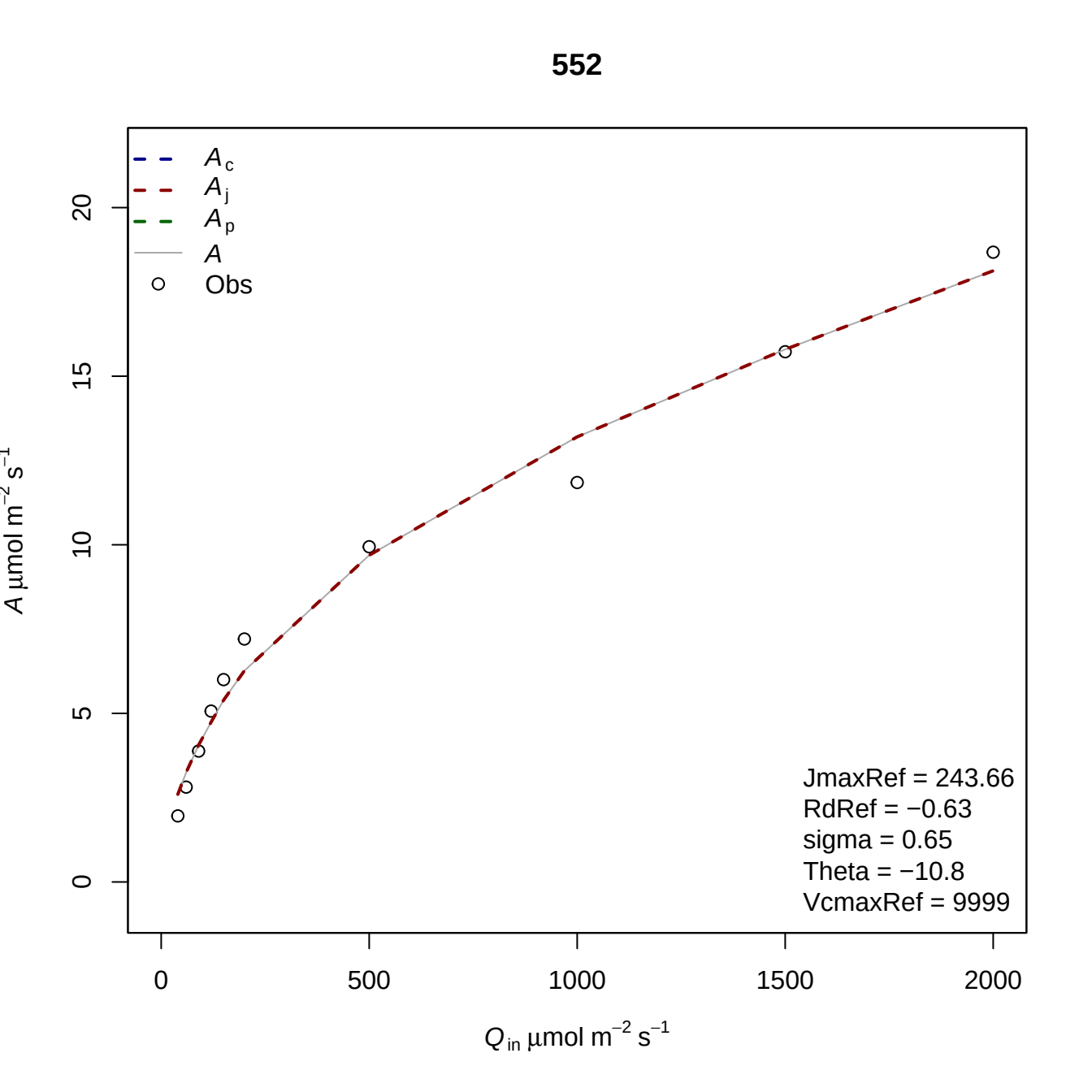

547
A_c
A_j
A_p
A
Obs
JmaxRef = 147.27
RdRef = 1.49
sigma = 0.28
Theta = 0.34
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

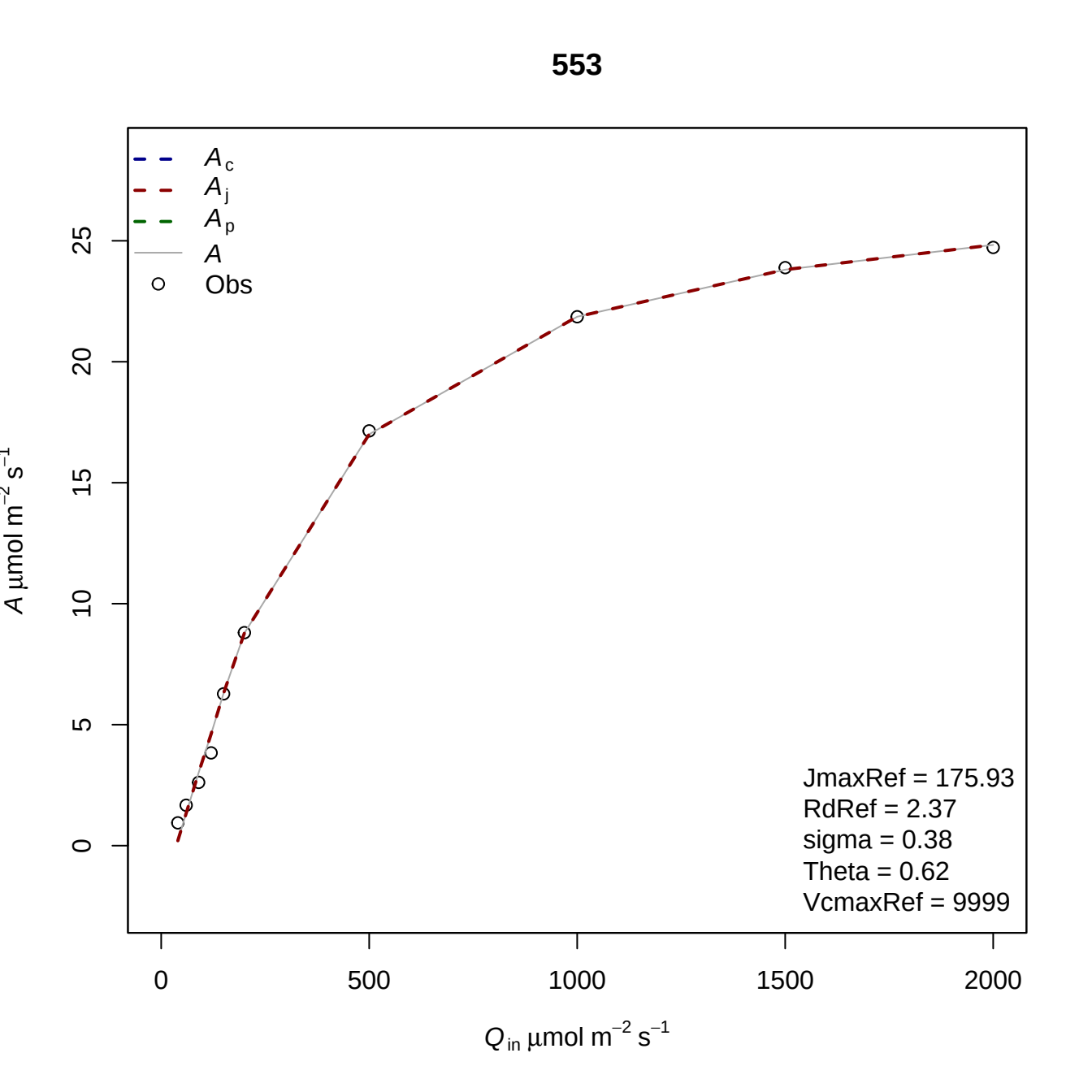

548
A_c
A_j
A_p
A
Obs
JmaxRef = 203.06
RdRef = 1.63
sigma = 0.18
Theta = 0.71
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

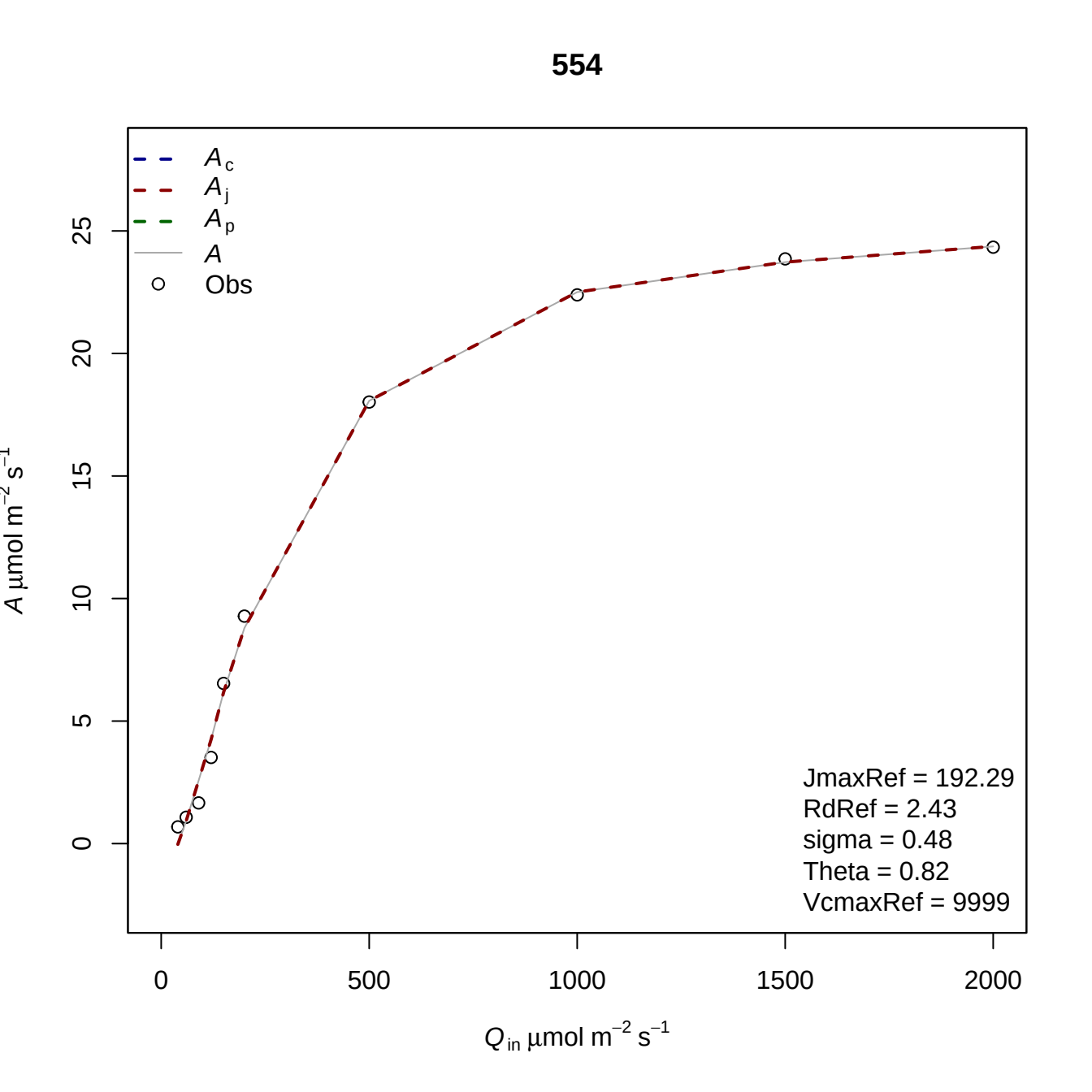

549
A_c
A_j
A_p
A
Obs
JmaxRef = 217.6
RdRef = 1.76
sigma = 0.14
Theta = 0.65
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

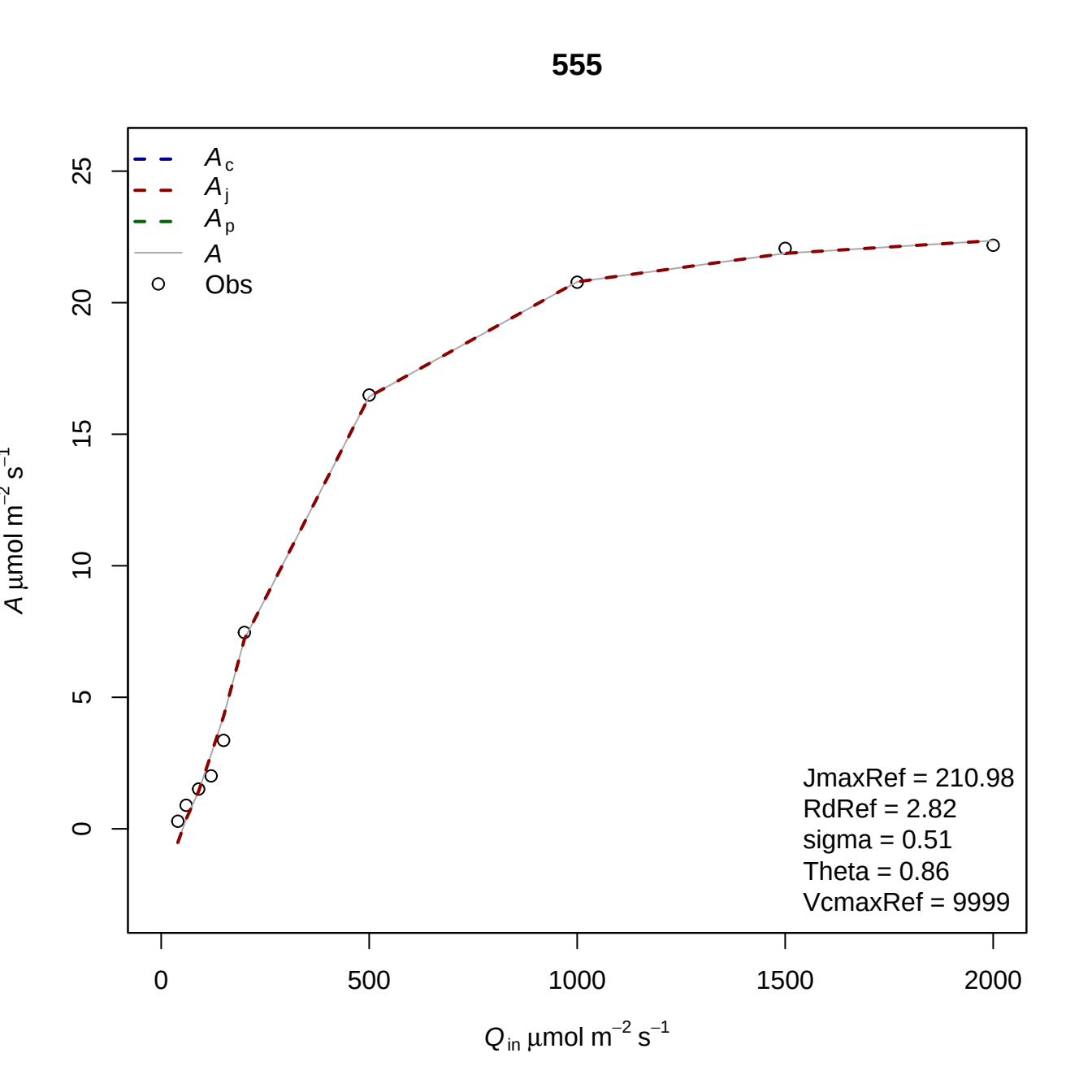

550
A_c
A_j
A_p
A
Obs
JmaxRef = 258.87
RdRef = 2.42
sigma = 0.14
Theta = 0.71
VcmaxRef = 217.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

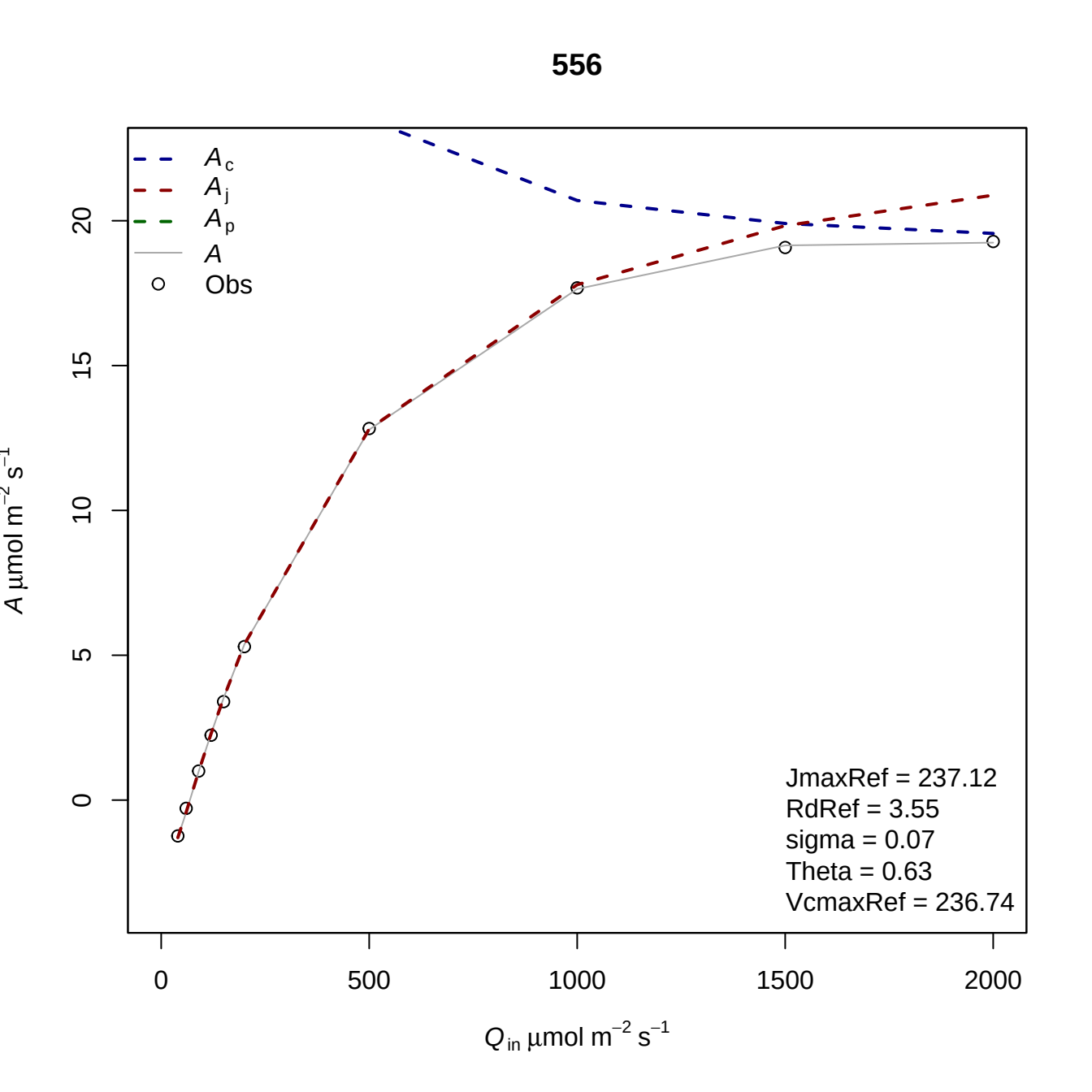

551
A_c
A_j
A_p
A
Obs
JmaxRef = 274.03
RdRef = 2.71
sigma = 0.05
Theta = 0.73
VcmaxRef = 292.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

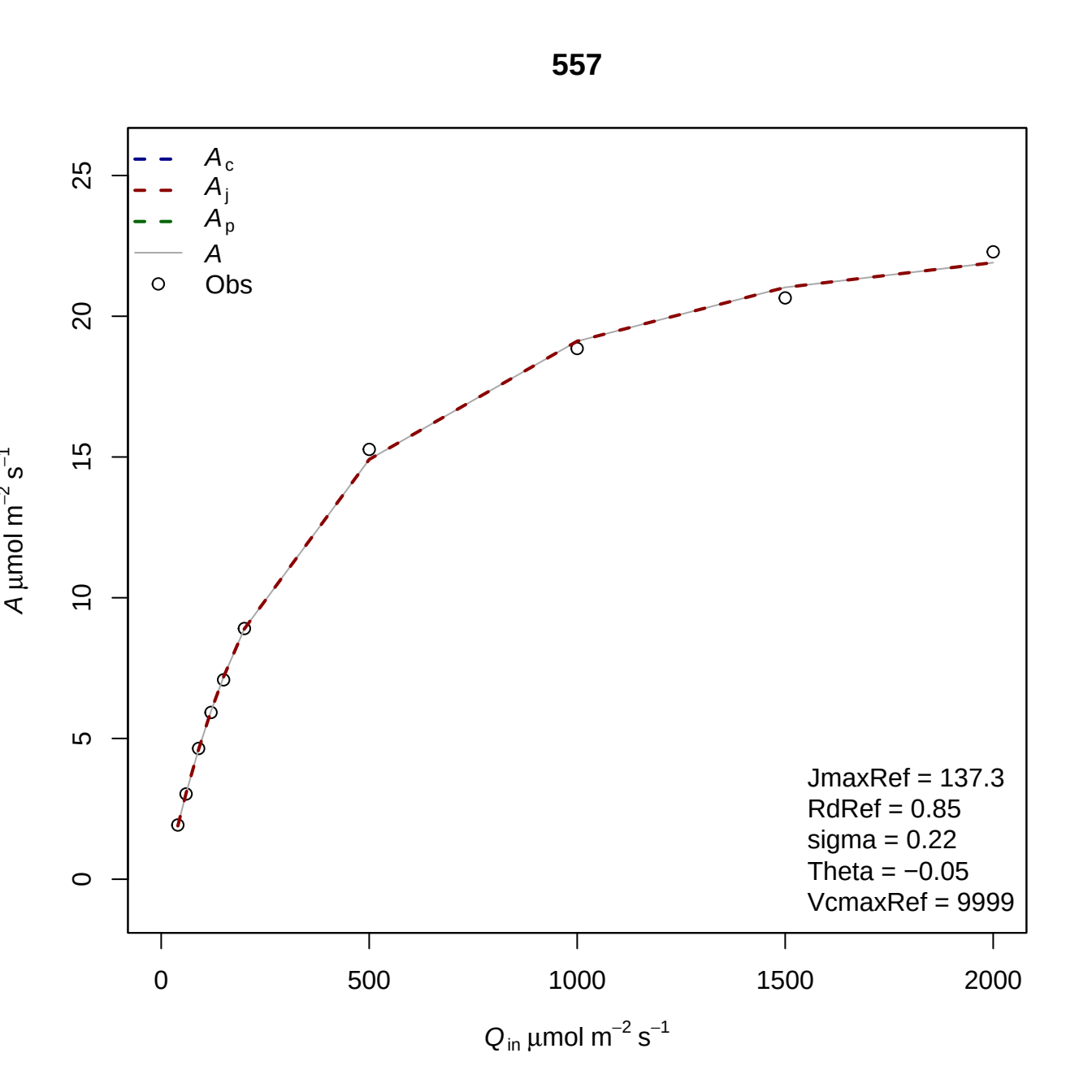

552
A_c
A_j
A_p
A
Obs
JmaxRef = 243.66
RdRef = −0.63
sigma = 0.65
Theta = −10.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

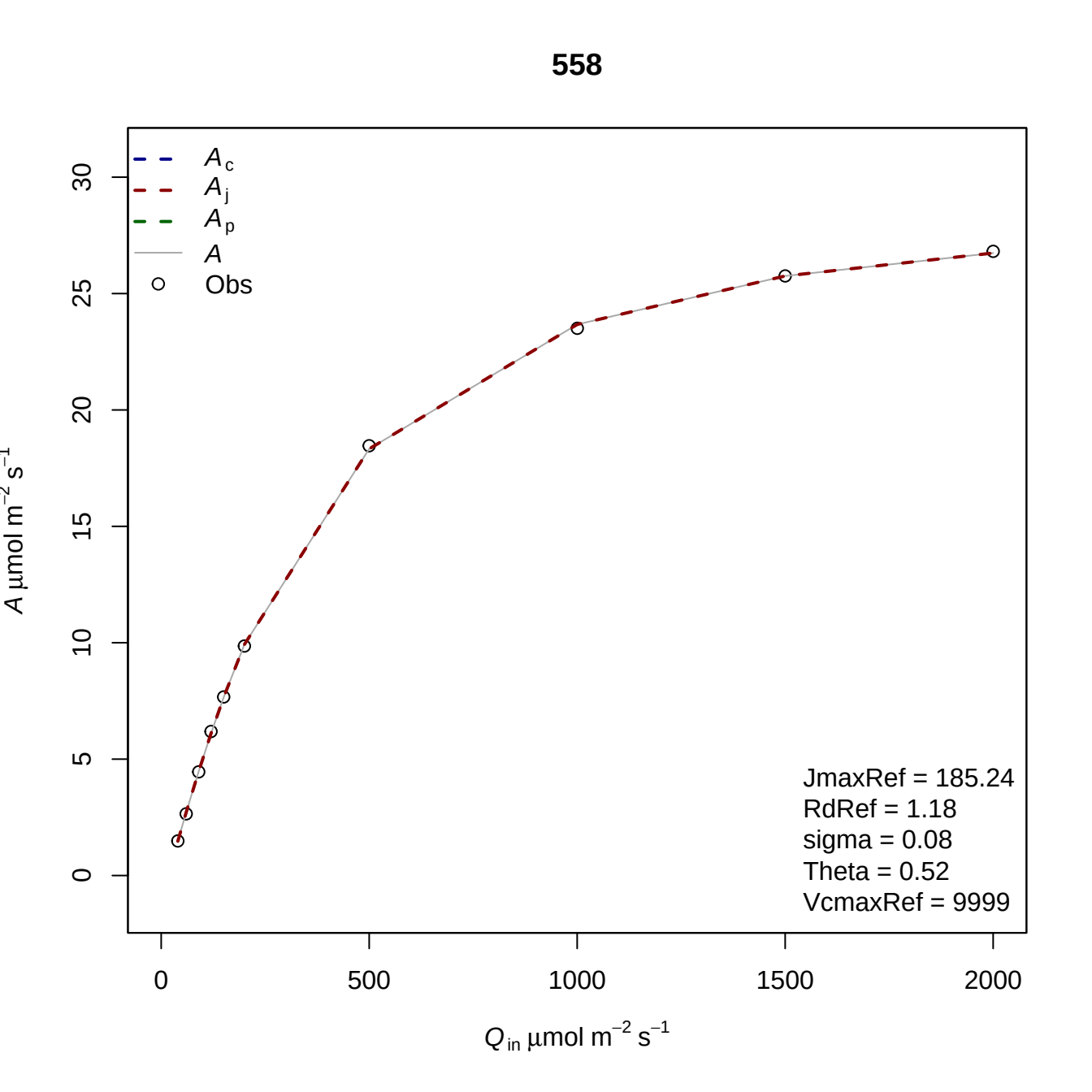

553
A_c
A_j
A_p
A
Obs
JmaxRef = 175.93
RdRef = 2.37
sigma = 0.38
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

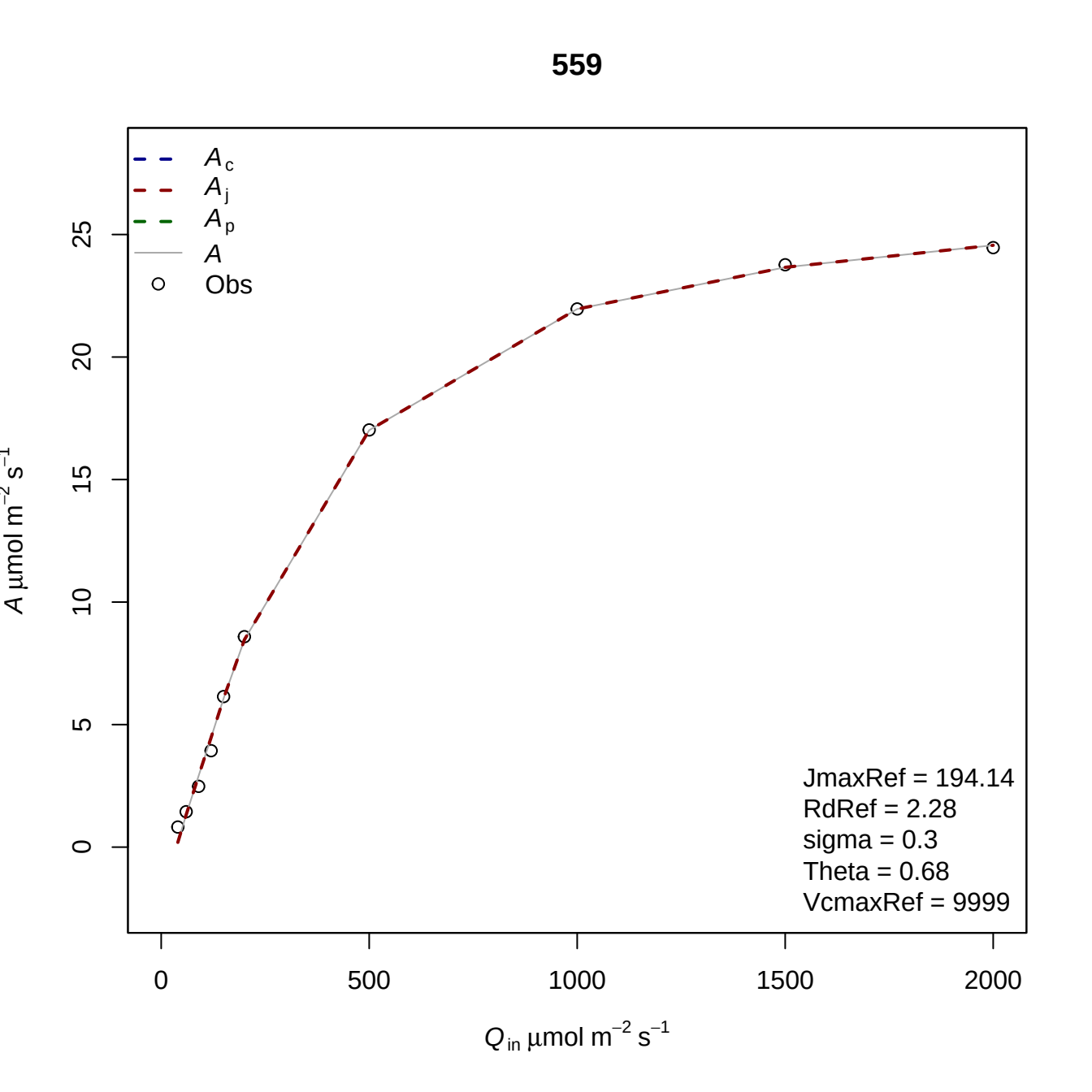

554
A_c
A_j
A_p
A
Obs
JmaxRef = 192.29
RdRef = 2.43
sigma = 0.48
Theta = 0.82
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

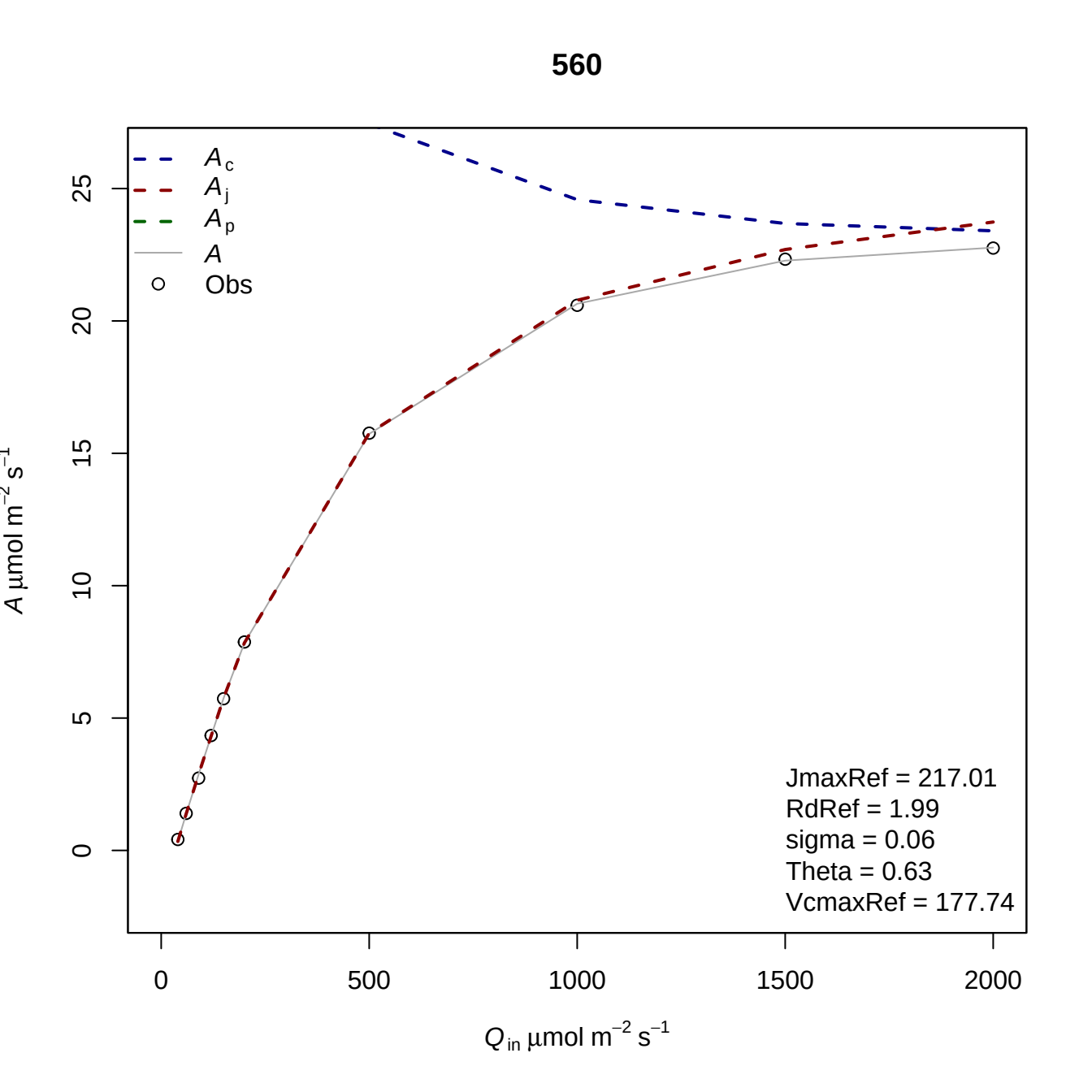

555
A_c
A_j
A_p
A
Obs
JmaxRef = 210.98
RdRef = 2.82
sigma = 0.51
Theta = 0.86
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

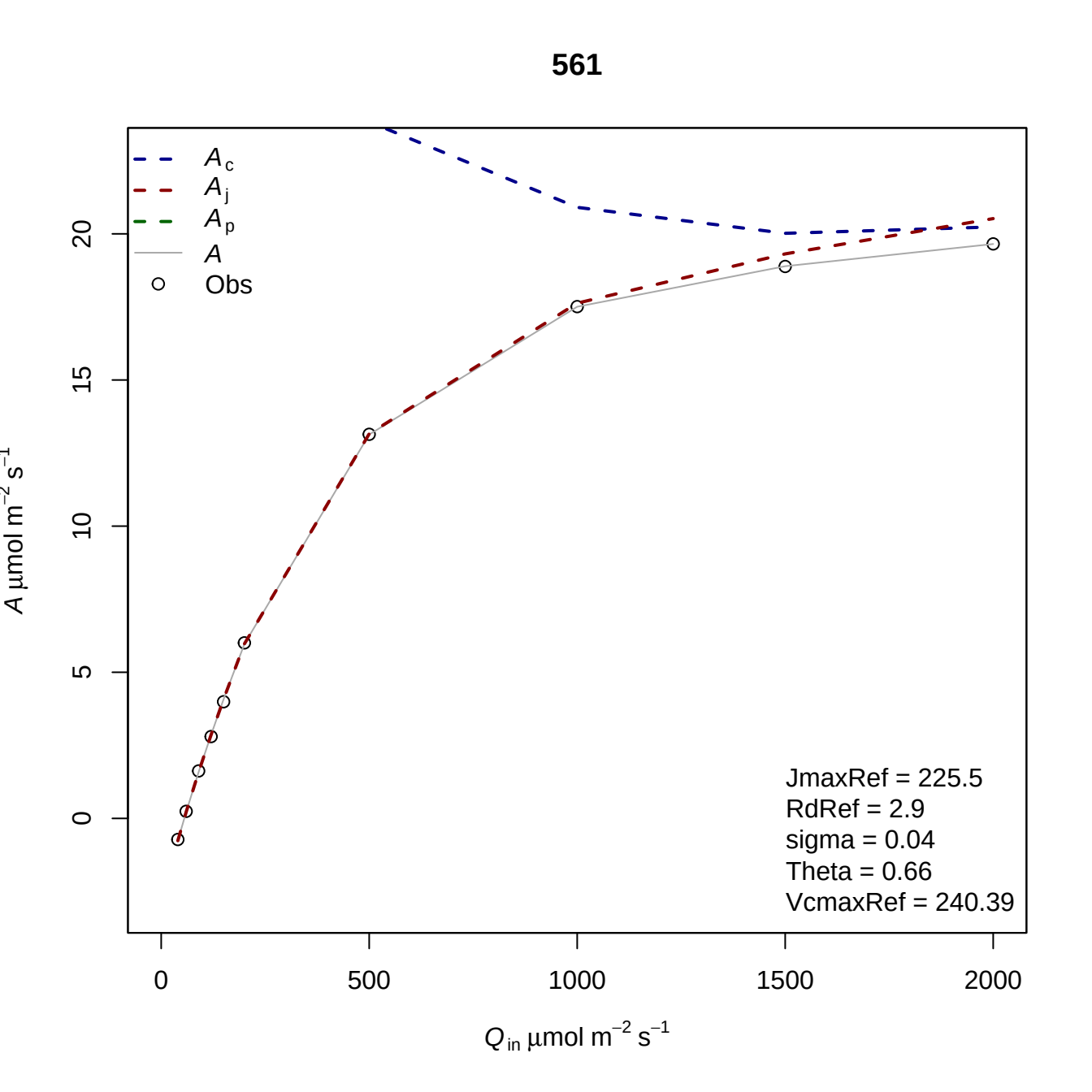

556
A_c
A_j
A_p
A
Obs
JmaxRef = 237.12
RdRef = 3.55
sigma = 0.07
Theta = 0.63
VcmaxRef = 236.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

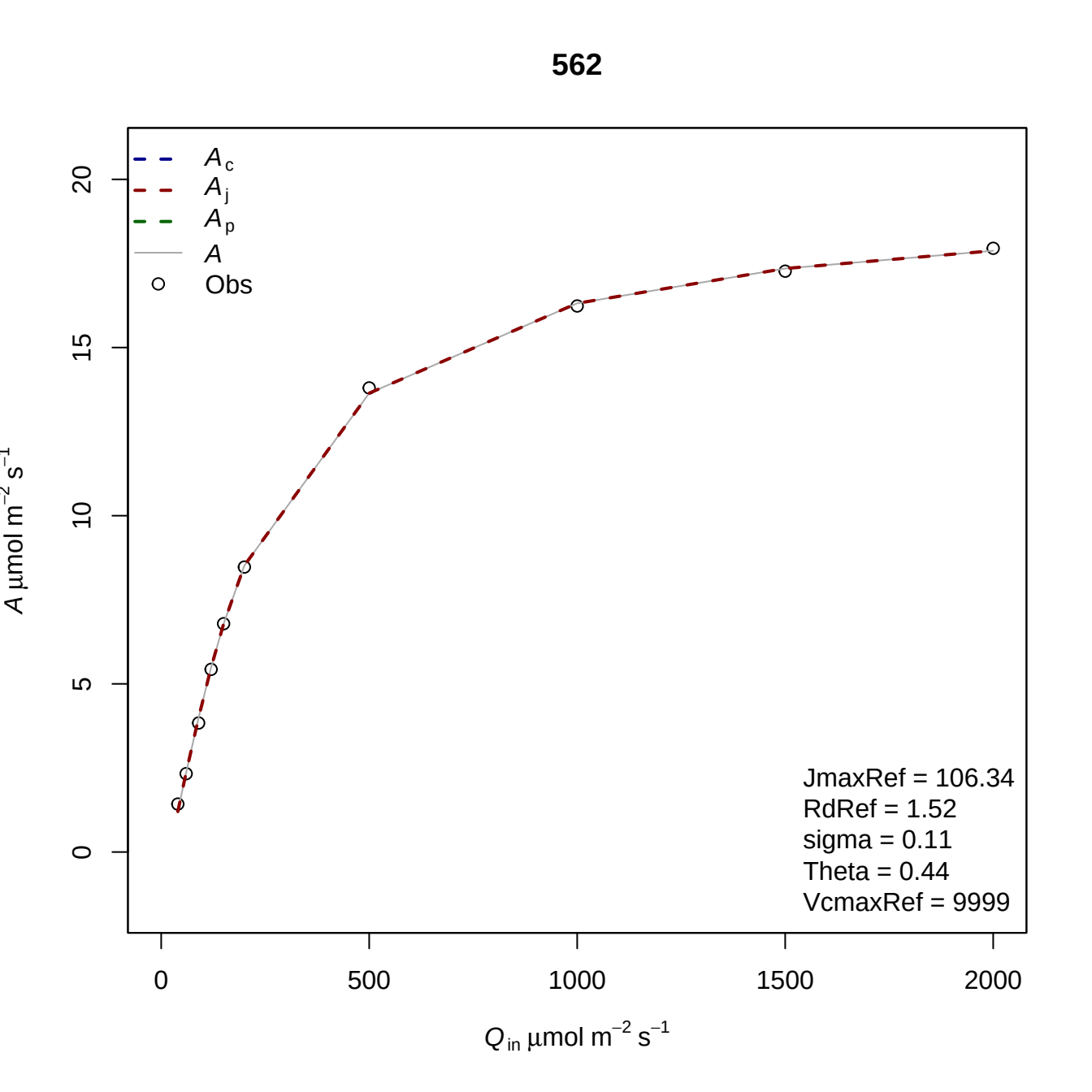

557
A_c
A_j
A_p
A
Obs
JmaxRef = 137.3
RdRef = 0.85
sigma = 0.22
Theta = −0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

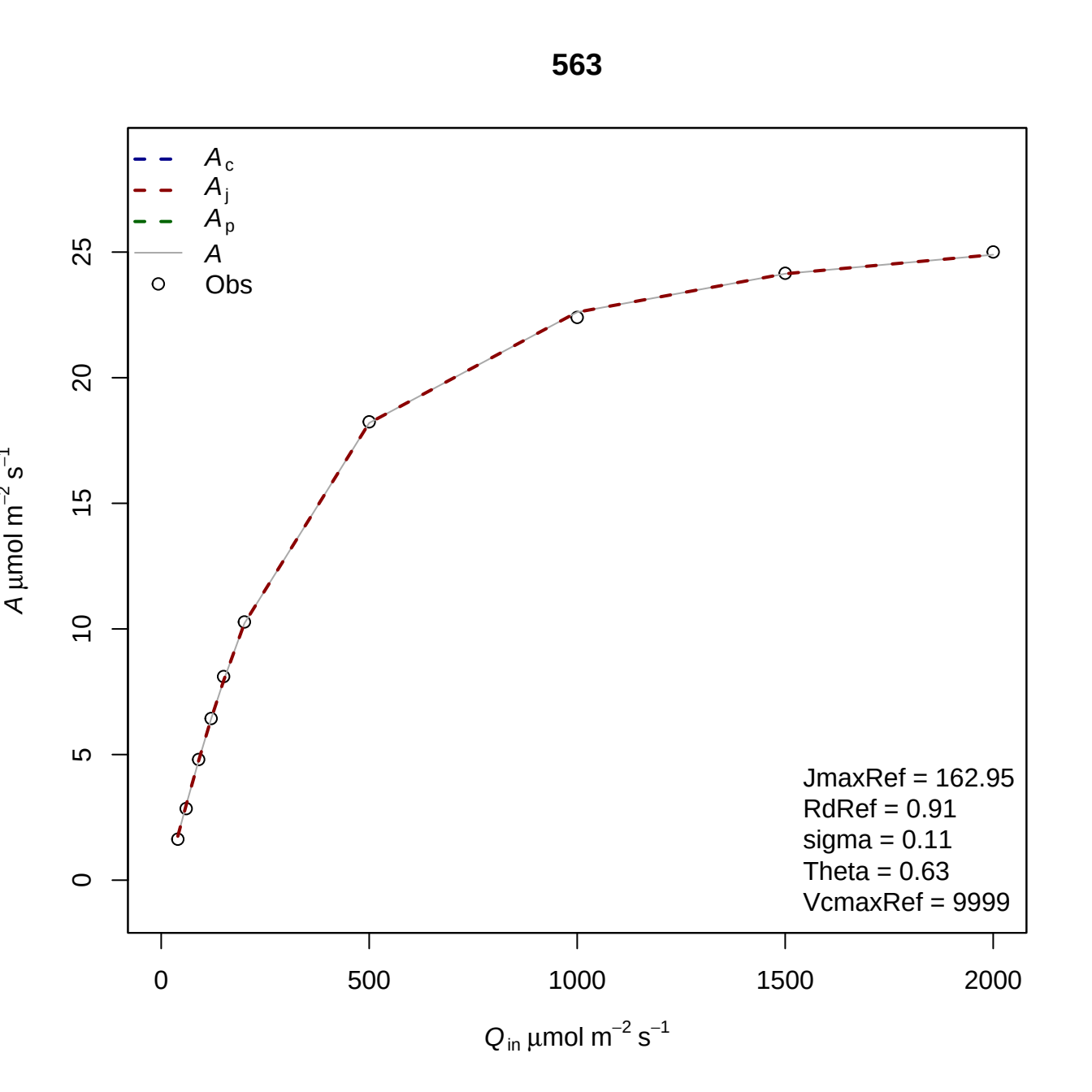

558
A_c
A_j
A_p
A
Obs
JmaxRef = 185.24
RdRef = 1.18
sigma = 0.08
Theta = 0.52
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

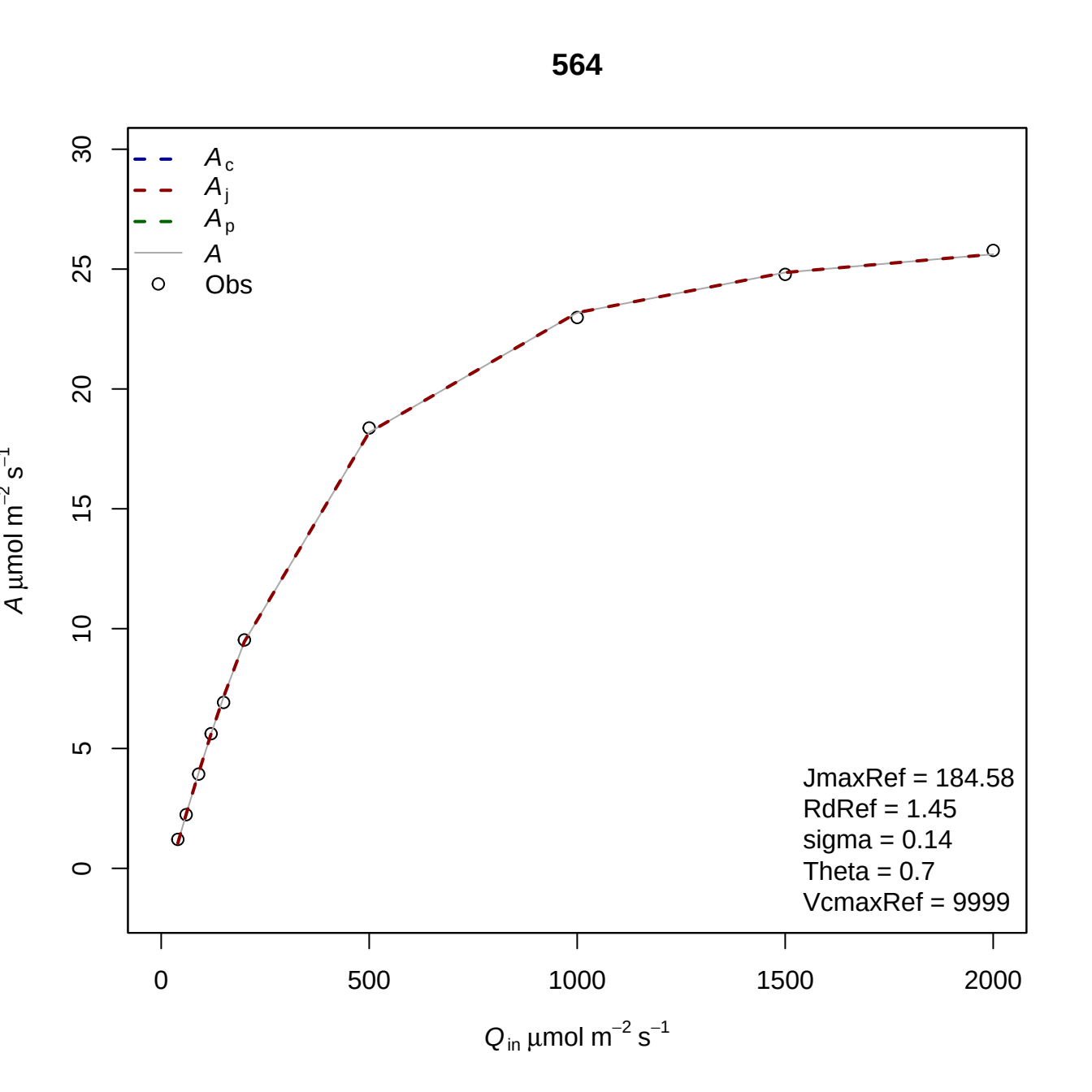

559
A_c
A_j
A_p
A
Obs
JmaxRef = 194.14
RdRef = 2.28
sigma = 0.3
Theta = 0.68
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

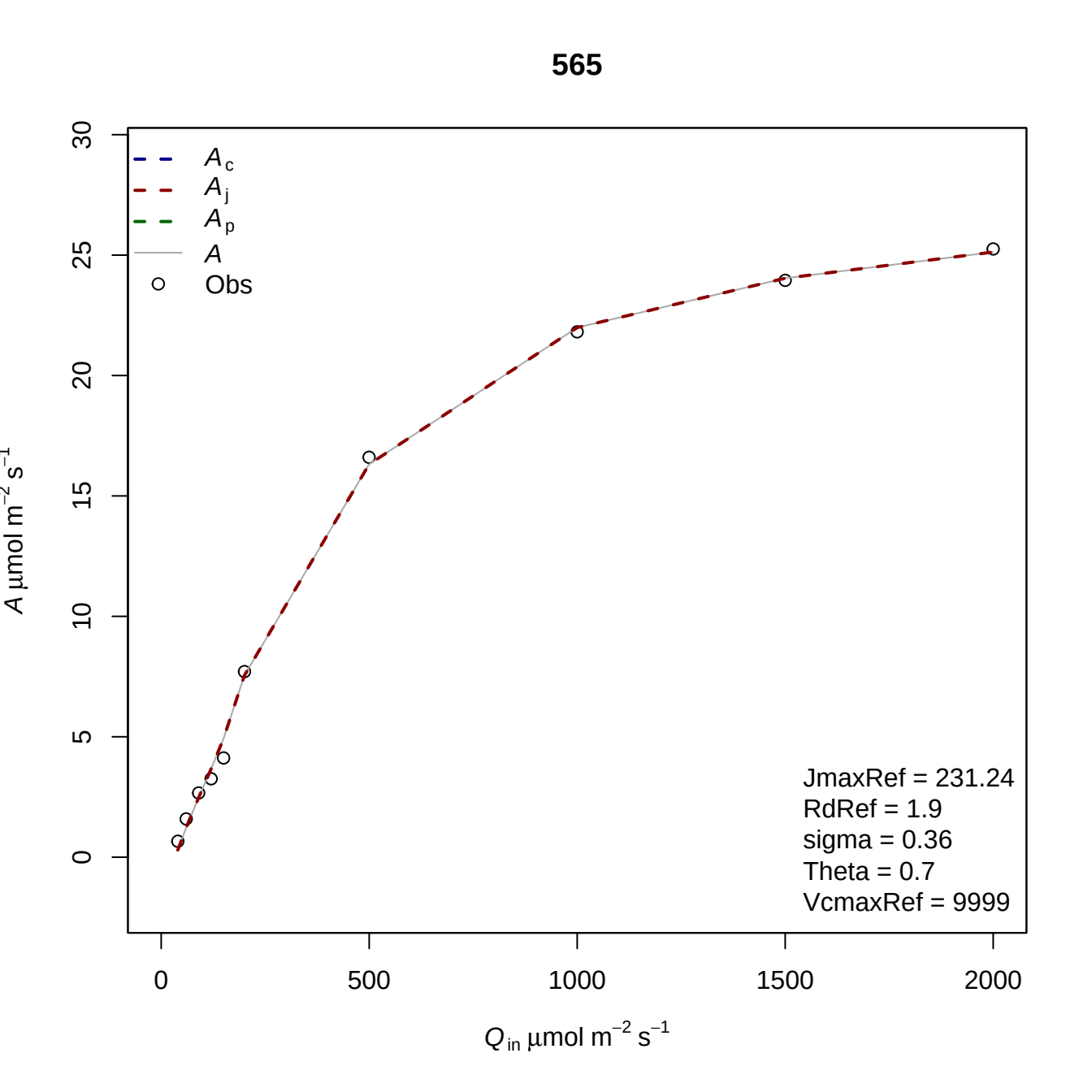

560
A_c
A_j
A_p
A
Obs
JmaxRef = 217.01
RdRef = 1.99
sigma = 0.06
Theta = 0.63
VcmaxRef = 177.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

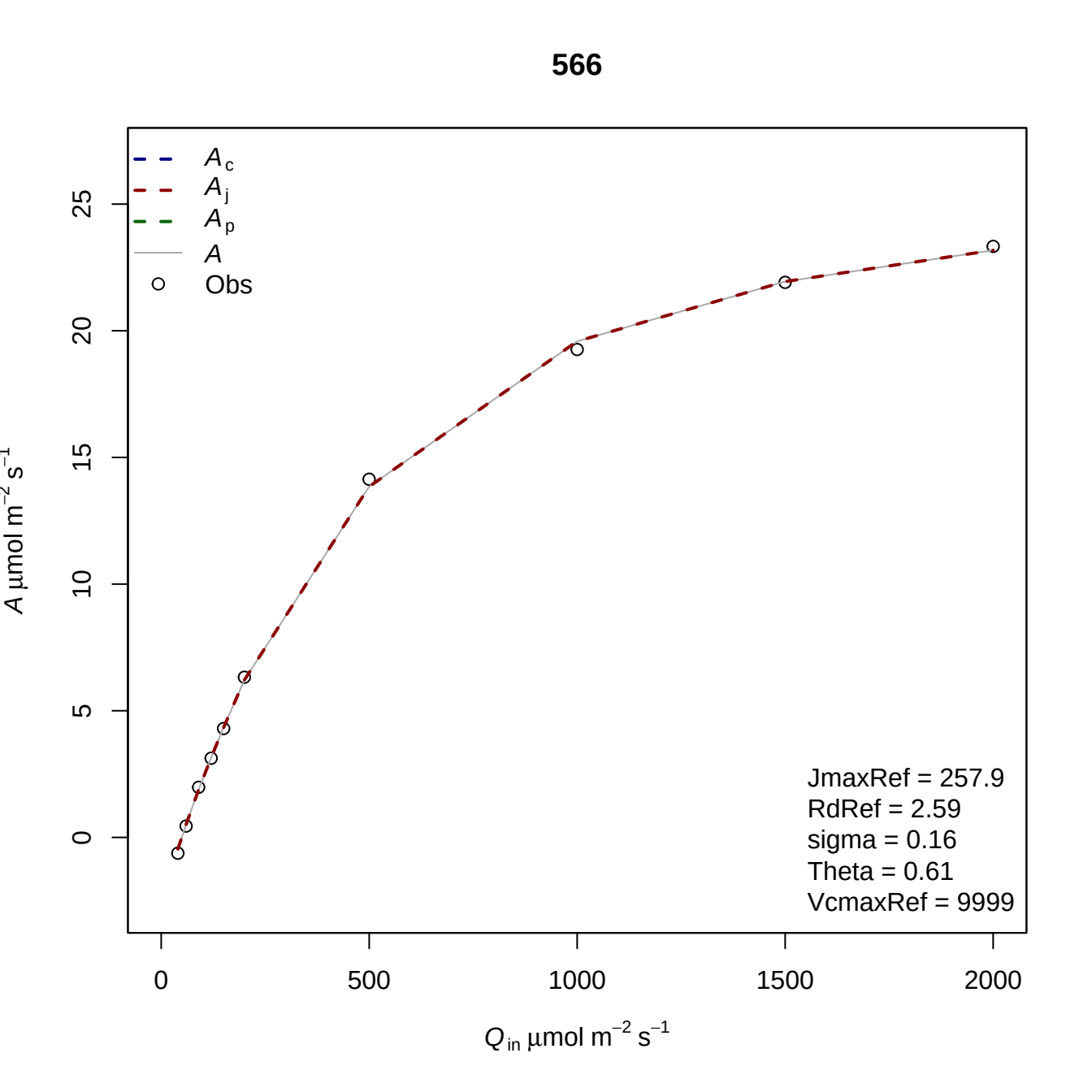

561
A_c
A_j
A_p
A
Obs
JmaxRef = 225.5
RdRef = 2.9
sigma = 0.04
Theta = 0.66
VcmaxRef = 240.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

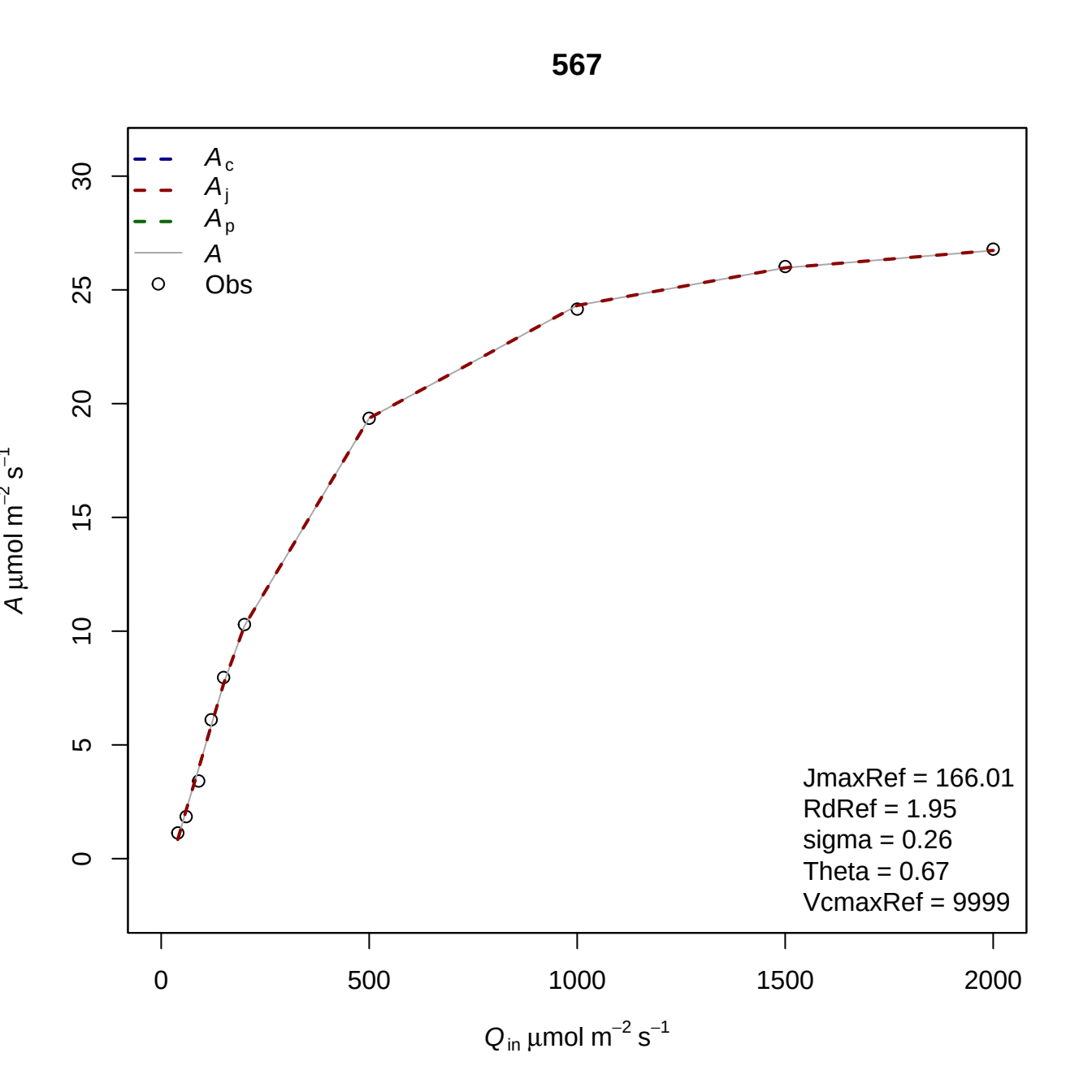

562
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 106.34
RdRef = 1.52
sigma = 0.11
Theta = 0.44
VcmaxRef = 9999

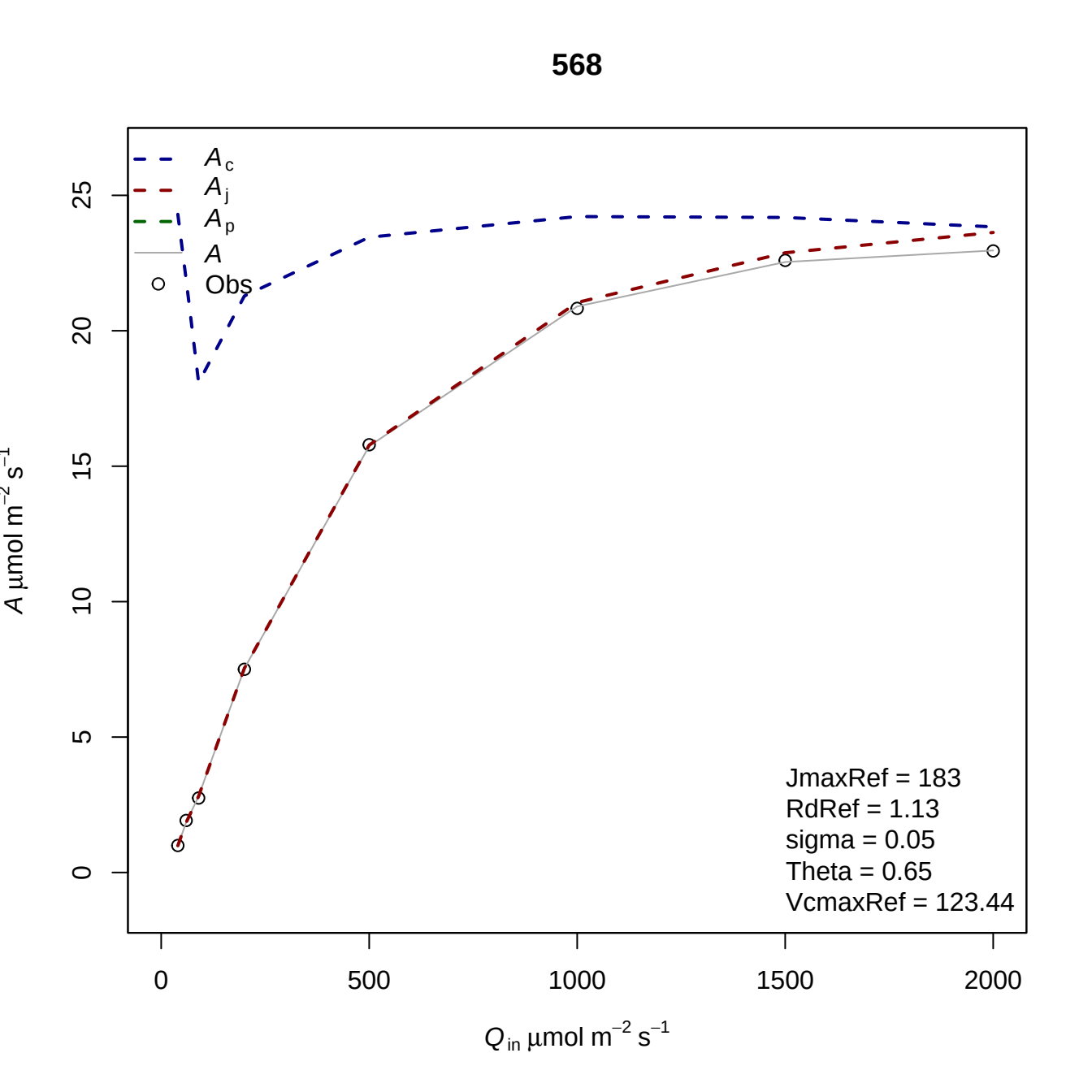

563
A_c
A_j
A_p
A
Obs
JmaxRef = 162.95
RdRef = 0.91
sigma = 0.11
Theta = 0.63
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

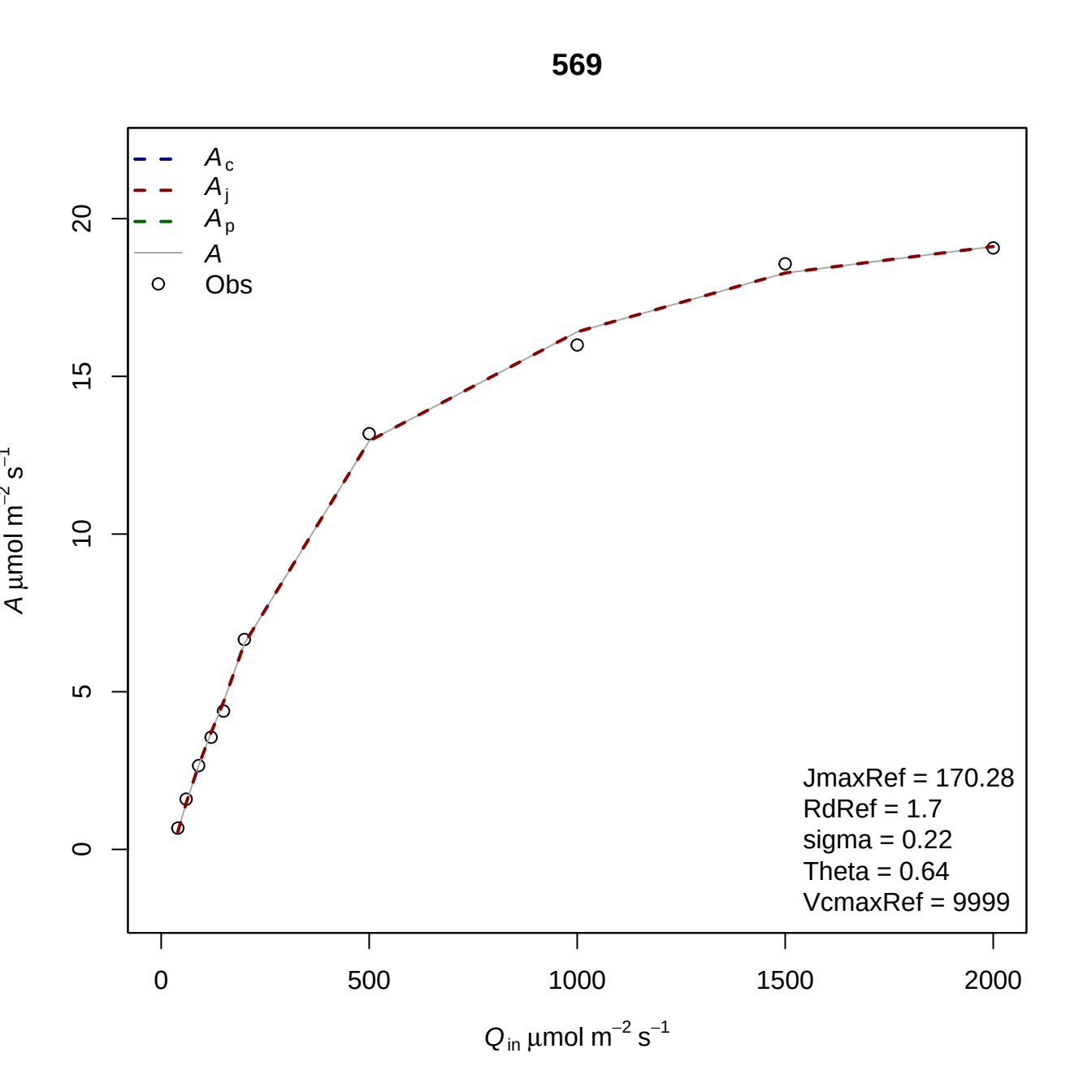

564
A_c
A_j
A_p
A
Obs
JmaxRef = 184.58
RdRef = 1.45
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

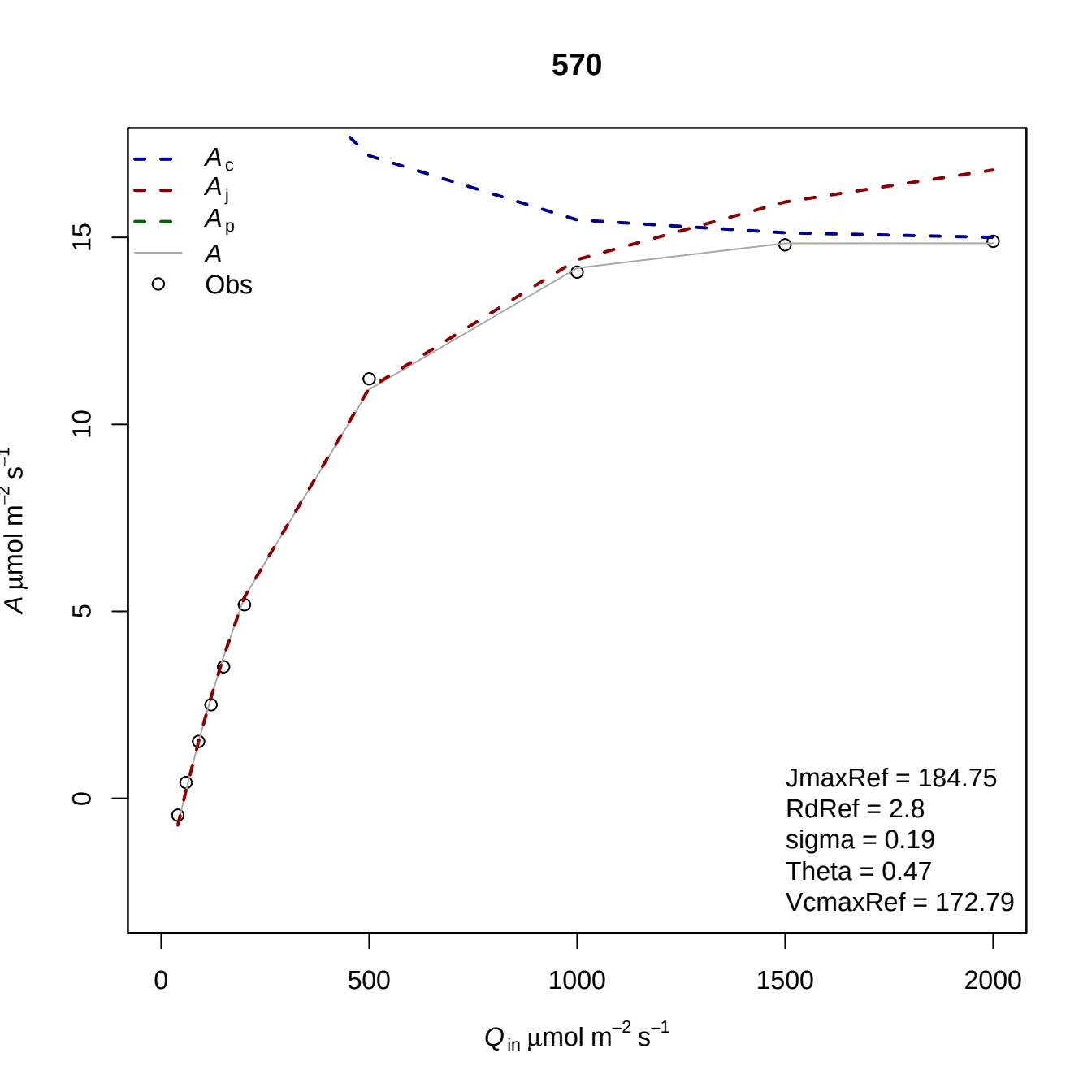

565
A_c
A_j
A_p
A
Obs
JmaxRef = 231.24
RdRef = 1.9
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

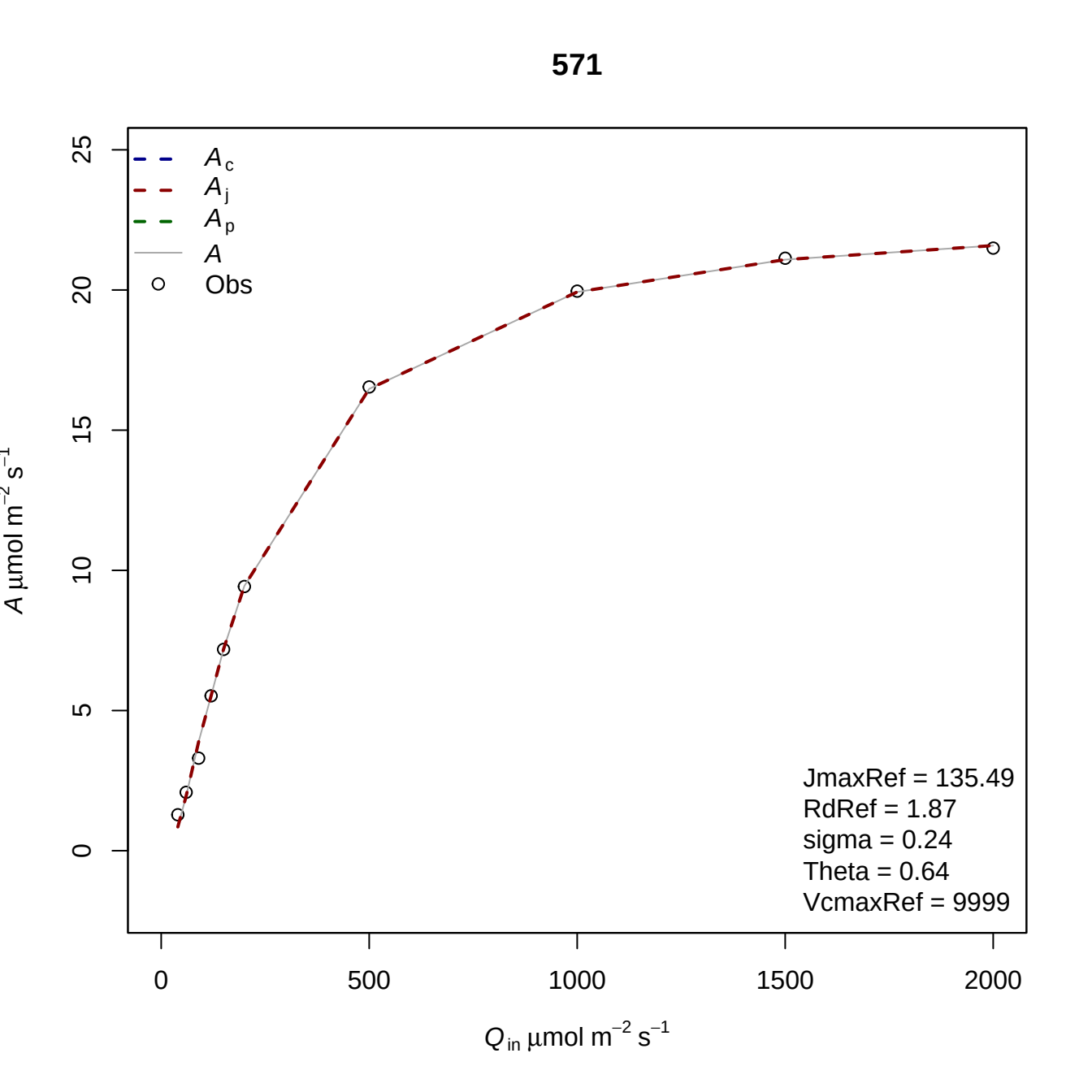

566
A_c
A_j
A_p
A
Obs
JmaxRef = 257.9
RdRef = 2.59
sigma = 0.16
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

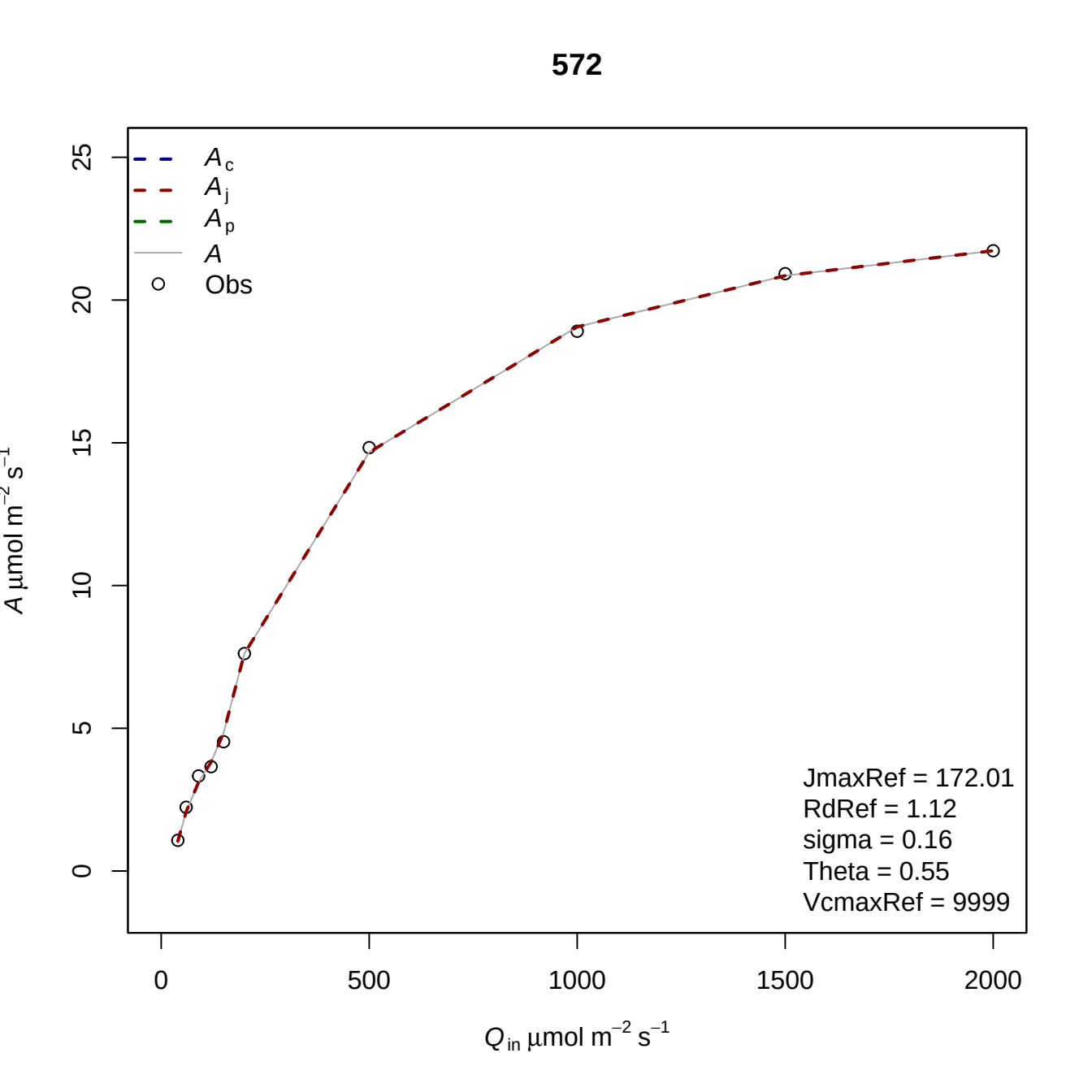

567
A_c
A_j
A_p
A
Obs
JmaxRef = 166.01
RdRef = 1.95
sigma = 0.26
Theta = 0.67
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

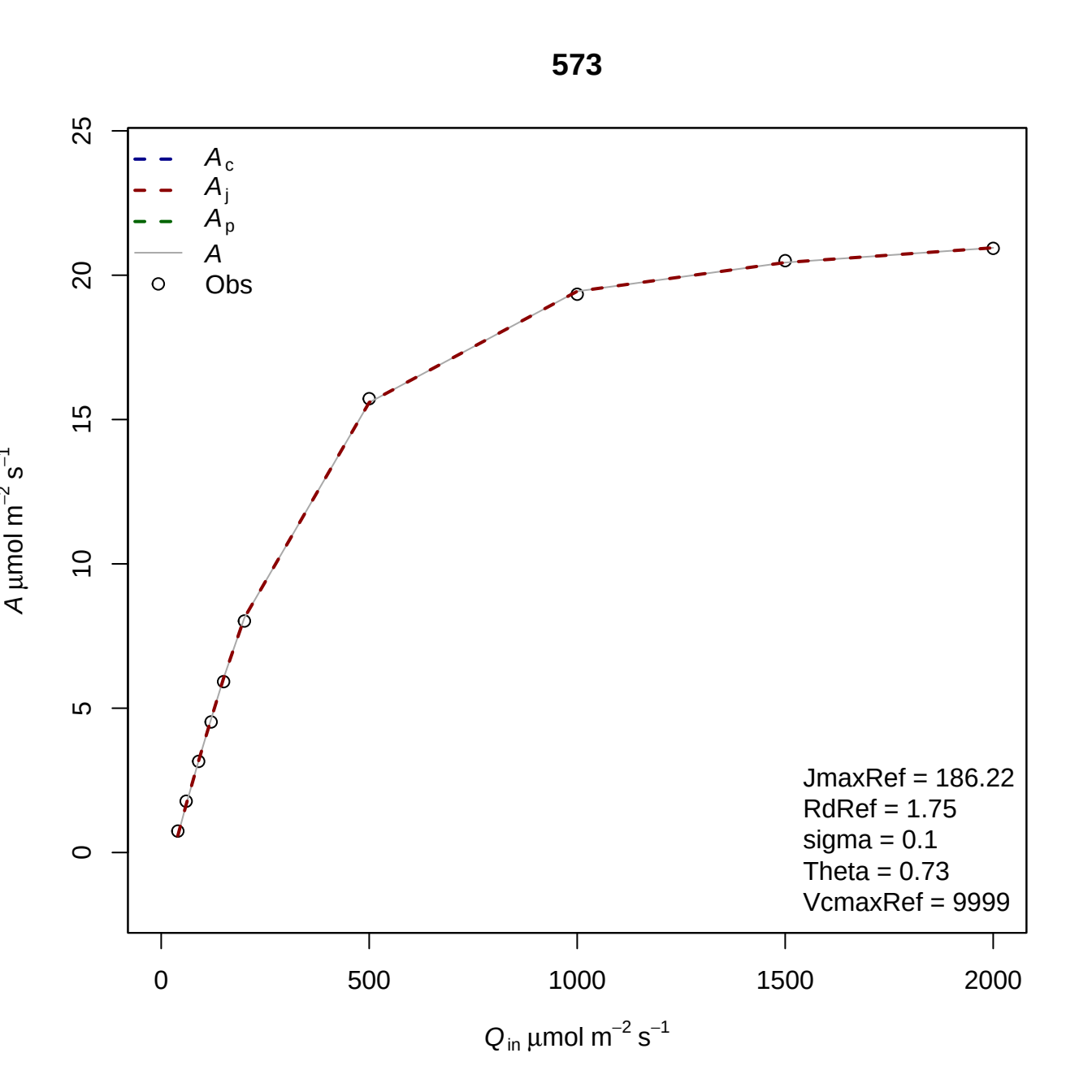

568
A_c
A_j
A_p
A
Obs
JmaxRef = 183
RdRef = 1.13
sigma = 0.05
Theta = 0.65
VcmaxRef = 123.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

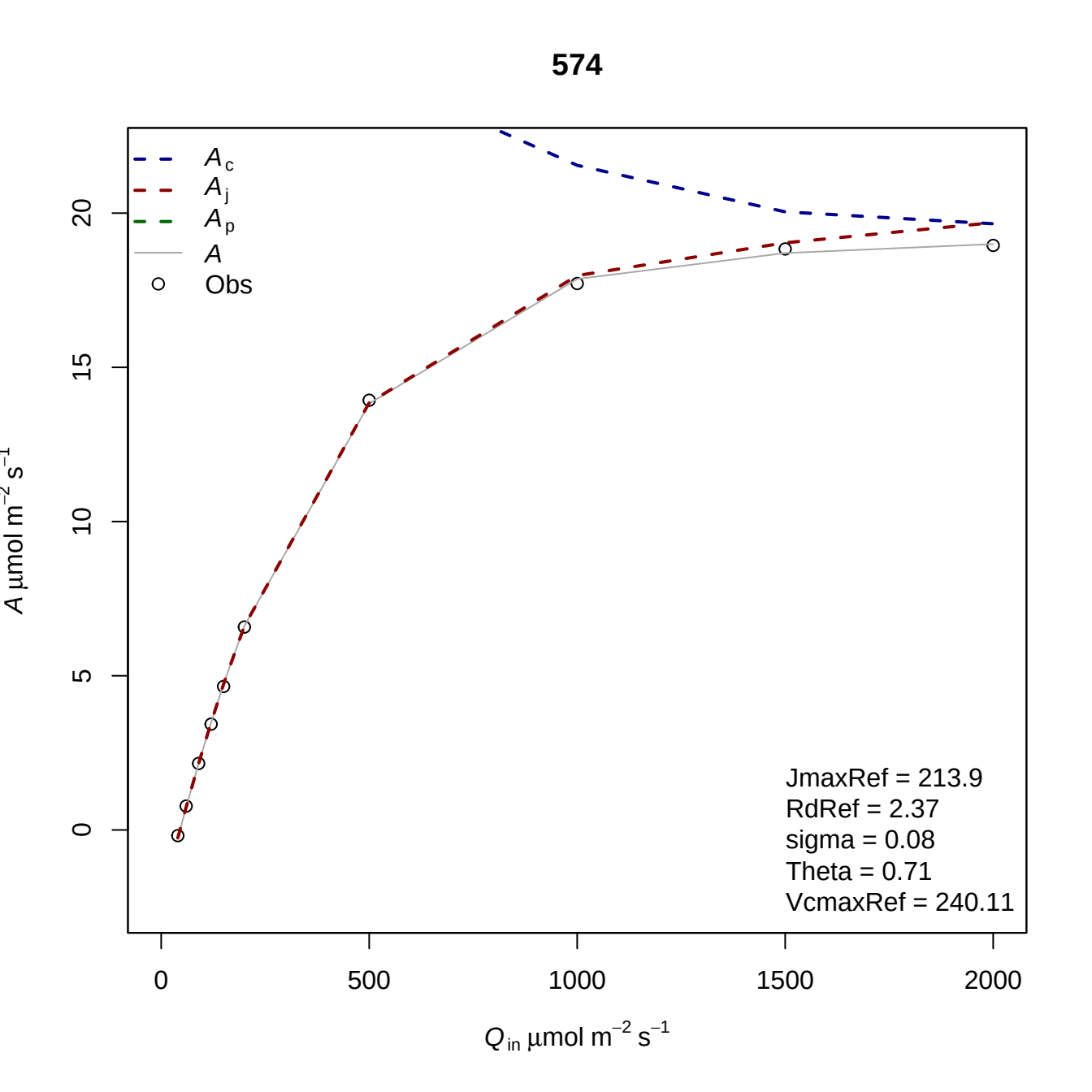

569
A_c
A_j
A_p
A
Obs
JmaxRef = 170.28
RdRef = 1.7
sigma = 0.22
Theta = 0.64
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

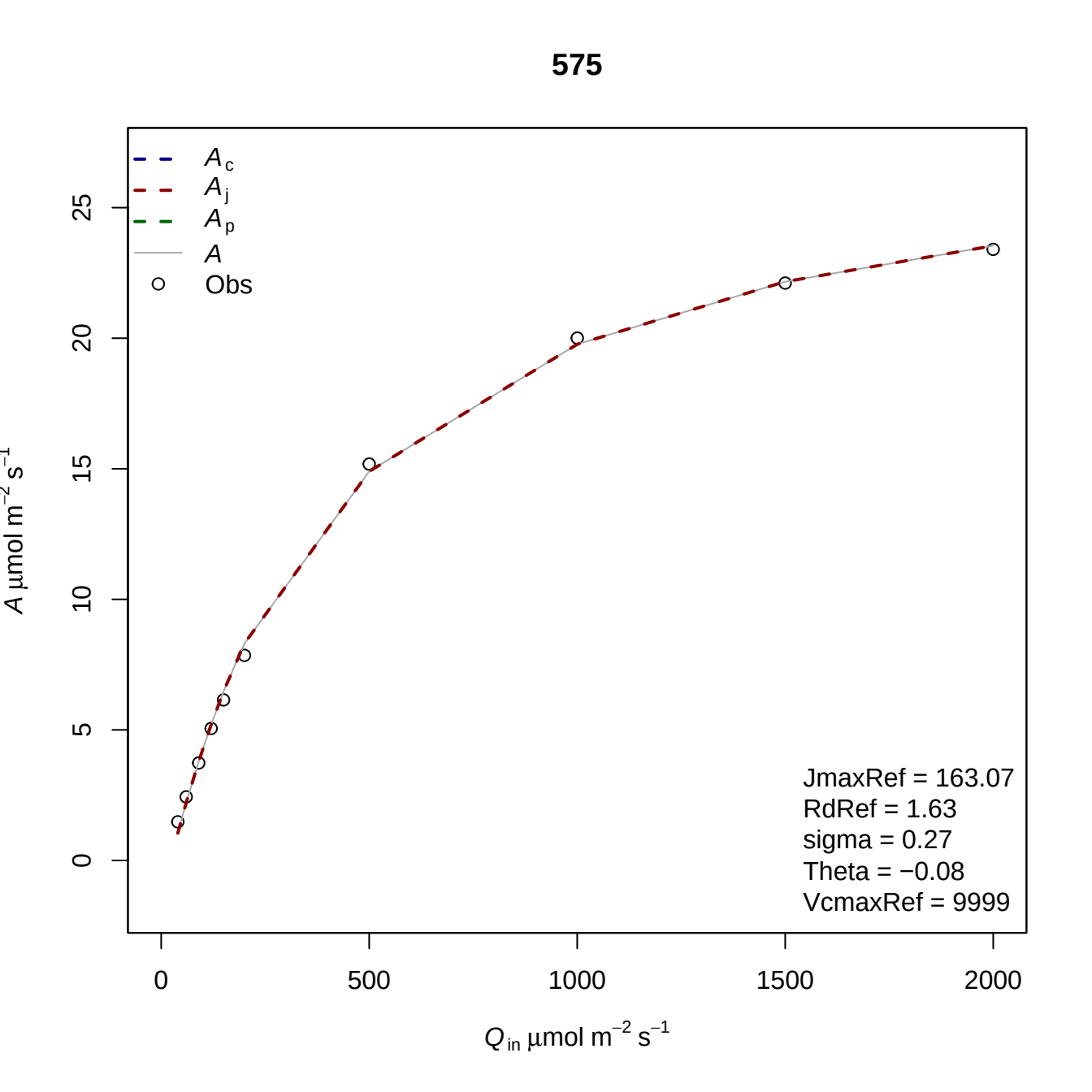

570
A_c
A_j
A_p
A
Obs
JmaxRef = 184.75
RdRef = 2.8
sigma = 0.19
Theta = 0.47
VcmaxRef = 172.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

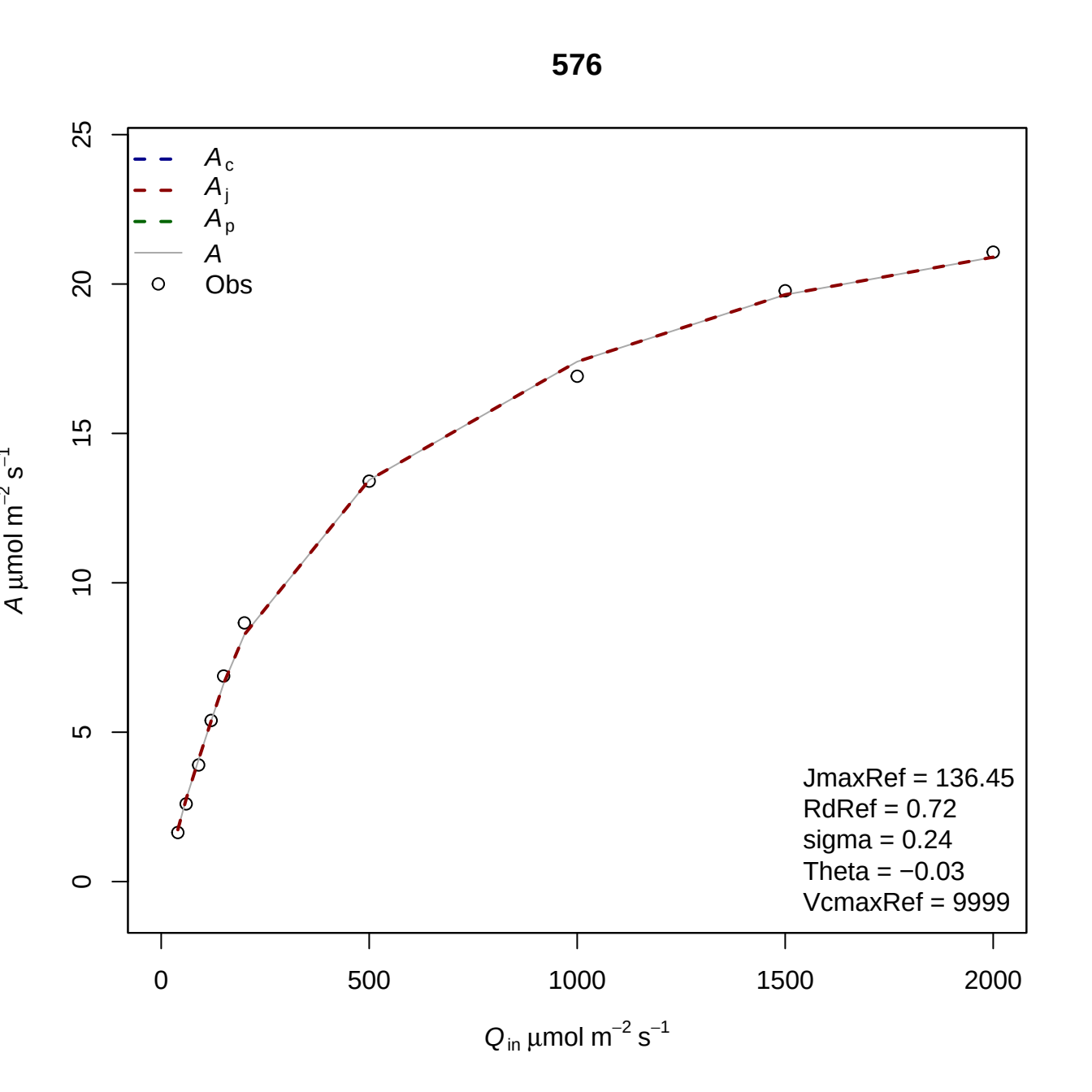

571
A_c
A_j
A_p
A
Obs
JmaxRef = 135.49
RdRef = 1.87
sigma = 0.24
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

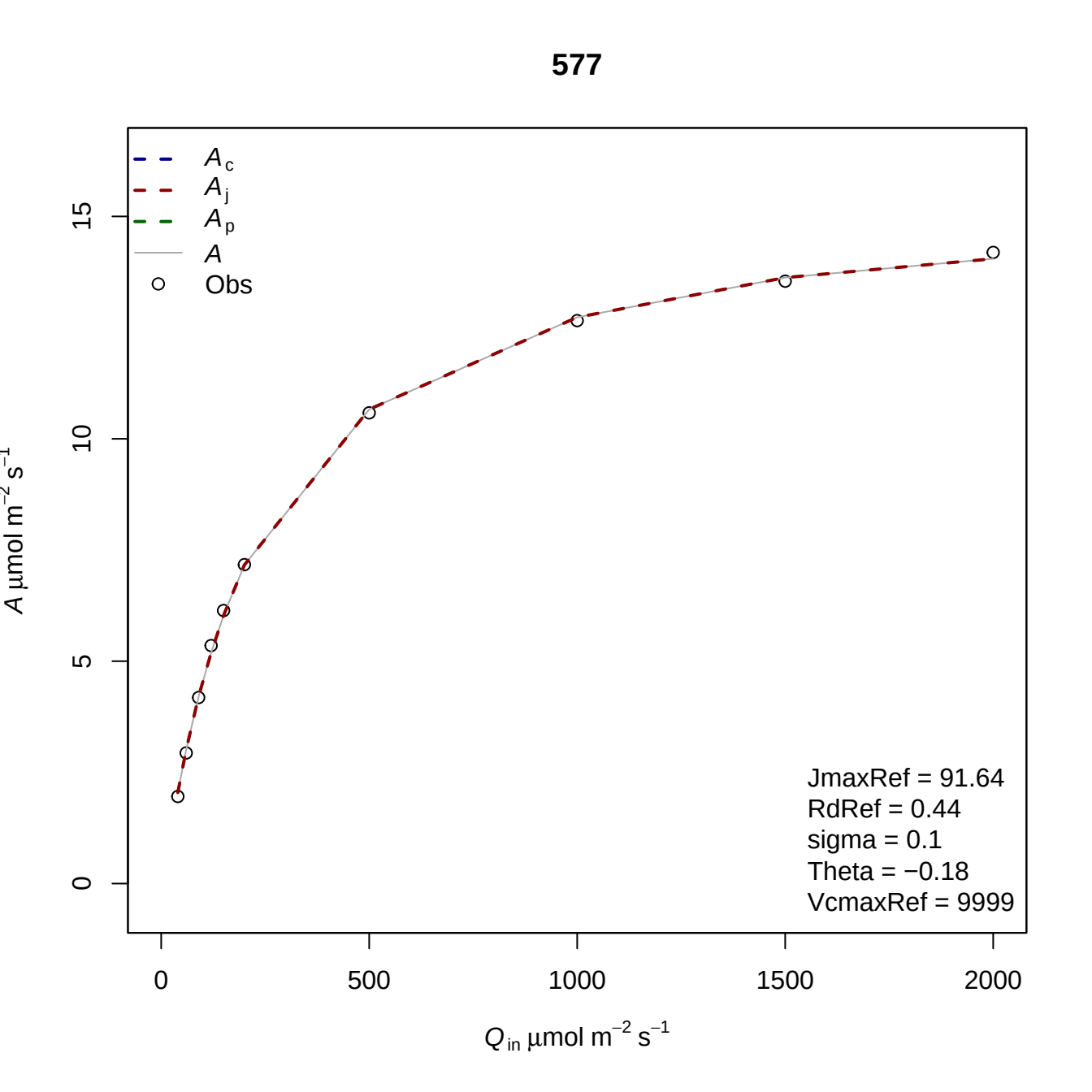

572
A_c
A_j
A_p
A
Obs
JmaxRef = 172.01
RdRef = 1.12
sigma = 0.16
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

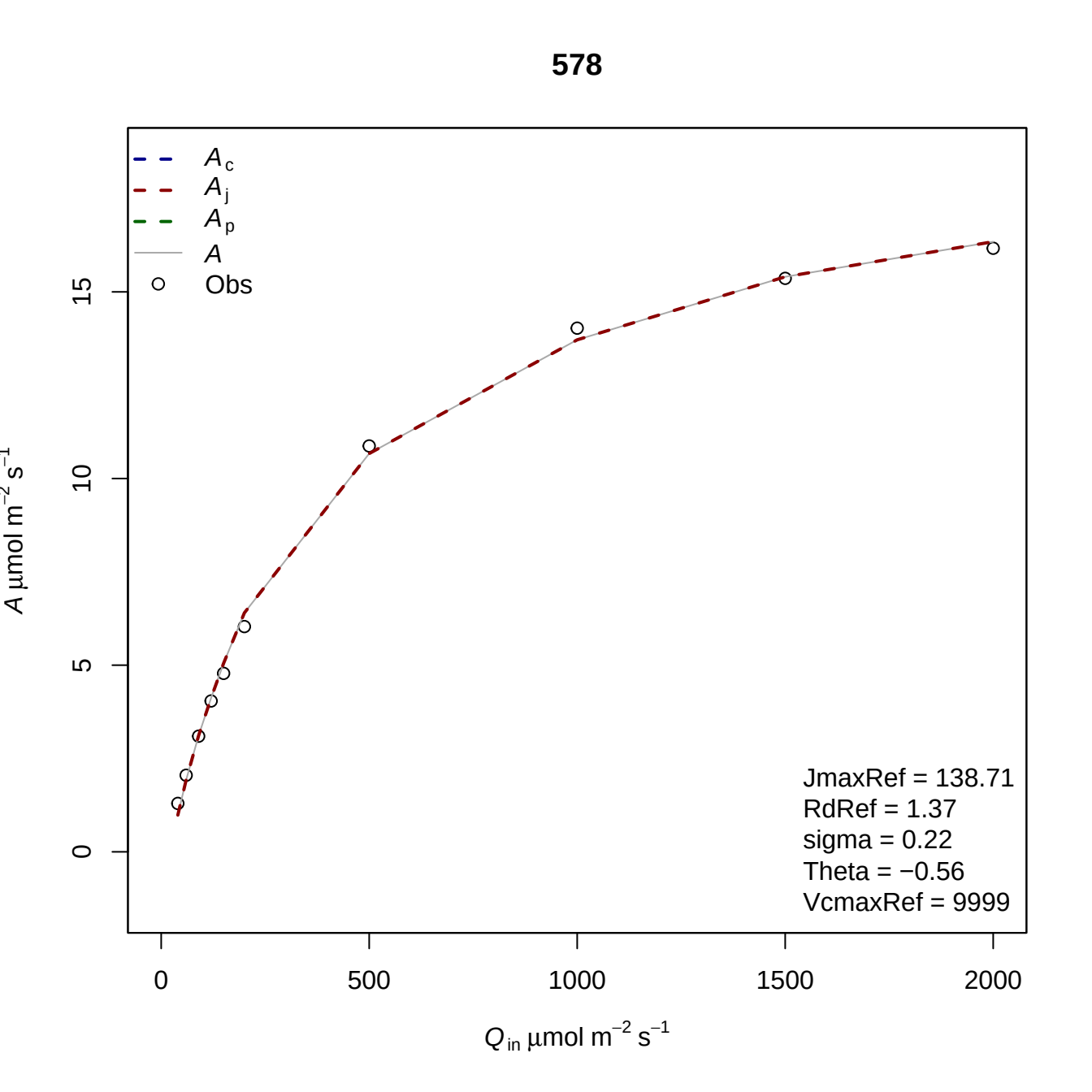

573
A_c
A_j
A_p
A
Obs
JmaxRef = 186.22
RdRef = 1.75
sigma = 0.1
Theta = 0.73
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

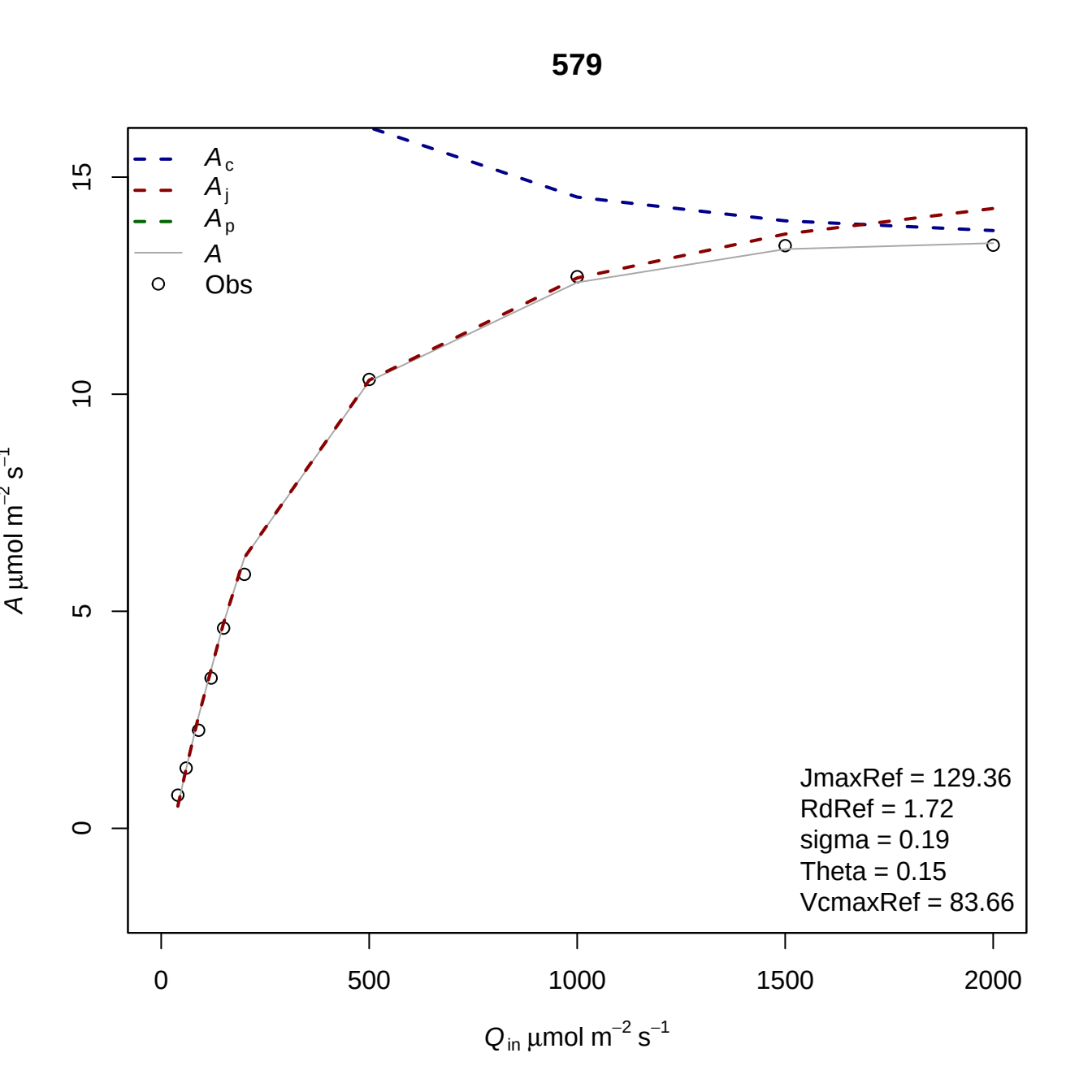

574
A_c
A_j
A_p
A
Obs
JmaxRef = 213.9
RdRef = 2.37
sigma = 0.08
Theta = 0.71
VcmaxRef = 240.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

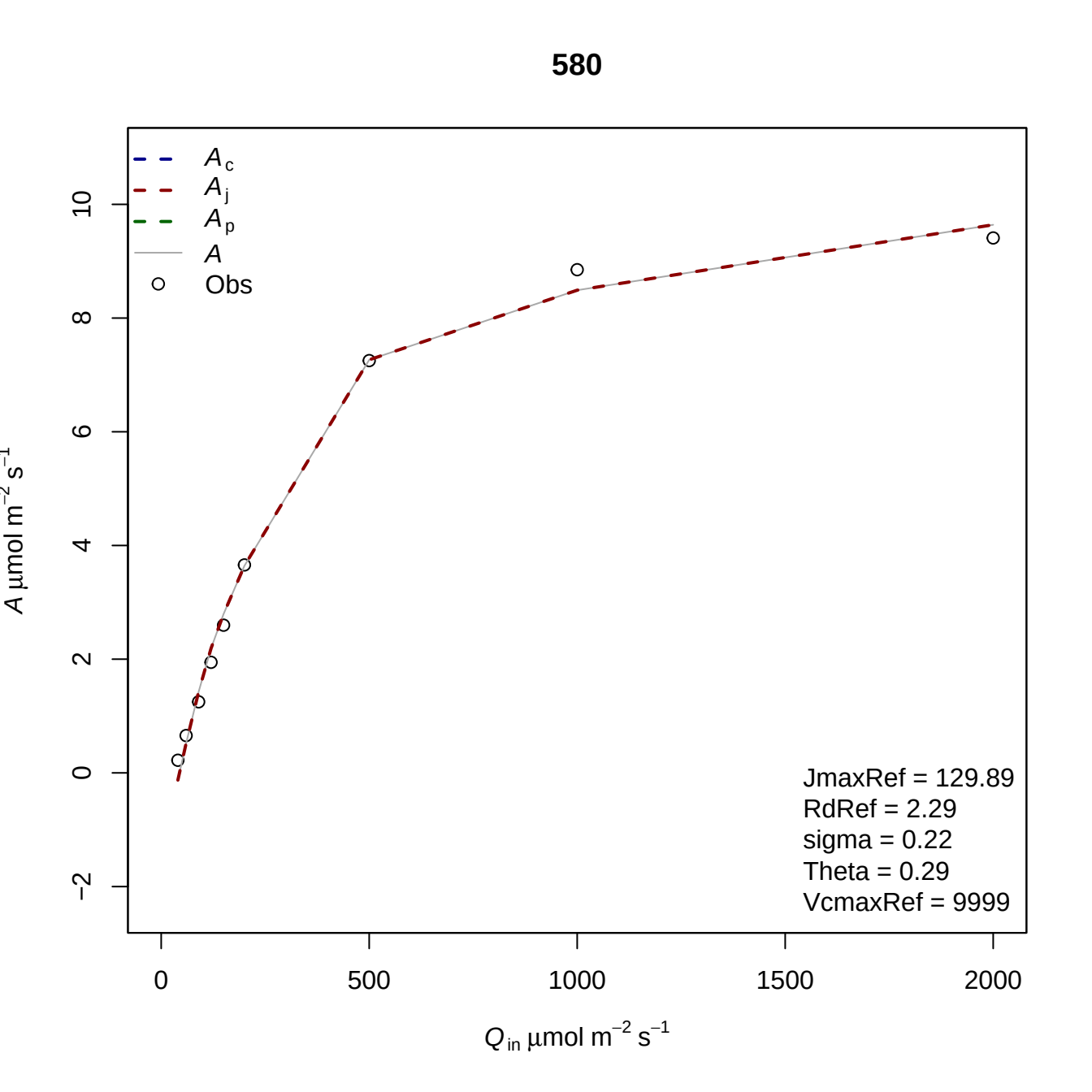

575
A_c
A_j
A_p
A
Obs
JmaxRef = 163.07
RdRef = 1.63
sigma = 0.27
Theta = −0.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

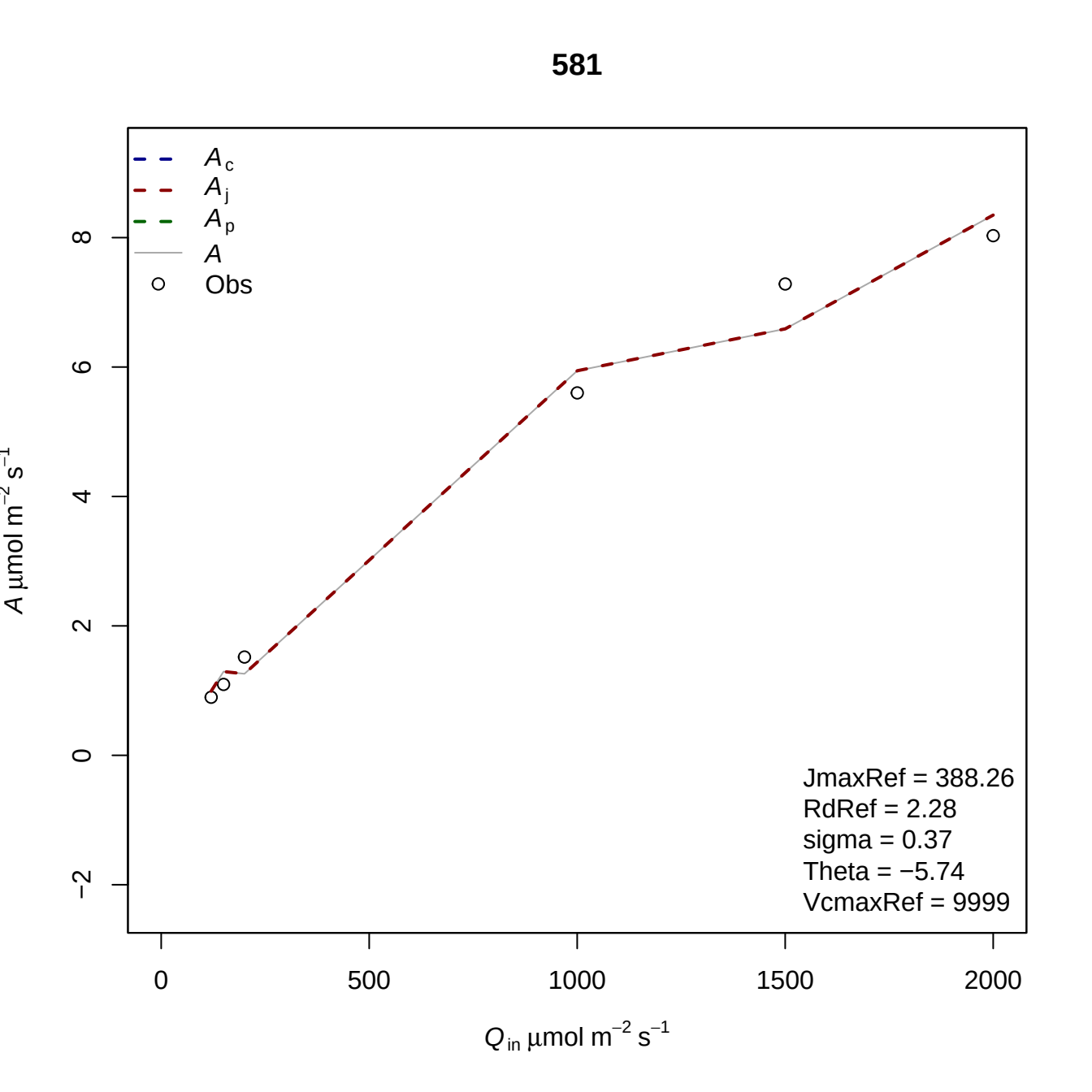

576
A_c
A_j
A_p
A
Obs
JmaxRef = 136.45
RdRef = 0.72
sigma = 0.24
Theta = −0.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

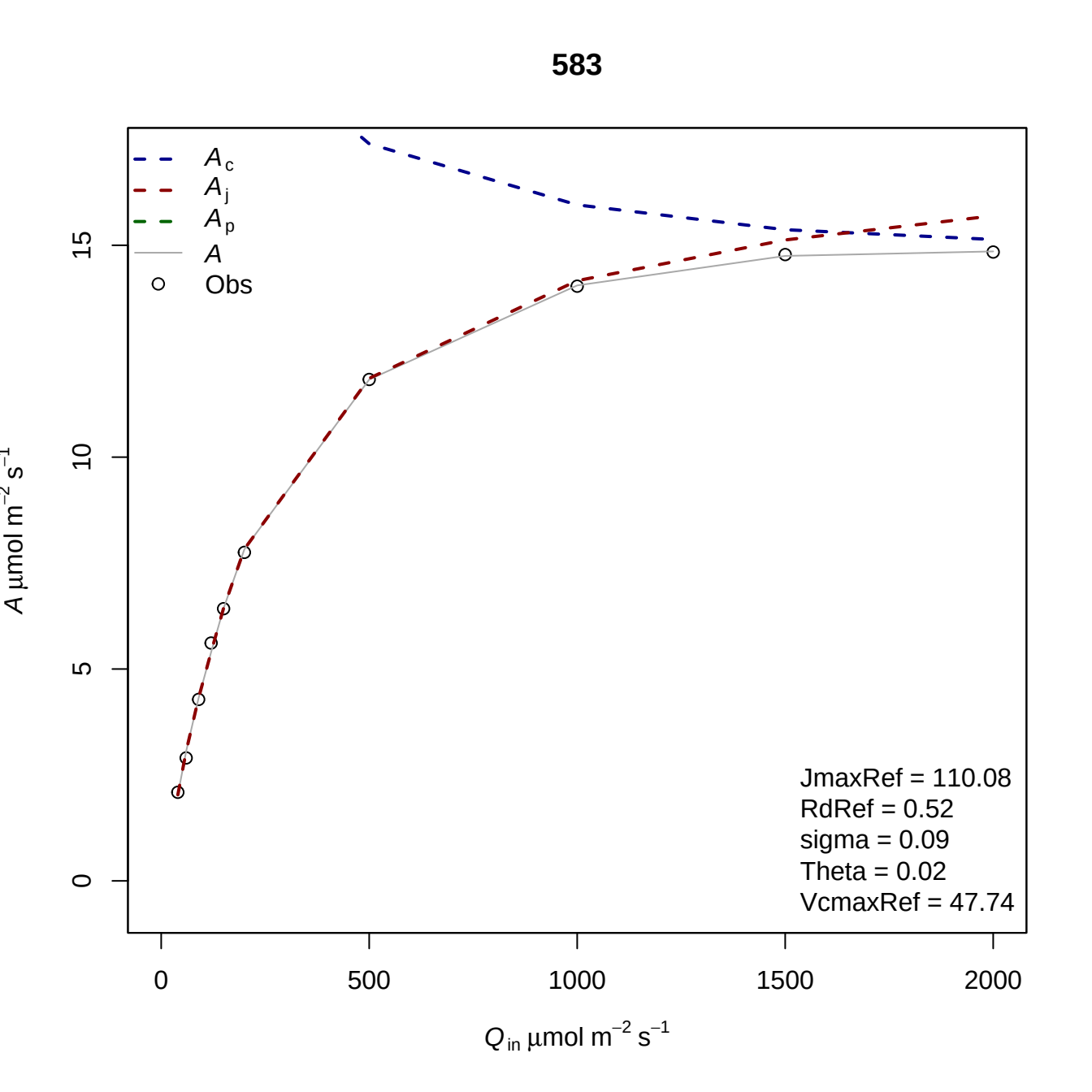

577
A_c
A_j
A_p
A
Obs
JmaxRef = 91.64
RdRef = 0.44
sigma = 0.1
Theta = −0.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

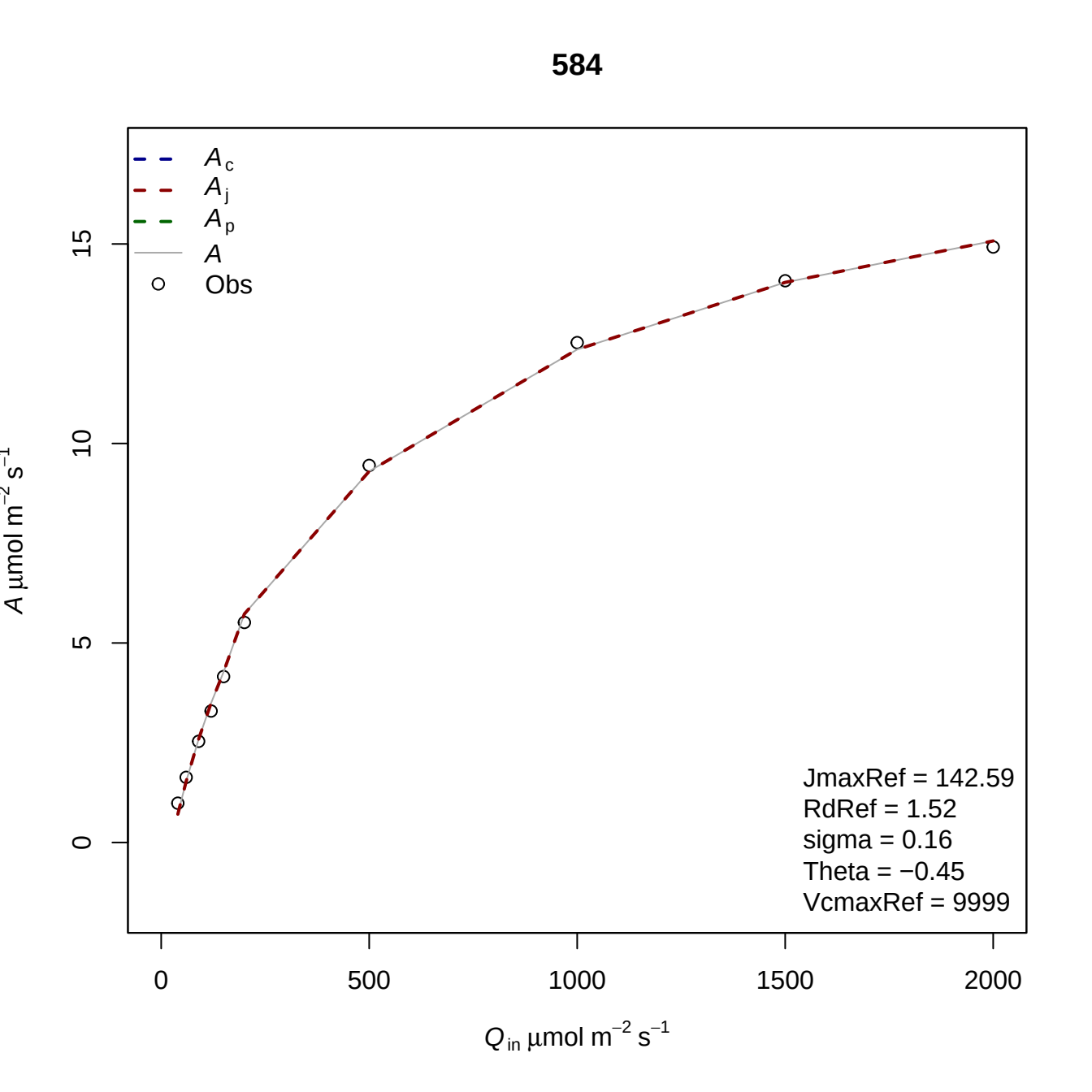

578
A_c
A_j
A_p
A
Obs
JmaxRef = 138.71
RdRef = 1.37
sigma = 0.22
Theta = −0.56
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

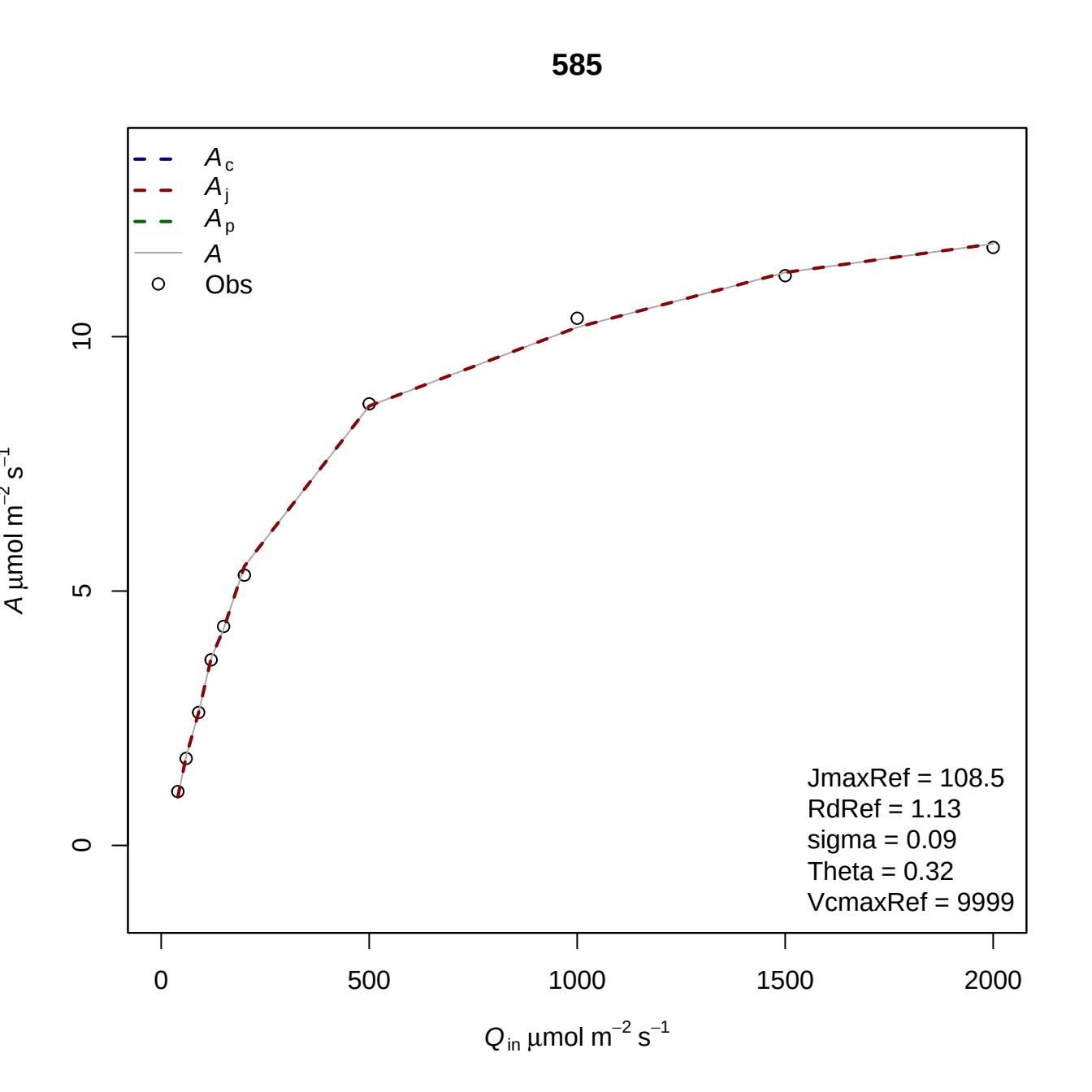

579
A_c
A_j
A_p
A
Obs
JmaxRef = 129.36
RdRef = 1.72
sigma = 0.19
Theta = 0.15
VcmaxRef = 83.66
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

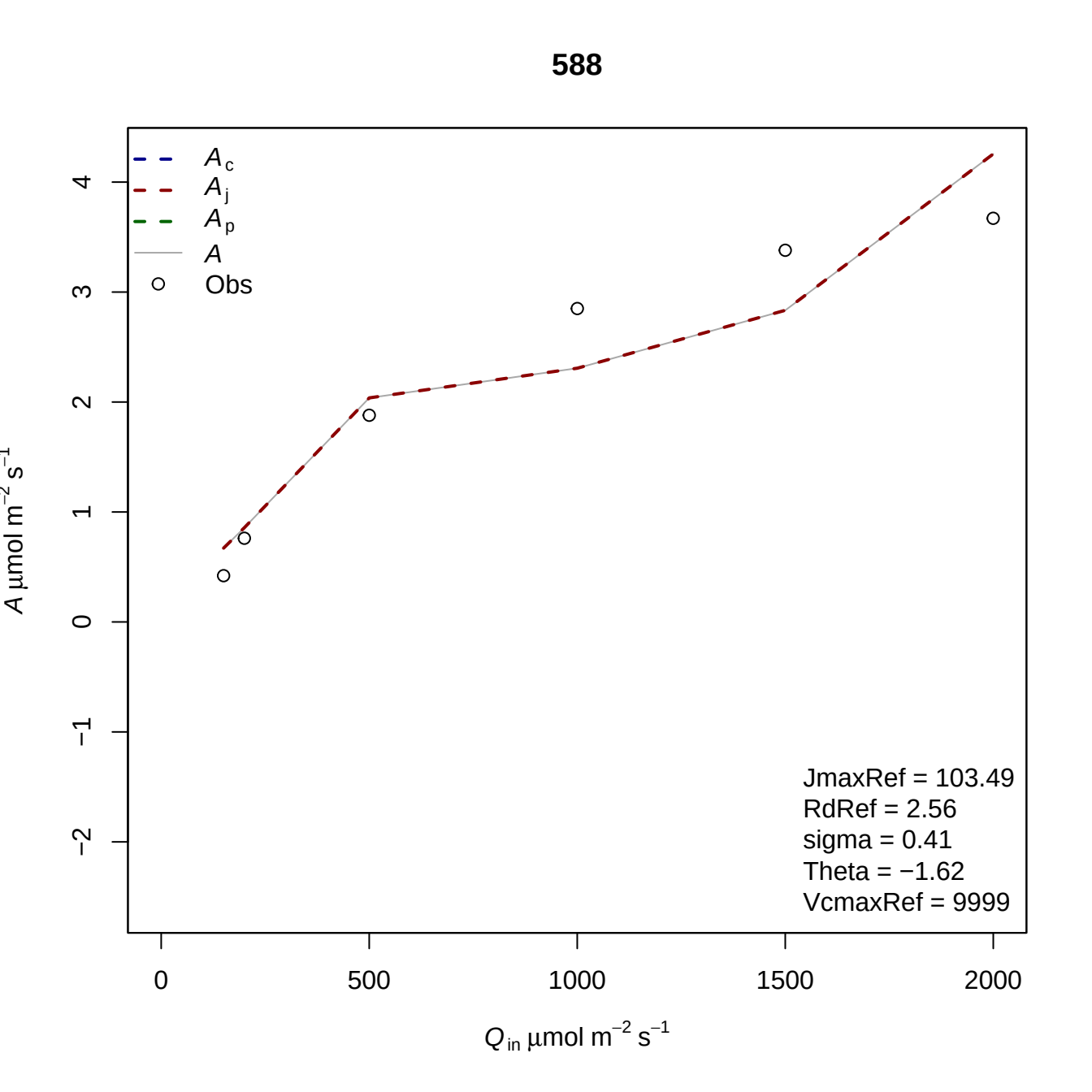

580
A_c
A_j
A_p
A
Obs
JmaxRef = 129.89
RdRef = 2.29
sigma = 0.22
Theta = 0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

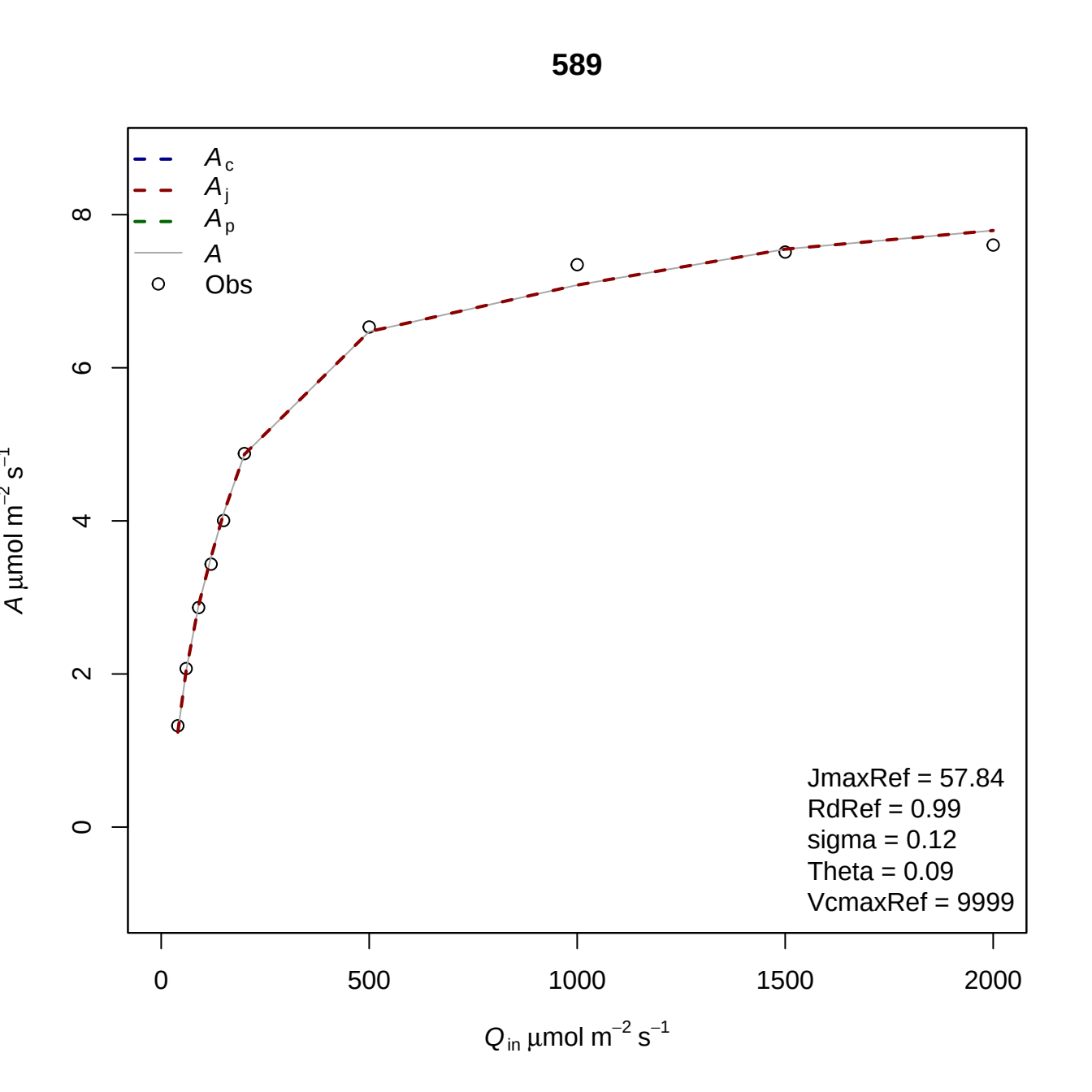

581
A_c
A_j
A_p
A
Obs
JmaxRef = 388.26
RdRef = 2.28
sigma = 0.37
Theta = −5.74
VcmaxRef = 9999
−2
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

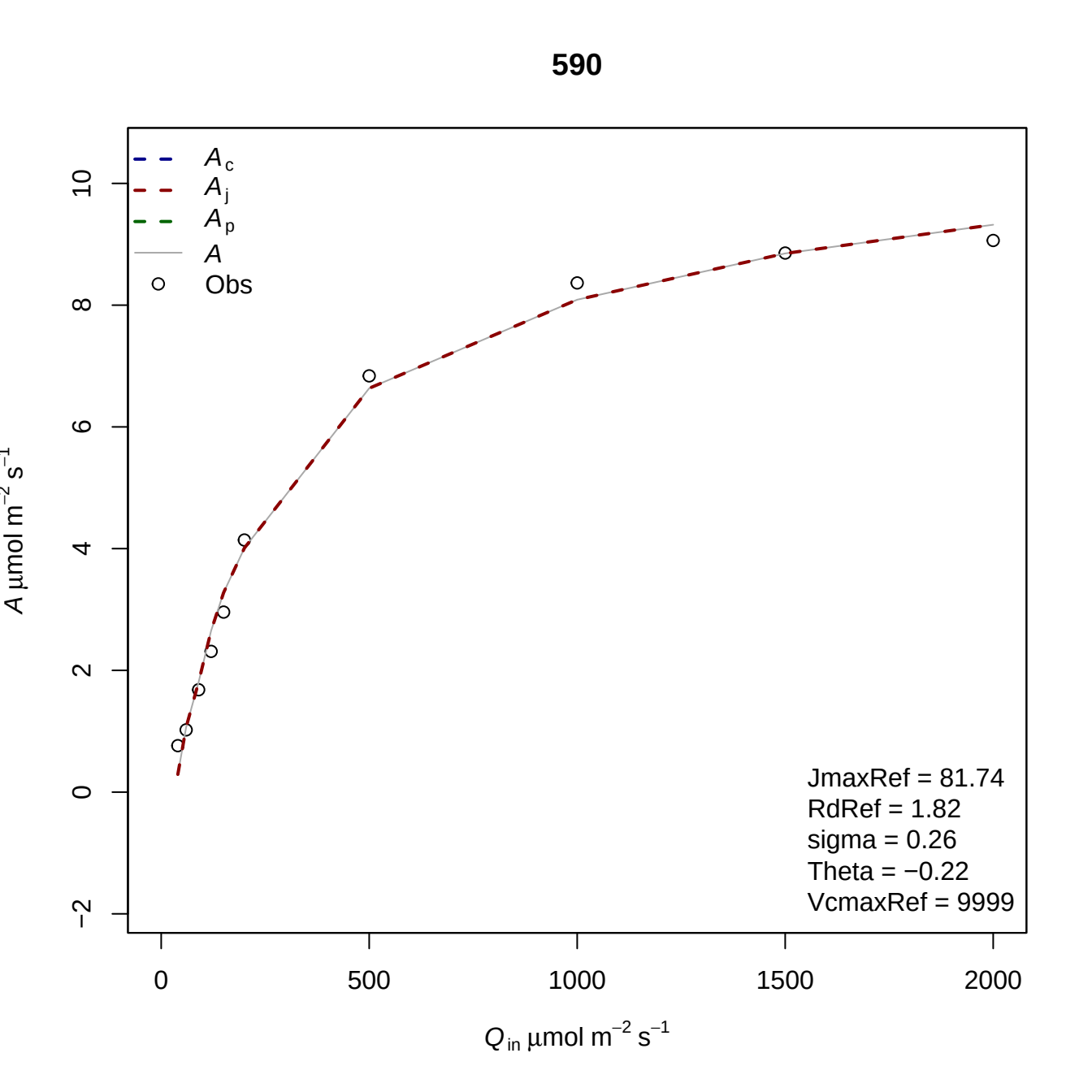

583
A_c
A_j
A_p
A
Obs
JmaxRef = 110.08
RdRef = 0.52
sigma = 0.09
Theta = 0.02
VcmaxRef = 47.74
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

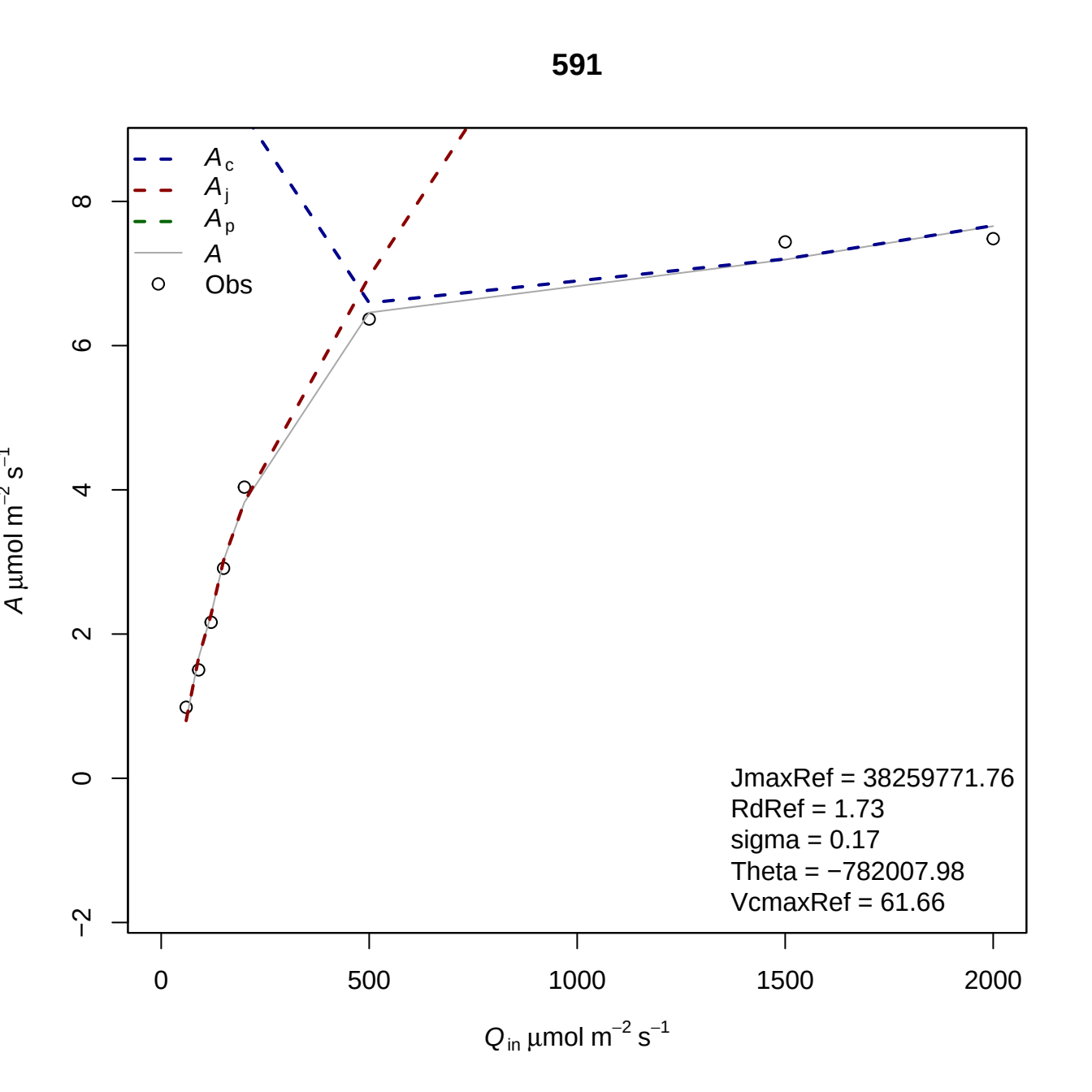

584
A_c
A_j
A_p
A
Obs
JmaxRef = 142.59
RdRef = 1.52
sigma = 0.16
Theta = −0.45
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

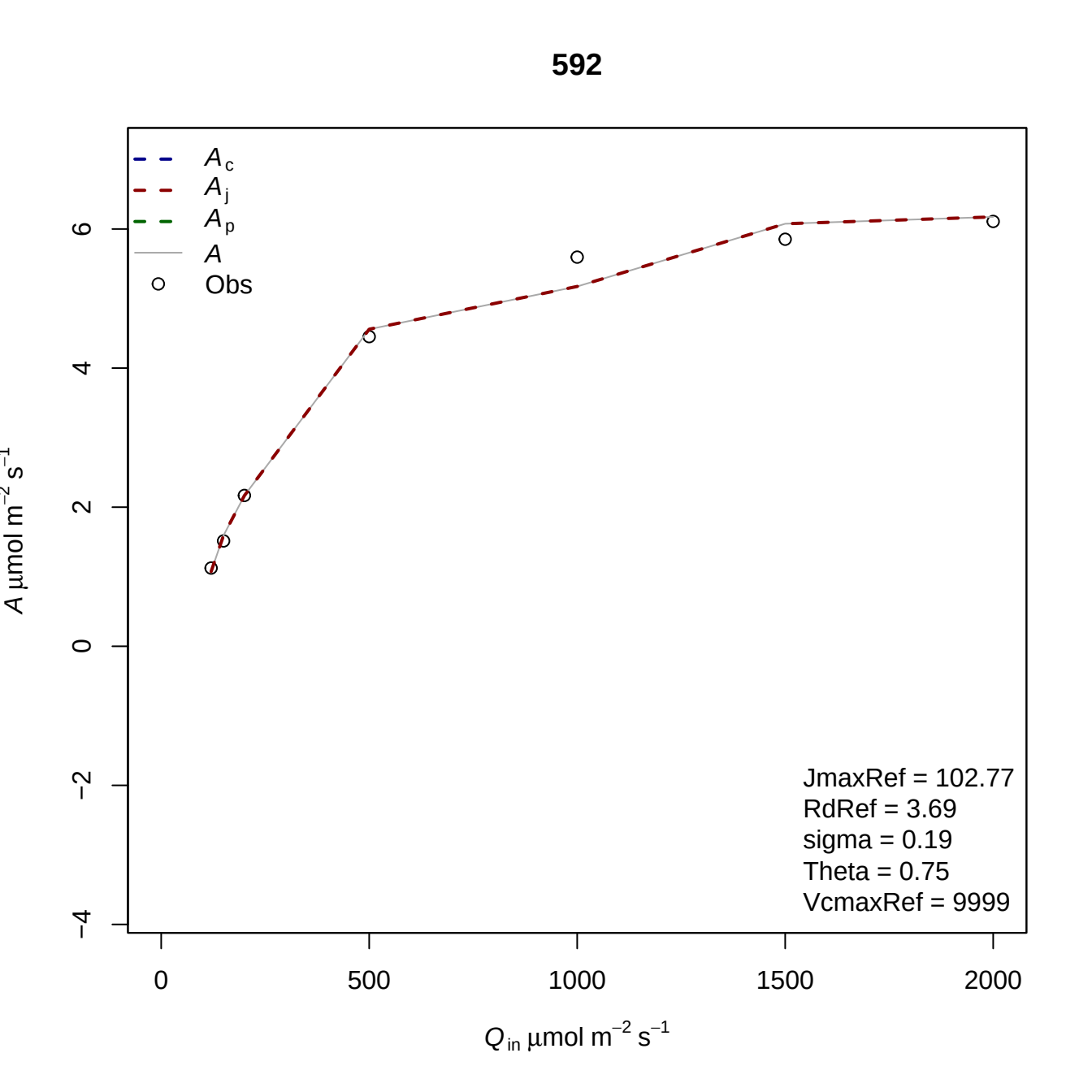

585
A_c
A_j
A_p
A
Obs
JmaxRef = 108.5
RdRef = 1.13
sigma = 0.09
Theta = 0.32
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

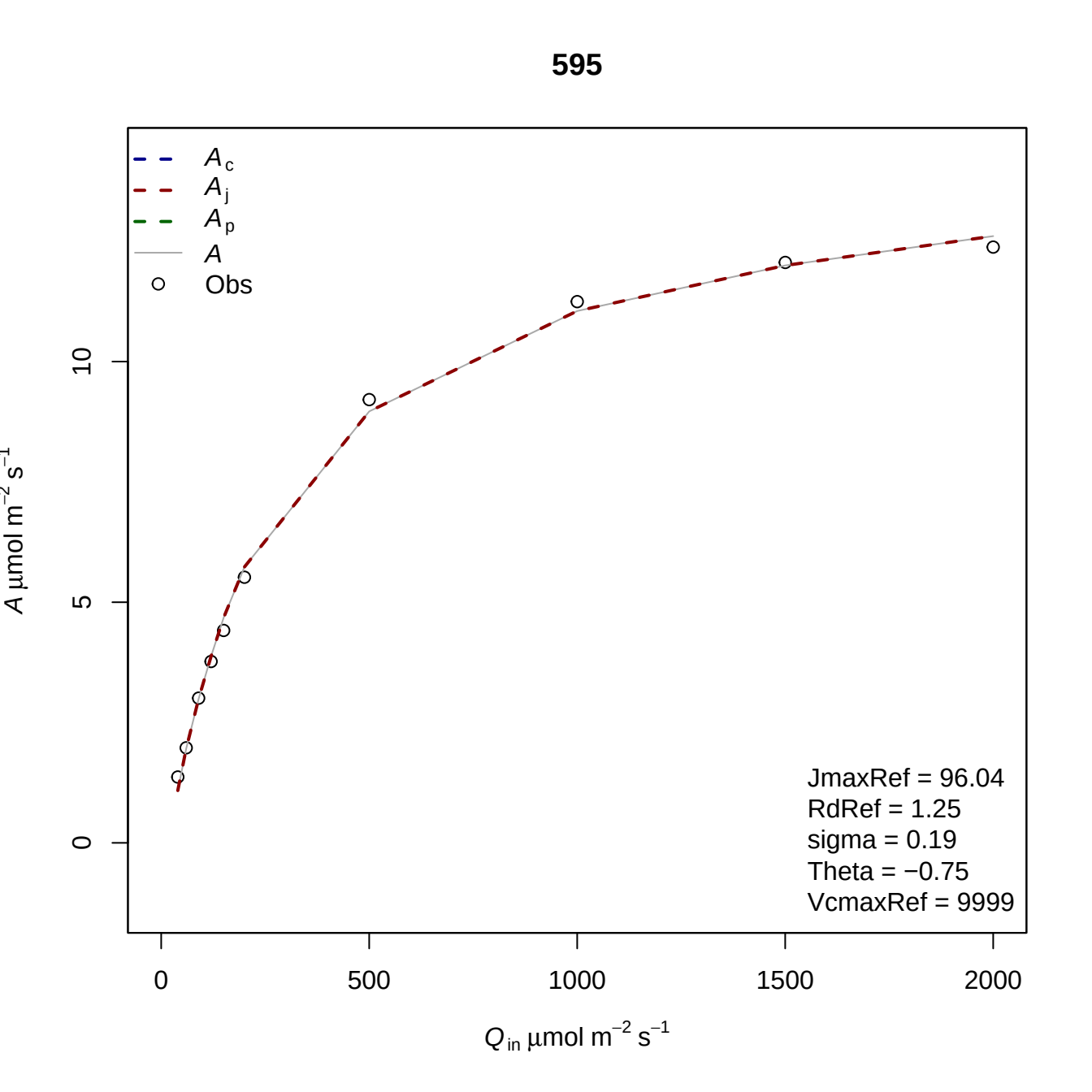

588
A_c
A_j
A_p
A
Obs
JmaxRef = 103.49
RdRef = 2.56
sigma = 0.41
Theta = −1.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

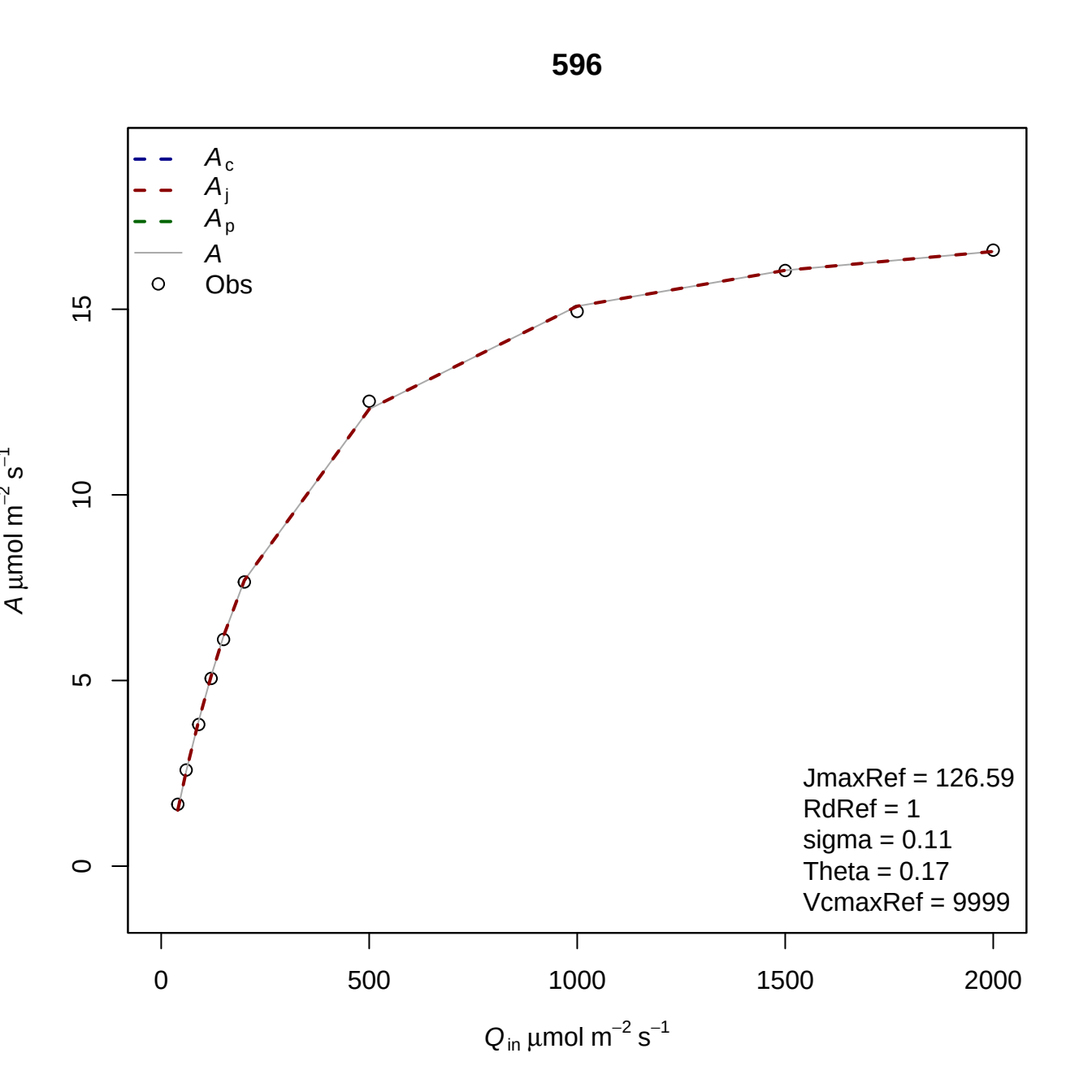

589
A_c
A_j
A_p
A
Obs
JmaxRef = 57.84
RdRef = 0.99
sigma = 0.12
Theta = 0.09
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

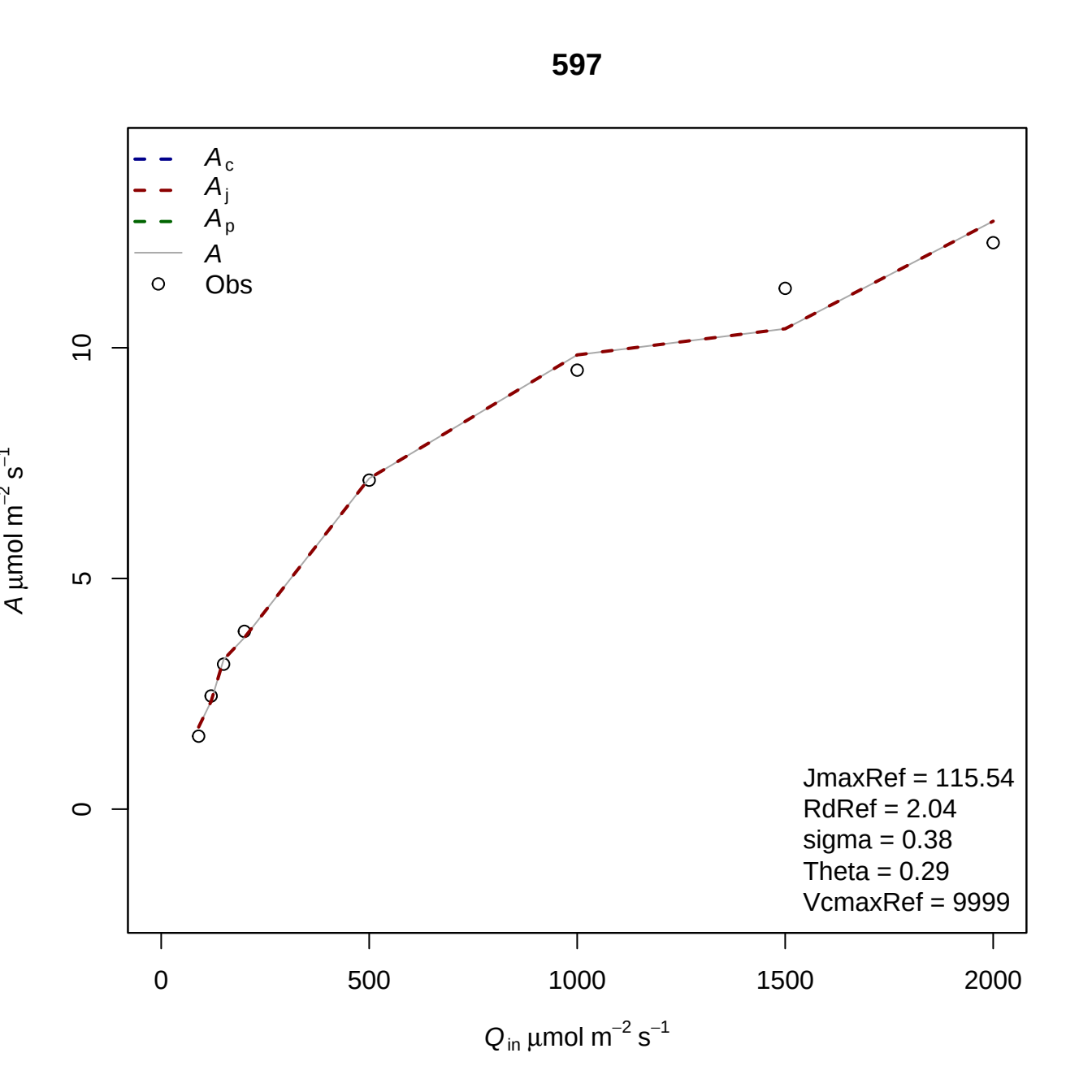

590
A_c
A_j
A_p
A
Obs
JmaxRef = 81.74
RdRef = 1.82
sigma = 0.26
Theta = −0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

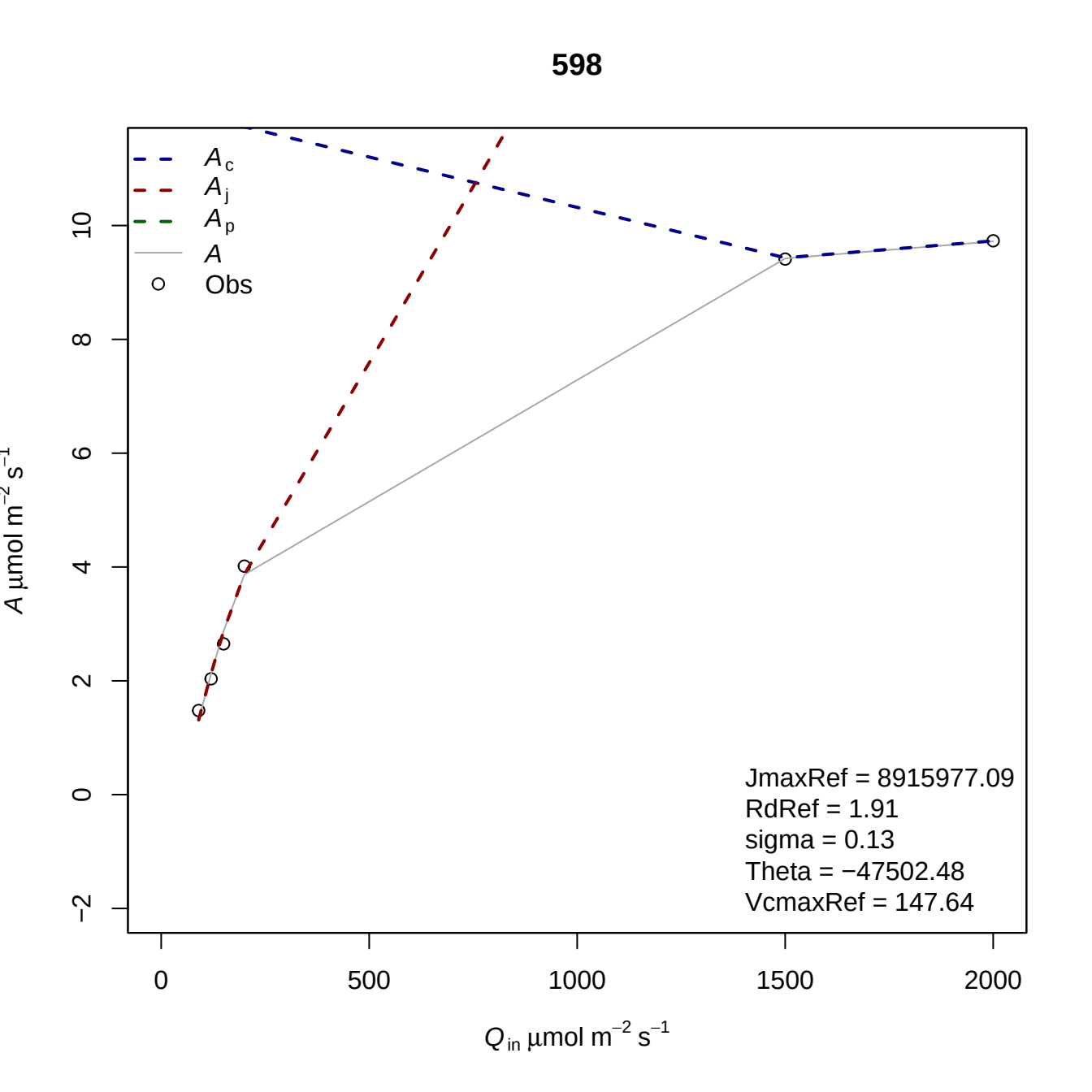

591
A_c
A_j
A_p
A
Obs
JmaxRef = 38259771.76
RdRef = 1.73
sigma = 0.17
Theta = −782007.98
VcmaxRef = 61.66
−2
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

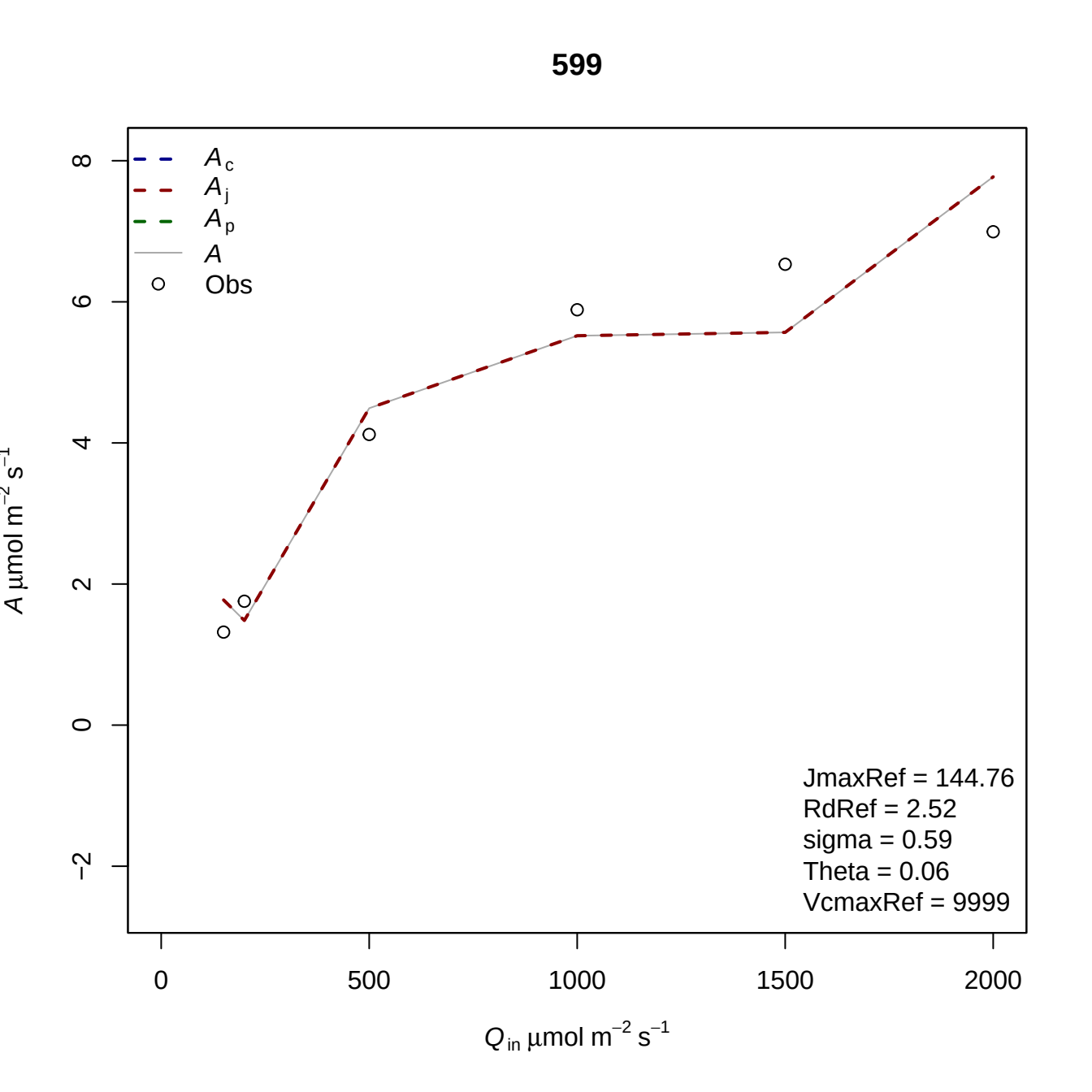

592
A_c
A_j
A_p
A
Obs
JmaxRef = 102.77
RdRef = 3.69
sigma = 0.19
Theta = 0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

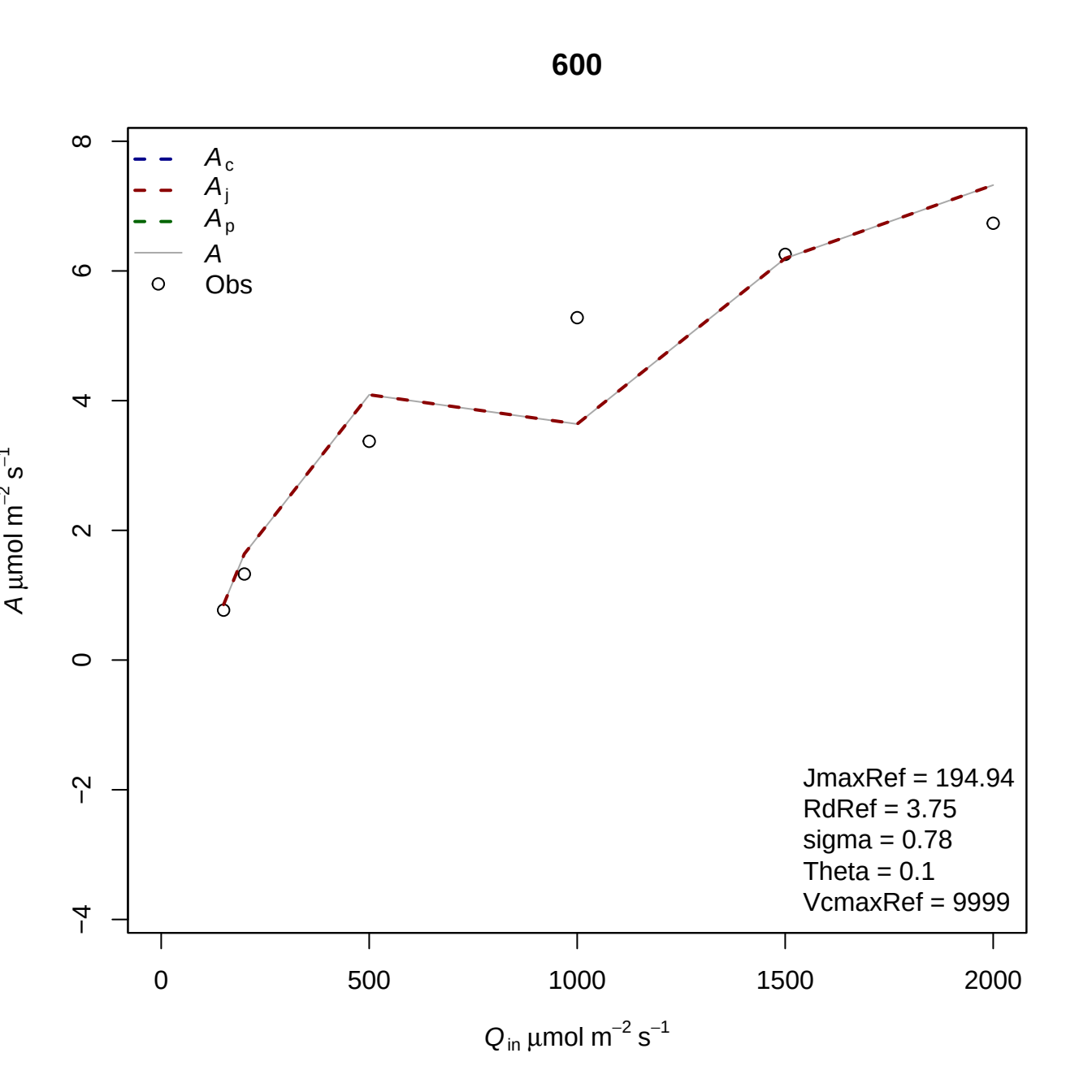

595
A_c
A_j
A_p
A
Obs
JmaxRef = 96.04
RdRef = 1.25
sigma = 0.19
Theta = −0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

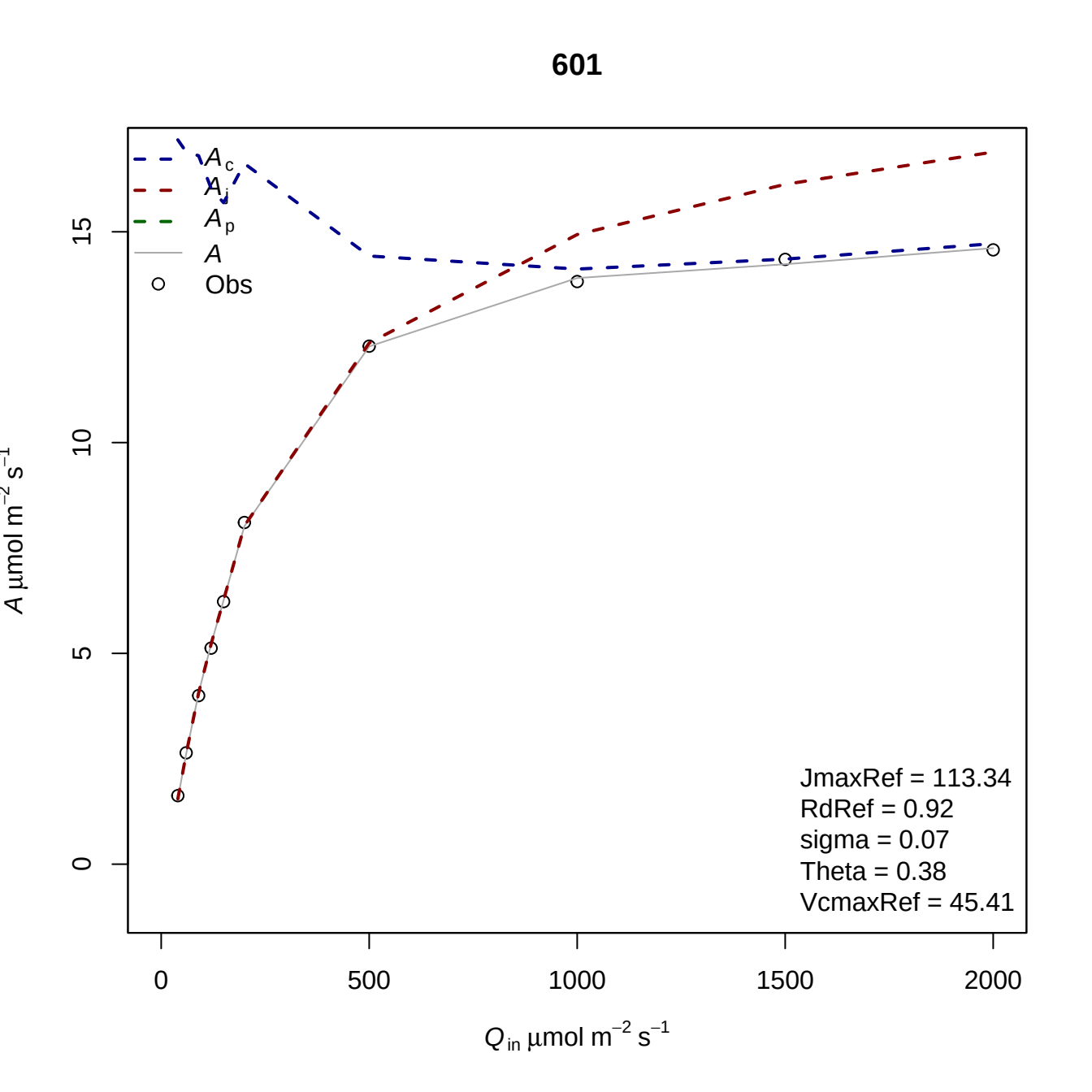

596
A_c
A_j
A_p
A
Obs
JmaxRef = 126.59
RdRef = 1
sigma = 0.11
Theta = 0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

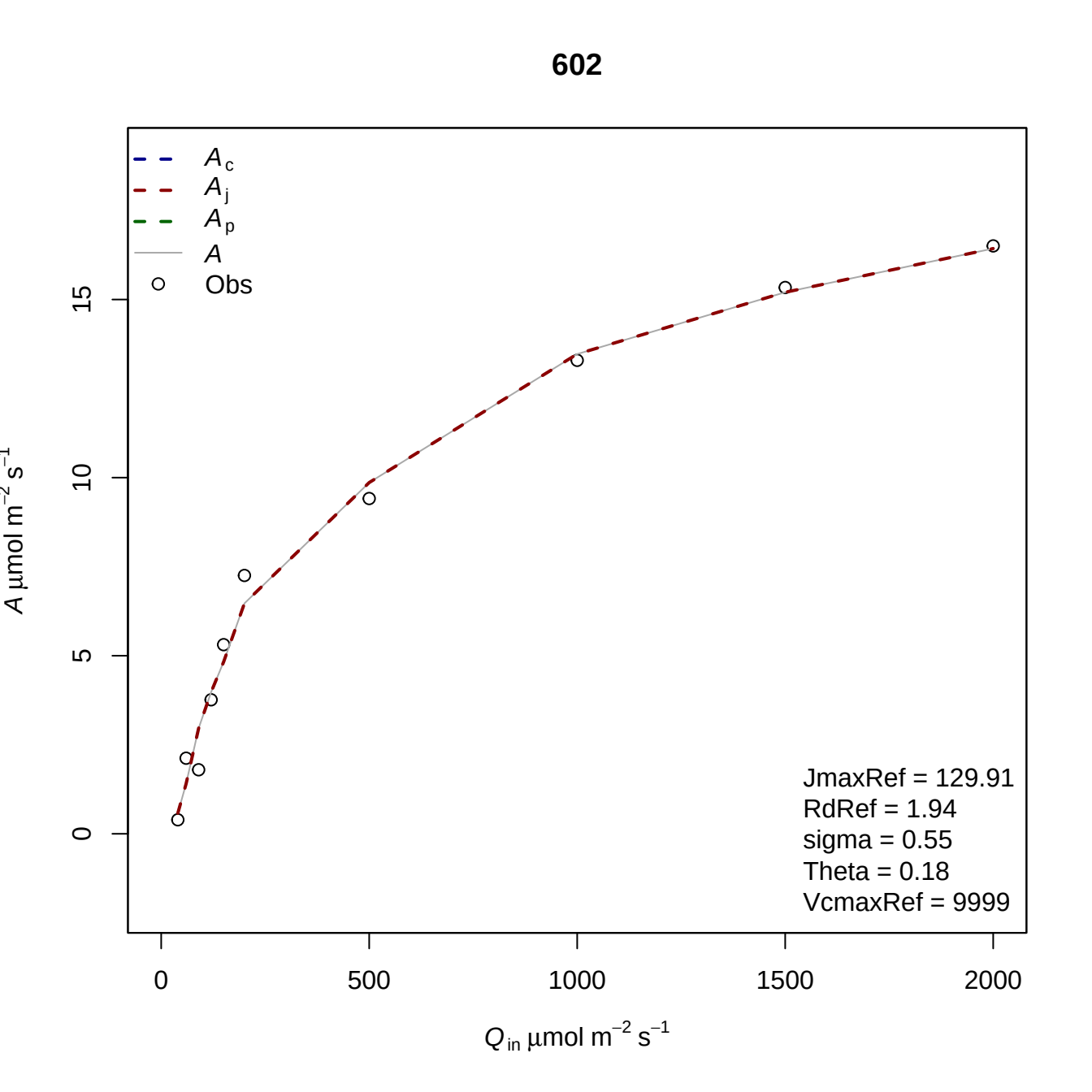

597
A_c
A_j
A_p
A
Obs
JmaxRef = 115.54
RdRef = 2.04
sigma = 0.38
Theta = 0.29
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

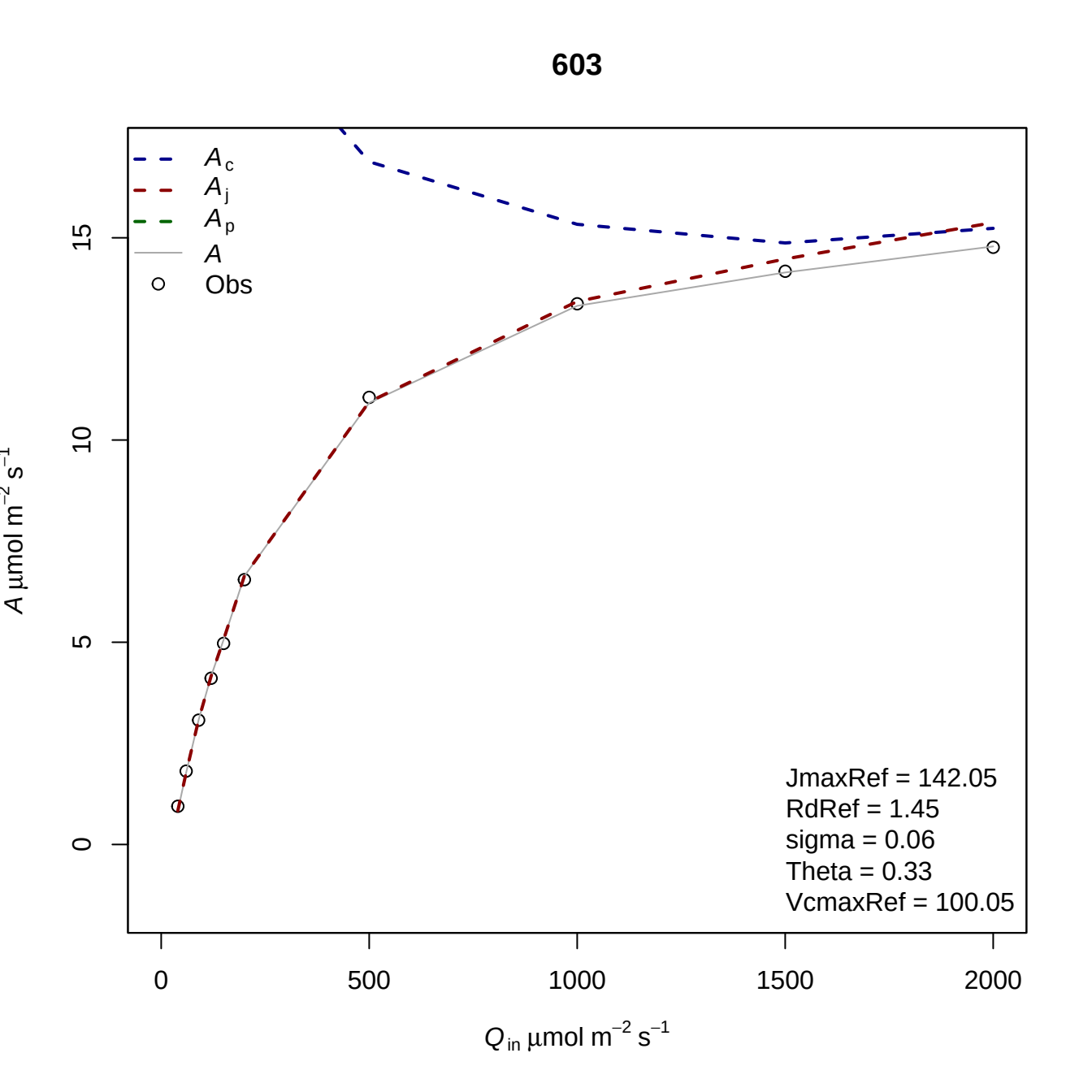

598
A_c
A_j
A_p
A
Obs
JmaxRef = 8915977.09
RdRef = 1.91
sigma = 0.13
Theta = −47502.48
VcmaxRef = 147.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

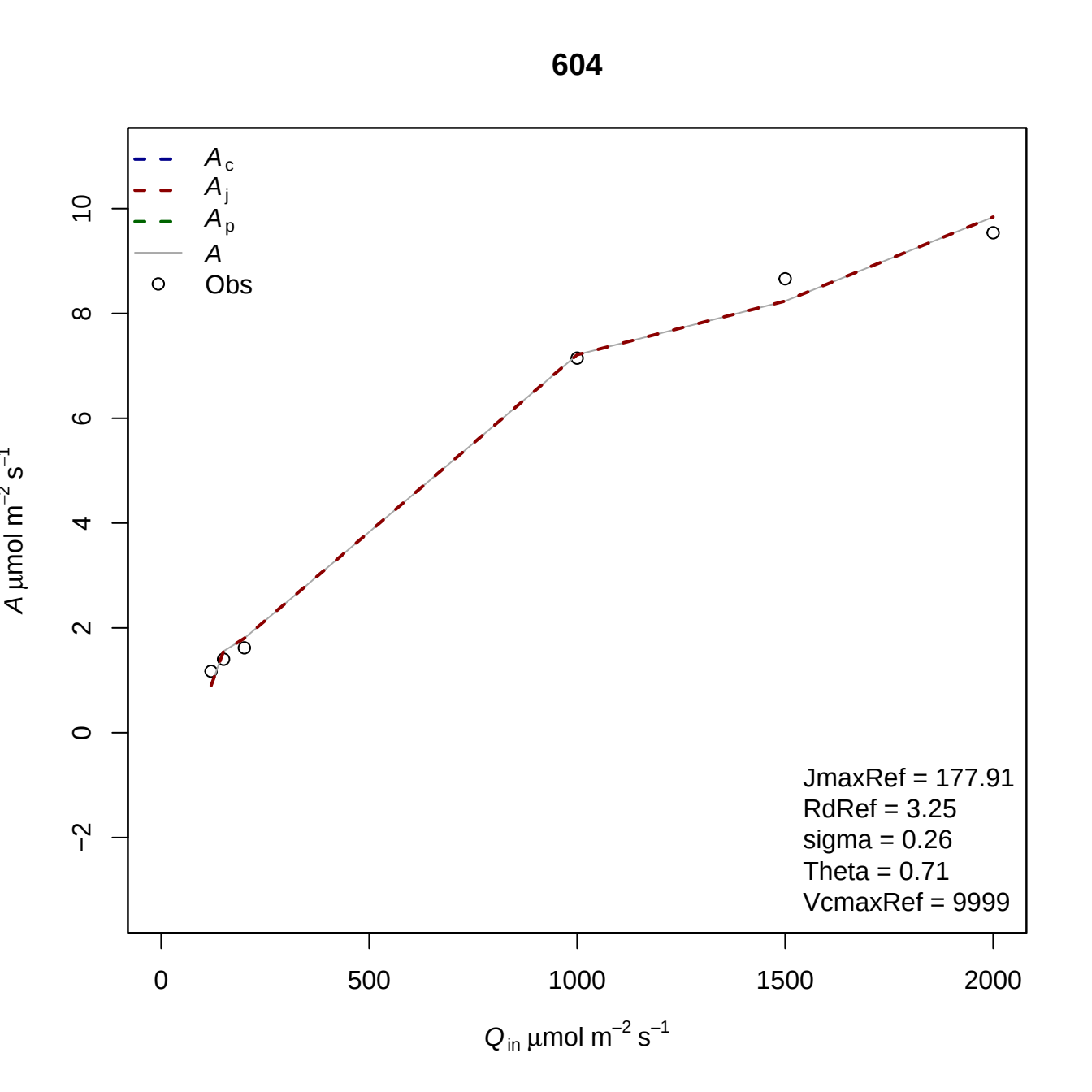

599
A_c
A_j
A_p
A
Obs
JmaxRef = 144.76
RdRef = 2.52
sigma = 0.59
Theta = 0.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

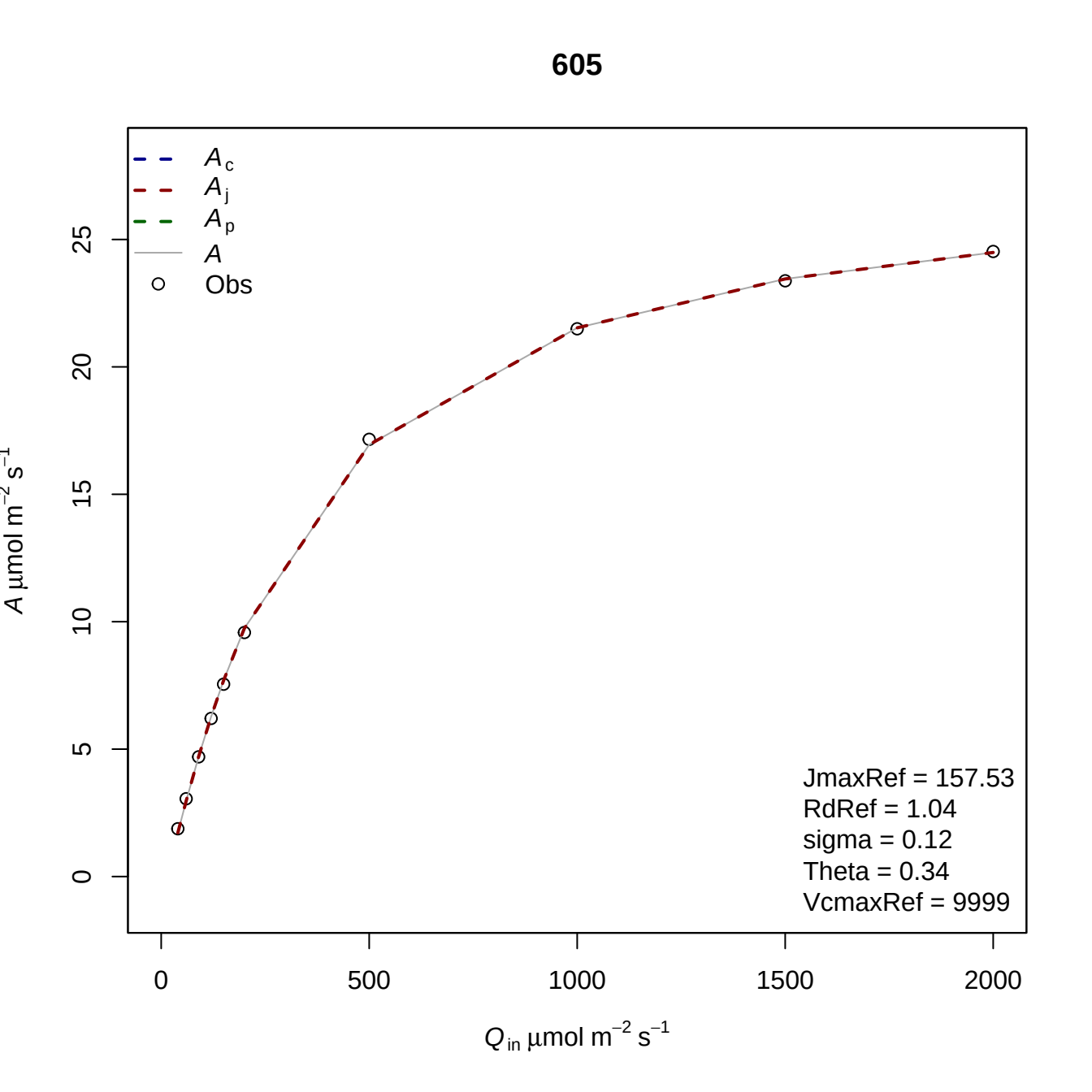

600
A_c
A_j
A_p
A
Obs
JmaxRef = 194.94
RdRef = 3.75
sigma = 0.78
Theta = 0.1
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

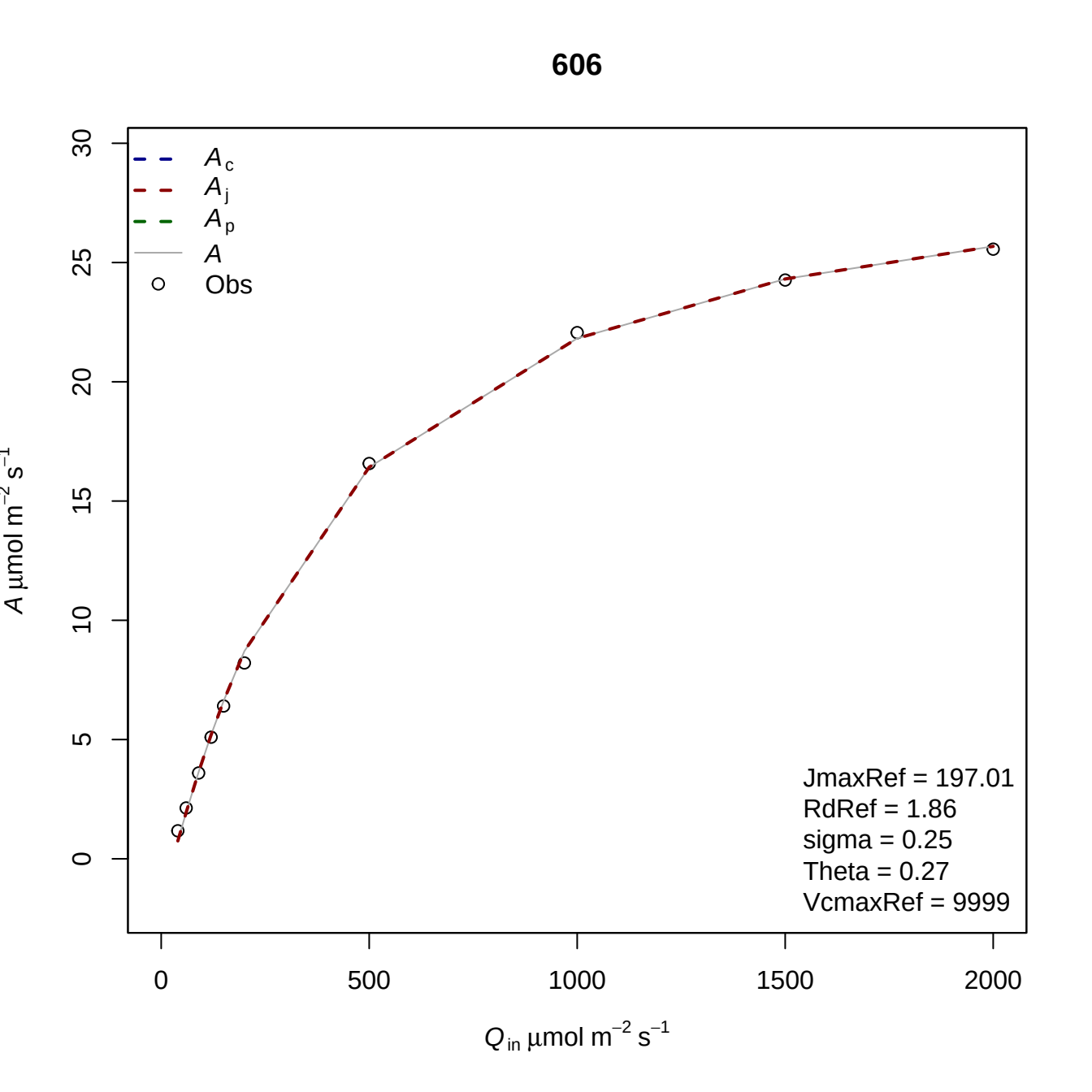

601
A_c
A_j
A_p
A
Obs
JmaxRef = 113.34
RdRef = 0.92
sigma = 0.07
Theta = 0.38
VcmaxRef = 45.41
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

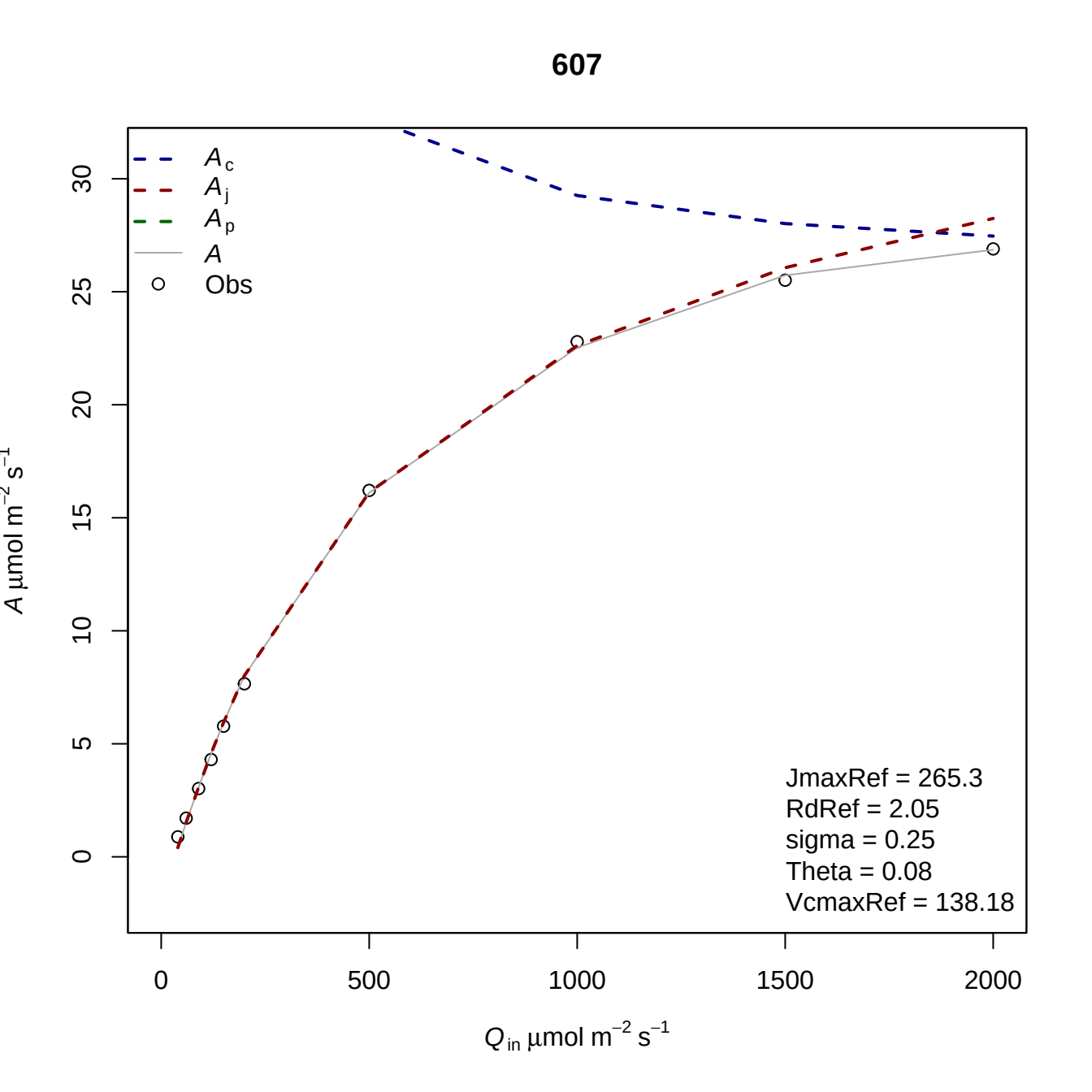

602
A_c
A_j
A_p
A
Obs
JmaxRef = 129.91
RdRef = 1.94
sigma = 0.55
Theta = 0.18
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

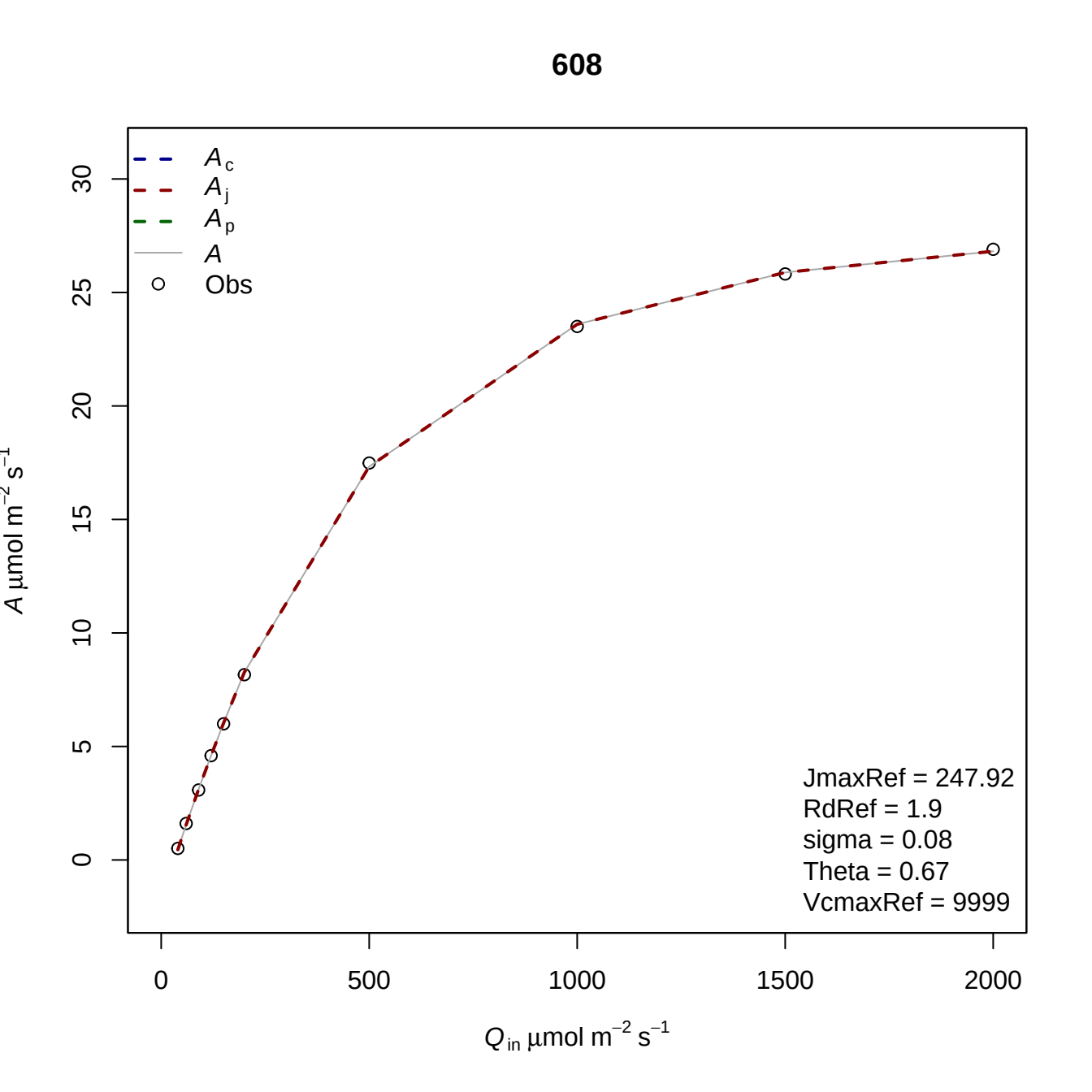

603
A_c
A_j
A_p
A
Obs
JmaxRef = 142.05
RdRef = 1.45
sigma = 0.06
Theta = 0.33
VcmaxRef = 100.05
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

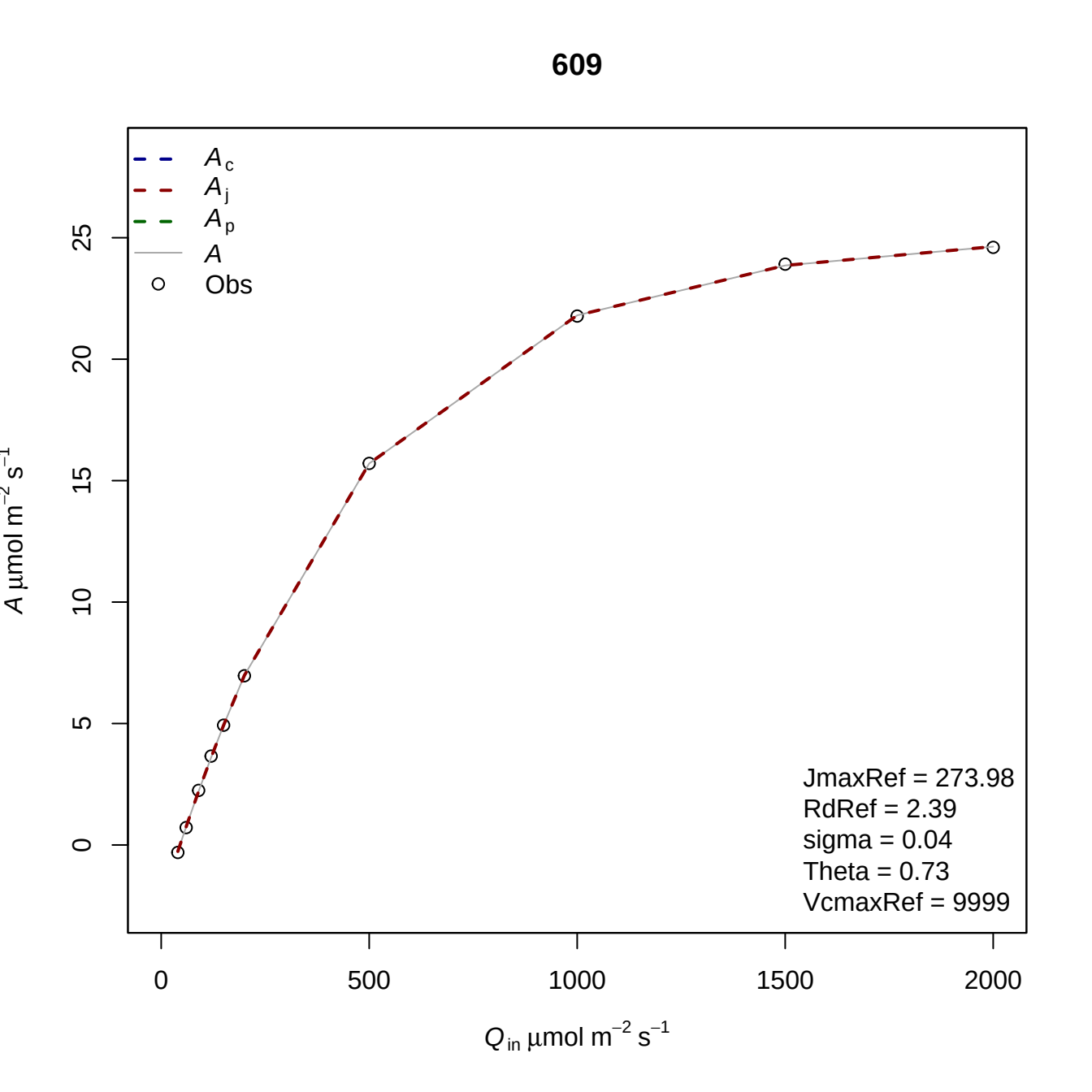

604
A_c
A_j
A_p
A
Obs
JmaxRef = 177.91
RdRef = 3.25
sigma = 0.26
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

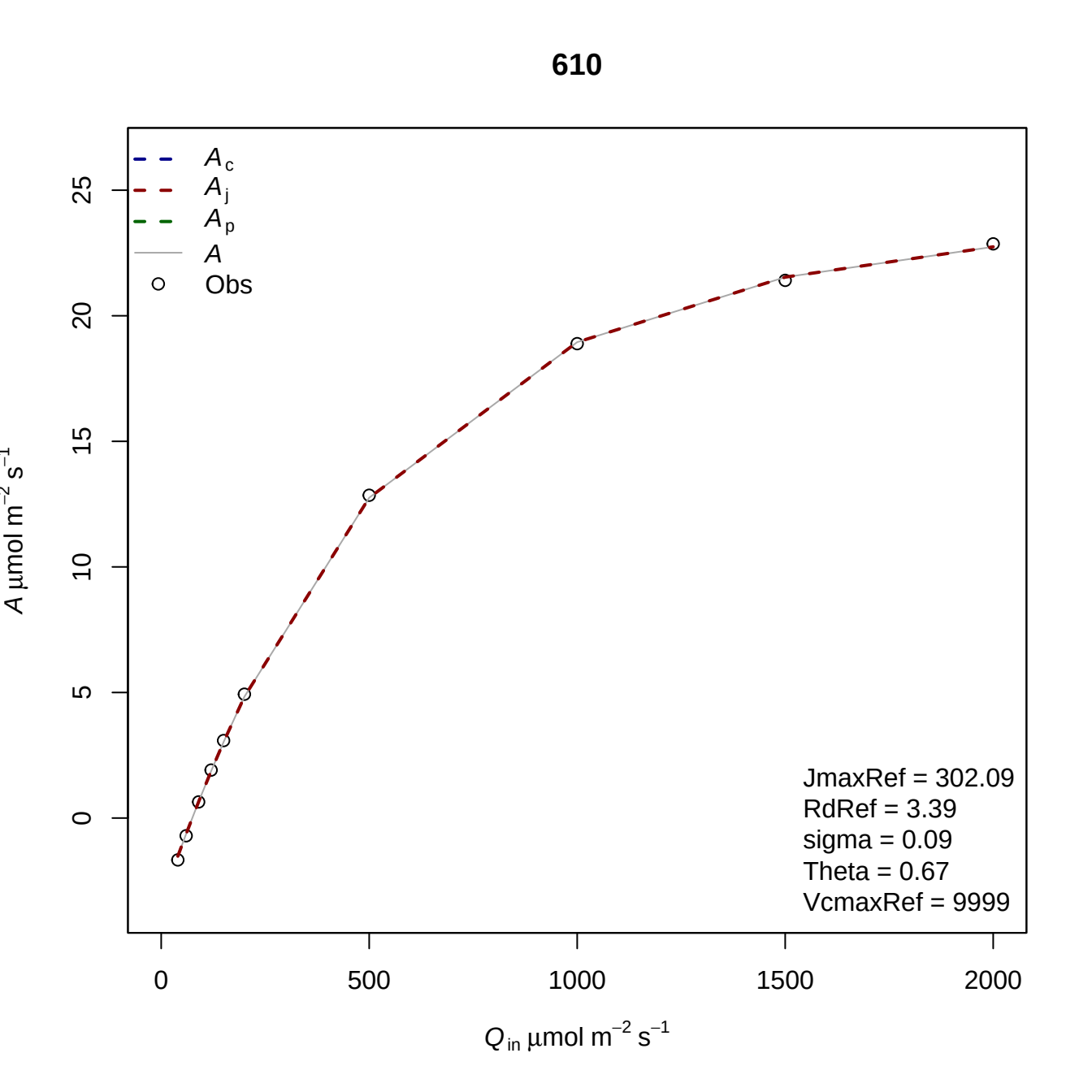

605
A_c
A_j
A_p
A
Obs
JmaxRef = 157.53
RdRef = 1.04
sigma = 0.12
Theta = 0.34
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

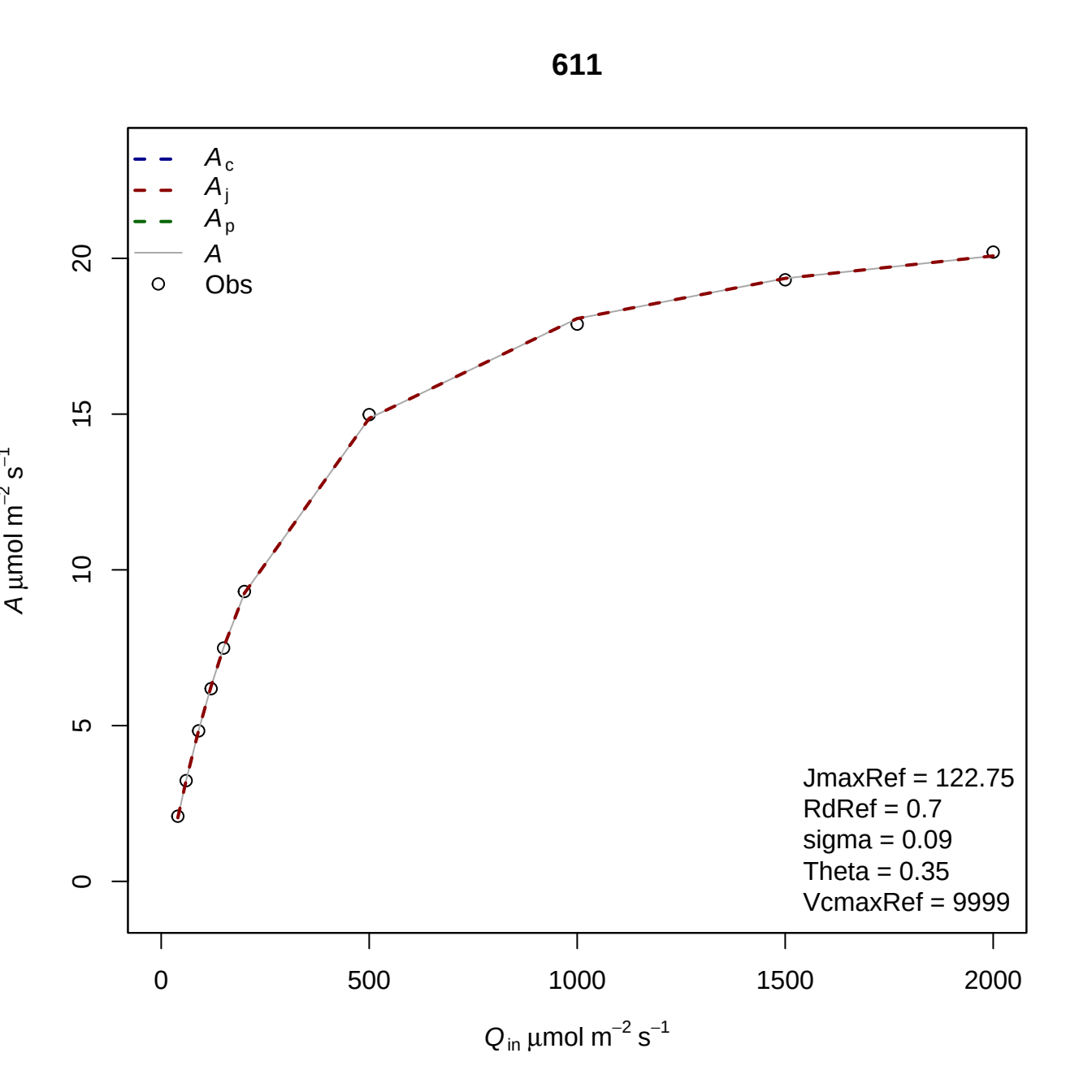

606
A_c
A_j
A_p
A
Obs
JmaxRef = 197.01
RdRef = 1.86
sigma = 0.25
Theta = 0.27
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

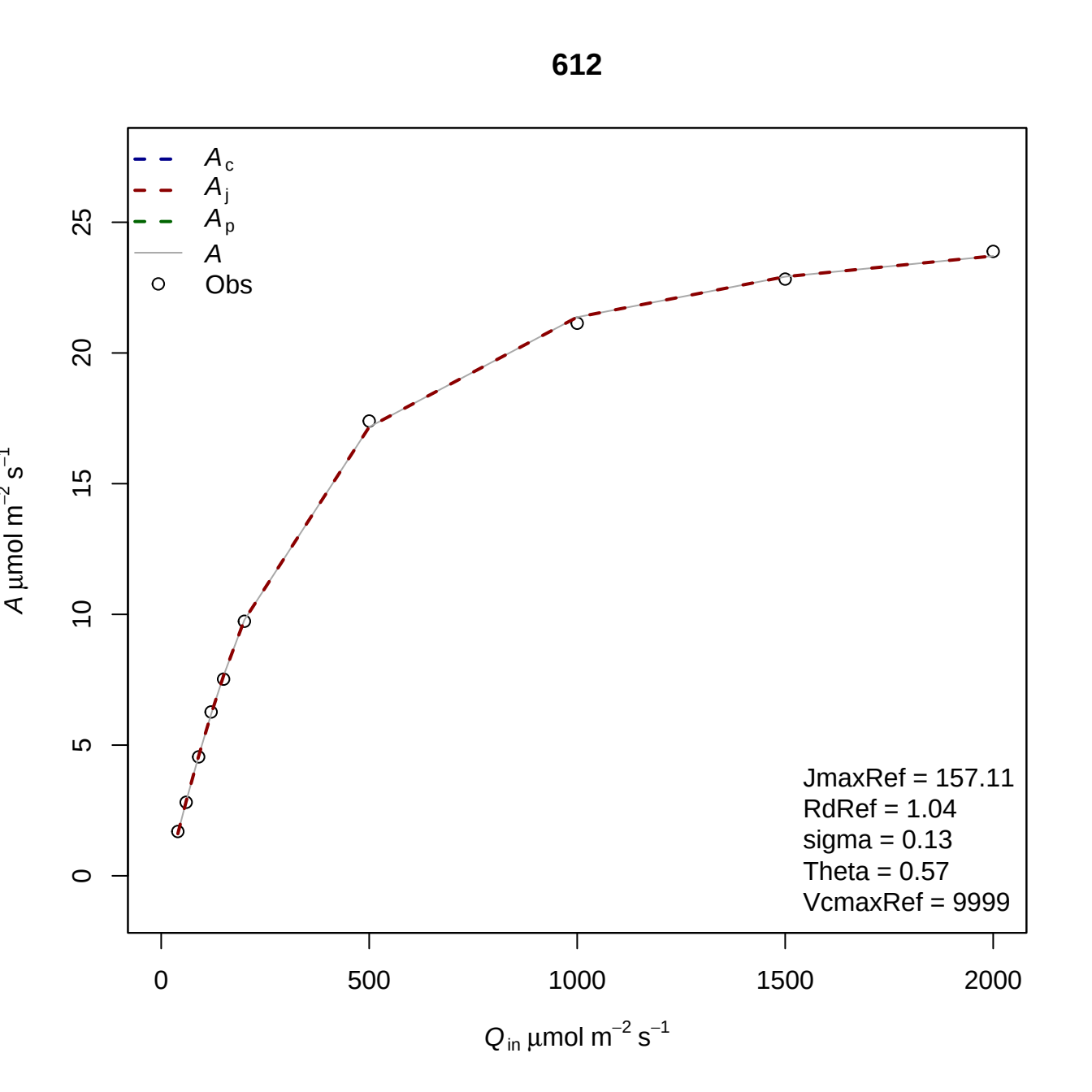

607
A_c
A_j
A_p
A
Obs
JmaxRef = 265.3
RdRef = 2.05
sigma = 0.25
Theta = 0.08
VcmaxRef = 138.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

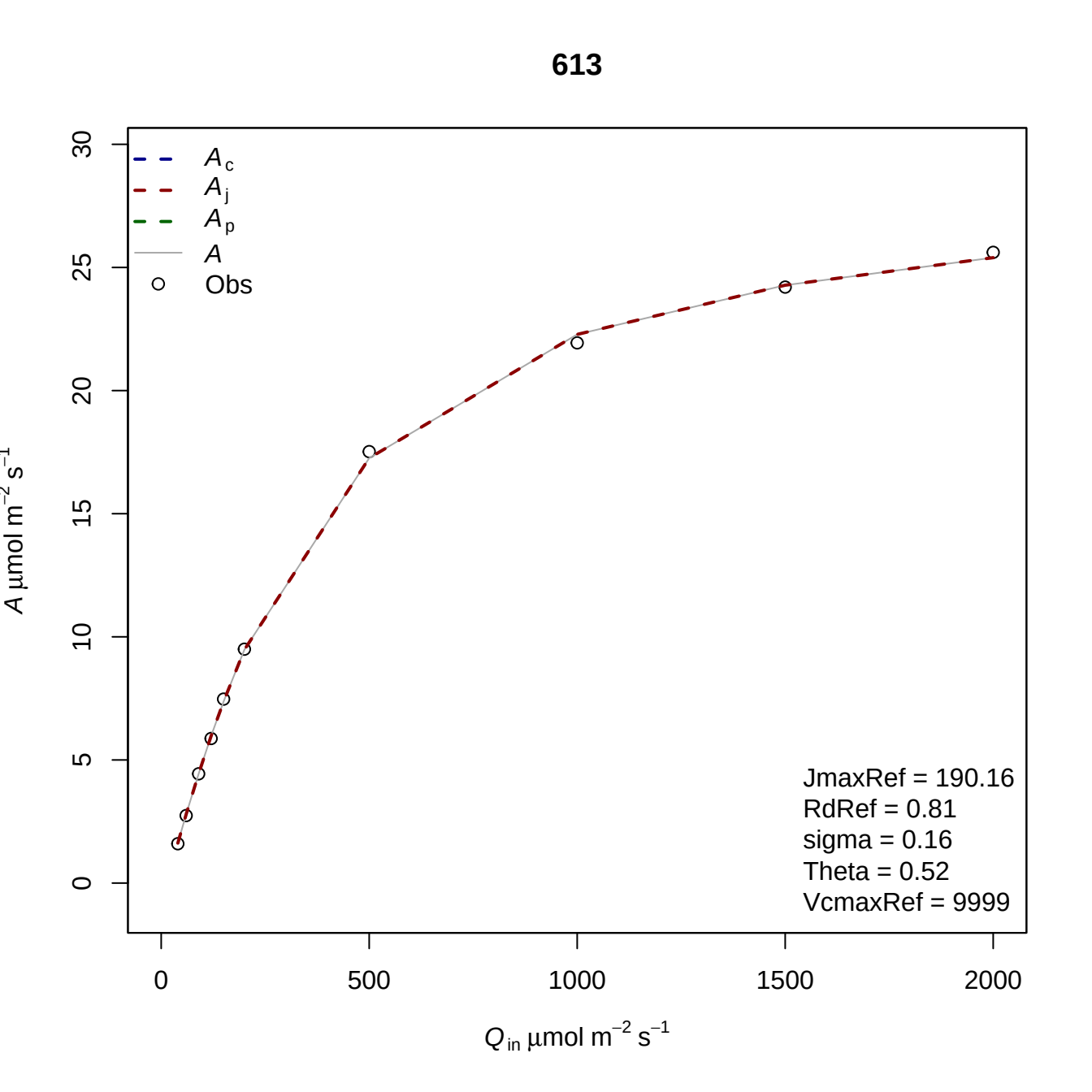

608
A_c
A_j
A_p
A
Obs
JmaxRef = 247.92
RdRef = 1.9
sigma = 0.08
Theta = 0.67
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

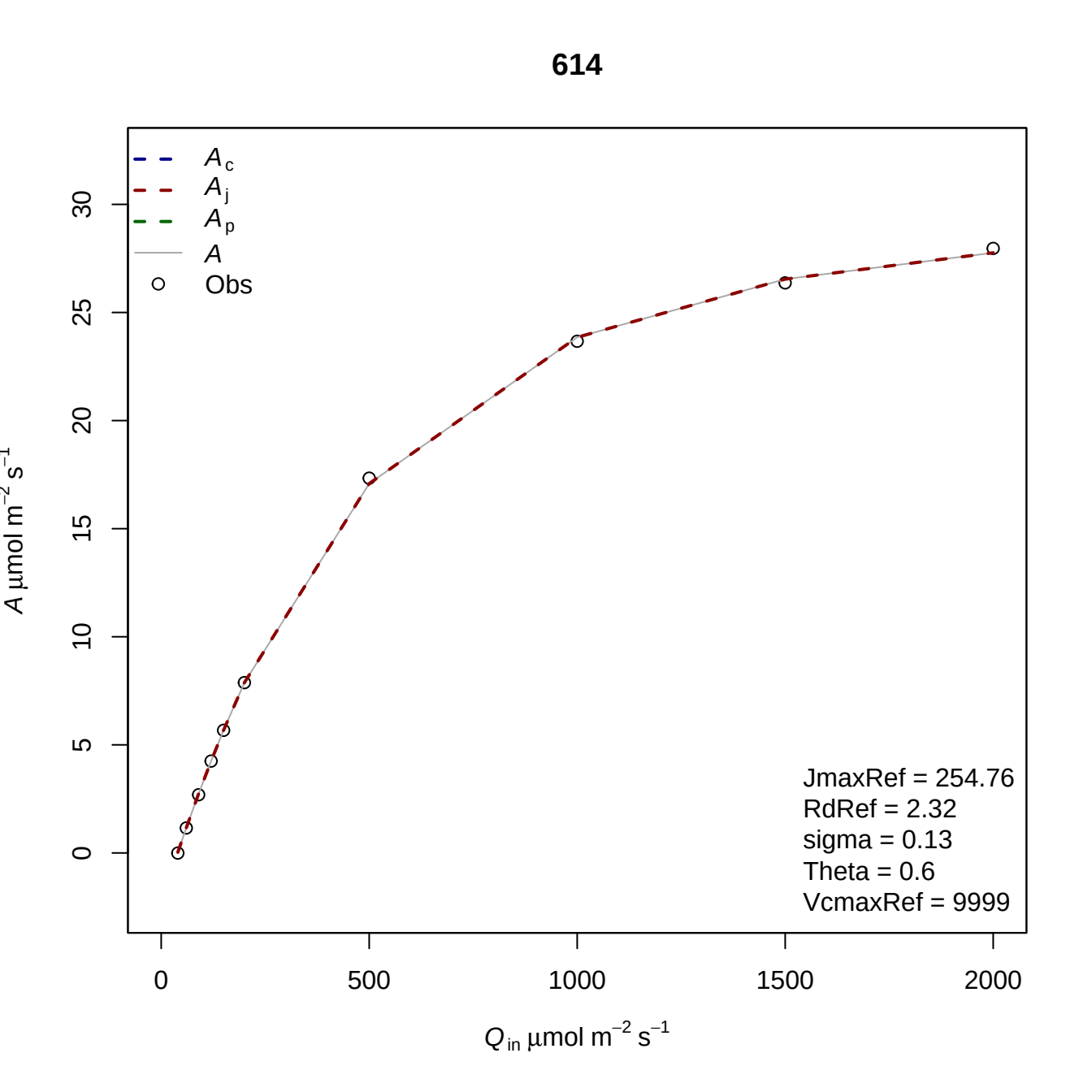

609
A_c
A_j
A_p
A
Obs
JmaxRef = 273.98
RdRef = 2.39
sigma = 0.04
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

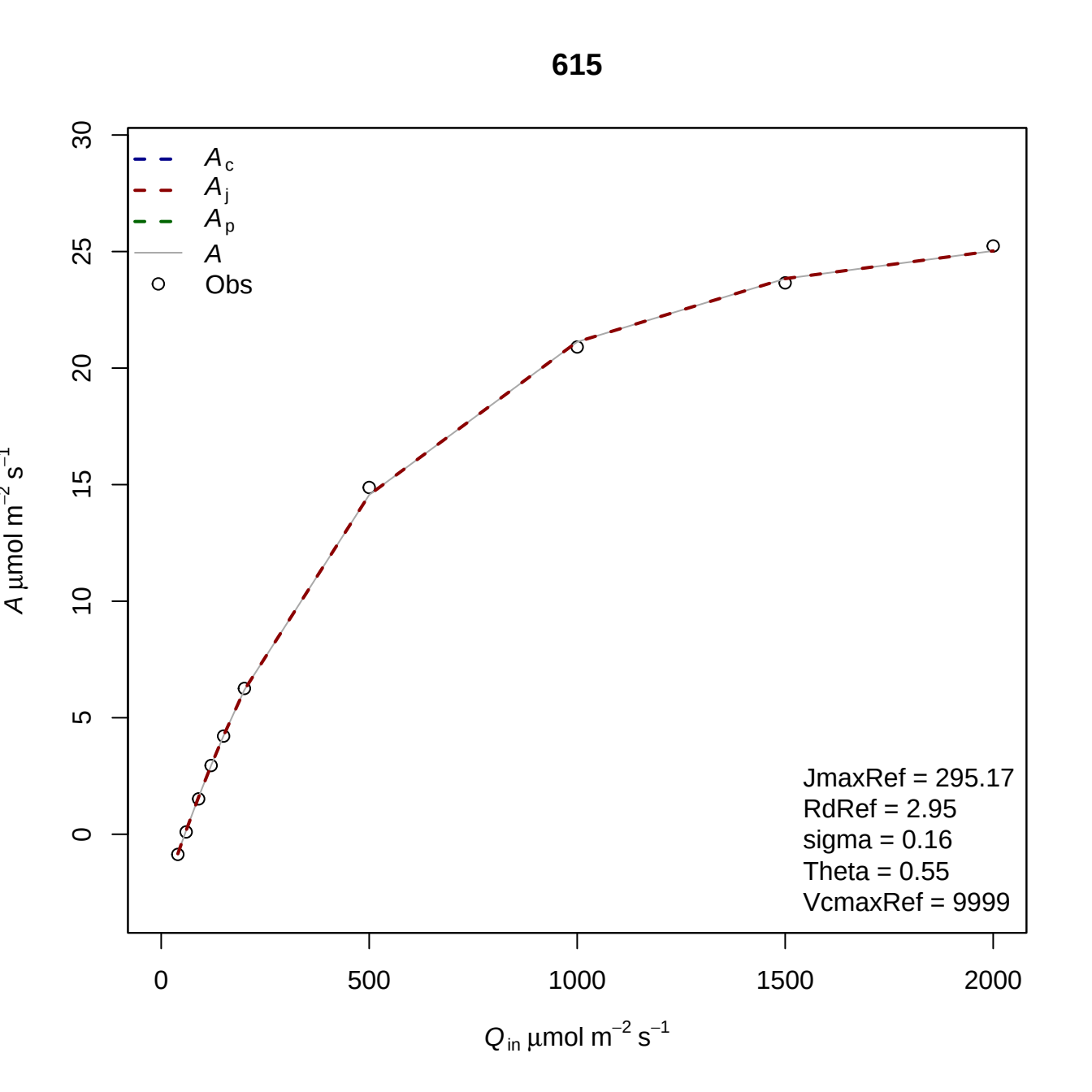

610
A_c
A_j
A_p
A
Obs
JmaxRef = 302.09
RdRef = 3.39
sigma = 0.09
Theta = 0.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

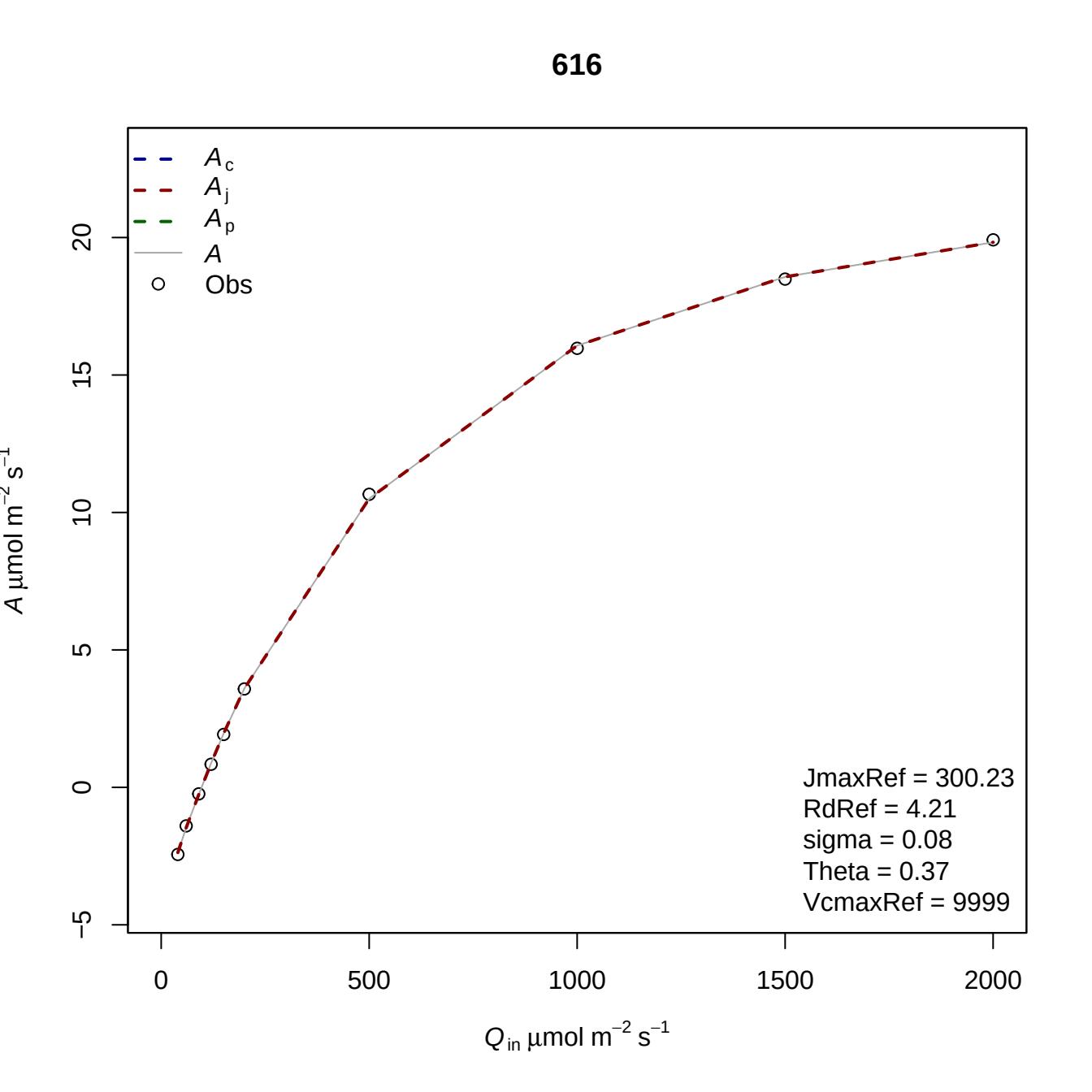

611
A_c
A_j
A_p
A
Obs
JmaxRef = 122.75
RdRef = 0.7
sigma = 0.09
Theta = 0.35
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

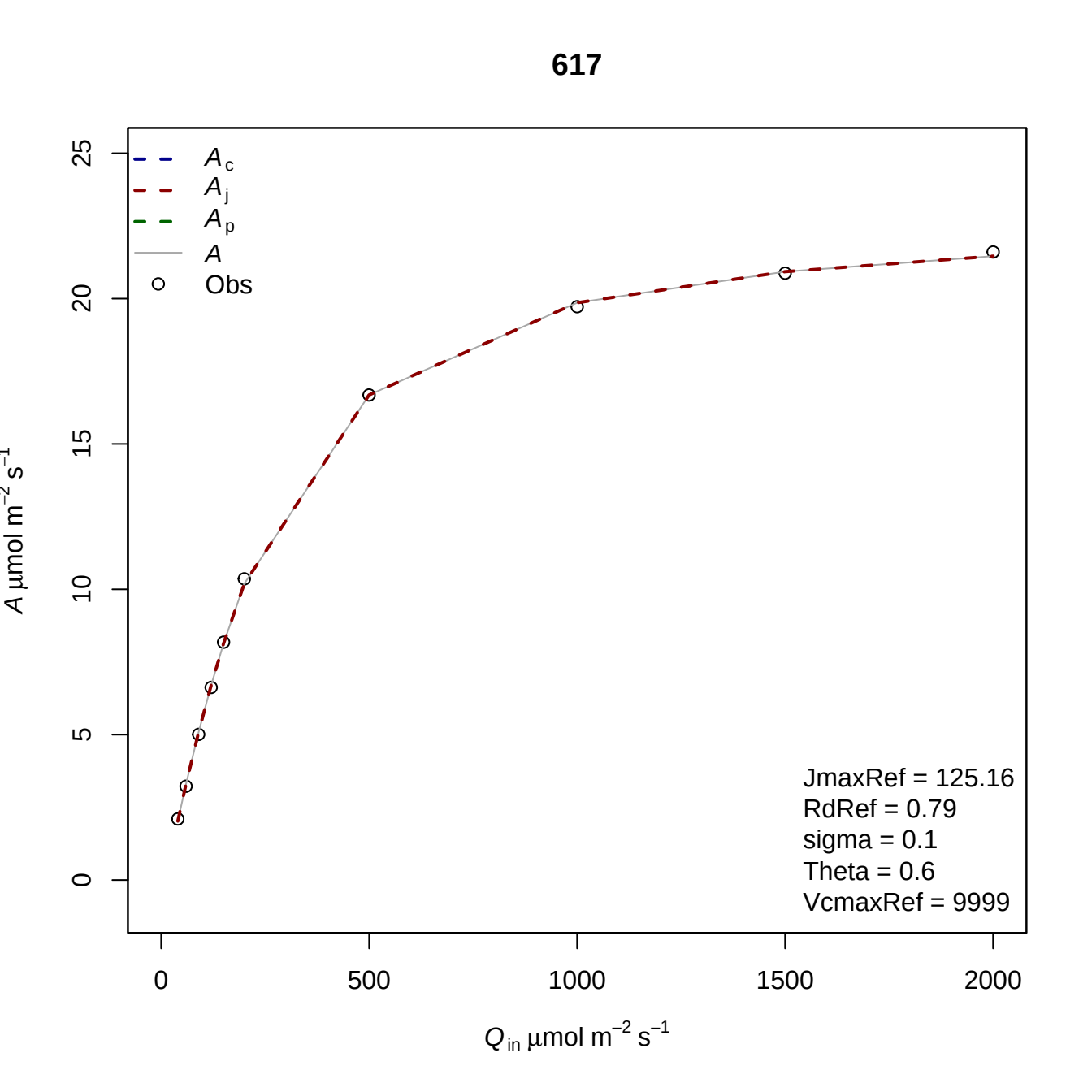

612
A_c
A_j
A_p
A
Obs
JmaxRef = 157.11
RdRef = 1.04
sigma = 0.13
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

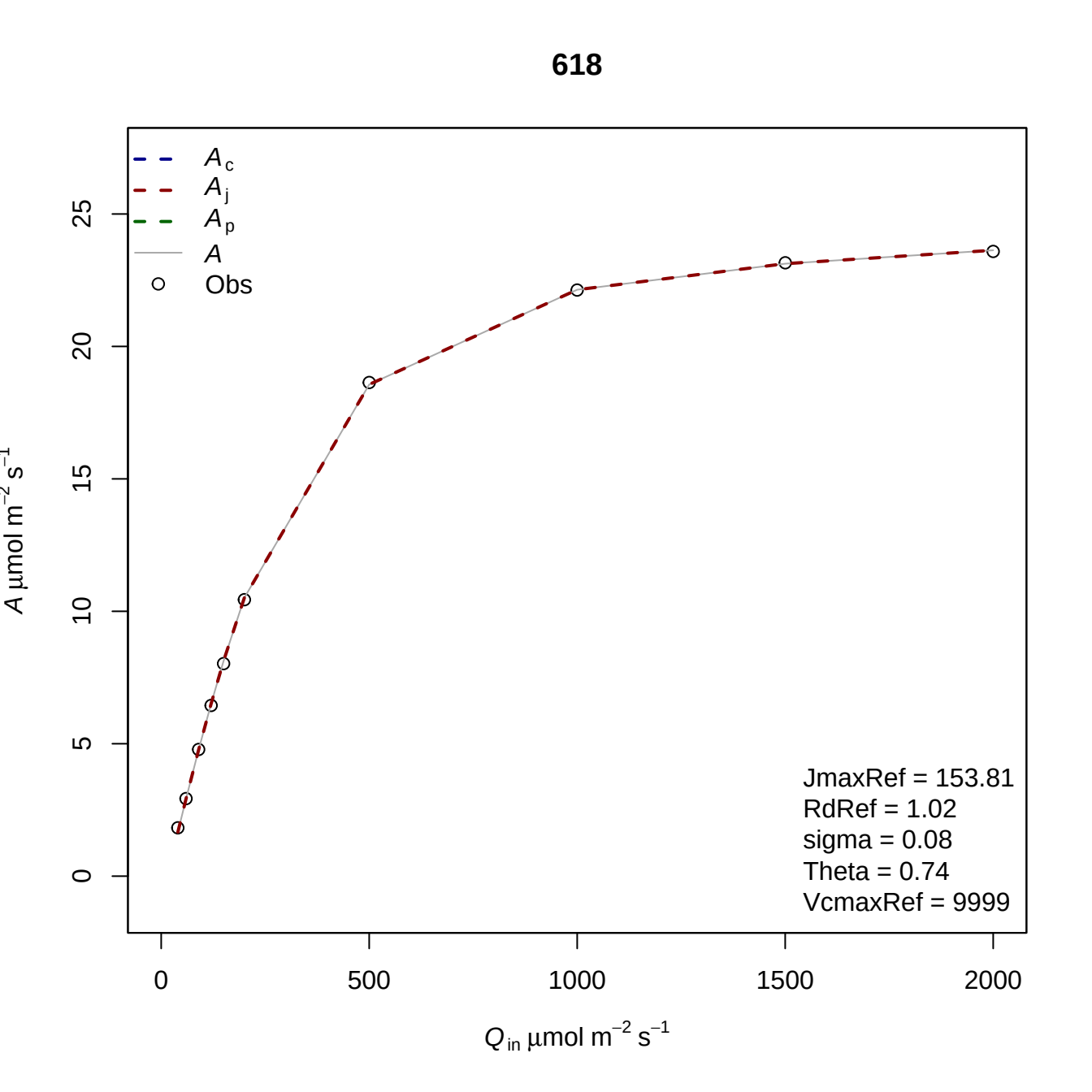

613
A_c
A_j
A_p
A
Obs
JmaxRef = 190.16
RdRef = 0.81
sigma = 0.16
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

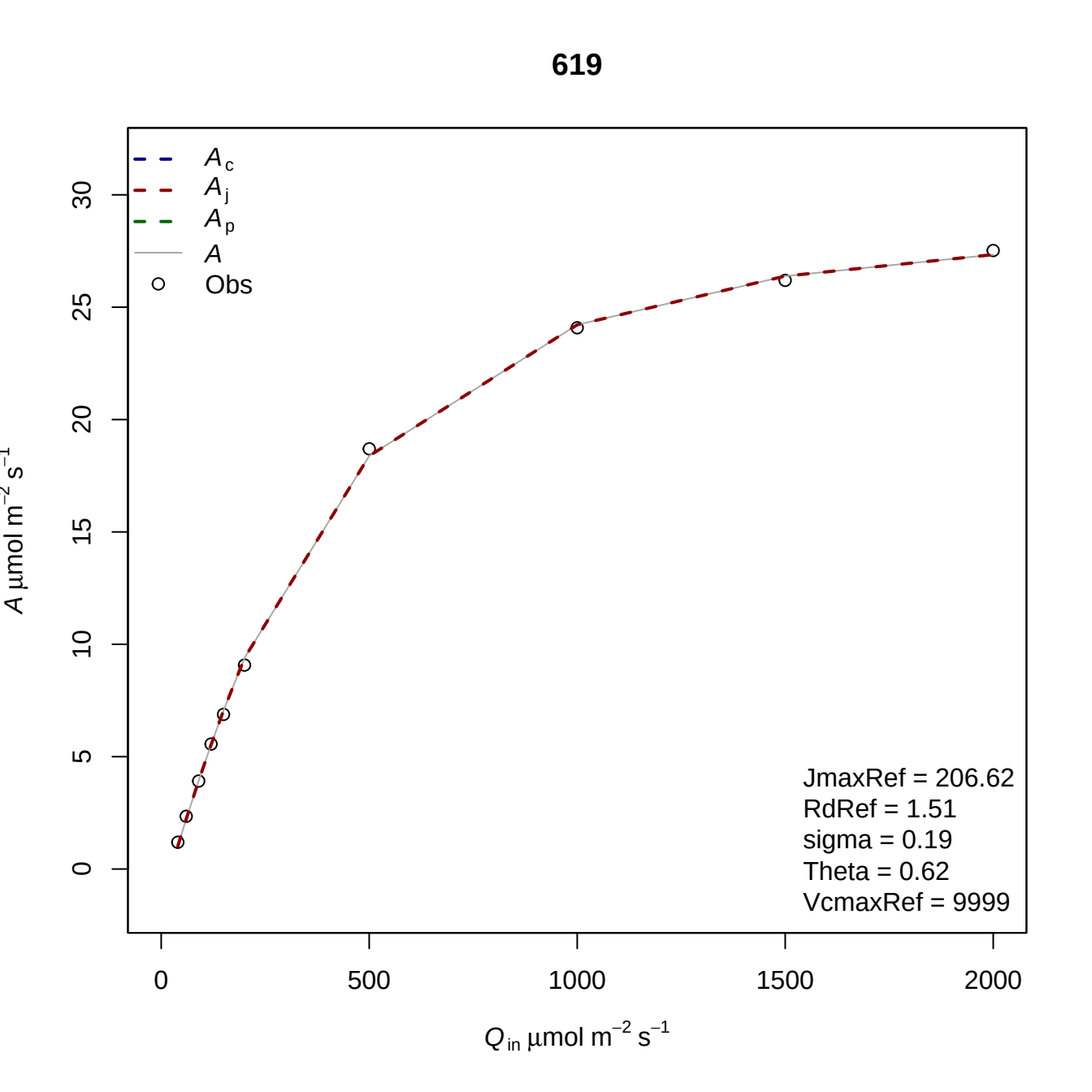

614
A_c
A_j
A_p
A
Obs
JmaxRef = 254.76
RdRef = 2.32
sigma = 0.13
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

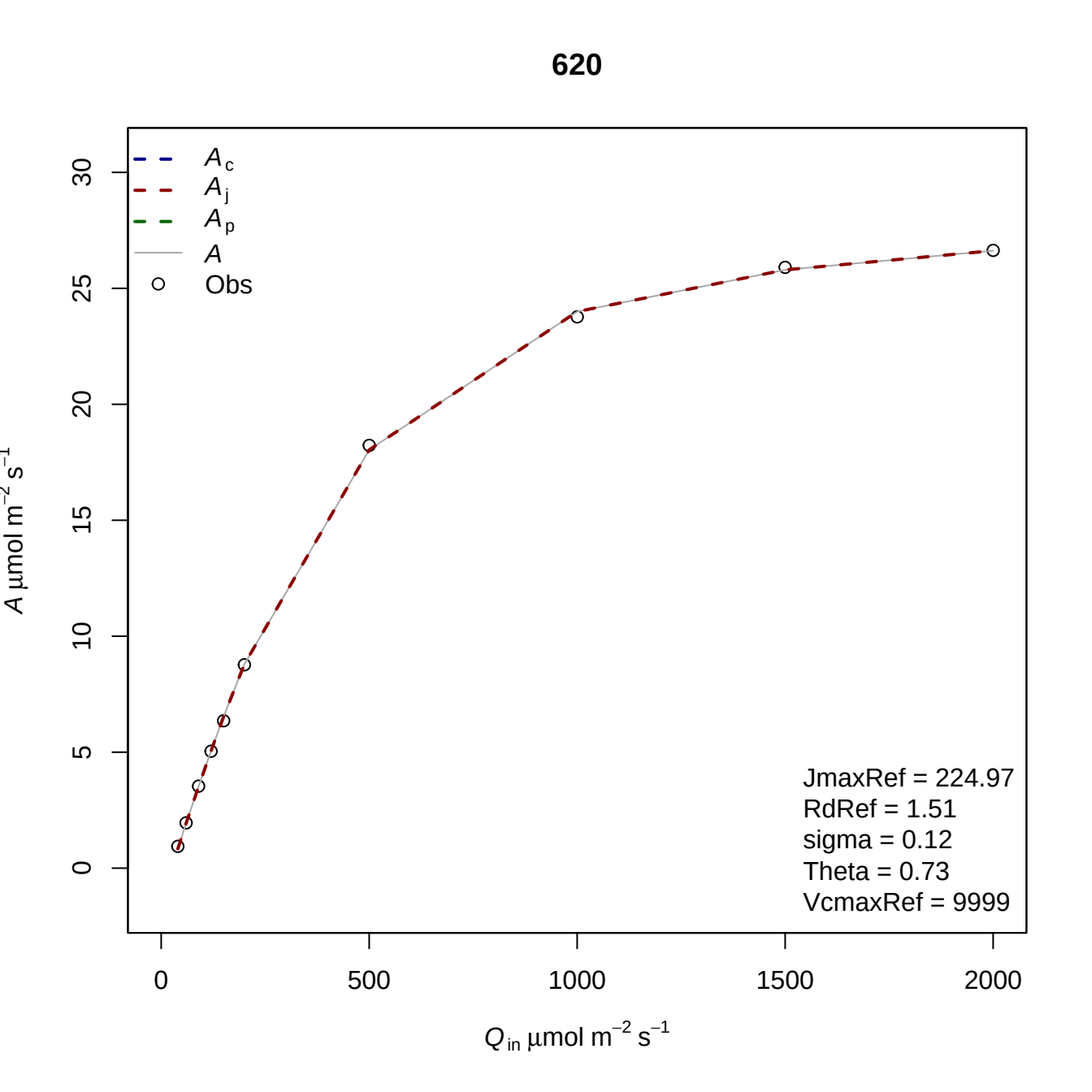

615
A_c
A_j
A_p
A
Obs
JmaxRef = 295.17
RdRef = 2.95
sigma = 0.16
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

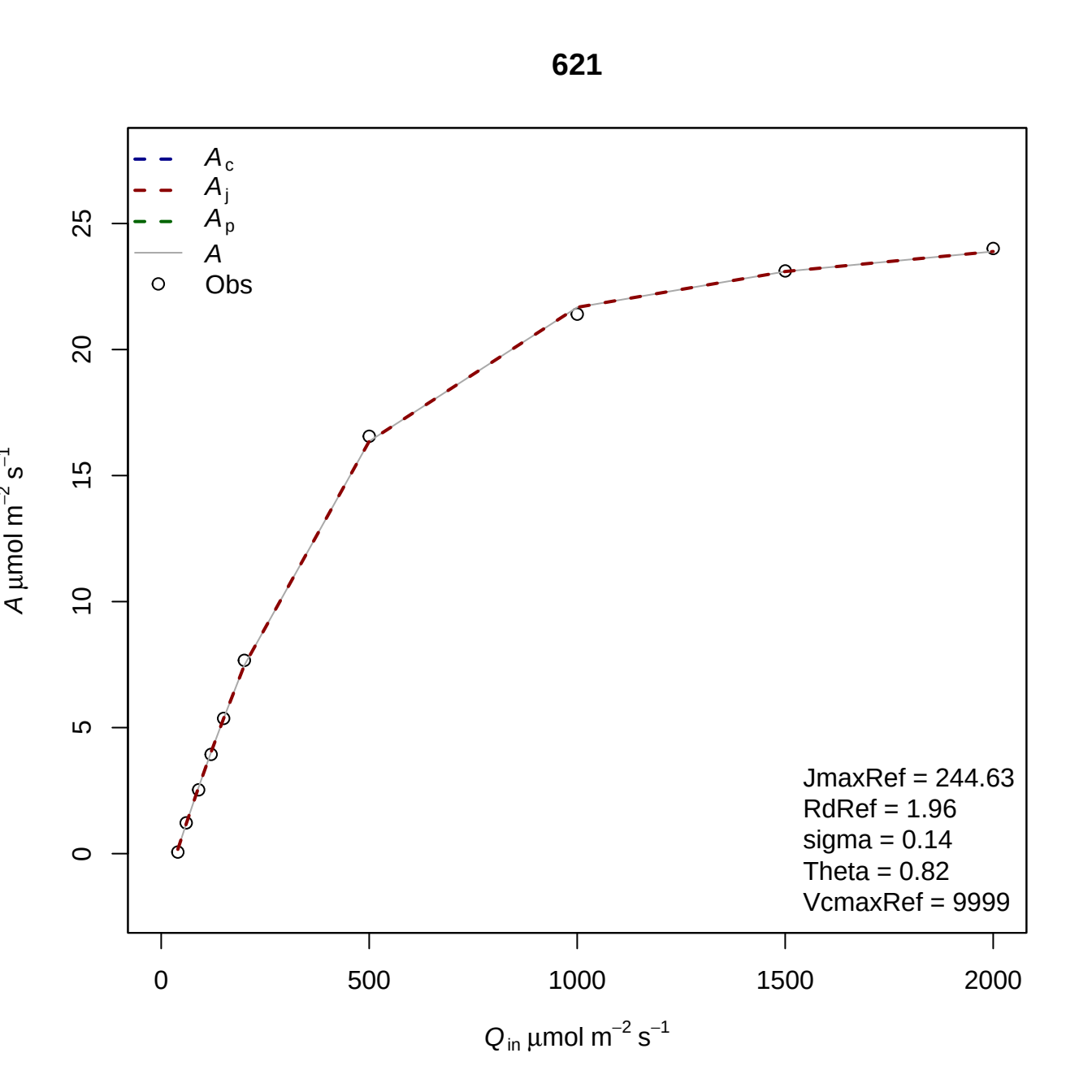

616
A_c
A_j
A_p
A
Obs
JmaxRef = 300.23
RdRef = 4.21
sigma = 0.08
Theta = 0.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

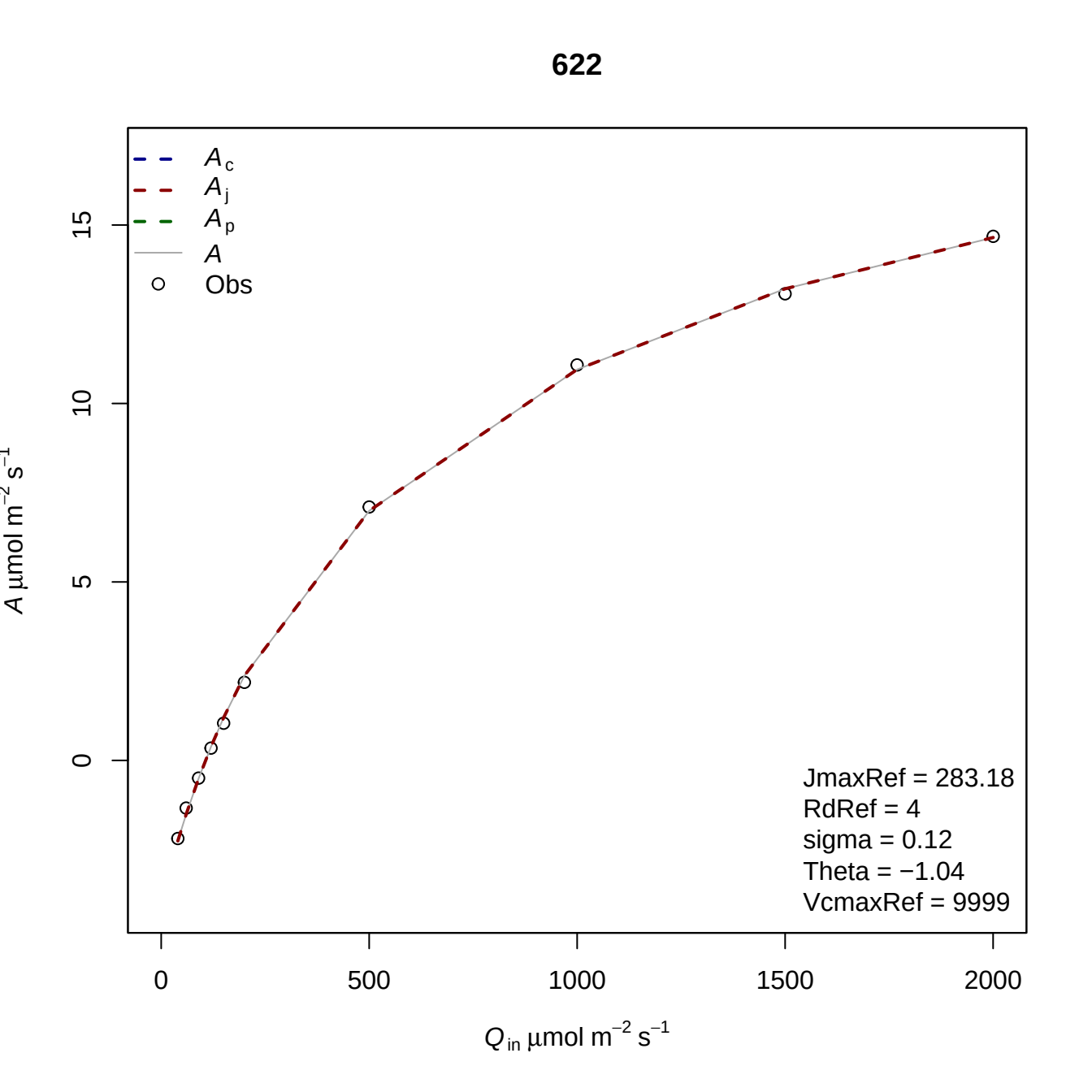

617
A_c
A_j
A_p
A
Obs
JmaxRef = 125.16
RdRef = 0.79
sigma = 0.1
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

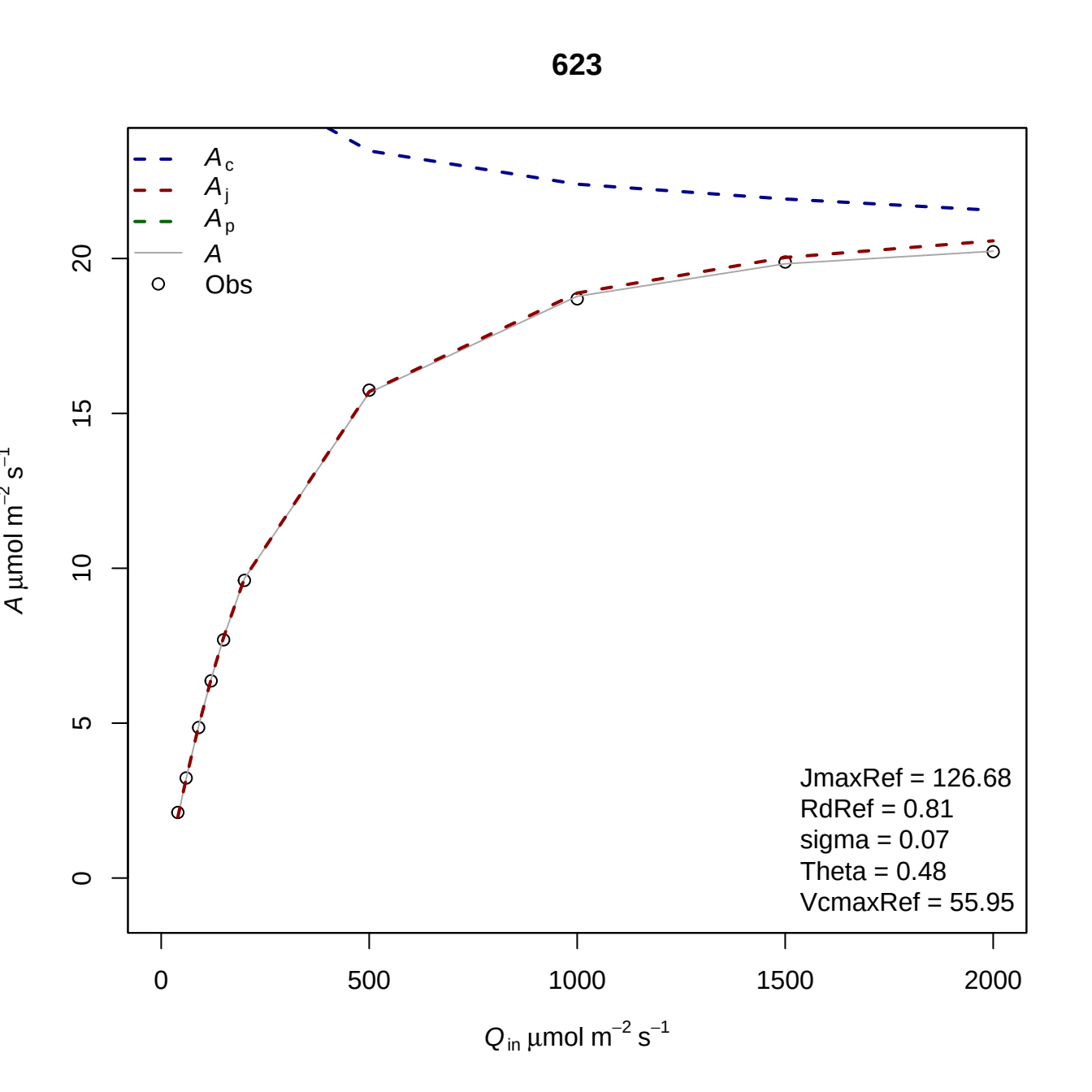

618
A_c
A_j
A_p
A
Obs
JmaxRef = 153.81
RdRef = 1.02
sigma = 0.08
Theta = 0.74
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

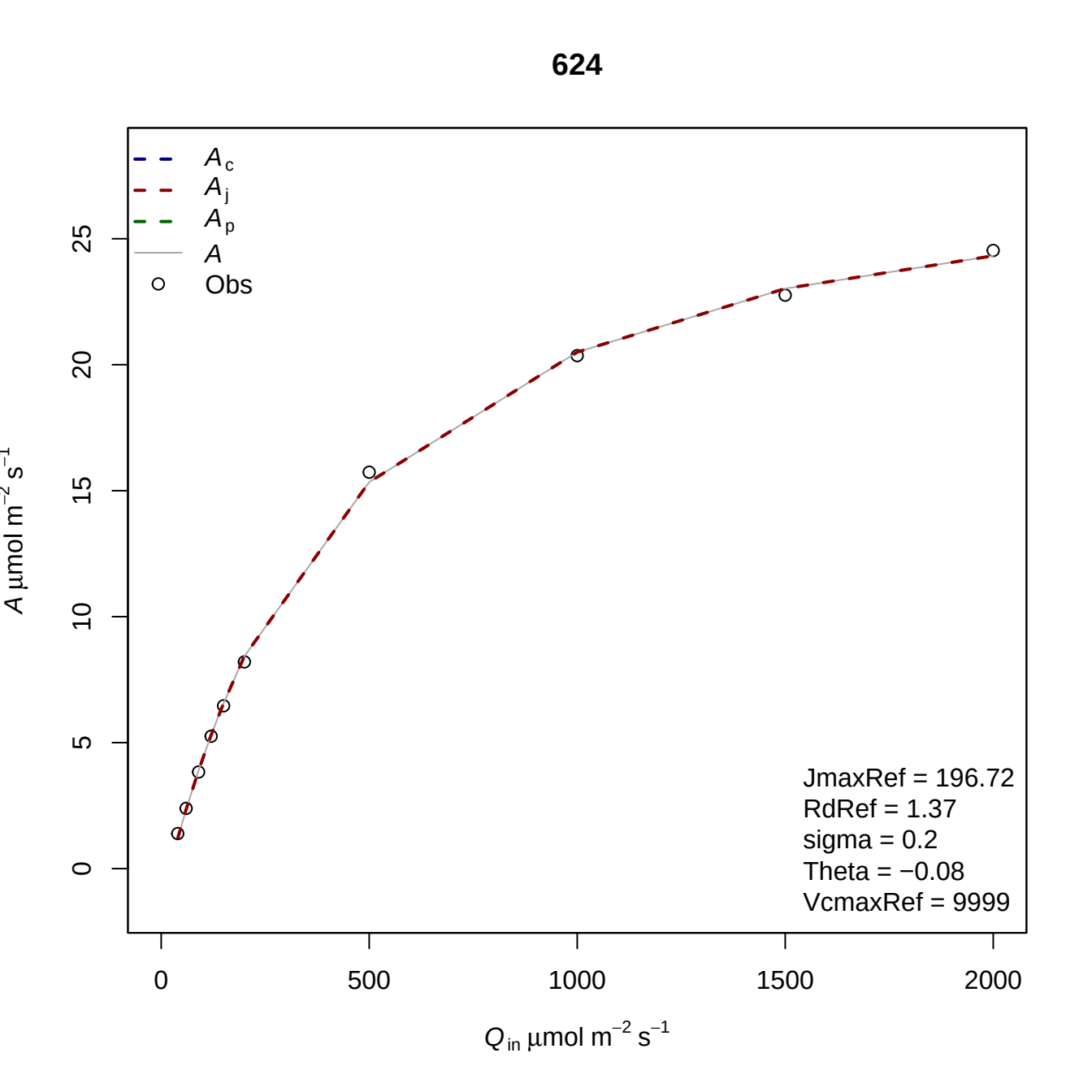

619
A_c
A_j
A_p
A
Obs
JmaxRef = 206.62
RdRef = 1.51
sigma = 0.19
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

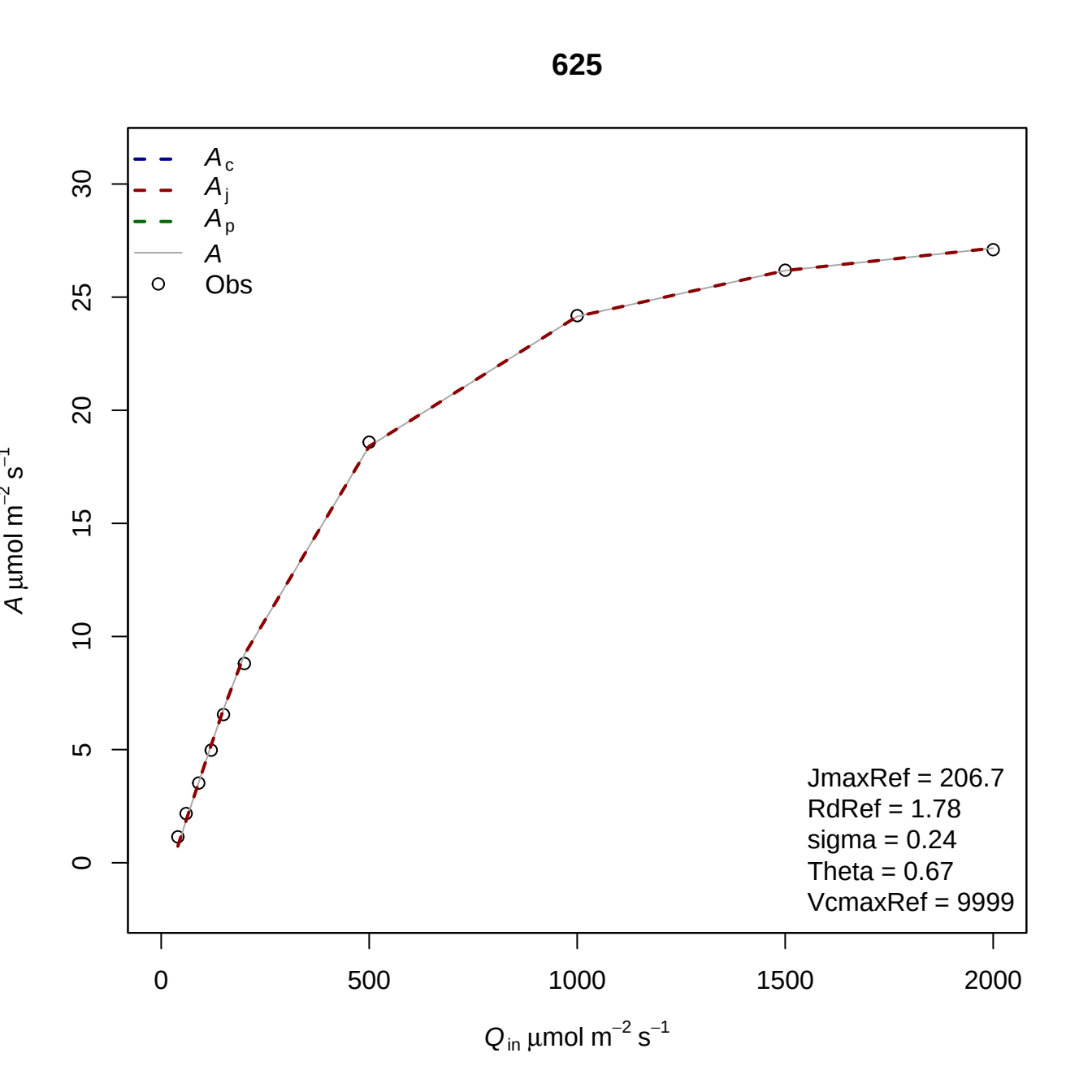

620
A_c
A_j
A_p
A
Obs
JmaxRef = 224.97
RdRef = 1.51
sigma = 0.12
Theta = 0.73
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

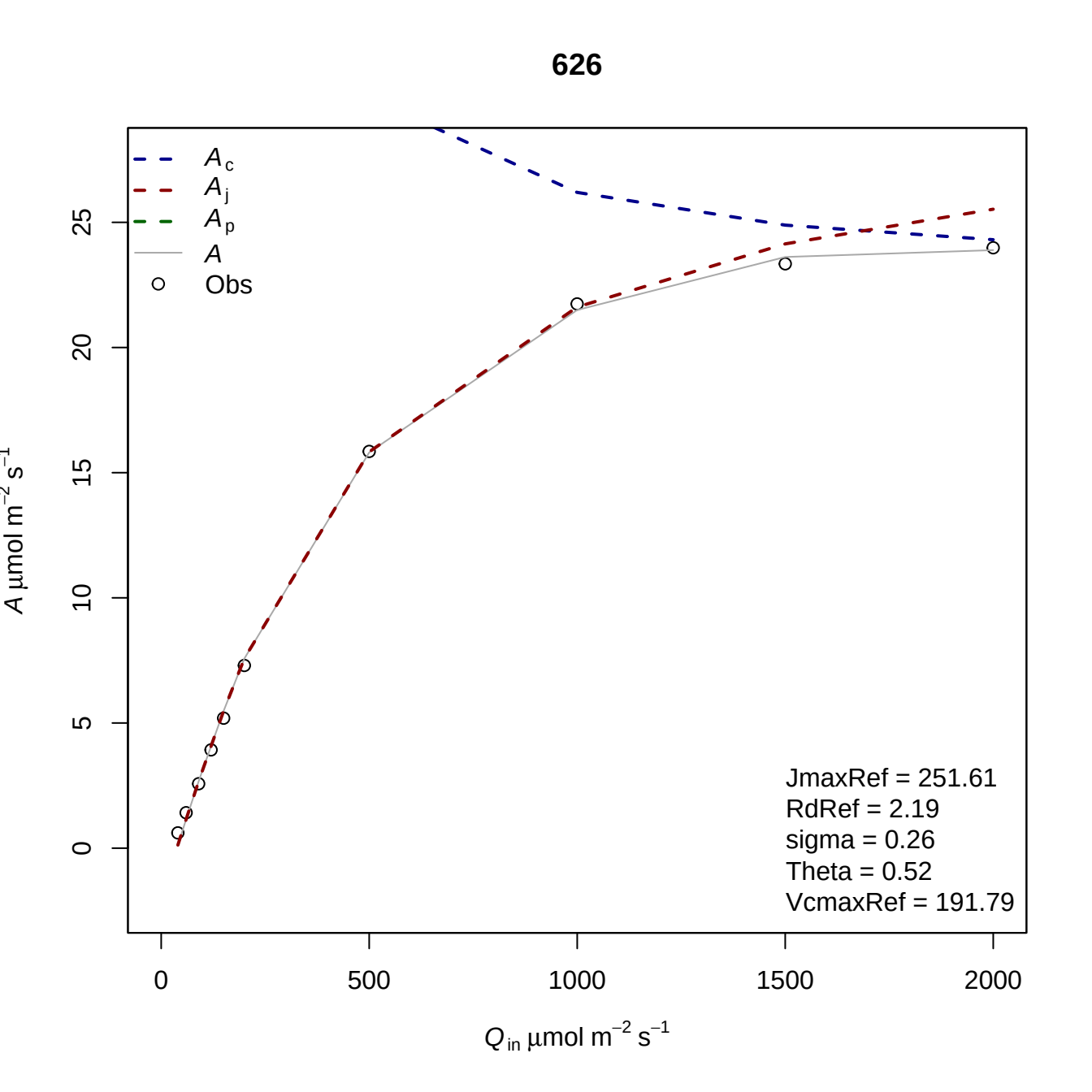

621
A_c
A_j
A_p
A
Obs
JmaxRef = 244.63
RdRef = 1.96
sigma = 0.14
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

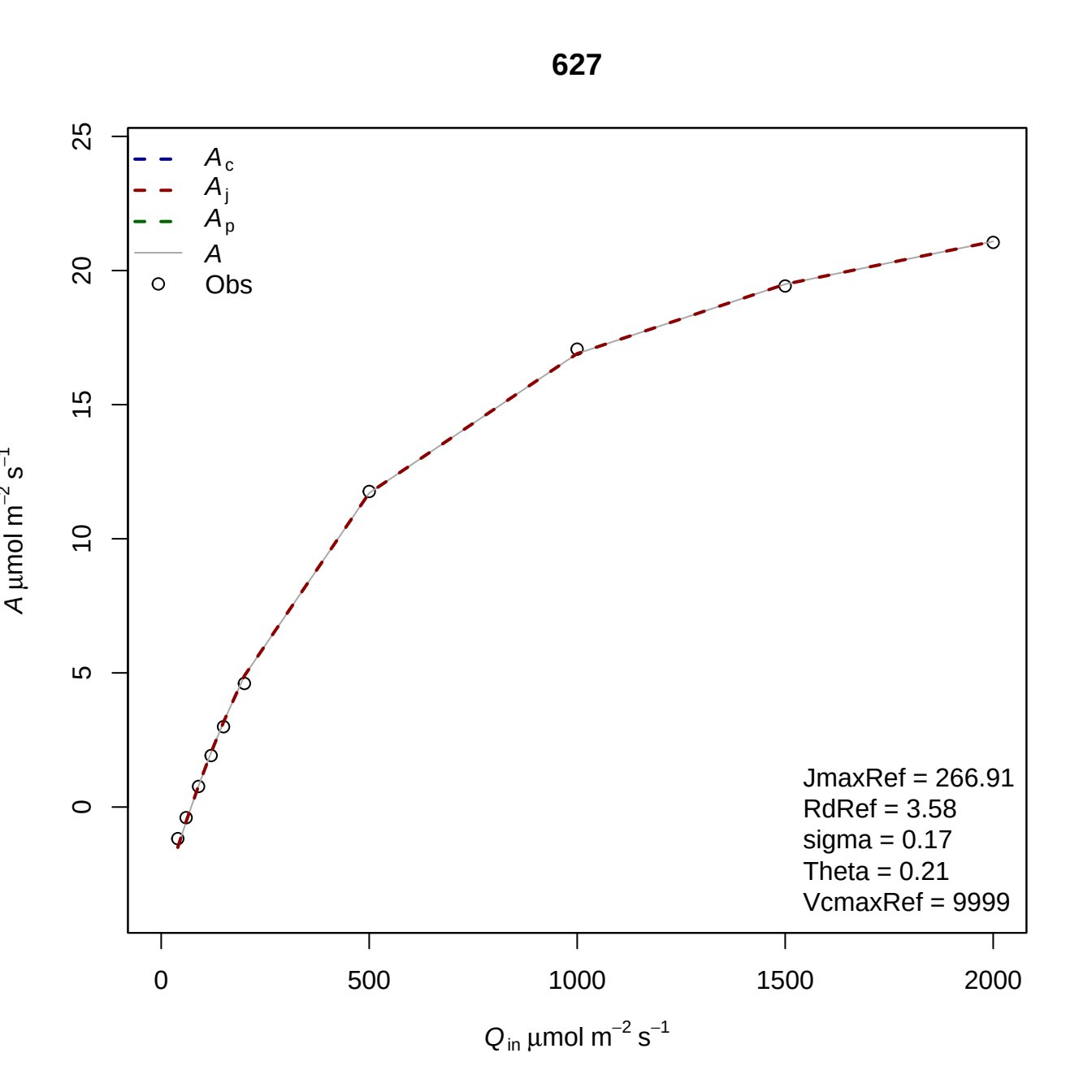

622
A_c
A_j
A_p
A
Obs
JmaxRef = 283.18
RdRef = 4
sigma = 0.12
Theta = −1.04
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

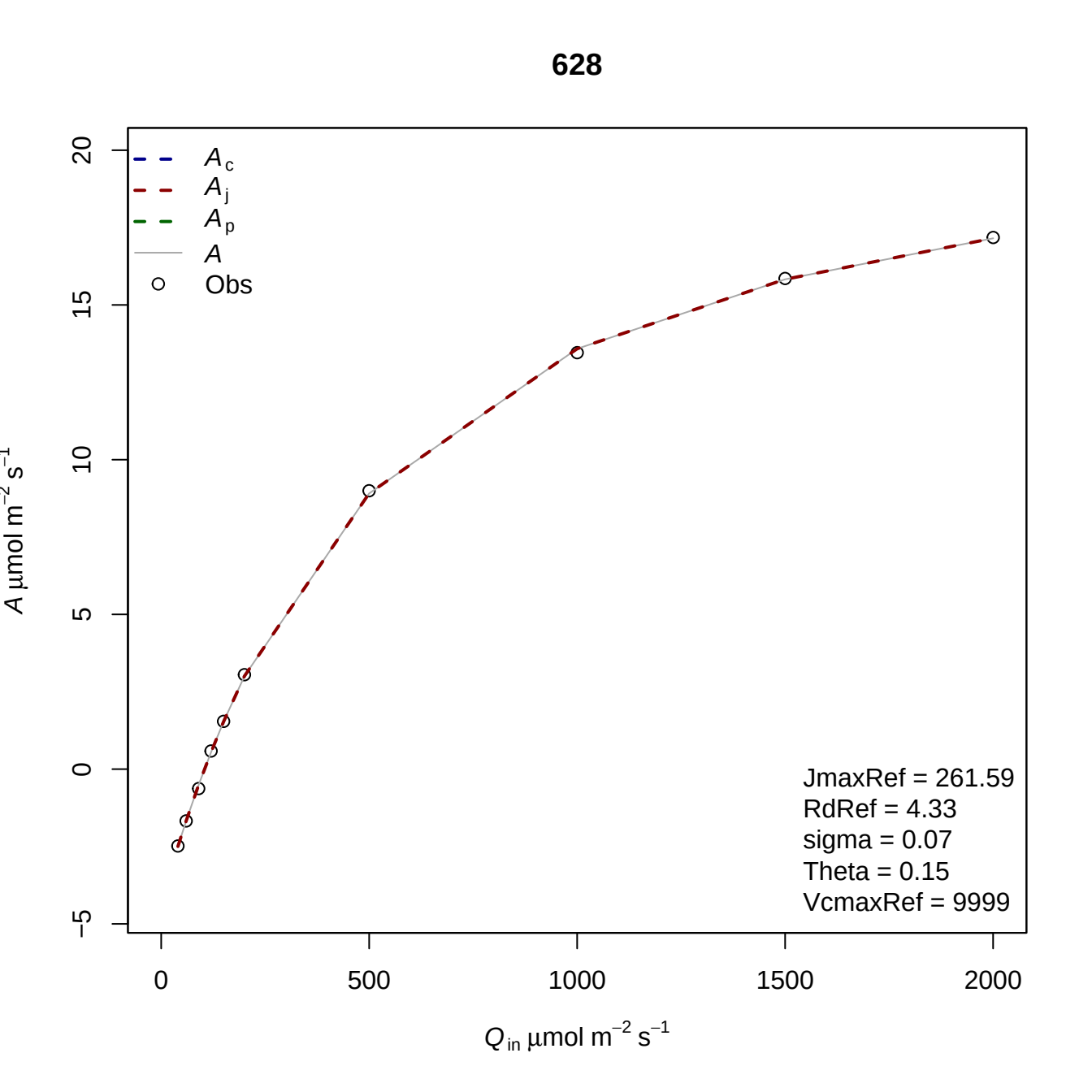

623
A_c
A_j
A_p
A
Obs
JmaxRef = 126.68
RdRef = 0.81
sigma = 0.07
Theta = 0.48
VcmaxRef = 55.95
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

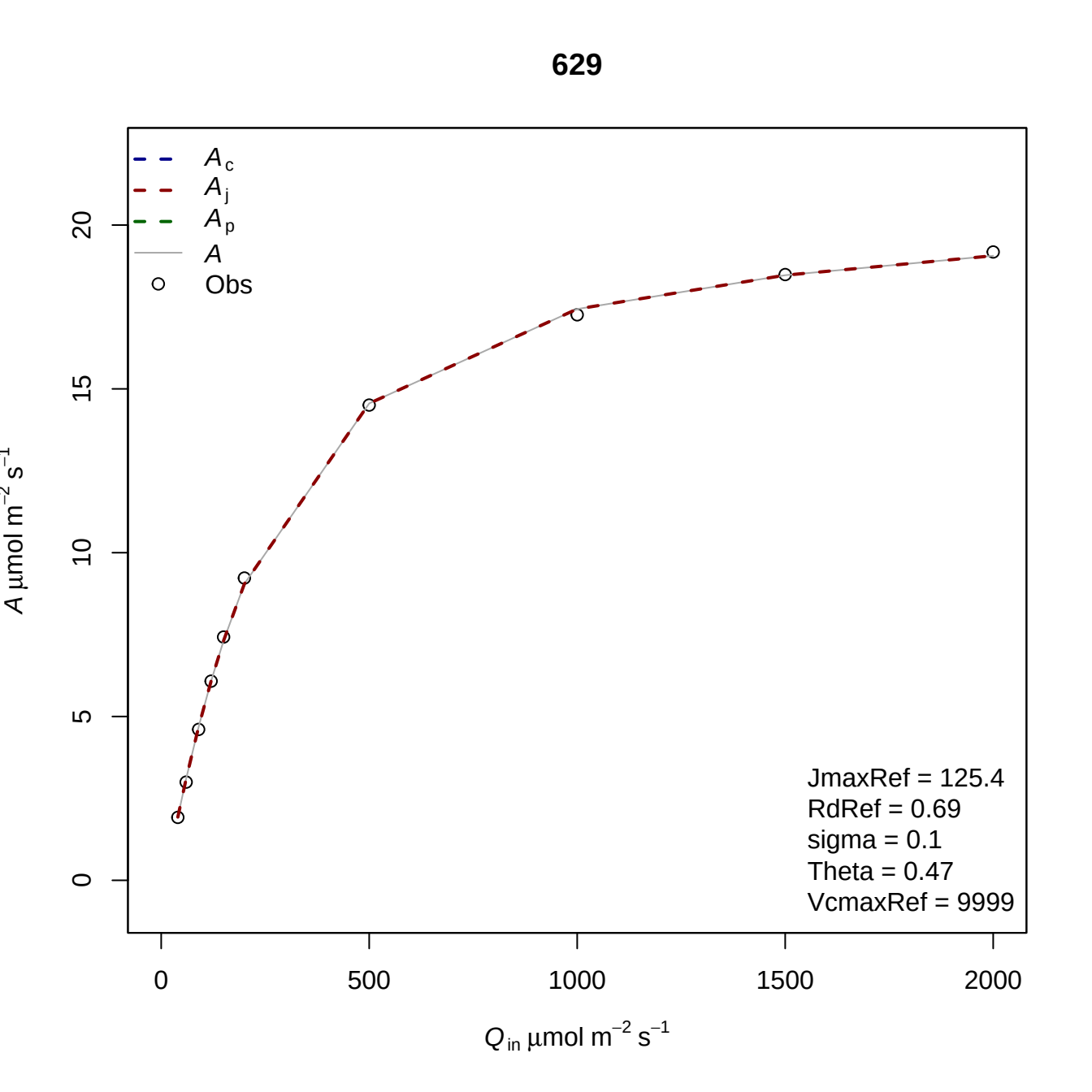

624
A_c
A_j
A_p
A
Obs
JmaxRef = 196.72
RdRef = 1.37
sigma = 0.2
Theta = −0.08
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

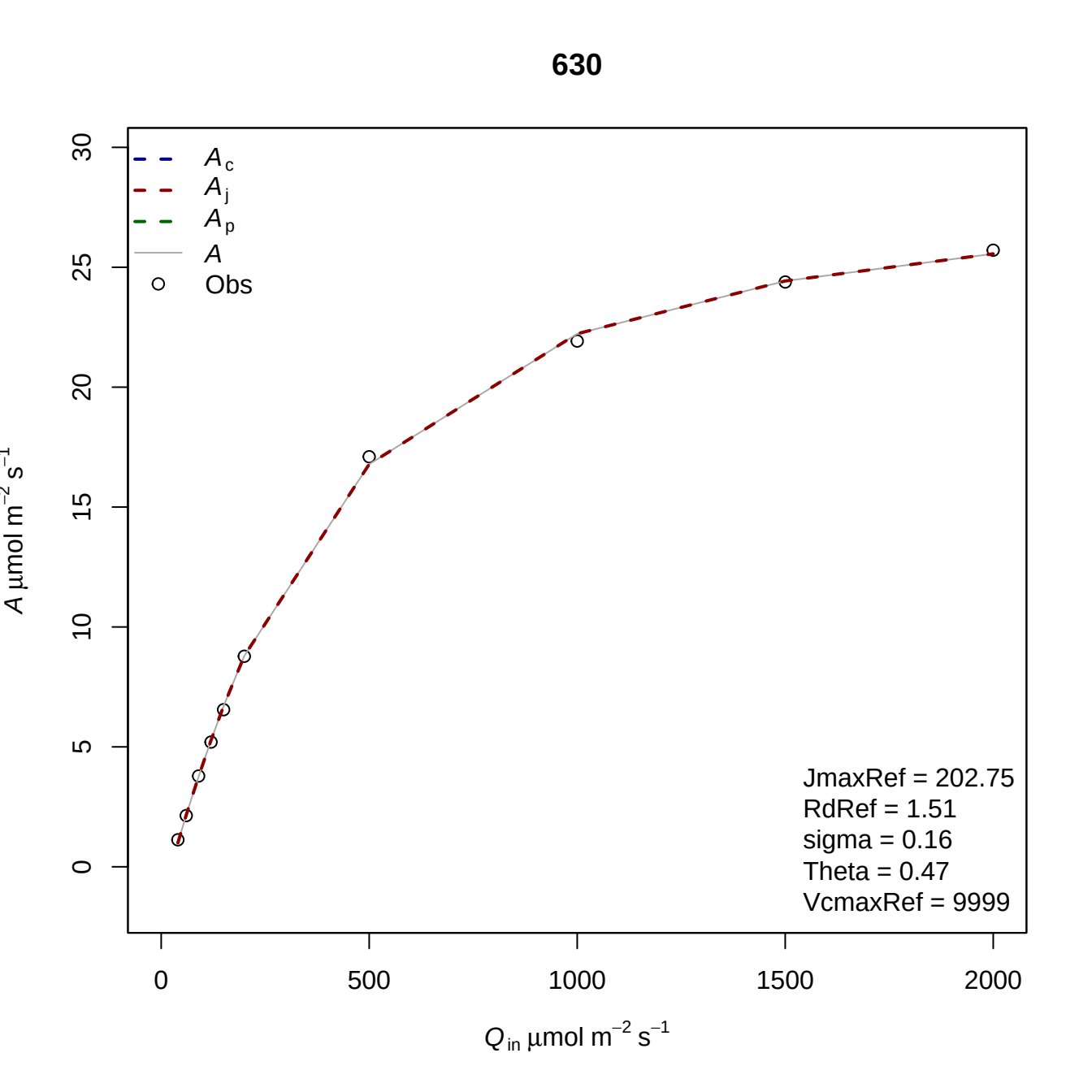

625
A_c
A_j
A_p
A
Obs
JmaxRef = 206.7
RdRef = 1.78
sigma = 0.24
Theta = 0.67
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

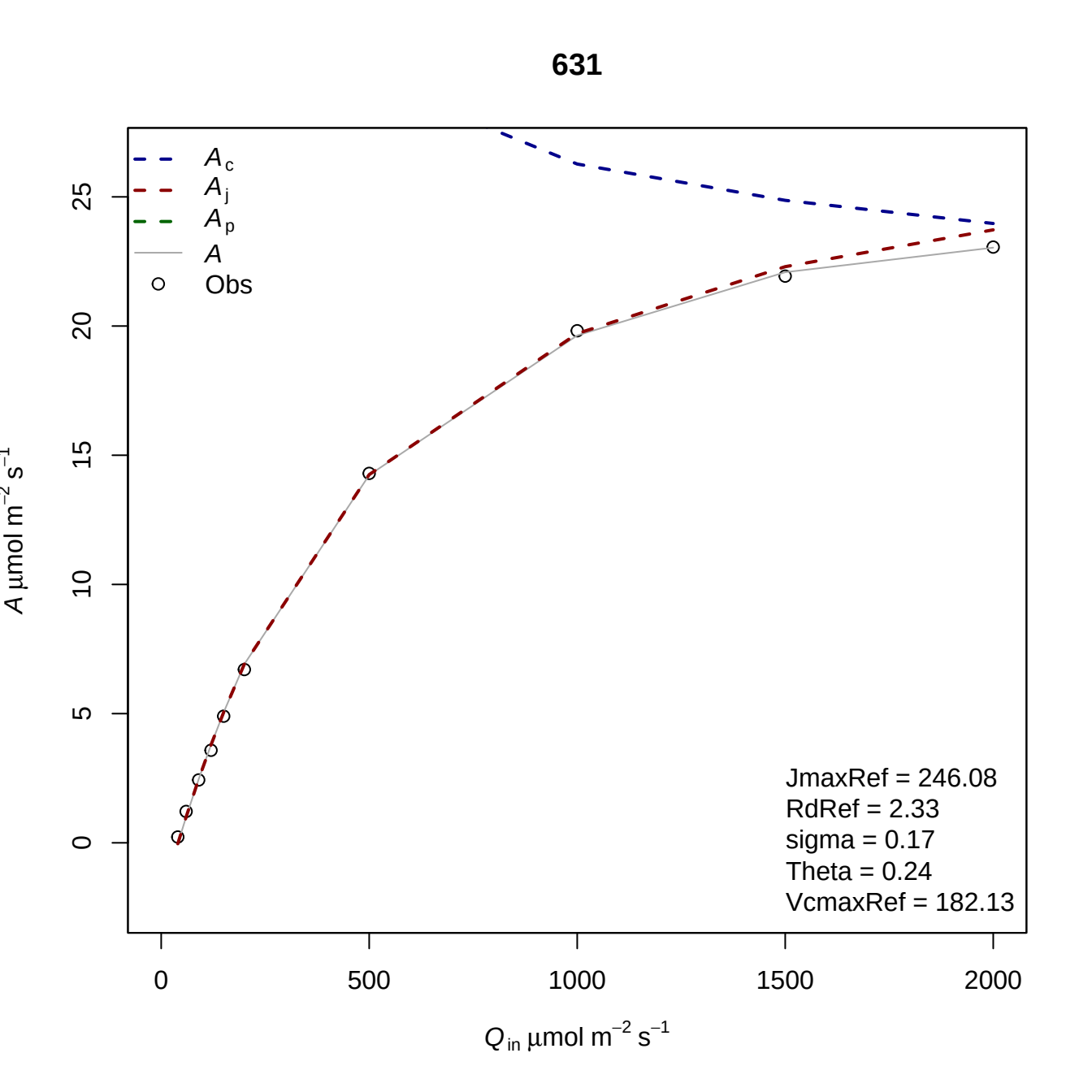

626
A_c
A_j
A_p
A
Obs
JmaxRef = 251.61
RdRef = 2.19
sigma = 0.26
Theta = 0.52
VcmaxRef = 191.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

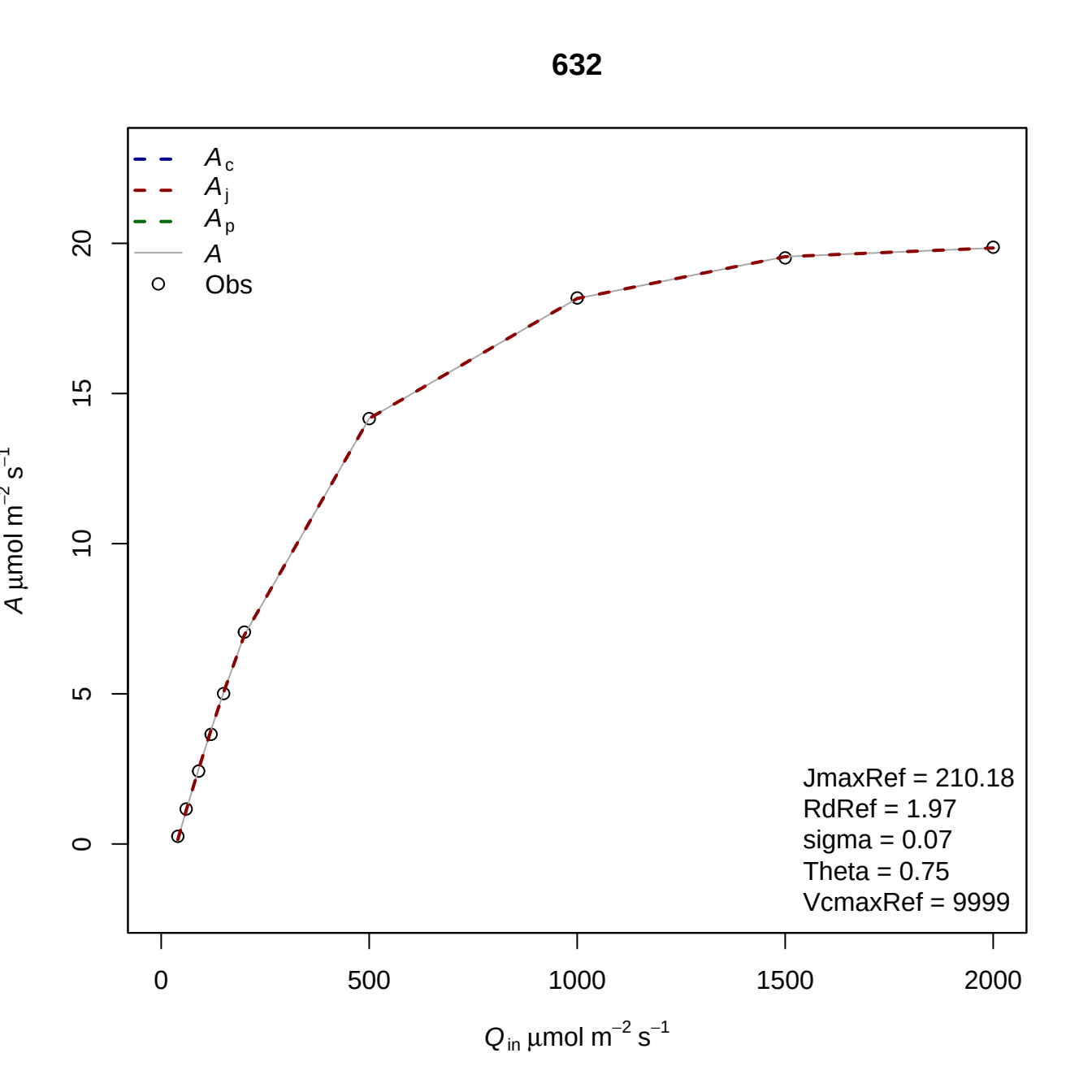

627
A_c
A_j
A_p
A
Obs
JmaxRef = 266.91
RdRef = 3.58
sigma = 0.17
Theta = 0.21
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

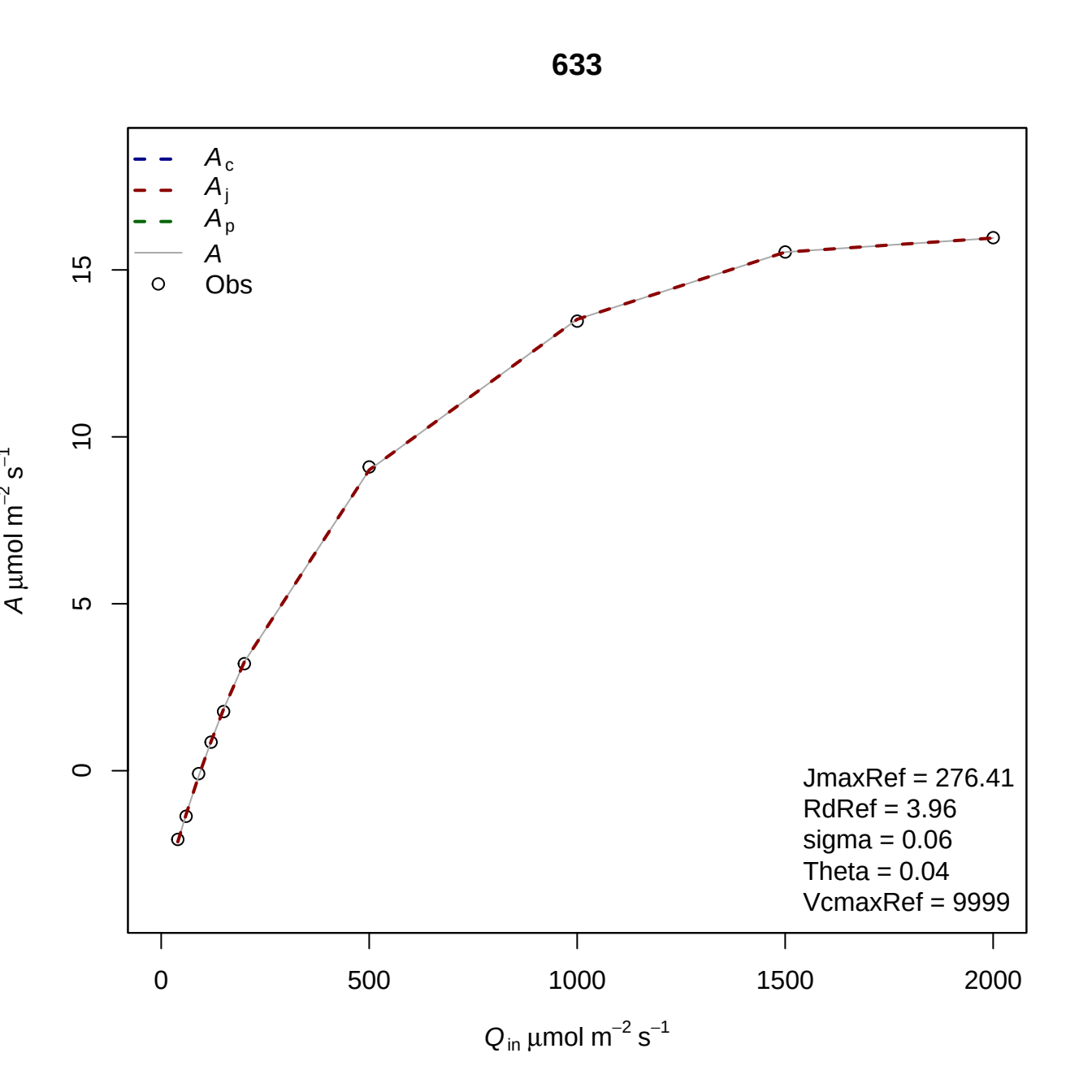

628
A_c
A_j
A_p
A
Obs
JmaxRef = 261.59
RdRef = 4.33
sigma = 0.07
Theta = 0.15
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

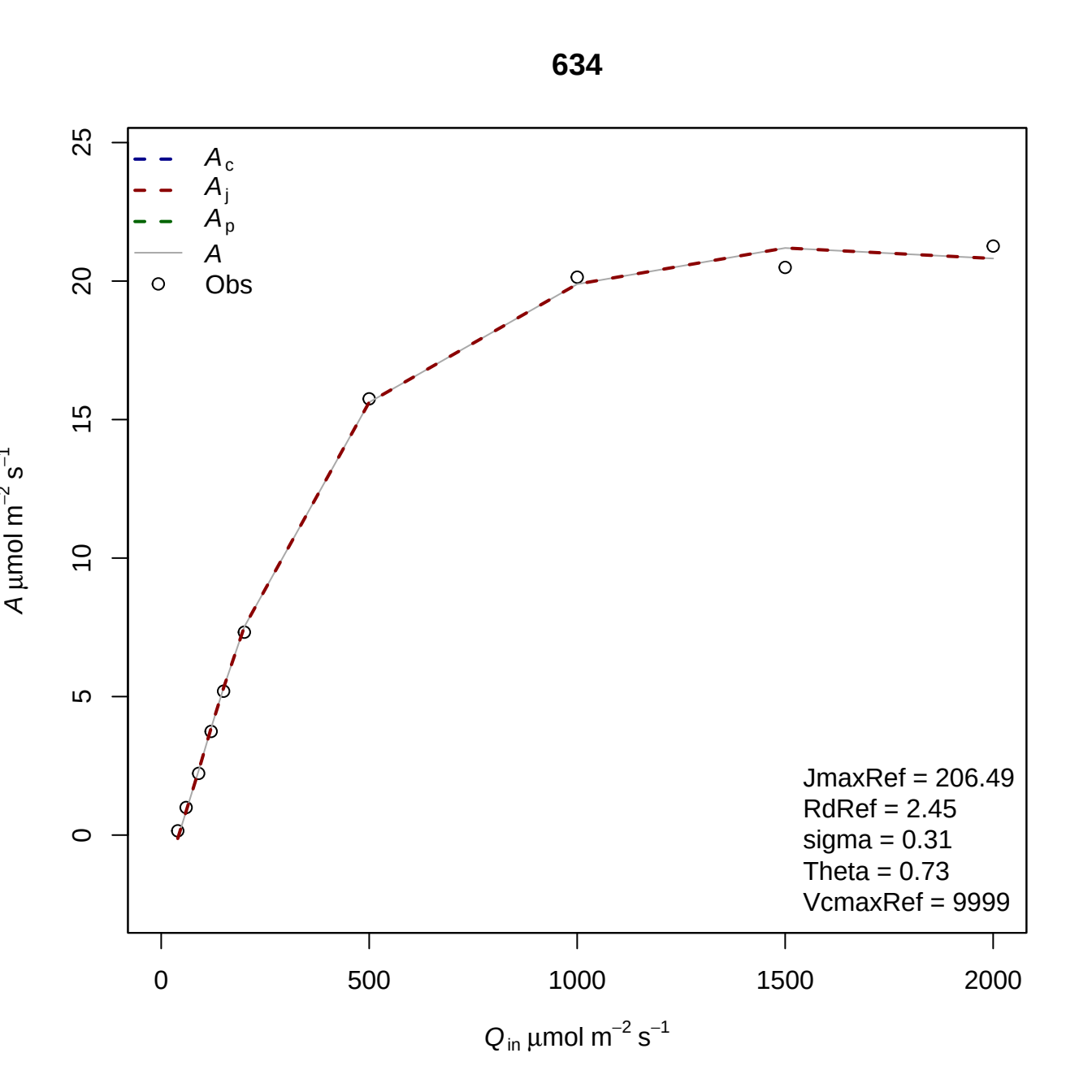

629
A_c
A_j
A_p
A
Obs
JmaxRef = 125.4
RdRef = 0.69
sigma = 0.1
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

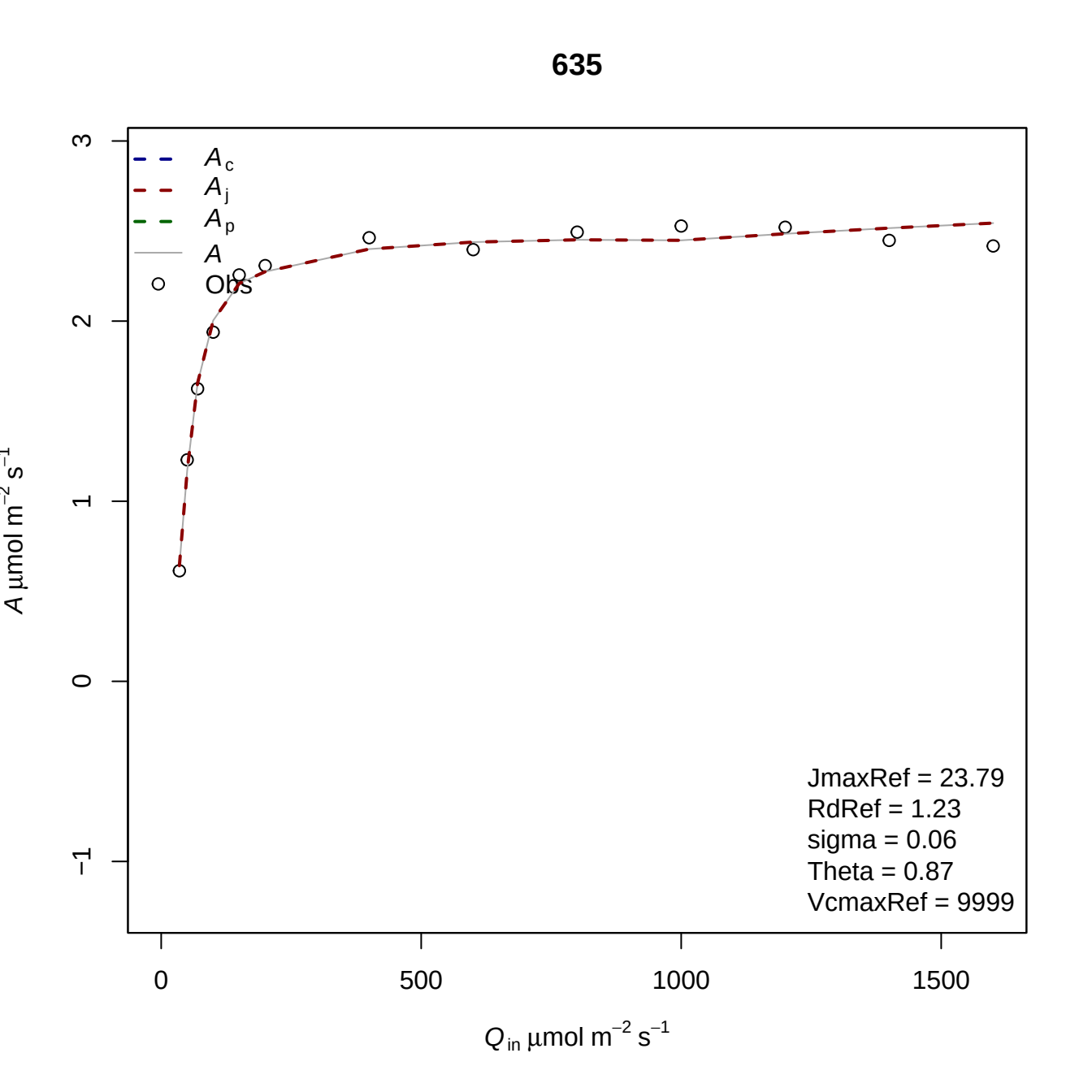

630
A_c
A_j
A_p
A
Obs
JmaxRef = 202.75
RdRef = 1.51
sigma = 0.16
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

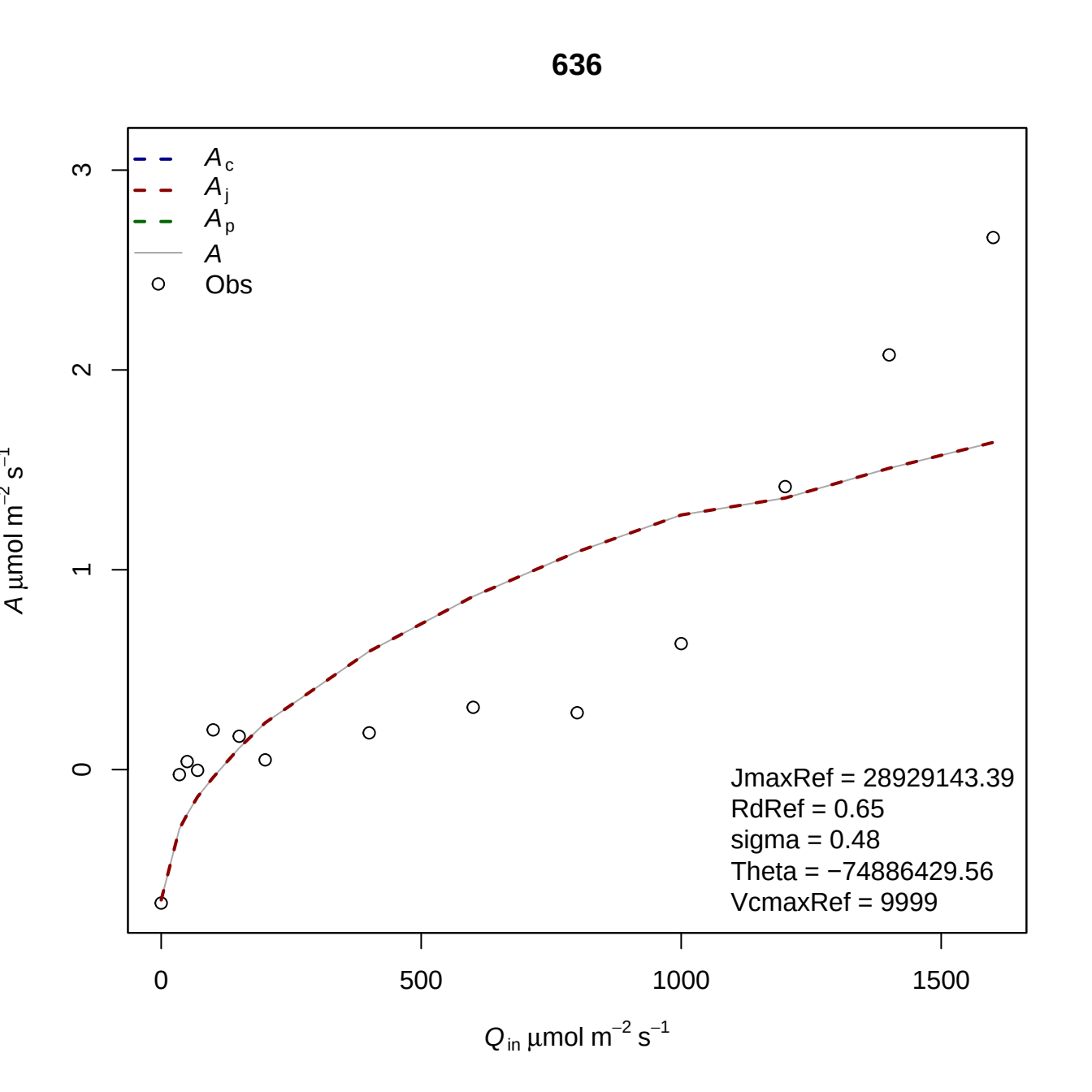

631
A_c
A_j
A_p
A
Obs
JmaxRef = 246.08
RdRef = 2.33
sigma = 0.17
Theta = 0.24
VcmaxRef = 182.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

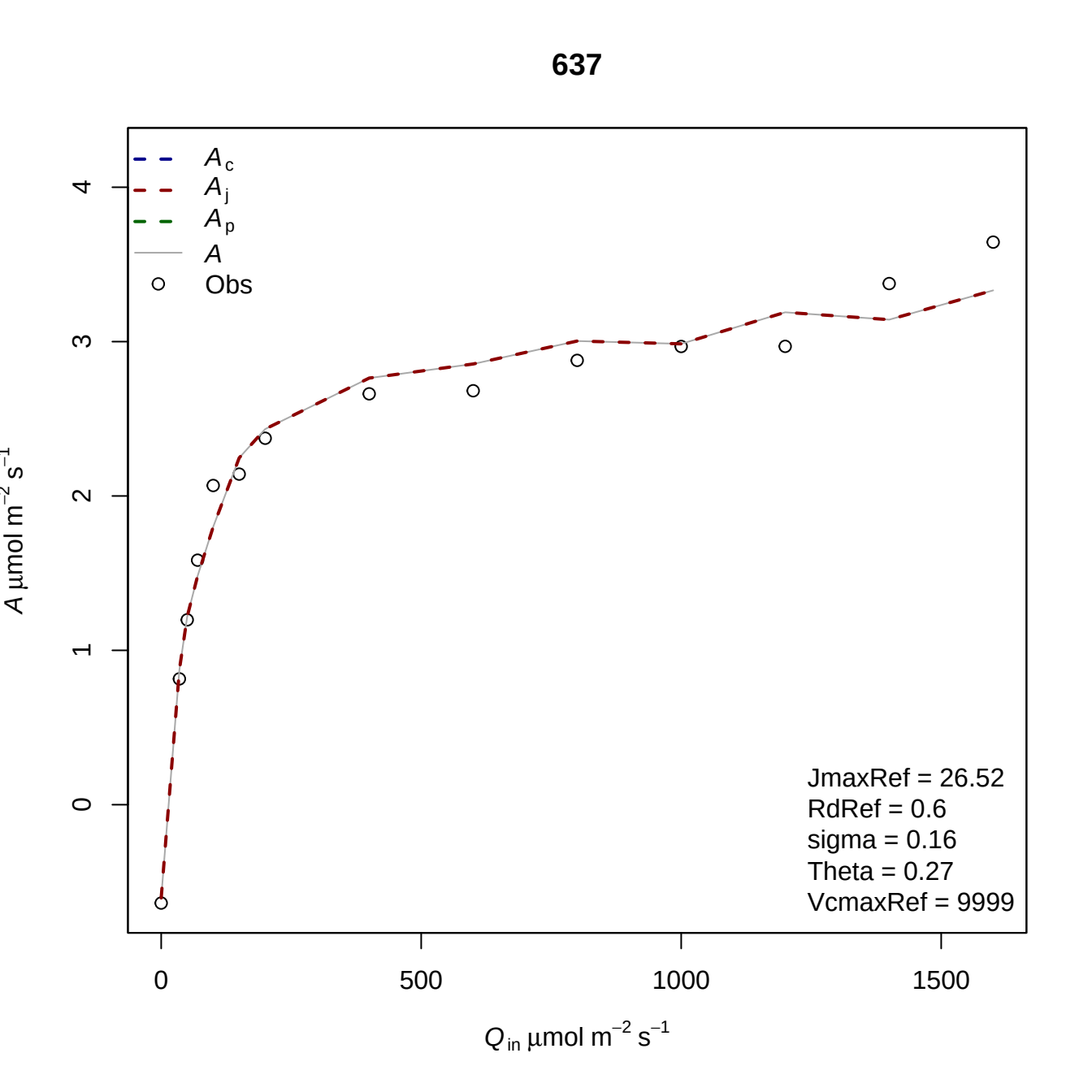

632
A_c
A_j
A_p
A
Obs
JmaxRef = 210.18
RdRef = 1.97
sigma = 0.07
Theta = 0.75
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

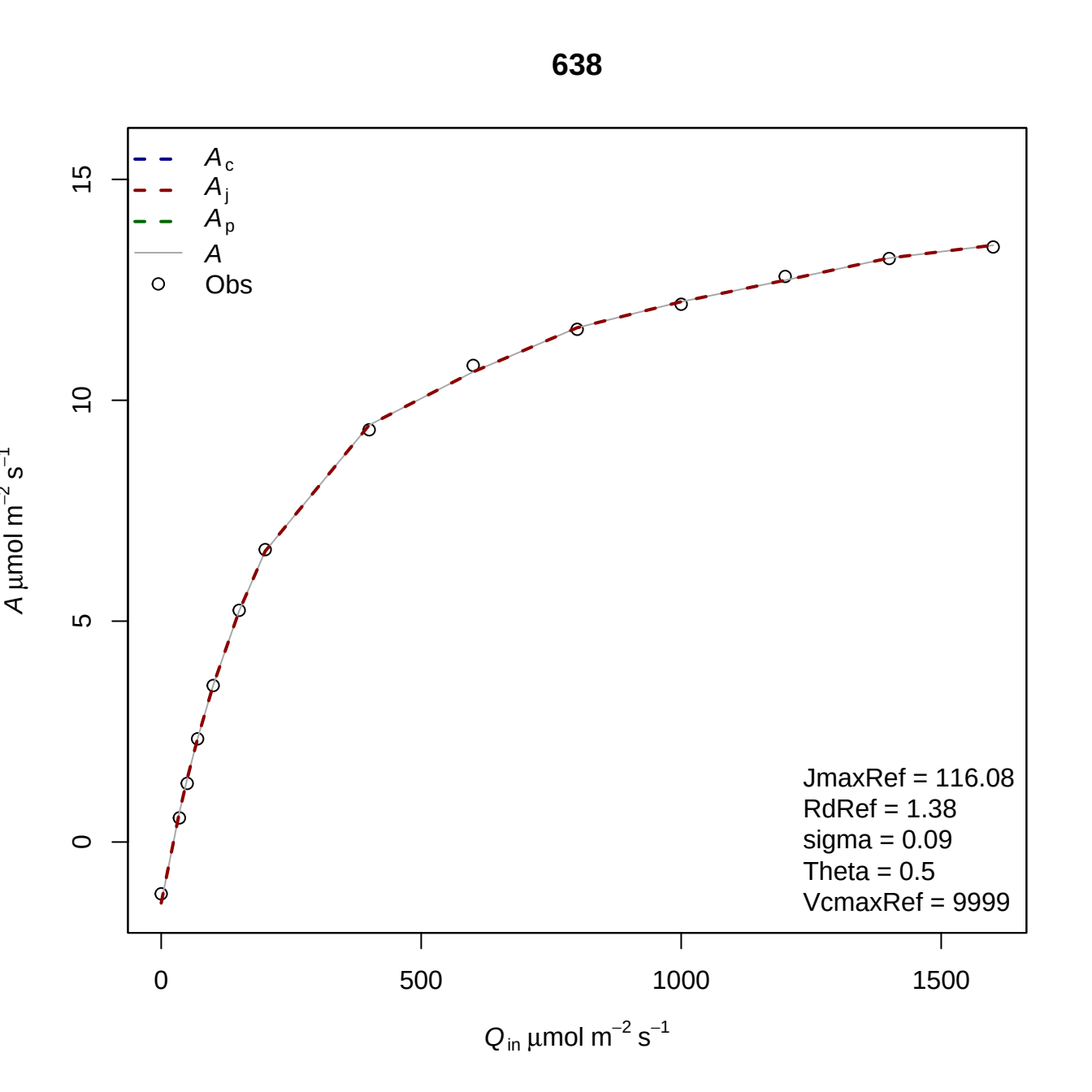

633
A_c
A_j
A_p
A
Obs
JmaxRef = 276.41
RdRef = 3.96
sigma = 0.06
Theta = 0.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

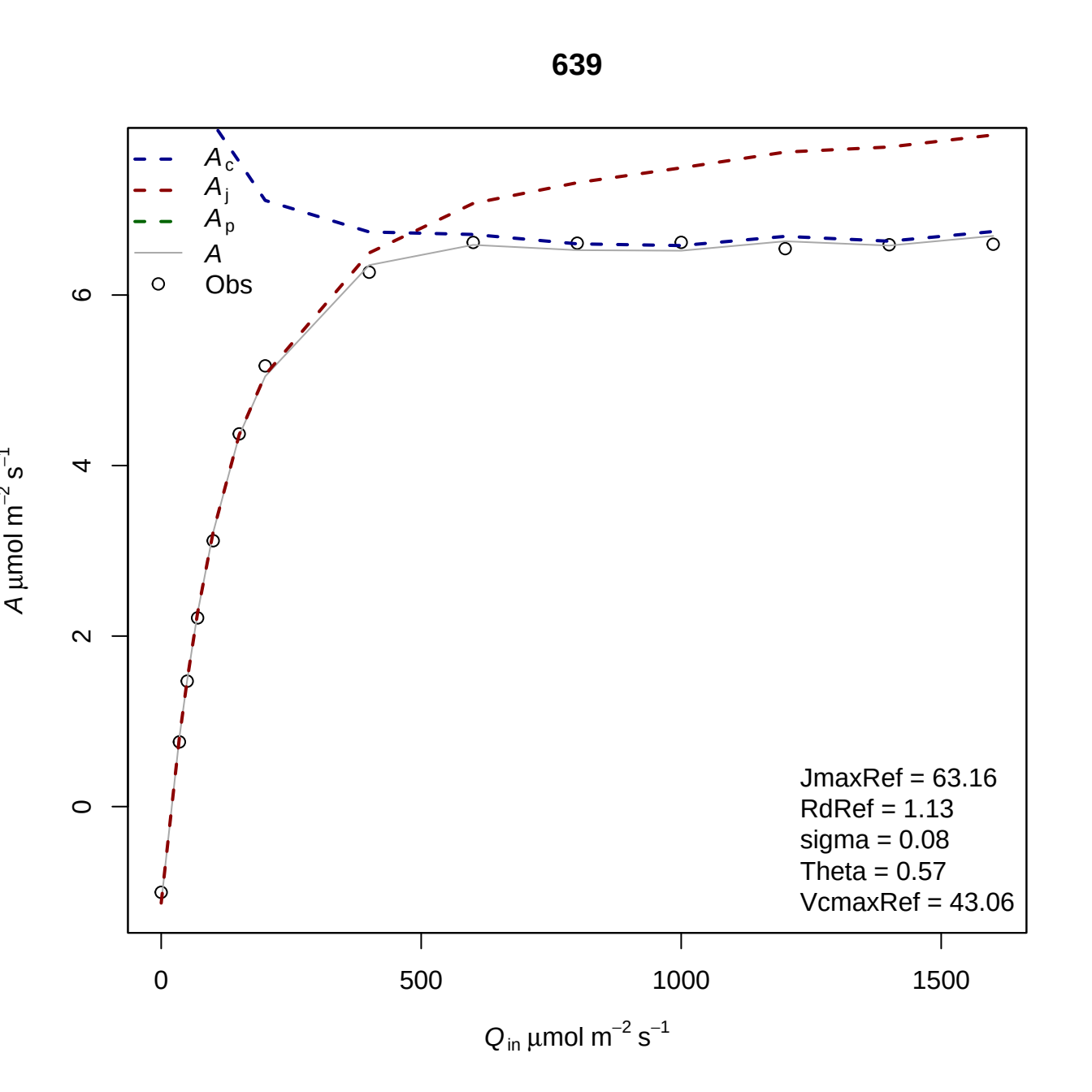

634
A_c
A_j
A_p
A
Obs
JmaxRef = 206.49
RdRef = 2.45
sigma = 0.31
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

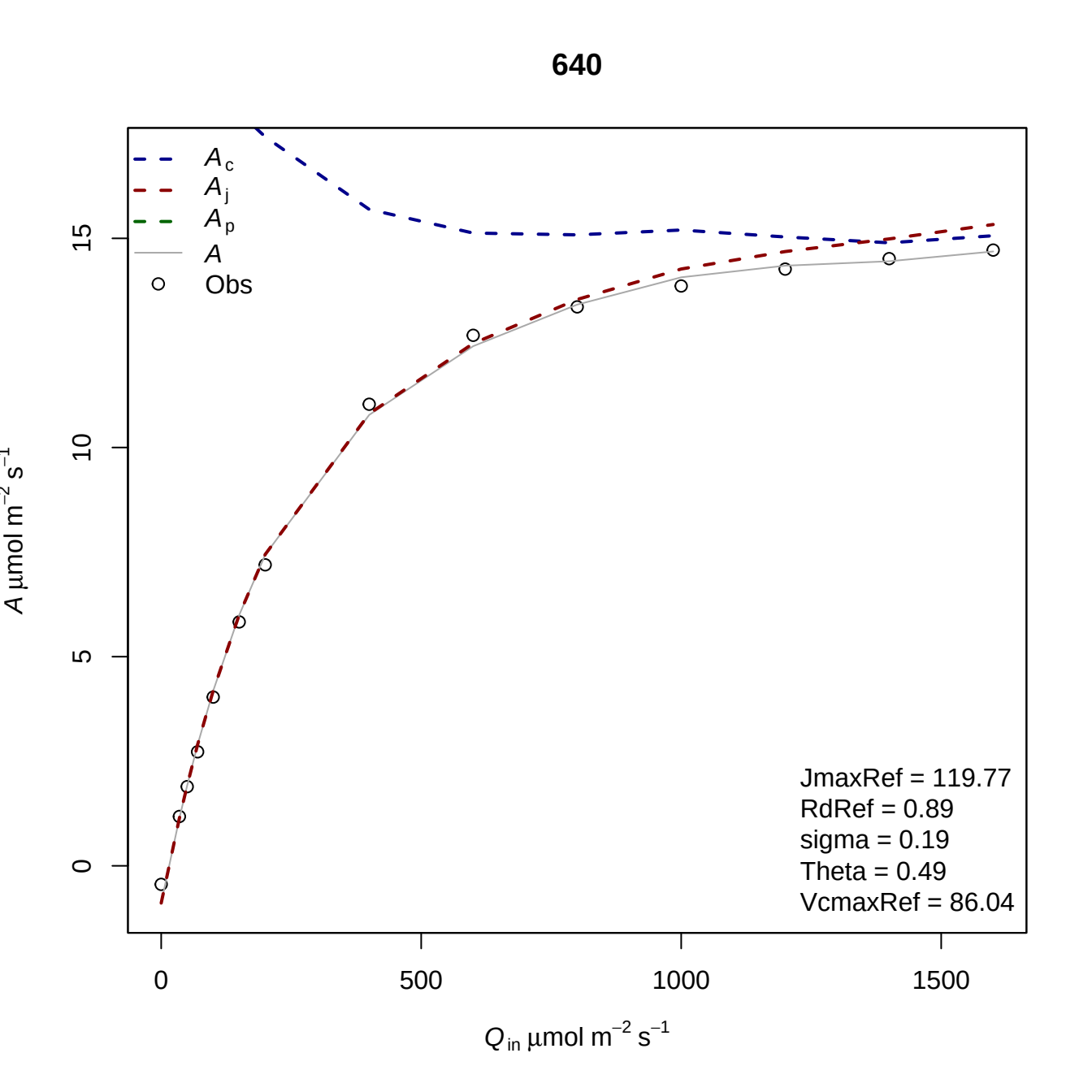

635
A_c
A_j
A_p
A
Obs
JmaxRef = 23.79
RdRef = 1.23
sigma = 0.06
Theta = 0.87
VcmaxRef = 9999
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

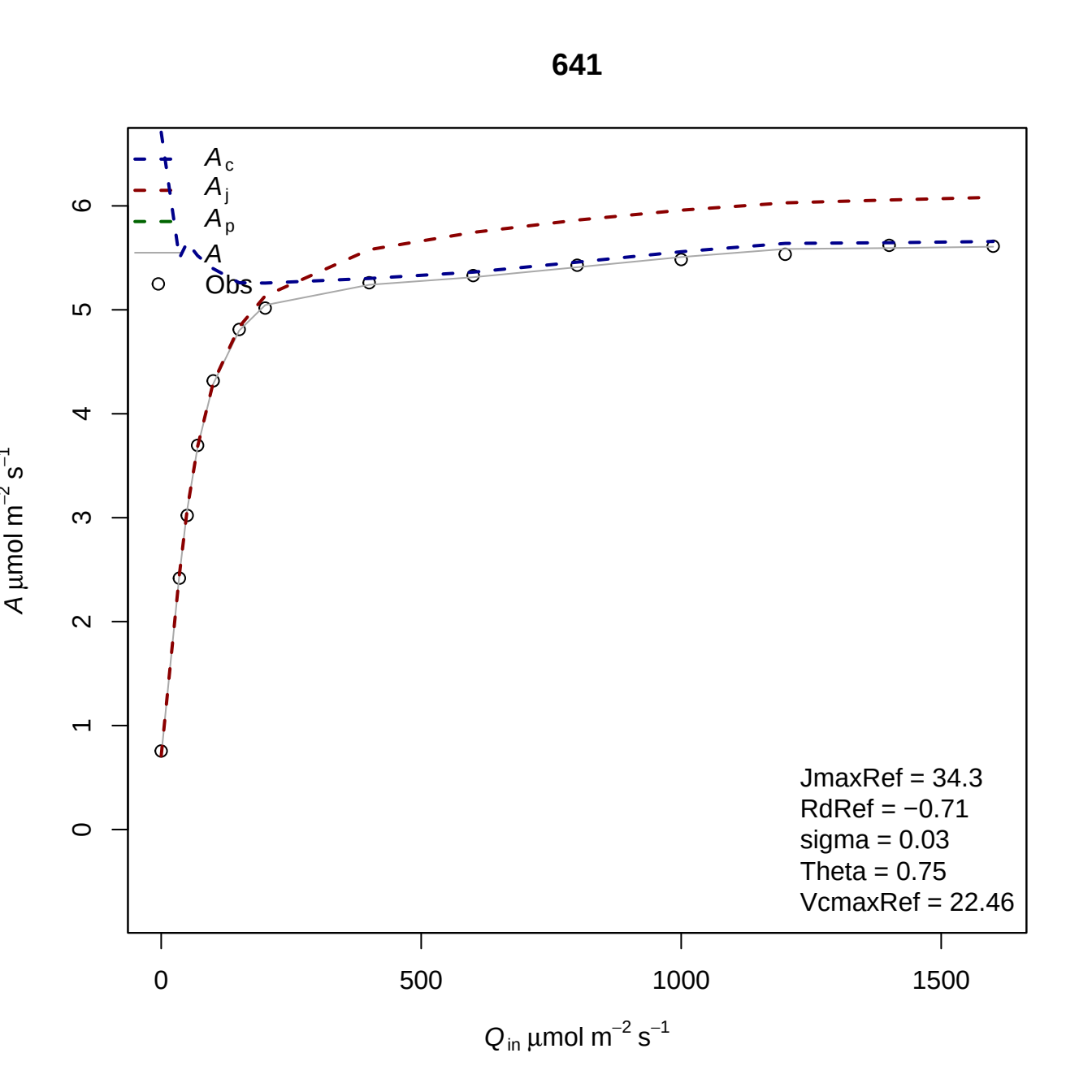

636
A_c
A_j
A_p
A
Obs
JmaxRef = 28929143.39
RdRef = 0.65
sigma = 0.48
Theta = −74886429.56
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

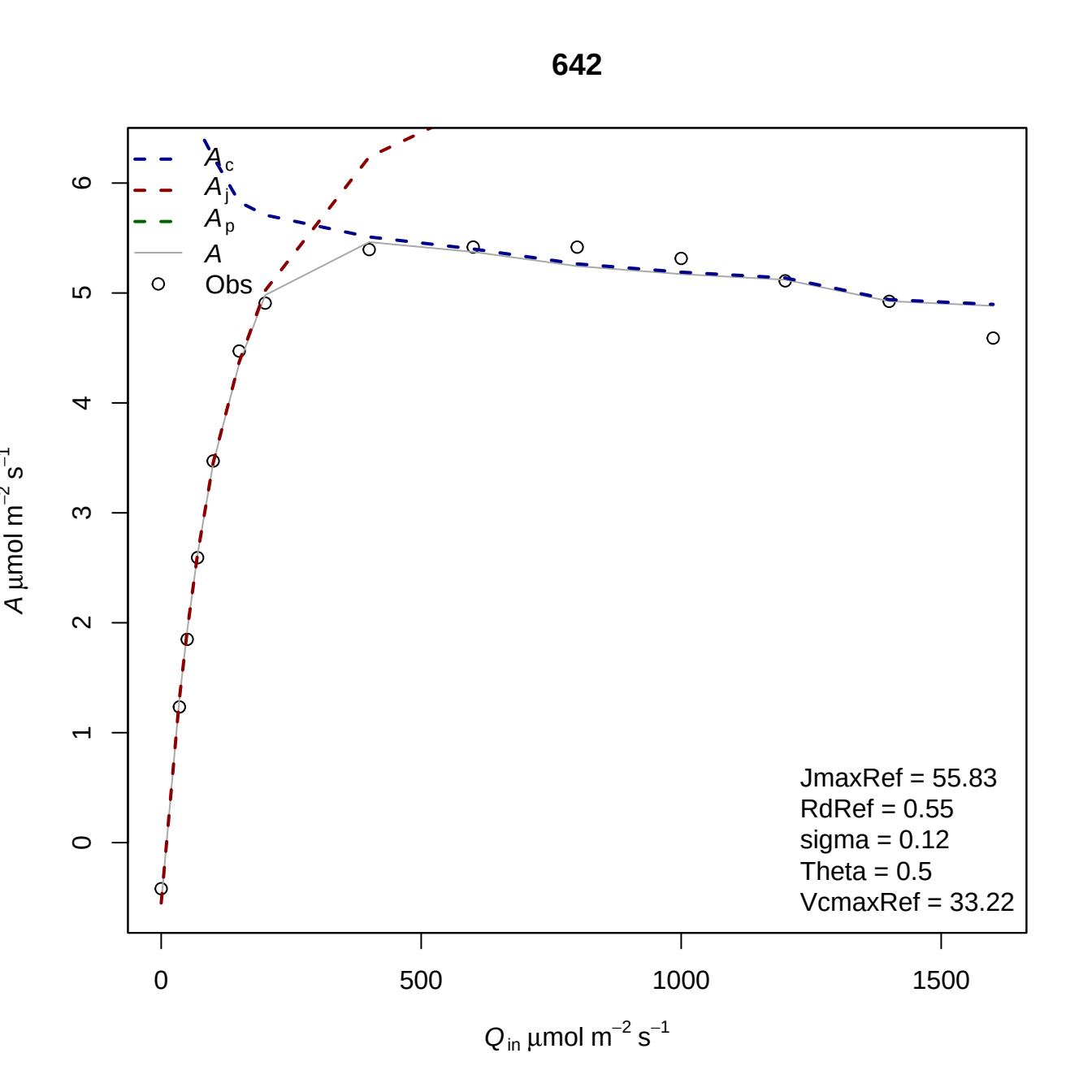

637
A_c
A_j
A_p
A
Obs
JmaxRef = 26.52
RdRef = 0.6
sigma = 0.16
Theta = 0.27
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

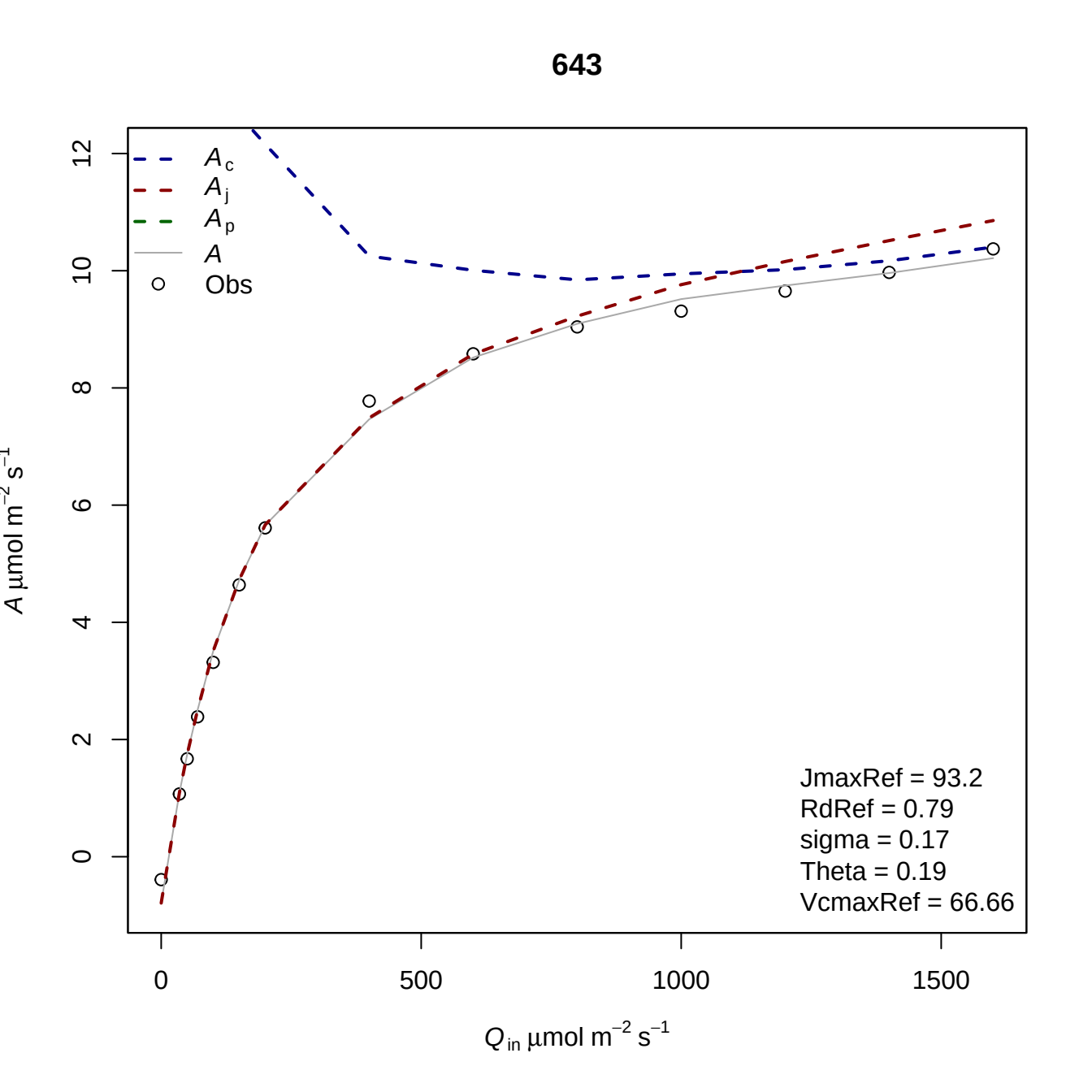

638
A_c
A_j
A_p
A
Obs
JmaxRef = 116.08
RdRef = 1.38
sigma = 0.09
Theta = 0.5
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

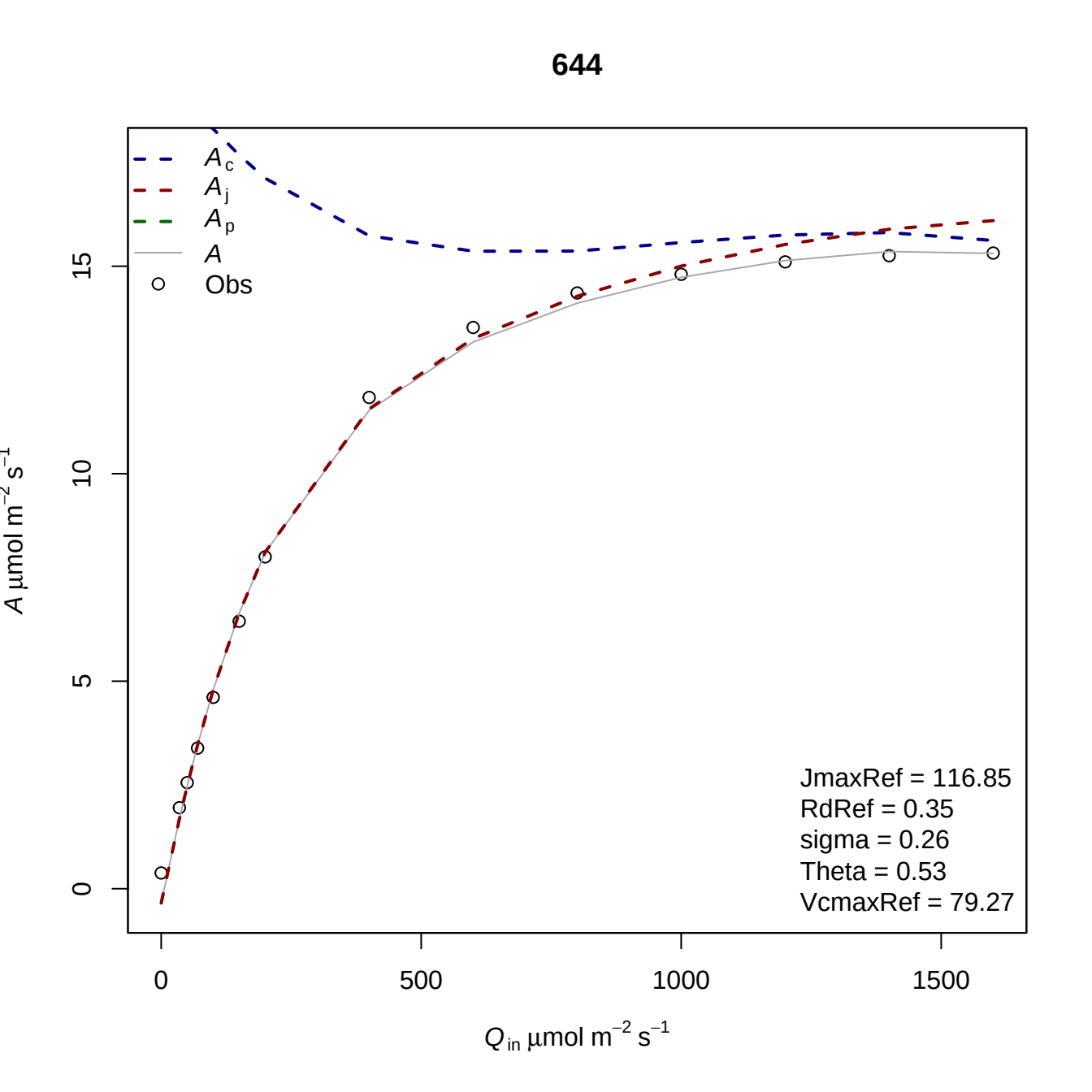

639
A_c
A_j
A_p
A
Obs
JmaxRef = 63.16
RdRef = 1.13
sigma = 0.08
Theta = 0.57
VcmaxRef = 43.06
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

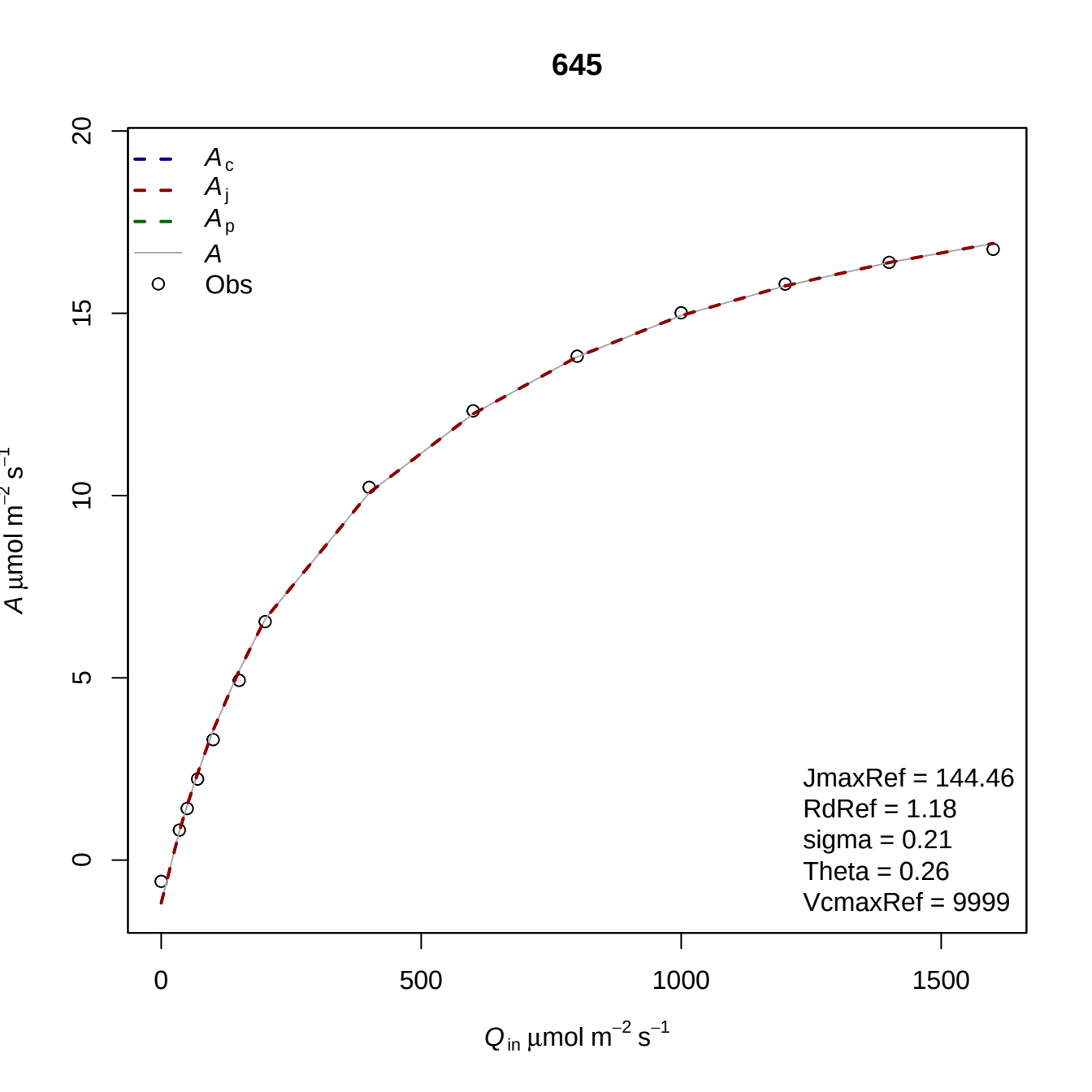

640
A_c
A_j
A_p
A
Obs
JmaxRef = 119.77
RdRef = 0.89
sigma = 0.19
Theta = 0.49
VcmaxRef = 86.04
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

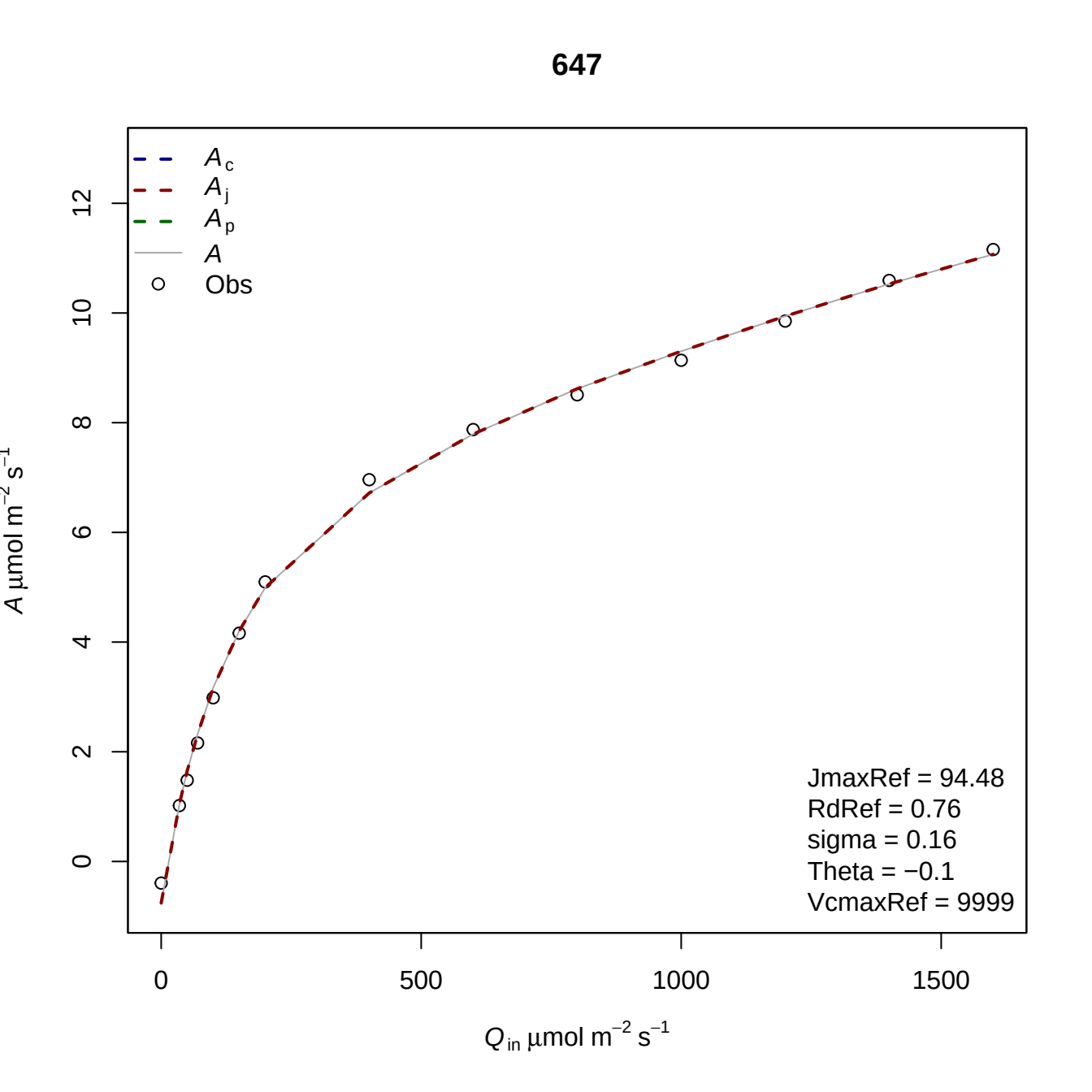

641
A_c
A_j
A_p
A
Obs
JmaxRef = 34.3
RdRef = −0.71
sigma = 0.03
Theta = 0.75
VcmaxRef = 22.46
0
1
2
3
4
5
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

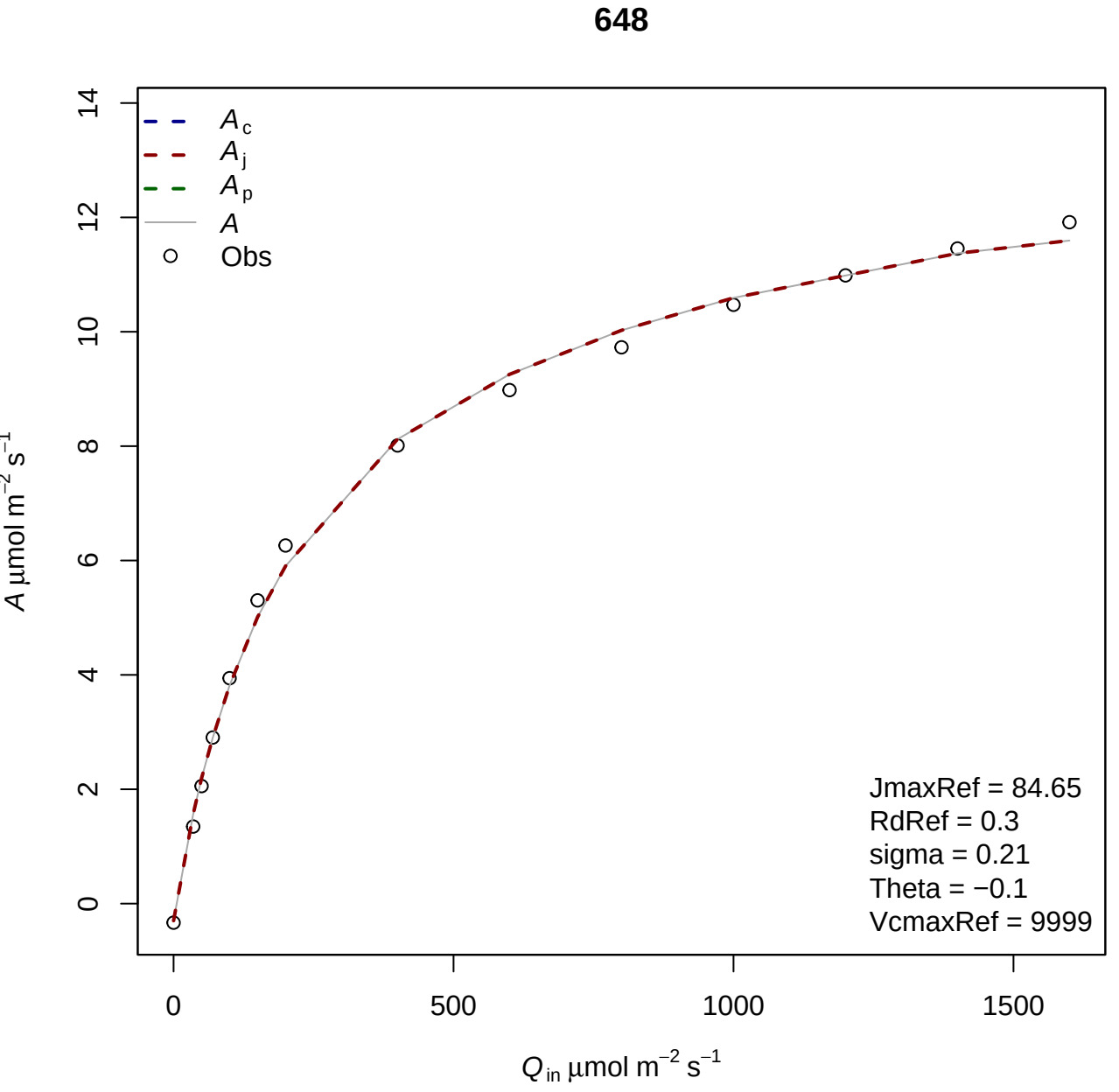

642
A_c
A_j
A_p
A
Obs
JmaxRef = 55.83
RdRef = 0.55
sigma = 0.12
Theta = 0.5
VcmaxRef = 33.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

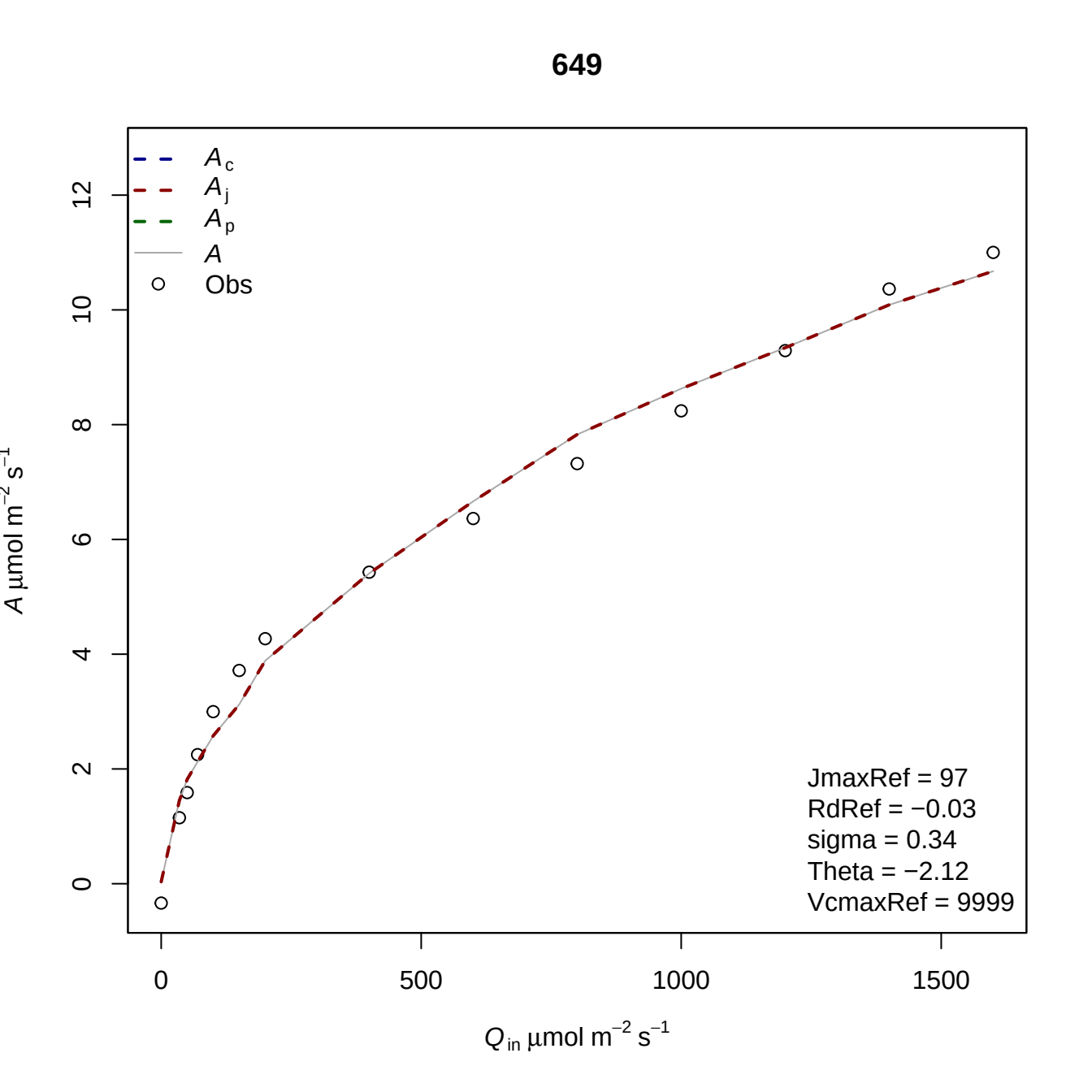

643
A_c
A_j
A_p
A
Obs
JmaxRef = 93.2
RdRef = 0.79
sigma = 0.17
Theta = 0.19
VcmaxRef = 66.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

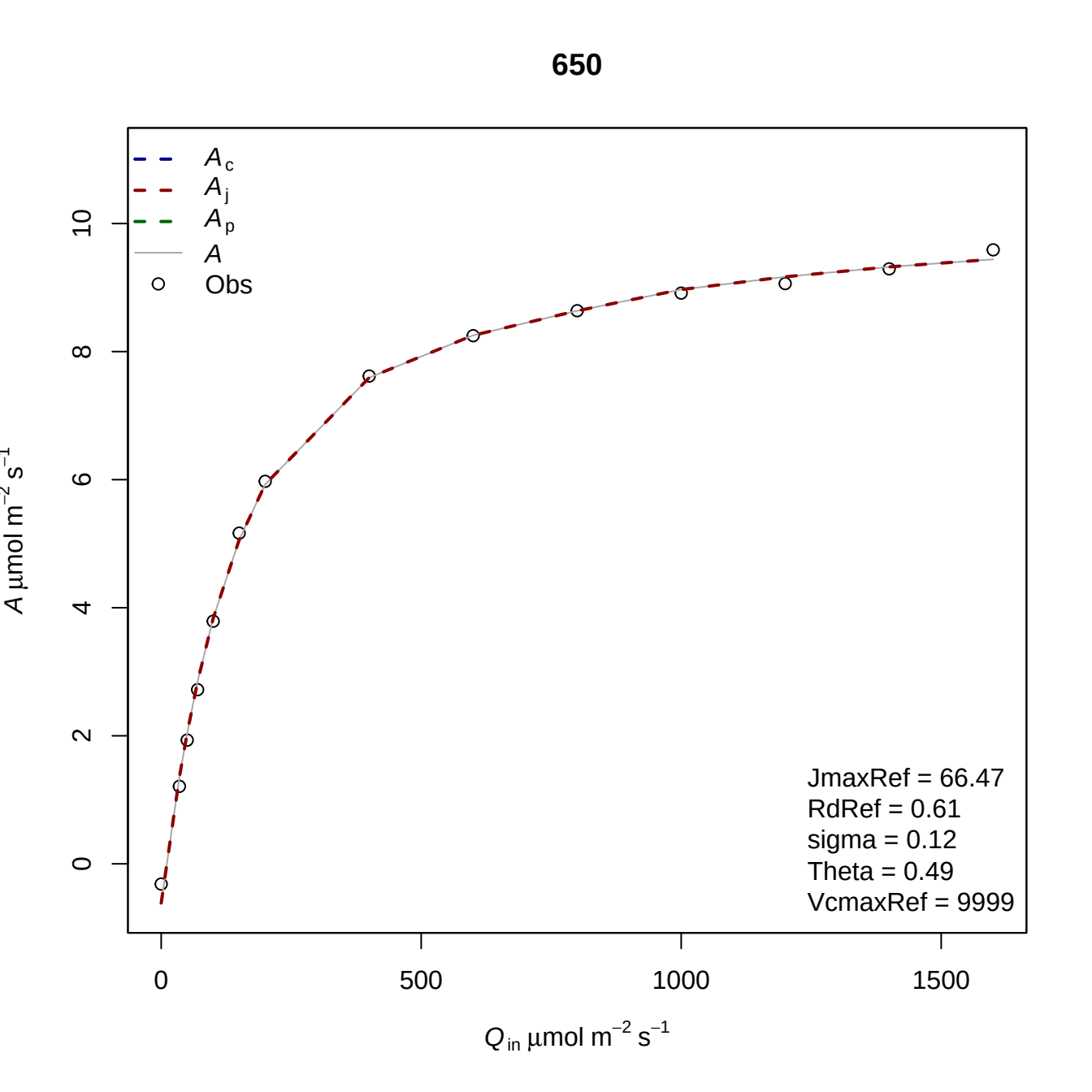

644
A_c
A_j
A_p
A
Obs
JmaxRef = 116.85
RdRef = 0.35
sigma = 0.26
Theta = 0.53
VcmaxRef = 79.27
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

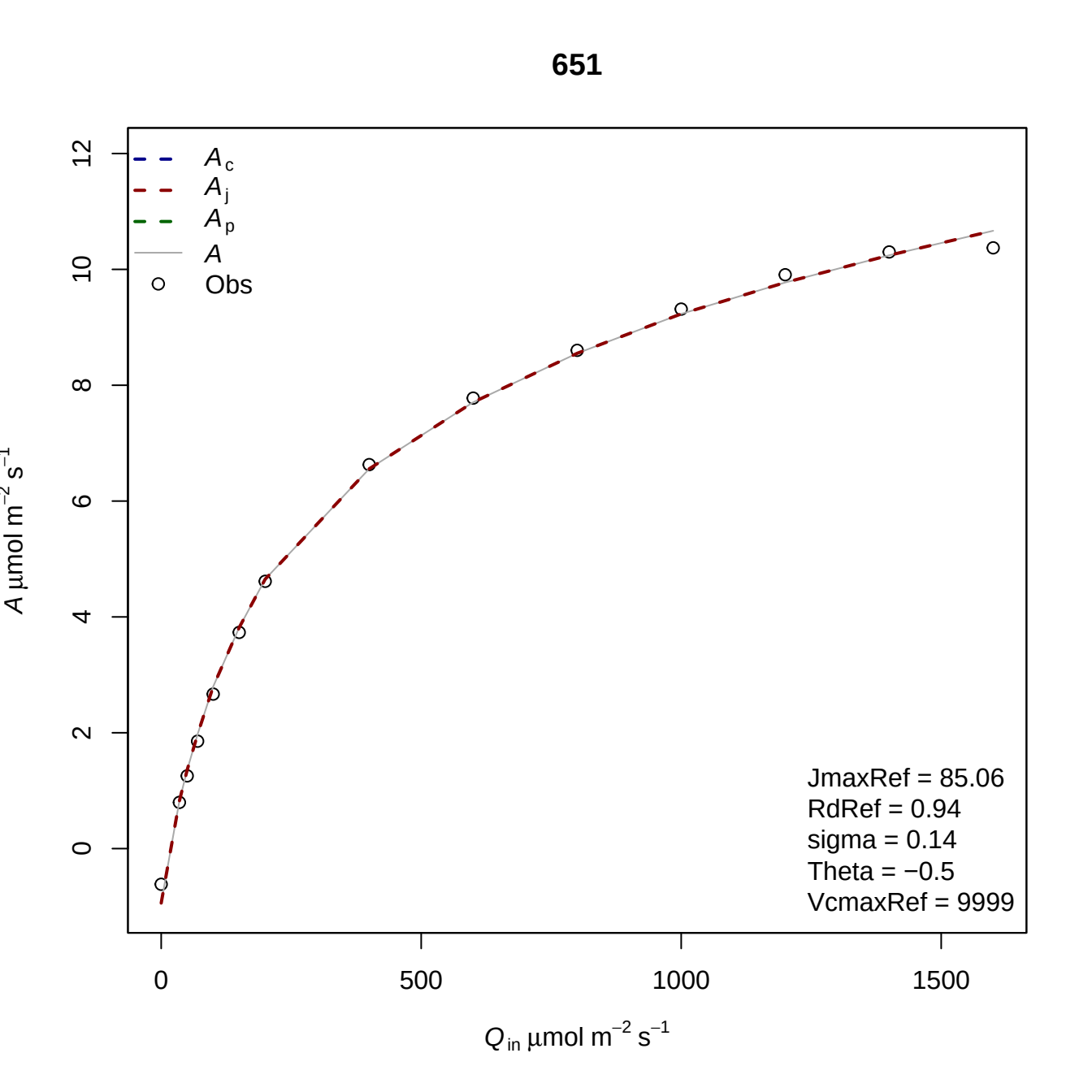

645
A_c
A_j
A_p
A
Obs
JmaxRef = 144.46
RdRef = 1.18
sigma = 0.21
Theta = 0.26
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

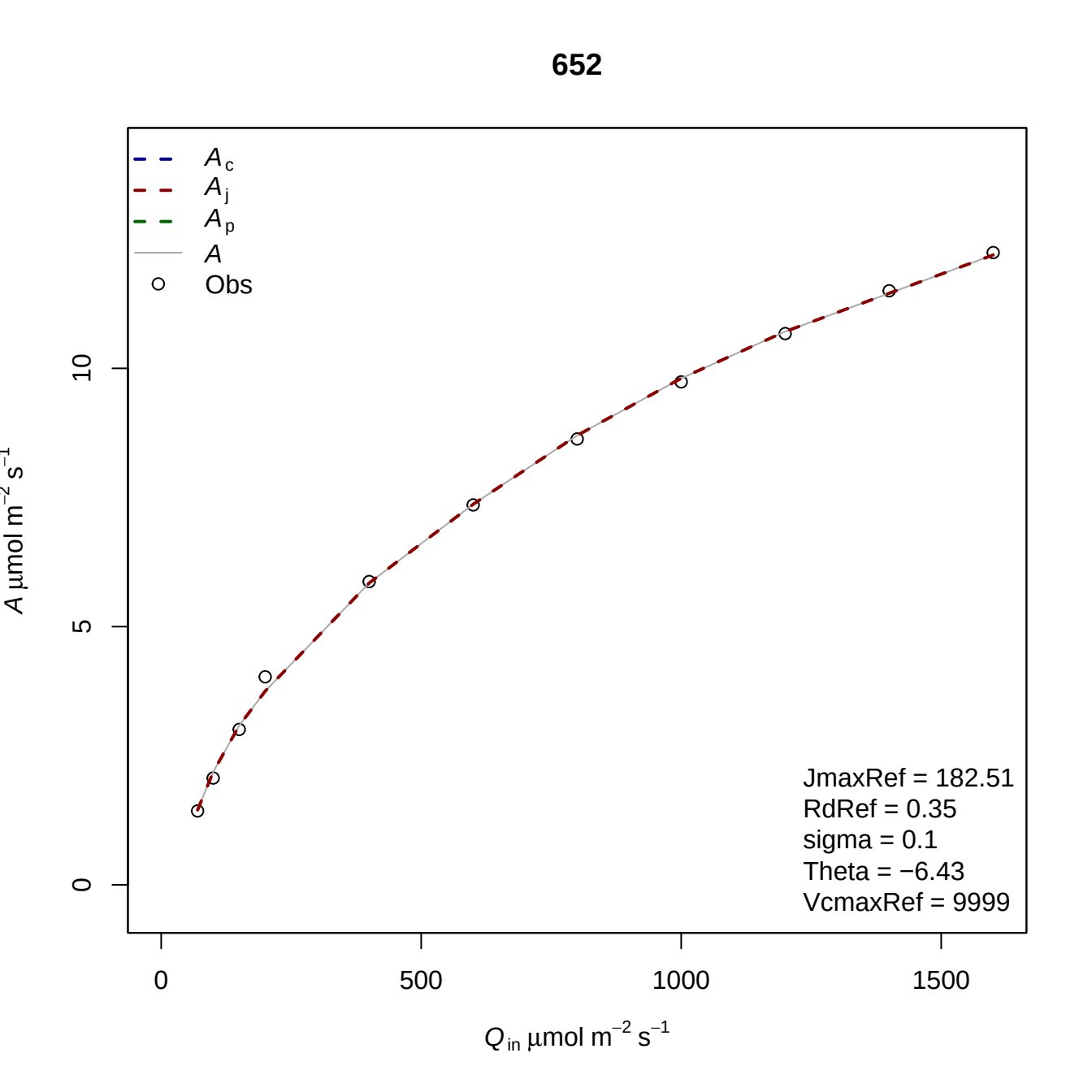

647
A_c
A_j
A_p
A
Obs
JmaxRef = 94.48
RdRef = 0.76
sigma = 0.16
Theta = −0.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

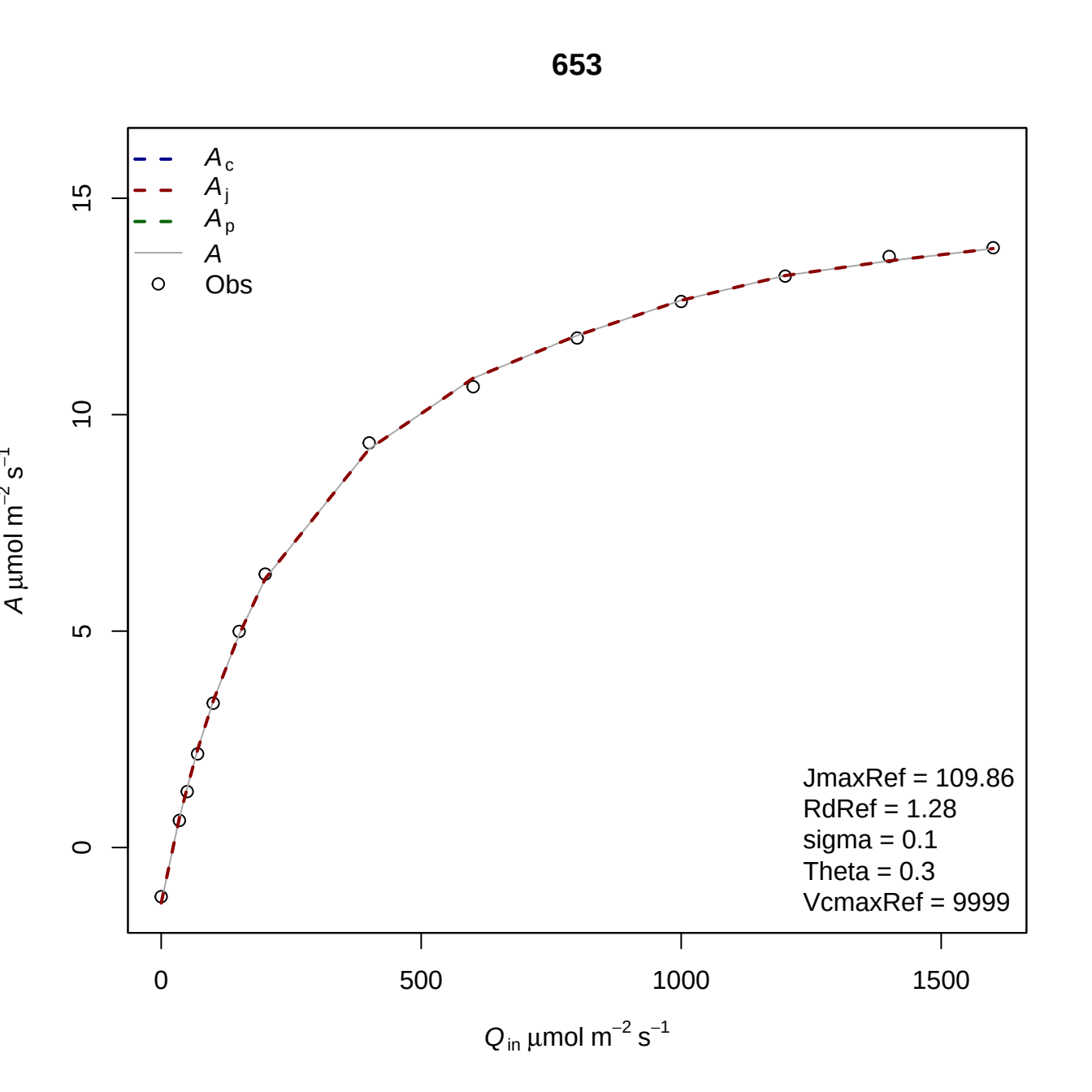

648
A_c
A_j
A_p
A
Obs
JmaxRef = 84.65
RdRef = 0.3
sigma = 0.21
Theta = −0.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

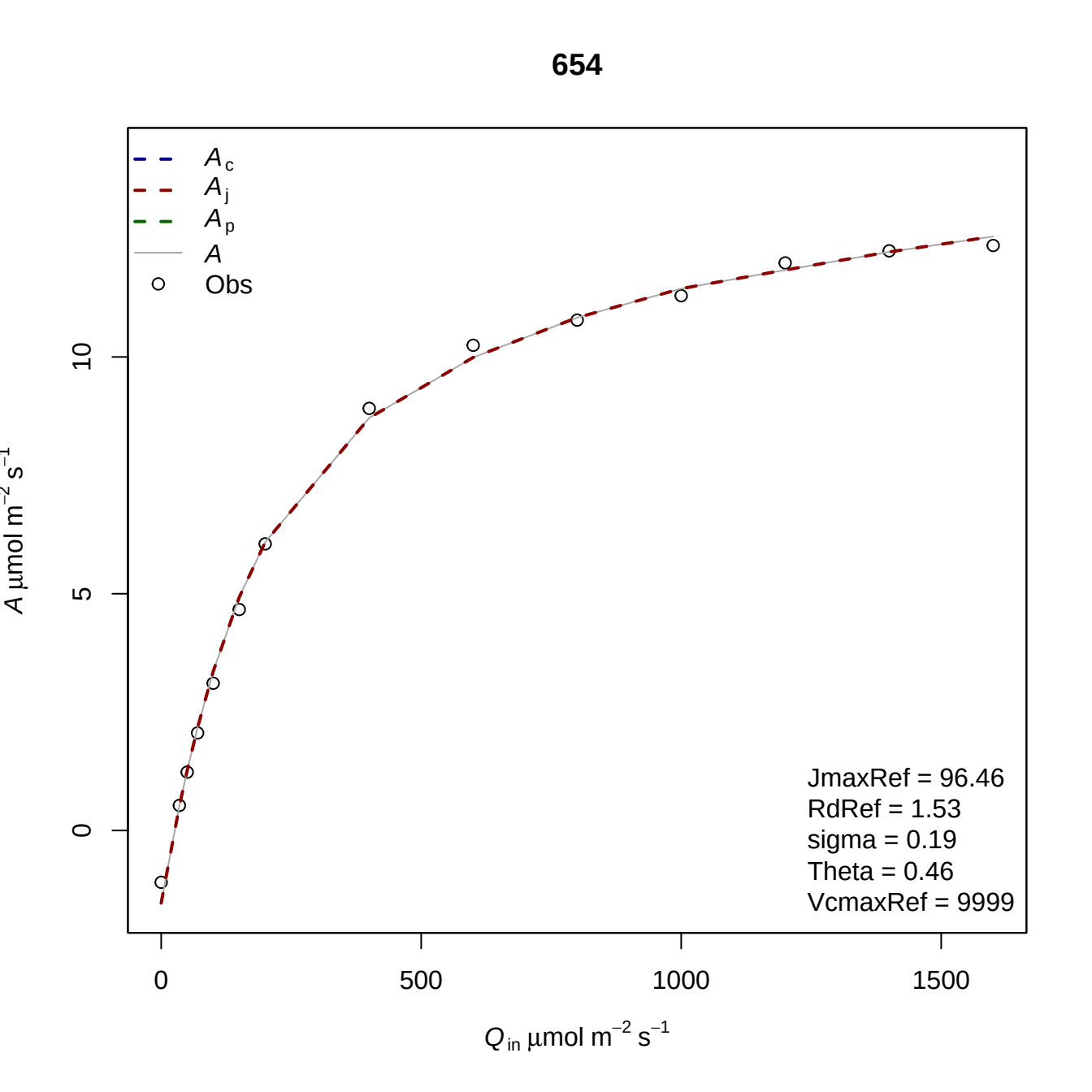

649
A_c
A_j
A_p
A
Obs
JmaxRef = 97
RdRef = −0.03
sigma = 0.34
Theta = −2.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

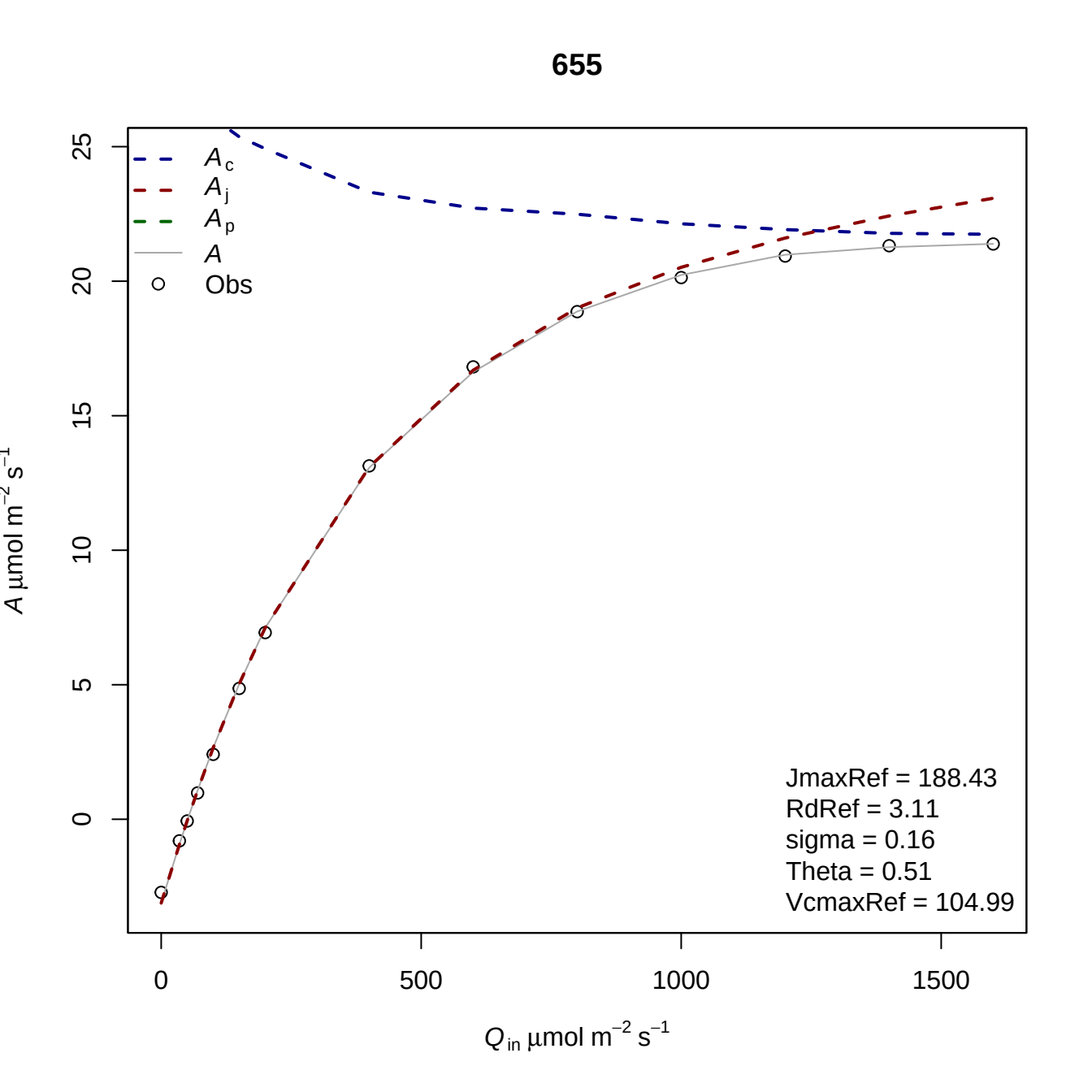

650
A_c
A_j
A_p
A
Obs
JmaxRef = 66.47
RdRef = 0.61
sigma = 0.12
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

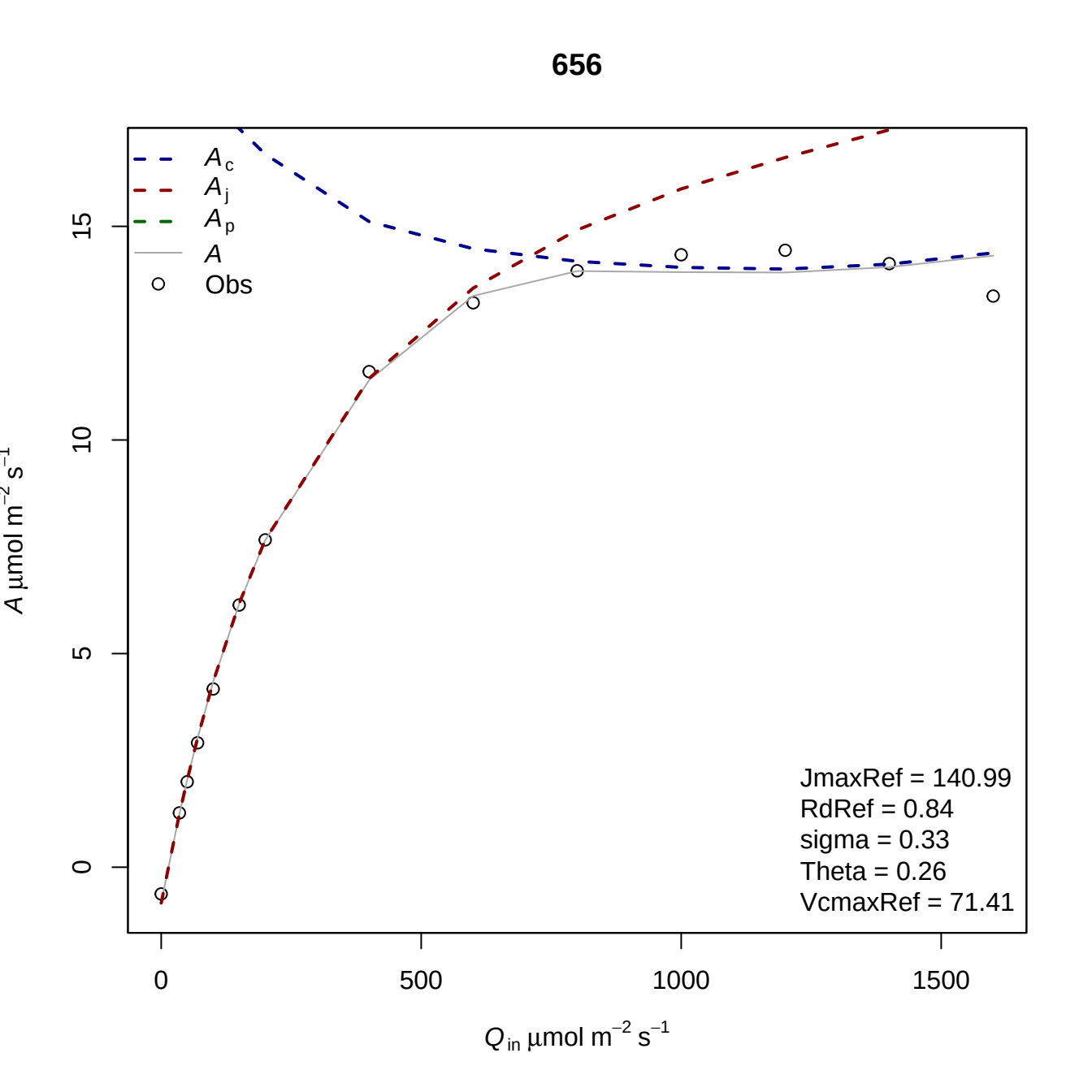

651
A_c
A_j
A_p
A
Obs
JmaxRef = 85.06
RdRef = 0.94
sigma = 0.14
Theta = −0.5
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

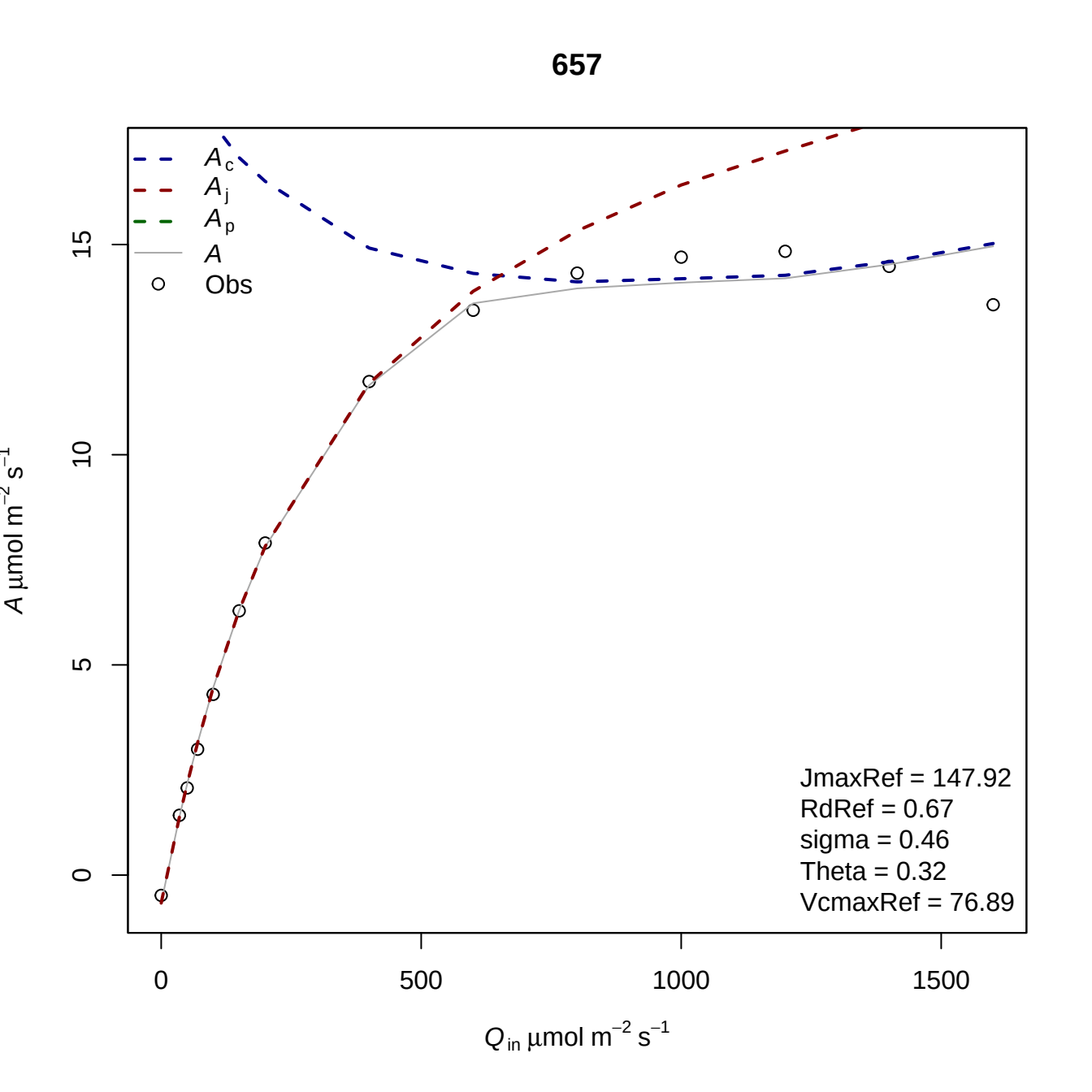

652
A_c
A_j
A_p
A
Obs
JmaxRef = 182.51
RdRef = 0.35
sigma = 0.1
Theta = −6.43
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

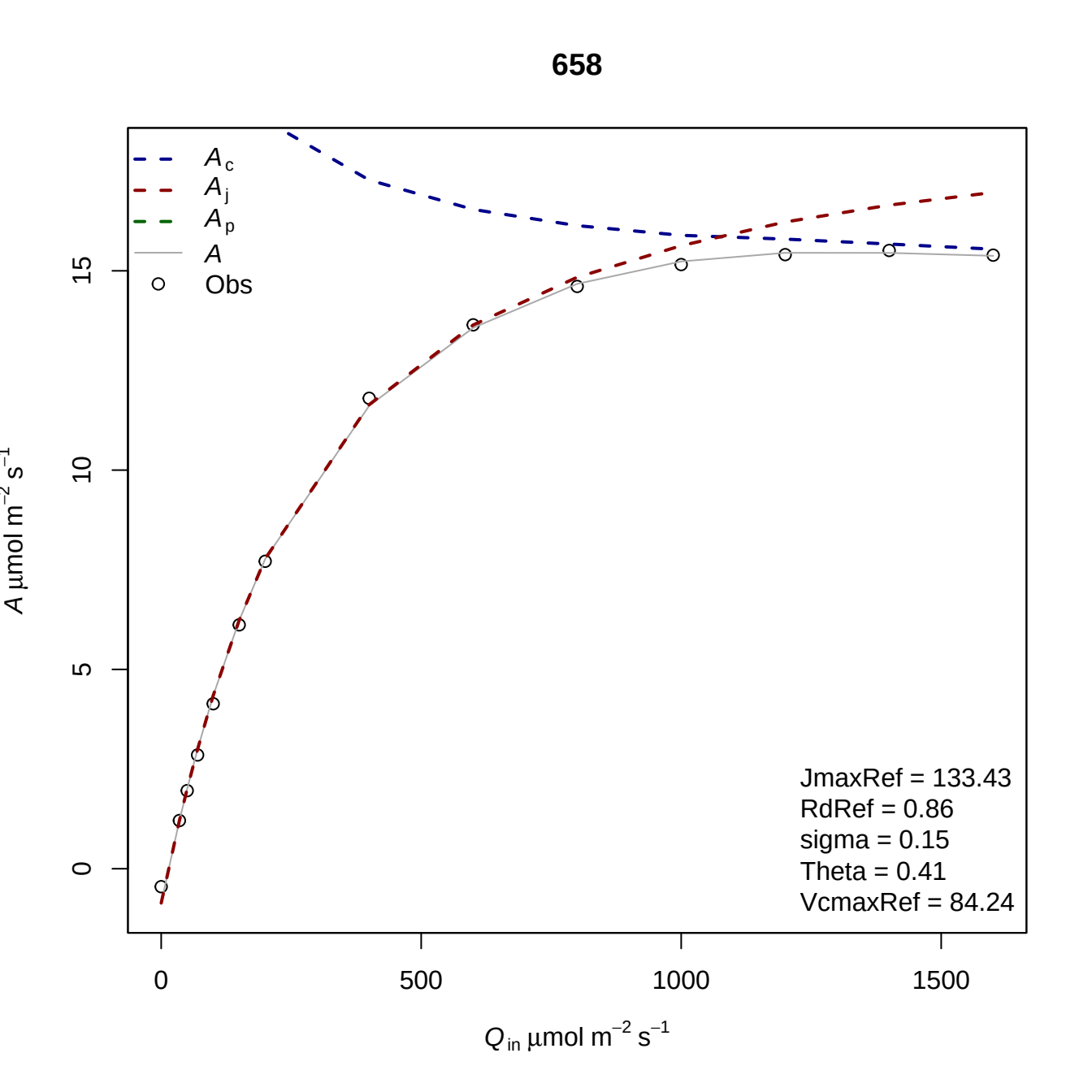

653
A_c
A_j
A_p
A
Obs
JmaxRef = 109.86
RdRef = 1.28
sigma = 0.1
Theta = 0.3
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

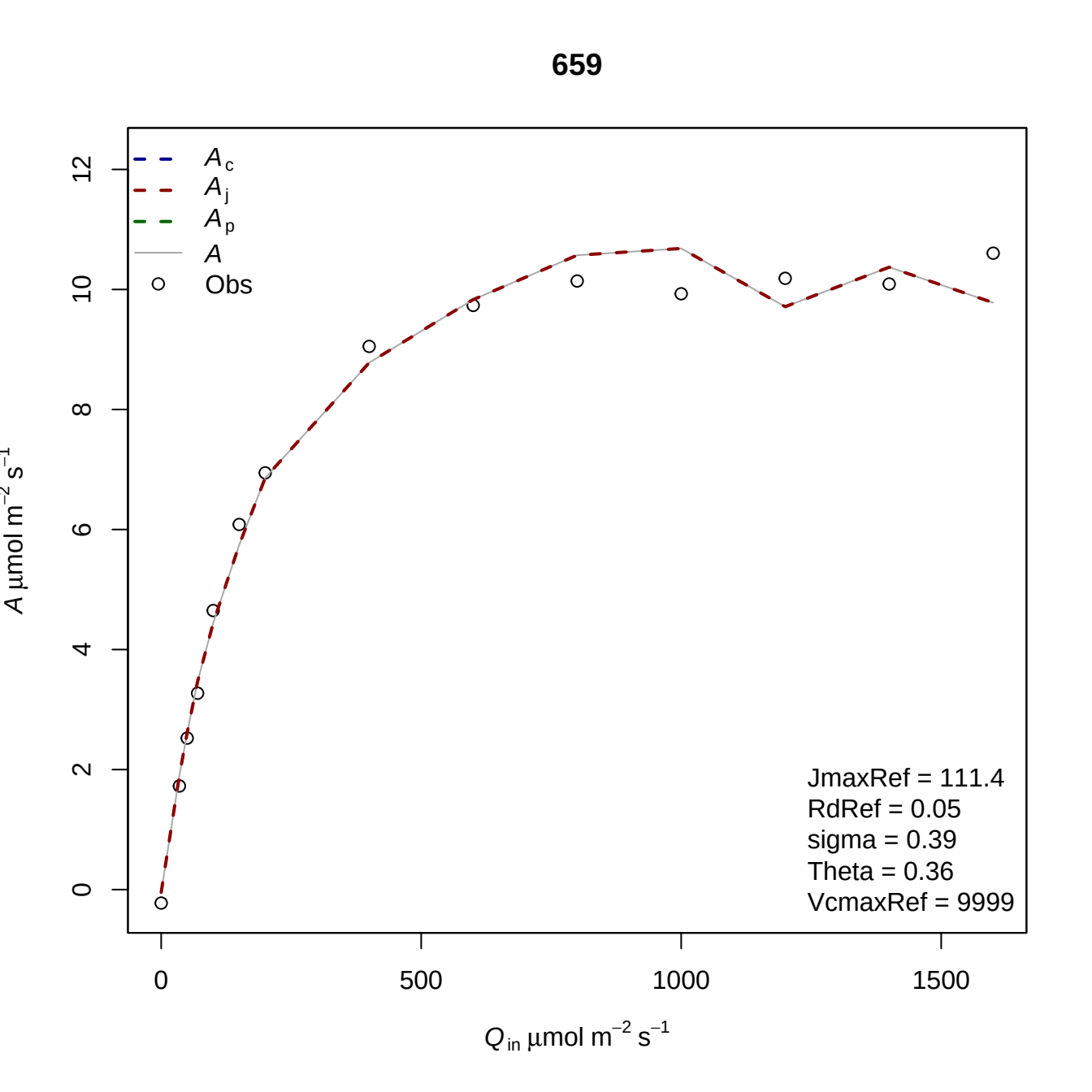

654
A_c
A_j
A_p
A
Obs
JmaxRef = 96.46
RdRef = 1.53
sigma = 0.19
Theta = 0.46
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

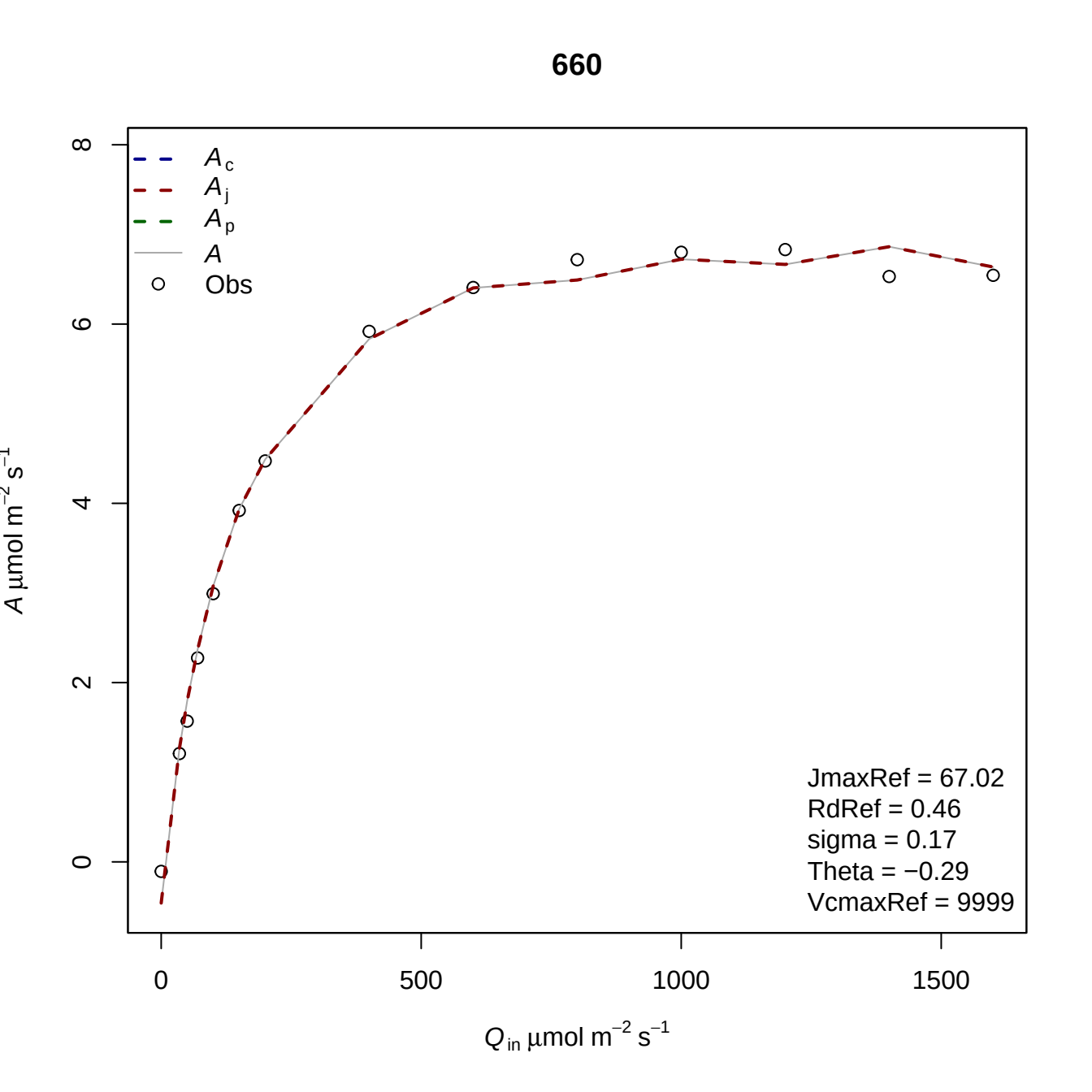

655
A_c
A_j
A_p
A
Obs
JmaxRef = 188.43
RdRef = 3.11
sigma = 0.16
Theta = 0.51
VcmaxRef = 104.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

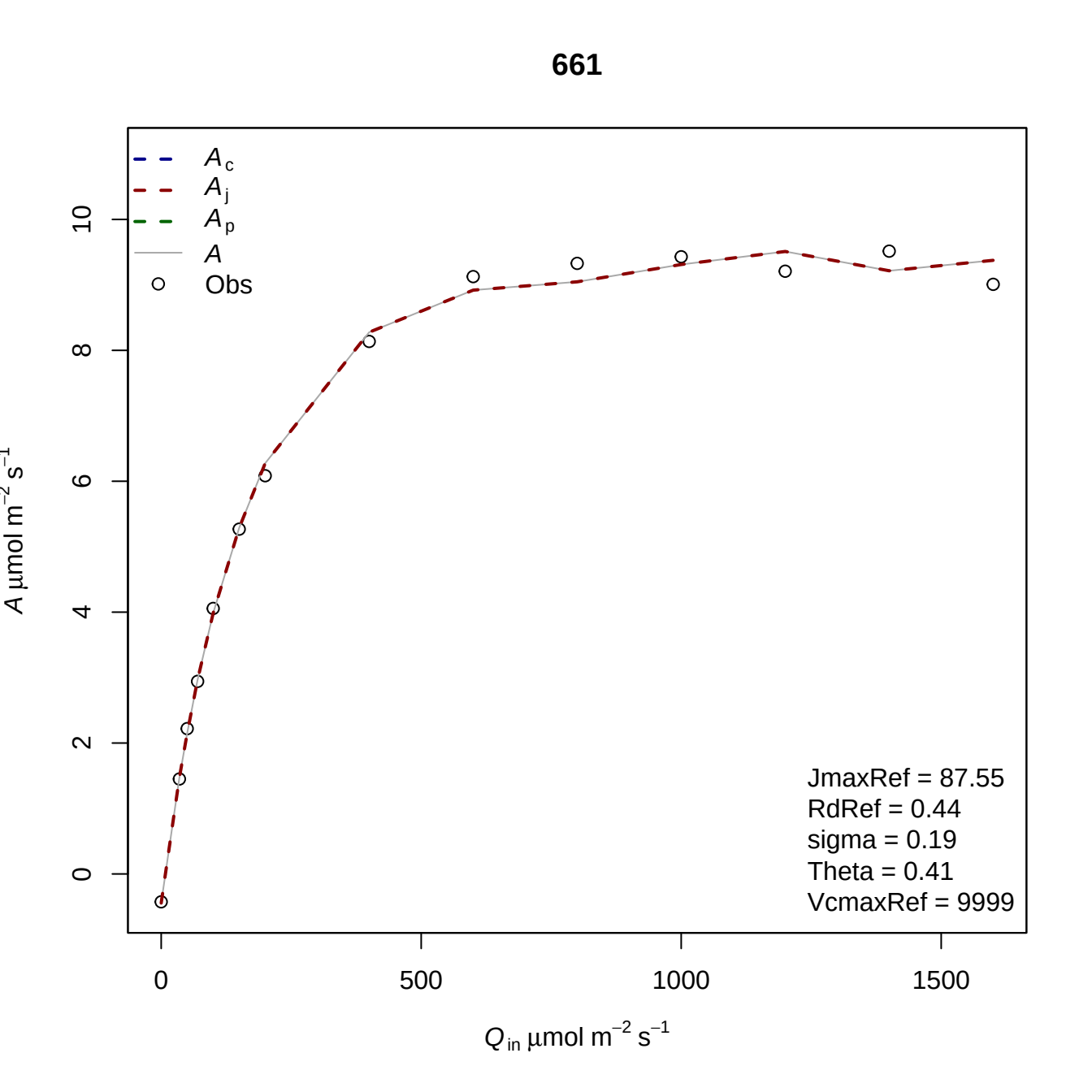

656
A_c
A_j
A_p
A
Obs
JmaxRef = 140.99
RdRef = 0.84
sigma = 0.33
Theta = 0.26
VcmaxRef = 71.41
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

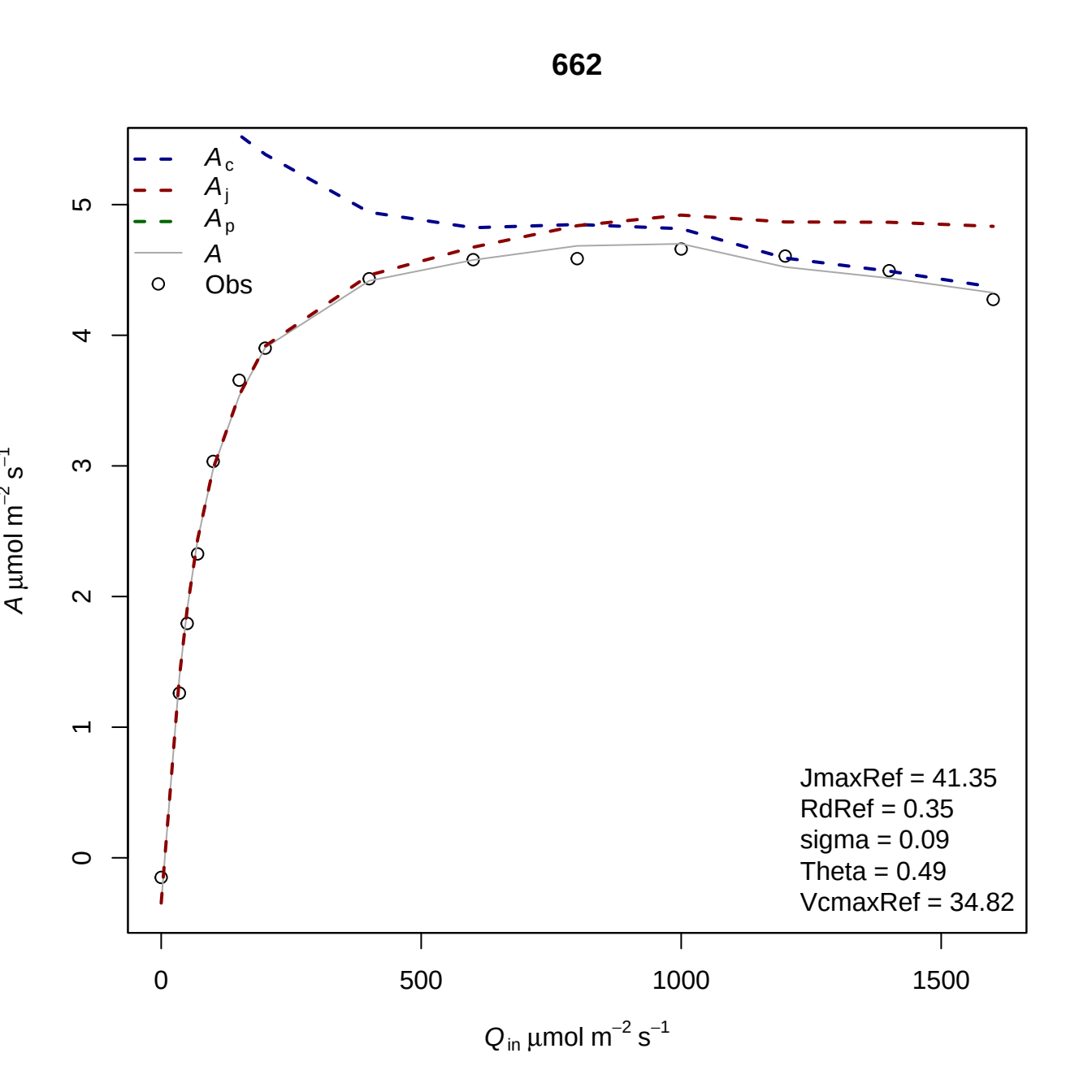

657
A_c
A_j
A_p
A
Obs
JmaxRef = 147.92
RdRef = 0.67
sigma = 0.46
Theta = 0.32
VcmaxRef = 76.89
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

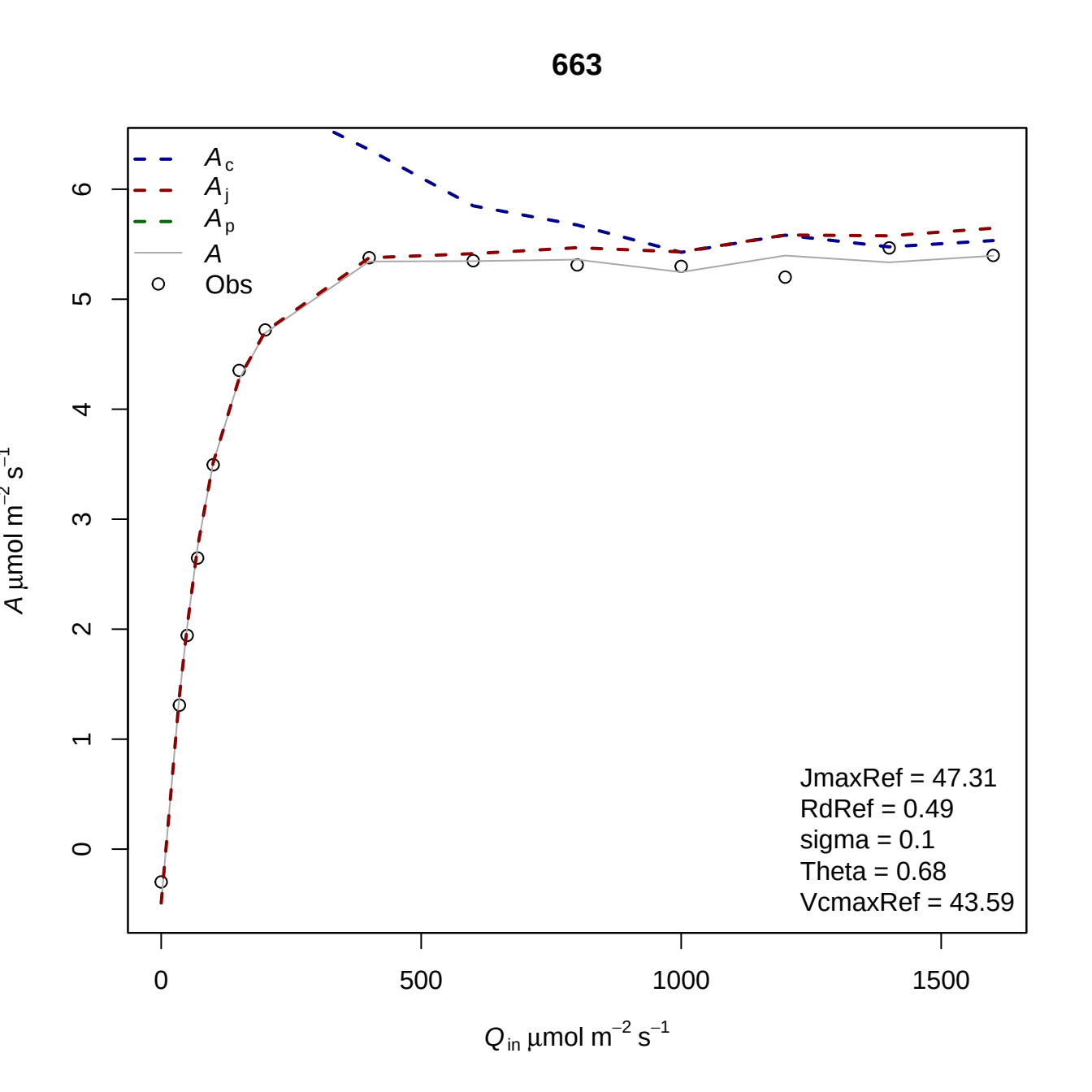

658
A_c
A_j
A_p
A
Obs
JmaxRef = 133.43
RdRef = 0.86
sigma = 0.15
Theta = 0.41
VcmaxRef = 84.24
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

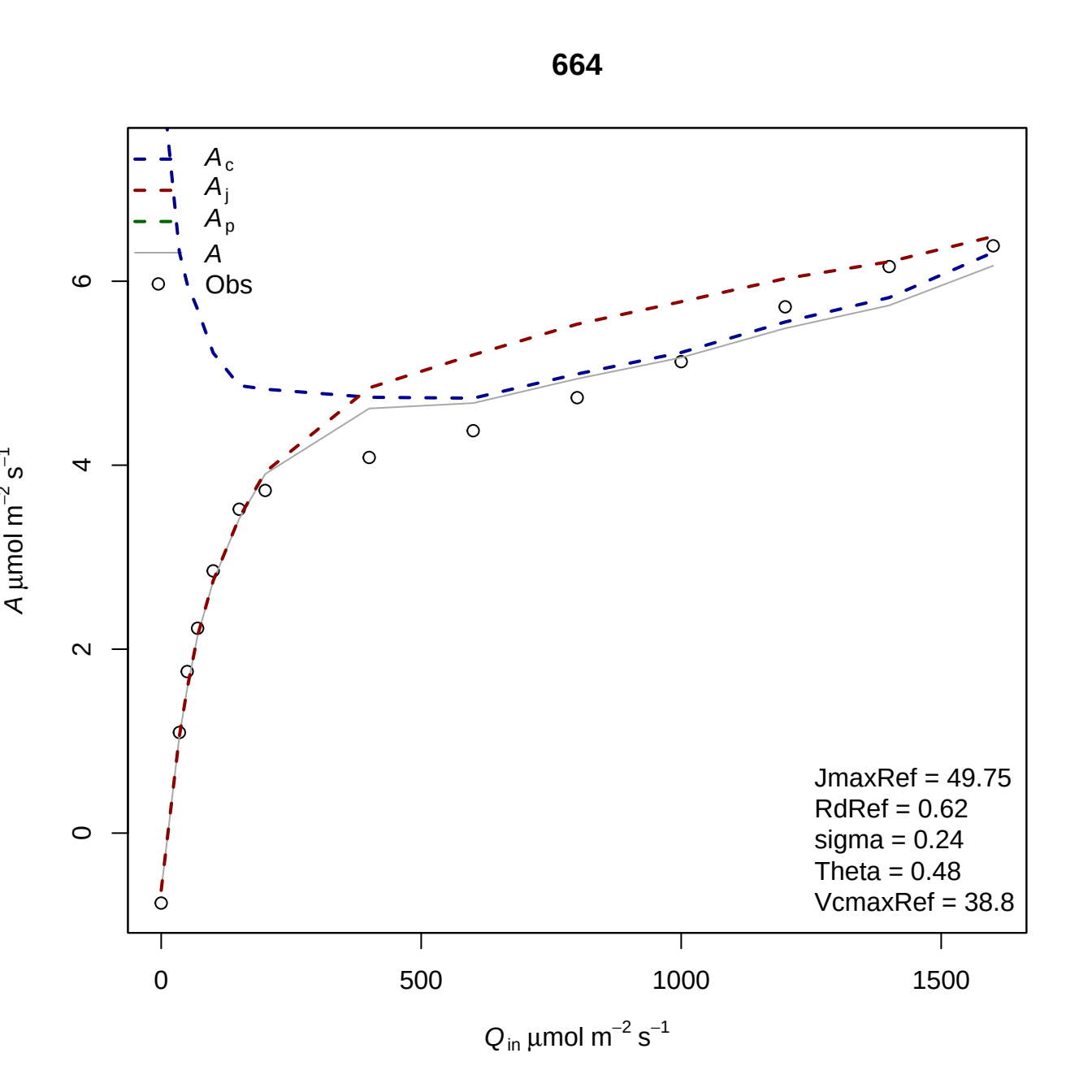

659
A_c
A_j
A_p
A
Obs
JmaxRef = 111.4
RdRef = 0.05
sigma = 0.39
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

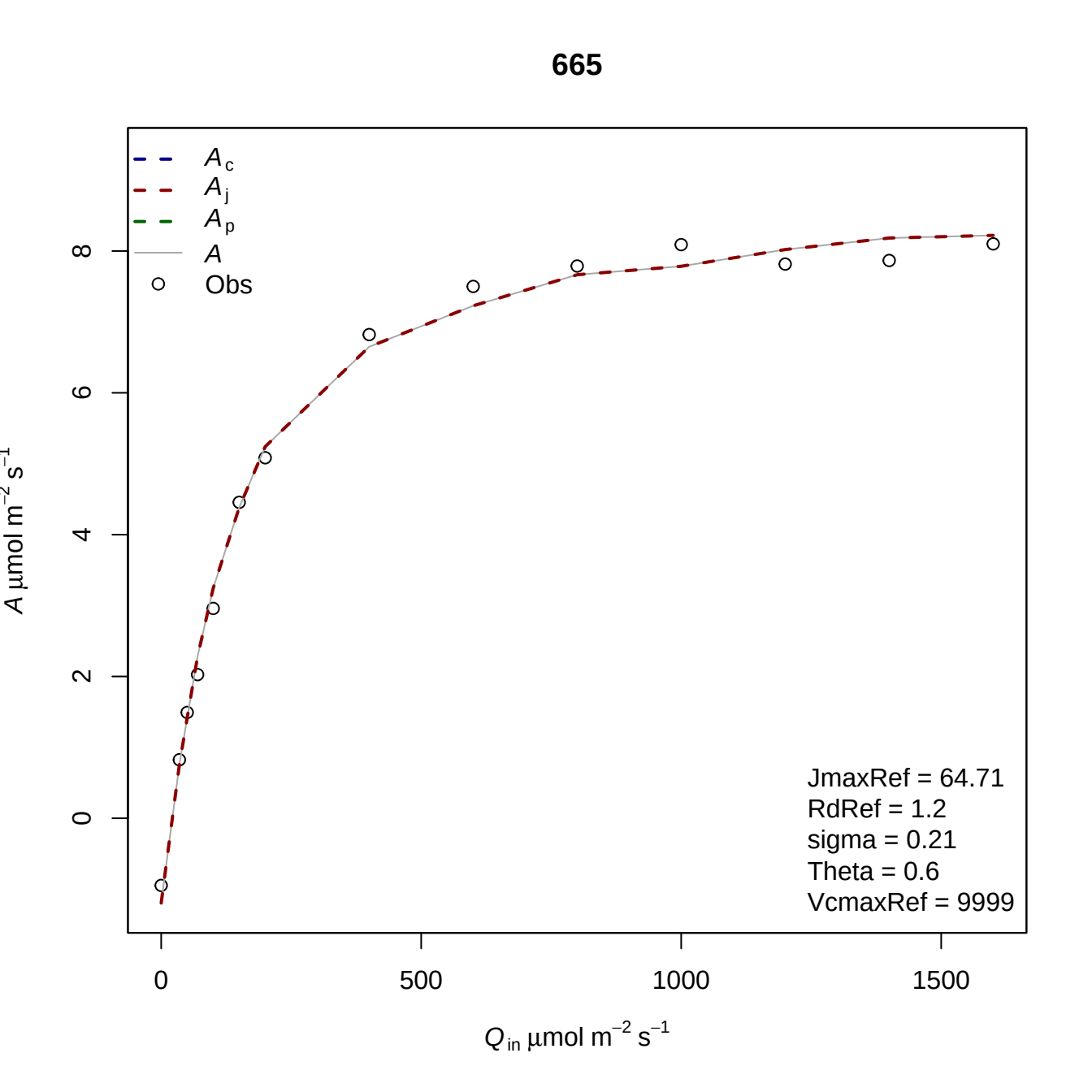

660
A_c
A_j
A_p
A
Obs
JmaxRef = 67.02
RdRef = 0.46
sigma = 0.17
Theta = −0.29
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

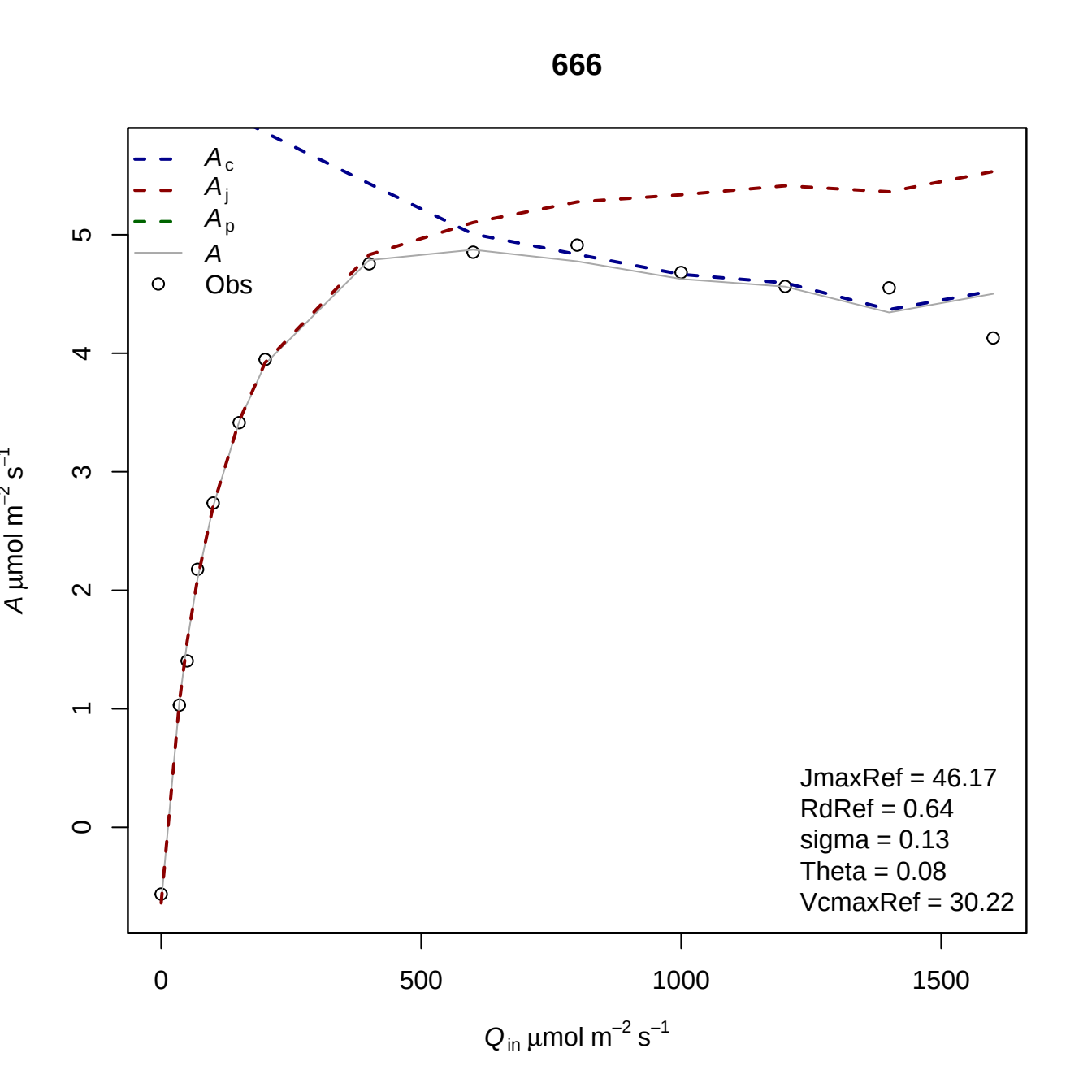

661
A_c
A_j
A_p
A
Obs
JmaxRef = 87.55
RdRef = 0.44
sigma = 0.19
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

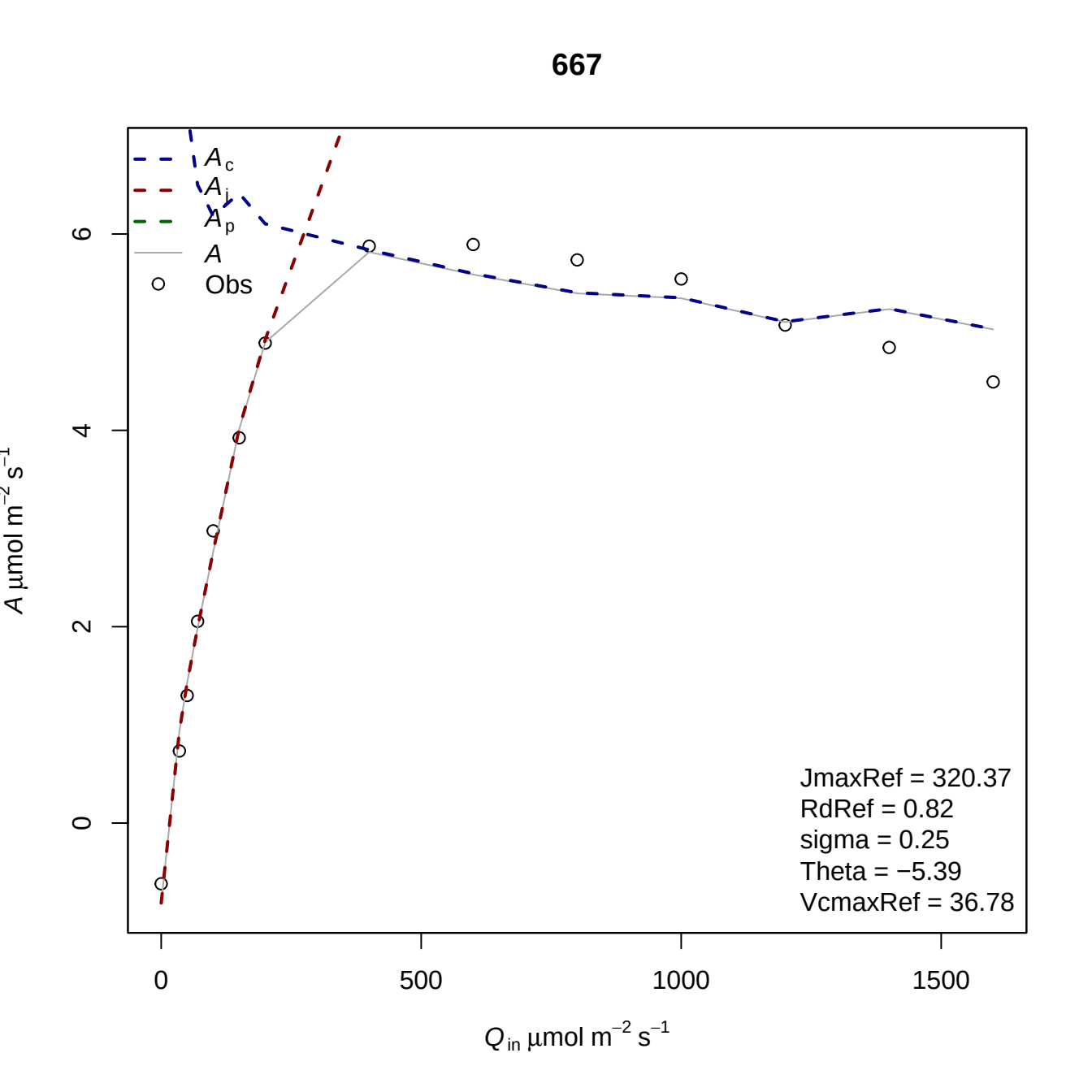

662
A_c
A_j
A_p
A
Obs
JmaxRef = 41.35
RdRef = 0.35
sigma = 0.09
Theta = 0.49
VcmaxRef = 34.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

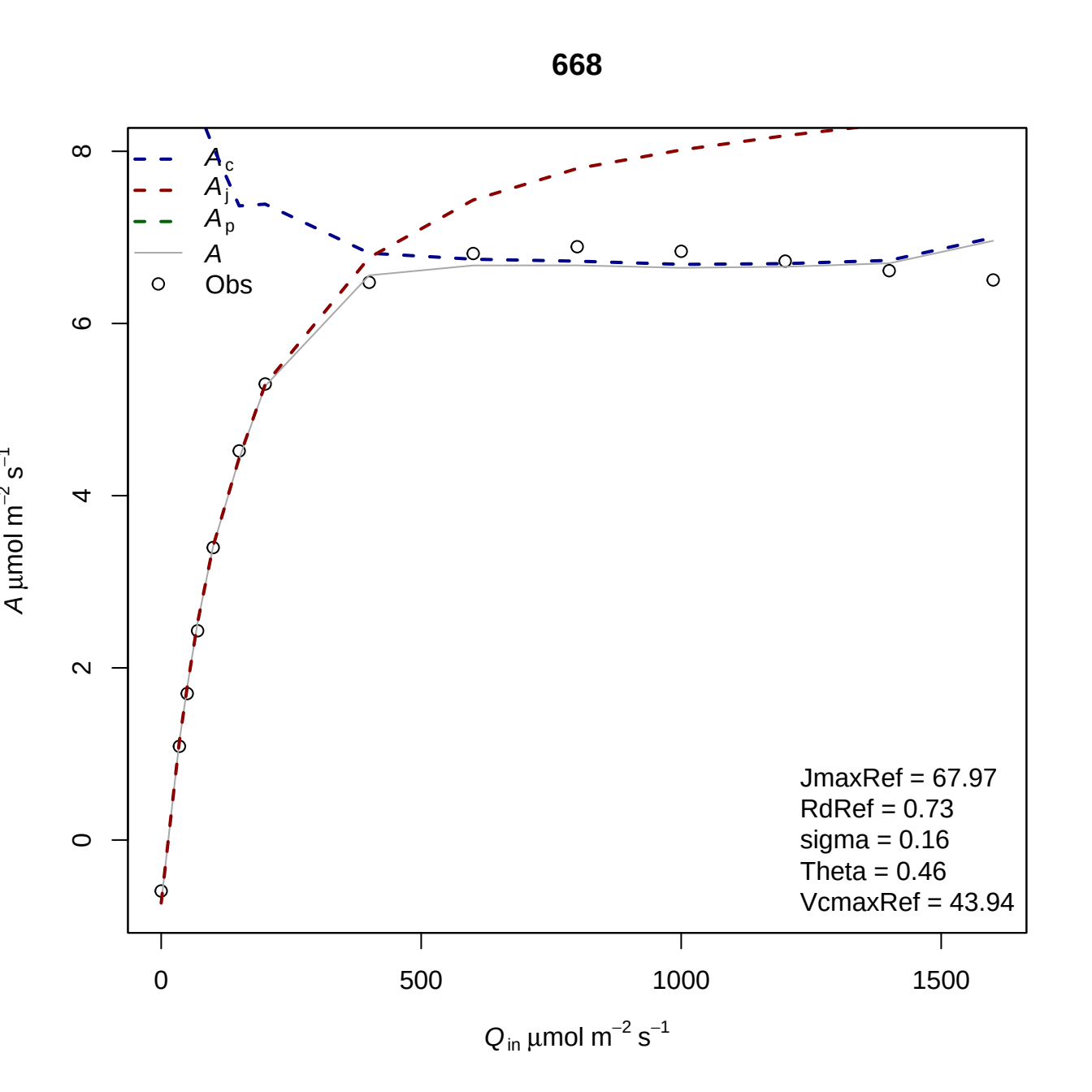

663
A_c
A_j
A_p
A
Obs
JmaxRef = 47.31
RdRef = 0.49
sigma = 0.1
Theta = 0.68
VcmaxRef = 43.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

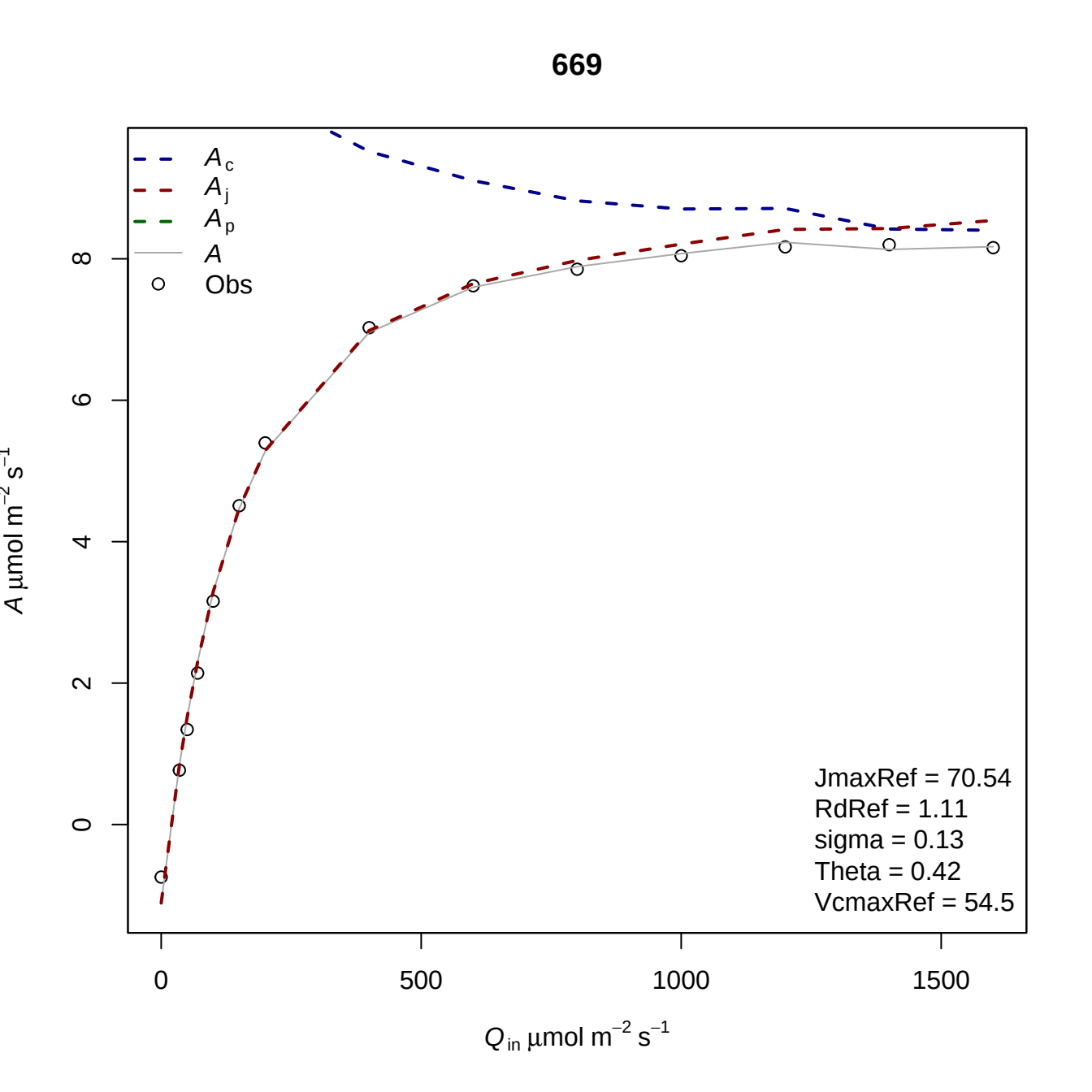

664
A_c
A_j
A_p
A
Obs
JmaxRef = 49.75
RdRef = 0.62
sigma = 0.24
Theta = 0.48
VcmaxRef = 38.8
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

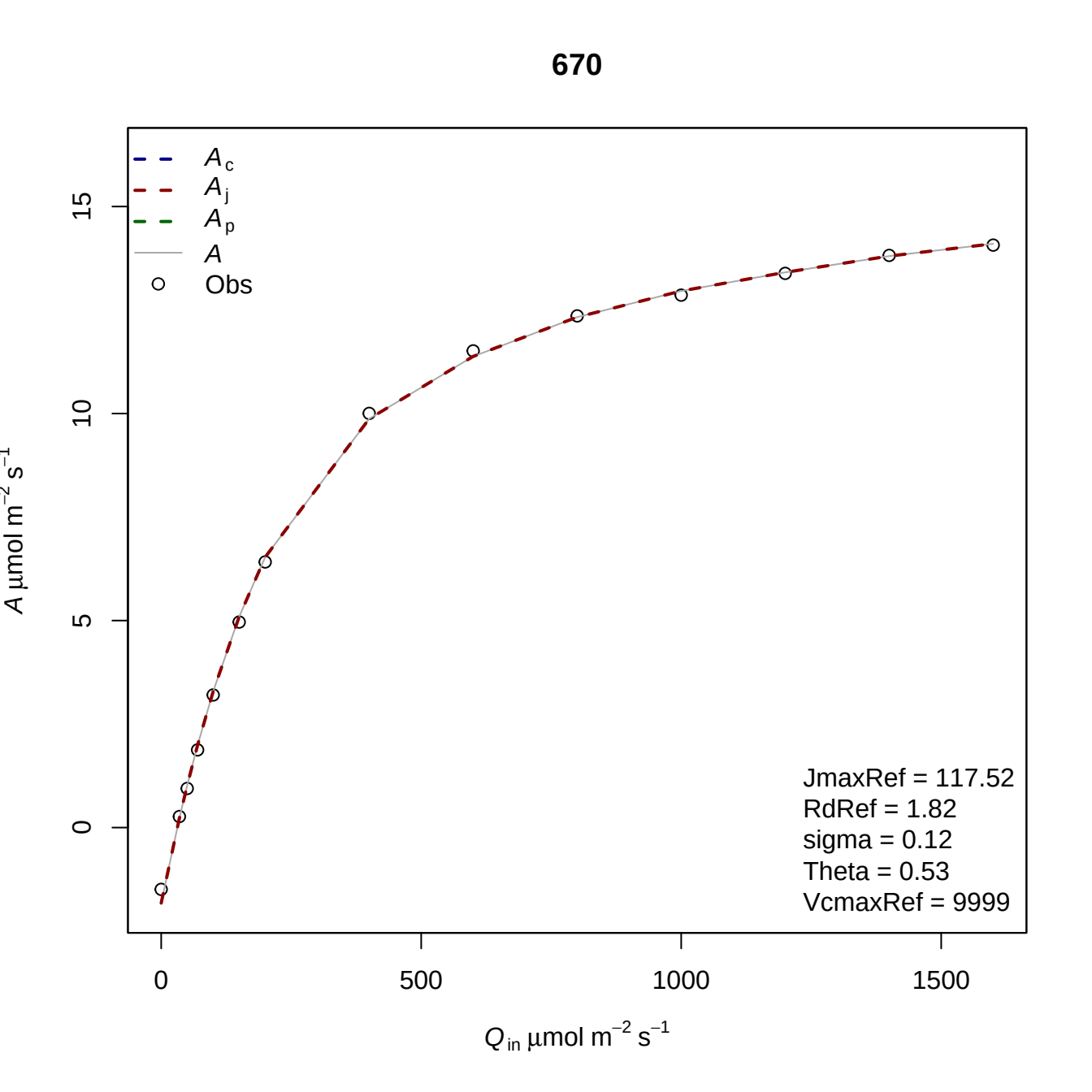

665
A_c
A_j
A_p
A
Obs
JmaxRef = 64.71
RdRef = 1.2
sigma = 0.21
Theta = 0.6
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

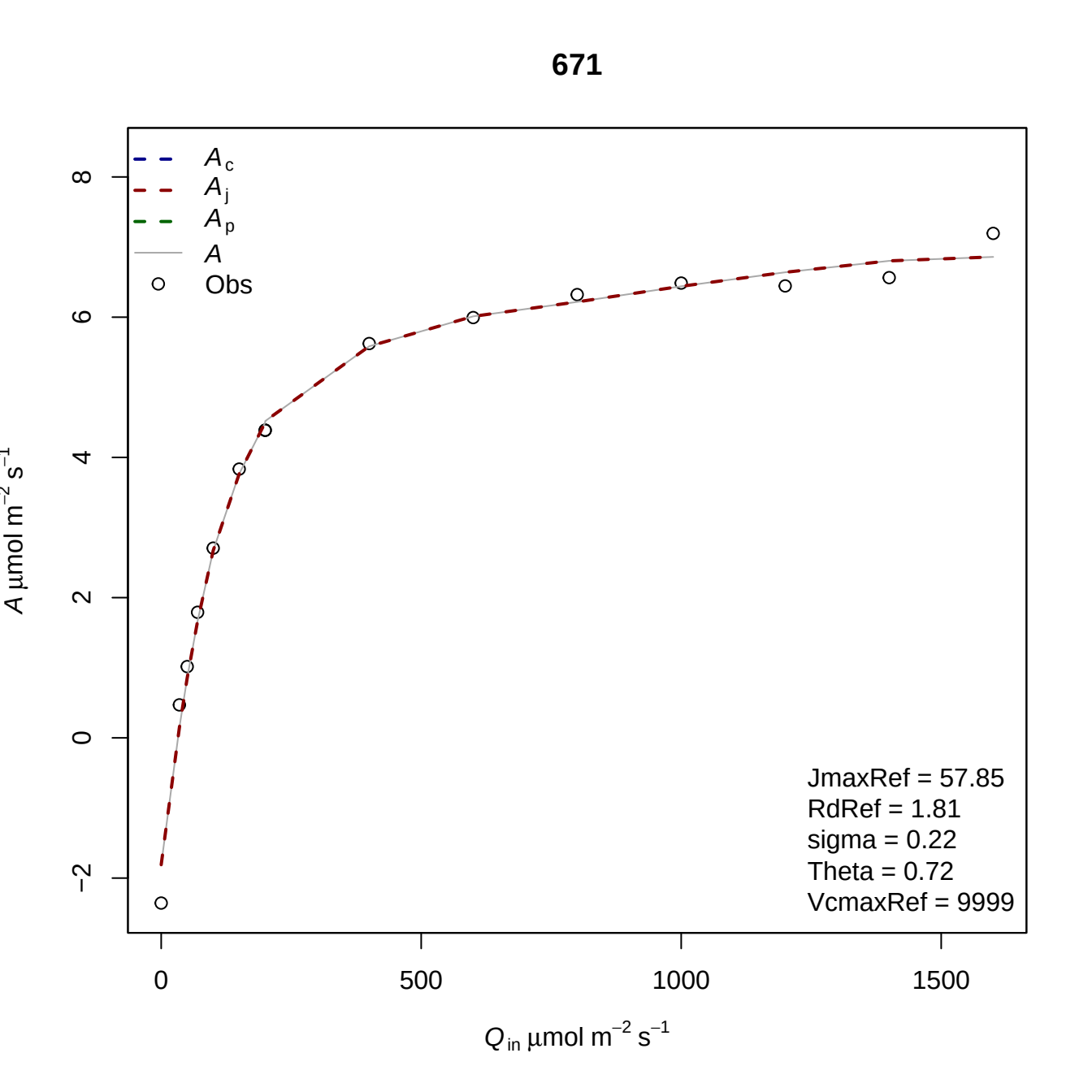

666
A_c
A_j
A_p
A
Obs
JmaxRef = 46.17
RdRef = 0.64
sigma = 0.13
Theta = 0.08
VcmaxRef = 30.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

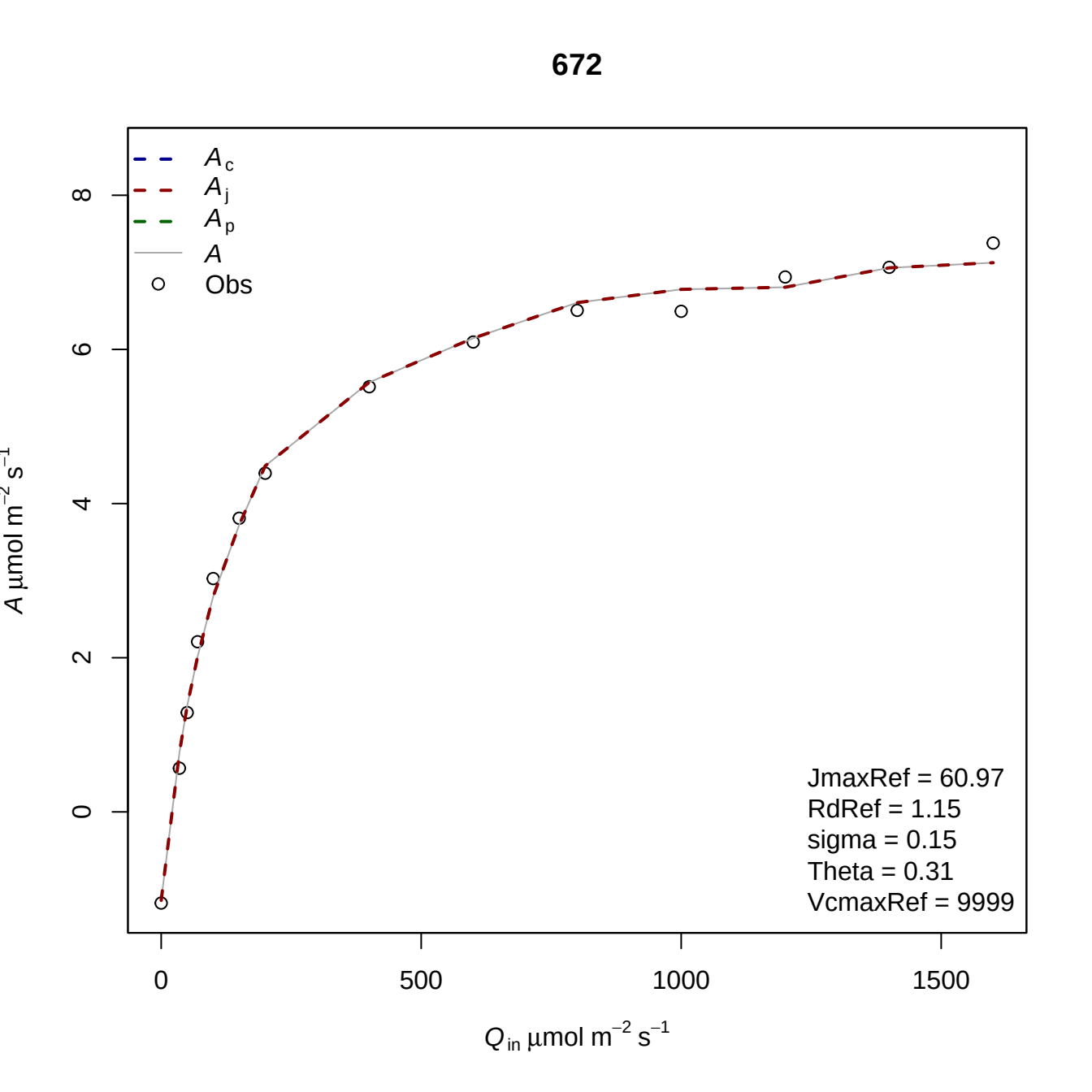

667
A_c
A_j
A_p
A
Obs
JmaxRef = 320.37
RdRef = 0.82
sigma = 0.25
Theta = −5.39
VcmaxRef = 36.78
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

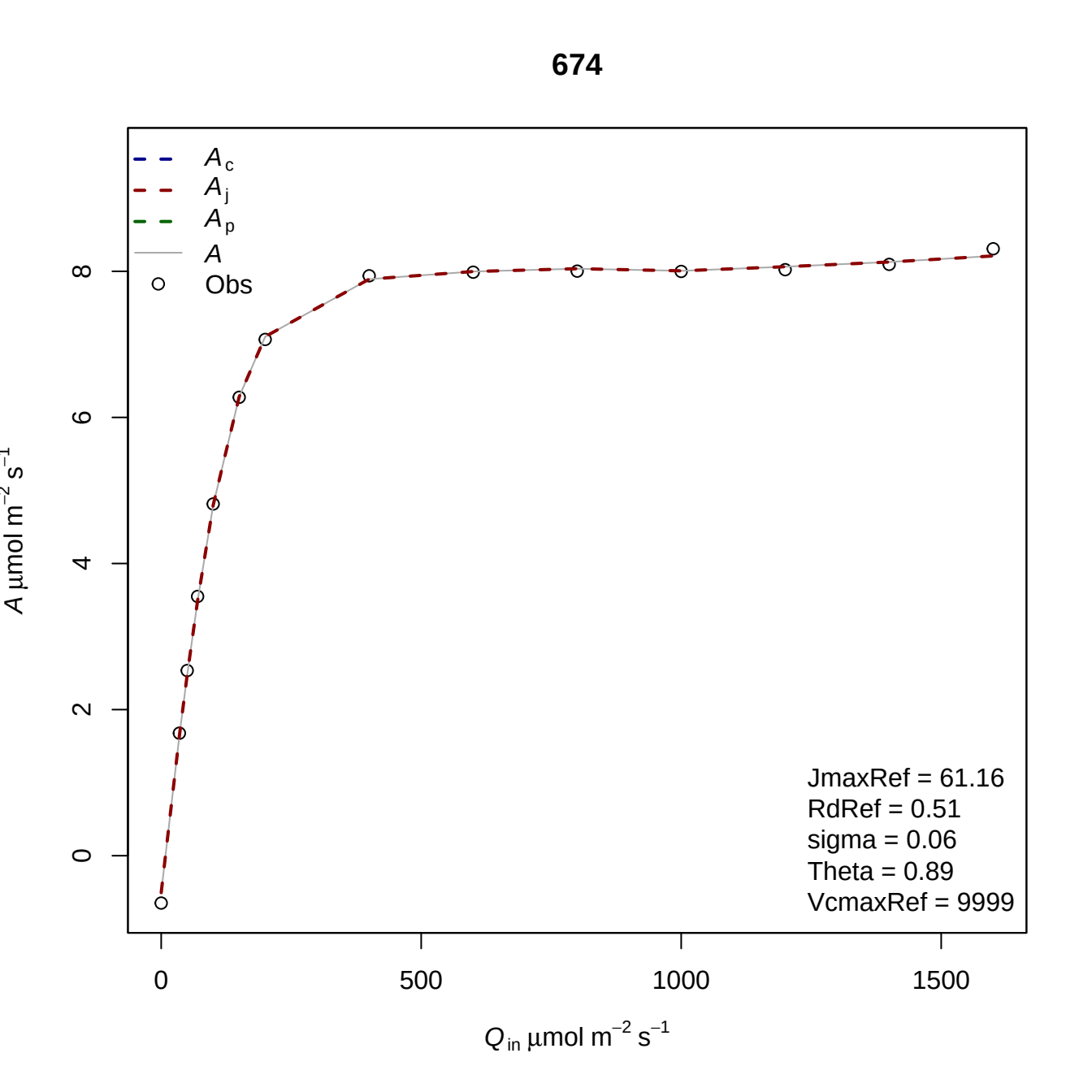

668
A_c
A_j
A_p
A
Obs
JmaxRef = 67.97
RdRef = 0.73
sigma = 0.16
Theta = 0.46
VcmaxRef = 43.94
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

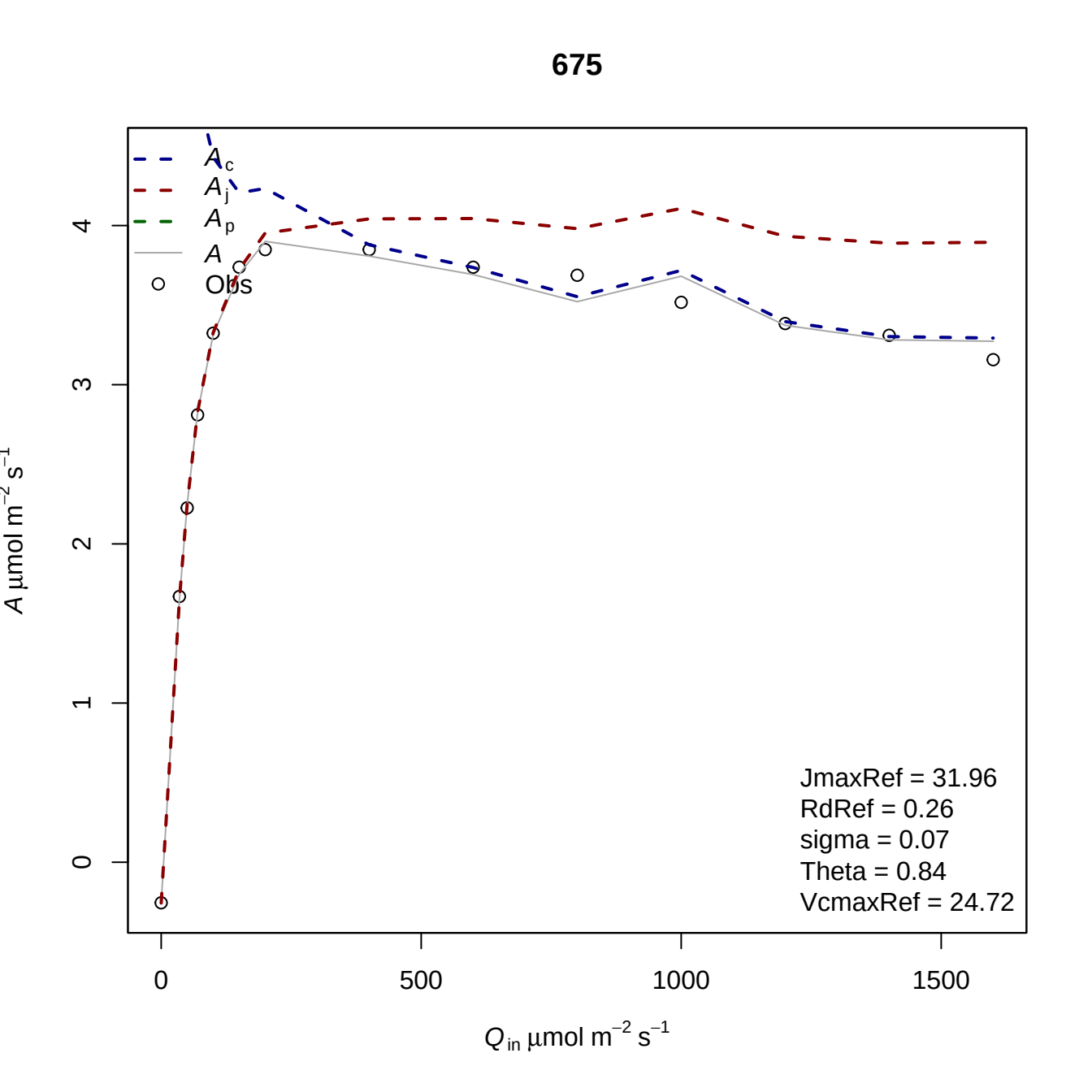

669
A_c
A_j
A_p
A
Obs
JmaxRef = 70.54
RdRef = 1.11
sigma = 0.13
Theta = 0.42
VcmaxRef = 54.5
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

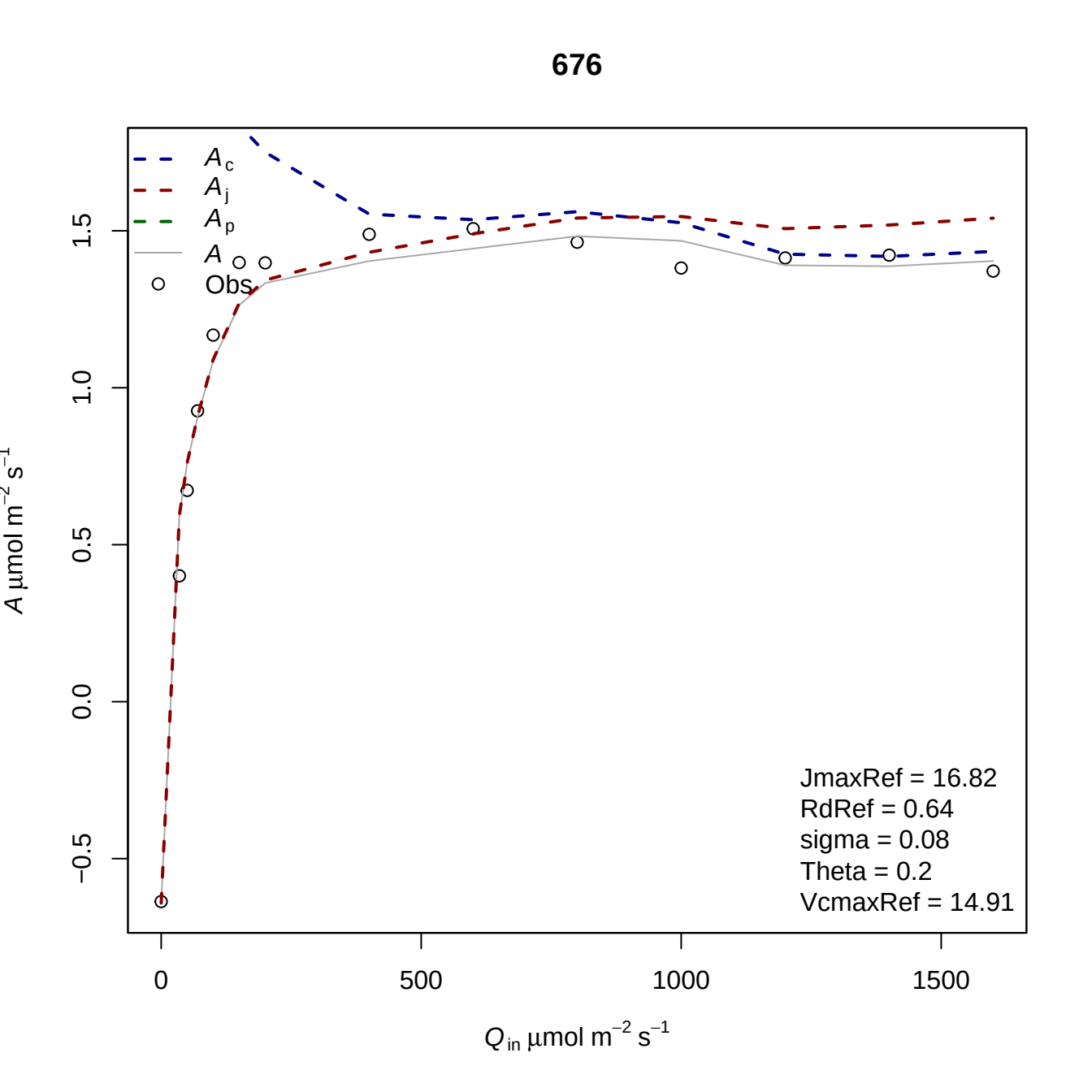

670
A_c
A_j
A_p
A
Obs
JmaxRef = 117.52
RdRef = 1.82
sigma = 0.12
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

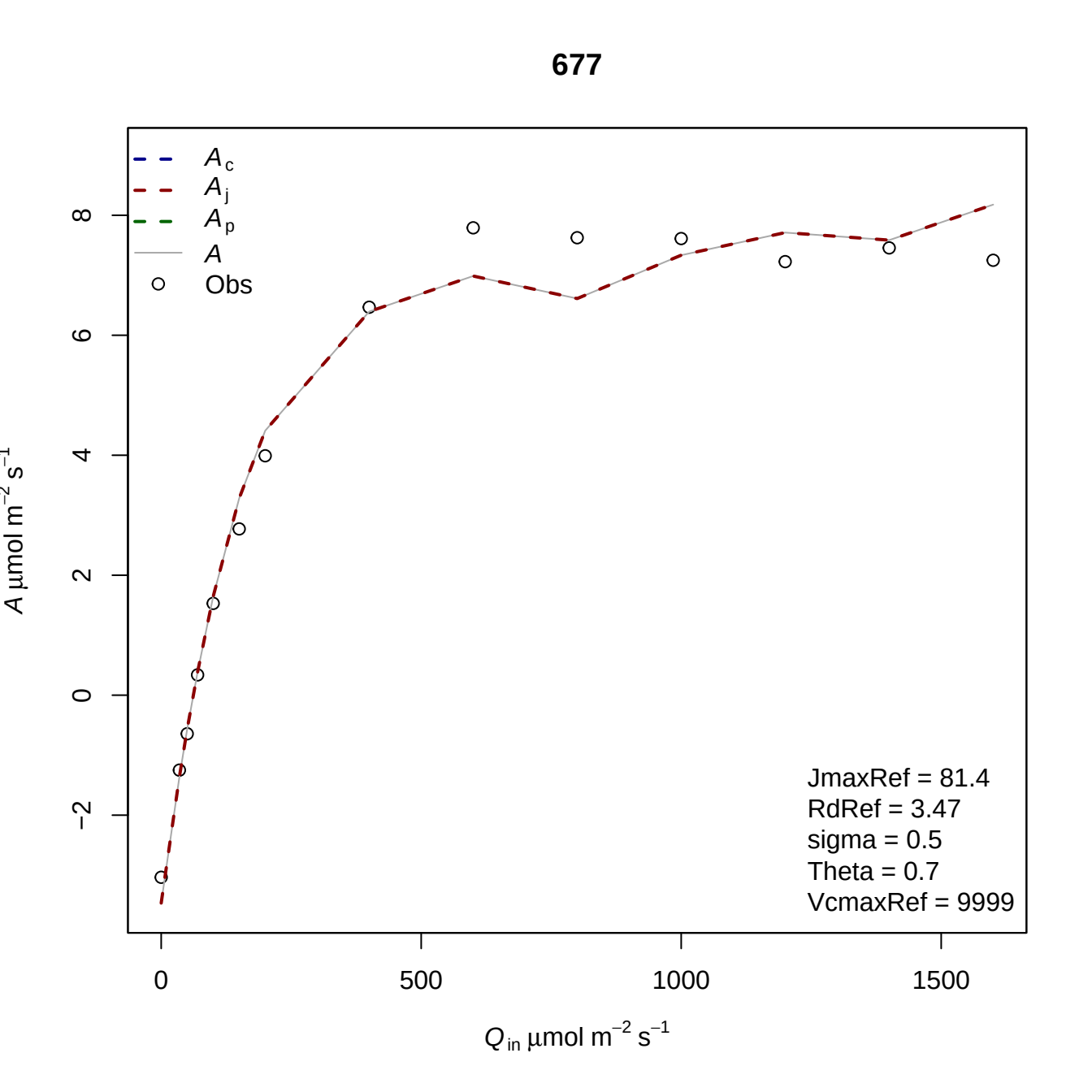

671
A_c
A_j
A_p
A
Obs
JmaxRef = 57.85
RdRef = 1.81
sigma = 0.22
Theta = 0.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

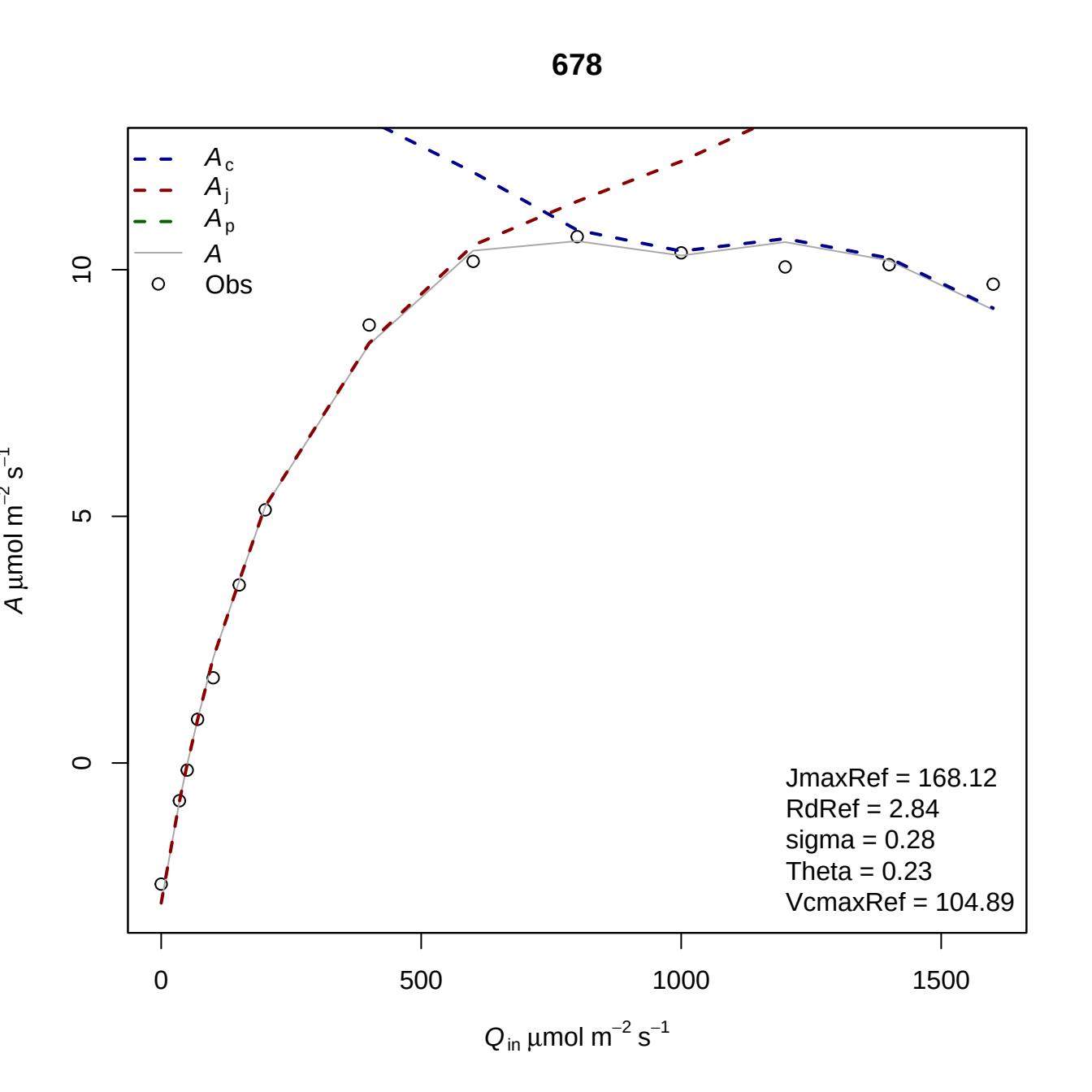

672
A_c
A_j
A_p
A
Obs
JmaxRef = 60.97
RdRef = 1.15
sigma = 0.15
Theta = 0.31
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

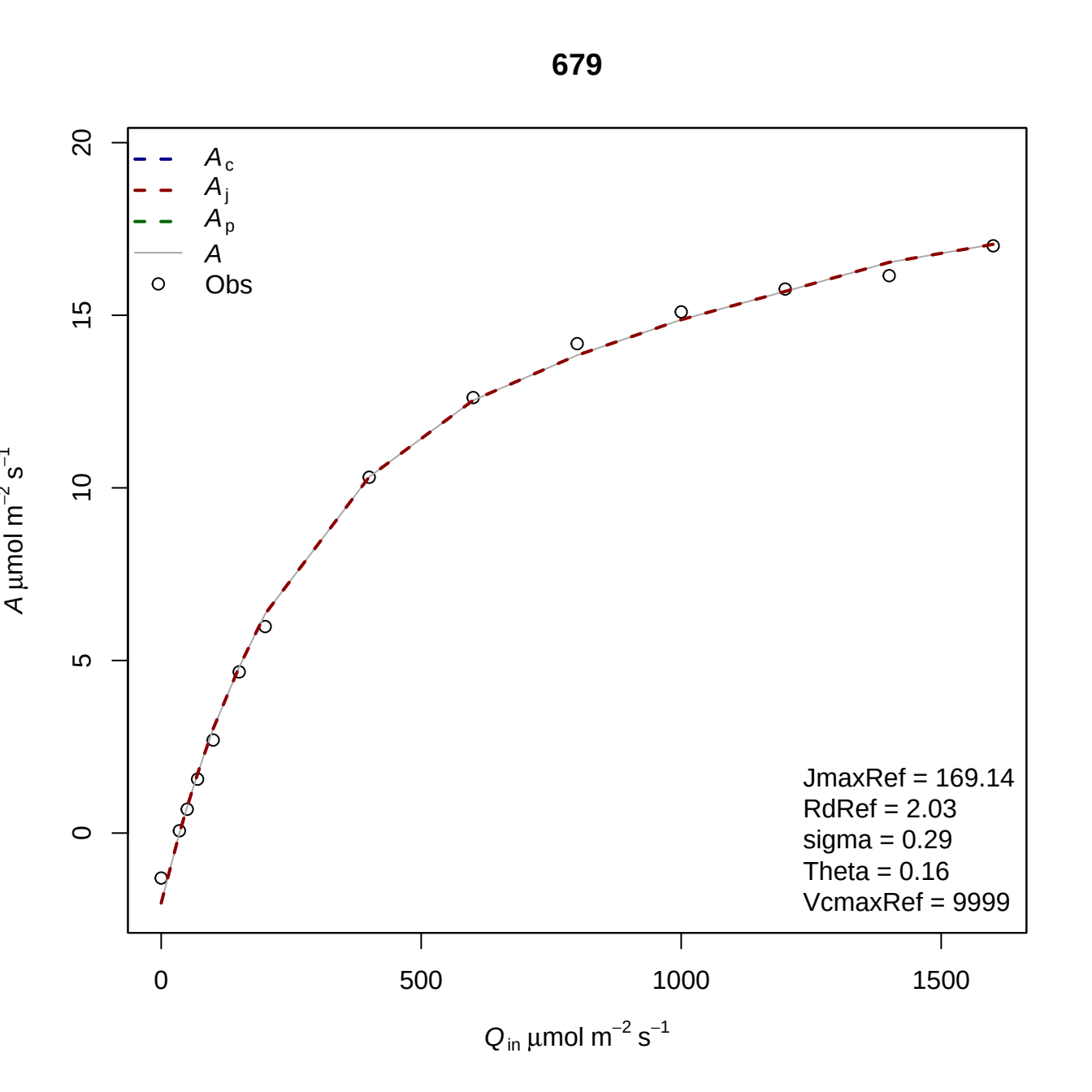

674
A_c
A_j
A_p
A
Obs
JmaxRef = 61.16
RdRef = 0.51
sigma = 0.06
Theta = 0.89
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

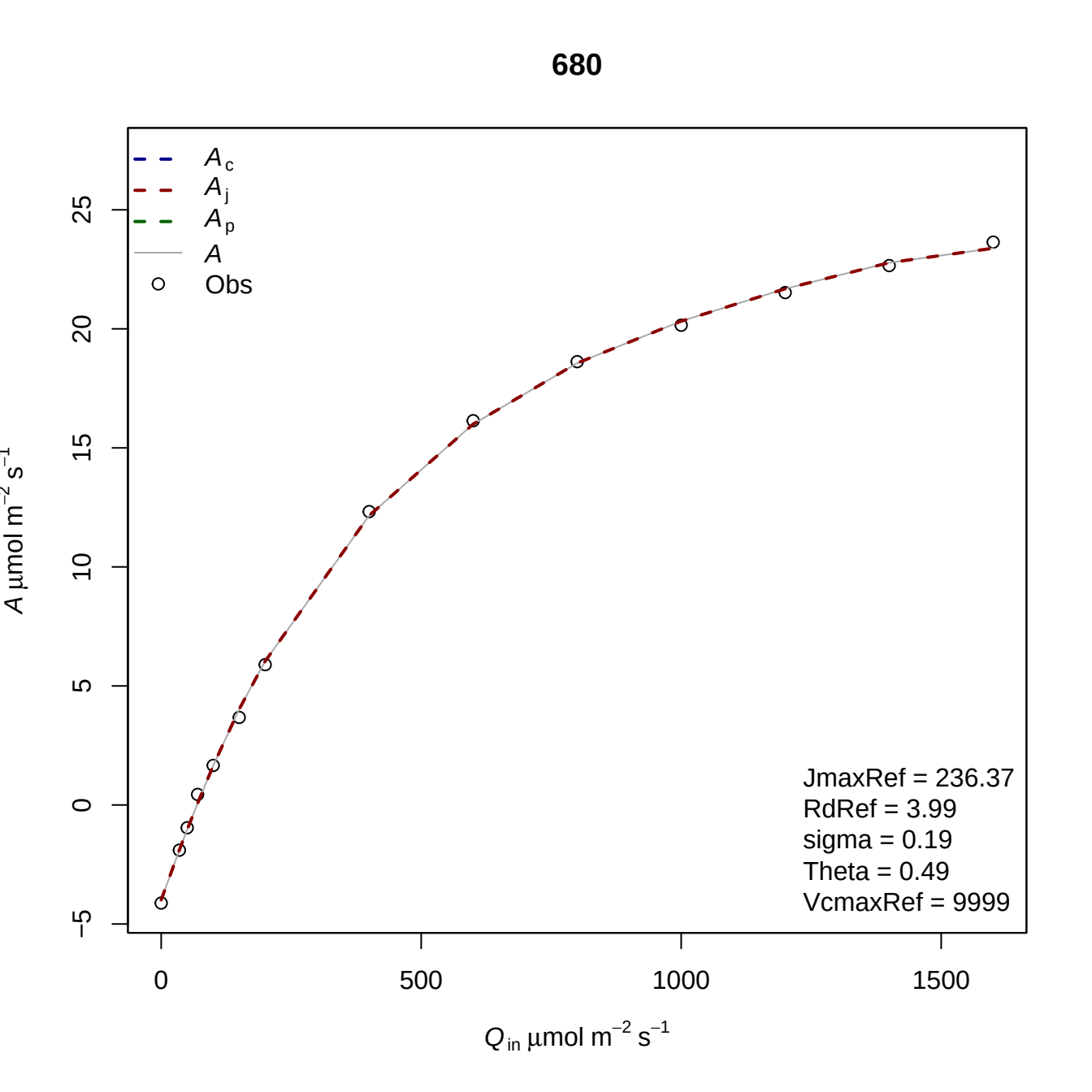

675
A_c
A_j
A_p
A
Obs
JmaxRef = 31.96
RdRef = 0.26
sigma = 0.07
Theta = 0.84
VcmaxRef = 24.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

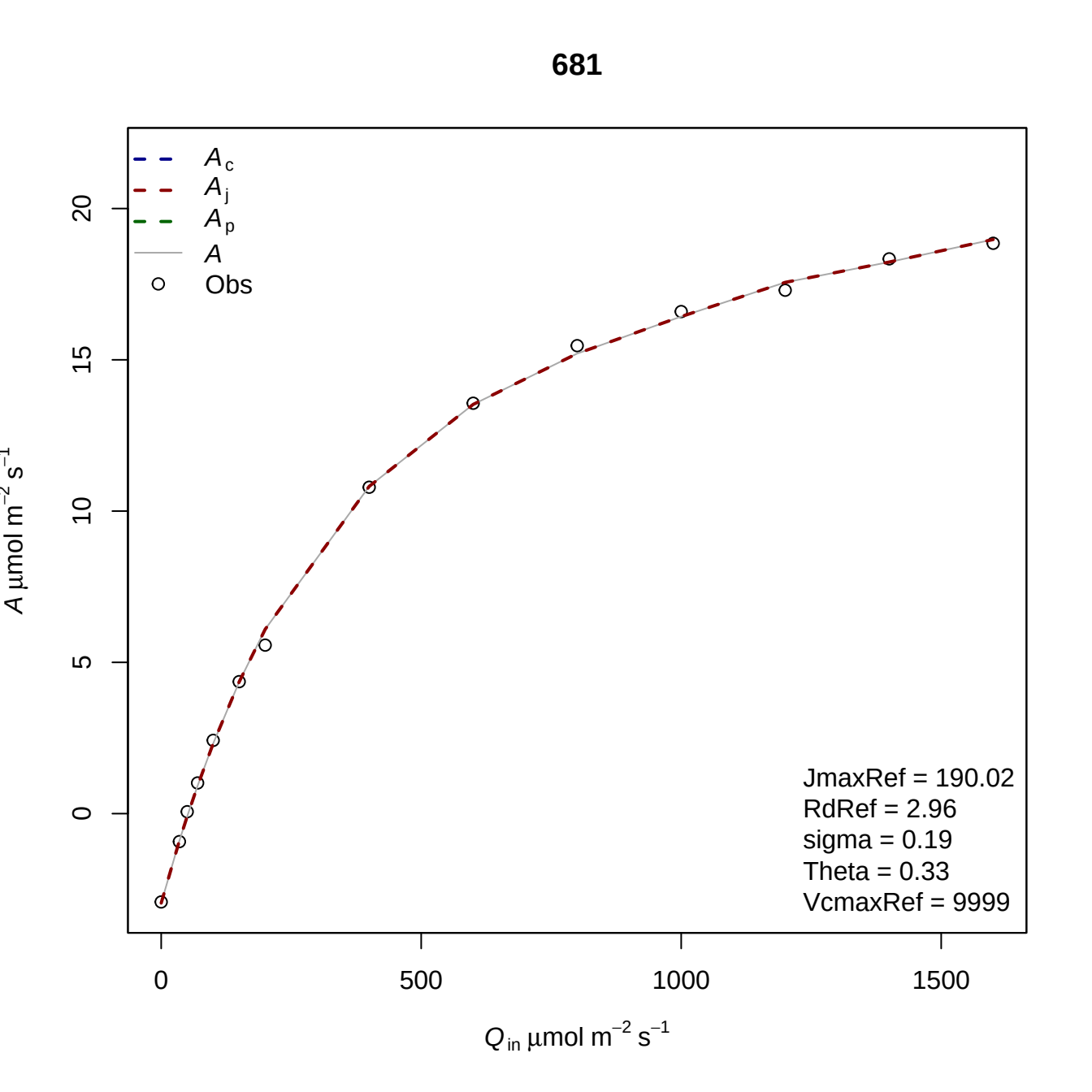

676
A_c
A_j
A_p
A
Obs
JmaxRef = 16.82
RdRef = 0.64
sigma = 0.08
Theta = 0.2
VcmaxRef = 14.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500

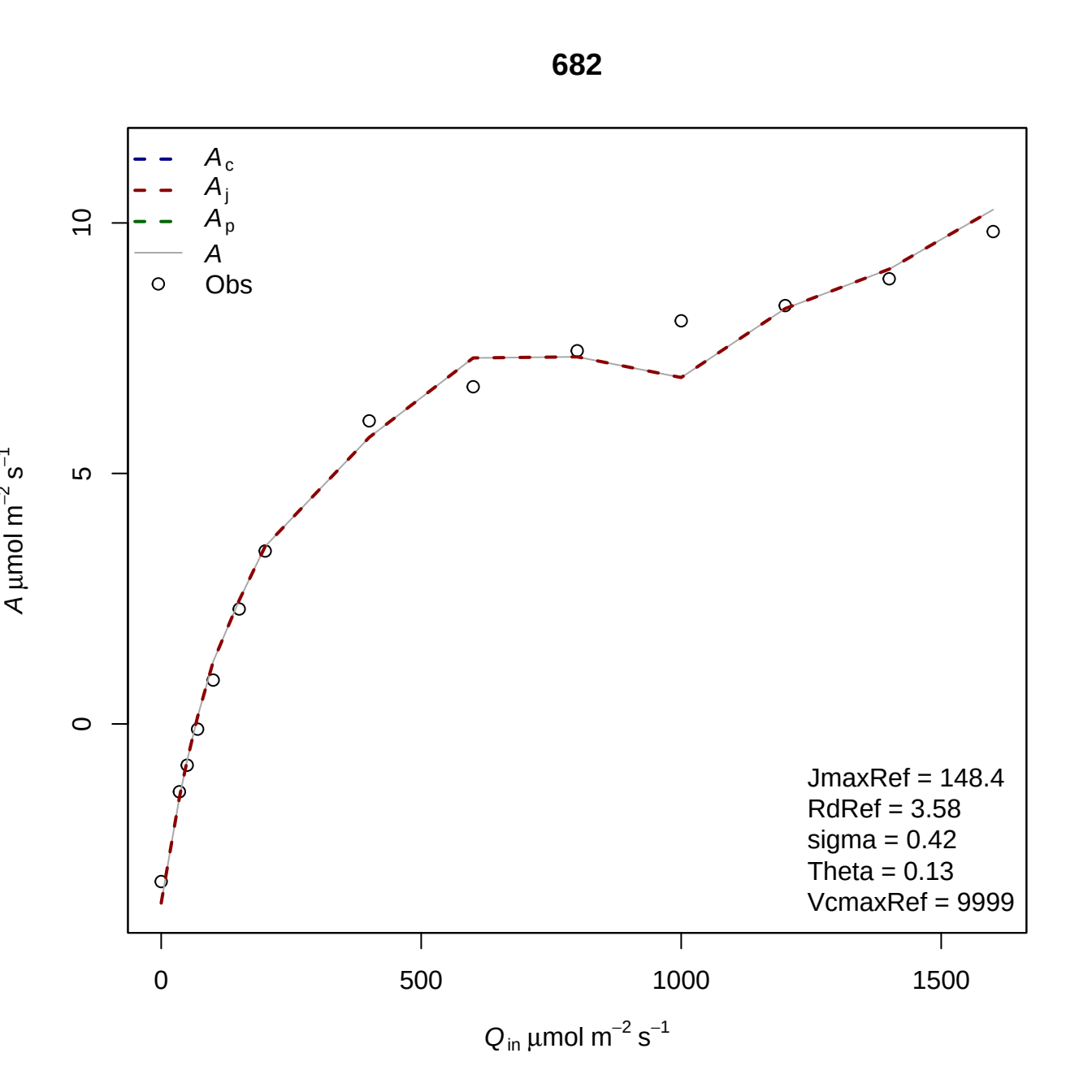

677
A_c
A_j
A_p
A
Obs
JmaxRef = 81.4
RdRef = 3.47
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

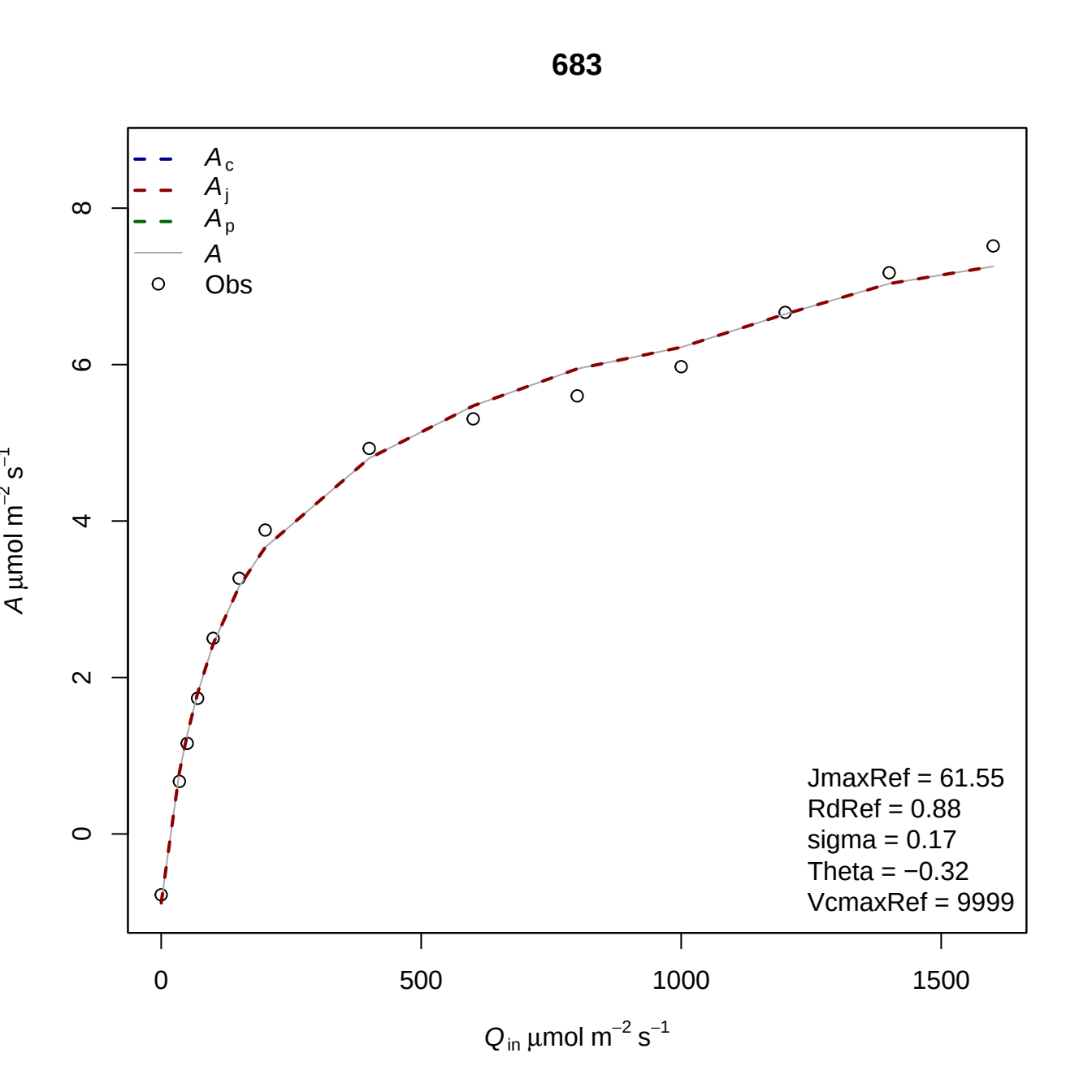

678
A_c
A_j
A_p
A
Obs
JmaxRef = 168.12
RdRef = 2.84
sigma = 0.28
Theta = 0.23
VcmaxRef = 104.89
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

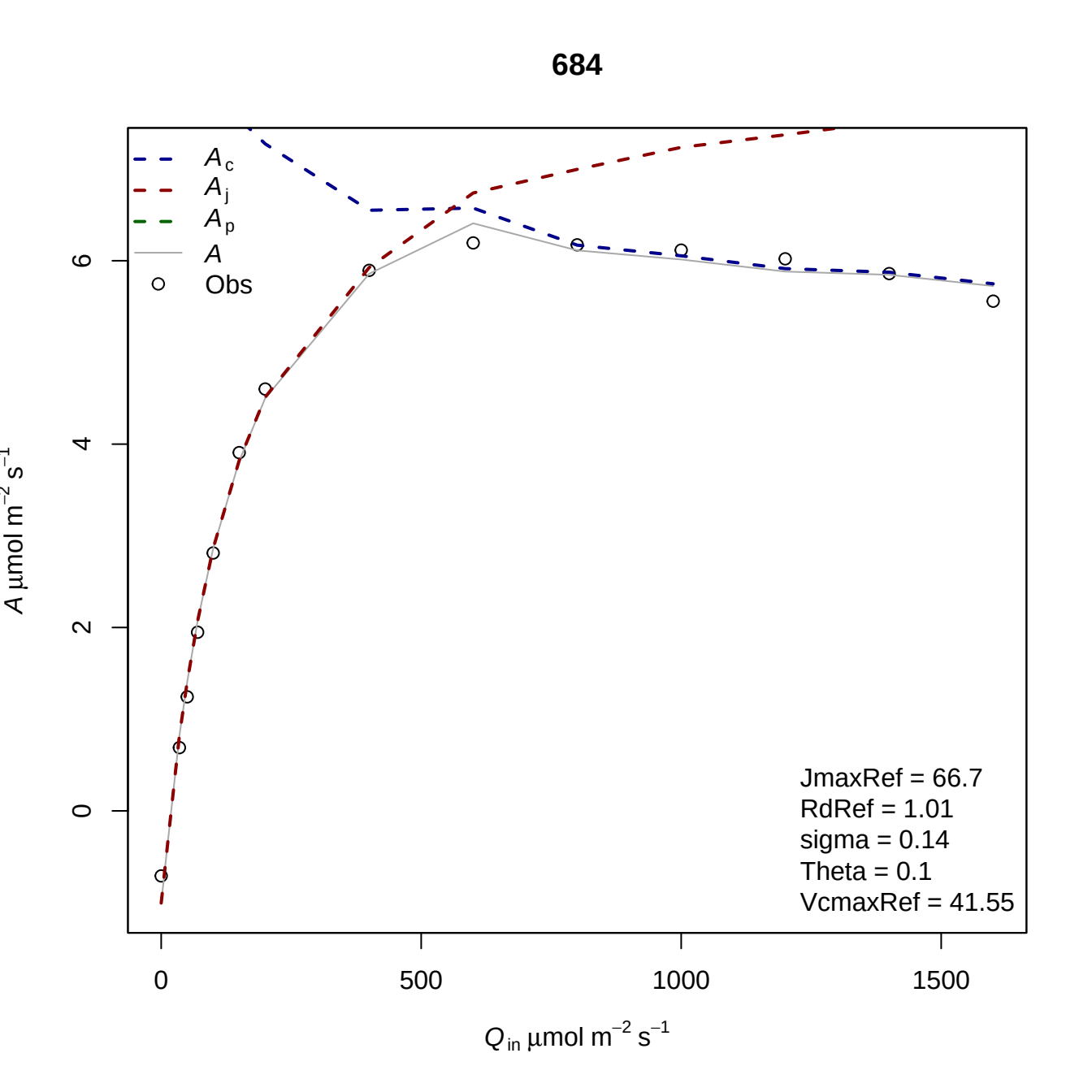

679
A_c
A_j
A_p
A
Obs
JmaxRef = 169.14
RdRef = 2.03
sigma = 0.29
Theta = 0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

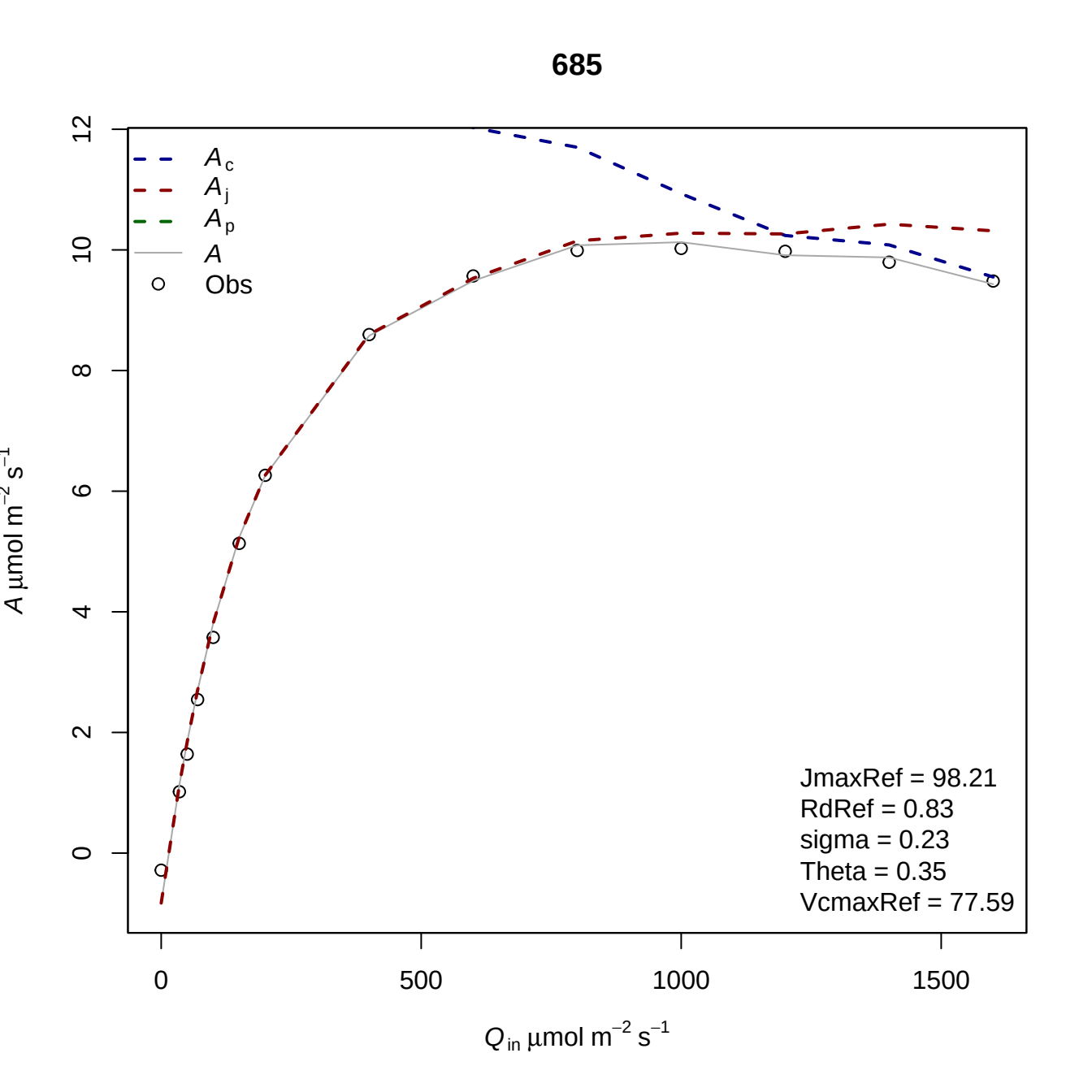

680
A_c
A_j
A_p
A
Obs
JmaxRef = 236.37
RdRef = 3.99
sigma = 0.19
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

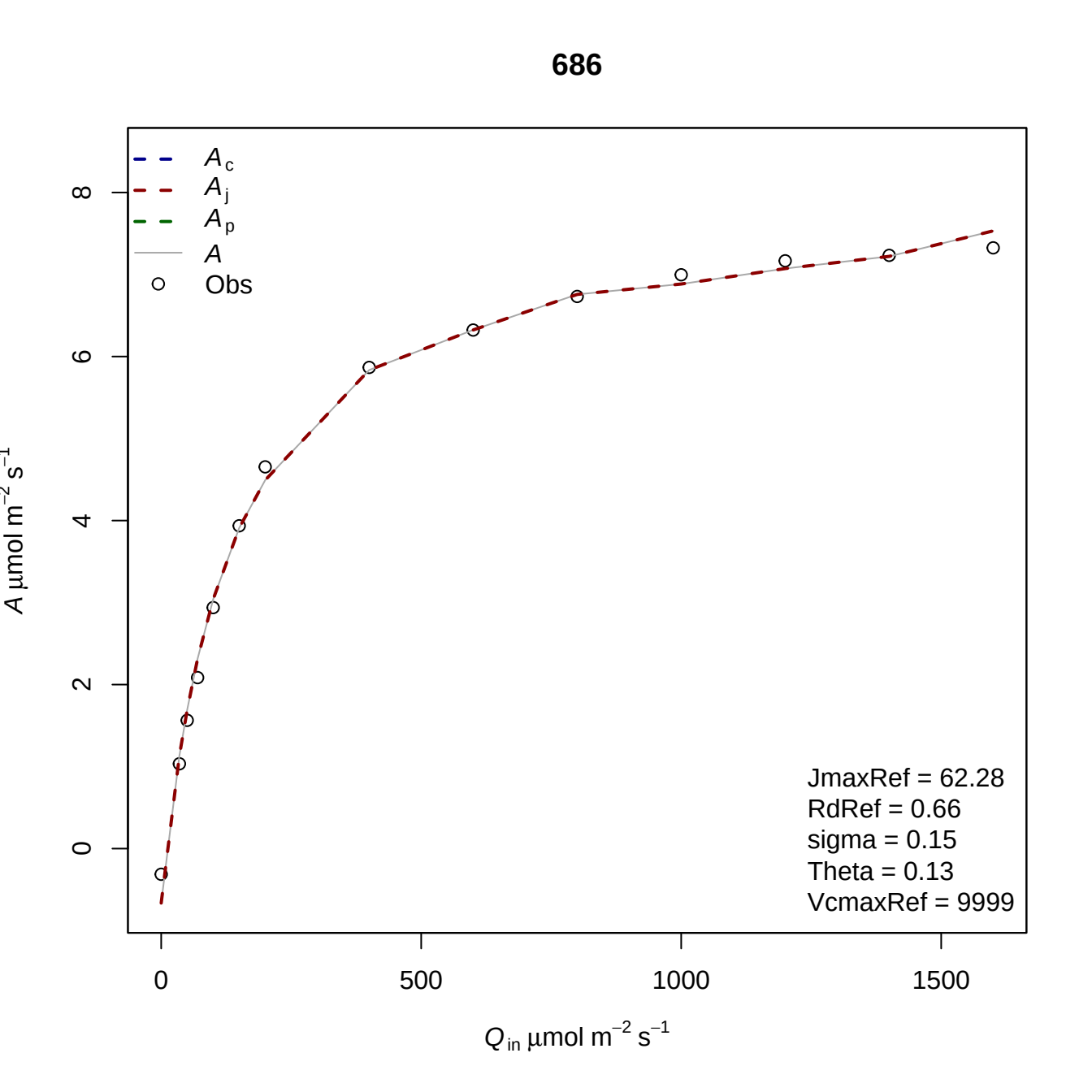

681
A_c
A_j
A_p
A
Obs
JmaxRef = 190.02
RdRef = 2.96
sigma = 0.19
Theta = 0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

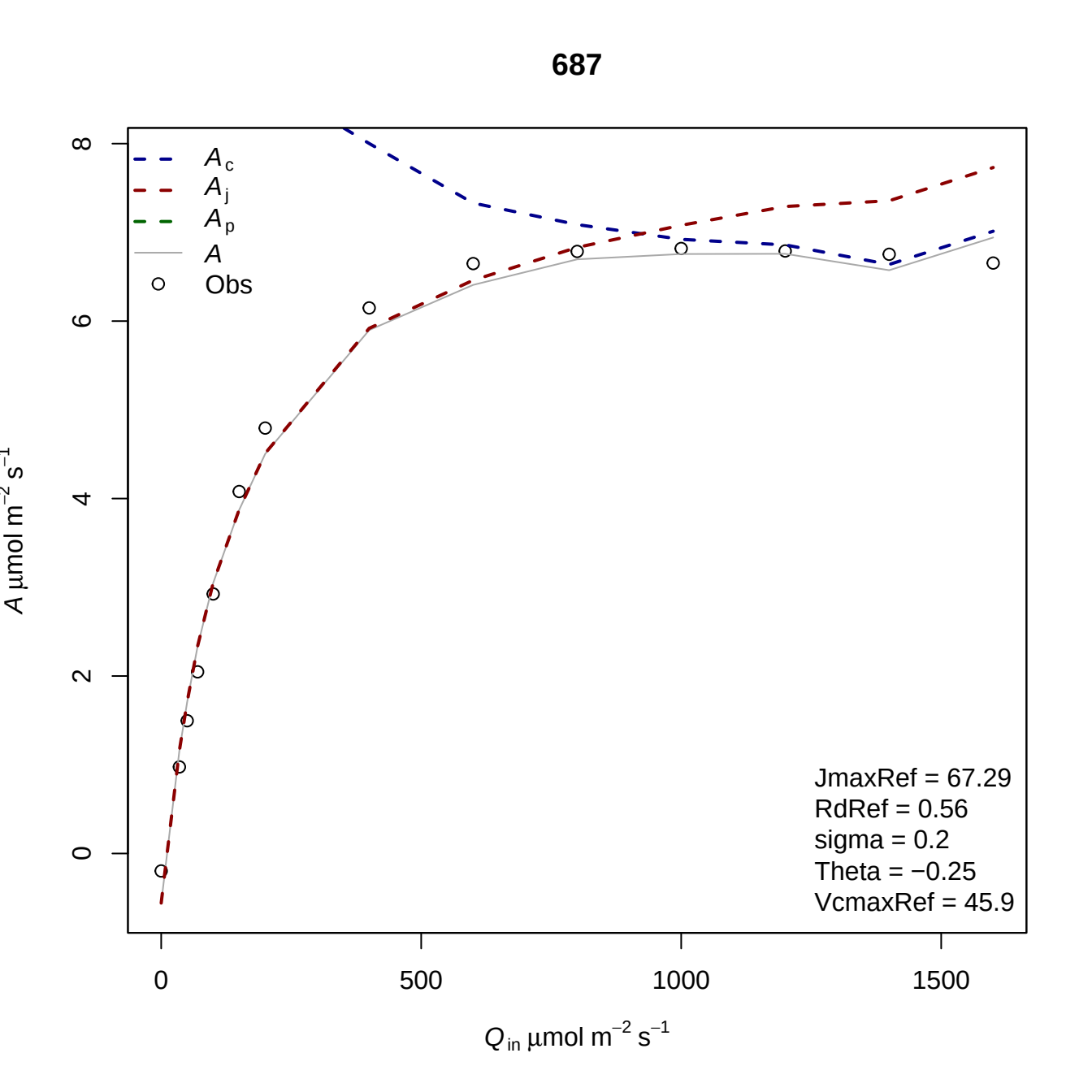

682
A_c
A_j
A_p
A
Obs
JmaxRef = 148.4
RdRef = 3.58
sigma = 0.42
Theta = 0.13
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

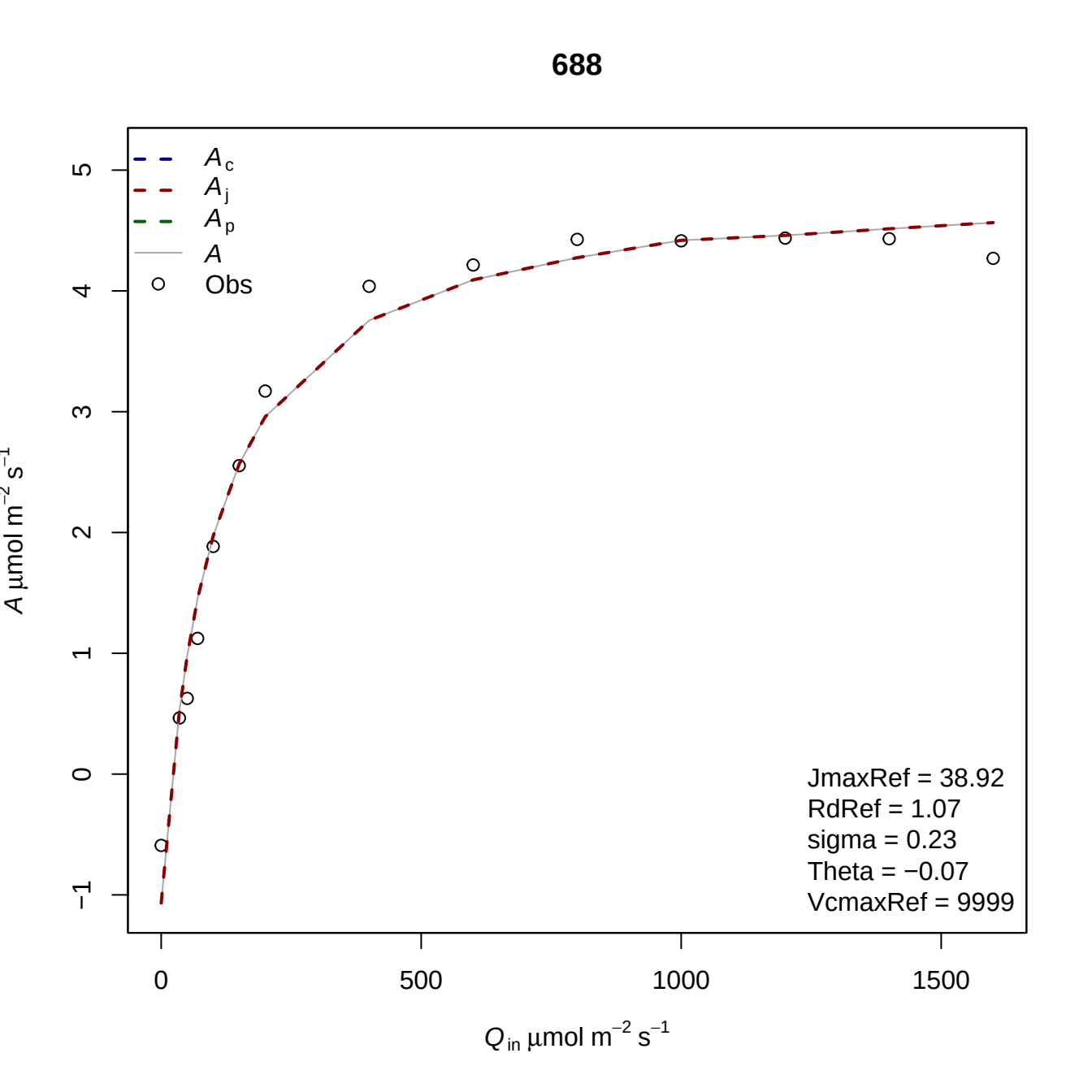

683
A_c
A_j
A_p
A
Obs
JmaxRef = 61.55
RdRef = 0.88
sigma = 0.17
Theta = −0.32
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

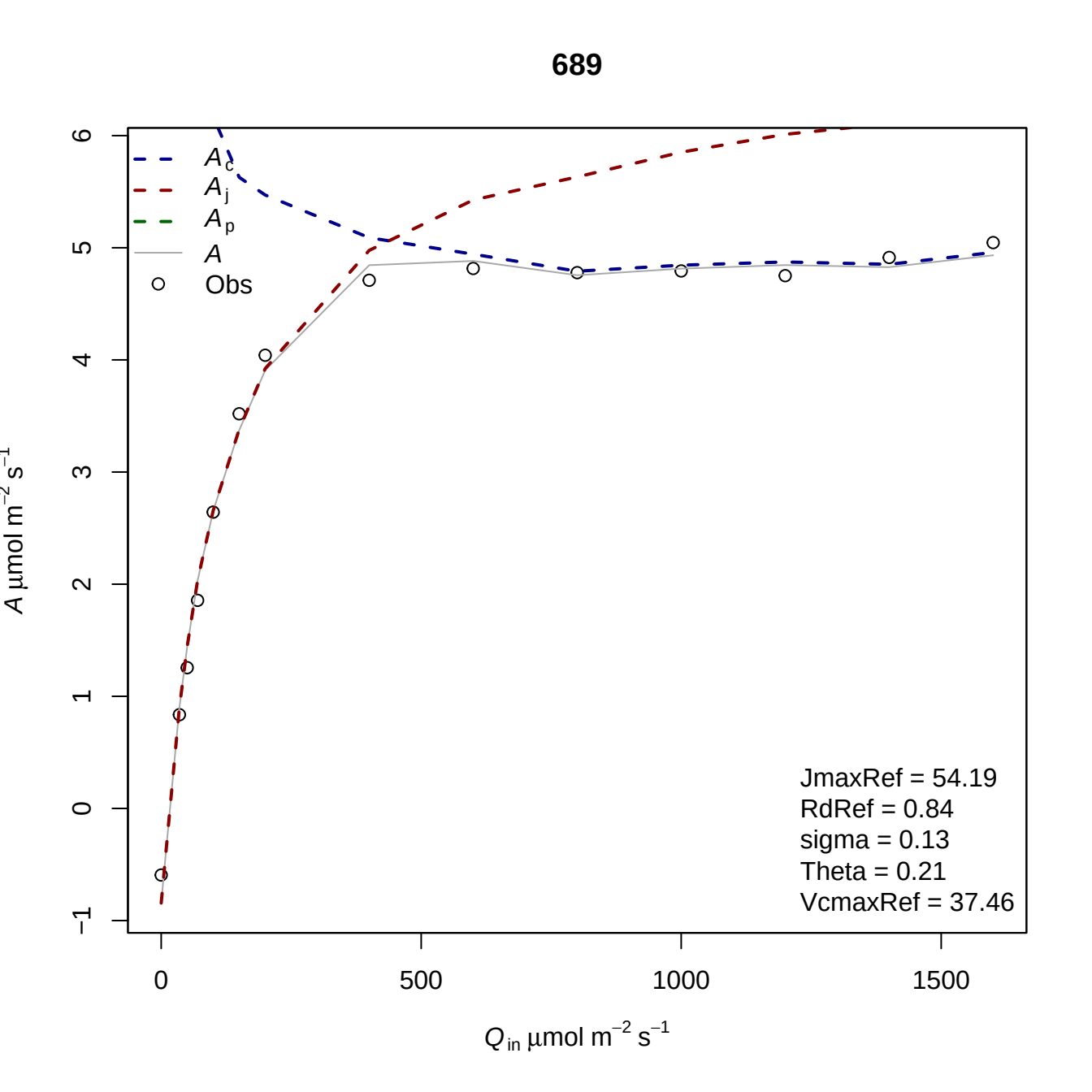

684
A_c
A_j
A_p
A
Obs
JmaxRef = 66.7
RdRef = 1.01
sigma = 0.14
Theta = 0.1
VcmaxRef = 41.55
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

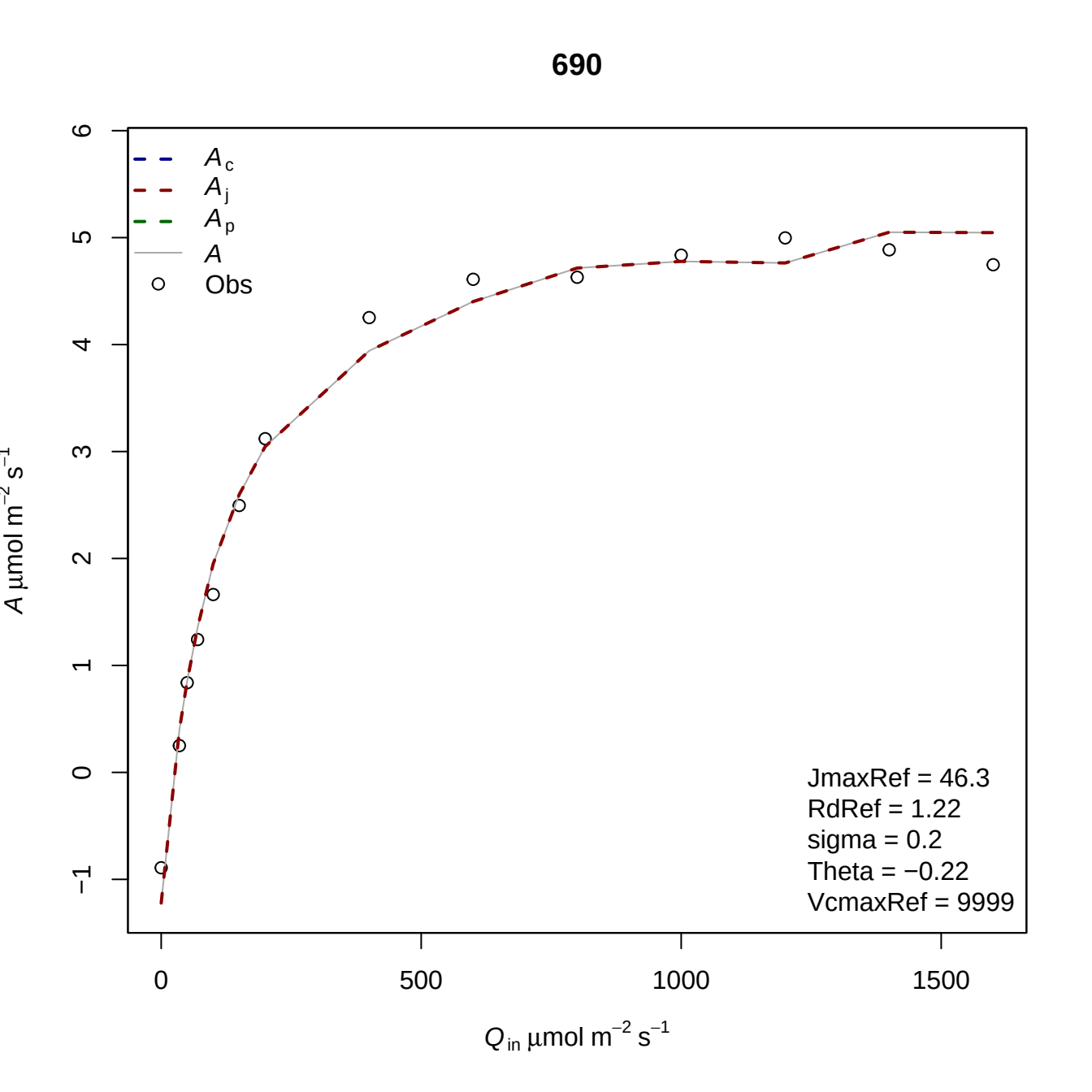

685
A_c
A_j
A_p
A
Obs
JmaxRef = 98.21
RdRef = 0.83
sigma = 0.23
Theta = 0.35
VcmaxRef = 77.59
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

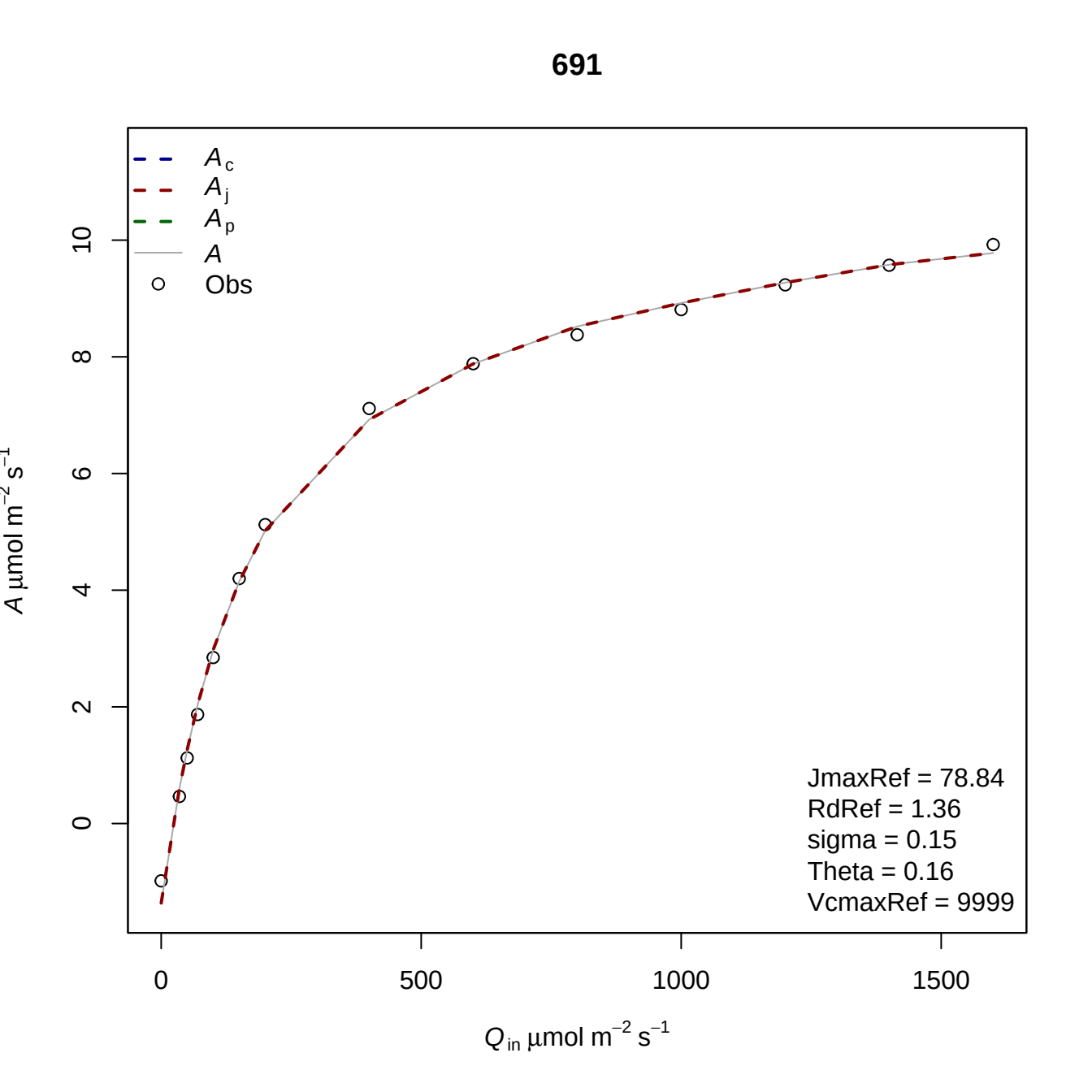

686
A_c
A_j
A_p
A
Obs
JmaxRef = 62.28
RdRef = 0.66
sigma = 0.15
Theta = 0.13
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

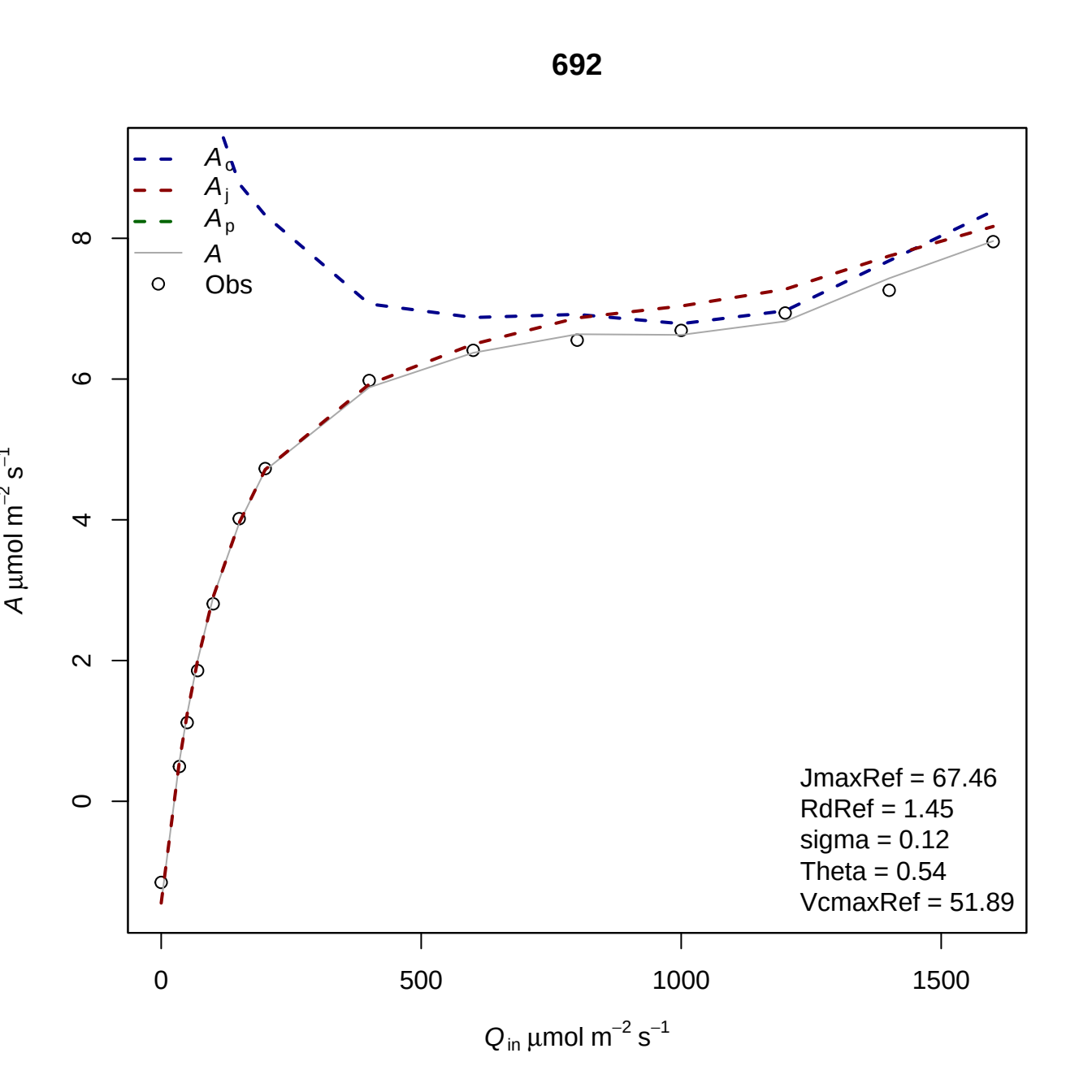

687
A_c
A_j
A_p
A
Obs
JmaxRef = 67.29
RdRef = 0.56
sigma = 0.2
Theta = −0.25
VcmaxRef = 45.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

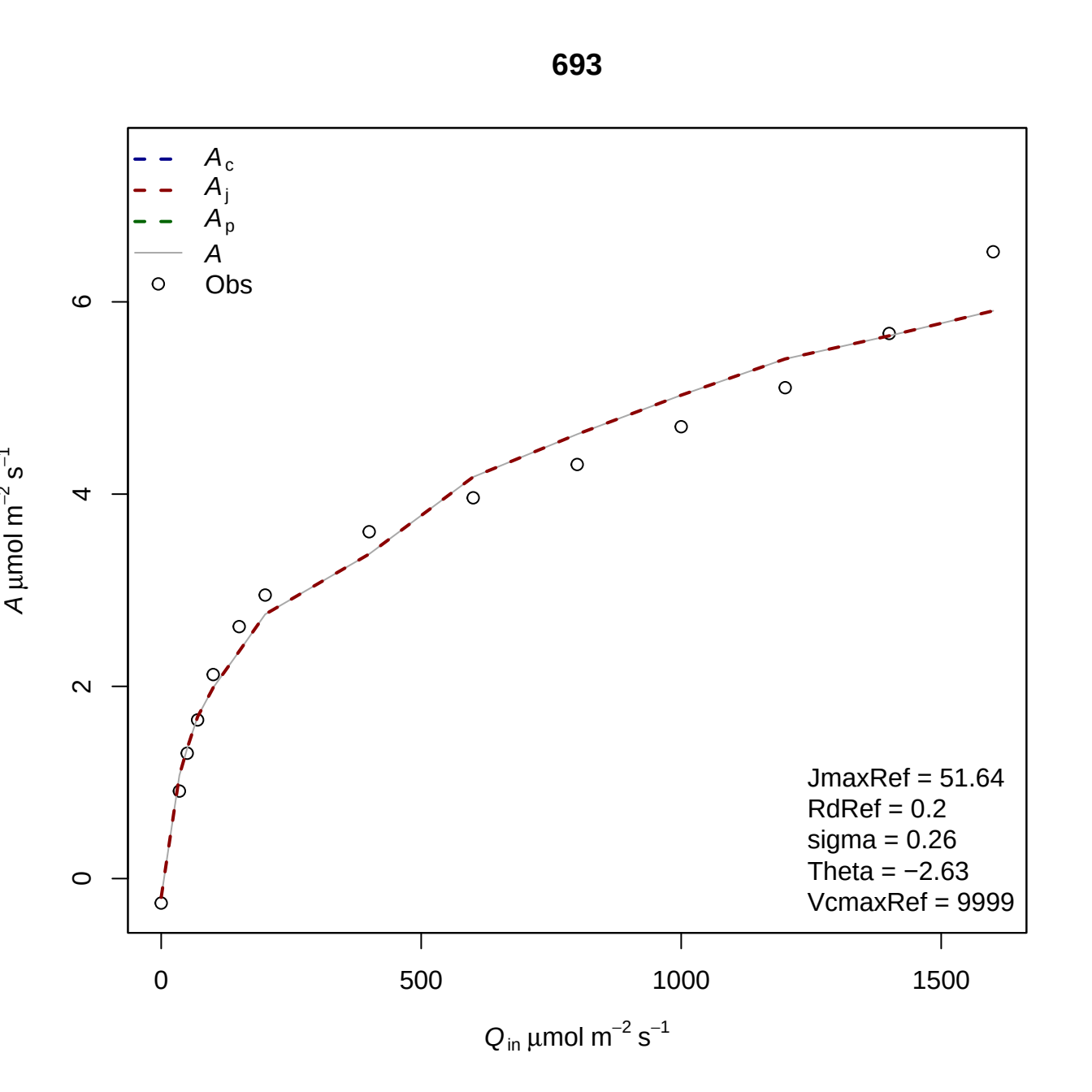

688
A_c
A_j
A_p
A
Obs
JmaxRef = 38.92
RdRef = 1.07
sigma = 0.23
Theta = −0.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

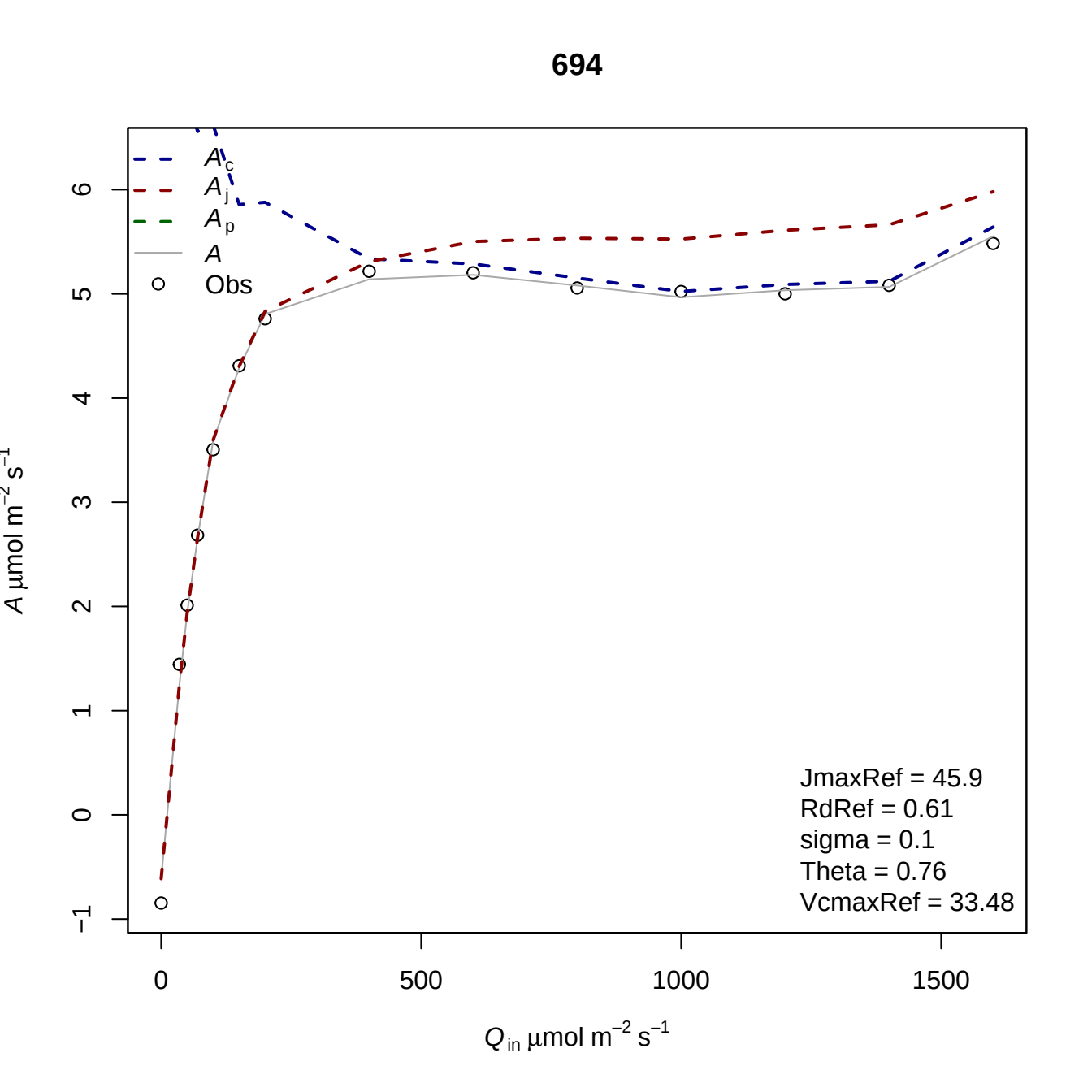

689
A_c
A_j
A_p
A
Obs
JmaxRef = 54.19
RdRef = 0.84
sigma = 0.13
Theta = 0.21
VcmaxRef = 37.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

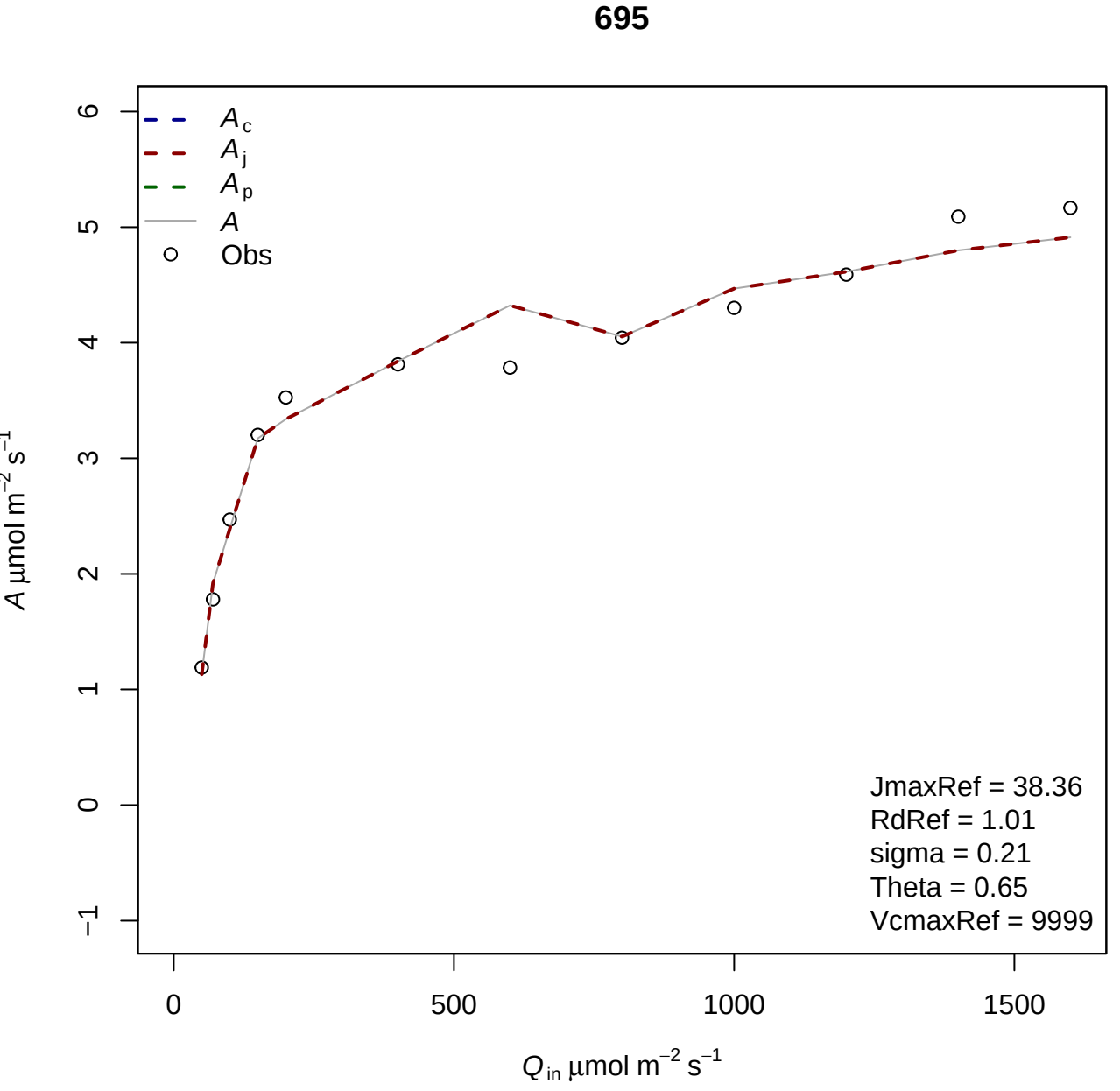

690
A_c
A_j
A_p
A
Obs
JmaxRef = 46.3
RdRef = 1.22
sigma = 0.2
Theta = −0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

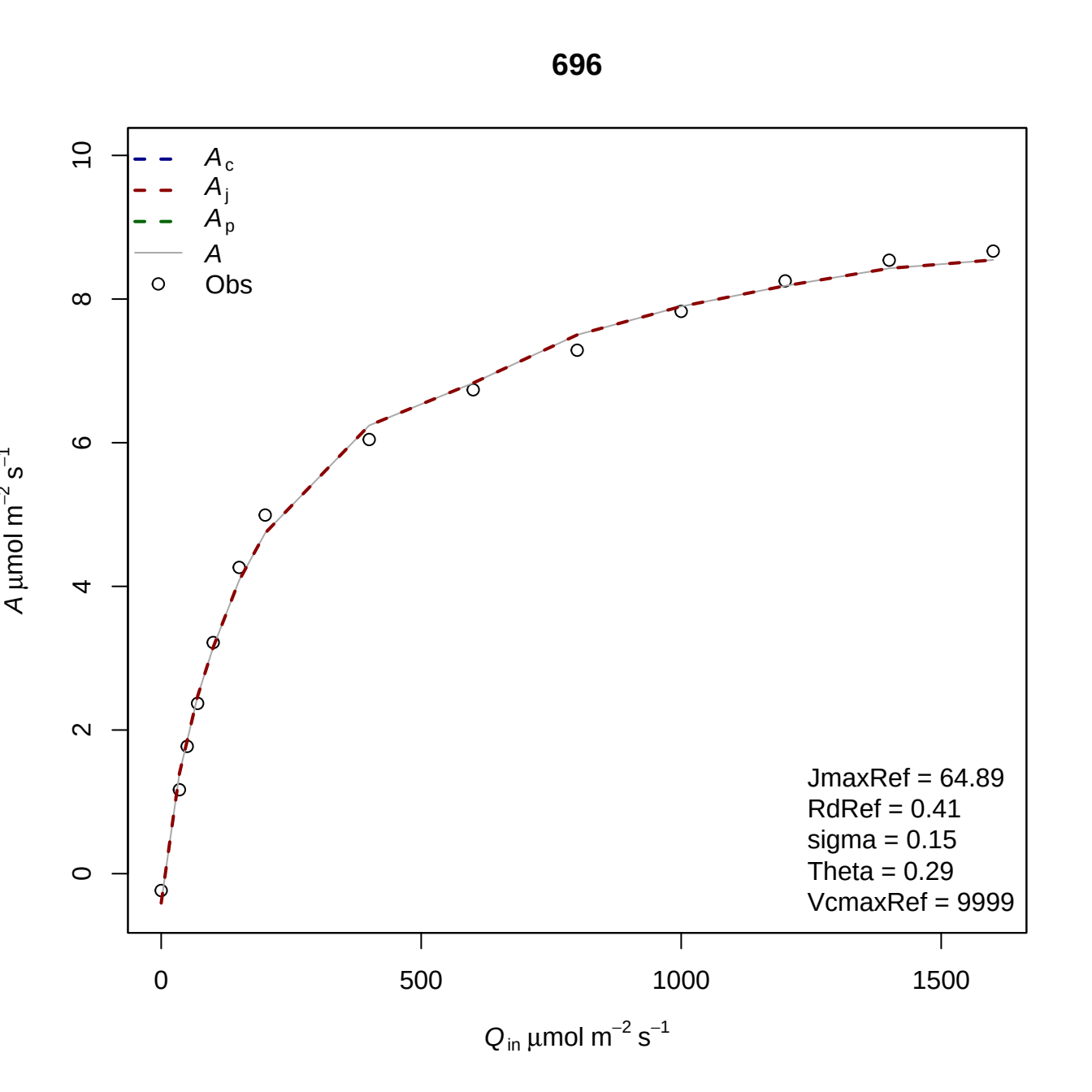

691
A_c
A_j
A_p
A
Obs
JmaxRef = 78.84
RdRef = 1.36
sigma = 0.15
Theta = 0.16
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

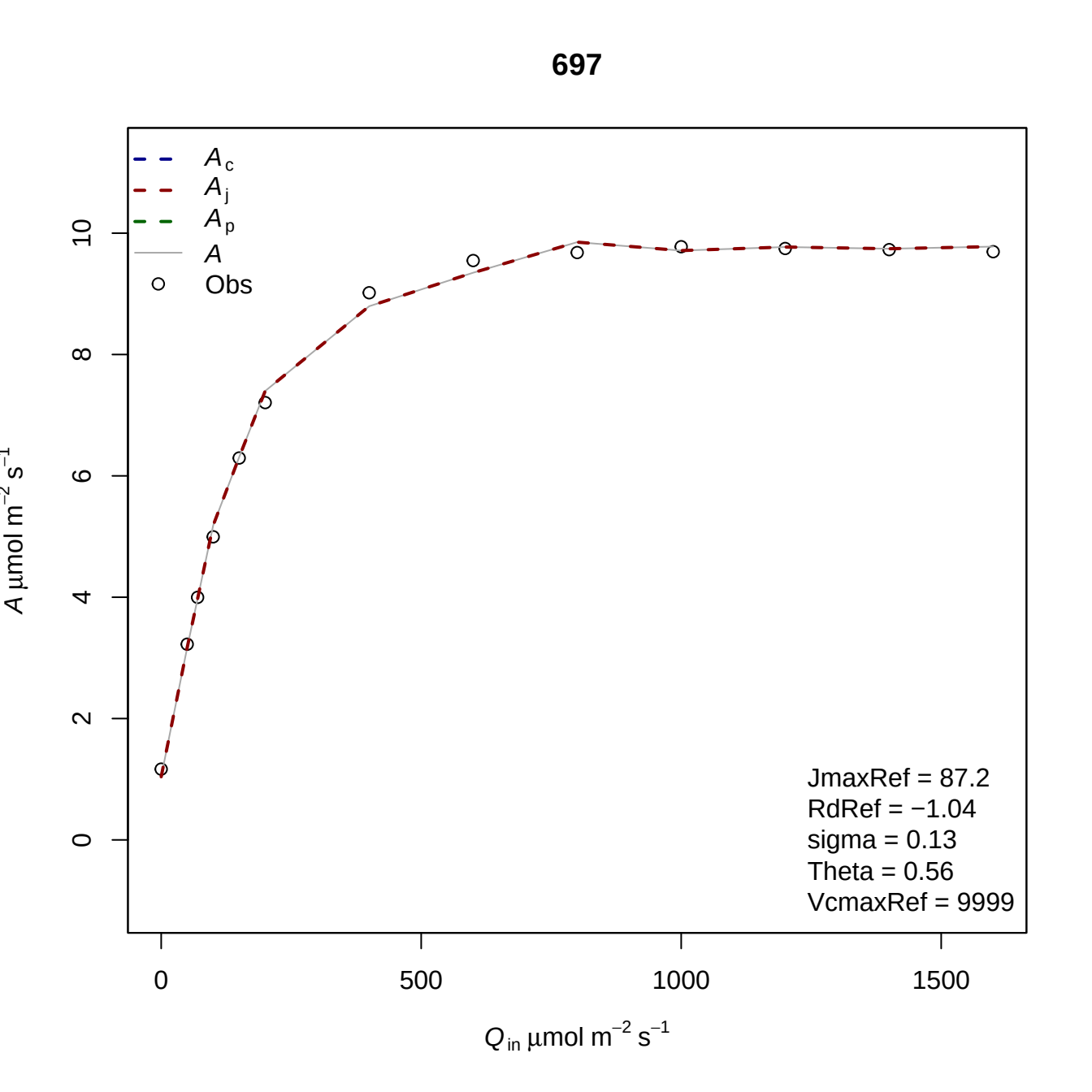

692
A_c
A_j
A_p
A
Obs
JmaxRef = 67.46
RdRef = 1.45
sigma = 0.12
Theta = 0.54
VcmaxRef = 51.89
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

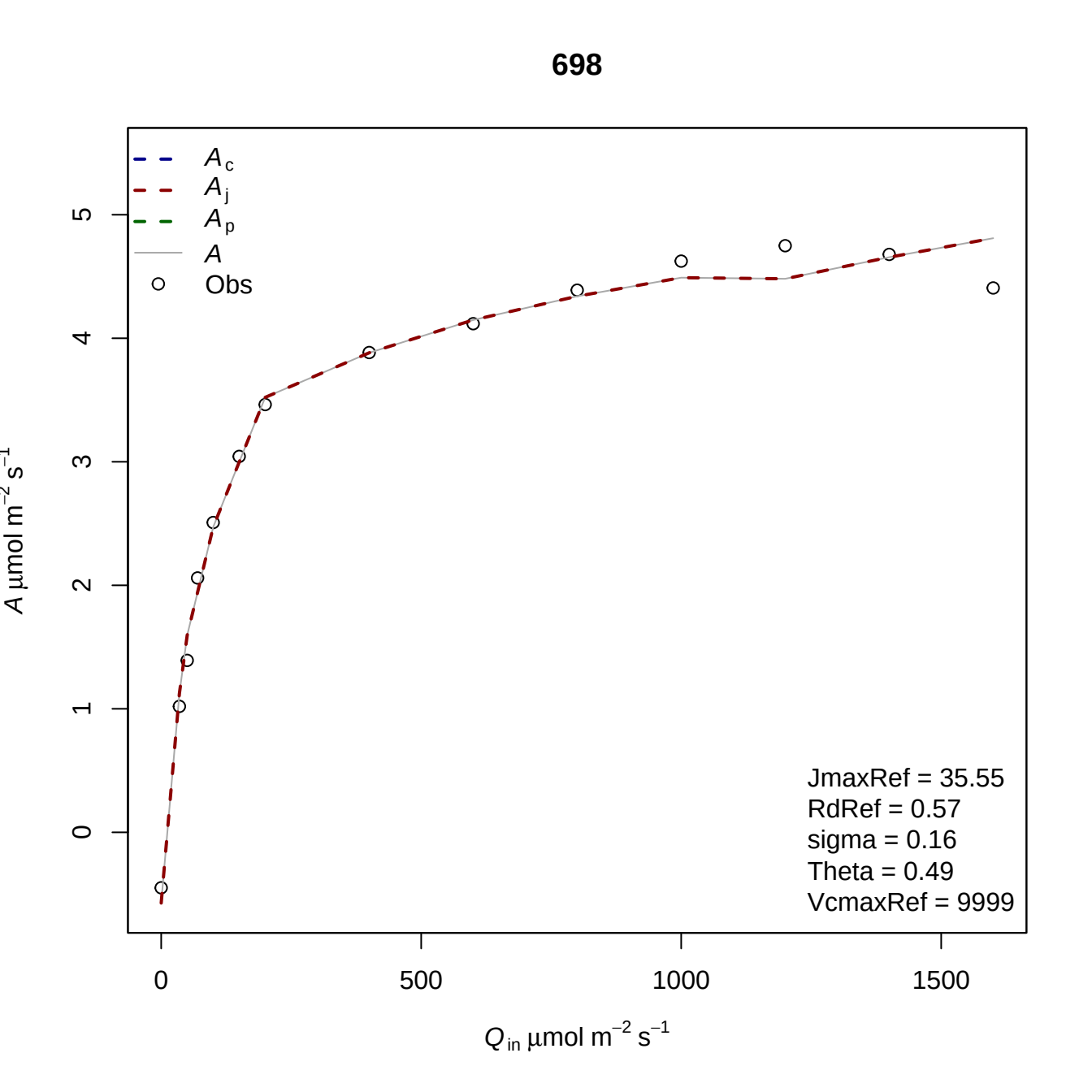

693
A_c
A_j
A_p
A
Obs
JmaxRef = 51.64
RdRef = 0.2
sigma = 0.26
Theta = −2.63
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

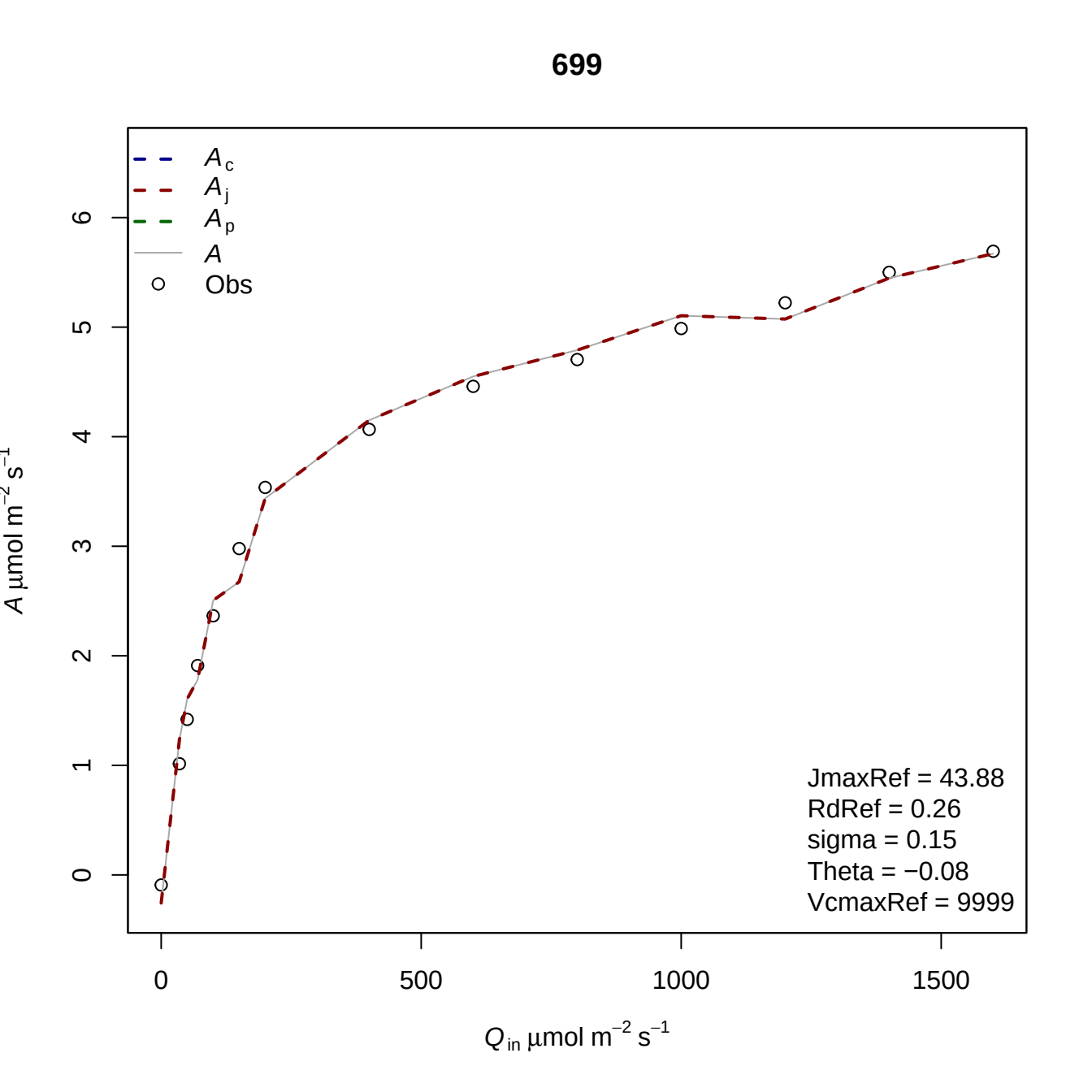

694
A_c
A_j
A_p
A
Obs
JmaxRef = 45.9
RdRef = 0.61
sigma = 0.1
Theta = 0.76
VcmaxRef = 33.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

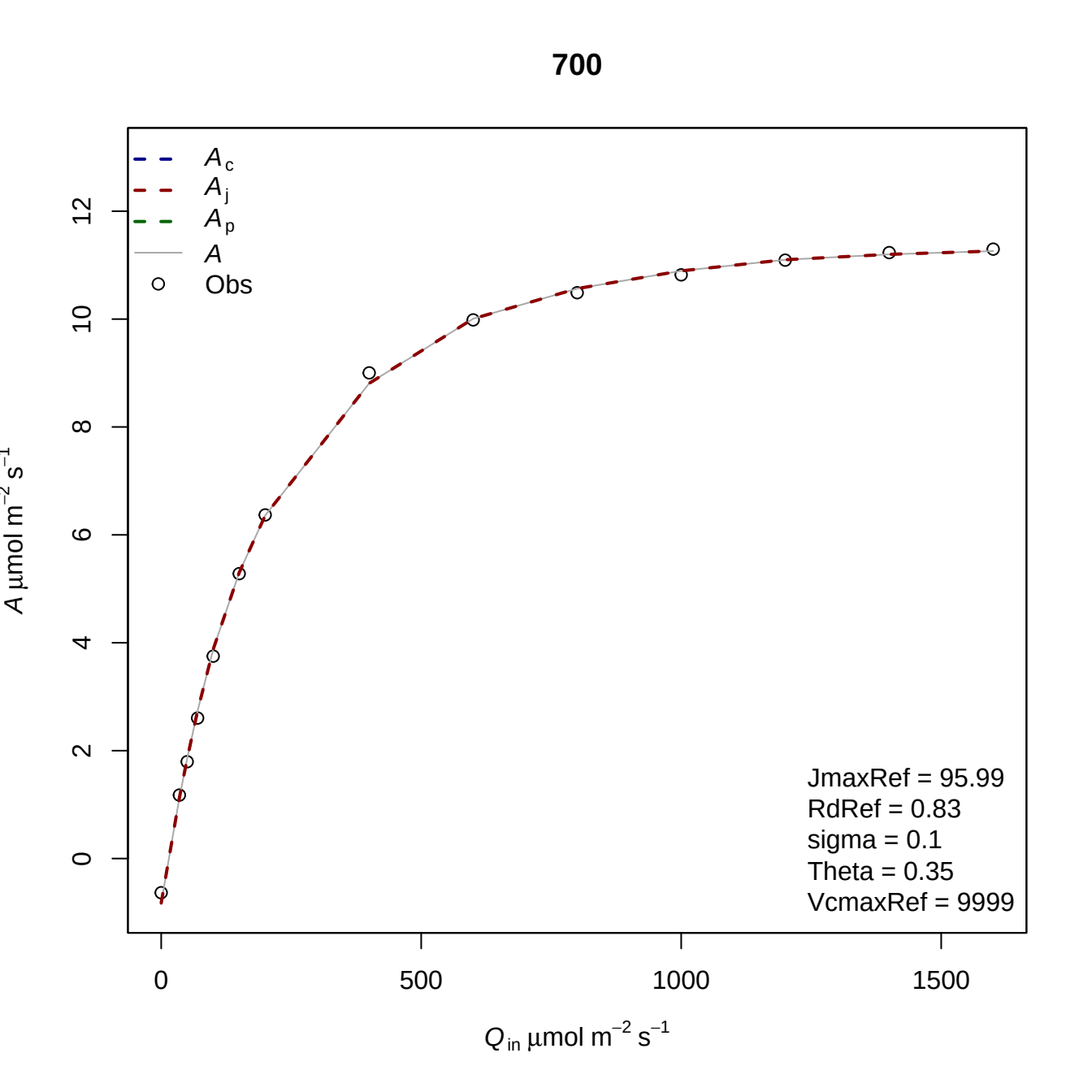

695
A_c
A_j
A_p
A
Obs
JmaxRef = 38.36
RdRef = 1.01
sigma = 0.21
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

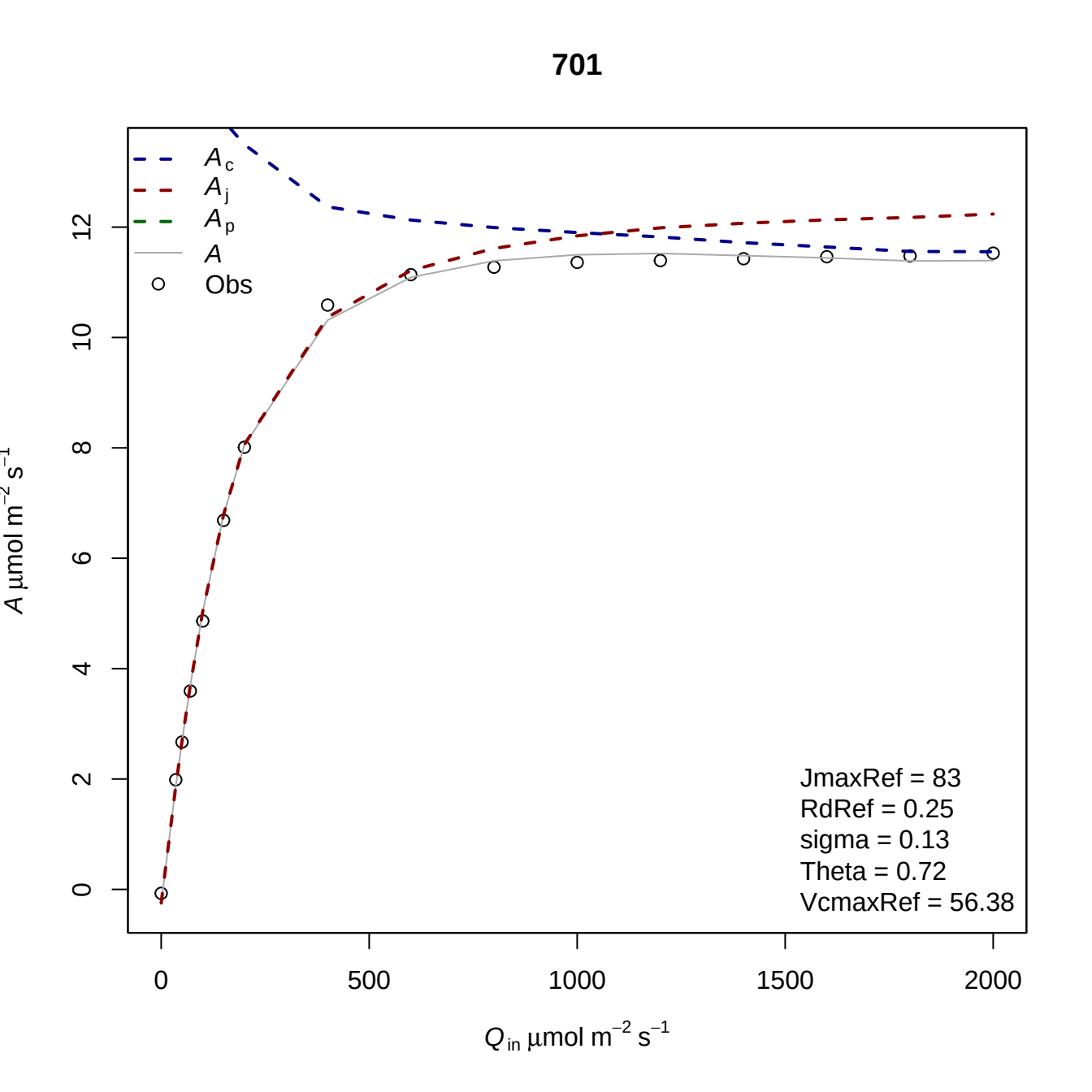

696
A_c
A_j
A_p
A
Obs
JmaxRef = 64.89
RdRef = 0.41
sigma = 0.15
Theta = 0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

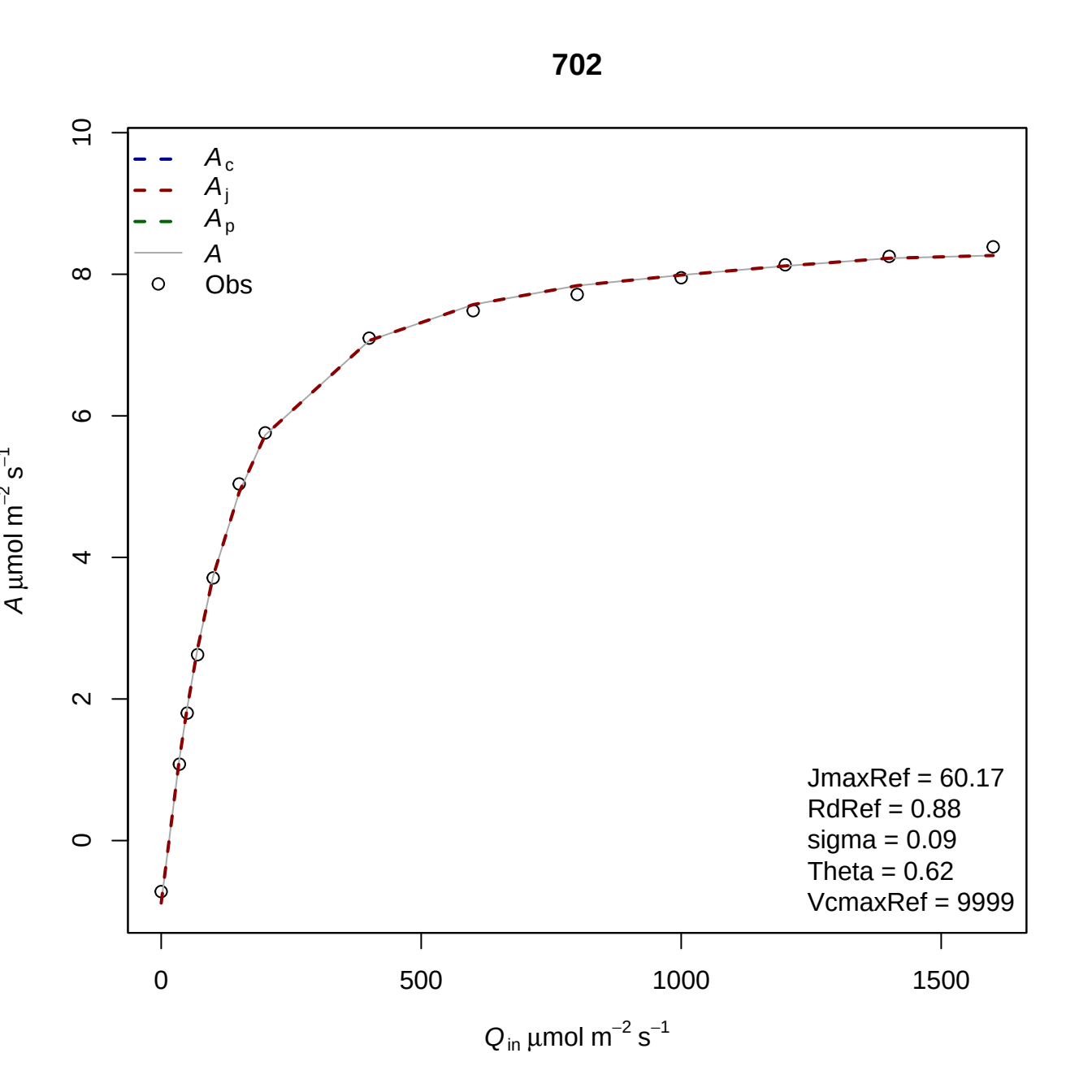

697
A_c
A_j
A_p
A
Obs
JmaxRef = 87.2
RdRef = −1.04
sigma = 0.13
Theta = 0.56
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

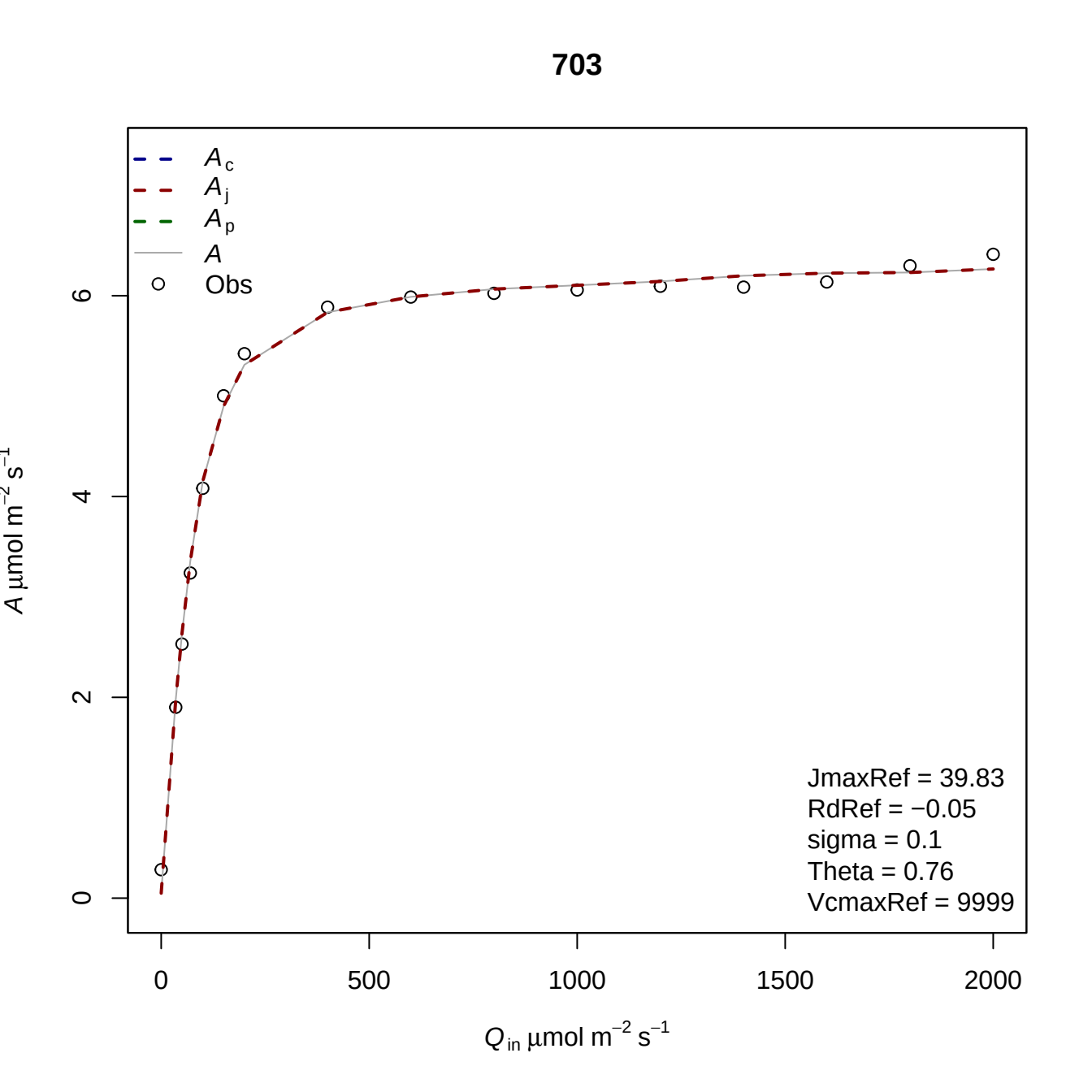

698
A_c
A_j
A_p
A
Obs
JmaxRef = 35.55
RdRef = 0.57
sigma = 0.16
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

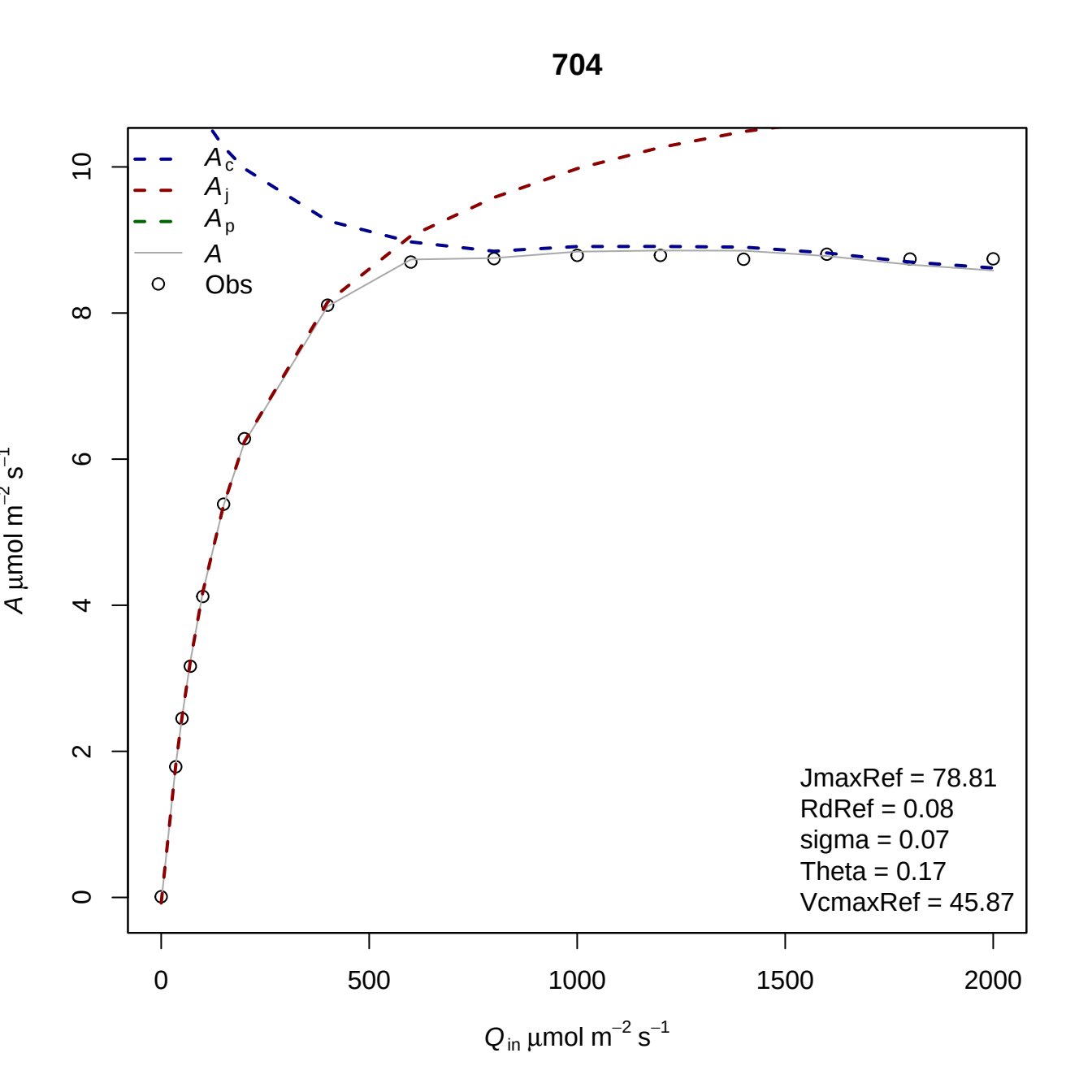

699
A_c
A_j
A_p
A
Obs
JmaxRef = 43.88
RdRef = 0.26
sigma = 0.15
Theta = −0.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

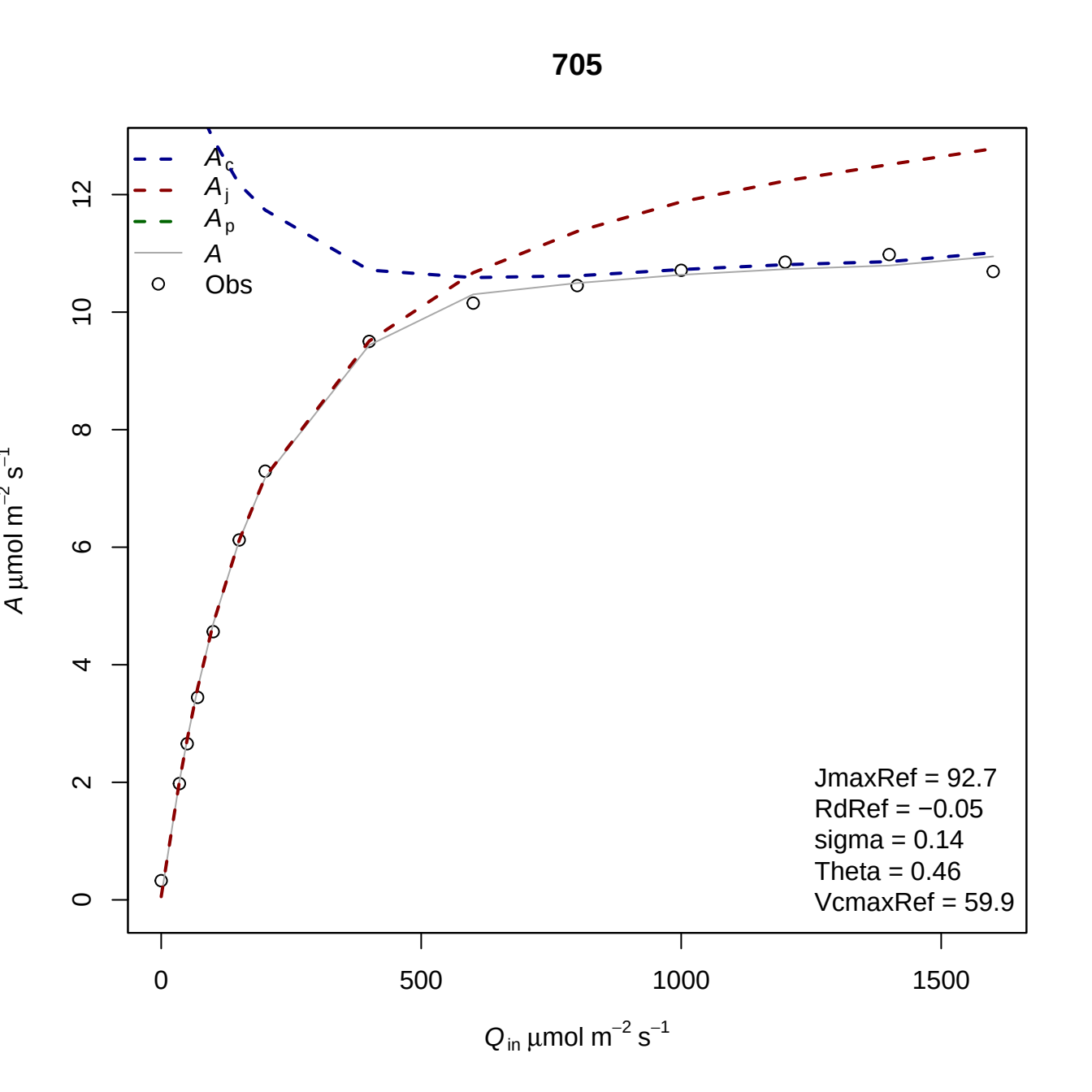

700
A_c
A_j
A_p
A
Obs
JmaxRef = 95.99
RdRef = 0.83
sigma = 0.1
Theta = 0.35
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

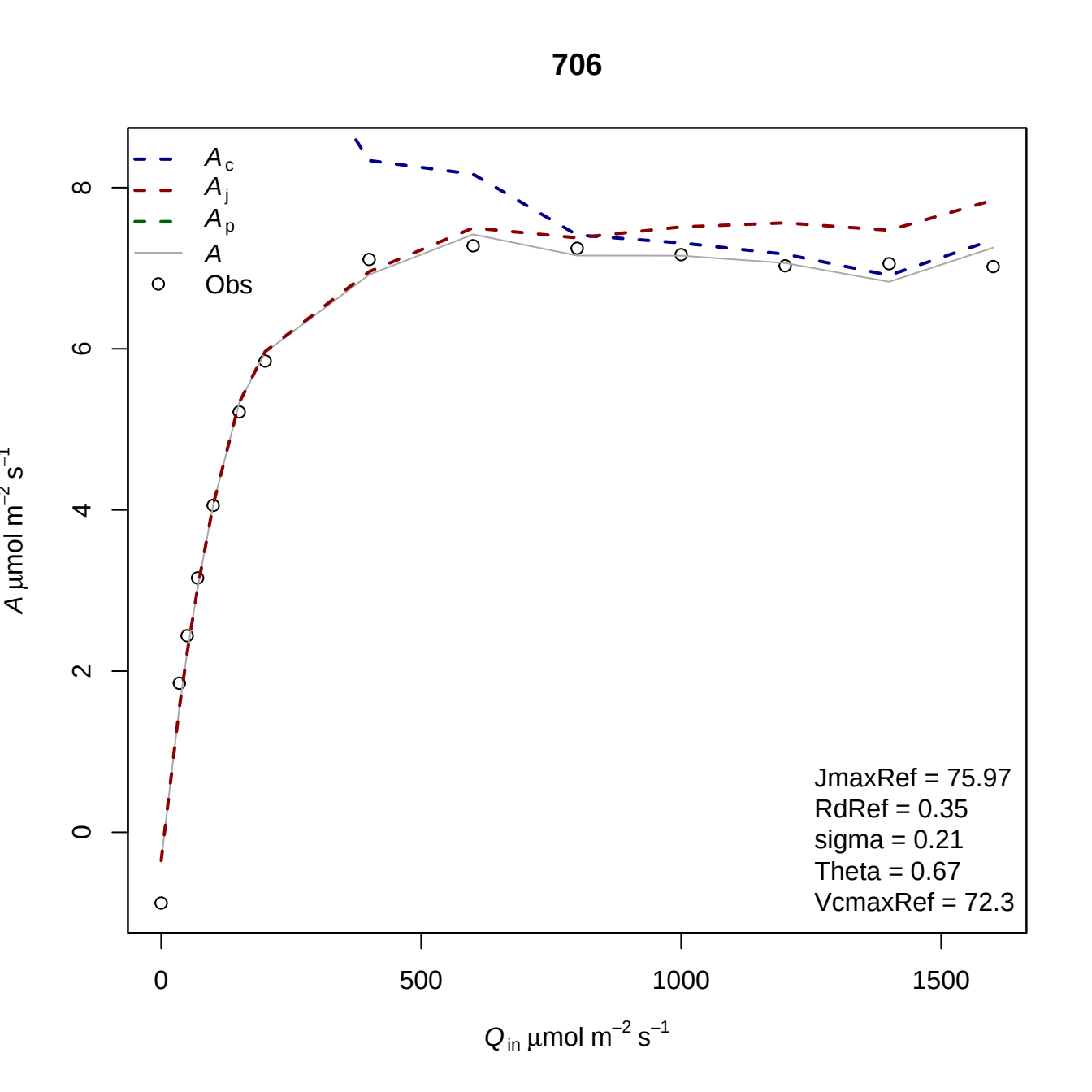

701
A_c
A_j
A_p
A
Obs
JmaxRef = 83
RdRef = 0.25
sigma = 0.13
Theta = 0.72
VcmaxRef = 56.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

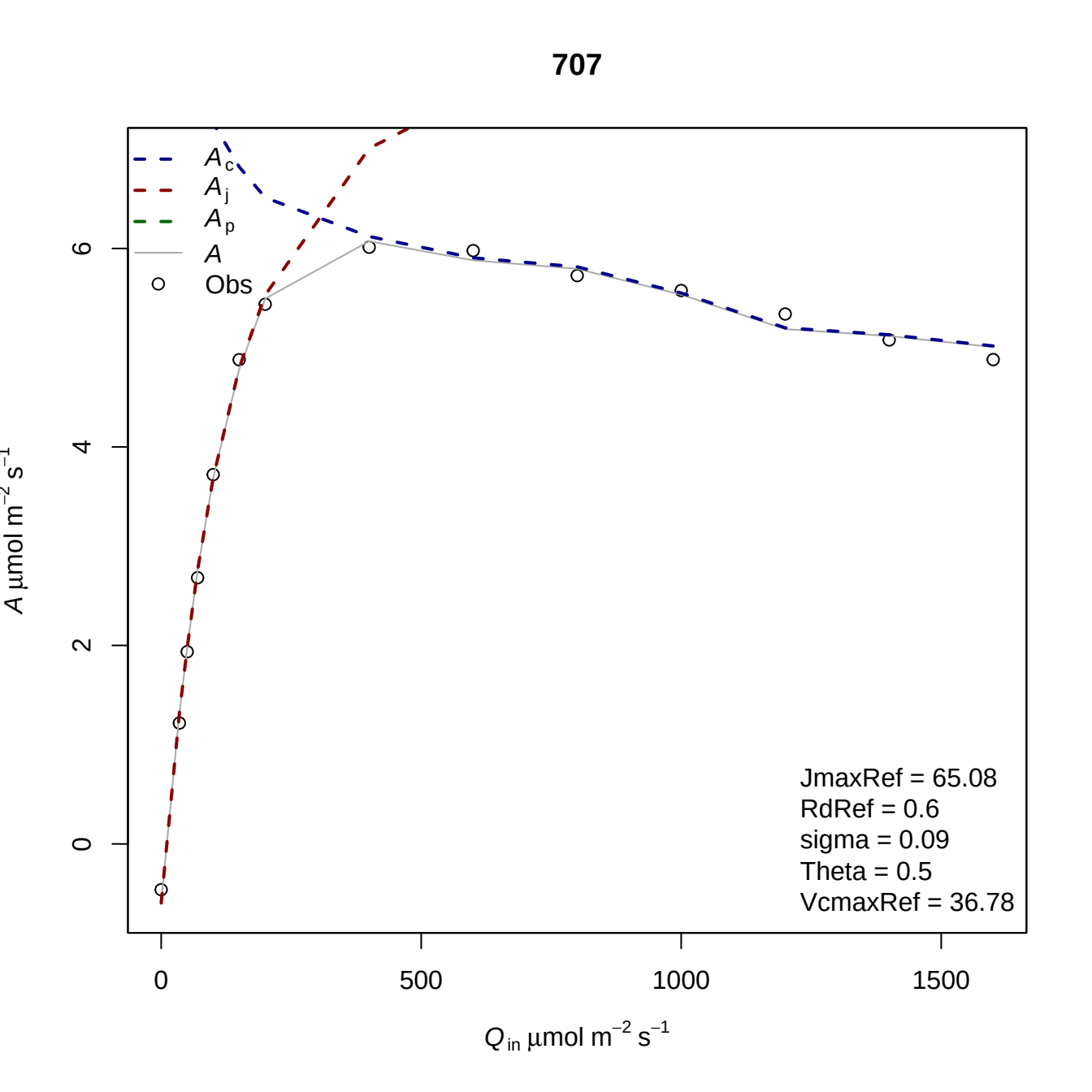

702
A_c
A_j
A_p
A
Obs
JmaxRef = 60.17
RdRef = 0.88
sigma = 0.09
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

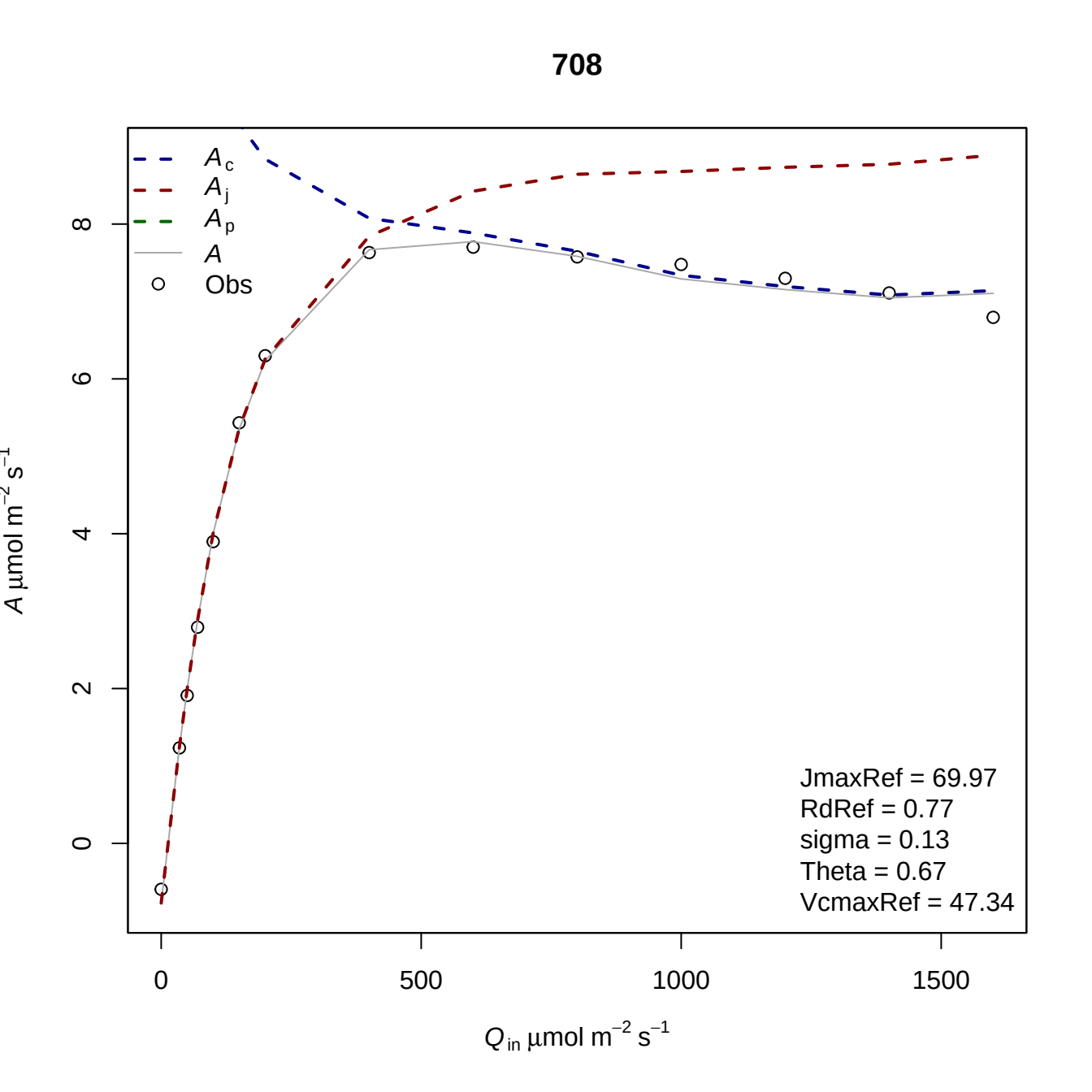

703
A_c
A_j
A_p
A
Obs
JmaxRef = 39.83
RdRef = −0.05
sigma = 0.1
Theta = 0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

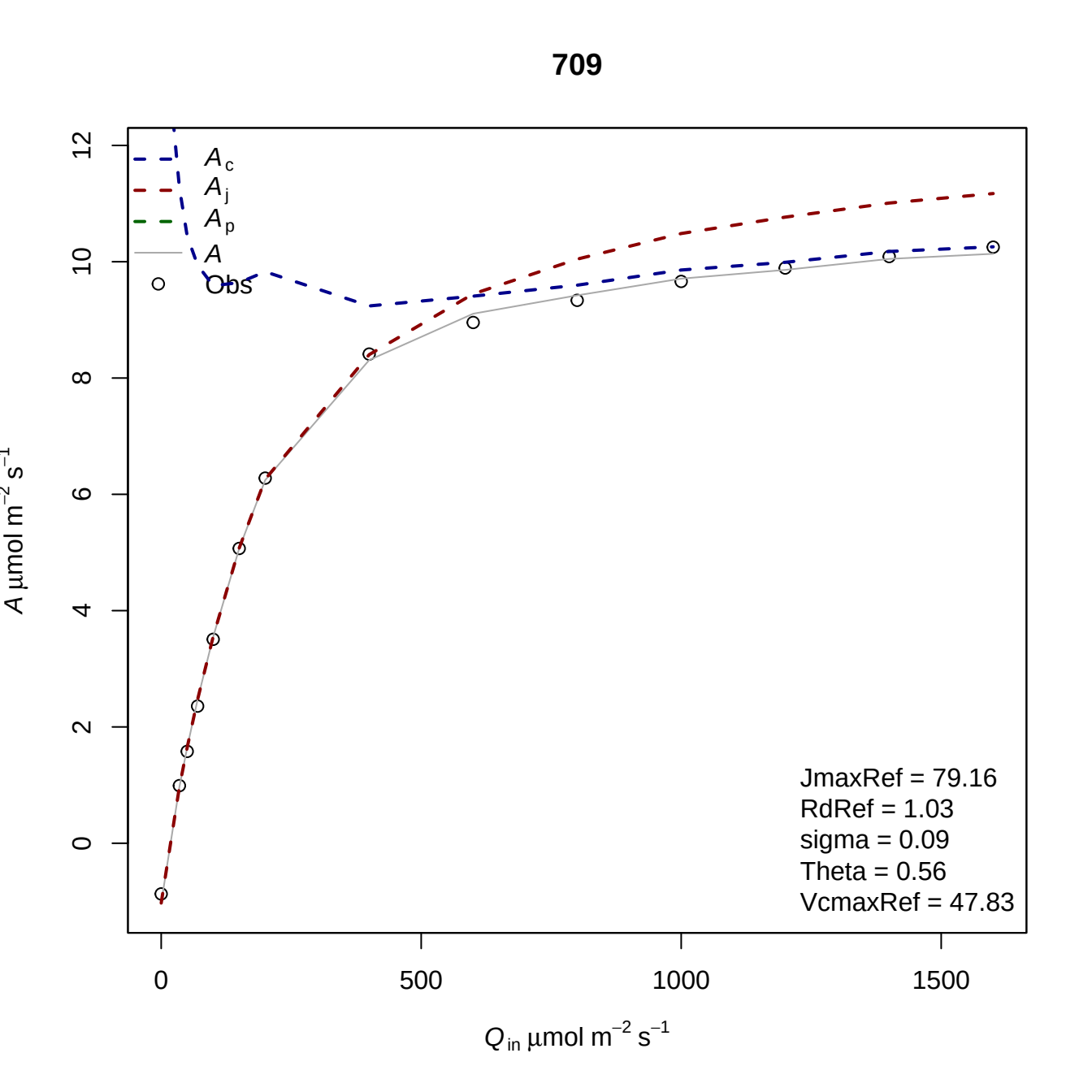

704
A_c
A_j
A_p
A
Obs
JmaxRef = 78.81
RdRef = 0.08
sigma = 0.07
Theta = 0.17
VcmaxRef = 45.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

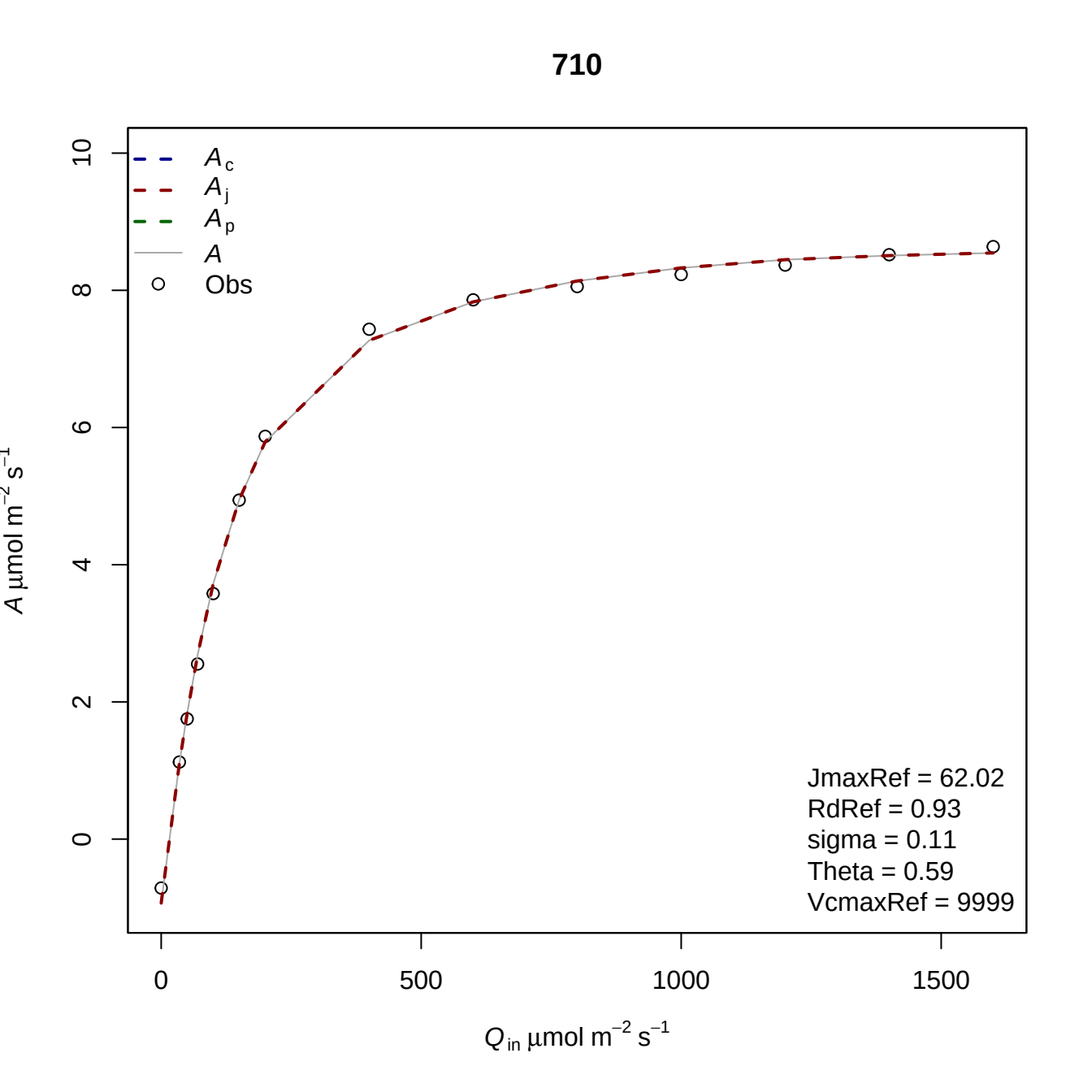

705
A_c
A_j
A_p
A
Obs
JmaxRef = 92.7
RdRef = −0.05
sigma = 0.14
Theta = 0.46
VcmaxRef = 59.9
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

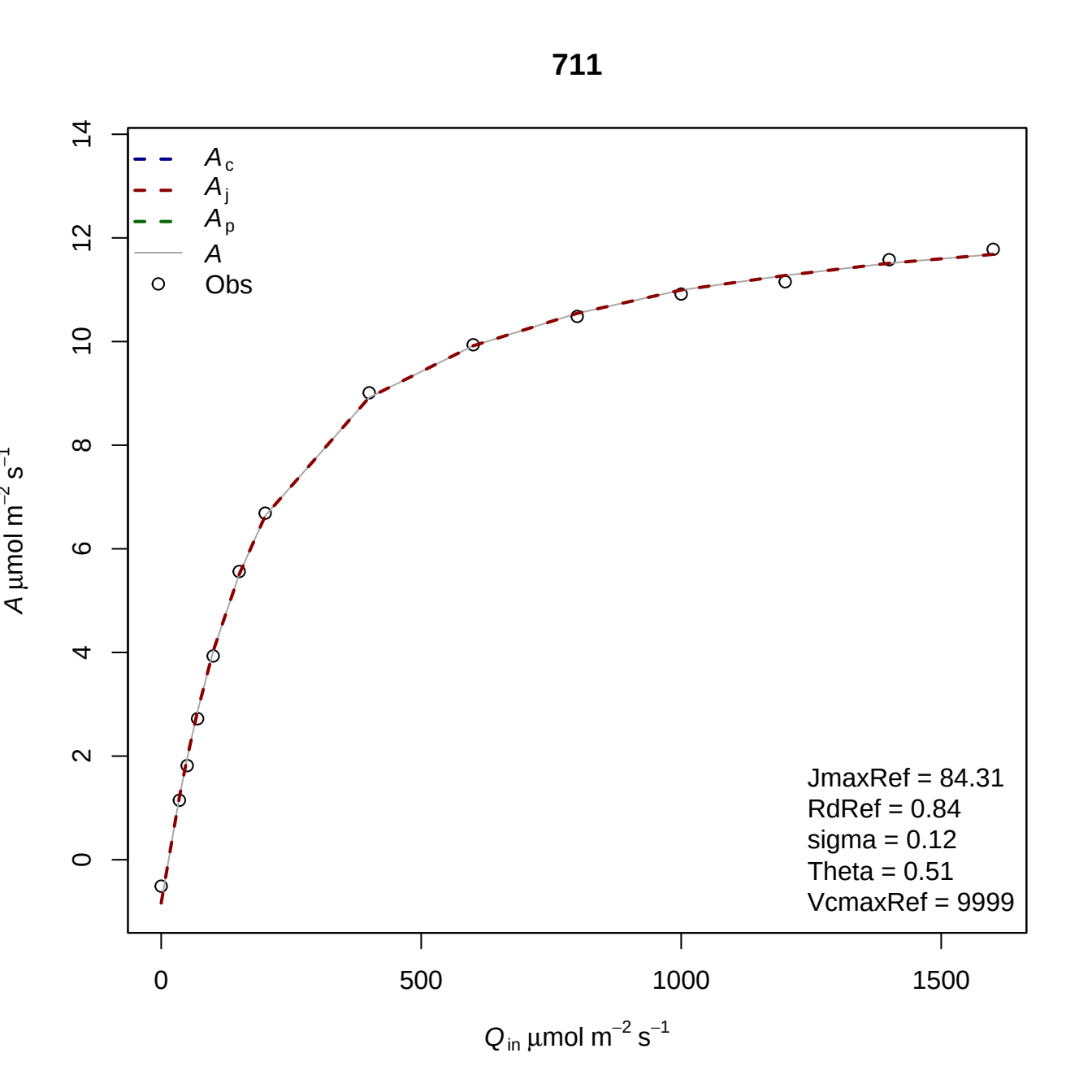

706
A_c
A_j
A_p
A
Obs
JmaxRef = 75.97
RdRef = 0.35
sigma = 0.21
Theta = 0.67
VcmaxRef = 72.3
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

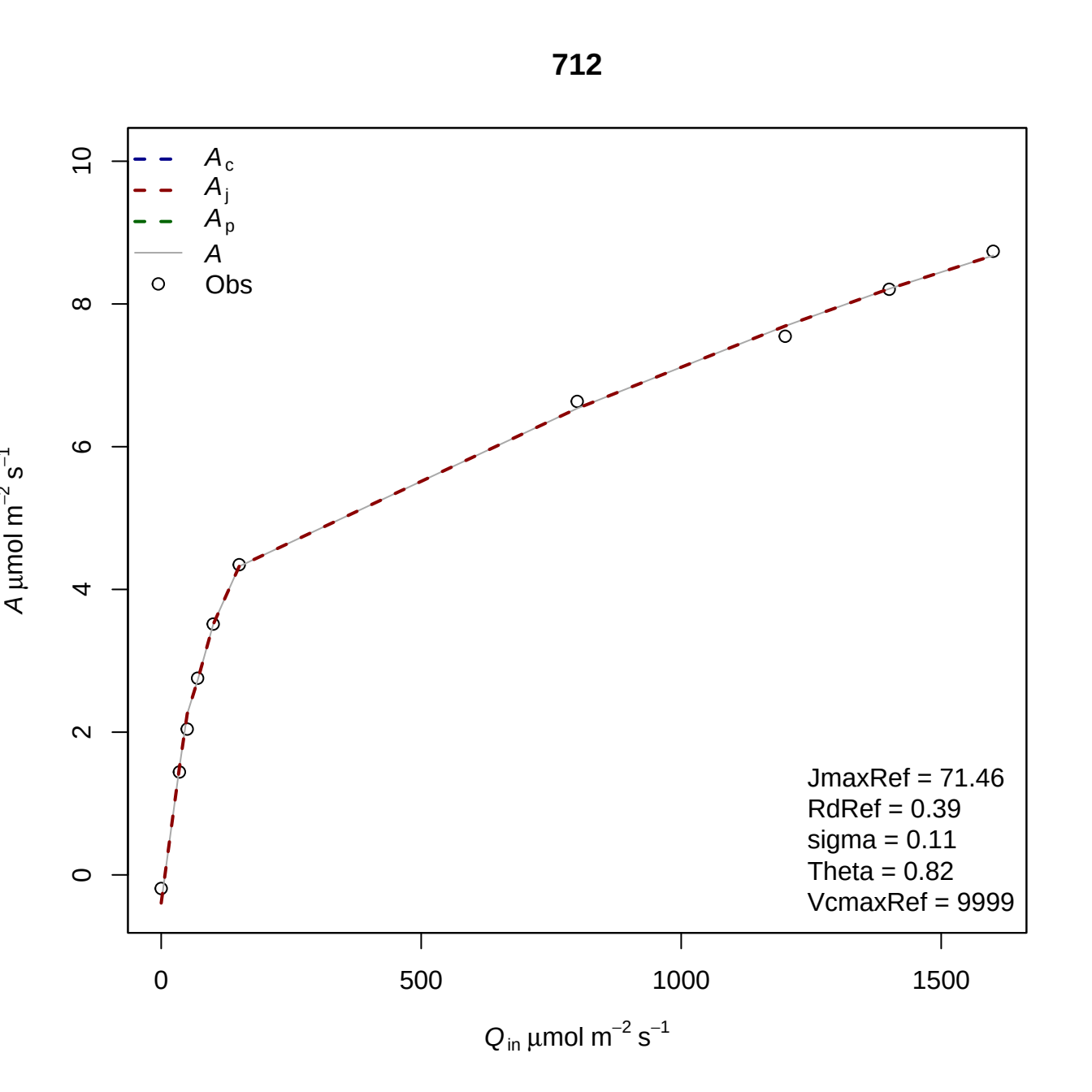

707
A_c
A_j
A_p
A
Obs
JmaxRef = 65.08
RdRef = 0.6
sigma = 0.09
Theta = 0.5
VcmaxRef = 36.78
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

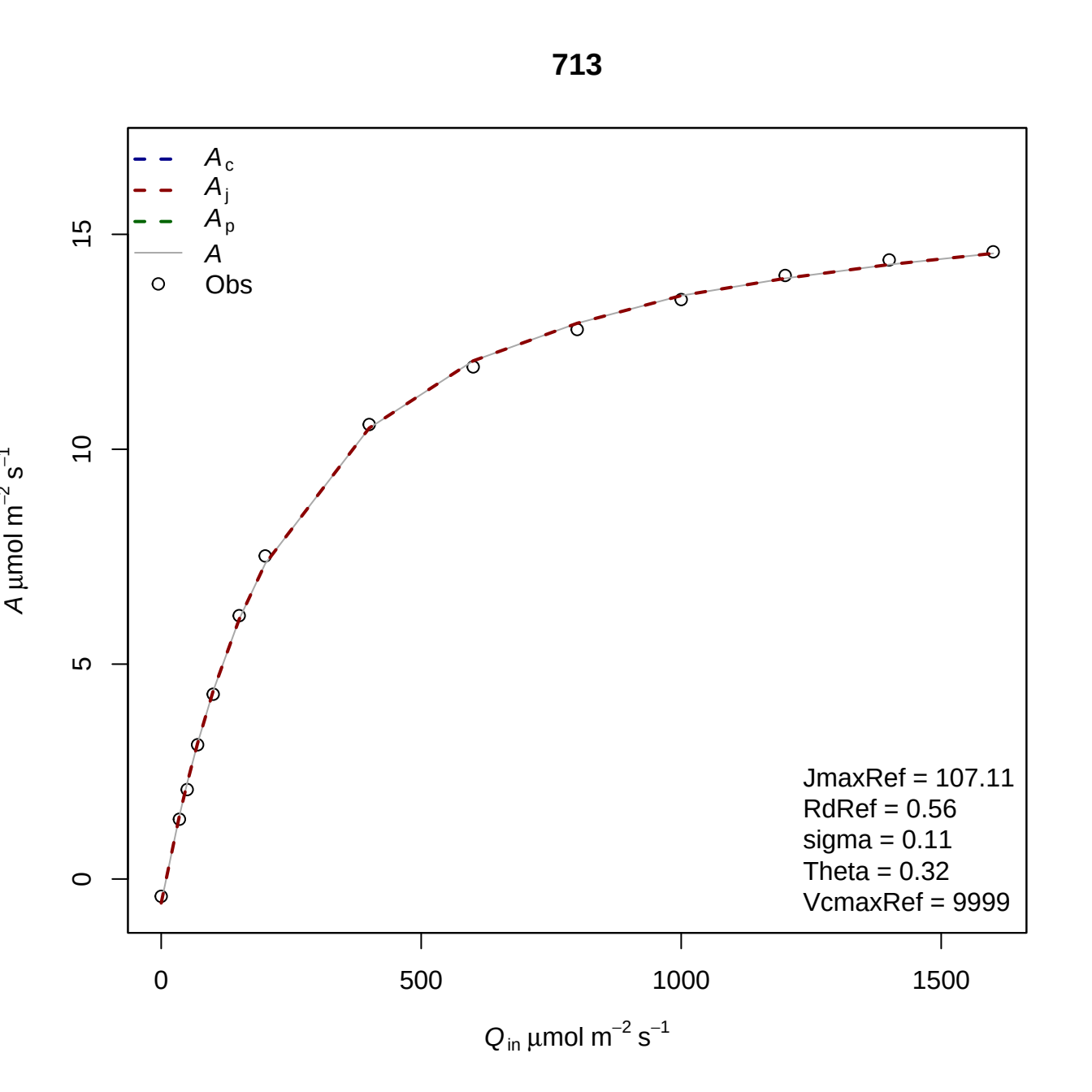

708
A_c
A_j
A_p
A
Obs
JmaxRef = 69.97
RdRef = 0.77
sigma = 0.13
Theta = 0.67
VcmaxRef = 47.34
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

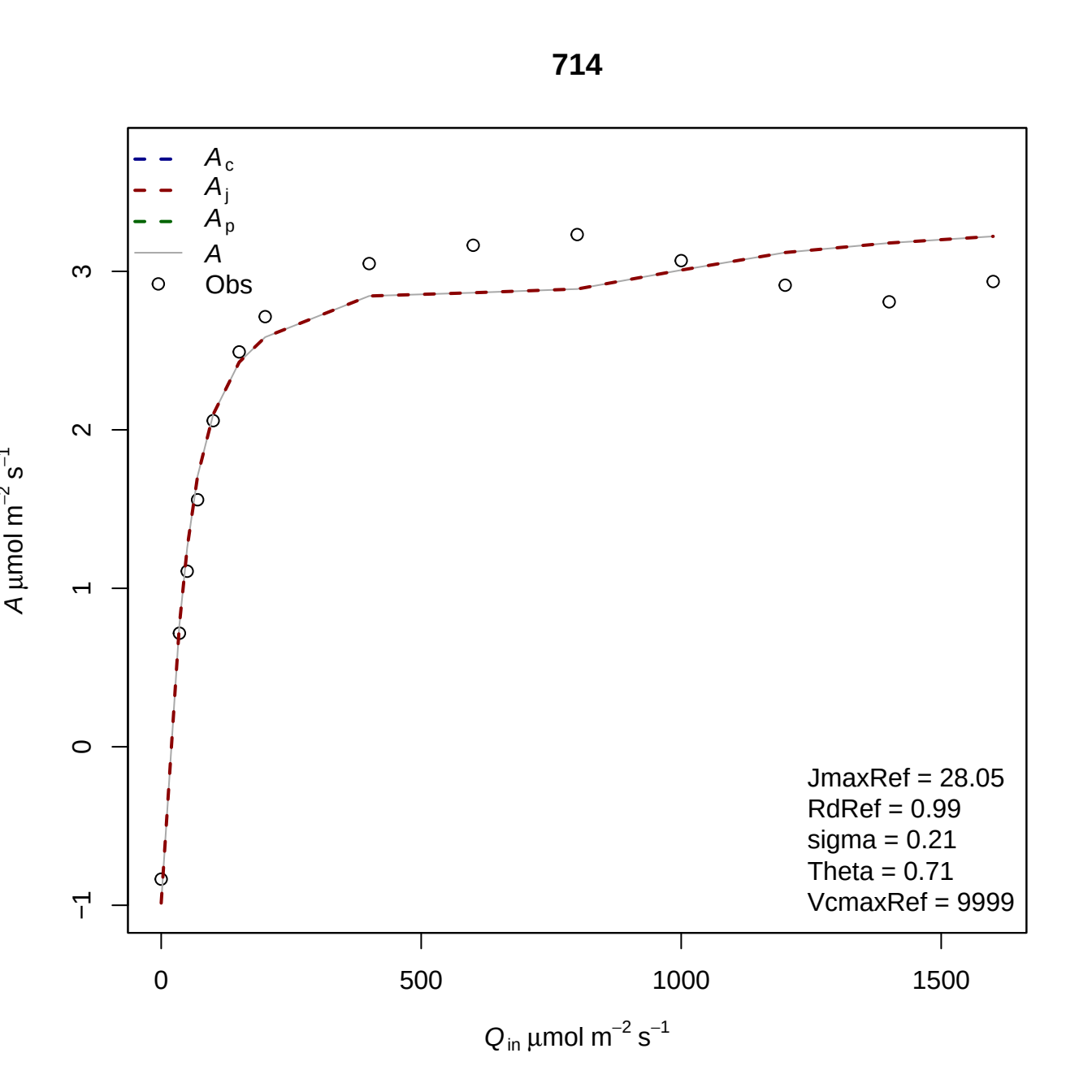

709
A_c
A_j
A_p
A
Obs
JmaxRef = 79.16
RdRef = 1.03
sigma = 0.09
Theta = 0.56
VcmaxRef = 47.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

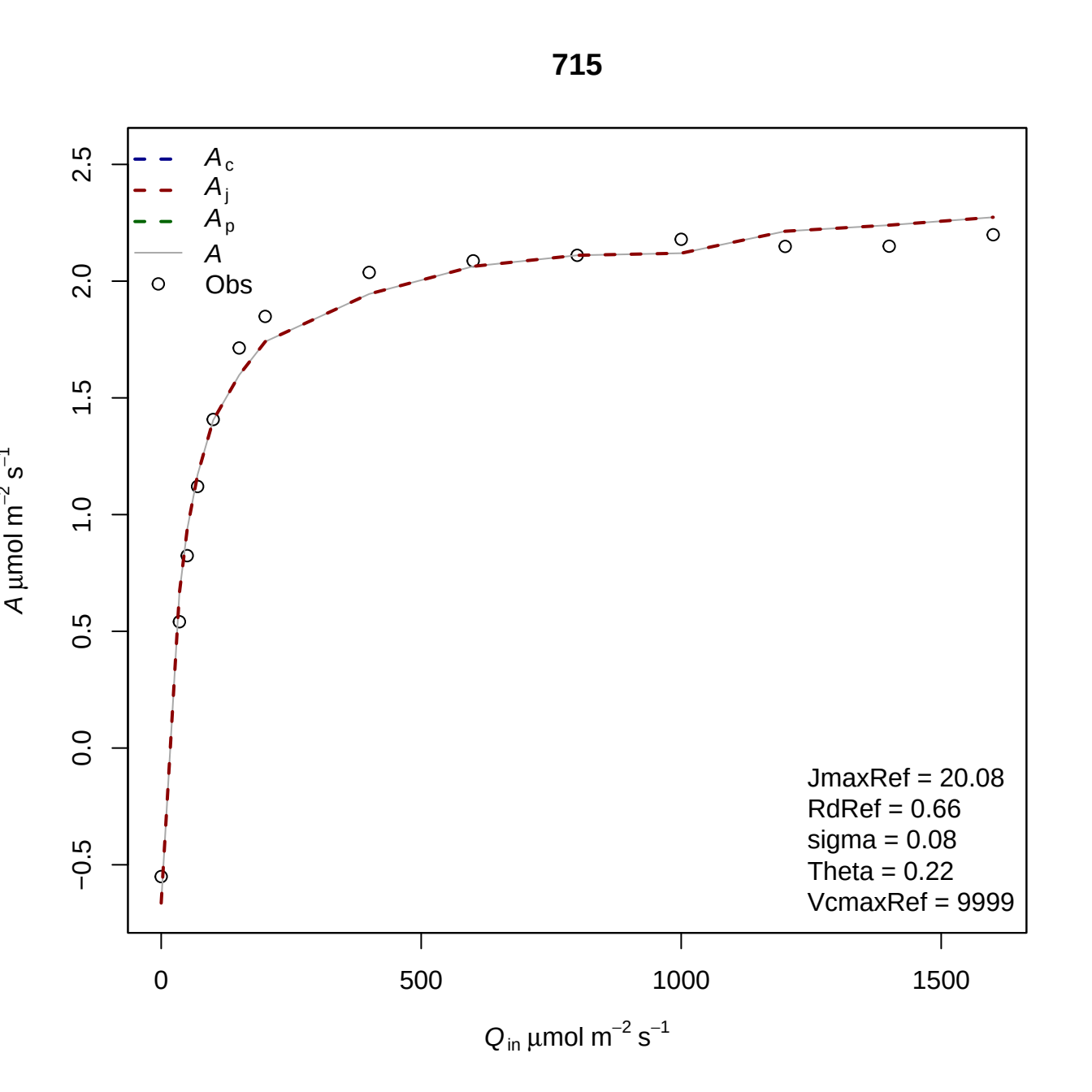

710
A_c
A_j
A_p
A
Obs
JmaxRef = 62.02
RdRef = 0.93
sigma = 0.11
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

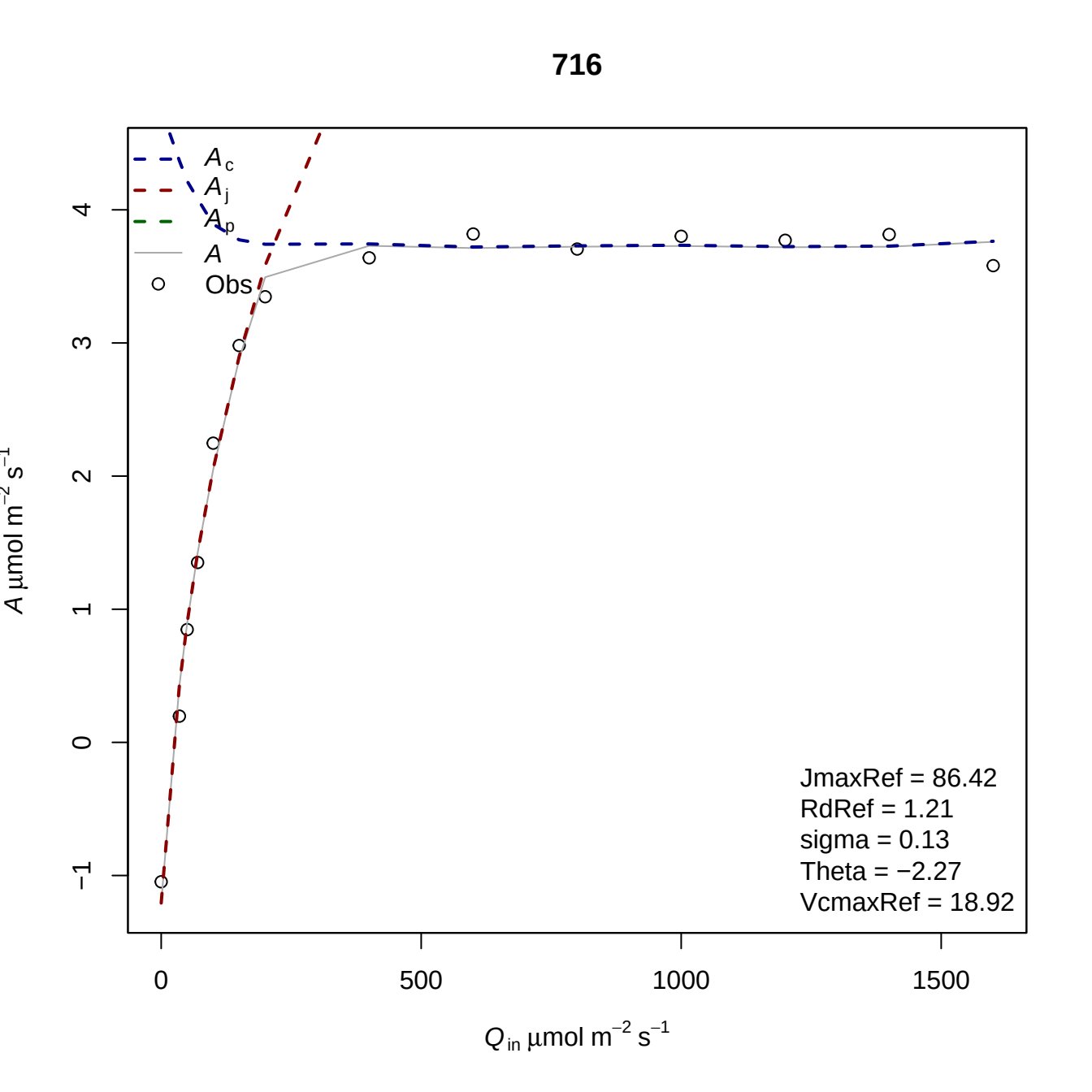

711
A_c
A_j
A_p
A
Obs
JmaxRef = 84.31
RdRef = 0.84
sigma = 0.12
Theta = 0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

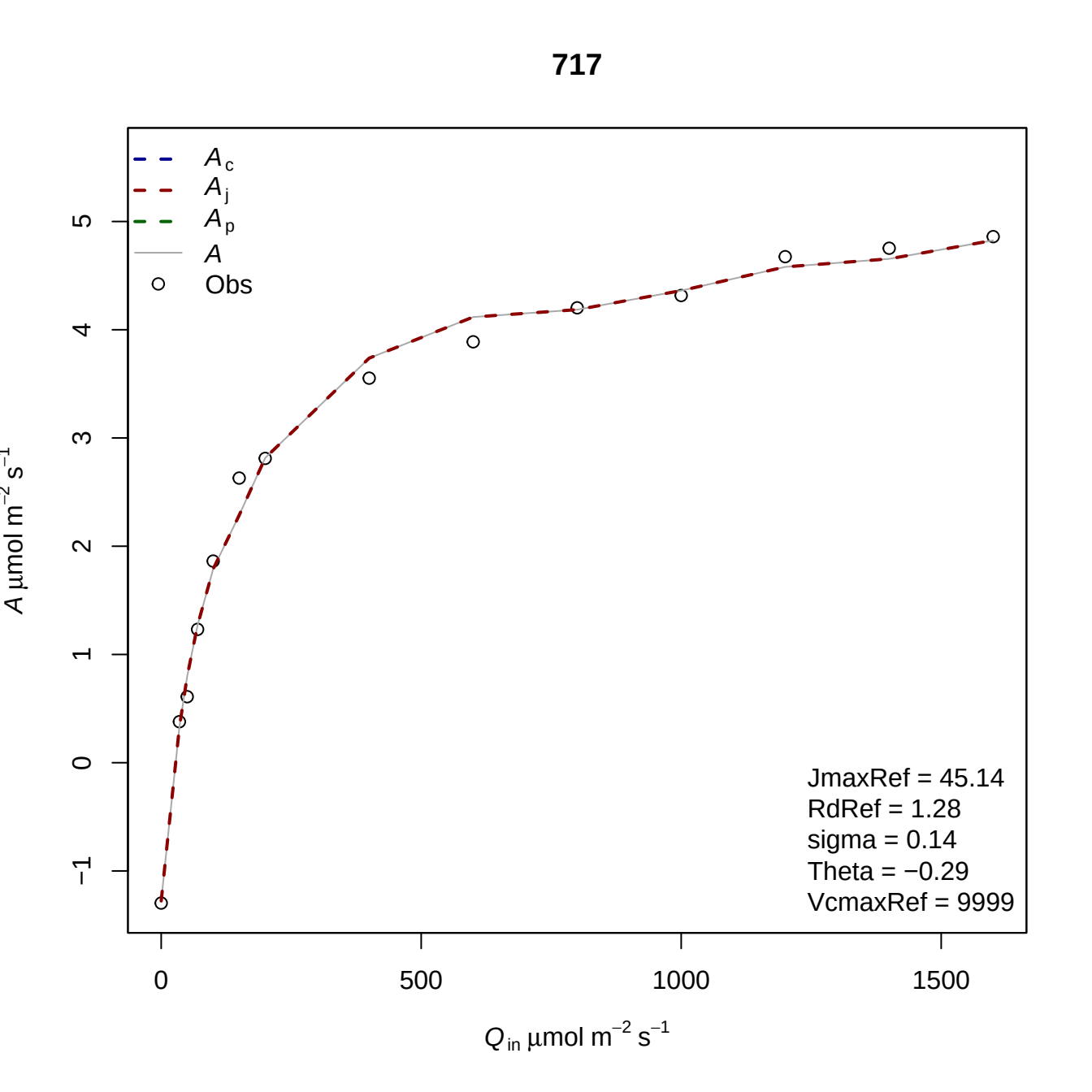

712
A_c
A_j
A_p
A
Obs
JmaxRef = 71.46
RdRef = 0.39
sigma = 0.11
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

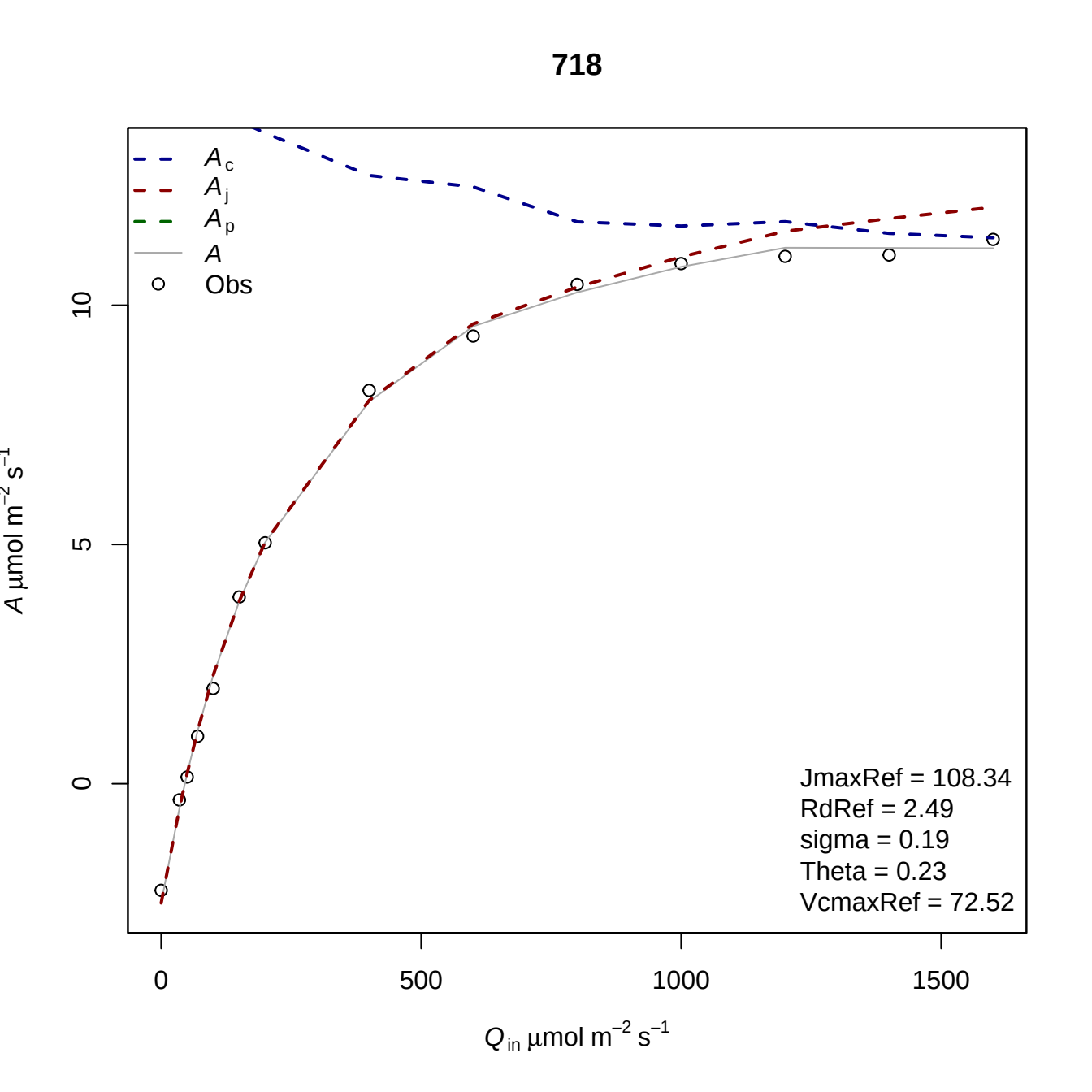

713
A_c
A_j
A_p
A
Obs
JmaxRef = 107.11
RdRef = 0.56
sigma = 0.11
Theta = 0.32
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

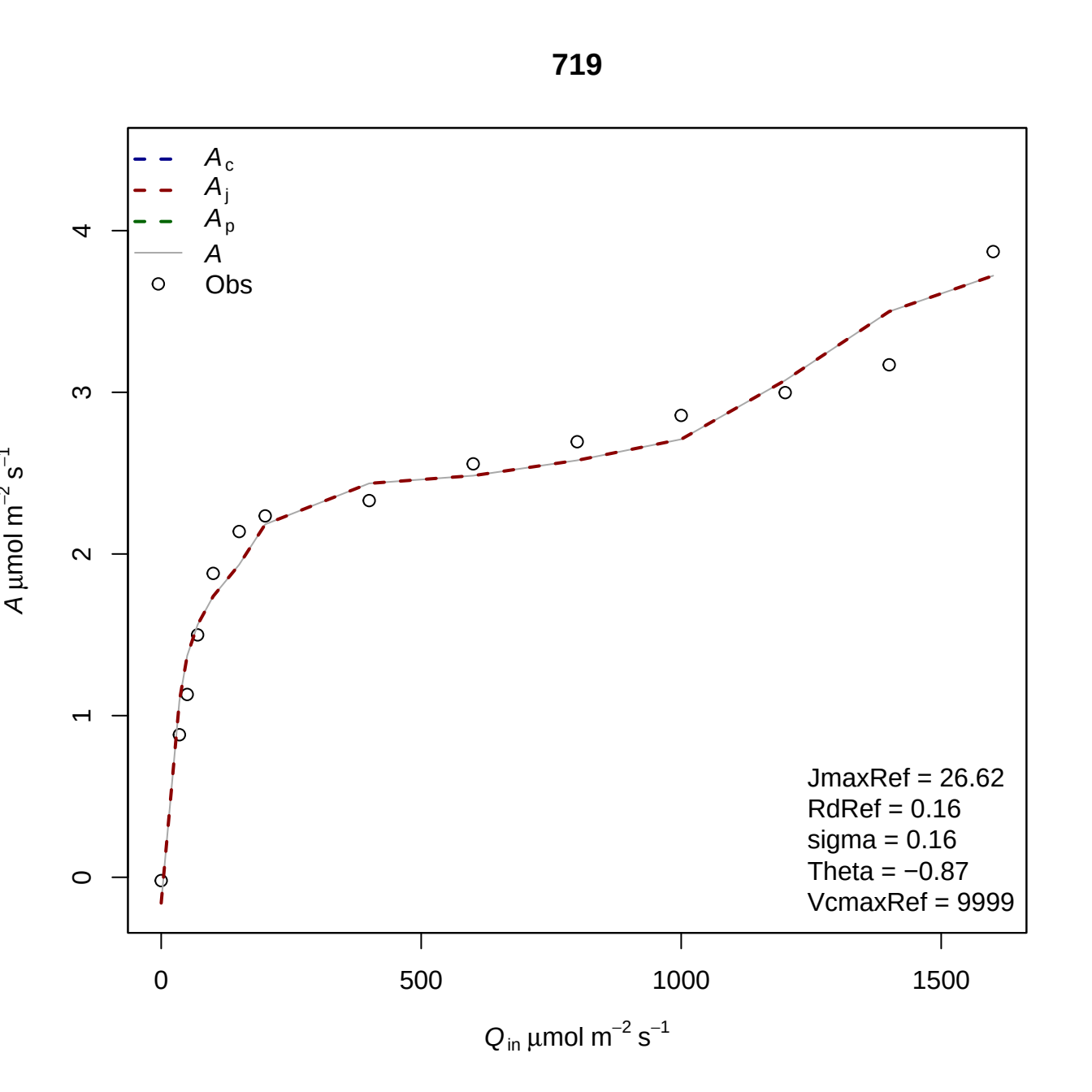

714
A_c
A_j
A_p
A
Obs
JmaxRef = 28.05
RdRef = 0.99
sigma = 0.21
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

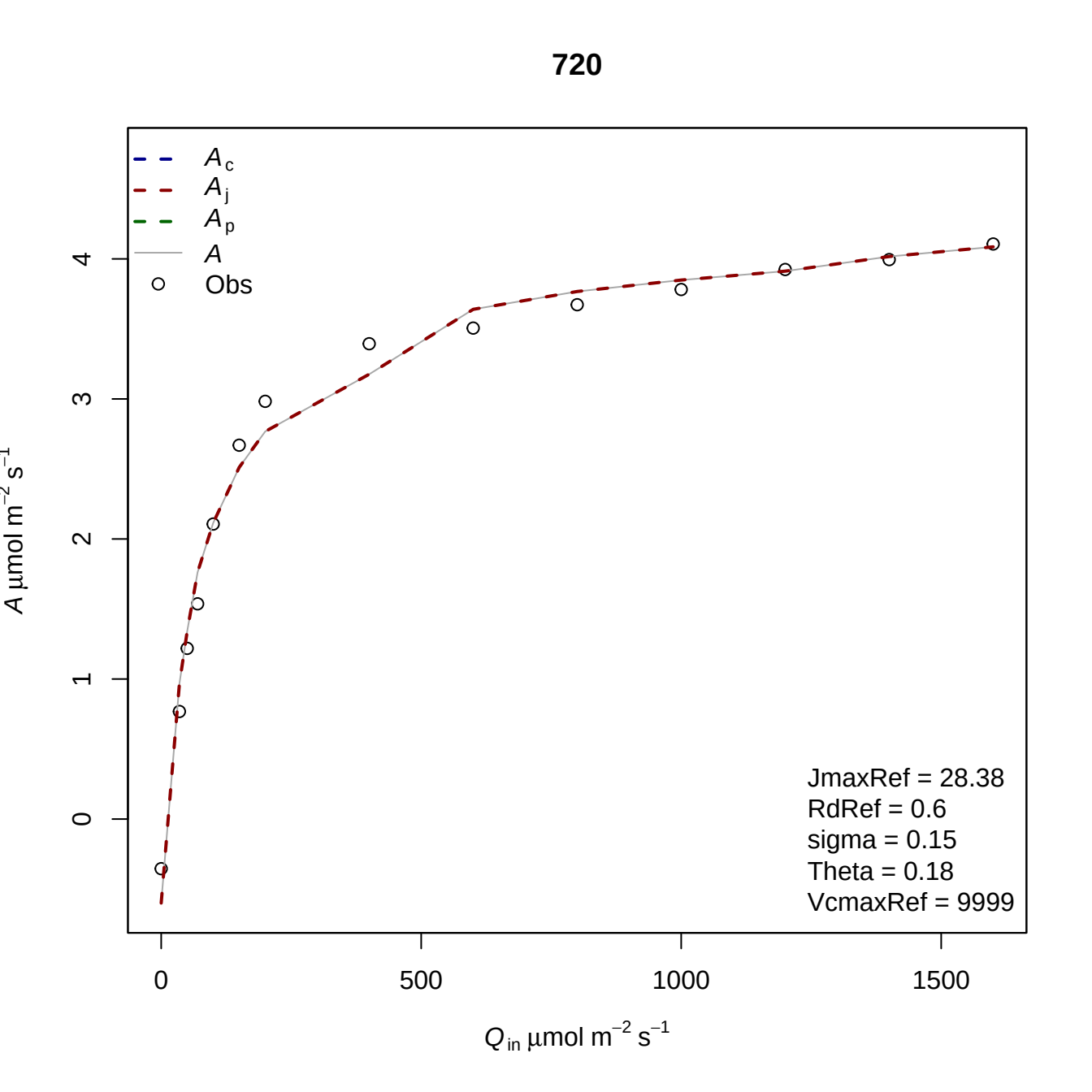

715
A_c
A_j
A_p
A
Obs
JmaxRef = 20.08
RdRef = 0.66
sigma = 0.08
Theta = 0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

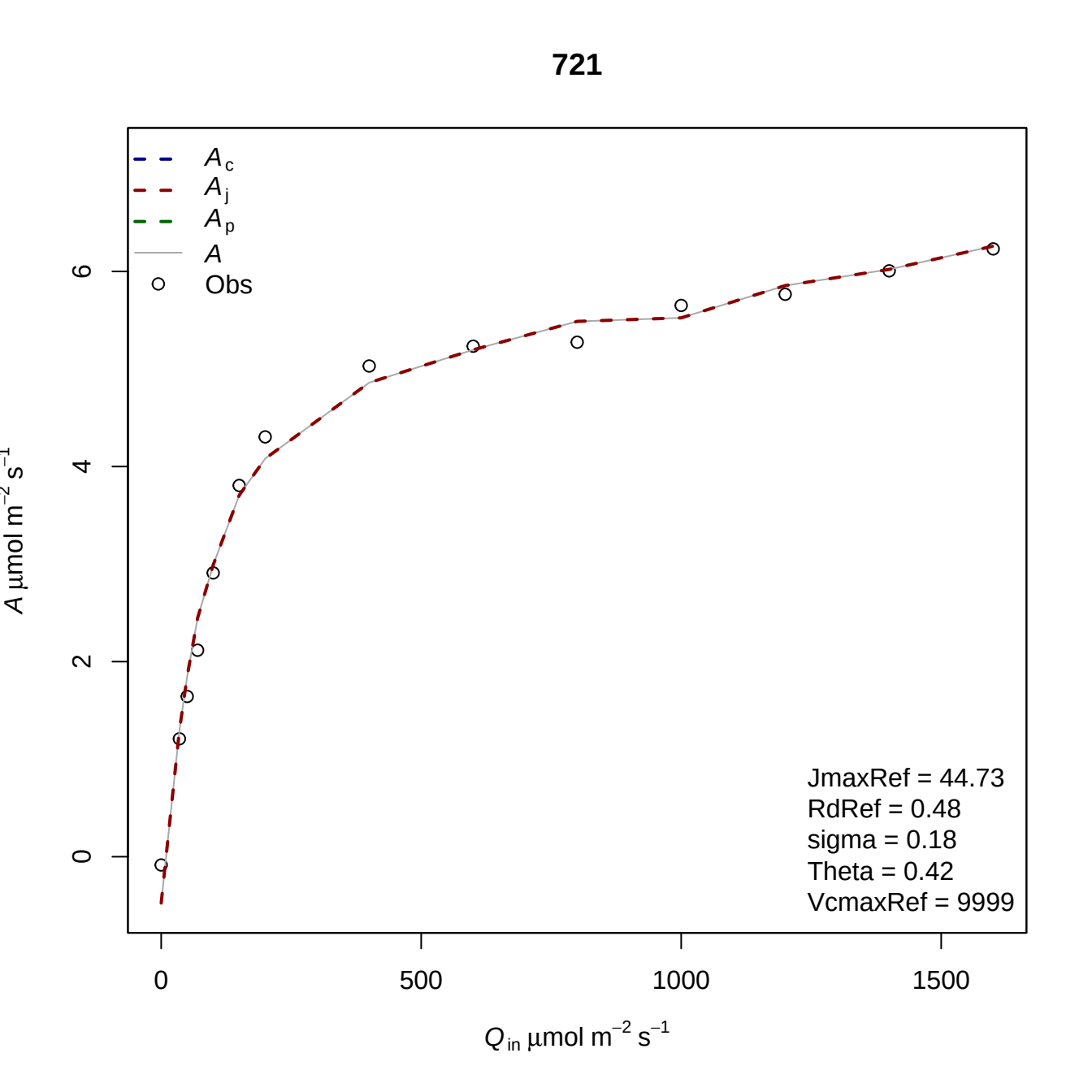

716
A_c
A_j
A_p
A
Obs
JmaxRef = 86.42
RdRef = 1.21
sigma = 0.13
Theta = −2.27
VcmaxRef = 18.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

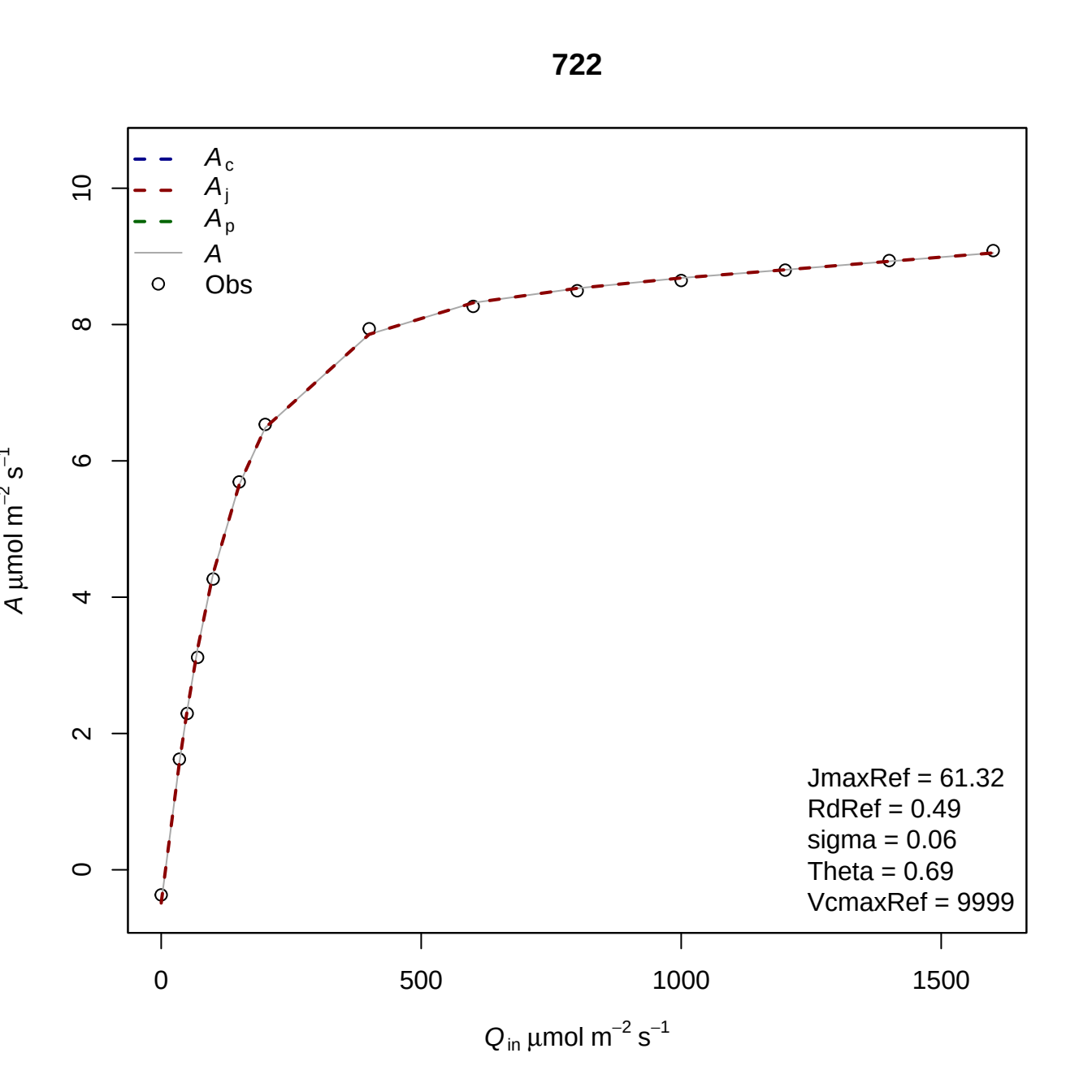

717
A_c
A_j
A_p
A
Obs
JmaxRef = 45.14
RdRef = 1.28
sigma = 0.14
Theta = −0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

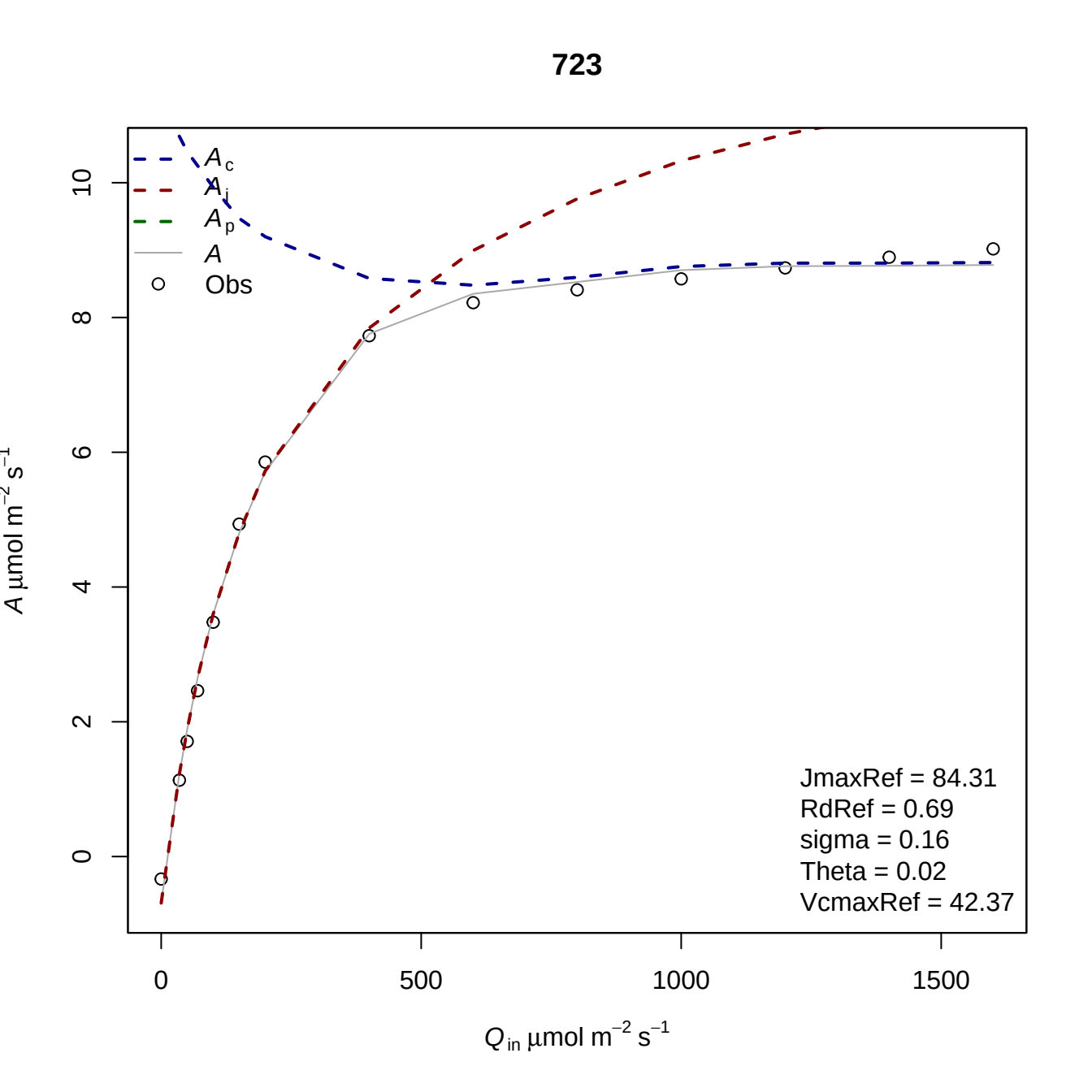

718
A_c
A_j
A_p
A
Obs
JmaxRef = 108.34
RdRef = 2.49
sigma = 0.19
Theta = 0.23
VcmaxRef = 72.52
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

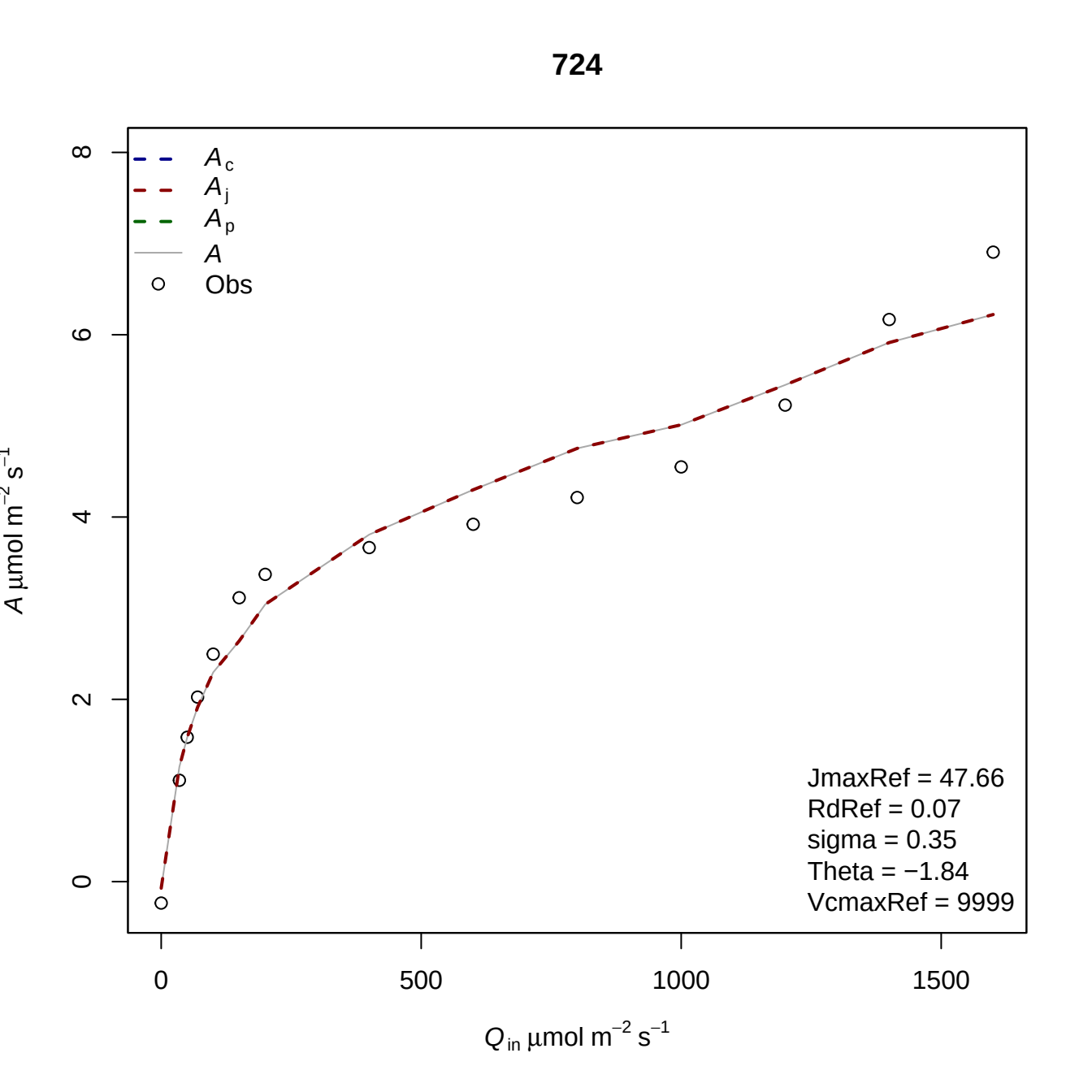

719
A_c
A_j
A_p
A
Obs
JmaxRef = 26.62
RdRef = 0.16
sigma = 0.16
Theta = −0.87
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

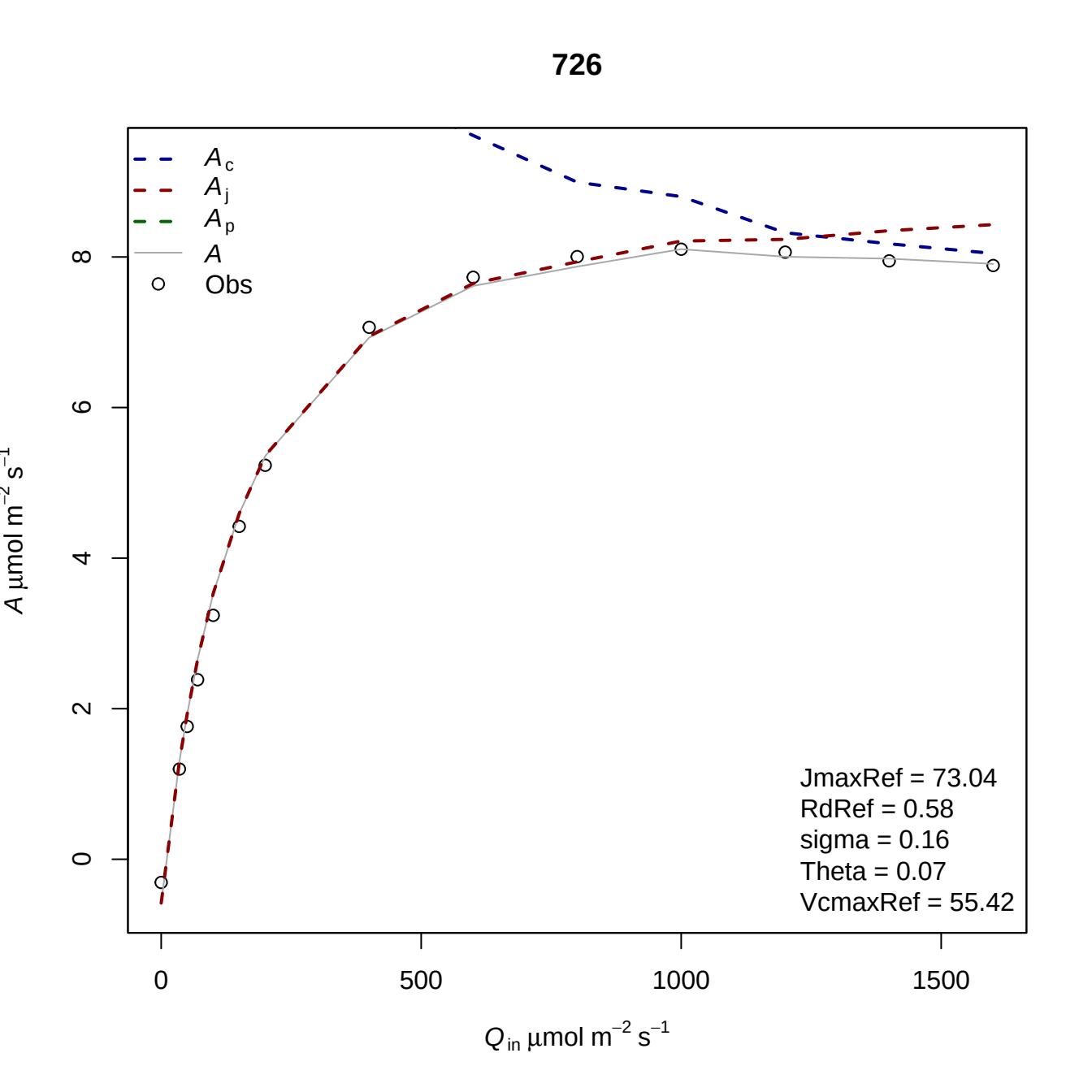

720
A_c
A_j
A_p
A
Obs
JmaxRef = 28.38
RdRef = 0.6
sigma = 0.15
Theta = 0.18
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

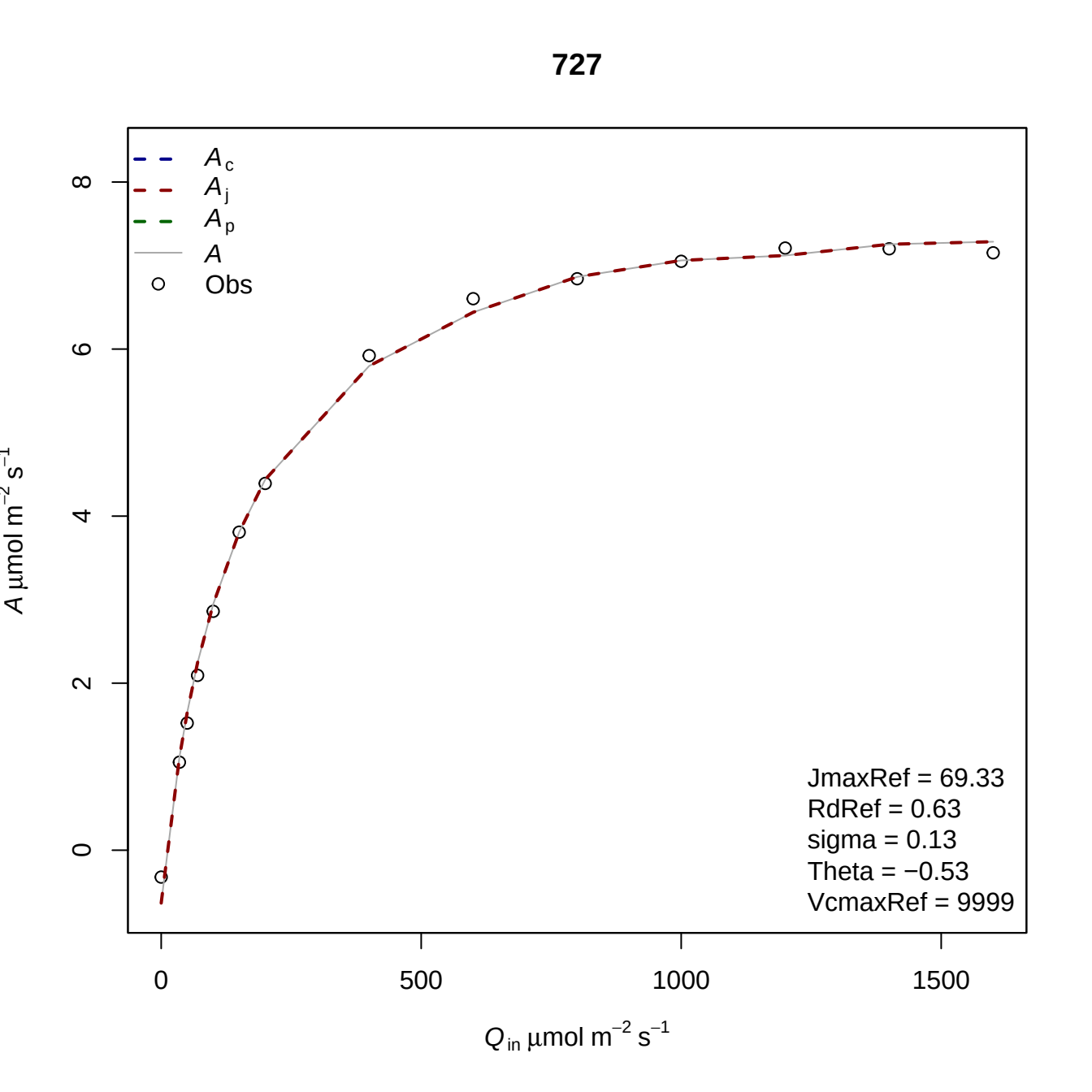

721
A_c
A_j
A_p
A
Obs
JmaxRef = 44.73
RdRef = 0.48
sigma = 0.18
Theta = 0.42
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

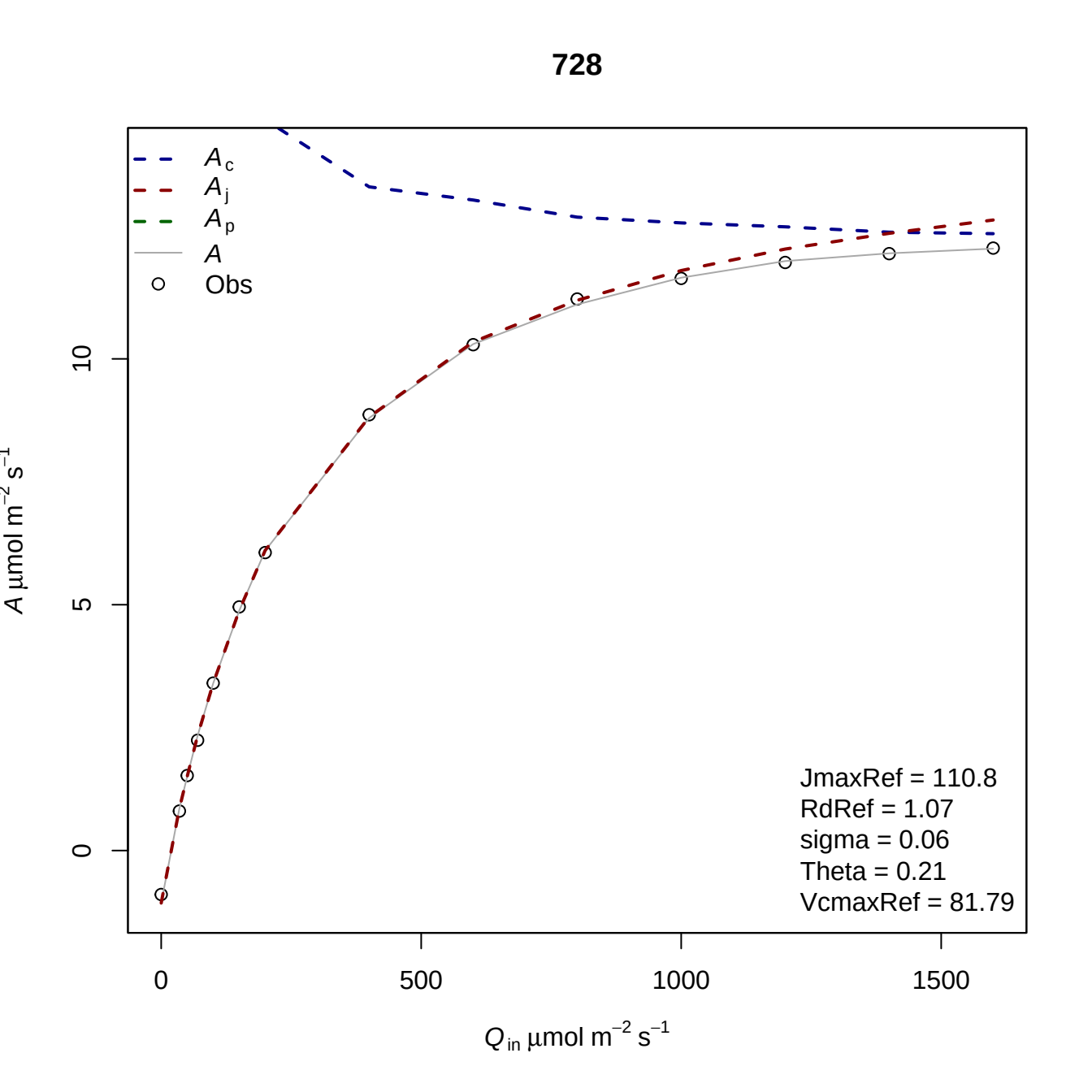

722
A_c
A_j
A_p
A
Obs
JmaxRef = 61.32
RdRef = 0.49
sigma = 0.06
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

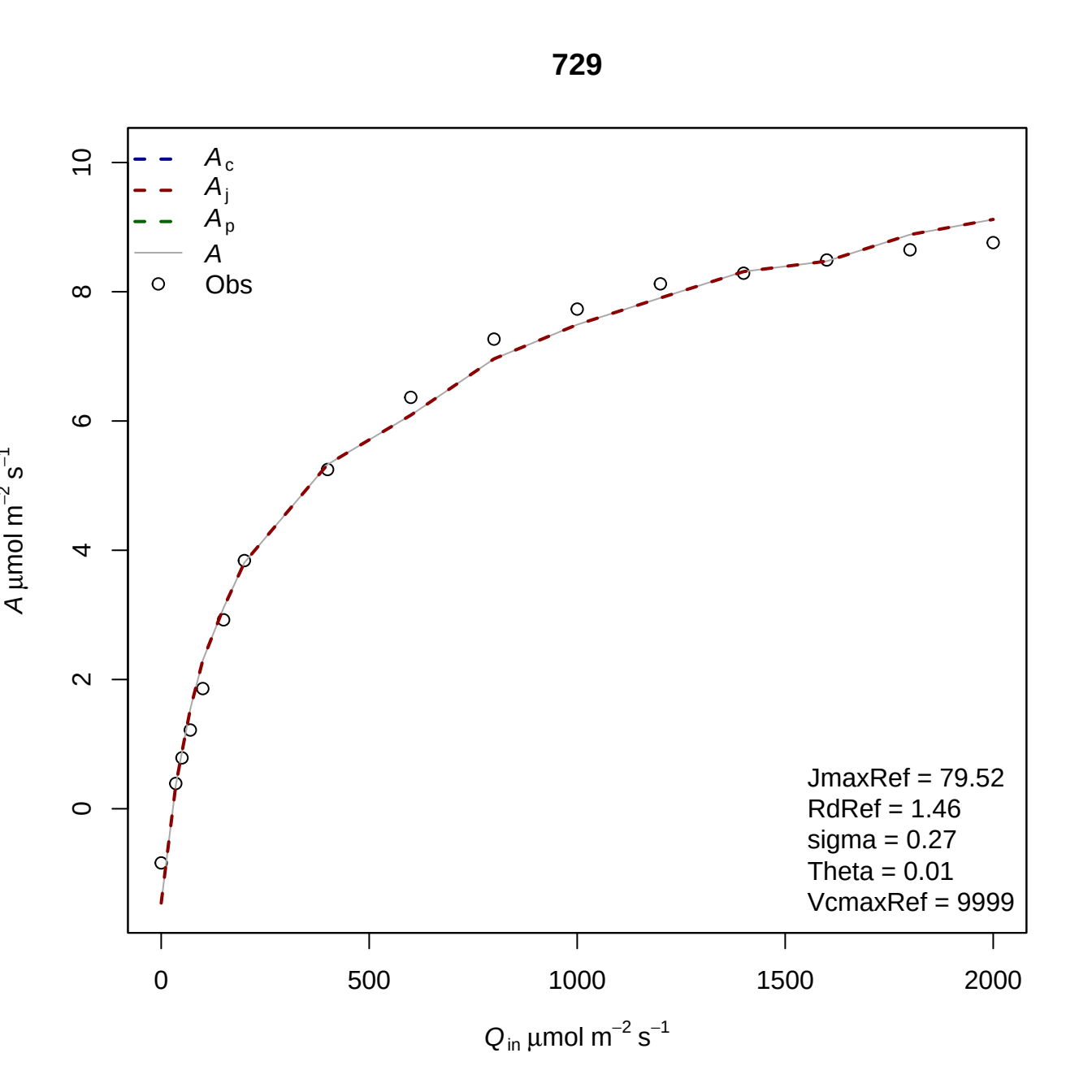

723
A_c
A_j
A_p
A
Obs
JmaxRef = 84.31
RdRef = 0.69
sigma = 0.16
Theta = 0.02
VcmaxRef = 42.37
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

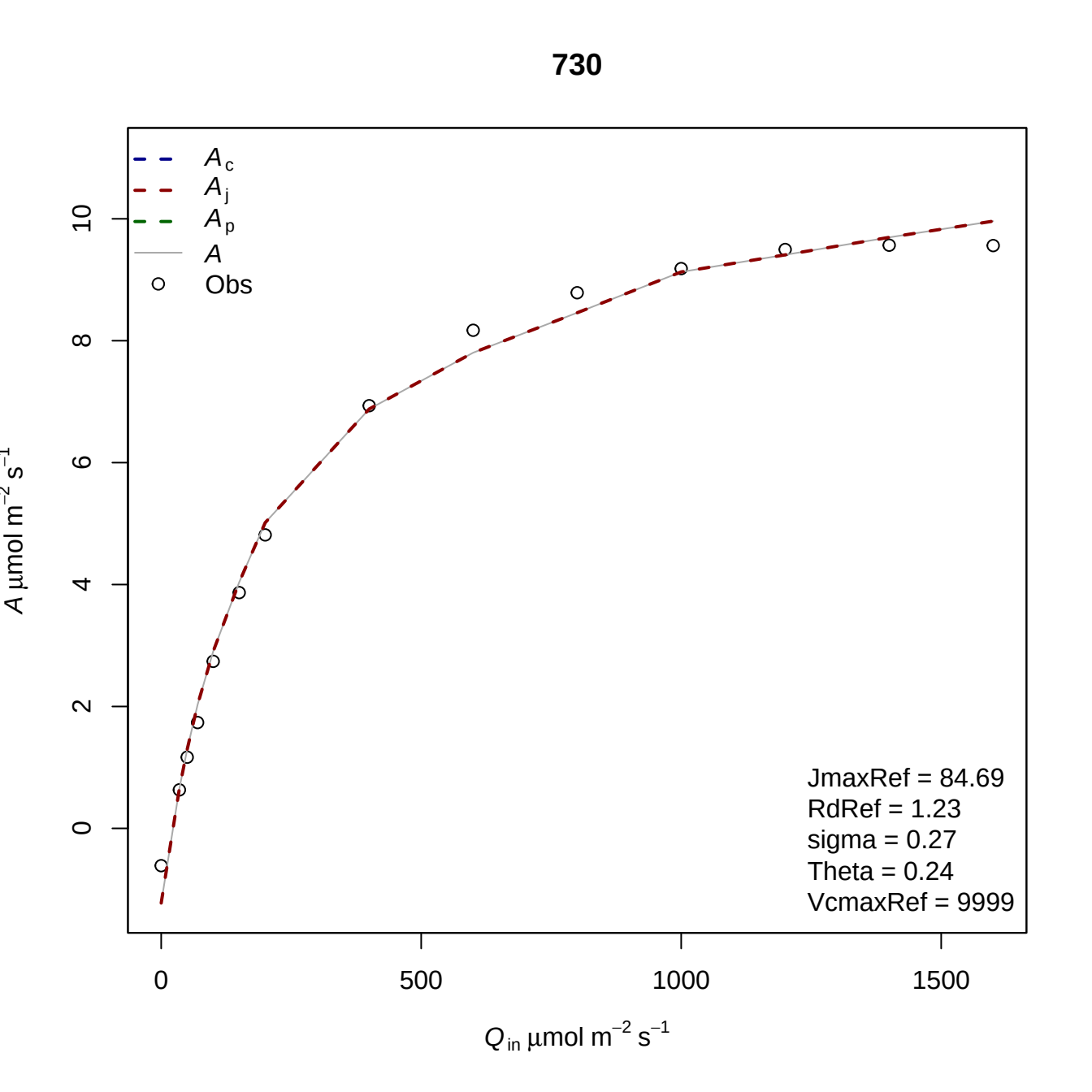

724
A_c
A_j
A_p
A
Obs
JmaxRef = 47.66
RdRef = 0.07
sigma = 0.35
Theta = −1.84
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

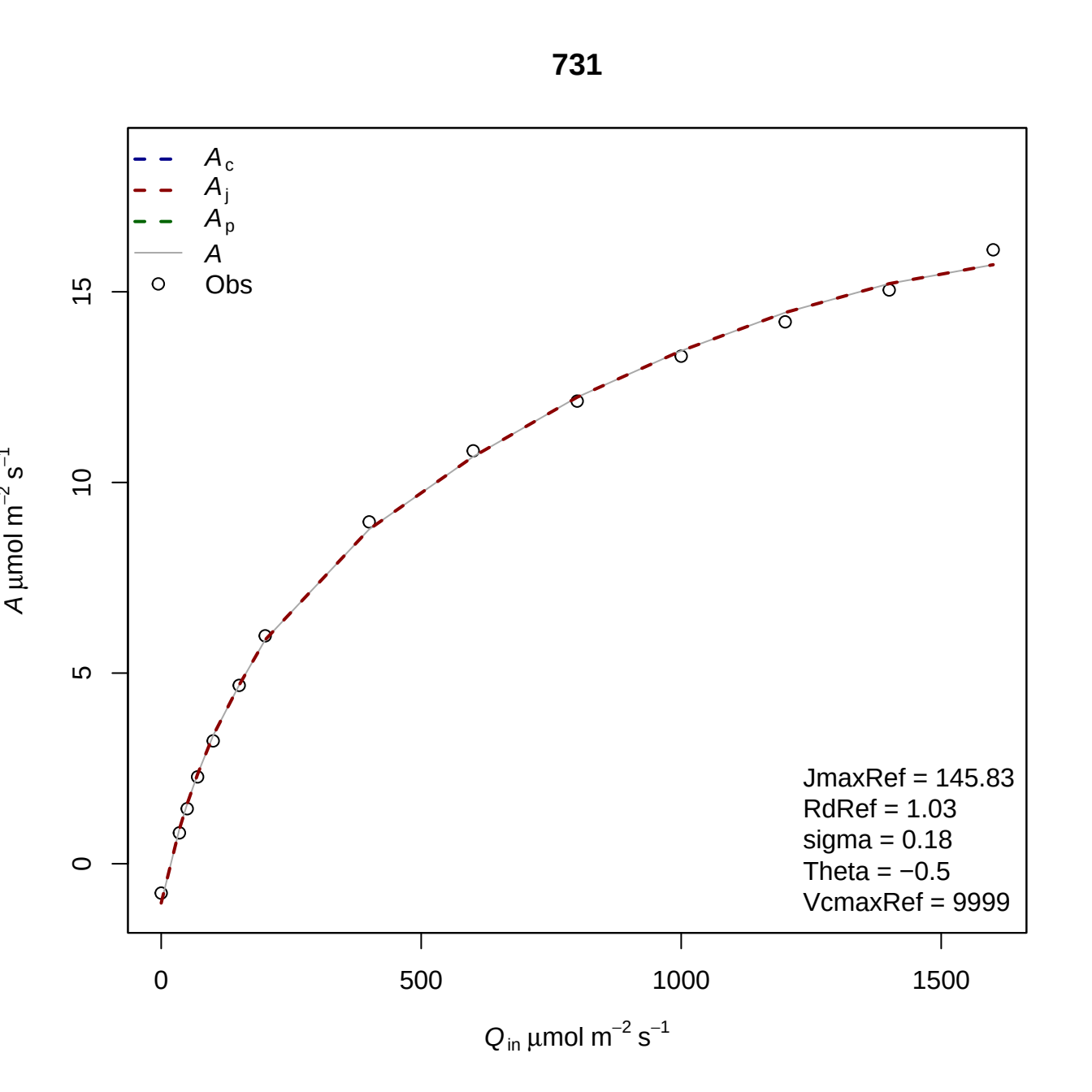

726
A_c
A_j
A_p
A
Obs
JmaxRef = 73.04
RdRef = 0.58
sigma = 0.16
Theta = 0.07
VcmaxRef = 55.42
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

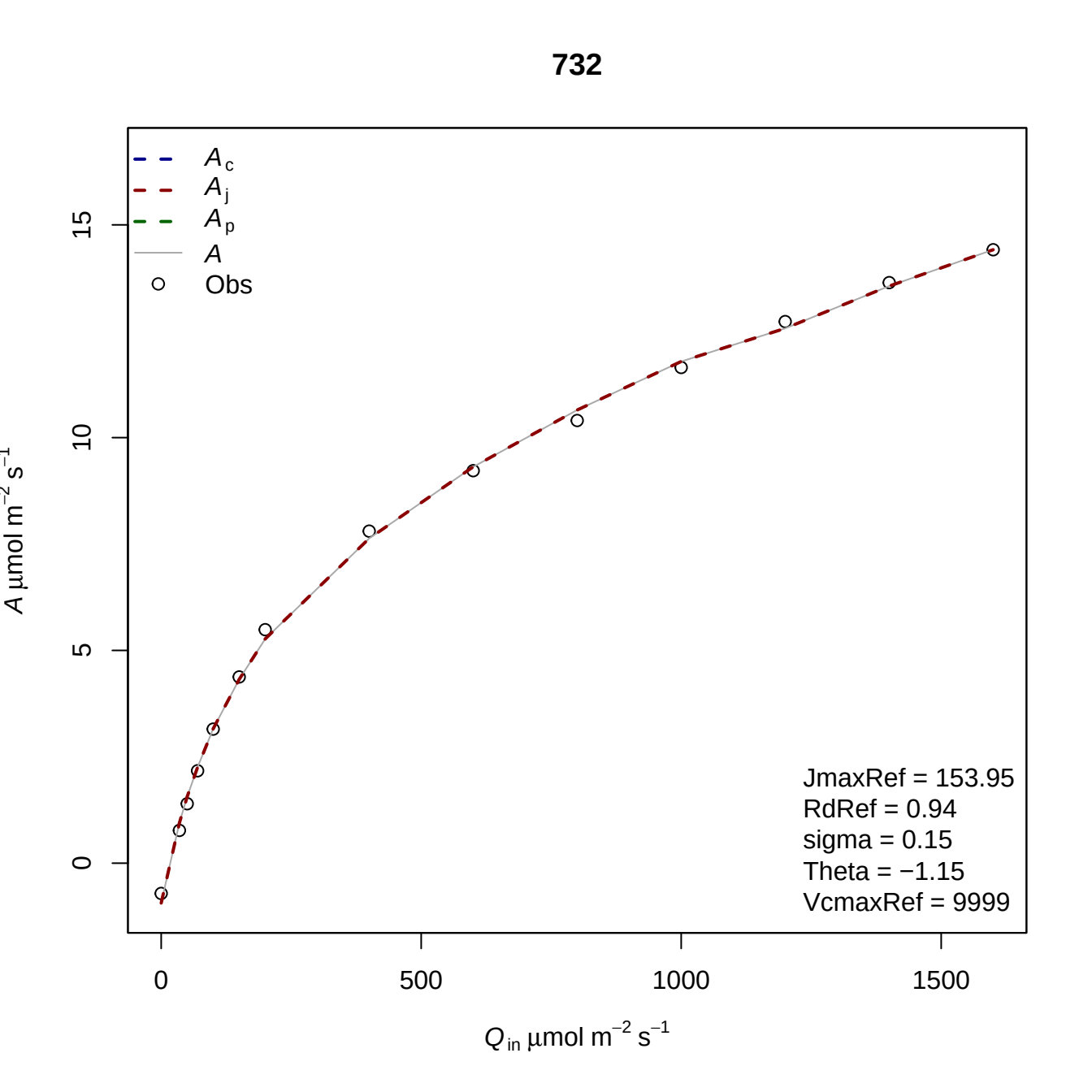

727
A_c
A_j
A_p
A
Obs
JmaxRef = 69.33
RdRef = 0.63
sigma = 0.13
Theta = −0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

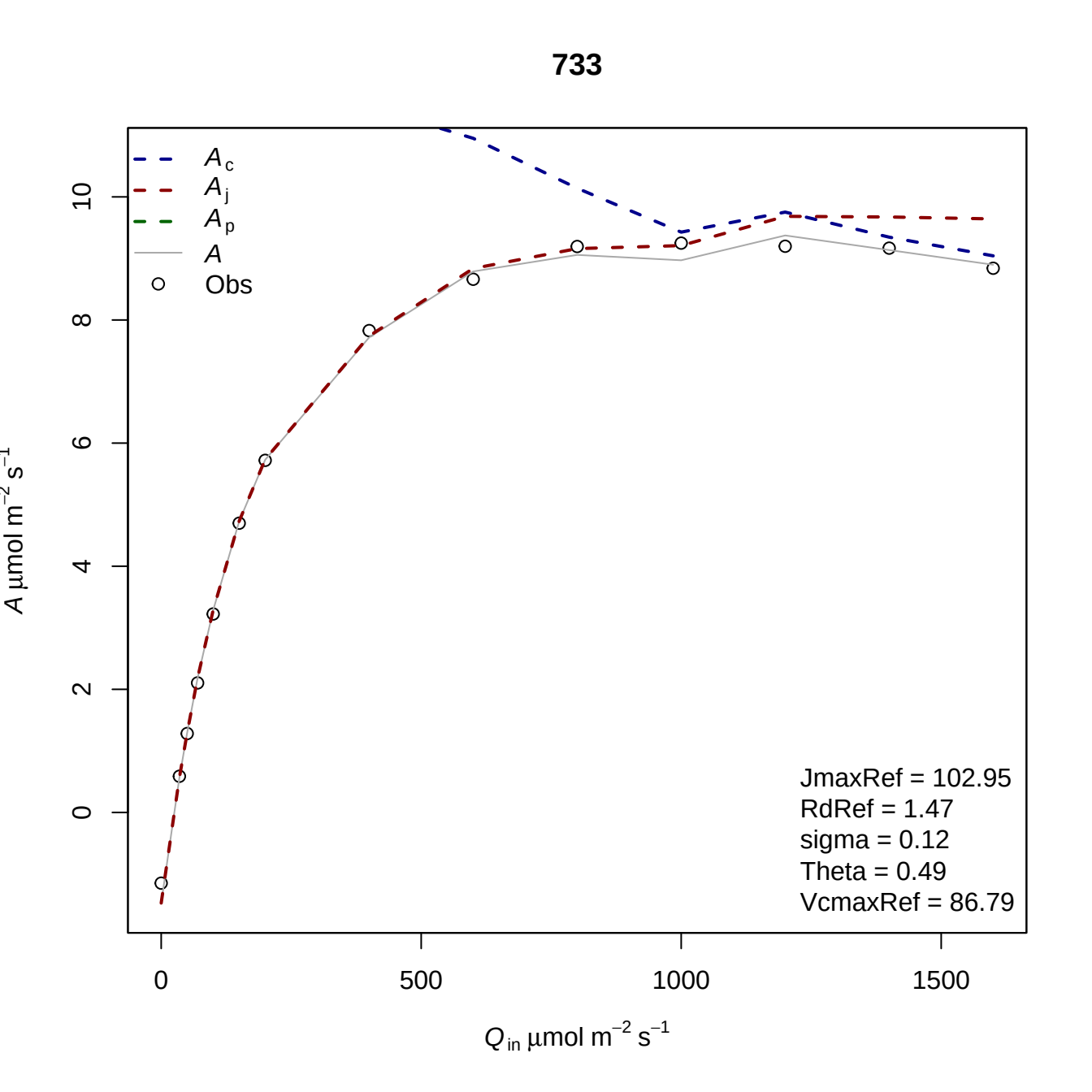

728
A_c
A_j
A_p
A
Obs
JmaxRef = 110.8
RdRef = 1.07
sigma = 0.06
Theta = 0.21
VcmaxRef = 81.79
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

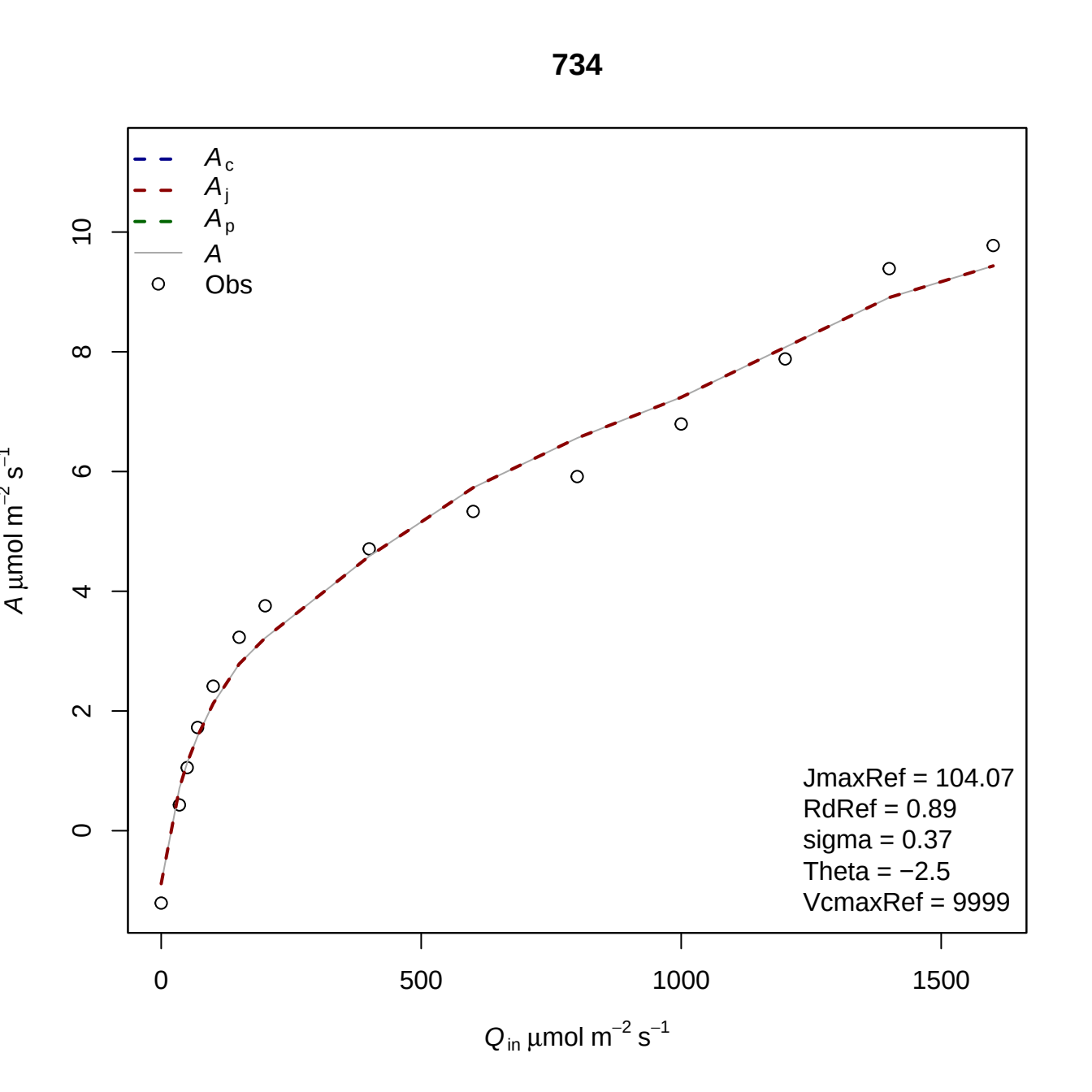

729
A_c
A_j
A_p
A
Obs
JmaxRef = 79.52
RdRef = 1.46
sigma = 0.27
Theta = 0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

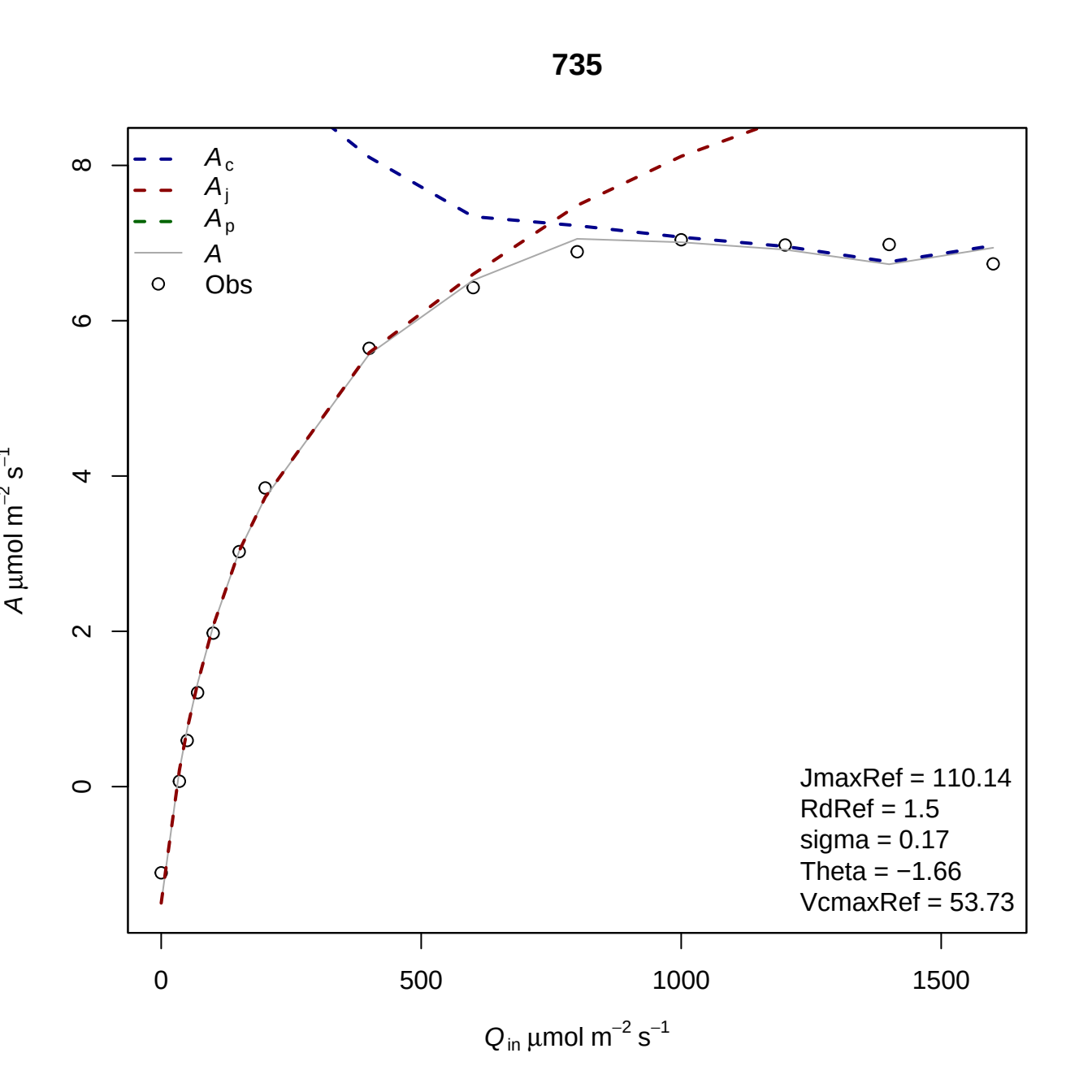

730
A_c
A_j
A_p
A
Obs
JmaxRef = 84.69
RdRef = 1.23
sigma = 0.27
Theta = 0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

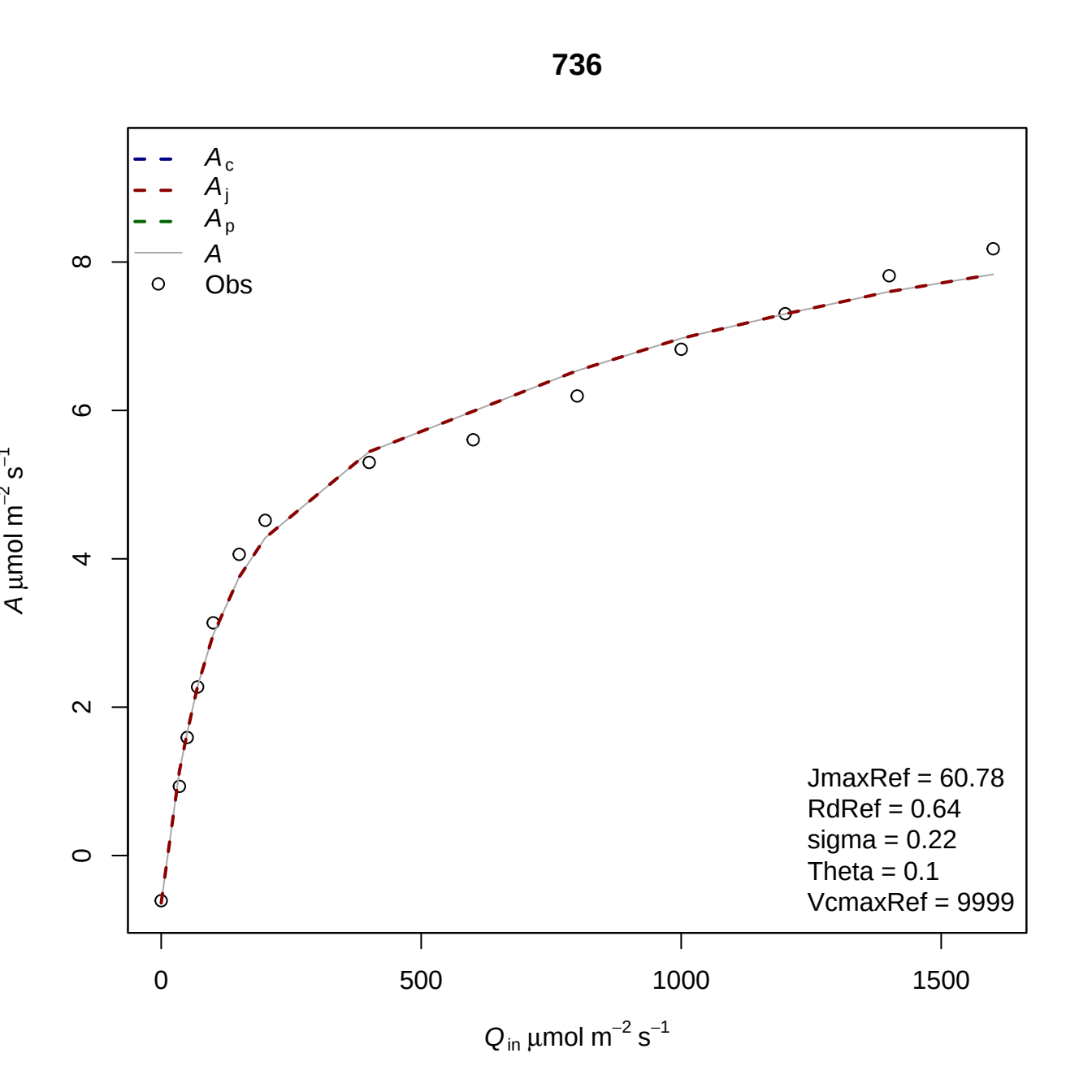

731
A_c
A_j
A_p
A
Obs
JmaxRef = 145.83
RdRef = 1.03
sigma = 0.18
Theta = −0.5
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

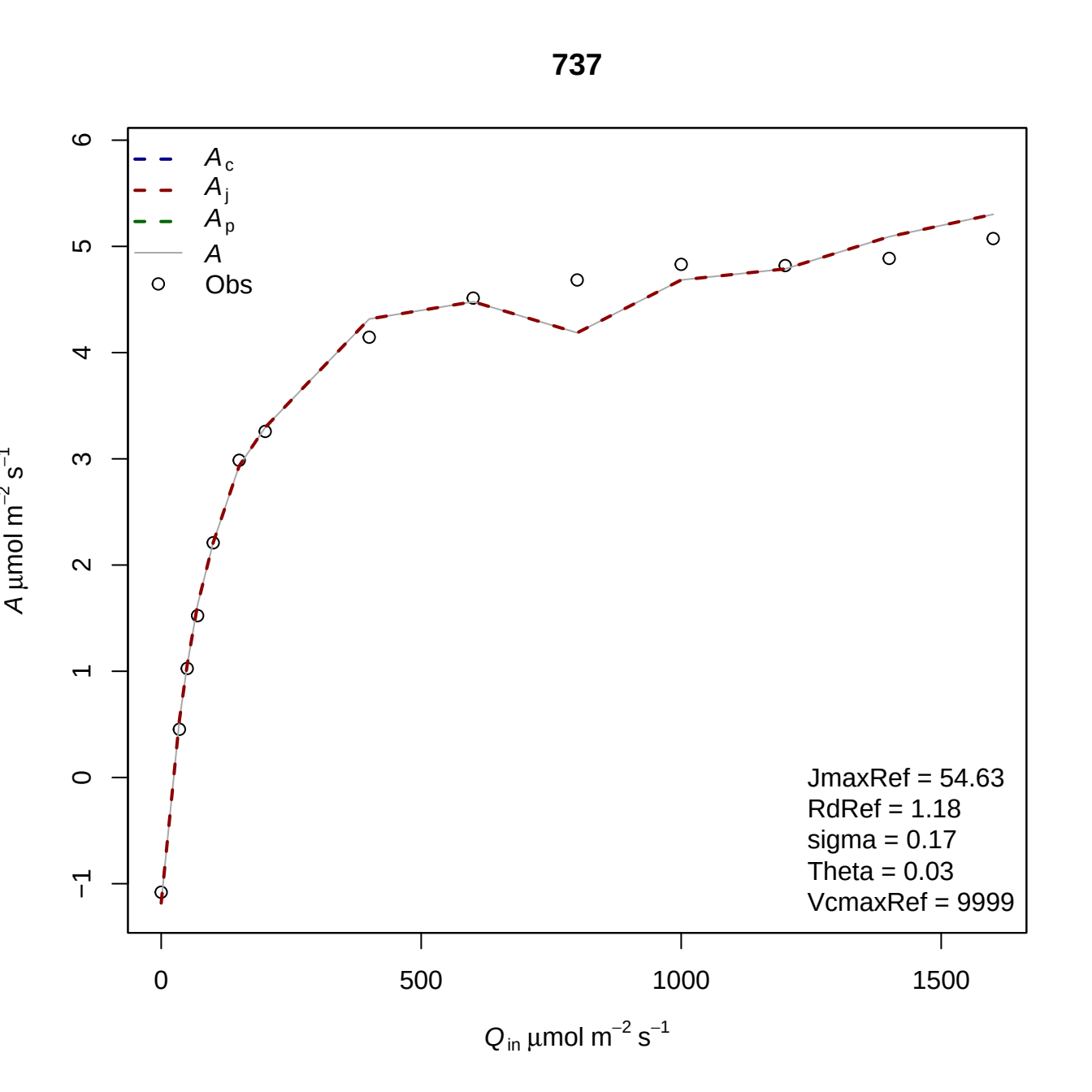

732
A_c
A_j
A_p
A
Obs
JmaxRef = 153.95
RdRef = 0.94
sigma = 0.15
Theta = −1.15
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

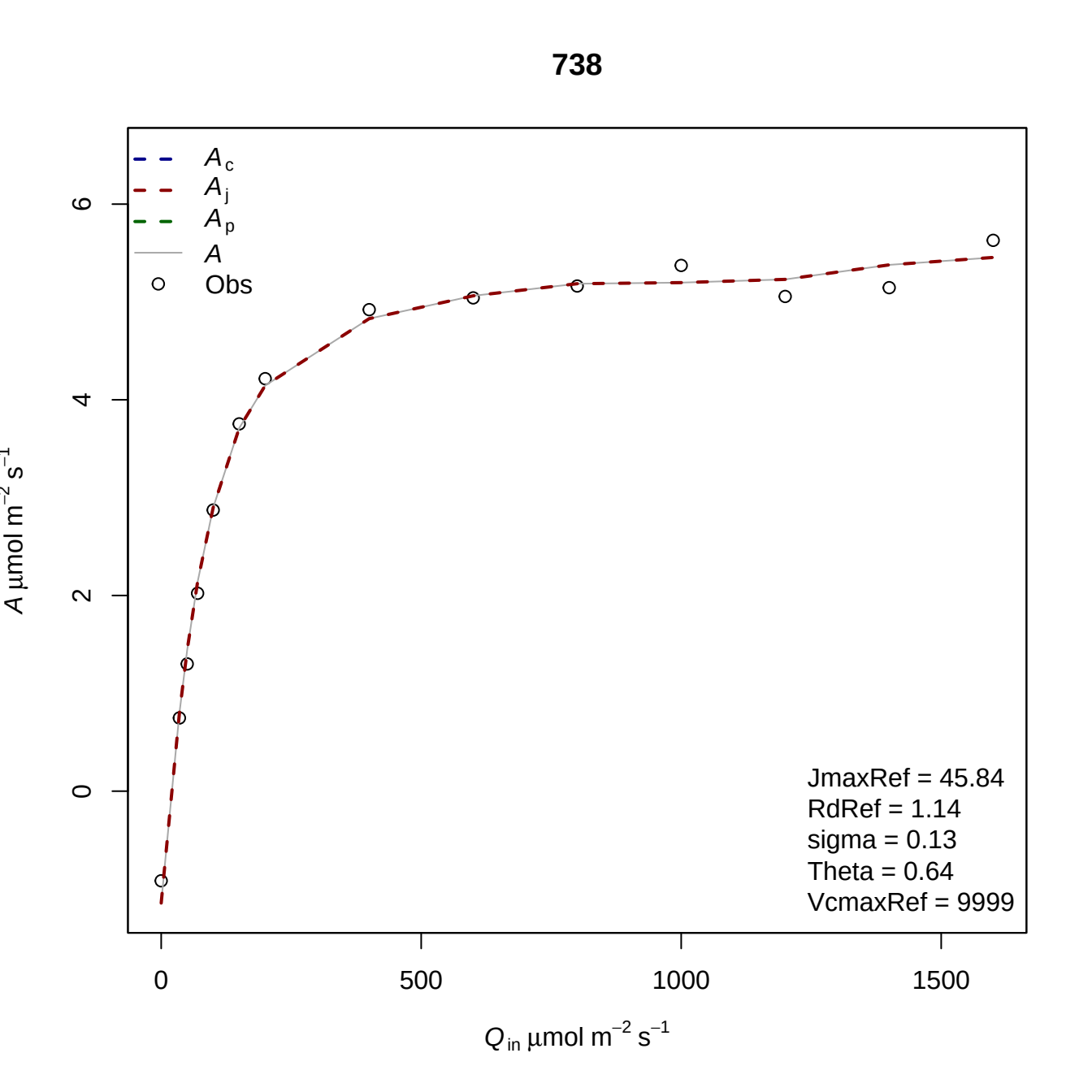

733
A_c
A_j
A_p
A
Obs
JmaxRef = 102.95
RdRef = 1.47
sigma = 0.12
Theta = 0.49
VcmaxRef = 86.79
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

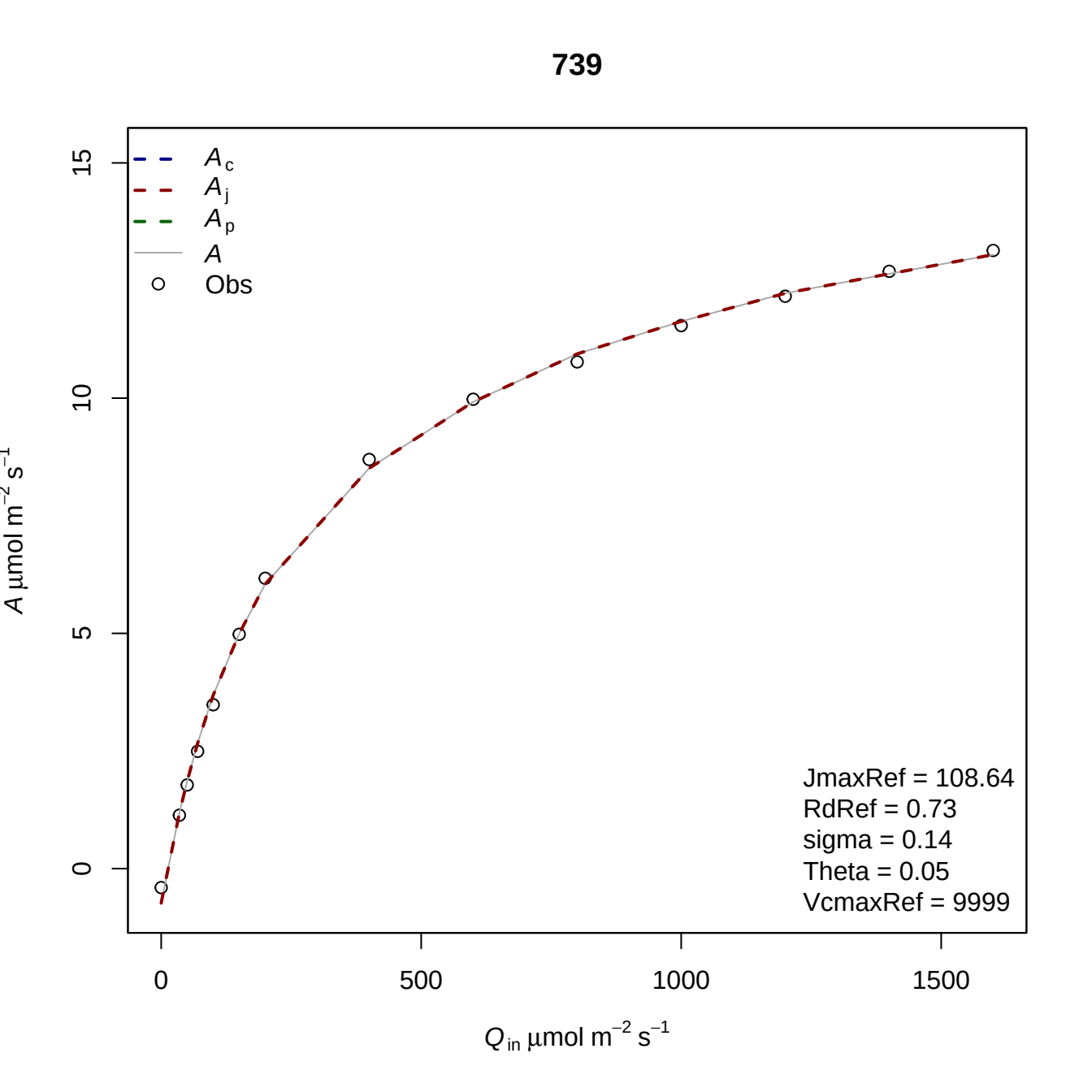

734
A_c
A_j
A_p
A
Obs
JmaxRef = 104.07
RdRef = 0.89
sigma = 0.37
Theta = −2.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

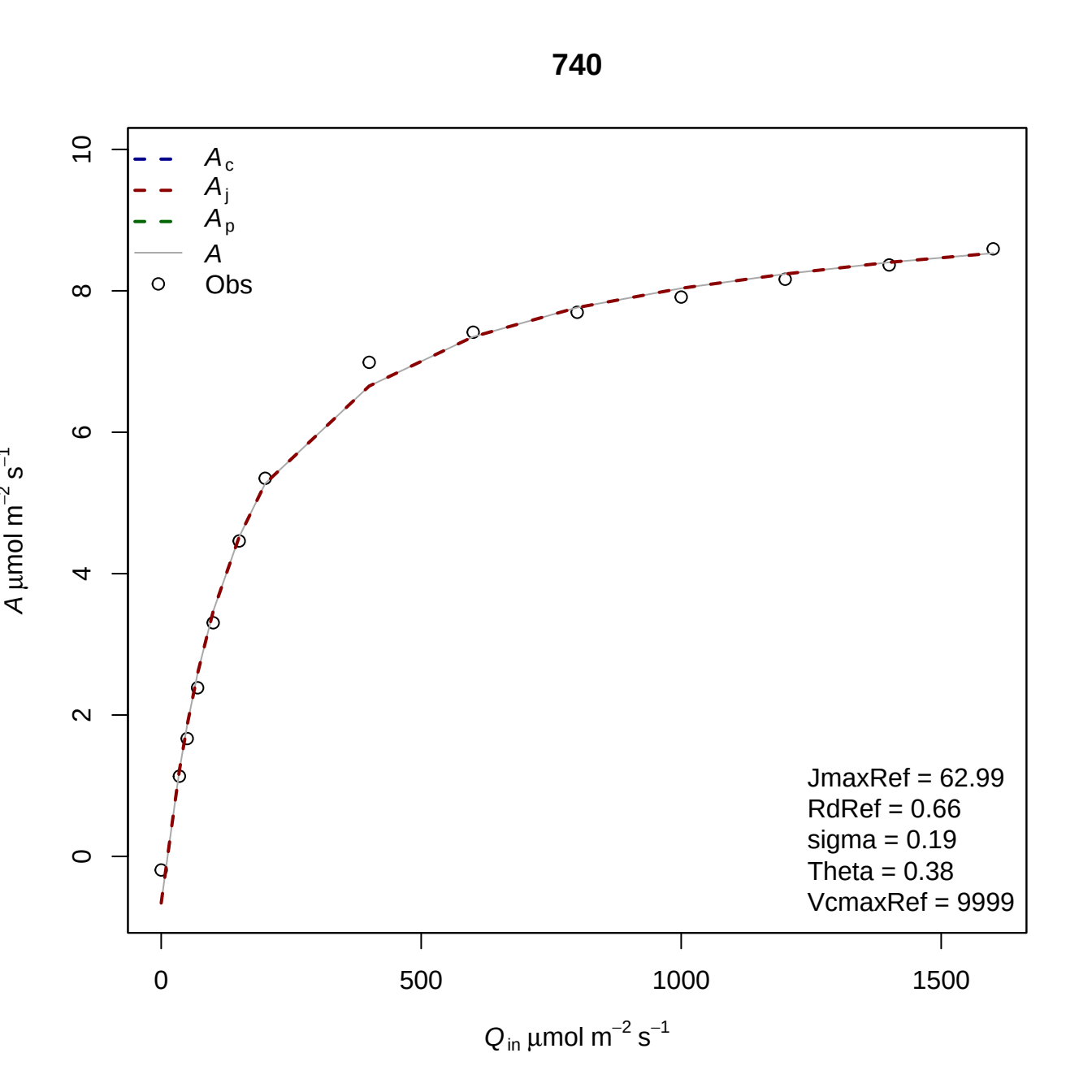

735
A_c
A_j
A_p
A
Obs
JmaxRef = 110.14
RdRef = 1.5
sigma = 0.17
Theta = −1.66
VcmaxRef = 53.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

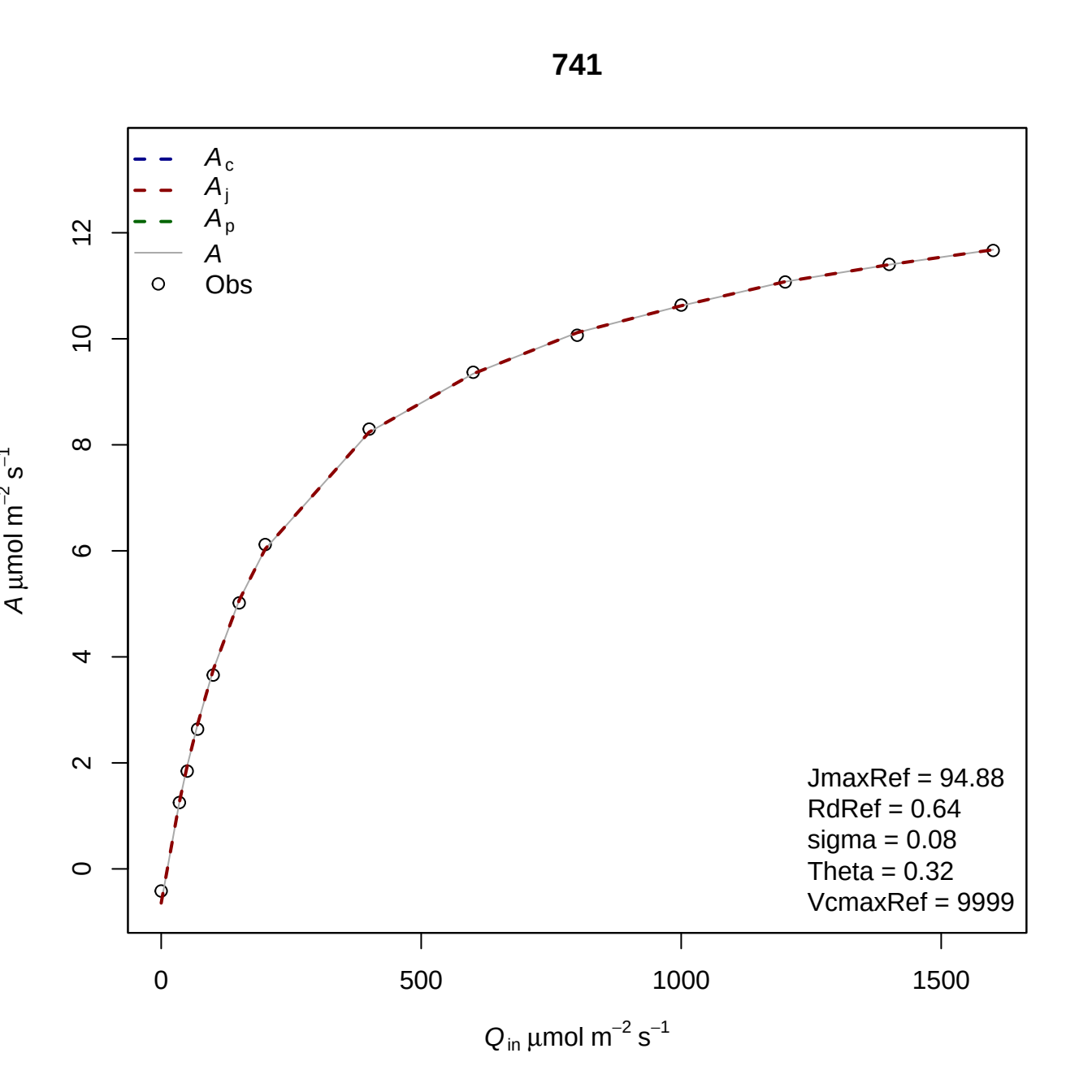

736
A_c
A_j
A_p
A
Obs
JmaxRef = 60.78
RdRef = 0.64
sigma = 0.22
Theta = 0.1
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

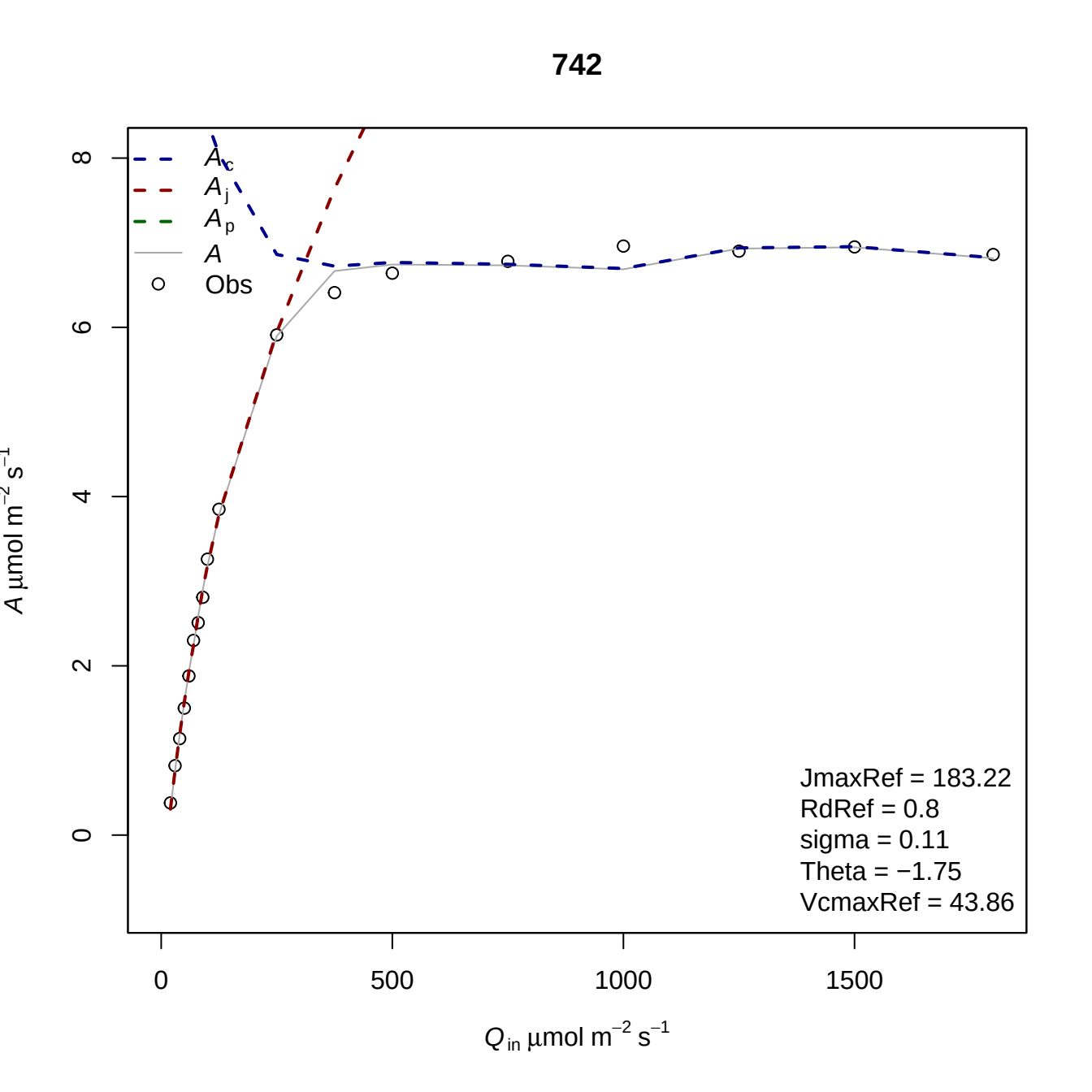

737
A_c
A_j
A_p
A
Obs
JmaxRef = 54.63
RdRef = 1.18
sigma = 0.17
Theta = 0.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

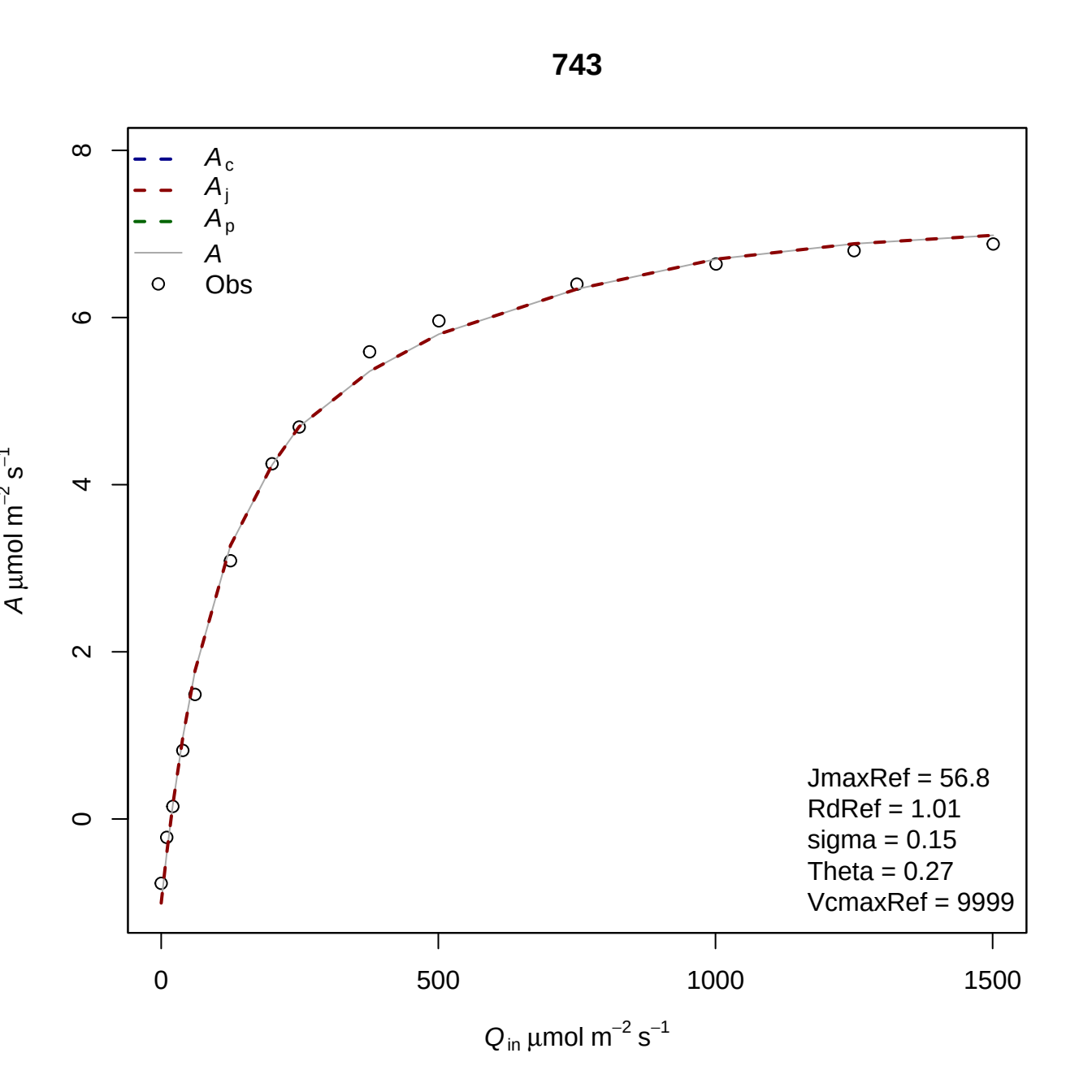

738
A_c
A_j
A_p
A
Obs
JmaxRef = 45.84
RdRef = 1.14
sigma = 0.13
Theta = 0.64
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

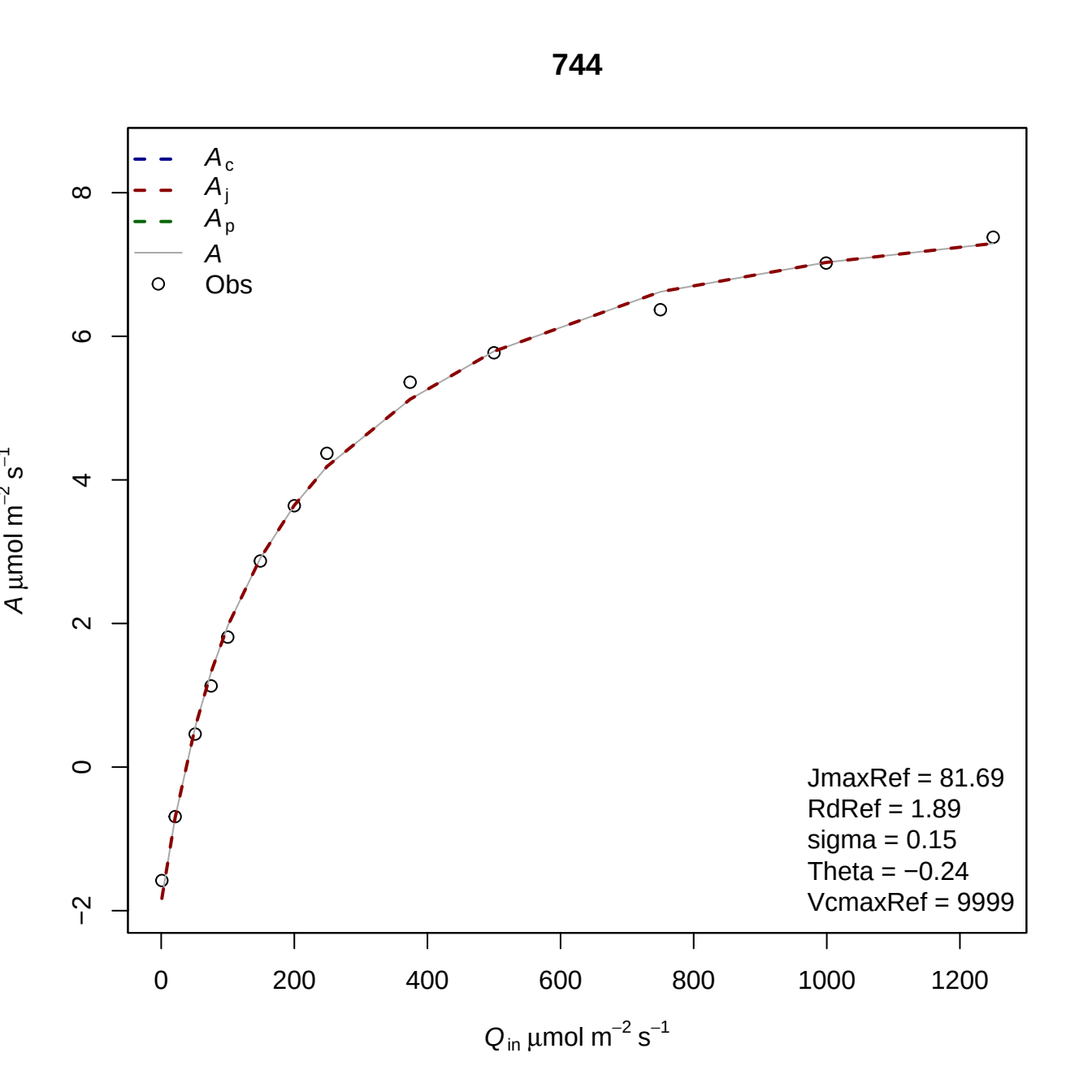

739
A_c
A_j
A_p
A
Obs
JmaxRef = 108.64
RdRef = 0.73
sigma = 0.14
Theta = 0.05
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

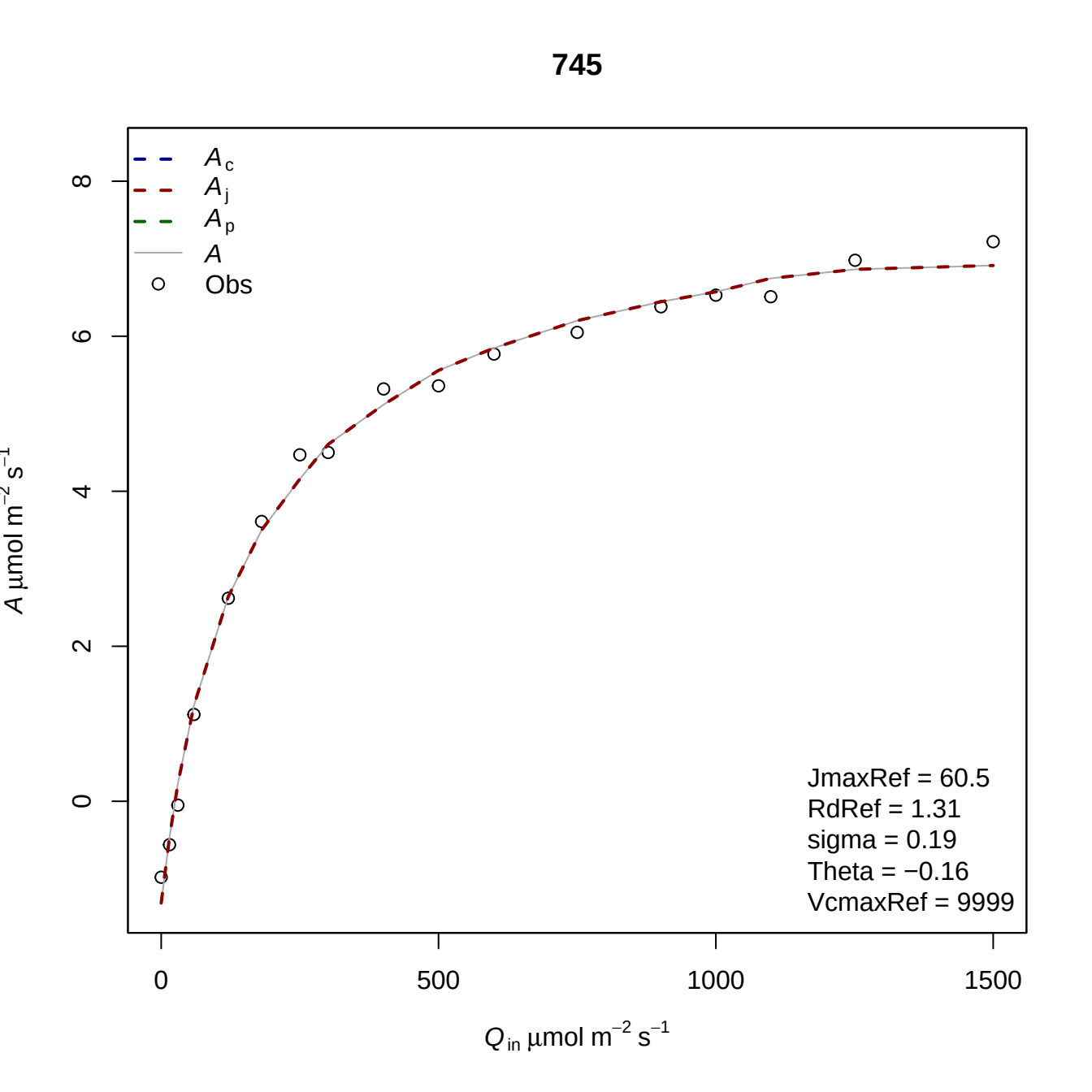

740
A_c
A_j
A_p
A
Obs
JmaxRef = 62.99
RdRef = 0.66
sigma = 0.19
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

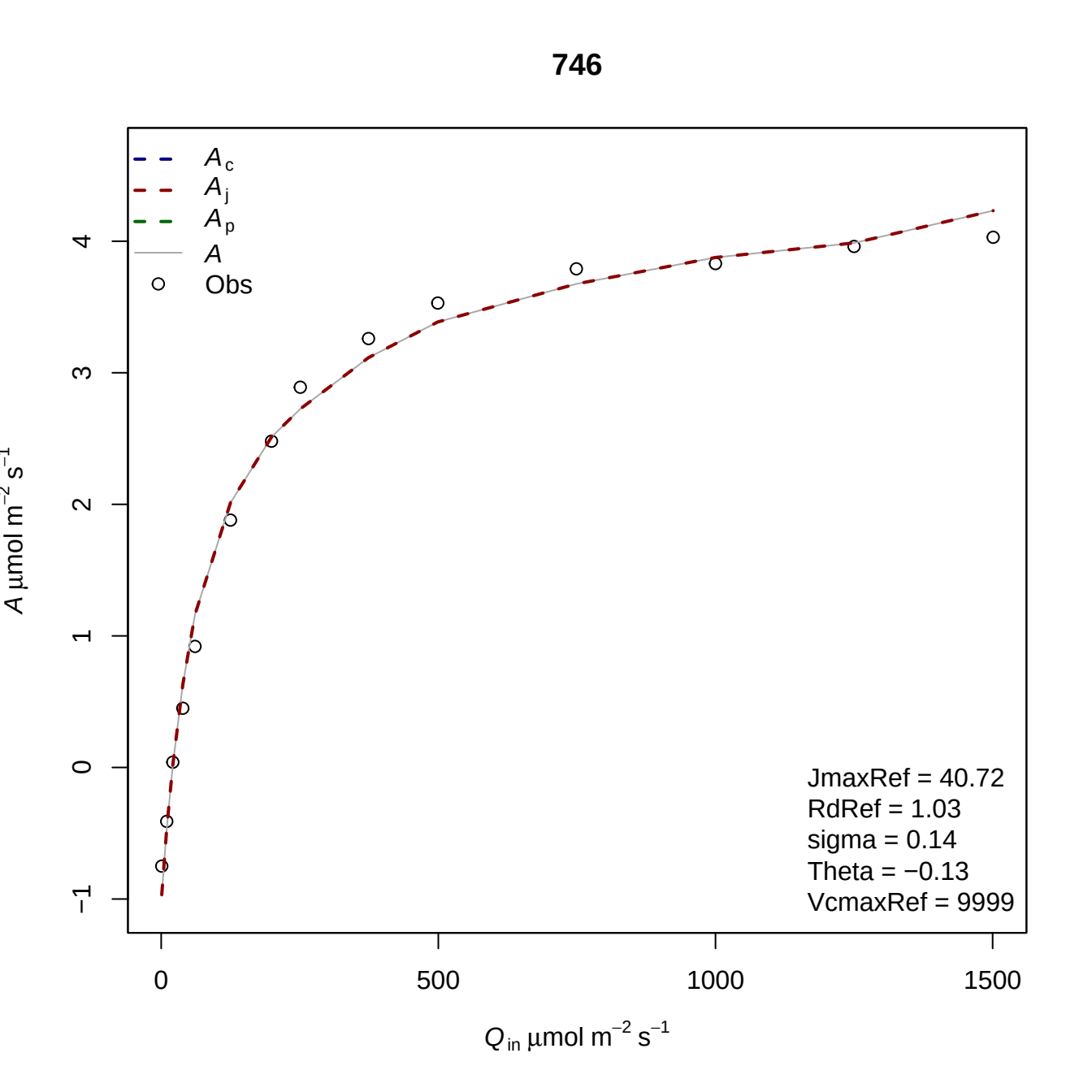

741
A_c
A_j
A_p
A
Obs
JmaxRef = 94.88
RdRef = 0.64
sigma = 0.08
Theta = 0.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

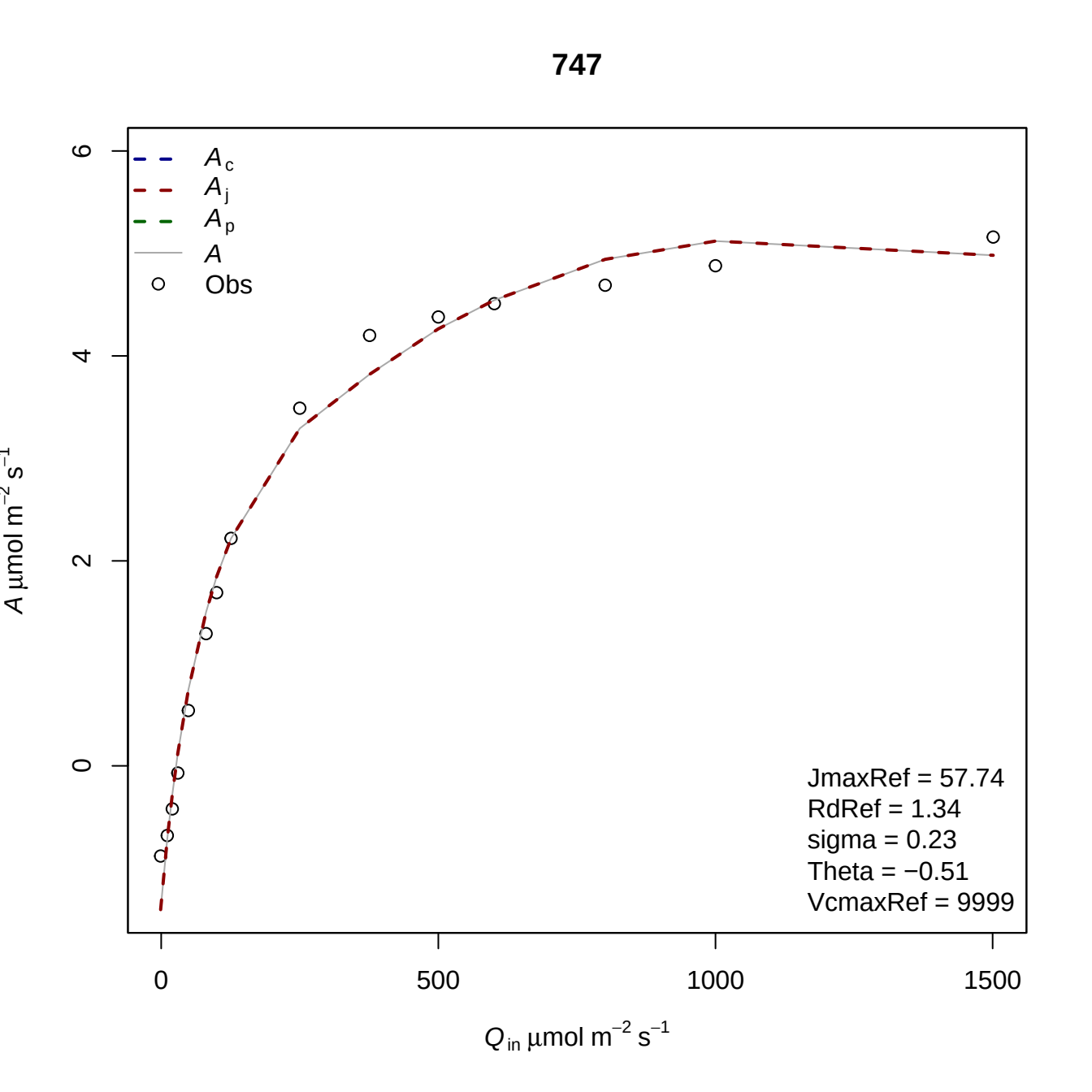

742
A_c
A_j
A_p
A
Obs
JmaxRef = 183.22
RdRef = 0.8
sigma = 0.11
Theta = −1.75
VcmaxRef = 43.86
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

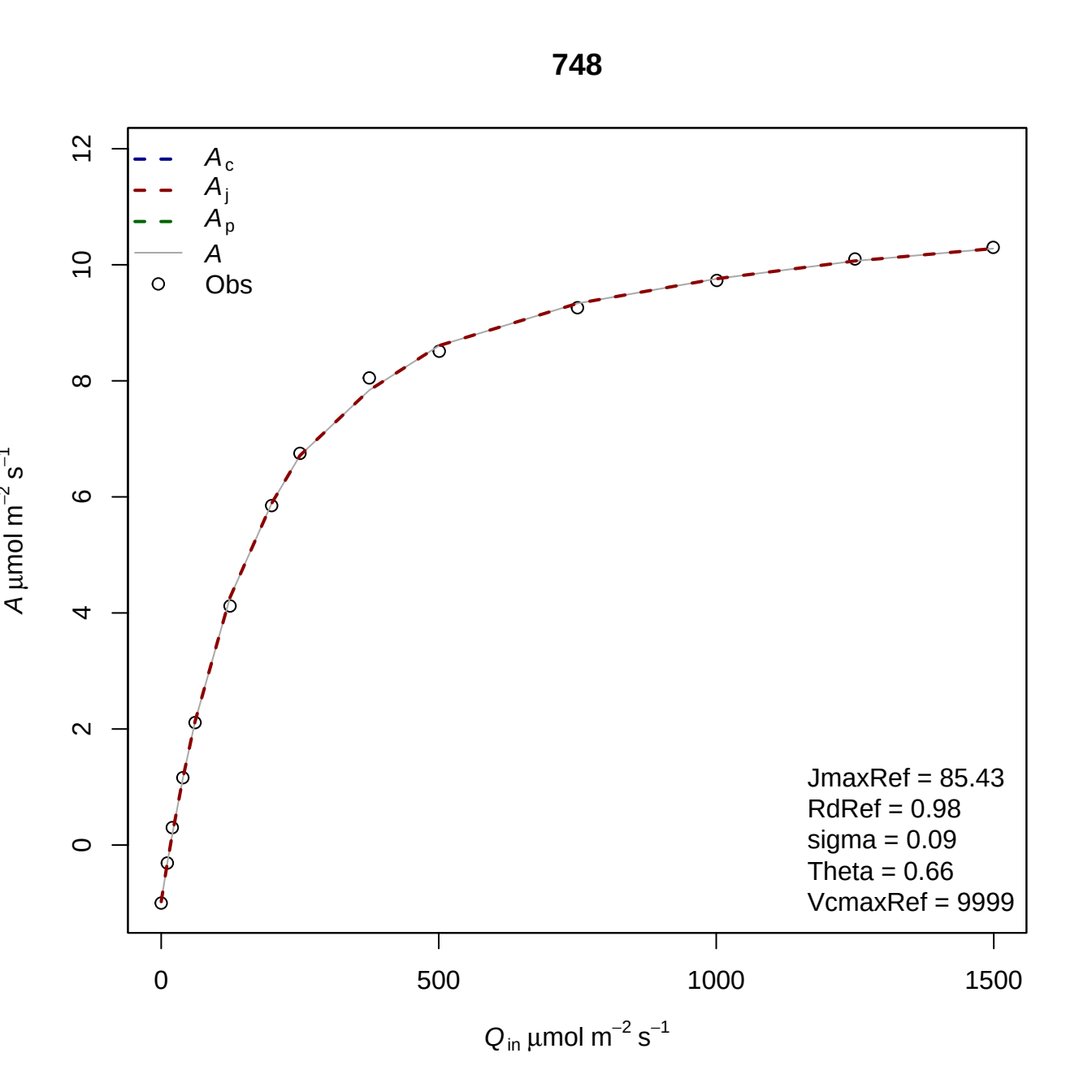

743
A_c
A_j
A_p
A
Obs
JmaxRef = 56.8
RdRef = 1.01
sigma = 0.15
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

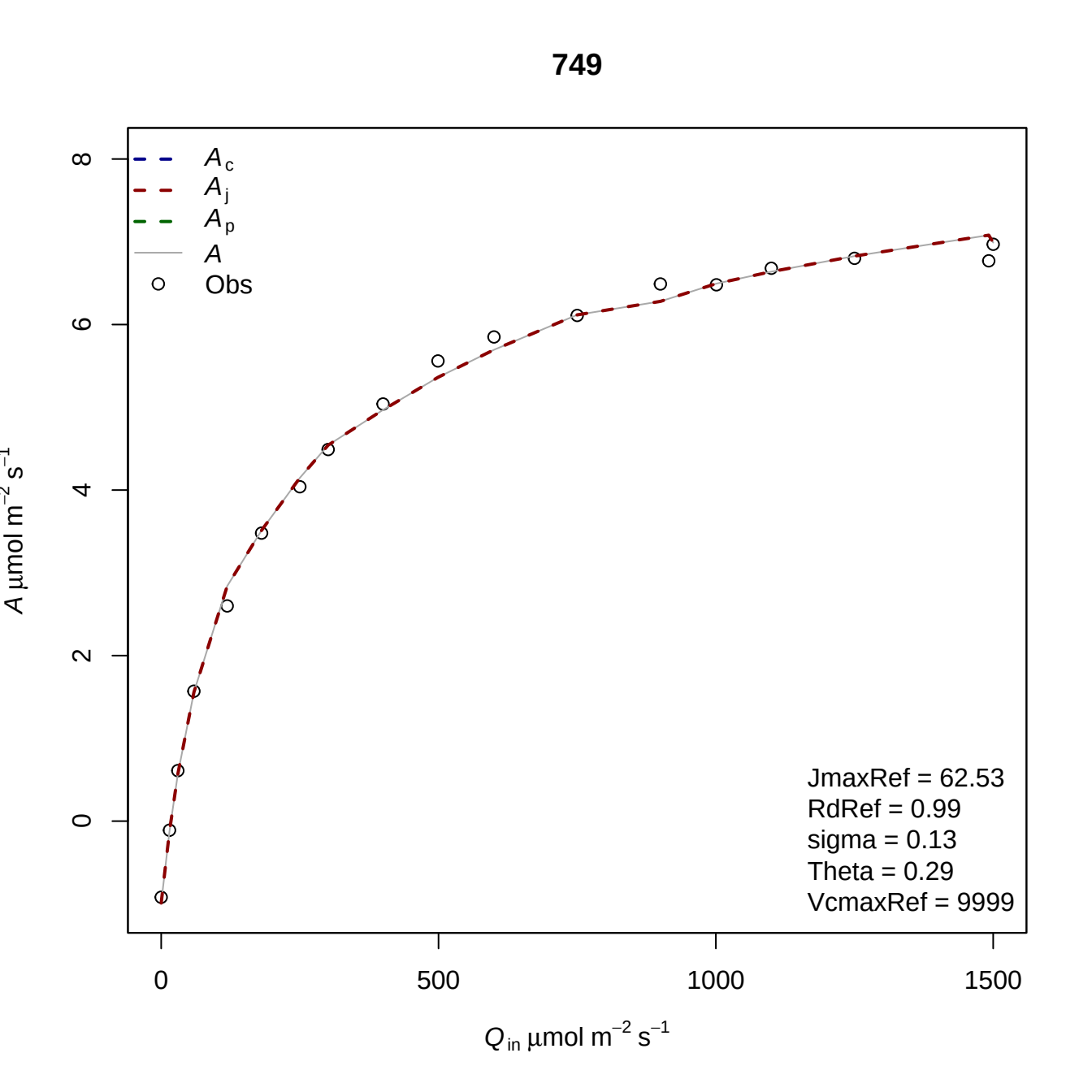

744
A_c
A_j
A_p
A
Obs
JmaxRef = 81.69
RdRef = 1.89
sigma = 0.15
Theta = −0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

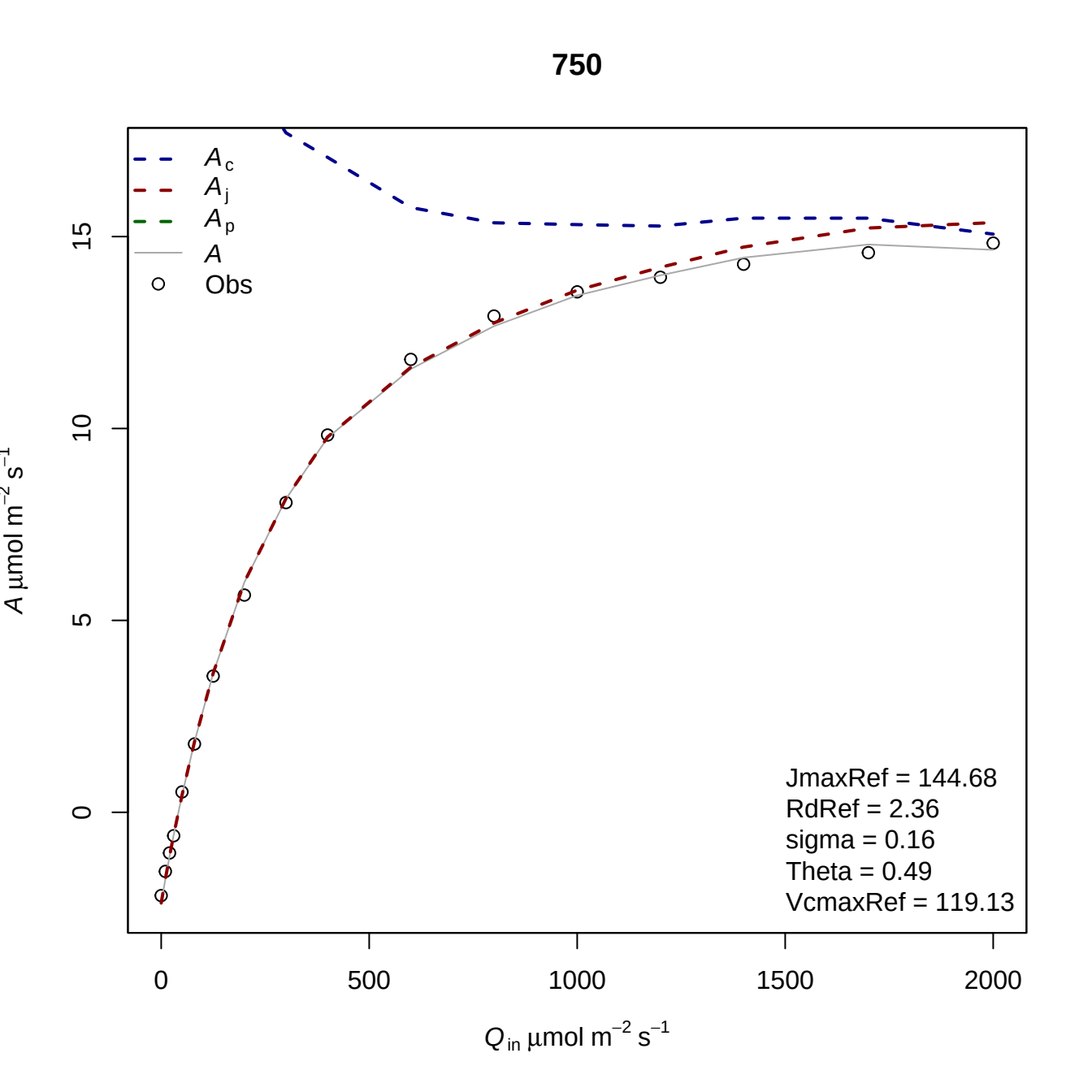

745
A_c
A_j
A_p
A
Obs
JmaxRef = 60.5
RdRef = 1.31
sigma = 0.19
Theta = −0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

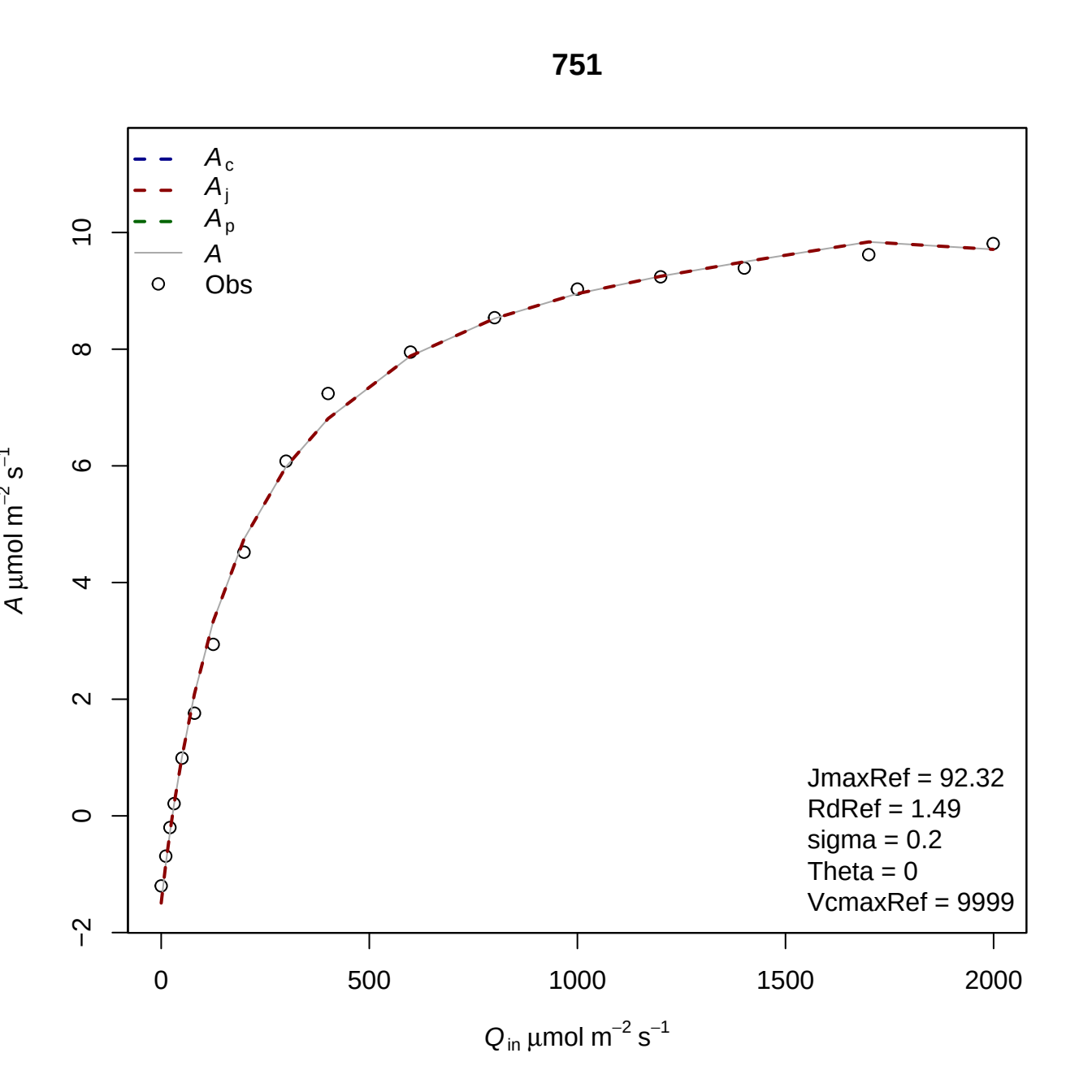

746
A_c
A_j
A_p
A
Obs
JmaxRef = 40.72
RdRef = 1.03
sigma = 0.14
Theta = −0.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

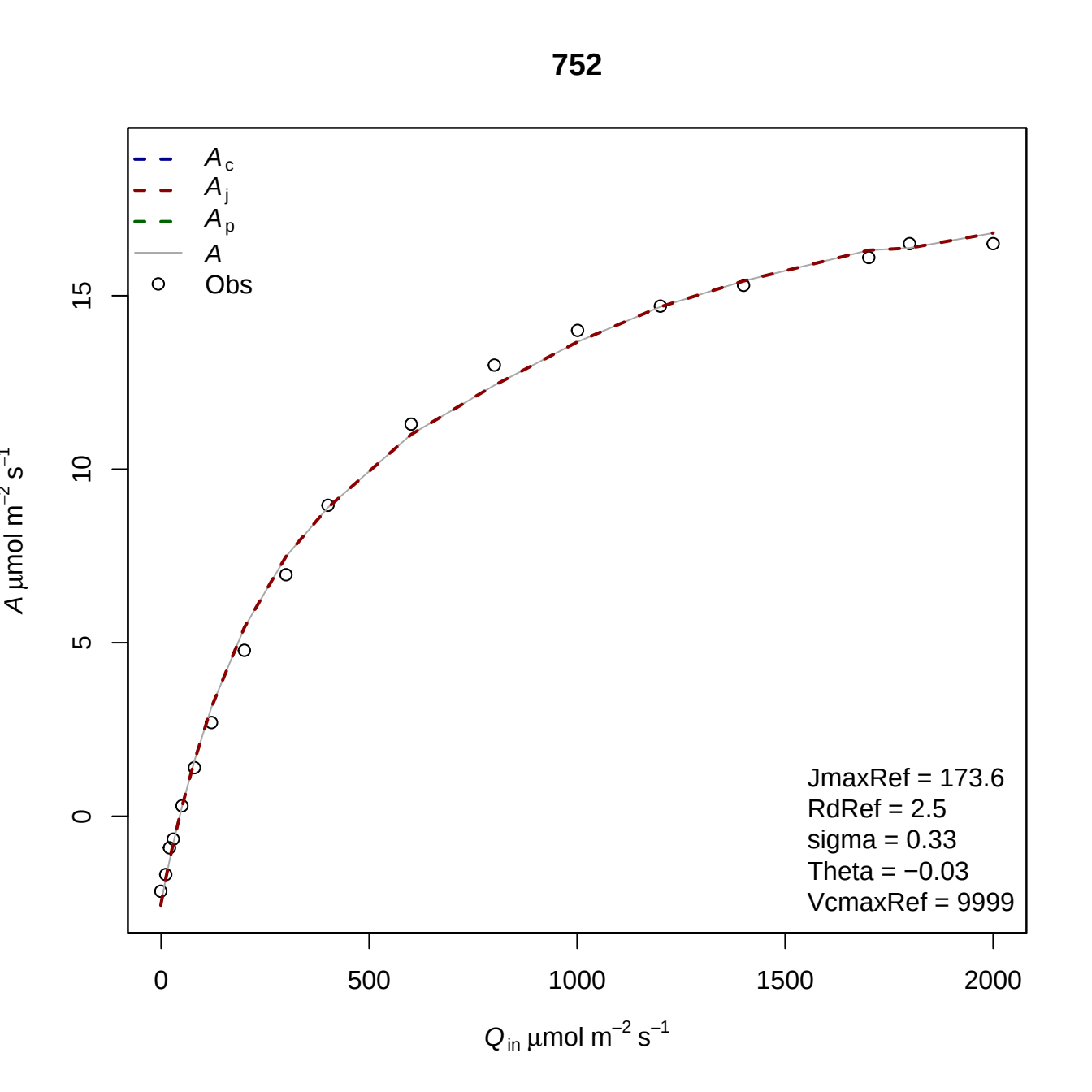

747
A_c
A_j
A_p
A
Obs
JmaxRef = 57.74
RdRef = 1.34
sigma = 0.23
Theta = −0.51
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

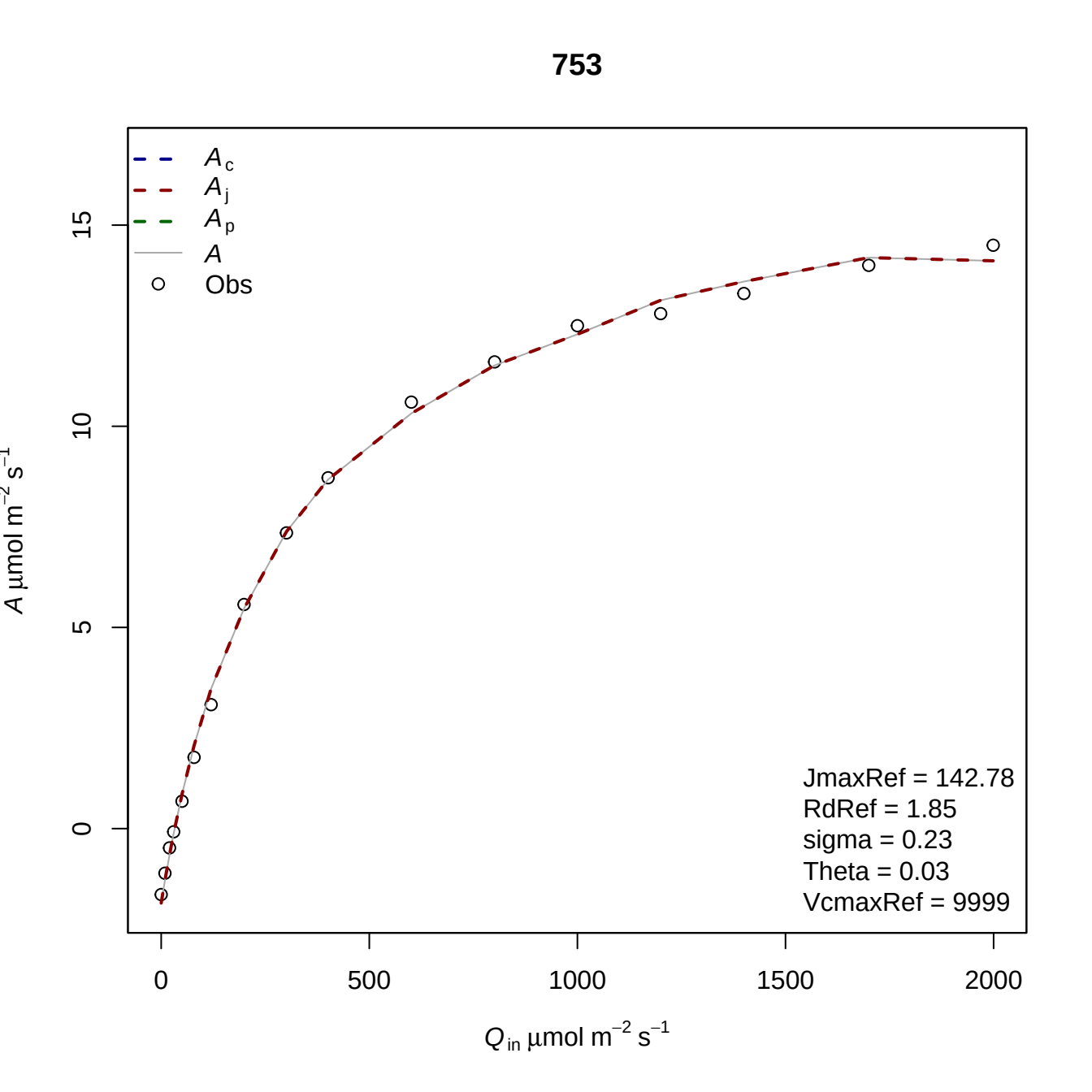

748
A_c
A_j
A_p
A
Obs
JmaxRef = 85.43
RdRef = 0.98
sigma = 0.09
Theta = 0.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

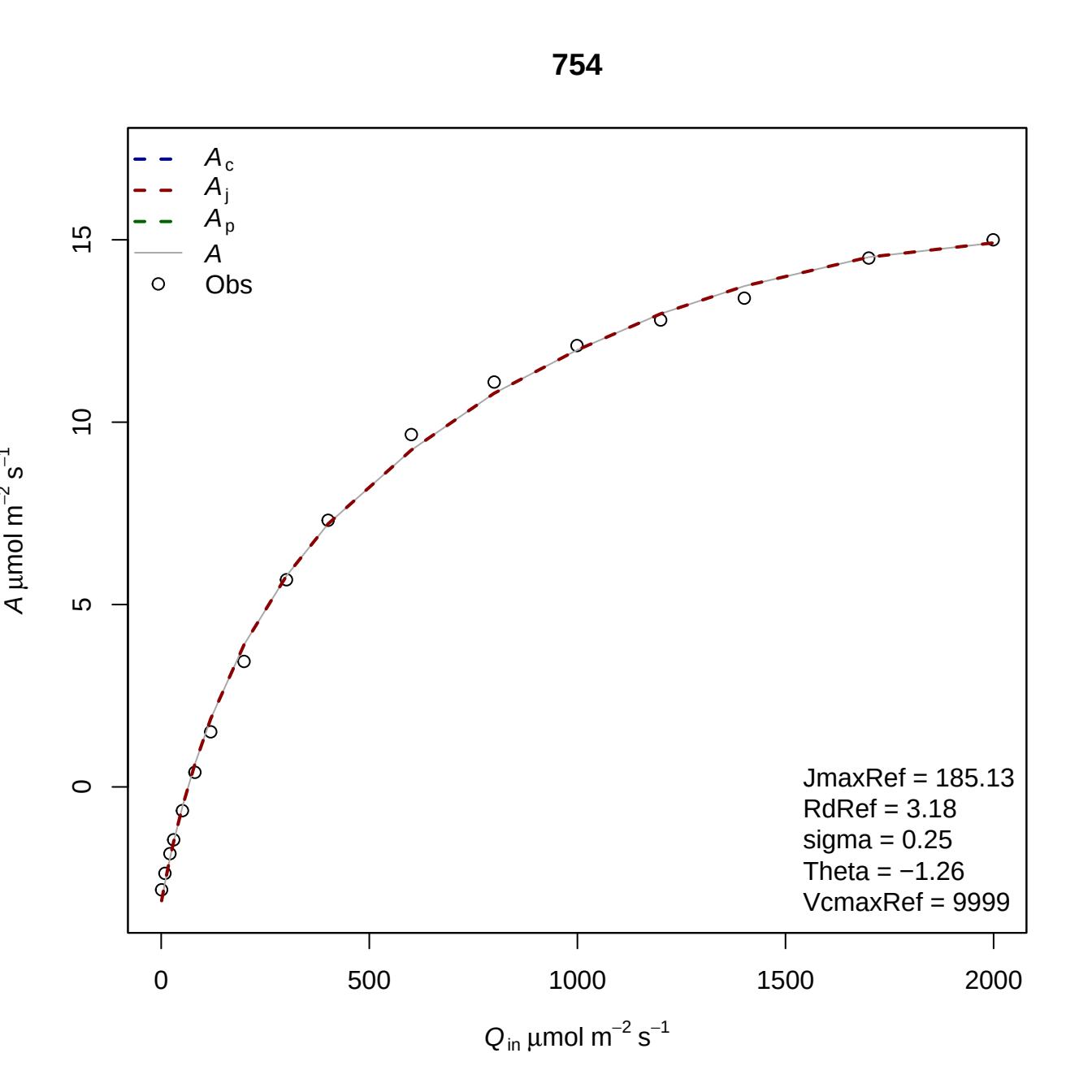

749
A_c
A_j
A_p
A
Obs
JmaxRef = 62.53
RdRef = 0.99
sigma = 0.13
Theta = 0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

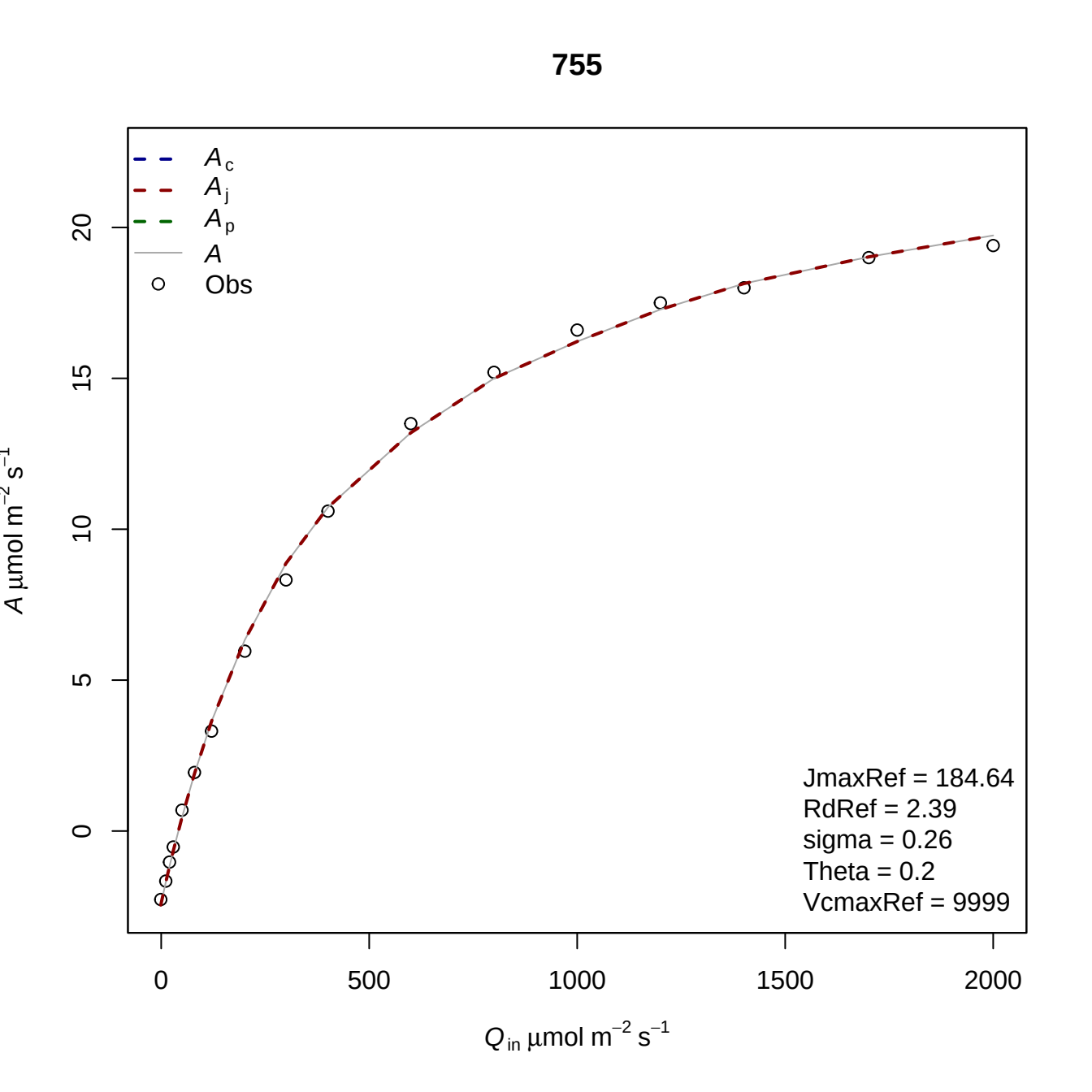

750
A_c
A_j
A_p
A
Obs
JmaxRef = 144.68
RdRef = 2.36
sigma = 0.16
Theta = 0.49
VcmaxRef = 119.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

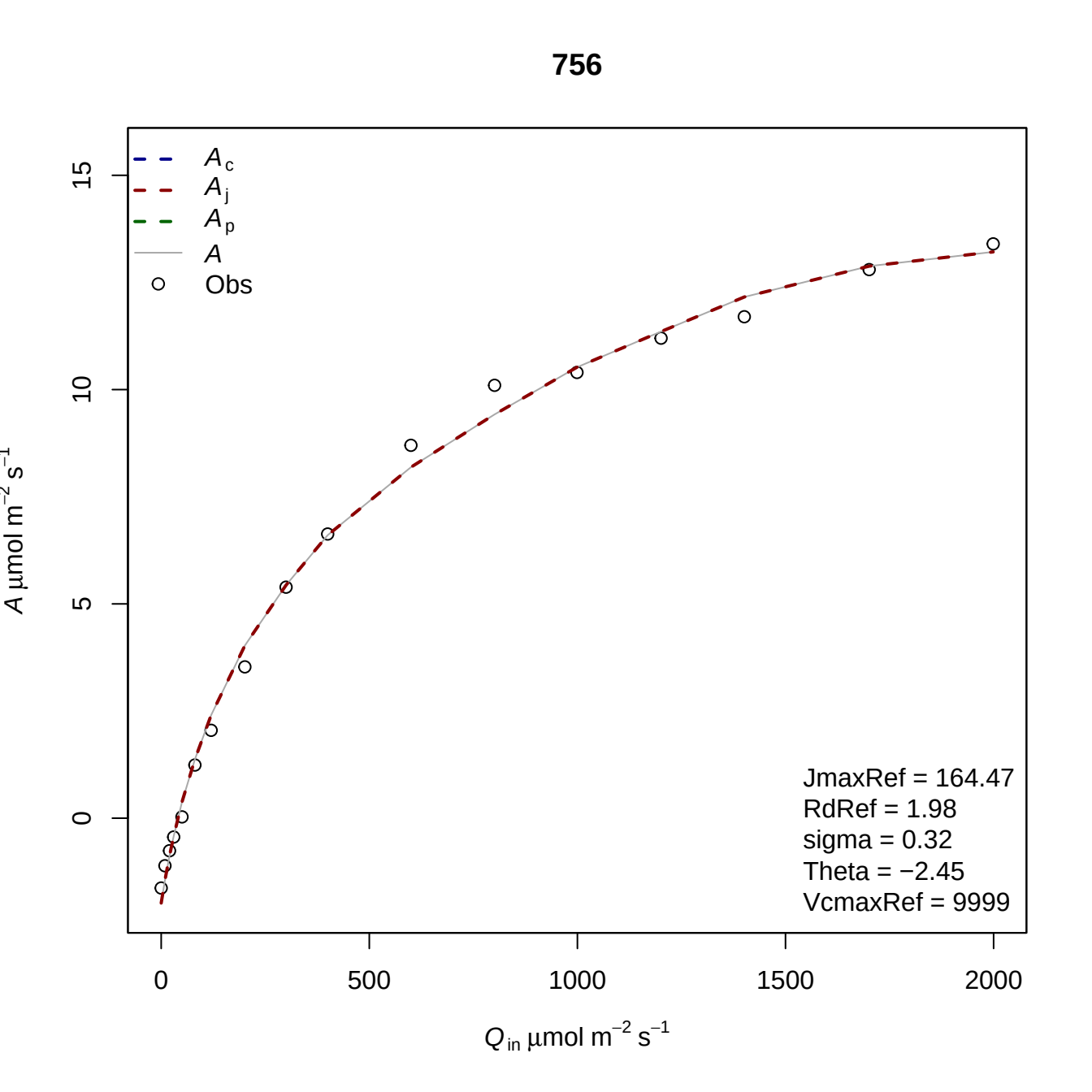

751
A_c
A_j
A_p
A
Obs
JmaxRef = 92.32
RdRef = 1.49
sigma = 0.2
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

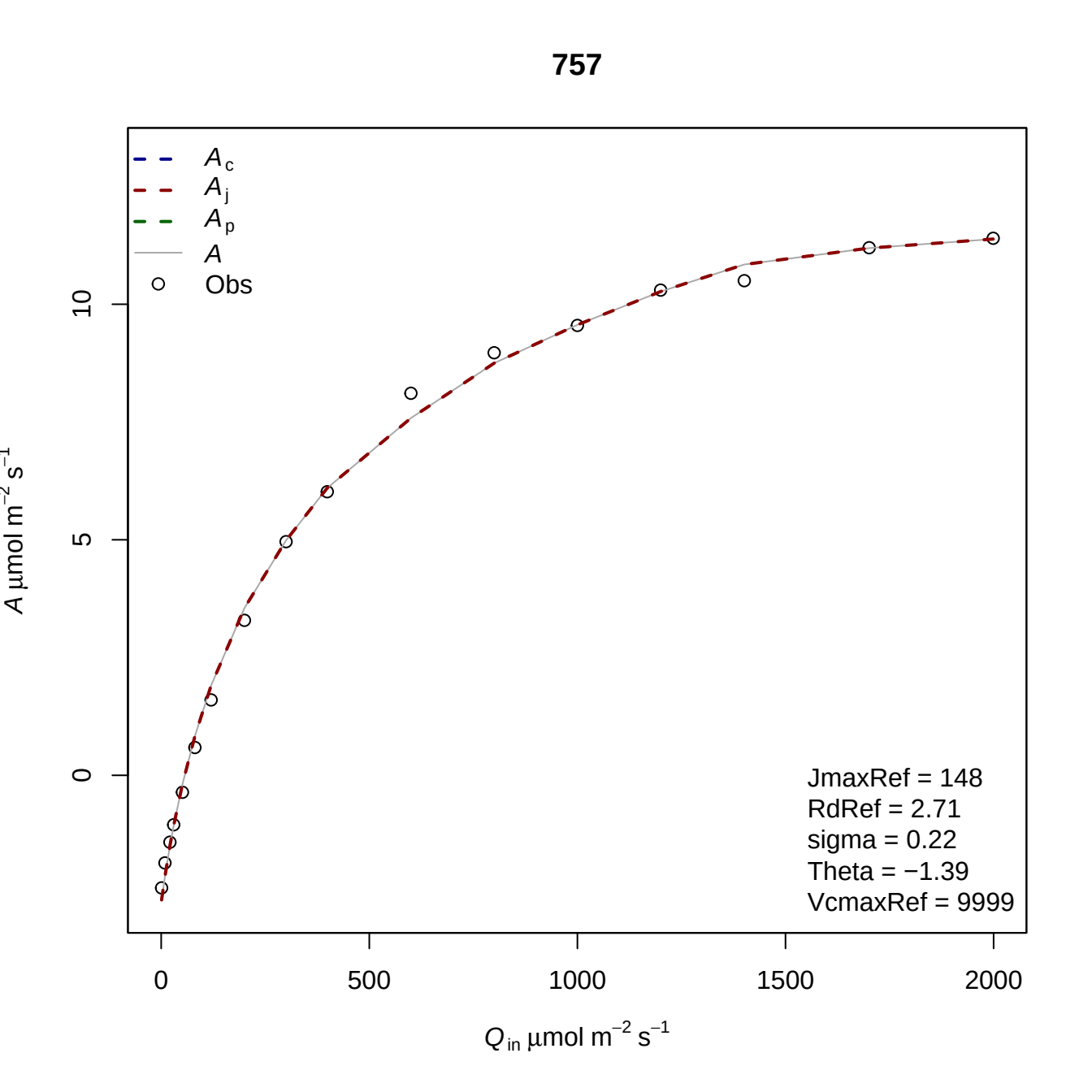

752
A_c
A_j
A_p
A
Obs
JmaxRef = 173.6
RdRef = 2.5
sigma = 0.33
Theta = −0.03
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

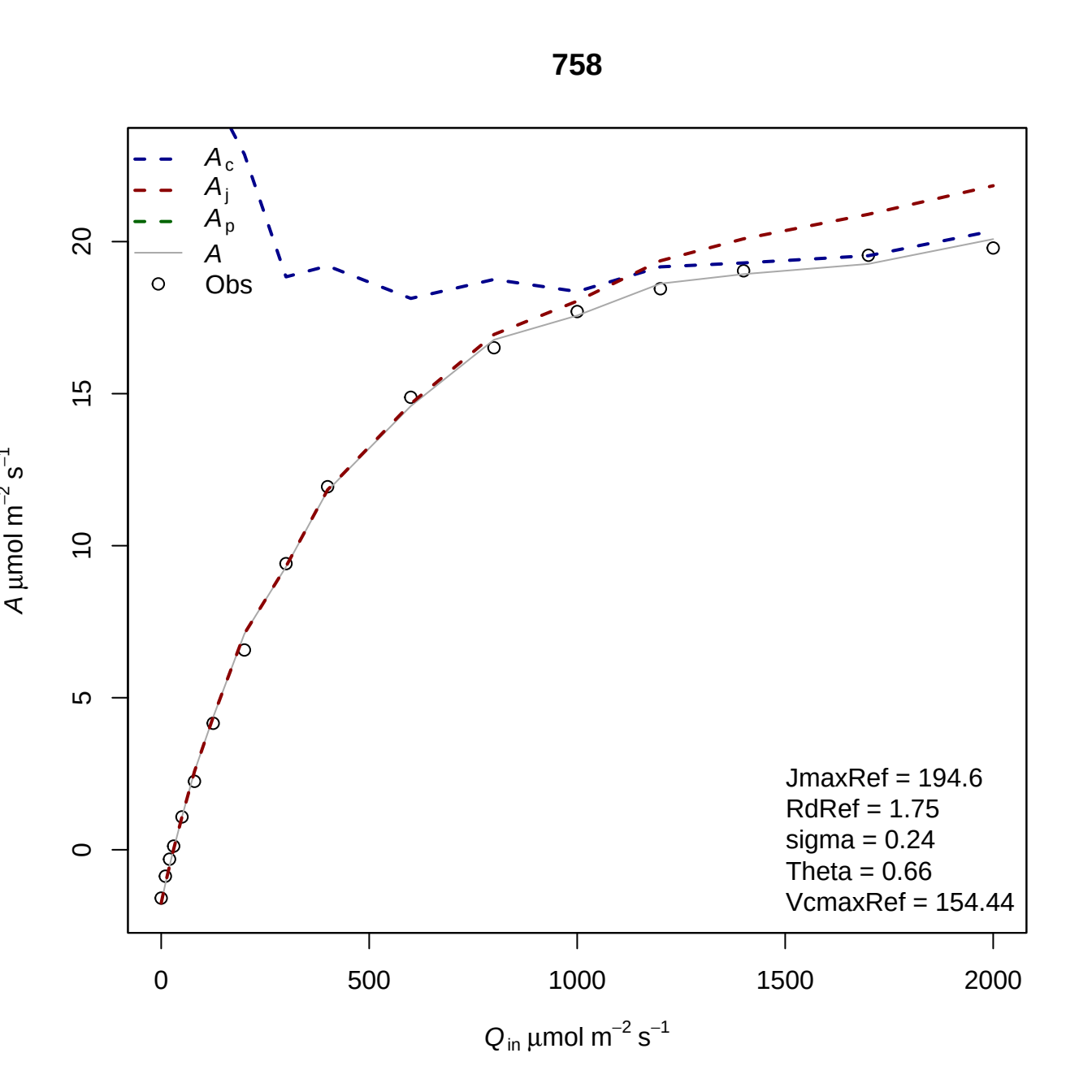

753
A_c
A_j
A_p
A
Obs
JmaxRef = 142.78
RdRef = 1.85
sigma = 0.23
Theta = 0.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

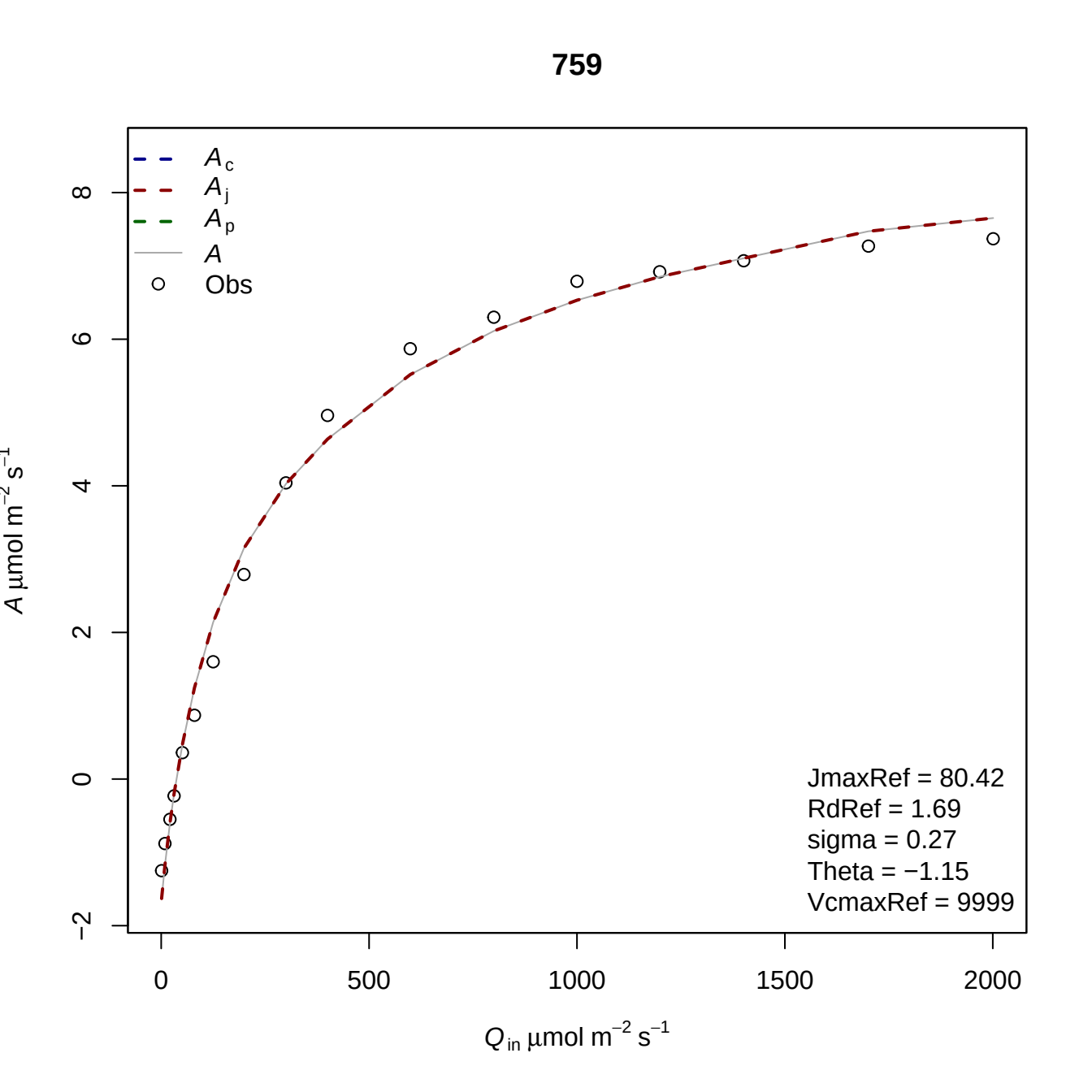

754
A_c
A_j
A_p
A
Obs
JmaxRef = 185.13
RdRef = 3.18
sigma = 0.25
Theta = −1.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

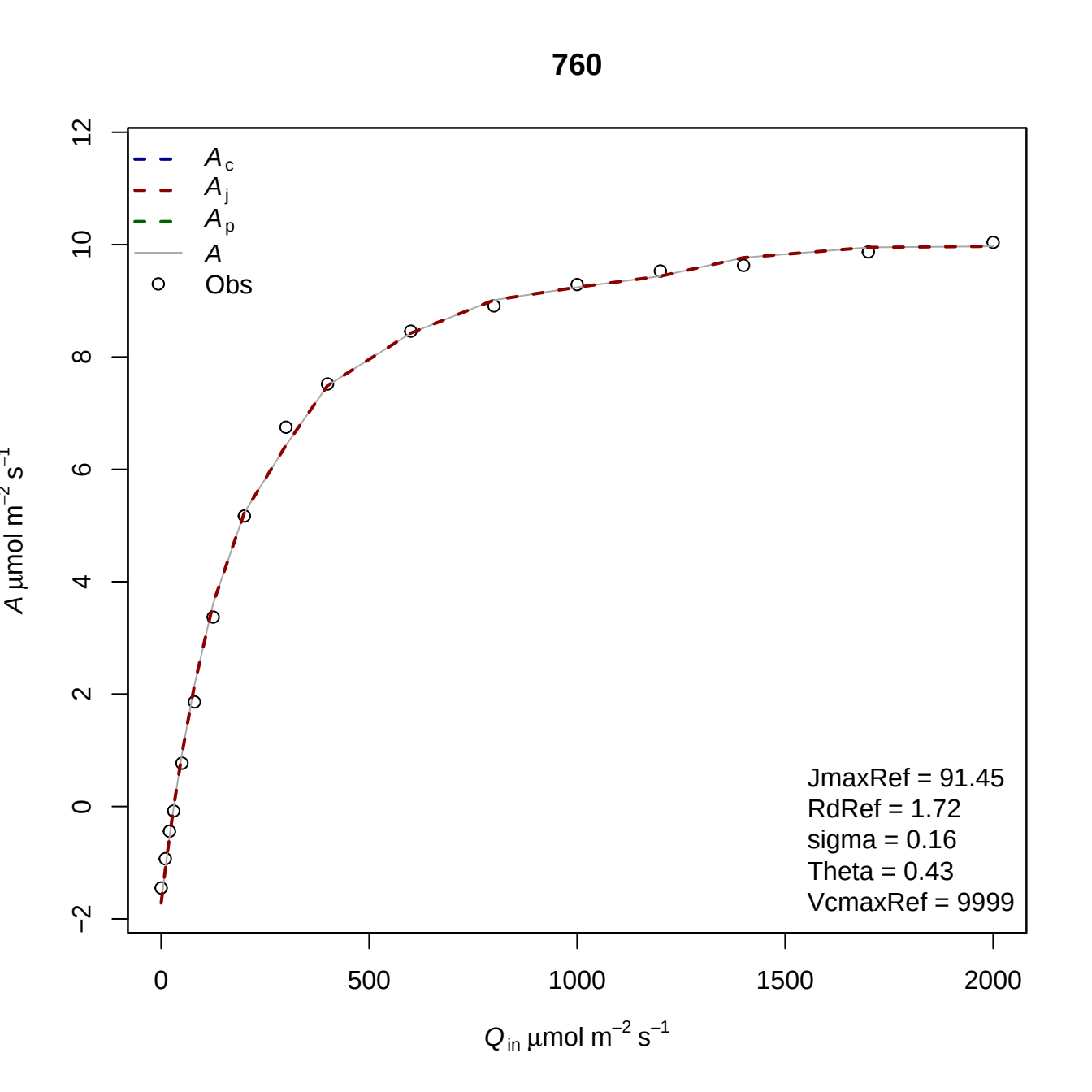

755
A_c
A_j
A_p
A
Obs
JmaxRef = 184.64
RdRef = 2.39
sigma = 0.26
Theta = 0.2
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

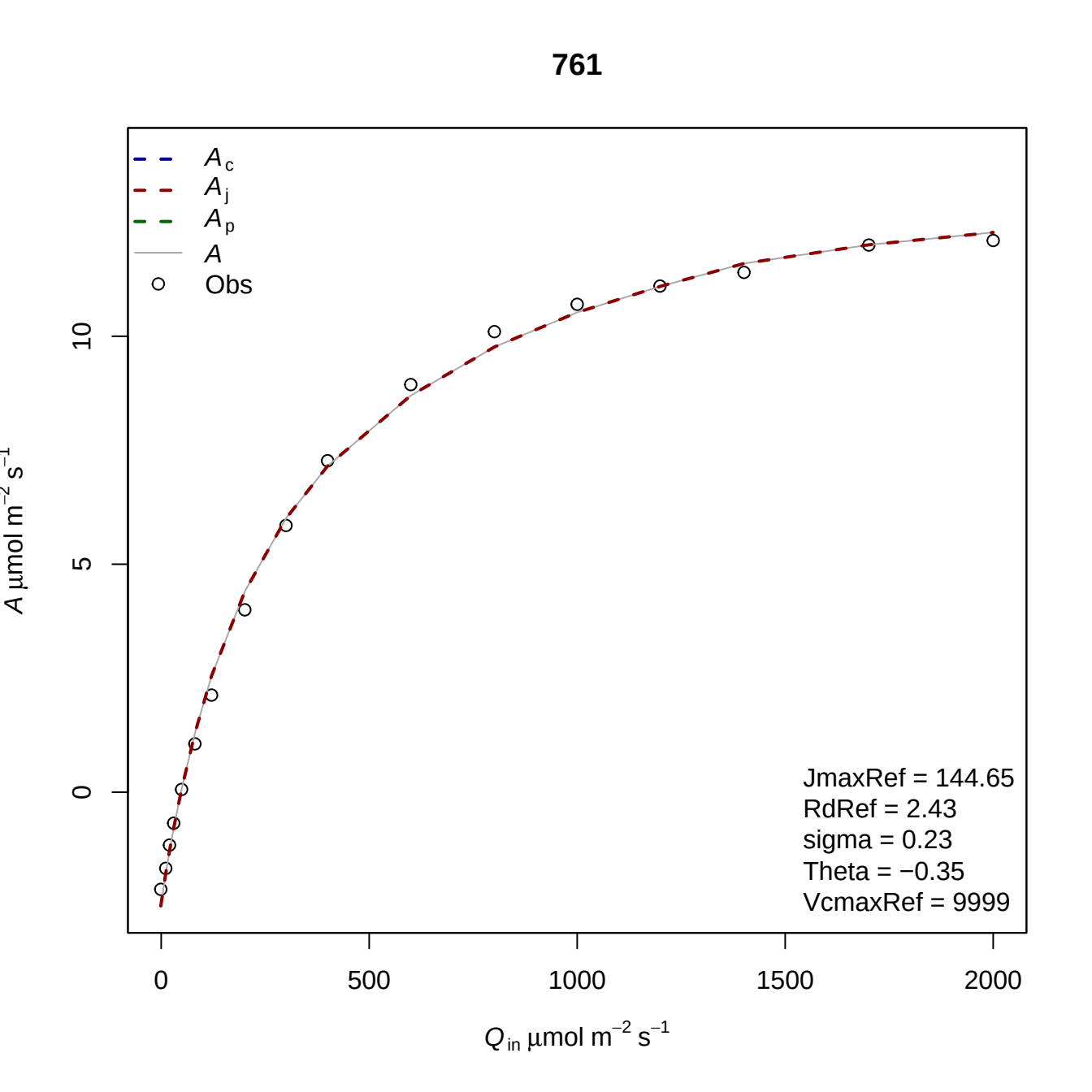

756
A_c
A_j
A_p
A
Obs
JmaxRef = 164.47
RdRef = 1.98
sigma = 0.32
Theta = −2.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

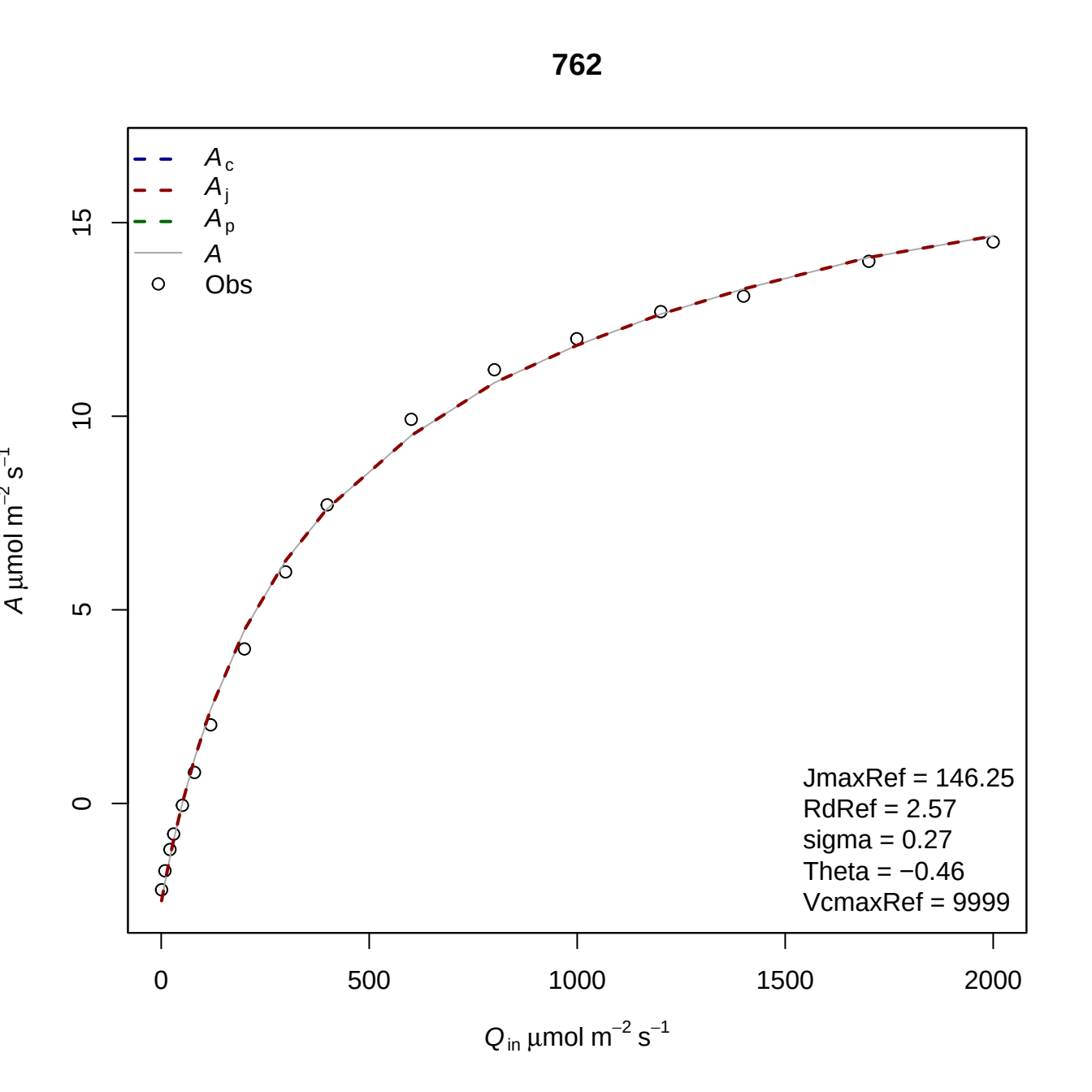

757
A_c
A_j
A_p
A
Obs
JmaxRef = 148
RdRef = 2.71
sigma = 0.22
Theta = −1.39
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

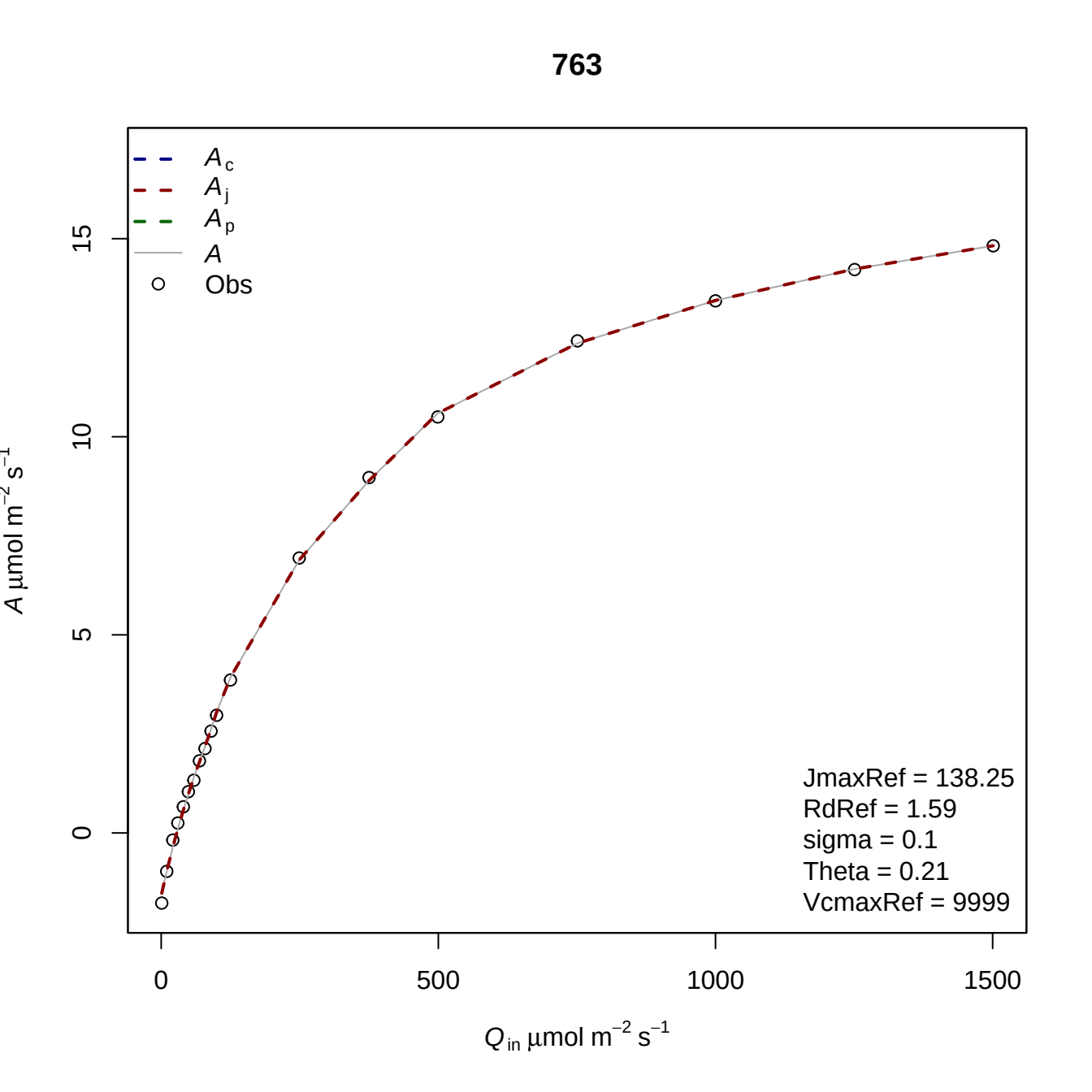

758
A_c
A_j
A_p
A
Obs
JmaxRef = 194.6
RdRef = 1.75
sigma = 0.24
Theta = 0.66
VcmaxRef = 154.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

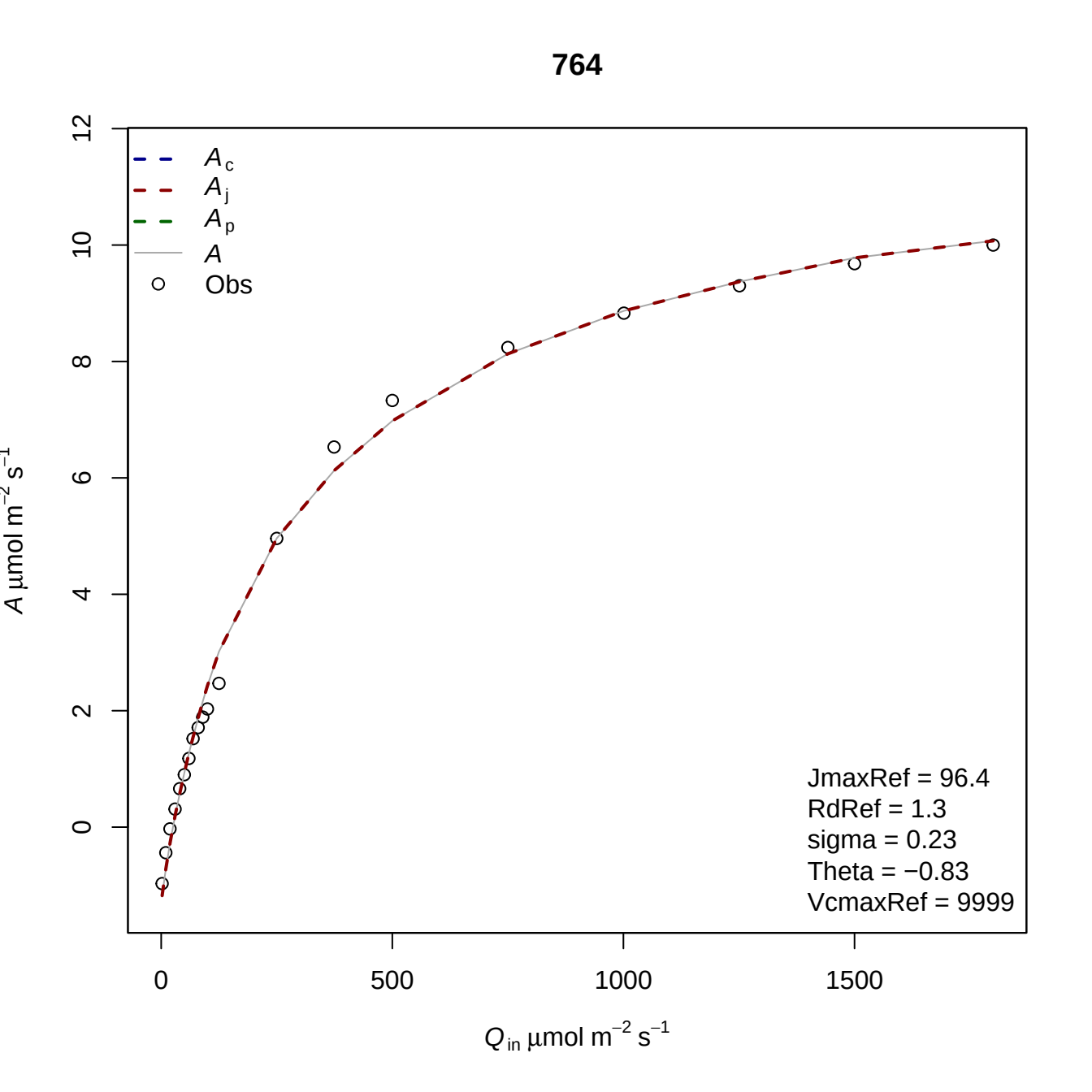

759
A_c
A_j
A_p
A
Obs
JmaxRef = 80.42
RdRef = 1.69
sigma = 0.27
Theta = −1.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

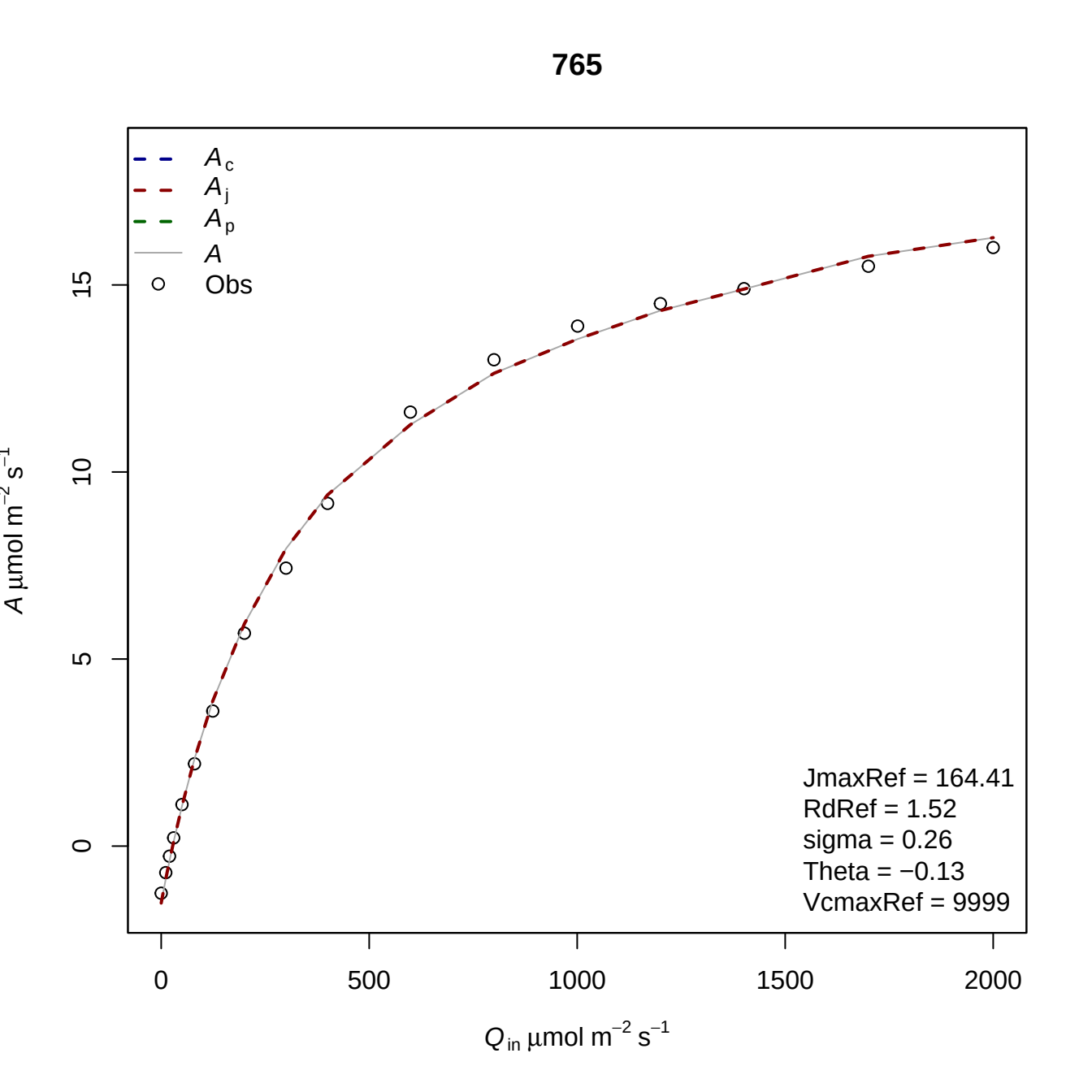

760
A_c
A_j
A_p
A
Obs
JmaxRef = 91.45
RdRef = 1.72
sigma = 0.16
Theta = 0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

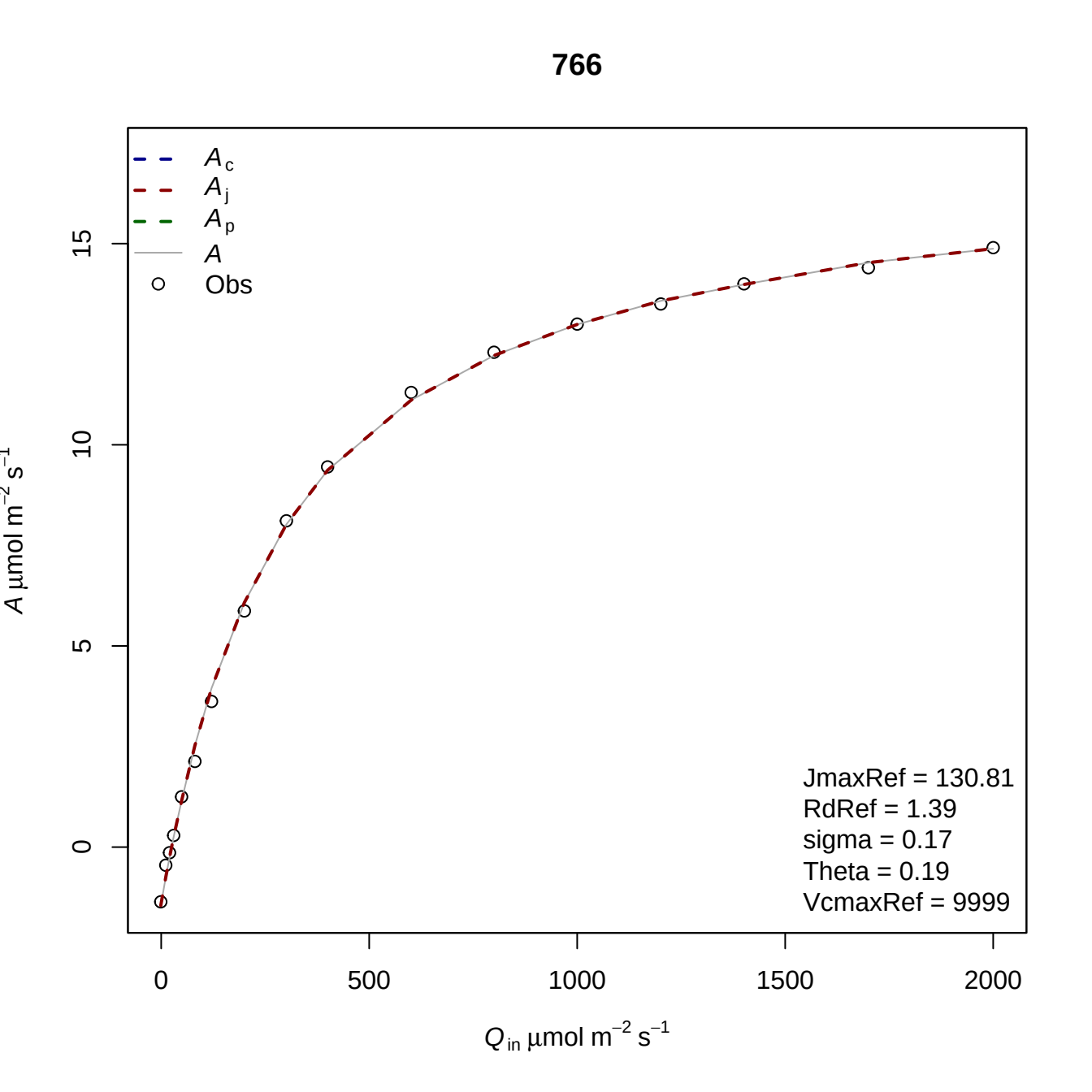

761
A_c
A_j
A_p
A
Obs
JmaxRef = 144.65
RdRef = 2.43
sigma = 0.23
Theta = −0.35
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

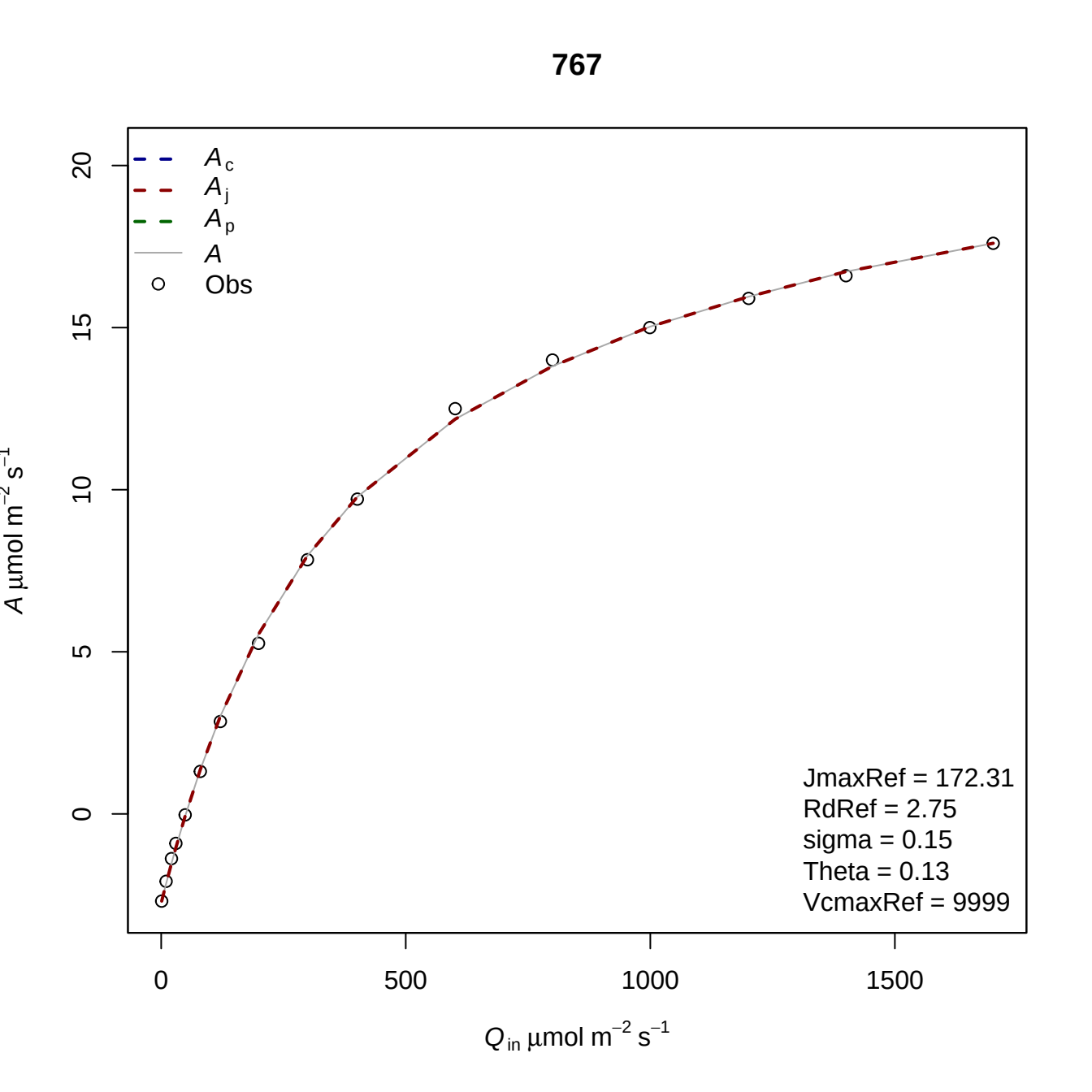

762
A_c
A_j
A_p
A
Obs
JmaxRef = 146.25
RdRef = 2.57
sigma = 0.27
Theta = −0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

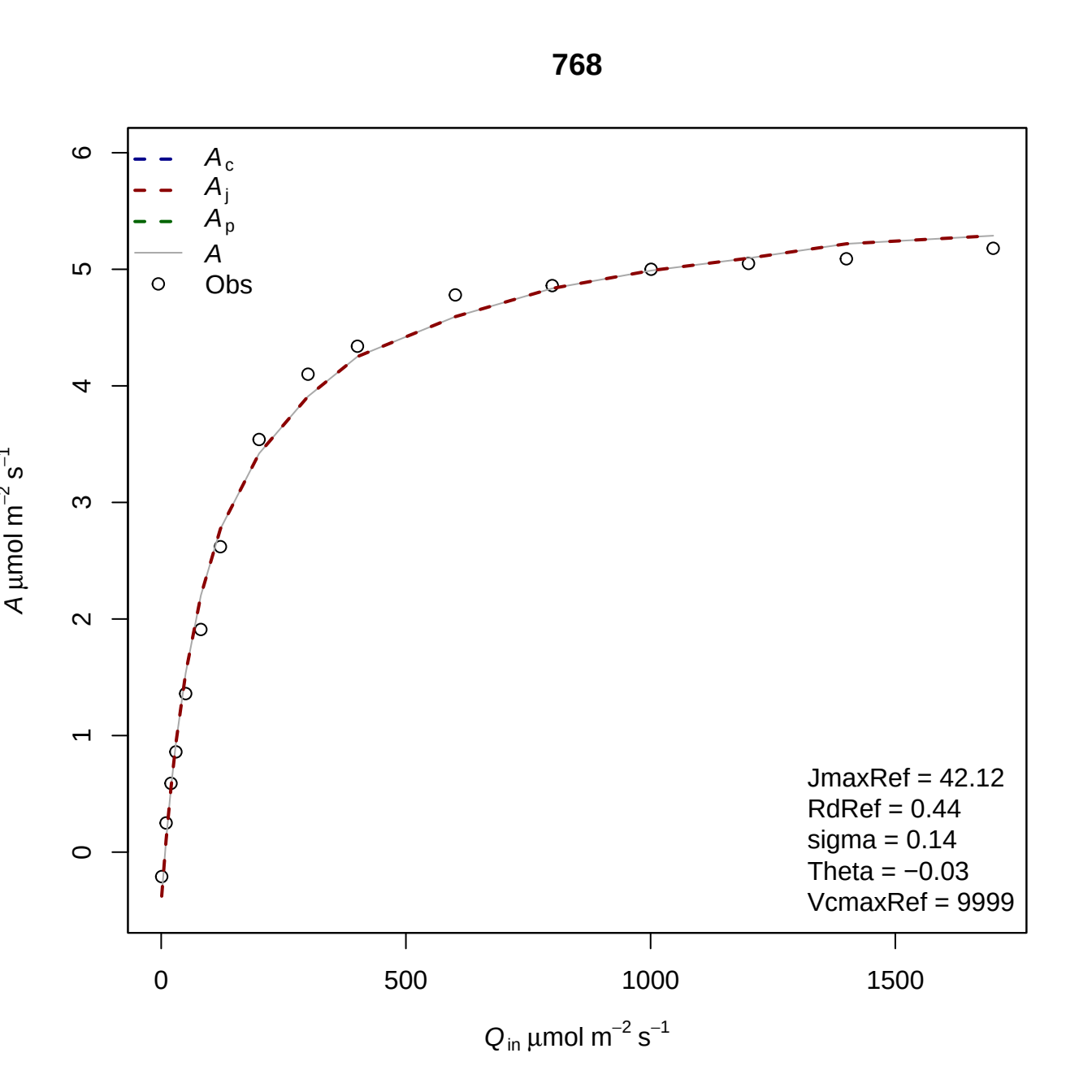

763
A_c
A_j
A_p
A
Obs
JmaxRef = 138.25
RdRef = 1.59
sigma = 0.1
Theta = 0.21
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

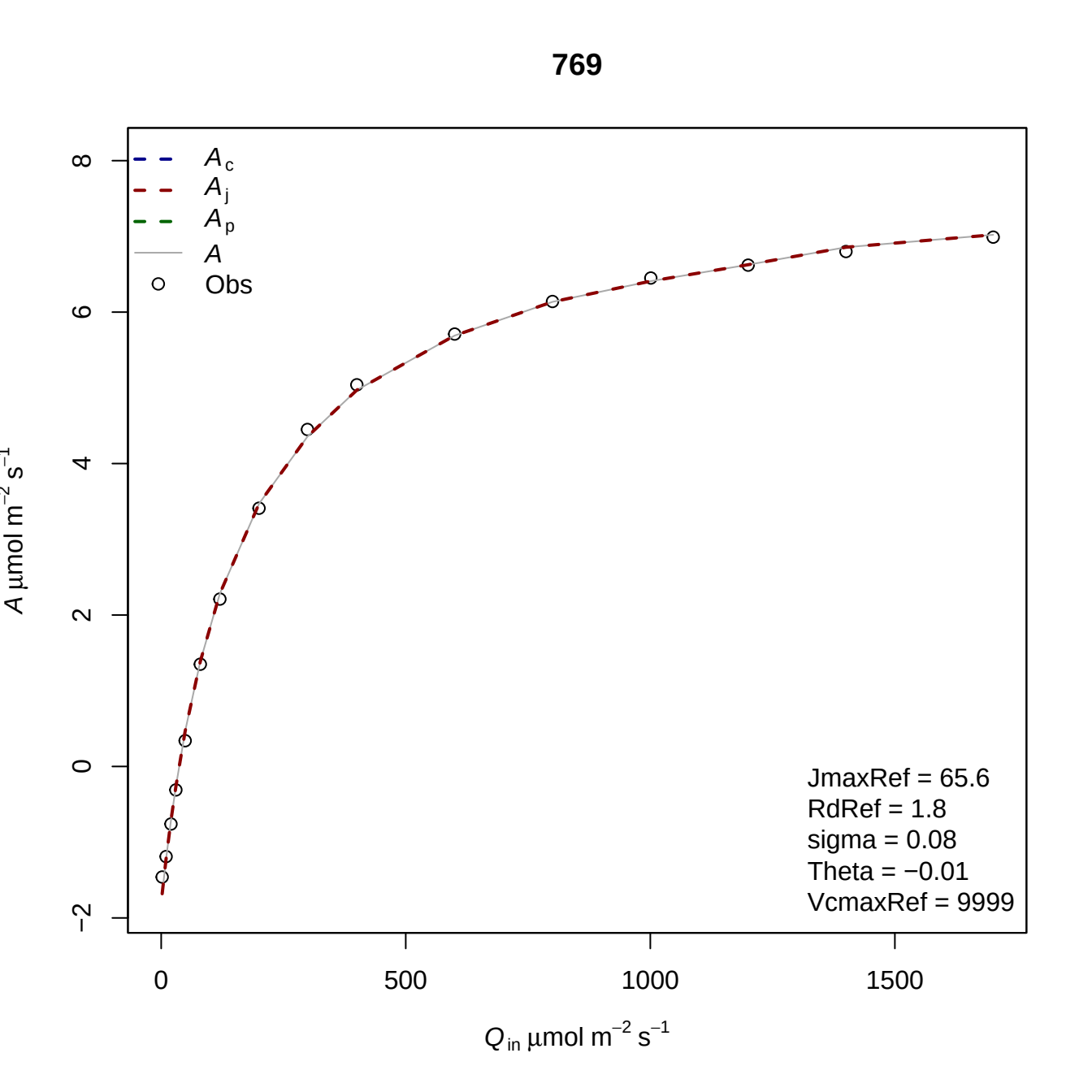

764
A_c
A_j
A_p
A
Obs
JmaxRef = 96.4
RdRef = 1.3
sigma = 0.23
Theta = −0.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

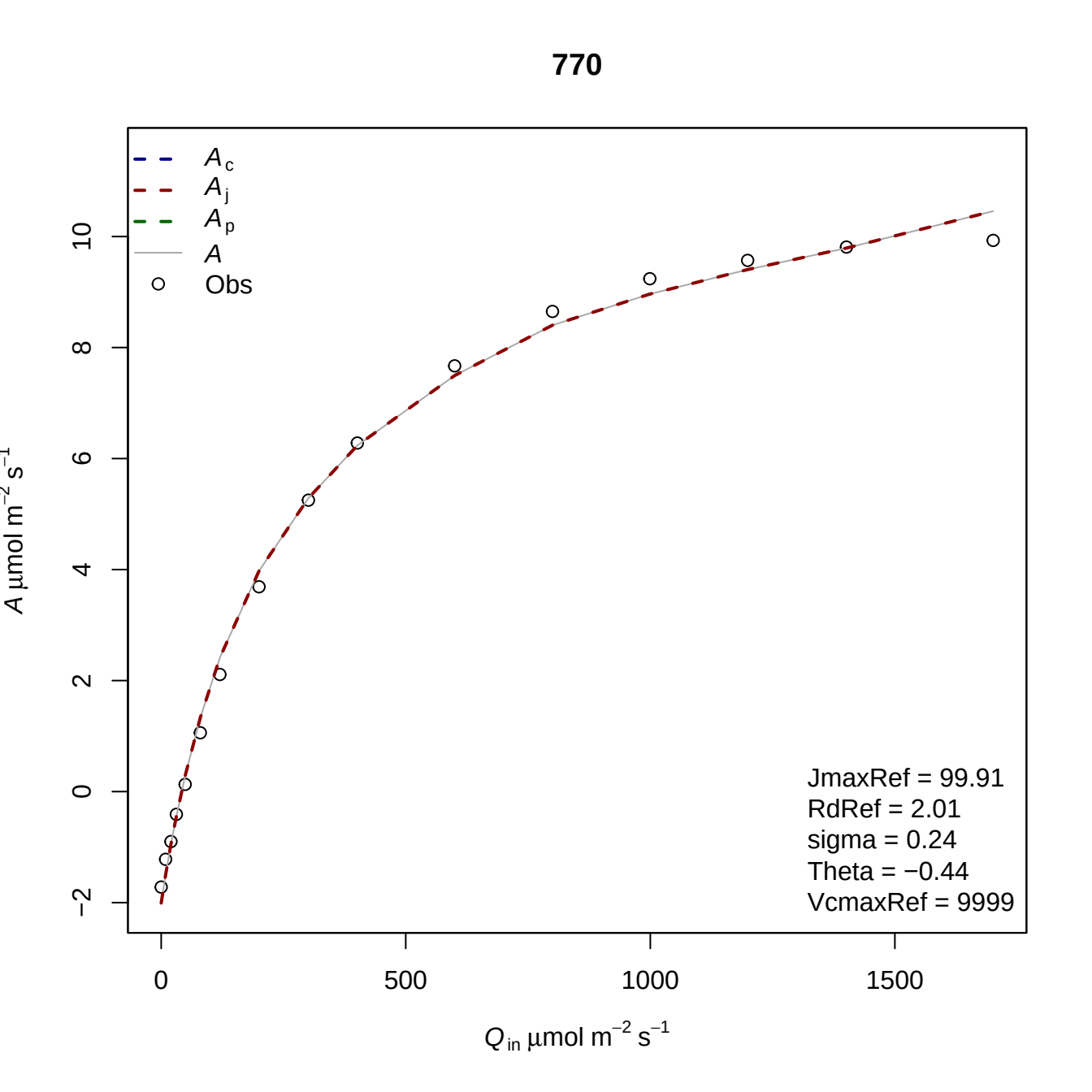

765
A_c
A_j
A_p
A
Obs
JmaxRef = 164.41
RdRef = 1.52
sigma = 0.26
Theta = −0.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

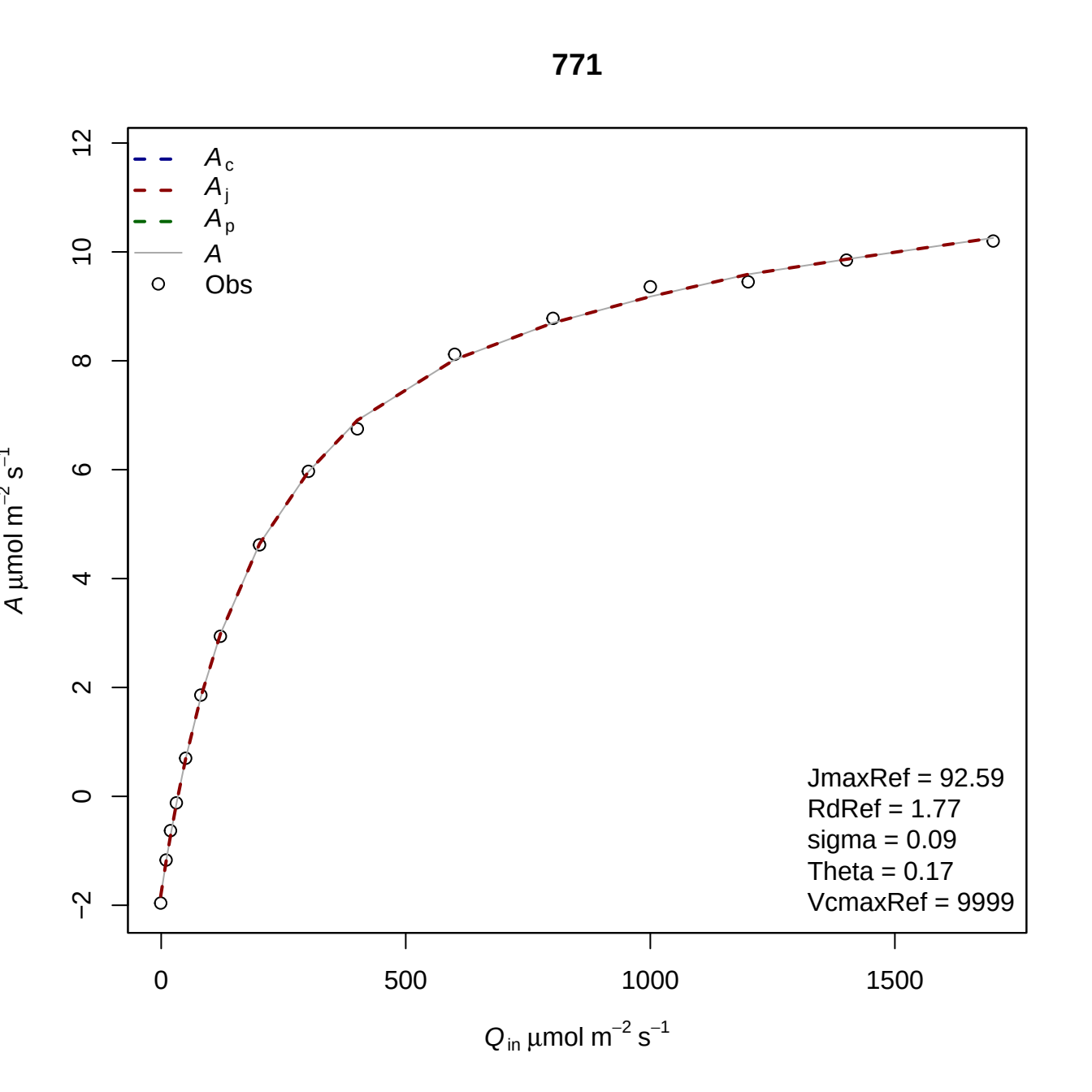

766
A_c
A_j
A_p
A
Obs
JmaxRef = 130.81
RdRef = 1.39
sigma = 0.17
Theta = 0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

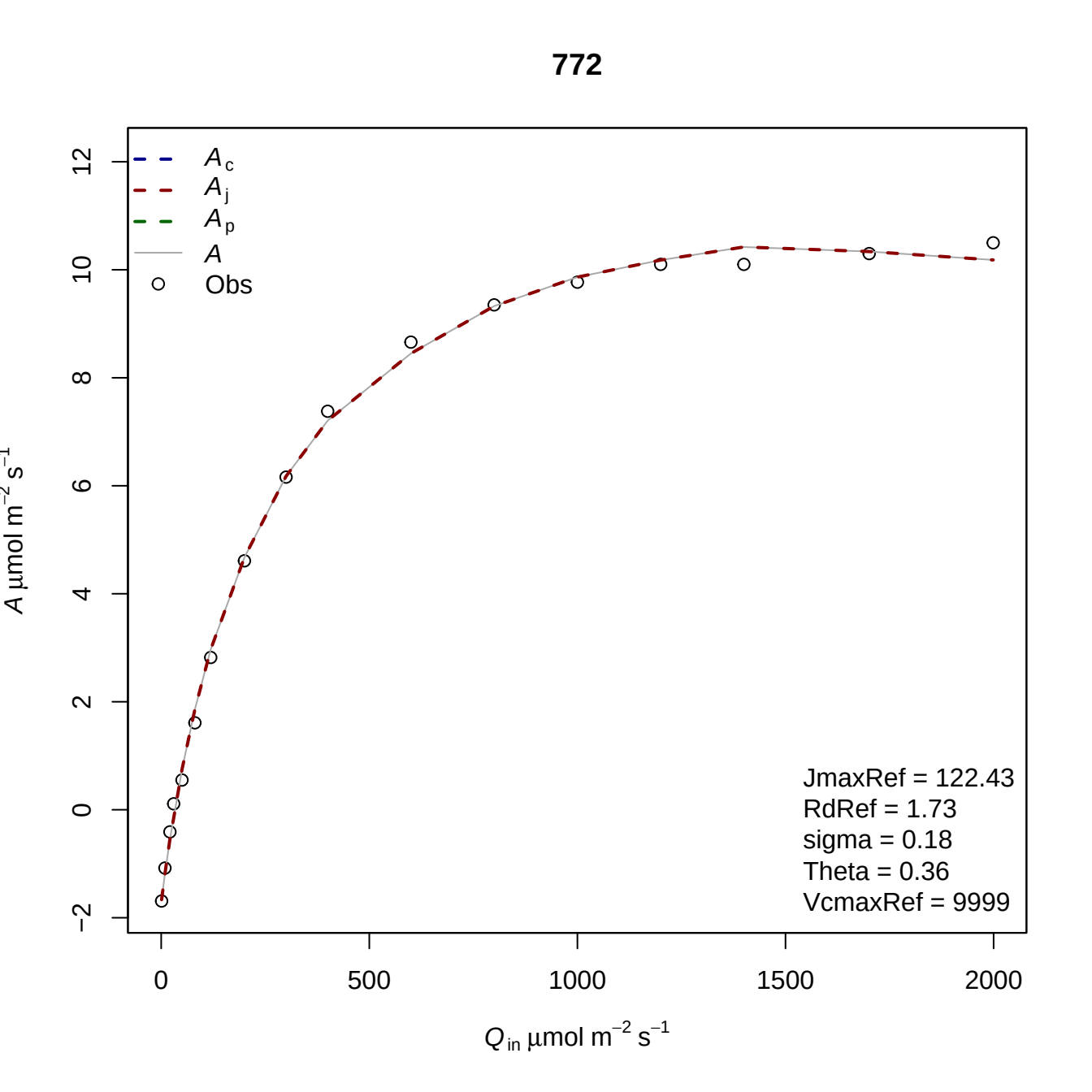

767
A_c
A_j
A_p
A
Obs
JmaxRef = 172.31
RdRef = 2.75
sigma = 0.15
Theta = 0.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

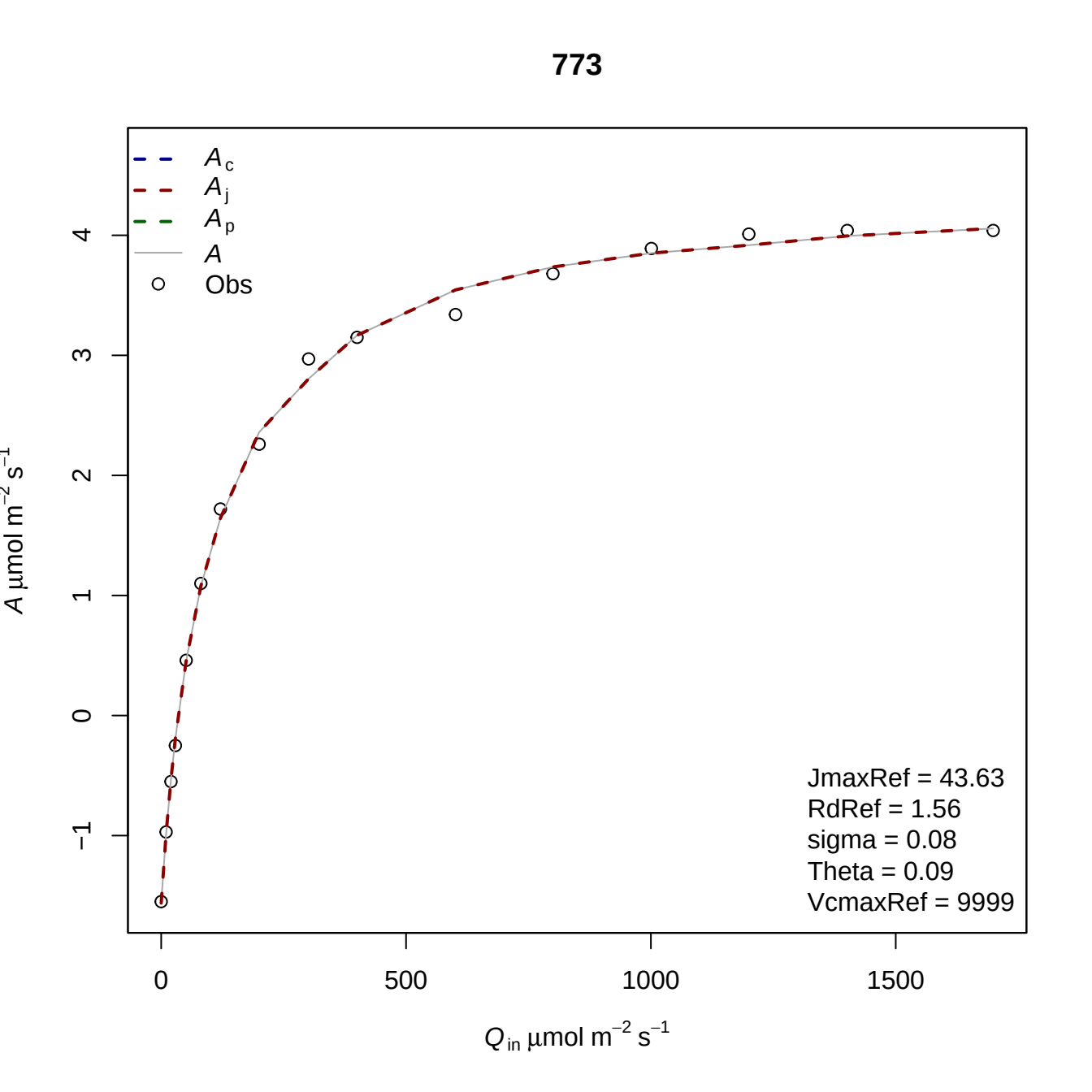

768
A_c
A_j
A_p
A
Obs
JmaxRef = 42.12
RdRef = 0.44
sigma = 0.14
Theta = −0.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

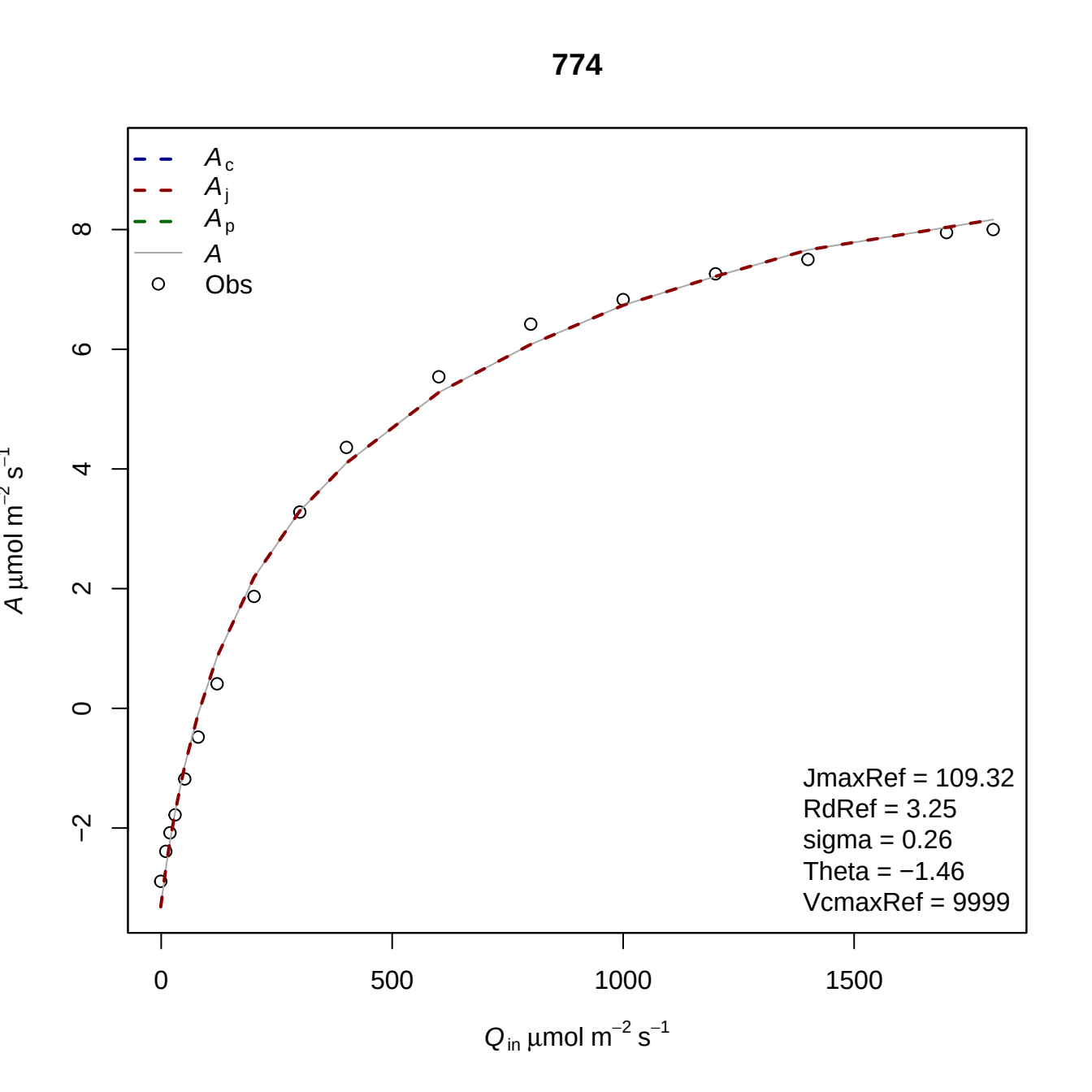

769
A_c
A_j
A_p
A
Obs
JmaxRef = 65.6
RdRef = 1.8
sigma = 0.08
Theta = −0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

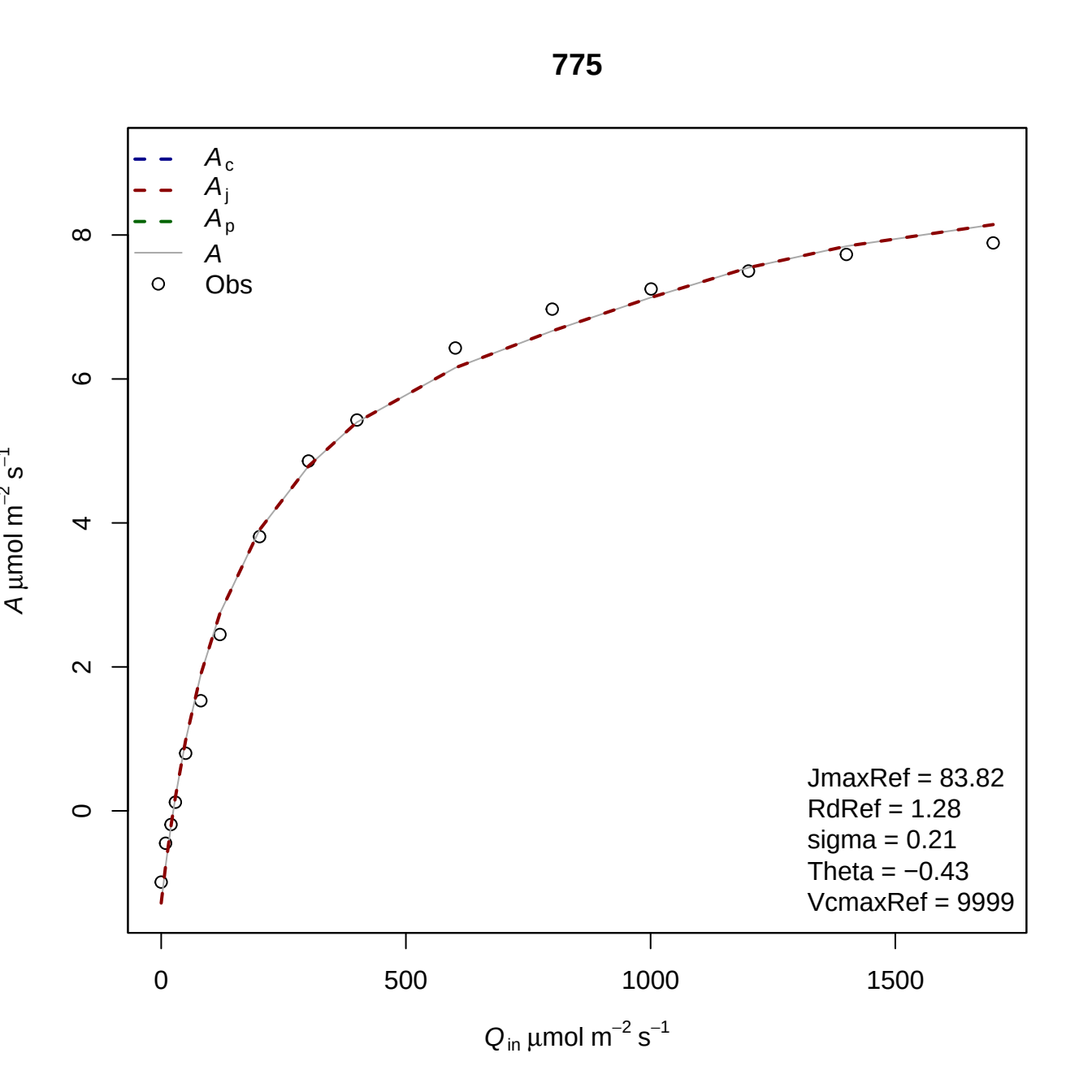

770
A_c
A_j
A_p
A
Obs
JmaxRef = 99.91
RdRef = 2.01
sigma = 0.24
Theta = −0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

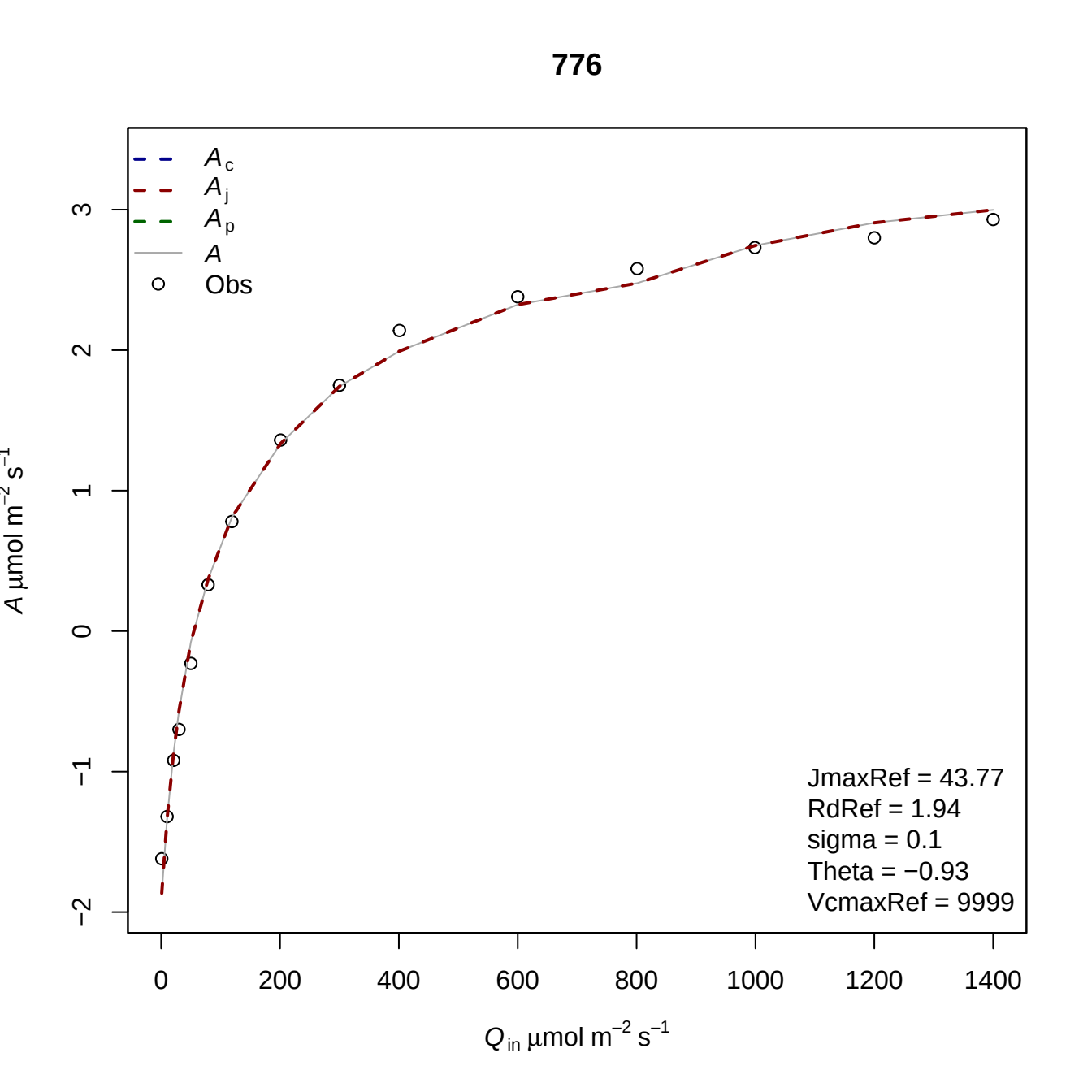

771
A_c
A_j
A_p
A
Obs
JmaxRef = 92.59
RdRef = 1.77
sigma = 0.09
Theta = 0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

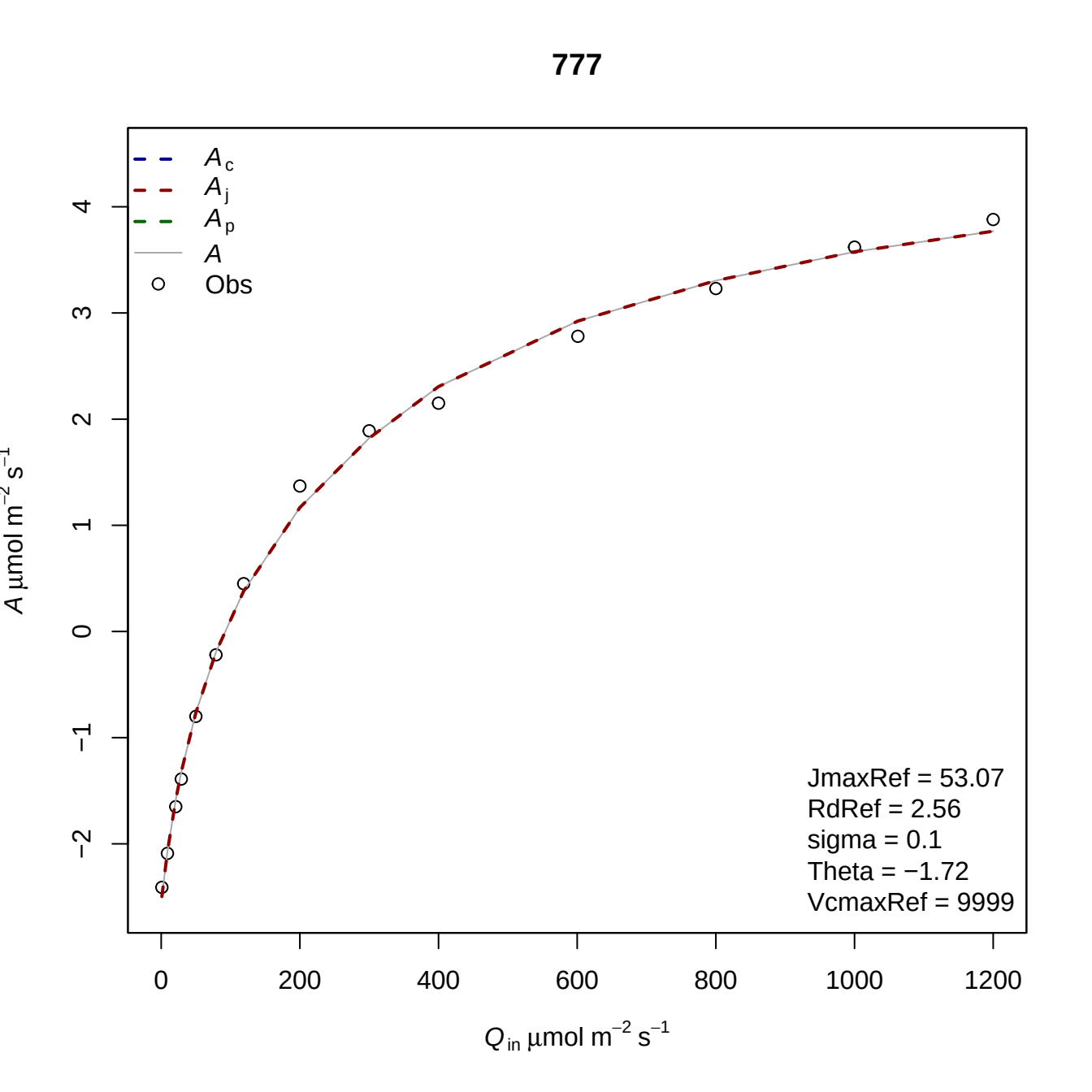

772
A_c
A_j
A_p
A
Obs
JmaxRef = 122.43
RdRef = 1.73
sigma = 0.18
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

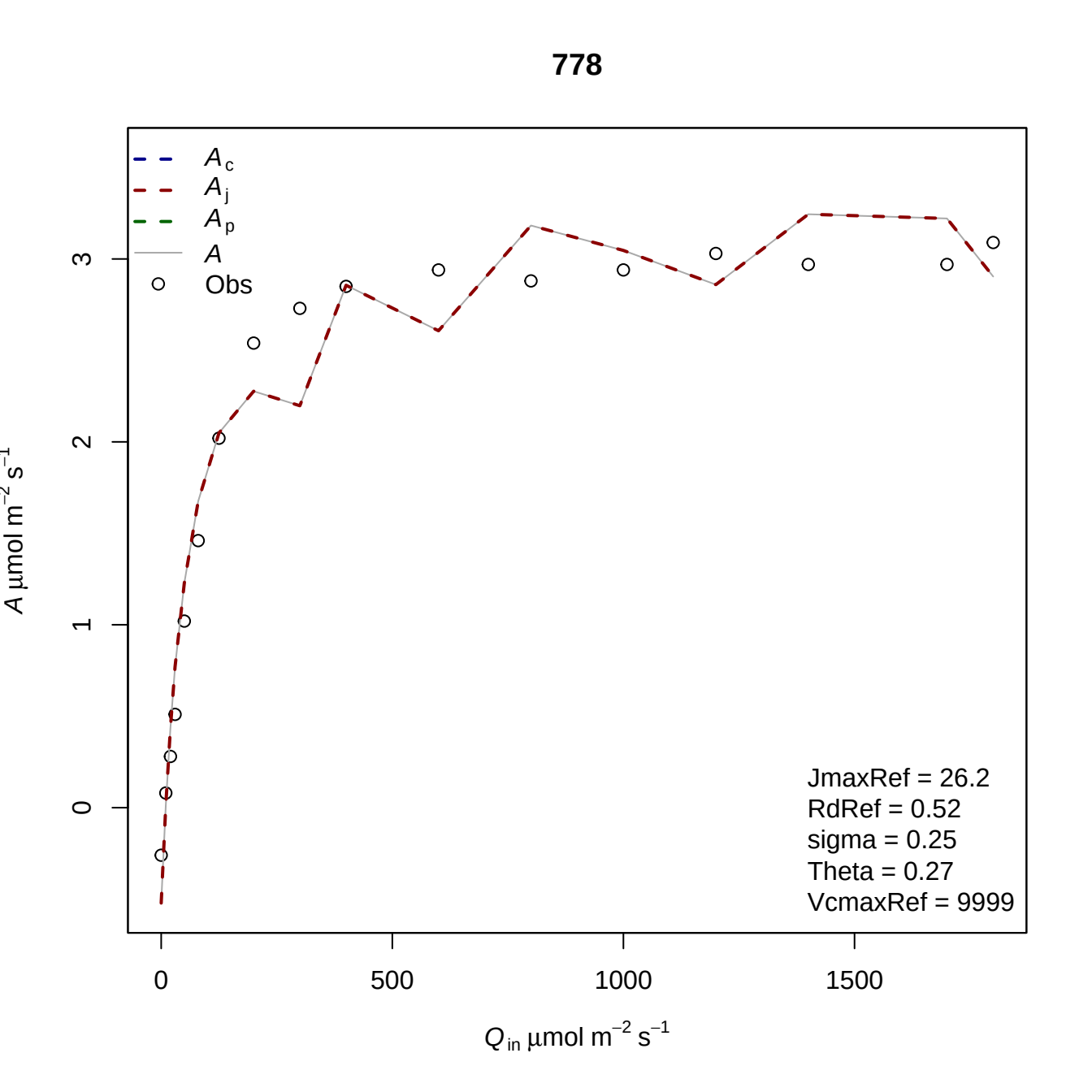

773
A_c
A_j
A_p
A
Obs
JmaxRef = 43.63
RdRef = 1.56
sigma = 0.08
Theta = 0.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

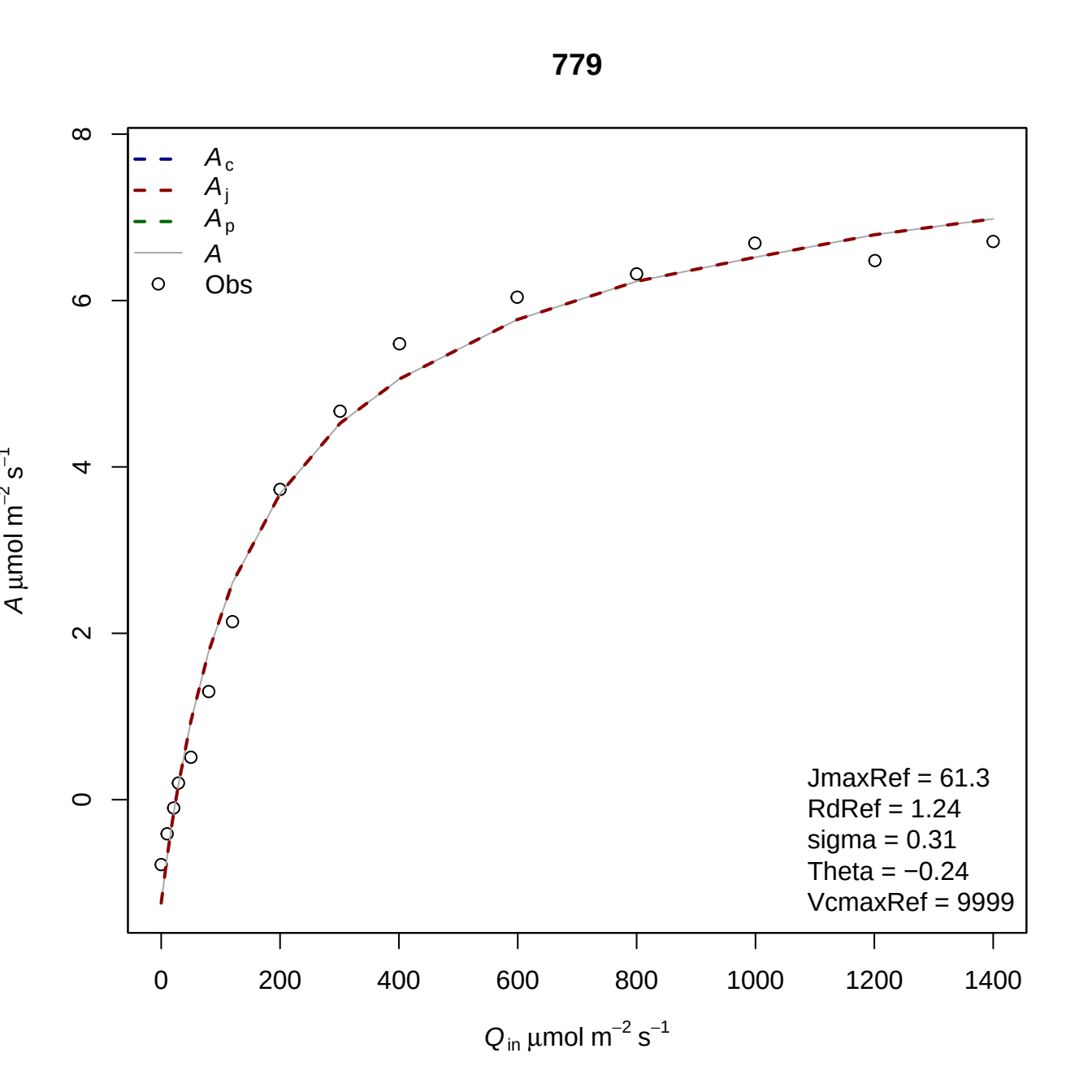

774
A_c
A_j
A_p
A
Obs
JmaxRef = 109.32
RdRef = 3.25
sigma = 0.26
Theta = −1.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

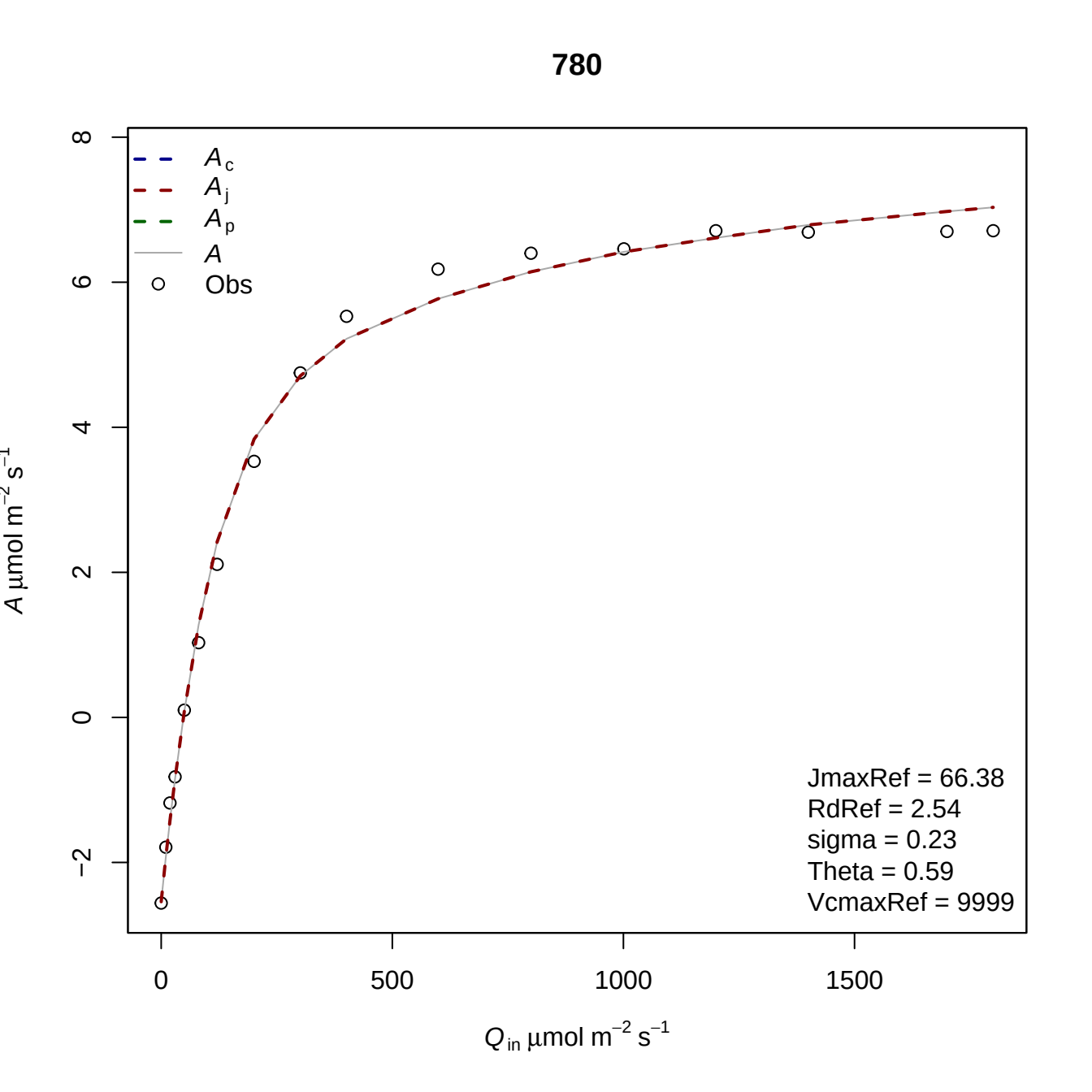

775
A_c
A_j
A_p
A
Obs
JmaxRef = 83.82
RdRef = 1.28
sigma = 0.21
Theta = −0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

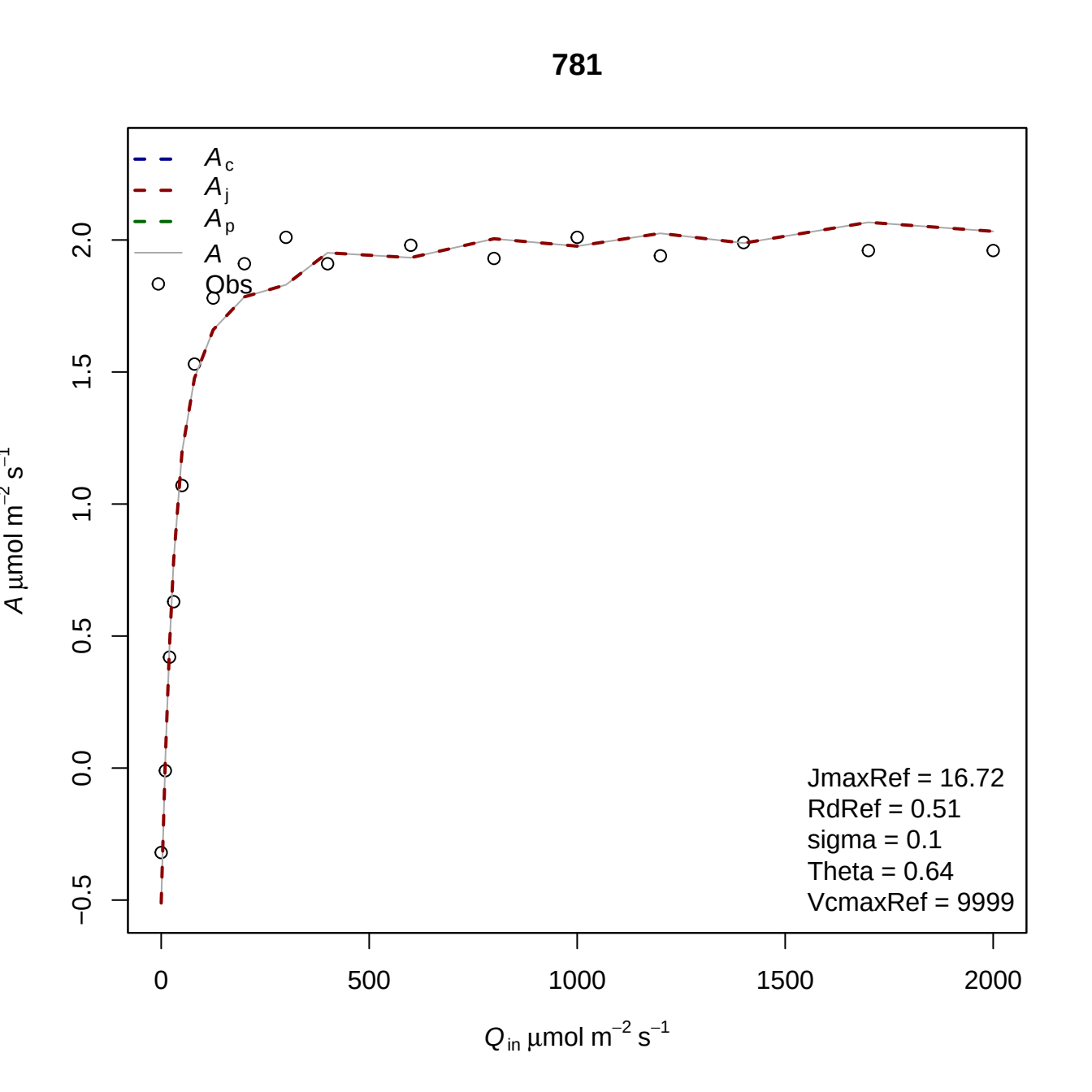

776
A_c
A_j
A_p
A
Obs
JmaxRef = 43.77
RdRef = 1.94
sigma = 0.1
Theta = −0.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

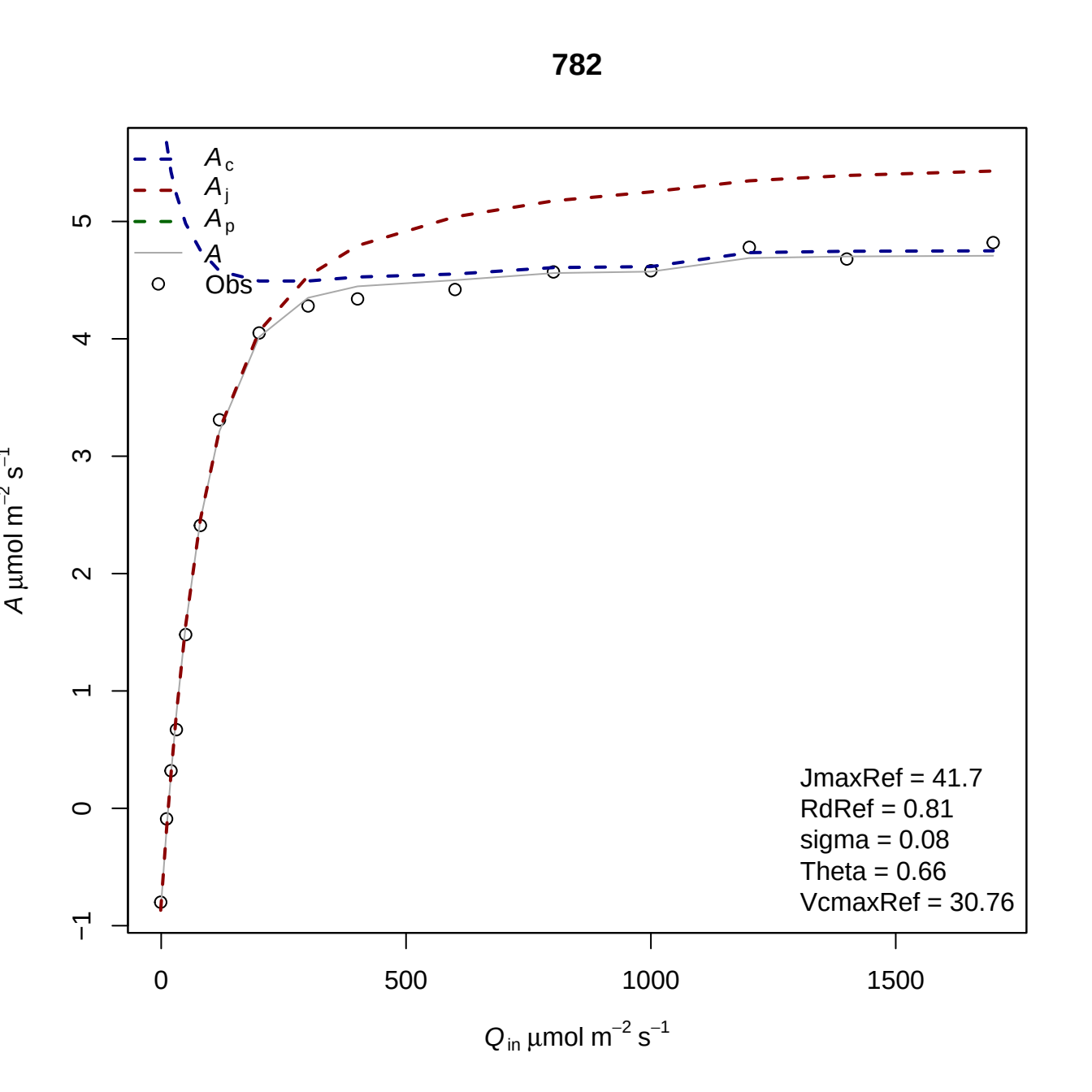

777
A_c
A_j
A_p
A
Obs
JmaxRef = 53.07
RdRef = 2.56
sigma = 0.1
Theta = −1.72
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
200
400
600
800
1000
1200

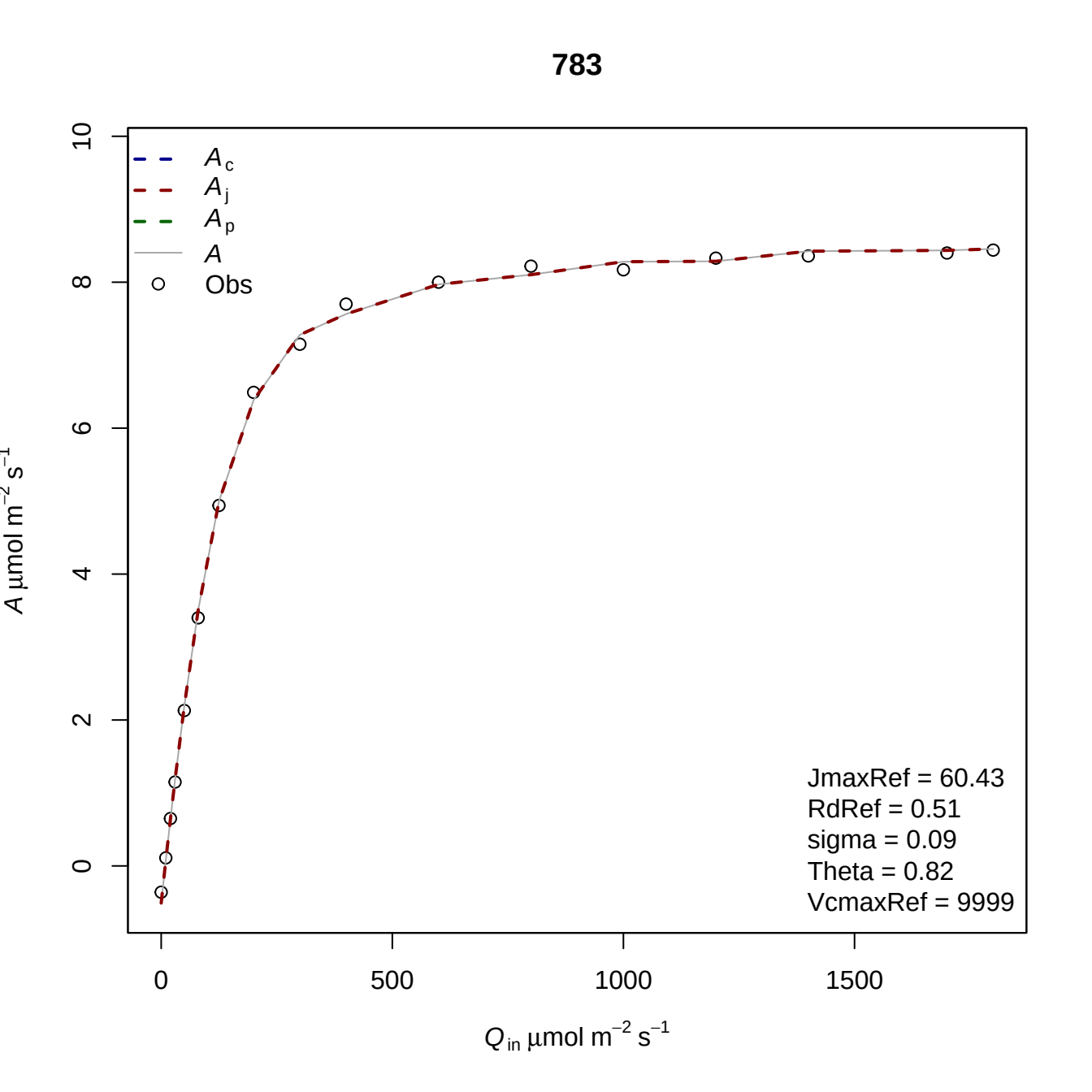

778
A_c
A_j
A_p
A
Obs
JmaxRef = 26.2
RdRef = 0.52
sigma = 0.25
Theta = 0.27
VcmaxRef = 9999
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

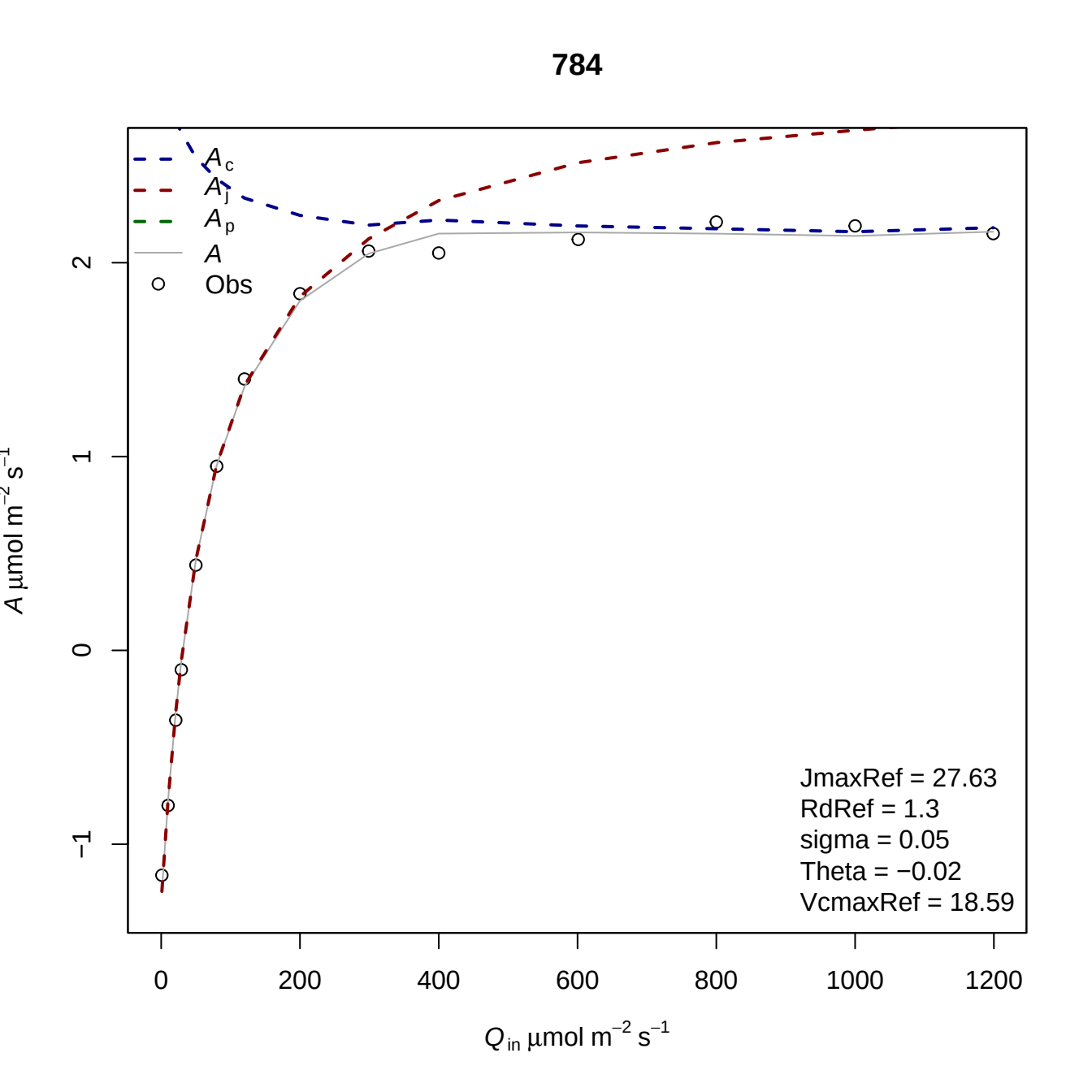

779
A_c
A_j
A_p
A
Obs
JmaxRef = 61.3
RdRef = 1.24
sigma = 0.31
Theta = −0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

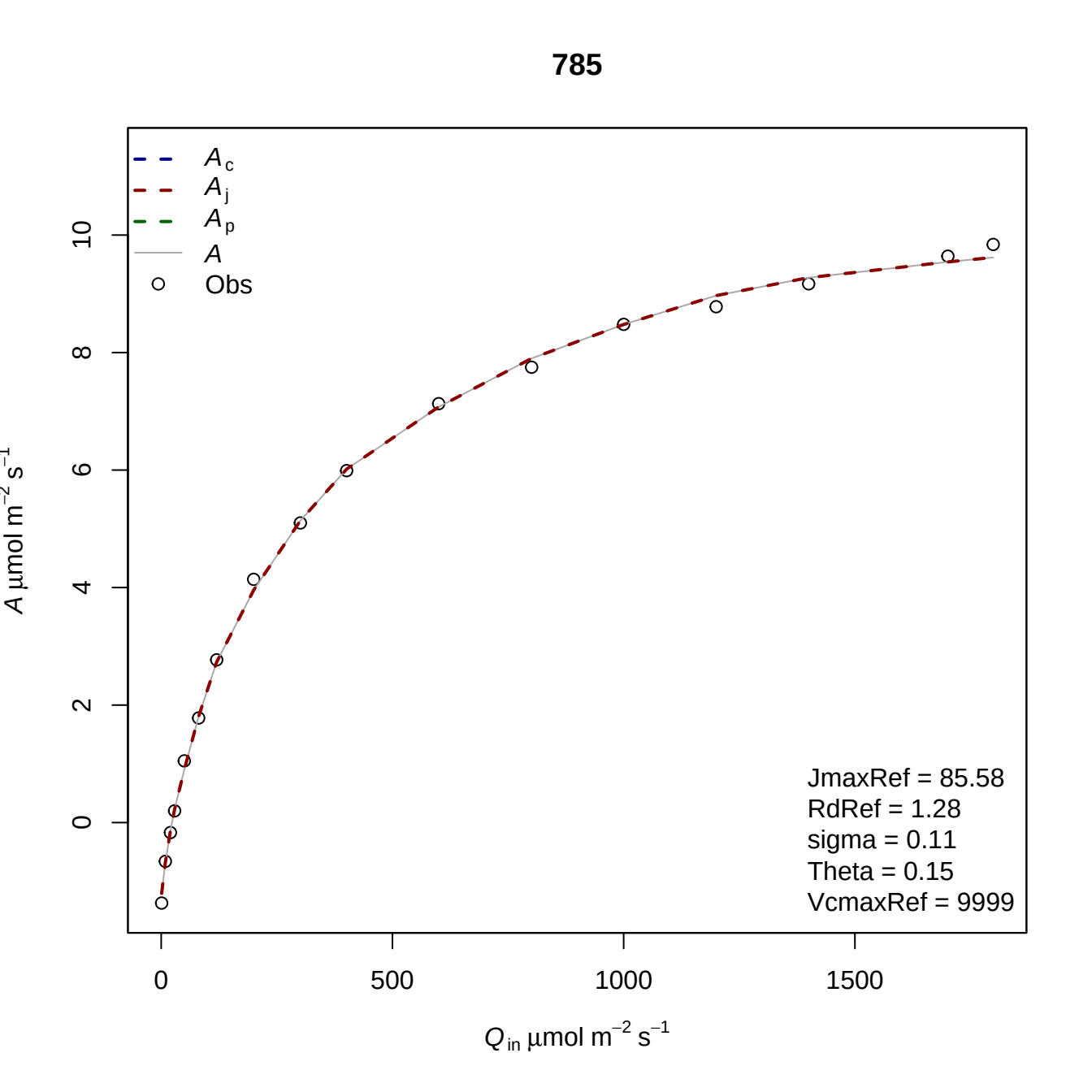

780
A_c
A_j
A_p
A
Obs
JmaxRef = 66.38
RdRef = 2.54
sigma = 0.23
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

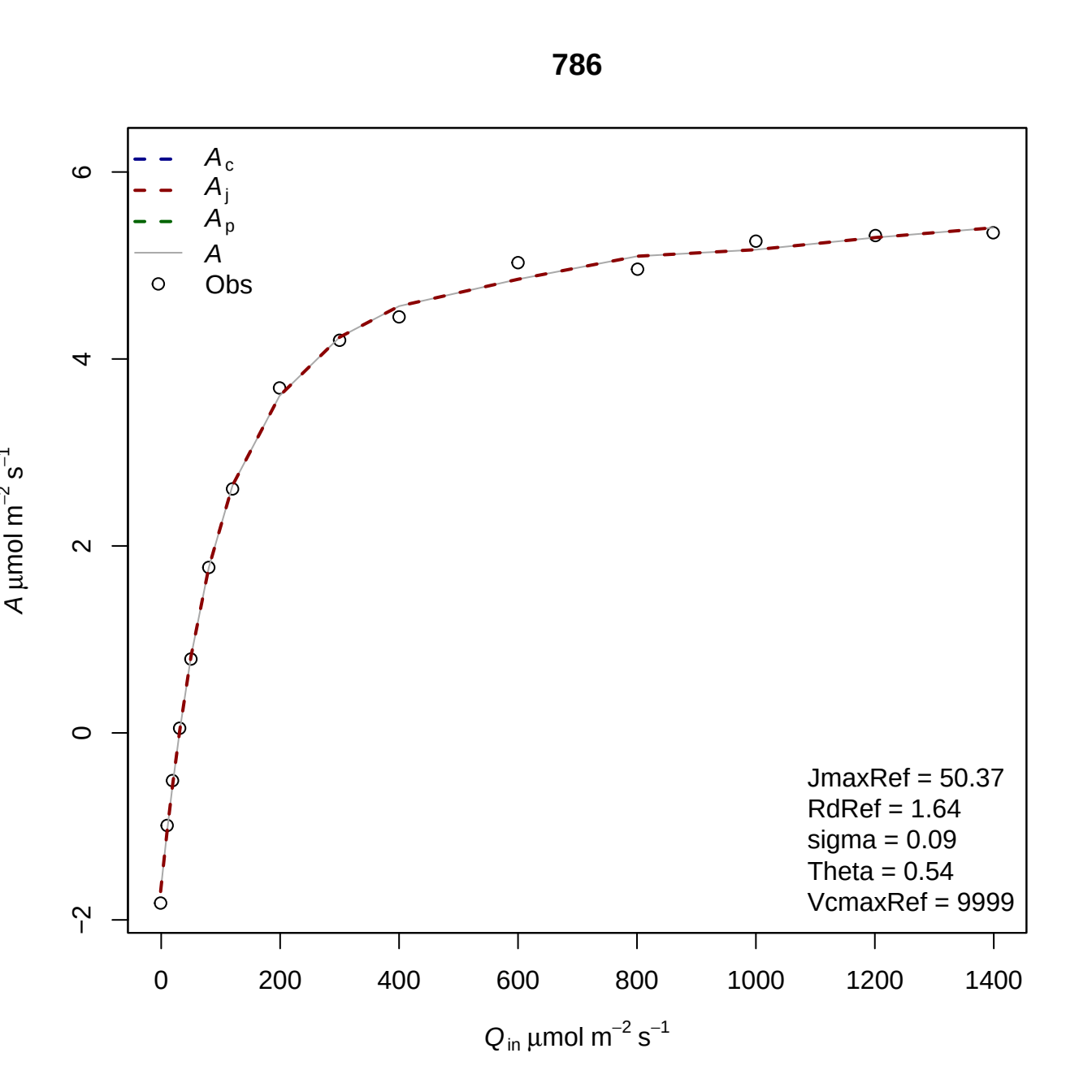

781
A_c
A_j
A_p
A
Obs
JmaxRef = 16.72
RdRef = 0.51
sigma = 0.1
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

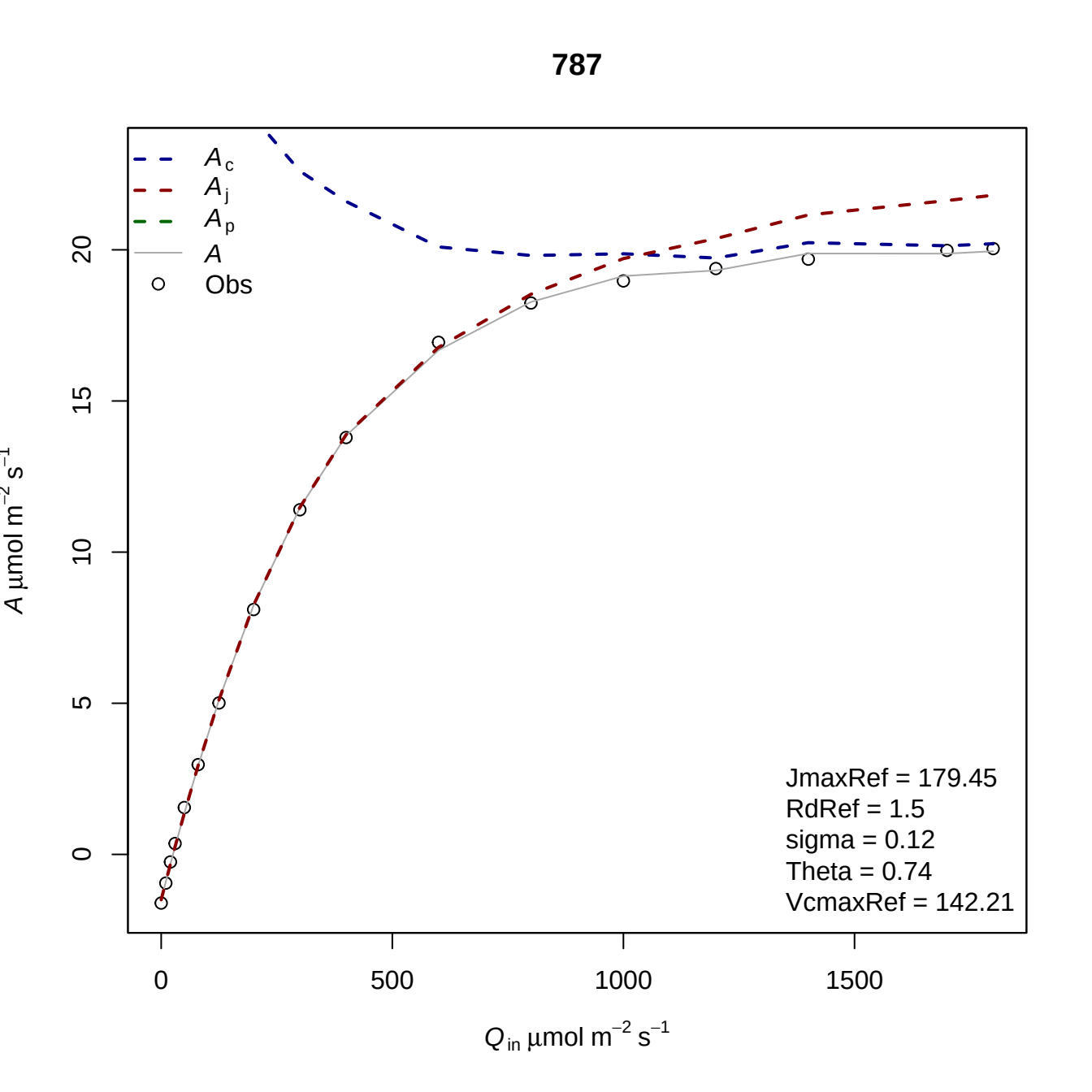

782
A_c
A_j
A_p
A
Obs
JmaxRef = 41.7
RdRef = 0.81
sigma = 0.08
Theta = 0.66
VcmaxRef = 30.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

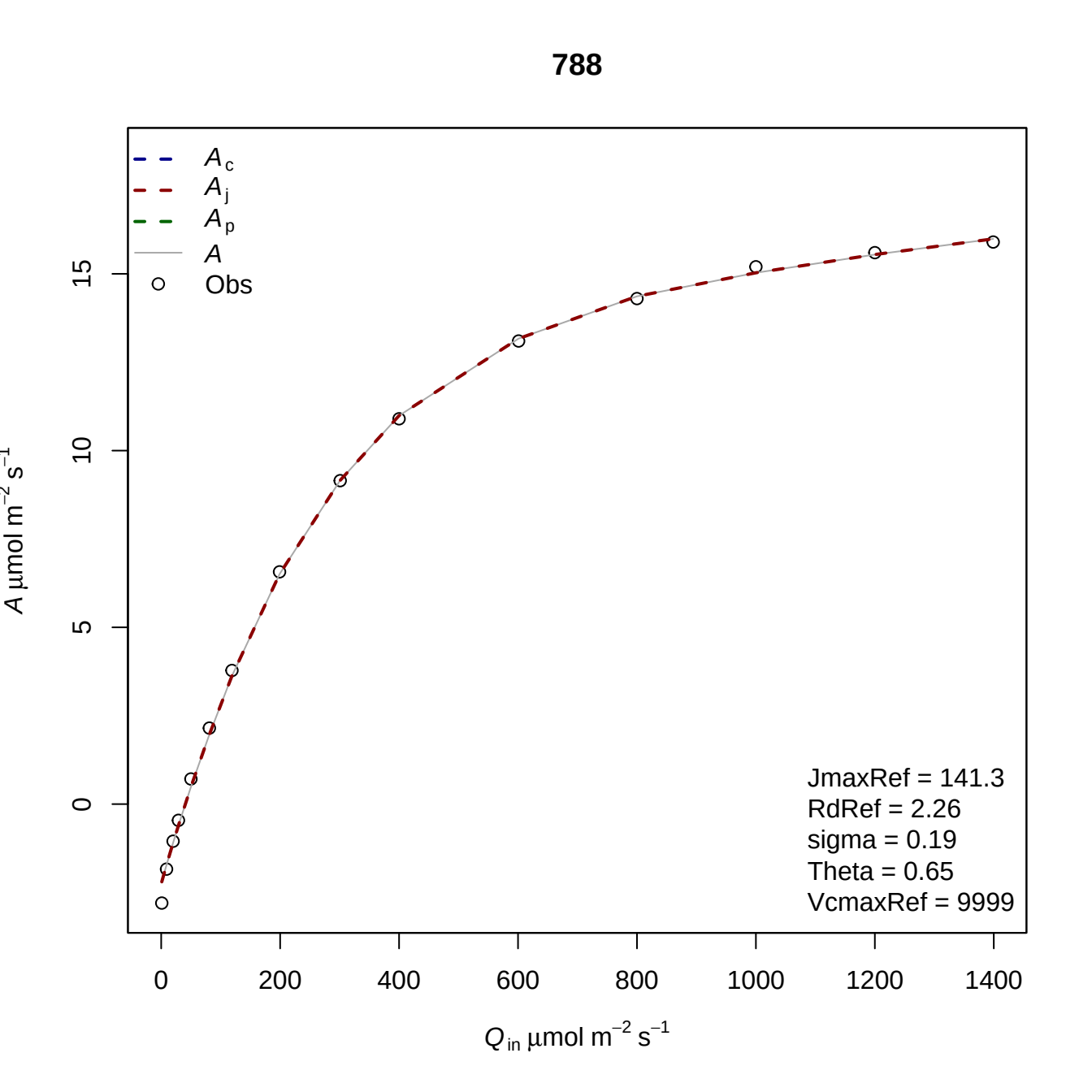

783
A_c
A_j
A_p
A
Obs
JmaxRef = 60.43
RdRef = 0.51
sigma = 0.09
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

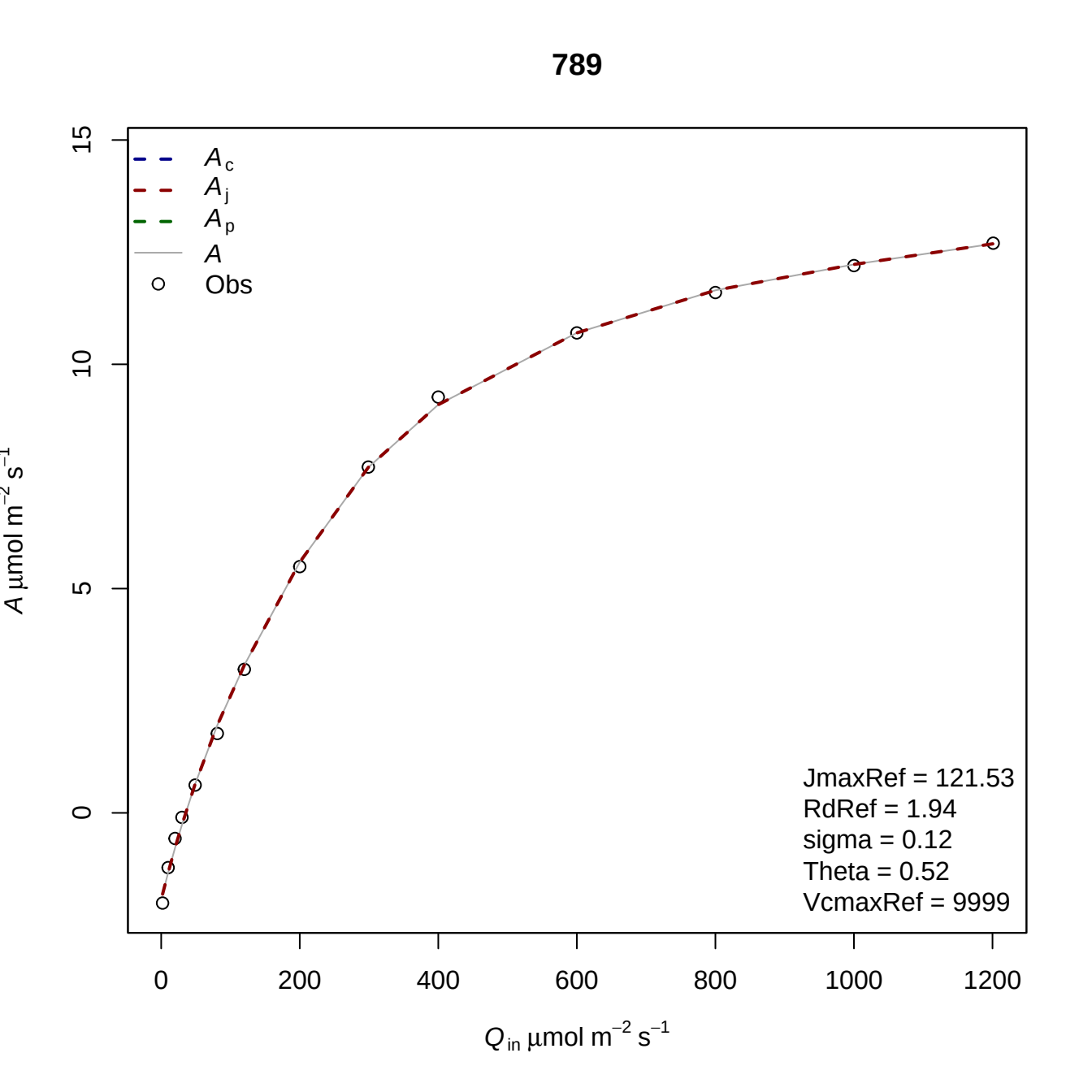

784
A_c
A_j
A_p
A
Obs
JmaxRef = 27.63
RdRef = 1.3
sigma = 0.05
Theta = −0.02
VcmaxRef = 18.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000
1200

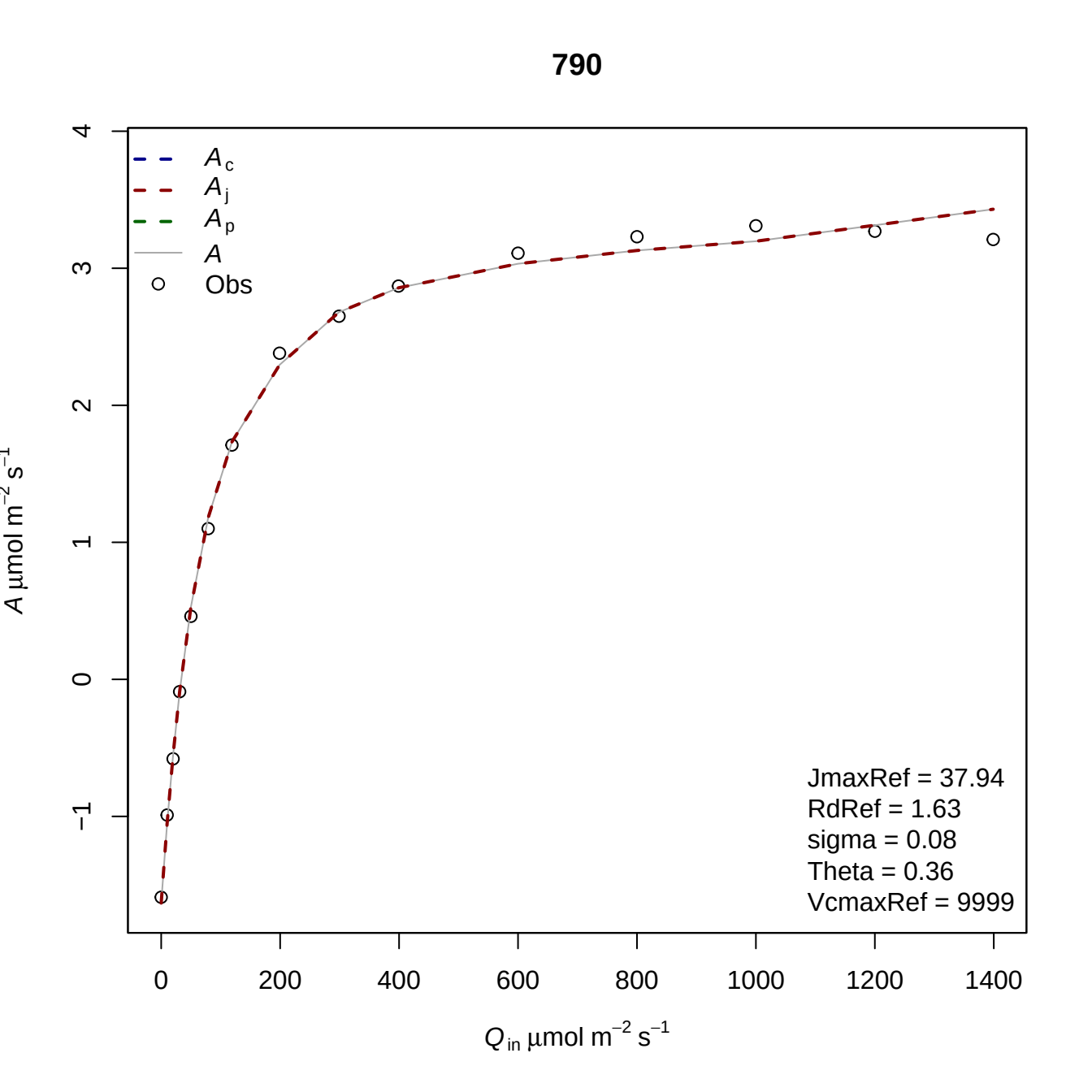

785
A_c
A_j
A_p
A
Obs
JmaxRef = 85.58
RdRef = 1.28
sigma = 0.11
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

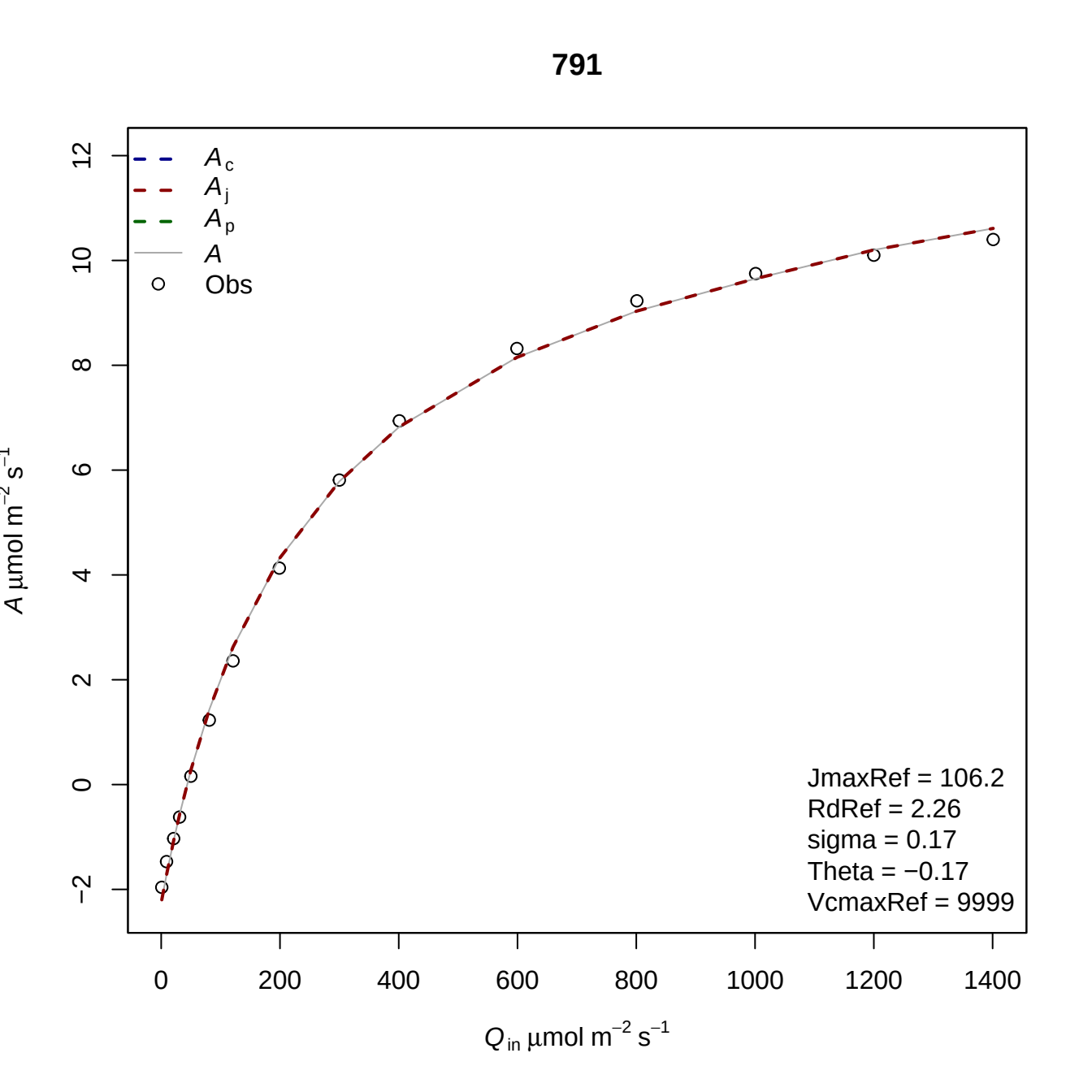

786
A_c
A_j
A_p
A
Obs
JmaxRef = 50.37
RdRef = 1.64
sigma = 0.09
Theta = 0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

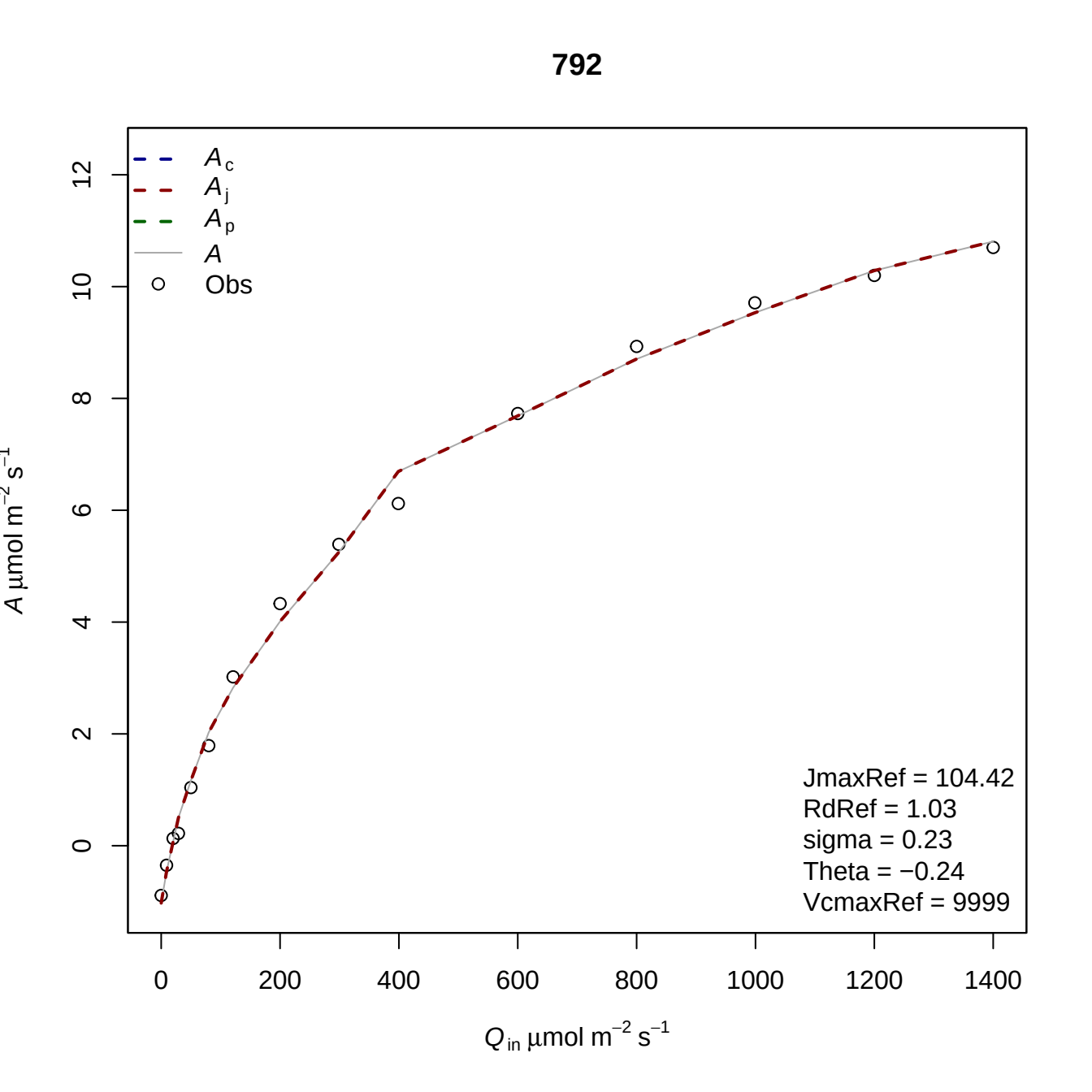

787
A_c
A_j
A_p
A
Obs
JmaxRef = 179.45
RdRef = 1.5
sigma = 0.12
Theta = 0.74
VcmaxRef = 142.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

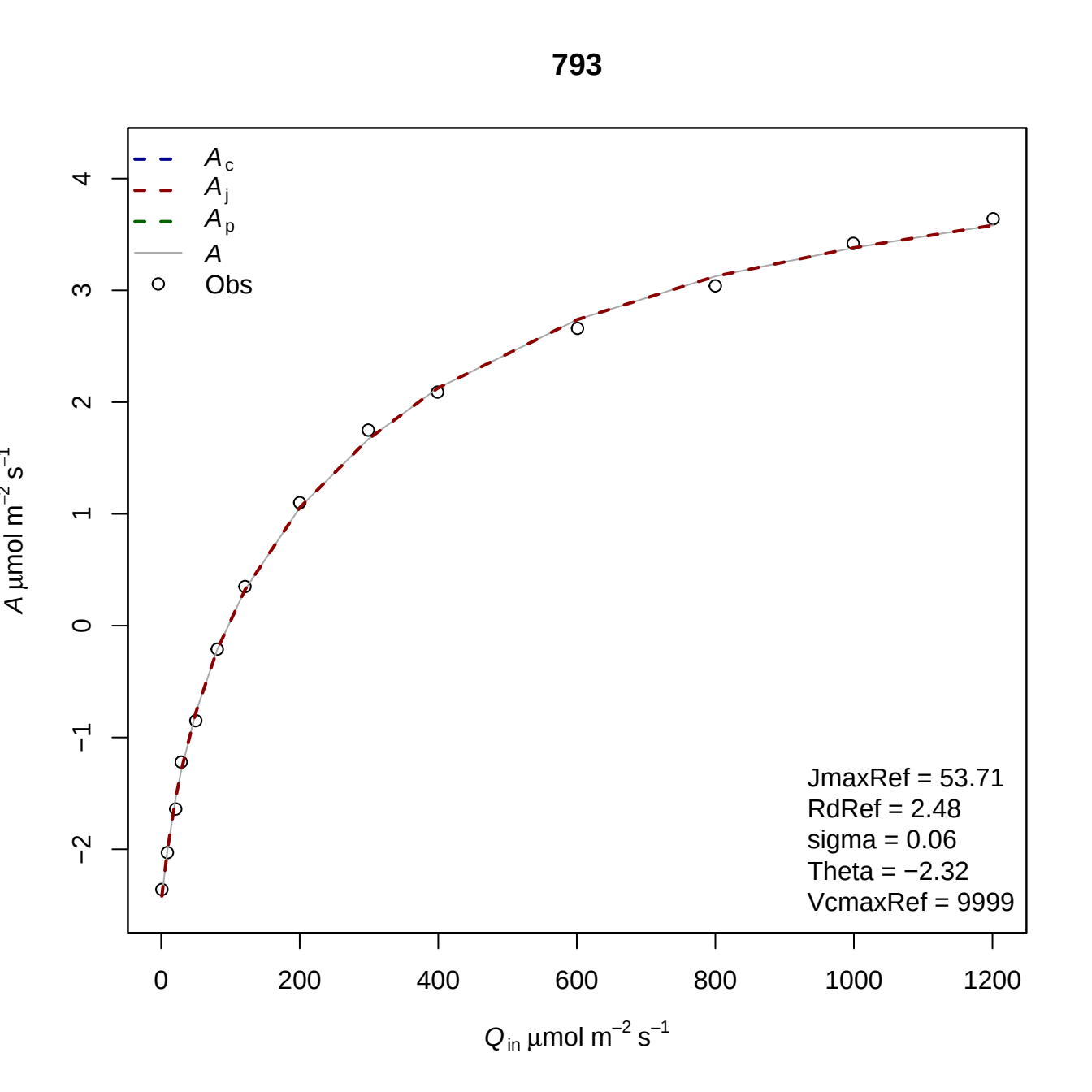

788
A_c
A_j
A_p
A
Obs
JmaxRef = 141.3
RdRef = 2.26
sigma = 0.19
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

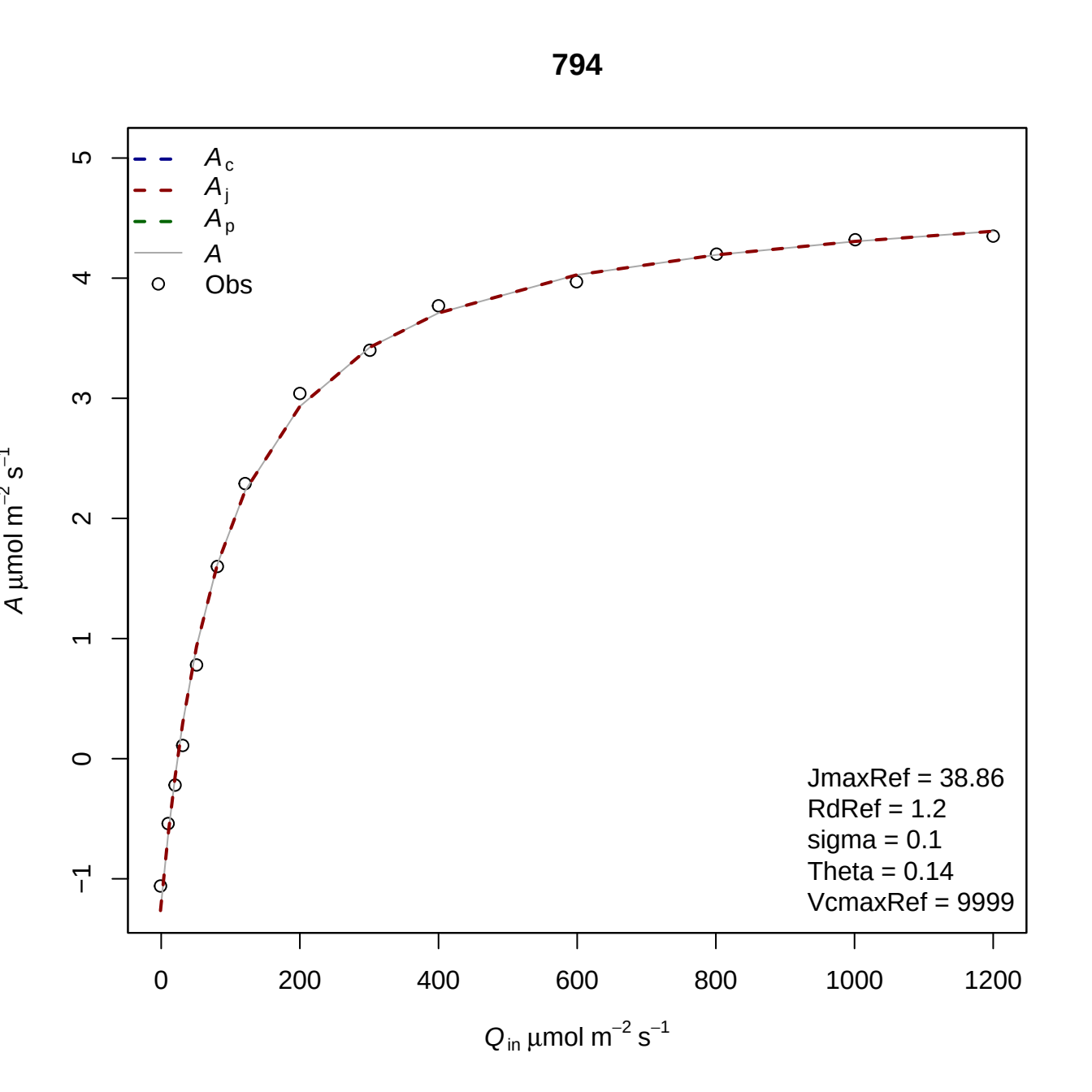

789
A_c
A_j
A_p
A
Obs
JmaxRef = 121.53
RdRef = 1.94
sigma = 0.12
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

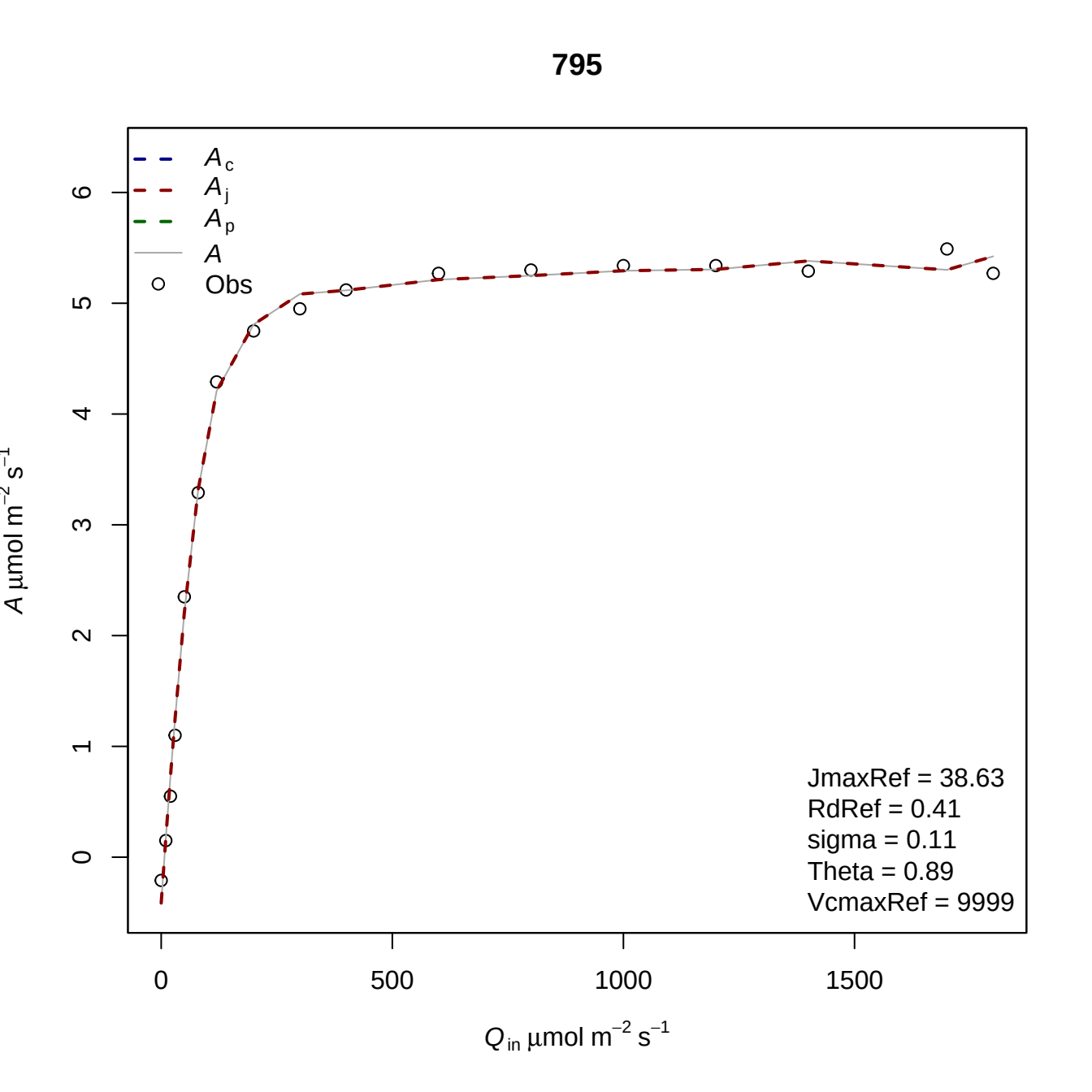

790
A_c
A_j
A_p
A
Obs
JmaxRef = 37.94
RdRef = 1.63
sigma = 0.08
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

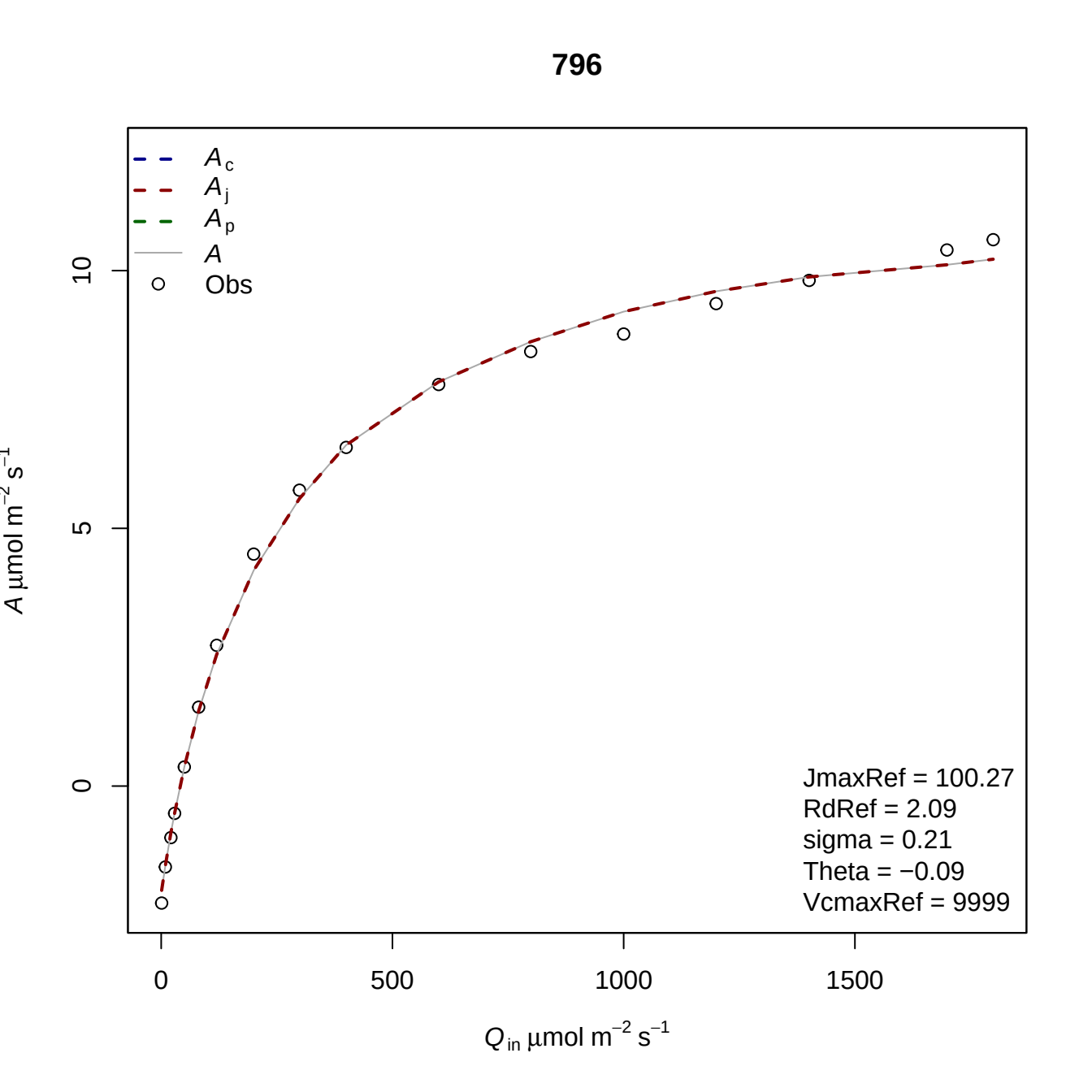

791
A_c
A_j
A_p
A
Obs
JmaxRef = 106.2
RdRef = 2.26
sigma = 0.17
Theta = −0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

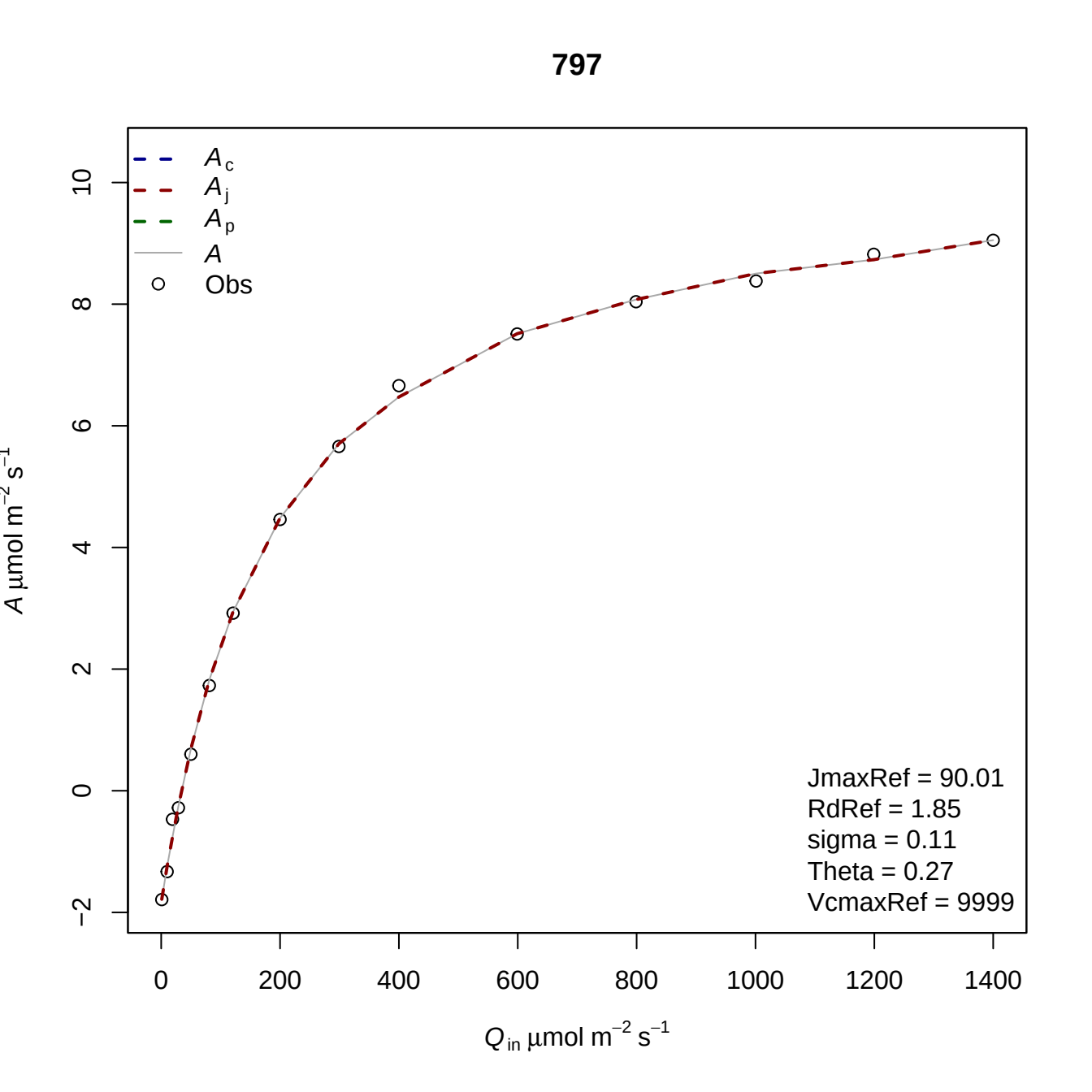

792
A_c
A_j
A_p
A
Obs
JmaxRef = 104.42
RdRef = 1.03
sigma = 0.23
Theta = −0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

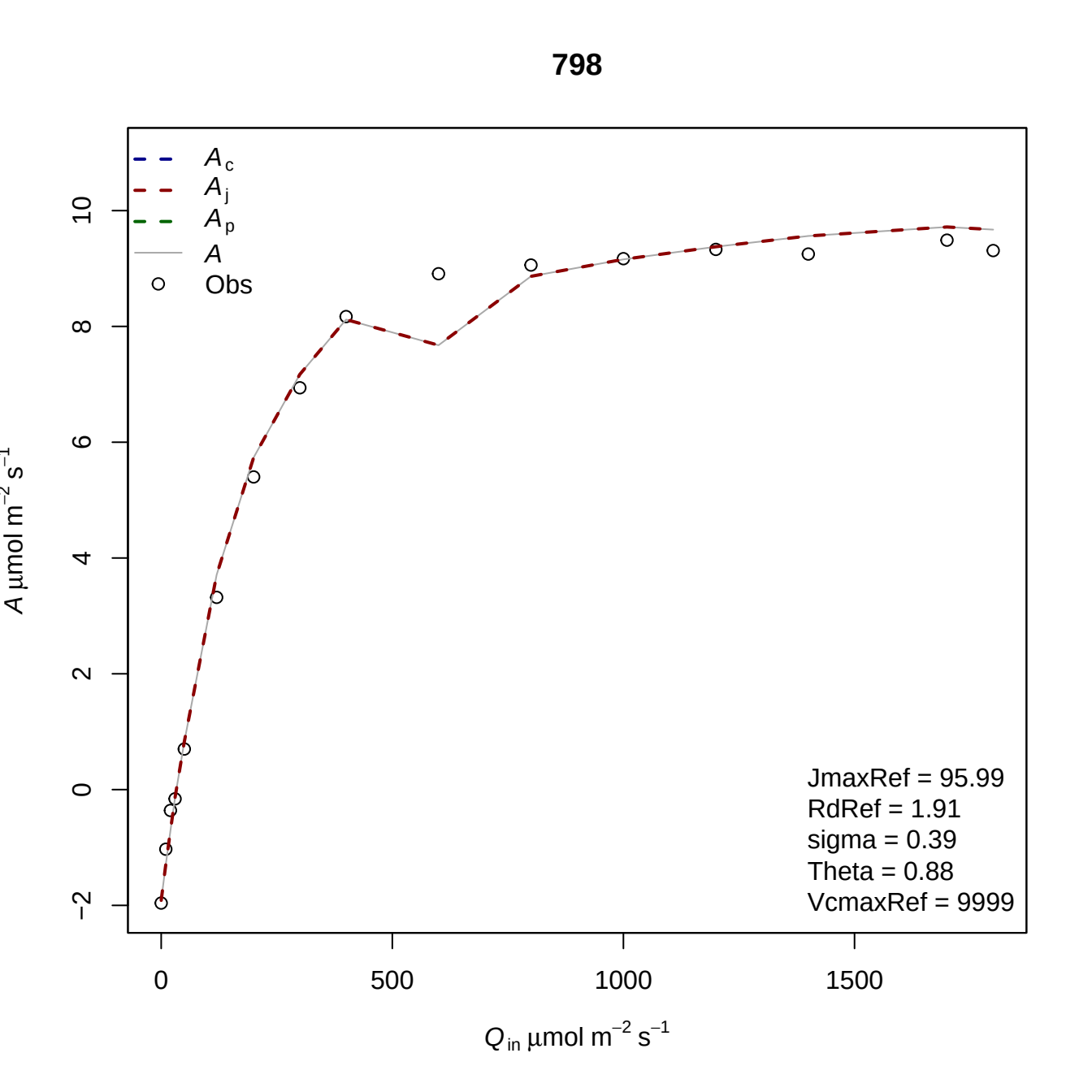

793
A_c
A_j
A_p
A
Obs
JmaxRef = 53.71
RdRef = 2.48
sigma = 0.06
Theta = −2.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200

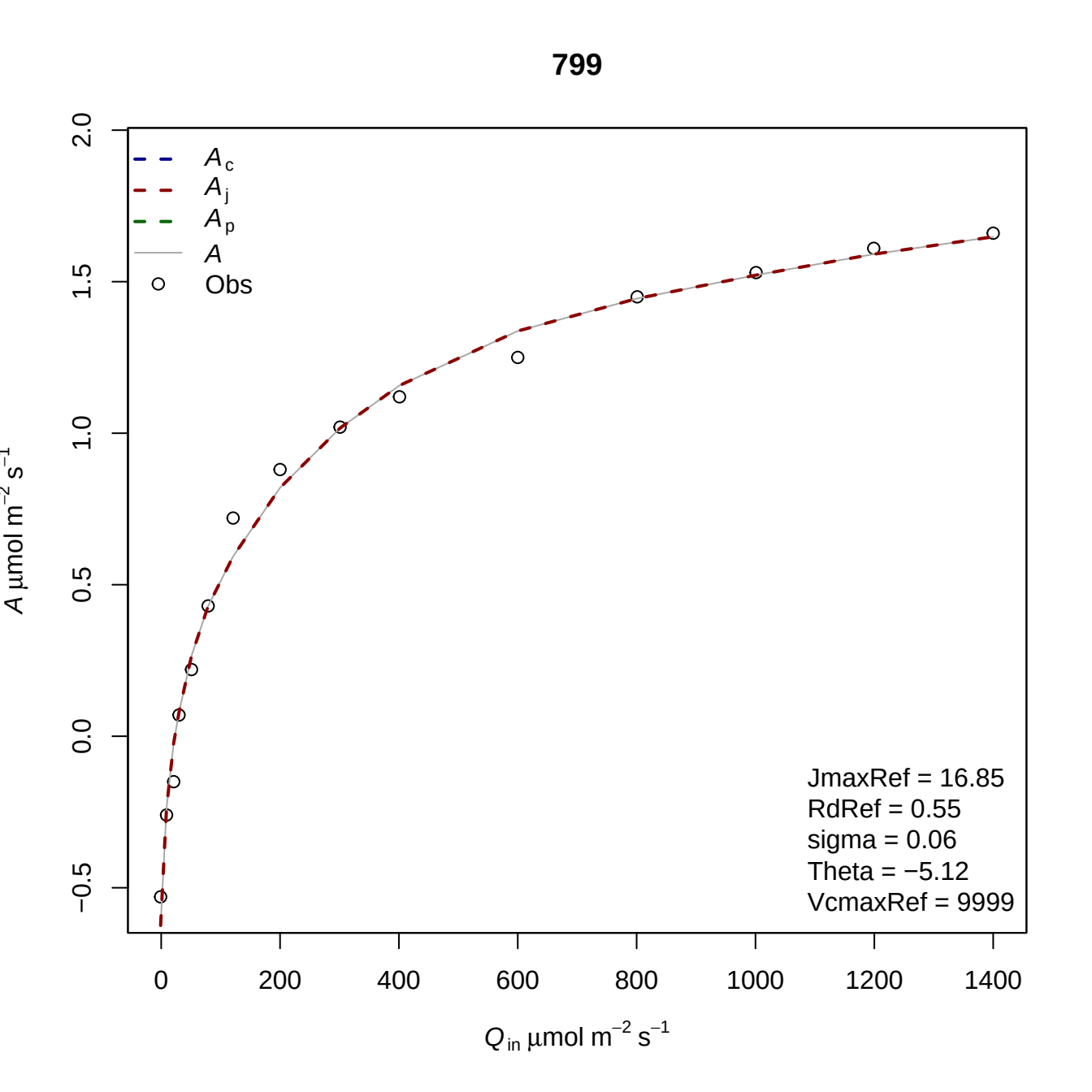

794
A_c
A_j
A_p
A
Obs
JmaxRef = 38.86
RdRef = 1.2
sigma = 0.1
Theta = 0.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

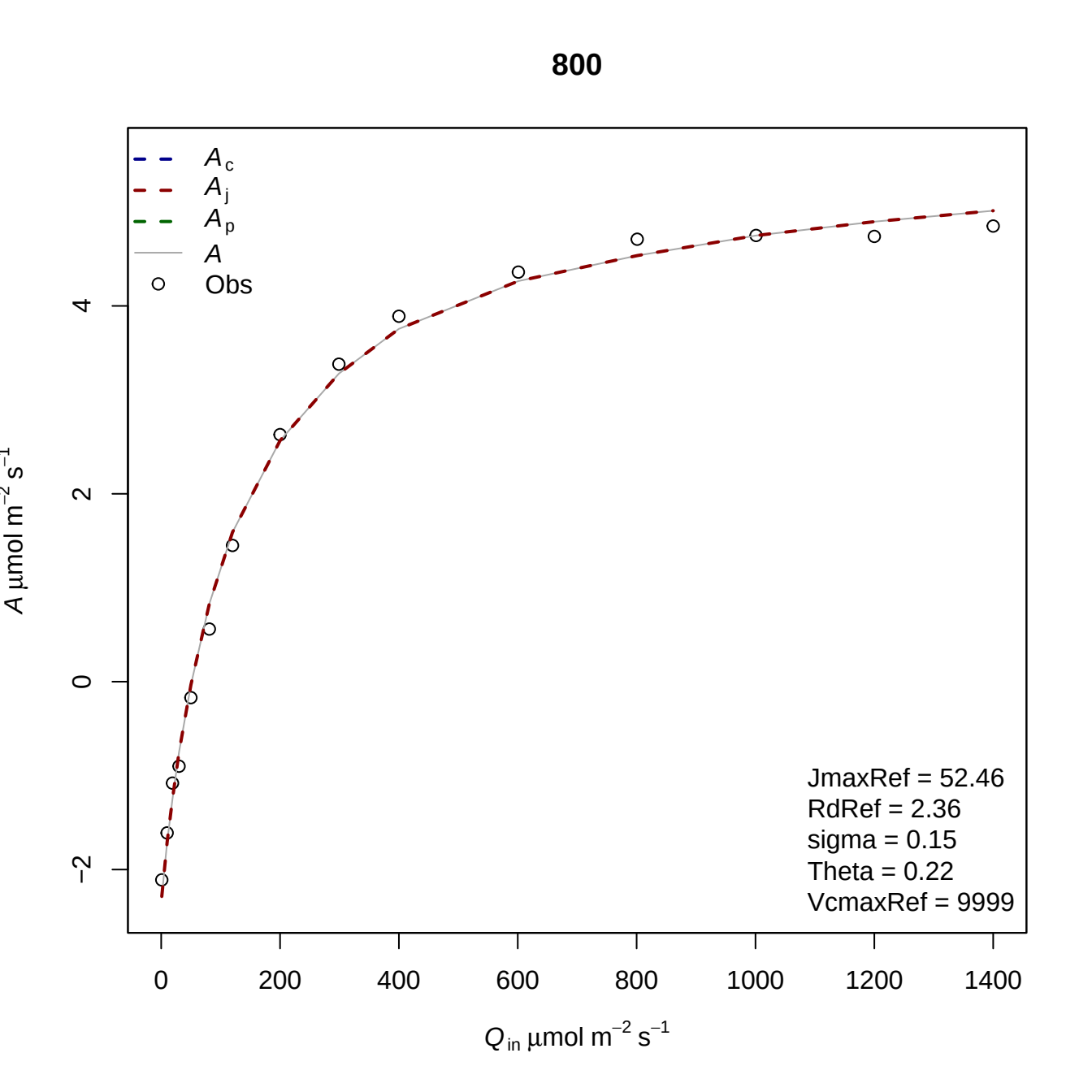

795
A_c
A_j
A_p
A
Obs
JmaxRef = 38.63
RdRef = 0.41
sigma = 0.11
Theta = 0.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

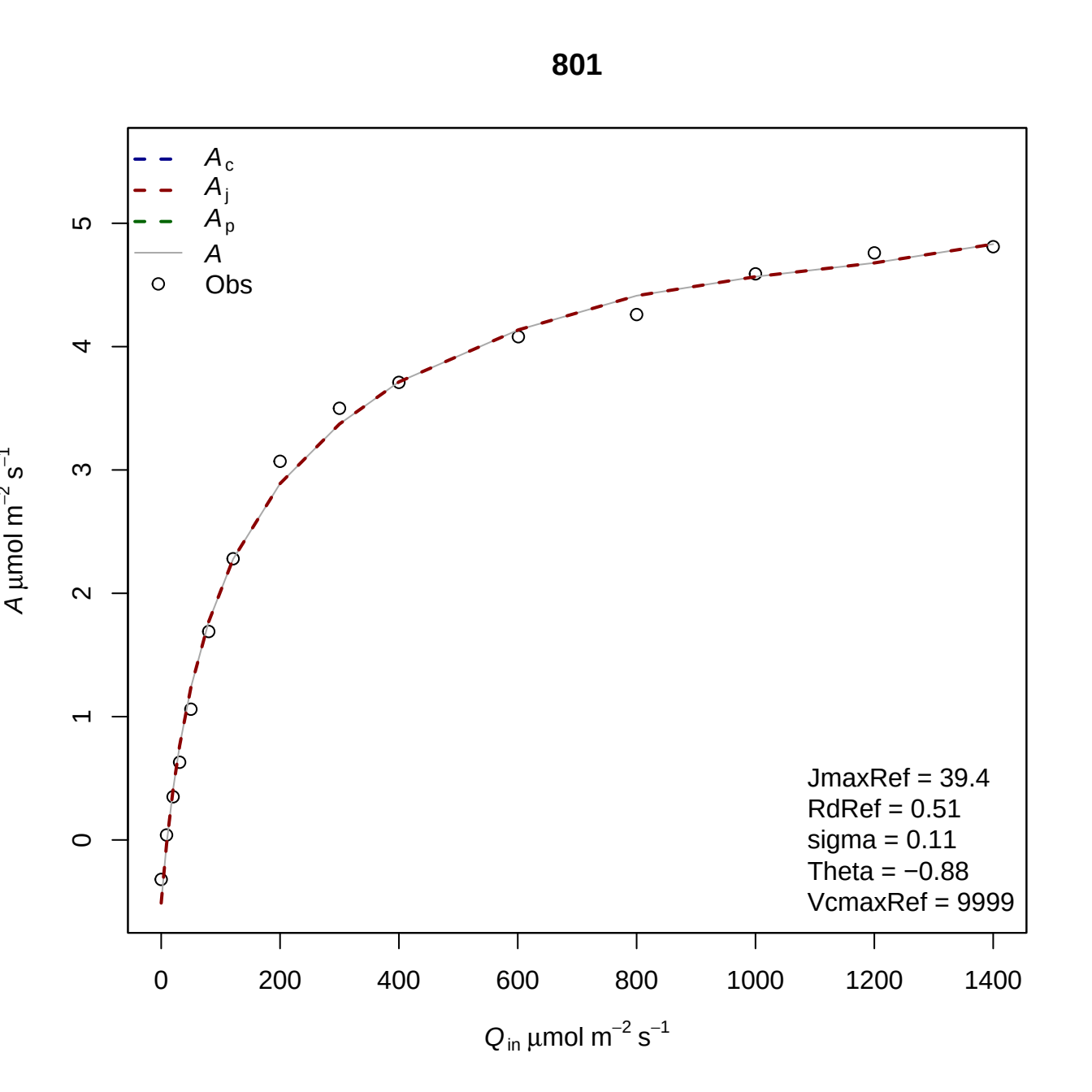

796
A_c
A_j
A_p
A
Obs
JmaxRef = 100.27
RdRef = 2.09
sigma = 0.21
Theta = −0.09
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

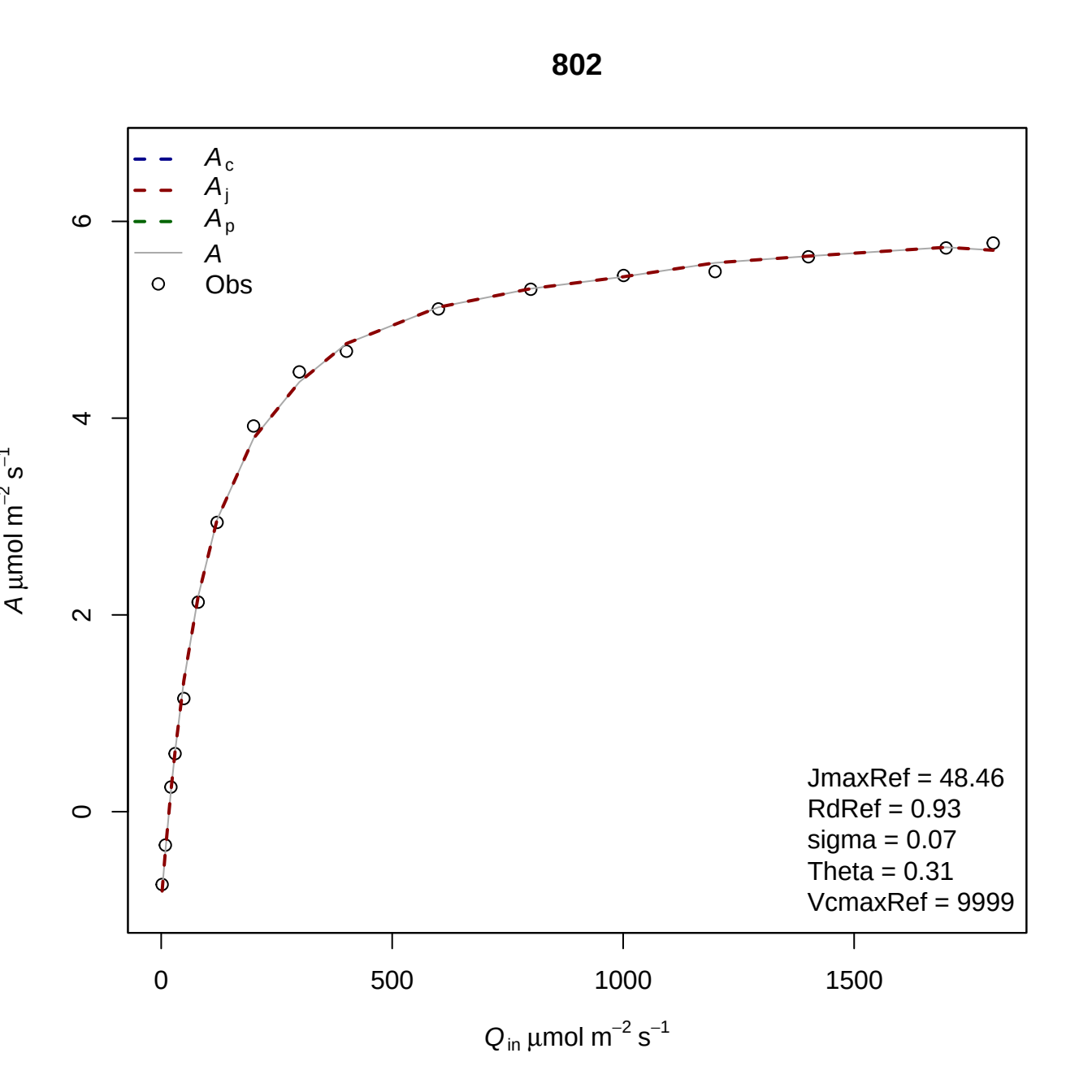

797
A_c
A_j
A_p
A
Obs
JmaxRef = 90.01
RdRef = 1.85
sigma = 0.11
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

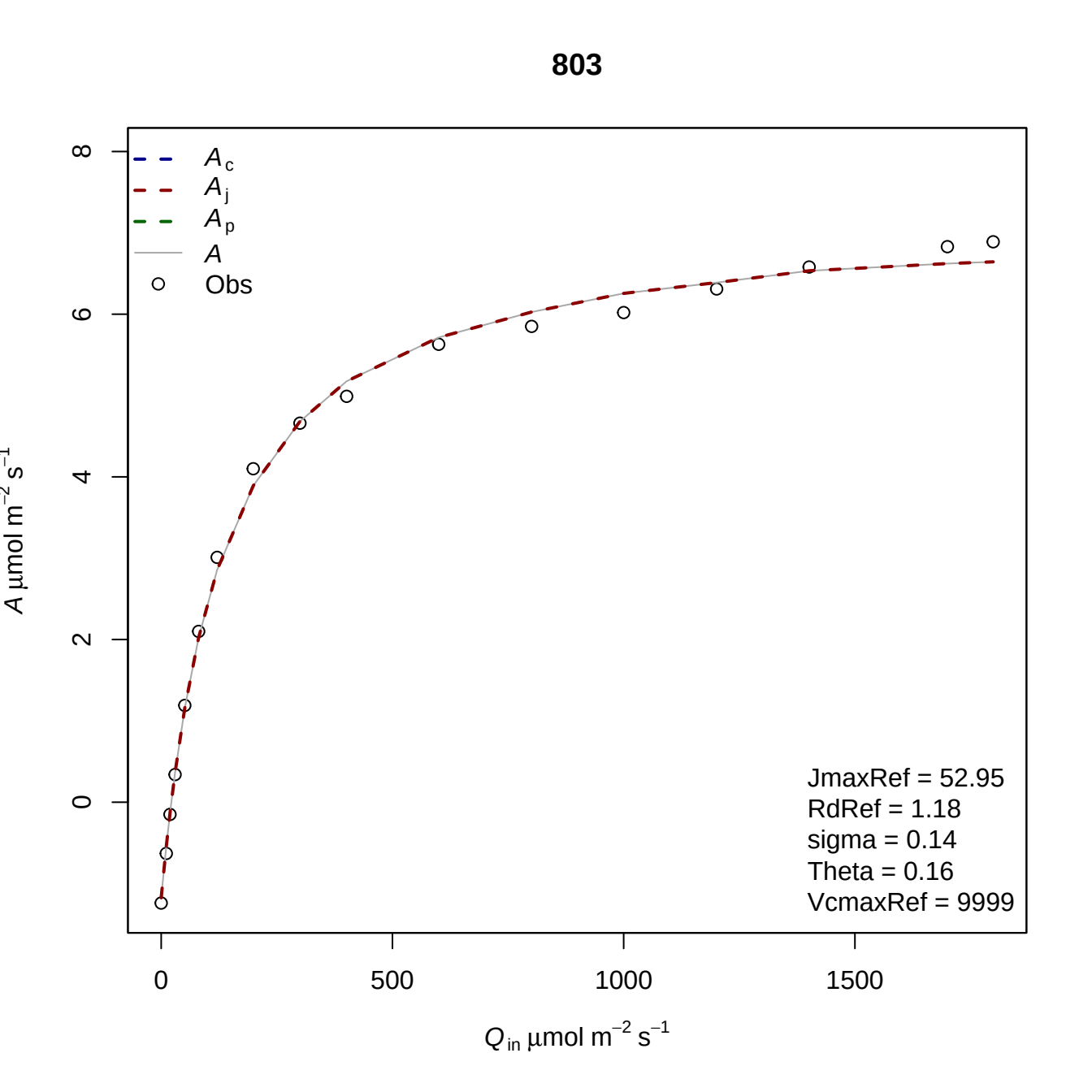

798
A_c
A_j
A_p
A
Obs
JmaxRef = 95.99
RdRef = 1.91
sigma = 0.39
Theta = 0.88
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

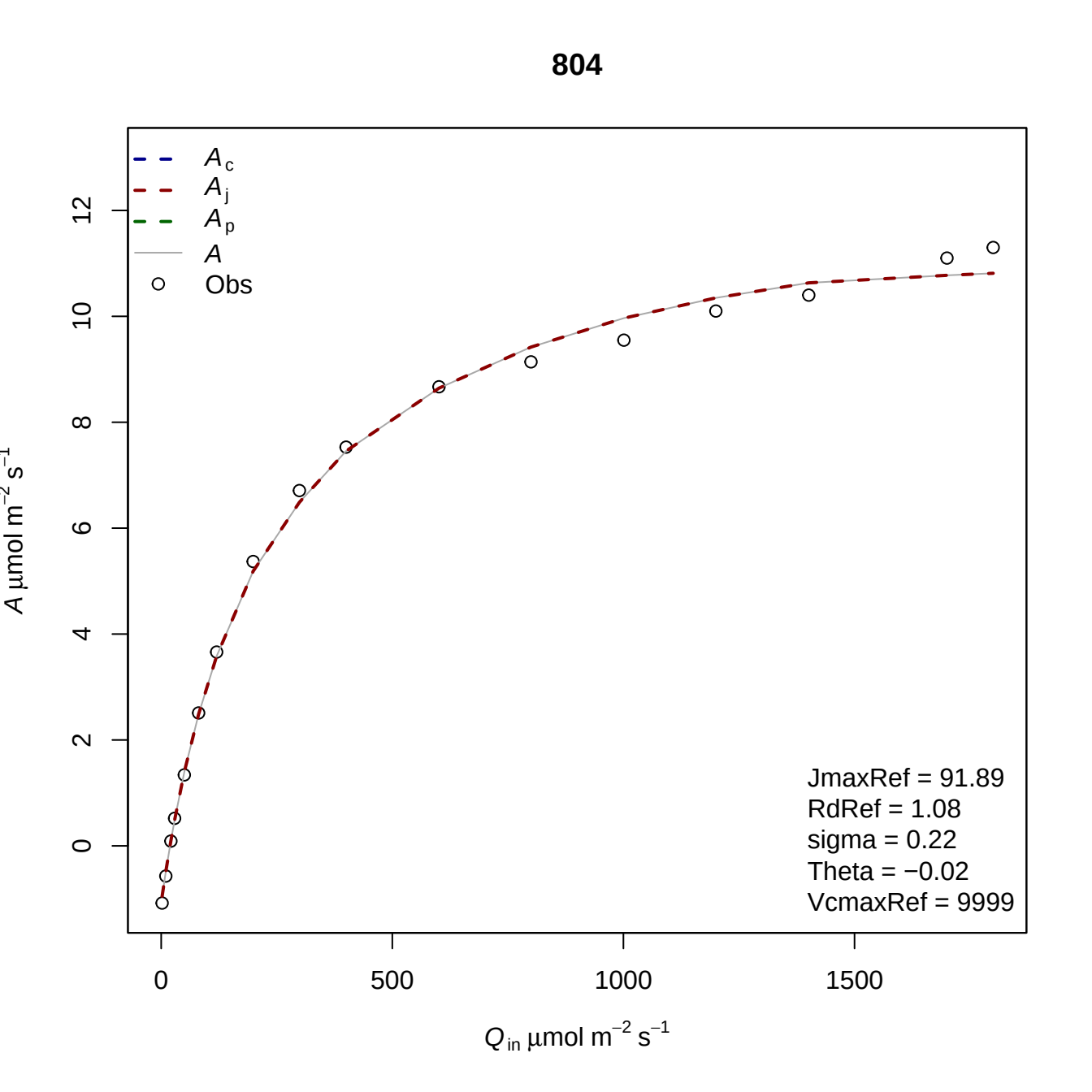

799
A_c
A_j
A_p
A
Obs
JmaxRef = 16.85
RdRef = 0.55
sigma = 0.06
Theta = −5.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.0
1.5
1.0
0.5
0.0
−0.5
0
200
400
600
800
1000
1200
1400

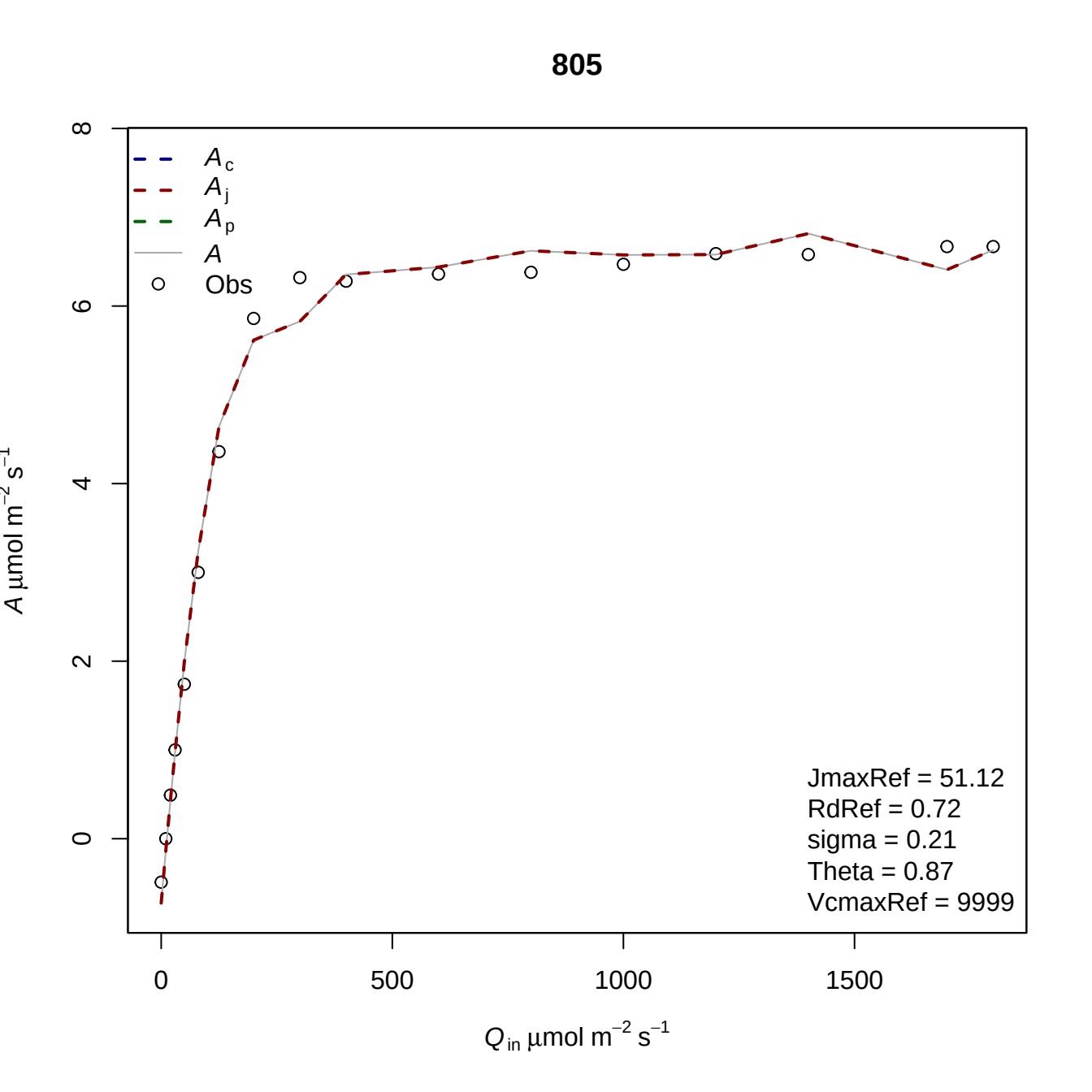

800
A_c
A_j
A_p
A
Obs
JmaxRef = 52.46
RdRef = 2.36
sigma = 0.15
Theta = 0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200
1400

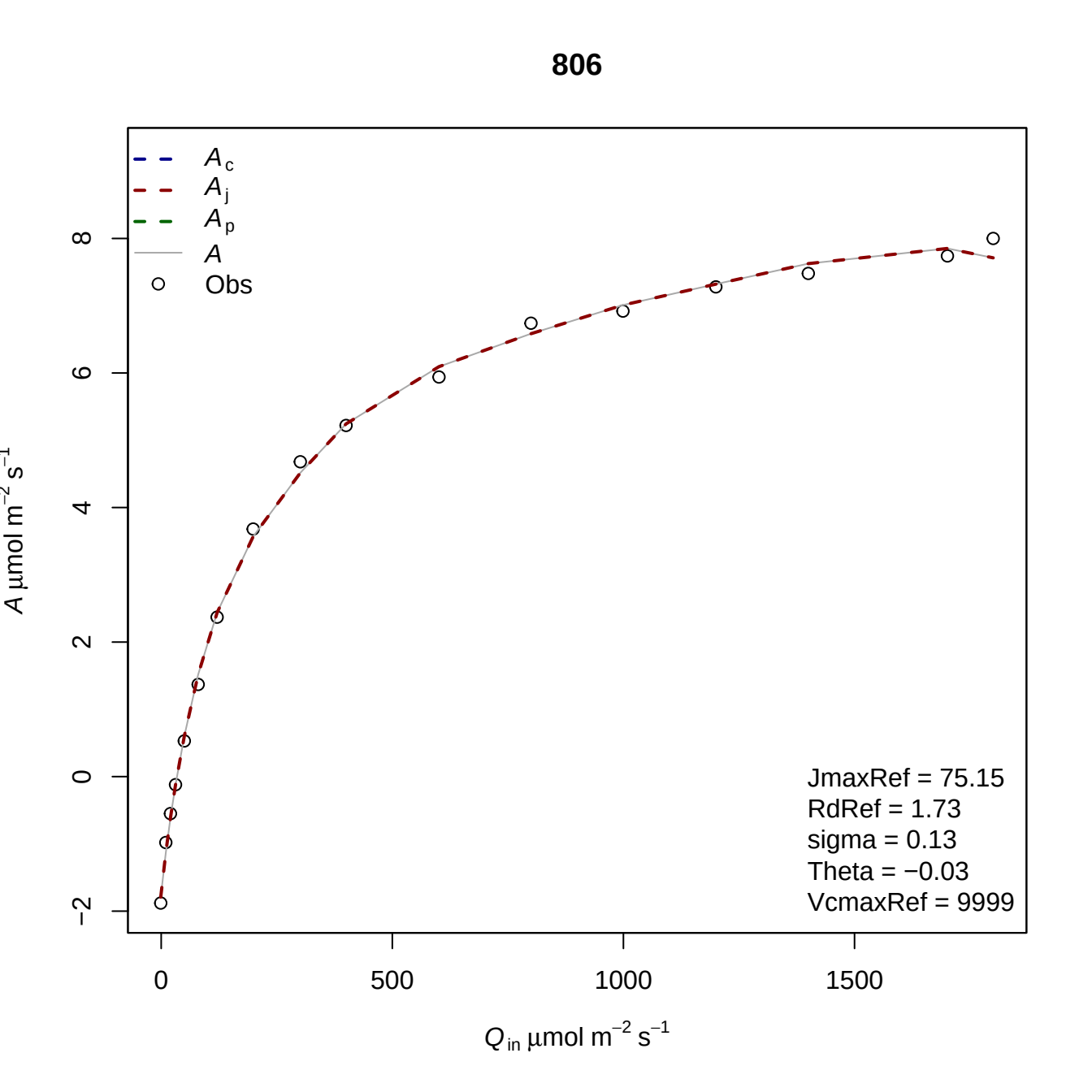

801
A_c
A_j
A_p
A
Obs
JmaxRef = 39.4
RdRef = 0.51
sigma = 0.11
Theta = −0.88
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

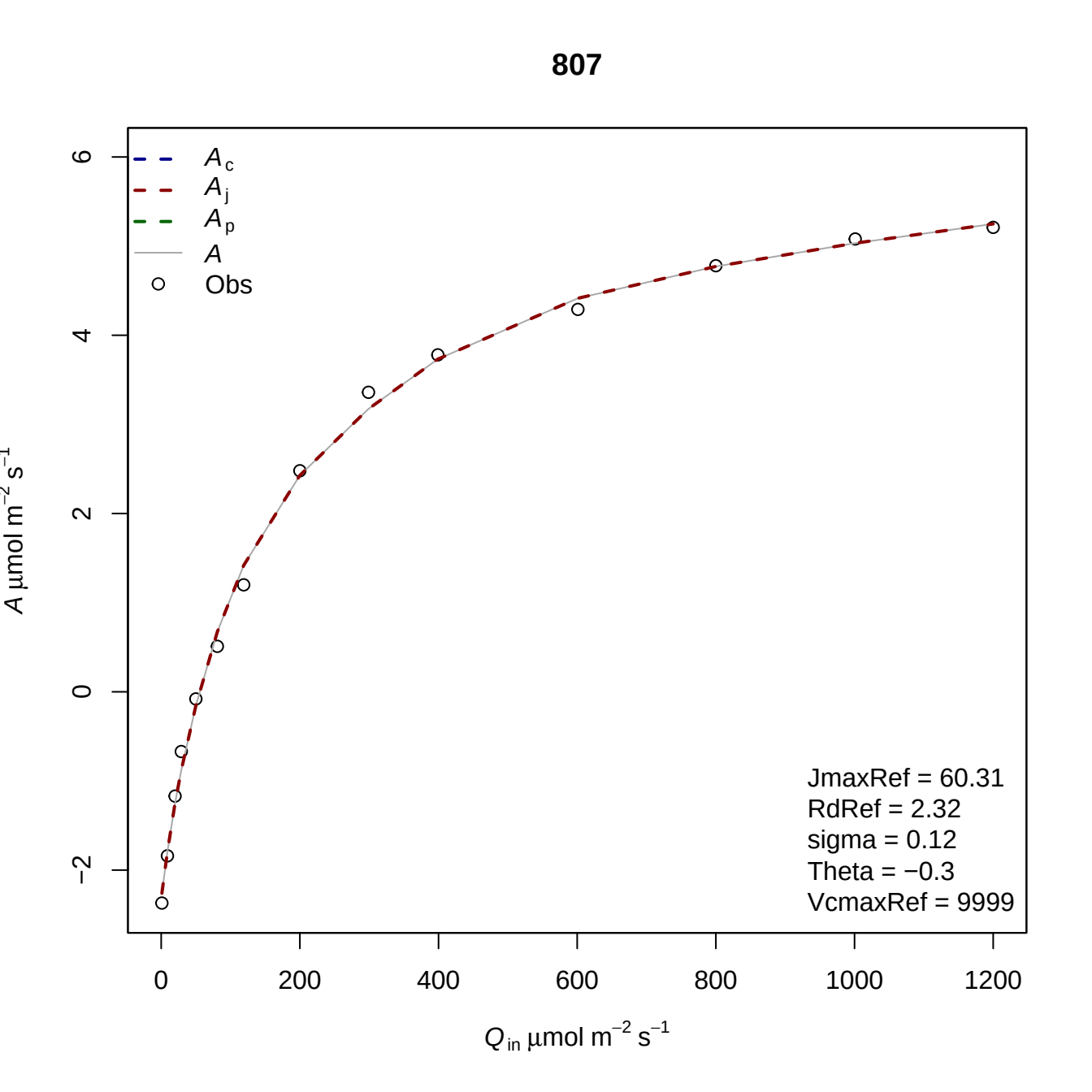

802
A_c
A_j
A_p
A
Obs
JmaxRef = 48.46
RdRef = 0.93
sigma = 0.07
Theta = 0.31
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

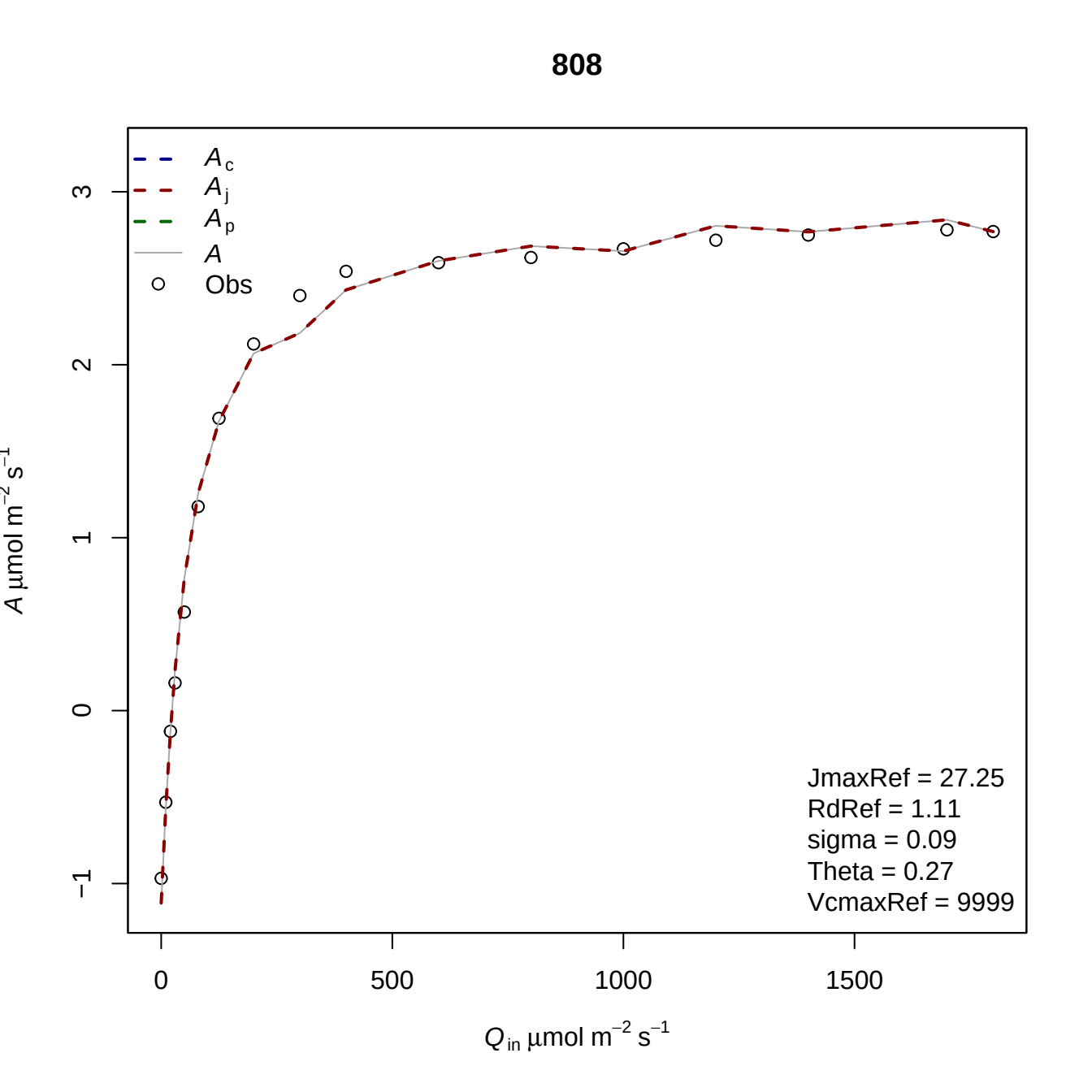

803
A_c
A_j
A_p
A
Obs
JmaxRef = 52.95
RdRef = 1.18
sigma = 0.14
Theta = 0.16
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

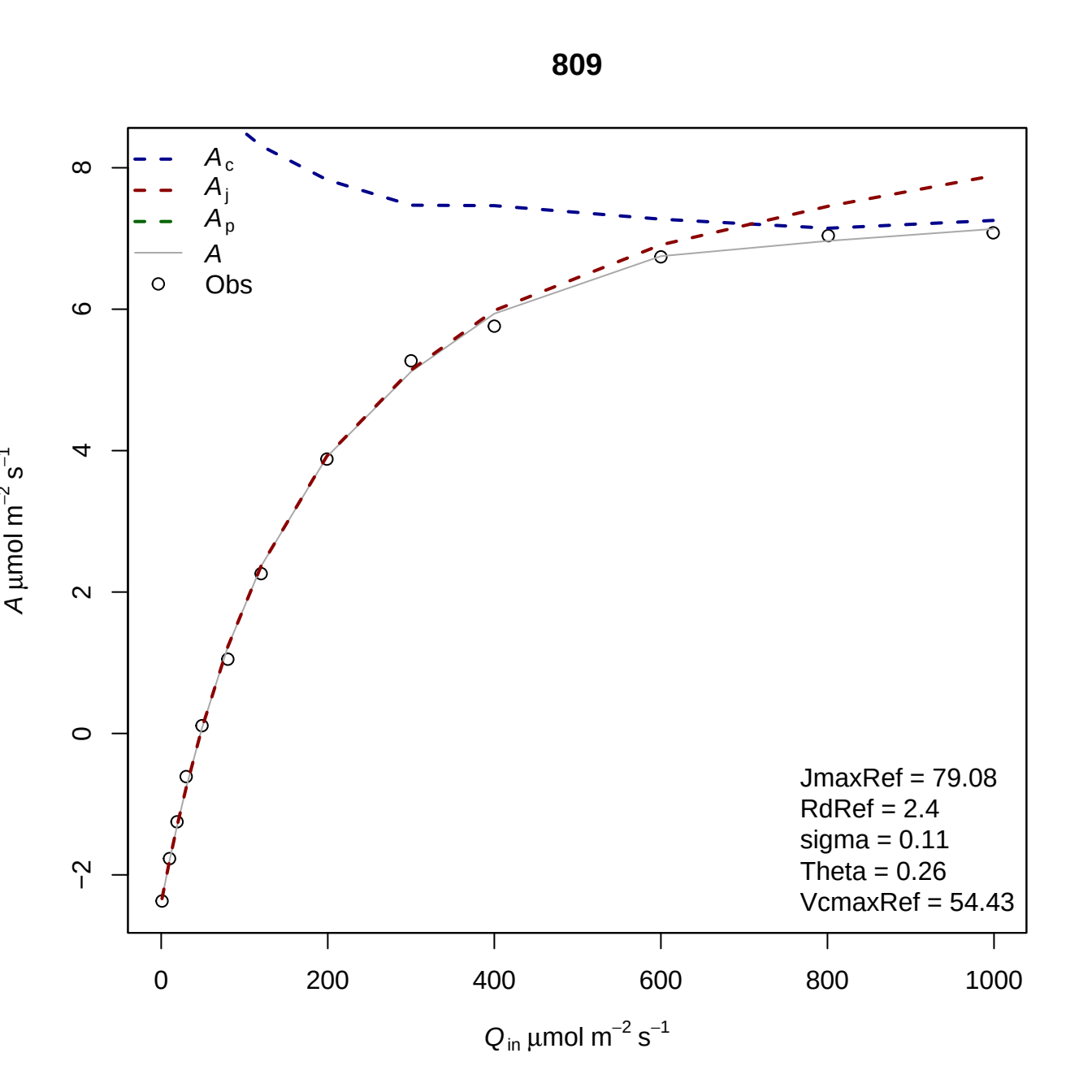

804
A_c
A_j
A_p
A
Obs
JmaxRef = 91.89
RdRef = 1.08
sigma = 0.22
Theta = −0.02
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

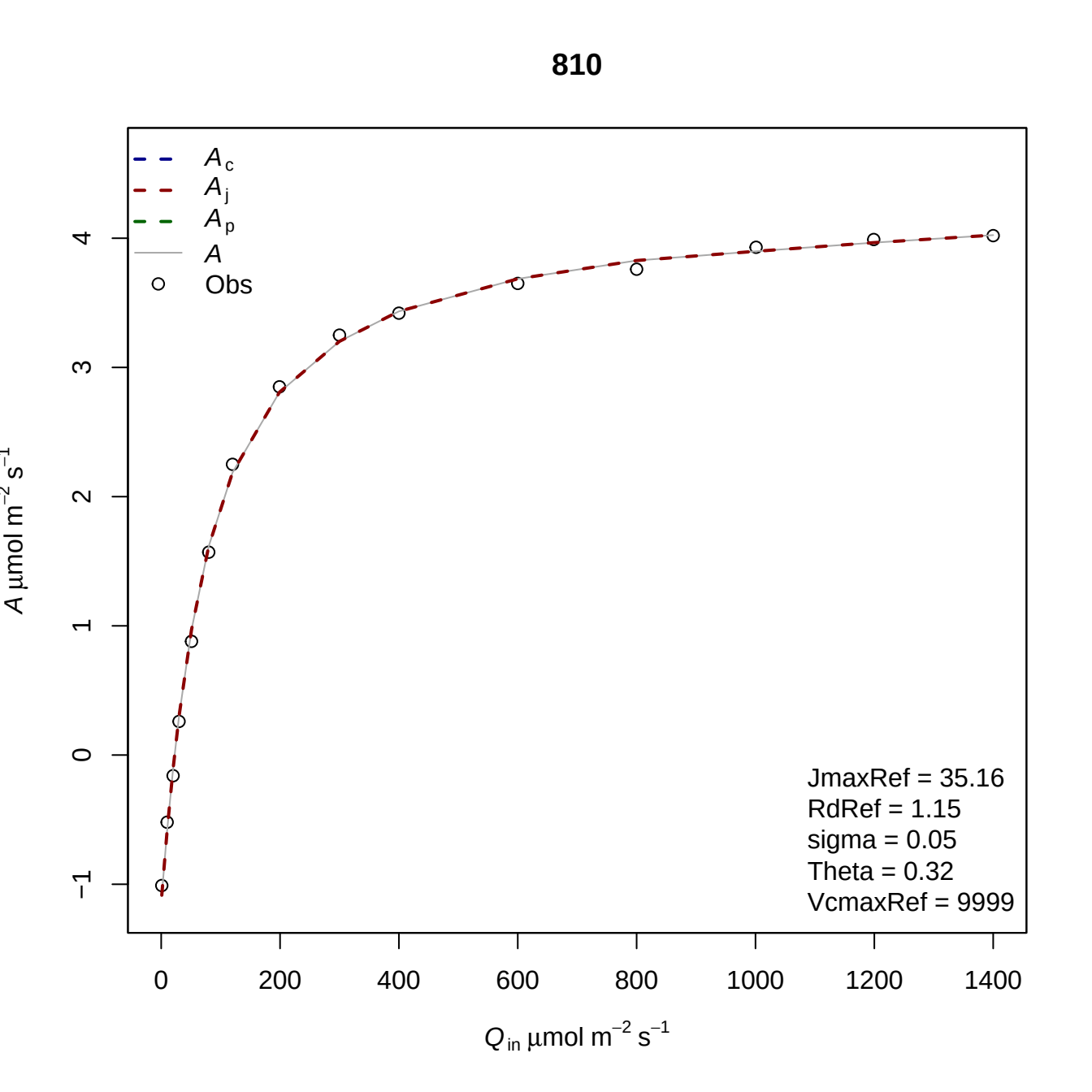

805
A_c
A_j
A_p
A
Obs
JmaxRef = 51.12
RdRef = 0.72
sigma = 0.21
Theta = 0.87
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

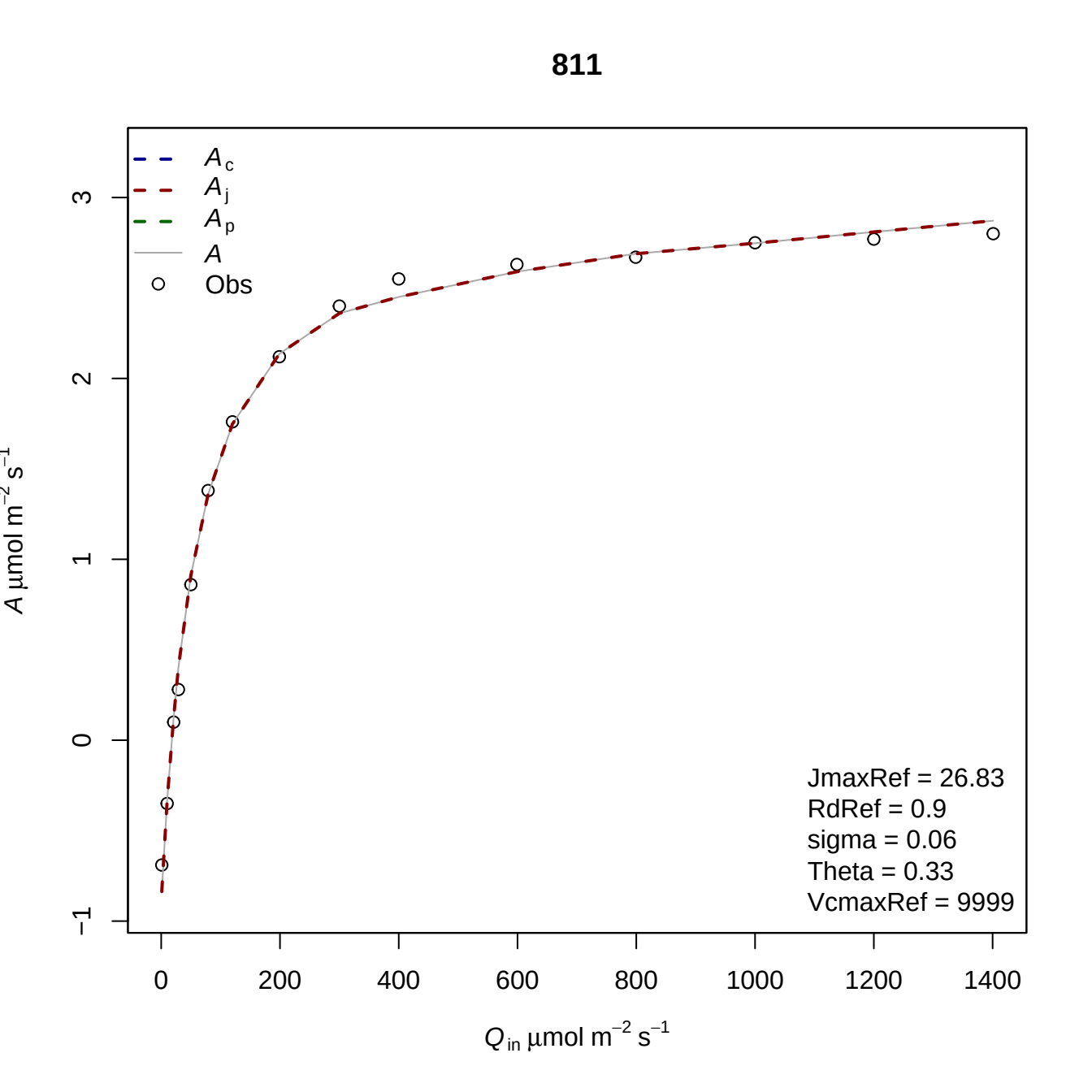

806
A_c
A_j
A_p
A
Obs
JmaxRef = 75.15
RdRef = 1.73
sigma = 0.13
Theta = −0.03
VcmaxRef = 9999
−2
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

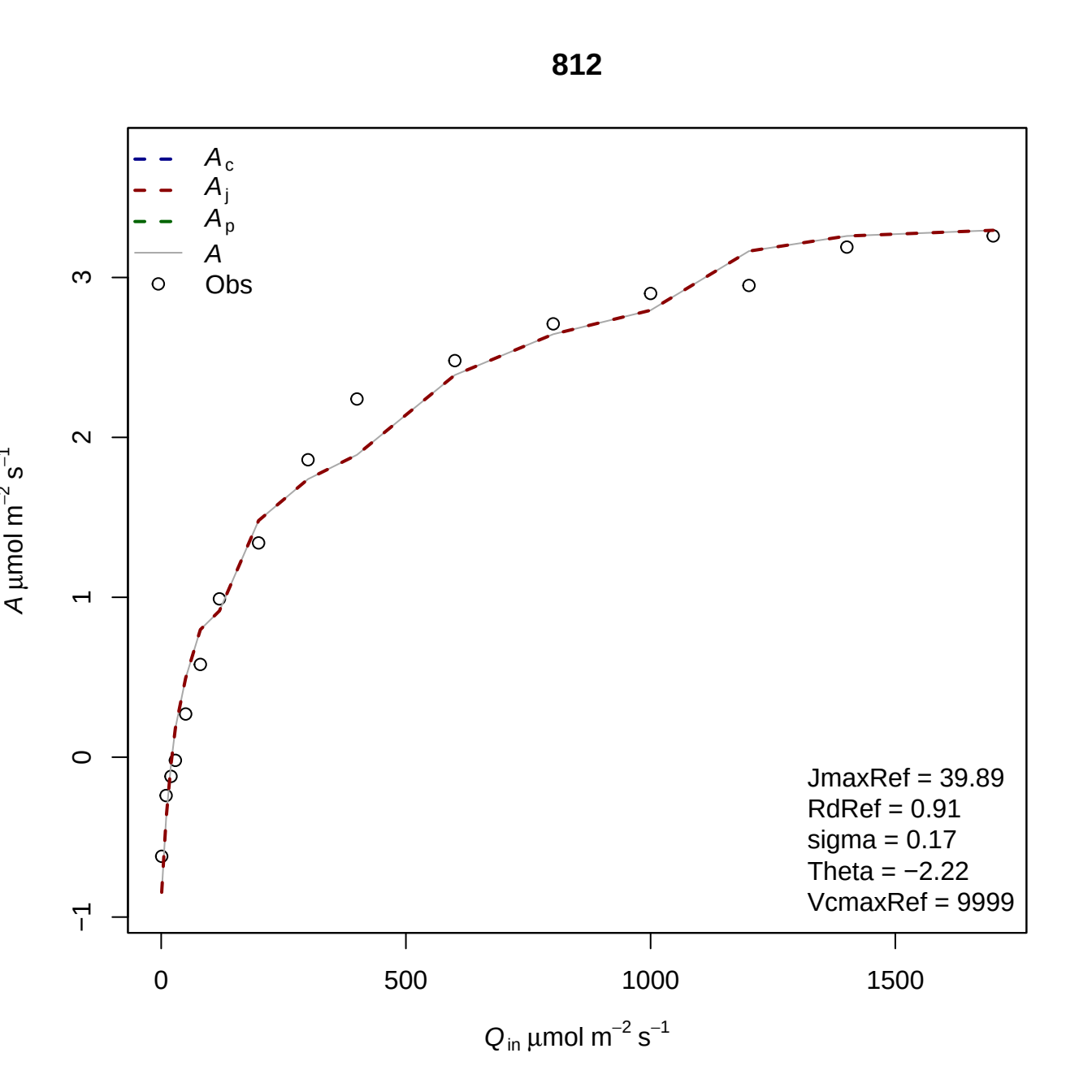

807
A_c
A_j
A_p
A
Obs
JmaxRef = 60.31
RdRef = 2.32
sigma = 0.12
Theta = −0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

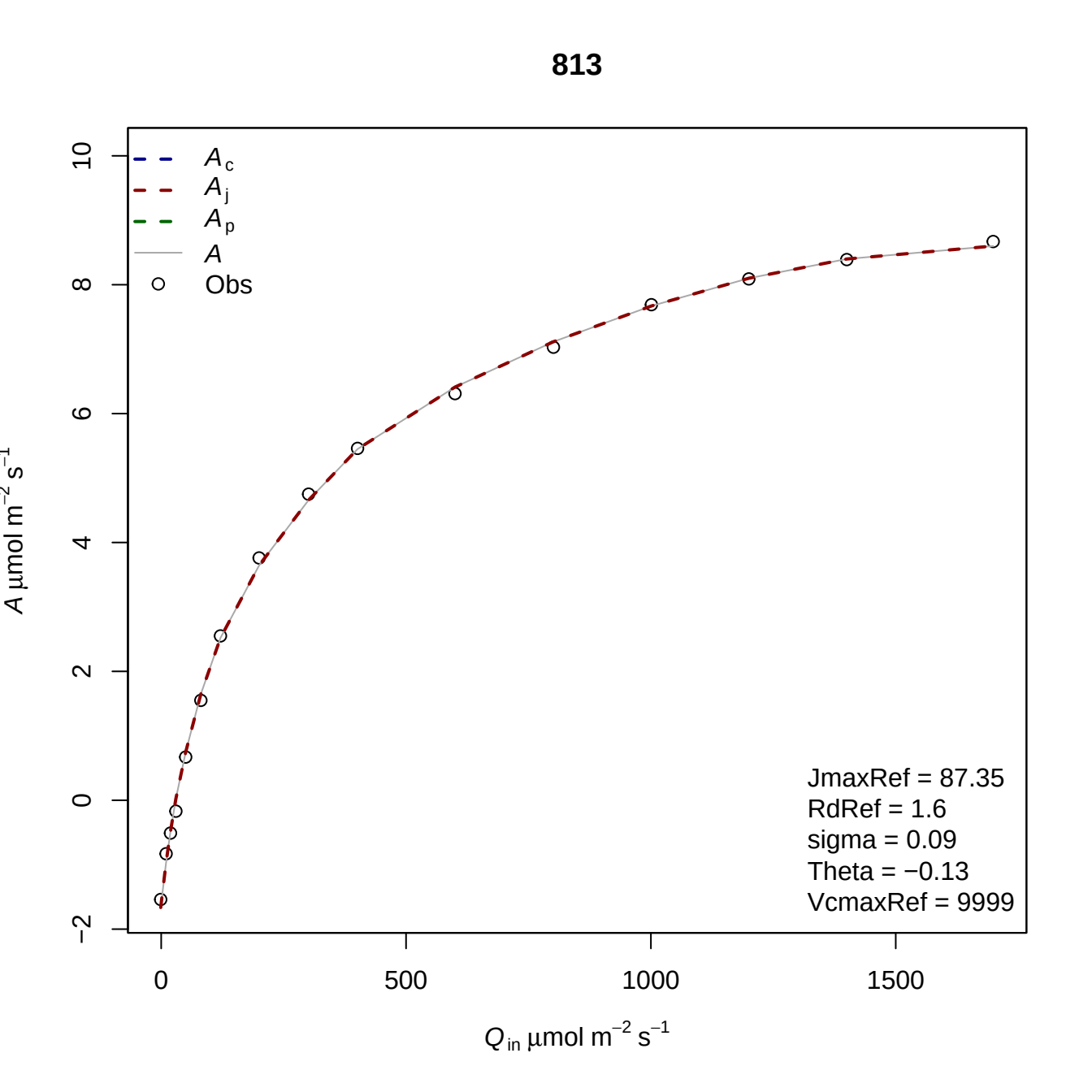

808
A_c
A_j
A_p
A
Obs
JmaxRef = 27.25
RdRef = 1.11
sigma = 0.09
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

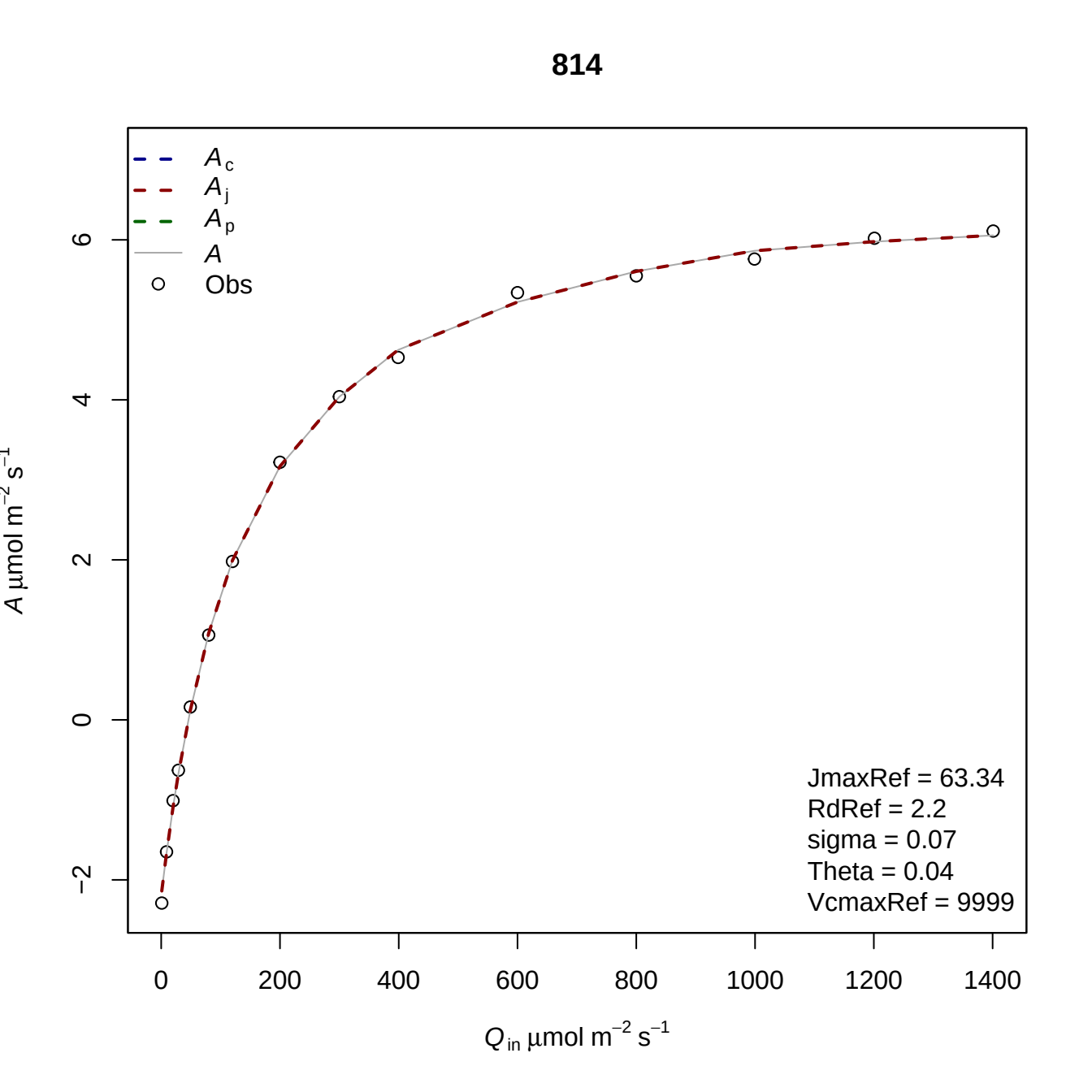

809
A_c
A_j
A_p
A
Obs
JmaxRef = 79.08
RdRef = 2.4
sigma = 0.11
Theta = 0.26
VcmaxRef = 54.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

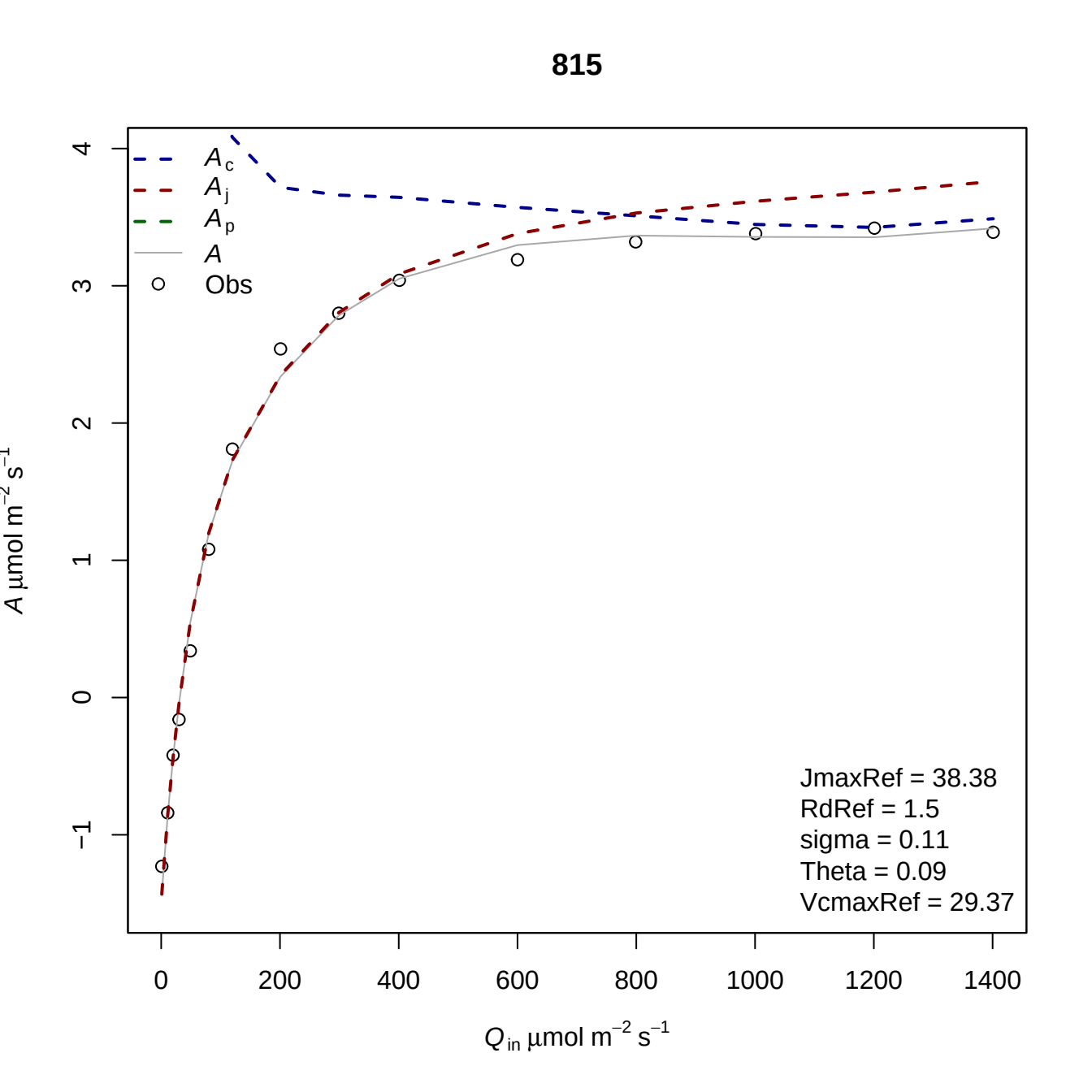

810
A_c
A_j
A_p
A
Obs
JmaxRef = 35.16
RdRef = 1.15
sigma = 0.05
Theta = 0.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

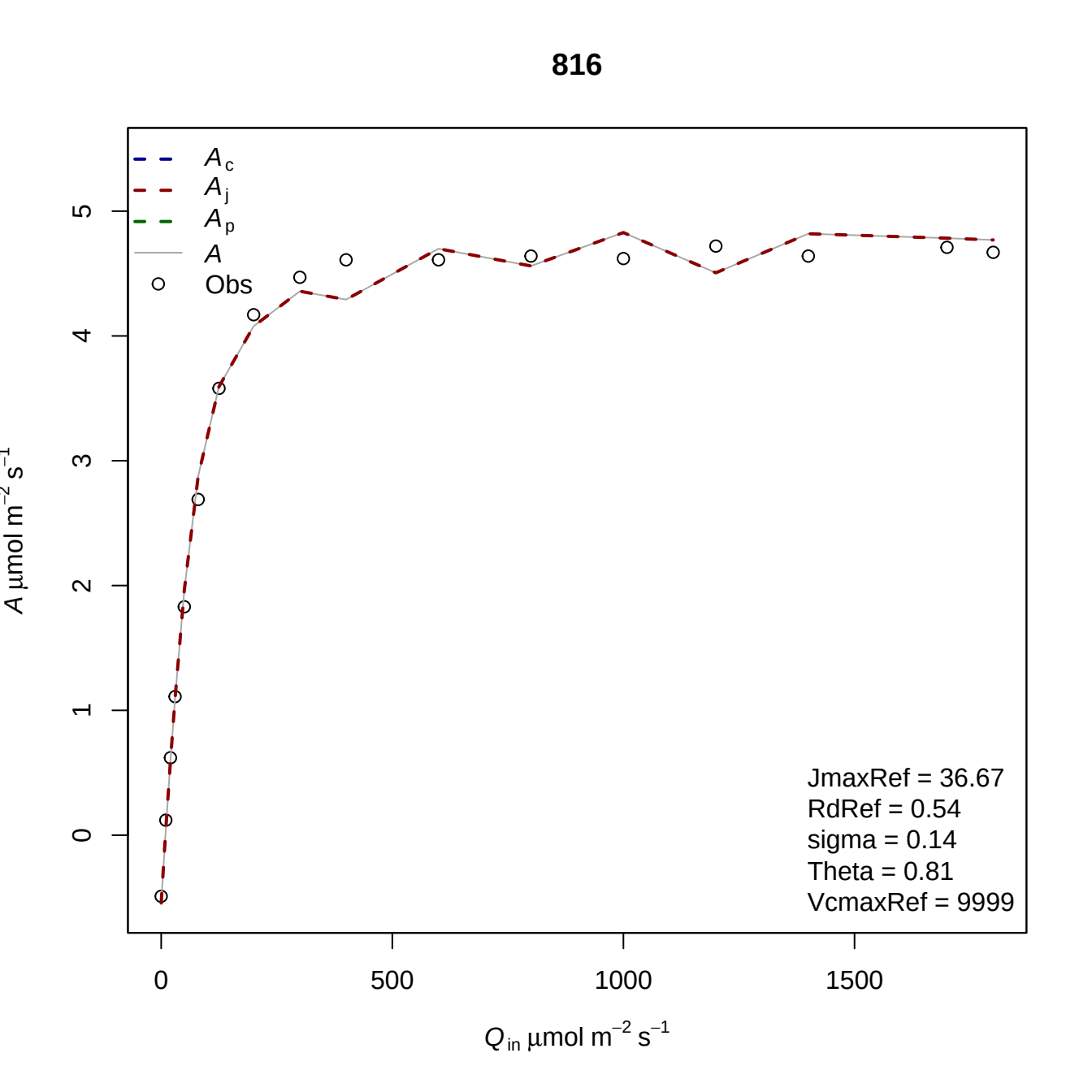

811
A_c
A_j
A_p
A
Obs
JmaxRef = 26.83
RdRef = 0.9
sigma = 0.06
Theta = 0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

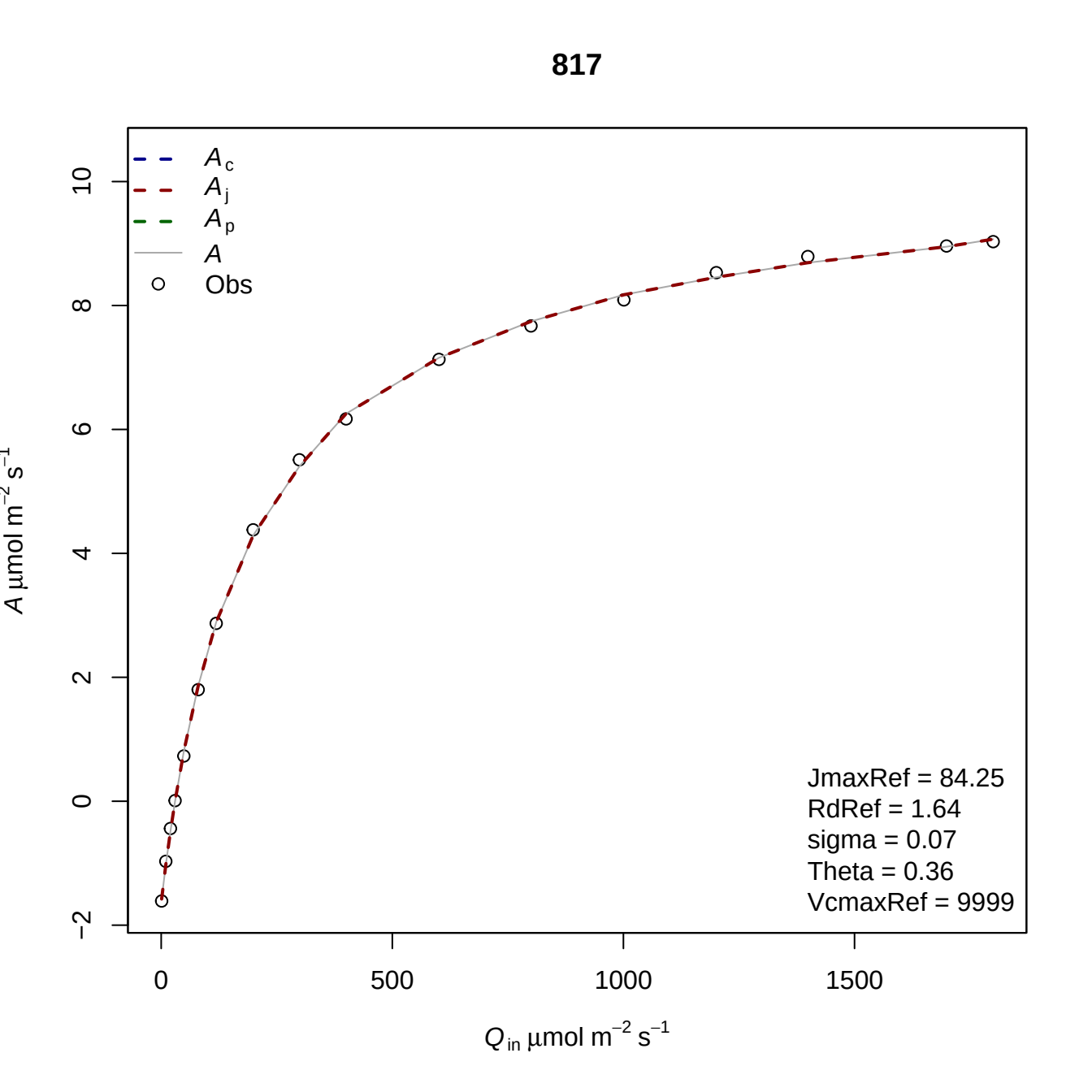

812
A_c
A_j
A_p
A
Obs
JmaxRef = 39.89
RdRef = 0.91
sigma = 0.17
Theta = −2.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

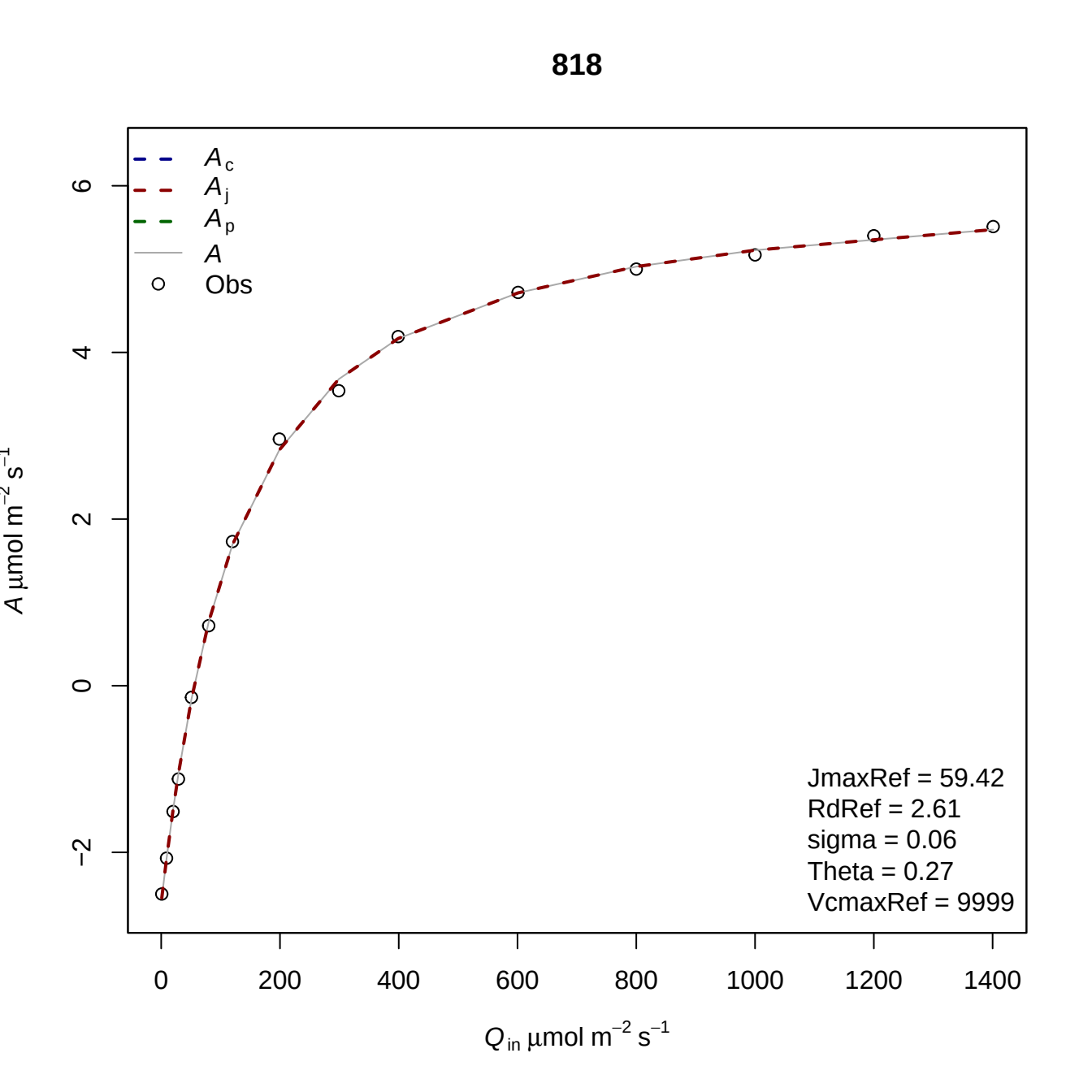

813
A_c
A_j
A_p
A
Obs
JmaxRef = 87.35
RdRef = 1.6
sigma = 0.09
Theta = −0.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

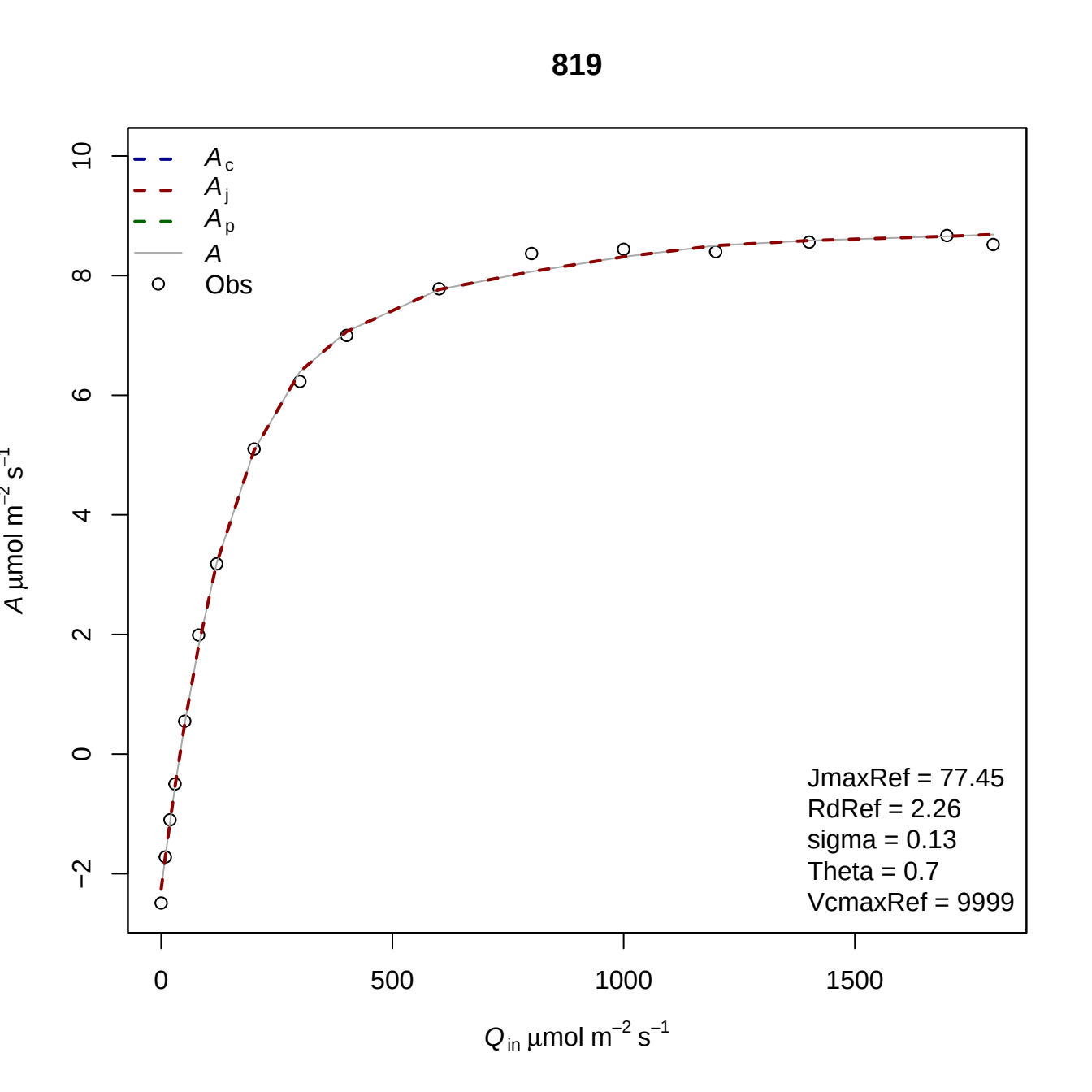

814
A_c
A_j
A_p
A
Obs
JmaxRef = 63.34
RdRef = 2.2
sigma = 0.07
Theta = 0.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

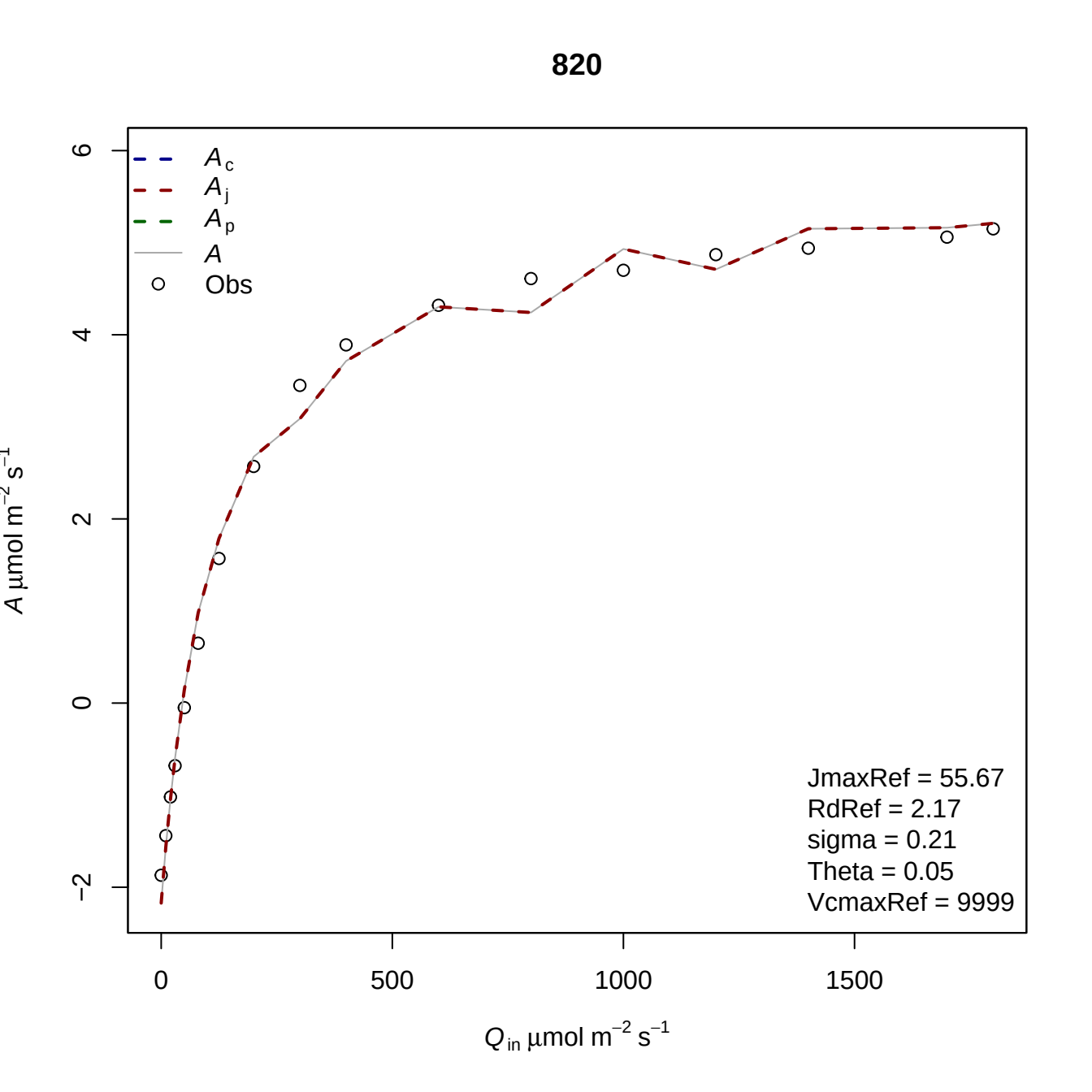

815
A_c
A_j
A_p
A
Obs
JmaxRef = 38.38
RdRef = 1.5
sigma = 0.11
Theta = 0.09
VcmaxRef = 29.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

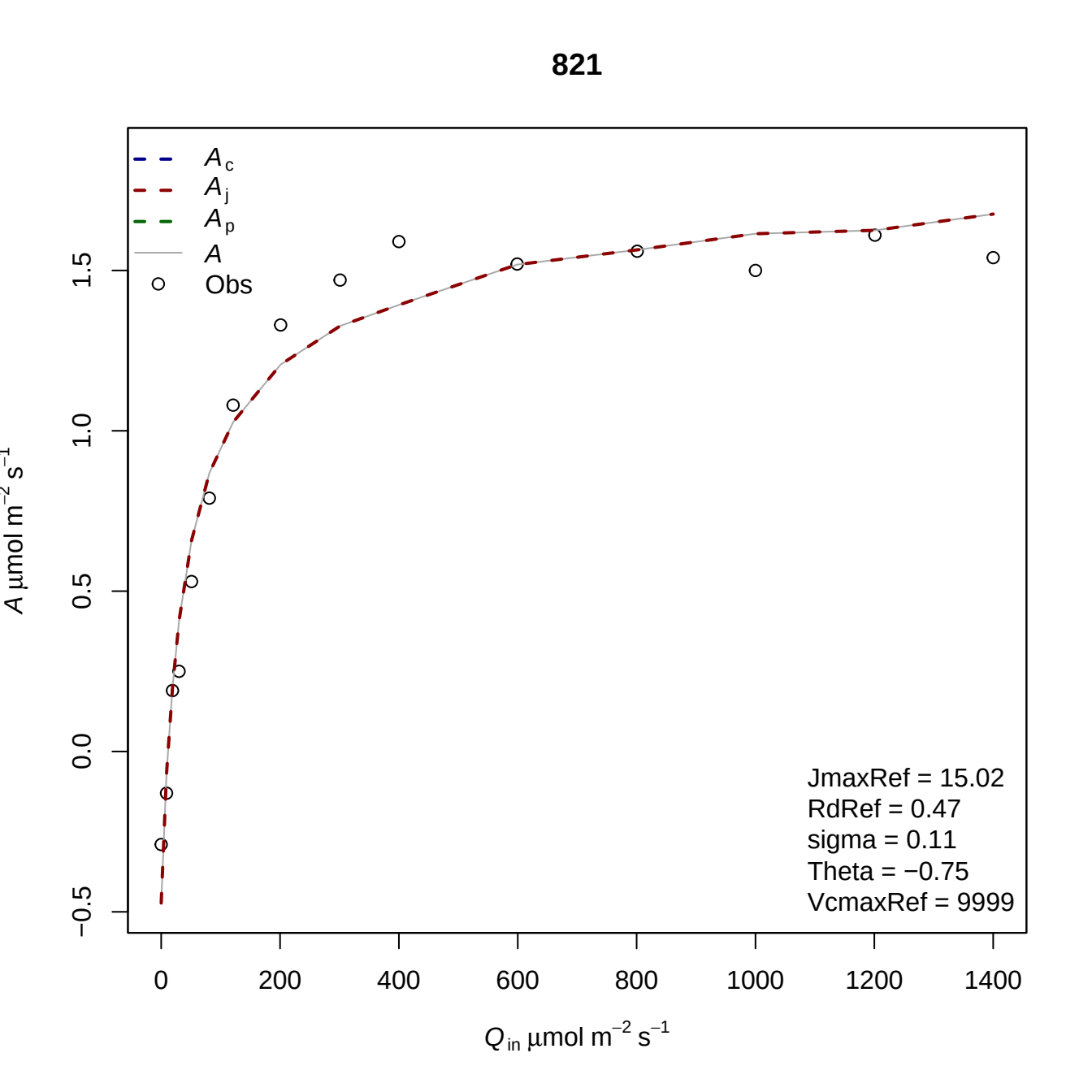

816
A_c
A_j
A_p
A
Obs
JmaxRef = 36.67
RdRef = 0.54
sigma = 0.14
Theta = 0.81
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

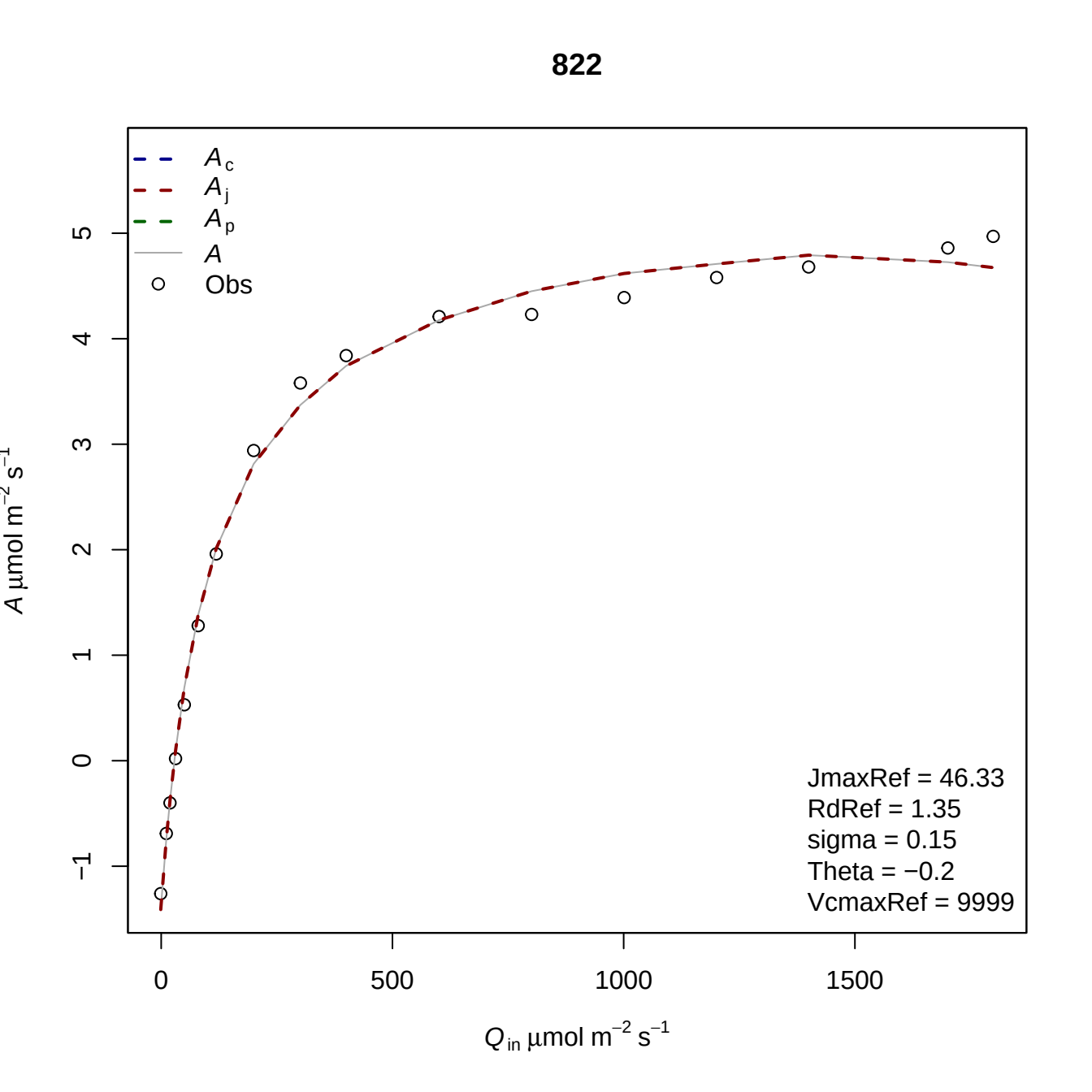

817
A_c
A_j
A_p
A
Obs
JmaxRef = 84.25
RdRef = 1.64
sigma = 0.07
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

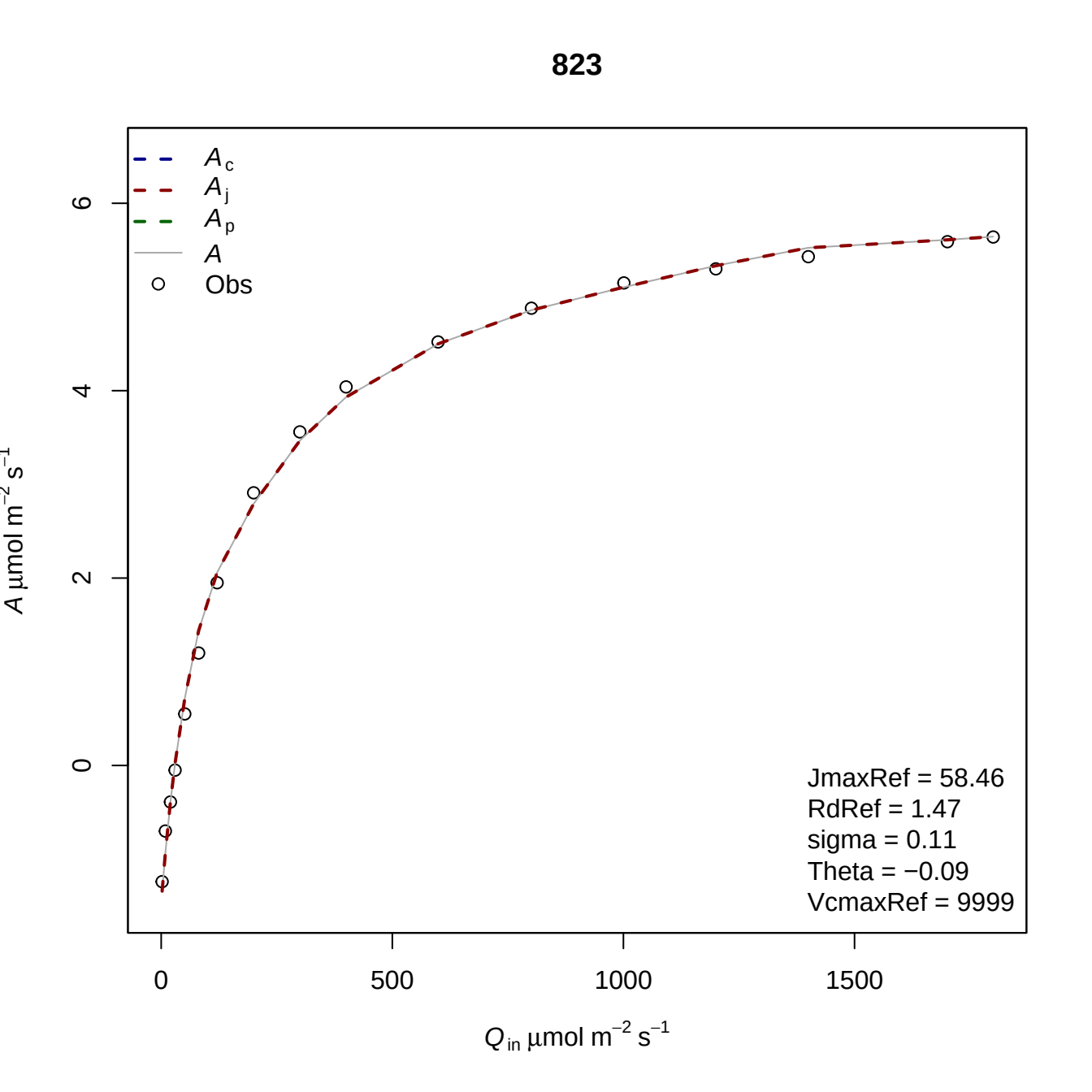

818
A_c
A_j
A_p
A
Obs
JmaxRef = 59.42
RdRef = 2.61
sigma = 0.06
Theta = 0.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

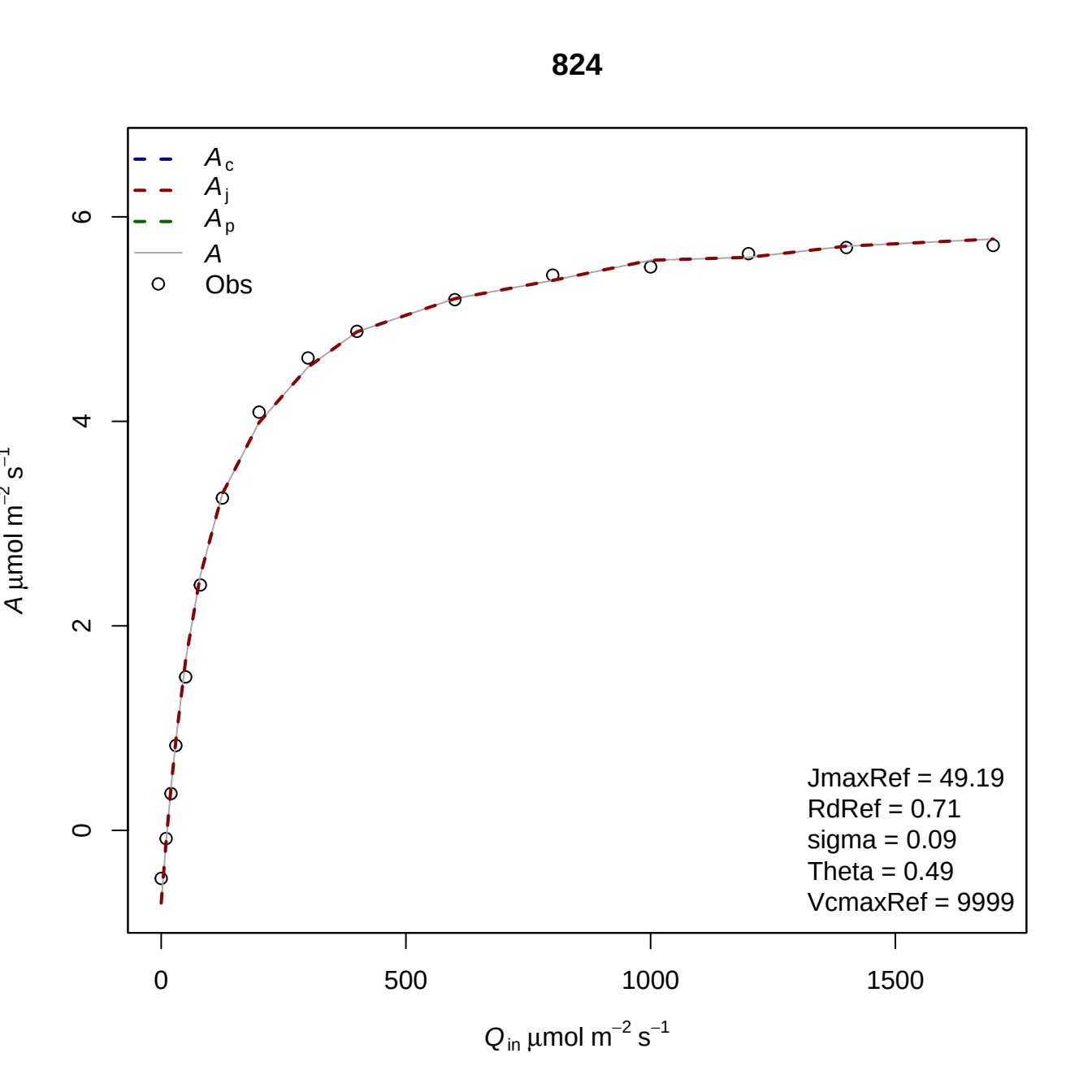

819
A_c
A_j
A_p
A
Obs
JmaxRef = 77.45
RdRef = 2.26
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

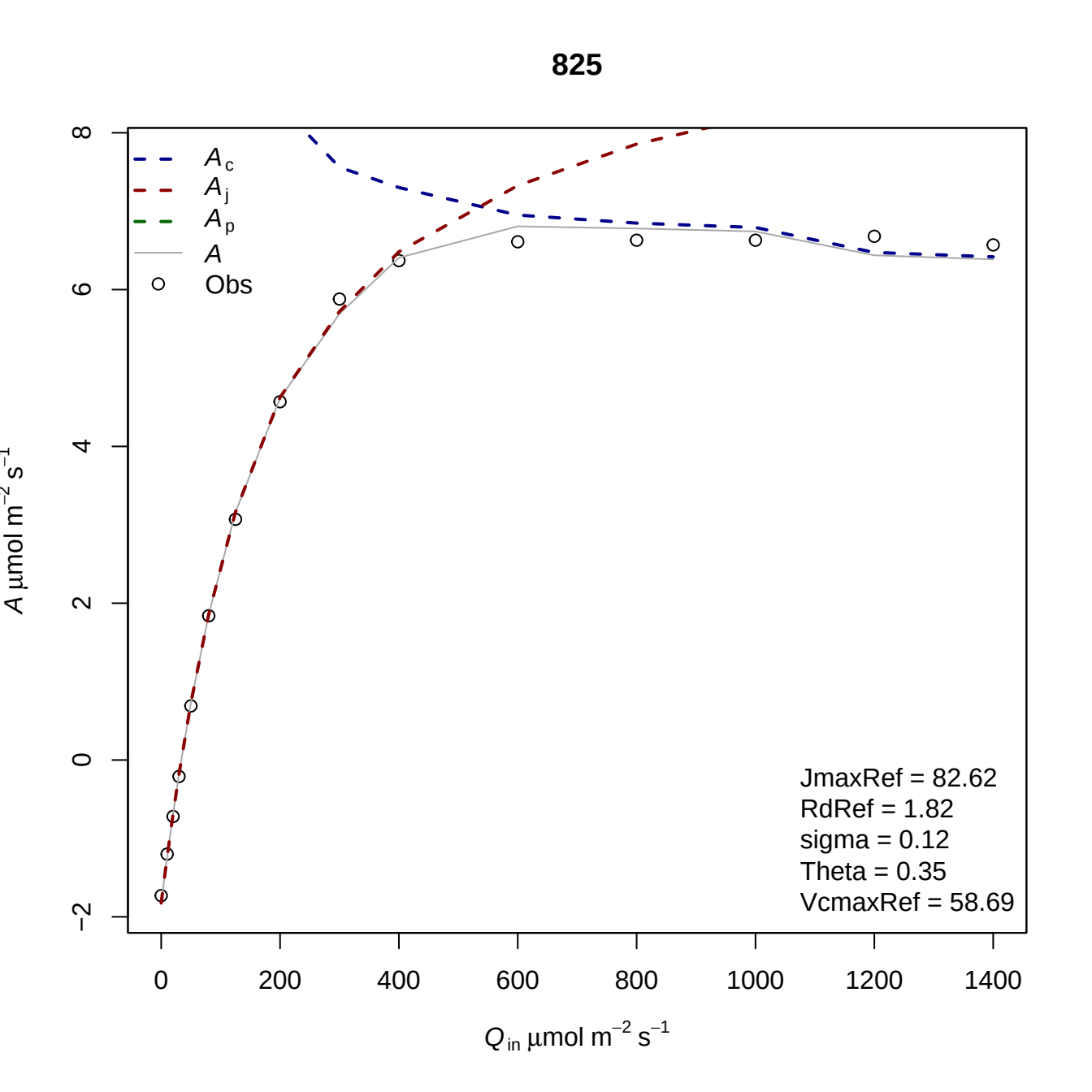

820
A_c
A_j
A_p
A
Obs
JmaxRef = 55.67
RdRef = 2.17
sigma = 0.21
Theta = 0.05
VcmaxRef = 9999
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

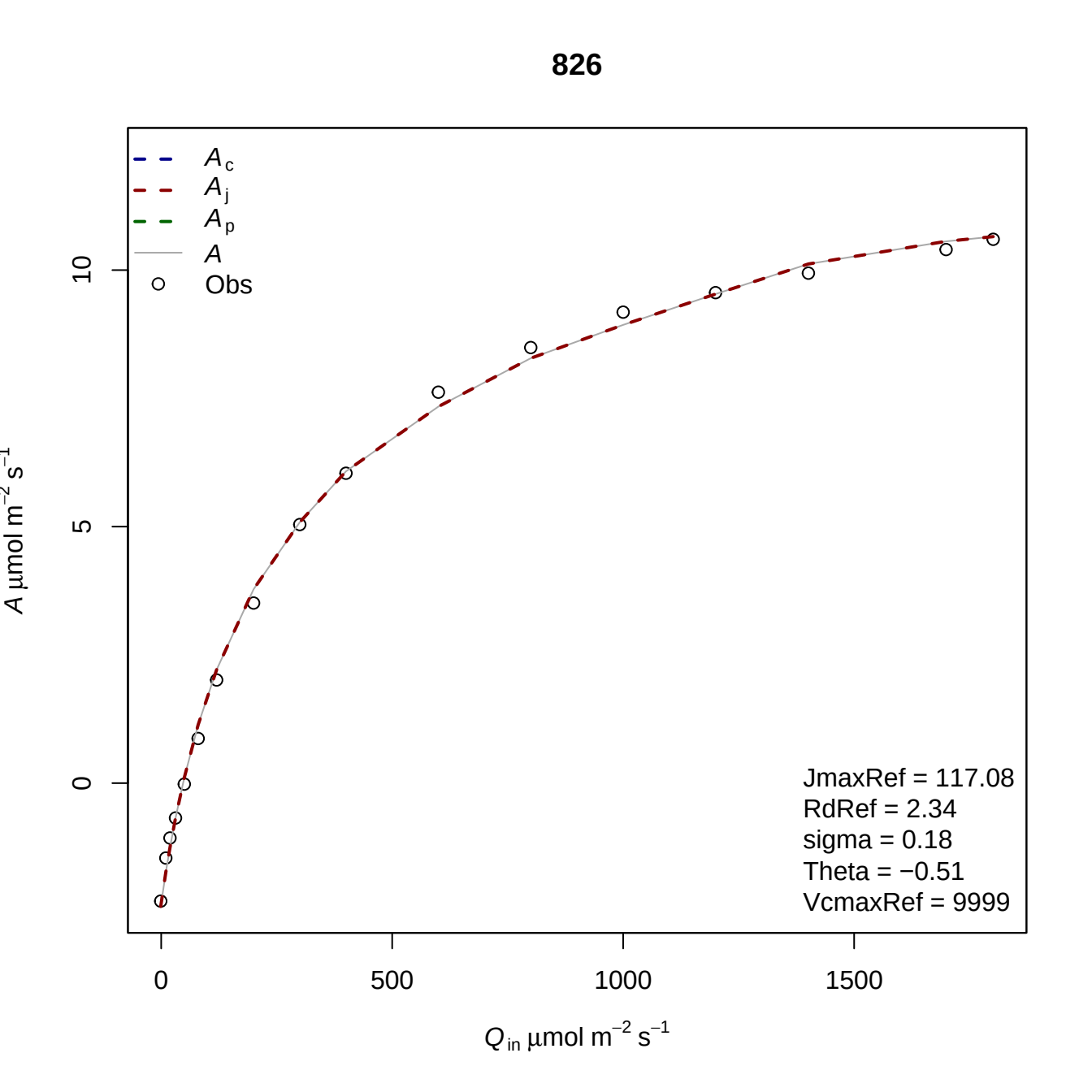

821
A_c
A_j
A_p
A
Obs
JmaxRef = 15.02
RdRef = 0.47
sigma = 0.11
Theta = −0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}

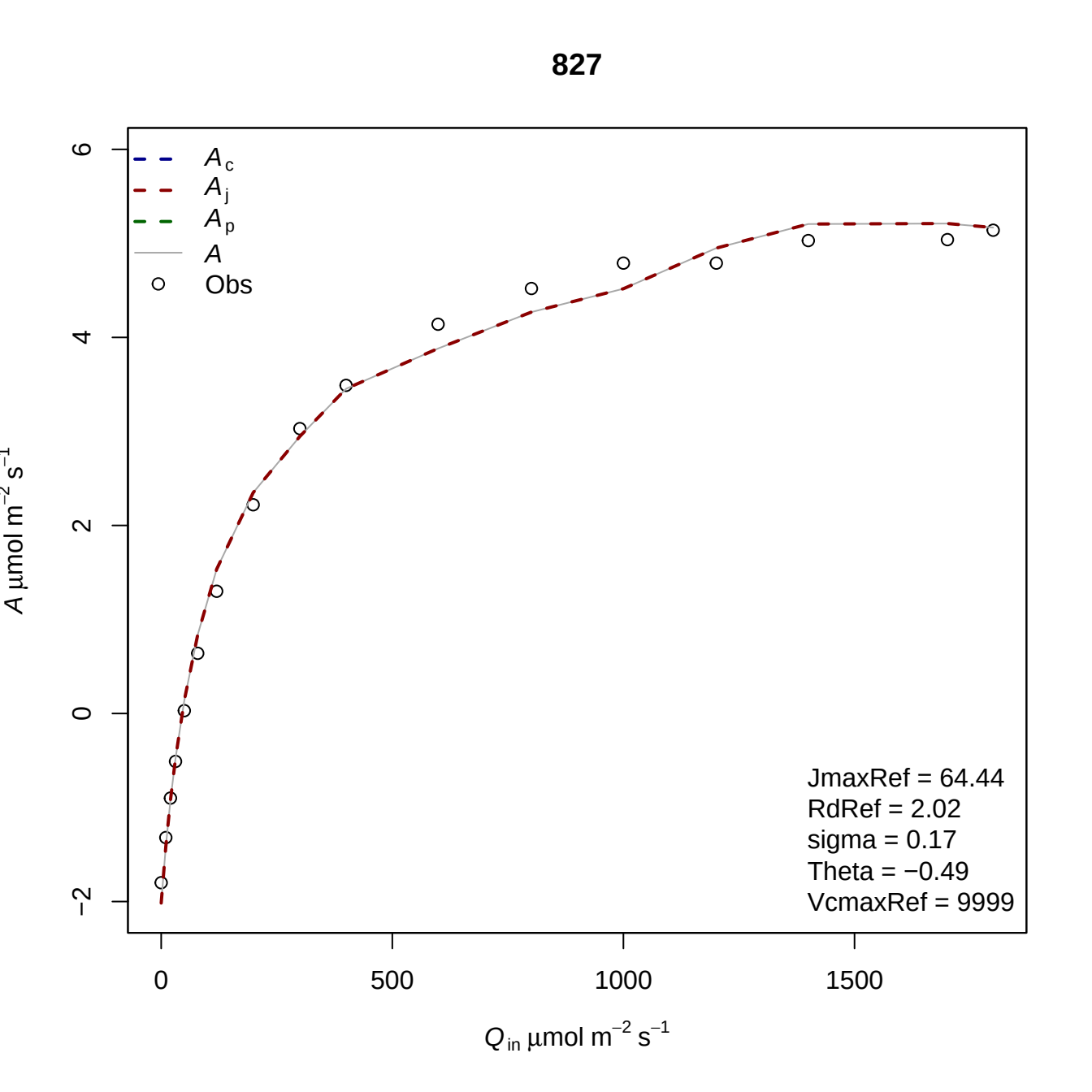

822
A_c
A_j
A_p
A
Obs
JmaxRef = 46.33
RdRef = 1.35
sigma = 0.15
Theta = −0.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

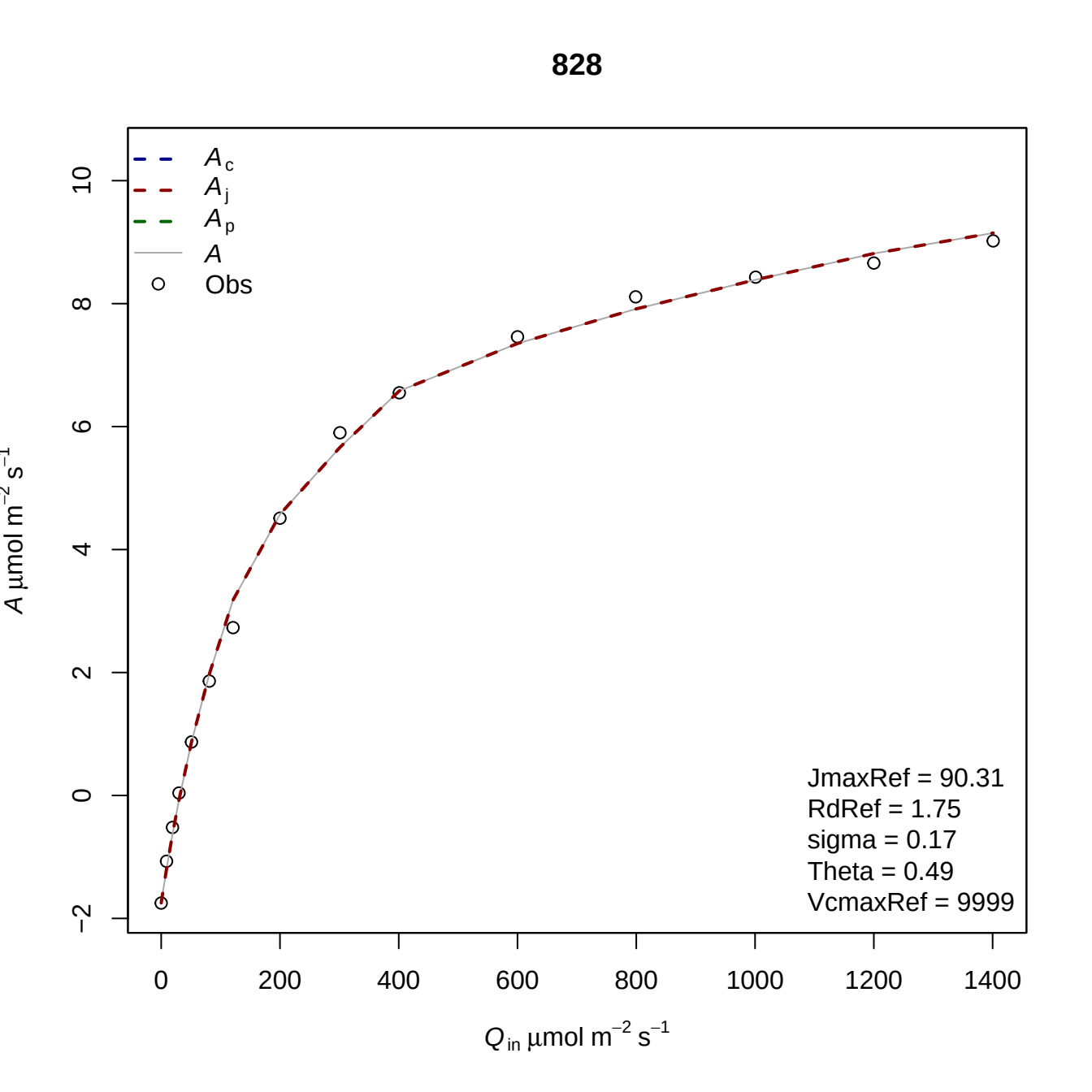

823
A_c
A_j
A_p
A
Obs
JmaxRef = 58.46
RdRef = 1.47
sigma = 0.11
Theta = −0.09
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

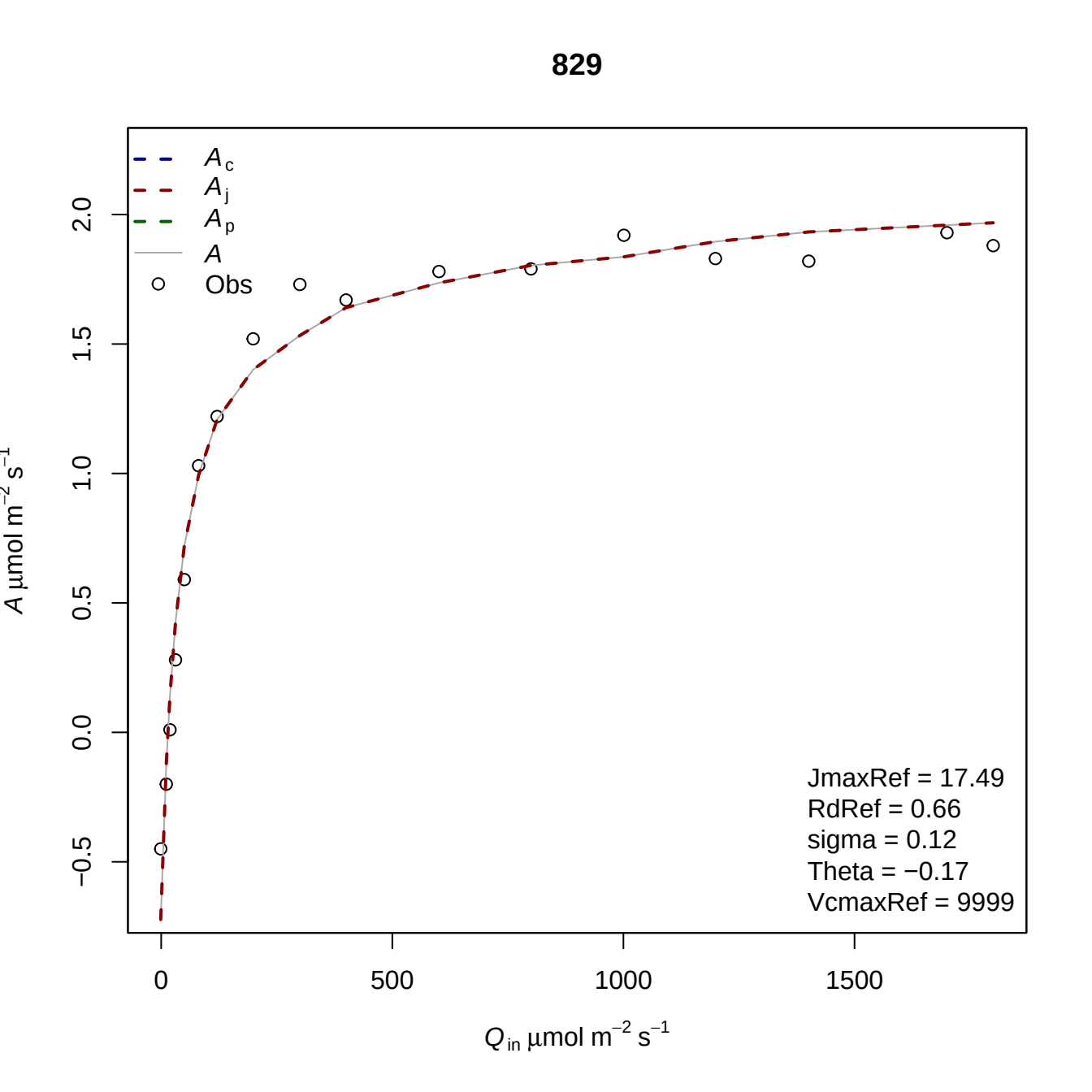

824
A_c
A_j
A_p
A
Obs
JmaxRef = 49.19
RdRef = 0.71
sigma = 0.09
Theta = 0.49
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

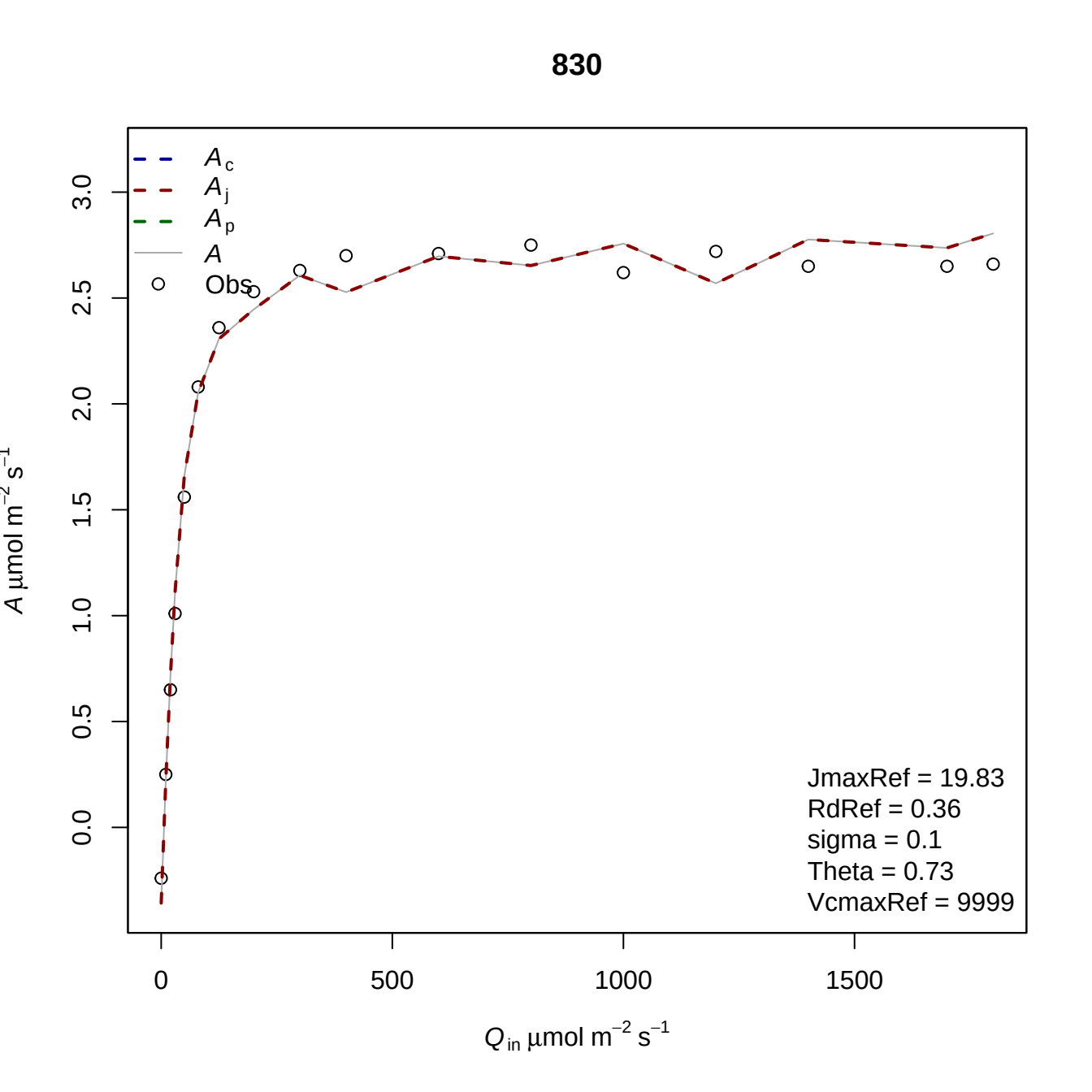

825
A_c
A_j
A_p
A
Obs
JmaxRef = 82.62
RdRef = 1.82
sigma = 0.12
Theta = 0.35
VcmaxRef = 58.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

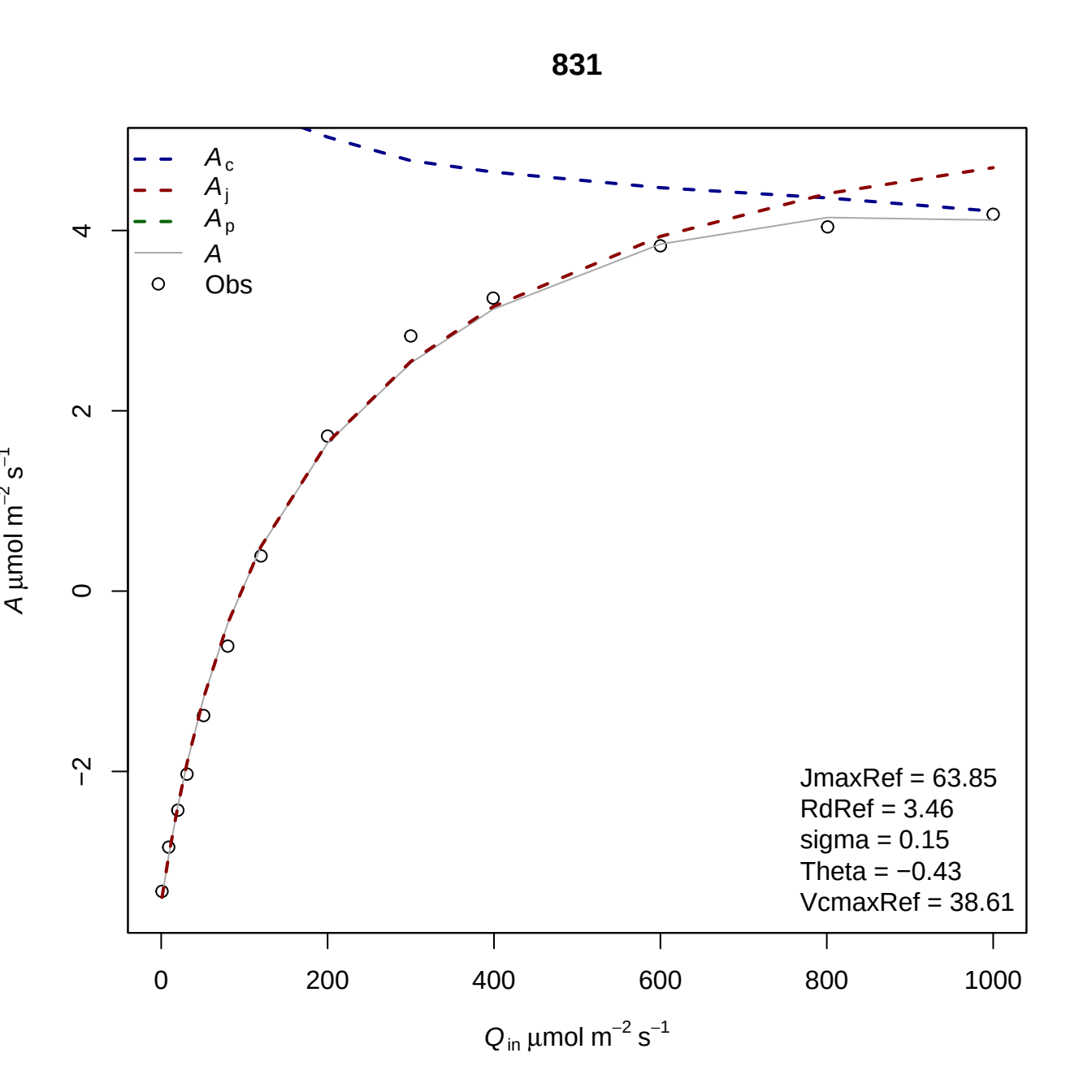

826
A_c
A_j
A_p
A
Obs
JmaxRef = 117.08
RdRef = 2.34
sigma = 0.18
Theta = −0.51
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

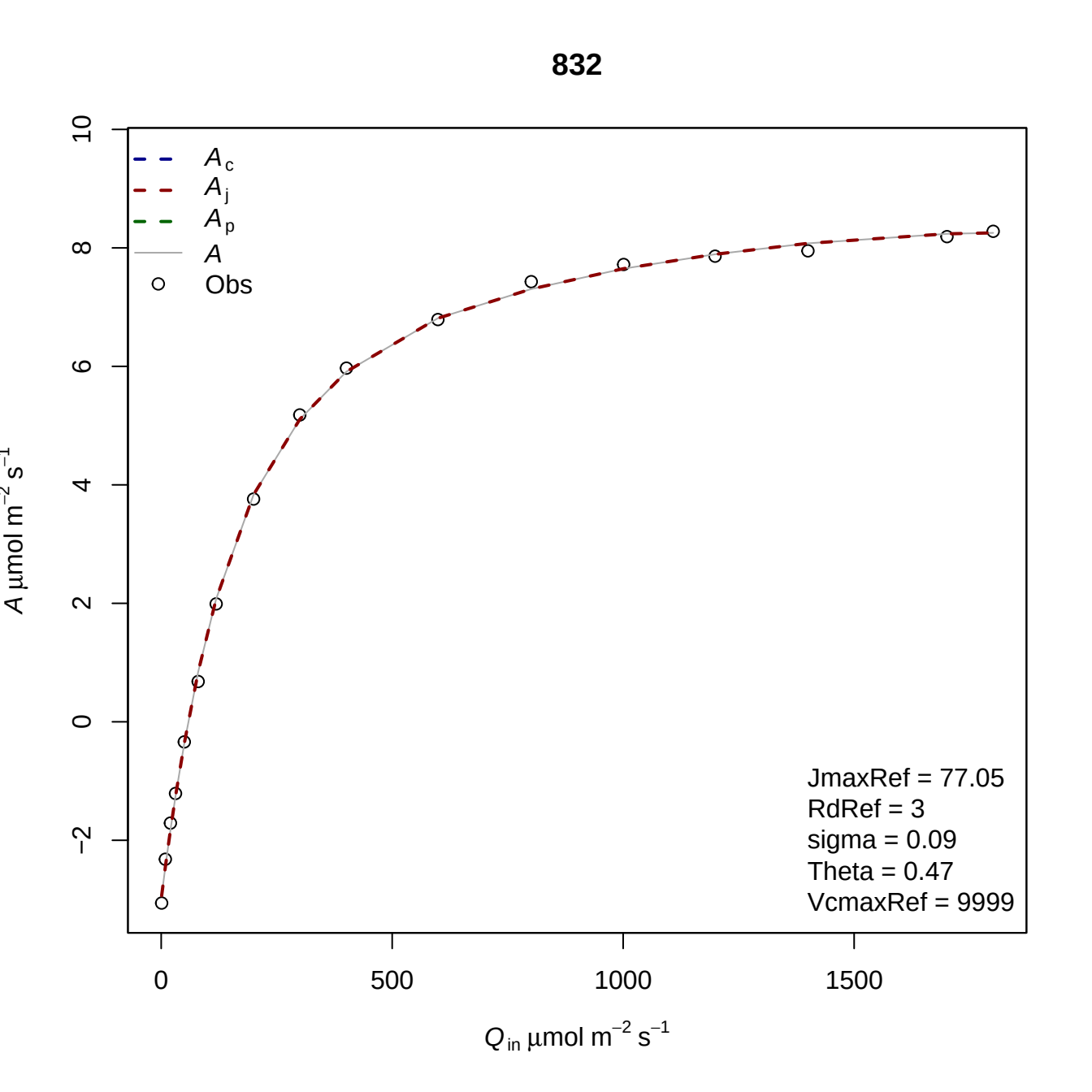

827
A_c
A_j
A_p
A
Obs
JmaxRef = 64.44
RdRef = 2.02
sigma = 0.17
Theta = −0.49
VcmaxRef = 9999
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

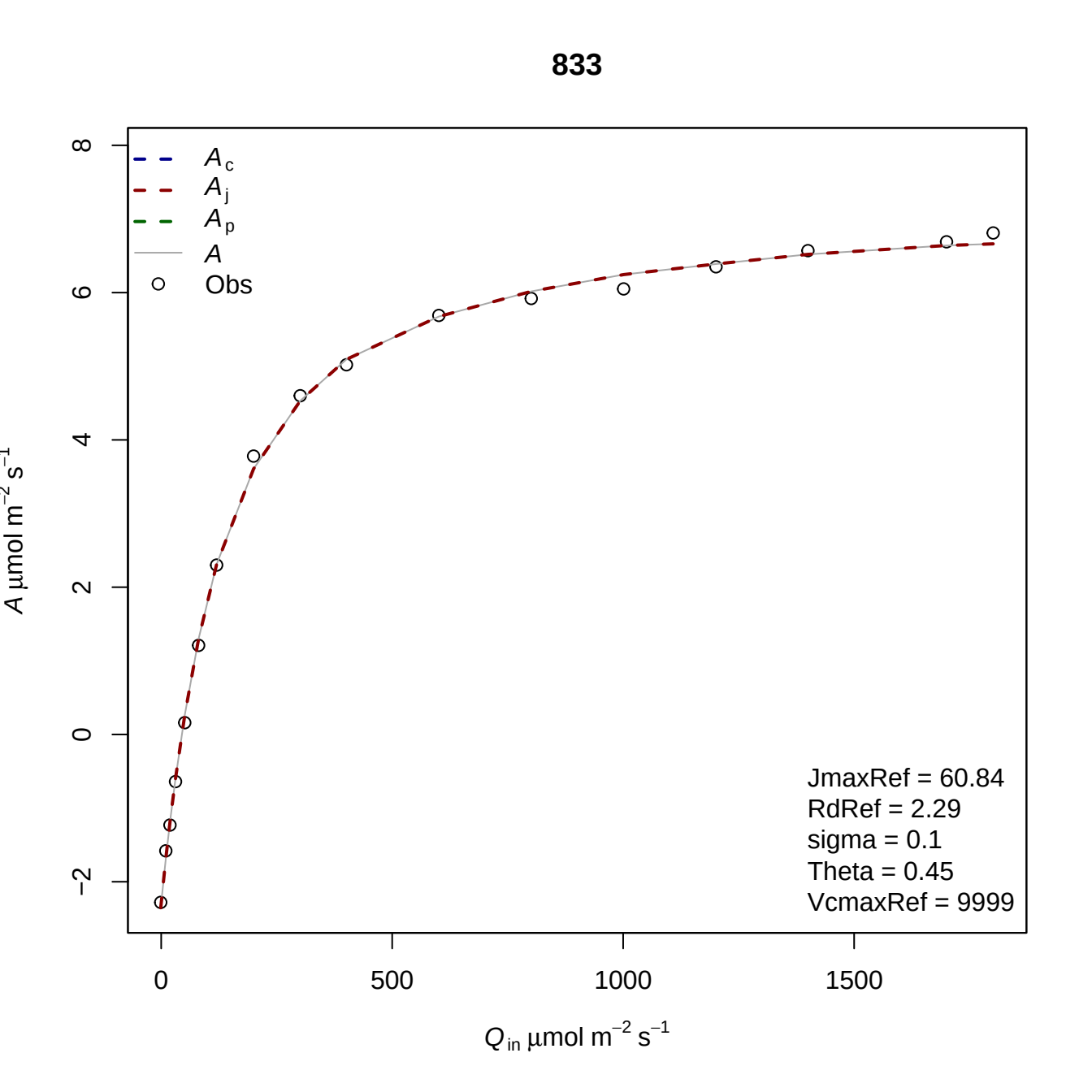

828
A_c
A_j
A_p
A
Obs
JmaxRef = 90.31
RdRef = 1.75
sigma = 0.17
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

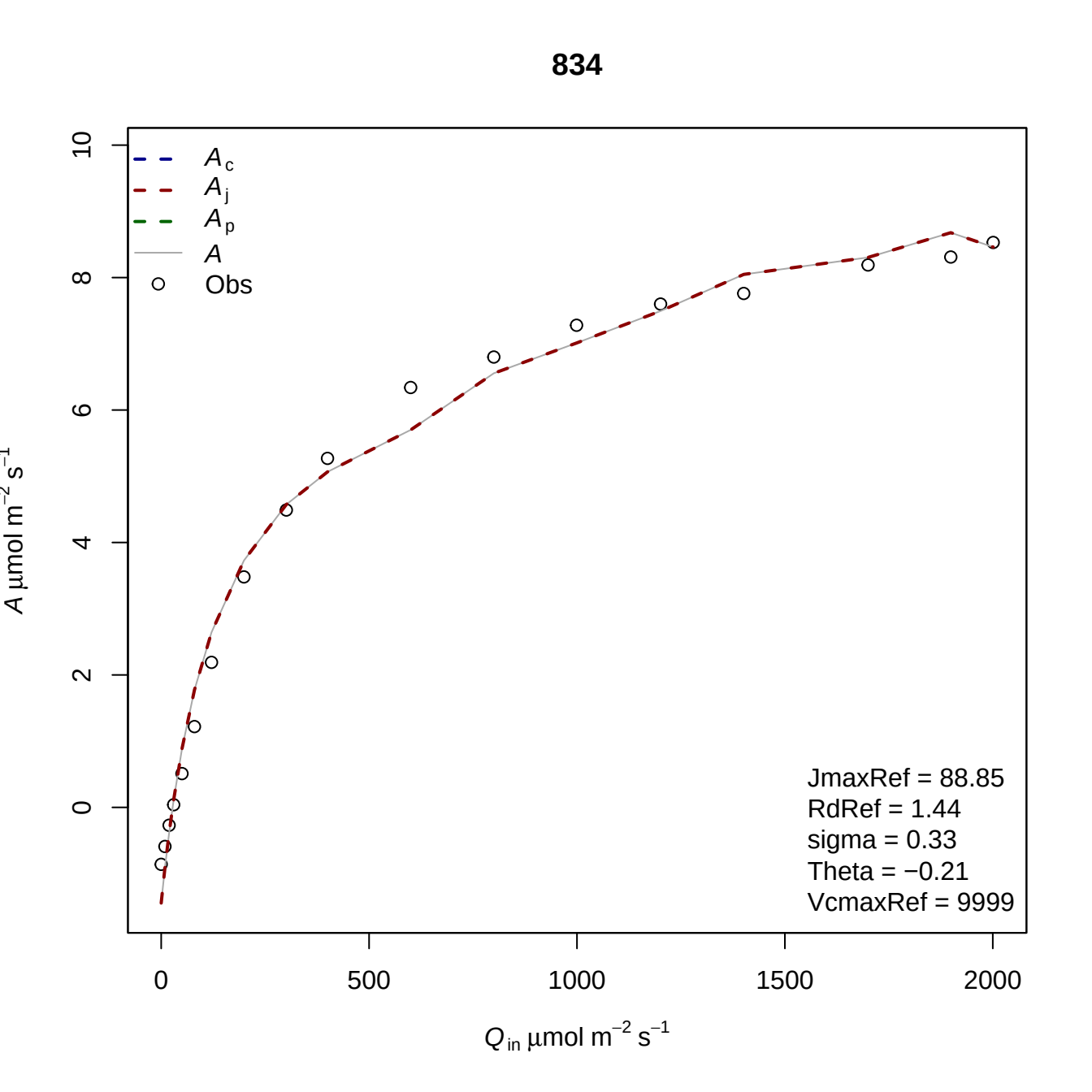

829
A_c
A_j
A_p
A
Obs
JmaxRef = 17.49
RdRef = 0.66
sigma = 0.12
Theta = −0.17
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

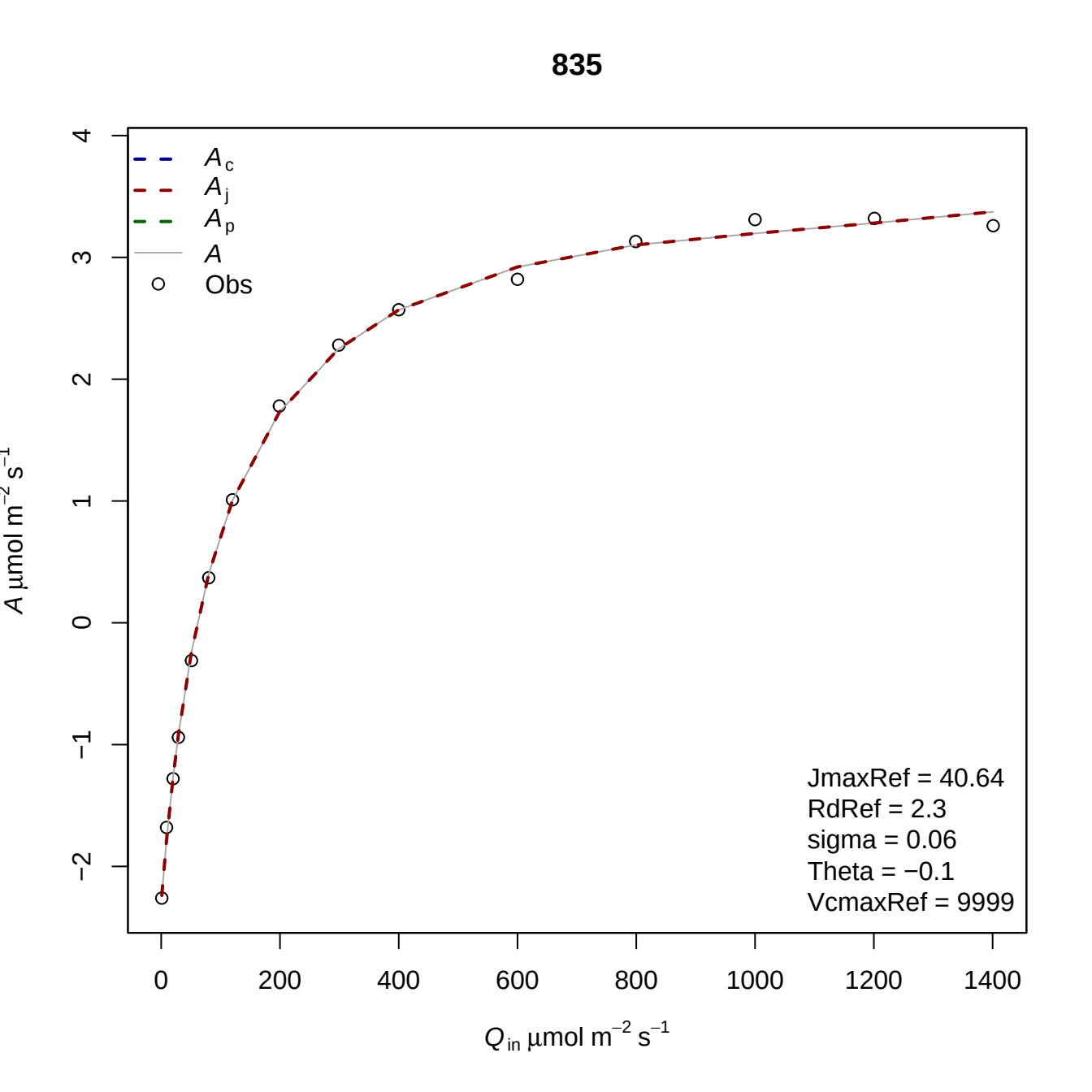

830
A_c
A_j
A_p
A
Obs
JmaxRef = 19.83
RdRef = 0.36
sigma = 0.1
Theta = 0.73
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

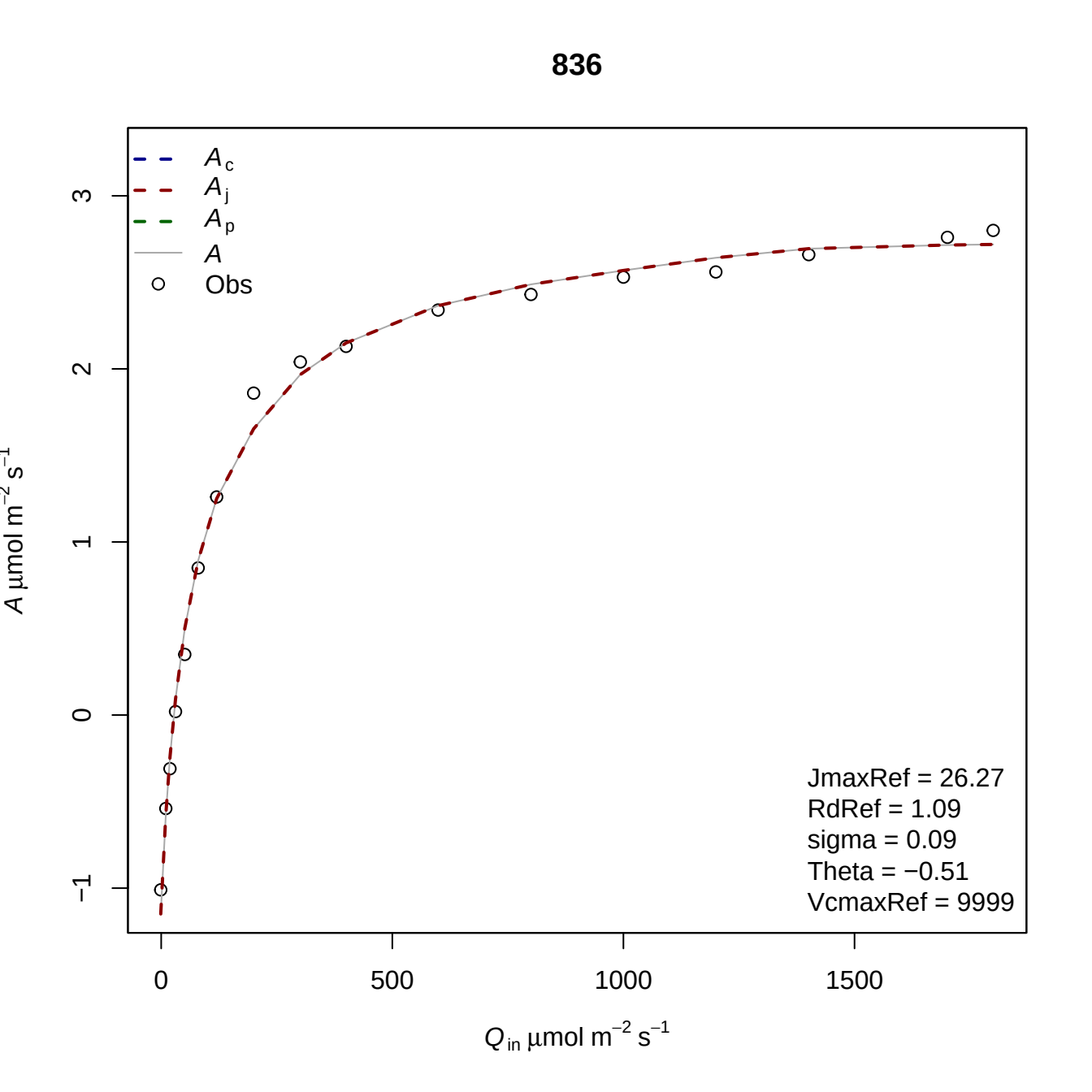

831
A_c
A_j
A_p
A
Obs
JmaxRef = 63.85
RdRef = 3.46
sigma = 0.15
Theta = −0.43
VcmaxRef = 38.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

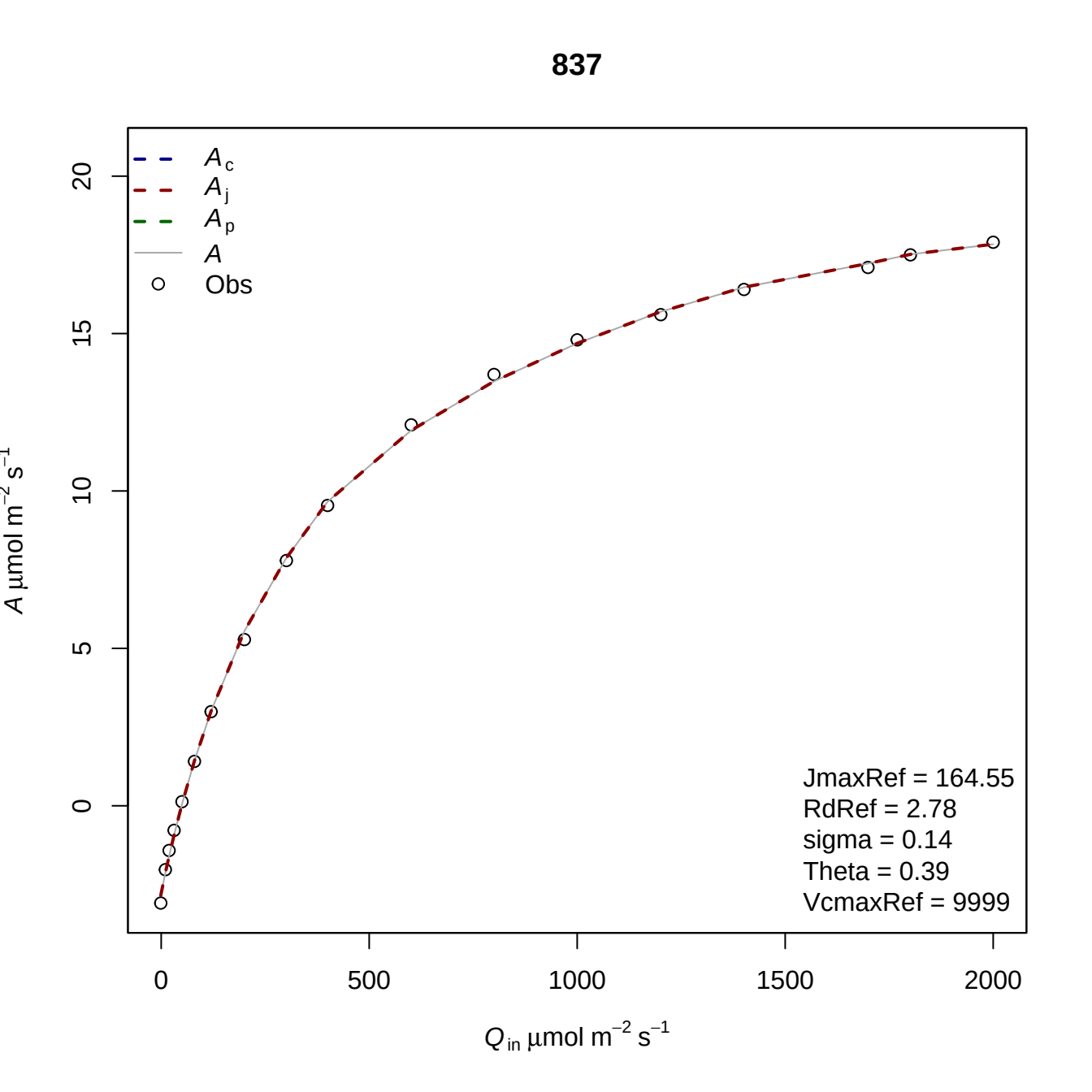

832
A_c
A_j
A_p
A
Obs
JmaxRef = 77.05
RdRef = 3
sigma = 0.09
Theta = 0.47
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

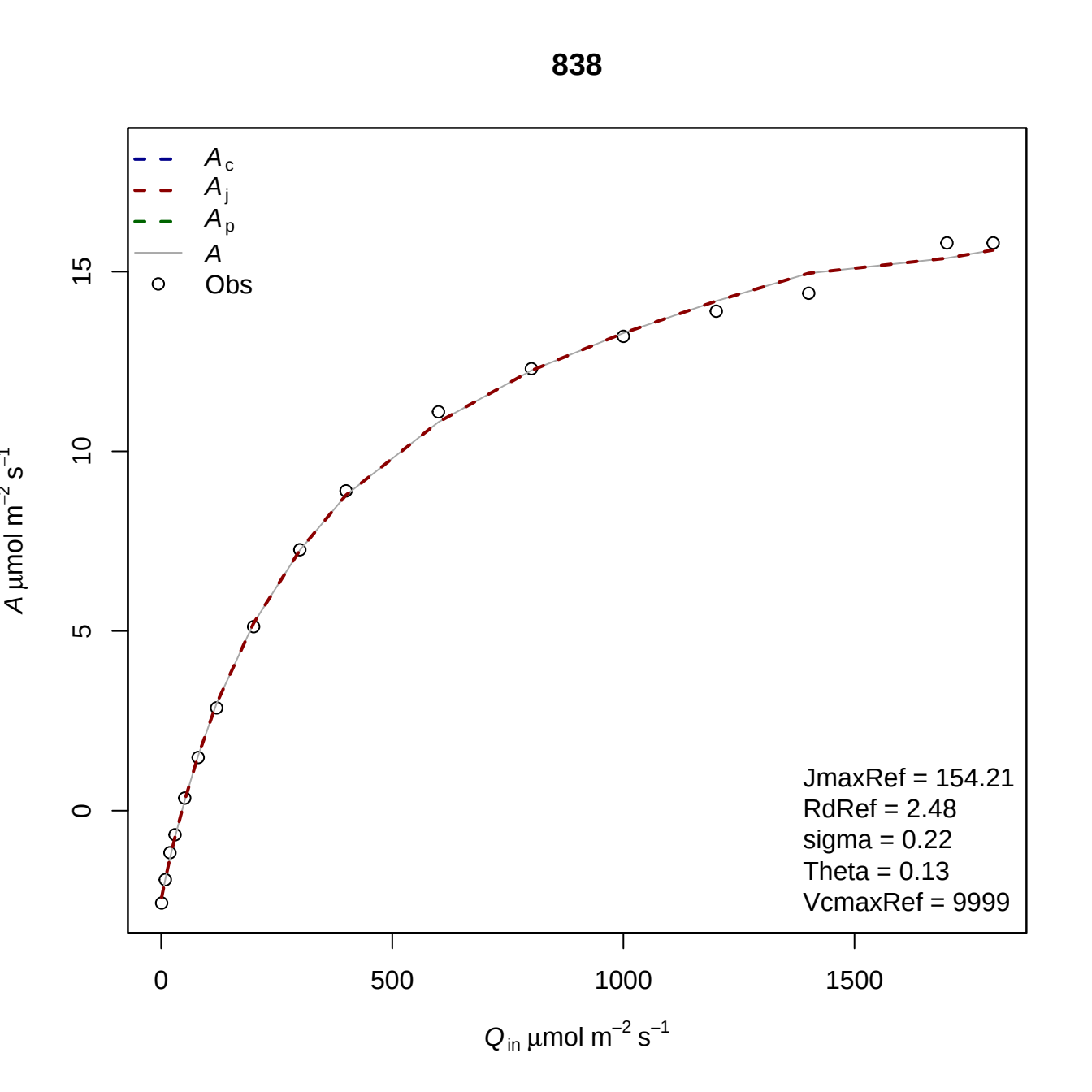

833
A_c
A_j
A_p
A
Obs
JmaxRef = 60.84
RdRef = 2.29
sigma = 0.1
Theta = 0.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

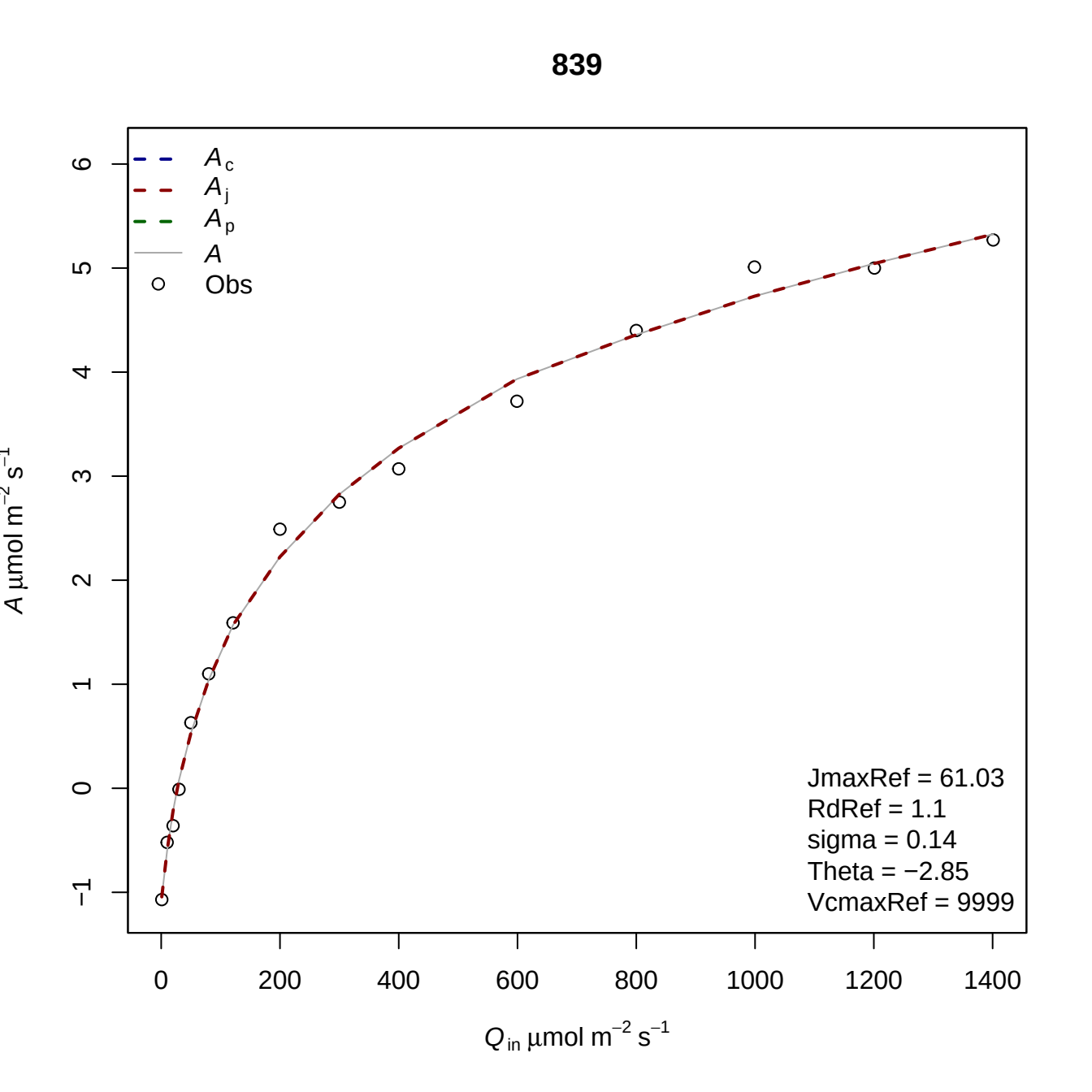

834
A_c
A_j
A_p
A
Obs
JmaxRef = 88.85
RdRef = 1.44
sigma = 0.33
Theta = −0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

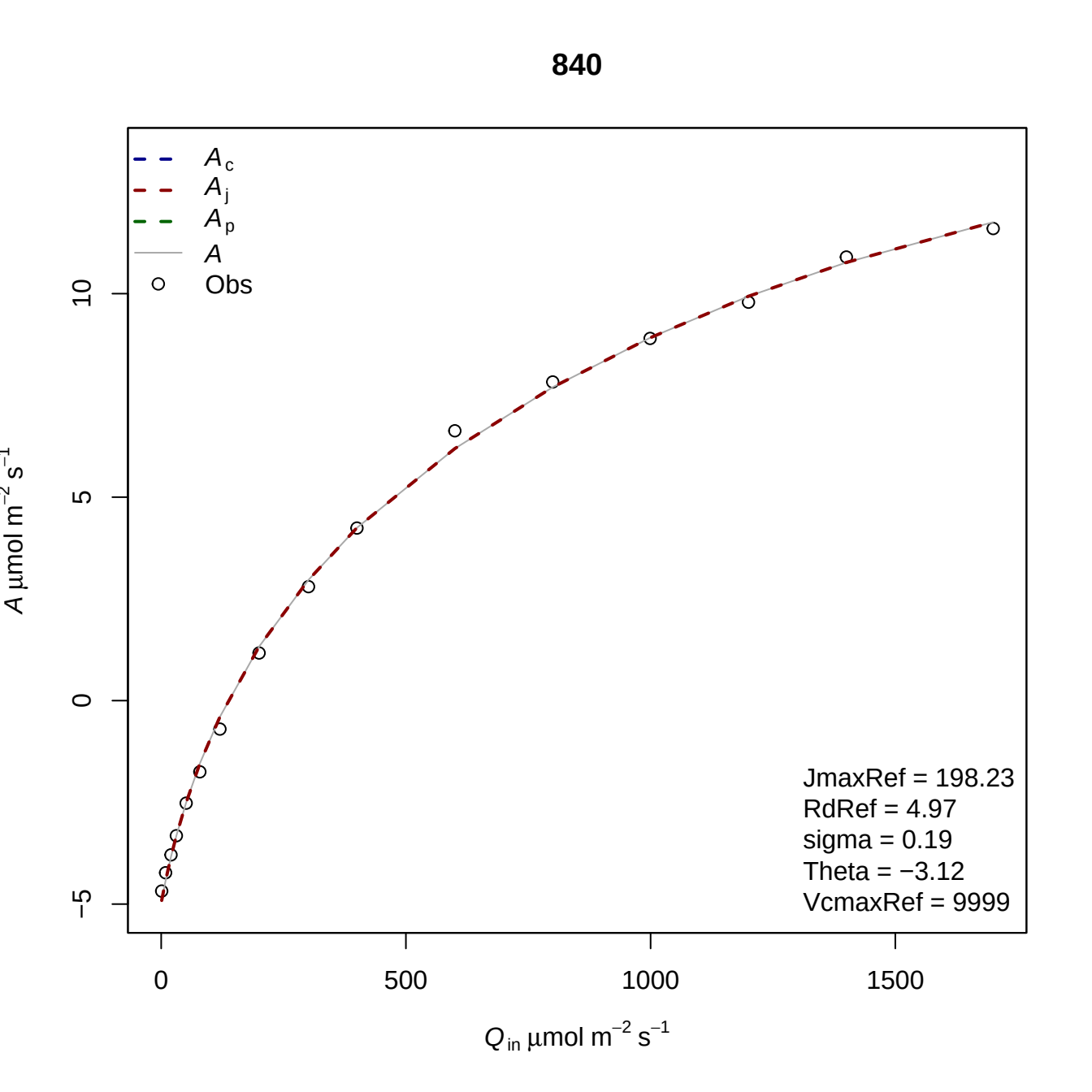

835
A_c
A_j
A_p
A
Obs
JmaxRef = 40.64
RdRef = 2.3
sigma = 0.06
Theta = −0.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

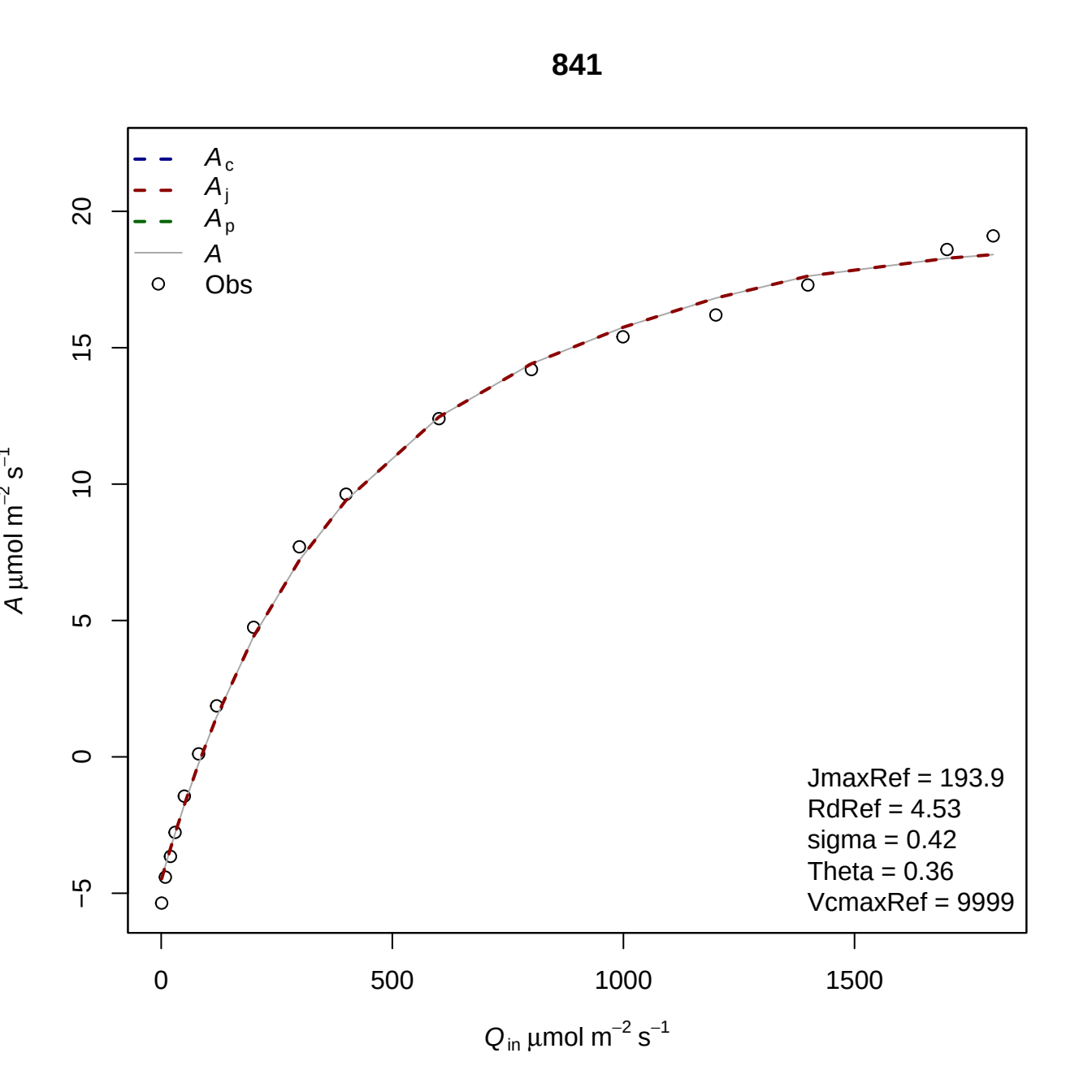

836
A_c
A_j
A_p
A
Obs
JmaxRef = 26.27
RdRef = 1.09
sigma = 0.09
Theta = −0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

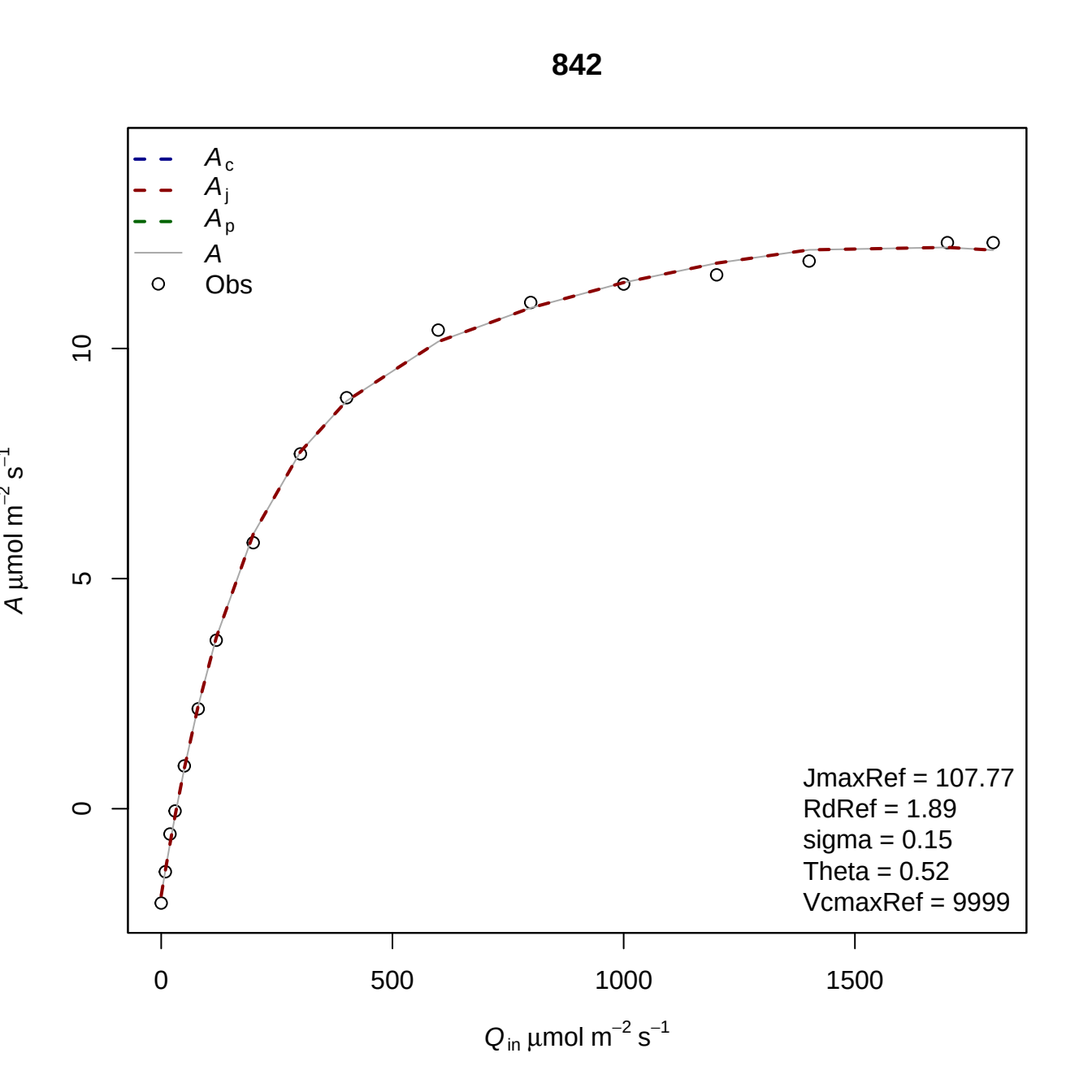

837
A_c
A_j
A_p
A
Obs
JmaxRef = 164.55
RdRef = 2.78
sigma = 0.14
Theta = 0.39
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

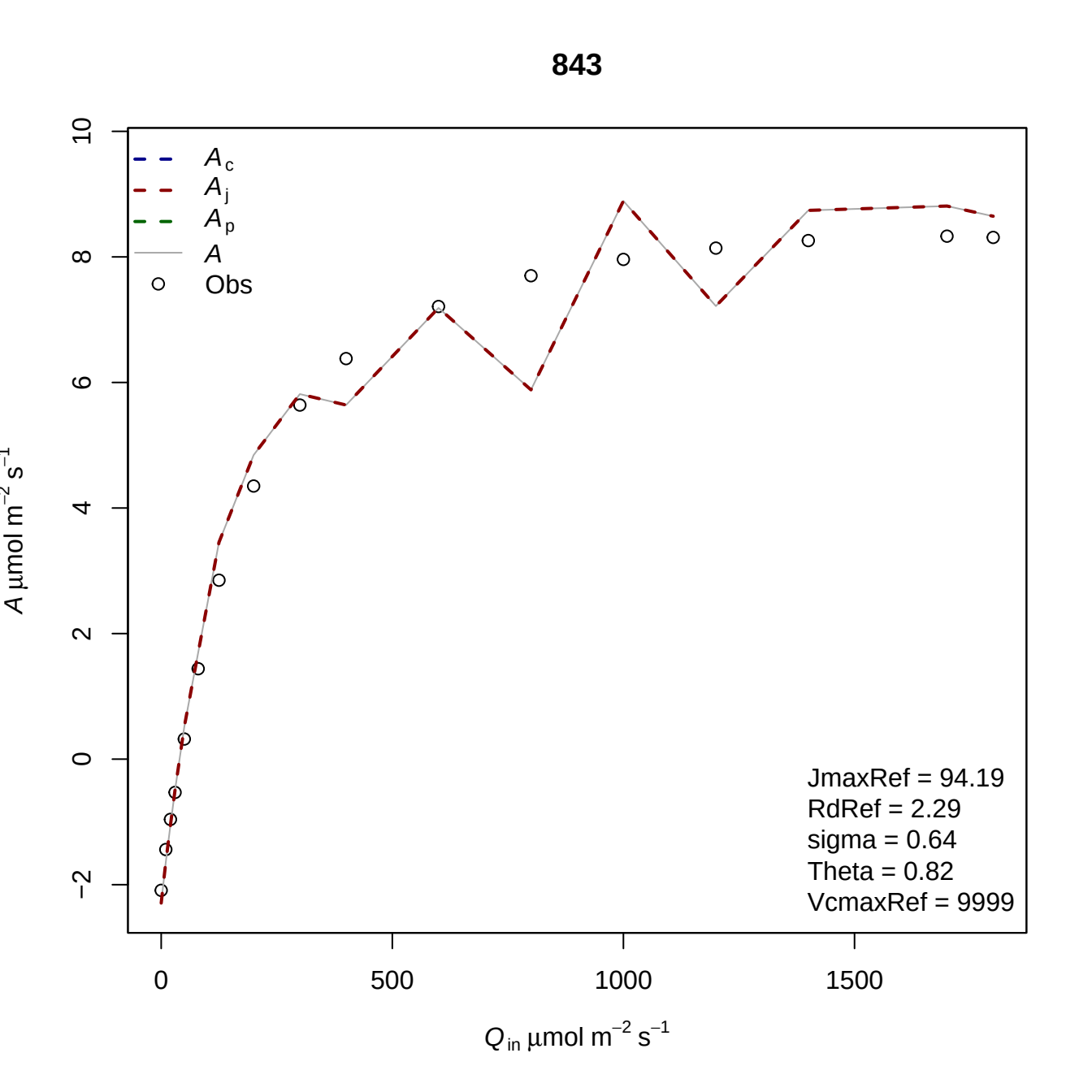

838
A_c
A_j
A_p
A
Obs
JmaxRef = 154.21
RdRef = 2.48
sigma = 0.22
Theta = 0.13
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

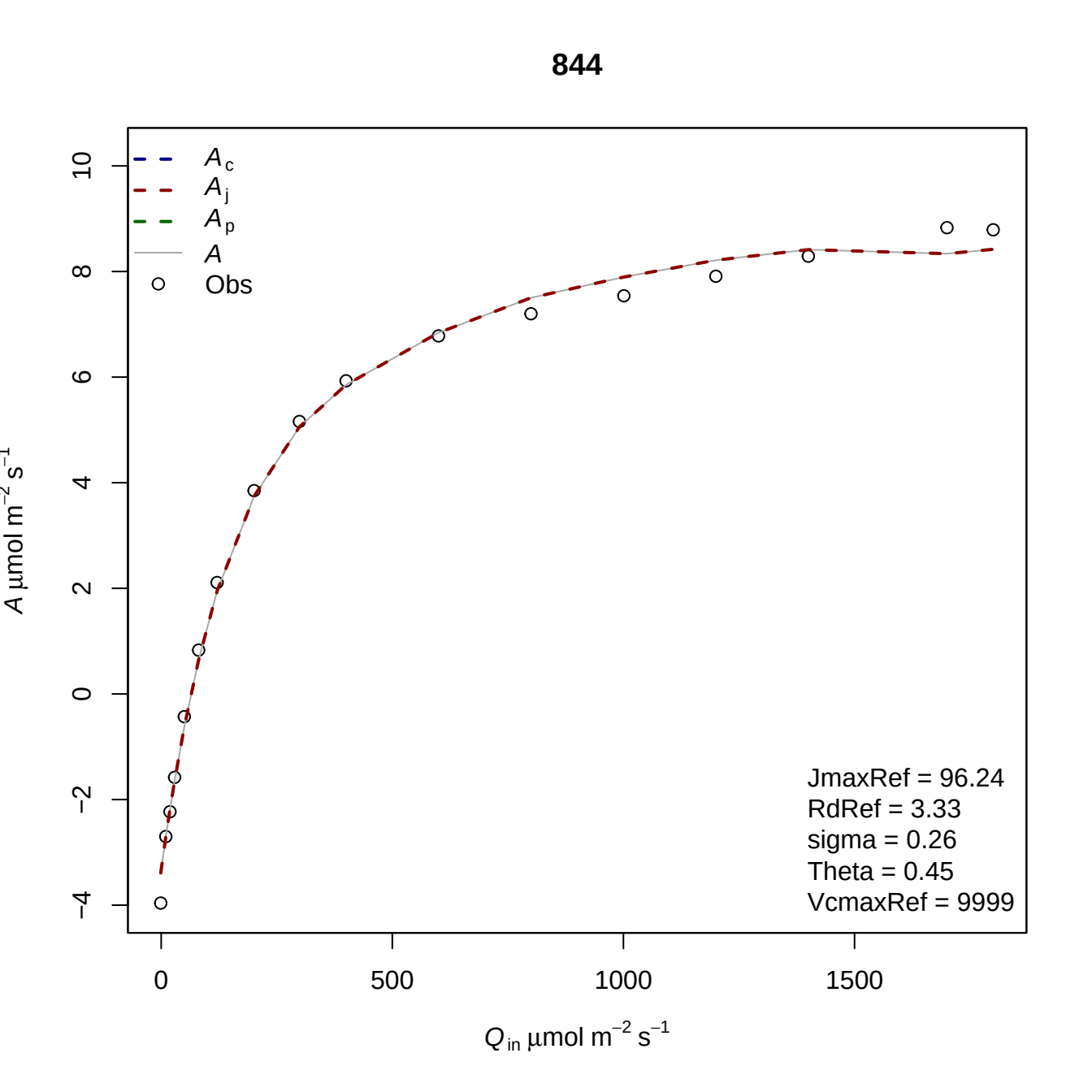

839
A_c
A_j
A_p
A
Obs
JmaxRef = 61.03
RdRef = 1.1
sigma = 0.14
Theta = −2.85
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
1400

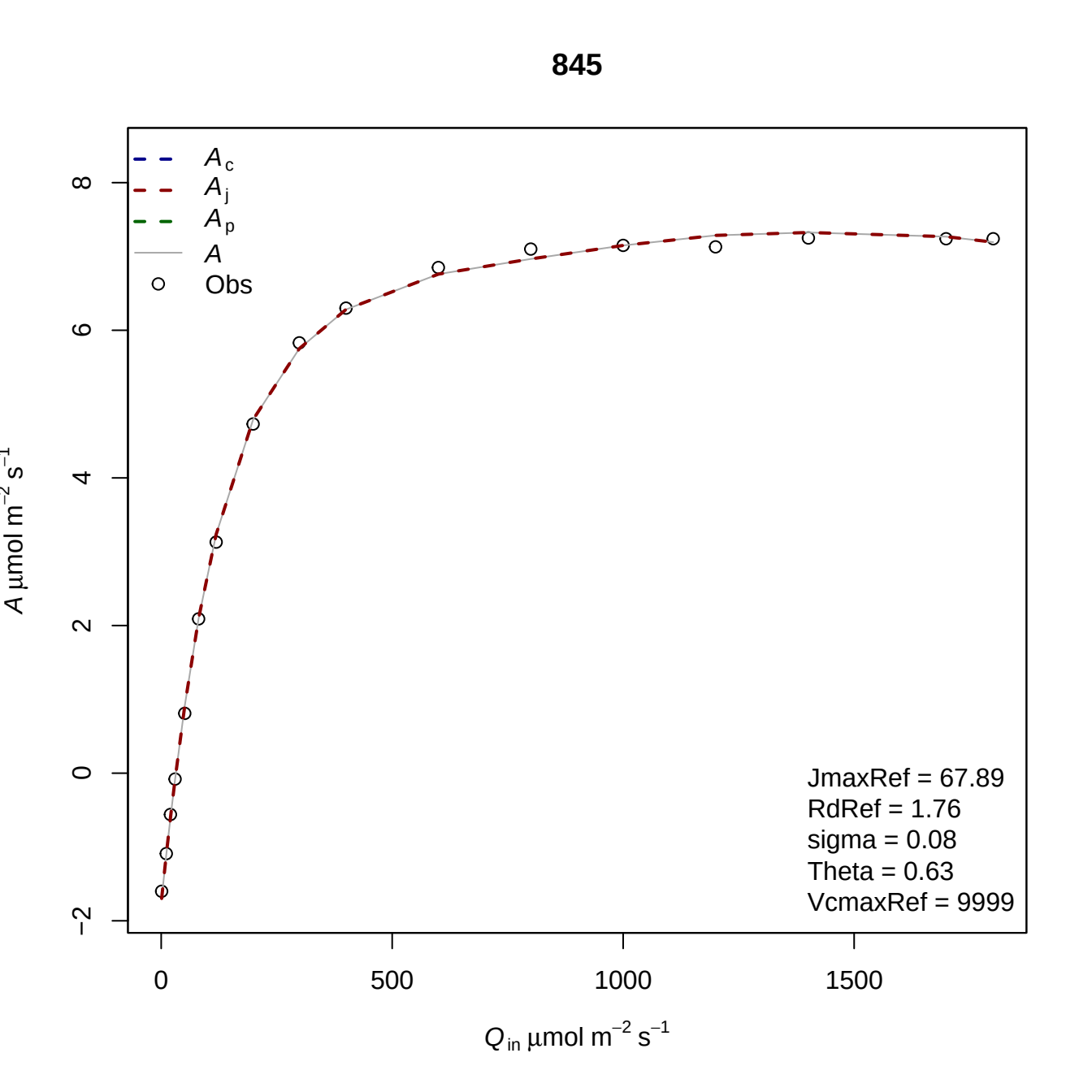

840
A_c
A_j
A_p
A
Obs
JmaxRef = 198.23
RdRef = 4.97
sigma = 0.19
Theta = −3.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

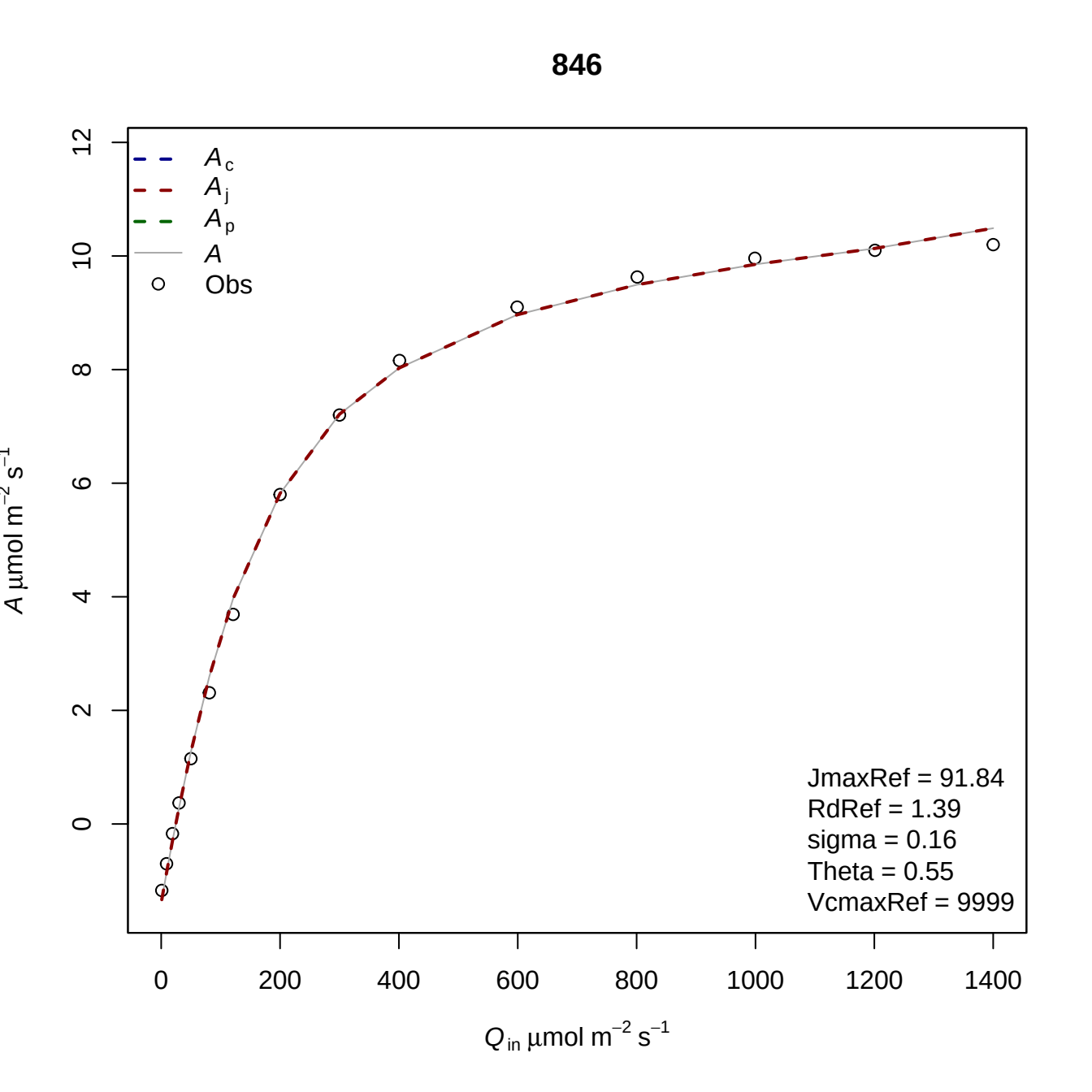

841
A_c
A_j
A_p
A
Obs
JmaxRef = 193.9
RdRef = 4.53
sigma = 0.42
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

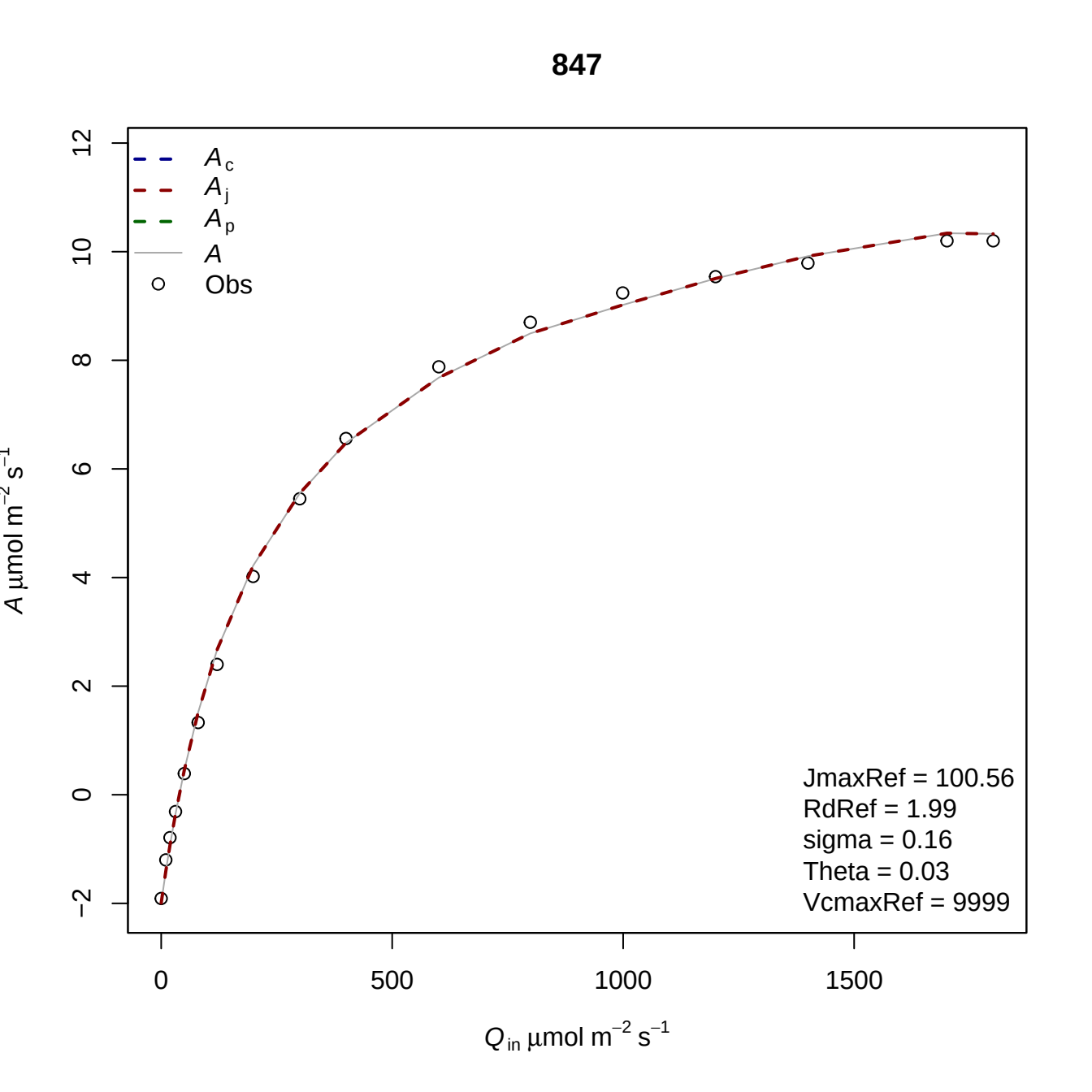

842
A_c
A_j
A_p
A
Obs
JmaxRef = 107.77
RdRef = 1.89
sigma = 0.15
Theta = 0.52
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

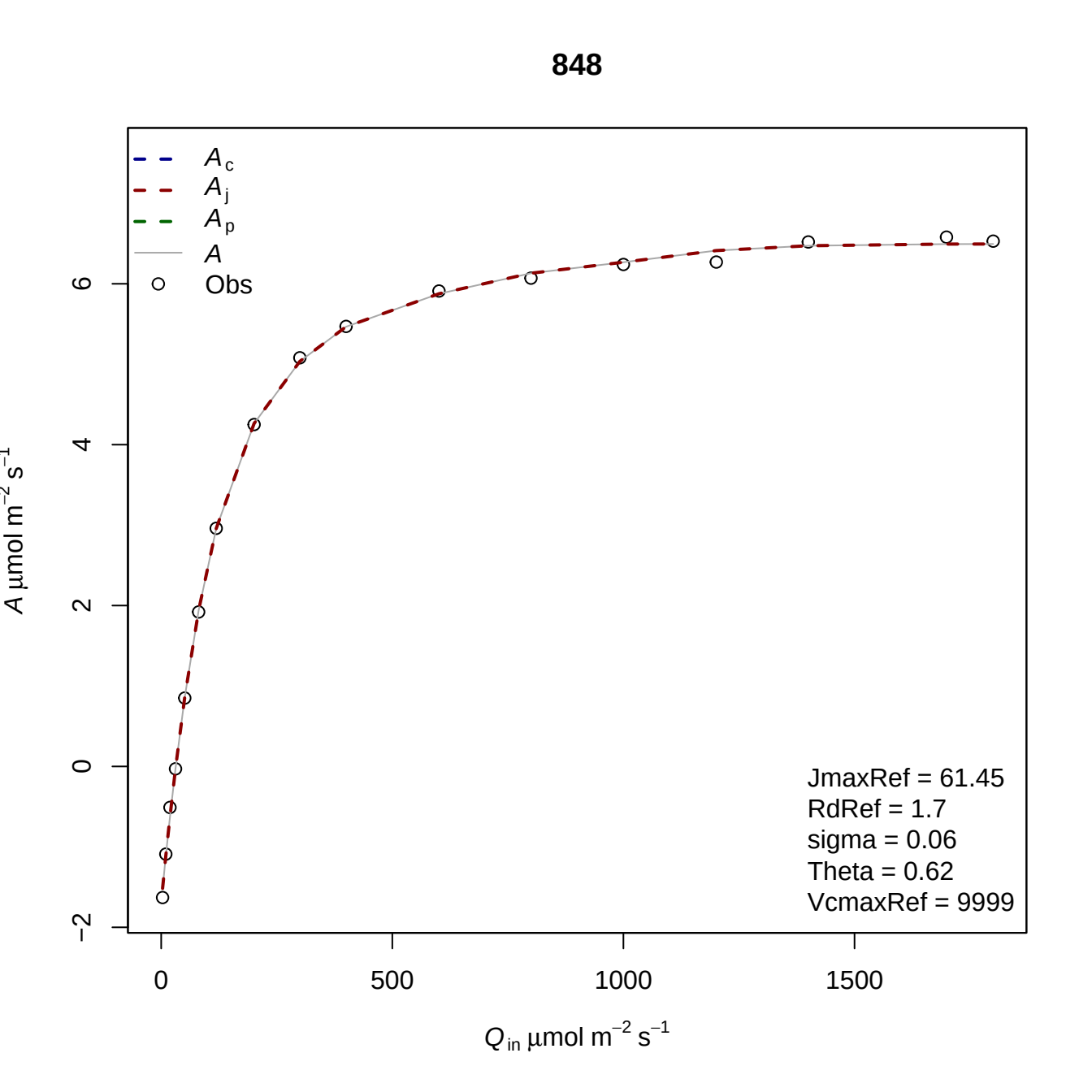

843
A_c
A_j
A_p
A
Obs
JmaxRef = 94.19
RdRef = 2.29
sigma = 0.64
Theta = 0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

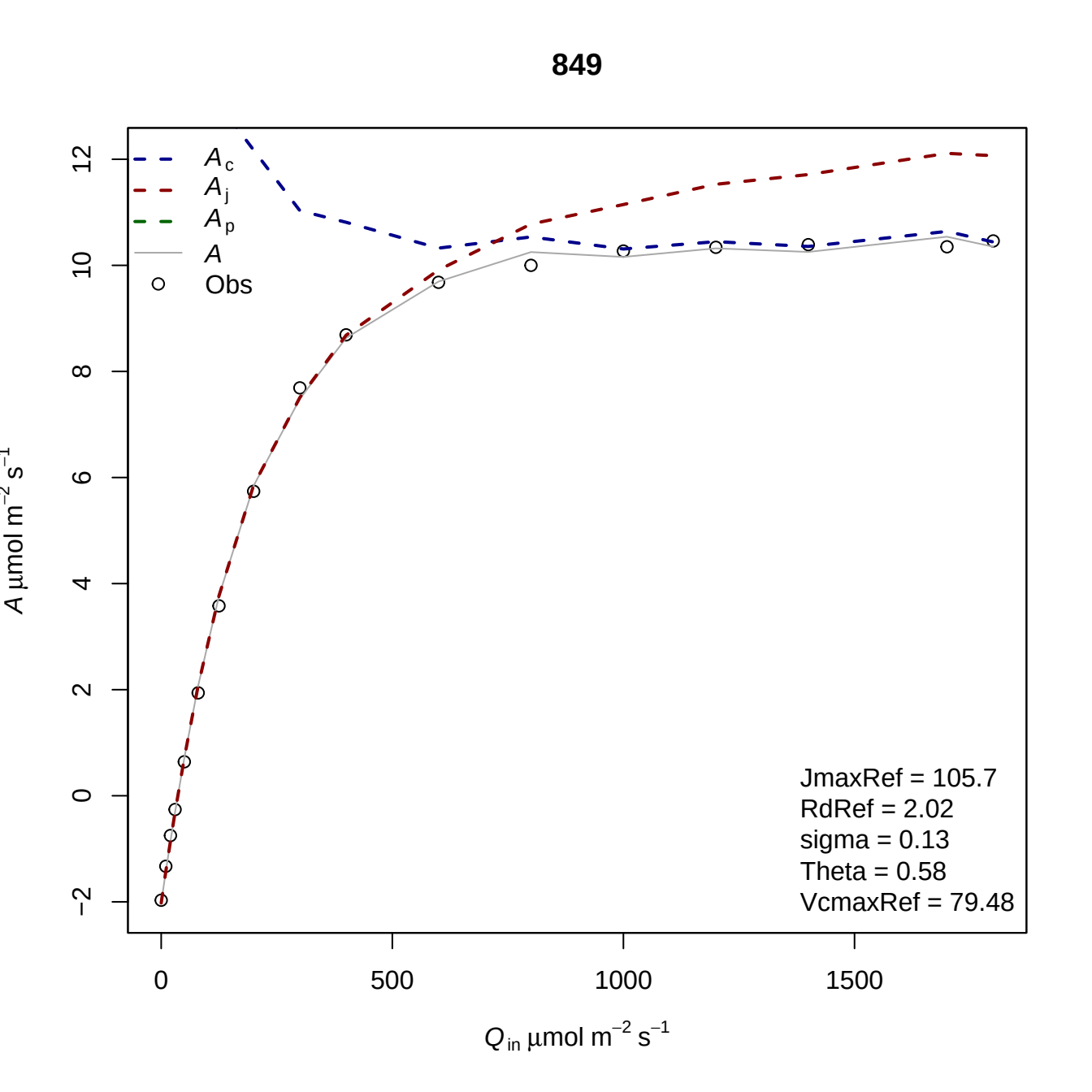

844
A_c
A_j
A_p
A
Obs
JmaxRef = 96.24
RdRef = 3.33
sigma = 0.26
Theta = 0.45
VcmaxRef = 9999
10
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

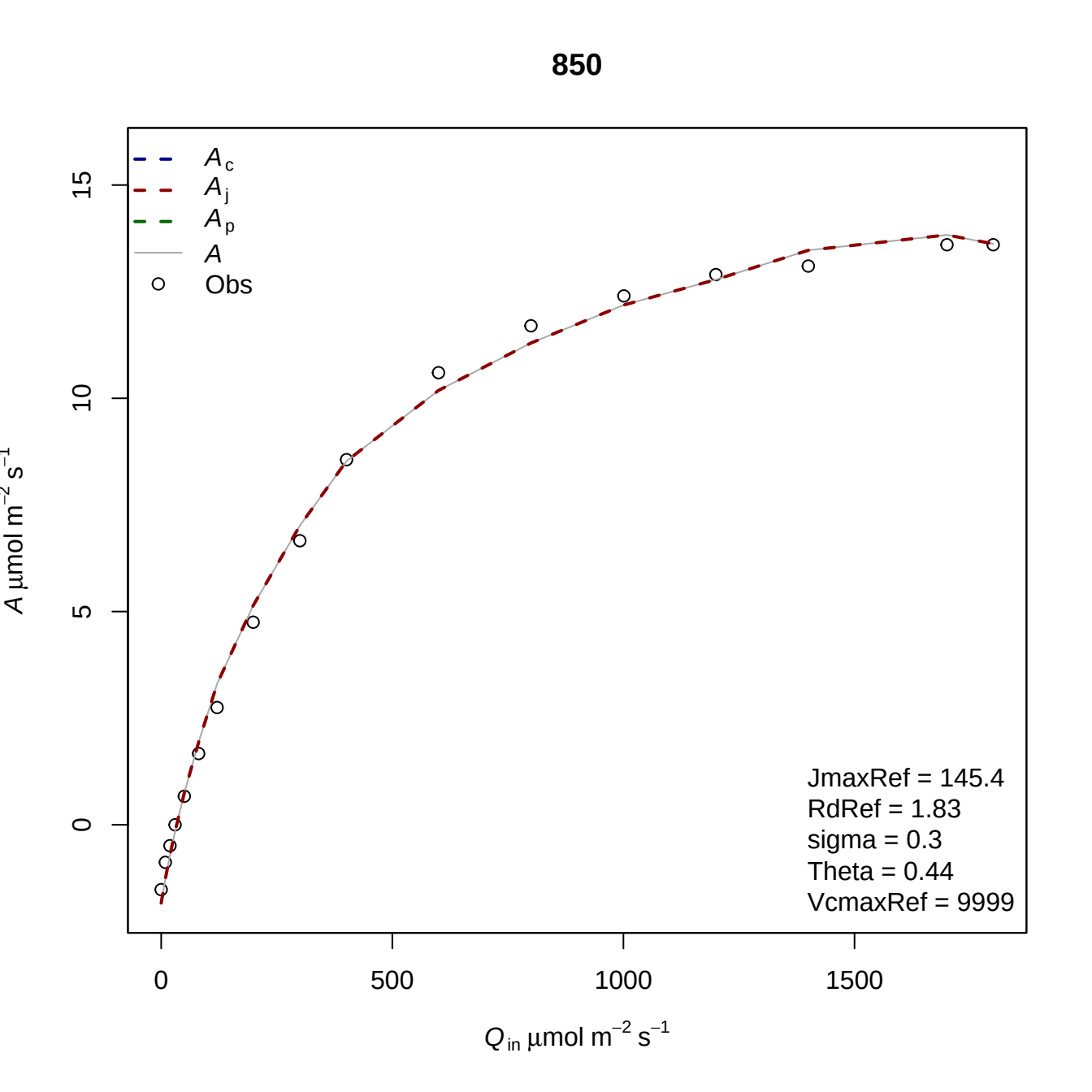

845
A_c
A_j
A_p
A
Obs
JmaxRef = 67.89
RdRef = 1.76
sigma = 0.08
Theta = 0.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

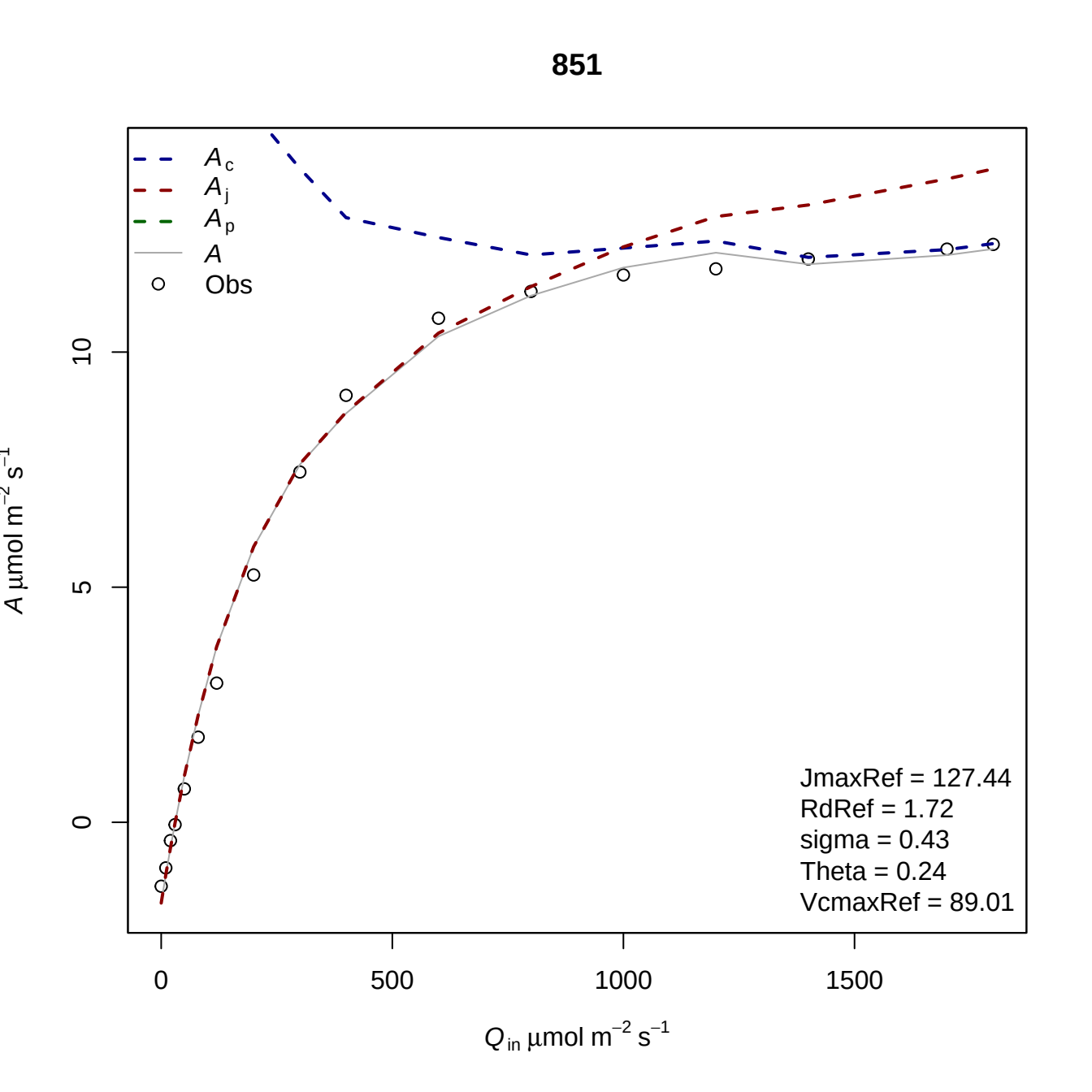

846
A_c
A_j
A_p
A
Obs
JmaxRef = 91.84
RdRef = 1.39
sigma = 0.16
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

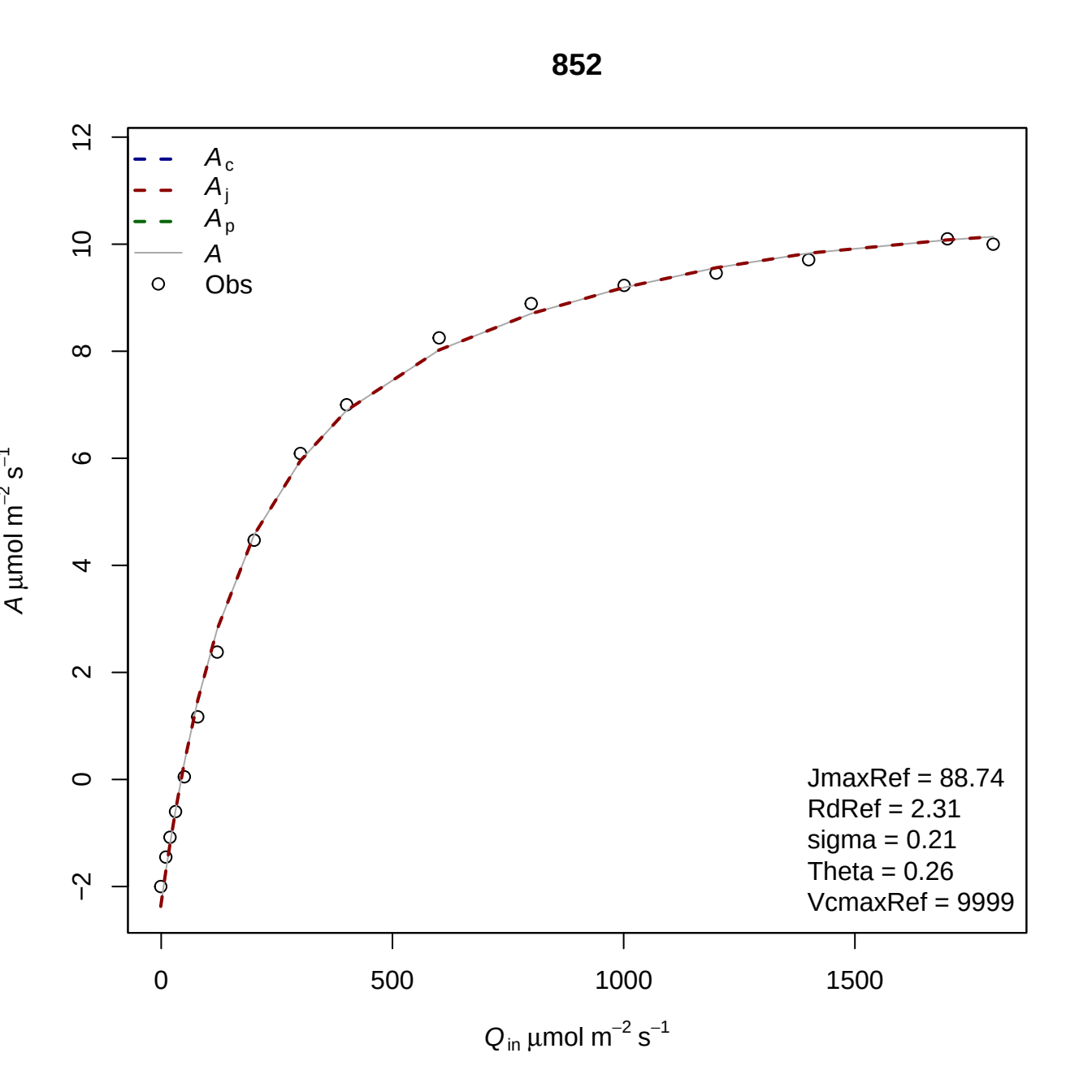

847
A_c
A_j
A_p
A
Obs
JmaxRef = 100.56
RdRef = 1.99
sigma = 0.16
Theta = 0.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

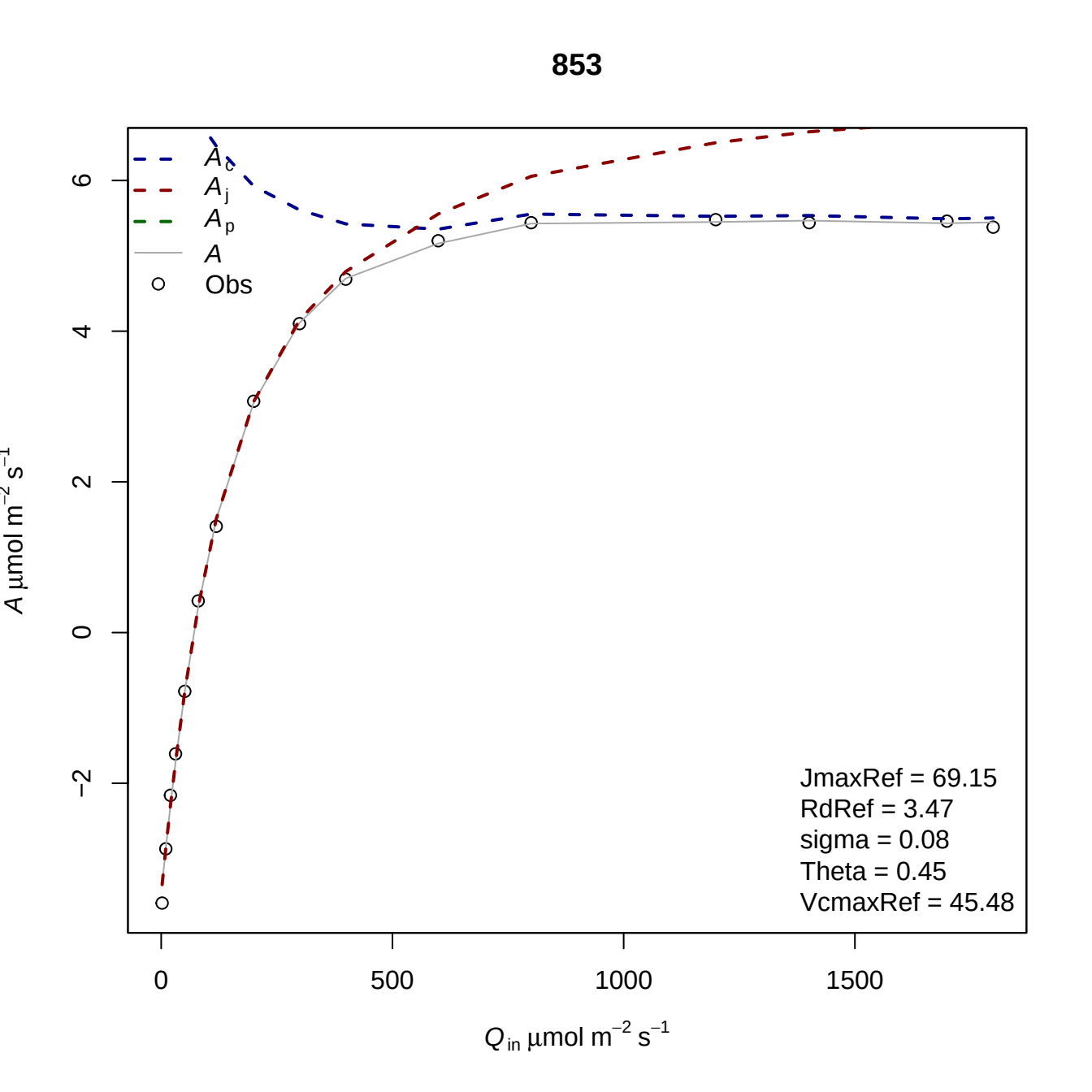

848
A_c
A_j
A_p
A
Obs
JmaxRef = 61.45
RdRef = 1.7
sigma = 0.06
Theta = 0.62
VcmaxRef = 9999
−2
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

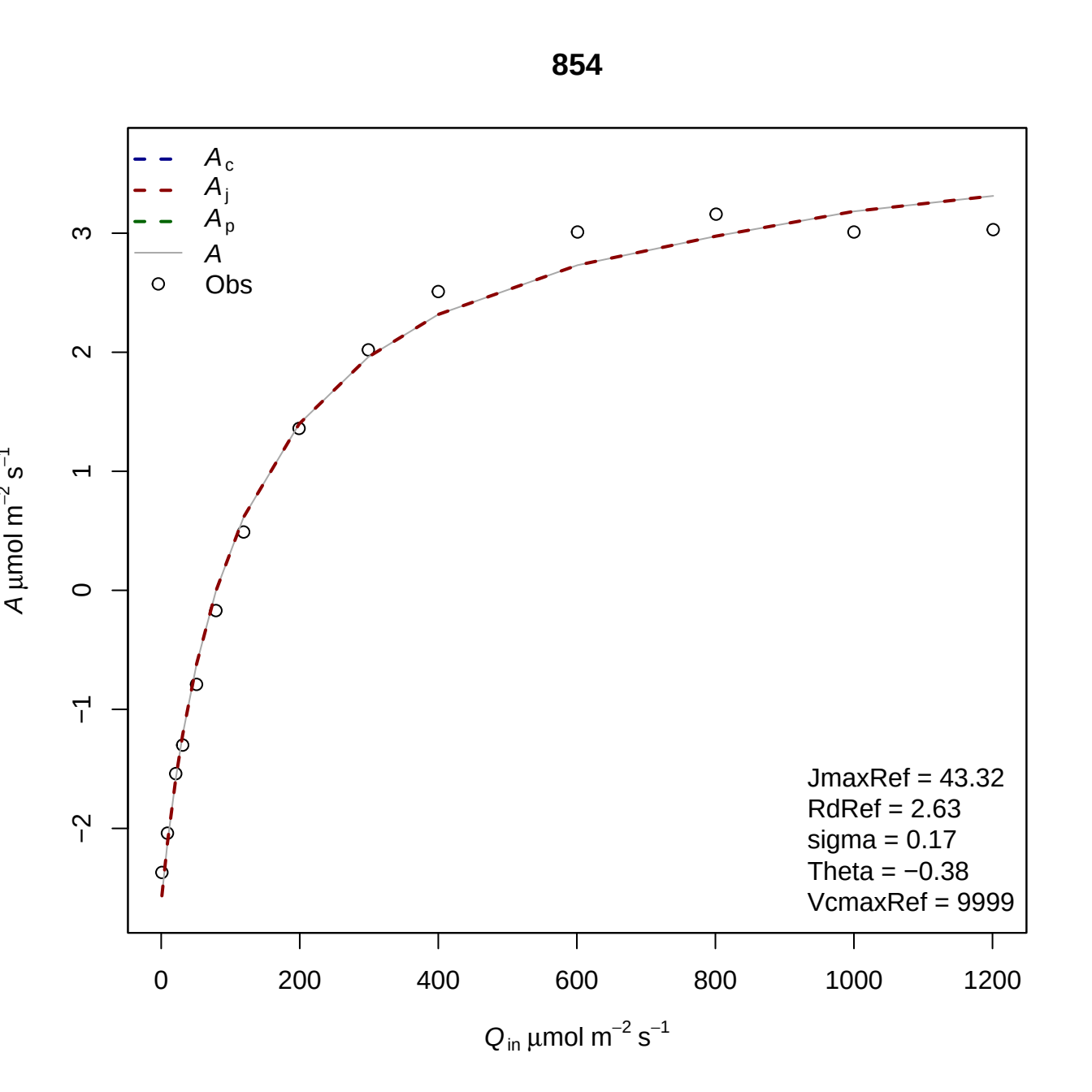

849
A_c
A_j
A_p
A
Obs
JmaxRef = 105.7
RdRef = 2.02
sigma = 0.13
Theta = 0.58
VcmaxRef = 79.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

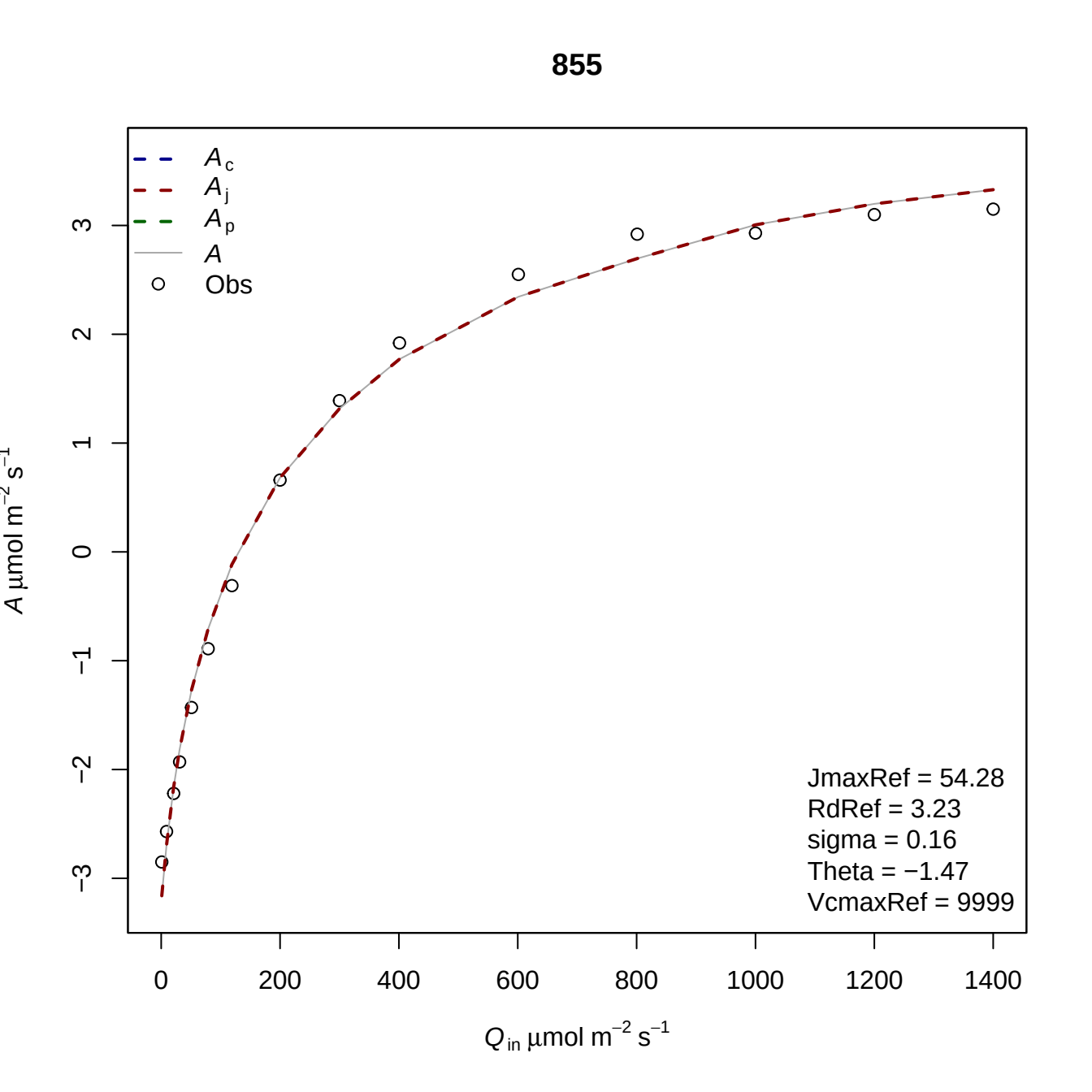

850
A_c
A_j
A_p
A
Obs
JmaxRef = 145.4
RdRef = 1.83
sigma = 0.3
Theta = 0.44
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

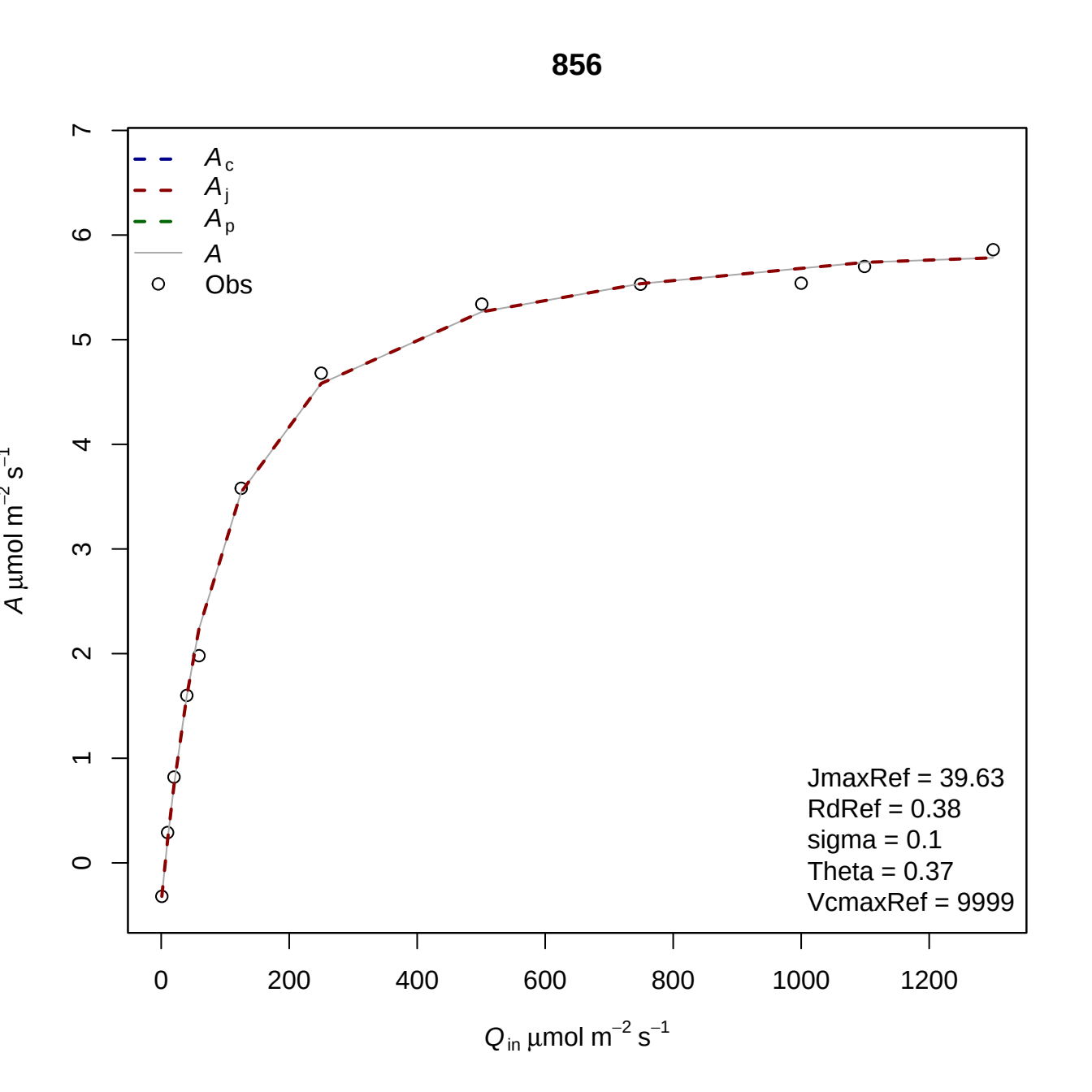

851
A_c
A_j
A_p
A
Obs
JmaxRef = 127.44
RdRef = 1.72
sigma = 0.43
Theta = 0.24
VcmaxRef = 89.01
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

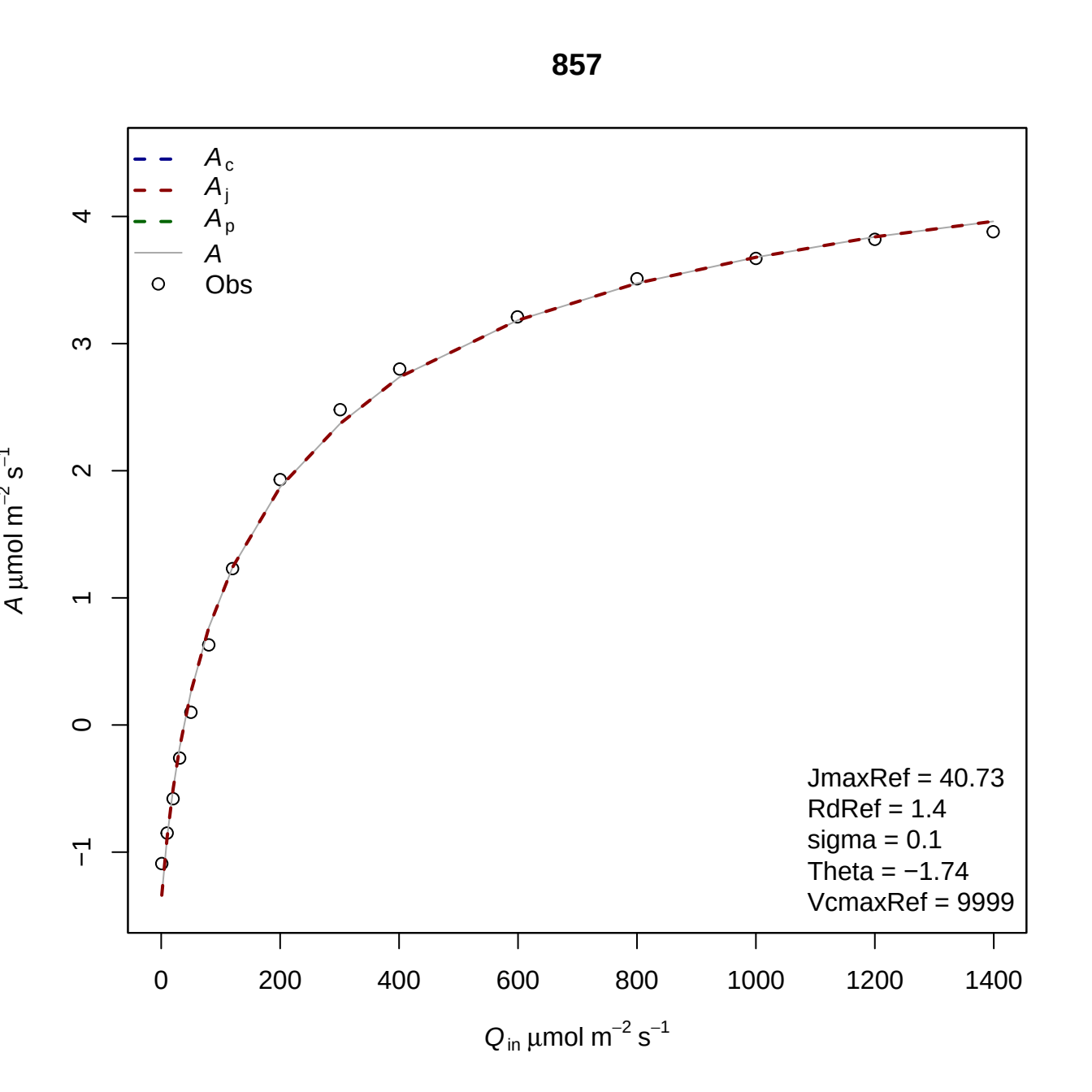

852
A_c
A_j
A_p
A
Obs
JmaxRef = 88.74
RdRef = 2.31
sigma = 0.21
Theta = 0.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

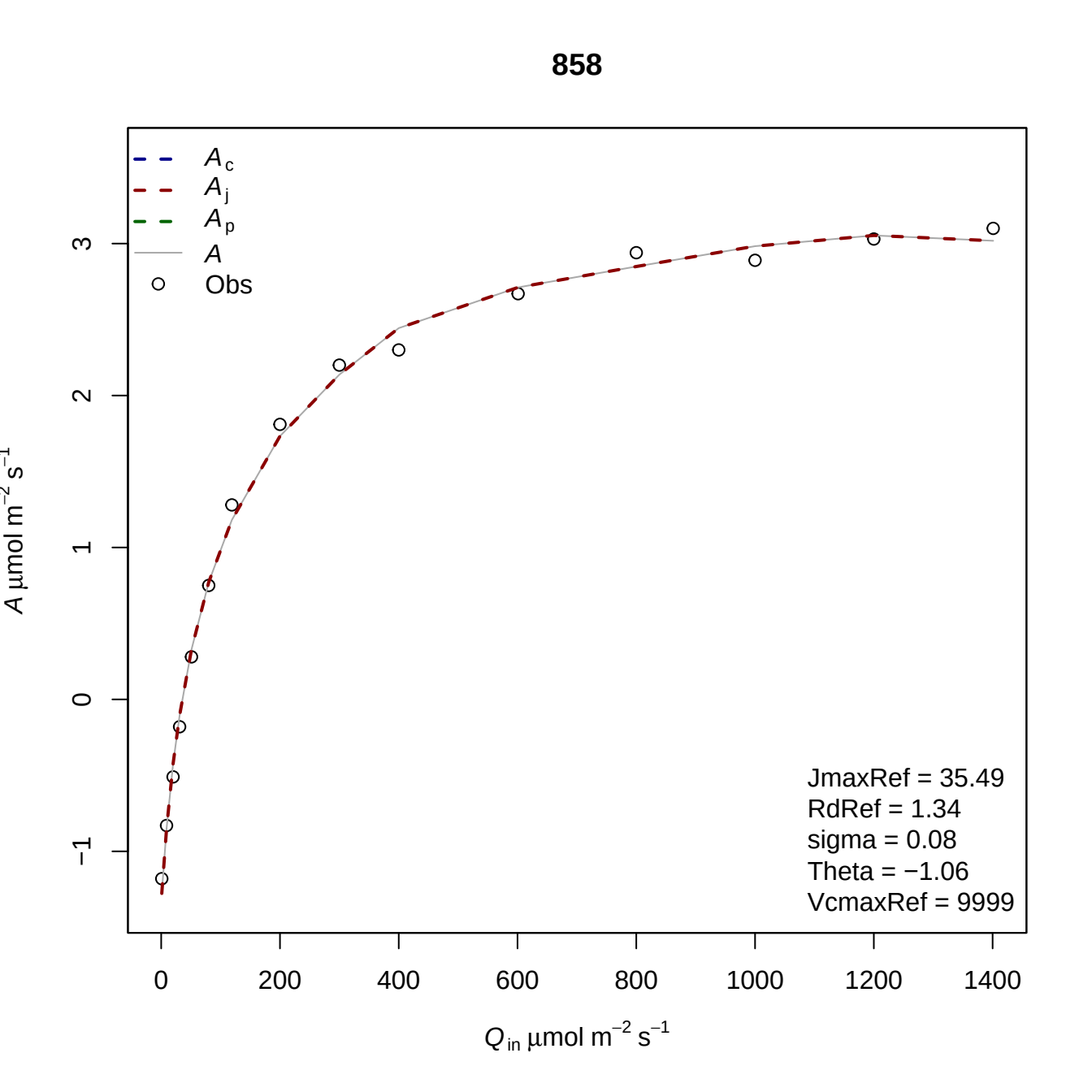

853
A_c
A_j
A_p
A
Obs
JmaxRef = 69.15
RdRef = 3.47
sigma = 0.08
Theta = 0.45
VcmaxRef = 45.48
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

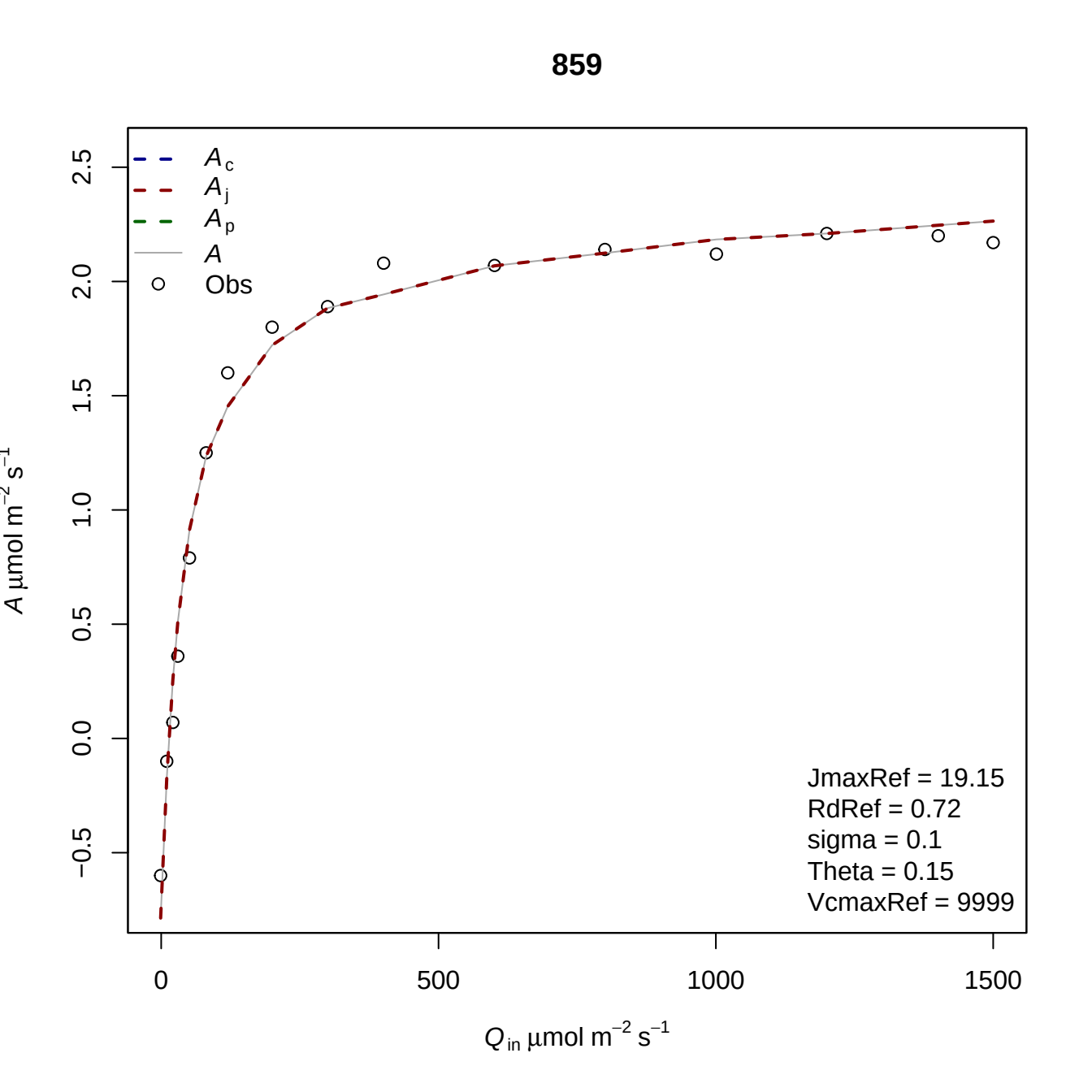

854
A_c
A_j
A_p
A
Obs
JmaxRef = 43.32
RdRef = 2.63
sigma = 0.17
Theta = −0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200

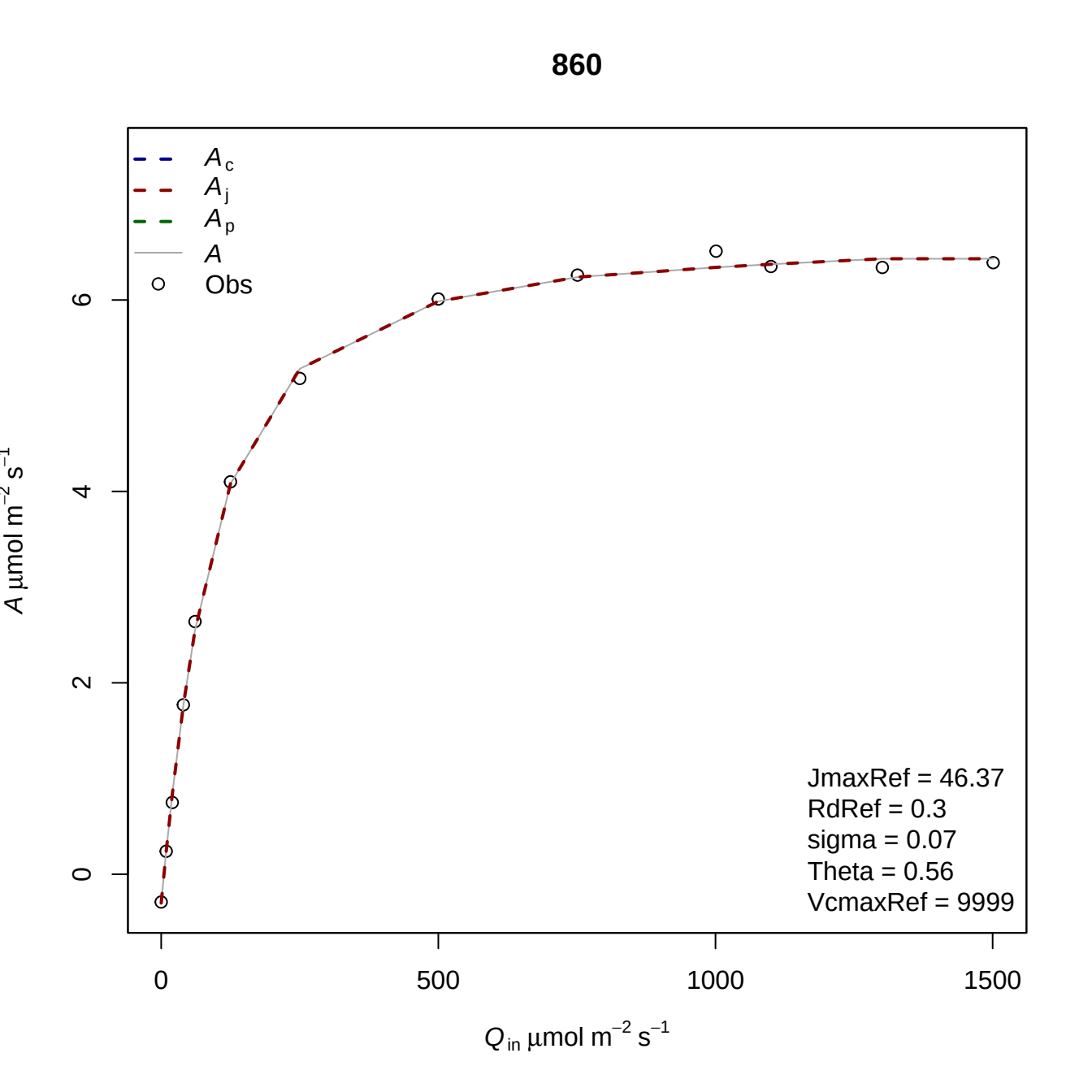

855
A_c
A_j
A_p
A
Obs
JmaxRef = 54.28
RdRef = 3.23
sigma = 0.16
Theta = −1.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

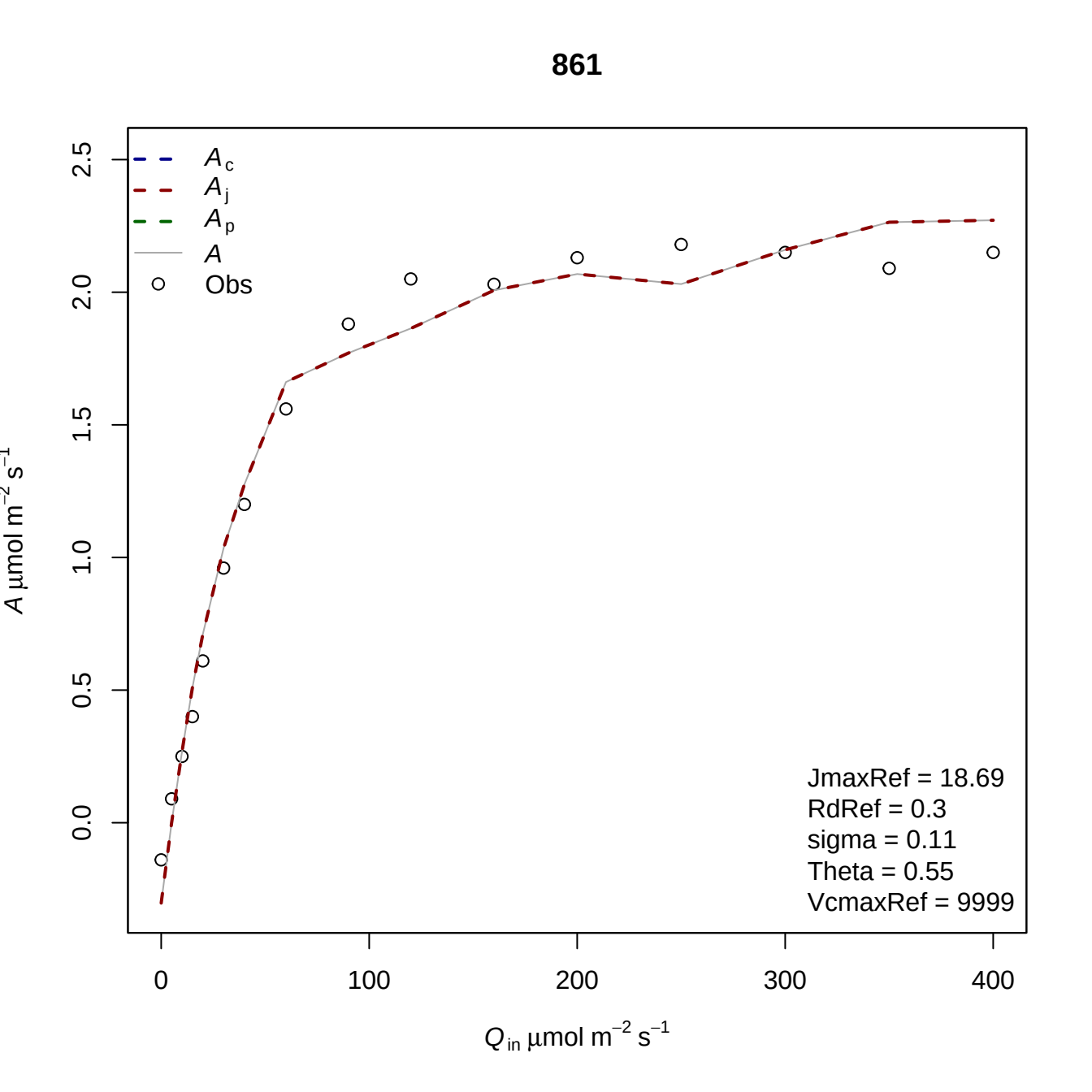

856
A_c
A_j
A_p
A
Obs
JmaxRef = 39.63
RdRef = 0.38
sigma = 0.1
Theta = 0.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200

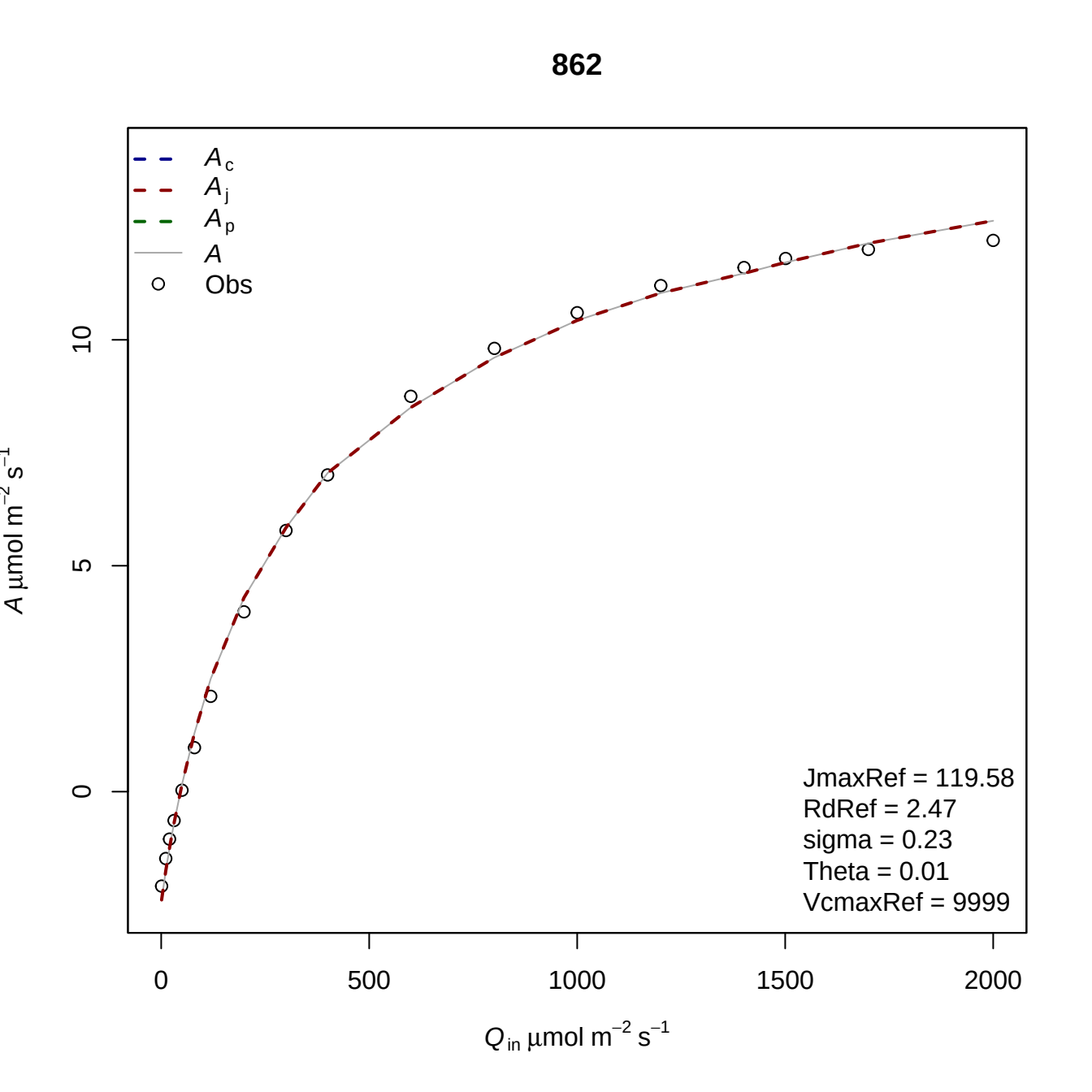

857
A_c
A_j
A_p
A
Obs
JmaxRef = 40.73
RdRef = 1.4
sigma = 0.1
Theta = −1.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}

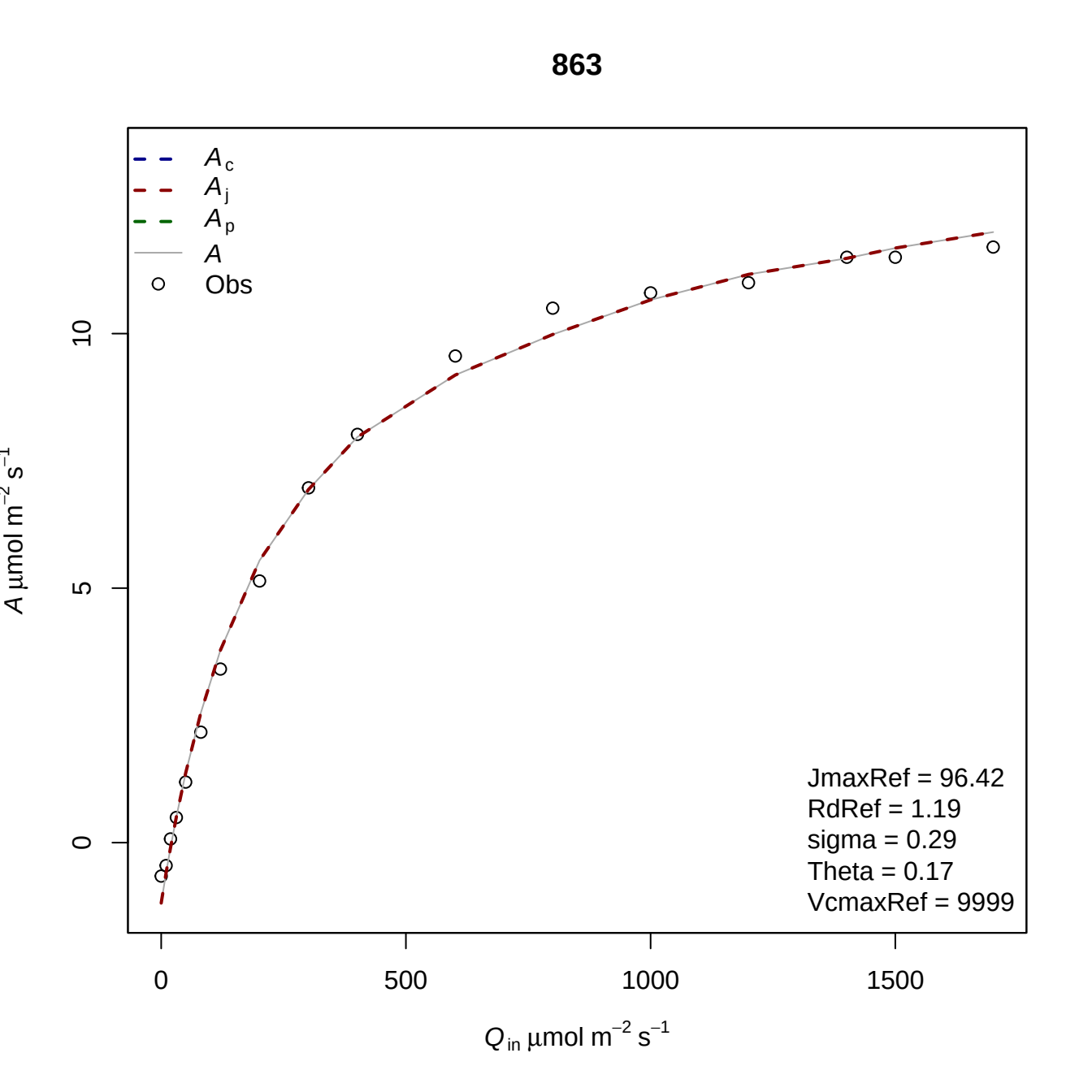

858
A_c
A_j
A_p
A
Obs
JmaxRef = 35.49
RdRef = 1.34
sigma = 0.08
Theta = −1.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

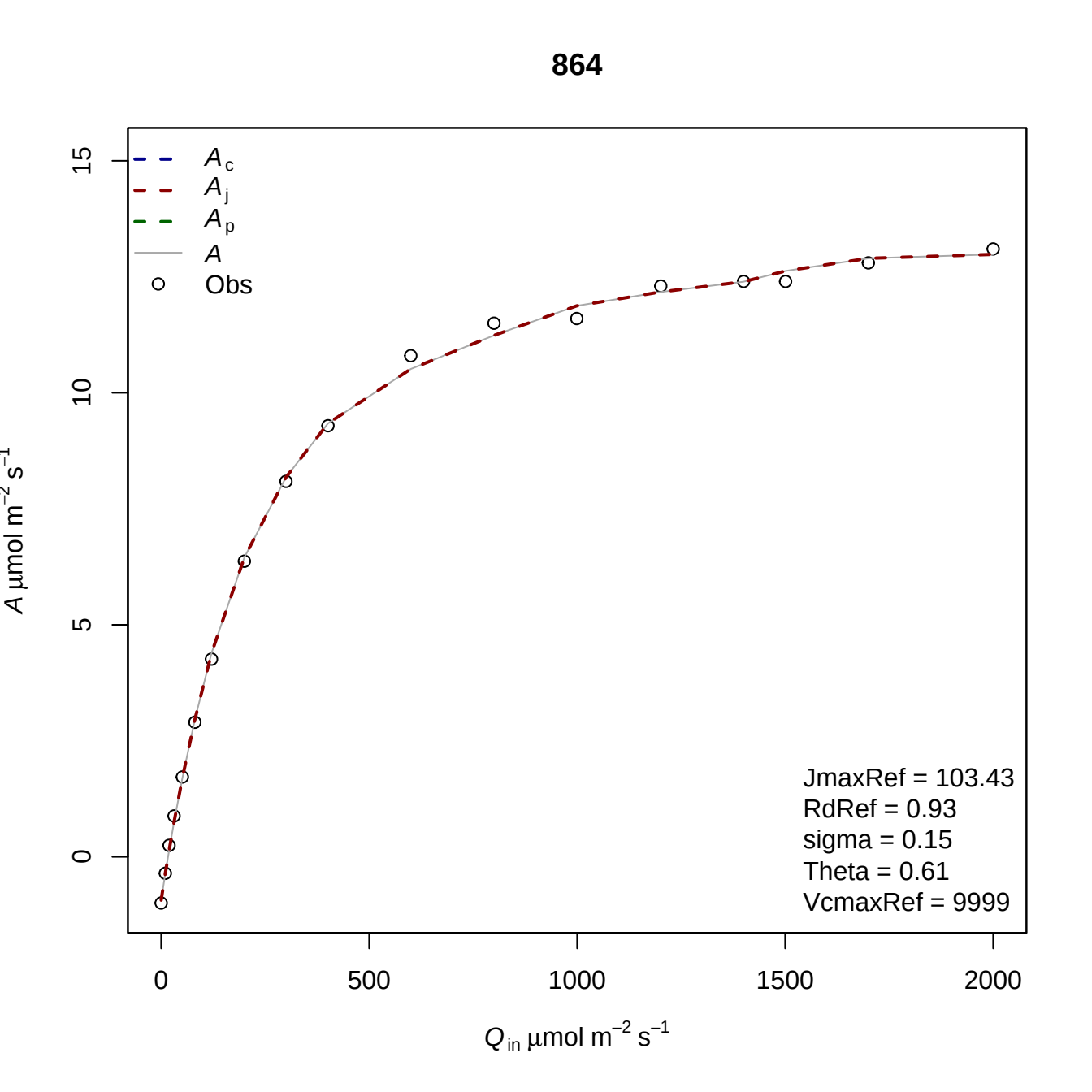

859
A_c
A_j
A_p
A
Obs
JmaxRef = 19.15
RdRef = 0.72
sigma = 0.1
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500

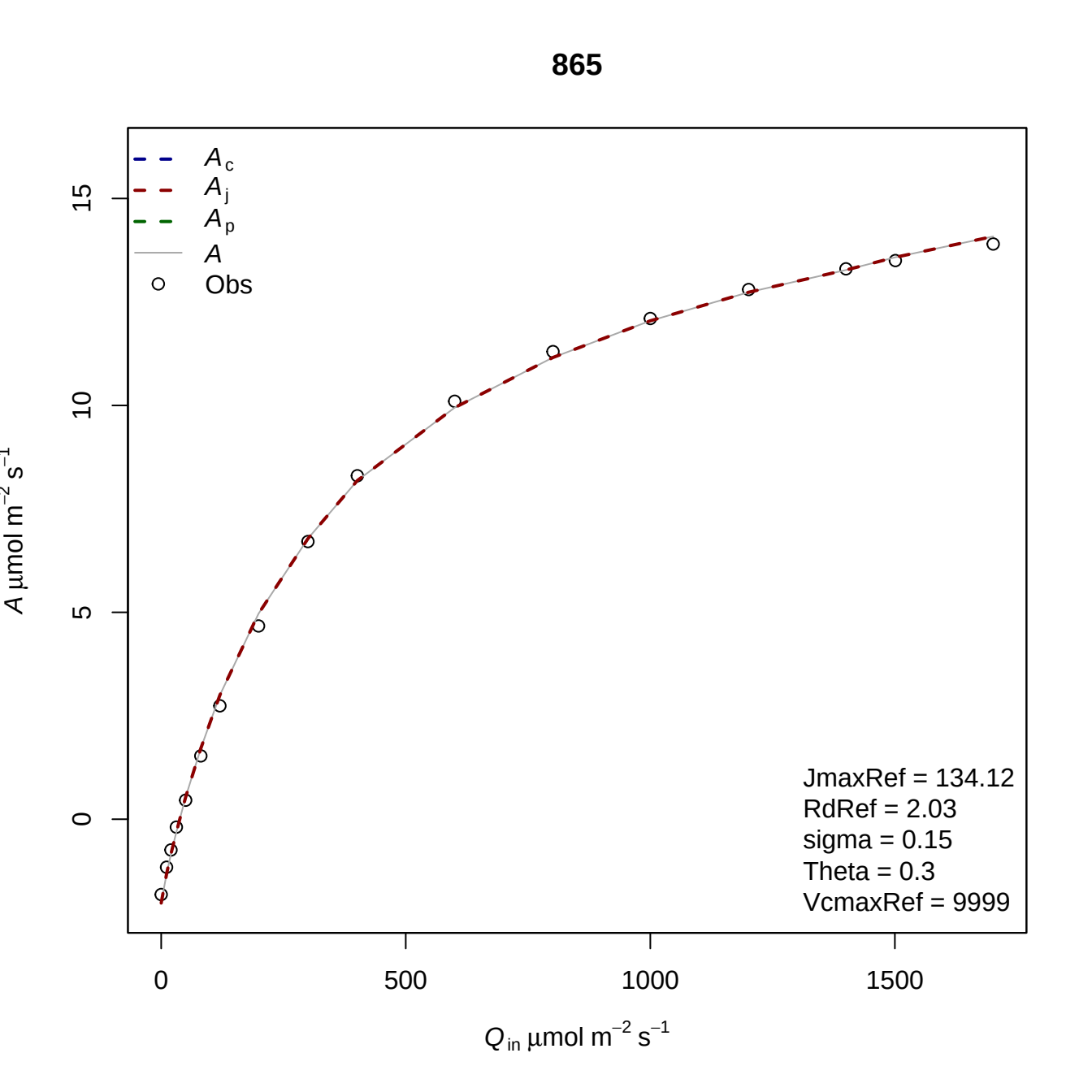

860
A_c
A_j
A_p
A
Obs
JmaxRef = 46.37
RdRef = 0.3
sigma = 0.07
Theta = 0.56
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

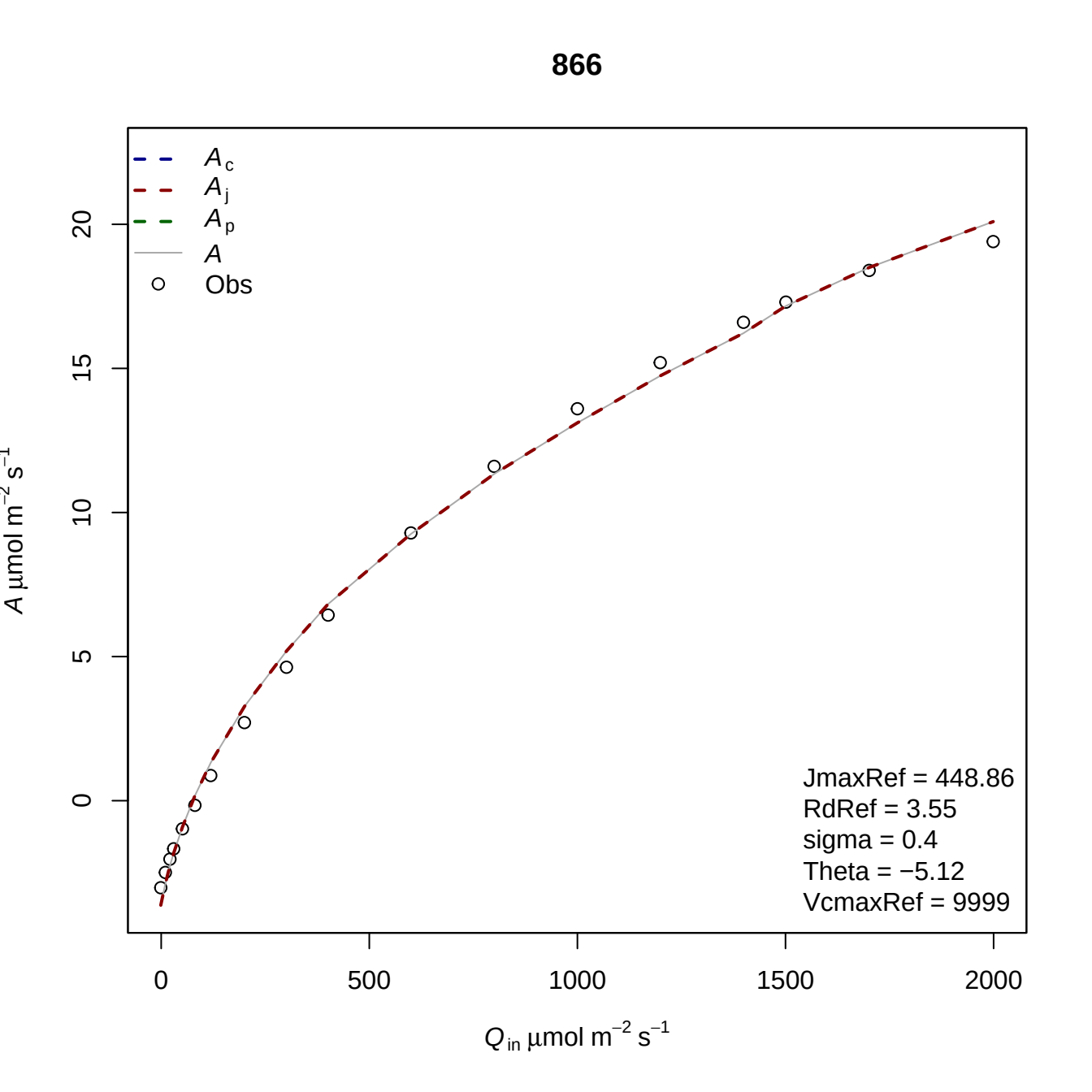

861
A_c
A_j
A_p
A
Obs
JmaxRef = 18.69
RdRef = 0.3
sigma = 0.11
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400

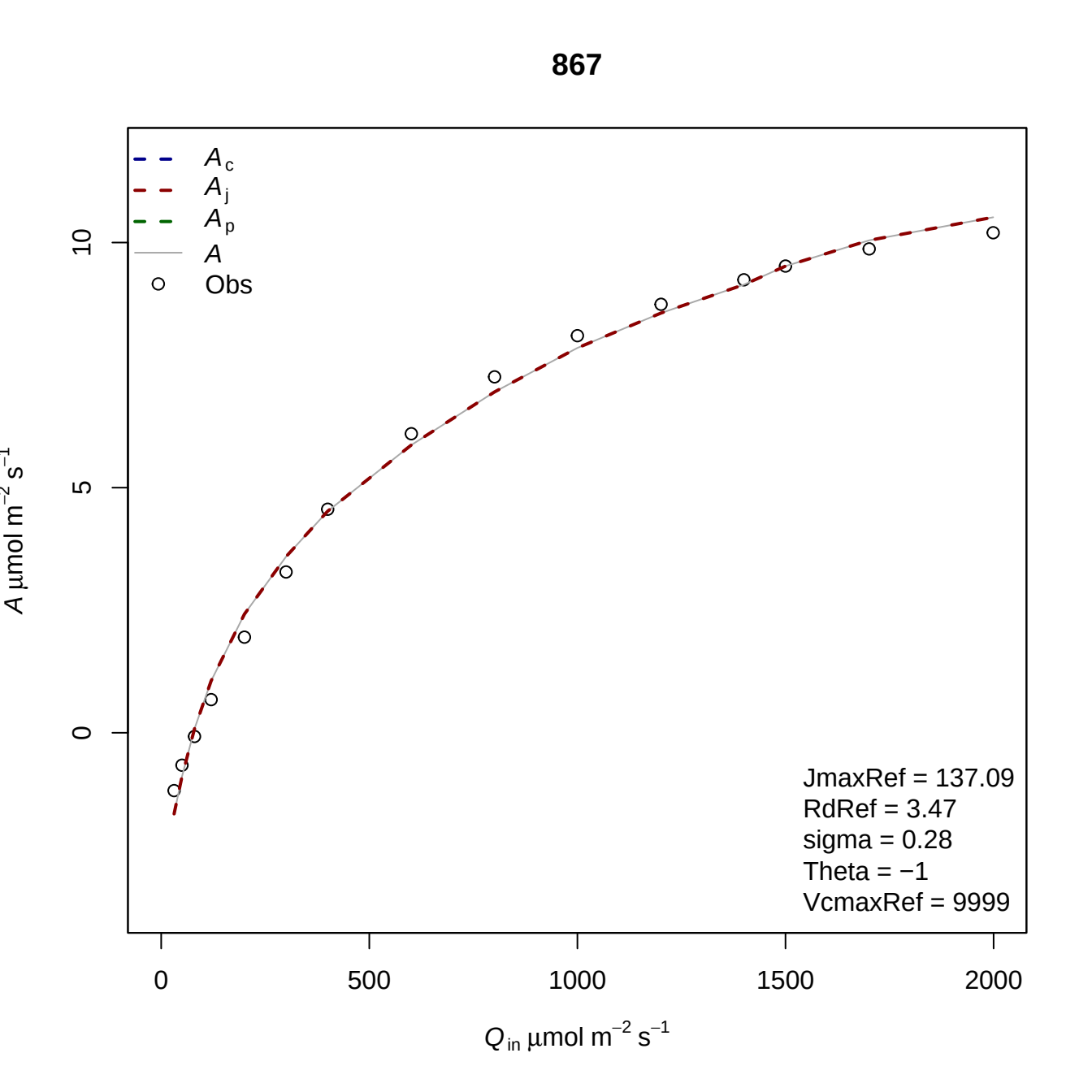

862
A_c
A_j
A_p
A
Obs
JmaxRef = 119.58
RdRef = 2.47
sigma = 0.23
Theta = 0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

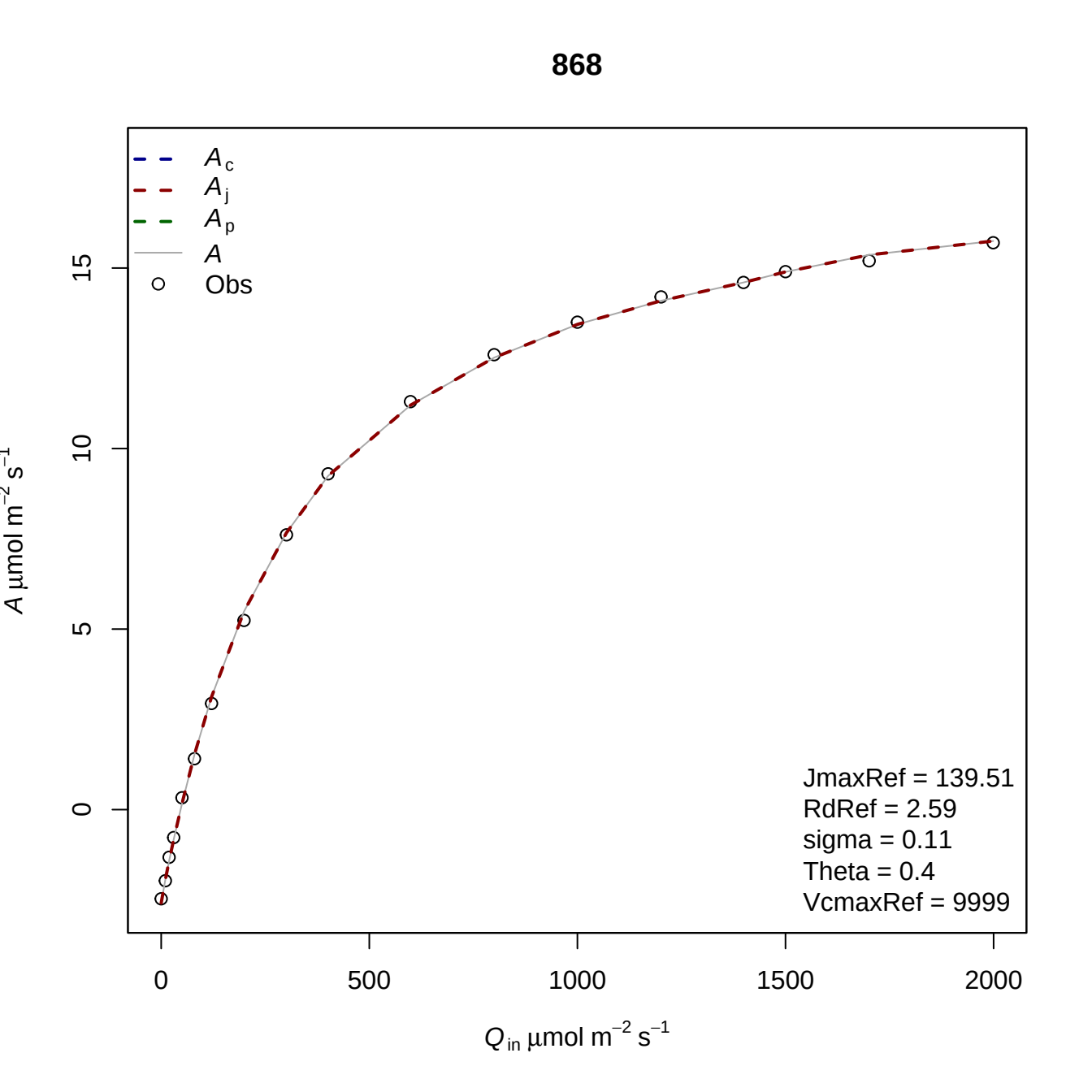

863
A_c
A_j
A_p
A
Obs
JmaxRef = 96.42
RdRef = 1.19
sigma = 0.29
Theta = 0.17
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

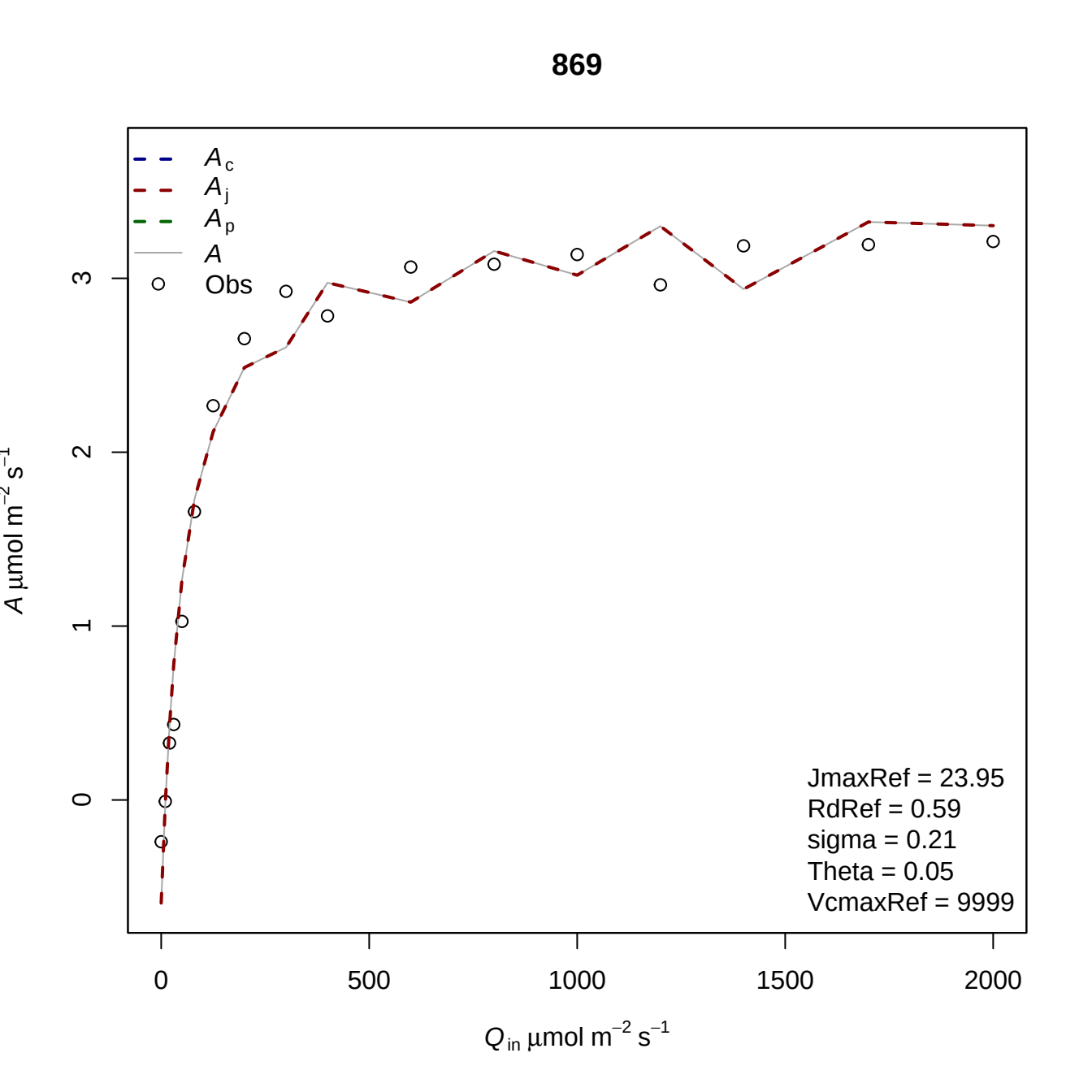

864
A_c
A_j
A_p
A
Obs
JmaxRef = 103.43
RdRef = 0.93
sigma = 0.15
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

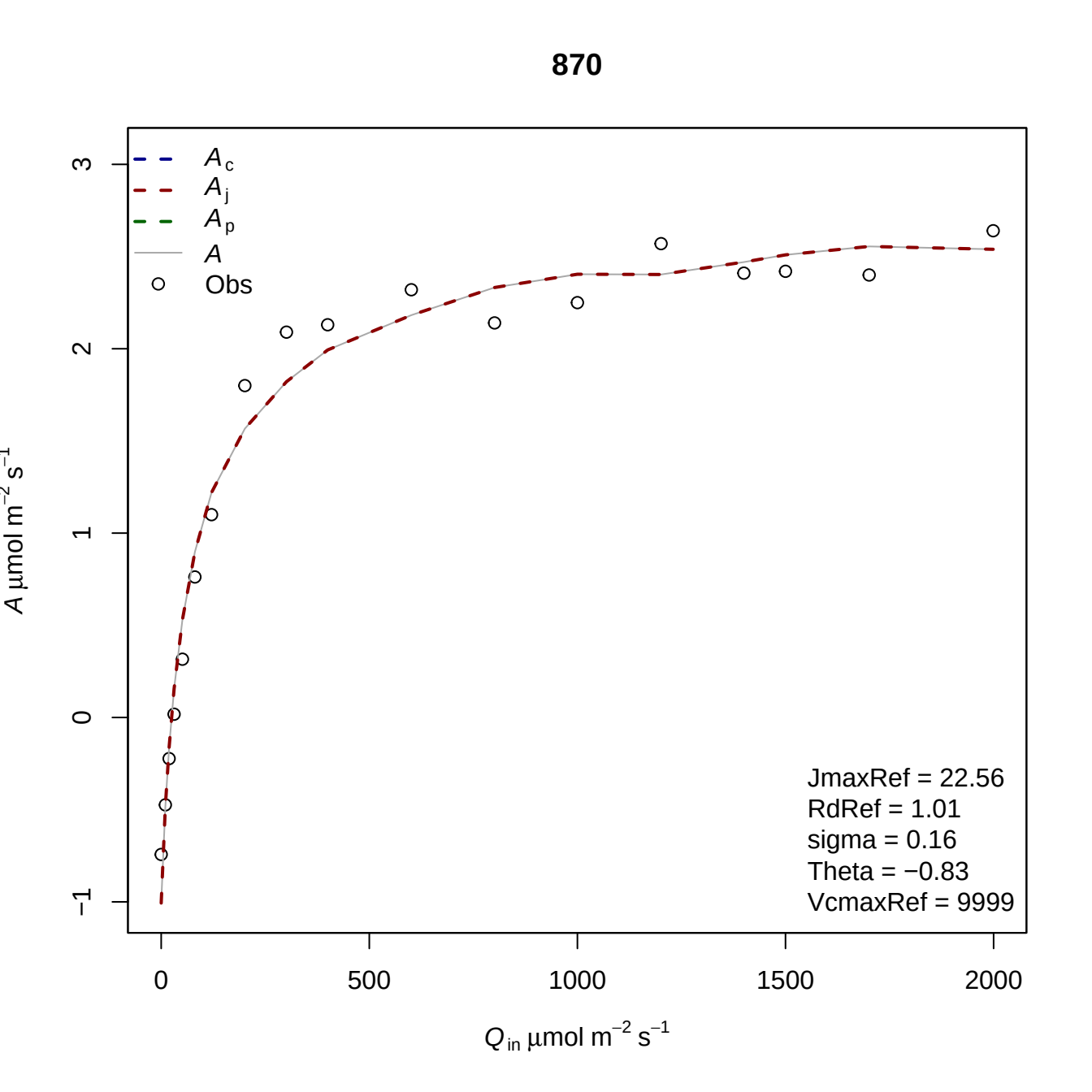

865
A_c
A_j
A_p
A
Obs
JmaxRef = 134.12
RdRef = 2.03
sigma = 0.15
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

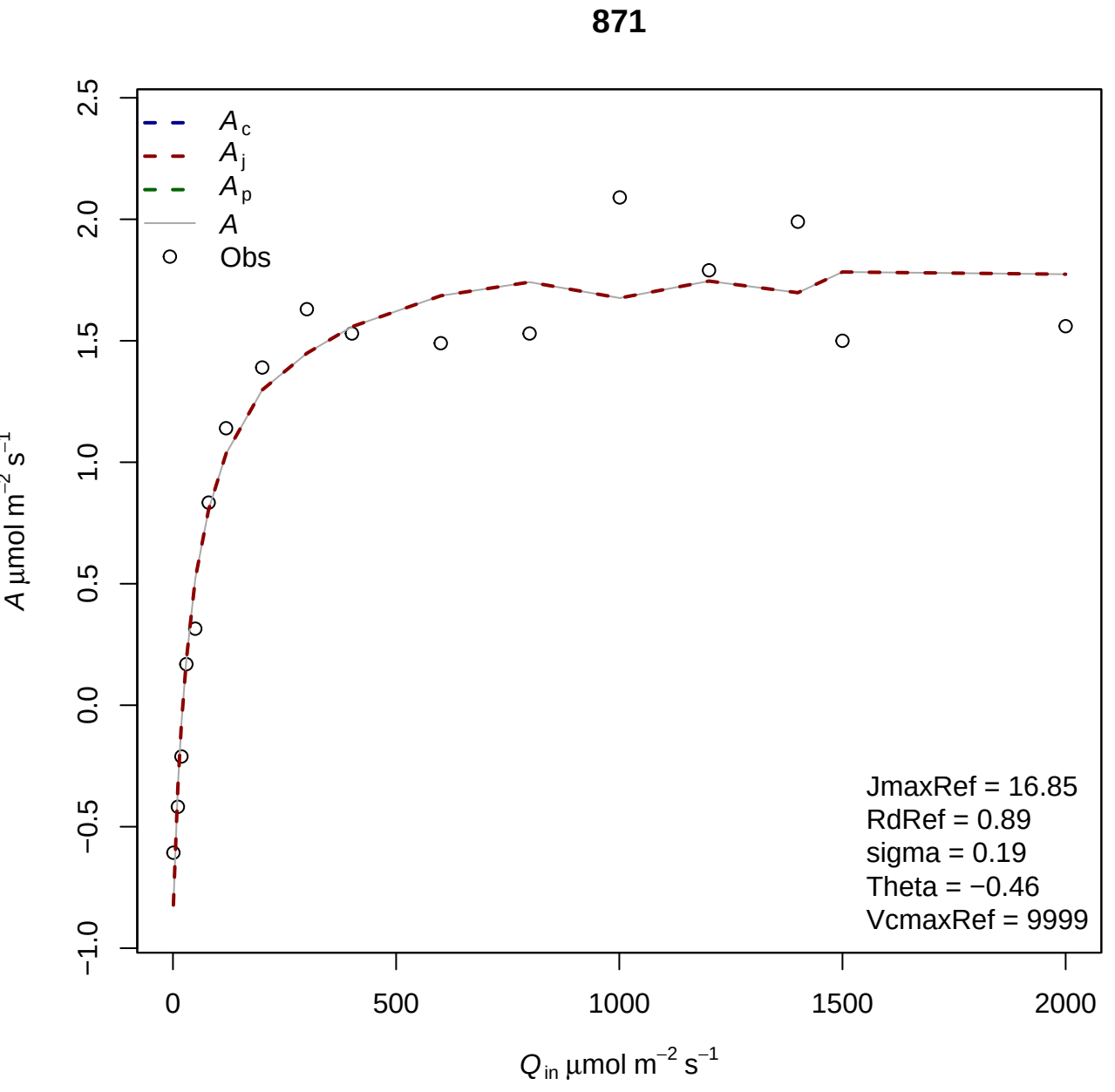

866
A_c
A_j
A_p
A
Obs
JmaxRef = 448.86
RdRef = 3.55
sigma = 0.4
Theta = −5.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

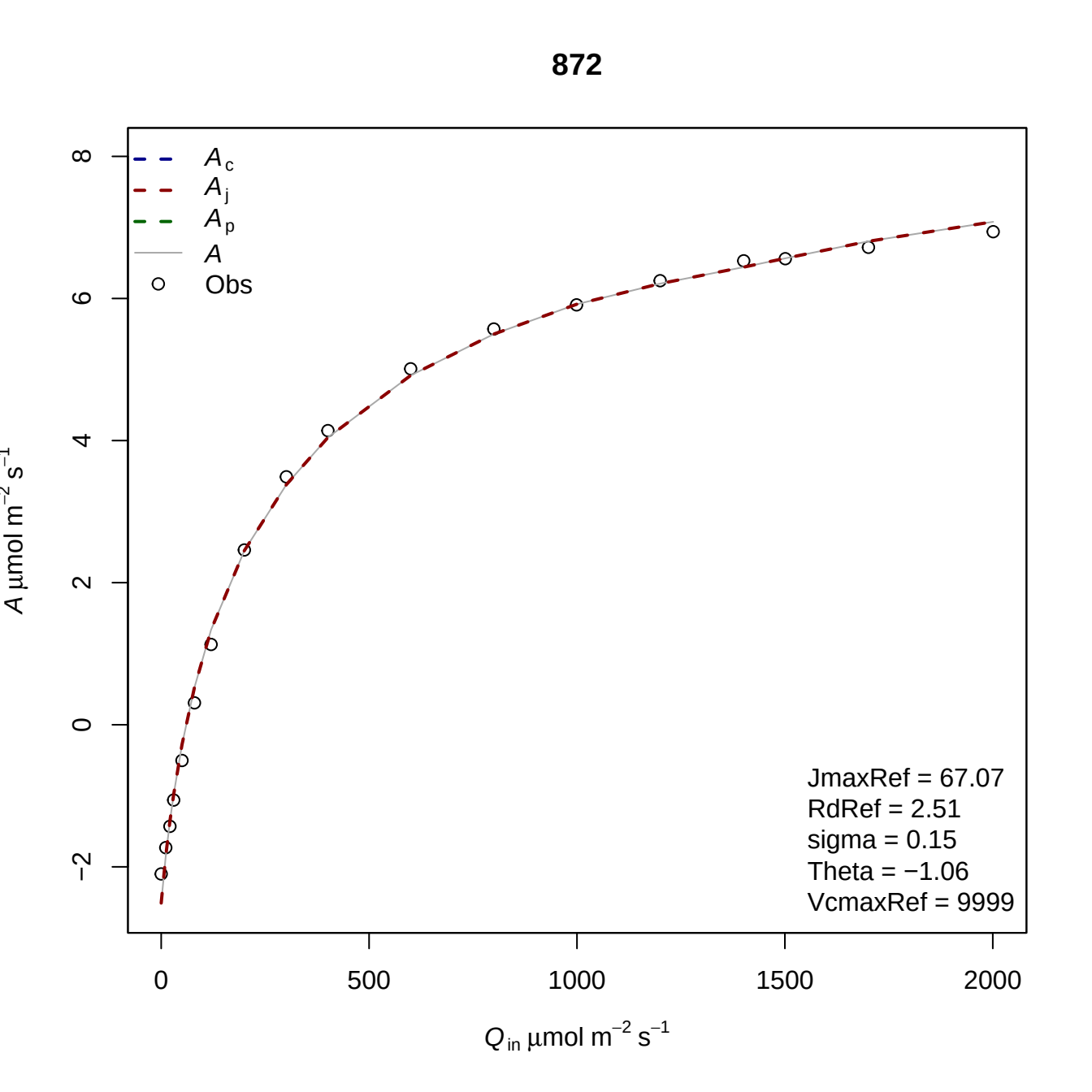

867
A_c
A_j
A_p
A
Obs
JmaxRef = 137.09
RdRef = 3.47
sigma = 0.28
Theta = −1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

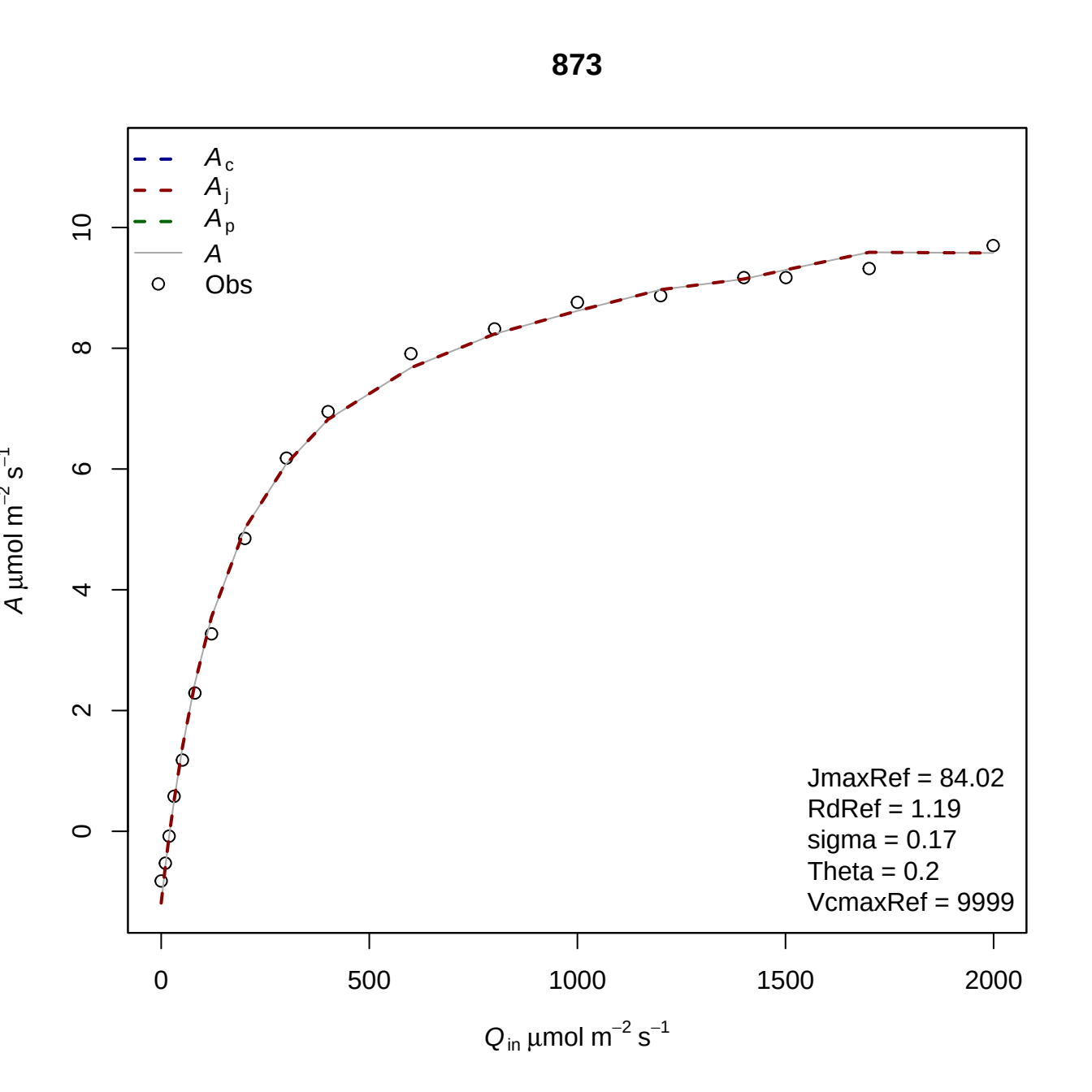

868
A_c
A_j
A_p
A
Obs
JmaxRef = 139.51
RdRef = 2.59
sigma = 0.11
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

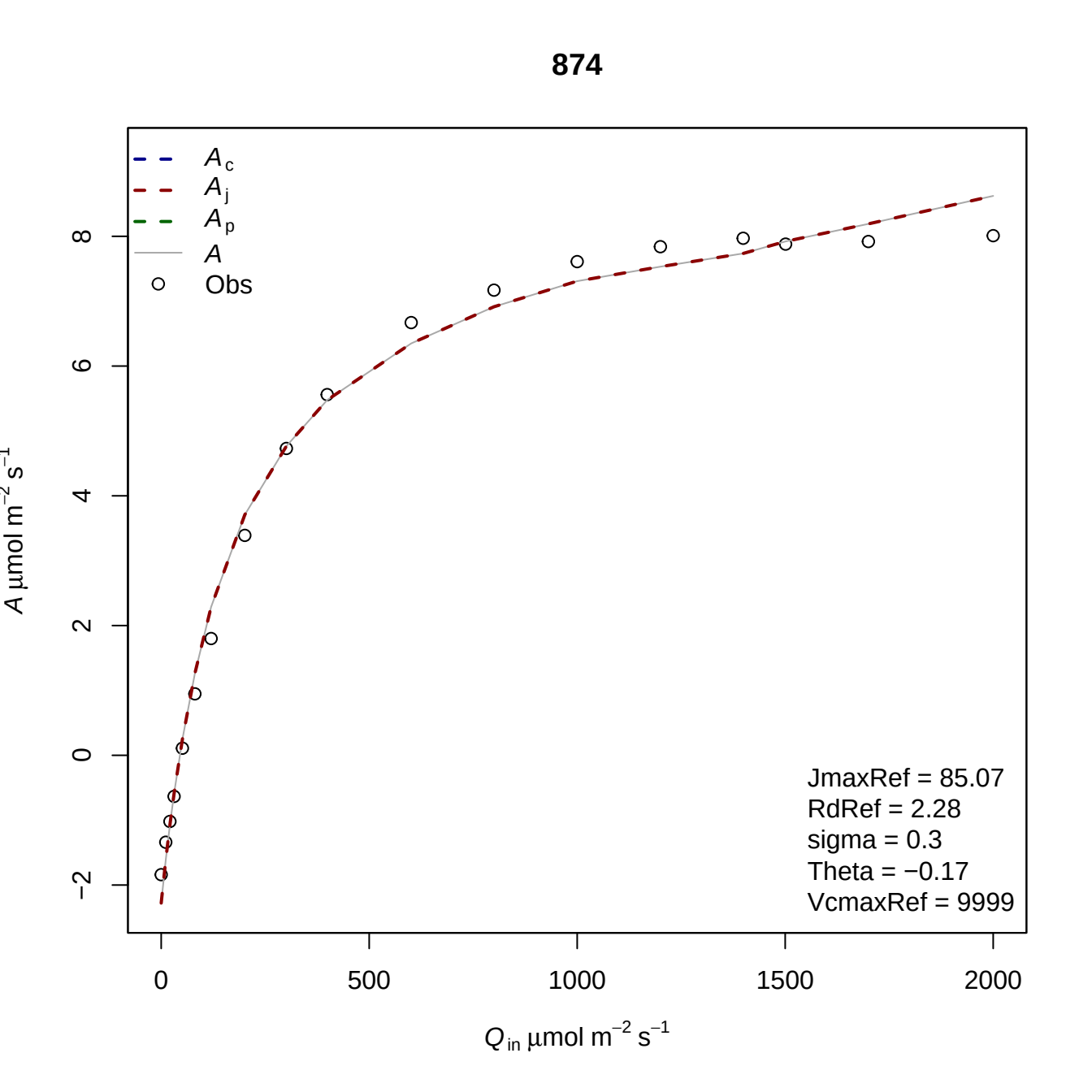

869
A_c
A_j
A_p
A
Obs
JmaxRef = 23.95
RdRef = 0.59
sigma = 0.21
Theta = 0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

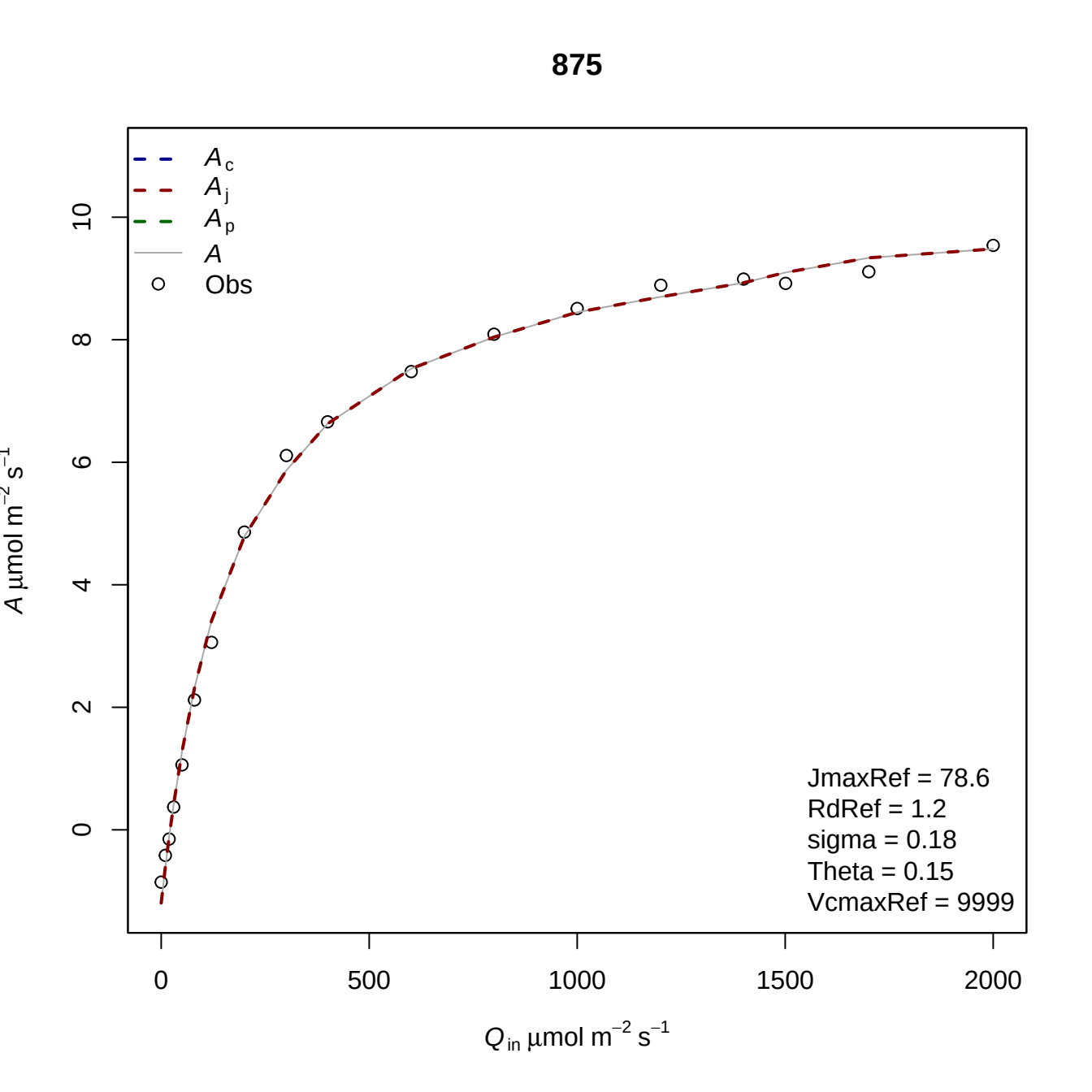

870
A_c
A_j
A_p
A
Obs
JmaxRef = 22.56
RdRef = 1.01
sigma = 0.16
Theta = −0.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

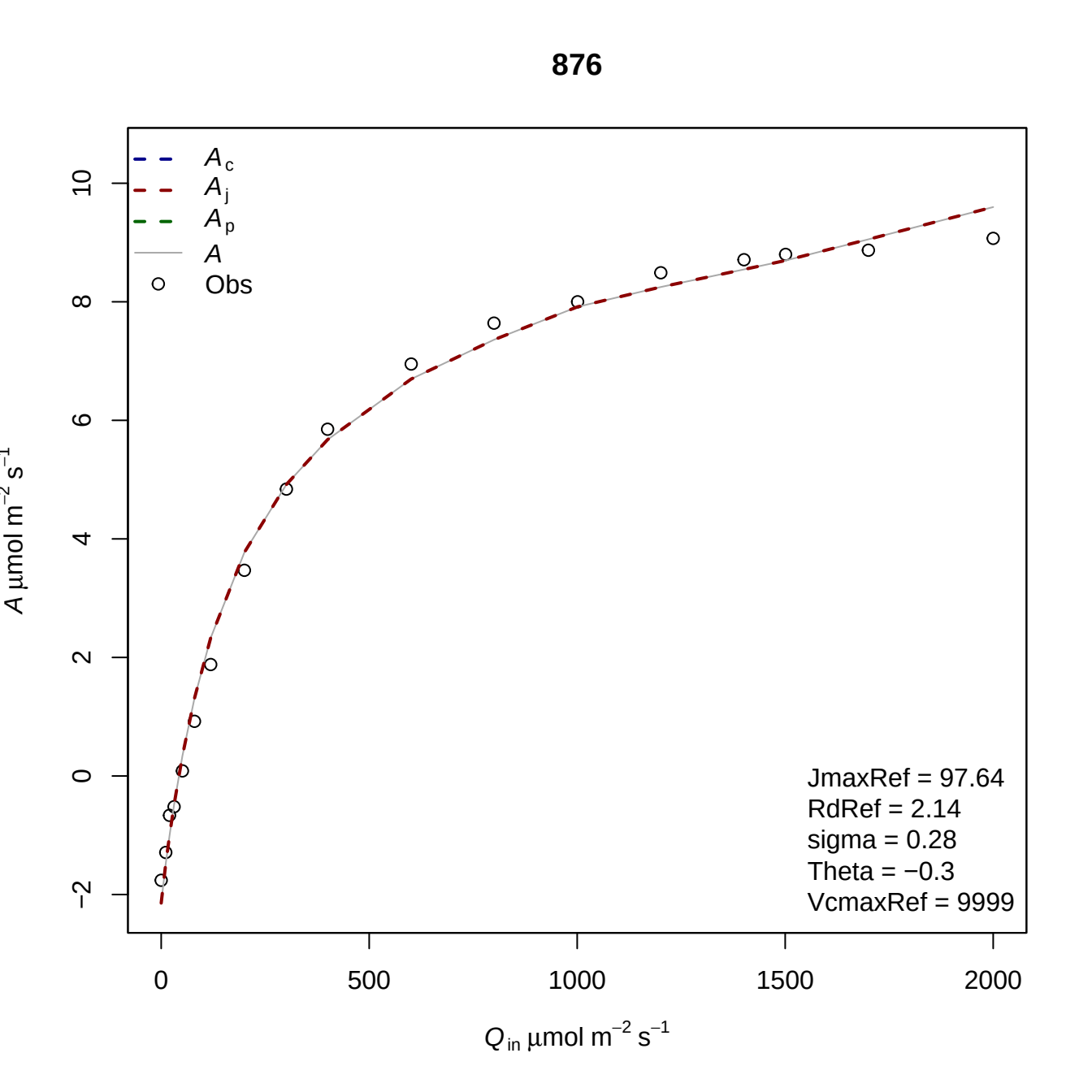

871
A_c
A_j
A_p
A
Obs
JmaxRef = 16.85
RdRef = 0.89
sigma = 0.19
Theta = −0.46
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

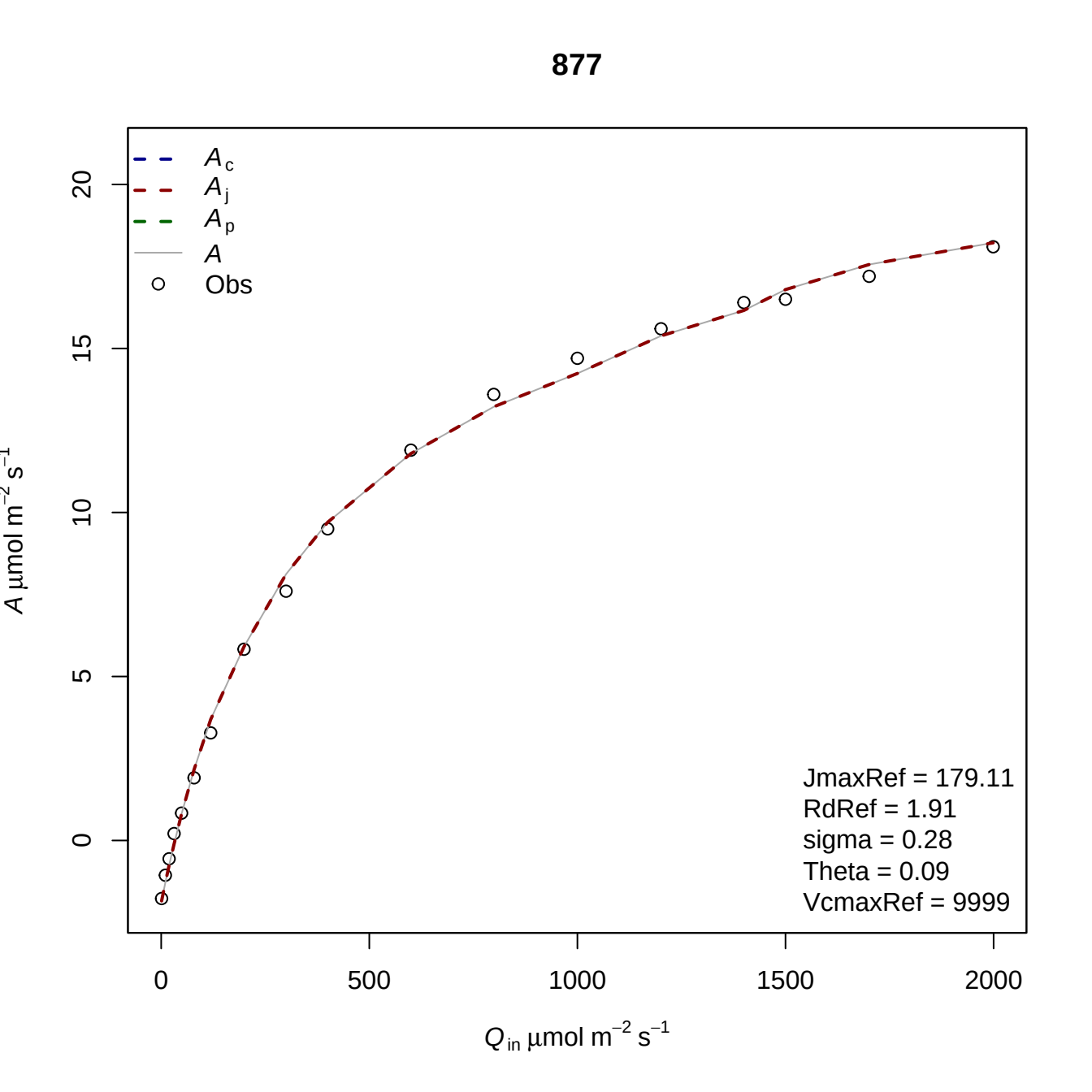

872
A_c
A_j
A_p
A
Obs
JmaxRef = 67.07
RdRef = 2.51
sigma = 0.15
Theta = −1.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

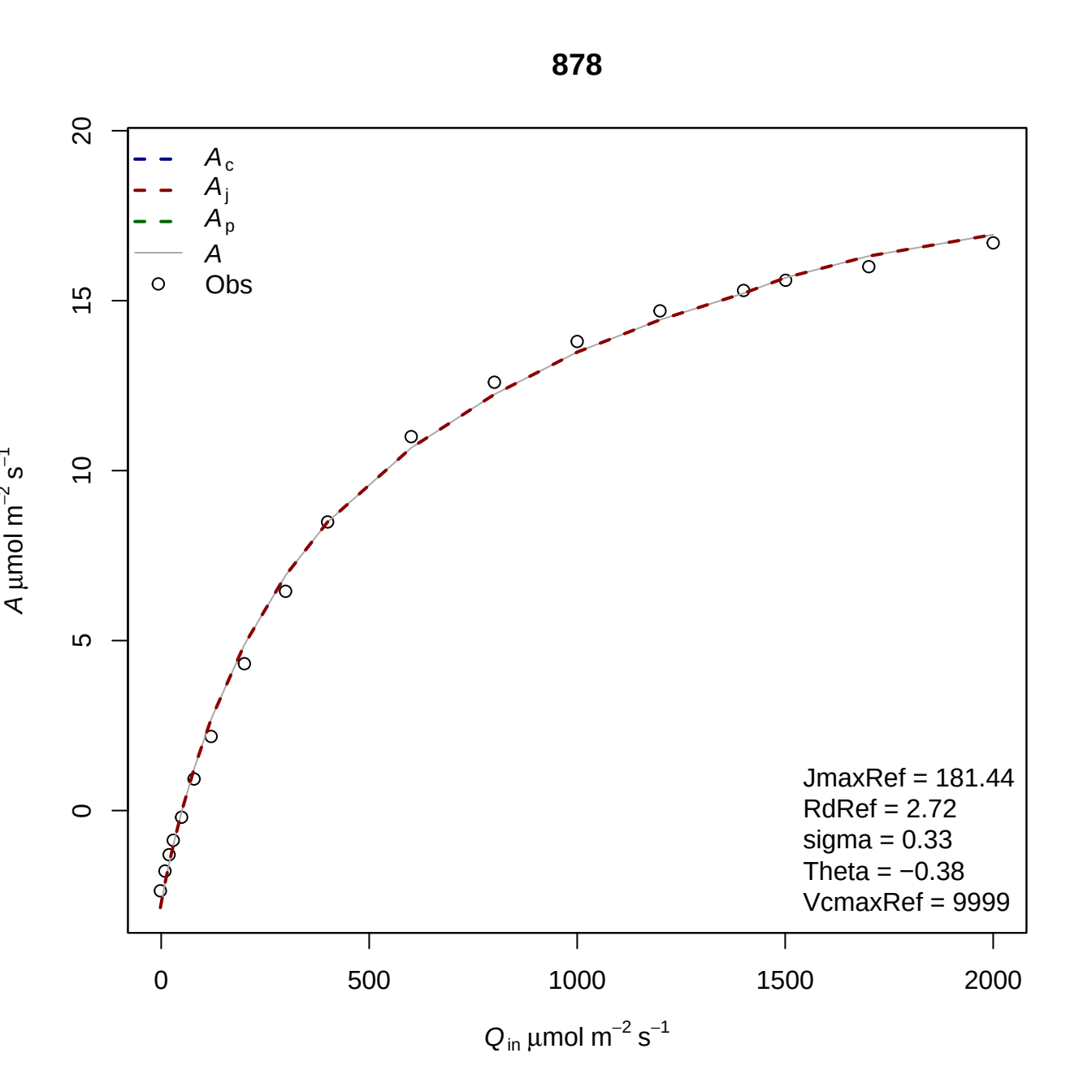

873
A_c
A_j
A_p
A
Obs
JmaxRef = 84.02
RdRef = 1.19
sigma = 0.17
Theta = 0.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

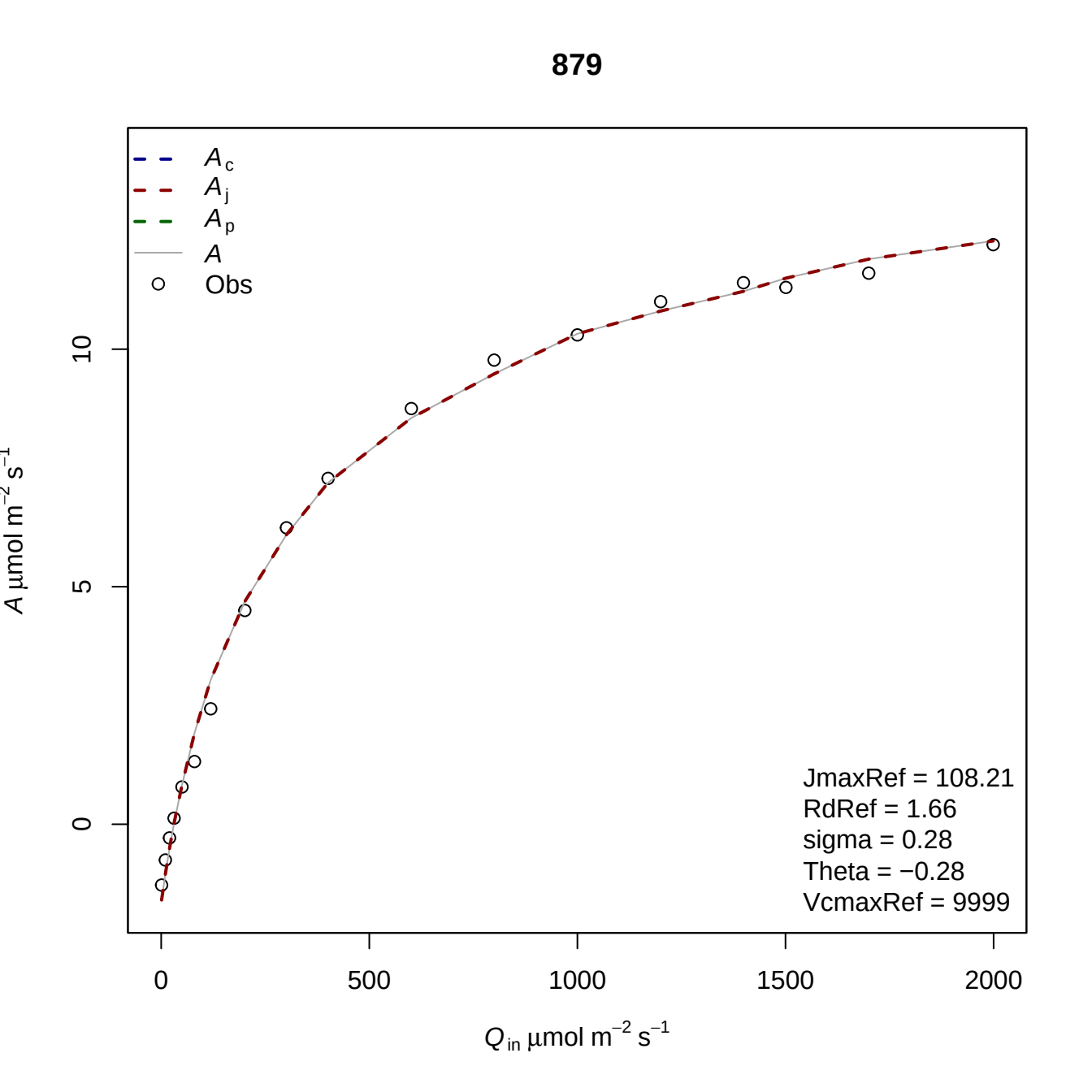

874
A_c
A_j
A_p
A
Obs
JmaxRef = 85.07
RdRef = 2.28
sigma = 0.3
Theta = −0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

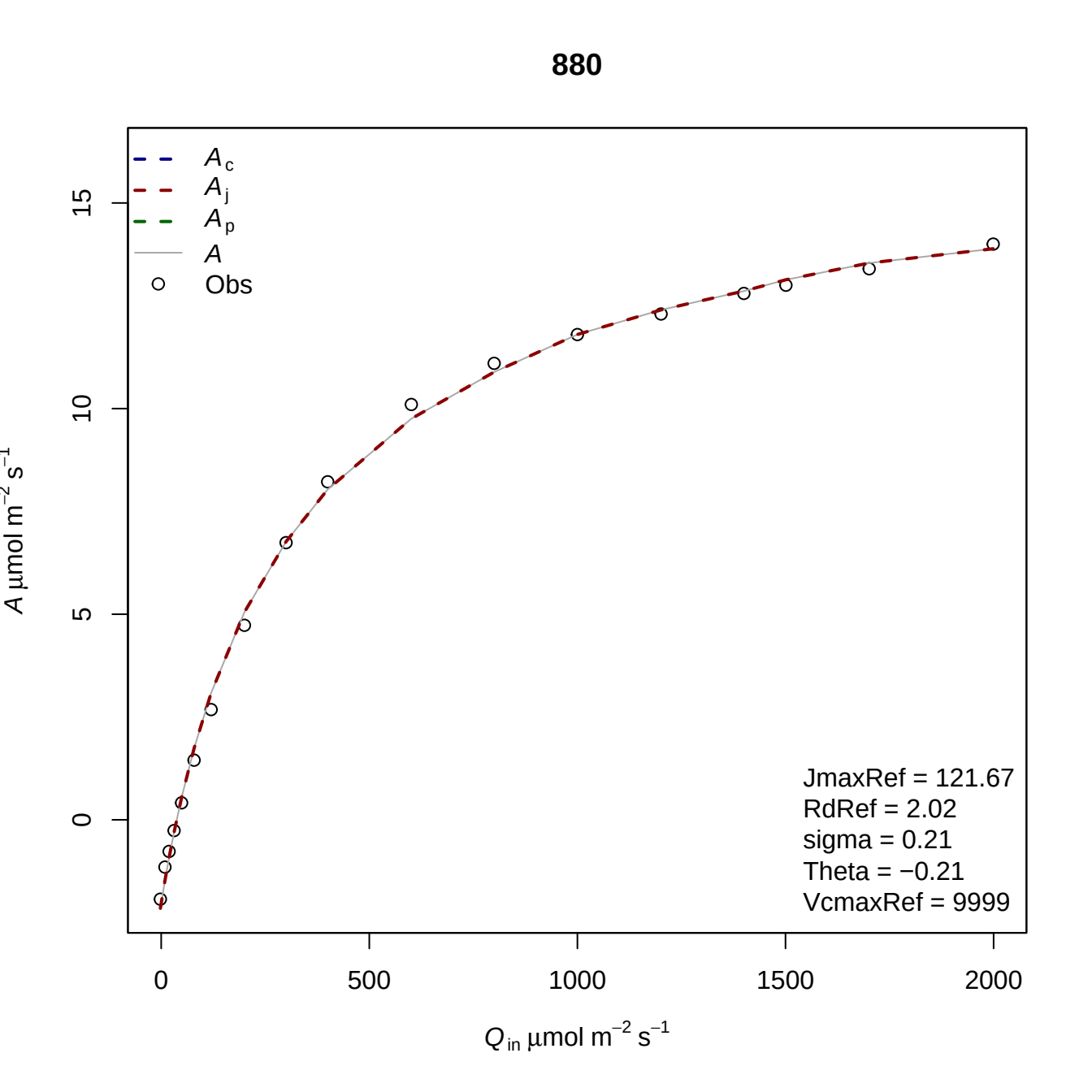

875
A_c
A_j
A_p
A
Obs
JmaxRef = 78.6
RdRef = 1.2
sigma = 0.18
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

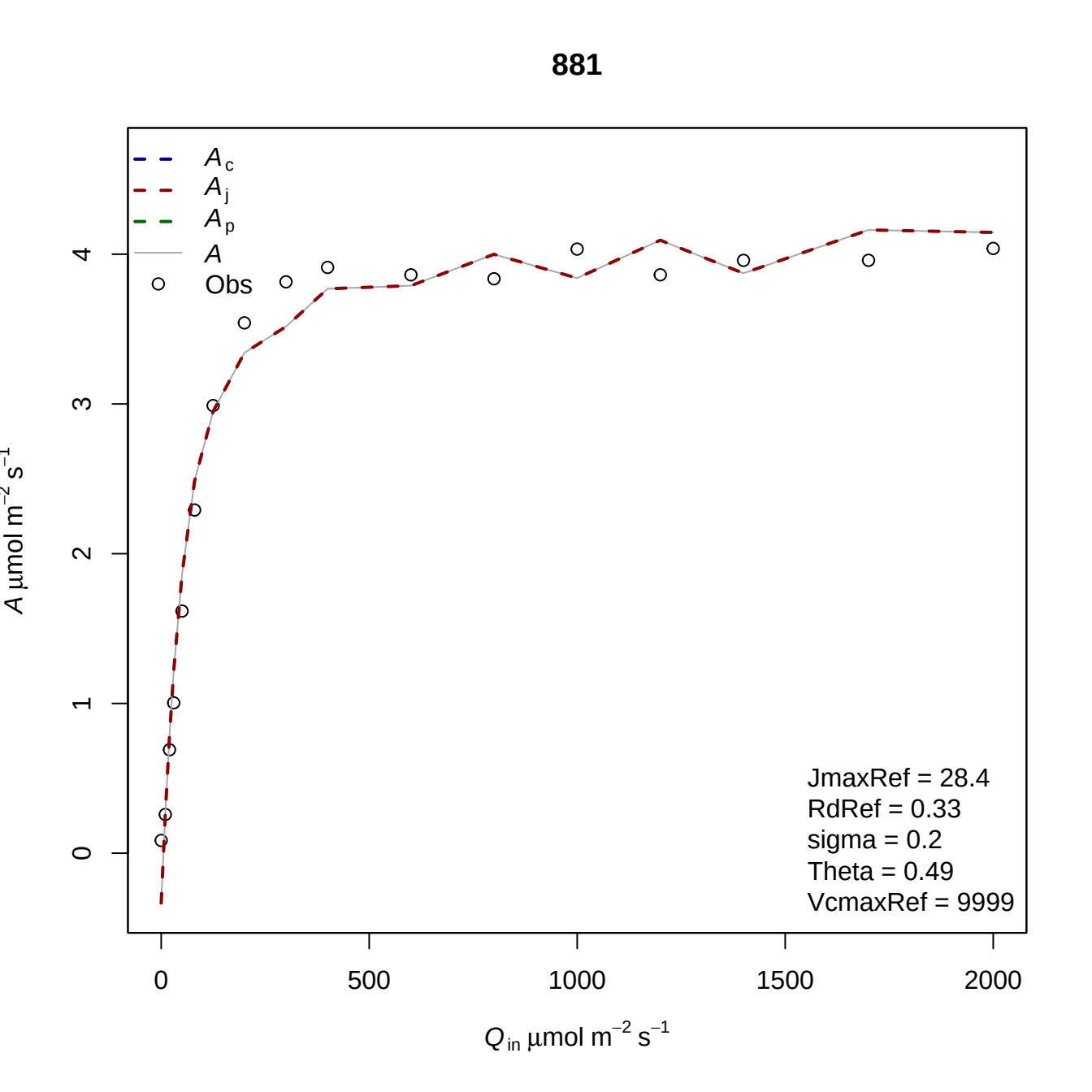

876
A_c
A_j
A_p
A
Obs
JmaxRef = 97.64
RdRef = 2.14
sigma = 0.28
Theta = −0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

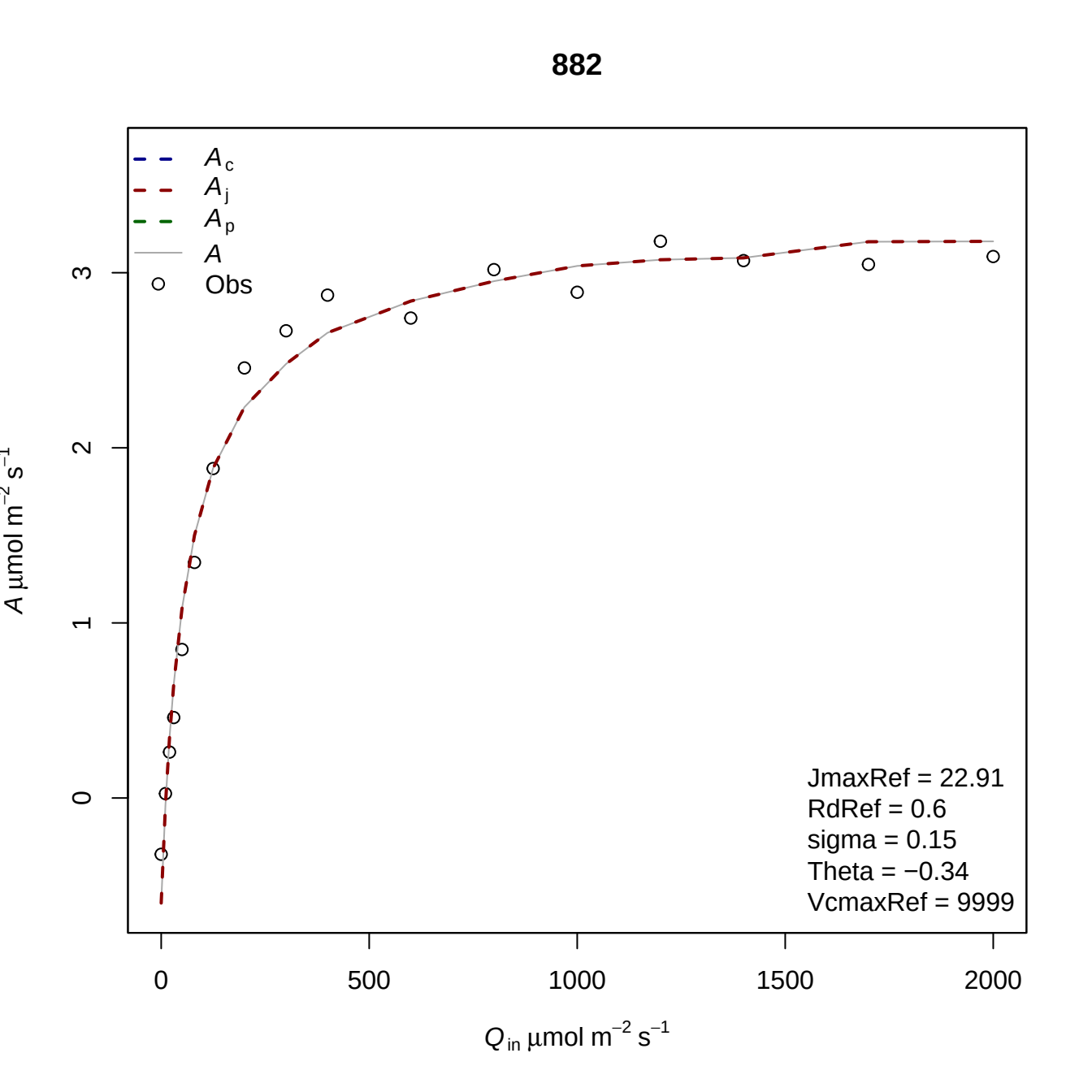

877
A_c
A_j
A_p
A
Obs
JmaxRef = 179.11
RdRef = 1.91
sigma = 0.28
Theta = 0.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

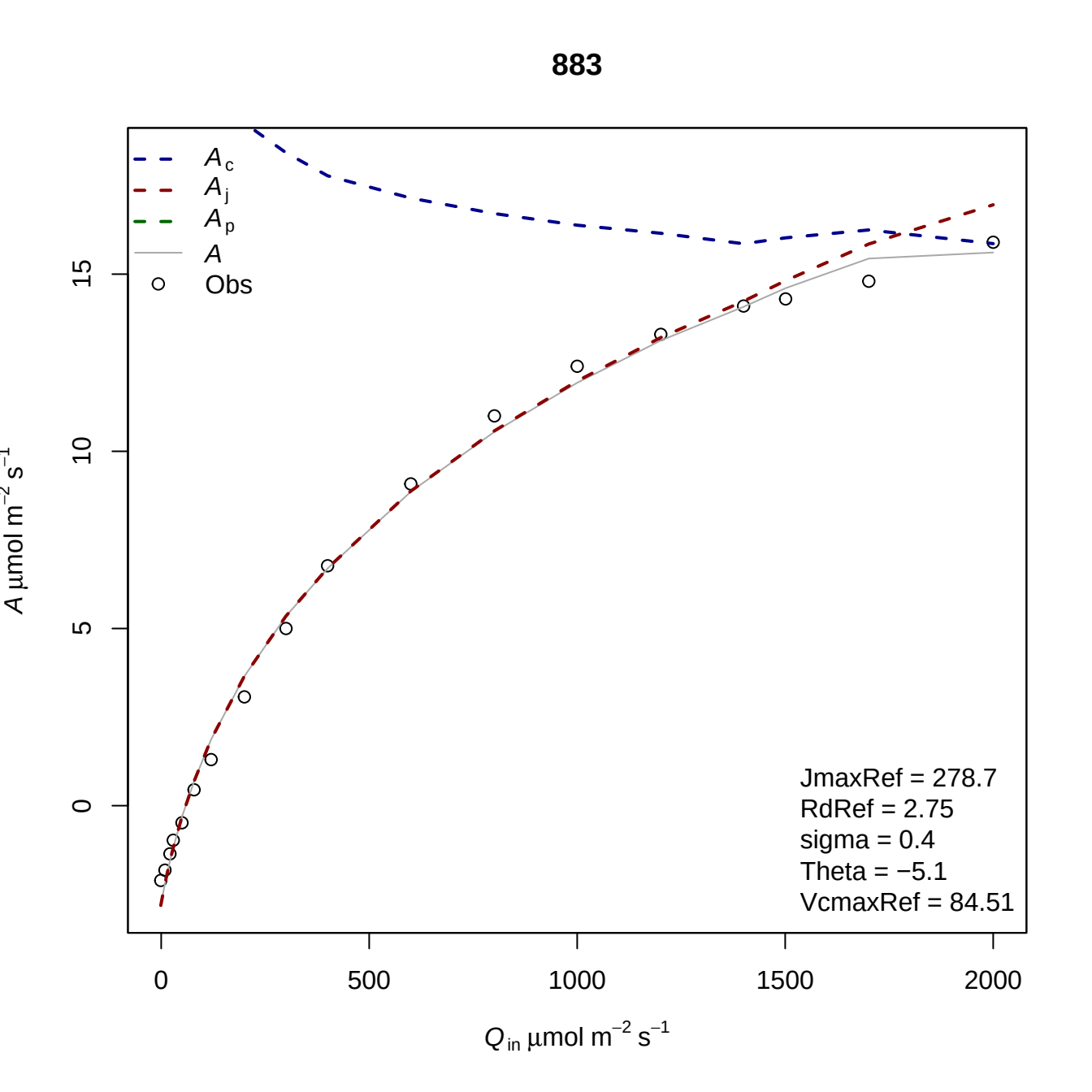

878
A_c
A_j
A_p
A
Obs
JmaxRef = 181.44
RdRef = 2.72
sigma = 0.33
Theta = −0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

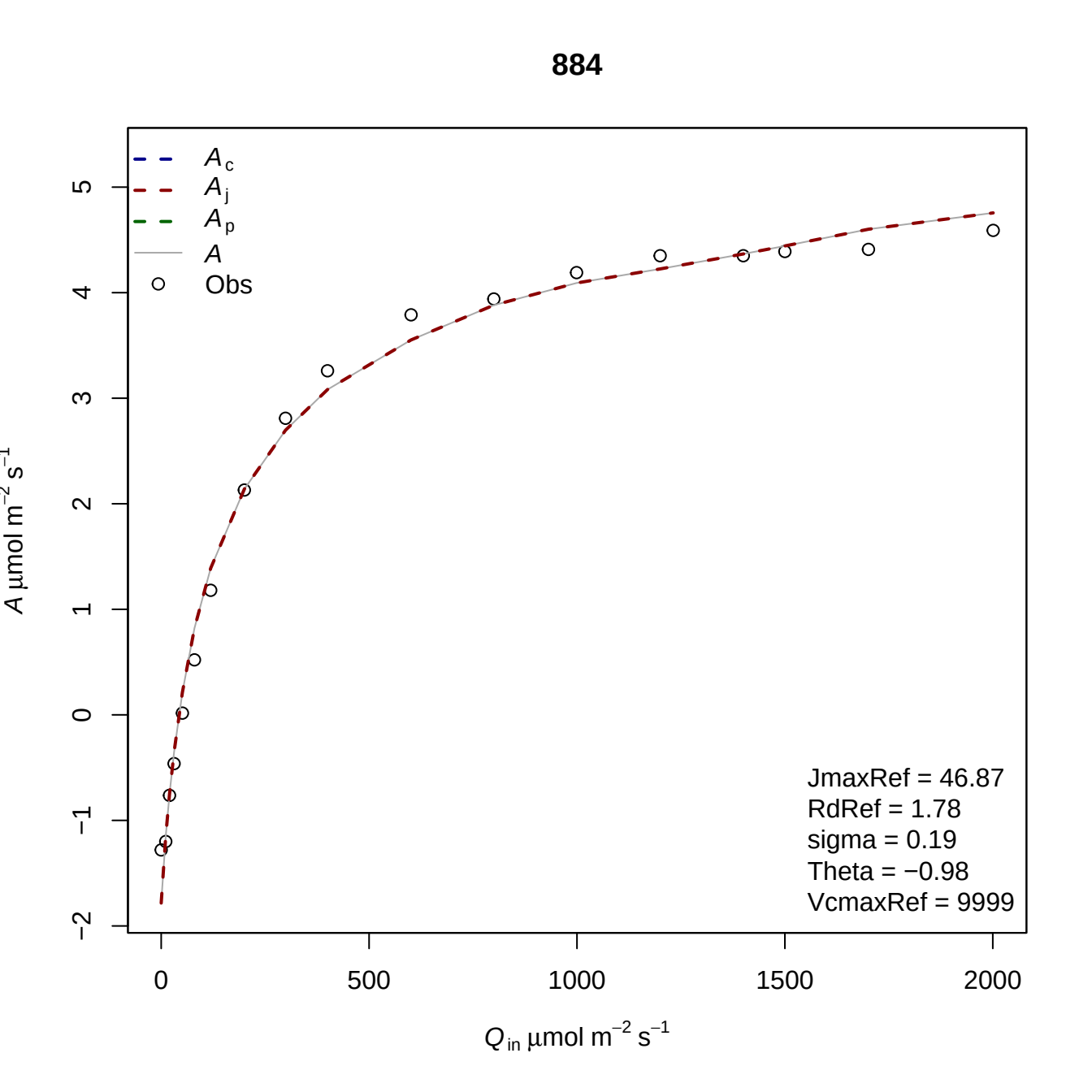

879
A_c
A_j
A_p
A
Obs
JmaxRef = 108.21
RdRef = 1.66
sigma = 0.28
Theta = −0.28
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

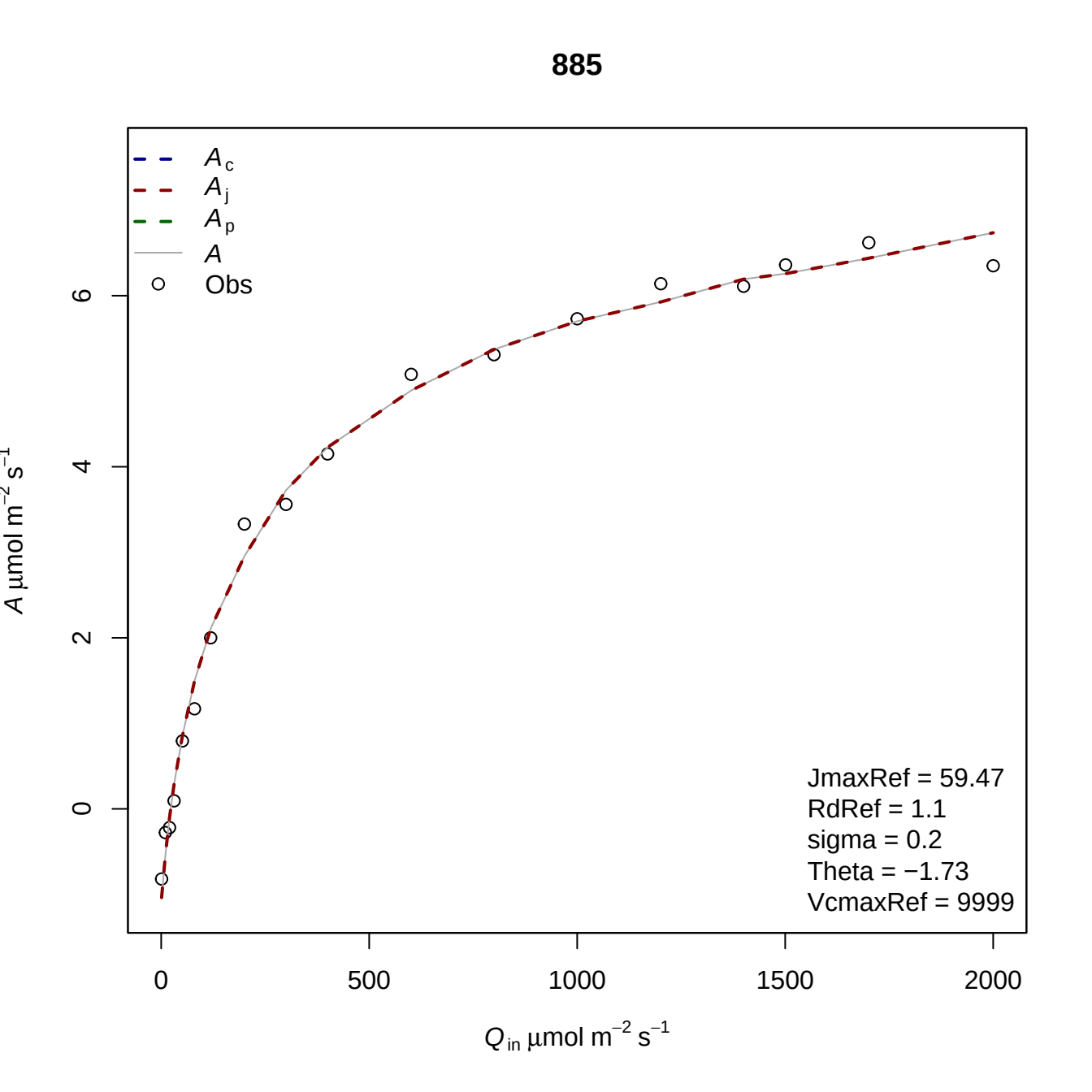

880
A_c
A_j
A_p
A
Obs
JmaxRef = 121.67
RdRef = 2.02
sigma = 0.21
Theta = −0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

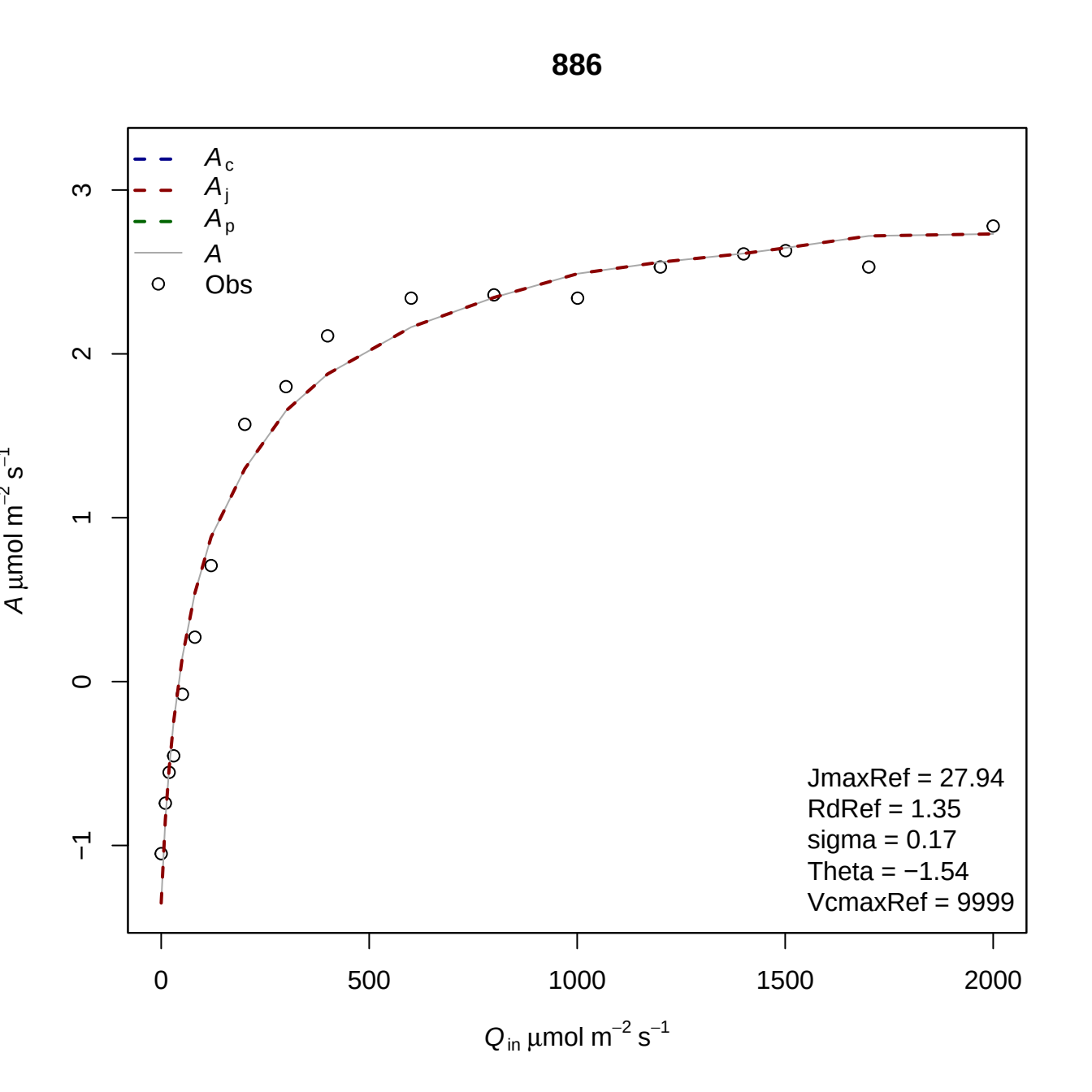

881
A_c
A_j
A_p
A
Obs
JmaxRef = 28.4
RdRef = 0.33
sigma = 0.2
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

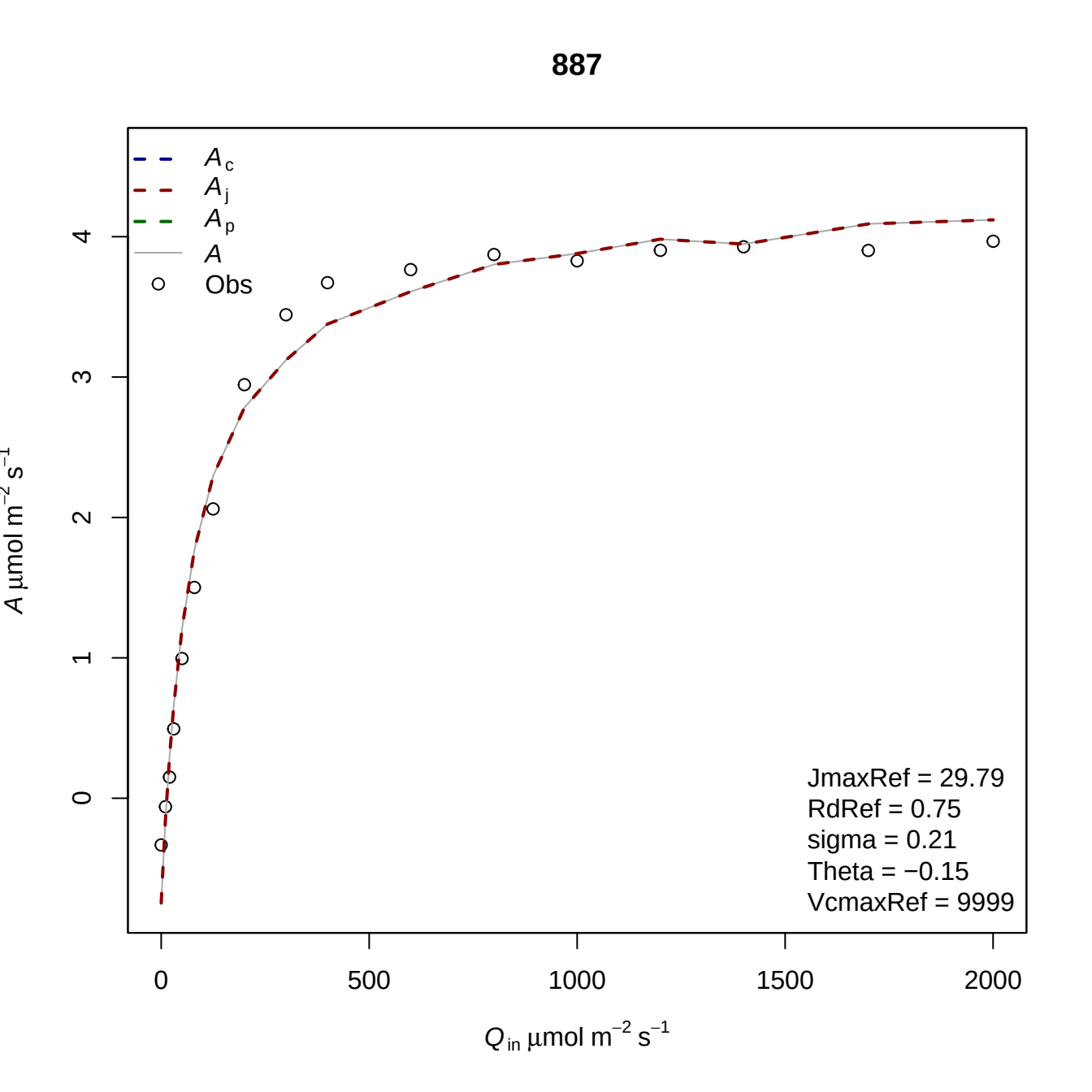

882
A_c
A_j
A_p
A
Obs
JmaxRef = 22.91
RdRef = 0.6
sigma = 0.15
Theta = −0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

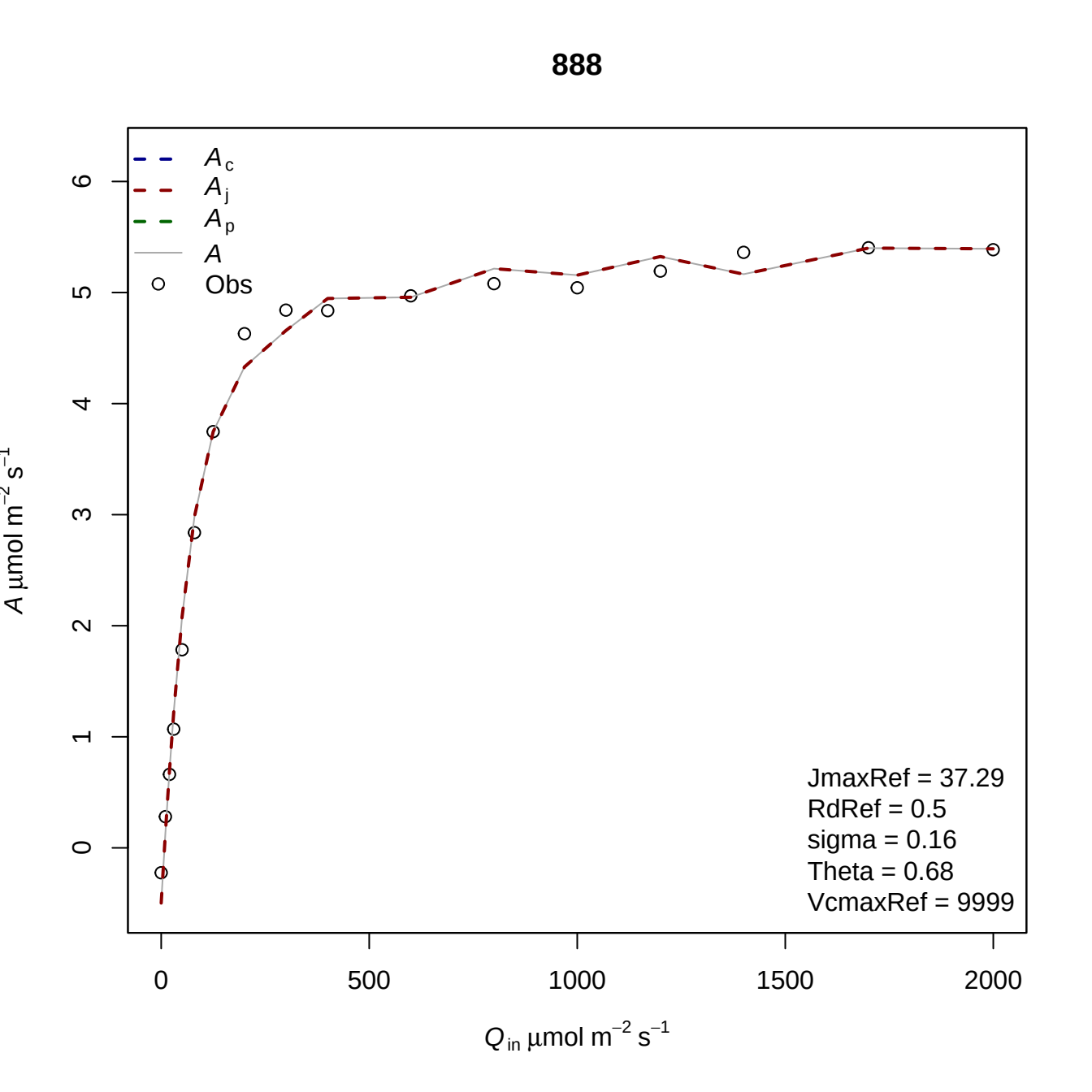

883
A_c
A_j
A_p
A
Obs
JmaxRef = 278.7
RdRef = 2.75
sigma = 0.4
Theta = −5.1
VcmaxRef = 84.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

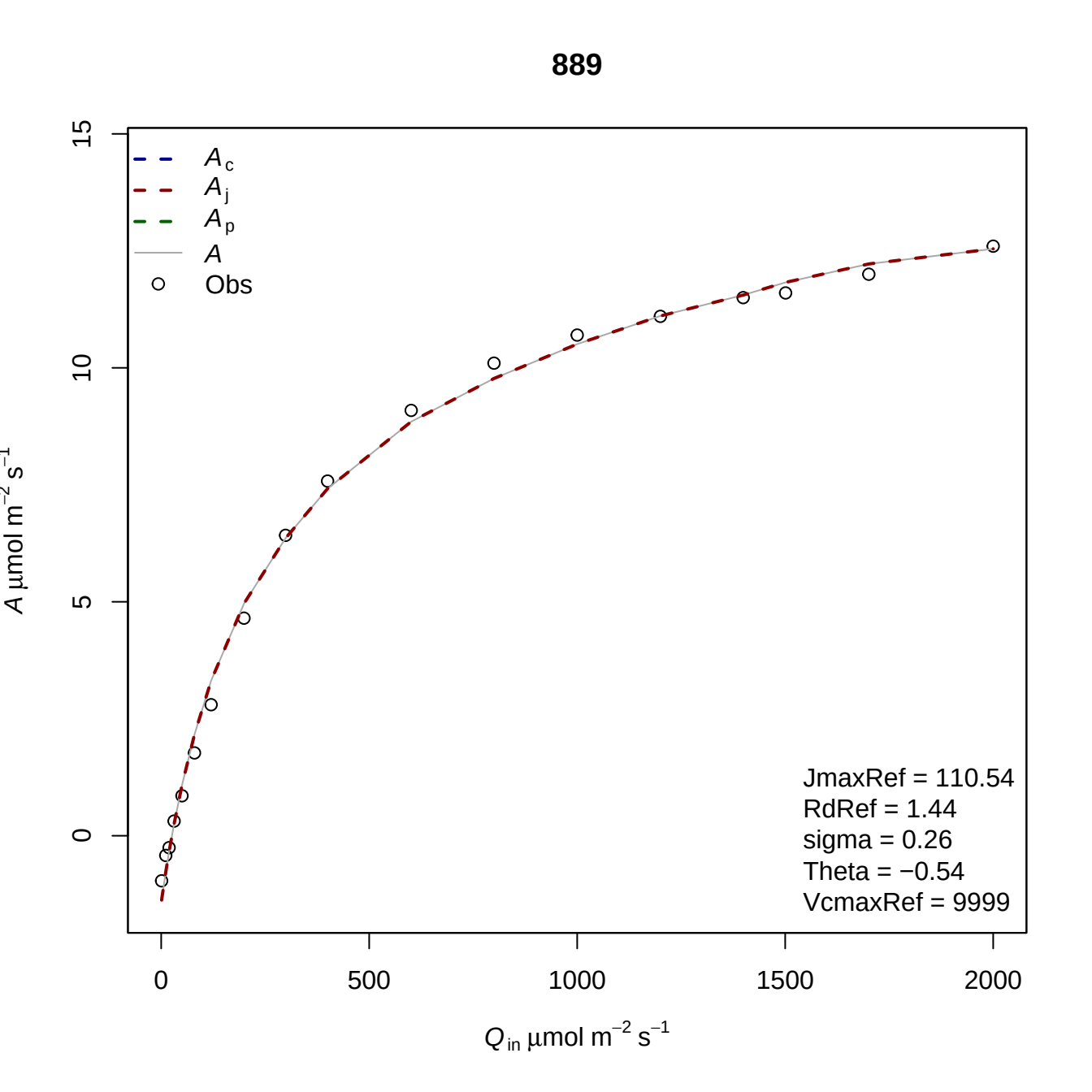

884
A_c
A_j
A_p
A
Obs
JmaxRef = 46.87
RdRef = 1.78
sigma = 0.19
Theta = −0.98
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

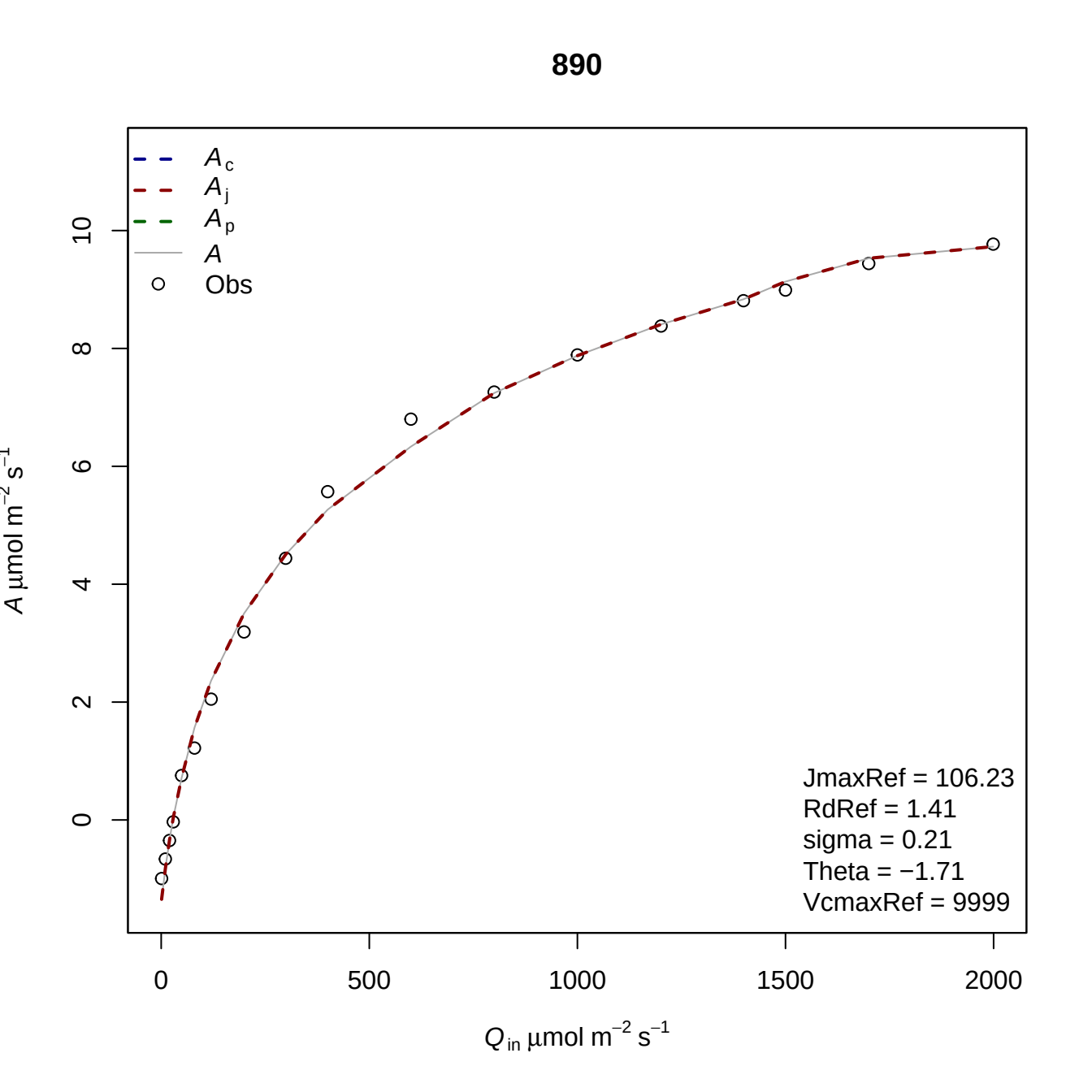

885
A_c
A_j
A_p
A
Obs
JmaxRef = 59.47
RdRef = 1.1
sigma = 0.2
Theta = −1.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

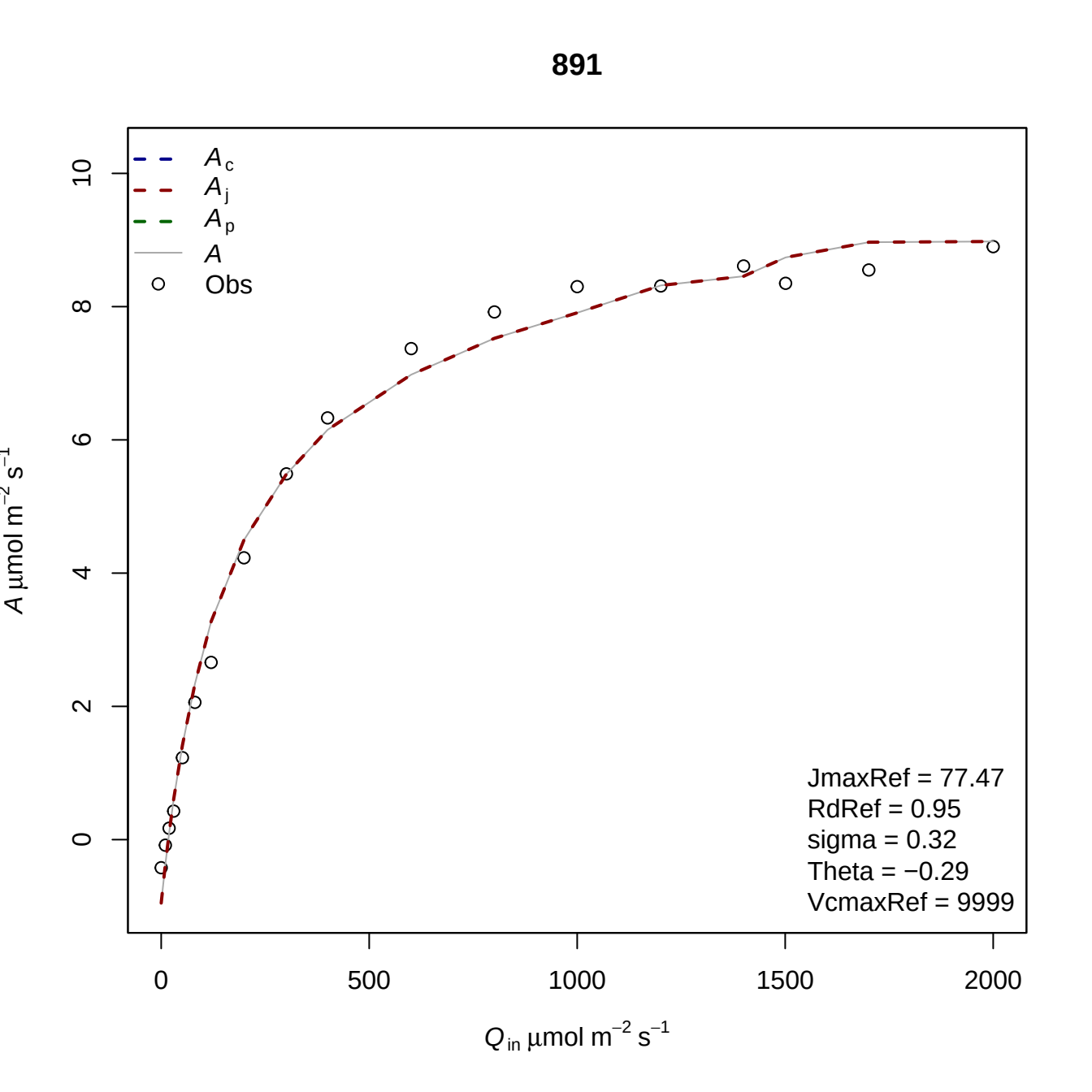

886
A_c
A_j
A_p
A
Obs
JmaxRef = 27.94
RdRef = 1.35
sigma = 0.17
Theta = −1.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

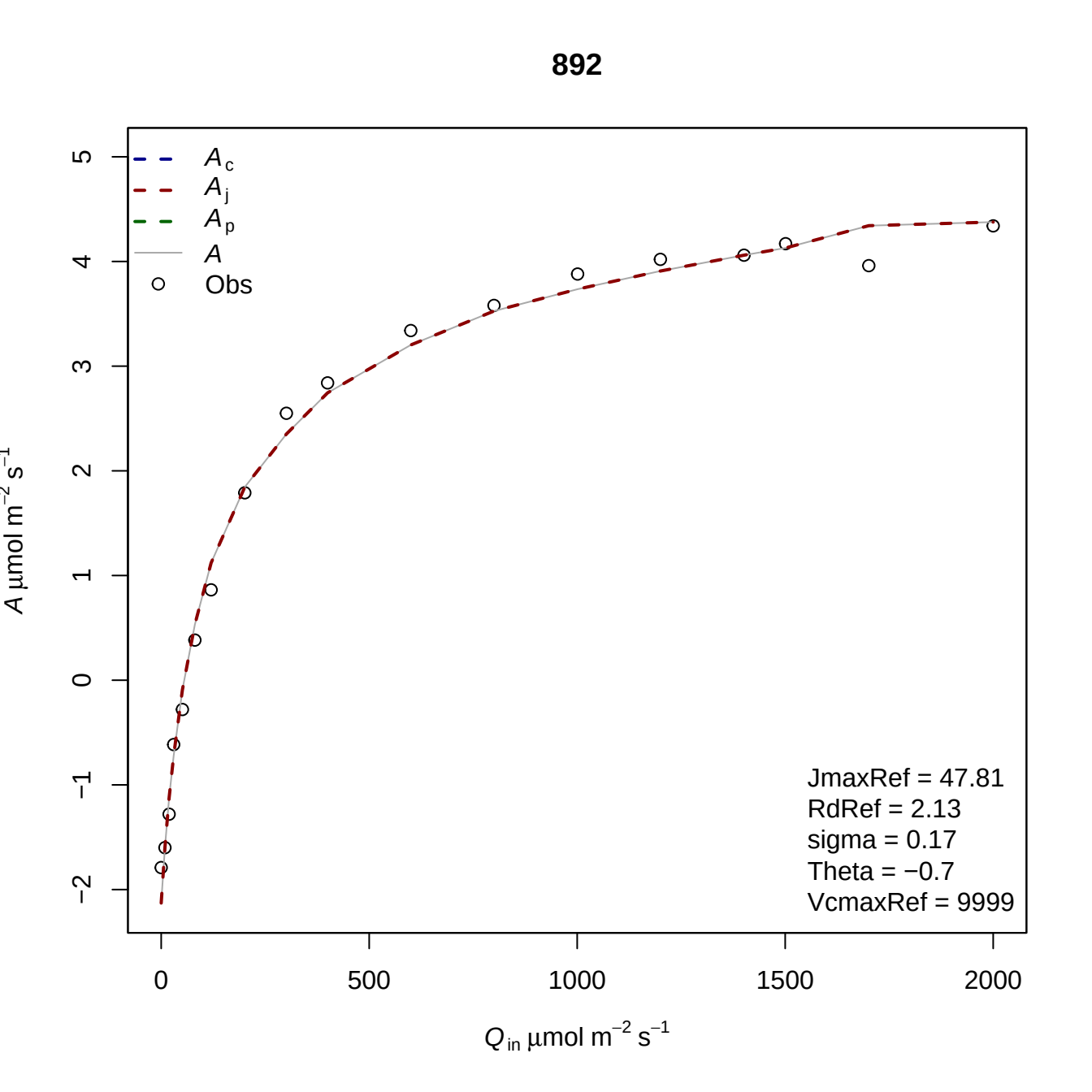

887
A_c
A_j
A_p
A
Obs
JmaxRef = 29.79
RdRef = 0.75
sigma = 0.21
Theta = −0.15
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

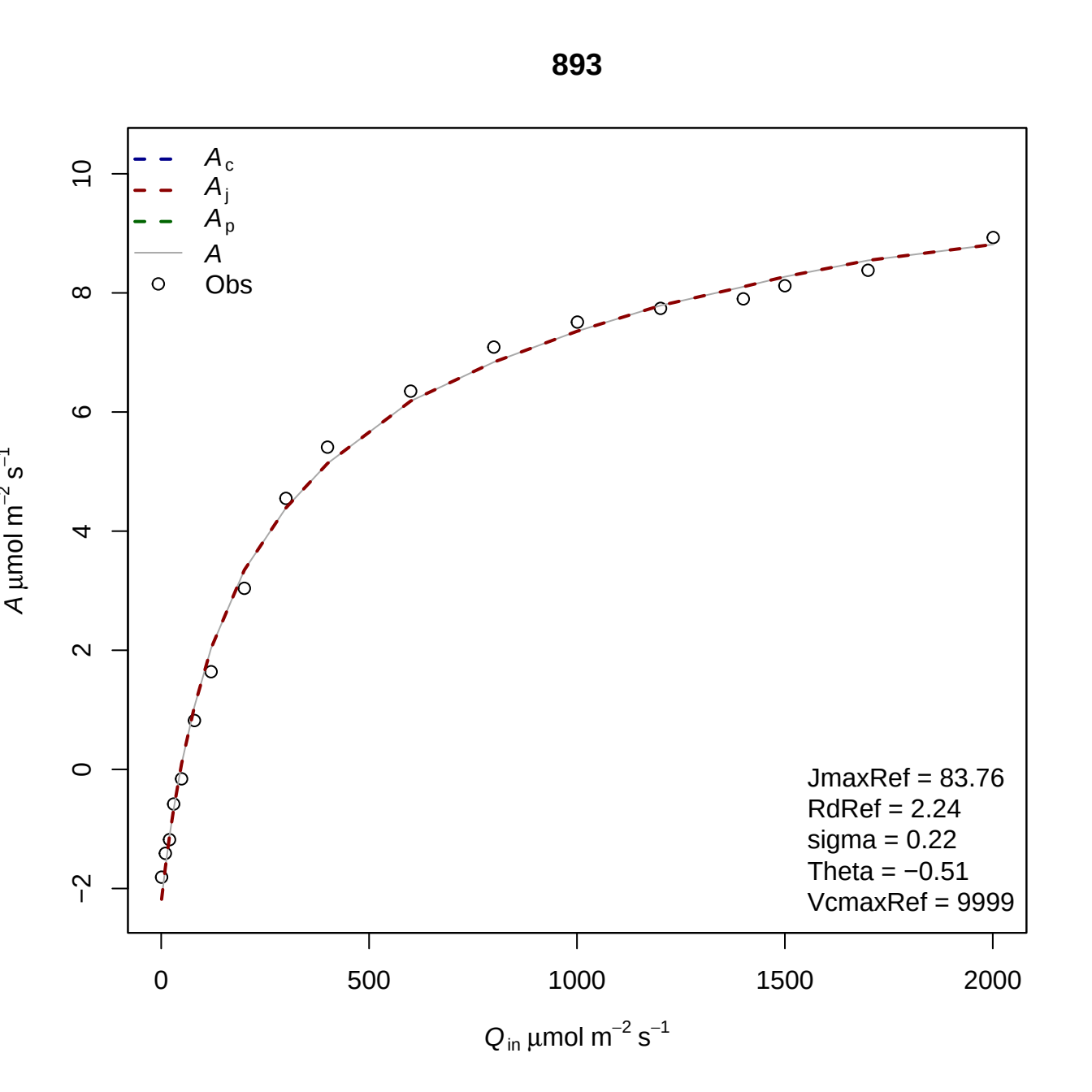

888
A_c
A_j
A_p
A
Obs
JmaxRef = 37.29
RdRef = 0.5
sigma = 0.16
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

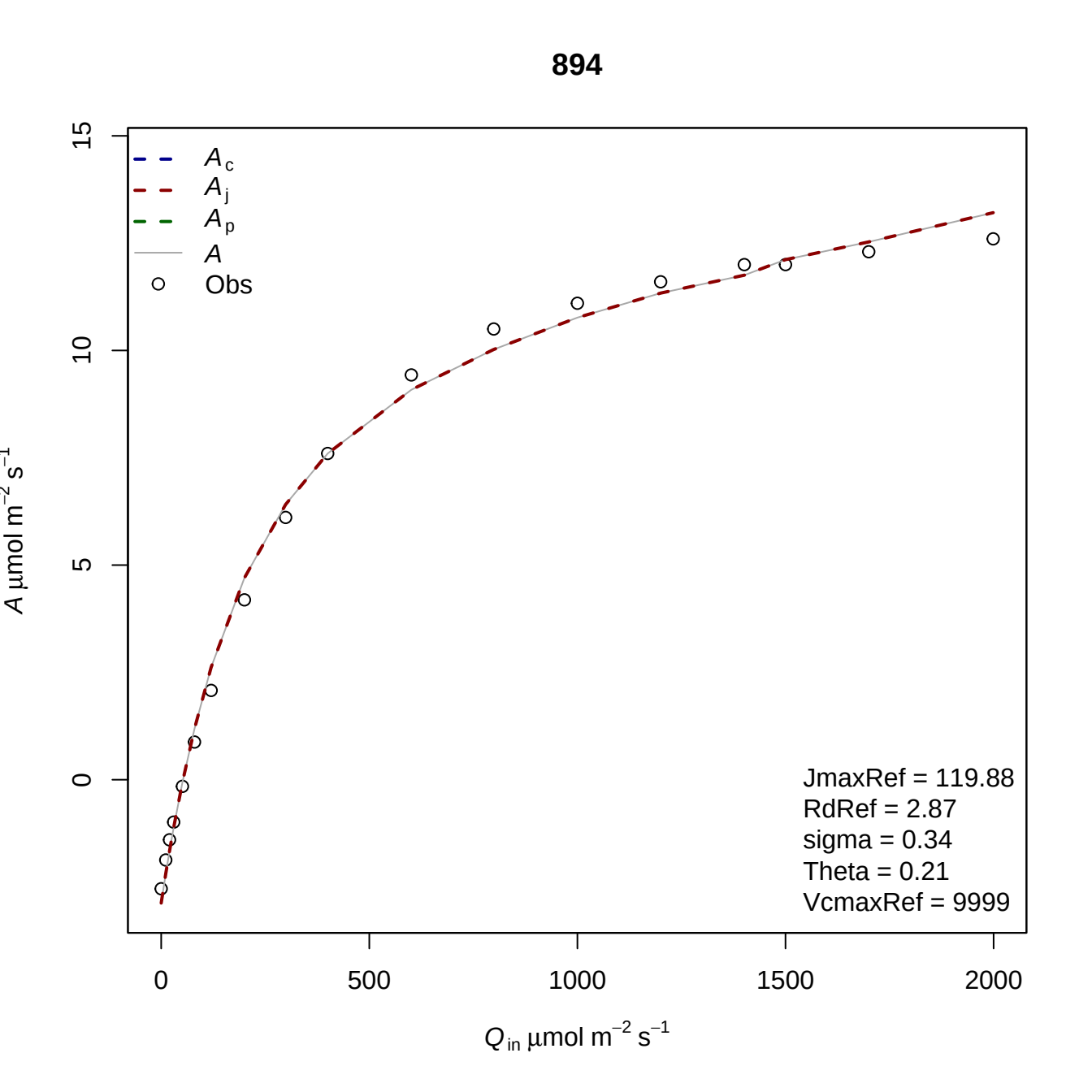

889
A_c
A_j
A_p
A
Obs
JmaxRef = 110.54
RdRef = 1.44
sigma = 0.26
Theta = −0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

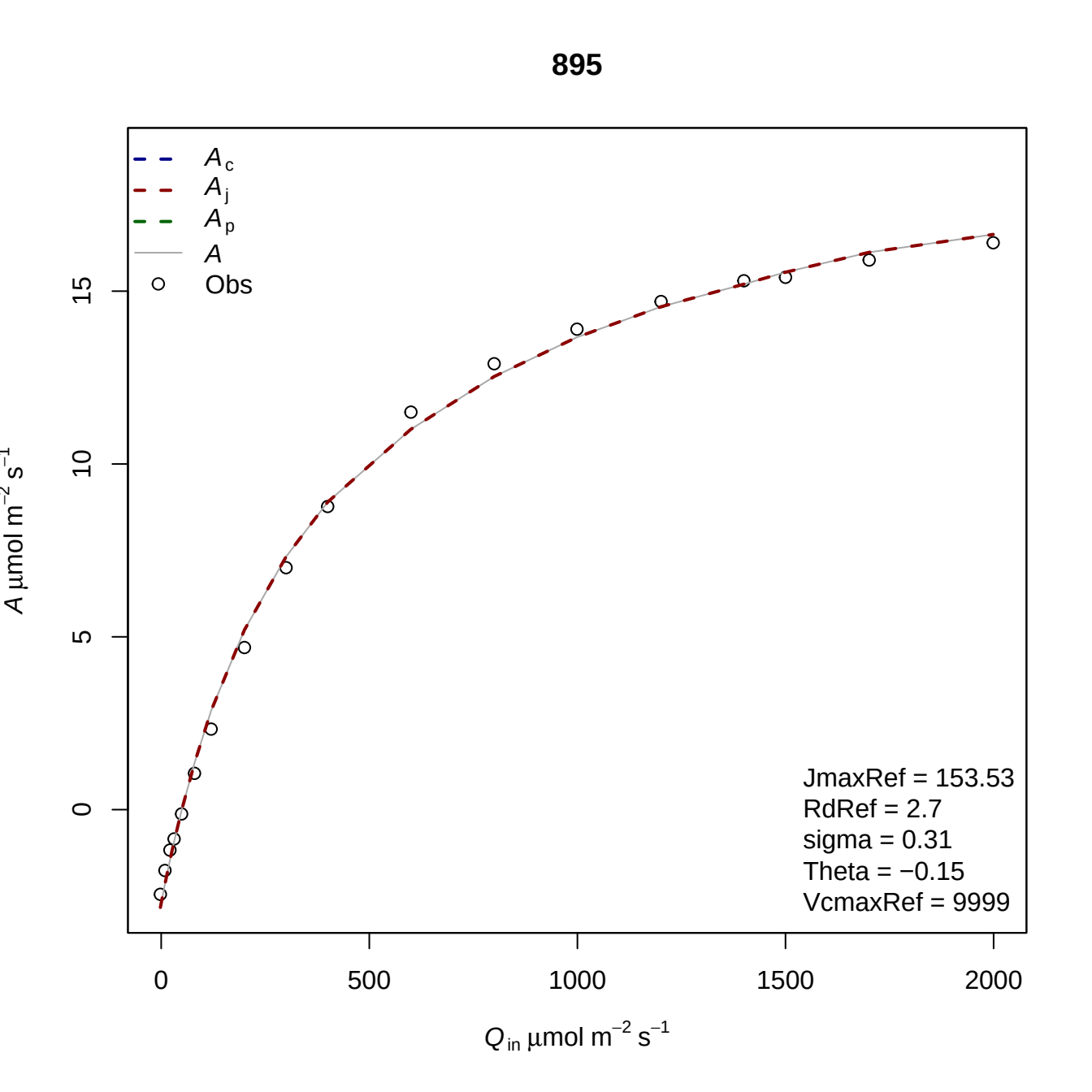

890
A_c
A_j
A_p
A
Obs
JmaxRef = 106.23
RdRef = 1.41
sigma = 0.21
Theta = −1.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

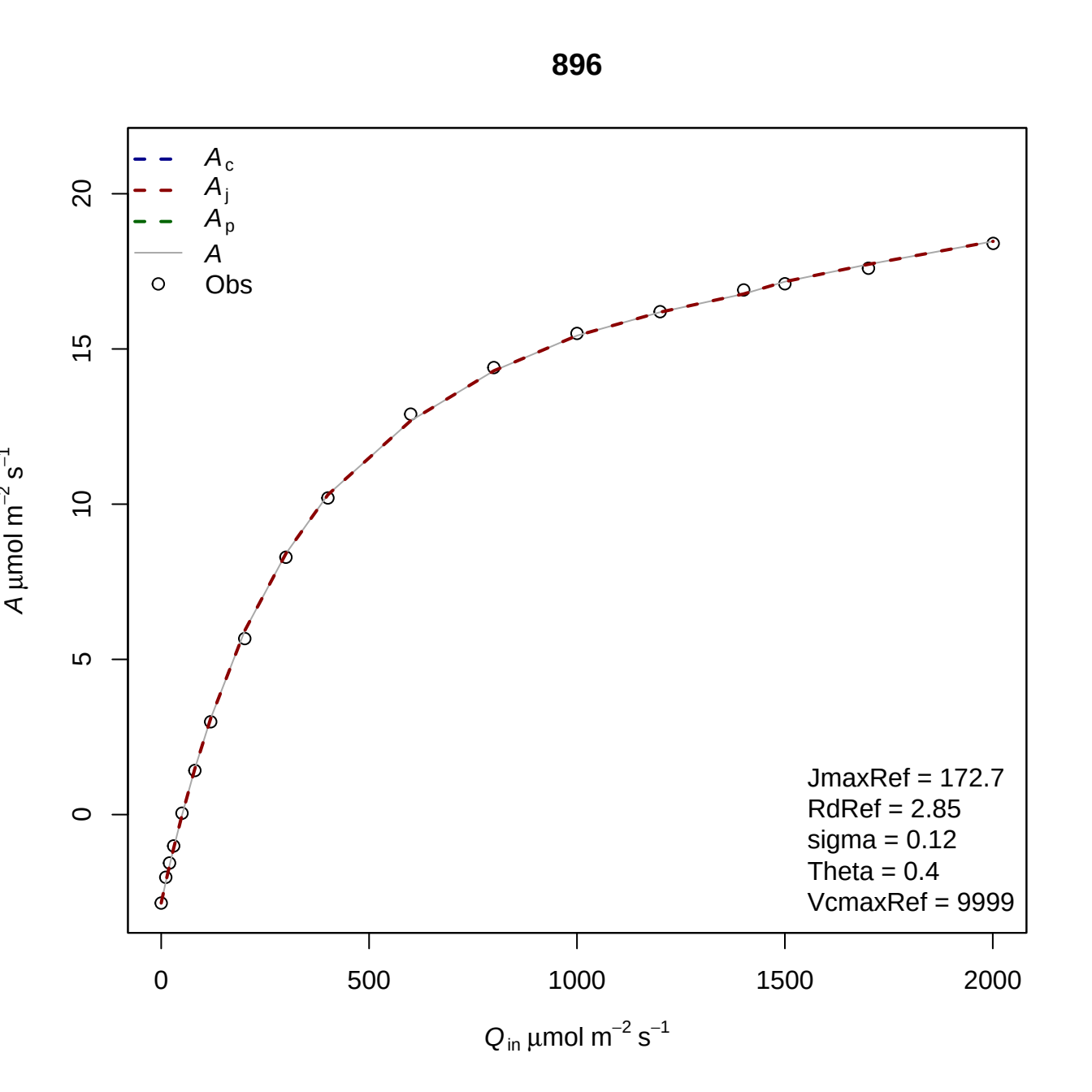

891
A_c
A_j
A_p
A
Obs
JmaxRef = 77.47
RdRef = 0.95
sigma = 0.32
Theta = −0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

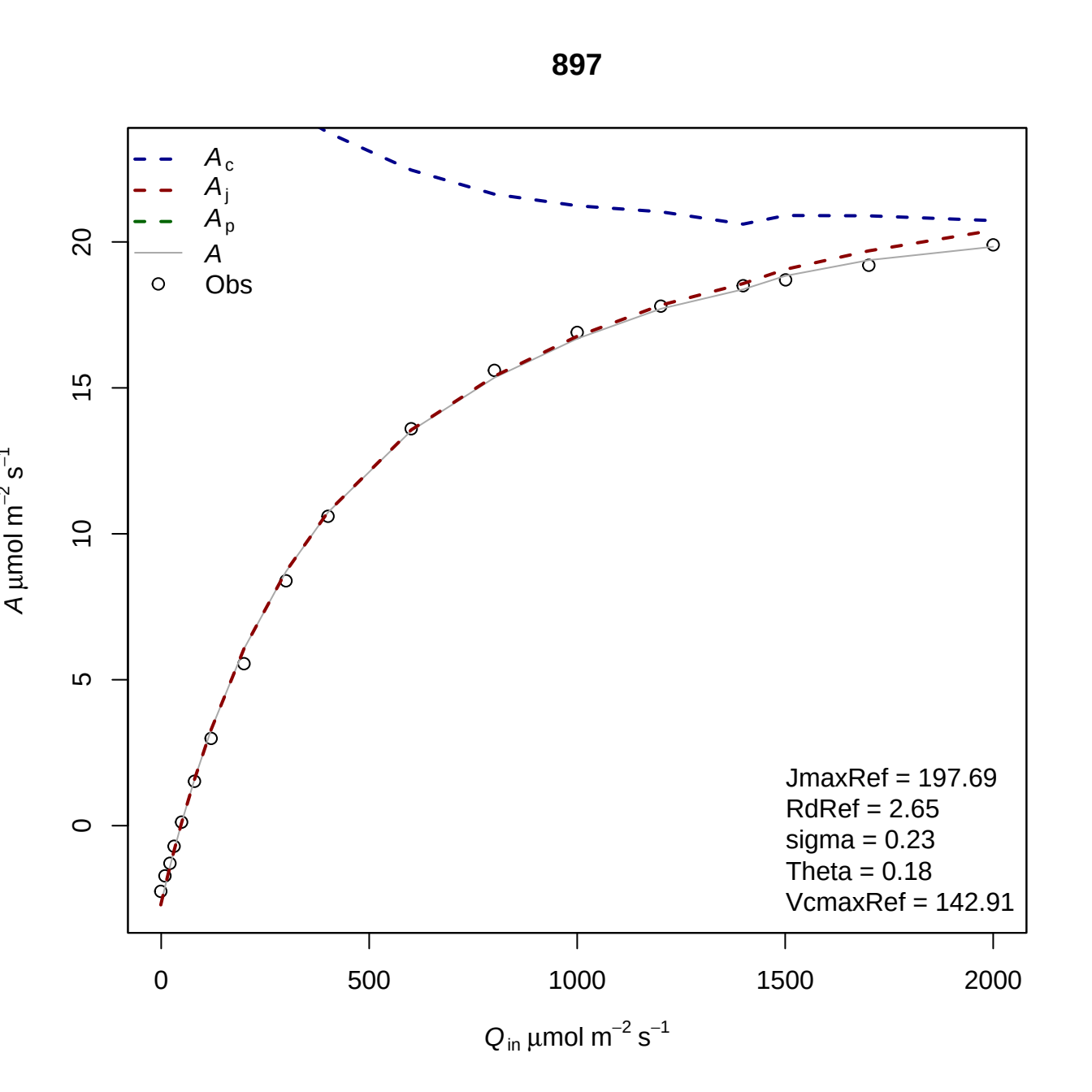

892
A_c
A_j
A_p
A
Obs
JmaxRef = 47.81
RdRef = 2.13
sigma = 0.17
Theta = −0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

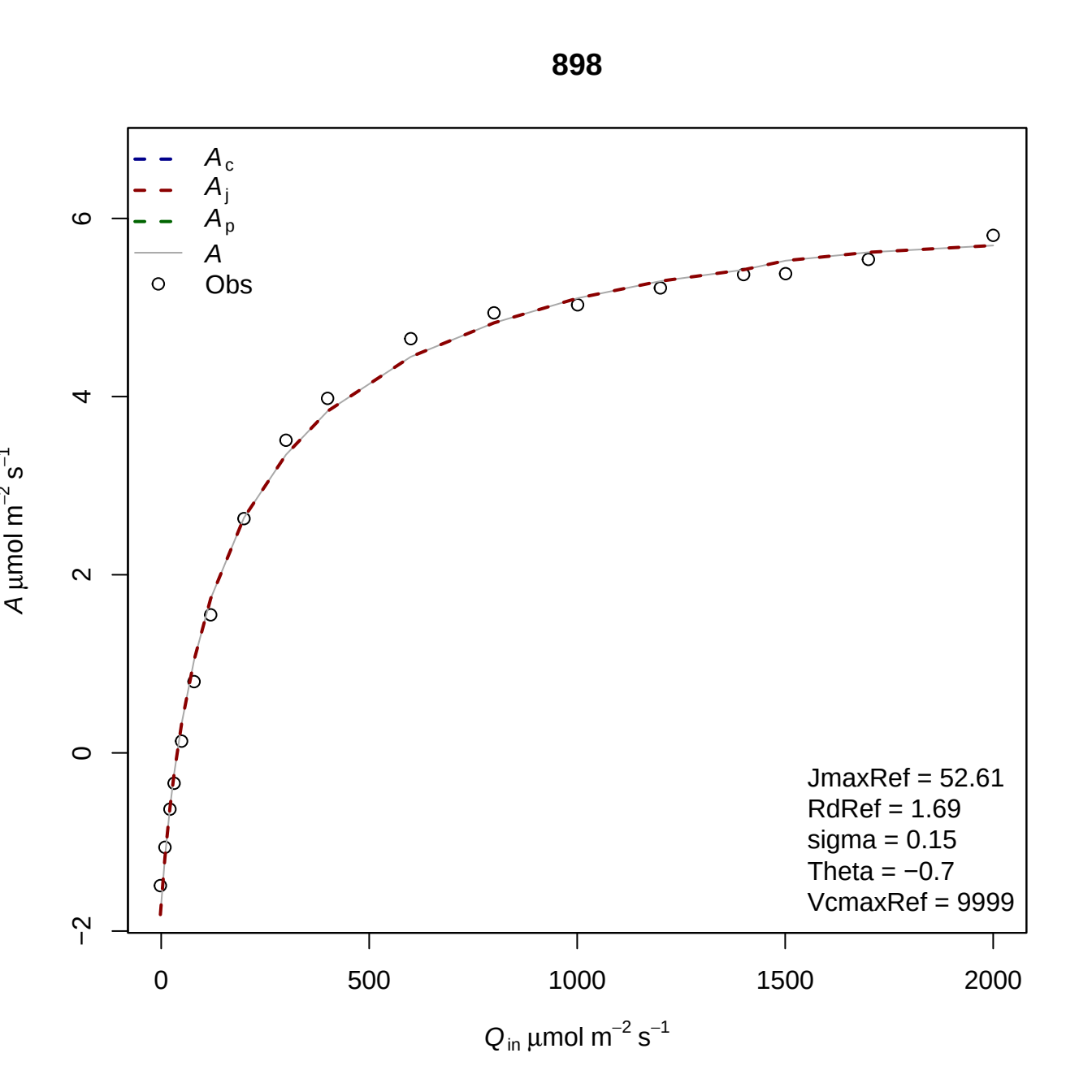

893
A_c
A_j
A_p
A
Obs
JmaxRef = 83.76
RdRef = 2.24
sigma = 0.22
Theta = −0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

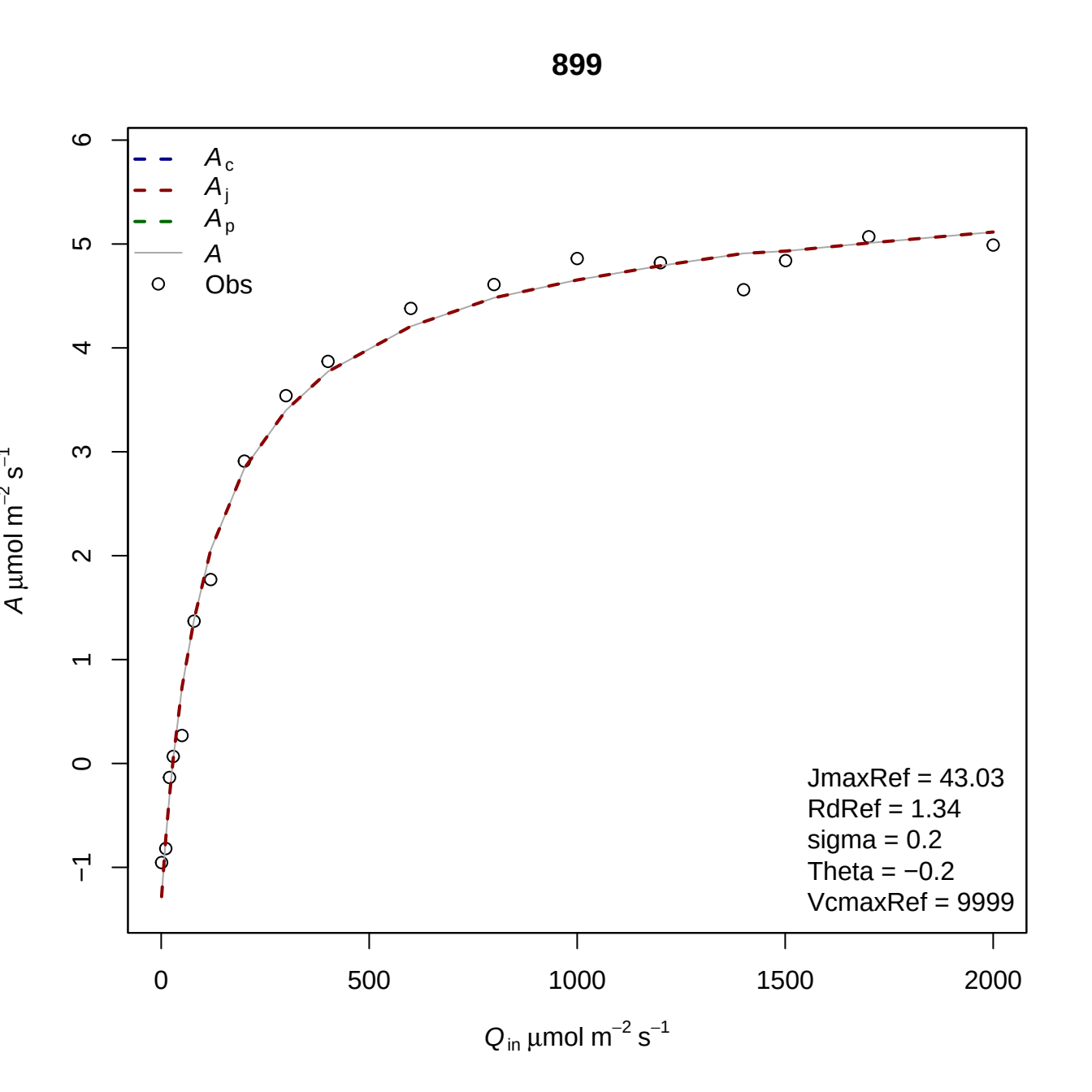

894
A_c
A_j
A_p
A
Obs
JmaxRef = 119.88
RdRef = 2.87
sigma = 0.34
Theta = 0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

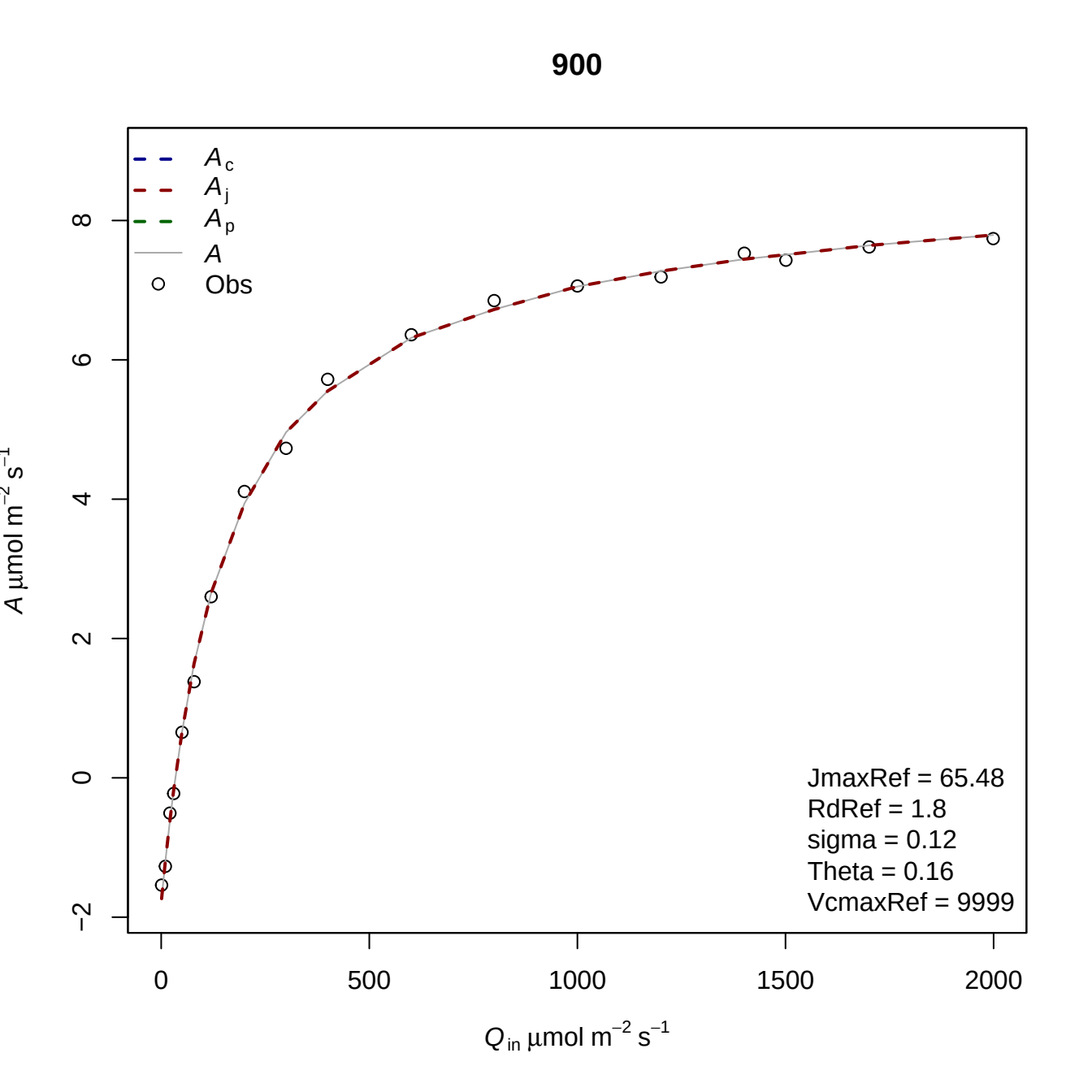

895
A_c
A_j
A_p
A
Obs
JmaxRef = 153.53
RdRef = 2.7
sigma = 0.31
Theta = −0.15
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

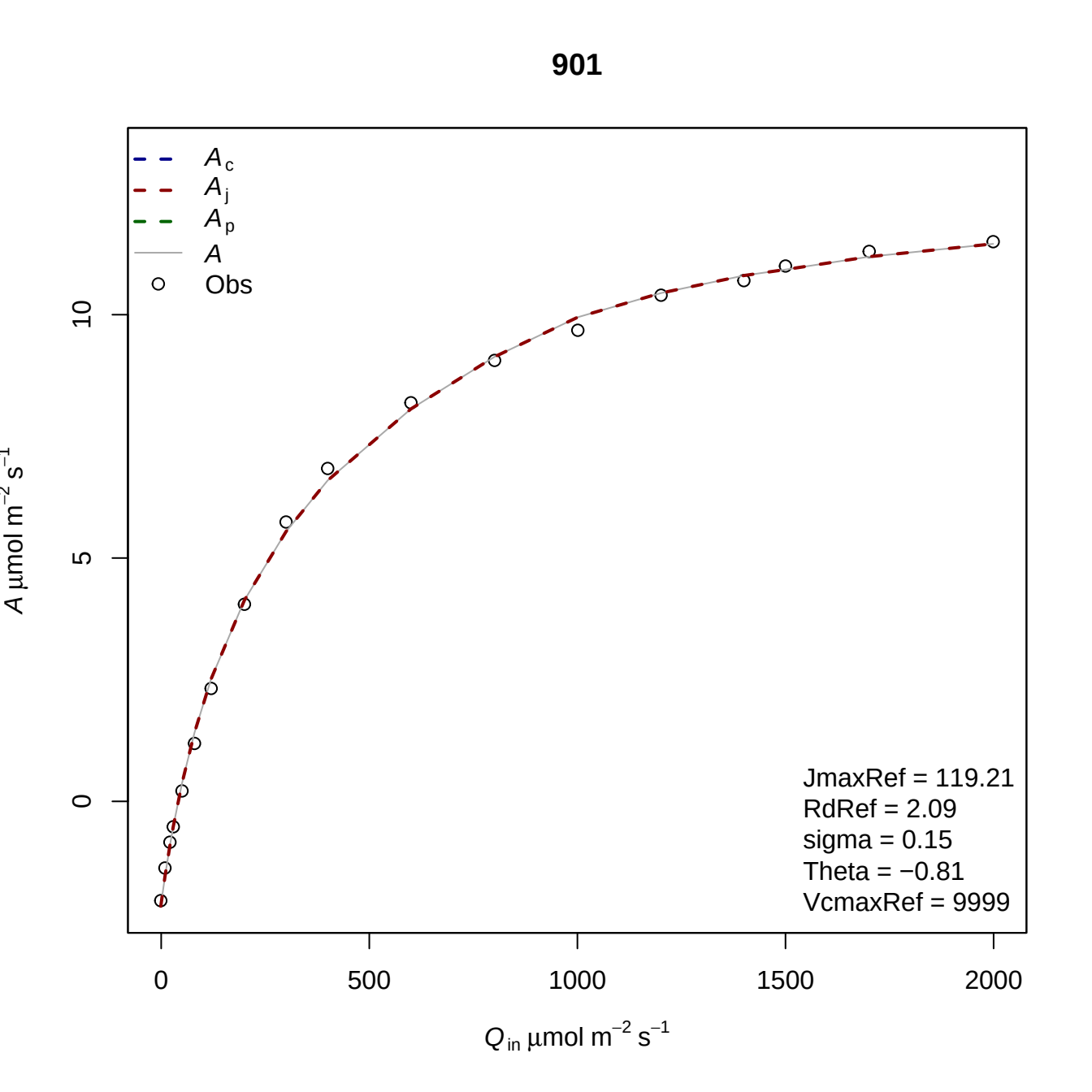

896
A_c
A_j
A_p
A
Obs
JmaxRef = 172.7
RdRef = 2.85
sigma = 0.12
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

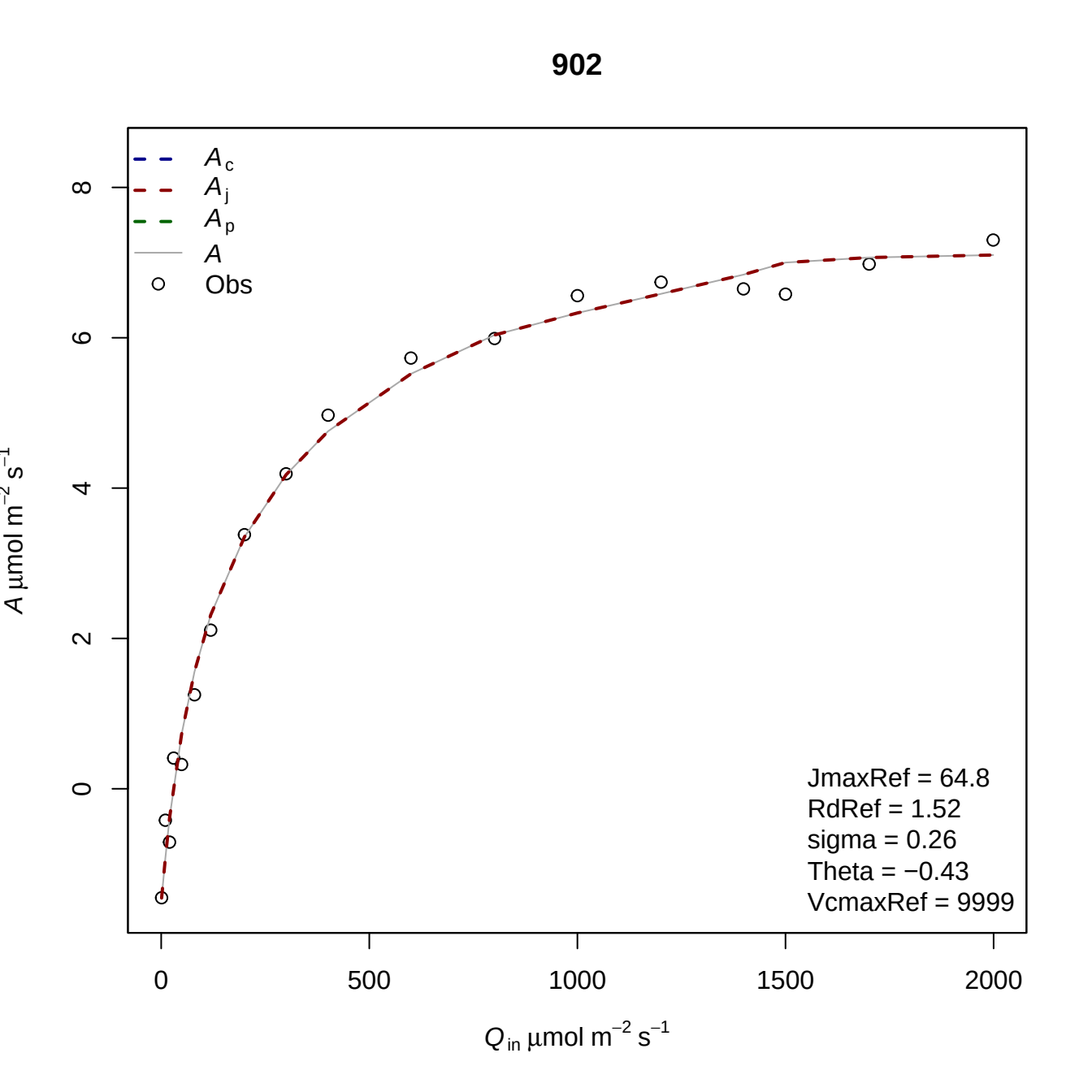

897
A_c
A_j
A_p
A
Obs
JmaxRef = 197.69
RdRef = 2.65
sigma = 0.23
Theta = 0.18
VcmaxRef = 142.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

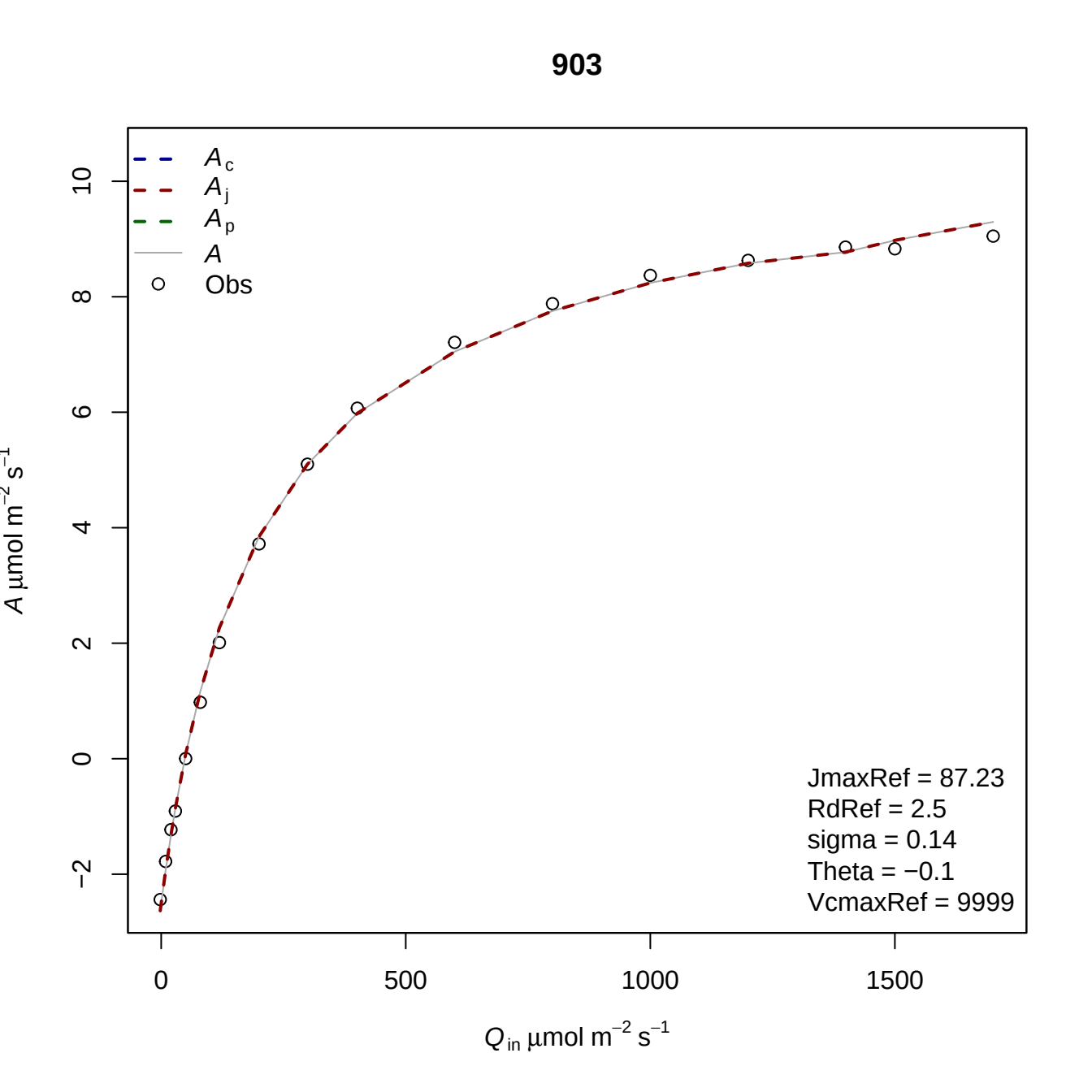

898
A_c
A_j
A_p
A
Obs
JmaxRef = 52.61
RdRef = 1.69
sigma = 0.15
Theta = −0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

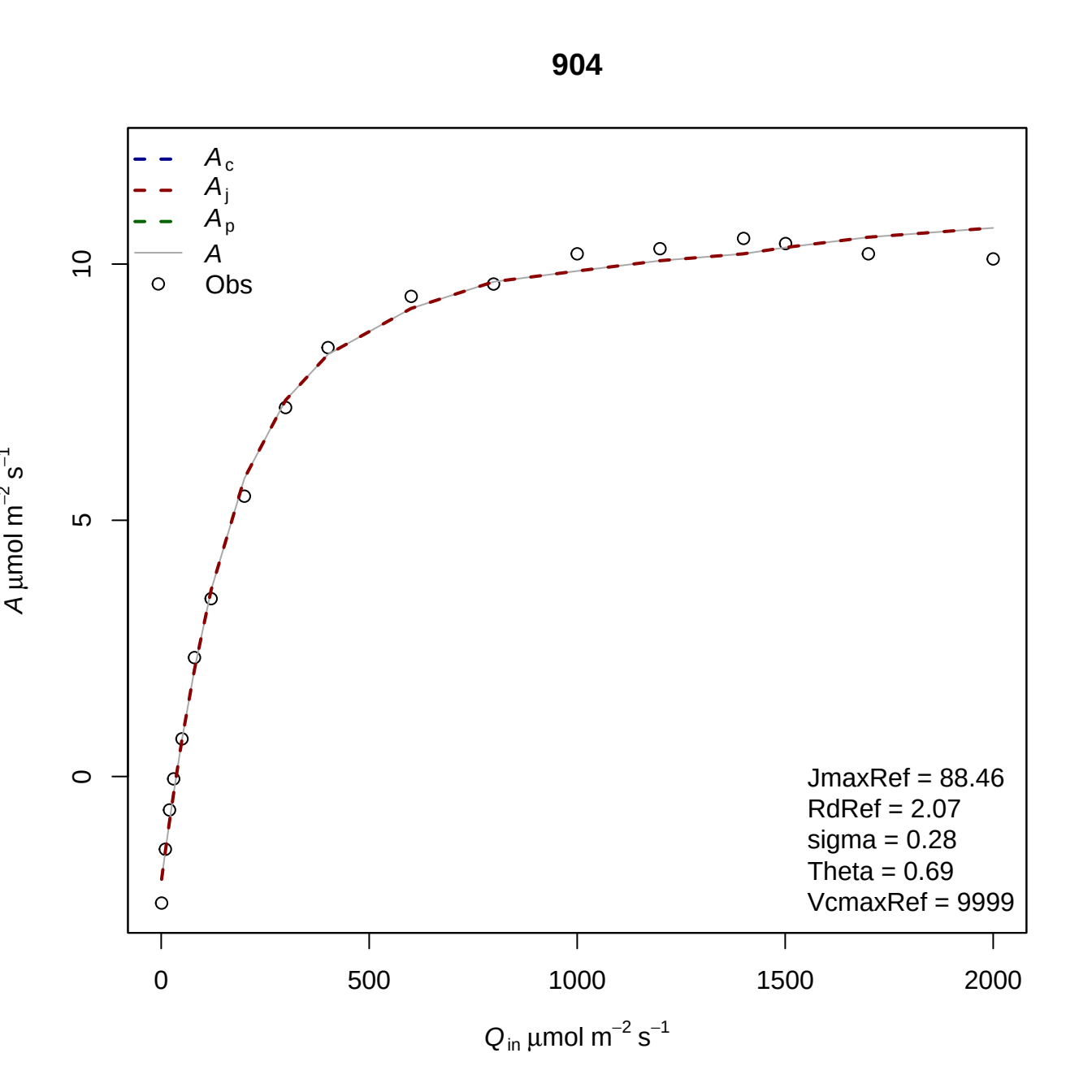

899
A_c
A_j
A_p
A
Obs
JmaxRef = 43.03
RdRef = 1.34
sigma = 0.2
Theta = −0.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

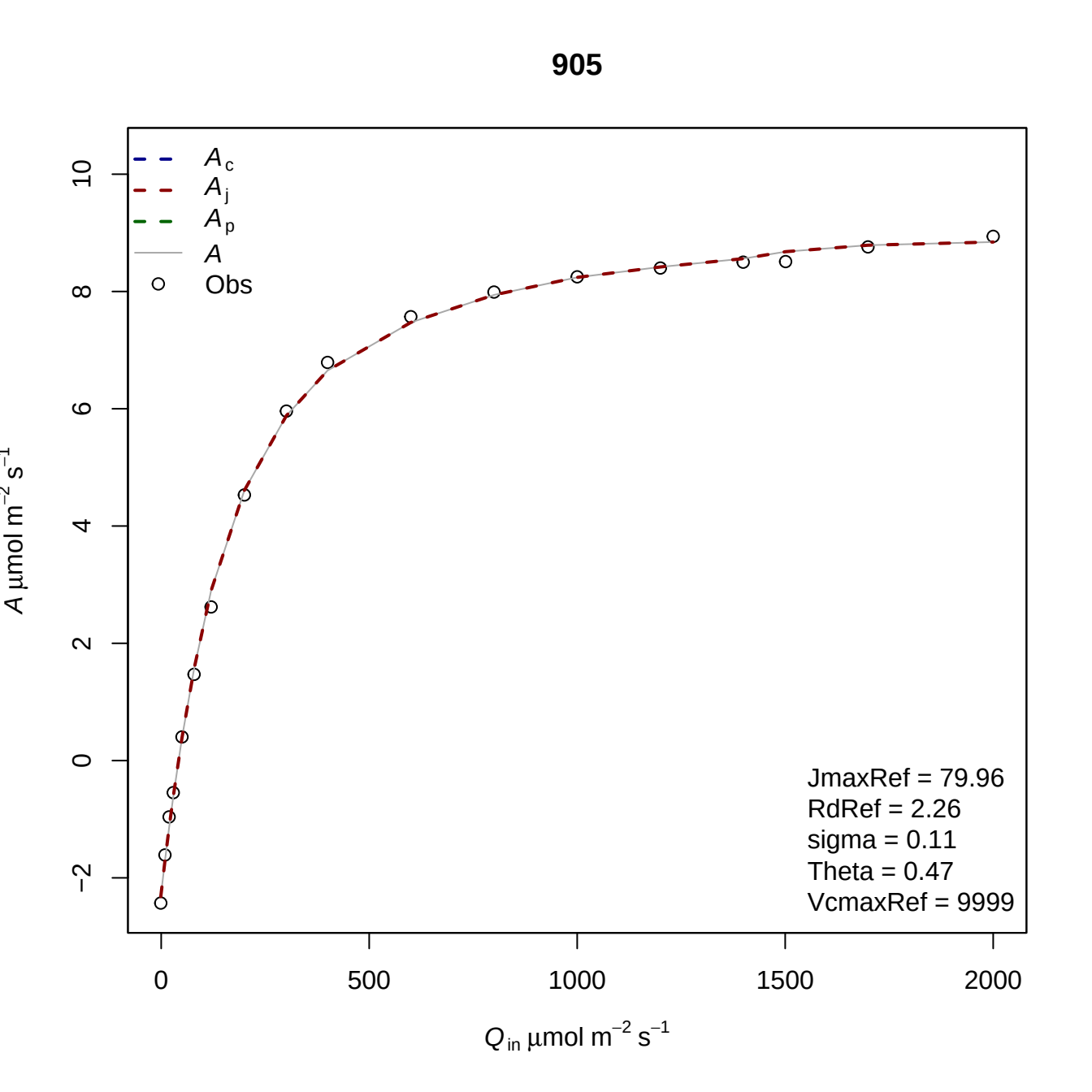

900
A_c
A_j
A_p
A
Obs
JmaxRef = 65.48
RdRef = 1.8
sigma = 0.12
Theta = 0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

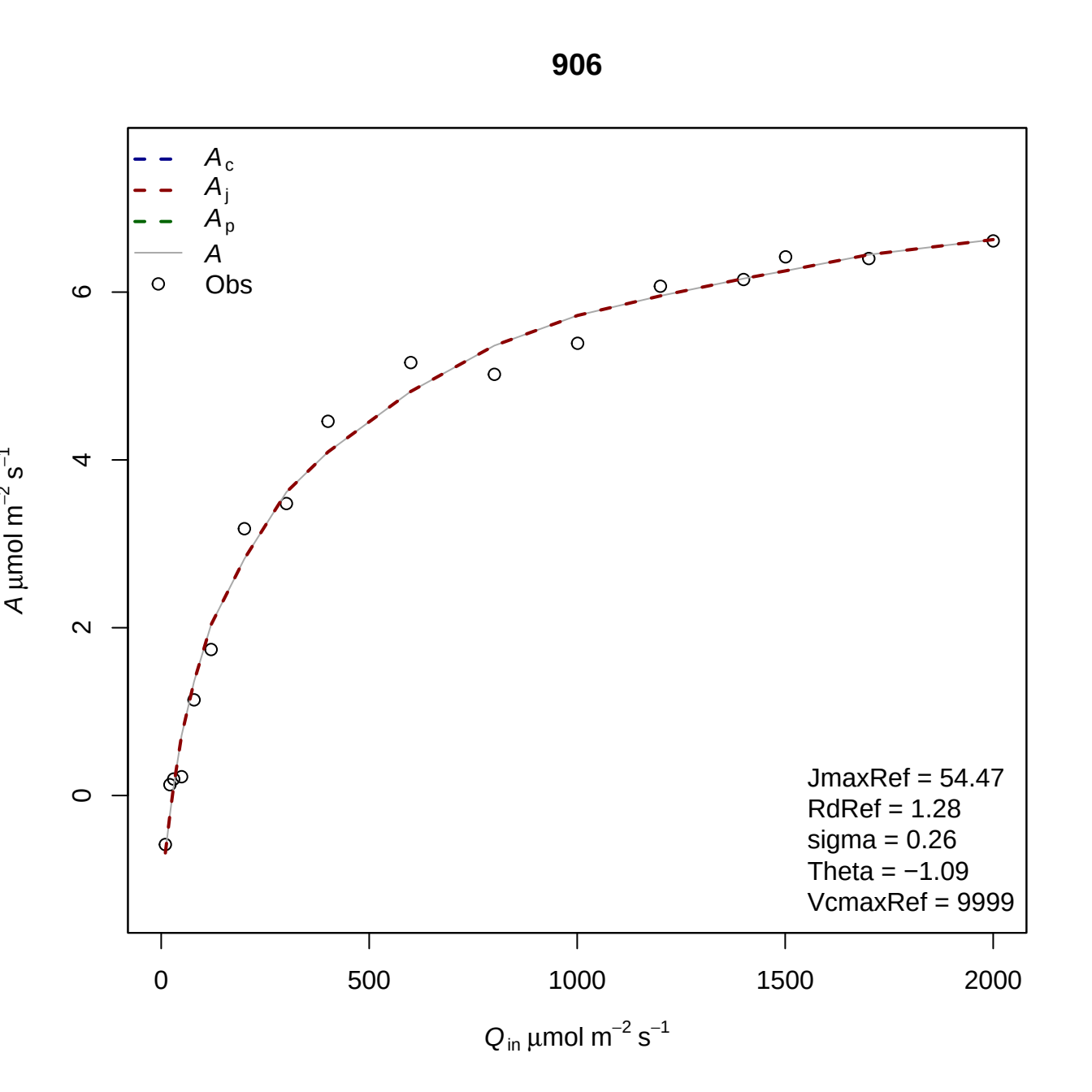

901
A_c
A_j
A_p
A
Obs
JmaxRef = 119.21
RdRef = 2.09
sigma = 0.15
Theta = −0.81
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

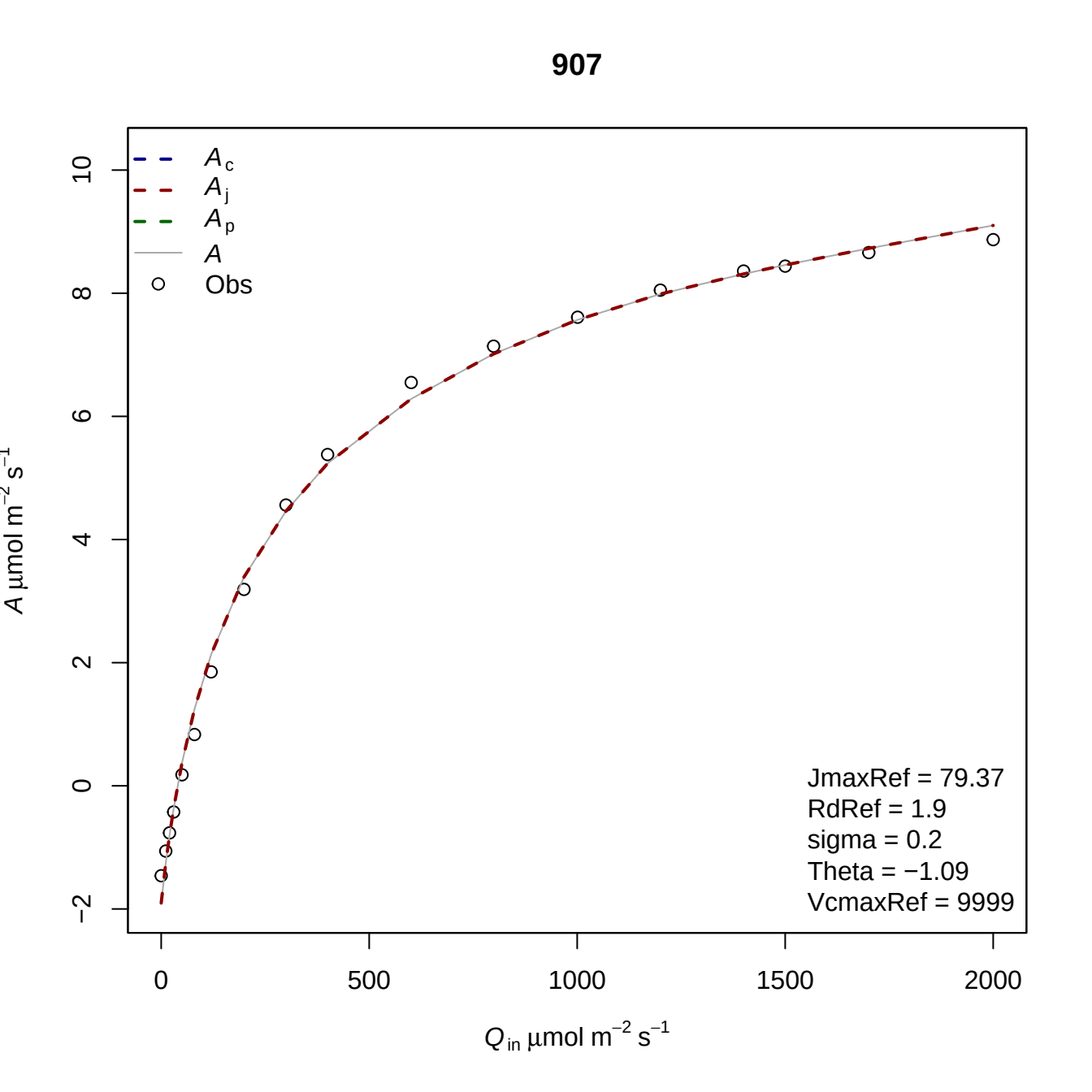

902
A_c
A_j
A_p
A
Obs
JmaxRef = 64.8
RdRef = 1.52
sigma = 0.26
Theta = −0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

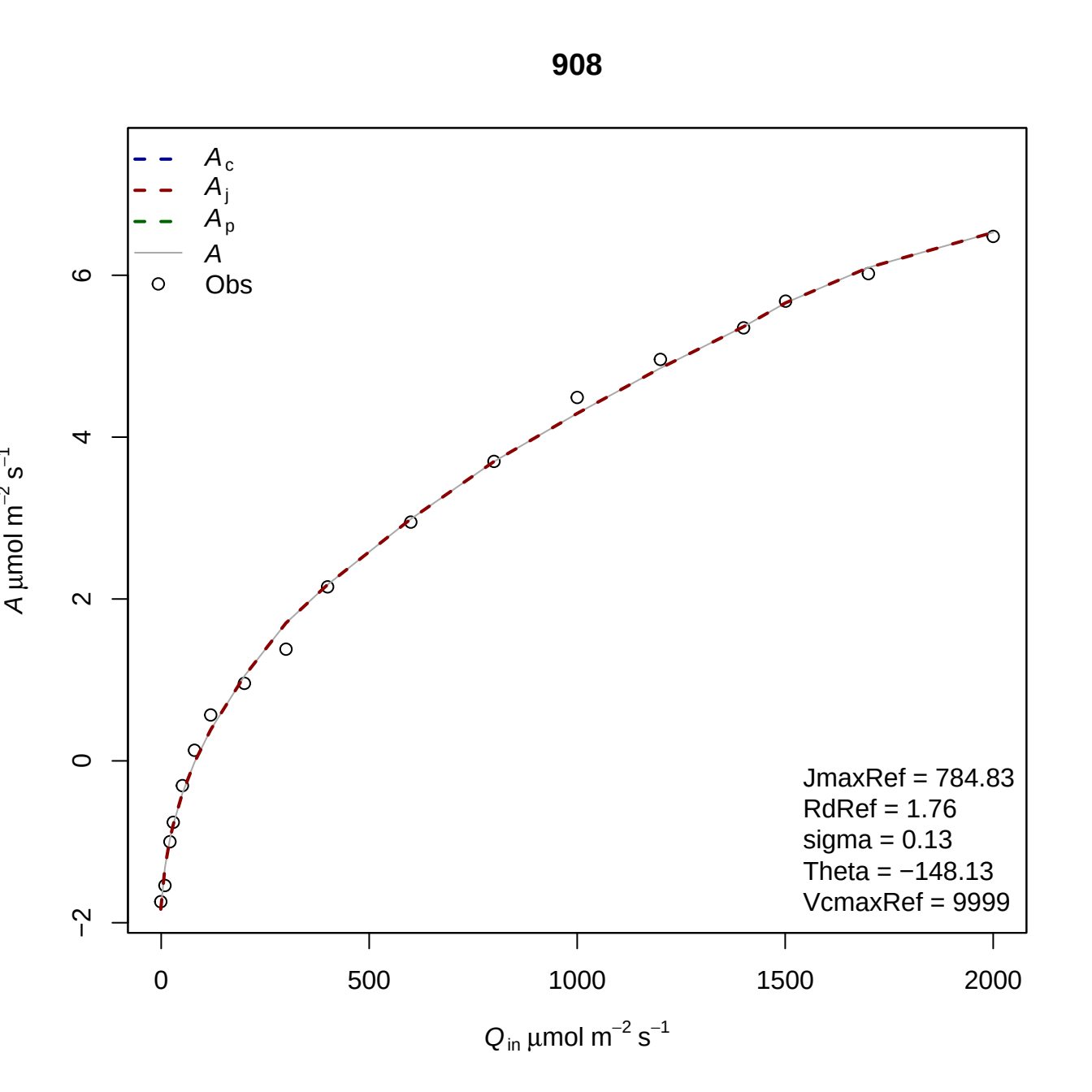

903
A_c
A_j
A_p
A
Obs
JmaxRef = 87.23
RdRef = 2.5
sigma = 0.14
Theta = −0.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

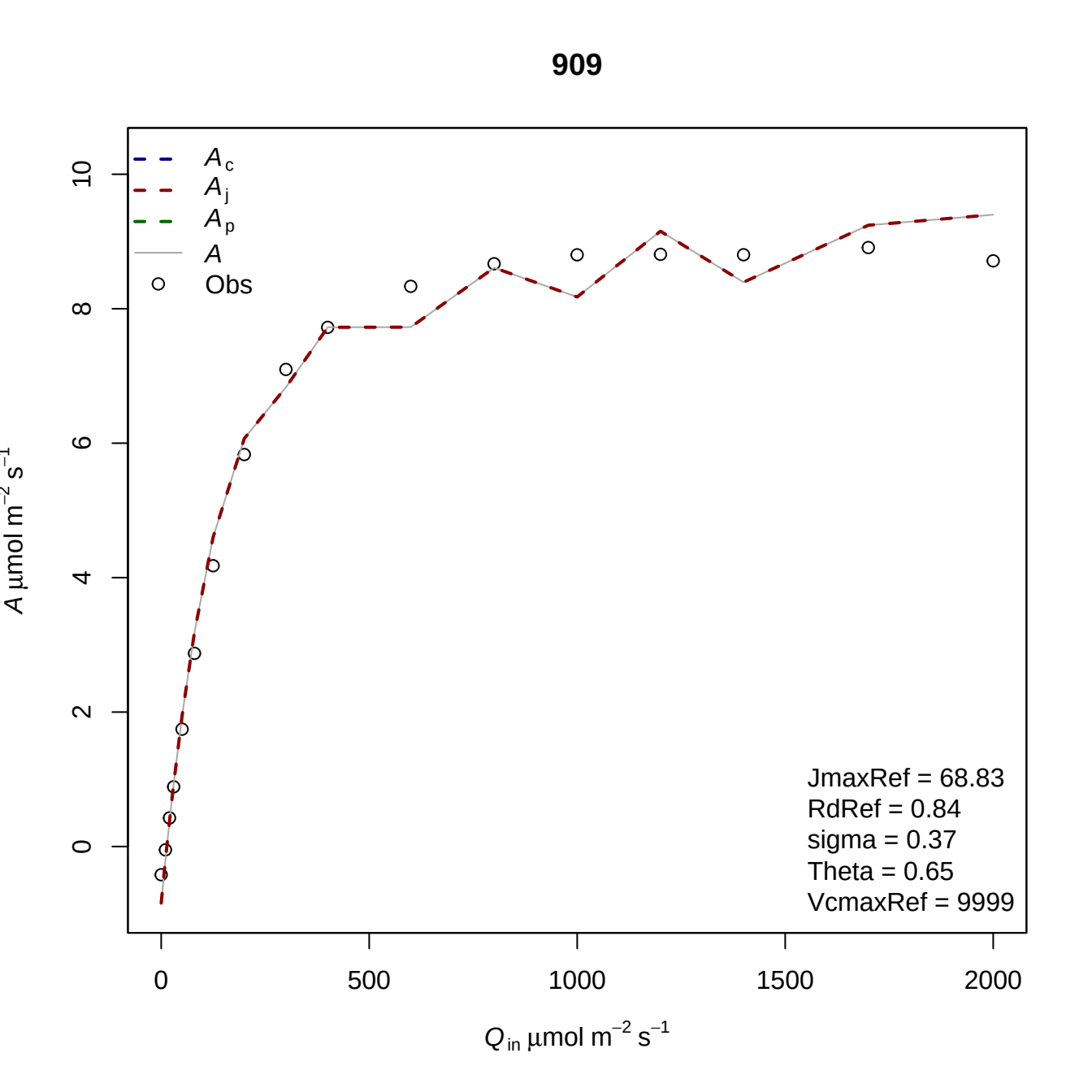

904
A_c
A_j
A_p
A
Obs
JmaxRef = 88.46
RdRef = 2.07
sigma = 0.28
Theta = 0.69
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

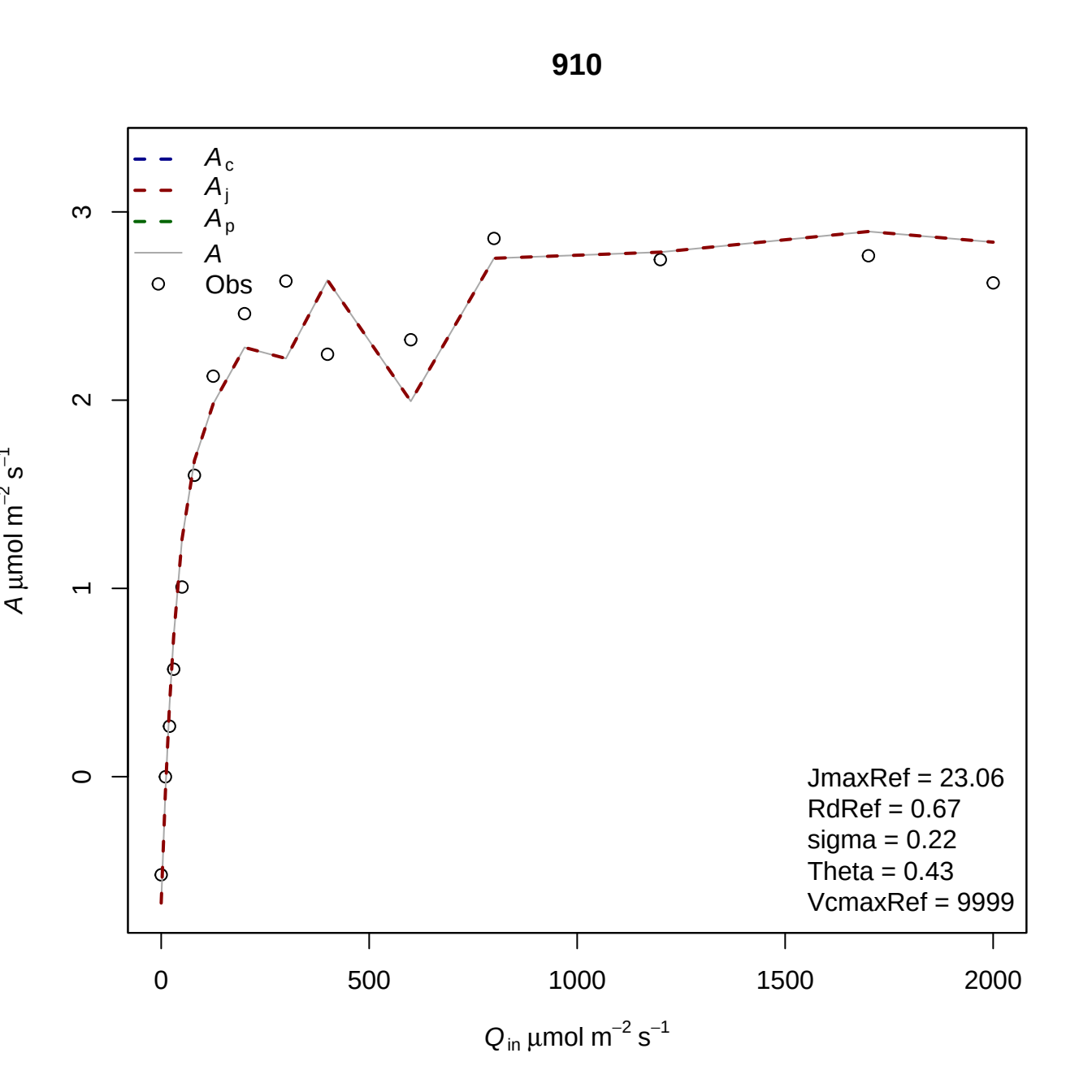

905
A_c
A_j
A_p
A
Obs
JmaxRef = 79.96
RdRef = 2.26
sigma = 0.11
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

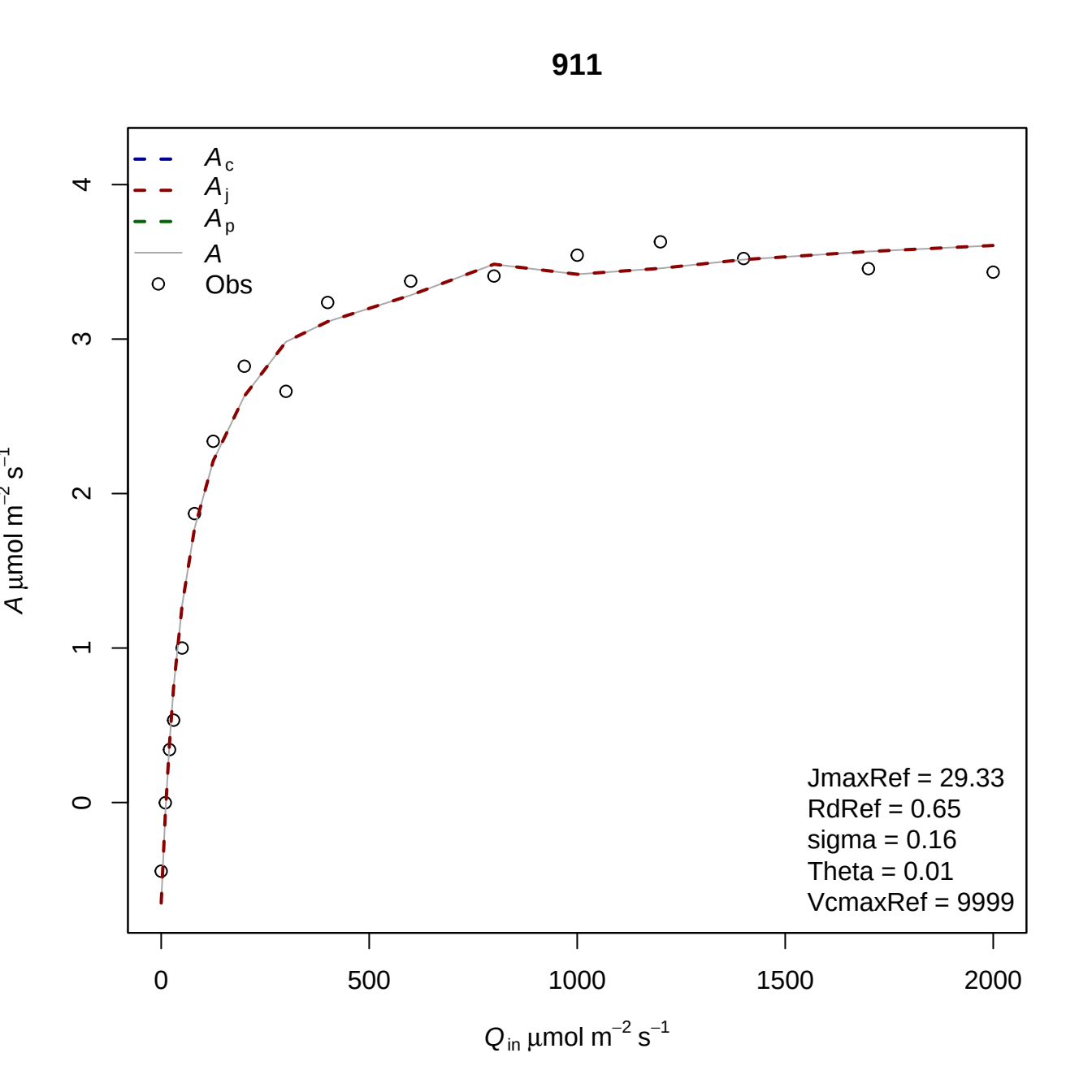

906
A_c
A_j
A_p
A
Obs
JmaxRef = 54.47
RdRef = 1.28
sigma = 0.26
Theta = −1.09
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

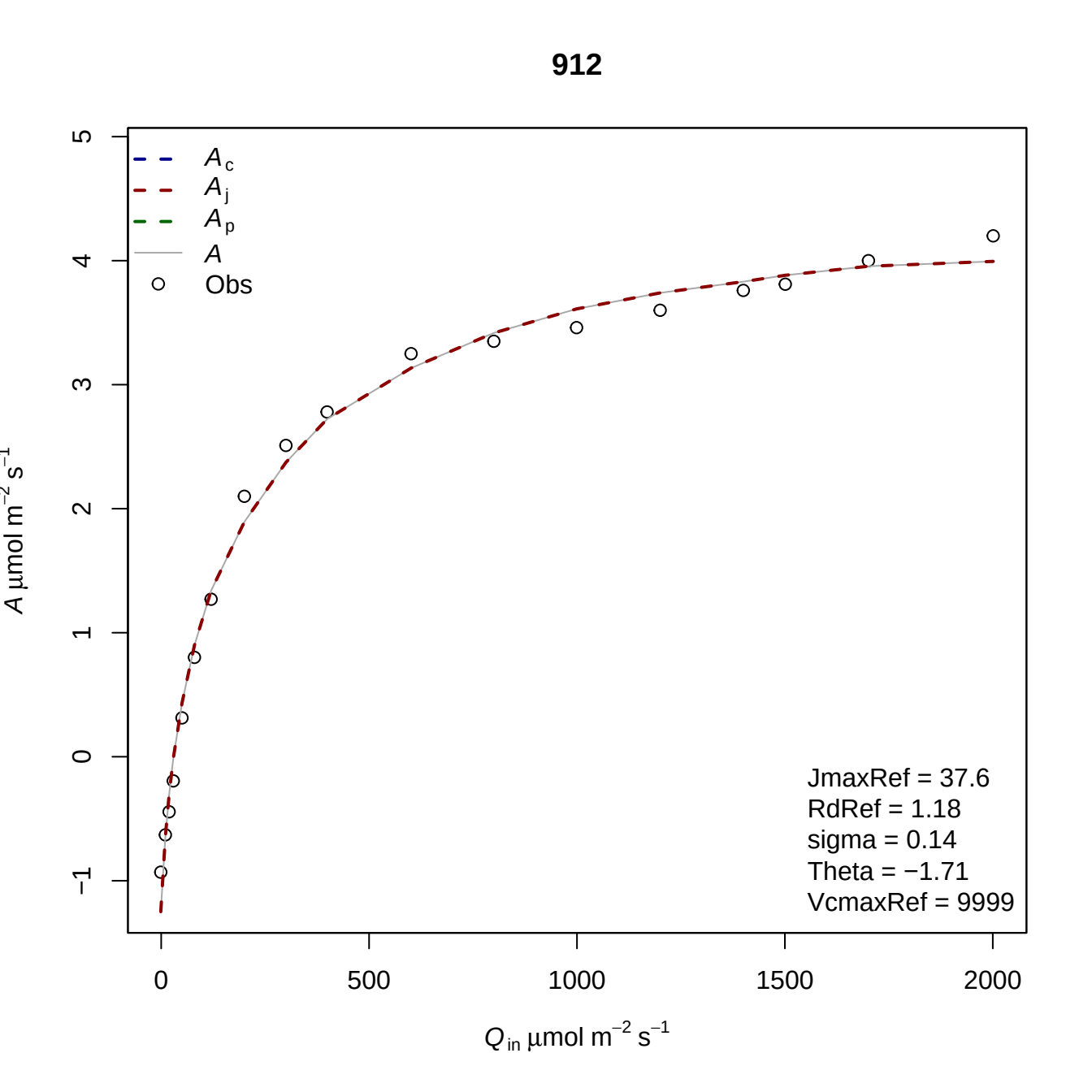

907
A_c
A_j
A_p
A
Obs
JmaxRef = 79.37
RdRef = 1.9
sigma = 0.2
Theta = −1.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

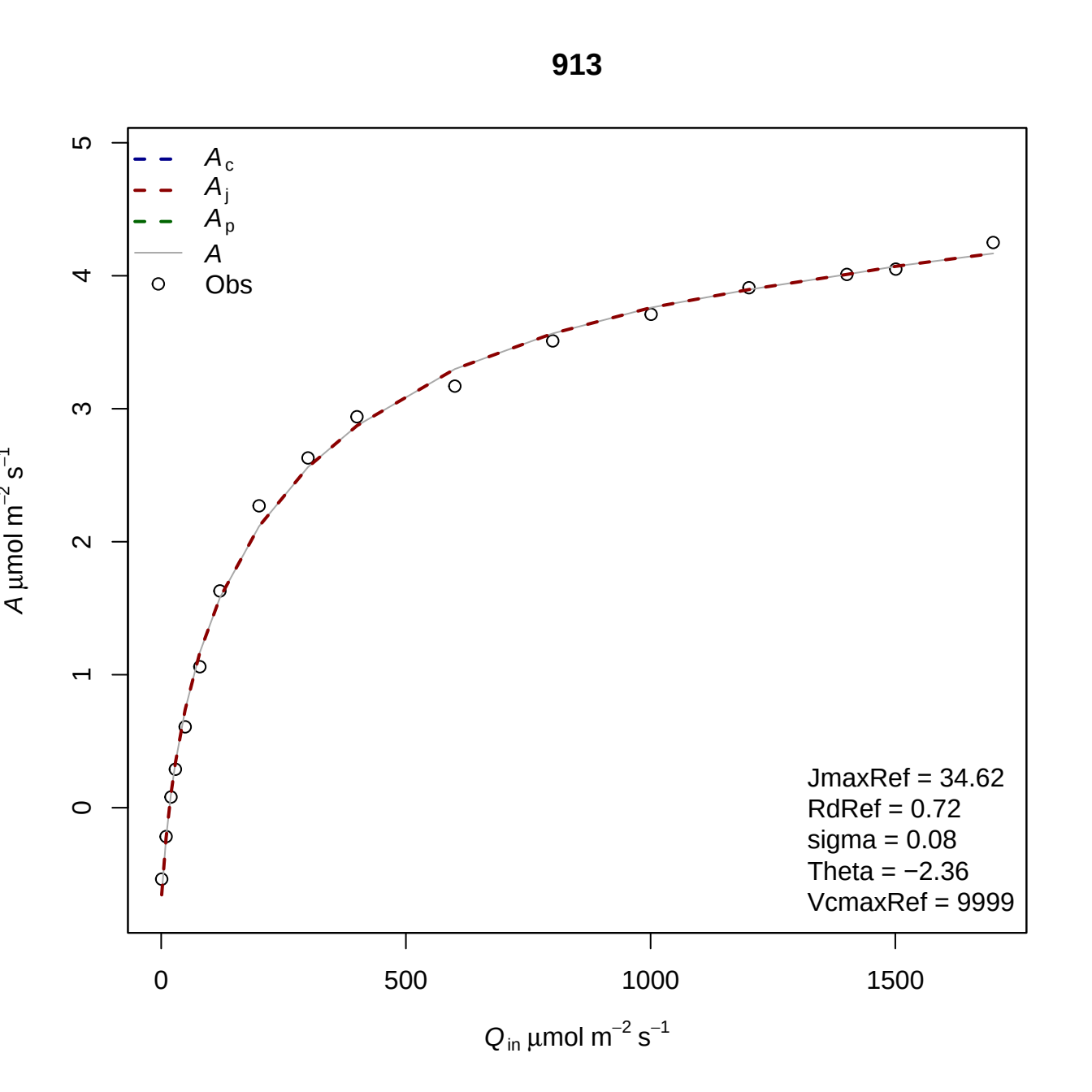

908
A_c
A_j
A_p
A
Obs
JmaxRef = 784.83
RdRef = 1.76
sigma = 0.13
Theta = −148.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

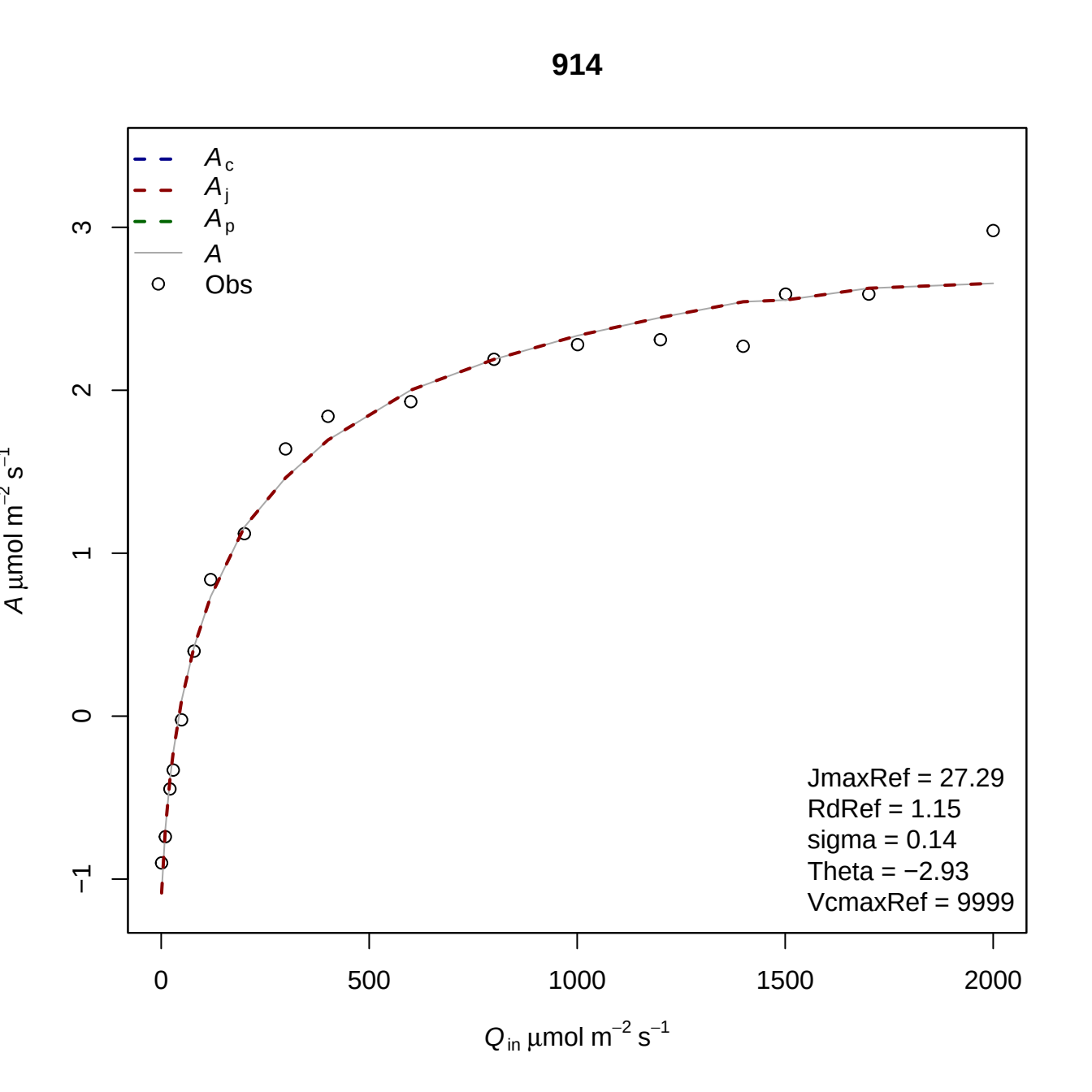

909
A_c
A_j
A_p
A
Obs
JmaxRef = 68.83
RdRef = 0.84
sigma = 0.37
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

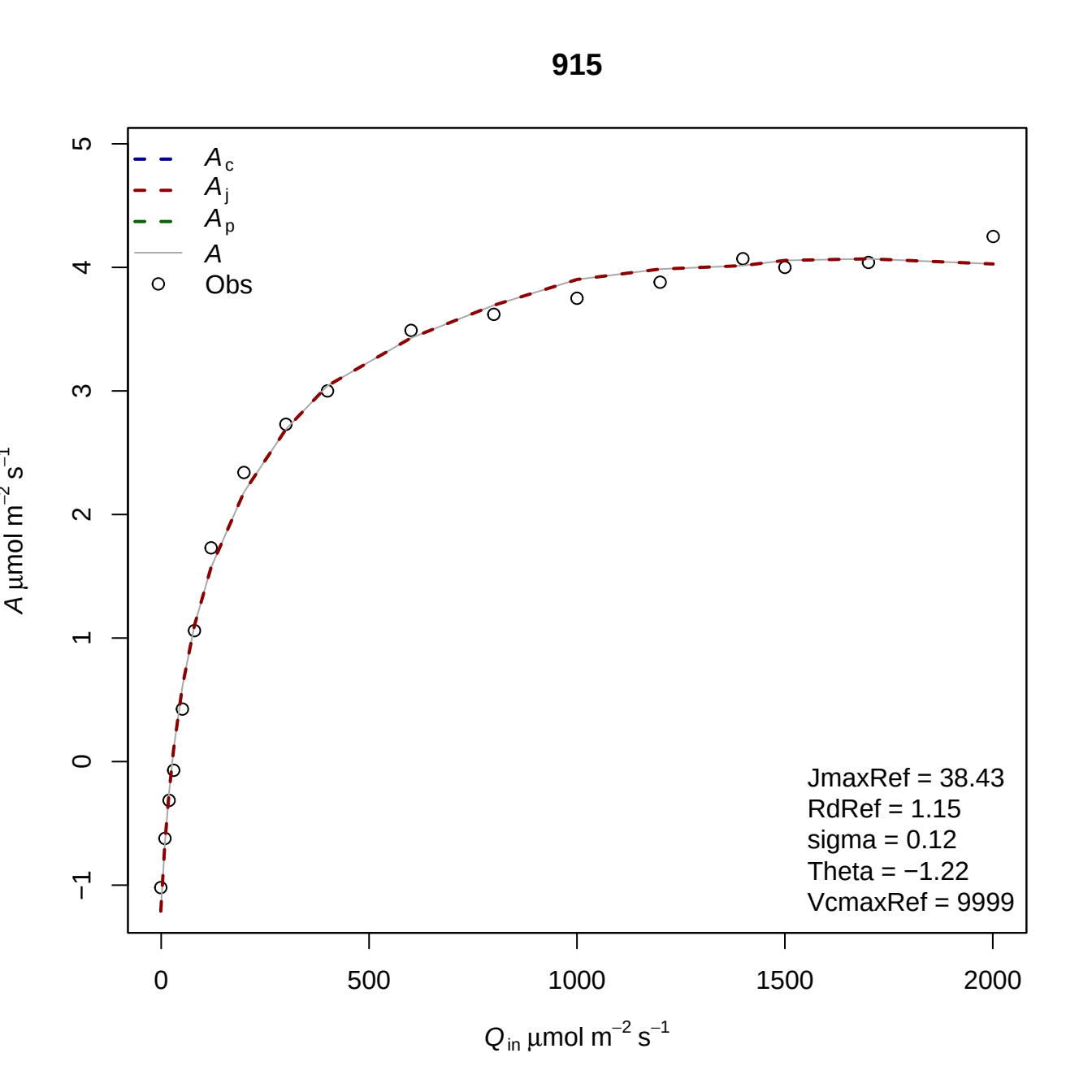

910
A_c
A_j
A_p
A
Obs
JmaxRef = 23.06
RdRef = 0.67
sigma = 0.22
Theta = 0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

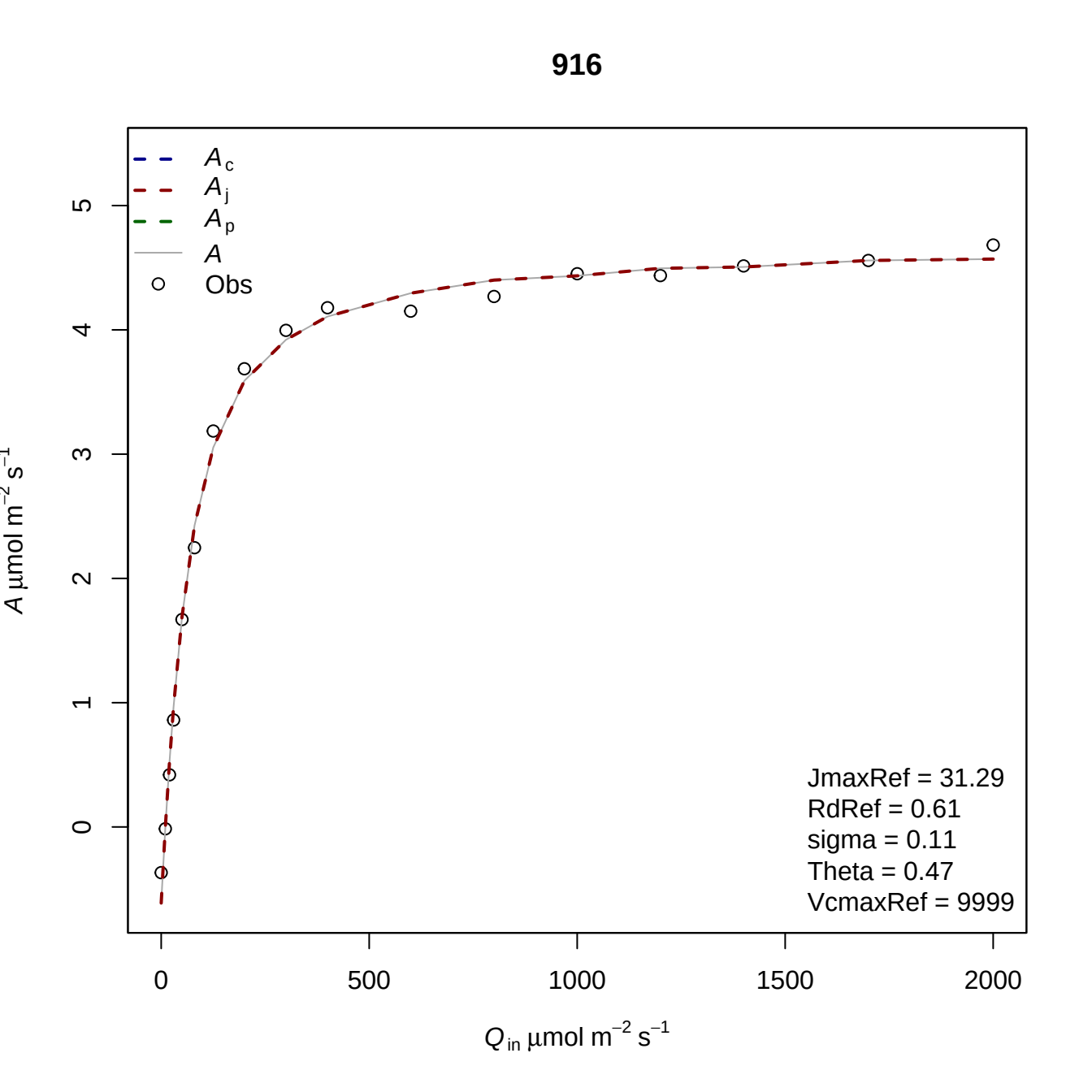

911
A_c
A_j
A_p
A
Obs
JmaxRef = 29.33
RdRef = 0.65
sigma = 0.16
Theta = 0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

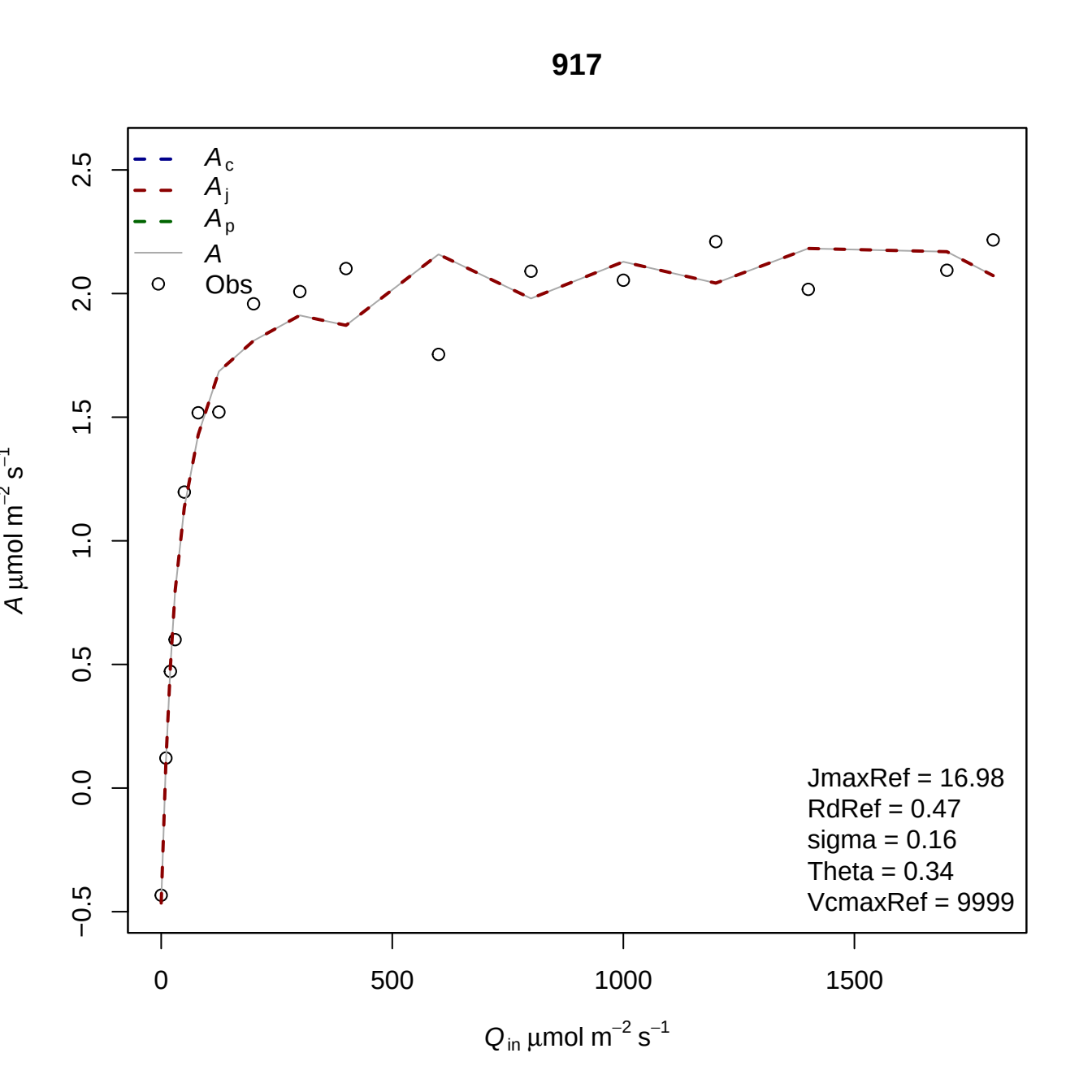

912
A_c
A_j
A_p
A
Obs
JmaxRef = 37.6
RdRef = 1.18
sigma = 0.14
Theta = −1.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

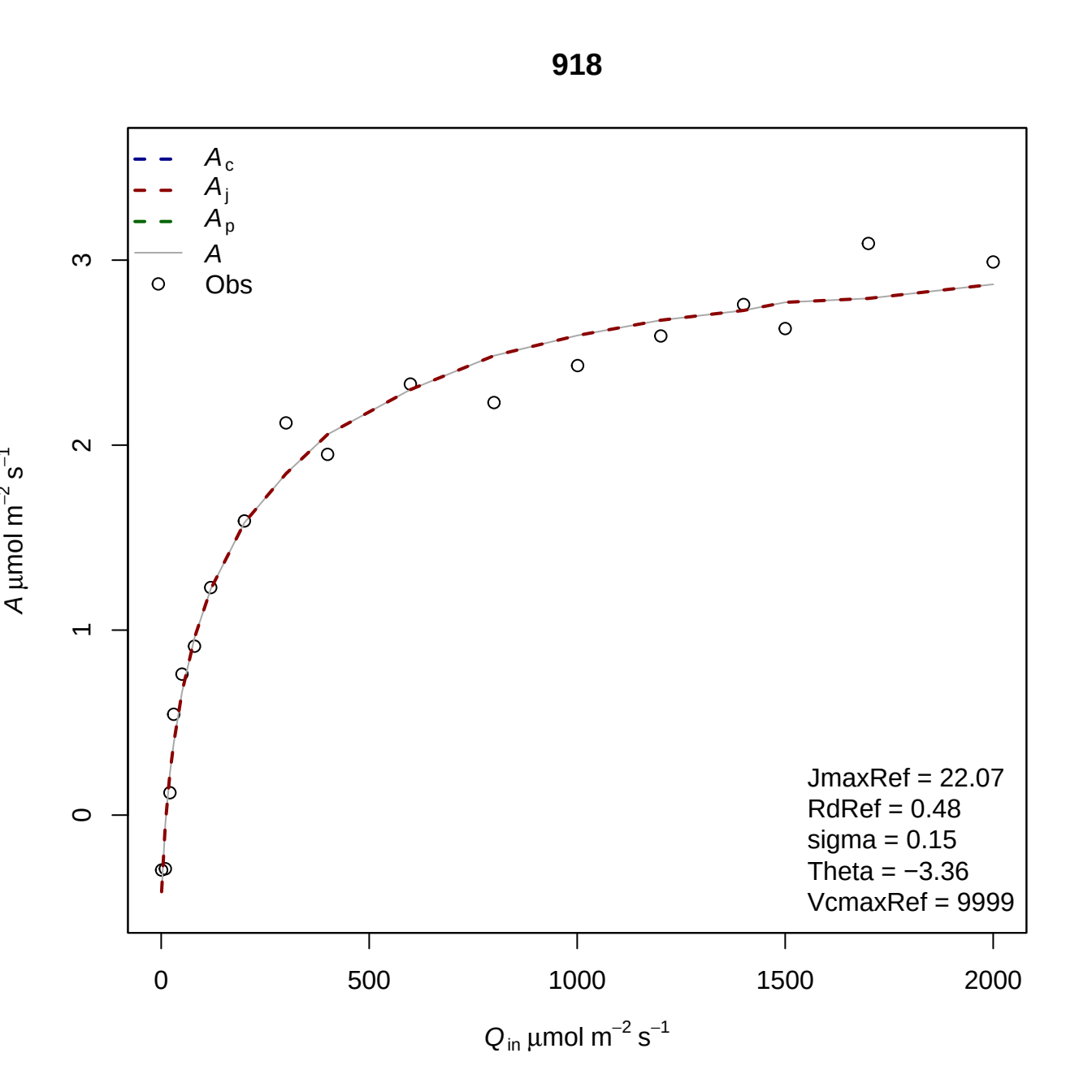

913
A_c
A_j
A_p
A
Obs
JmaxRef = 34.62
RdRef = 0.72
sigma = 0.08
Theta = −2.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

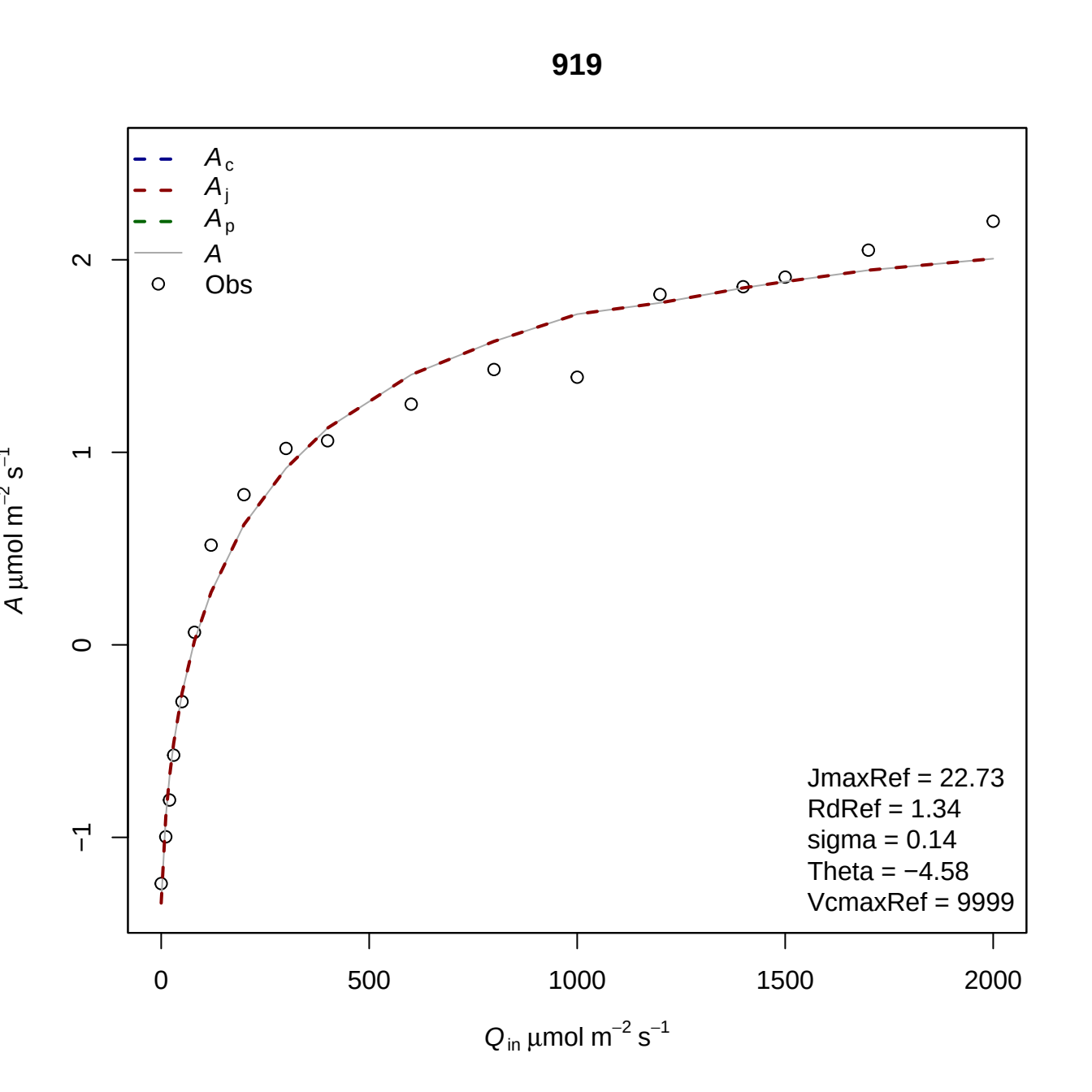

914
A_c
A_j
A_p
A
Obs
JmaxRef = 27.29
RdRef = 1.15
sigma = 0.14
Theta = −2.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

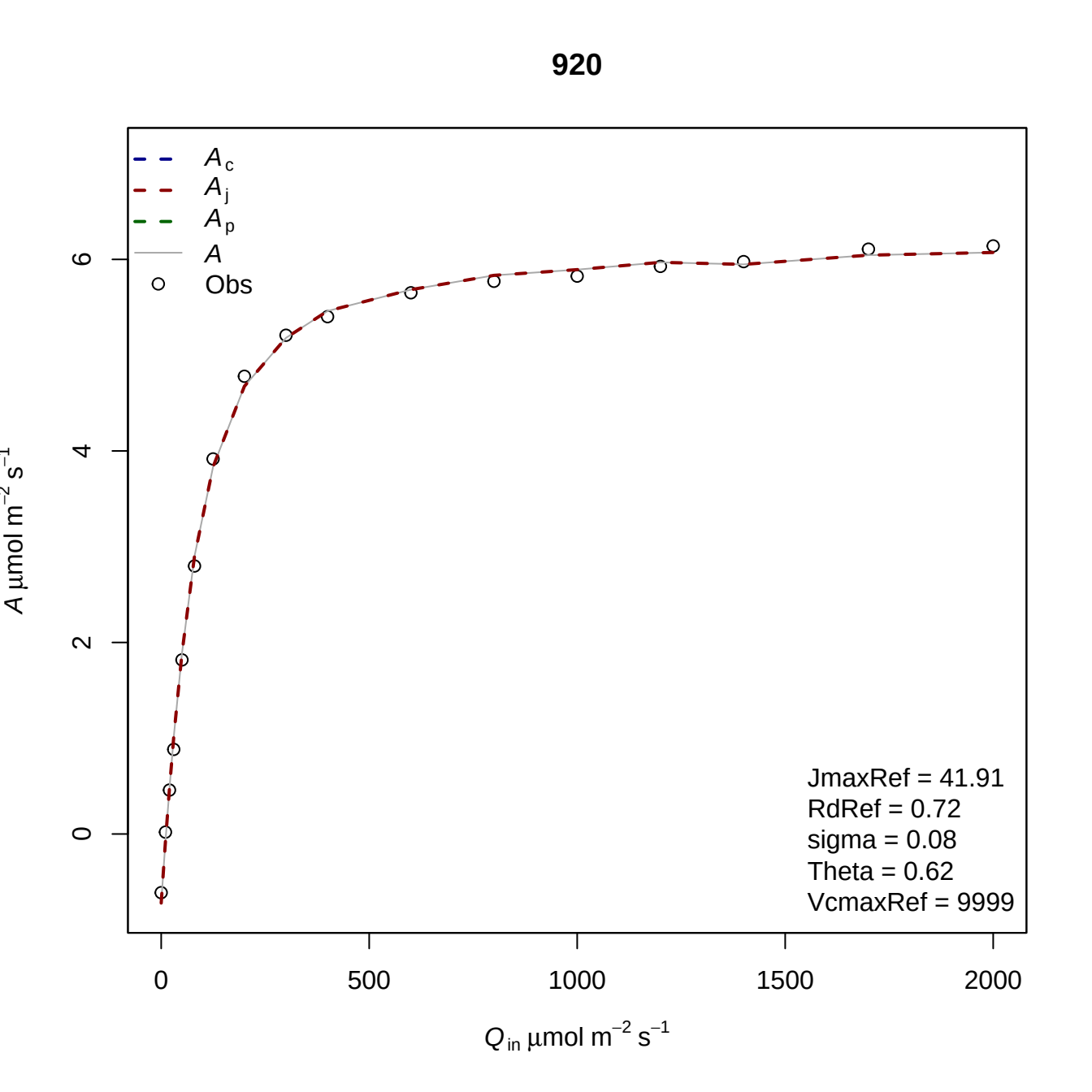

915
A_c
A_j
A_p
A
Obs
JmaxRef = 38.43
RdRef = 1.15
sigma = 0.12
Theta = −1.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

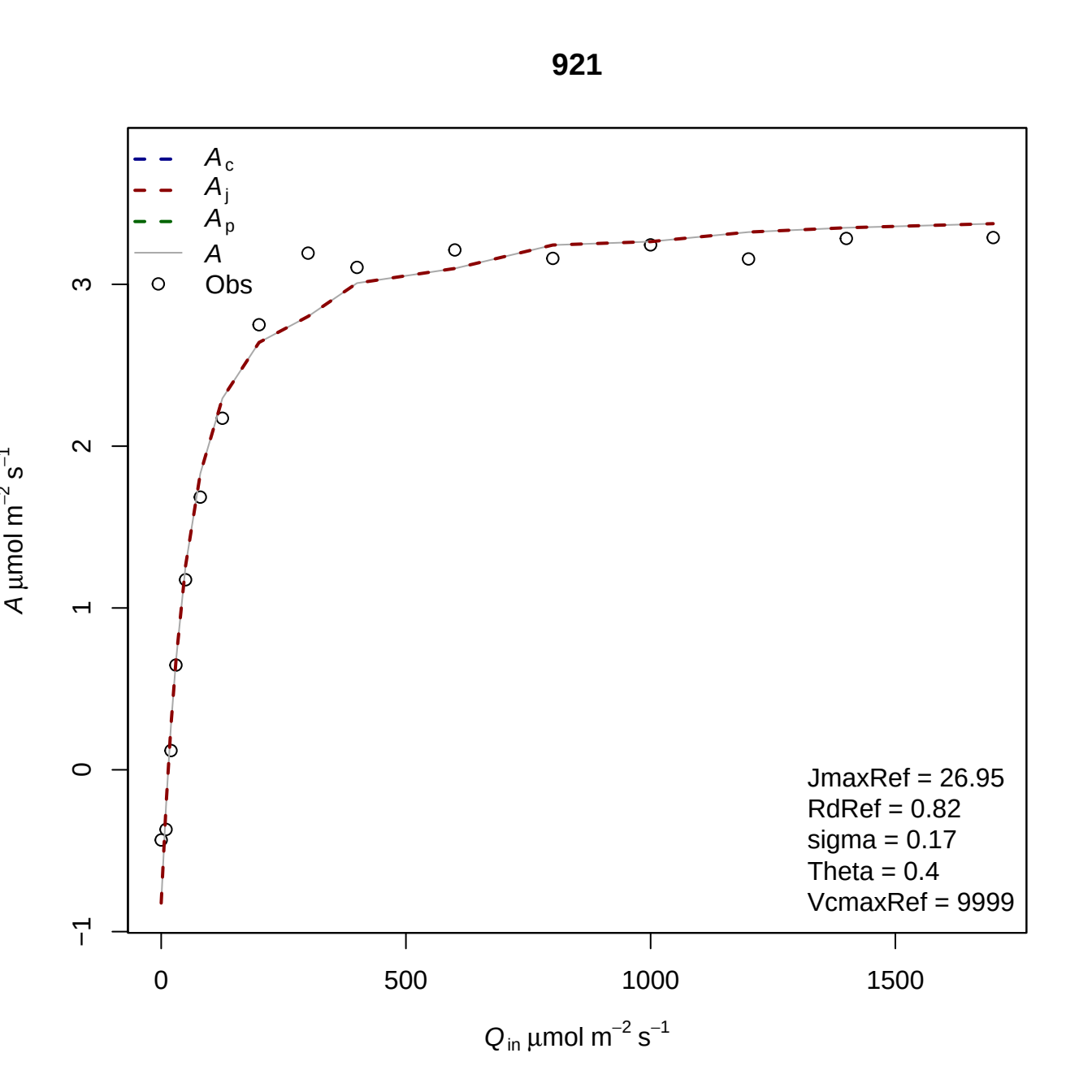

916
A_c
A_j
A_p
A
Obs
JmaxRef = 31.29
RdRef = 0.61
sigma = 0.11
Theta = 0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

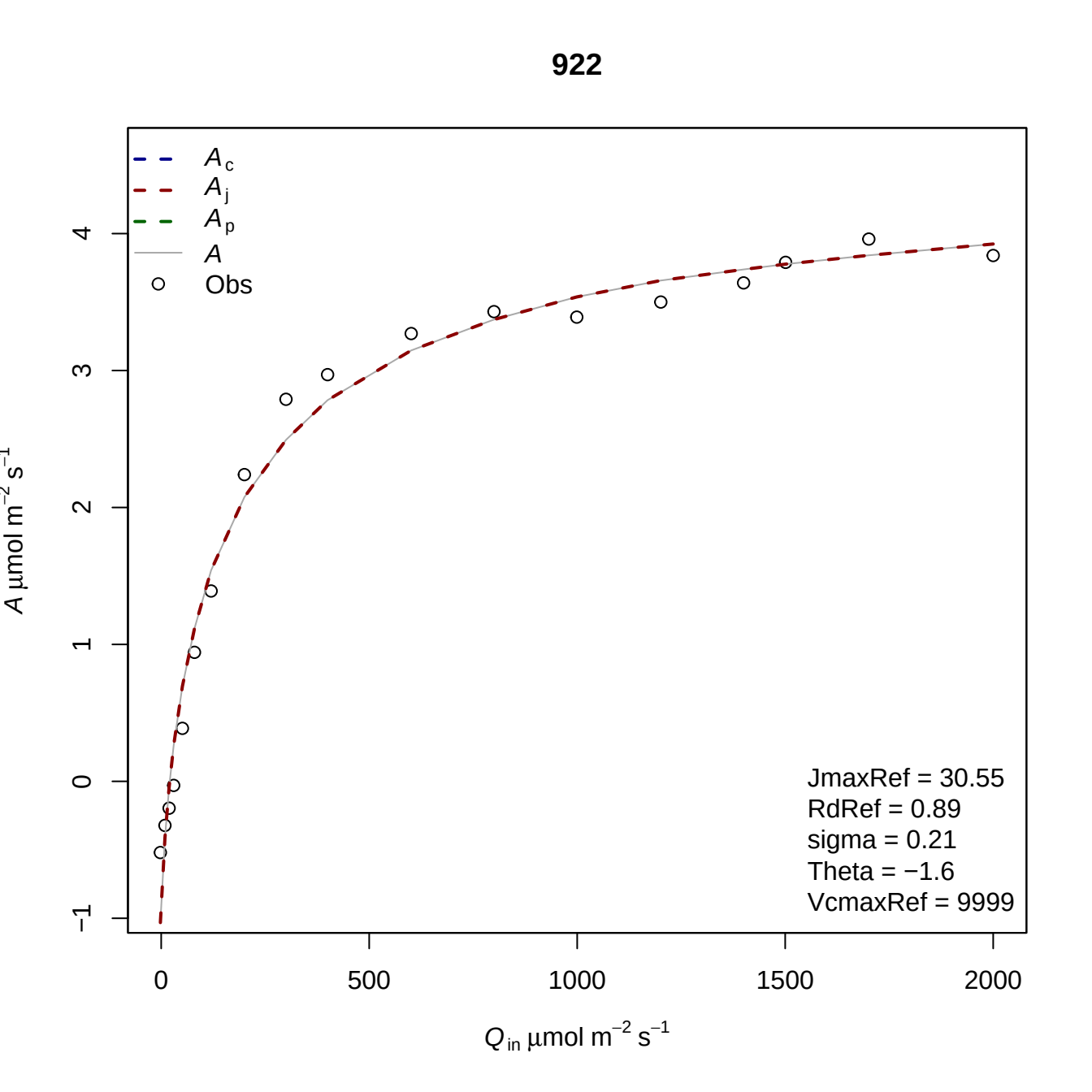

917
A_c
A_j
A_p
A
Obs
JmaxRef = 16.98
RdRef = 0.47
sigma = 0.16
Theta = 0.34
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

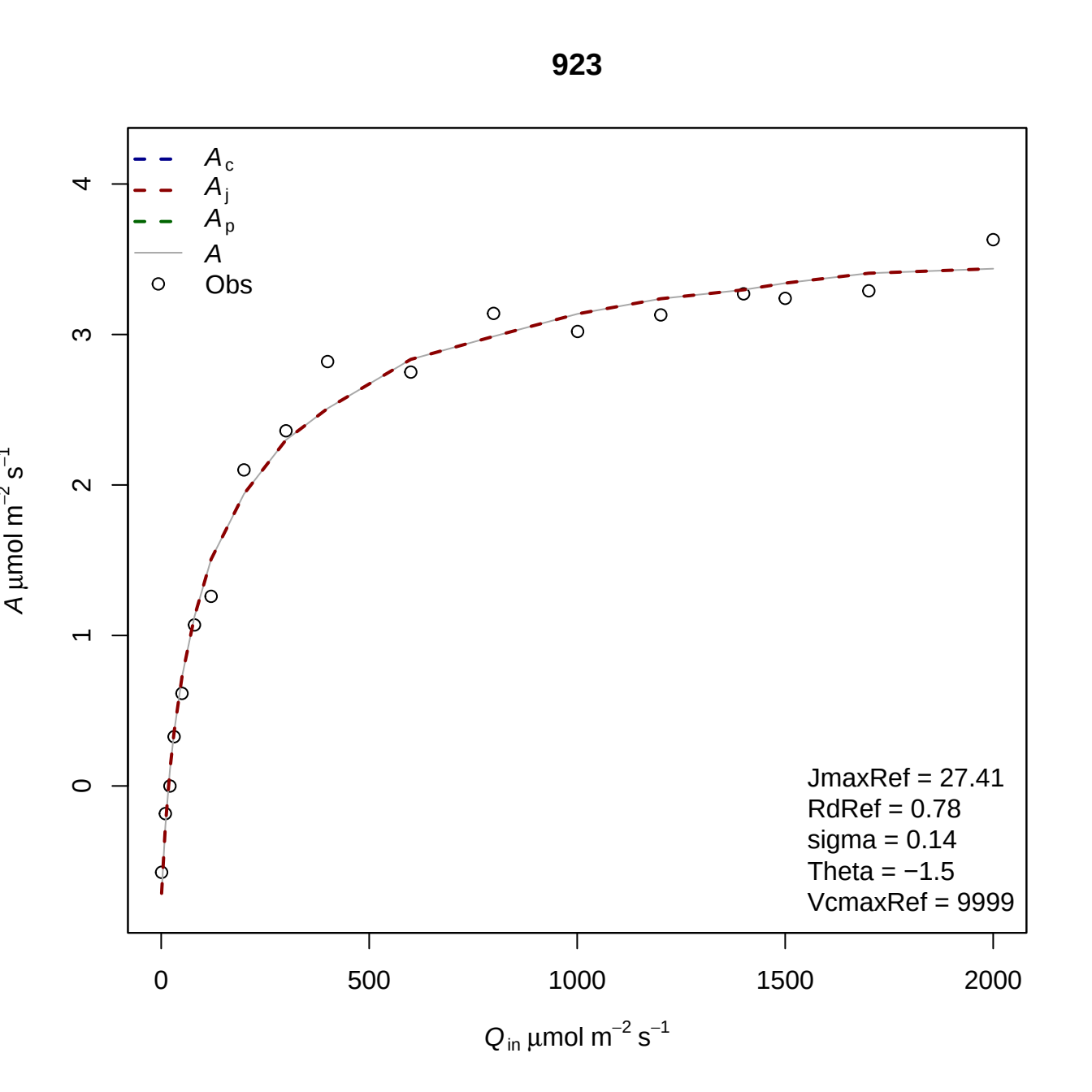

918
A_c
A_j
A_p
A
Obs
JmaxRef = 22.07
RdRef = 0.48
sigma = 0.15
Theta = −3.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

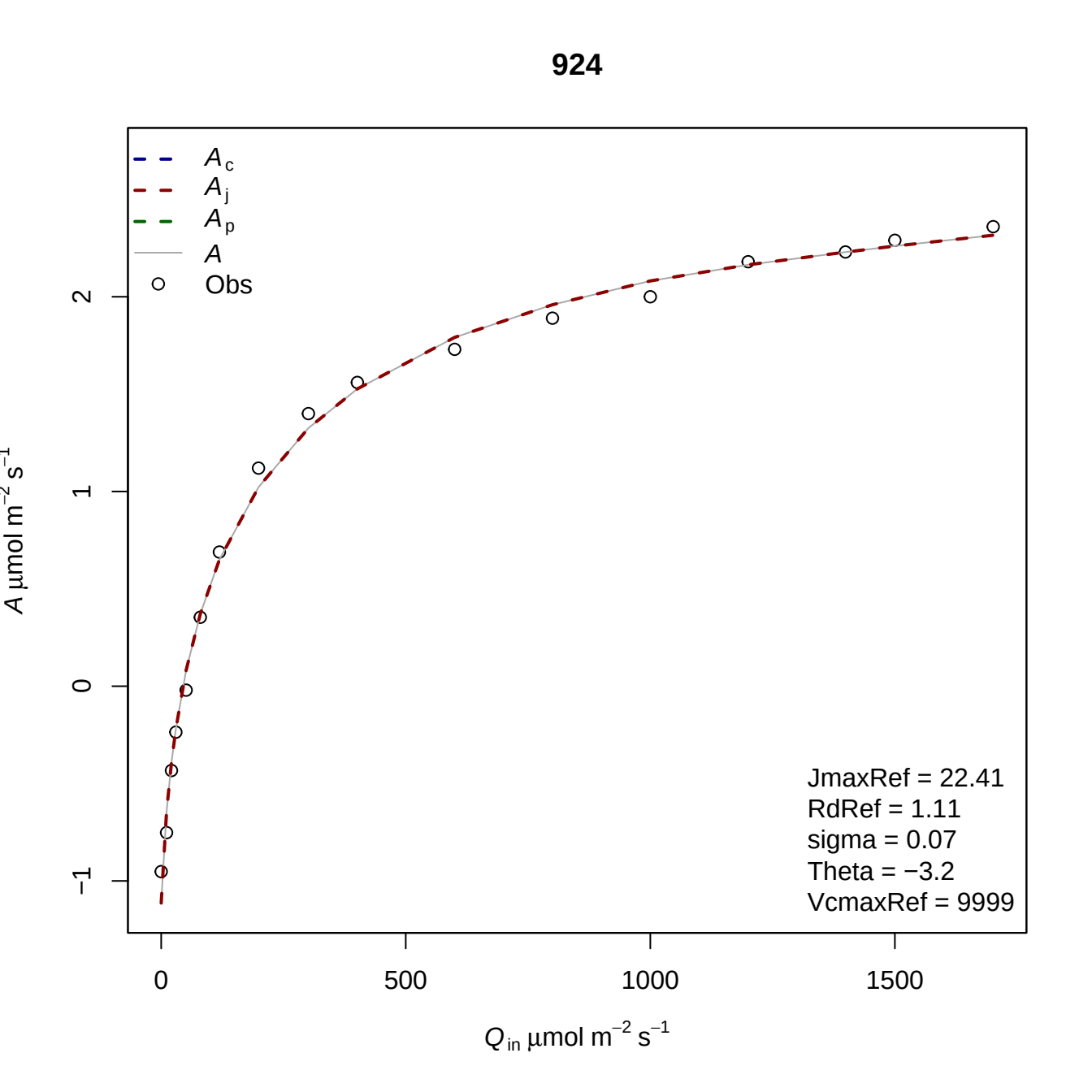

919
A_c
A_j
A_p
A
Obs
JmaxRef = 22.73
RdRef = 1.34
sigma = 0.14
Theta = −4.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

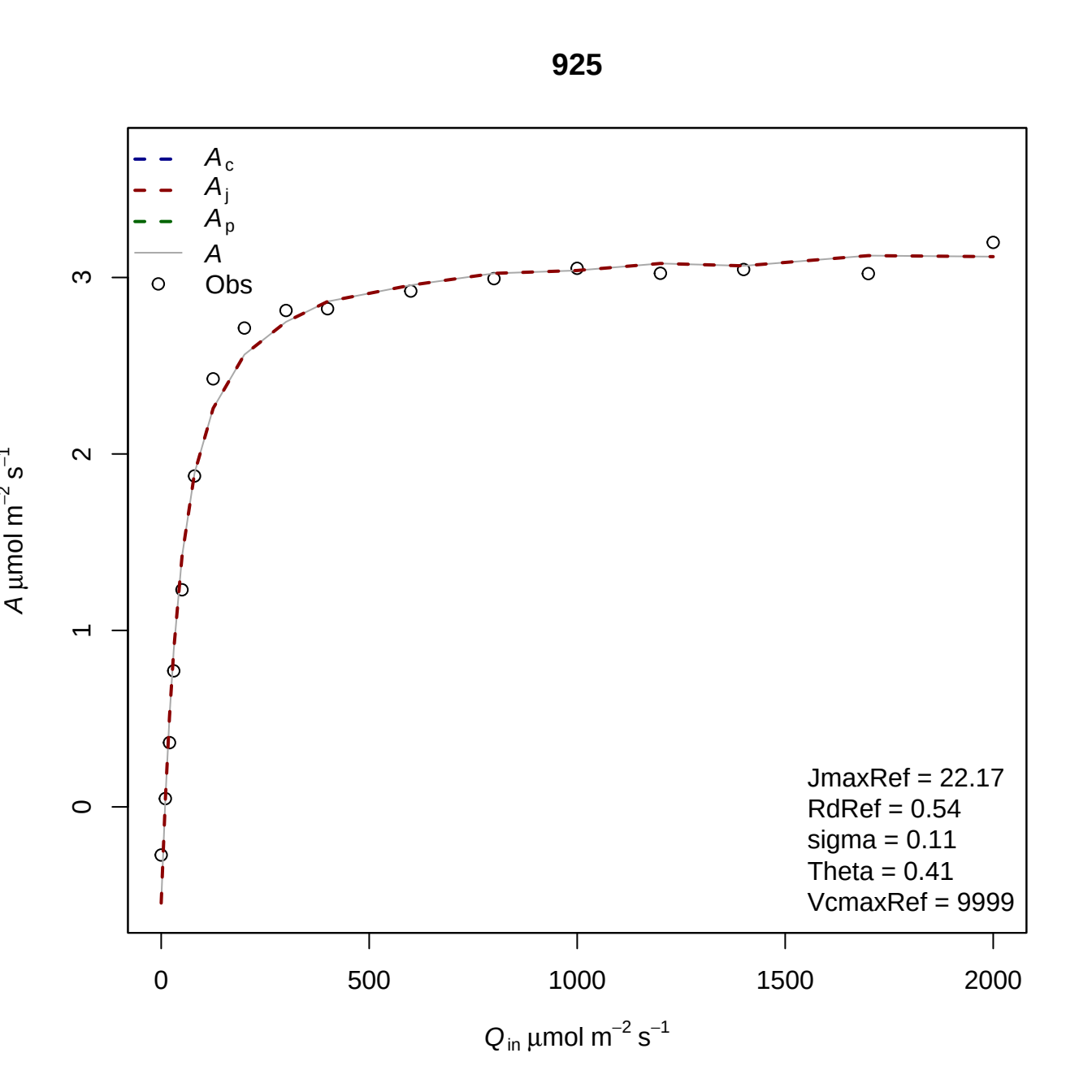

920
A_c
A_j
A_p
A
Obs
JmaxRef = 41.91
RdRef = 0.72
sigma = 0.08
Theta = 0.62
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

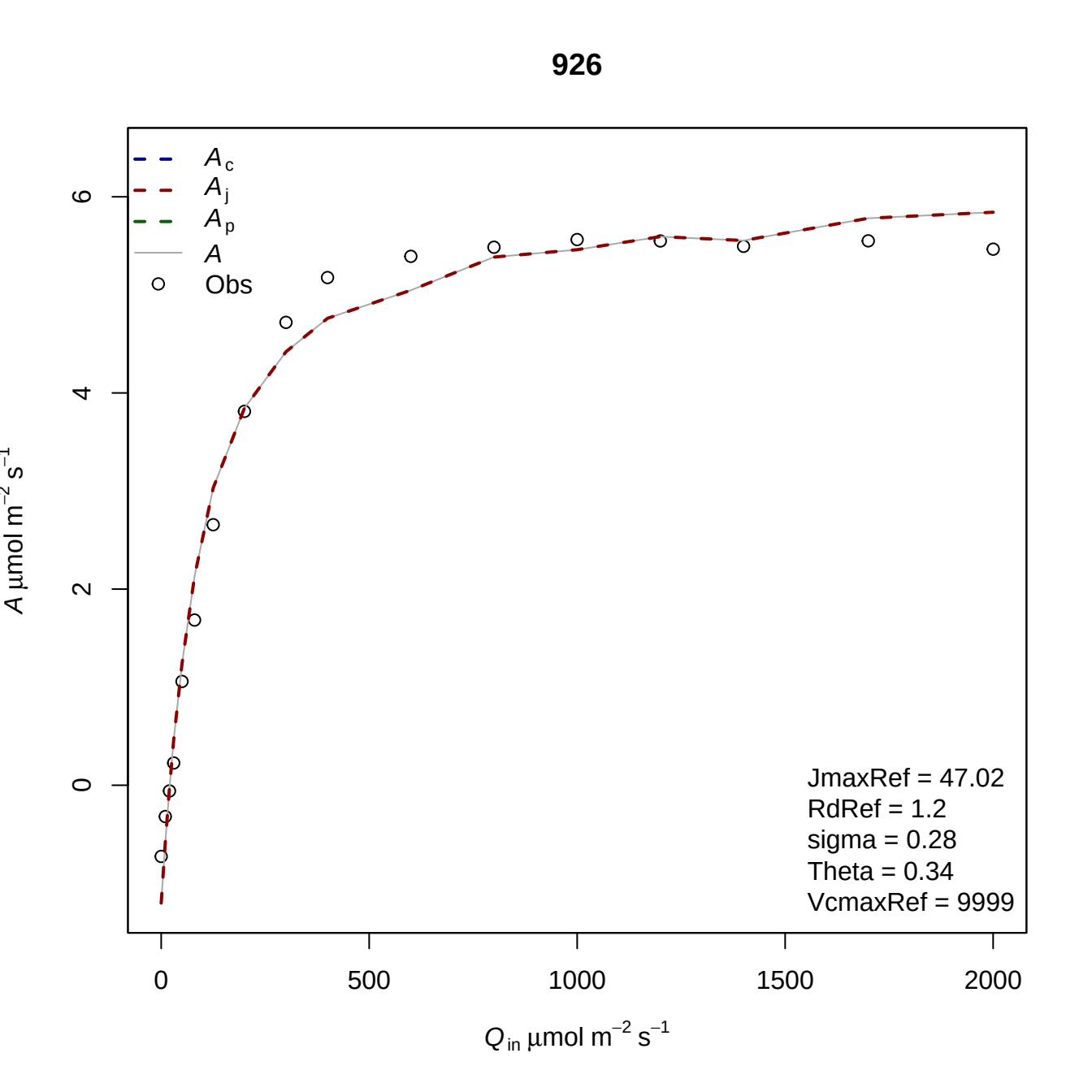

921
A_c
A_j
A_p
A
Obs
JmaxRef = 26.95
RdRef = 0.82
sigma = 0.17
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

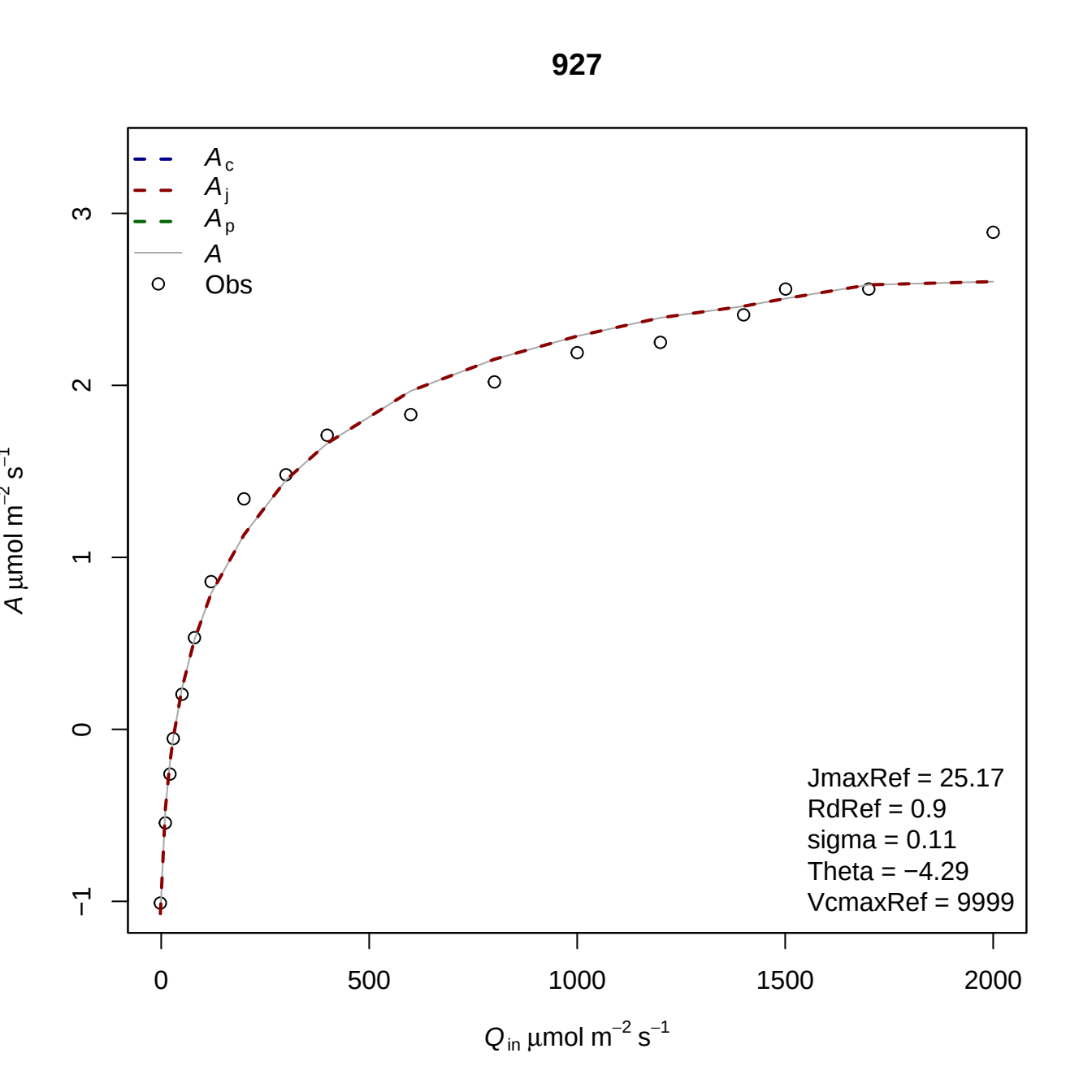

922
A_c
A_j
A_p
A
Obs
JmaxRef = 30.55
RdRef = 0.89
sigma = 0.21
Theta = −1.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

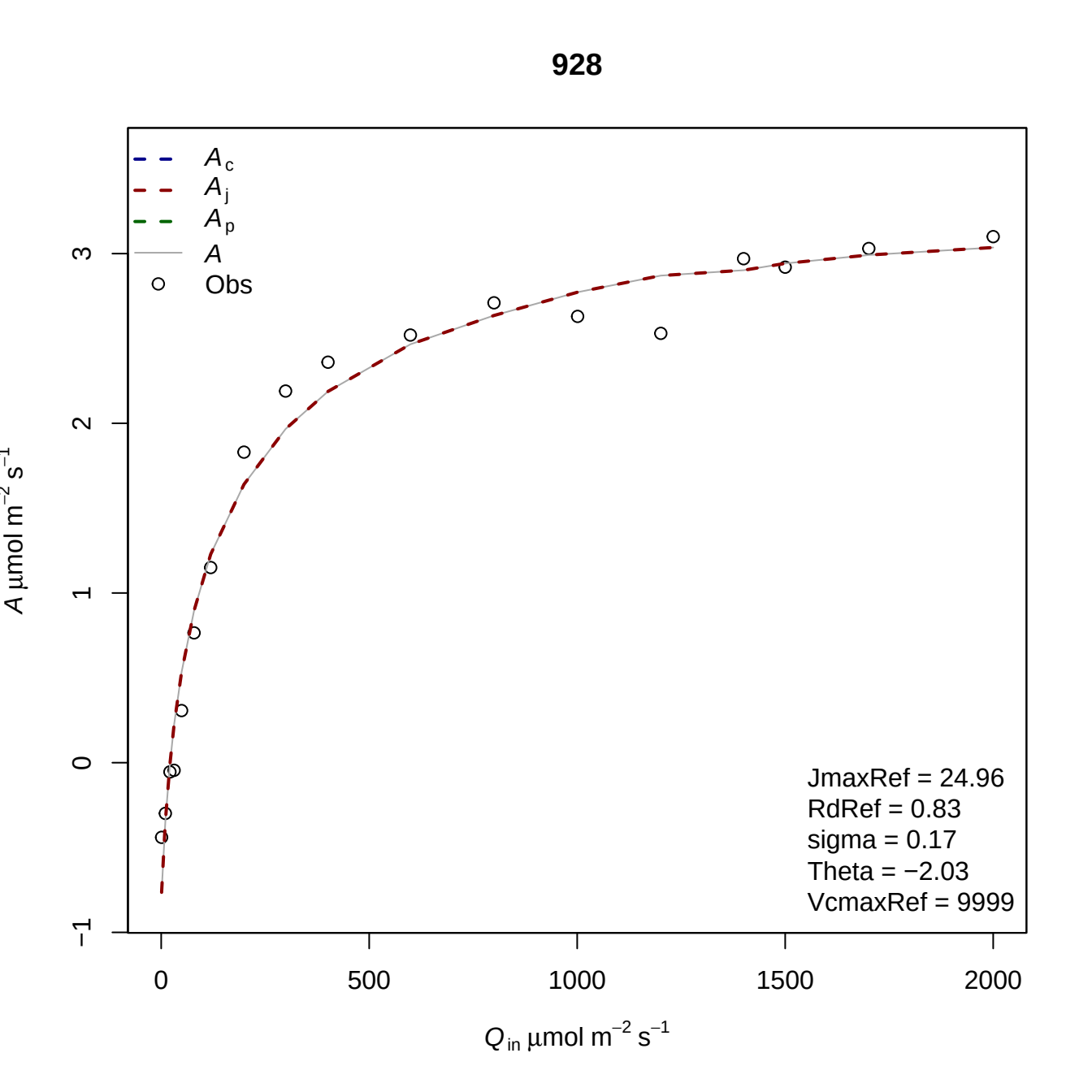

923
A_c
A_j
A_p
A
Obs
JmaxRef = 27.41
RdRef = 0.78
sigma = 0.14
Theta = −1.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

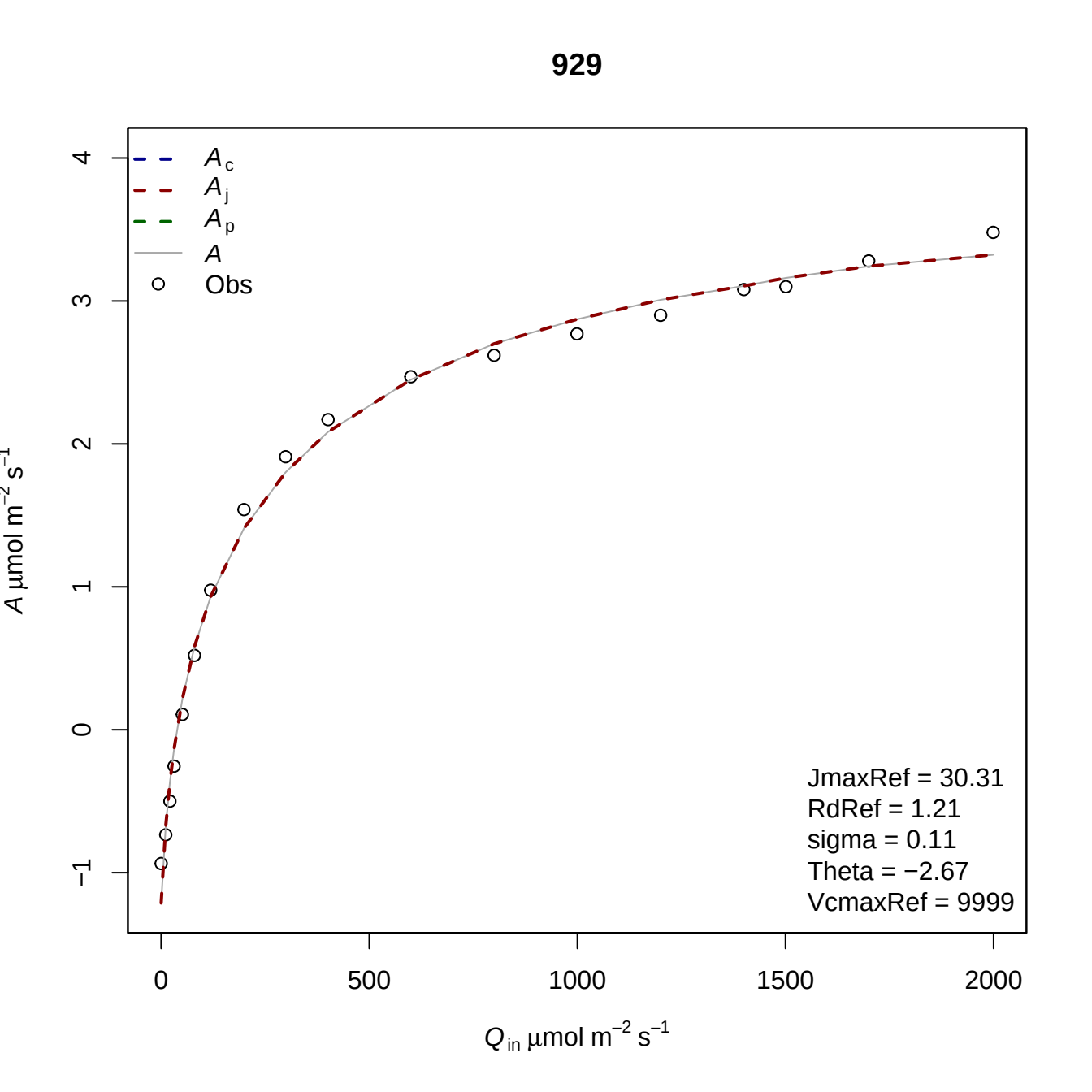

924
A_c
A_j
A_p
A
Obs
JmaxRef = 22.41
RdRef = 1.11
sigma = 0.07
Theta = −3.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

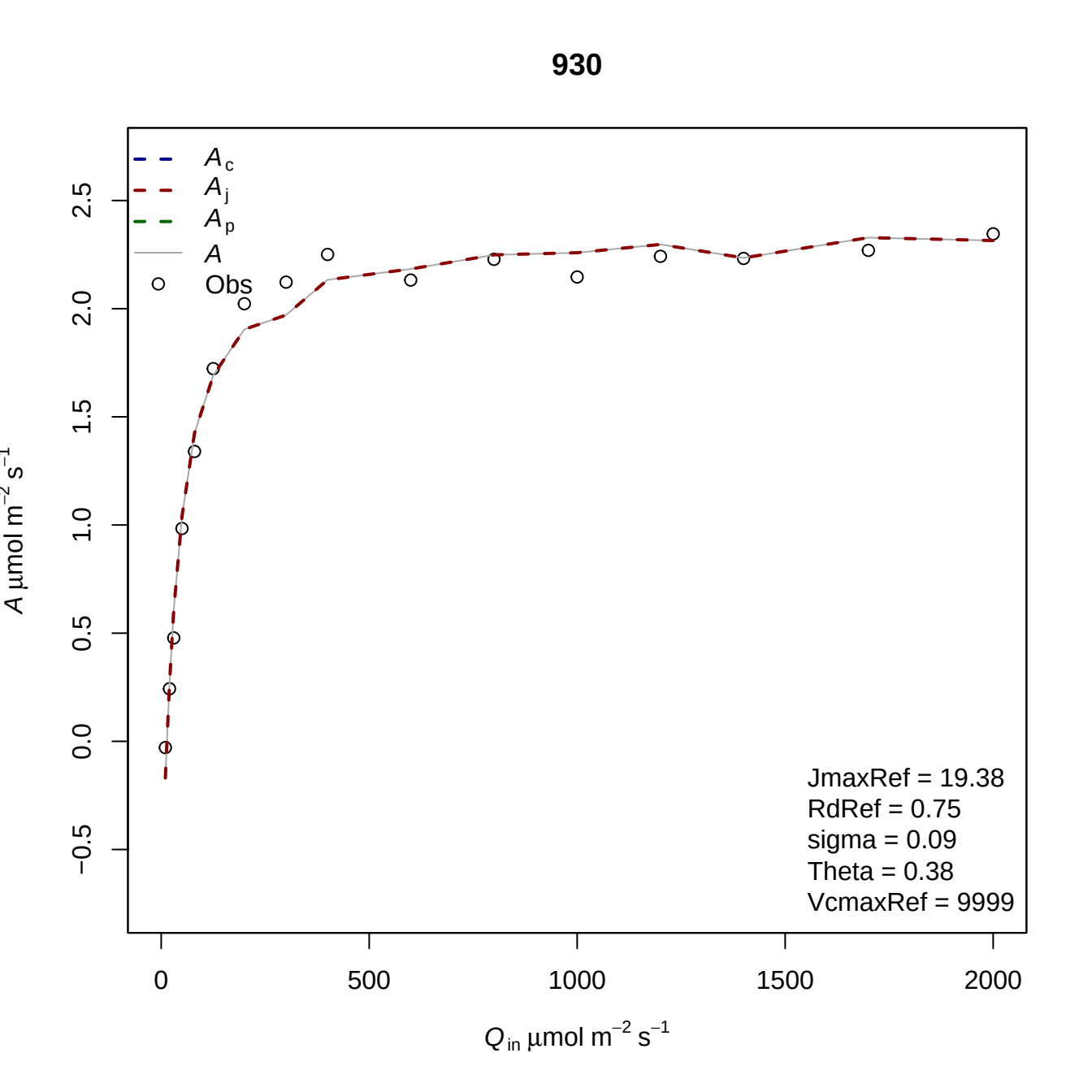

925
A_c
A_j
A_p
A
Obs
JmaxRef = 22.17
RdRef = 0.54
sigma = 0.11
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

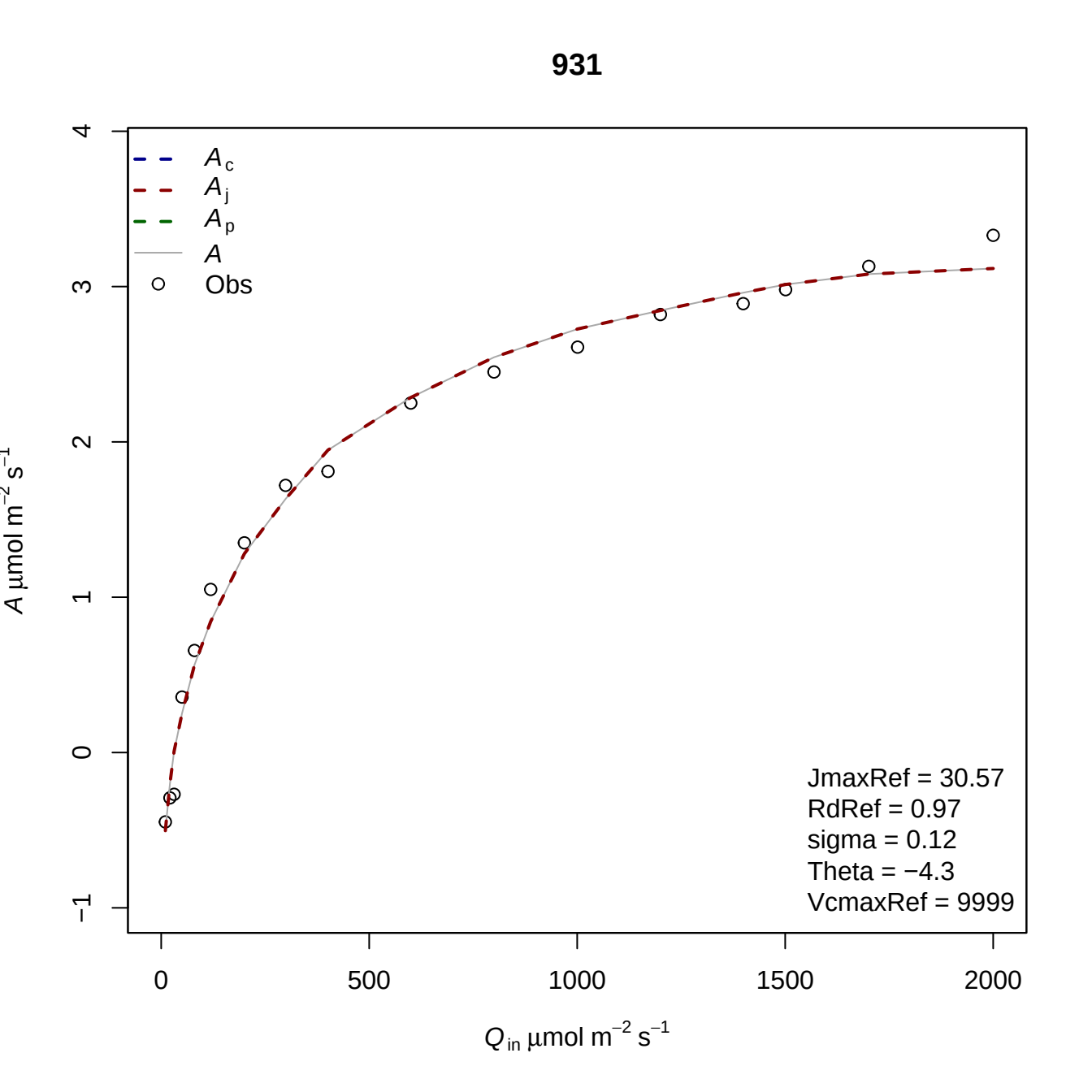

926
A_c
A_j
A_p
A
Obs
JmaxRef = 47.02
RdRef = 1.2
sigma = 0.28
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

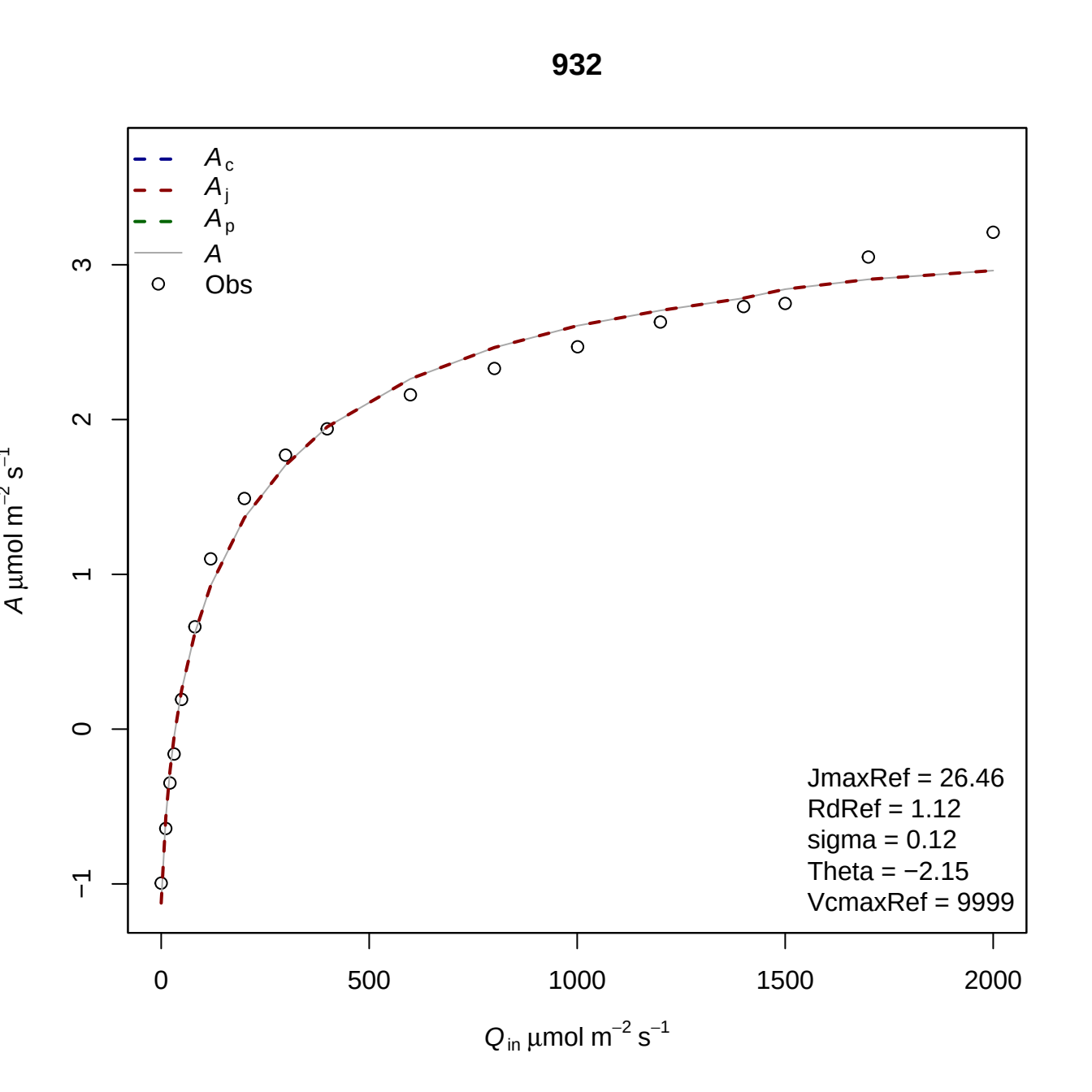

927
A_c
A_j
A_p
A
Obs
JmaxRef = 25.17
RdRef = 0.9
sigma = 0.11
Theta = −4.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

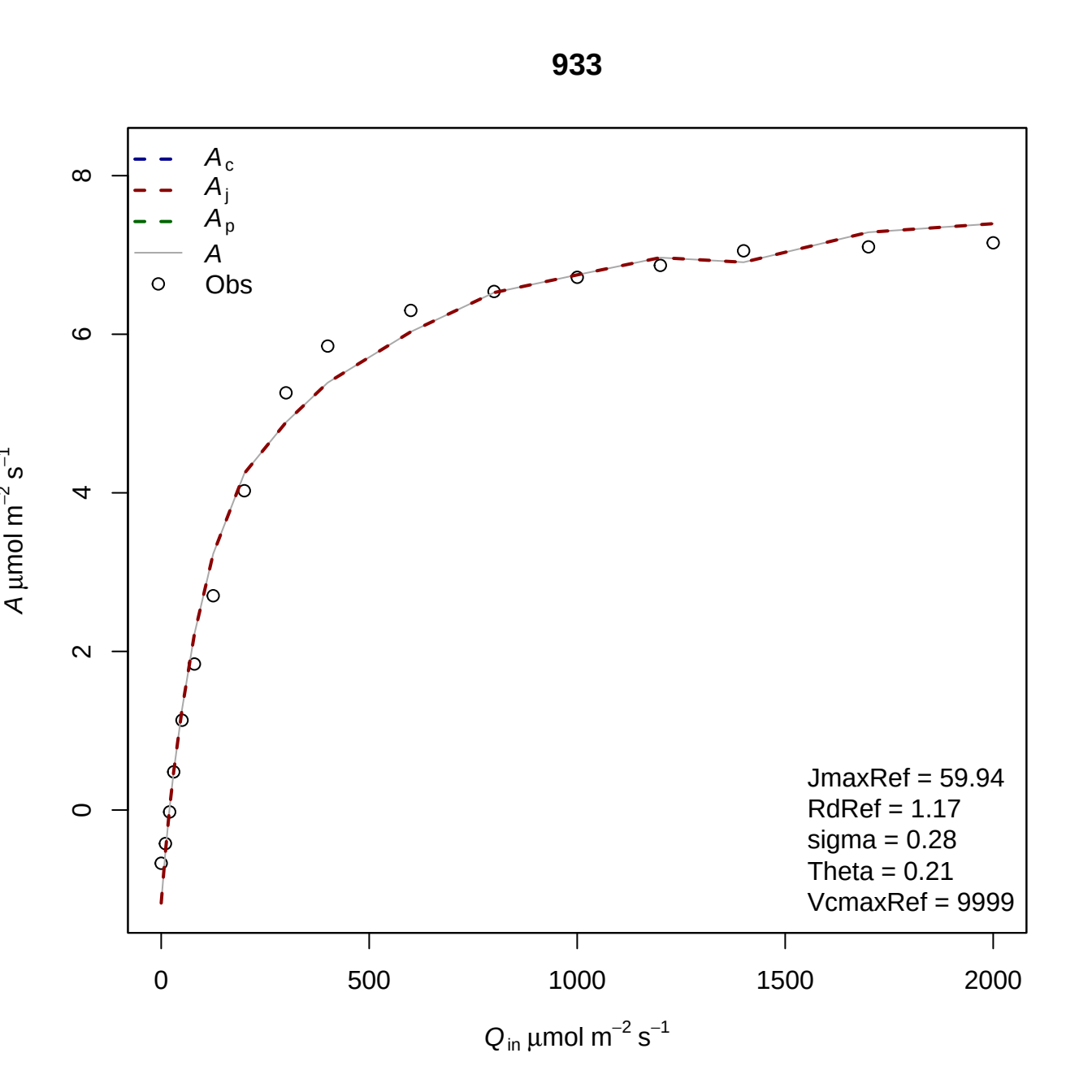

928
A_c
A_j
A_p
A
Obs
JmaxRef = 24.96
RdRef = 0.83
sigma = 0.17
Theta = −2.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

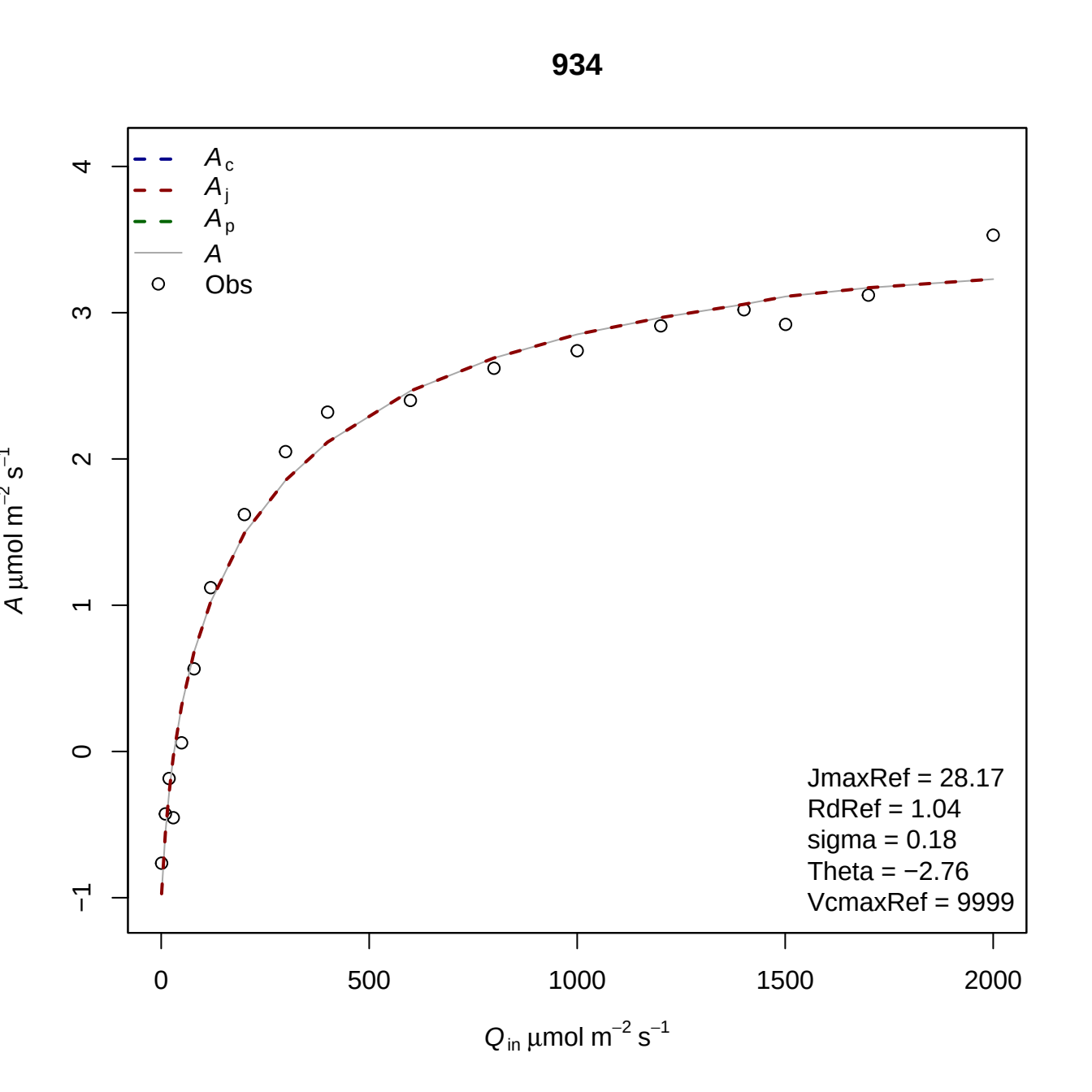

929
A_c
A_j
A_p
A
Obs
JmaxRef = 30.31
RdRef = 1.21
sigma = 0.11
Theta = −2.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

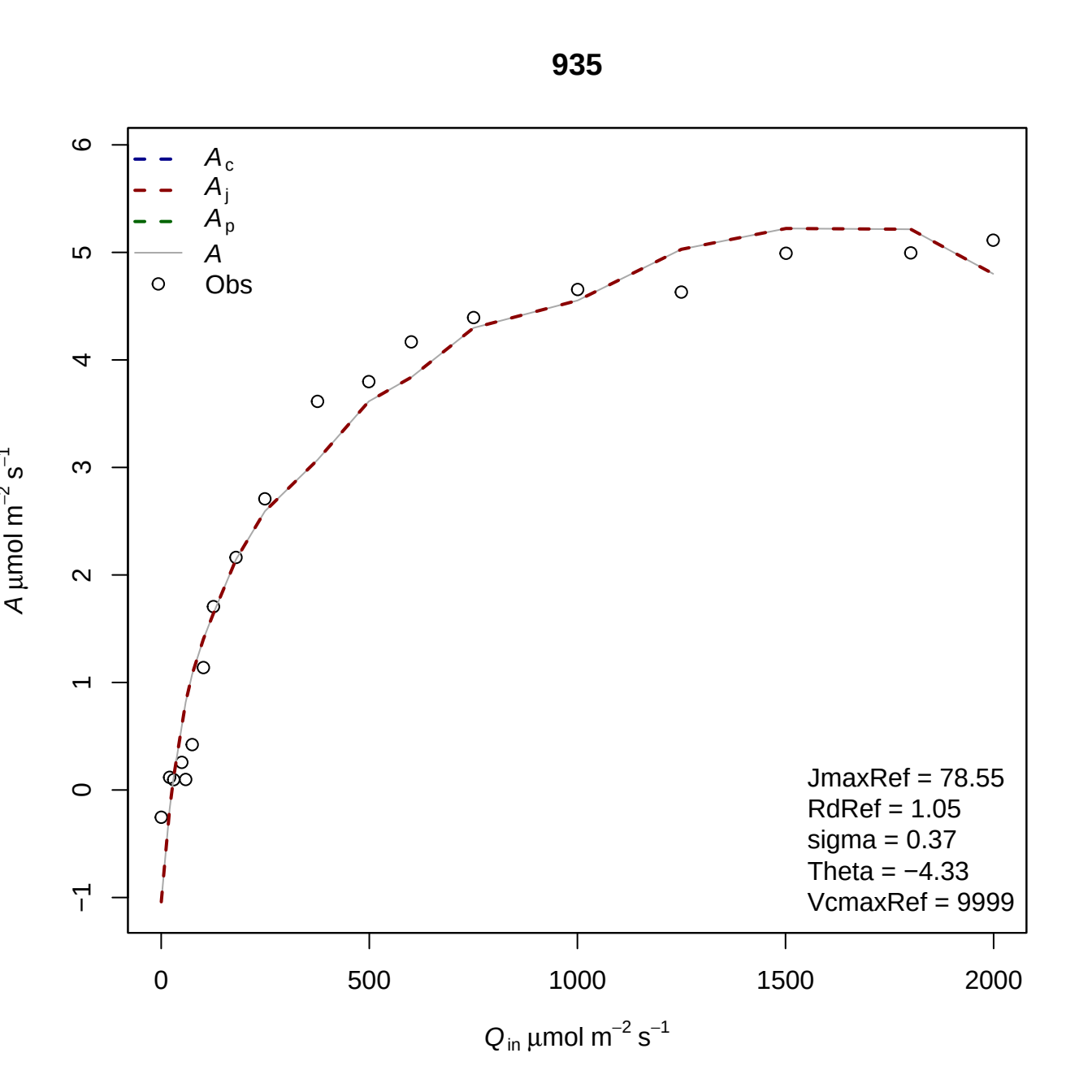

930
A_c
A_j
A_p
A
Obs
JmaxRef = 19.38
RdRef = 0.75
sigma = 0.09
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
2000

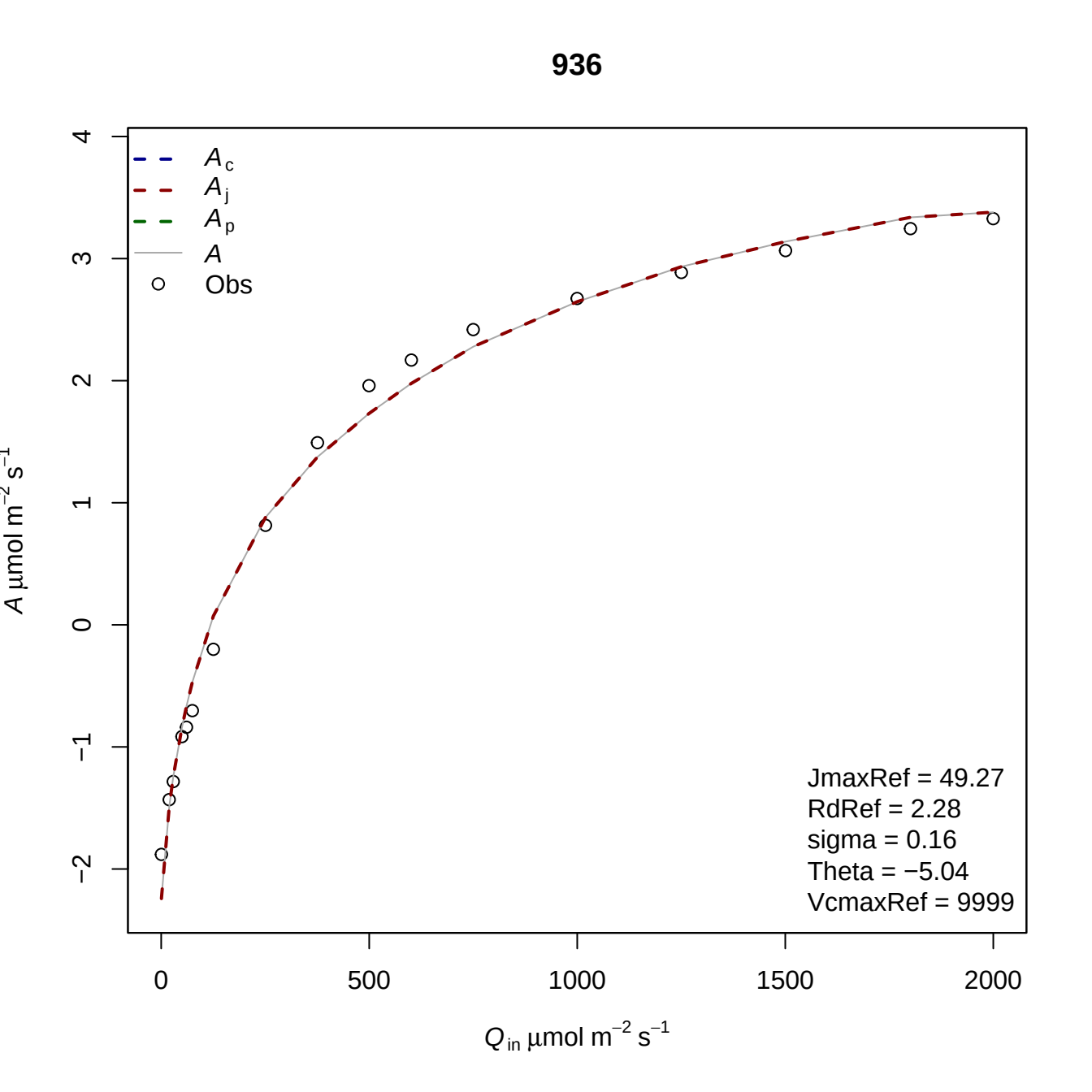

931
A_c
A_j
A_p
A
Obs
JmaxRef = 30.57
RdRef = 0.97
sigma = 0.12
Theta = −4.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500
2000

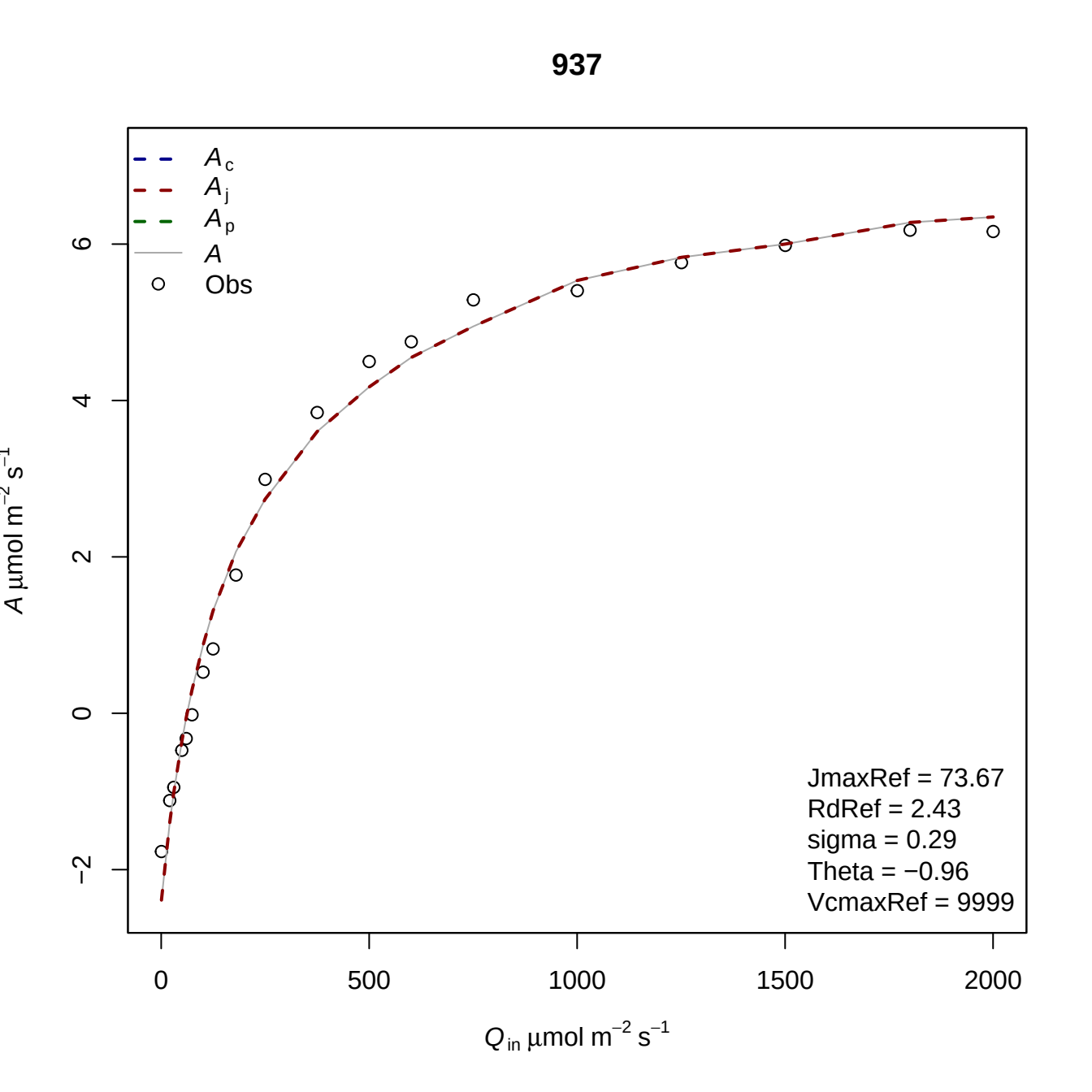

932
A_c
A_j
A_p
A
Obs
JmaxRef = 26.46
RdRef = 1.12
sigma = 0.12
Theta = −2.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

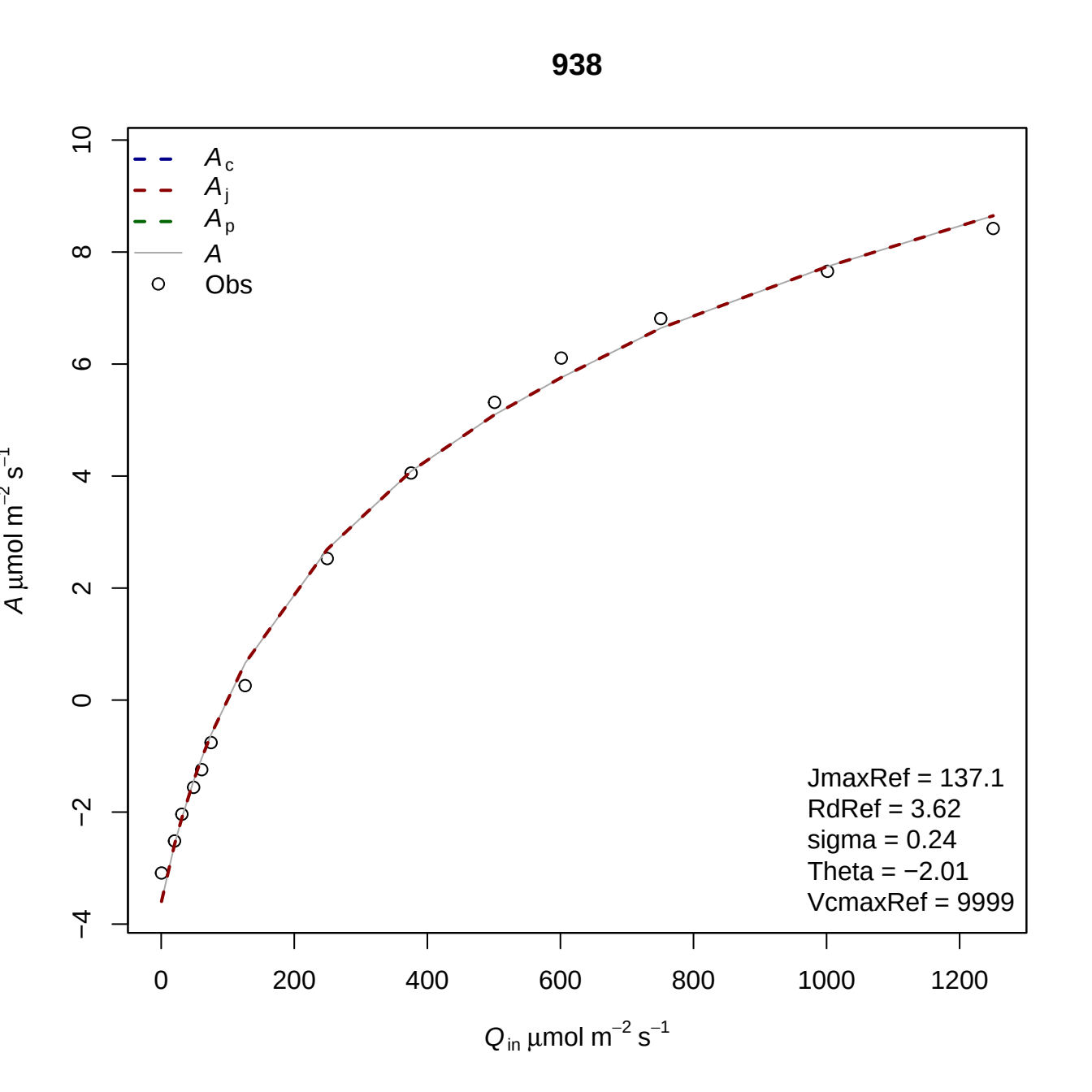

933
A_c
A_j
A_p
A
Obs
JmaxRef = 59.94
RdRef = 1.17
sigma = 0.28
Theta = 0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

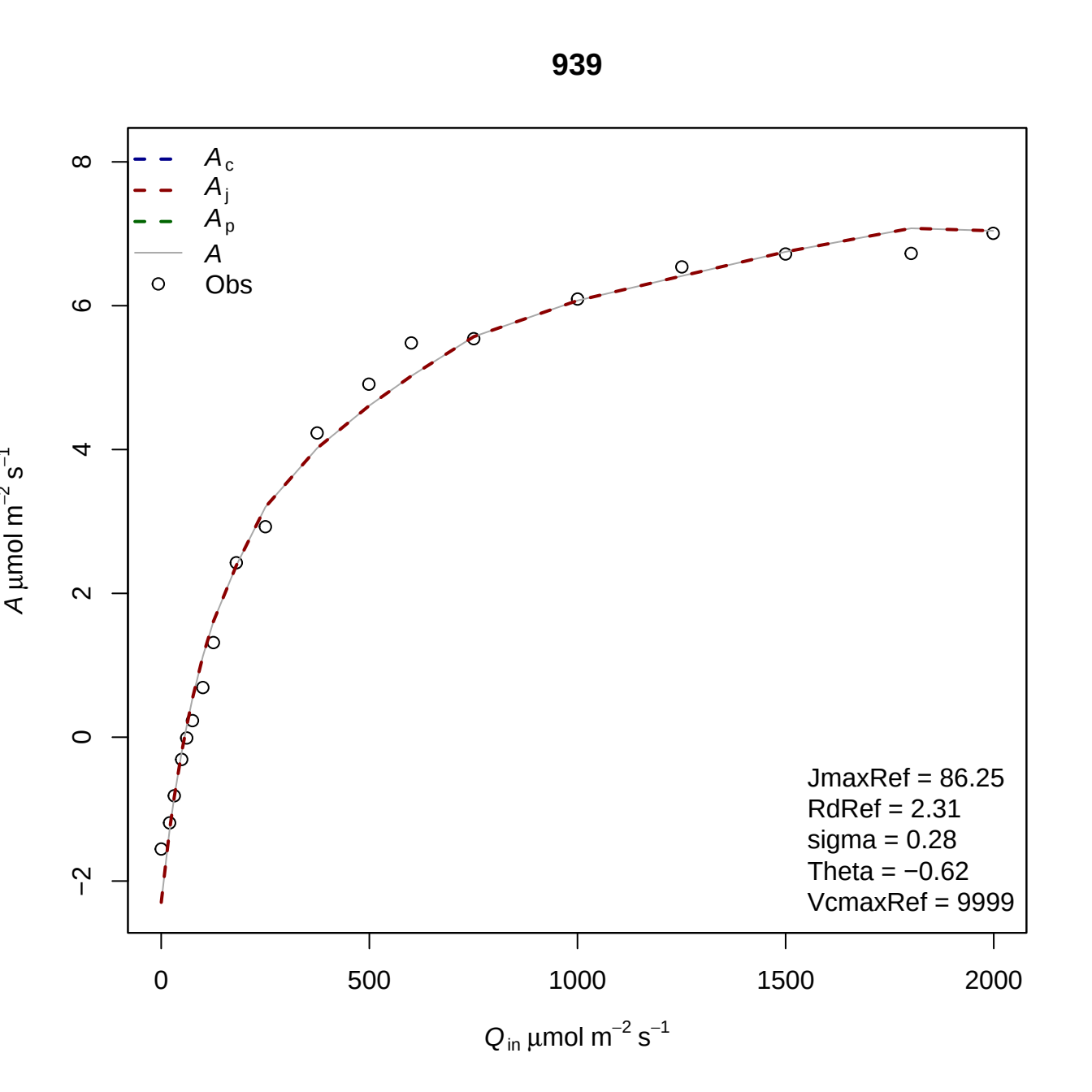

934
A_c
A_j
A_p
A
Obs
JmaxRef = 28.17
RdRef = 1.04
sigma = 0.18
Theta = −2.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

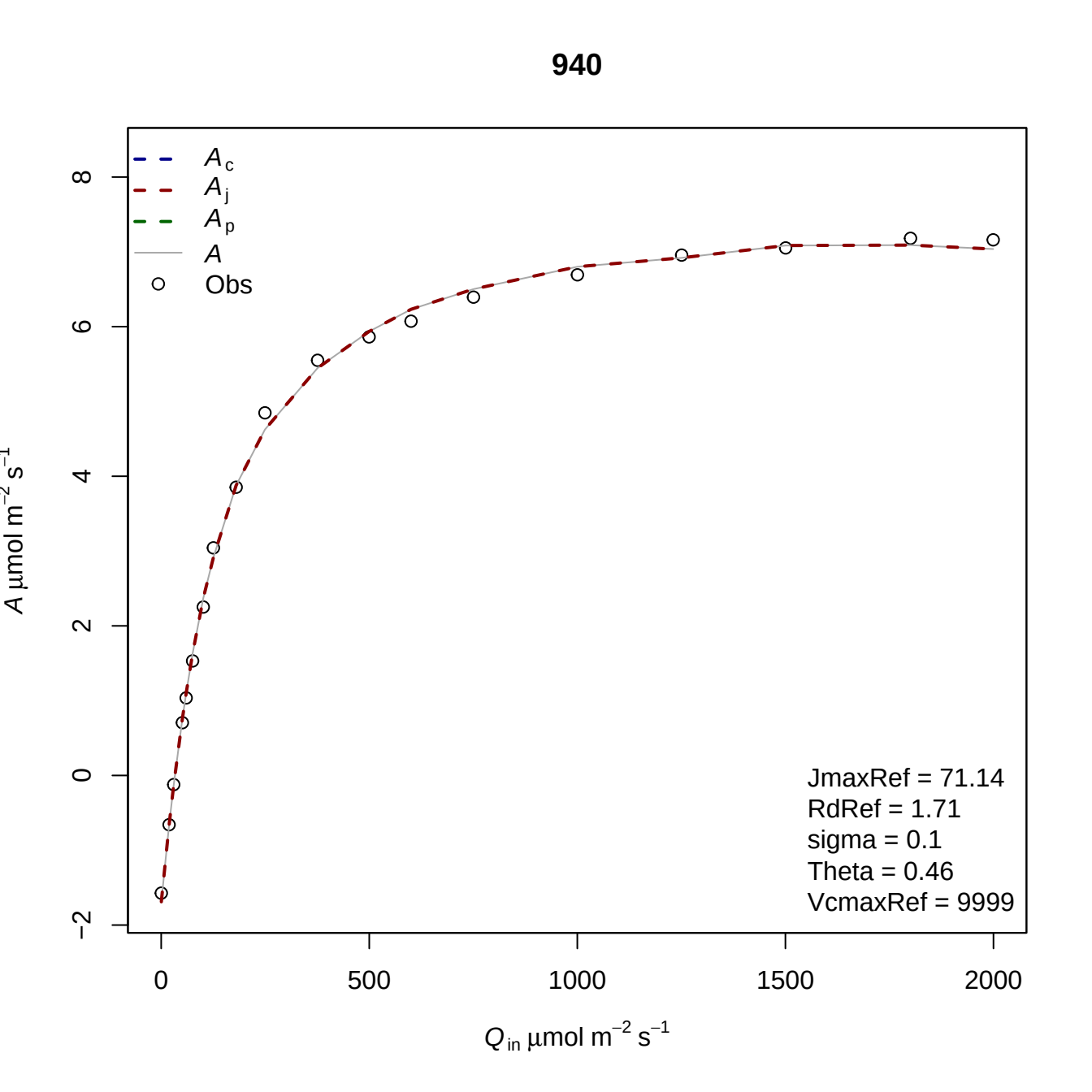

935
A_c
A_j
A_p
A
Obs
JmaxRef = 78.55
RdRef = 1.05
sigma = 0.37
Theta = −4.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

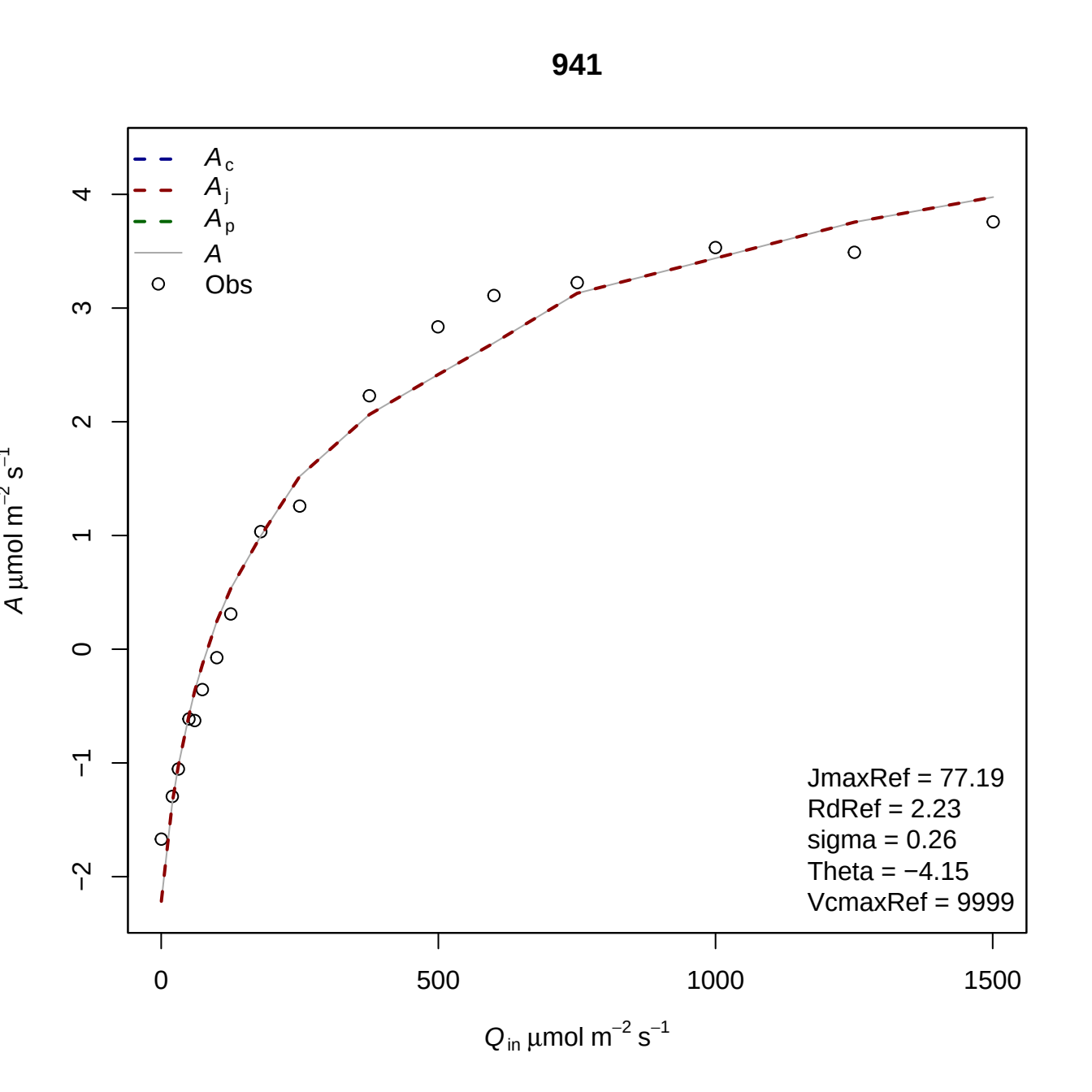

936
A_c
A_j
A_p
A
Obs
JmaxRef = 49.27
RdRef = 2.28
sigma = 0.16
Theta = −5.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

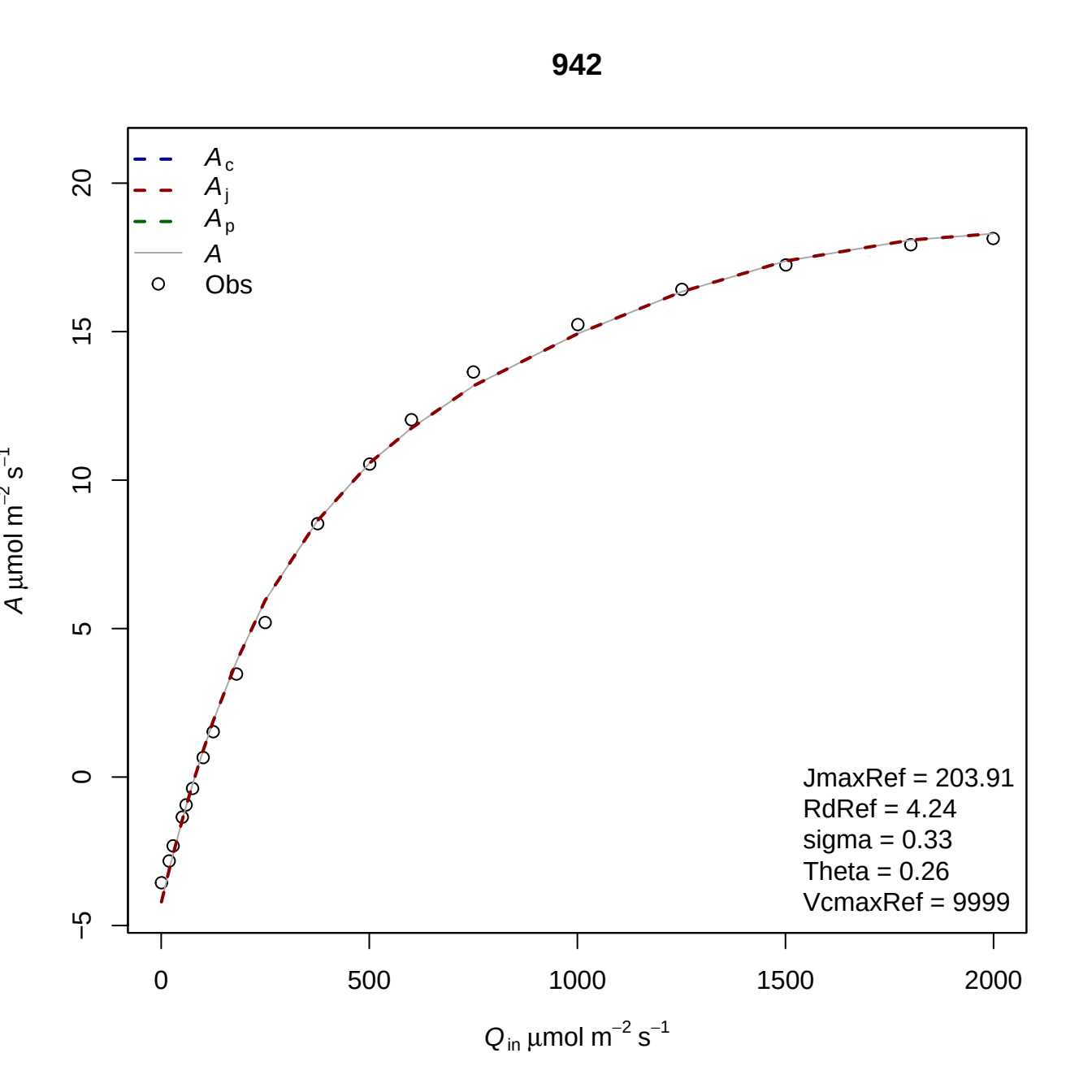

937
A_c
A_j
A_p
A
Obs
JmaxRef = 73.67
RdRef = 2.43
sigma = 0.29
Theta = −0.96
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

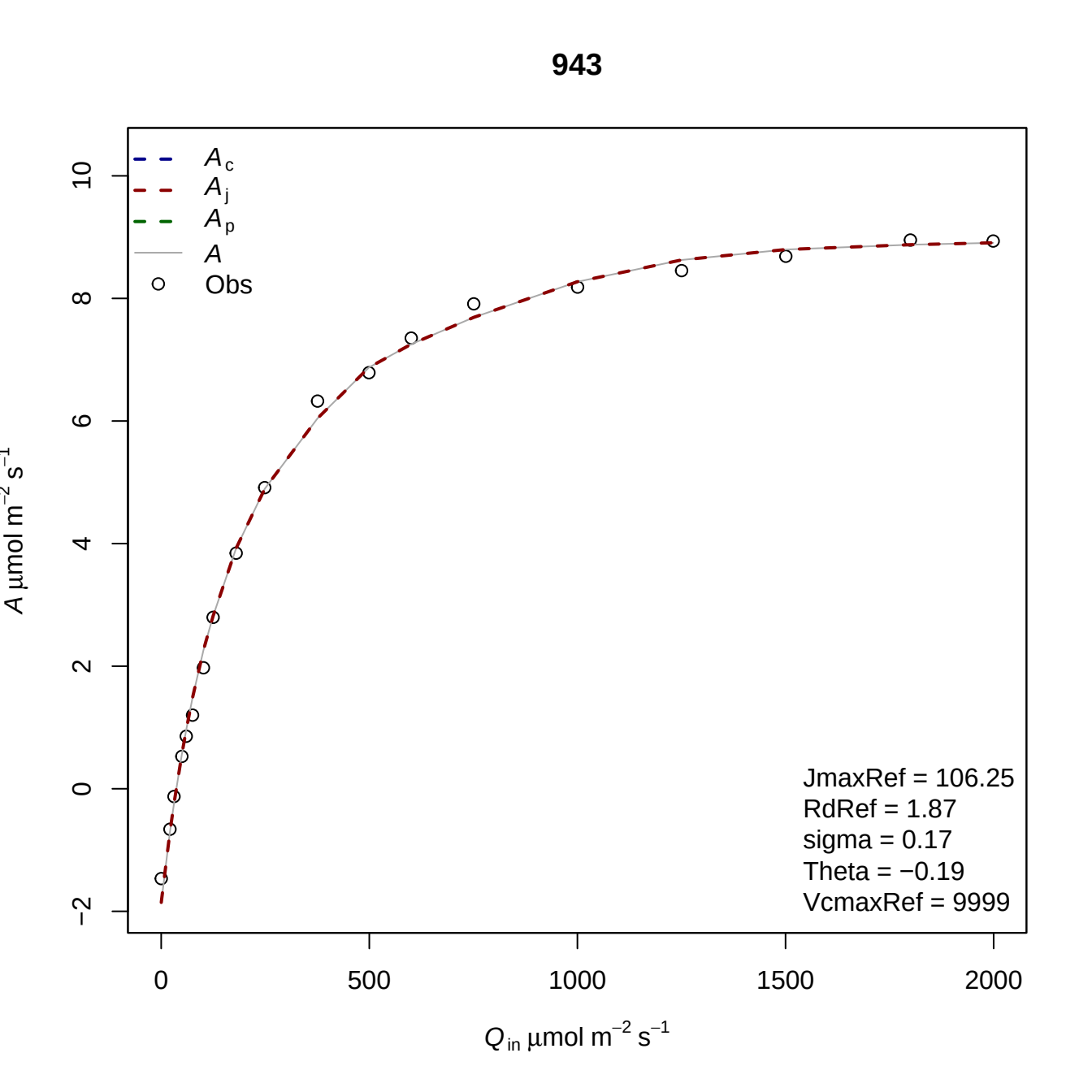

938
A_c
A_j
A_p
A
Obs
JmaxRef = 137.1
RdRef = 3.62
sigma = 0.24
Theta = −2.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

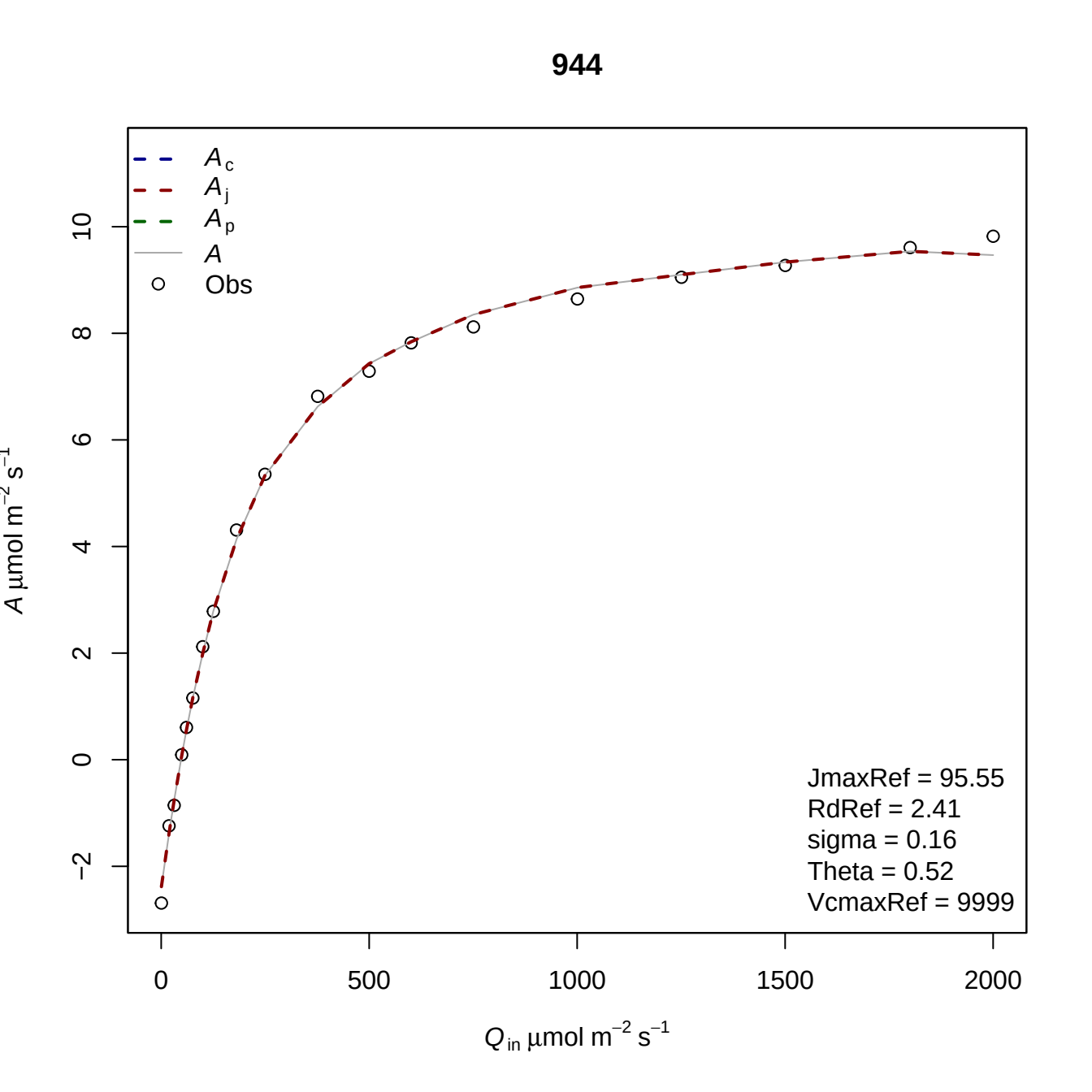

939
A_c
A_j
A_p
A
Obs
JmaxRef = 86.25
RdRef = 2.31
sigma = 0.28
Theta = −0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

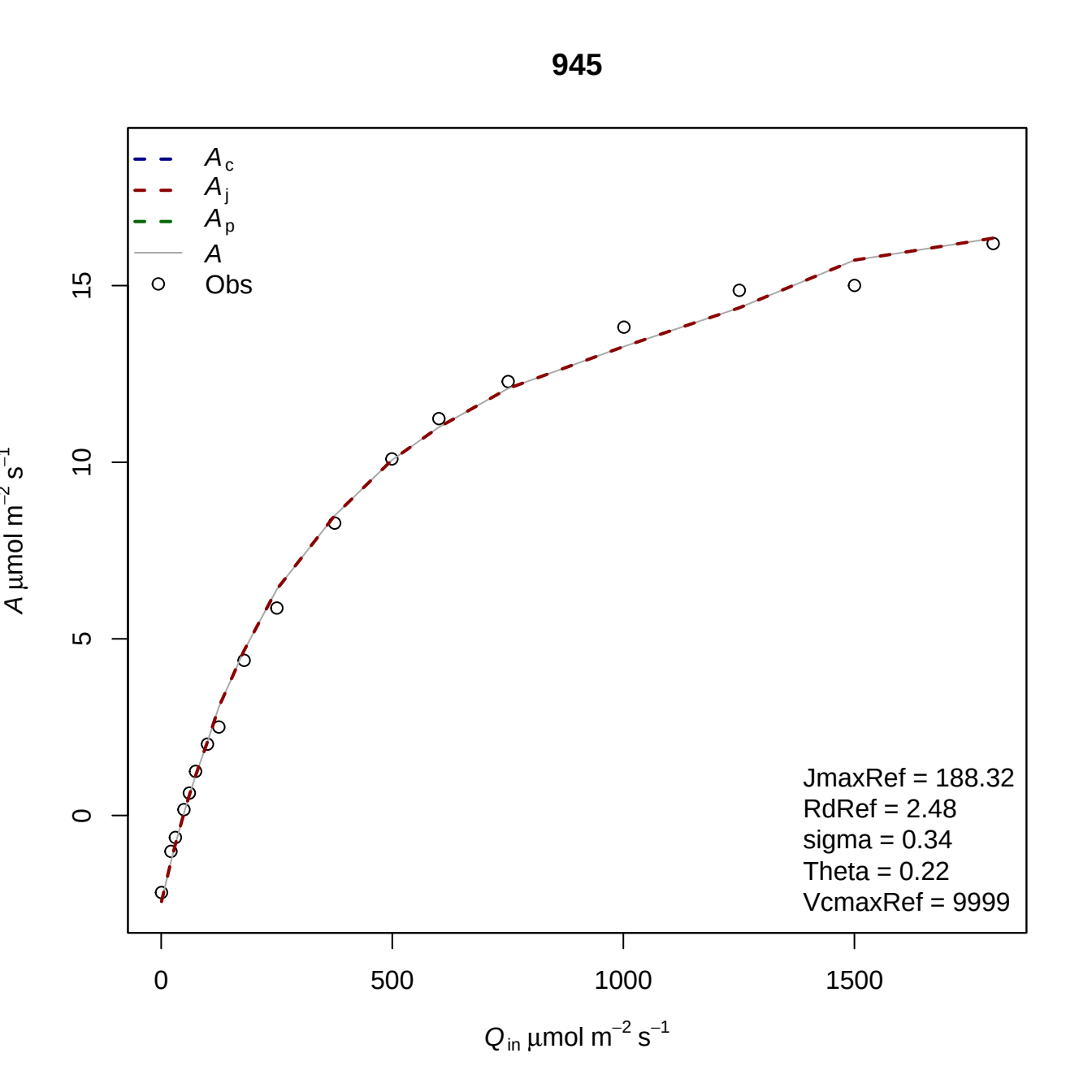

940
A_c
A_j
A_p
A
Obs
JmaxRef = 71.14
RdRef = 1.71
sigma = 0.1
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

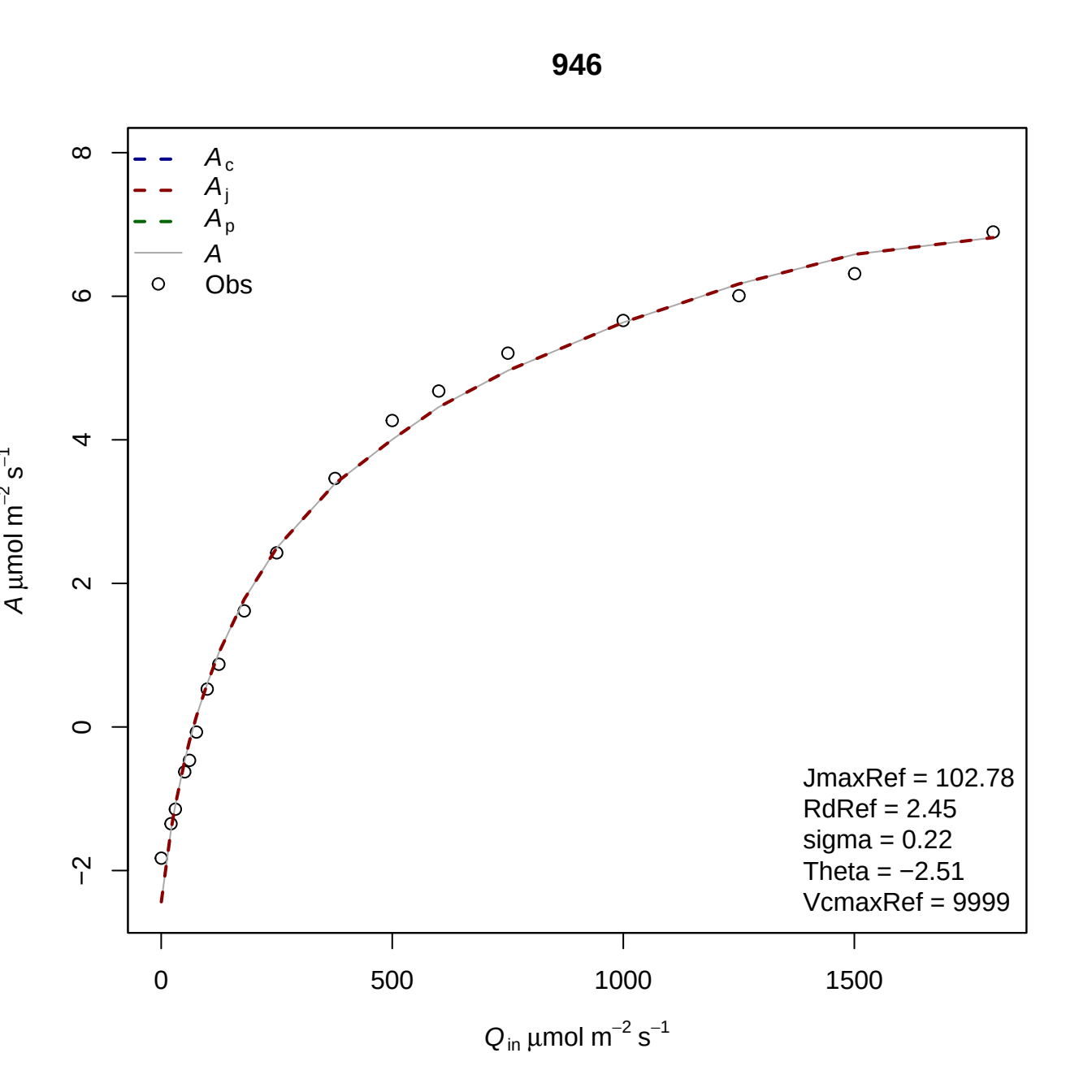

941
A_c
A_j
A_p
A
Obs
JmaxRef = 77.19
RdRef = 2.23
sigma = 0.26
Theta = −4.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

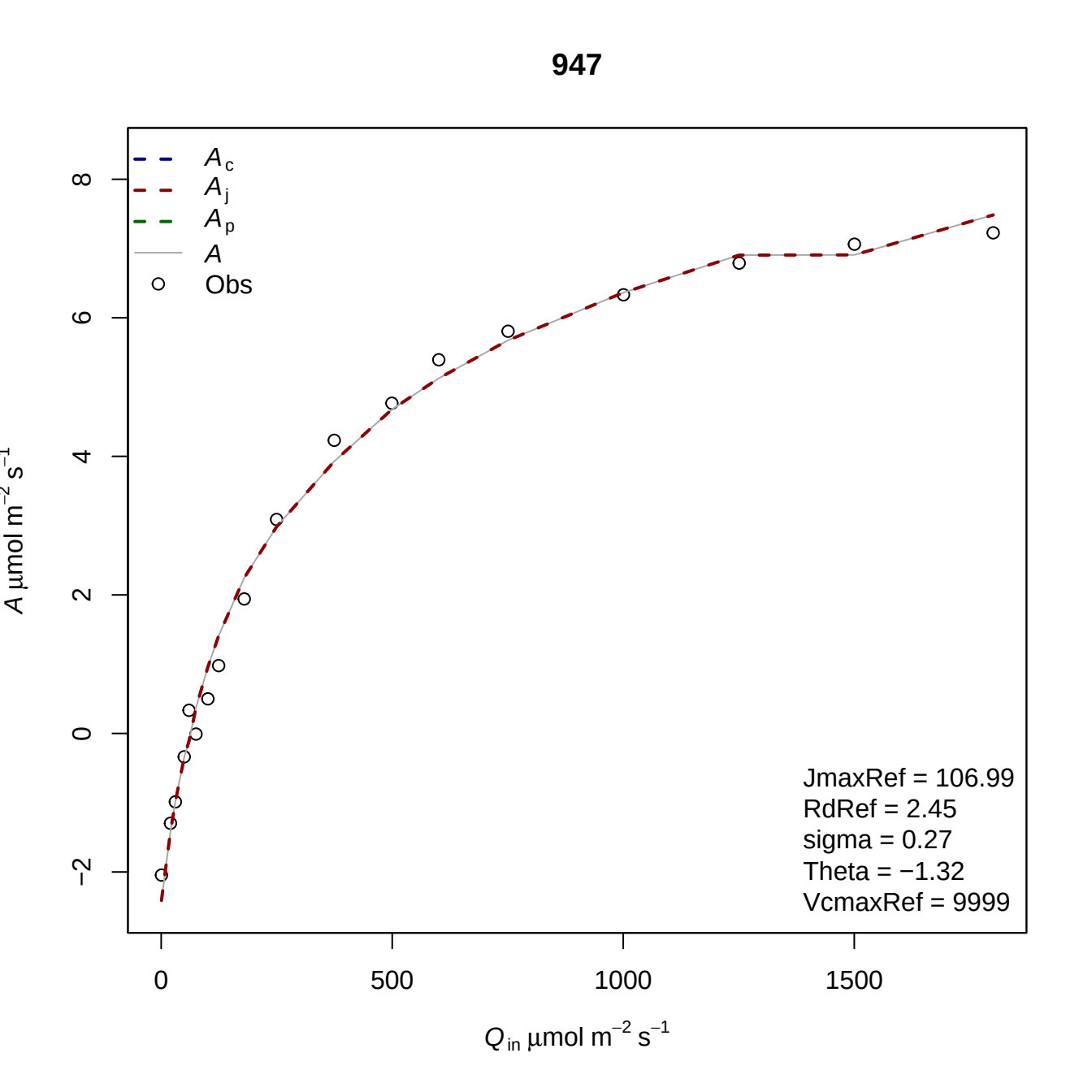

942
A_c
A_j
A_p
A
Obs
JmaxRef = 203.91
RdRef = 4.24
sigma = 0.33
Theta = 0.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

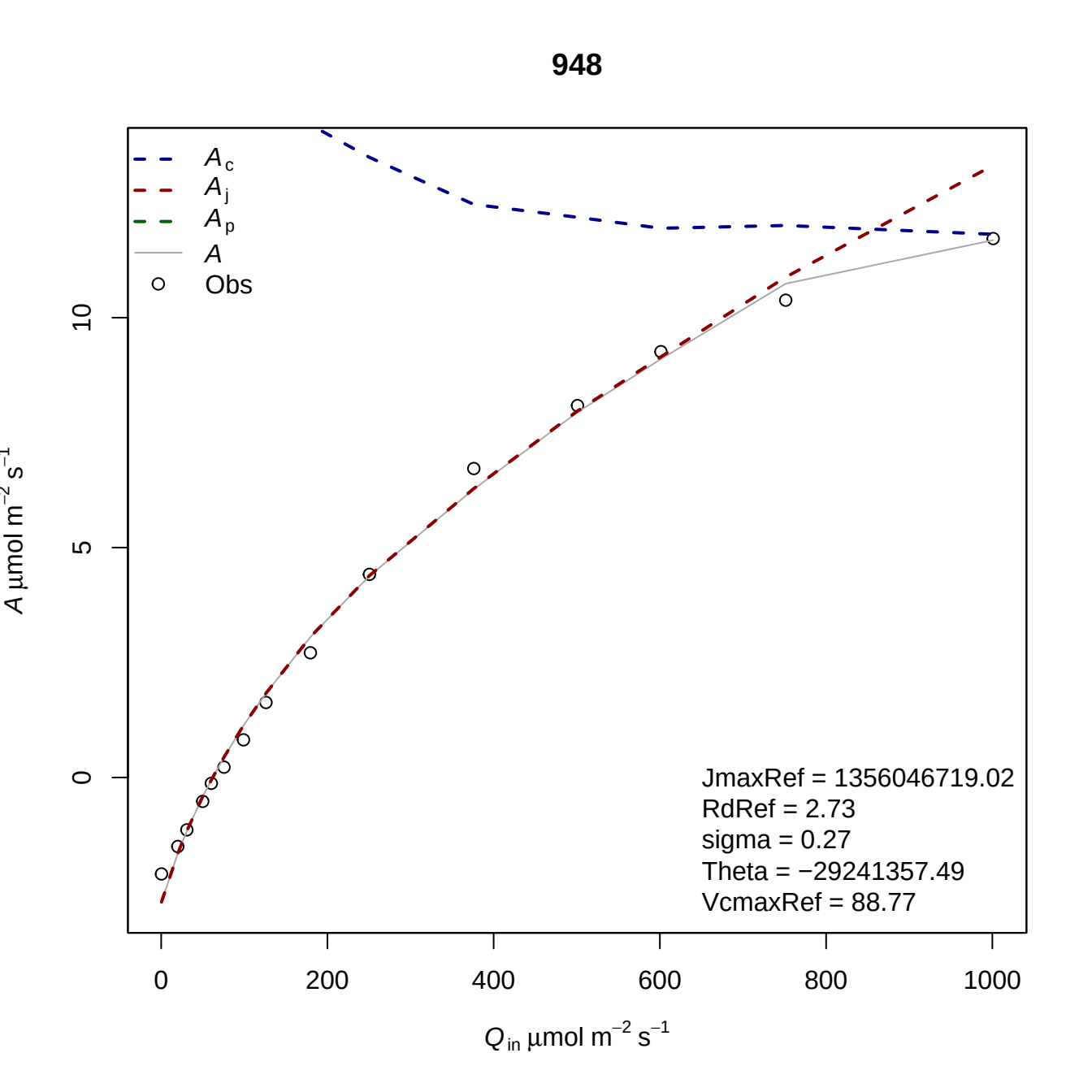

943
A_c
A_j
A_p
A
Obs
JmaxRef = 106.25
RdRef = 1.87
sigma = 0.17
Theta = −0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

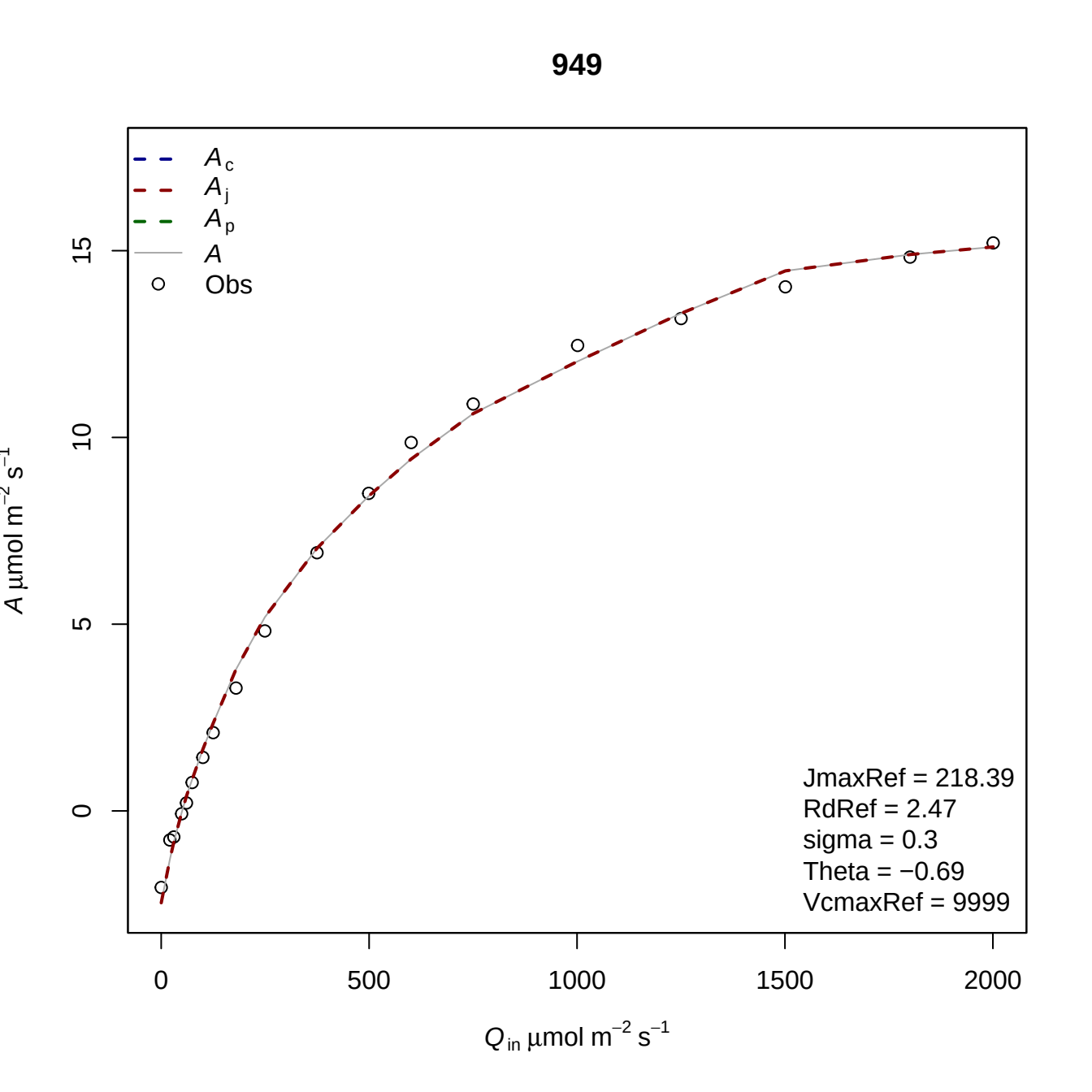

944
A_c
A_j
A_p
A
Obs
JmaxRef = 95.55
RdRef = 2.41
sigma = 0.16
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

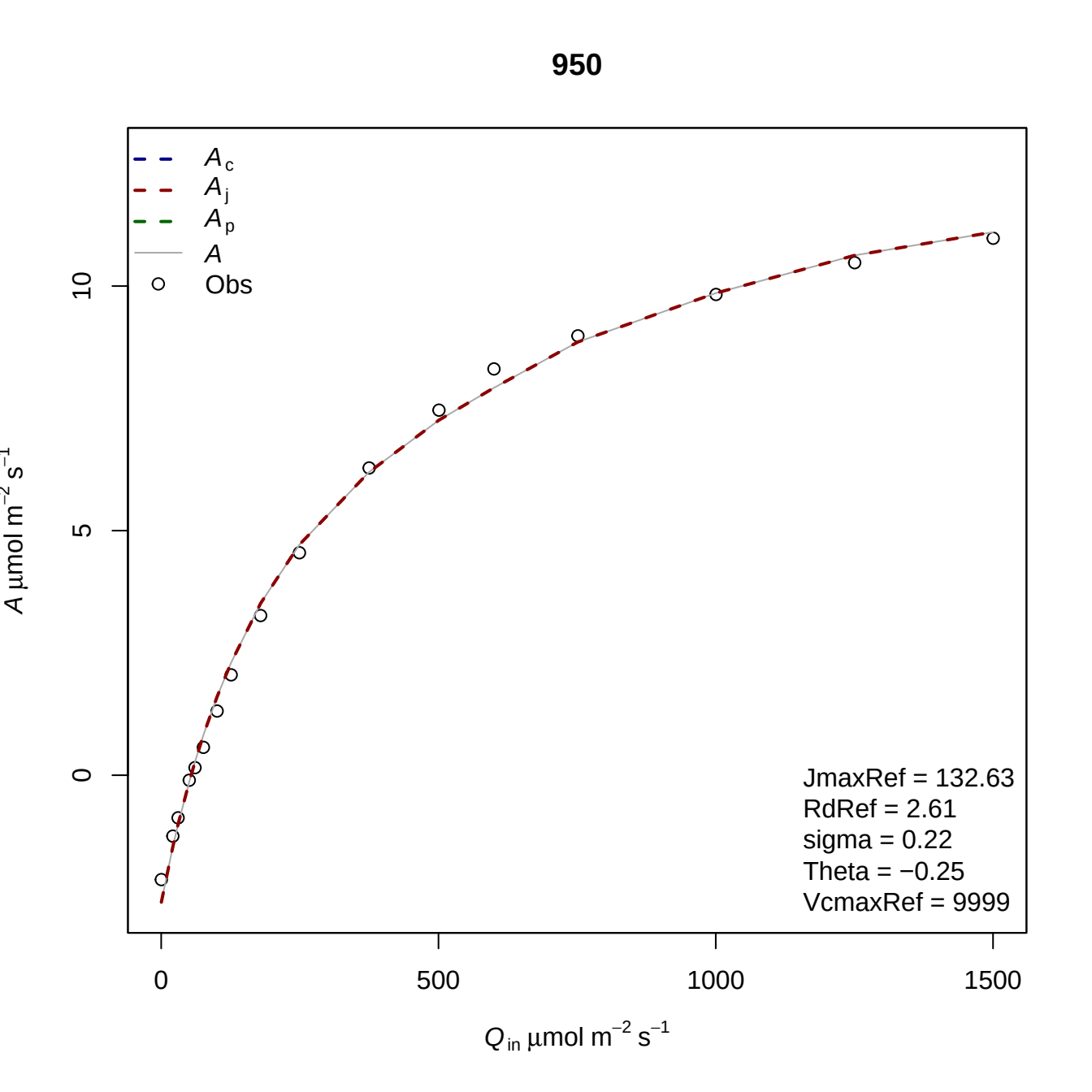

945
A_c
A_j
A_p
A
Obs
JmaxRef = 188.32
RdRef = 2.48
sigma = 0.34
Theta = 0.22
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

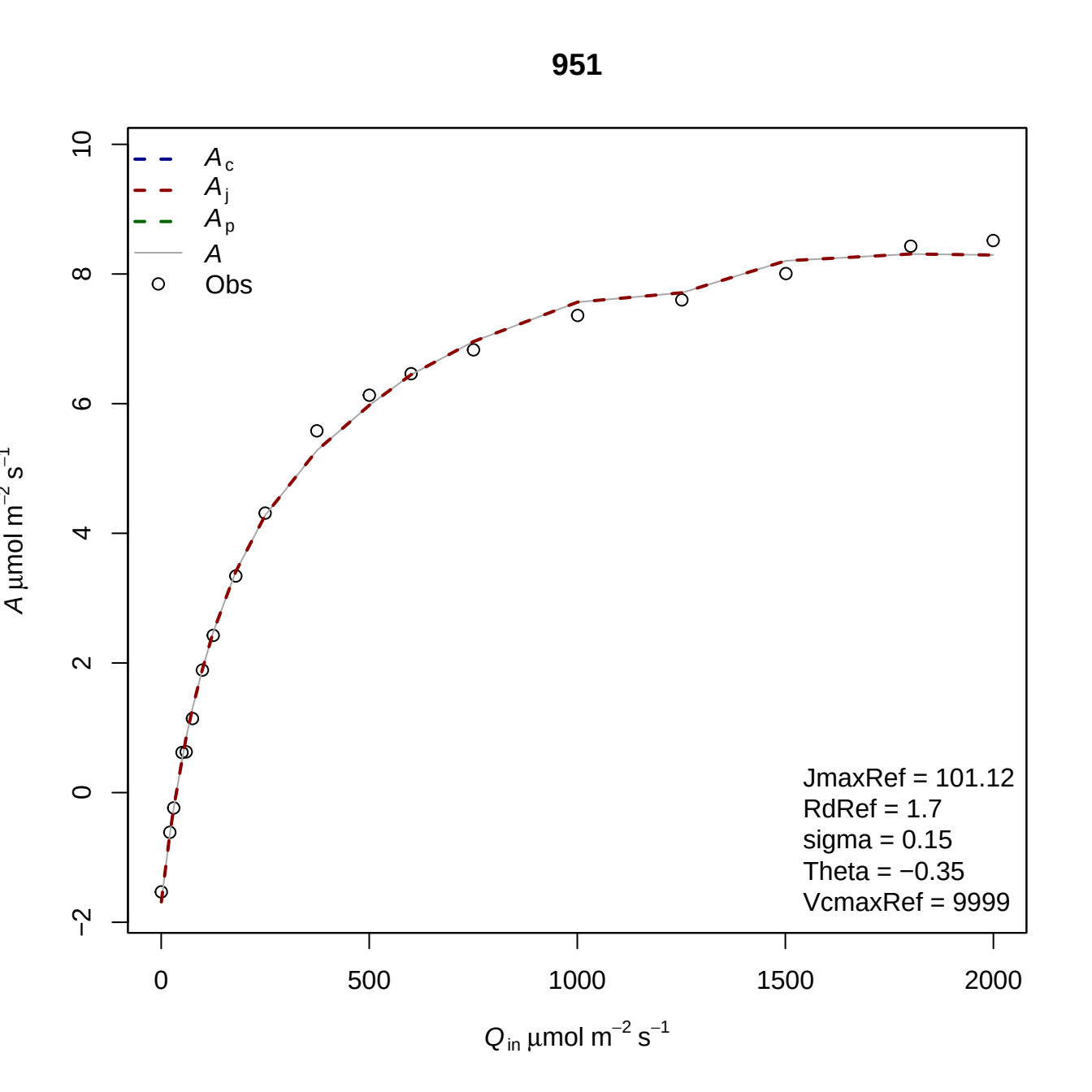

946
A_c
A_j
A_p
A
Obs
JmaxRef = 102.78
RdRef = 2.45
sigma = 0.22
Theta = −2.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

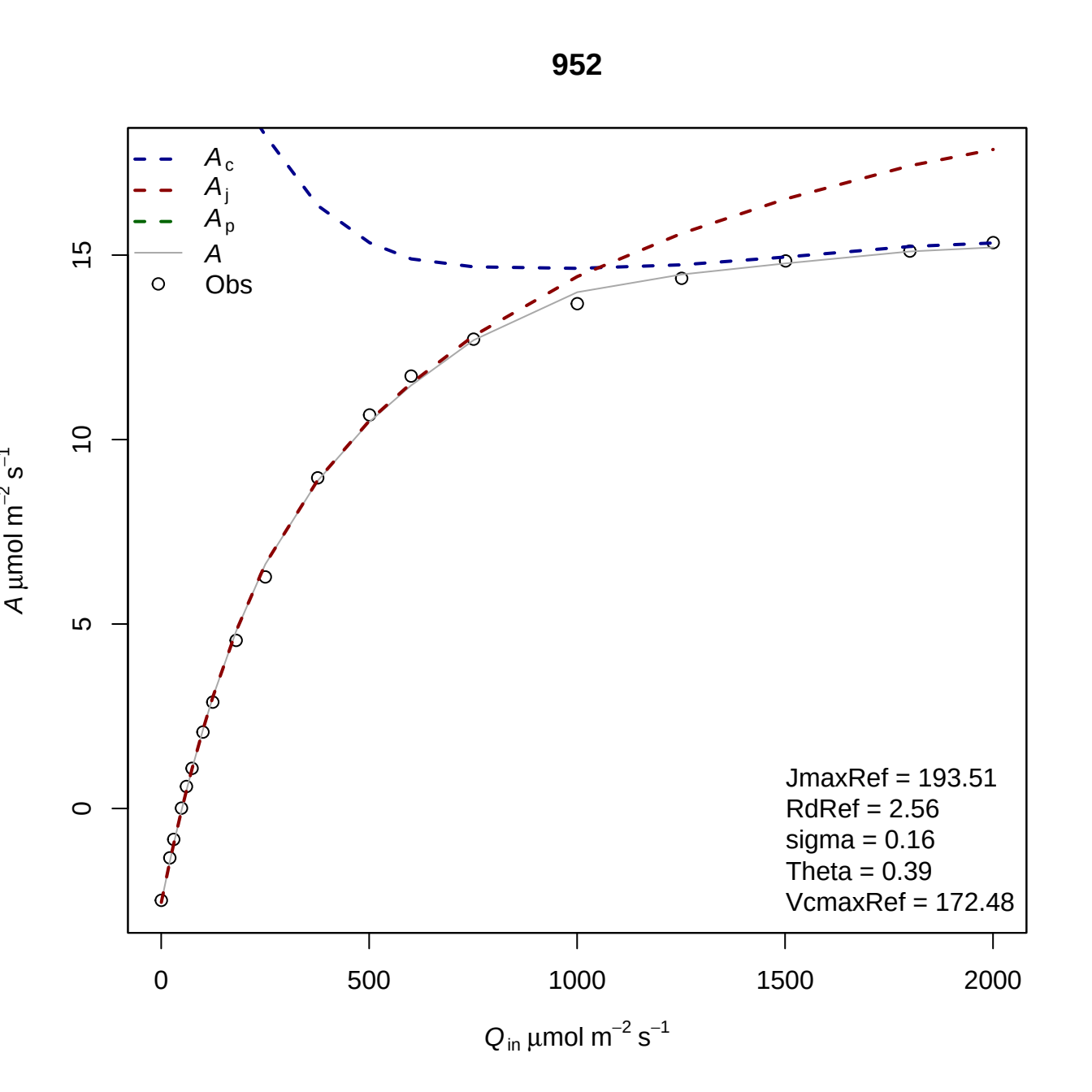

947
A_c
A_j
A_p
A
Obs
JmaxRef = 106.99
RdRef = 2.45
sigma = 0.27
Theta = −1.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

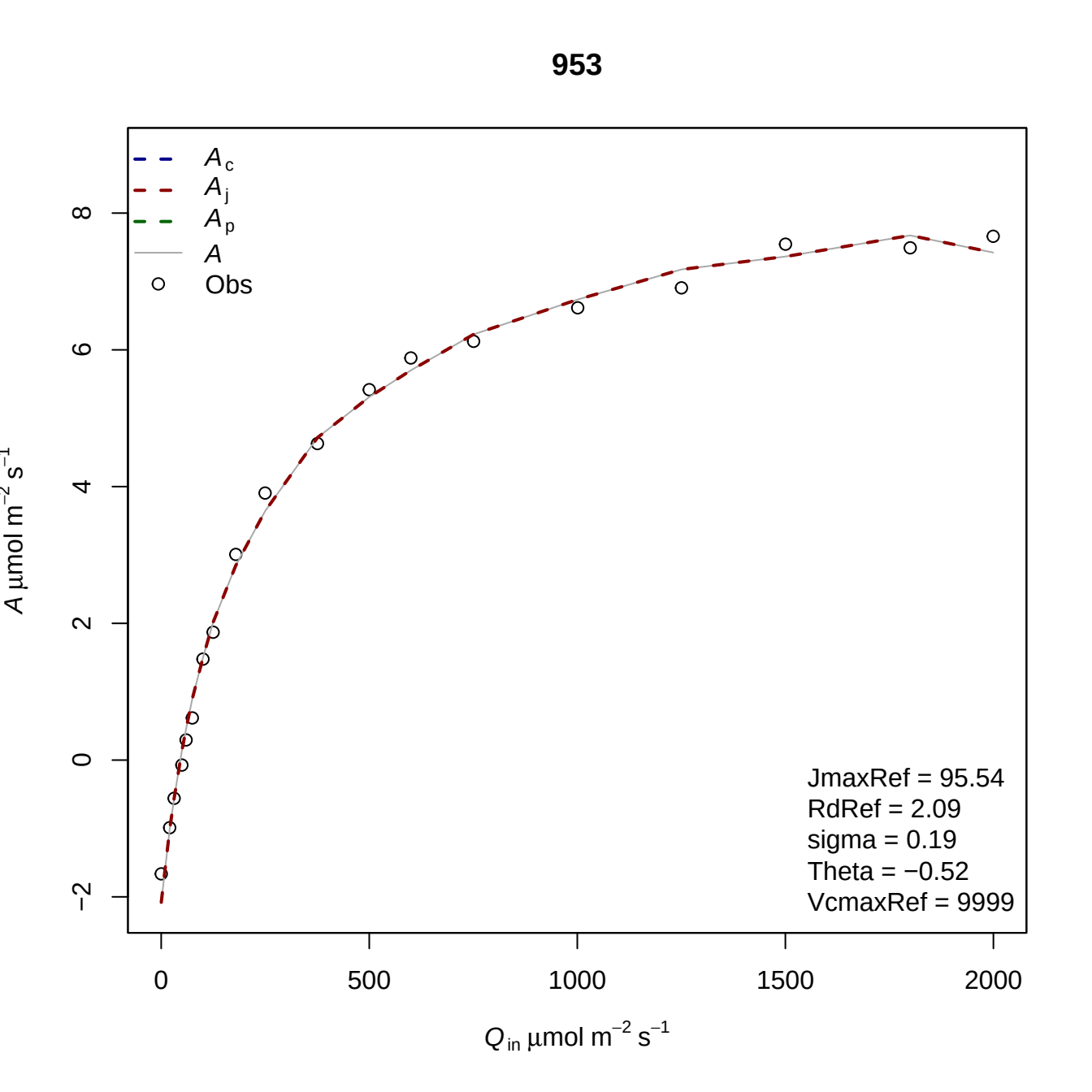

948
A_c
A_j
A_p
A
Obs
JmaxRef = 1356046719.02
RdRef = 2.73
sigma = 0.27
Theta = −29241357.49
VcmaxRef = 88.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

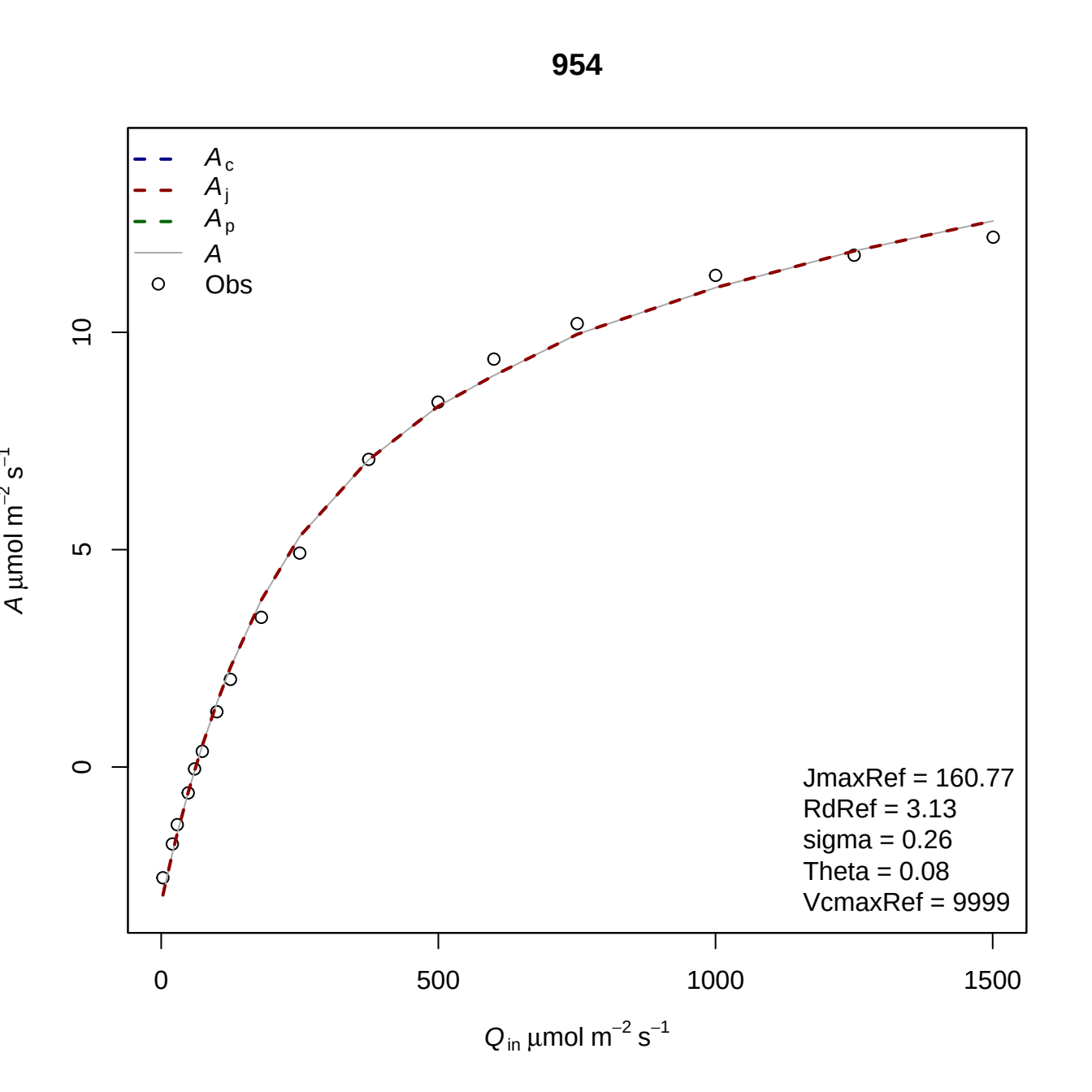

949
A_c
A_j
A_p
A
Obs
JmaxRef = 218.39
RdRef = 2.47
sigma = 0.3
Theta = −0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

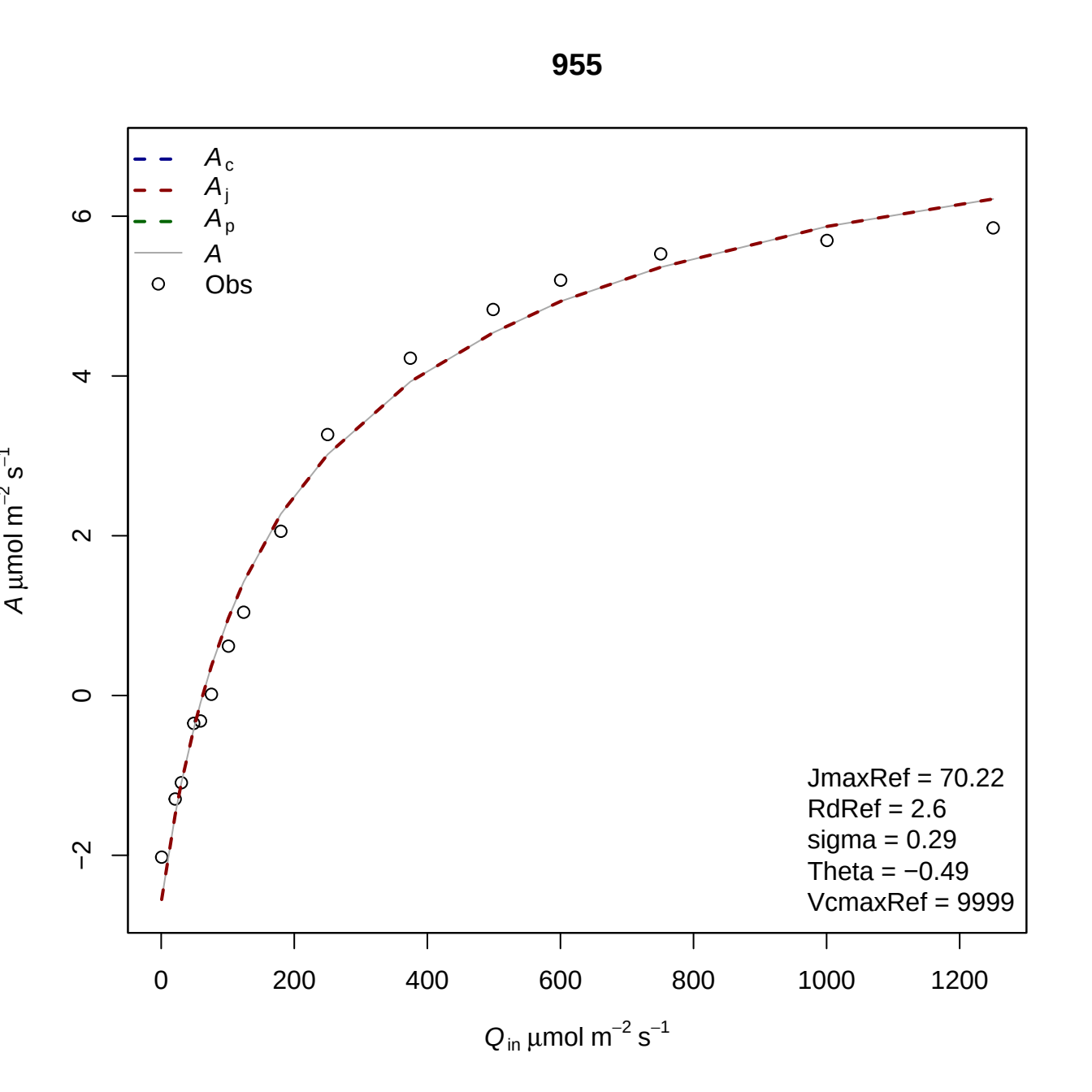

950
A_c
A_j
A_p
A
Obs
JmaxRef = 132.63
RdRef = 2.61
sigma = 0.22
Theta = −0.25
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

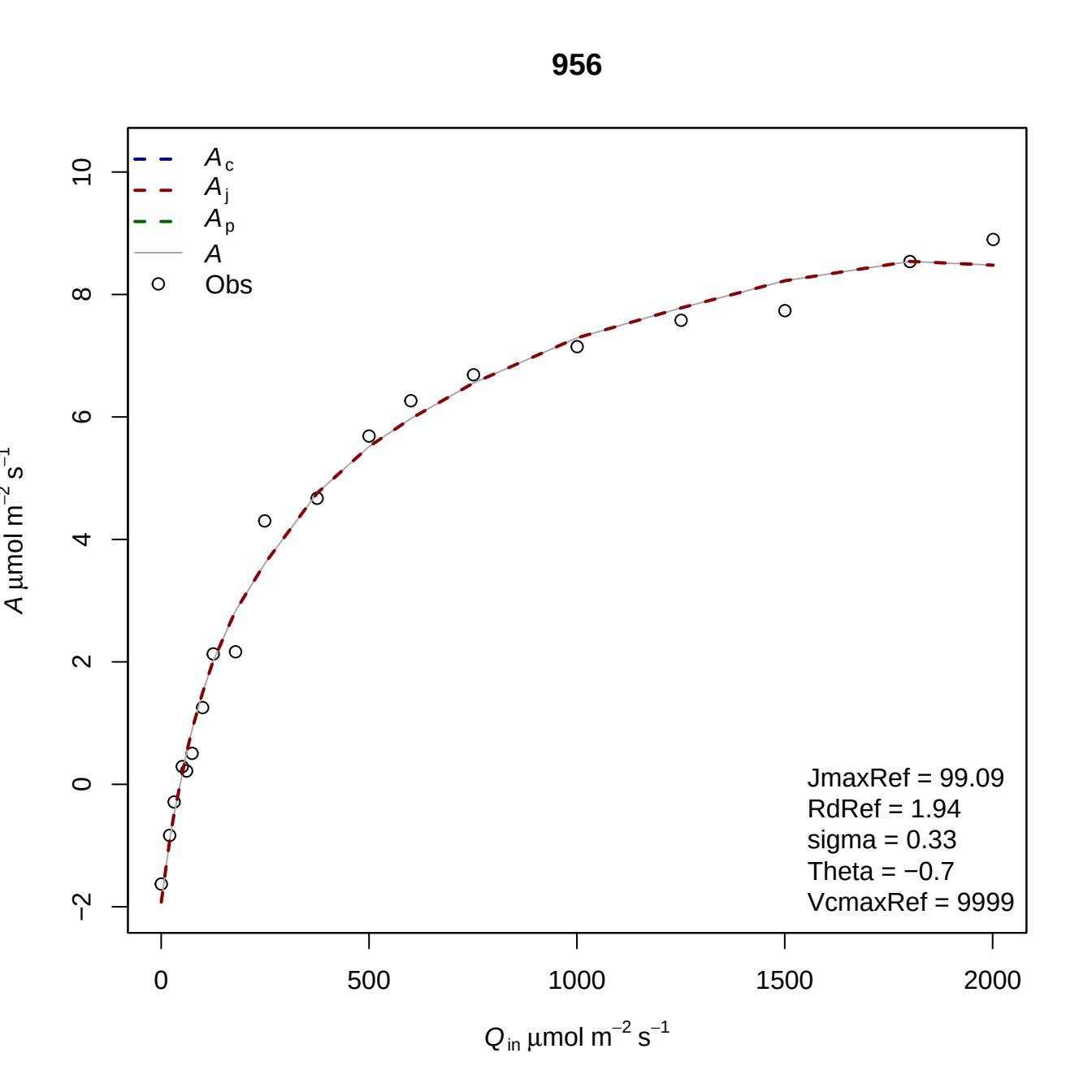

951
A_c
A_j
A_p
A
Obs
JmaxRef = 101.12
RdRef = 1.7
sigma = 0.15
Theta = −0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

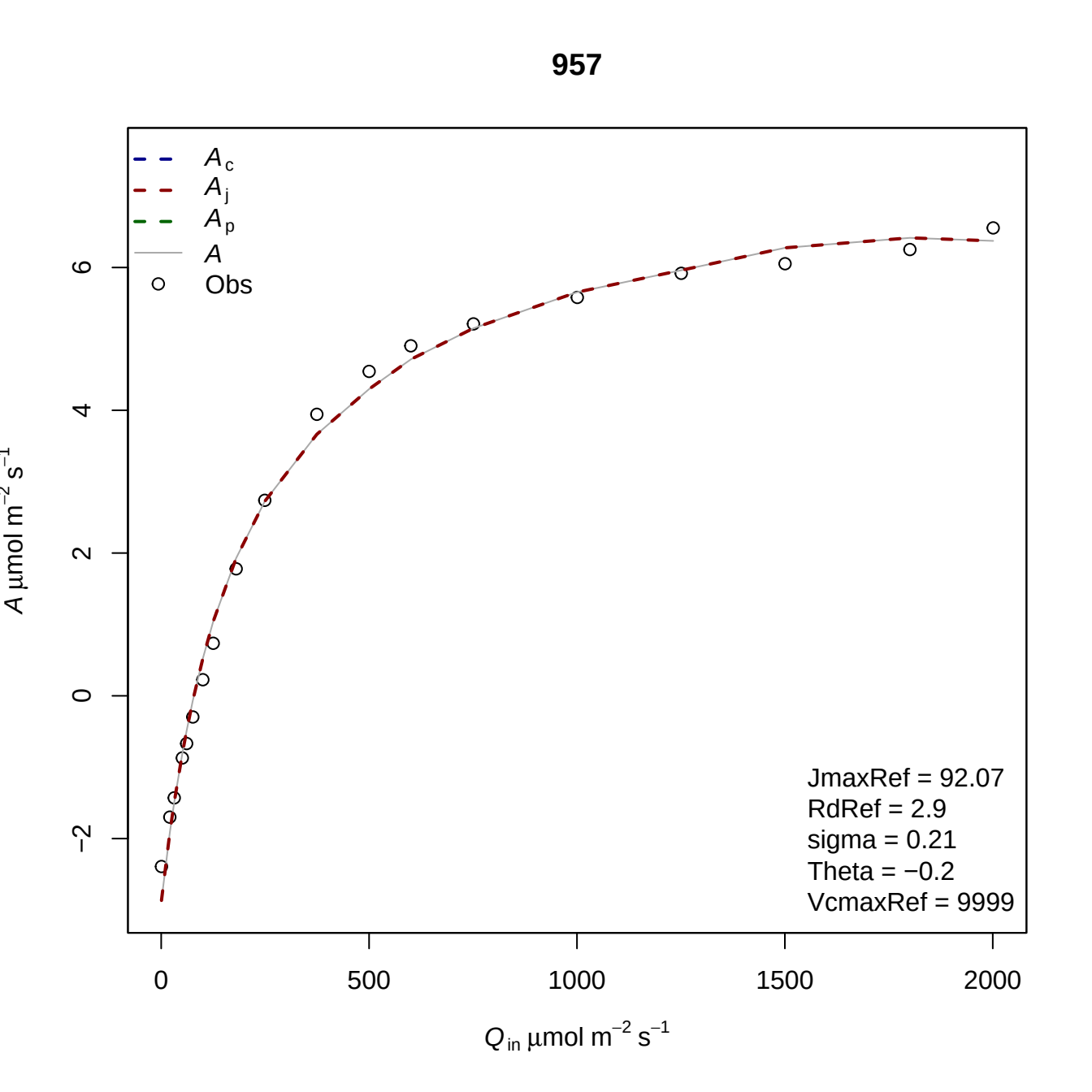

952
A_c
A_j
A_p
A
Obs
JmaxRef = 193.51
RdRef = 2.56
sigma = 0.16
Theta = 0.39
VcmaxRef = 172.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

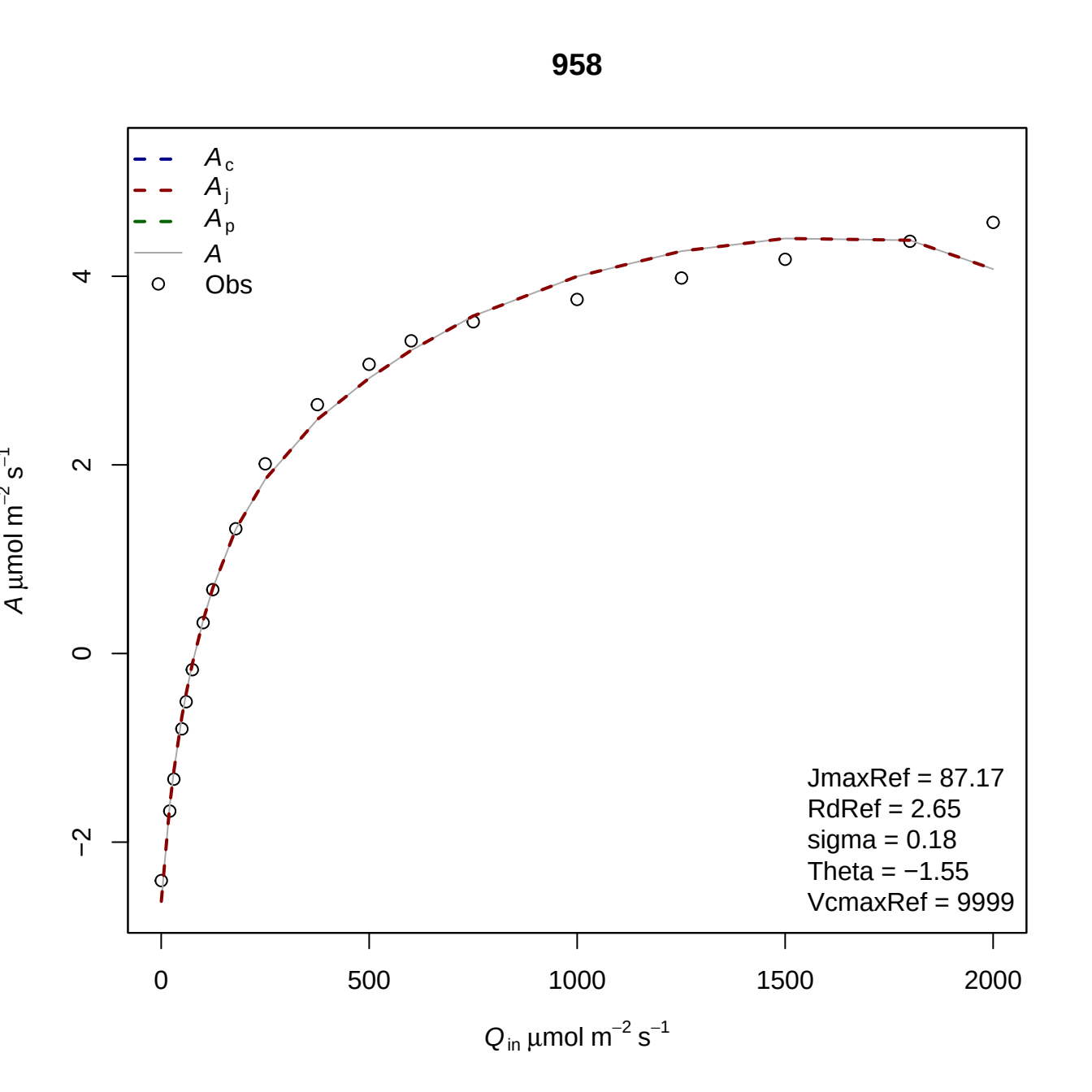

953
A_c
A_j
A_p
A
Obs
JmaxRef = 95.54
RdRef = 2.09
sigma = 0.19
Theta = −0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

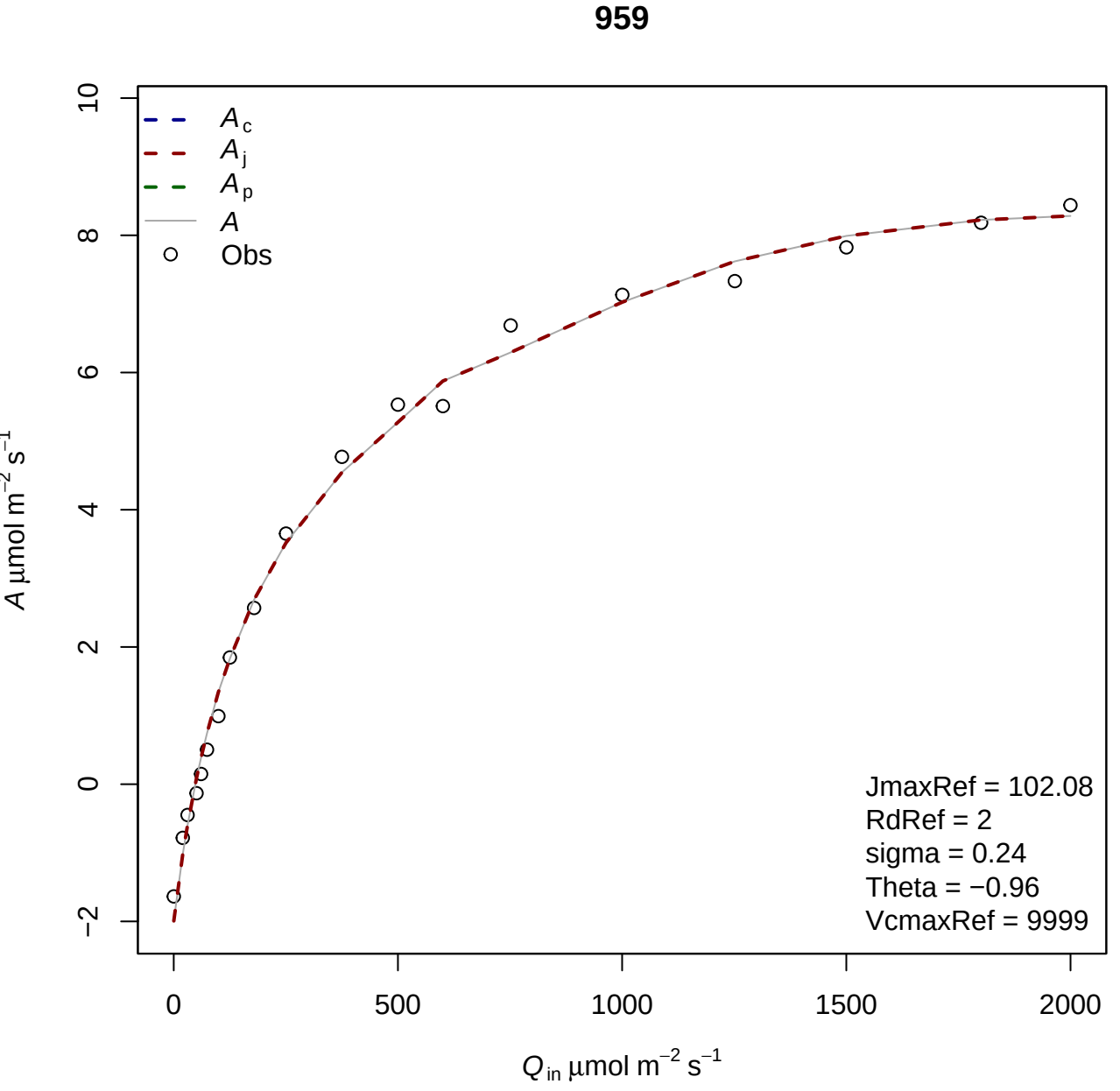

954
A_c
A_j
A_p
A
Obs
JmaxRef = 160.77
RdRef = 3.13
sigma = 0.26
Theta = 0.08
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

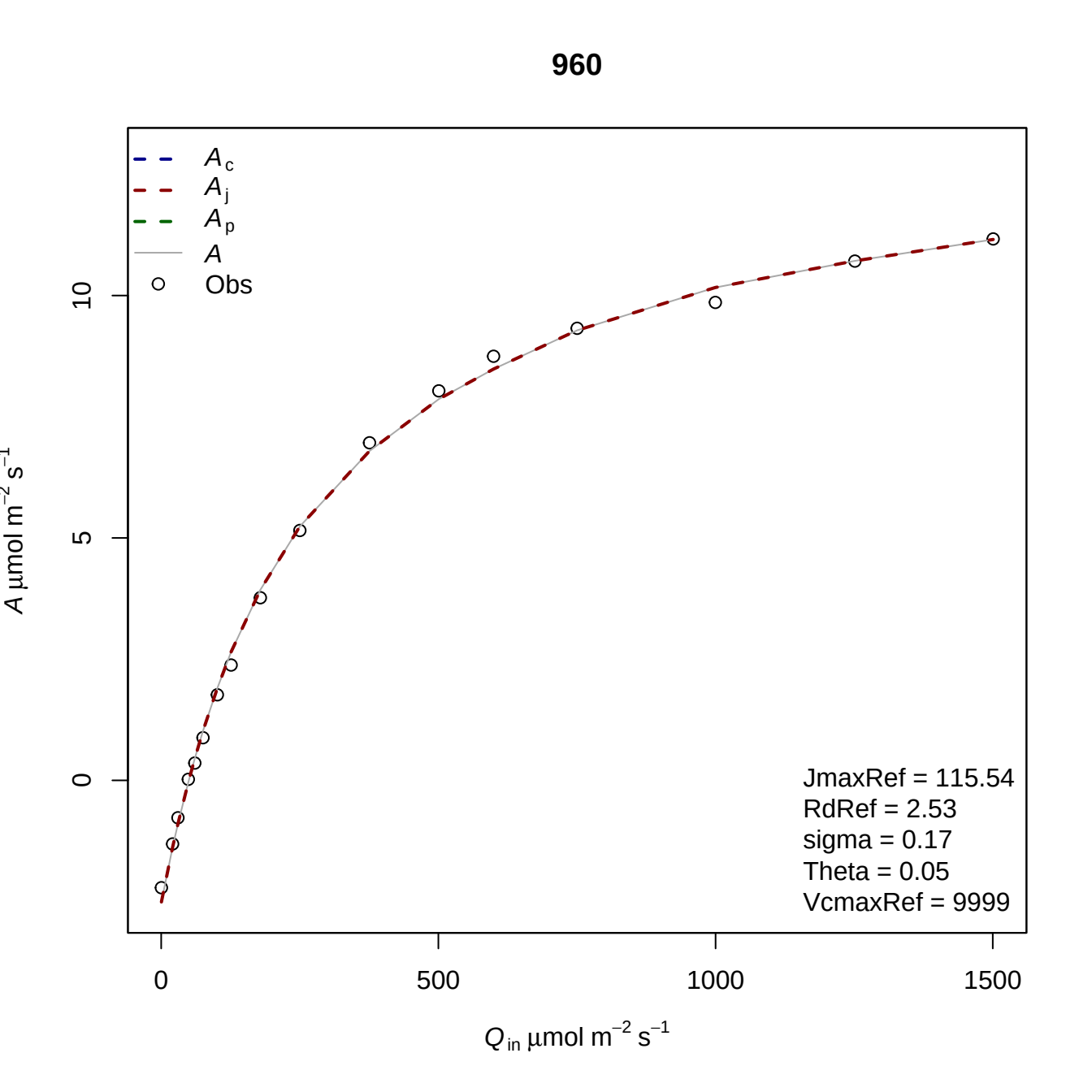

955
A_c
A_j
A_p
A
Obs
JmaxRef = 70.22
RdRef = 2.6
sigma = 0.29
Theta = −0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

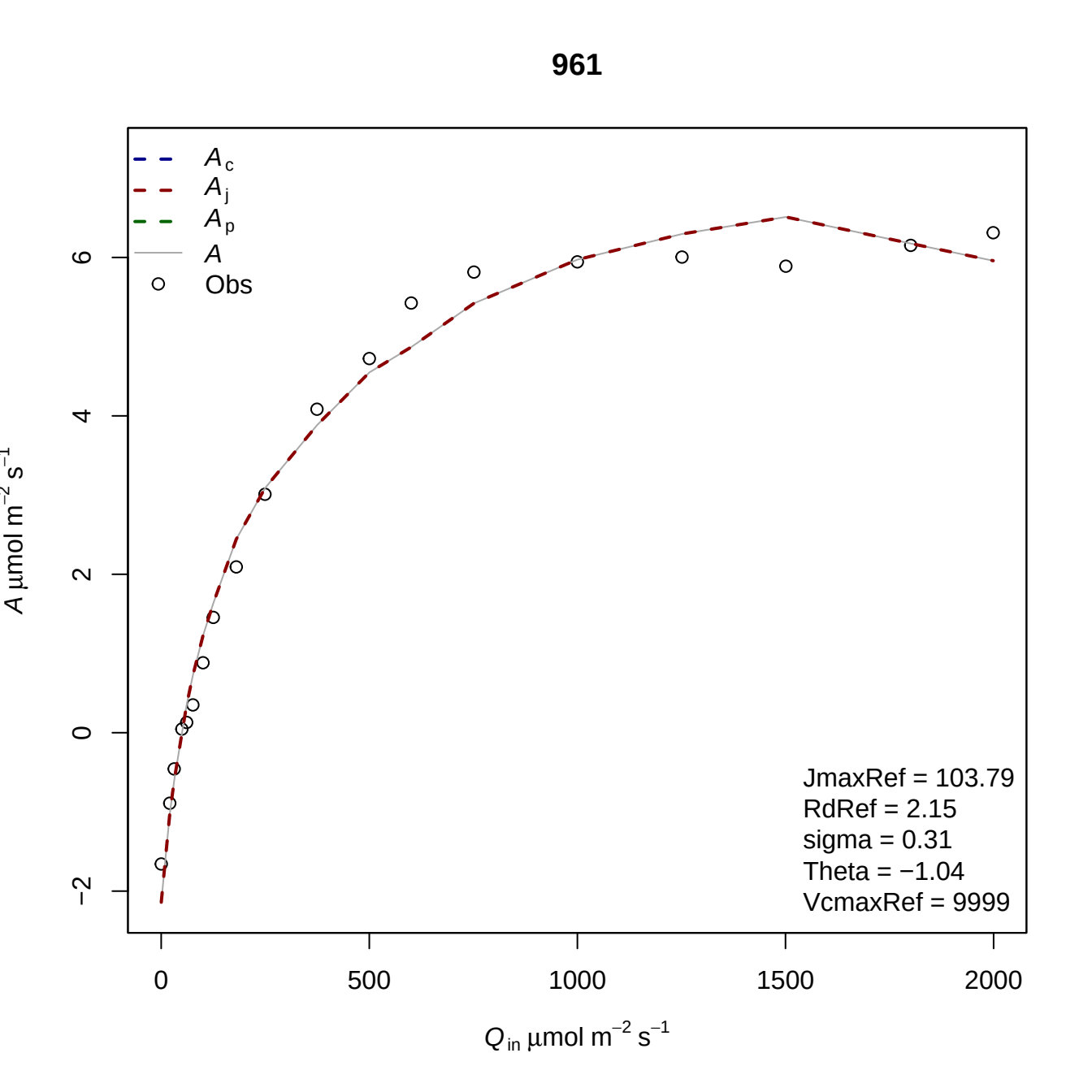

956
A_c
A_j
A_p
A
Obs
JmaxRef = 99.09
RdRef = 1.94
sigma = 0.33
Theta = −0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

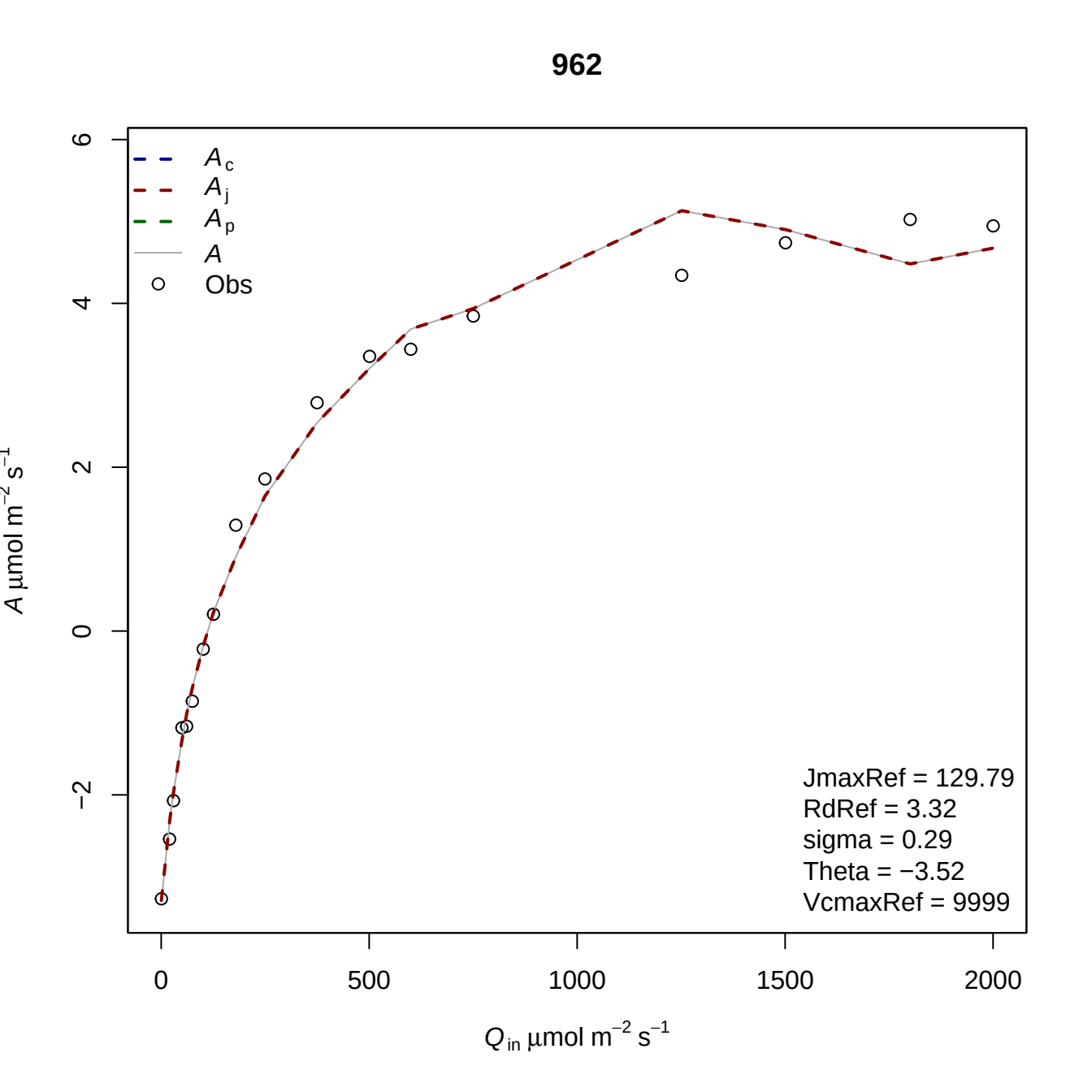

957
A_c
A_j
A_p
A
Obs
JmaxRef = 92.07
RdRef = 2.9
sigma = 0.21
Theta = −0.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

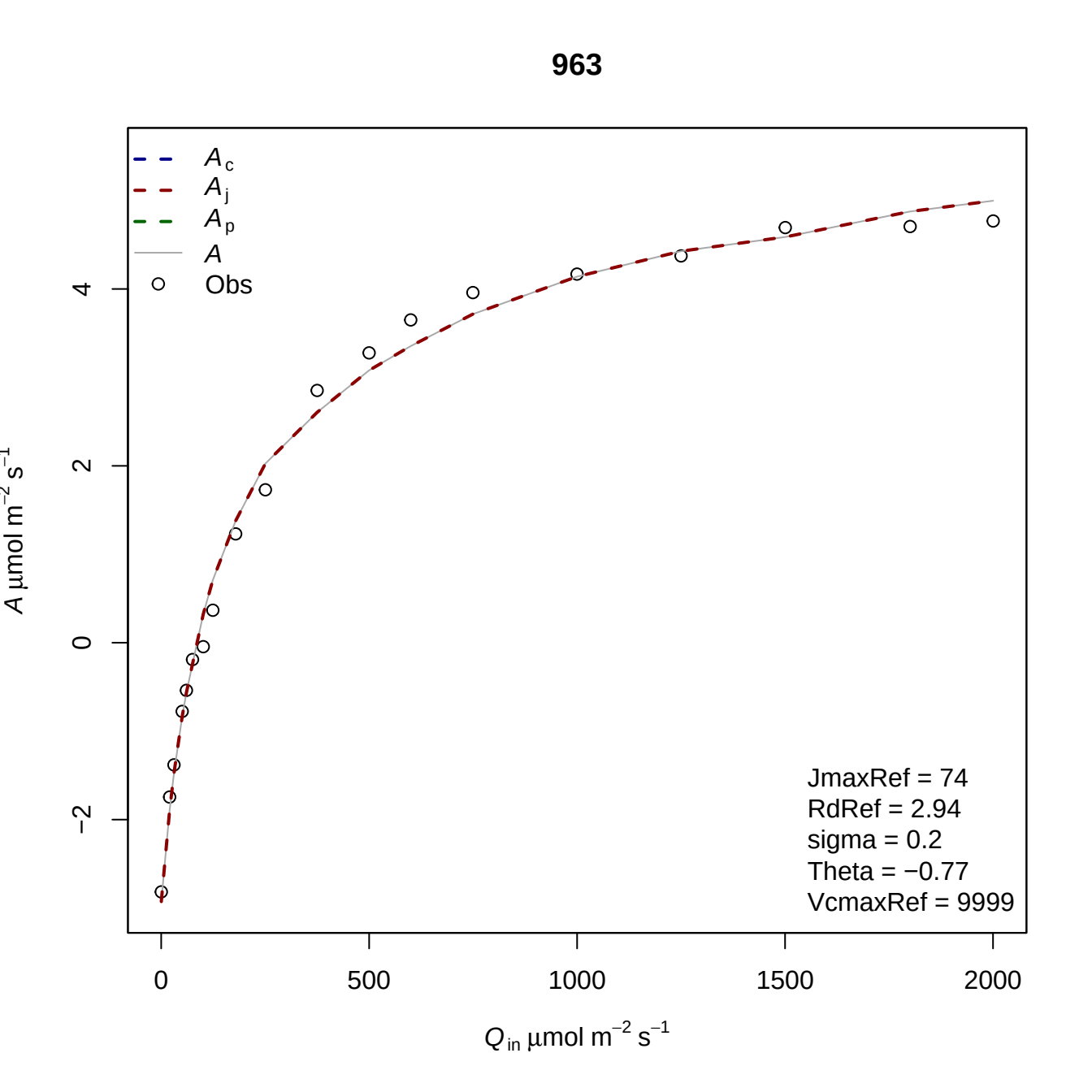

958
A_c
A_j
A_p
A
Obs
JmaxRef = 87.17
RdRef = 2.65
sigma = 0.18
Theta = −1.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

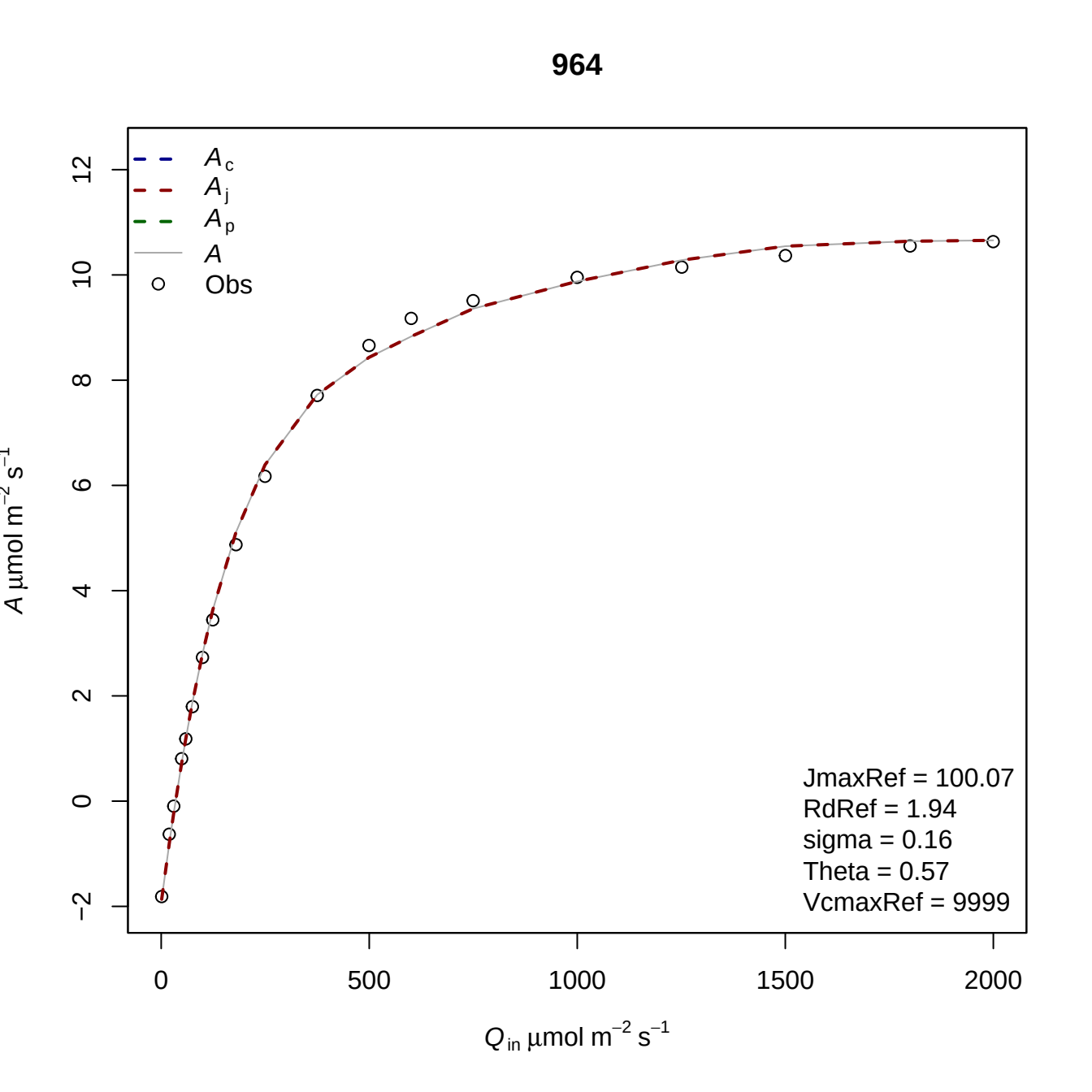

959
A_c
A_j
A_p
A
Obs
JmaxRef = 102.08
RdRef = 2
sigma = 0.24
Theta = −0.96
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

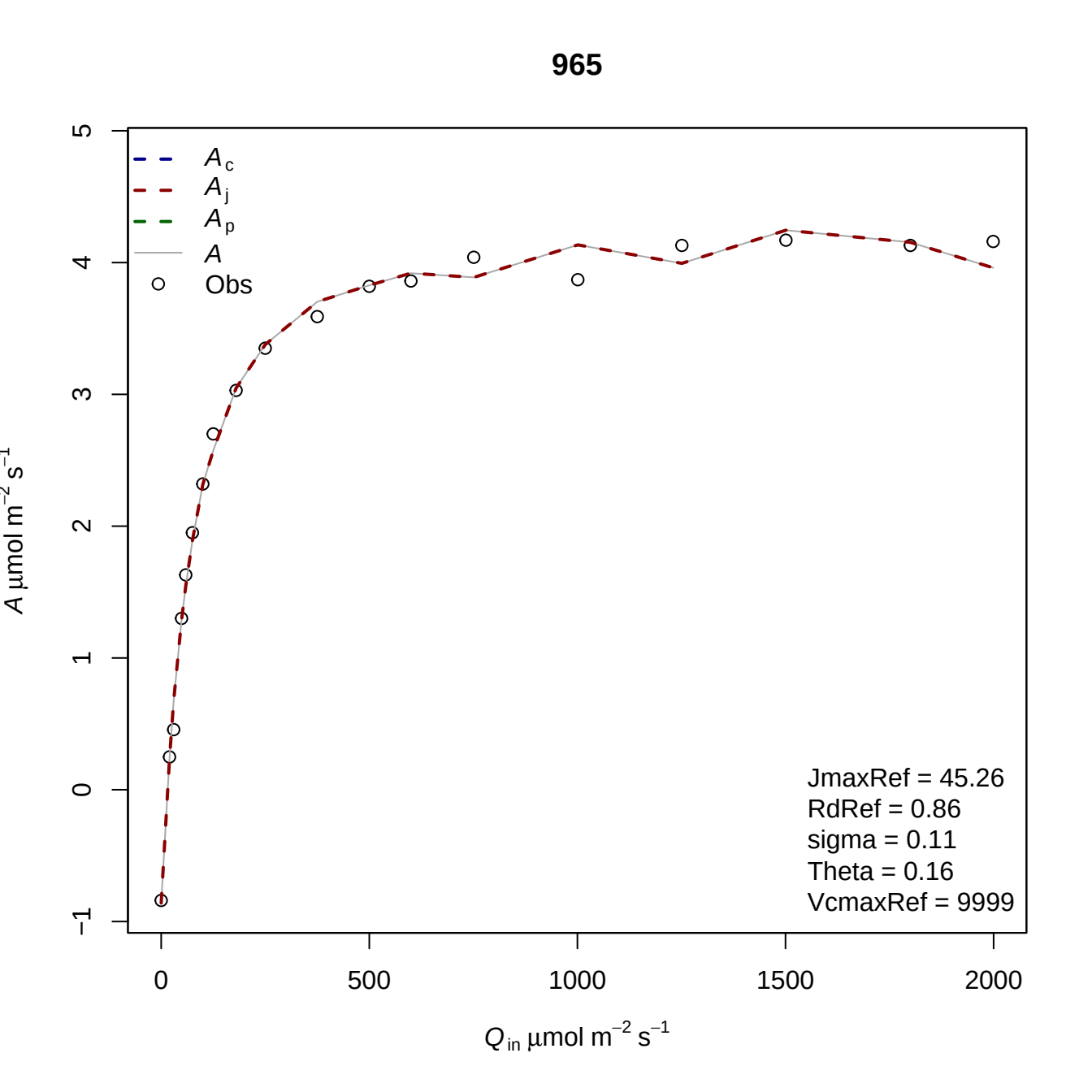

960
A_c
A_j
A_p
A
Obs
JmaxRef = 115.54
RdRef = 2.53
sigma = 0.17
Theta = 0.05
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

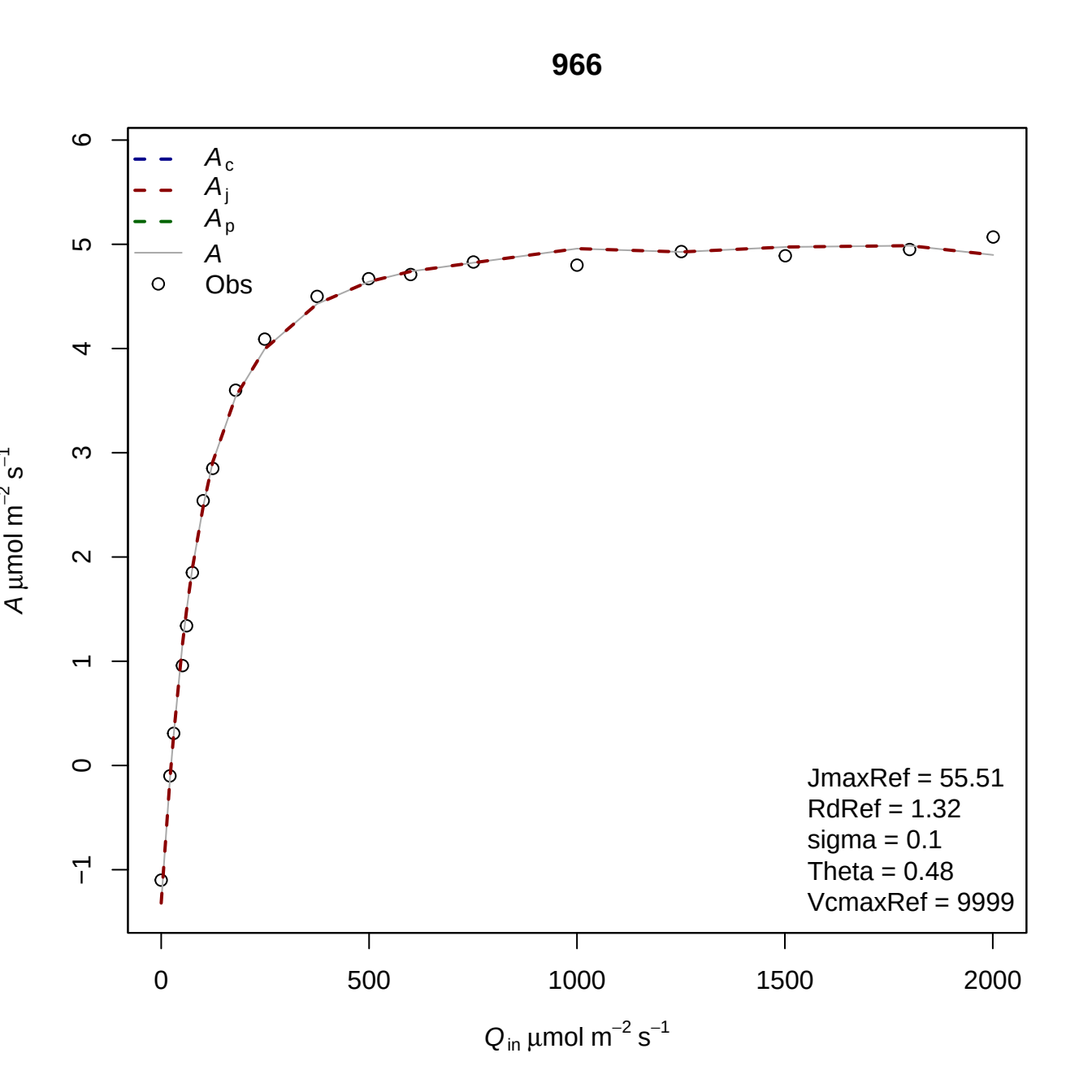

961
A_c
A_j
A_p
A
Obs
JmaxRef = 103.79
RdRef = 2.15
sigma = 0.31
Theta = −1.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

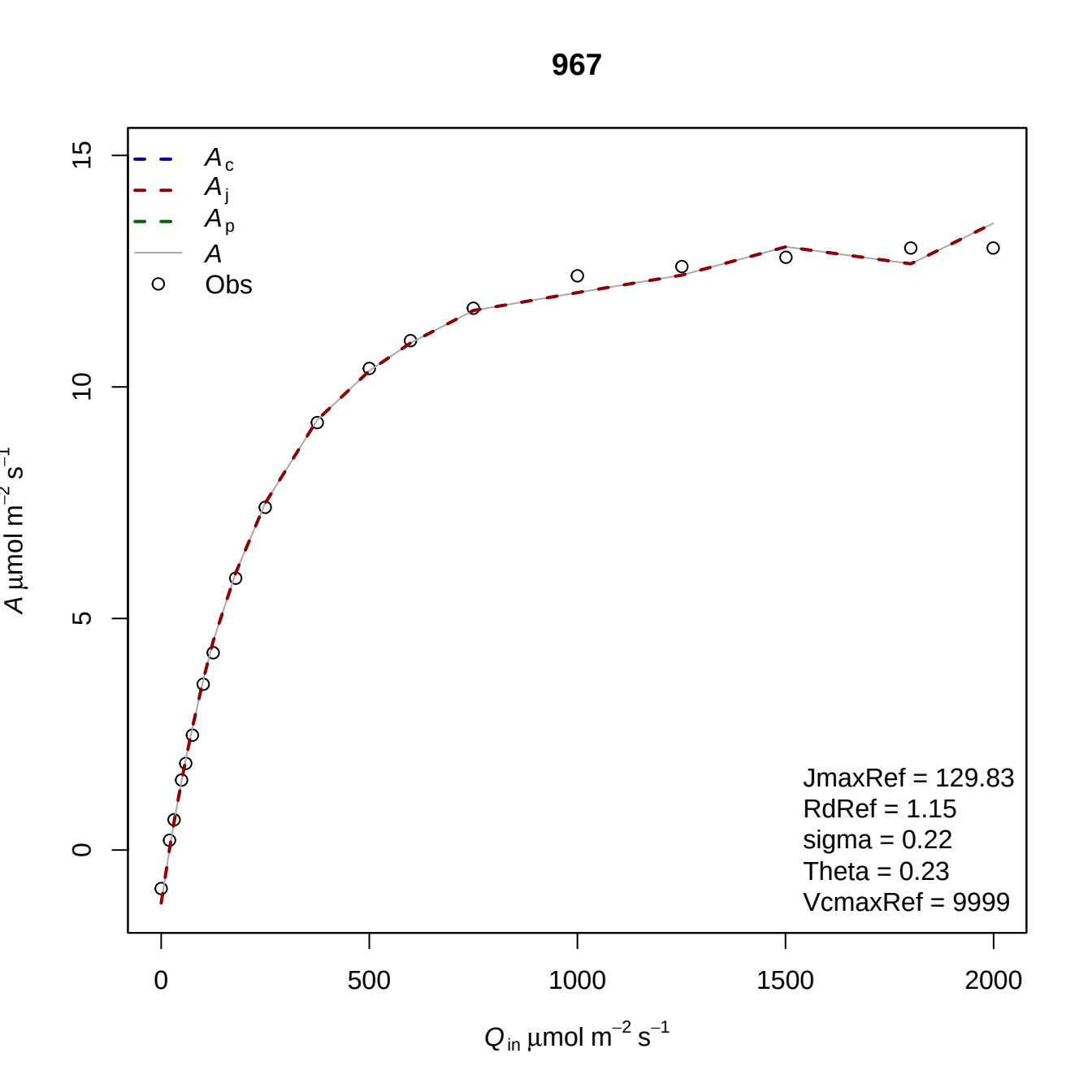

962
A_c
A_j
A_p
A
Obs
JmaxRef = 129.79
RdRef = 3.32
sigma = 0.29
Theta = −3.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

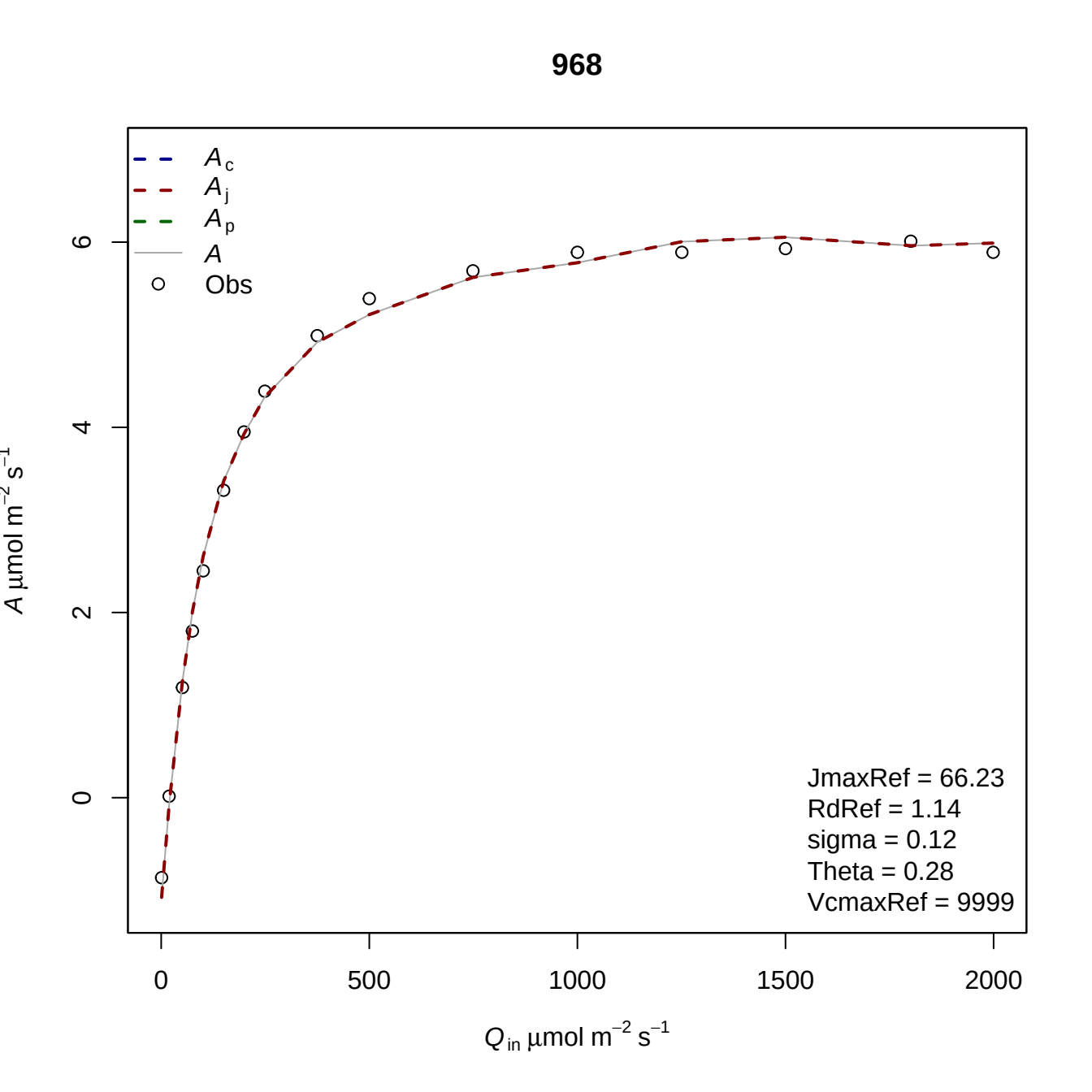

963
A_c
A_j
A_p
A
Obs
JmaxRef = 74
RdRef = 2.94
sigma = 0.2
Theta = −0.77
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

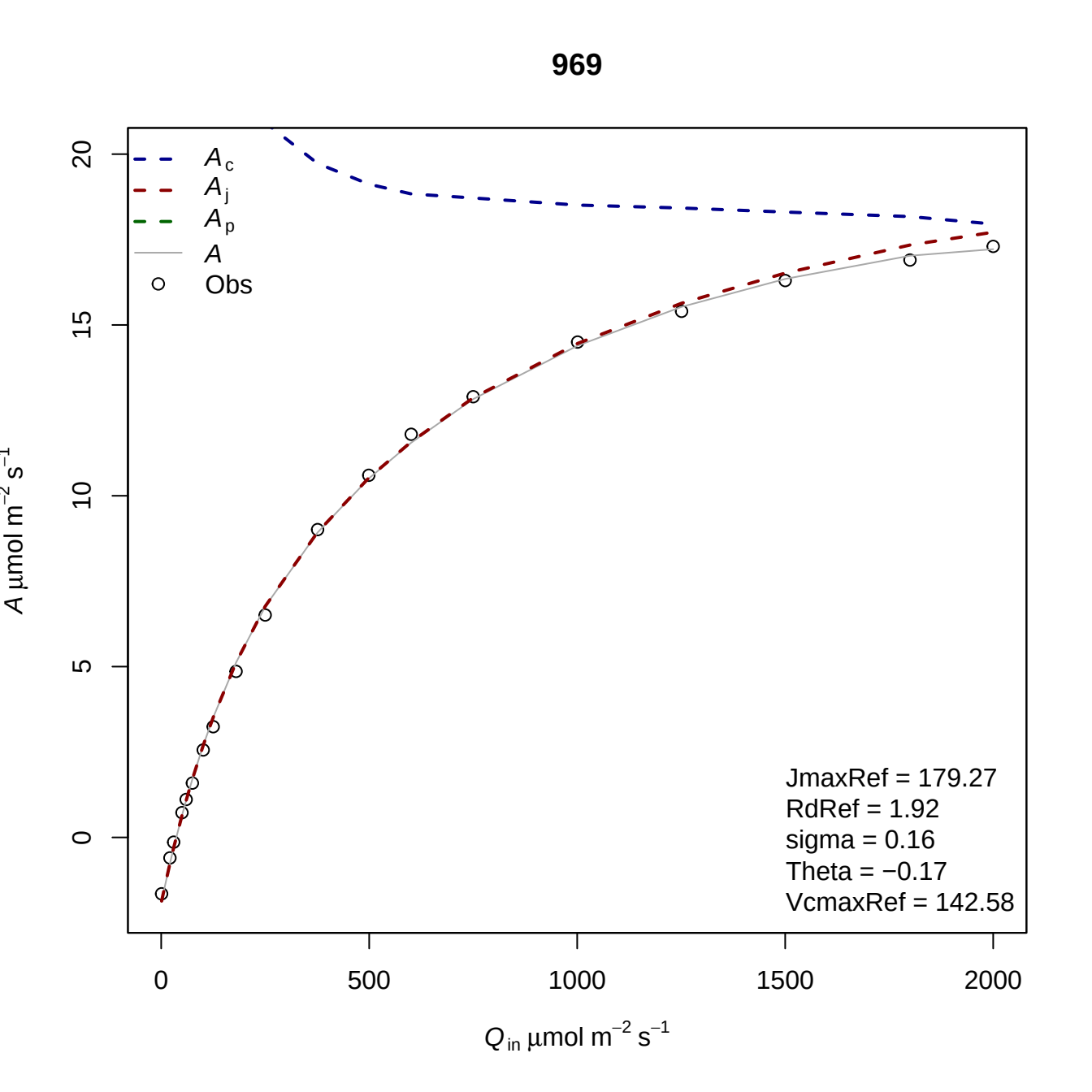

964
A_c
A_j
A_p
A
Obs
JmaxRef = 100.07
RdRef = 1.94
sigma = 0.16
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

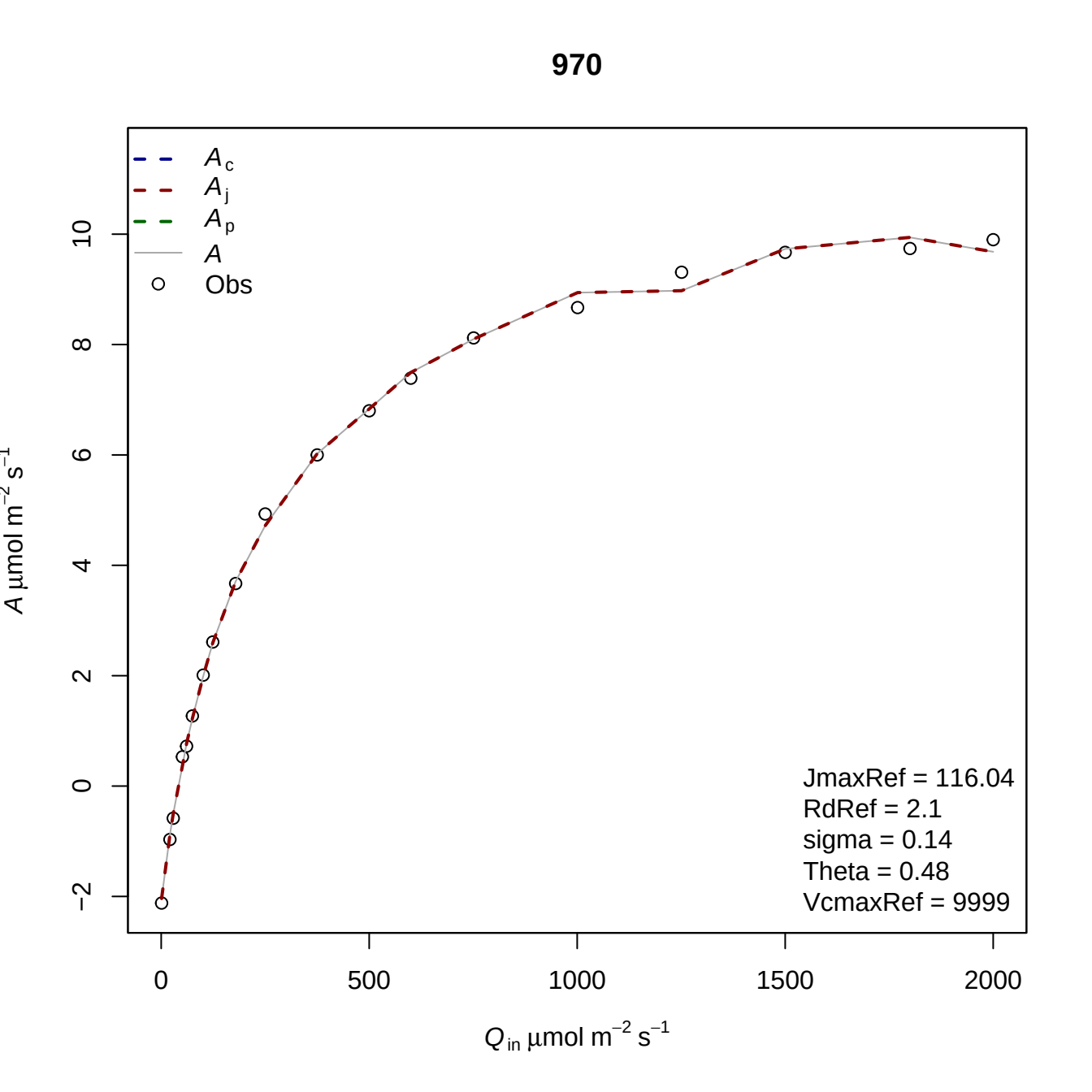

965
A_c
A_j
A_p
A
Obs
JmaxRef = 45.26
RdRef = 0.86
sigma = 0.11
Theta = 0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

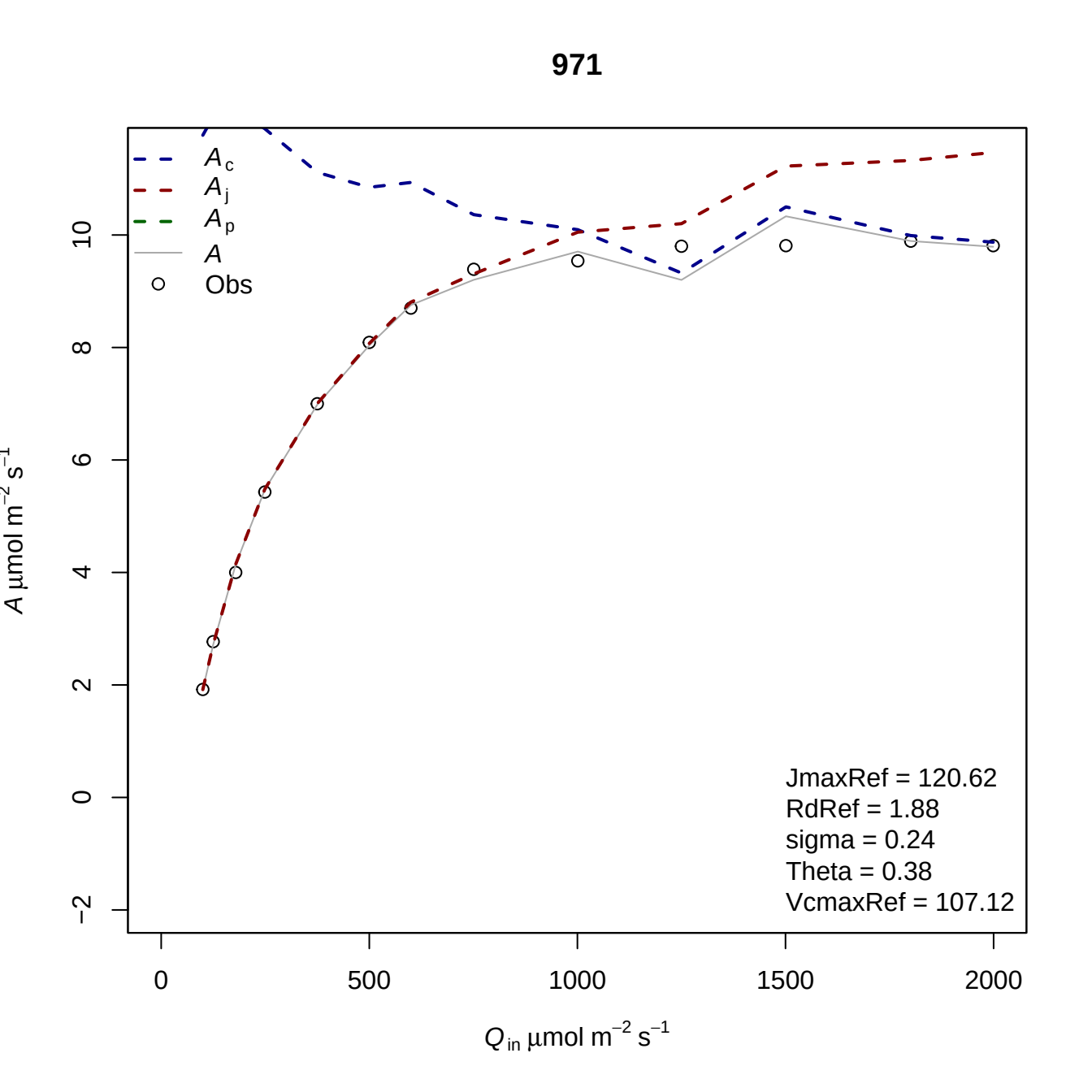

966
A_c
A_j
A_p
A
Obs
JmaxRef = 55.51
RdRef = 1.32
sigma = 0.1
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

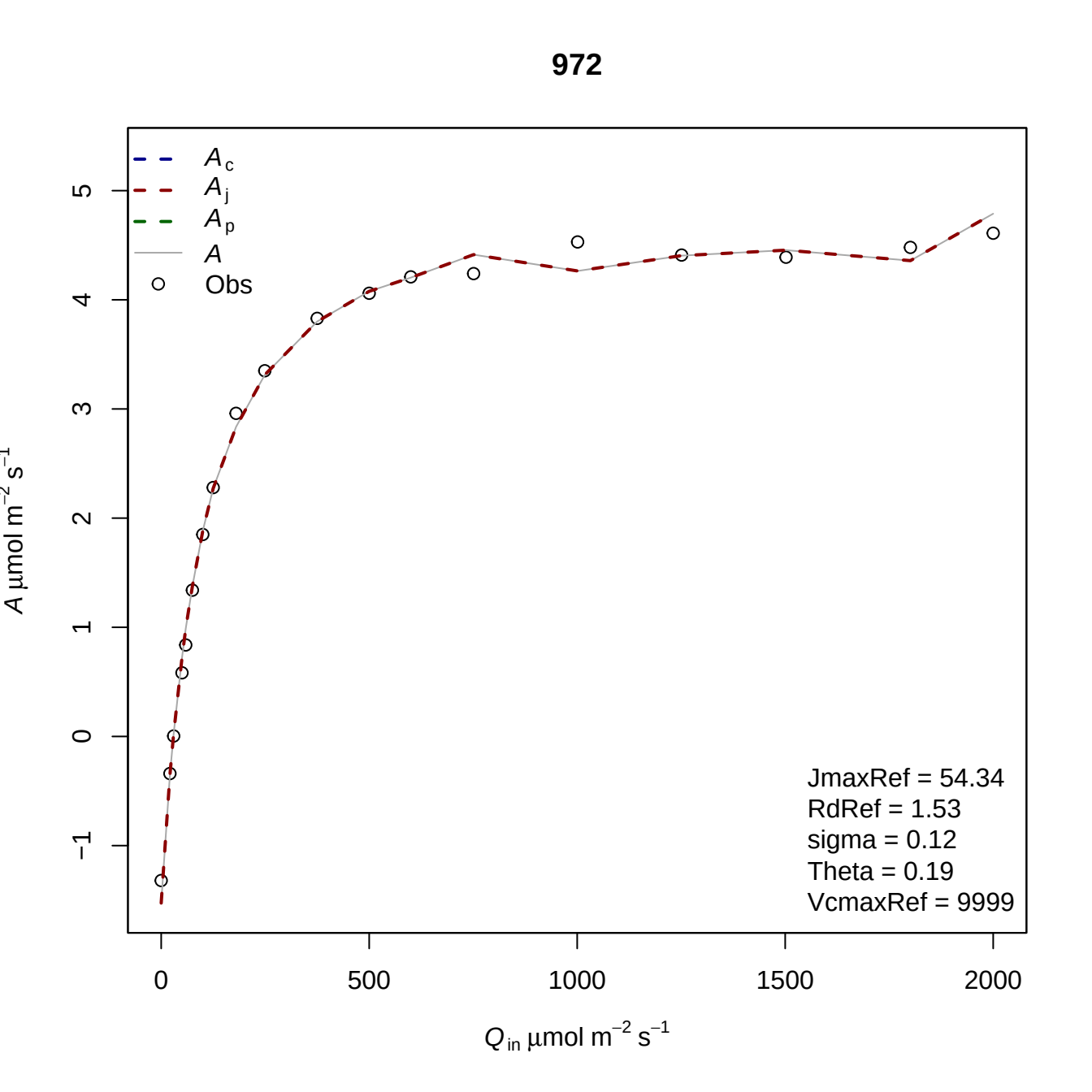

967
A_c
A_j
A_p
A
Obs
JmaxRef = 129.83
RdRef = 1.15
sigma = 0.22
Theta = 0.23
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

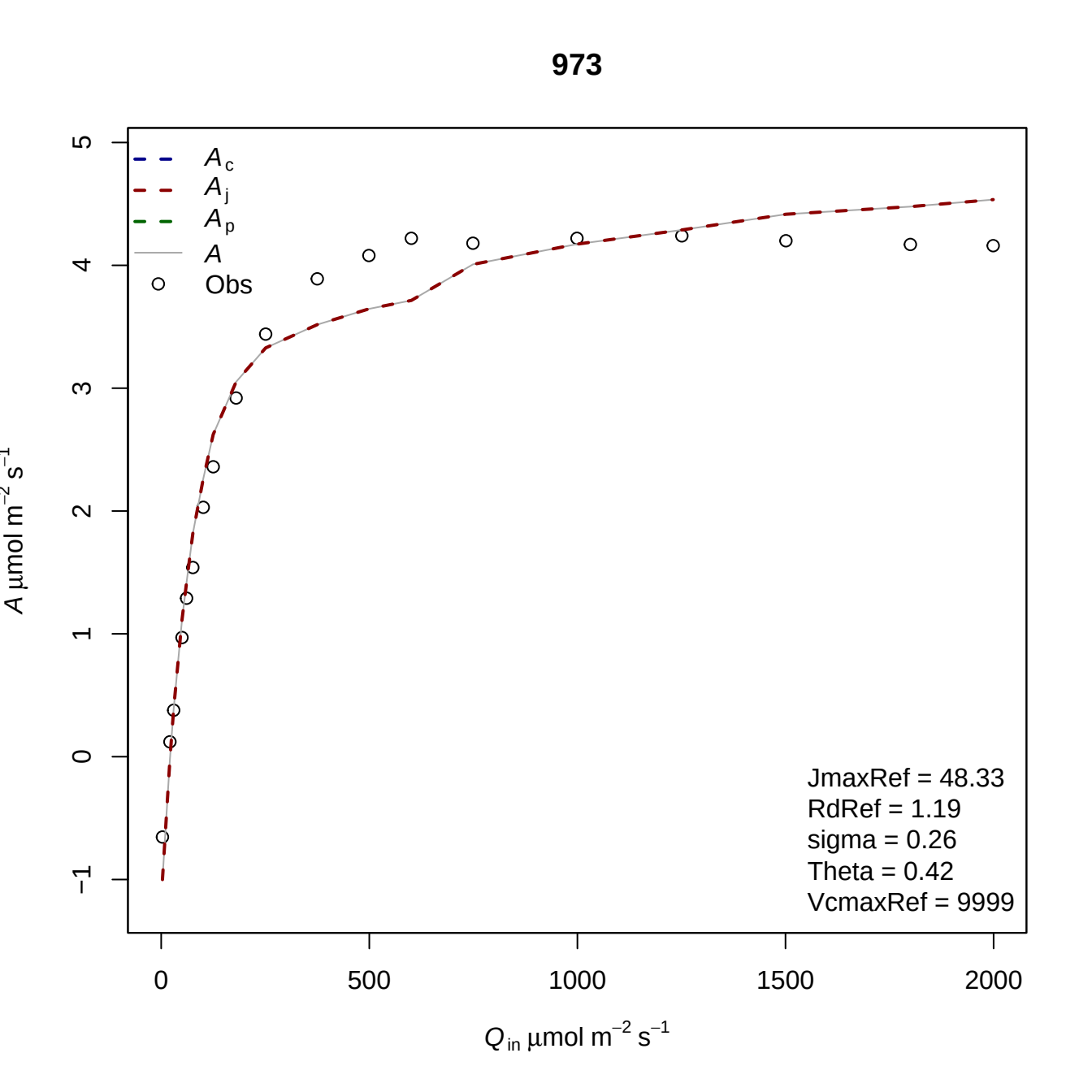

968
A_c
A_j
A_p
A
Obs
JmaxRef = 66.23
RdRef = 1.14
sigma = 0.12
Theta = 0.28
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

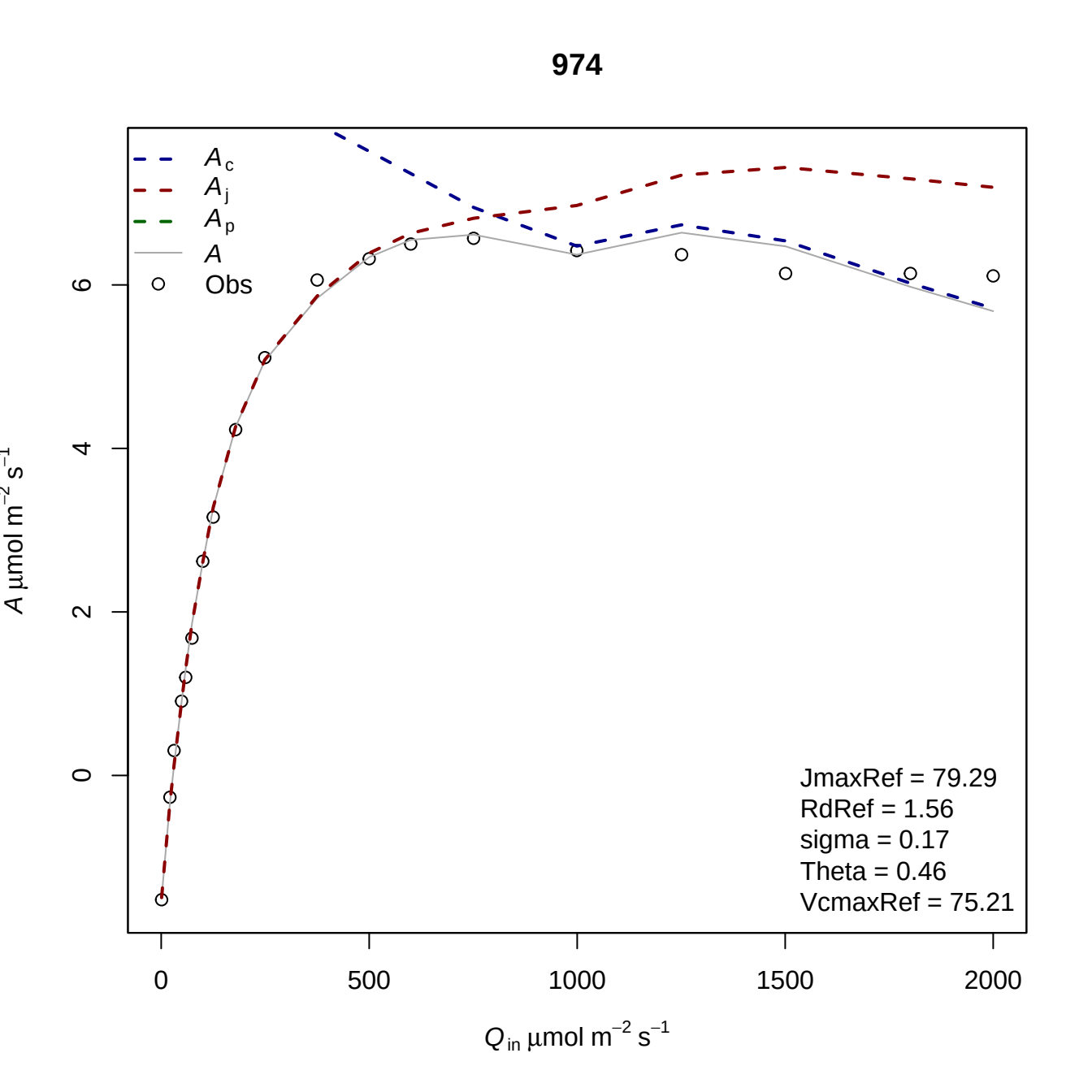

969
A_c
A_j
A_p
A
Obs
JmaxRef = 179.27
RdRef = 1.92
sigma = 0.16
Theta = −0.17
VcmaxRef = 142.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

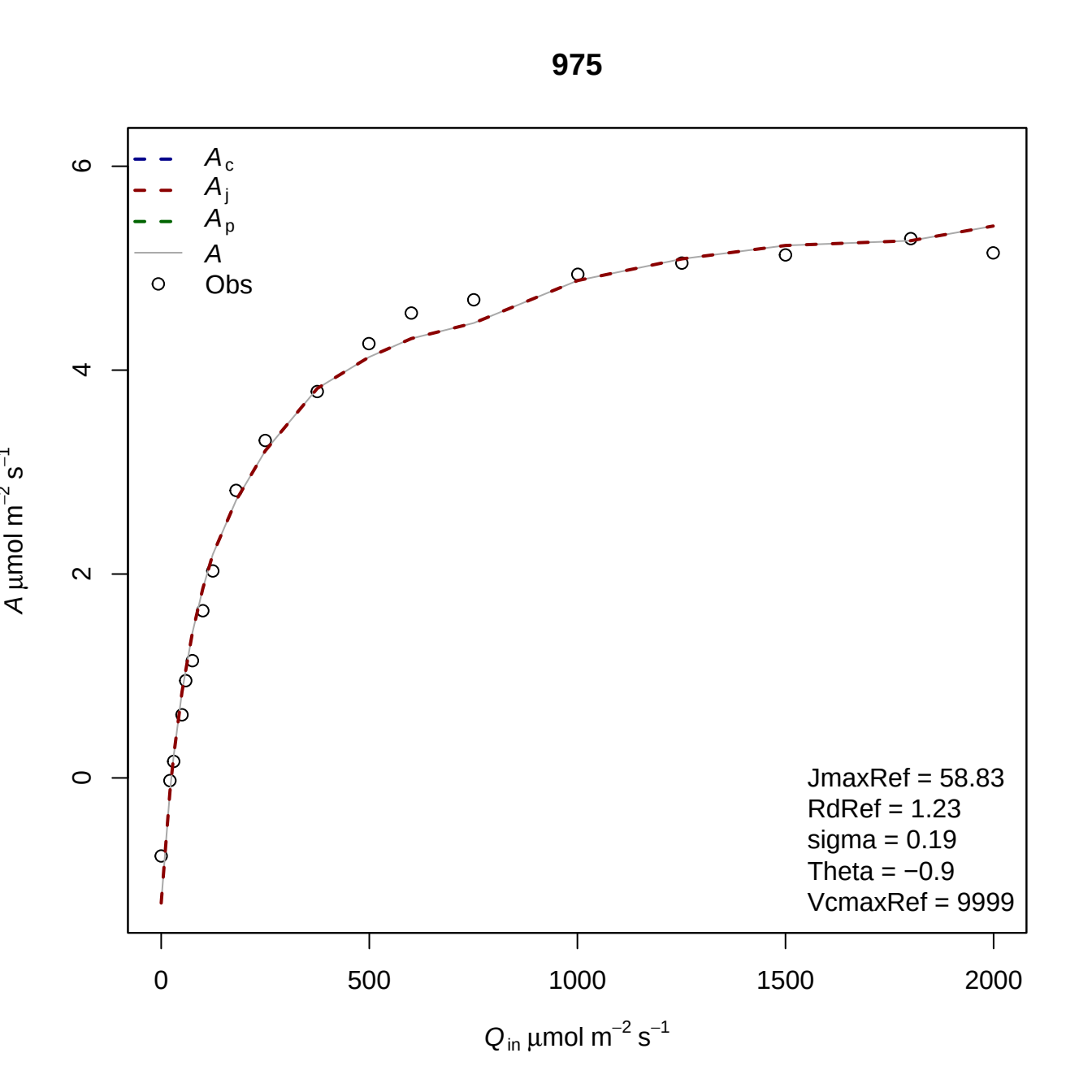

970
A_c
A_j
A_p
A
Obs
JmaxRef = 116.04
RdRef = 2.1
sigma = 0.14
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

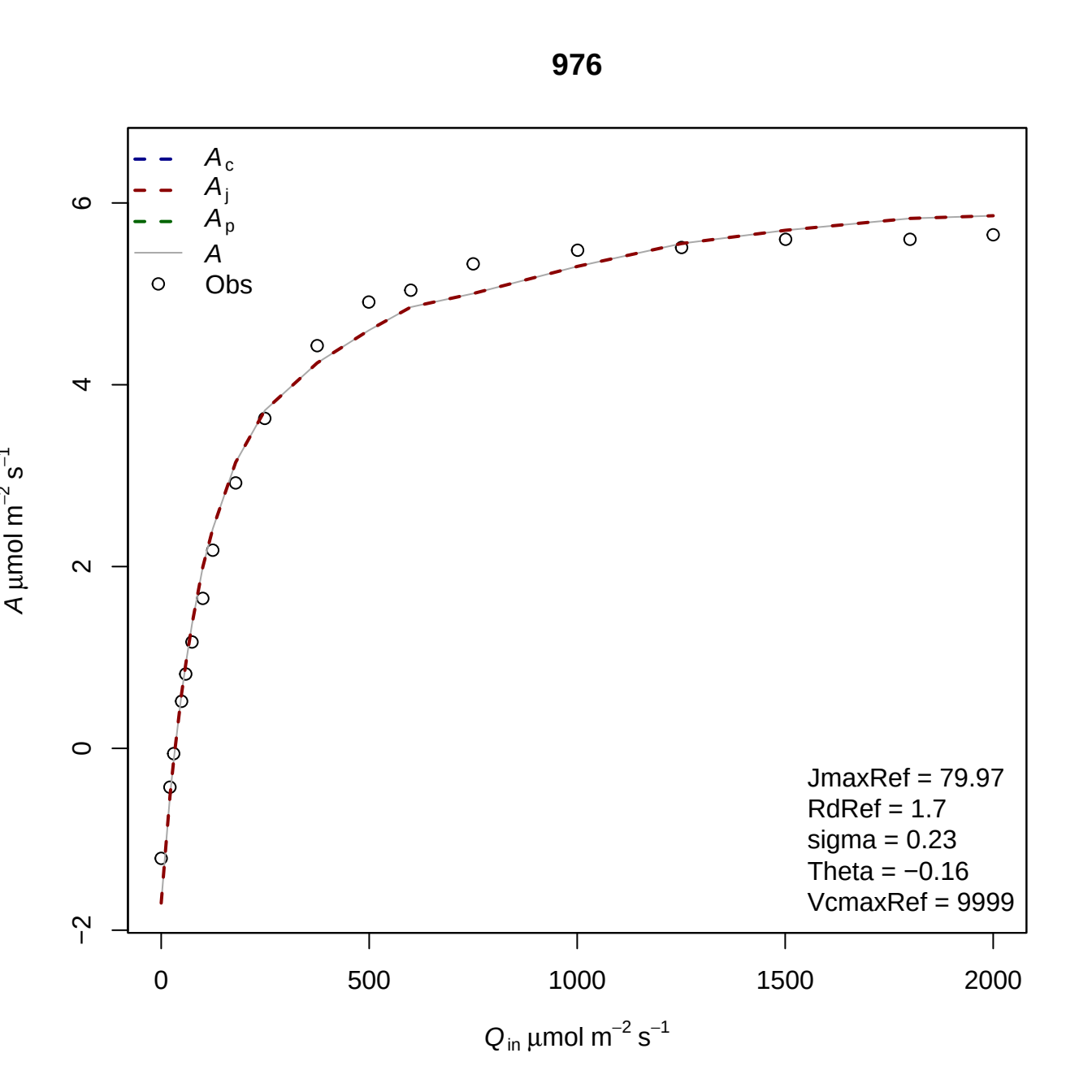

971
A_c
A_j
A_p
A
Obs
JmaxRef = 120.62
RdRef = 1.88
sigma = 0.24
Theta = 0.38
VcmaxRef = 107.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

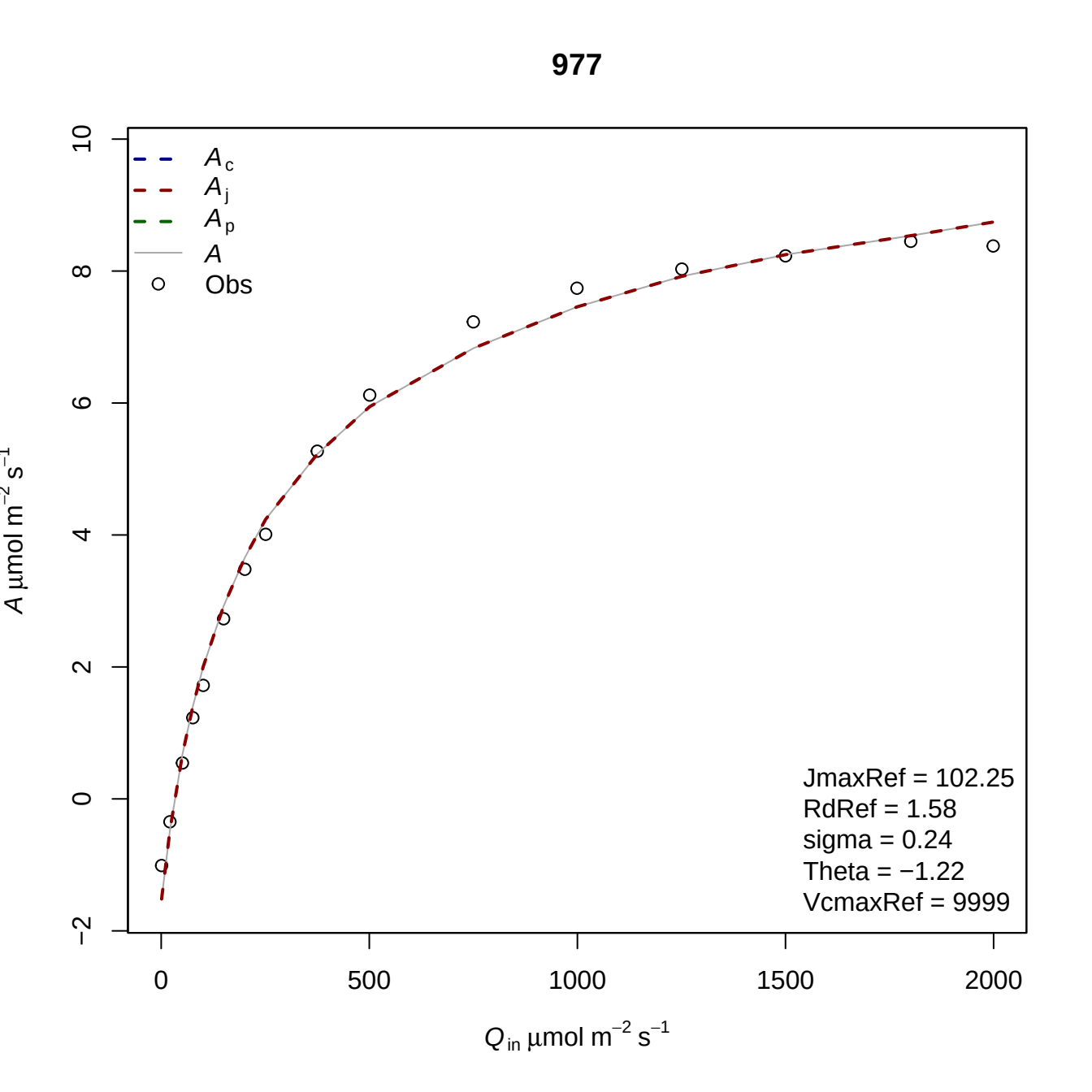

972
A_c
A_j
A_p
A
Obs
JmaxRef = 54.34
RdRef = 1.53
sigma = 0.12
Theta = 0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

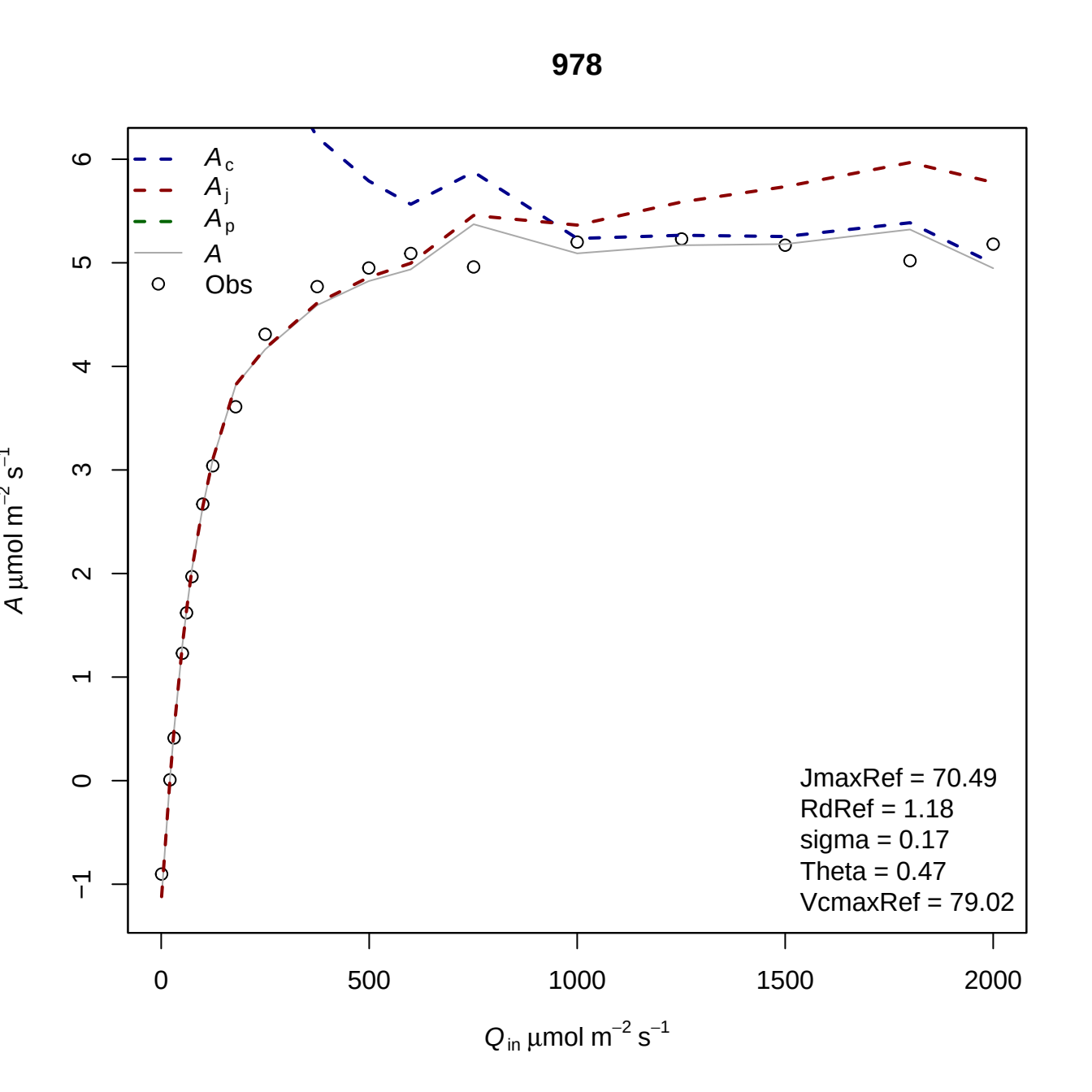

973
A_c
A_j
A_p
A
Obs
JmaxRef = 48.33
RdRef = 1.19
sigma = 0.26
Theta = 0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

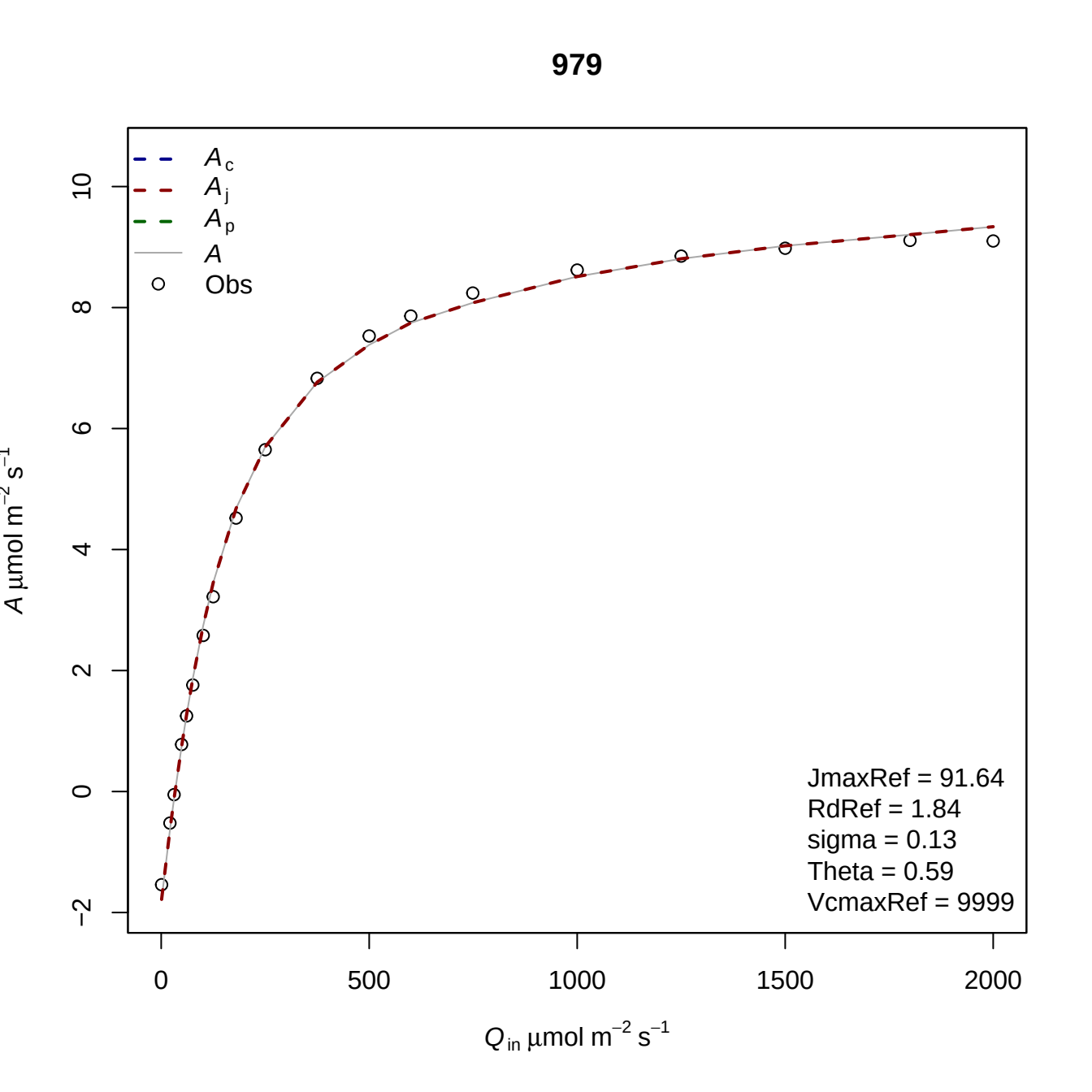

974
A_c
A_j
A_p
A
Obs
JmaxRef = 79.29
RdRef = 1.56
sigma = 0.17
Theta = 0.46
VcmaxRef = 75.21
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

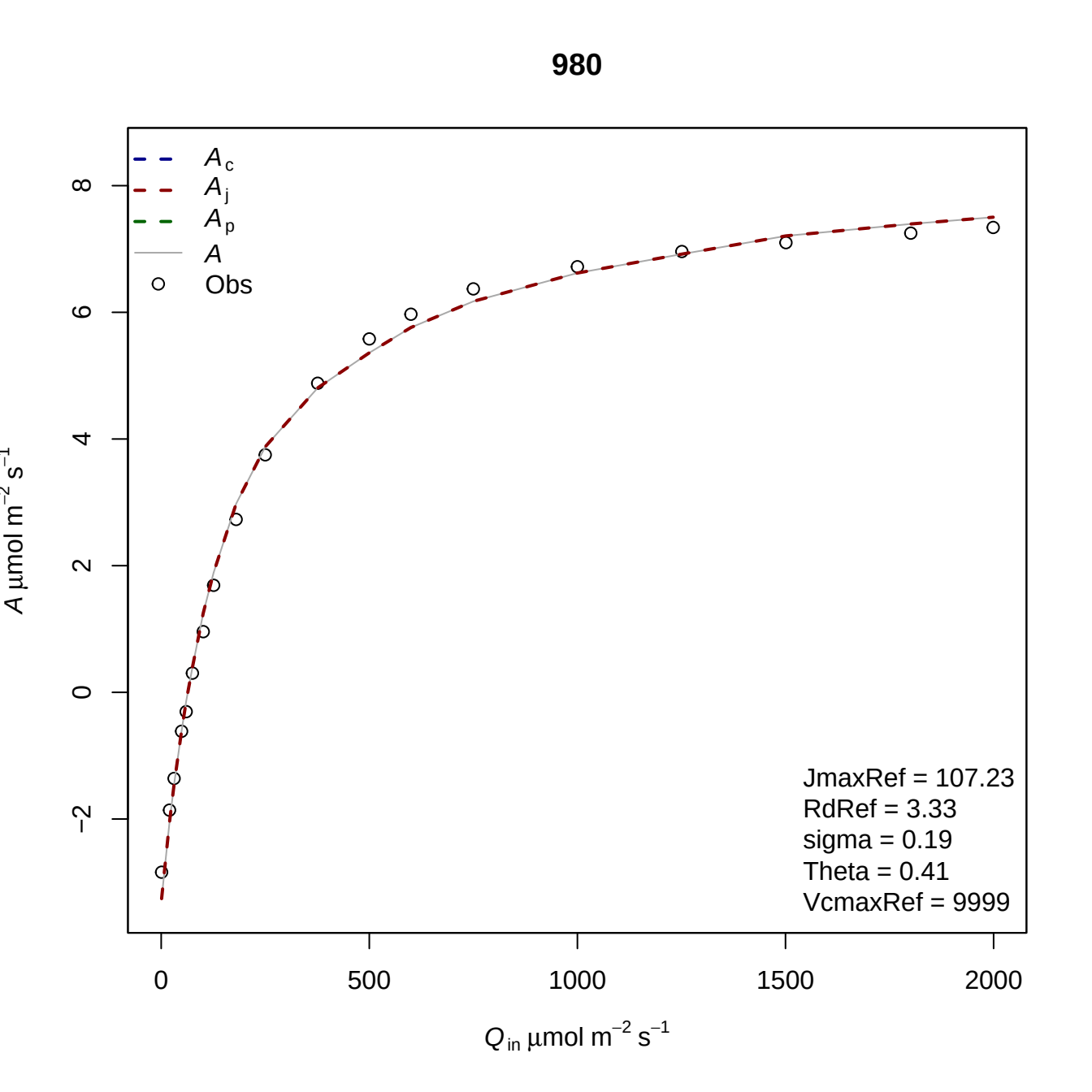

975
A_c
A_j
A_p
A
Obs
JmaxRef = 58.83
RdRef = 1.23
sigma = 0.19
Theta = −0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

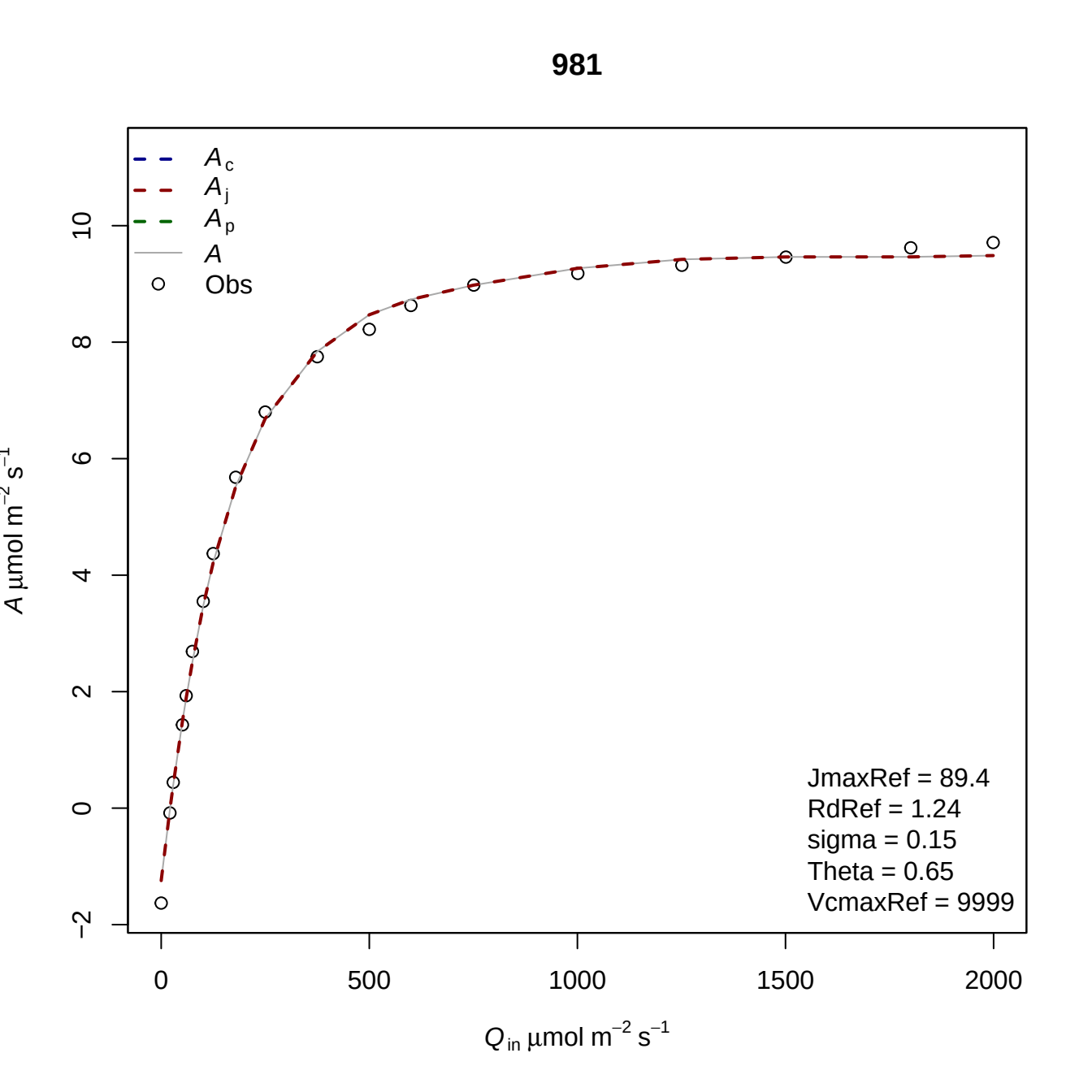

976
A_c
A_j
A_p
A
Obs
JmaxRef = 79.97
RdRef = 1.7
sigma = 0.23
Theta = −0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

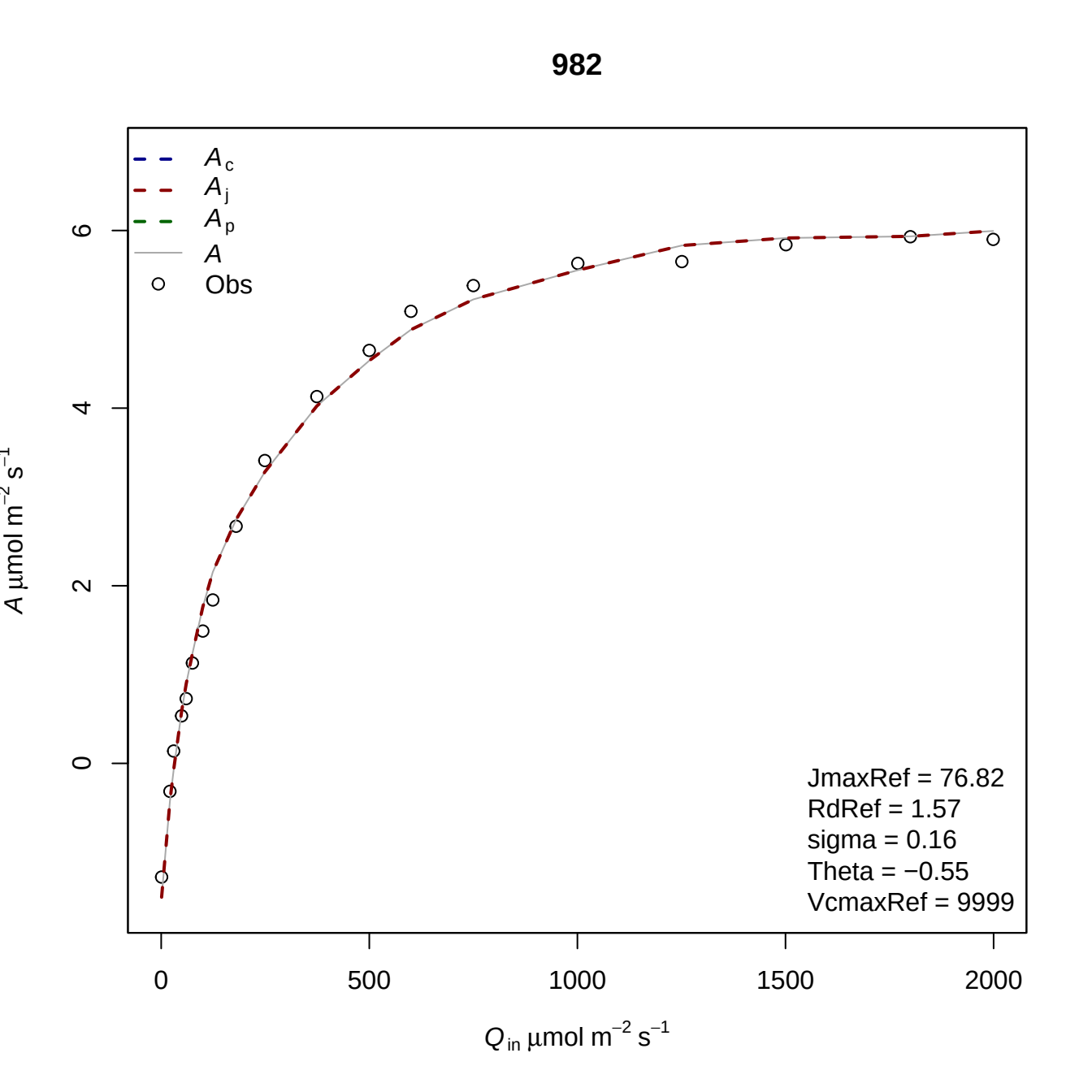

977
A_c
A_j
A_p
A
Obs
JmaxRef = 102.25
RdRef = 1.58
sigma = 0.24
Theta = −1.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

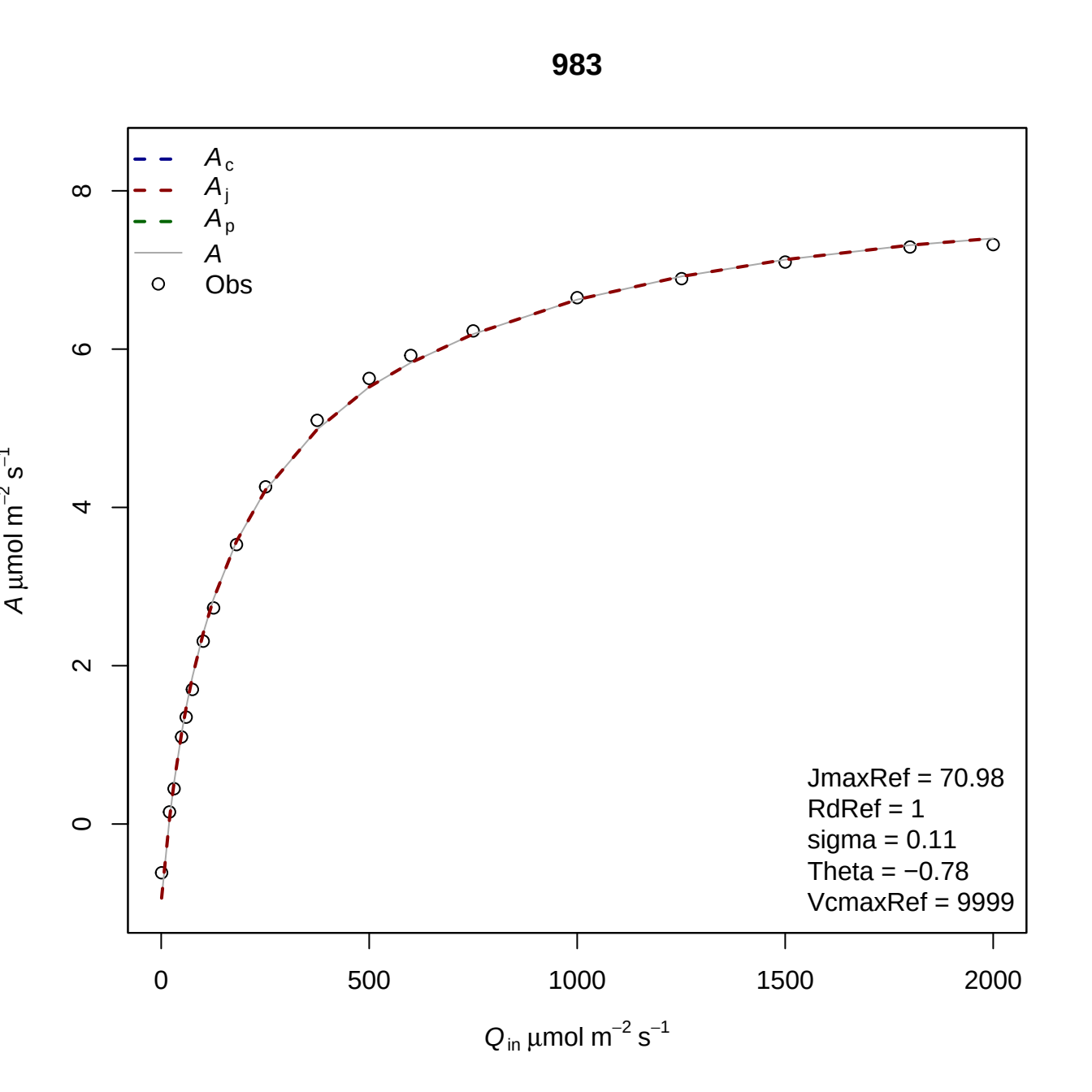

978
A_c
A_j
A_p
A
Obs
JmaxRef = 70.49
RdRef = 1.18
sigma = 0.17
Theta = 0.47
VcmaxRef = 79.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

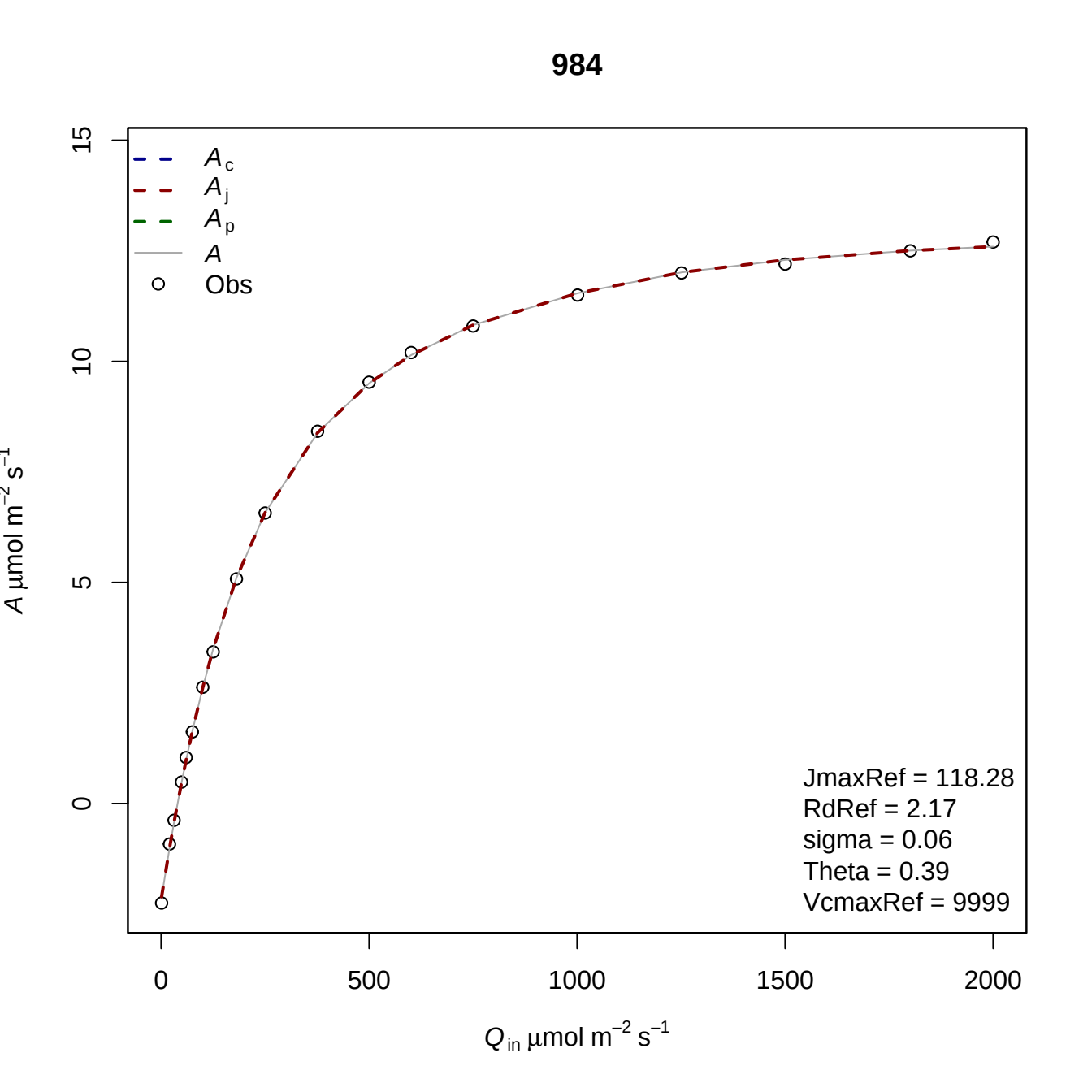

979
A_c
A_j
A_p
A
Obs
JmaxRef = 91.64
RdRef = 1.84
sigma = 0.13
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

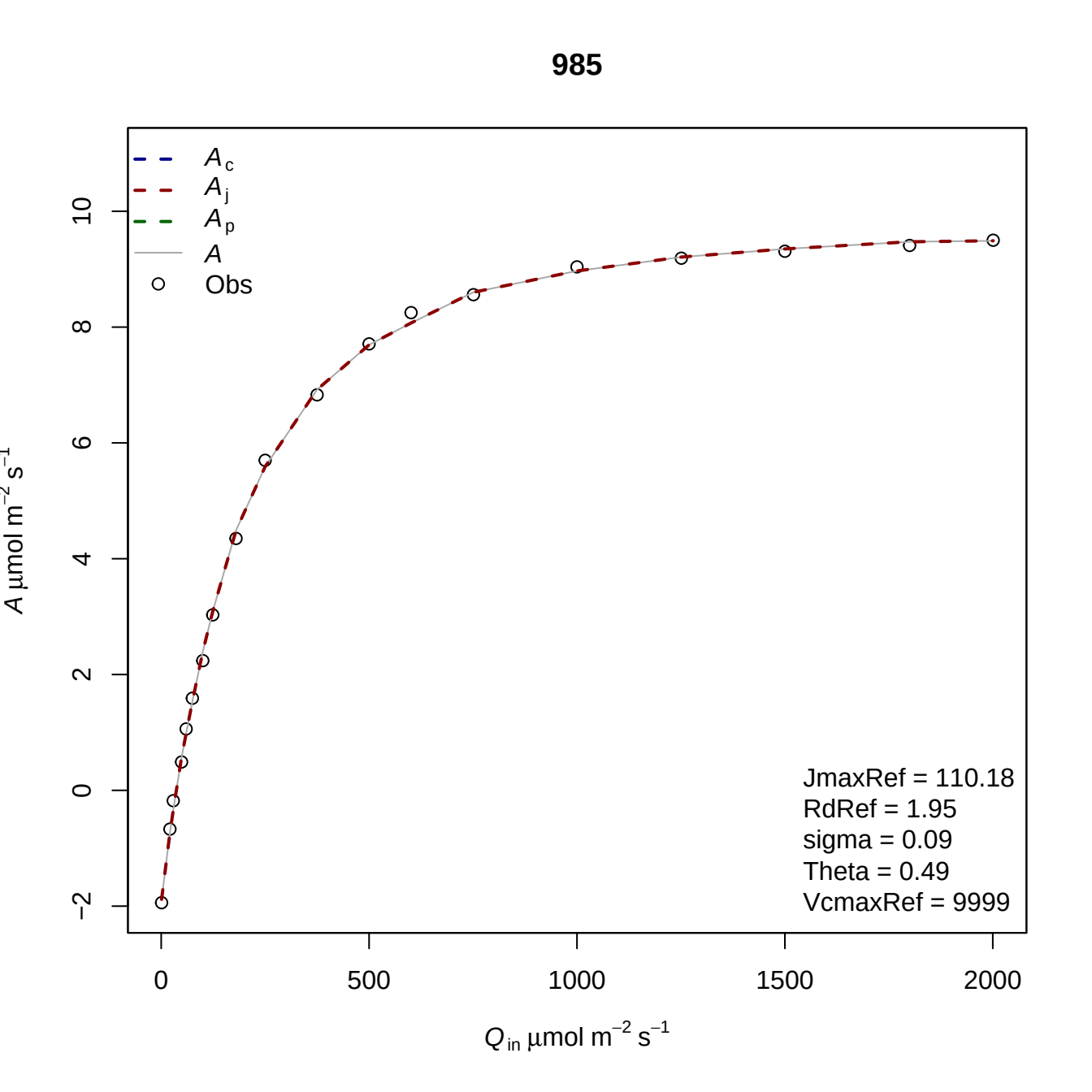

980
A_c
A_j
A_p
A
Obs
JmaxRef = 107.23
RdRef = 3.33
sigma = 0.19
Theta = 0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

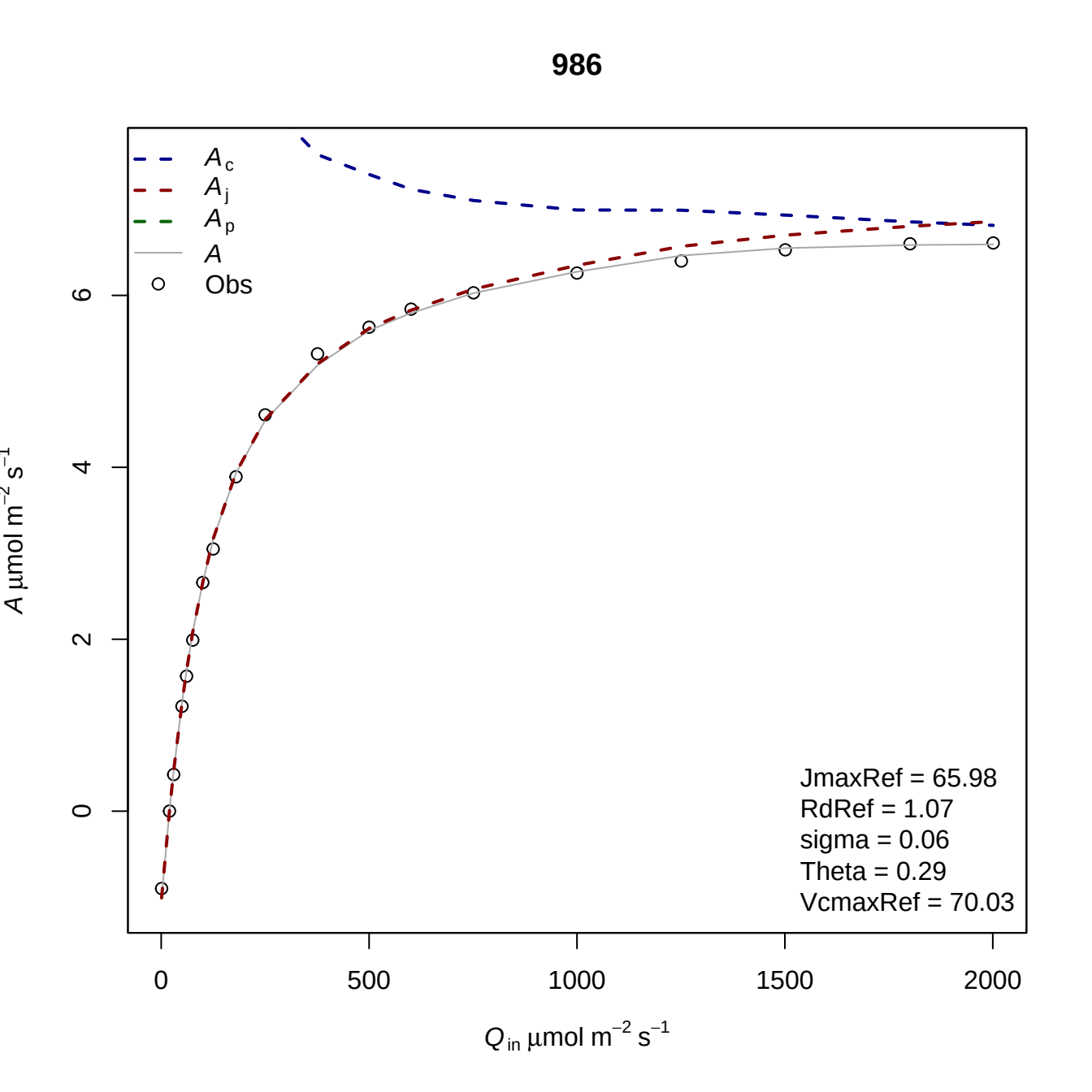

981
A_c
A_j
A_p
A
Obs
JmaxRef = 89.4
RdRef = 1.24
sigma = 0.15
Theta = 0.65
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

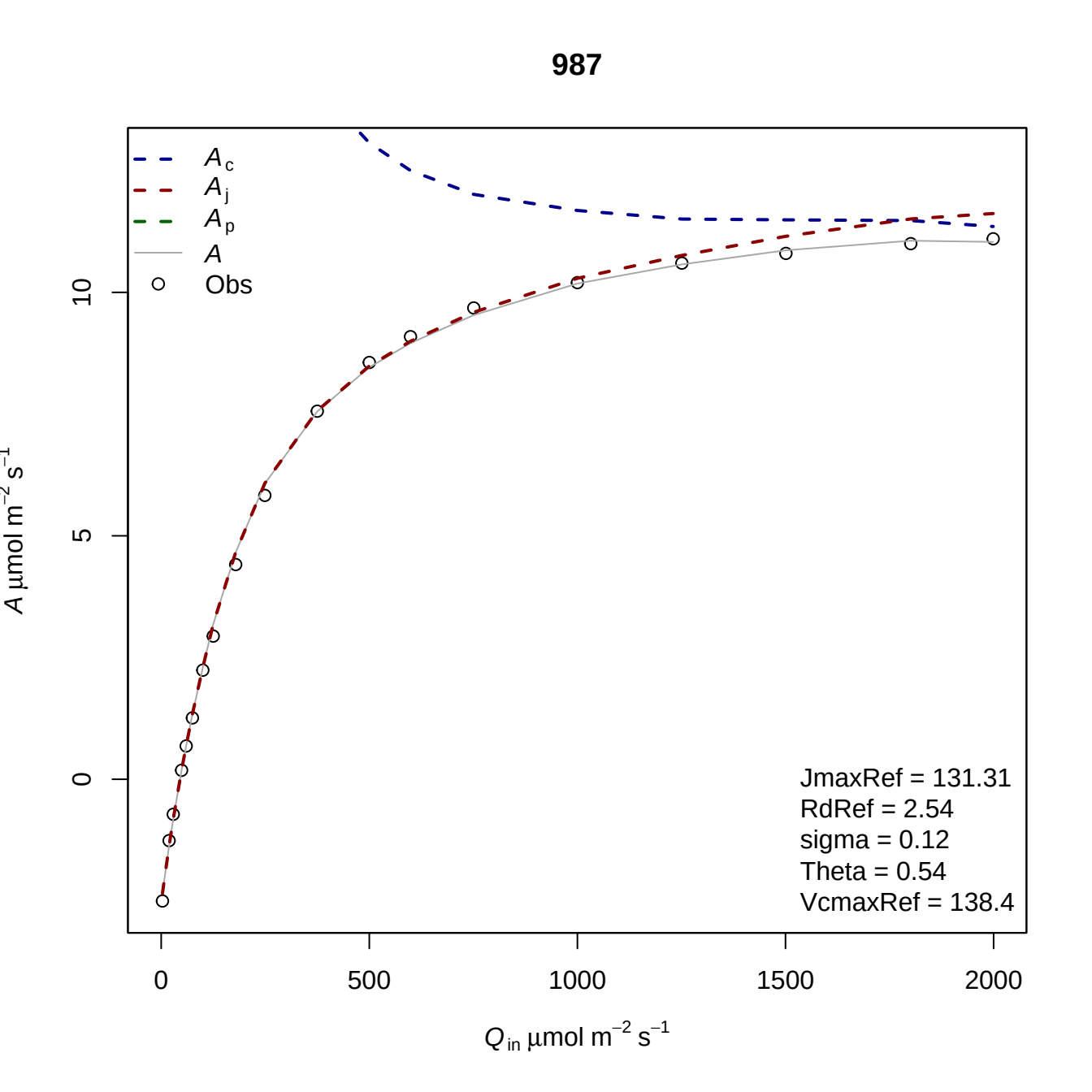

982
A_c
A_j
A_p
A
Obs
JmaxRef = 76.82
RdRef = 1.57
sigma = 0.16
Theta = −0.55
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

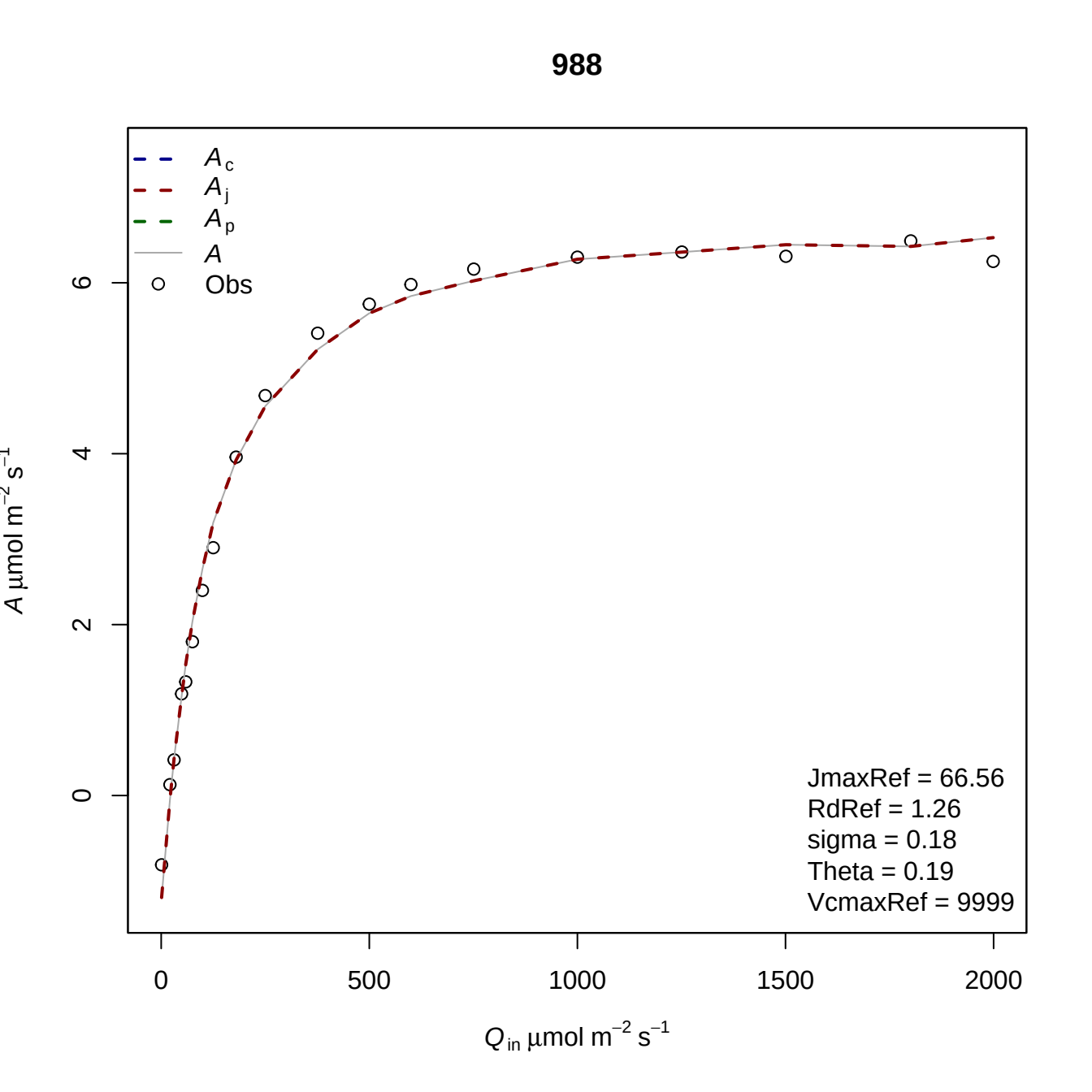

983
A_c
A_j
A_p
A
Obs
JmaxRef = 70.98
RdRef = 1
sigma = 0.11
Theta = −0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

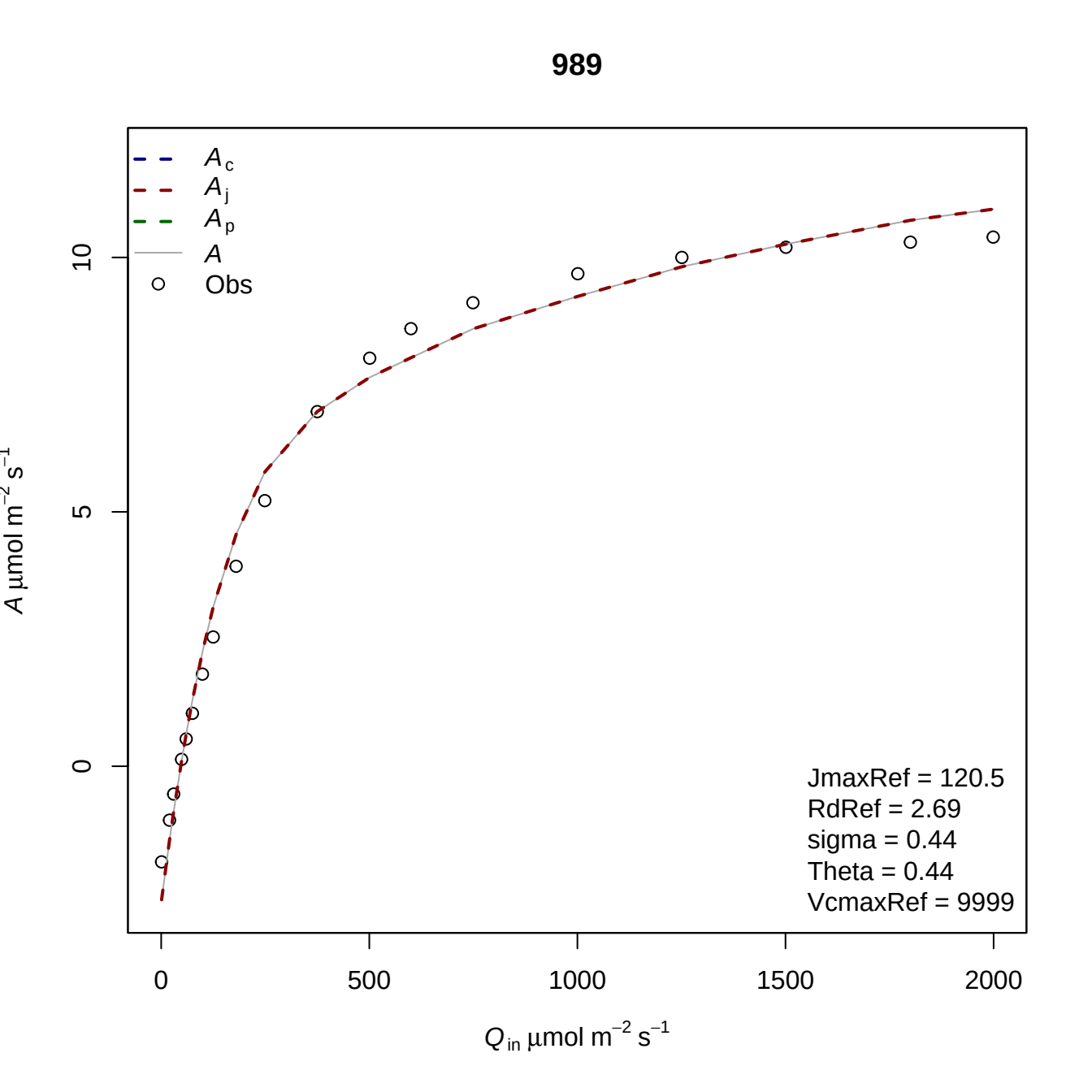

984
A_c
A_j
A_p
A
Obs
JmaxRef = 118.28
RdRef = 2.17
sigma = 0.06
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

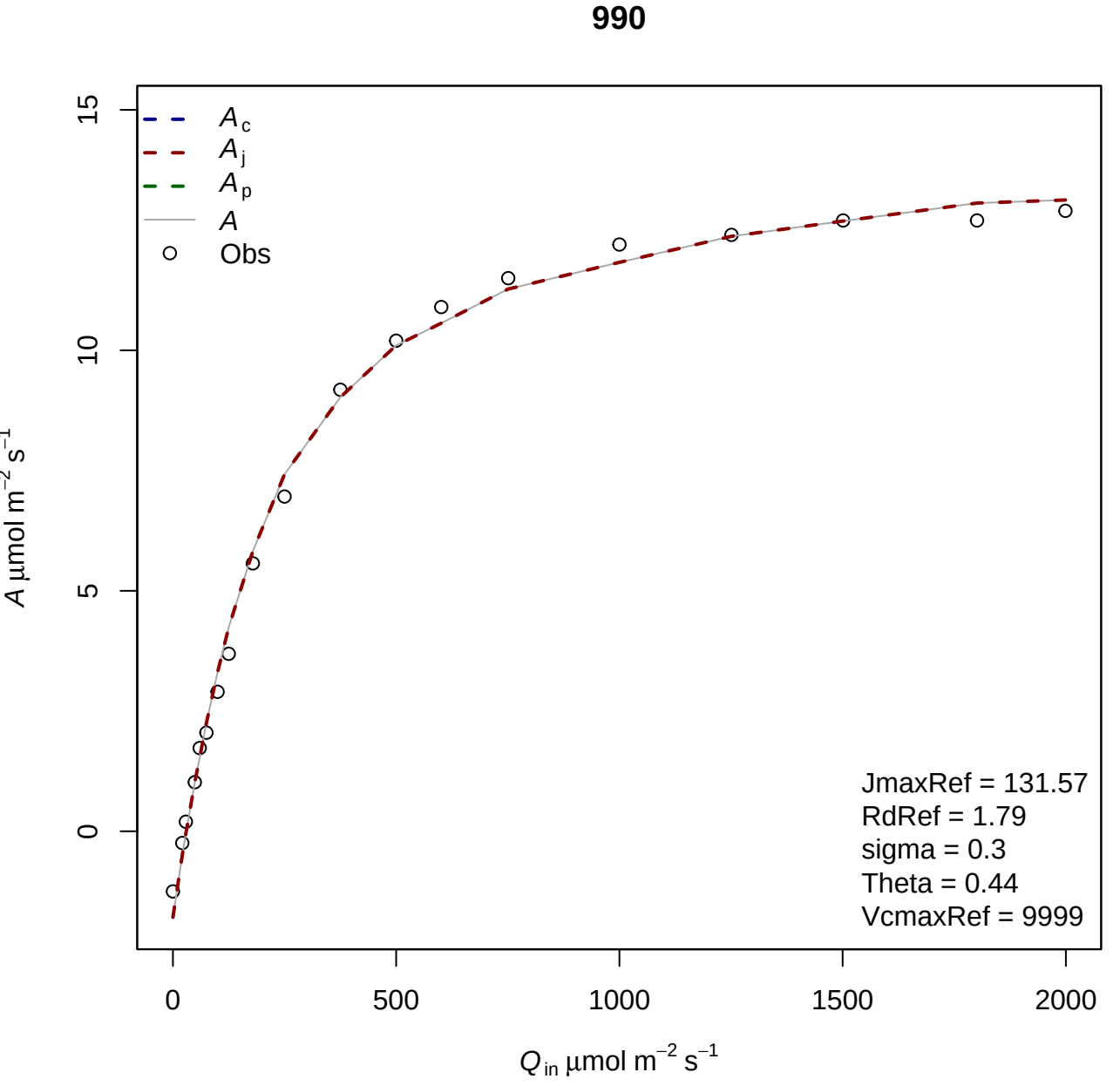

985
A_c
A_j
A_p
A
Obs
JmaxRef = 110.18
RdRef = 1.95
sigma = 0.09
Theta = 0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

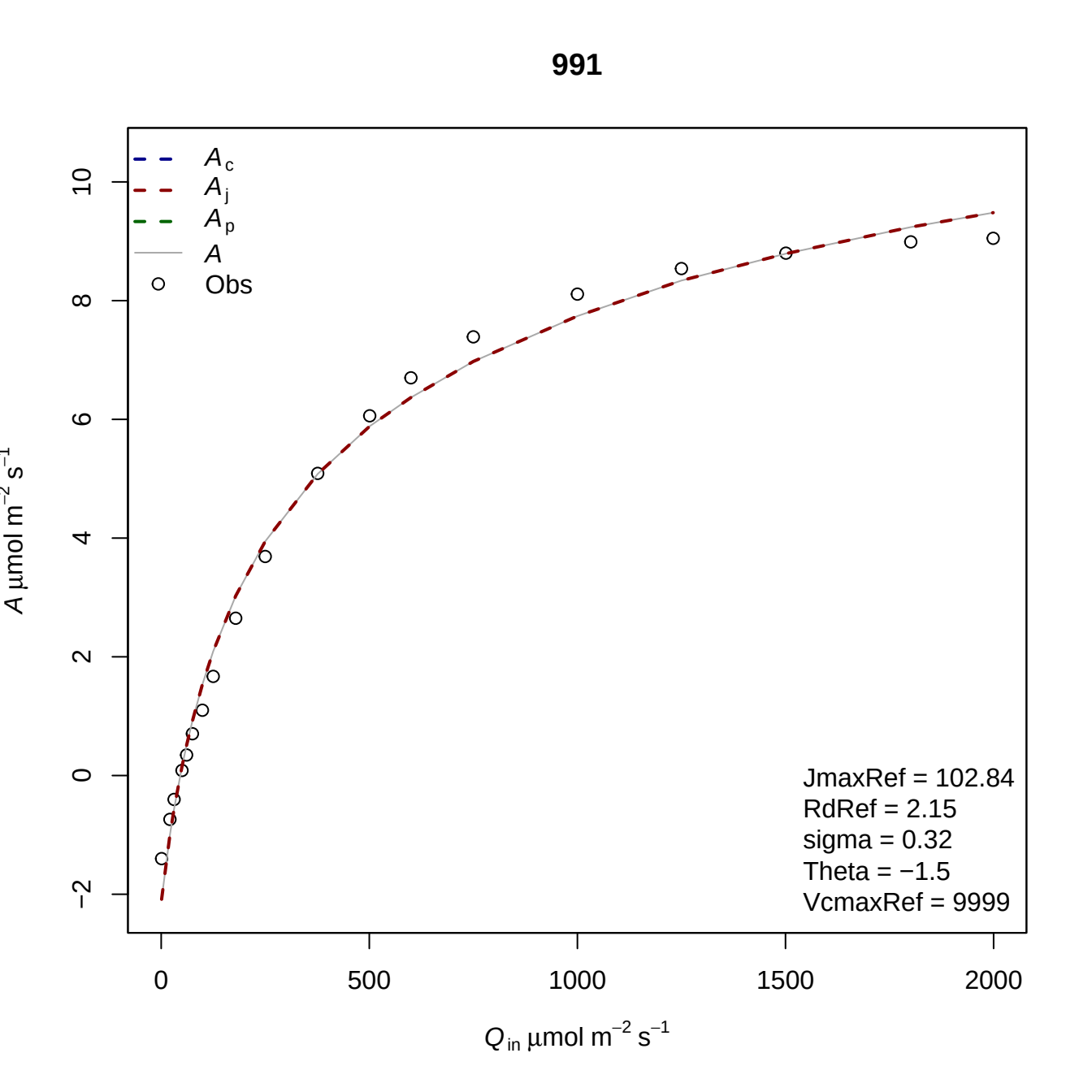

986
A_c
A_j
A_p
A
Obs
JmaxRef = 65.98
RdRef = 1.07
sigma = 0.06
Theta = 0.29
VcmaxRef = 70.03
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

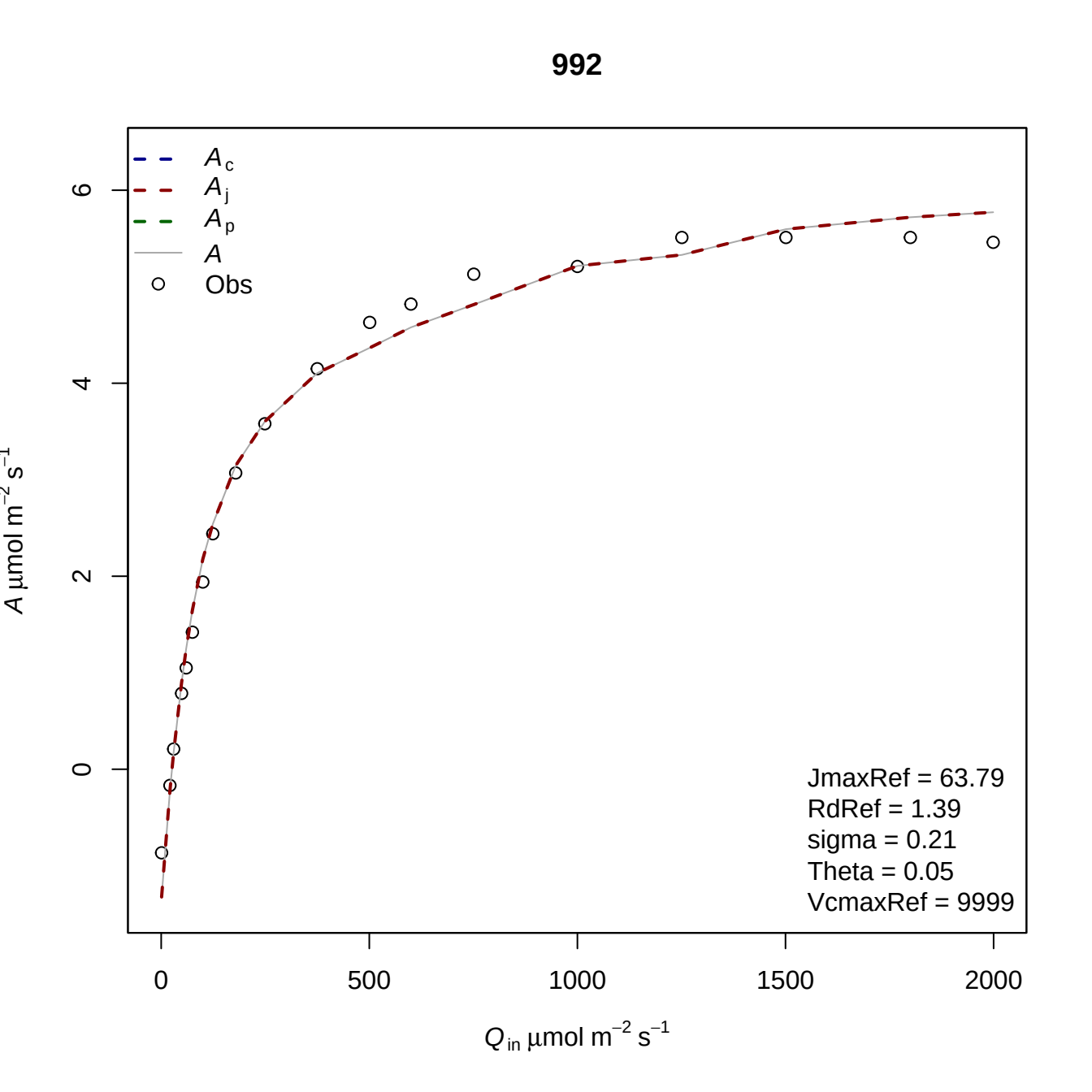

987
A_c
A_j
A_p
A
Obs
JmaxRef = 131.31
RdRef = 2.54
sigma = 0.12
Theta = 0.54
VcmaxRef = 138.4
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

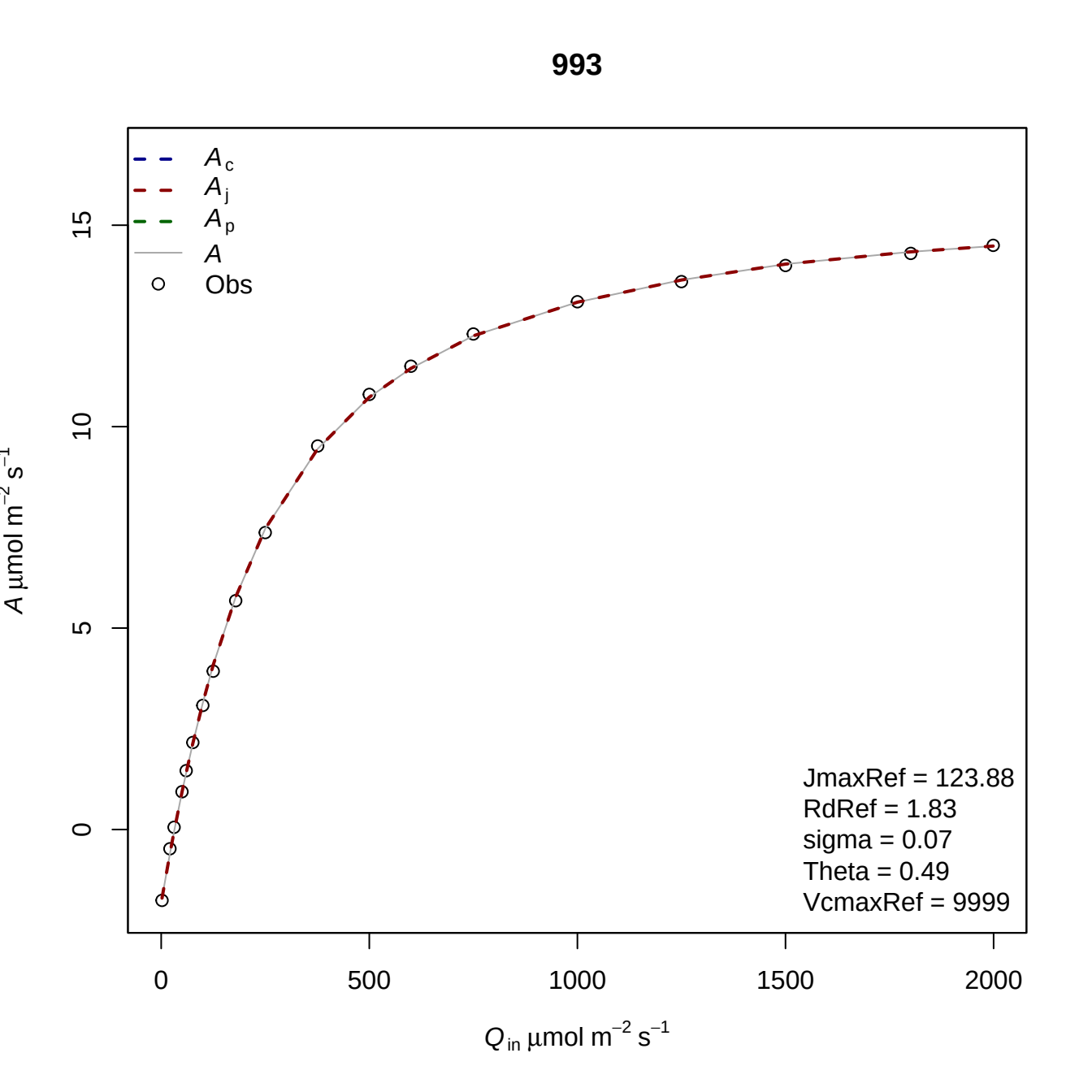

988
A_c
A_j
A_p
A
Obs
JmaxRef = 66.56
RdRef = 1.26
sigma = 0.18
Theta = 0.19
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

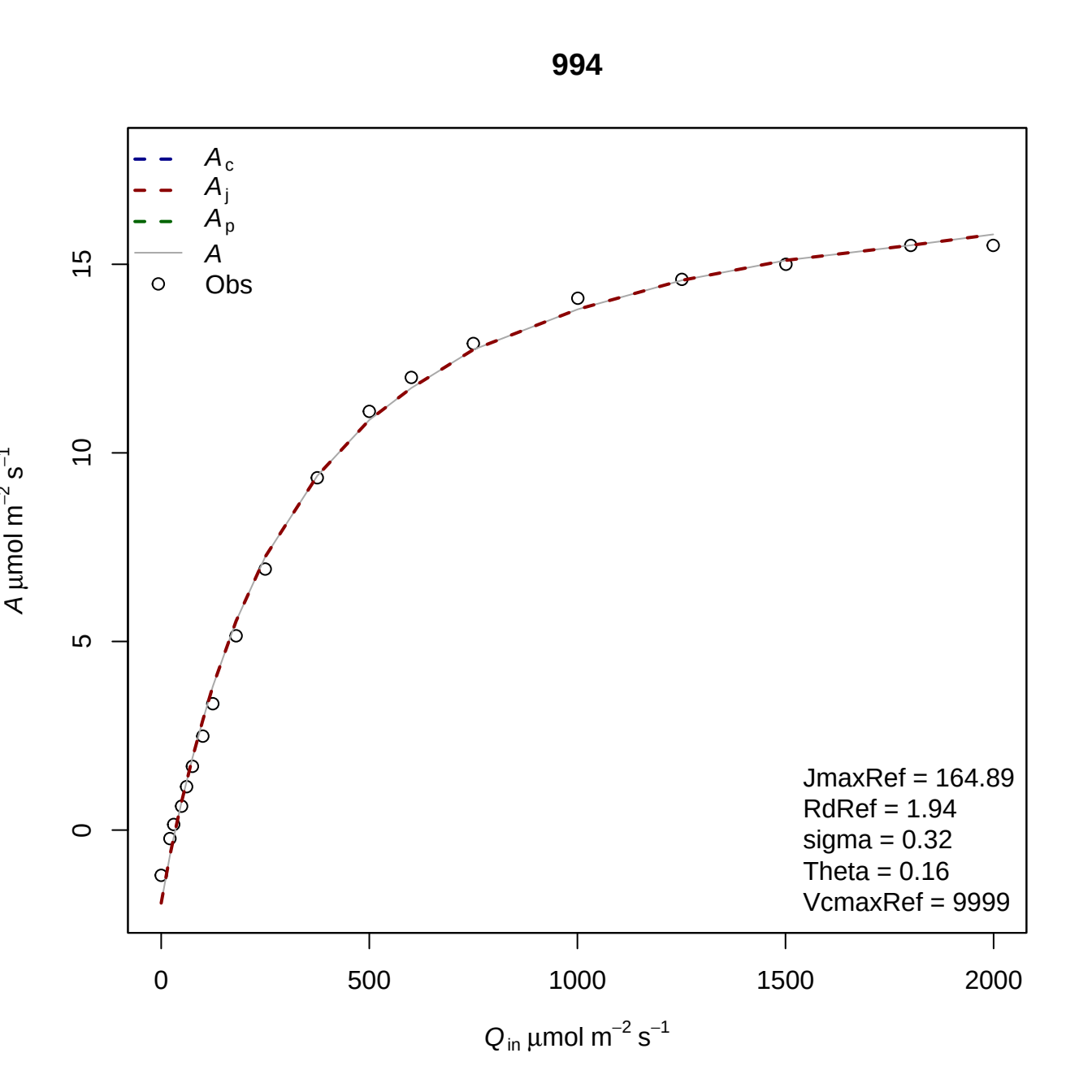

989
A_c
A_j
A_p
A
Obs
JmaxRef = 120.5
RdRef = 2.69
sigma = 0.44
Theta = 0.44
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

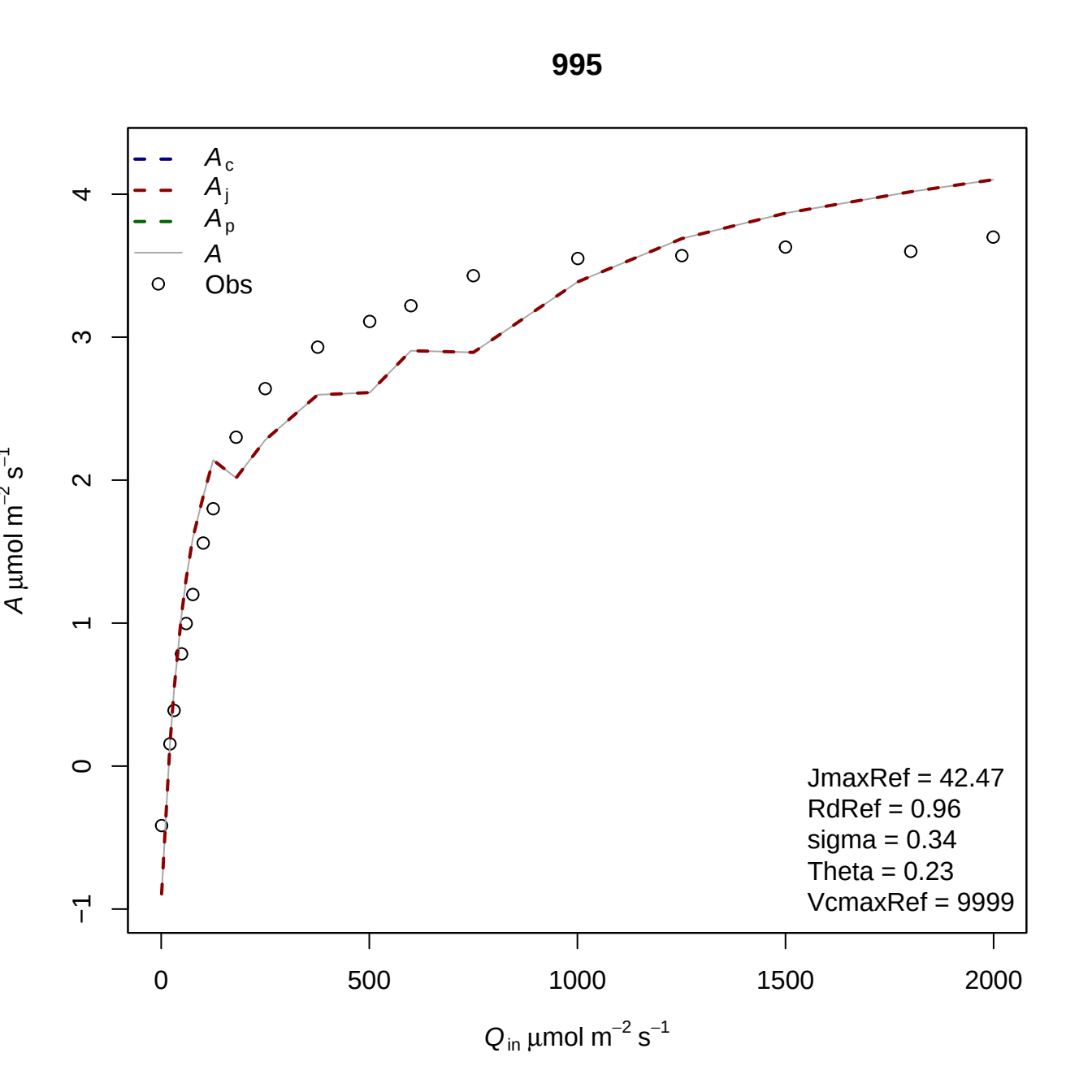

990
A_c
A_j
A_p
A
Obs
JmaxRef = 131.57
RdRef = 1.79
sigma = 0.3
Theta = 0.44
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

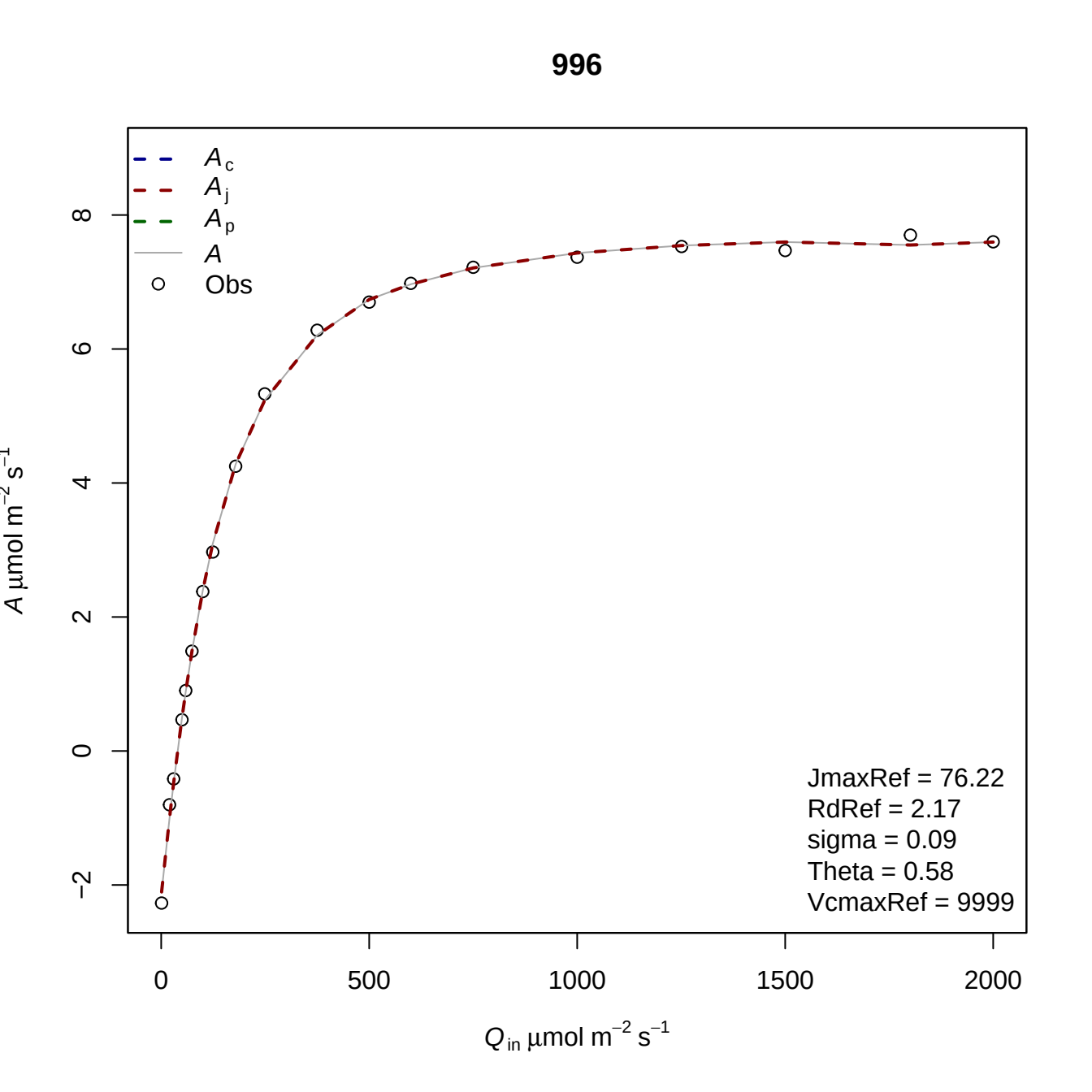

991
A_c
A_j
A_p
A
Obs
JmaxRef = 102.84
RdRef = 2.15
sigma = 0.32
Theta = −1.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

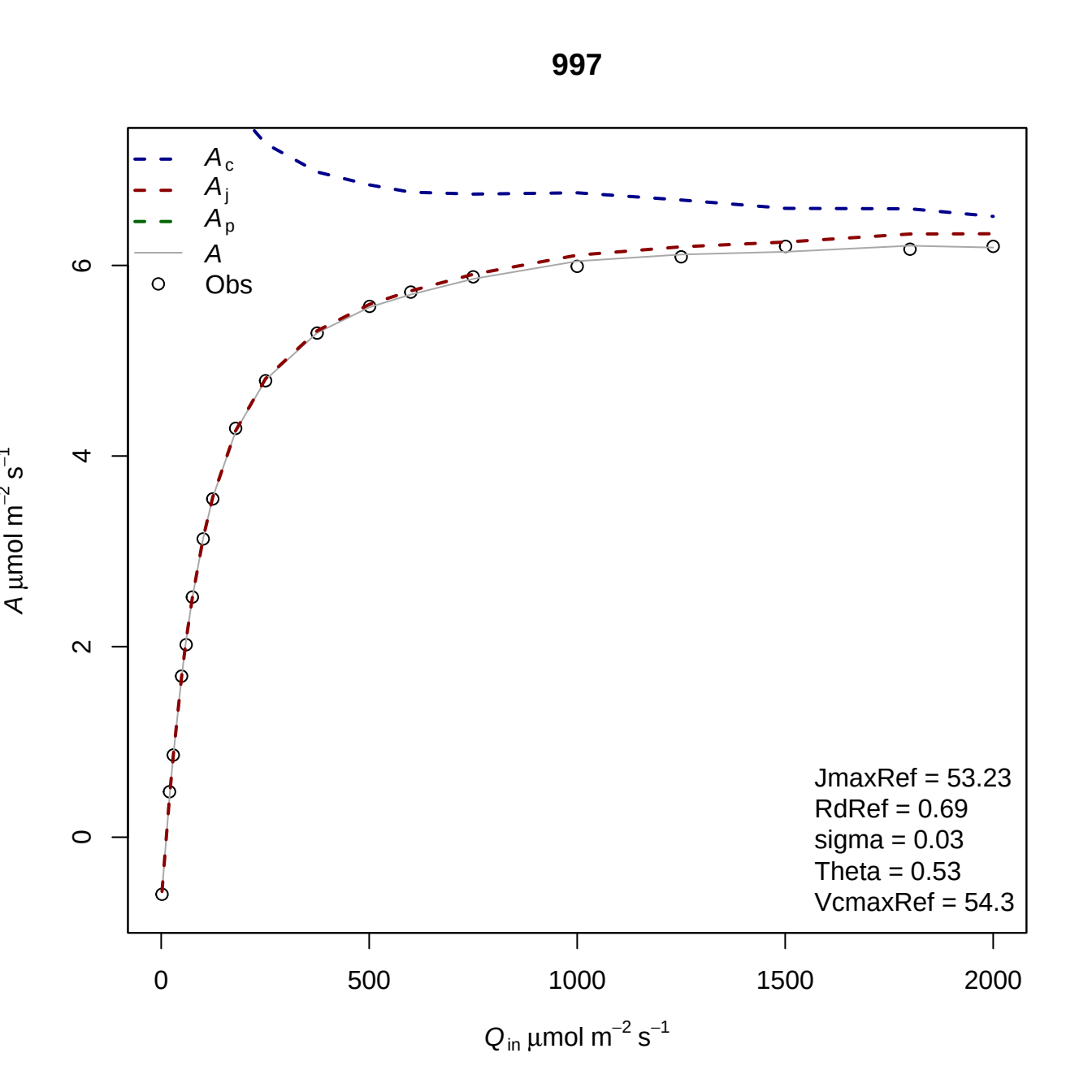

992
A_c
A_j
A_p
A
Obs
JmaxRef = 63.79
RdRef = 1.39
sigma = 0.21
Theta = 0.05
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

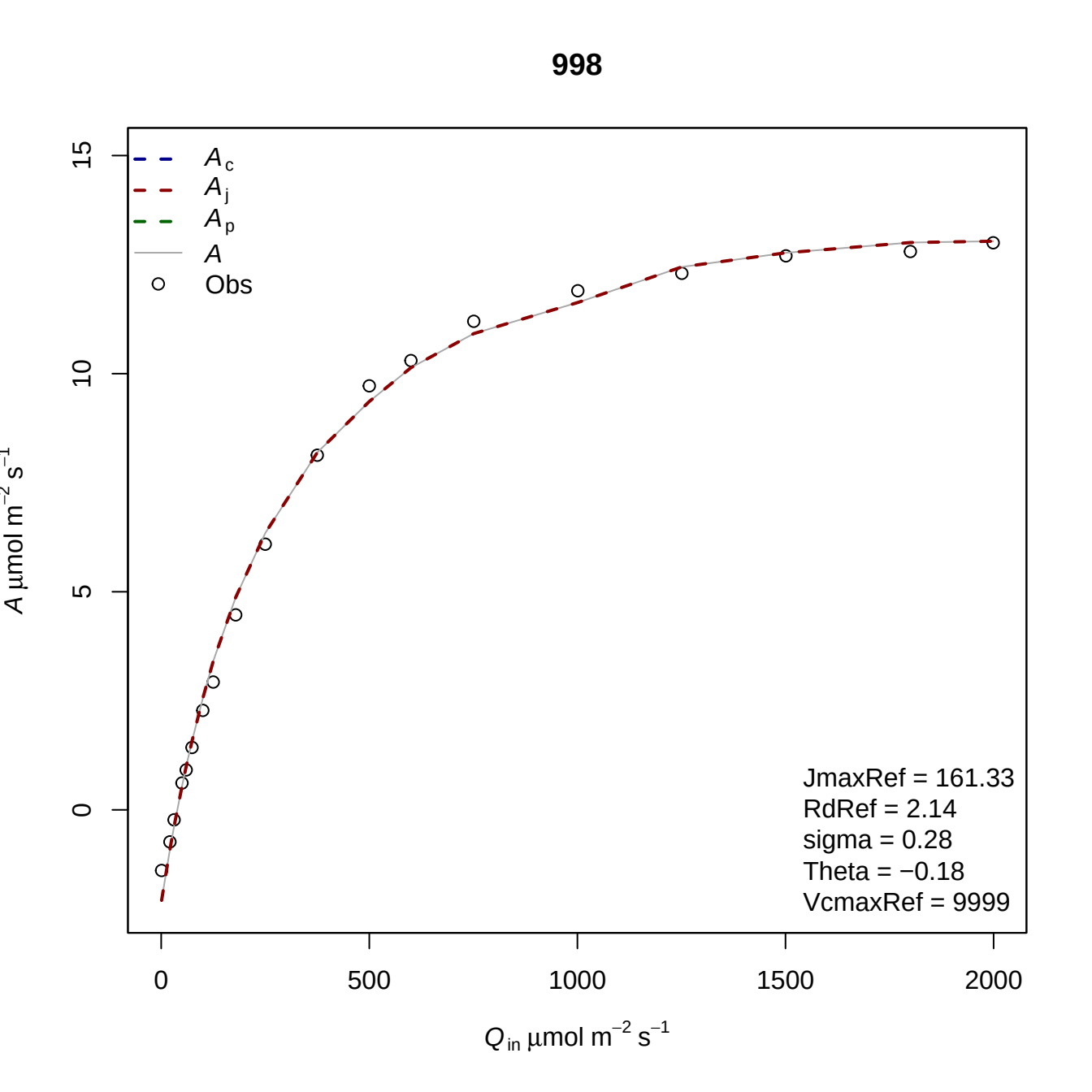

993
A_c
A_j
A_p
A
Obs
JmaxRef = 123.88
RdRef = 1.83
sigma = 0.07
Theta = 0.49
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

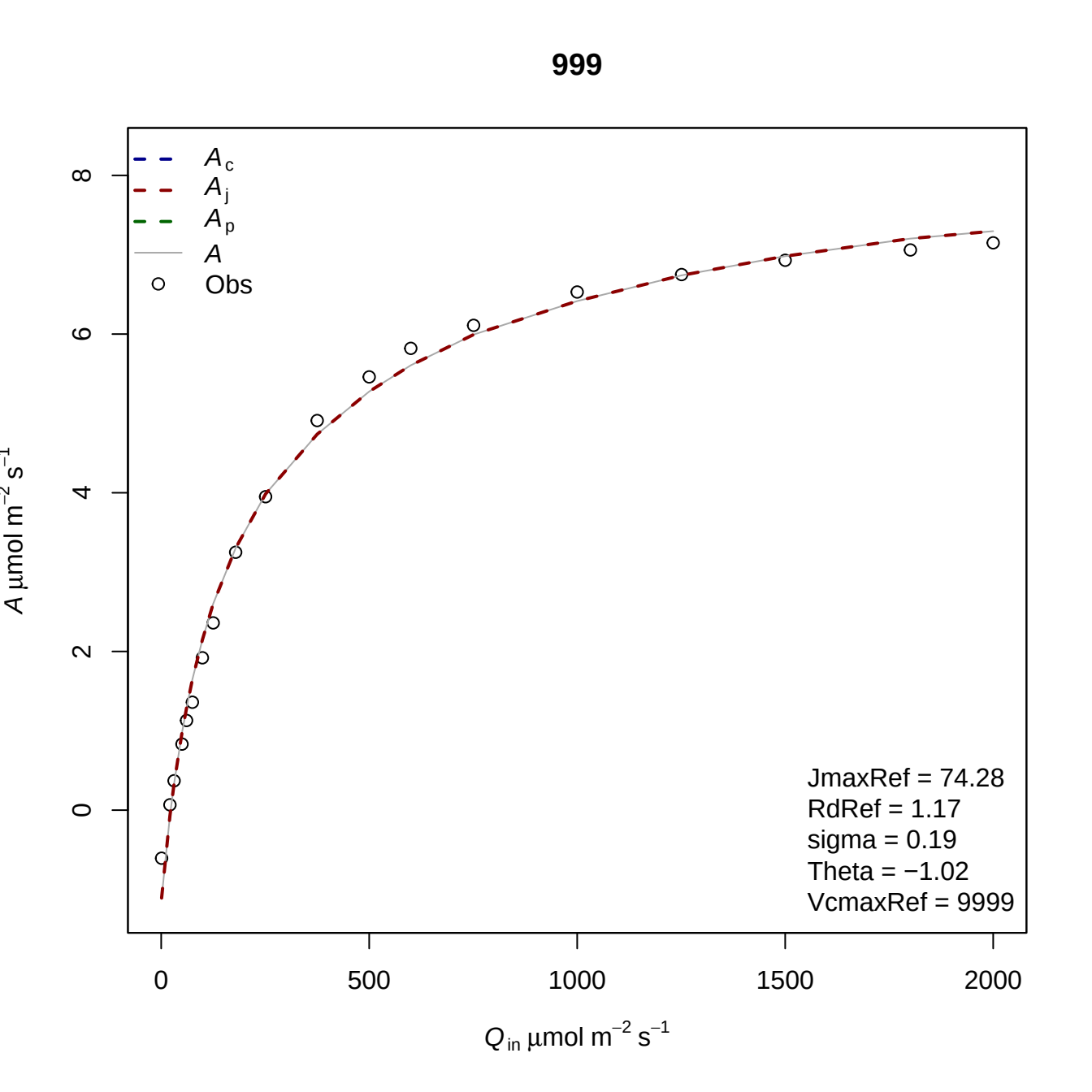

994
A_c
A_j
A_p
A
Obs
JmaxRef = 164.89
RdRef = 1.94
sigma = 0.32
Theta = 0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

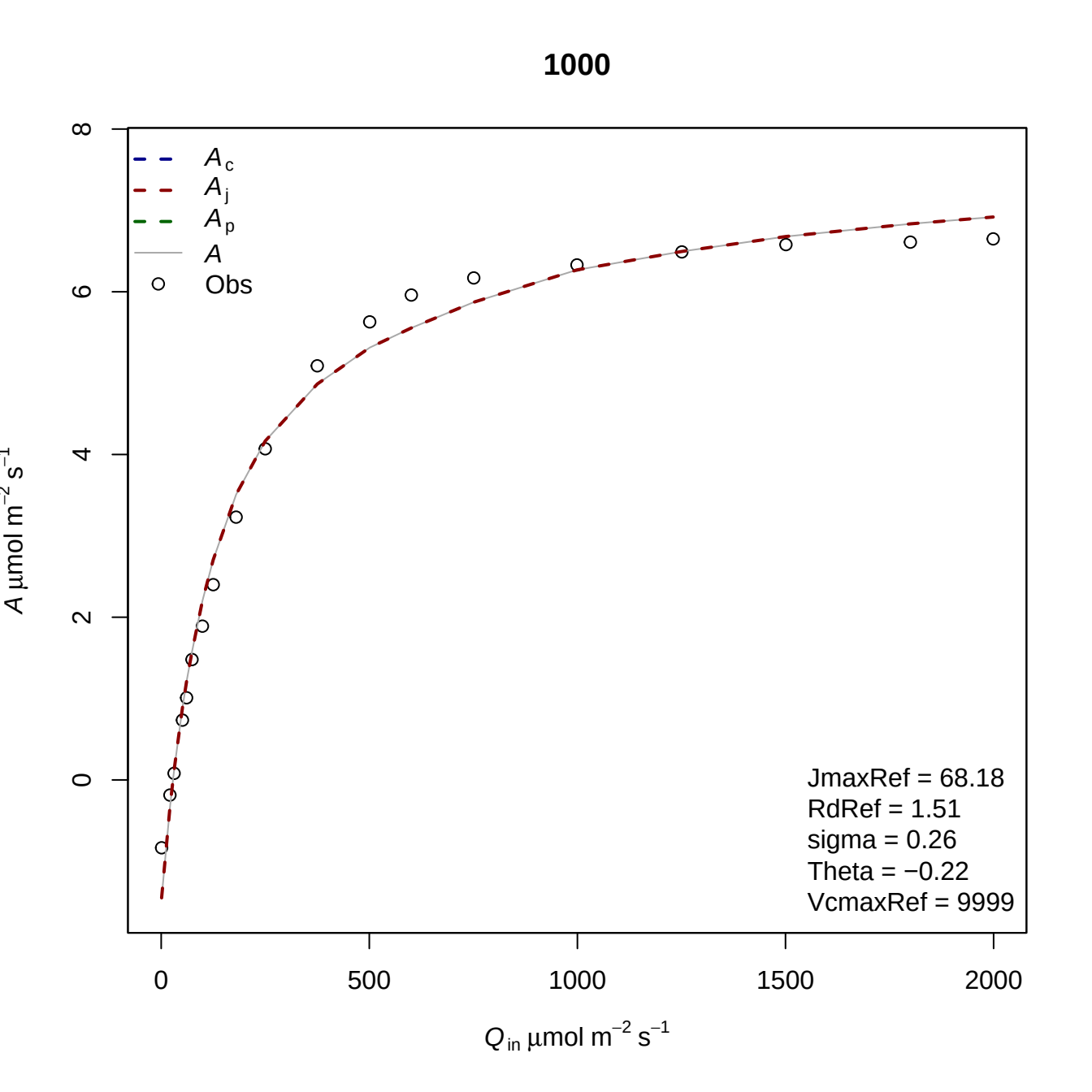

995
A_c
A_j
A_p
A
Obs
JmaxRef = 42.47
RdRef = 0.96
sigma = 0.34
Theta = 0.23
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

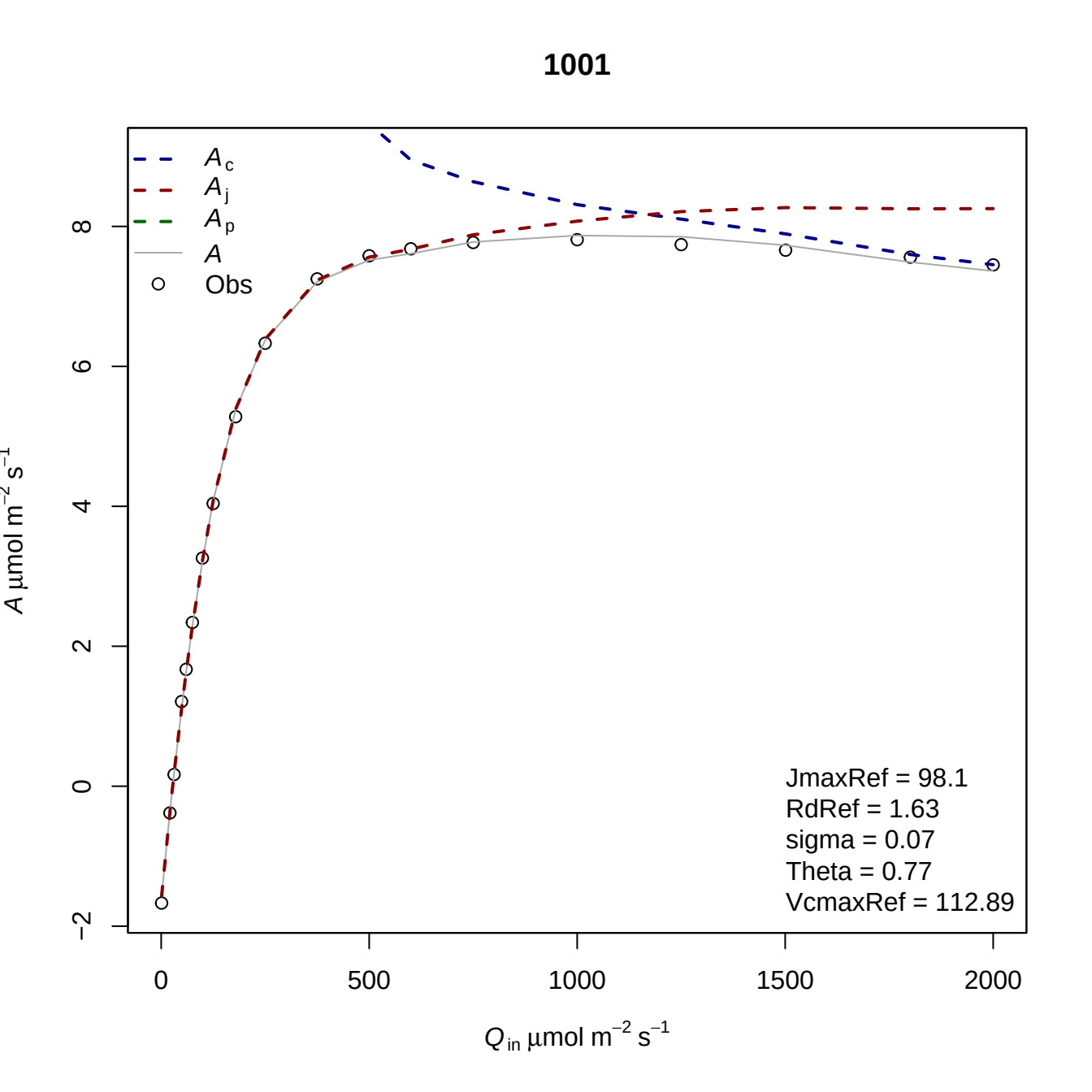

996
A_c
A_j
A_p
A
Obs
JmaxRef = 76.22
RdRef = 2.17
sigma = 0.09
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

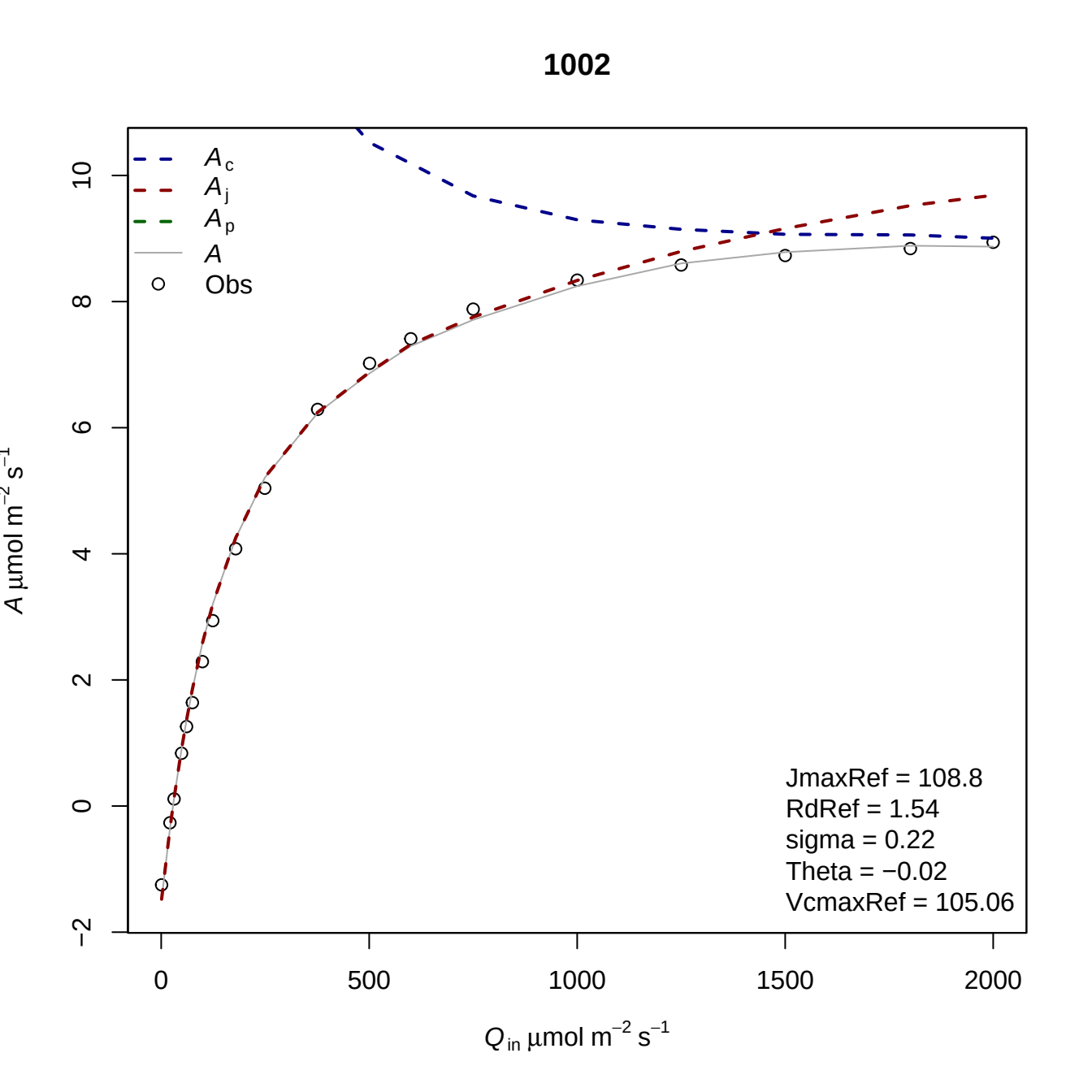

997
A_c
A_j
A_p
A
Obs
JmaxRef = 53.23
RdRef = 0.69
sigma = 0.03
Theta = 0.53
VcmaxRef = 54.3
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

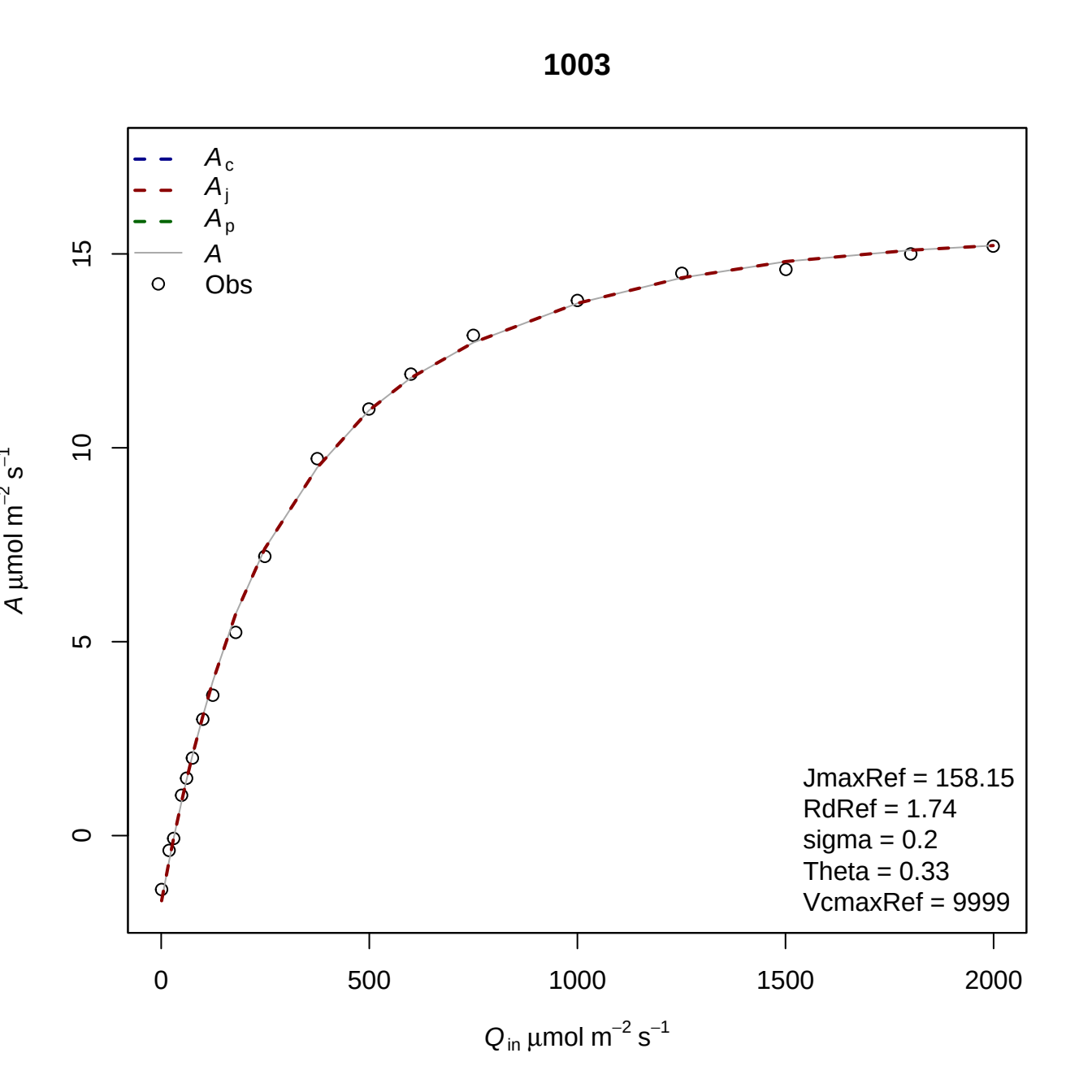

998
A_c
A_j
A_p
A
Obs
JmaxRef = 161.33
RdRef = 2.14
sigma = 0.28
Theta = −0.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

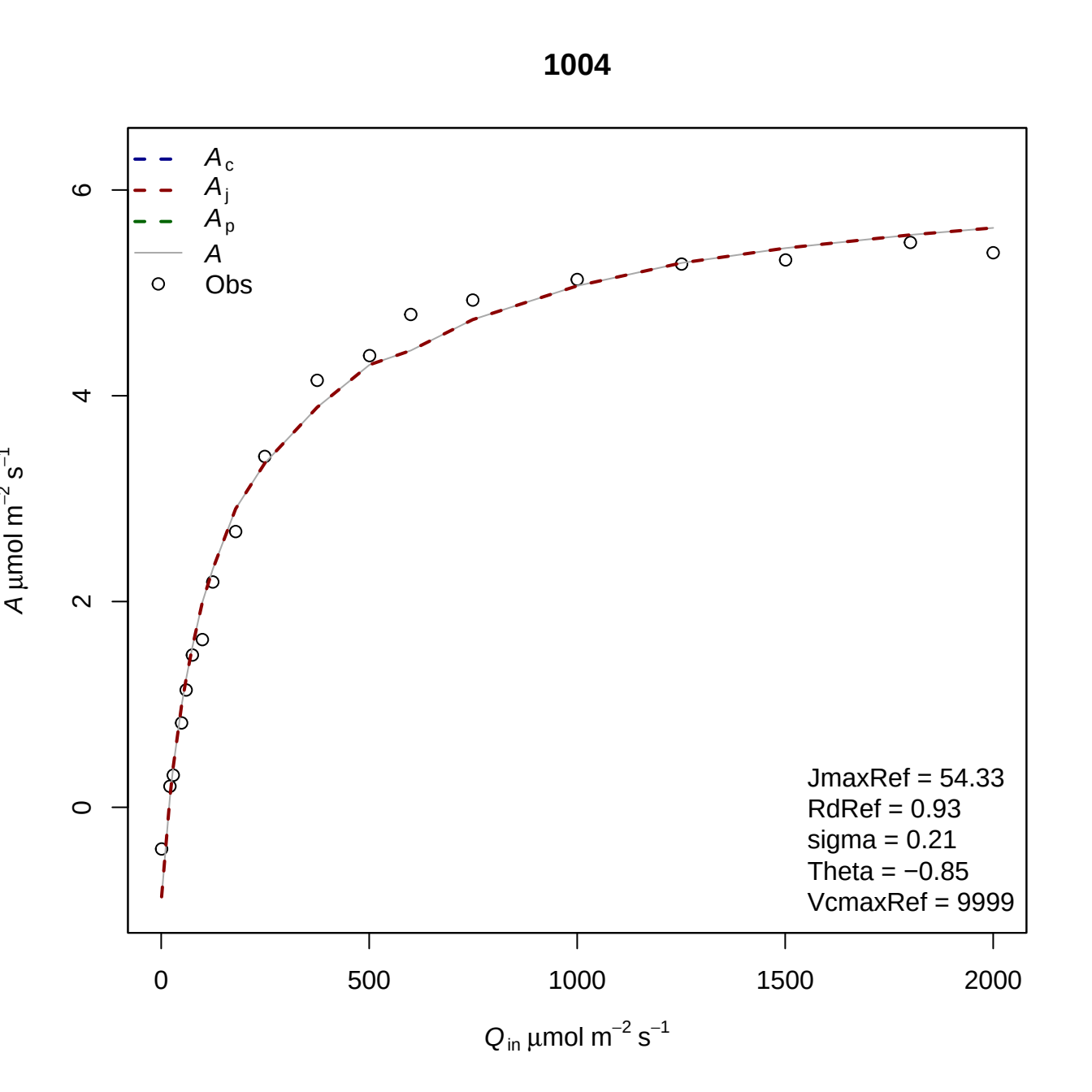

999
A_c
A_j
A_p
A
Obs
JmaxRef = 74.28
RdRef = 1.17
sigma = 0.19
Theta = −1.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

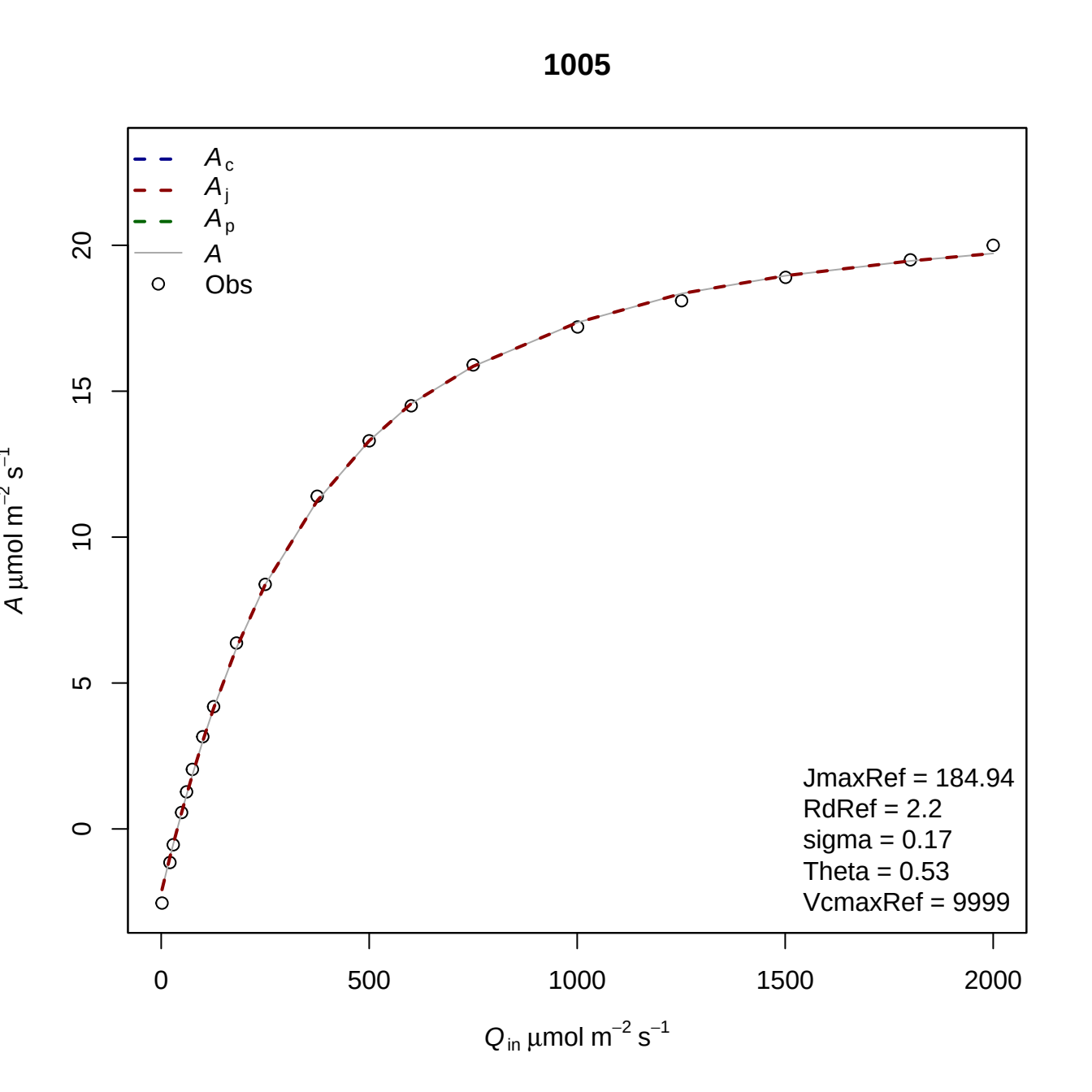

1000
A_c
A_j
A_p
A
Obs
JmaxRef = 68.18
RdRef = 1.51
sigma = 0.26
Theta = −0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

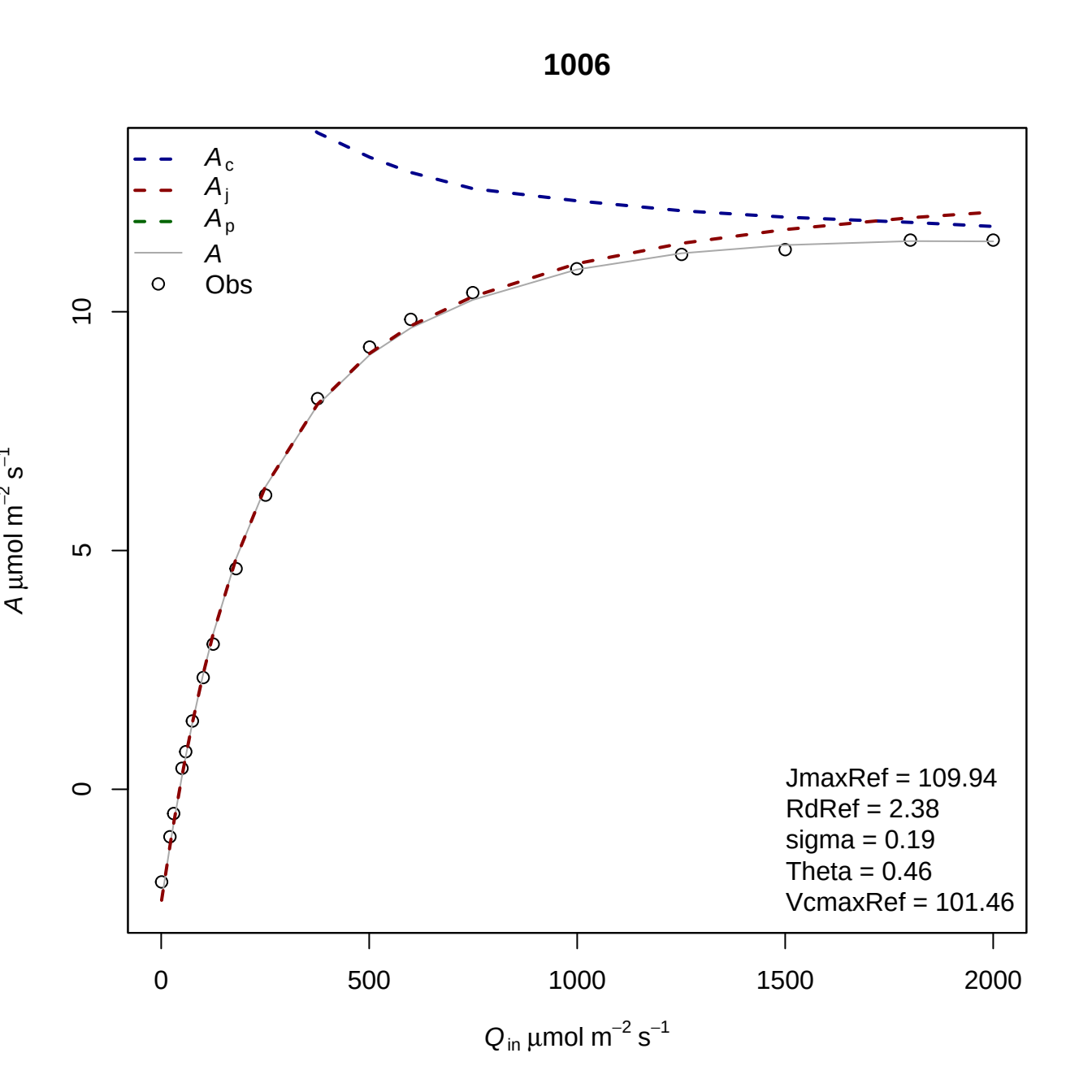

1001
A_c
A_j
A_p
A
Obs
JmaxRef = 98.1
RdRef = 1.63
sigma = 0.07
Theta = 0.77
VcmaxRef = 112.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

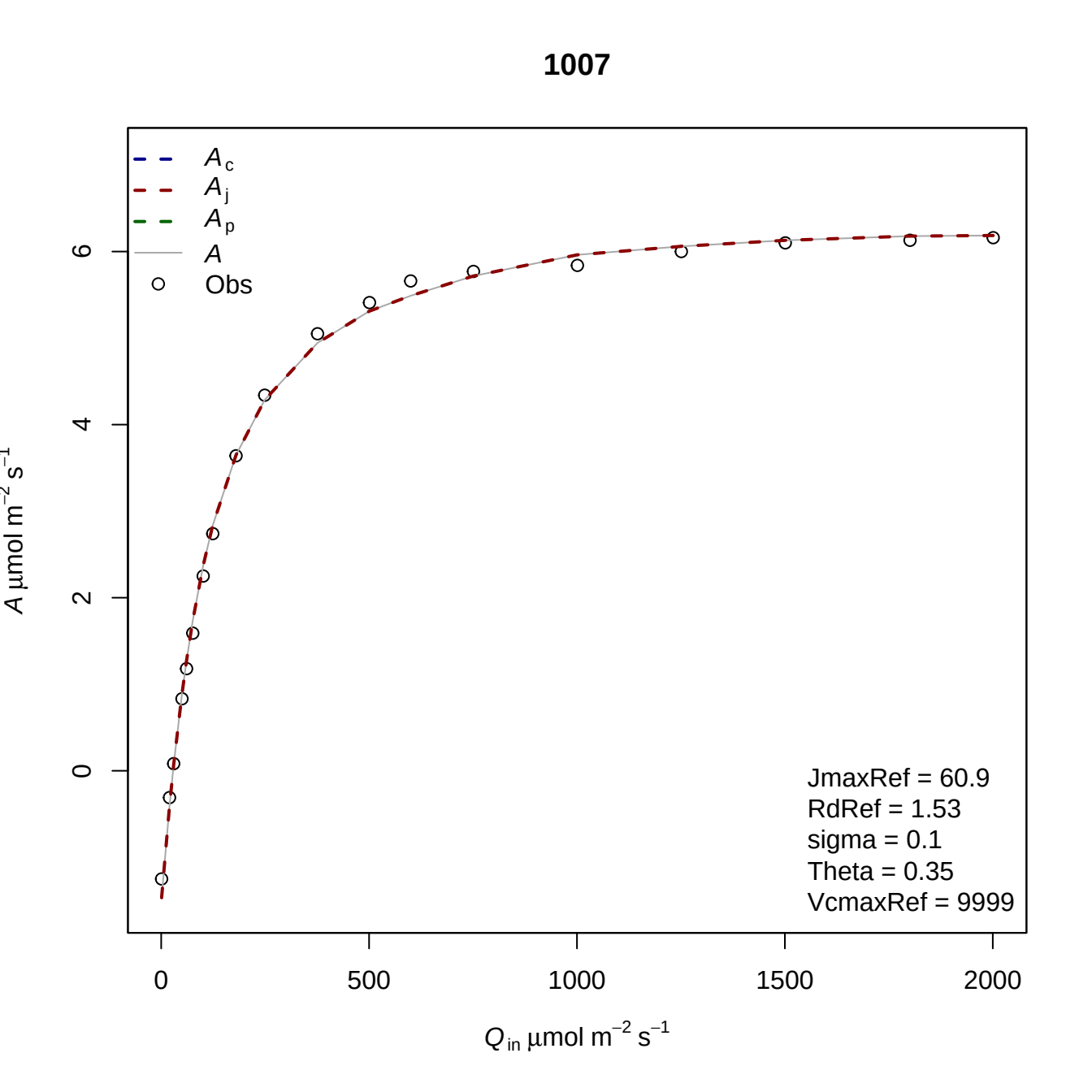

1002
A_c
A_j
A_p
A
Obs
JmaxRef = 108.8
RdRef = 1.54
sigma = 0.22
Theta = −0.02
VcmaxRef = 105.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

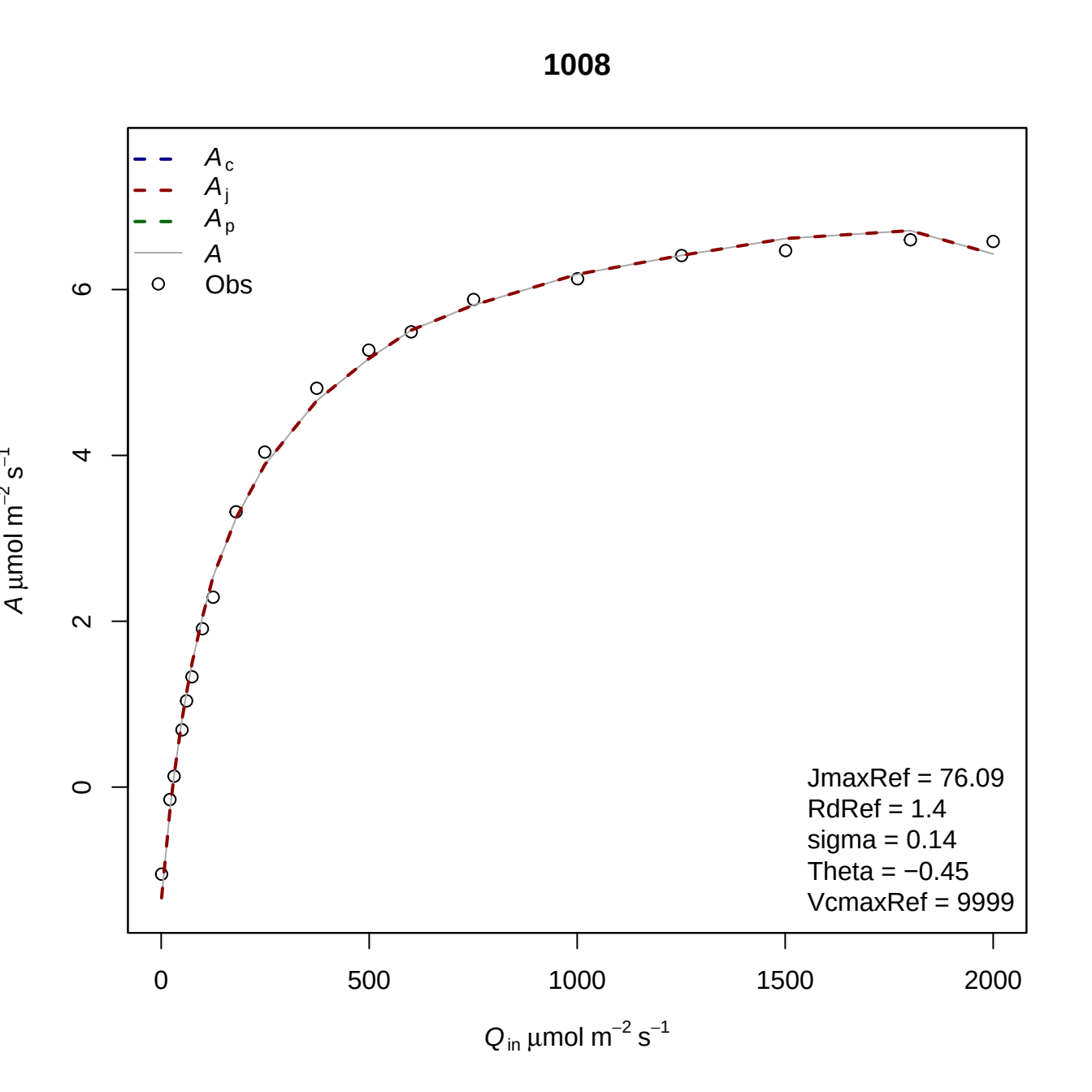

1003
A_c
A_j
A_p
A
Obs
JmaxRef = 158.15
RdRef = 1.74
sigma = 0.2
Theta = 0.33
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

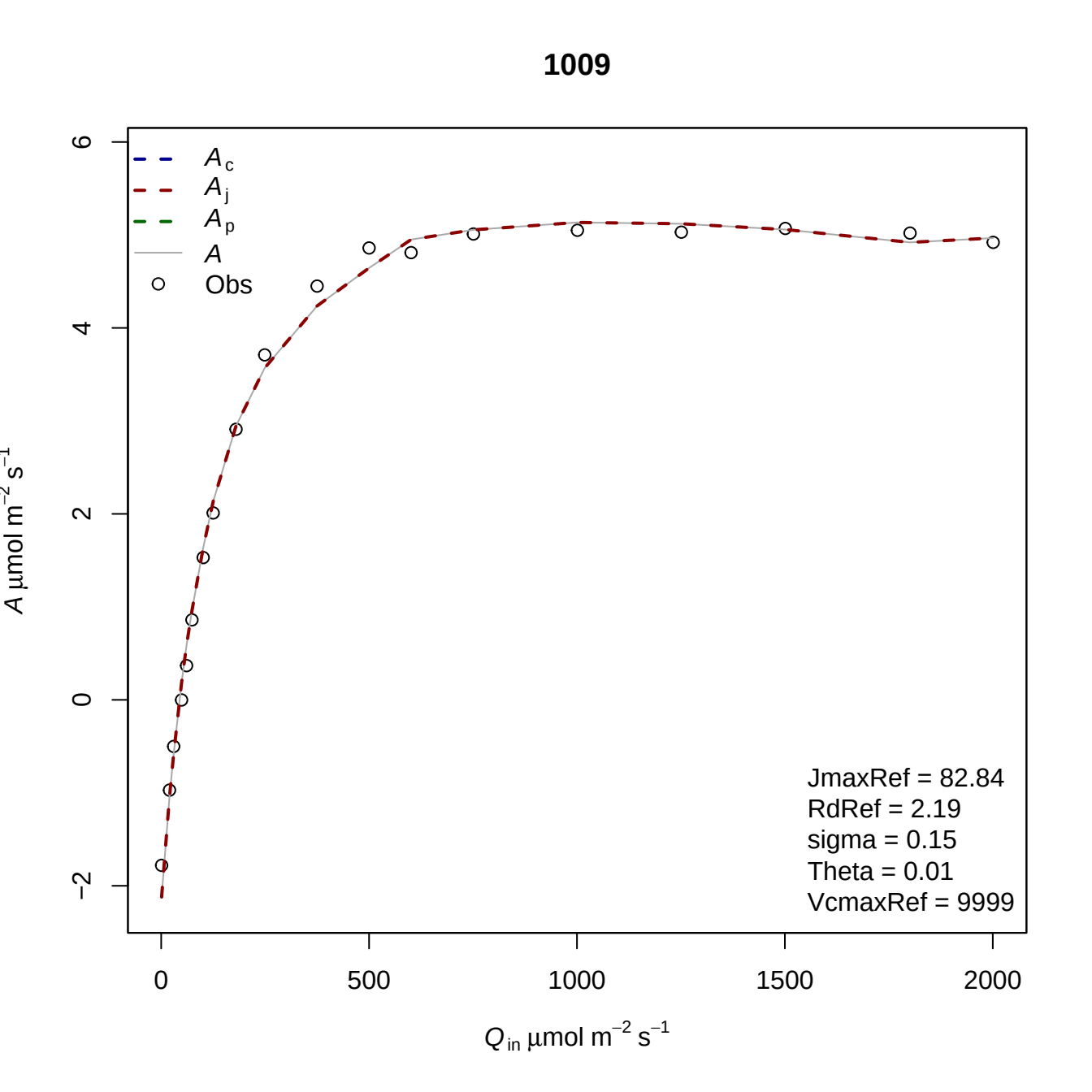

1004
A_c
A_j
A_p
A
Obs
JmaxRef = 54.33
RdRef = 0.93
sigma = 0.21
Theta = −0.85
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

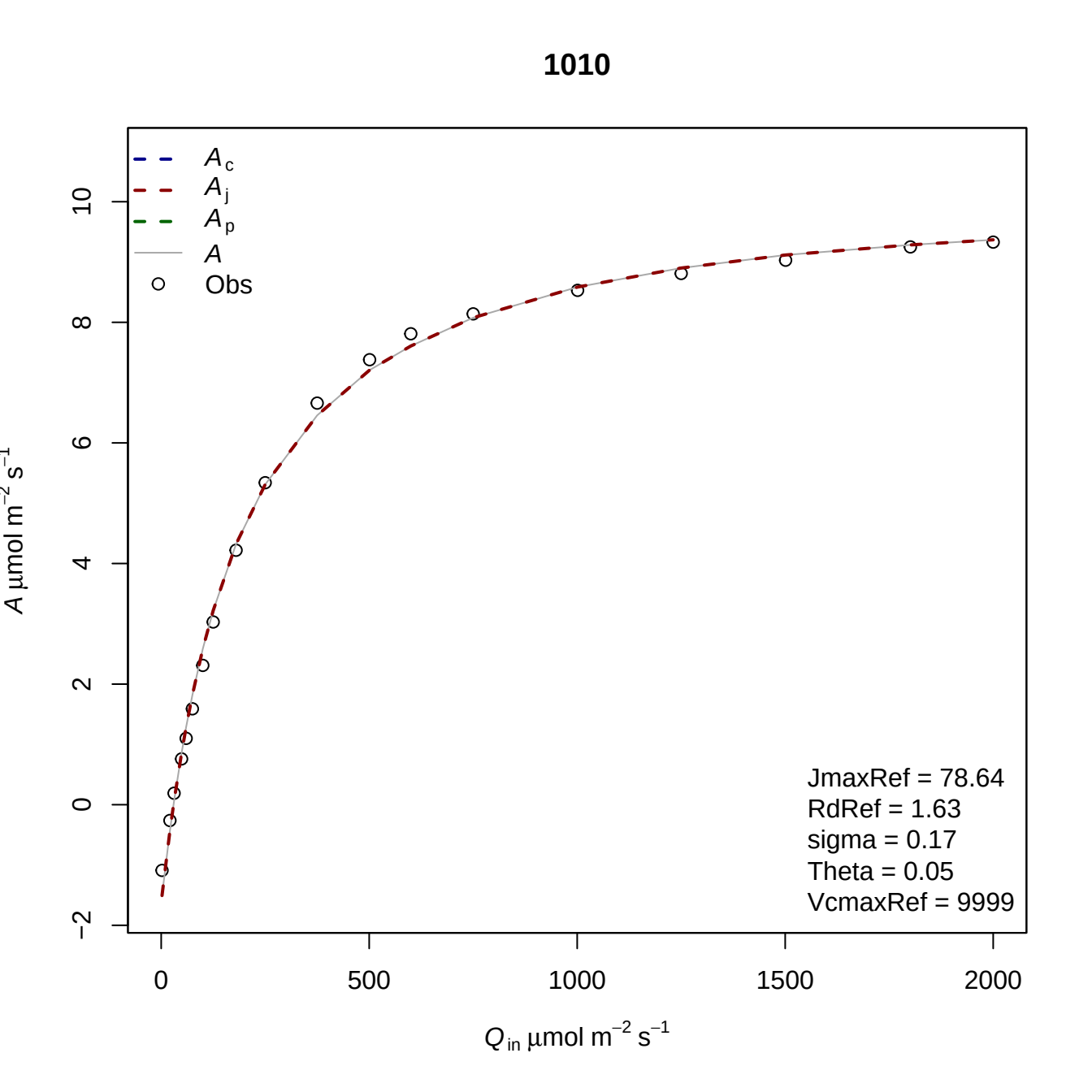

1005
A_c
A_j
A_p
A
Obs
JmaxRef = 184.94
RdRef = 2.2
sigma = 0.17
Theta = 0.53
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

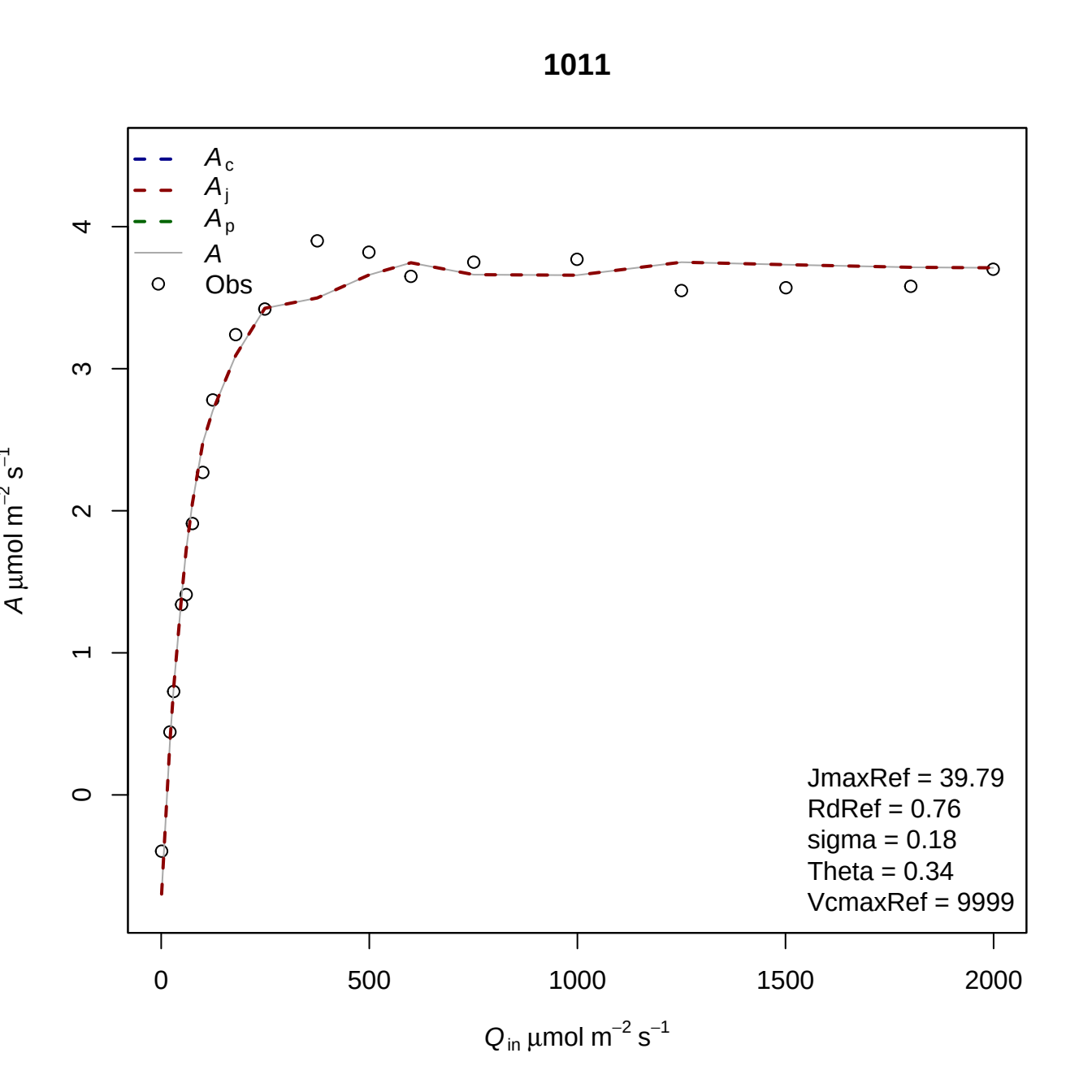

1006
A_c
A_j
A_p
A
Obs
JmaxRef = 109.94
RdRef = 2.38
sigma = 0.19
Theta = 0.46
VcmaxRef = 101.46
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

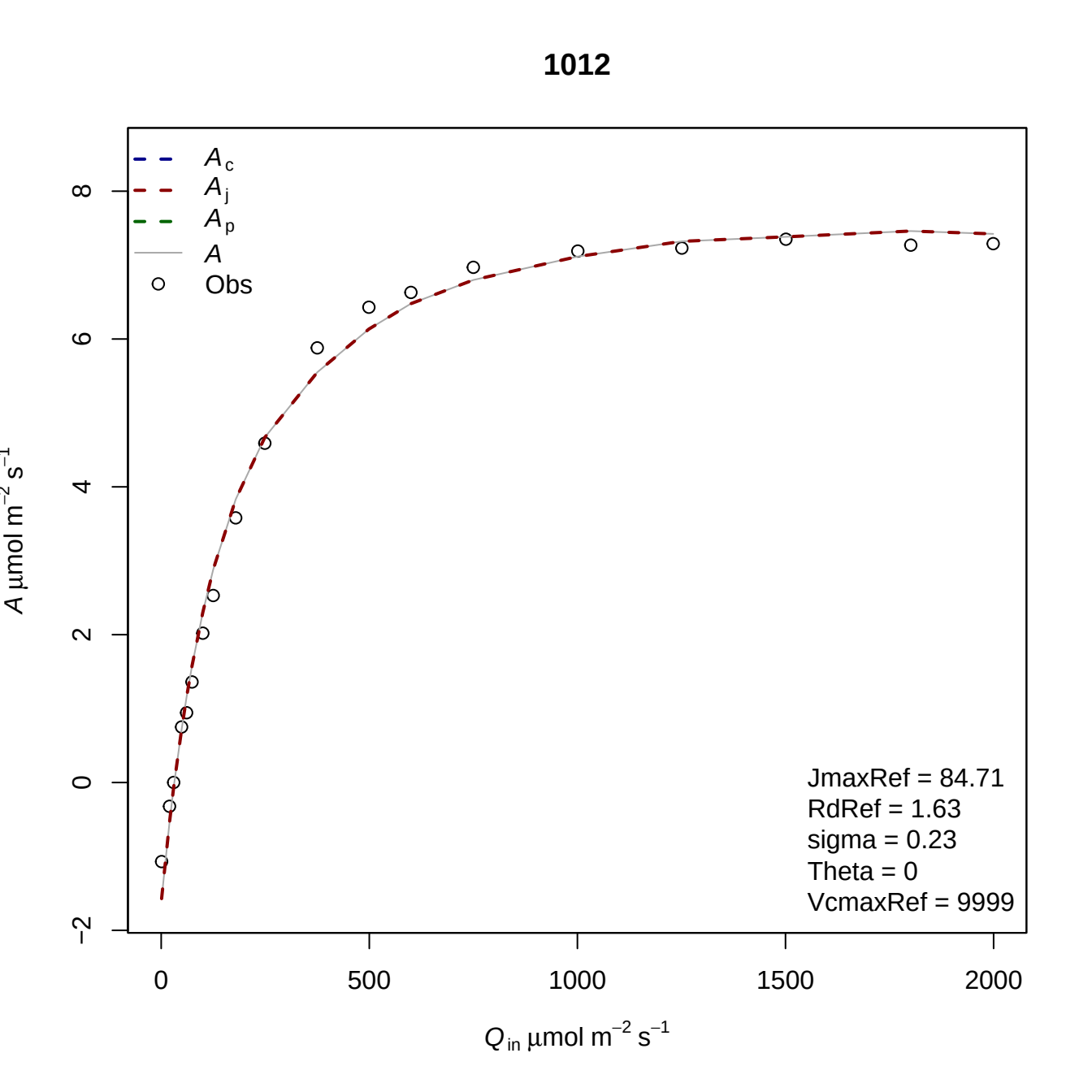

1007
A_c
A_j
A_p
A
Obs
JmaxRef = 60.9
RdRef = 1.53
sigma = 0.1
Theta = 0.35
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

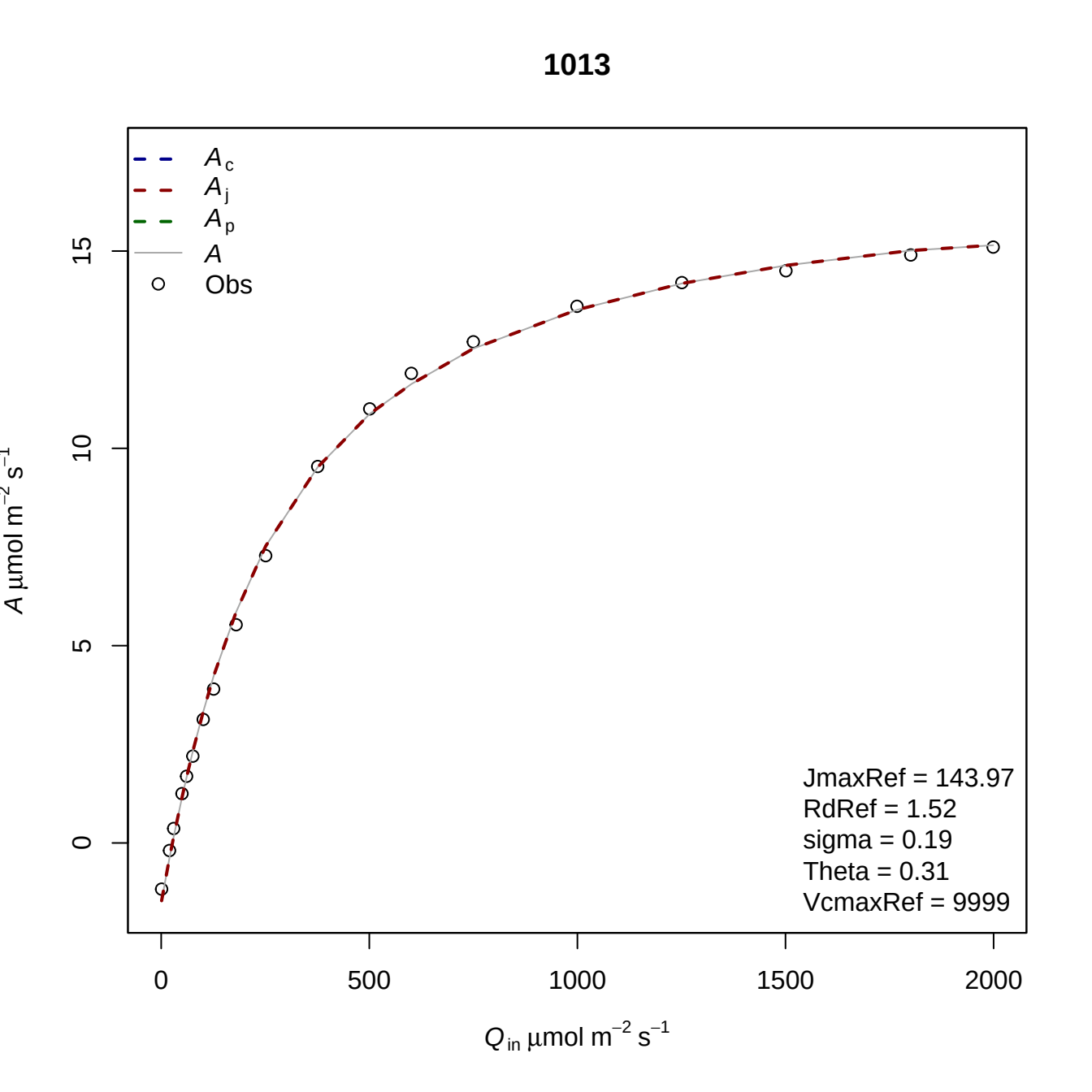

1008
A_c
A_j
A_p
A
Obs
JmaxRef = 76.09
RdRef = 1.4
sigma = 0.14
Theta = −0.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

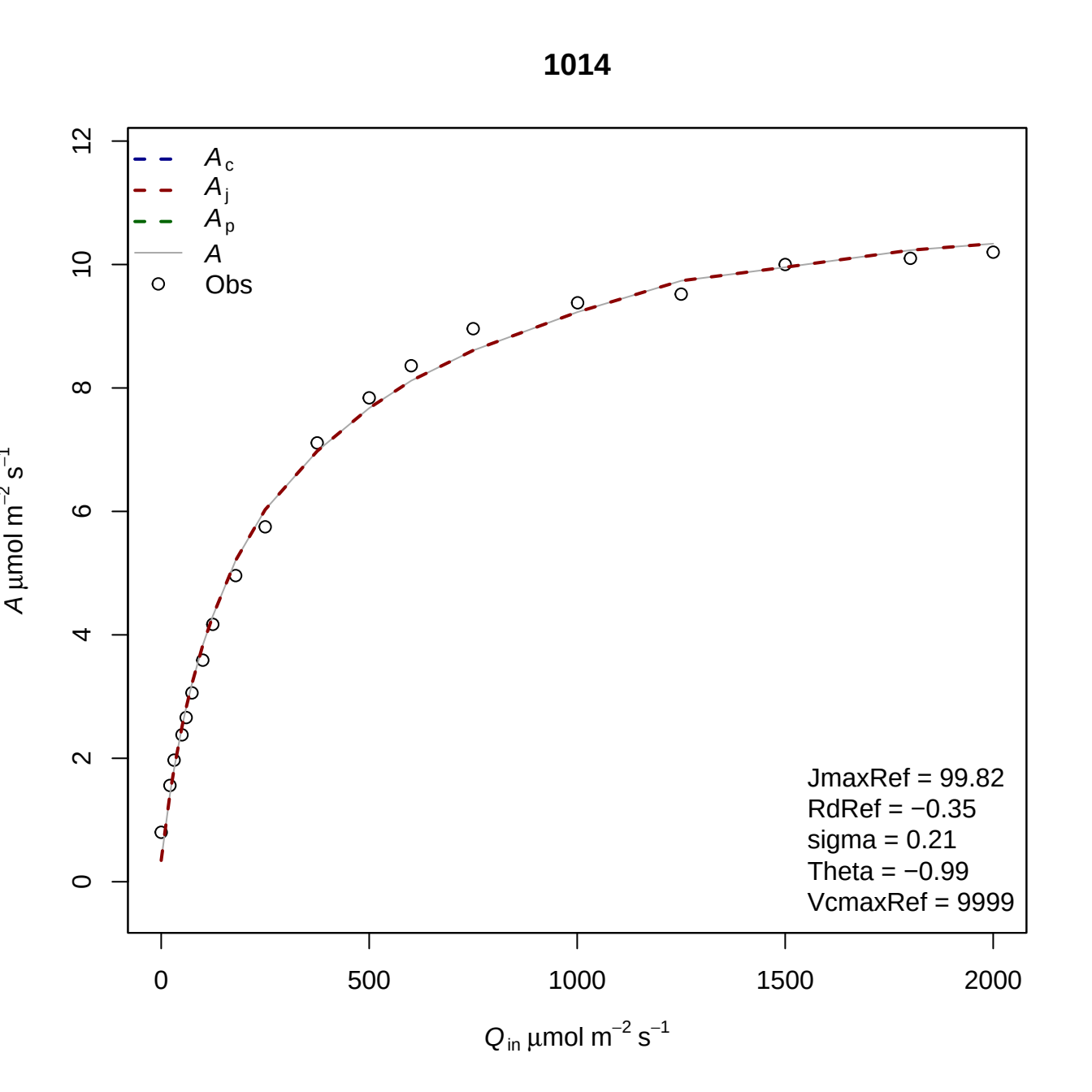

1009
A_c
A_j
A_p
A
Obs
JmaxRef = 82.84
RdRef = 2.19
sigma = 0.15
Theta = 0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

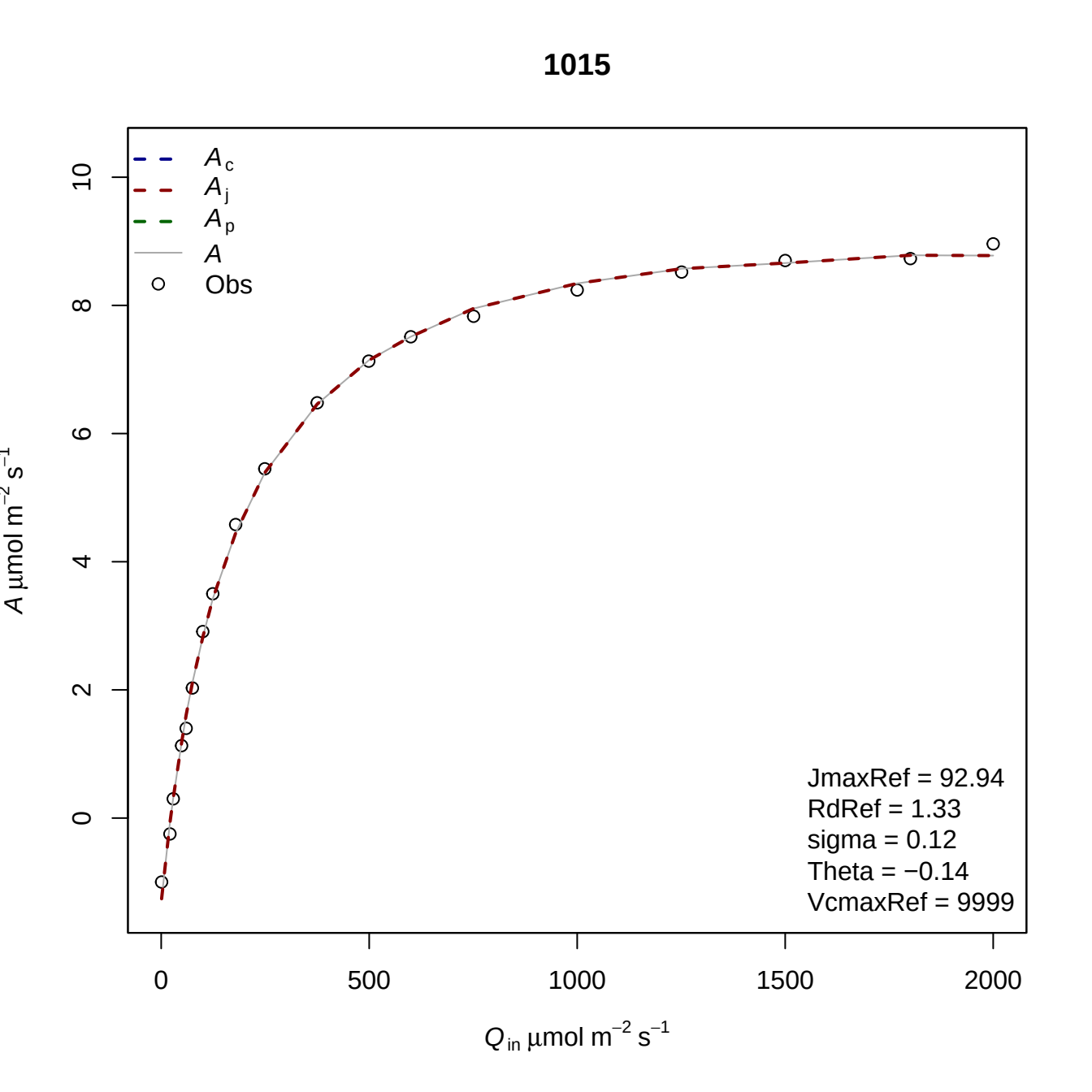

1010
A_c
A_j
A_p
A
Obs
JmaxRef = 78.64
RdRef = 1.63
sigma = 0.17
Theta = 0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

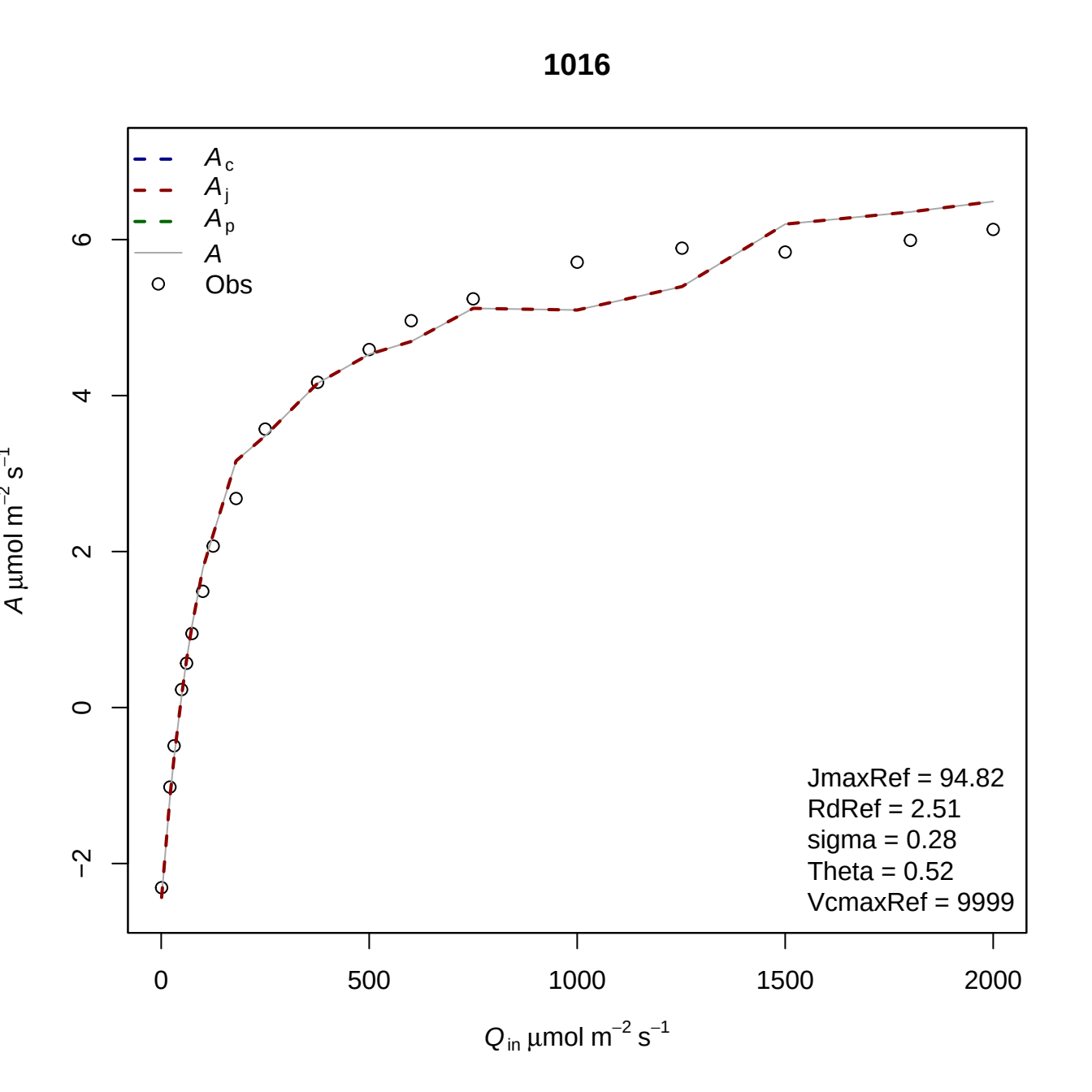

1011
A_c
A_j
A_p
A
Obs
JmaxRef = 39.79
RdRef = 0.76
sigma = 0.18
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

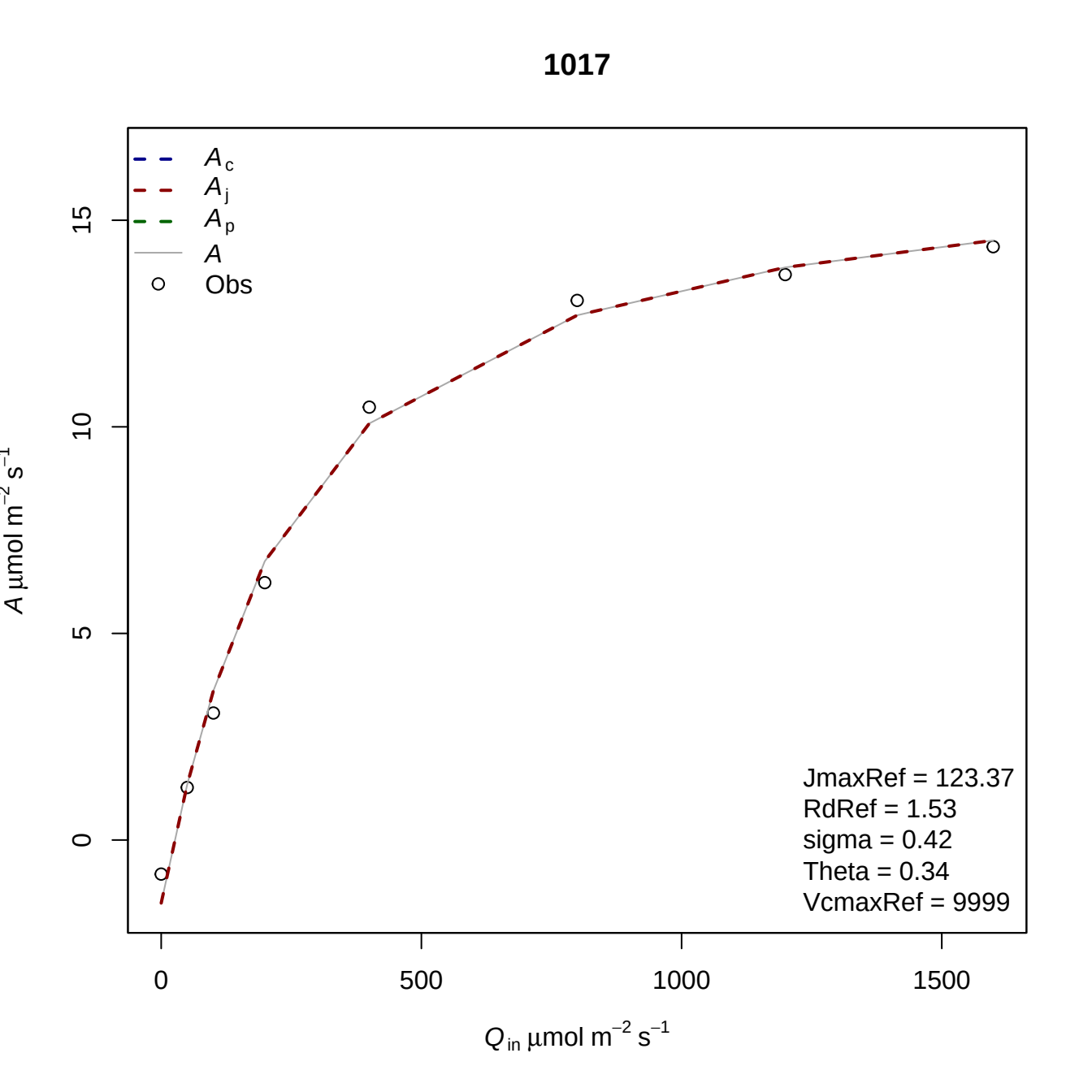

1012
A_c
A_j
A_p
A
Obs
JmaxRef = 84.71
RdRef = 1.63
sigma = 0.23
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

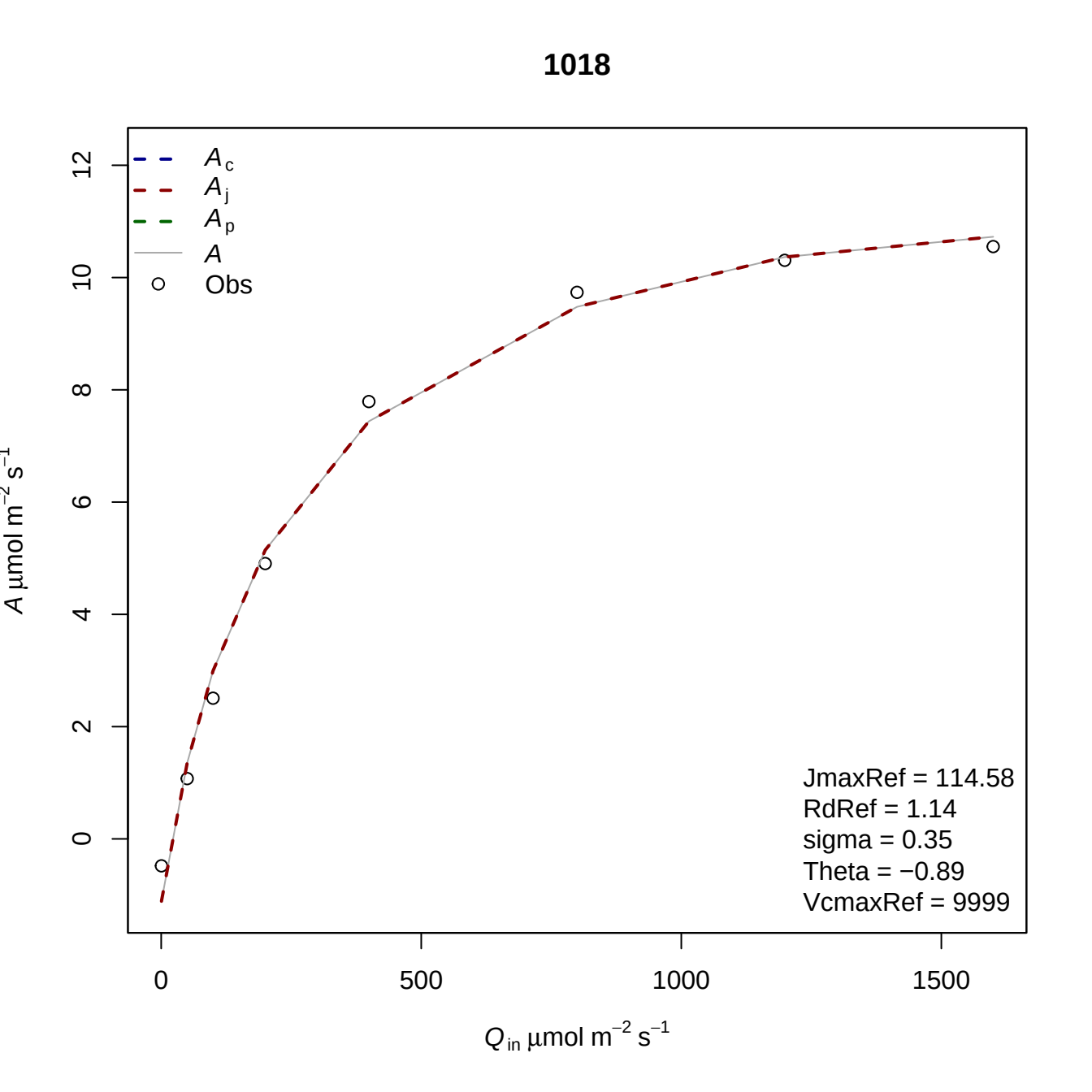

1013
A_c
A_j
A_p
A
Obs
JmaxRef = 143.97
RdRef = 1.52
sigma = 0.19
Theta = 0.31
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

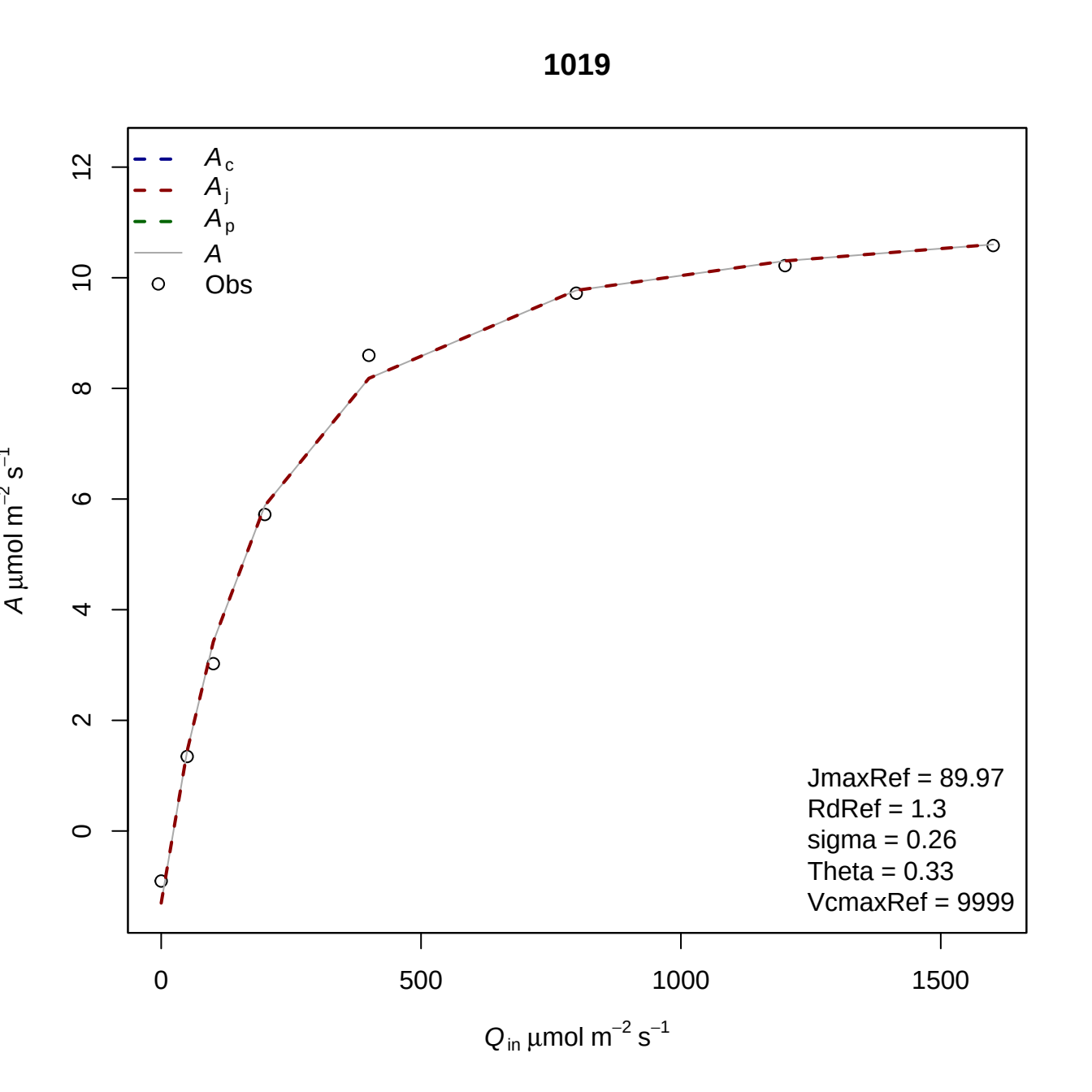

1014
A_c
A_j
A_p
A
Obs
JmaxRef = 99.82
RdRef = −0.35
sigma = 0.21
Theta = −0.99
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

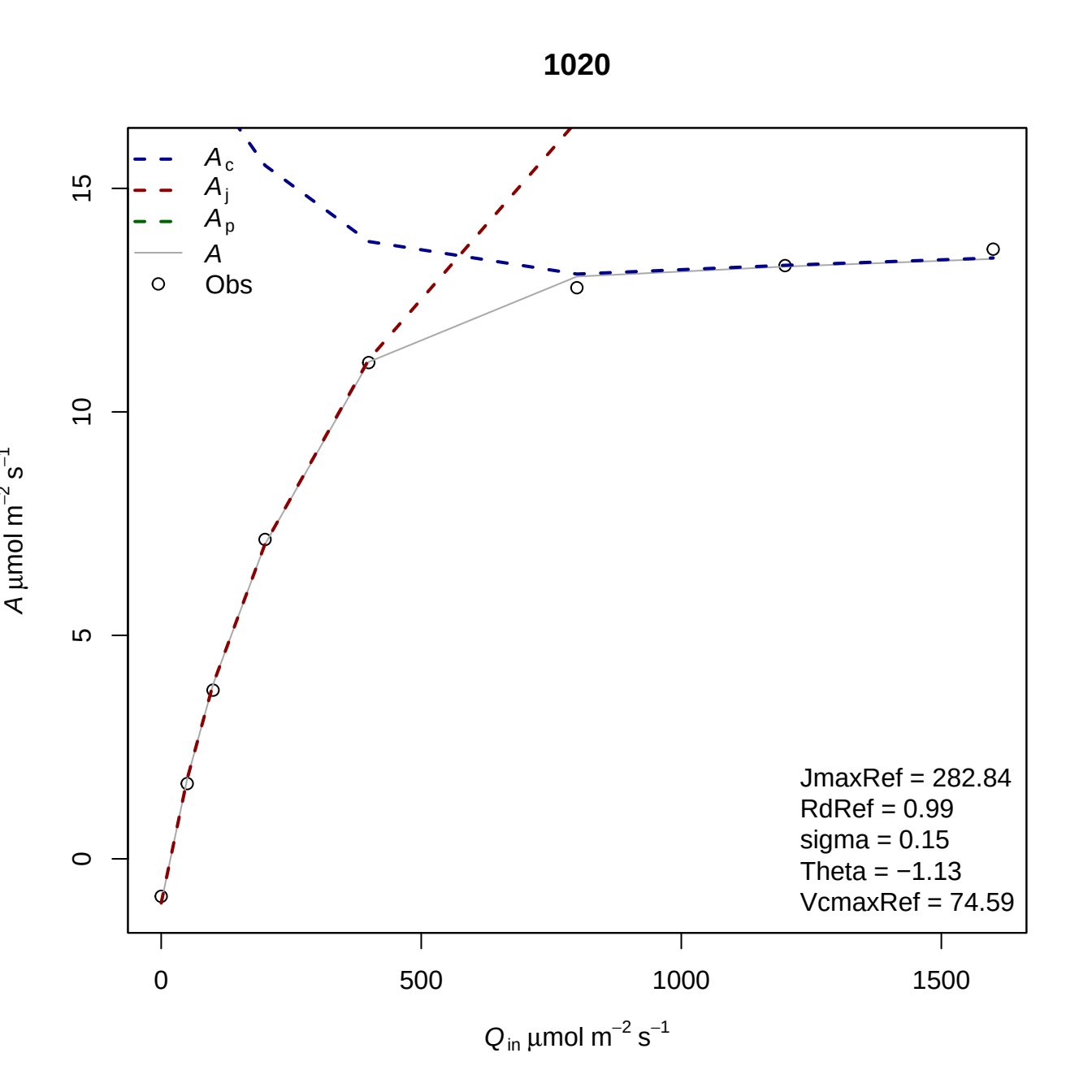

1015
A_c
A_j
A_p
A
Obs
JmaxRef = 92.94
RdRef = 1.33
sigma = 0.12
Theta = −0.14
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

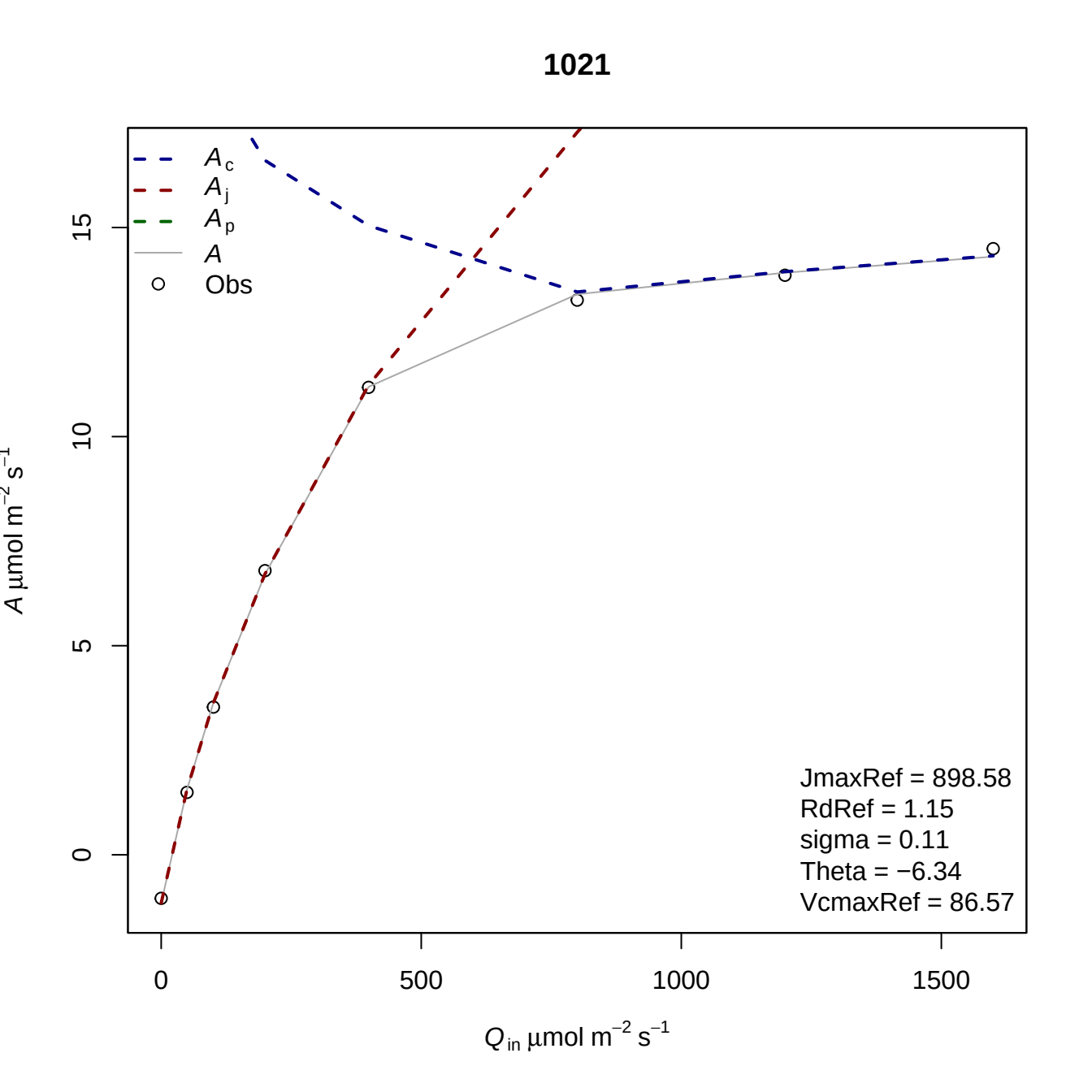

1016
A_c
A_j
A_p
A
Obs
JmaxRef = 94.82
RdRef = 2.51
sigma = 0.28
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

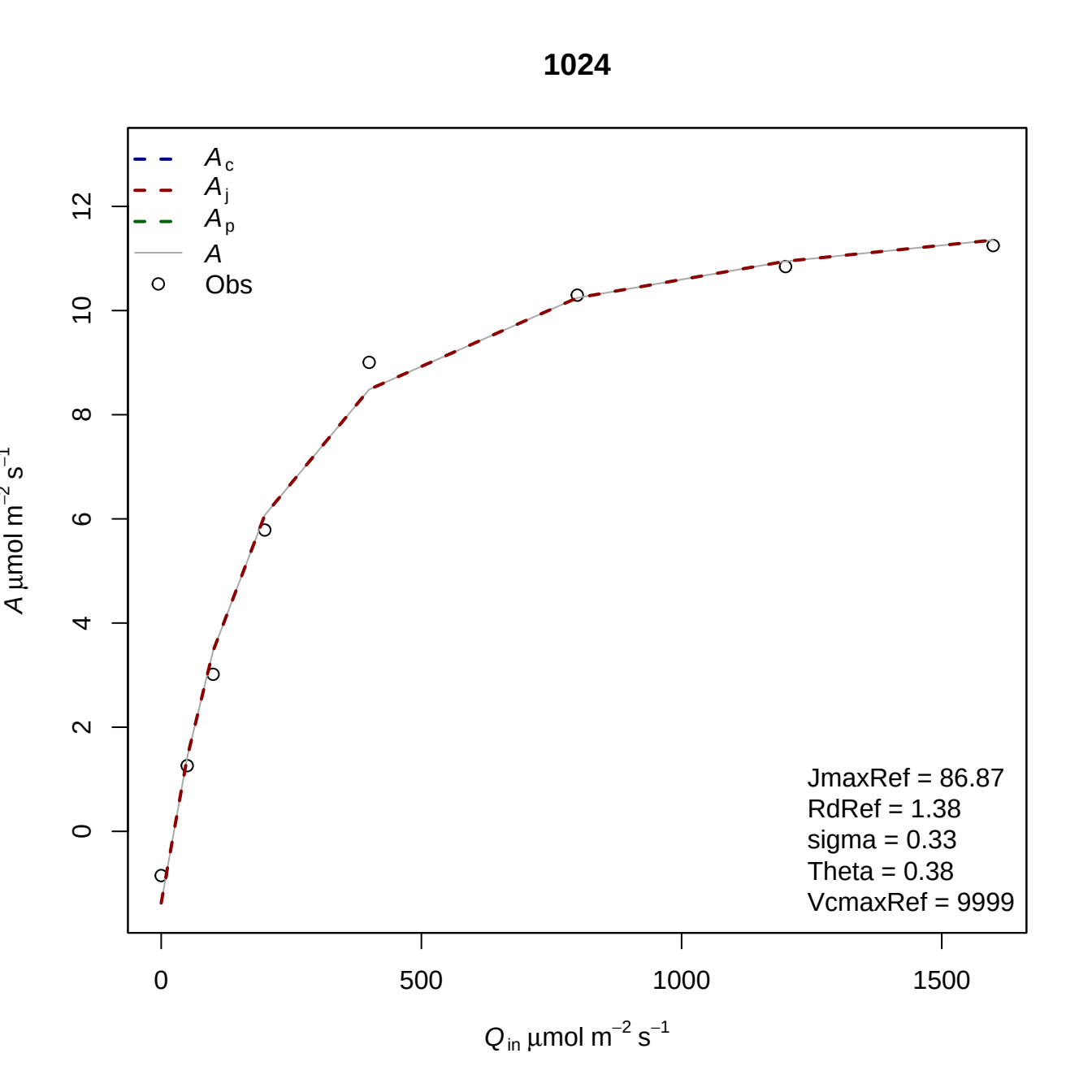

1017
A_c
A_j
A_p
A
Obs
JmaxRef = 123.37
RdRef = 1.53
sigma = 0.42
Theta = 0.34
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

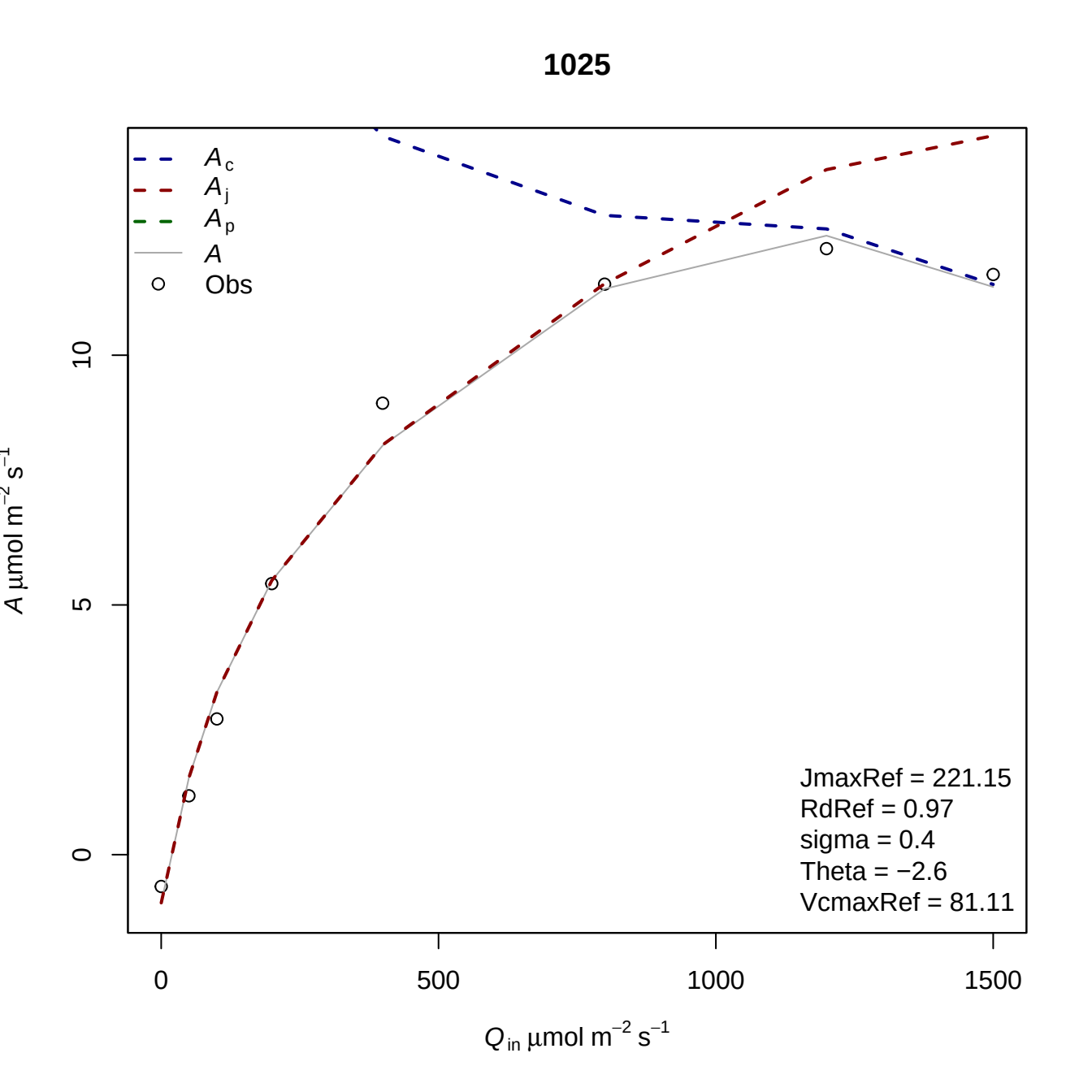

1018
A_c
A_j
A_p
A
Obs
JmaxRef = 114.58
RdRef = 1.14
sigma = 0.35
Theta = −0.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

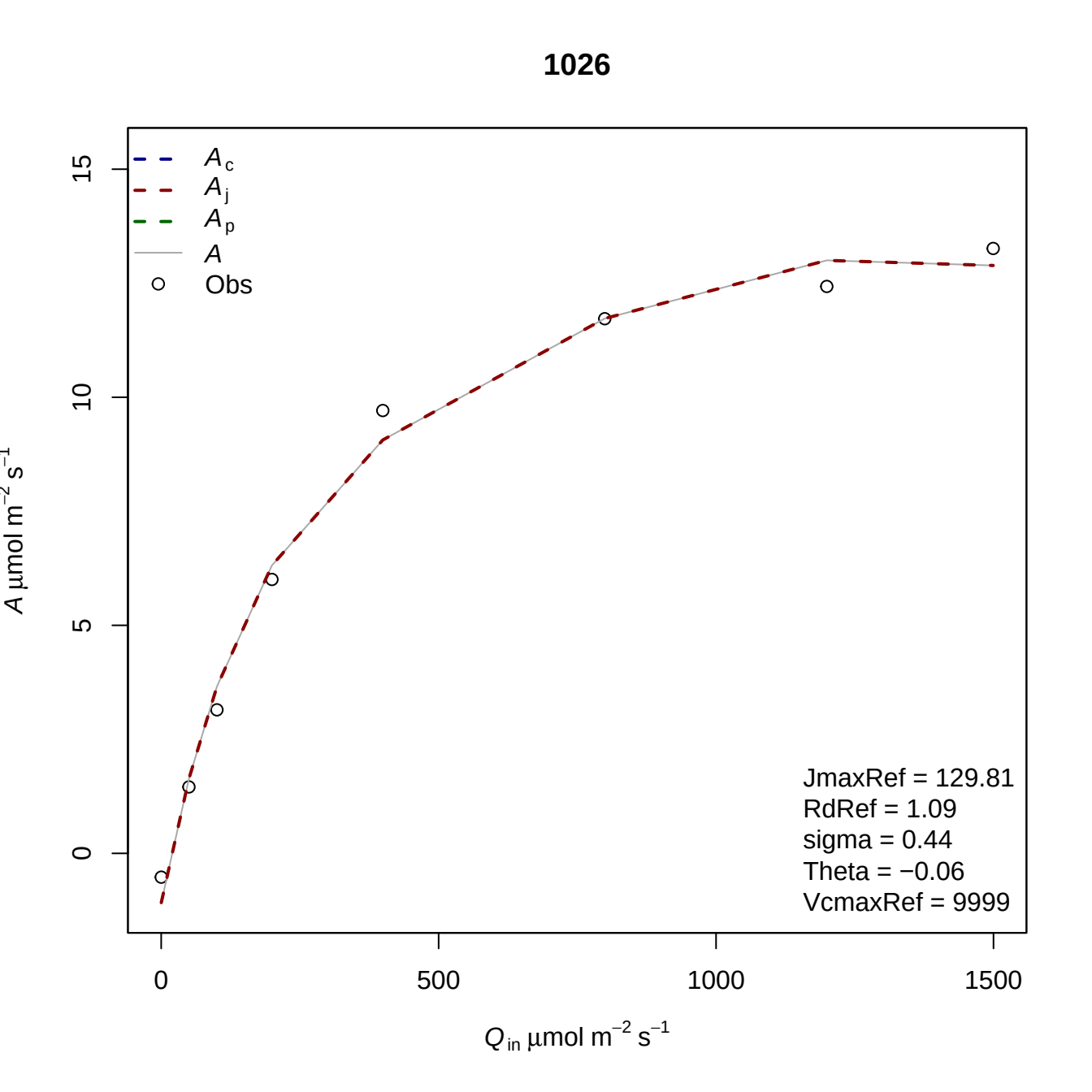

1019
A_c
A_j
A_p
A
Obs
JmaxRef = 89.97
RdRef = 1.3
sigma = 0.26
Theta = 0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

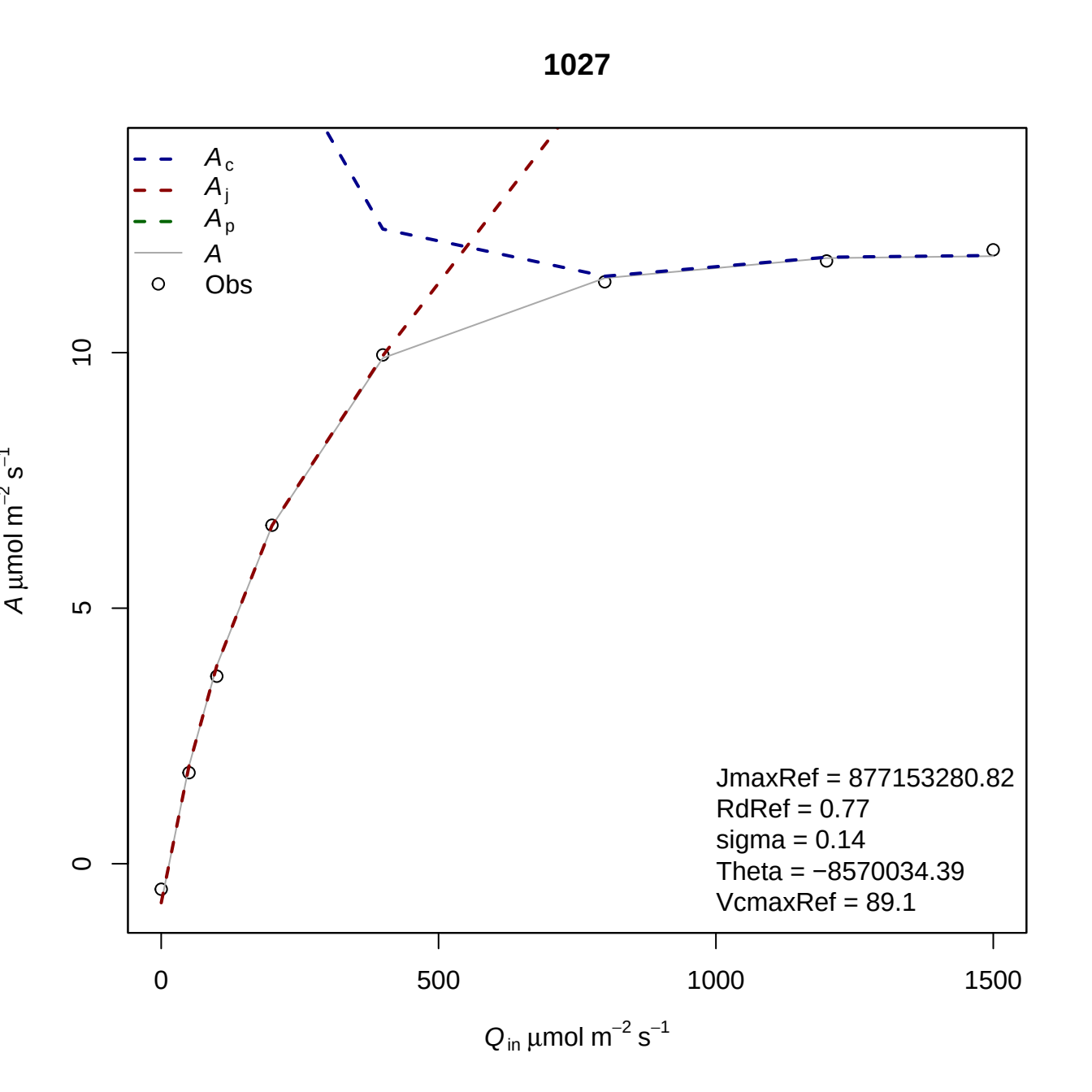

1020
A_c
A_j
A_p
A
Obs
JmaxRef = 282.84
RdRef = 0.99
sigma = 0.15
Theta = −1.13
VcmaxRef = 74.59
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

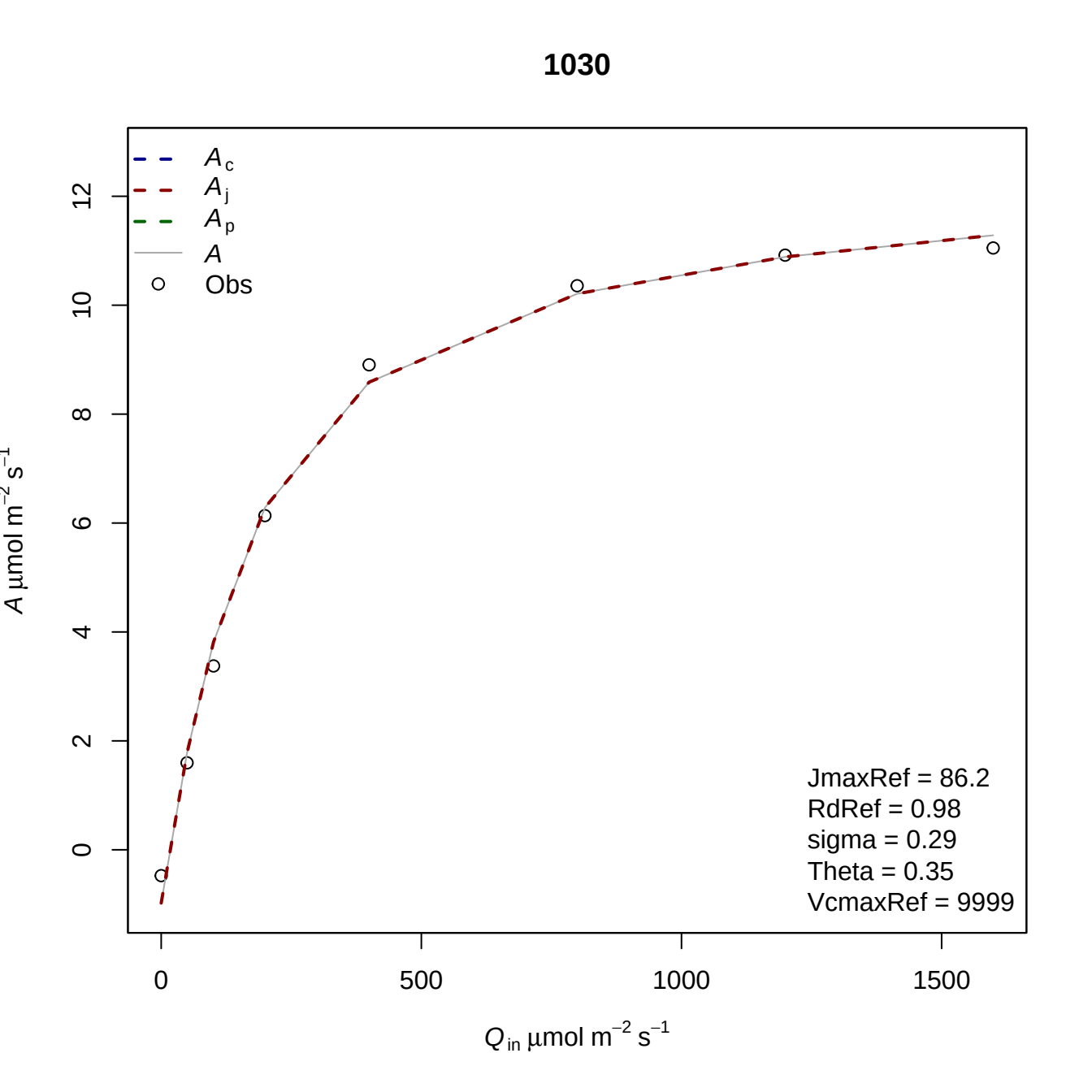

1021
A_c
A_j
A_p
A
Obs
JmaxRef = 898.58
RdRef = 1.15
sigma = 0.11
Theta = −6.34
VcmaxRef = 86.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

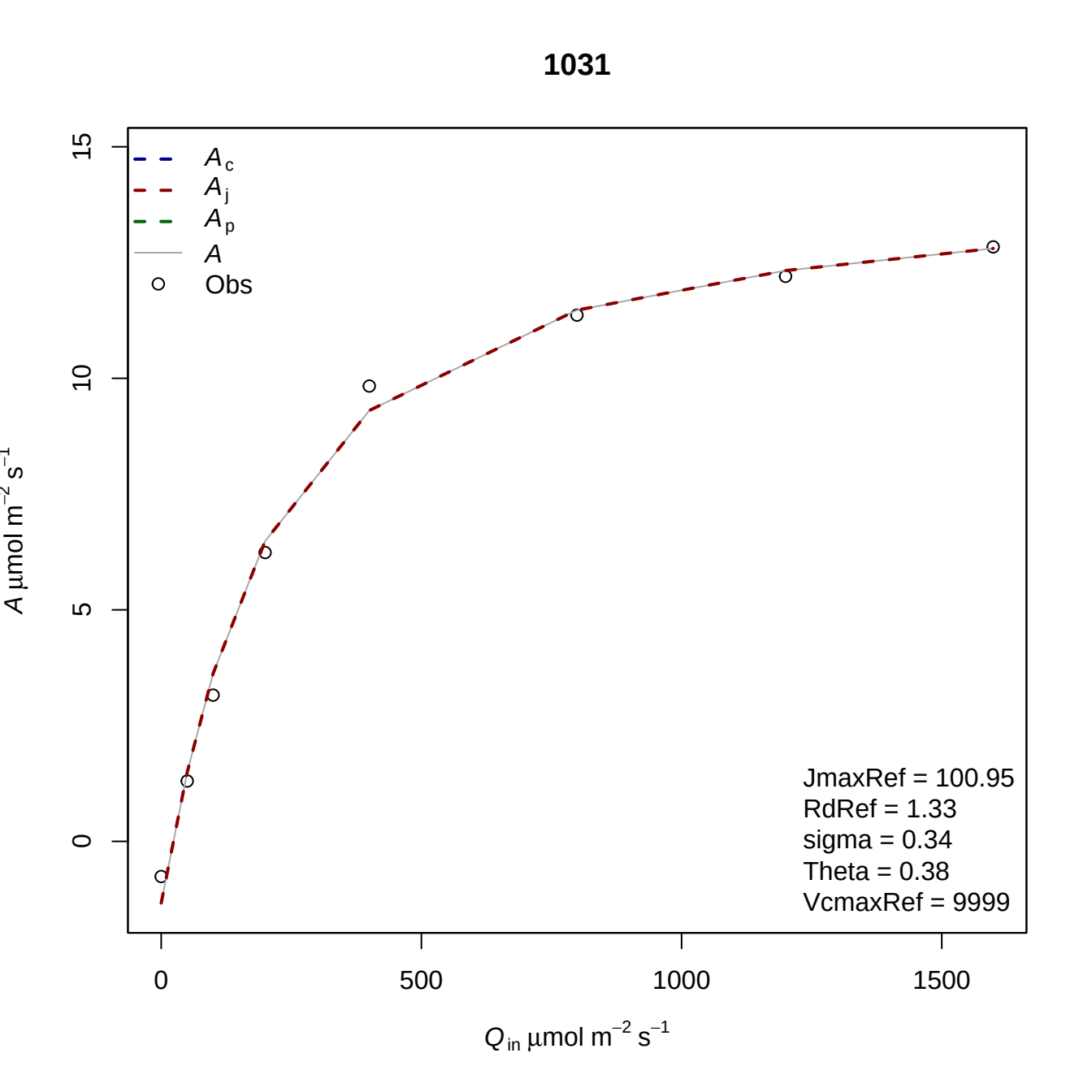

1024
A_c
A_j
A_p
A
Obs
JmaxRef = 86.87
RdRef = 1.38
sigma = 0.33
Theta = 0.38
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

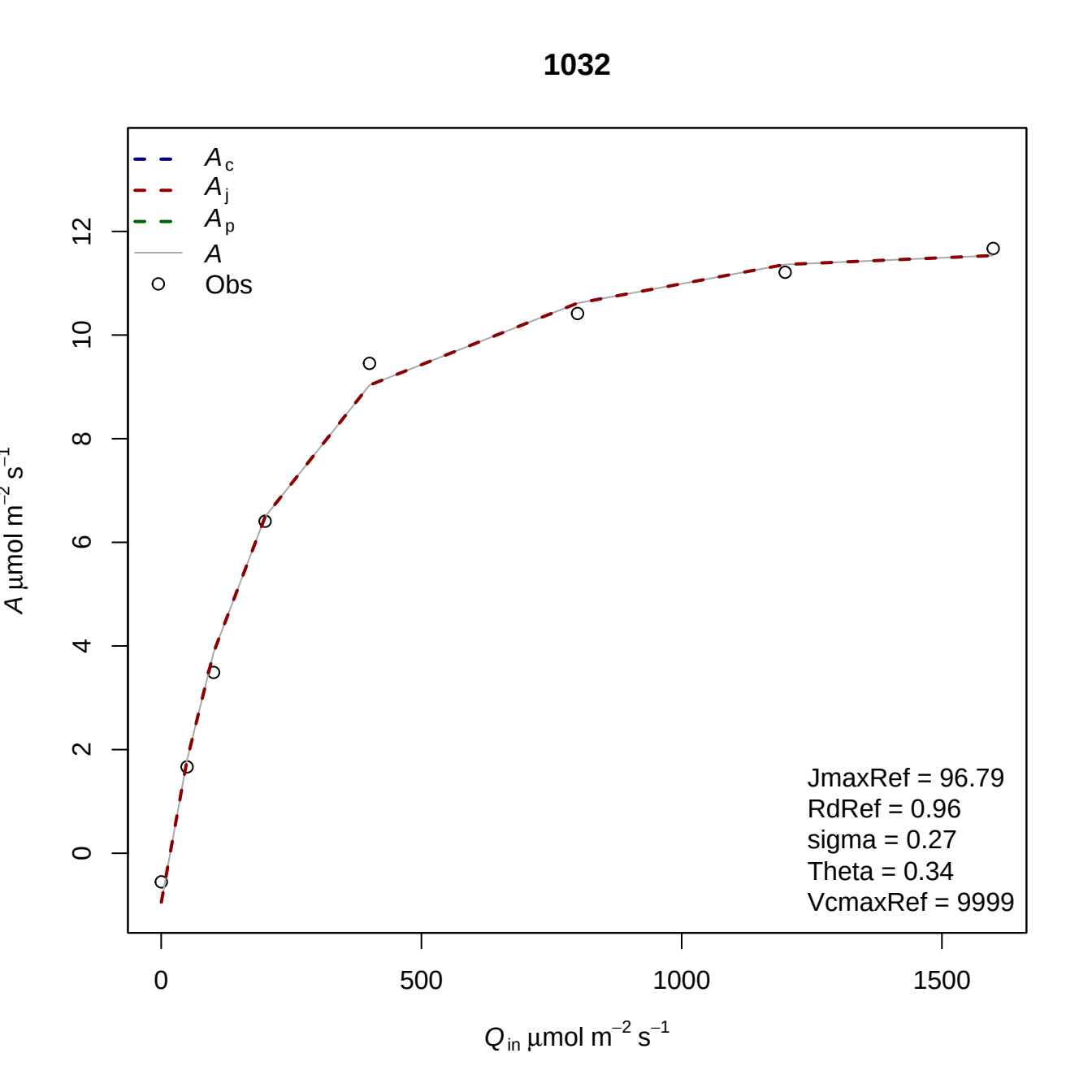

1025
A_c
A_j
A_p
A
Obs
JmaxRef = 221.15
RdRef = 0.97
sigma = 0.4
Theta = −2.6
VcmaxRef = 81.11
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

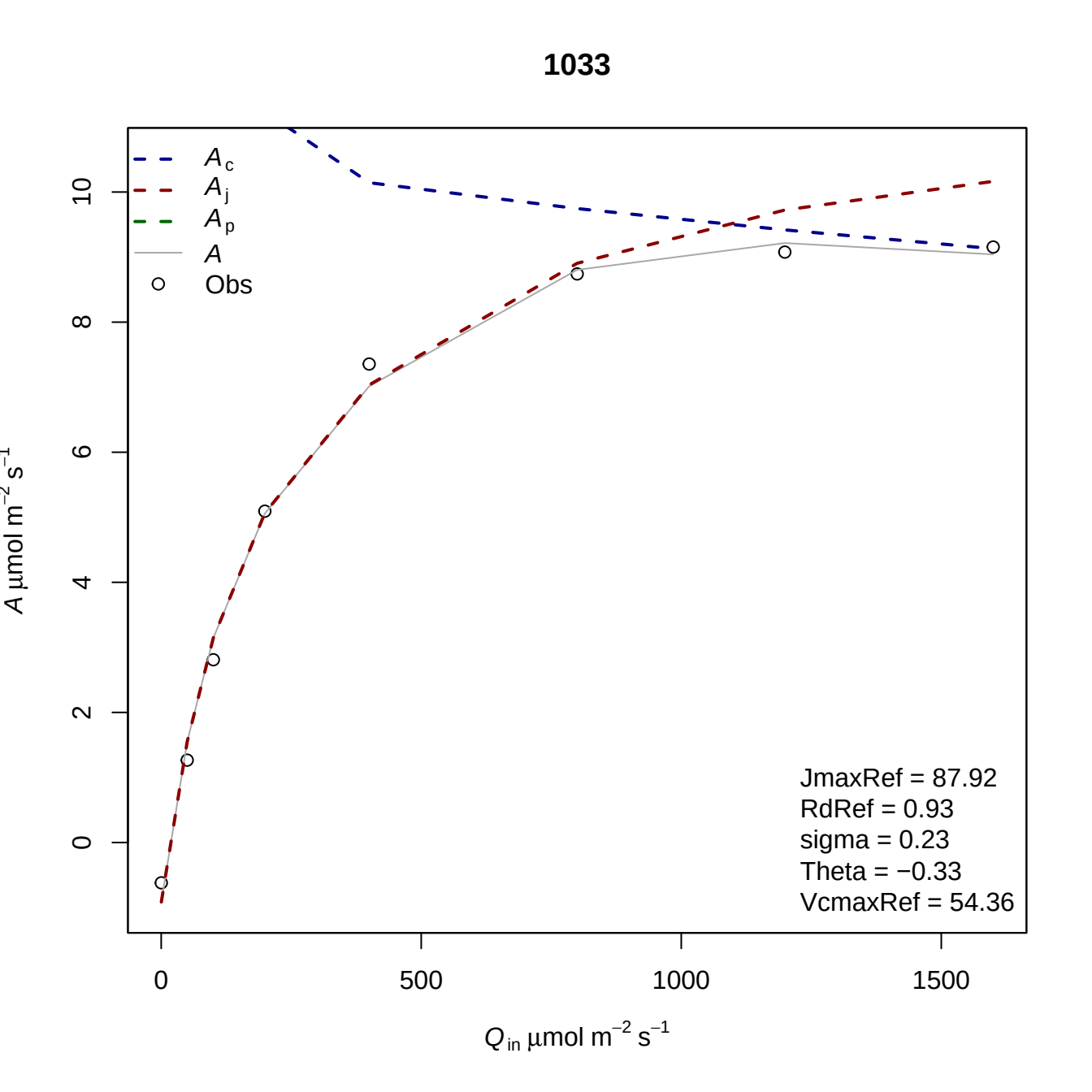

1026
A_c
A_j
A_p
A
Obs
JmaxRef = 129.81
RdRef = 1.09
sigma = 0.44
Theta = −0.06
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

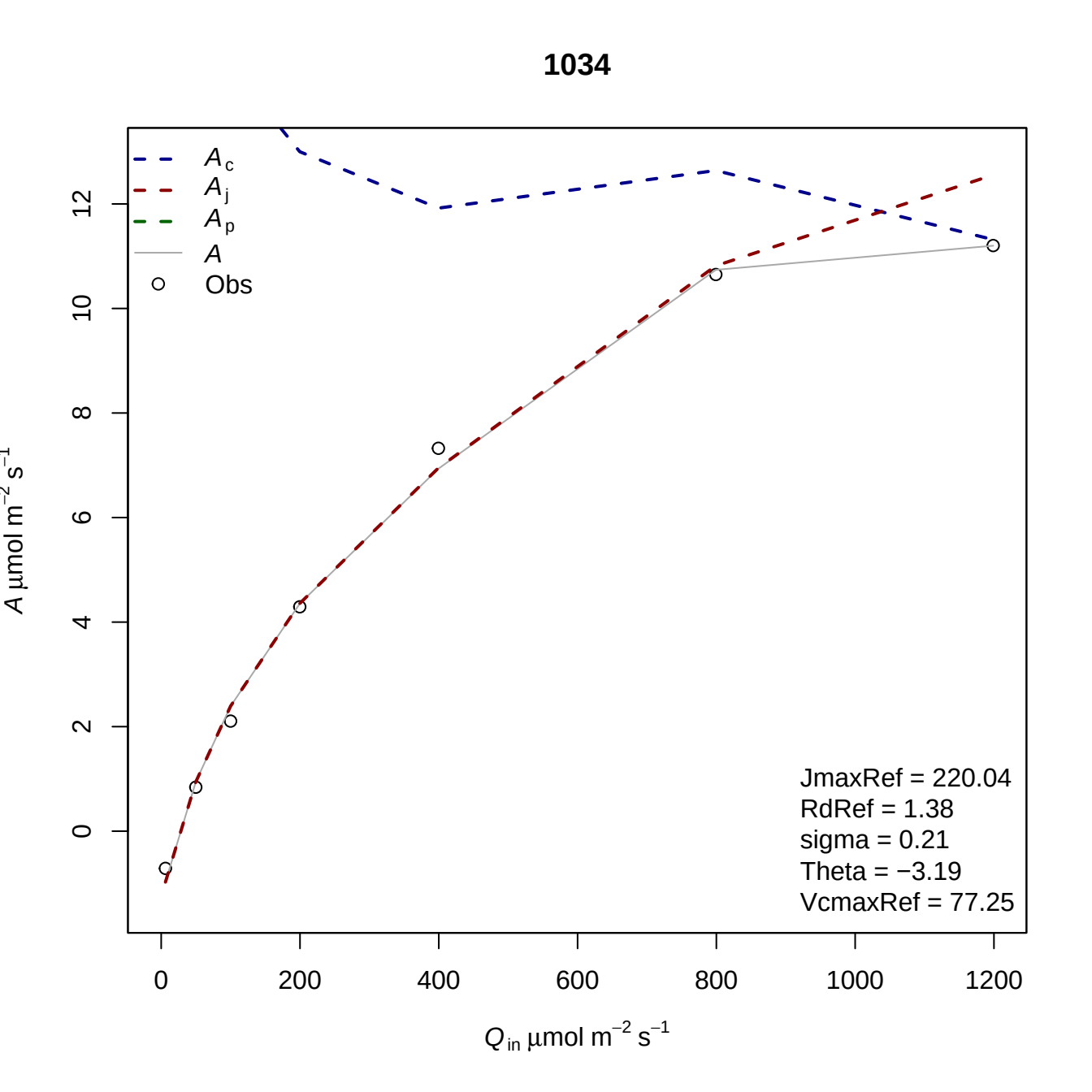

1027
A_c
A_j
A_p
A
Obs
JmaxRef = 877153280.82
RdRef = 0.77
sigma = 0.14
Theta = −8570034.39
VcmaxRef = 89.1
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

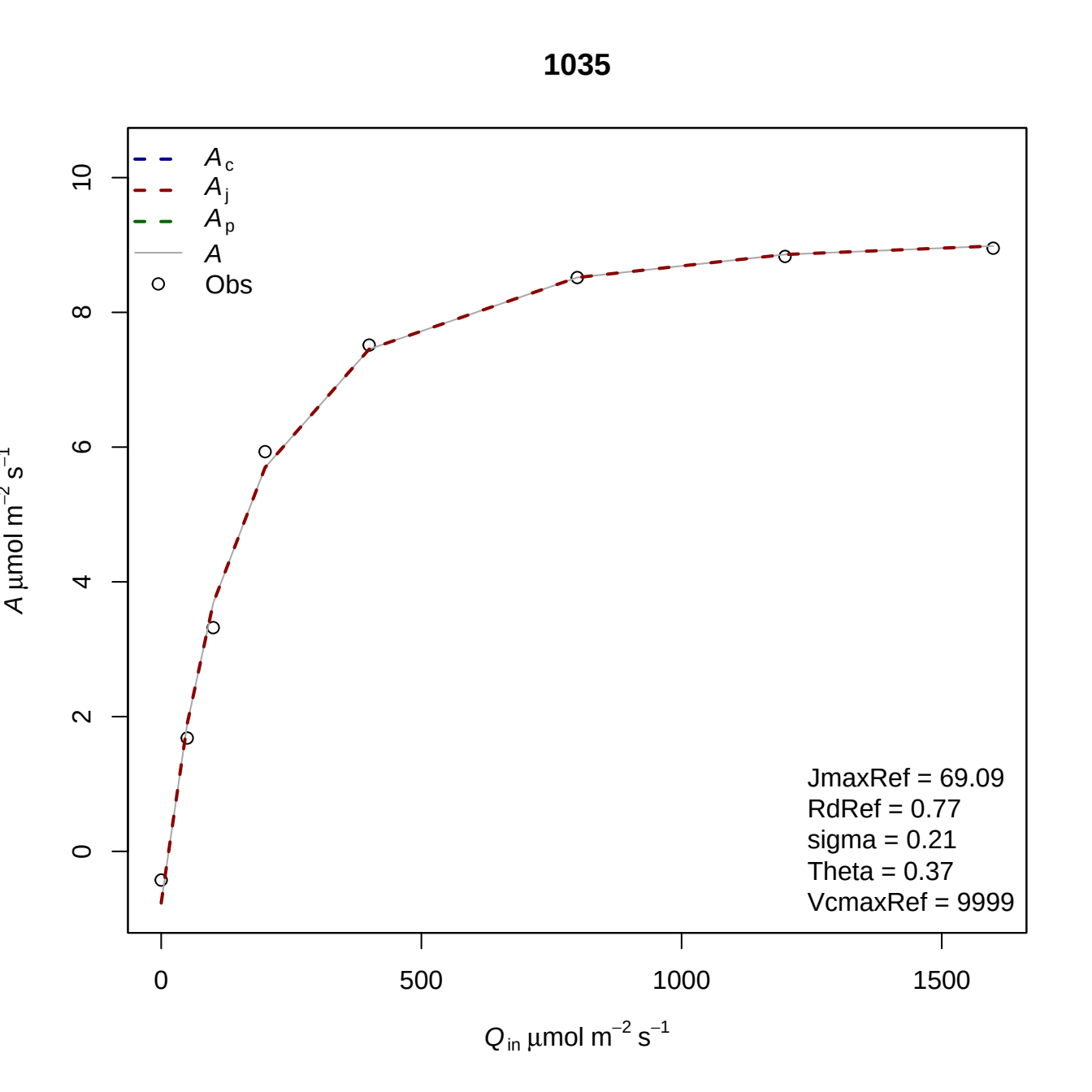

1030
A_c
A_j
A_p
A
Obs
JmaxRef = 86.2
RdRef = 0.98
sigma = 0.29
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

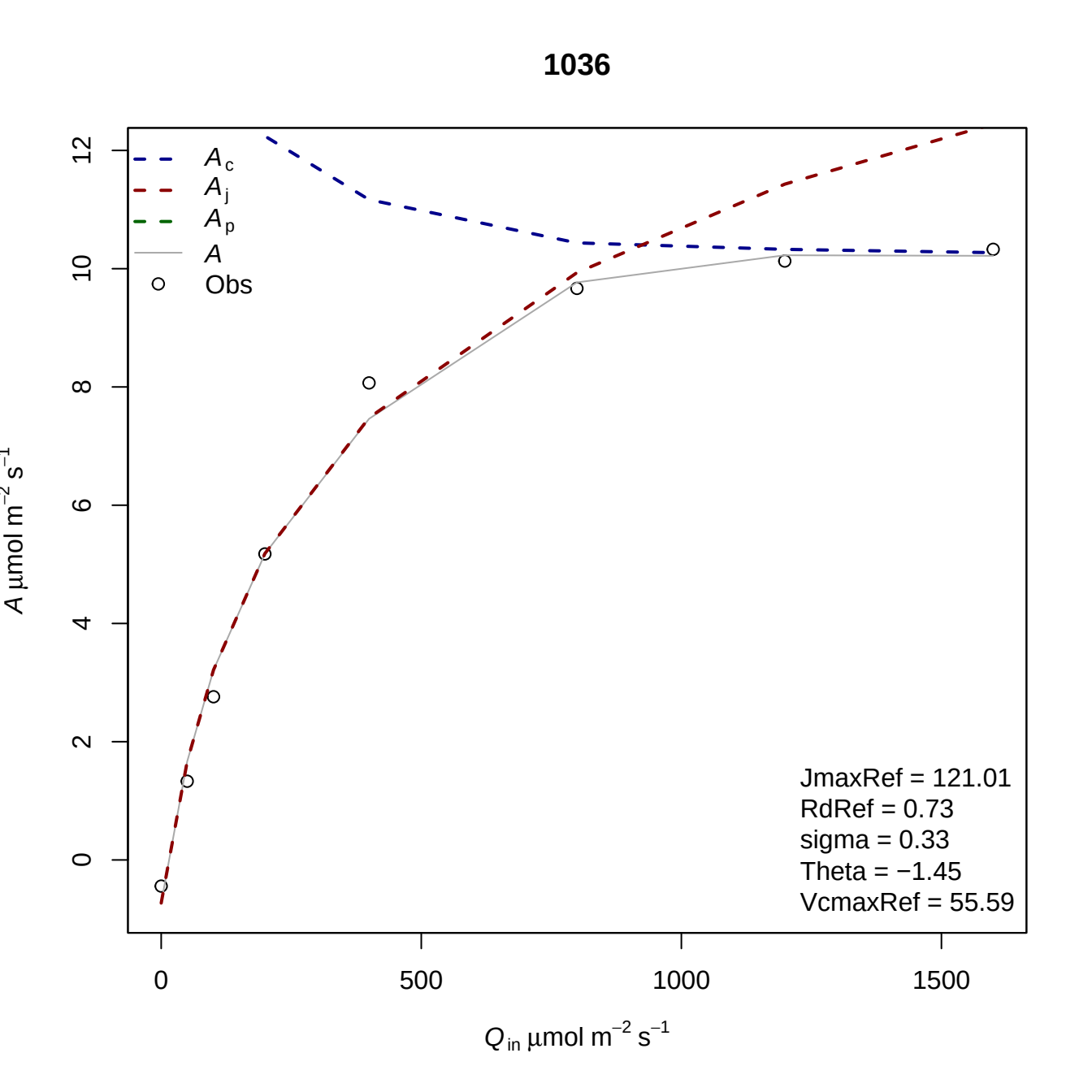

1031
A_c
A_j
A_p
A
Obs
JmaxRef = 100.95
RdRef = 1.33
sigma = 0.34
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

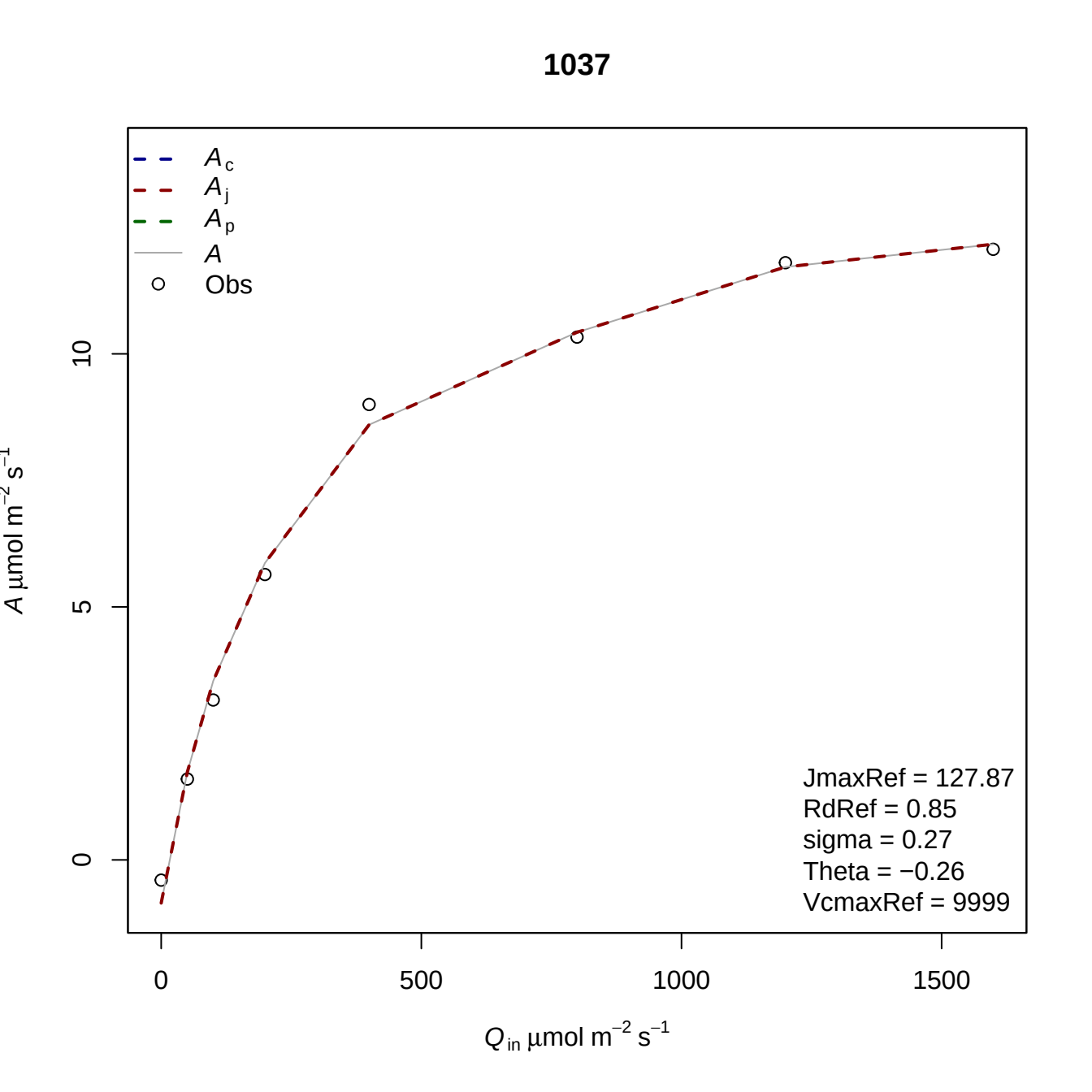

1032
A_c
A_j
A_p
A
Obs
JmaxRef = 96.79
RdRef = 0.96
sigma = 0.27
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

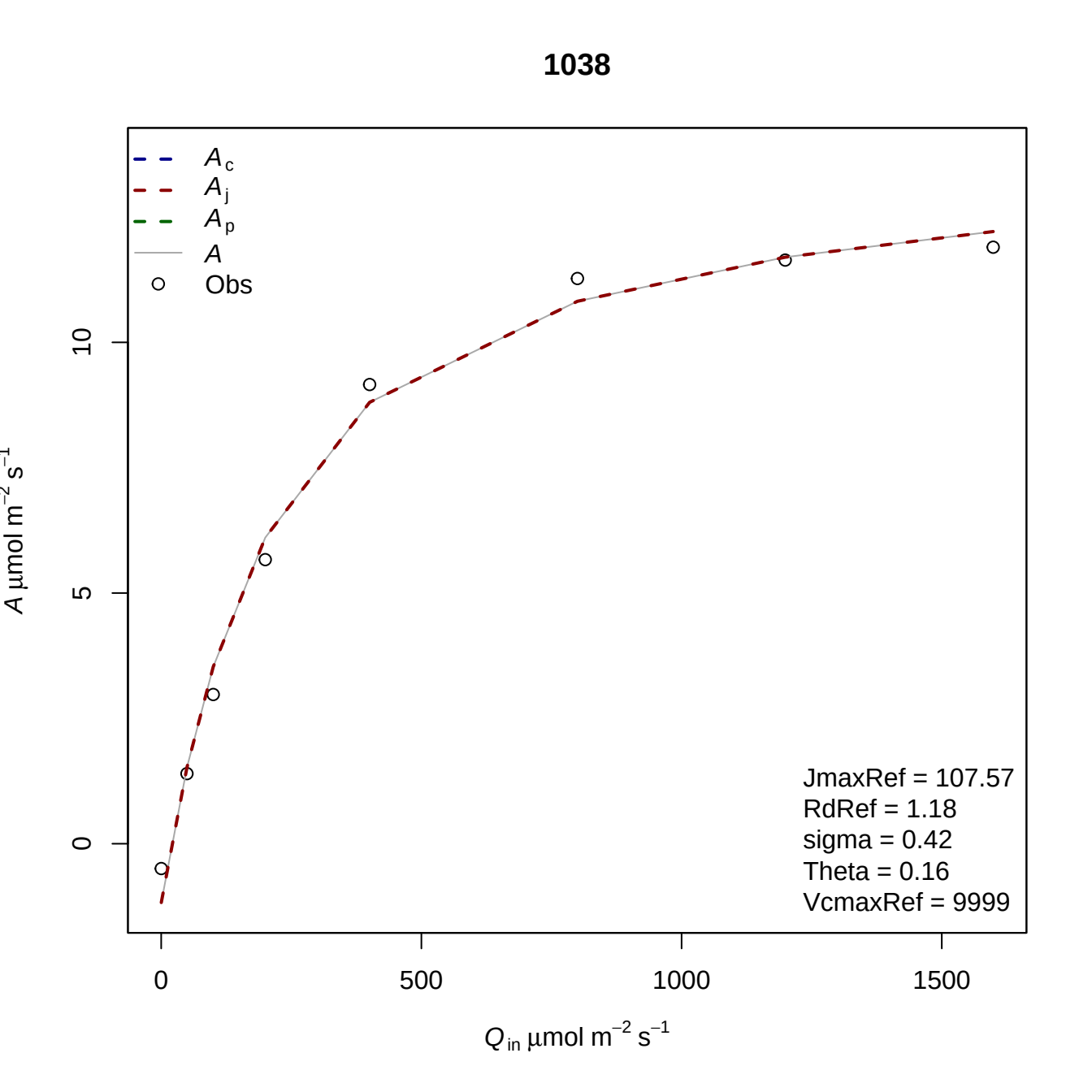

1033
A_c
A_j
A_p
A
Obs
JmaxRef = 87.92
RdRef = 0.93
sigma = 0.23
Theta = −0.33
VcmaxRef = 54.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

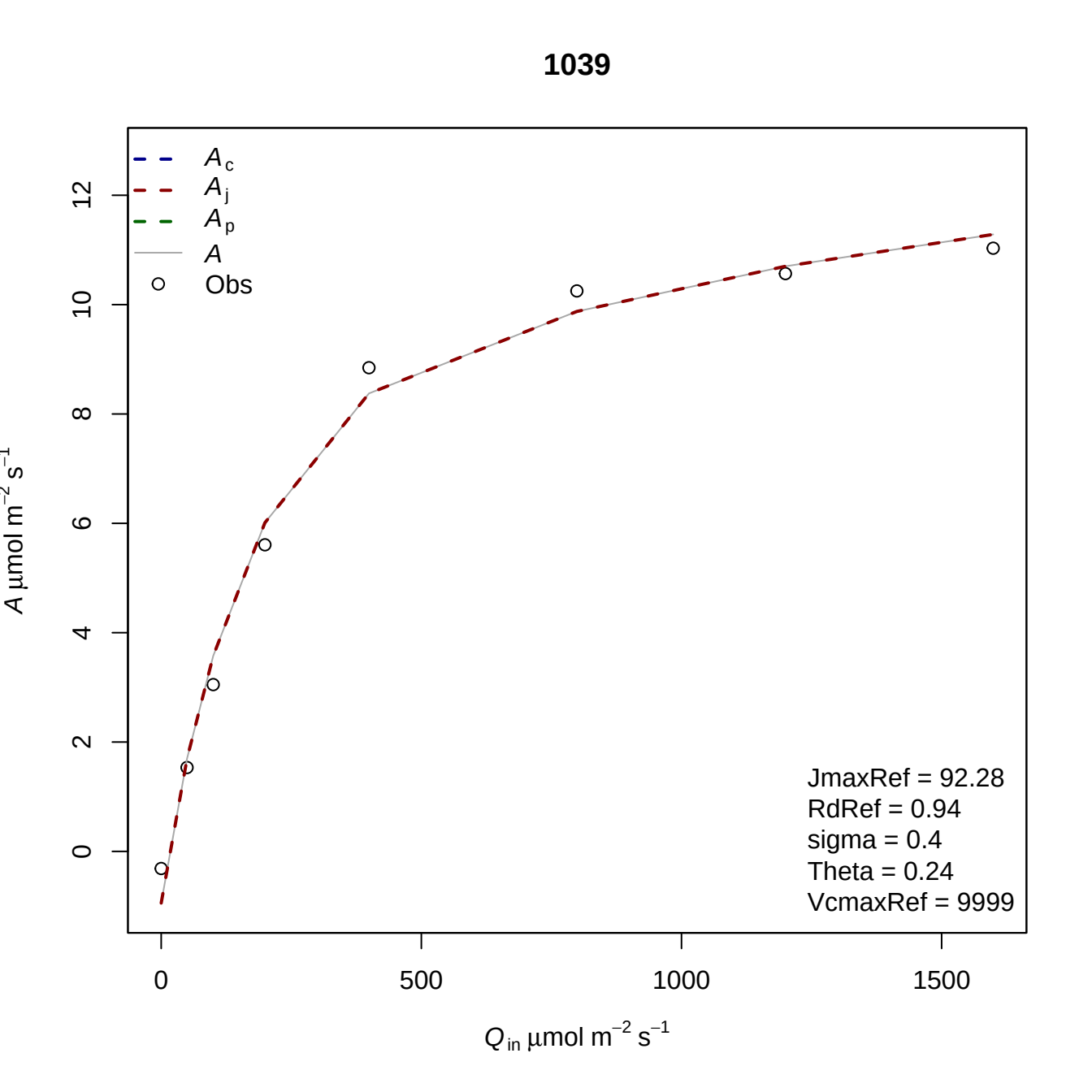

1034
A_c
A_j
A_p
A
Obs
JmaxRef = 220.04
RdRef = 1.38
sigma = 0.21
Theta = −3.19
VcmaxRef = 77.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

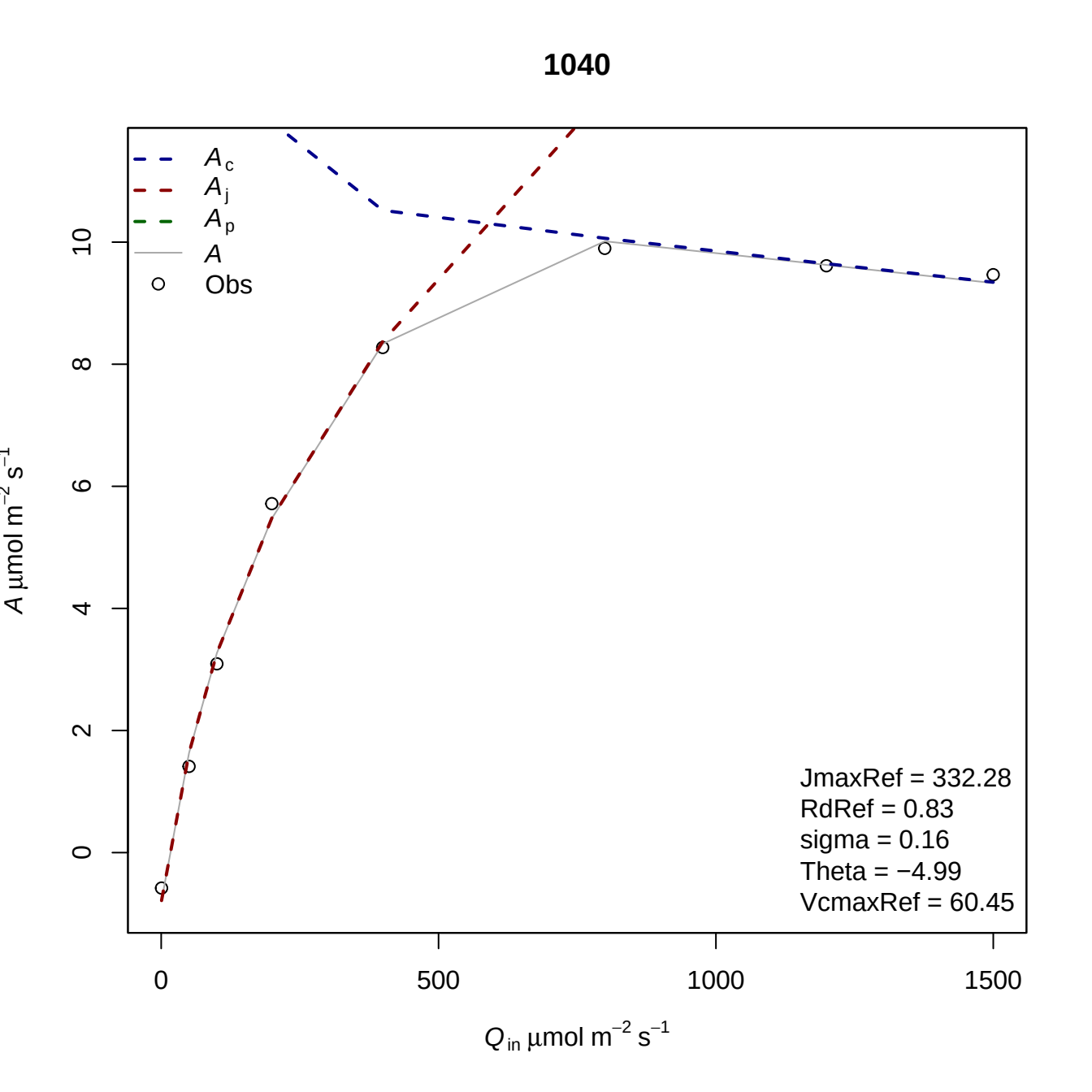

1035
A_c
A_j
A_p
A
Obs
JmaxRef = 69.09
RdRef = 0.77
sigma = 0.21
Theta = 0.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

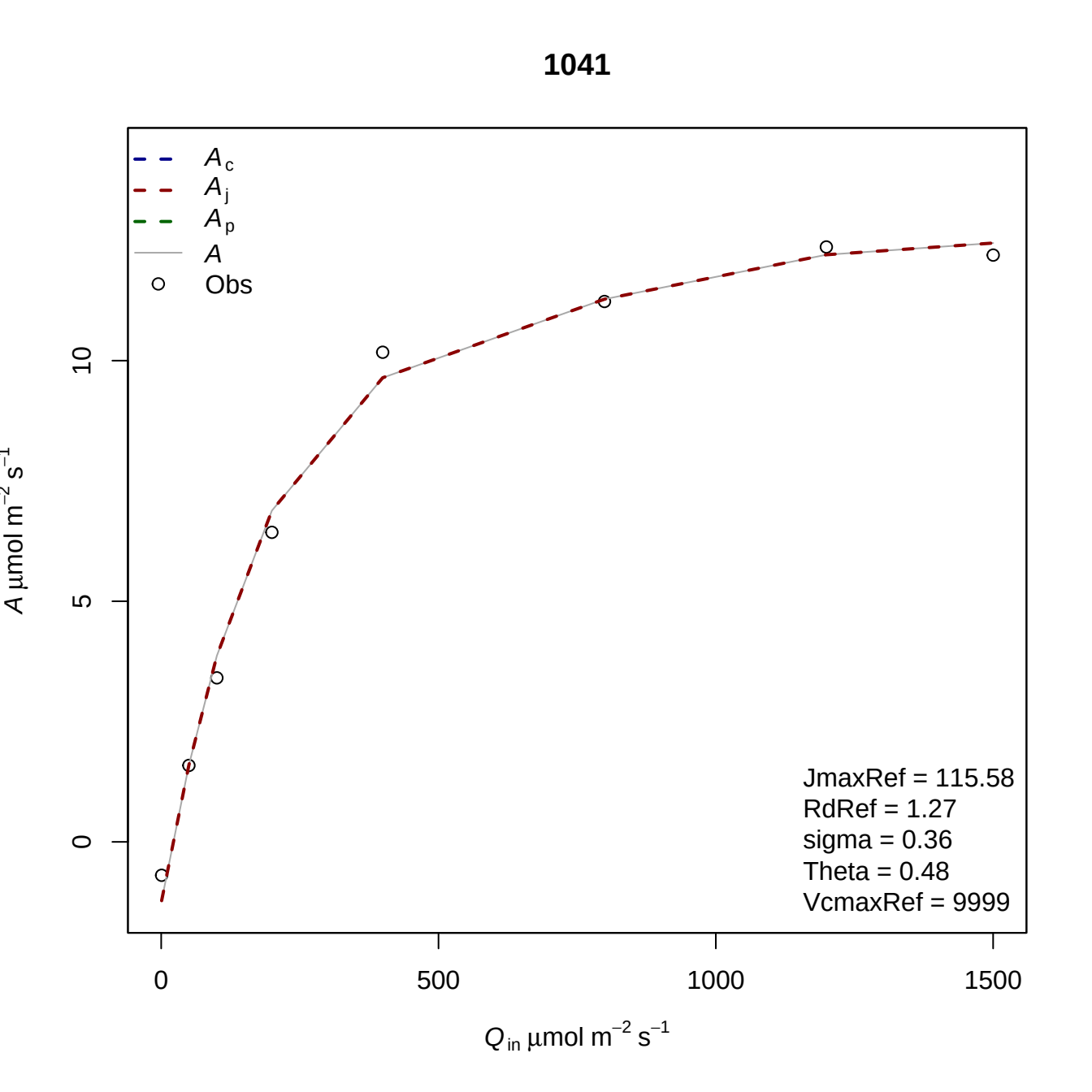

1036
A_c
A_j
A_p
A
Obs
JmaxRef = 121.01
RdRef = 0.73
sigma = 0.33
Theta = −1.45
VcmaxRef = 55.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

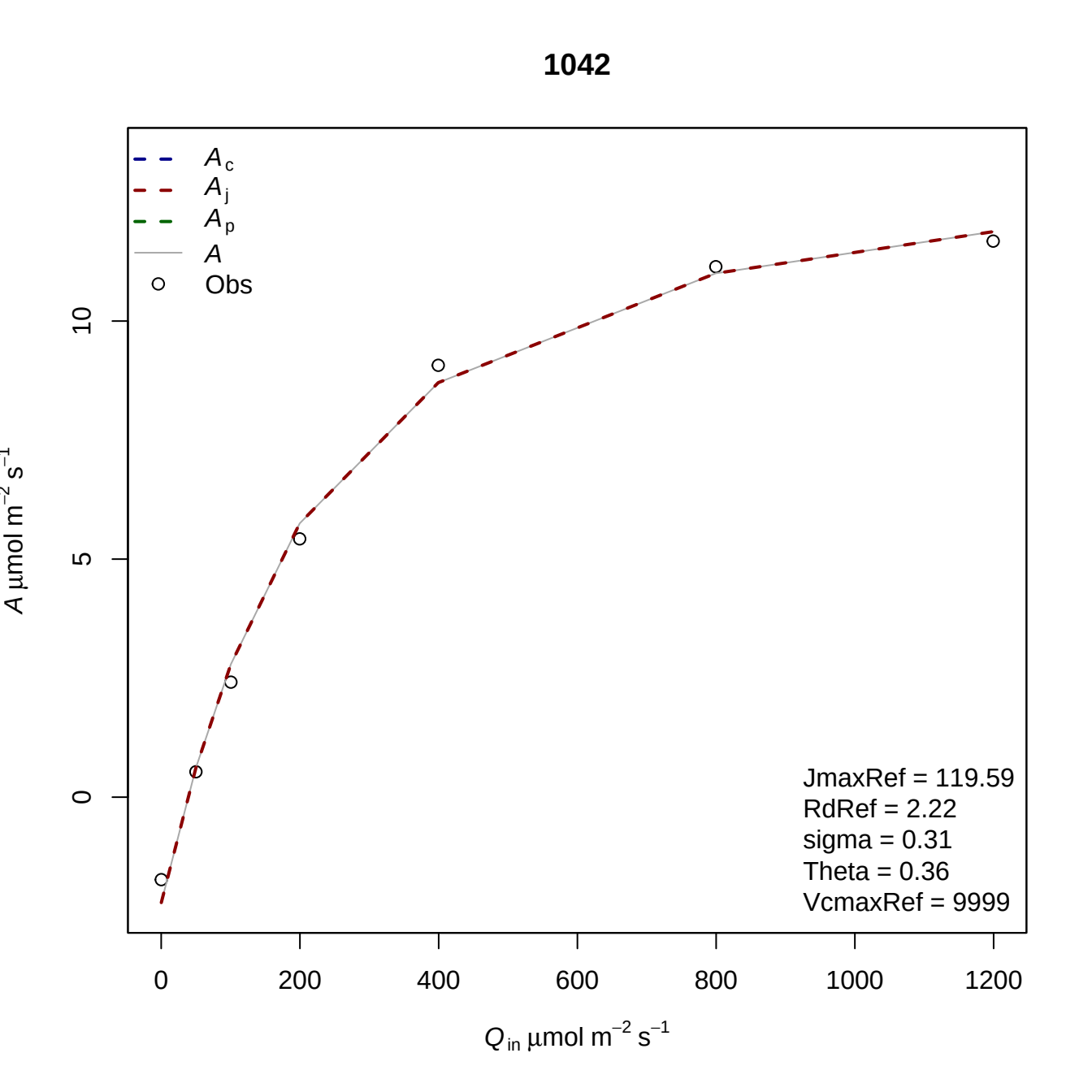

1037
A_c
A_j
A_p
A
Obs
JmaxRef = 127.87
RdRef = 0.85
sigma = 0.27
Theta = −0.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

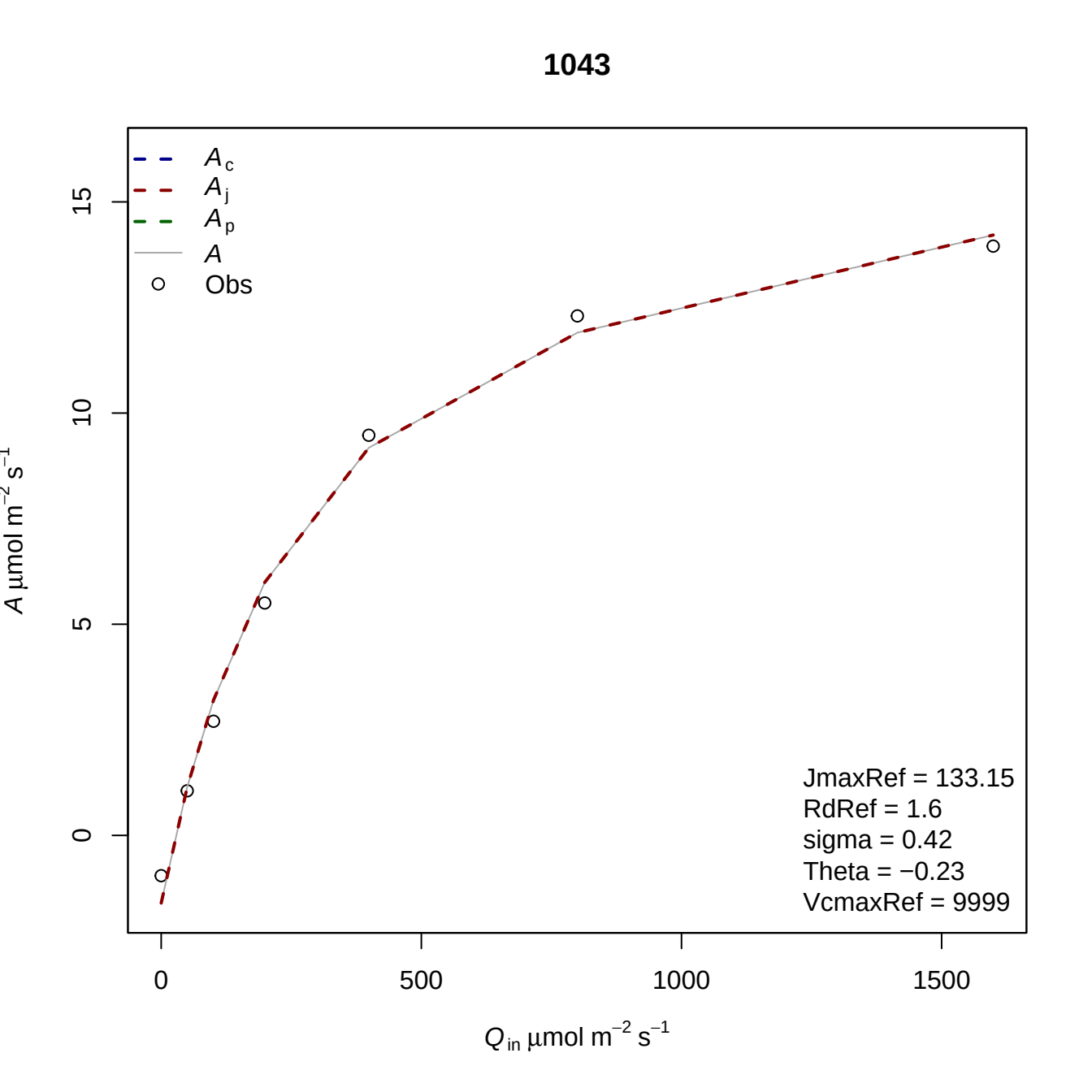

1038
A_c
A_j
A_p
A
Obs
JmaxRef = 107.57
RdRef = 1.18
sigma = 0.42
Theta = 0.16
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

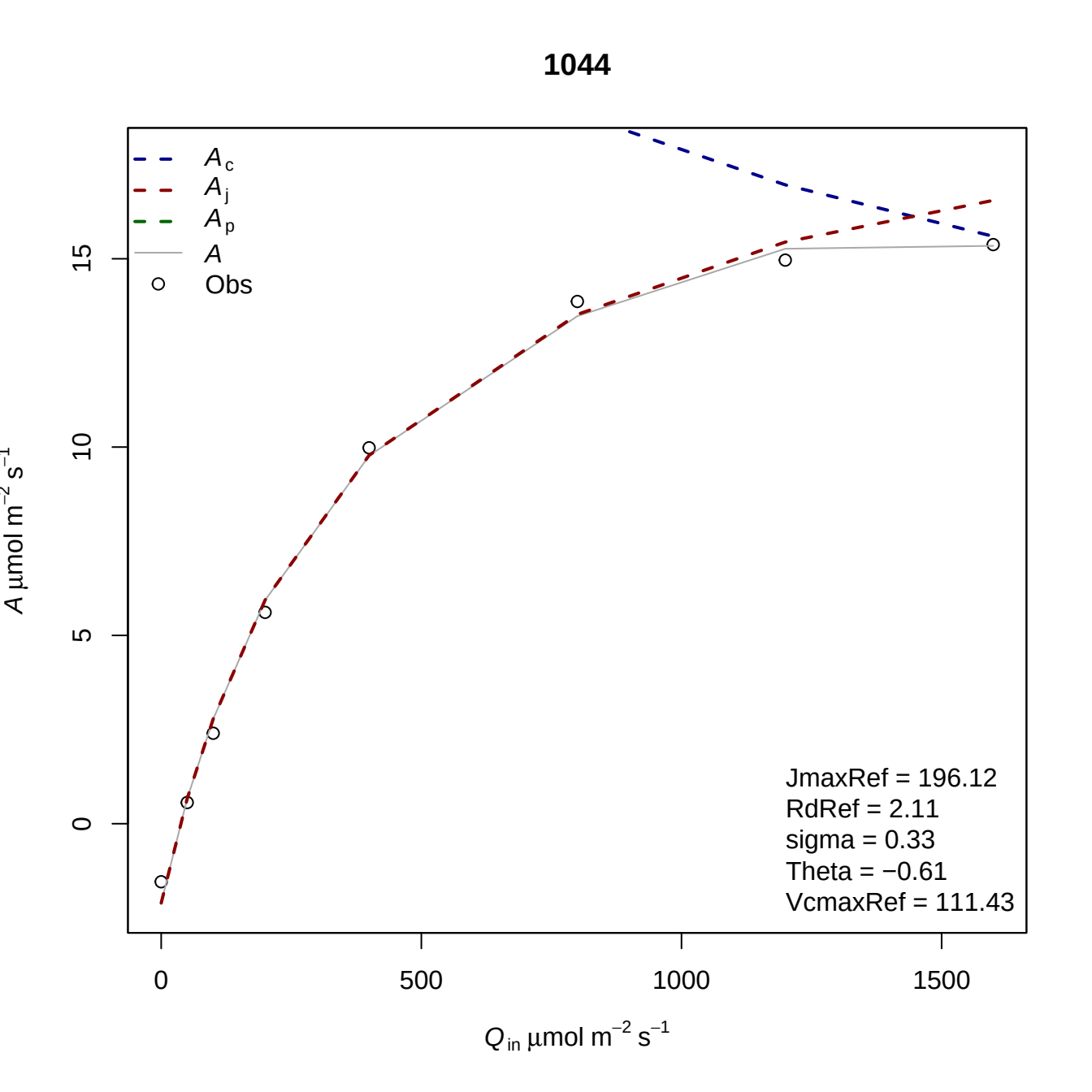

1039
A_c
A_j
A_p
A
Obs
JmaxRef = 92.28
RdRef = 0.94
sigma = 0.4
Theta = 0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

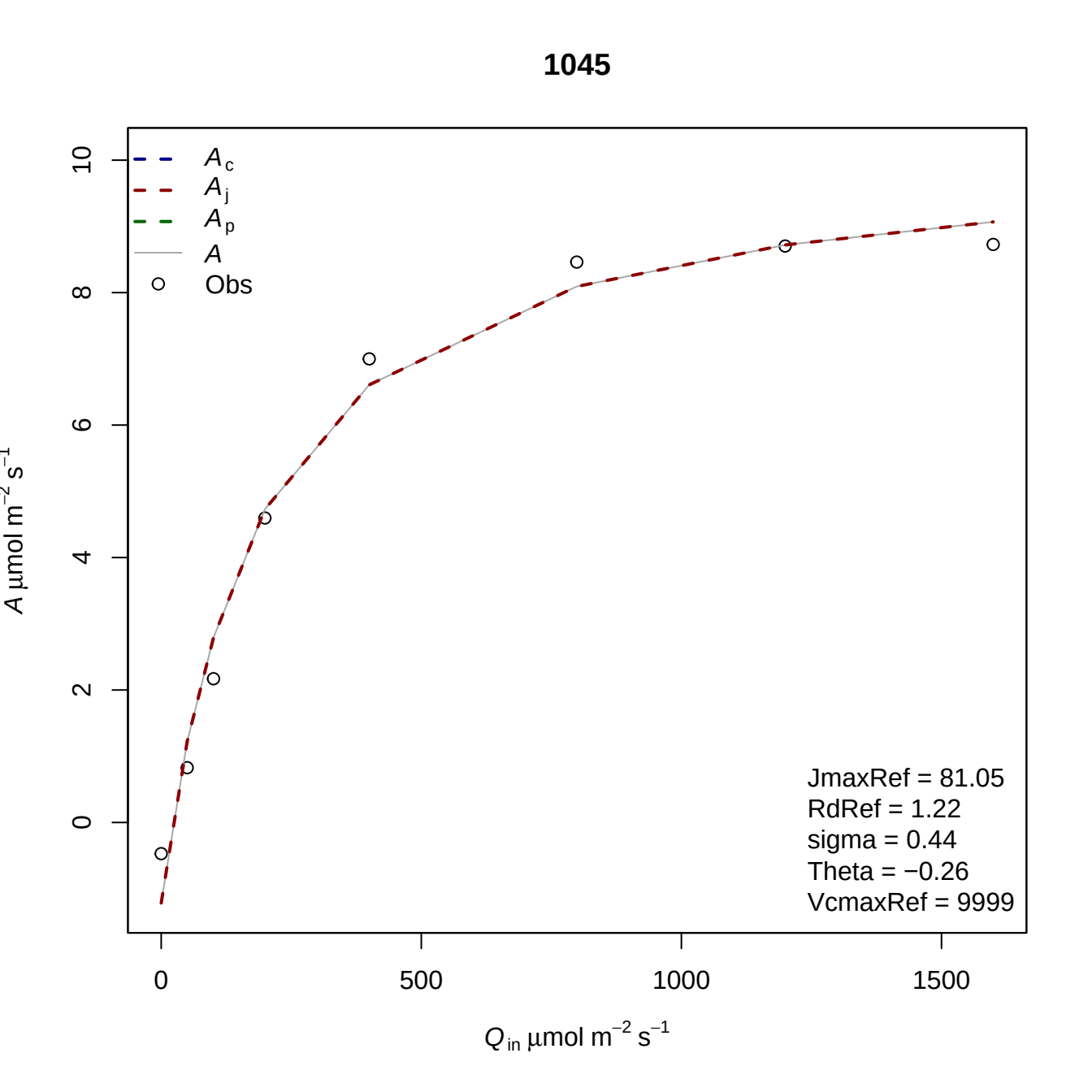

1040
A_c
A_j
A_p
A
Obs
JmaxRef = 332.28
RdRef = 0.83
sigma = 0.16
Theta = −4.99
VcmaxRef = 60.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

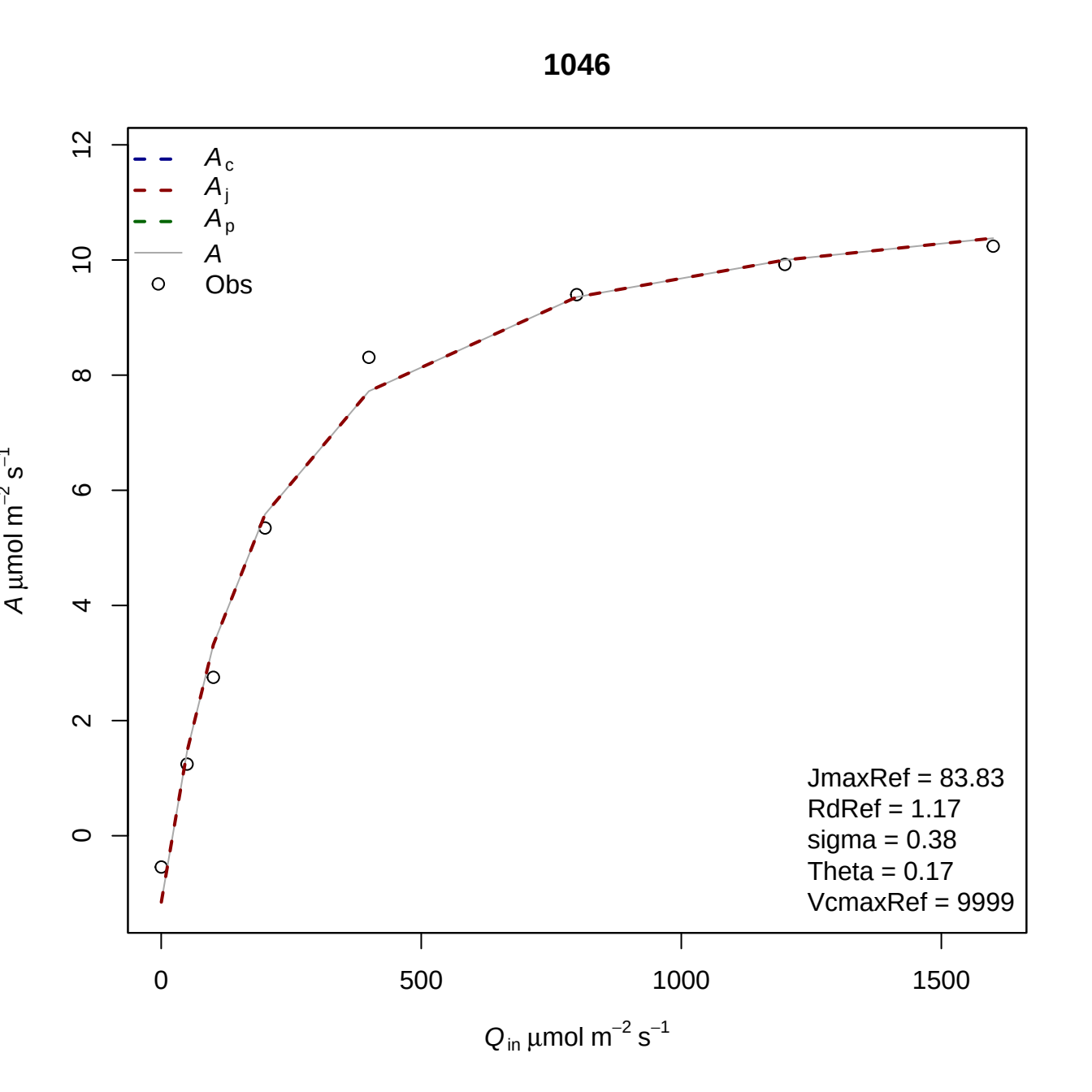

1041
A_c
A_j
A_p
A
Obs
JmaxRef = 115.58
RdRef = 1.27
sigma = 0.36
Theta = 0.48
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

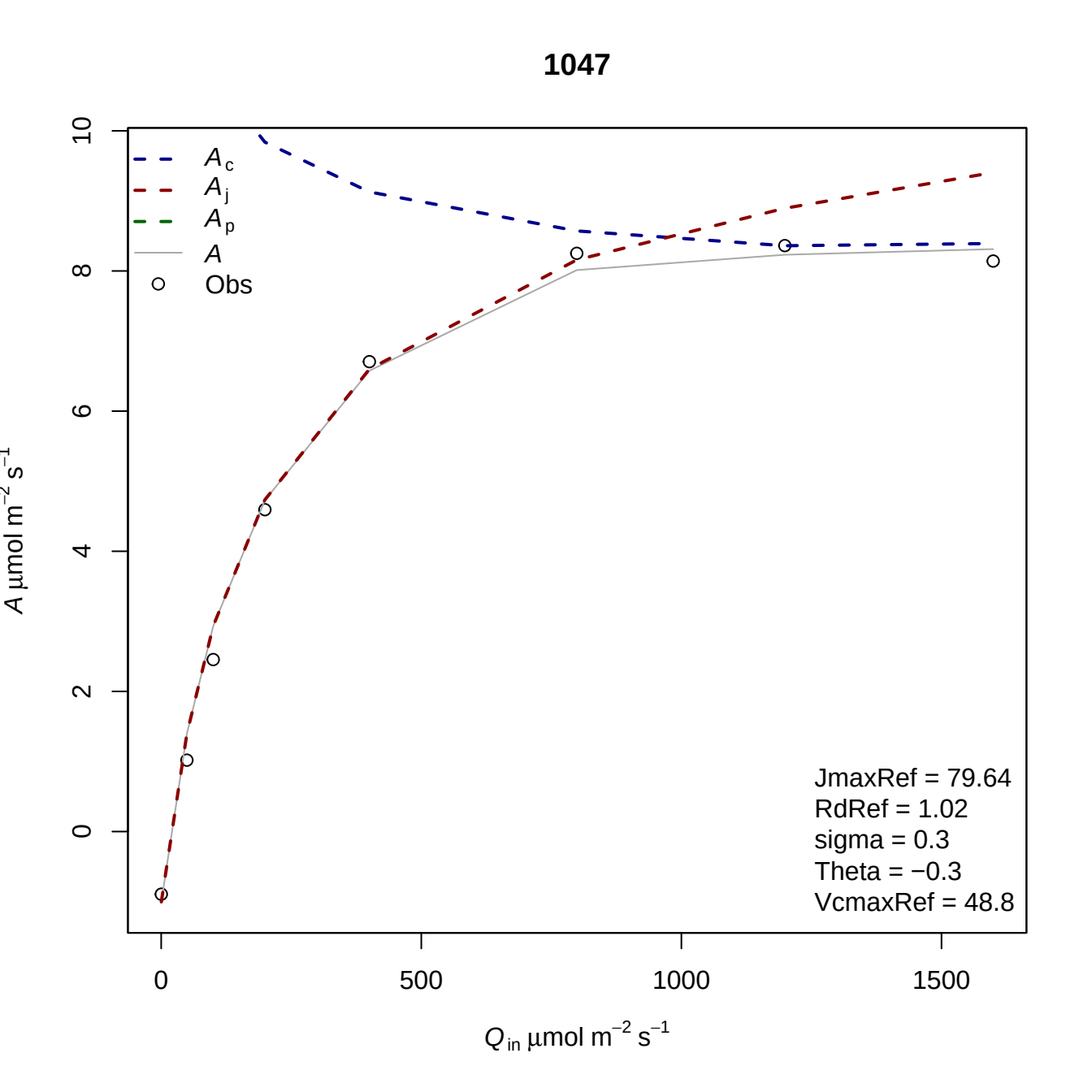

1042
A_c
A_j
A_p
A
Obs
JmaxRef = 119.59
RdRef = 2.22
sigma = 0.31
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

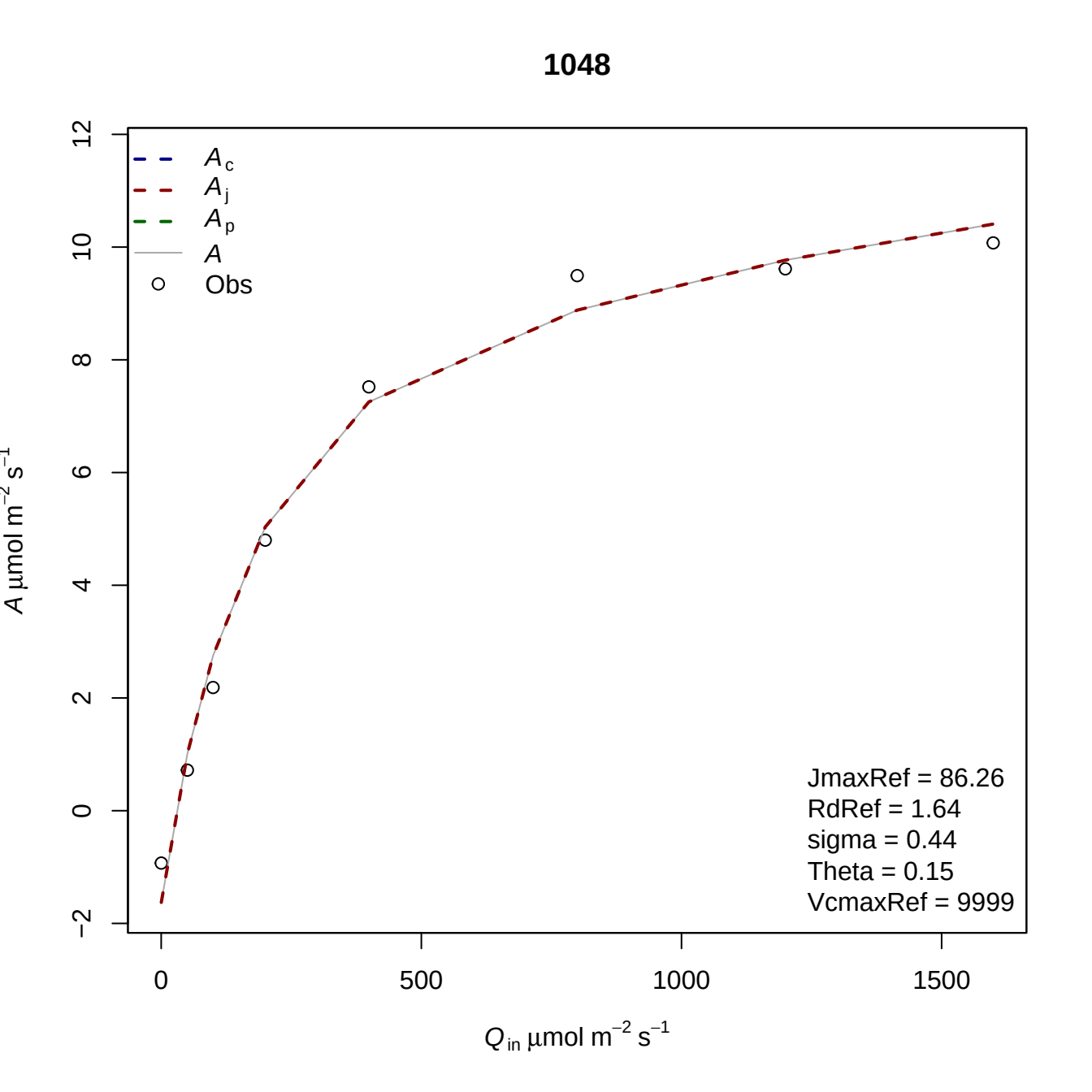

1043
A_c
A_j
A_p
A
Obs
JmaxRef = 133.15
RdRef = 1.6
sigma = 0.42
Theta = −0.23
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

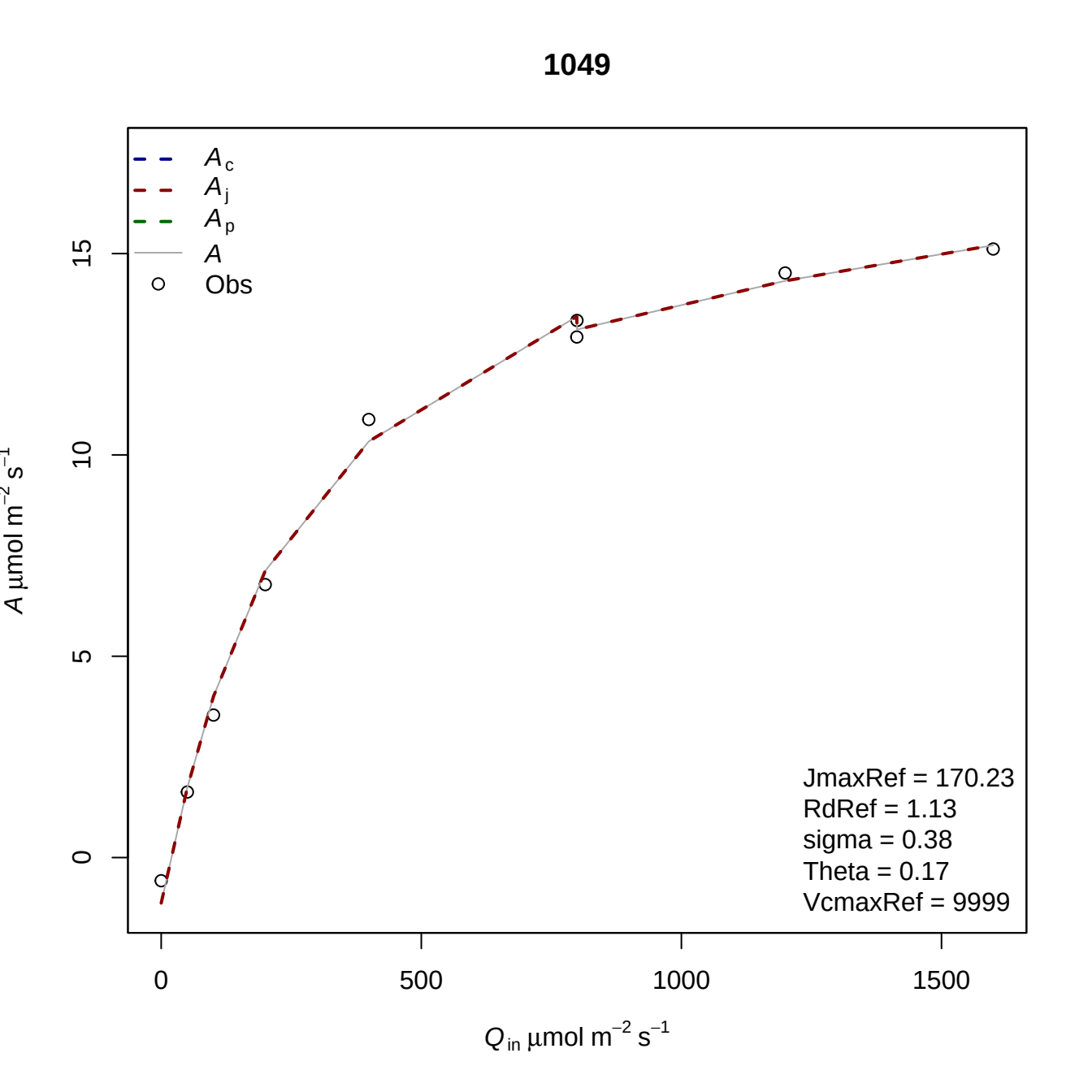

1044
A_c
A_j
A_p
A
Obs
JmaxRef = 196.12
RdRef = 2.11
sigma = 0.33
Theta = −0.61
VcmaxRef = 111.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

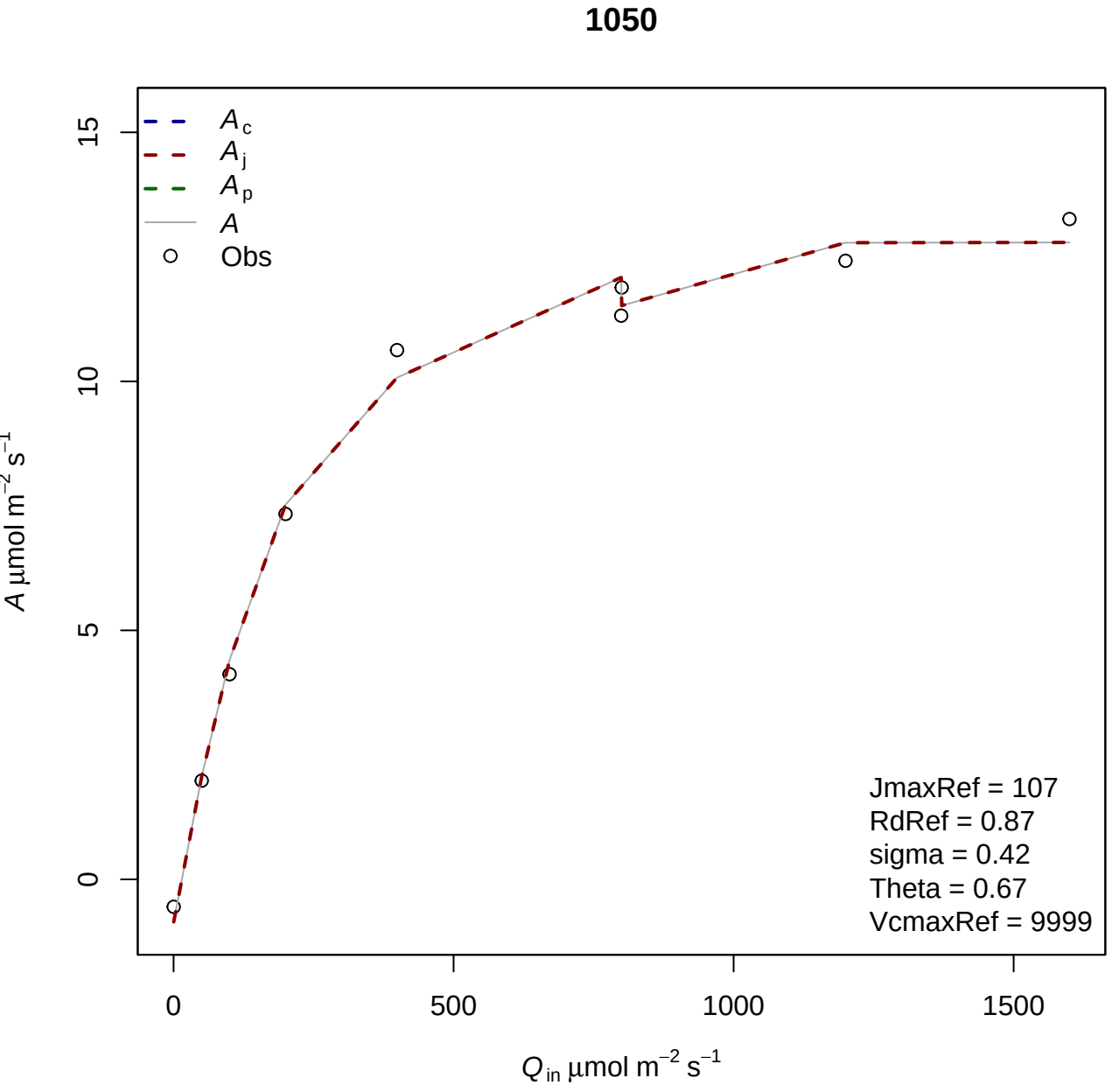

1045
A_c
A_j
A_p
A
Obs
JmaxRef = 81.05
RdRef = 1.22
sigma = 0.44
Theta = −0.26
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

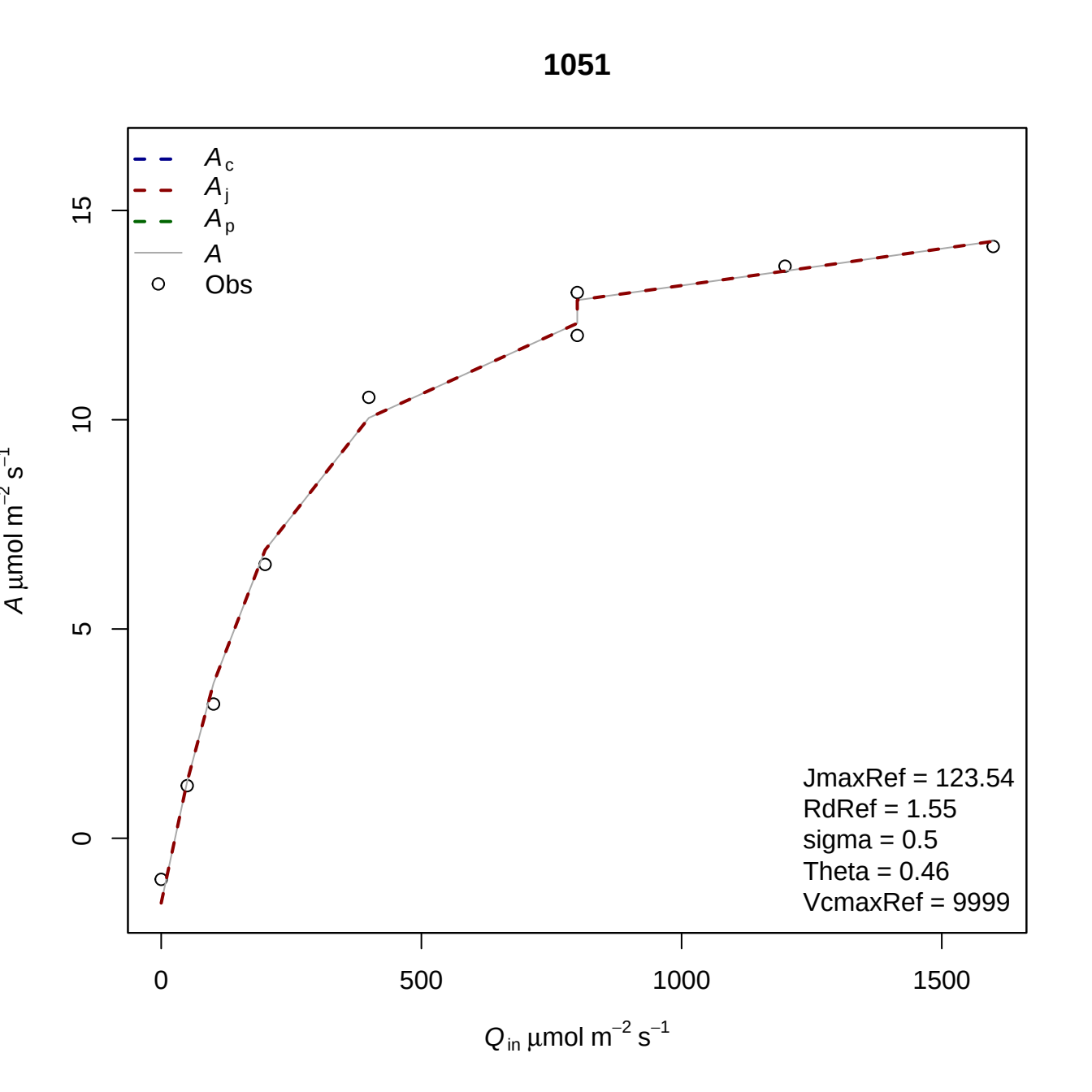

1046
A_c
A_j
A_p
A
Obs
JmaxRef = 83.83
RdRef = 1.17
sigma = 0.38
Theta = 0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

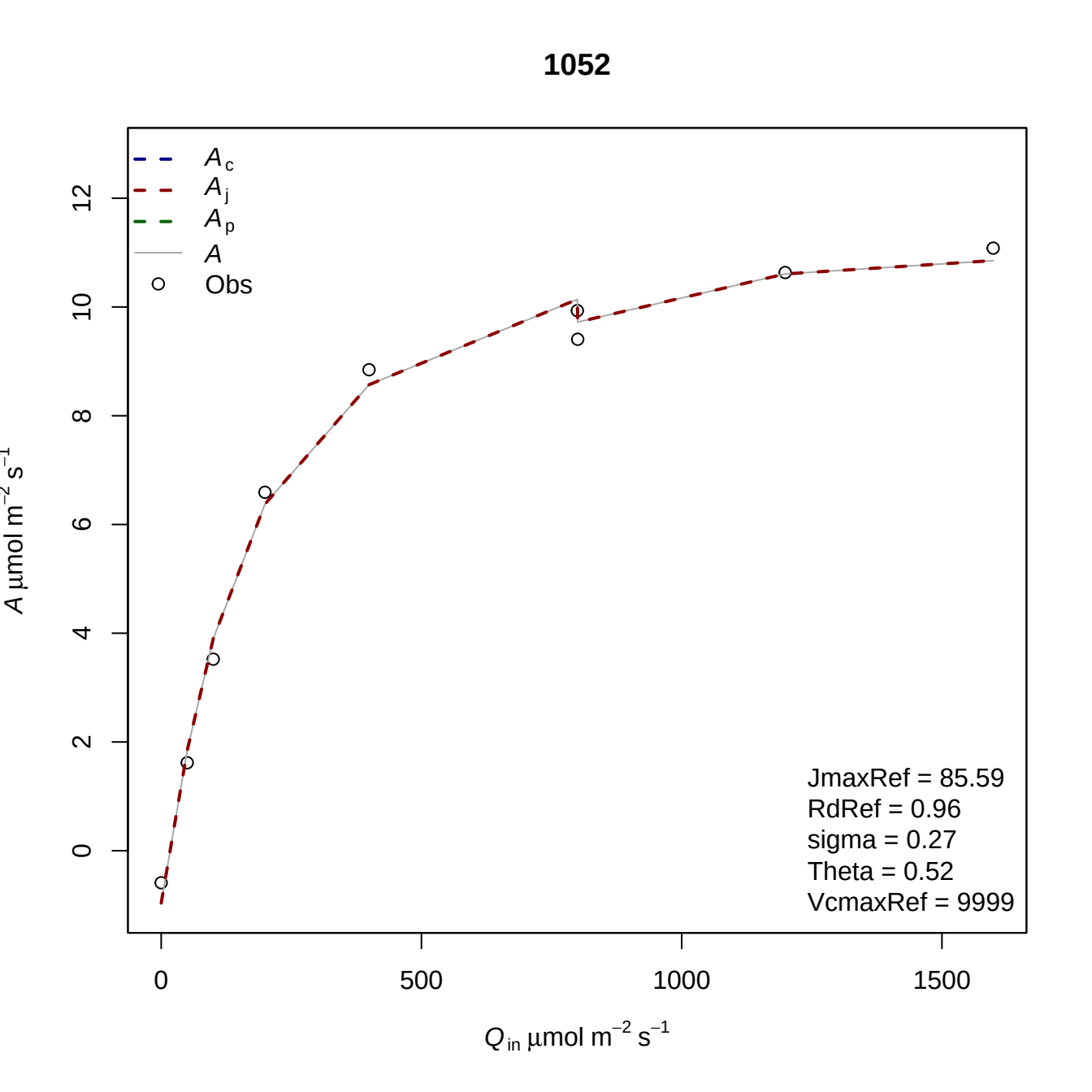

1047
A_c
A_j
A_p
A
Obs
JmaxRef = 79.64
RdRef = 1.02
sigma = 0.3
Theta = −0.3
VcmaxRef = 48.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

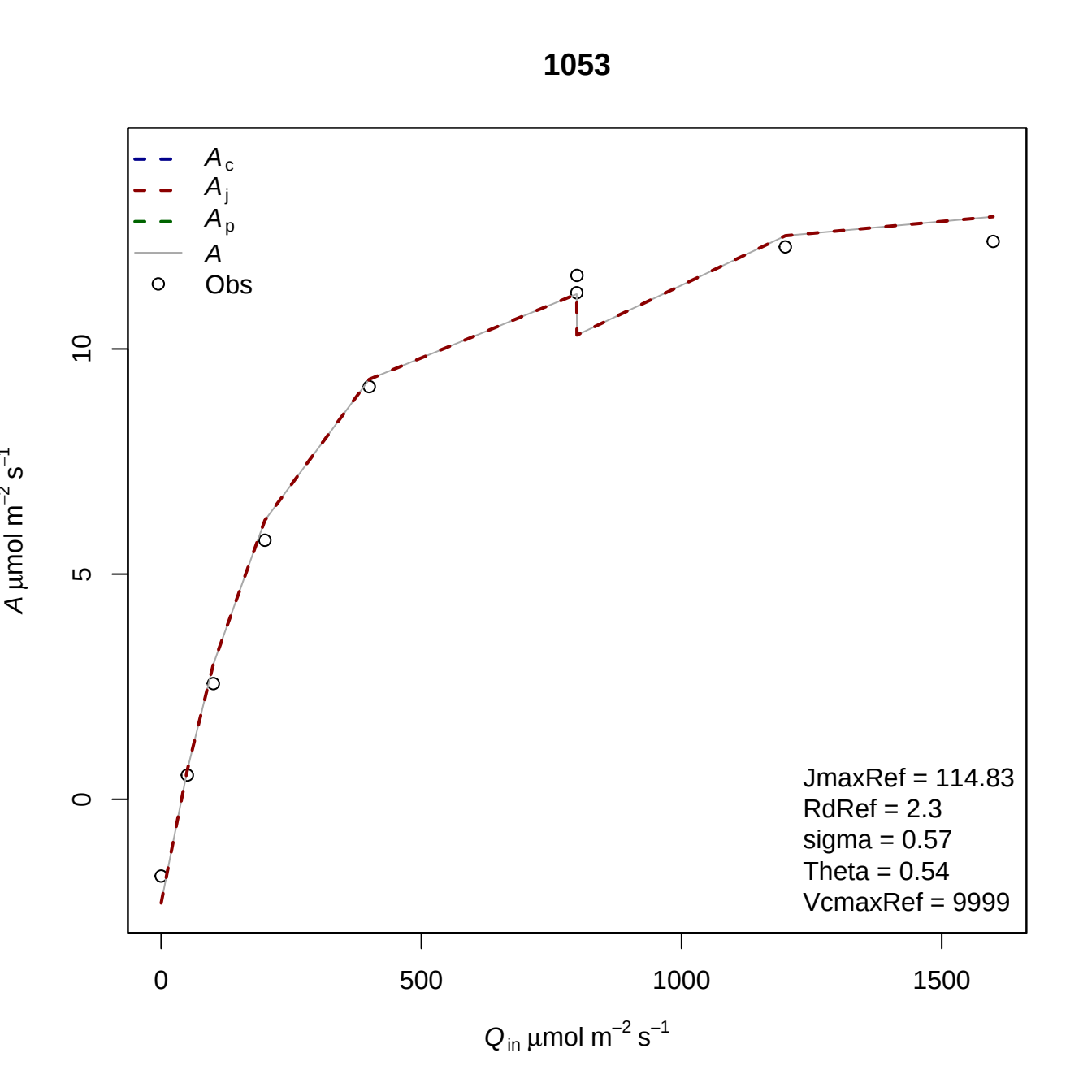

1048
A_c
A_j
A_p
A
Obs
JmaxRef = 86.26
RdRef = 1.64
sigma = 0.44
Theta = 0.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

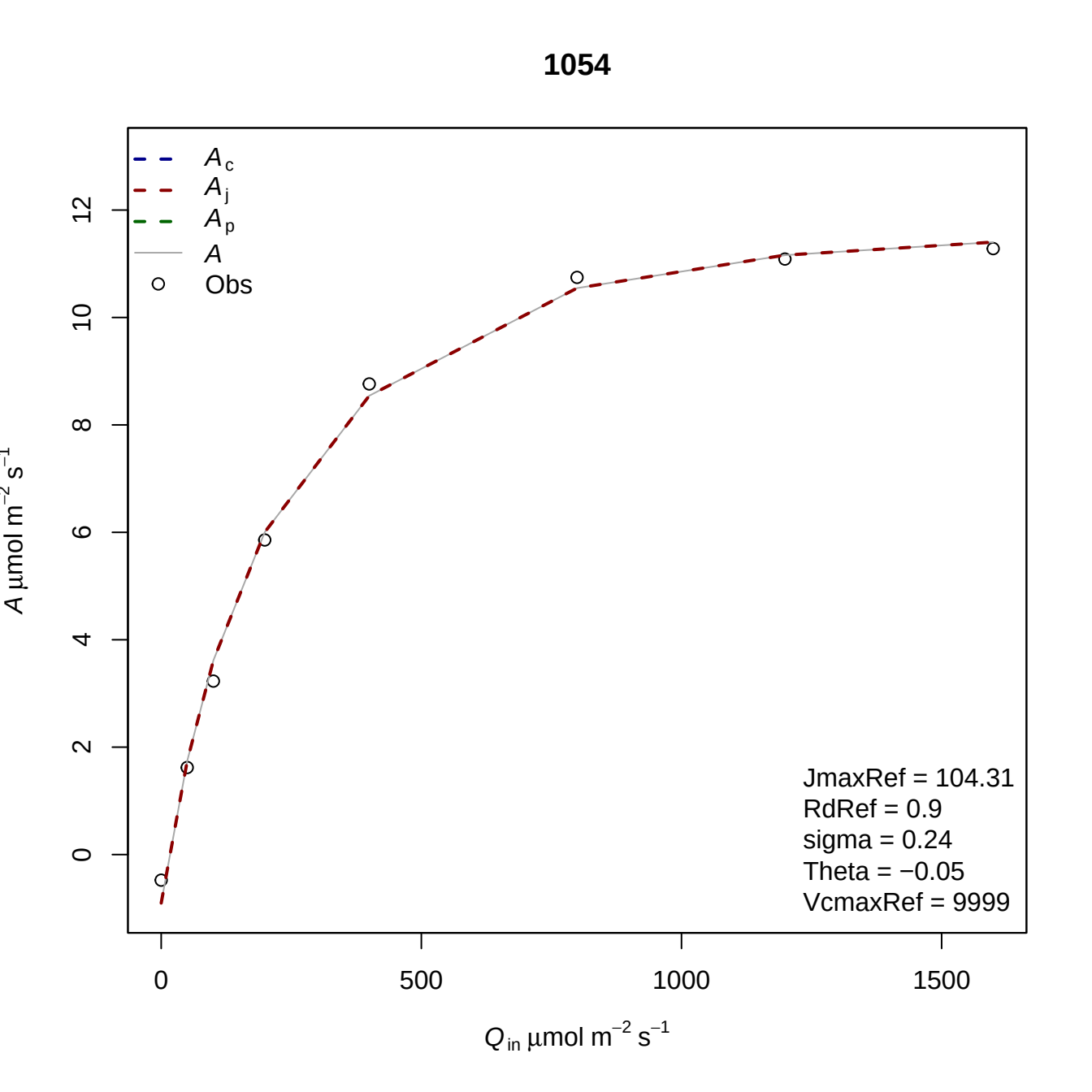

1049
A_c
A_j
A_p
A
Obs
JmaxRef = 170.23
RdRef = 1.13
sigma = 0.38
Theta = 0.17
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

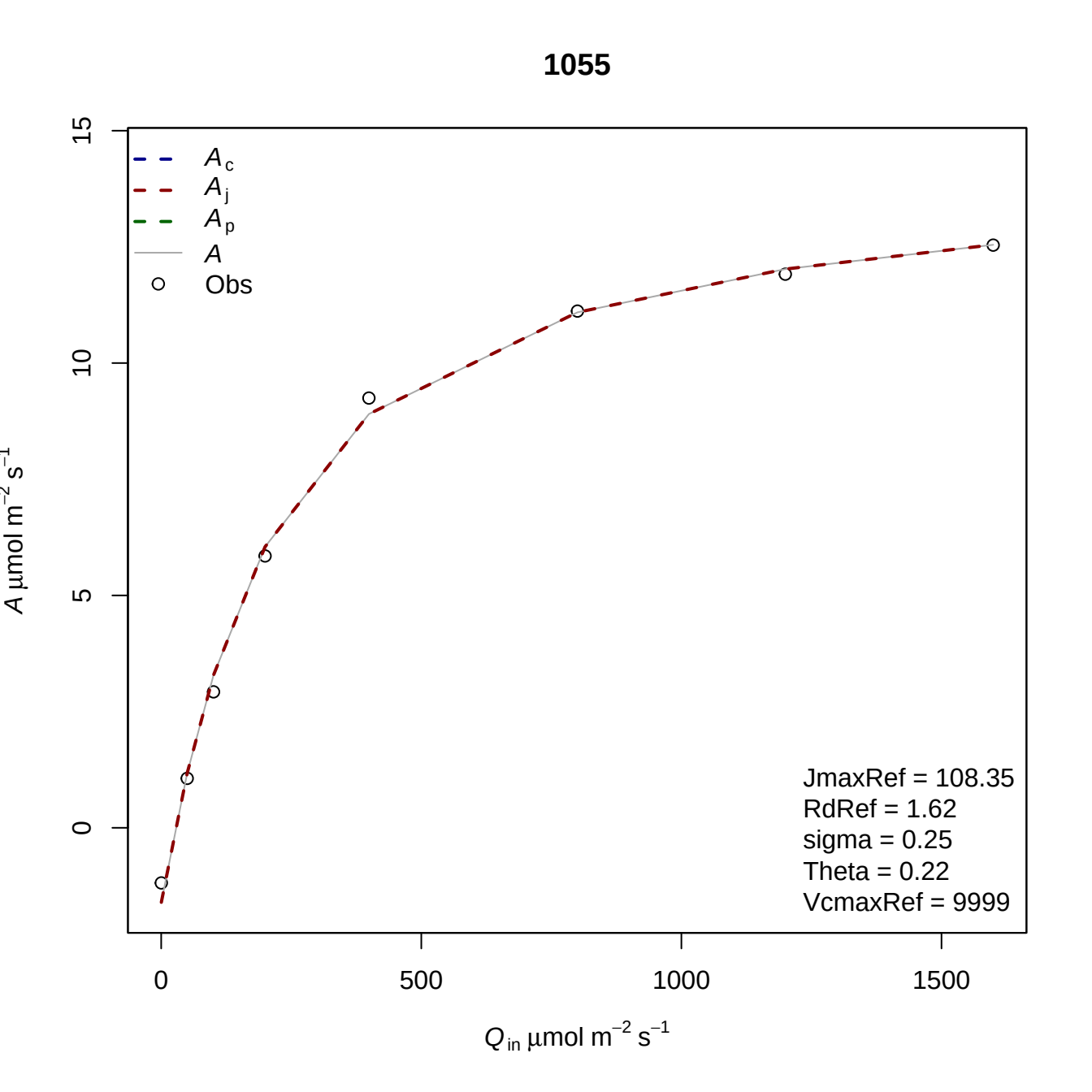

1050
A_c
A_j
A_p
A
Obs
JmaxRef = 107
RdRef = 0.87
sigma = 0.42
Theta = 0.67
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

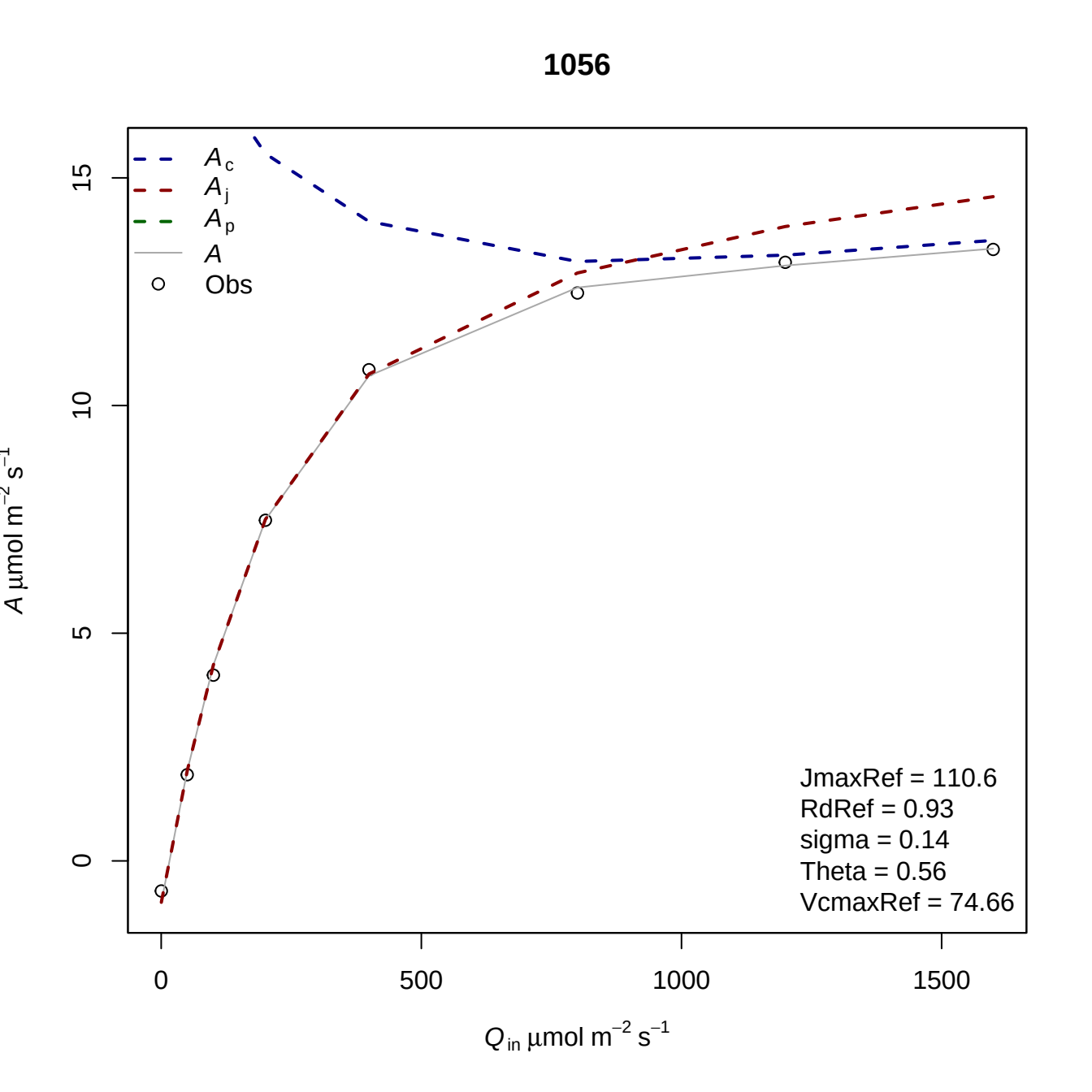

1051
A_c
A_j
A_p
A
Obs
JmaxRef = 123.54
RdRef = 1.55
sigma = 0.5
Theta = 0.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

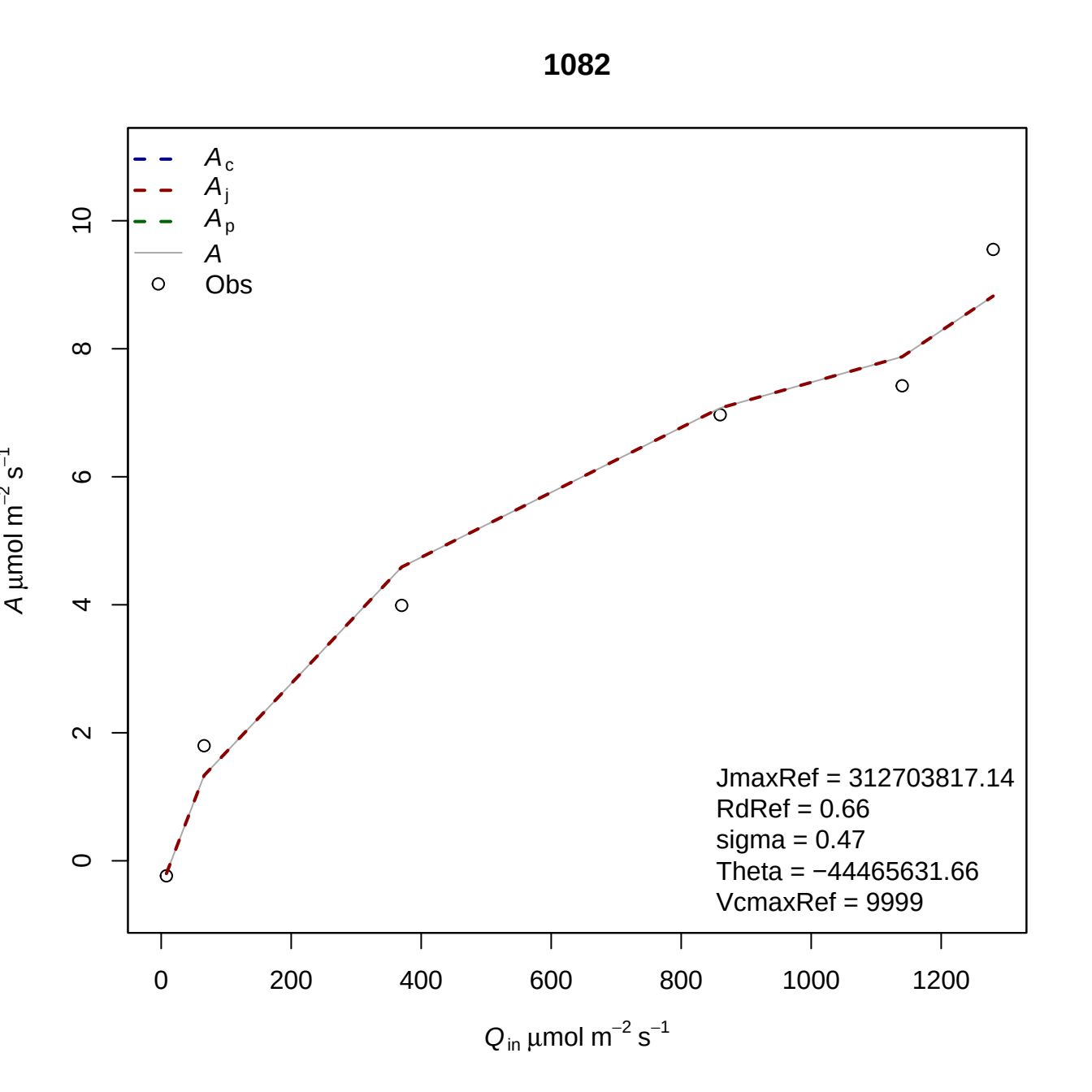

1052
A_c
A_j
A_p
A
Obs
JmaxRef = 85.59
RdRef = 0.96
sigma = 0.27
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

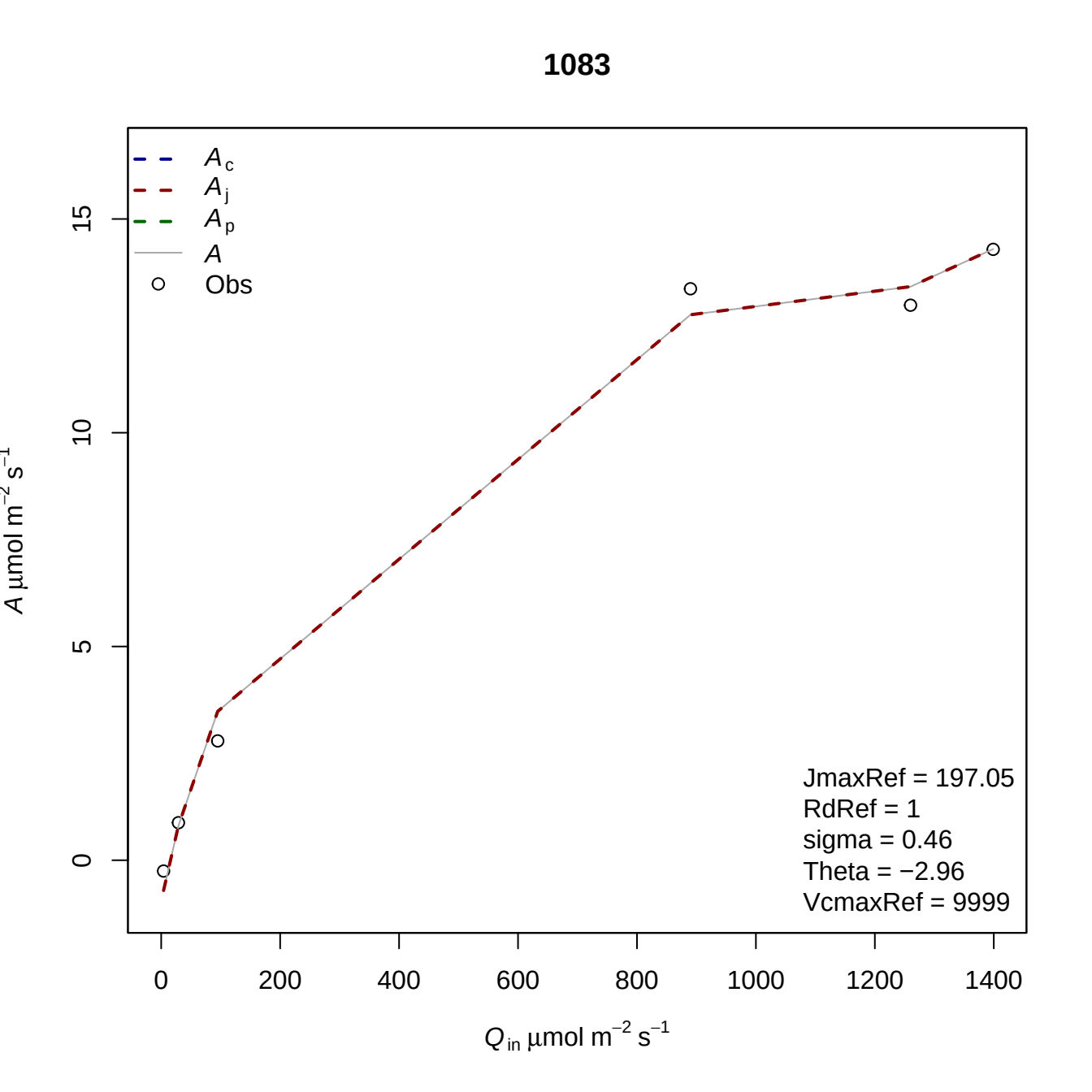

1053
A_c
A_j
A_p
A
Obs
JmaxRef = 114.83
RdRef = 2.3
sigma = 0.57
Theta = 0.54
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

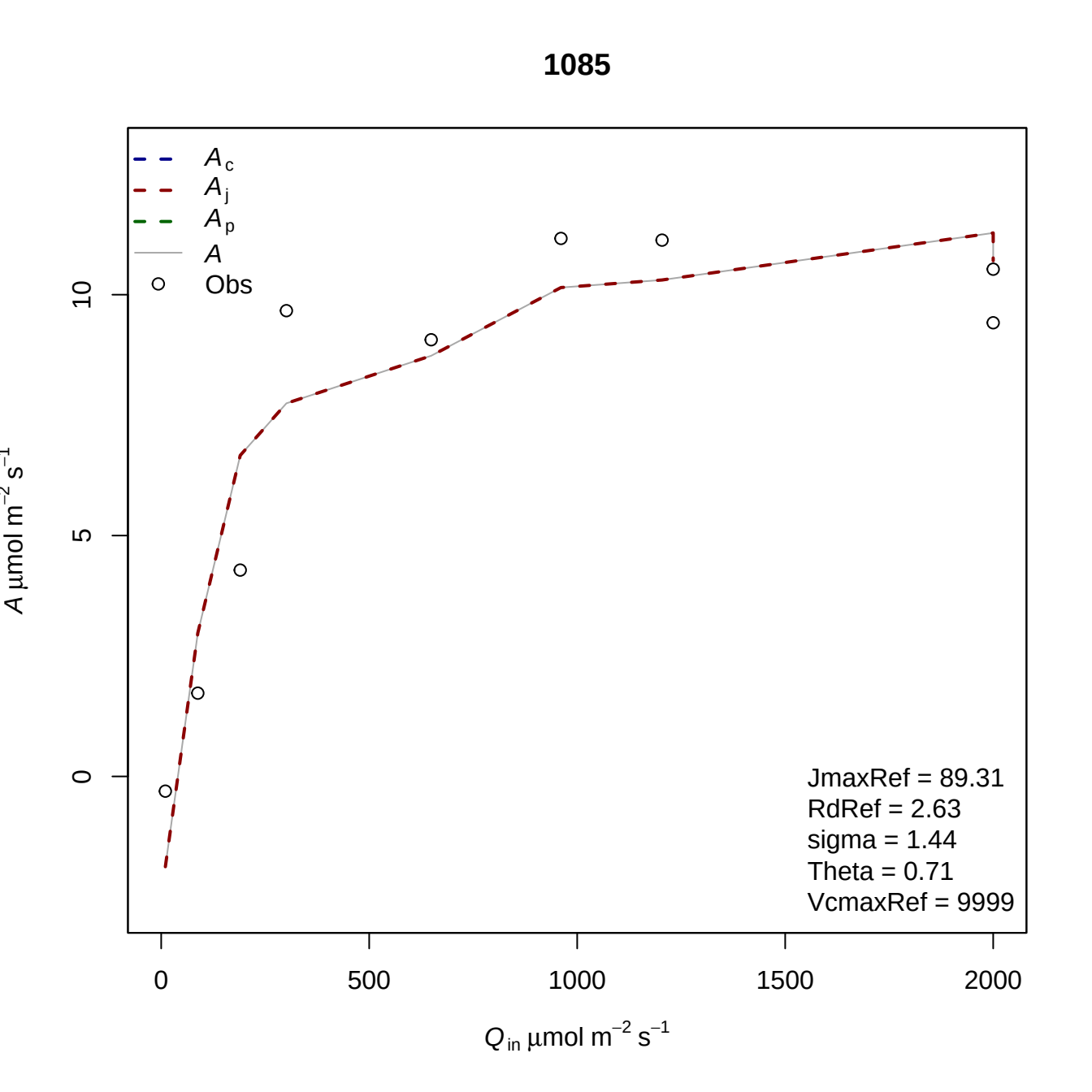

1054
A_c
A_j
A_p
A
Obs
JmaxRef = 104.31
RdRef = 0.9
sigma = 0.24
Theta = −0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

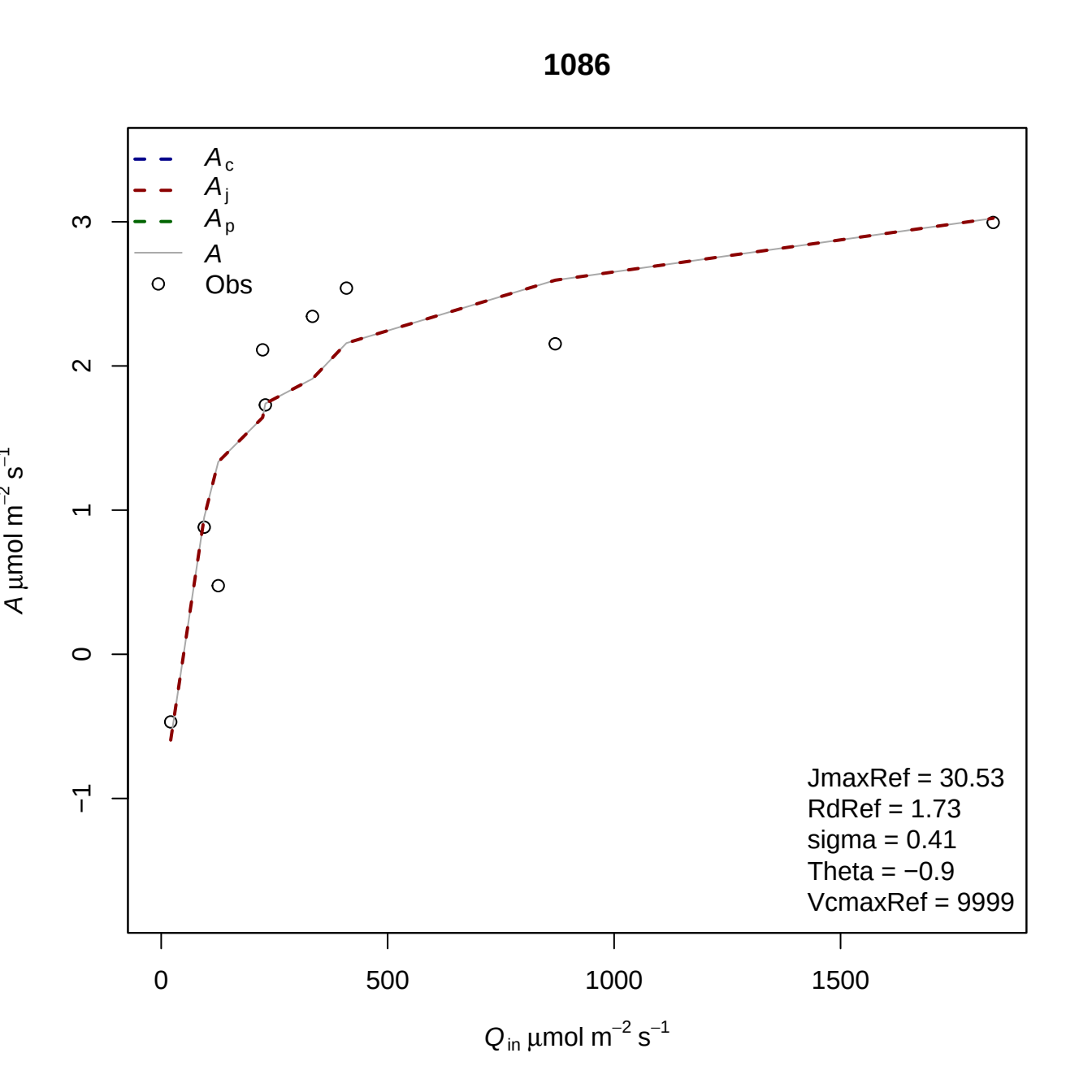

1055
A_c
A_j
A_p
A
Obs
JmaxRef = 108.35
RdRef = 1.62
sigma = 0.25
Theta = 0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

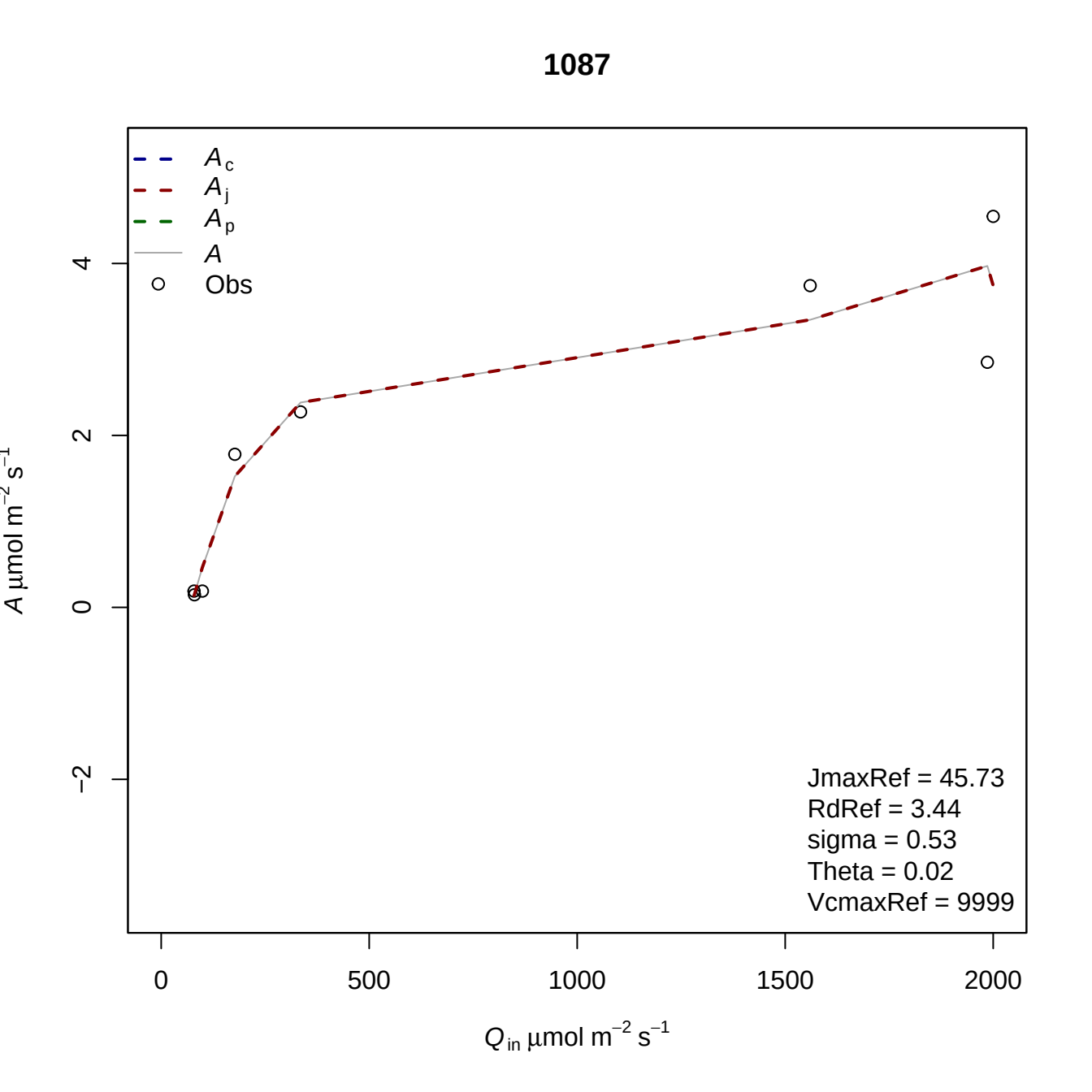

1056
A_c
A_j
A_p
A
Obs
JmaxRef = 110.6
RdRef = 0.93
sigma = 0.14
Theta = 0.56
VcmaxRef = 74.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

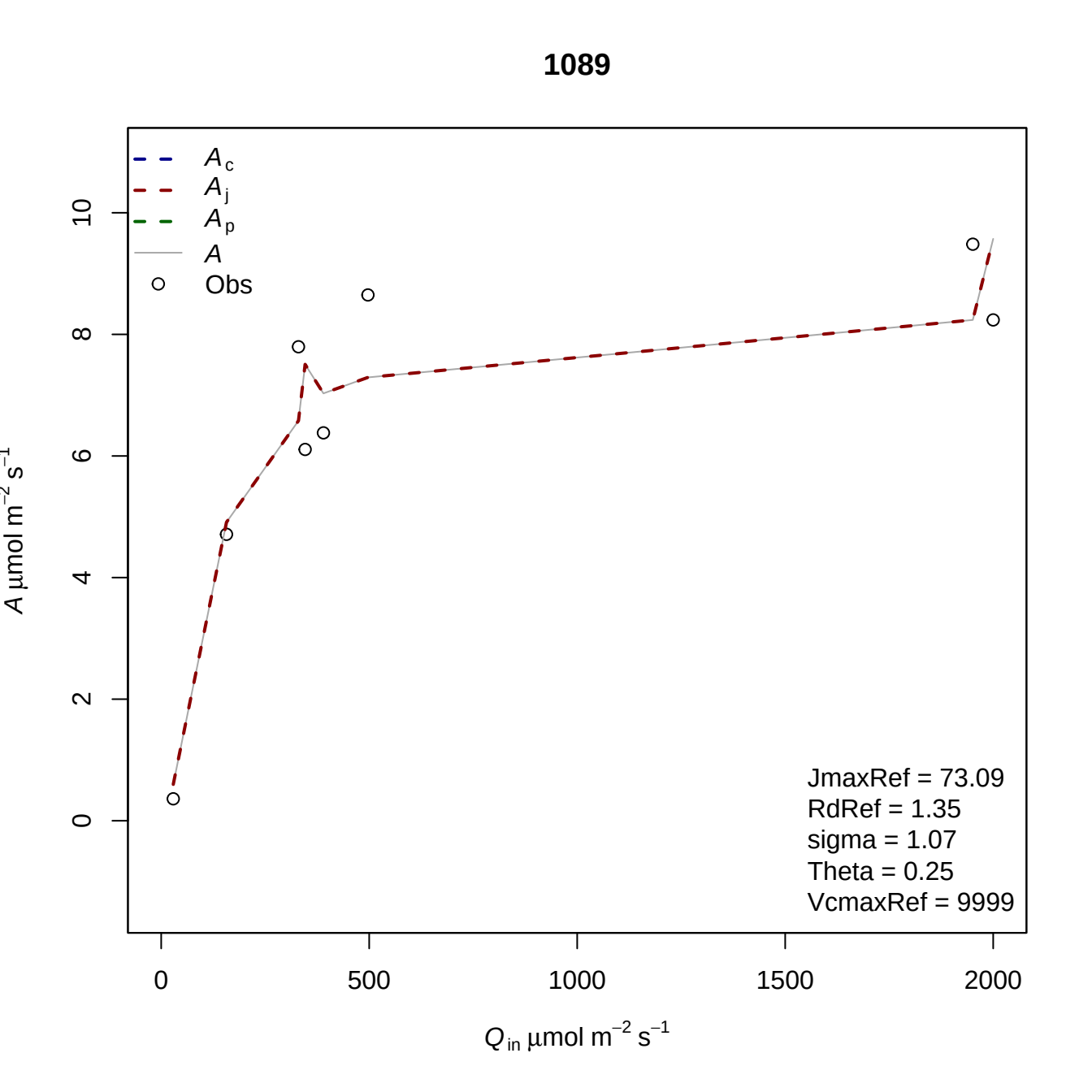

1082
A_c
A_j
A_p
A
Obs
JmaxRef = 312703817.14
RdRef = 0.66
sigma = 0.47
Theta = −44465631.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

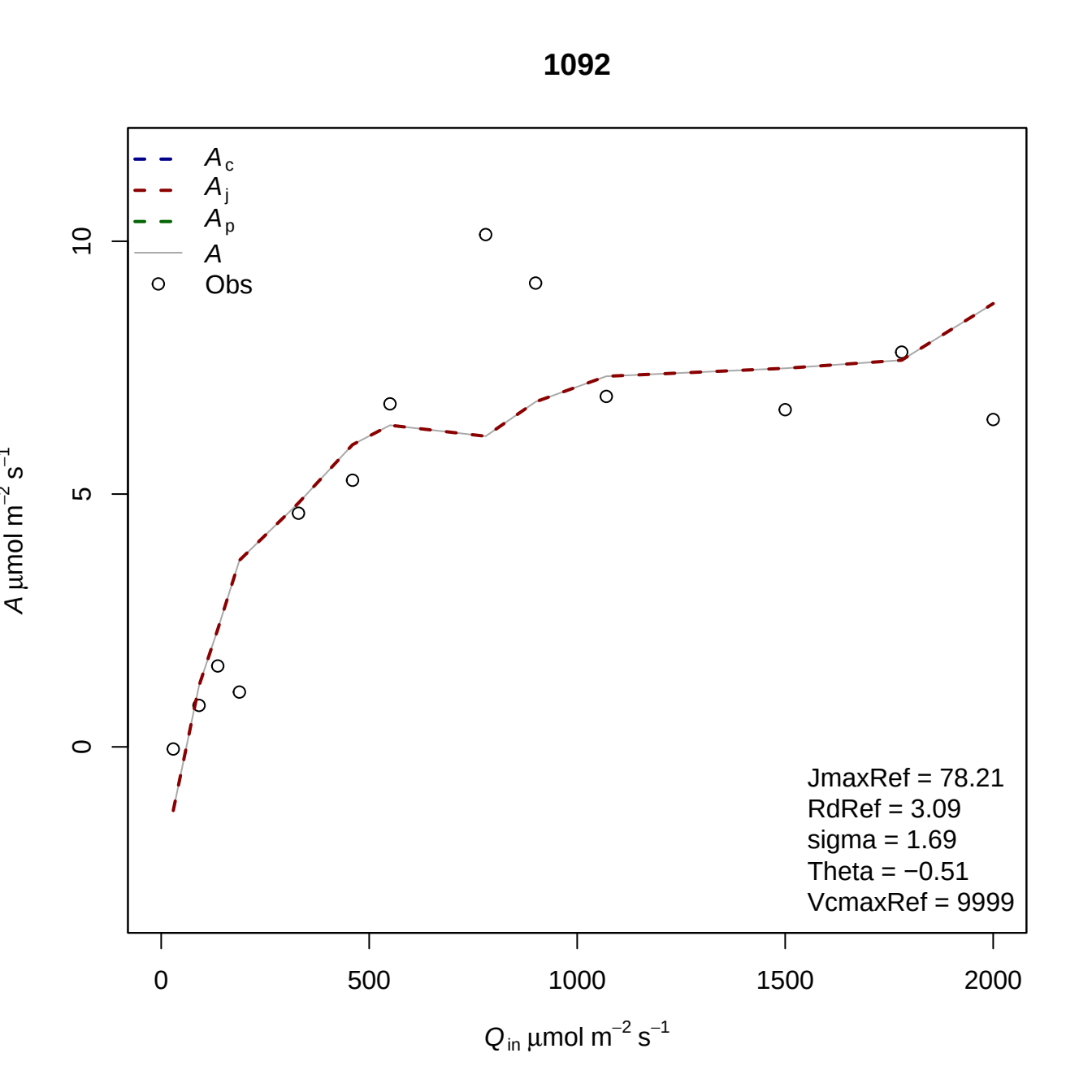

1083
A_c
A_j
A_p
A
Obs
JmaxRef = 197.05
RdRef = 1
sigma = 0.46
Theta = −2.96
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

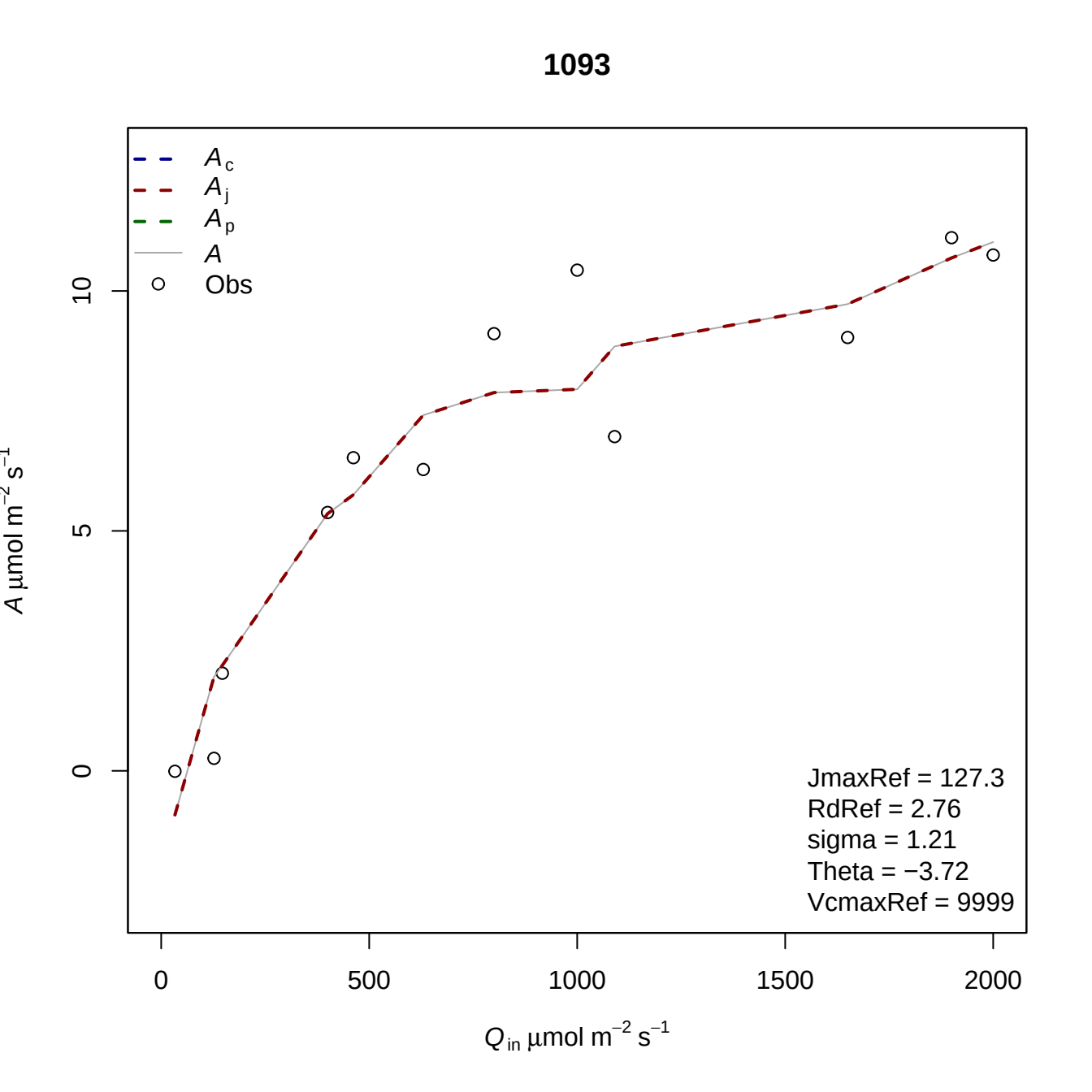

1085
A_c
A_j
A_p
A
Obs
JmaxRef = 89.31
RdRef = 2.63
sigma = 1.44
Theta = 0.71
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

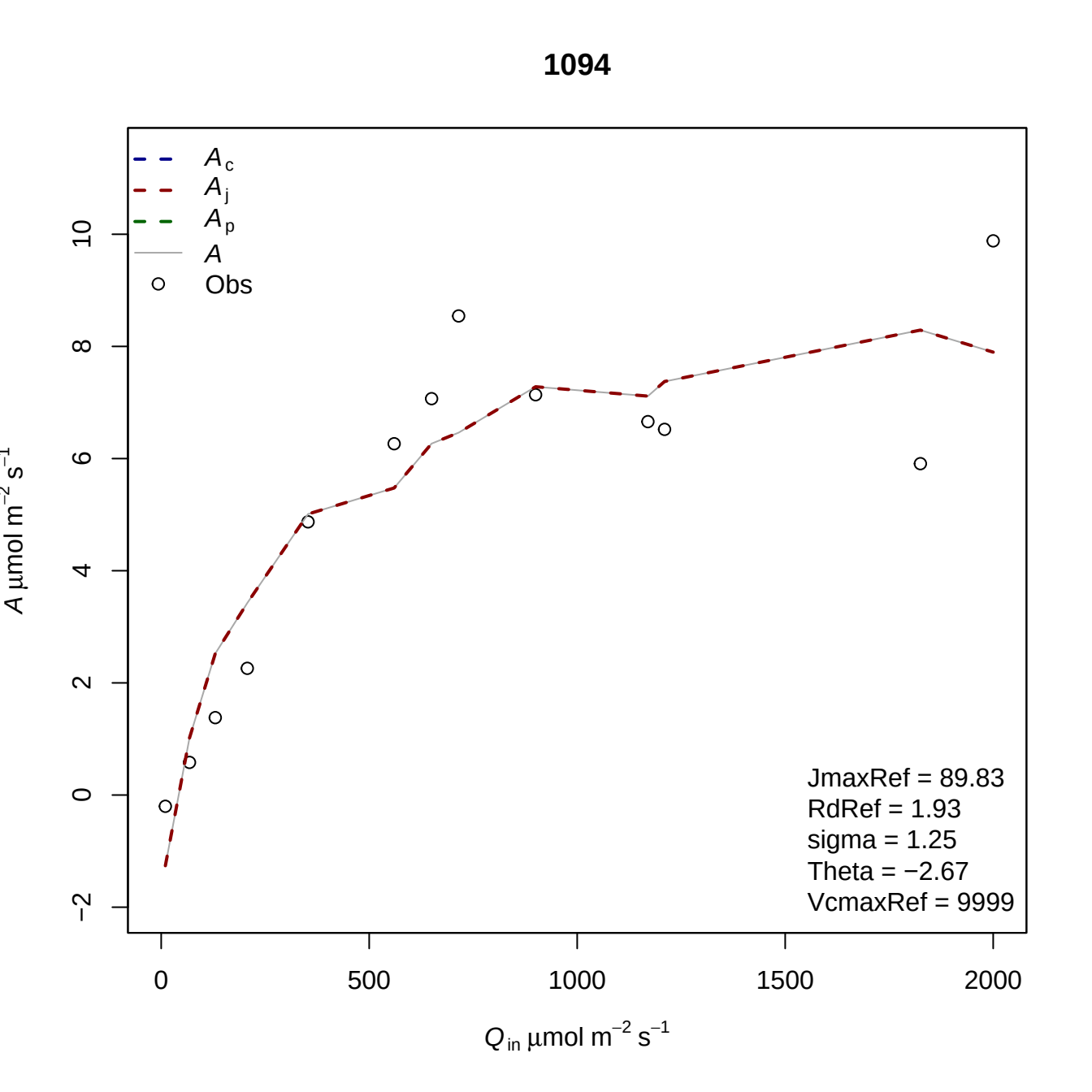

1086
A_c
A_j
A_p
A
Obs
JmaxRef = 30.53
RdRef = 1.73
sigma = 0.41
Theta = −0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

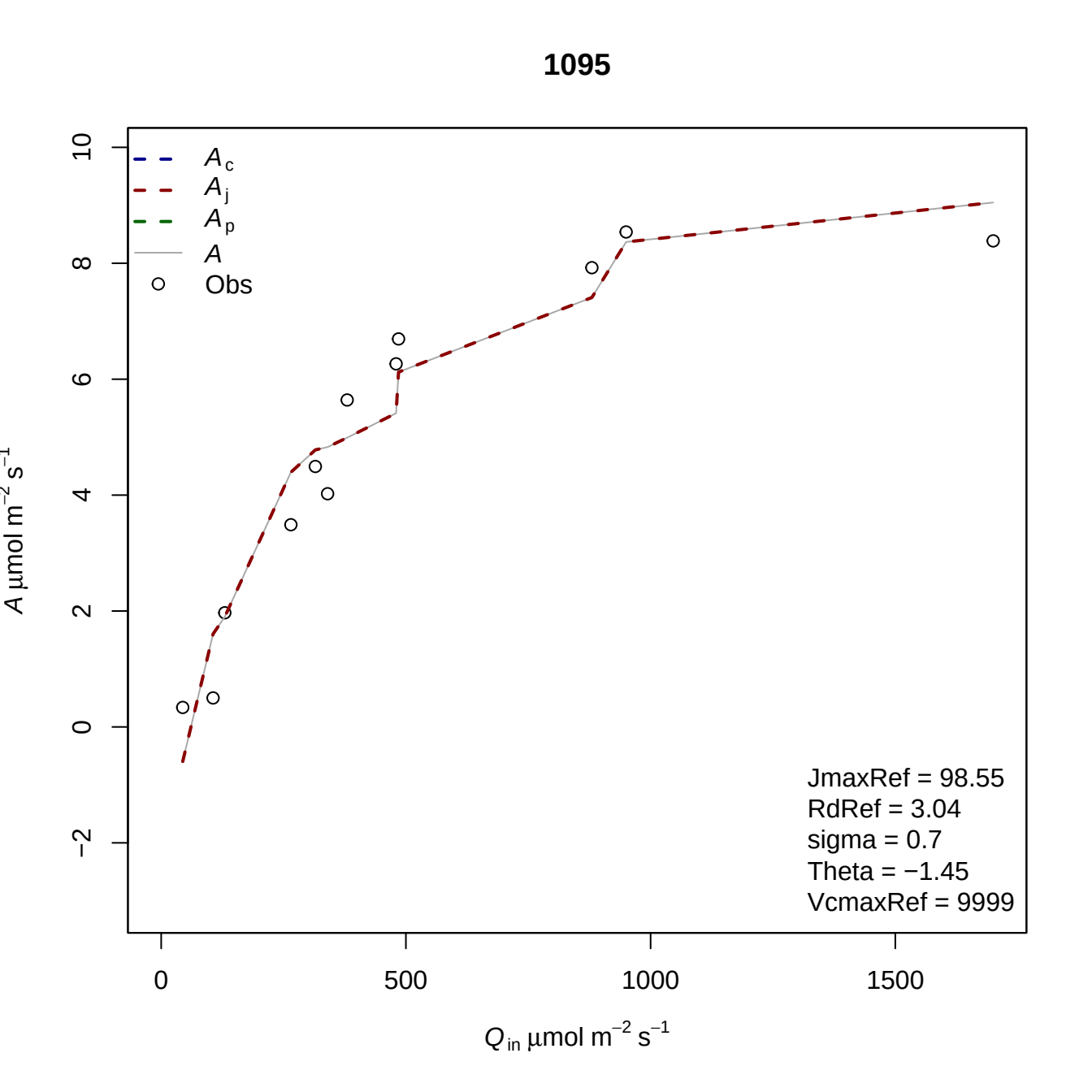

1087
A_c
A_j
A_p
A
Obs
JmaxRef = 45.73
RdRef = 3.44
sigma = 0.53
Theta = 0.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

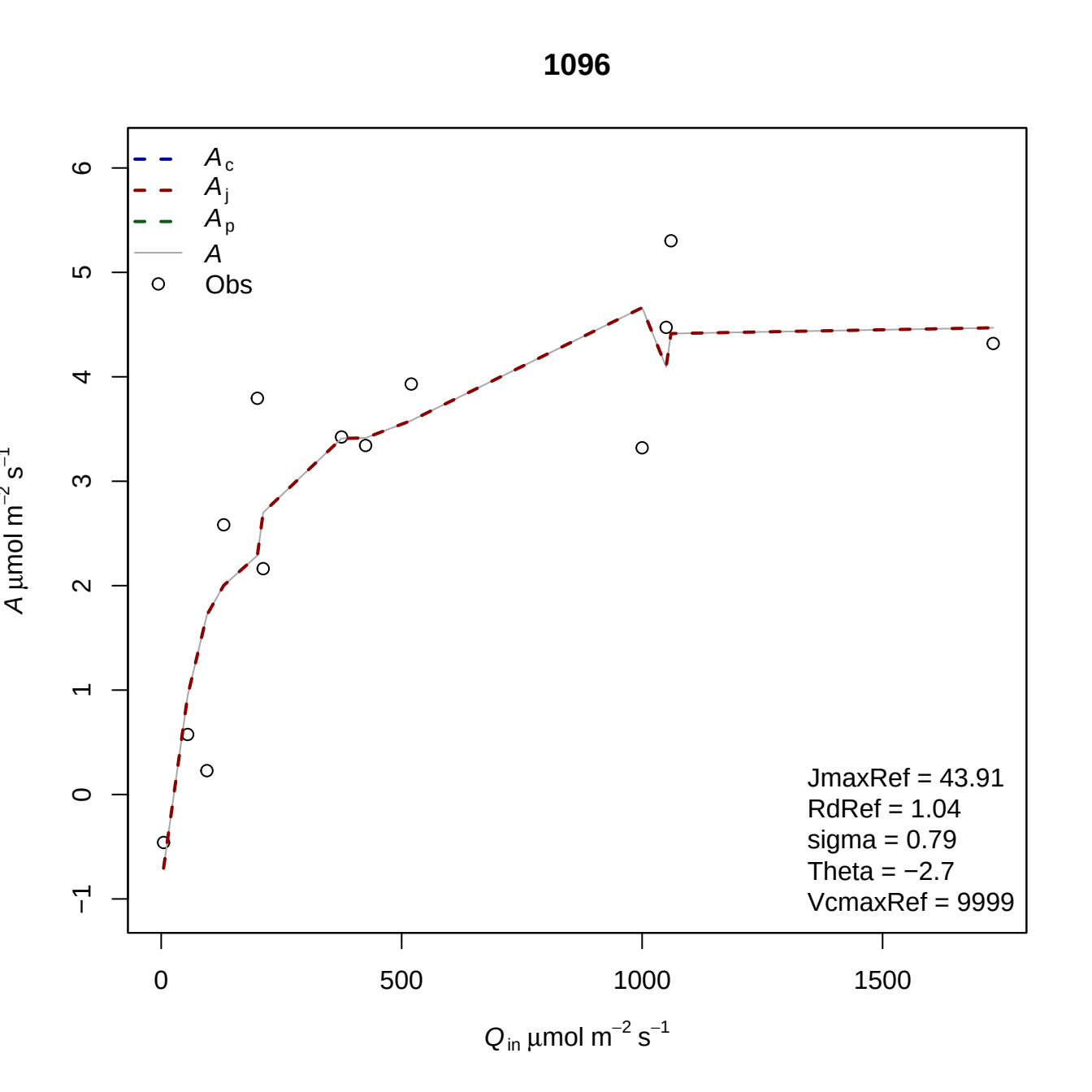

1089
A_c
A_j
A_p
A
Obs
JmaxRef = 73.09
RdRef = 1.35
sigma = 1.07
Theta = 0.25
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

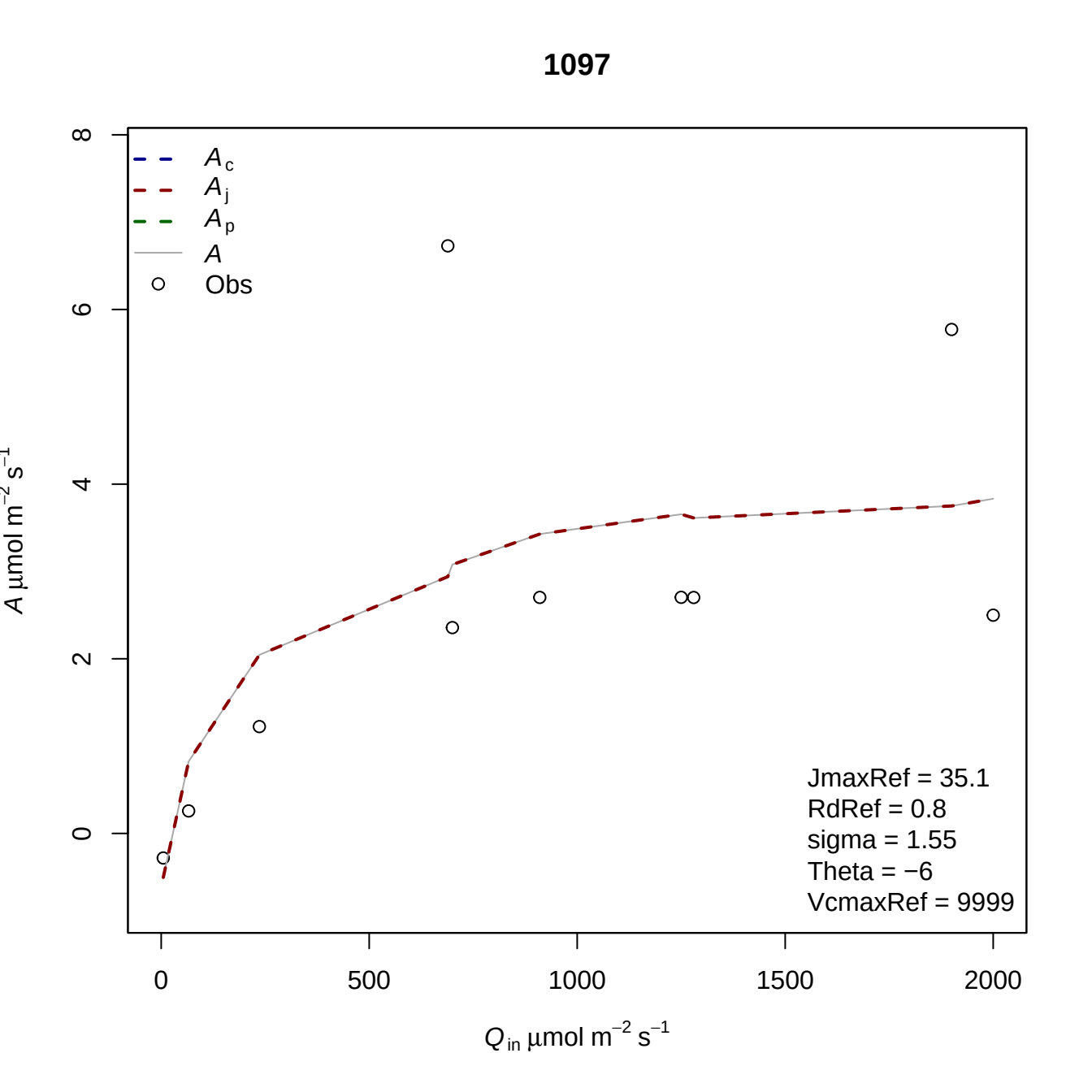

1092
A_c
A_j
A_p
A
Obs
JmaxRef = 78.21
RdRef = 3.09
sigma = 1.69
Theta = −0.51
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

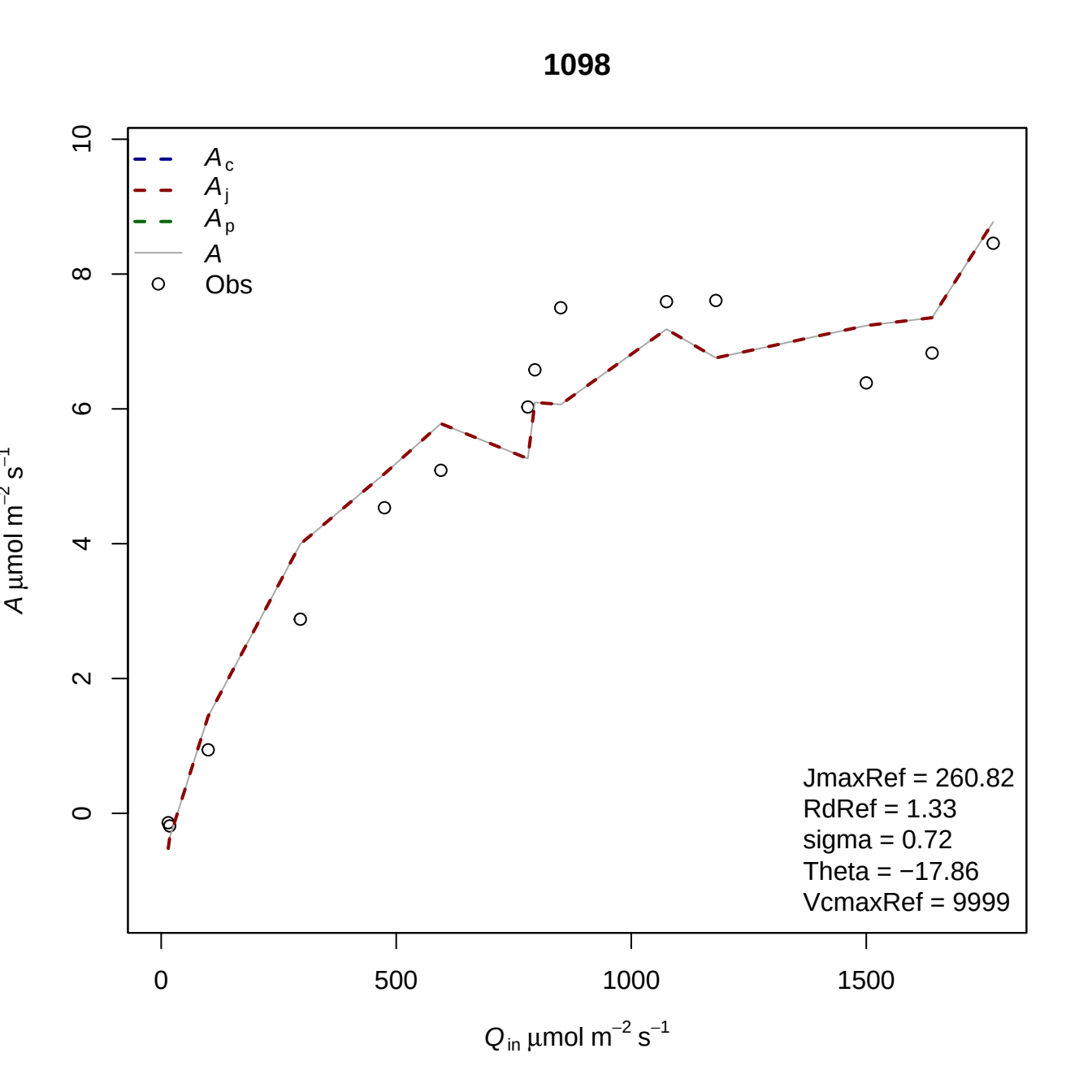

1093
A_c
A_j
A_p
A
Obs
JmaxRef = 127.3
RdRef = 2.76
sigma = 1.21
Theta = −3.72
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

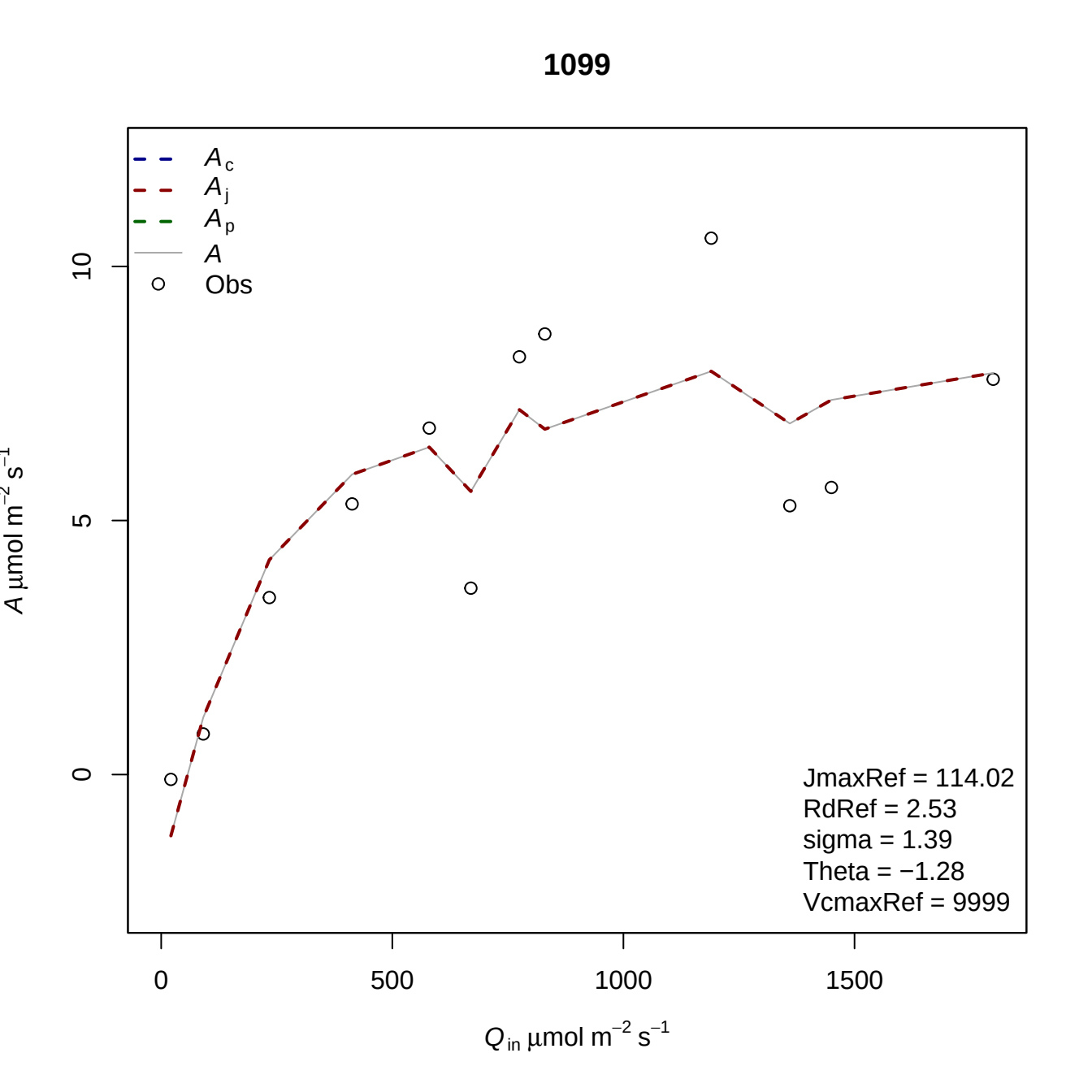

1094
A_c
A_j
A_p
A
Obs
JmaxRef = 89.83
RdRef = 1.93
sigma = 1.25
Theta = −2.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

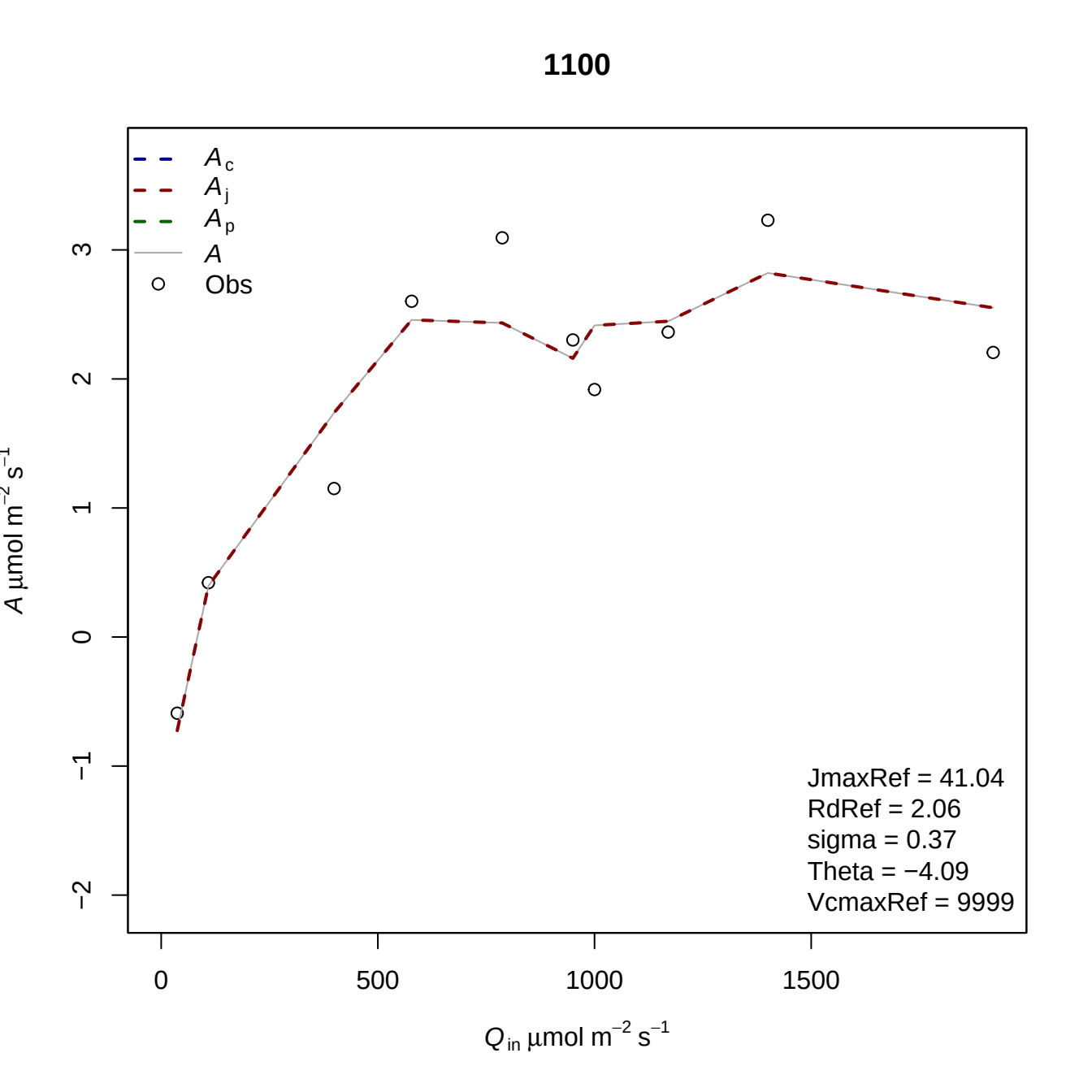

1095
A_c
A_j
A_p
A
Obs
JmaxRef = 98.55
RdRef = 3.04
sigma = 0.7
Theta = −1.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

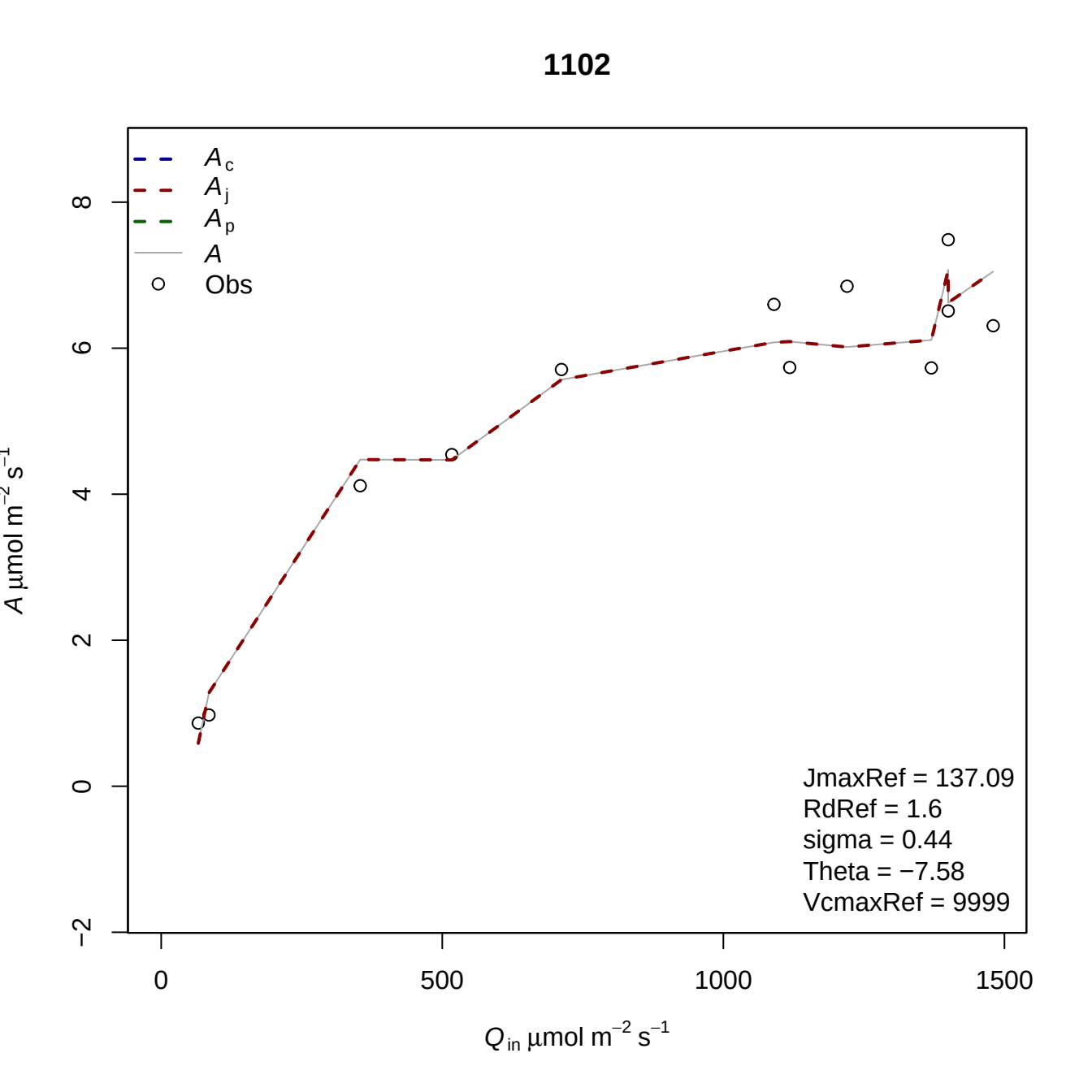

1096
A_c
A_j
A_p
A
Obs
JmaxRef = 43.91
RdRef = 1.04
sigma = 0.79
Theta = −2.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

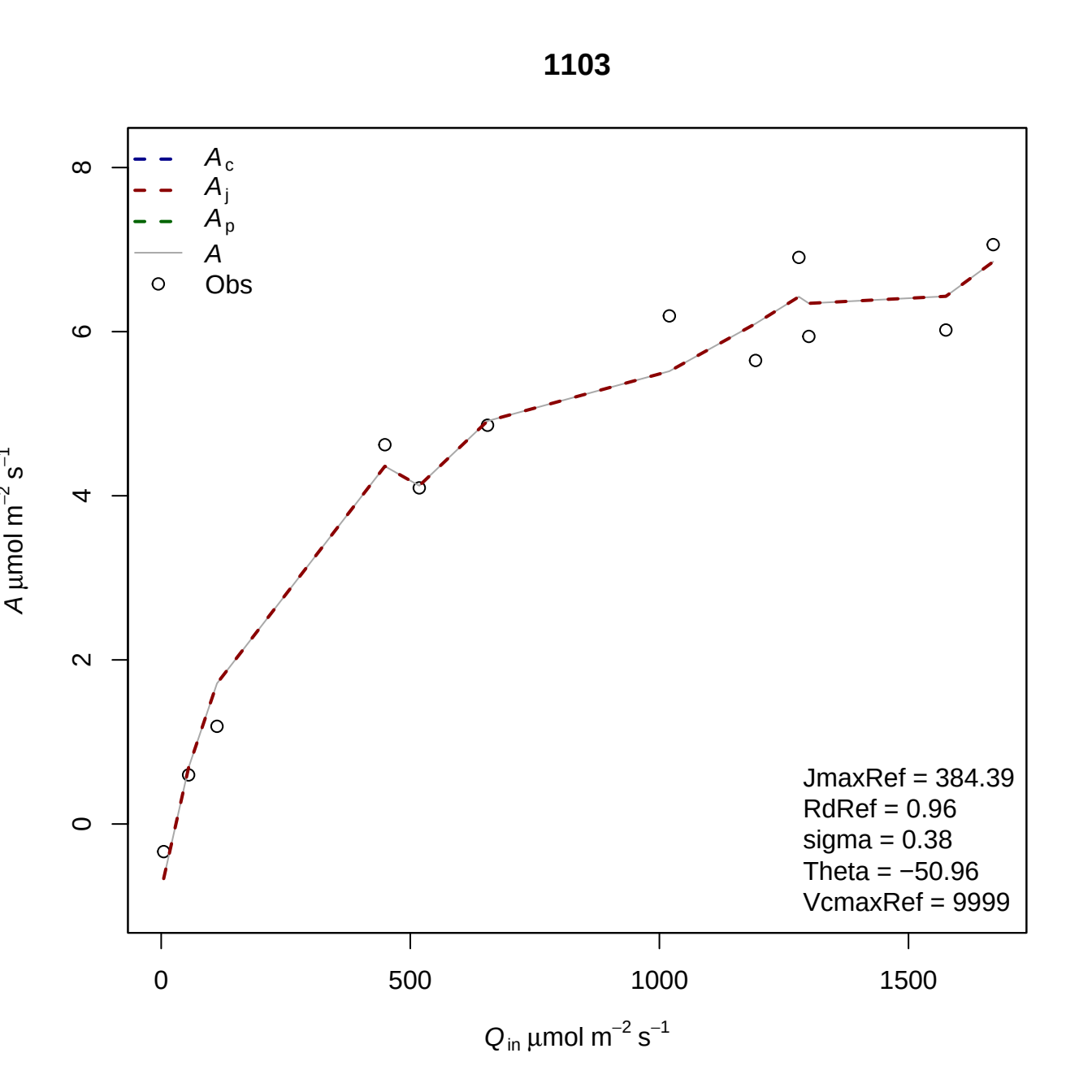

1097
A_c
A_j
A_p
A
Obs
JmaxRef = 35.1
RdRef = 0.8
sigma = 1.55
Theta = −6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

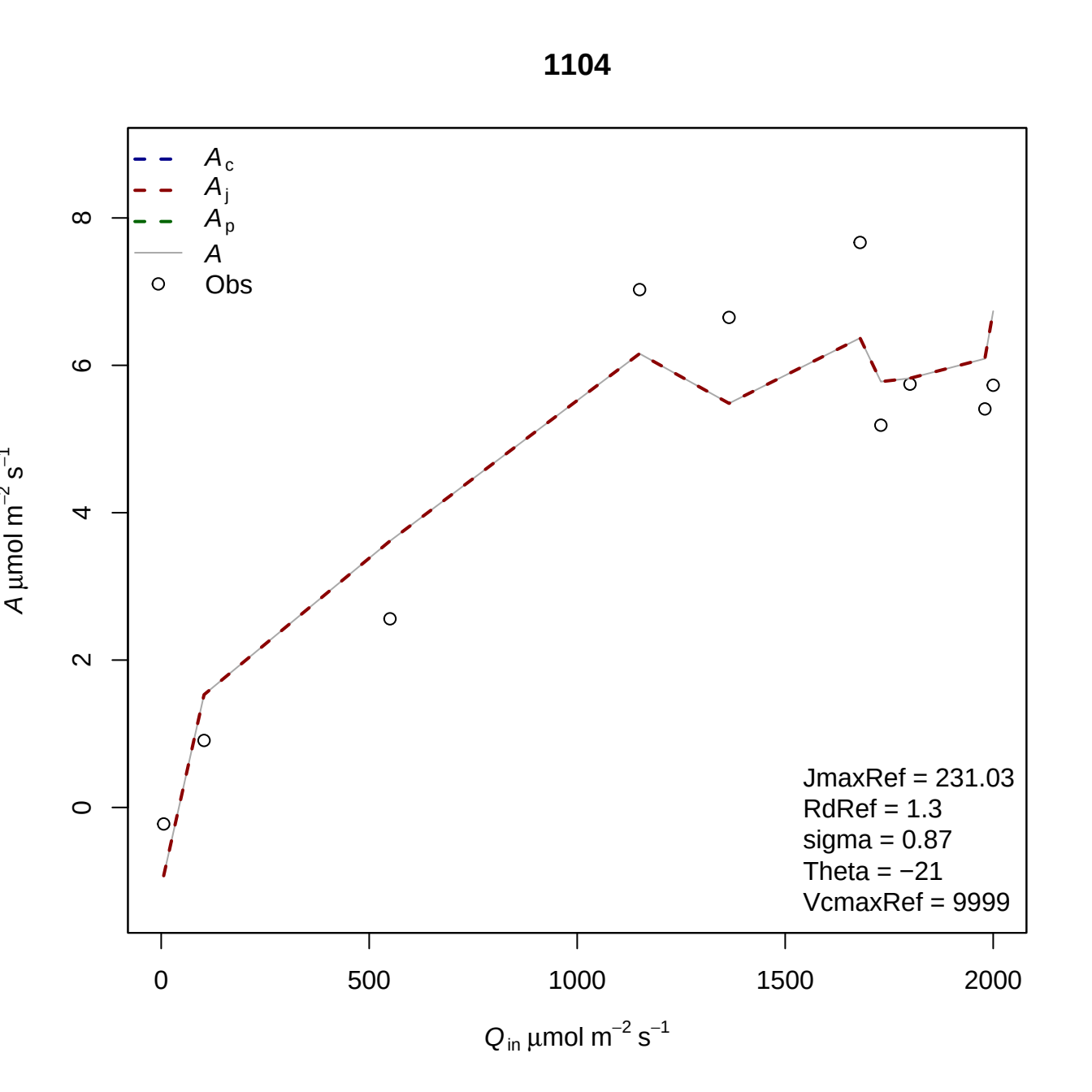

1098
A_c
A_j
A_p
A
Obs
JmaxRef = 260.82
RdRef = 1.33
sigma = 0.72
Theta = −17.86
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

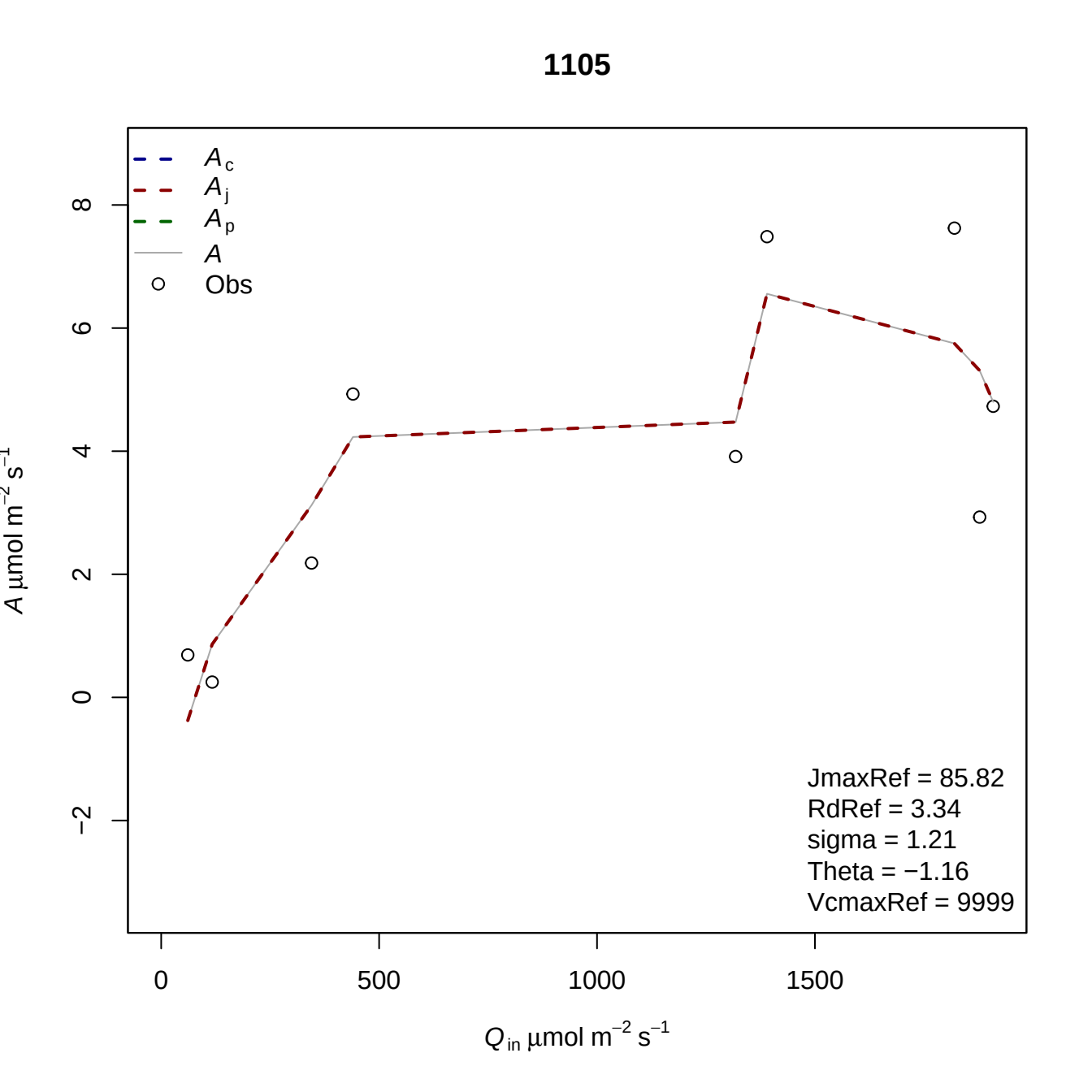

1099
A_c
A_j
A_p
A
Obs
JmaxRef = 114.02
RdRef = 2.53
sigma = 1.39
Theta = −1.28
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

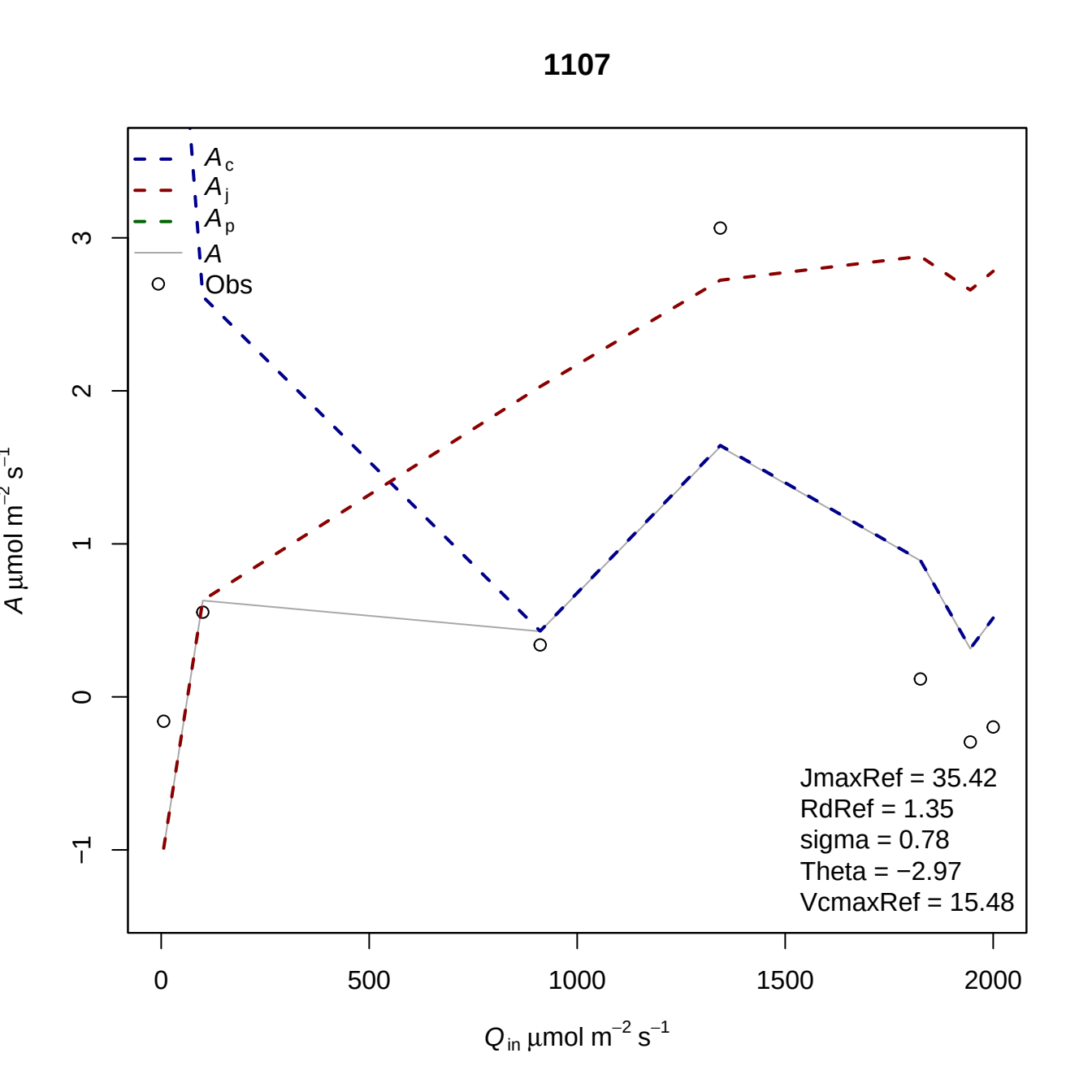

1100
A_c
A_j
A_p
A
Obs
JmaxRef = 41.04
RdRef = 2.06
sigma = 0.37
Theta = −4.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
500
1000
1500

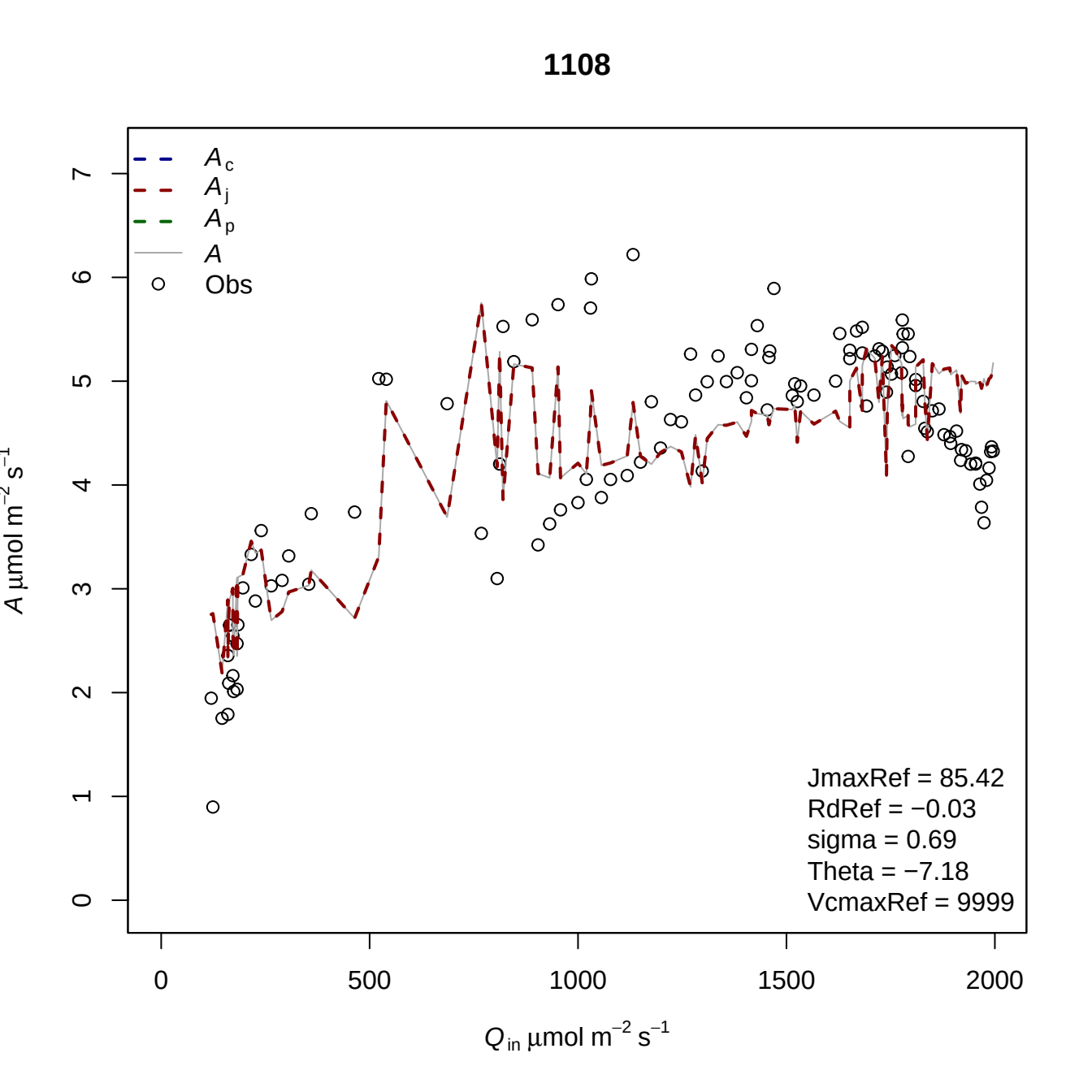

1102
A_c
A_j
A_p
A
Obs
JmaxRef = 137.09
RdRef = 1.6
sigma = 0.44
Theta = −7.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

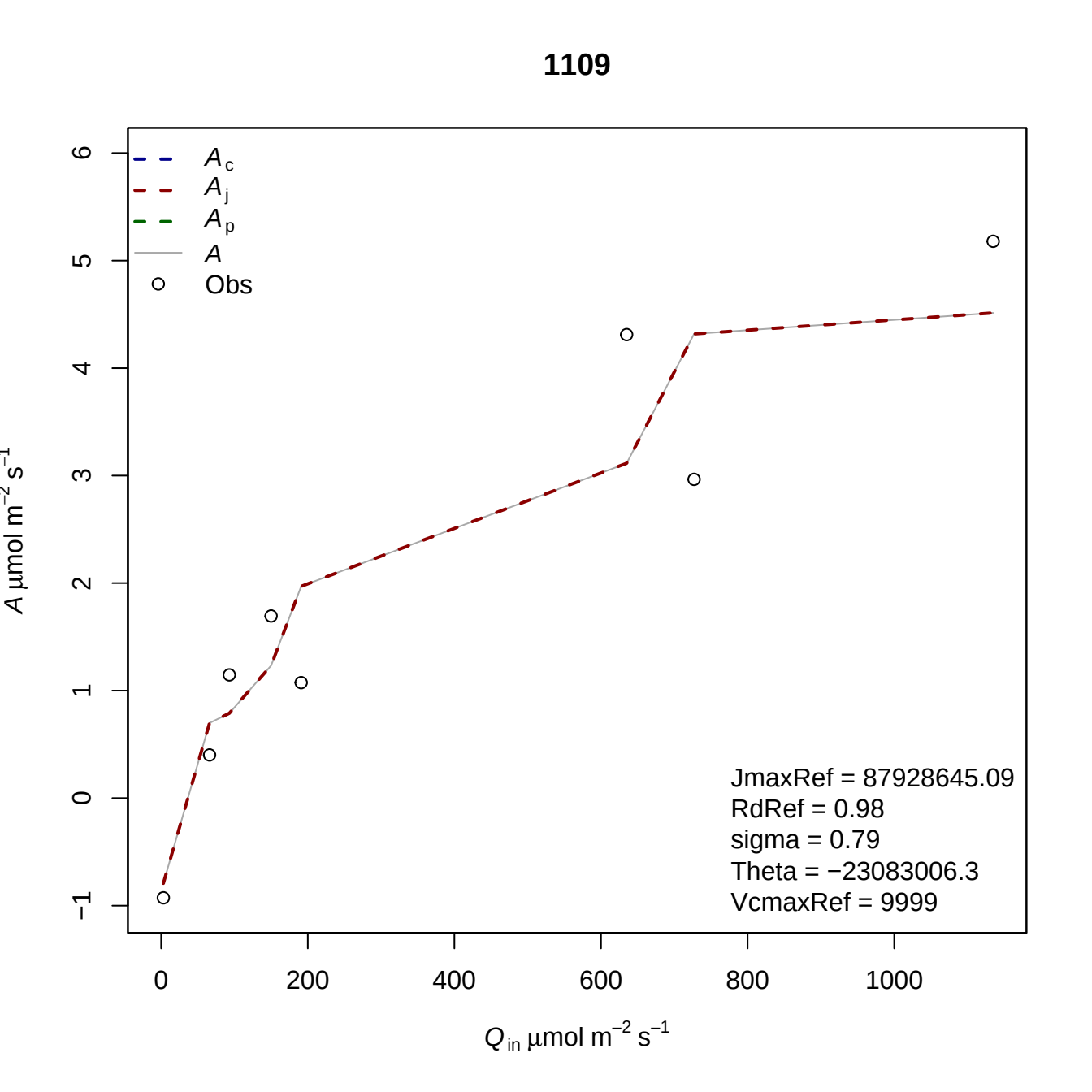

1103
A_c
A_j
A_p
A
Obs
JmaxRef = 384.39
RdRef = 0.96
sigma = 0.38
Theta = −50.96
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

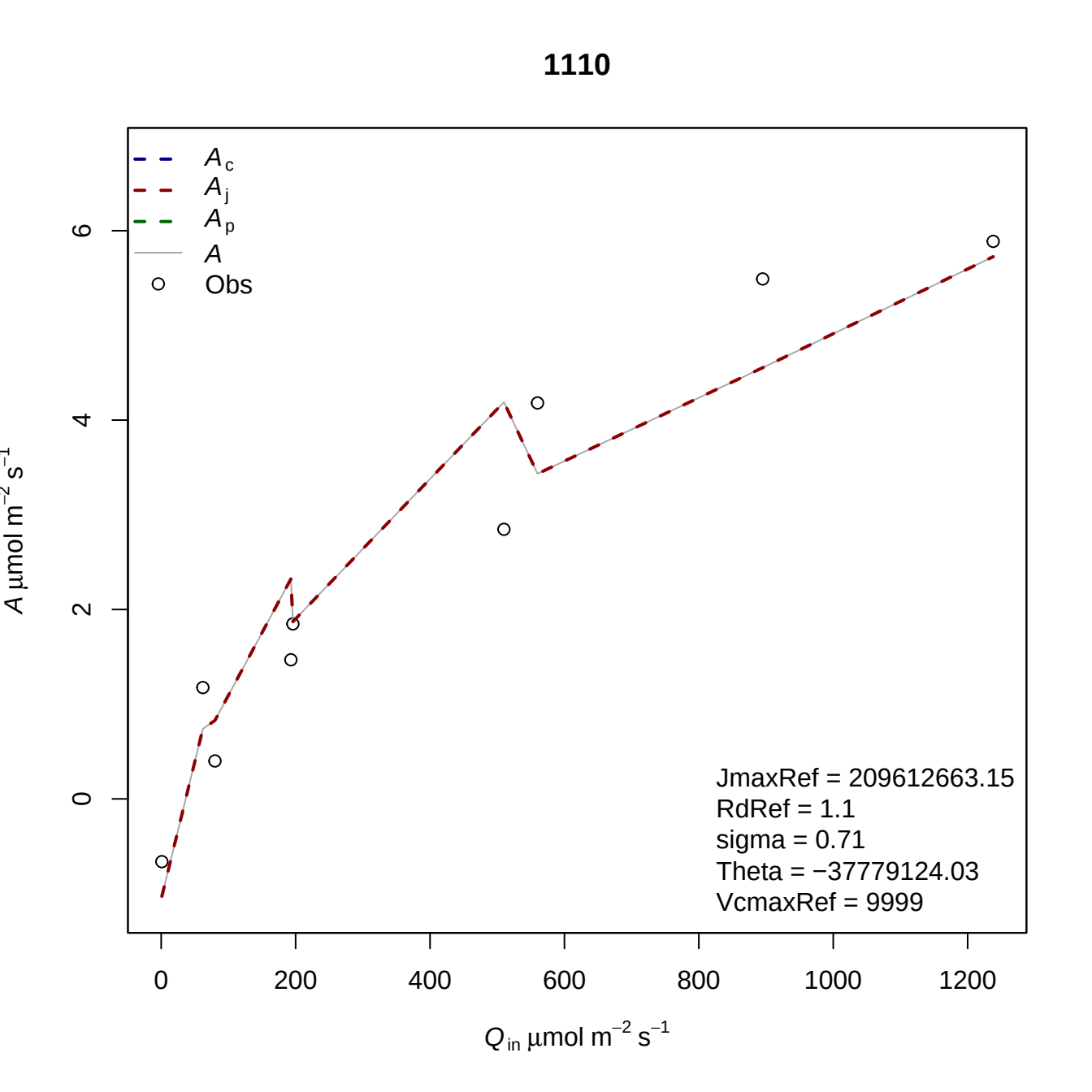

1104
A_c
A_j
A_p
A
Obs
JmaxRef = 231.03
RdRef = 1.3
sigma = 0.87
Theta = −21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

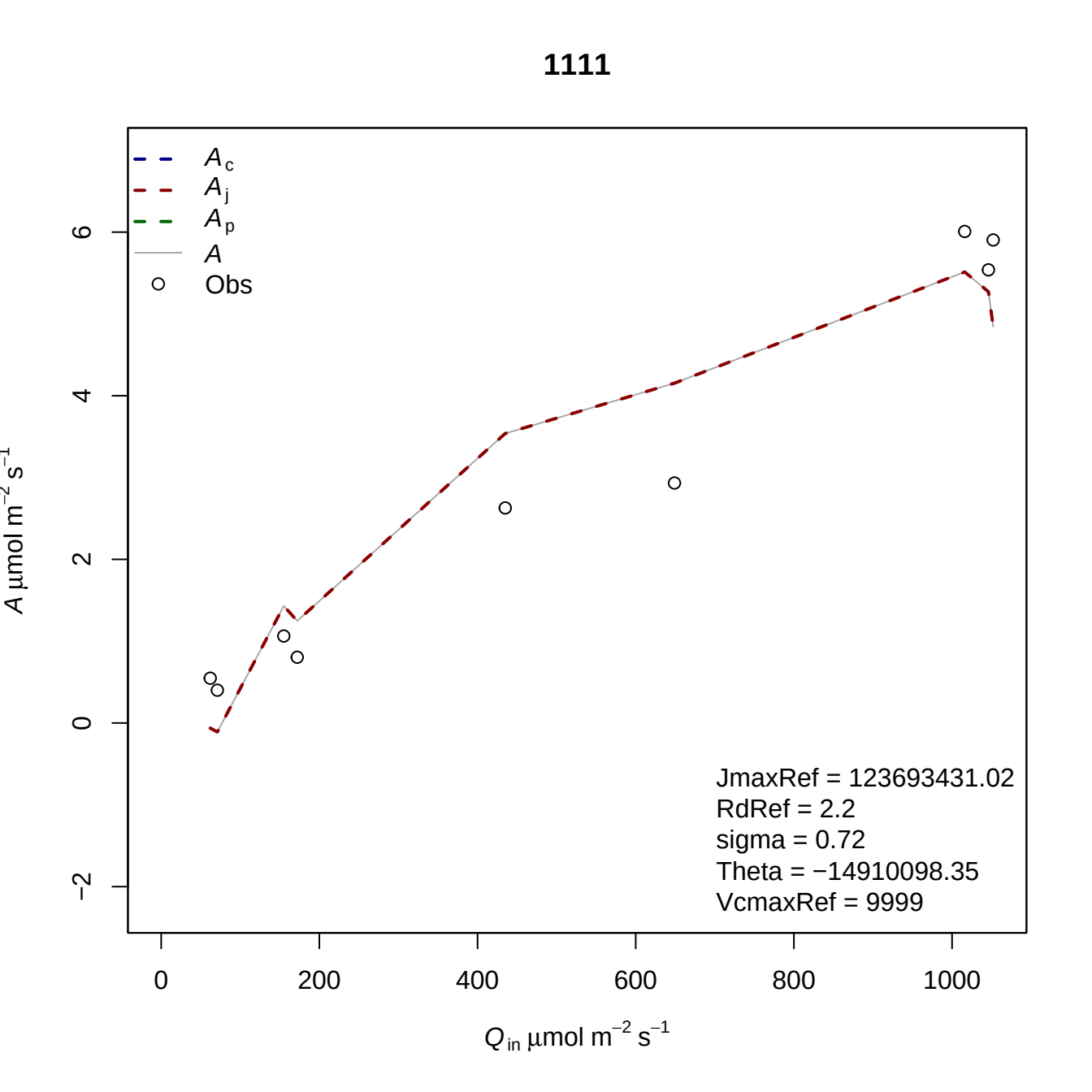

1105
A_c
A_j
A_p
A
Obs
JmaxRef = 85.82
RdRef = 3.34
sigma = 1.21
Theta = −1.16
VcmaxRef = 9999
8
6
4
2
0
−2
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

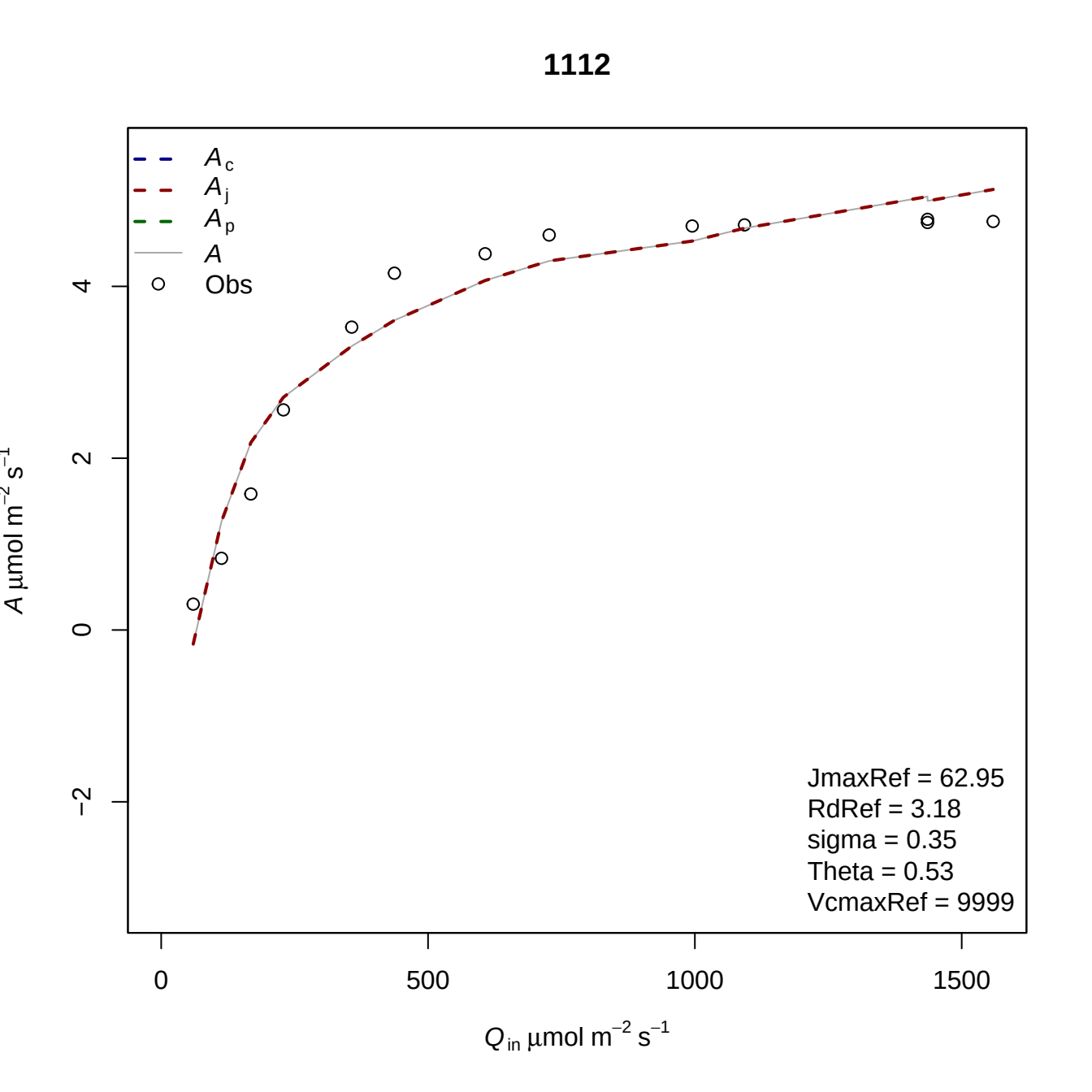

1107
A_c
A_j
A_p
A
Obs
JmaxRef = 35.42
RdRef = 1.35
sigma = 0.78
Theta = −2.97
VcmaxRef = 15.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

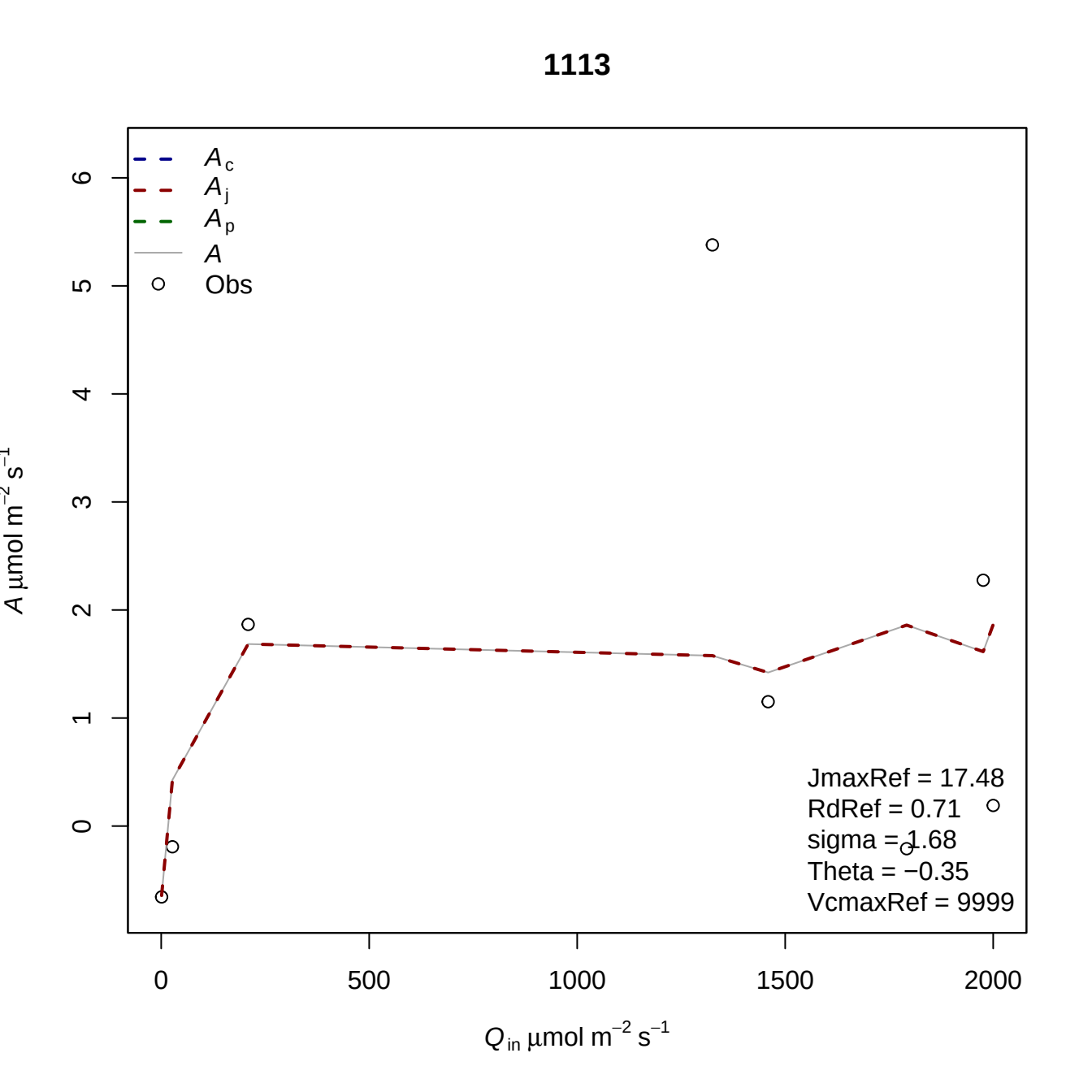

1108
A_c
A_j
A_p
A
Obs
JmaxRef = 85.42
RdRef = −0.03
sigma = 0.69
Theta = −7.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
500
1000
1500
2000

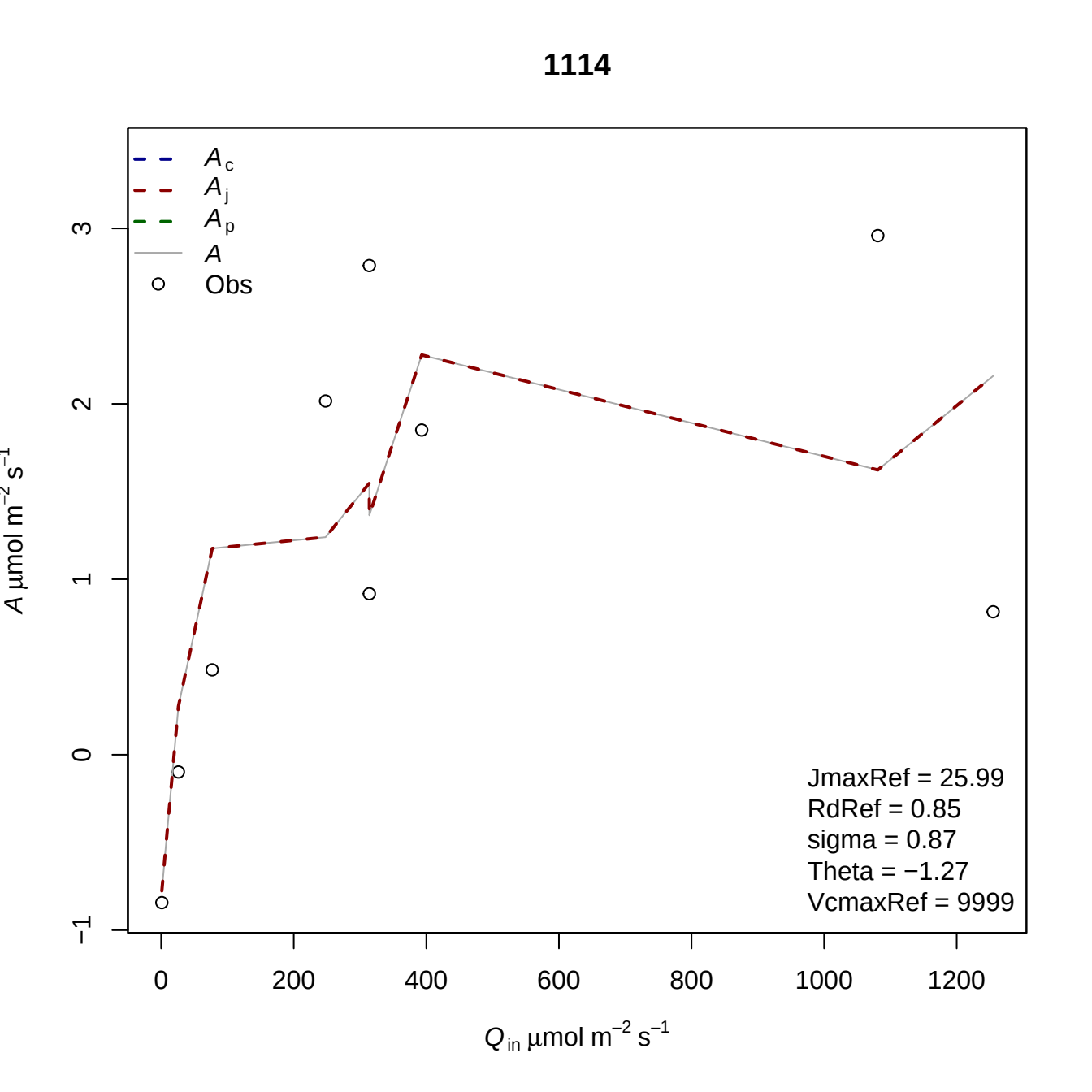

1109
A_c
A_j
A_p
A
Obs
JmaxRef = 87928645.09
RdRef = 0.98
sigma = 0.79
Theta = −23083006.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000

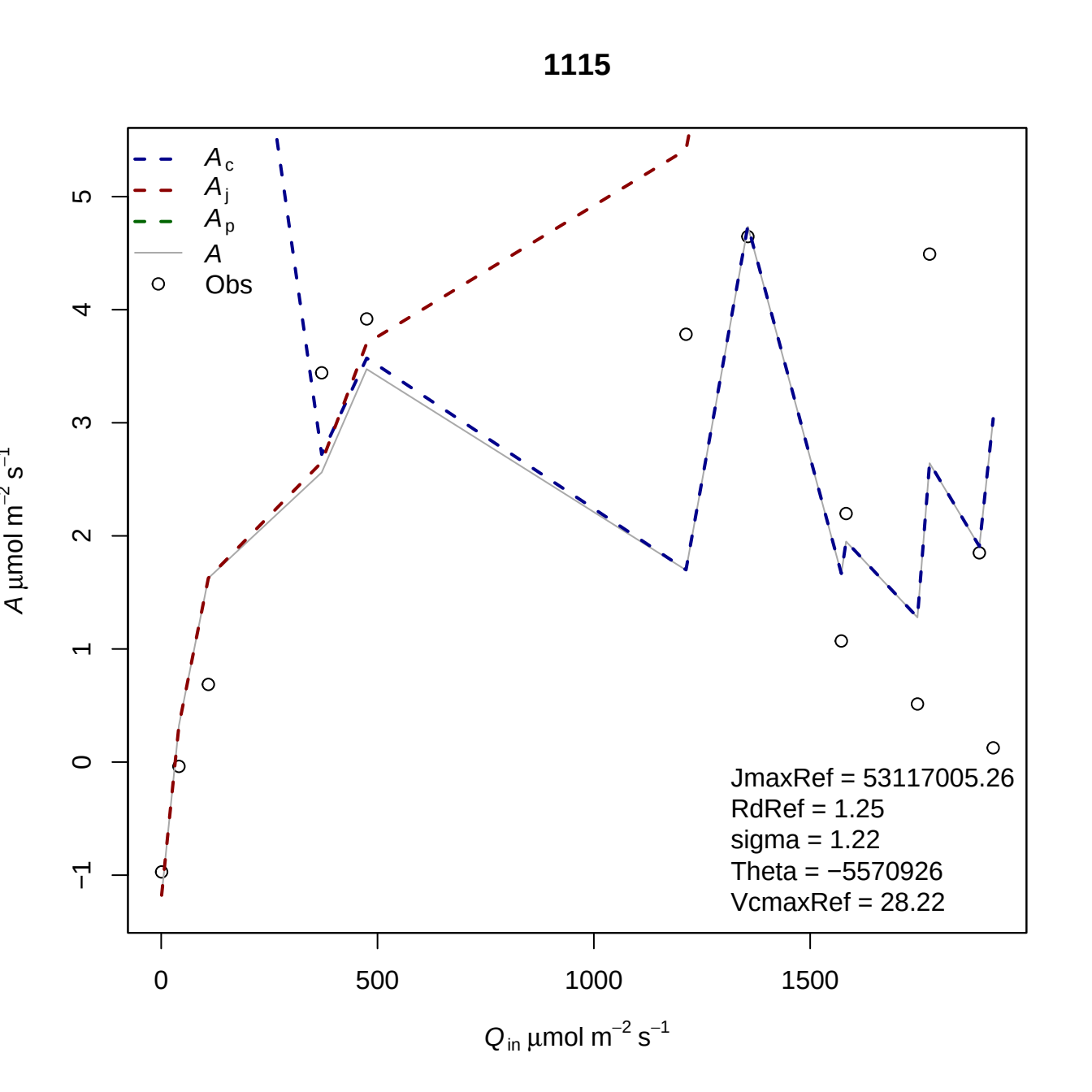

1110
A_c
A_j
A_p
A
Obs
JmaxRef = 209612663.15
RdRef = 1.1
sigma = 0.71
Theta = −37779124.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000
1200

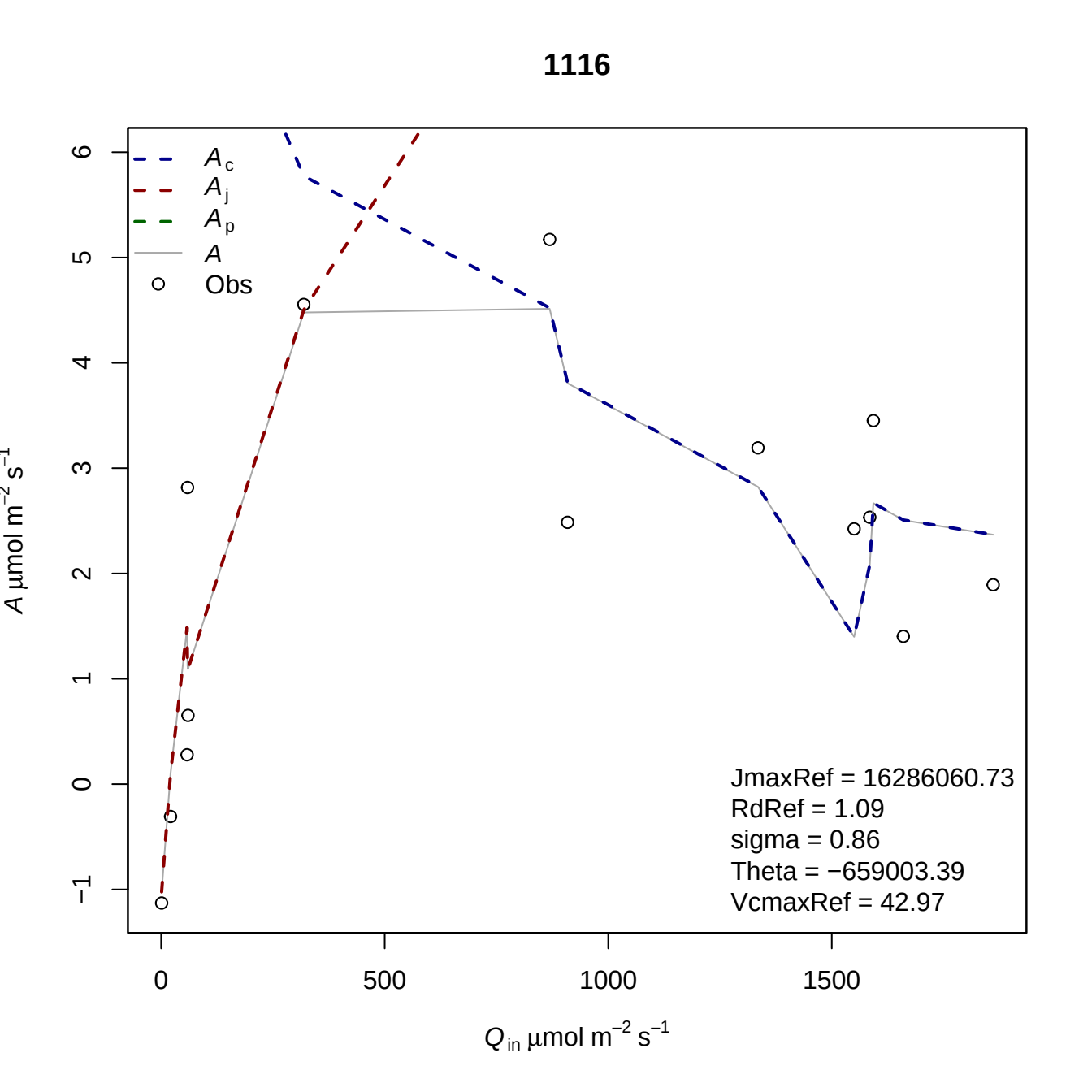

1111
A_c
A_j
A_p
A
Obs
JmaxRef = 123693431.02
RdRef = 2.2
sigma = 0.72
Theta = −14910098.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

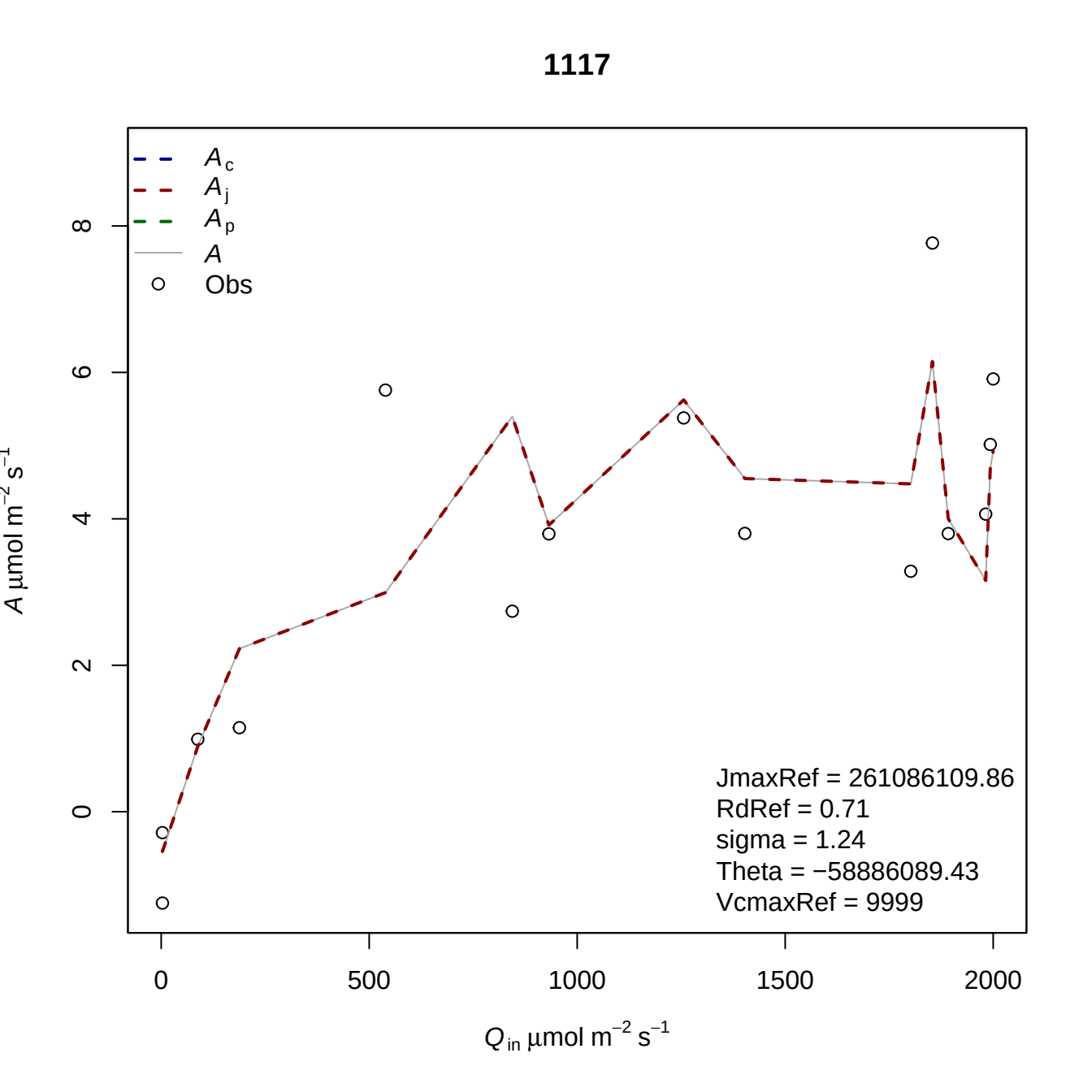

1112
A_c
A_j
A_p
A
Obs
JmaxRef = 62.95
RdRef = 3.18
sigma = 0.35
Theta = 0.53
VcmaxRef = 9999
−2
0
2
4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

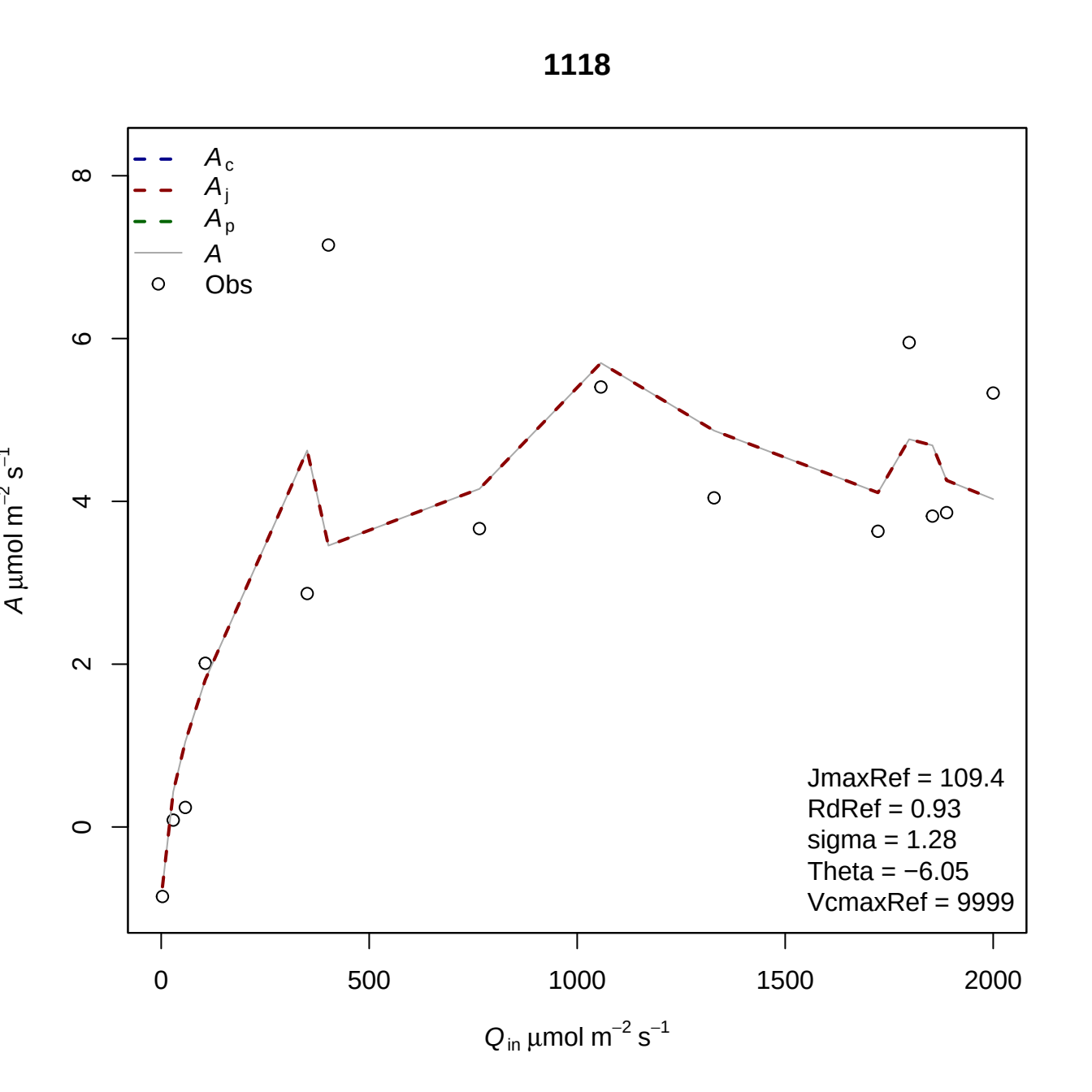

1113
A_c
A_j
A_p
A
Obs
JmaxRef = 17.48
RdRef = 0.71
sigma = 1.68
Theta = −0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

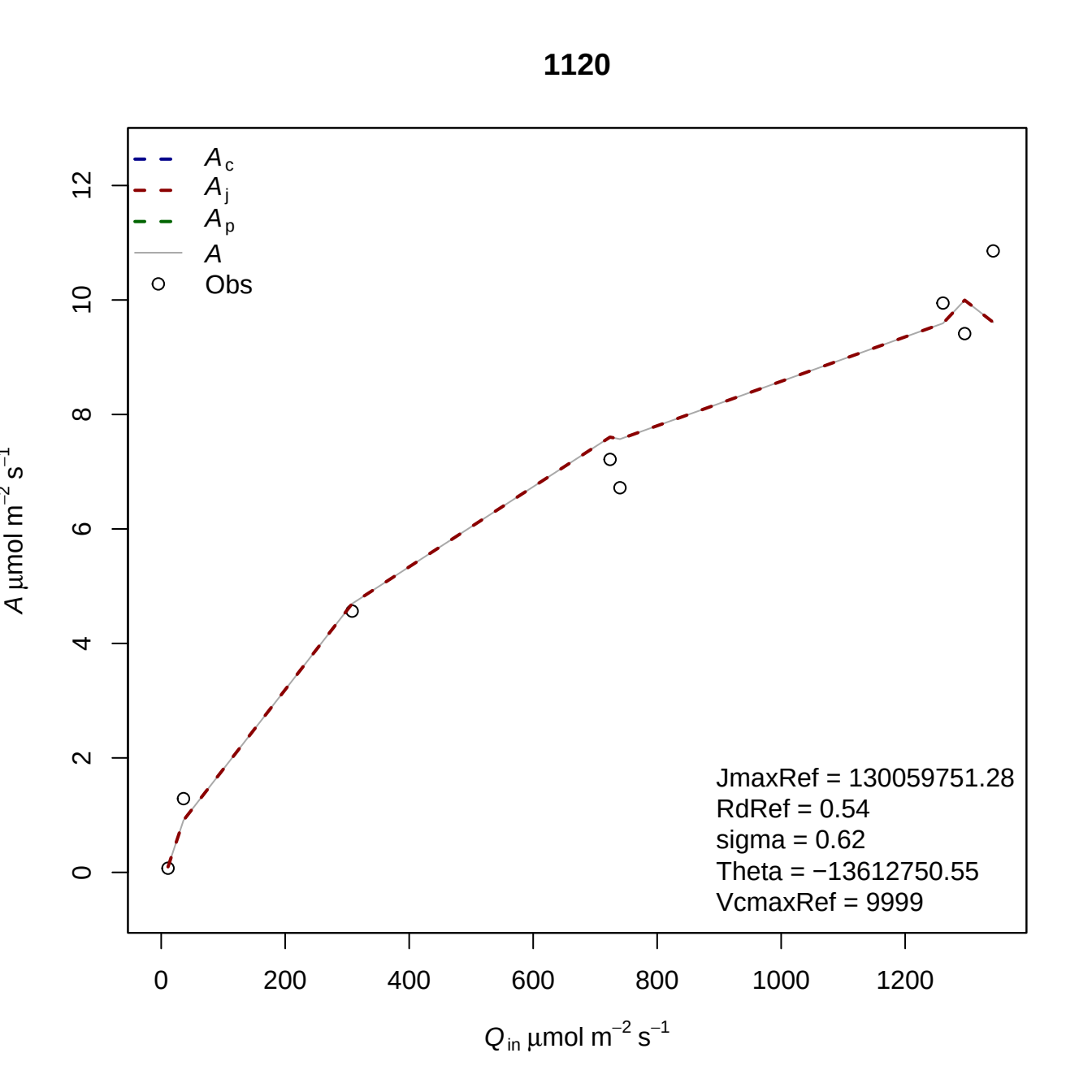

1114
A_c
A_j
A_p
A
Obs
JmaxRef = 25.99
RdRef = 0.85
sigma = 0.87
Theta = −1.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200

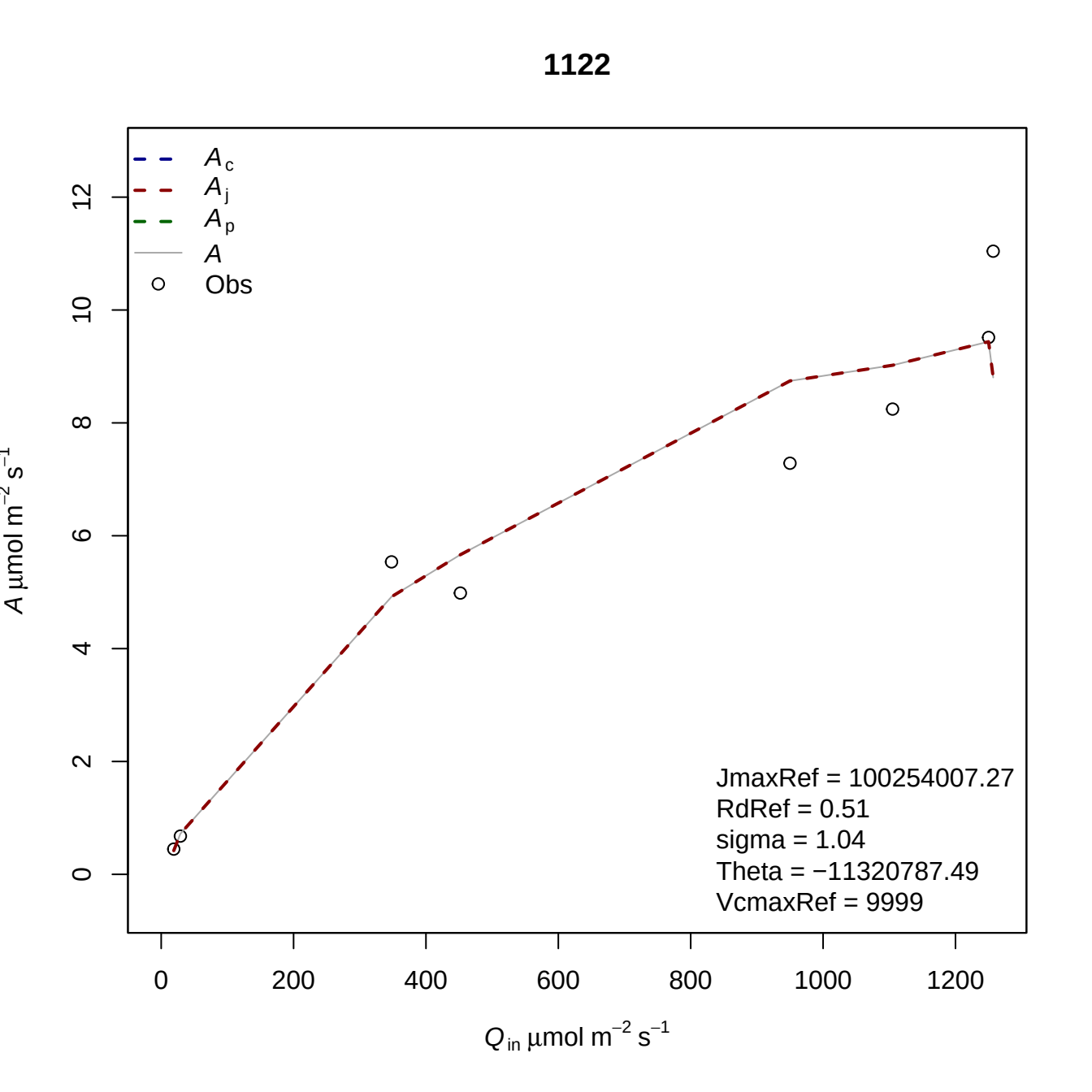

1115
A_c
A_j
A_p
A
Obs
JmaxRef = 53117005.26
RdRef = 1.25
sigma = 1.22
Theta = −5570926
VcmaxRef = 28.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

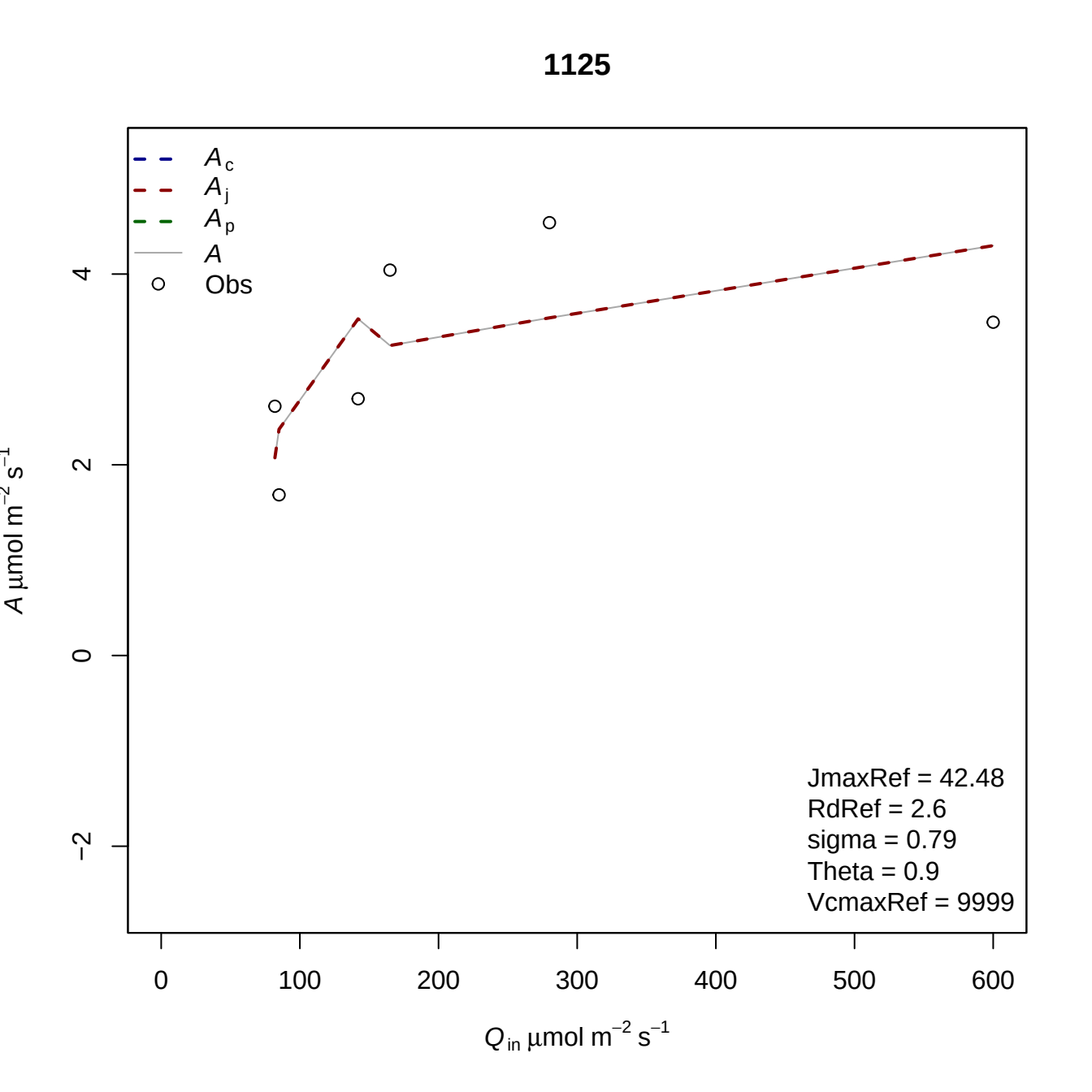

1116
A_c
A_j
A_p
A
Obs
JmaxRef = 16286060.73
RdRef = 1.09
sigma = 0.86
Theta = −659003.39
VcmaxRef = 42.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

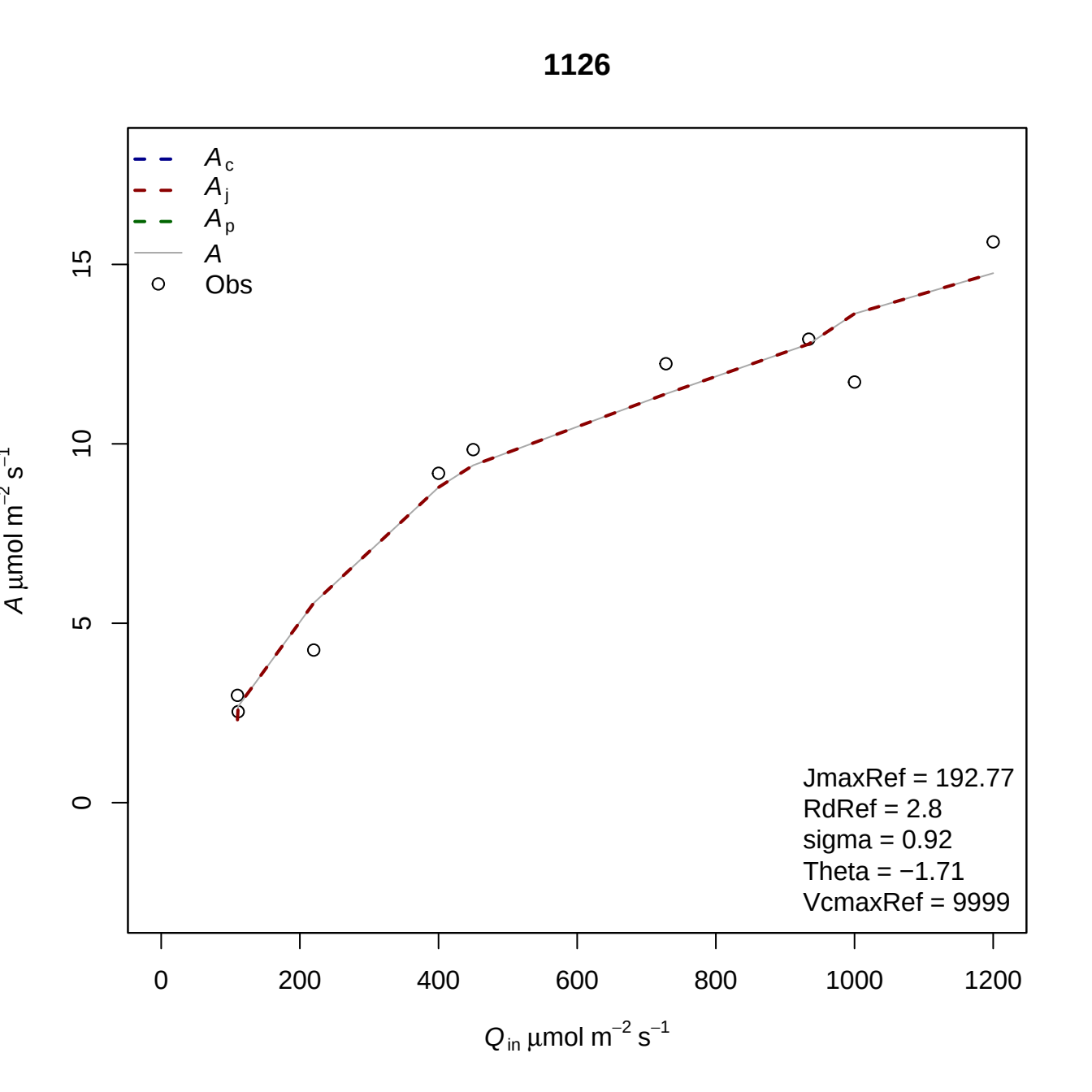

1117
A_c
A_j
A_p
A
Obs
JmaxRef = 261086109.86
RdRef = 0.71
sigma = 1.24
Theta = −58886089.43
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

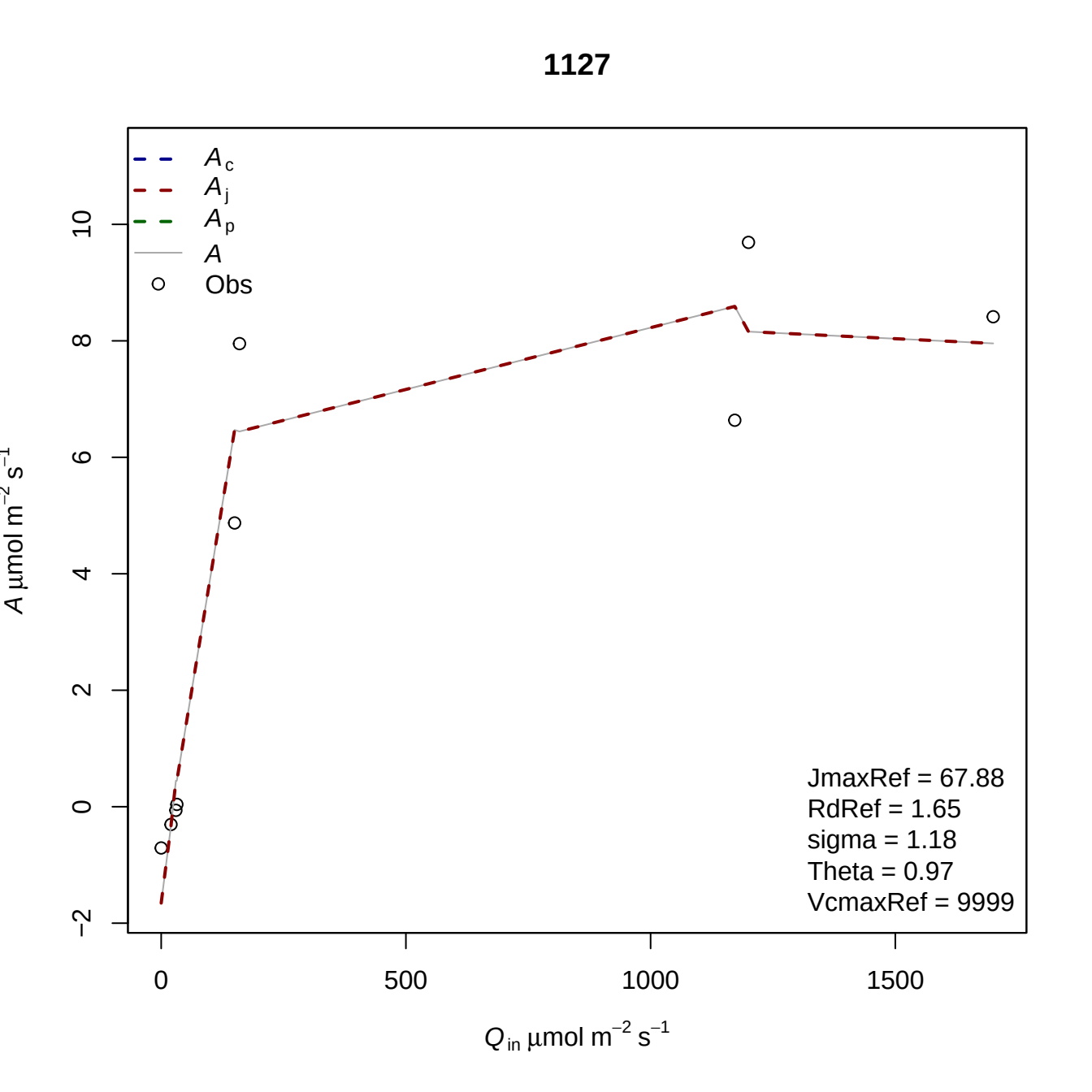

1118
A_c
A_j
A_p
A
Obs
JmaxRef = 109.4
RdRef = 0.93
sigma = 1.28
Theta = −6.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

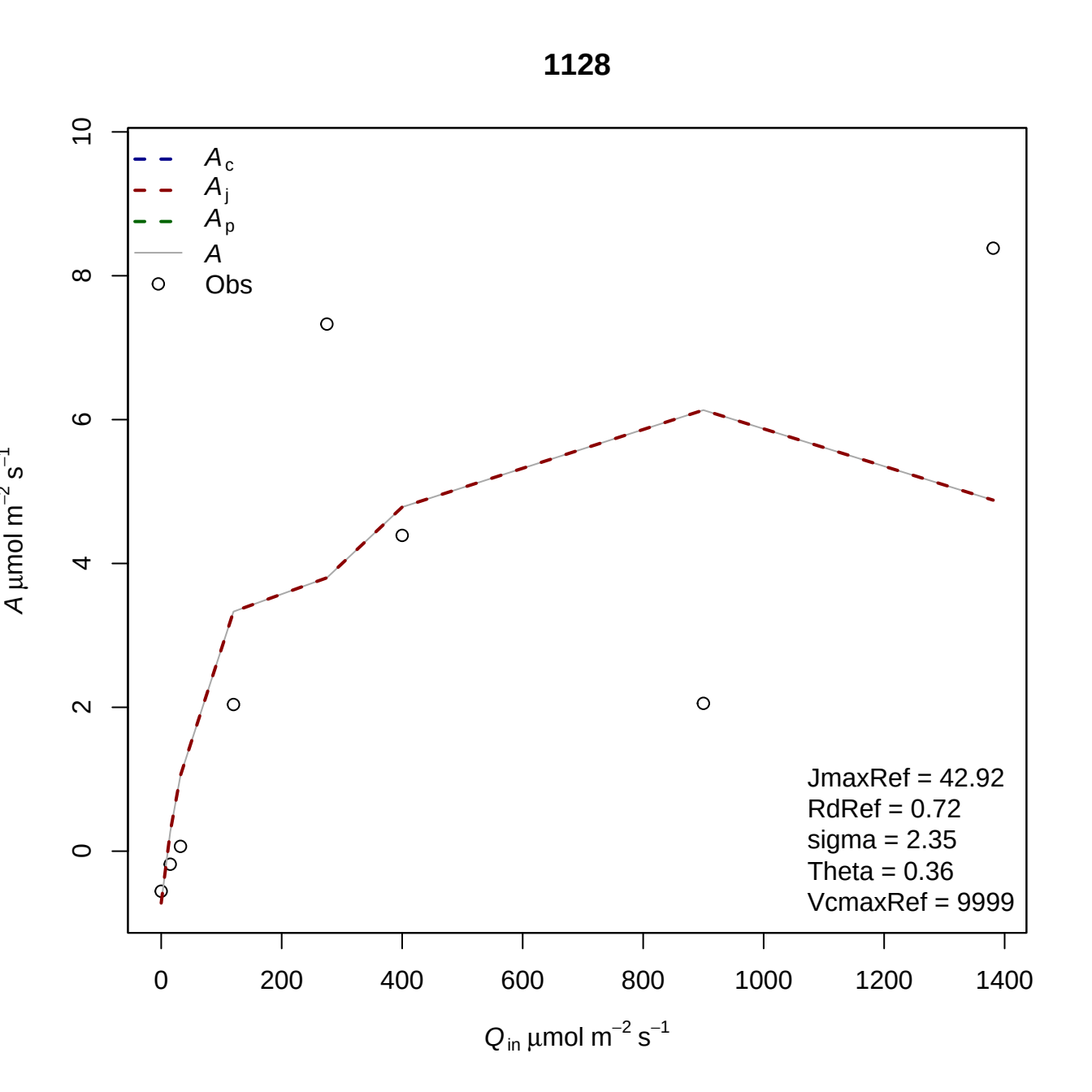

1120
A_c
A_j
A_p
A
Obs
JmaxRef = 130059751.28
RdRef = 0.54
sigma = 0.62
Theta = −13612750.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

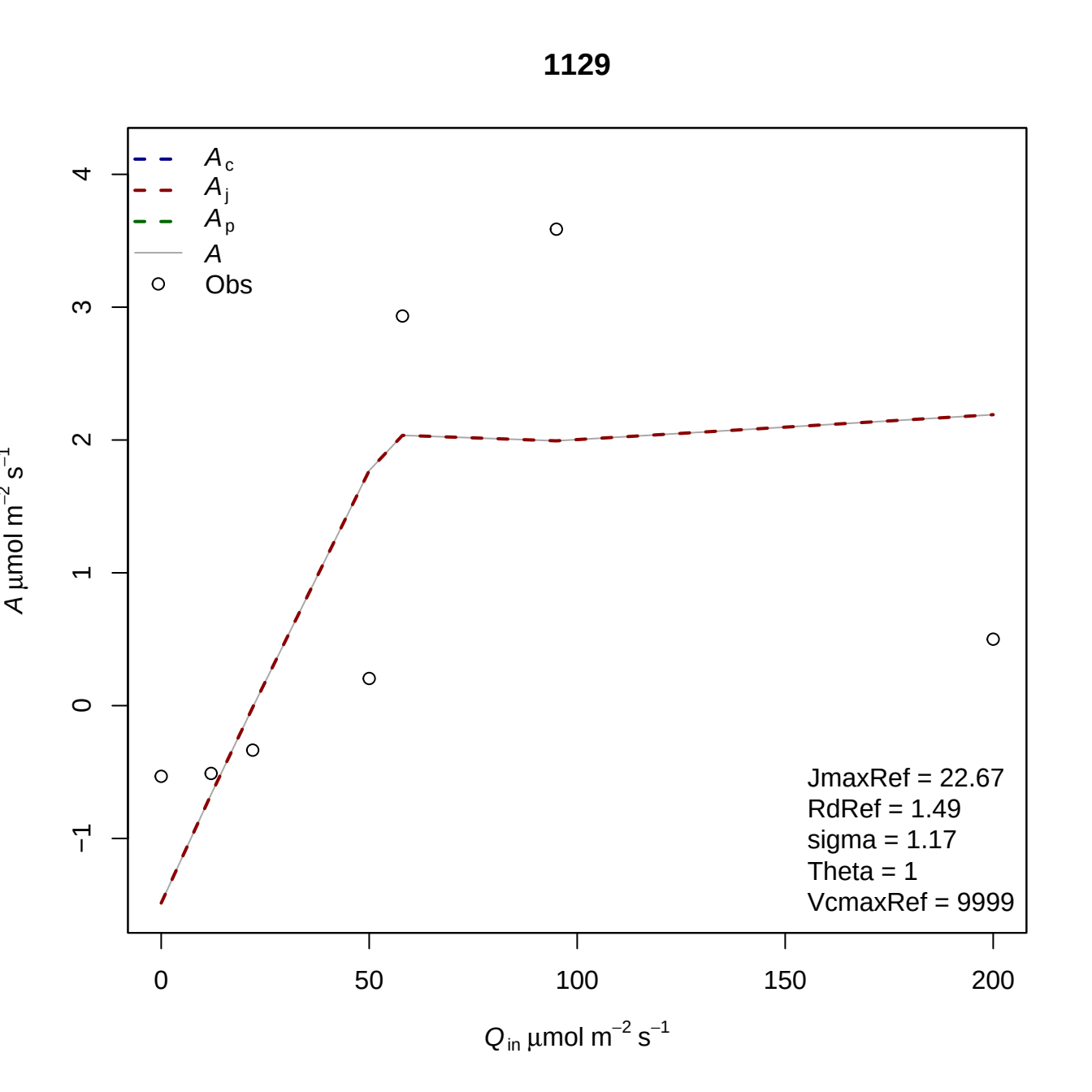

1122
A_c
A_j
A_p
A
Obs
JmaxRef = 100254007.27
RdRef = 0.51
sigma = 1.04
Theta = −11320787.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

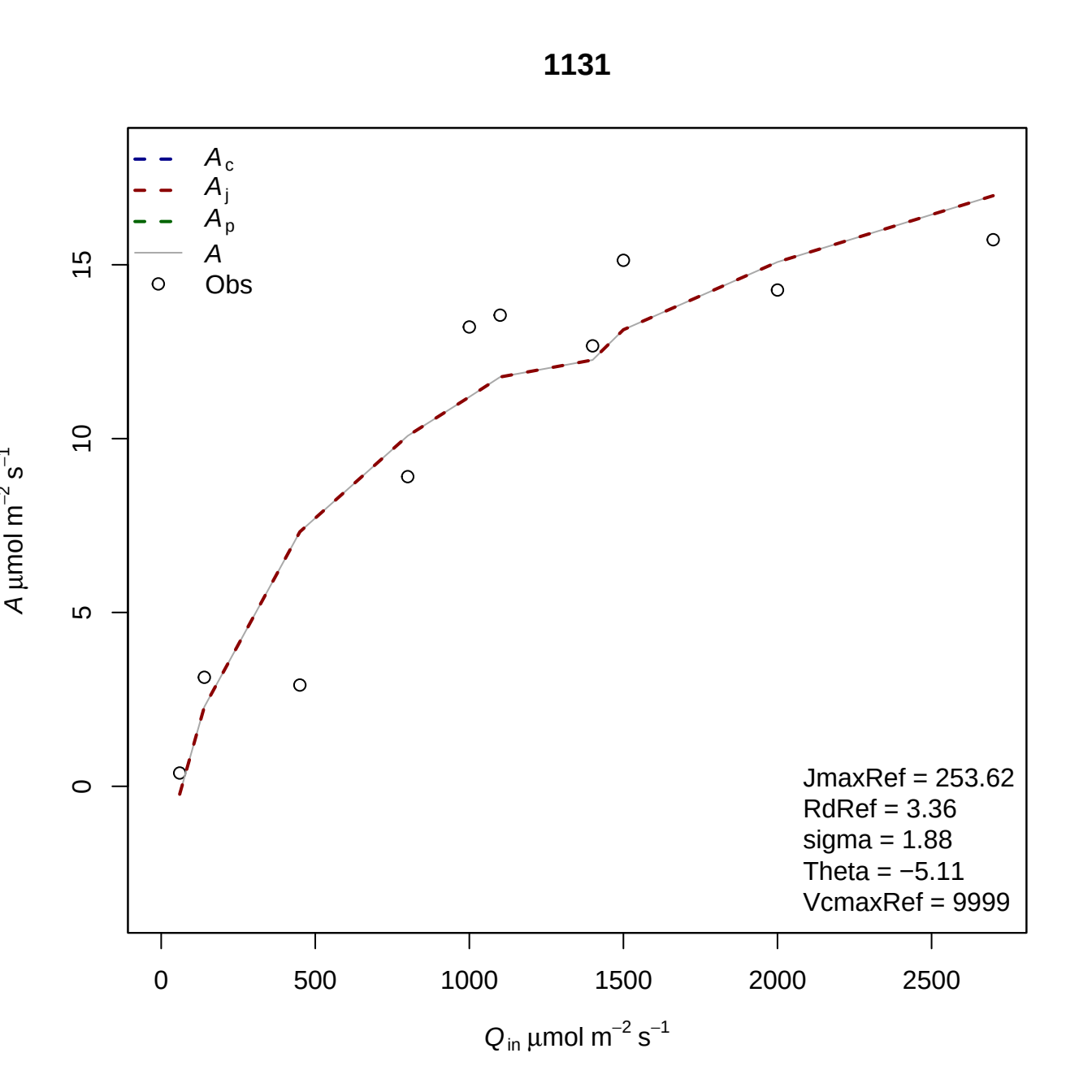

1125
A_c
A_j
A_p
A
Obs
JmaxRef = 42.48
RdRef = 2.6
sigma = 0.79
Theta = 0.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
100
200
300
400
500
600

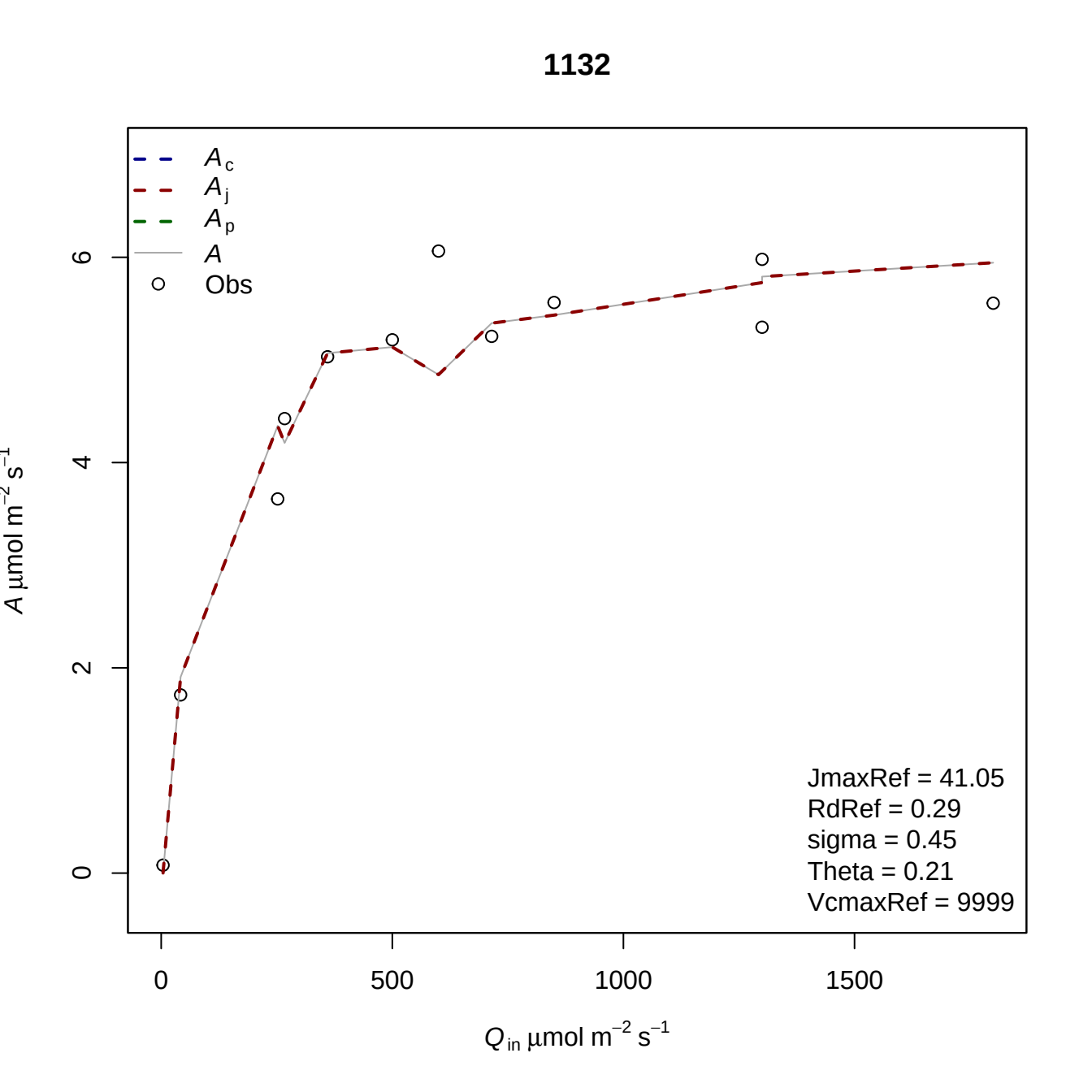

1126
A_c
A_j
A_p
A
Obs
JmaxRef = 192.77
RdRef = 2.8
sigma = 0.92
Theta = −1.71
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

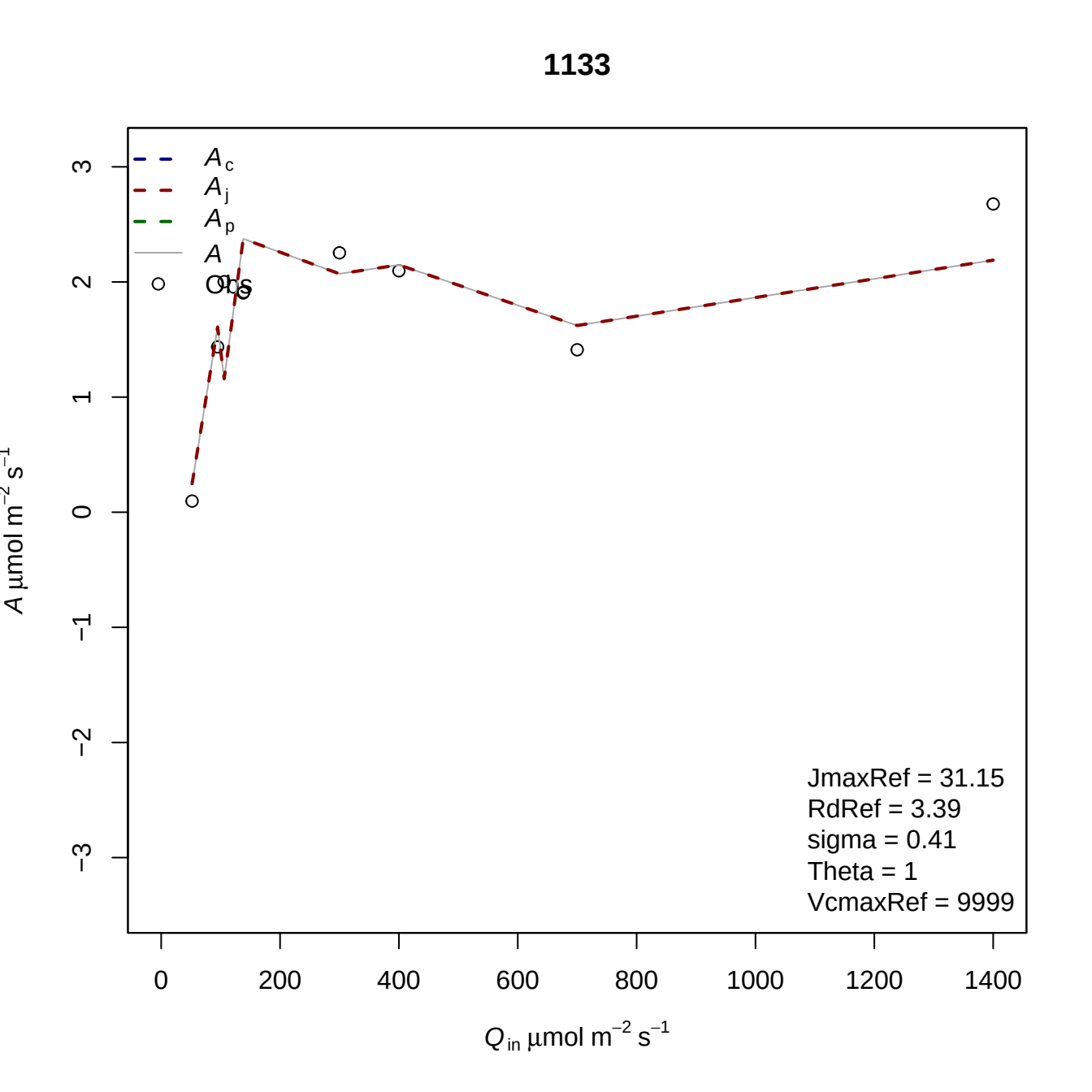

1127
A_c
A_j
A_p
A
Obs
JmaxRef = 67.88
RdRef = 1.65
sigma = 1.18
Theta = 0.97
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

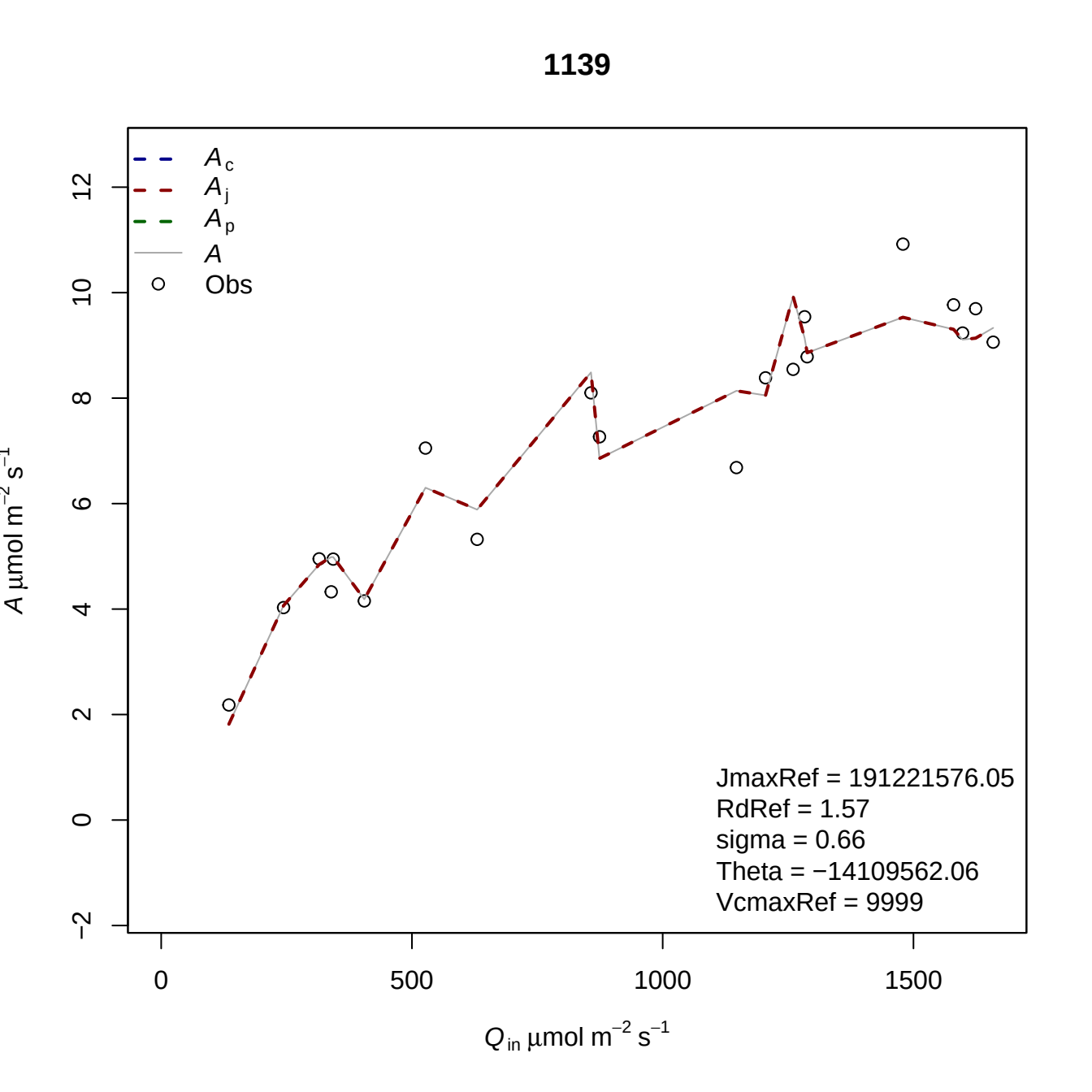

1128
A_c
A_j
A_p
A
Obs
JmaxRef = 42.92
RdRef = 0.72
sigma = 2.35
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

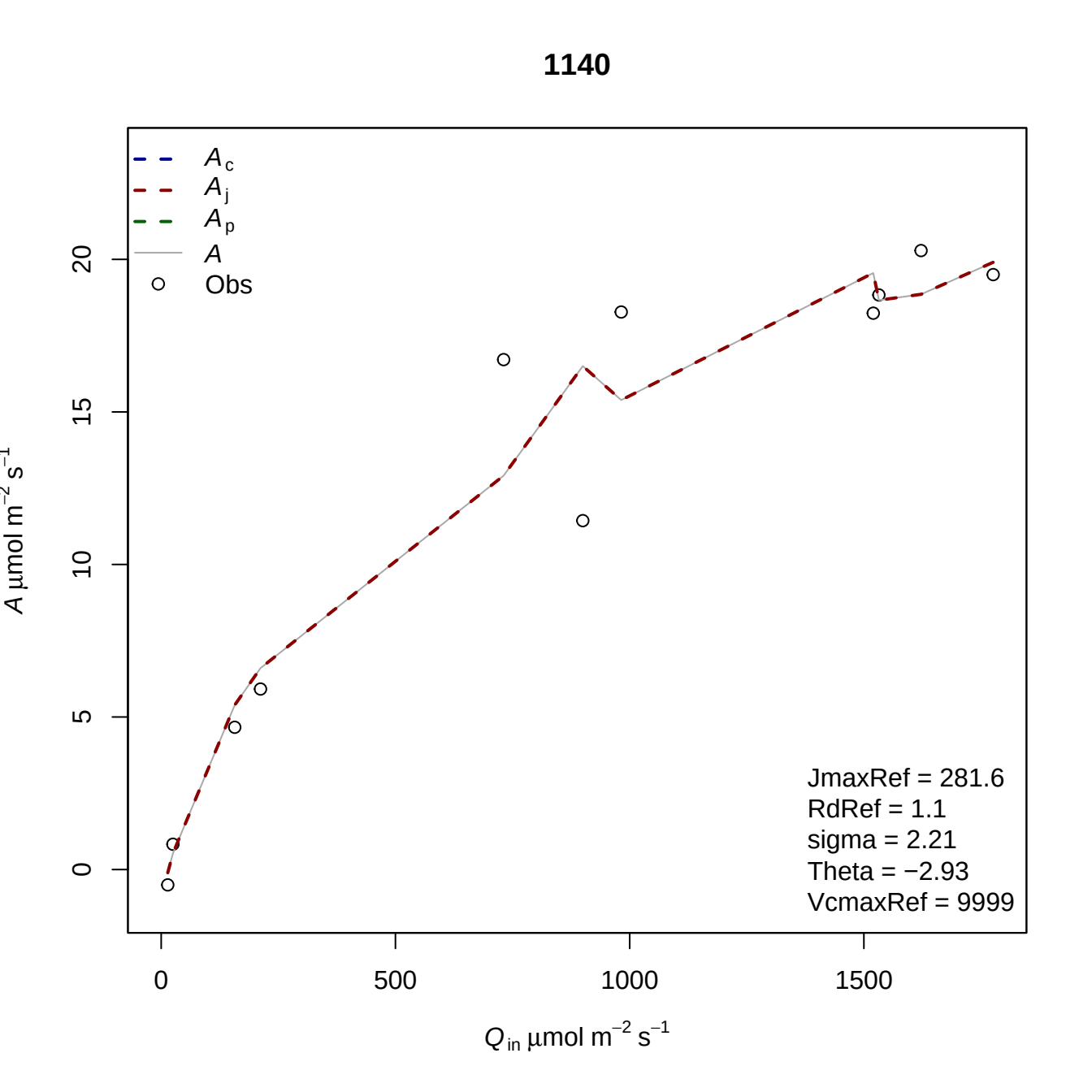

1129
A_c
A_j
A_p
A
Obs
JmaxRef = 22.67
RdRef = 1.49
sigma = 1.17
Theta = 1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
50
100
150
200

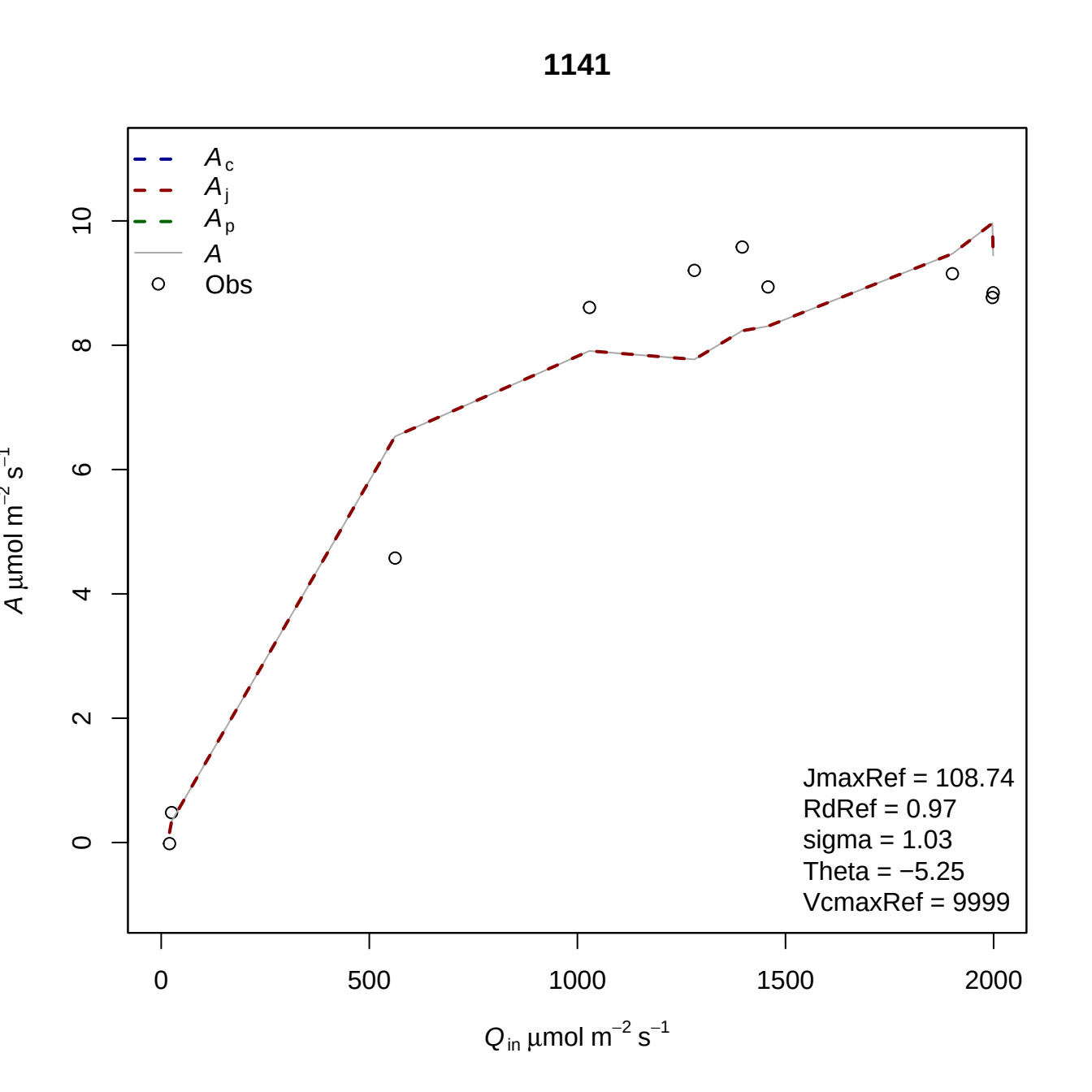

1131
A_c
A_j
A_p
A
Obs
JmaxRef = 253.62
RdRef = 3.36
sigma = 1.88
Theta = −5.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000
2500
Q_{in} µmol m^{-2} s^{-1}

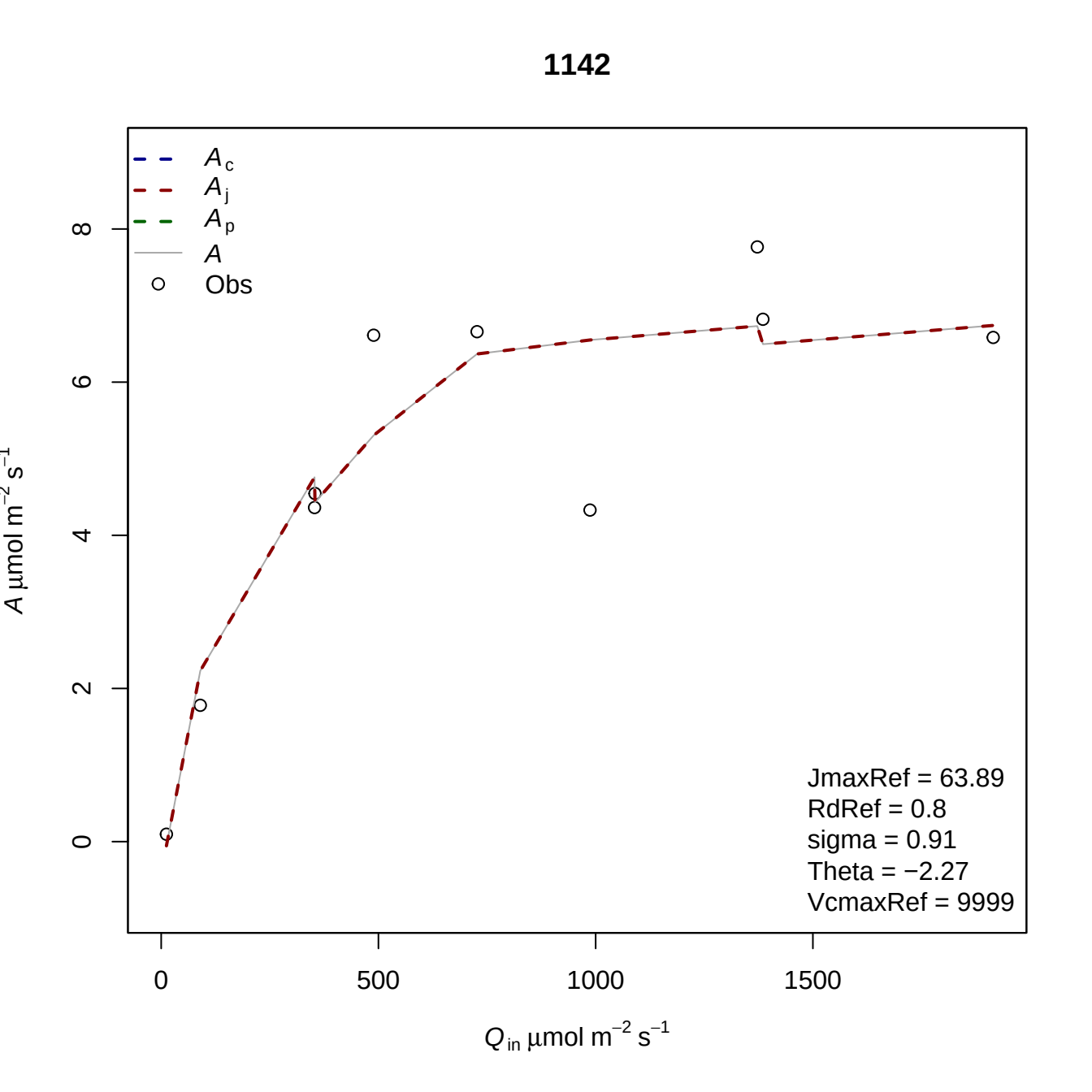

1132
A_c
A_j
A_p
A
Obs
JmaxRef = 41.05
RdRef = 0.29
sigma = 0.45
Theta = 0.21
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

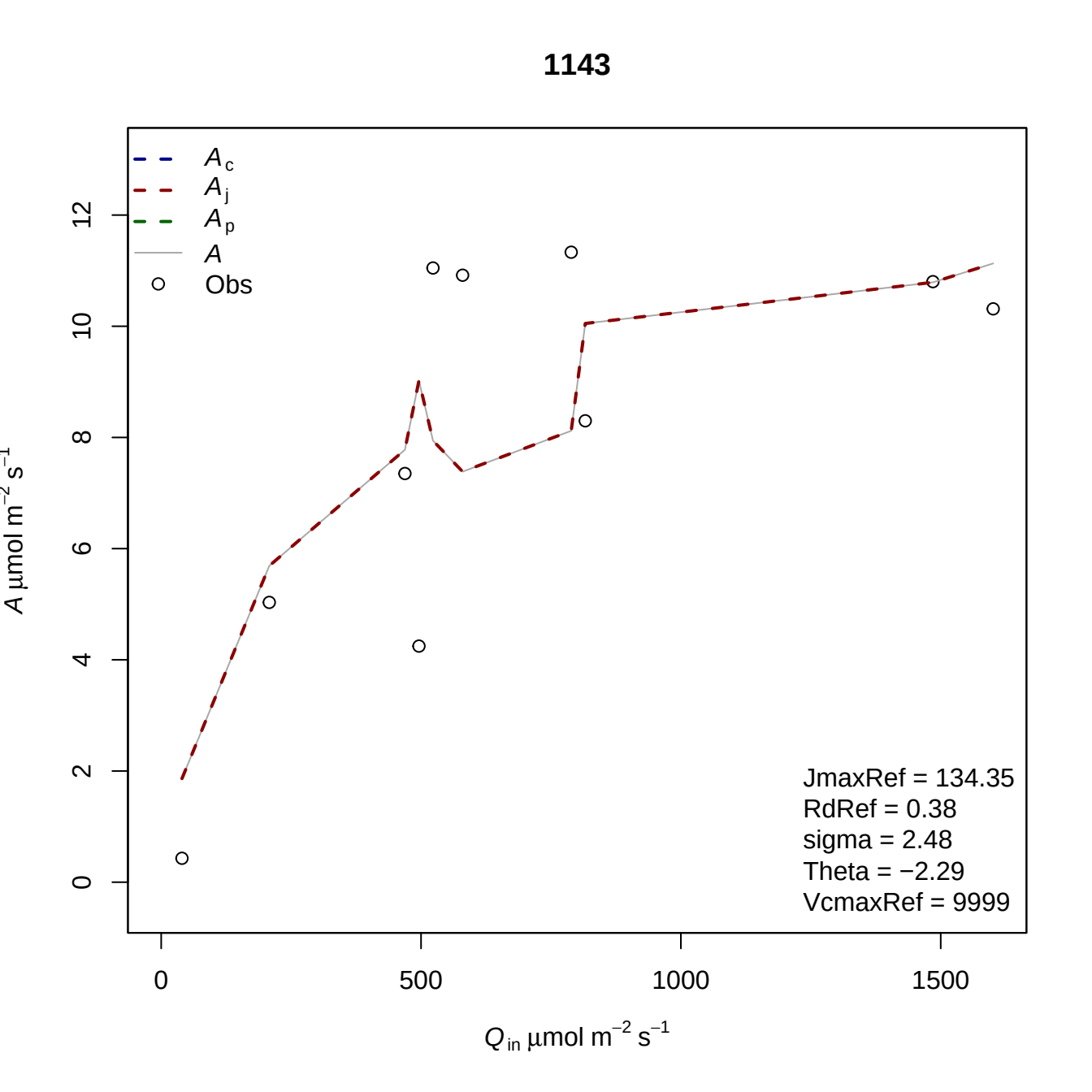

1133
A_c
A_j
A_p
A
Obs
3
2
1
0
−1
−2
−3
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 31.15
RdRef = 3.39
sigma = 0.41
Theta = 1
VcmaxRef = 9999

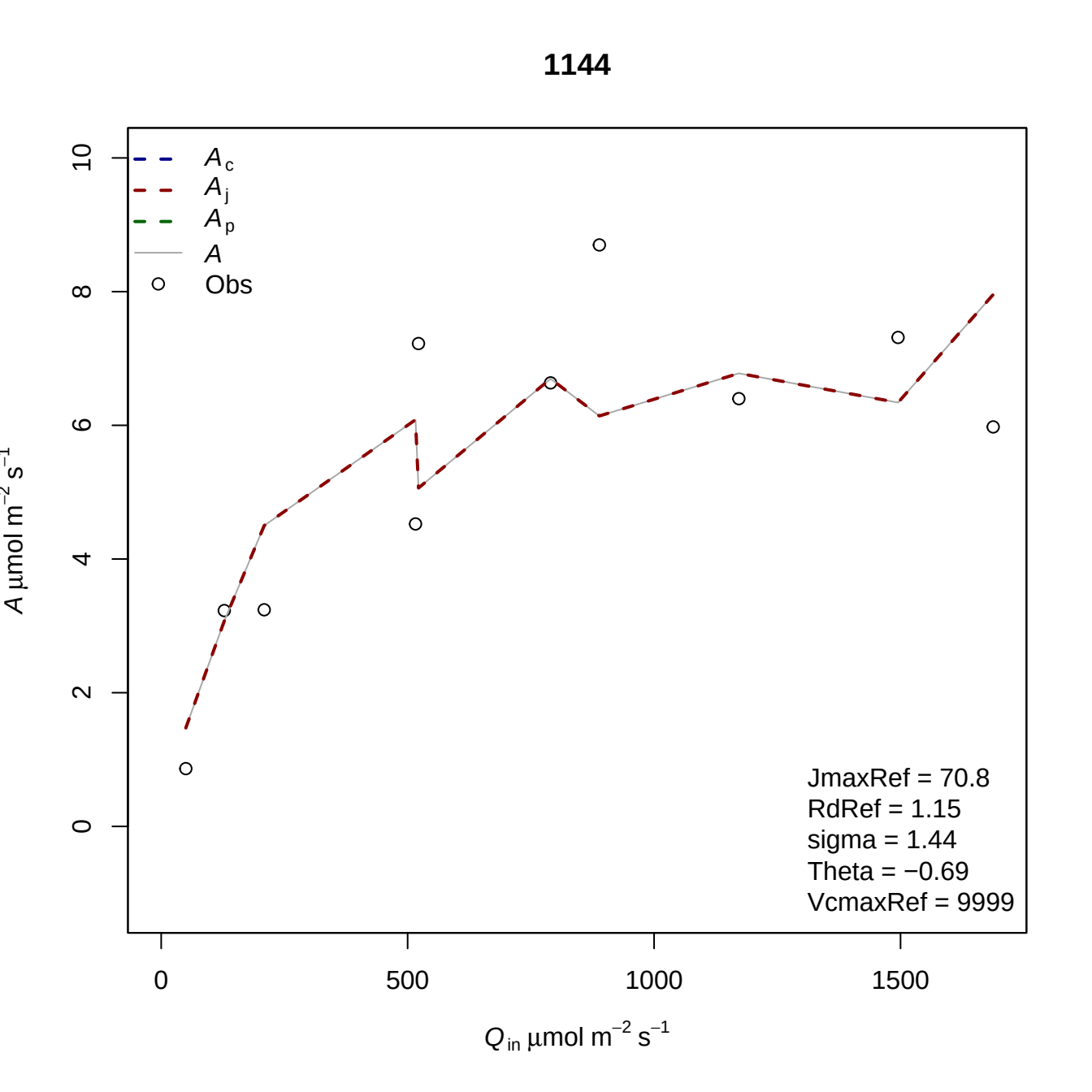

1139
A_c
A_j
A_p
A
Obs
JmaxRef = 191221576.05
RdRef = 1.57
sigma = 0.66
Theta = −14109562.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

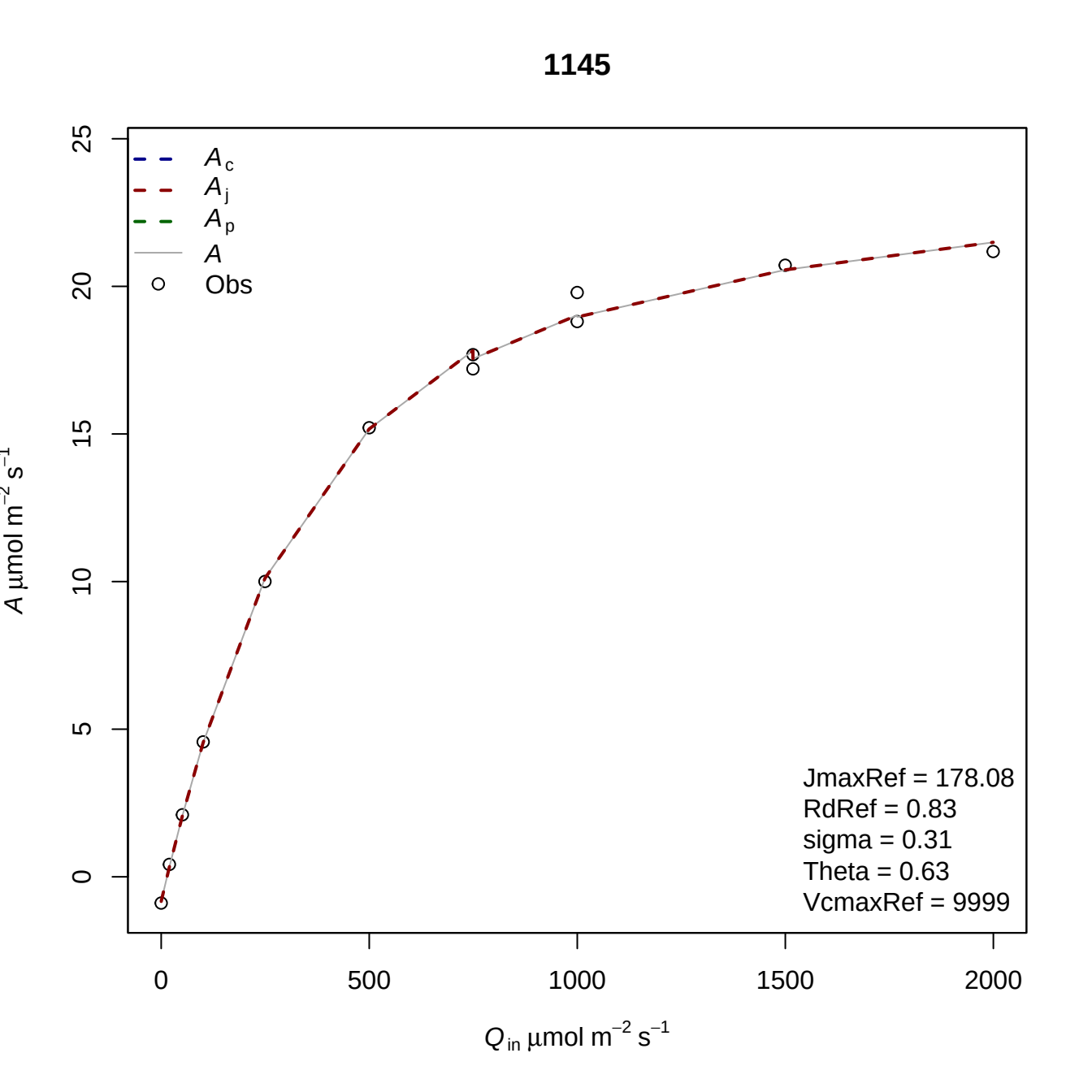

1140
A_c
A_j
A_p
A
Obs
JmaxRef = 281.6
RdRef = 1.1
sigma = 2.21
Theta = −2.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

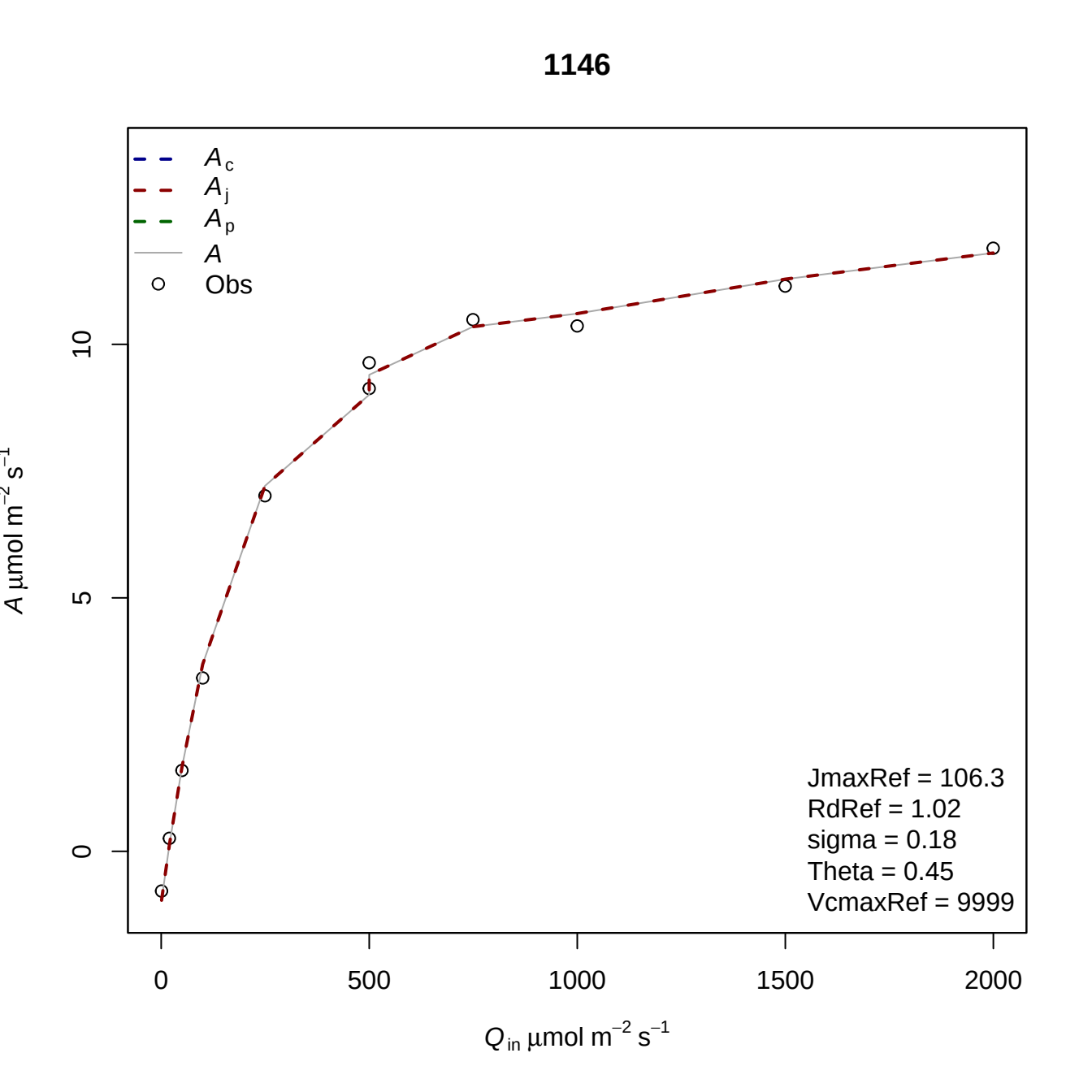

1141
A_c
A_j
A_p
A
Obs
JmaxRef = 108.74
RdRef = 0.97
sigma = 1.03
Theta = −5.25
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

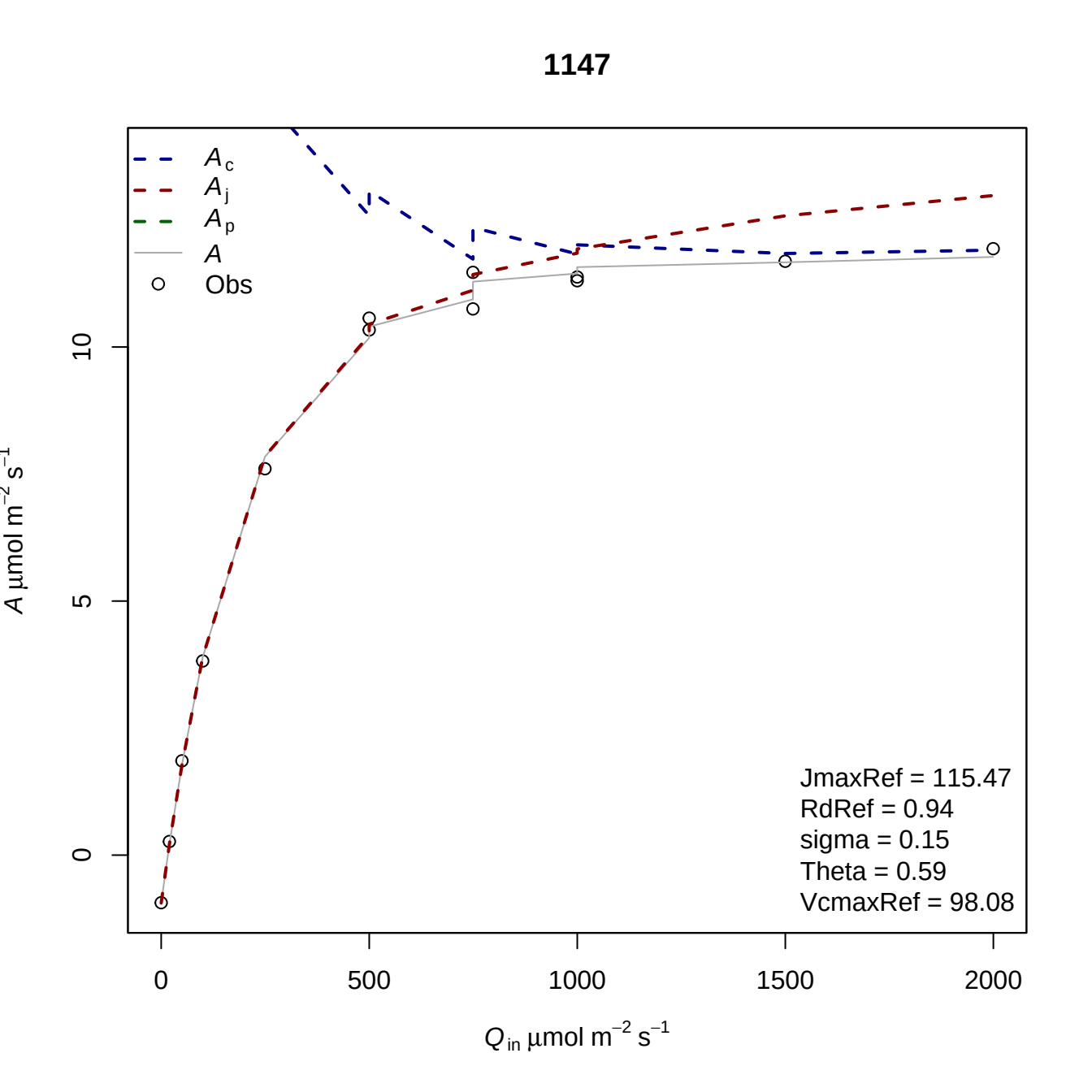

1142
A_c
A_j
A_p
A
Obs
JmaxRef = 63.89
RdRef = 0.8
sigma = 0.91
Theta = −2.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

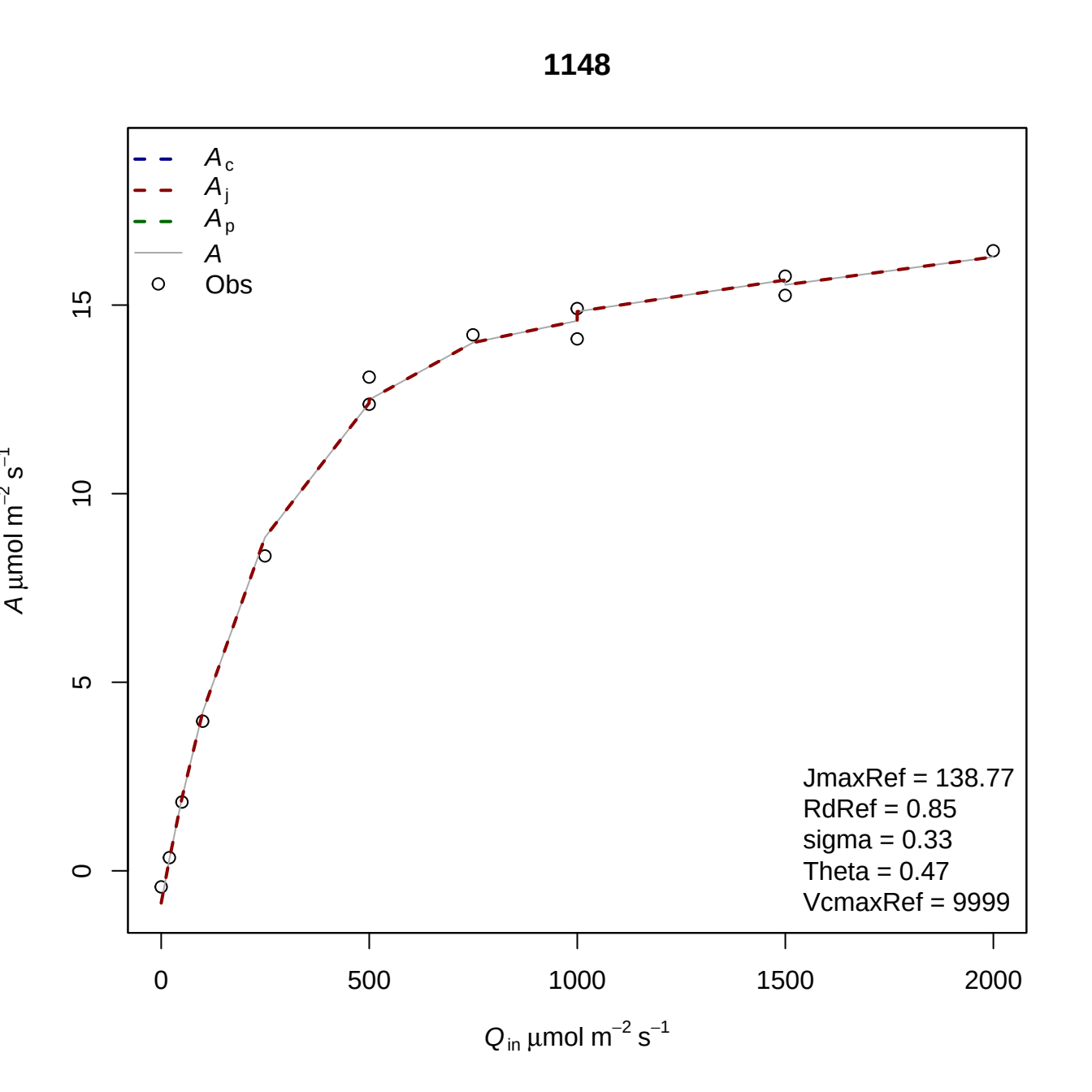

1143
A_c
A_j
A_p
A
Obs
JmaxRef = 134.35
RdRef = 0.38
sigma = 2.48
Theta = −2.29
VcmaxRef = 9999
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

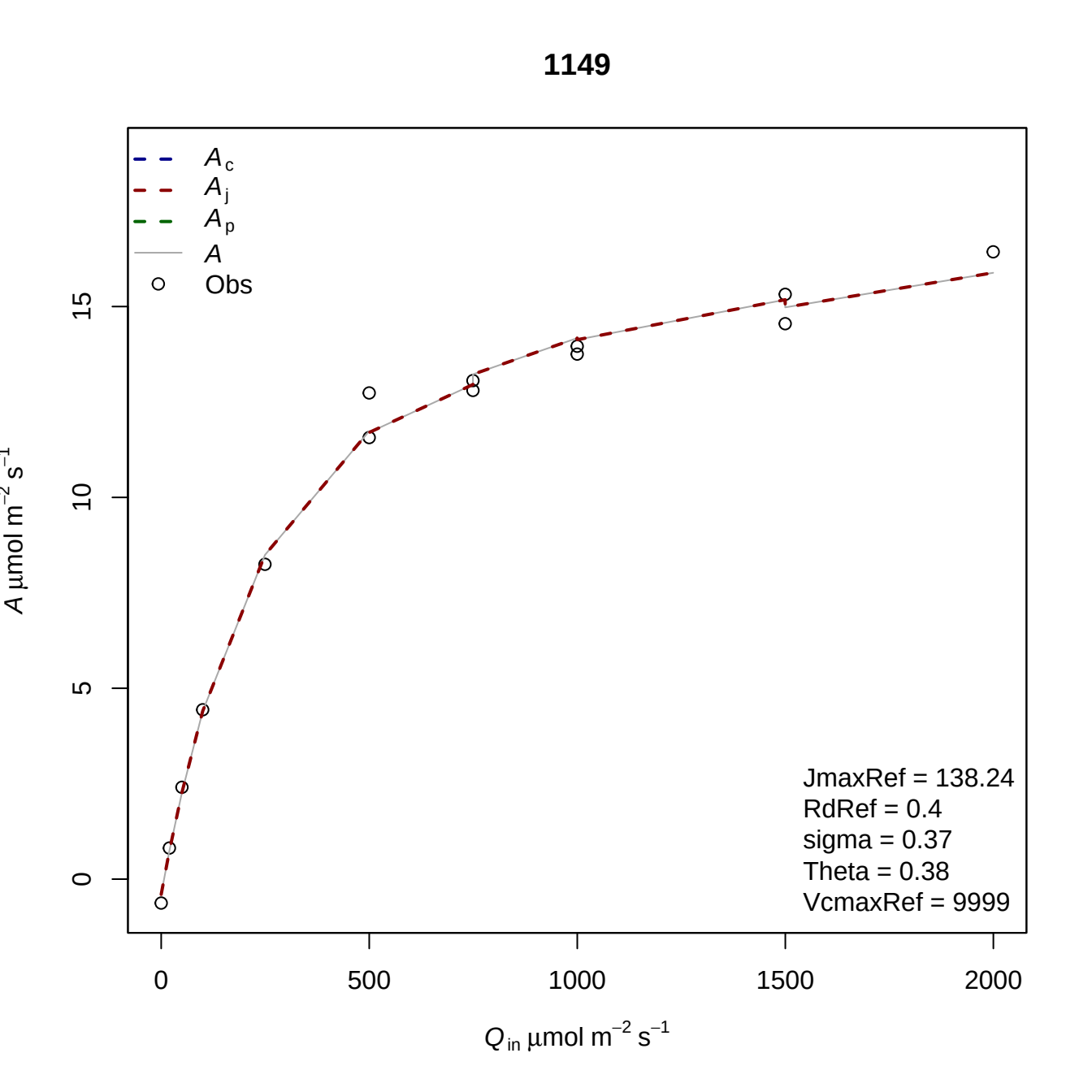

1144
A_c
A_j
A_p
A
Obs
JmaxRef = 70.8
RdRef = 1.15
sigma = 1.44
Theta = −0.69
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

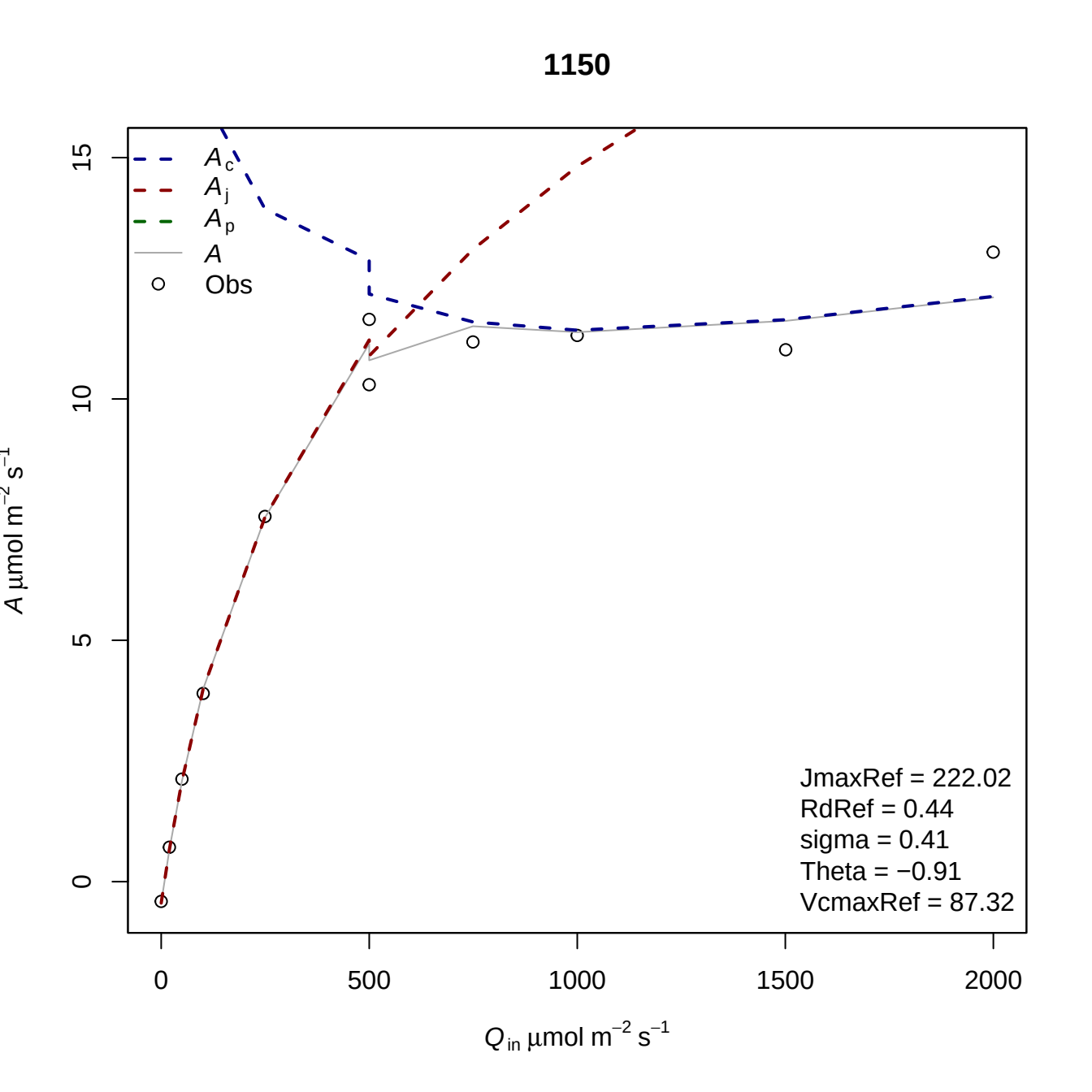

1145
A_c
A_j
A_p
A
Obs
JmaxRef = 178.08
RdRef = 0.83
sigma = 0.31
Theta = 0.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

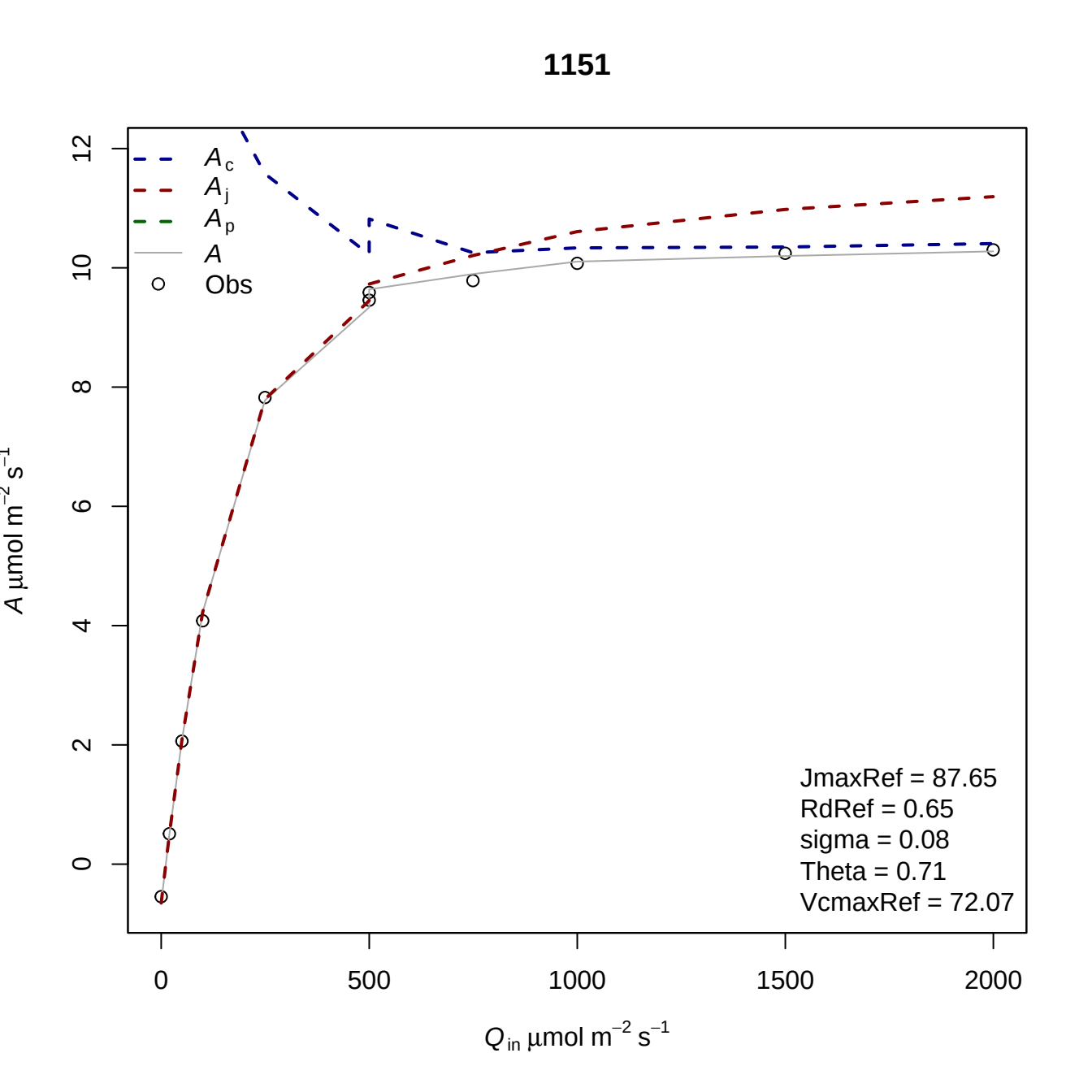

1146
A_c
A_j
A_p
A
Obs
JmaxRef = 106.3
RdRef = 1.02
sigma = 0.18
Theta = 0.45
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

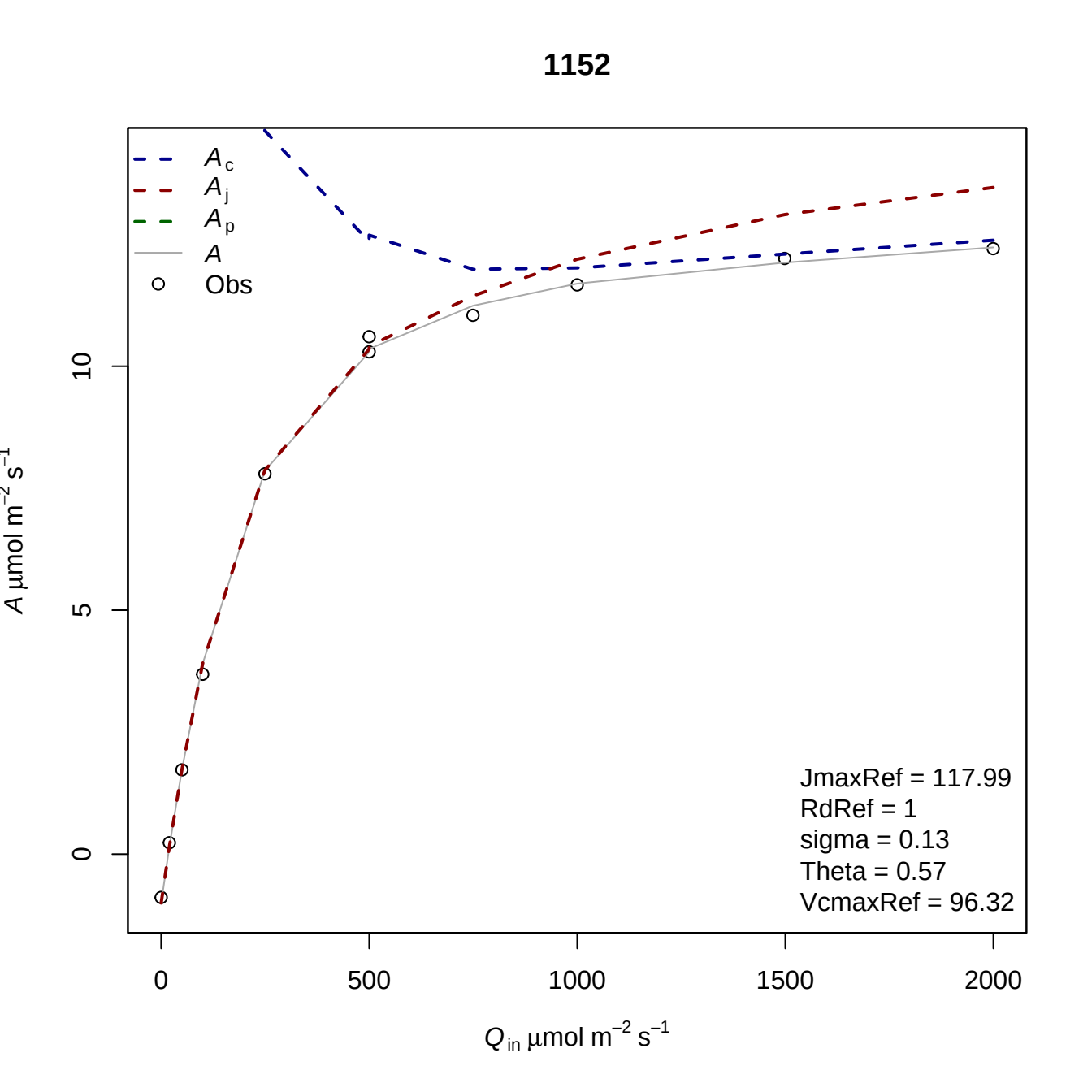

1147
A_c
A_j
A_p
A
Obs
JmaxRef = 115.47
RdRef = 0.94
sigma = 0.15
Theta = 0.59
VcmaxRef = 98.08
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

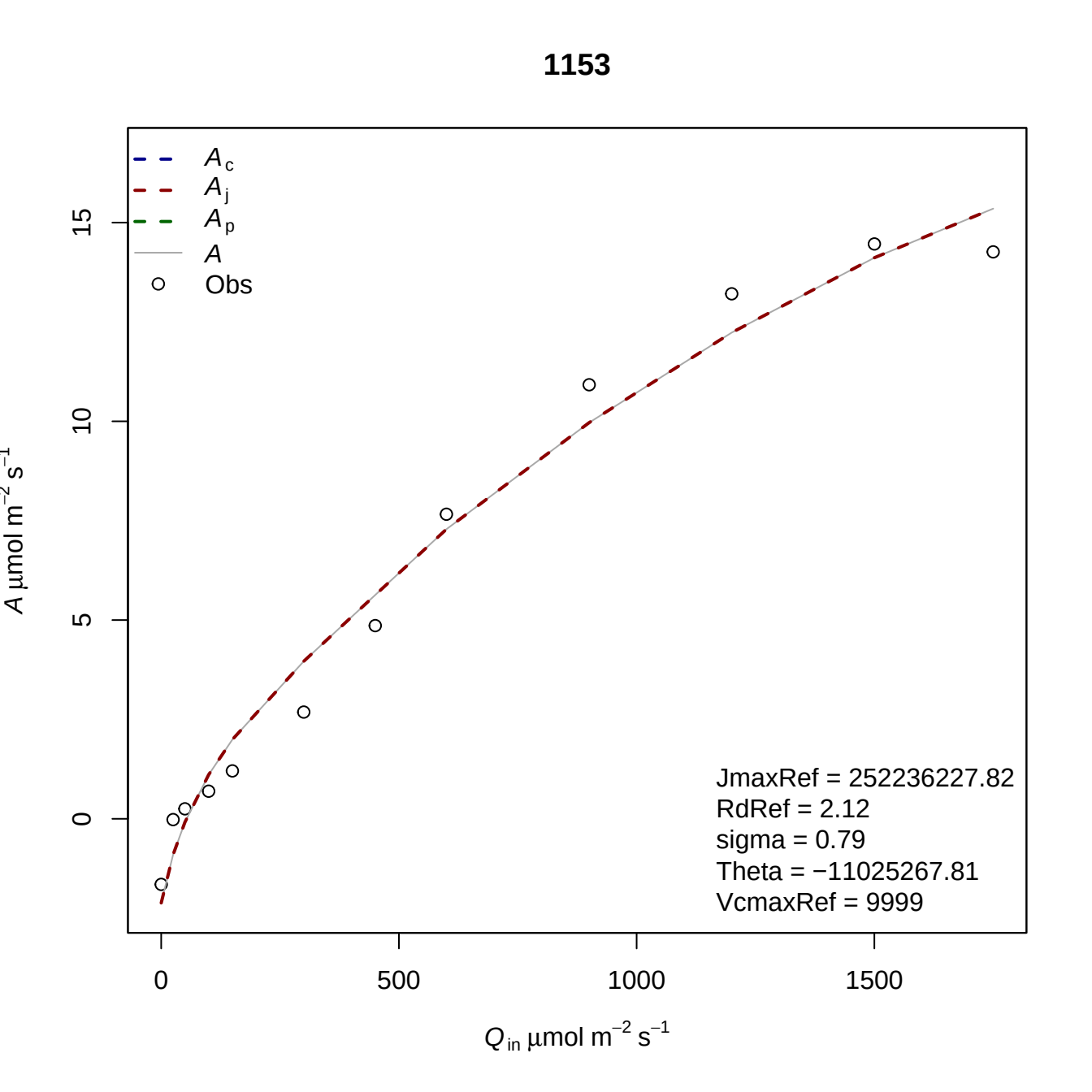

1148
A_c
A_j
A_p
A
Obs
JmaxRef = 138.77
RdRef = 0.85
sigma = 0.33
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

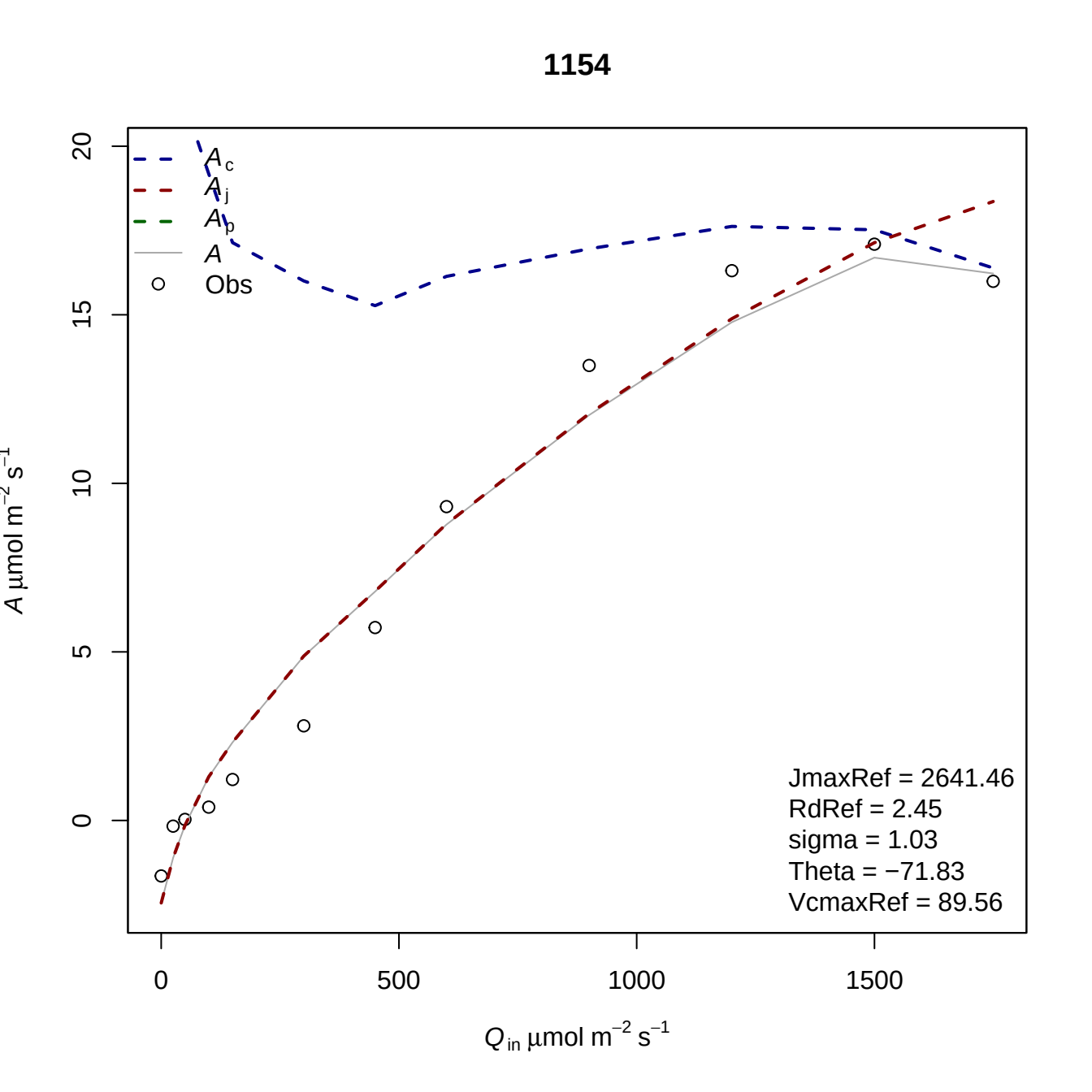

1149
A_c
A_j
A_p
A
Obs
JmaxRef = 138.24
RdRef = 0.4
sigma = 0.37
Theta = 0.38
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

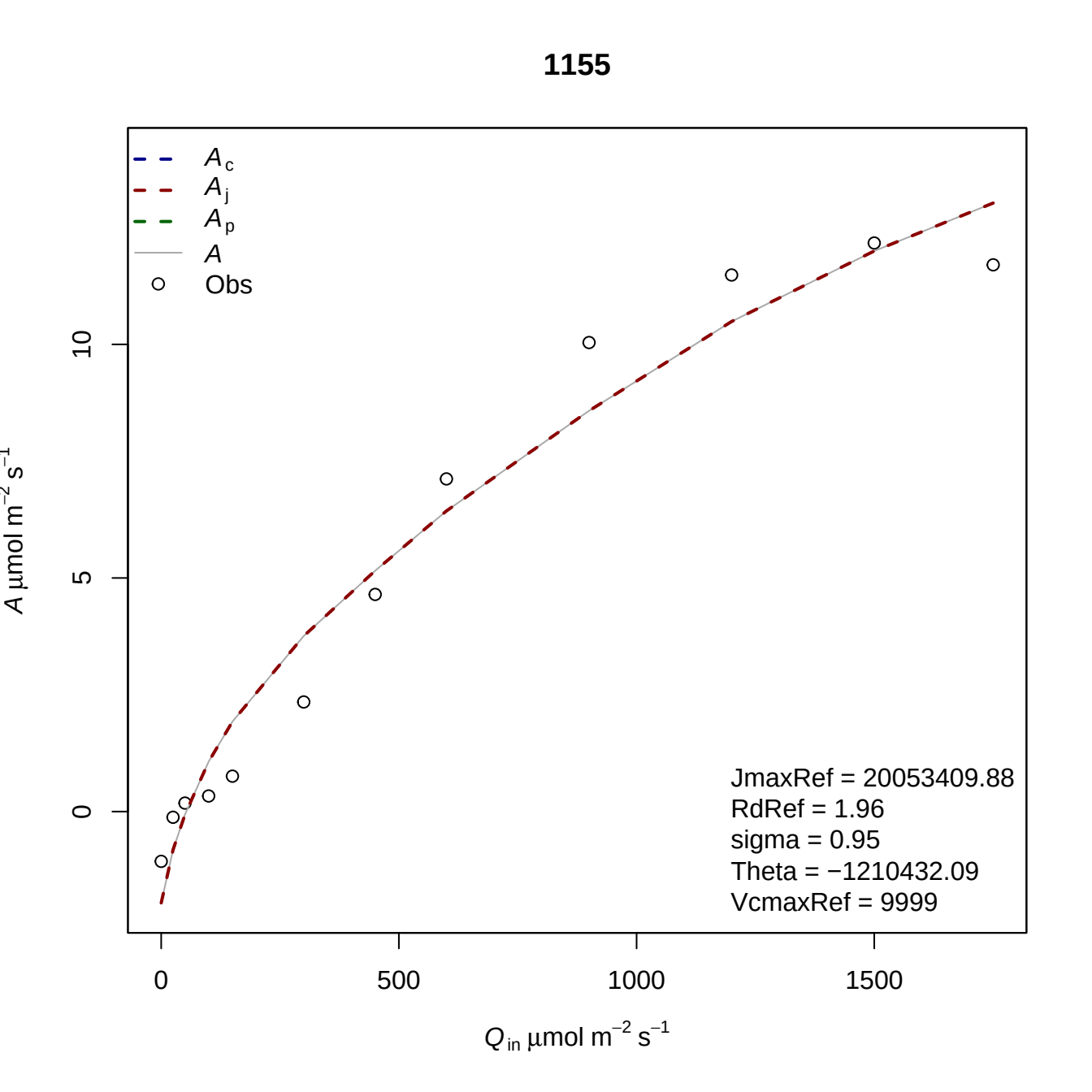

1150
A_c
A_j
A_p
A
Obs
JmaxRef = 222.02
RdRef = 0.44
sigma = 0.41
Theta = −0.91
VcmaxRef = 87.32
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

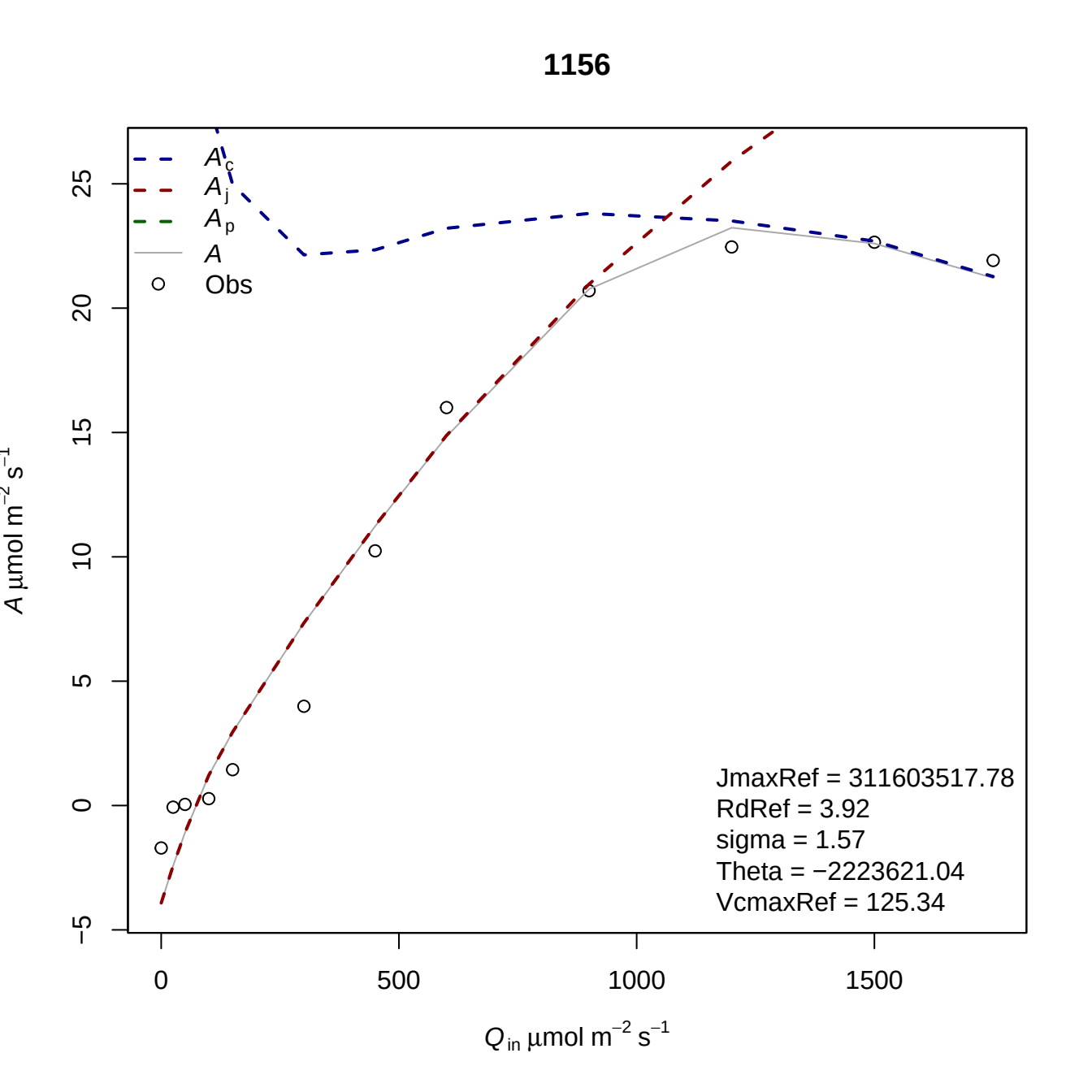

1151
A_c
A_j
A_p
A
Obs
JmaxRef = 87.65
RdRef = 0.65
sigma = 0.08
Theta = 0.71
VcmaxRef = 72.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

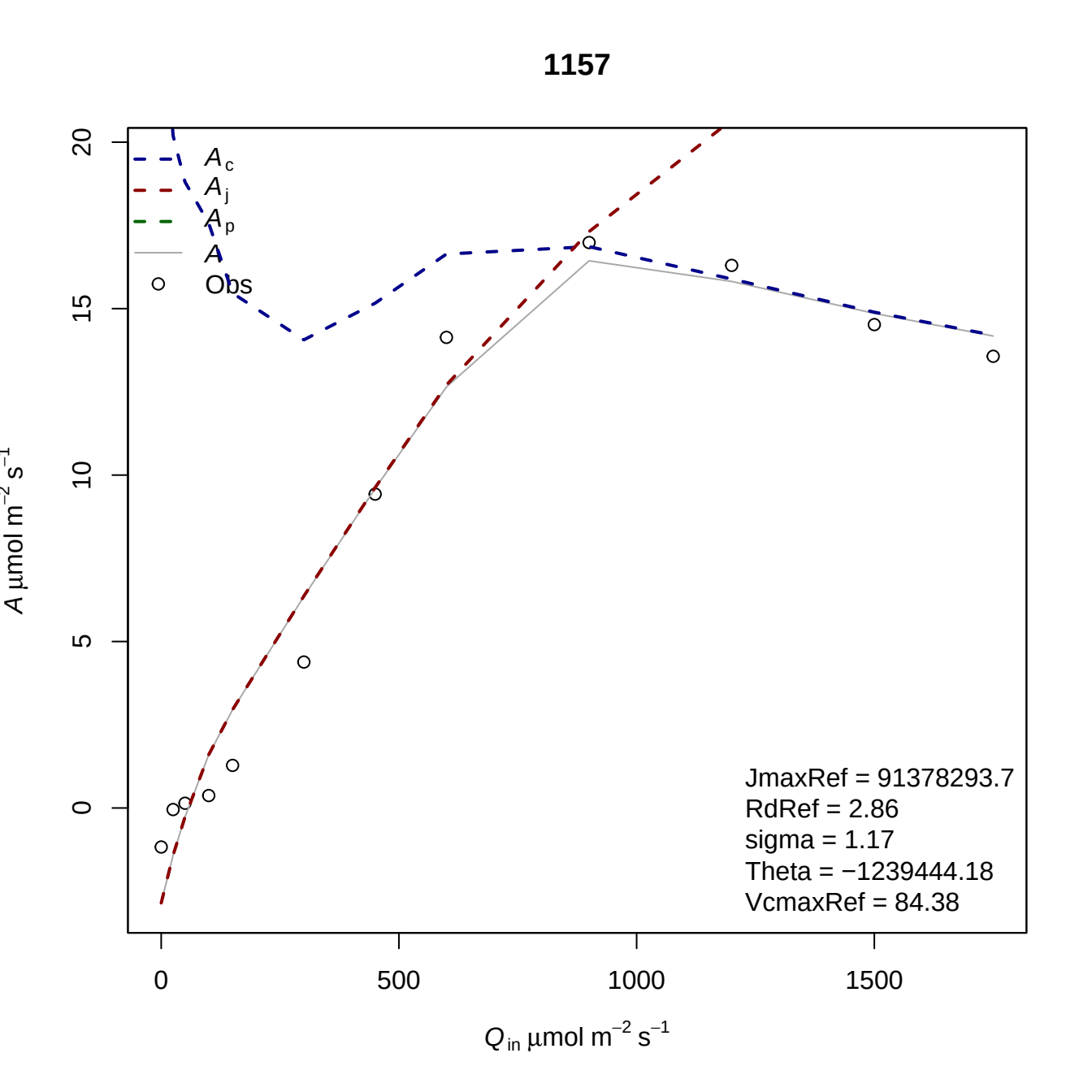

1152
A_c
A_j
A_p
A
Obs
JmaxRef = 117.99
RdRef = 1
sigma = 0.13
Theta = 0.57
VcmaxRef = 96.32
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

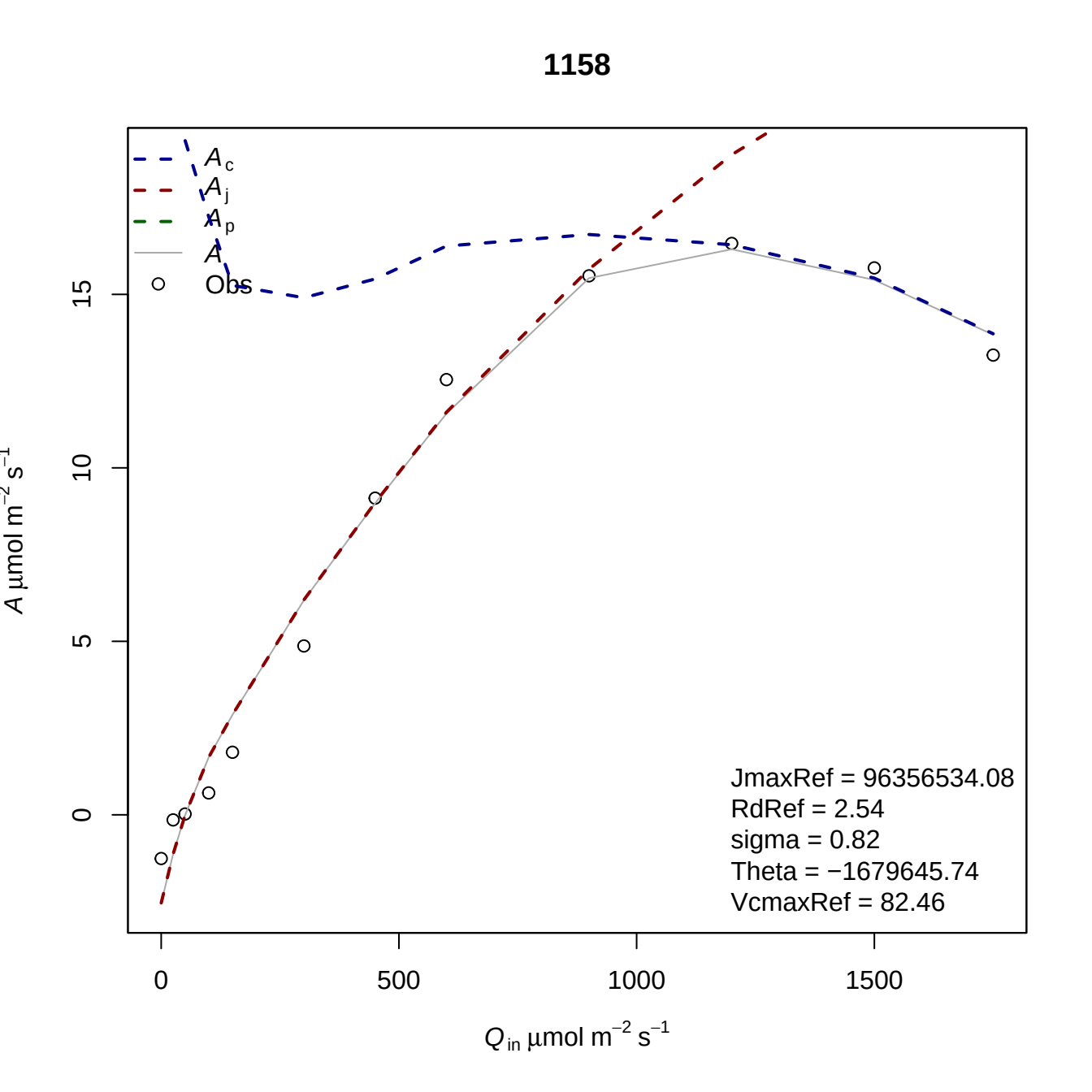

1153
A_c
A_j
A_p
A
Obs
JmaxRef = 252236227.82
RdRef = 2.12
sigma = 0.79
Theta = −11025267.81
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

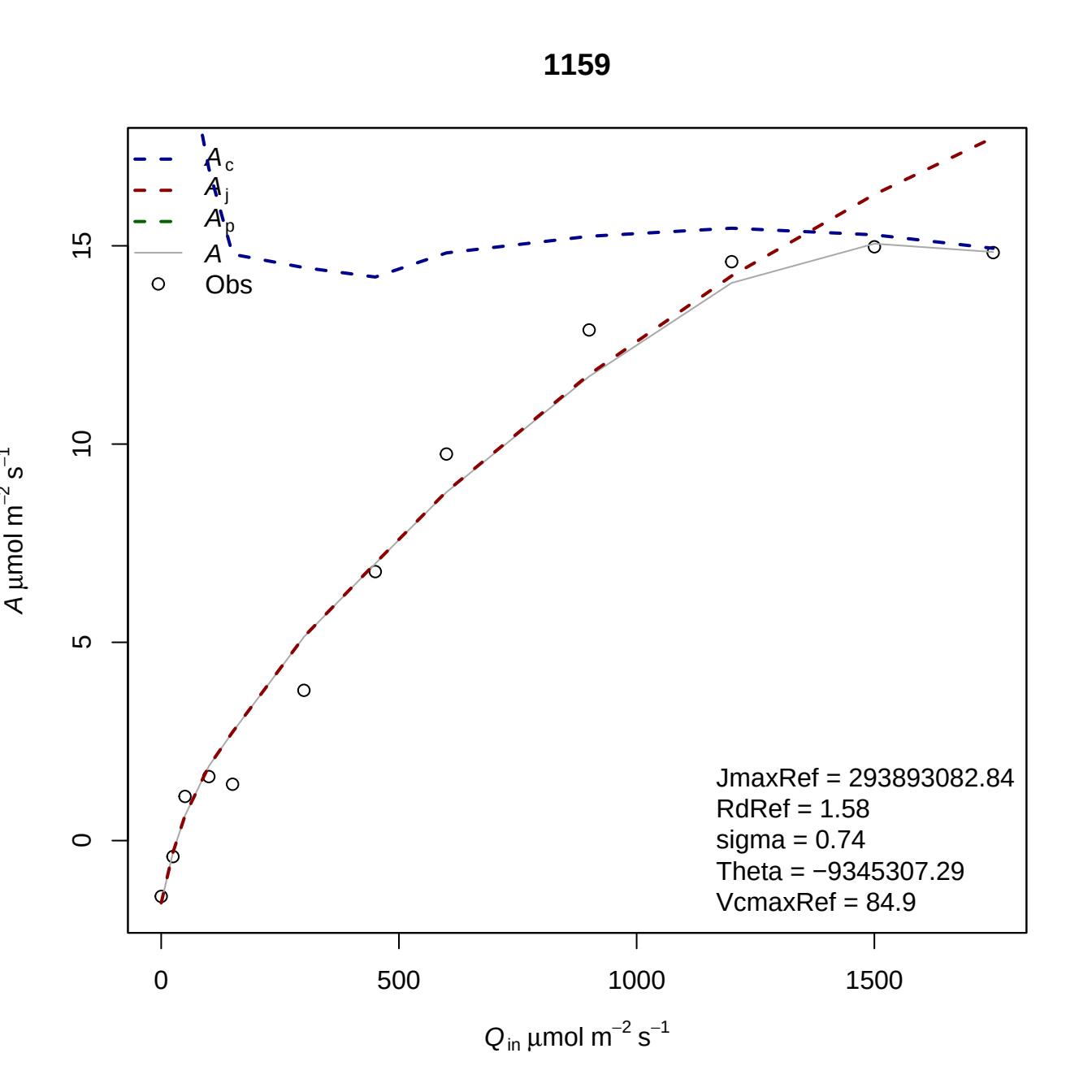

1154
A_c
A_j
A_p
A
Obs
JmaxRef = 2641.46
RdRef = 2.45
sigma = 1.03
Theta = −71.83
VcmaxRef = 89.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

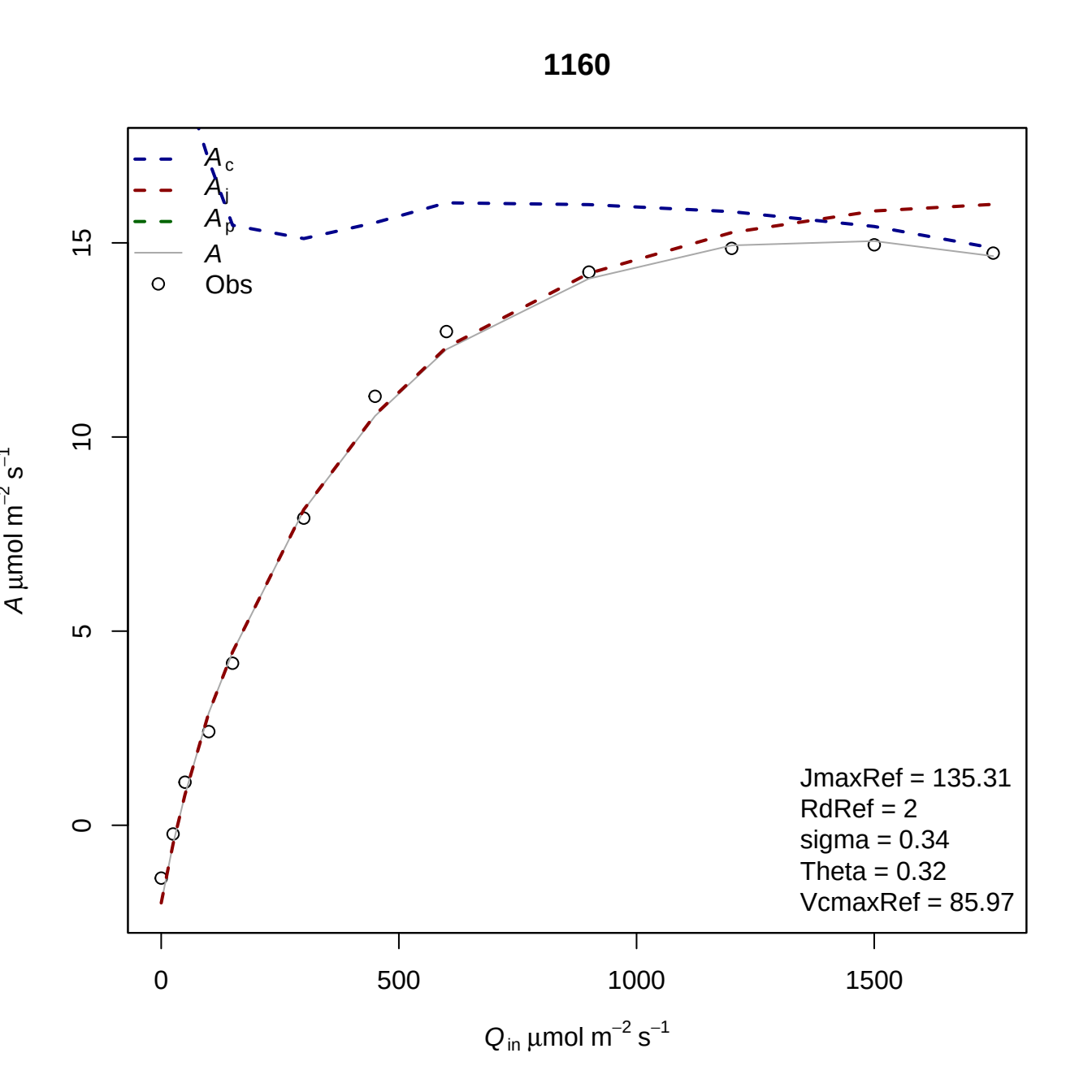

1155
A_c
A_j
A_p
A
Obs
JmaxRef = 20053409.88
RdRef = 1.96
sigma = 0.95
Theta = −1210432.09
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

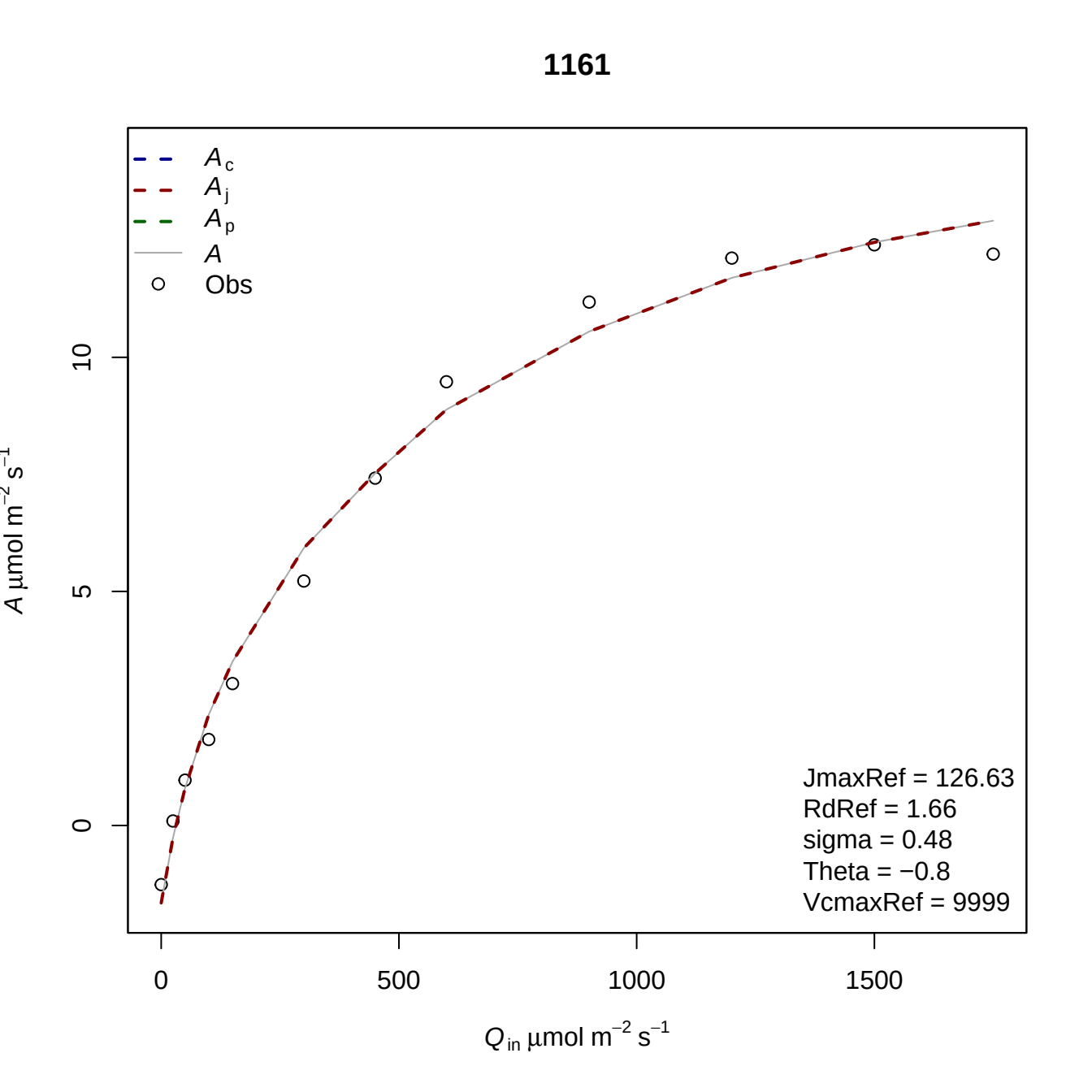

1156
A_c
A_j
A_p
A
Obs
JmaxRef = 311603517.78
RdRef = 3.92
sigma = 1.57
Theta = −2223621.04
VcmaxRef = 125.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

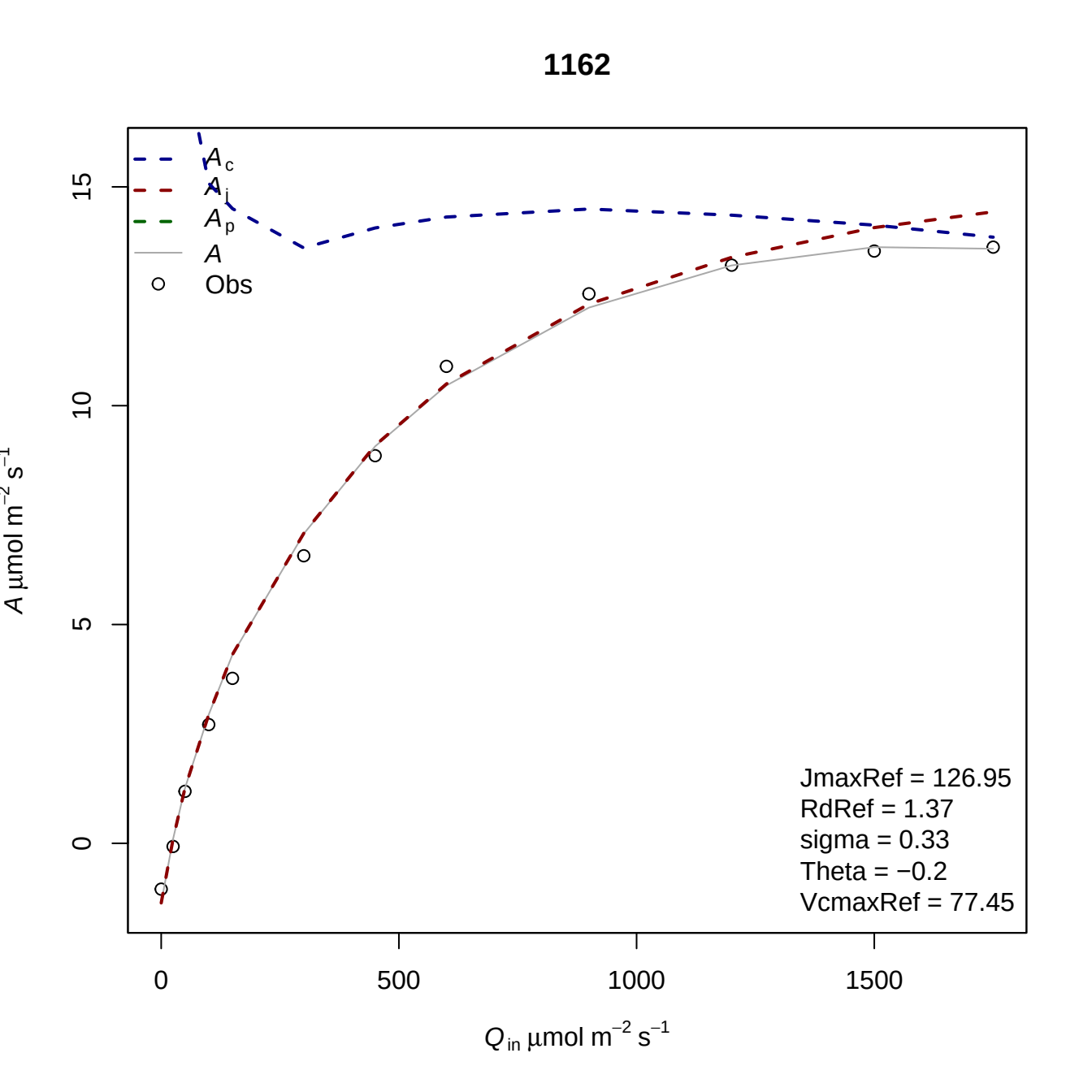

1157
A_c
A_j
A_p
A
Obs
JmaxRef = 91378293.7
RdRef = 2.86
sigma = 1.17
Theta = −1239444.18
VcmaxRef = 84.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

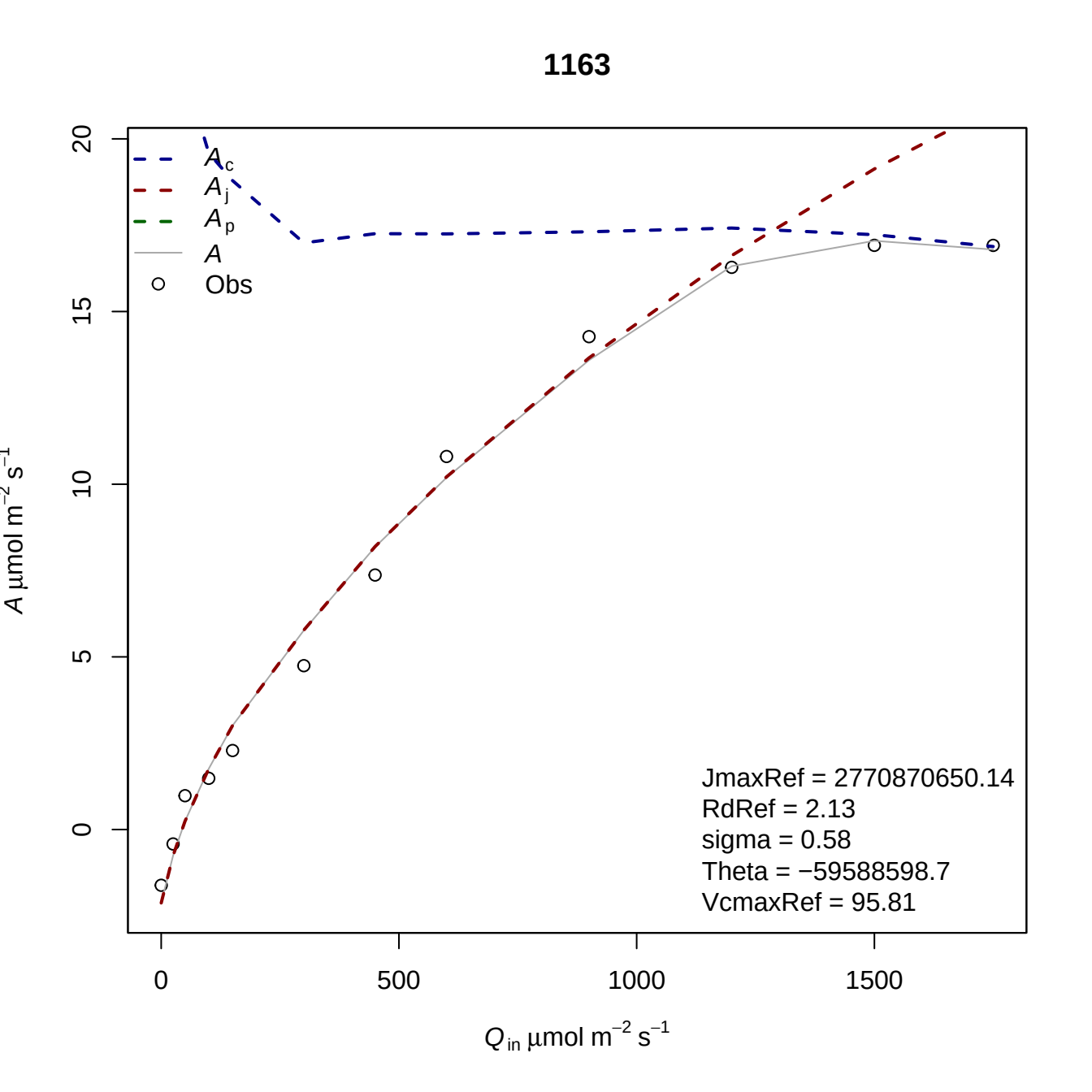

1158
A_c
A_j
A_p
A
Obs
JmaxRef = 96356534.08
RdRef = 2.54
sigma = 0.82
Theta = −1679645.74
VcmaxRef = 82.46
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

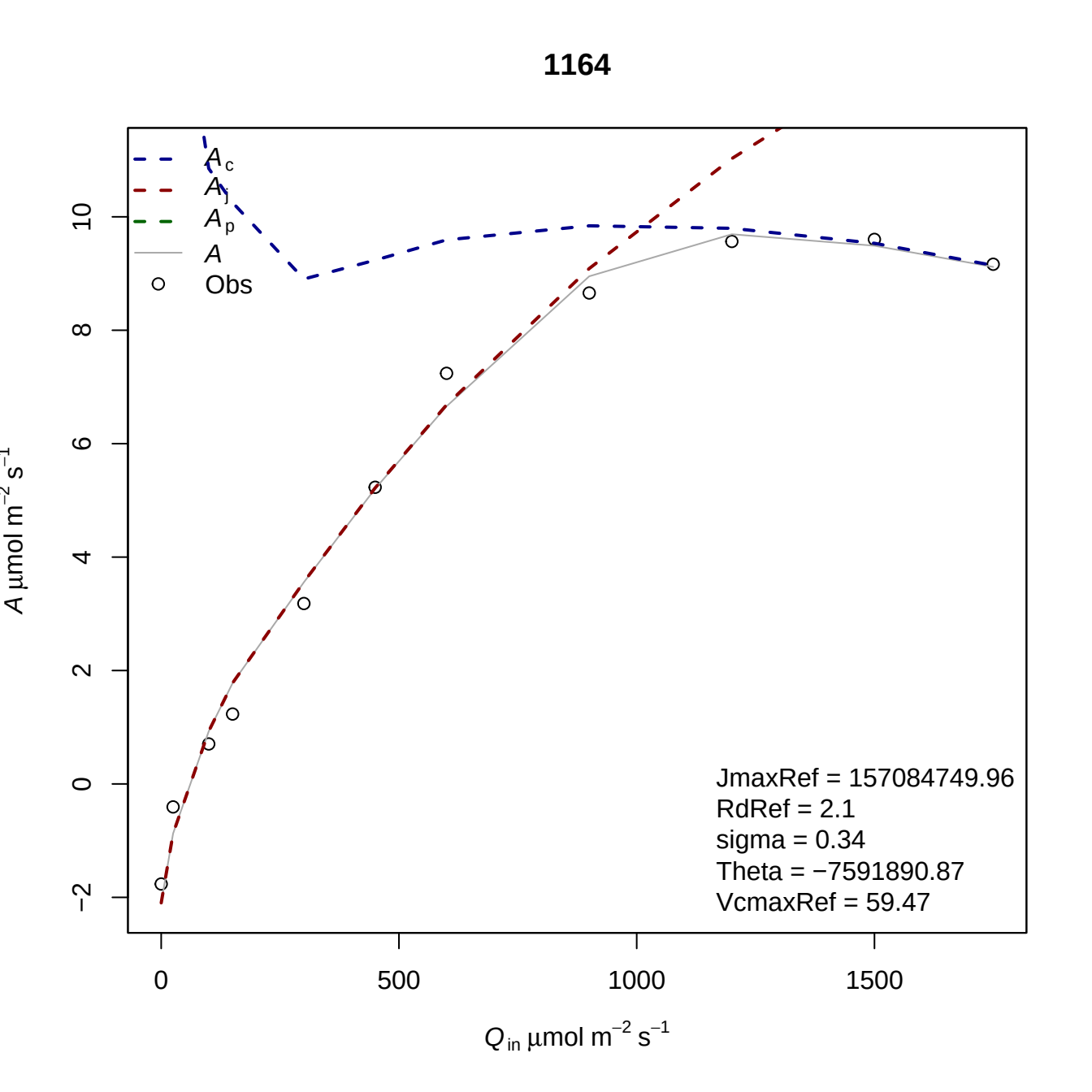

1159
A_c
A_j
A_p
A
Obs
JmaxRef = 293893082.84
RdRef = 1.58
sigma = 0.74
Theta = −9345307.29
VcmaxRef = 84.9
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

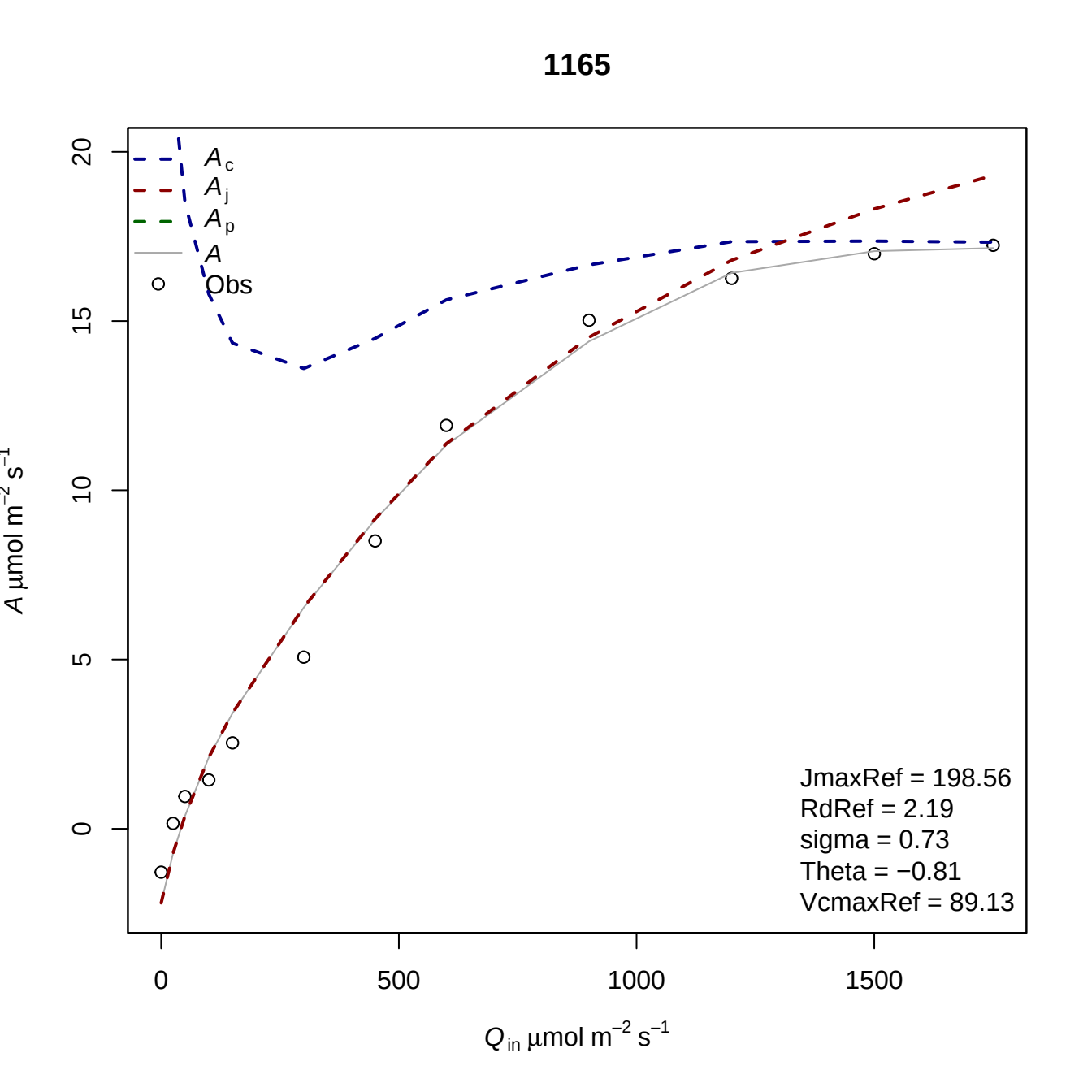

1160
A_c
A_j
A_p
A
Obs
JmaxRef = 135.31
RdRef = 2
sigma = 0.34
Theta = 0.32
VcmaxRef = 85.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

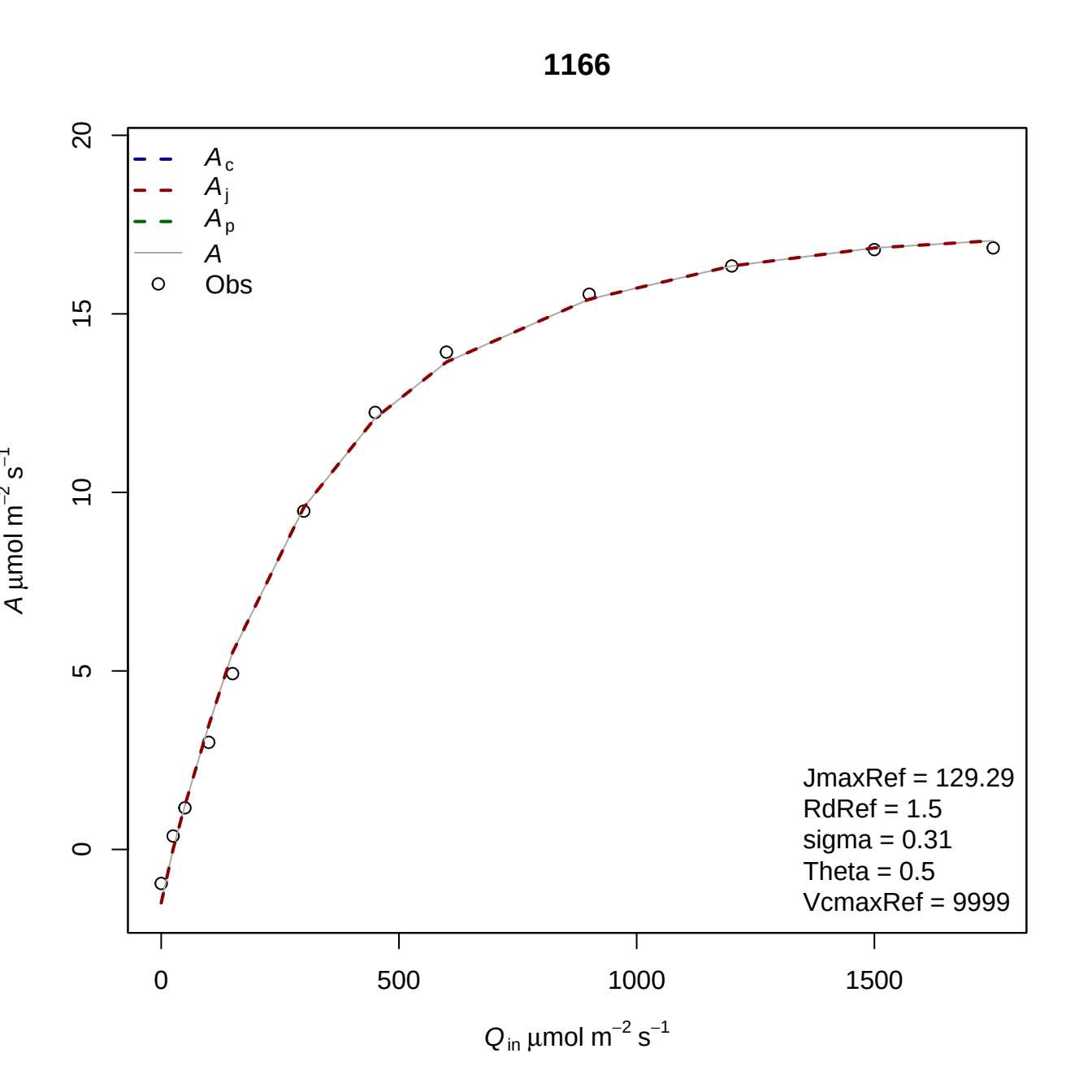

1161
A_c
A_j
A_p
A
Obs
JmaxRef = 126.63
RdRef = 1.66
sigma = 0.48
Theta = −0.8
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

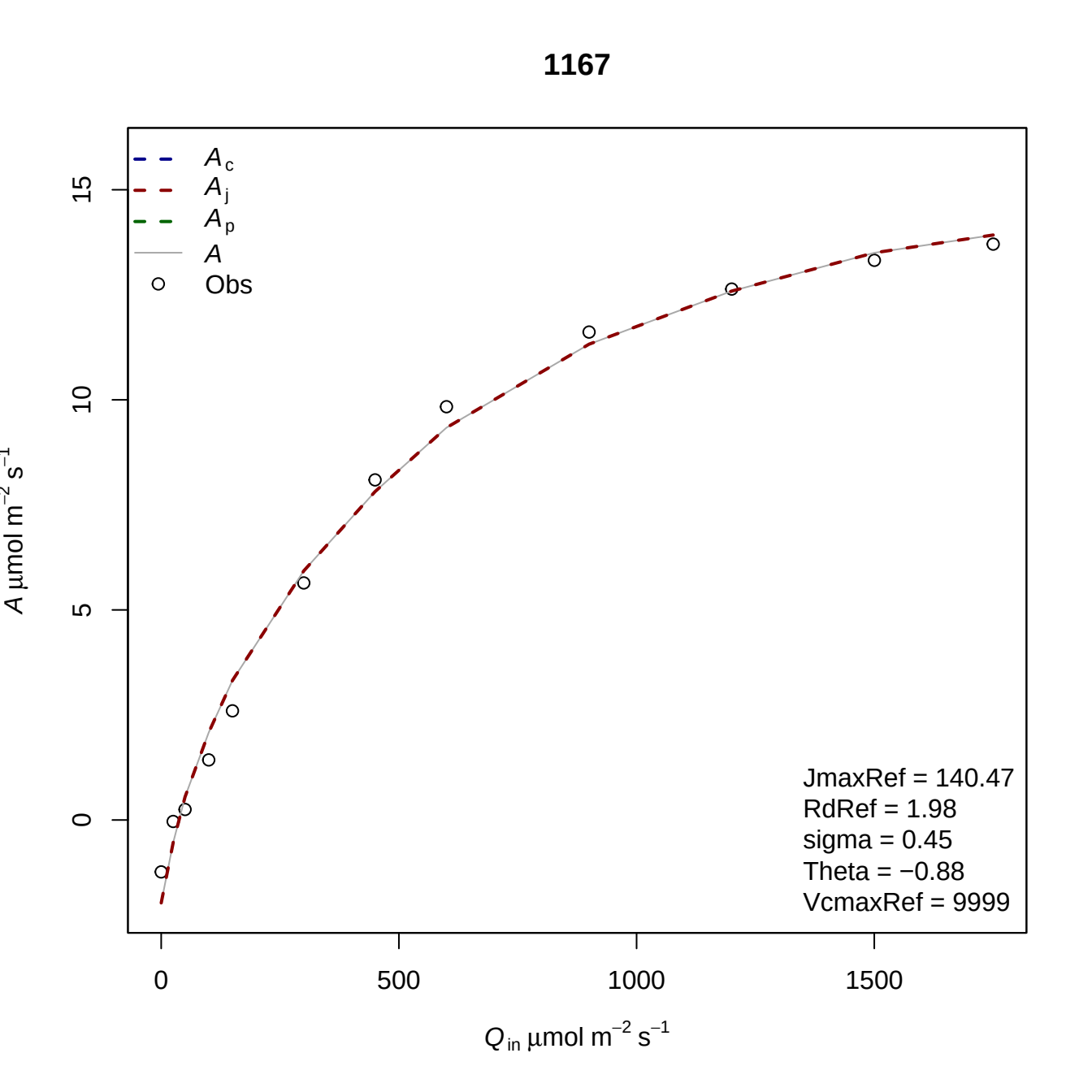

1162
A_c
A_j
A_p
A
Obs
JmaxRef = 126.95
RdRef = 1.37
sigma = 0.33
Theta = −0.2
VcmaxRef = 77.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

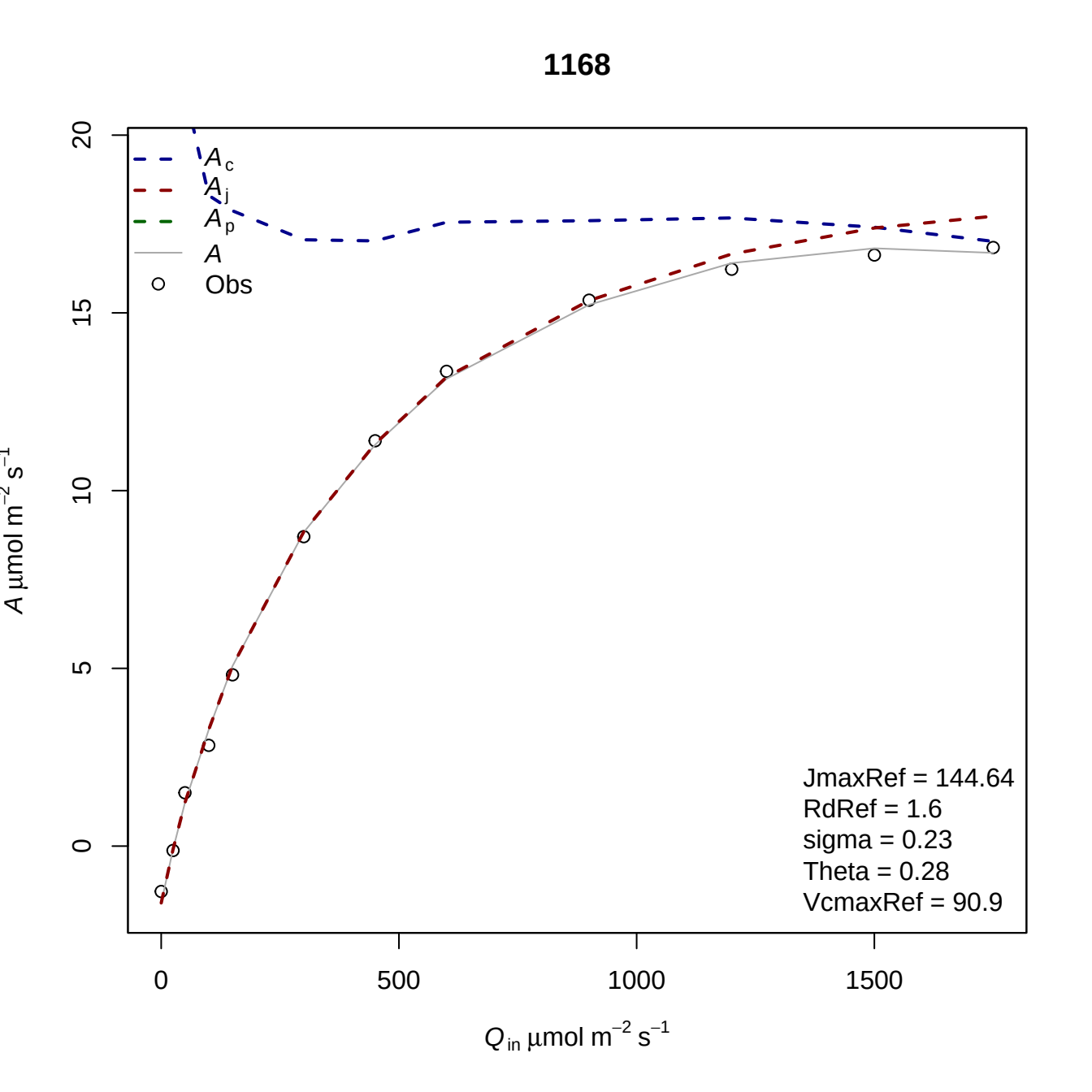

1163
A_c
A_j
A_p
A
Obs
JmaxRef = 2770870650.14
RdRef = 2.13
sigma = 0.58
Theta = −59588598.7
VcmaxRef = 95.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

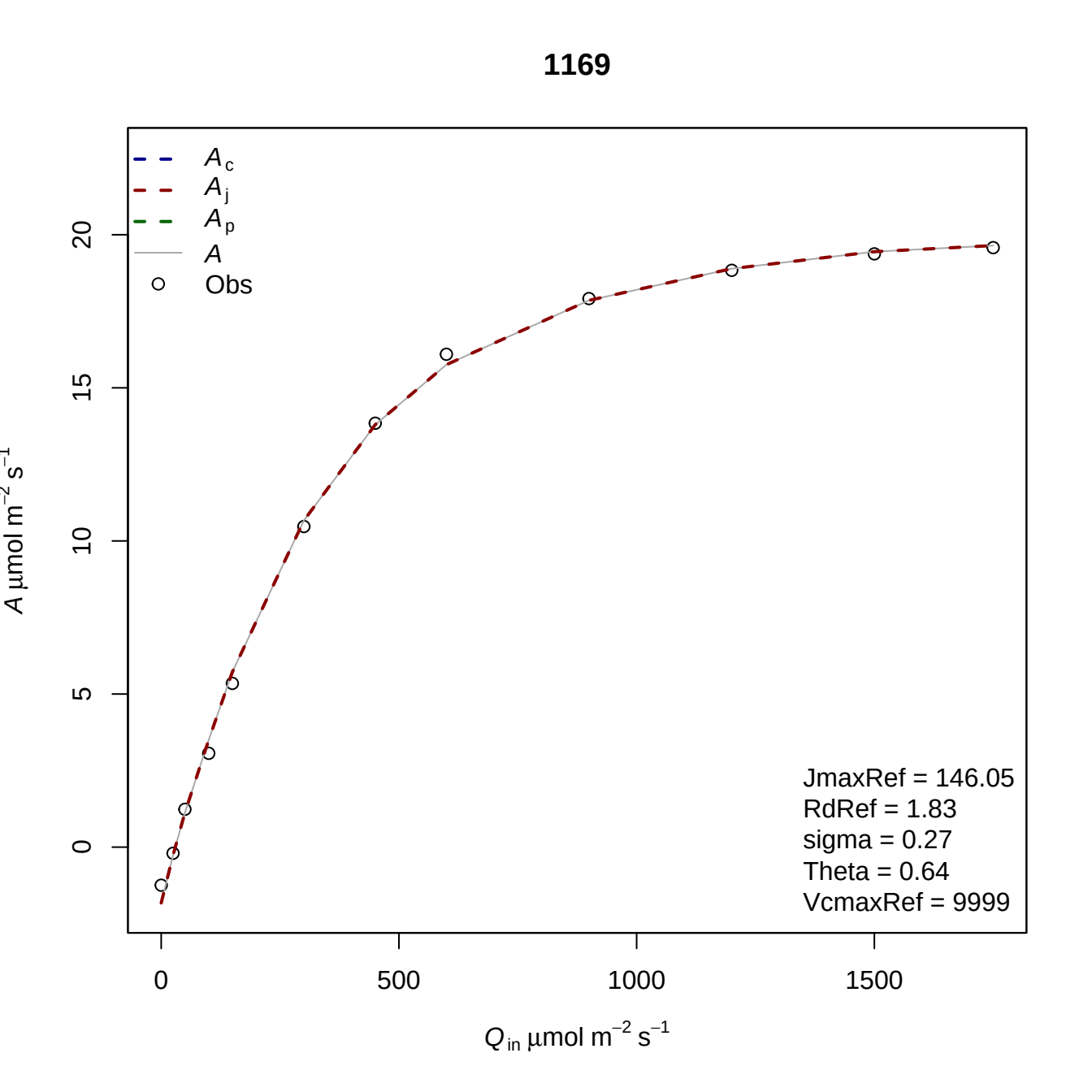

1164
A_c
A_j
A_p
A
Obs
JmaxRef = 157084749.96
RdRef = 2.1
sigma = 0.34
Theta = −7591890.87
VcmaxRef = 59.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

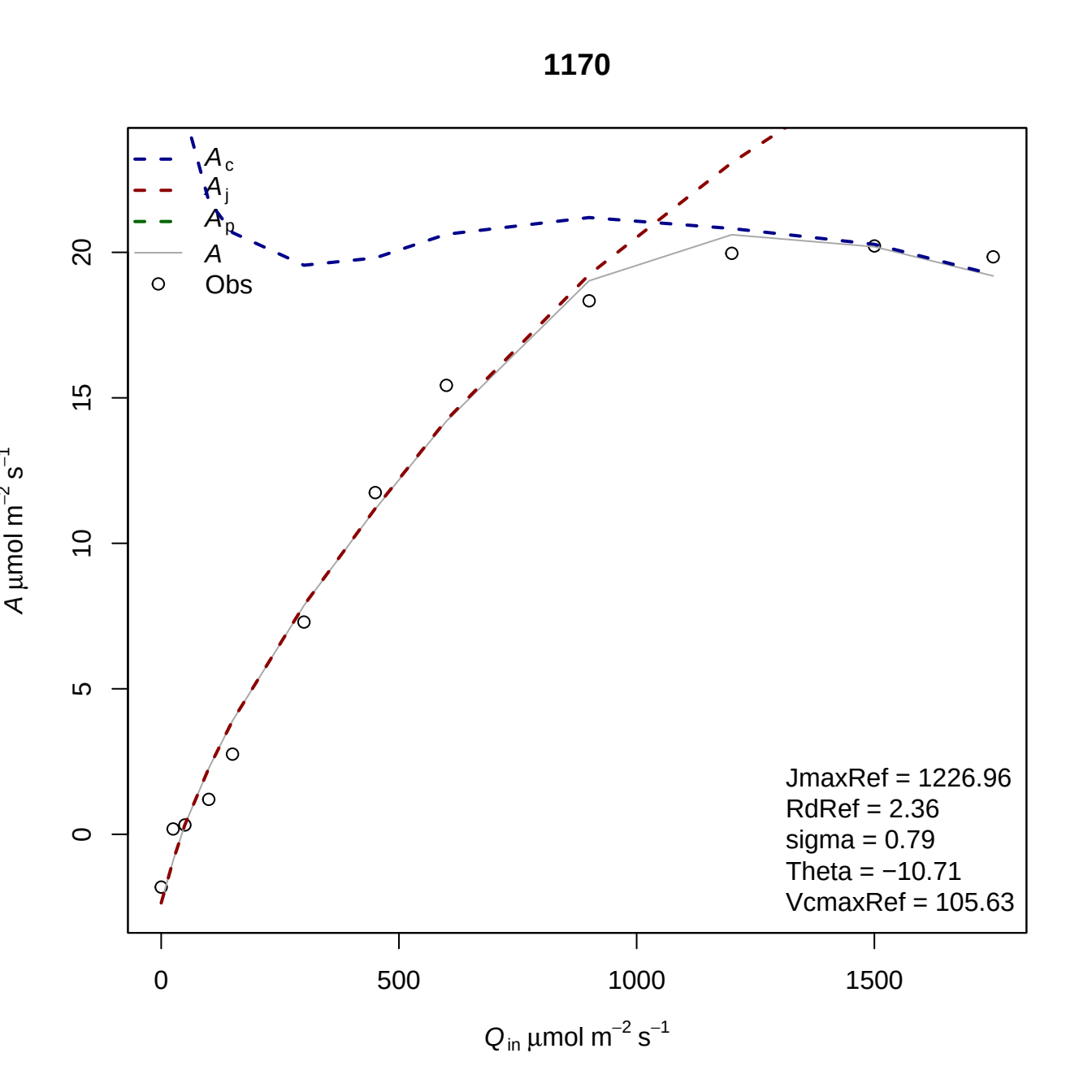

1165
A_c
A_j
A_p
A
Obs
JmaxRef = 198.56
RdRef = 2.19
sigma = 0.73
Theta = −0.81
VcmaxRef = 89.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

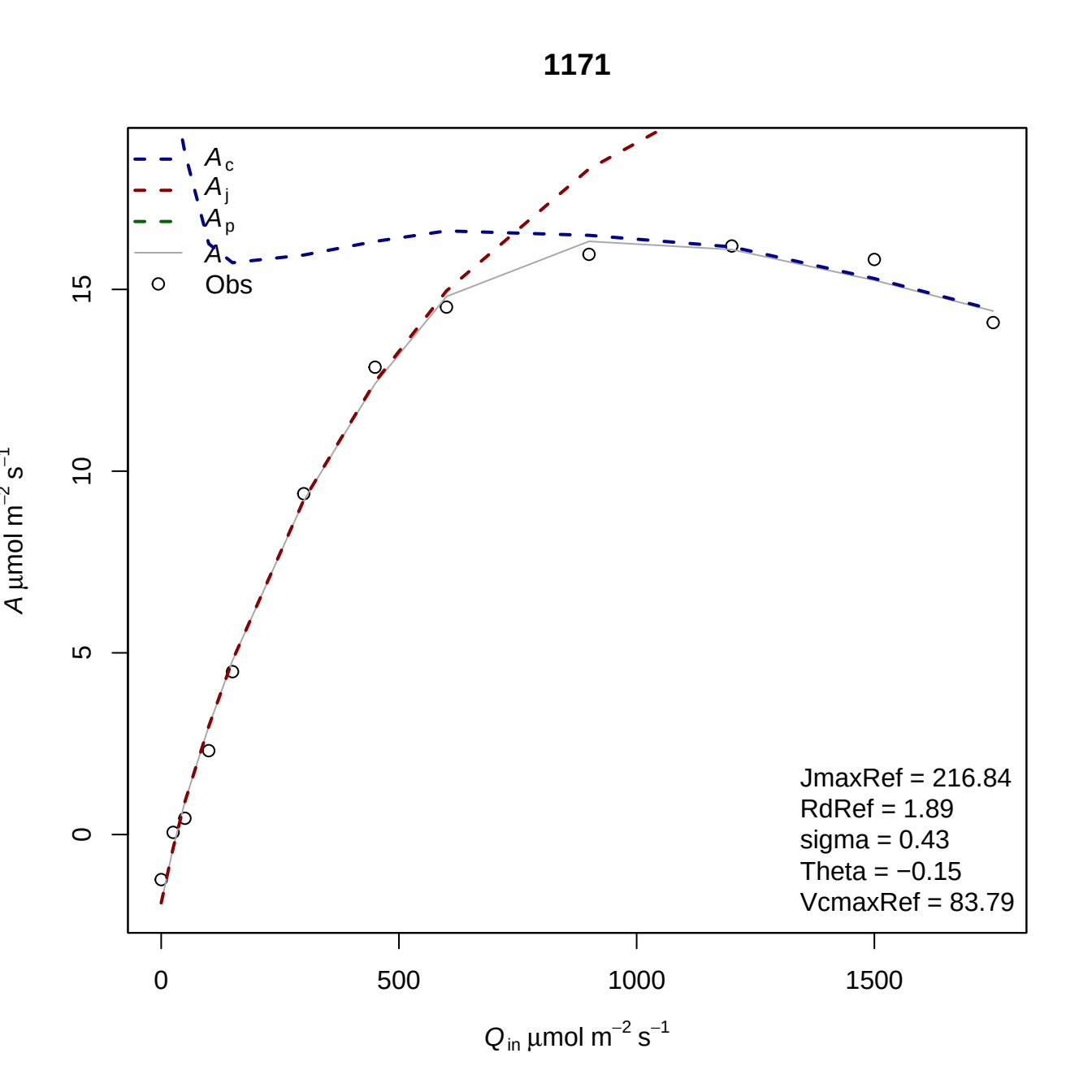

1166
A_c
A_j
A_p
A
Obs
JmaxRef = 129.29
RdRef = 1.5
sigma = 0.31
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

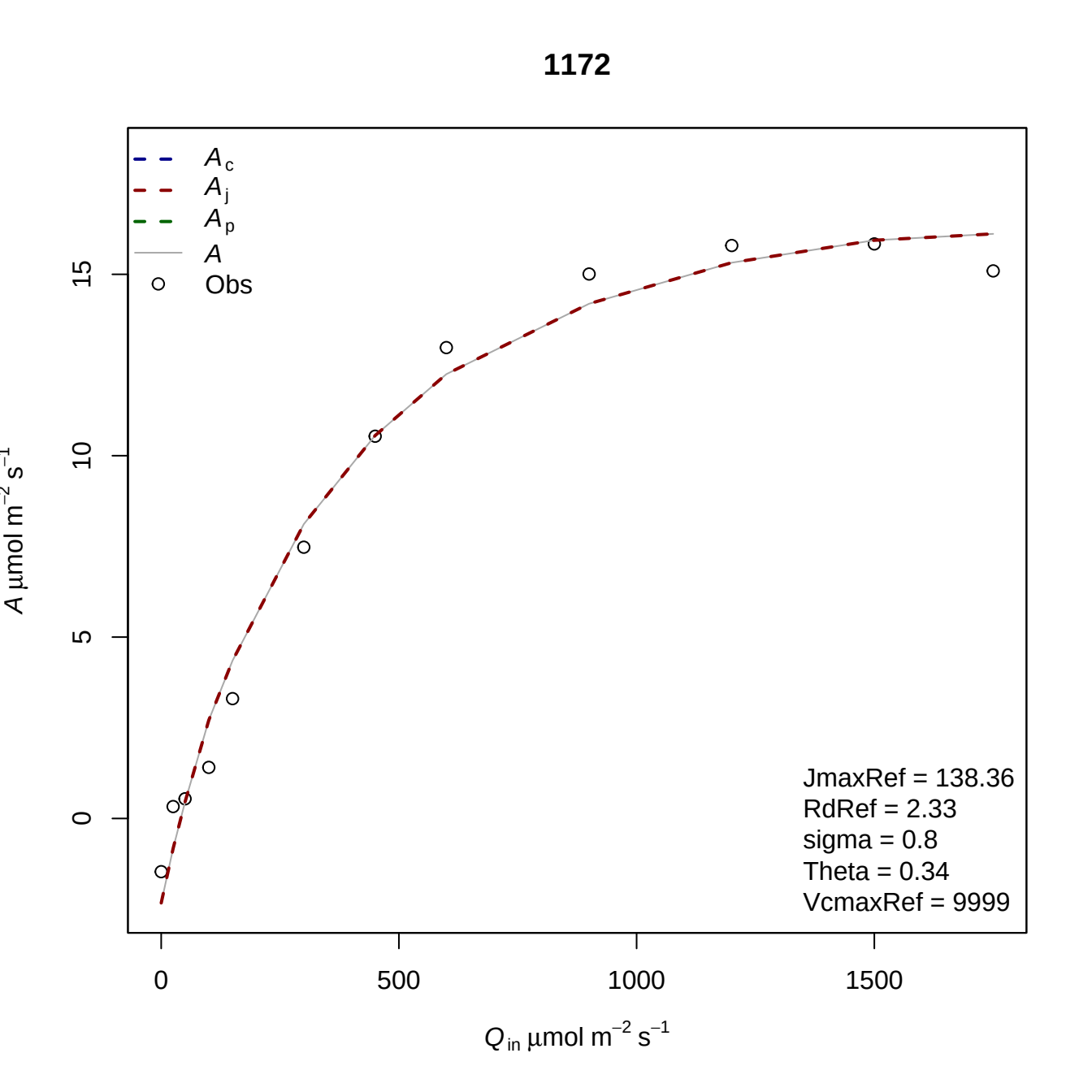

1167
A_c
A_j
A_p
A
Obs
JmaxRef = 140.47
RdRef = 1.98
sigma = 0.45
Theta = −0.88
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

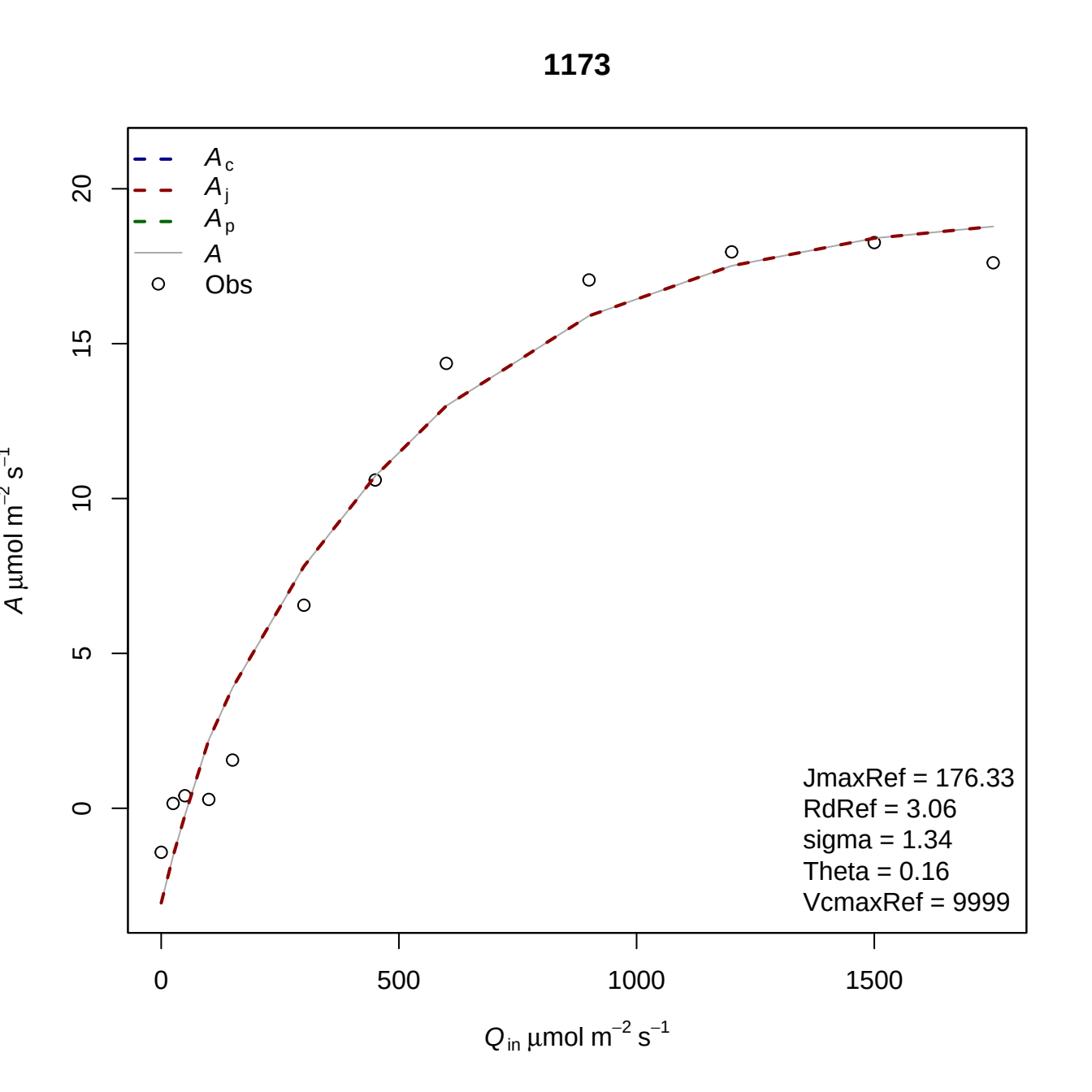

1168
A_c
A_j
A_p
A
Obs
JmaxRef = 144.64
RdRef = 1.6
sigma = 0.23
Theta = 0.28
VcmaxRef = 90.9
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

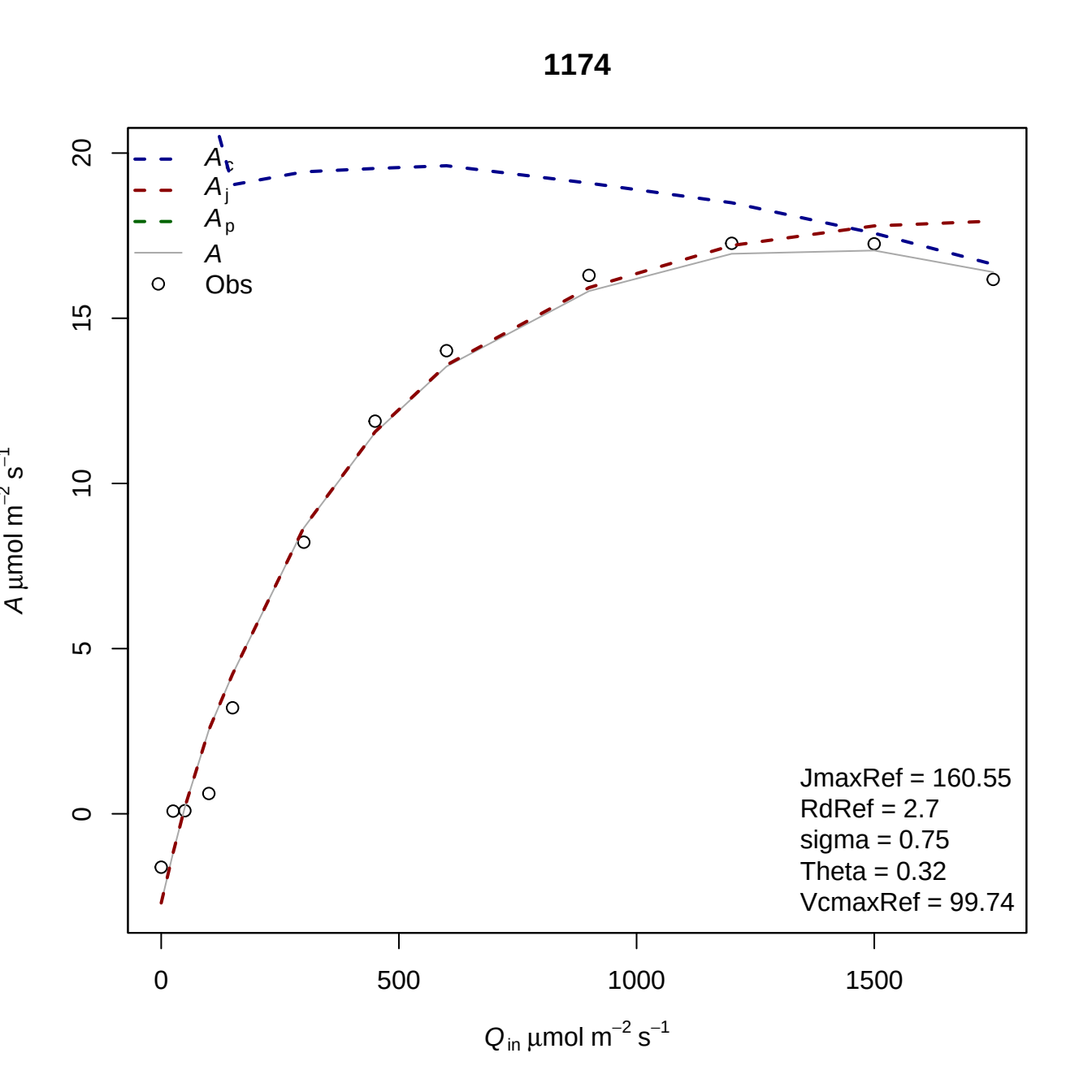

1169
A_c
A_j
A_p
A
Obs
JmaxRef = 146.05
RdRef = 1.83
sigma = 0.27
Theta = 0.64
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

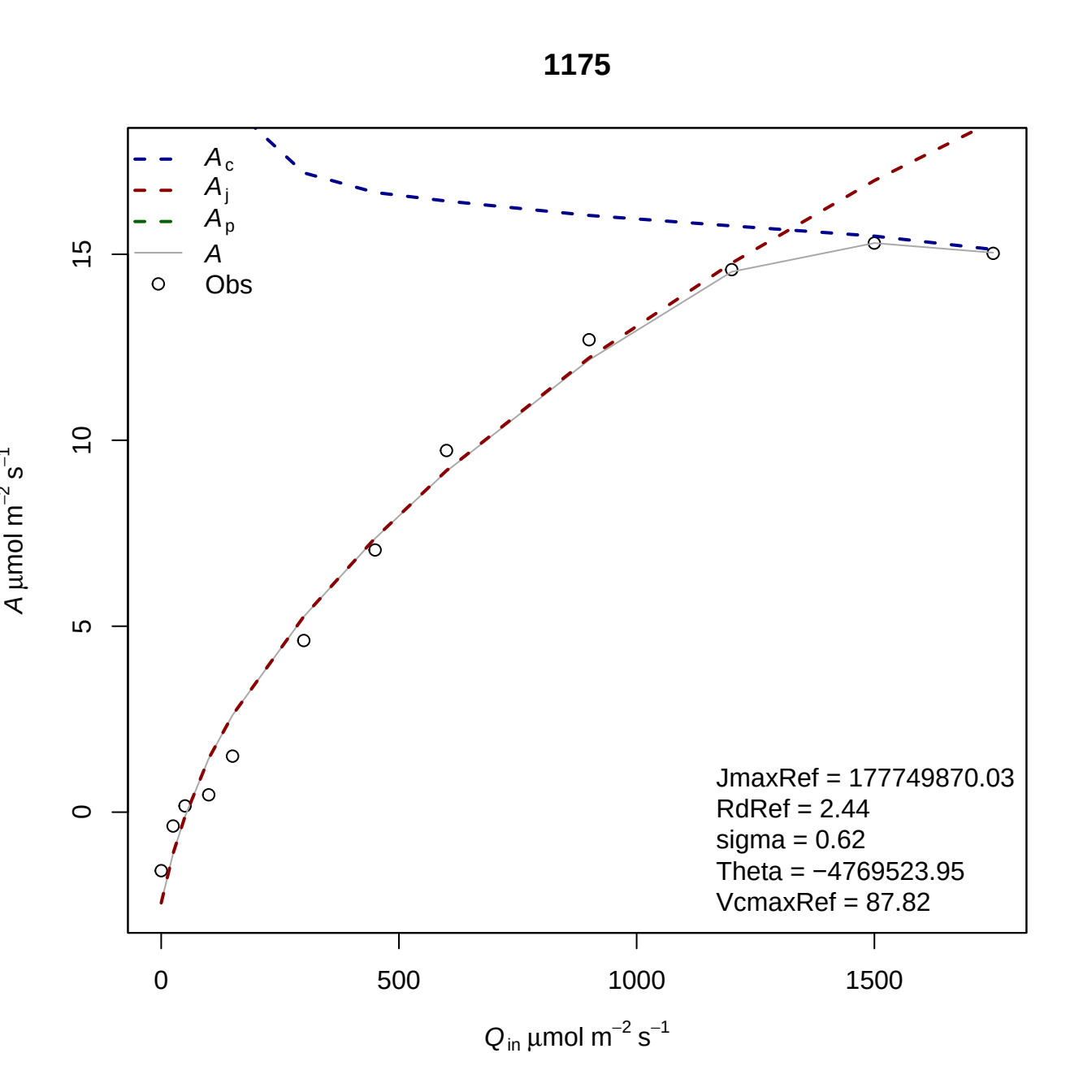

1170
A_c
A_j
A_p
A
Obs
JmaxRef = 1226.96
RdRef = 2.36
sigma = 0.79
Theta = −10.71
VcmaxRef = 105.63
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

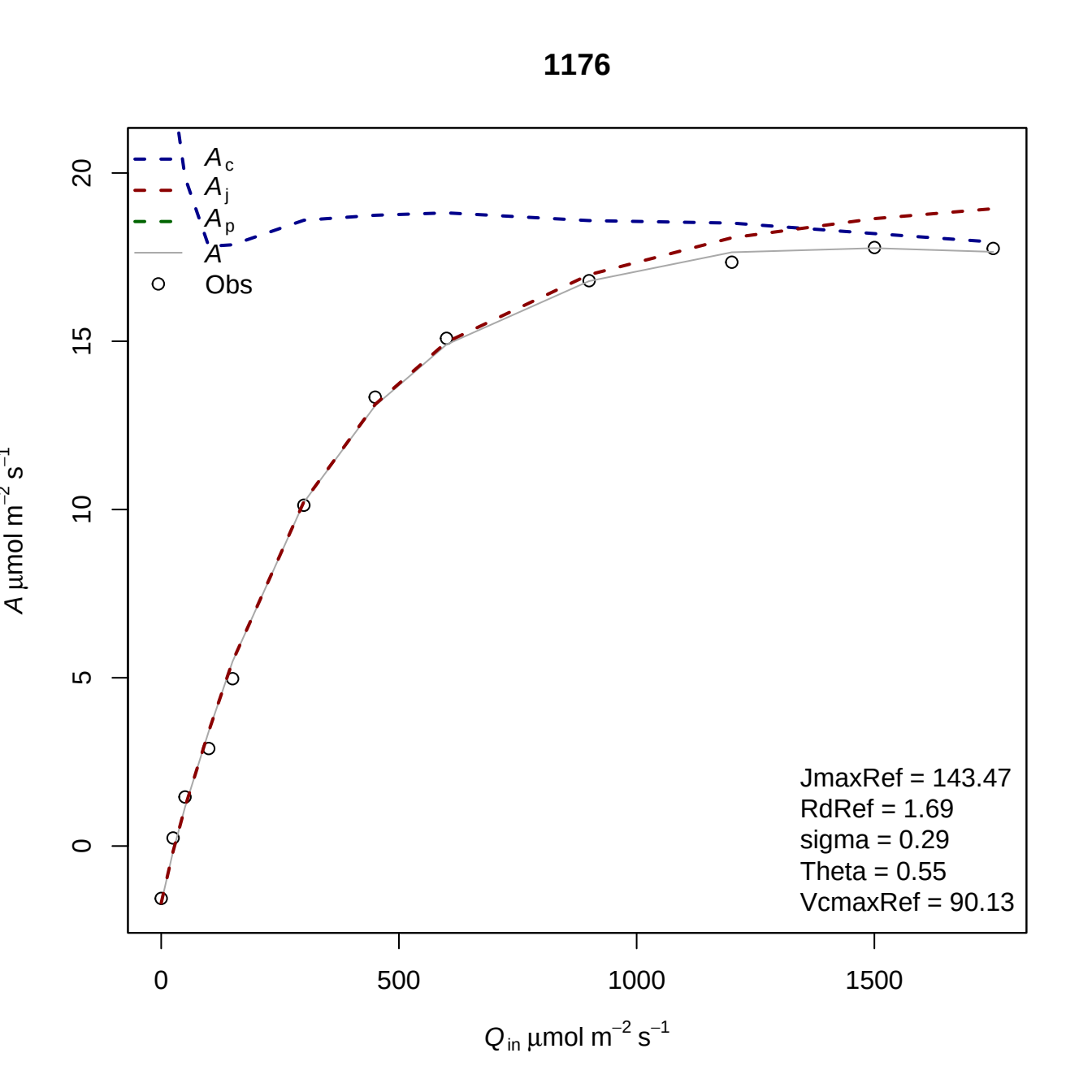

1171
A_c
A_j
A_p
A
Obs
JmaxRef = 216.84
RdRef = 1.89
sigma = 0.43
Theta = −0.15
VcmaxRef = 83.79
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

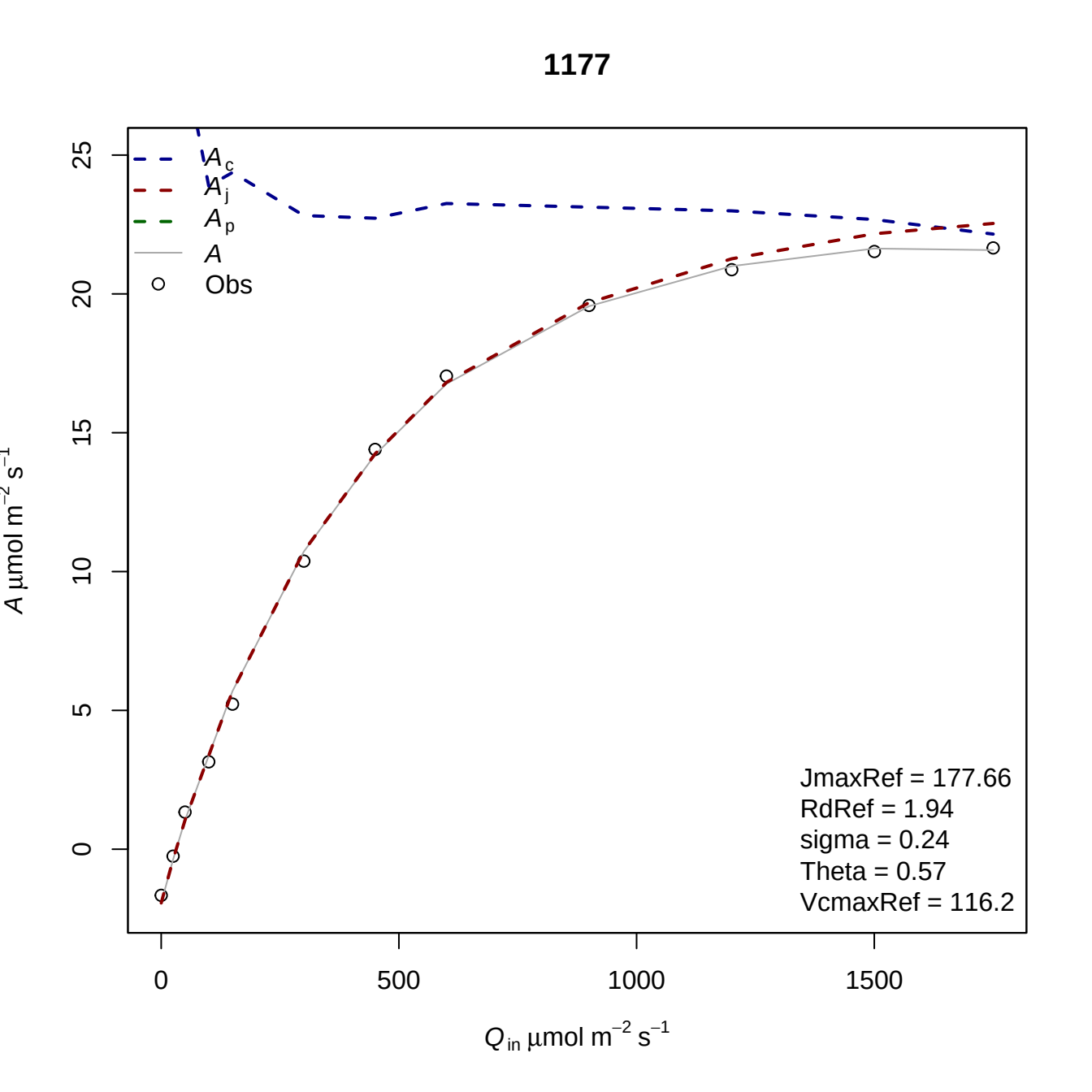

1172
A_c
A_j
A_p
A
Obs
JmaxRef = 138.36
RdRef = 2.33
sigma = 0.8
Theta = 0.34
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

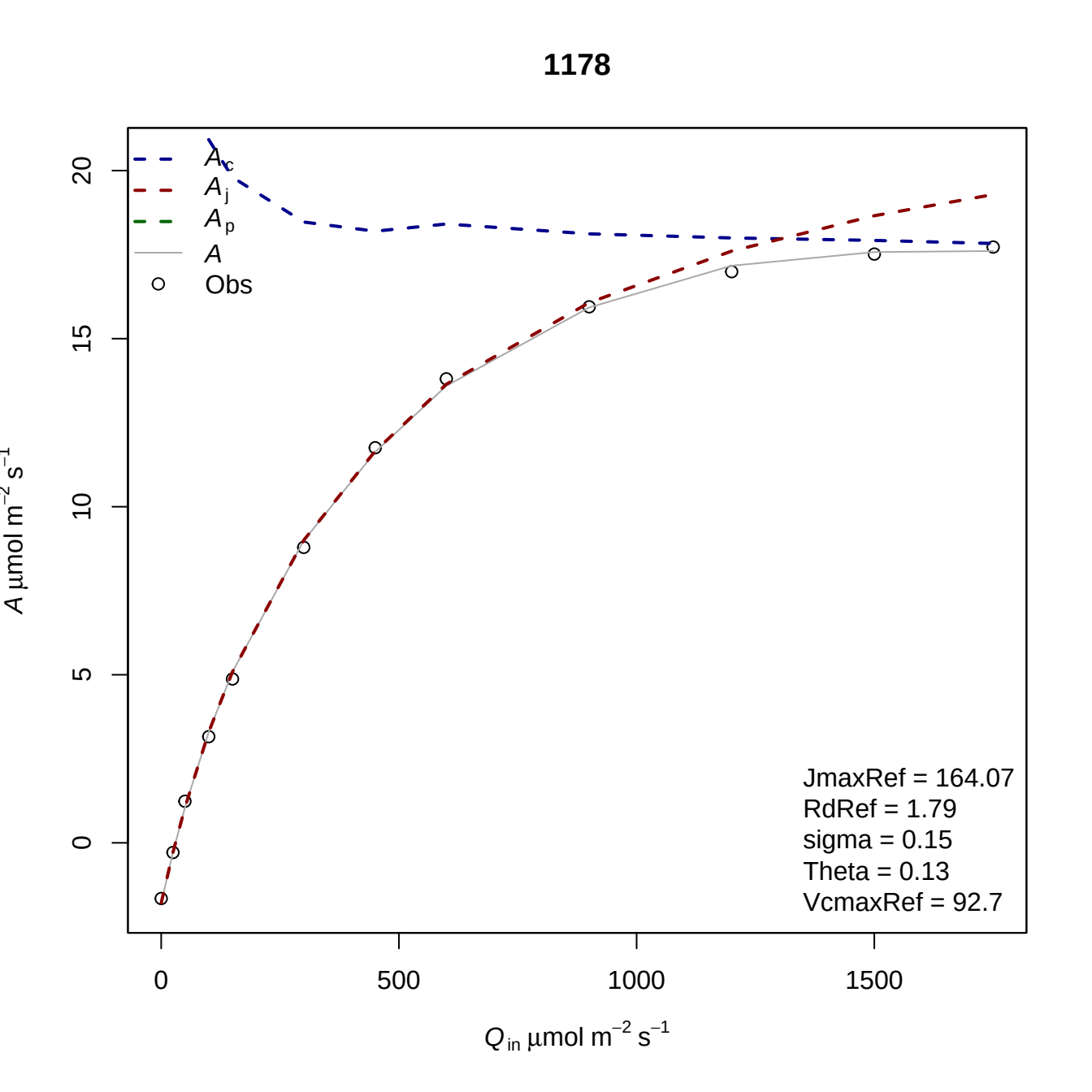

1173
A_c
A_j
A_p
A
Obs
JmaxRef = 176.33
RdRef = 3.06
sigma = 1.34
Theta = 0.16
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

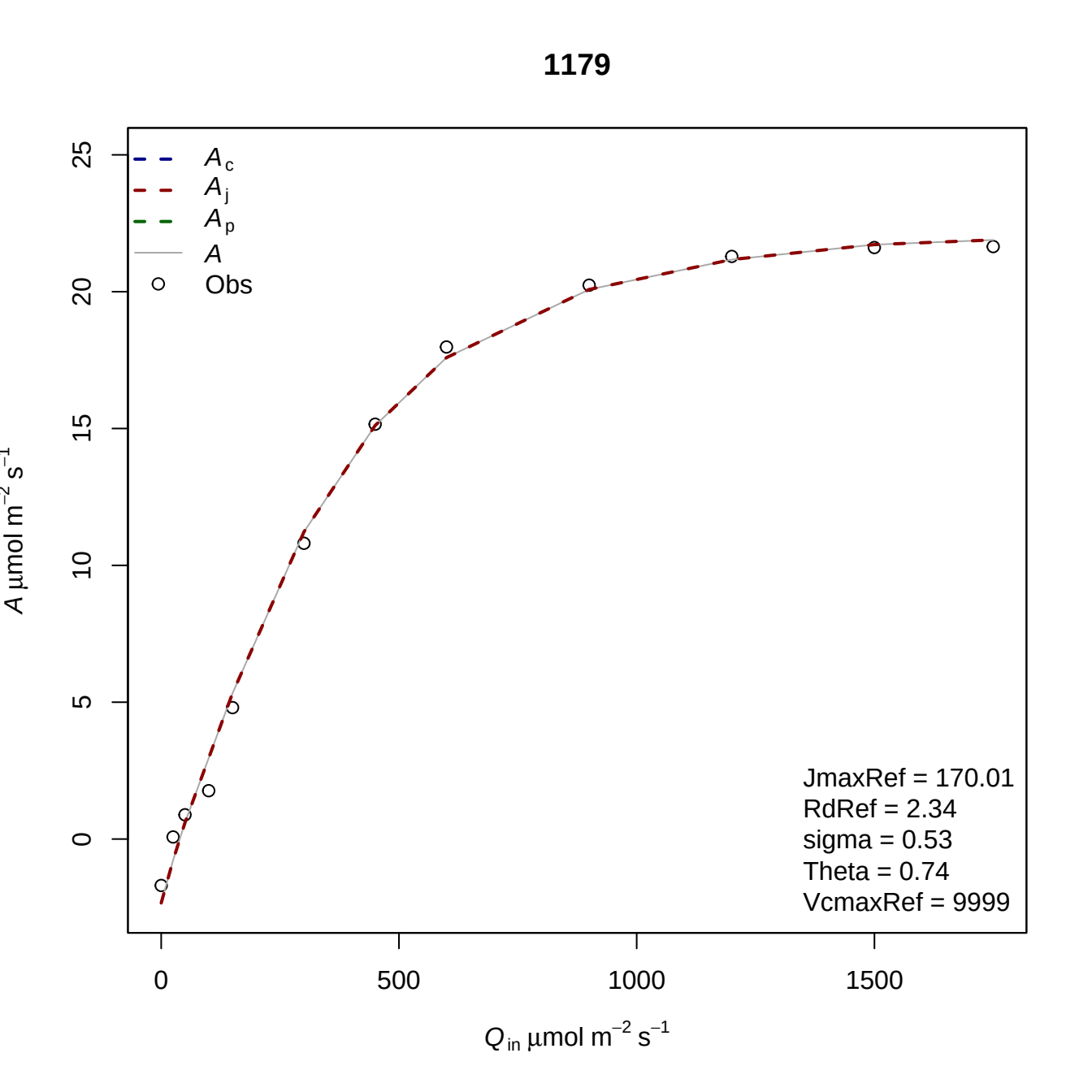

1174
A_c
A_j
A_p
A
Obs
JmaxRef = 160.55
RdRef = 2.7
sigma = 0.75
Theta = 0.32
VcmaxRef = 99.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

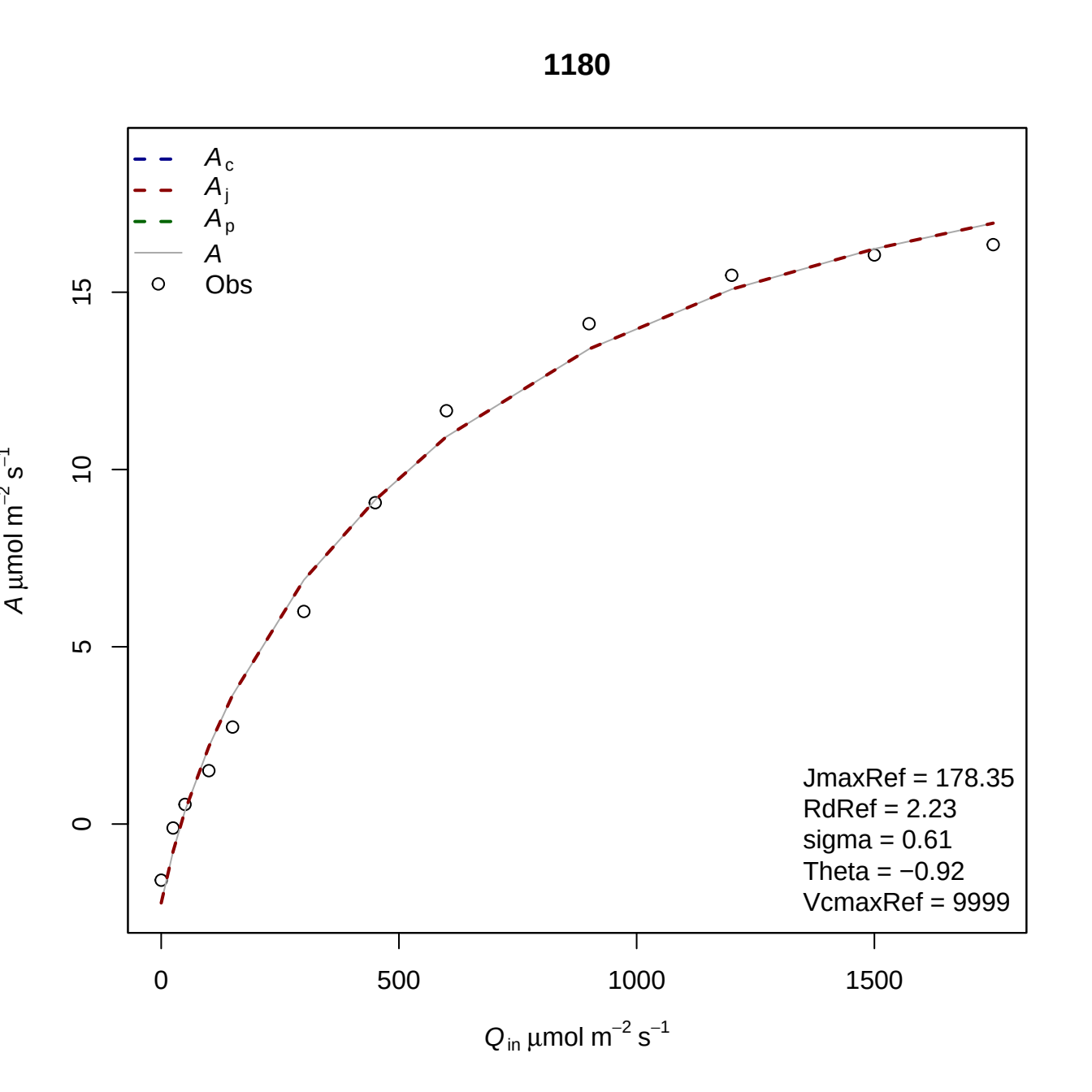

1175
A_c
A_j
A_p
A
Obs
JmaxRef = 177749870.03
RdRef = 2.44
sigma = 0.62
Theta = −4769523.95
VcmaxRef = 87.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

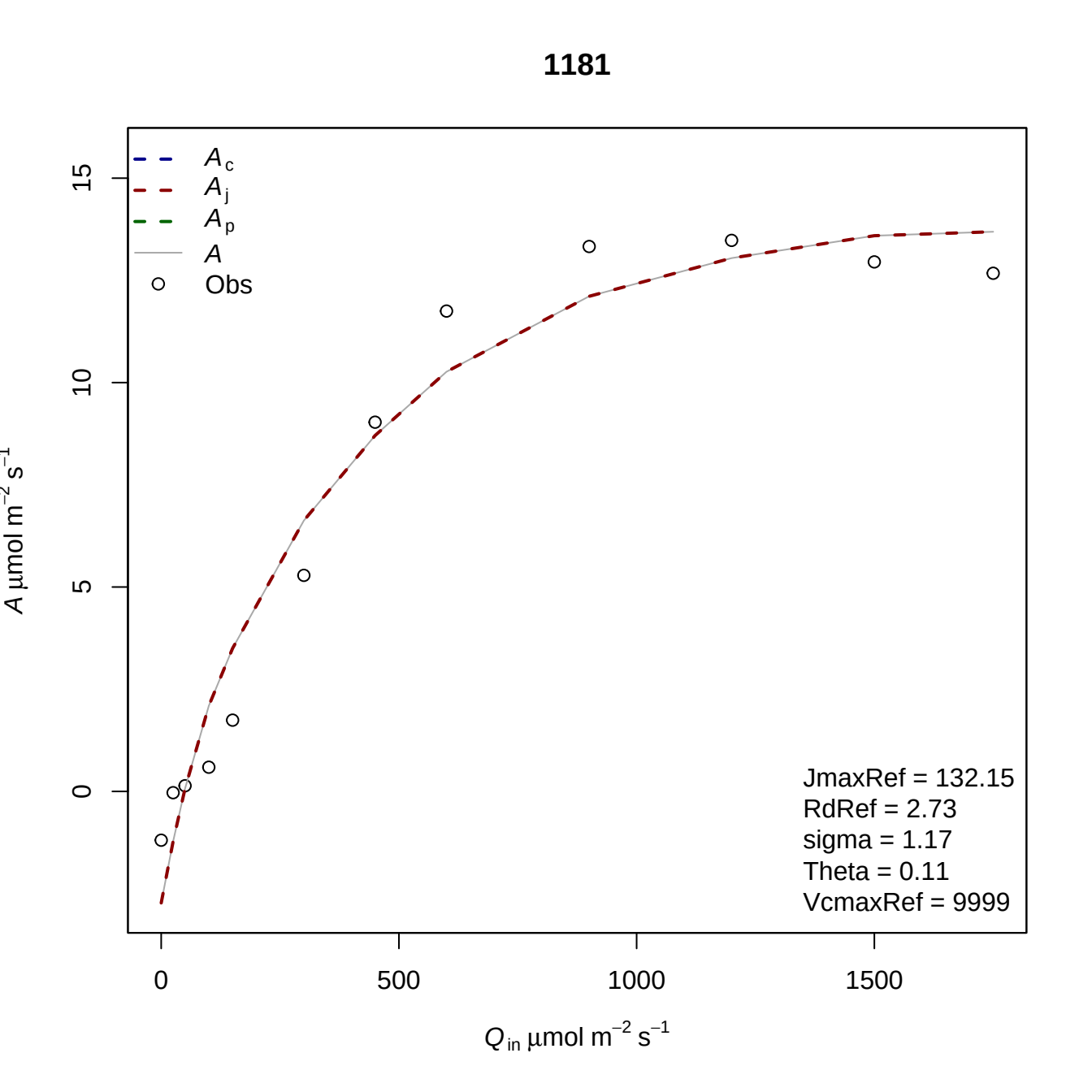

1176
A_c
A_j
A_p
A
Obs
JmaxRef = 143.47
RdRef = 1.69
sigma = 0.29
Theta = 0.55
VcmaxRef = 90.13
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

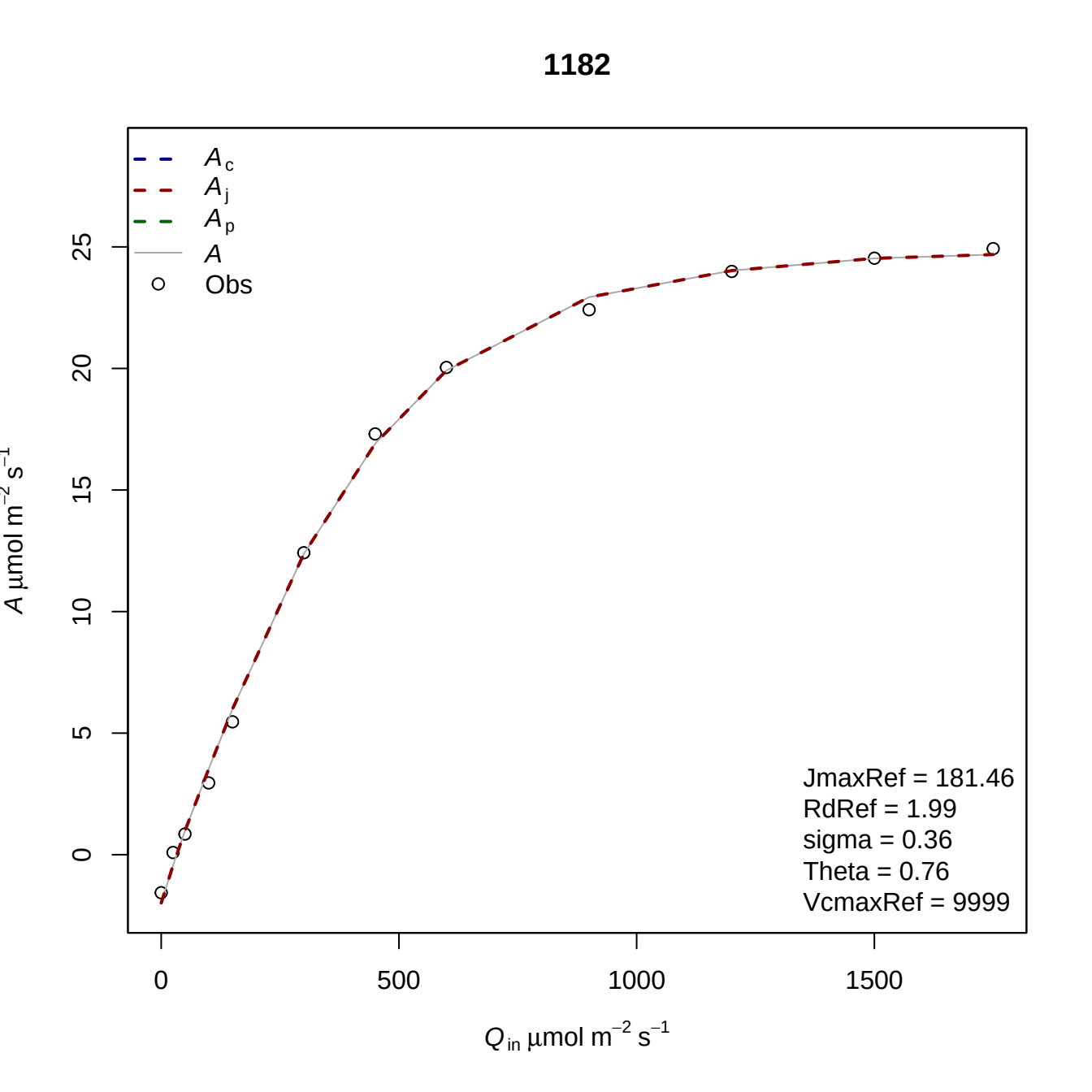

1177
A_c
A_j
A_p
A
Obs
JmaxRef = 177.66
RdRef = 1.94
sigma = 0.24
Theta = 0.57
VcmaxRef = 116.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

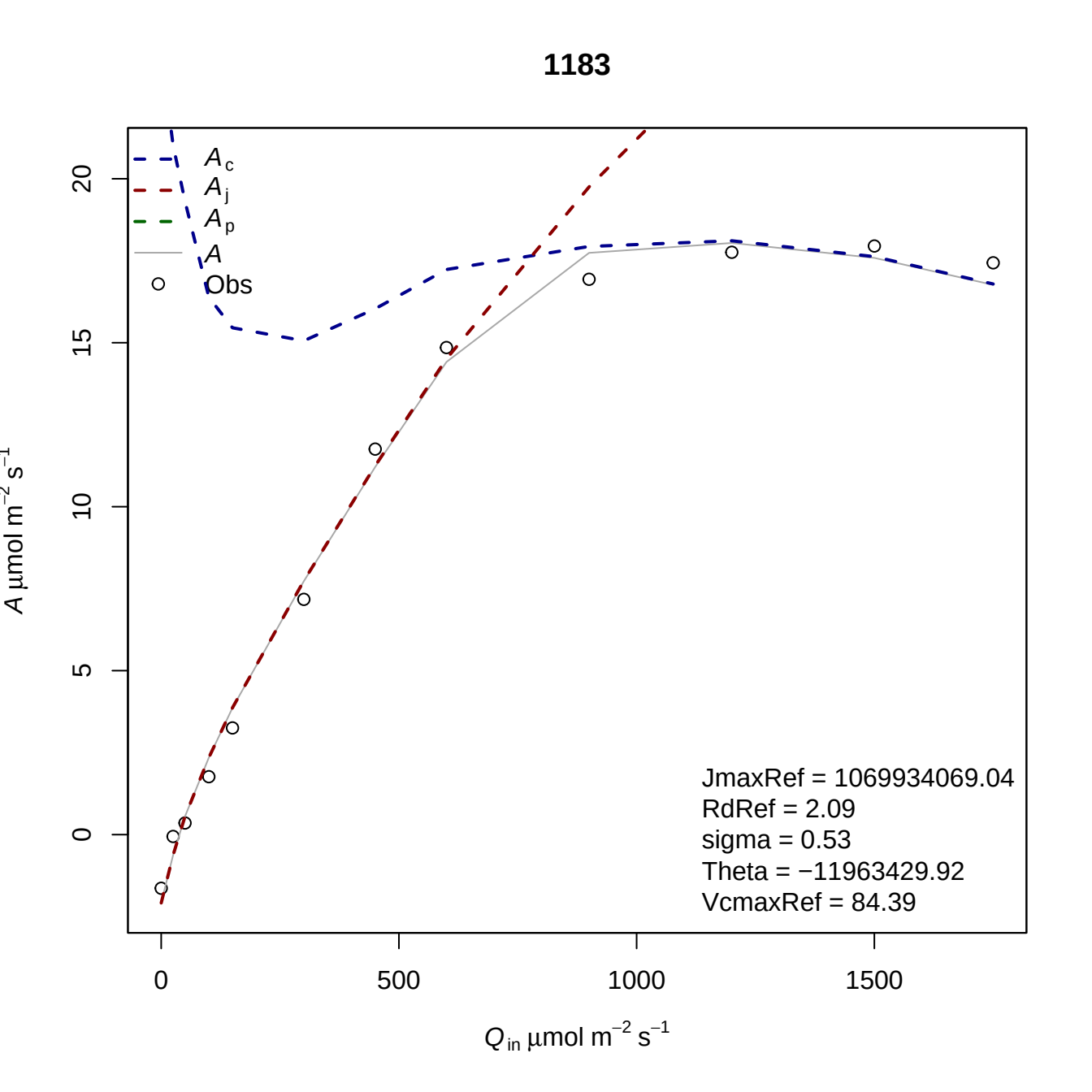

1178
A_c
A_j
A_p
A
Obs
JmaxRef = 164.07
RdRef = 1.79
sigma = 0.15
Theta = 0.13
VcmaxRef = 92.7
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

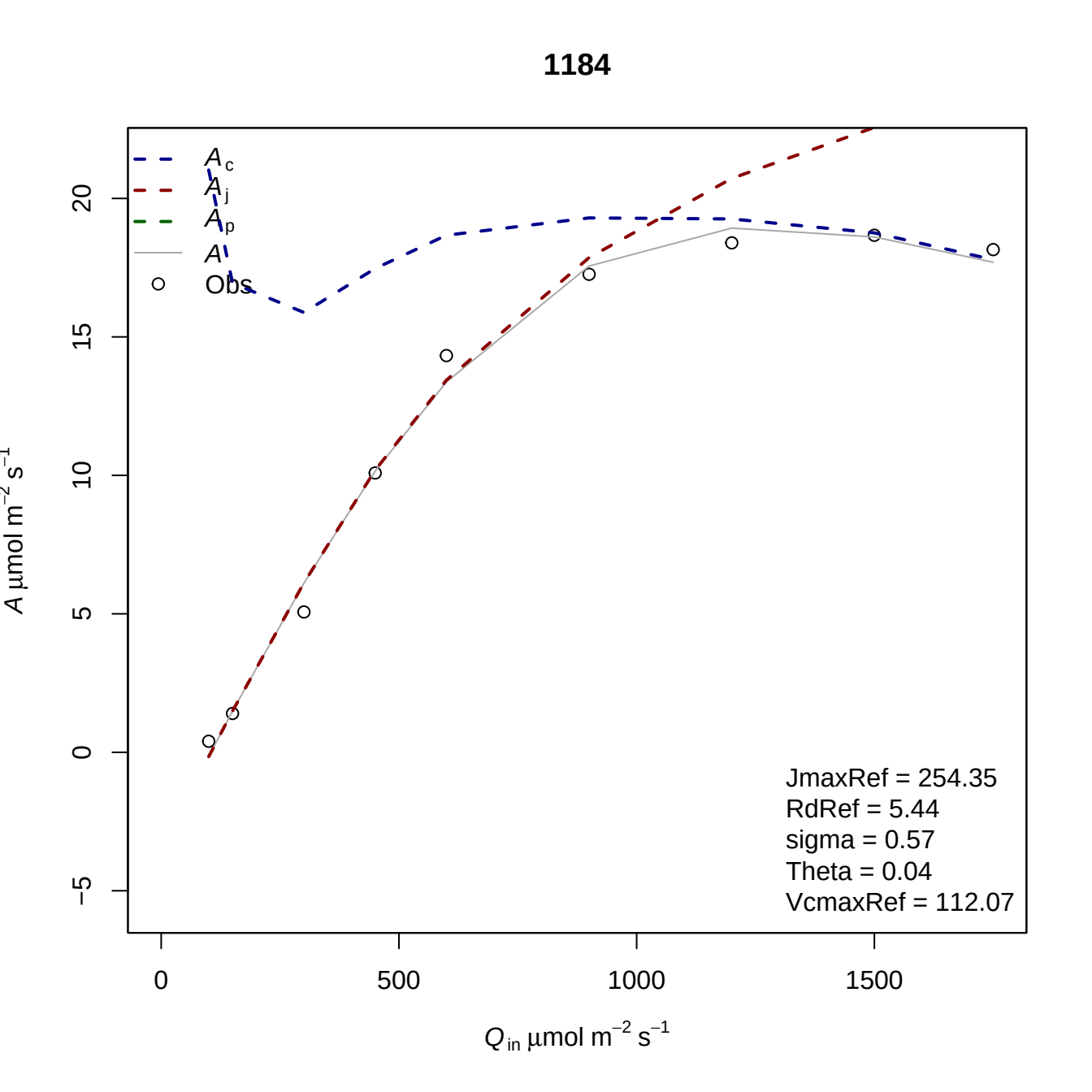

1179
A_c
A_j
A_p
A
Obs
JmaxRef = 170.01
RdRef = 2.34
sigma = 0.53
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

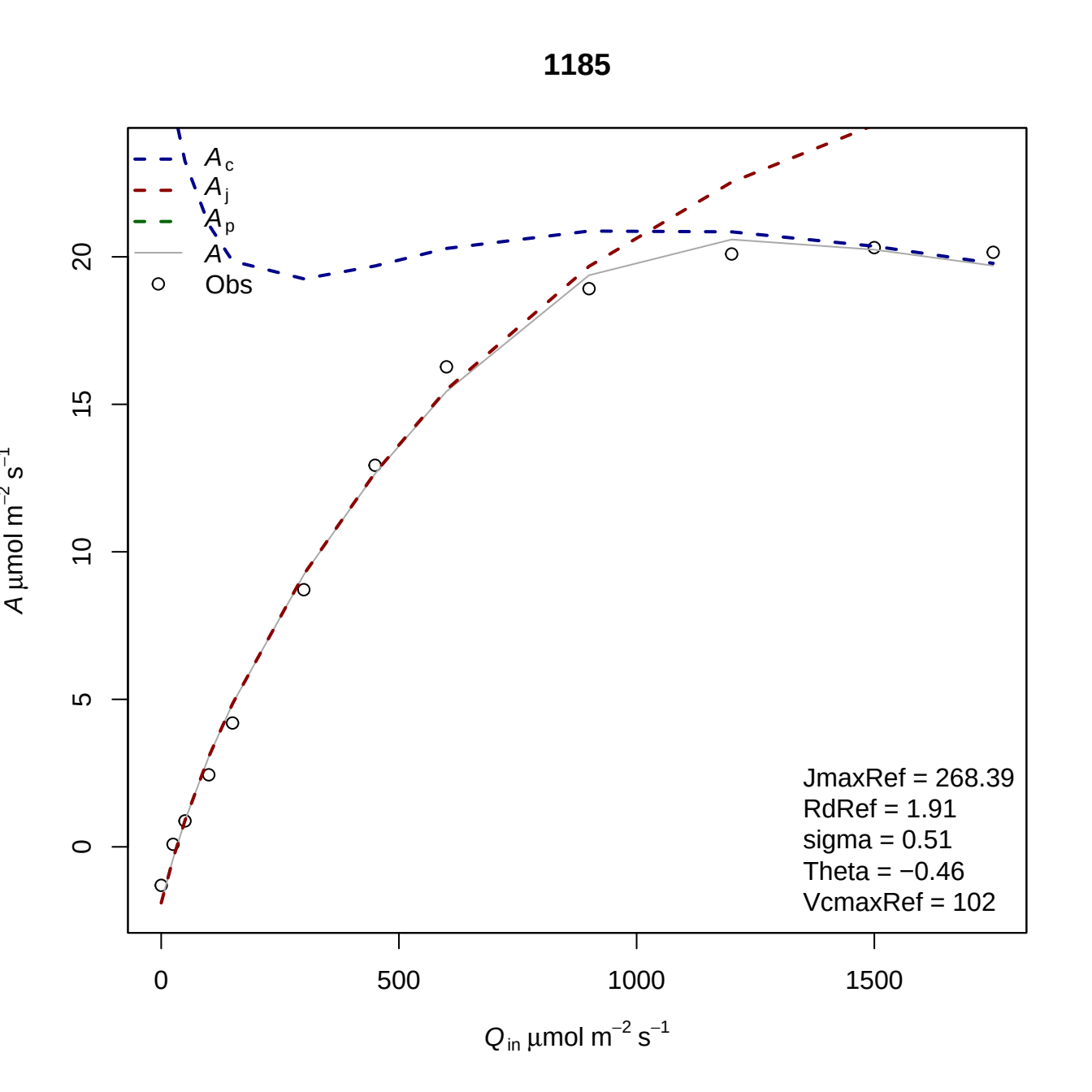

1180
A_c
A_j
A_p
A
Obs
JmaxRef = 178.35
RdRef = 2.23
sigma = 0.61
Theta = −0.92
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

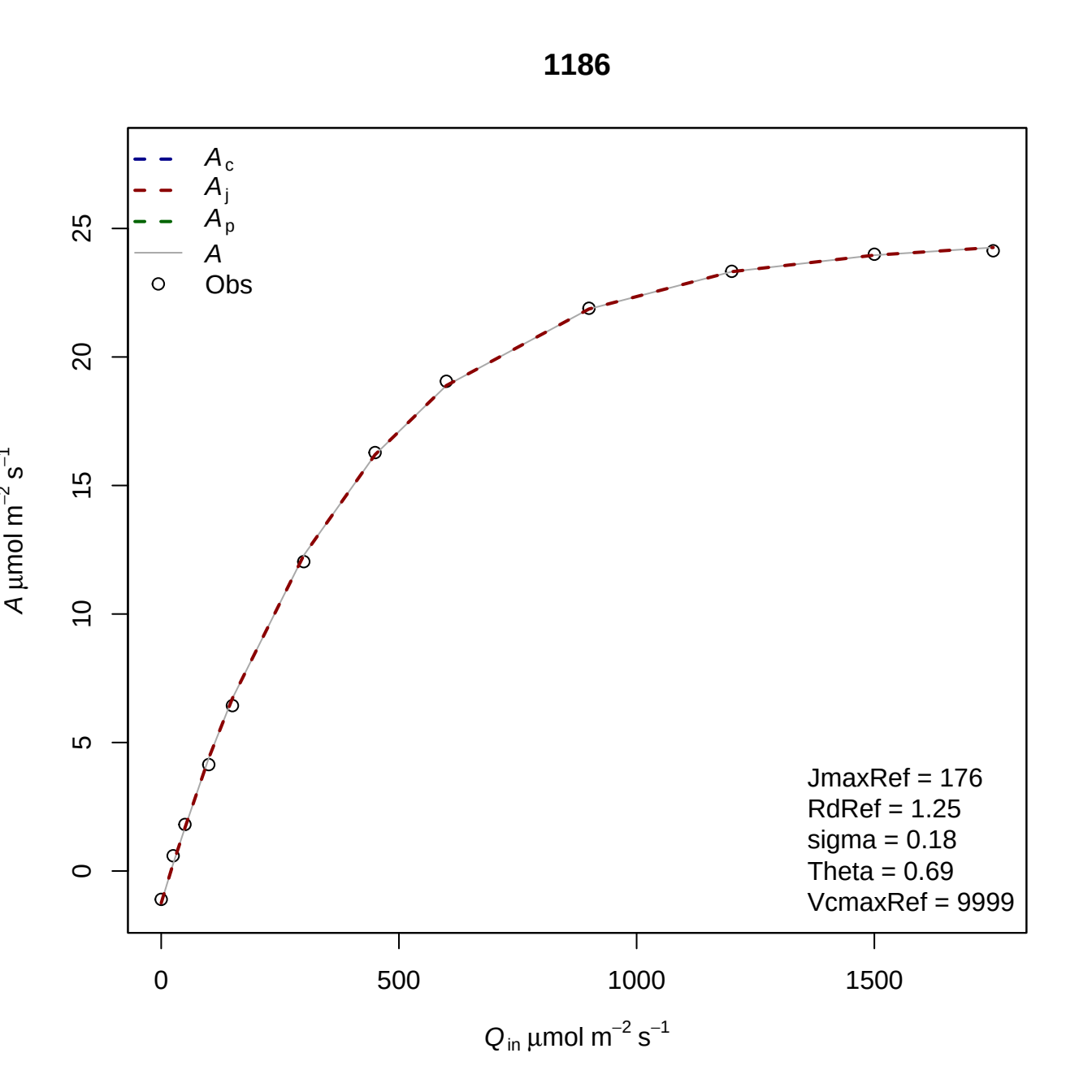

1181
A_c
A_j
A_p
A
Obs
JmaxRef = 132.15
RdRef = 2.73
sigma = 1.17
Theta = 0.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

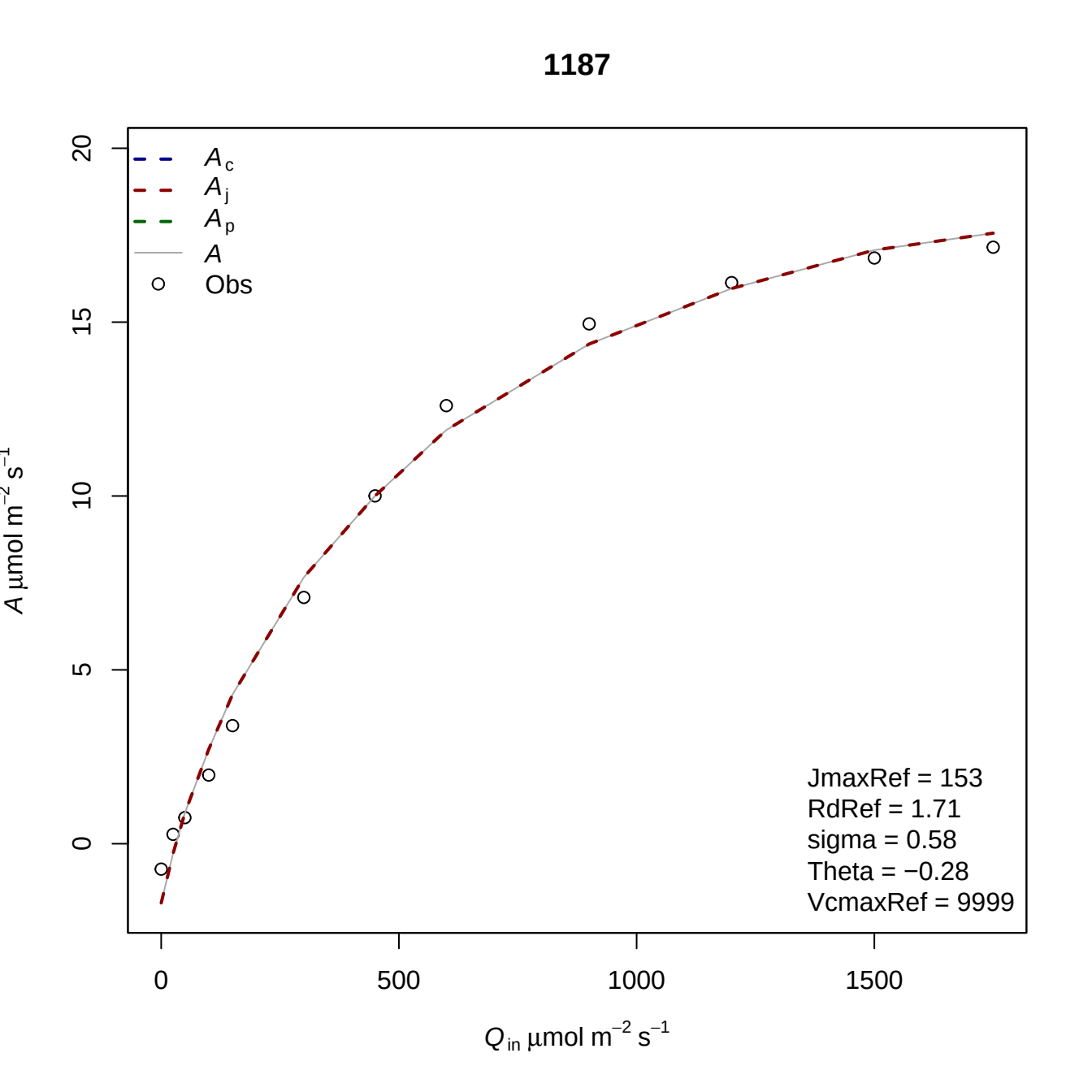

1182
A_c
A_j
A_p
A
Obs
JmaxRef = 181.46
RdRef = 1.99
sigma = 0.36
Theta = 0.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

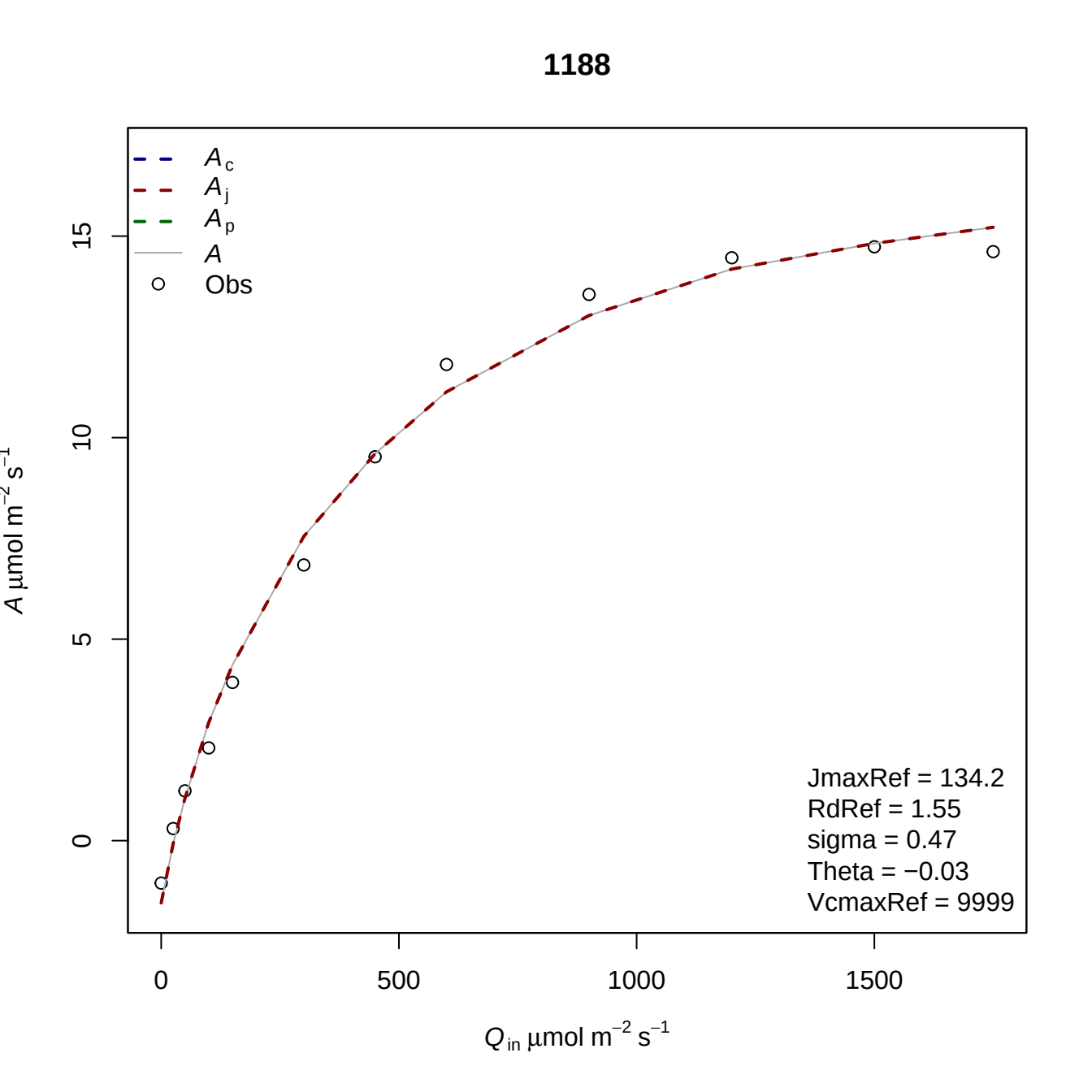

1183
A_c
A_j
A_p
A
Obs
JmaxRef = 1069934069.04
RdRef = 2.09
sigma = 0.53
Theta = −11963429.92
VcmaxRef = 84.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

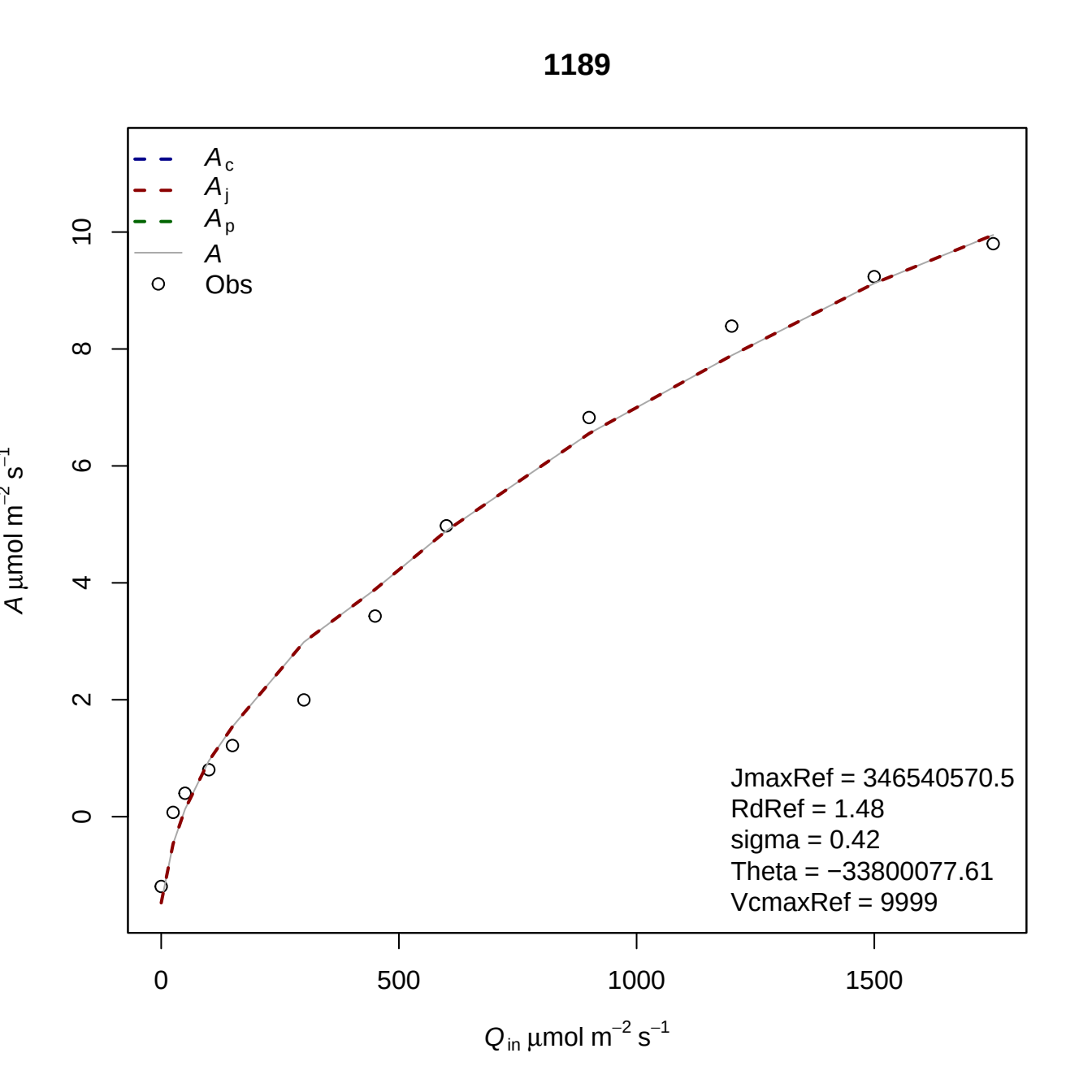

1184
A_c
A_j
A_p
A
Obs
JmaxRef = 254.35
RdRef = 5.44
sigma = 0.57
Theta = 0.04
VcmaxRef = 112.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

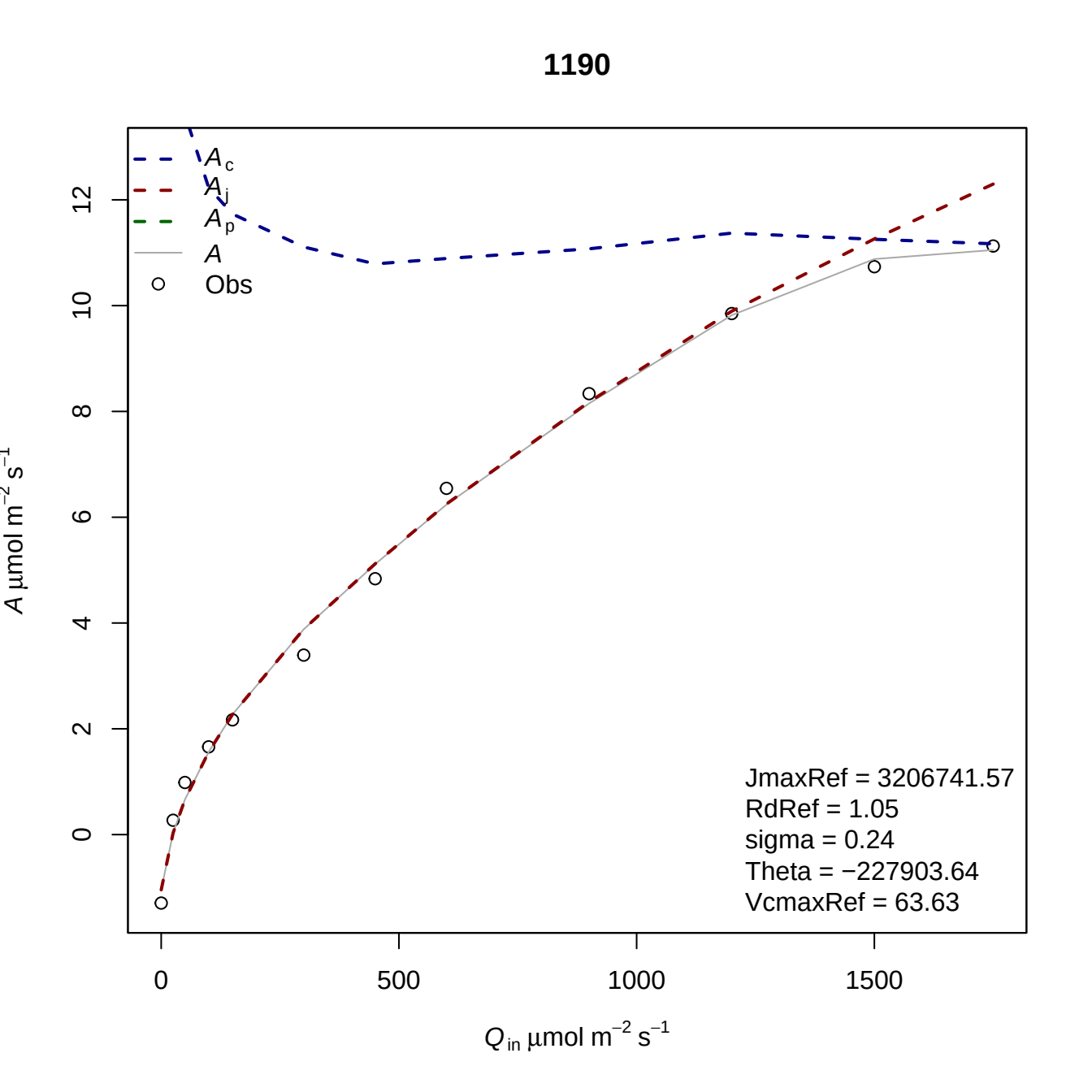

1185
A_c
A_j
A_p
A
Obs
JmaxRef = 268.39
RdRef = 1.91
sigma = 0.51
Theta = −0.46
VcmaxRef = 102
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

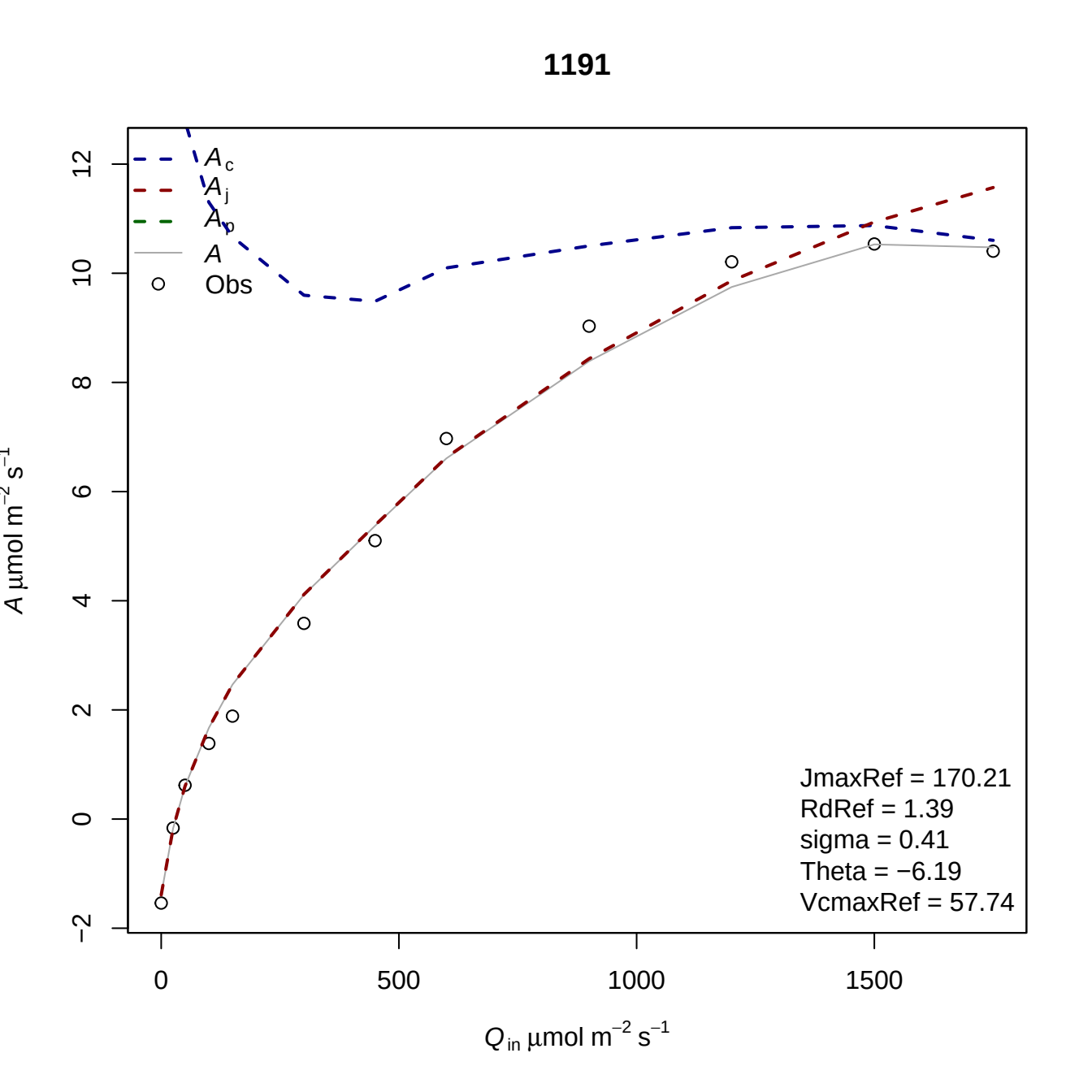

1186
A_c
A_j
A_p
A
Obs
JmaxRef = 176
RdRef = 1.25
sigma = 0.18
Theta = 0.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

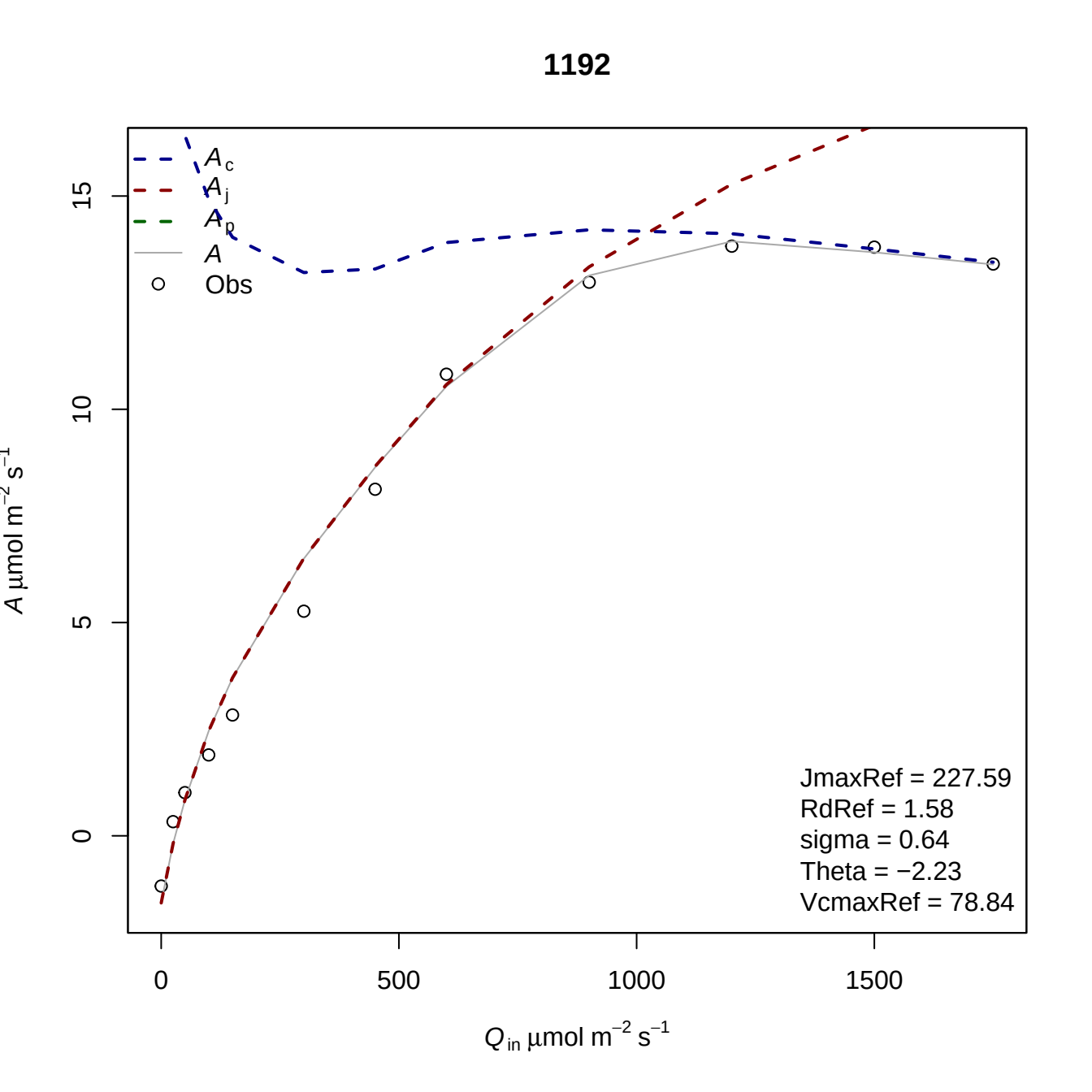

1187
A_c
A_j
A_p
A
Obs
JmaxRef = 153
RdRef = 1.71
sigma = 0.58
Theta = −0.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

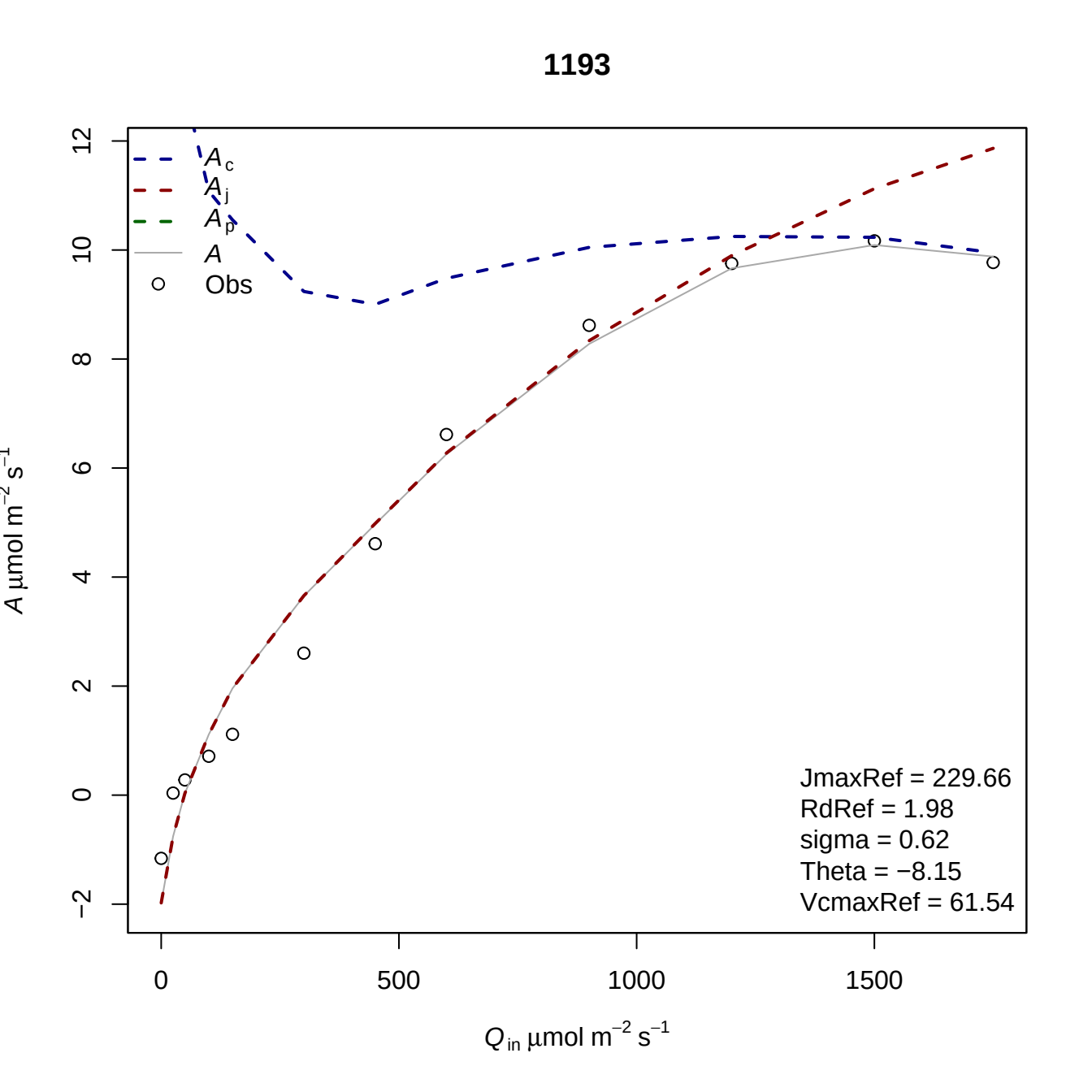

1188
A_c
A_j
A_p
A
Obs
JmaxRef = 134.2
RdRef = 1.55
sigma = 0.47
Theta = −0.03
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

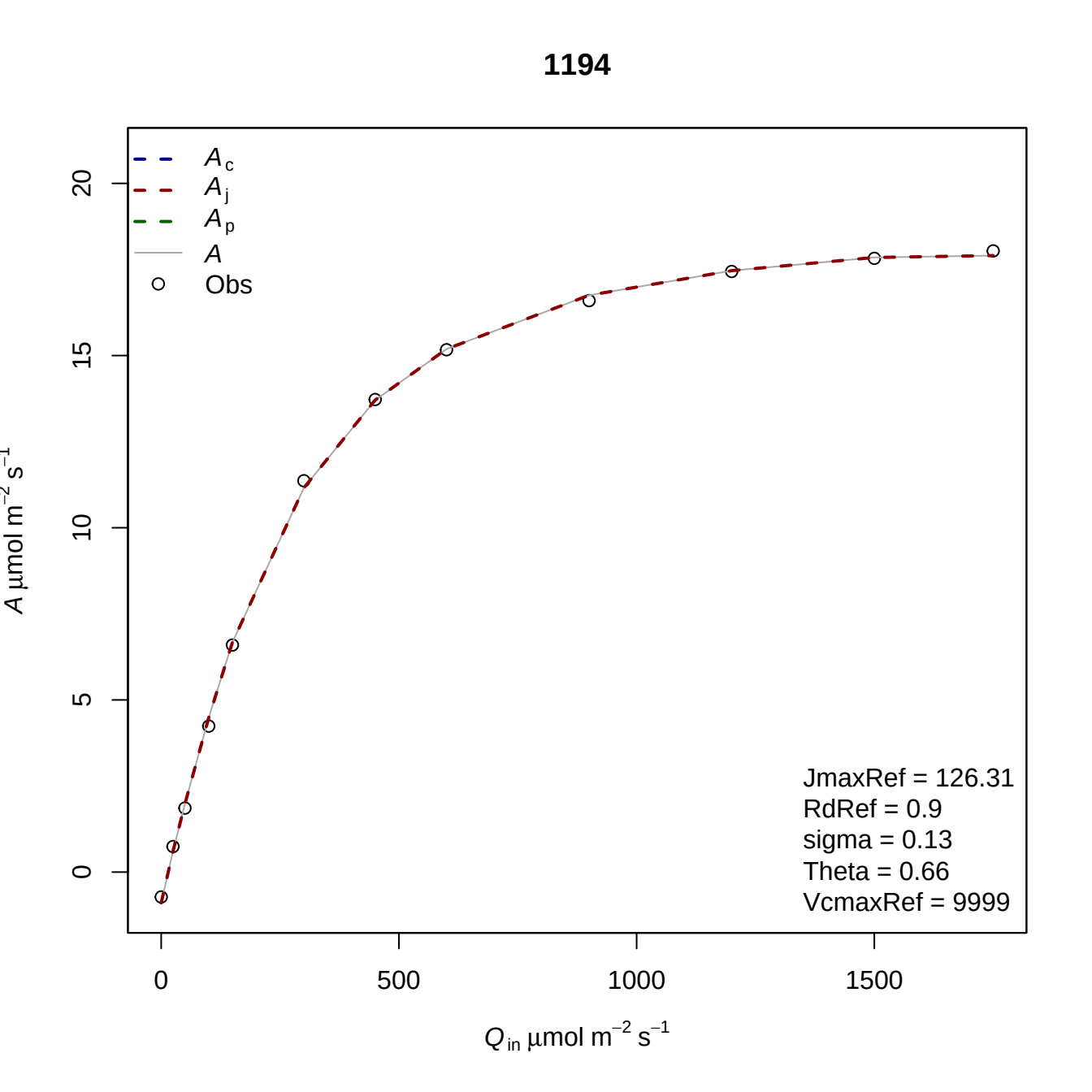

1189
A_c
A_j
A_p
A
Obs
JmaxRef = 346540570.5
RdRef = 1.48
sigma = 0.42
Theta = −33800077.61
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

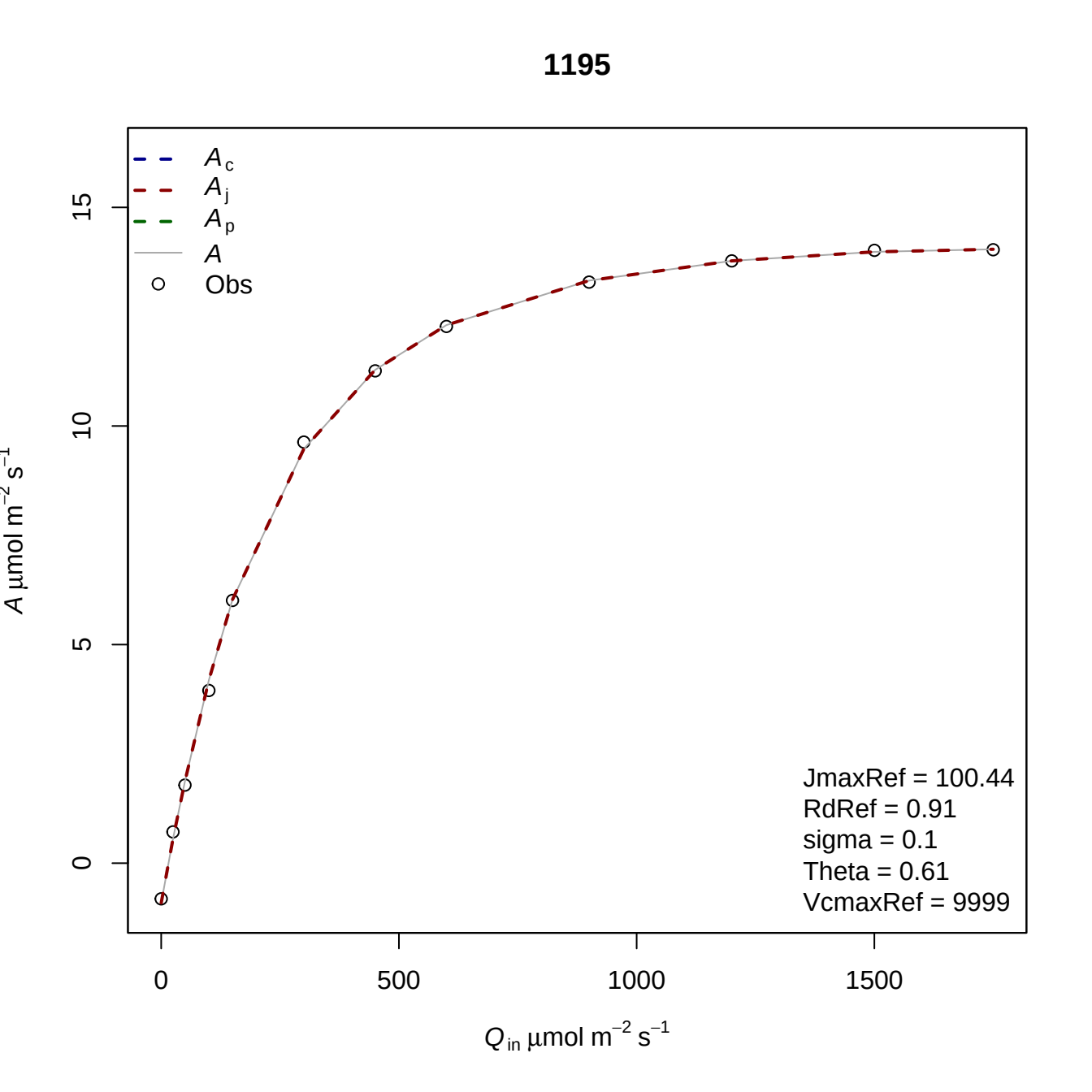

1190
A_c
A_j
A_p
A
Obs
JmaxRef = 3206741.57
RdRef = 1.05
sigma = 0.24
Theta = −227903.64
VcmaxRef = 63.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

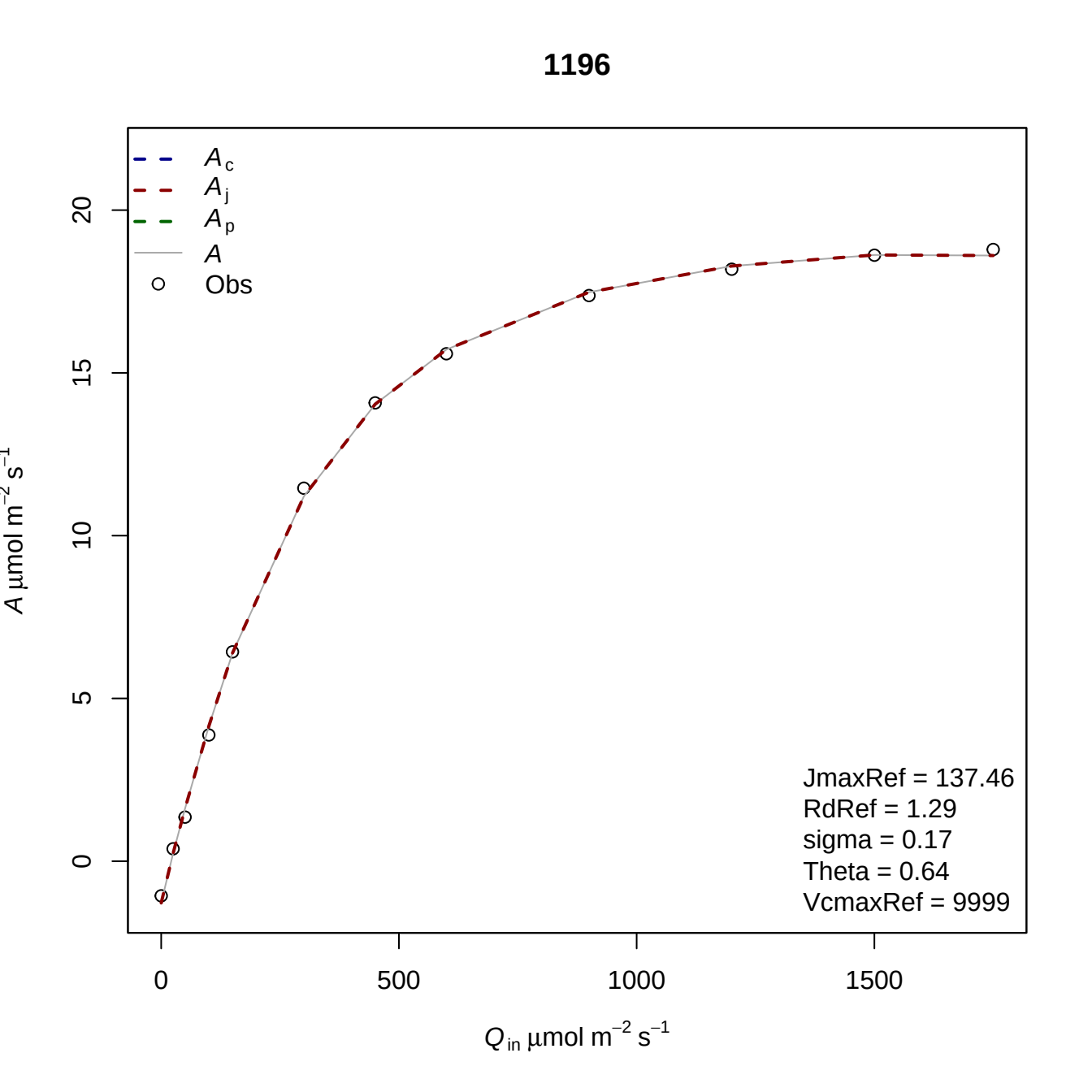

1191
A_c
A_j
A_p
A
Obs
JmaxRef = 170.21
RdRef = 1.39
sigma = 0.41
Theta = −6.19
VcmaxRef = 57.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

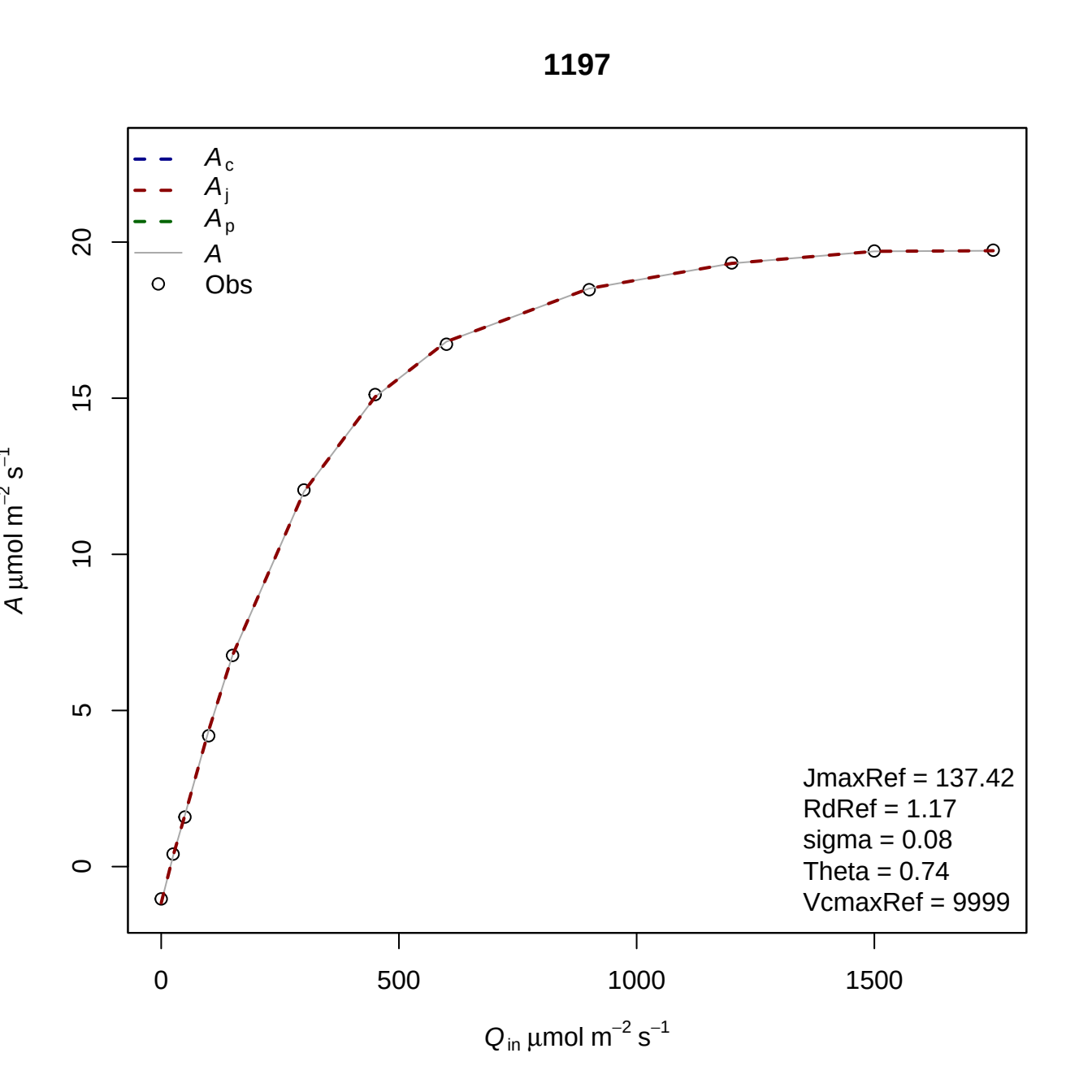

1192
A_c
A_j
A_p
A
Obs
JmaxRef = 227.59
RdRef = 1.58
sigma = 0.64
Theta = −2.23
VcmaxRef = 78.84
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

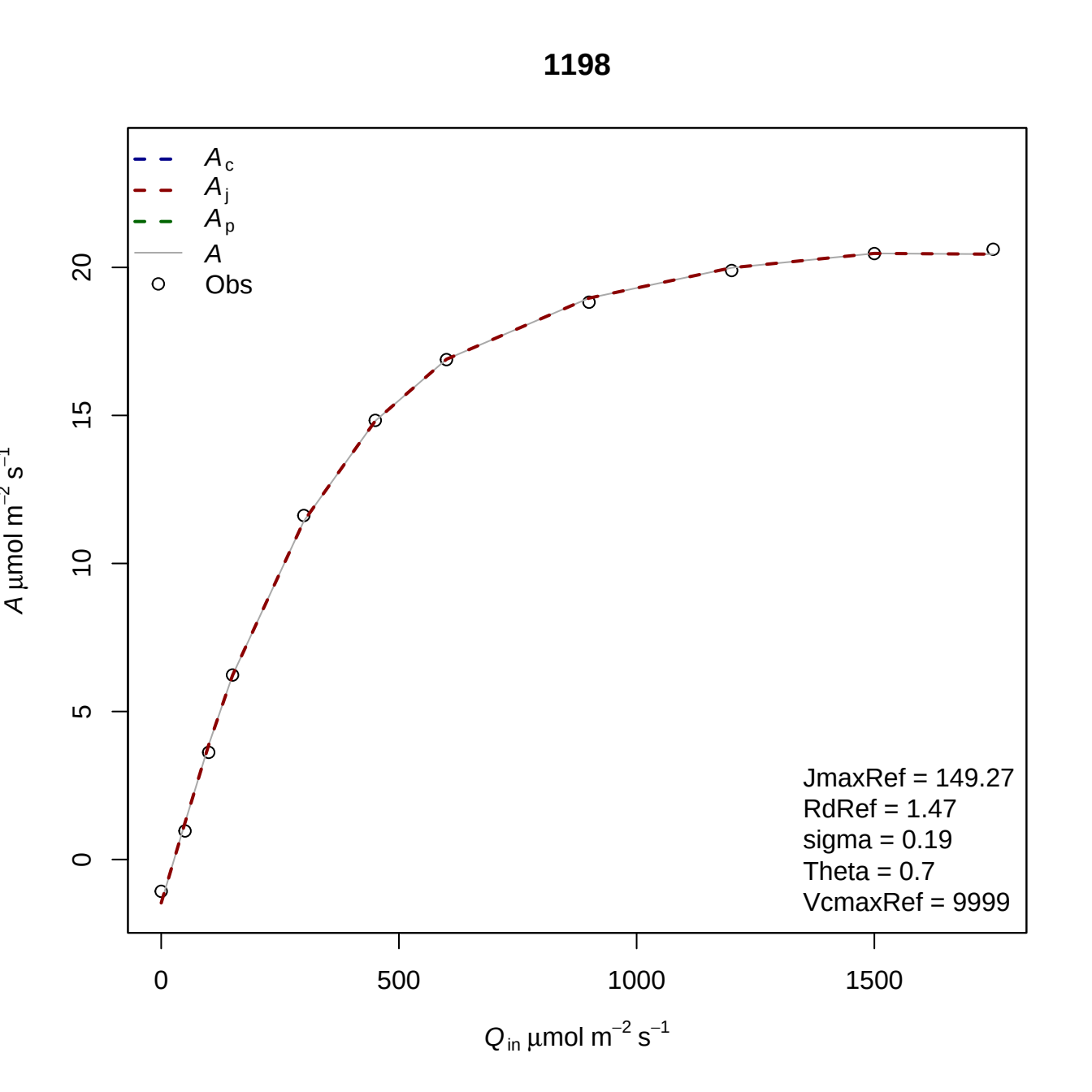

1193
A_c
A_j
A_p
A
Obs
JmaxRef = 229.66
RdRef = 1.98
sigma = 0.62
Theta = −8.15
VcmaxRef = 61.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

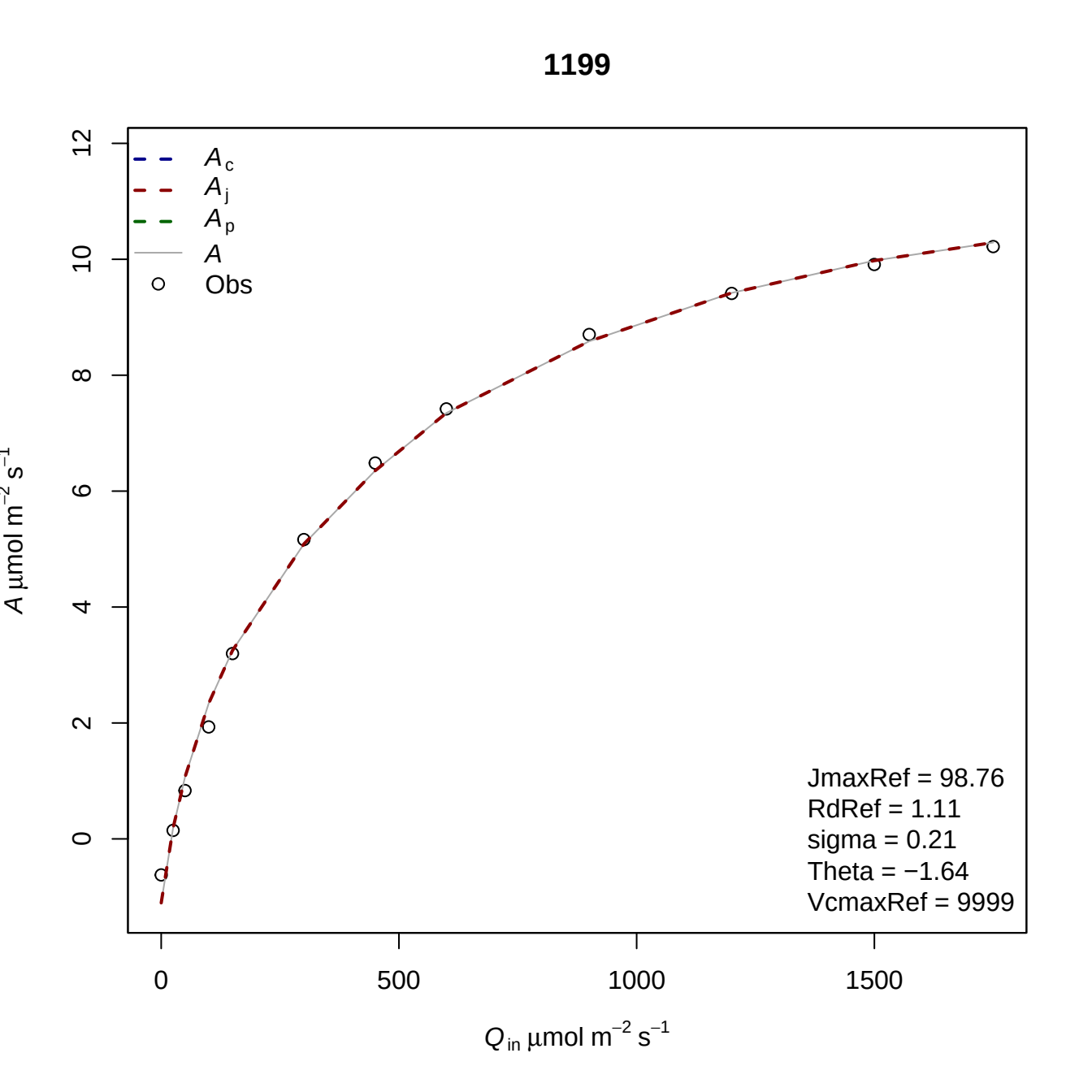

1194
A_c
A_j
A_p
A
Obs
JmaxRef = 126.31
RdRef = 0.9
sigma = 0.13
Theta = 0.66
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

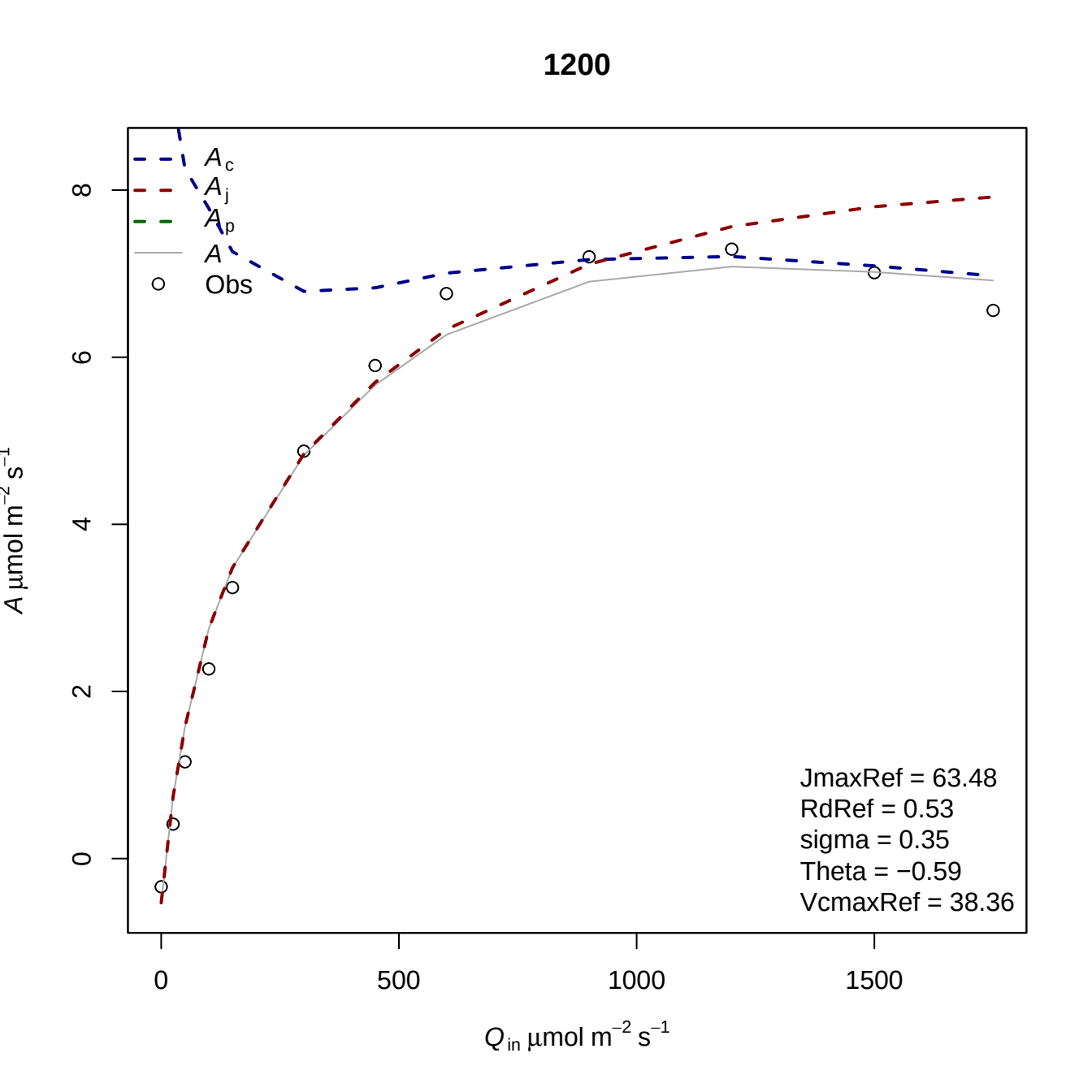

1195
A_c
A_j
A_p
A
Obs
JmaxRef = 100.44
RdRef = 0.91
sigma = 0.1
Theta = 0.61
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

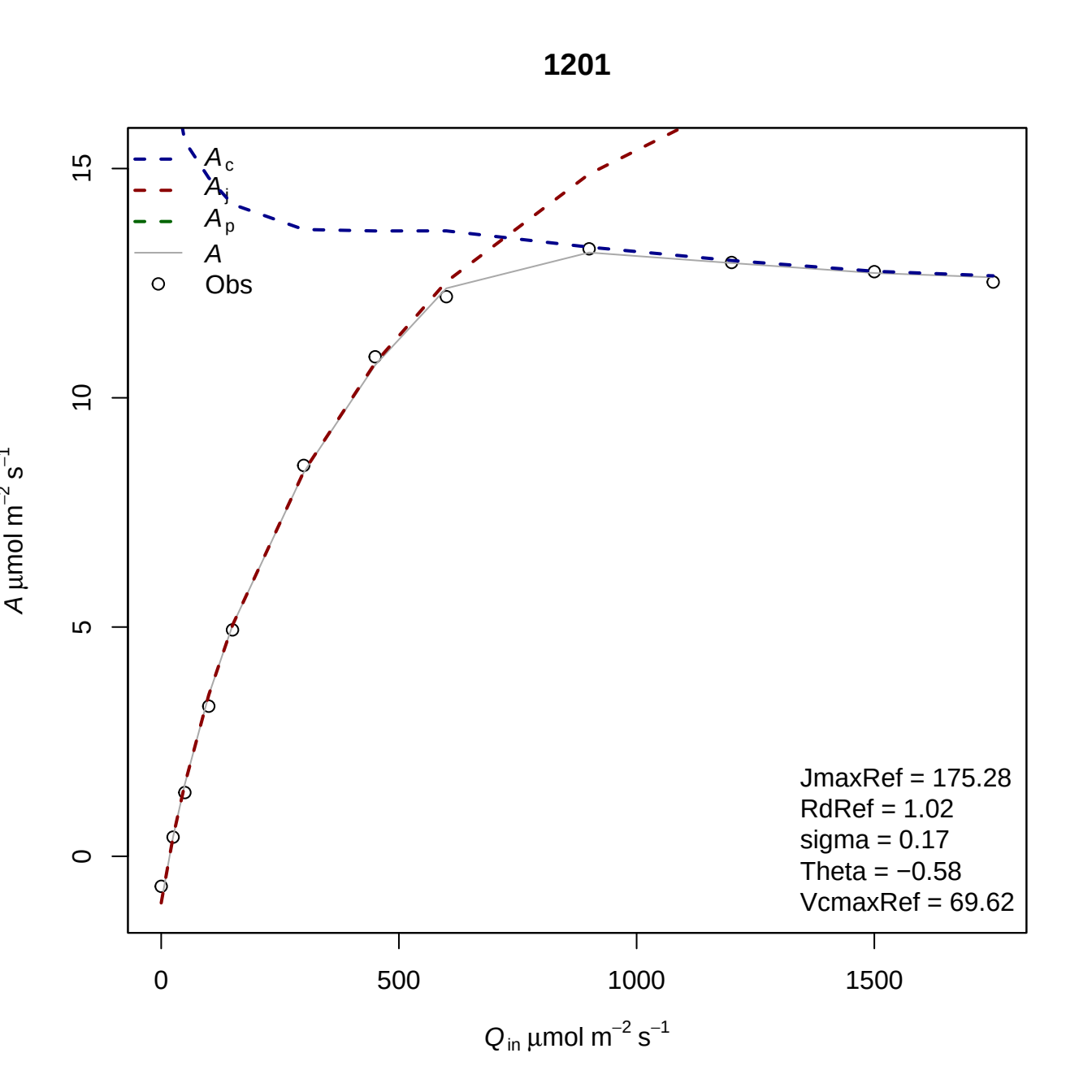

1196
A_c
A_j
A_p
A
Obs
JmaxRef = 137.46
RdRef = 1.29
sigma = 0.17
Theta = 0.64
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

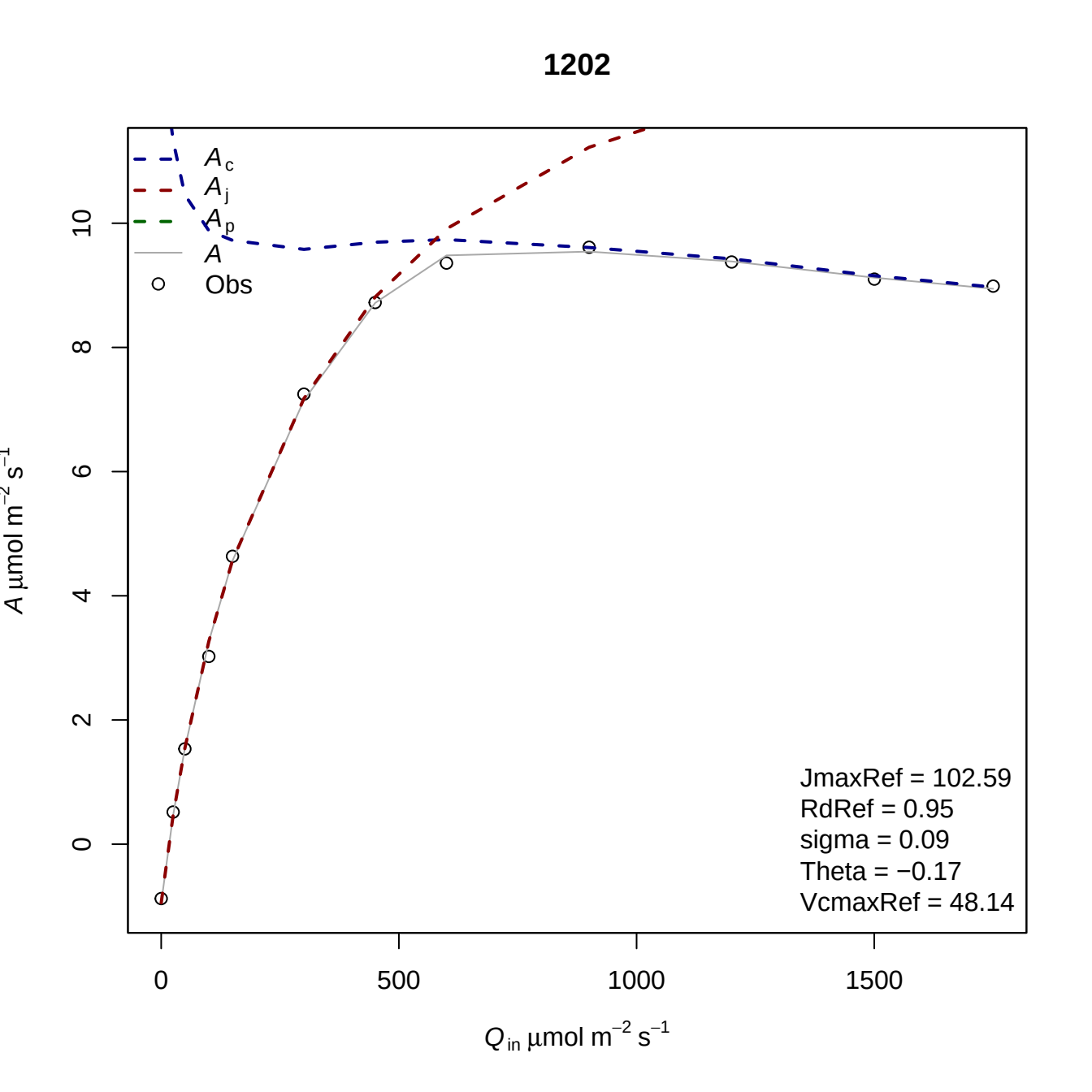

1197
A_c
A_j
A_p
A
Obs
JmaxRef = 137.42
RdRef = 1.17
sigma = 0.08
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

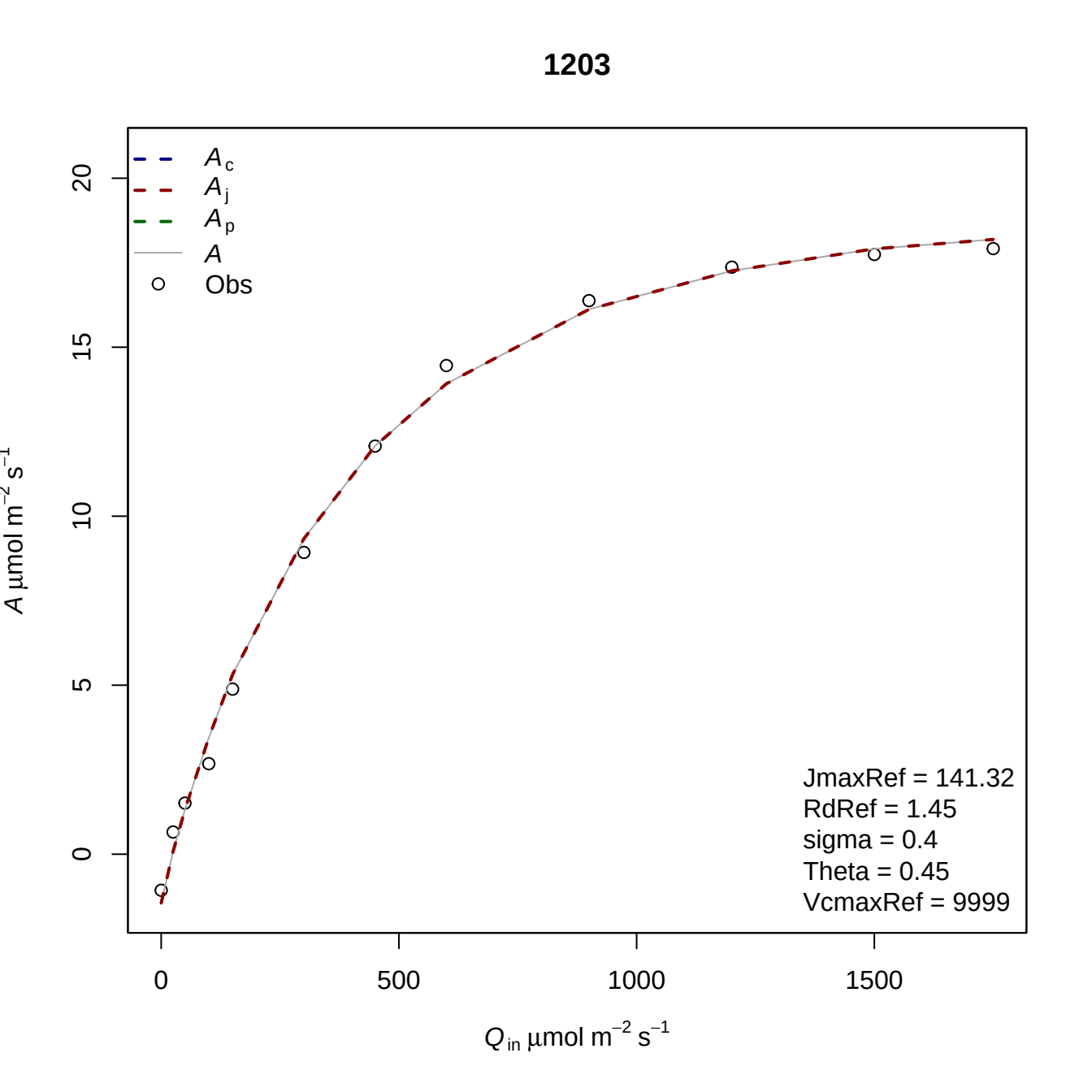

1198
A_c
A_j
A_p
A
Obs
JmaxRef = 149.27
RdRef = 1.47
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

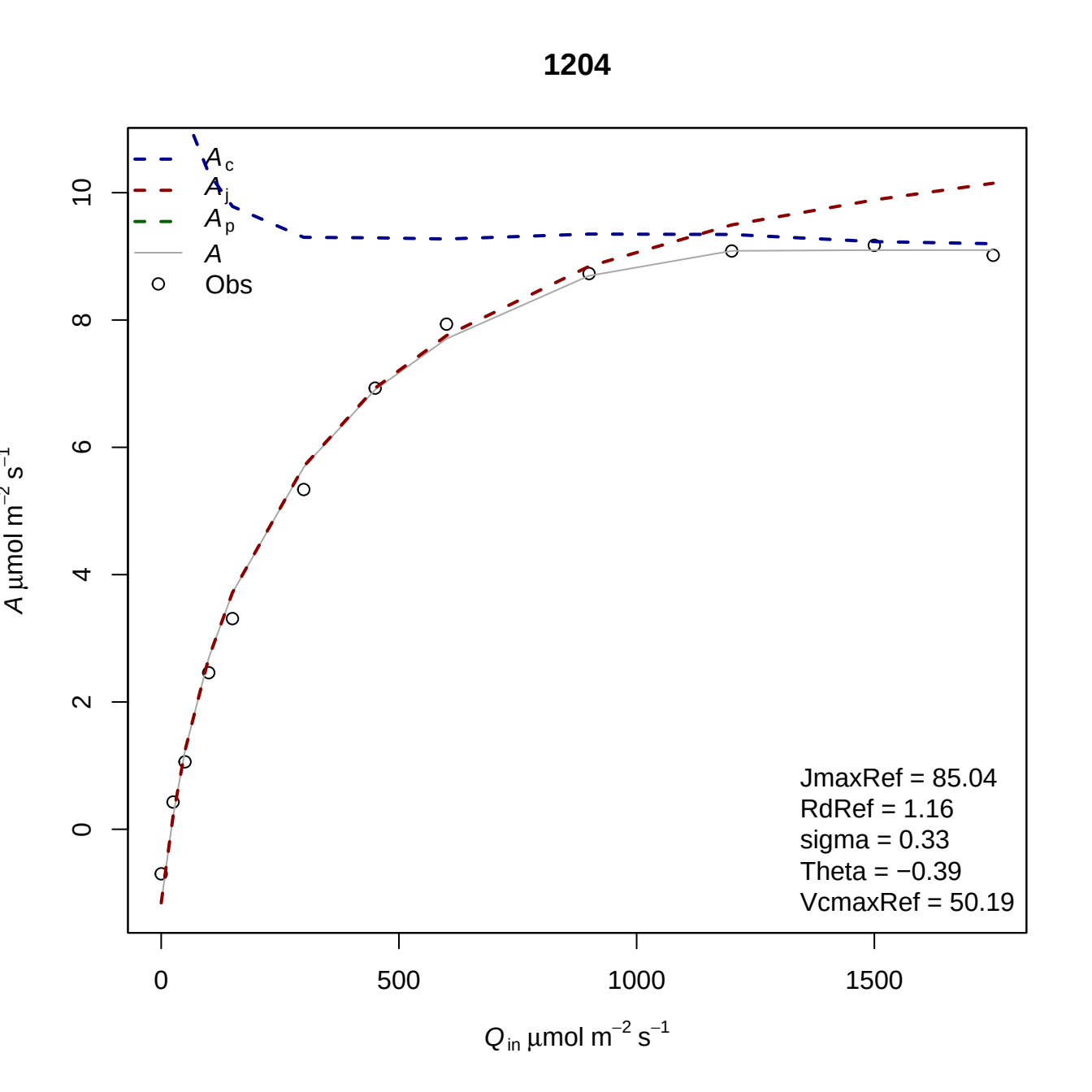

1199
A_c
A_j
A_p
A
Obs
JmaxRef = 98.76
RdRef = 1.11
sigma = 0.21
Theta = −1.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

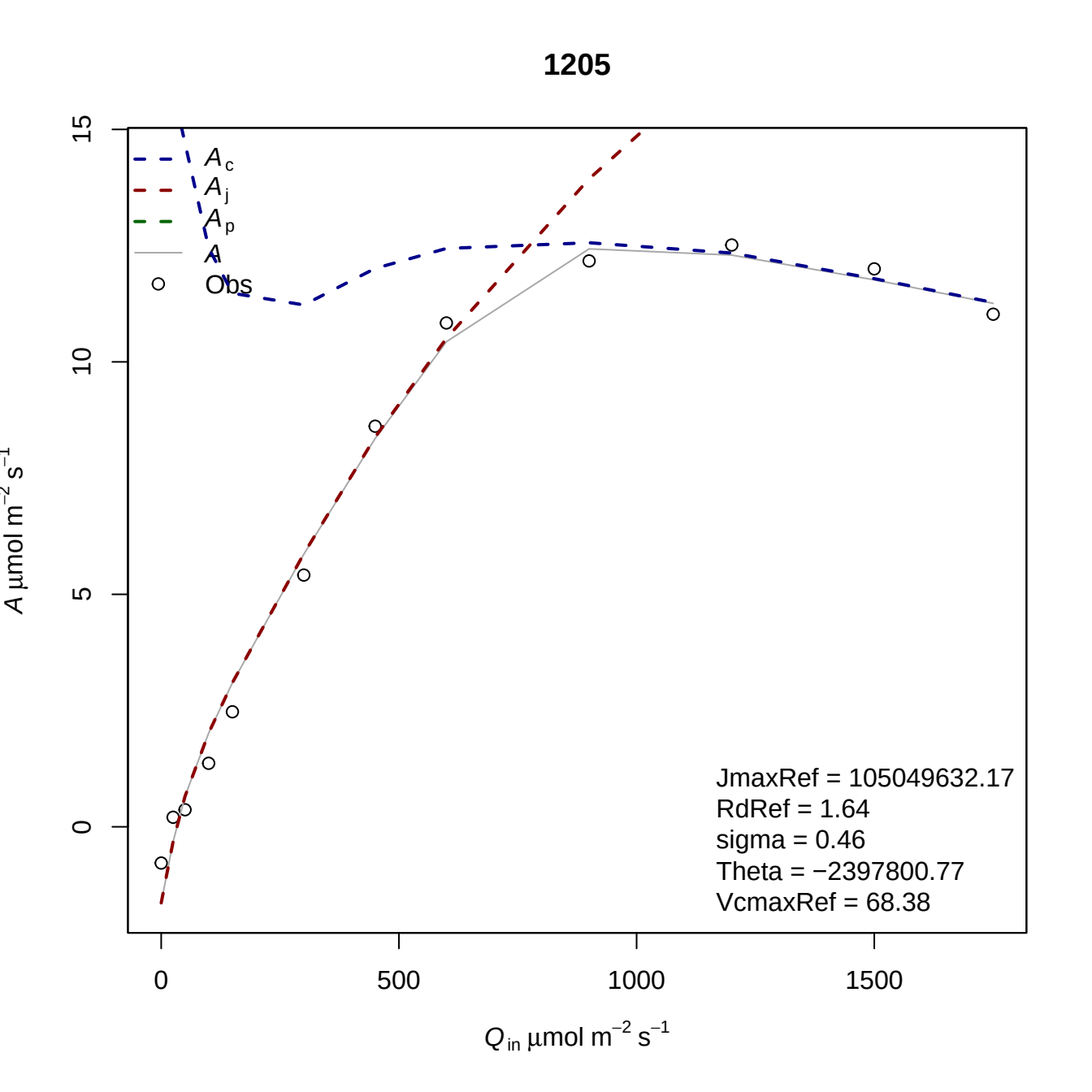

1200
A_c
A_j
A_p
A
Obs
JmaxRef = 63.48
RdRef = 0.53
sigma = 0.35
Theta = −0.59
VcmaxRef = 38.36
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

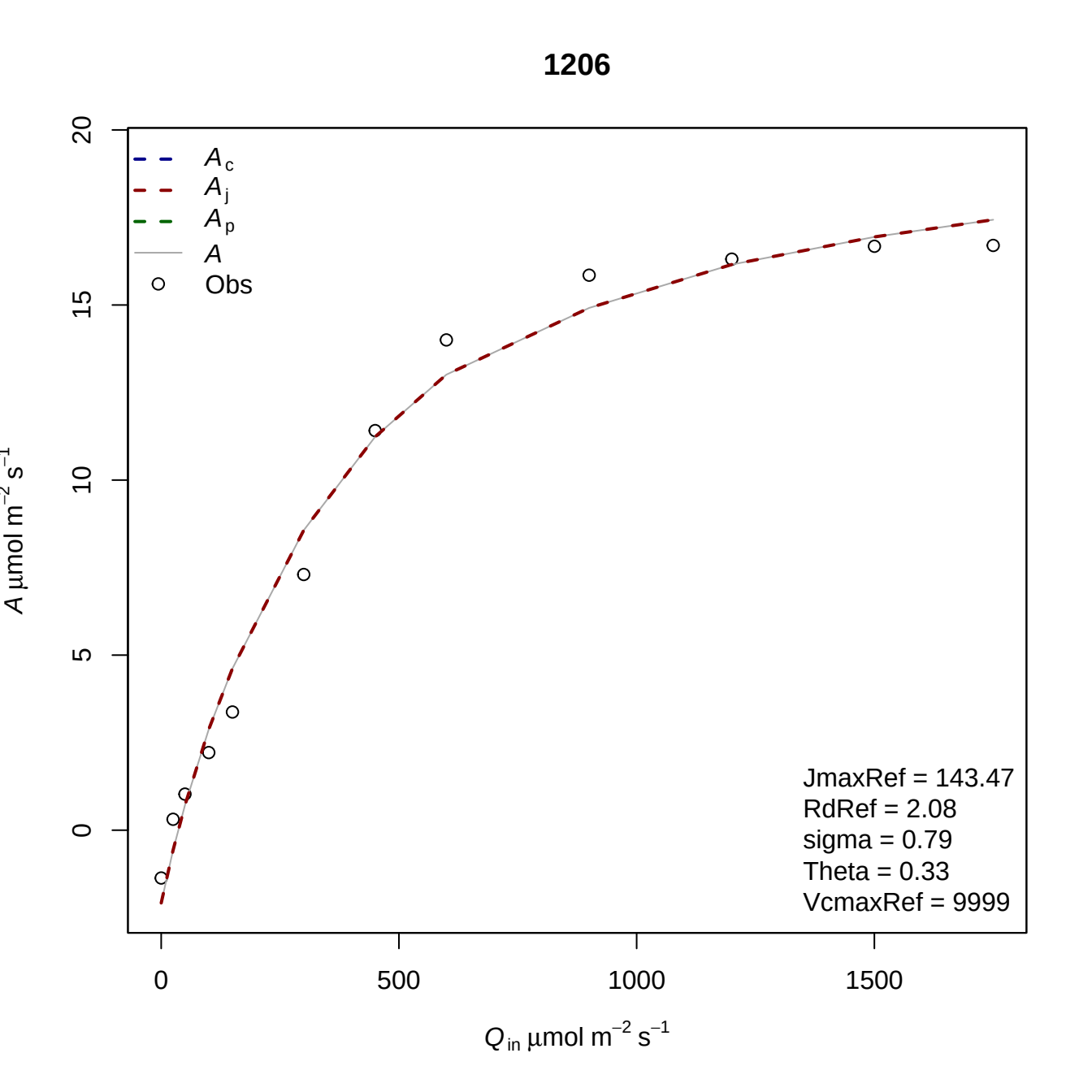

1201
A_c
A_j
A_p
A
Obs
JmaxRef = 175.28
RdRef = 1.02
sigma = 0.17
Theta = −0.58
VcmaxRef = 69.62
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

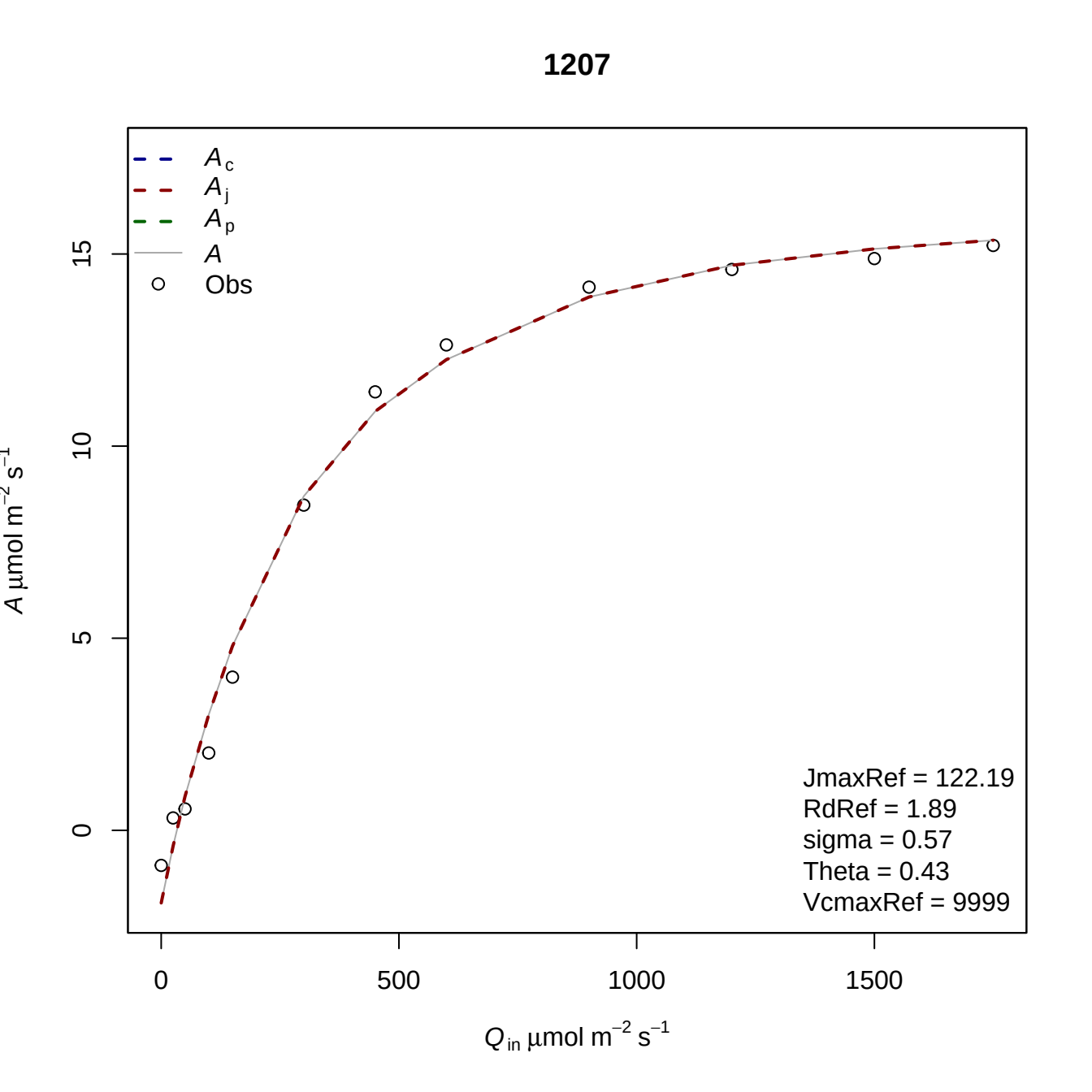

1202
A_c
A_j
A_p
A
Obs
JmaxRef = 102.59
RdRef = 0.95
sigma = 0.09
Theta = −0.17
VcmaxRef = 48.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

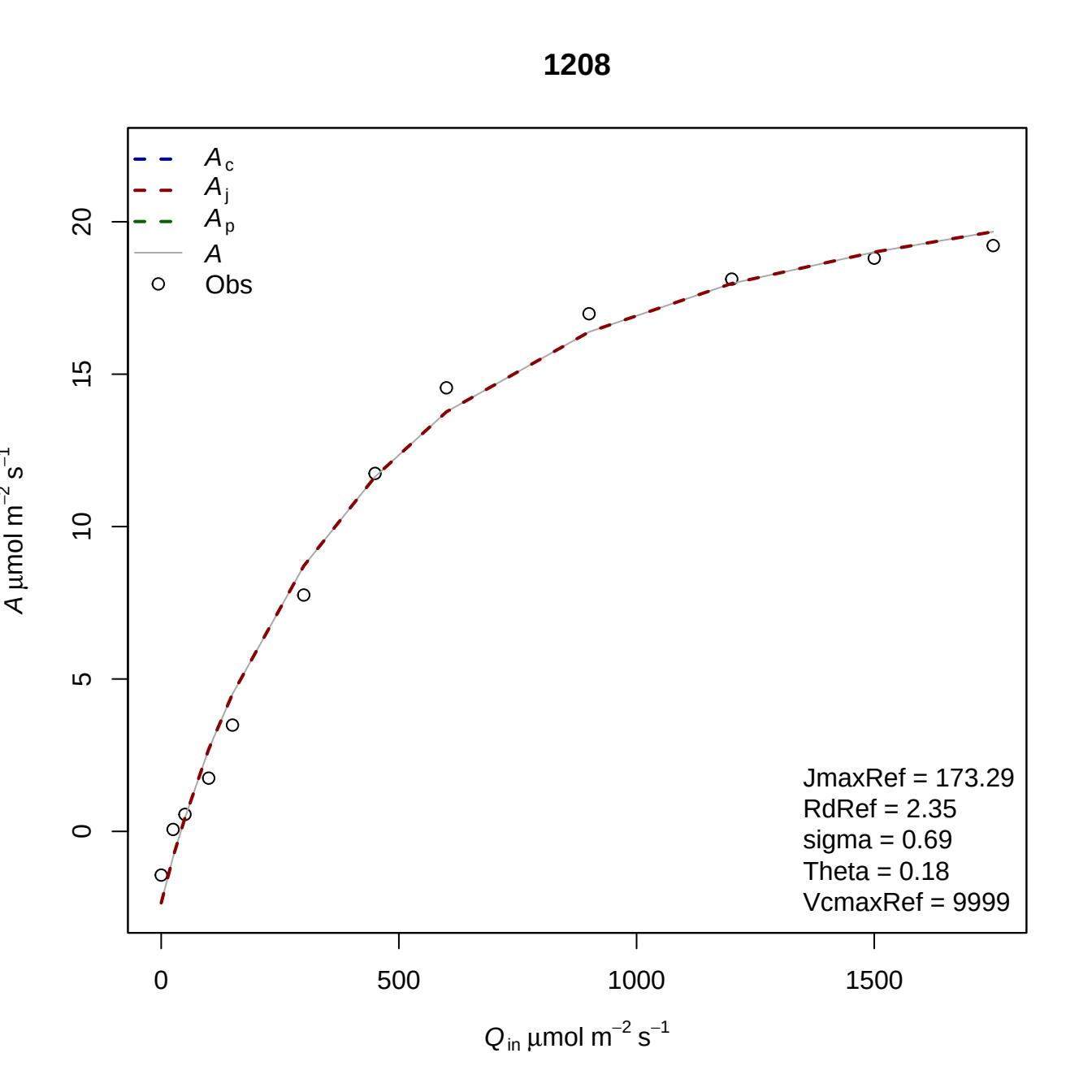

1203
A_c
A_j
A_p
A
Obs
JmaxRef = 141.32
RdRef = 1.45
sigma = 0.4
Theta = 0.45
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

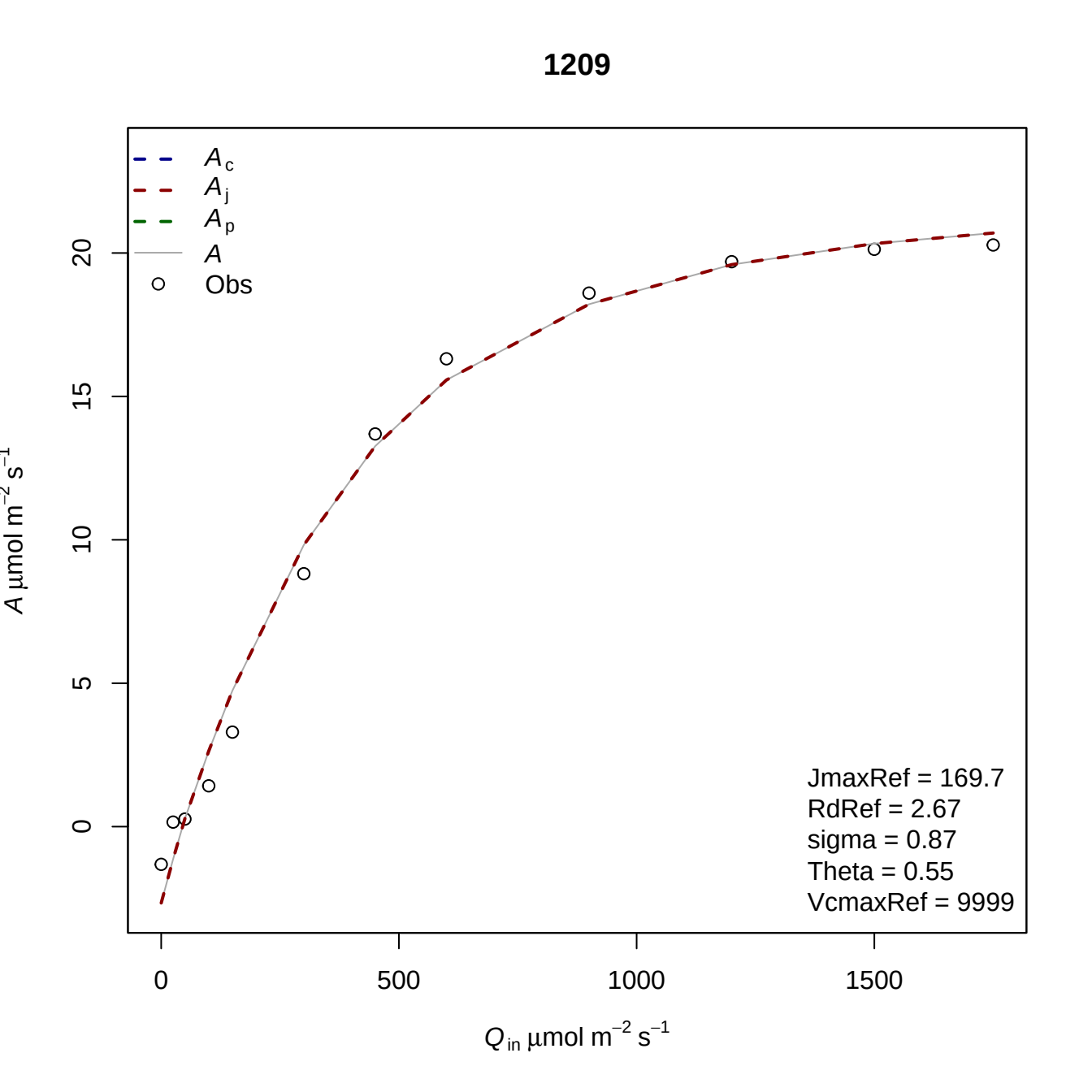

1204
A_c
A_j
A_p
A
Obs
JmaxRef = 85.04
RdRef = 1.16
sigma = 0.33
Theta = −0.39
VcmaxRef = 50.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

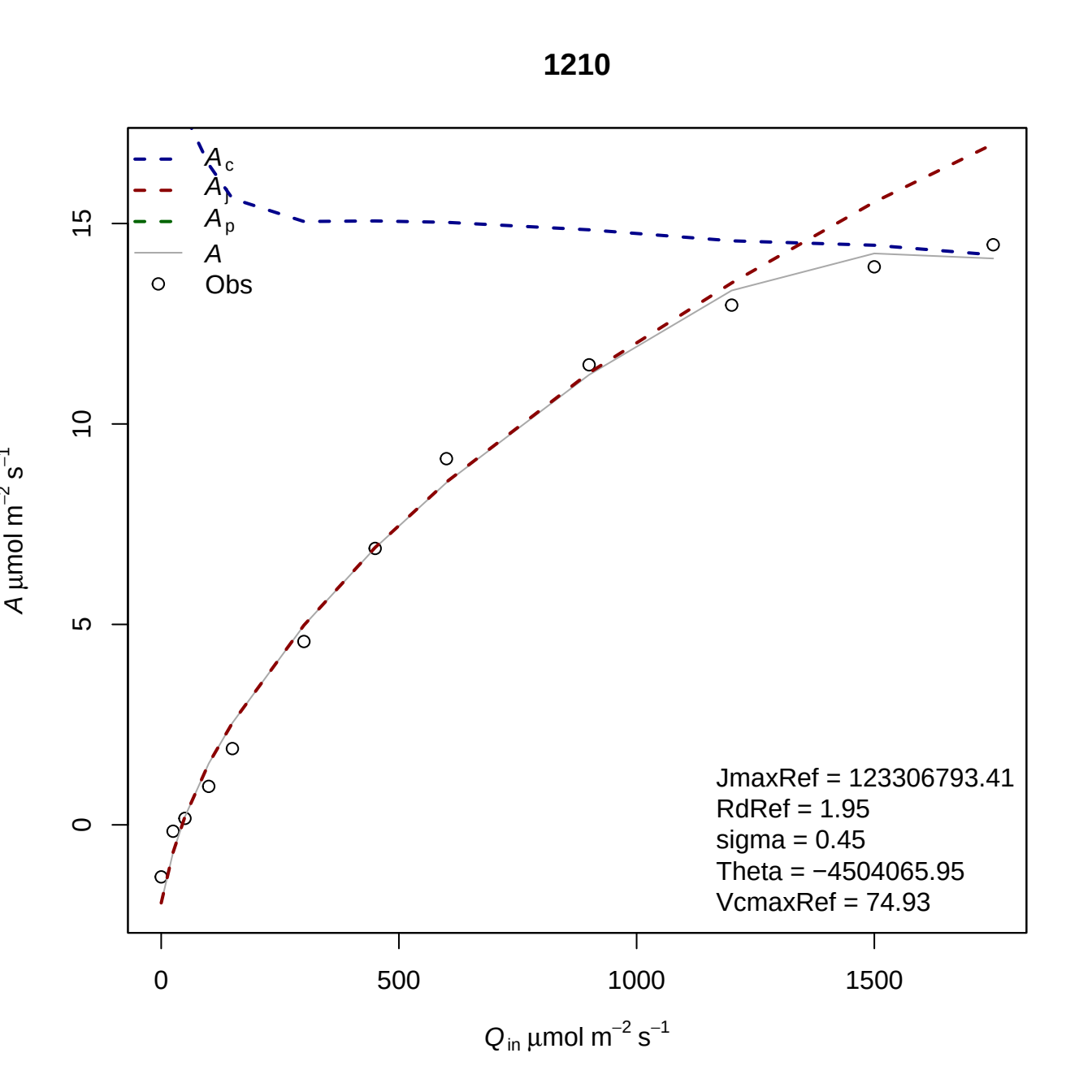

1205
A_c
A_j
A_p
A
Obs
JmaxRef = 105049632.17
RdRef = 1.64
sigma = 0.46
Theta = −2397800.77
VcmaxRef = 68.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

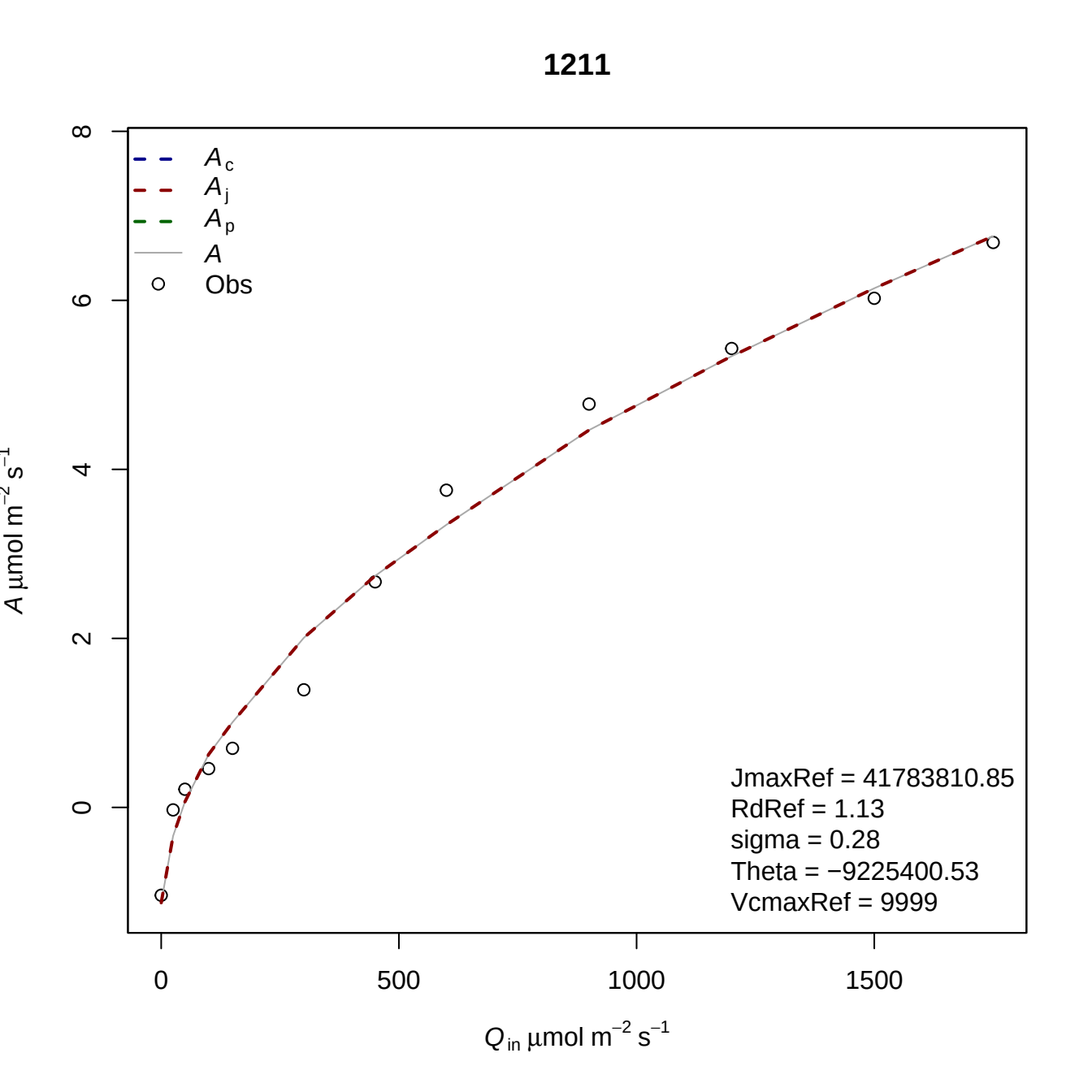

1206
A_c
A_j
A_p
A
Obs
JmaxRef = 143.47
RdRef = 2.08
sigma = 0.79
Theta = 0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

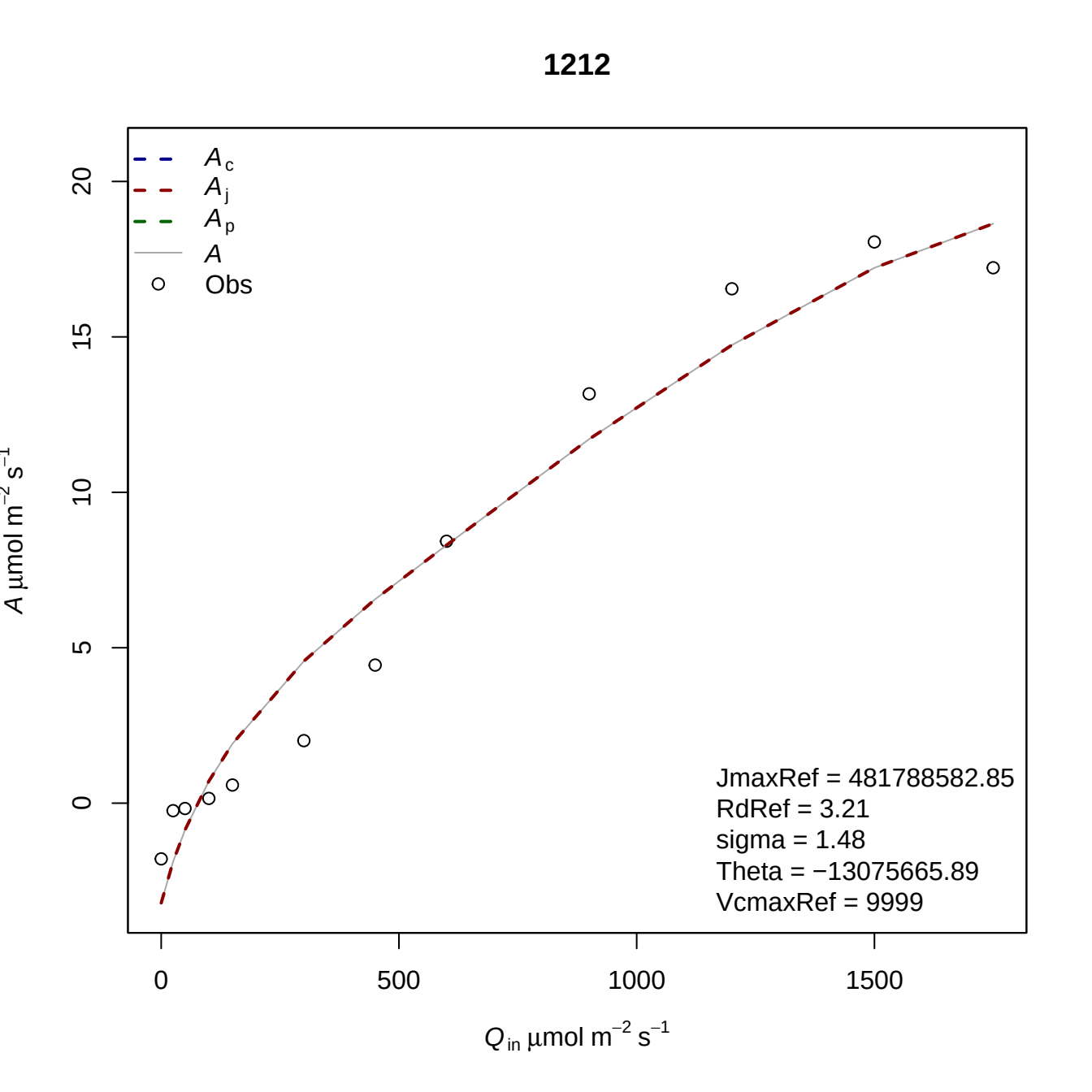

1207
A_c
A_j
A_p
A
Obs
JmaxRef = 122.19
RdRef = 1.89
sigma = 0.57
Theta = 0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

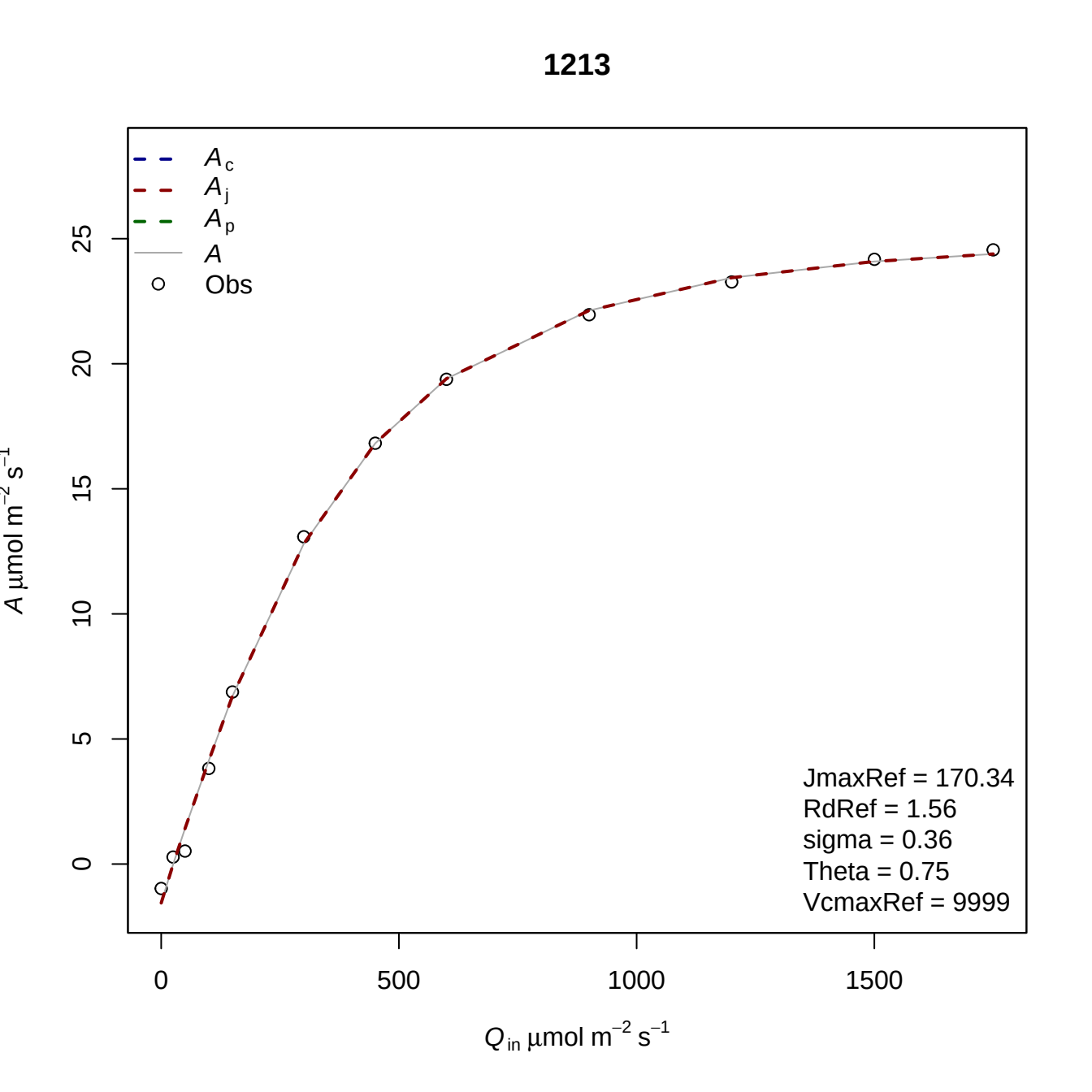

1208
A_c
A_j
A_p
A
Obs
JmaxRef = 173.29
RdRef = 2.35
sigma = 0.69
Theta = 0.18
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

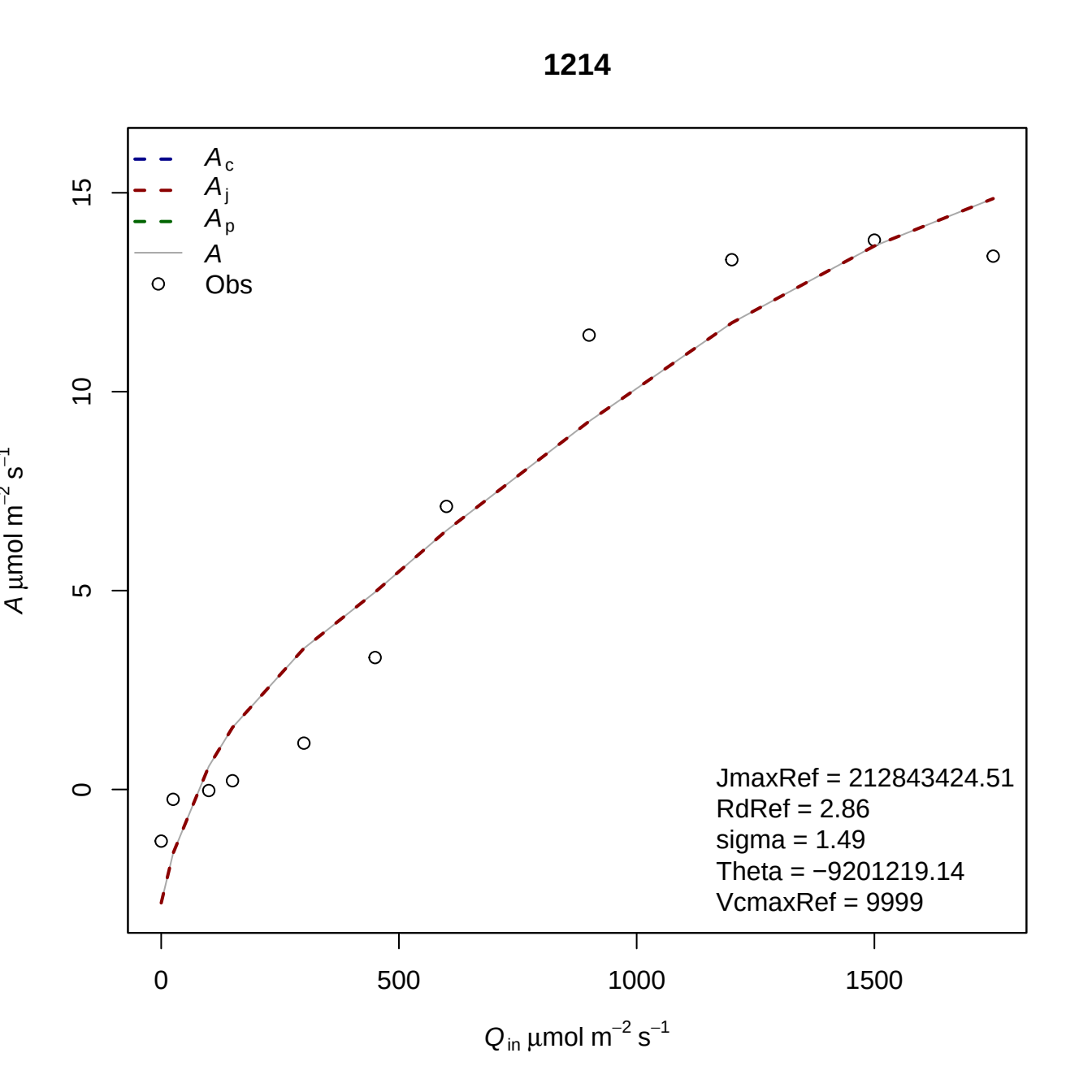

1209
A_c
A_j
A_p
A
Obs
JmaxRef = 169.7
RdRef = 2.67
sigma = 0.87
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

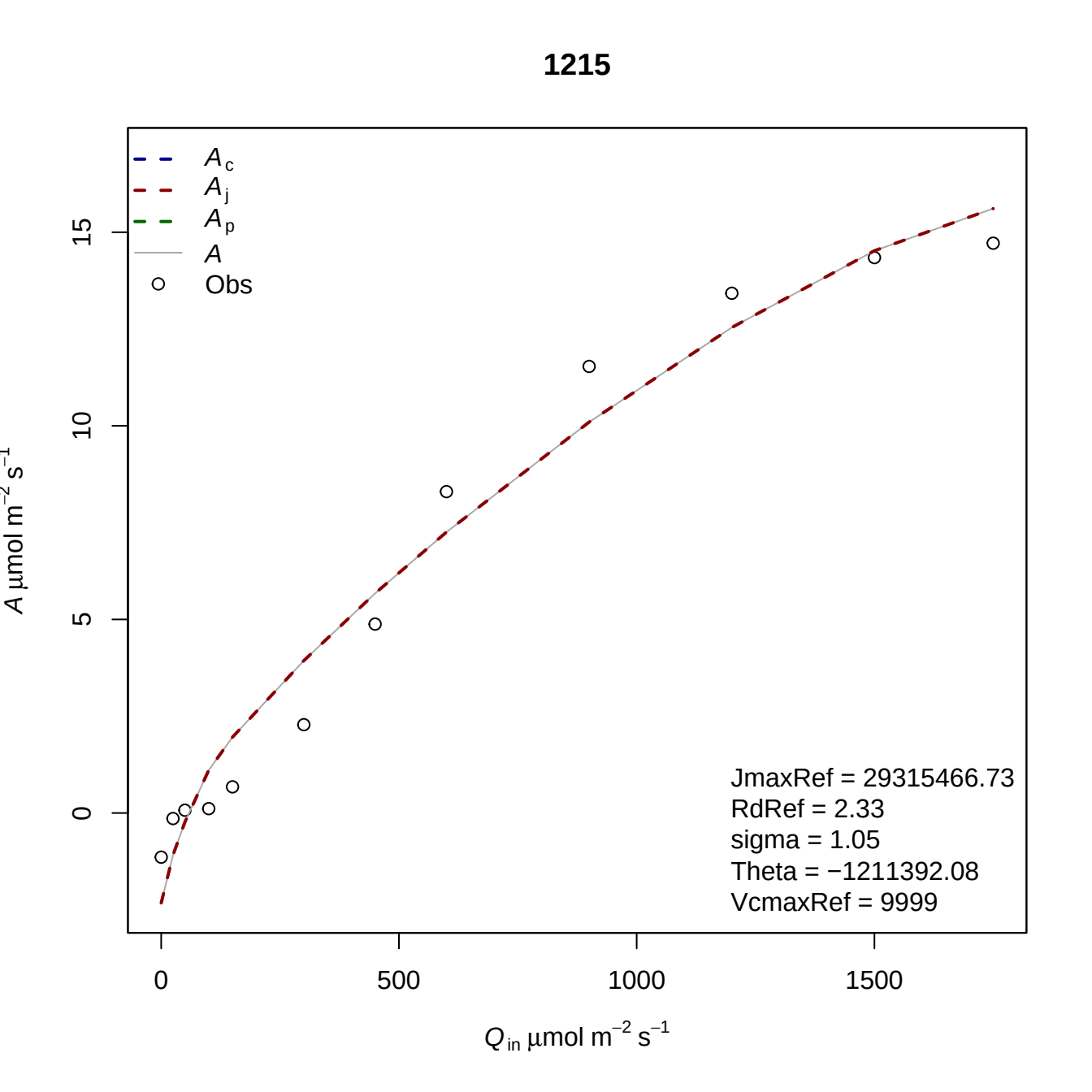

1210
A_c
A_j
A_p
A
Obs
JmaxRef = 123306793.41
RdRef = 1.95
sigma = 0.45
Theta = −4504065.95
VcmaxRef = 74.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

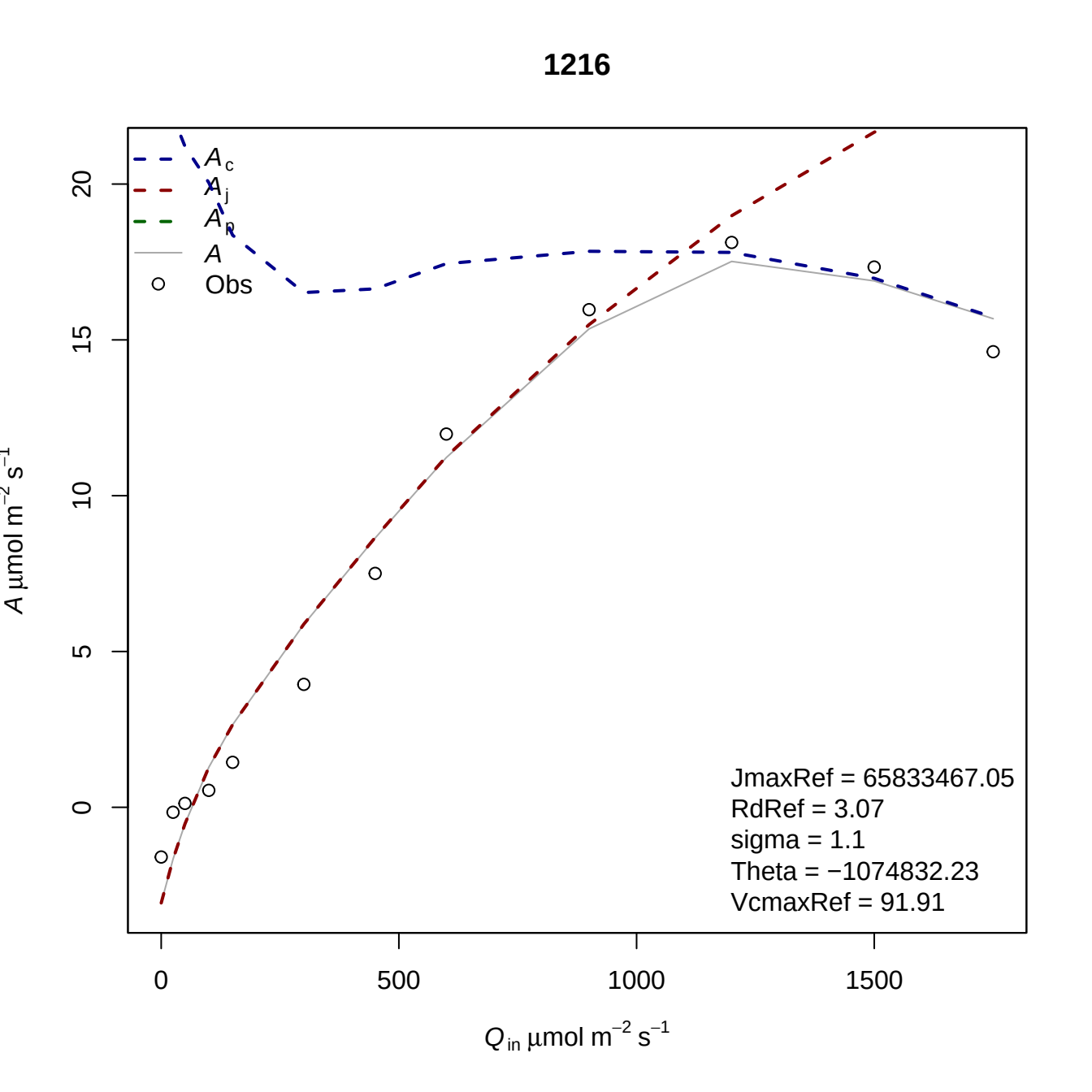

1211
A_c
A_j
A_p
A
Obs
JmaxRef = 41783810.85
RdRef = 1.13
sigma = 0.28
Theta = −9225400.53
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

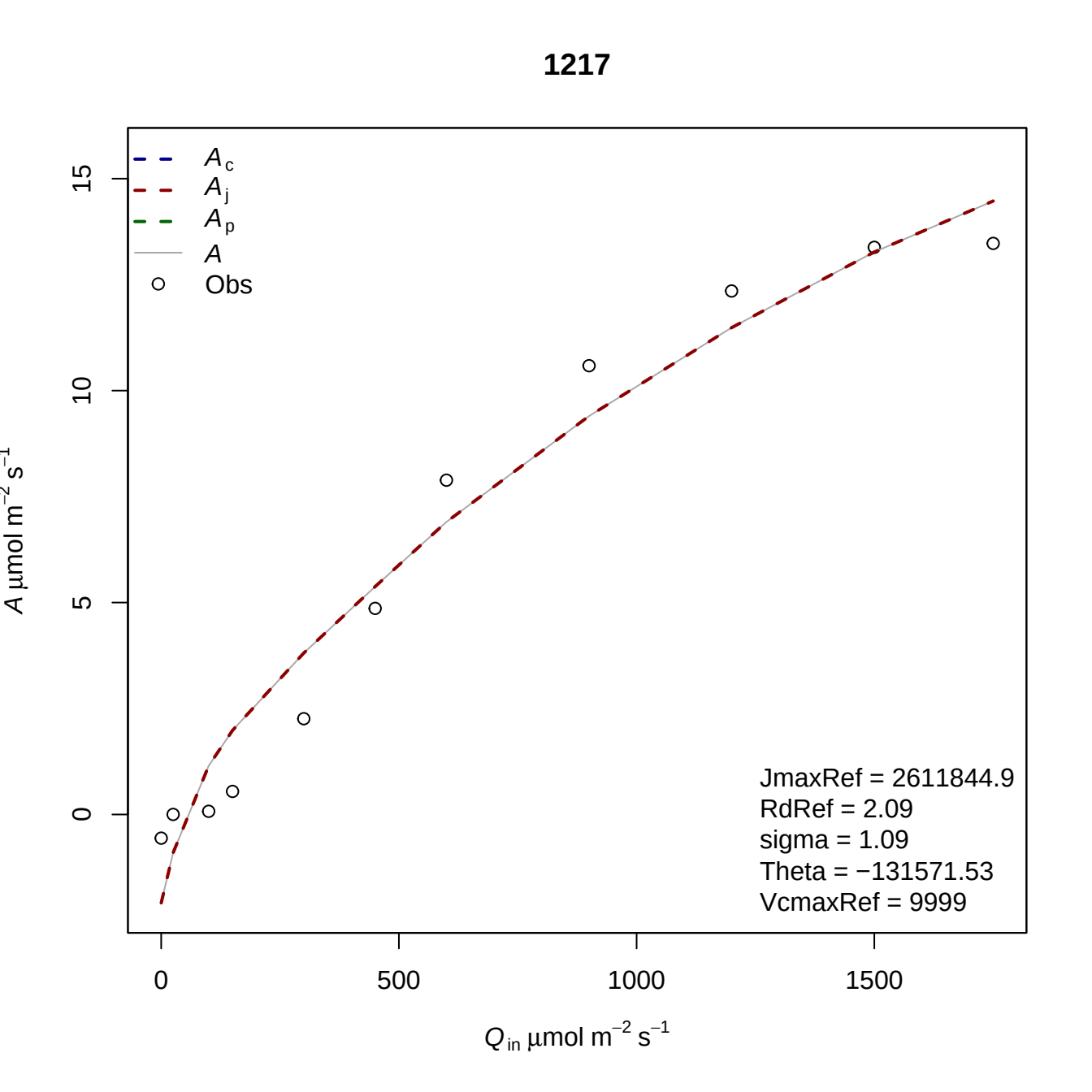

1212
A_c
A_j
A_p
A
Obs
JmaxRef = 481788582.85
RdRef = 3.21
sigma = 1.48
Theta = −13075665.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

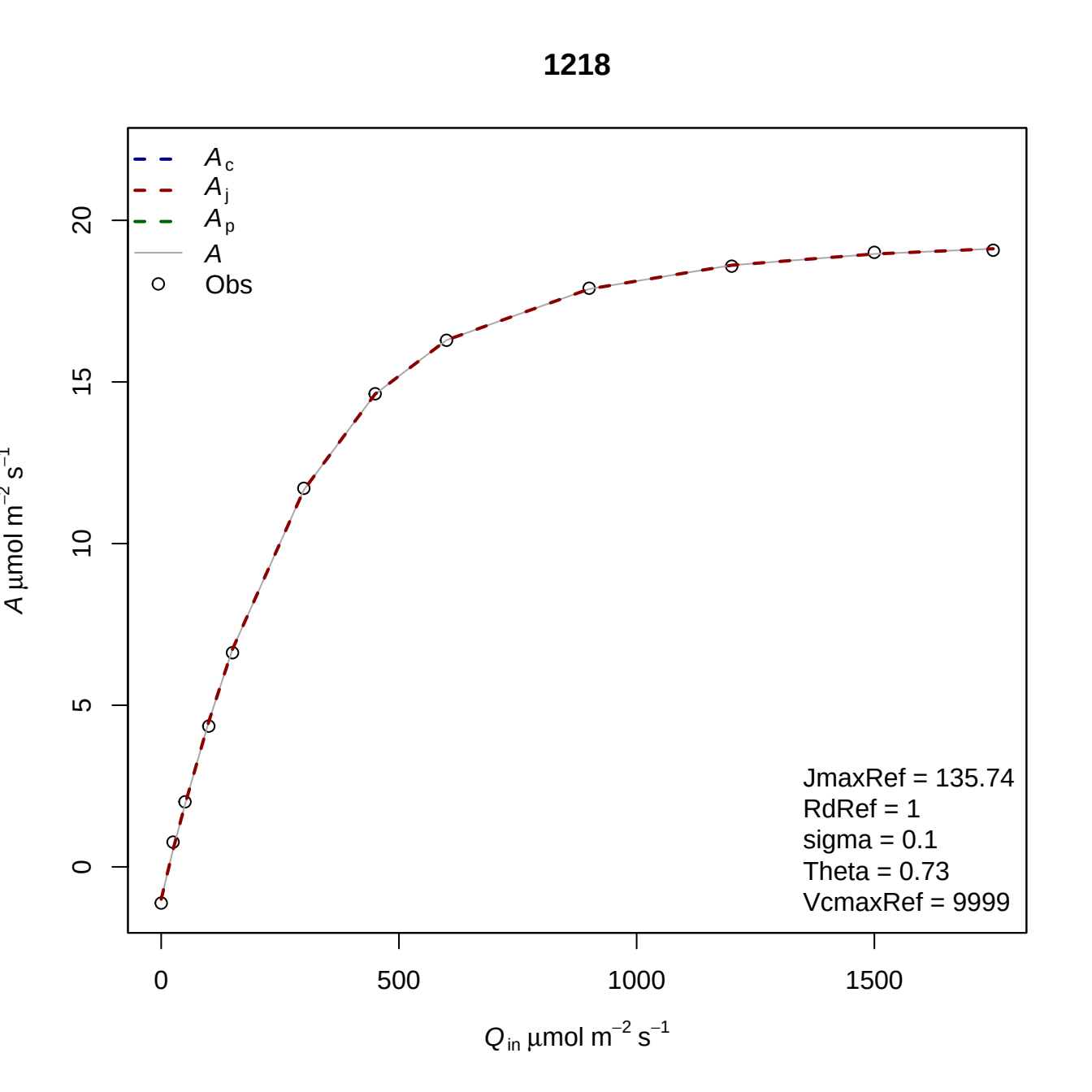

1213
A_c
A_j
A_p
A
Obs
JmaxRef = 170.34
RdRef = 1.56
sigma = 0.36
Theta = 0.75
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

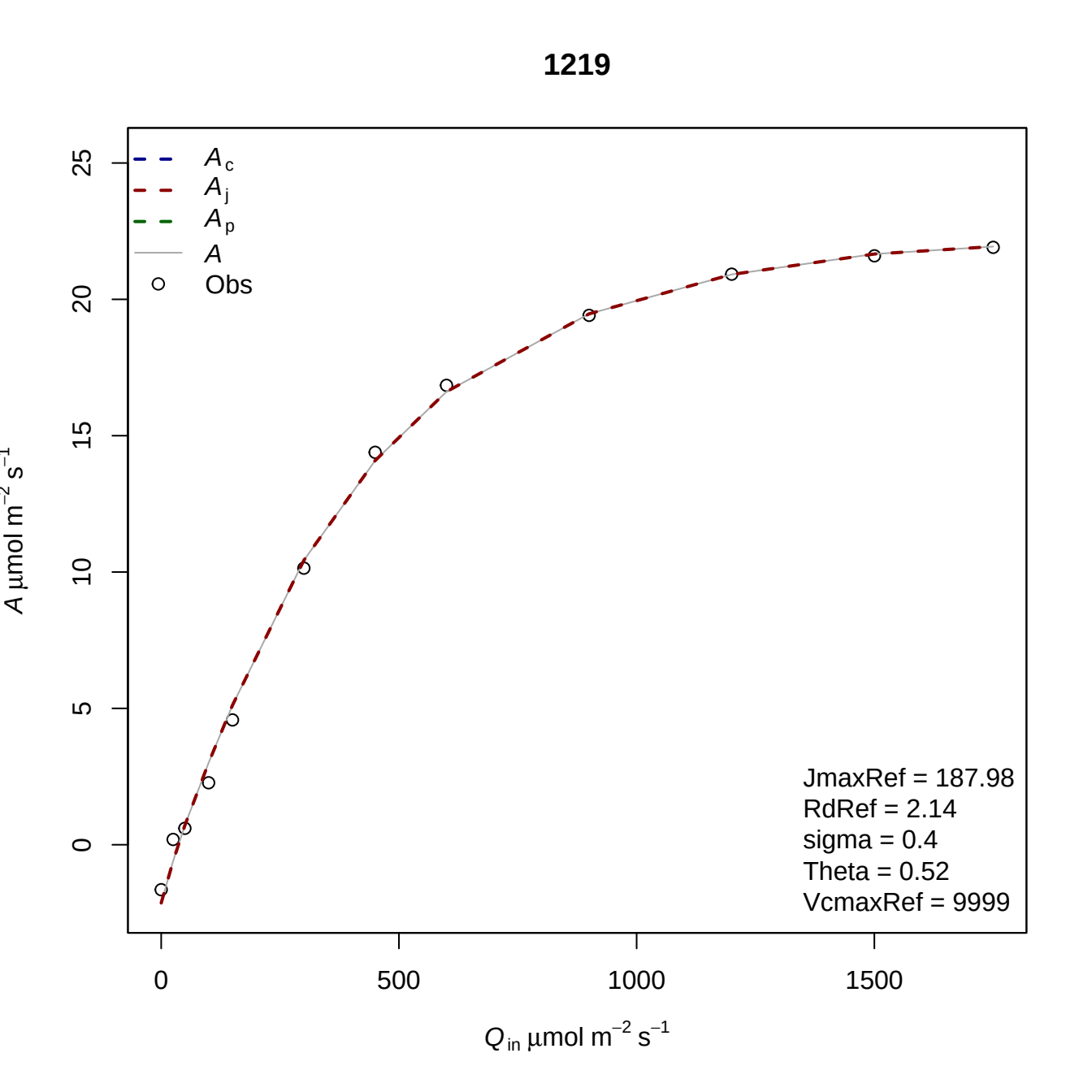

1214
A_c
A_j
A_p
A
Obs
JmaxRef = 212843424.51
RdRef = 2.86
sigma = 1.49
Theta = −9201219.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

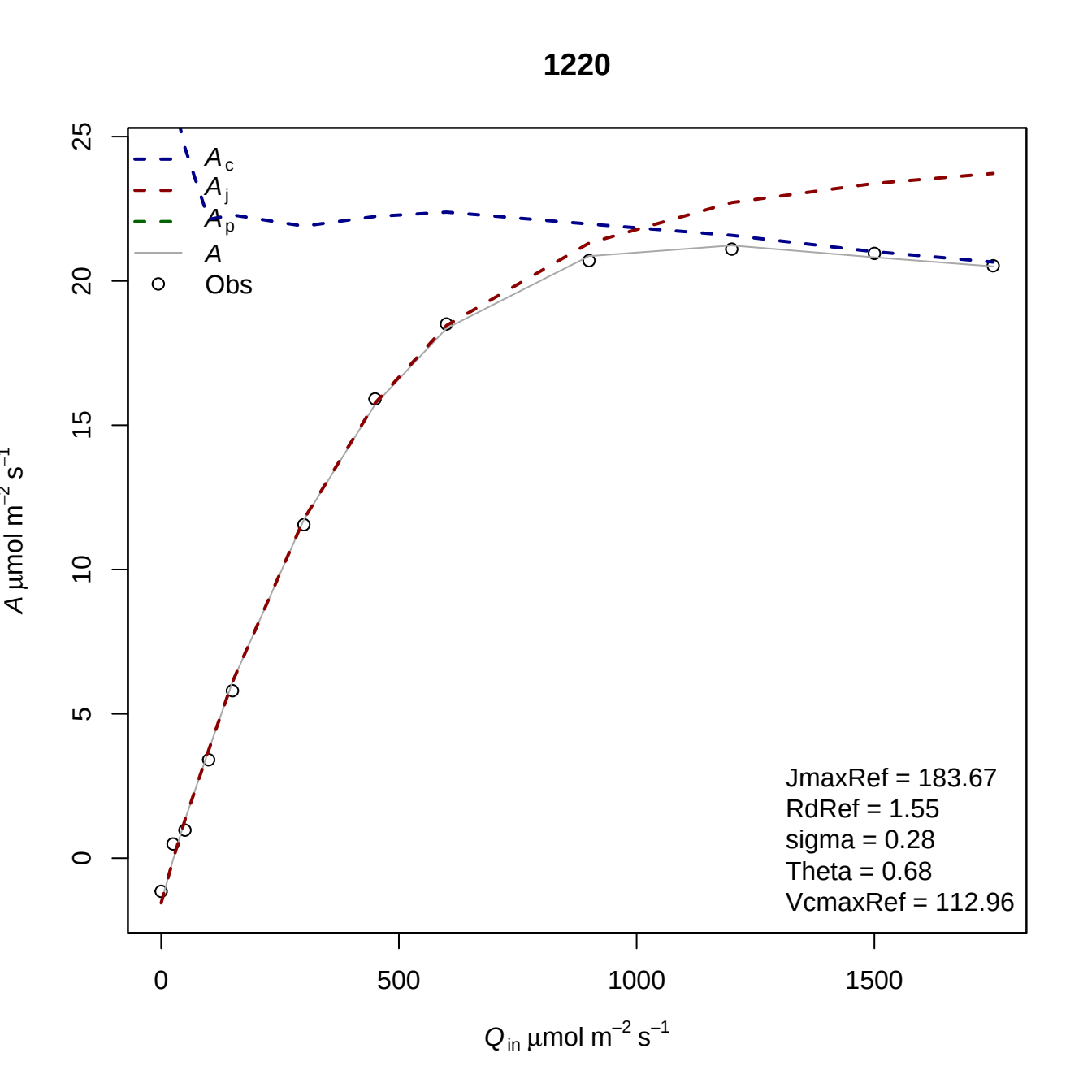

1215
A_c
A_j
A_p
A
Obs
JmaxRef = 29315466.73
RdRef = 2.33
sigma = 1.05
Theta = −1211392.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

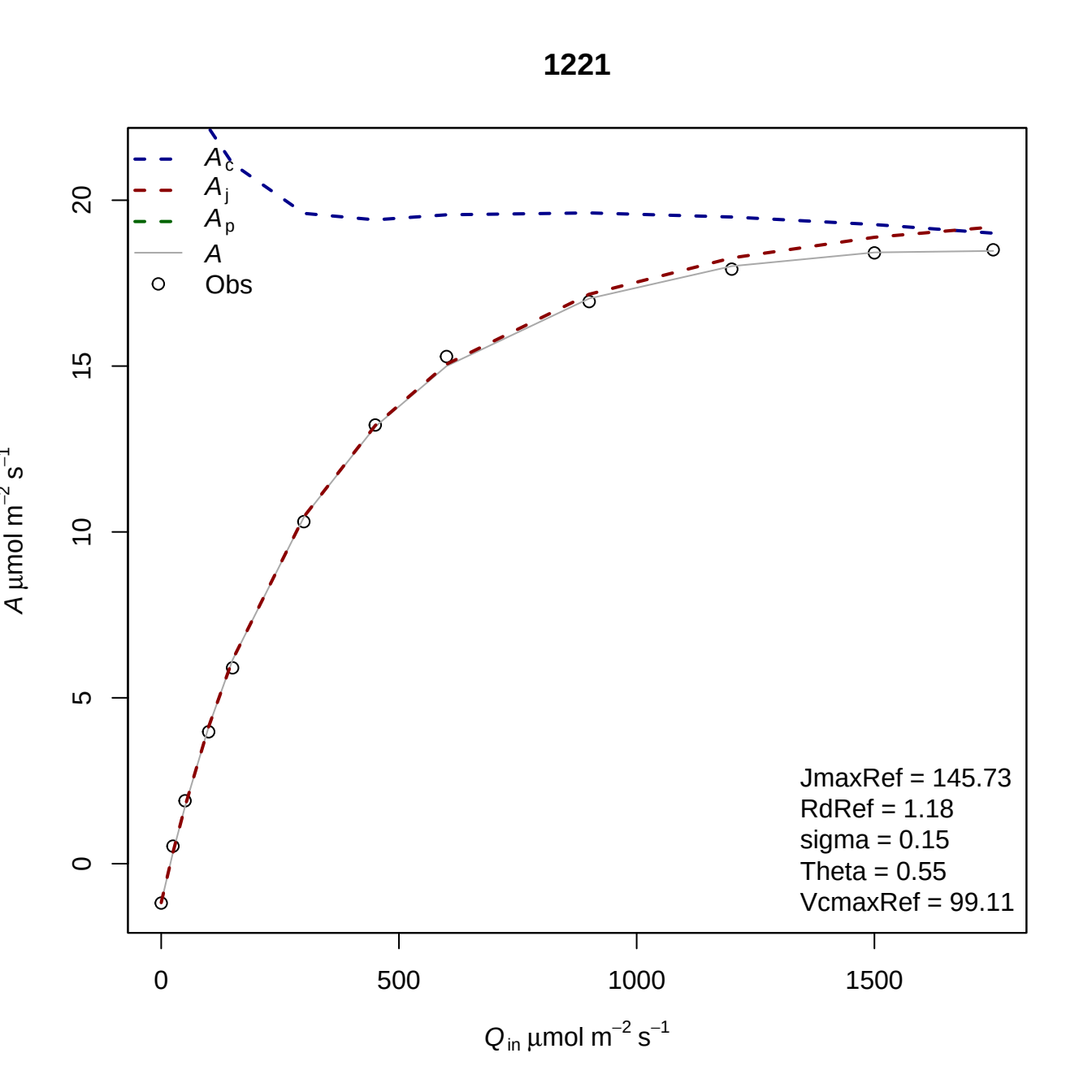

1216
A_c
A_j
A_p
A
Obs
JmaxRef = 65833467.05
RdRef = 3.07
sigma = 1.1
Theta = −1074832.23
VcmaxRef = 91.91
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

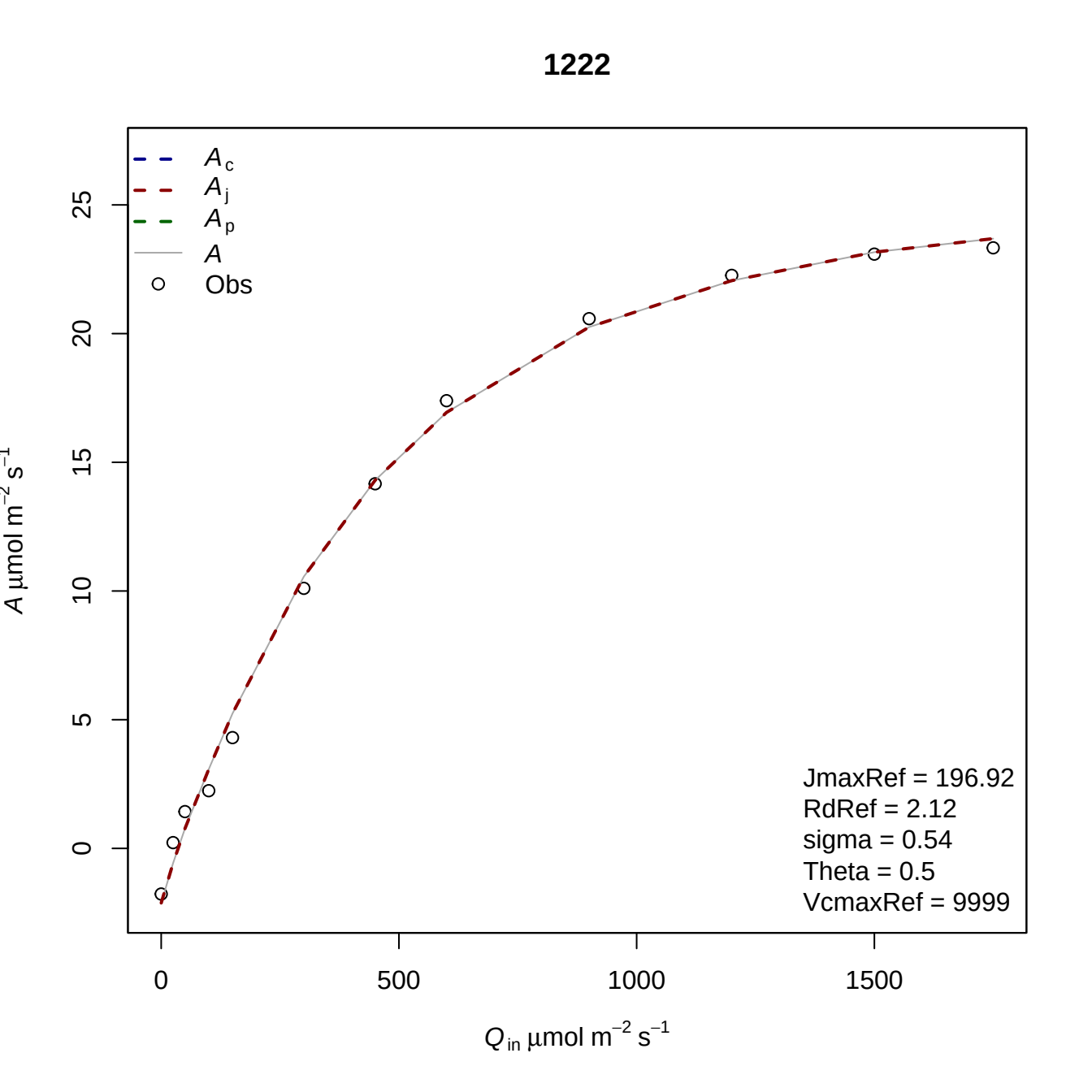

1217
A_c
A_j
A_p
A
Obs
JmaxRef = 2611844.9
RdRef = 2.09
sigma = 1.09
Theta = −131571.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

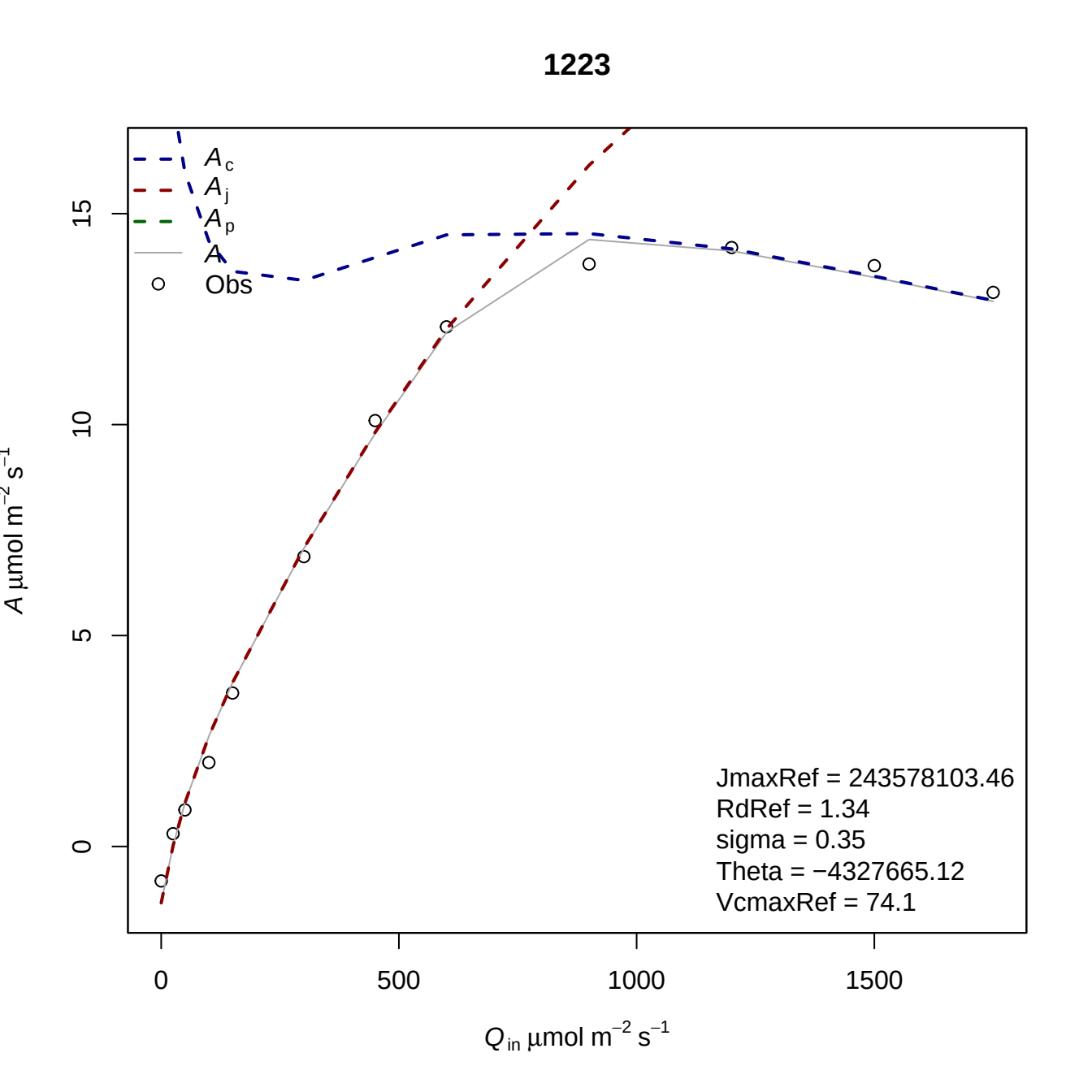

1218
A_c
A_j
A_p
A
Obs
JmaxRef = 135.74
RdRef = 1
sigma = 0.1
Theta = 0.73
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

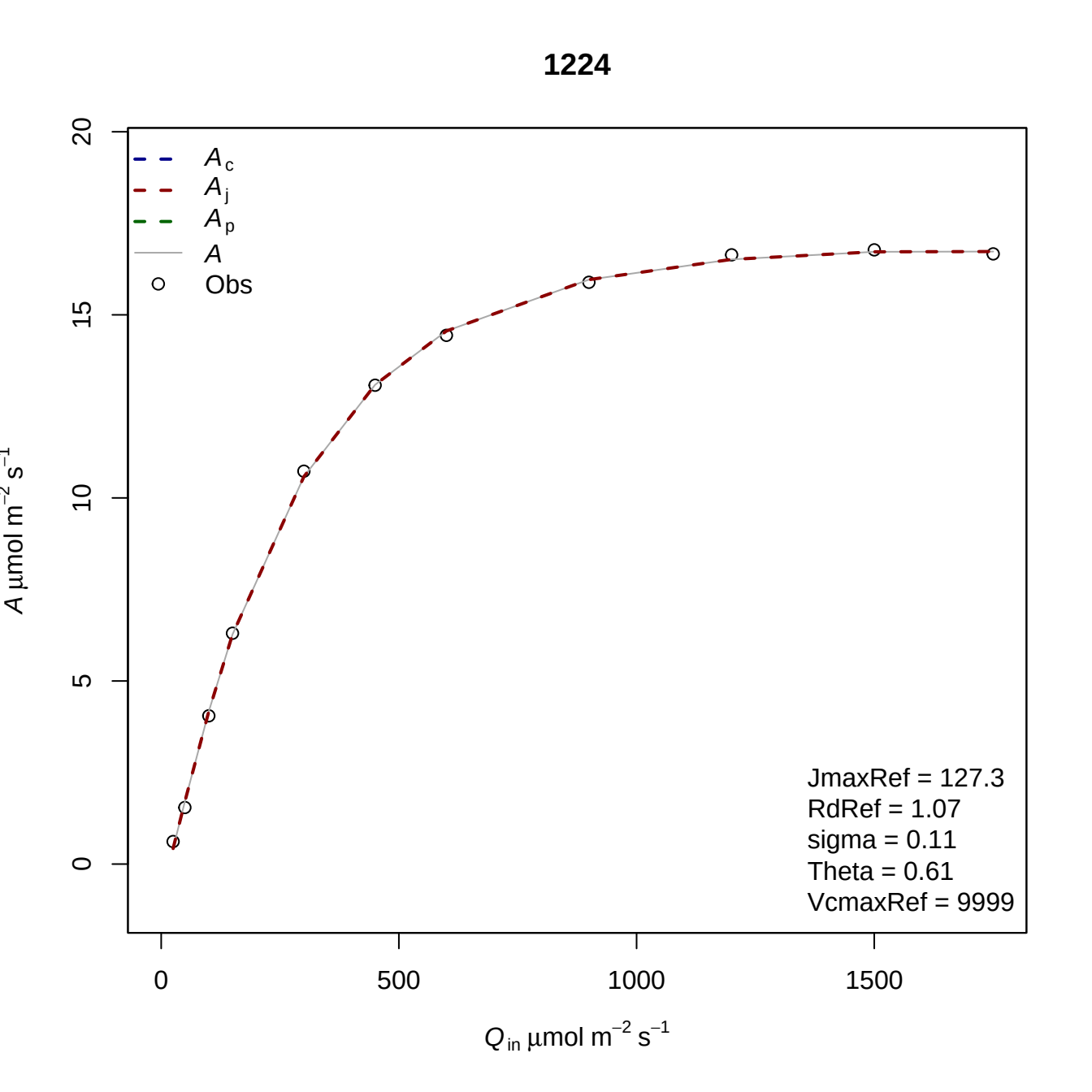

1219
A_c
A_j
A_p
A
Obs
JmaxRef = 187.98
RdRef = 2.14
sigma = 0.4
Theta = 0.52
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

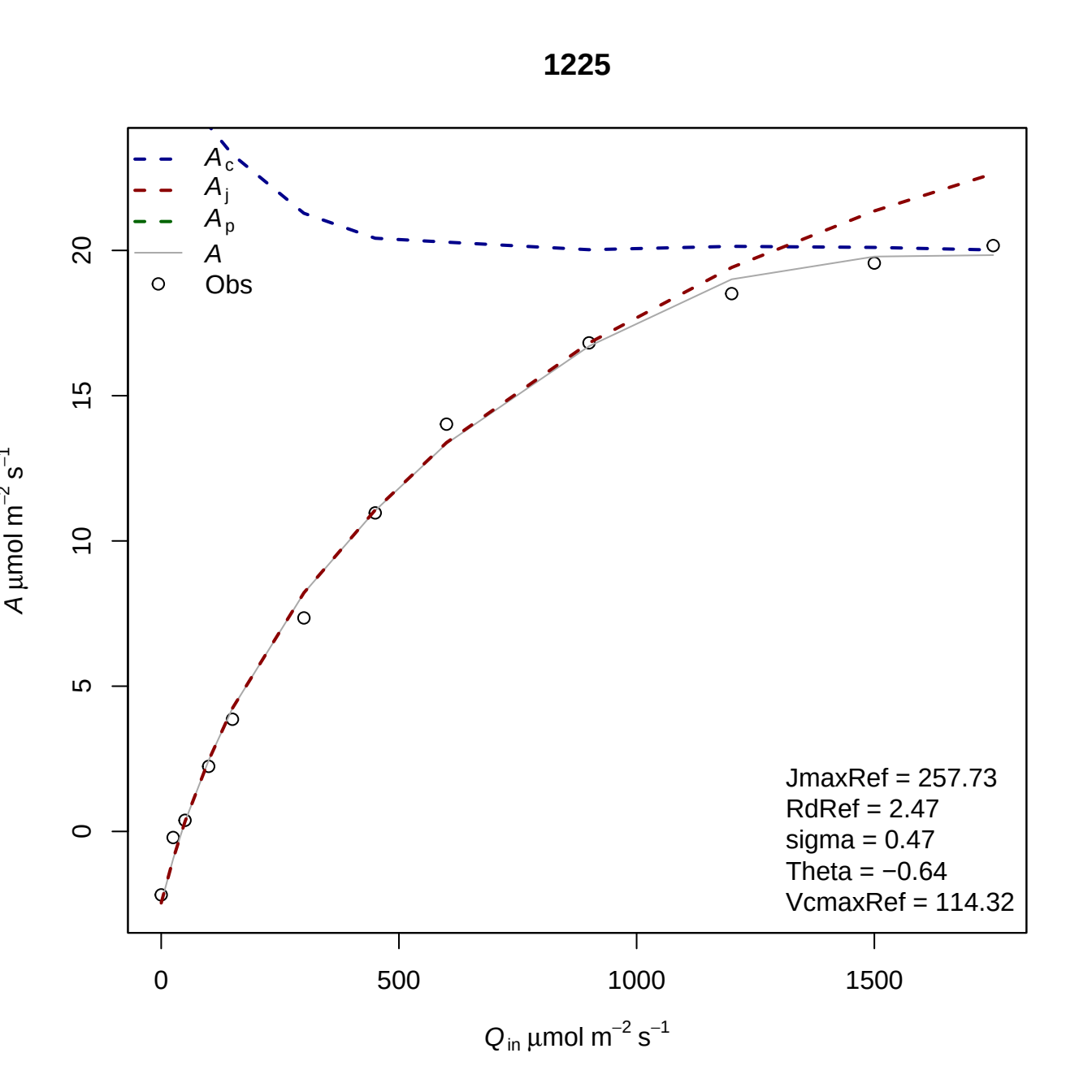

1220
A_c
A_j
A_p
A
Obs
JmaxRef = 183.67
RdRef = 1.55
sigma = 0.28
Theta = 0.68
VcmaxRef = 112.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

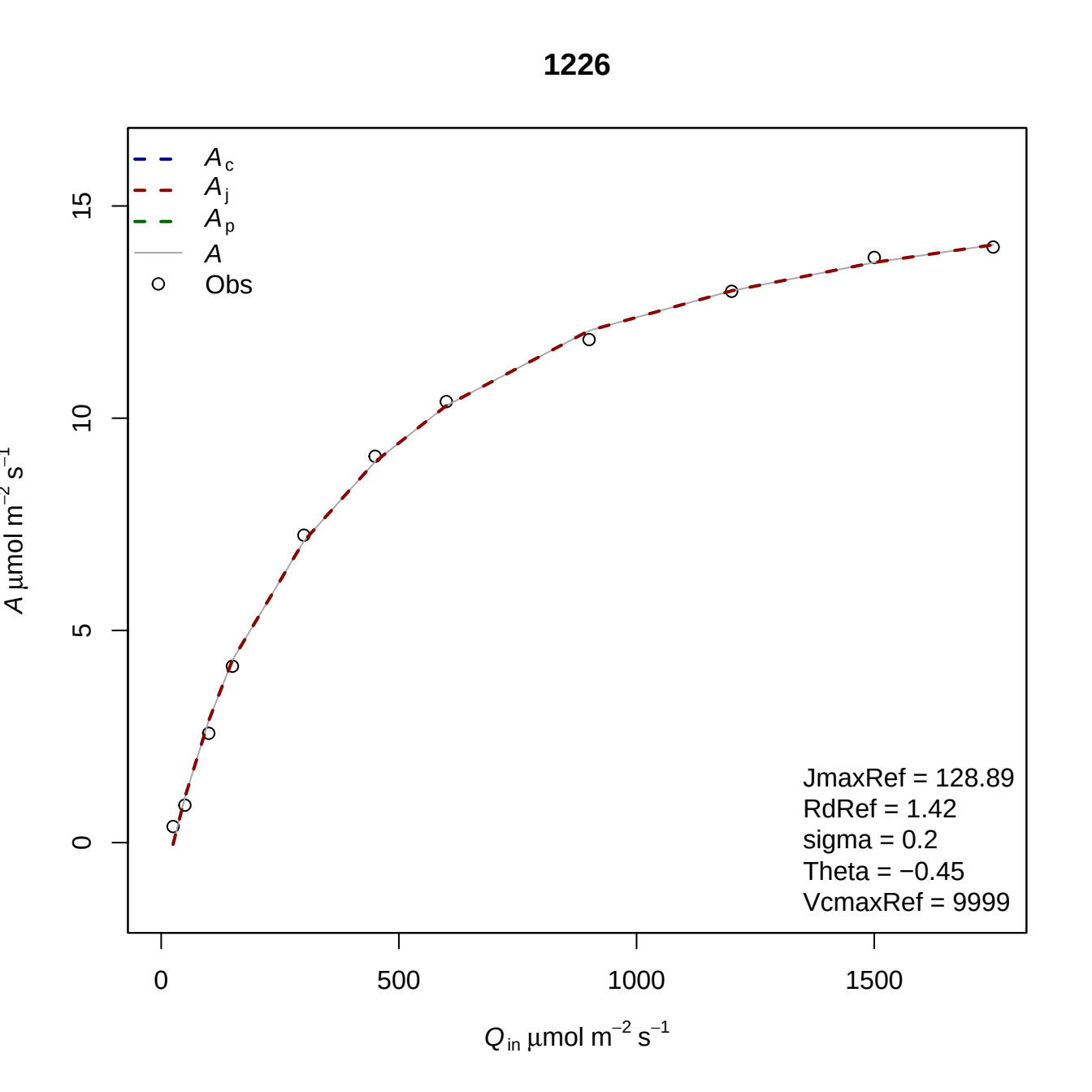

1221
A_c
A_j
A_p
A
Obs
JmaxRef = 145.73
RdRef = 1.18
sigma = 0.15
Theta = 0.55
VcmaxRef = 99.11
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

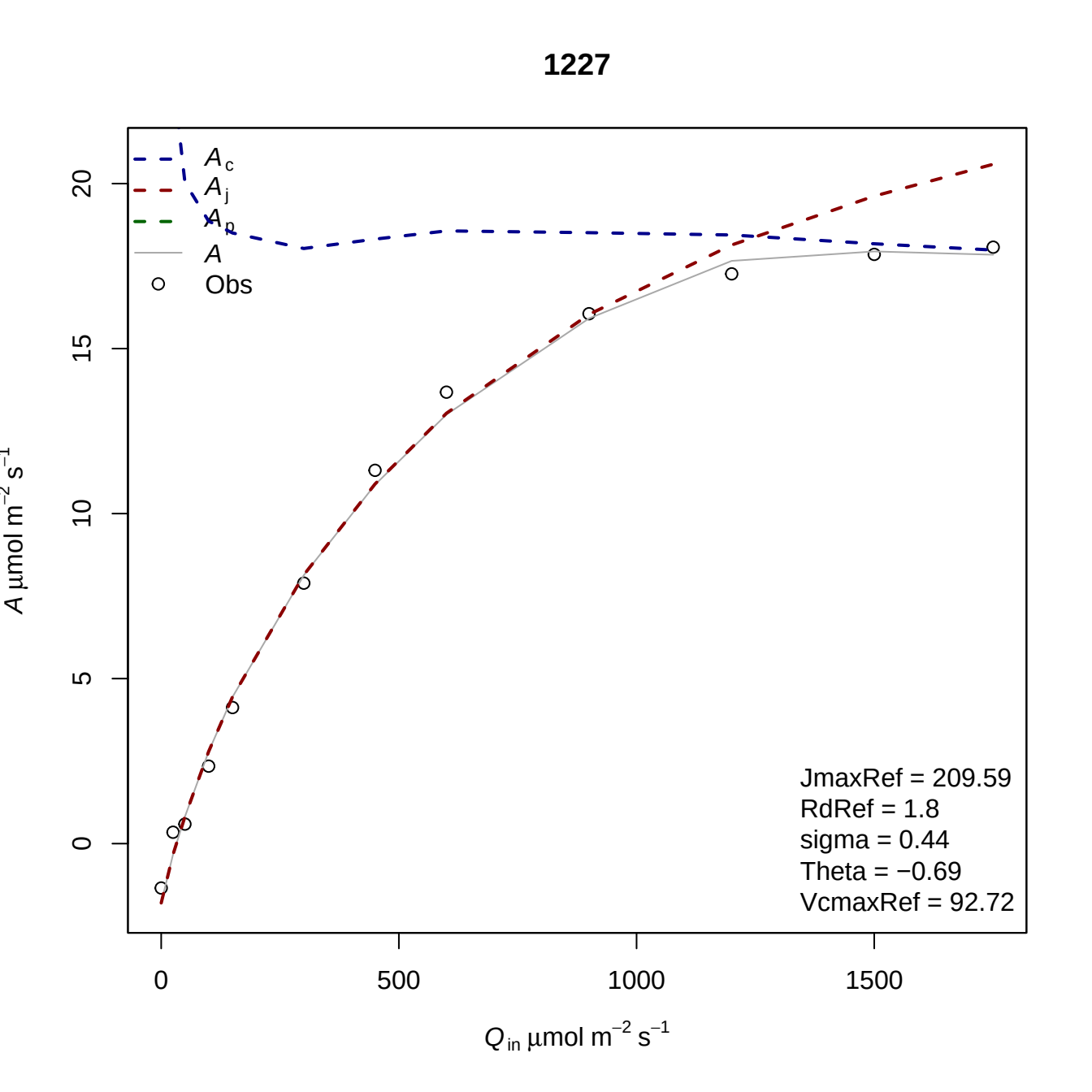

1222
A_c
A_j
A_p
A
Obs
JmaxRef = 196.92
RdRef = 2.12
sigma = 0.54
Theta = 0.5
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

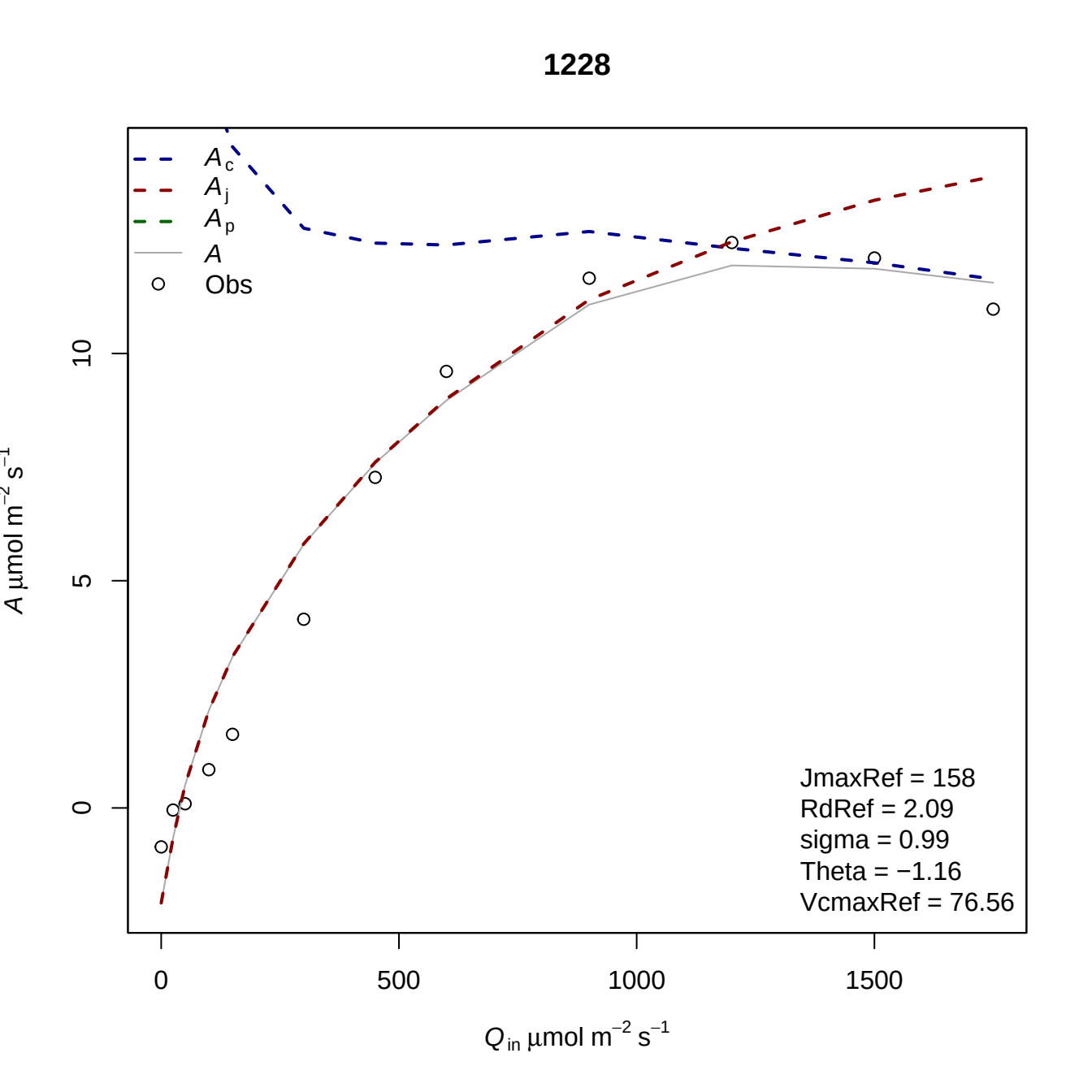

1223
A_c
A_j
A_p
A
Obs
JmaxRef = 243578103.46
RdRef = 1.34
sigma = 0.35
Theta = −4327665.12
VcmaxRef = 74.1
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

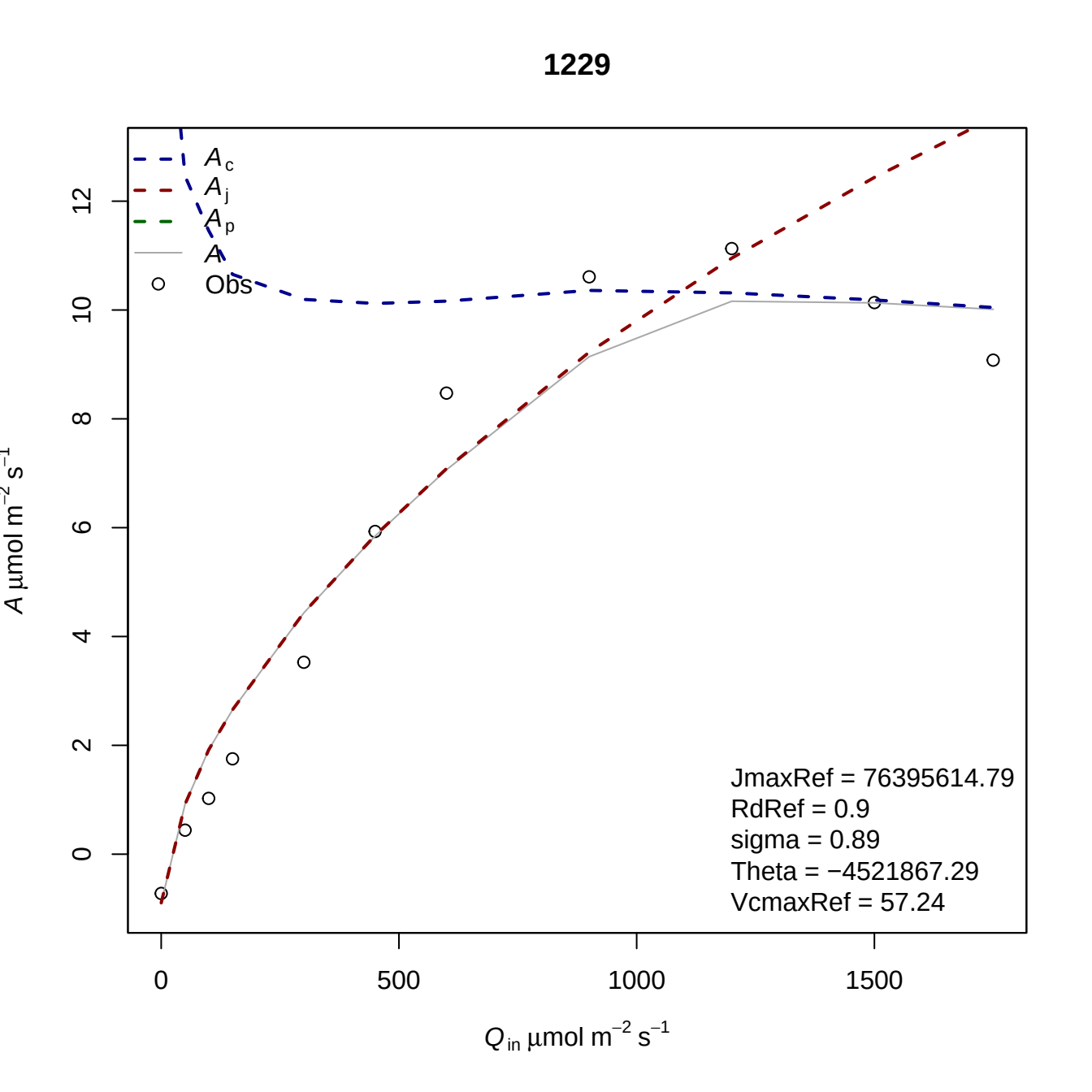

1224
A_c
A_j
A_p
A
Obs
JmaxRef = 127.3
RdRef = 1.07
sigma = 0.11
Theta = 0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

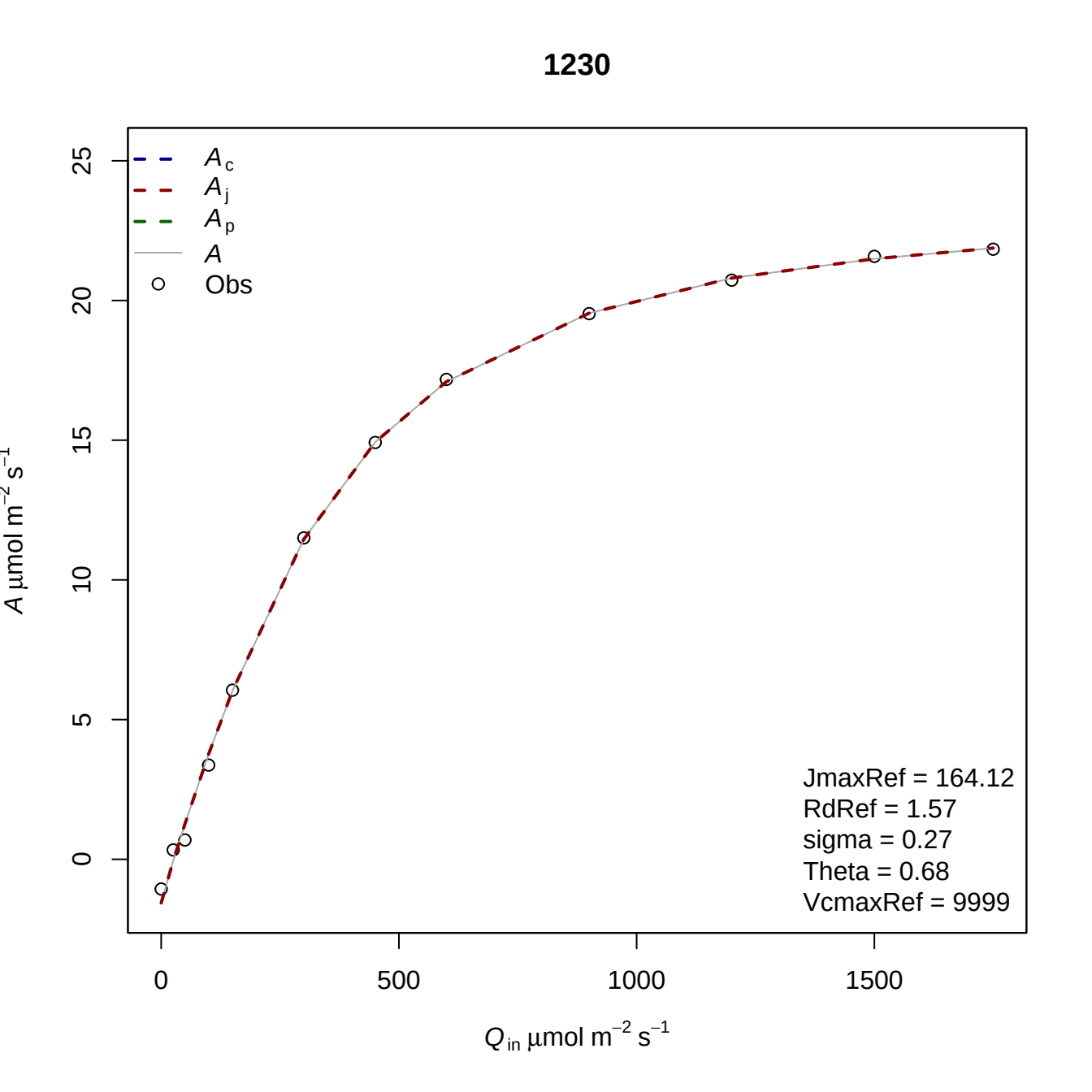

1225
A_c
A_j
A_p
A
Obs
JmaxRef = 257.73
RdRef = 2.47
sigma = 0.47
Theta = −0.64
VcmaxRef = 114.32
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

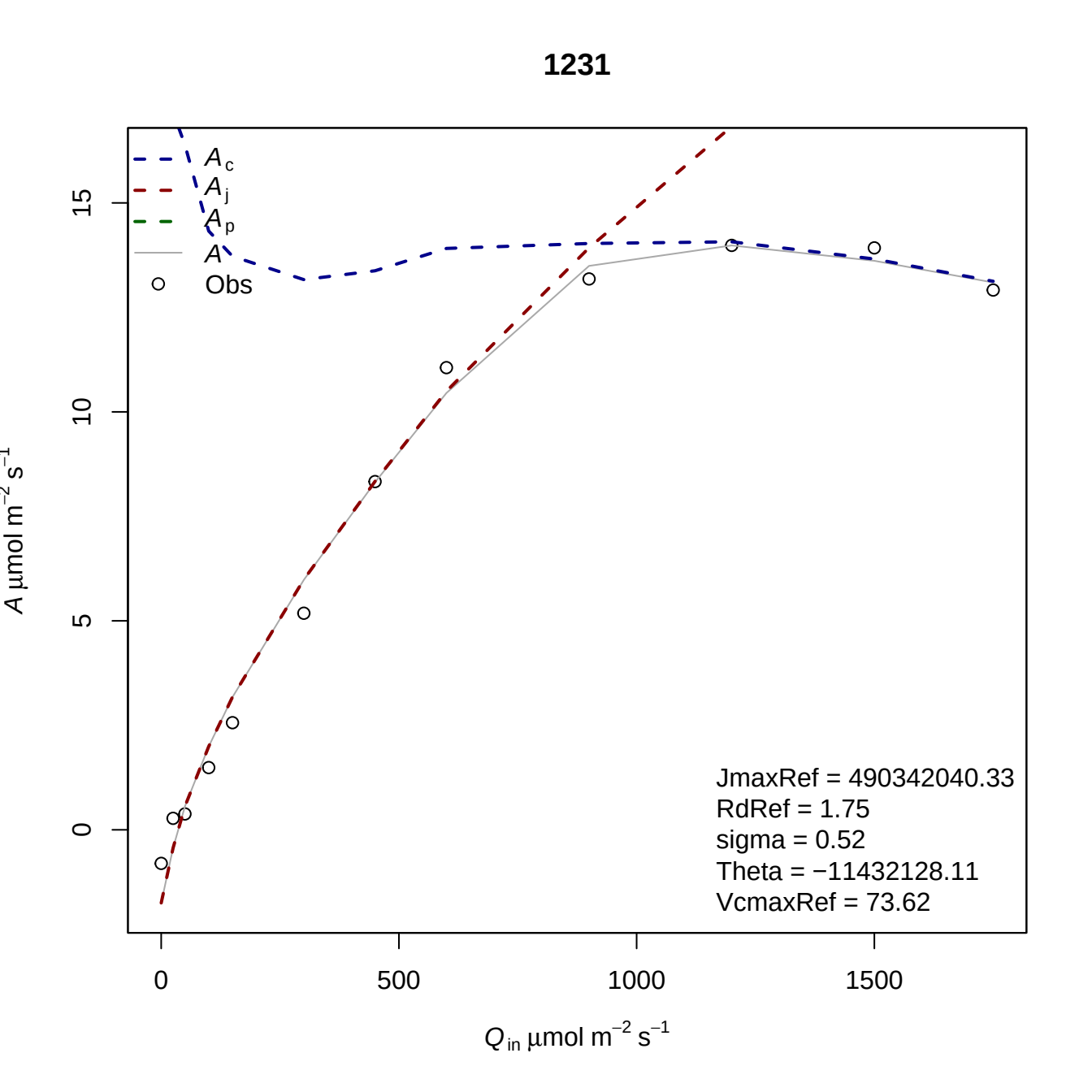

1226
A_c
A_j
A_p
A
Obs
JmaxRef = 128.89
RdRef = 1.42
sigma = 0.2
Theta = −0.45
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

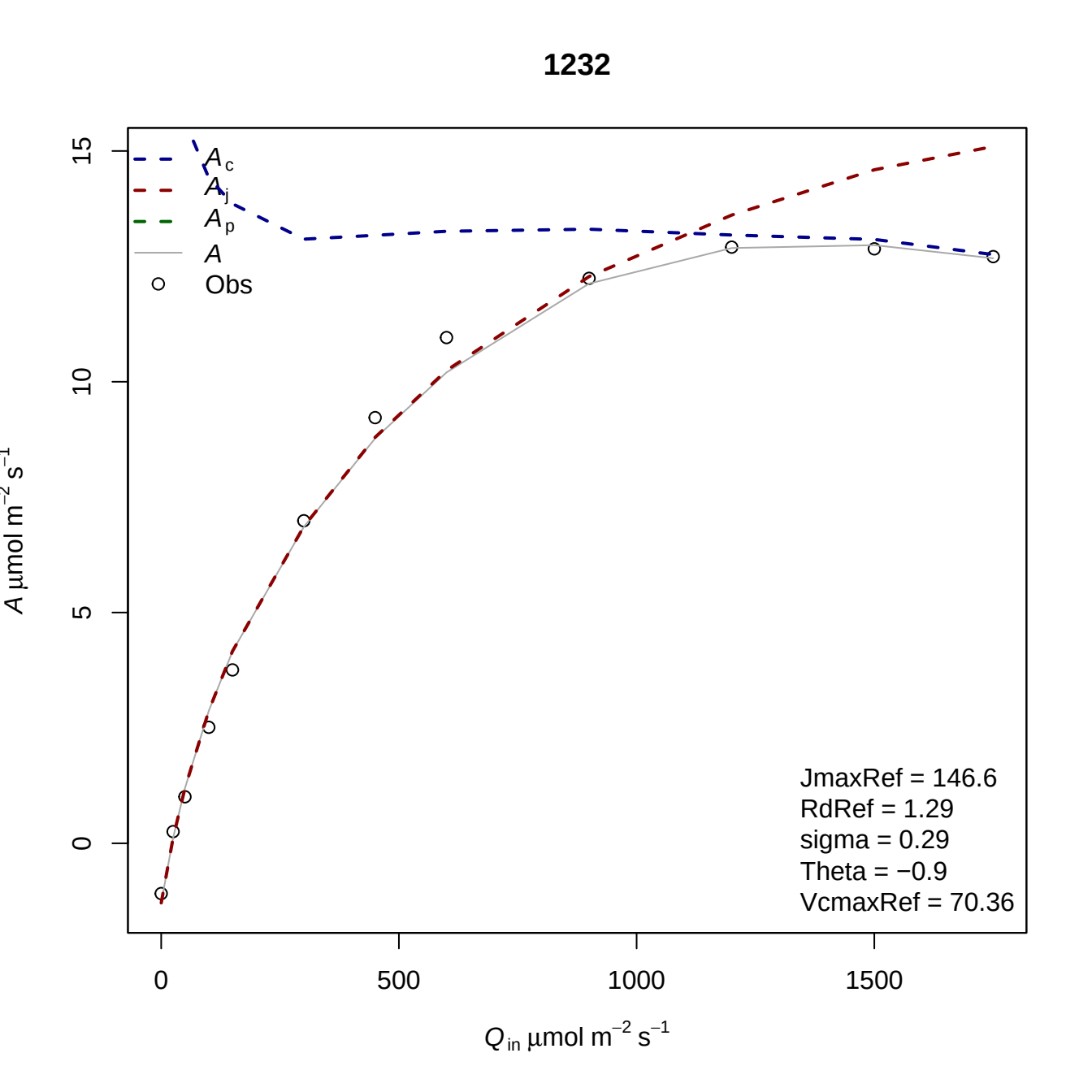

1227
A_c
A_j
A_p
A
Obs
JmaxRef = 209.59
RdRef = 1.8
sigma = 0.44
Theta = −0.69
VcmaxRef = 92.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

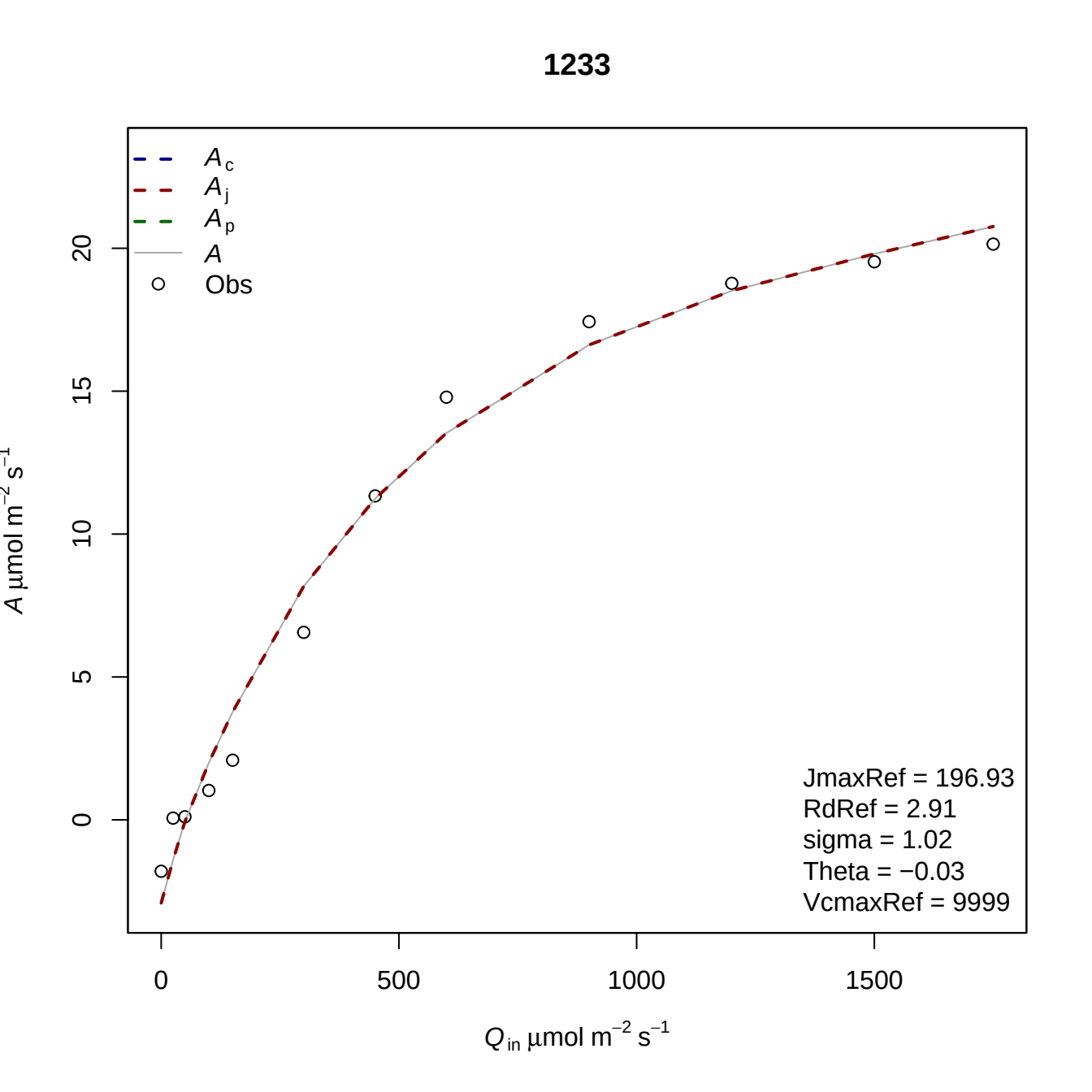

1228
A_c
A_j
A_p
A
Obs
JmaxRef = 158
RdRef = 2.09
sigma = 0.99
Theta = −1.16
VcmaxRef = 76.56
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

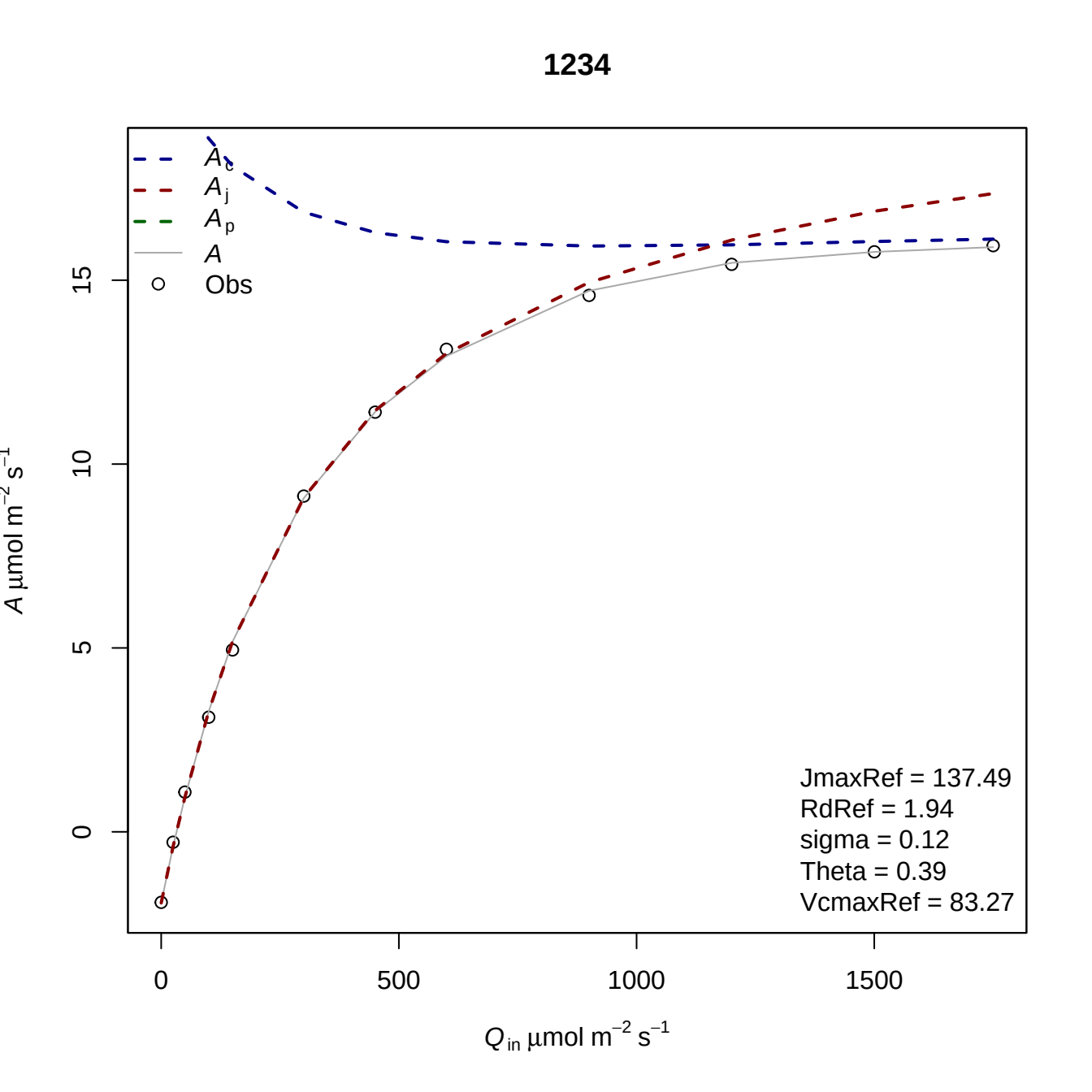

1229
A_c
A_j
A_p
A
Obs
JmaxRef = 76395614.79
RdRef = 0.9
sigma = 0.89
Theta = −4521867.29
VcmaxRef = 57.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

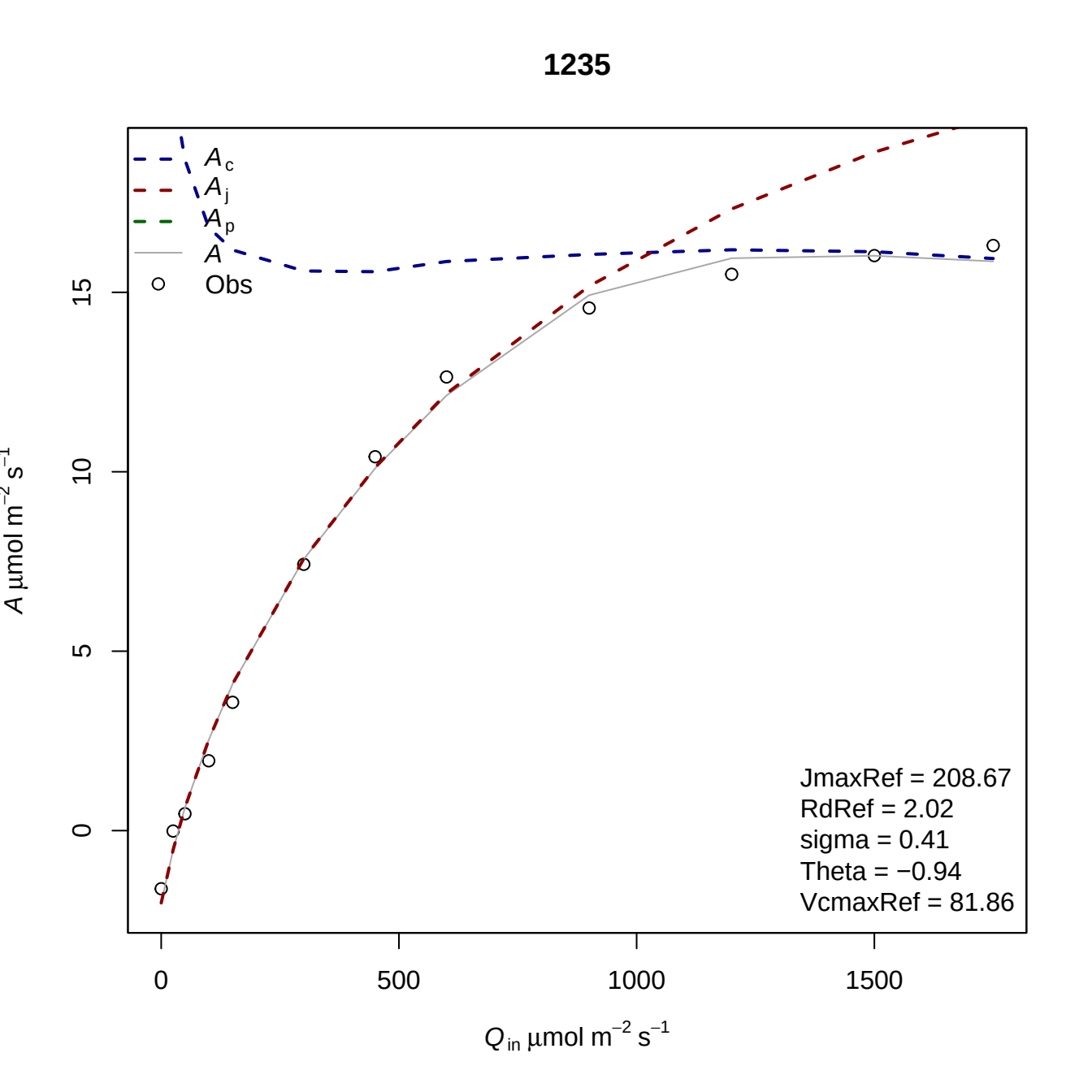

1230
A_c
A_j
A_p
A
Obs
JmaxRef = 164.12
RdRef = 1.57
sigma = 0.27
Theta = 0.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

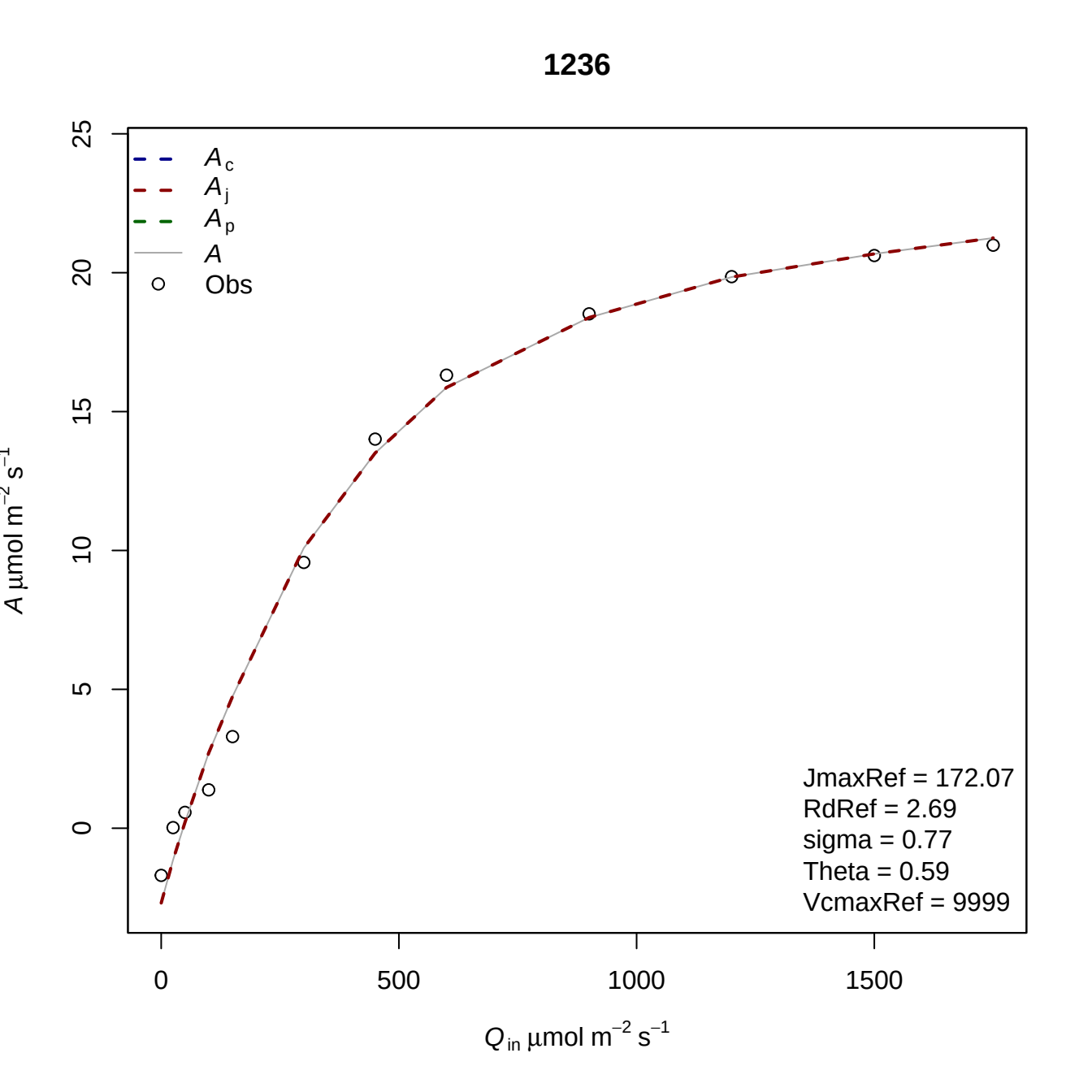

1231
A_c
A_j
A_p
A
Obs
JmaxRef = 490342040.33
RdRef = 1.75
sigma = 0.52
Theta = −11432128.11
VcmaxRef = 73.62
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

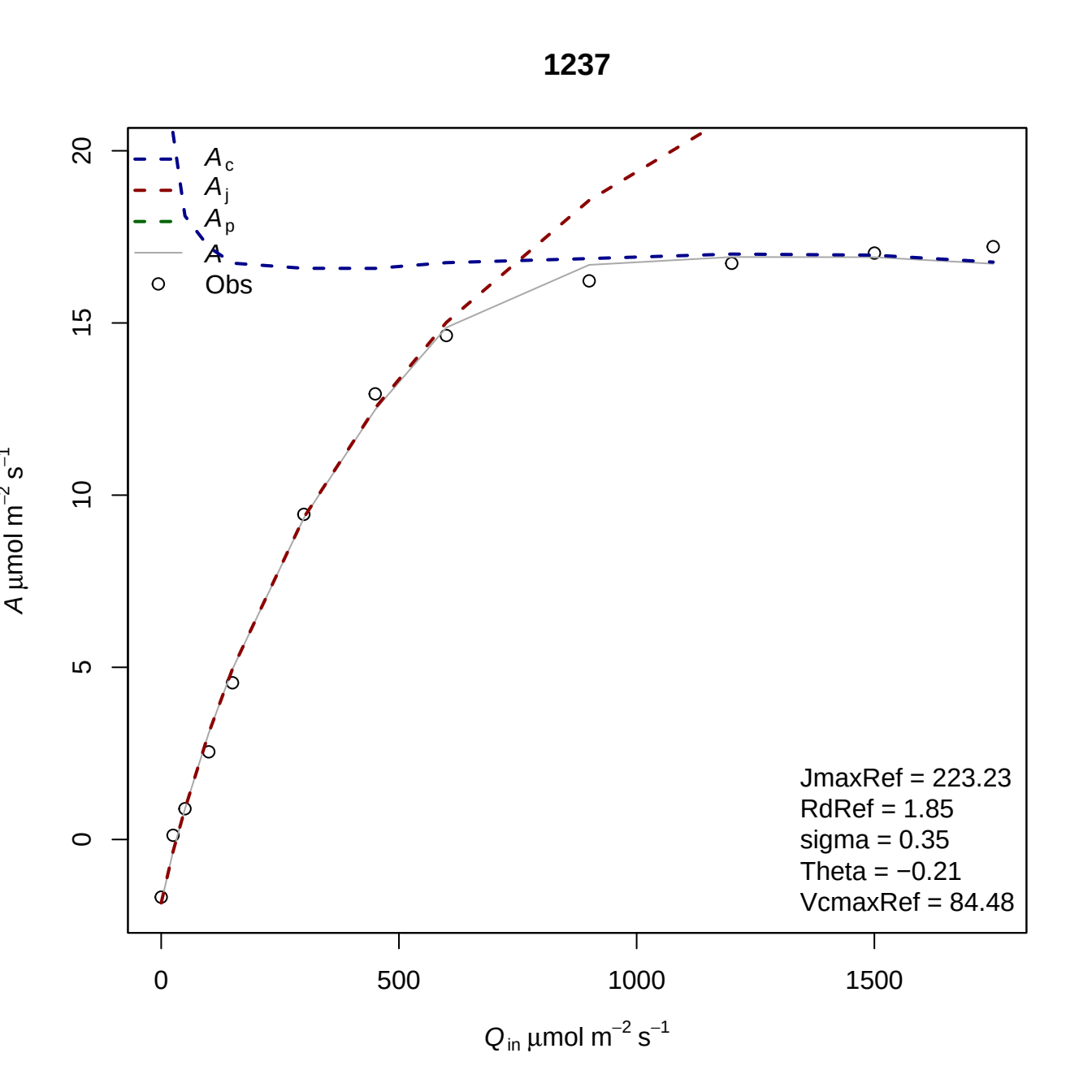

1232
A_c
A_j
A_p
A
Obs
JmaxRef = 146.6
RdRef = 1.29
sigma = 0.29
Theta = −0.9
VcmaxRef = 70.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

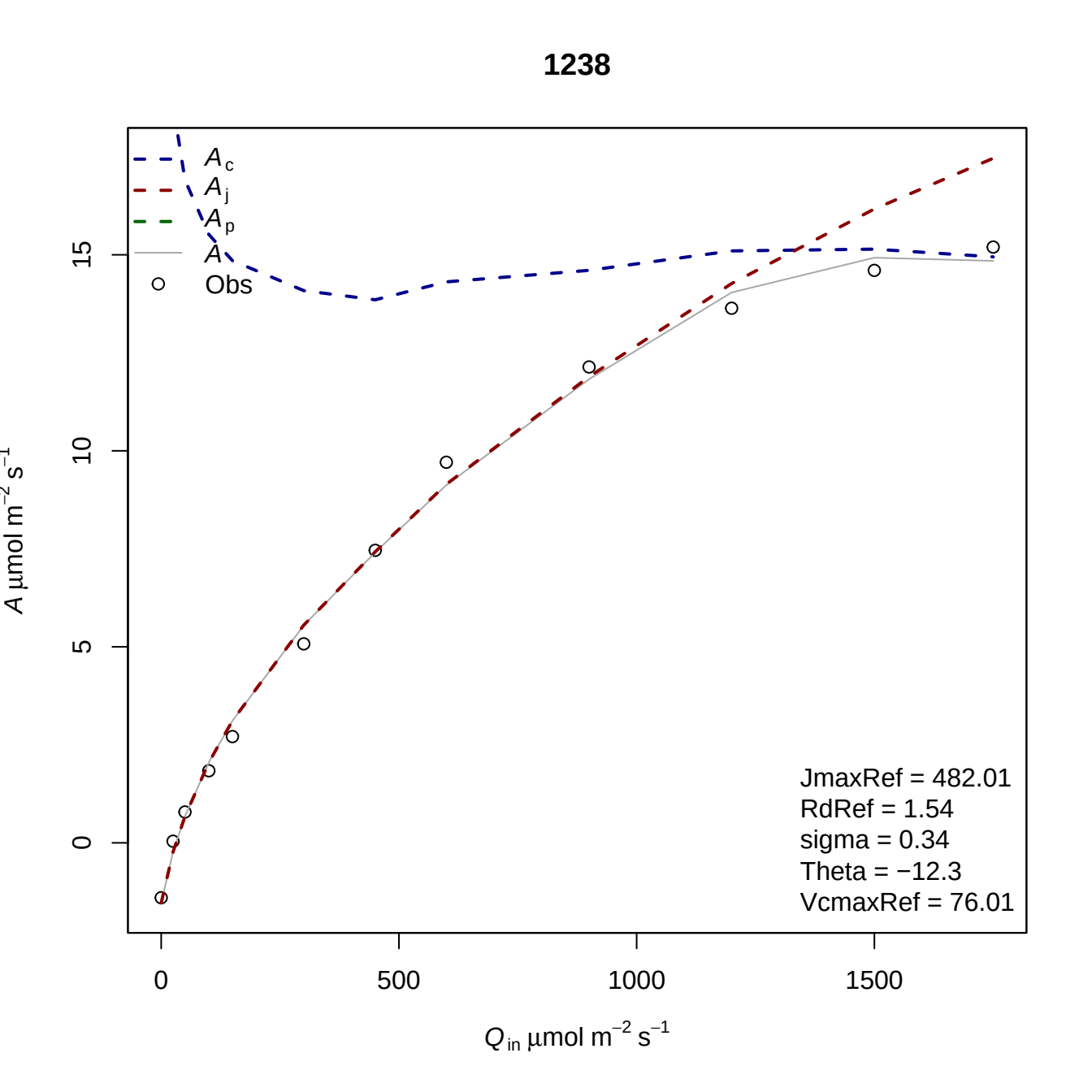

1233
A_c
A_j
A_p
A
Obs
JmaxRef = 196.93
RdRef = 2.91
sigma = 1.02
Theta = −0.03
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

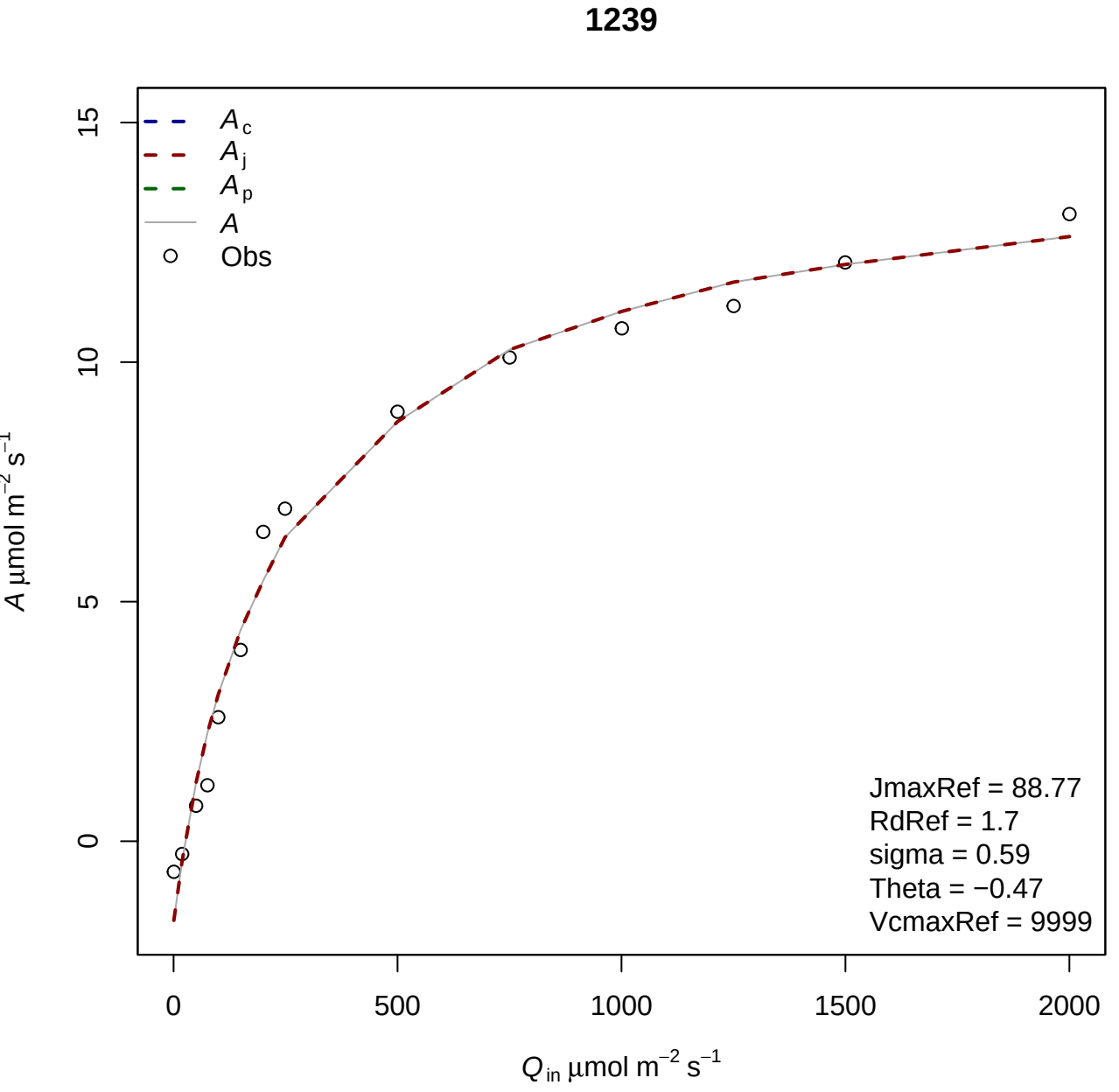

1234
A_c
A_j
A_p
A
Obs
JmaxRef = 137.49
RdRef = 1.94
sigma = 0.12
Theta = 0.39
VcmaxRef = 83.27
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

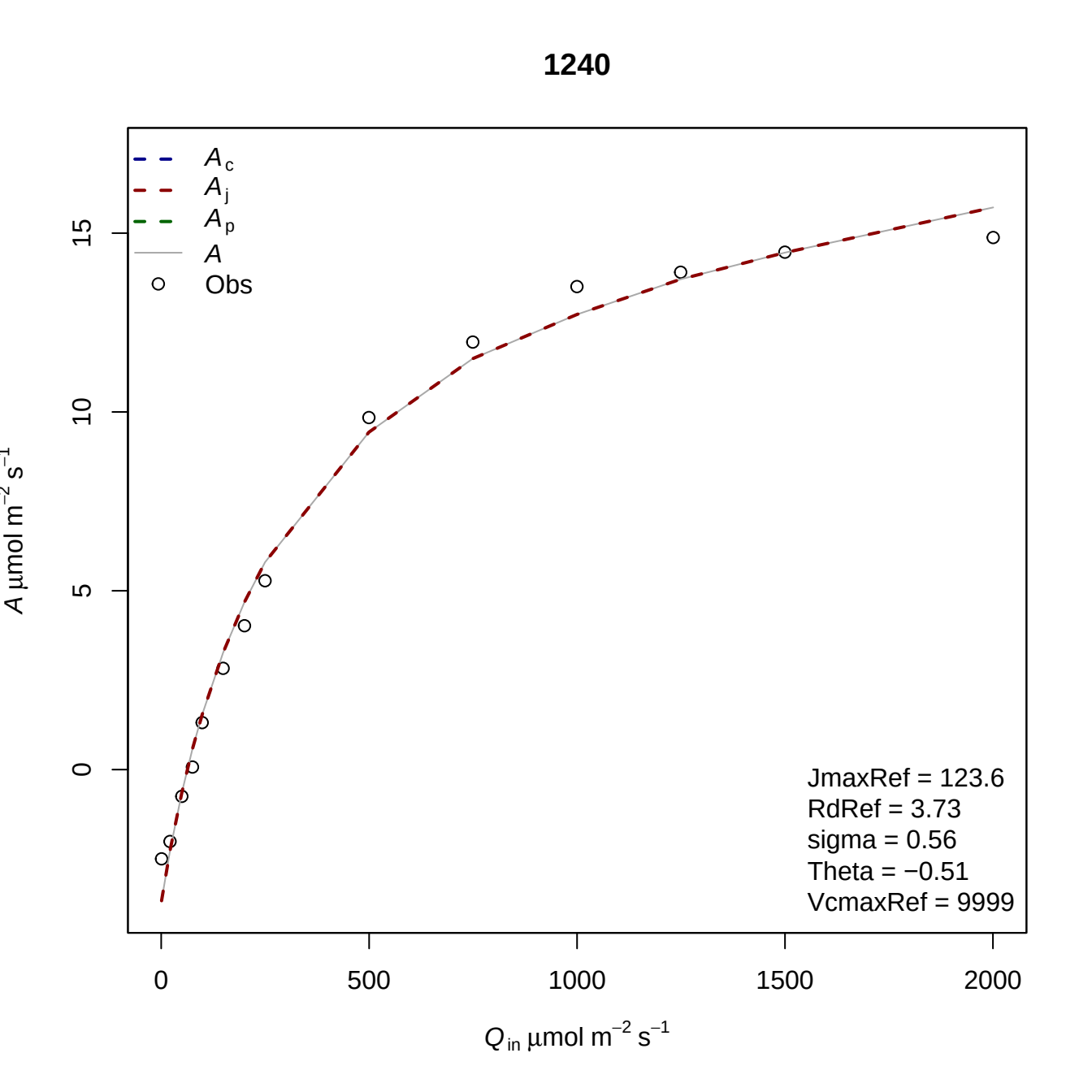

1235
A_c
A_j
A_p
A
Obs
JmaxRef = 208.67
RdRef = 2.02
sigma = 0.41
Theta = −0.94
VcmaxRef = 81.86
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

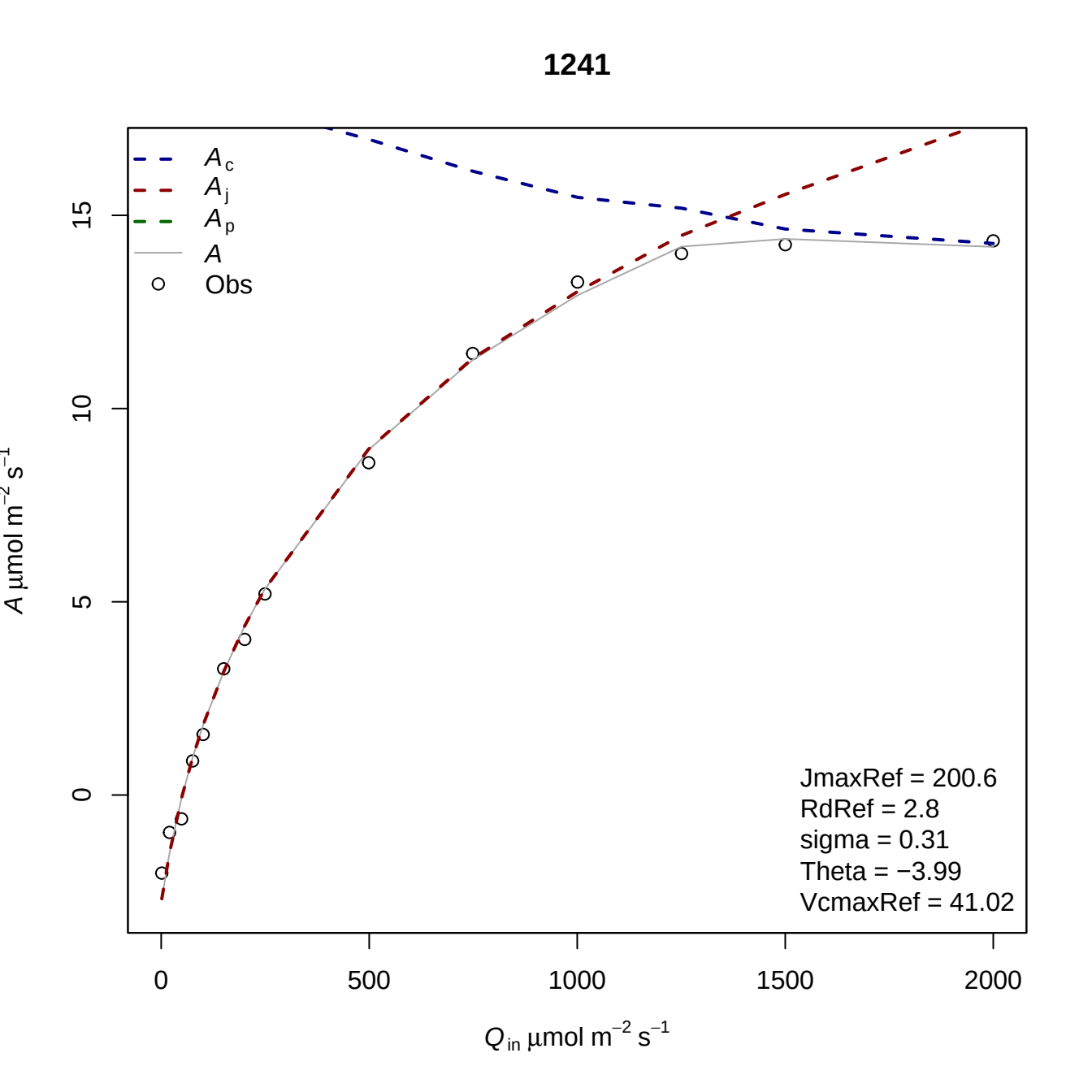

1236
A_c
A_j
A_p
A
Obs
JmaxRef = 172.07
RdRef = 2.69
sigma = 0.77
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

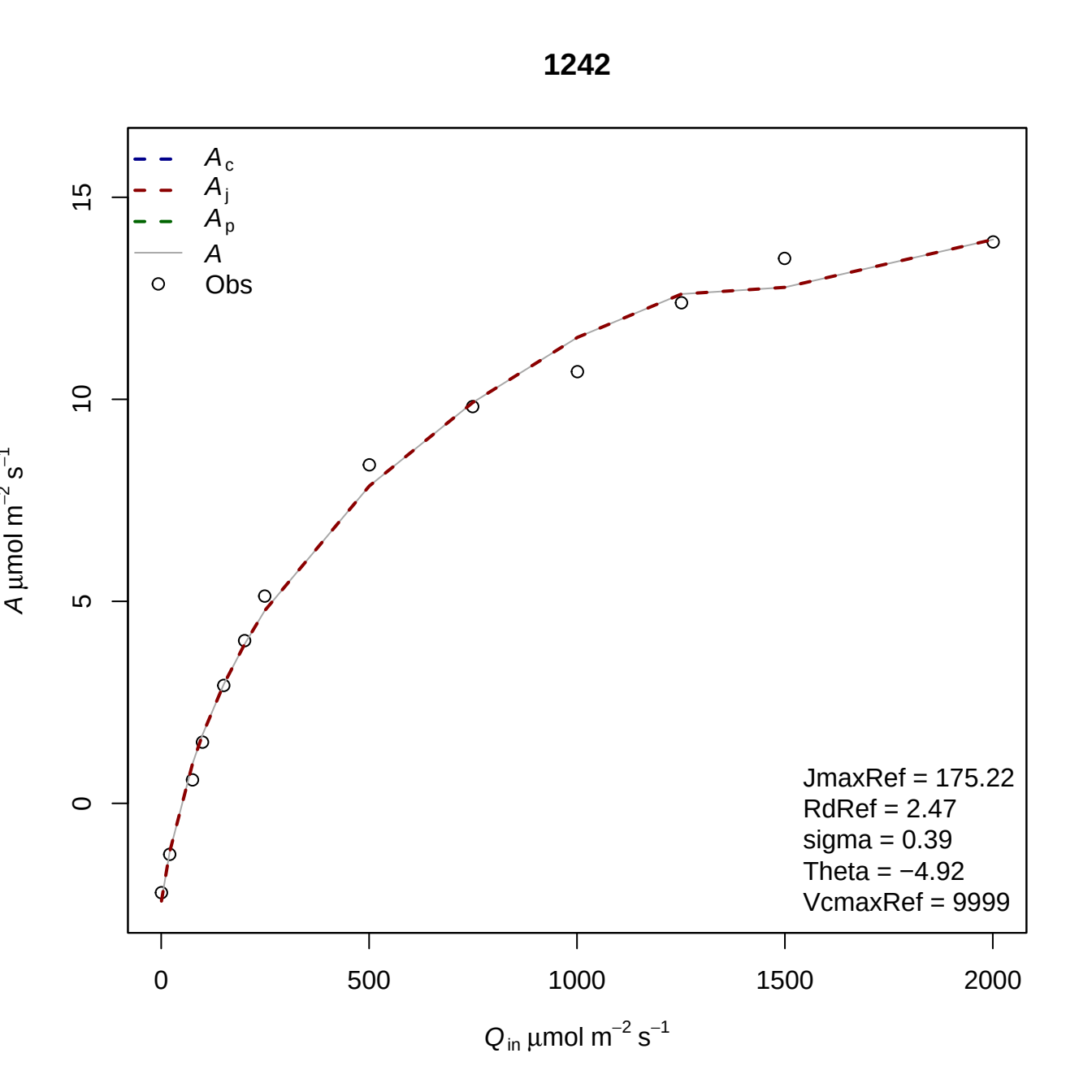

1237
A_c
A_j
A_p
A
Obs
JmaxRef = 223.23
RdRef = 1.85
sigma = 0.35
Theta = −0.21
VcmaxRef = 84.48
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

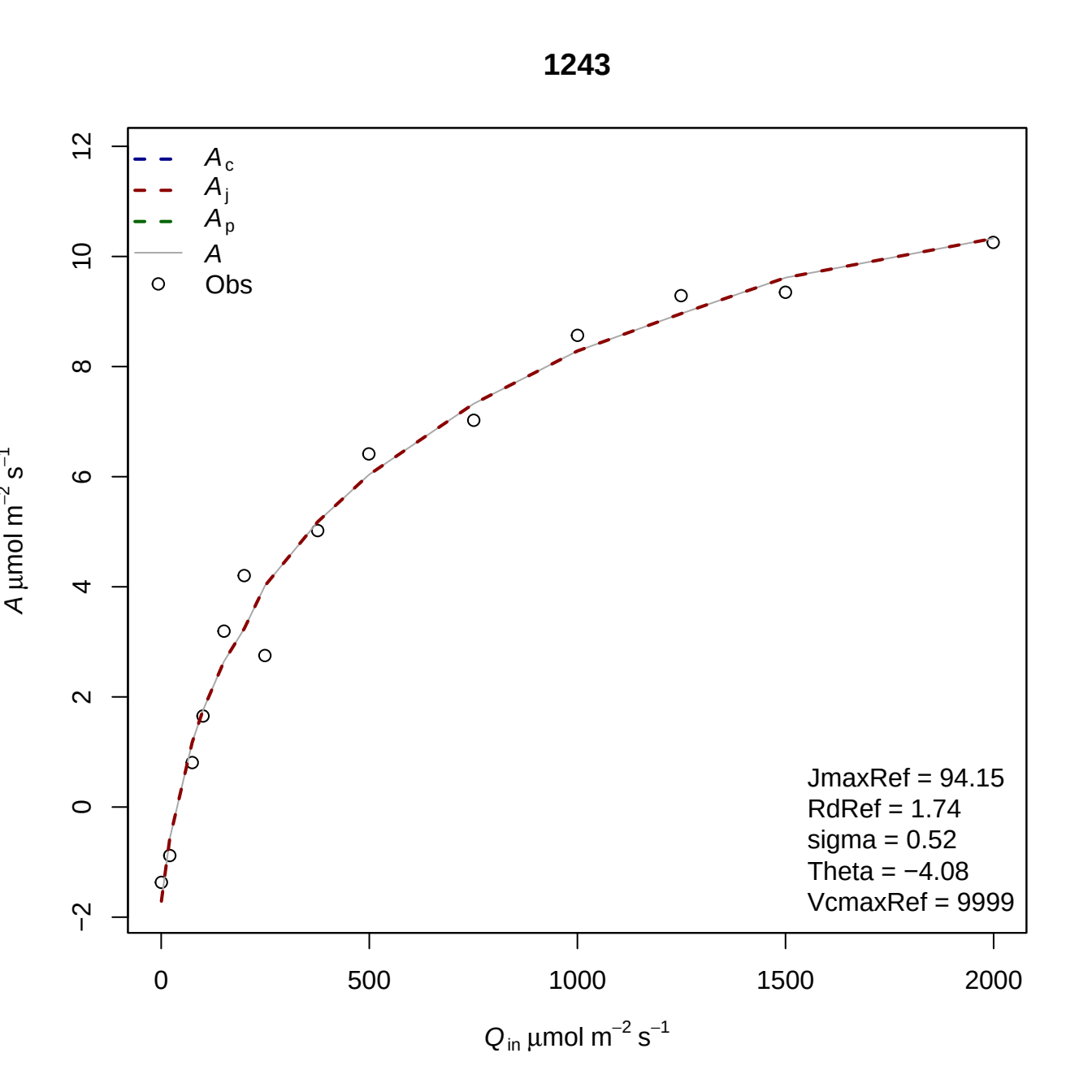

1238
A_c
A_j
A_p
A
Obs
JmaxRef = 482.01
RdRef = 1.54
sigma = 0.34
Theta = −12.3
VcmaxRef = 76.01
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

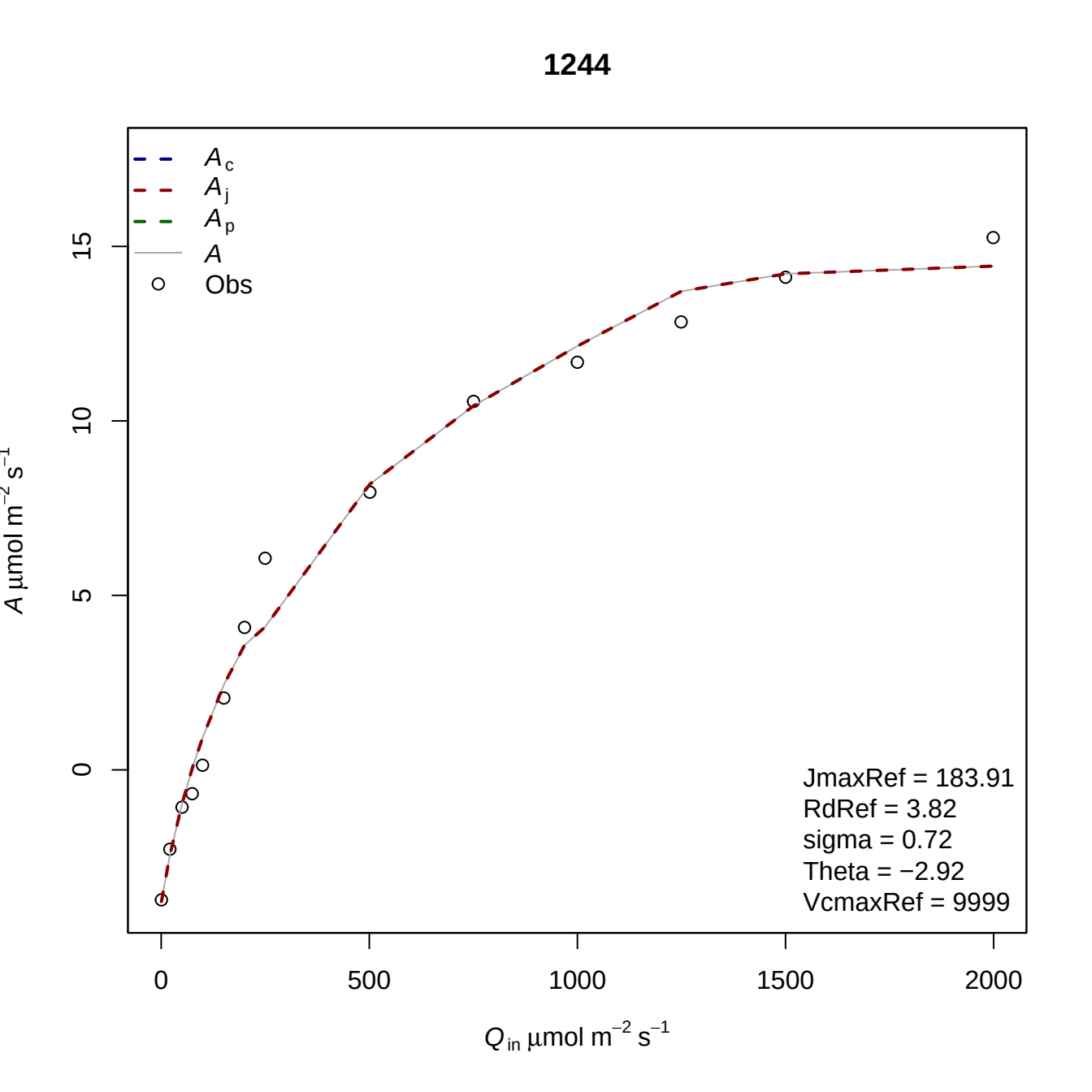

1239
A_c
A_j
A_p
A
Obs
JmaxRef = 88.77
RdRef = 1.7
sigma = 0.59
Theta = −0.47
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

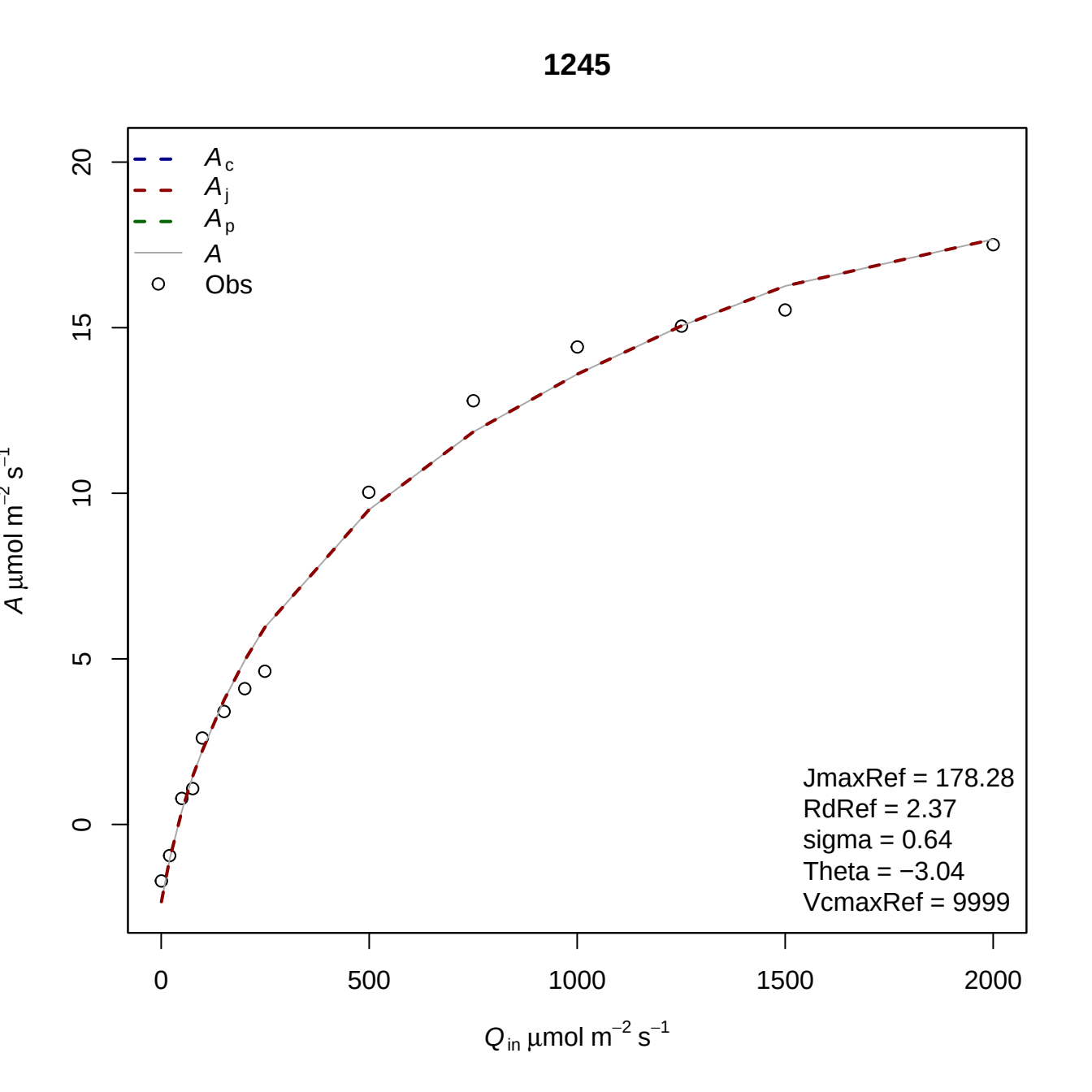

1240
A_c
A_j
A_p
A
Obs
JmaxRef = 123.6
RdRef = 3.73
sigma = 0.56
Theta = −0.51
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

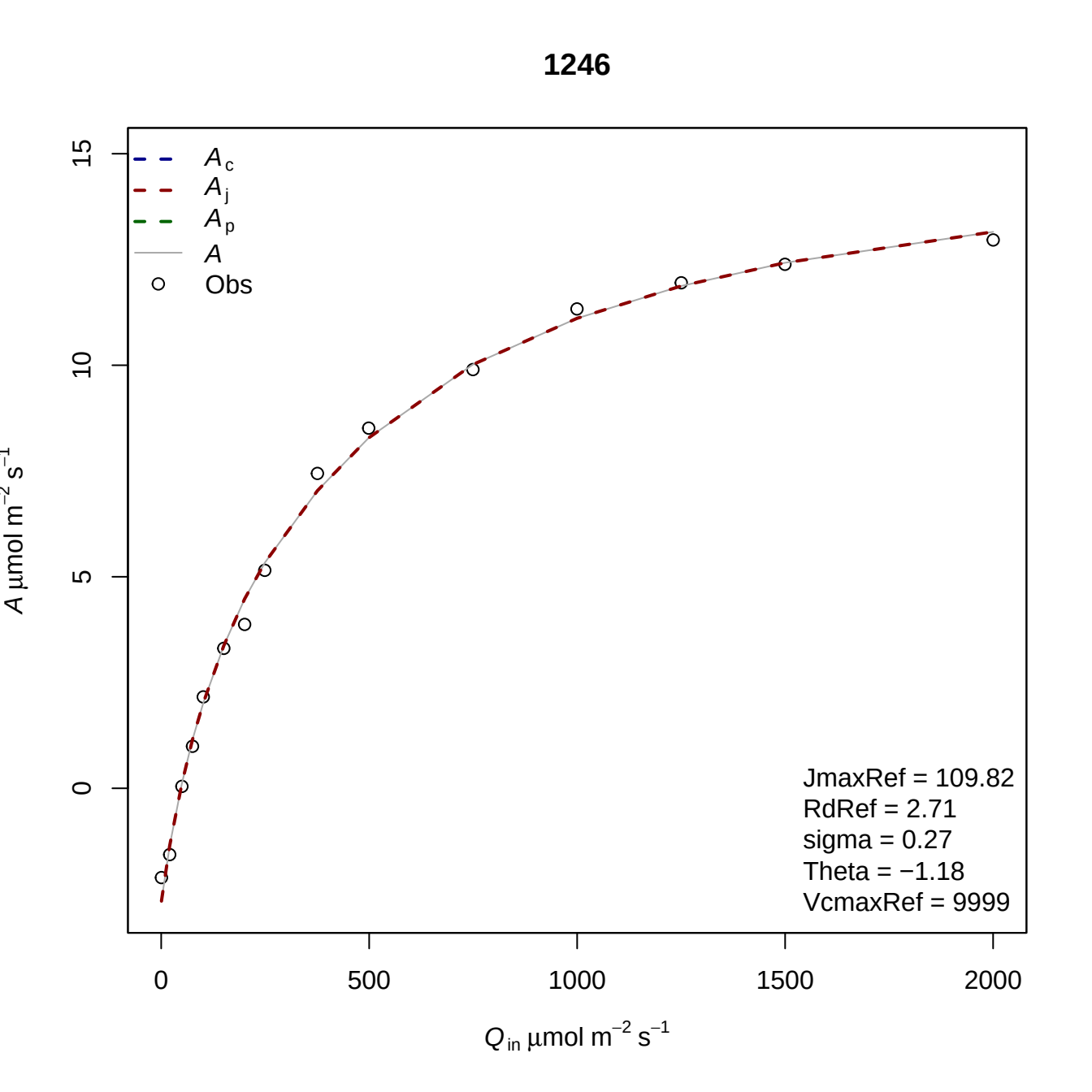

1241
A_c
A_j
A_p
A
Obs
JmaxRef = 200.6
RdRef = 2.8
sigma = 0.31
Theta = −3.99
VcmaxRef = 41.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

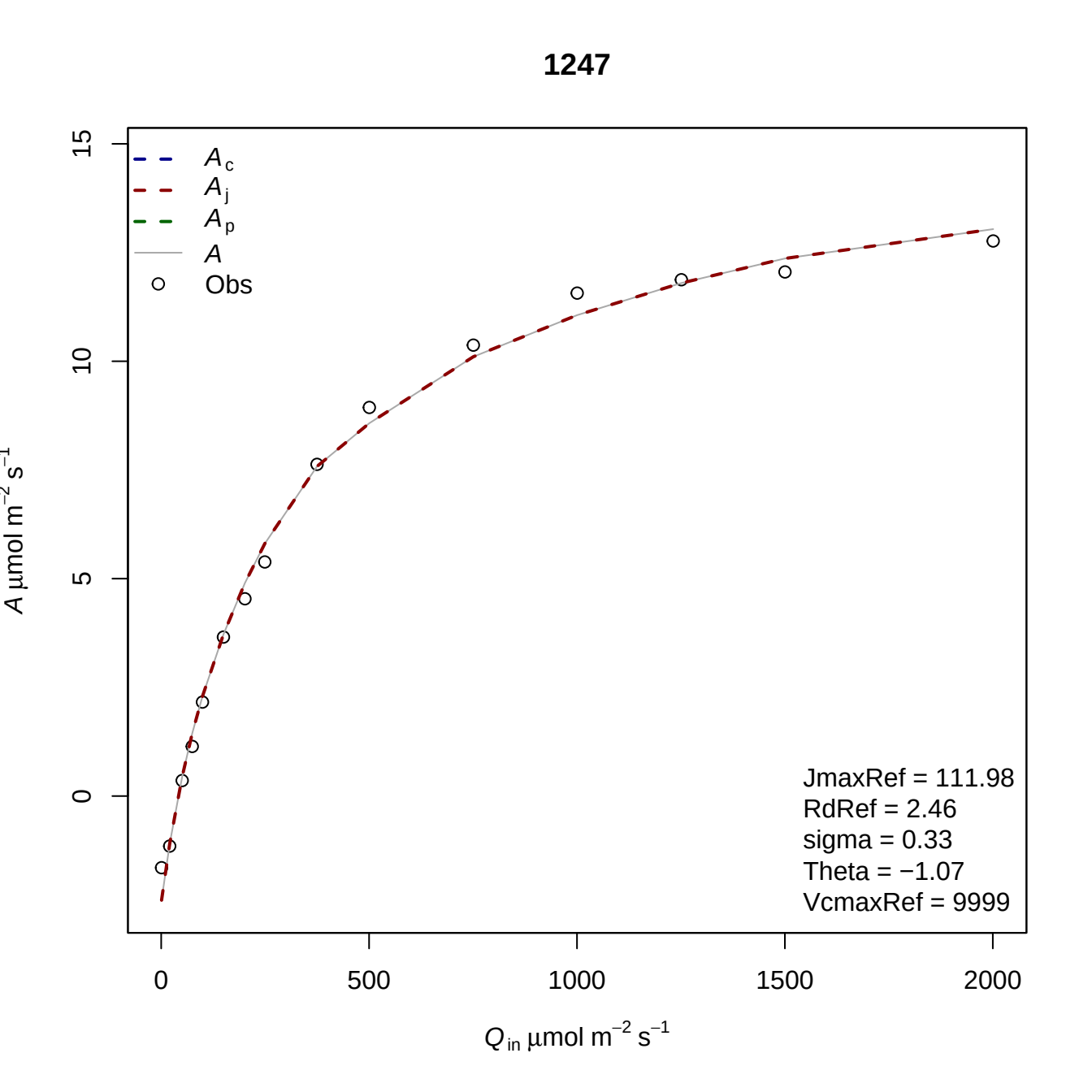

1242
A_c
A_j
A_p
A
Obs
JmaxRef = 175.22
RdRef = 2.47
sigma = 0.39
Theta = −4.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

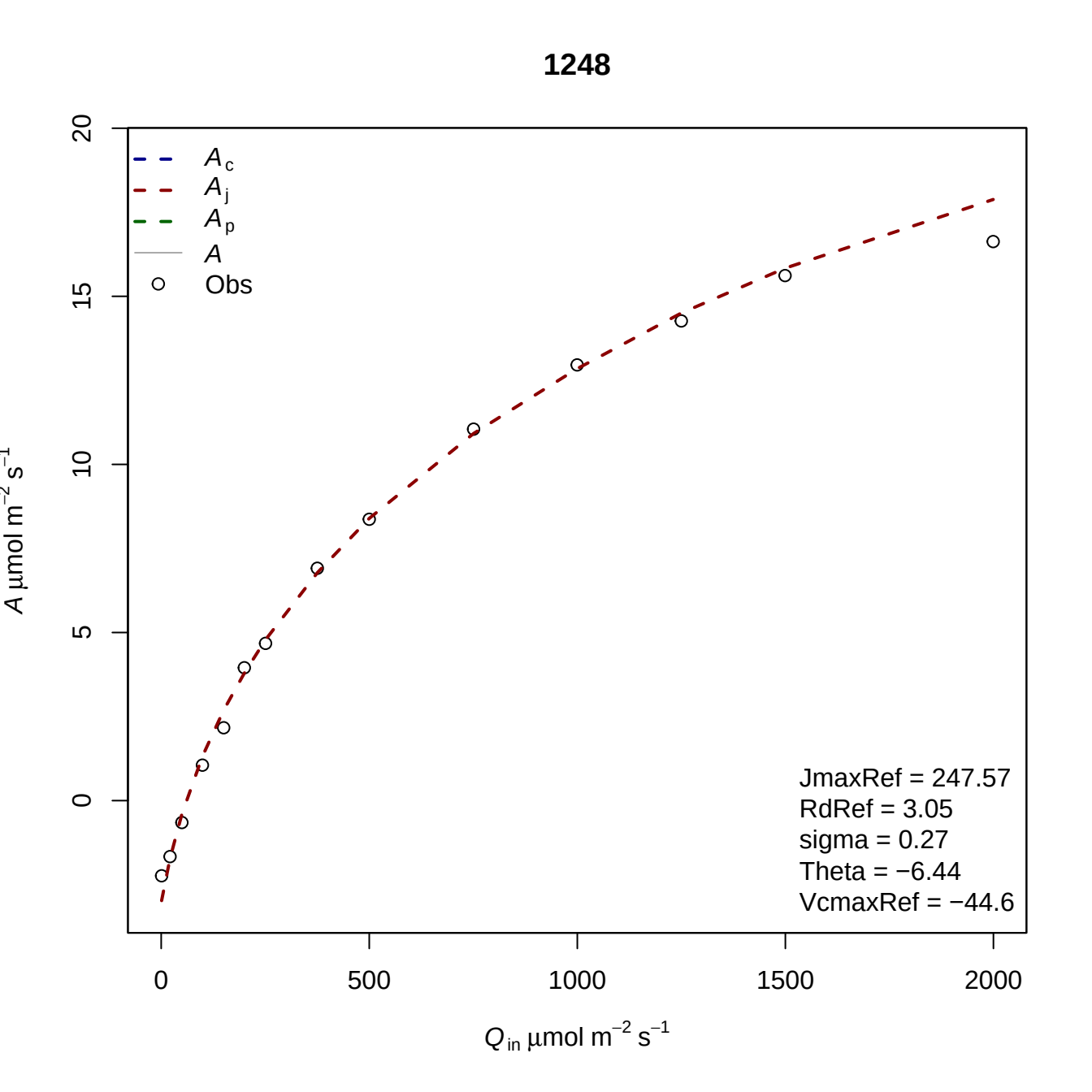

1243
A_c
A_j
A_p
A
Obs
JmaxRef = 94.15
RdRef = 1.74
sigma = 0.52
Theta = −4.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

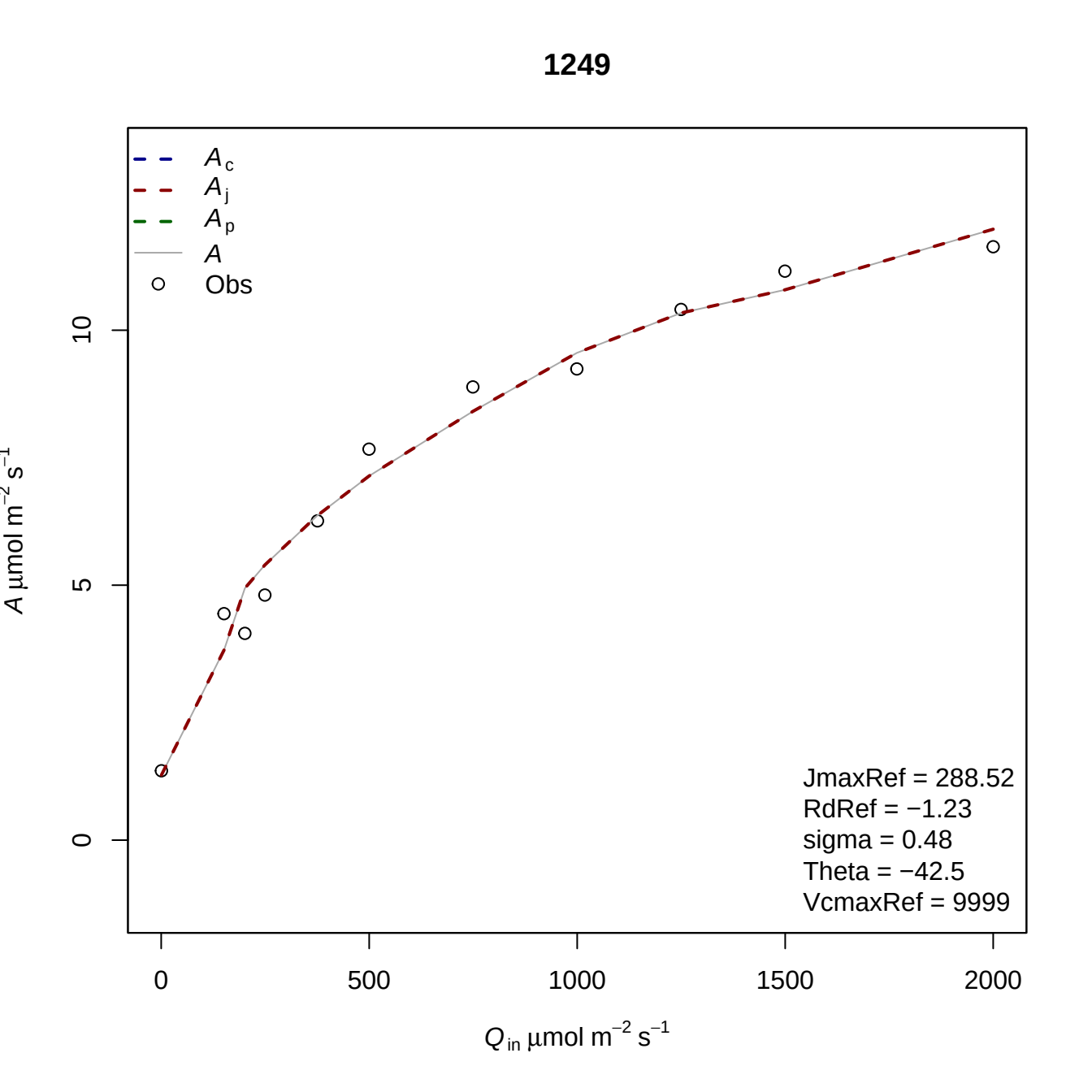

1244
A_c
A_j
A_p
A
Obs
JmaxRef = 183.91
RdRef = 3.82
sigma = 0.72
Theta = −2.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

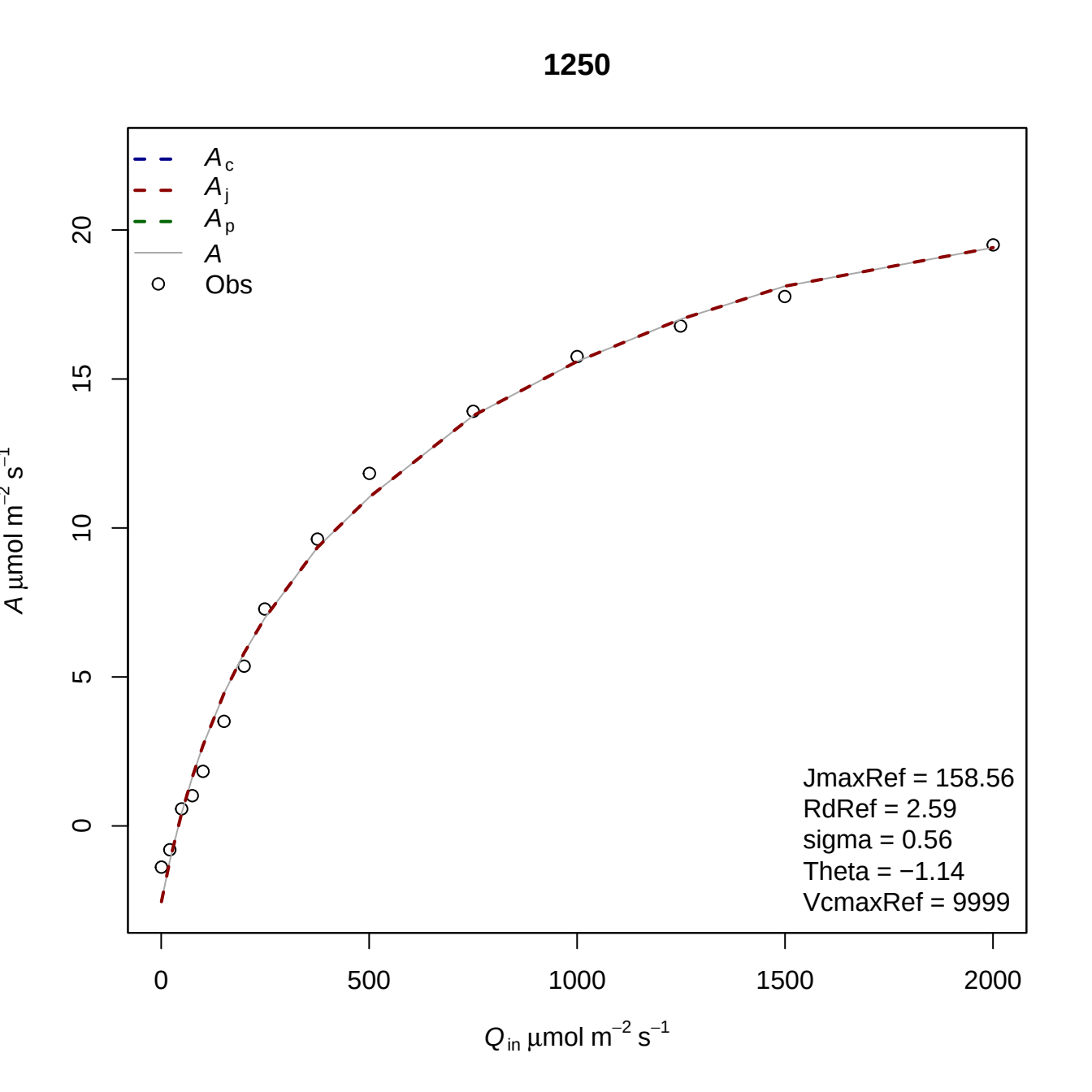

1245
A_c
A_j
A_p
A
Obs
JmaxRef = 178.28
RdRef = 2.37
sigma = 0.64
Theta = −3.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

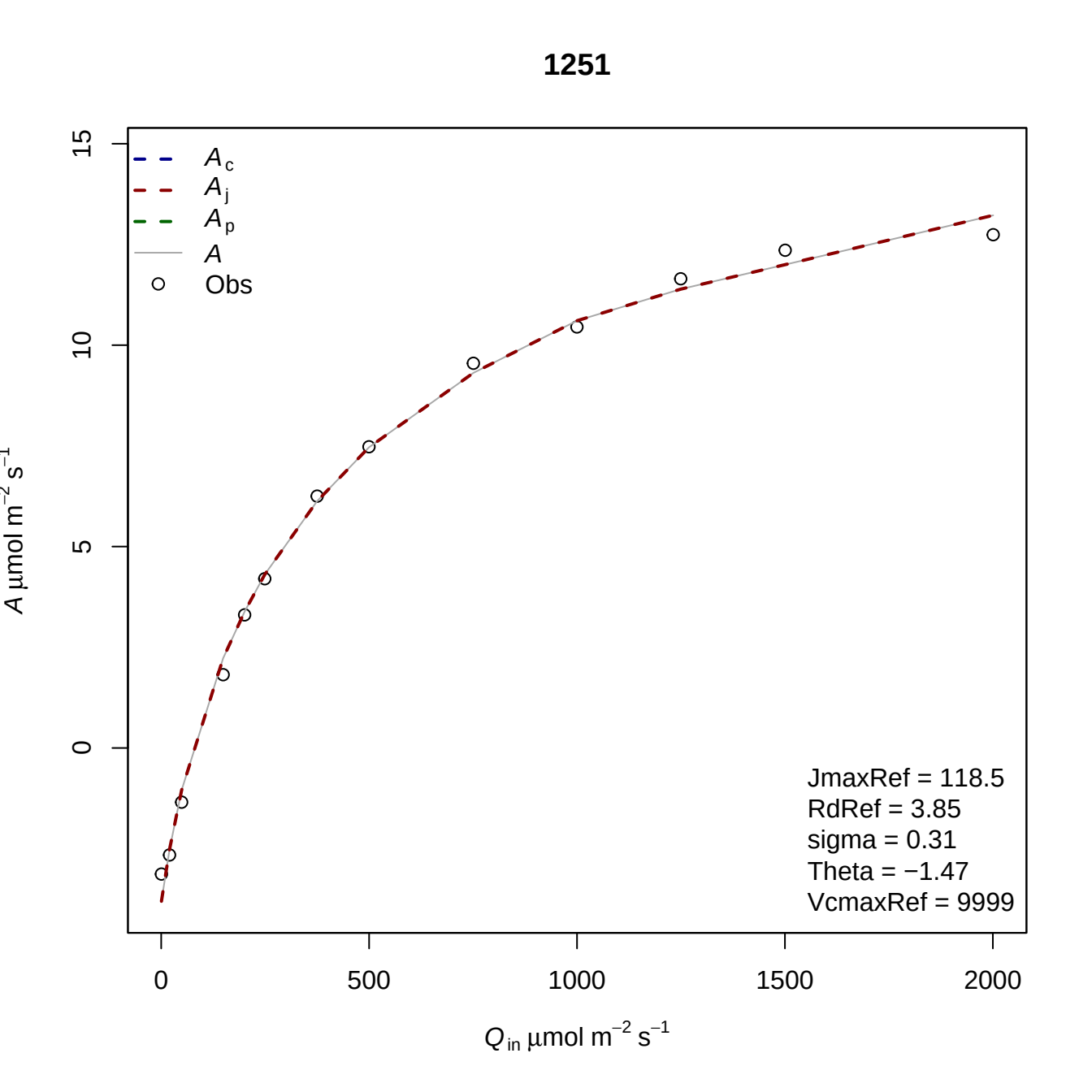

1246
A_c
A_j
A_p
A
Obs
JmaxRef = 109.82
RdRef = 2.71
sigma = 0.27
Theta = −1.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

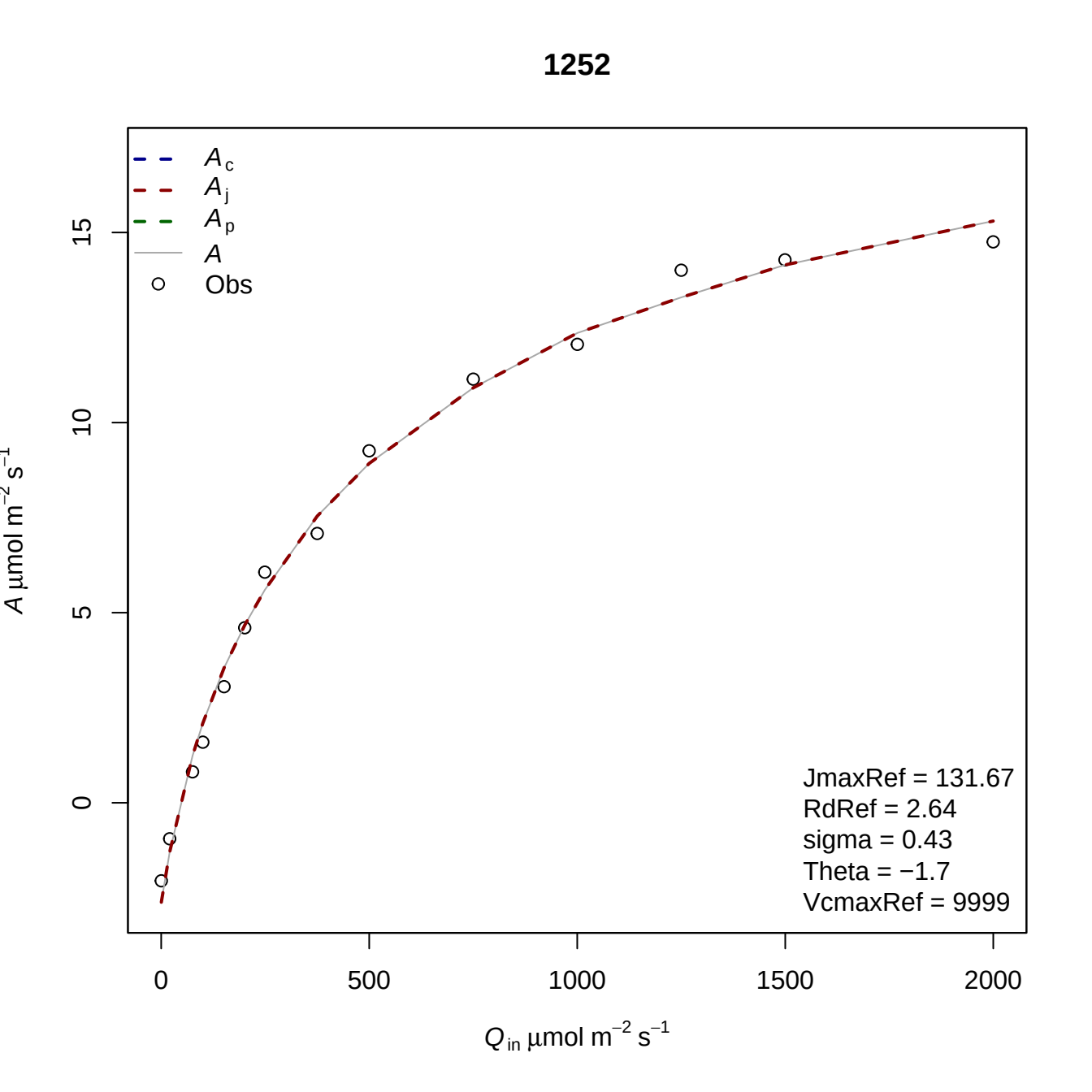

1247
A_c
A_j
A_p
A
Obs
JmaxRef = 111.98
RdRef = 2.46
sigma = 0.33
Theta = −1.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

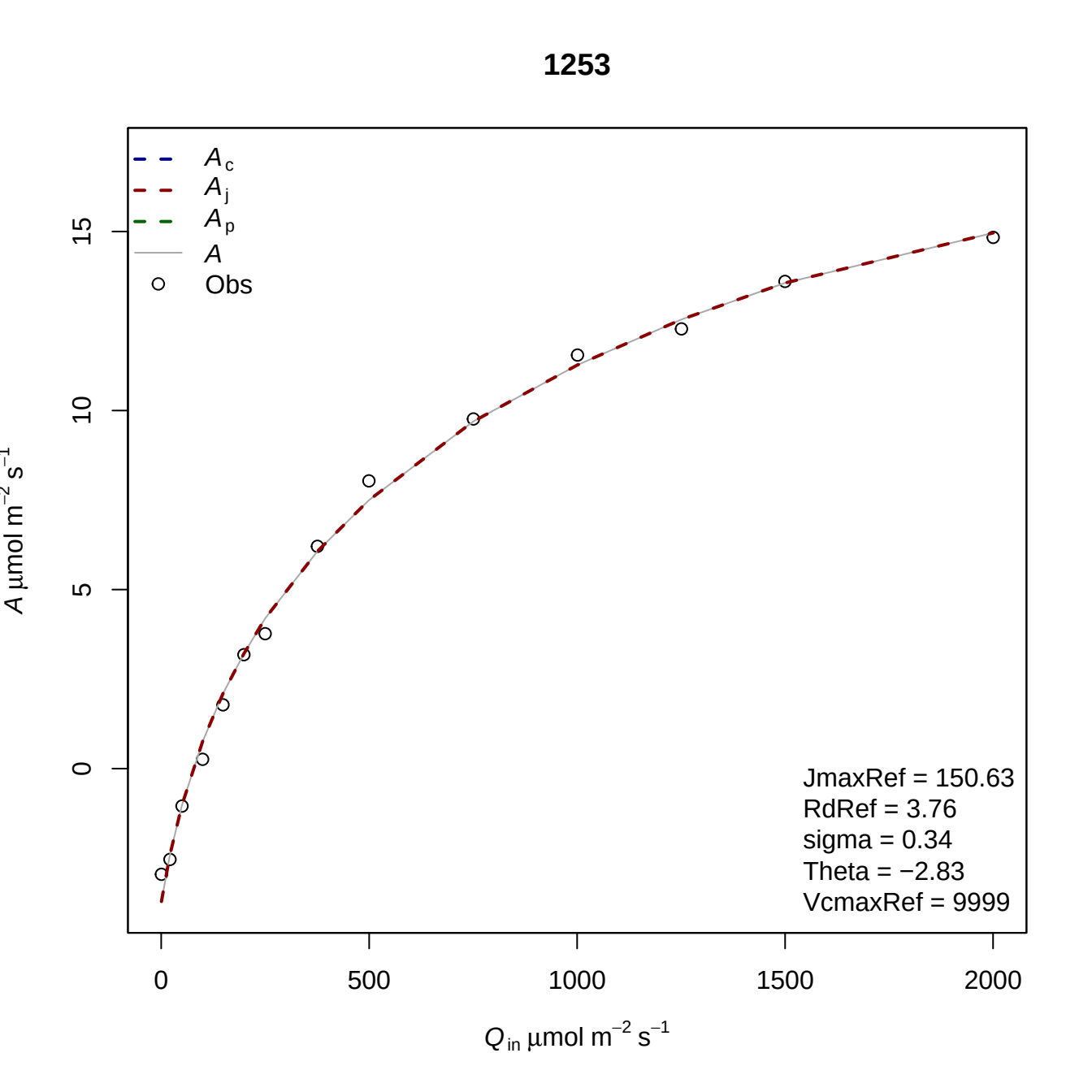

1248
A_c
A_j
A_p
A
Obs
JmaxRef = 247.57
RdRef = 3.05
sigma = 0.27
Theta = −6.44
VcmaxRef = −44.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

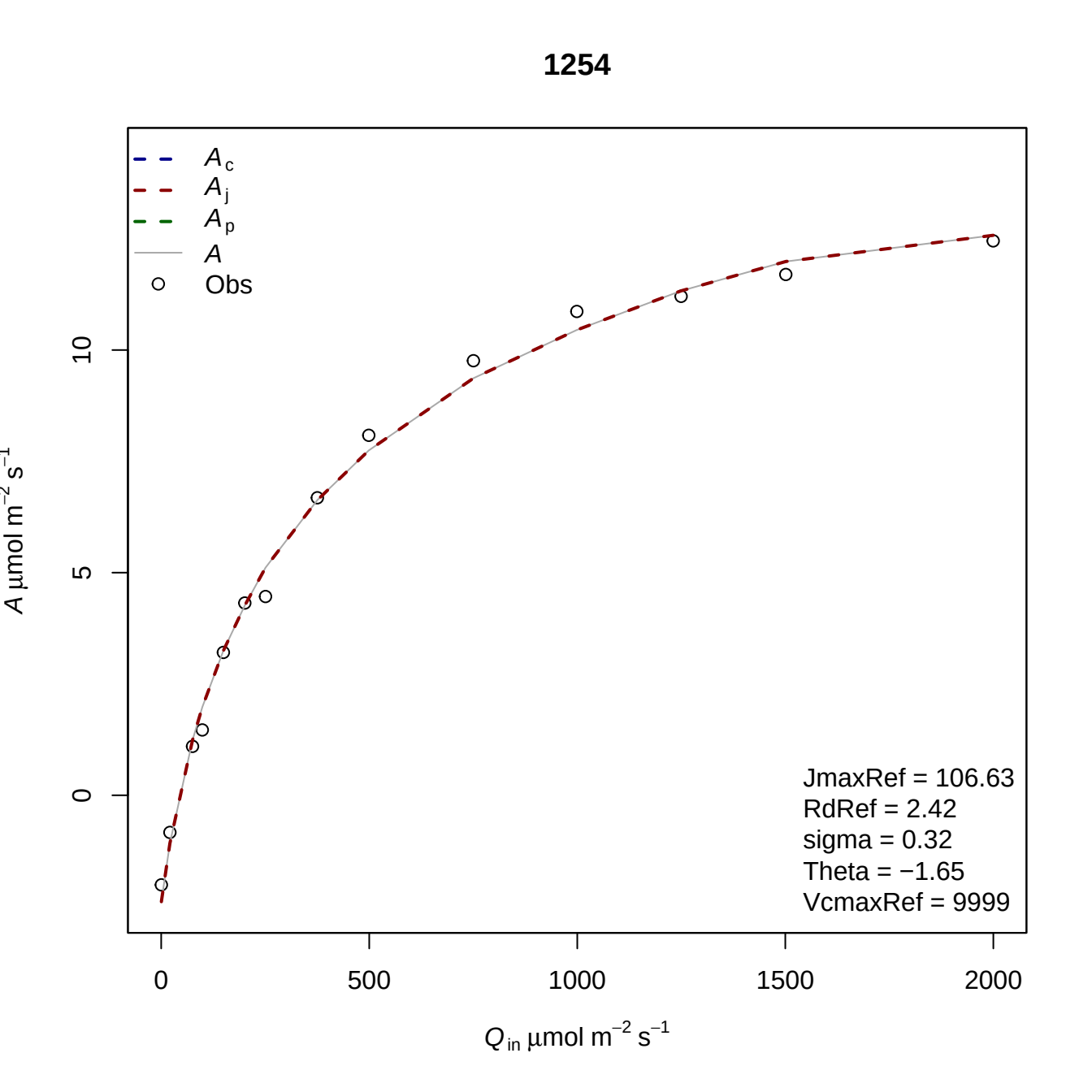

1249
A_c
A_j
A_p
A
Obs
JmaxRef = 288.52
RdRef = −1.23
sigma = 0.48
Theta = −42.5
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

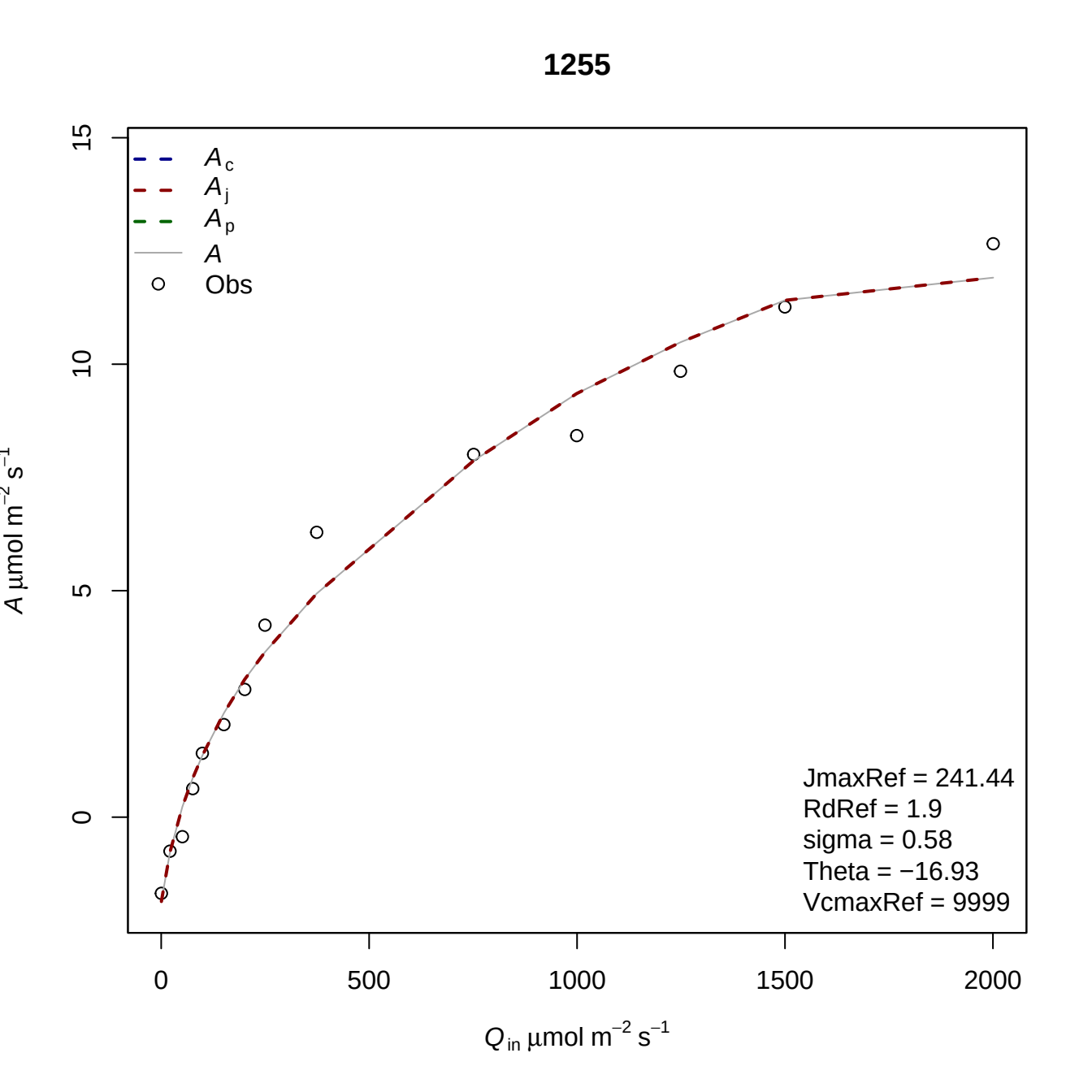

1250
A_c
A_j
A_p
A
Obs
JmaxRef = 158.56
RdRef = 2.59
sigma = 0.56
Theta = −1.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

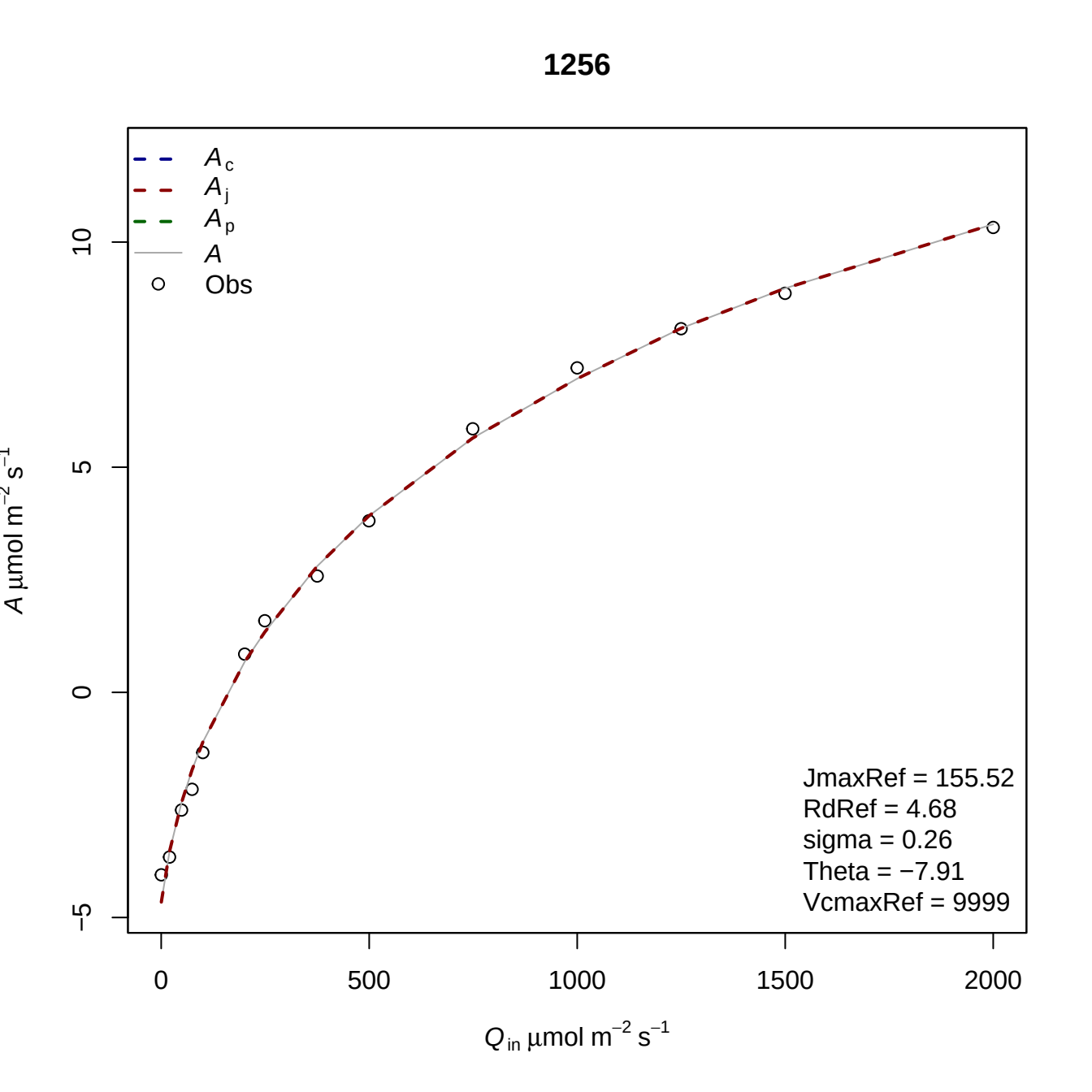

1251
A_c
A_j
A_p
A
Obs
JmaxRef = 118.5
RdRef = 3.85
sigma = 0.31
Theta = −1.47
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

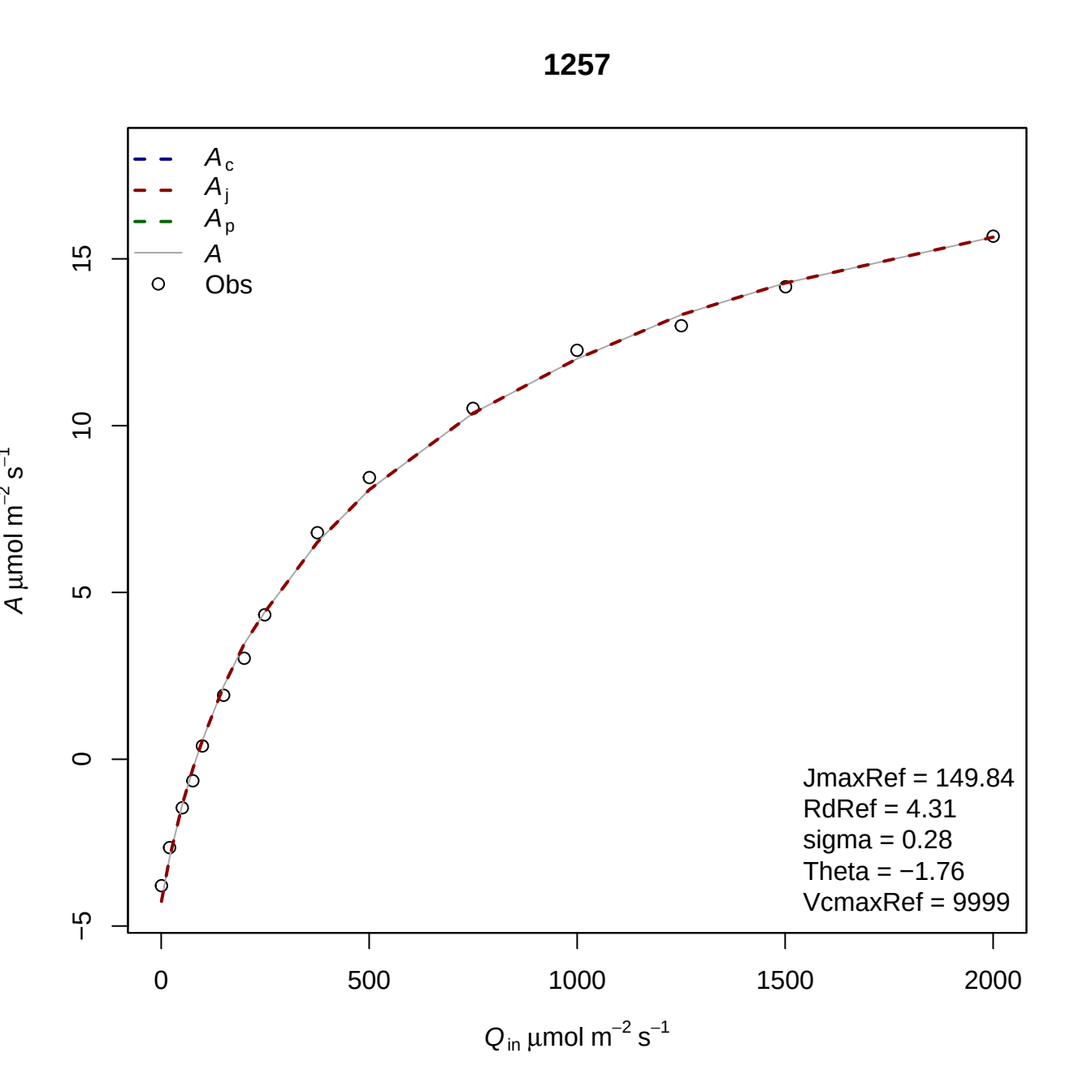

1252
A_c
A_j
A_p
A
Obs
JmaxRef = 131.67
RdRef = 2.64
sigma = 0.43
Theta = −1.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

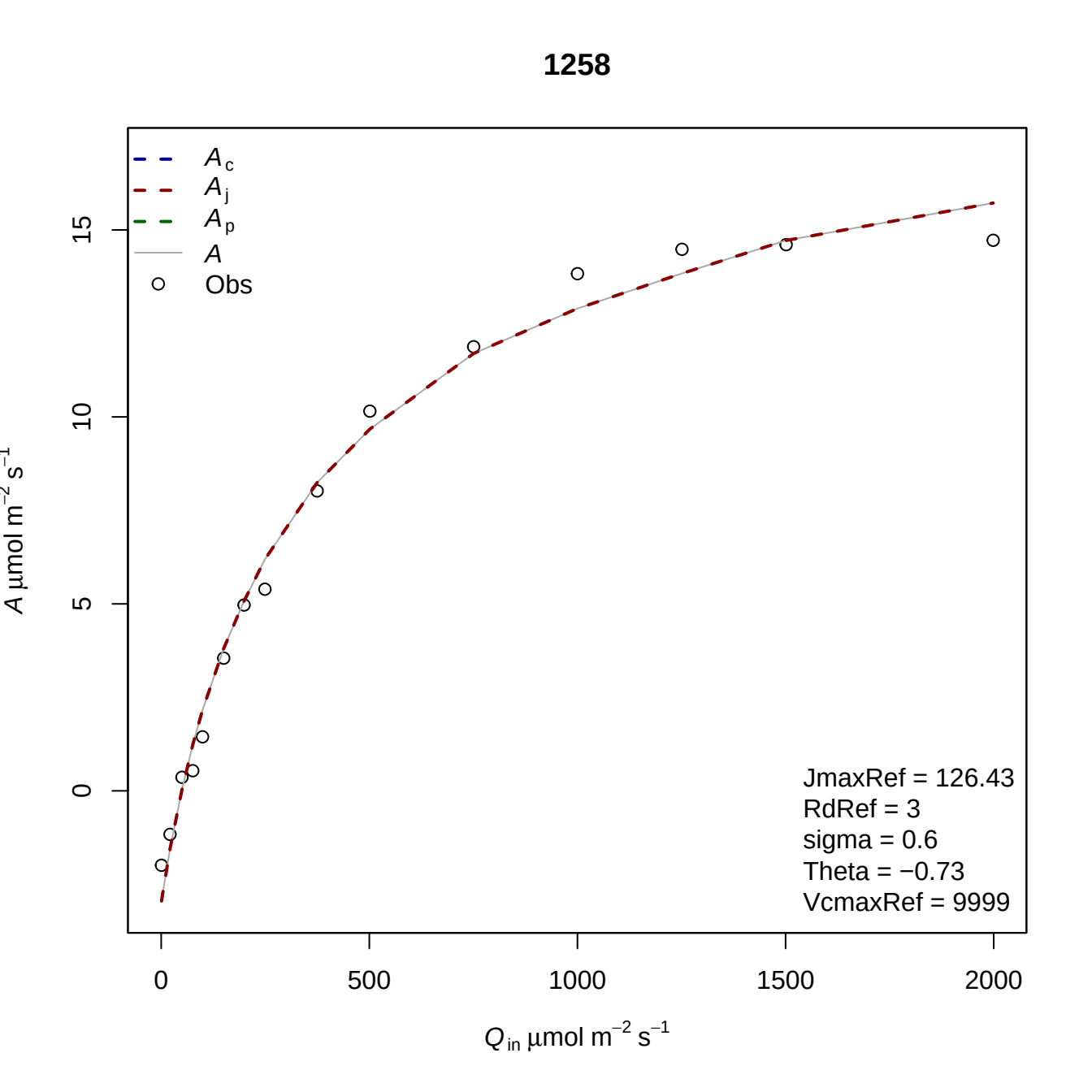

1253
A_c
A_j
A_p
A
Obs
JmaxRef = 150.63
RdRef = 3.76
sigma = 0.34
Theta = −2.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

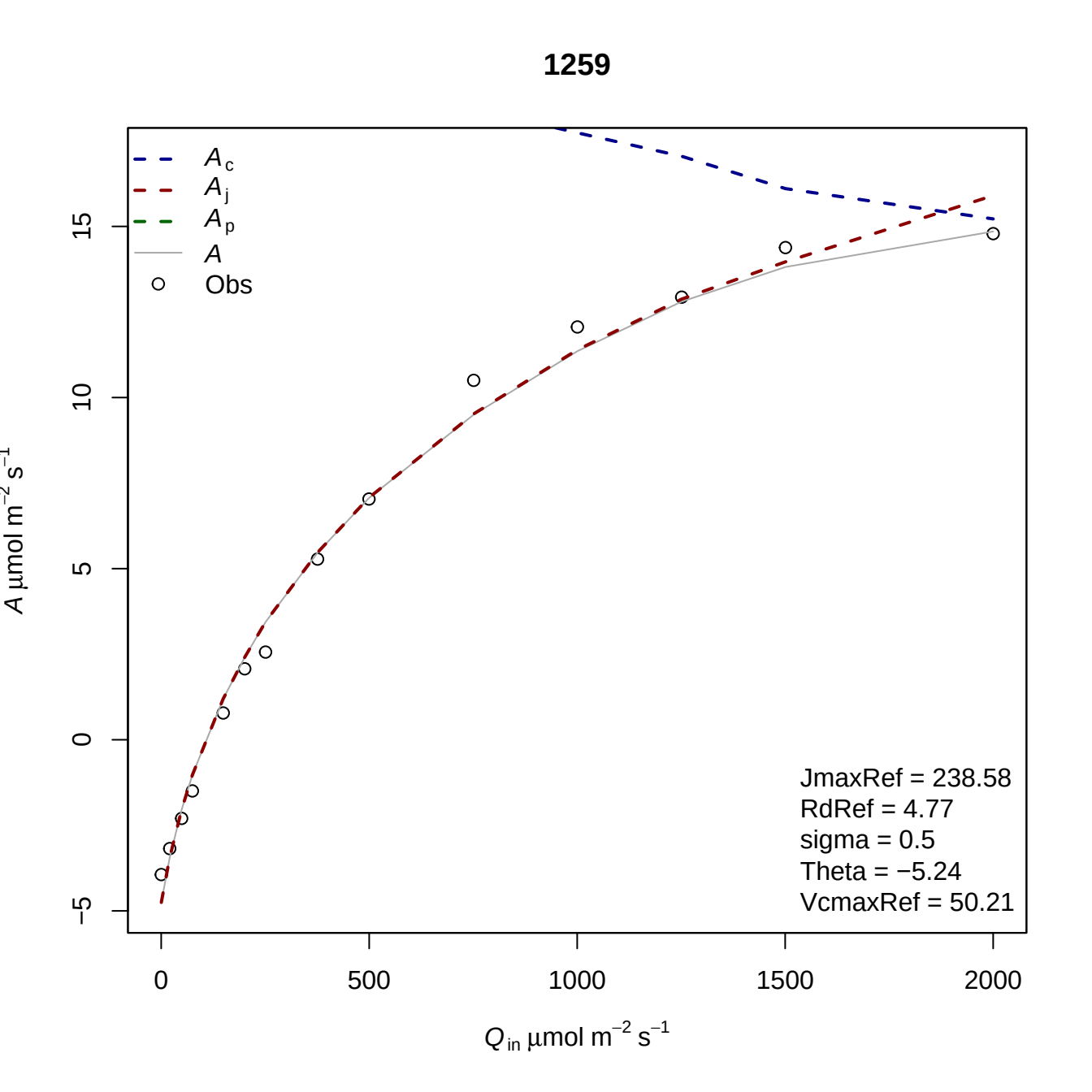

1254
A_c
A_j
A_p
A
Obs
JmaxRef = 106.63
RdRef = 2.42
sigma = 0.32
Theta = −1.65
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

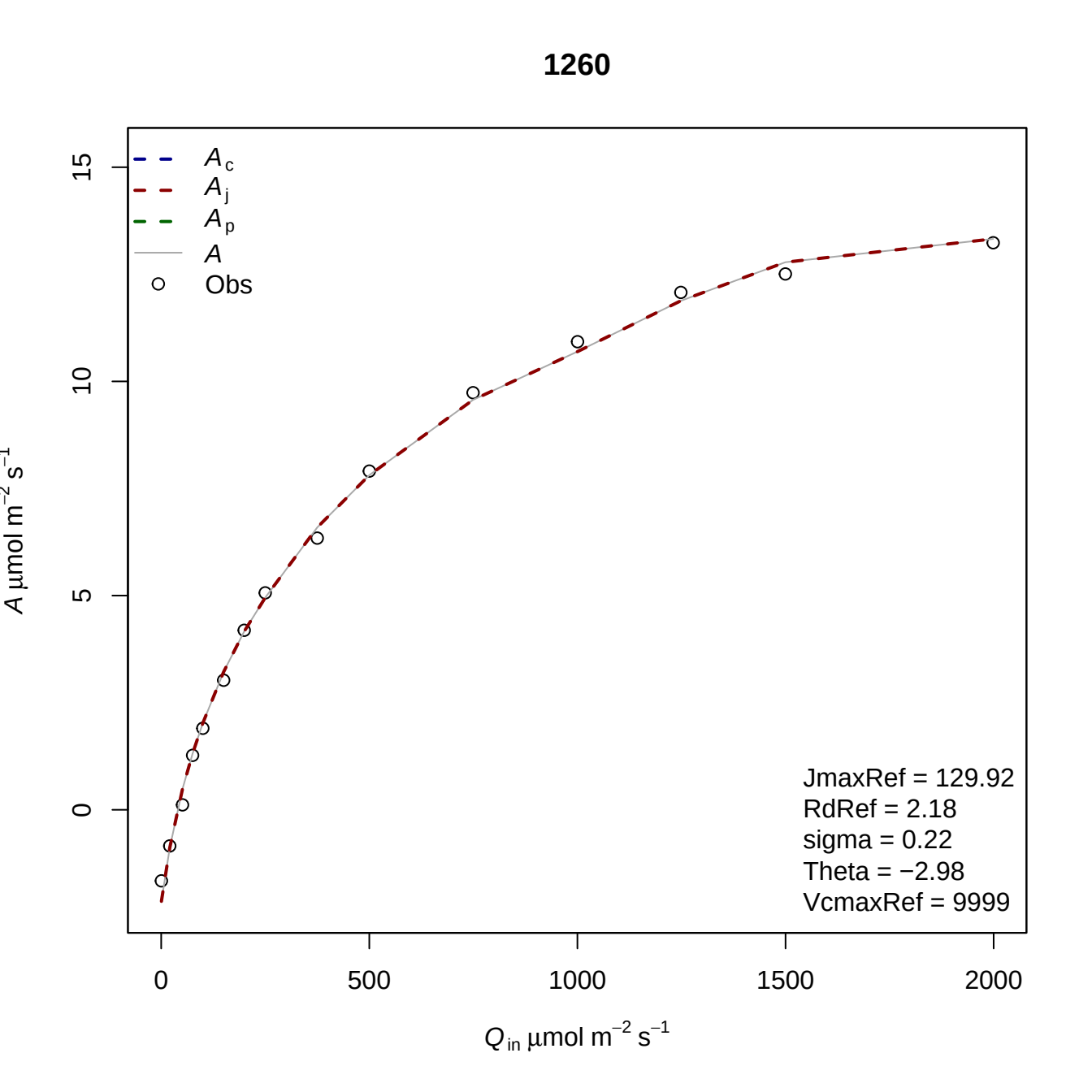

1255
A_c
A_j
A_p
A
Obs
JmaxRef = 241.44
RdRef = 1.9
sigma = 0.58
Theta = −16.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

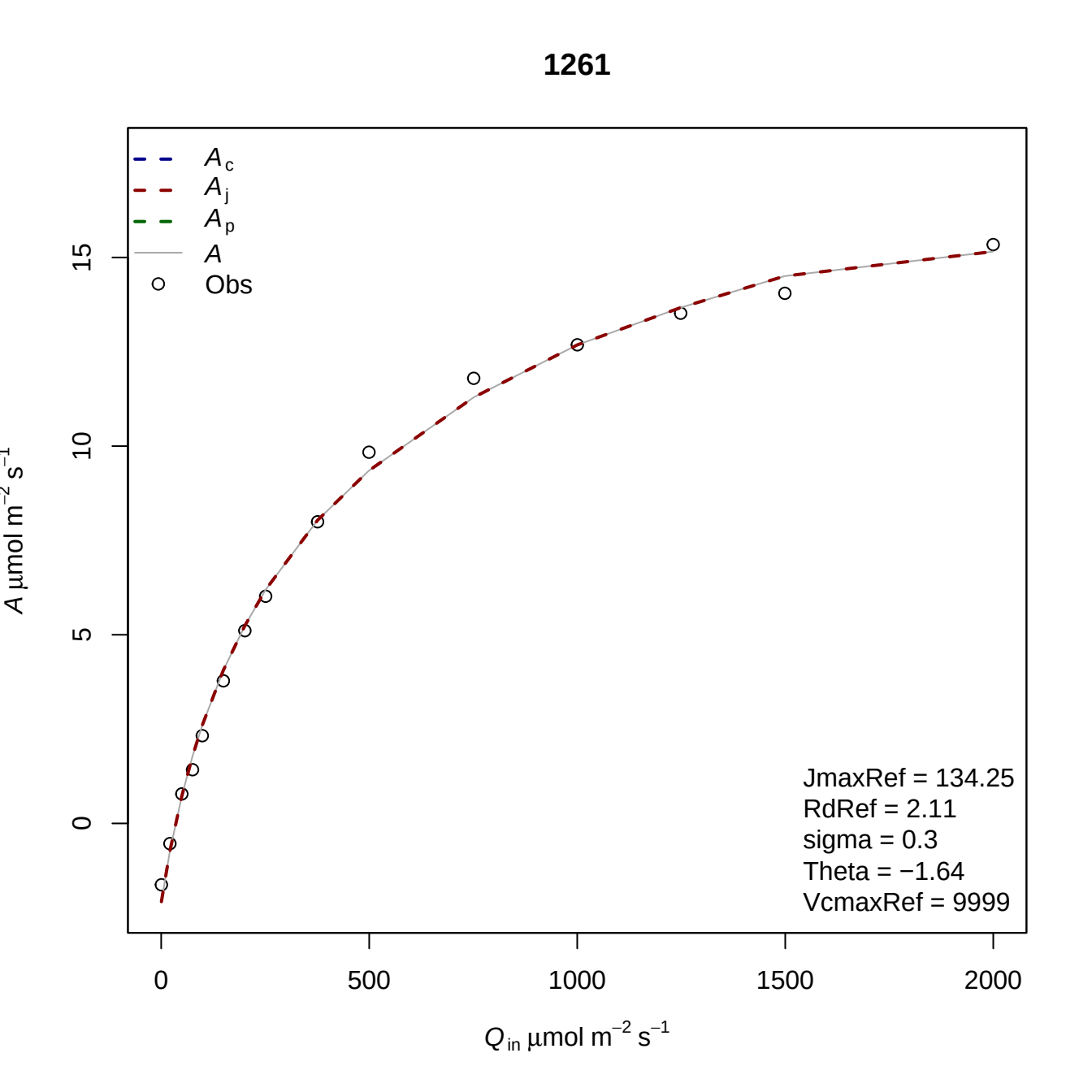

1256
A_c
A_j
A_p
A
Obs
JmaxRef = 155.52
RdRef = 4.68
sigma = 0.26
Theta = −7.91
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

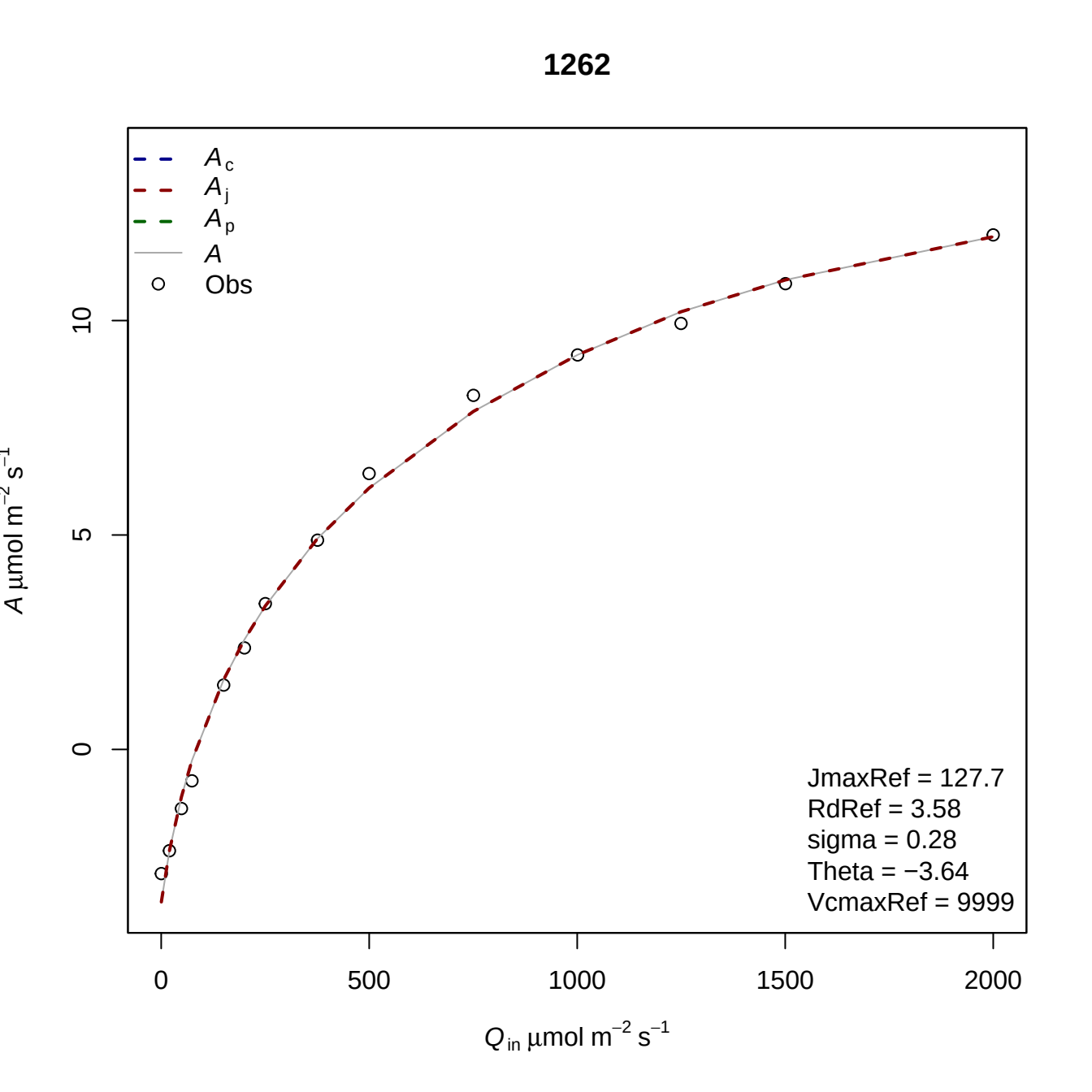

1257
A_c
A_j
A_p
A
Obs
JmaxRef = 149.84
RdRef = 4.31
sigma = 0.28
Theta = −1.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

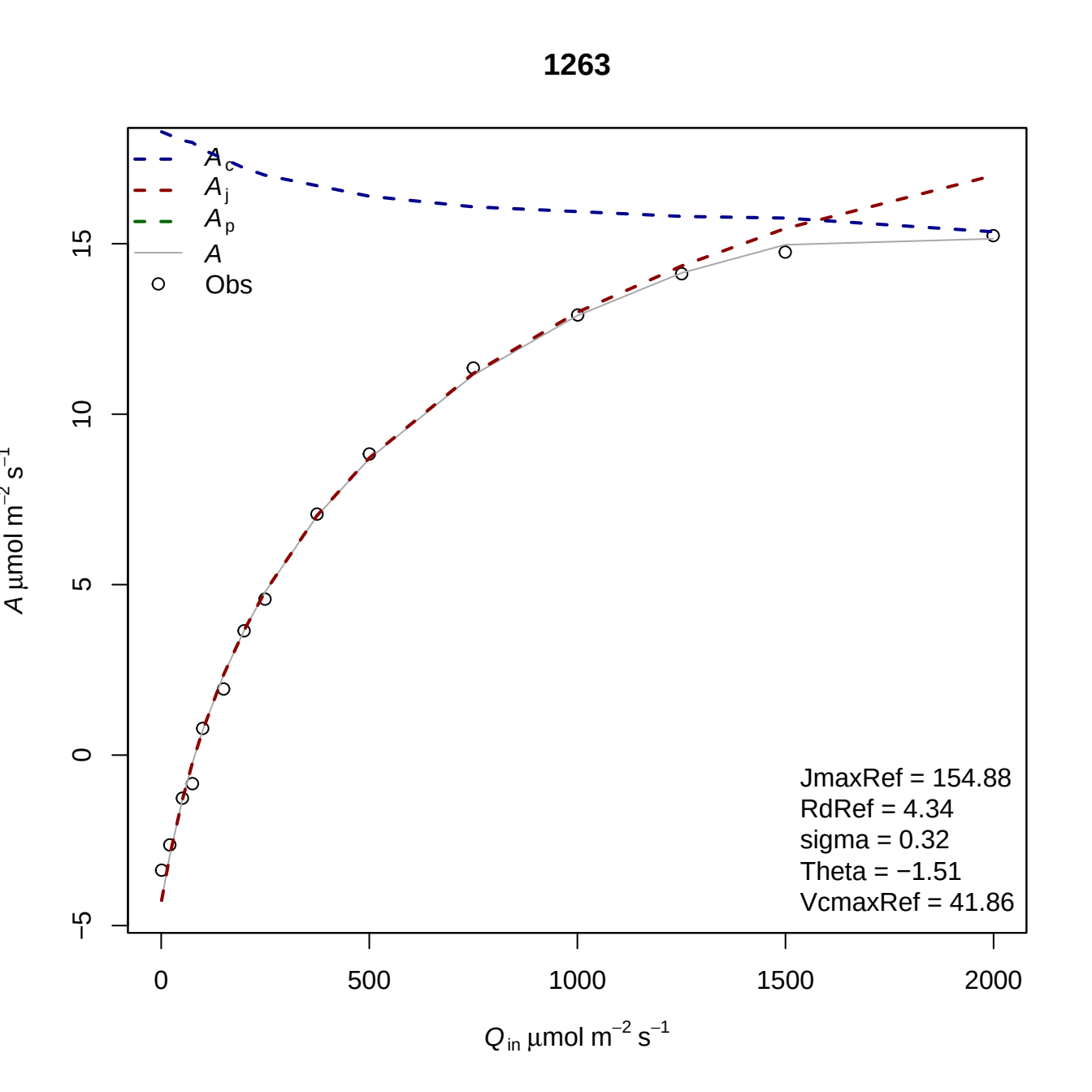

1258
A_c
A_j
A_p
A
Obs
JmaxRef = 126.43
RdRef = 3
sigma = 0.6
Theta = −0.73
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

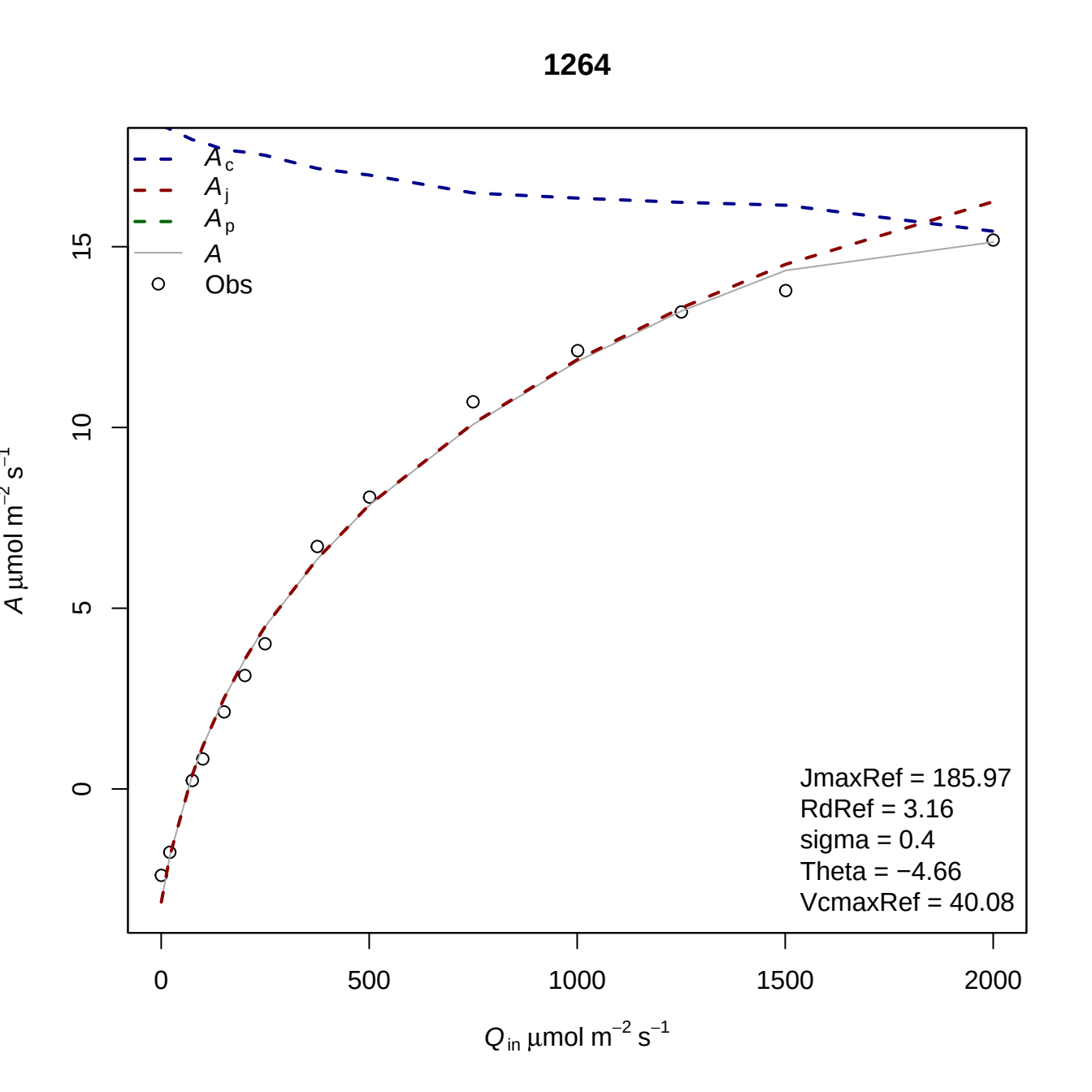

1259
A_c
A_j
A_p
A
Obs
JmaxRef = 238.58
RdRef = 4.77
sigma = 0.5
Theta = −5.24
VcmaxRef = 50.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

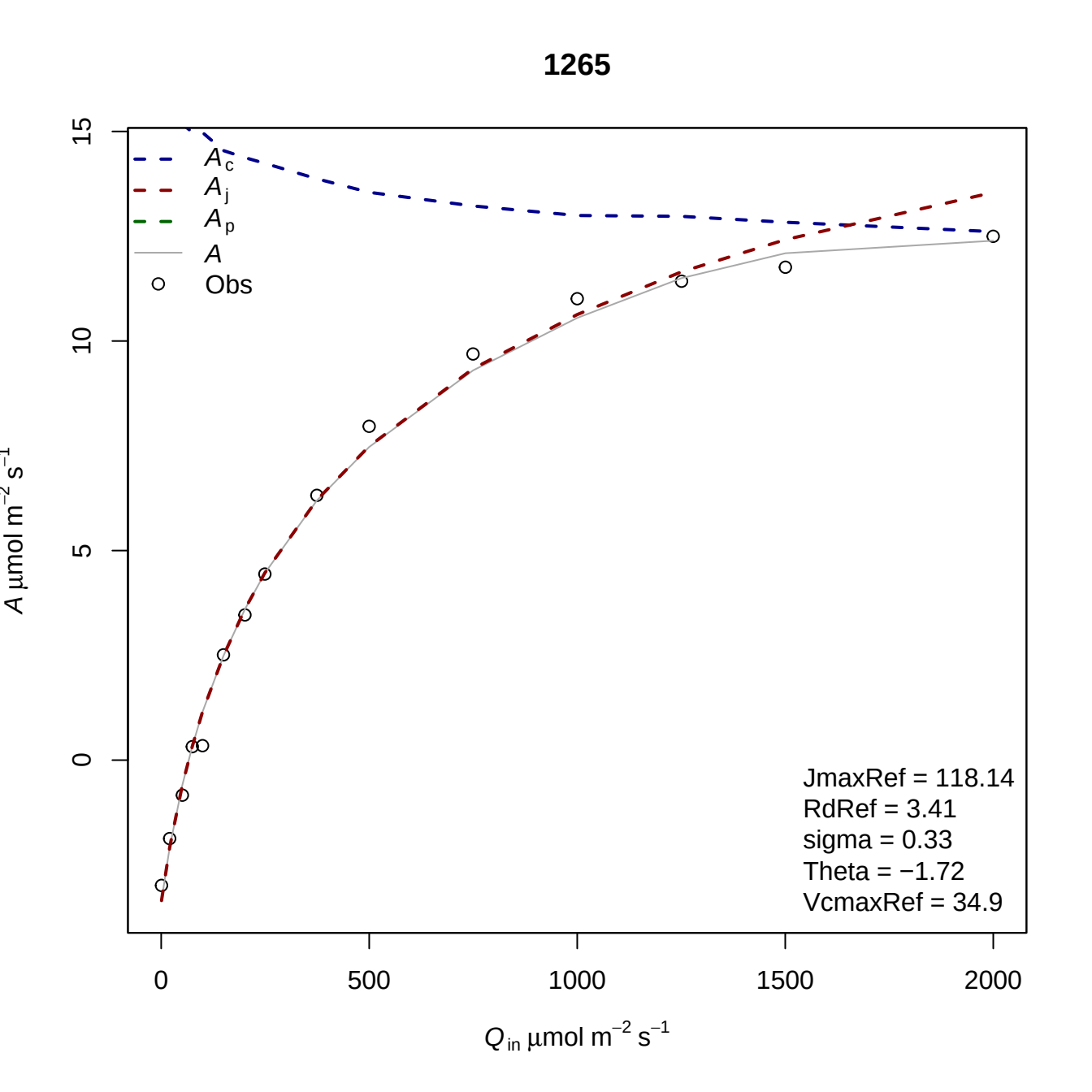

1260
A_c
A_j
A_p
A
Obs
JmaxRef = 129.92
RdRef = 2.18
sigma = 0.22
Theta = −2.98
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

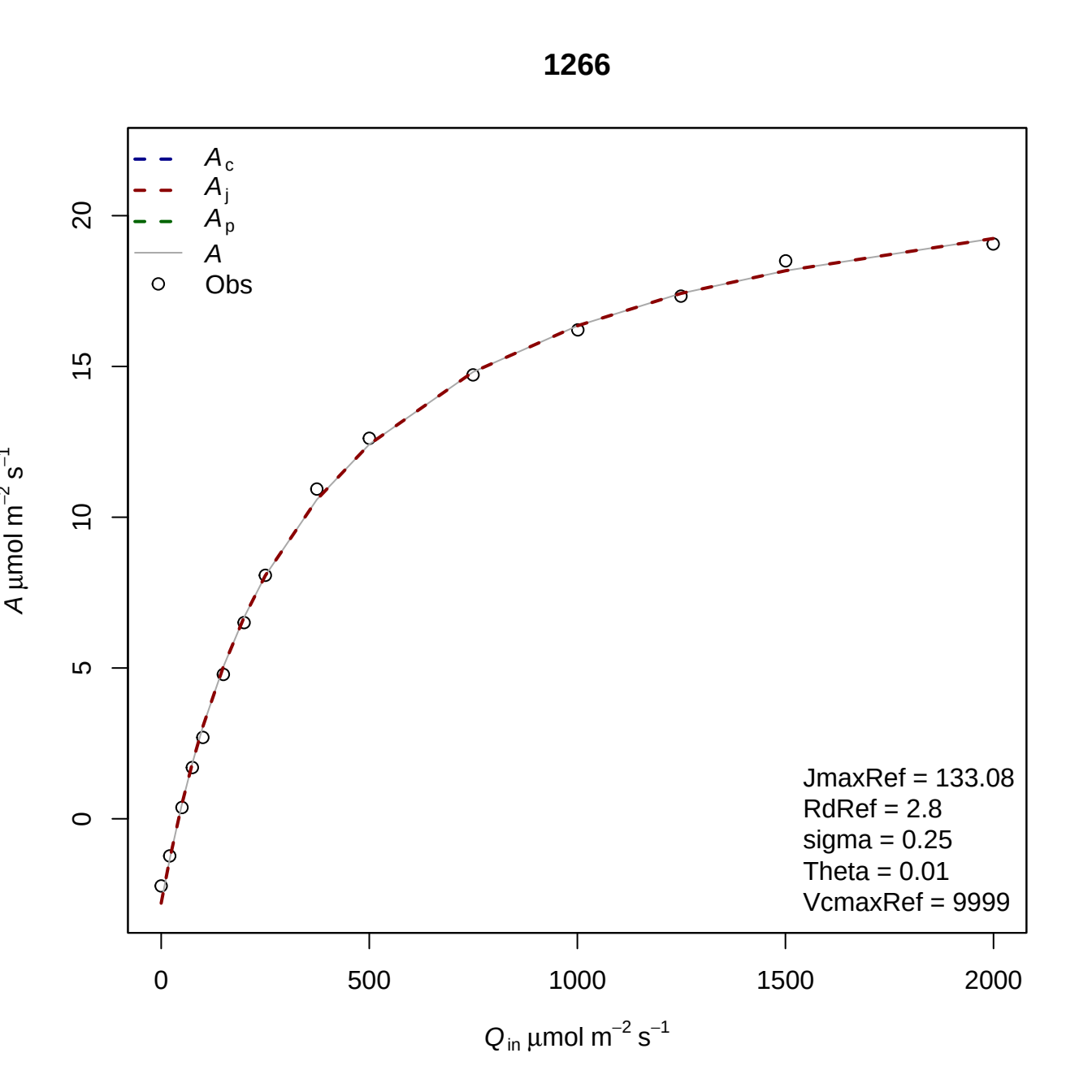

1261
A_c
A_j
A_p
A
Obs
JmaxRef = 134.25
RdRef = 2.11
sigma = 0.3
Theta = −1.64
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

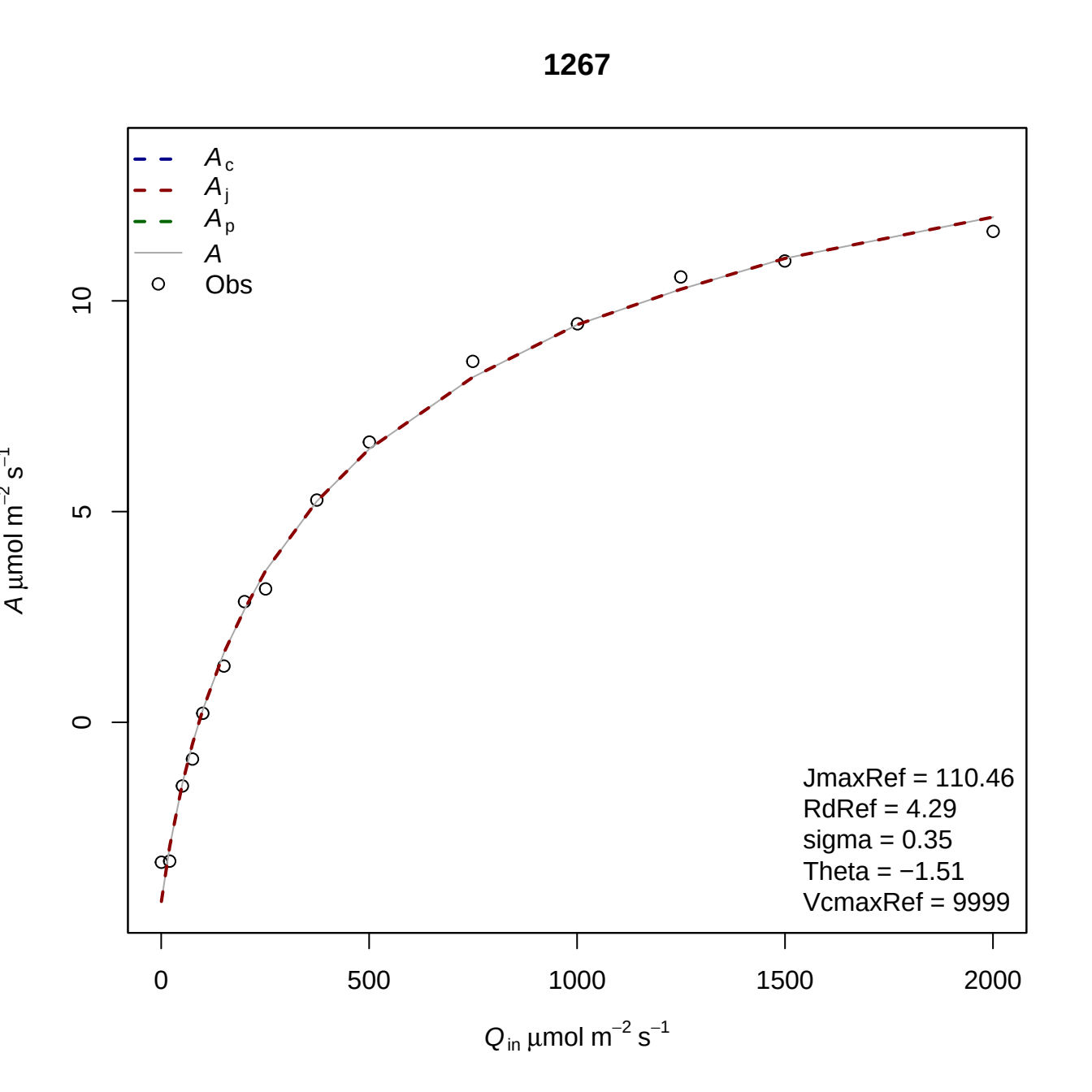

1262
A_c
A_j
A_p
A
Obs
JmaxRef = 127.7
RdRef = 3.58
sigma = 0.28
Theta = −3.64
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

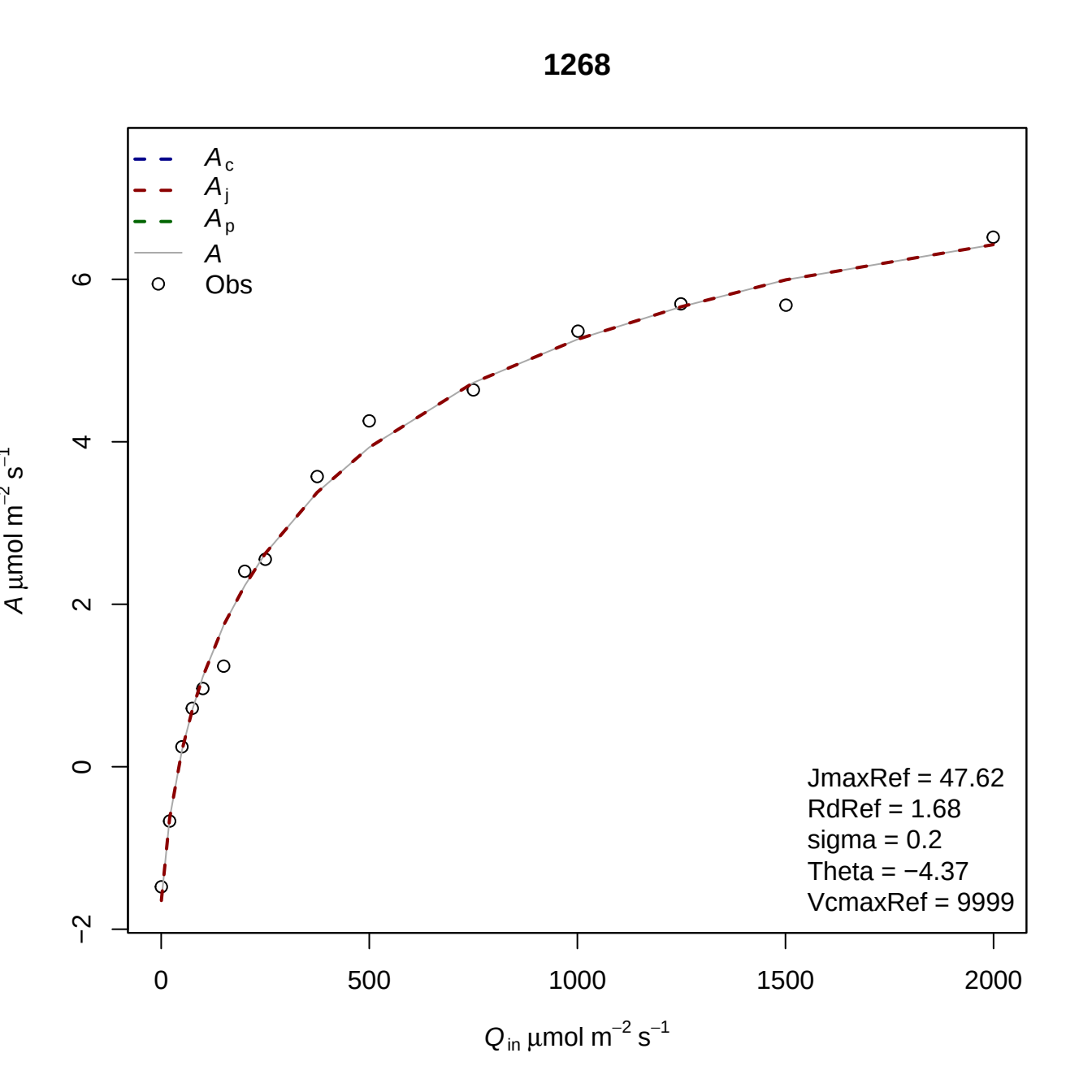

1263
A_c
A_j
A_p
A
Obs
JmaxRef = 154.88
RdRef = 4.34
sigma = 0.32
Theta = −1.51
VcmaxRef = 41.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

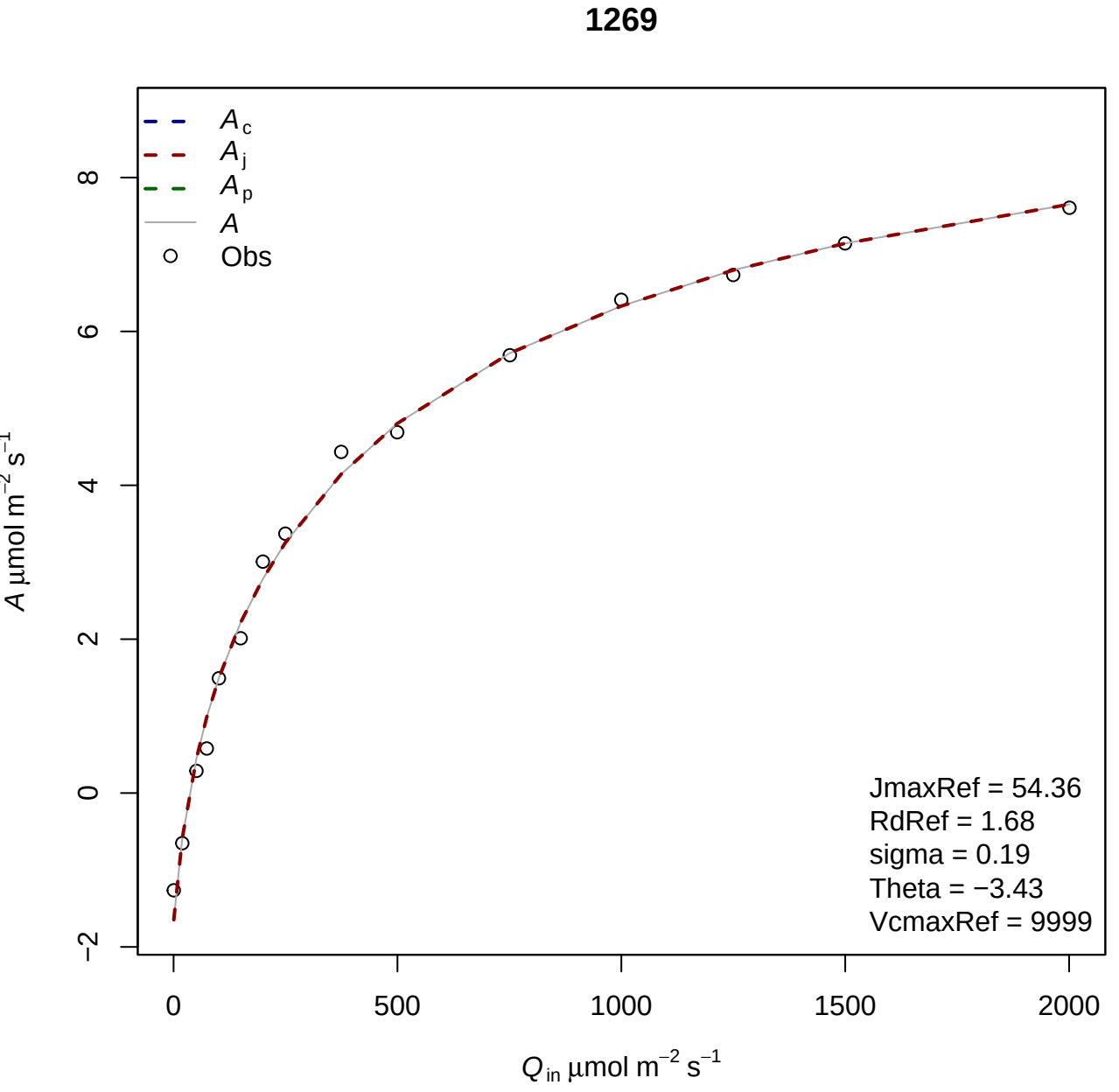

1264
A_c
A_j
A_p
A
Obs
JmaxRef = 185.97
RdRef = 3.16
sigma = 0.4
Theta = −4.66
VcmaxRef = 40.08
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

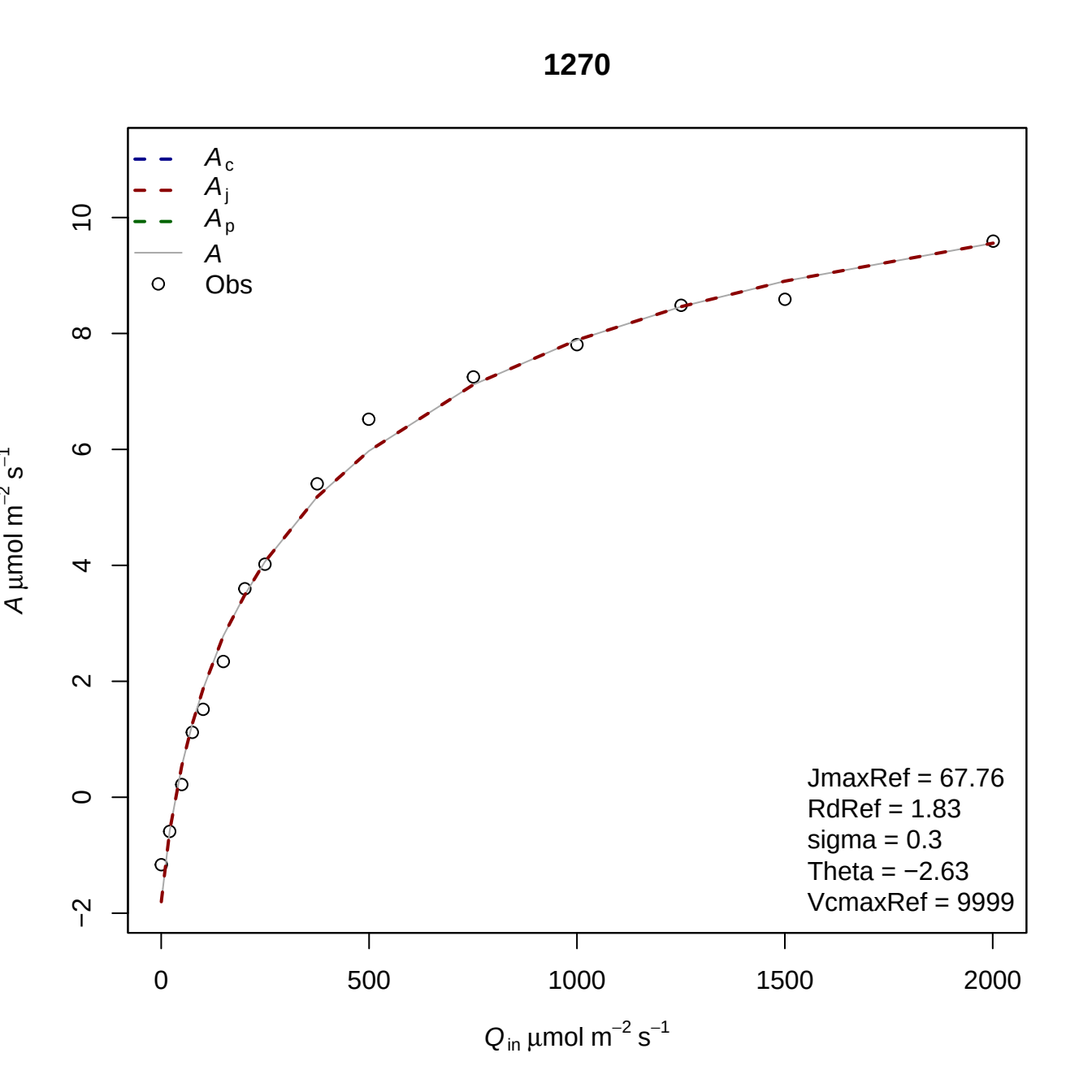

1265
A_c
A_j
A_p
A
Obs
JmaxRef = 118.14
RdRef = 3.41
sigma = 0.33
Theta = −1.72
VcmaxRef = 34.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

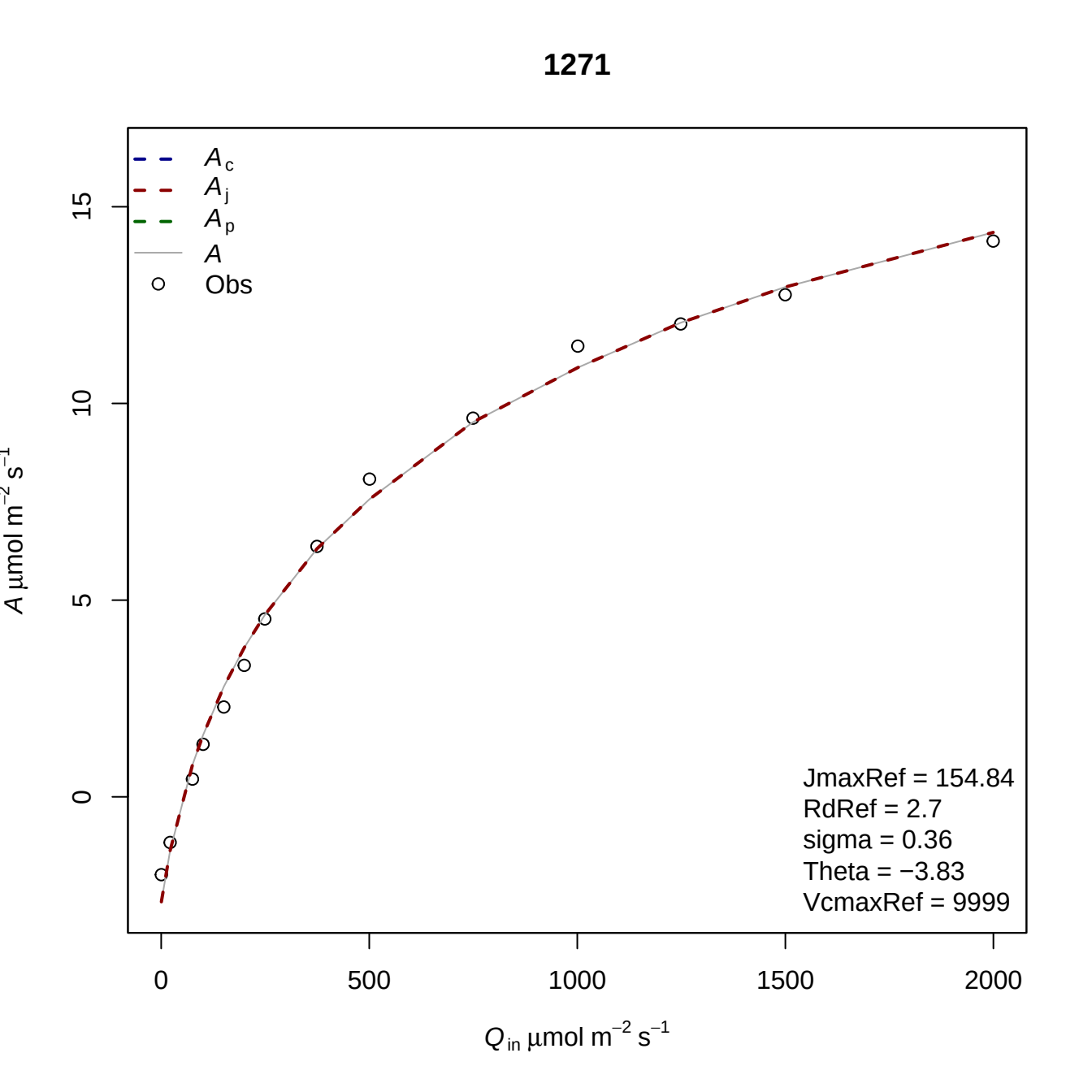

1266
A_c
A_j
A_p
A
Obs
JmaxRef = 133.08
RdRef = 2.8
sigma = 0.25
Theta = 0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

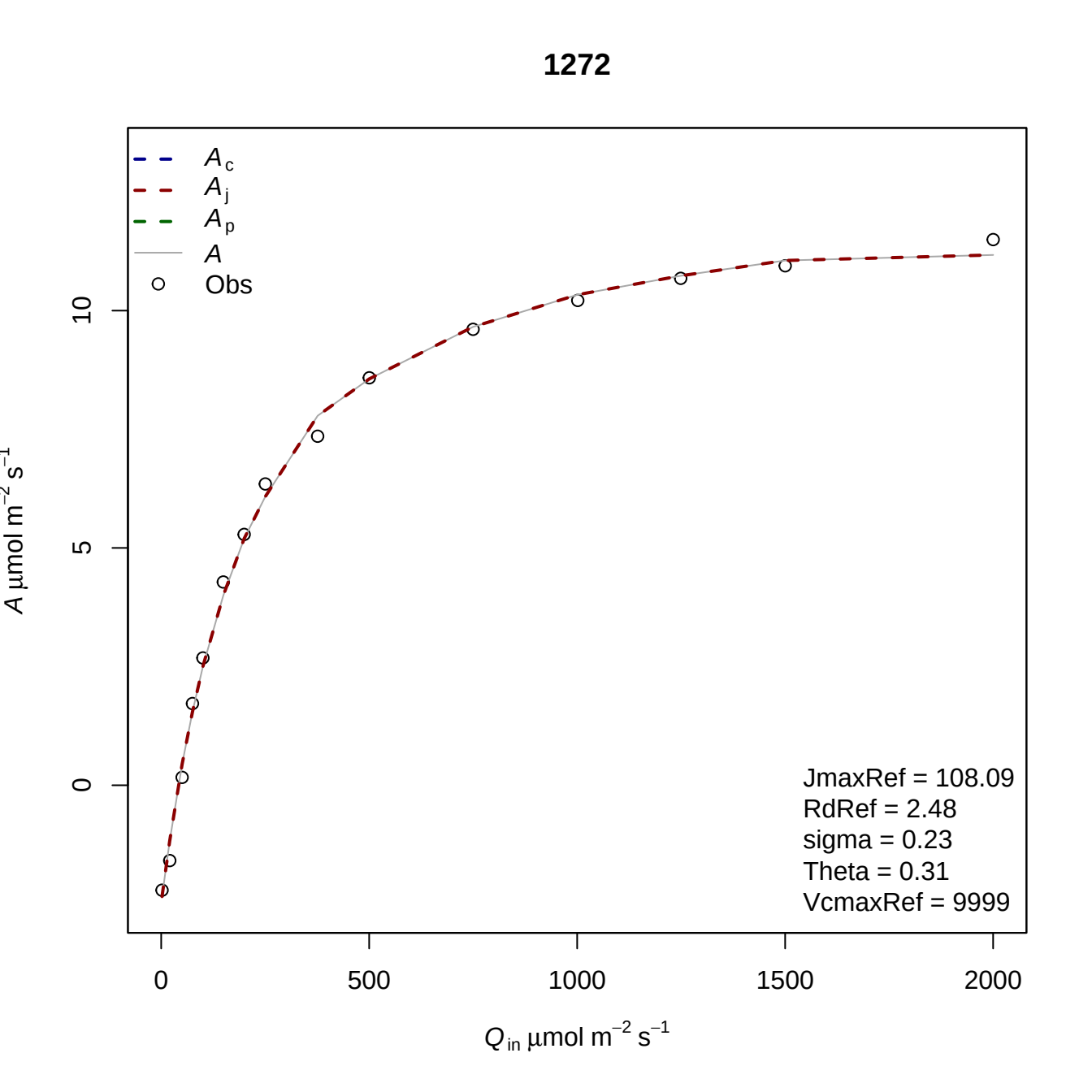

1267
A_c
A_j
A_p
A
Obs
JmaxRef = 110.46
RdRef = 4.29
sigma = 0.35
Theta = −1.51
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

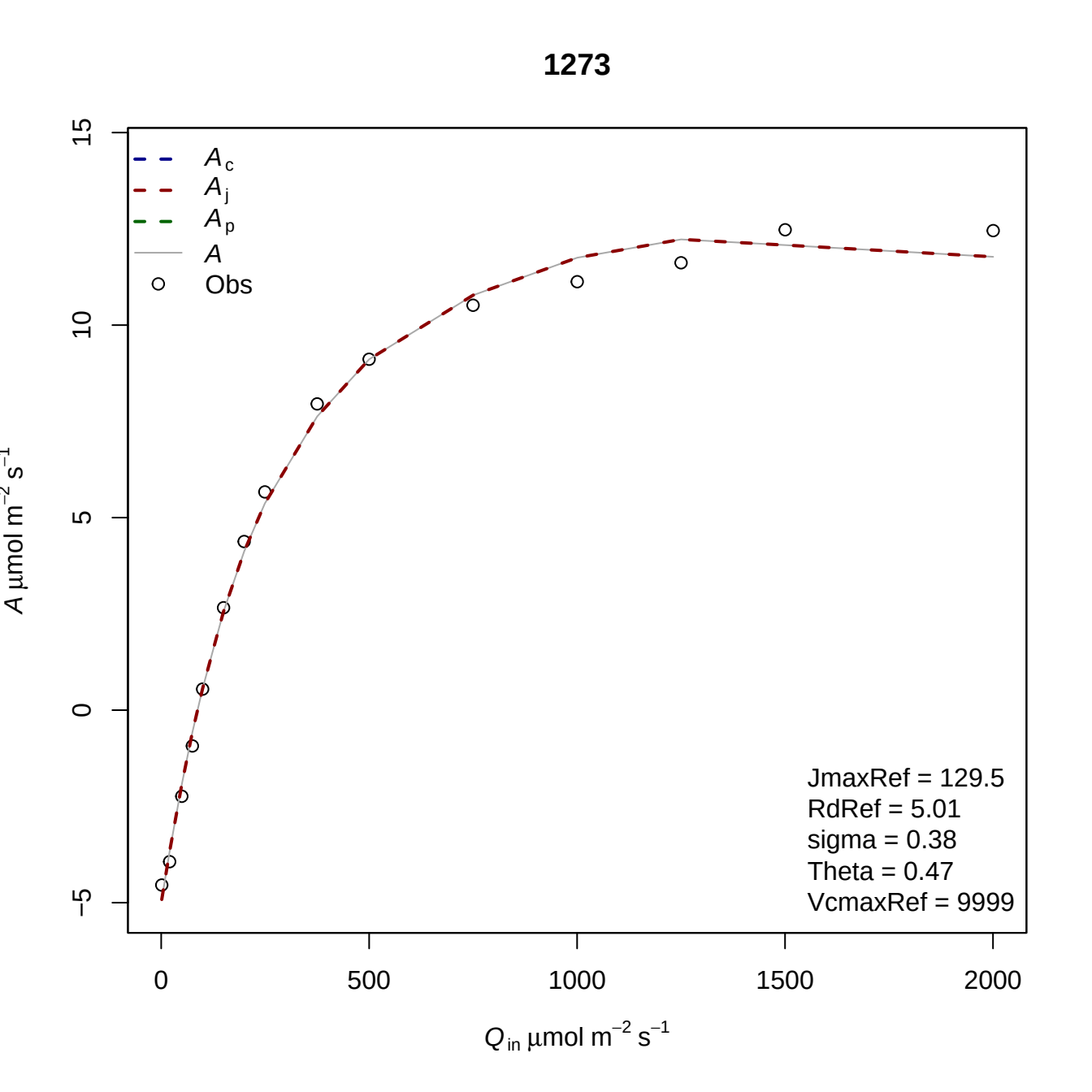

1268
A_c
A_j
A_p
A
Obs
JmaxRef = 47.62
RdRef = 1.68
sigma = 0.2
Theta = −4.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

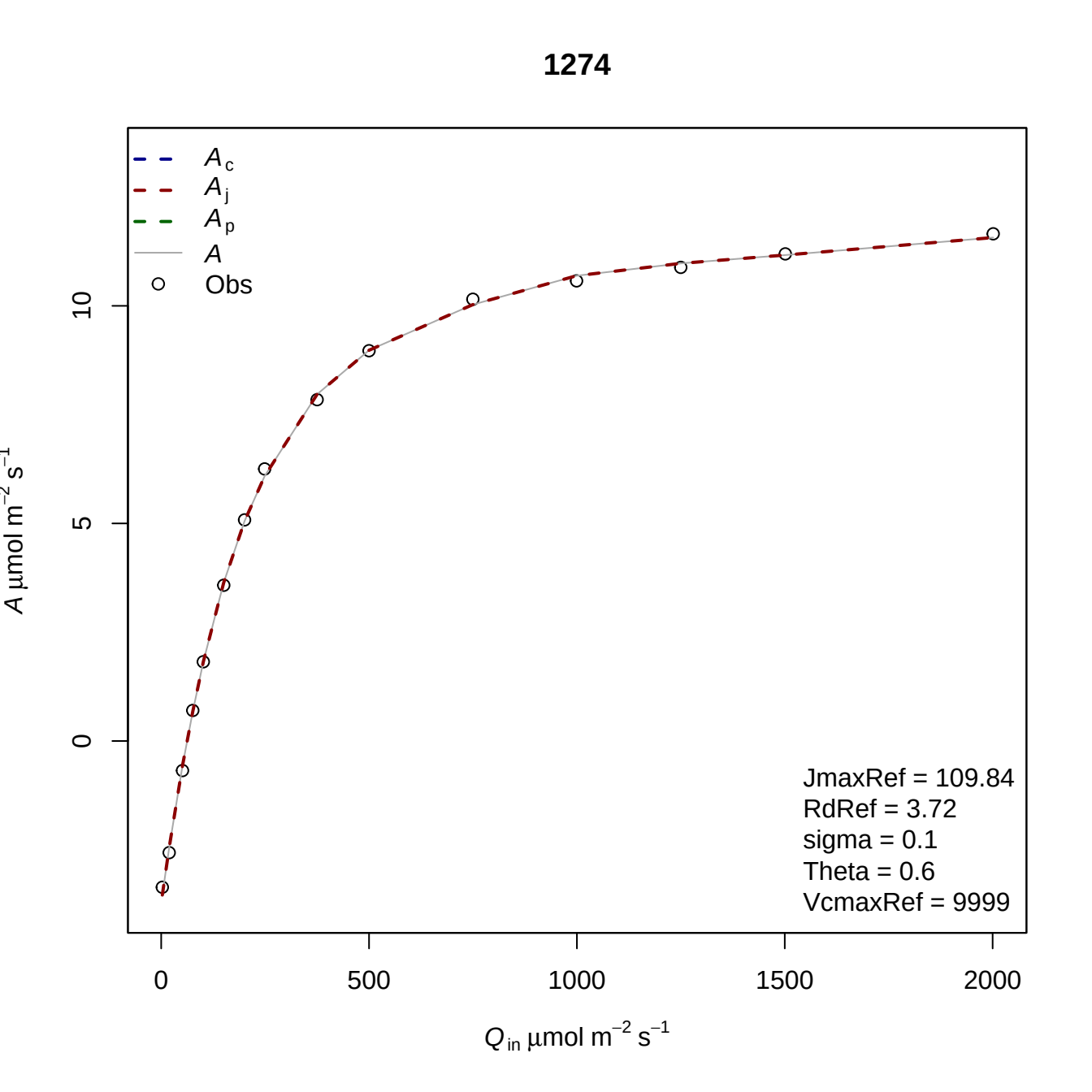

1269
A_c
A_j
A_p
A
Obs
JmaxRef = 54.36
RdRef = 1.68
sigma = 0.19
Theta = −3.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

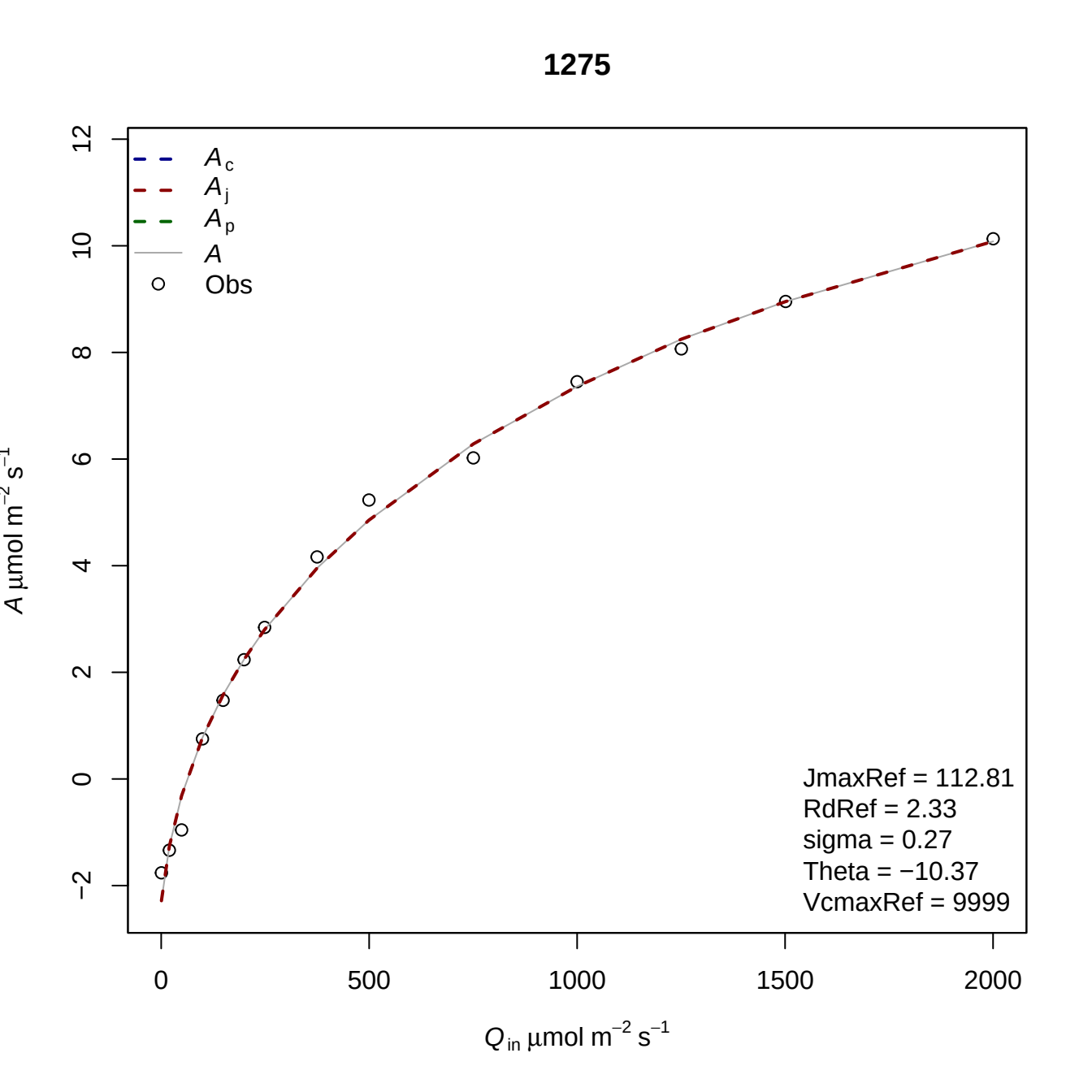

1270
A_c
A_j
A_p
A
Obs
JmaxRef = 67.76
RdRef = 1.83
sigma = 0.3
Theta = −2.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

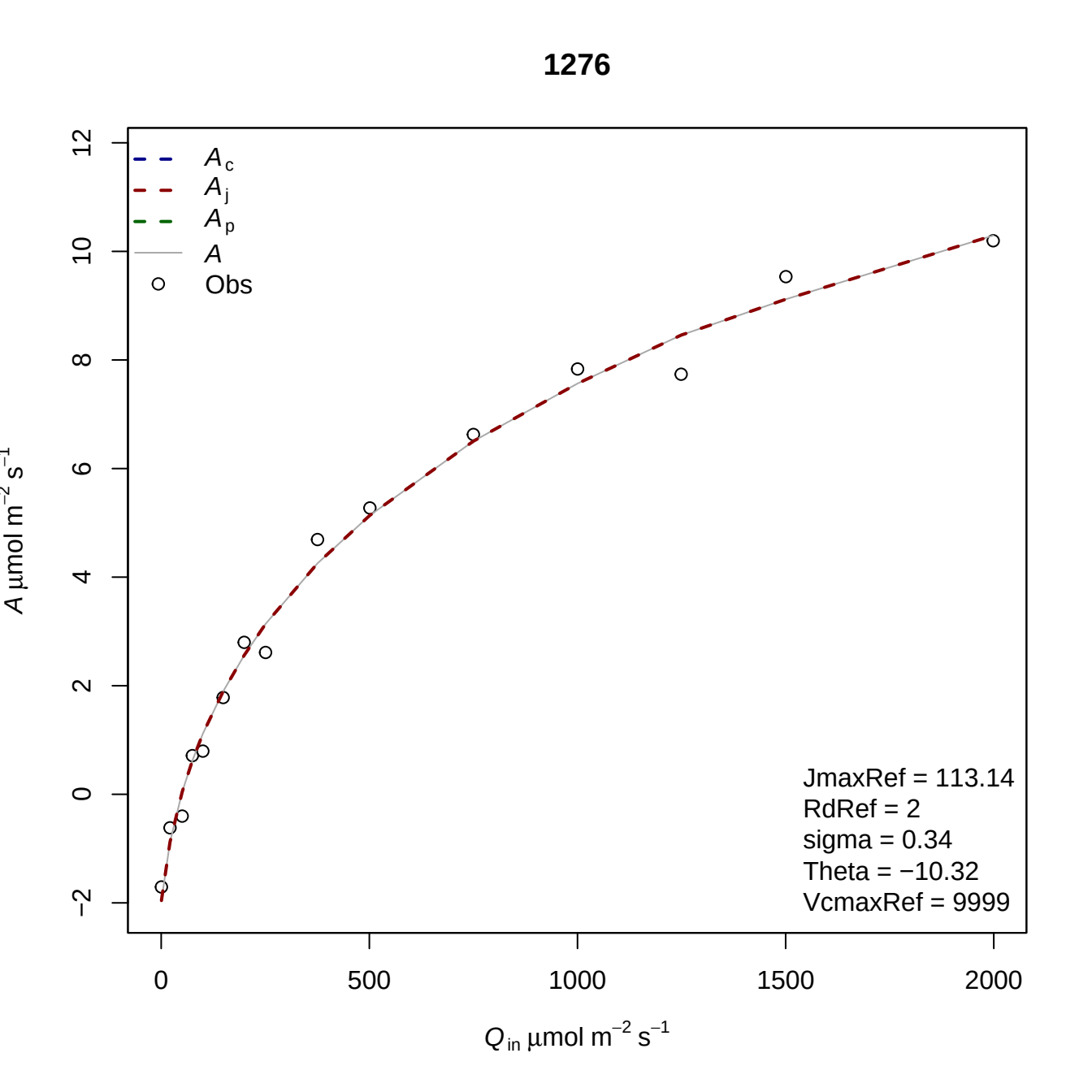

1271
A_c
A_j
A_p
A
Obs
JmaxRef = 154.84
RdRef = 2.7
sigma = 0.36
Theta = −3.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

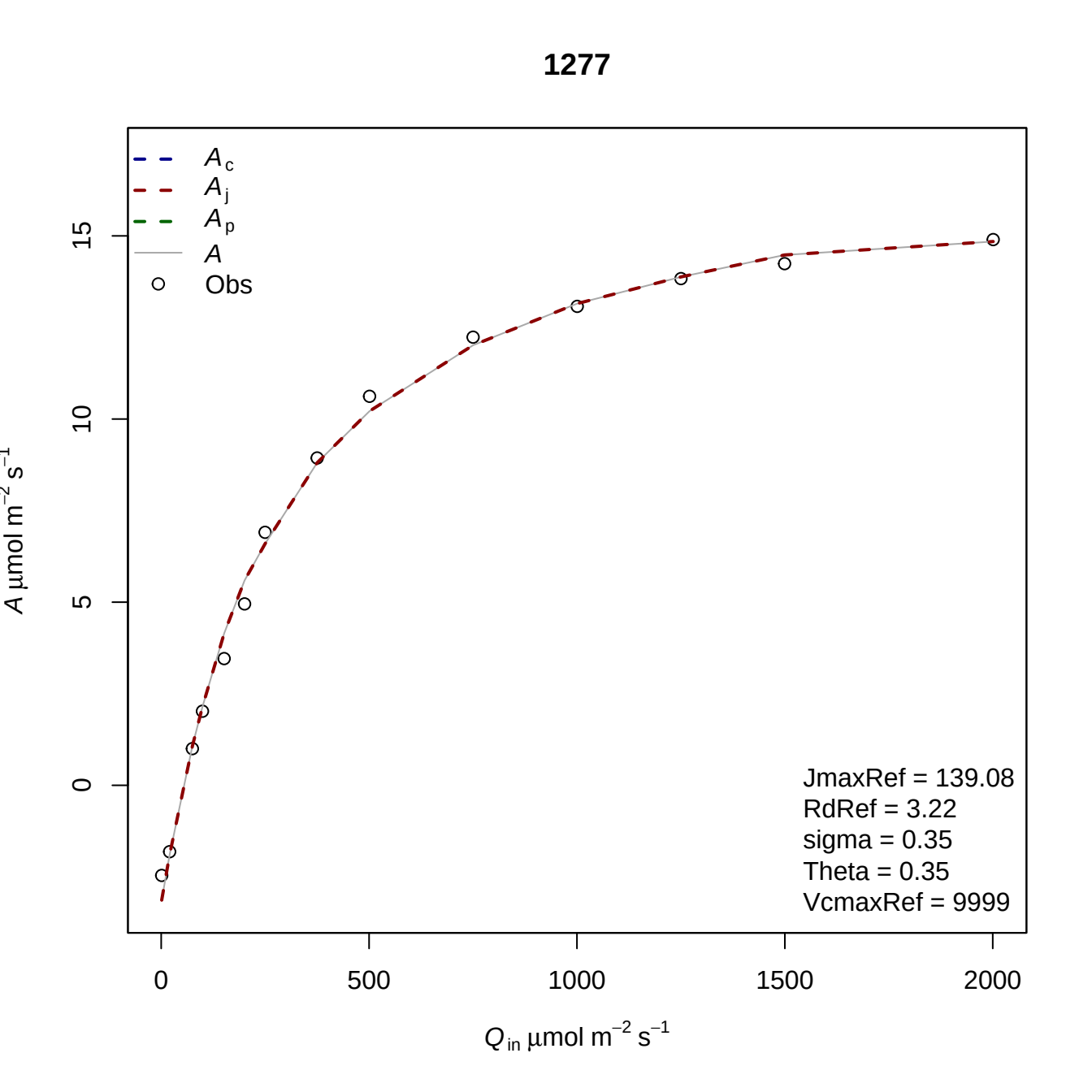

1272
A_c
A_j
A_p
A
Obs
JmaxRef = 108.09
RdRef = 2.48
sigma = 0.23
Theta = 0.31
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

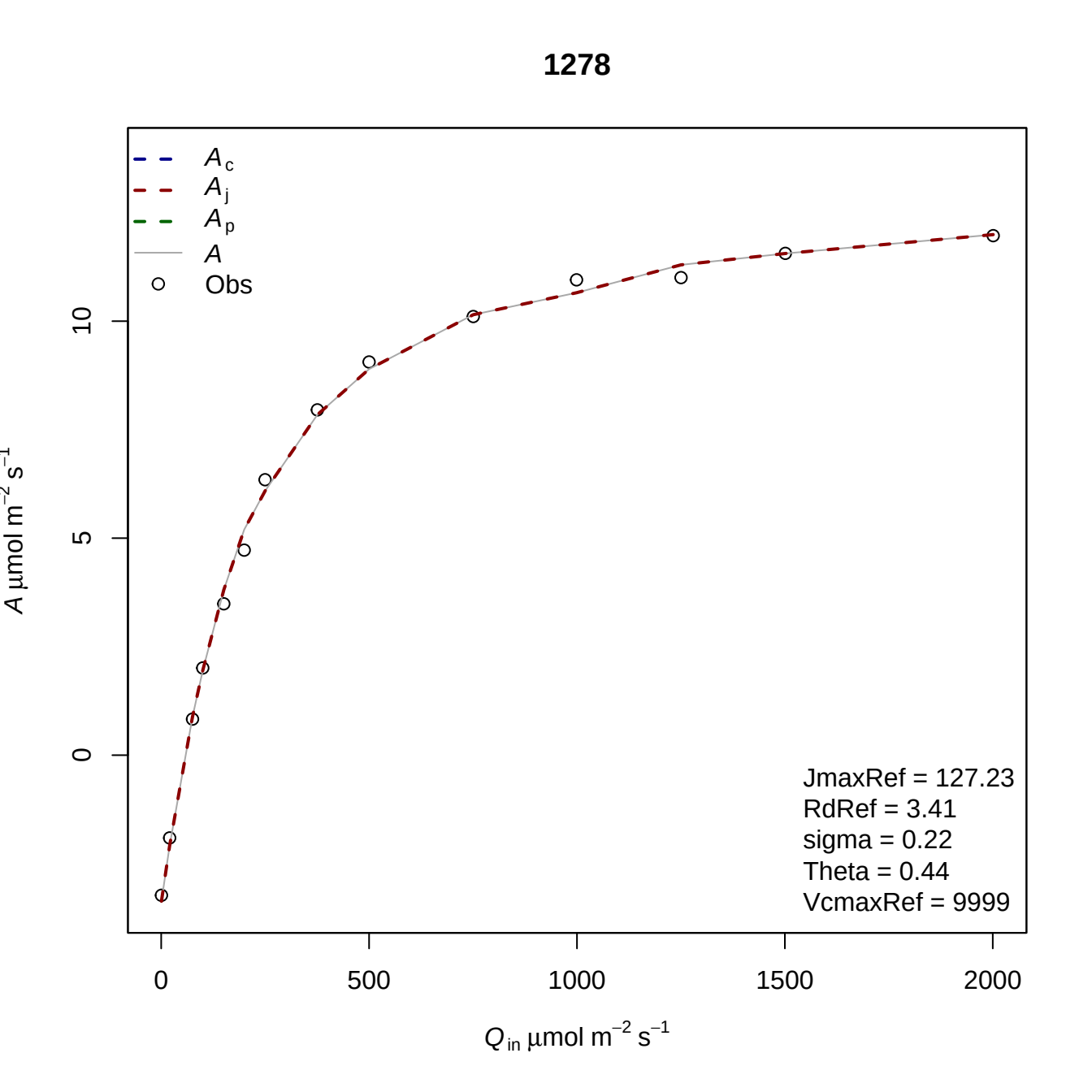

1273
A_c
A_j
A_p
A
Obs
JmaxRef = 129.5
RdRef = 5.01
sigma = 0.38
Theta = 0.47
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

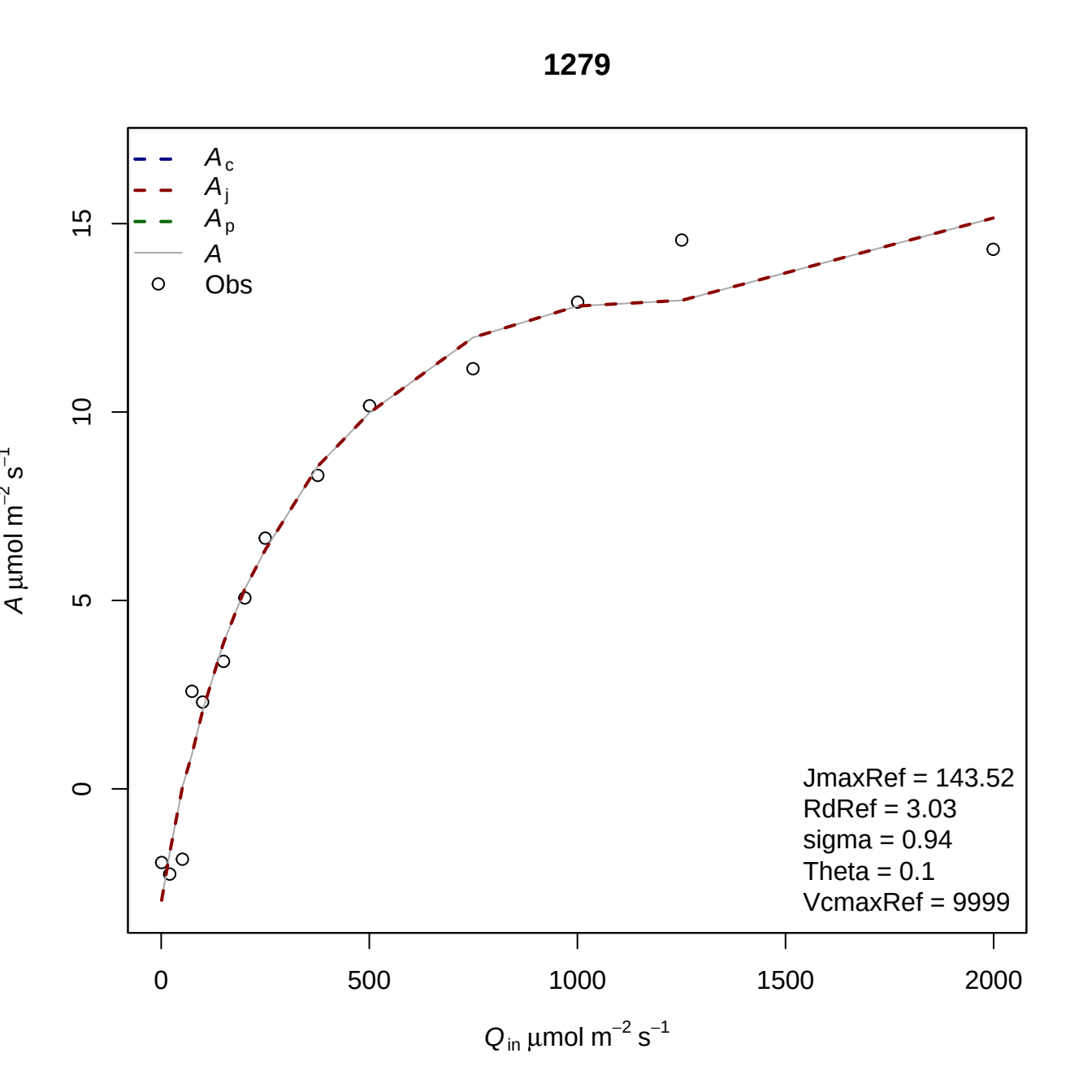

1274
A_c
A_j
A_p
A
Obs
JmaxRef = 109.84
RdRef = 3.72
sigma = 0.1
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

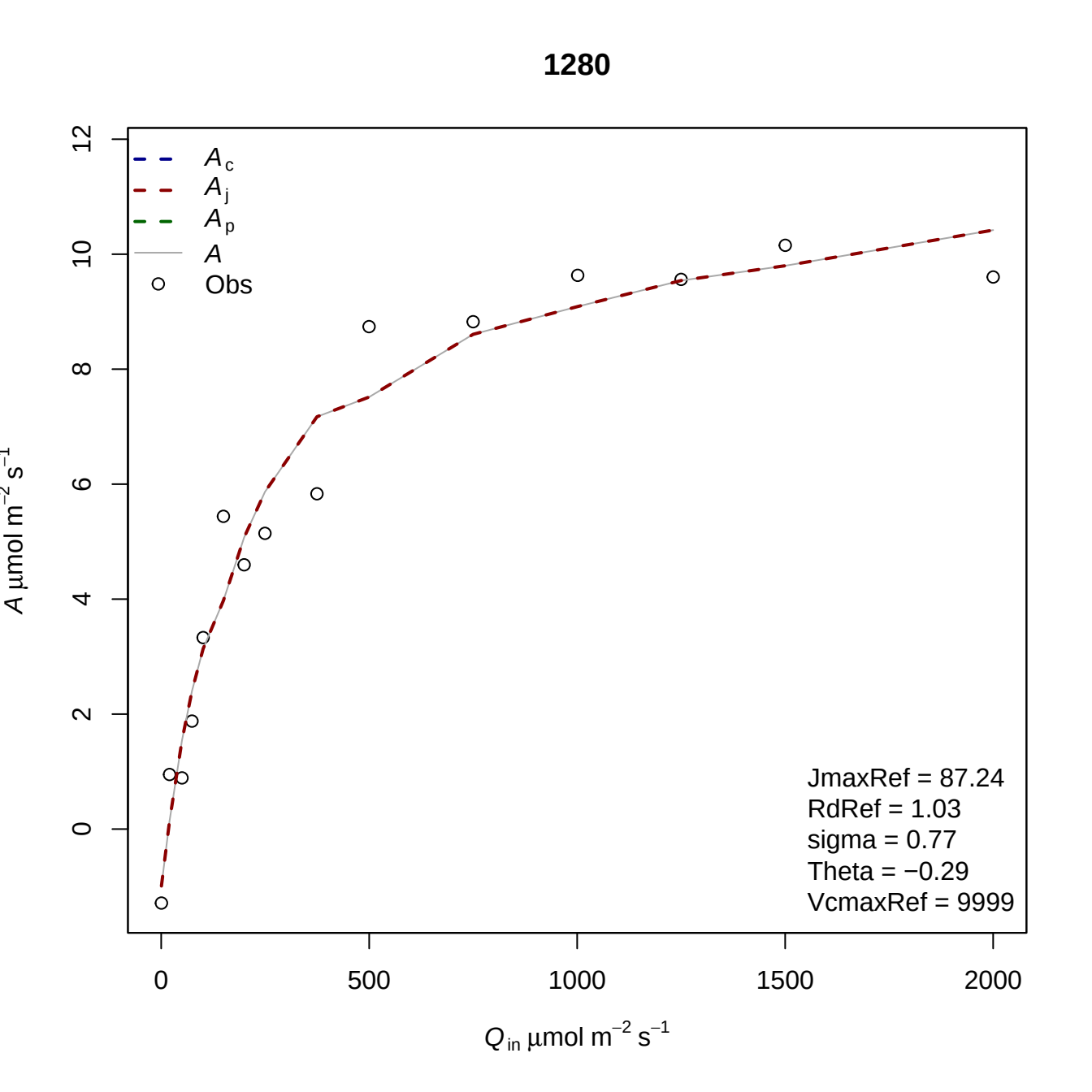

1275
A_c
A_j
A_p
A
Obs
JmaxRef = 112.81
RdRef = 2.33
sigma = 0.27
Theta = −10.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

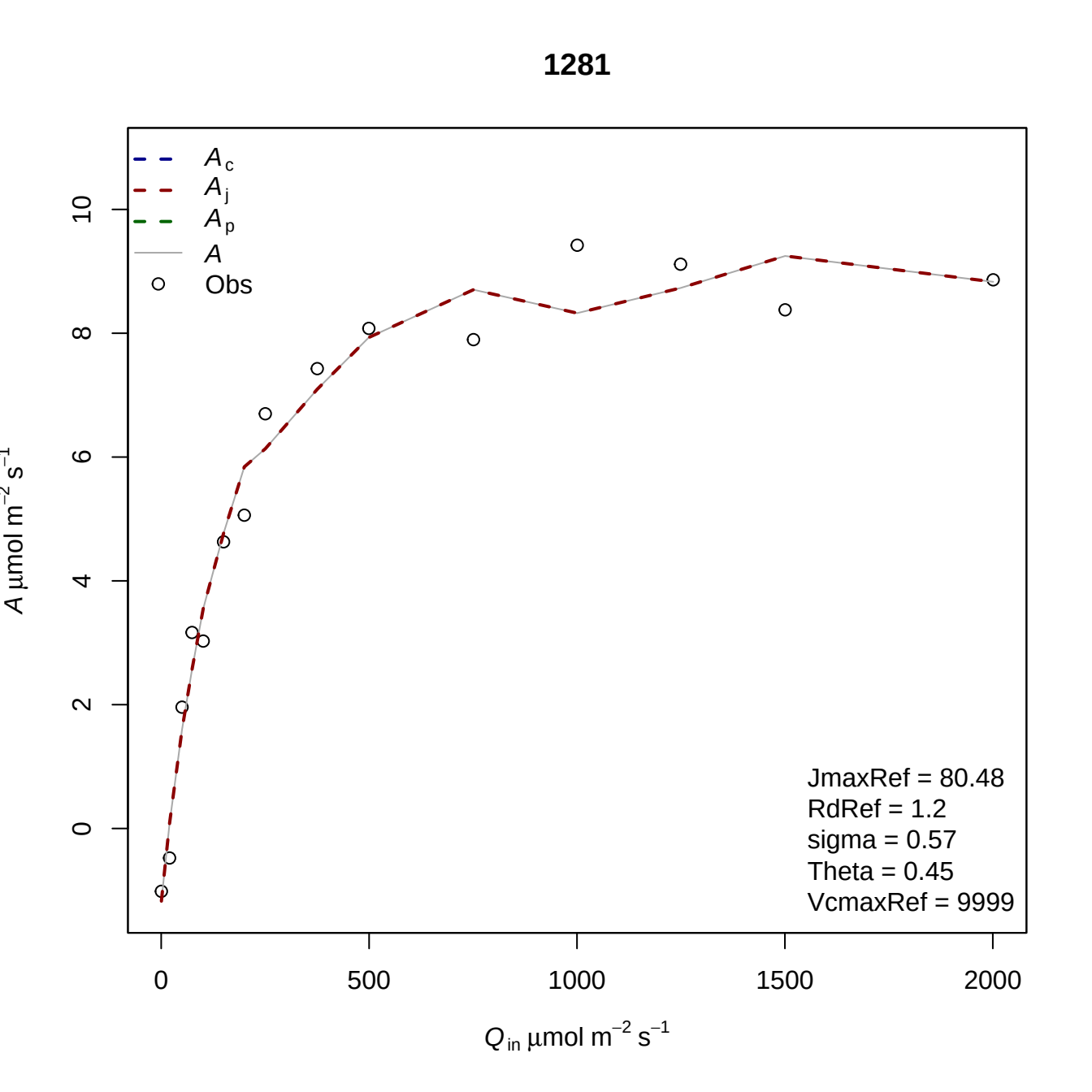

1276
A_c
A_j
A_p
A
Obs
JmaxRef = 113.14
RdRef = 2
sigma = 0.34
Theta = −10.32
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

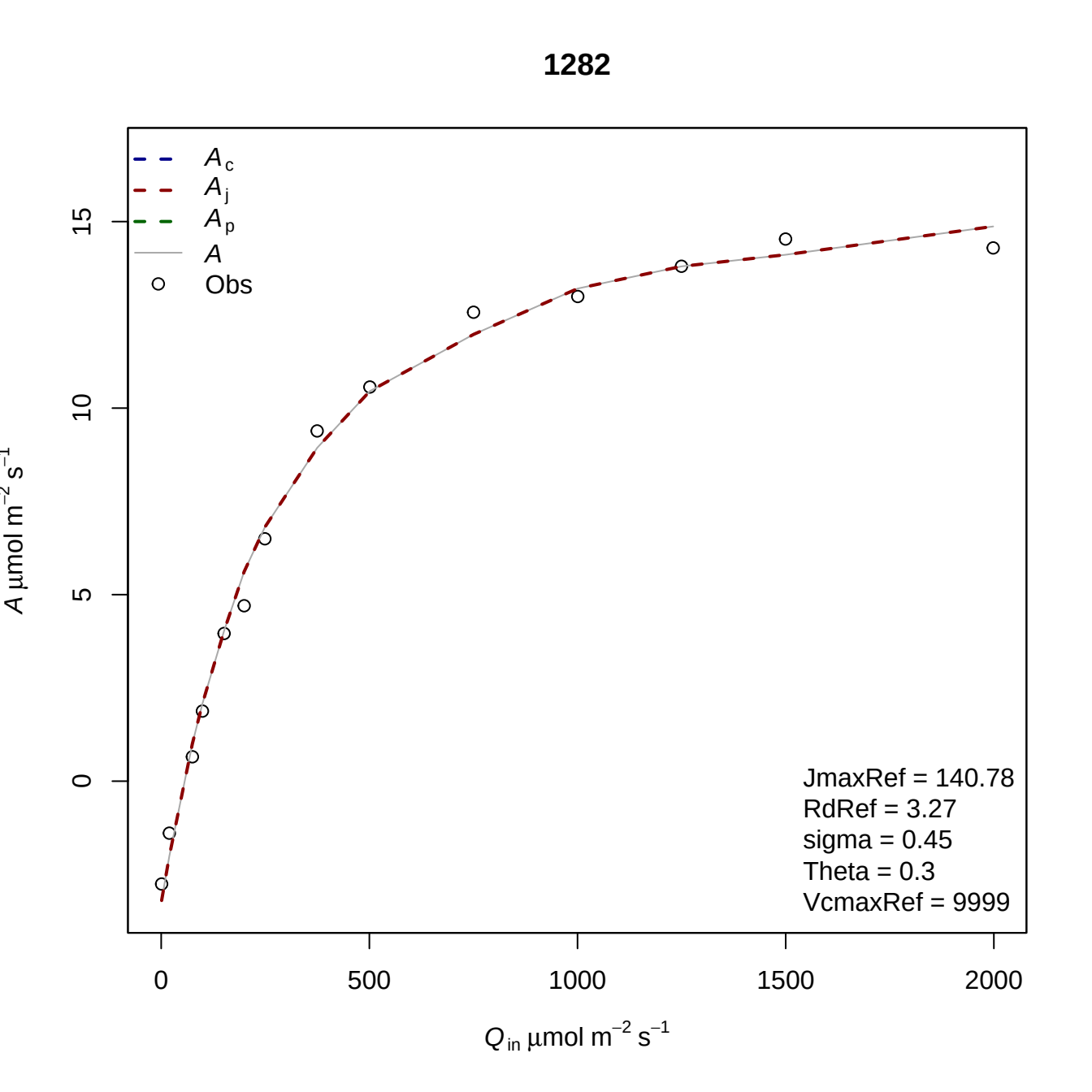

1277
A_c
A_j
A_p
A
Obs
JmaxRef = 139.08
RdRef = 3.22
sigma = 0.35
Theta = 0.35
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

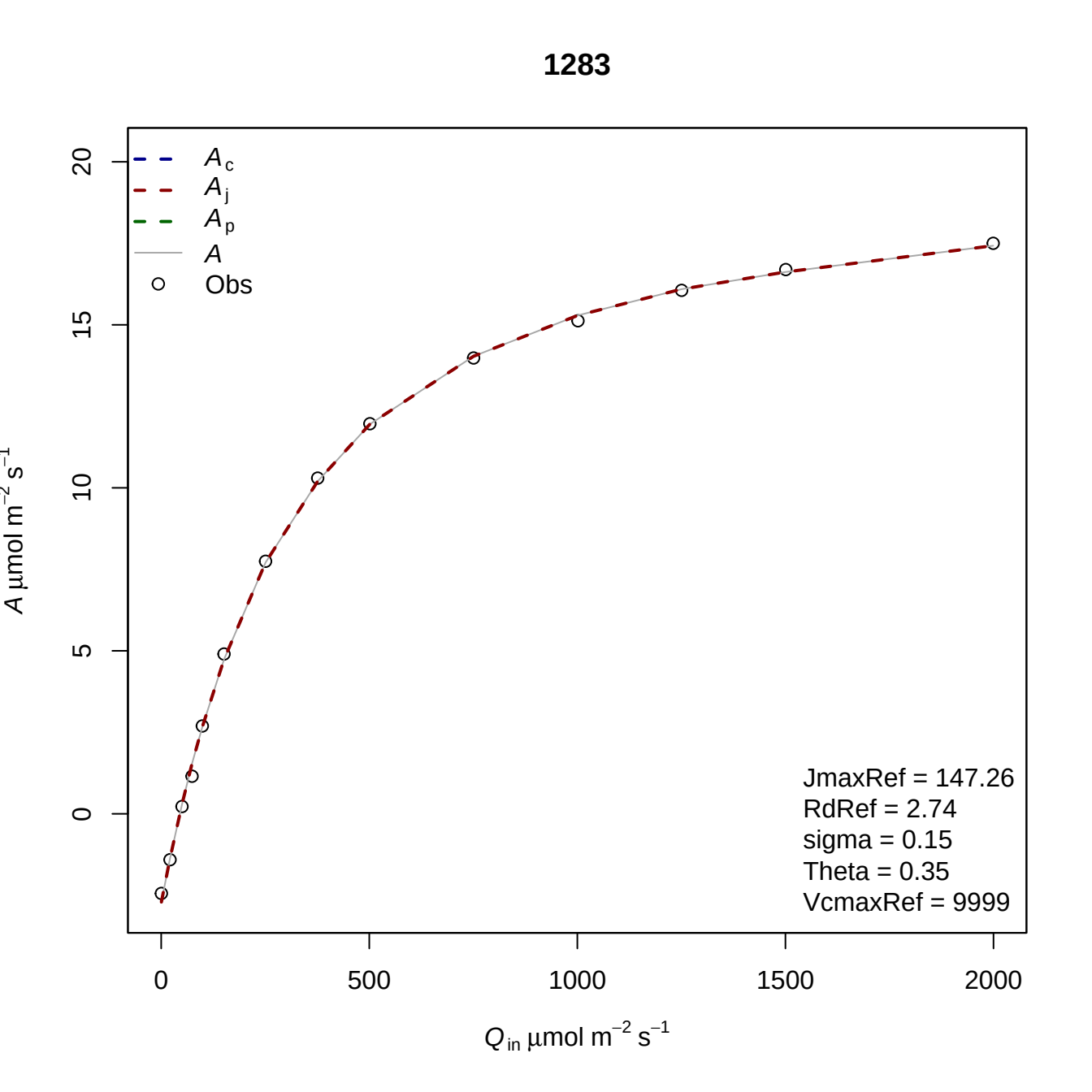

1278
A_c
A_j
A_p
A
Obs
JmaxRef = 127.23
RdRef = 3.41
sigma = 0.22
Theta = 0.44
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

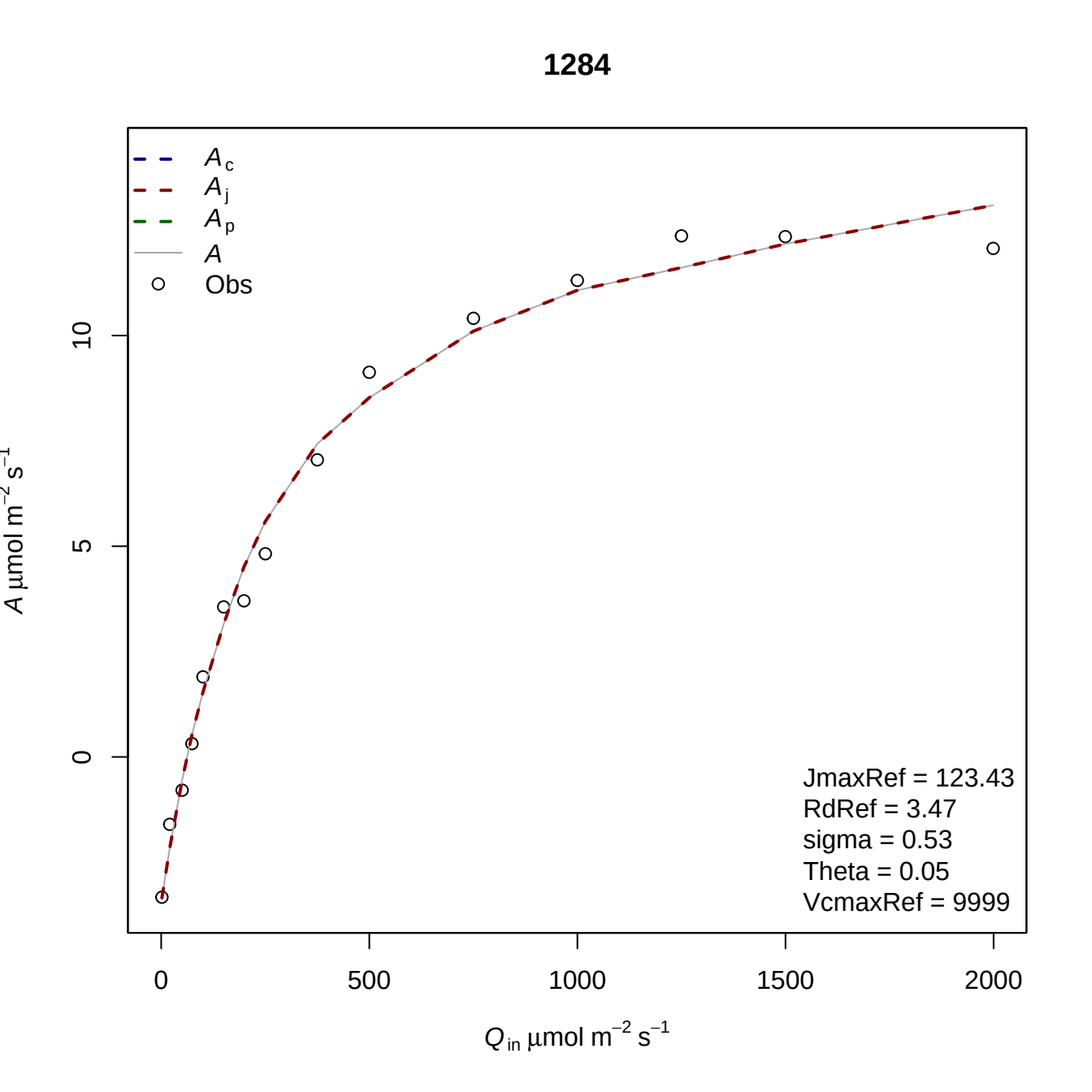

1279
A_c
A_j
A_p
A
Obs
JmaxRef = 143.52
RdRef = 3.03
sigma = 0.94
Theta = 0.1
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

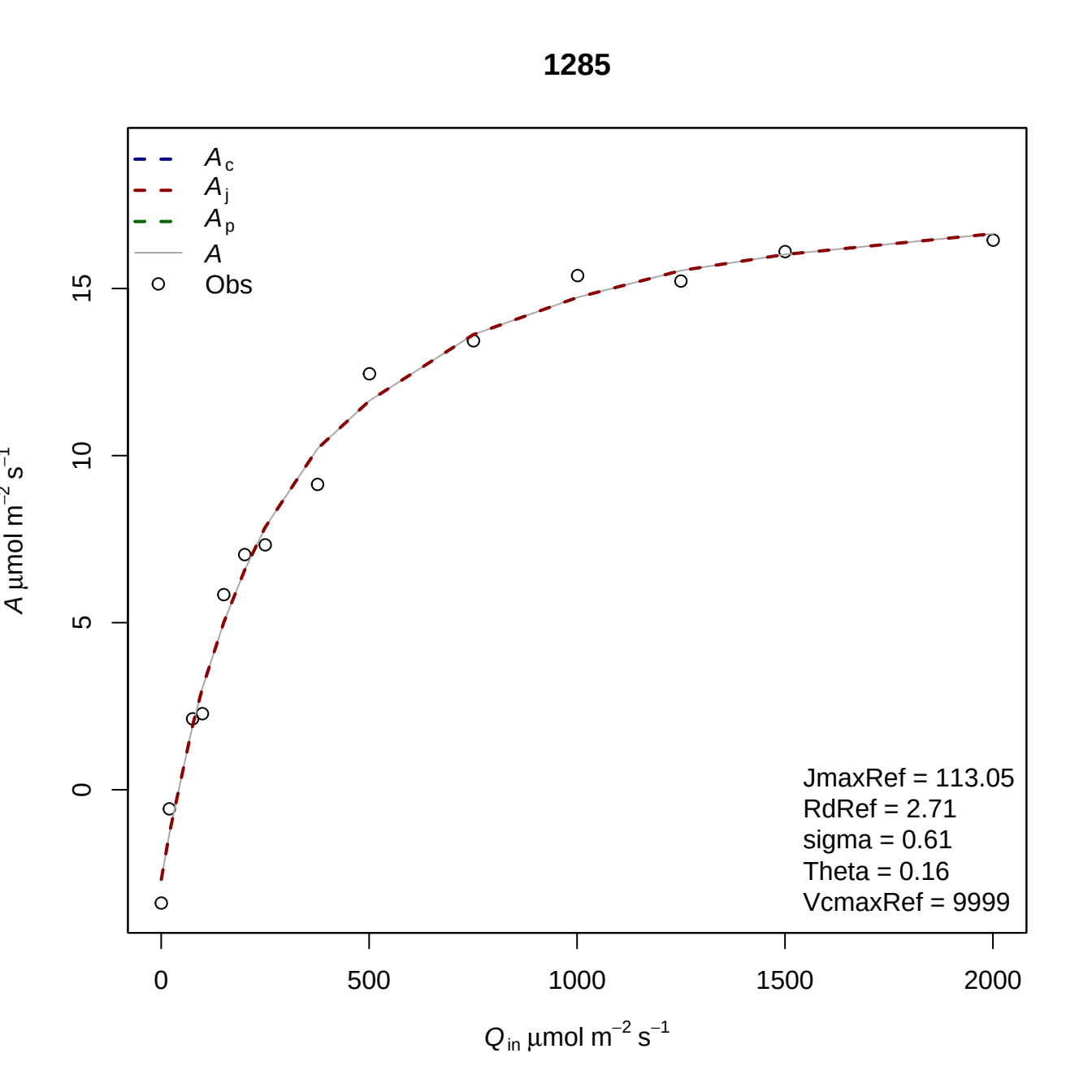

1280
A_c
A_j
A_p
A
Obs
JmaxRef = 87.24
RdRef = 1.03
sigma = 0.77
Theta = −0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

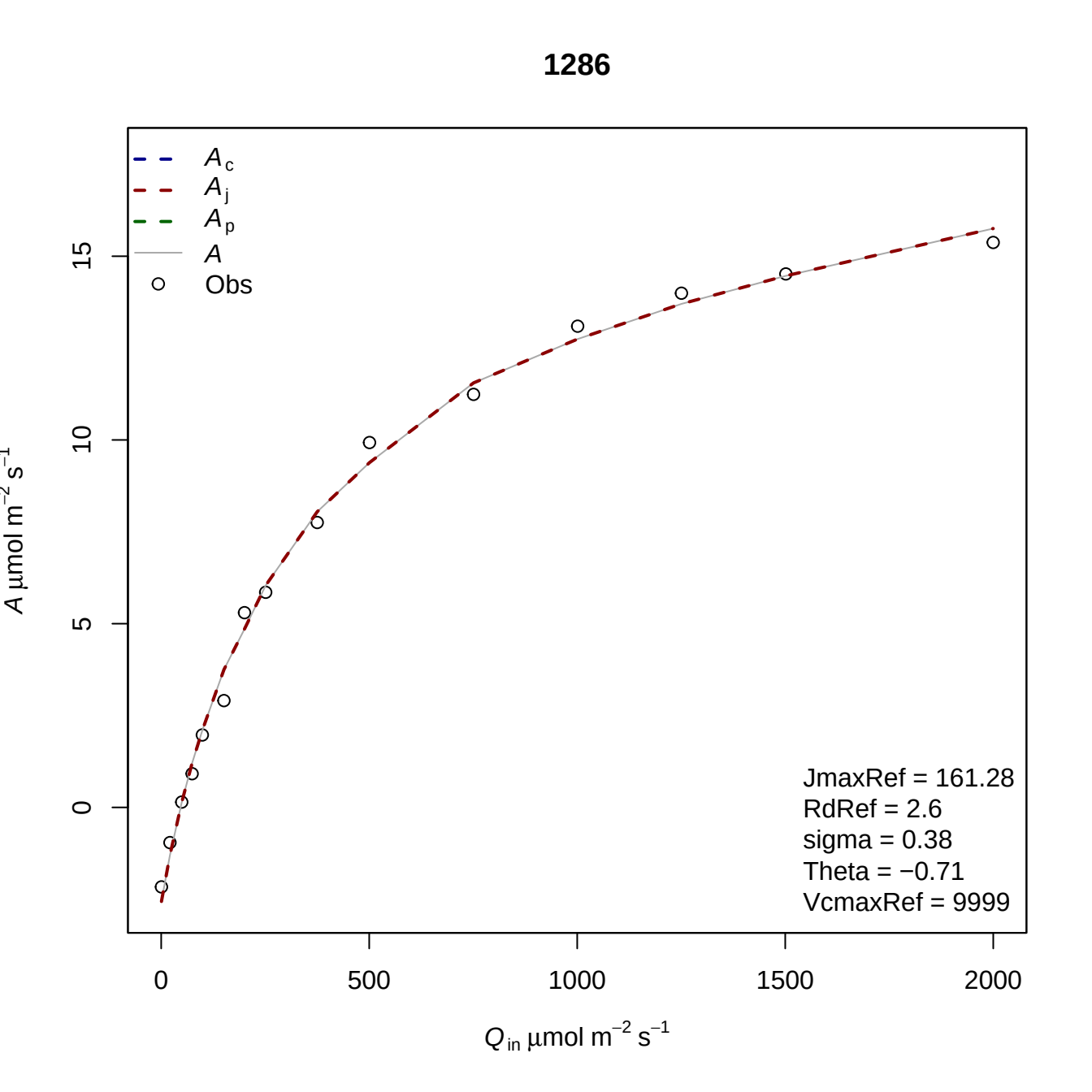

1281
A_c
A_j
A_p
A
Obs
JmaxRef = 80.48
RdRef = 1.2
sigma = 0.57
Theta = 0.45
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

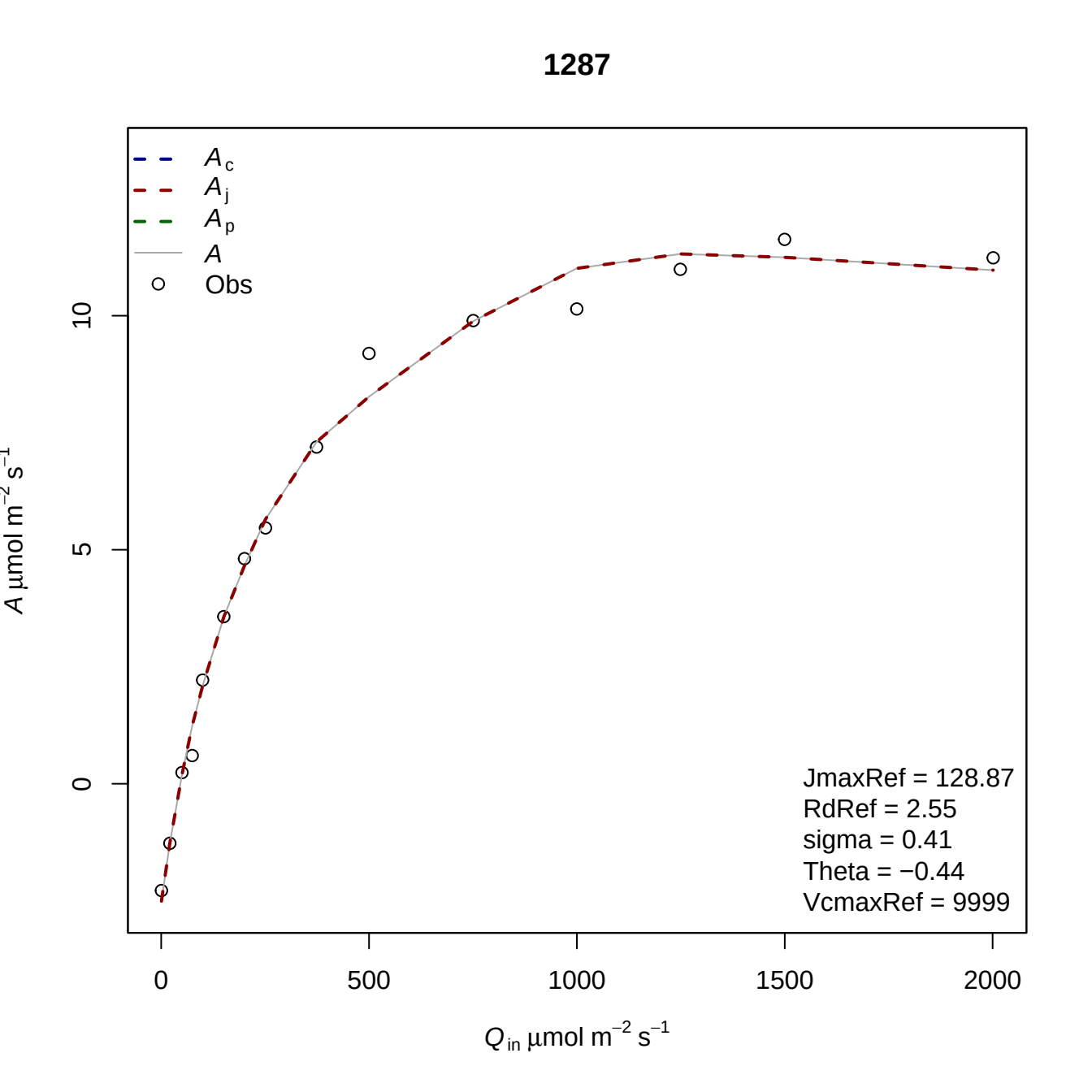

1282
A_c
A_j
A_p
A
Obs
JmaxRef = 140.78
RdRef = 3.27
sigma = 0.45
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

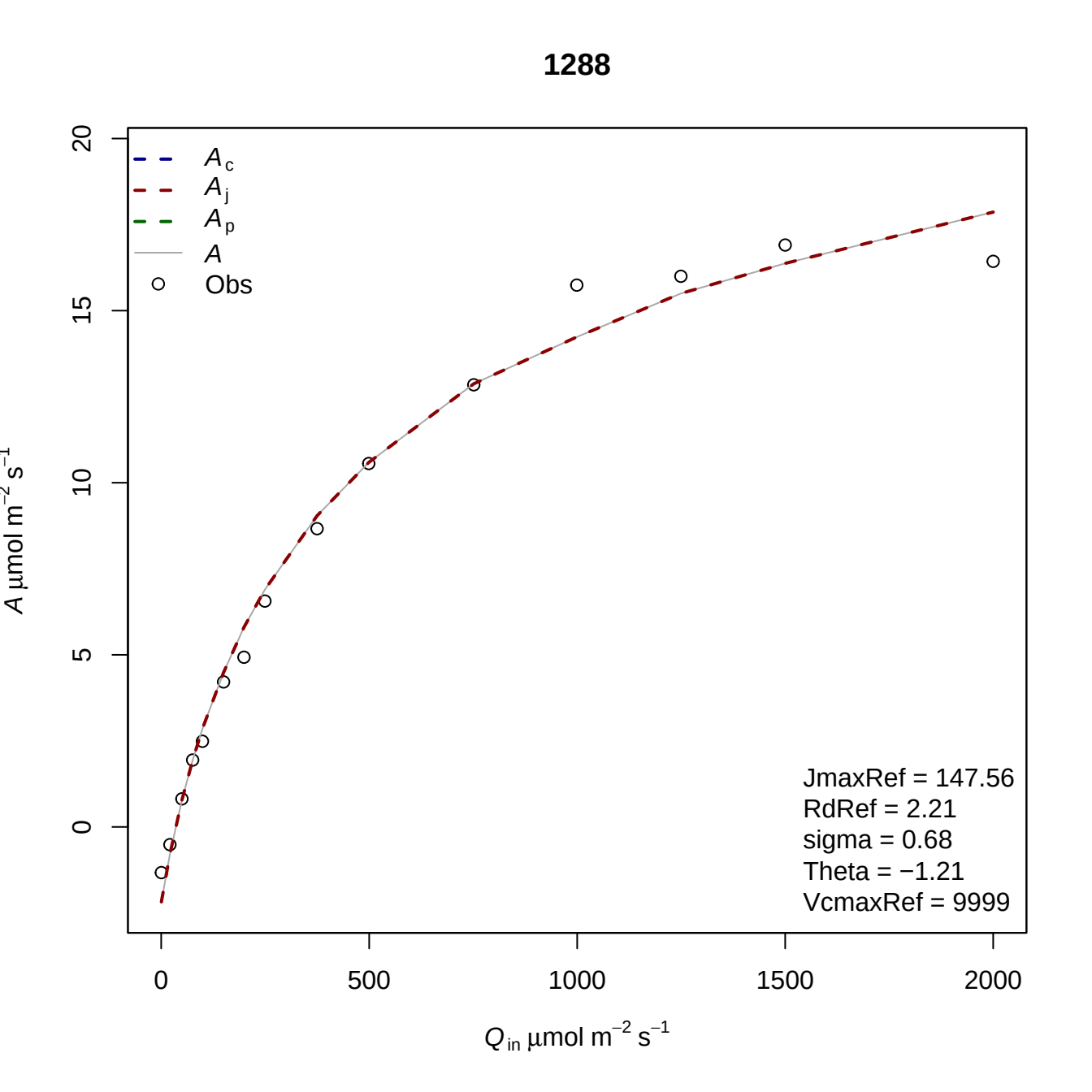

1283
A_c
A_j
A_p
A
Obs
JmaxRef = 147.26
RdRef = 2.74
sigma = 0.15
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

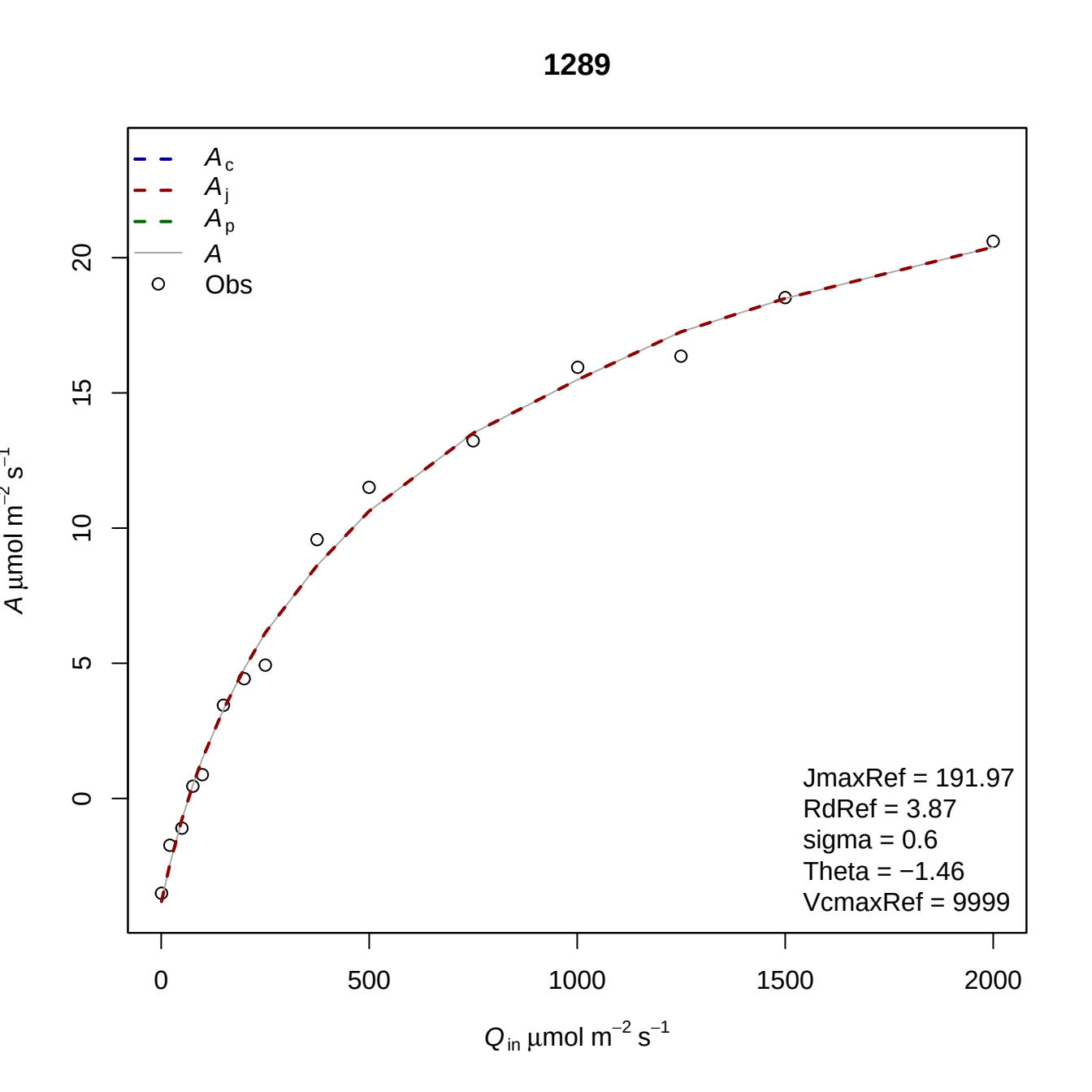

1284
A_c
A_j
A_p
A
Obs
JmaxRef = 123.43
RdRef = 3.47
sigma = 0.53
Theta = 0.05
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

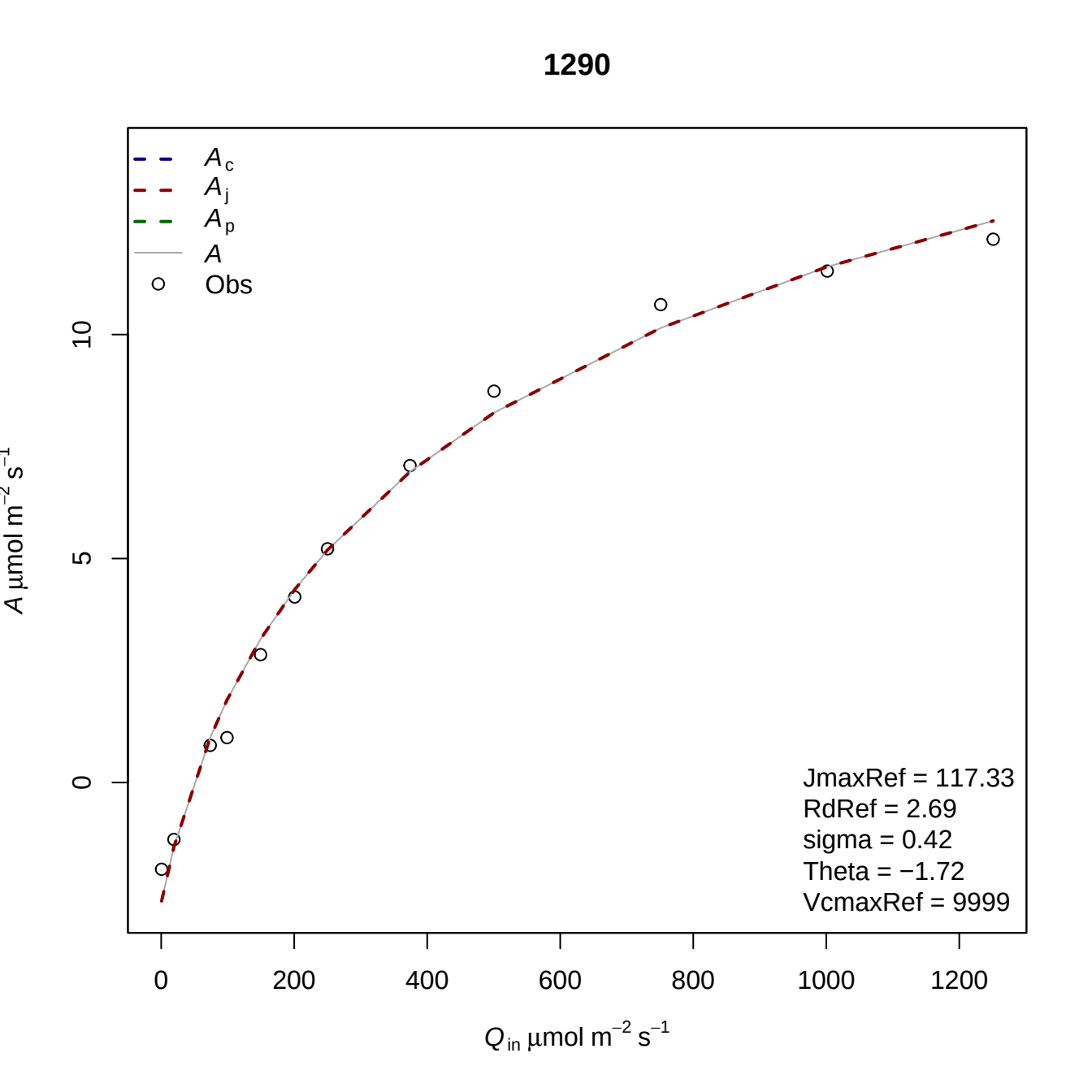

1285
A_c
A_j
A_p
A
Obs
JmaxRef = 113.05
RdRef = 2.71
sigma = 0.61
Theta = 0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

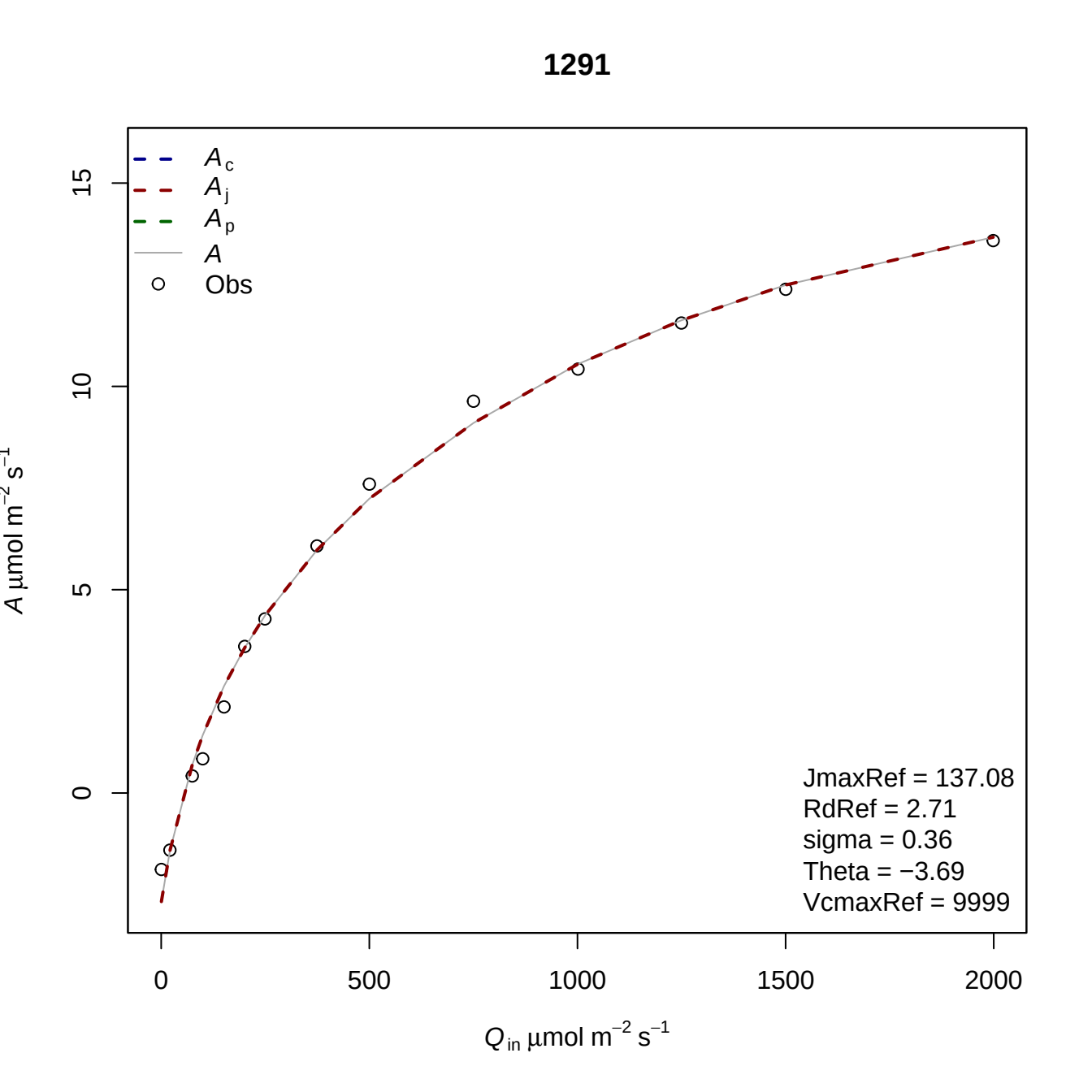

1286
A_c
A_j
A_p
A
Obs
JmaxRef = 161.28
RdRef = 2.6
sigma = 0.38
Theta = −0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

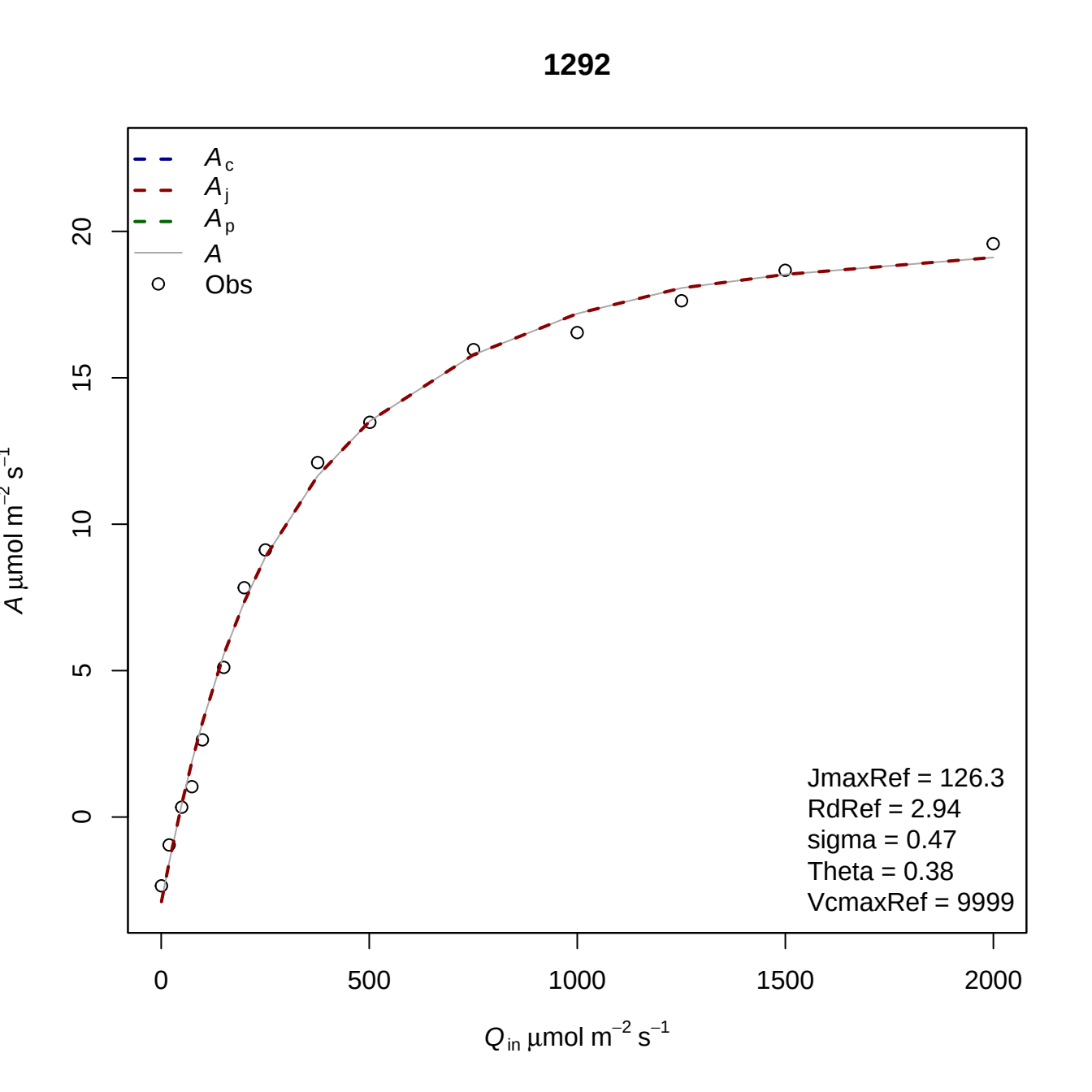

1287
A_c
A_j
A_p
A
Obs
JmaxRef = 128.87
RdRef = 2.55
sigma = 0.41
Theta = −0.44
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

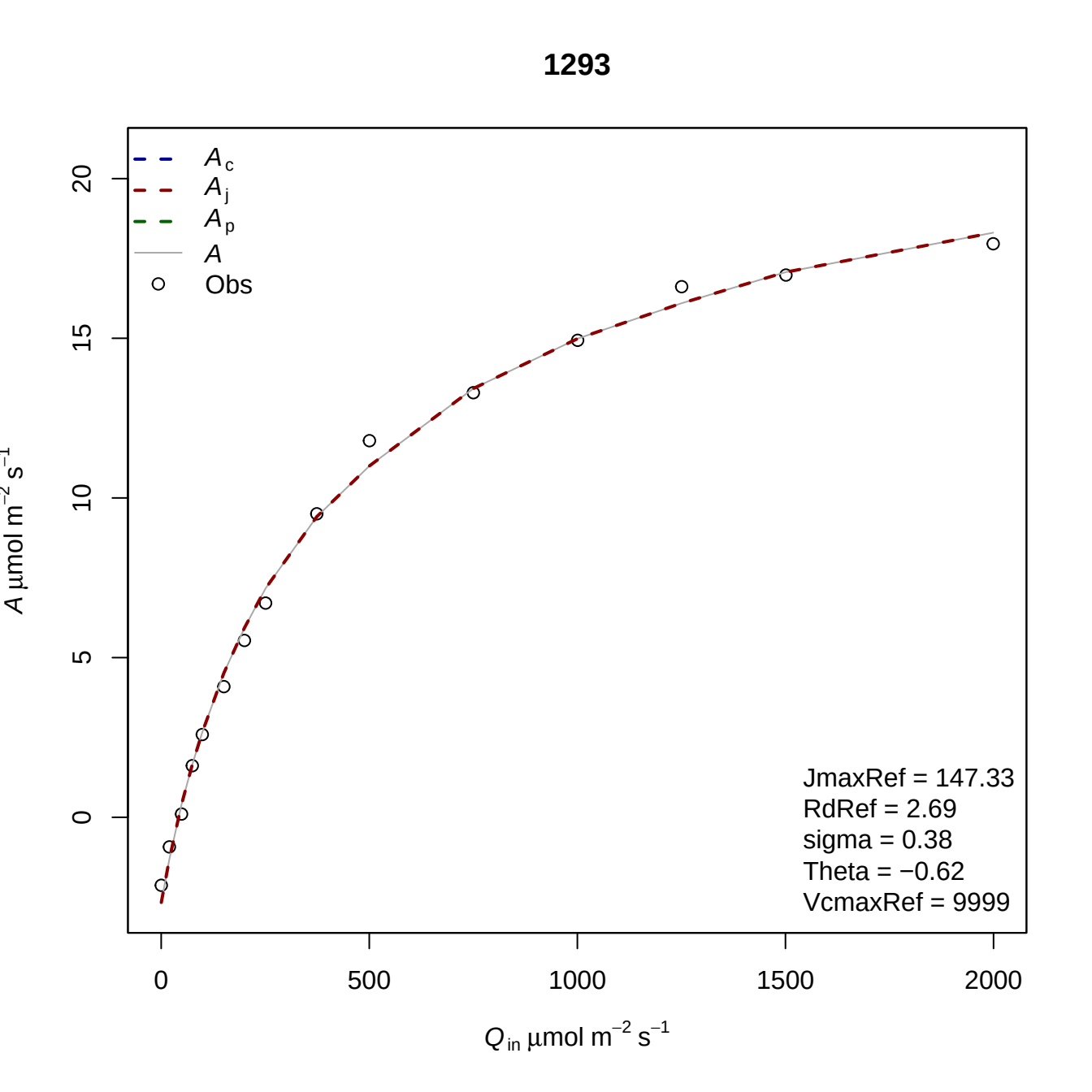

1288
A_c
A_j
A_p
A
Obs
JmaxRef = 147.56
RdRef = 2.21
sigma = 0.68
Theta = −1.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

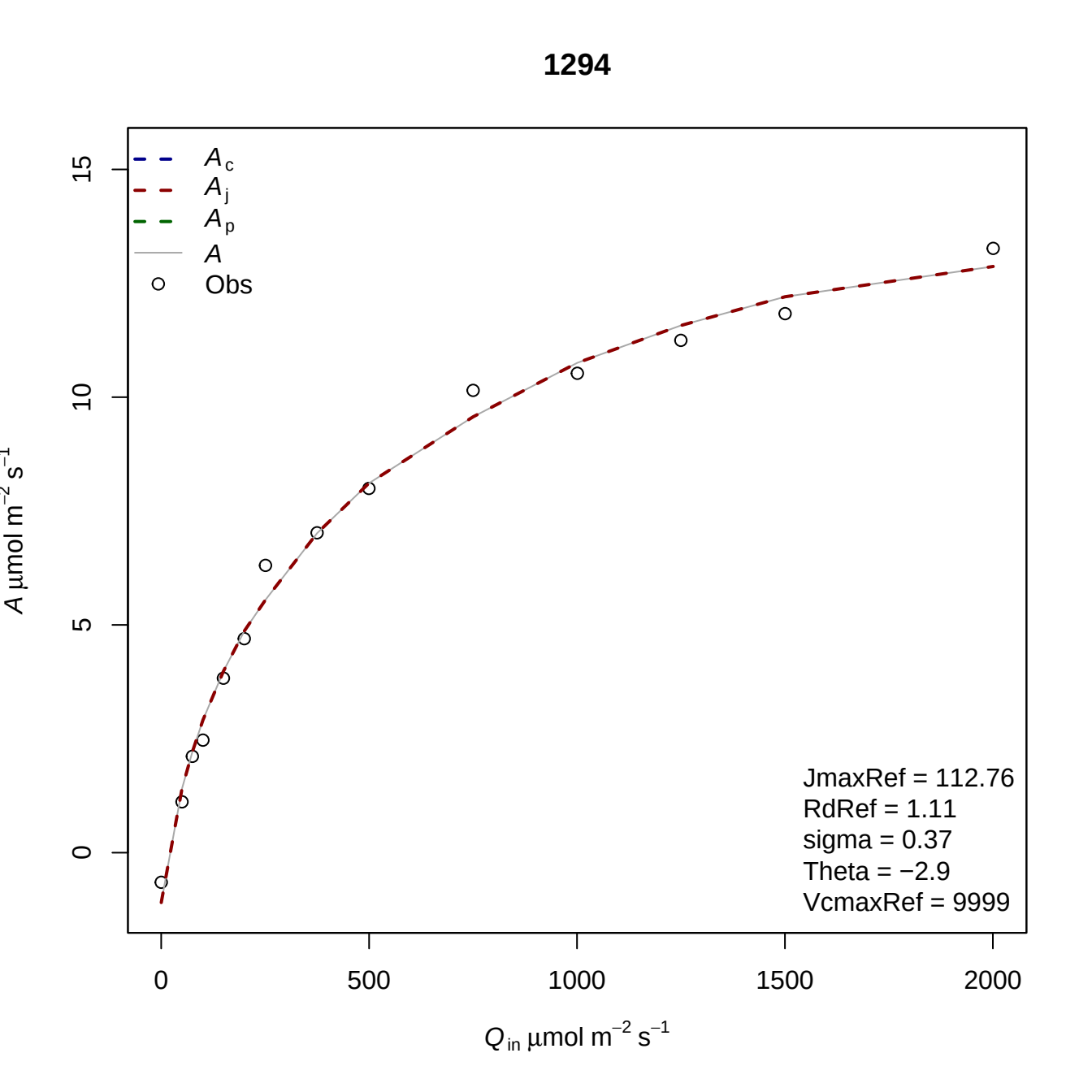

1289
A_c
A_j
A_p
A
Obs
JmaxRef = 191.97
RdRef = 3.87
sigma = 0.6
Theta = −1.46
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

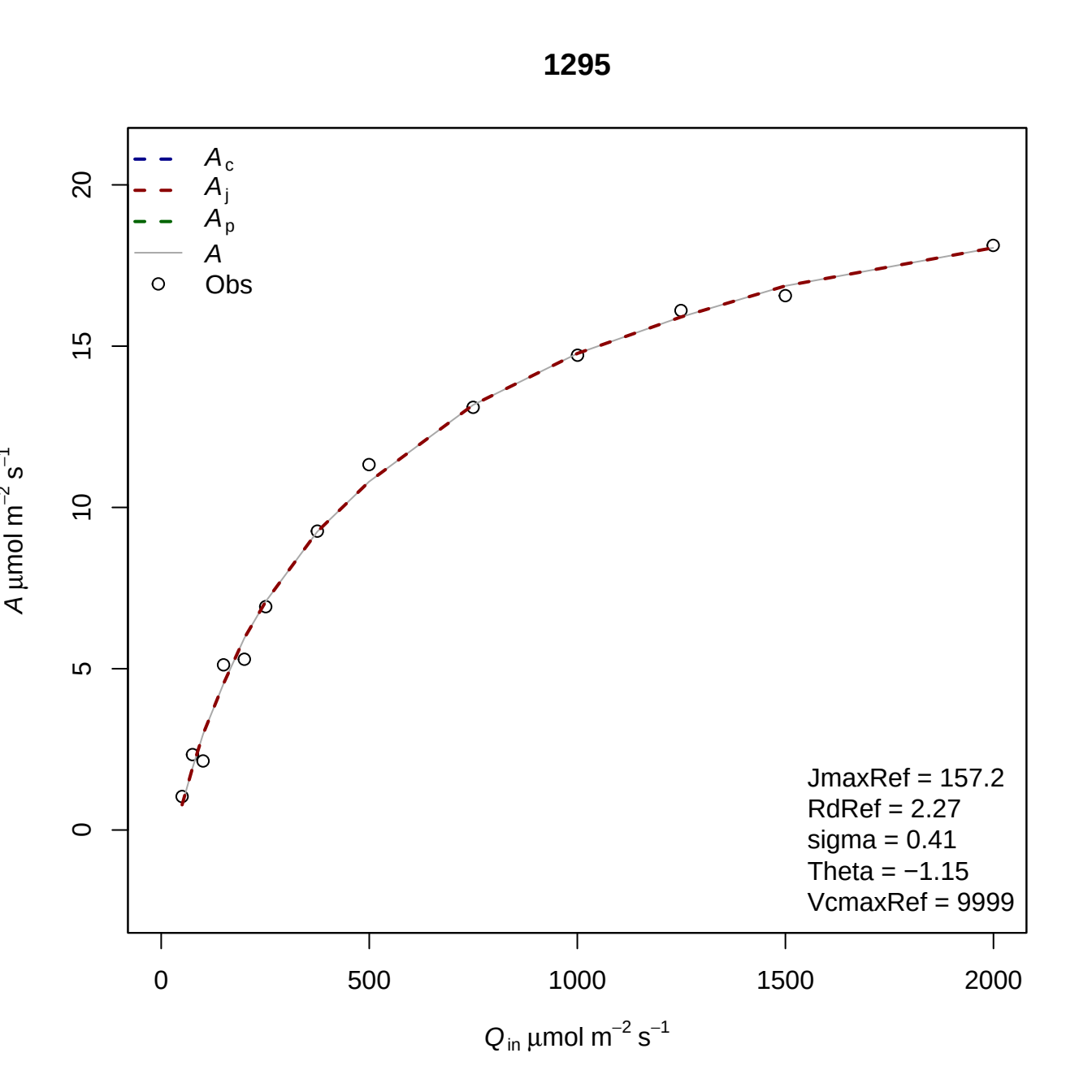

1290
A_c
A_j
A_p
A
Obs
JmaxRef = 117.33
RdRef = 2.69
sigma = 0.42
Theta = −1.72
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

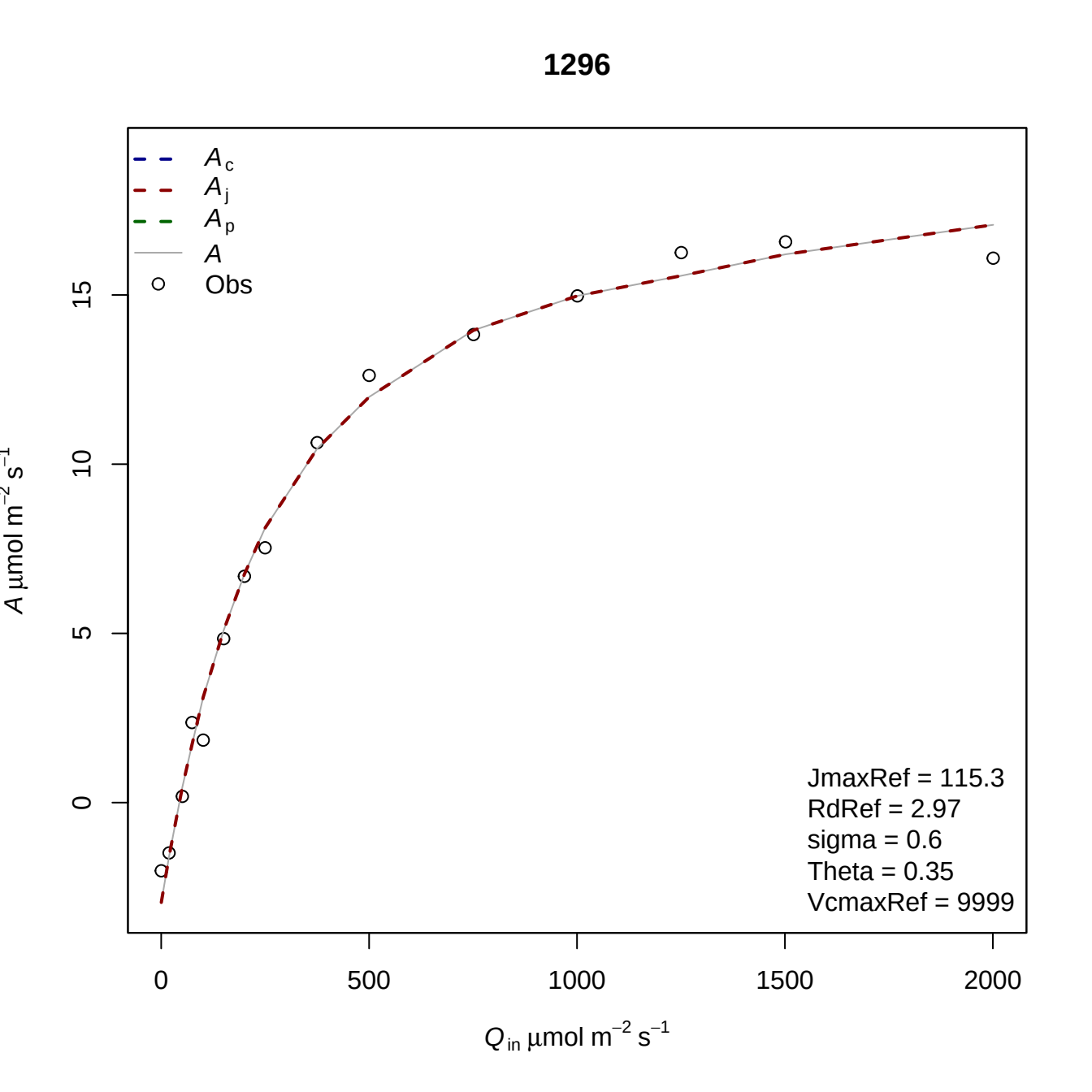

1291
A_c
A_j
A_p
A
Obs
JmaxRef = 137.08
RdRef = 2.71
sigma = 0.36
Theta = −3.69
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

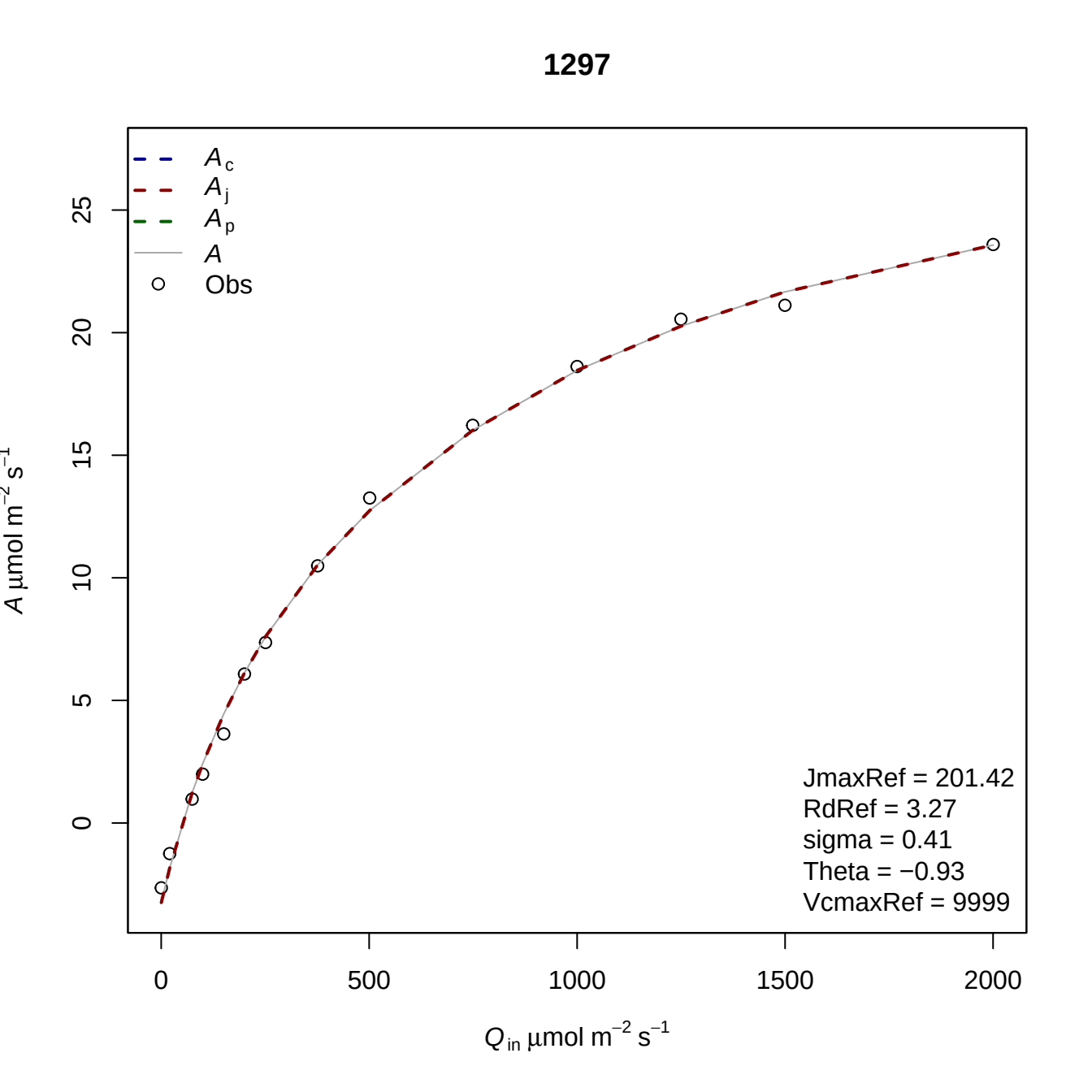

1292
A_c
A_j
A_p
A
Obs
JmaxRef = 126.3
RdRef = 2.94
sigma = 0.47
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

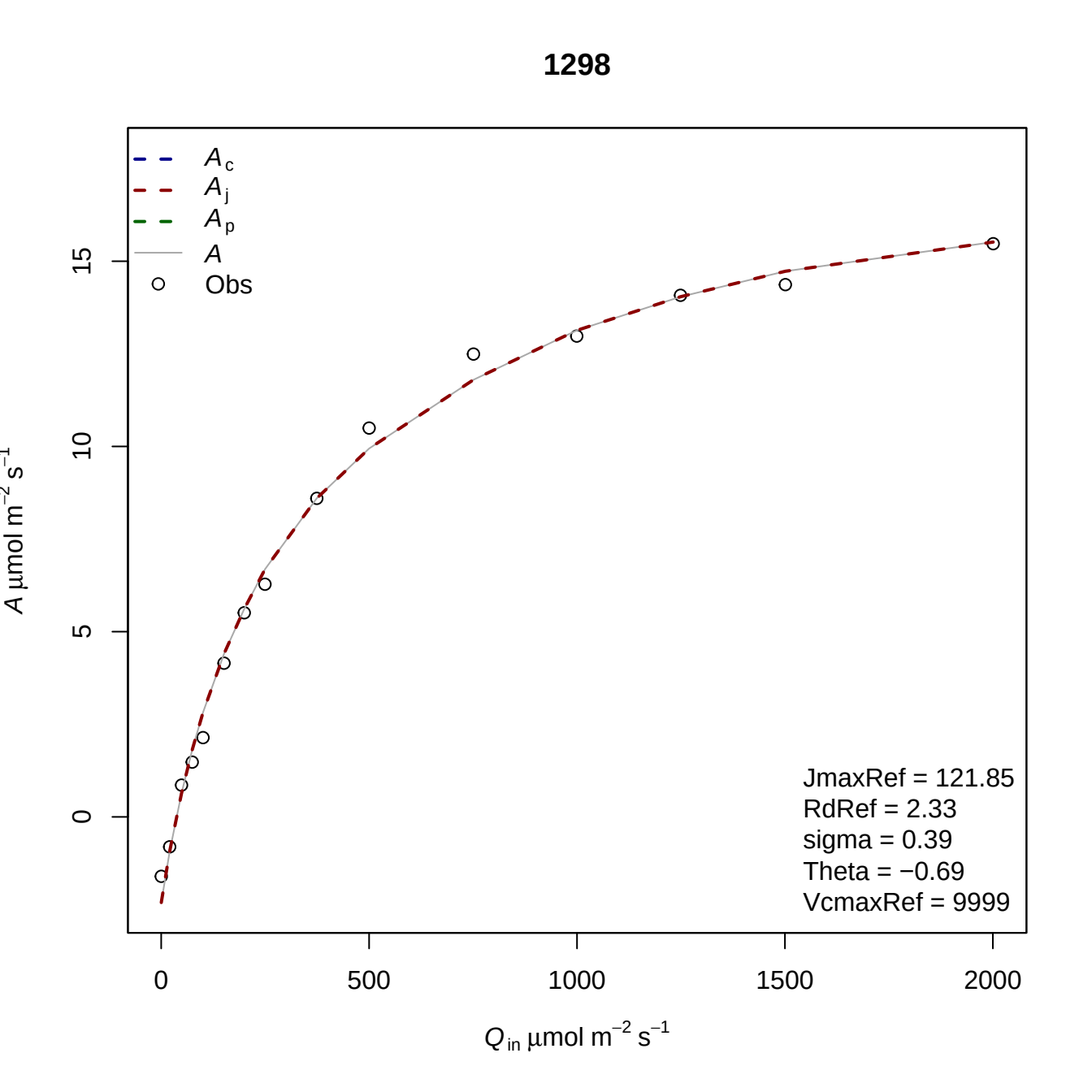

1293
A_c
A_j
A_p
A
Obs
JmaxRef = 147.33
RdRef = 2.69
sigma = 0.38
Theta = −0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

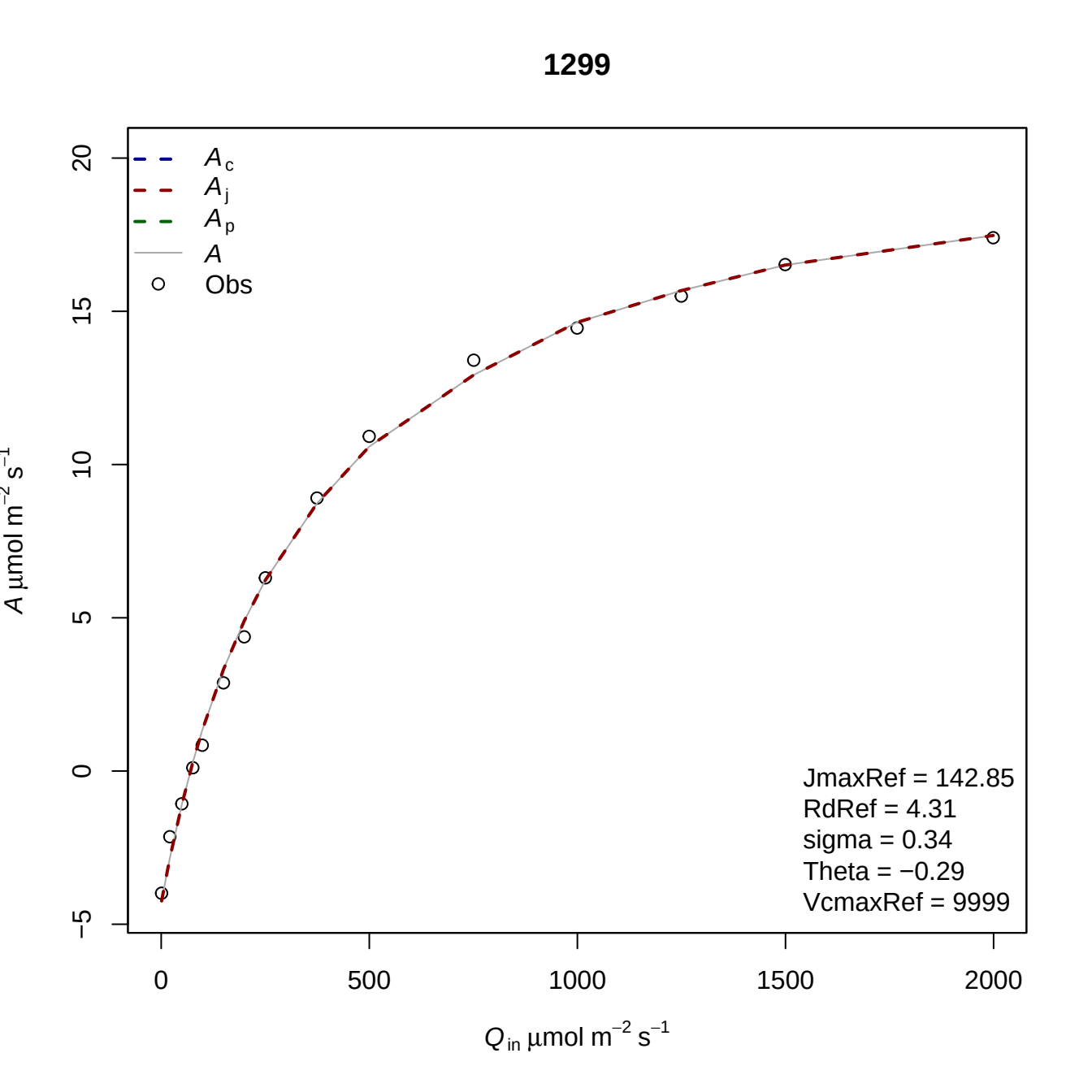

1294
A_c
A_j
A_p
A
Obs
JmaxRef = 112.76
RdRef = 1.11
sigma = 0.37
Theta = −2.9
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

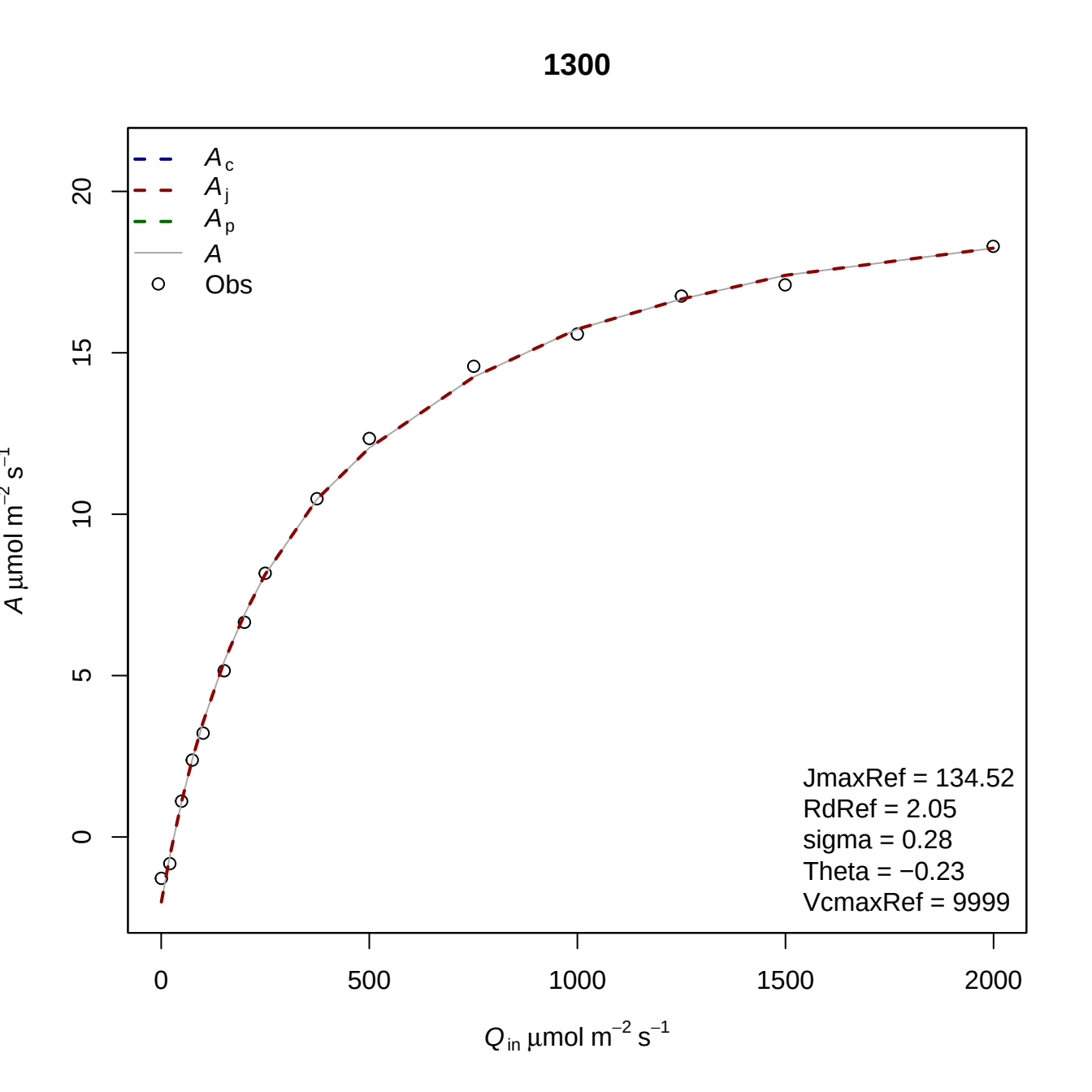

1295
A_c
A_j
A_p
A
Obs
JmaxRef = 157.2
RdRef = 2.27
sigma = 0.41
Theta = −1.15
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

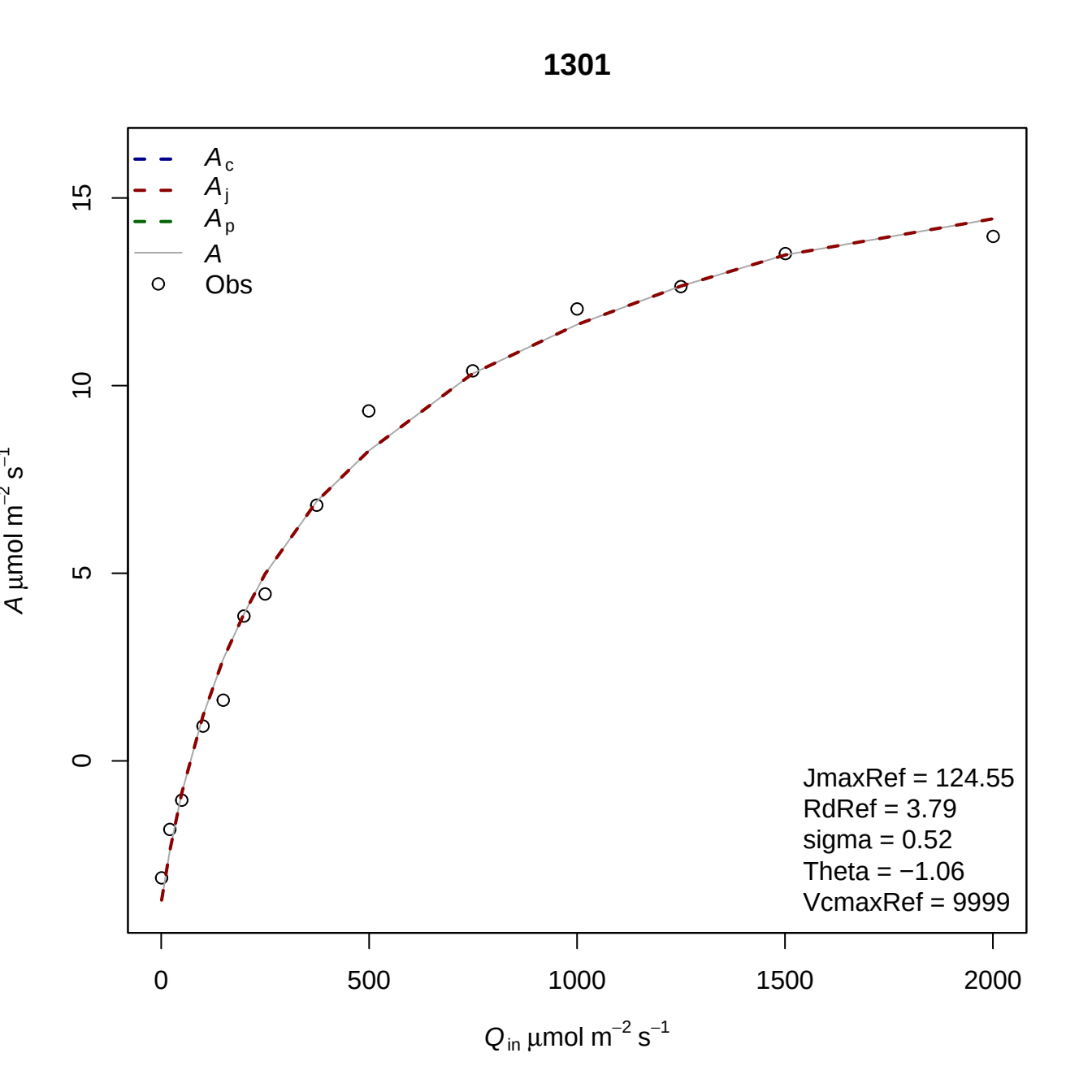

1296
A_c
A_j
A_p
A
Obs
JmaxRef = 115.3
RdRef = 2.97
sigma = 0.6
Theta = 0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

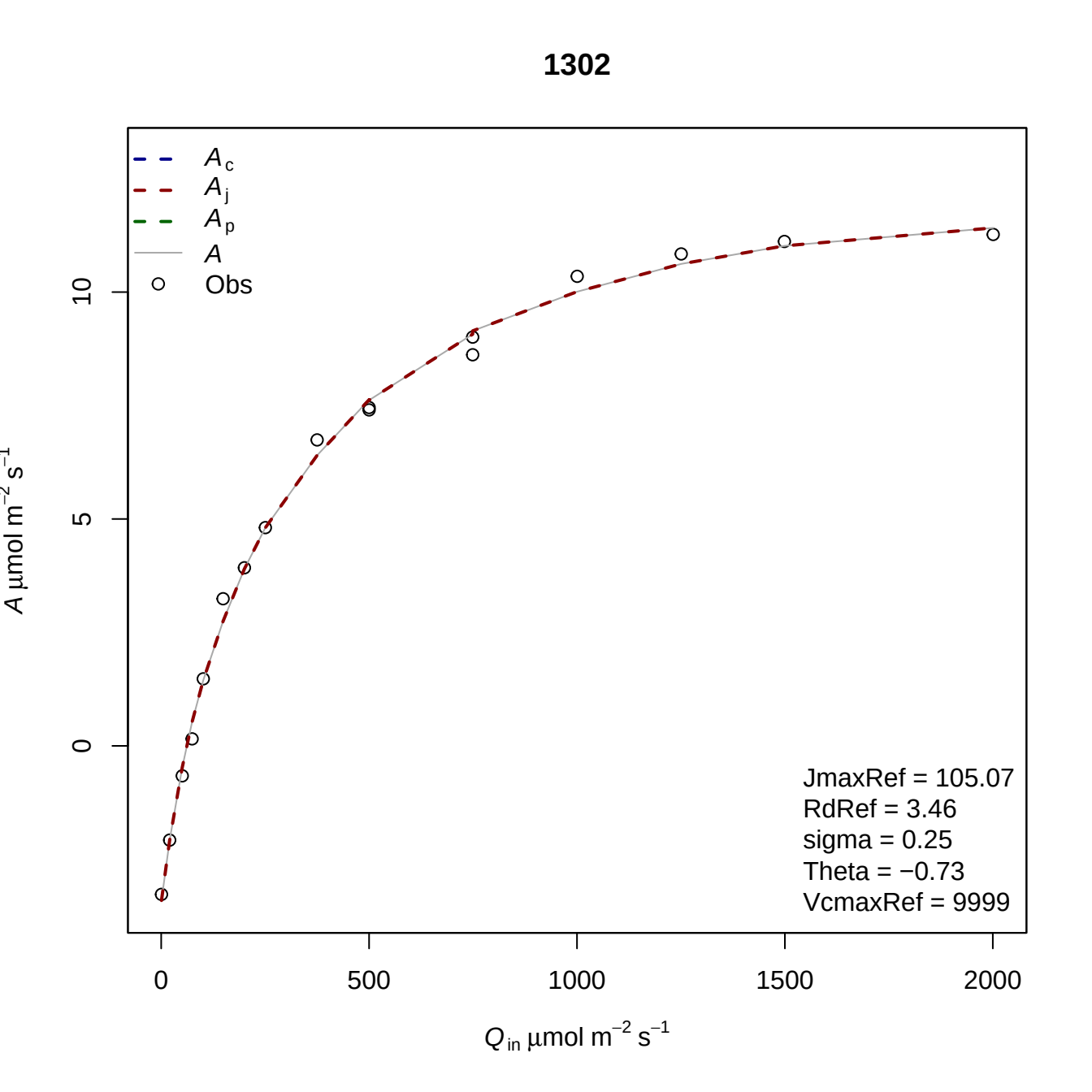

1297
A_c
A_j
A_p
A
Obs
JmaxRef = 201.42
RdRef = 3.27
sigma = 0.41
Theta = −0.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

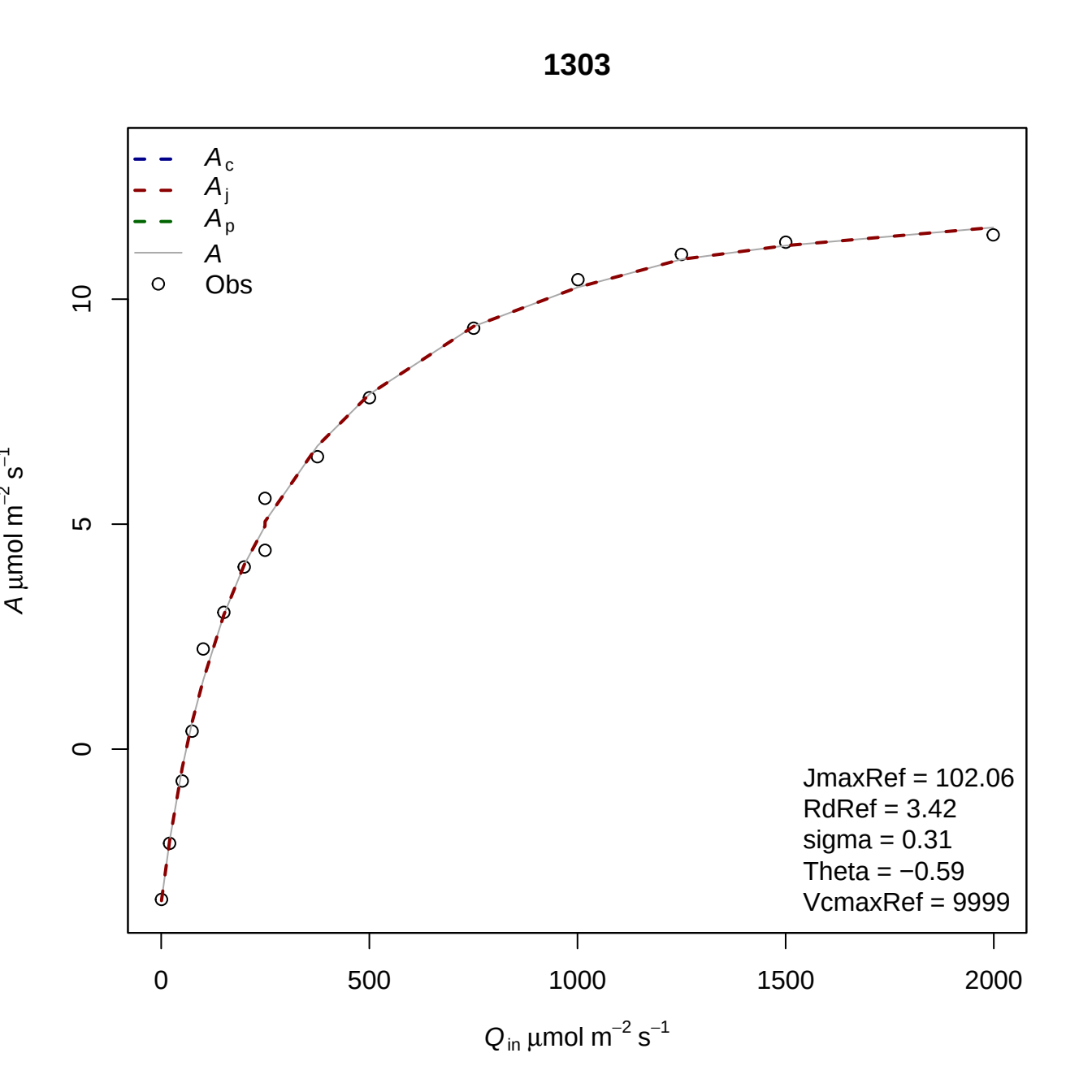

1298
A_c
A_j
A_p
A
Obs
JmaxRef = 121.85
RdRef = 2.33
sigma = 0.39
Theta = −0.69
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

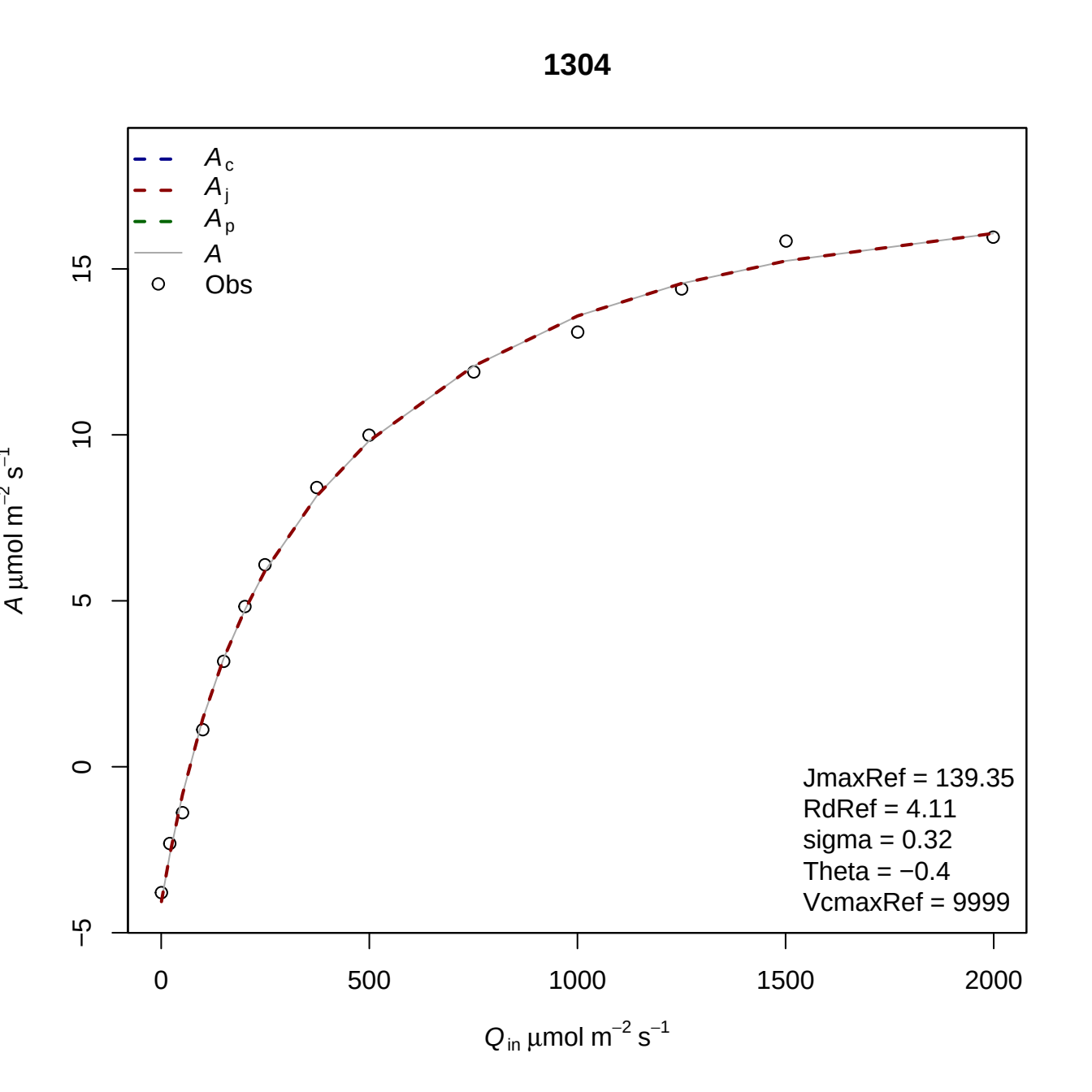

1299
A_c
A_j
A_p
A
Obs
JmaxRef = 142.85
RdRef = 4.31
sigma = 0.34
Theta = −0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

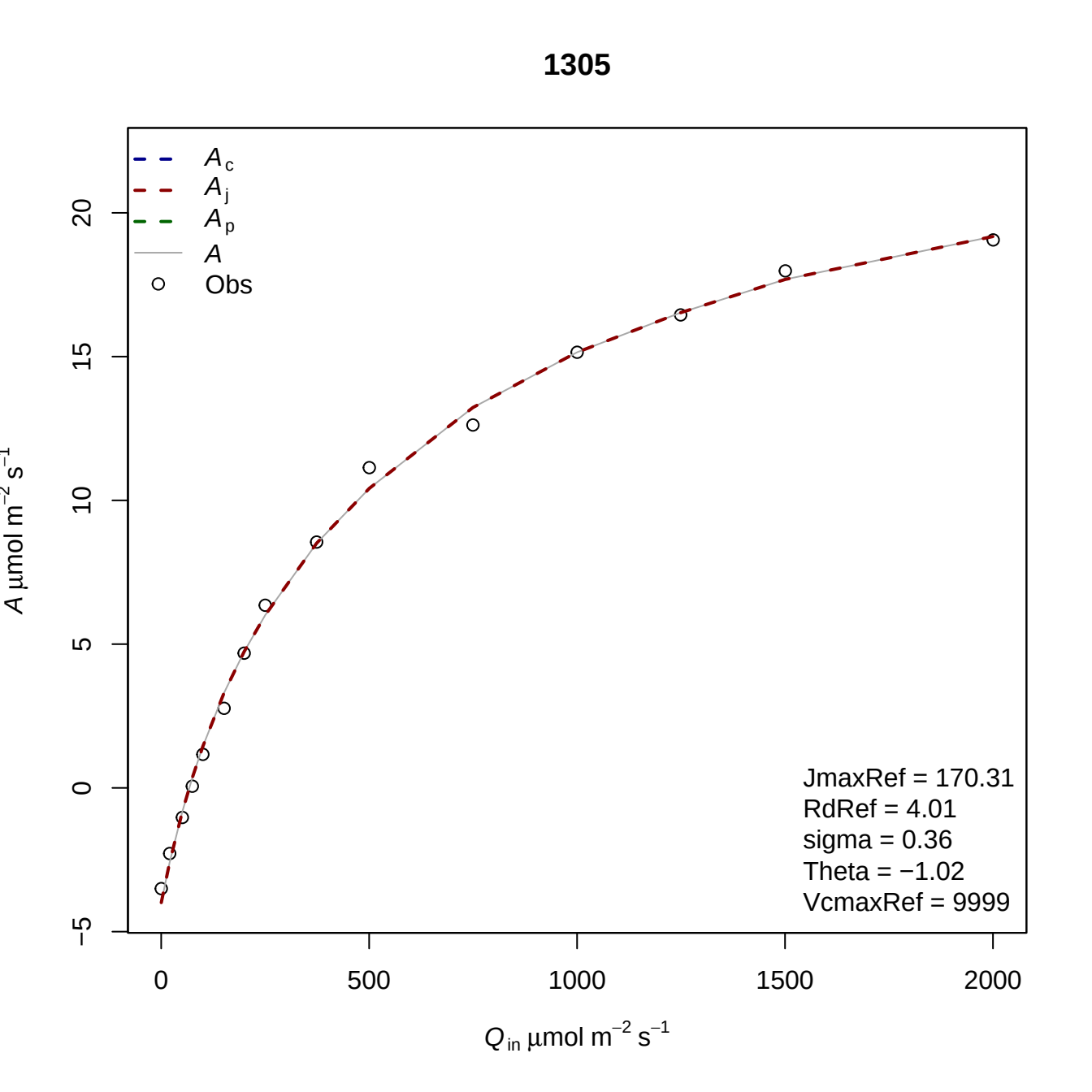

1300
A_c
A_j
A_p
A
Obs
JmaxRef = 134.52
RdRef = 2.05
sigma = 0.28
Theta = −0.23
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

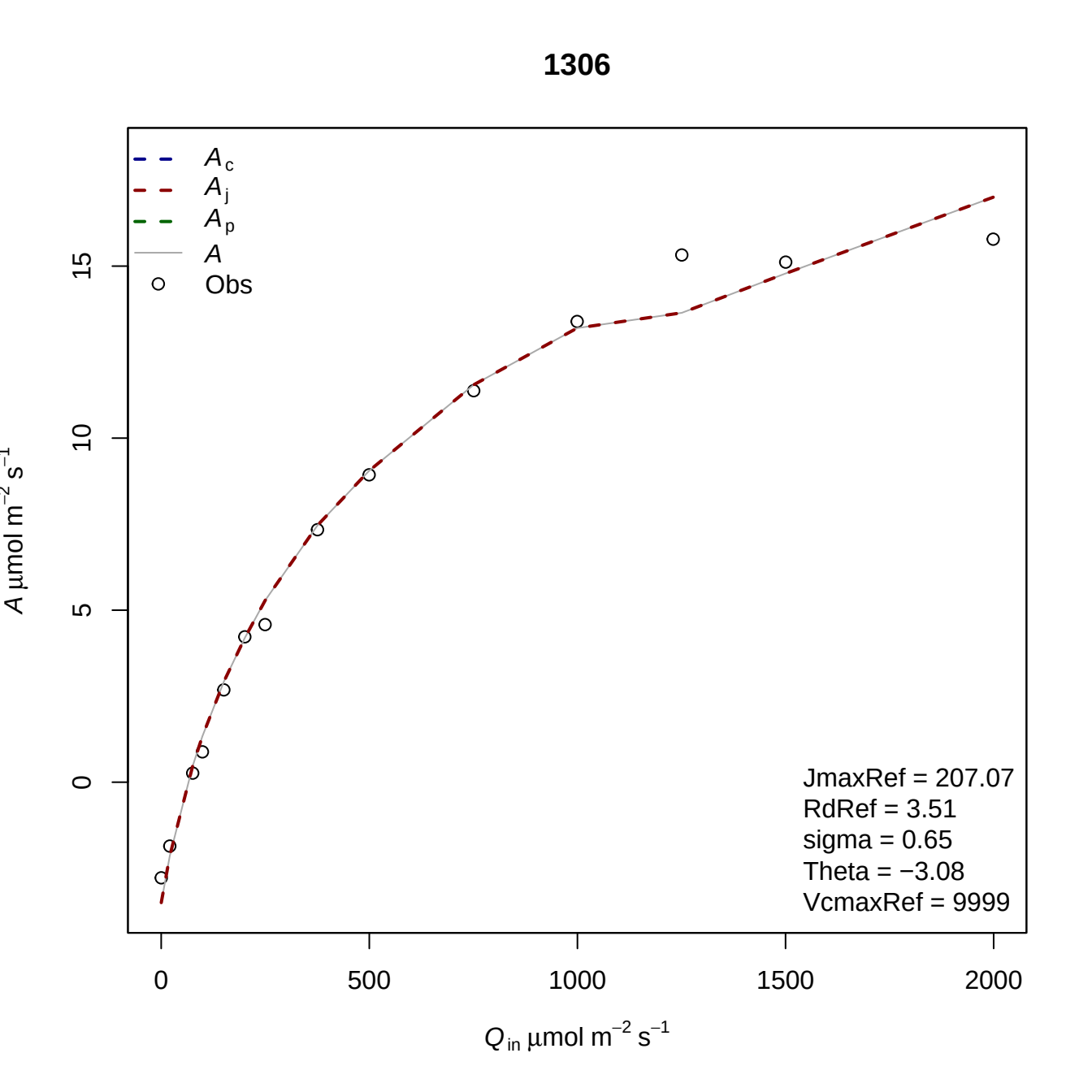

1301
A_c
A_j
A_p
A
Obs
JmaxRef = 124.55
RdRef = 3.79
sigma = 0.52
Theta = −1.06
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

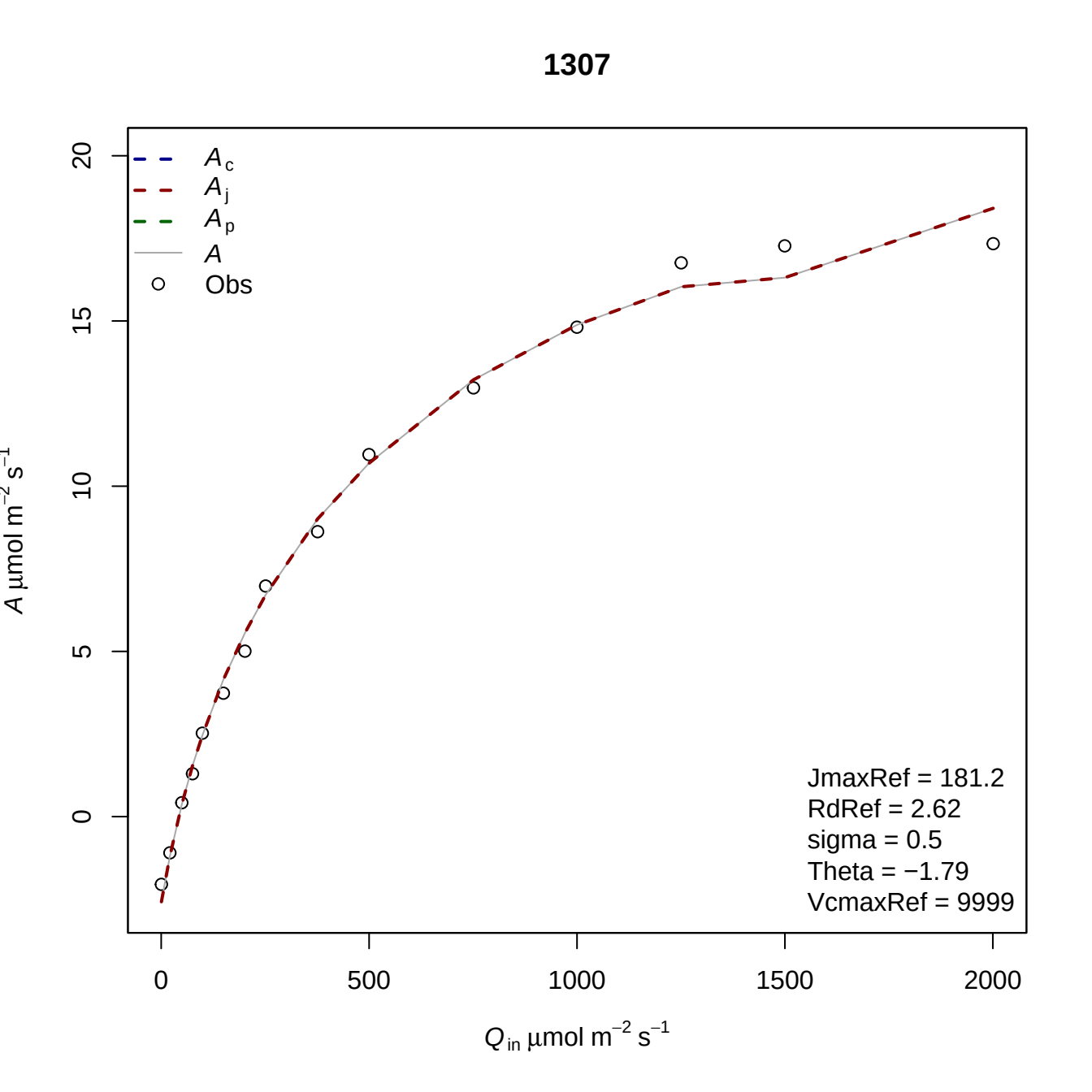

1302
A_c
A_j
A_p
A
Obs
JmaxRef = 105.07
RdRef = 3.46
sigma = 0.25
Theta = −0.73
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

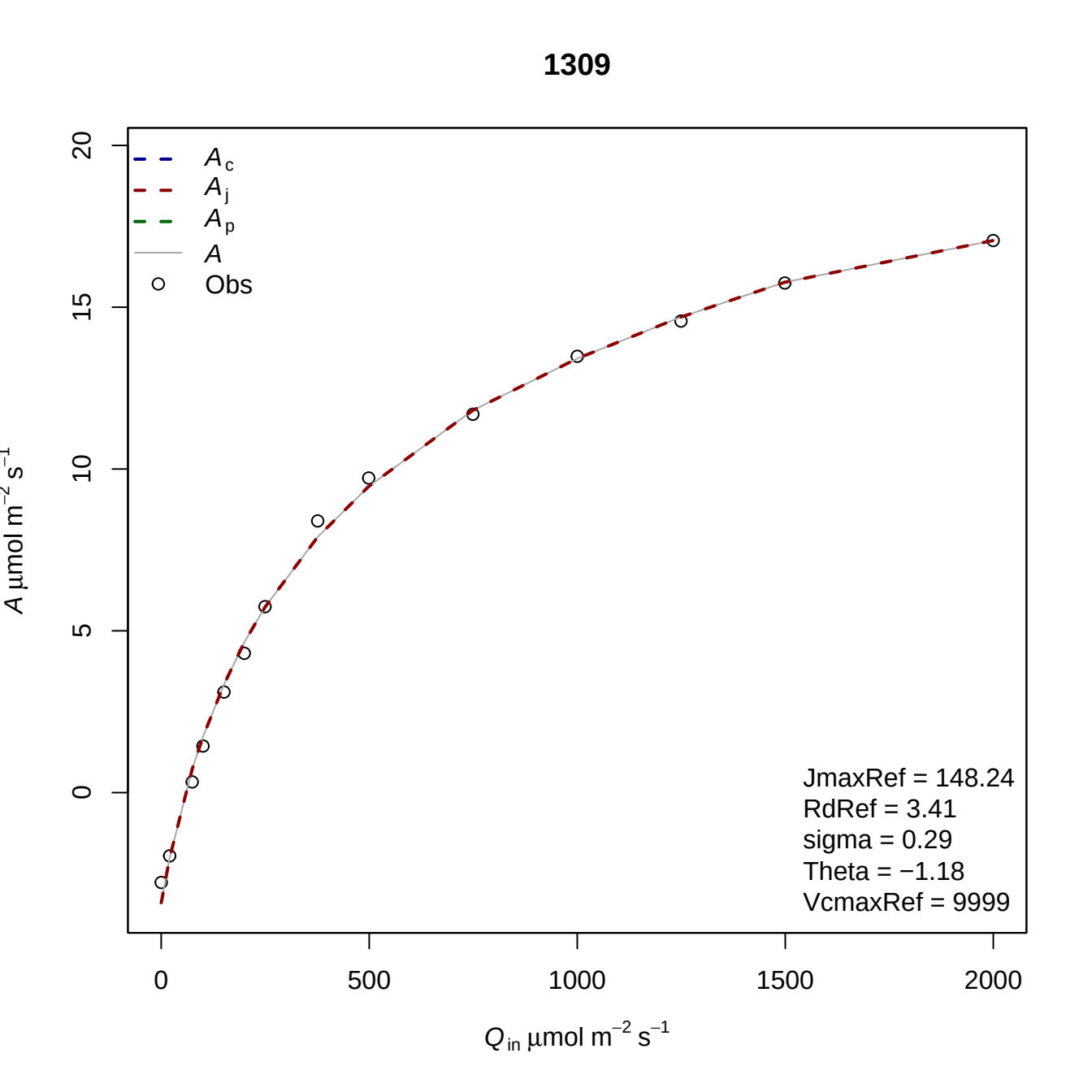

1303
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 102.06
RdRef = 3.42
sigma = 0.31
Theta = −0.59
VcmaxRef = 9999

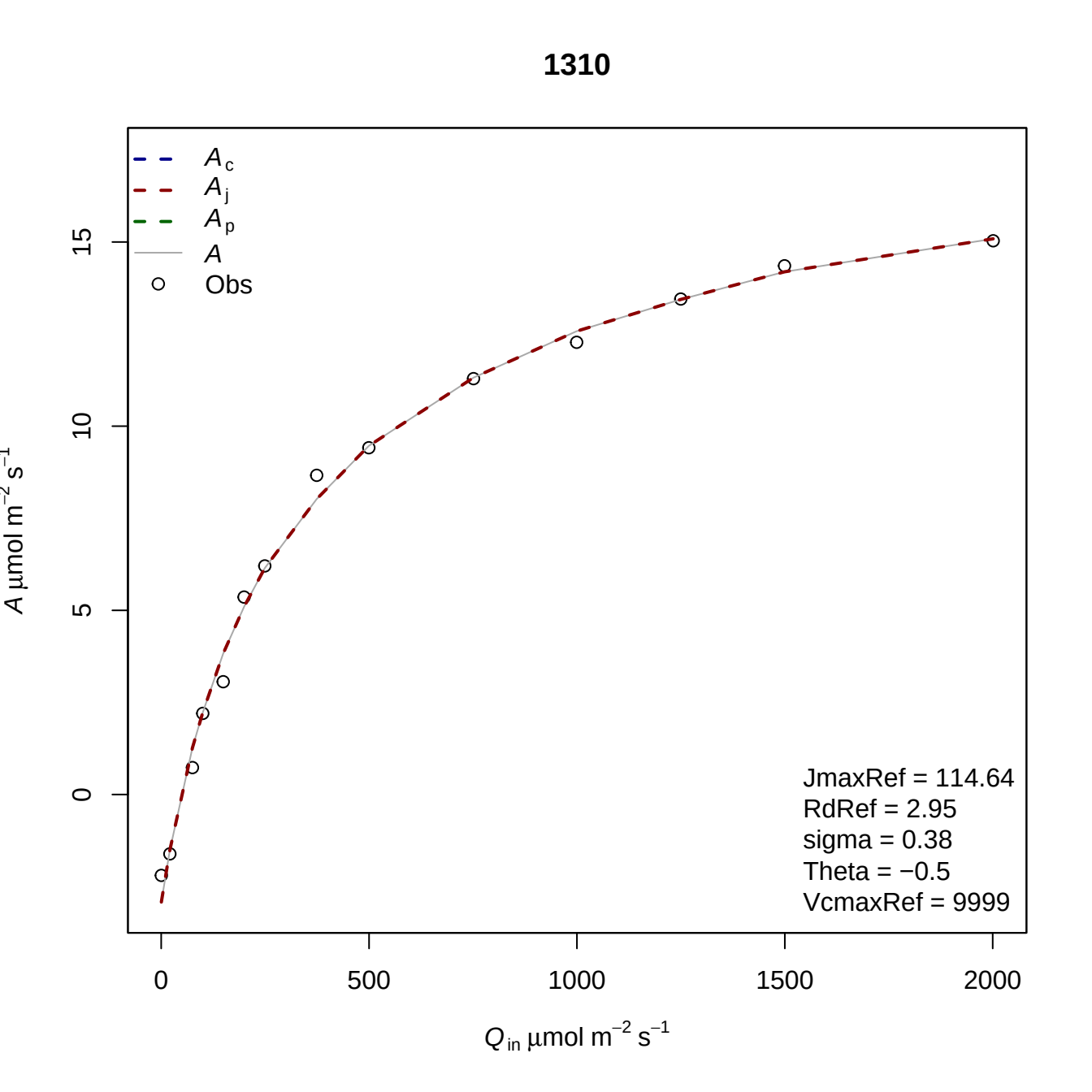

1304
A_c
A_j
A_p
A
Obs
JmaxRef = 139.35
RdRef = 4.11
sigma = 0.32
Theta = −0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

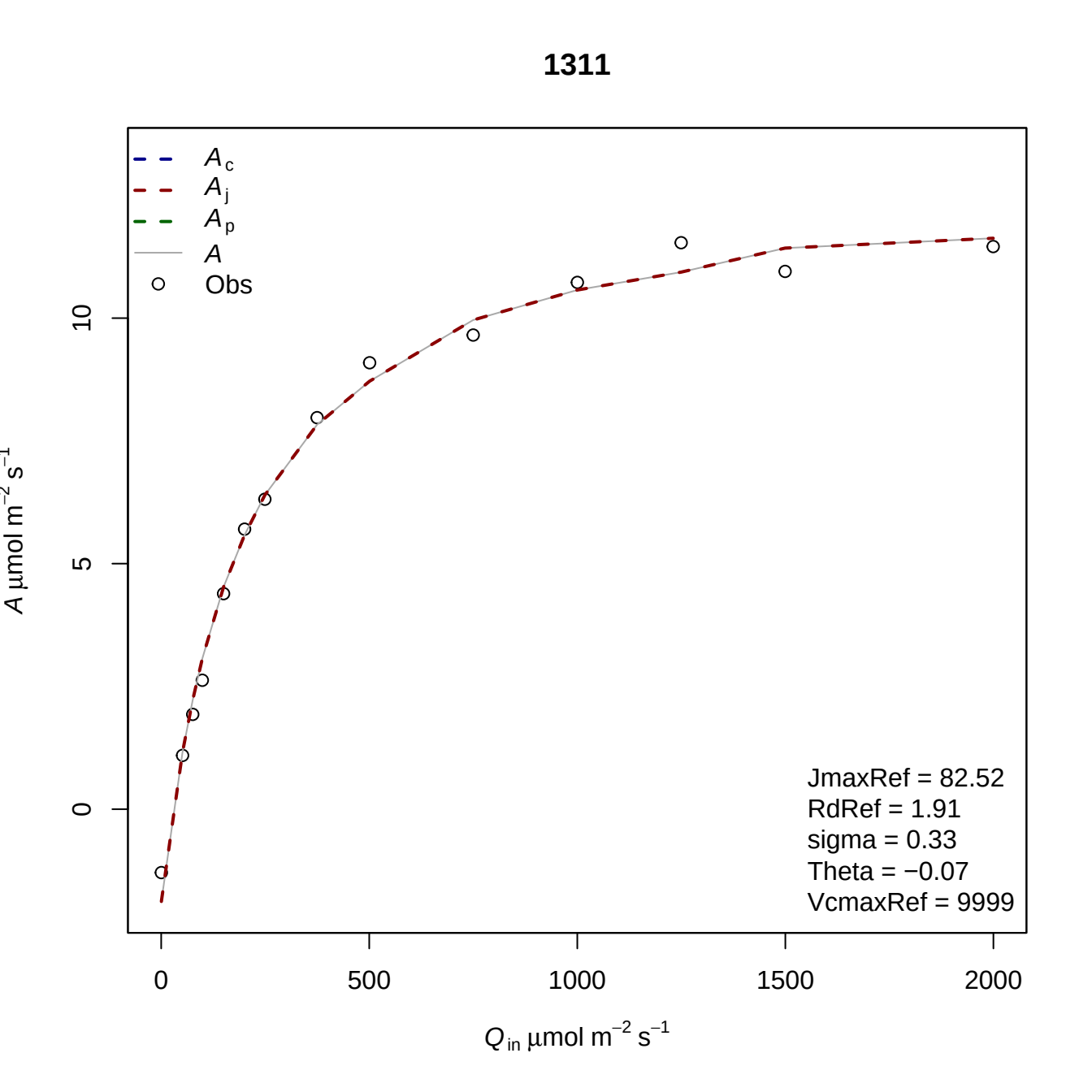

1305
A_c
A_j
A_p
A
Obs
JmaxRef = 170.31
RdRef = 4.01
sigma = 0.36
Theta = −1.02
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

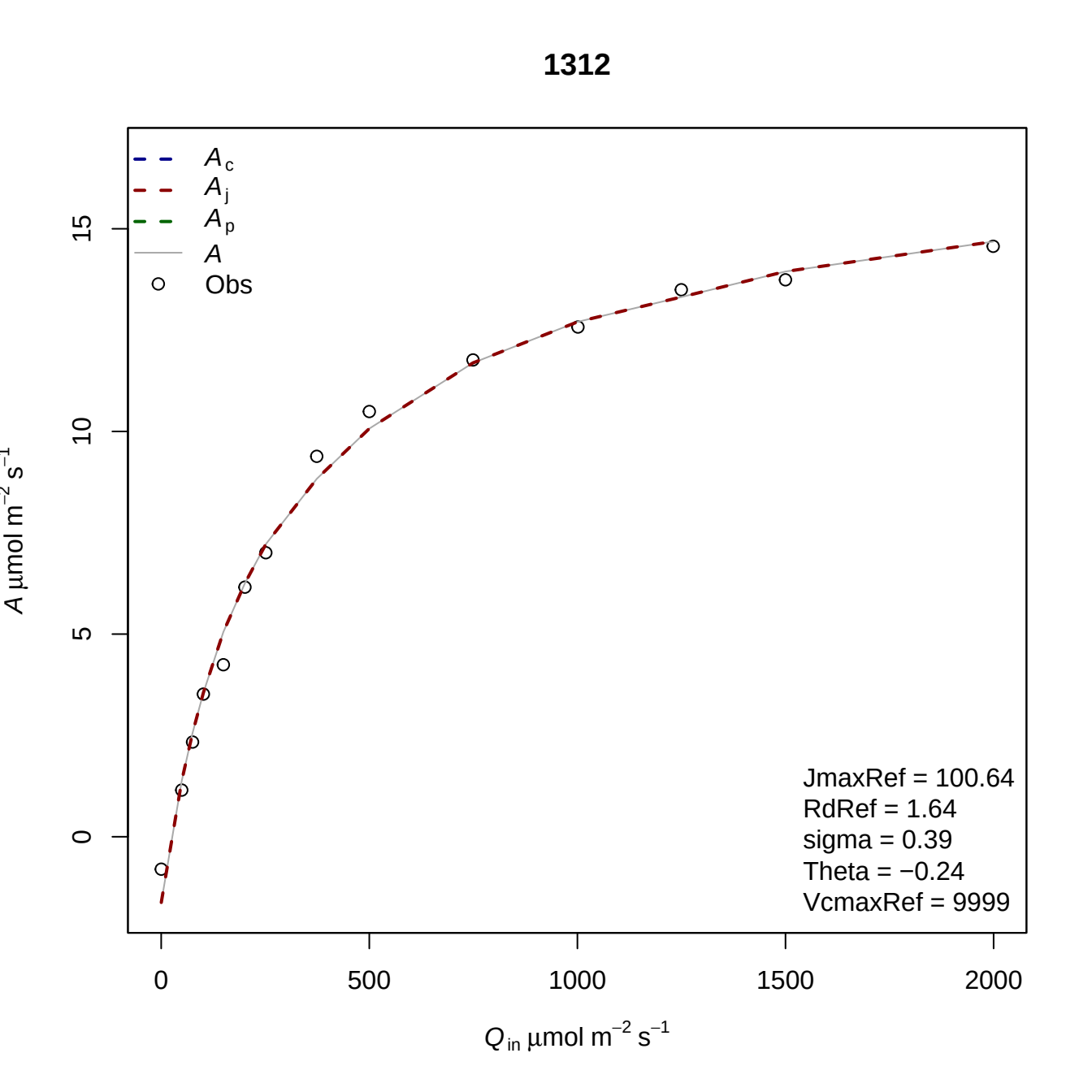

1306
A_c
A_j
A_p
A
Obs
JmaxRef = 207.07
RdRef = 3.51
sigma = 0.65
Theta = −3.08
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

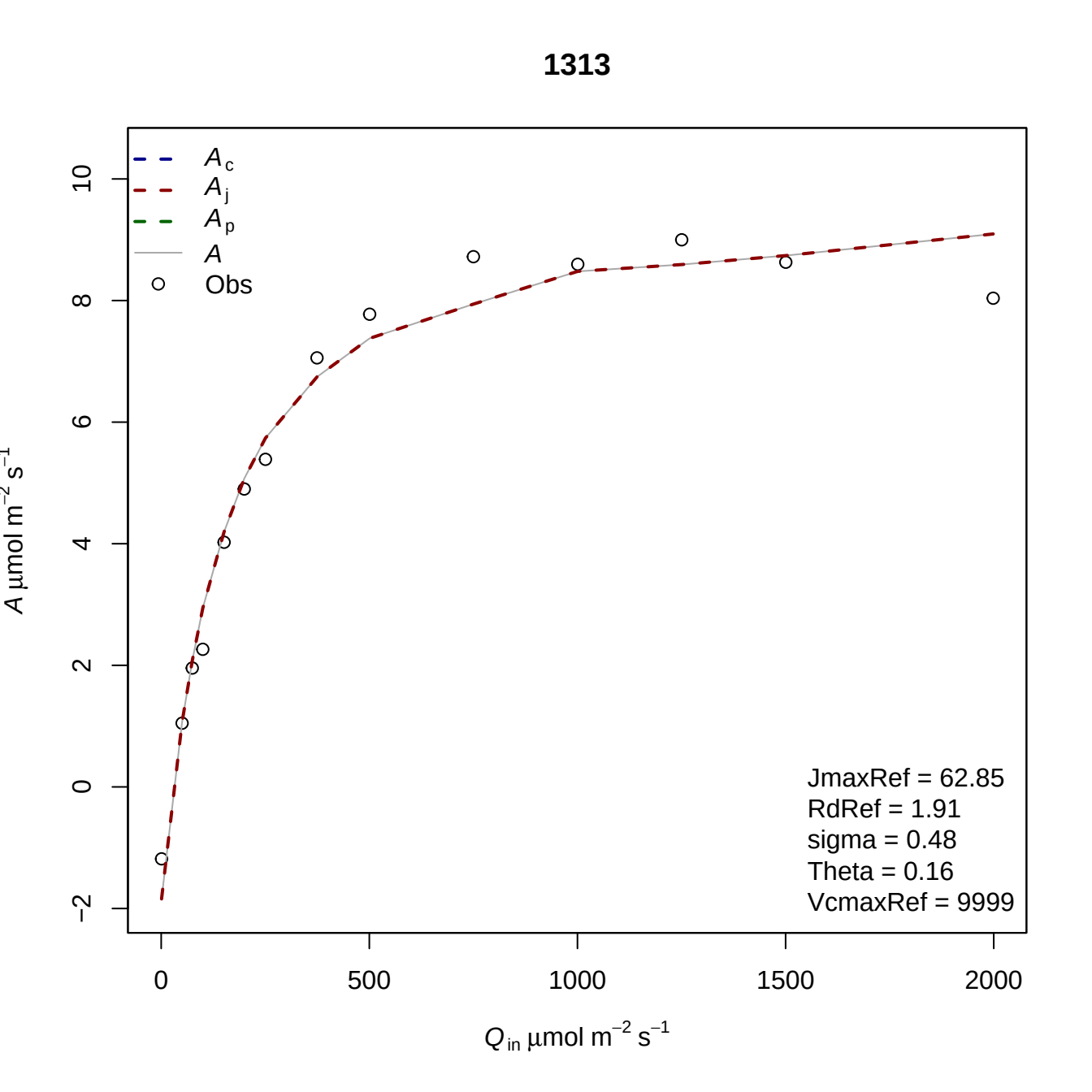

1307
A_c
A_j
A_p
A
Obs
JmaxRef = 181.2
RdRef = 2.62
sigma = 0.5
Theta = −1.79
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

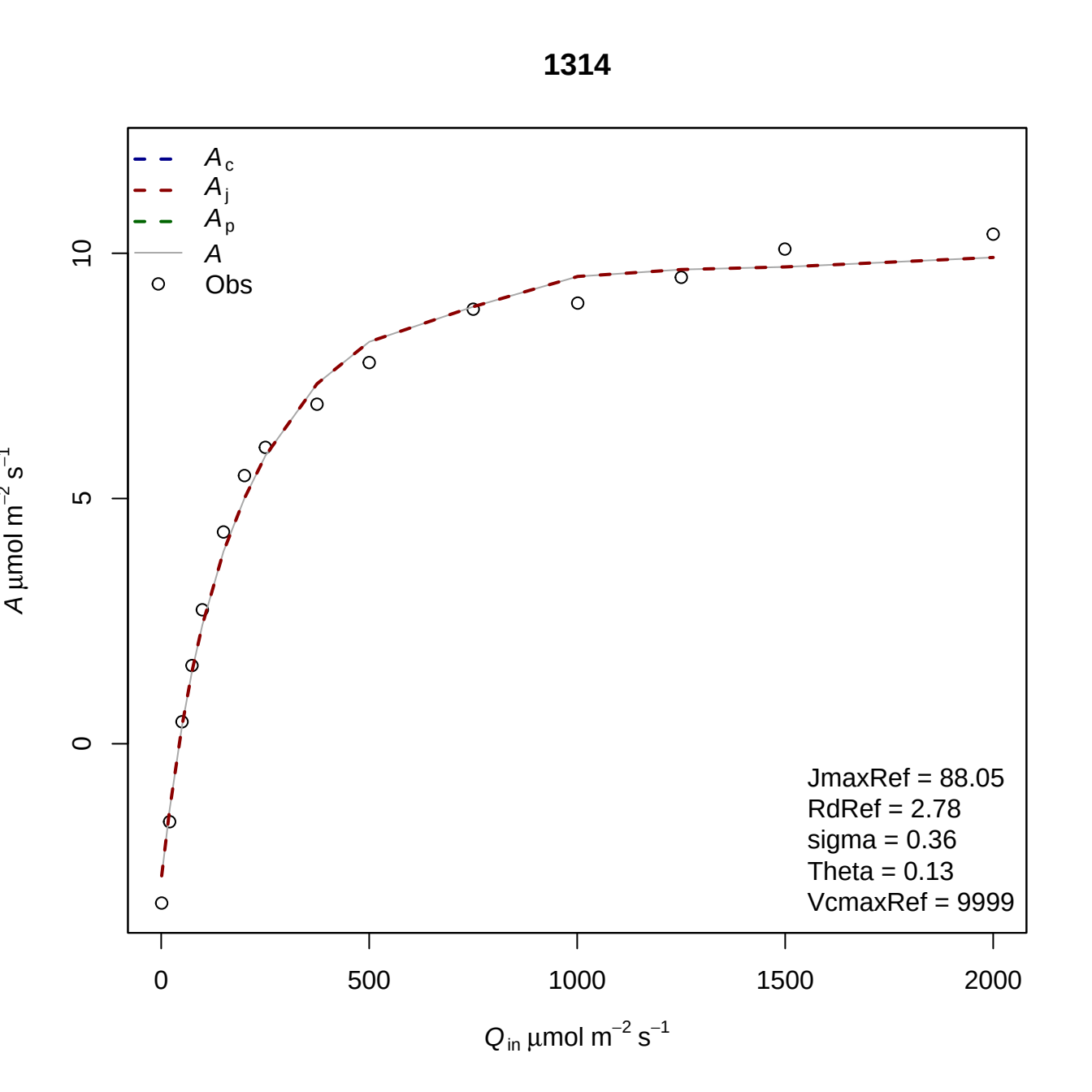

1309
A_c
A_j
A_p
A
Obs
JmaxRef = 148.24
RdRef = 3.41
sigma = 0.29
Theta = −1.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

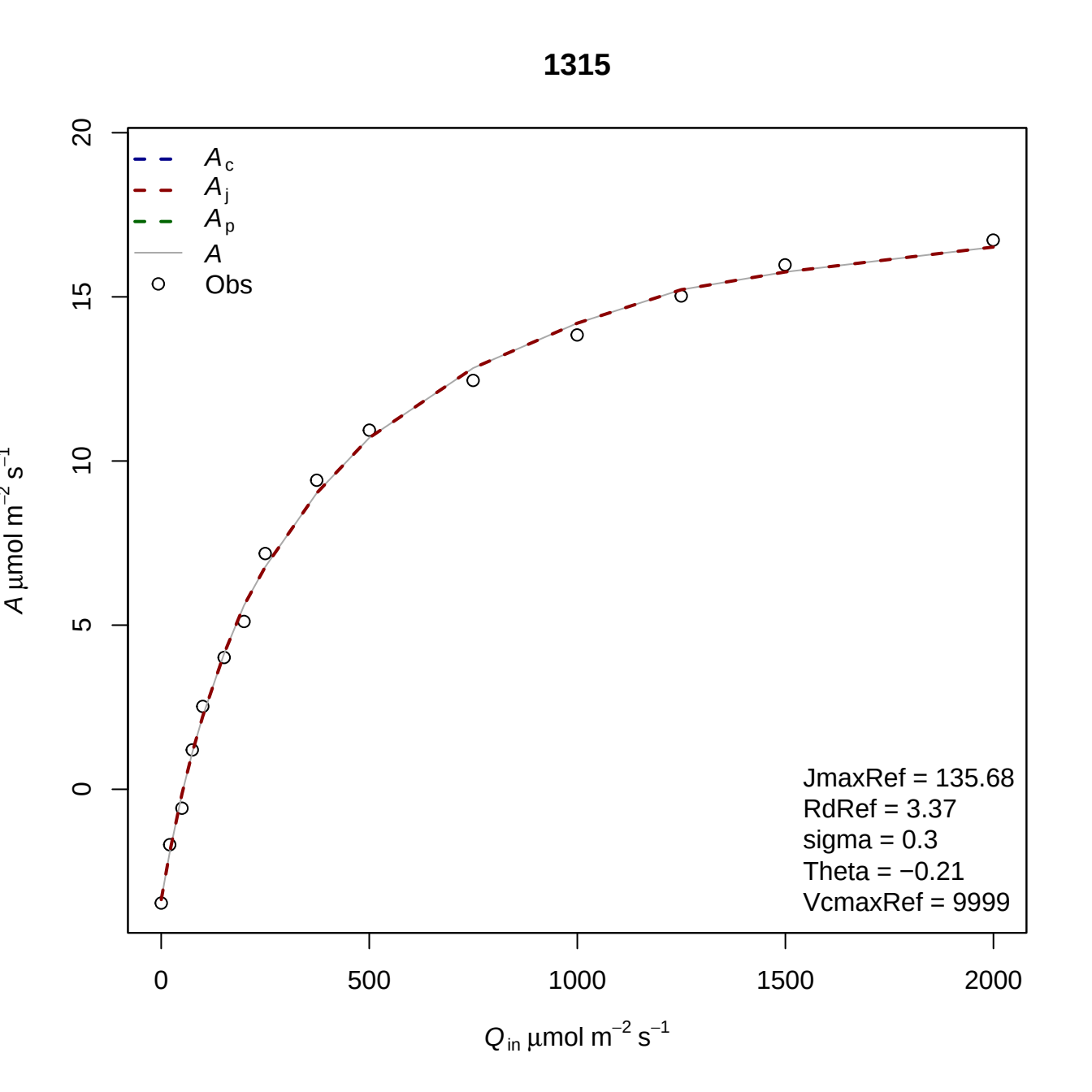

1310
A_c
A_j
A_p
A
Obs
JmaxRef = 114.64
RdRef = 2.95
sigma = 0.38
Theta = −0.5
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

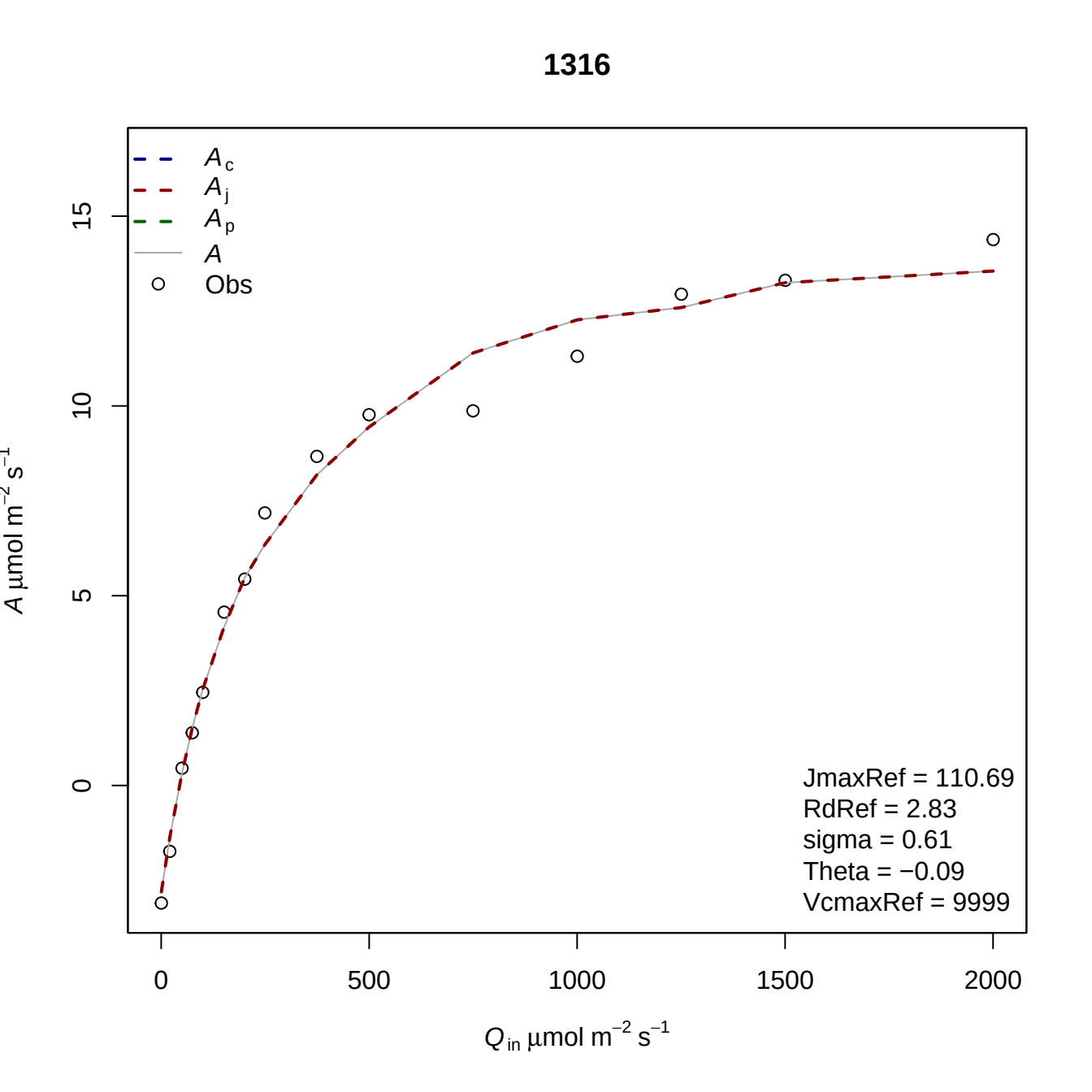

1311
A_c
A_j
A_p
A
Obs
JmaxRef = 82.52
RdRef = 1.91
sigma = 0.33
Theta = −0.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

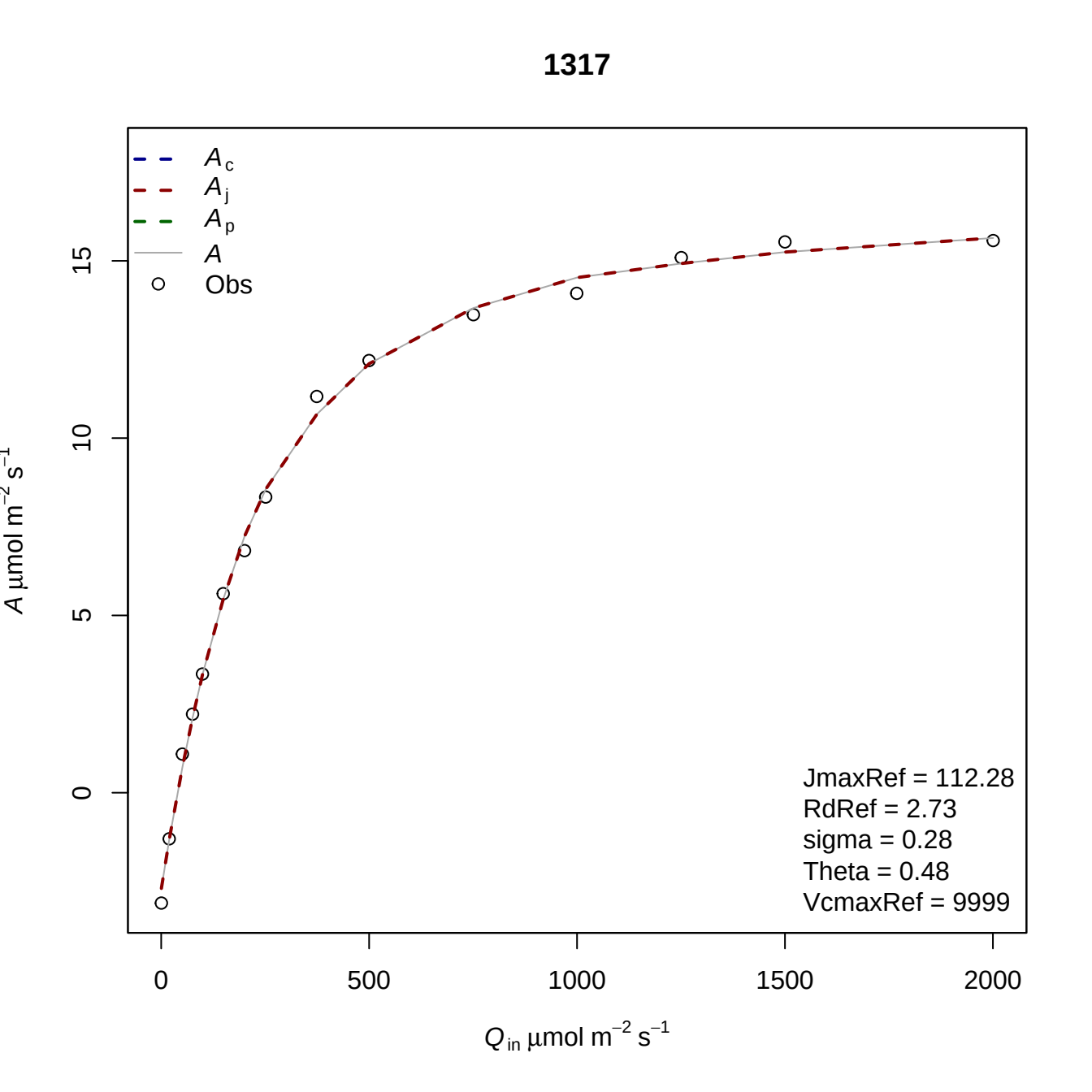

1312
A_c
A_j
A_p
A
Obs
JmaxRef = 100.64
RdRef = 1.64
sigma = 0.39
Theta = −0.24
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

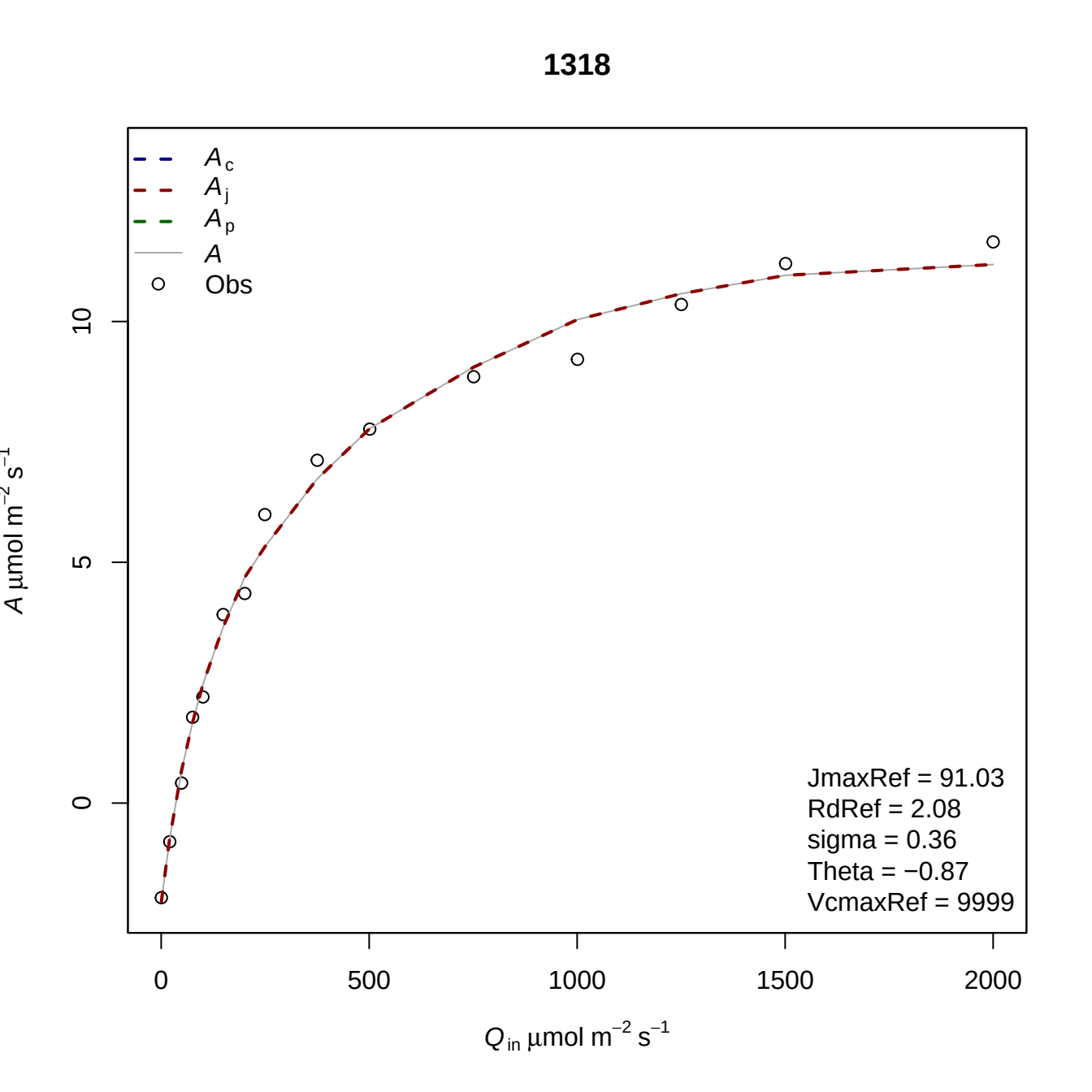

1313
A_c
A_j
A_p
A
Obs
JmaxRef = 62.85
RdRef = 1.91
sigma = 0.48
Theta = 0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

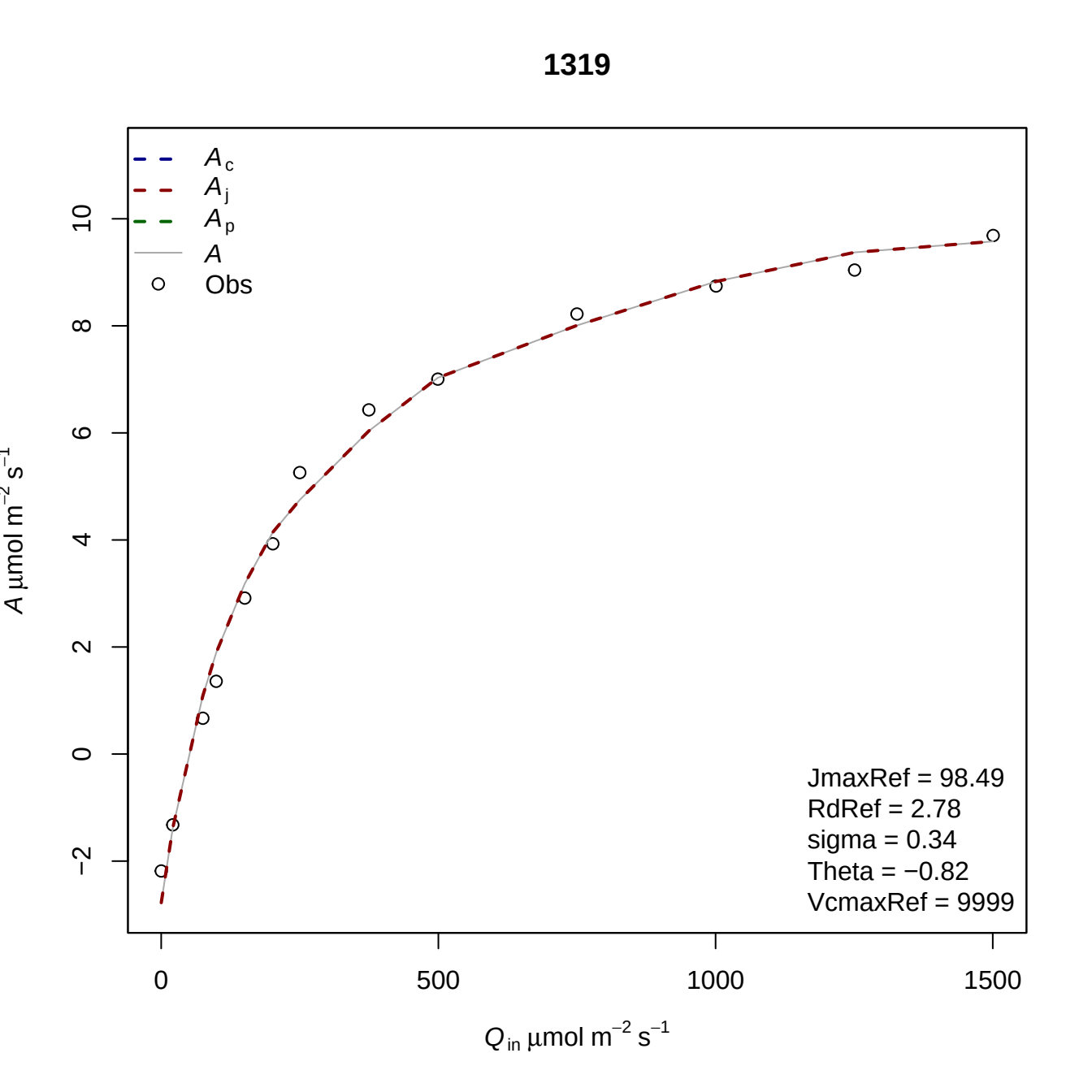

1314
A_c
A_j
A_p
A
Obs
JmaxRef = 88.05
RdRef = 2.78
sigma = 0.36
Theta = 0.13
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

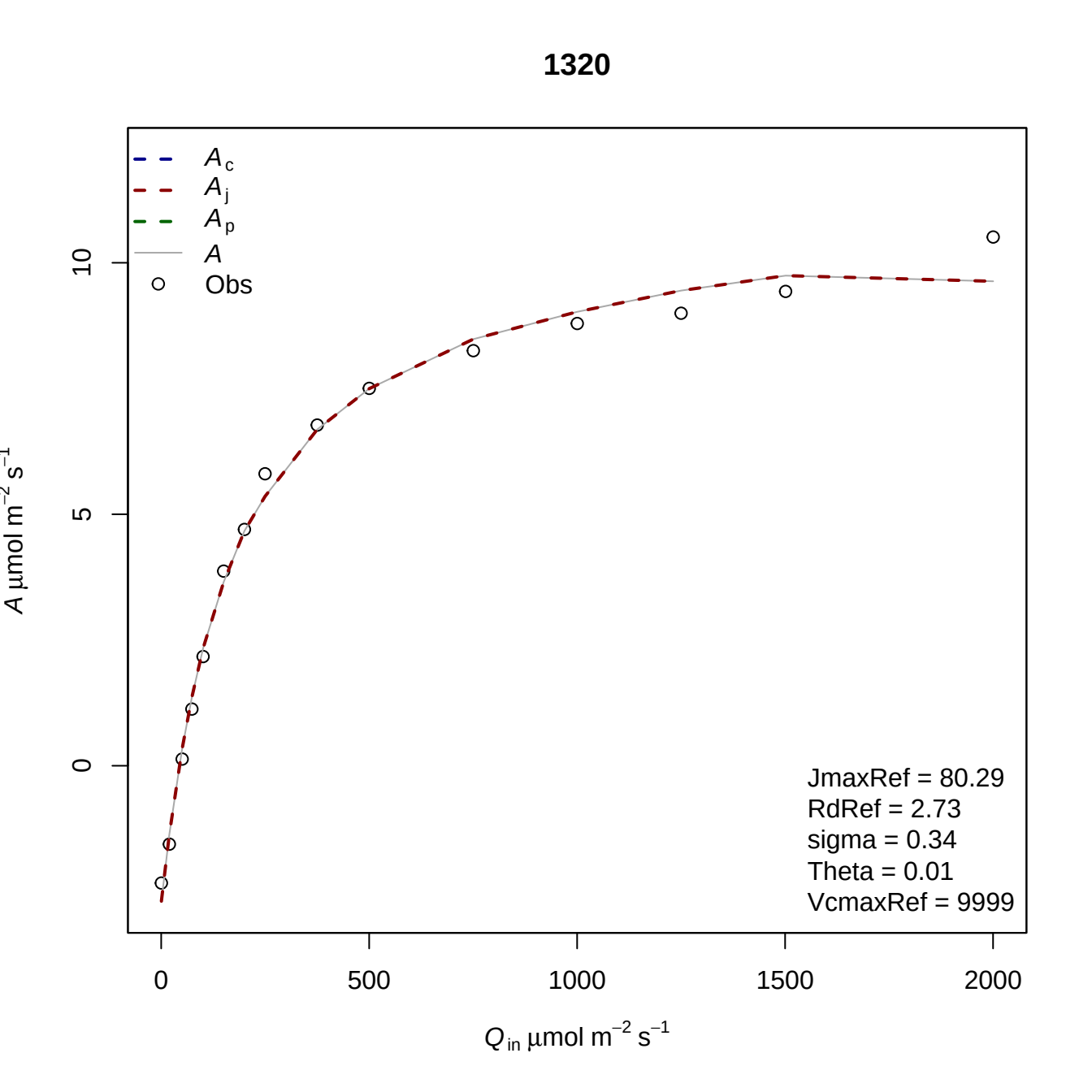

1315
A_c
A_j
A_p
A
Obs
JmaxRef = 135.68
RdRef = 3.37
sigma = 0.3
Theta = −0.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

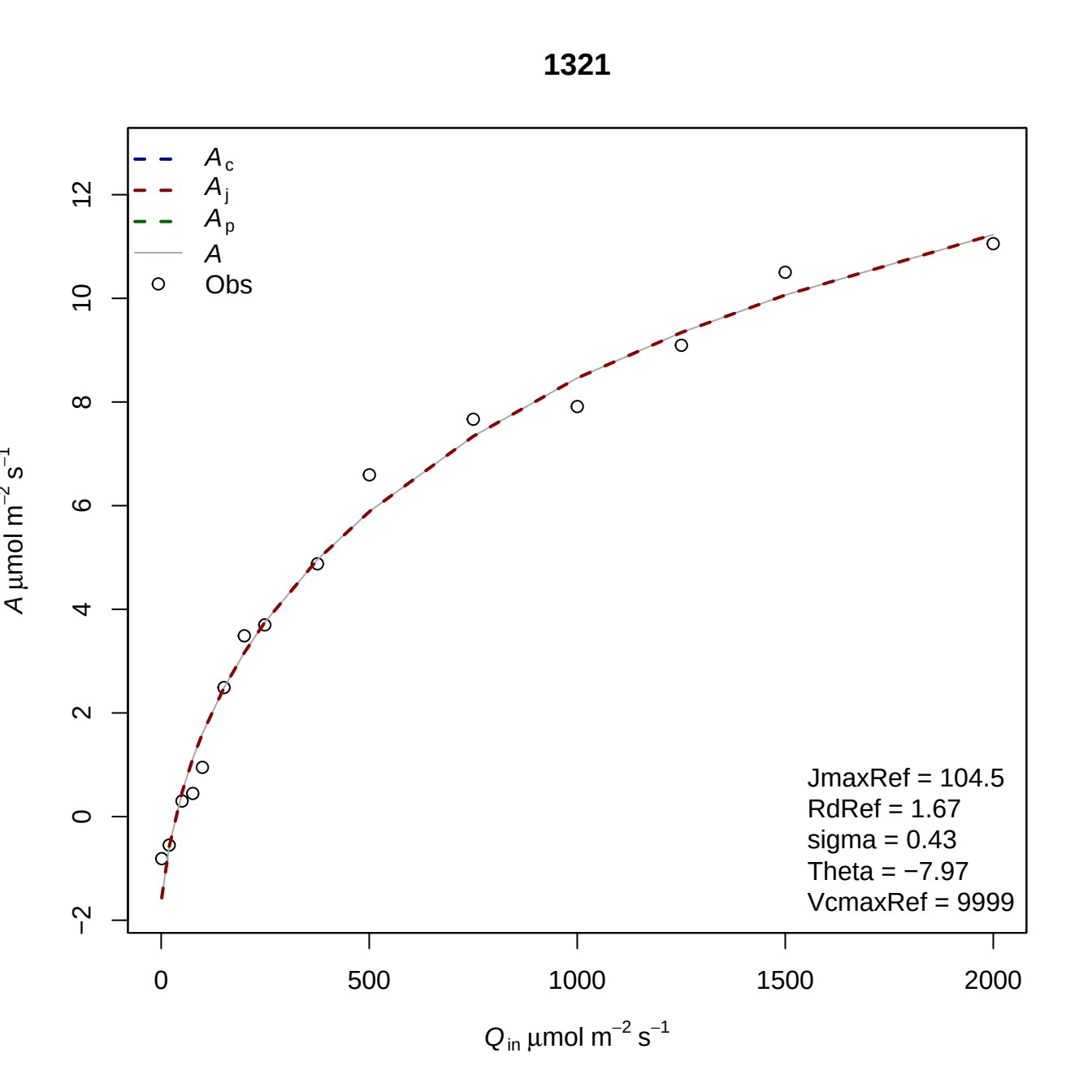

1316
A_c
A_j
A_p
A
Obs
JmaxRef = 110.69
RdRef = 2.83
sigma = 0.61
Theta = −0.09
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

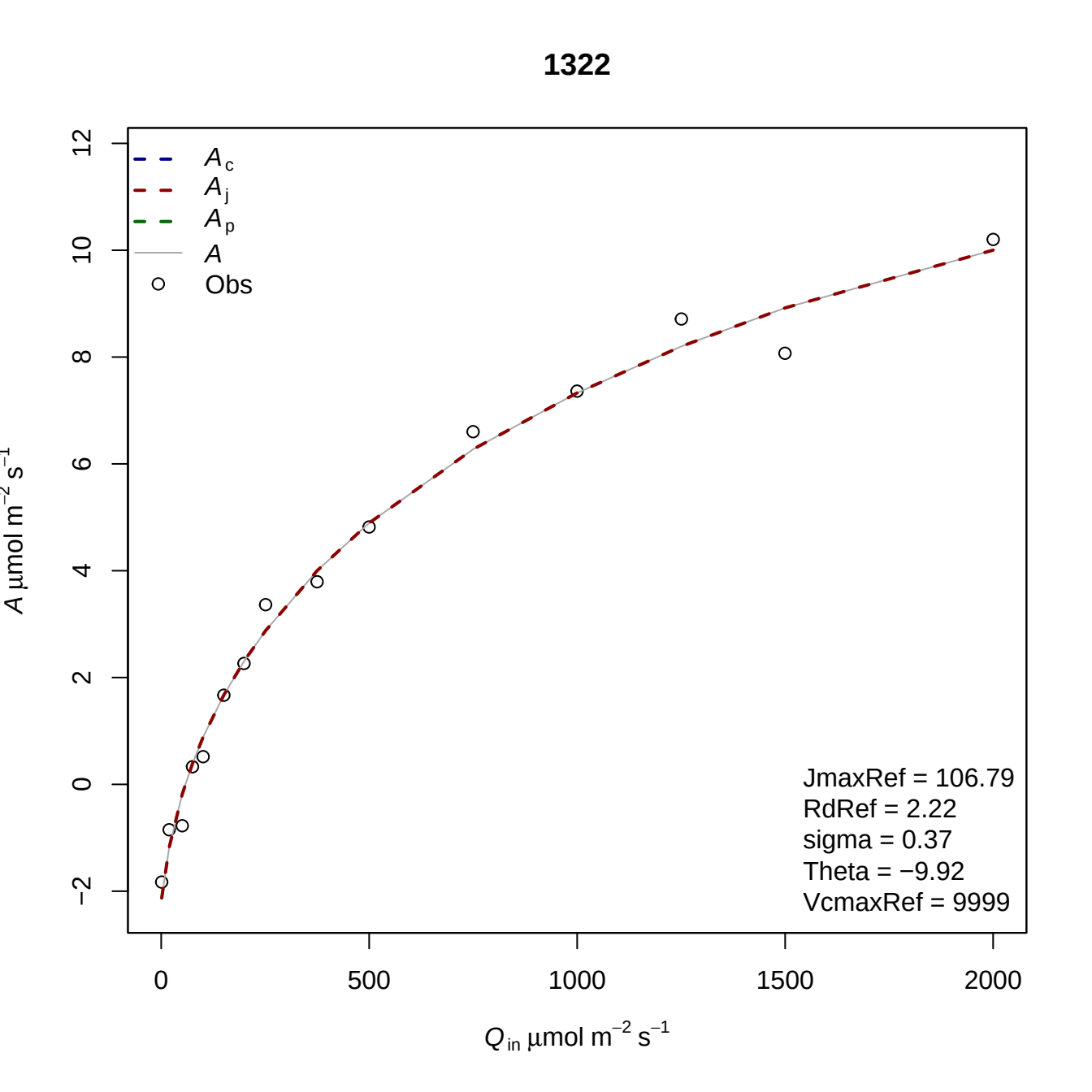

1317
A_c
A_j
A_p
A
Obs
JmaxRef = 112.28
RdRef = 2.73
sigma = 0.28
Theta = 0.48
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

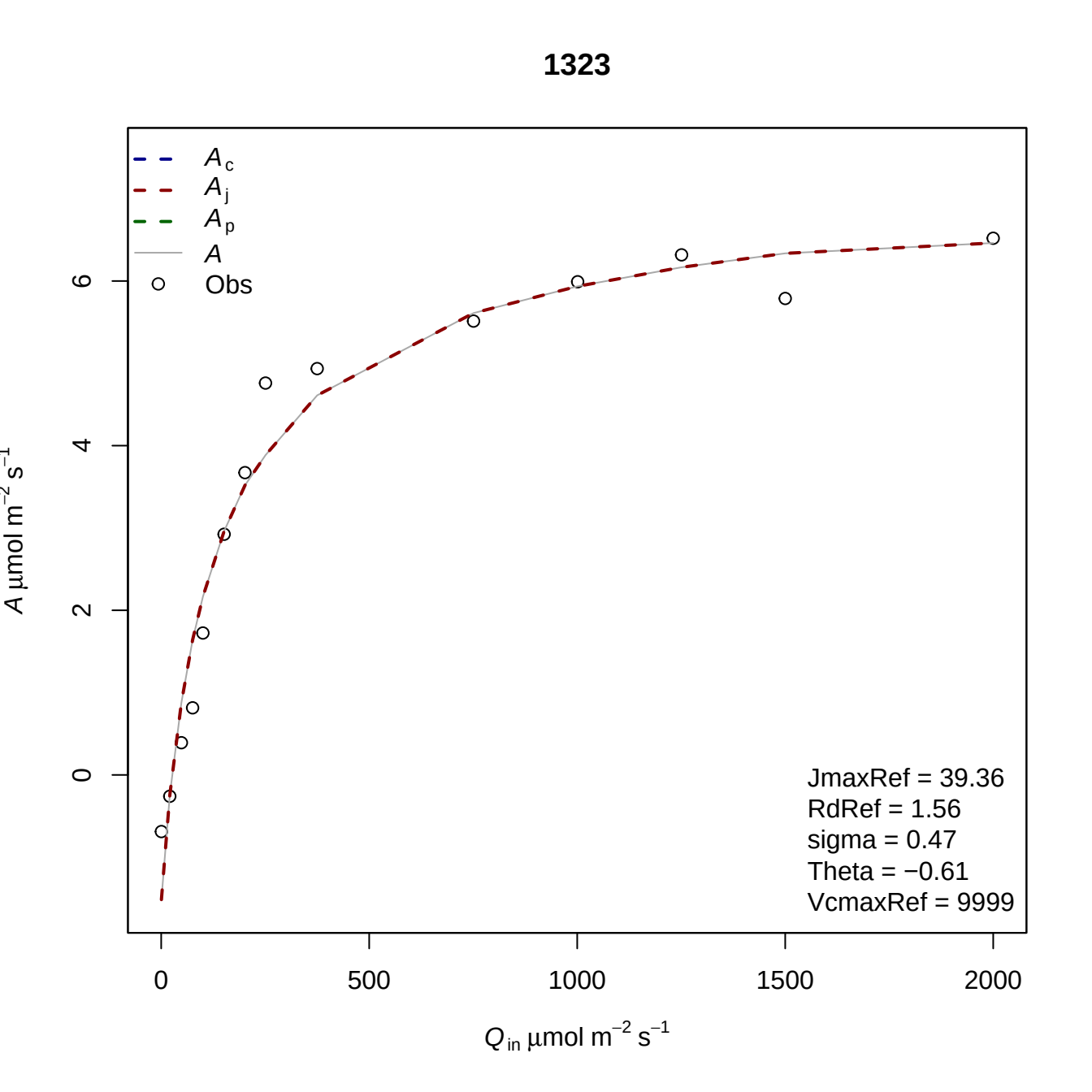

1318
A_c
A_j
A_p
A
Obs
JmaxRef = 91.03
RdRef = 2.08
sigma = 0.36
Theta = −0.87
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

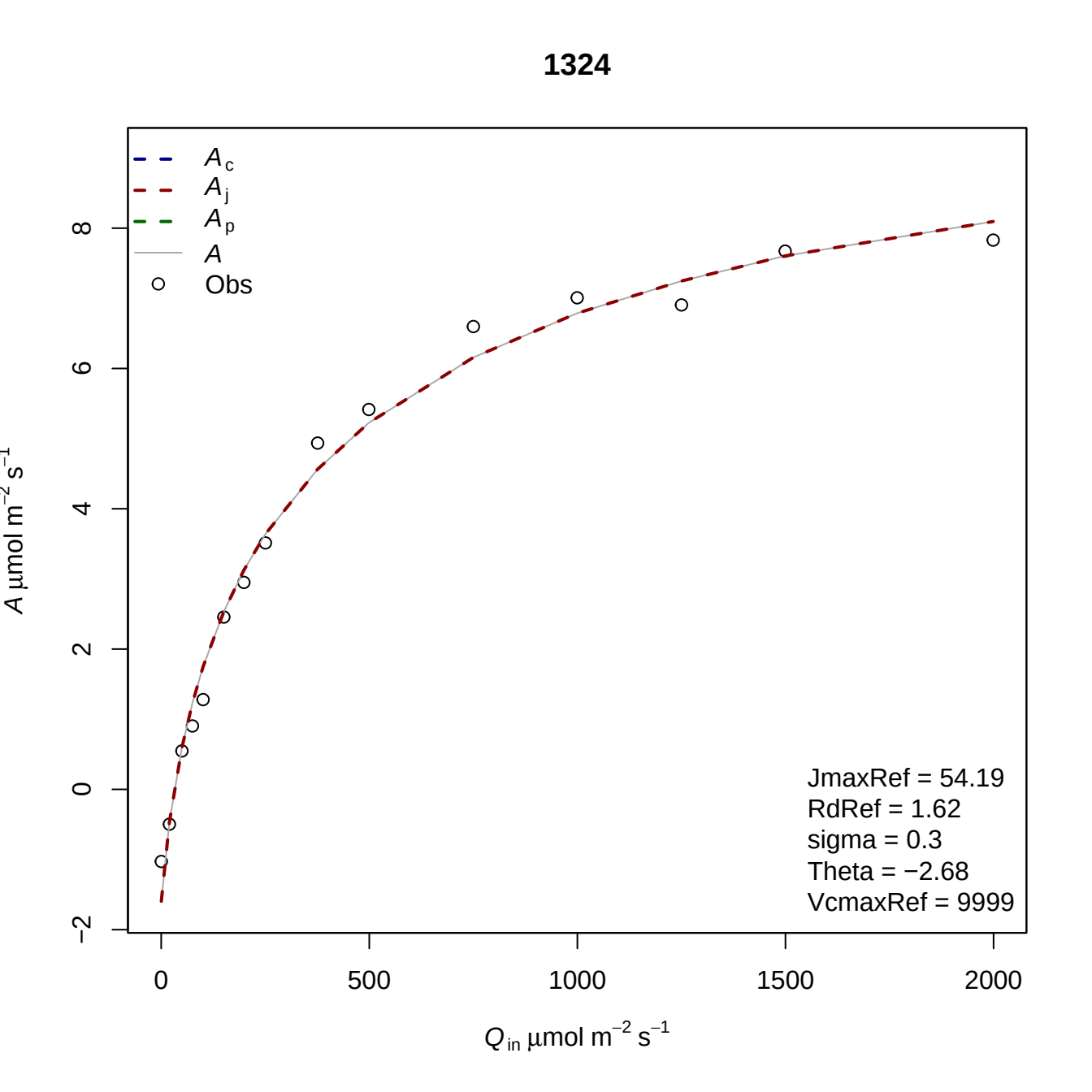

1319
A_c
A_j
A_p
A
Obs
JmaxRef = 98.49
RdRef = 2.78
sigma = 0.34
Theta = −0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

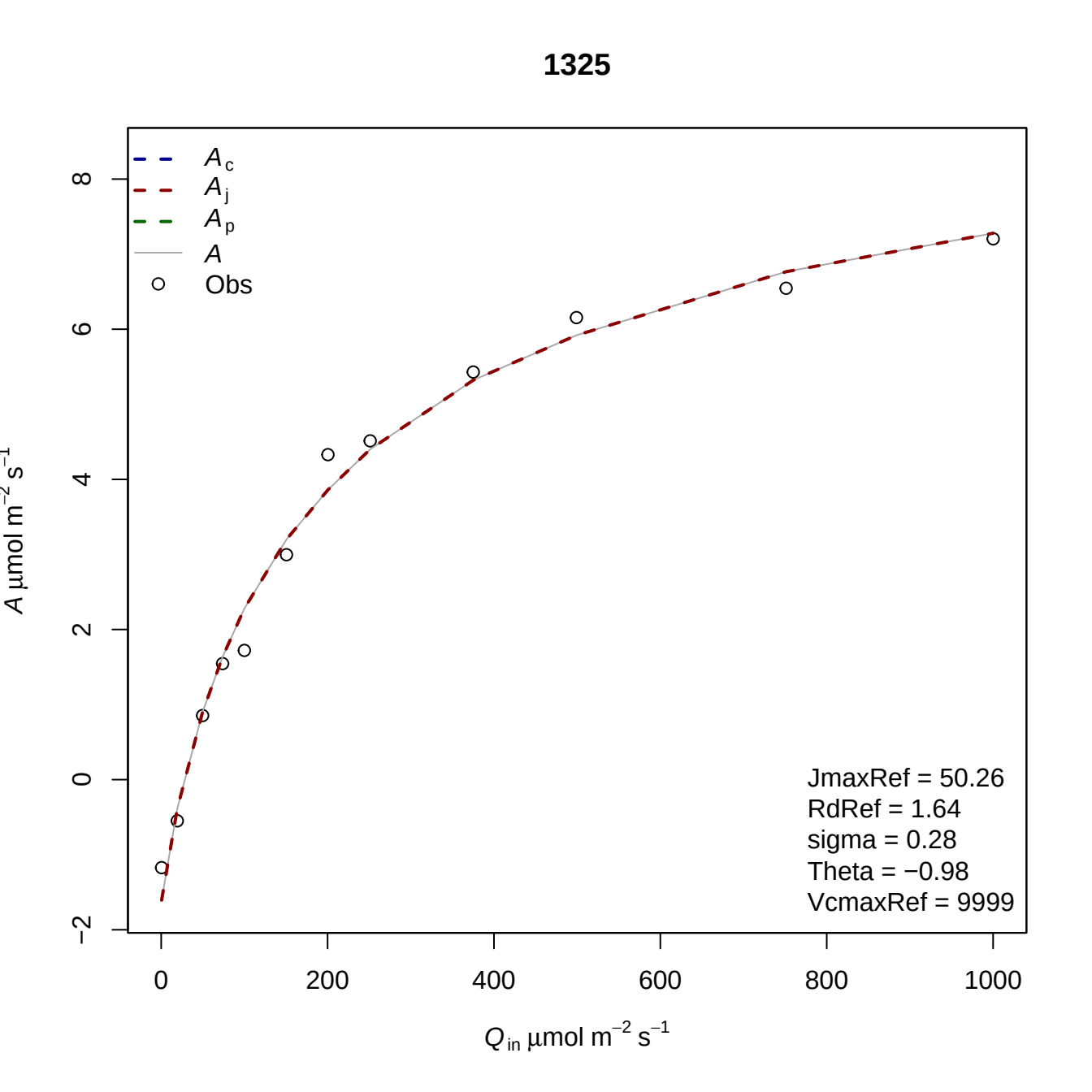

1320
A_c
A_j
A_p
A
Obs
JmaxRef = 80.29
RdRef = 2.73
sigma = 0.34
Theta = 0.01
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

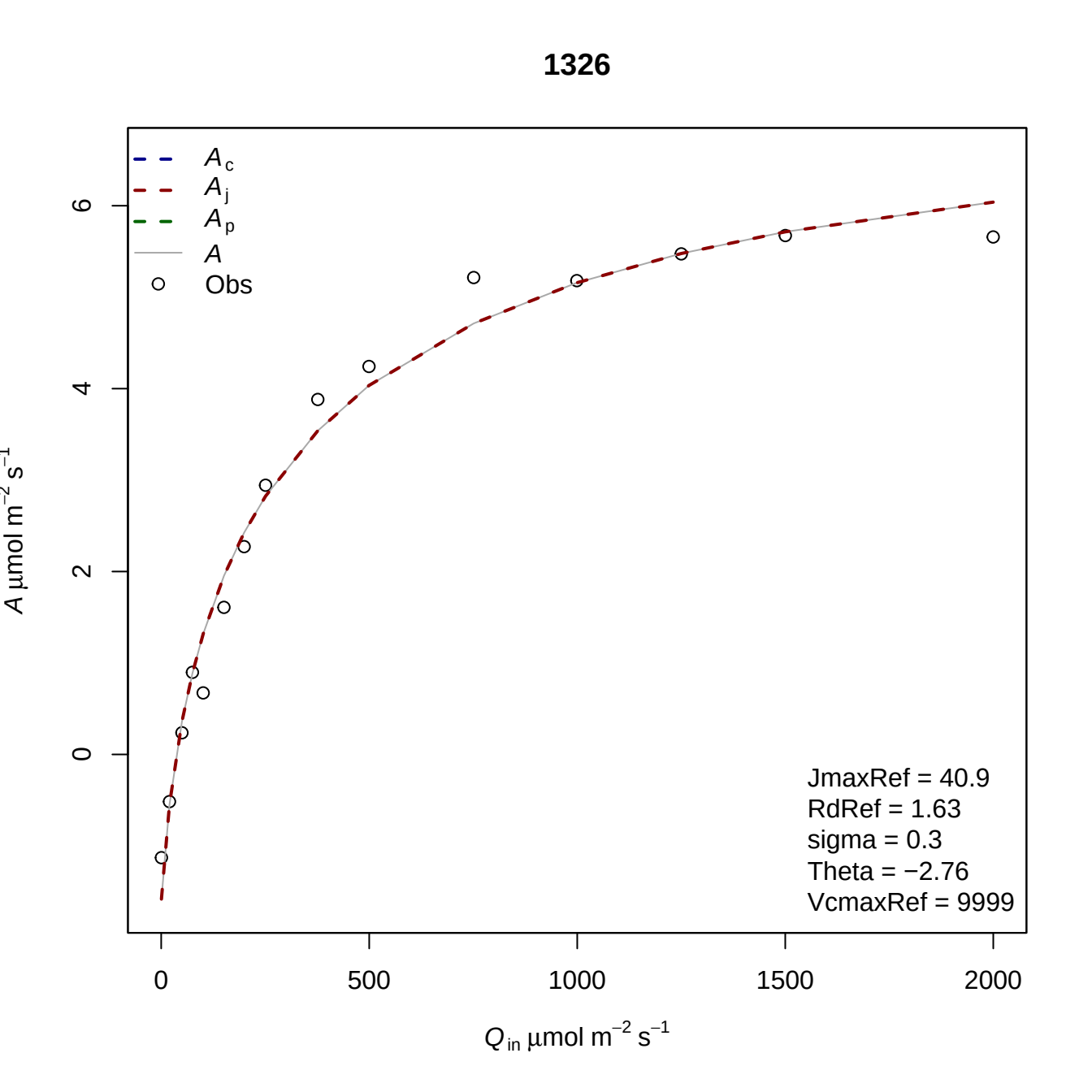

1321
A_c
A_j
A_p
A
Obs
JmaxRef = 104.5
RdRef = 1.67
sigma = 0.43
Theta = −7.97
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

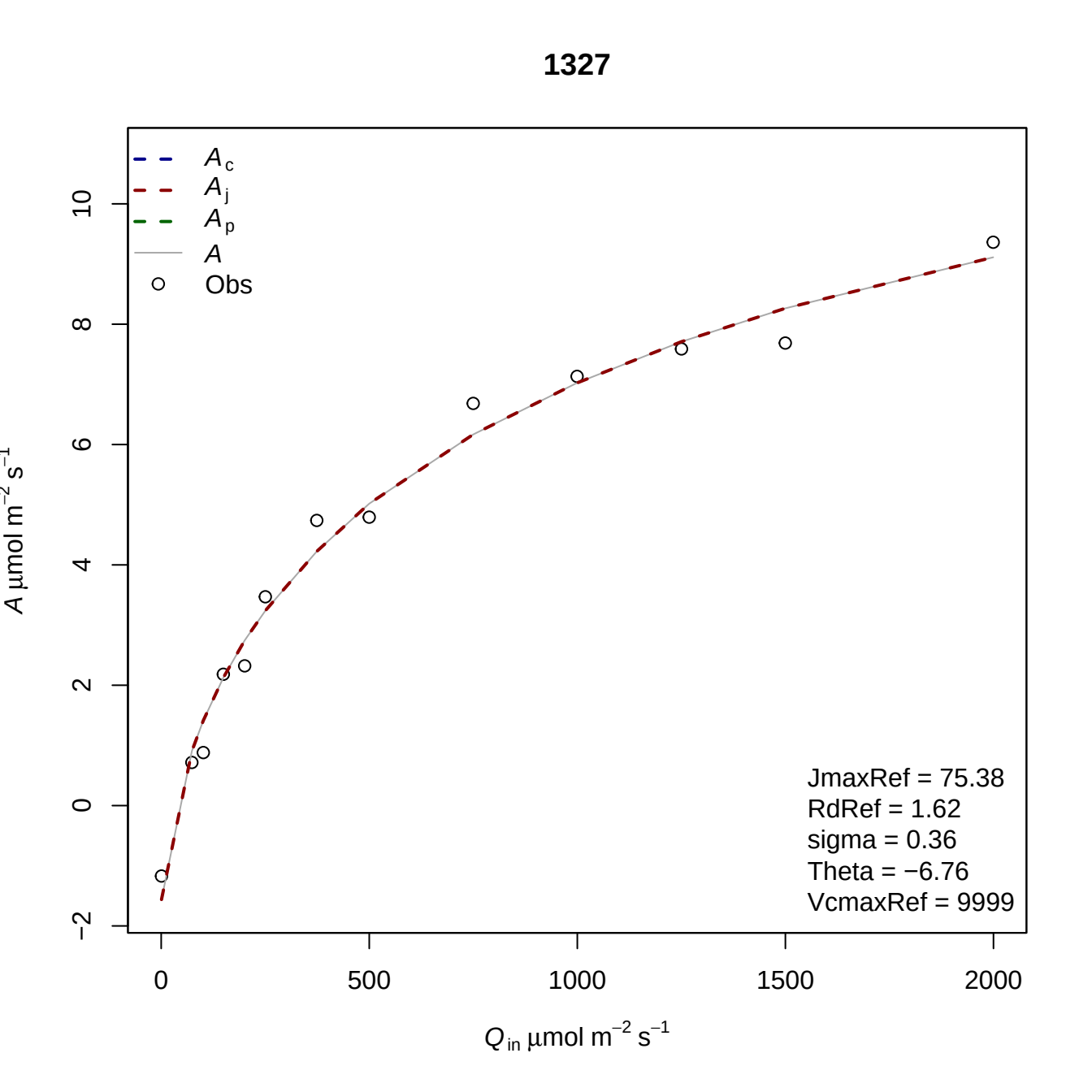

1322
A_c
A_j
A_p
A
Obs
JmaxRef = 106.79
RdRef = 2.22
sigma = 0.37
Theta = −9.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

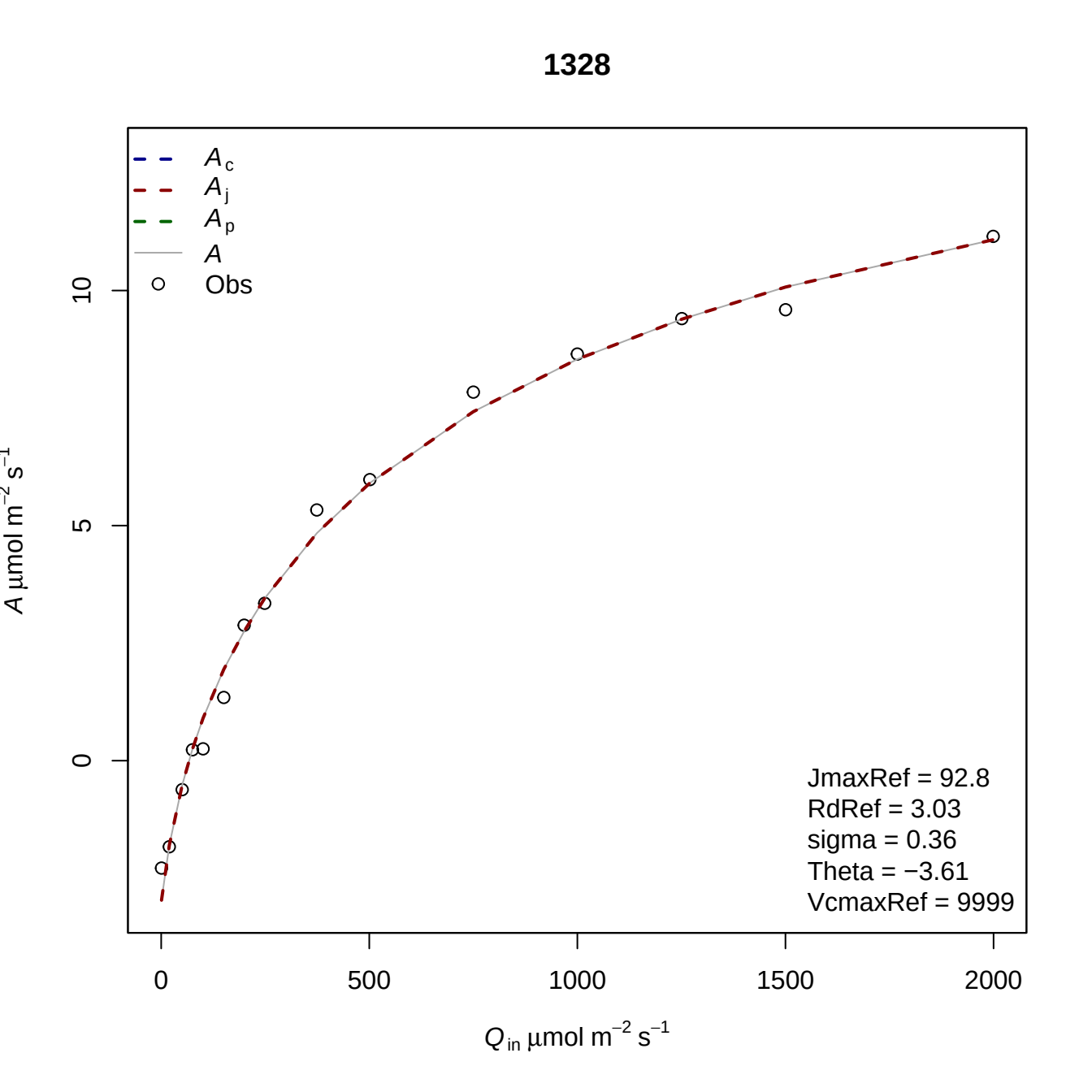

1323
A_c
A_j
A_p
A
Obs
JmaxRef = 39.36
RdRef = 1.56
sigma = 0.47
Theta = −0.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

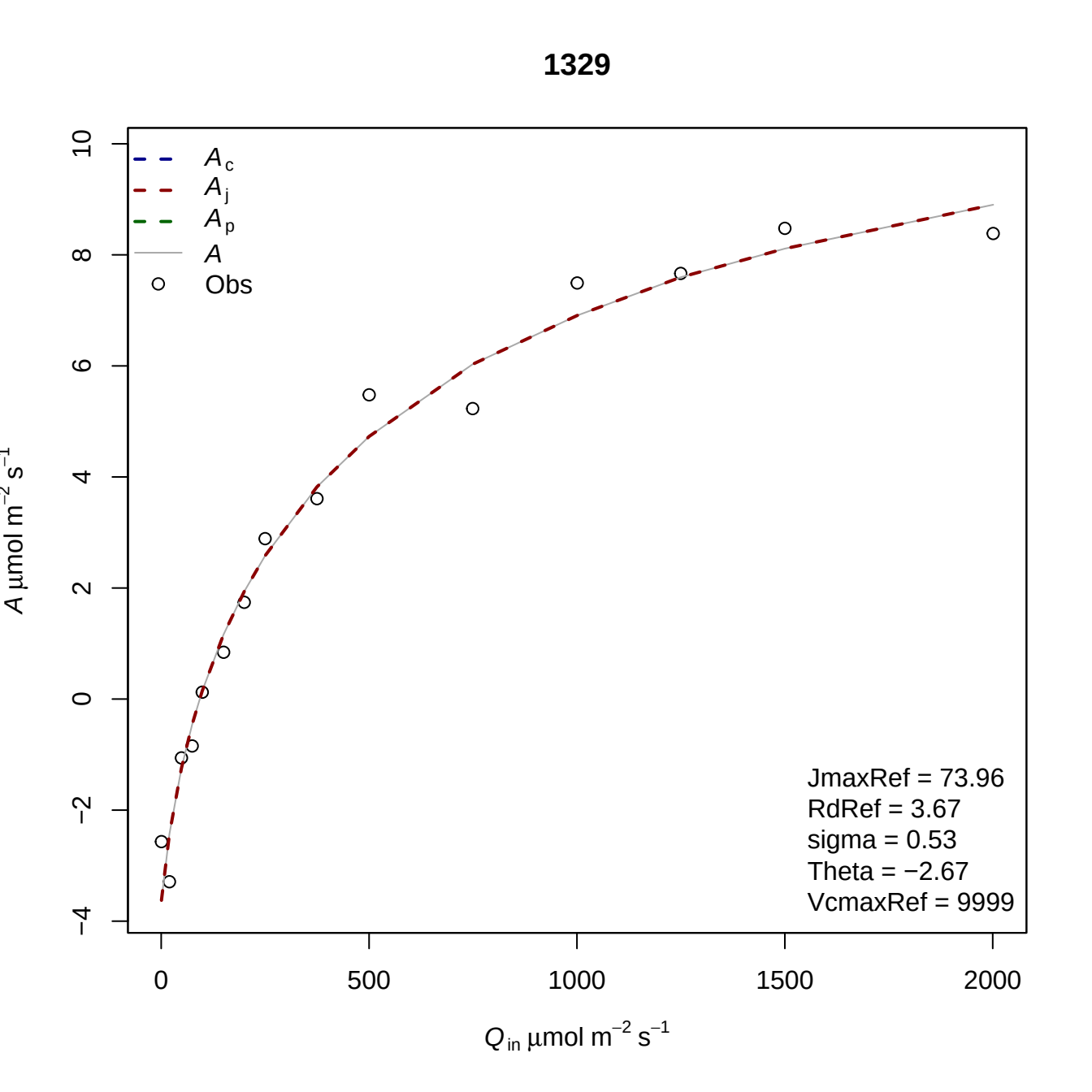

1324
A_c
A_j
A_p
A
Obs
JmaxRef = 54.19
RdRef = 1.62
sigma = 0.3
Theta = −2.68
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

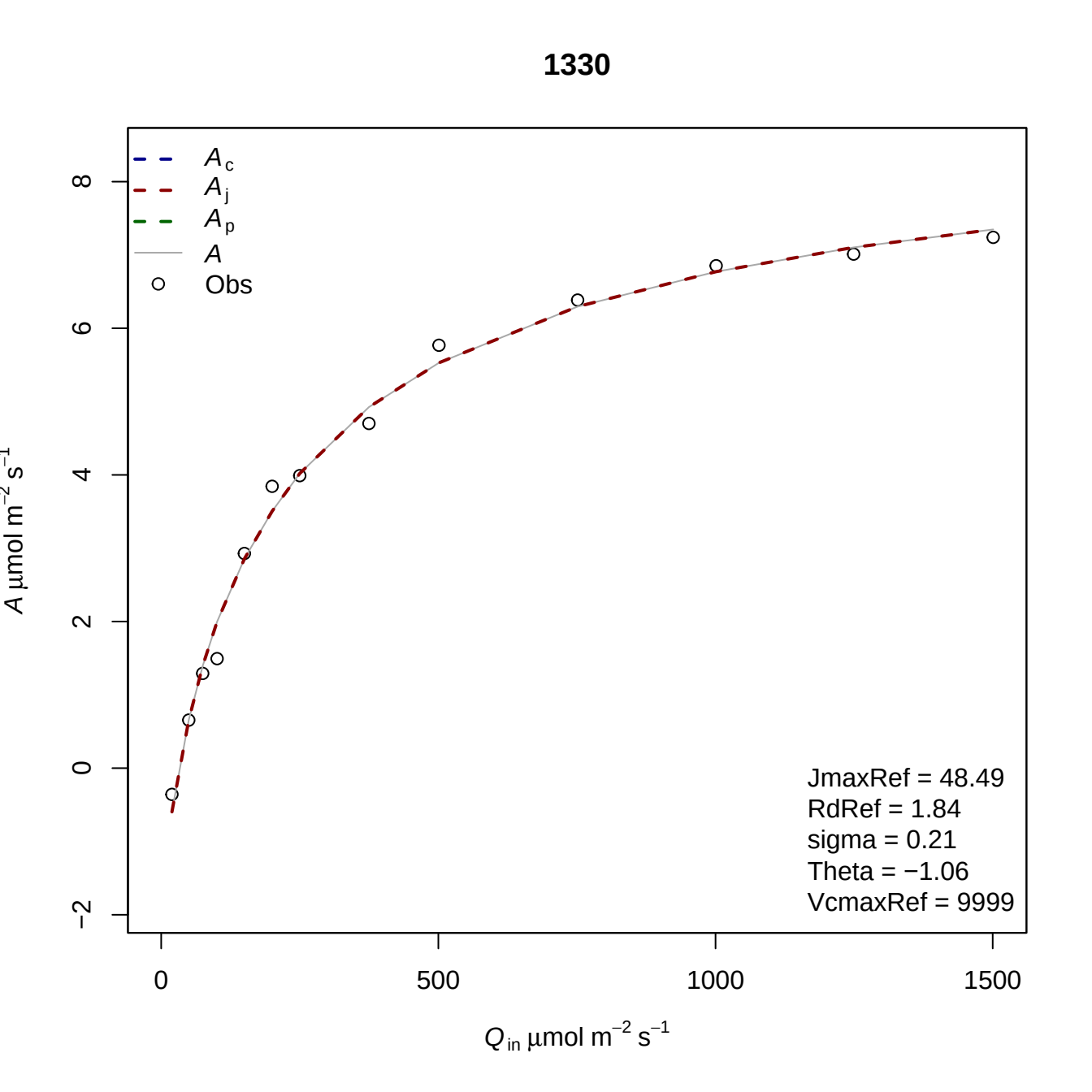

1325
A_c
A_j
A_p
A
Obs
JmaxRef = 50.26
RdRef = 1.64
sigma = 0.28
Theta = −0.98
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

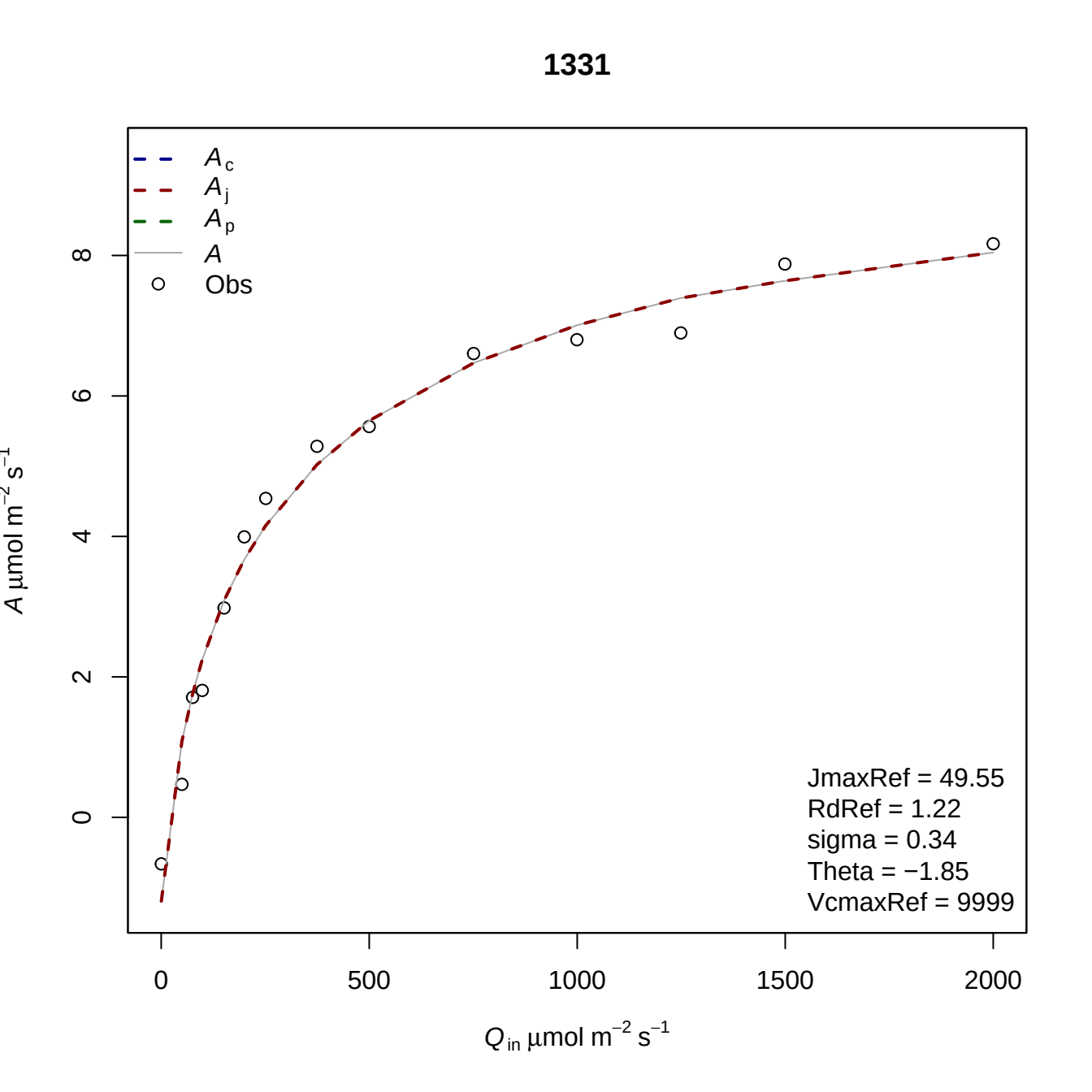

1326
A_c
A_j
A_p
A
Obs
JmaxRef = 40.9
RdRef = 1.63
sigma = 0.3
Theta = −2.76
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

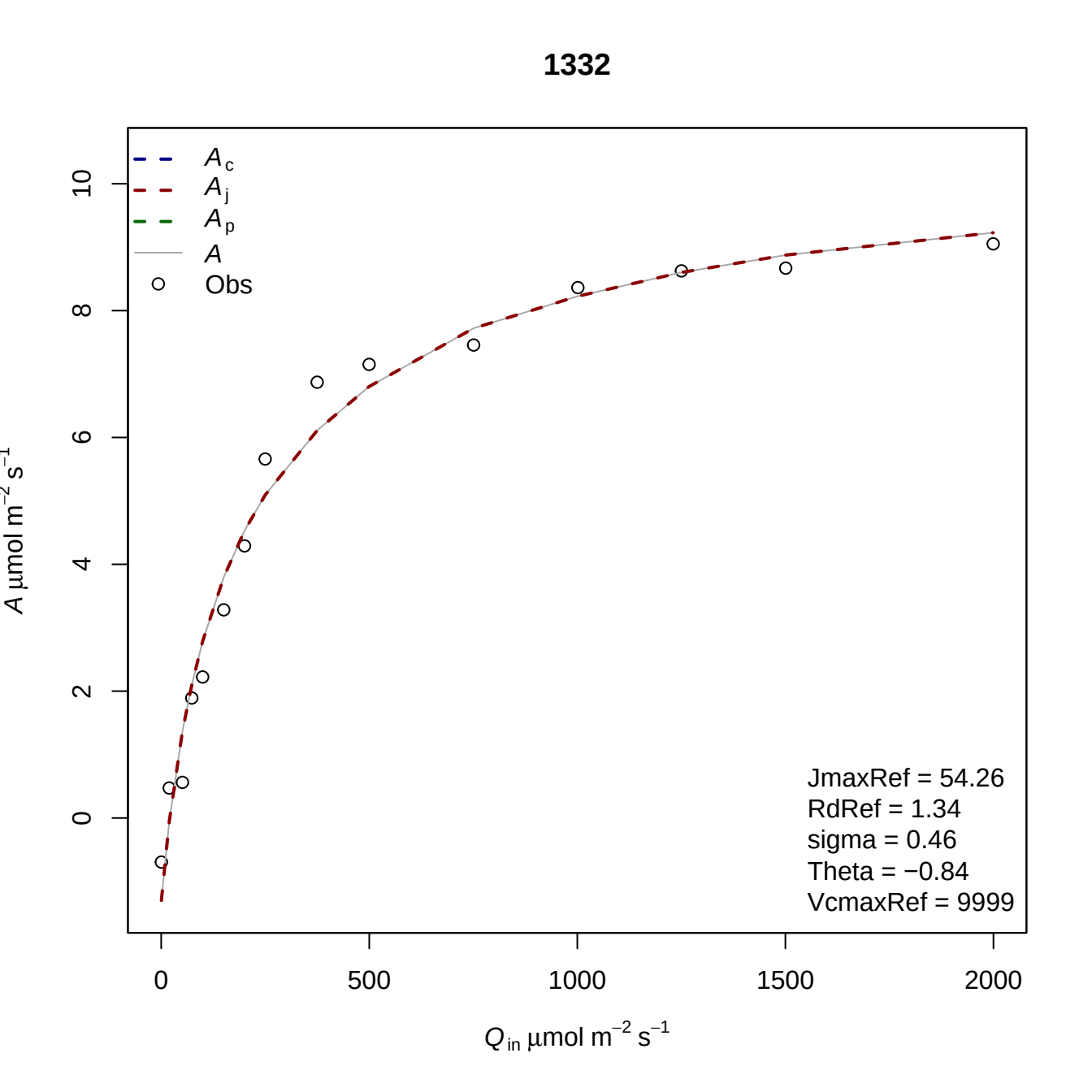

1327
A_c
A_j
A_p
A
Obs
JmaxRef = 75.38
RdRef = 1.62
sigma = 0.36
Theta = −6.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

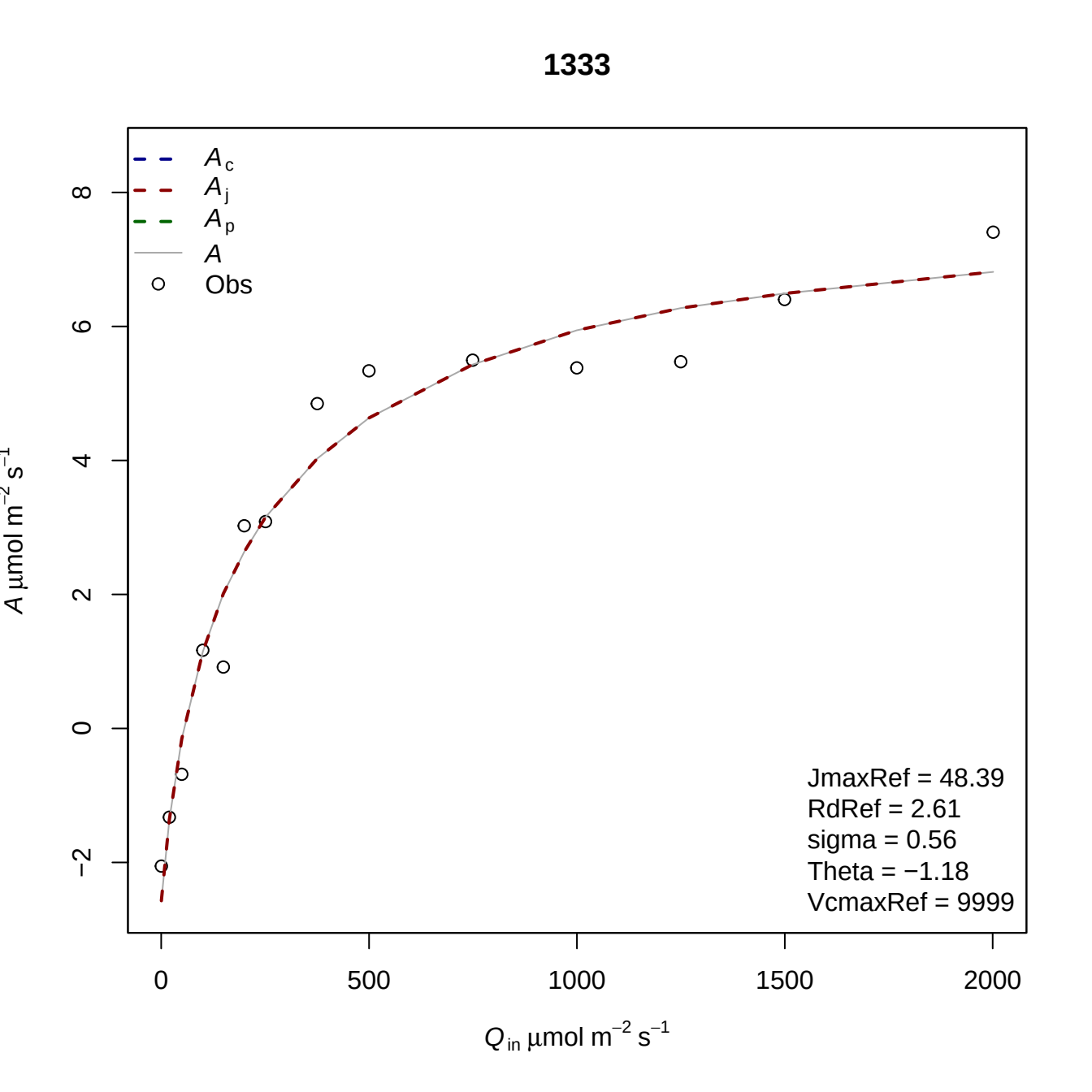

1328
A_c
A_j
A_p
A
Obs
JmaxRef = 92.8
RdRef = 3.03
sigma = 0.36
Theta = −3.61
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

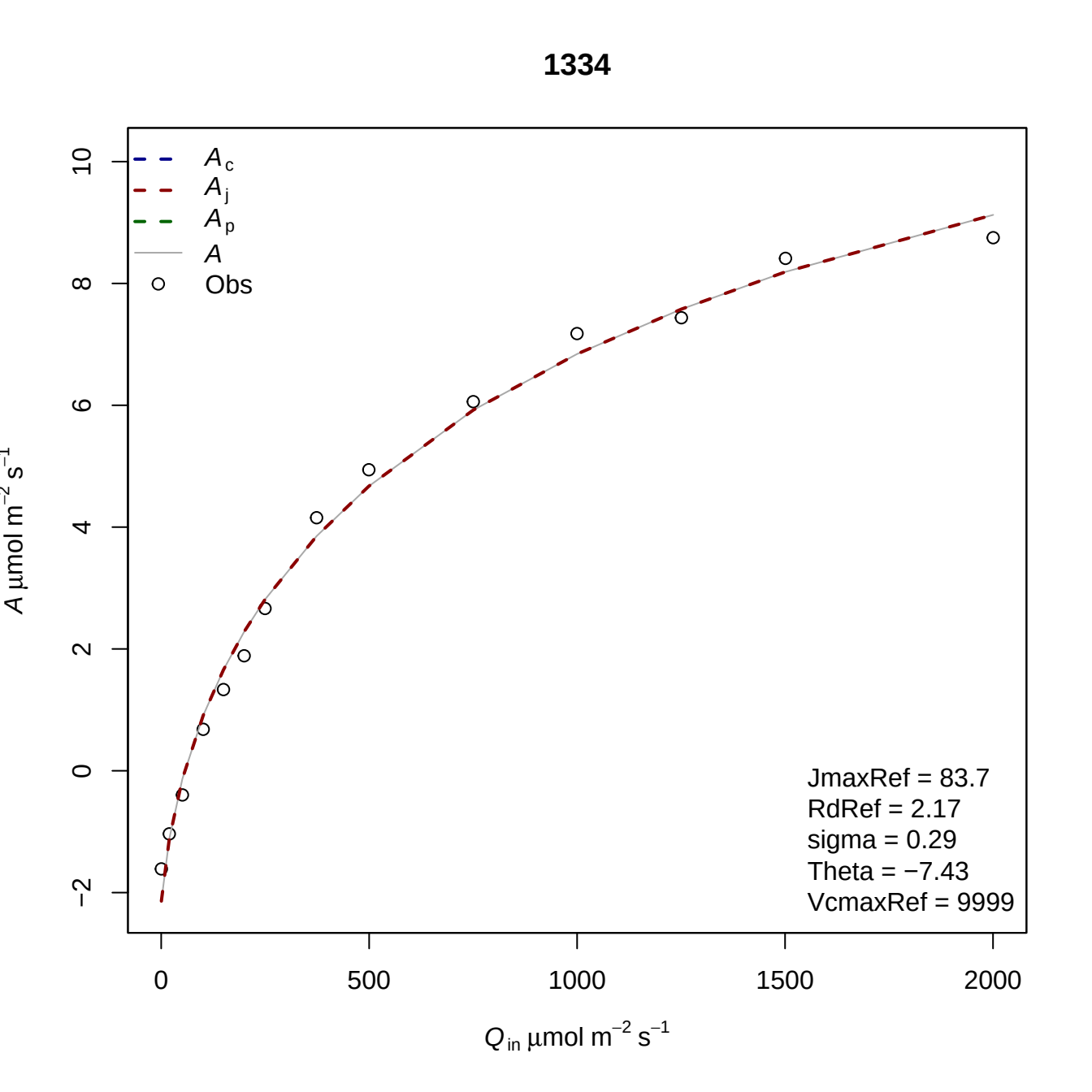

1329
A_c
A_j
A_p
A
Obs
JmaxRef = 73.96
RdRef = 3.67
sigma = 0.53
Theta = −2.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

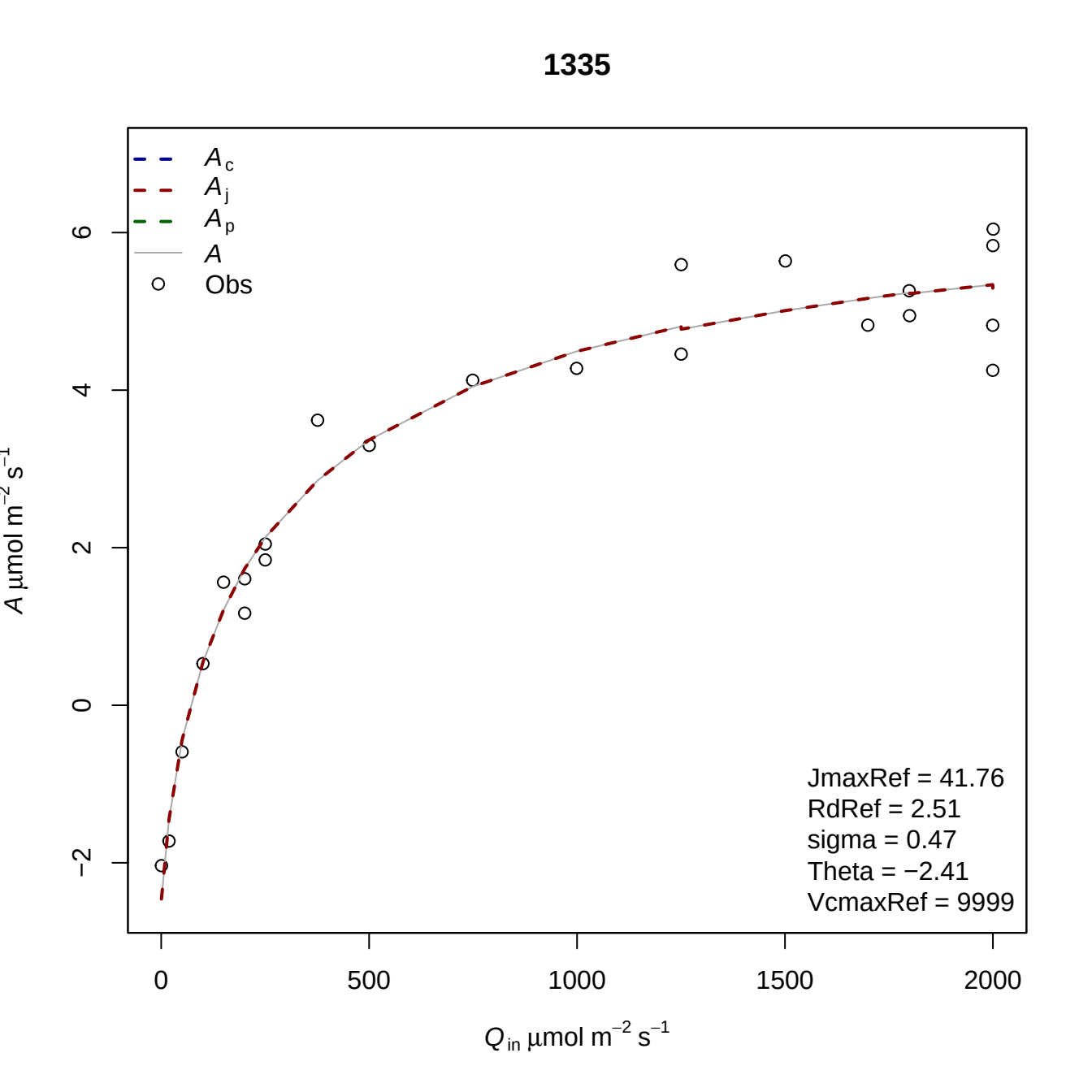

1330
A_c
A_j
A_p
A
Obs
JmaxRef = 48.49
RdRef = 1.84
sigma = 0.21
Theta = −1.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

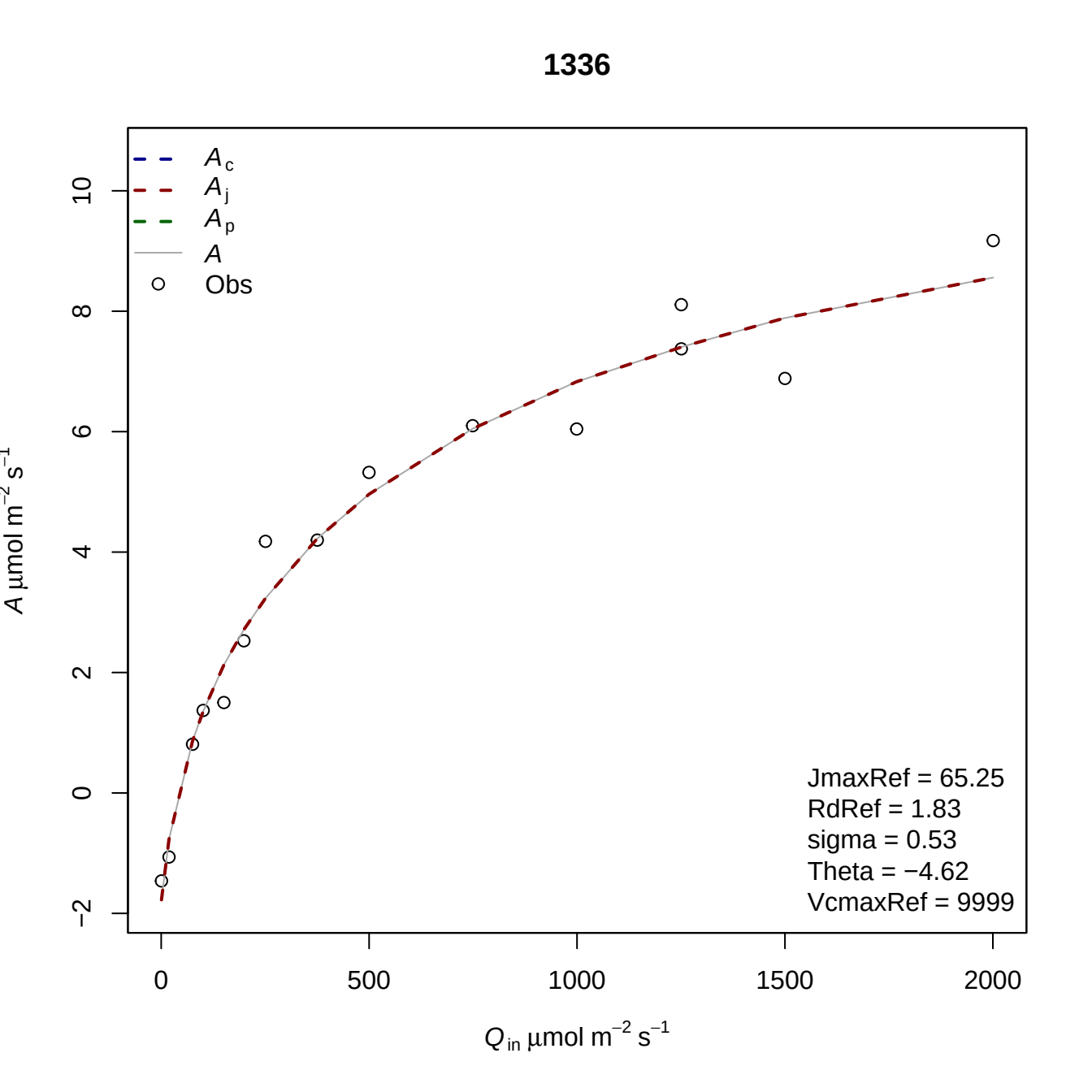

1331
A_c
A_j
A_p
A
Obs
JmaxRef = 49.55
RdRef = 1.22
sigma = 0.34
Theta = −1.85
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

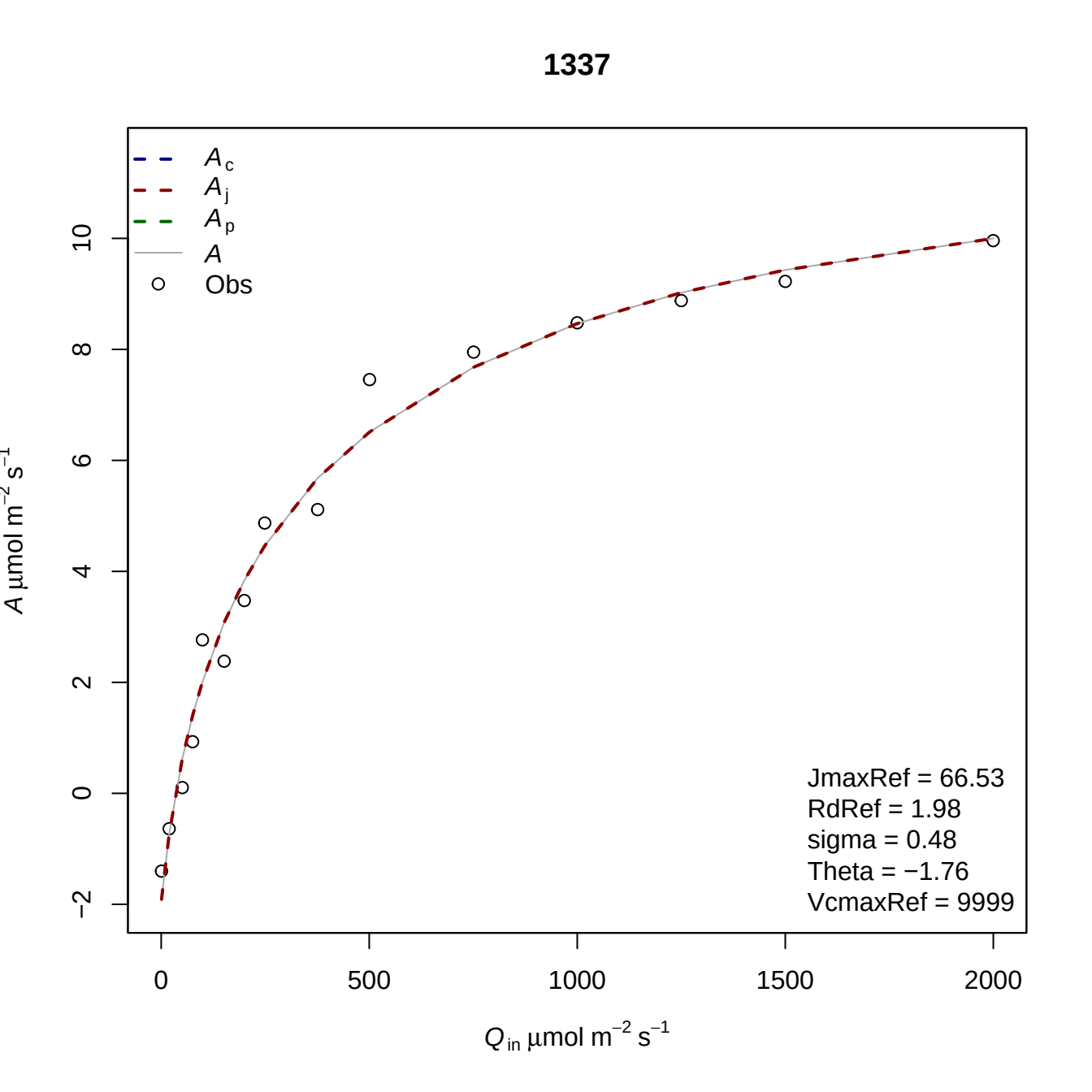

1332
A_c
A_j
A_p
A
Obs
JmaxRef = 54.26
RdRef = 1.34
sigma = 0.46
Theta = −0.84
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

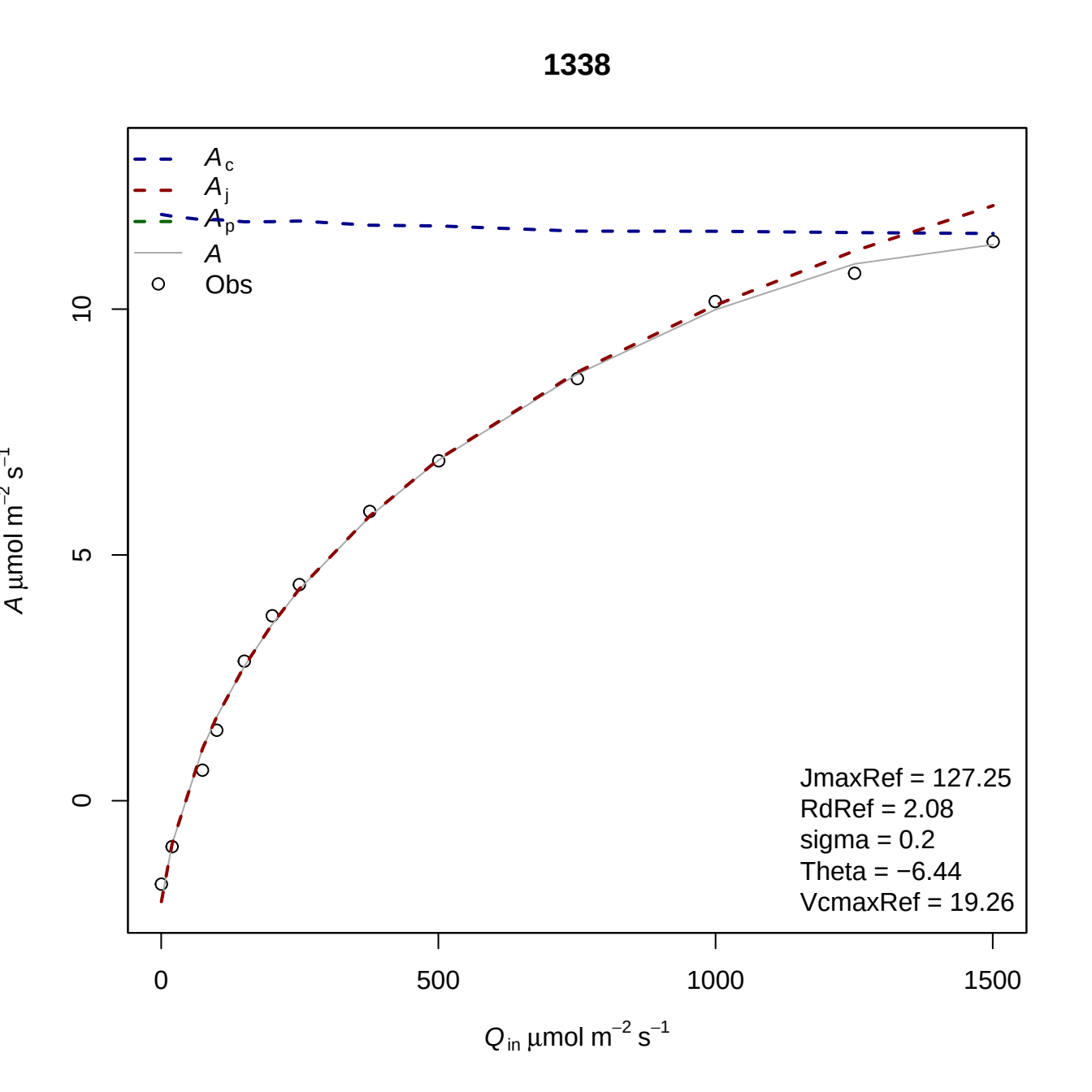

1333
A_c
A_j
A_p
A
Obs
JmaxRef = 48.39
RdRef = 2.61
sigma = 0.56
Theta = −1.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

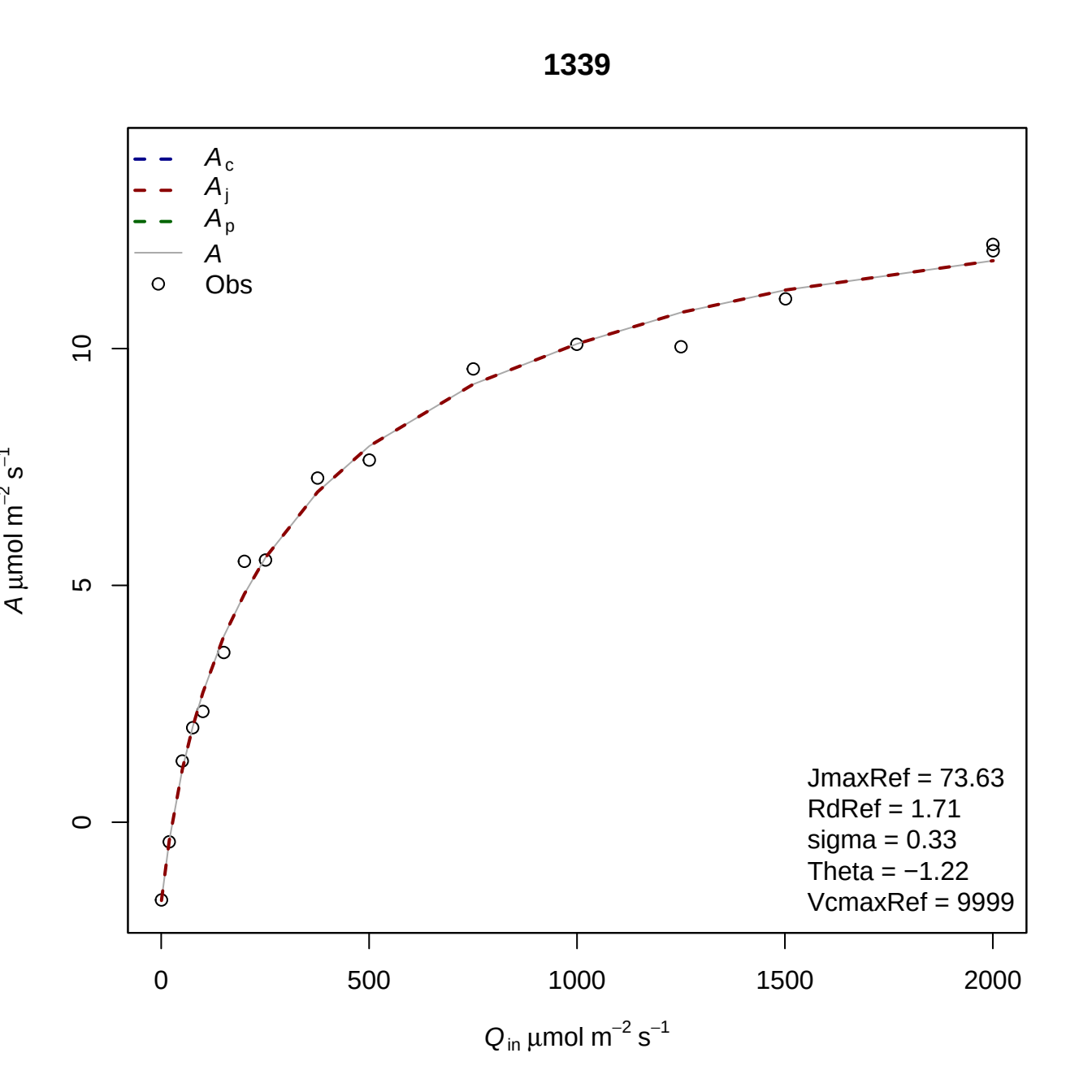

1334
A_c
A_j
A_p
A
Obs
JmaxRef = 83.7
RdRef = 2.17
sigma = 0.29
Theta = −7.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

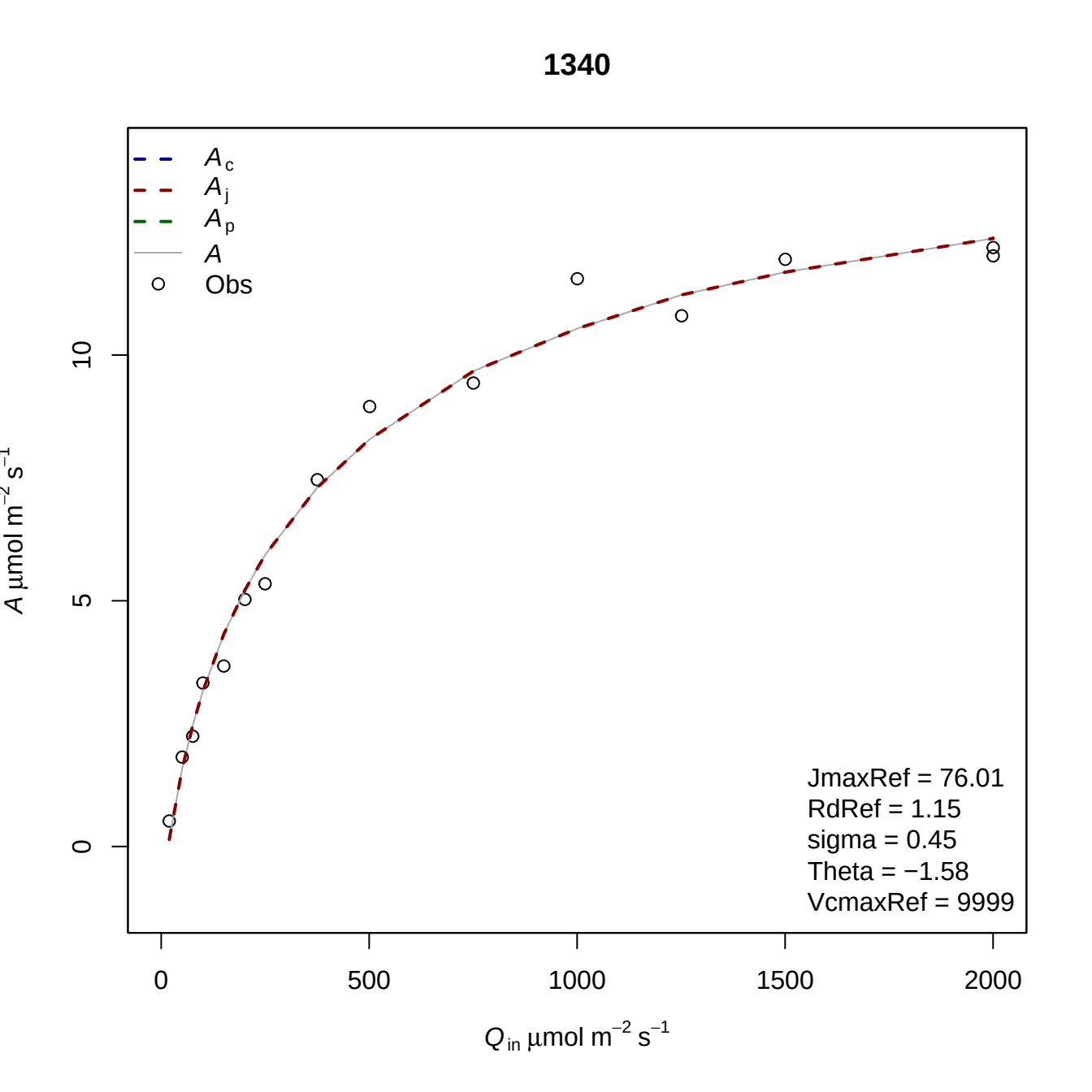

1335
A_c
A_j
A_p
A
Obs
JmaxRef = 41.76
RdRef = 2.51
sigma = 0.47
Theta = −2.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

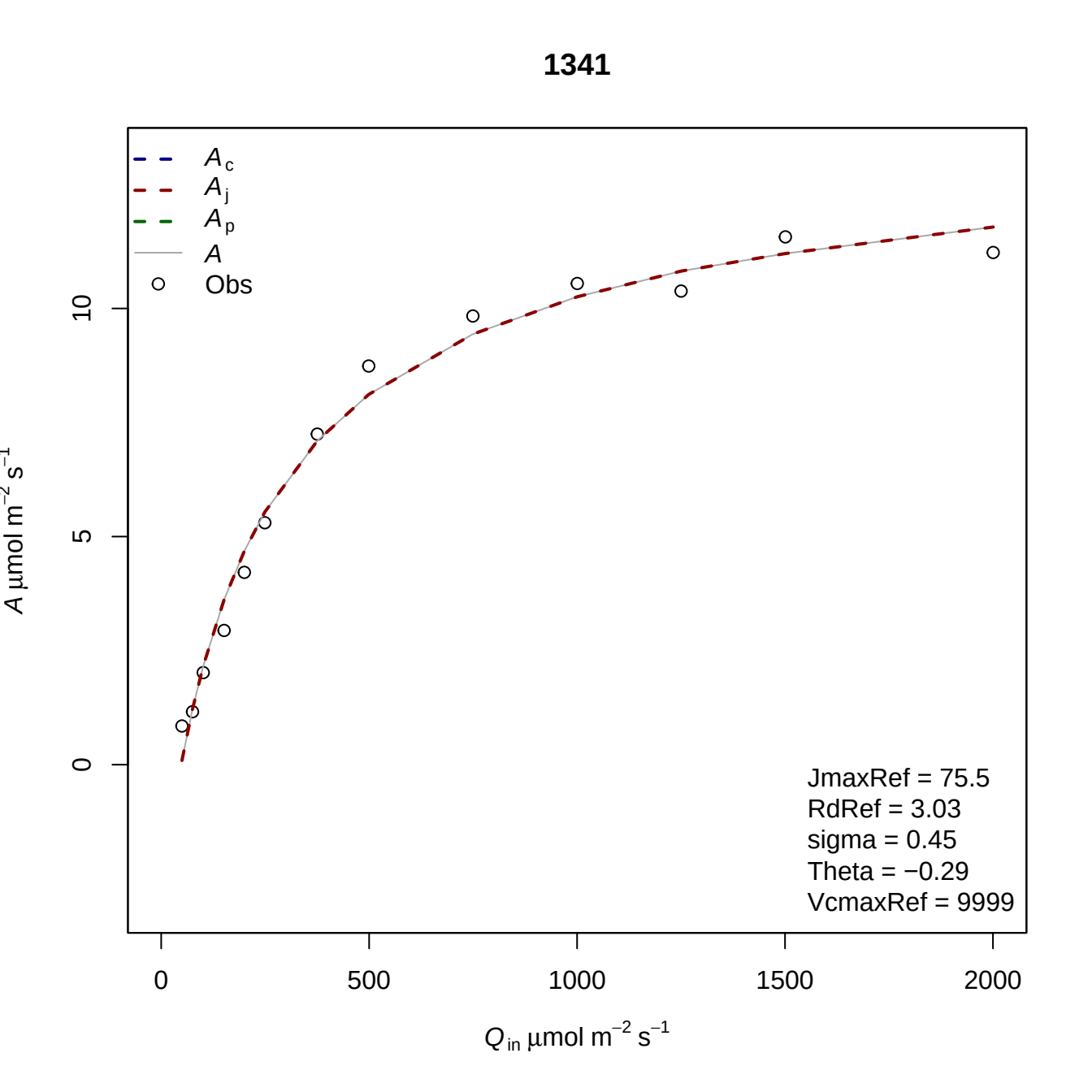

1336
A_c
A_j
A_p
A
Obs
JmaxRef = 65.25
RdRef = 1.83
sigma = 0.53
Theta = −4.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

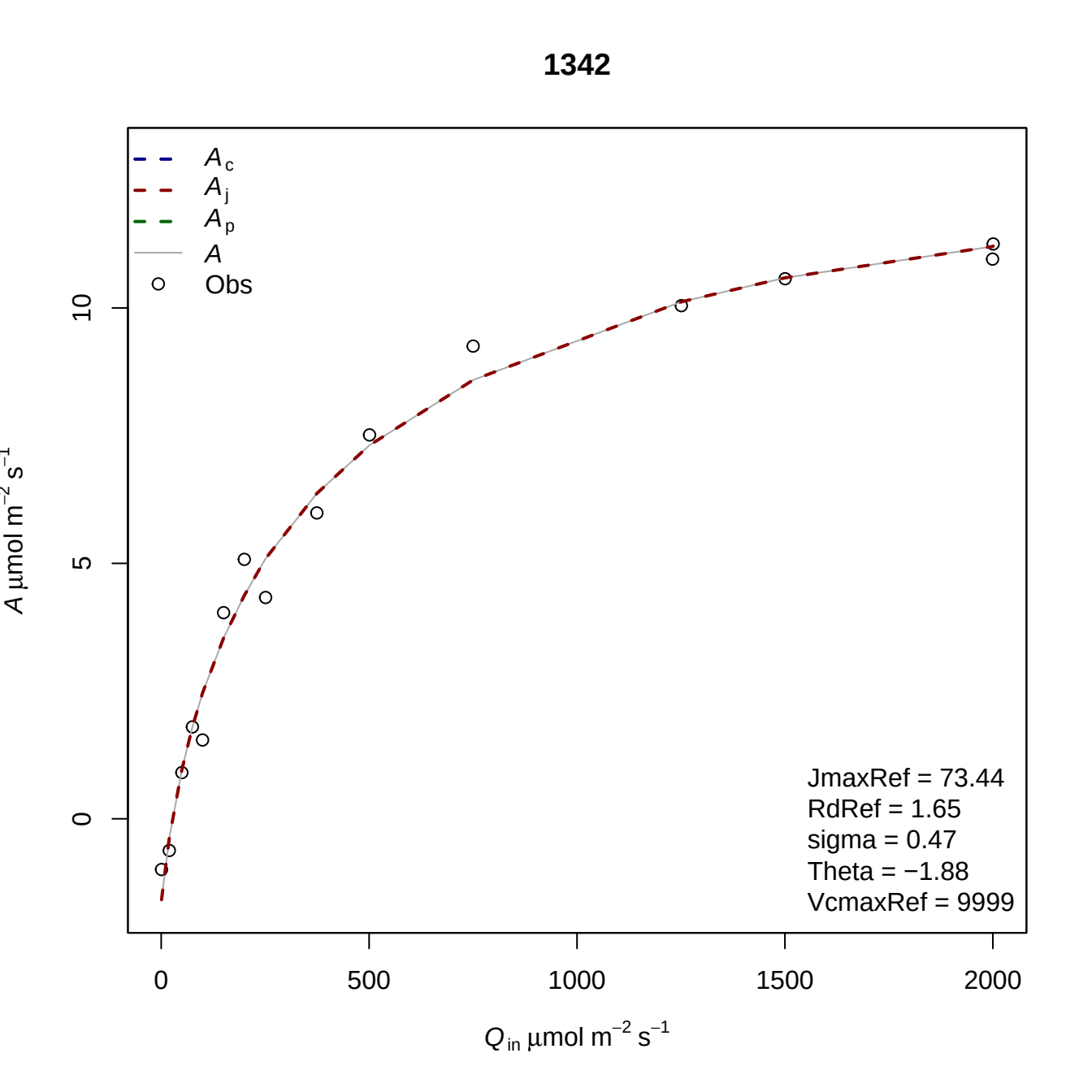

1337
A_c
A_j
A_p
A
Obs
JmaxRef = 66.53
RdRef = 1.98
sigma = 0.48
Theta = −1.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

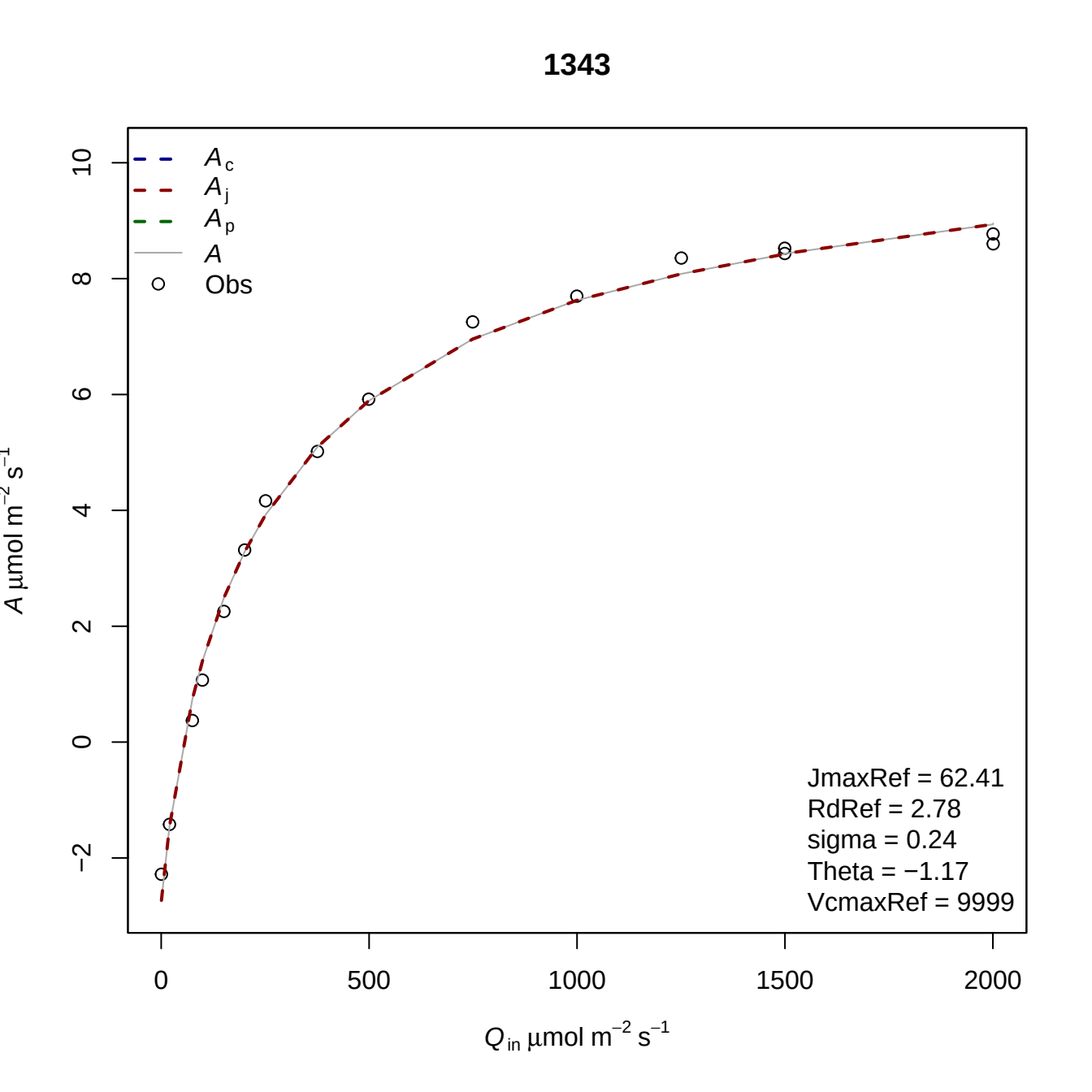

1338
A_c
A_j
A_p
A
Obs
JmaxRef = 127.25
RdRef = 2.08
sigma = 0.2
Theta = −6.44
VcmaxRef = 19.26
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

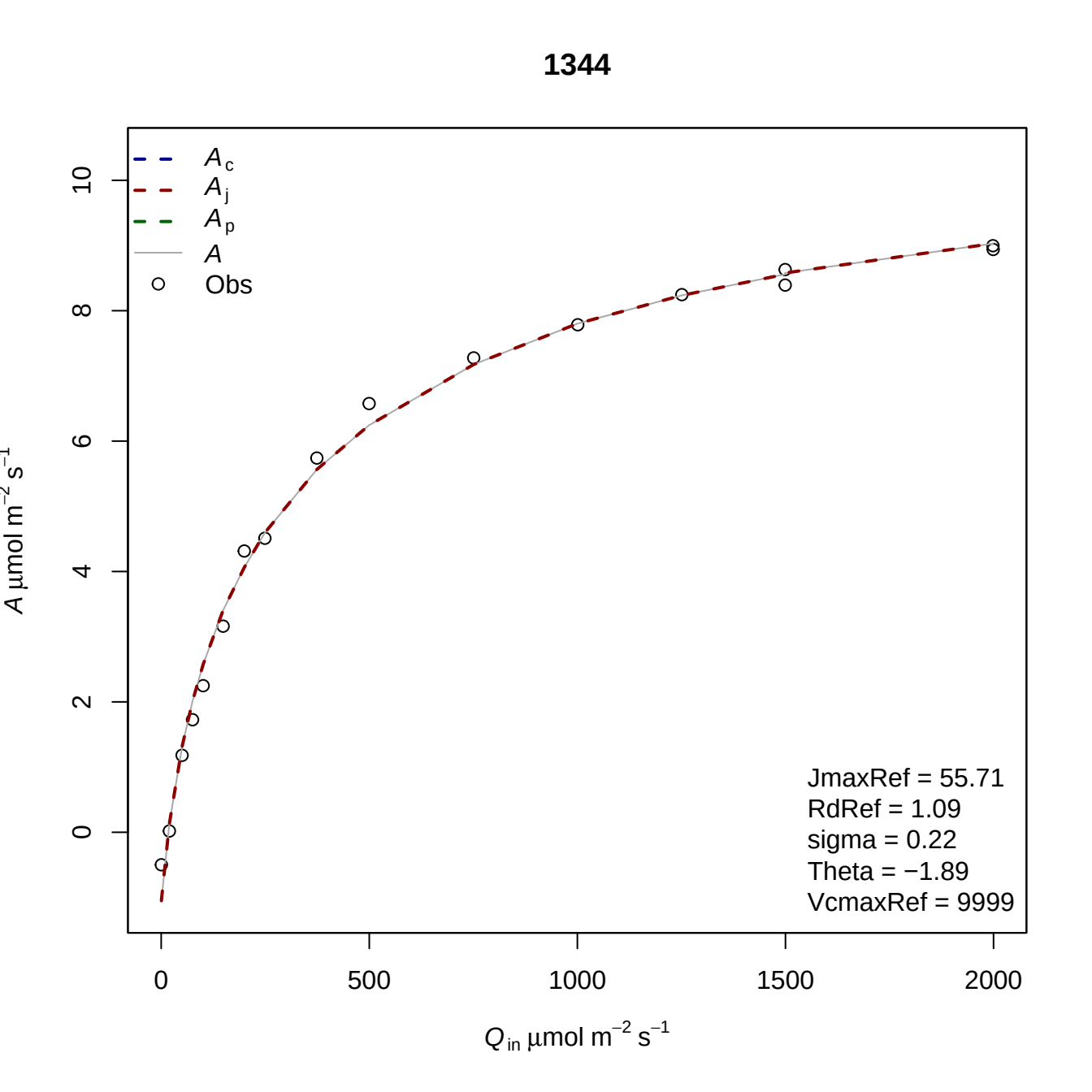

1339
A_c
A_j
A_p
A
Obs
JmaxRef = 73.63
RdRef = 1.71
sigma = 0.33
Theta = −1.22
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

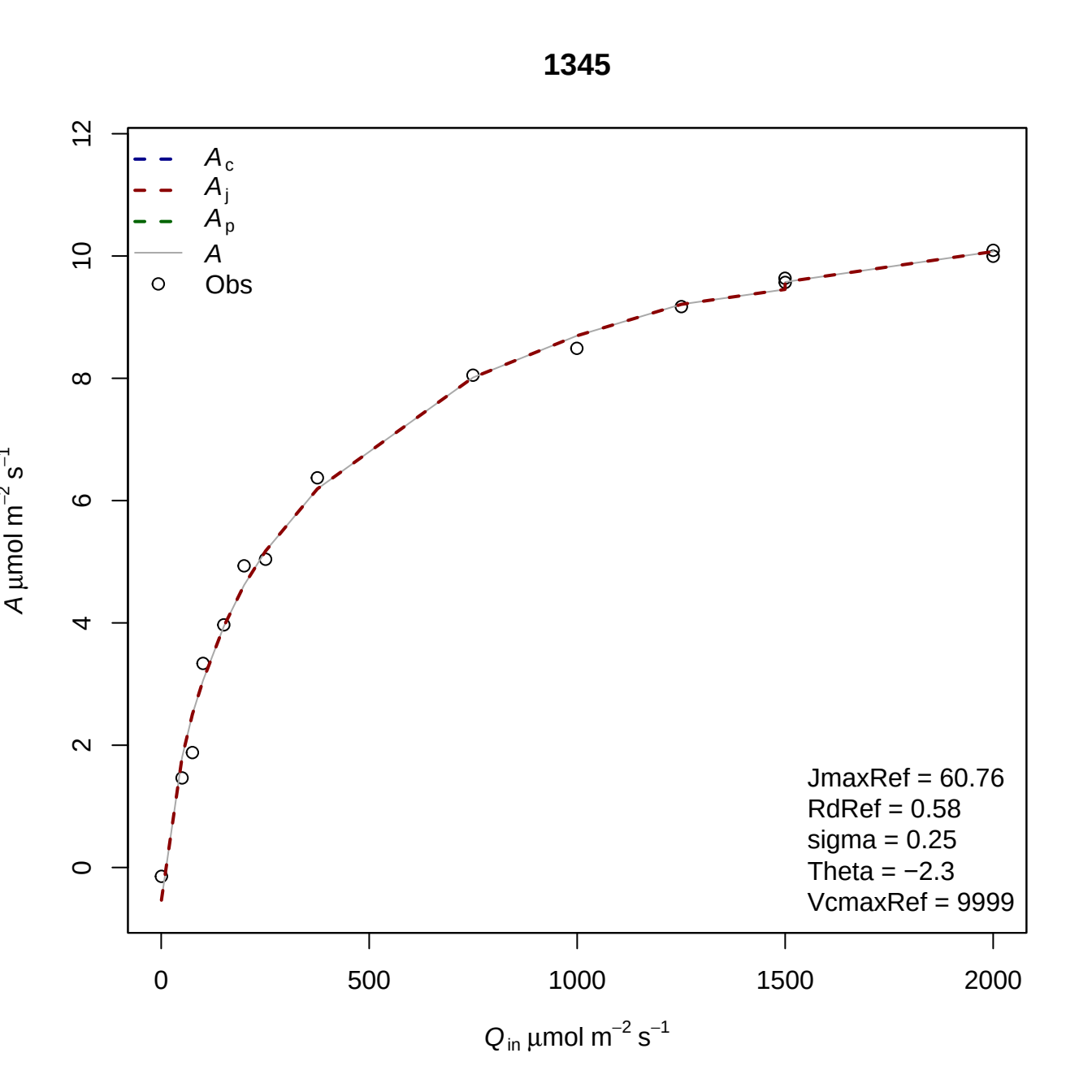

1340
A_c
A_j
A_p
A
Obs
JmaxRef = 76.01
RdRef = 1.15
sigma = 0.45
Theta = −1.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

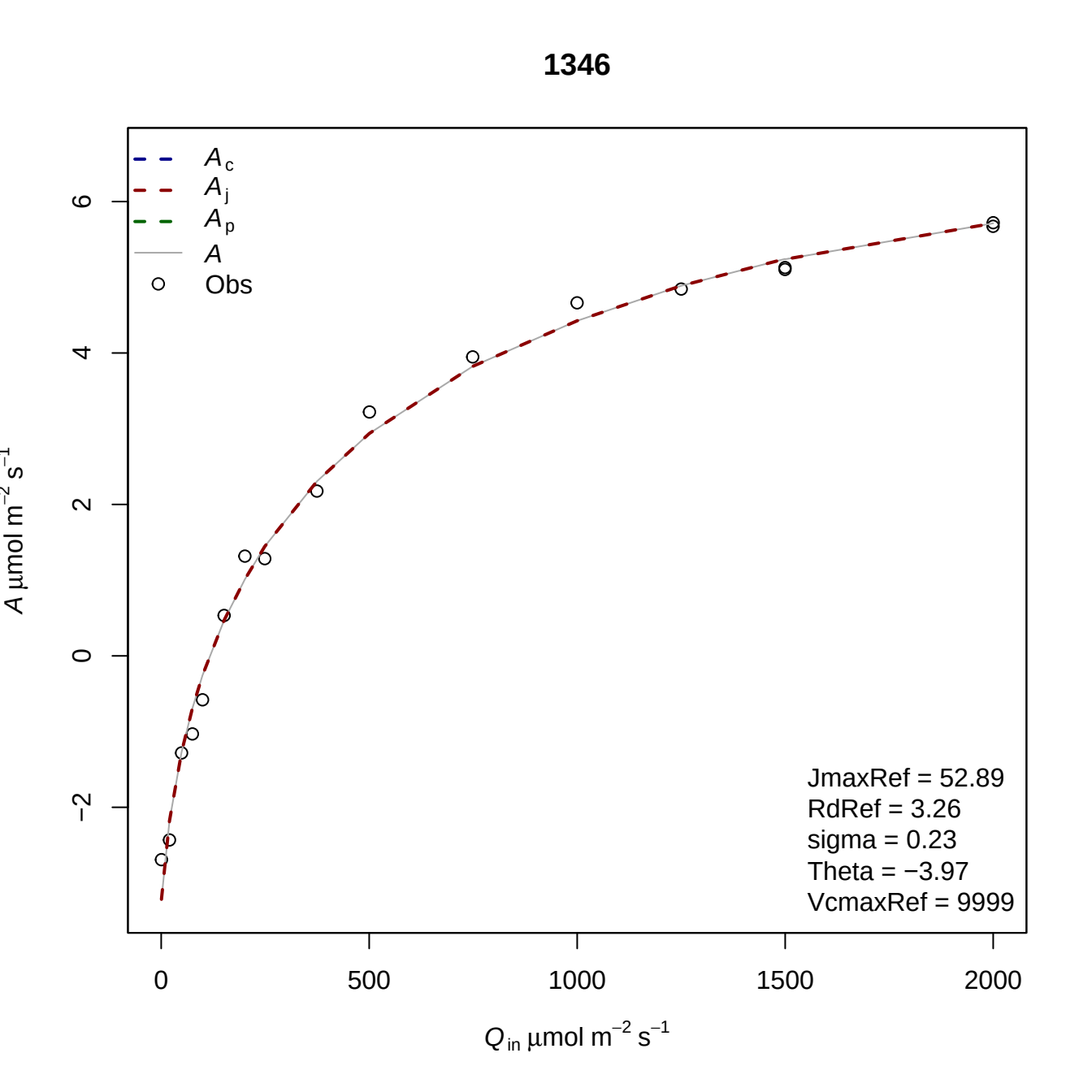

1341
A_c
A_j
A_p
A
Obs
JmaxRef = 75.5
RdRef = 3.03
sigma = 0.45
Theta = −0.29
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

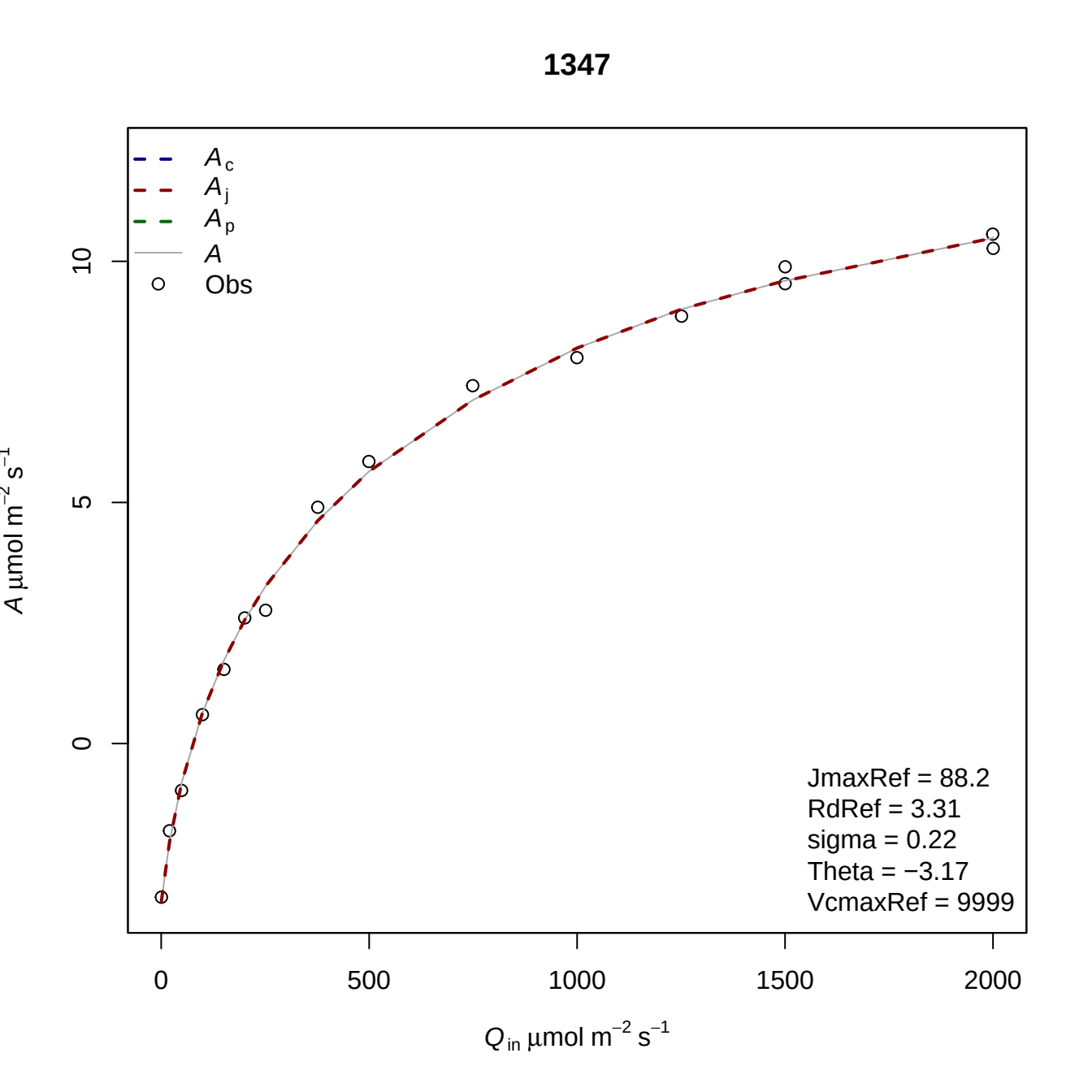

1342
A_c
A_j
A_p
A
Obs
JmaxRef = 73.44
RdRef = 1.65
sigma = 0.47
Theta = −1.88
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

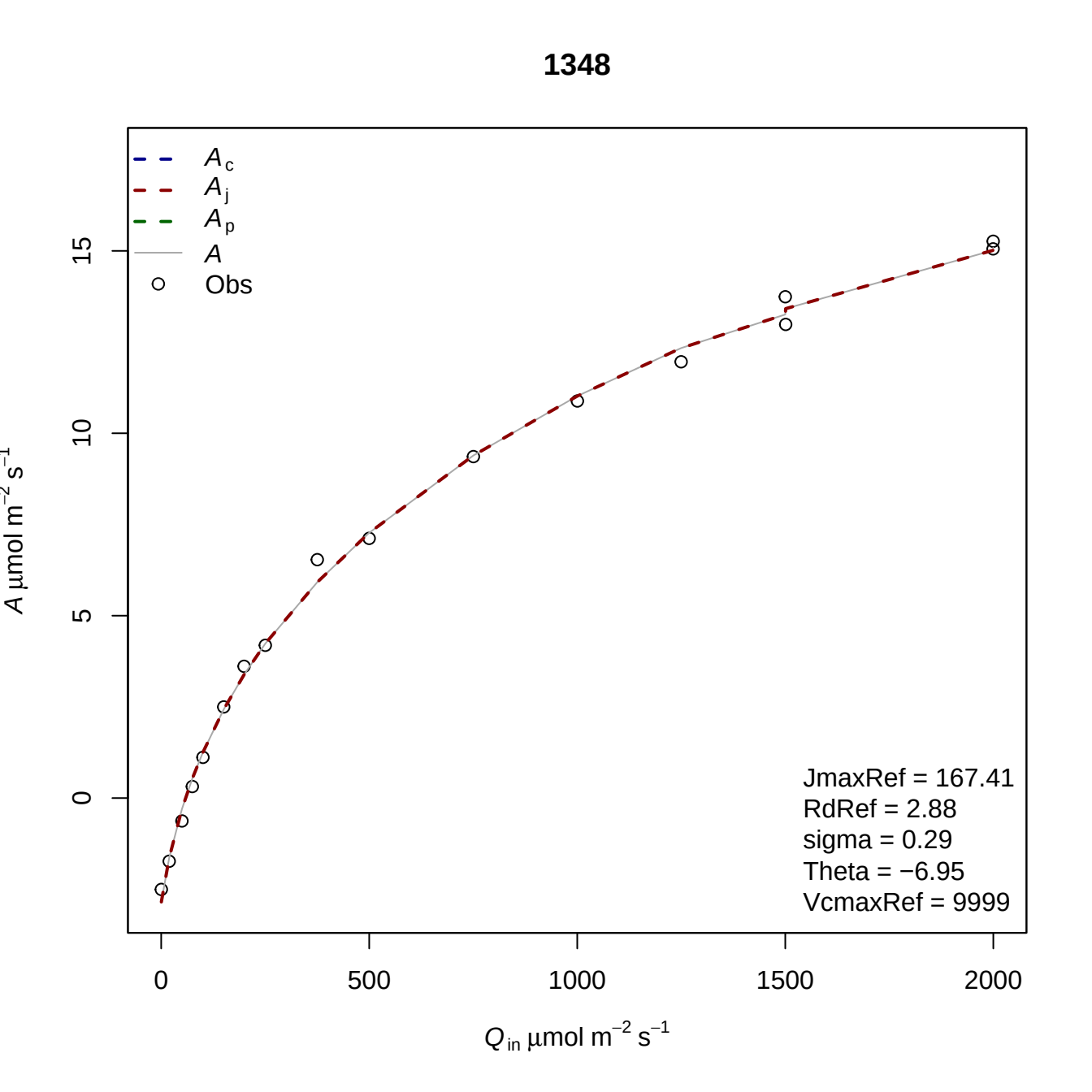

1343
A_c
A_j
A_p
A
Obs
JmaxRef = 62.41
RdRef = 2.78
sigma = 0.24
Theta = −1.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

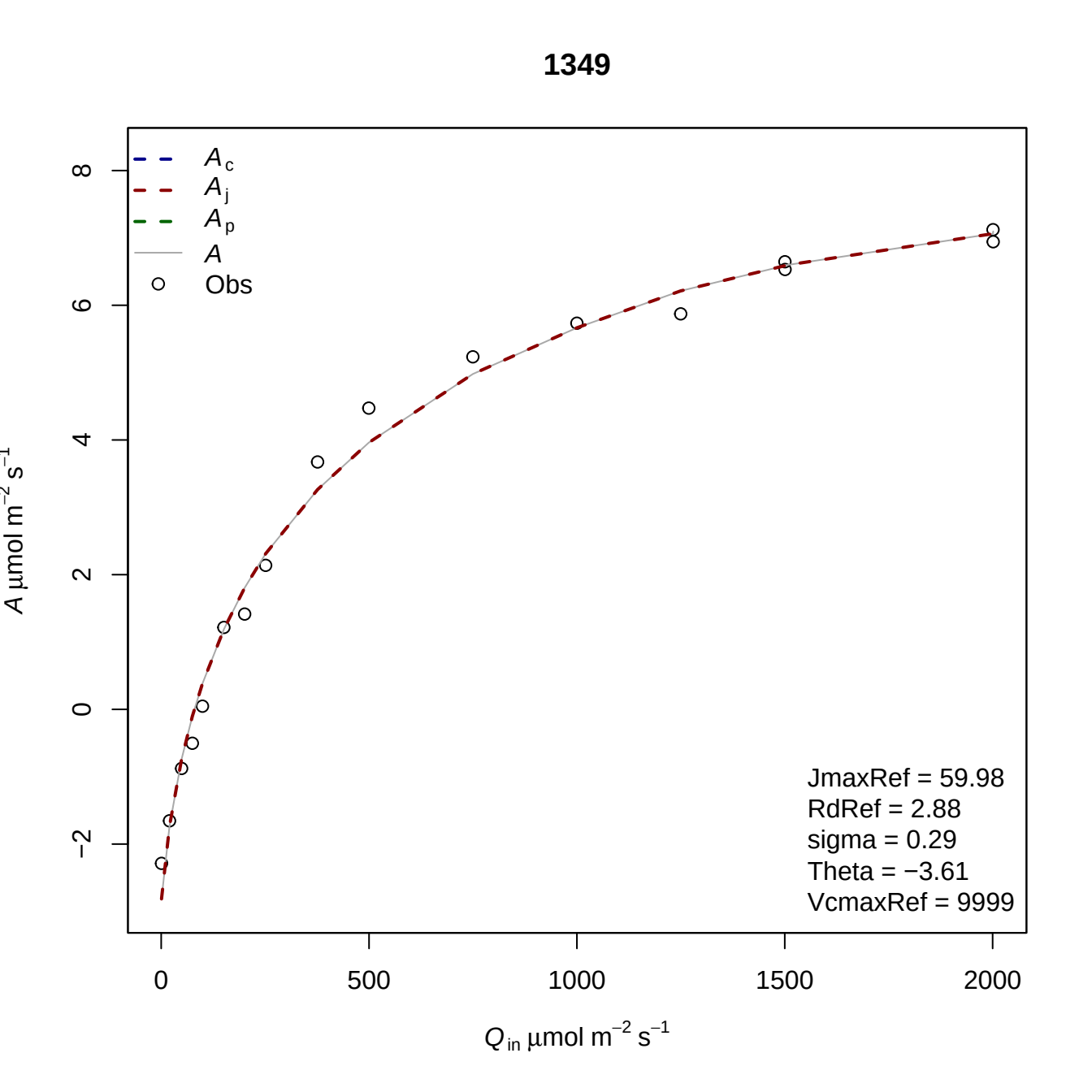

1344
A_c
A_j
A_p
A
Obs
JmaxRef = 55.71
RdRef = 1.09
sigma = 0.22
Theta = −1.89
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

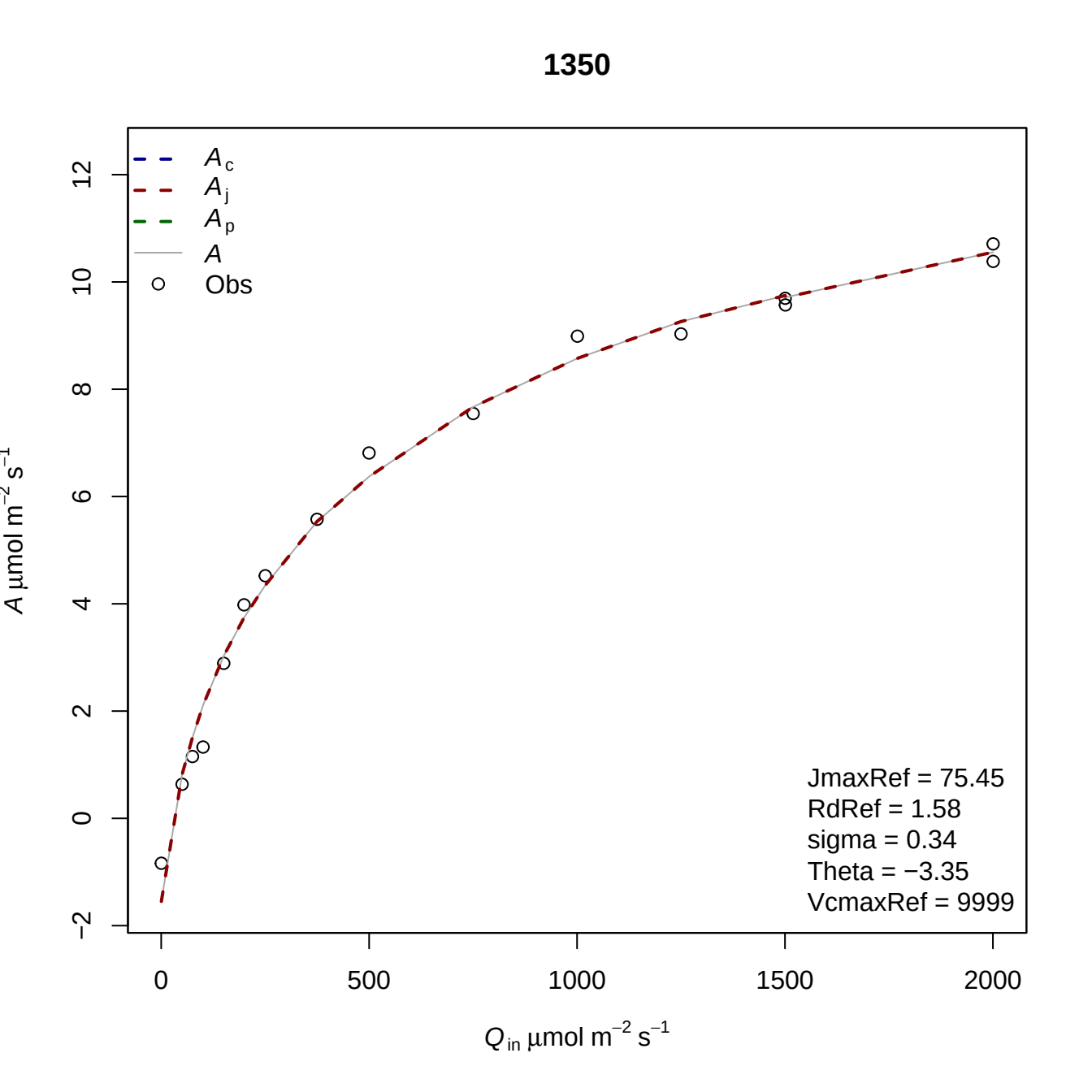

1345
A_c
A_j
A_p
A
Obs
JmaxRef = 60.76
RdRef = 0.58
sigma = 0.25
Theta = −2.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

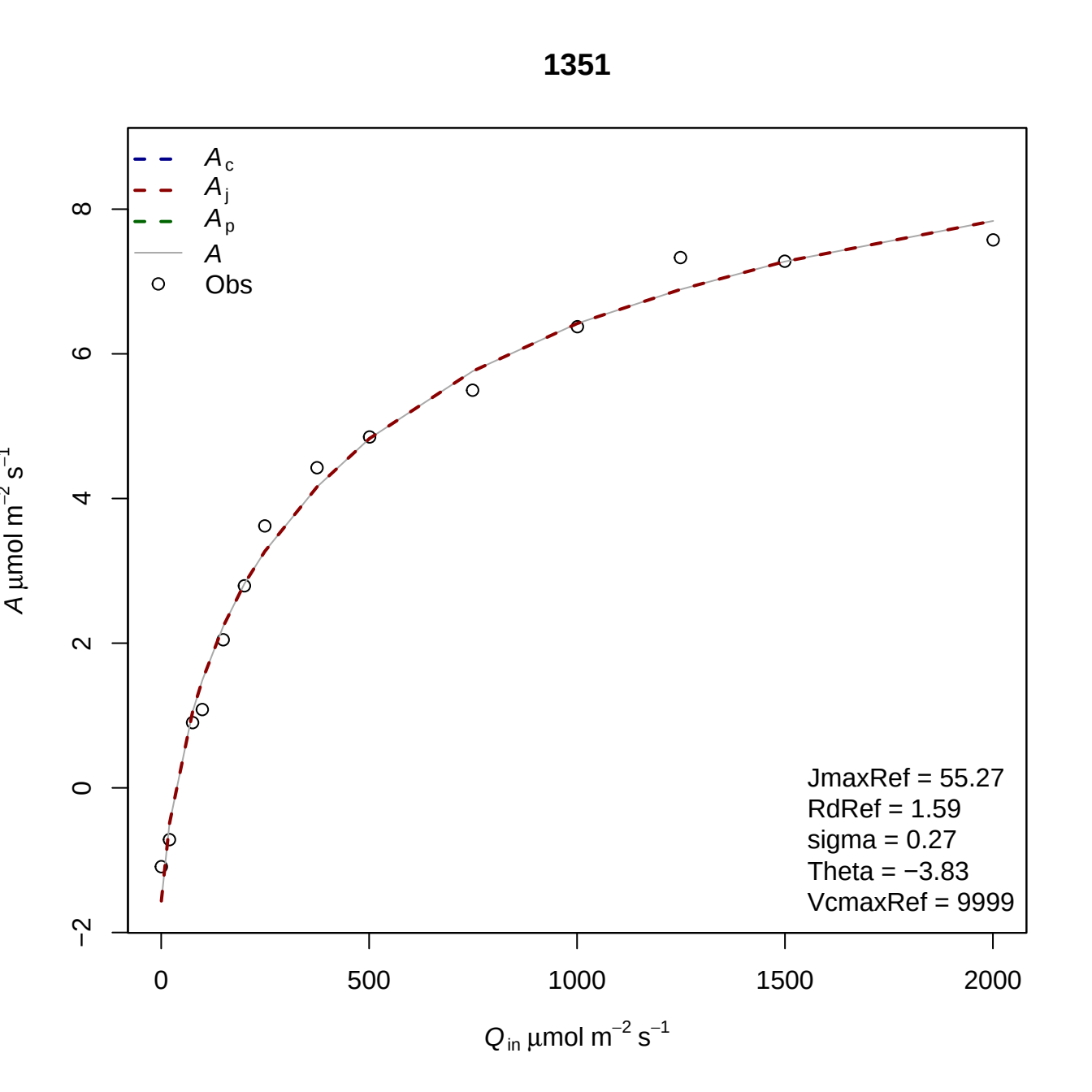

1346
A_c
A_j
A_p
A
Obs
JmaxRef = 52.89
RdRef = 3.26
sigma = 0.23
Theta = −3.97
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

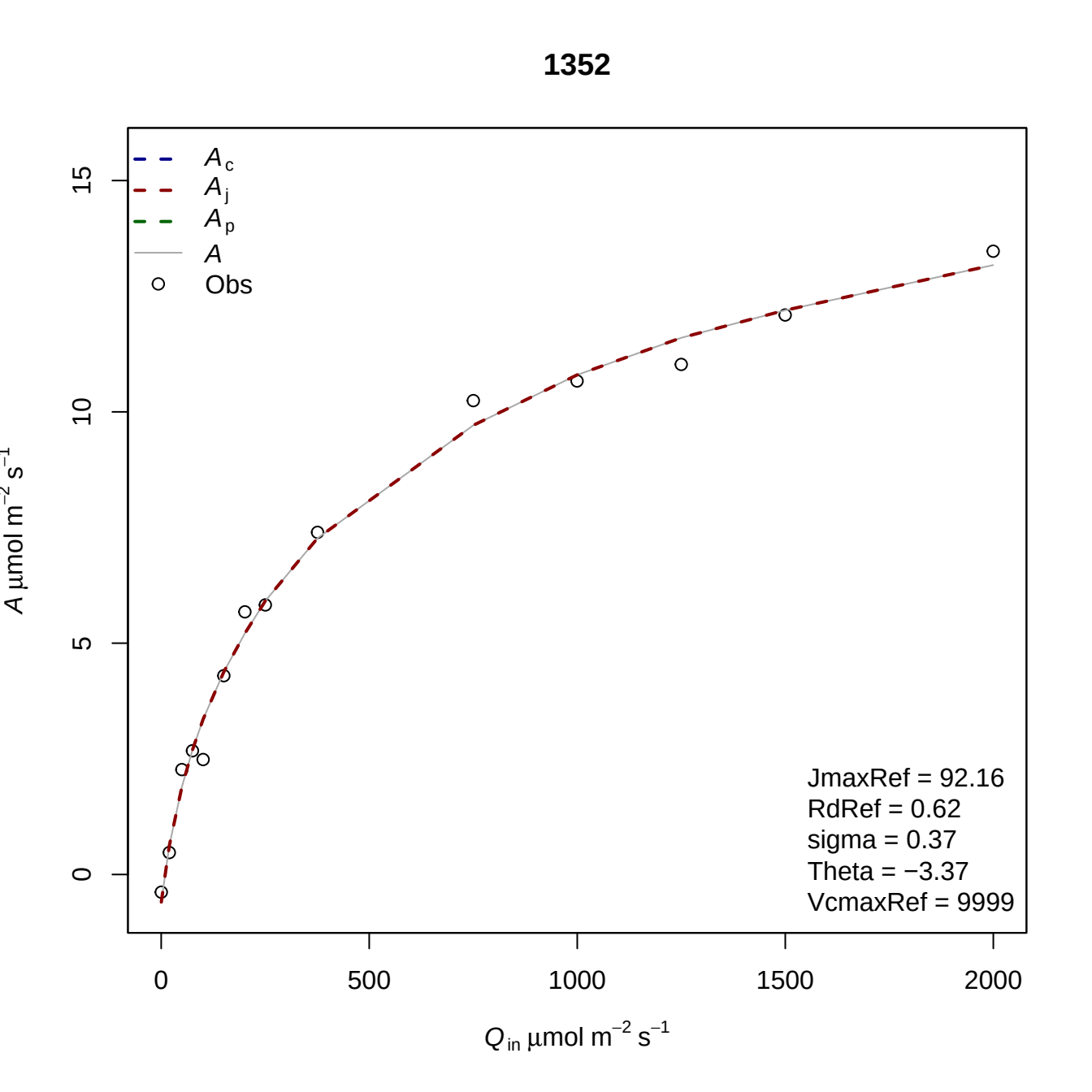

1347
A_c
A_j
A_p
A
Obs
JmaxRef = 88.2
RdRef = 3.31
sigma = 0.22
Theta = −3.17
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

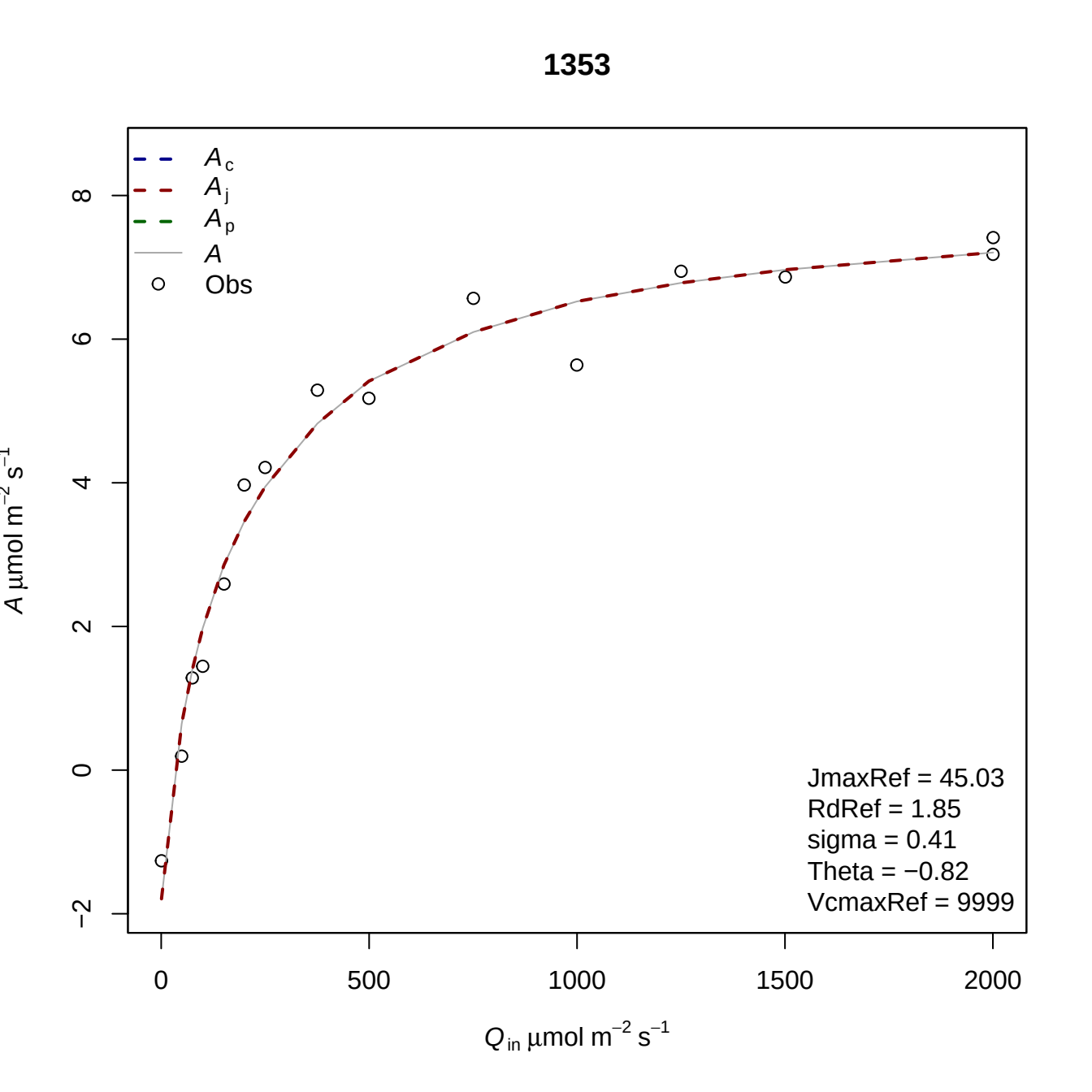

1348
A_c
A_j
A_p
A
Obs
JmaxRef = 167.41
RdRef = 2.88
sigma = 0.29
Theta = −6.95
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

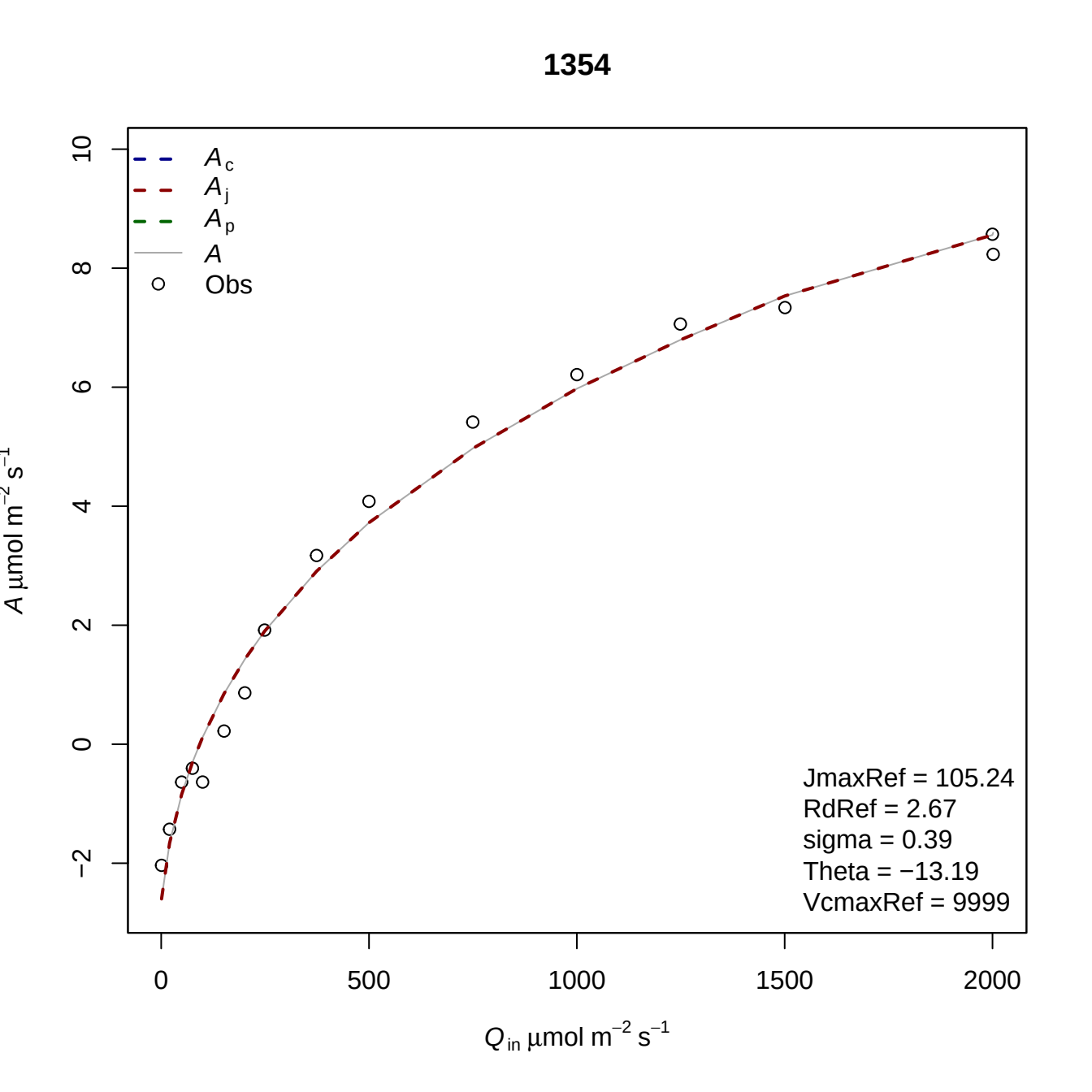

1349
A_c
A_j
A_p
A
Obs
JmaxRef = 59.98
RdRef = 2.88
sigma = 0.29
Theta = −3.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

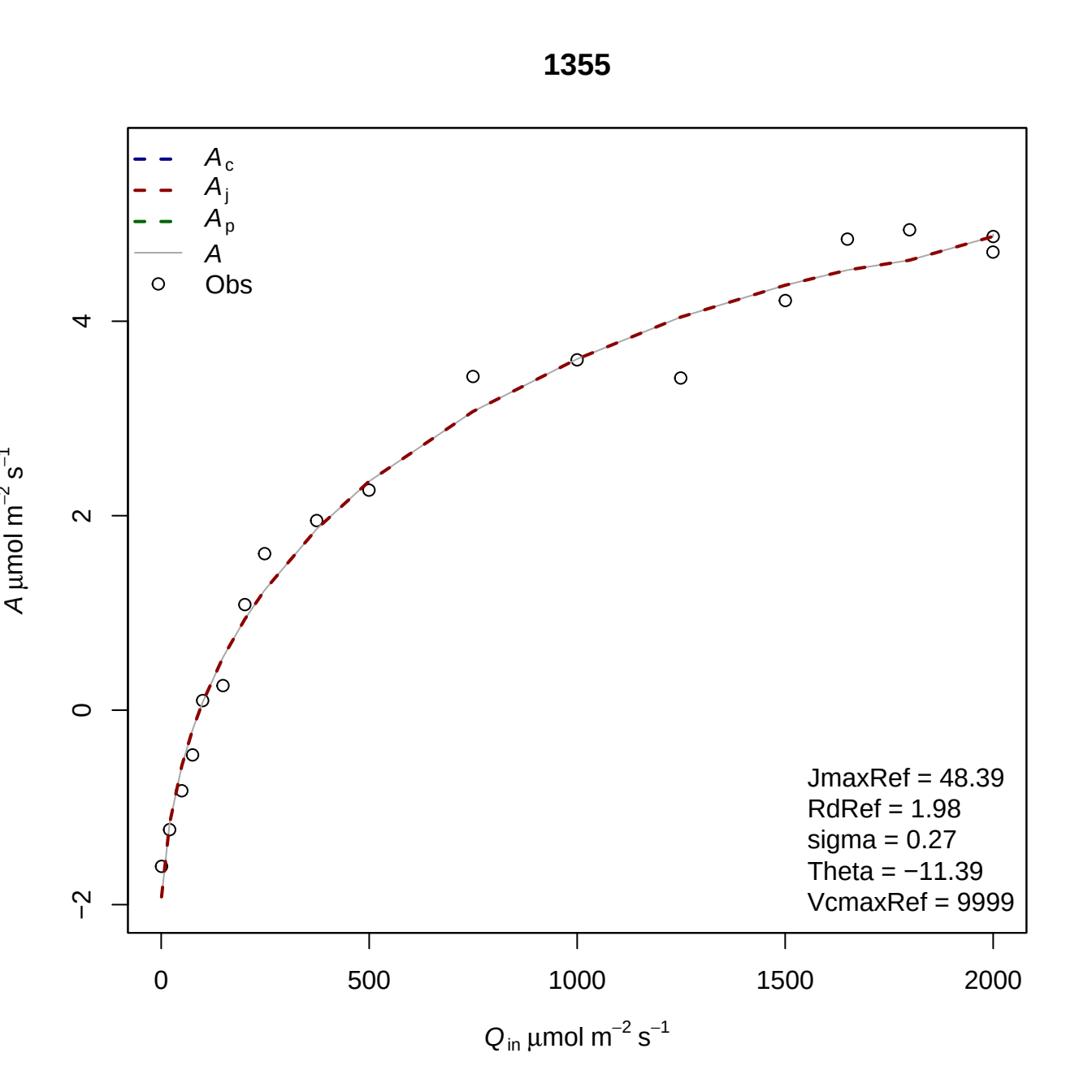

1350
A_c
A_j
A_p
A
Obs
JmaxRef = 75.45
RdRef = 1.58
sigma = 0.34
Theta = −3.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

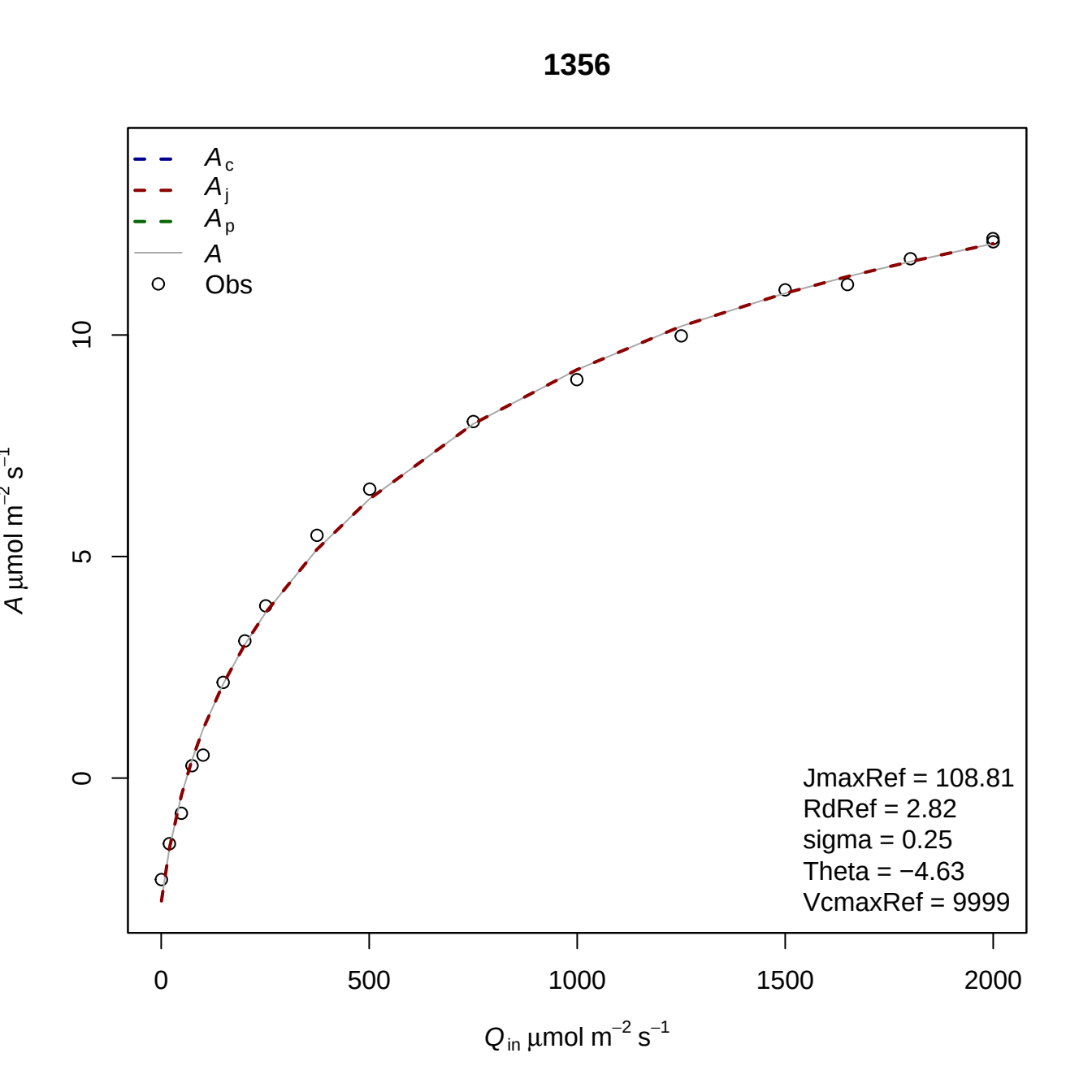

1351
A_c
A_j
A_p
A
Obs
JmaxRef = 55.27
RdRef = 1.59
sigma = 0.27
Theta = −3.83
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

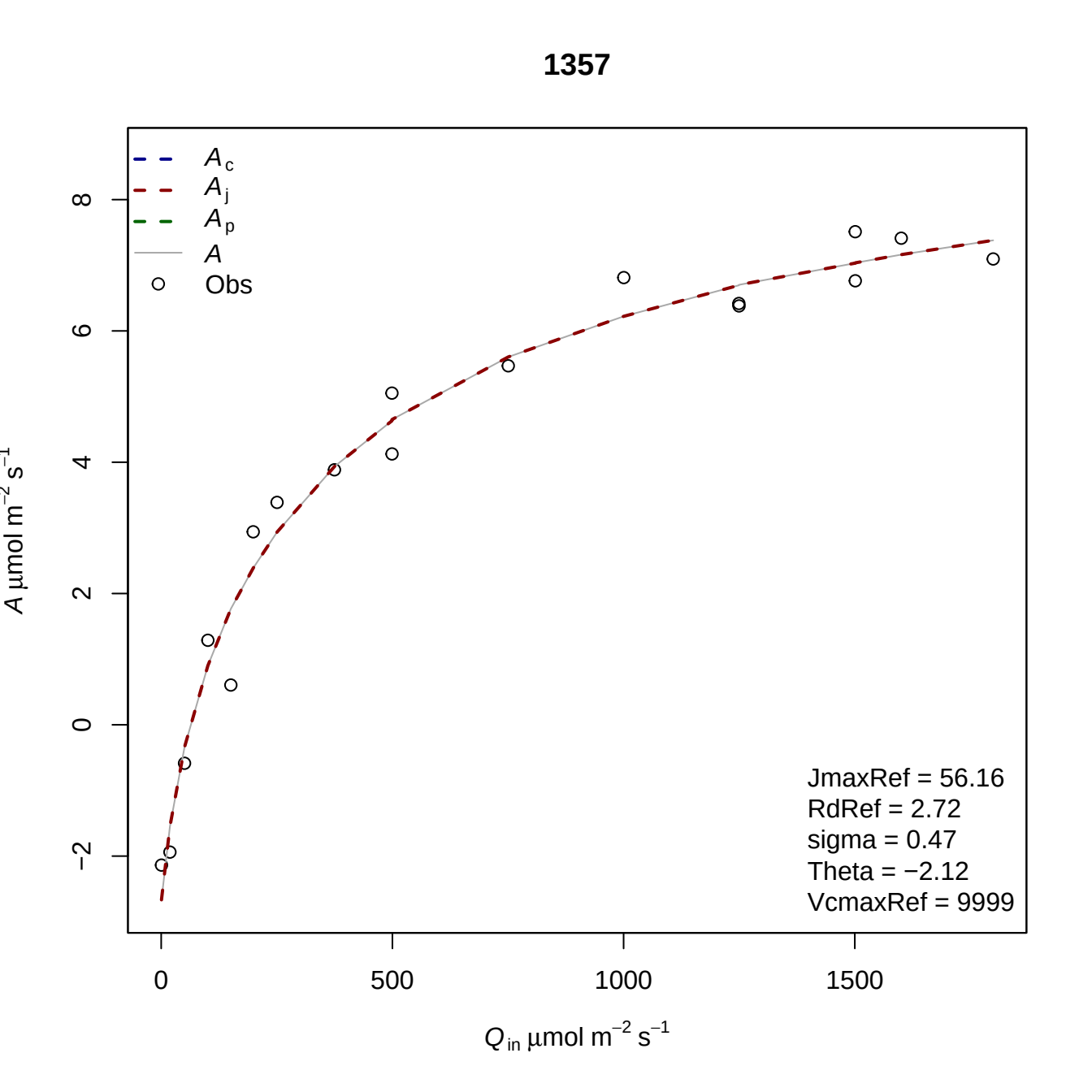

1352
A_c
A_j
A_p
A
Obs
JmaxRef = 92.16
RdRef = 0.62
sigma = 0.37
Theta = −3.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

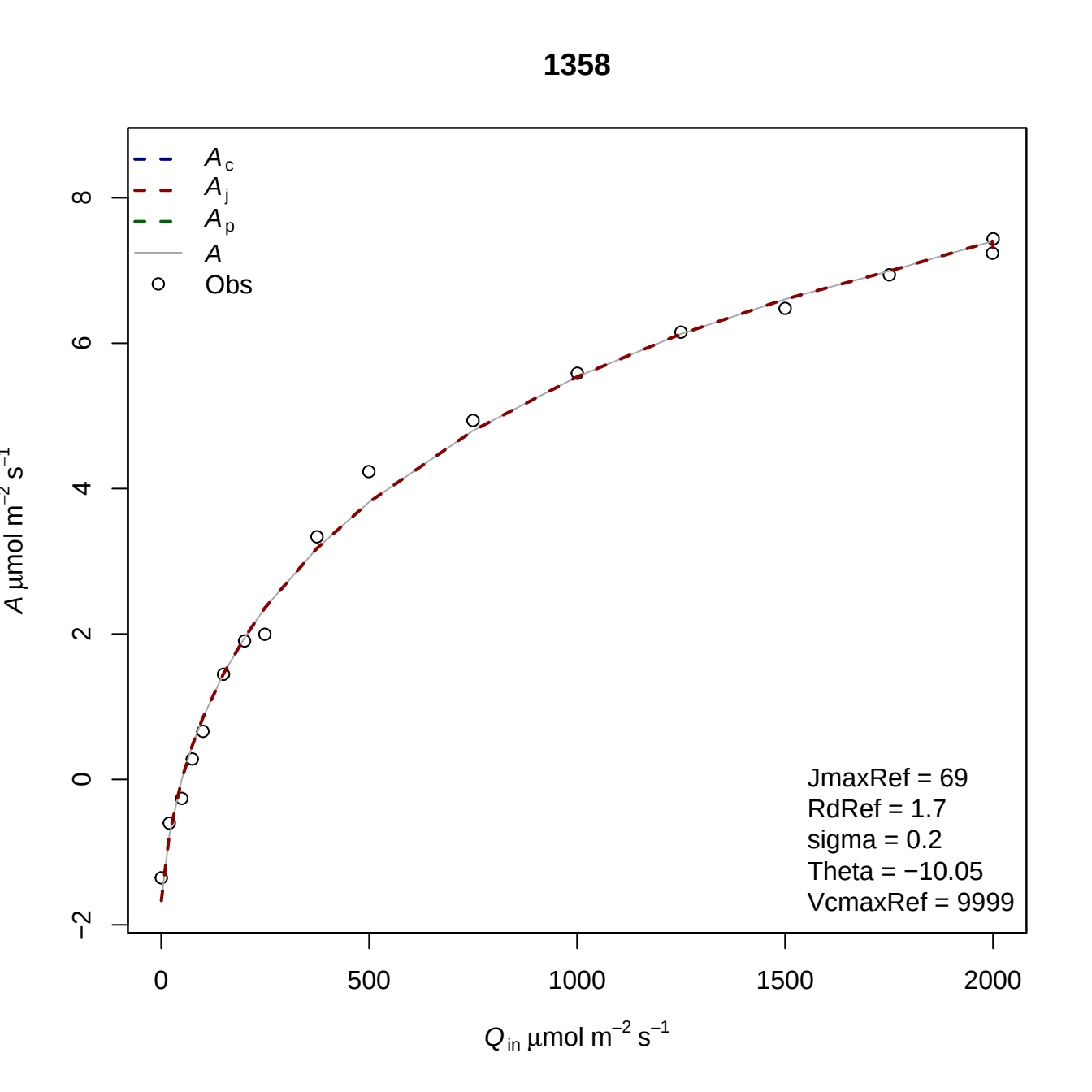

1353
A_c
A_j
A_p
A
Obs
JmaxRef = 45.03
RdRef = 1.85
sigma = 0.41
Theta = −0.82
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

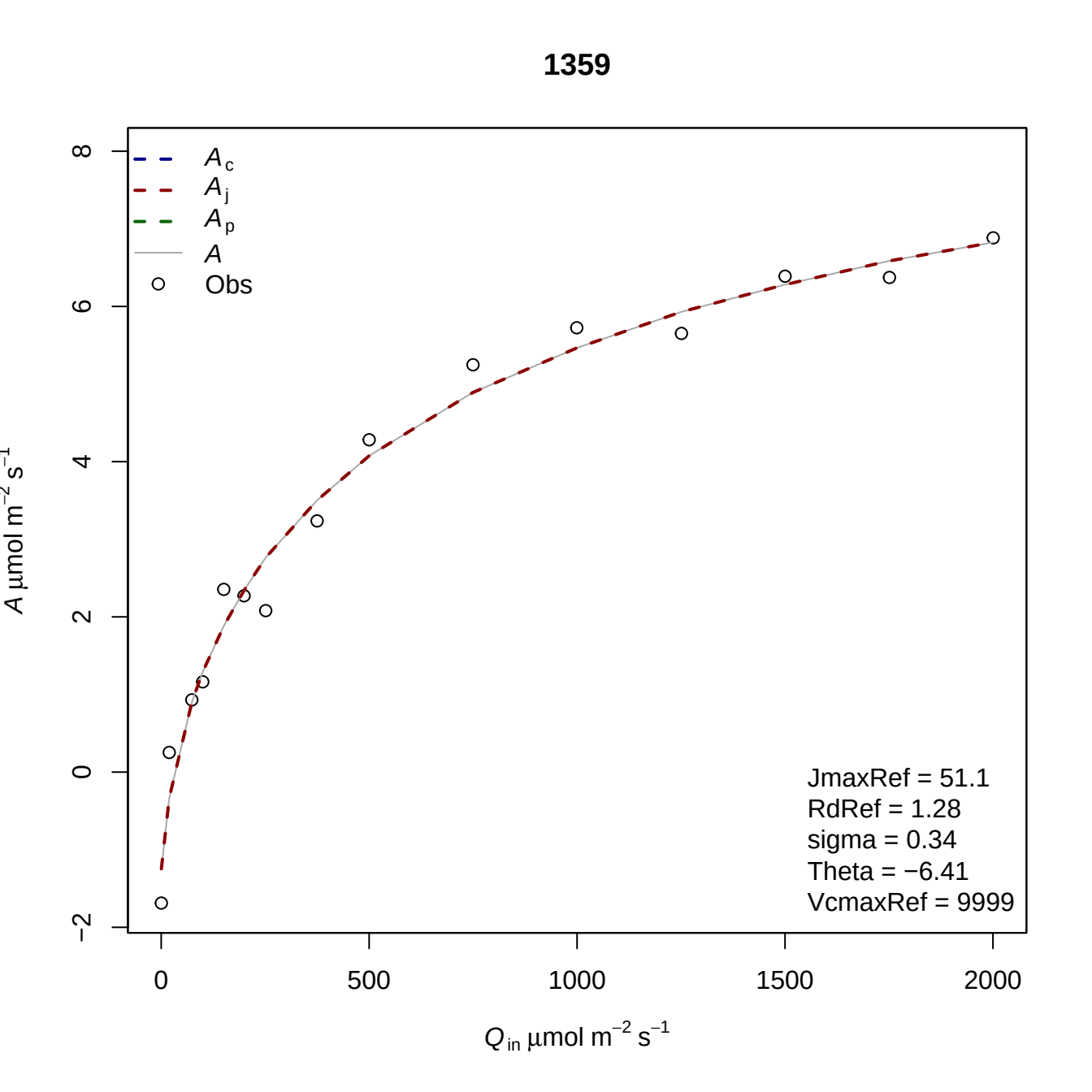

1354
A_c
A_j
A_p
A
Obs
JmaxRef = 105.24
RdRef = 2.67
sigma = 0.39
Theta = −13.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

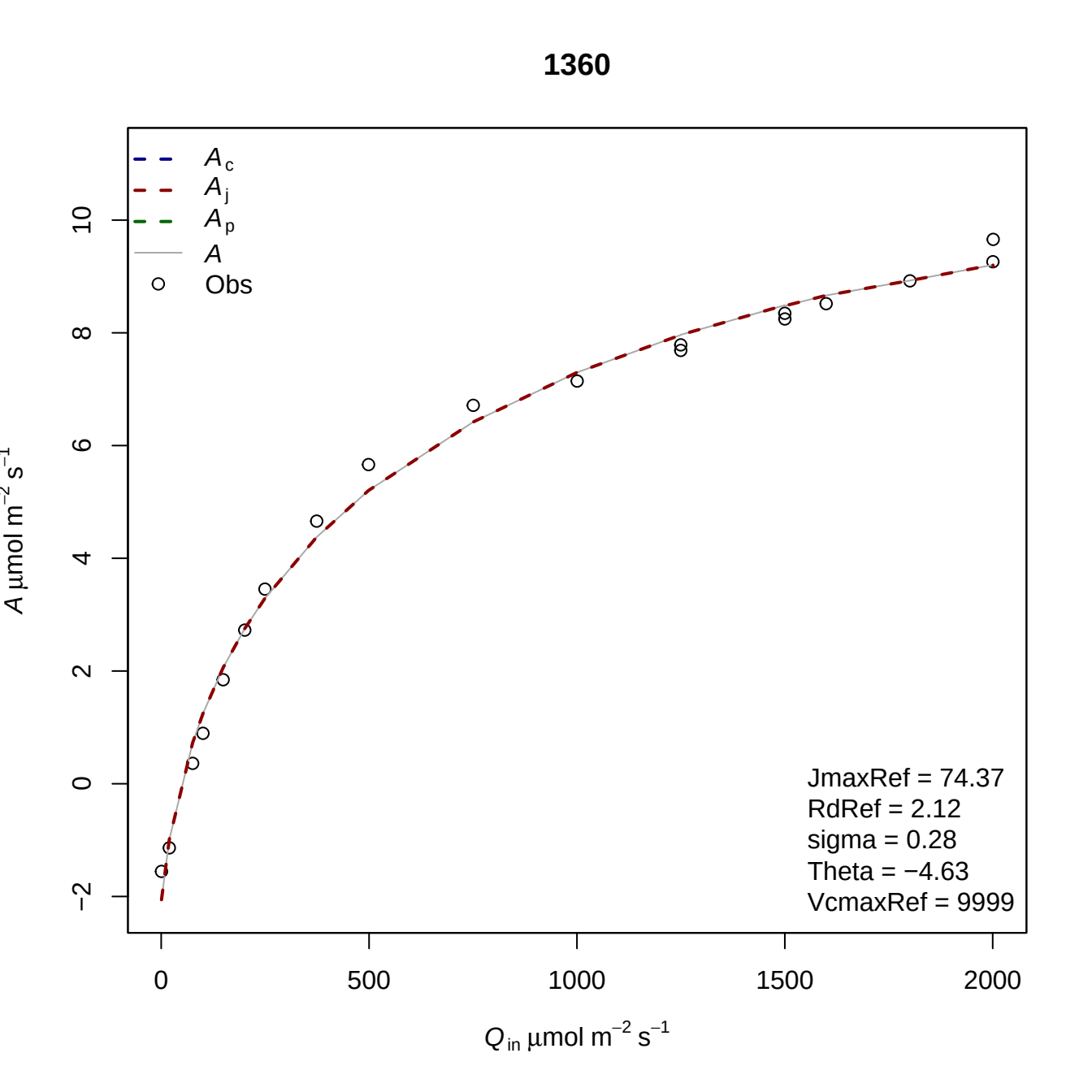

1355
A_c
A_j
A_p
A
Obs
JmaxRef = 48.39
RdRef = 1.98
sigma = 0.27
Theta = −11.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

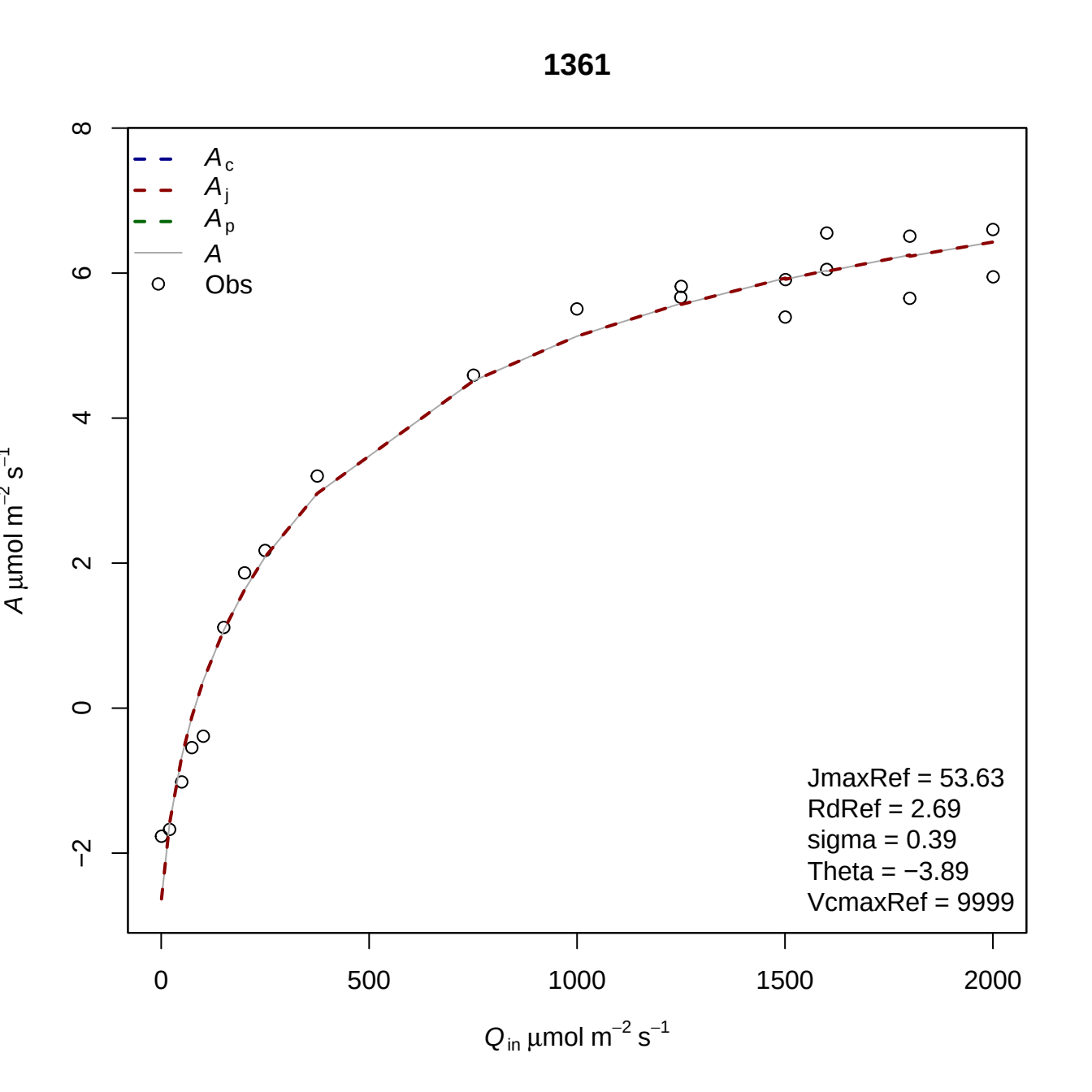

1356
A_c
A_j
A_p
A
Obs
JmaxRef = 108.81
RdRef = 2.82
sigma = 0.25
Theta = −4.63
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

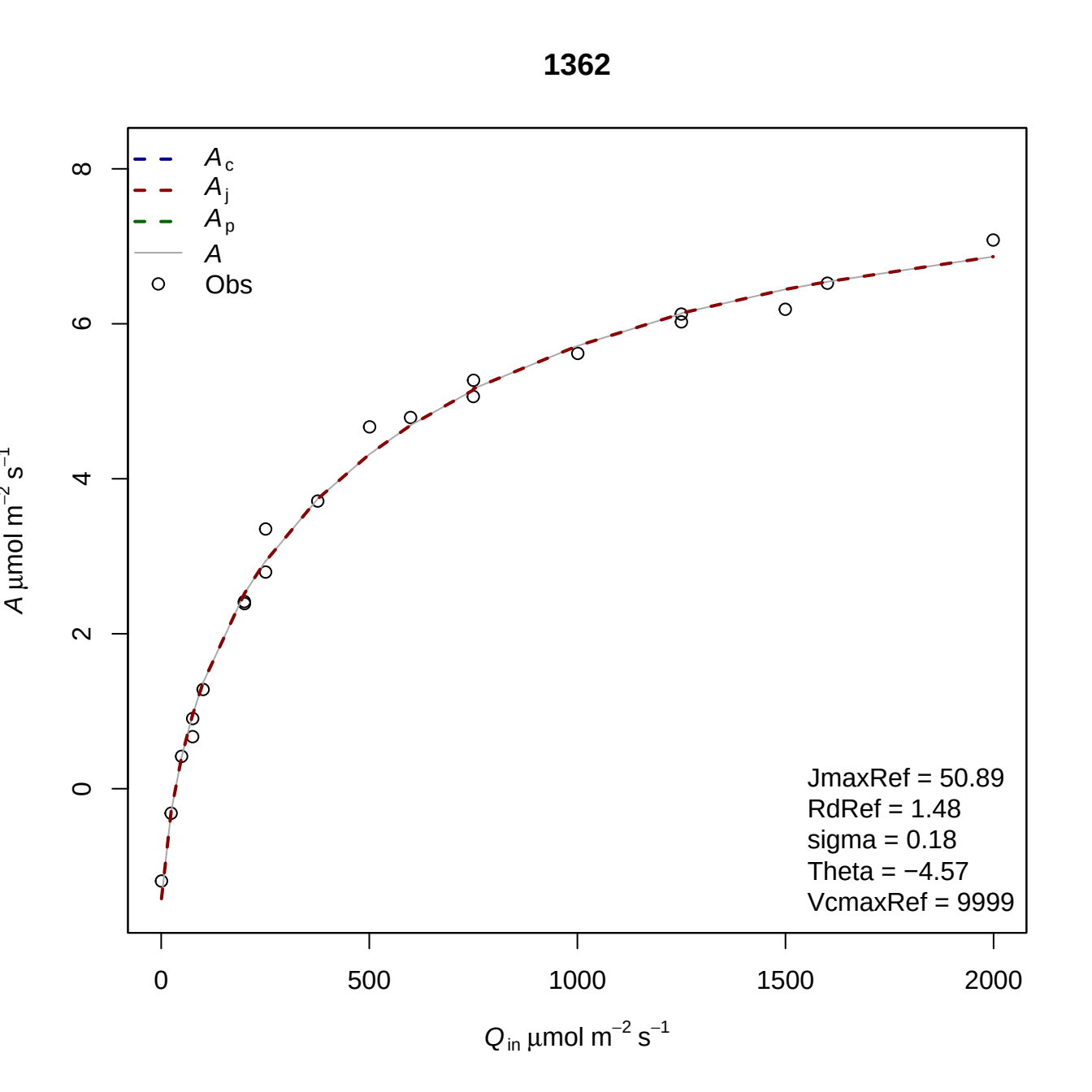

1357
A_c
A_j
A_p
A
Obs
JmaxRef = 56.16
RdRef = 2.72
sigma = 0.47
Theta = −2.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

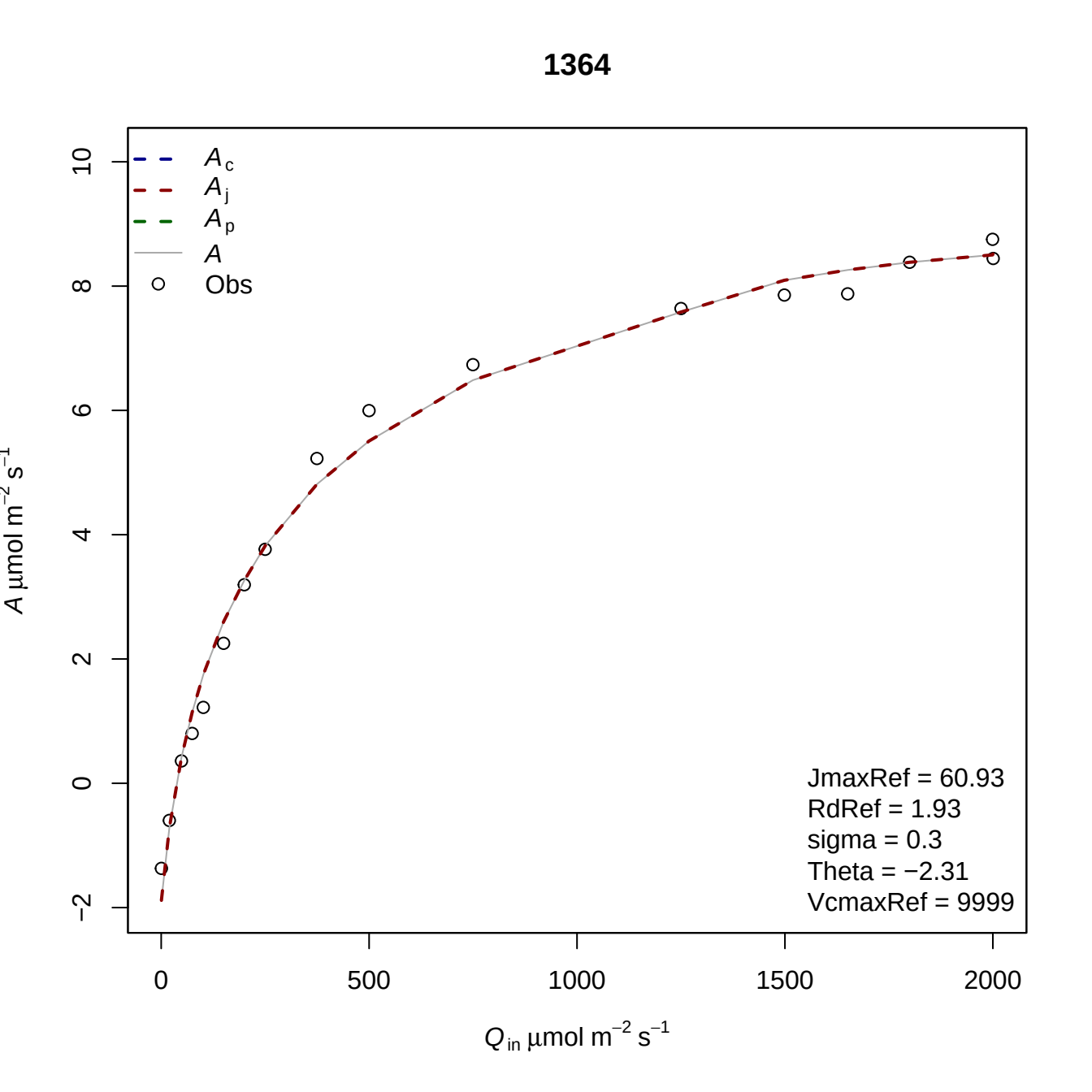

1358
A_c
A_j
A_p
A
Obs
JmaxRef = 69
RdRef = 1.7
sigma = 0.2
Theta = −10.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

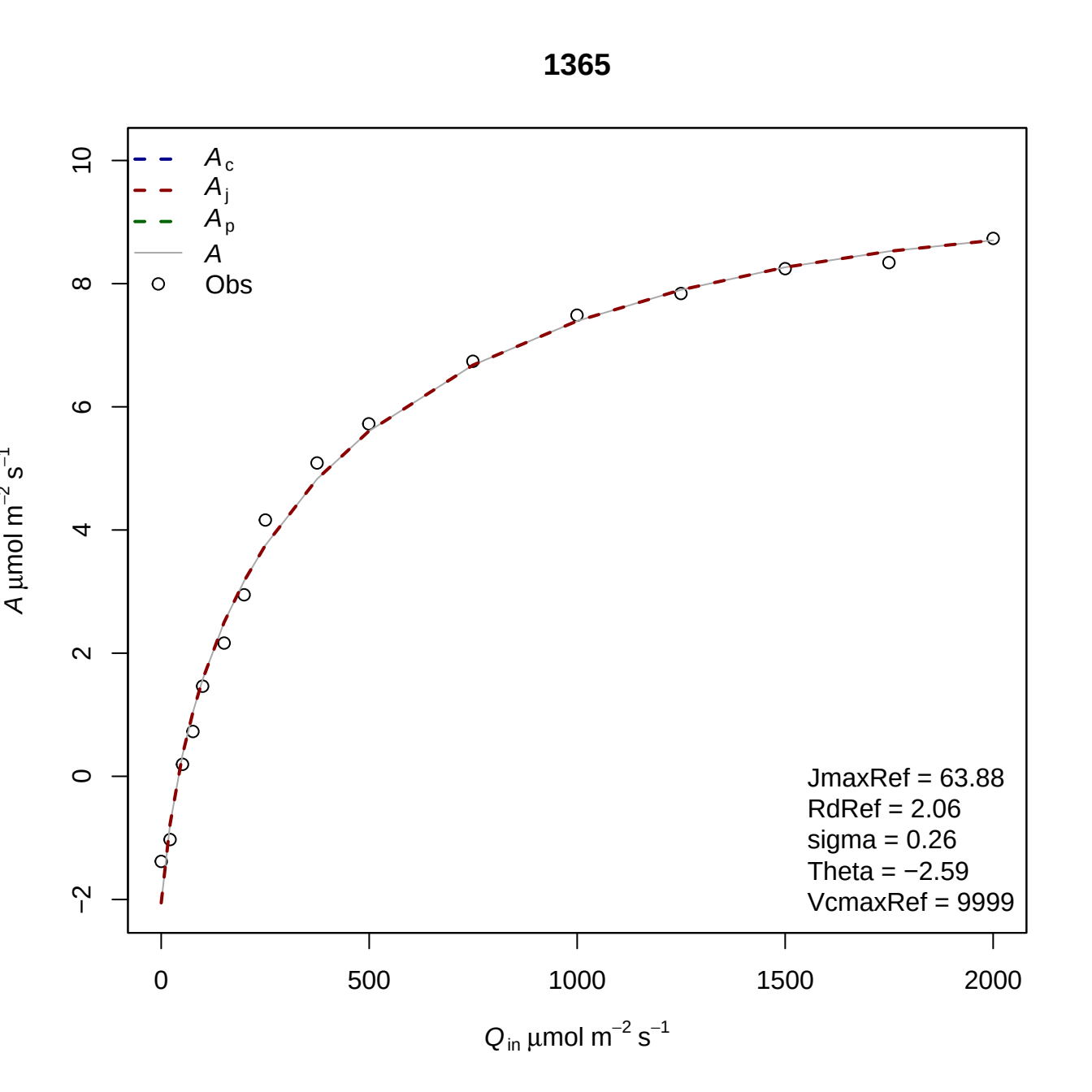

1359
A_c
A_j
A_p
A
Obs
JmaxRef = 51.1
RdRef = 1.28
sigma = 0.34
Theta = −6.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

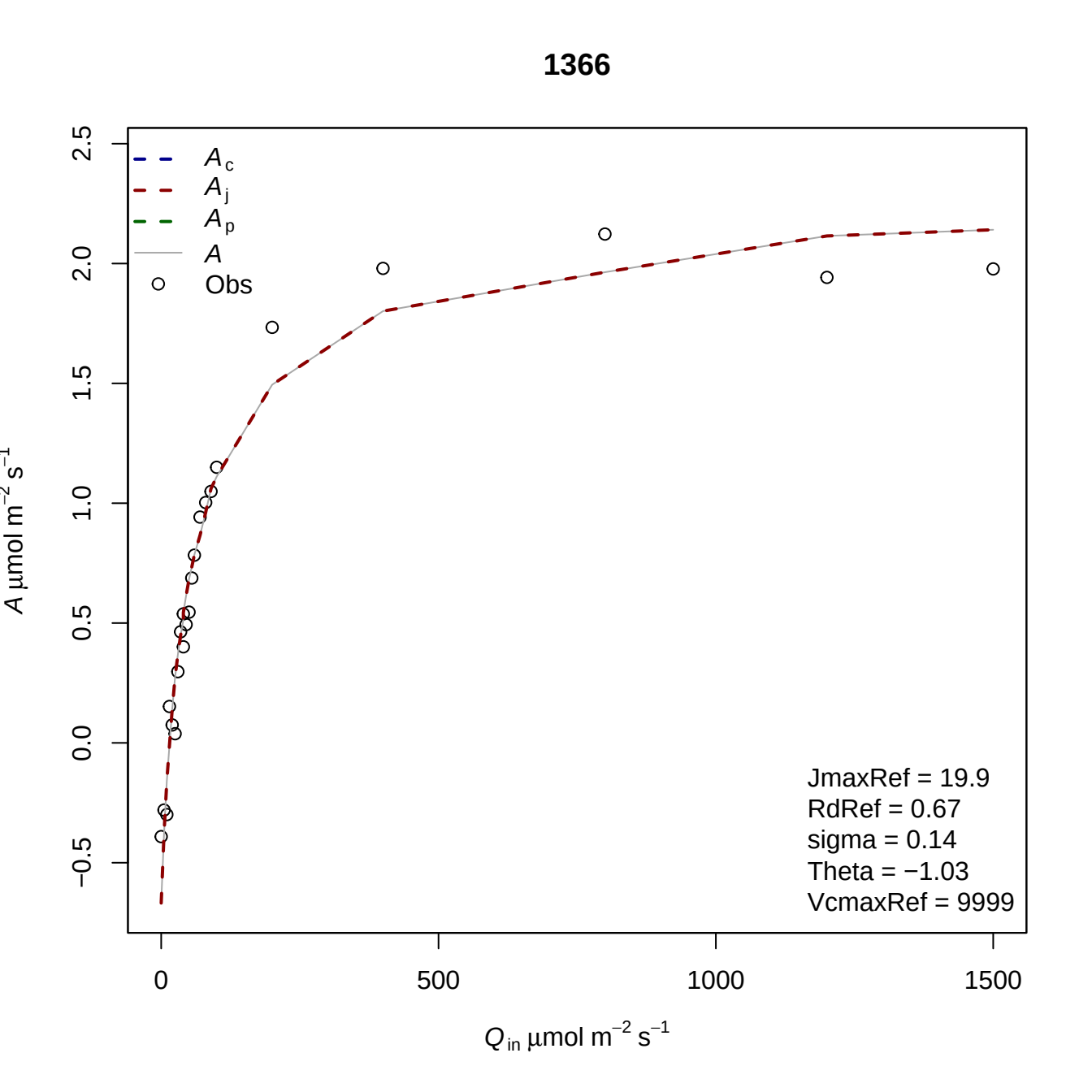

1360
A_c
A_j
A_p
A
Obs
JmaxRef = 74.37
RdRef = 2.12
sigma = 0.28
Theta = −4.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

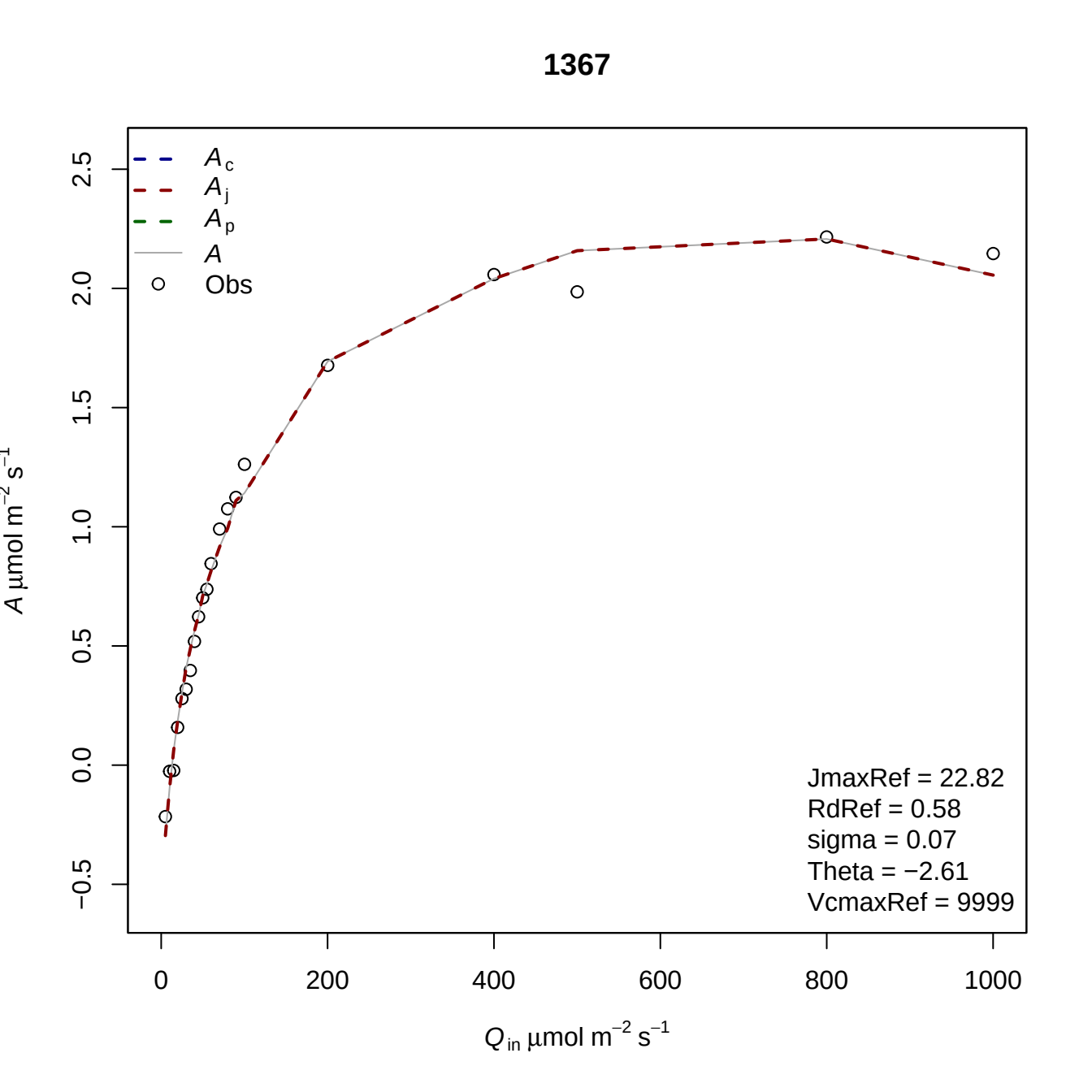

1361
A_c
A_j
A_p
A
Obs
JmaxRef = 53.63
RdRef = 2.69
sigma = 0.39
Theta = −3.89
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

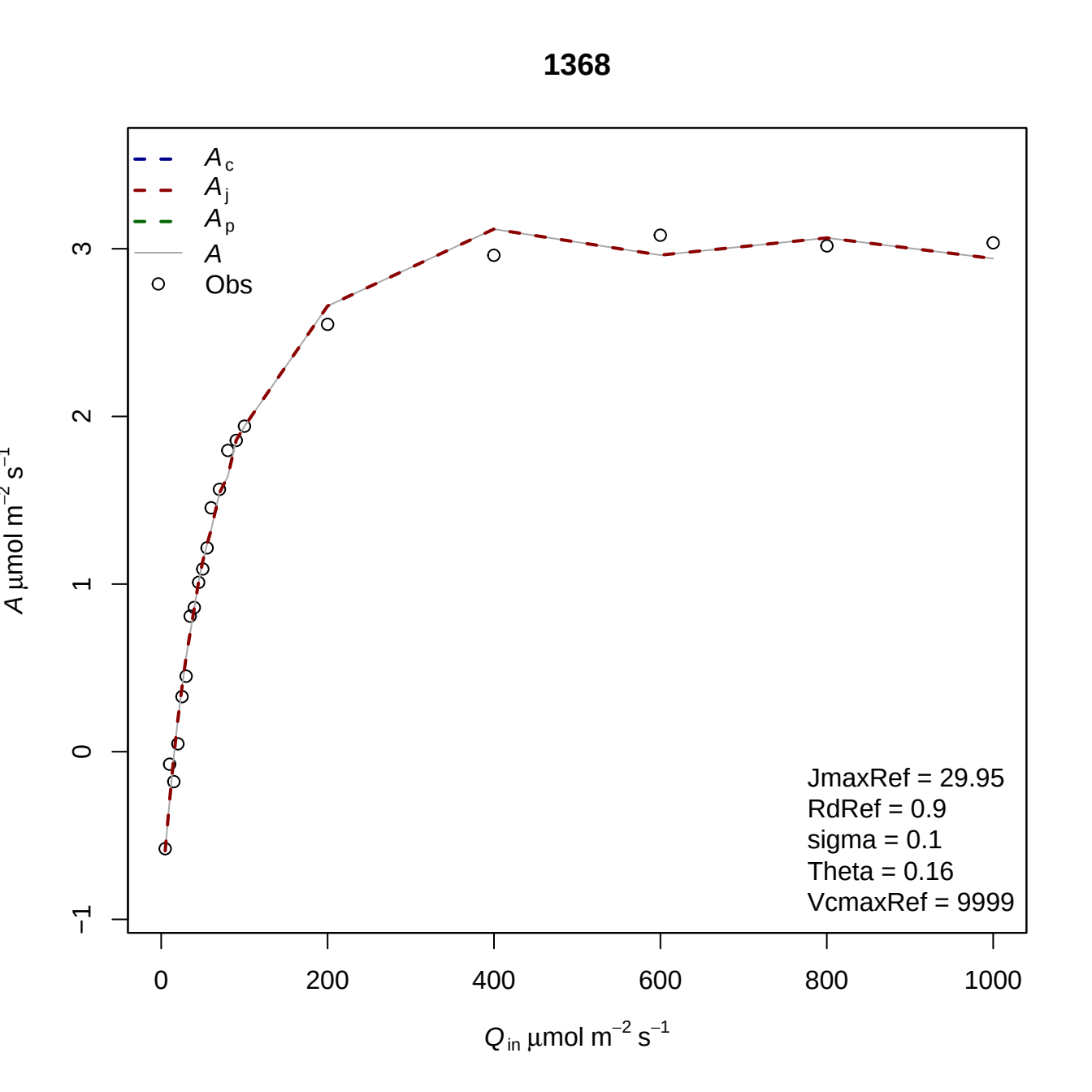

1362
A_c
A_j
A_p
A
Obs
JmaxRef = 50.89
RdRef = 1.48
sigma = 0.18
Theta = −4.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

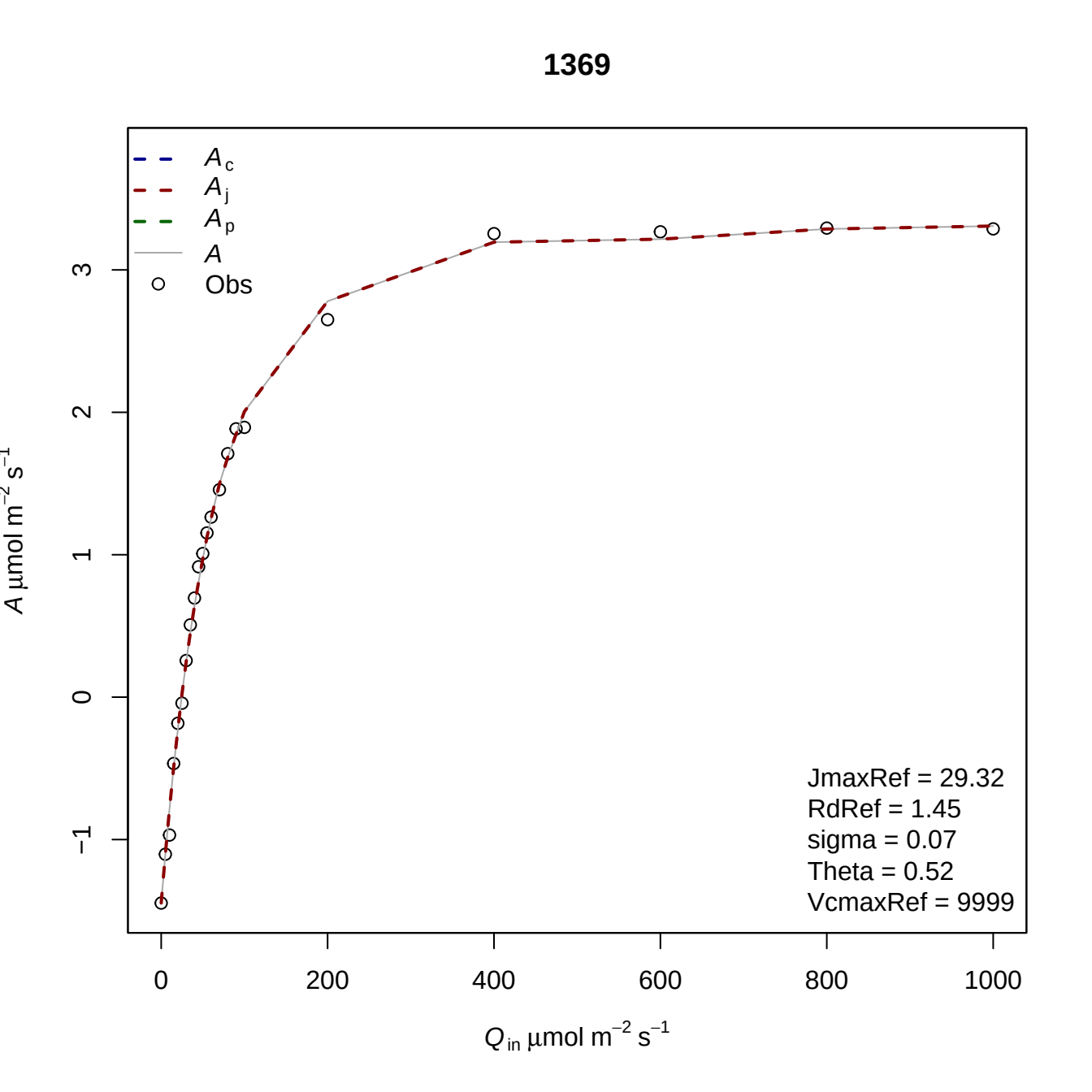

1364
A_c
A_j
A_p
A
Obs
JmaxRef = 60.93
RdRef = 1.93
sigma = 0.3
Theta = −2.31
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

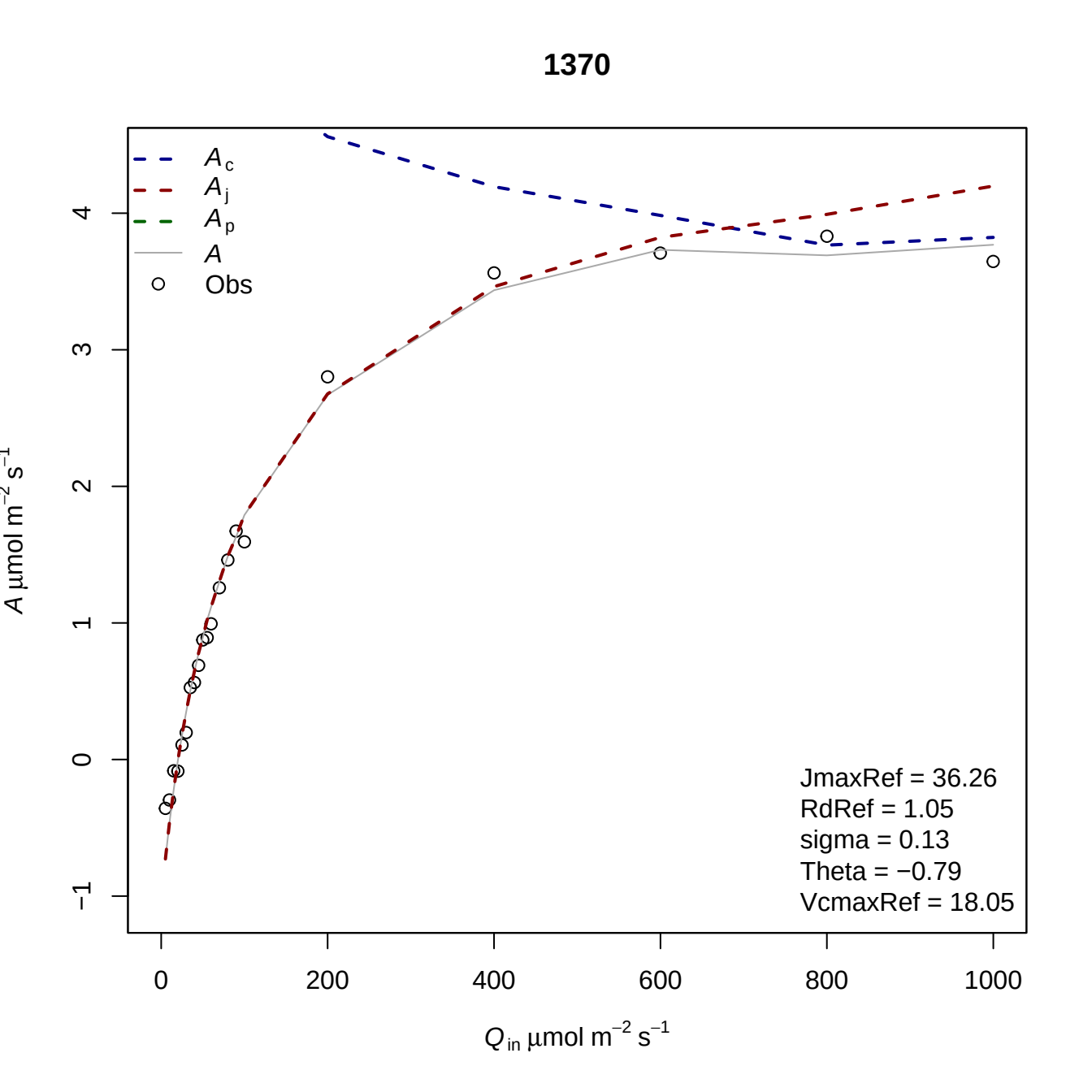

1365
A_c
A_j
A_p
A
Obs
JmaxRef = 63.88
RdRef = 2.06
sigma = 0.26
Theta = −2.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

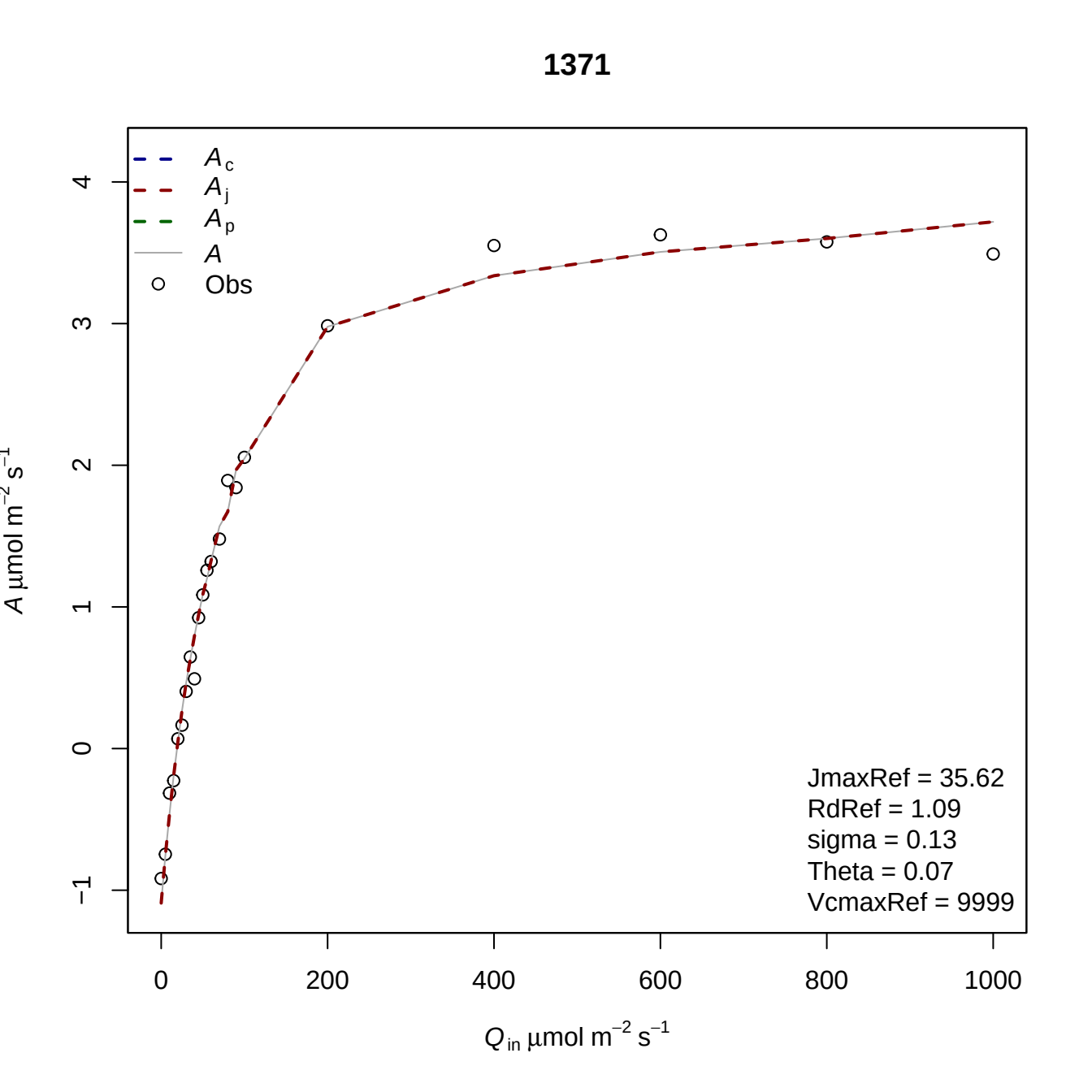

1366
A_c
A_j
A_p
A
Obs
JmaxRef = 19.9
RdRef = 0.67
sigma = 0.14
Theta = −1.03
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

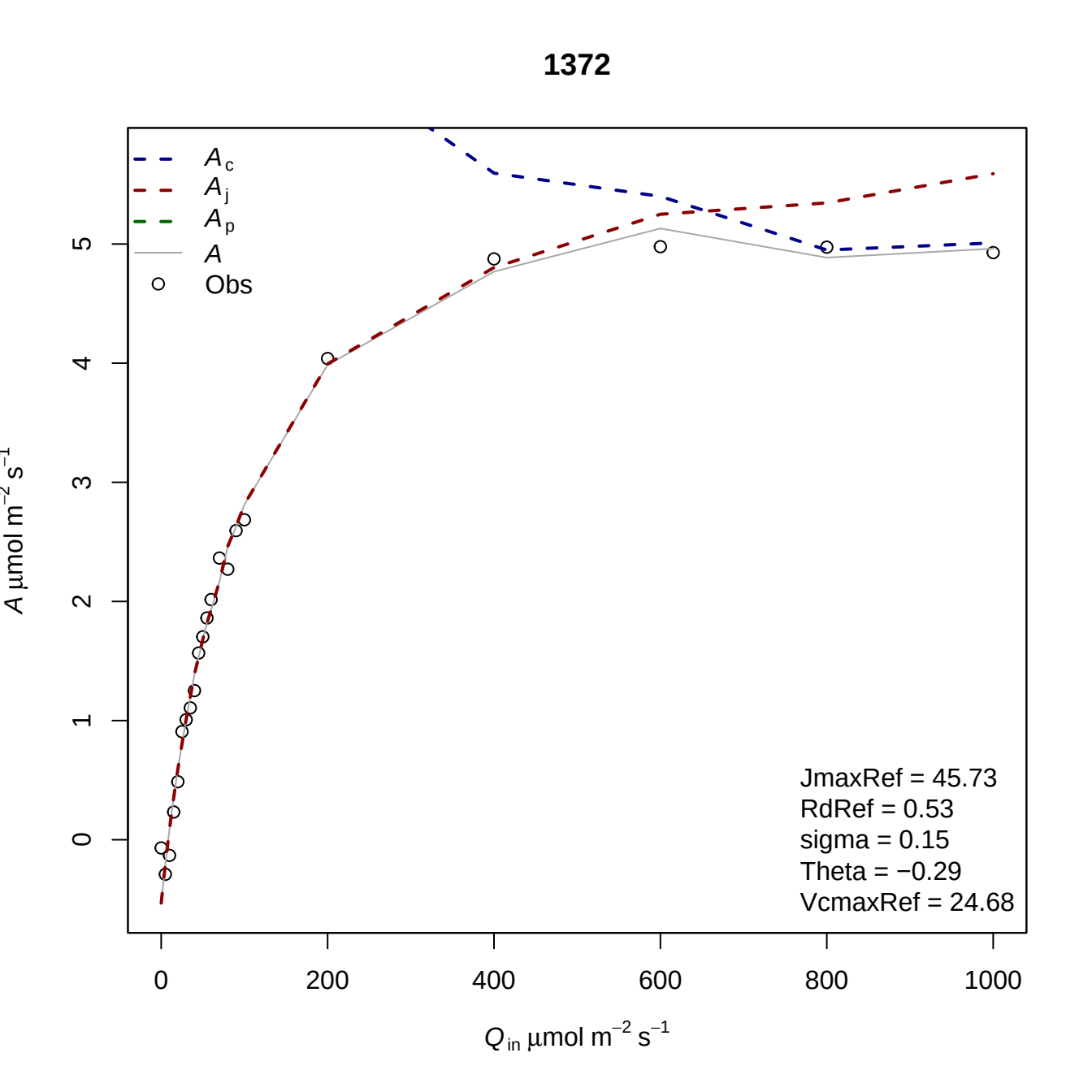

1367
A_c
A_j
A_p
A
Obs
JmaxRef = 22.82
RdRef = 0.58
sigma = 0.07
Theta = −2.61
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
200
400
600
800
1000

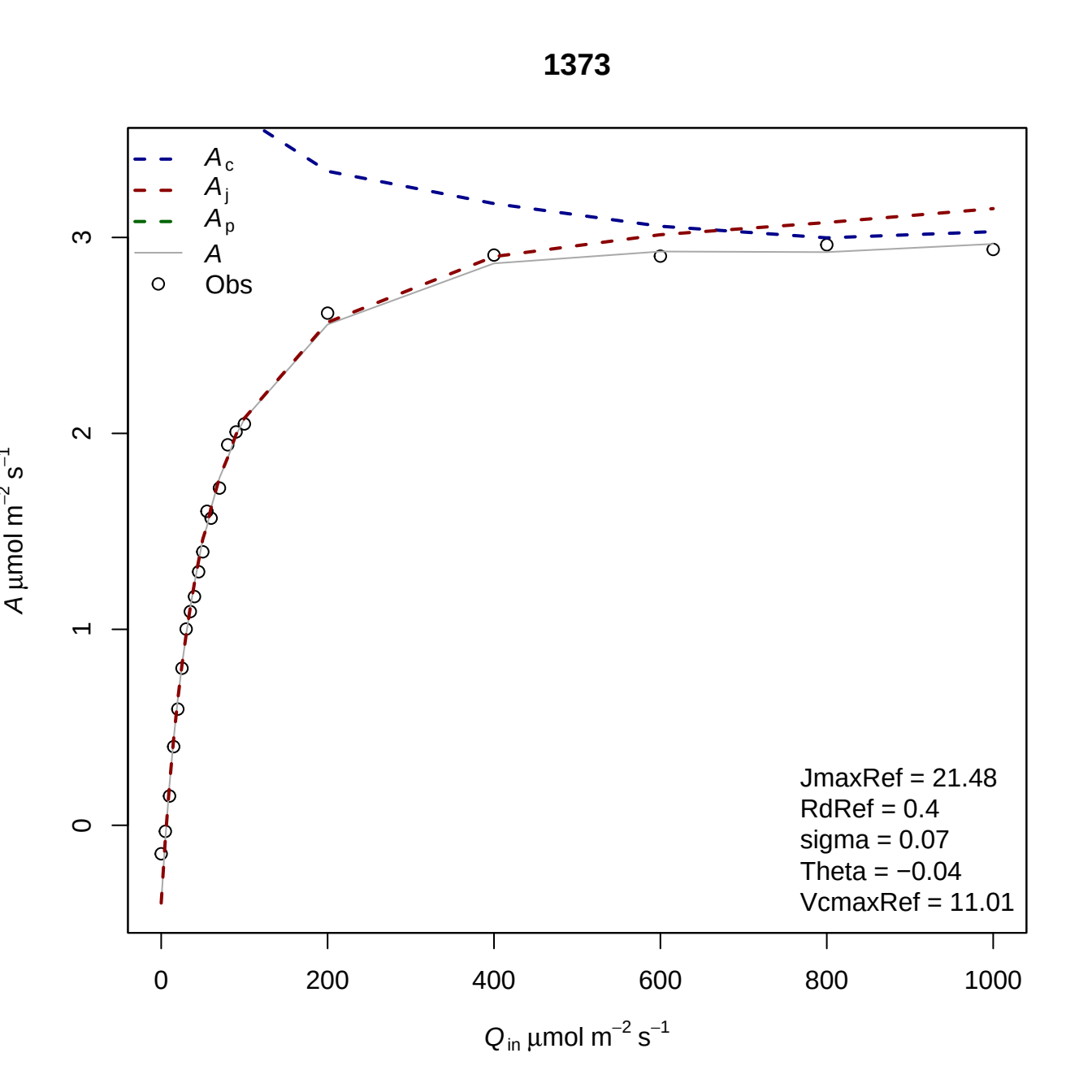

1368
A_c
A_j
A_p
A
Obs
JmaxRef = 29.95
RdRef = 0.9
sigma = 0.1
Theta = 0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

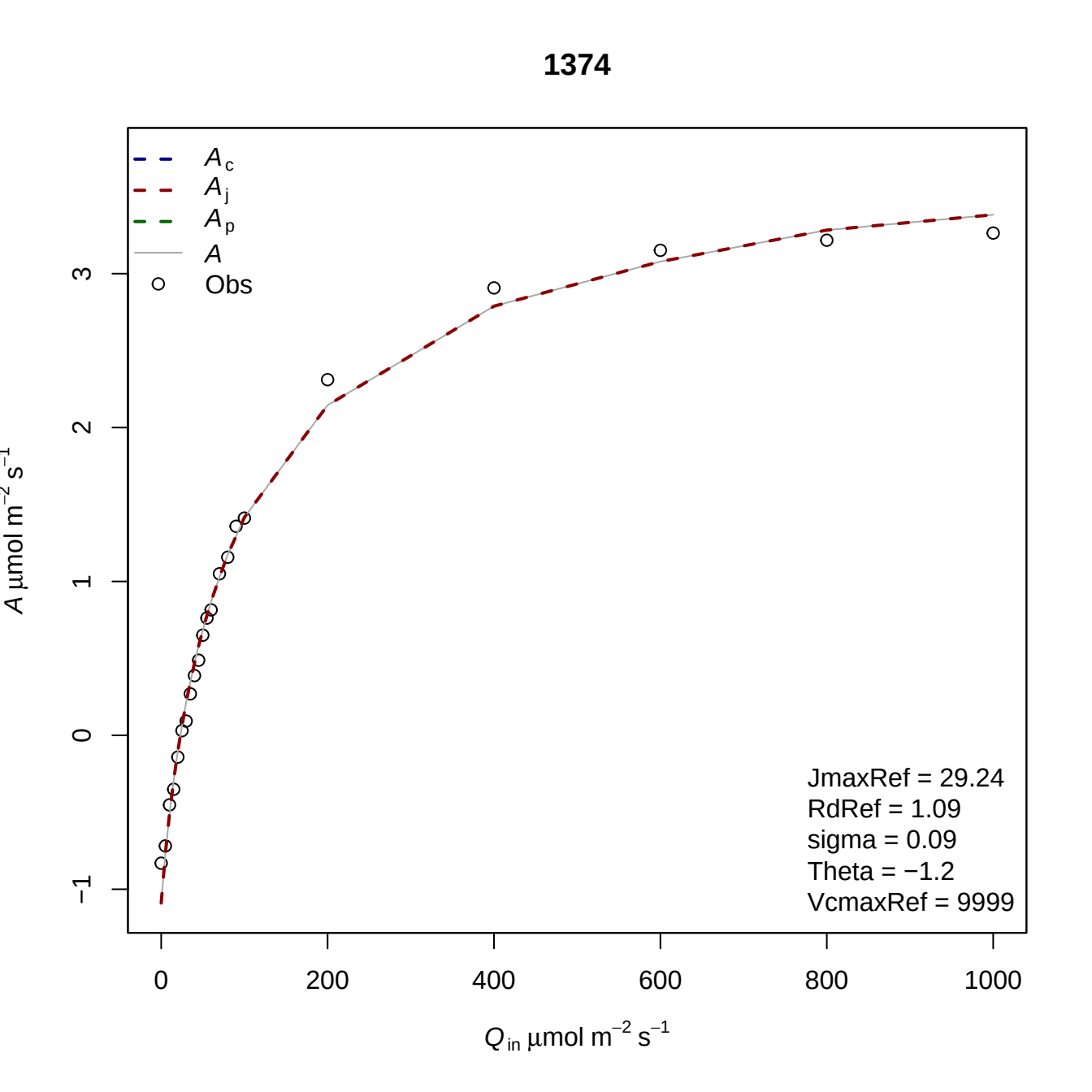

1369
A_c
A_j
A_p
A
Obs
JmaxRef = 29.32
RdRef = 1.45
sigma = 0.07
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

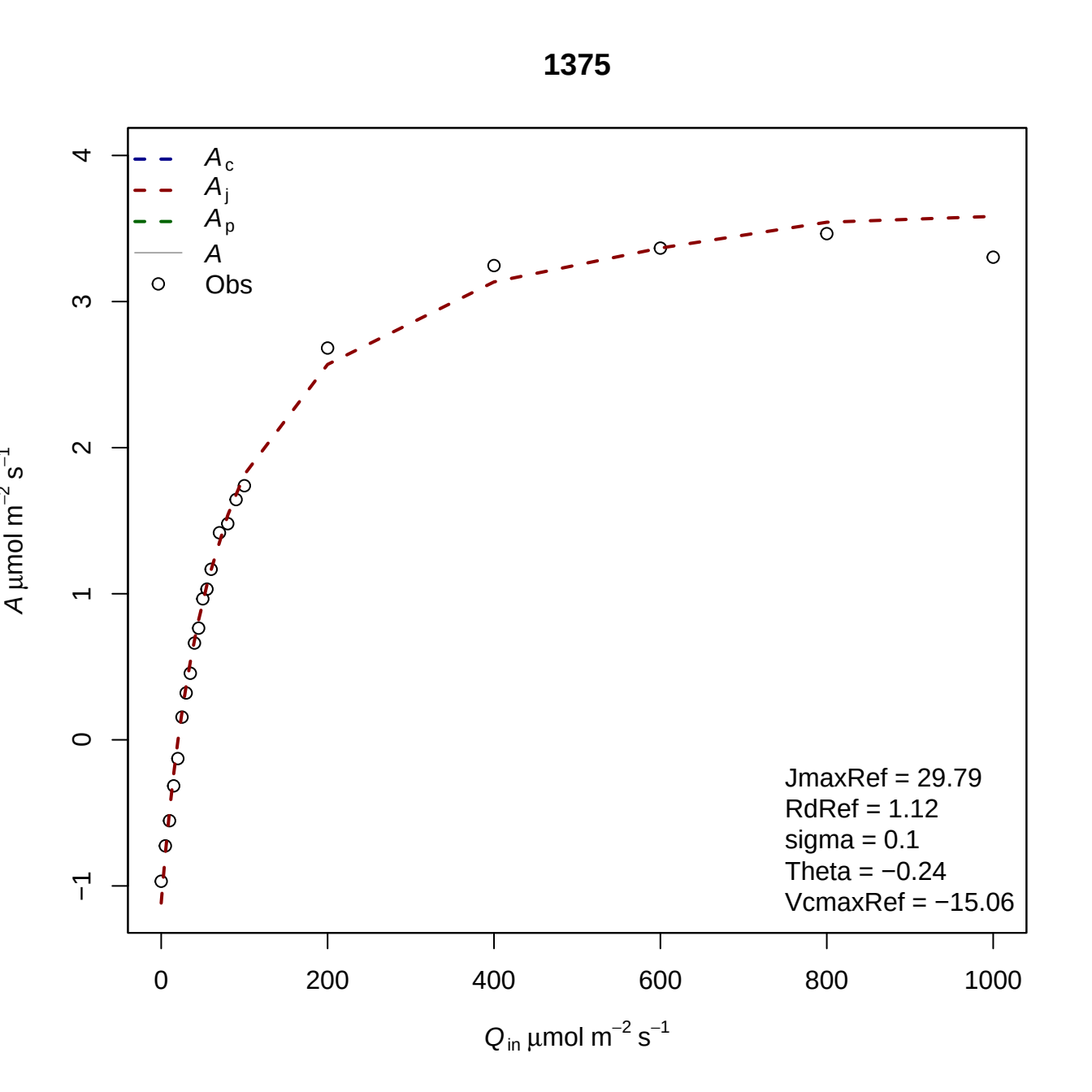

1370
A_c
A_j
A_p
A
Obs
JmaxRef = 36.26
RdRef = 1.05
sigma = 0.13
Theta = −0.79
VcmaxRef = 18.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

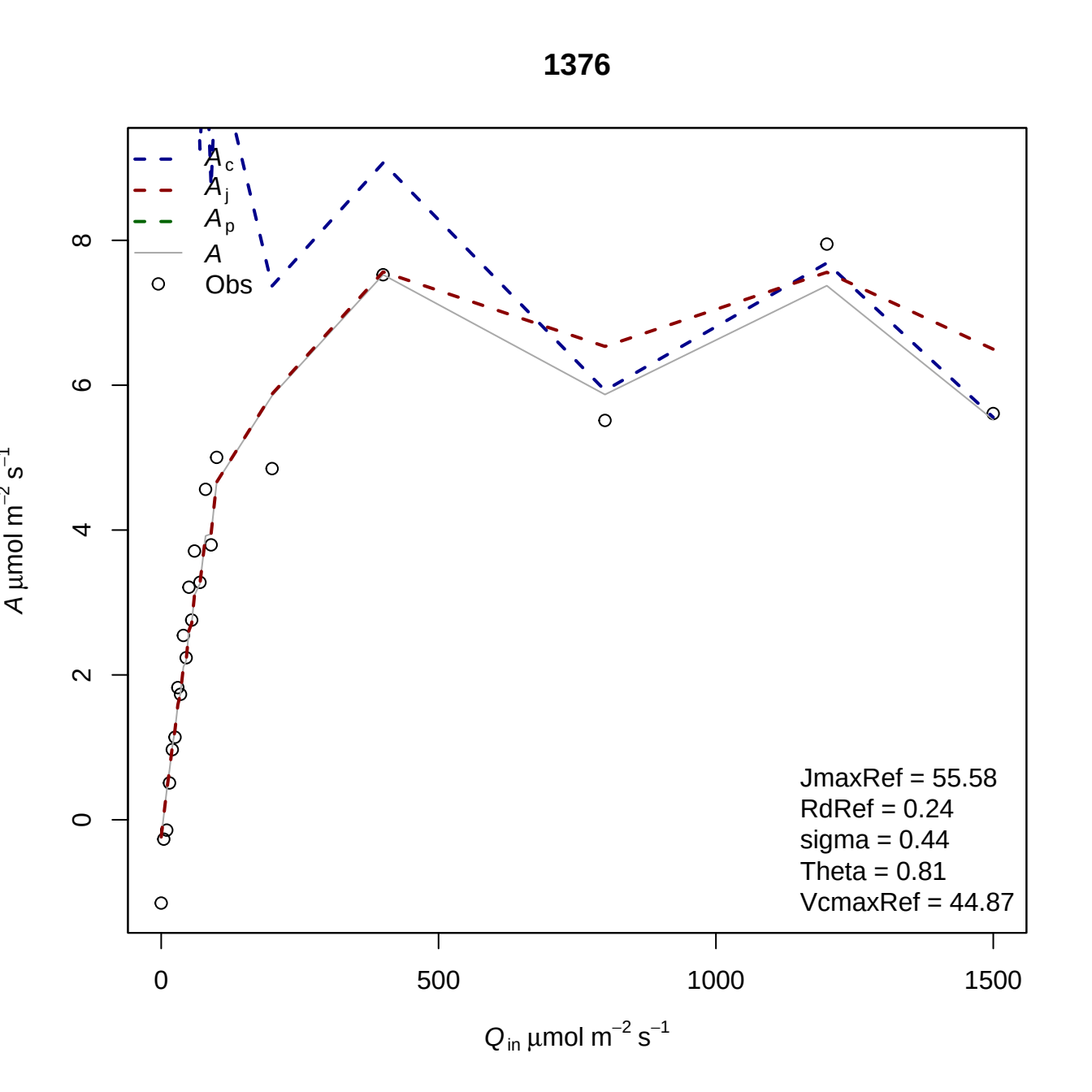

1371
A_c
A_j
A_p
A
Obs
JmaxRef = 35.62
RdRef = 1.09
sigma = 0.13
Theta = 0.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

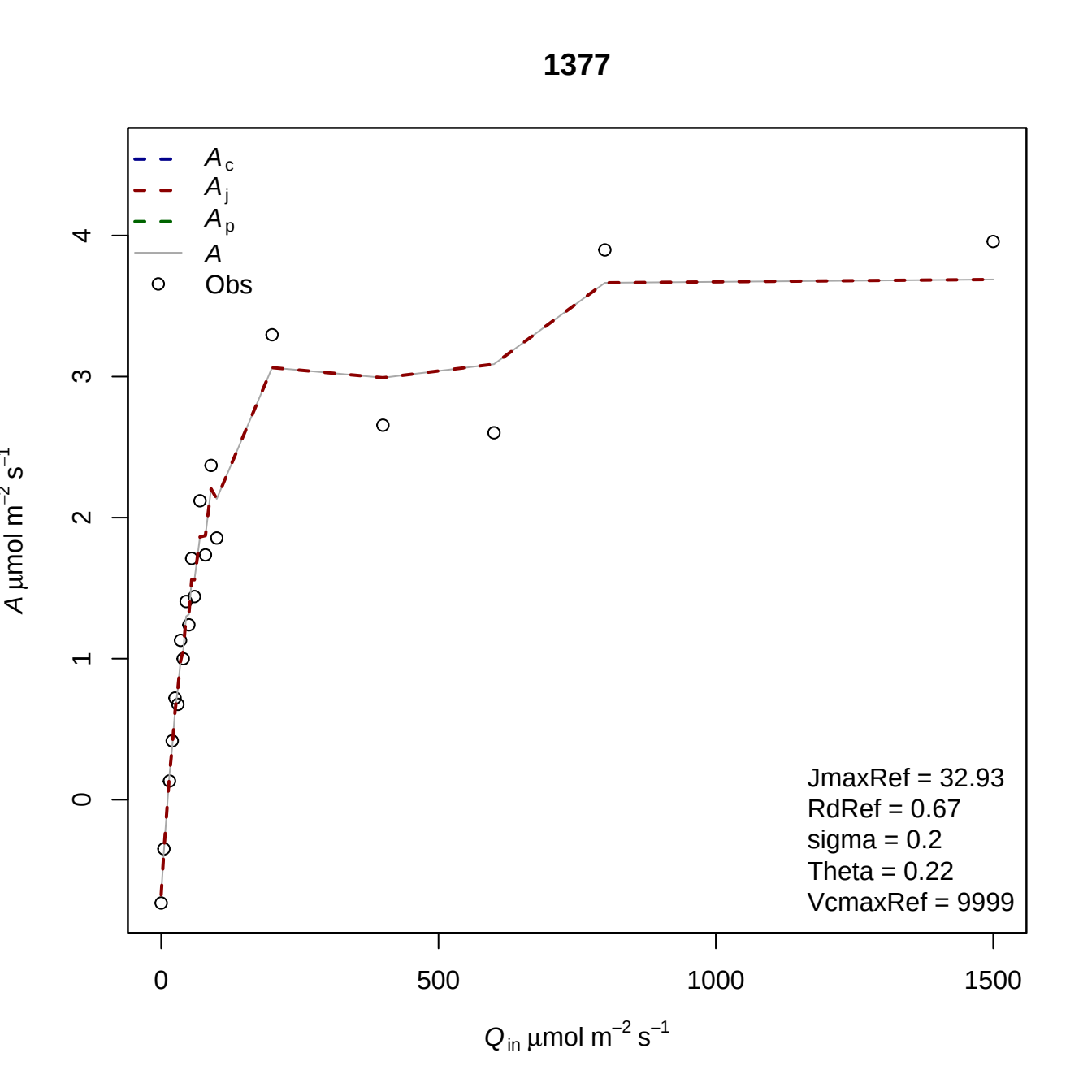

1372
A_c
A_j
A_p
A
Obs
JmaxRef = 45.73
RdRef = 0.53
sigma = 0.15
Theta = −0.29
VcmaxRef = 24.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

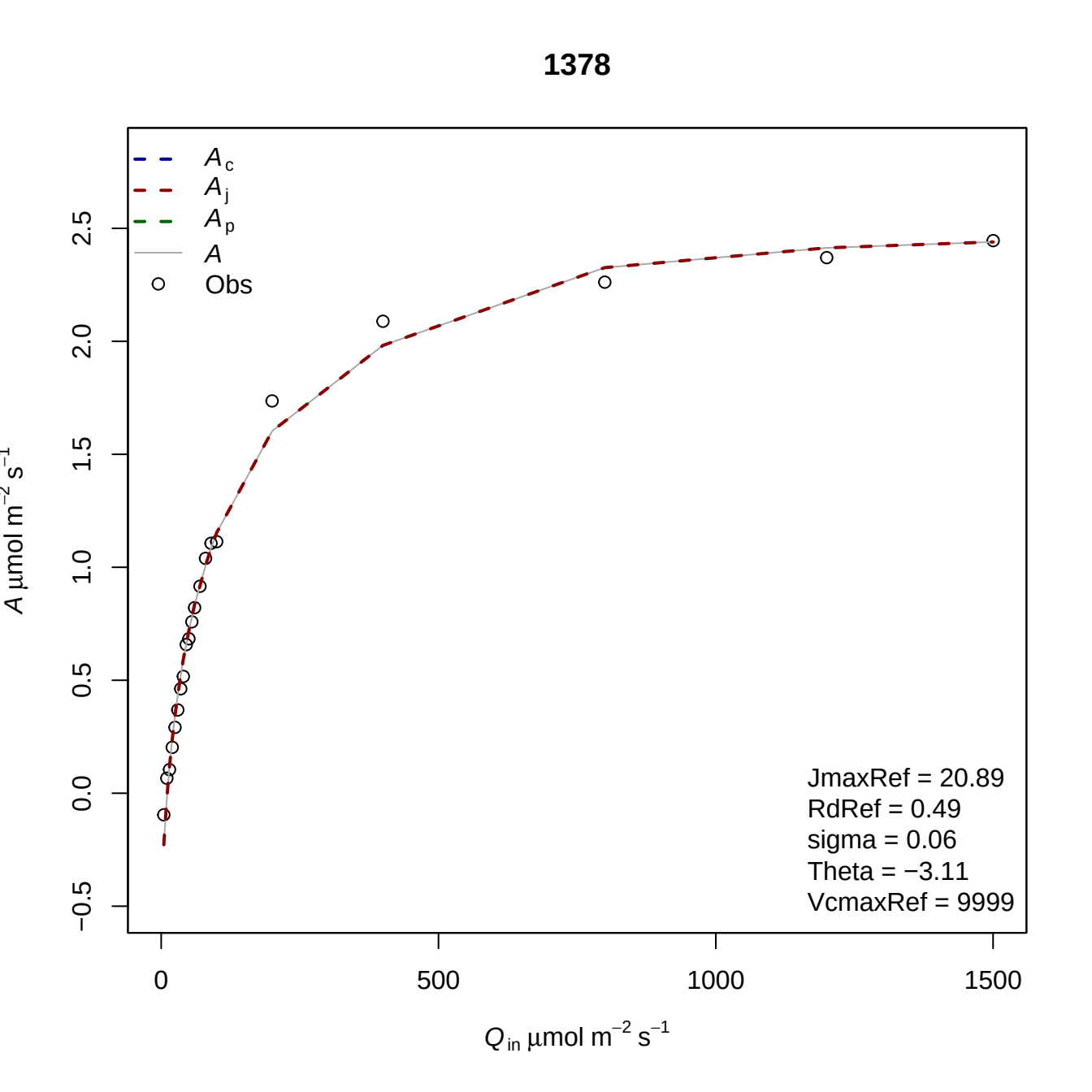

1373
A_c
A_j
A_p
A
Obs
JmaxRef = 21.48
RdRef = 0.4
sigma = 0.07
Theta = −0.04
VcmaxRef = 11.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

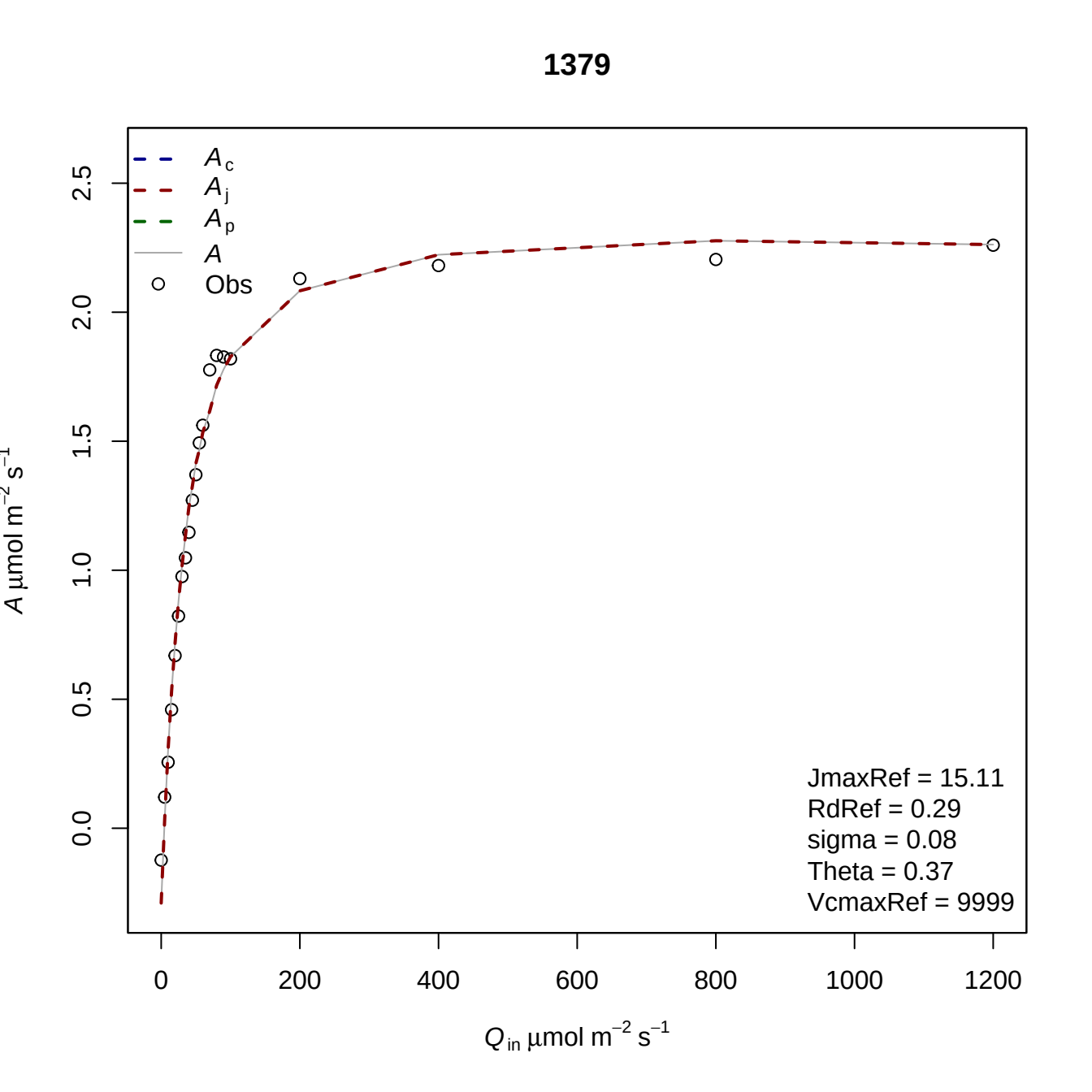

1374
A_c
A_j
A_p
A
Obs
JmaxRef = 29.24
RdRef = 1.09
sigma = 0.09
Theta = −1.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

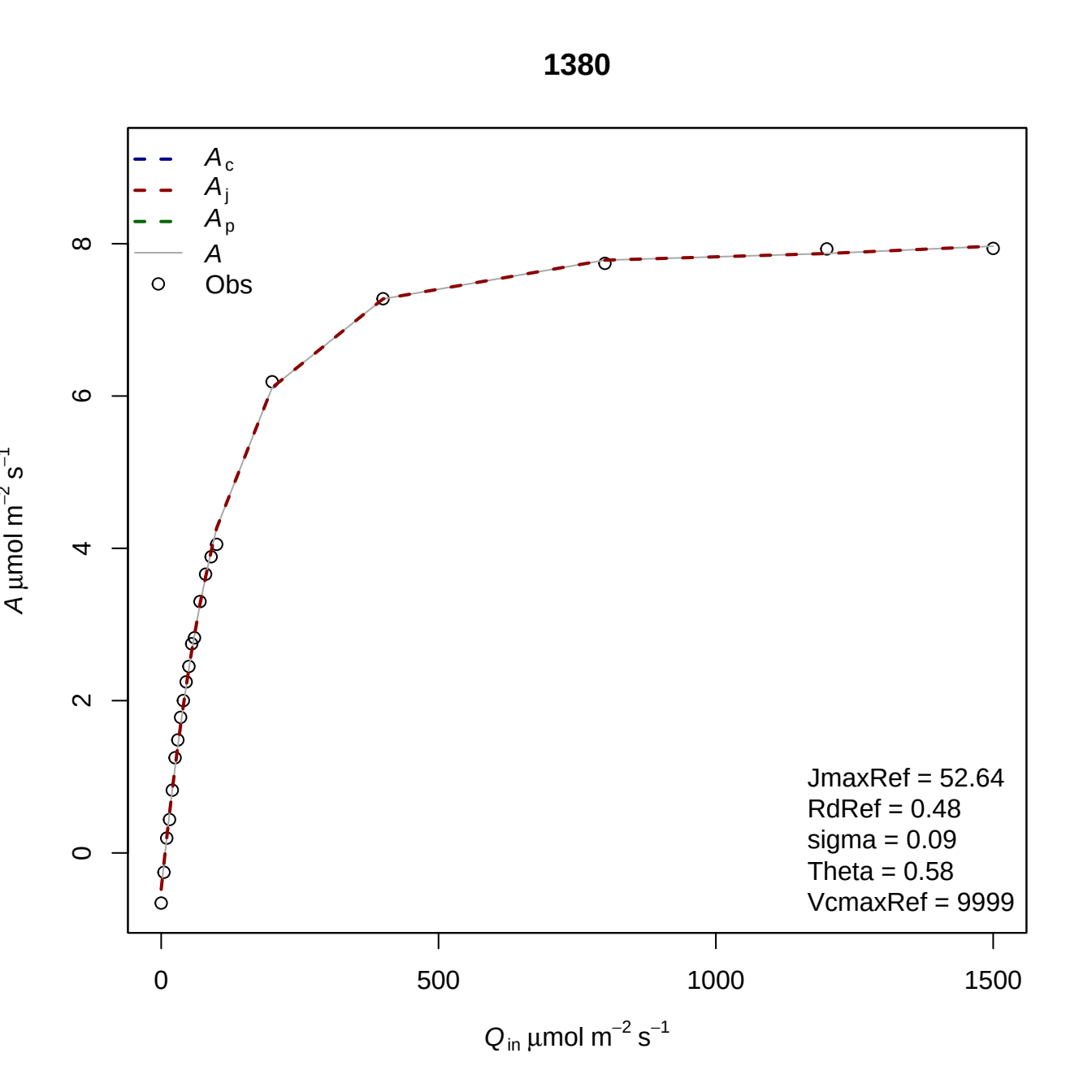

1375
A_c
A_j
A_p
A
Obs
JmaxRef = 29.79
RdRef = 1.12
sigma = 0.1
Theta = −0.24
VcmaxRef = −15.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

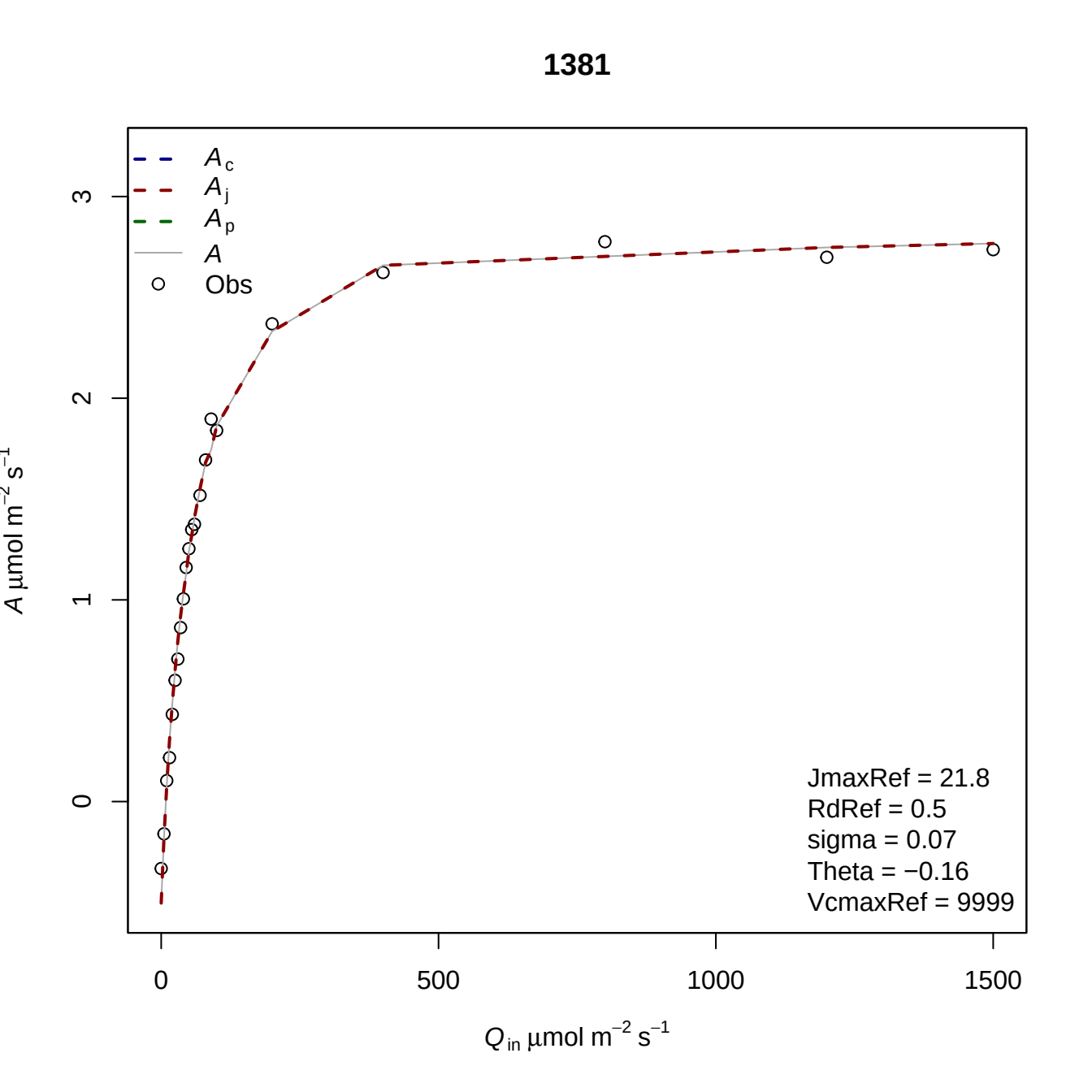

1376
A_c
A_j
A_p
A
Obs
JmaxRef = 55.58
RdRef = 0.24
sigma = 0.44
Theta = 0.81
VcmaxRef = 44.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

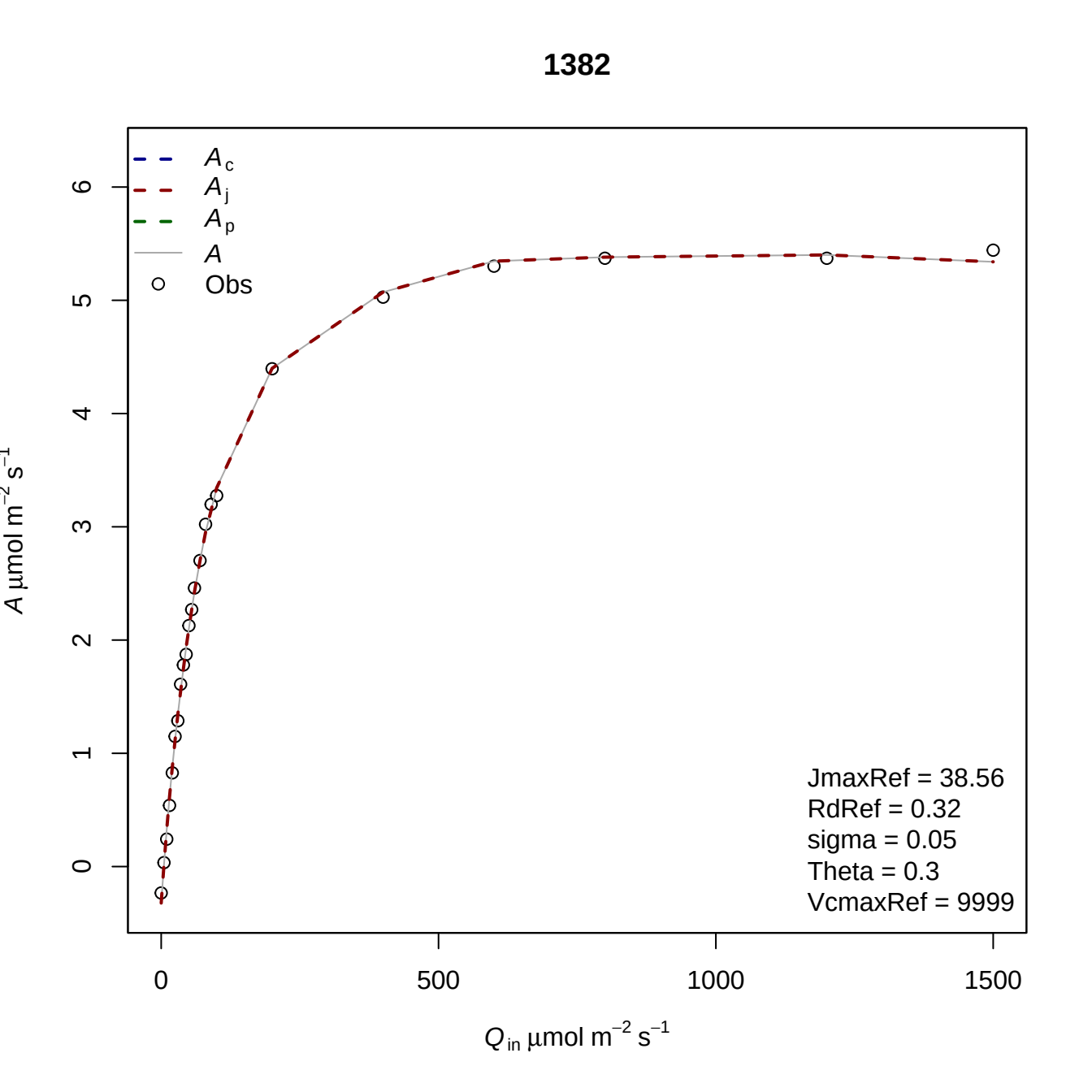

1377
A_c
A_j
A_p
A
Obs
JmaxRef = 32.93
RdRef = 0.67
sigma = 0.2
Theta = 0.22
VcmaxRef = 9999
0
1
2
3
4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

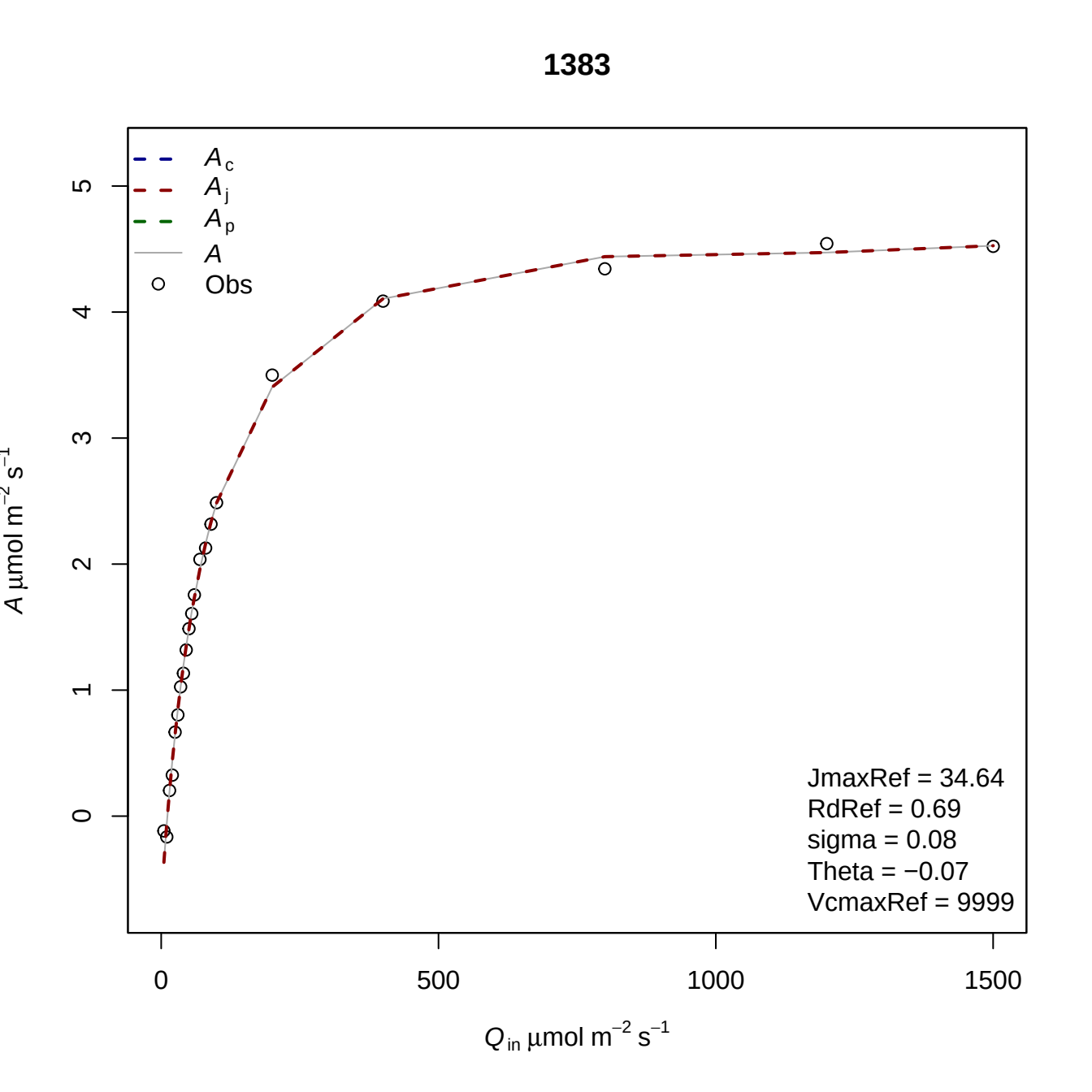

1378
A_c
A_j
A_p
A
Obs
JmaxRef = 20.89
RdRef = 0.49
sigma = 0.06
Theta = −3.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

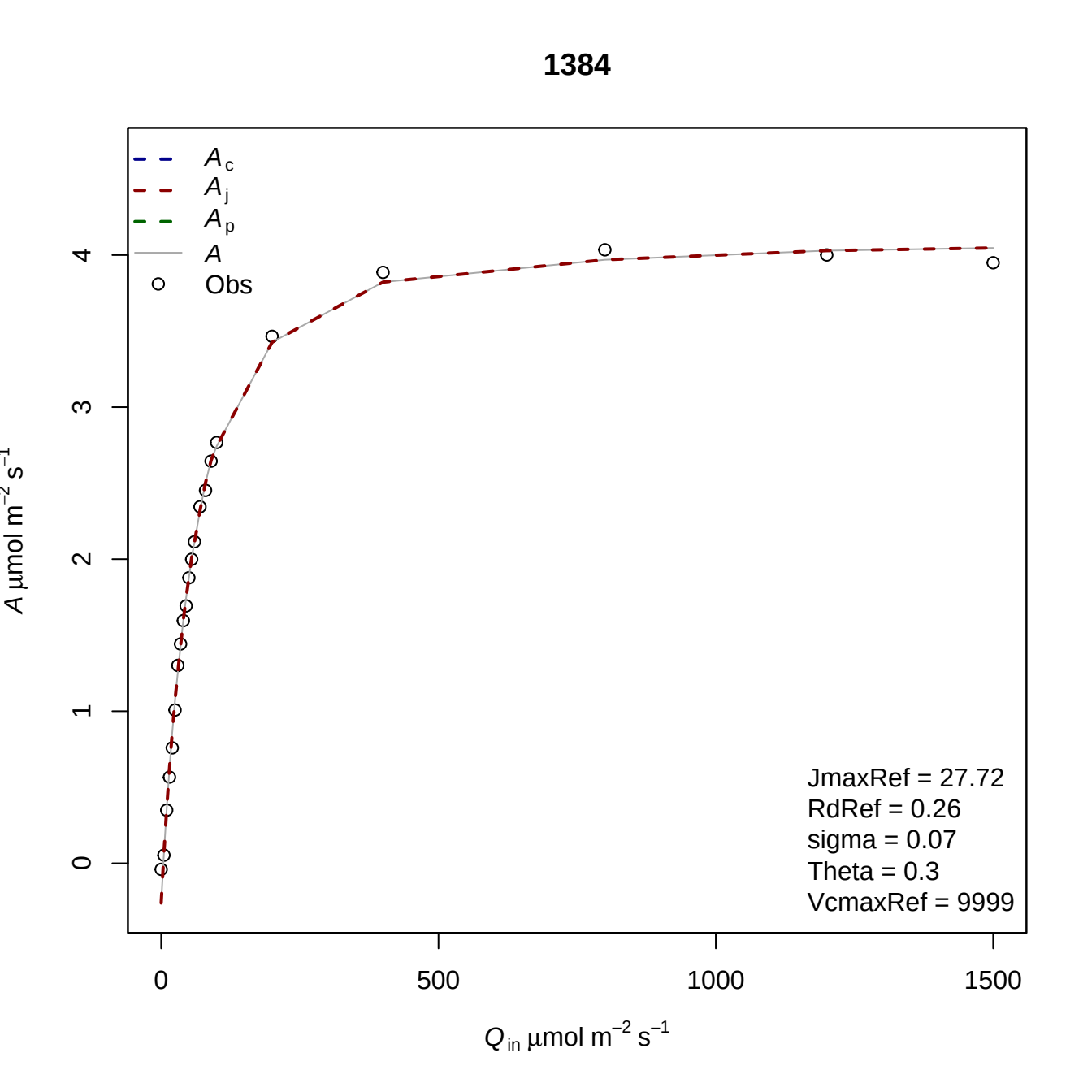

1379
A_c
A_j
A_p
A
Obs
JmaxRef = 15.11
RdRef = 0.29
sigma = 0.08
Theta = 0.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200

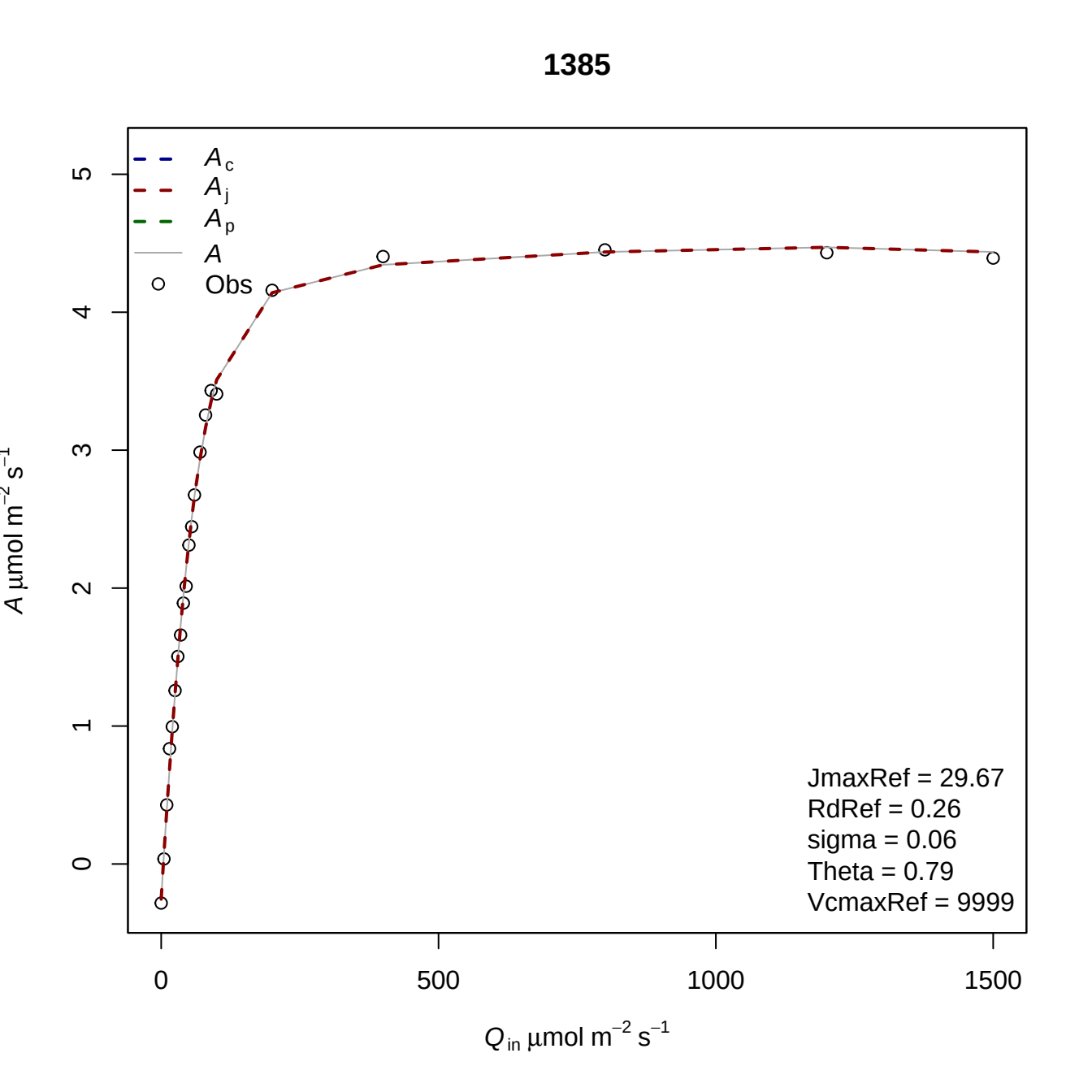

1380
A_c
A_j
A_p
A
Obs
JmaxRef = 52.64
RdRef = 0.48
sigma = 0.09
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

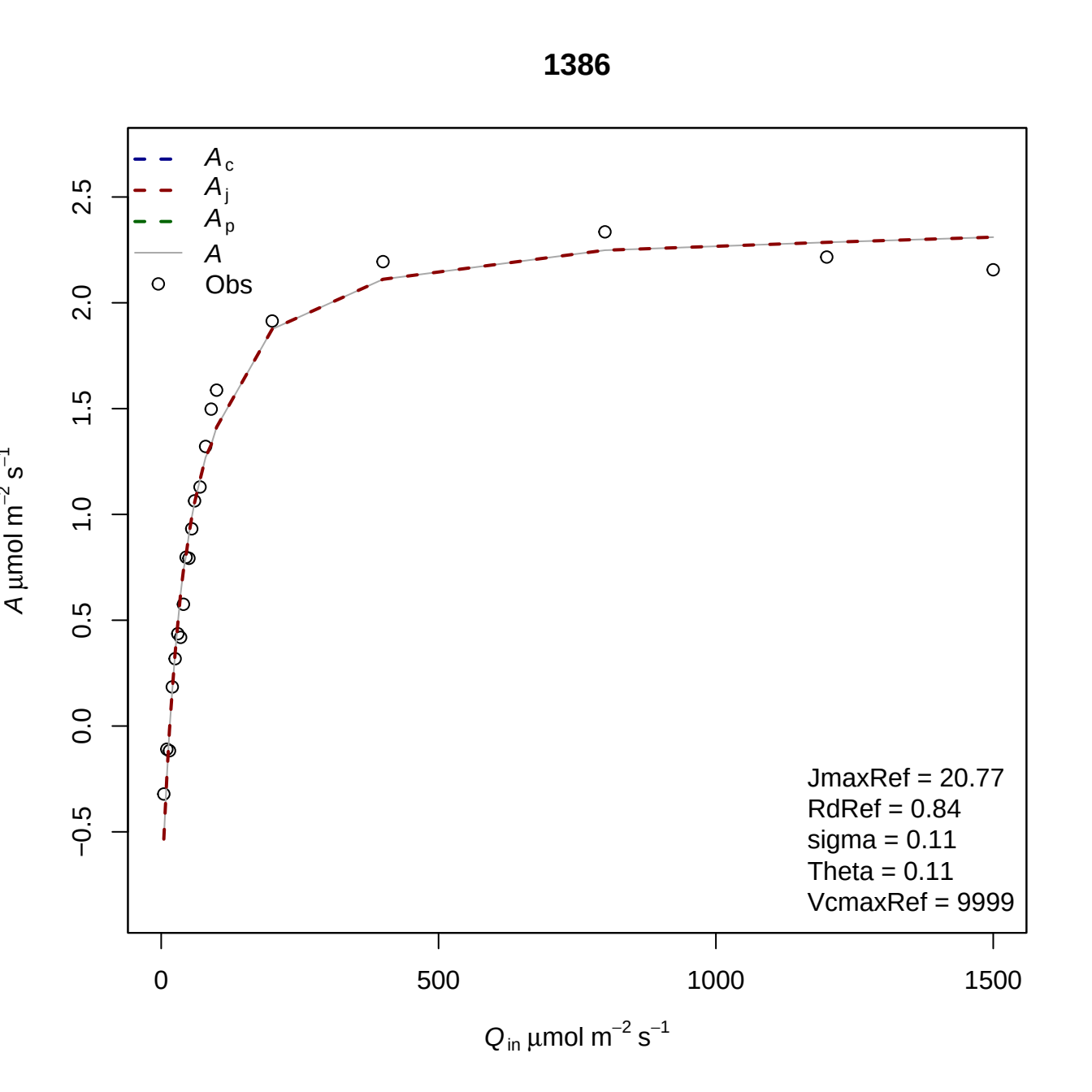

1381
A_c
A_j
A_p
A
Obs
JmaxRef = 21.8
RdRef = 0.5
sigma = 0.07
Theta = −0.16
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

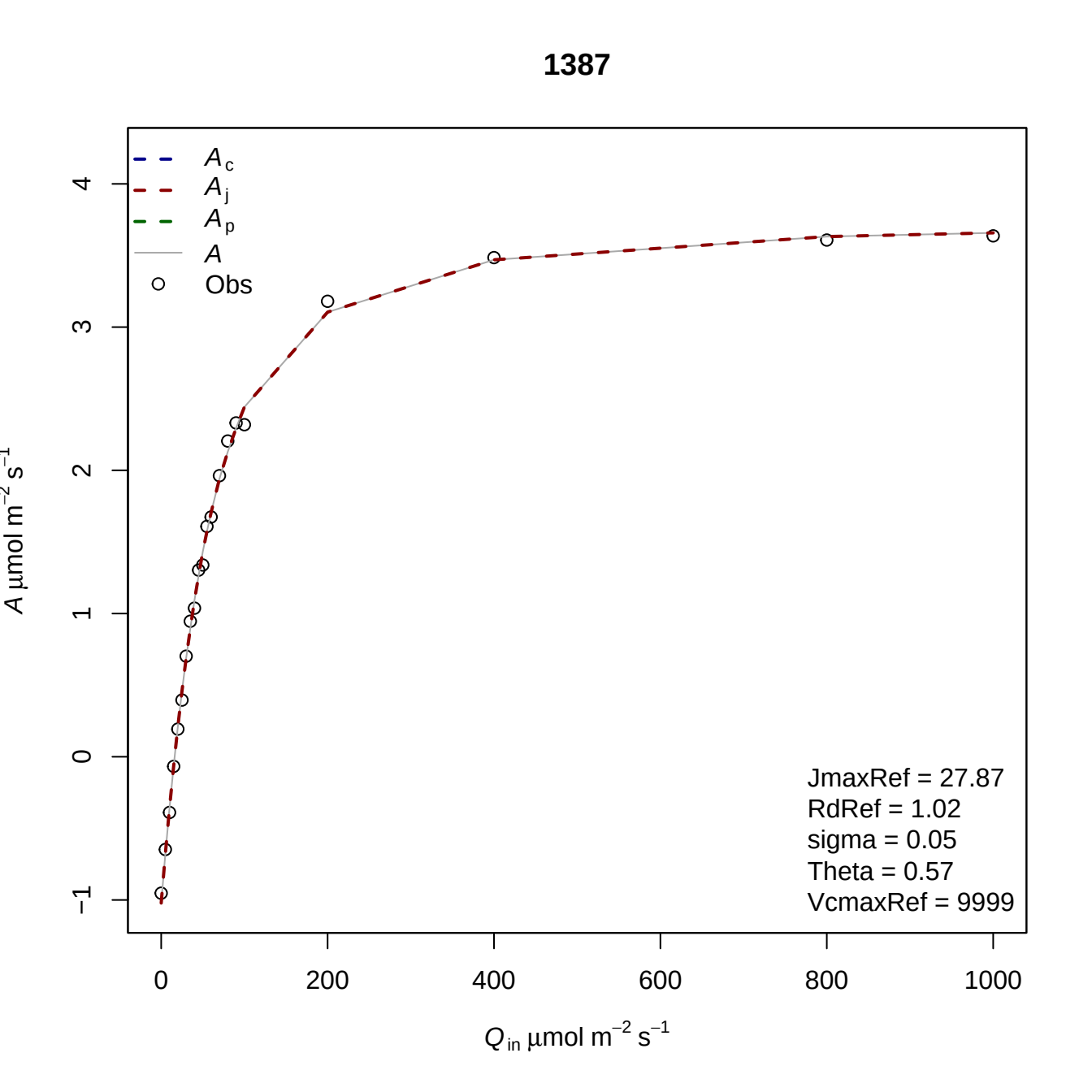

1382
A_c
A_j
A_p
A
Obs
JmaxRef = 38.56
RdRef = 0.32
sigma = 0.05
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

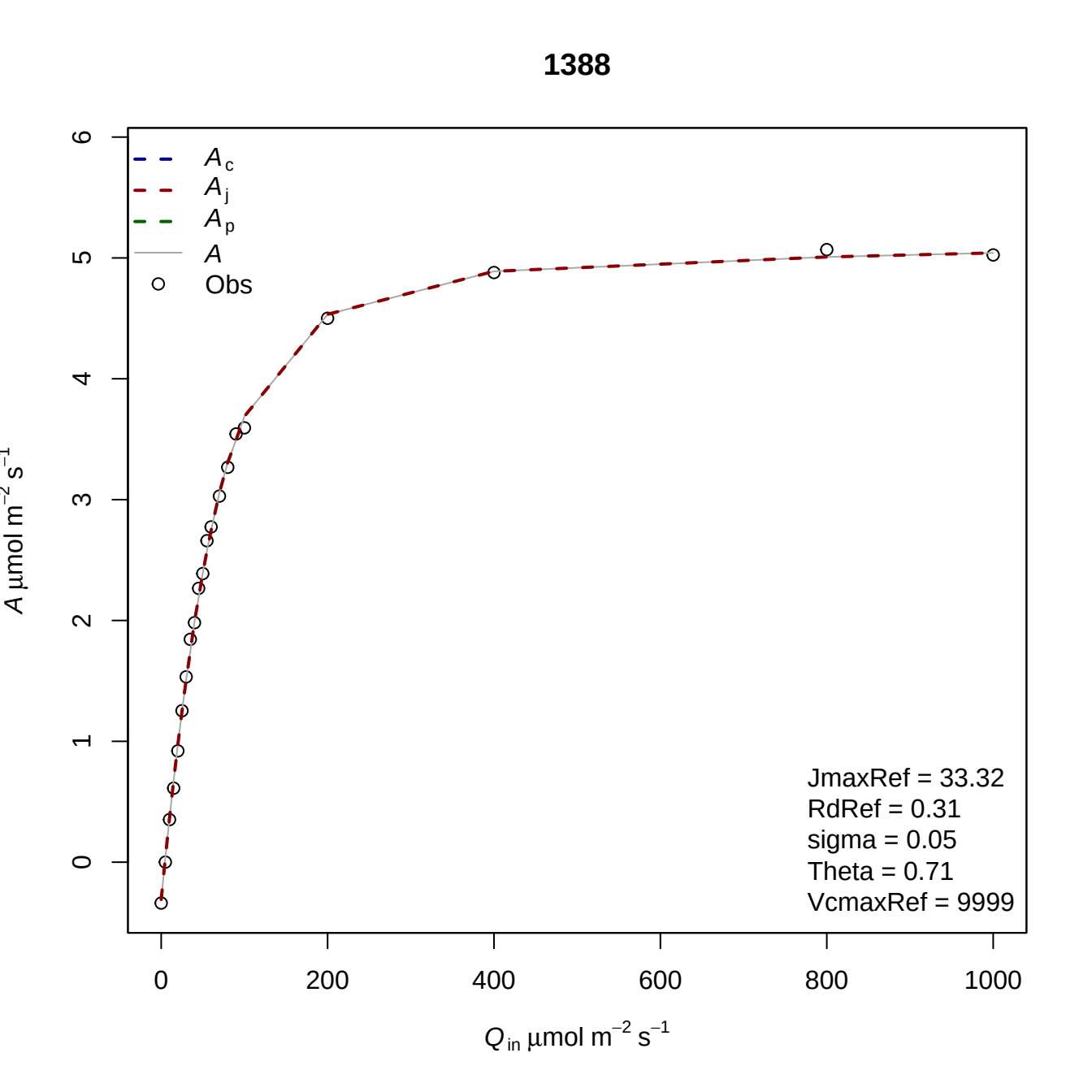

1383
A_c
A_j
A_p
A
Obs
JmaxRef = 34.64
RdRef = 0.69
sigma = 0.08
Theta = −0.07
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

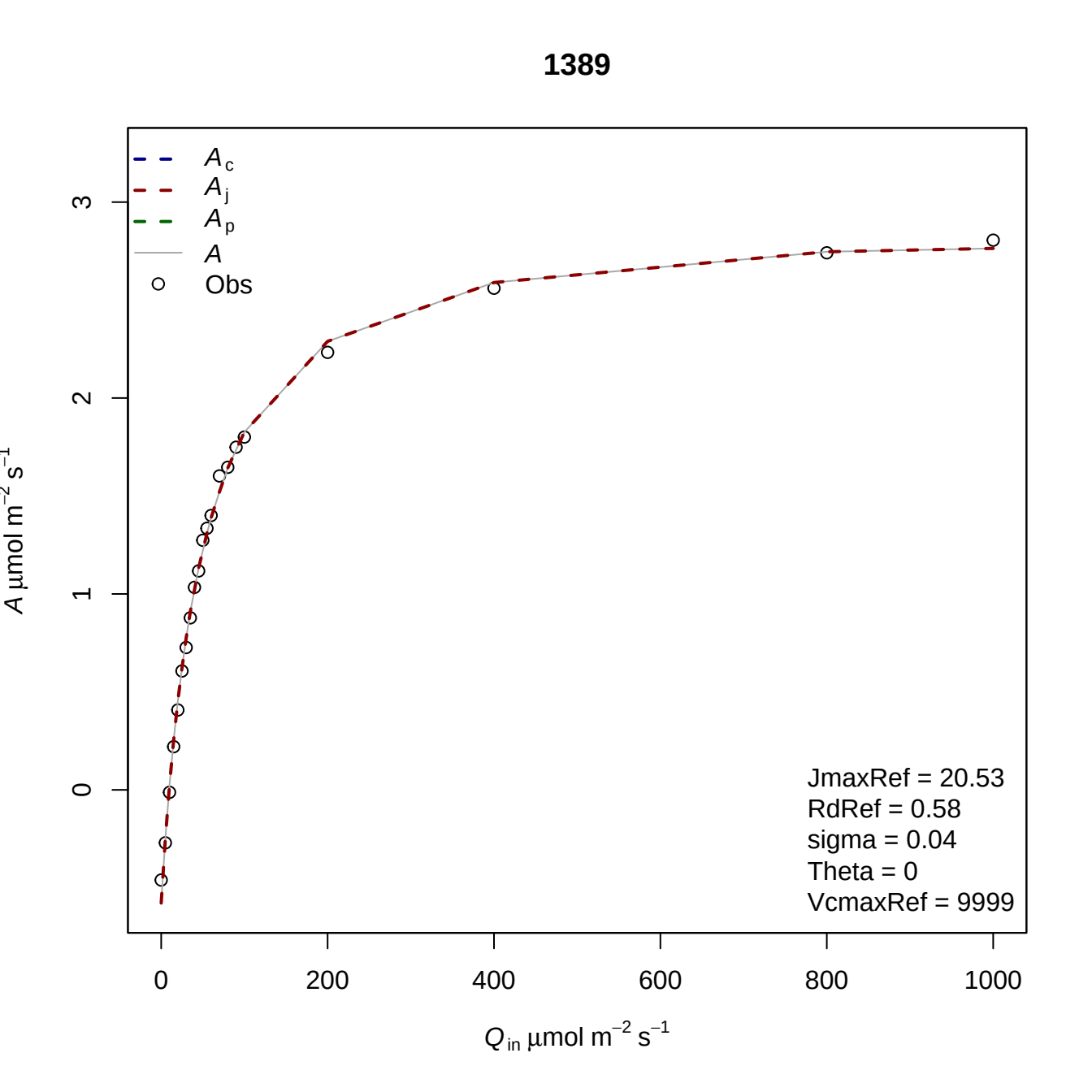

1384
A_c
A_j
A_p
A
Obs
JmaxRef = 27.72
RdRef = 0.26
sigma = 0.07
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

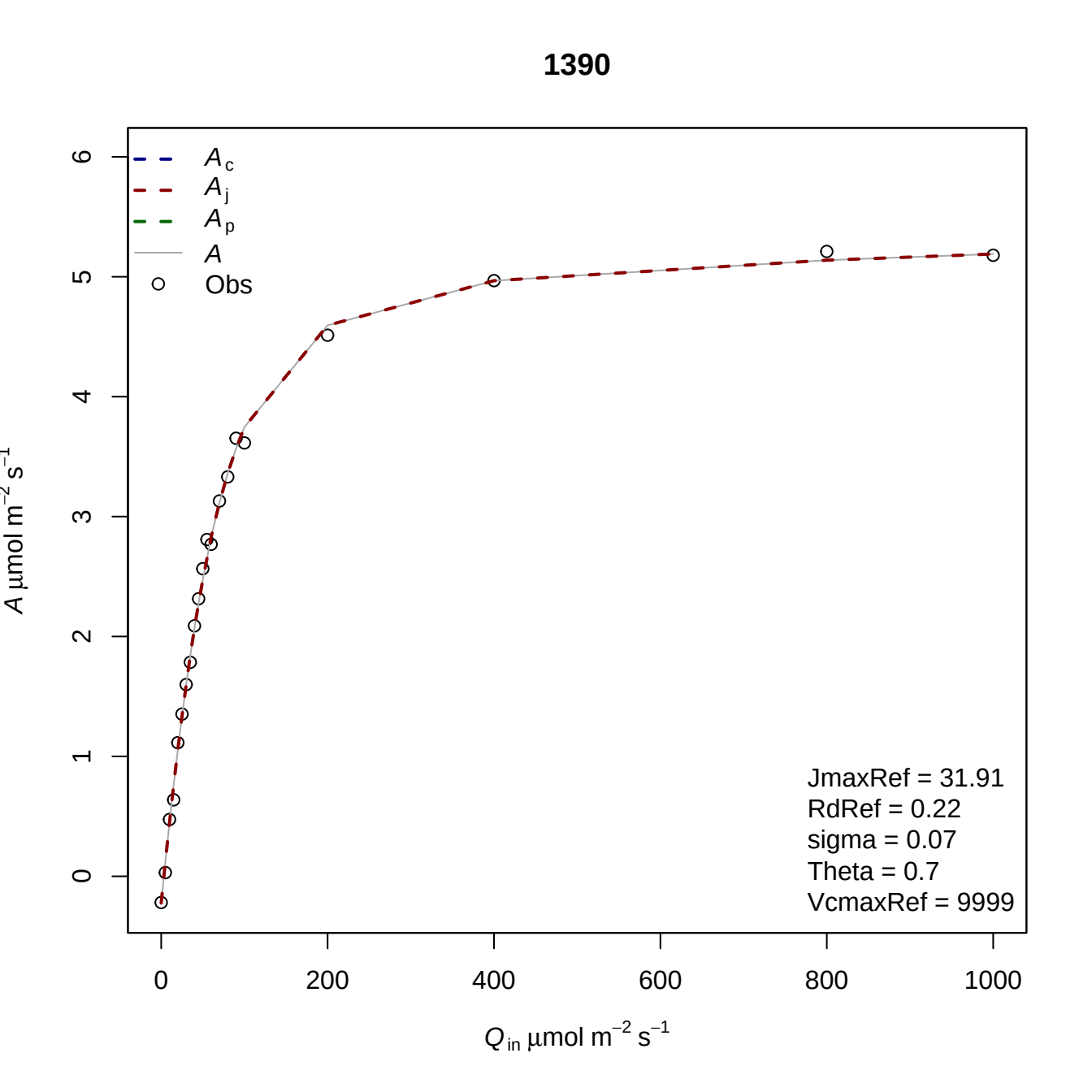

1385
A_c
A_j
A_p
A
Obs
JmaxRef = 29.67
RdRef = 0.26
sigma = 0.06
Theta = 0.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

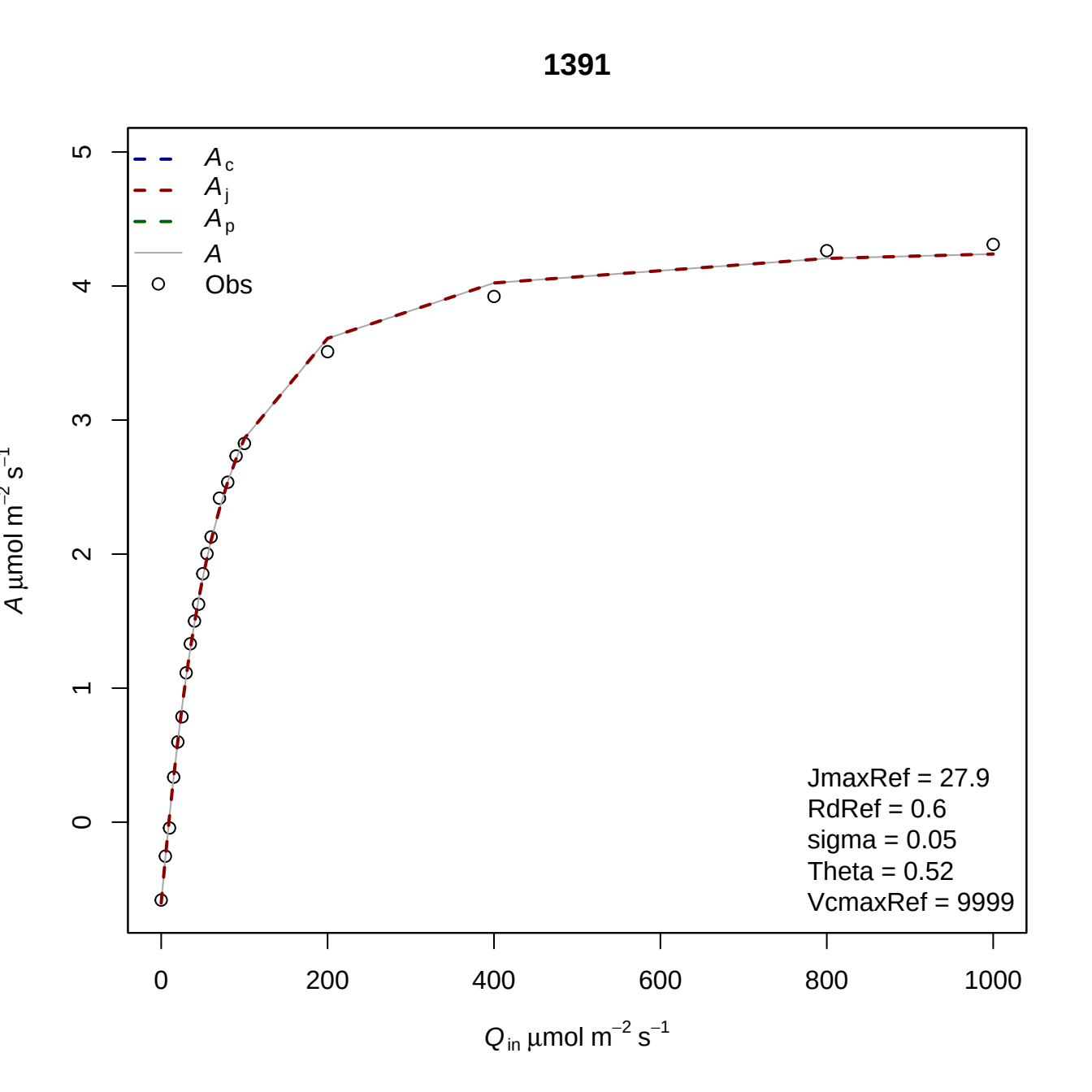

1386
A_c
A_j
A_p
A
Obs
JmaxRef = 20.77
RdRef = 0.84
sigma = 0.11
Theta = 0.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

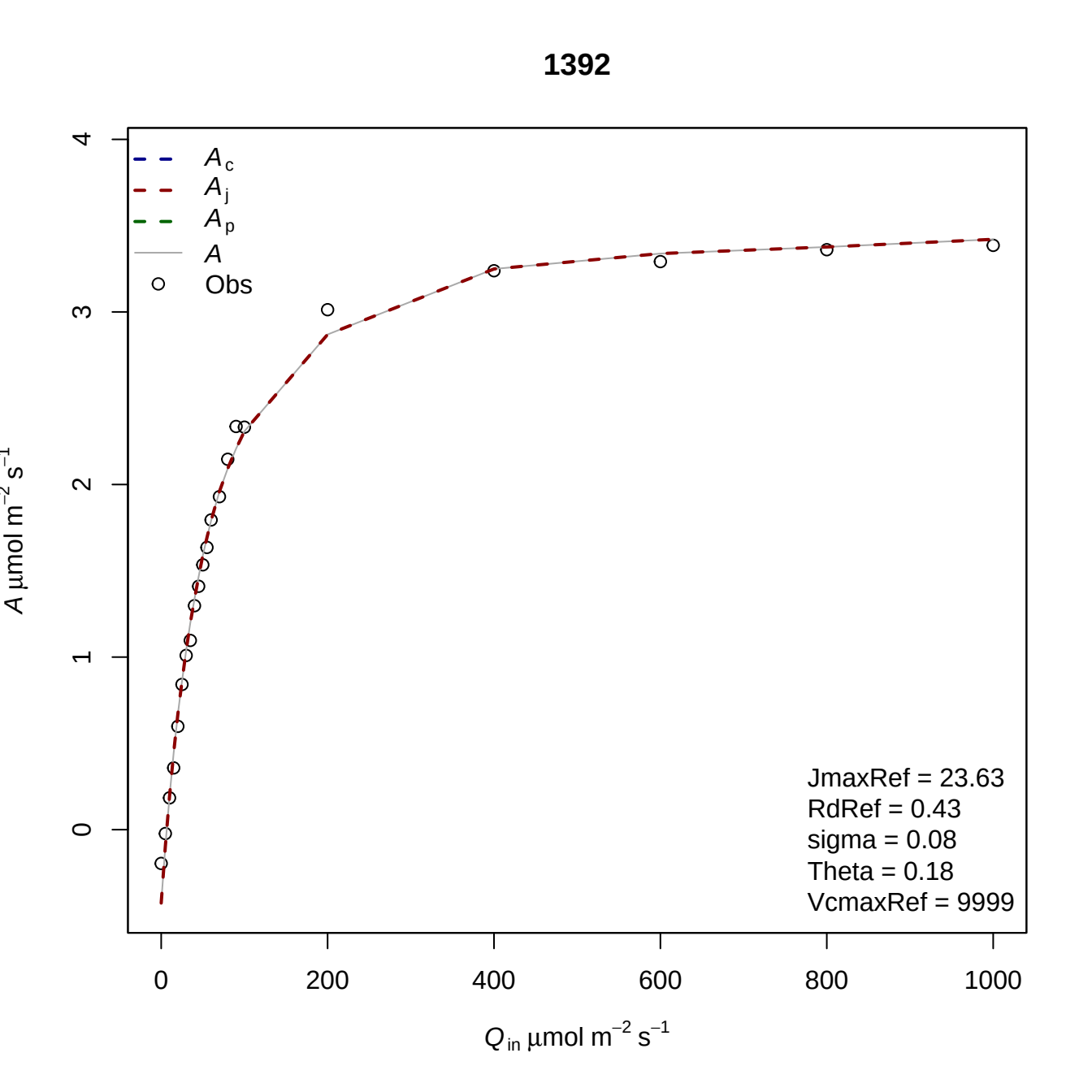

1387
A_c
A_j
A_p
A
Obs
JmaxRef = 27.87
RdRef = 1.02
sigma = 0.05
Theta = 0.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

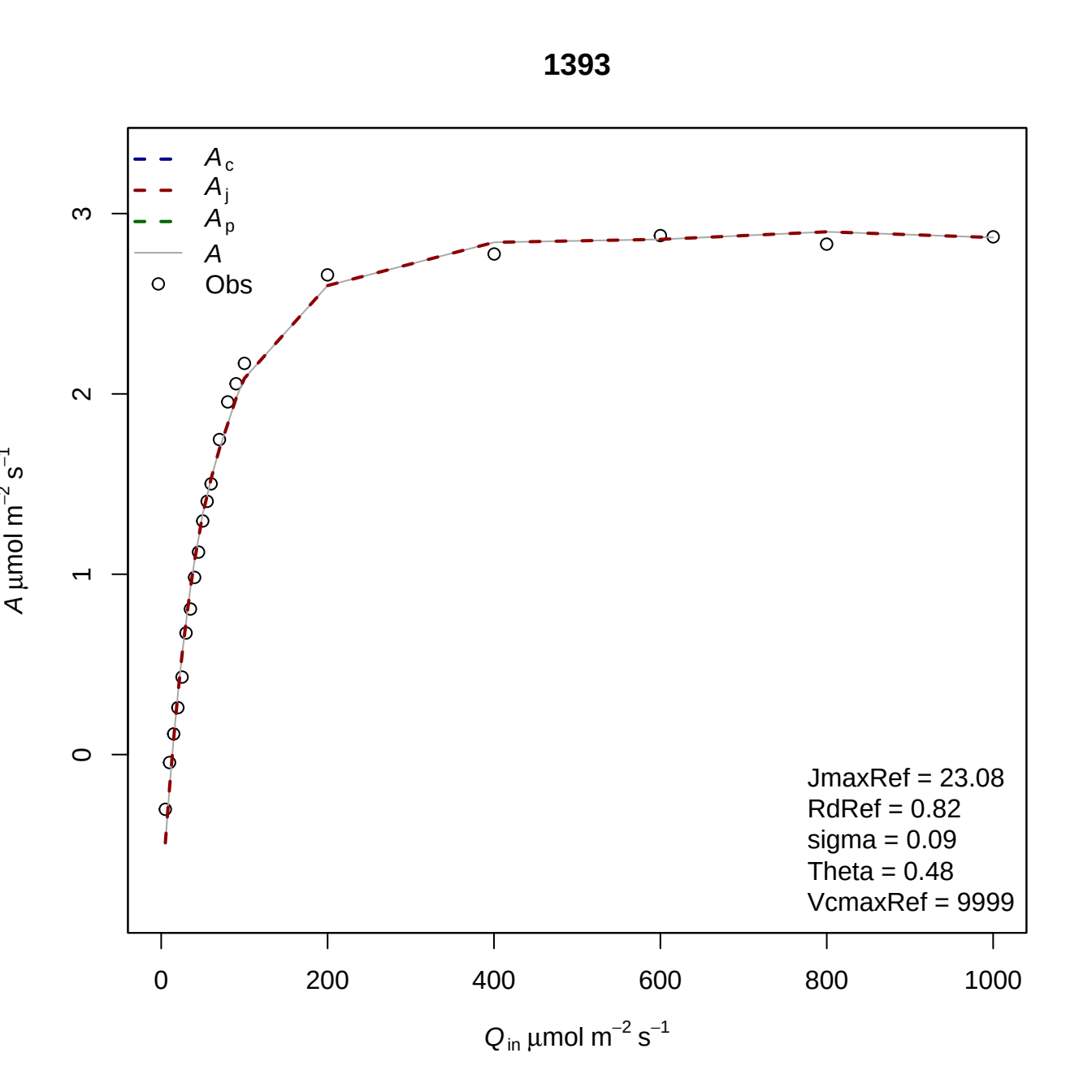

1388
A_c
A_j
A_p
A
Obs
JmaxRef = 33.32
RdRef = 0.31
sigma = 0.05
Theta = 0.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

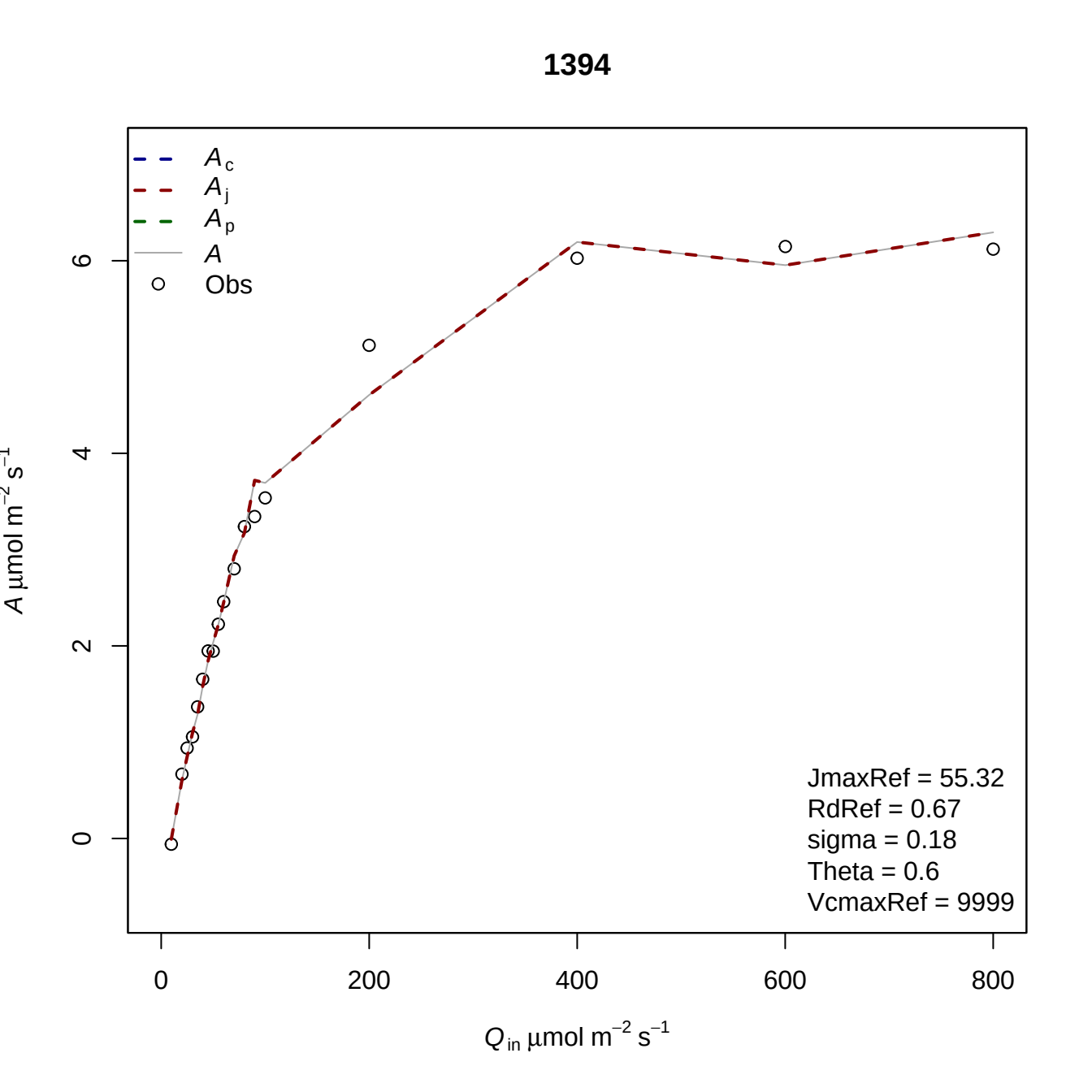

1389
A_c
A_j
A_p
A
Obs
JmaxRef = 20.53
RdRef = 0.58
sigma = 0.04
Theta = 0
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

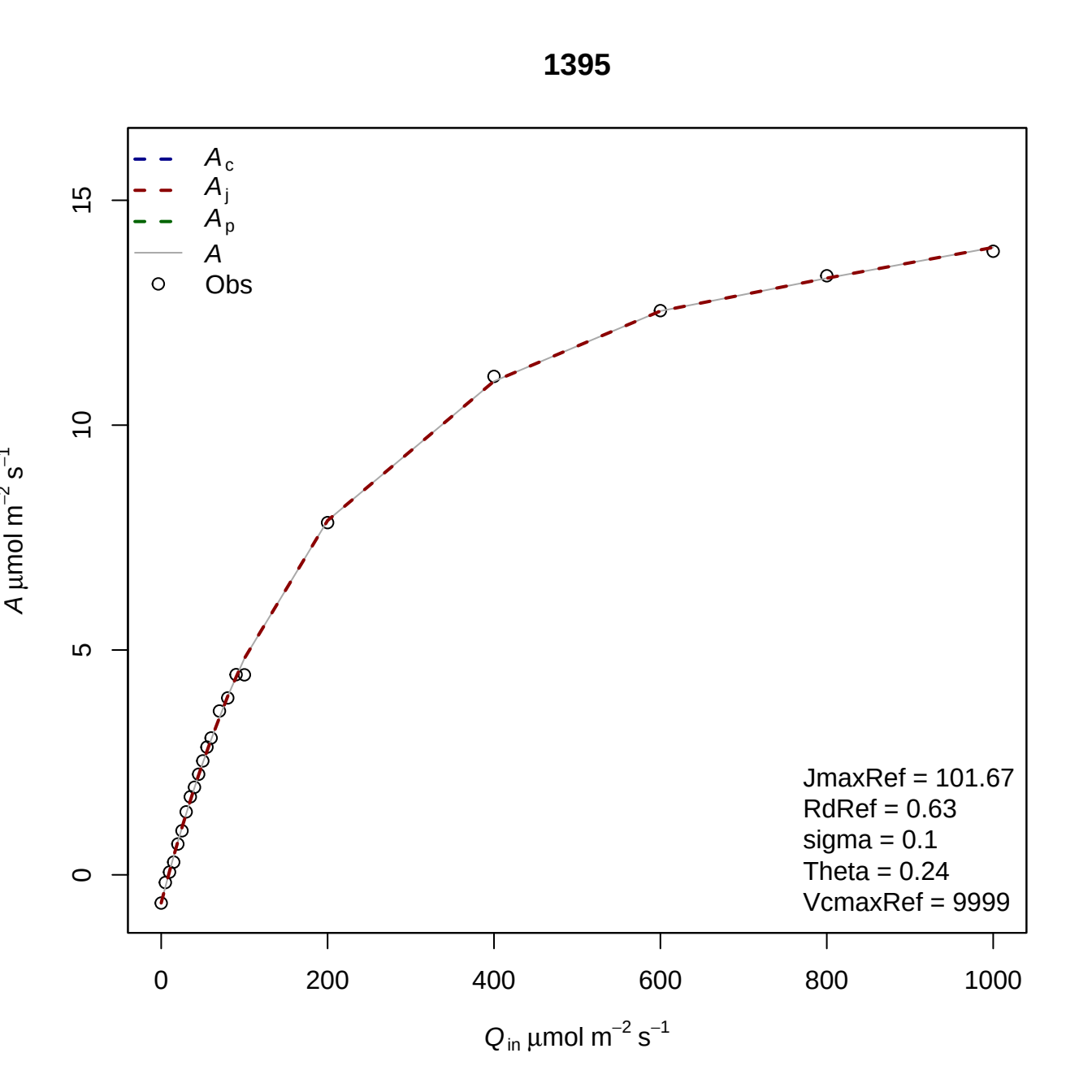

1390
A_c
A_j
A_p
A
Obs
JmaxRef = 31.91
RdRef = 0.22
sigma = 0.07
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

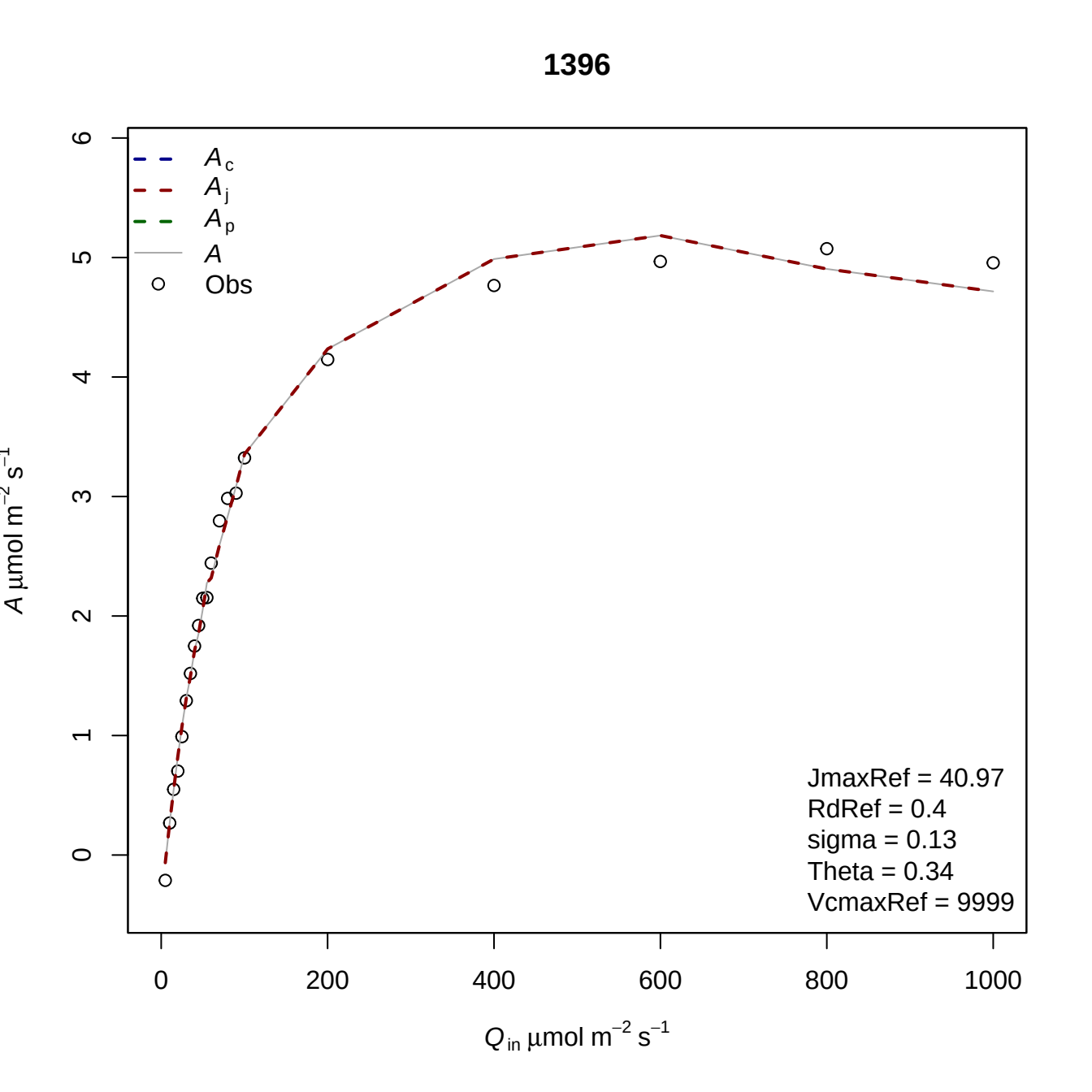

1391
A_c
A_j
A_p
A
Obs
JmaxRef = 27.9
RdRef = 0.6
sigma = 0.05
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

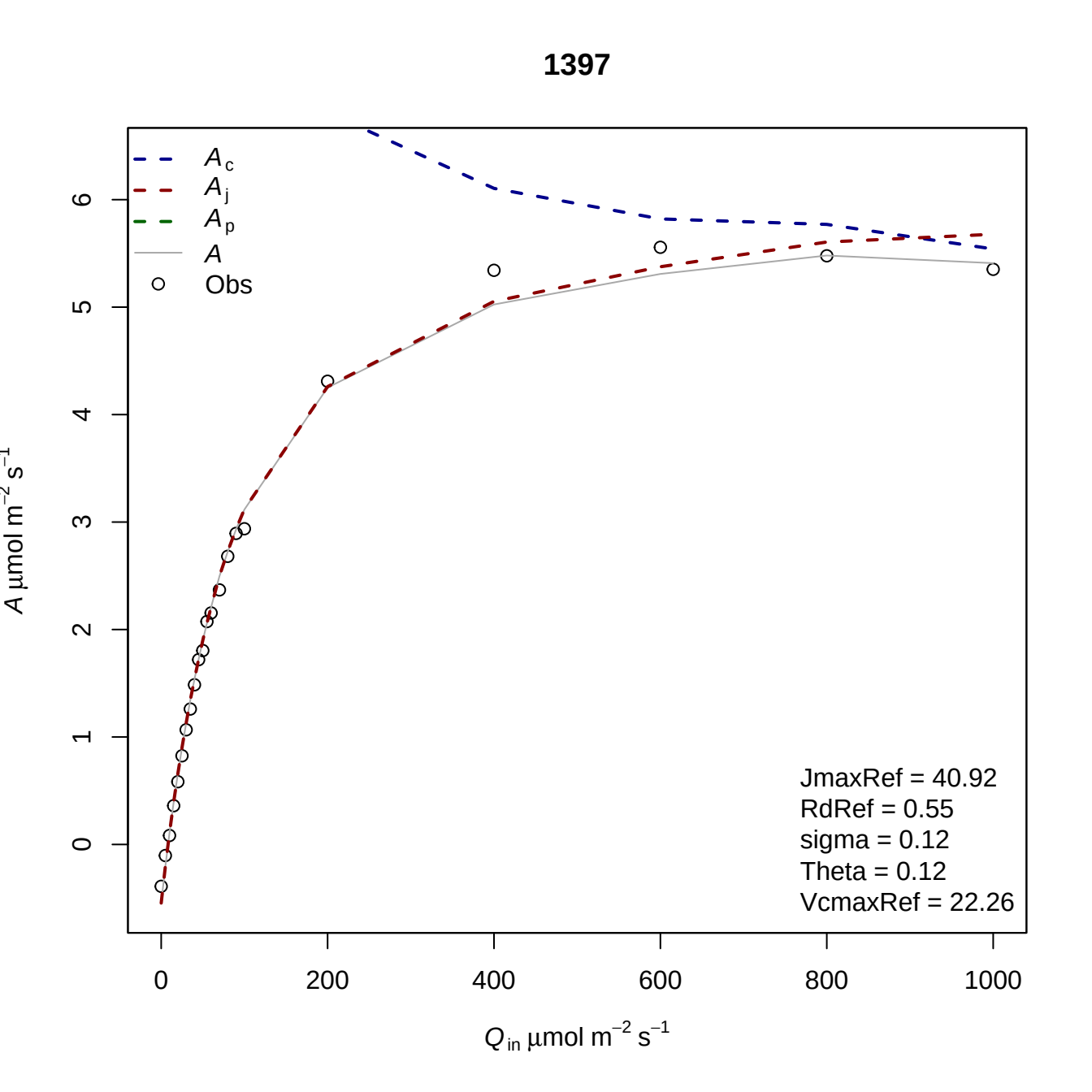

1392
A_c
A_j
A_p
A
Obs
JmaxRef = 23.63
RdRef = 0.43
sigma = 0.08
Theta = 0.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
200
400
600
800
1000

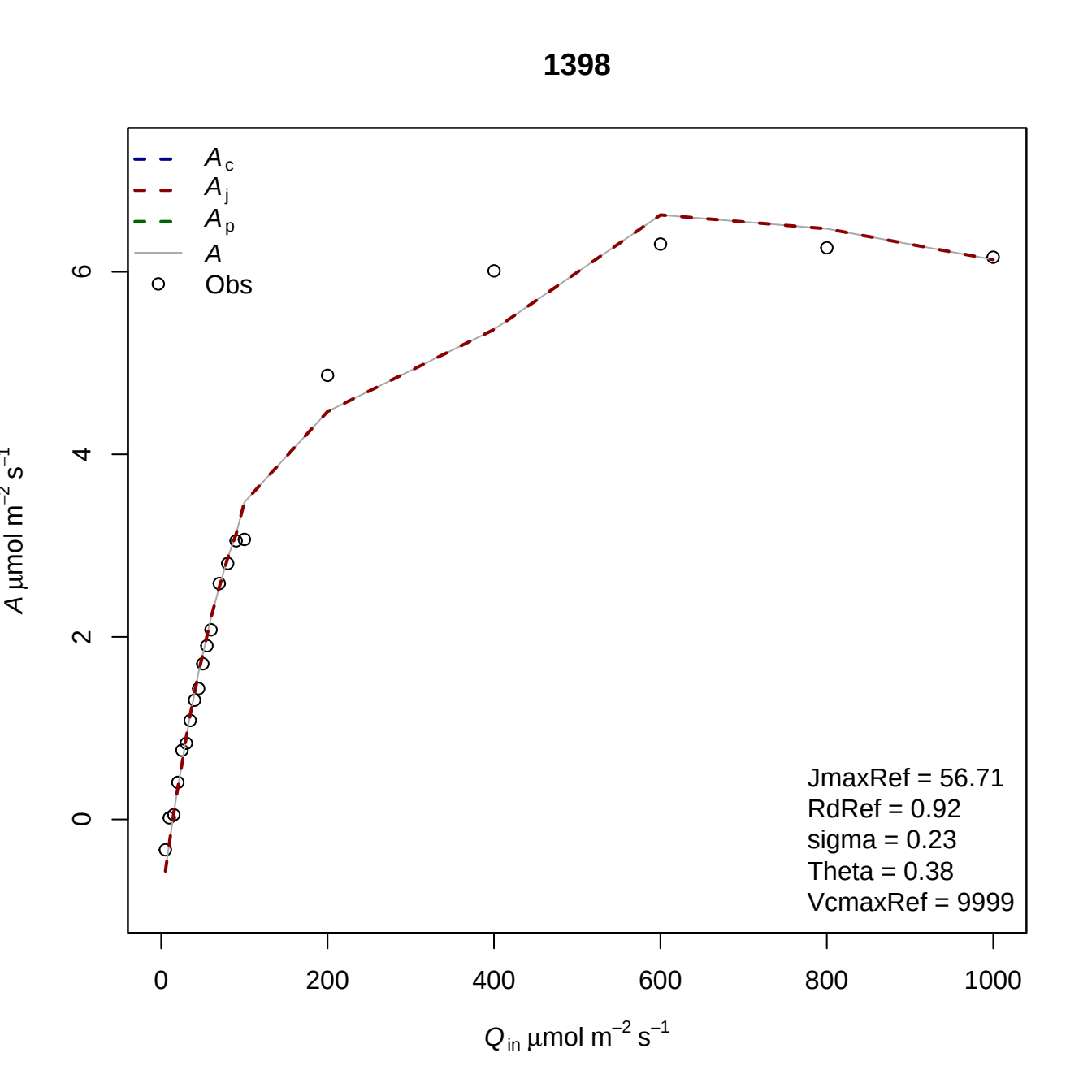

1393
A_c
A_j
A_p
A
Obs
JmaxRef = 23.08
RdRef = 0.82
sigma = 0.09
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

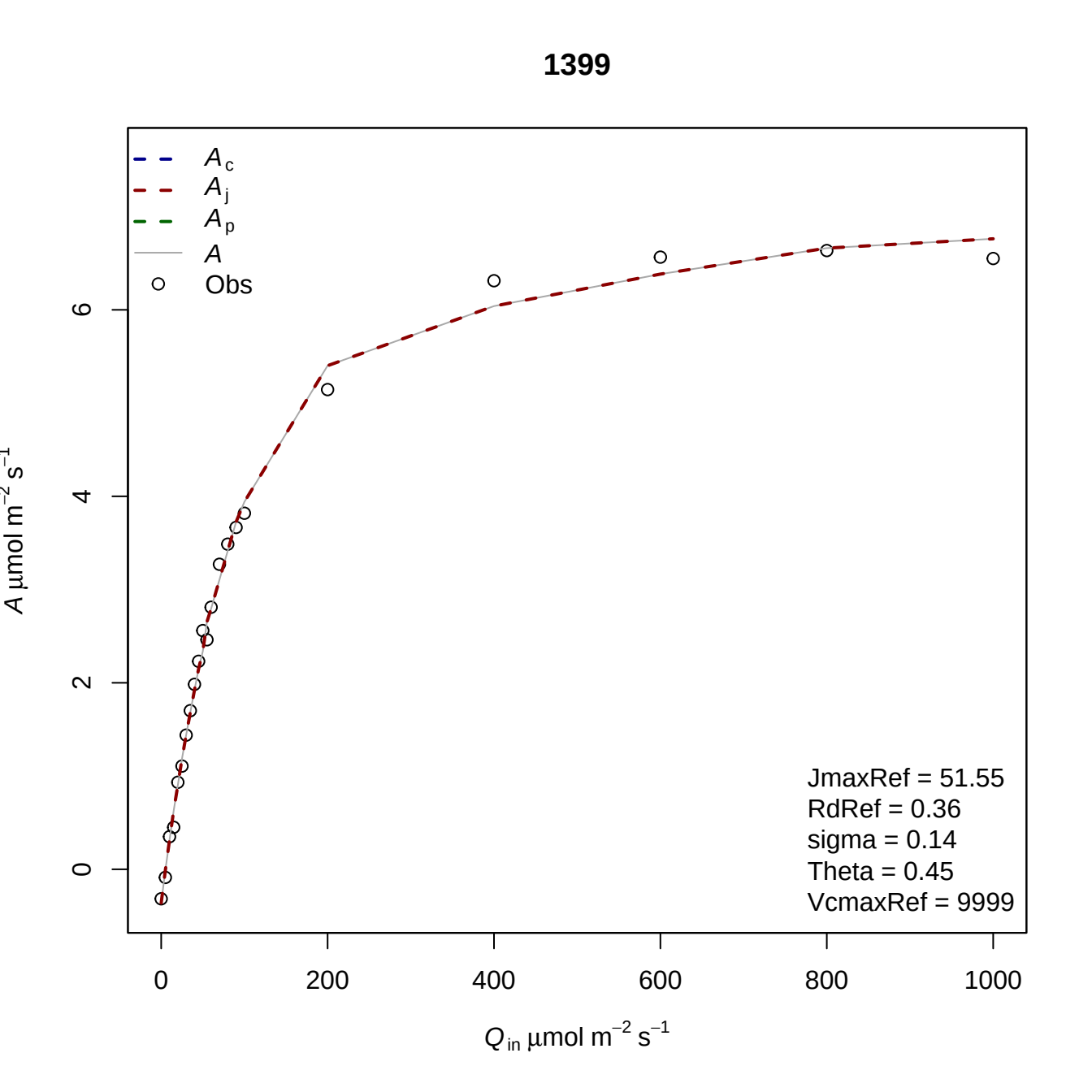

1394
A_c
A_j
A_p
A
Obs
JmaxRef = 55.32
RdRef = 0.67
sigma = 0.18
Theta = 0.6
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800

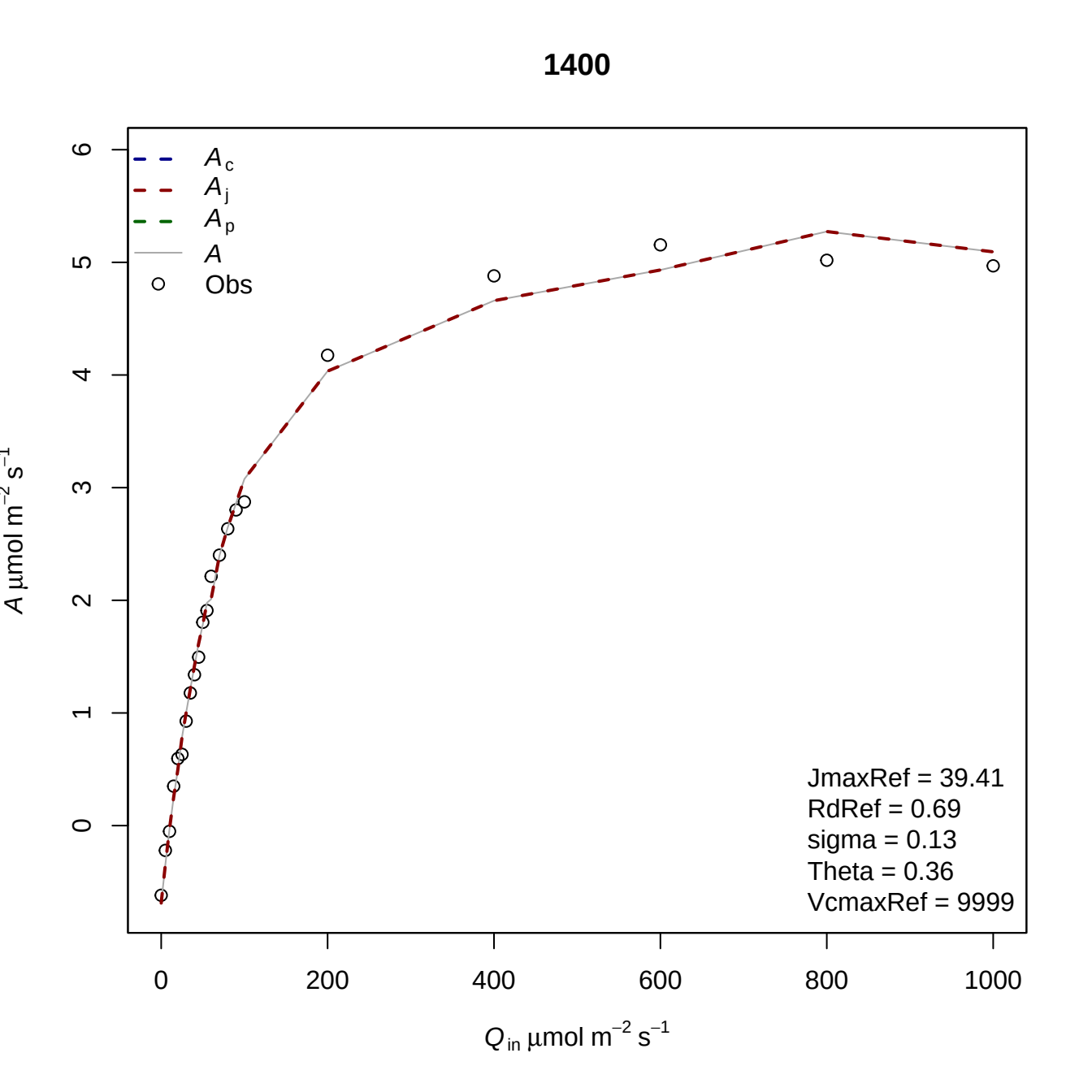

1395
A_c
A_j
A_p
A
Obs
JmaxRef = 101.67
RdRef = 0.63
sigma = 0.1
Theta = 0.24
VcmaxRef = 9999
0
5
10
15
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

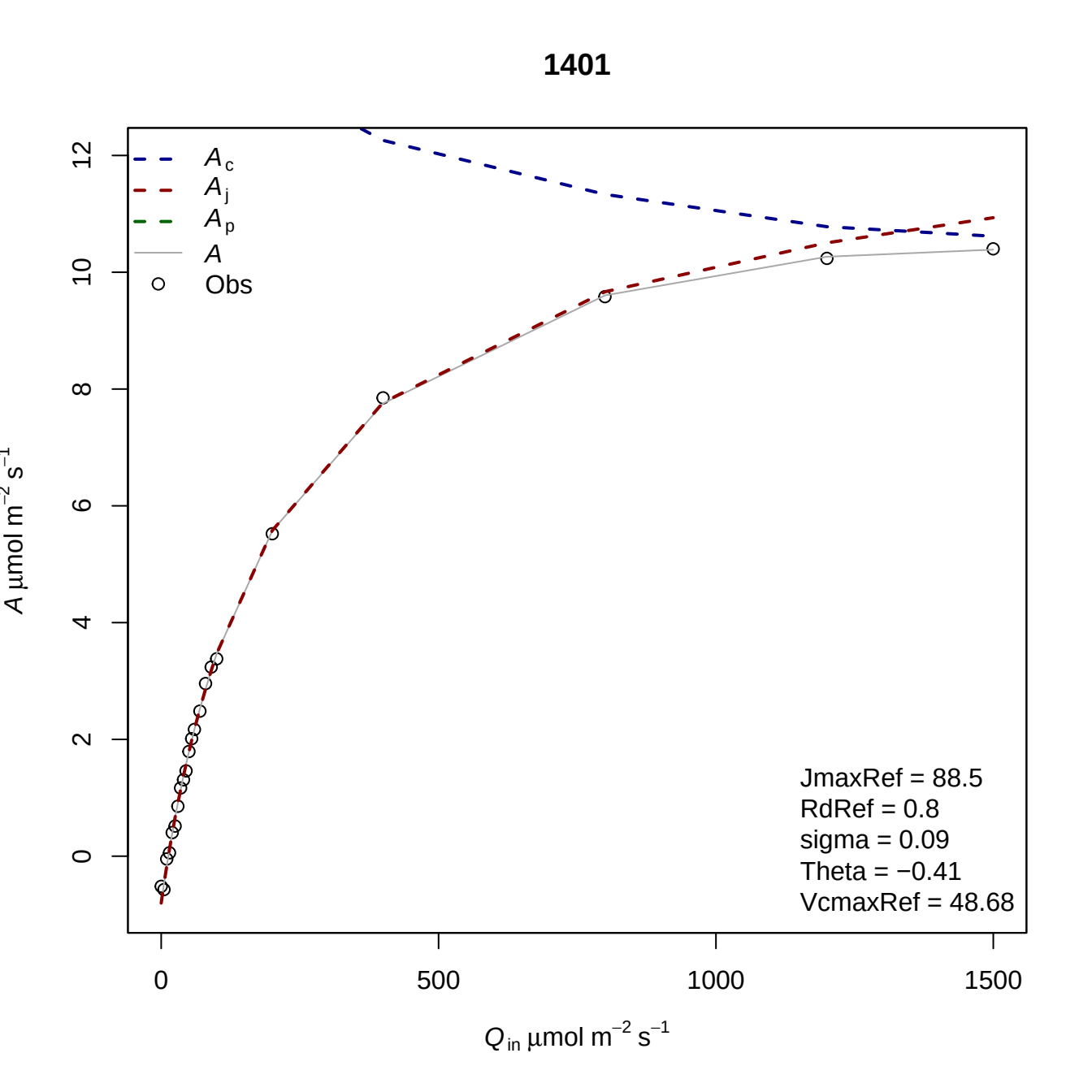

1396
A_c
A_j
A_p
A
Obs
JmaxRef = 40.97
RdRef = 0.4
sigma = 0.13
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

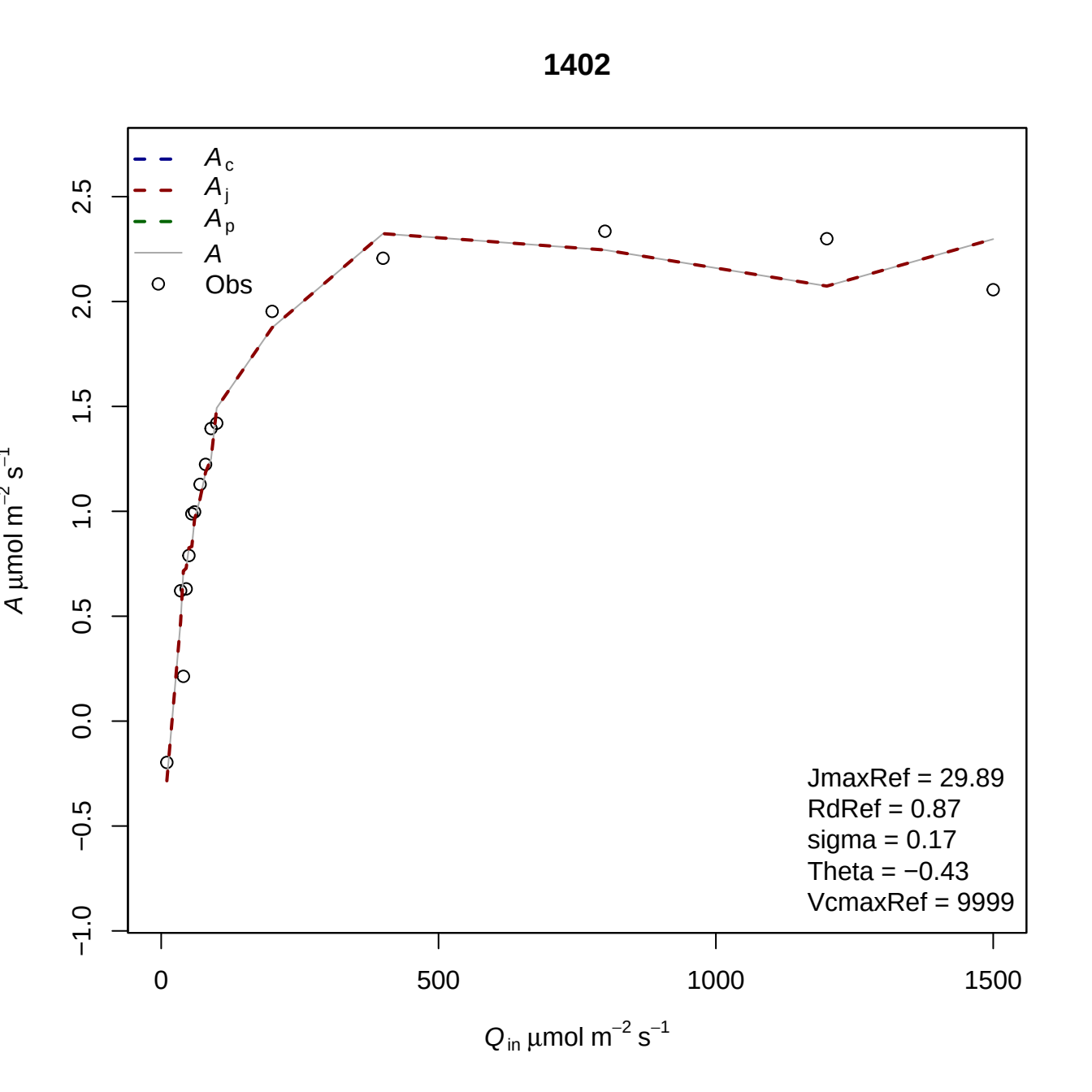

1397
A_c
A_j
A_p
A
Obs
JmaxRef = 40.92
RdRef = 0.55
sigma = 0.12
Theta = 0.12
VcmaxRef = 22.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

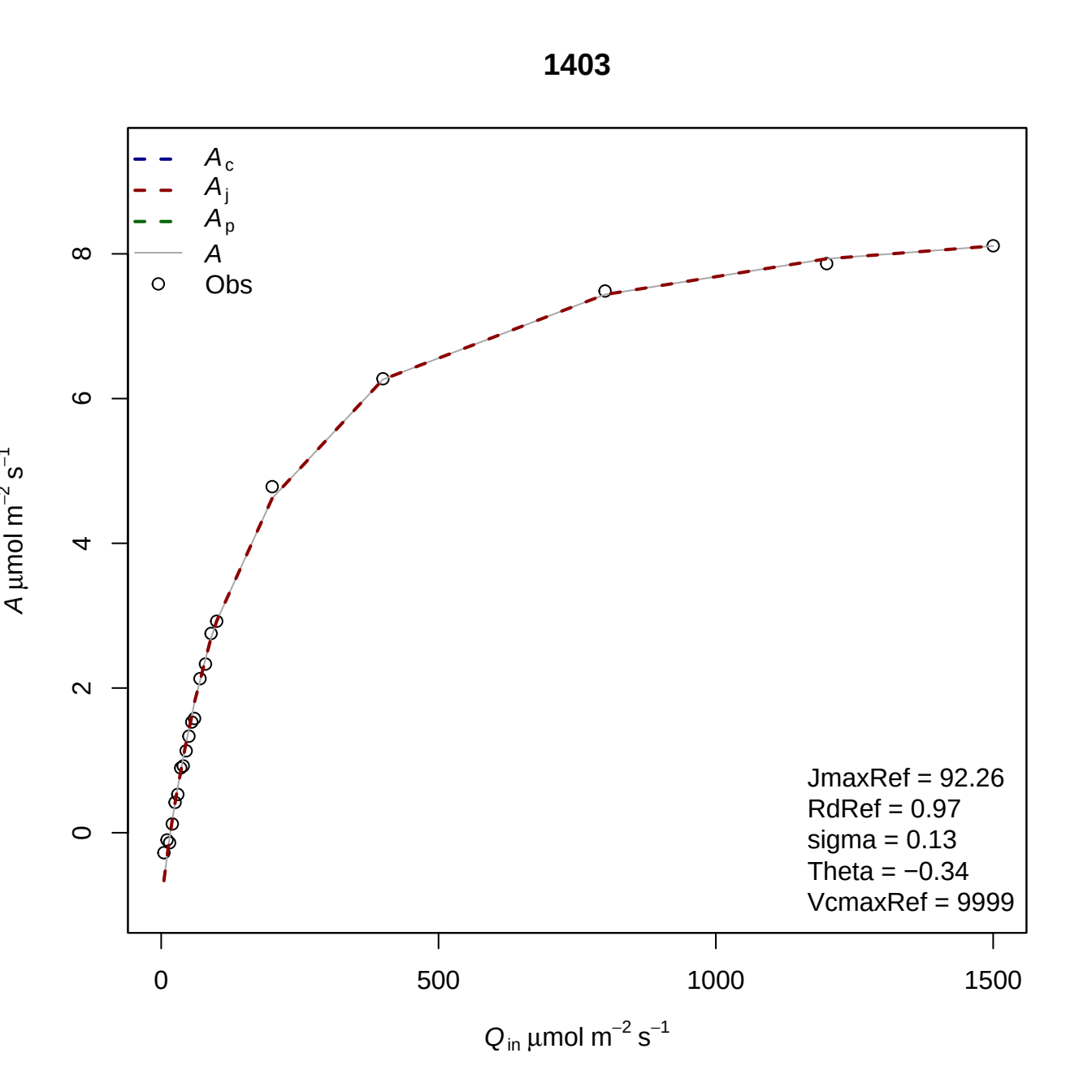

1398
A_c
A_j
A_p
A
Obs
JmaxRef = 56.71
RdRef = 0.92
sigma = 0.23
Theta = 0.38
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

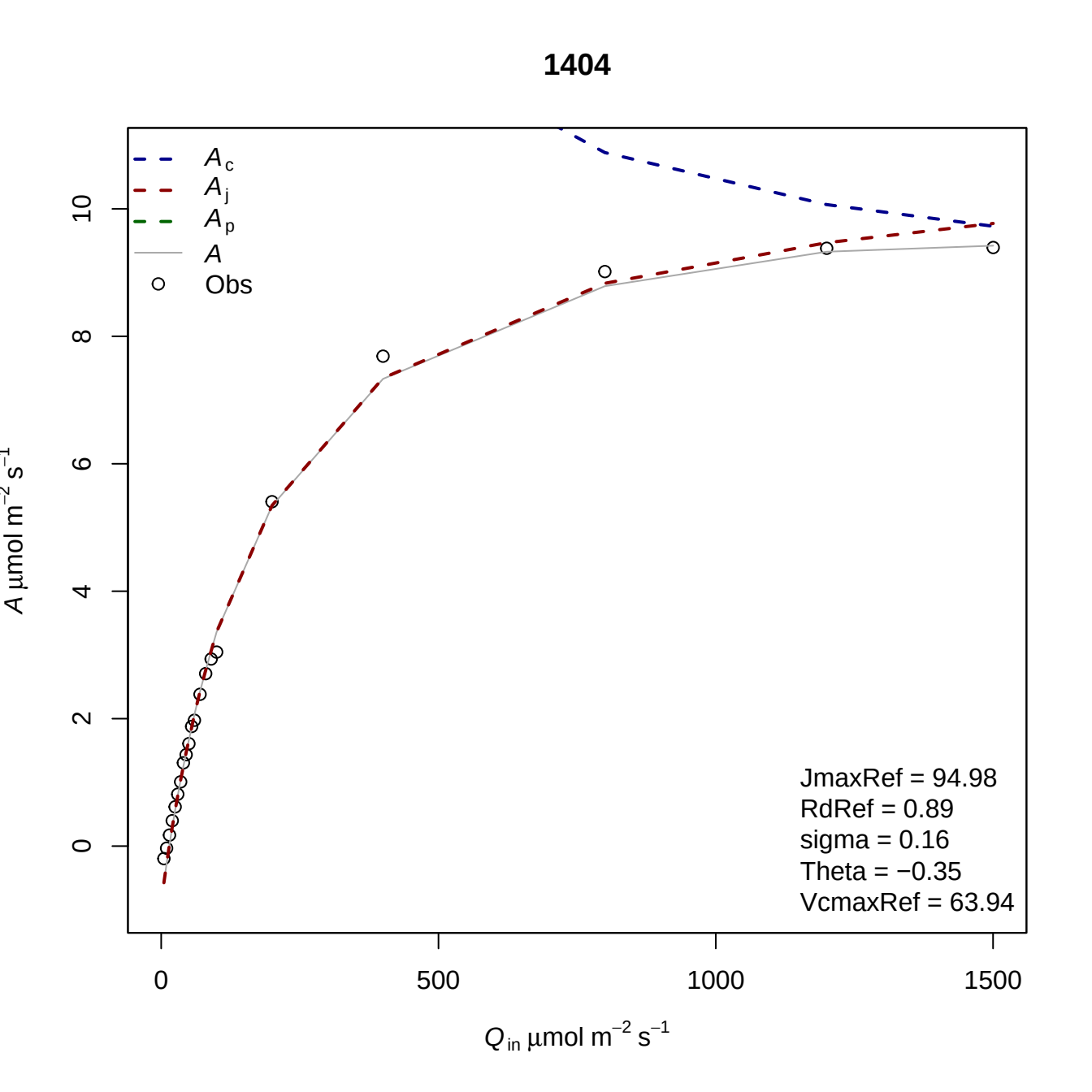

1399
A_c
A_j
A_p
A
Obs
JmaxRef = 51.55
RdRef = 0.36
sigma = 0.14
Theta = 0.45
VcmaxRef = 9999
0
2
4
6
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

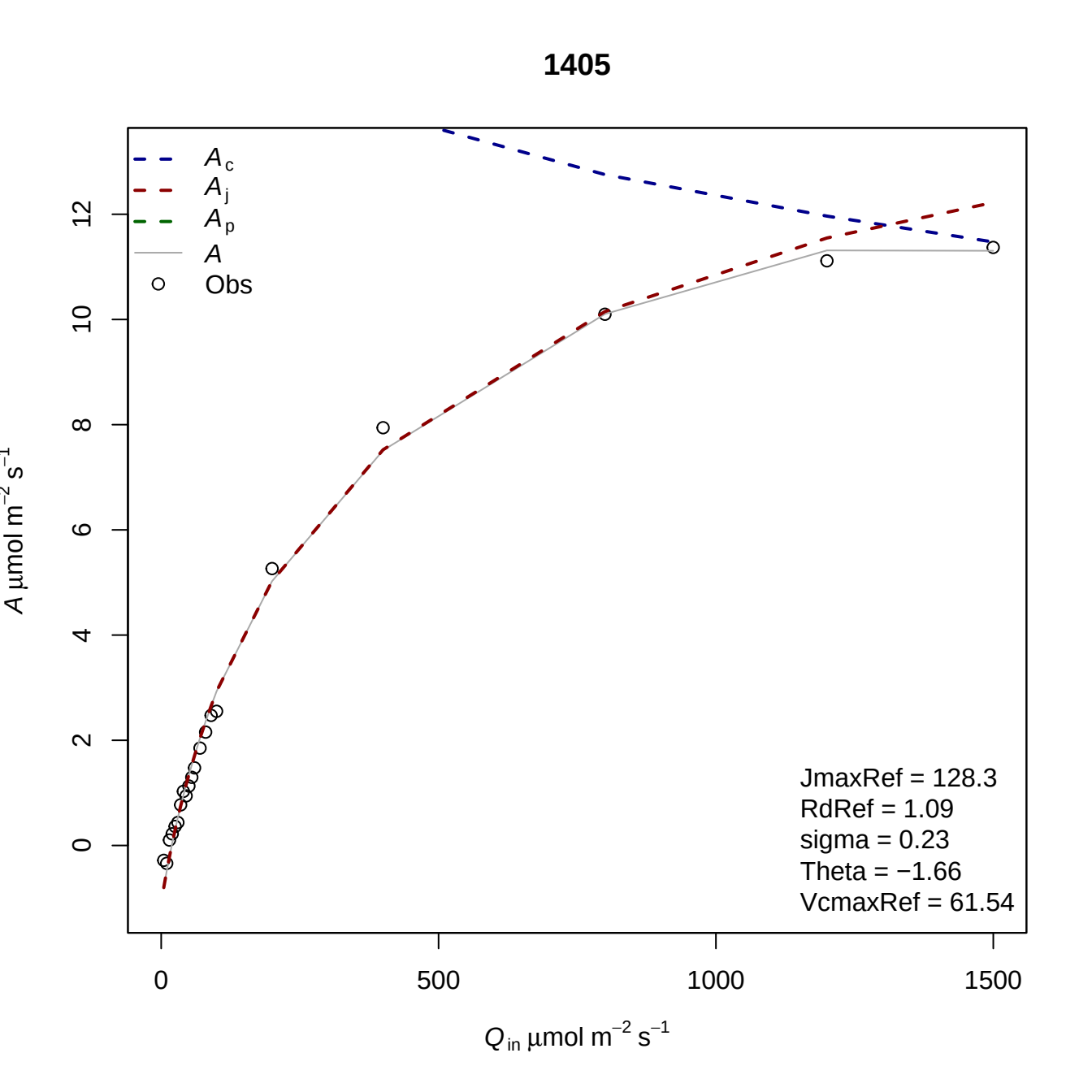

1400
A_c
A_j
A_p
A
Obs
JmaxRef = 39.41
RdRef = 0.69
sigma = 0.13
Theta = 0.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

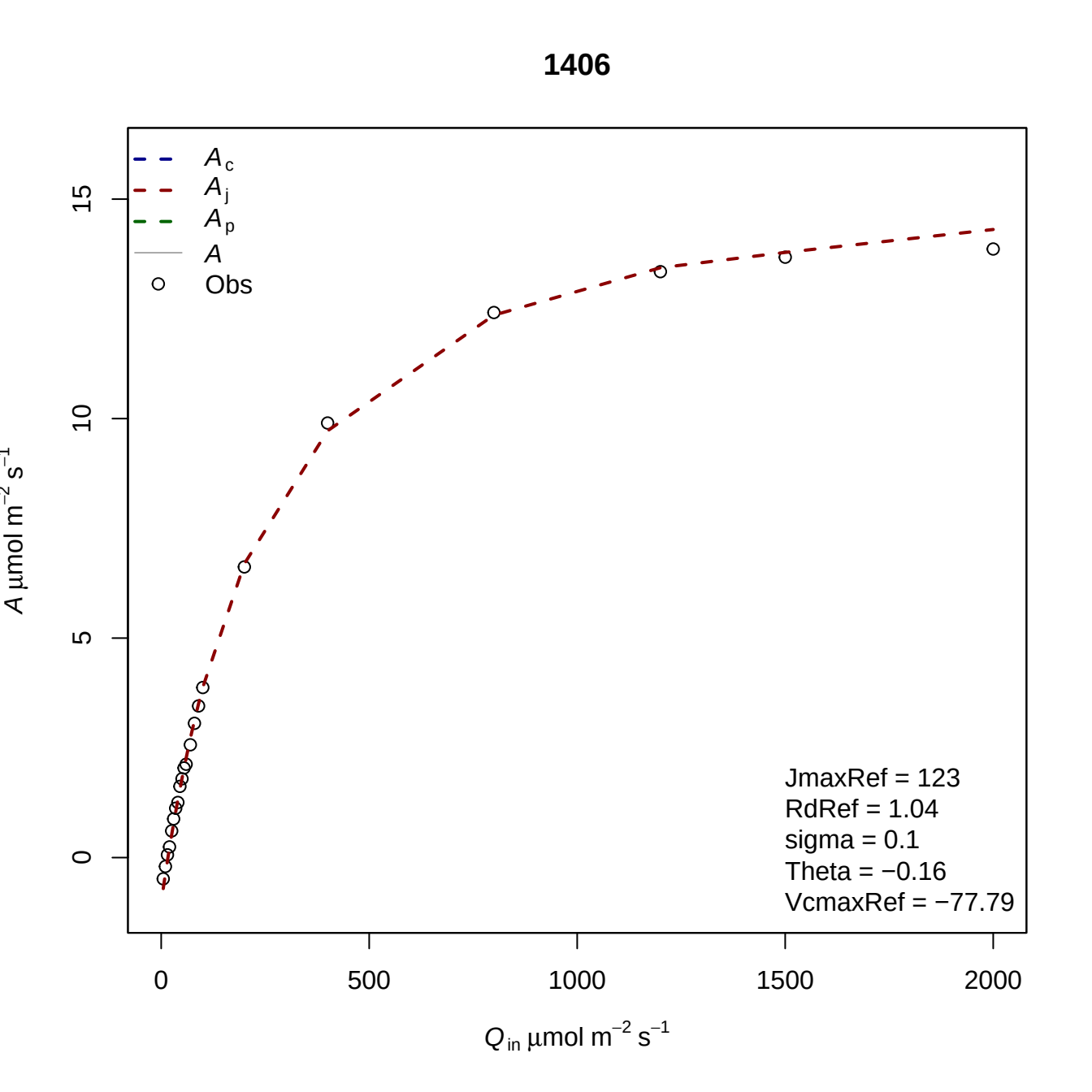

1401
A_c
A_j
A_p
A
Obs
JmaxRef = 88.5
RdRef = 0.8
sigma = 0.09
Theta = −0.41
VcmaxRef = 48.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

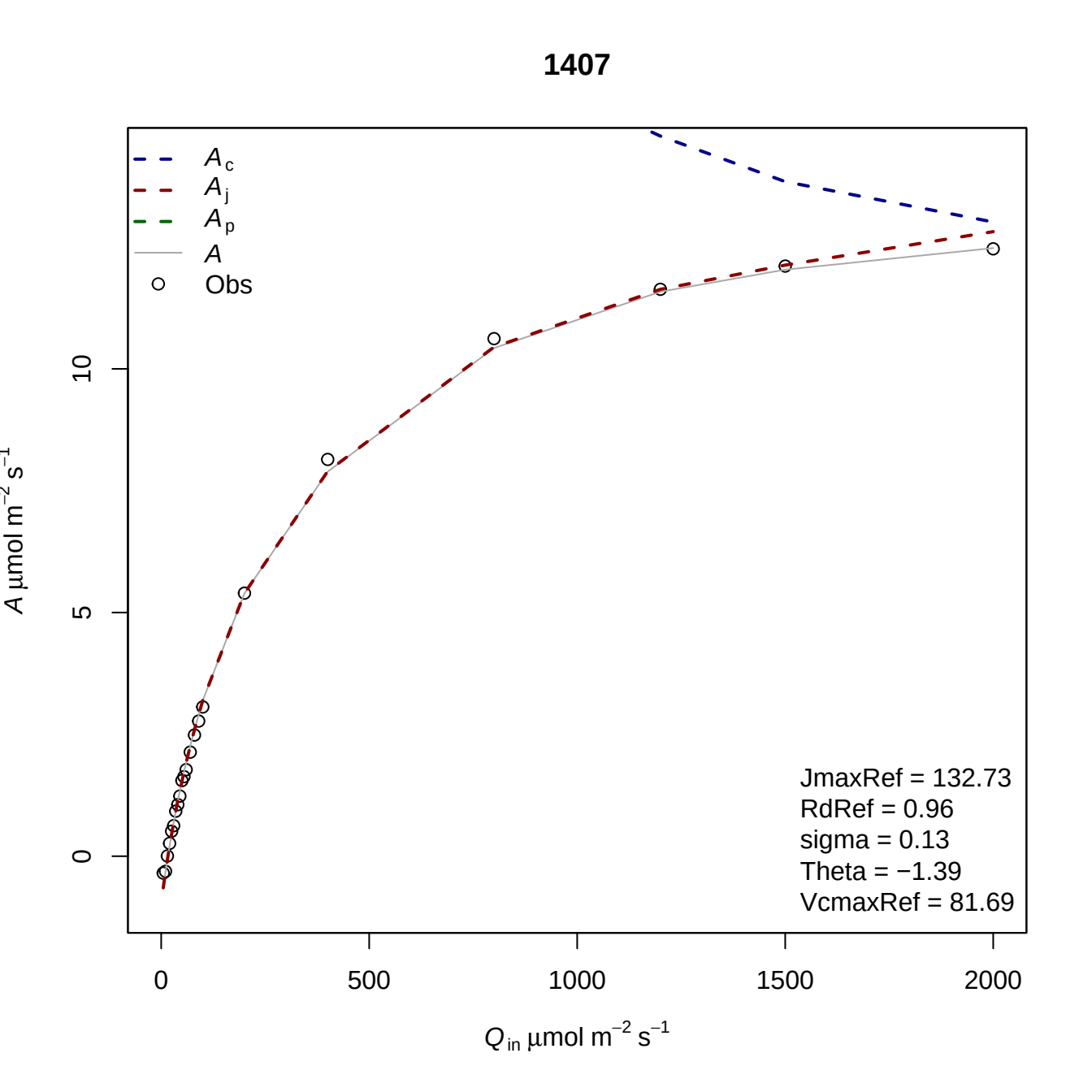

1402
A_c
A_j
A_p
A
Obs
JmaxRef = 29.89
RdRef = 0.87
sigma = 0.17
Theta = −0.43
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

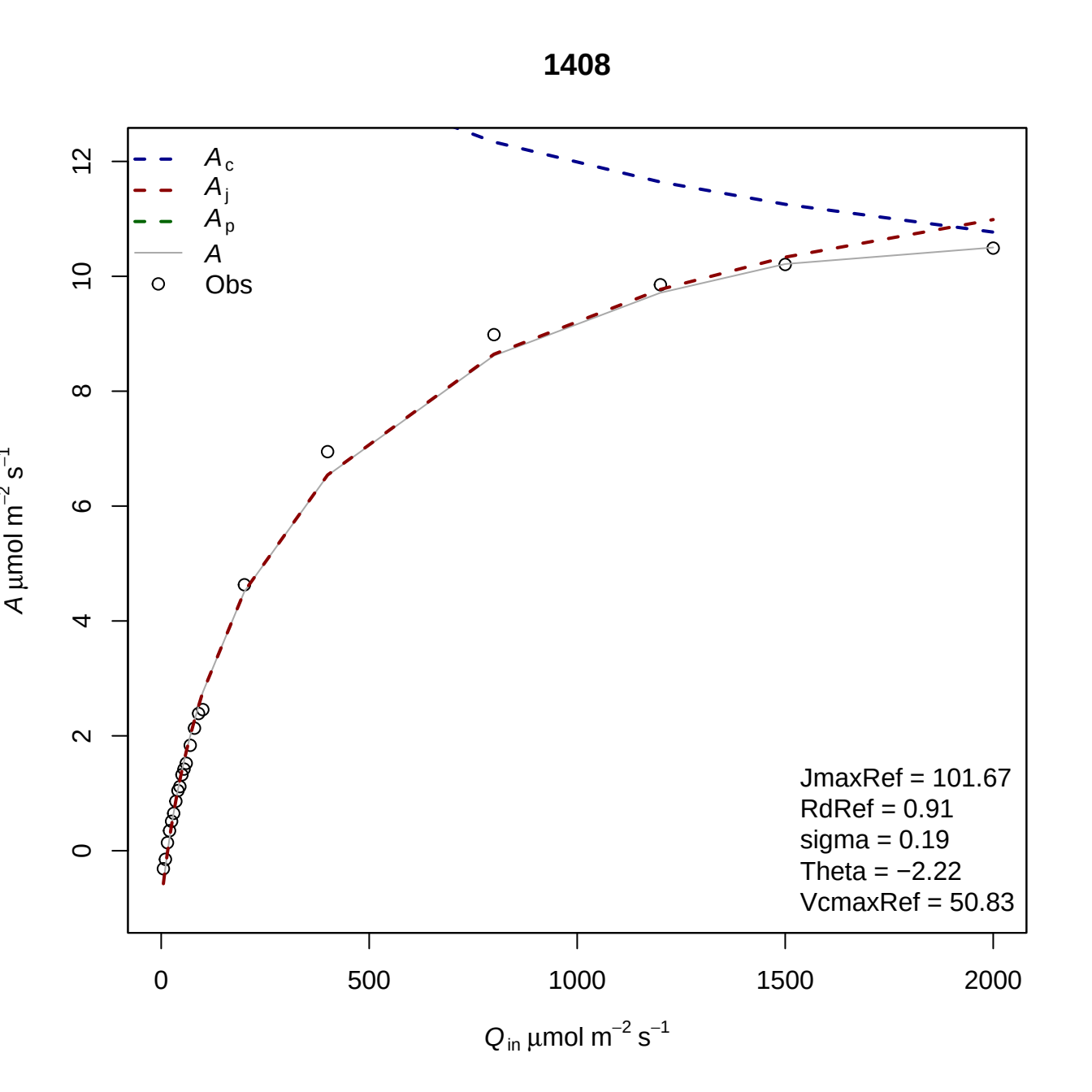

1403
A_c
A_j
A_p
A
Obs
JmaxRef = 92.26
RdRef = 0.97
sigma = 0.13
Theta = −0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

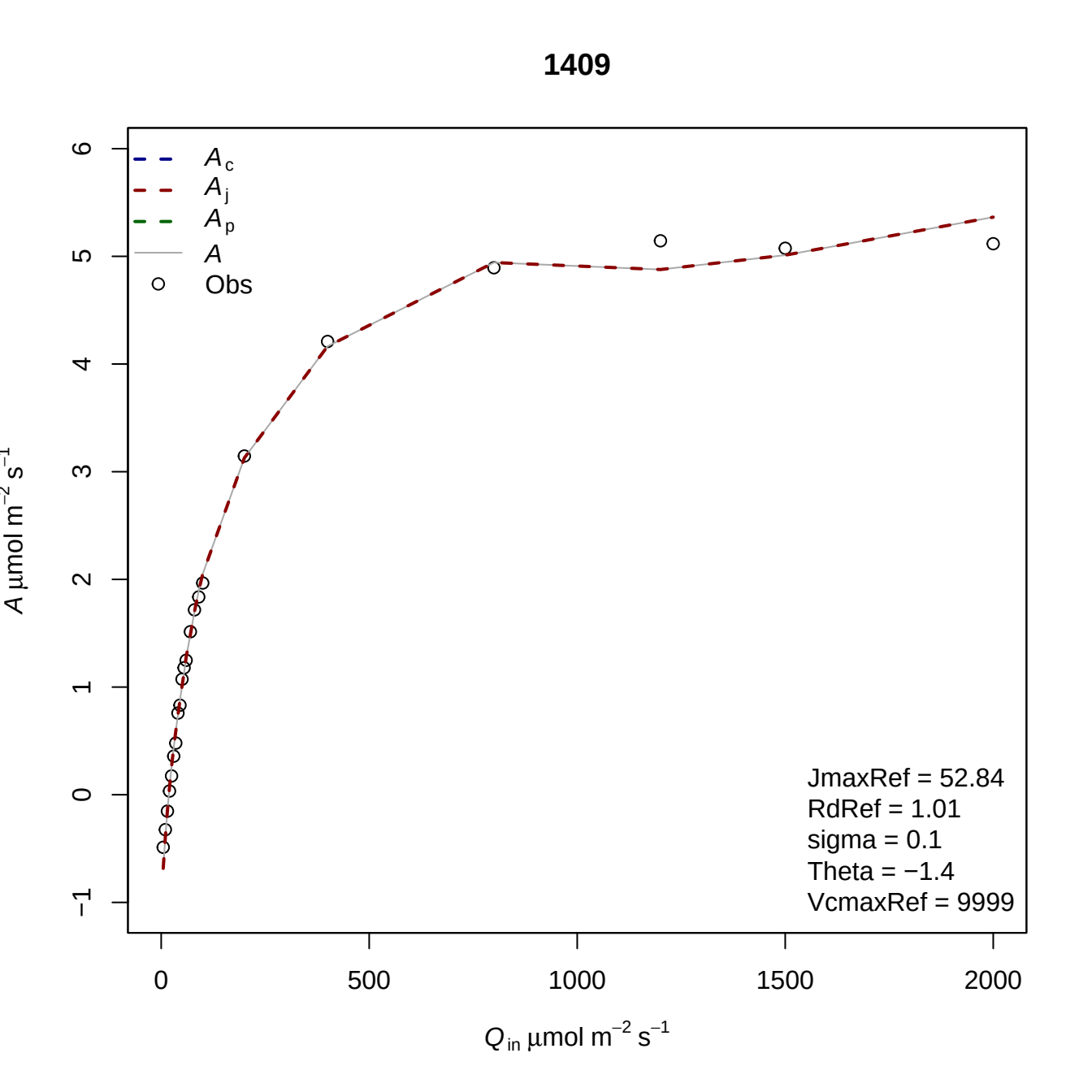

1404
A_c
A_j
A_p
A
Obs
JmaxRef = 94.98
RdRef = 0.89
sigma = 0.16
Theta = −0.35
VcmaxRef = 63.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

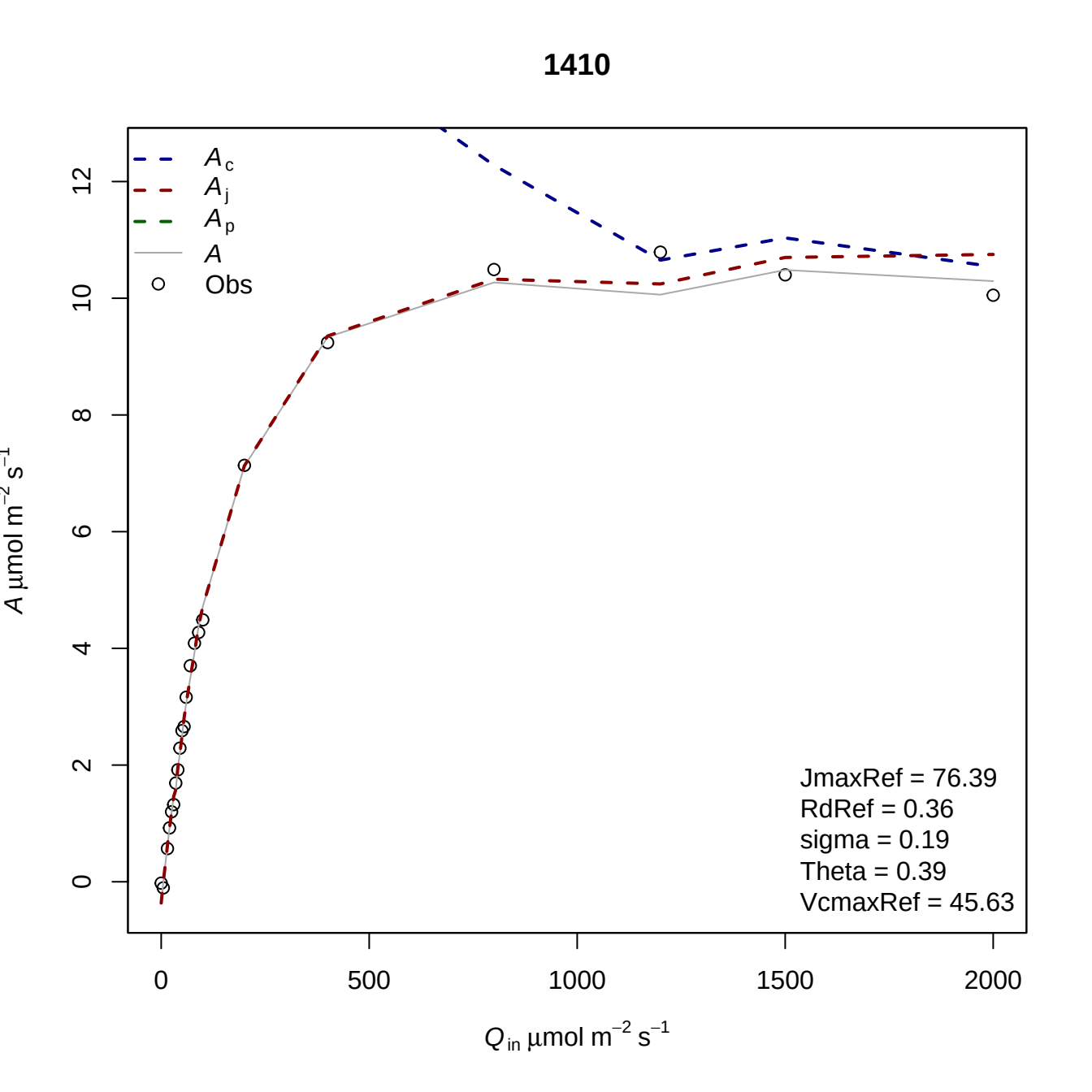

1405
A_c
A_j
A_p
A
Obs
JmaxRef = 128.3
RdRef = 1.09
sigma = 0.23
Theta = −1.66
VcmaxRef = 61.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

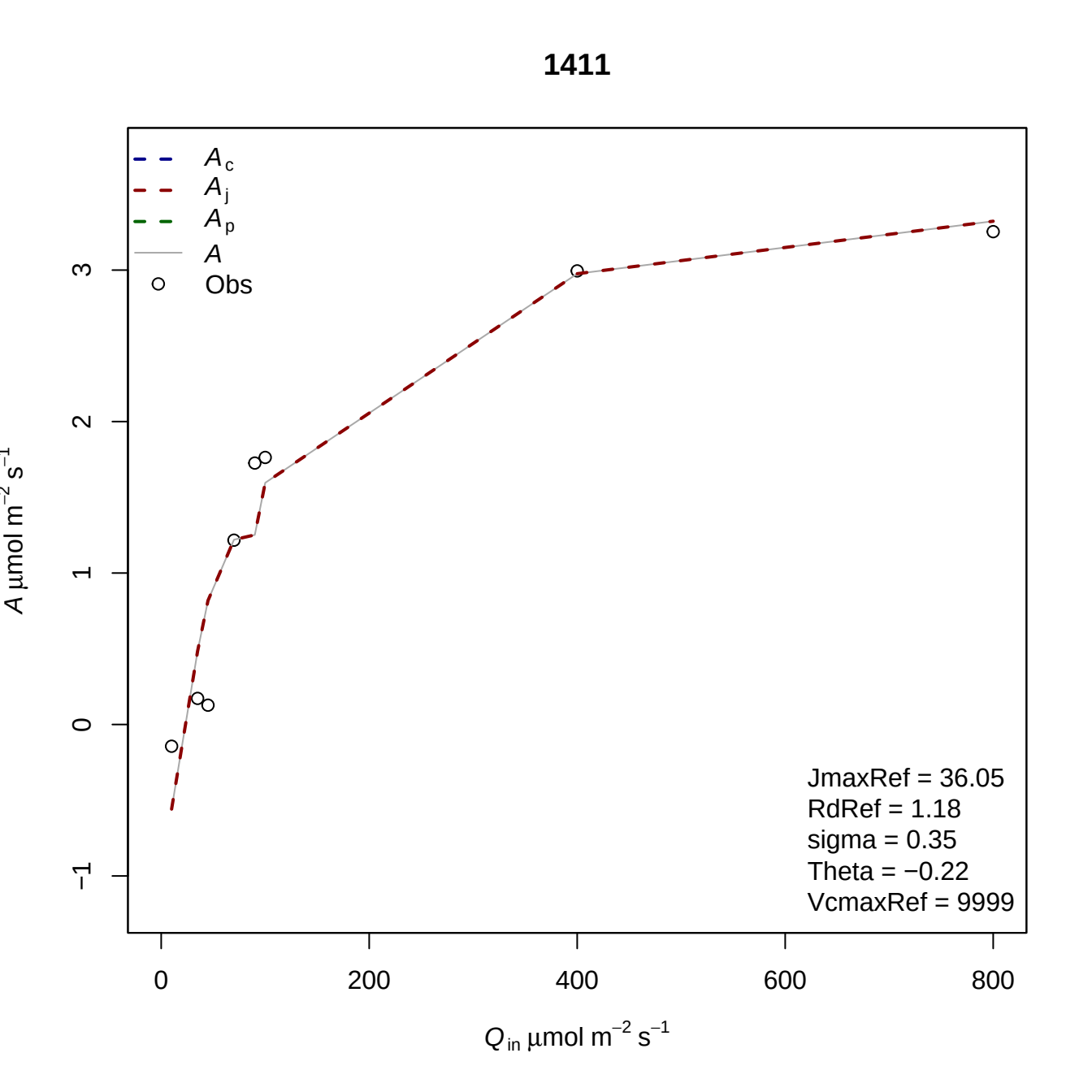

1406
A_c
A_j
A_p
A
Obs
JmaxRef = 123
RdRef = 1.04
sigma = 0.1
Theta = −0.16
VcmaxRef = −77.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

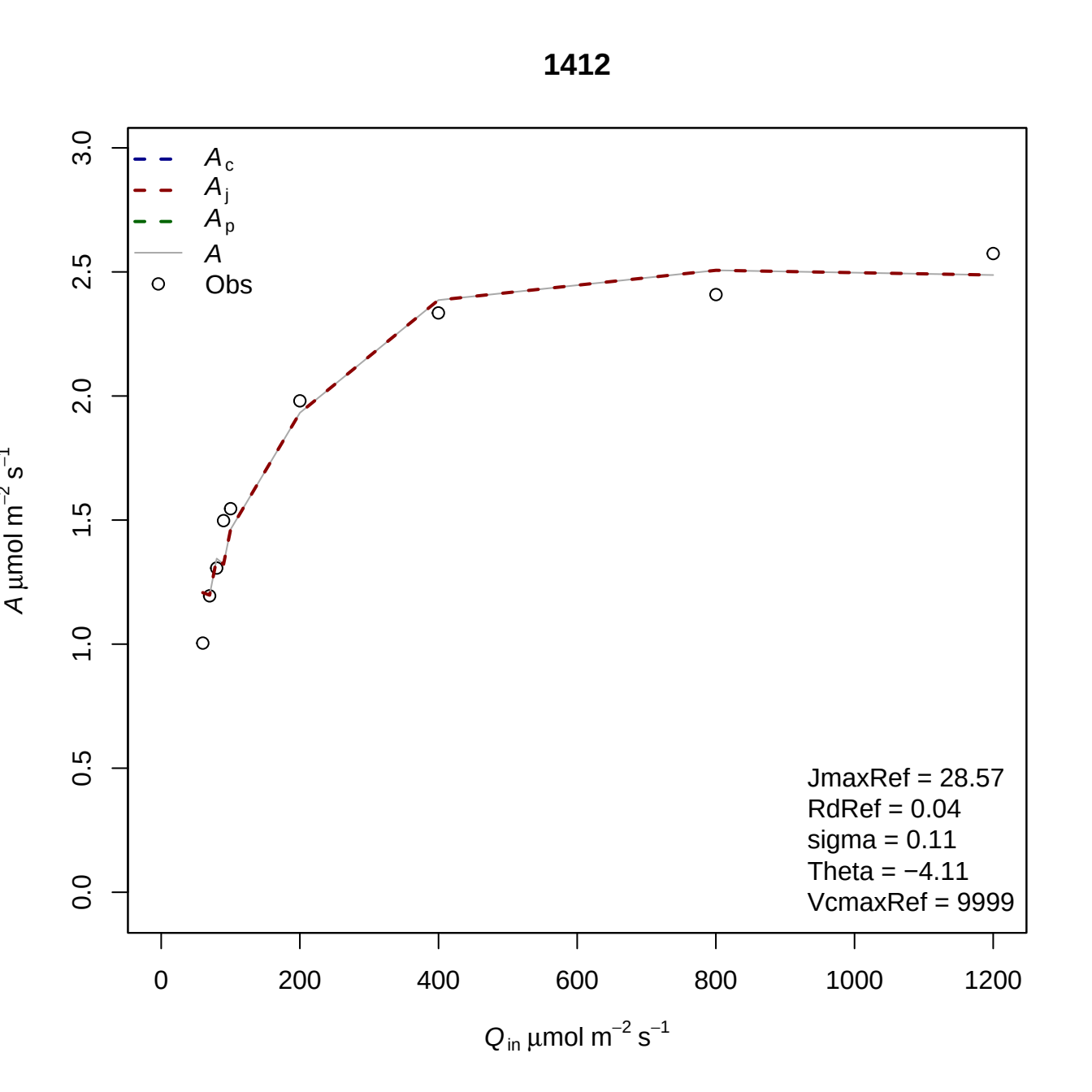

1407
A_c
A_j
A_p
A
Obs
JmaxRef = 132.73
RdRef = 0.96
sigma = 0.13
Theta = −1.39
VcmaxRef = 81.69
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

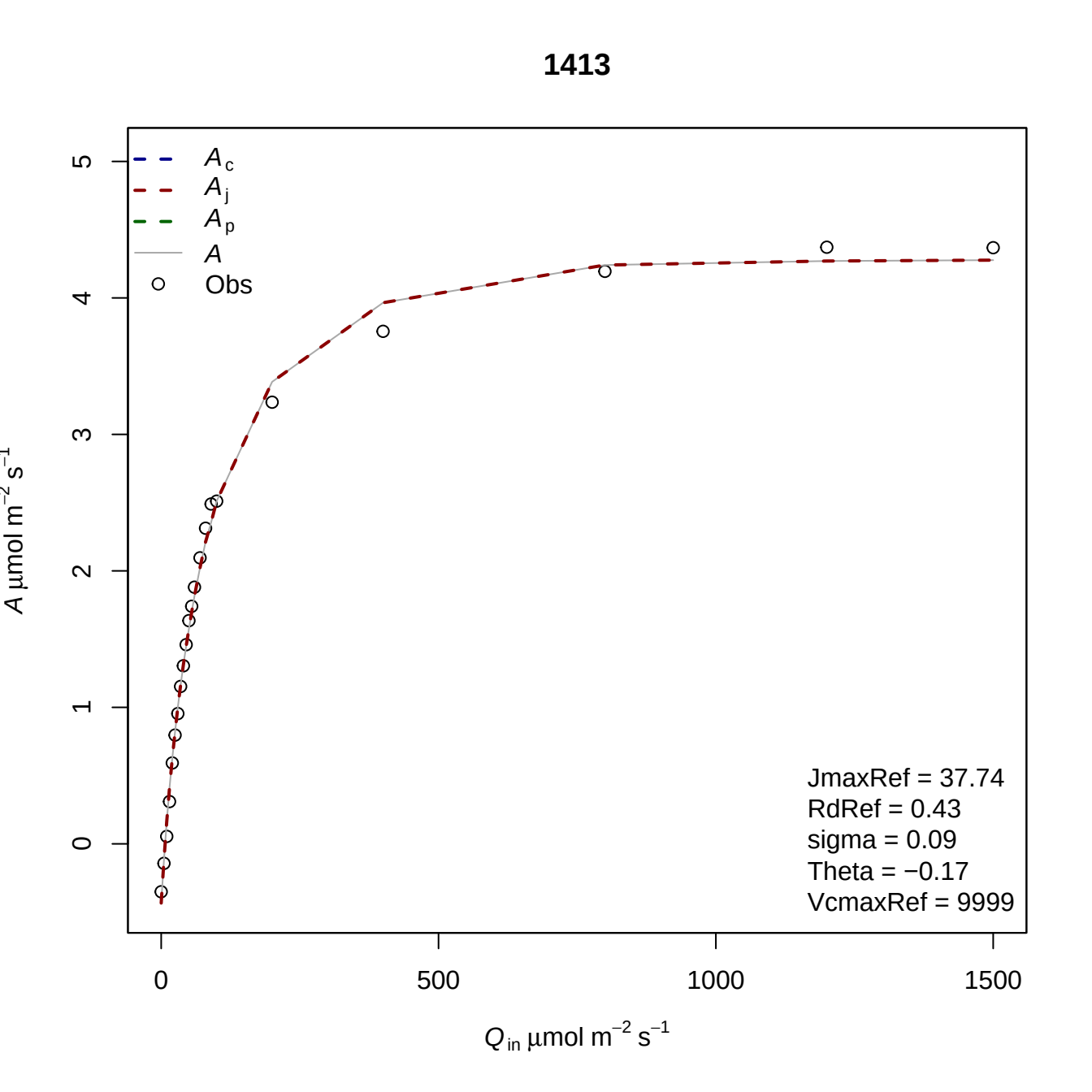

1408
A_c
A_j
A_p
A
Obs
JmaxRef = 101.67
RdRef = 0.91
sigma = 0.19
Theta = −2.22
VcmaxRef = 50.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

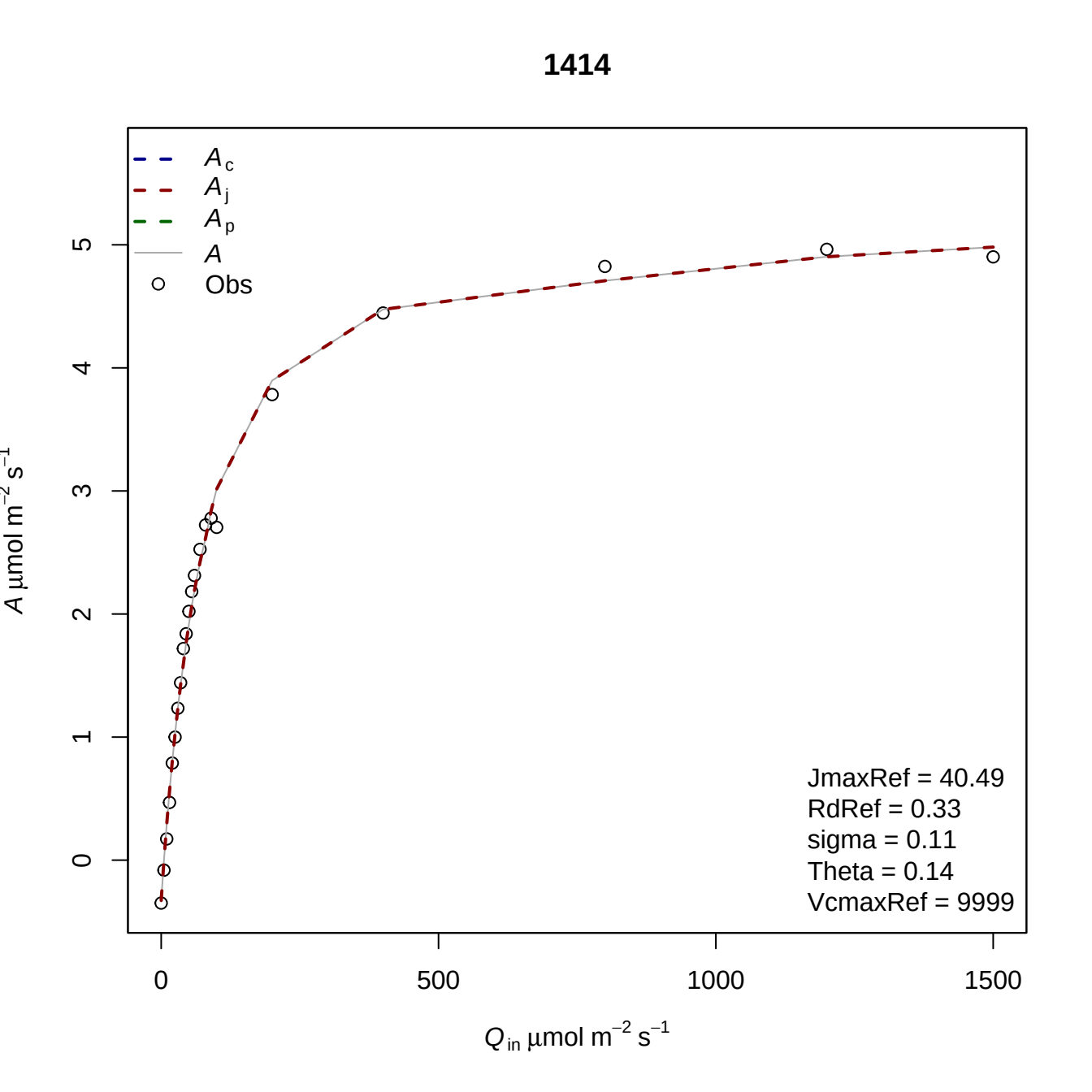

1409
A_c
A_j
A_p
A
Obs
JmaxRef = 52.84
RdRef = 1.01
sigma = 0.1
Theta = −1.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

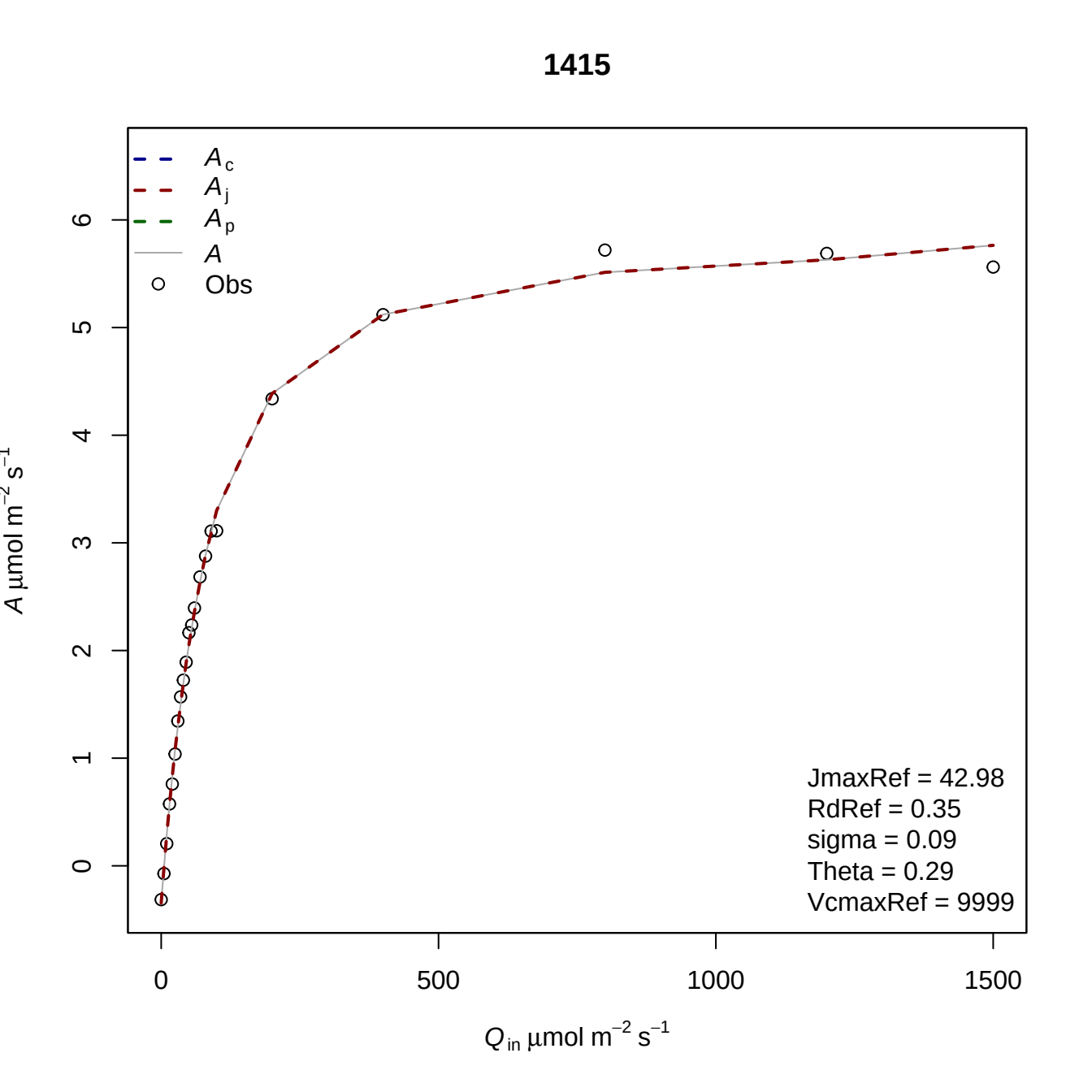

1410
A_c
A_j
A_p
A
Obs
JmaxRef = 76.39
RdRef = 0.36
sigma = 0.19
Theta = 0.39
VcmaxRef = 45.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

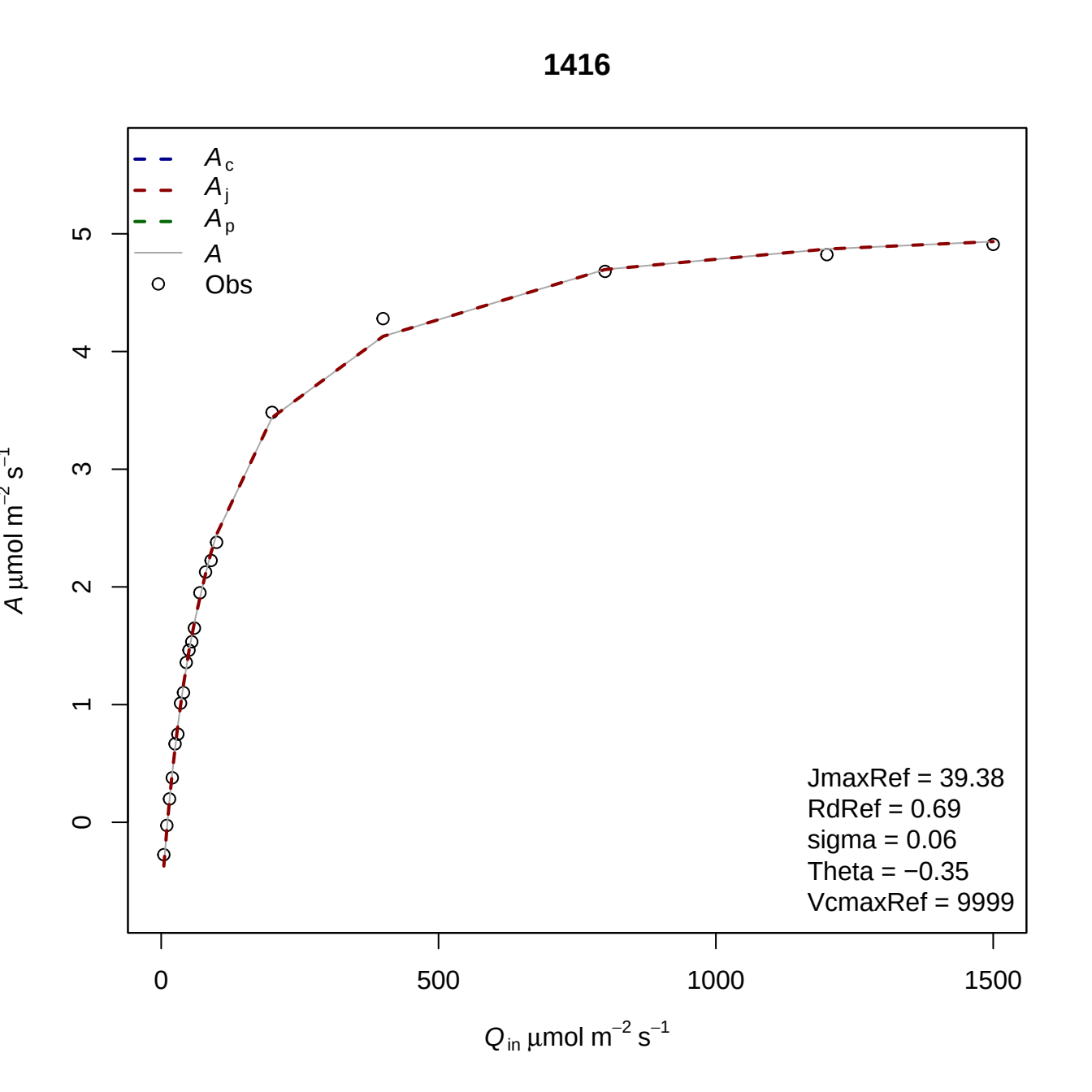

1411
A_c
A_j
A_p
A
Obs
JmaxRef = 36.05
RdRef = 1.18
sigma = 0.35
Theta = −0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800

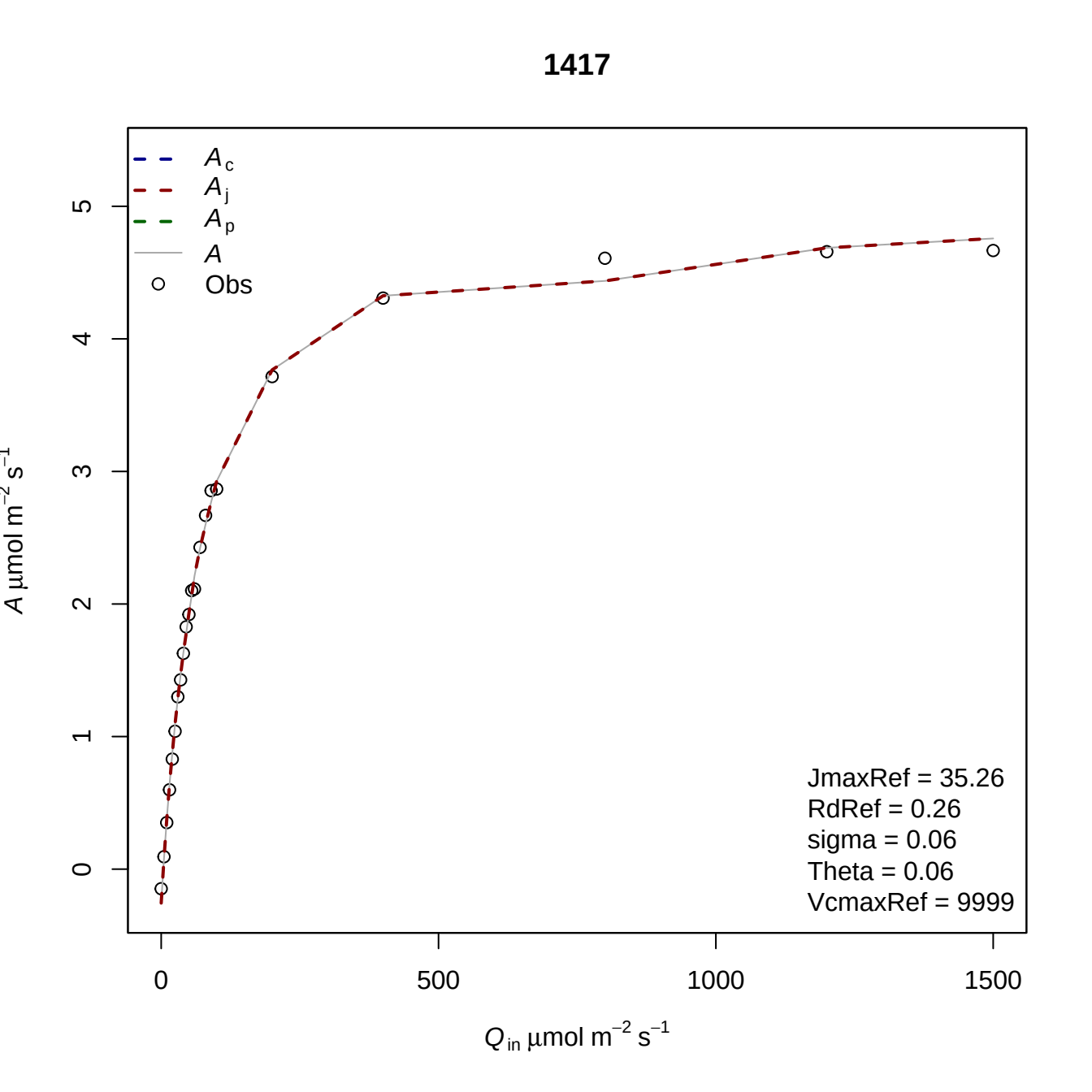

1412
A_c
A_j
A_p
A
Obs
JmaxRef = 28.57
RdRef = 0.04
sigma = 0.11
Theta = −4.11
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200

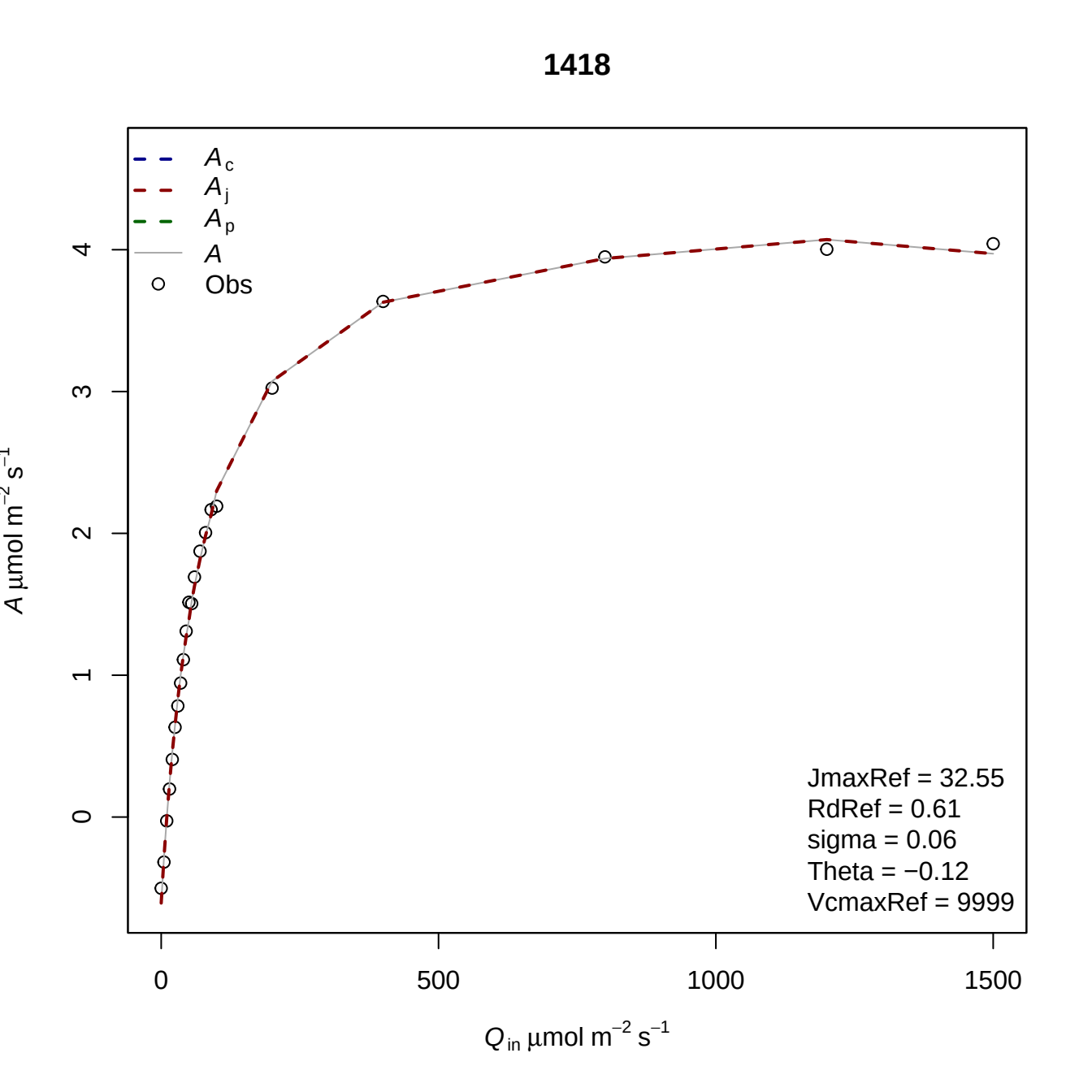

1413
A_c
A_j
A_p
A
Obs
JmaxRef = 37.74
RdRef = 0.43
sigma = 0.09
Theta = −0.17
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

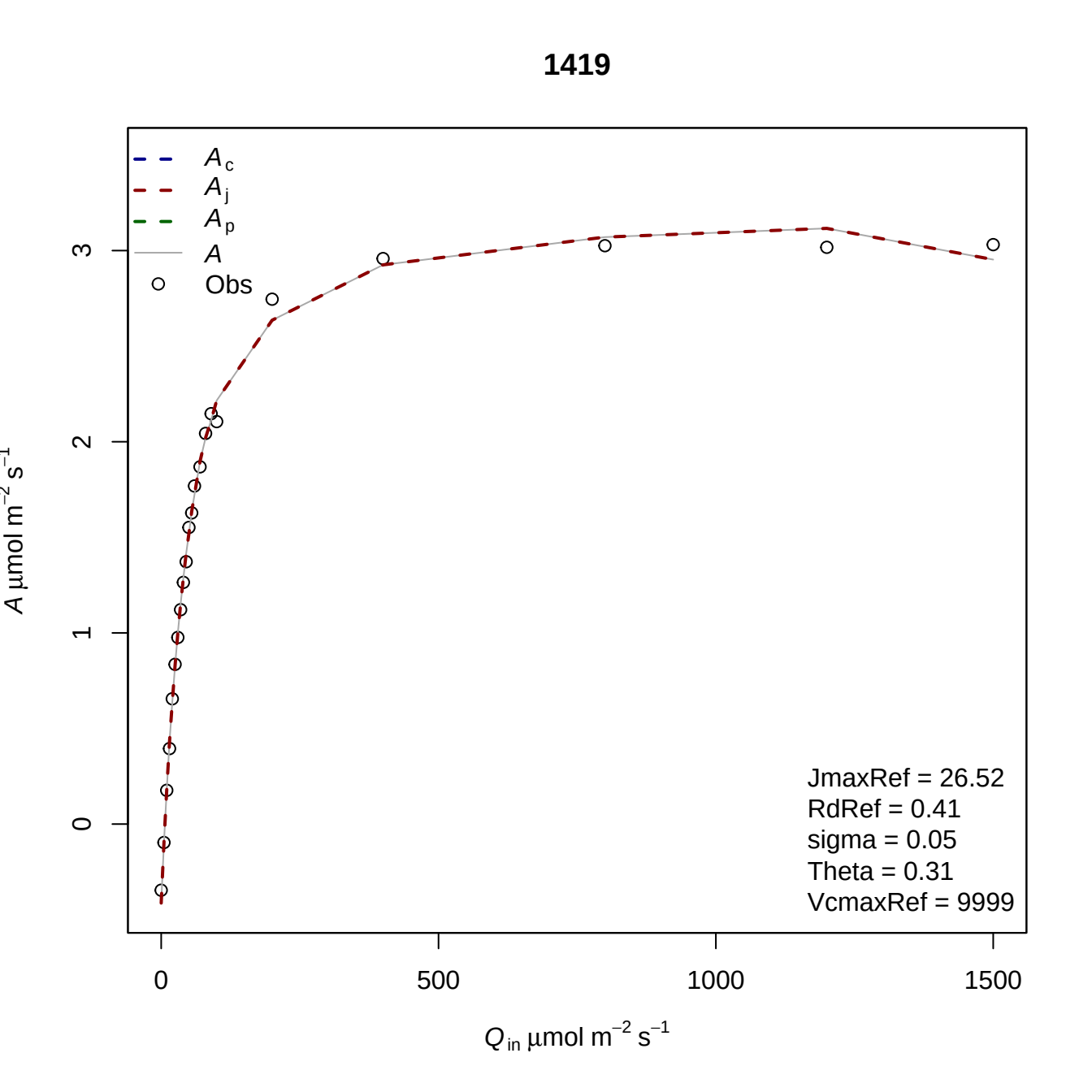

1414
A_c
A_j
A_p
A
Obs
JmaxRef = 40.49
RdRef = 0.33
sigma = 0.11
Theta = 0.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

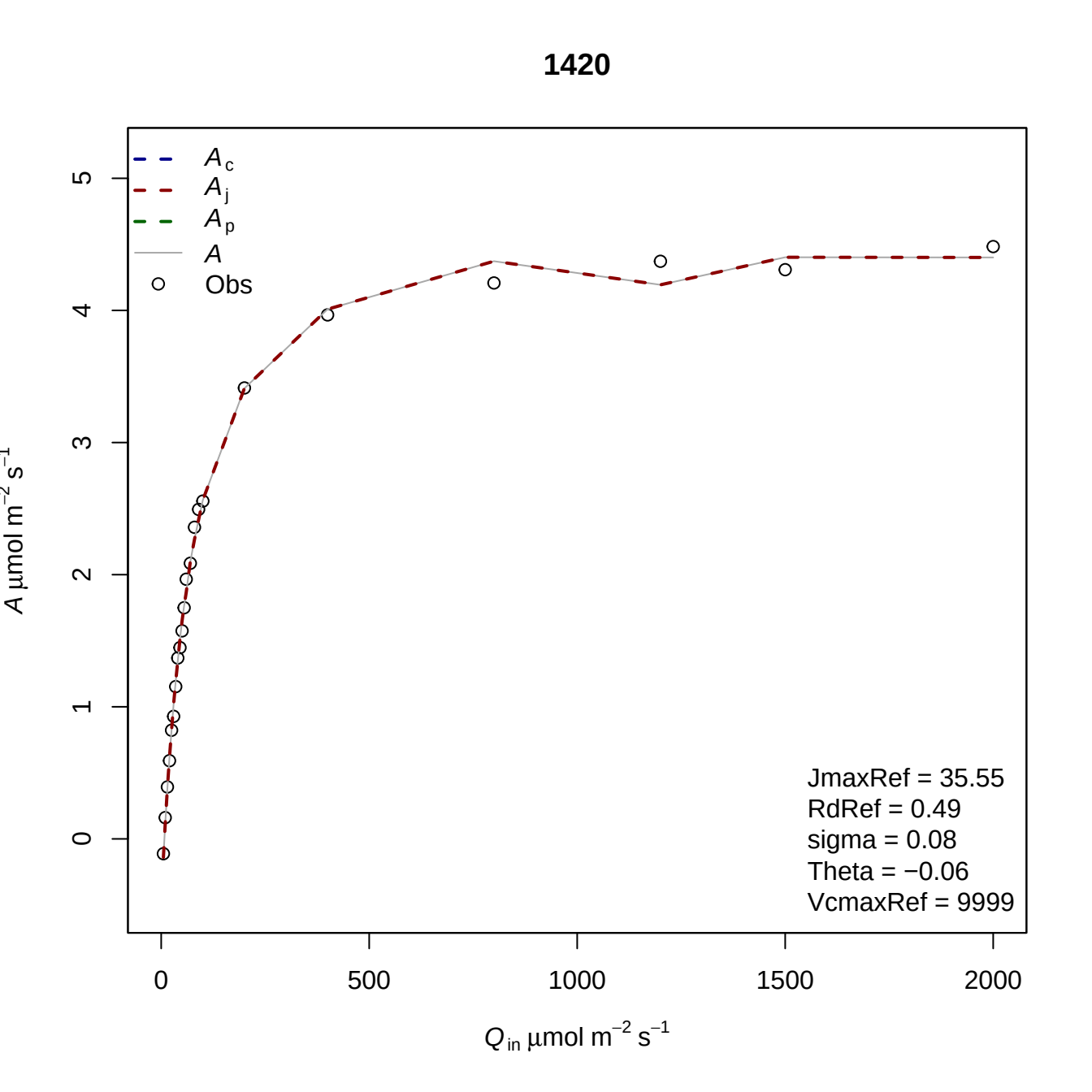

1415
A_c
A_j
A_p
A
Obs
JmaxRef = 42.98
RdRef = 0.35
sigma = 0.09
Theta = 0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

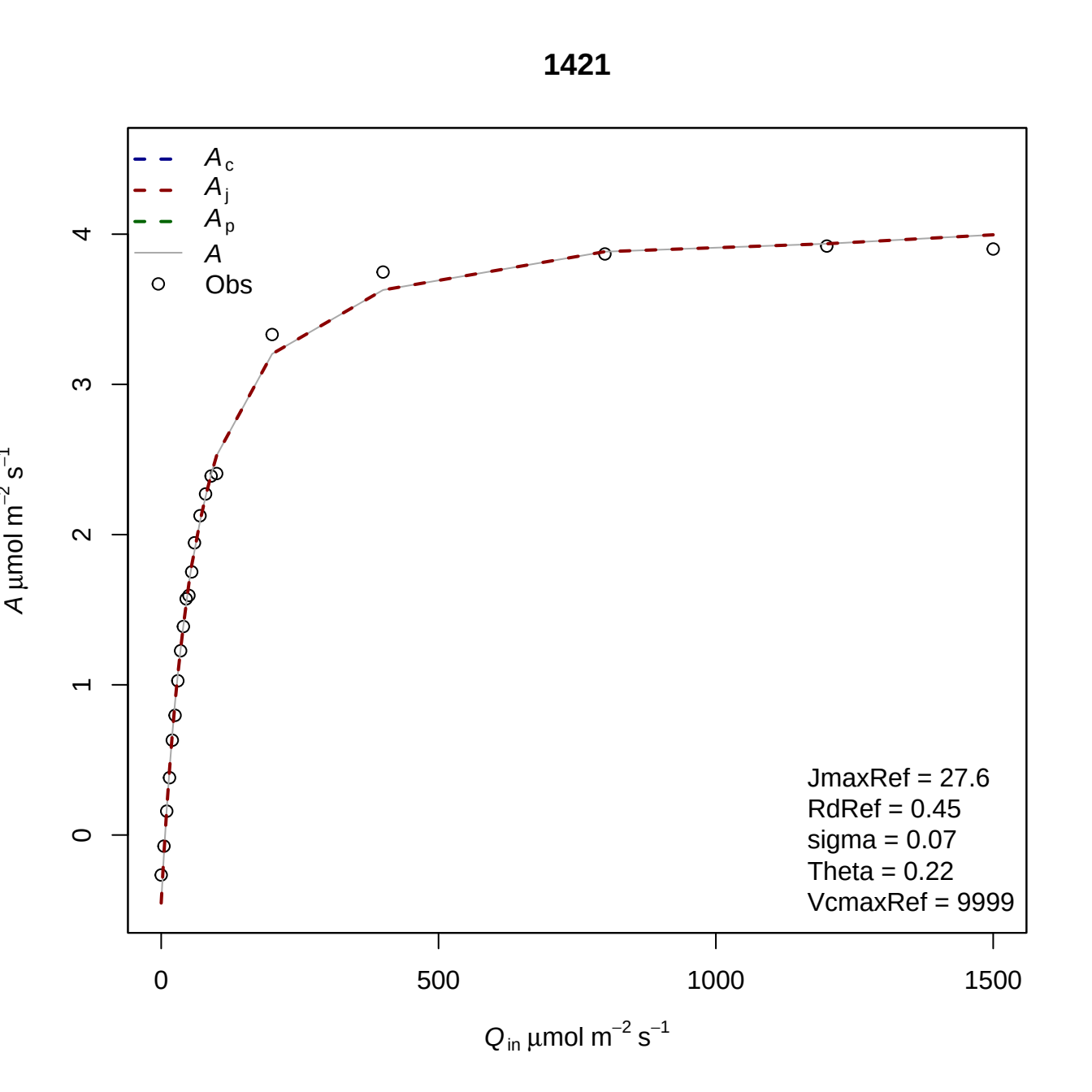

1416
A_c
A_j
A_p
A
Obs
JmaxRef = 39.38
RdRef = 0.69
sigma = 0.06
Theta = −0.35
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

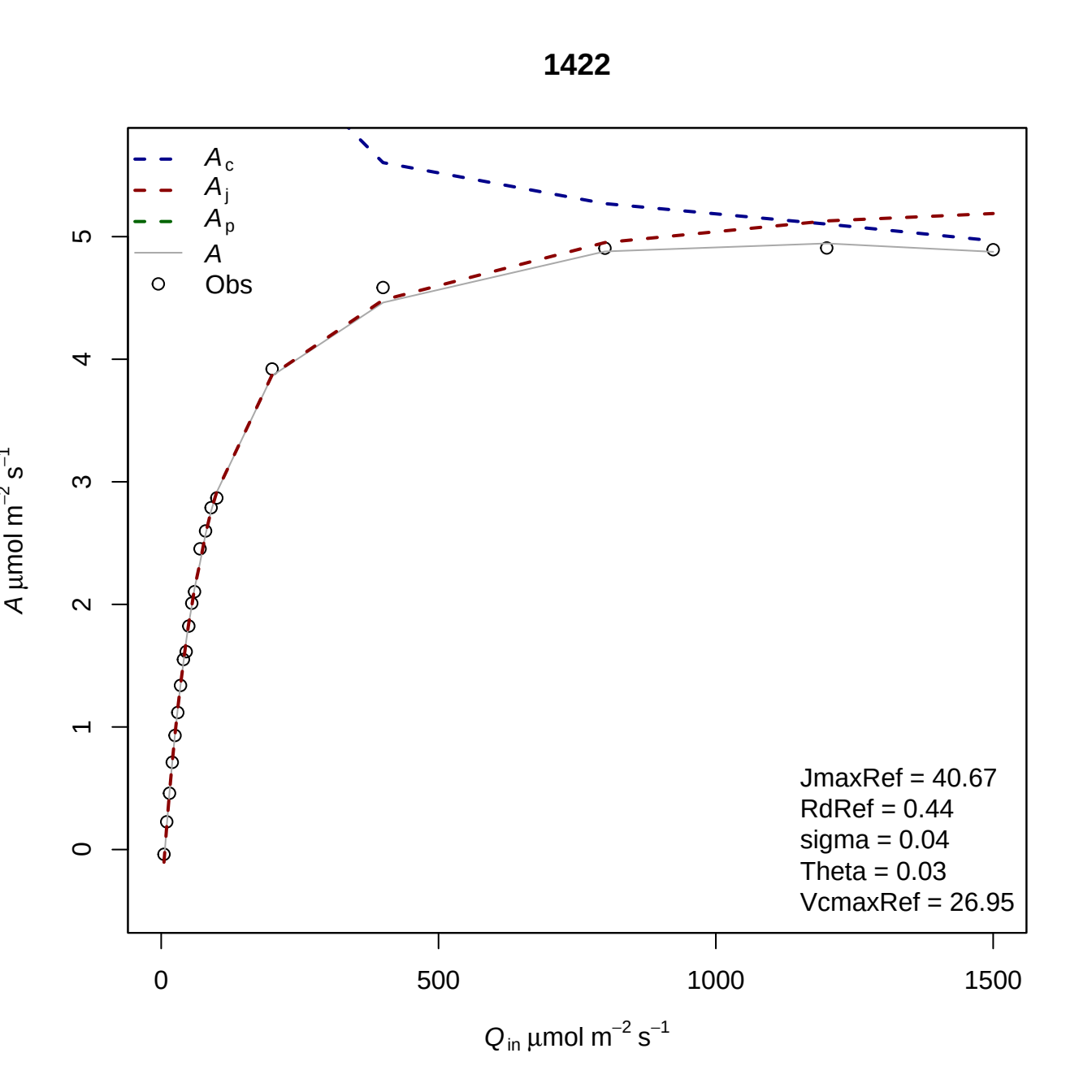

1417
A_c
A_j
A_p
A
Obs
JmaxRef = 35.26
RdRef = 0.26
sigma = 0.06
Theta = 0.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

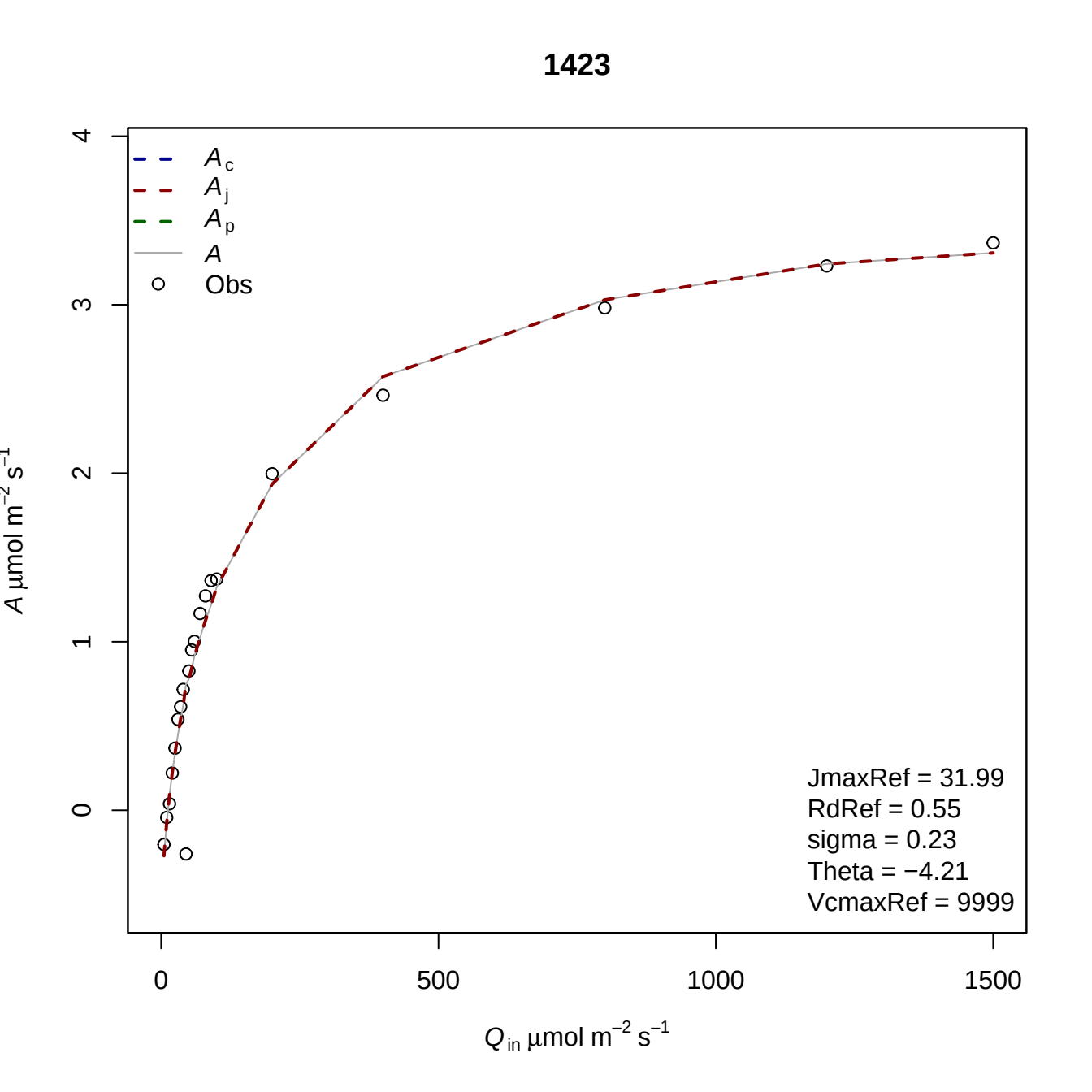

1418
A_c
A_j
A_p
A
Obs
JmaxRef = 32.55
RdRef = 0.61
sigma = 0.06
Theta = −0.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

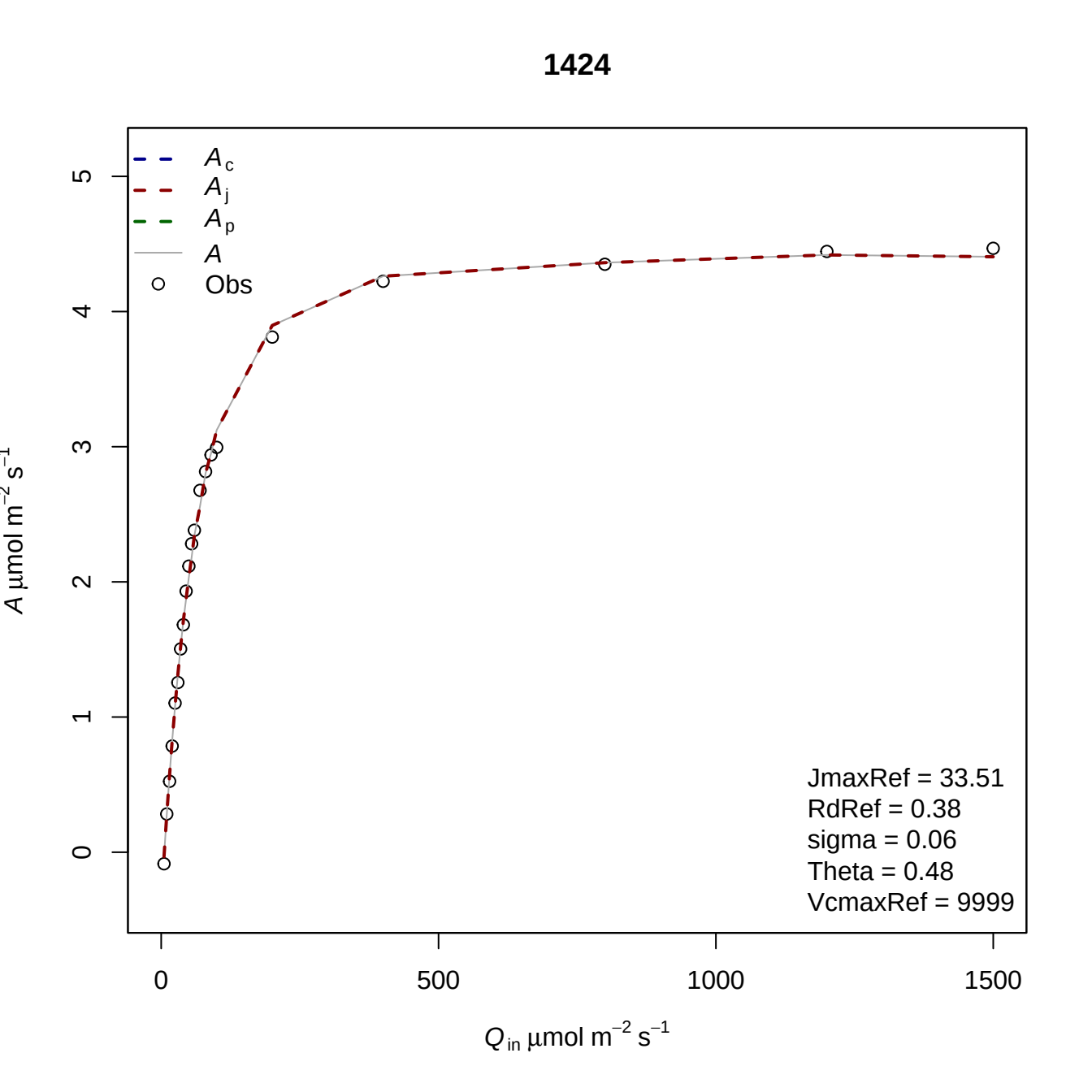

1419
A_c
A_j
A_p
A
Obs
JmaxRef = 26.52
RdRef = 0.41
sigma = 0.05
Theta = 0.31
VcmaxRef = 9999
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

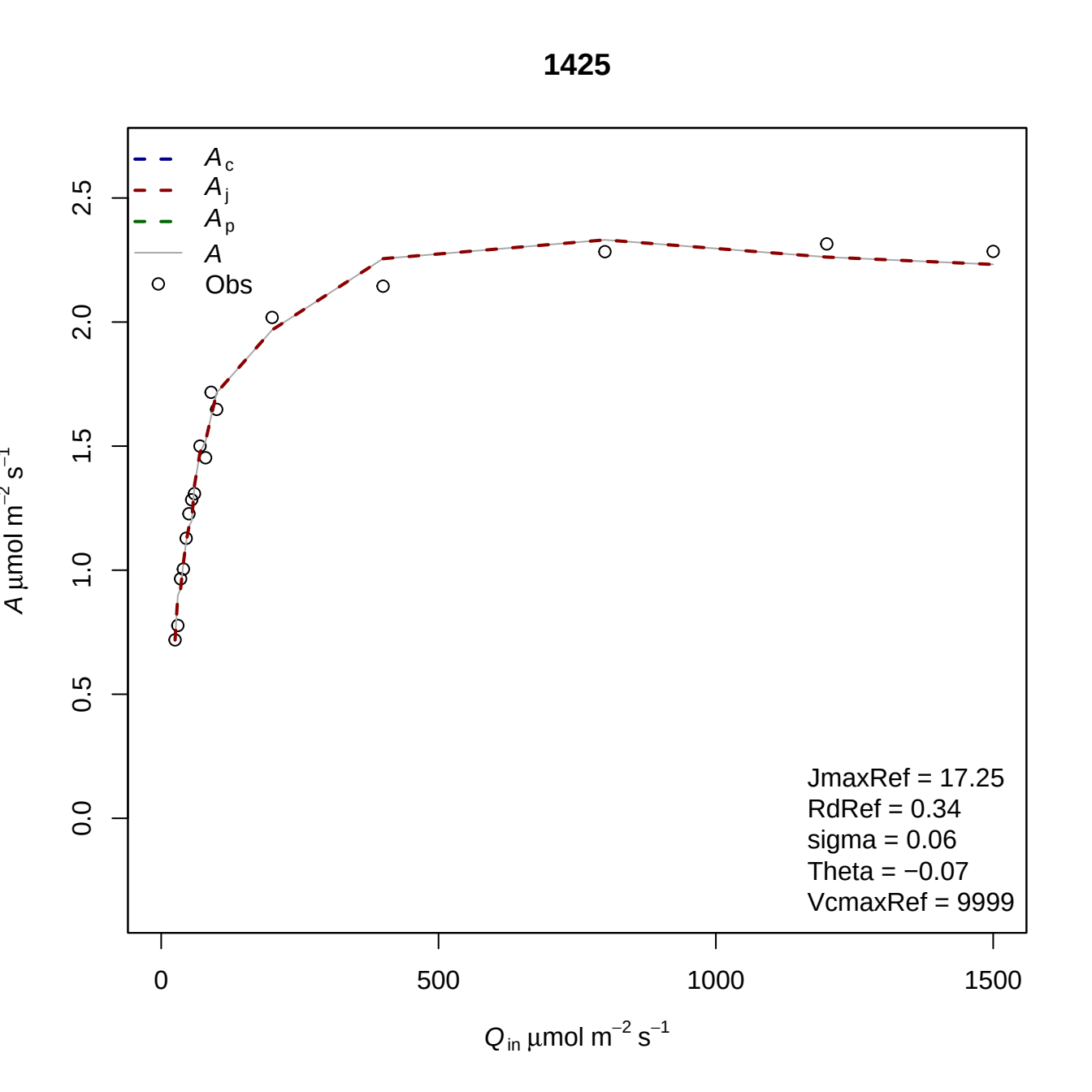

1420
A_c
A_j
A_p
A
Obs
JmaxRef = 35.55
RdRef = 0.49
sigma = 0.08
Theta = −0.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

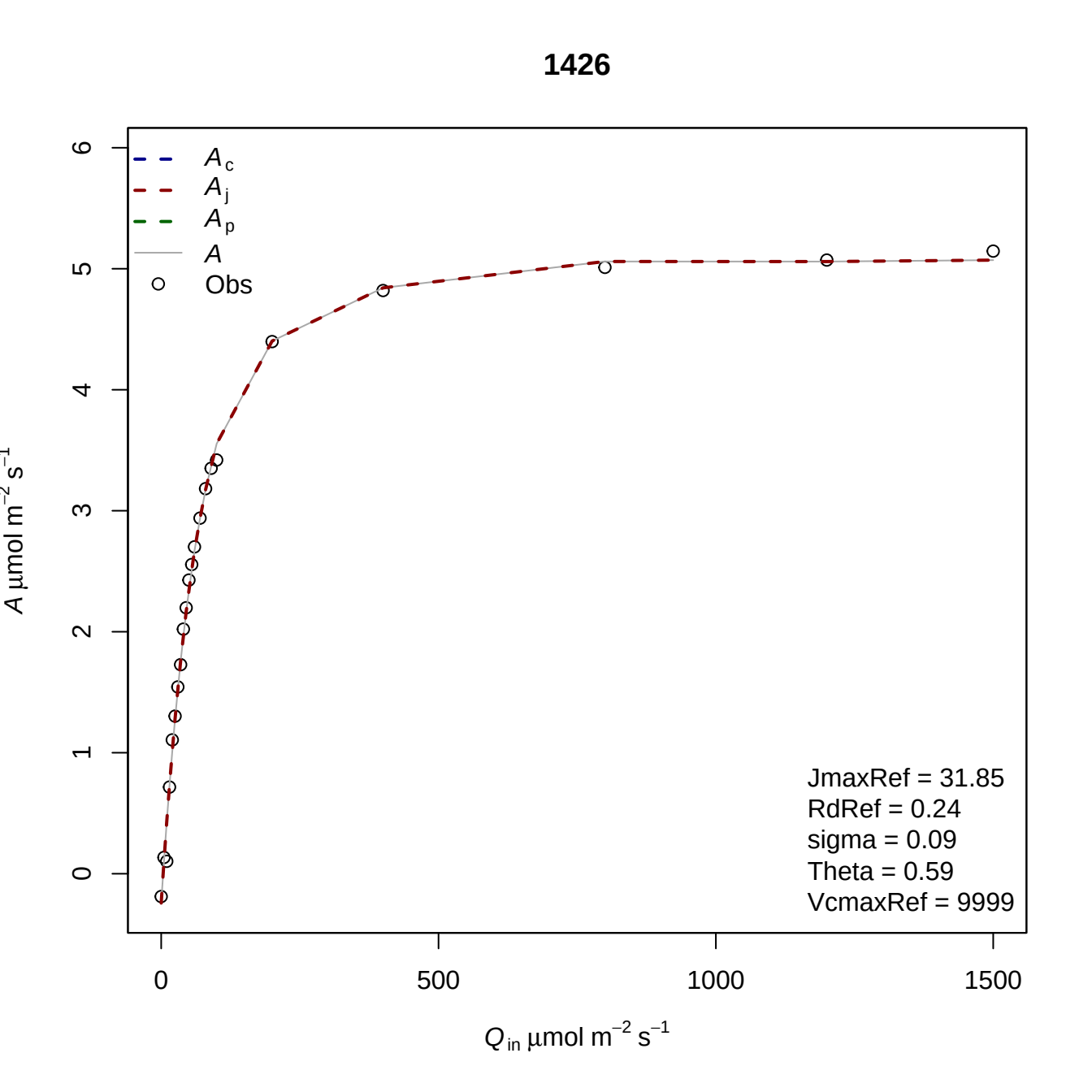

1421
A_c
A_j
A_p
A
Obs
JmaxRef = 27.6
RdRef = 0.45
sigma = 0.07
Theta = 0.22
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

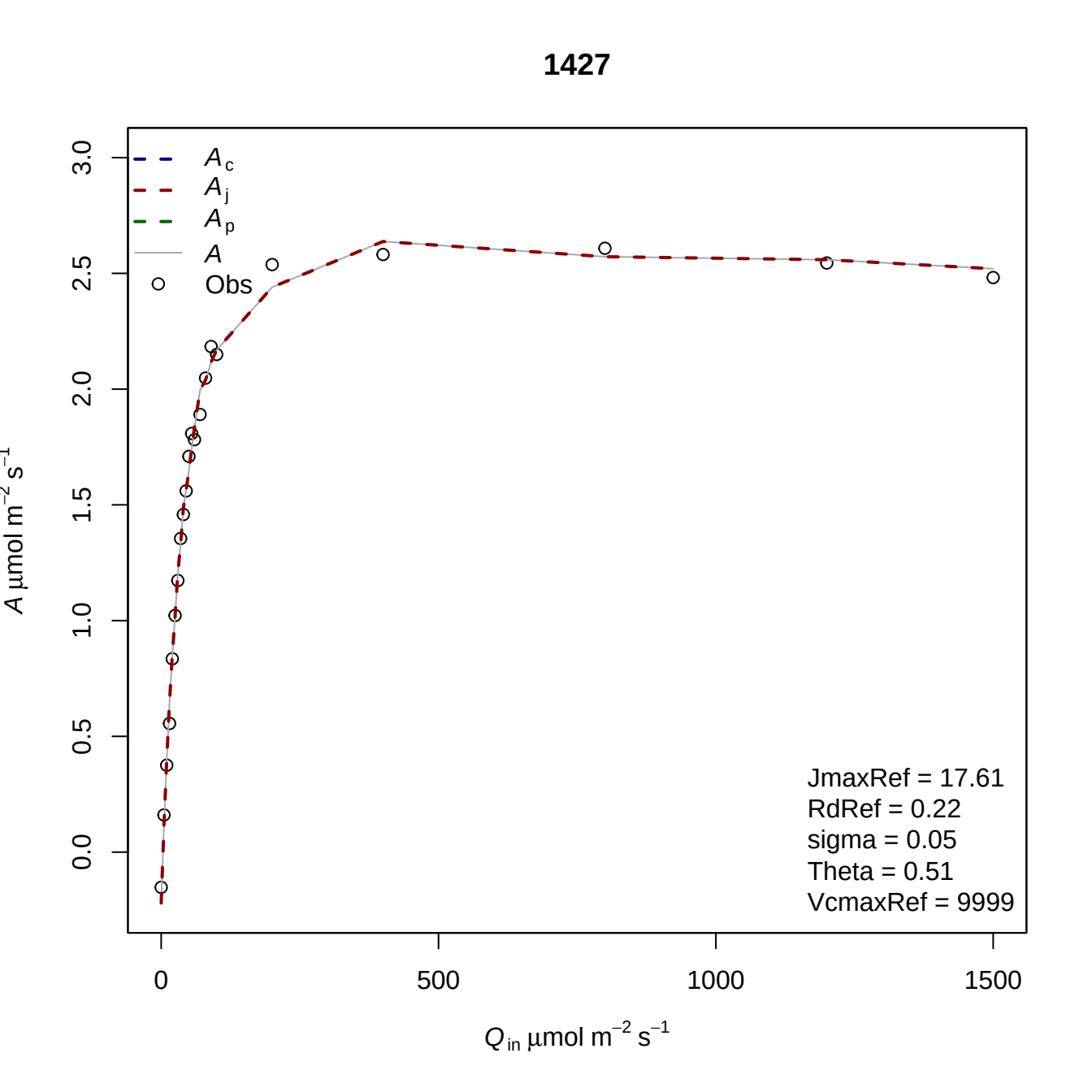

1422
A_c
A_j
A_p
A
Obs
JmaxRef = 40.67
RdRef = 0.44
sigma = 0.04
Theta = 0.03
VcmaxRef = 26.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

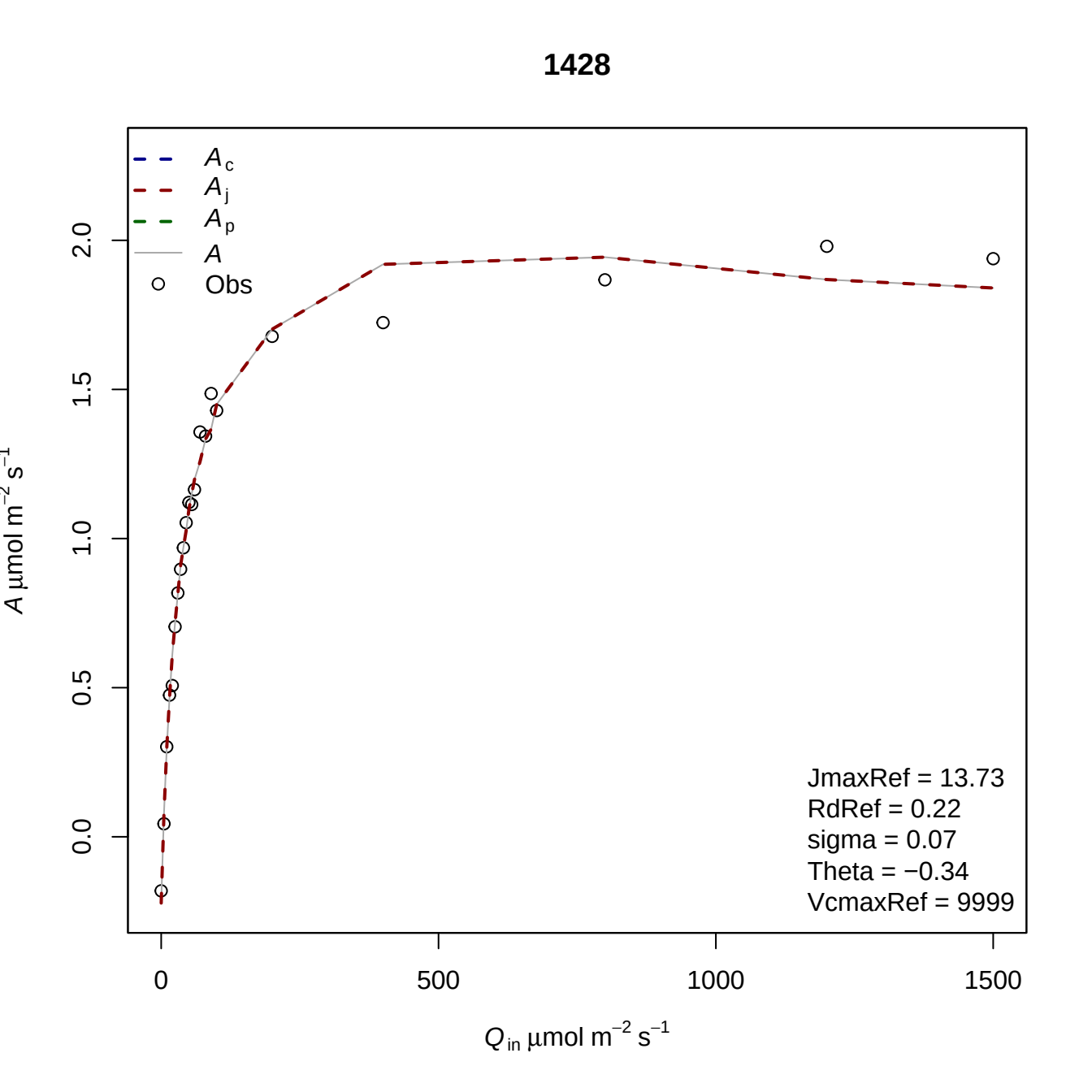

1423
A_c
A_j
A_p
A
Obs
JmaxRef = 31.99
RdRef = 0.55
sigma = 0.23
Theta = −4.21
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

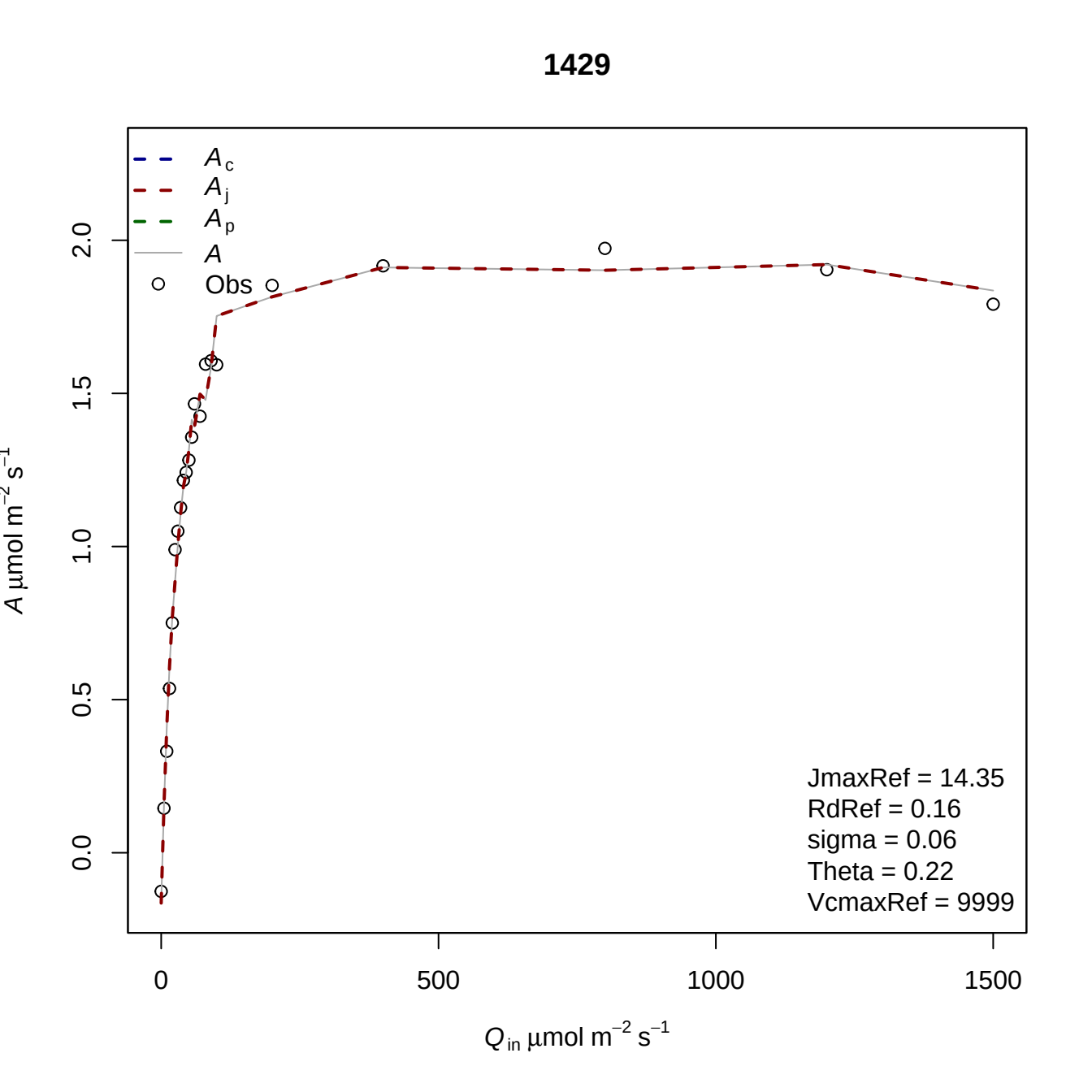

1424
A_c
A_j
A_p
A
Obs
JmaxRef = 33.51
RdRef = 0.38
sigma = 0.06
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

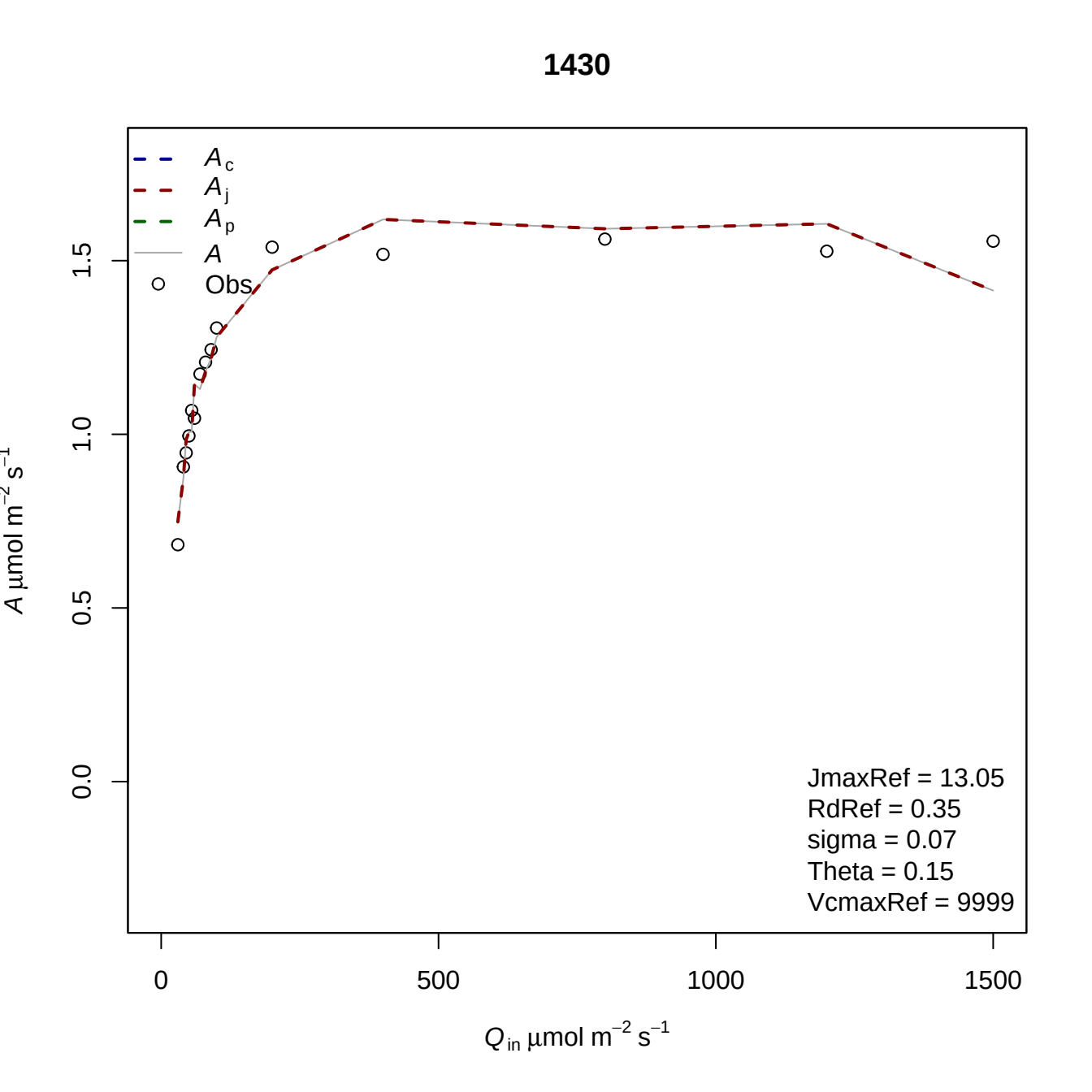

1425
A_c
A_j
A_p
A
Obs
JmaxRef = 17.25
RdRef = 0.34
sigma = 0.06
Theta = −0.07
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

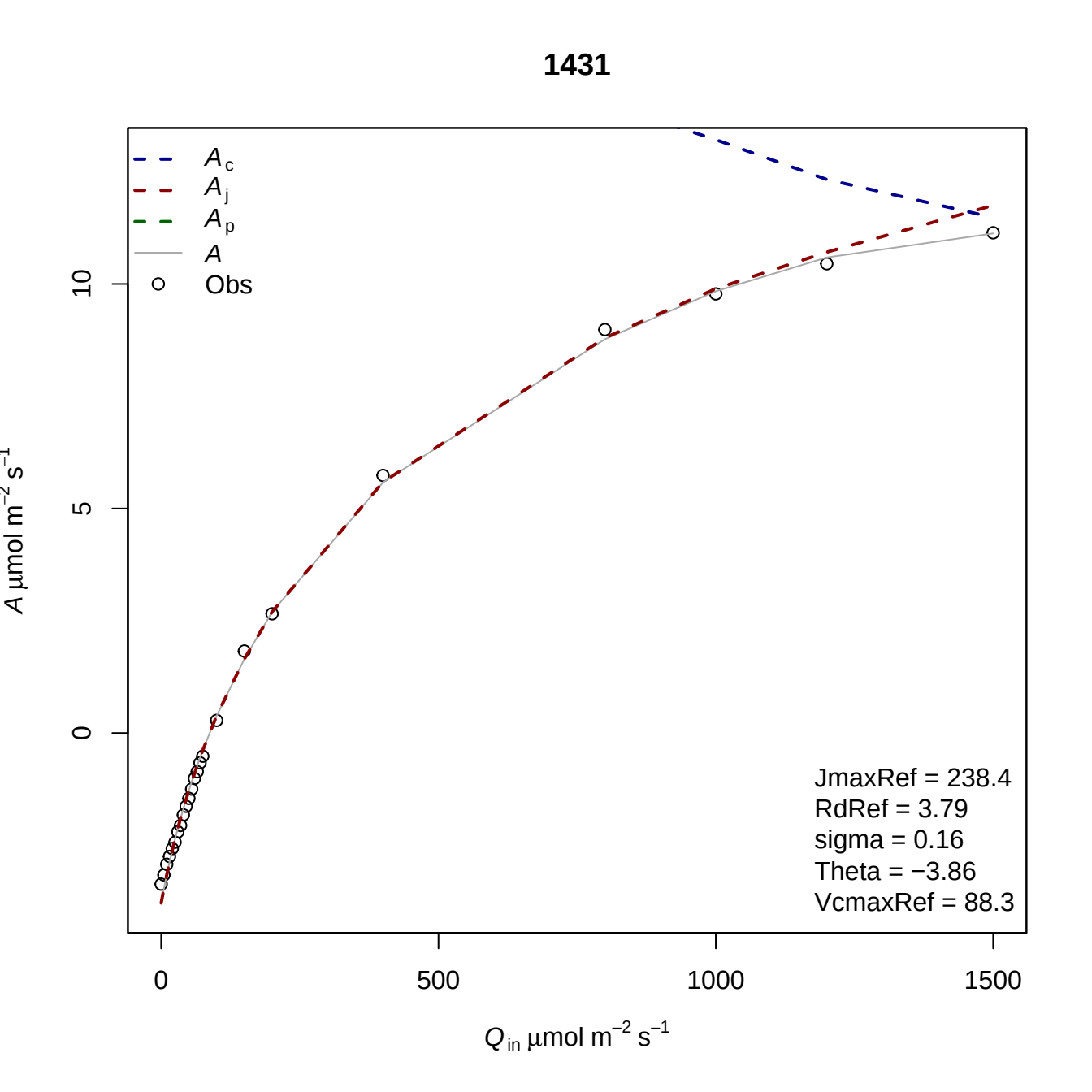

1426
A_c
A_j
A_p
A
Obs
JmaxRef = 31.85
RdRef = 0.24
sigma = 0.09
Theta = 0.59
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

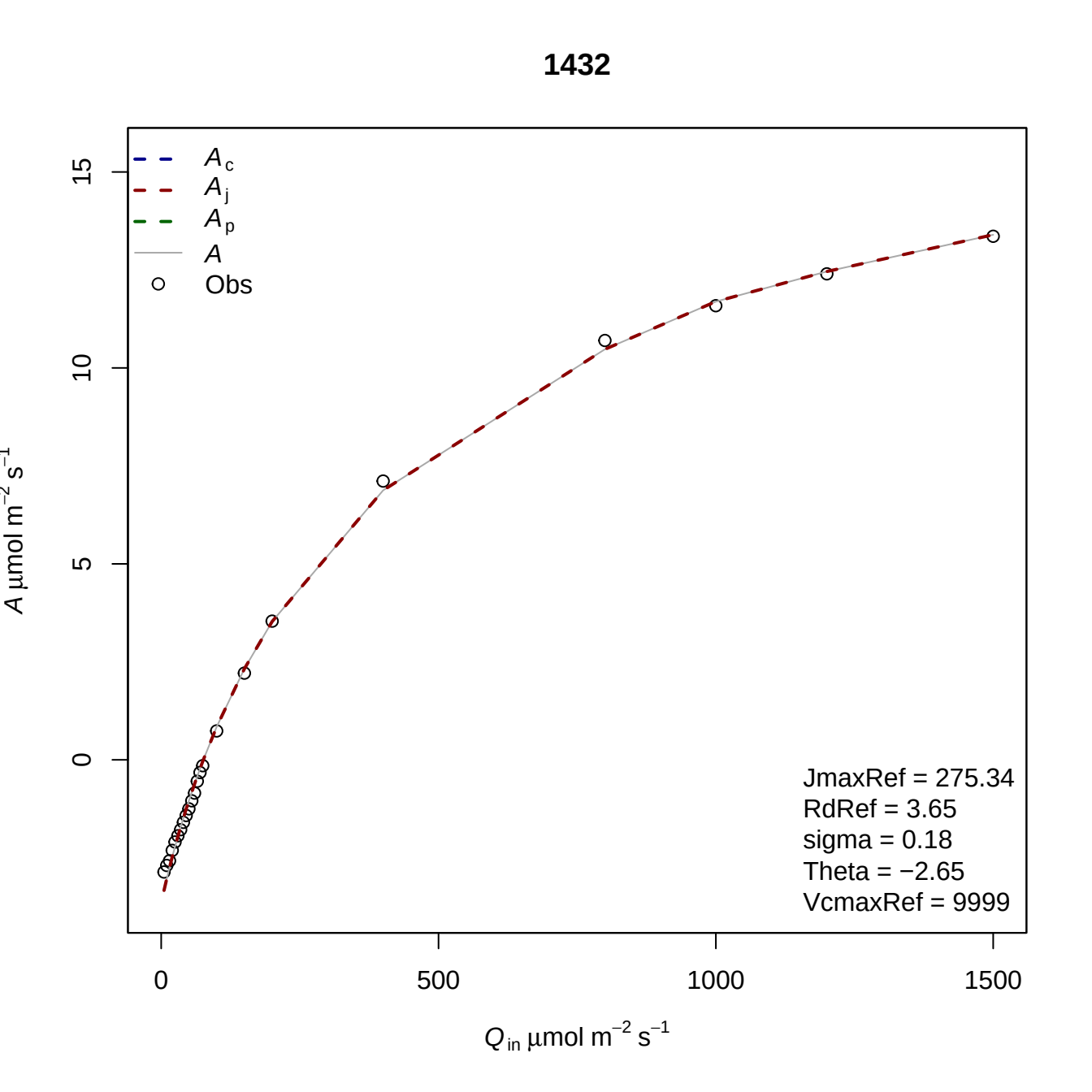

1427
A_c
A_j
A_p
A
Obs
JmaxRef = 17.61
RdRef = 0.22
sigma = 0.05
Theta = 0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

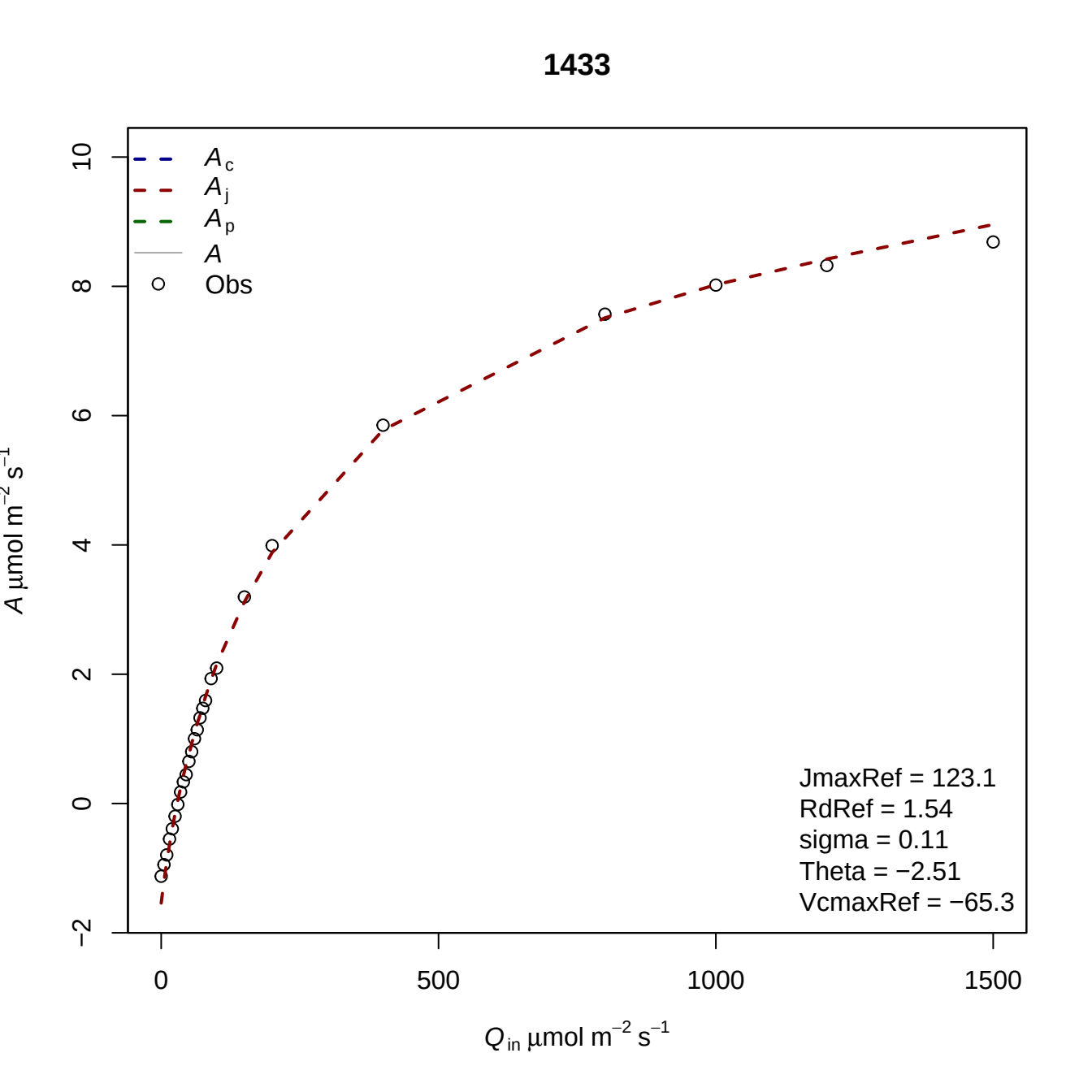

1428
A_c
A_j
A_p
A
Obs
JmaxRef = 13.73
RdRef = 0.22
sigma = 0.07
Theta = −0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500

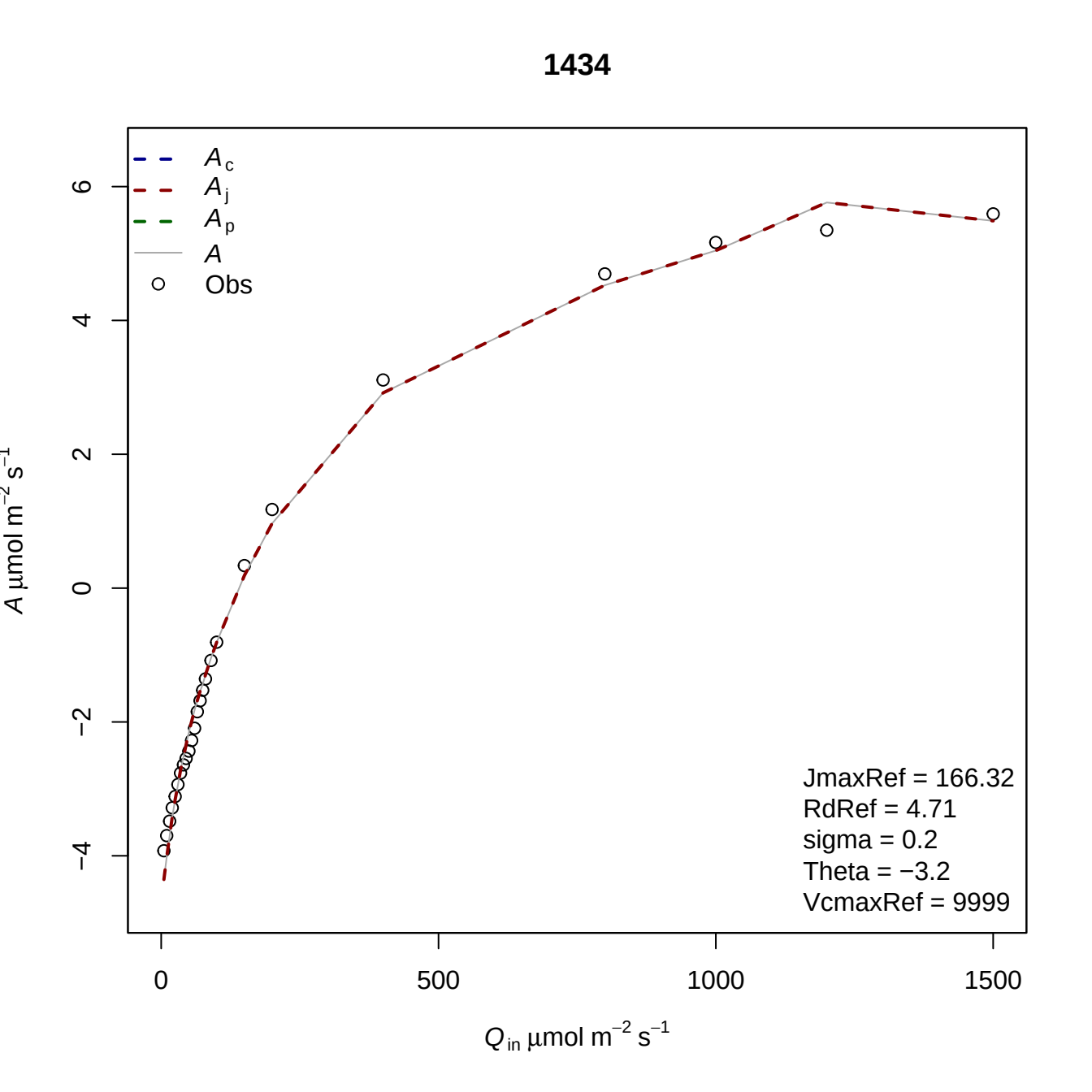

1429
A_c
A_j
A_p
A
Obs
JmaxRef = 14.35
RdRef = 0.16
sigma = 0.06
Theta = 0.22
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

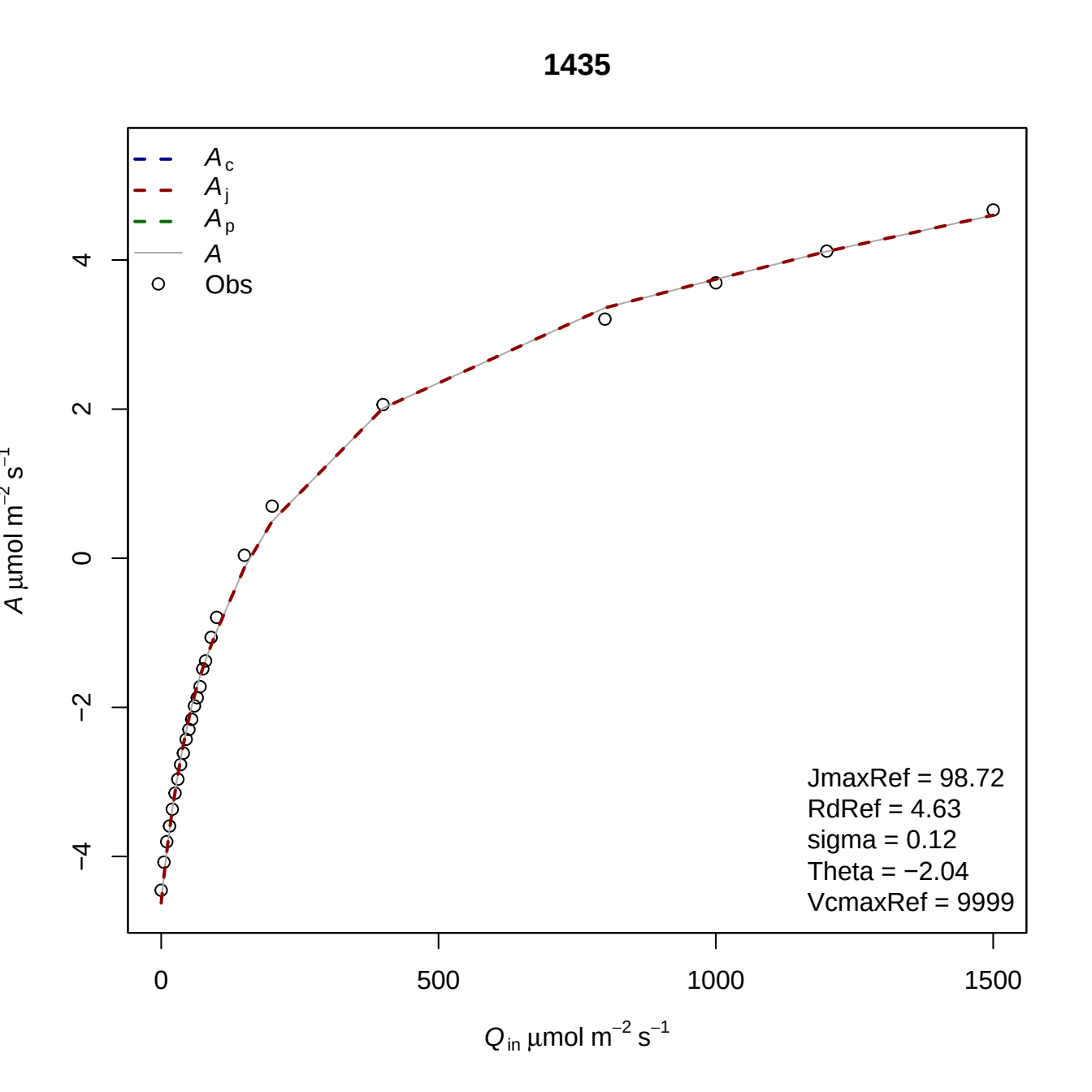

1430
A_c
A_j
A_p
A
Obs
JmaxRef = 13.05
RdRef = 0.35
sigma = 0.07
Theta = 0.15
VcmaxRef = 9999
0.0
0.5
1.0
1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

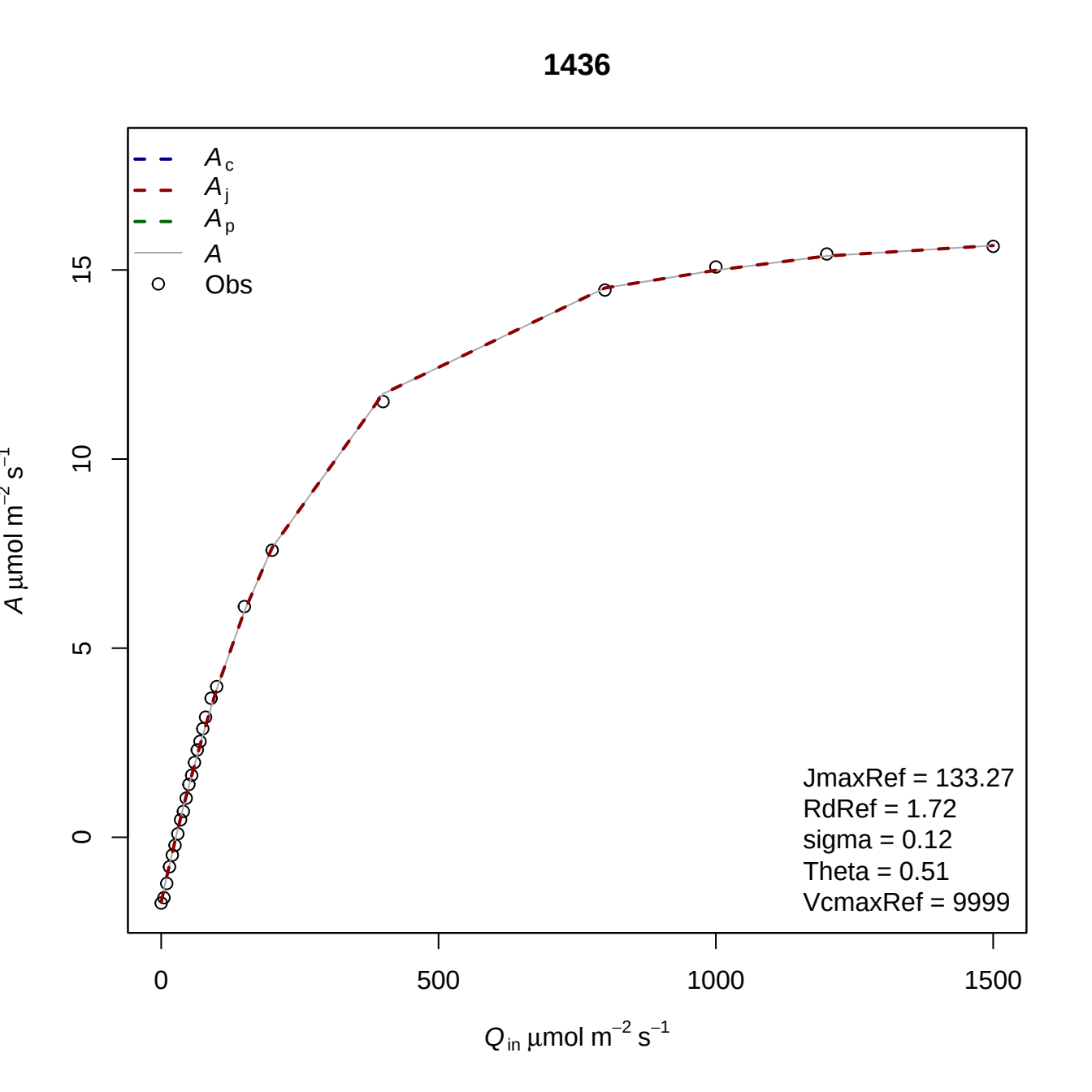

1431
A_c
A_j
A_p
A
Obs
JmaxRef = 238.4
RdRef = 3.79
sigma = 0.16
Theta = −3.86
VcmaxRef = 88.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

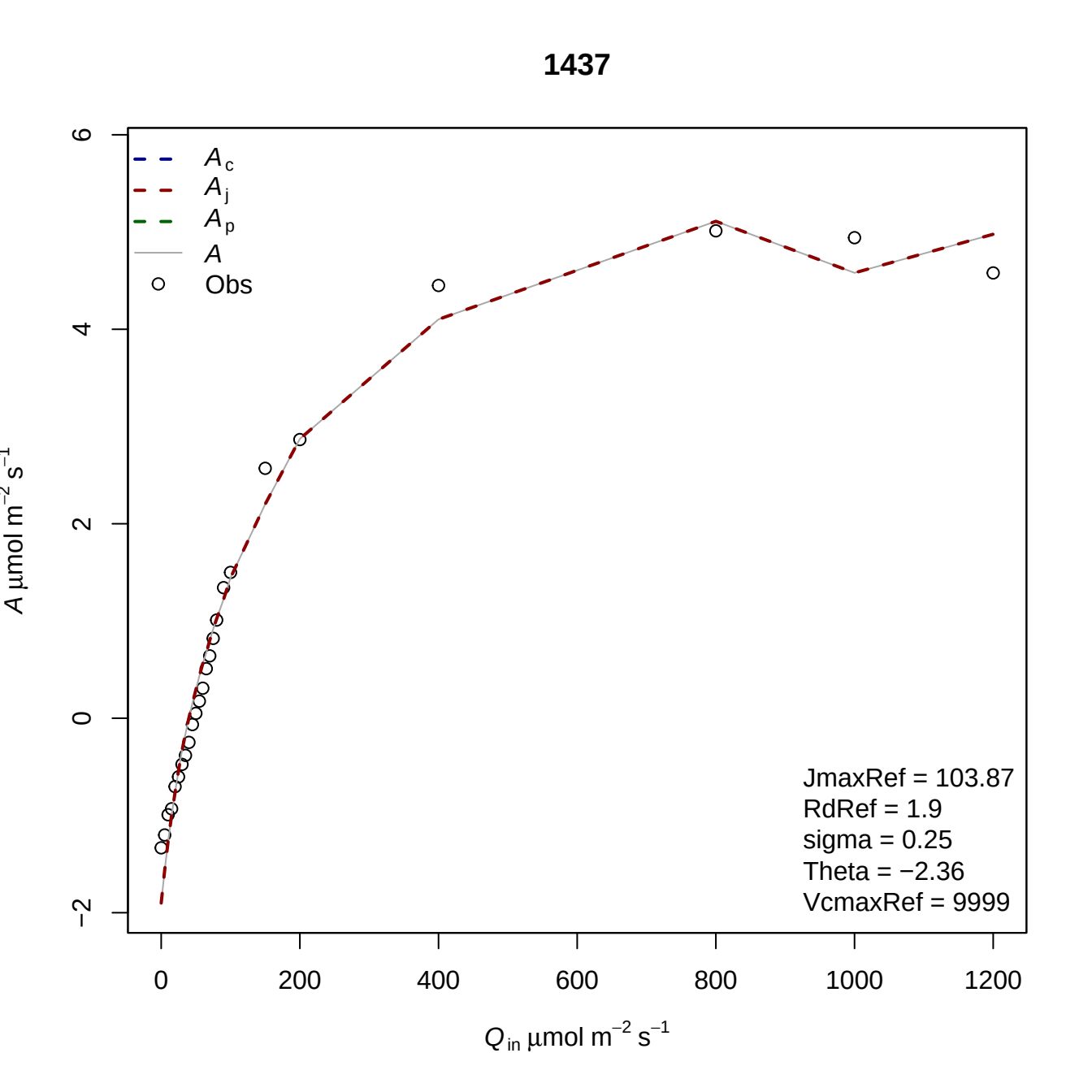

1432
A_c
A_j
A_p
A
Obs
JmaxRef = 275.34
RdRef = 3.65
sigma = 0.18
Theta = −2.65
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

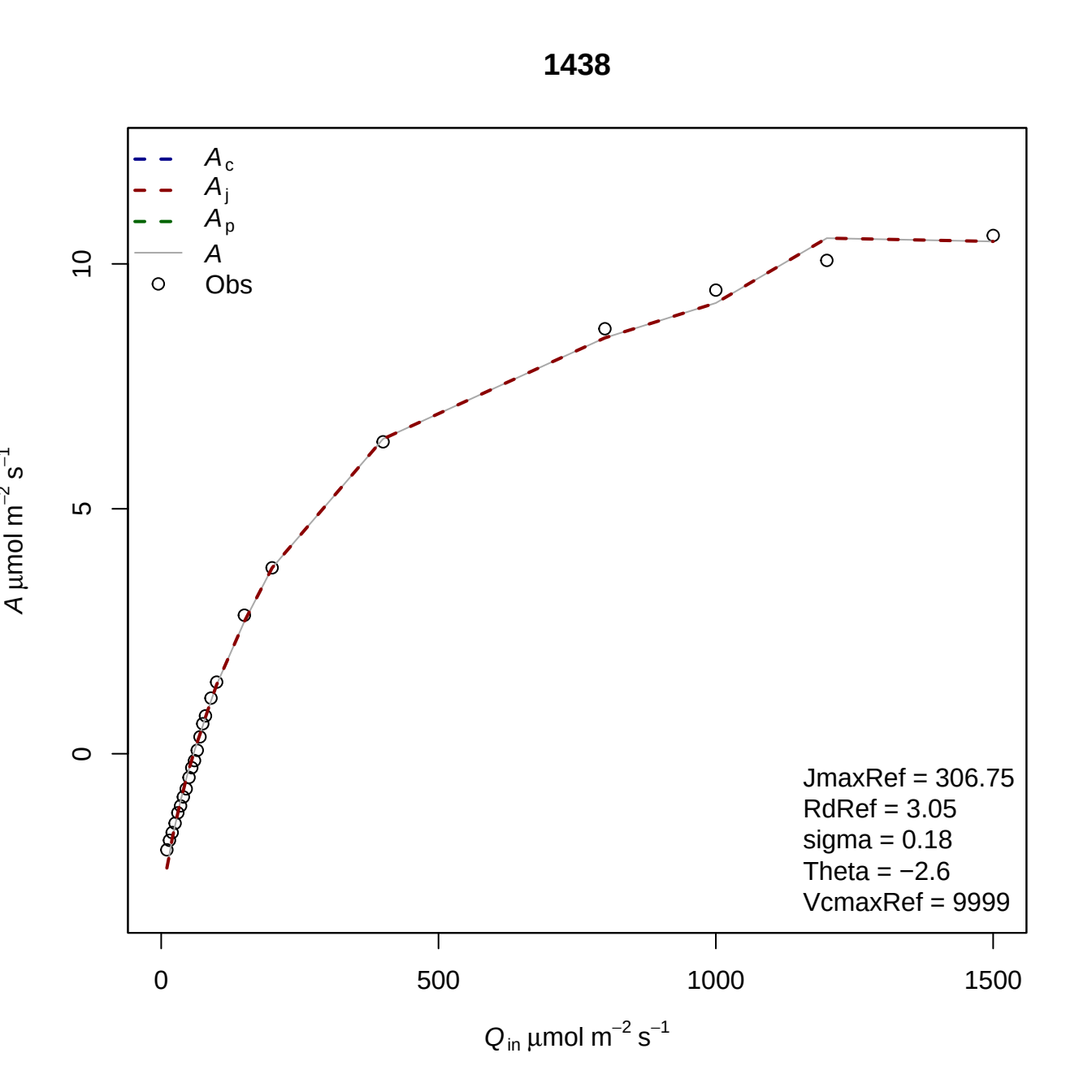

1433
A_c
A_j
A_p
A
Obs
JmaxRef = 123.1
RdRef = 1.54
sigma = 0.11
Theta = −2.51
VcmaxRef = −65.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

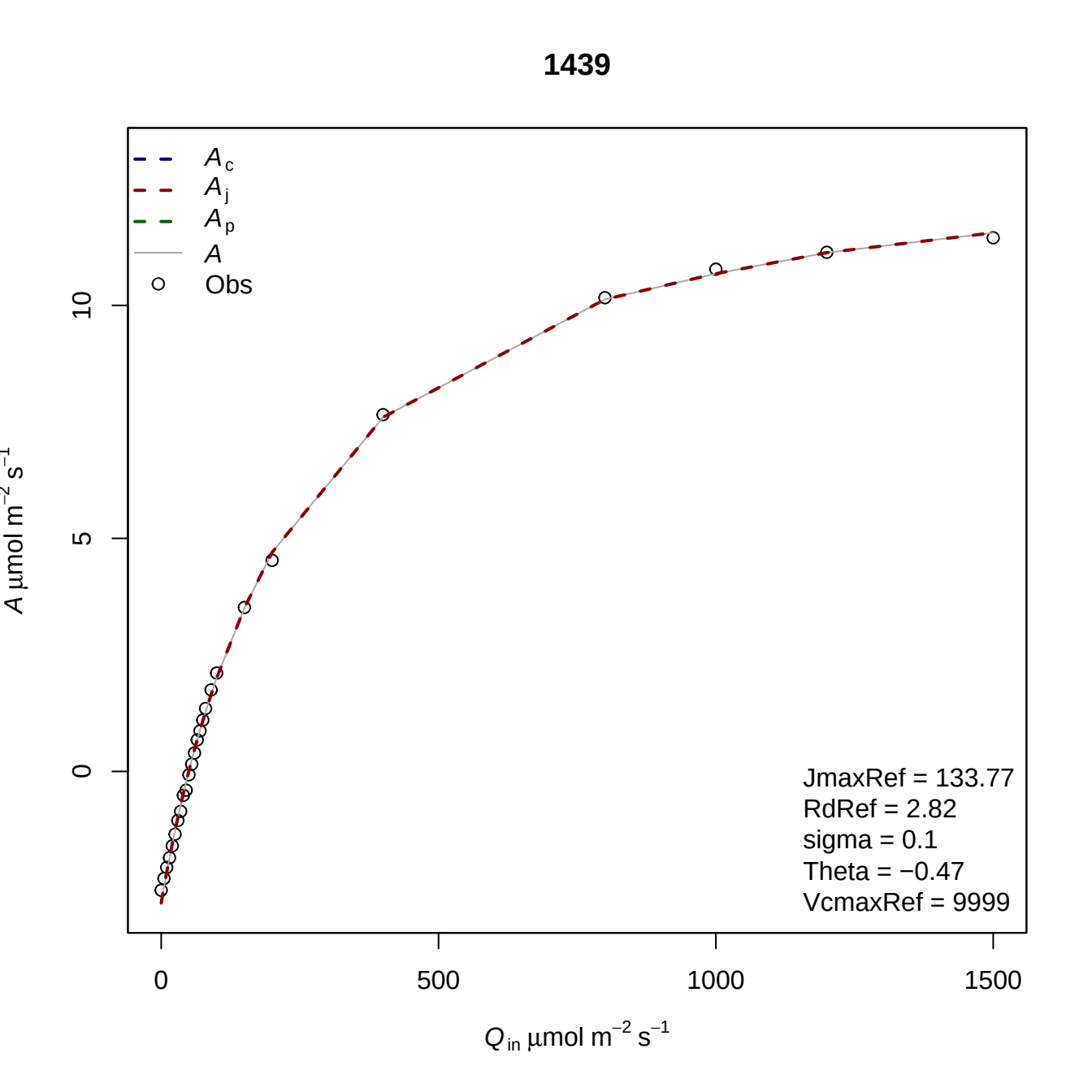

1434
A_c
A_j
A_p
A
Obs
JmaxRef = 166.32
RdRef = 4.71
sigma = 0.2
Theta = −3.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

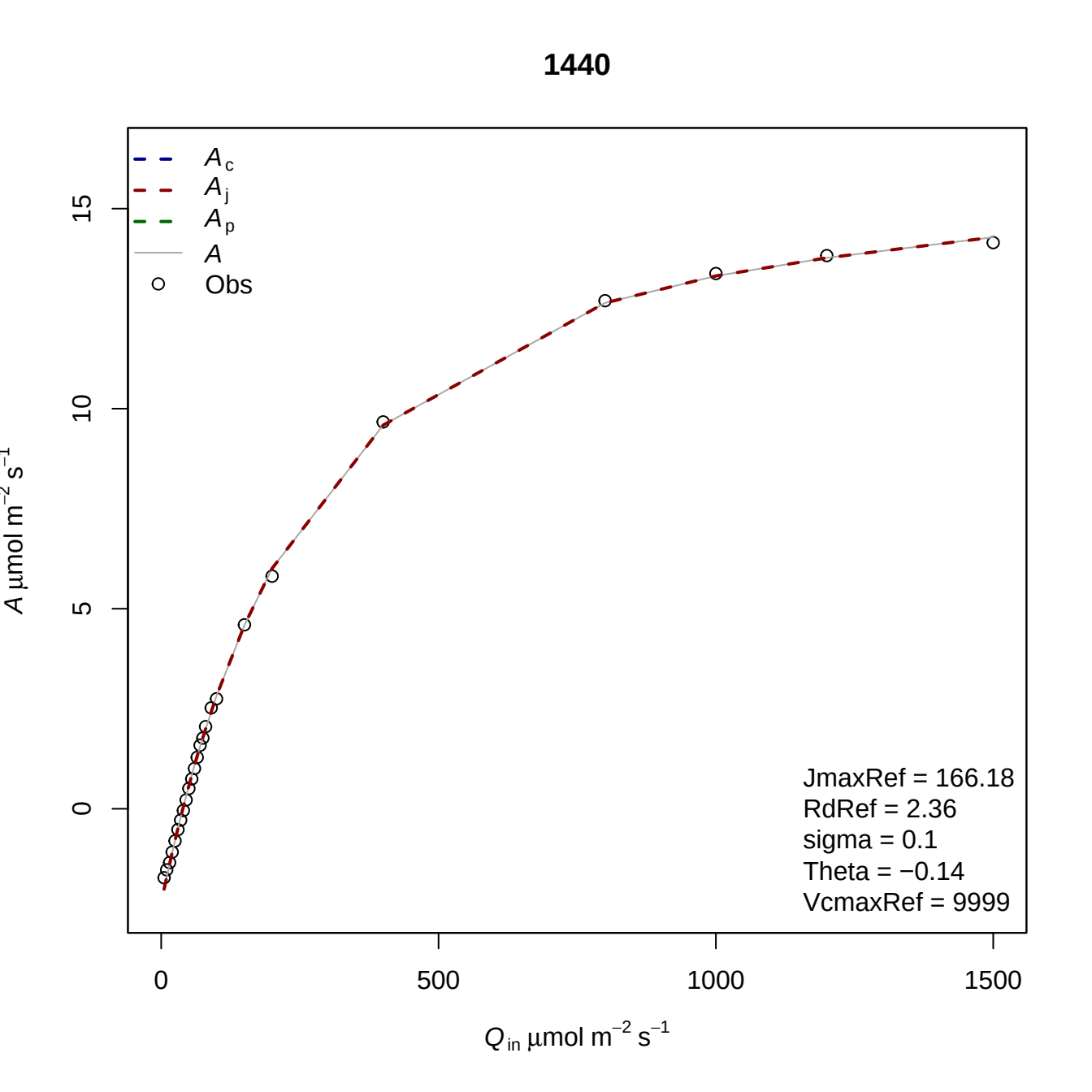

1435
A_c
A_j
A_p
A
Obs
JmaxRef = 98.72
RdRef = 4.63
sigma = 0.12
Theta = −2.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

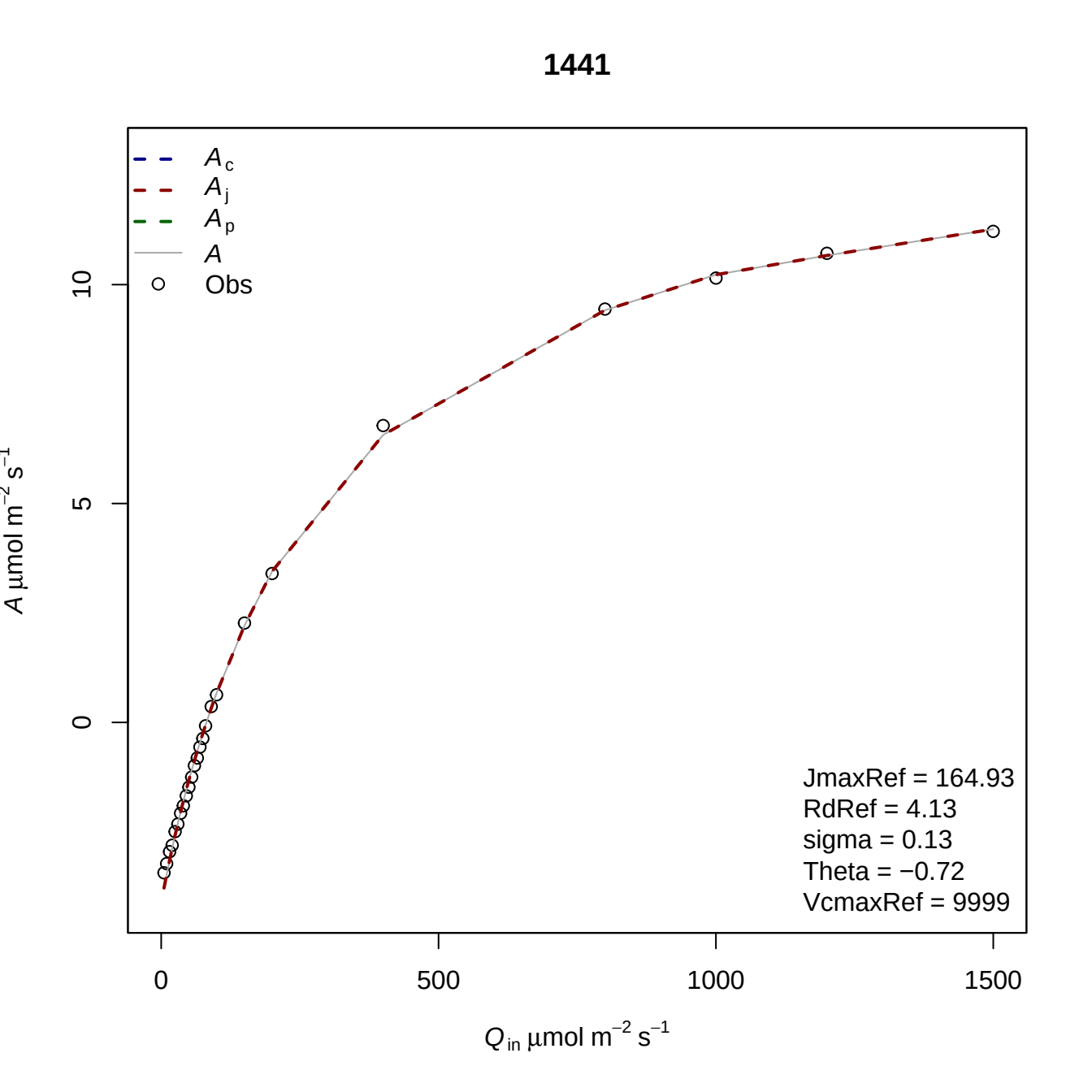

1436
A_c
A_j
A_p
A
Obs
JmaxRef = 133.27
RdRef = 1.72
sigma = 0.12
Theta = 0.51
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

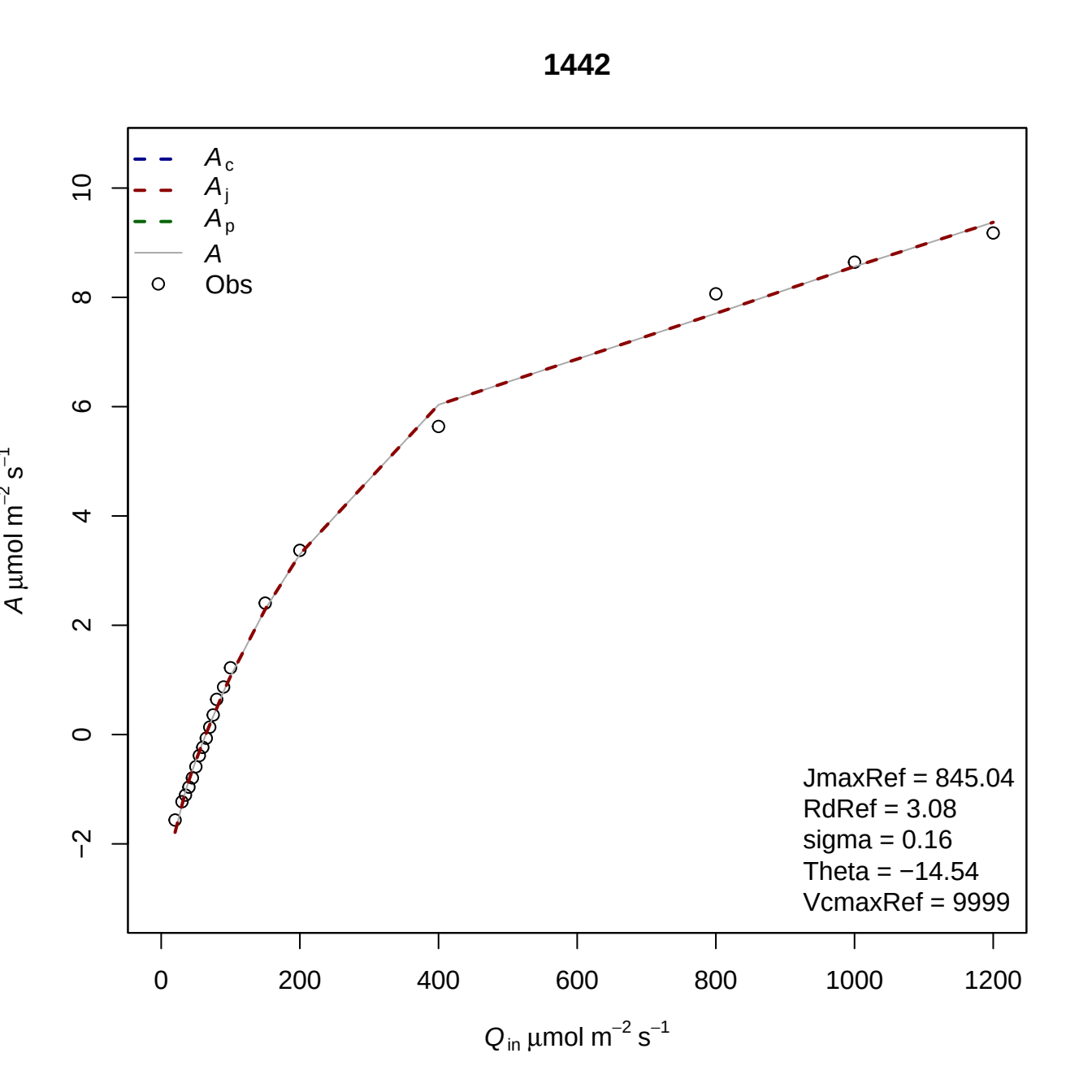

1437
A_c
A_j
A_p
A
Obs
JmaxRef = 103.87
RdRef = 1.9
sigma = 0.25
Theta = −2.36
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

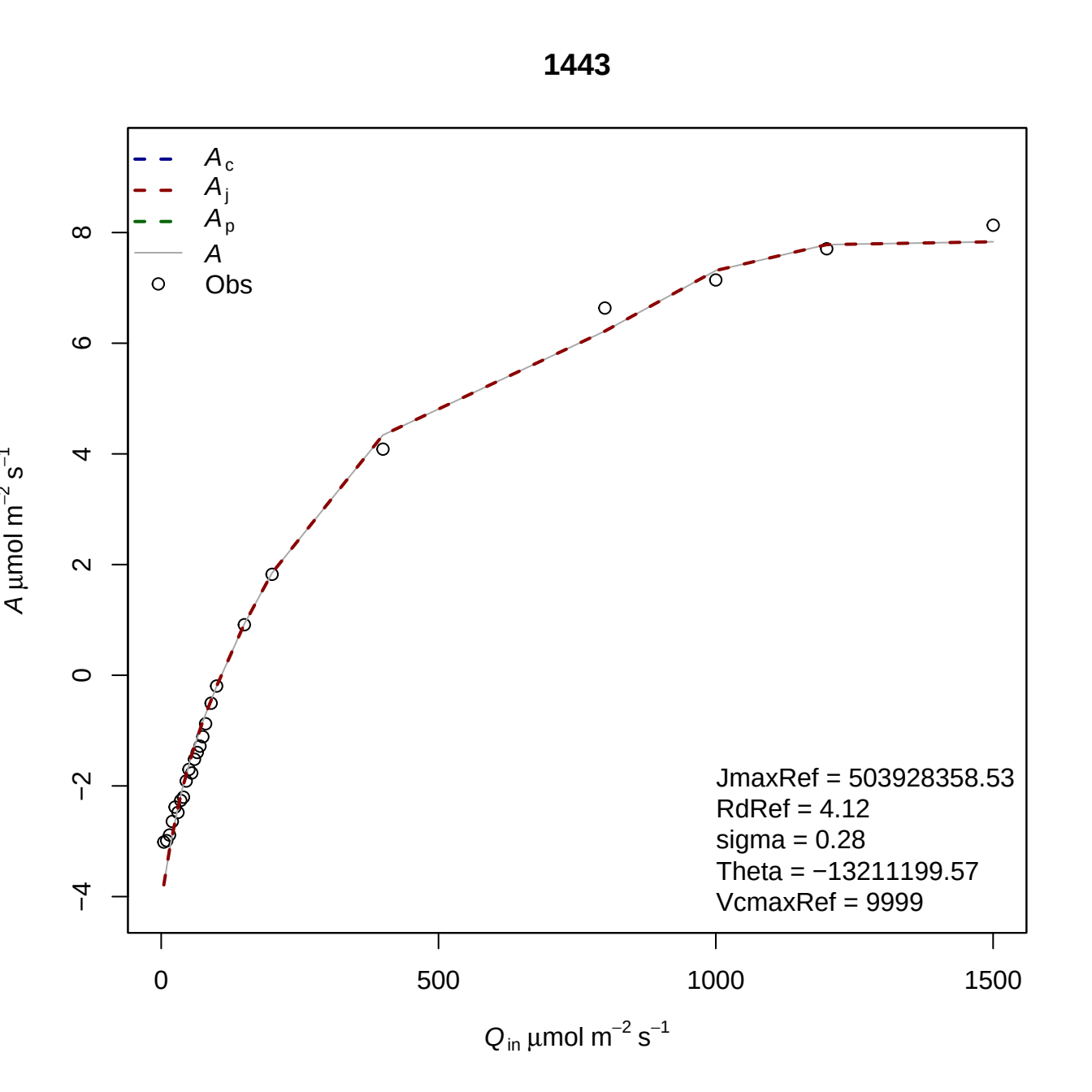

1438
A_c
A_j
A_p
A
Obs
JmaxRef = 306.75
RdRef = 3.05
sigma = 0.18
Theta = −2.6
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

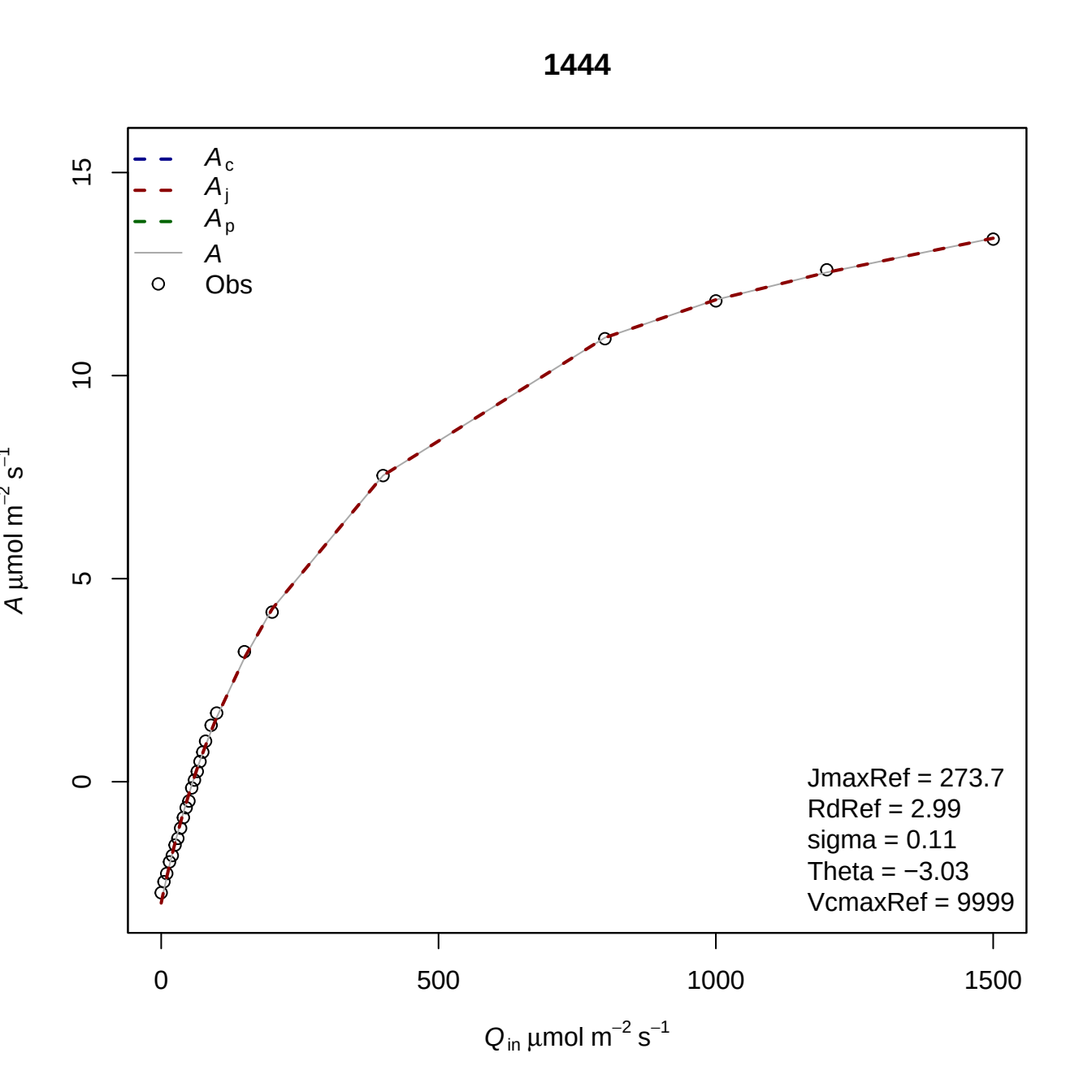

1439
A_c
A_j
A_p
A
Obs
JmaxRef = 133.77
RdRef = 2.82
sigma = 0.1
Theta = −0.47
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

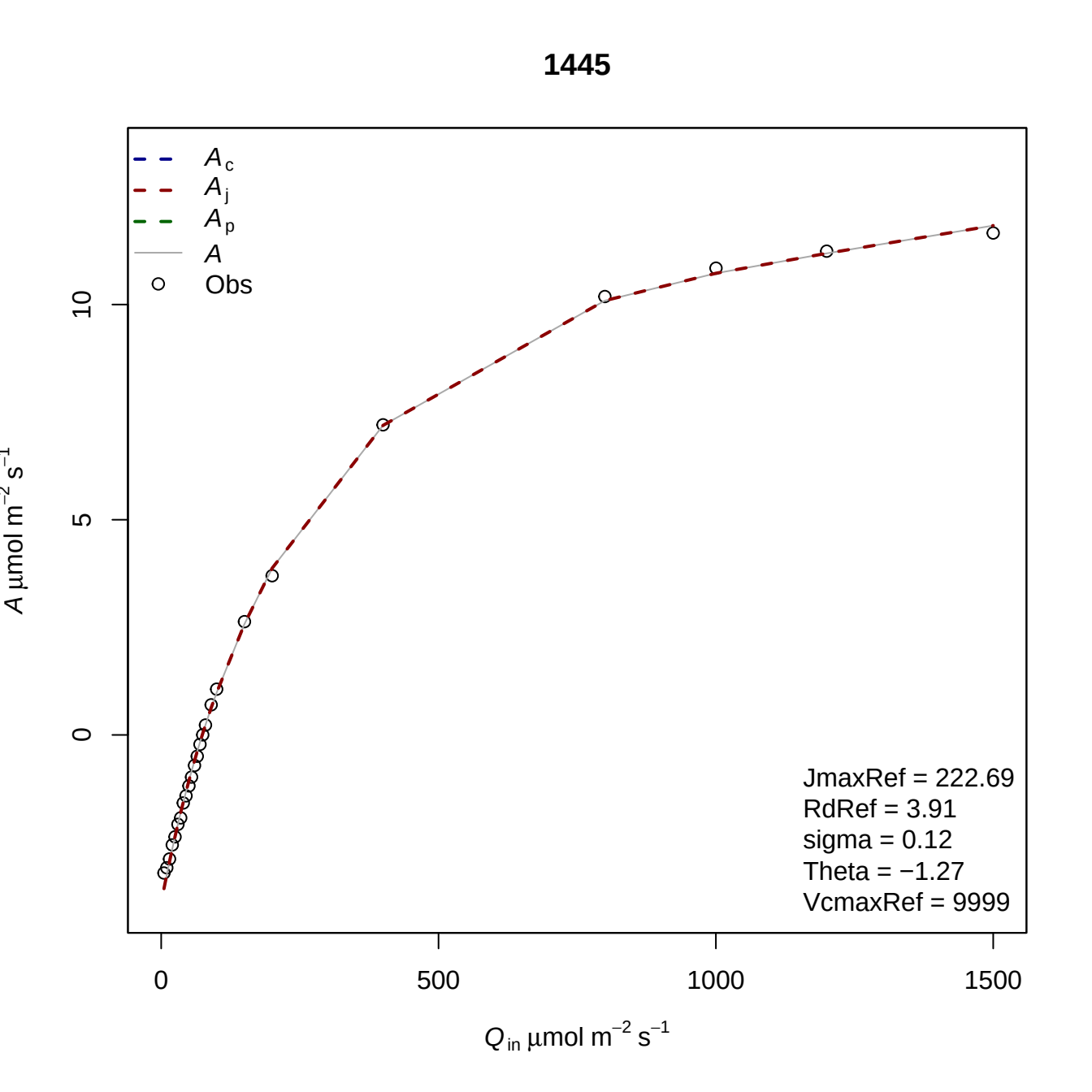

1440
A_c
A_j
A_p
A
Obs
JmaxRef = 166.18
RdRef = 2.36
sigma = 0.1
Theta = −0.14
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

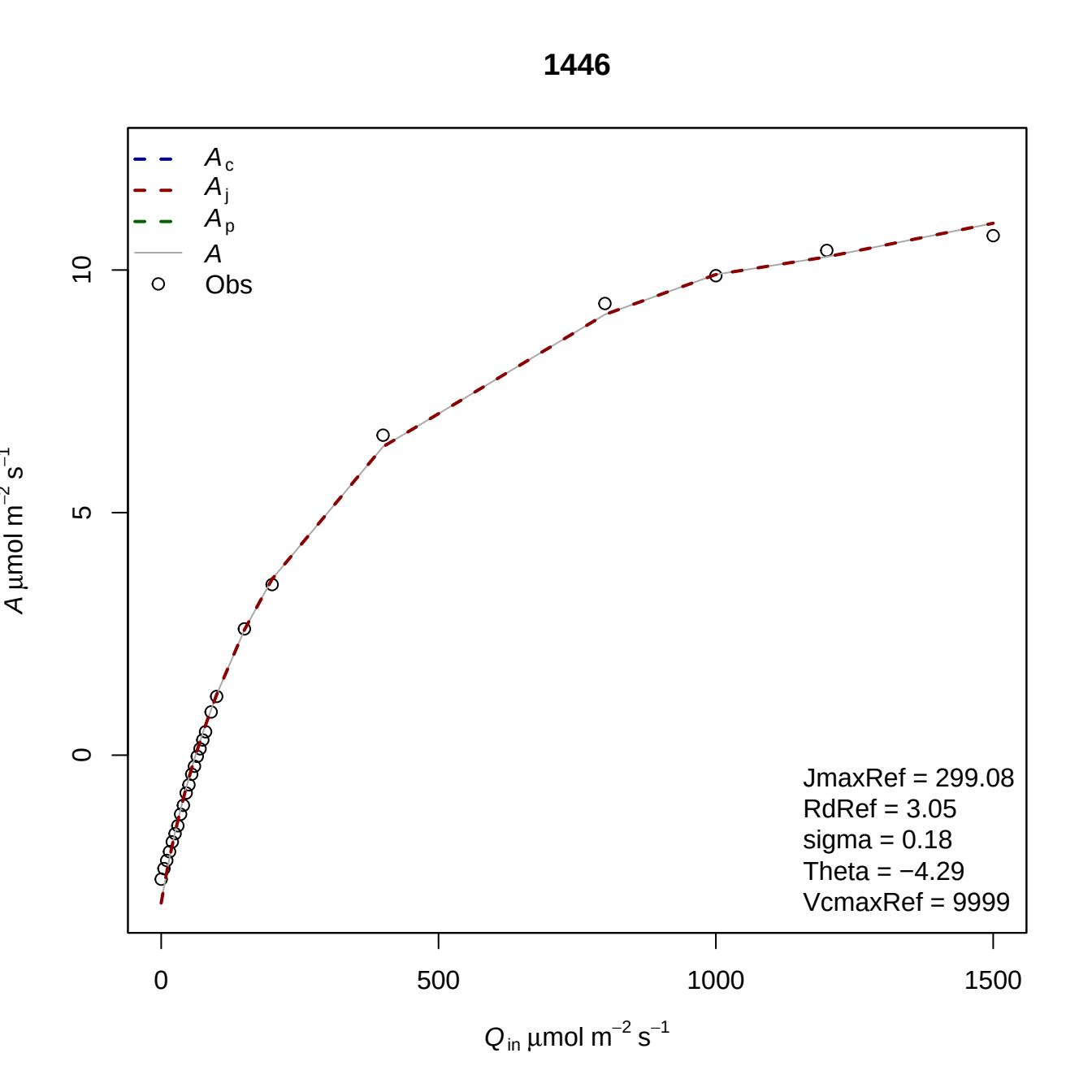

1441
A_c
A_j
A_p
A
Obs
JmaxRef = 164.93
RdRef = 4.13
sigma = 0.13
Theta = −0.72
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

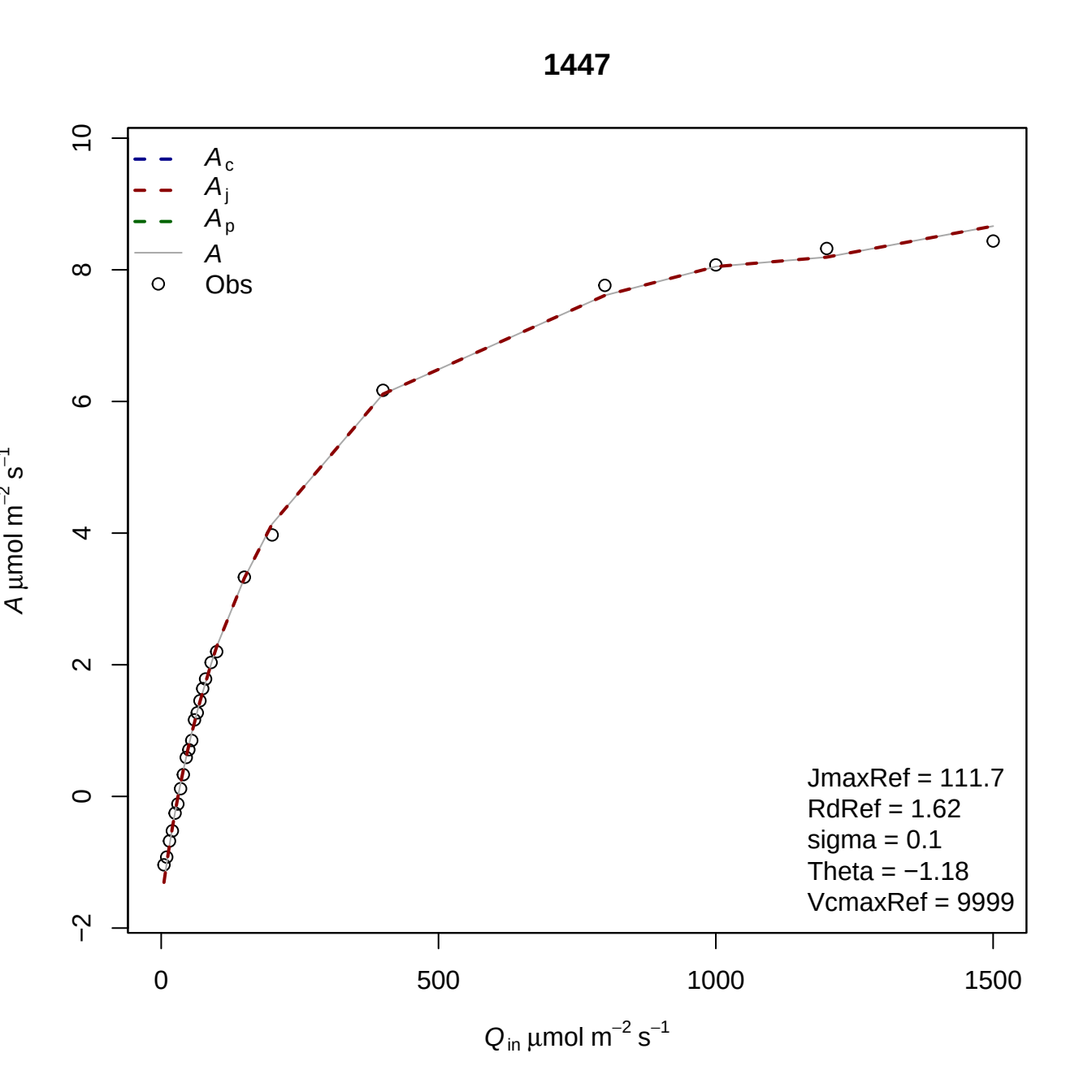

1442
A_c
A_j
A_p
A
Obs
JmaxRef = 845.04
RdRef = 3.08
sigma = 0.16
Theta = −14.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

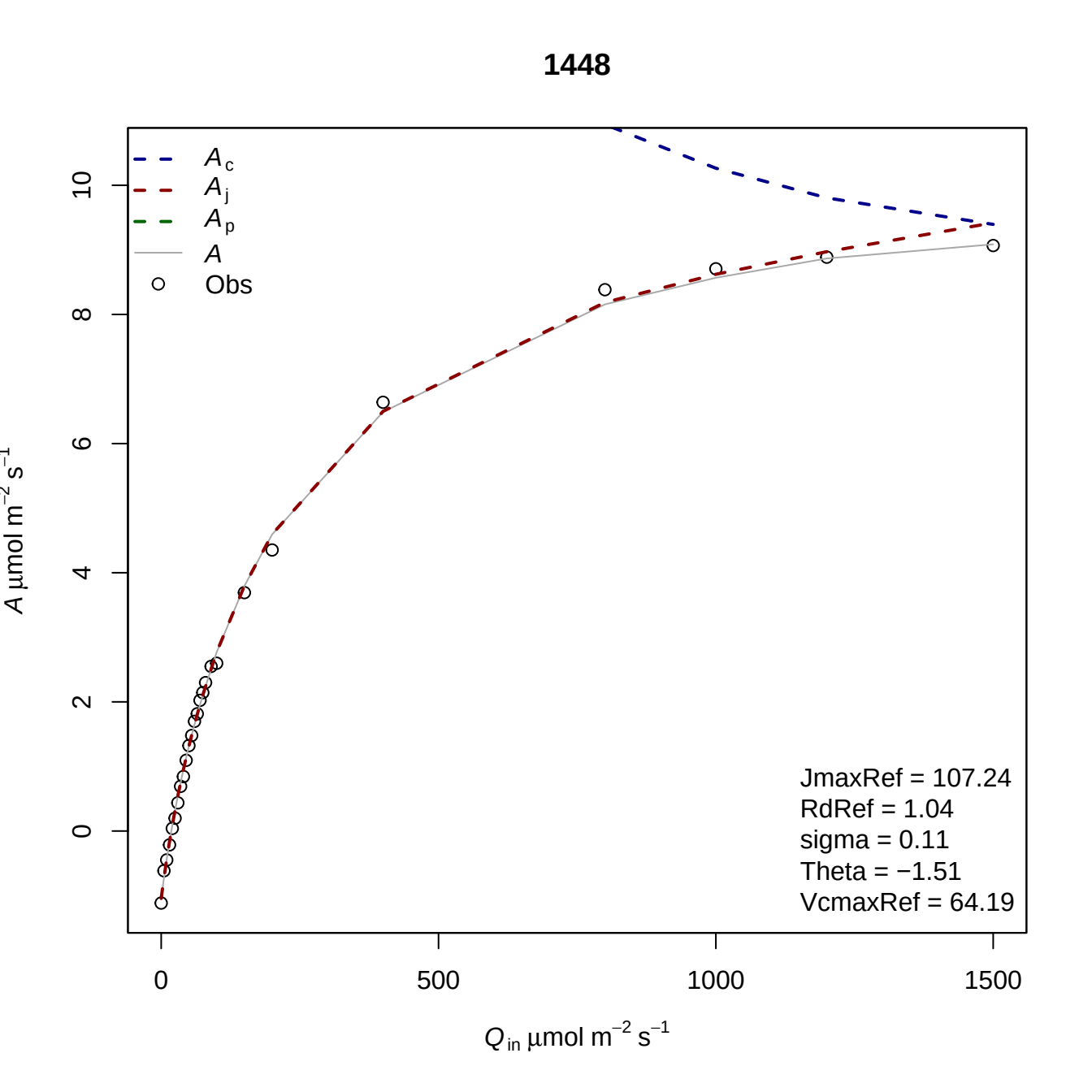

1443
A_c
A_j
A_p
A
Obs
JmaxRef = 503928358.53
RdRef = 4.12
sigma = 0.28
Theta = −13211199.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

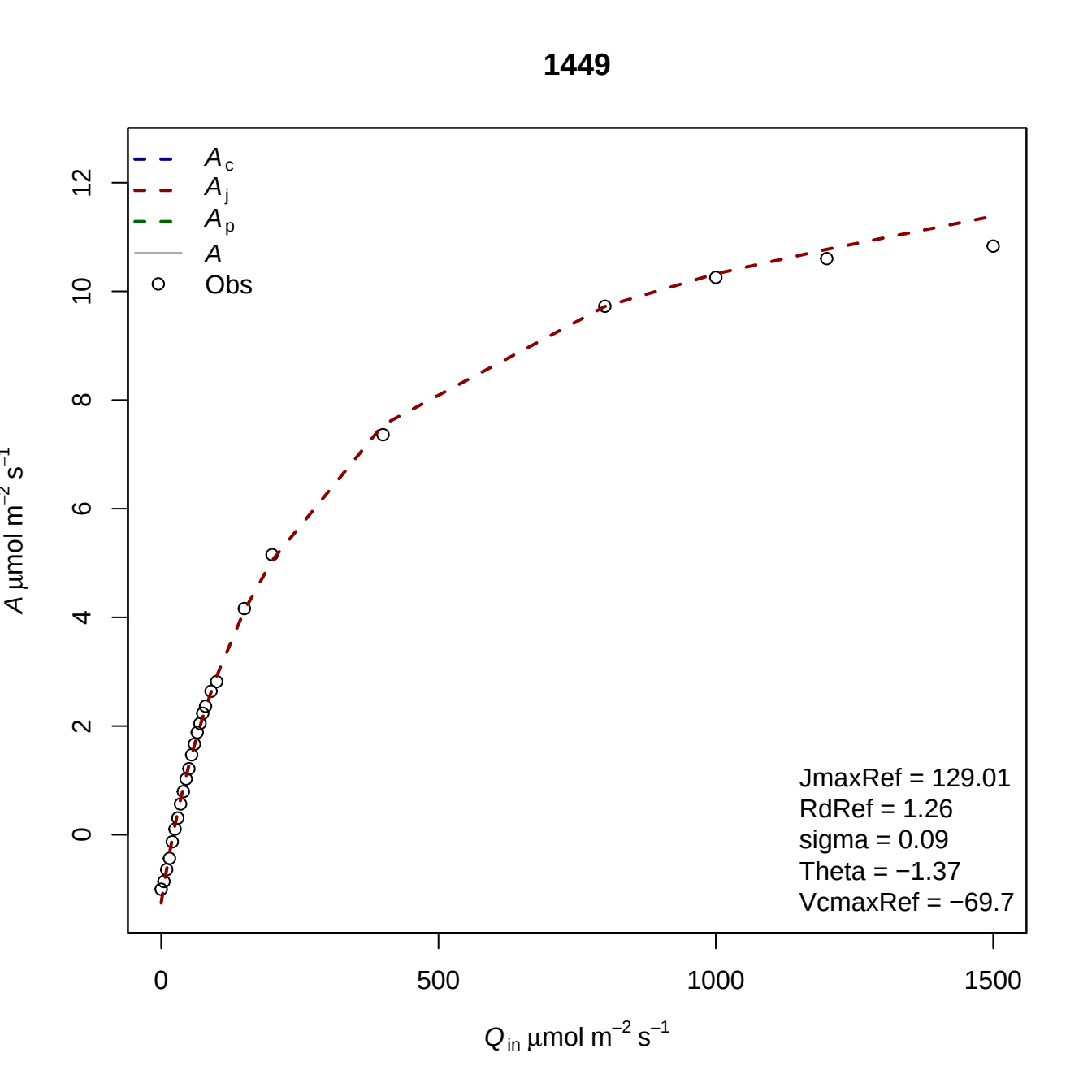

1444
A_c
A_j
A_p
A
Obs
JmaxRef = 273.7
RdRef = 2.99
sigma = 0.11
Theta = −3.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

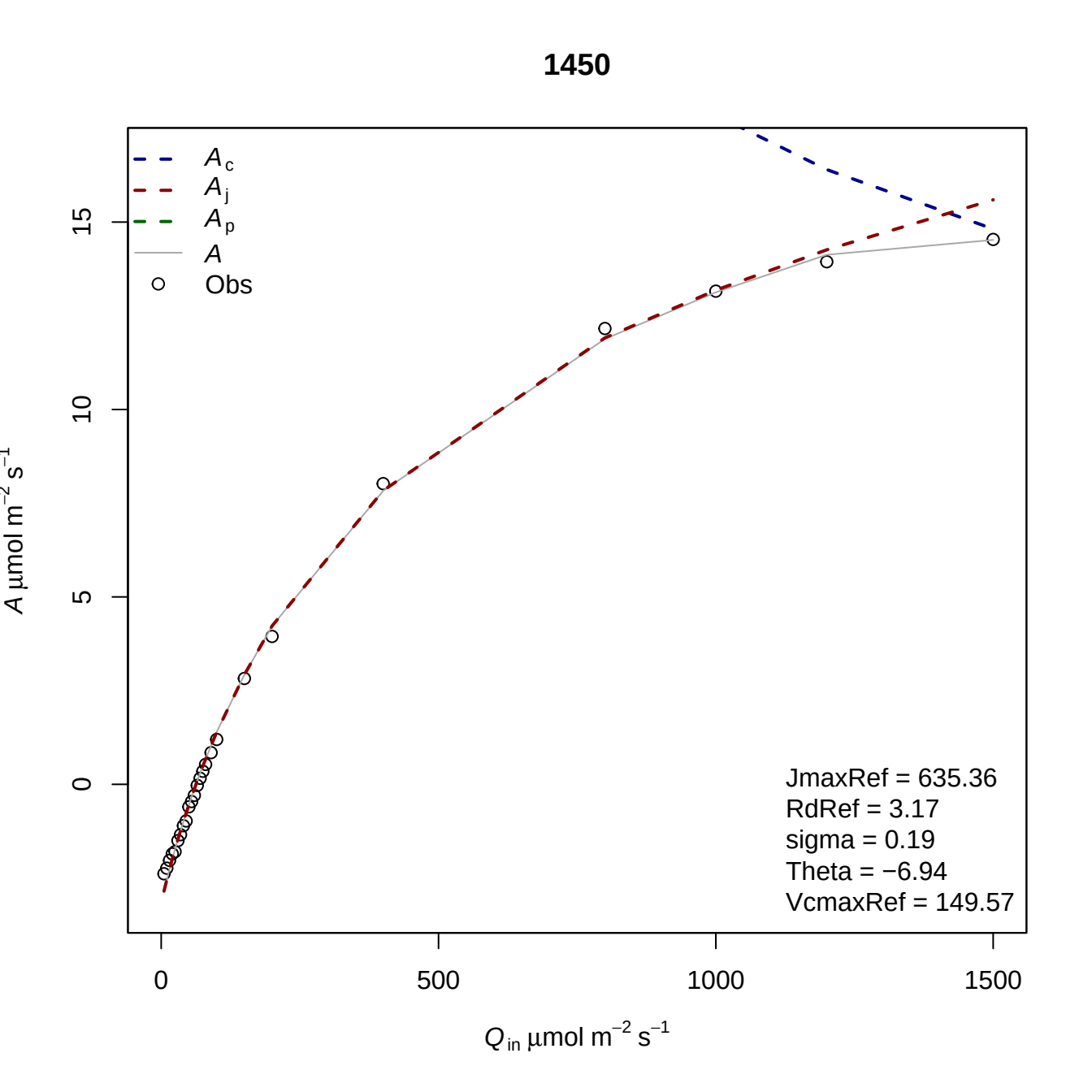

1445
A_c
A_j
A_p
A
Obs
JmaxRef = 222.69
RdRef = 3.91
sigma = 0.12
Theta = −1.27
VcmaxRef = 9999
0
5
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

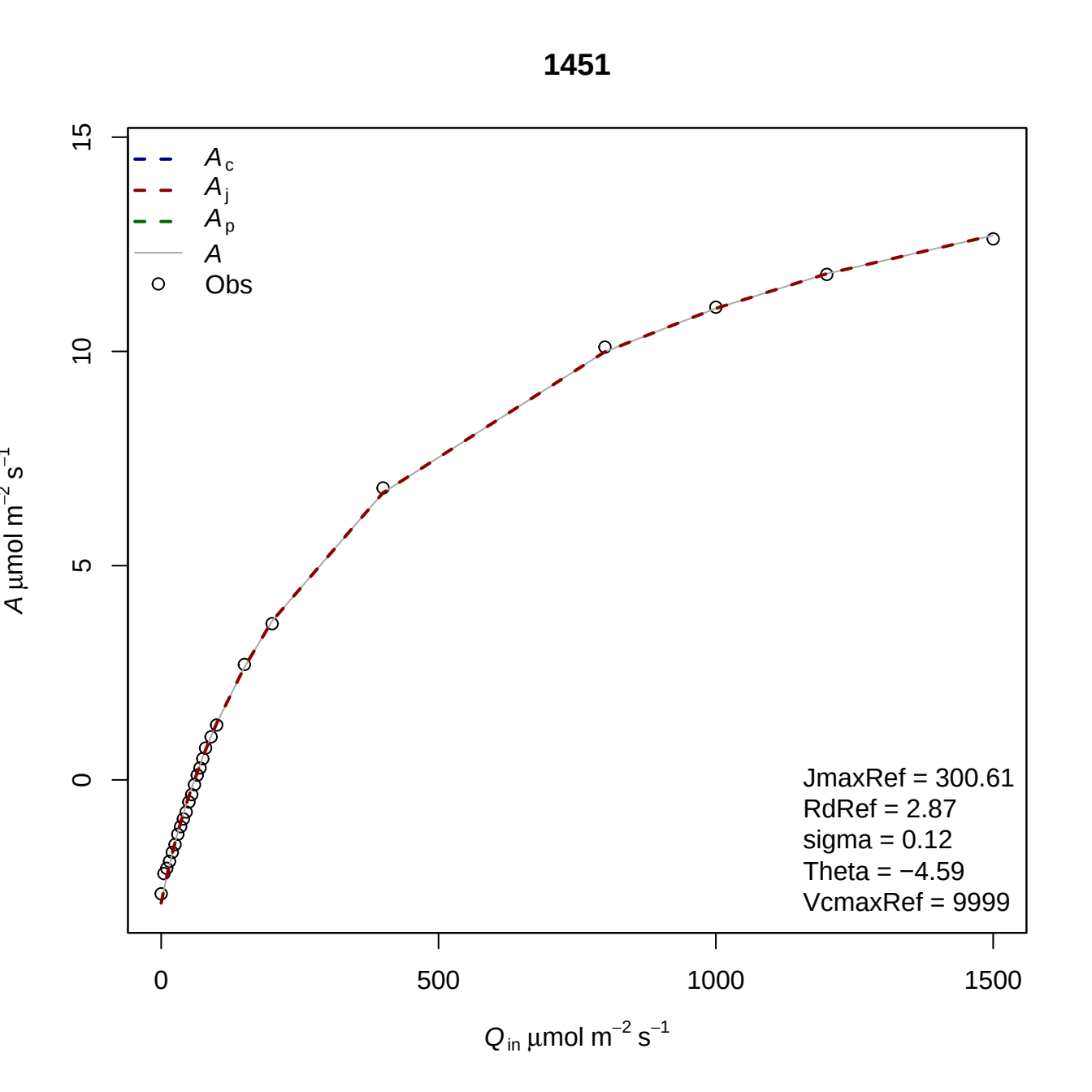

1446
A_c
A_j
A_p
A
Obs
JmaxRef = 299.08
RdRef = 3.05
sigma = 0.18
Theta = −4.29
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

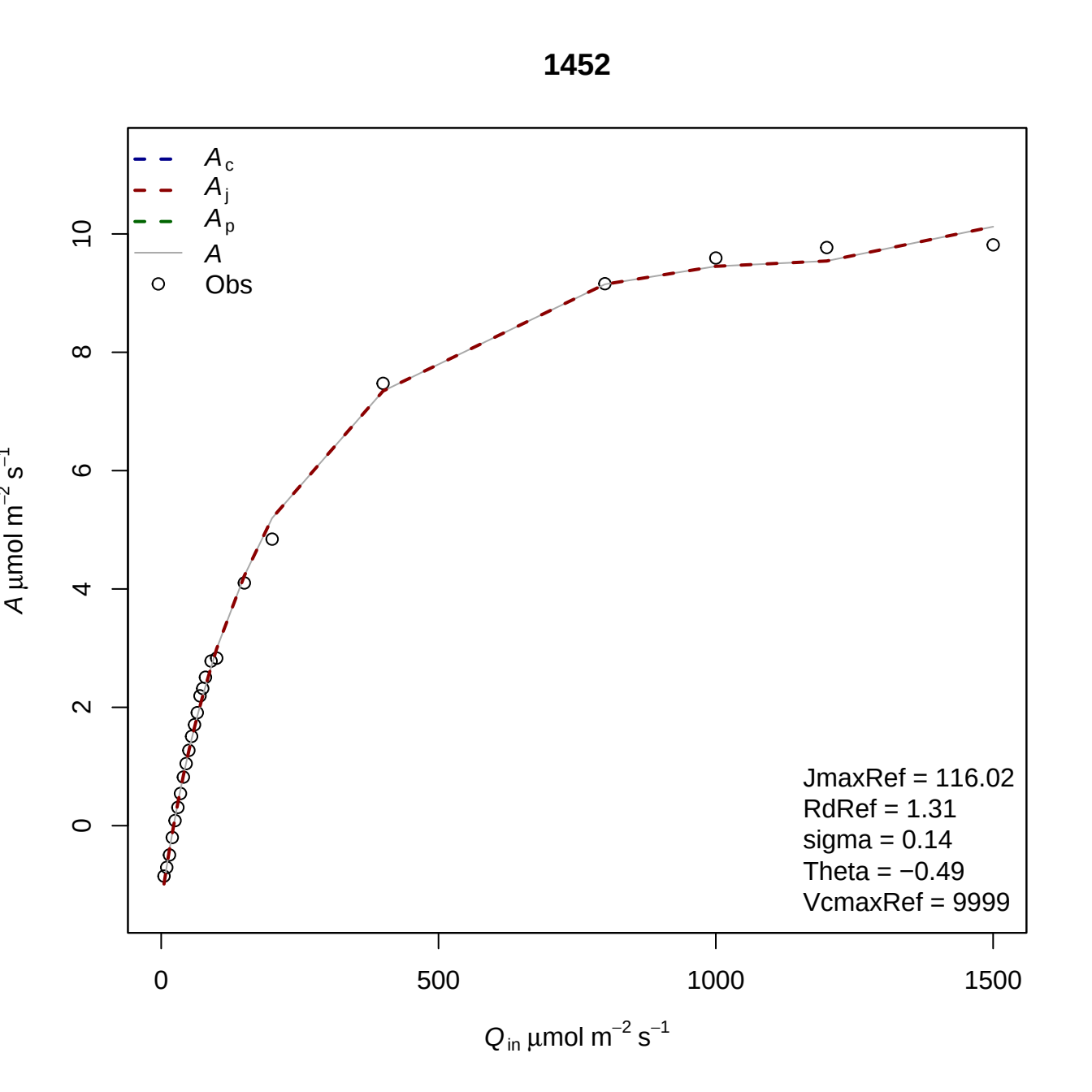

1447
A_c
A_j
A_p
A
Obs
JmaxRef = 111.7
RdRef = 1.62
sigma = 0.1
Theta = −1.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

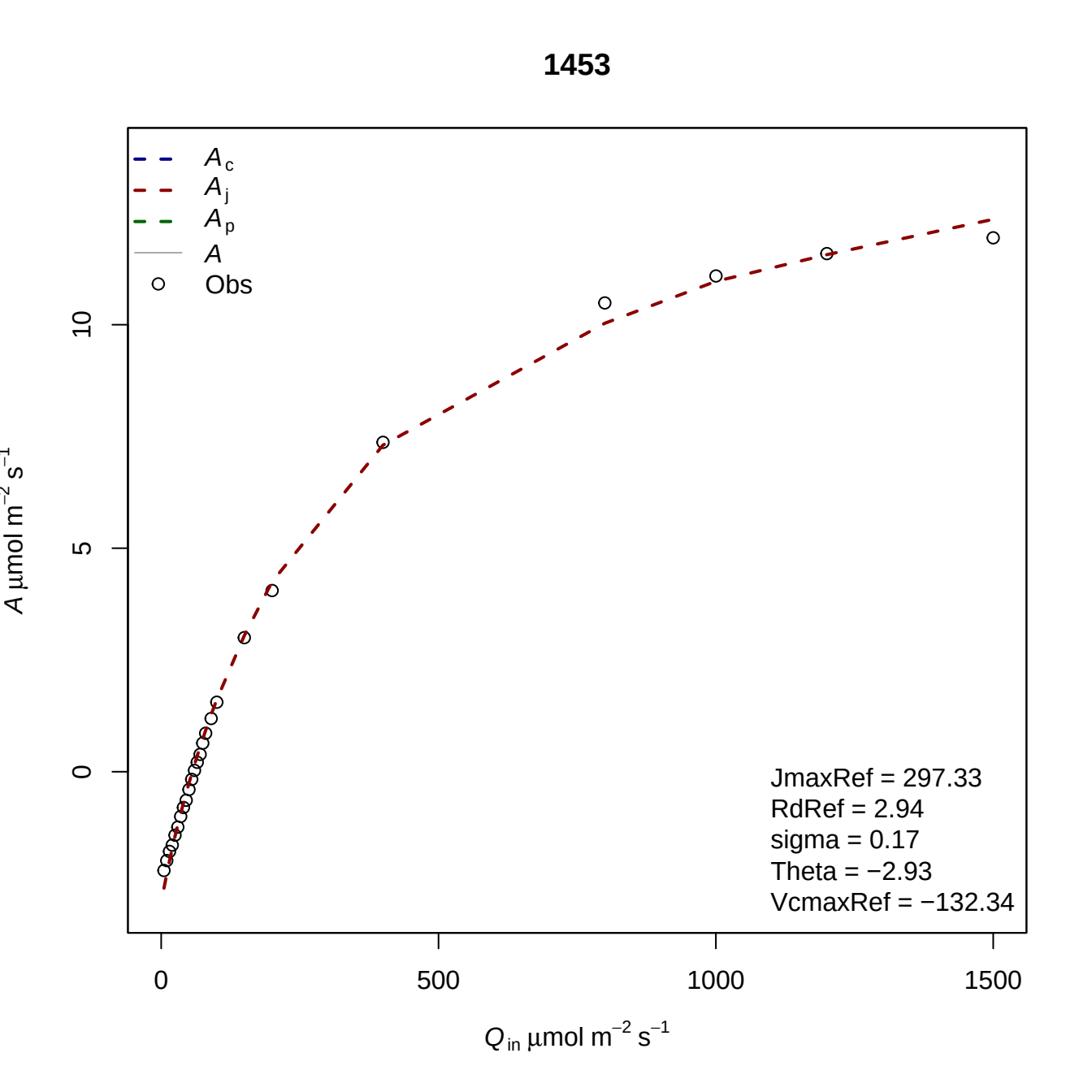

1448
A_c
A_j
A_p
A
Obs
JmaxRef = 107.24
RdRef = 1.04
sigma = 0.11
Theta = −1.51
VcmaxRef = 64.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

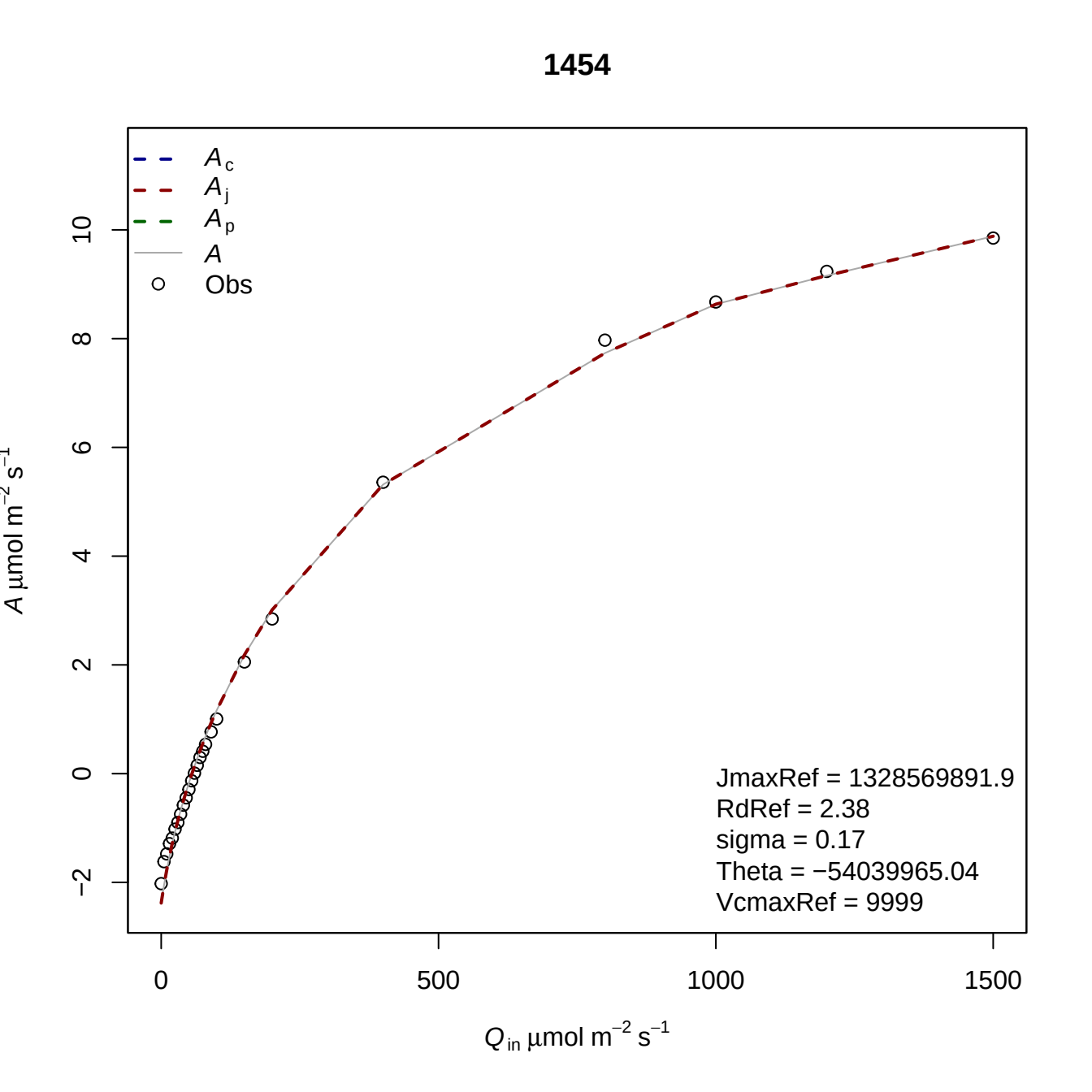

1449
A_c
A_j
A_p
A
Obs
JmaxRef = 129.01
RdRef = 1.26
sigma = 0.09
Theta = −1.37
VcmaxRef = −69.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

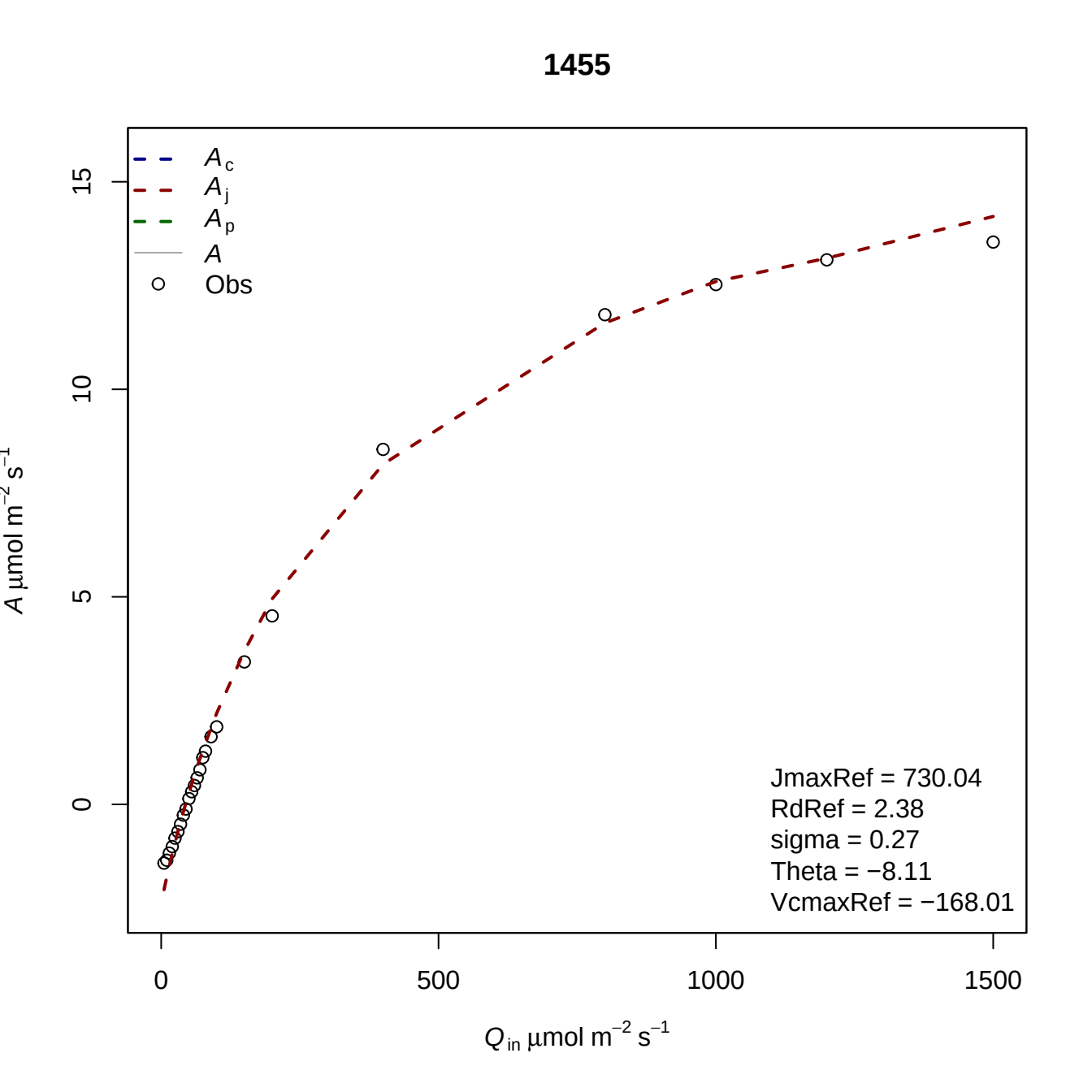

1450
A_c
A_j
A_p
A
Obs
JmaxRef = 635.36
RdRef = 3.17
sigma = 0.19
Theta = −6.94
VcmaxRef = 149.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

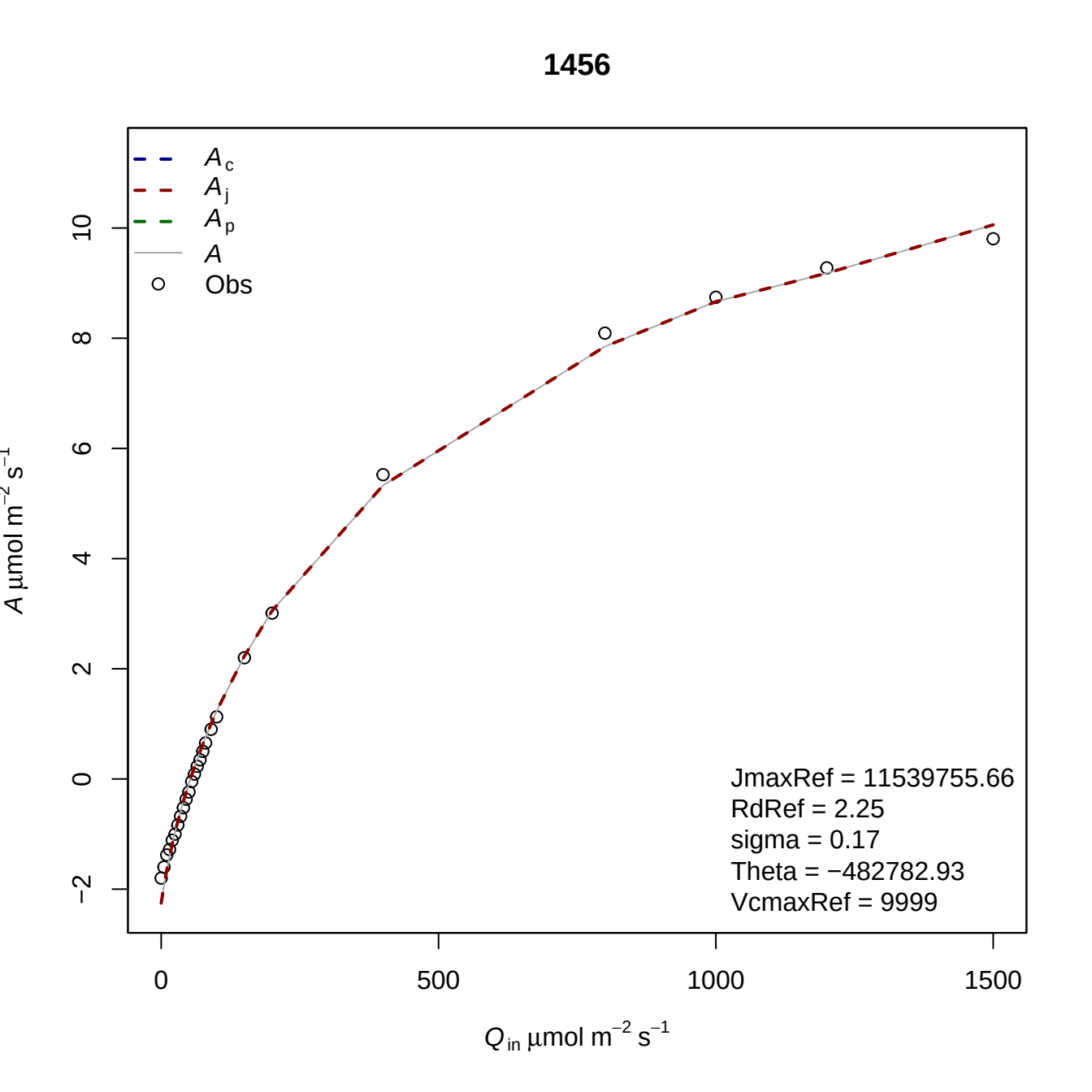

1451
A_c
A_j
A_p
A
Obs
JmaxRef = 300.61
RdRef = 2.87
sigma = 0.12
Theta = −4.59
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

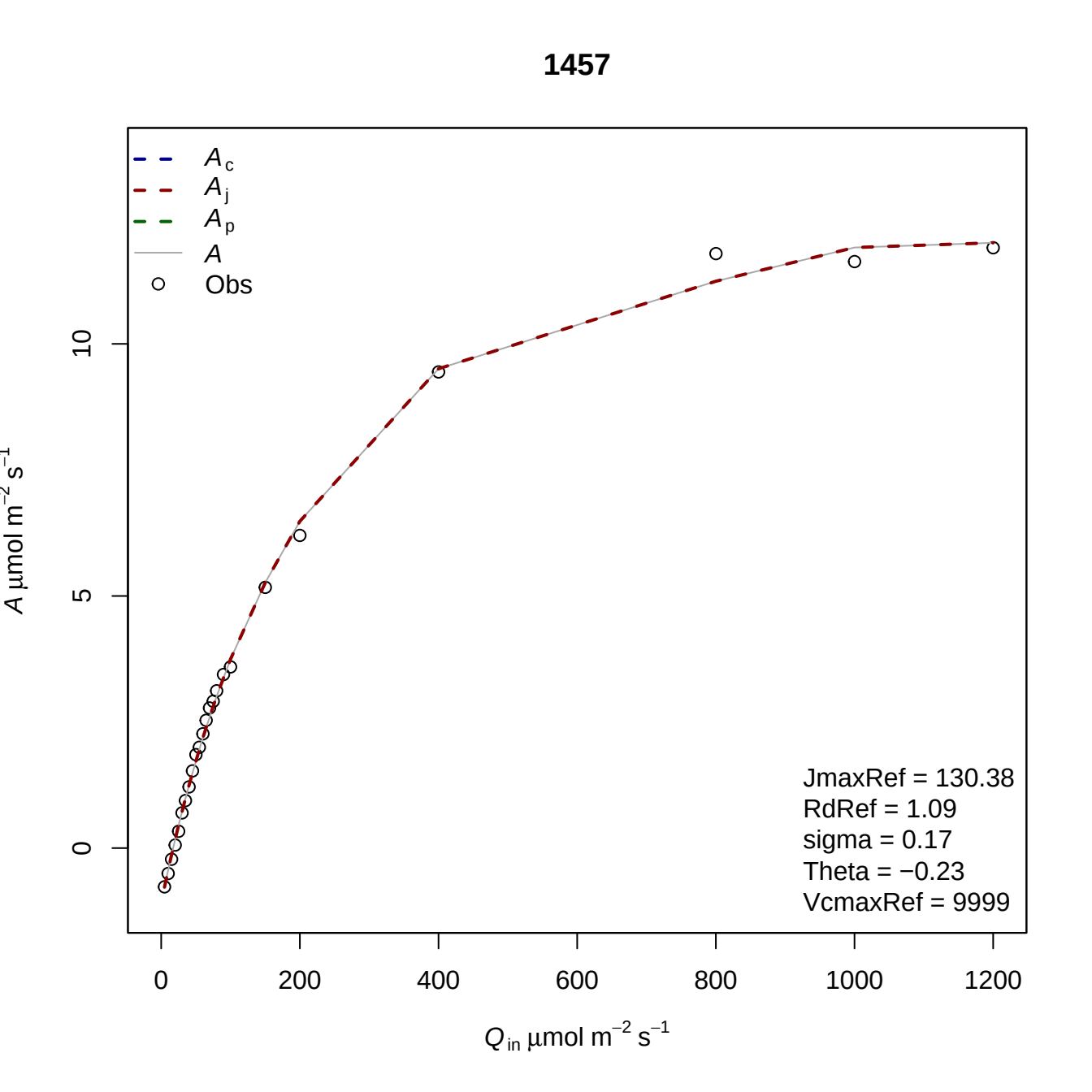

1452
A_c
A_j
A_p
A
Obs
JmaxRef = 116.02
RdRef = 1.31
sigma = 0.14
Theta = −0.49
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

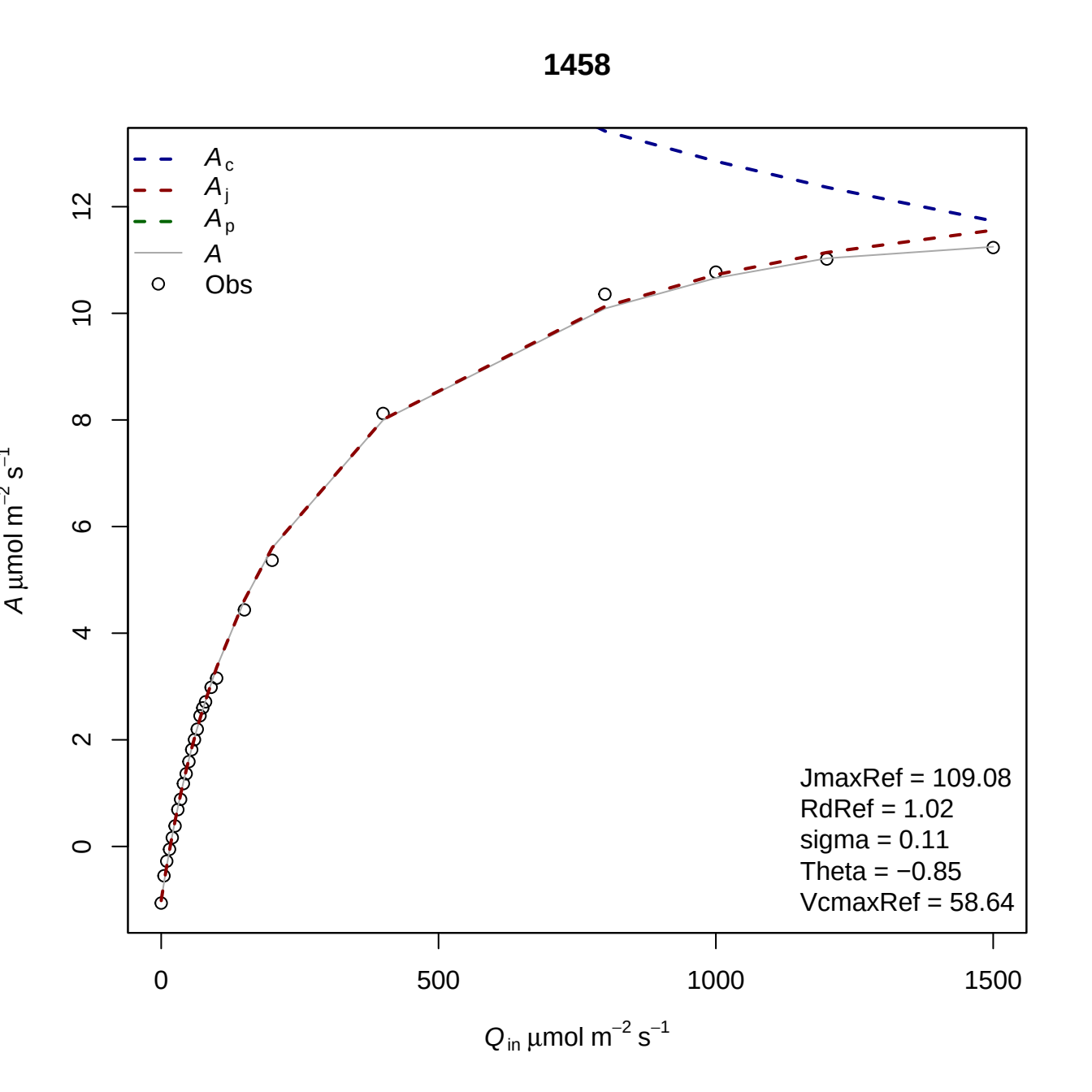

1453
A_c
A_j
A_p
A
Obs
JmaxRef = 297.33
RdRef = 2.94
sigma = 0.17
Theta = −2.93
VcmaxRef = −132.34
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

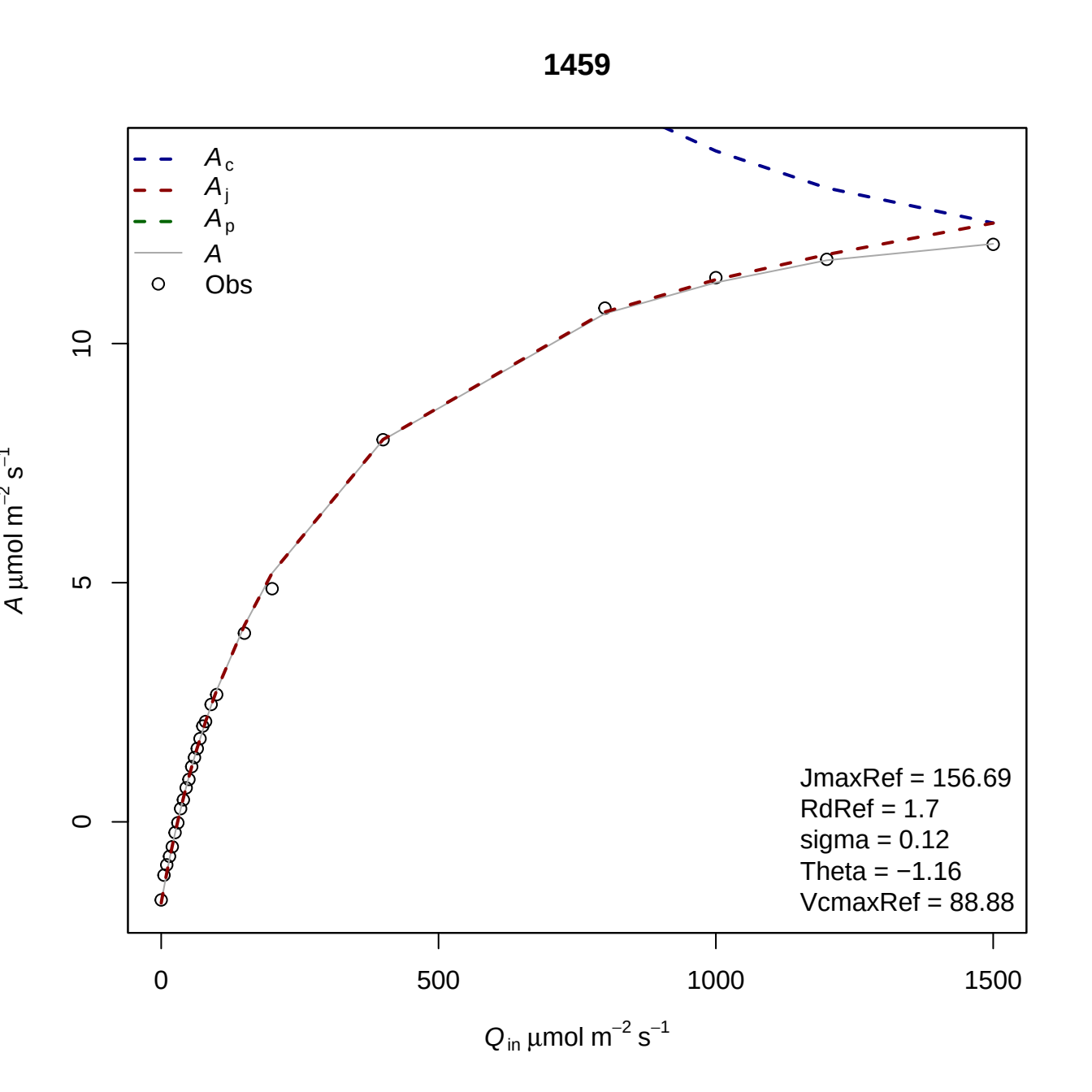

1454
A_c
A_j
A_p
A
Obs
JmaxRef = 1328569891.9
RdRef = 2.38
sigma = 0.17
Theta = −54039965.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

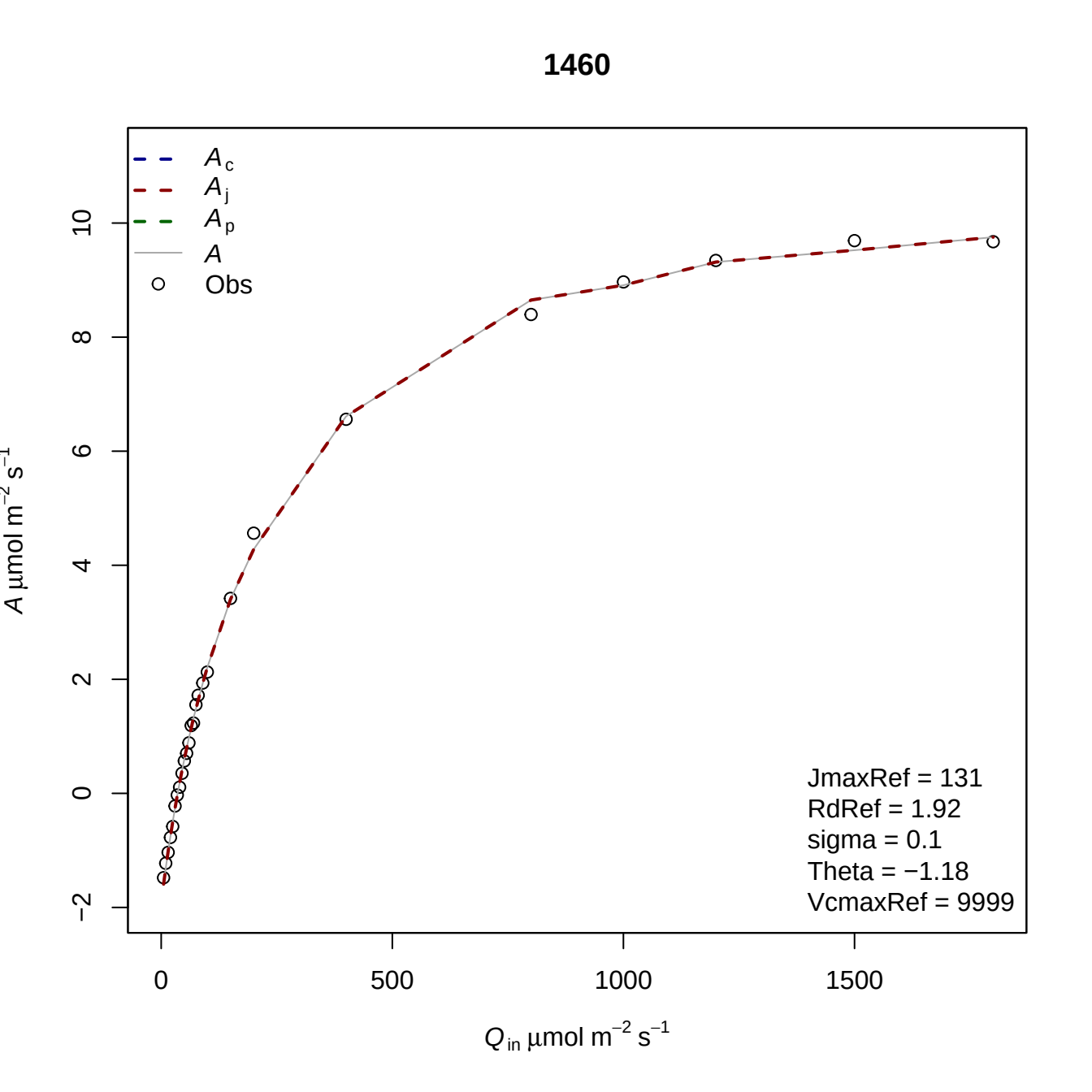

1455
A_c
A_j
A_p
A
Obs
JmaxRef = 730.04
RdRef = 2.38
sigma = 0.27
Theta = −8.11
VcmaxRef = −168.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

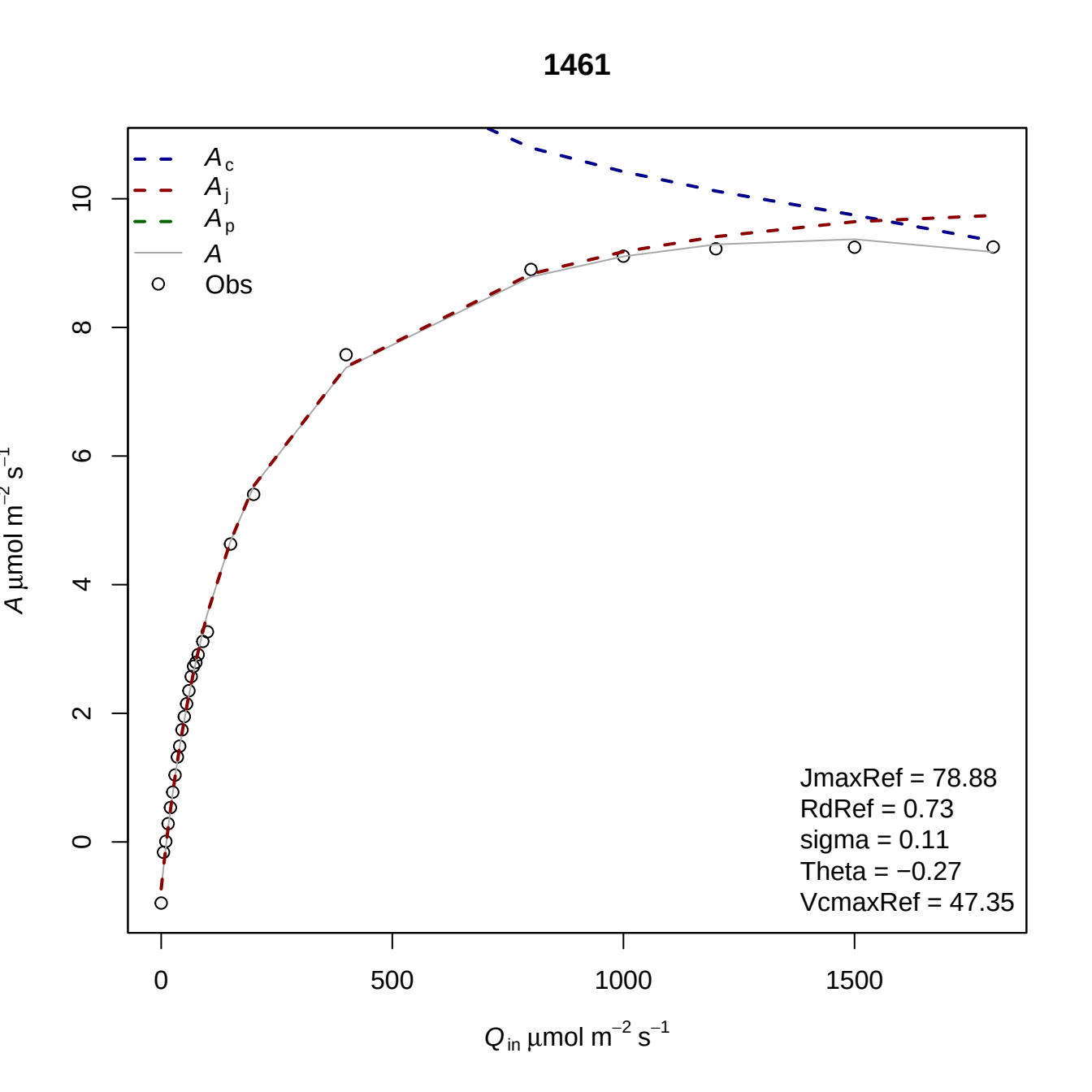

1456
A_c
A_j
A_p
A
Obs
JmaxRef = 11539755.66
RdRef = 2.25
sigma = 0.17
Theta = −482782.93
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

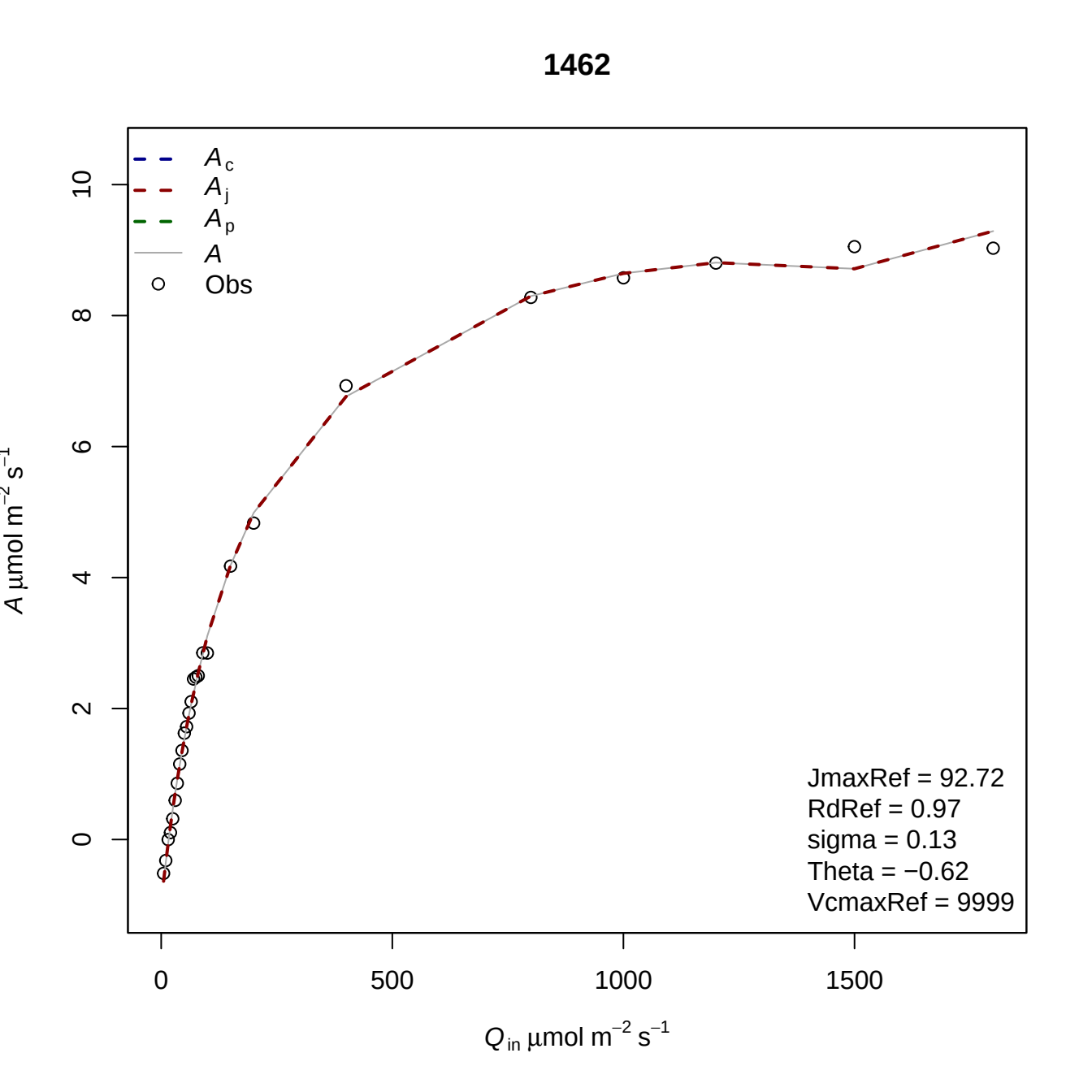

1457
A_c
A_j
A_p
A
Obs
JmaxRef = 130.38
RdRef = 1.09
sigma = 0.17
Theta = −0.23
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

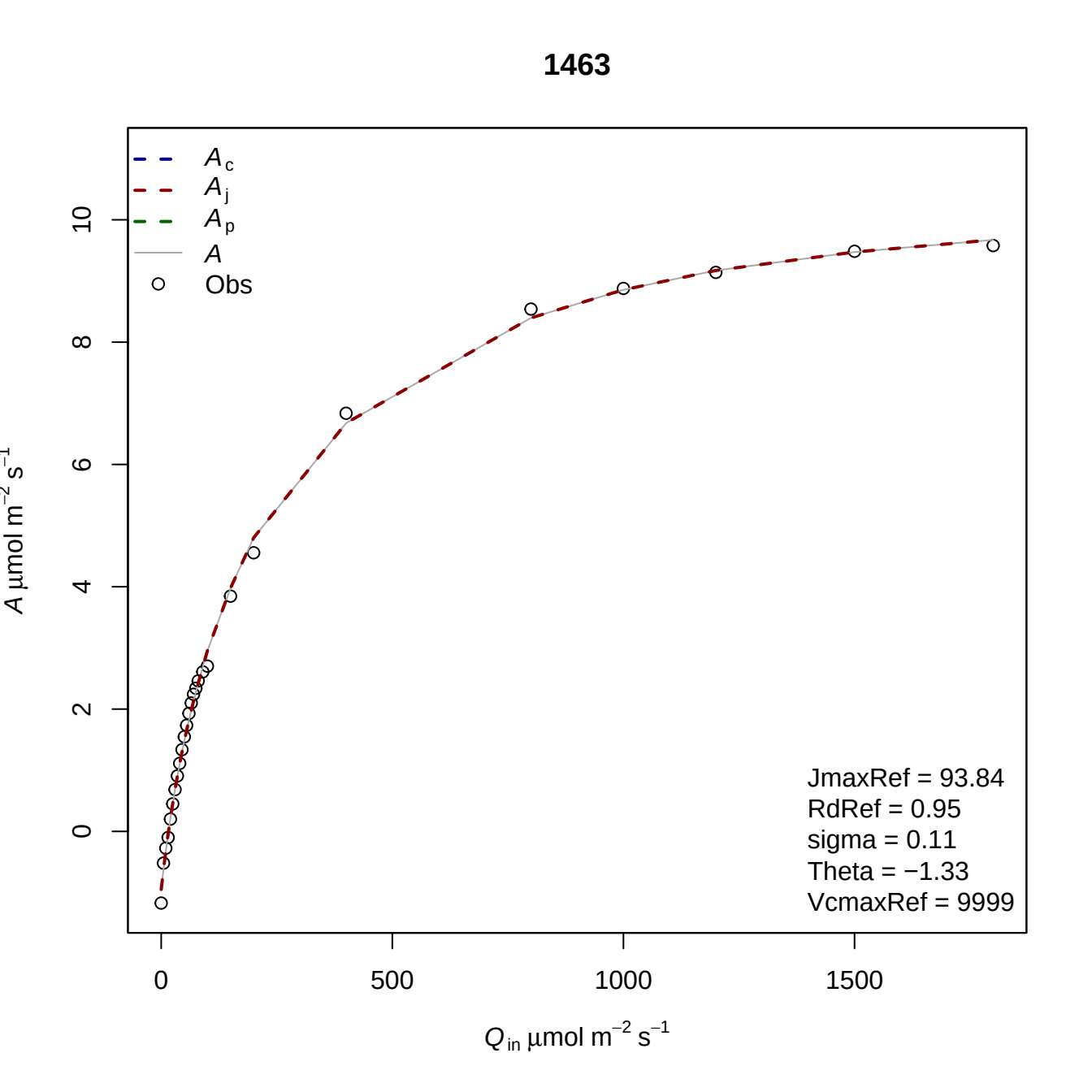

1458
A_c
A_j
A_p
A
Obs
JmaxRef = 109.08
RdRef = 1.02
sigma = 0.11
Theta = −0.85
VcmaxRef = 58.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

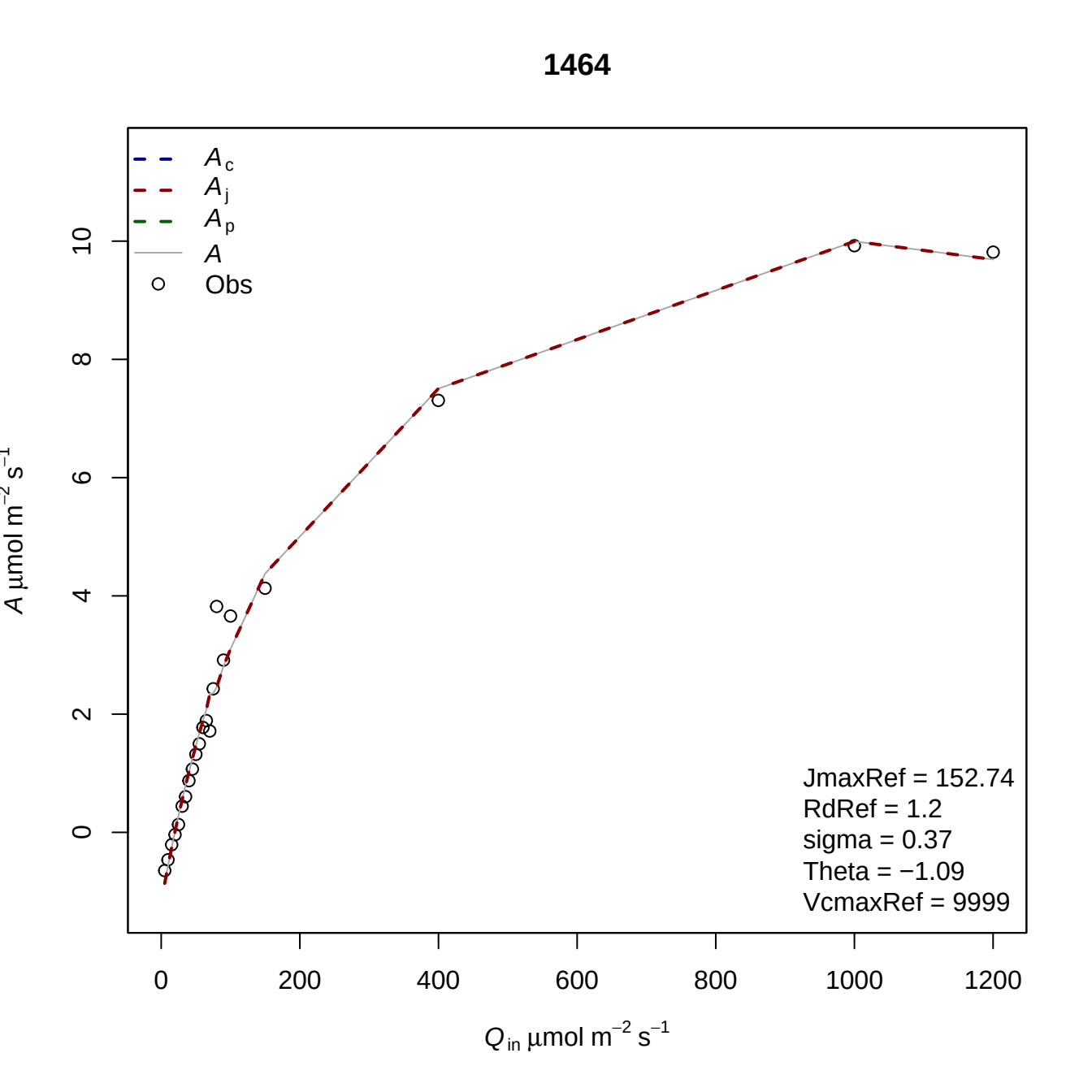

1459
A_c
A_j
A_p
A
Obs
JmaxRef = 156.69
RdRef = 1.7
sigma = 0.12
Theta = −1.16
VcmaxRef = 88.88
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

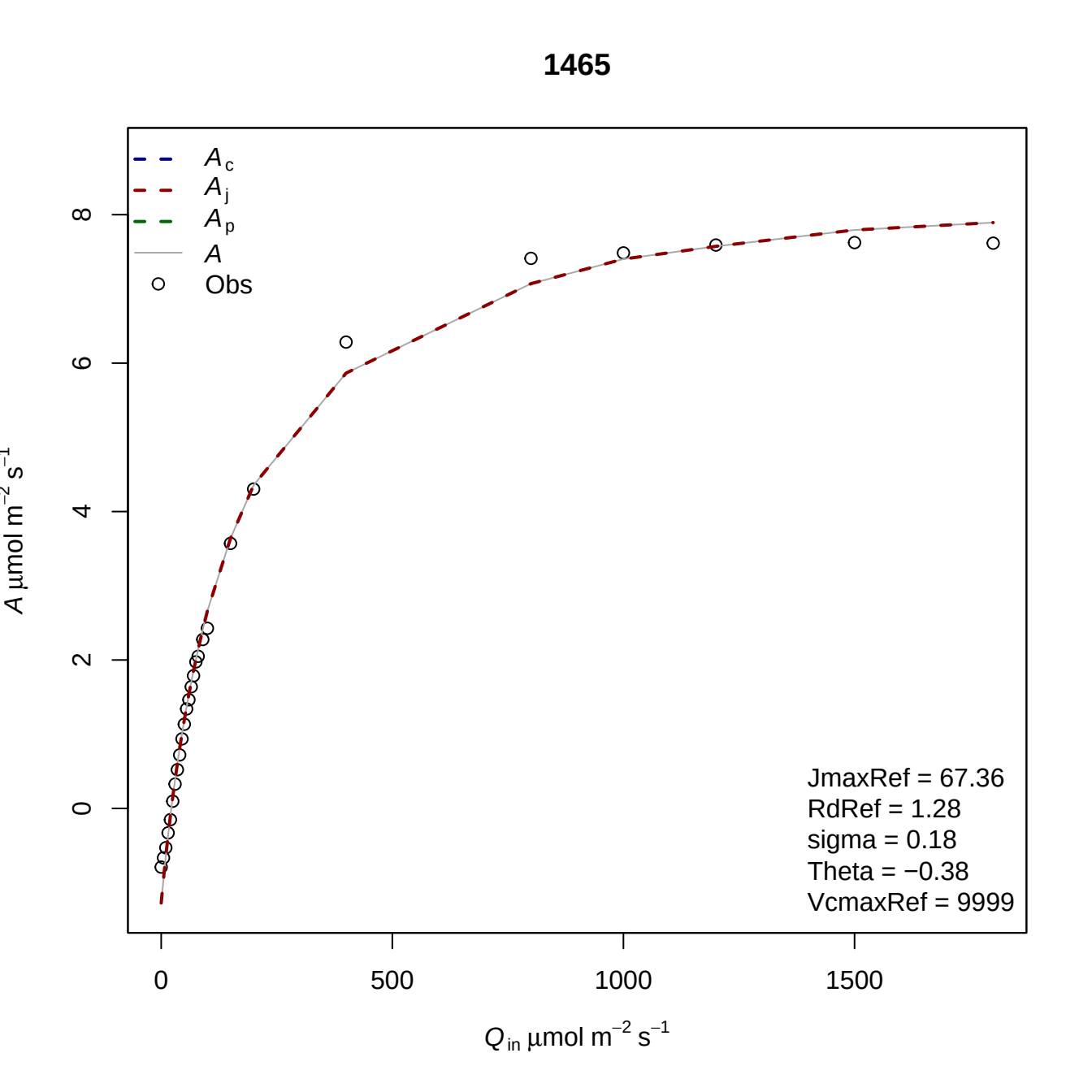

1460
A_c
A_j
A_p
A
Obs
JmaxRef = 131
RdRef = 1.92
sigma = 0.1
Theta = −1.18
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

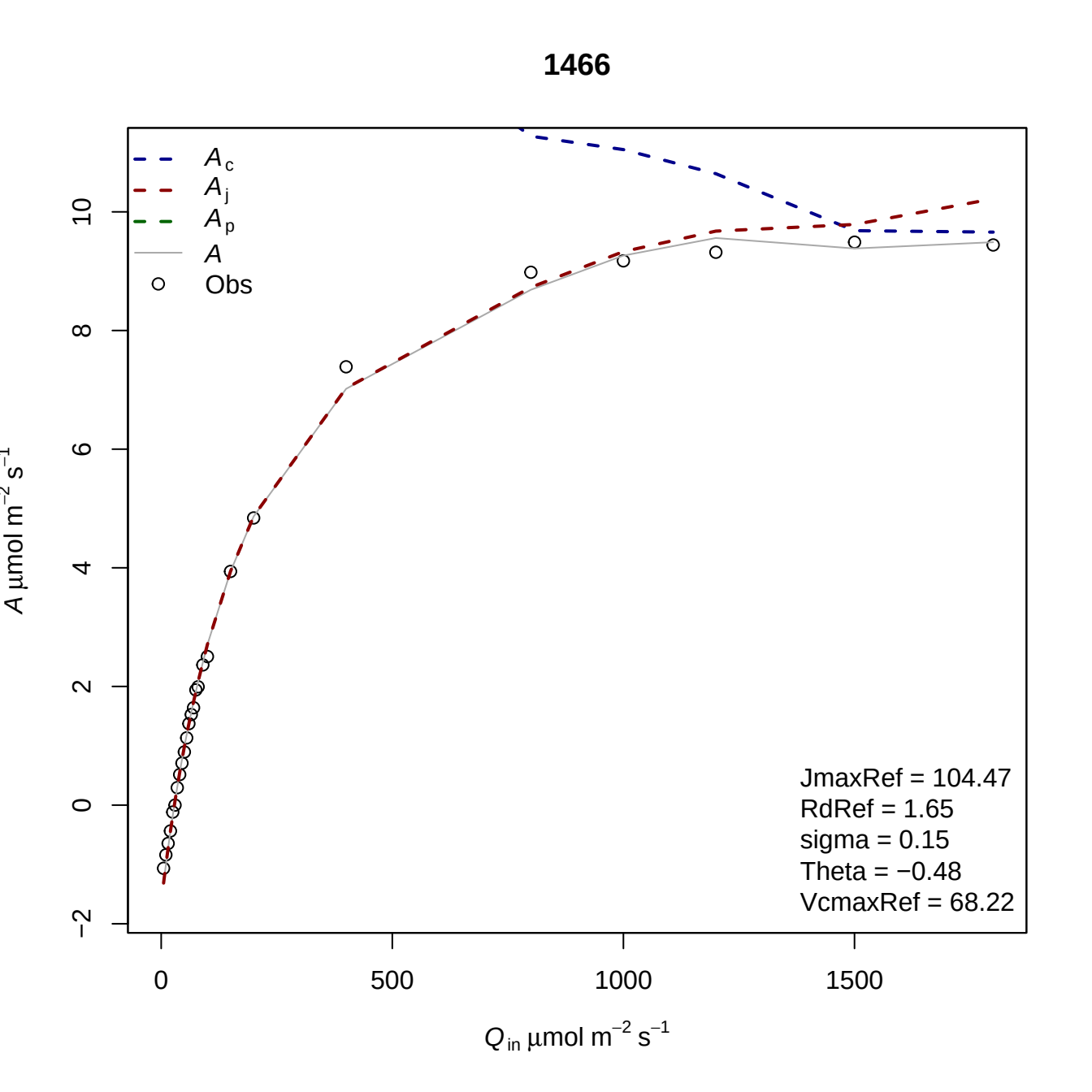

1461
A_c
A_j
A_p
A
Obs
JmaxRef = 78.88
RdRef = 0.73
sigma = 0.11
Theta = −0.27
VcmaxRef = 47.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

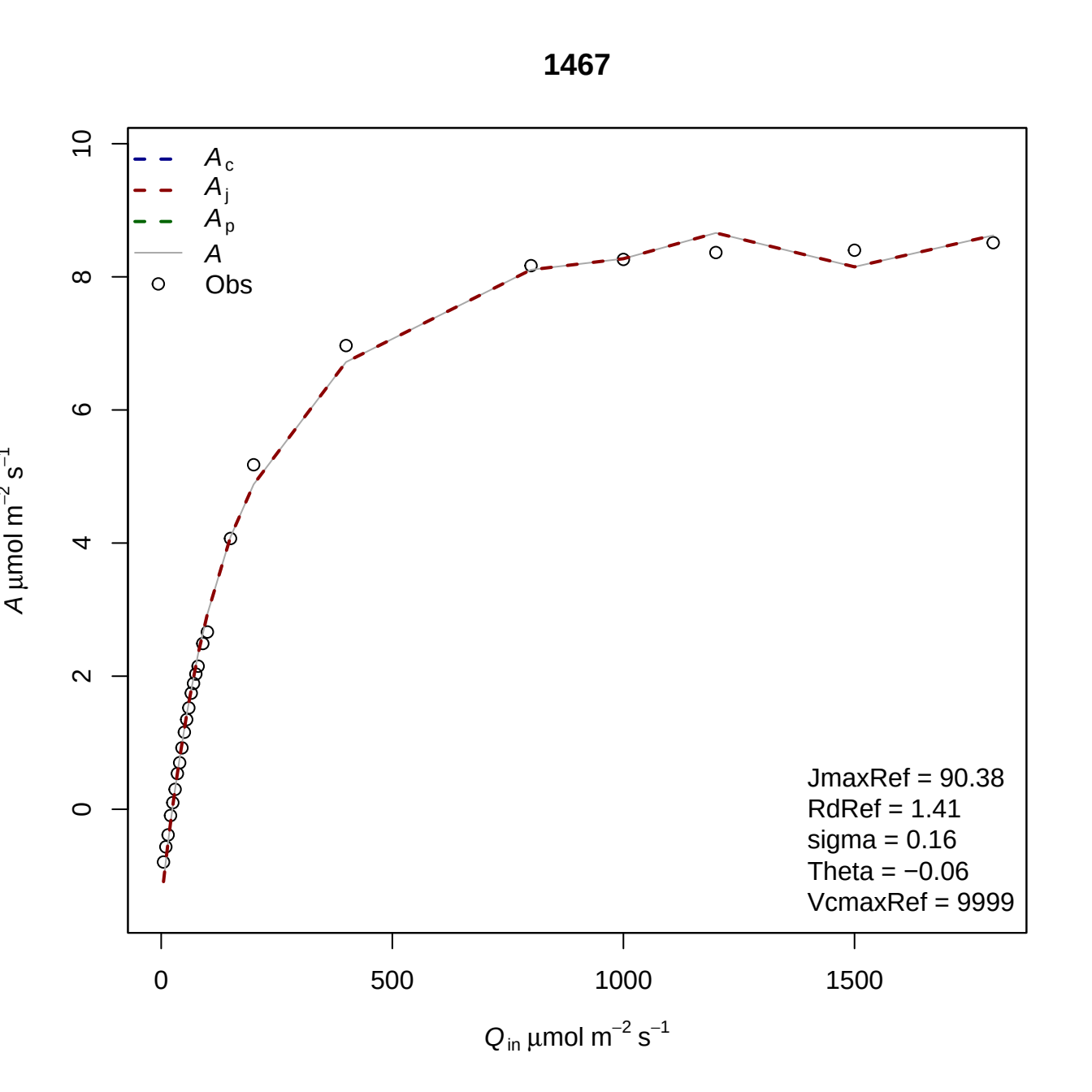

1462
A_c
A_j
A_p
A
Obs
JmaxRef = 92.72
RdRef = 0.97
sigma = 0.13
Theta = −0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

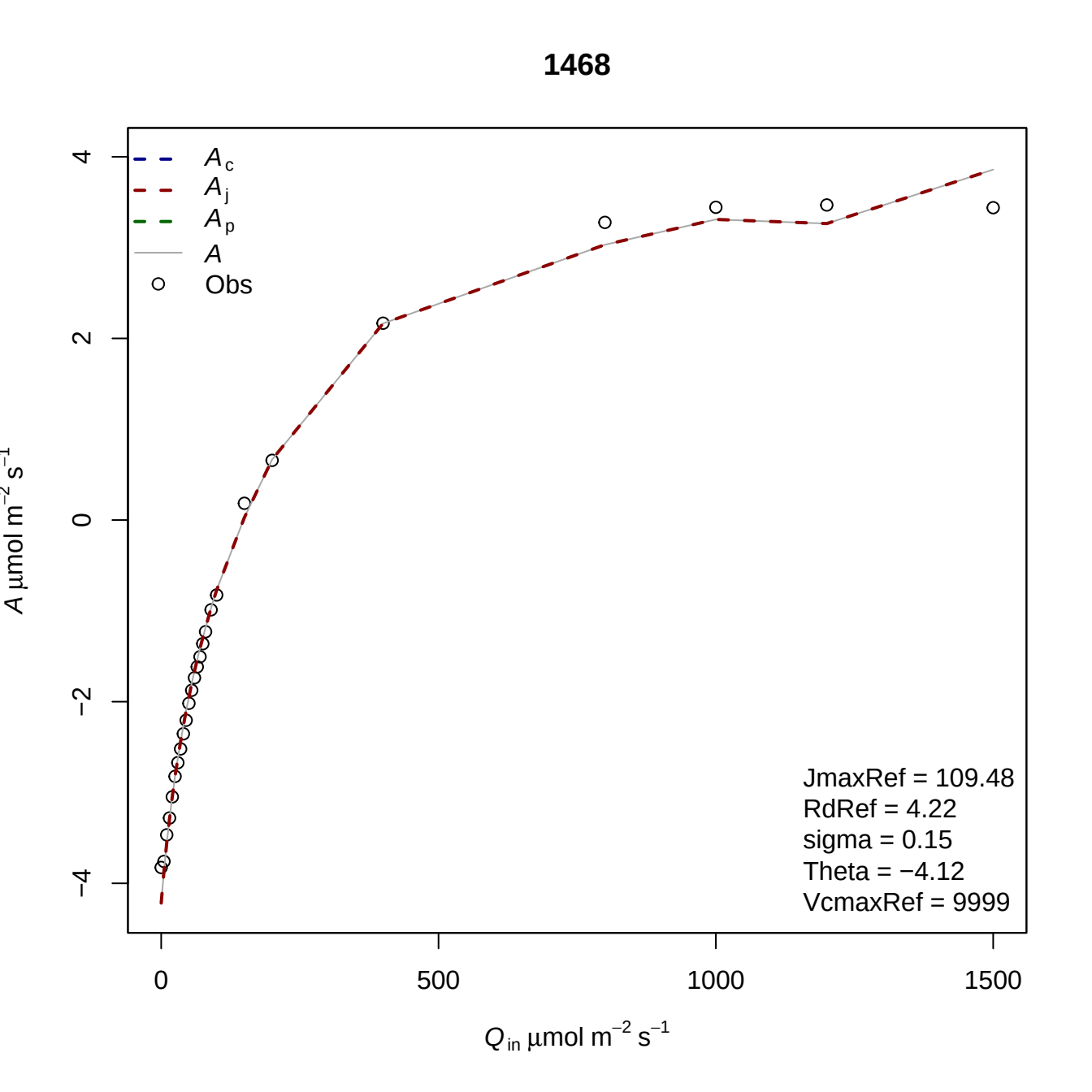

1463
A_c
A_j
A_p
A
Obs
JmaxRef = 93.84
RdRef = 0.95
sigma = 0.11
Theta = −1.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

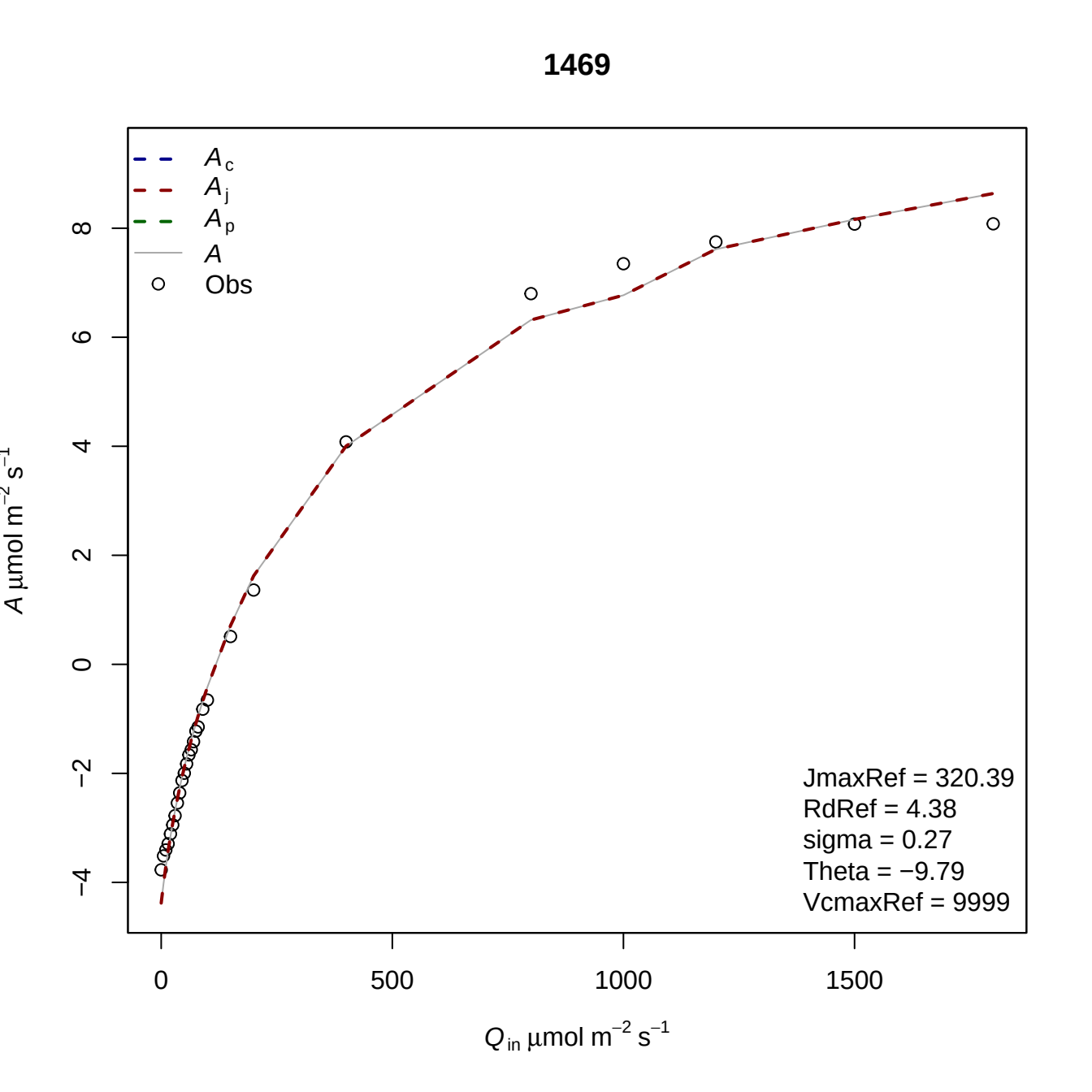

1464
A_c
A_j
A_p
A
Obs
JmaxRef = 152.74
RdRef = 1.2
sigma = 0.37
Theta = −1.09
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

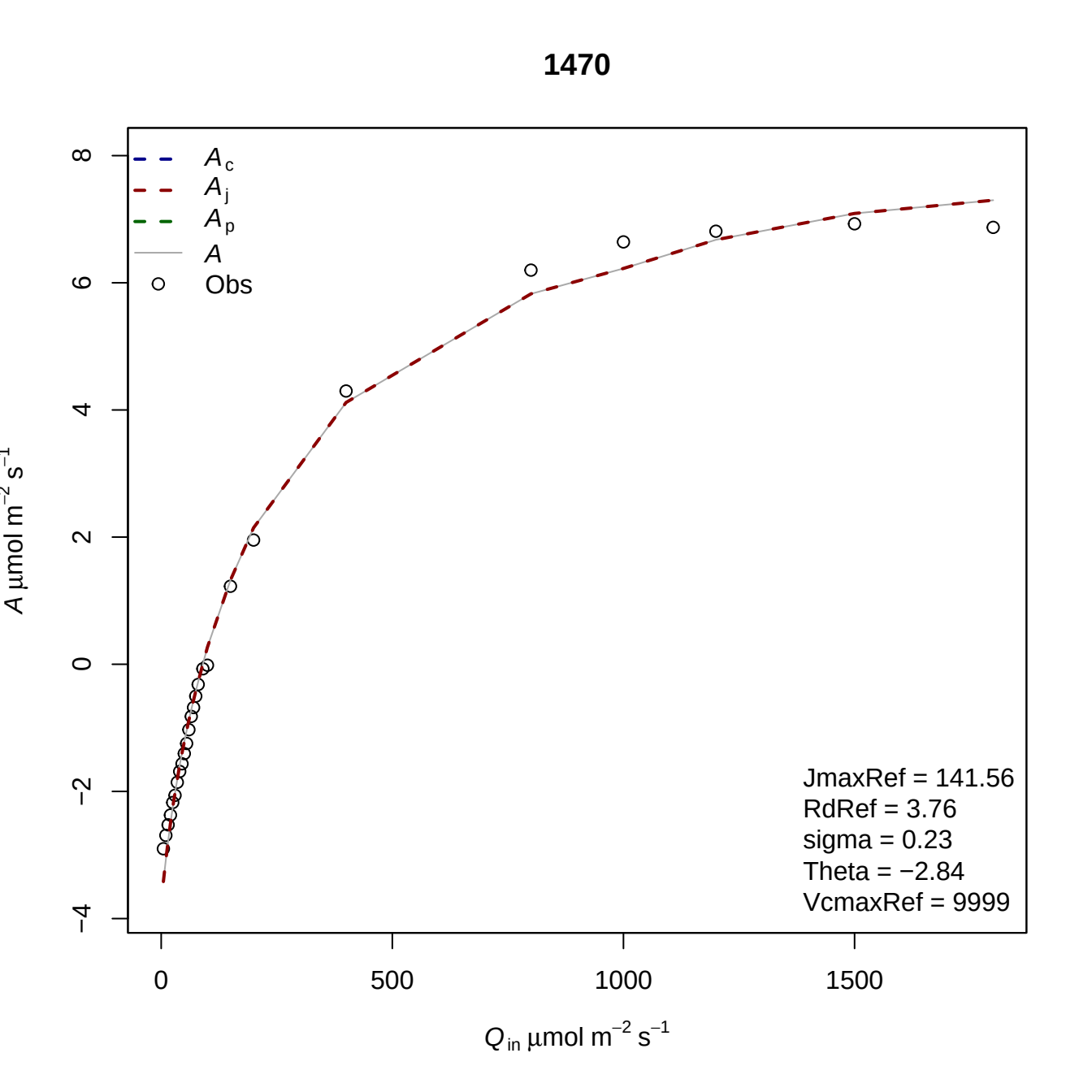

1465
A_c
A_j
A_p
A
Obs
JmaxRef = 67.36
RdRef = 1.28
sigma = 0.18
Theta = −0.38
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

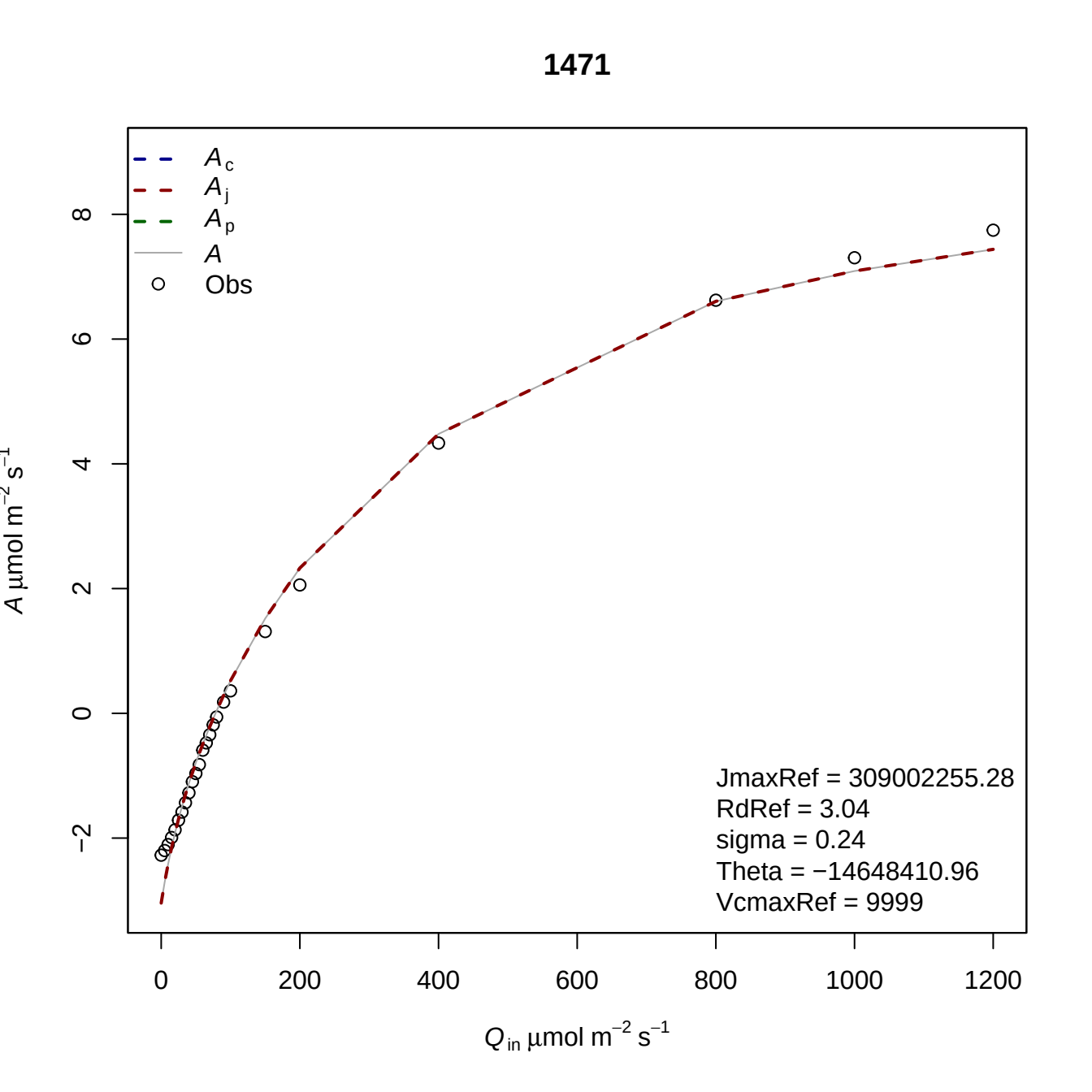

1466
A_c
A_j
A_p
A
Obs
JmaxRef = 104.47
RdRef = 1.65
sigma = 0.15
Theta = −0.48
VcmaxRef = 68.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

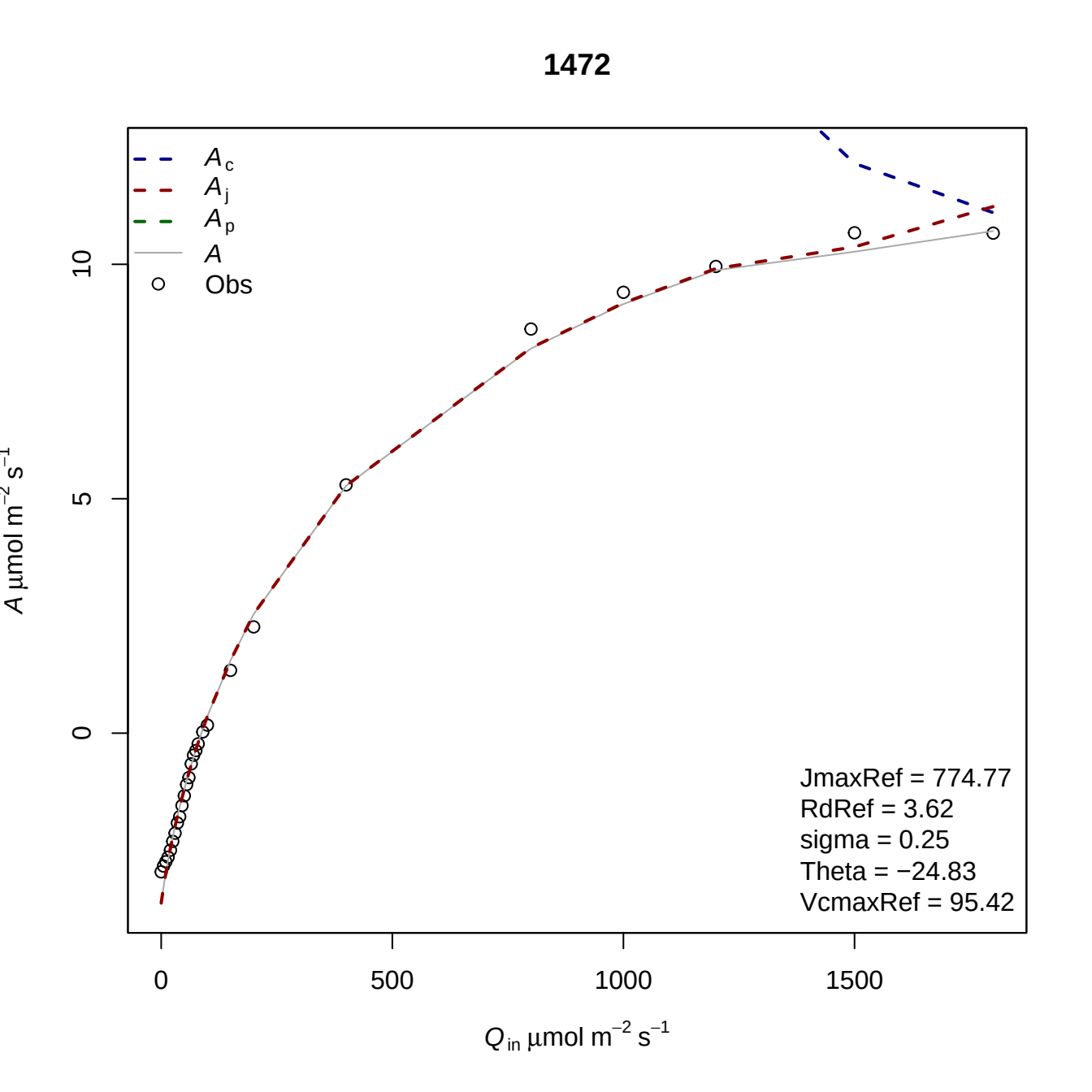

1467
A_c
A_j
A_p
A
Obs
JmaxRef = 90.38
RdRef = 1.41
sigma = 0.16
Theta = −0.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

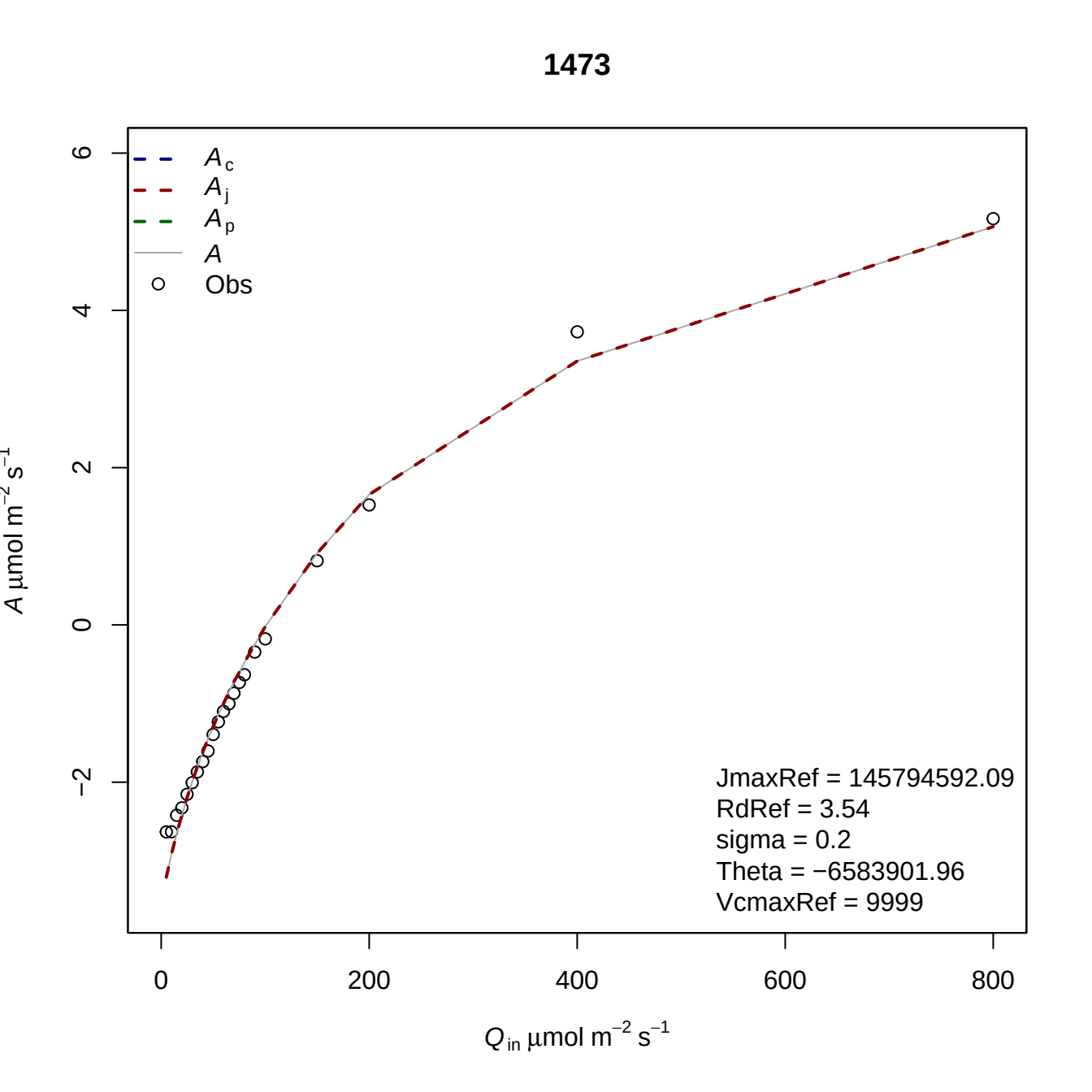

1468
A_c
A_j
A_p
A
Obs
JmaxRef = 109.48
RdRef = 4.22
sigma = 0.15
Theta = −4.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

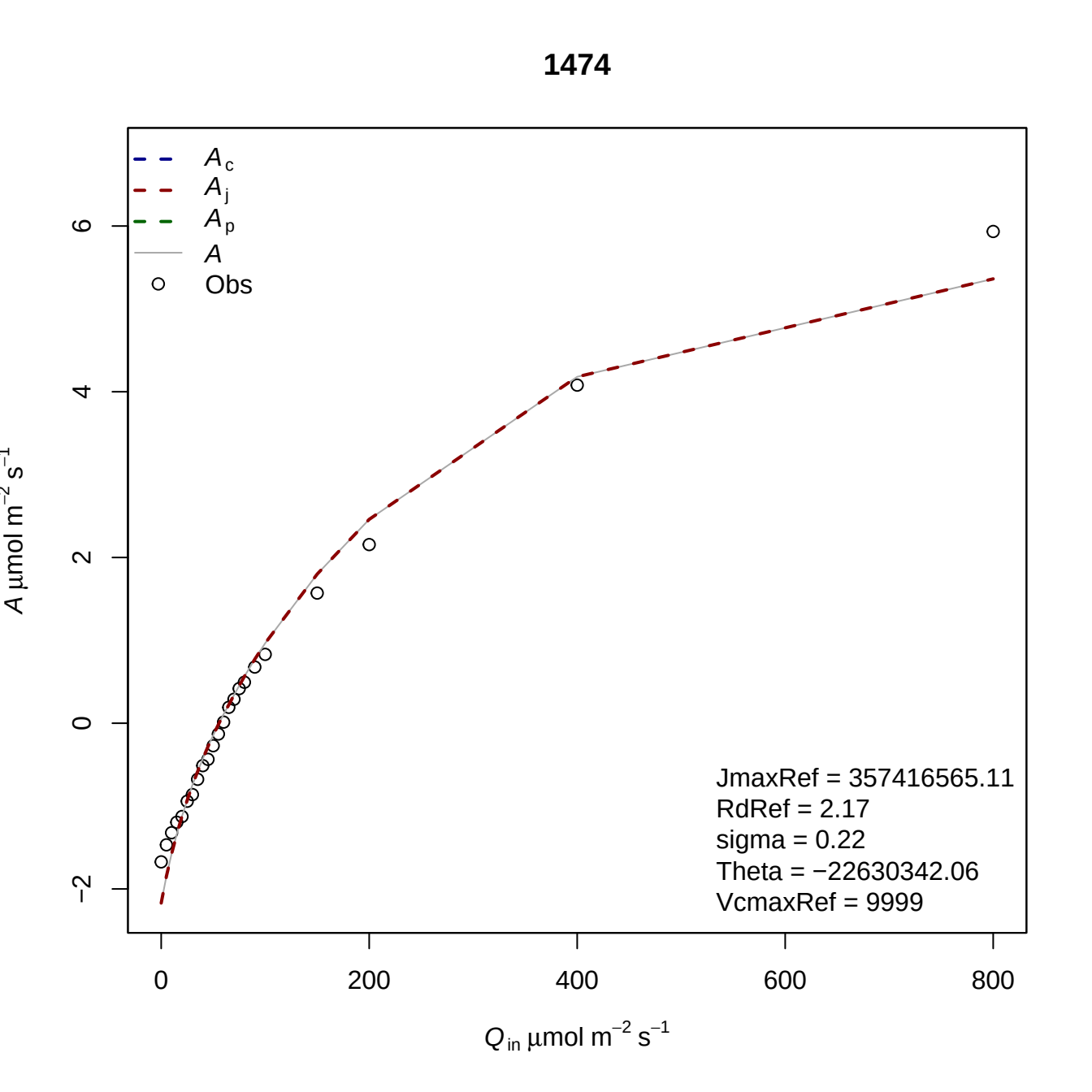

1469
A_c
A_j
A_p
A
Obs
JmaxRef = 320.39
RdRef = 4.38
sigma = 0.27
Theta = −9.79
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
500
1000
1500

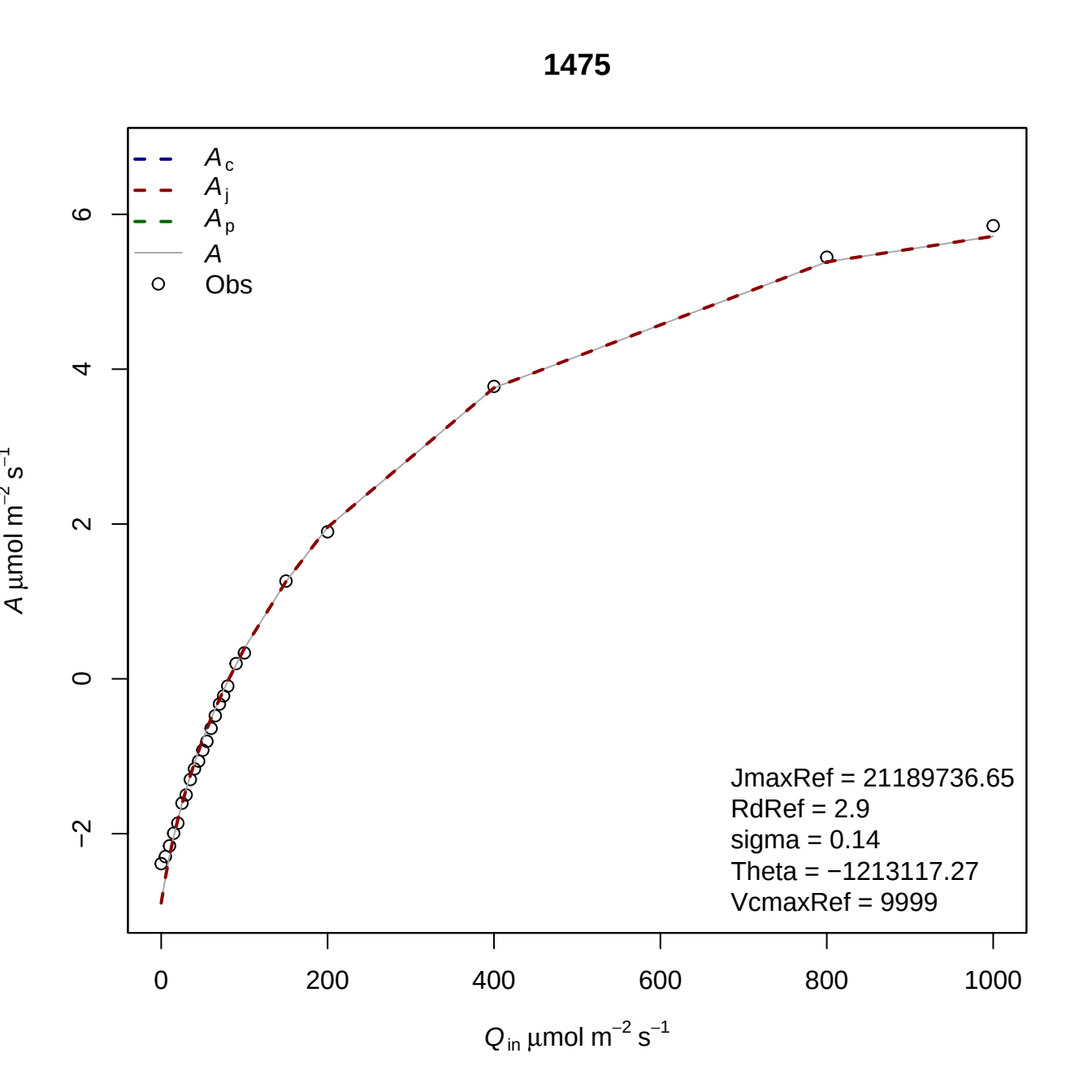

1470
A_c
A_j
A_p
A
Obs
JmaxRef = 141.56
RdRef = 3.76
sigma = 0.23
Theta = −2.84
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

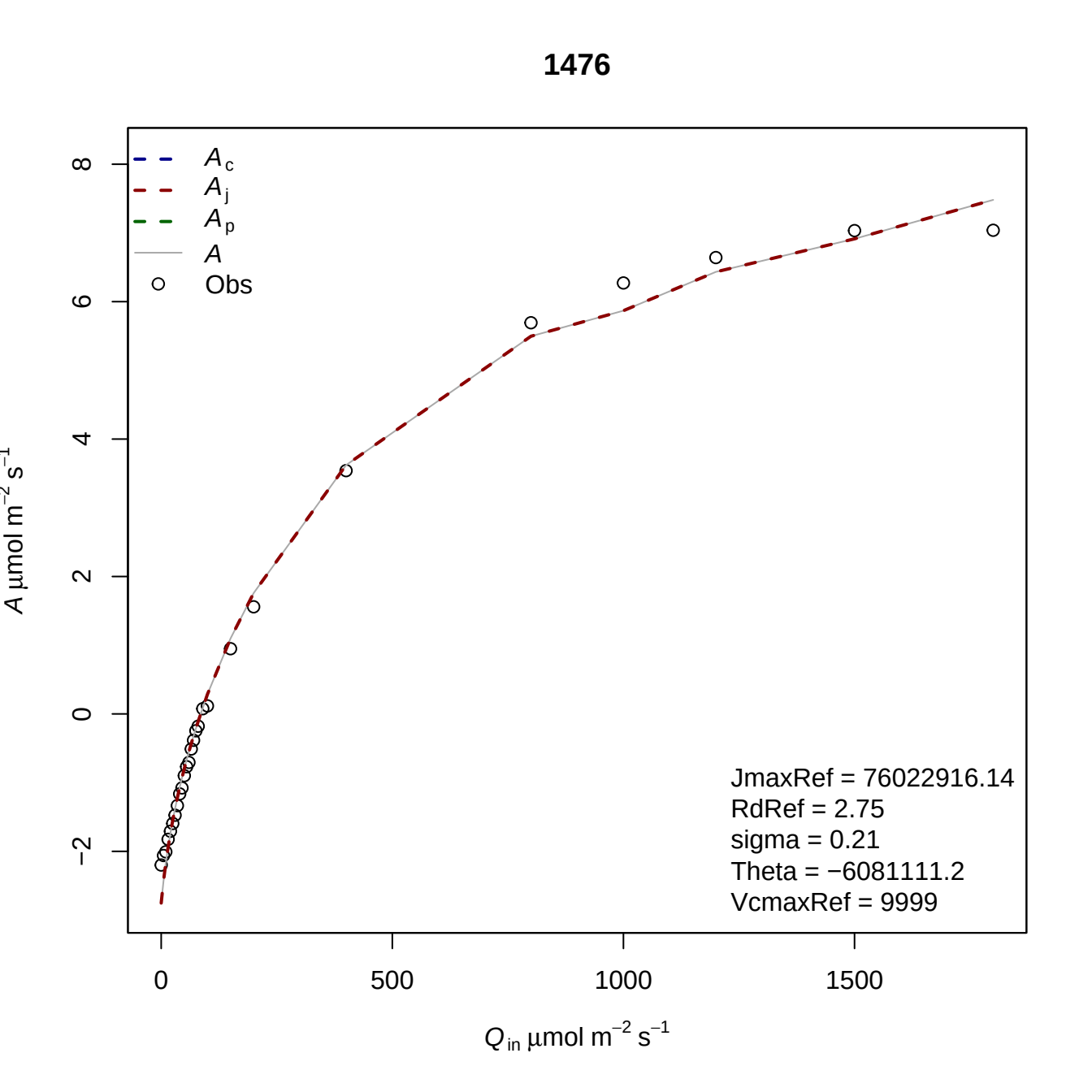

1471
A_c
A_j
A_p
A
Obs
JmaxRef = 309002255.28
RdRef = 3.04
sigma = 0.24
Theta = −14648410.96
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

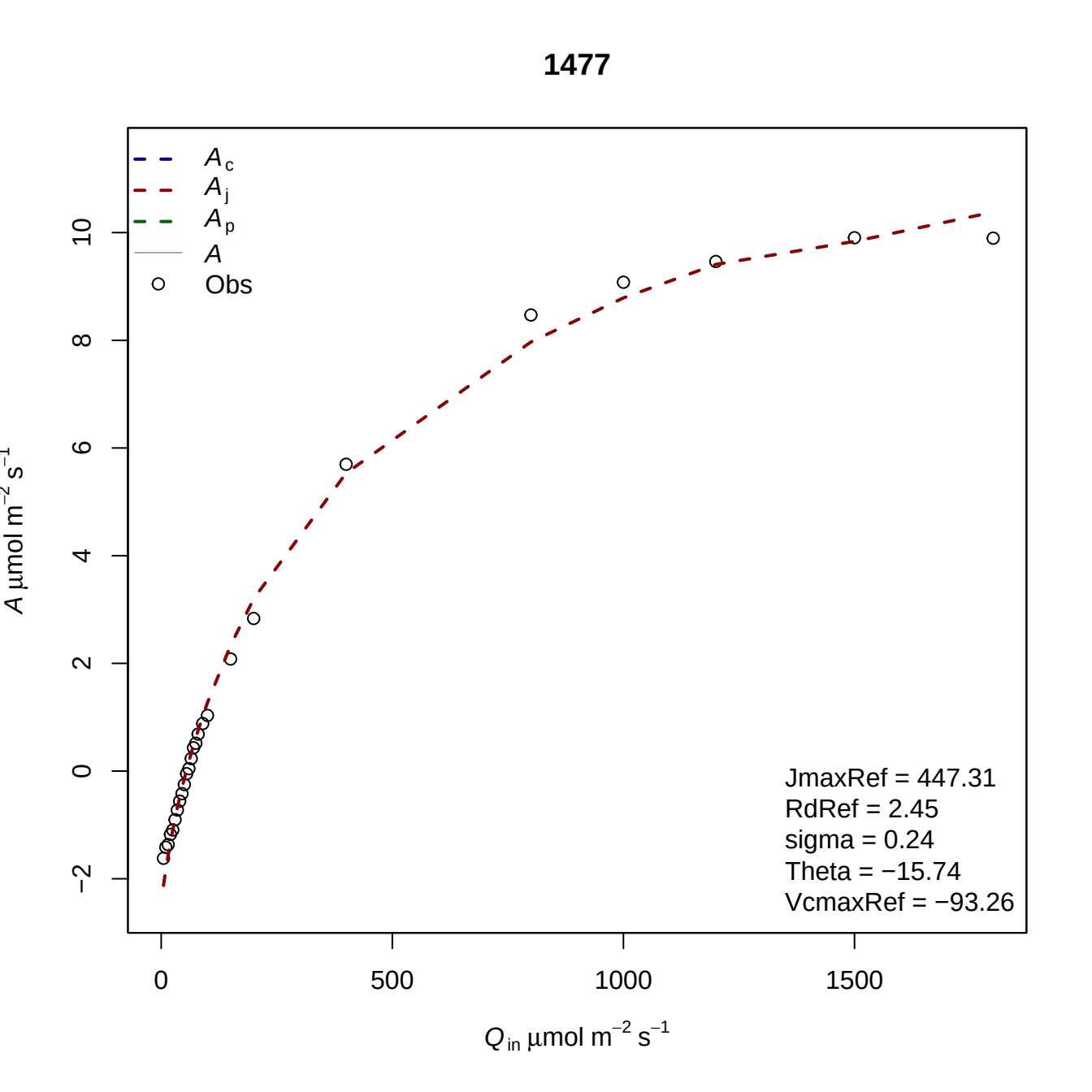

1472
A_c
A_j
A_p
A
Obs
JmaxRef = 774.77
RdRef = 3.62
sigma = 0.25
Theta = −24.83
VcmaxRef = 95.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

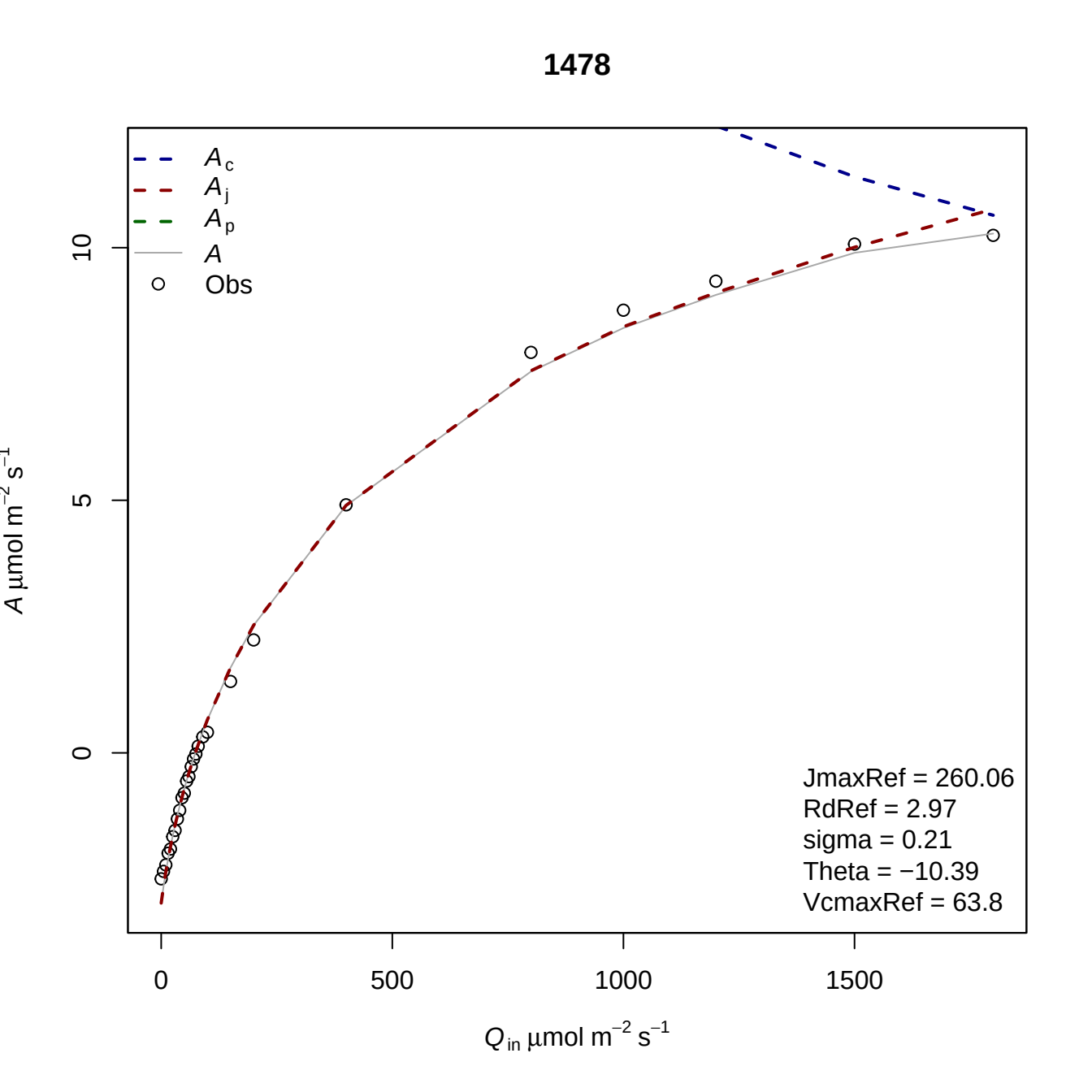

1473
A_c
A_j
A_p
A
Obs
JmaxRef = 145794592.09
RdRef = 3.54
sigma = 0.2
Theta = −6583901.96
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800

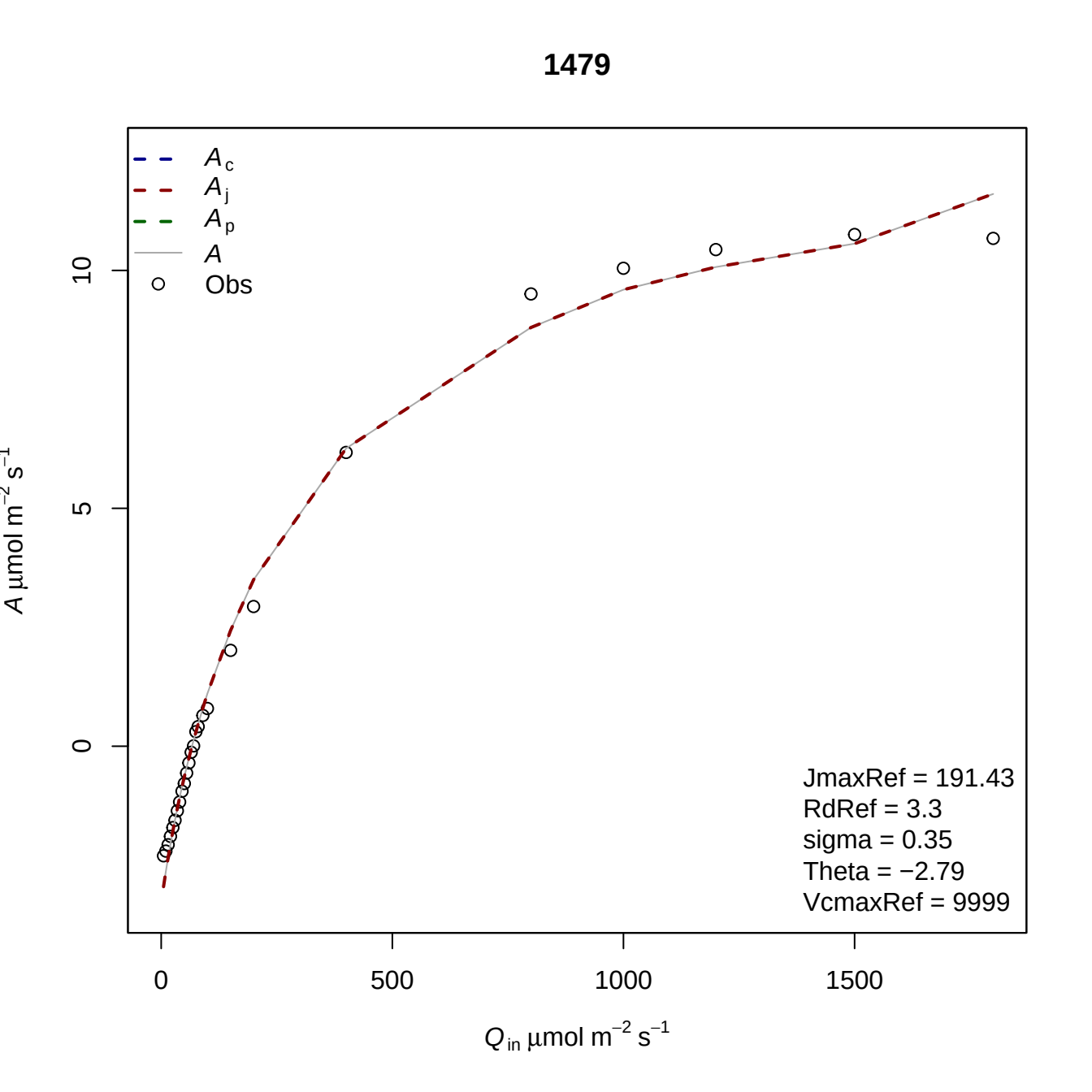

1474
A_c
A_j
A_p
A
Obs
JmaxRef = 357416565.11
RdRef = 2.17
sigma = 0.22
Theta = −22630342.06
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

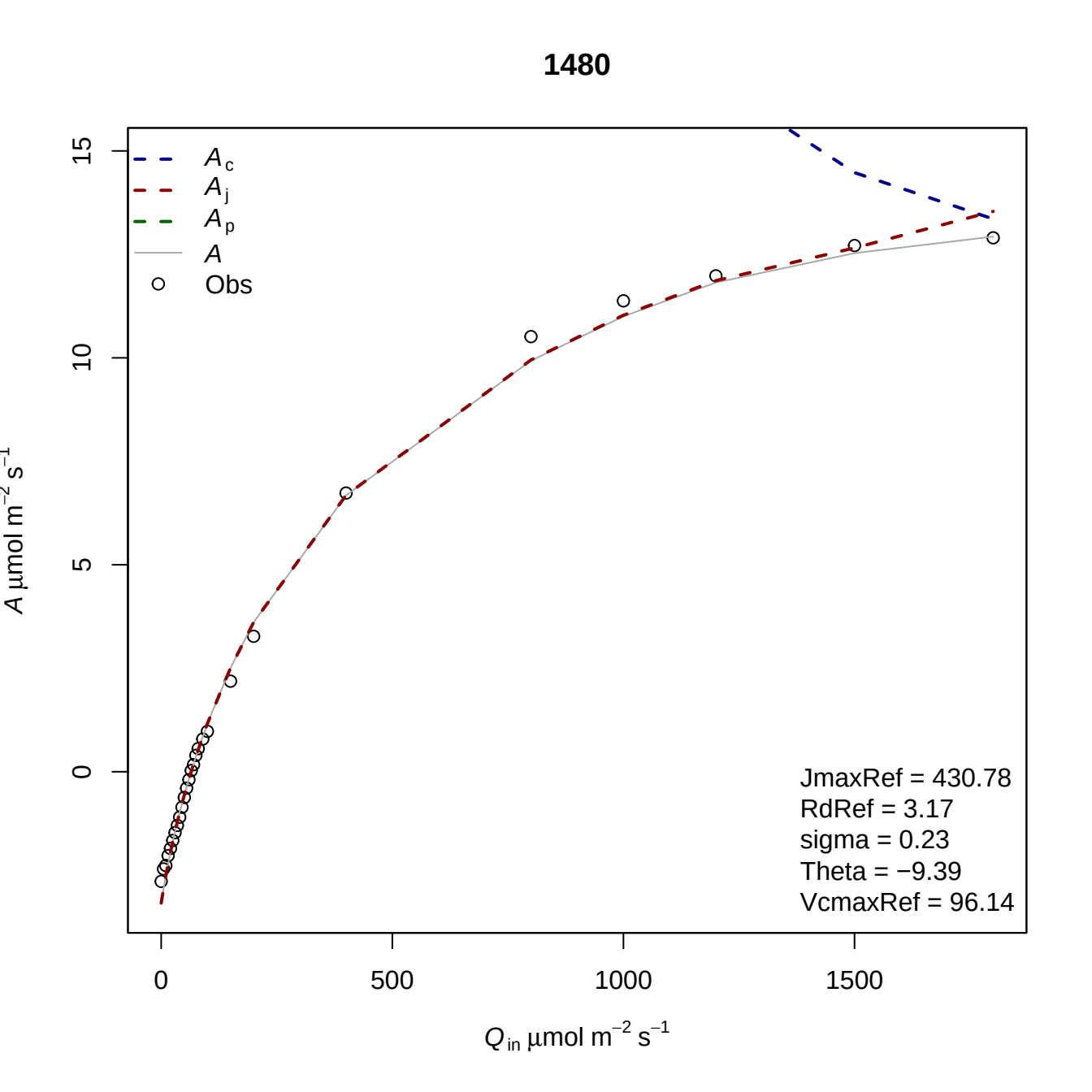

1475
A_c
A_j
A_p
A
Obs
JmaxRef = 21189736.65
RdRef = 2.9
sigma = 0.14
Theta = −1213117.27
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

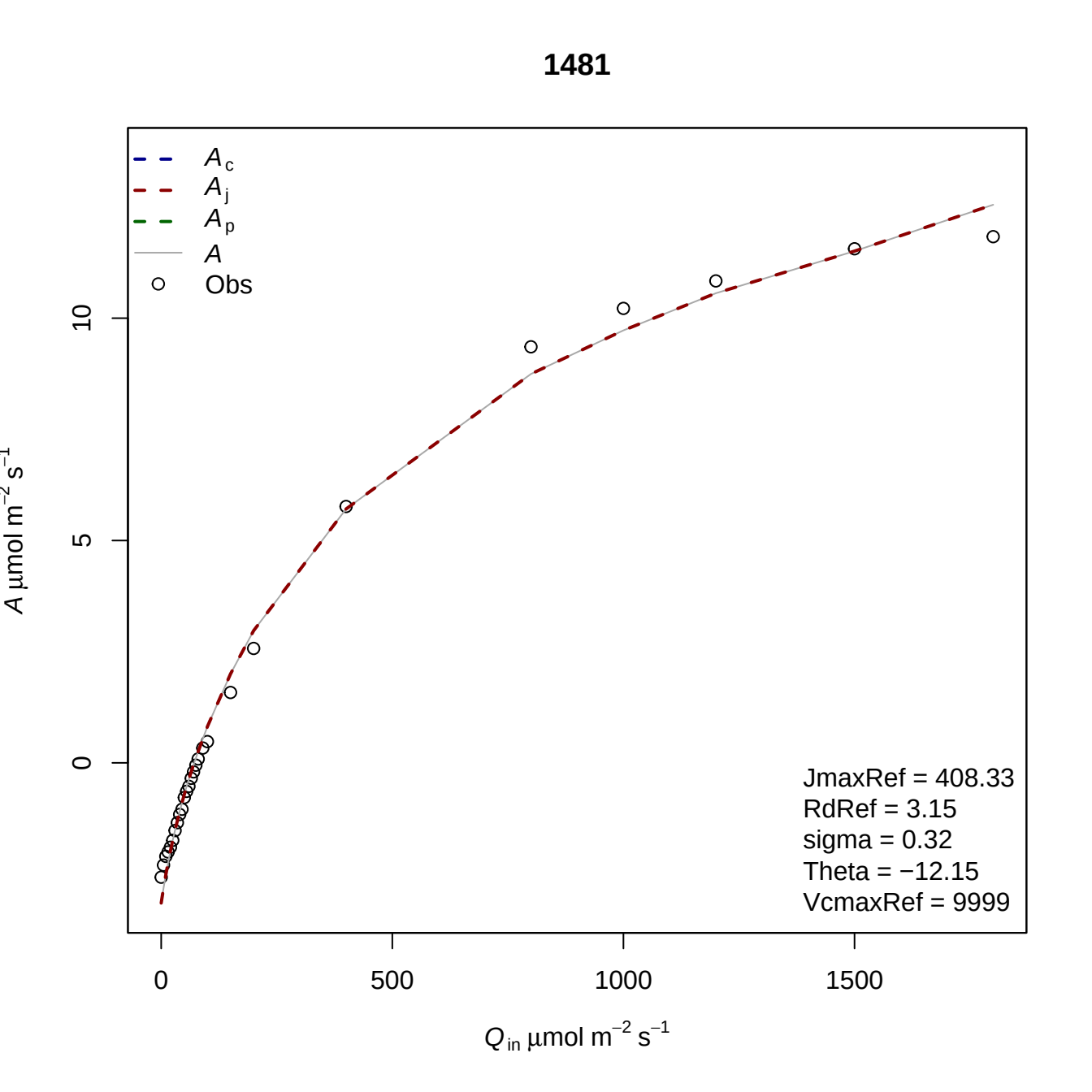

1476
A_c
A_j
A_p
A
Obs
JmaxRef = 76022916.14
RdRef = 2.75
sigma = 0.21
Theta = −6081111.2
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

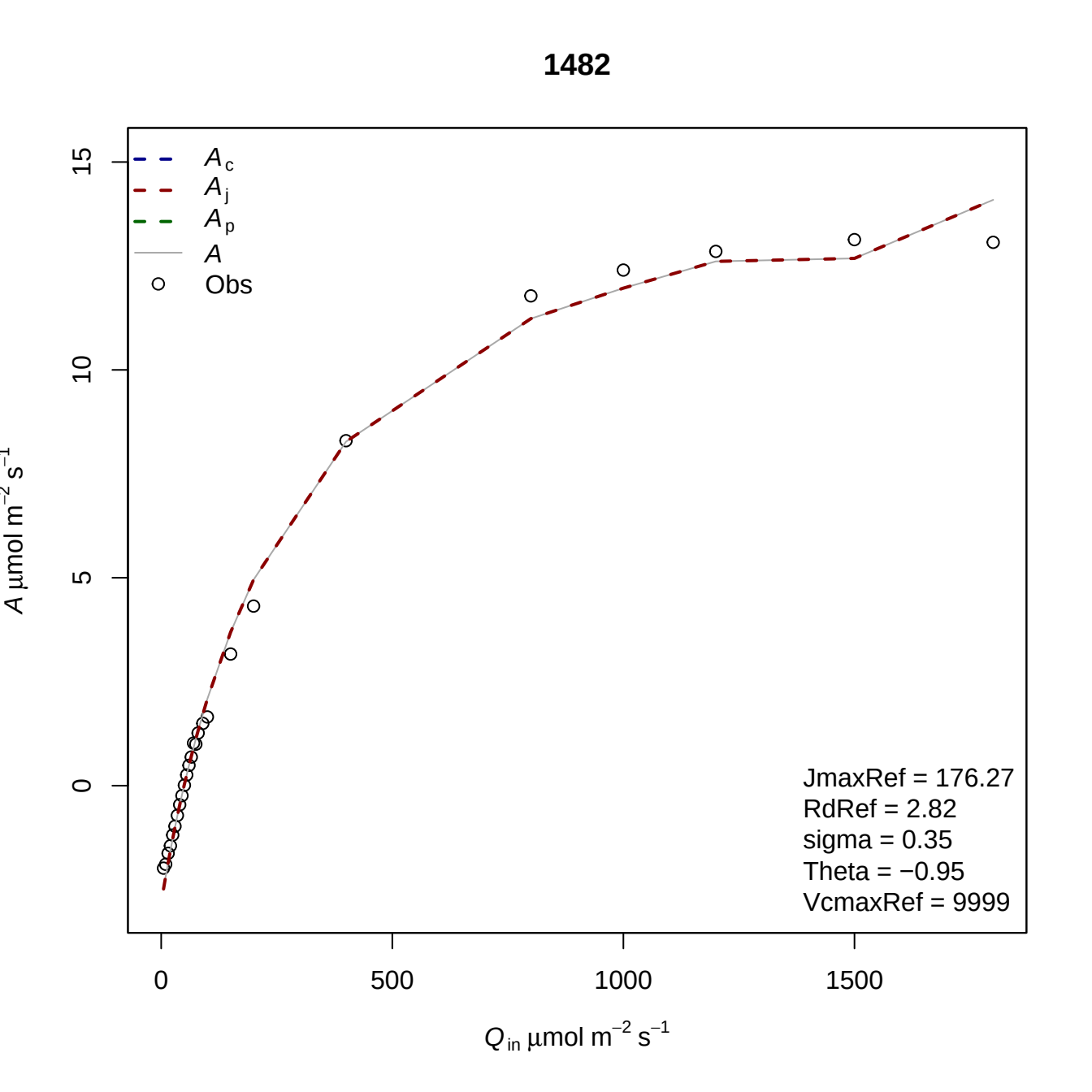

1477
A_c
A_j
A_p
A
Obs
JmaxRef = 447.31
RdRef = 2.45
sigma = 0.24
Theta = −15.74
VcmaxRef = −93.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

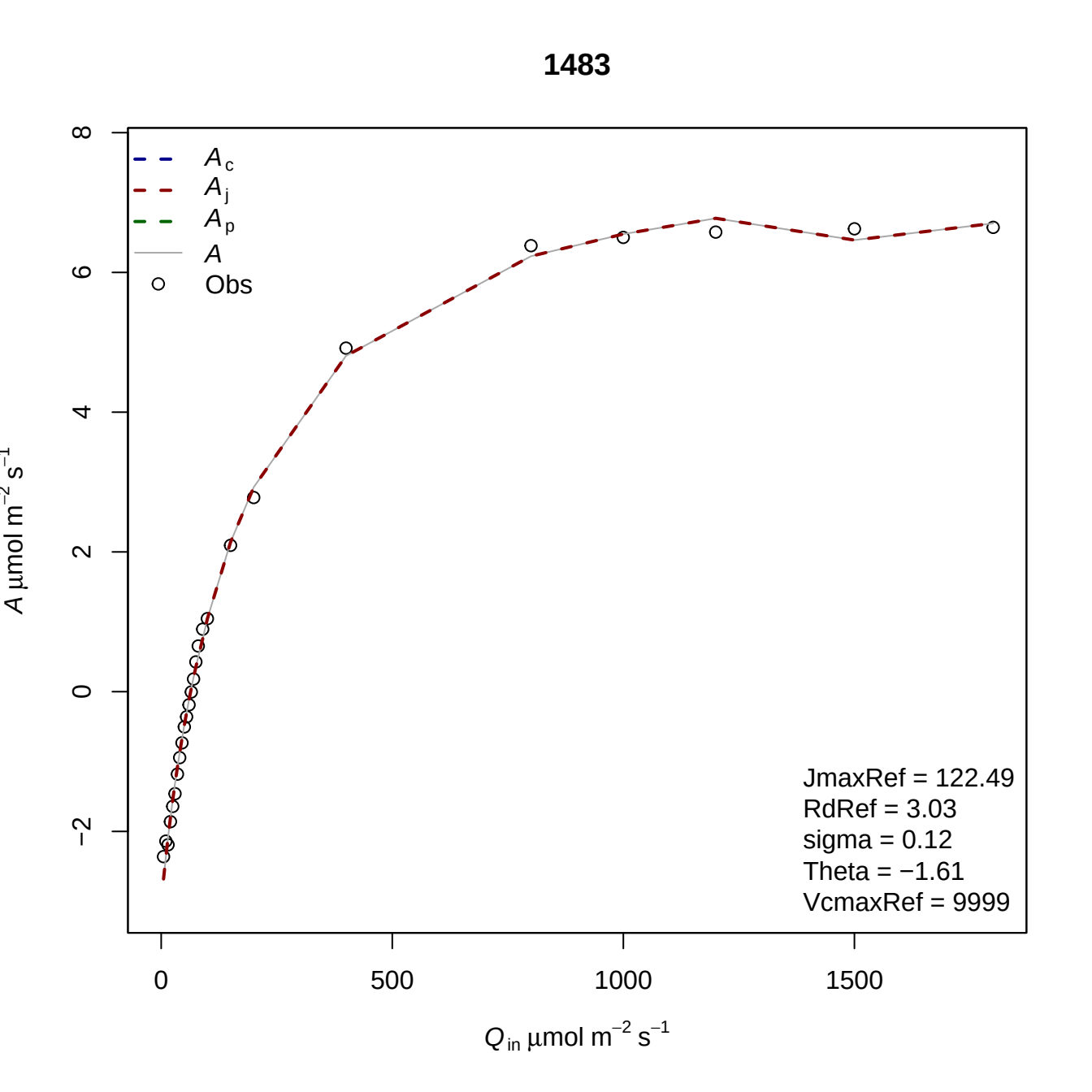

1478
A_c
A_j
A_p
A
Obs
JmaxRef = 260.06
RdRef = 2.97
sigma = 0.21
Theta = −10.39
VcmaxRef = 63.8
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

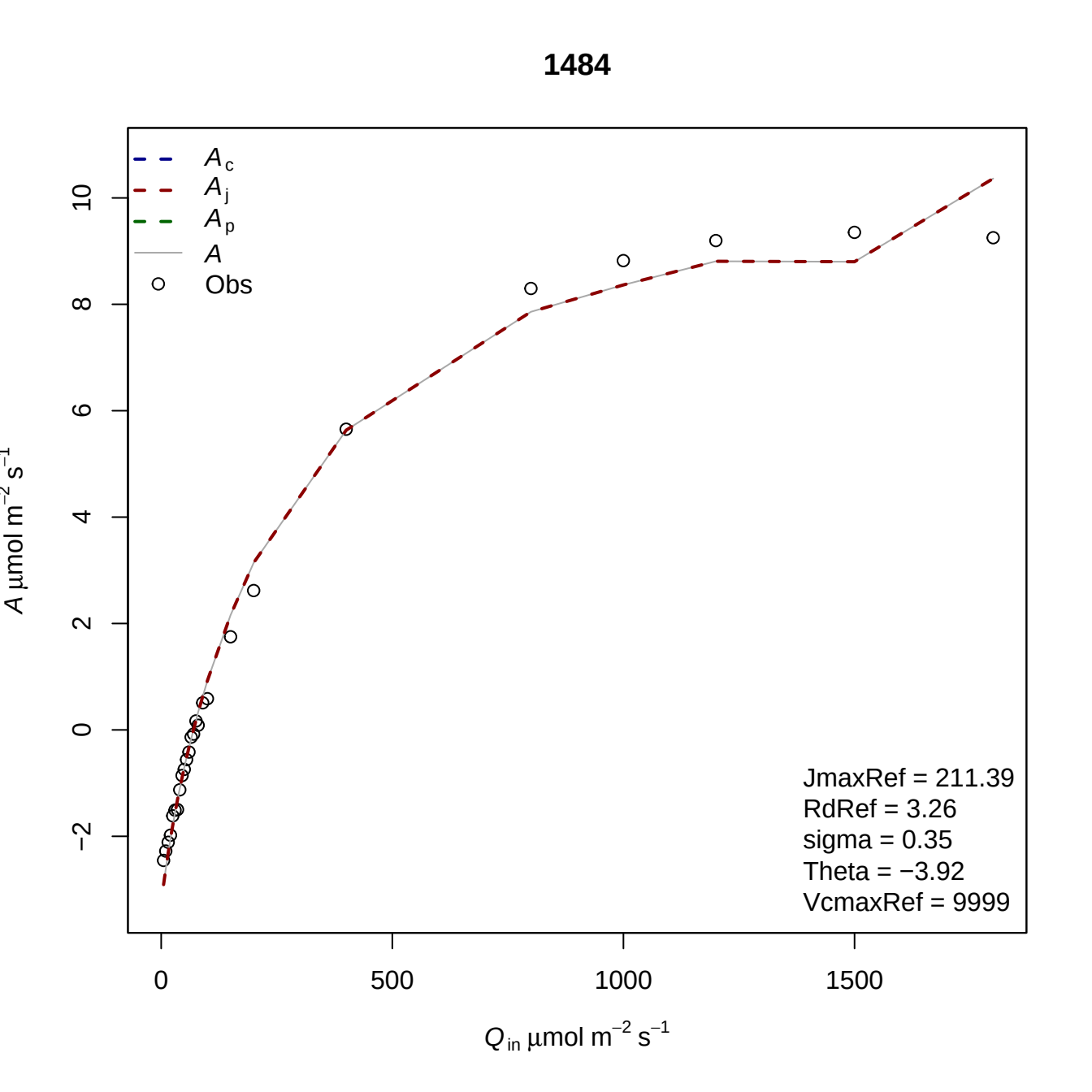

1479
A_c
A_j
A_p
A
Obs
JmaxRef = 191.43
RdRef = 3.3
sigma = 0.35
Theta = −2.79
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

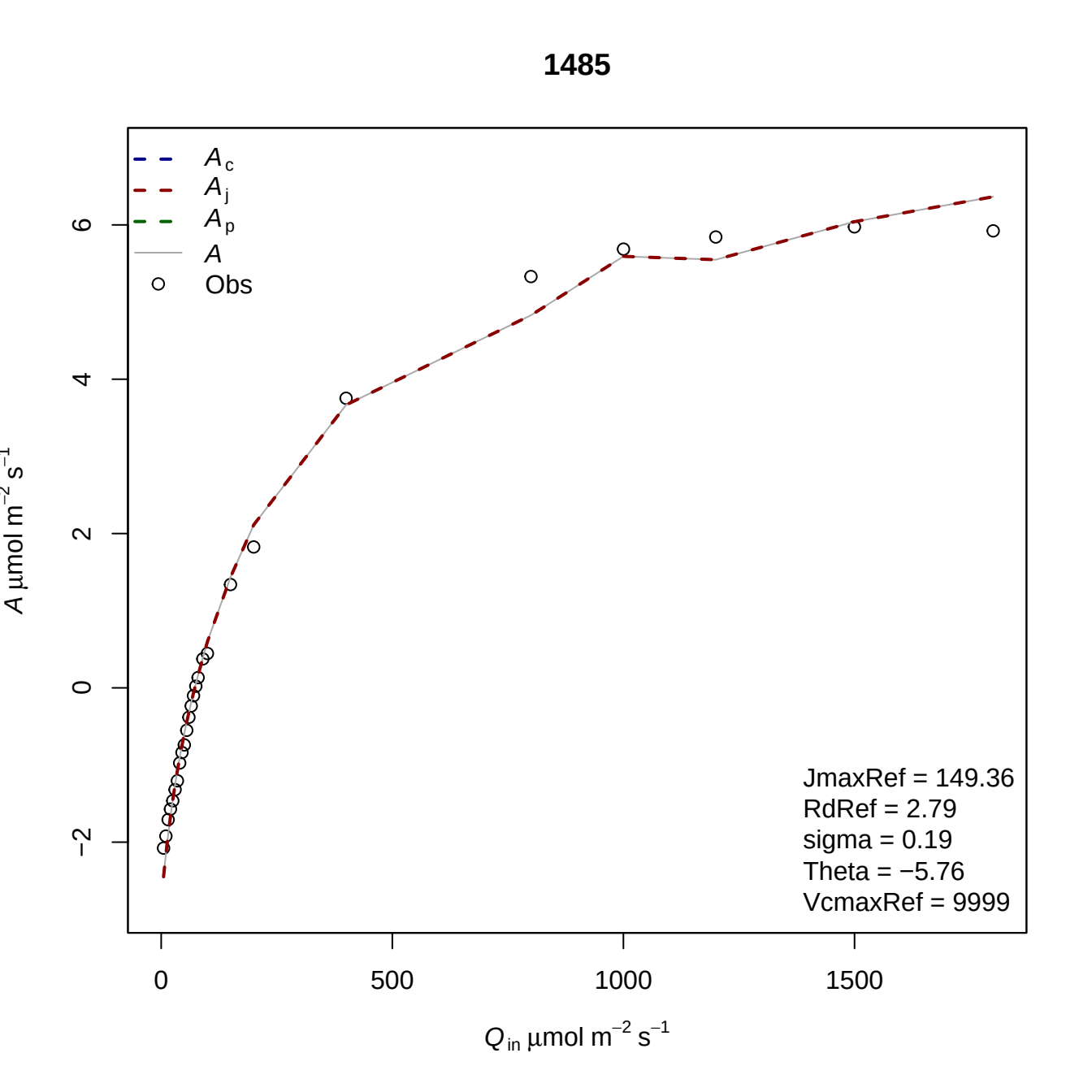

1480
A_c
A_j
A_p
A
Obs
JmaxRef = 430.78
RdRef = 3.17
sigma = 0.23
Theta = −9.39
VcmaxRef = 96.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

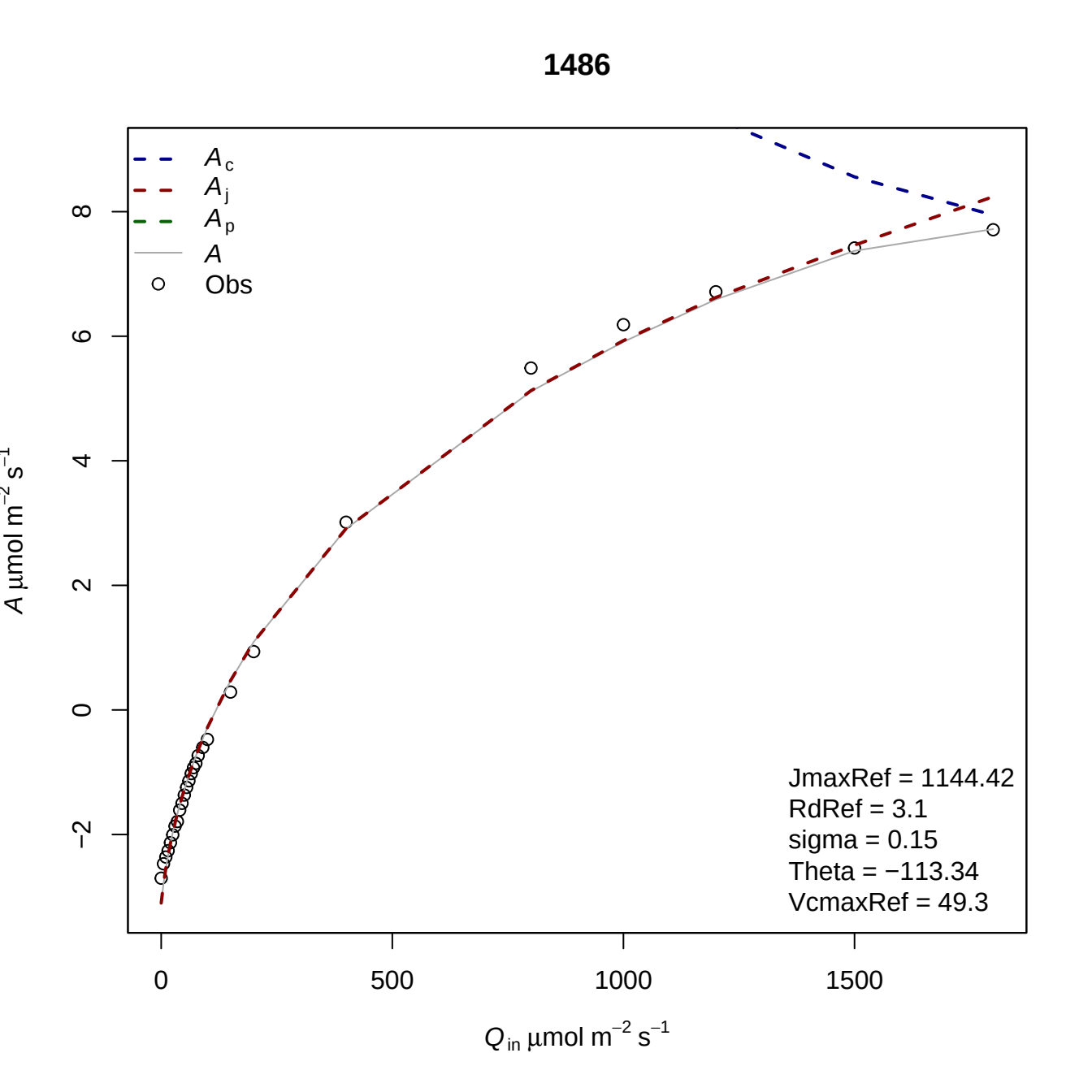

1481
A_c
A_j
A_p
A
Obs
JmaxRef = 408.33
RdRef = 3.15
sigma = 0.32
Theta = −12.15
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

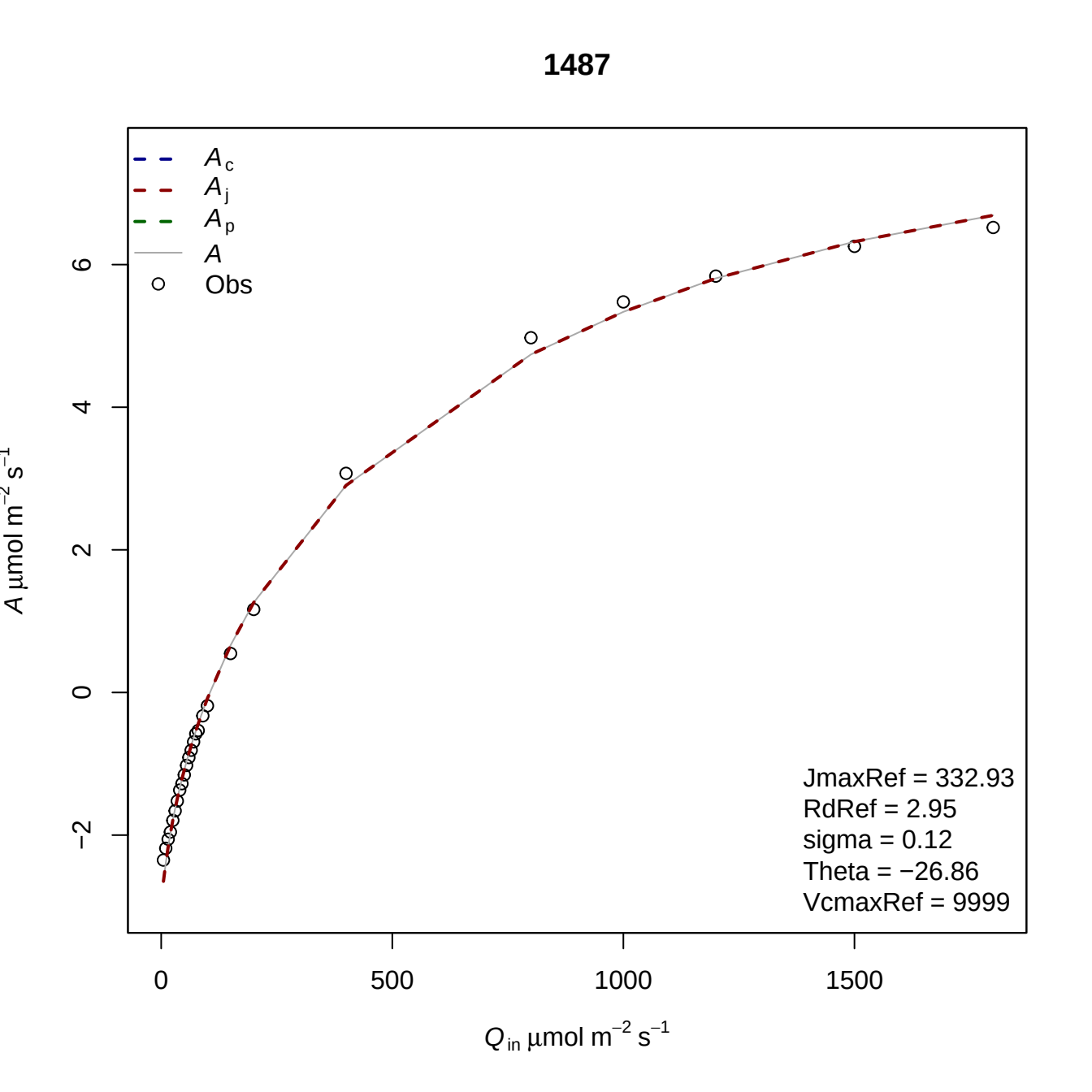

1482
A_c
A_j
A_p
A
Obs
JmaxRef = 176.27
RdRef = 2.82
sigma = 0.35
Theta = −0.95
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

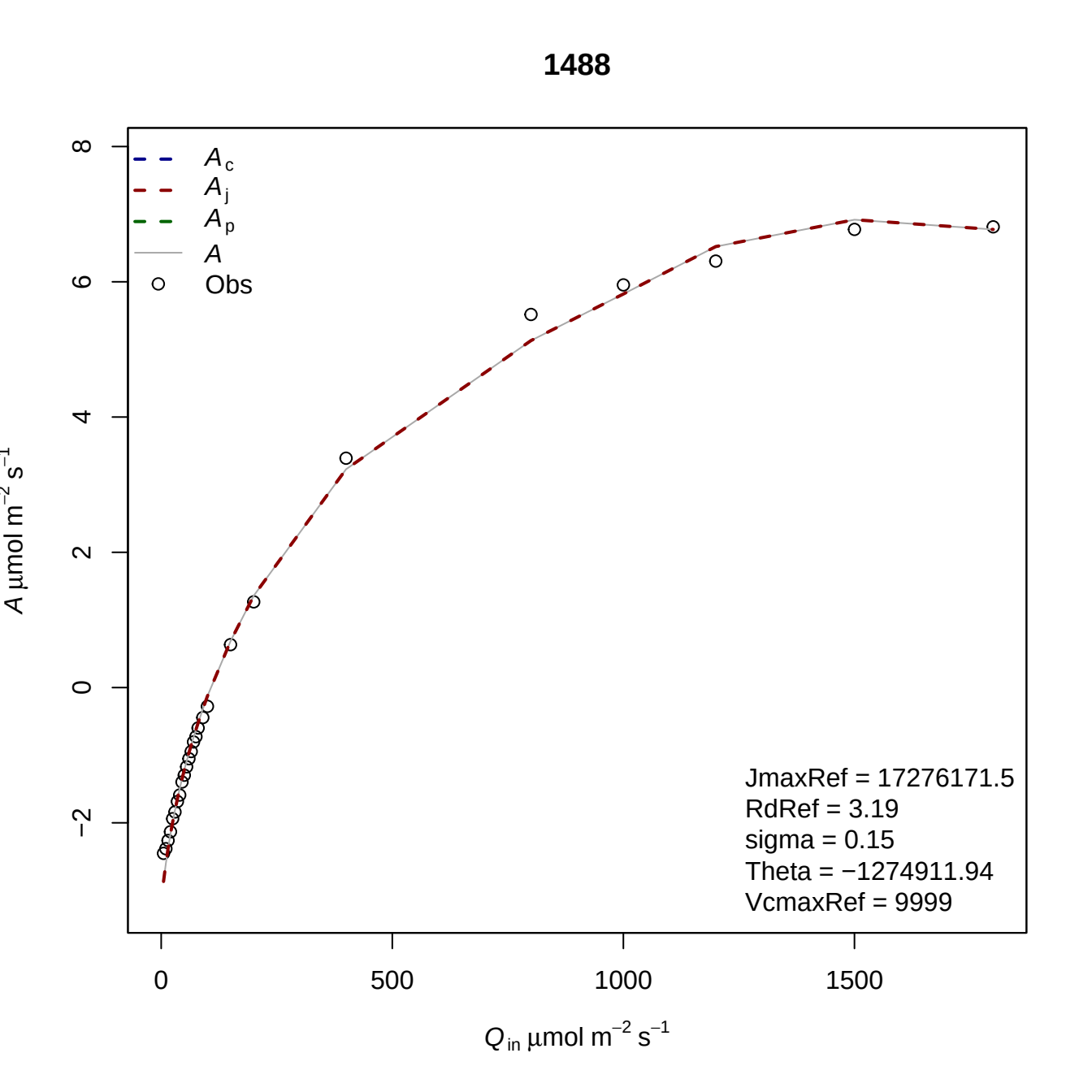

1483
A_c
A_j
A_p
A
Obs
JmaxRef = 122.49
RdRef = 3.03
sigma = 0.12
Theta = −1.61
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

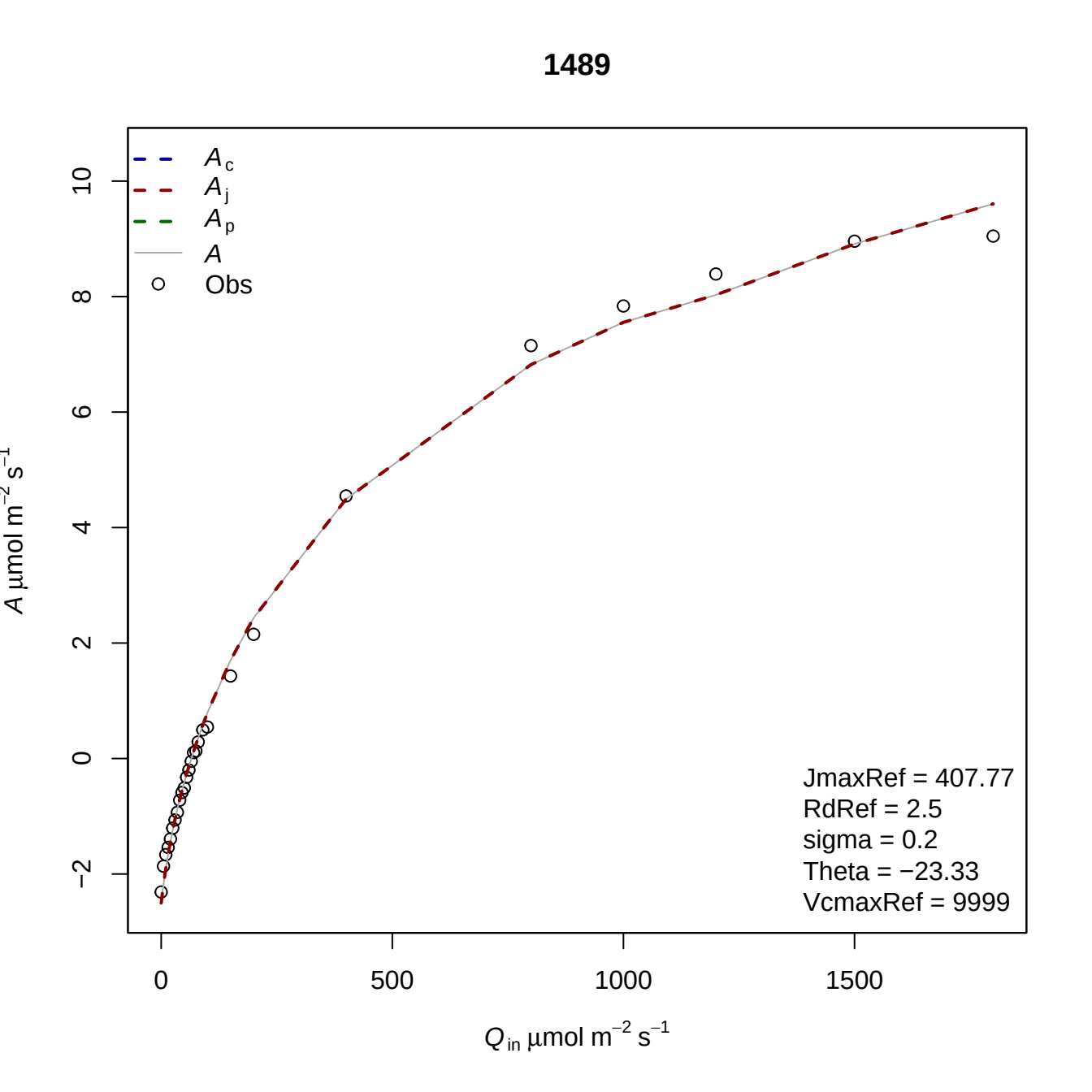

1484
A_c
A_j
A_p
A
Obs
JmaxRef = 211.39
RdRef = 3.26
sigma = 0.35
Theta = −3.92
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

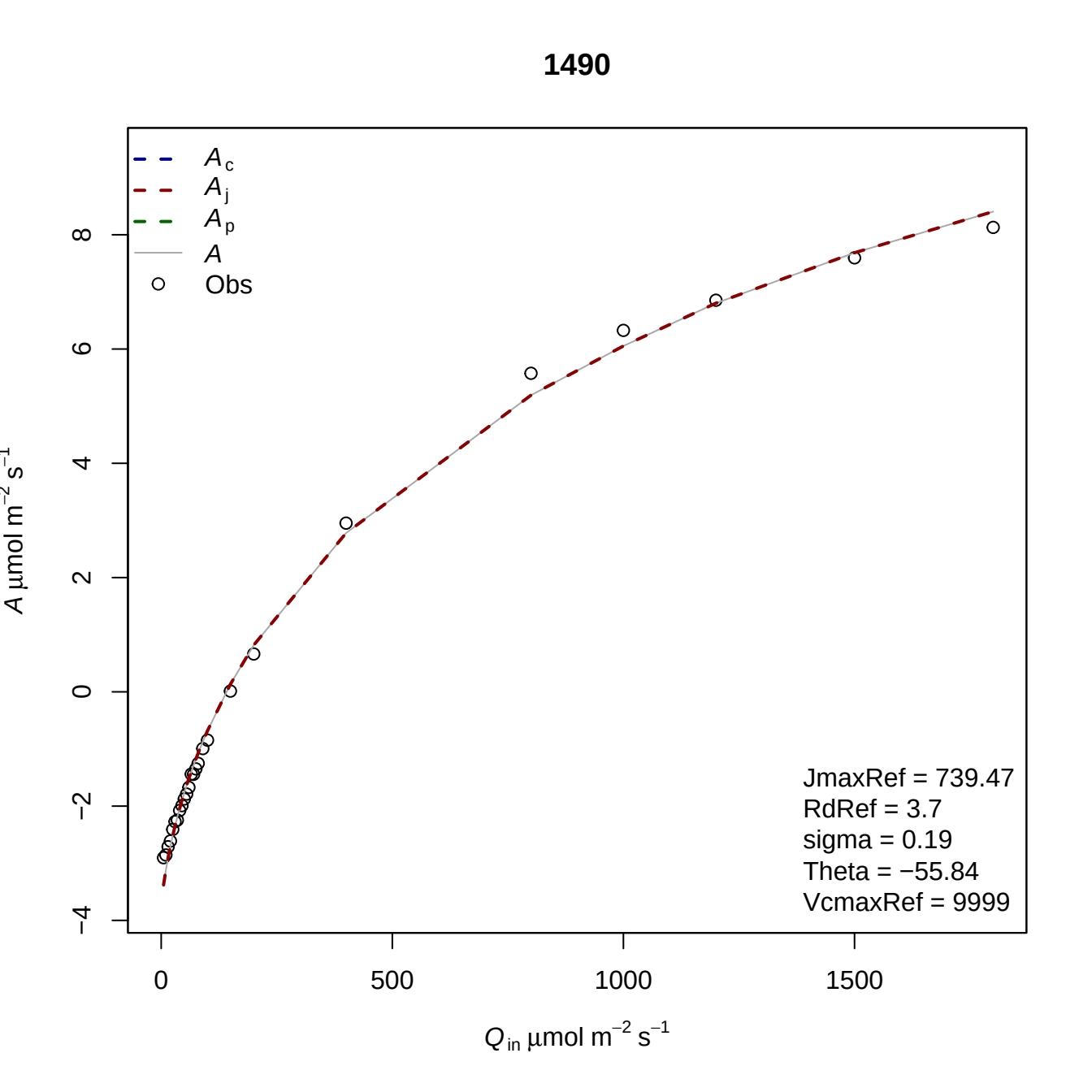

1485
A_c
A_j
A_p
A
Obs
JmaxRef = 149.36
RdRef = 2.79
sigma = 0.19
Theta = −5.76
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

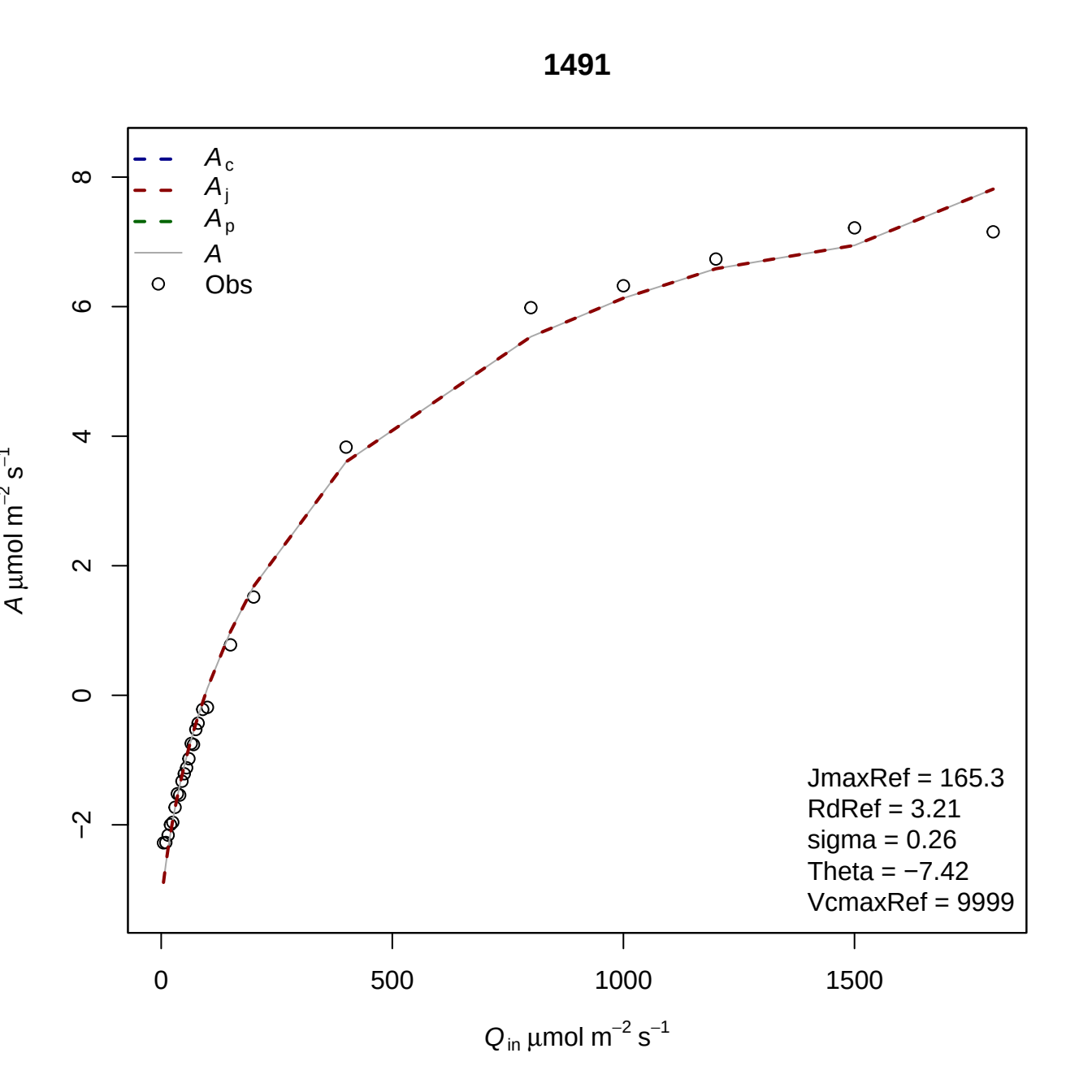

1486
A_c
A_j
A_p
A
Obs
JmaxRef = 1144.42
RdRef = 3.1
sigma = 0.15
Theta = −113.34
VcmaxRef = 49.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

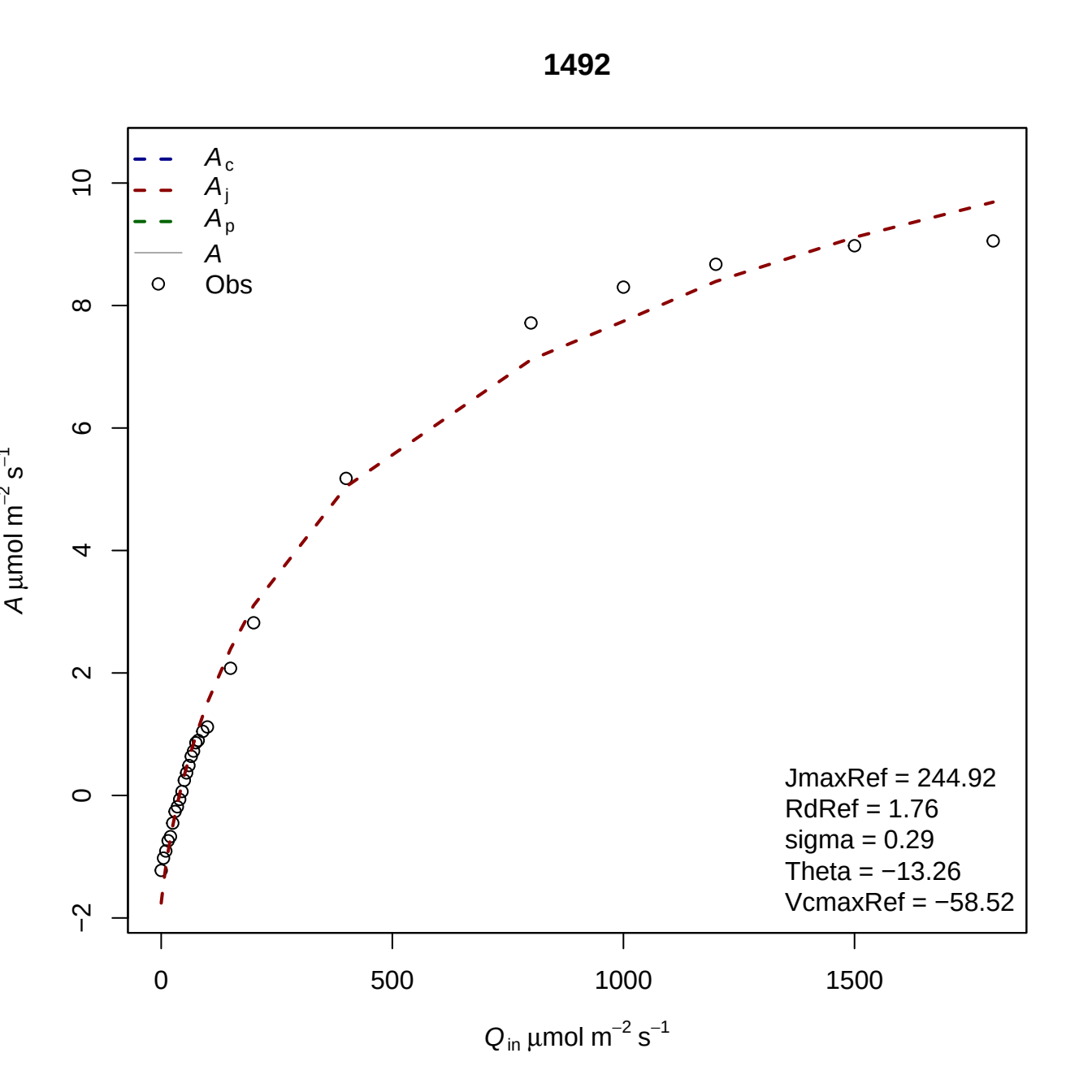

1487
A_c
A_j
A_p
A
Obs
JmaxRef = 332.93
RdRef = 2.95
sigma = 0.12
Theta = −26.86
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

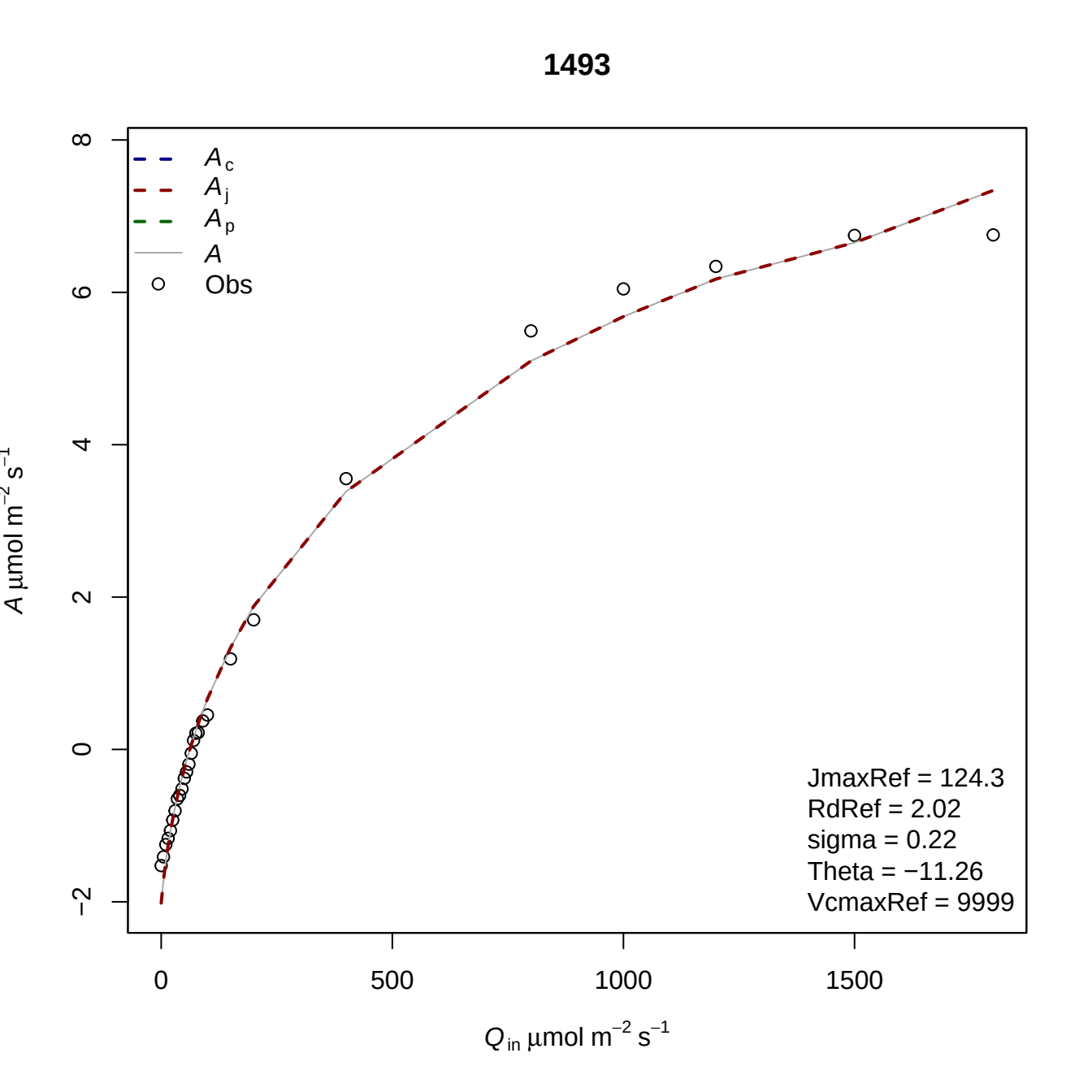

1488
A_c
A_j
A_p
A
Obs
JmaxRef = 17276171.5
RdRef = 3.19
sigma = 0.15
Theta = −1274911.94
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

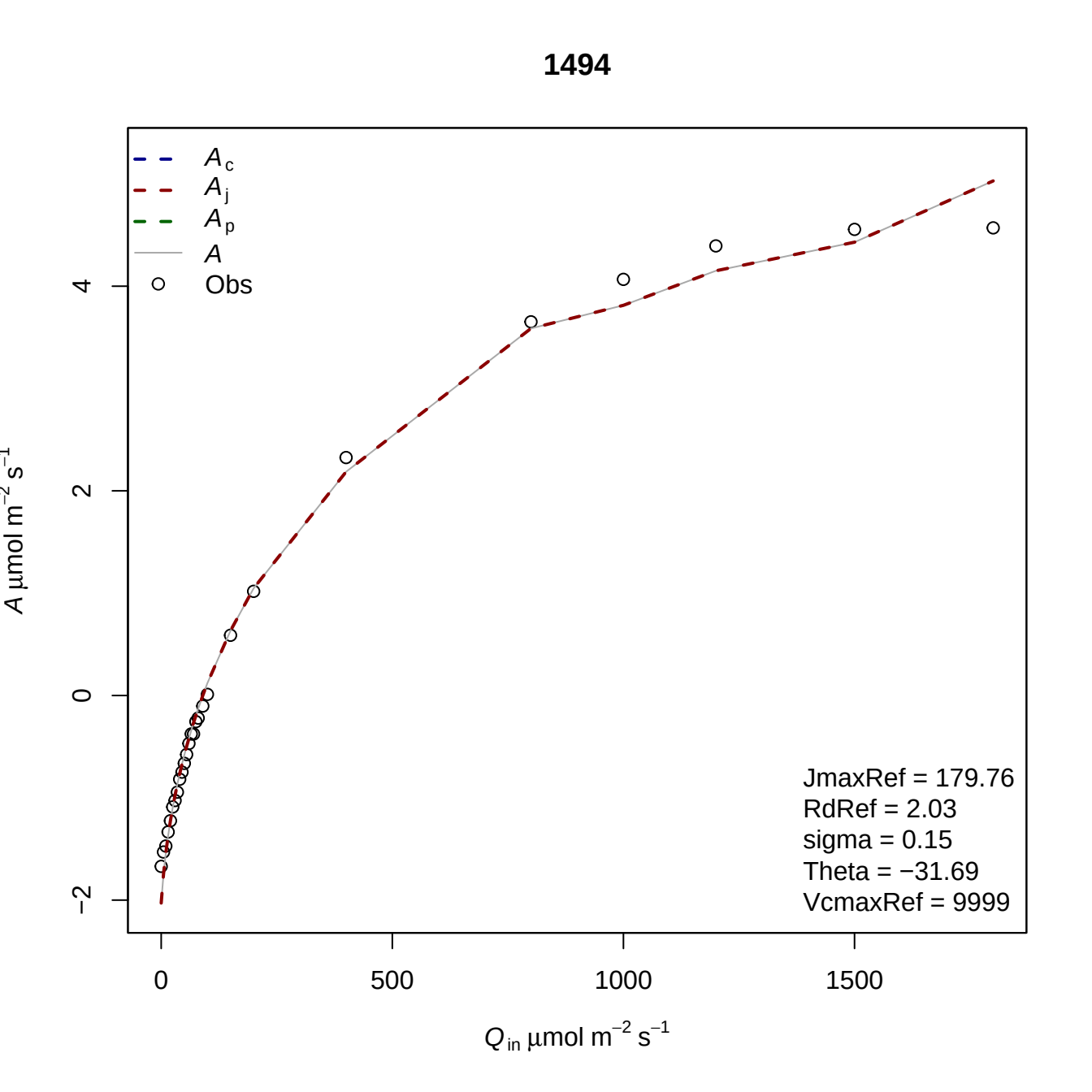

1489
A_c
A_j
A_p
A
Obs
JmaxRef = 407.77
RdRef = 2.5
sigma = 0.2
Theta = −23.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

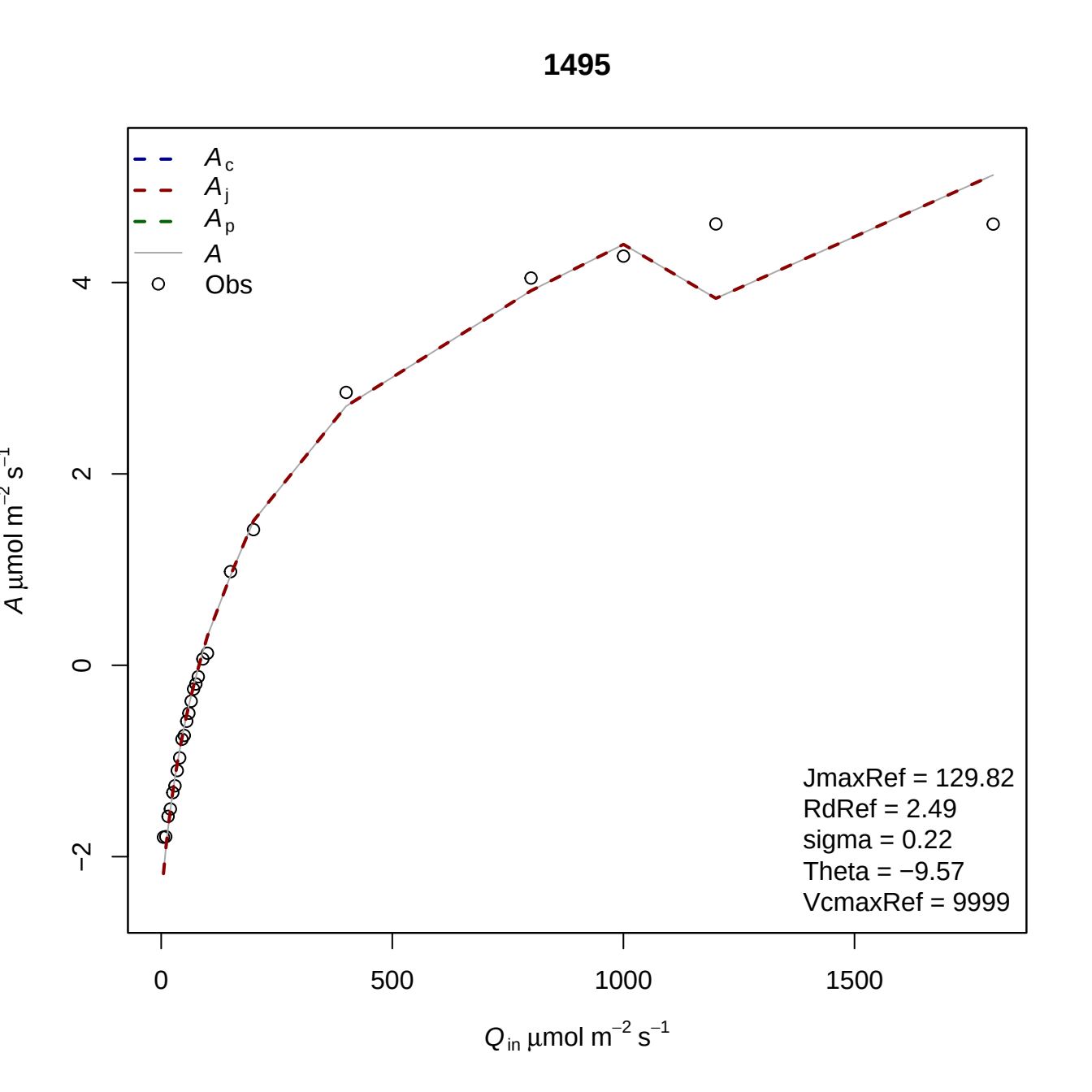

1490
A_c
A_j
A_p
A
Obs
JmaxRef = 739.47
RdRef = 3.7
sigma = 0.19
Theta = −55.84
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
500
1000
1500

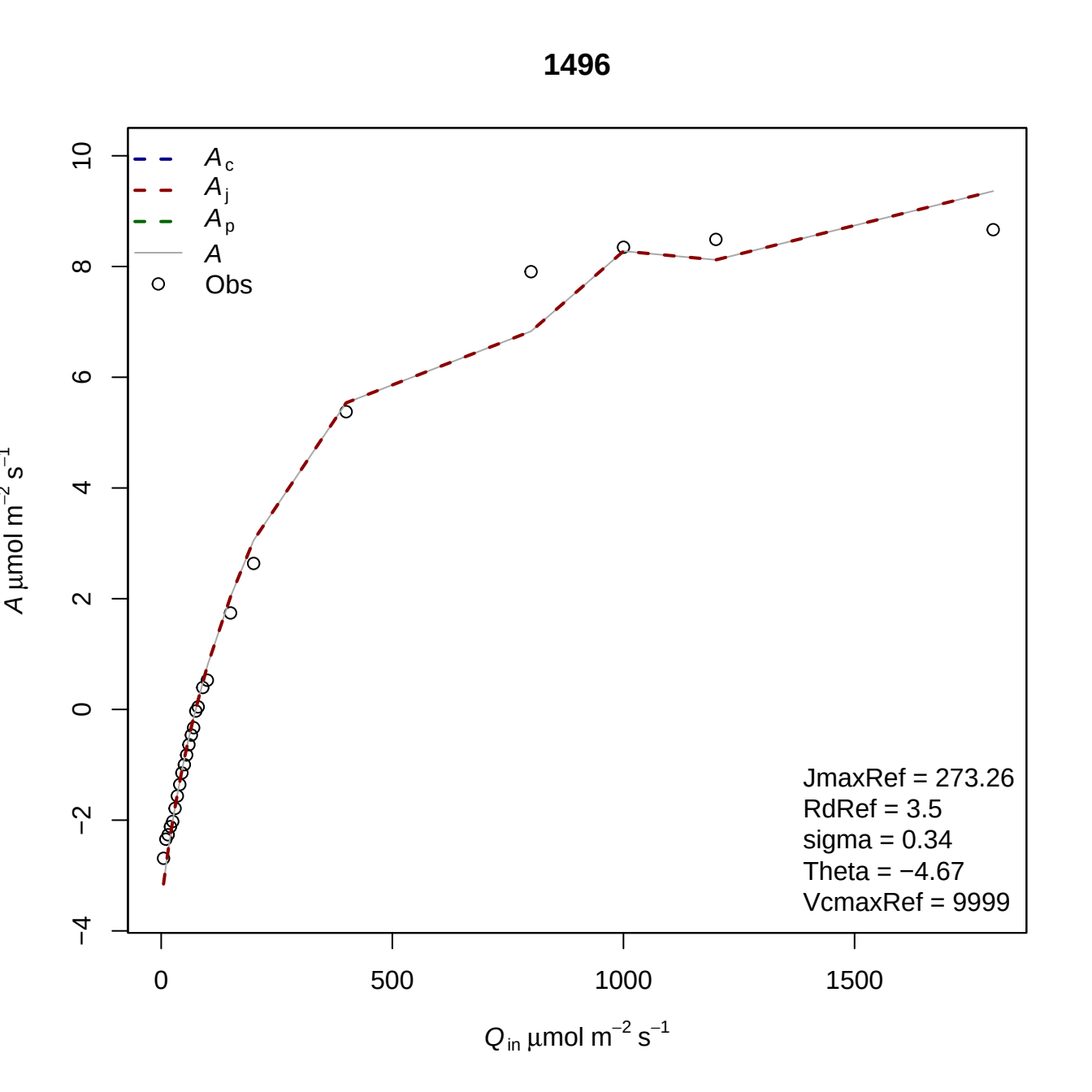

1491
A_c
A_j
A_p
A
Obs
JmaxRef = 165.3
RdRef = 3.21
sigma = 0.26
Theta = −7.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

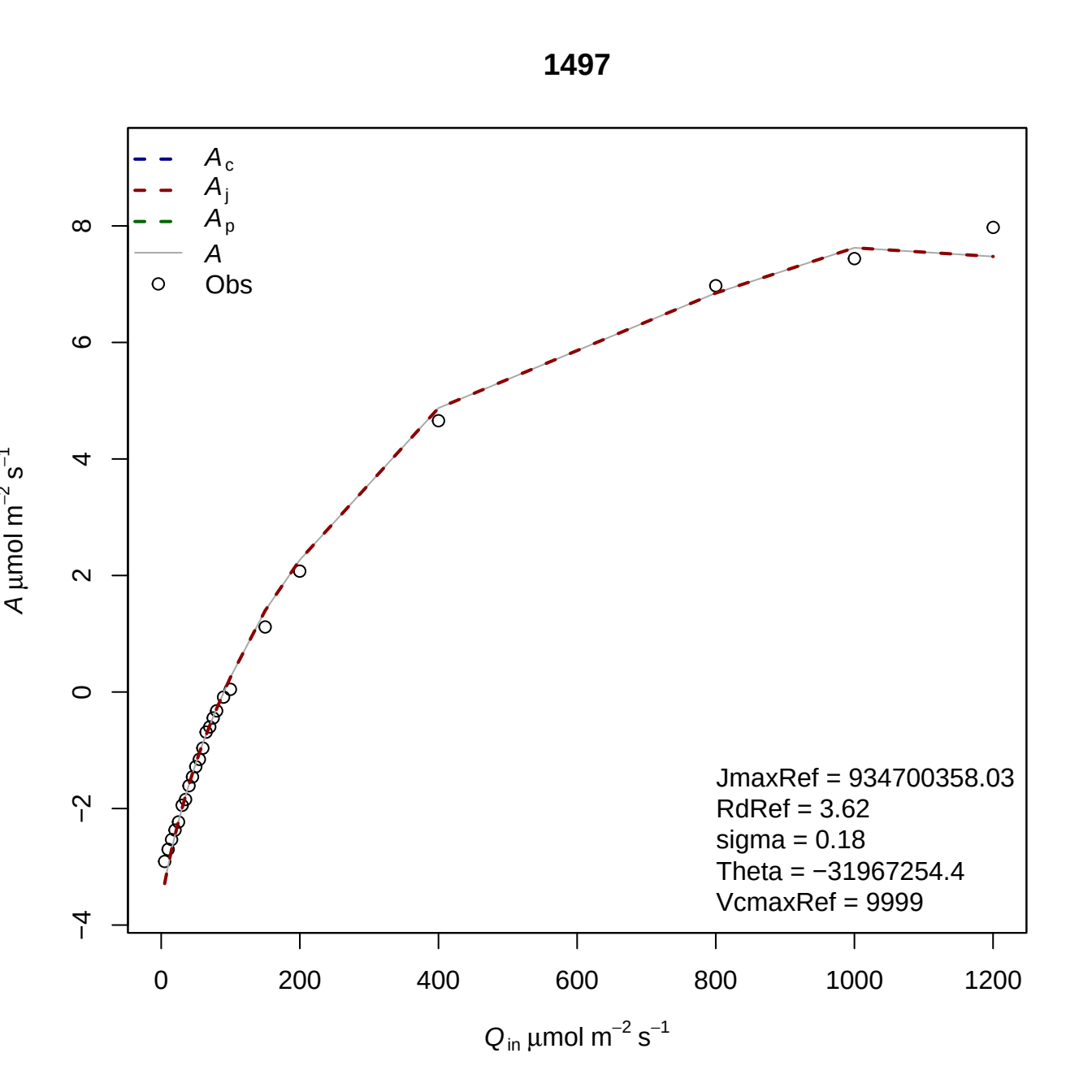

1492
A_c
A_j
A_p
A
Obs
JmaxRef = 244.92
RdRef = 1.76
sigma = 0.29
Theta = −13.26
VcmaxRef = −58.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

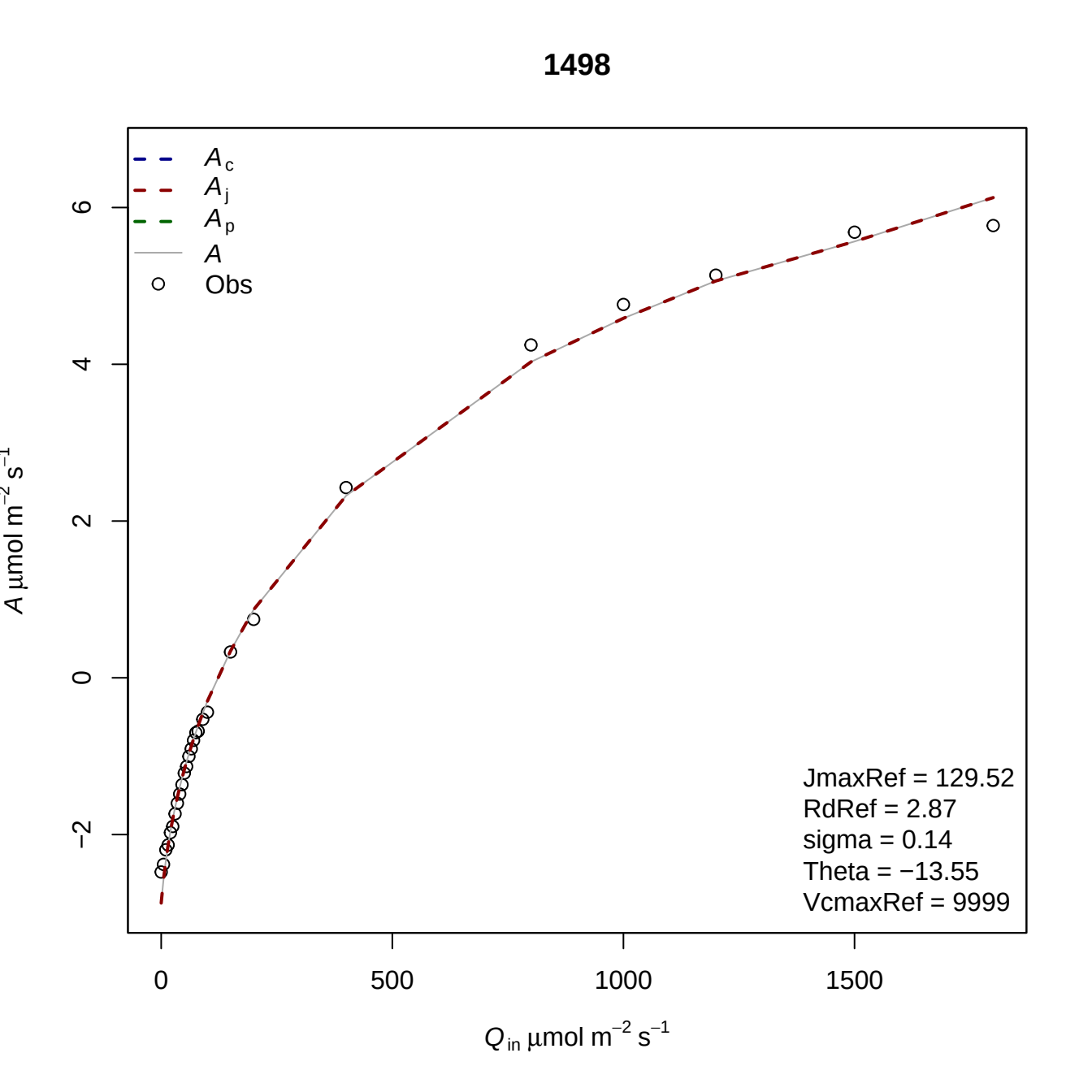

1493
A_c
A_j
A_p
A
Obs
JmaxRef = 124.3
RdRef = 2.02
sigma = 0.22
Theta = −11.26
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

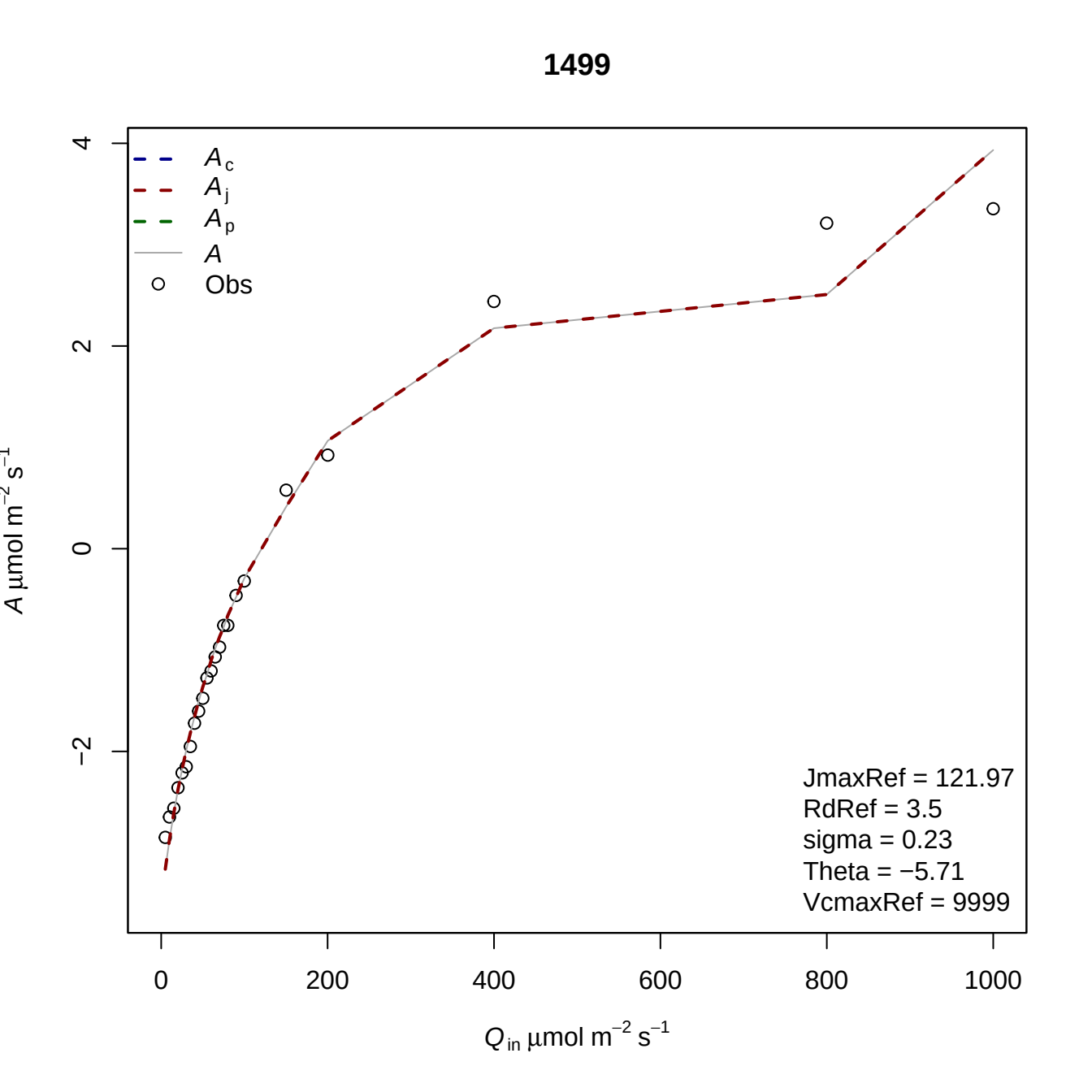

1494
A_c
A_j
A_p
A
Obs
JmaxRef = 179.76
RdRef = 2.03
sigma = 0.15
Theta = −31.69
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

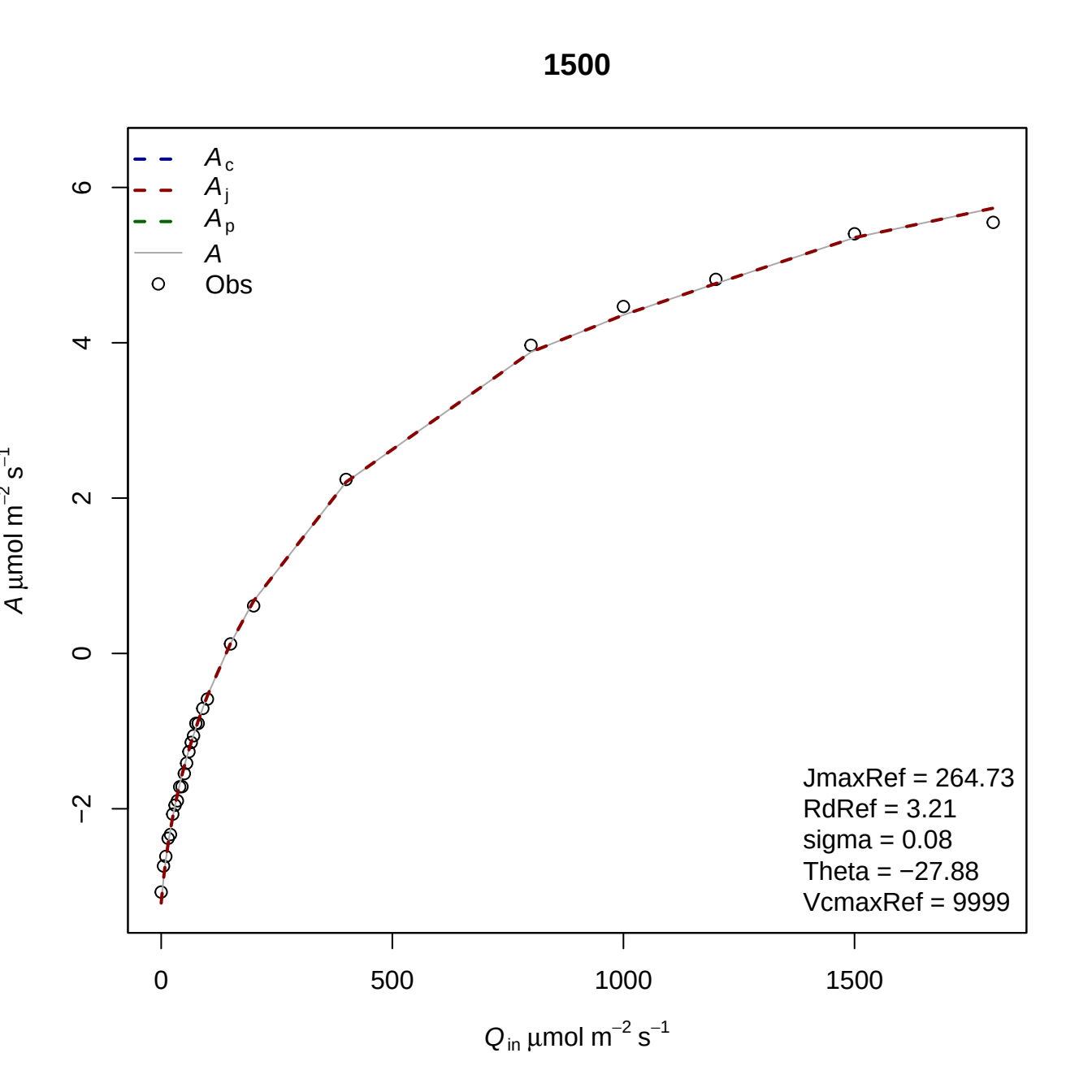

1495
A_c
A_j
A_p
A
Obs
JmaxRef = 129.82
RdRef = 2.49
sigma = 0.22
Theta = −9.57
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

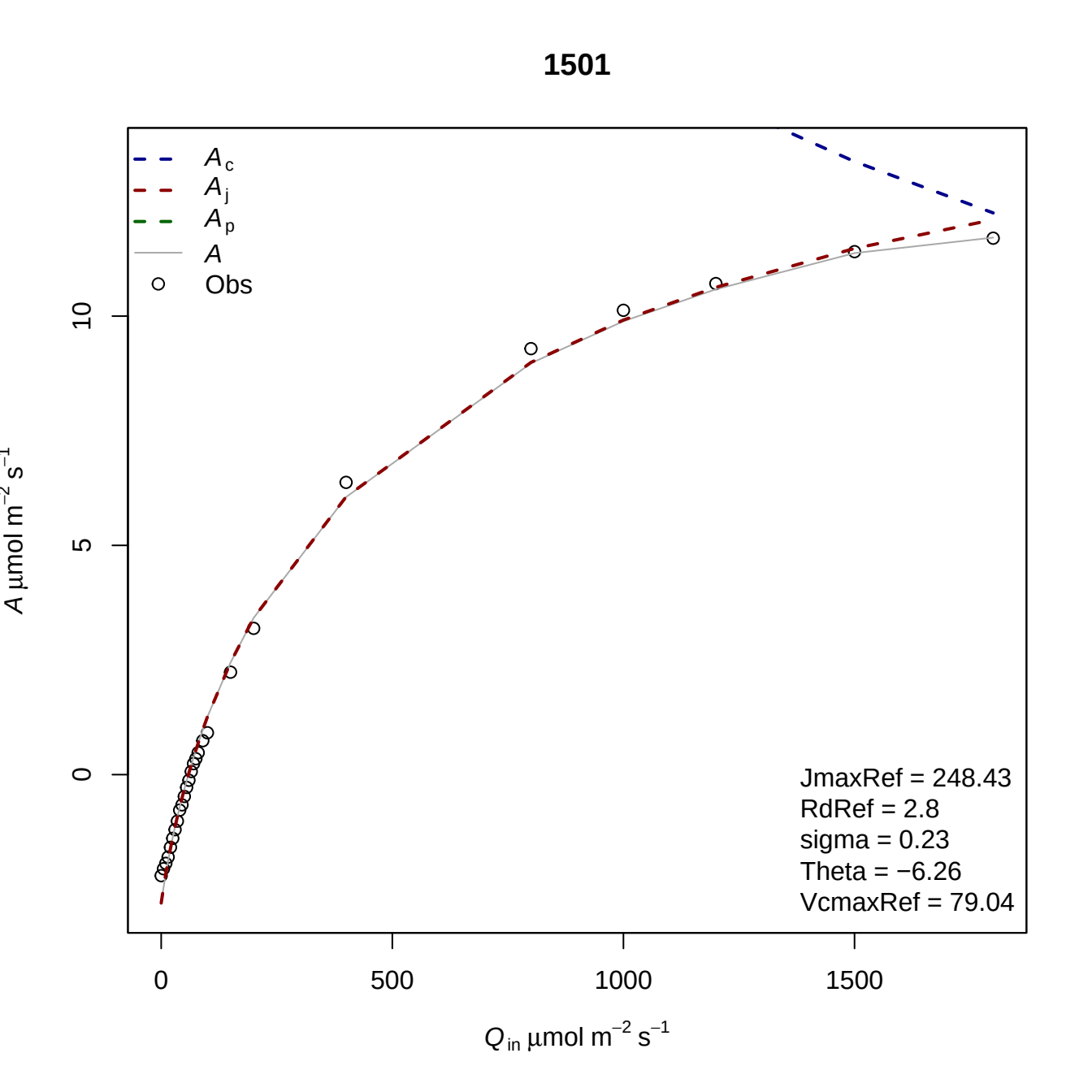

1496
A_c
A_j
A_p
A
Obs
JmaxRef = 273.26
RdRef = 3.5
sigma = 0.34
Theta = −4.67
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

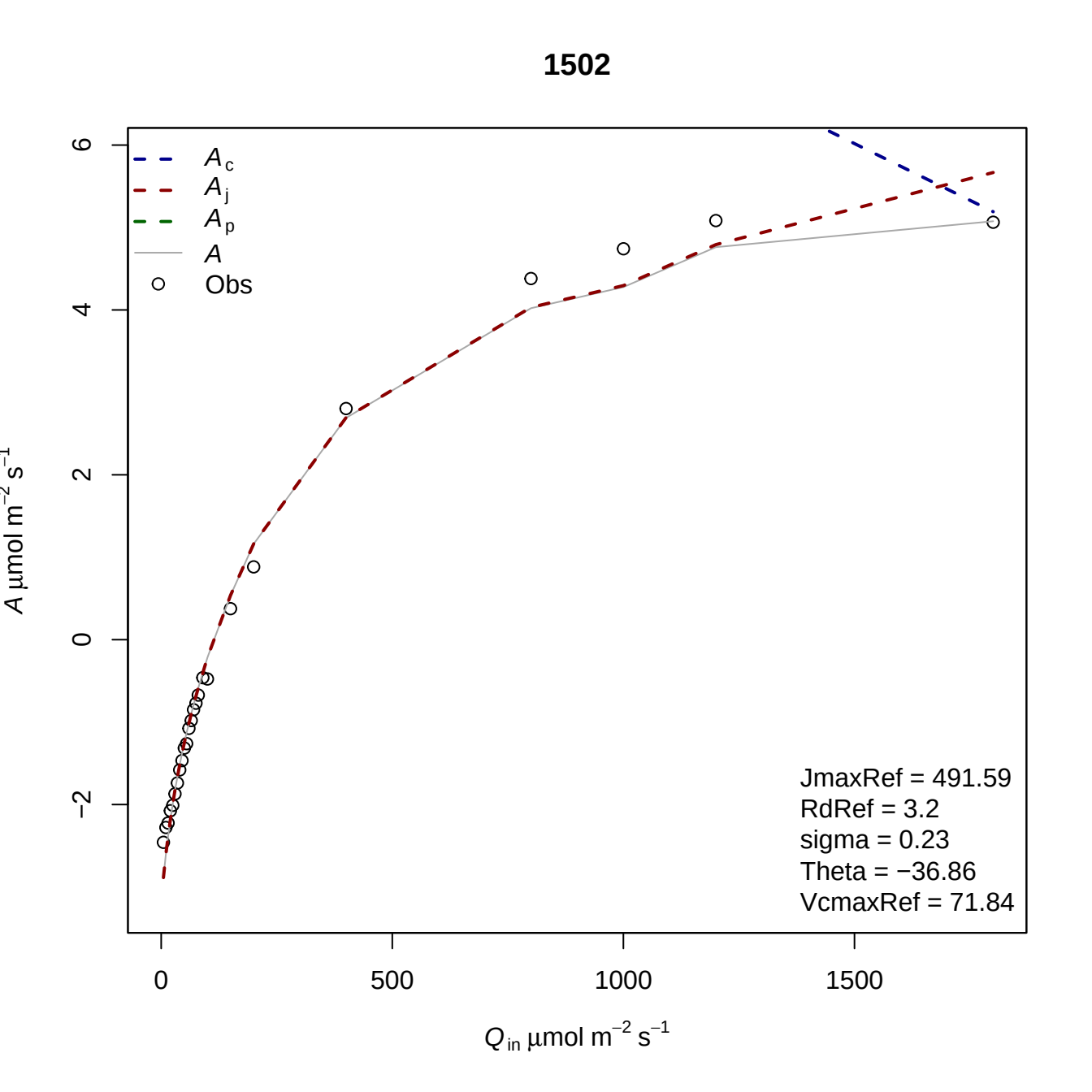

1497
A_c
A_j
A_p
A
Obs
JmaxRef = 934700358.03
RdRef = 3.62
sigma = 0.18
Theta = −31967254.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

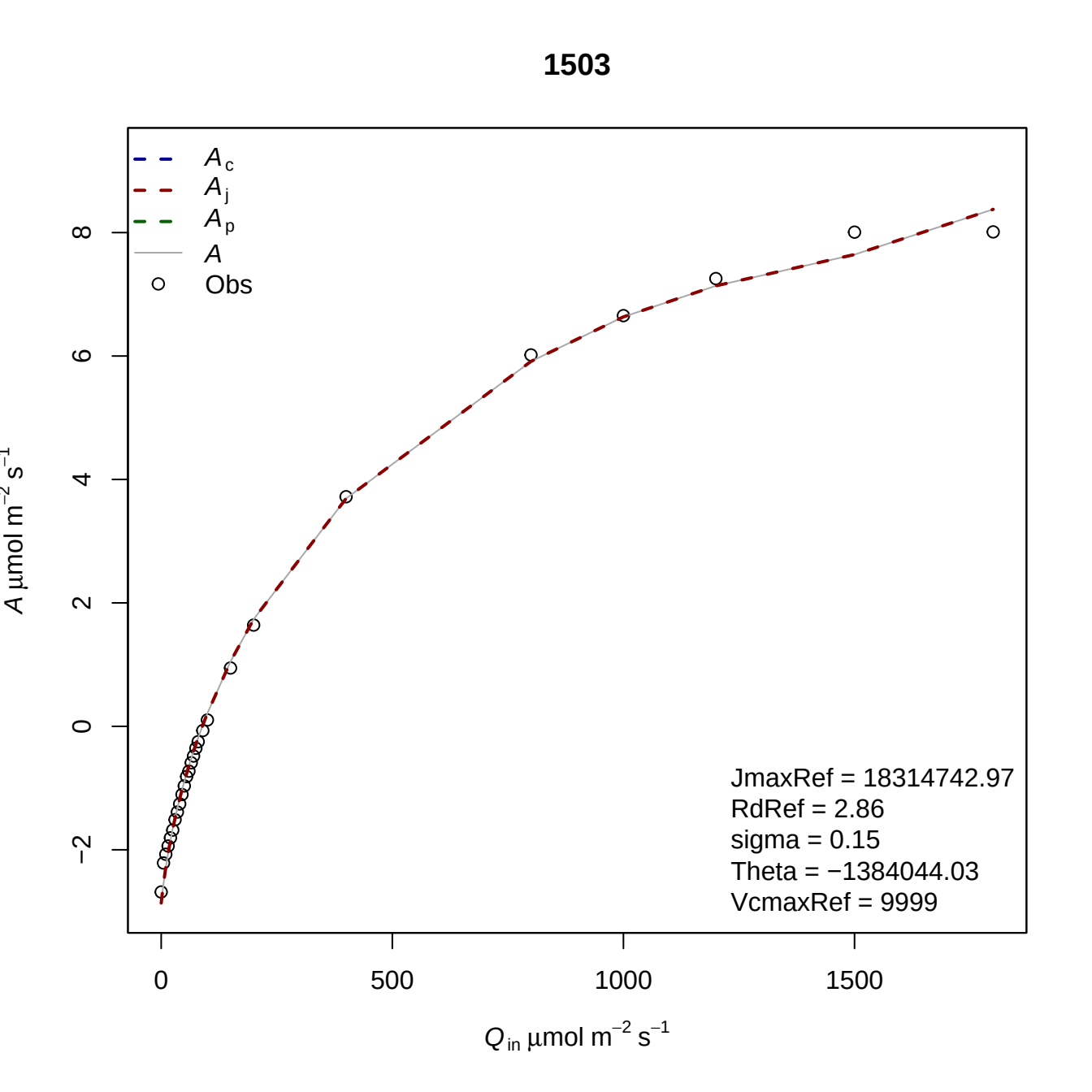

1498
A_c
A_j
A_p
A
Obs
JmaxRef = 129.52
RdRef = 2.87
sigma = 0.14
Theta = −13.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

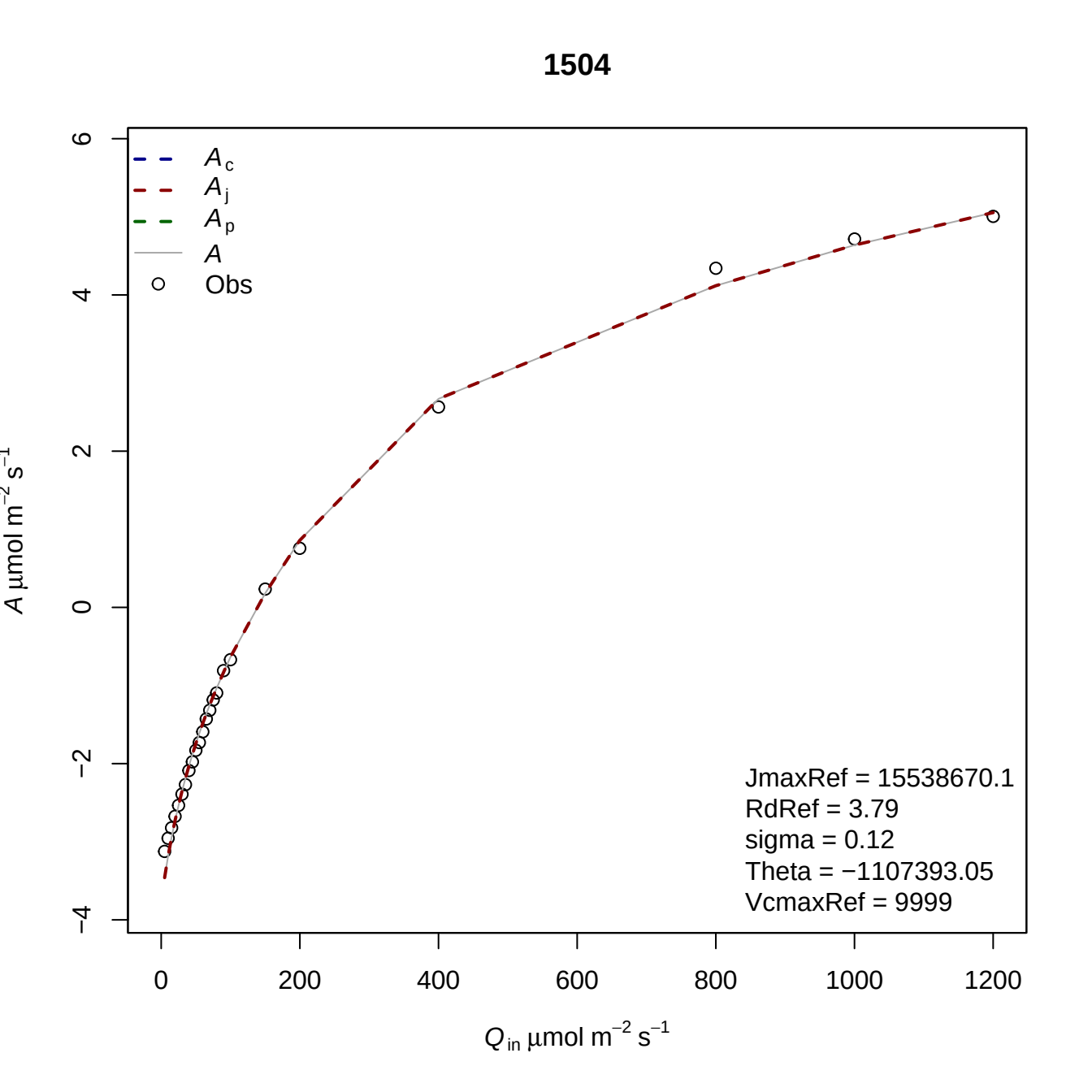

1499
A_c
A_j
A_p
A
Obs
JmaxRef = 121.97
RdRef = 3.5
sigma = 0.23
Theta = −5.71
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
200
400
600
800
1000

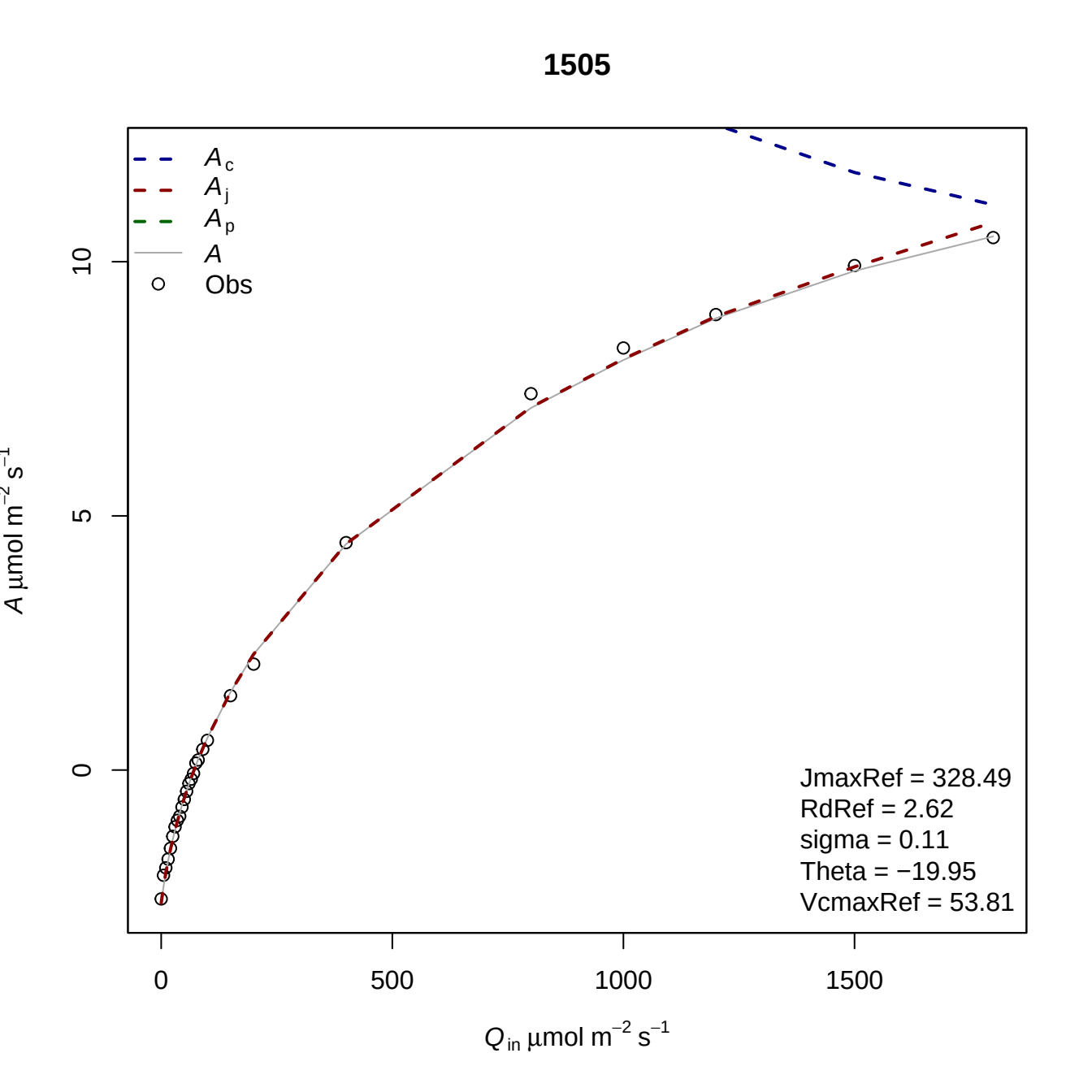

1500
A_c
A_j
A_p
A
Obs
JmaxRef = 264.73
RdRef = 3.21
sigma = 0.08
Theta = −27.88
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

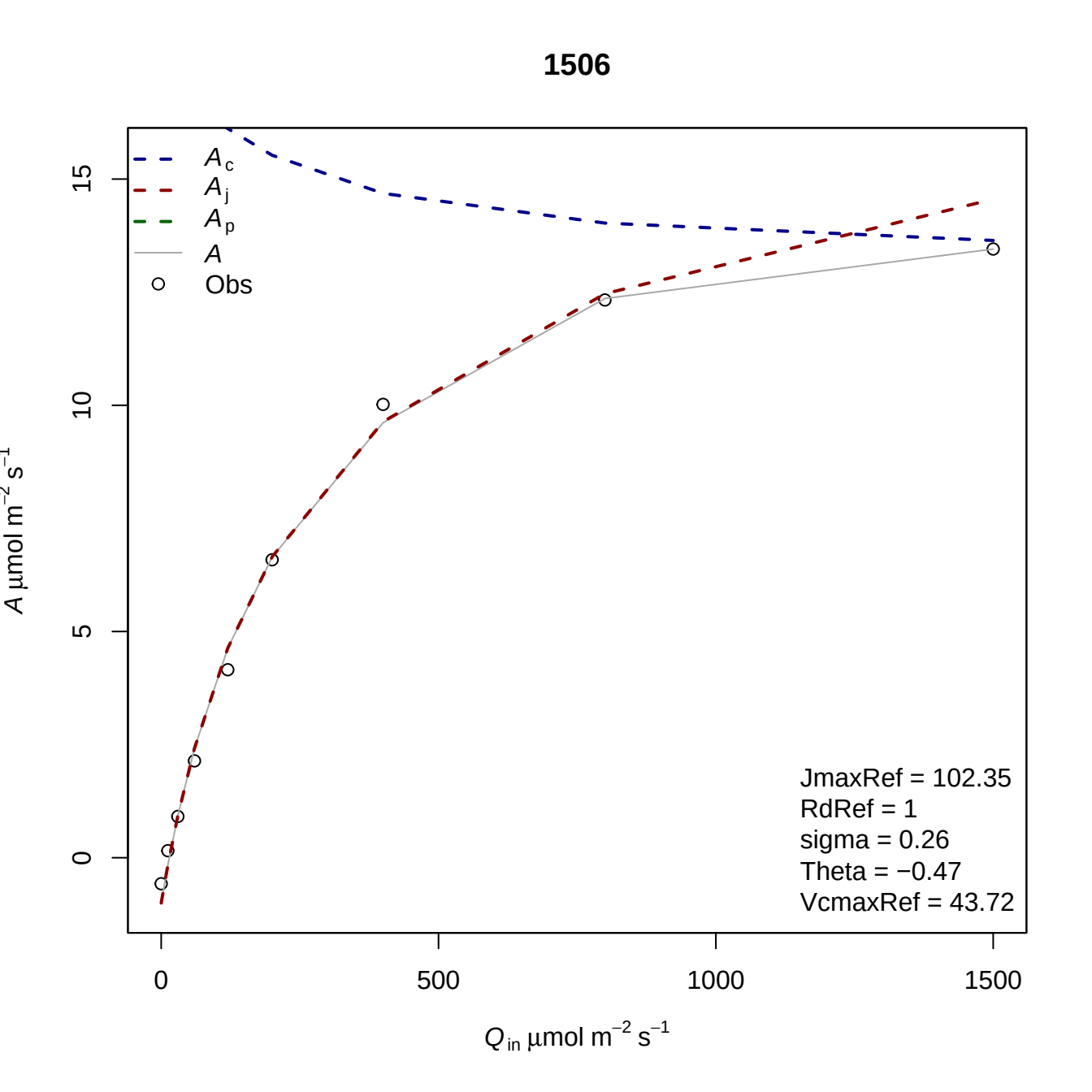

1501
A_c
A_j
A_p
A
Obs
JmaxRef = 248.43
RdRef = 2.8
sigma = 0.23
Theta = −6.26
VcmaxRef = 79.04
0
5
10
A µmol m−2 s−1
0
500
1000
1500
Qin µmol m−2 s−1

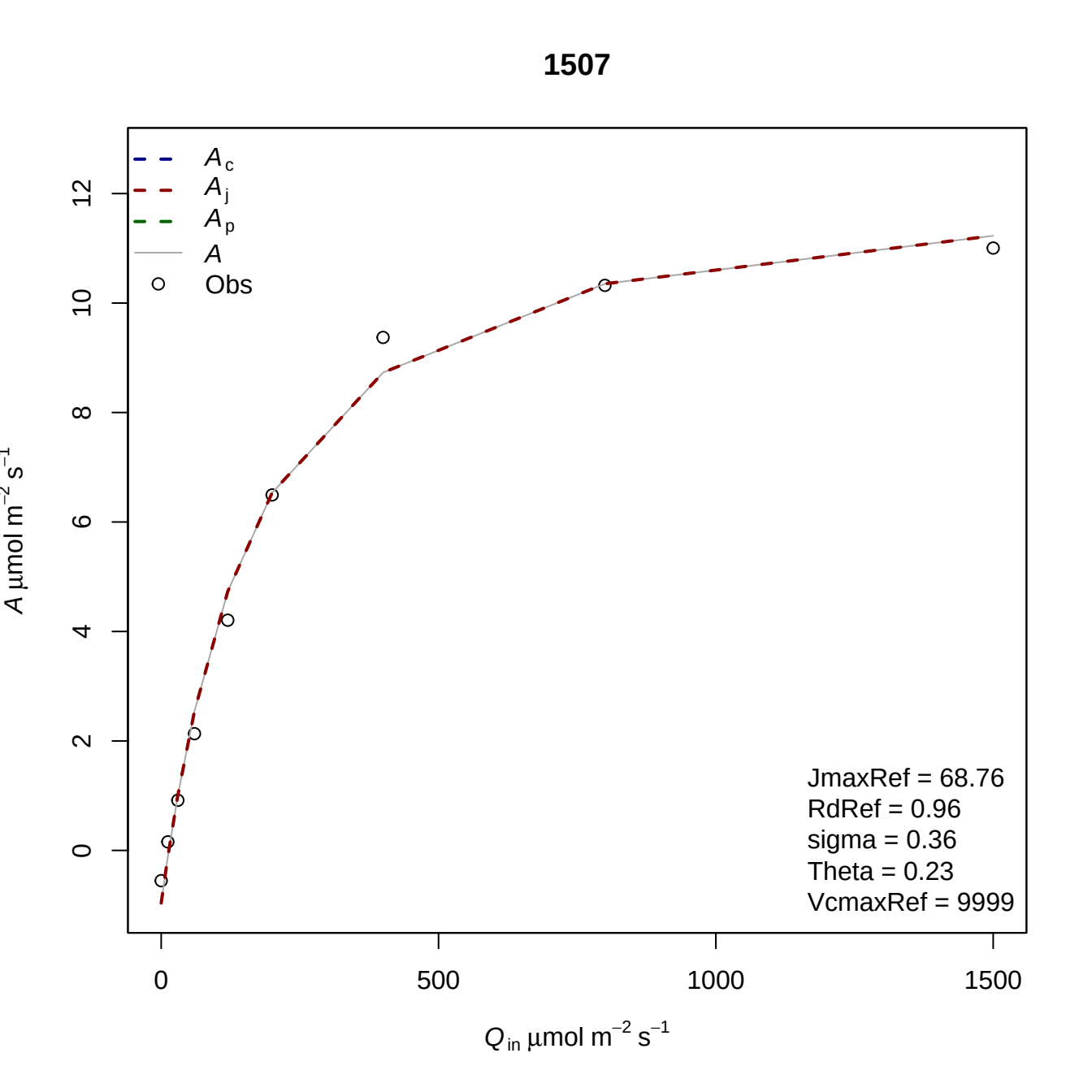

1502
A_c
A_j
A_p
A
Obs
JmaxRef = 491.59
RdRef = 3.2
sigma = 0.23
Theta = −36.86
VcmaxRef = 71.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

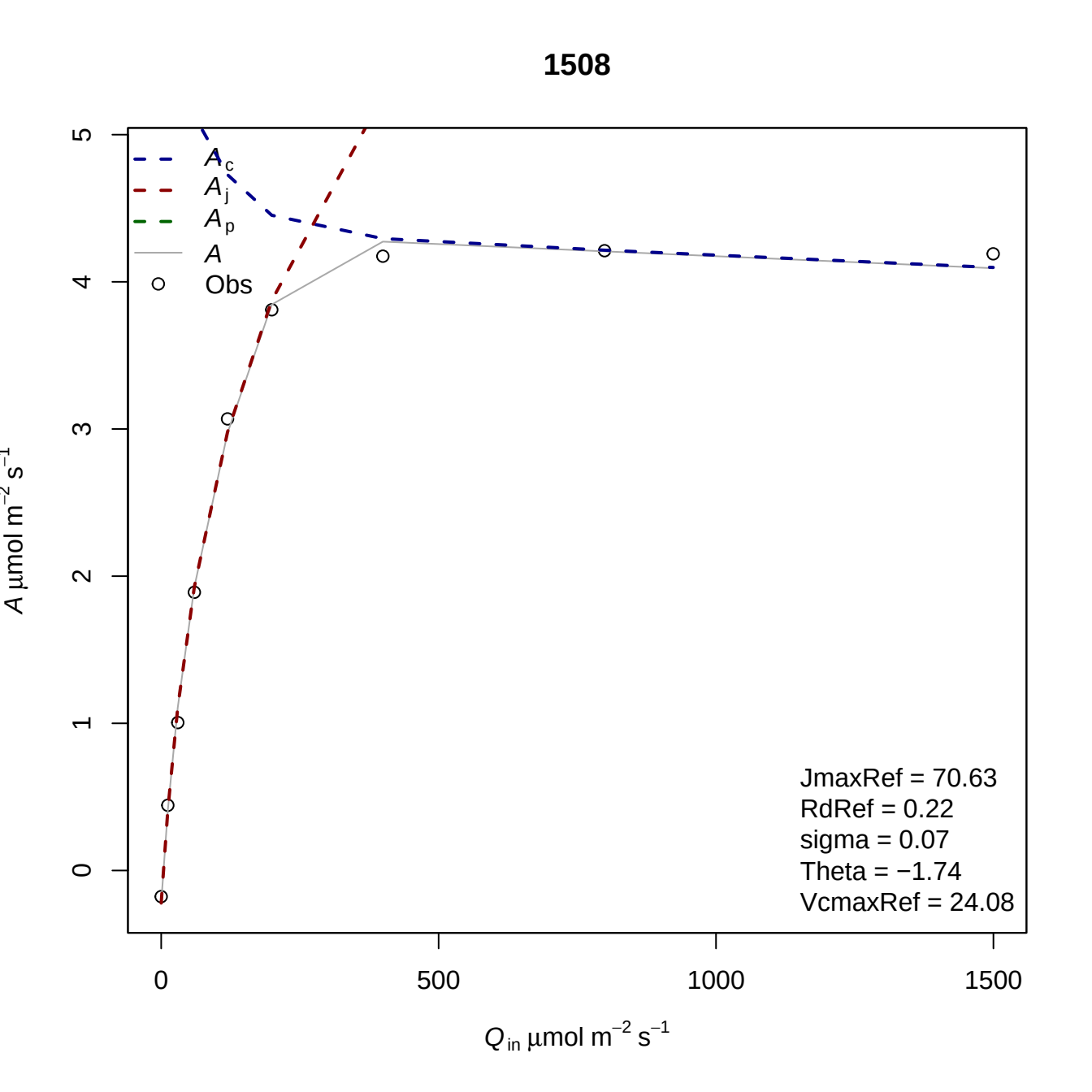

1503
A_c
A_j
A_p
A
Obs
JmaxRef = 18314742.97
RdRef = 2.86
sigma = 0.15
Theta = −1384044.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

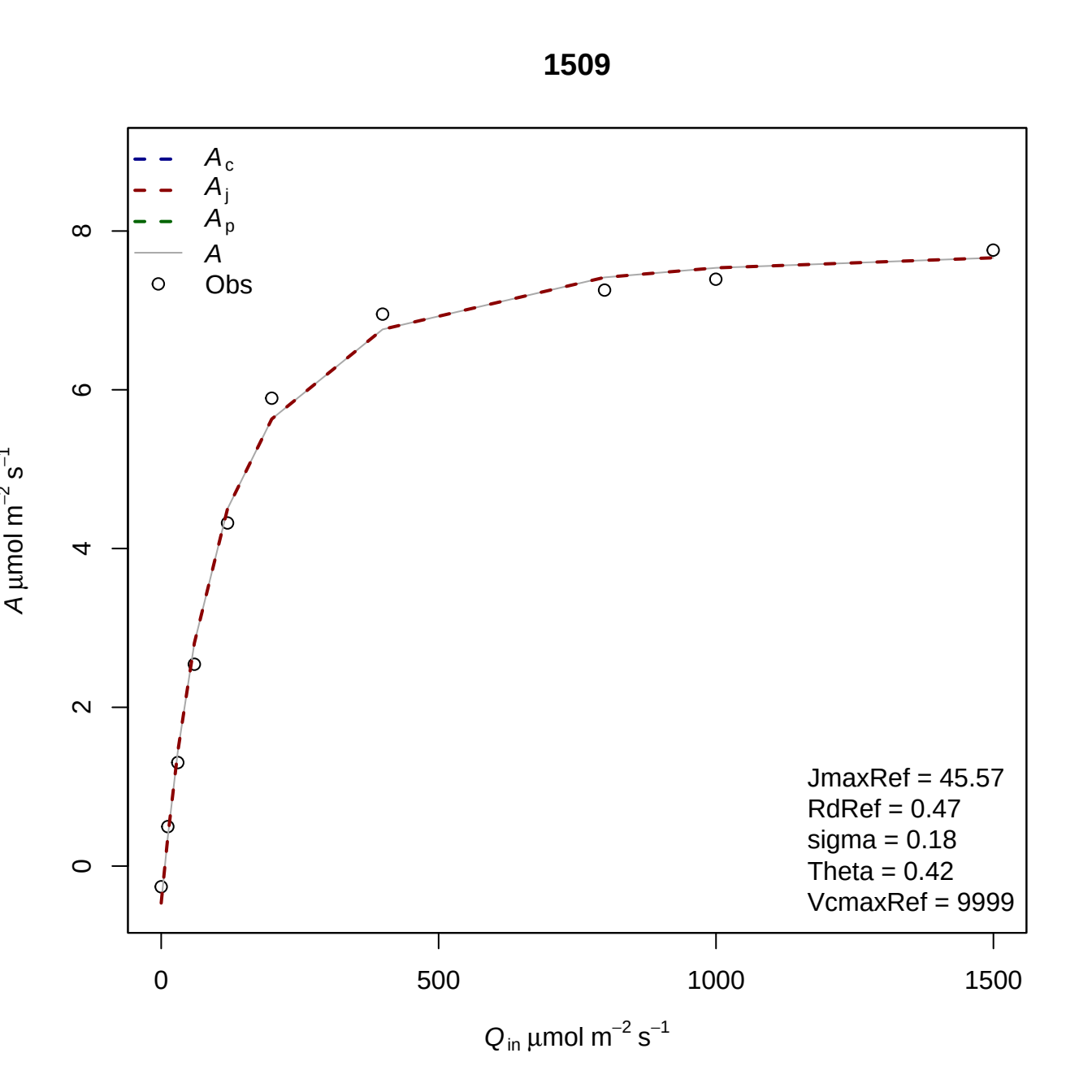

1504
A_c
A_j
A_p
A
Obs
JmaxRef = 15538670.1
RdRef = 3.79
sigma = 0.12
Theta = −1107393.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

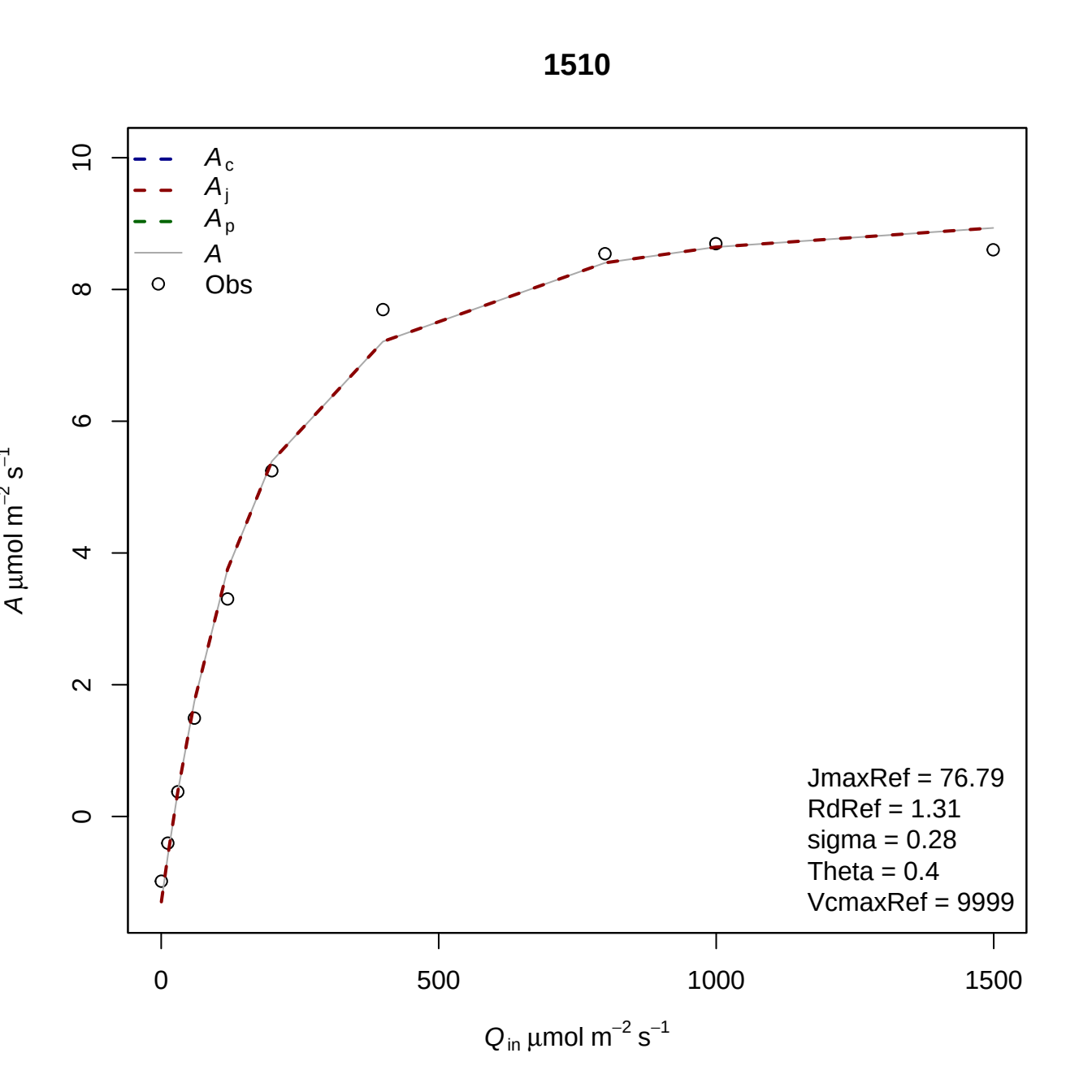

1505
A_c
A_j
A_p
A
Obs
JmaxRef = 328.49
RdRef = 2.62
sigma = 0.11
Theta = −19.95
VcmaxRef = 53.81
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

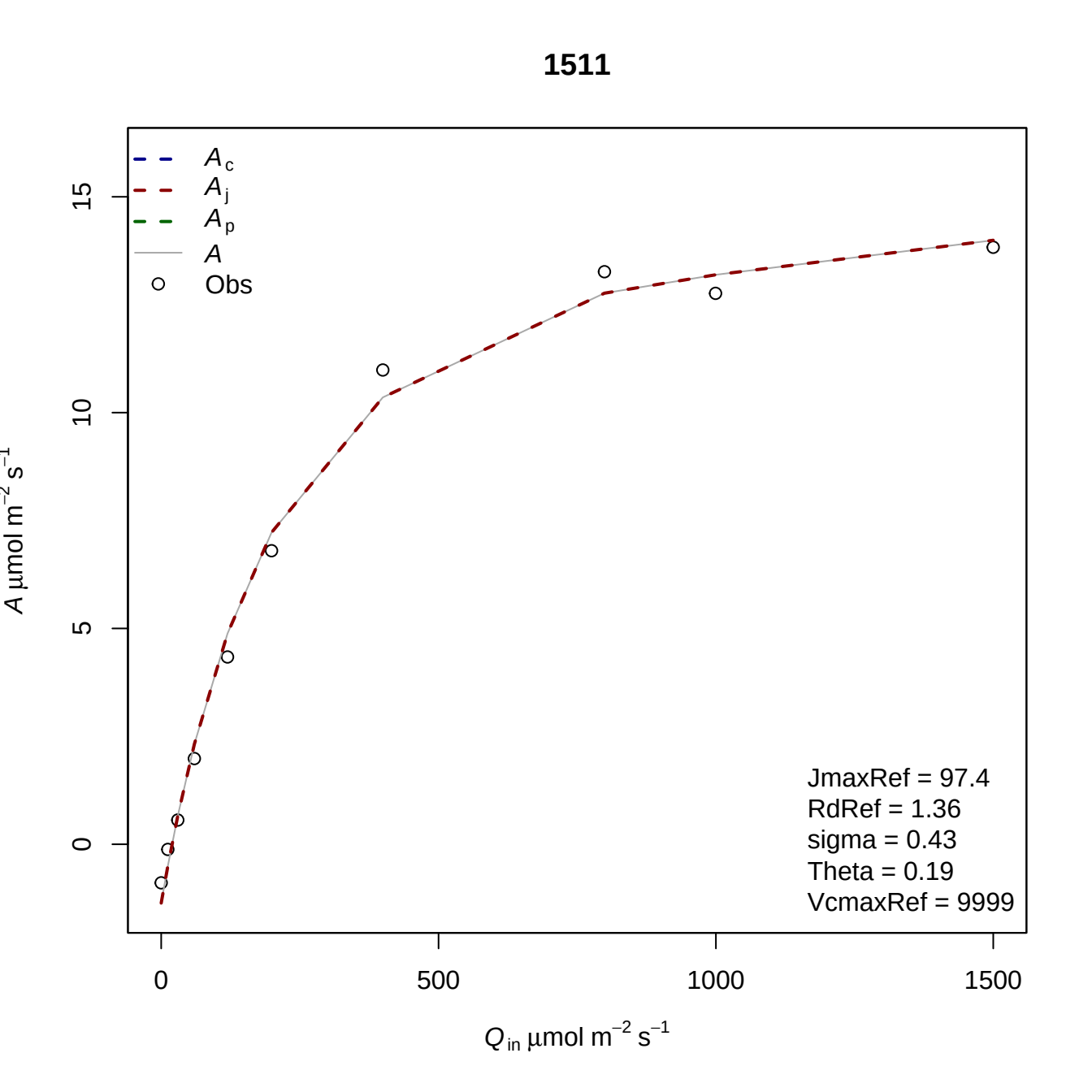

1506
A_c
A_j
A_p
A
Obs
JmaxRef = 102.35
RdRef = 1
sigma = 0.26
Theta = −0.47
VcmaxRef = 43.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

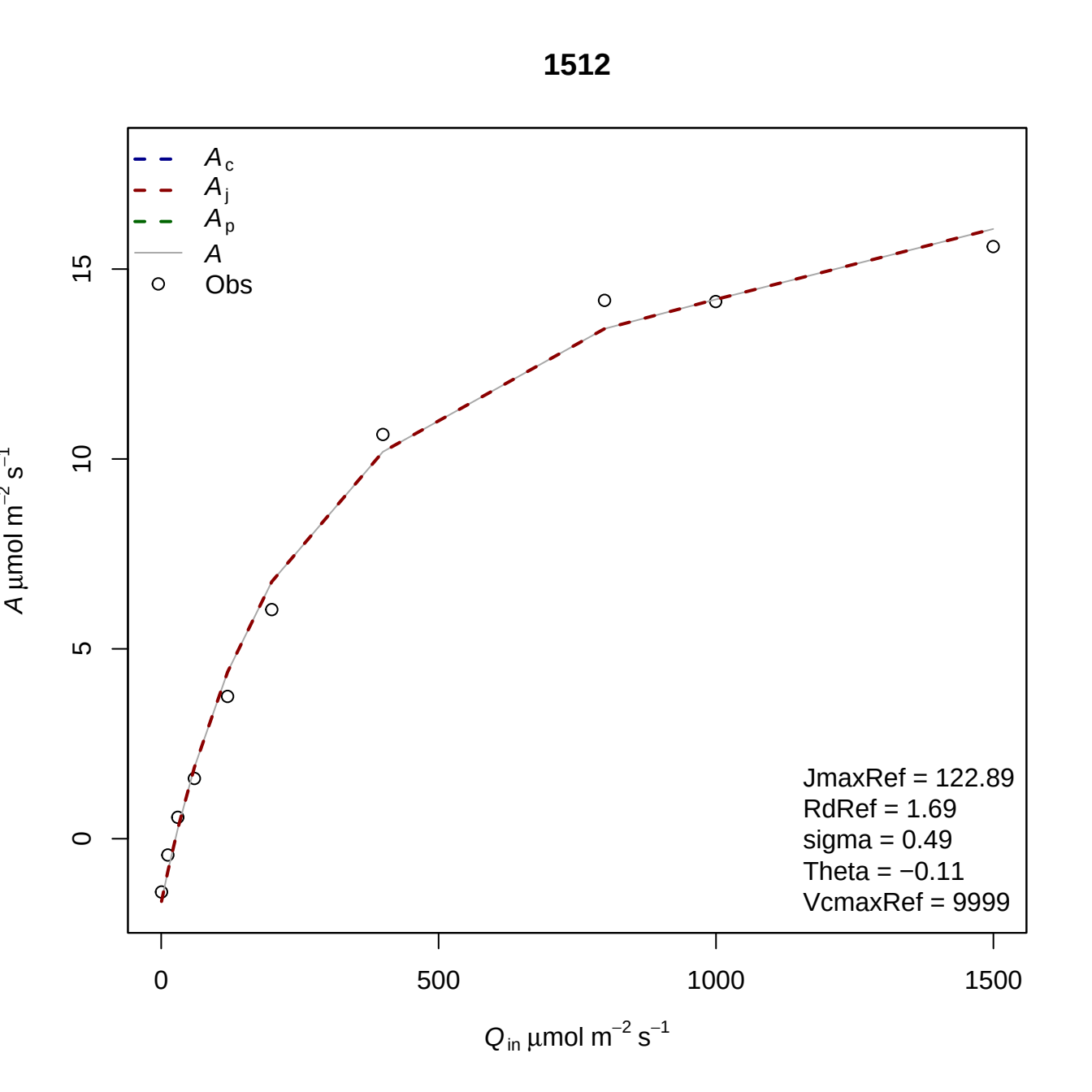

1507
A_c
A_j
A_p
A
Obs
JmaxRef = 68.76
RdRef = 0.96
sigma = 0.36
Theta = 0.23
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

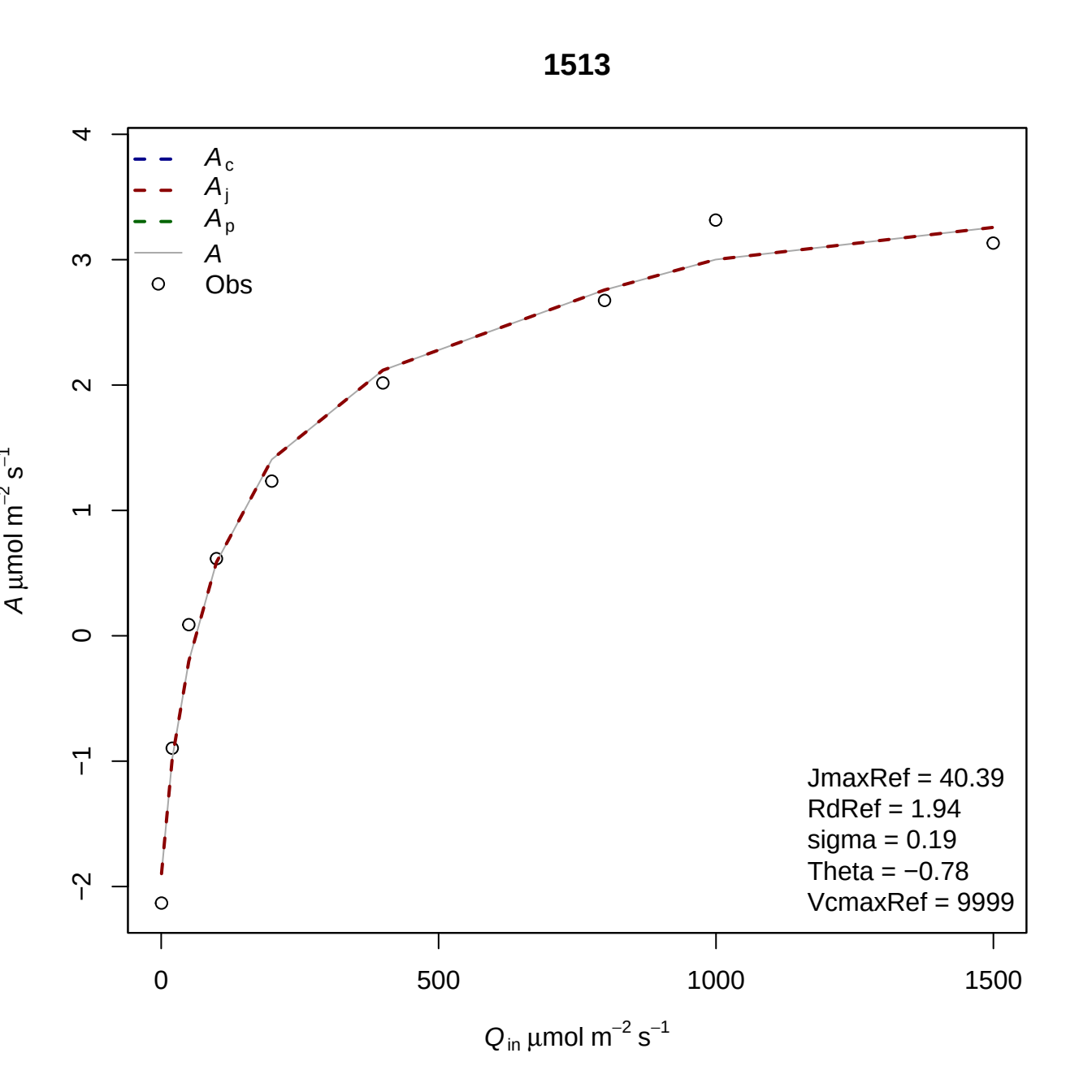

1508
A_c
A_j
A_p
A
Obs
JmaxRef = 70.63
RdRef = 0.22
sigma = 0.07
Theta = −1.74
VcmaxRef = 24.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

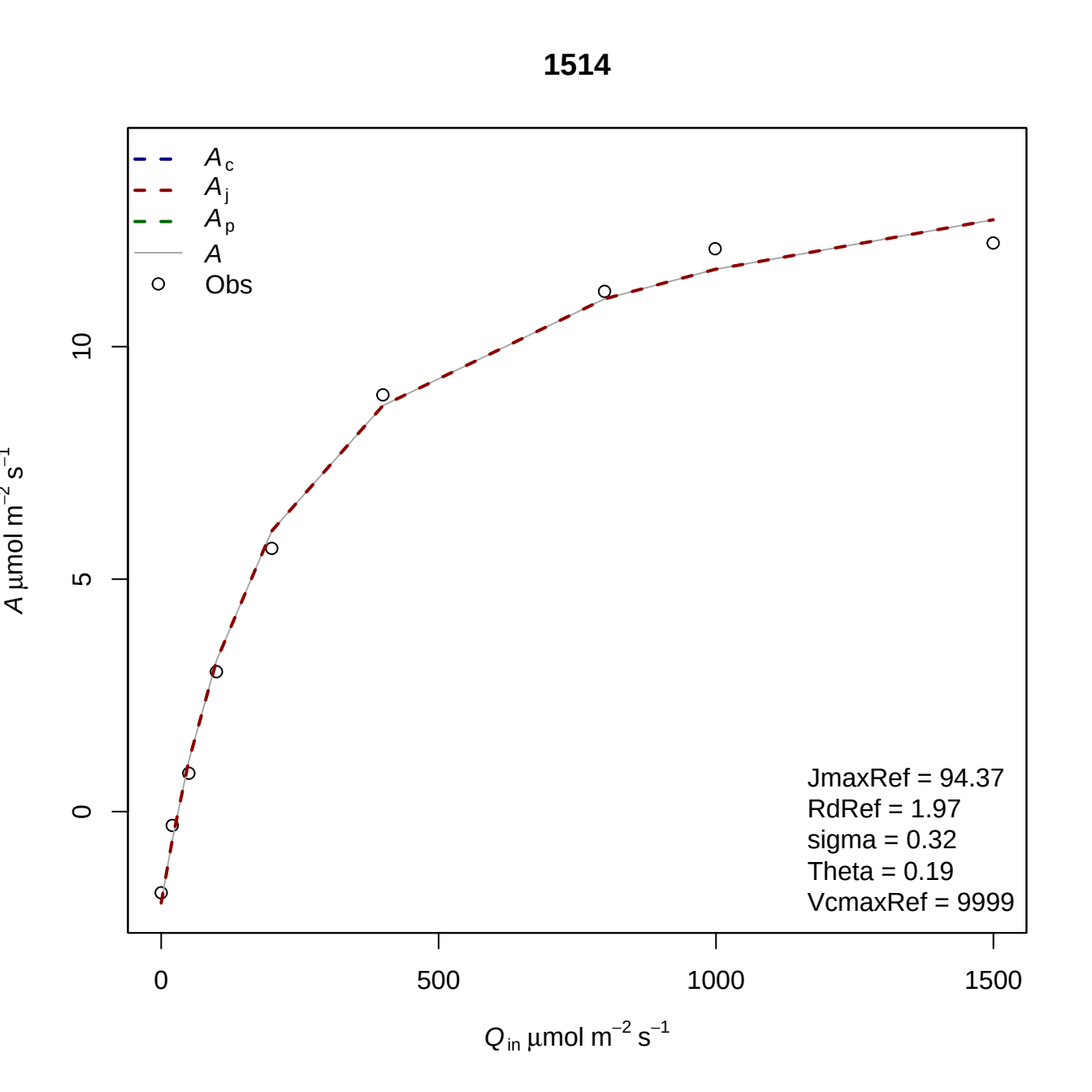

1509
A_c
A_j
A_p
A
Obs
JmaxRef = 45.57
RdRef = 0.47
sigma = 0.18
Theta = 0.42
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

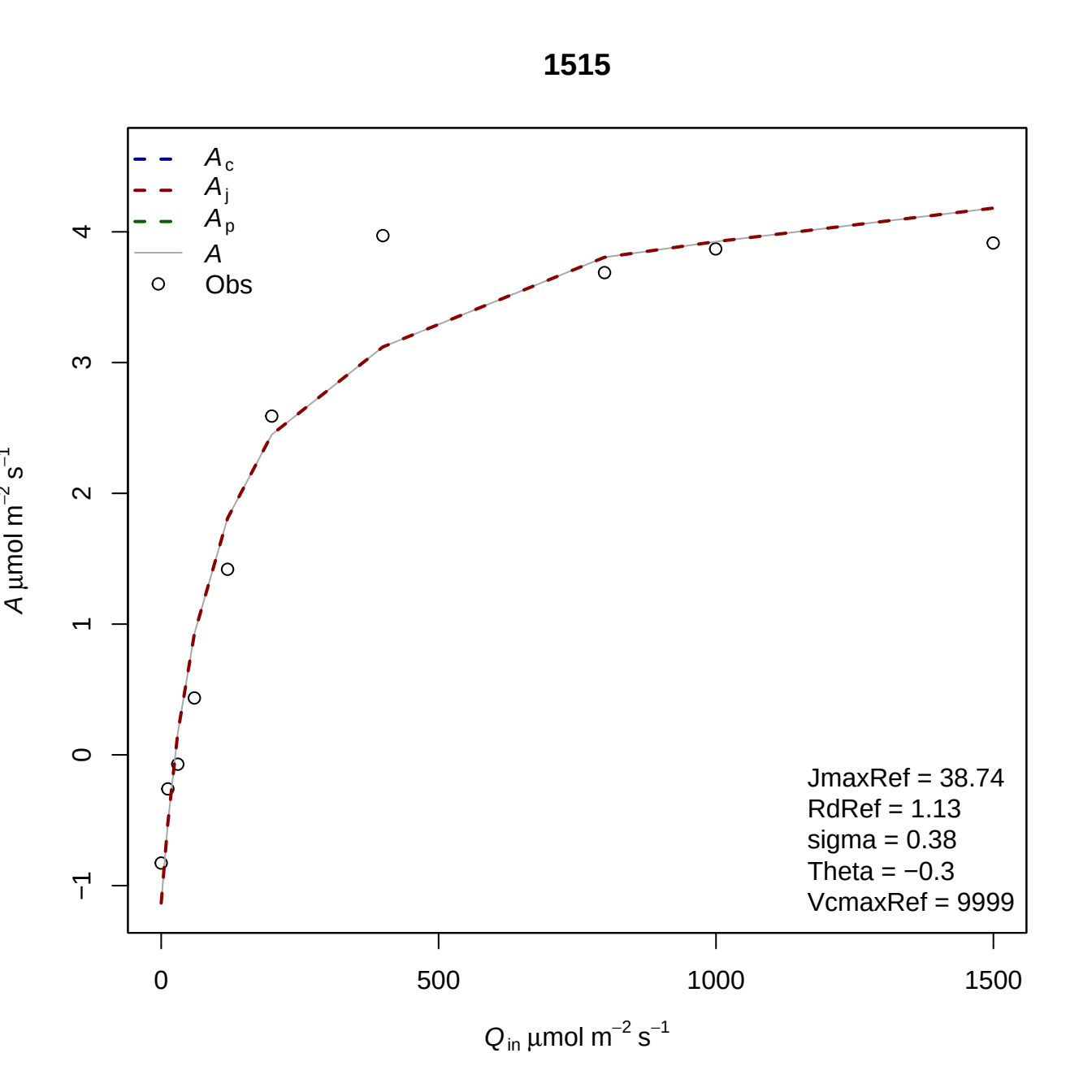

1510
A_c
A_j
A_p
A
Obs
JmaxRef = 76.79
RdRef = 1.31
sigma = 0.28
Theta = 0.4
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

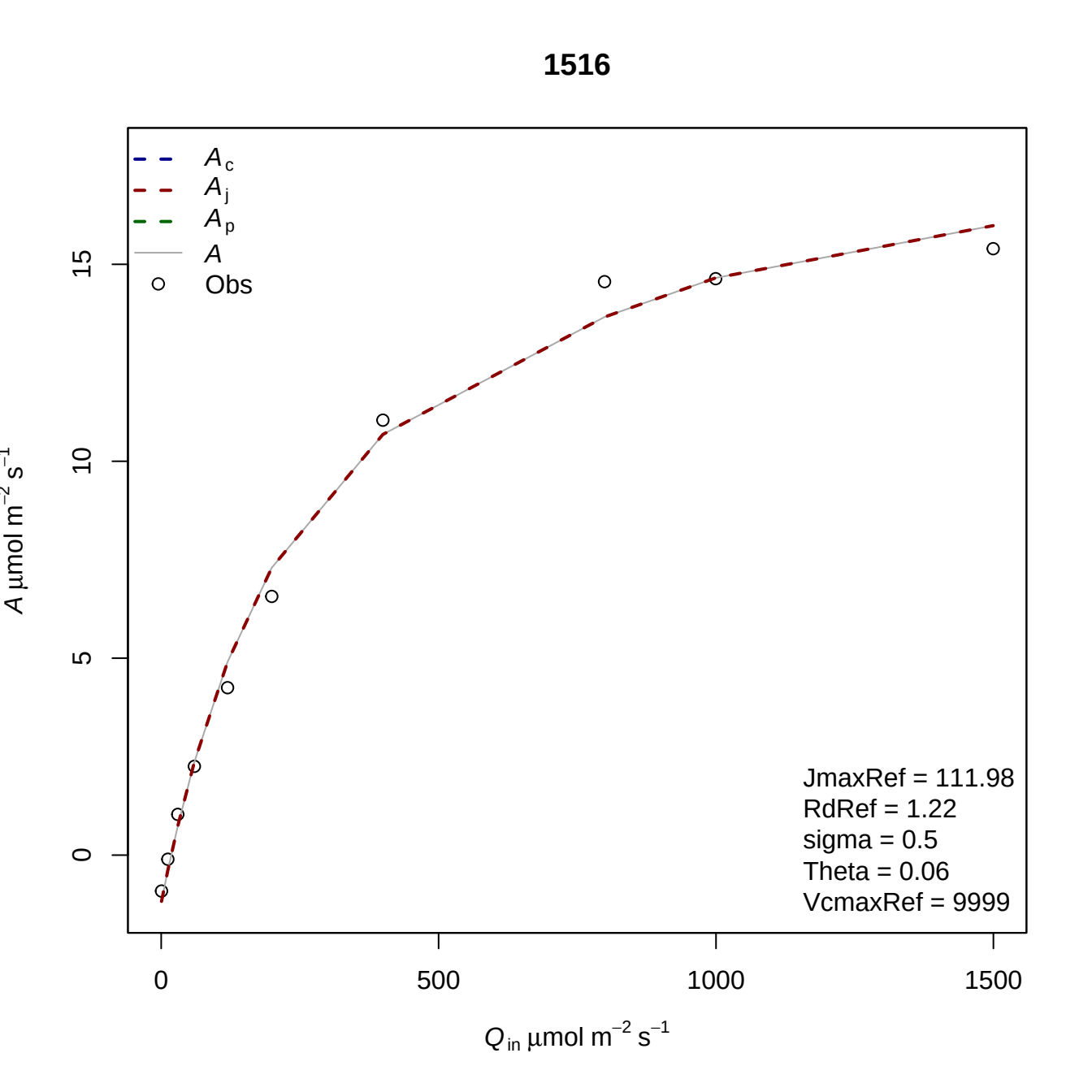

1511
A_c
A_j
A_p
A
Obs
JmaxRef = 97.4
RdRef = 1.36
sigma = 0.43
Theta = 0.19
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

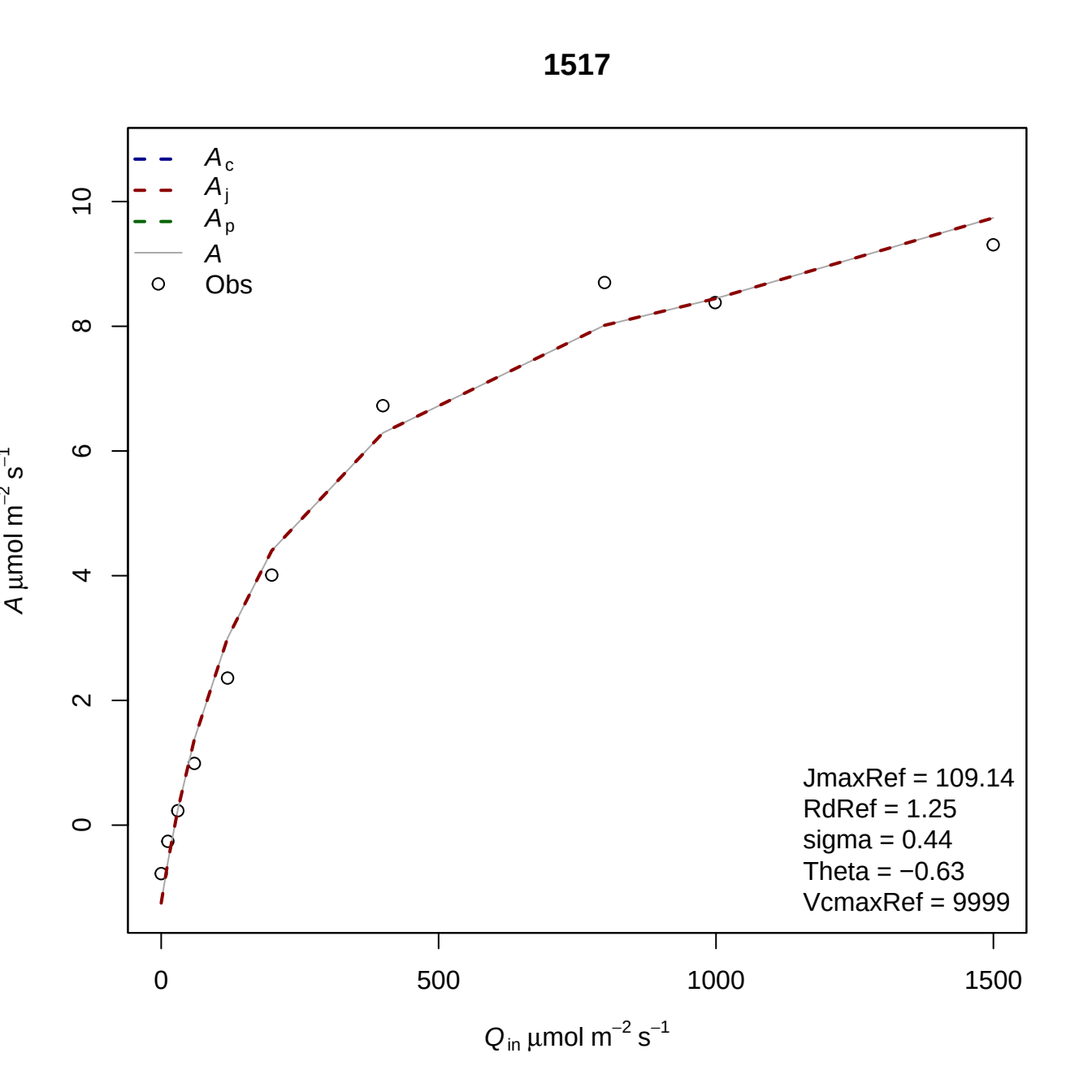

1512
A_c
A_j
A_p
A
Obs
JmaxRef = 122.89
RdRef = 1.69
sigma = 0.49
Theta = −0.11
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

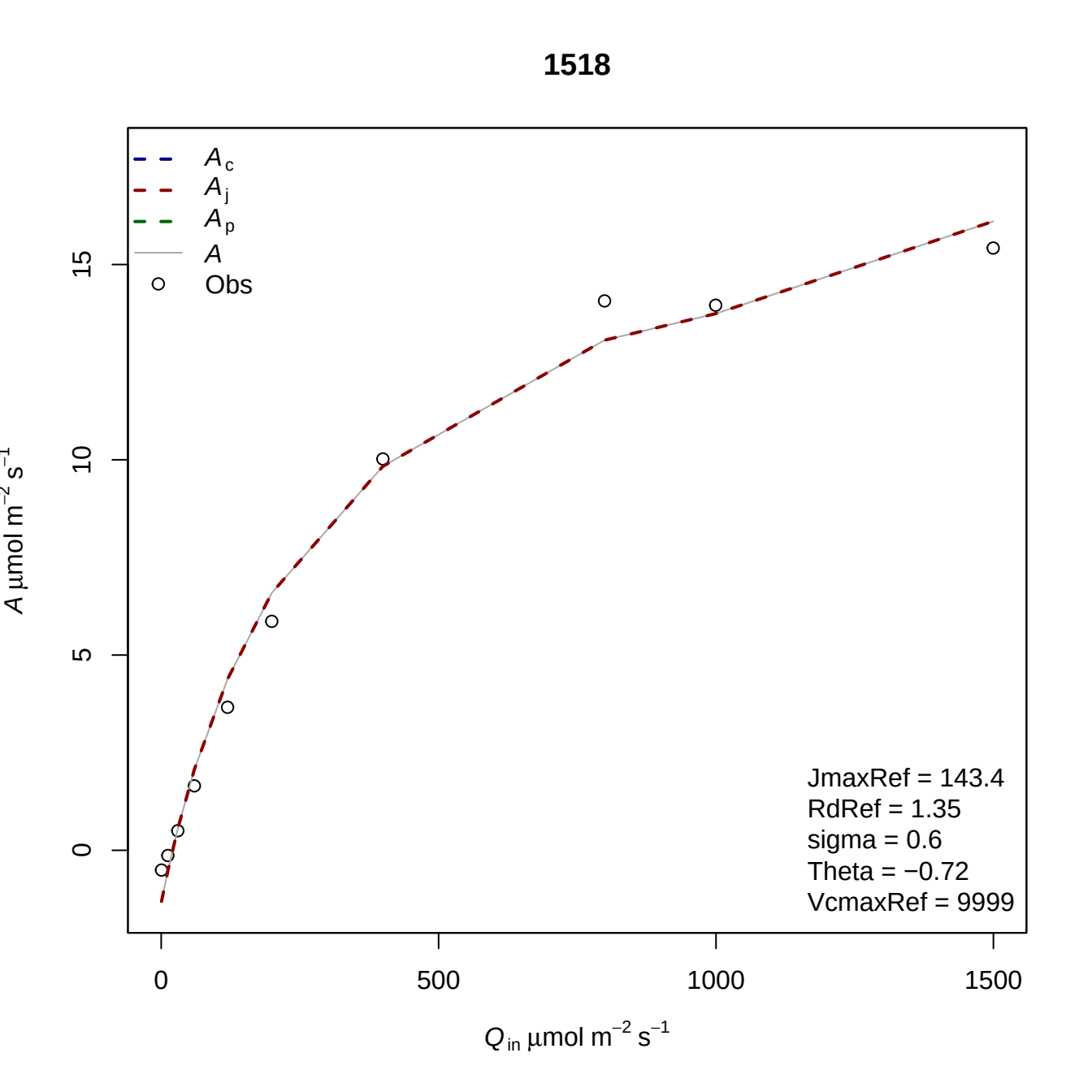

1513
A_c
A_j
A_p
A
Obs
JmaxRef = 40.39
RdRef = 1.94
sigma = 0.19
Theta = −0.78
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

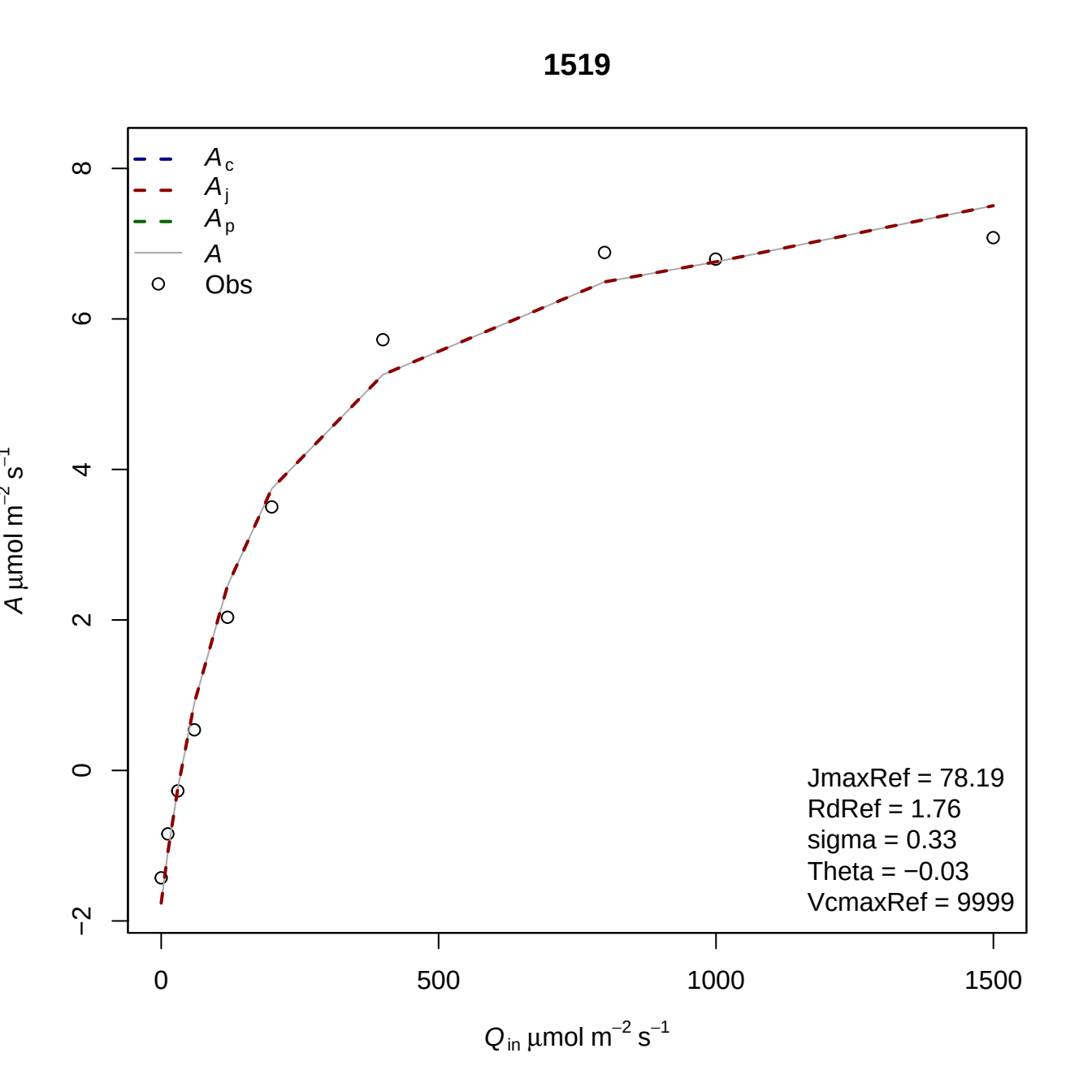

1514
A_c
A_j
A_p
A
Obs
JmaxRef = 94.37
RdRef = 1.97
sigma = 0.32
Theta = 0.19
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

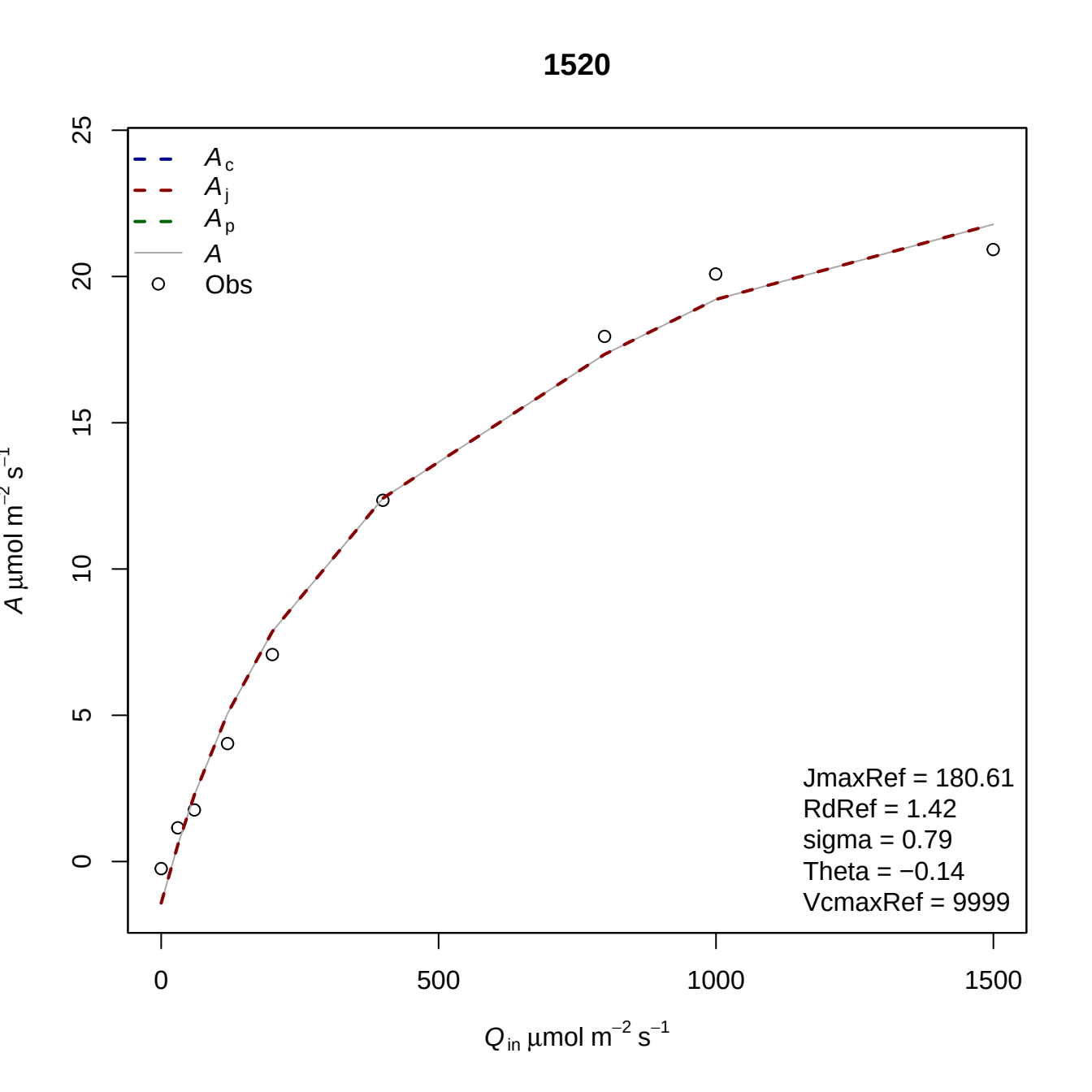

1515
A_c
A_j
A_p
A
Obs
JmaxRef = 38.74
RdRef = 1.13
sigma = 0.38
Theta = −0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

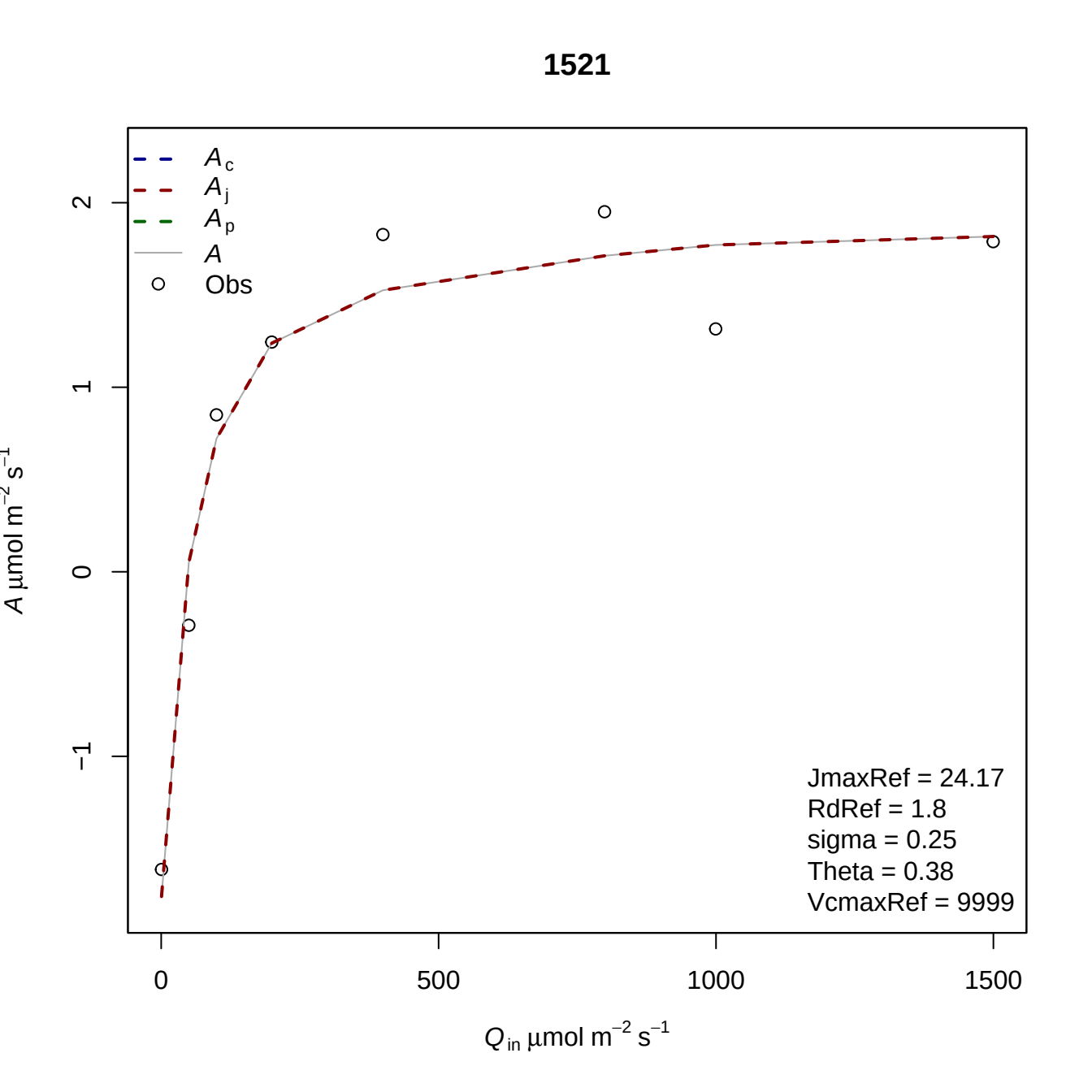

1516
A_c
A_j
A_p
A
Obs
JmaxRef = 111.98
RdRef = 1.22
sigma = 0.5
Theta = 0.06
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

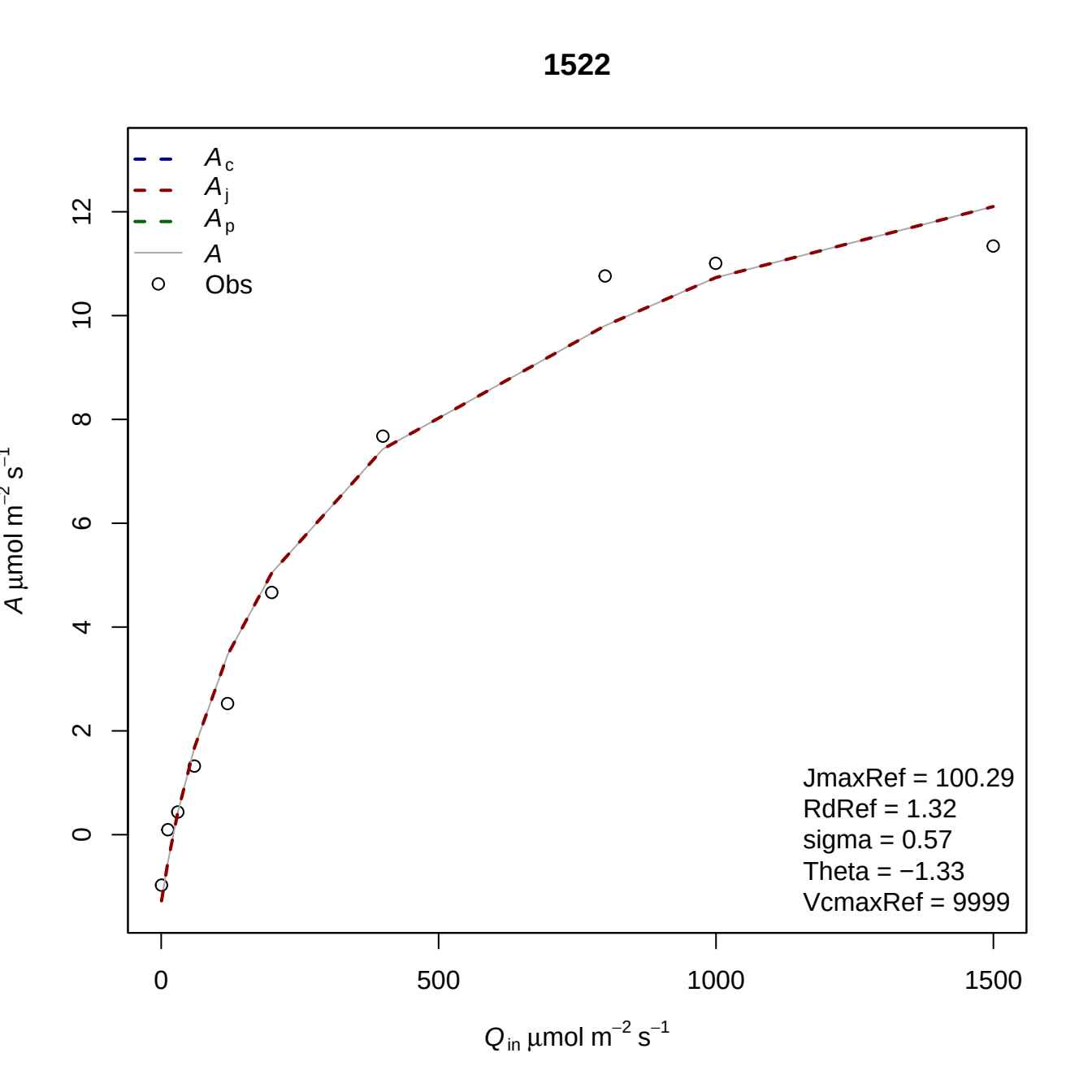

1517
A_c
A_j
A_p
A
Obs
JmaxRef = 109.14
RdRef = 1.25
sigma = 0.44
Theta = −0.63
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

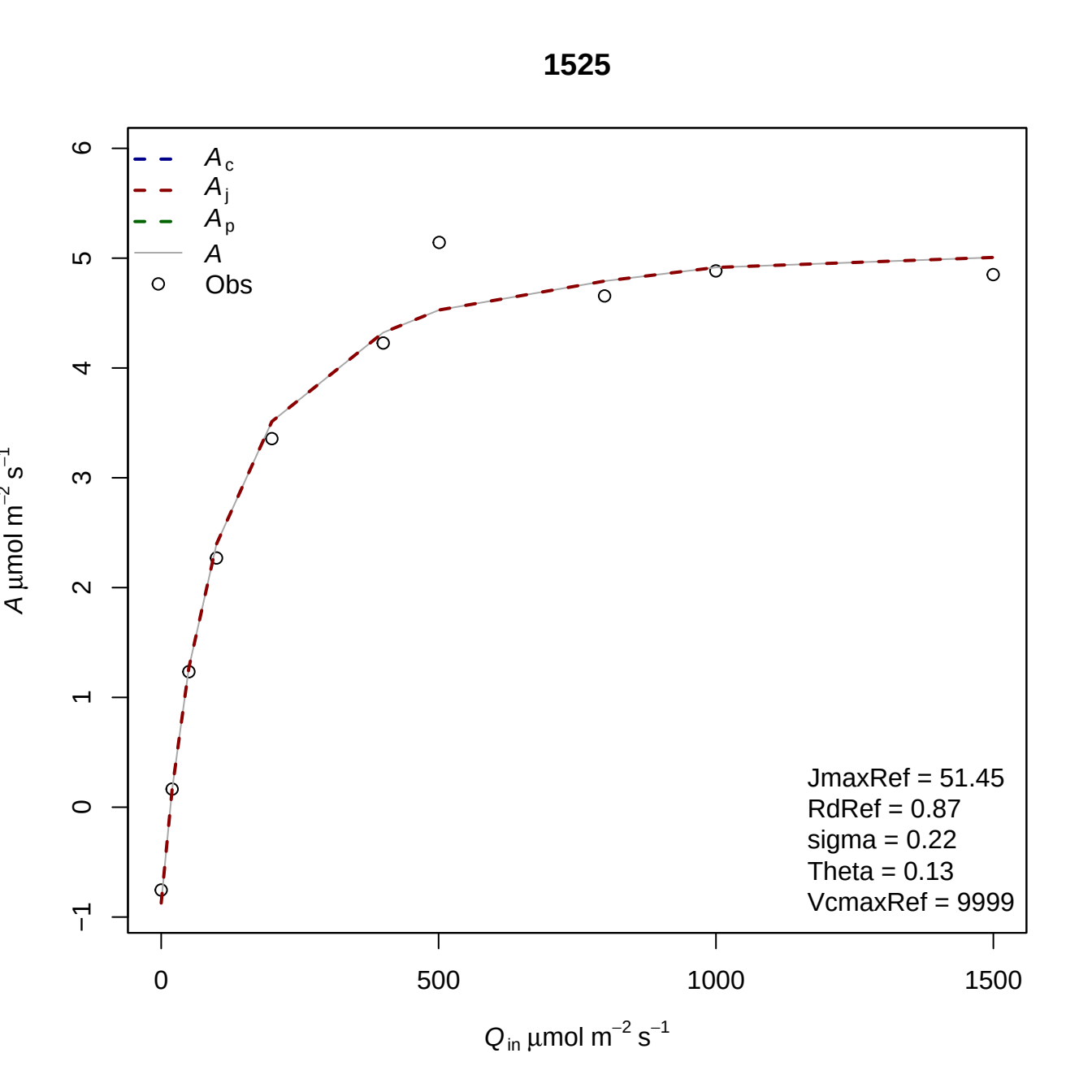

1518
A_c
A_j
A_p
A
Obs
JmaxRef = 143.4
RdRef = 1.35
sigma = 0.6
Theta = −0.72
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

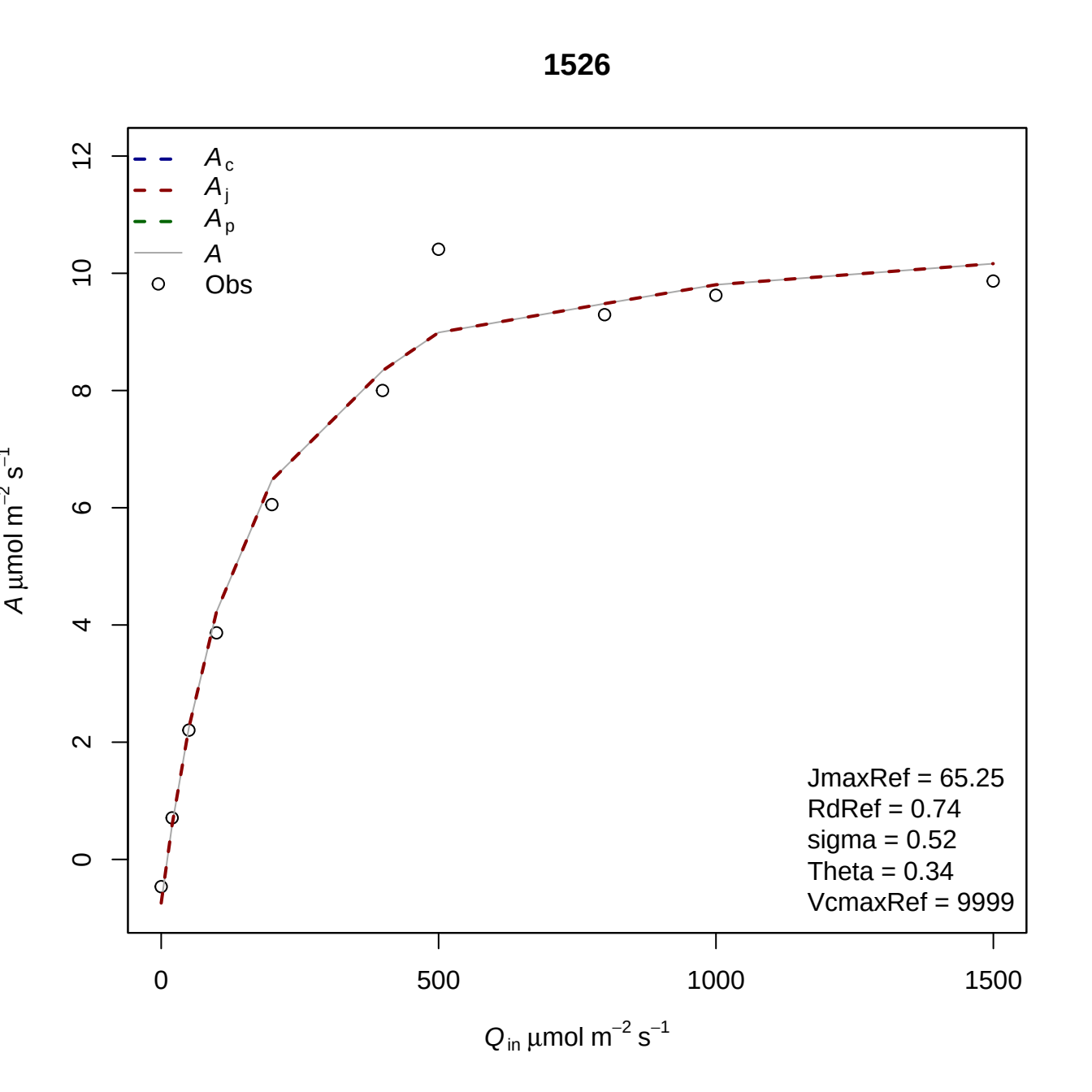

1519
A_c
A_j
A_p
A
Obs
JmaxRef = 78.19
RdRef = 1.76
sigma = 0.33
Theta = −0.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

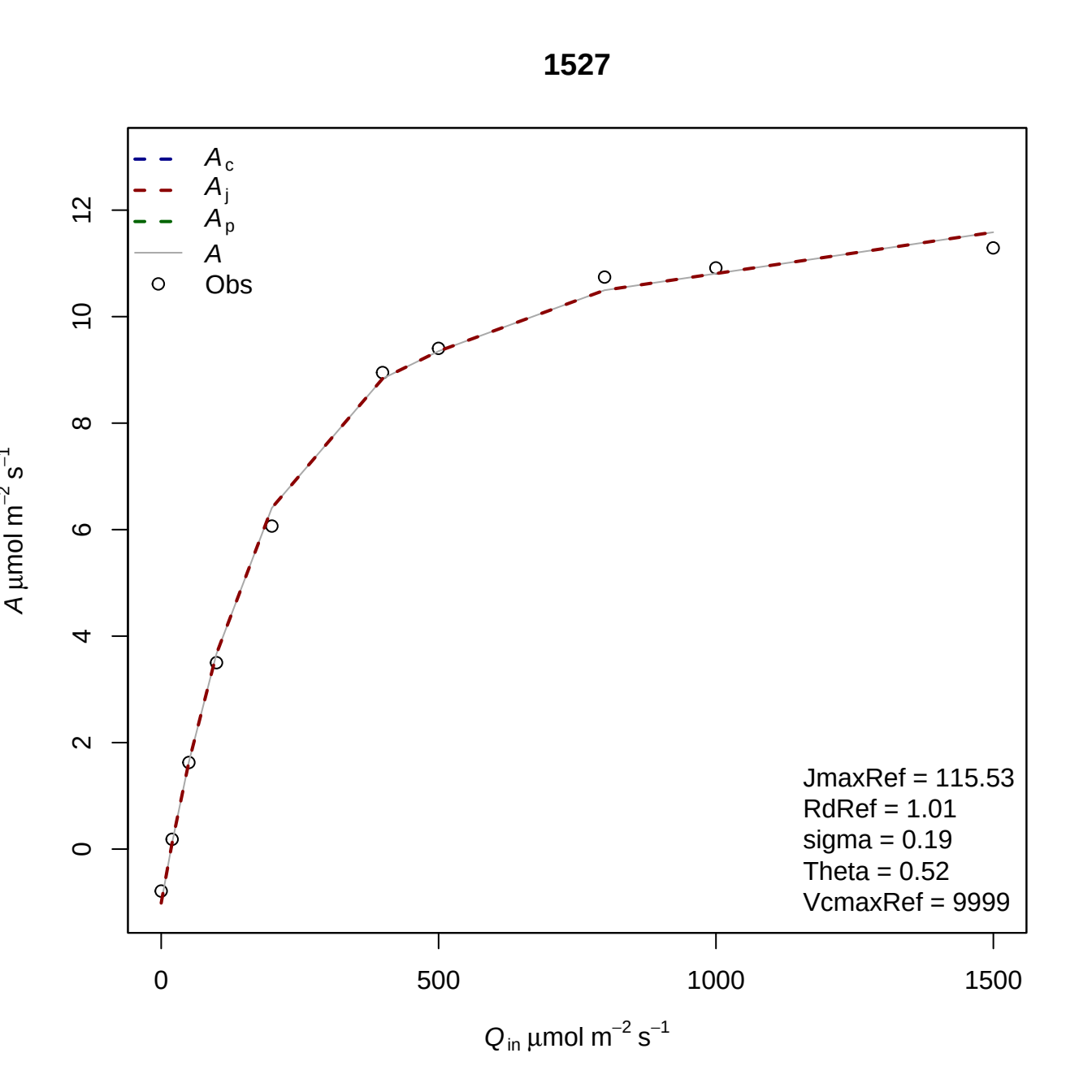

1520
A_c
A_j
A_p
A
Obs
JmaxRef = 180.61
RdRef = 1.42
sigma = 0.79
Theta = −0.14
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

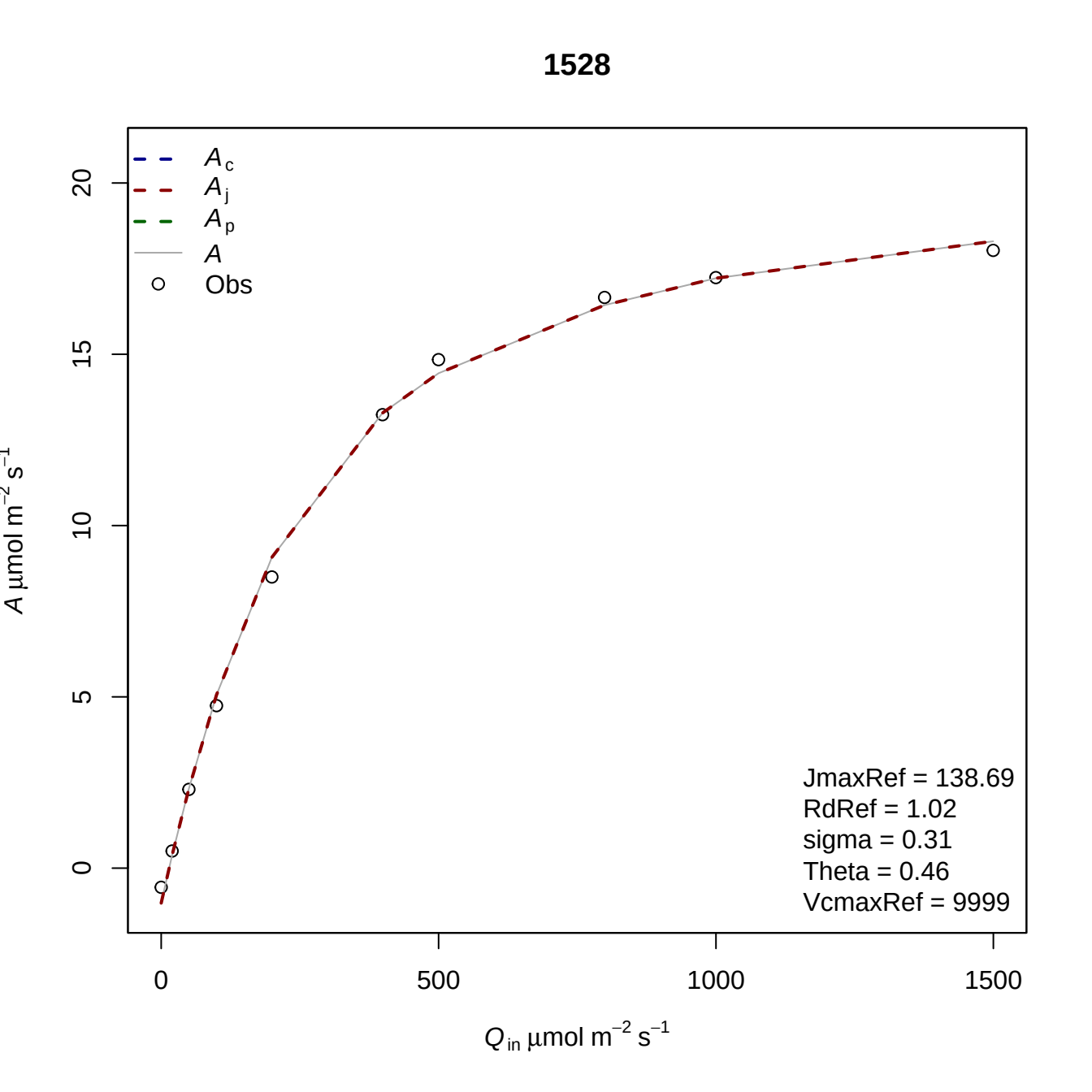

1521
A_c
A_j
A_p
A
Obs
JmaxRef = 24.17
RdRef = 1.8
sigma = 0.25
Theta = 0.38
VcmaxRef = 9999
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

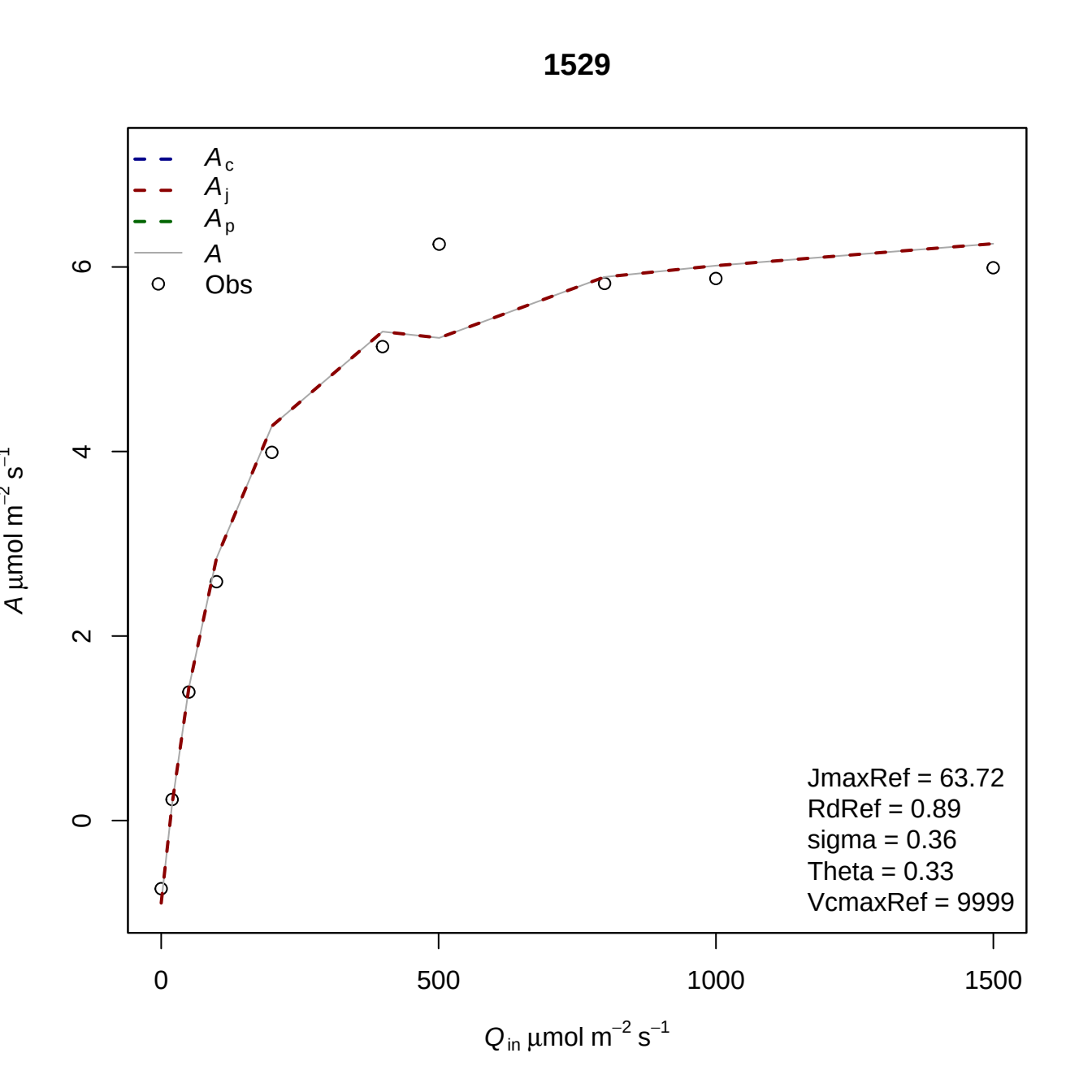

1522
A_c
A_j
A_p
A
Obs
JmaxRef = 100.29
RdRef = 1.32
sigma = 0.57
Theta = −1.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

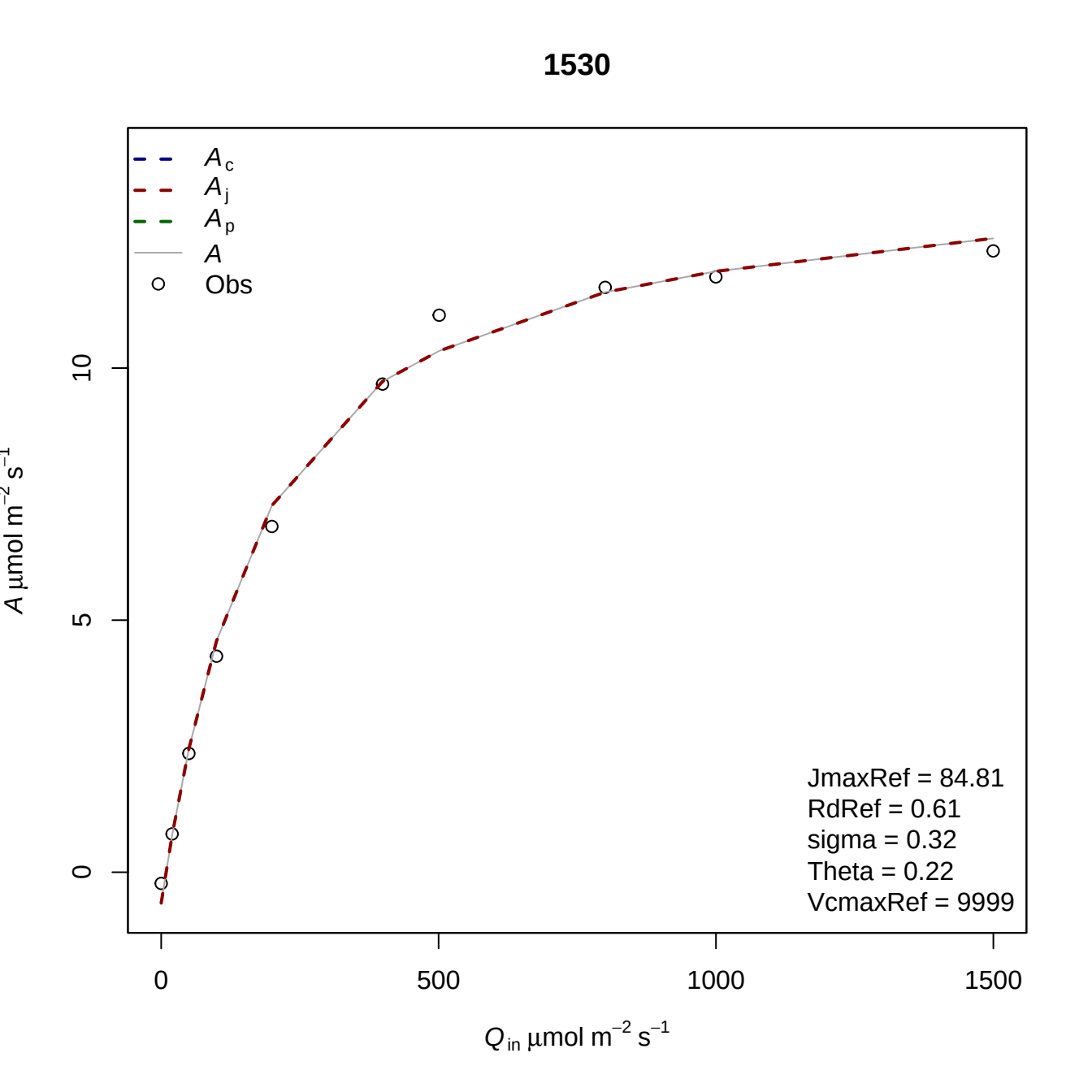

1525
A_c
A_j
A_p
A
Obs
JmaxRef = 51.45
RdRef = 0.87
sigma = 0.22
Theta = 0.13
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

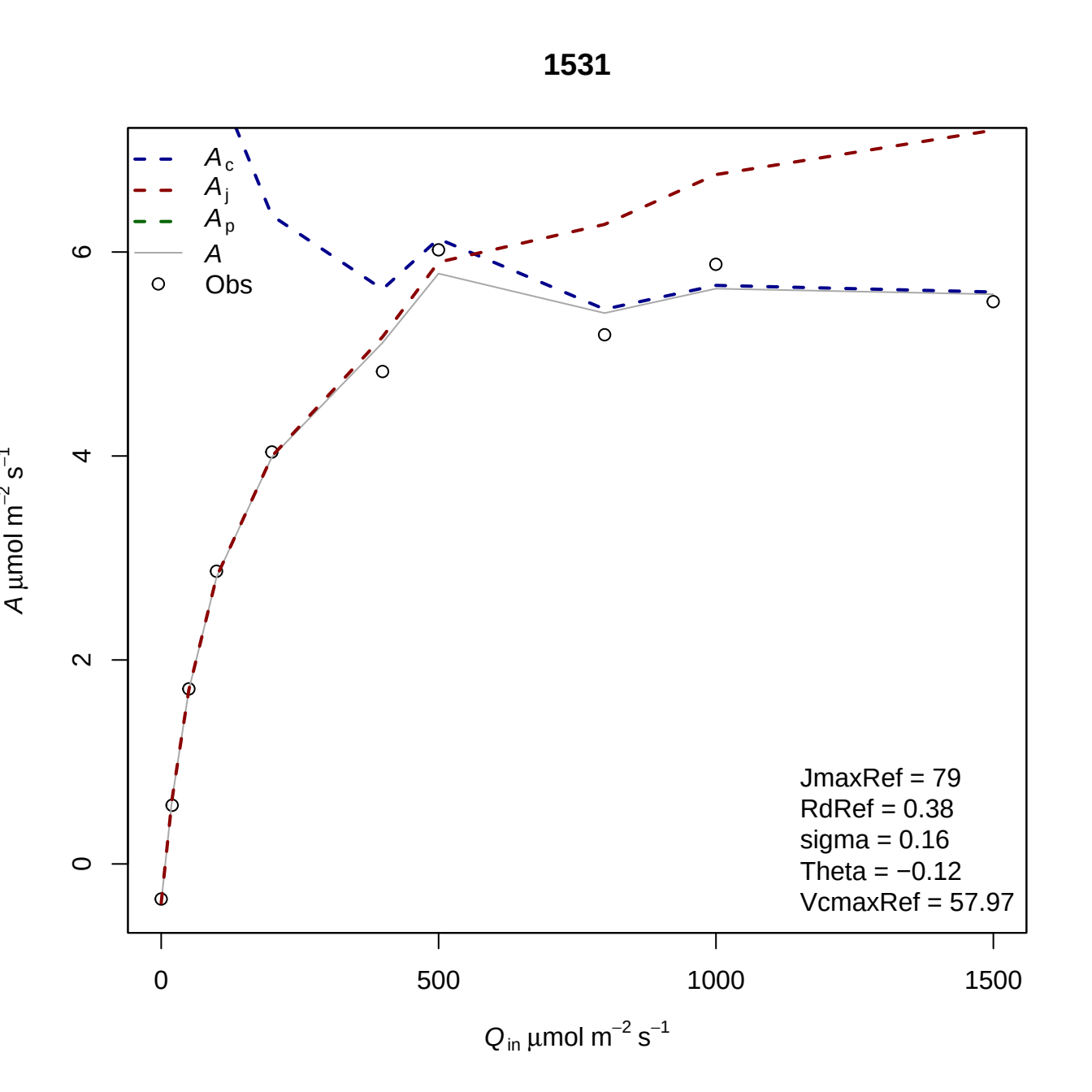

1526
A_c
A_j
A_p
A
Obs
JmaxRef = 65.25
RdRef = 0.74
sigma = 0.52
Theta = 0.34
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

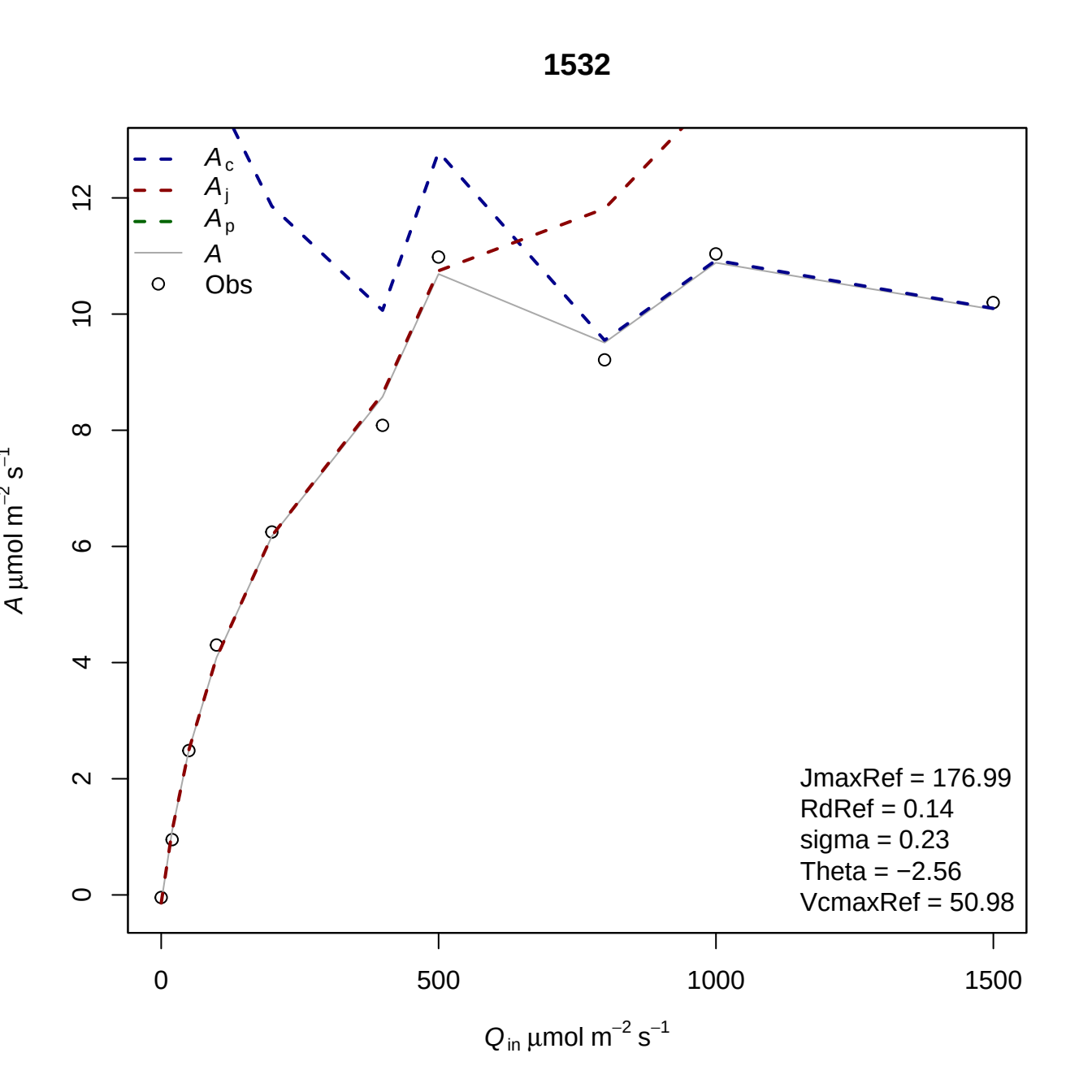

1527
A_c
A_j
A_p
A
Obs
JmaxRef = 115.53
RdRef = 1.01
sigma = 0.19
Theta = 0.52
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

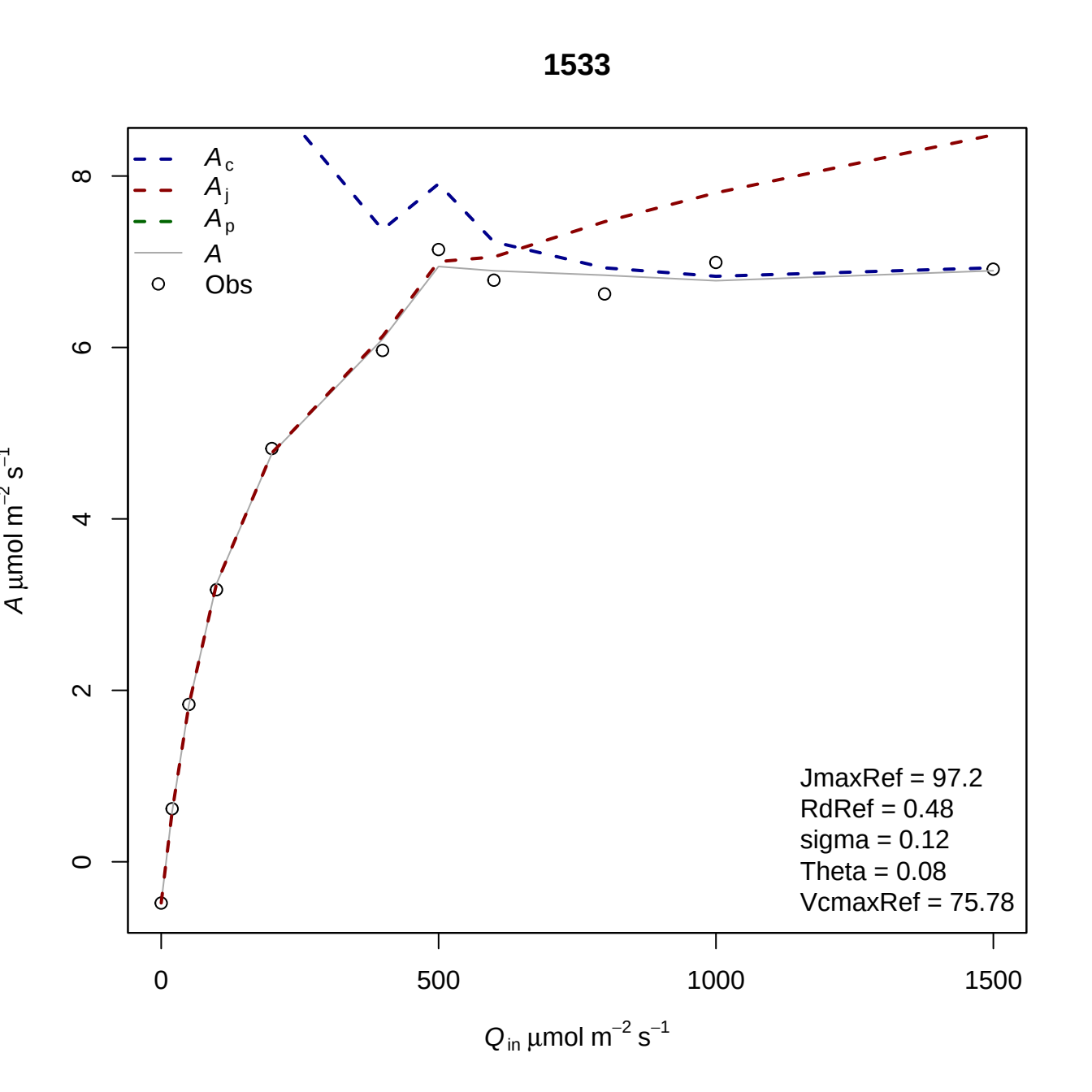

1528
A_c
A_j
A_p
A
Obs
JmaxRef = 138.69
RdRef = 1.02
sigma = 0.31
Theta = 0.46
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

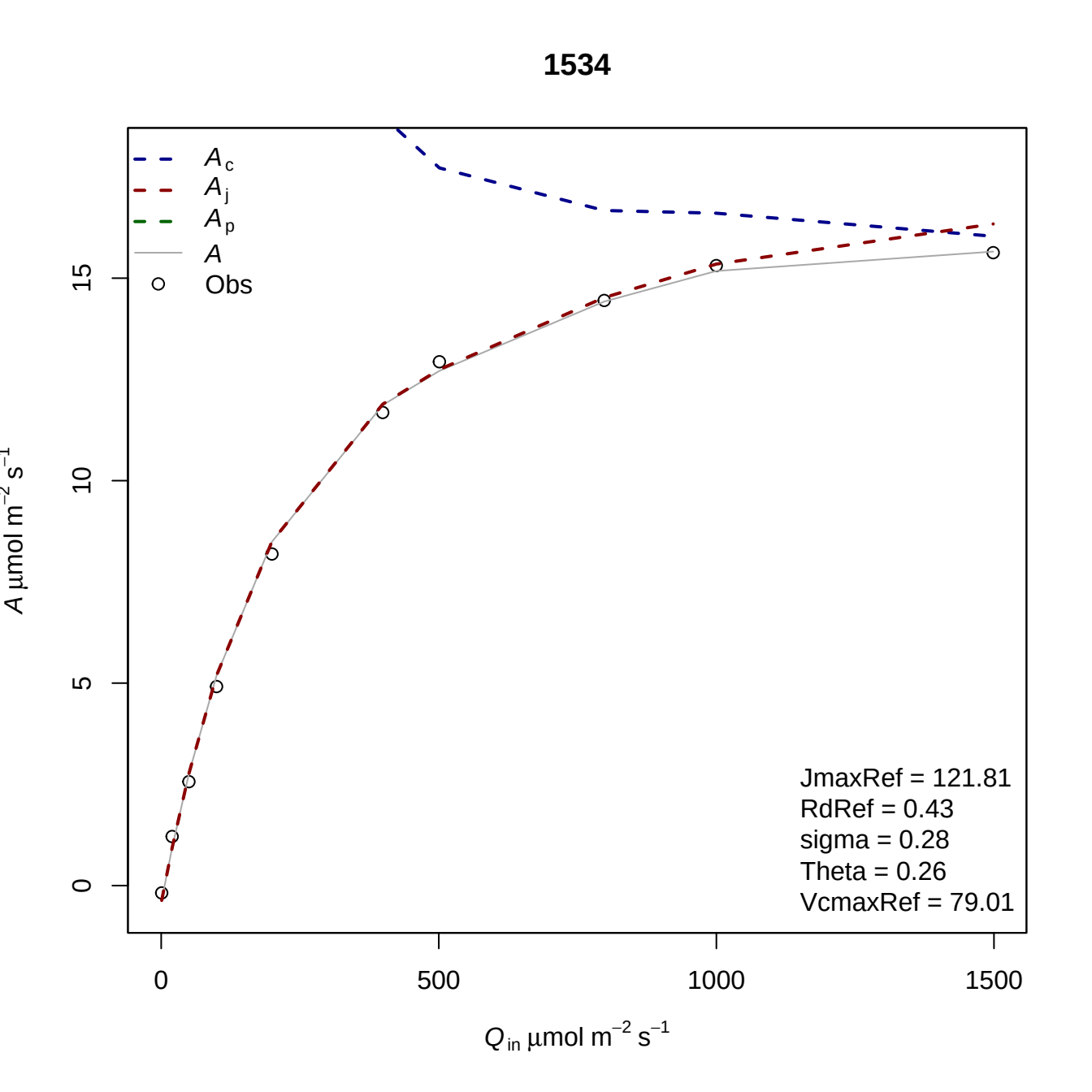

1529
A_c
A_j
A_p
A
Obs
JmaxRef = 63.72
RdRef = 0.89
sigma = 0.36
Theta = 0.33
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

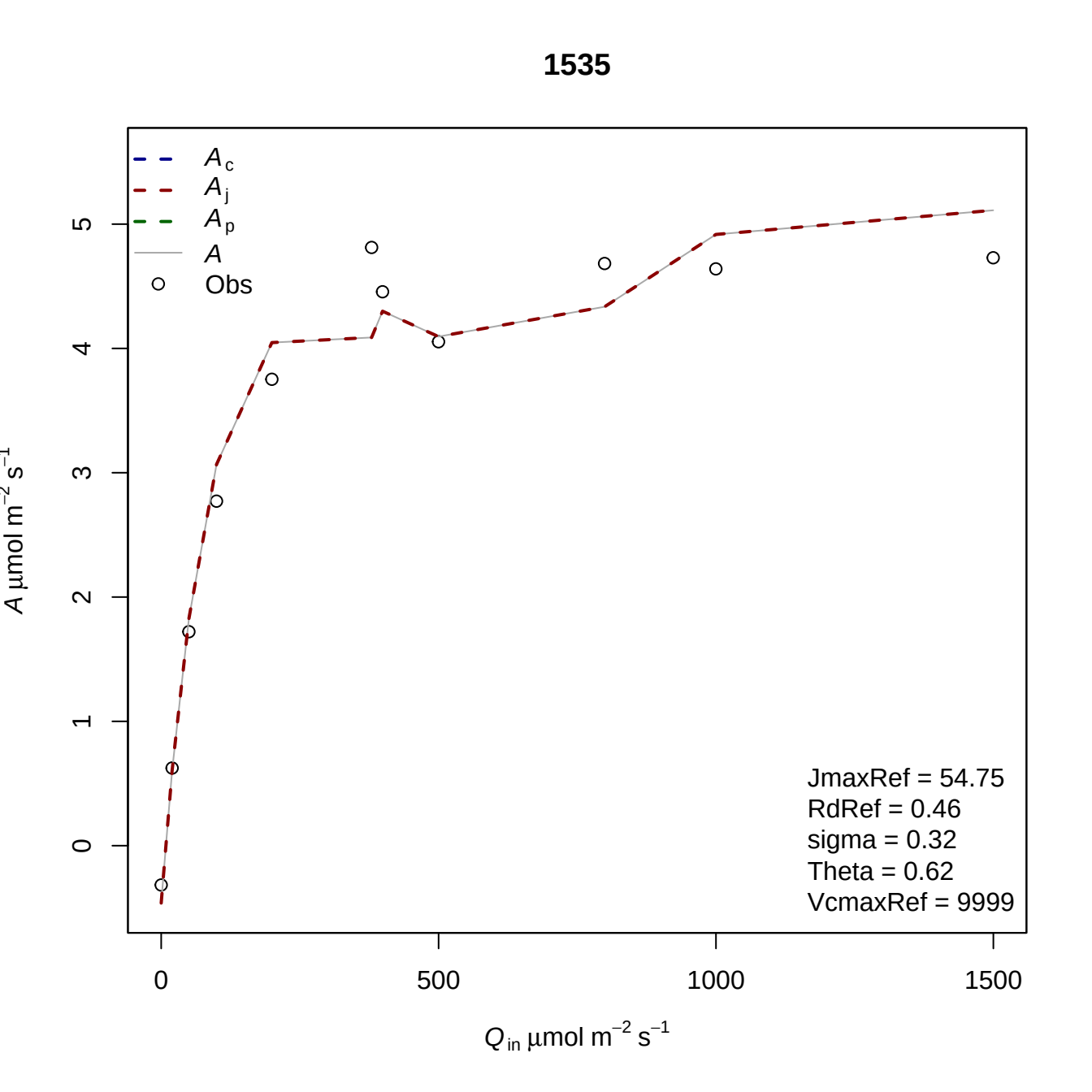

1530
A_c
A_j
A_p
A
Obs
JmaxRef = 84.81
RdRef = 0.61
sigma = 0.32
Theta = 0.22
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

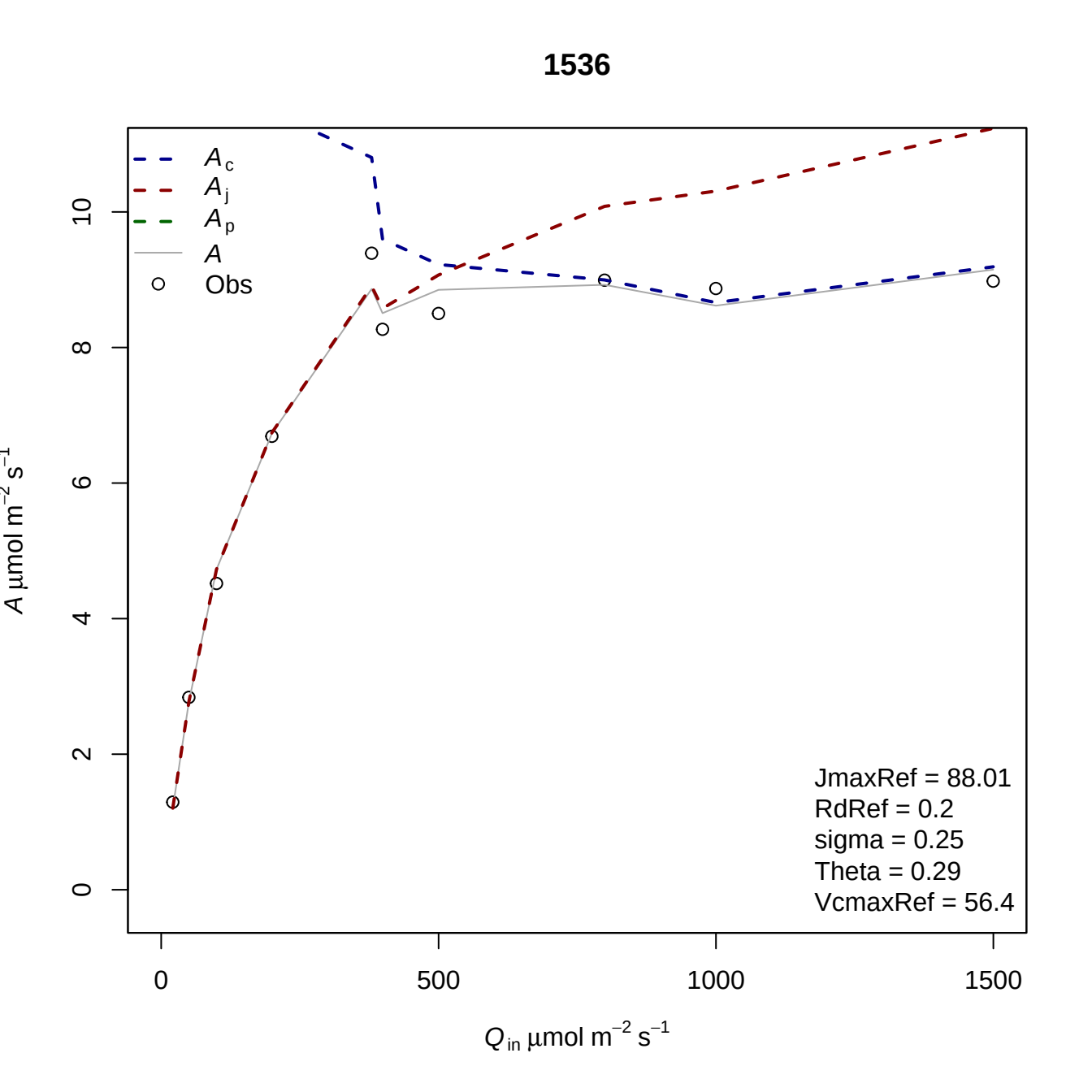

1531
A_c
A_j
A_p
A
Obs
JmaxRef = 79
RdRef = 0.38
sigma = 0.16
Theta = −0.12
VcmaxRef = 57.97
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

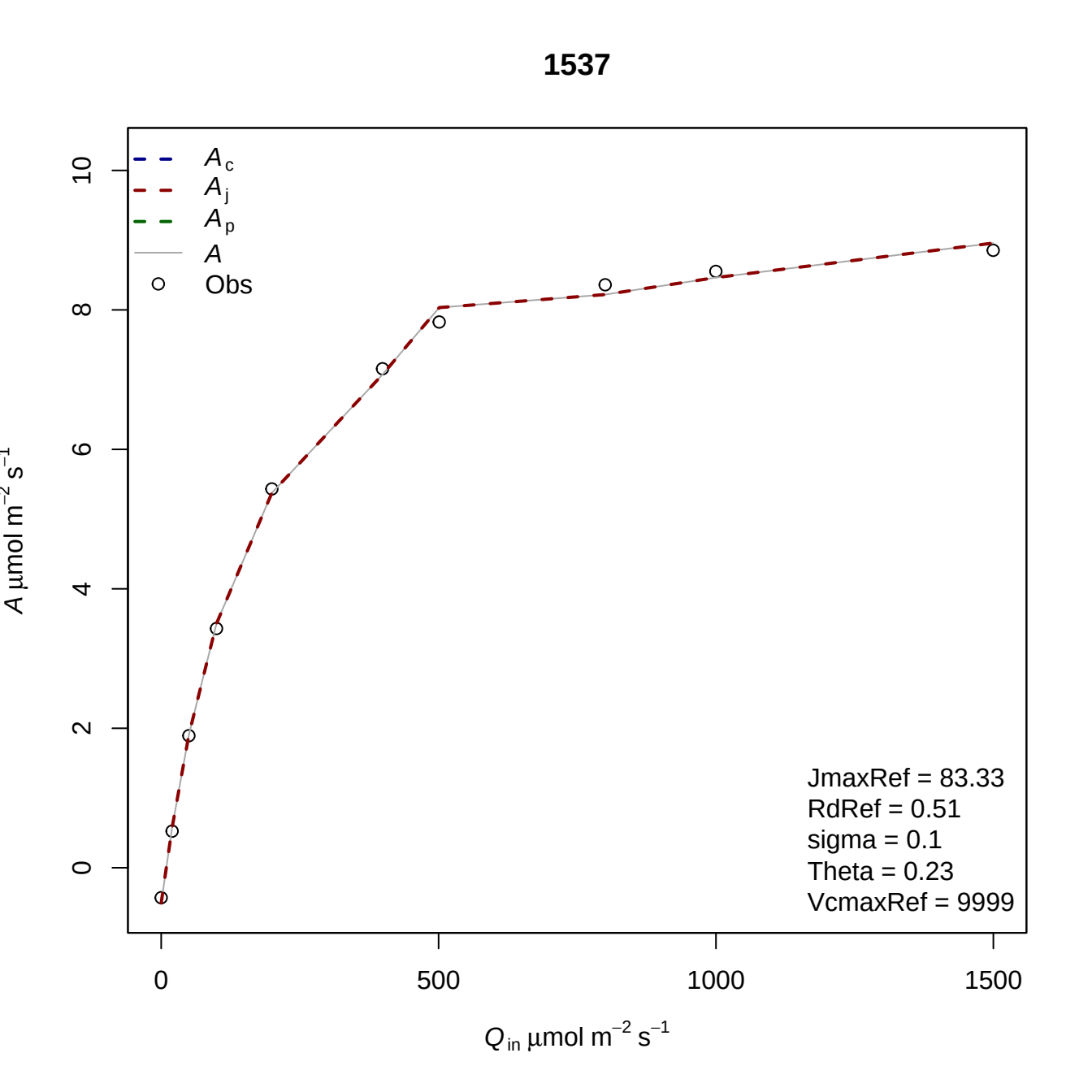

1532
A_c
A_j
A_p
A
Obs
JmaxRef = 176.99
RdRef = 0.14
sigma = 0.23
Theta = −2.56
VcmaxRef = 50.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

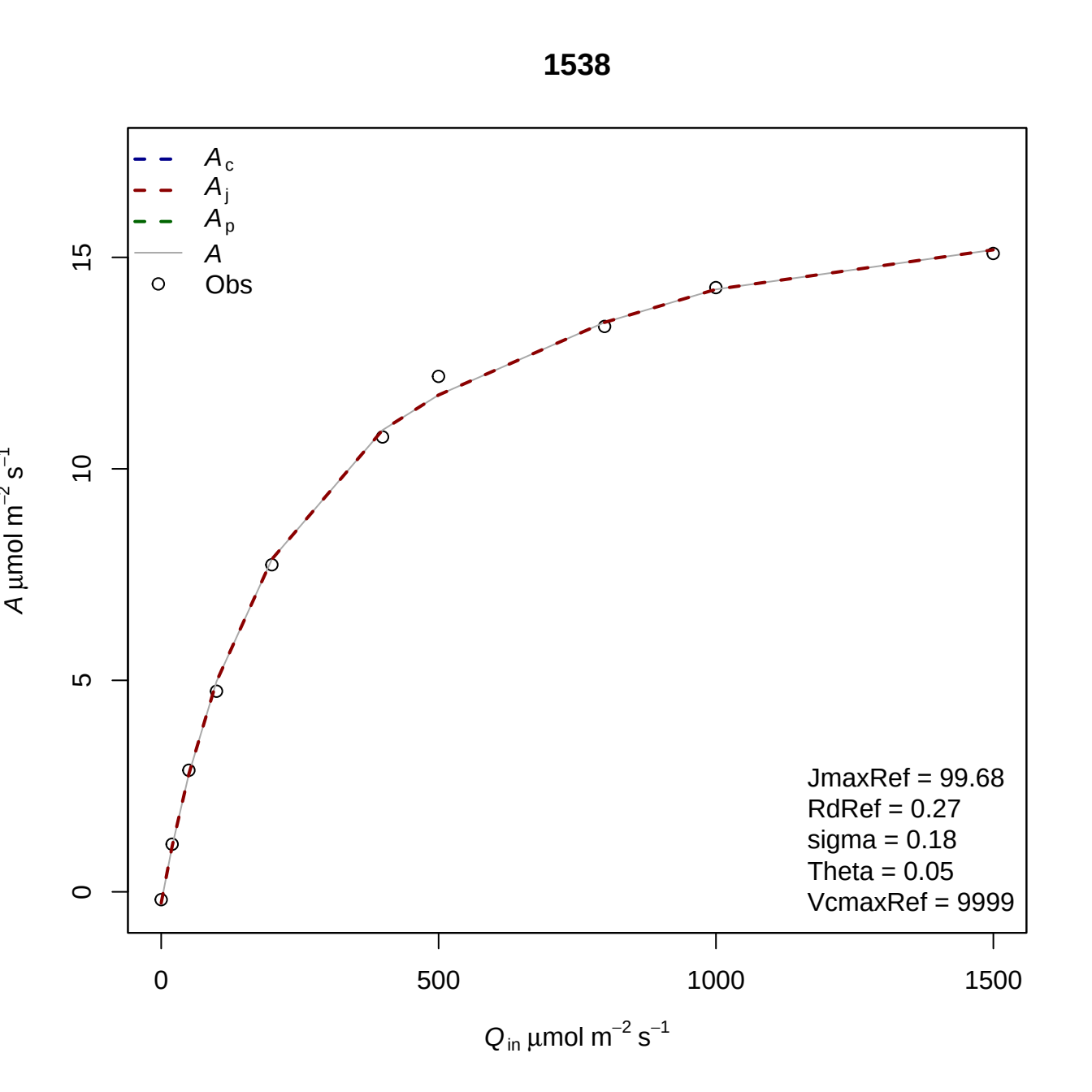

1533
A_c
A_j
A_p
A
Obs
JmaxRef = 97.2
RdRef = 0.48
sigma = 0.12
Theta = 0.08
VcmaxRef = 75.78
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

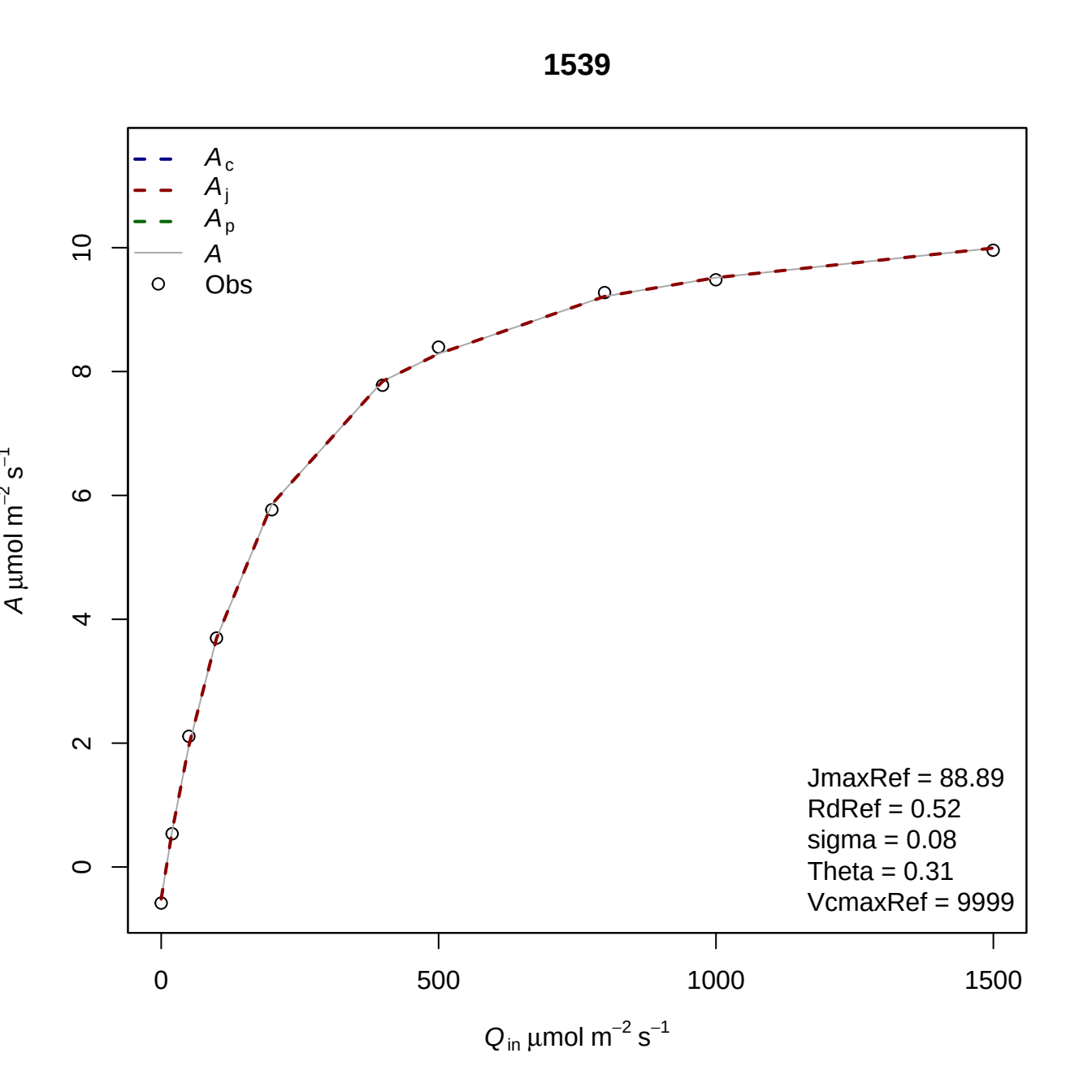

1534
A_c
A_j
A_p
A
Obs
JmaxRef = 121.81
RdRef = 0.43
sigma = 0.28
Theta = 0.26
VcmaxRef = 79.01
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

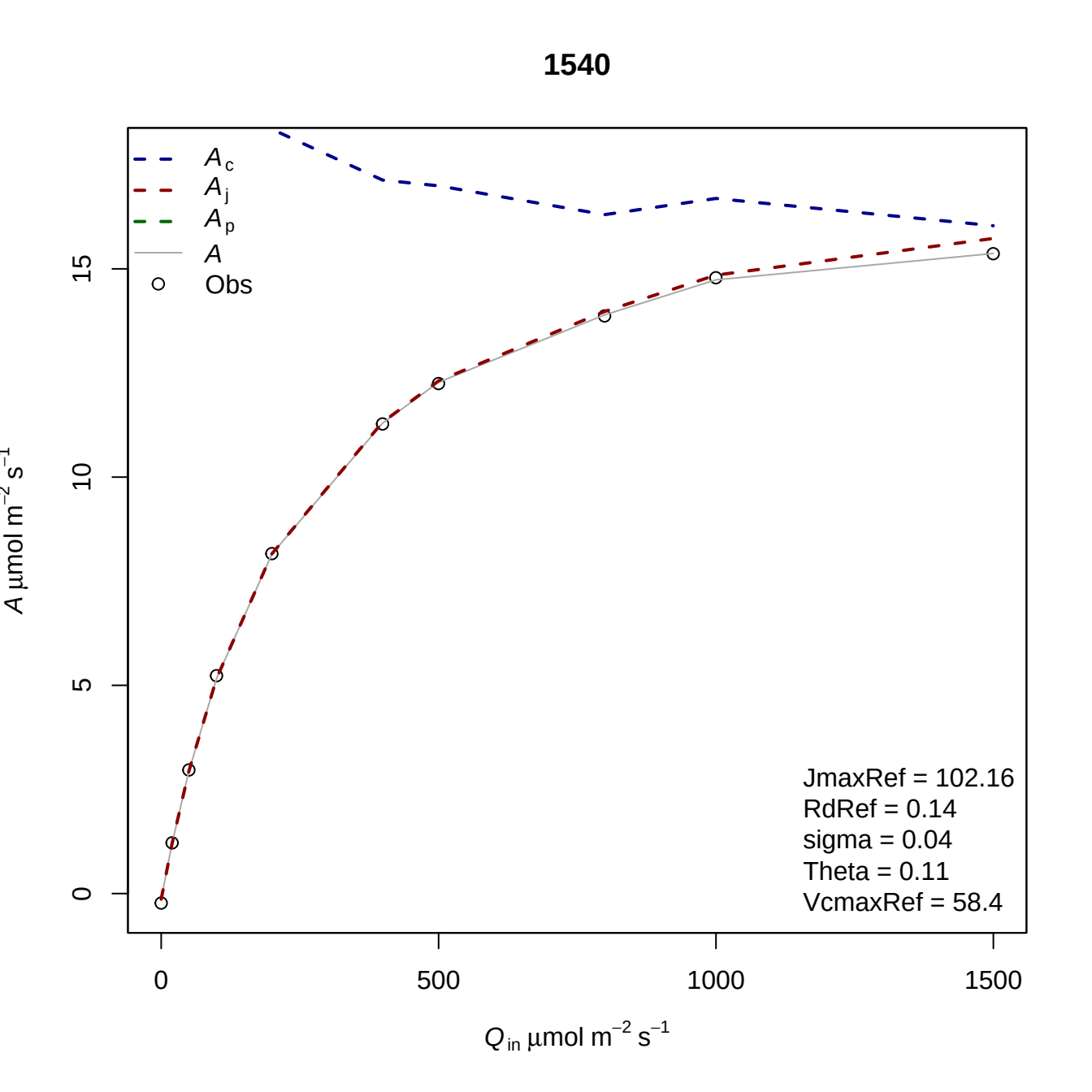

1535
A_c
A_j
A_p
A
Obs
JmaxRef = 54.75
RdRef = 0.46
sigma = 0.32
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

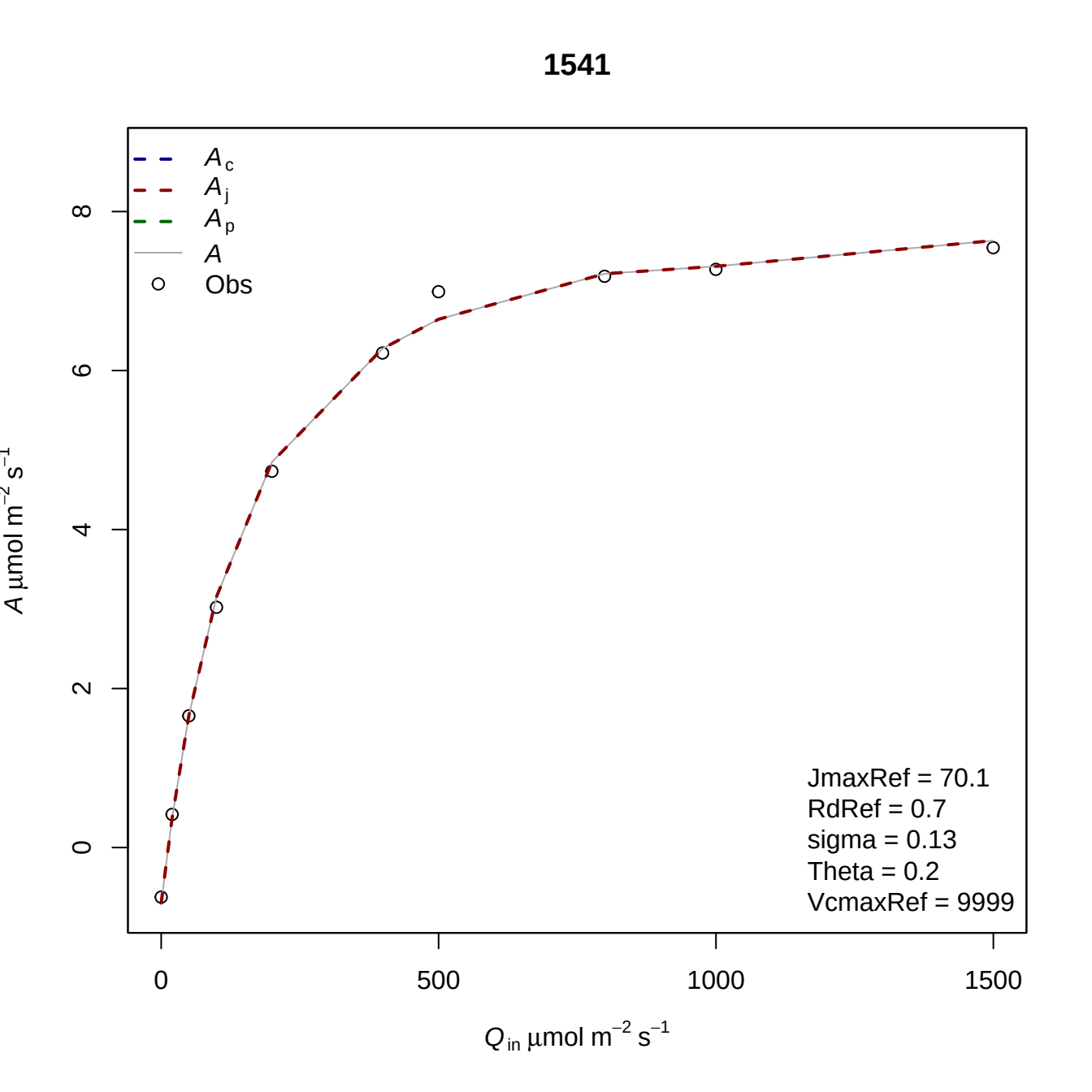

1536
A_c
A_j
A_p
A
Obs
JmaxRef = 88.01
RdRef = 0.2
sigma = 0.25
Theta = 0.29
VcmaxRef = 56.4
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

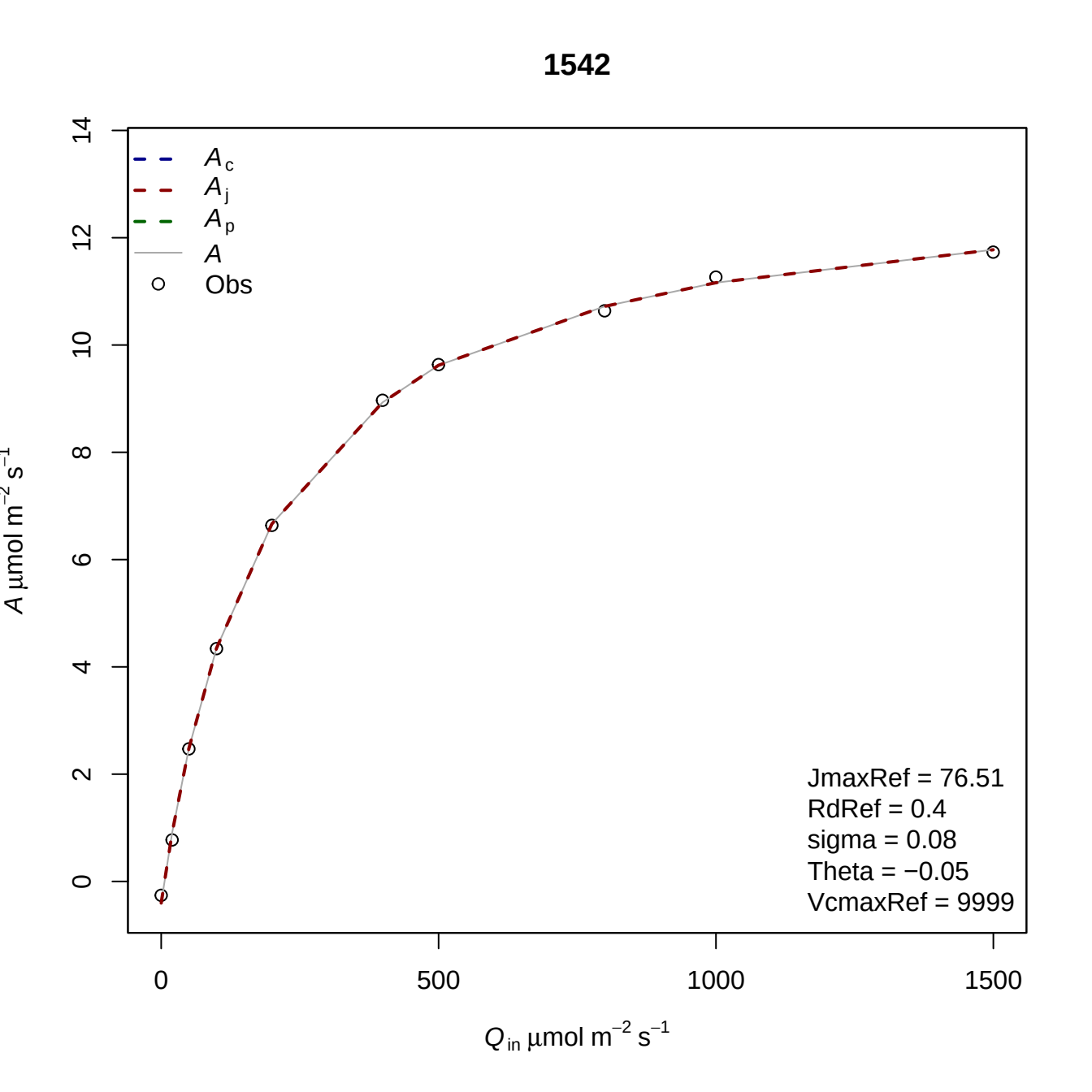

1537
A_c
A_j
A_p
A
Obs
JmaxRef = 83.33
RdRef = 0.51
sigma = 0.1
Theta = 0.23
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

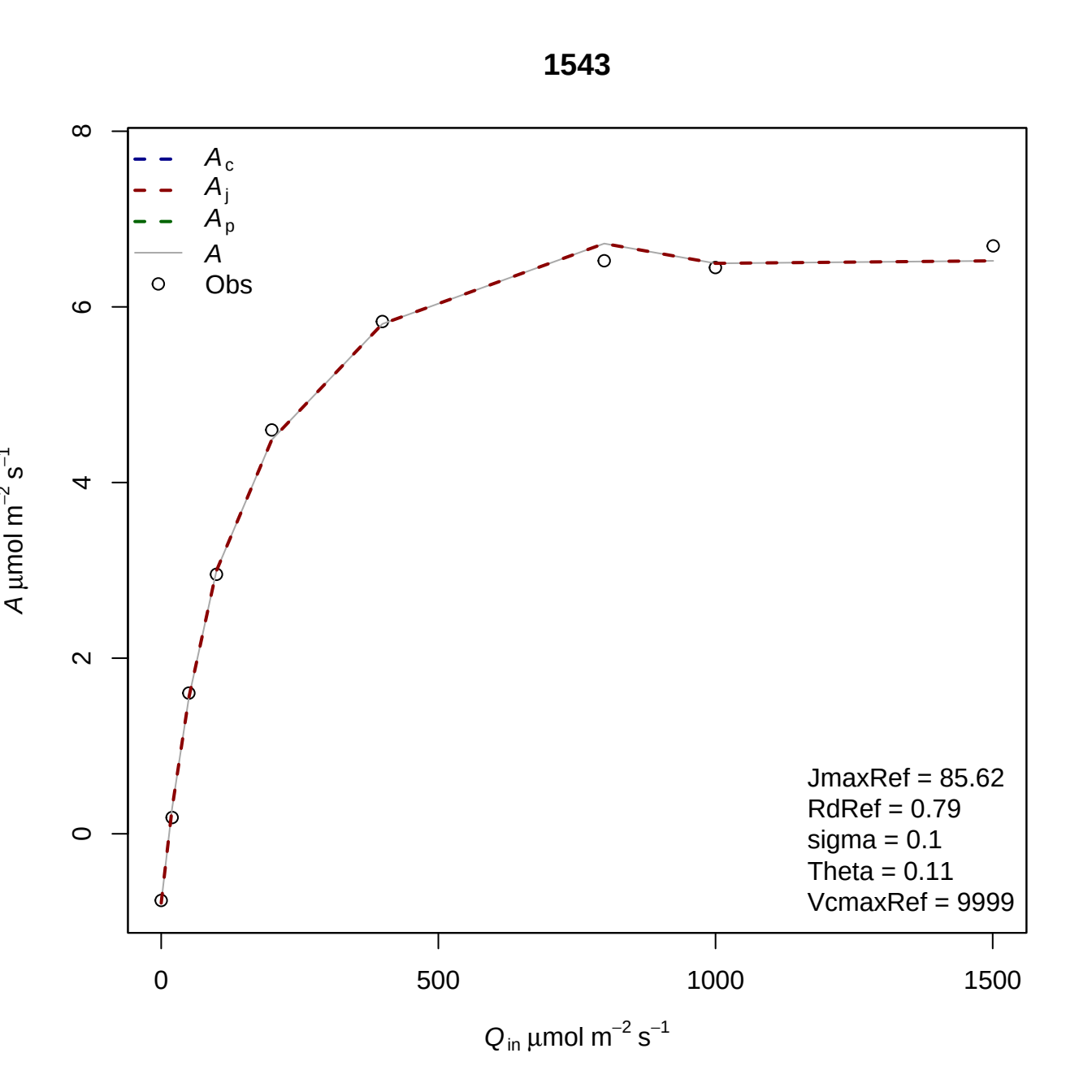

1538
A_c
A_j
A_p
A
Obs
JmaxRef = 99.68
RdRef = 0.27
sigma = 0.18
Theta = 0.05
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

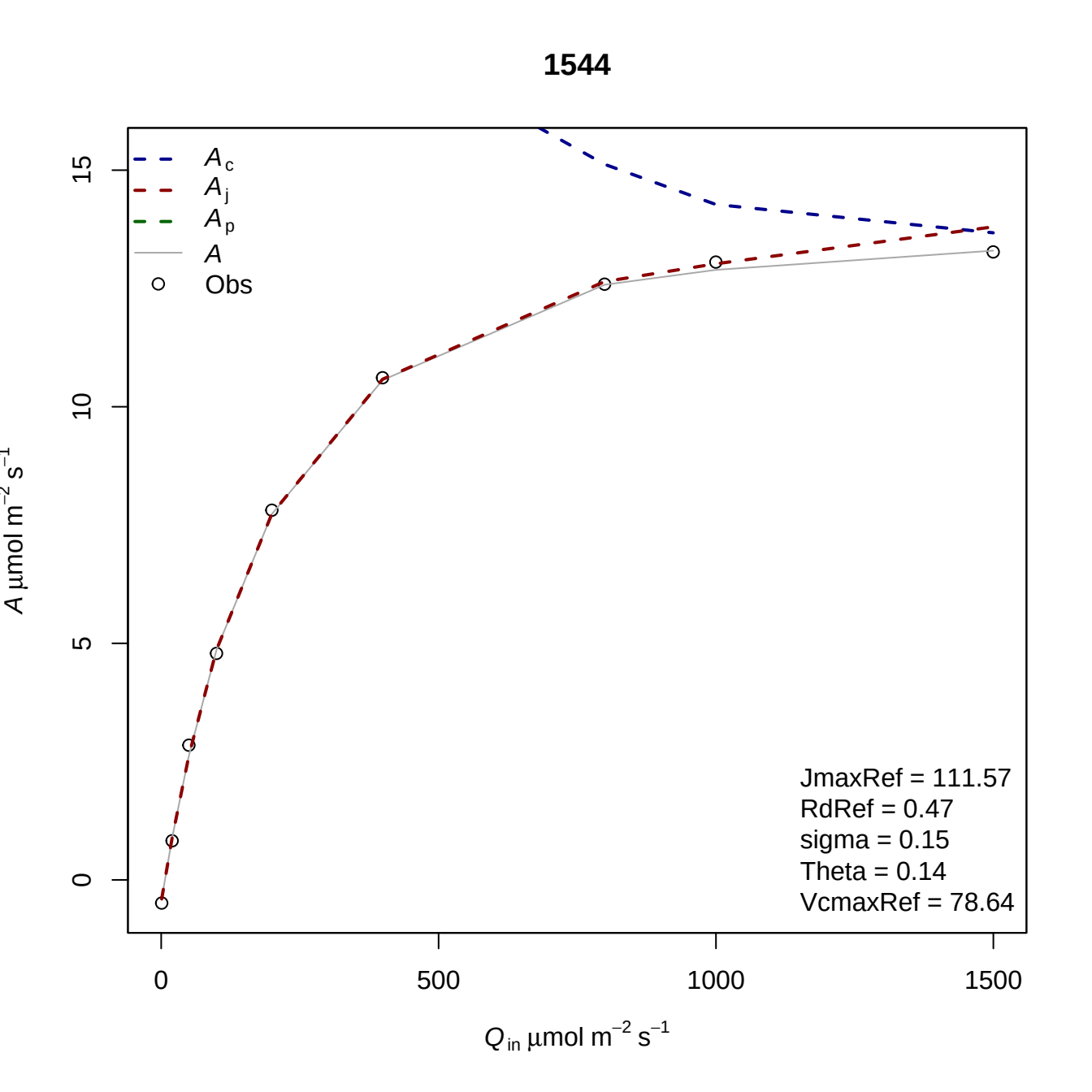

1539
A_c
A_j
A_p
A
Obs
JmaxRef = 88.89
RdRef = 0.52
sigma = 0.08
Theta = 0.31
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

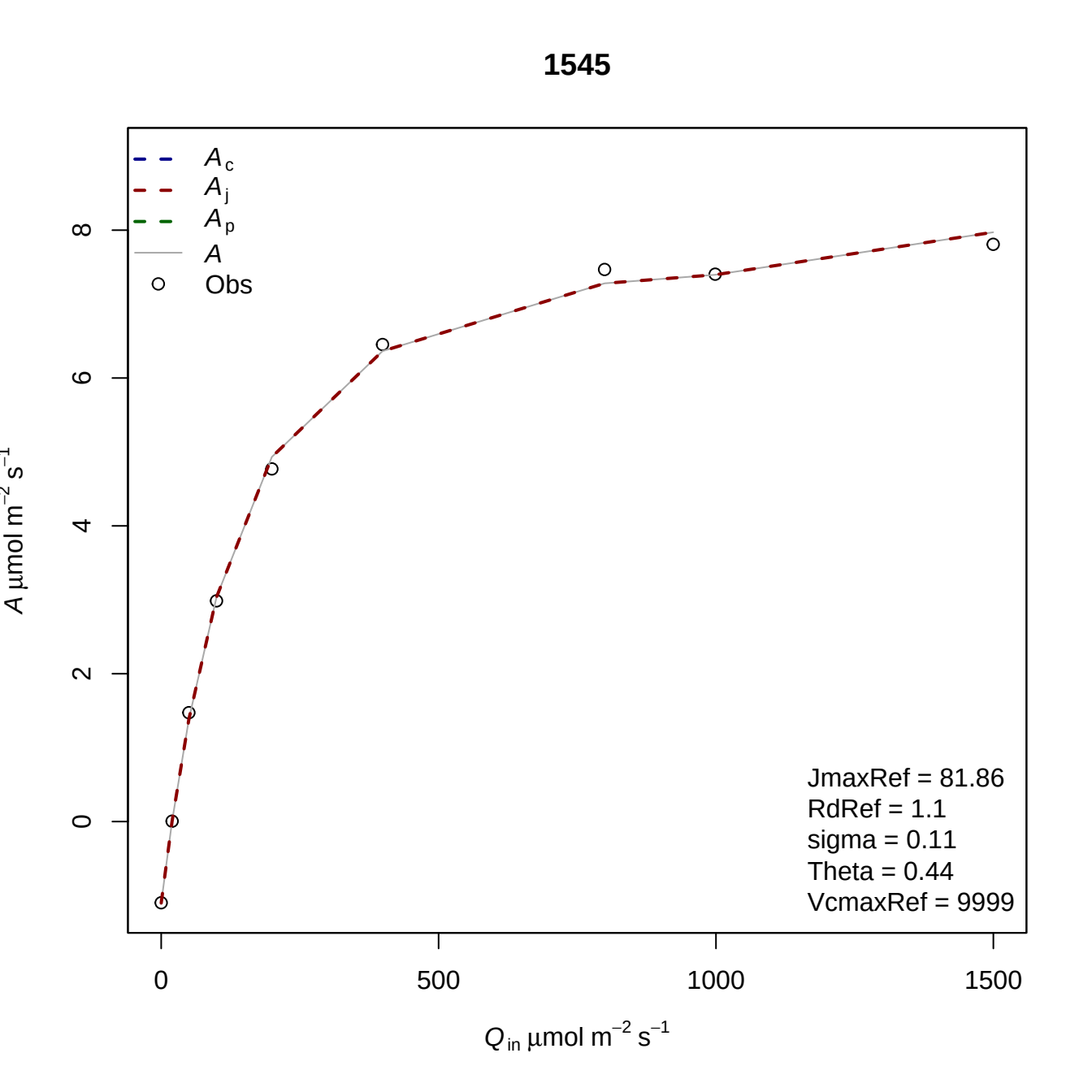

1540
A_c
A_j
A_p
A
Obs
JmaxRef = 102.16
RdRef = 0.14
sigma = 0.04
Theta = 0.11
VcmaxRef = 58.4
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

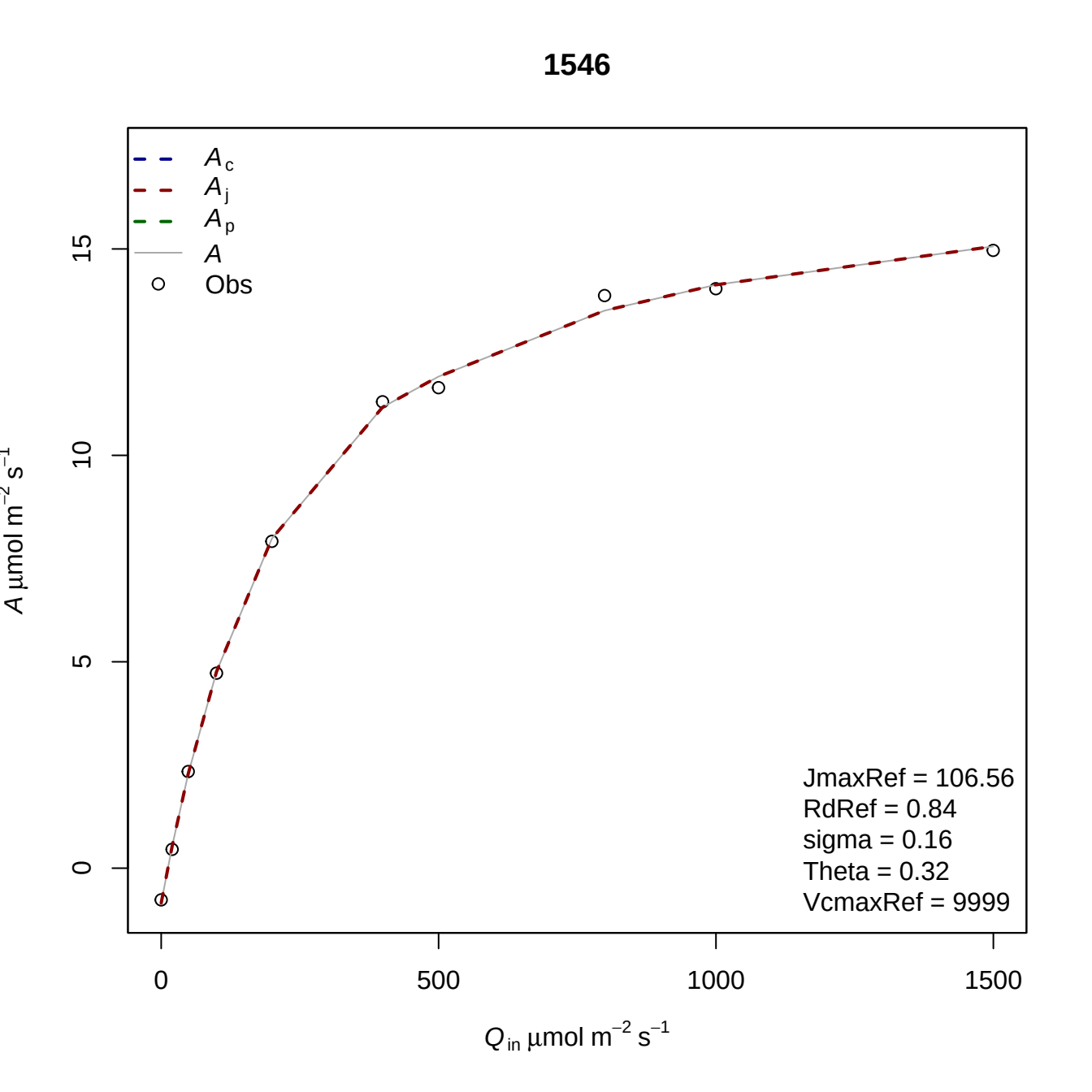

1541
A_c
A_j
A_p
A
Obs
JmaxRef = 70.1
RdRef = 0.7
sigma = 0.13
Theta = 0.2
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

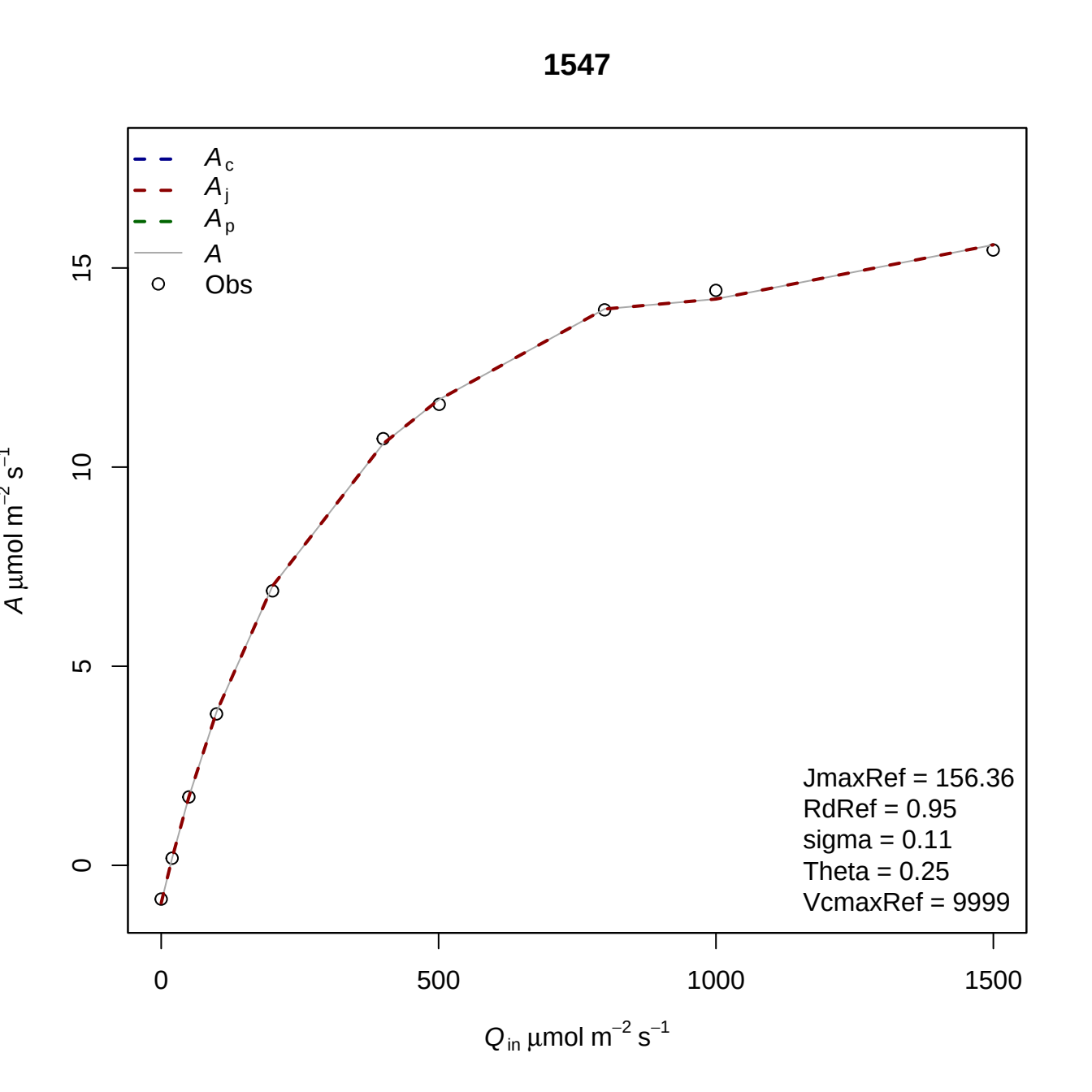

1542
A_c
A_j
A_p
A
Obs
JmaxRef = 76.51
RdRef = 0.4
sigma = 0.08
Theta = −0.05
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

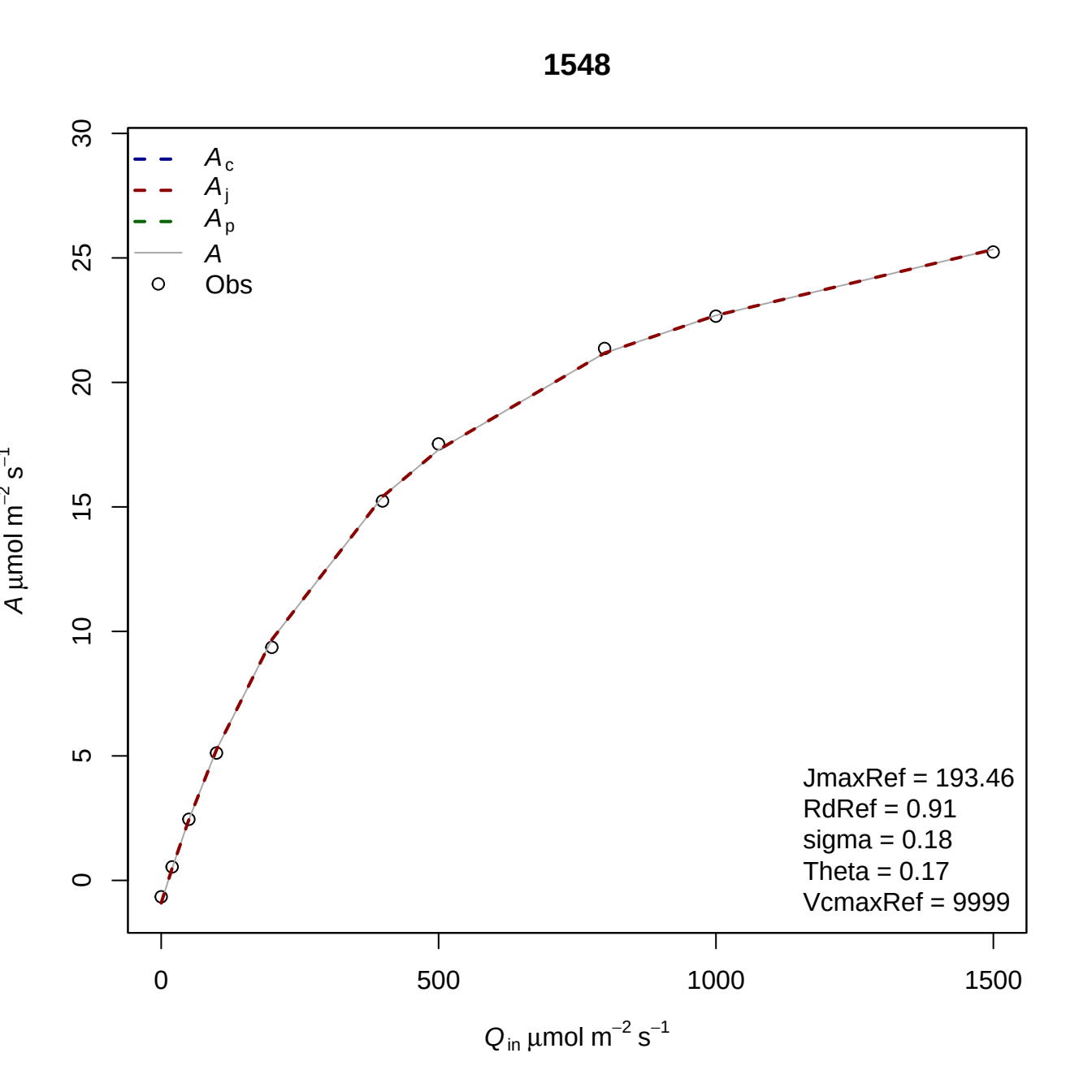

1543
A_c
A_j
A_p
A
Obs
JmaxRef = 85.62
RdRef = 0.79
sigma = 0.1
Theta = 0.11
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

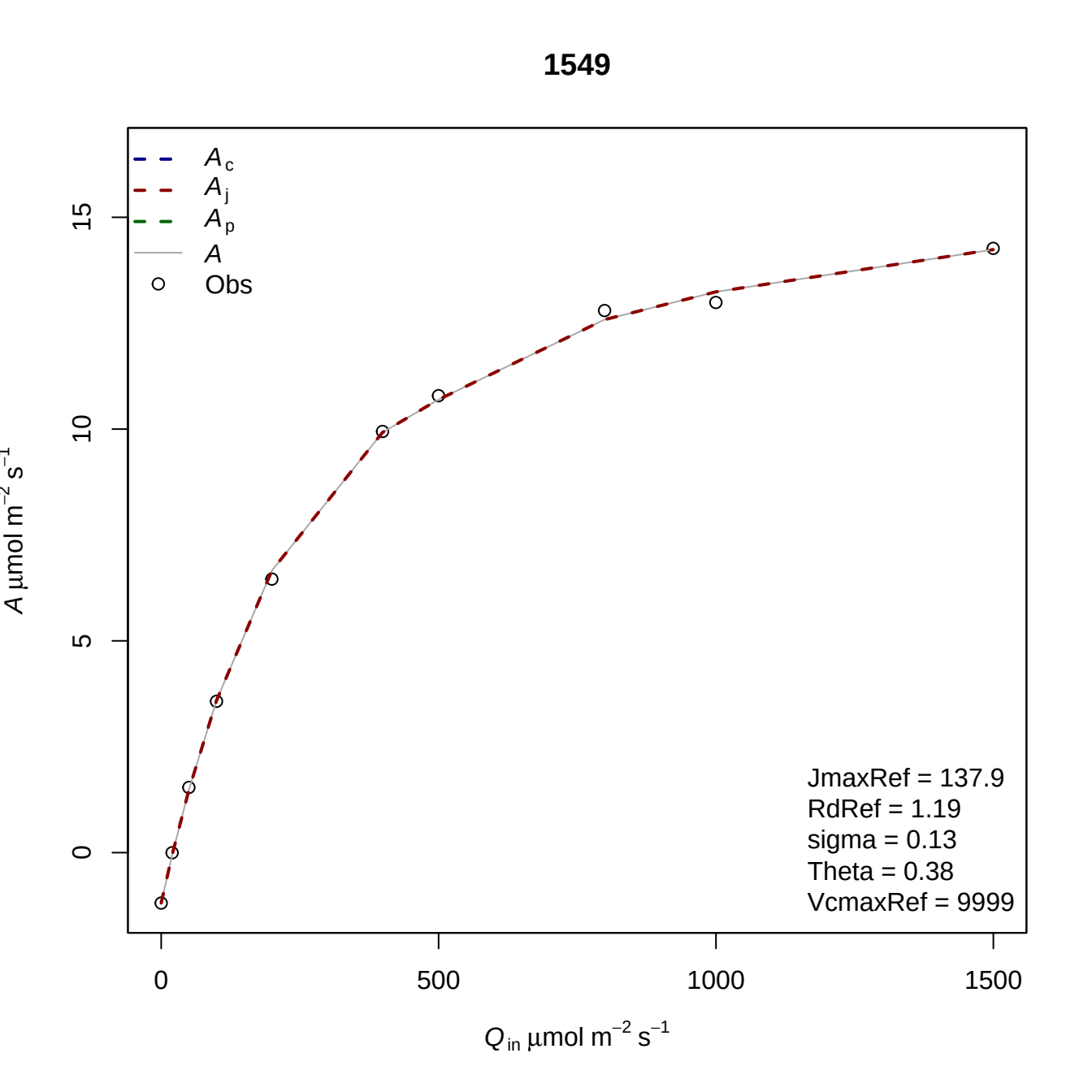

1544
A_c
A_j
A_p
A
Obs
JmaxRef = 111.57
RdRef = 0.47
sigma = 0.15
Theta = 0.14
VcmaxRef = 78.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

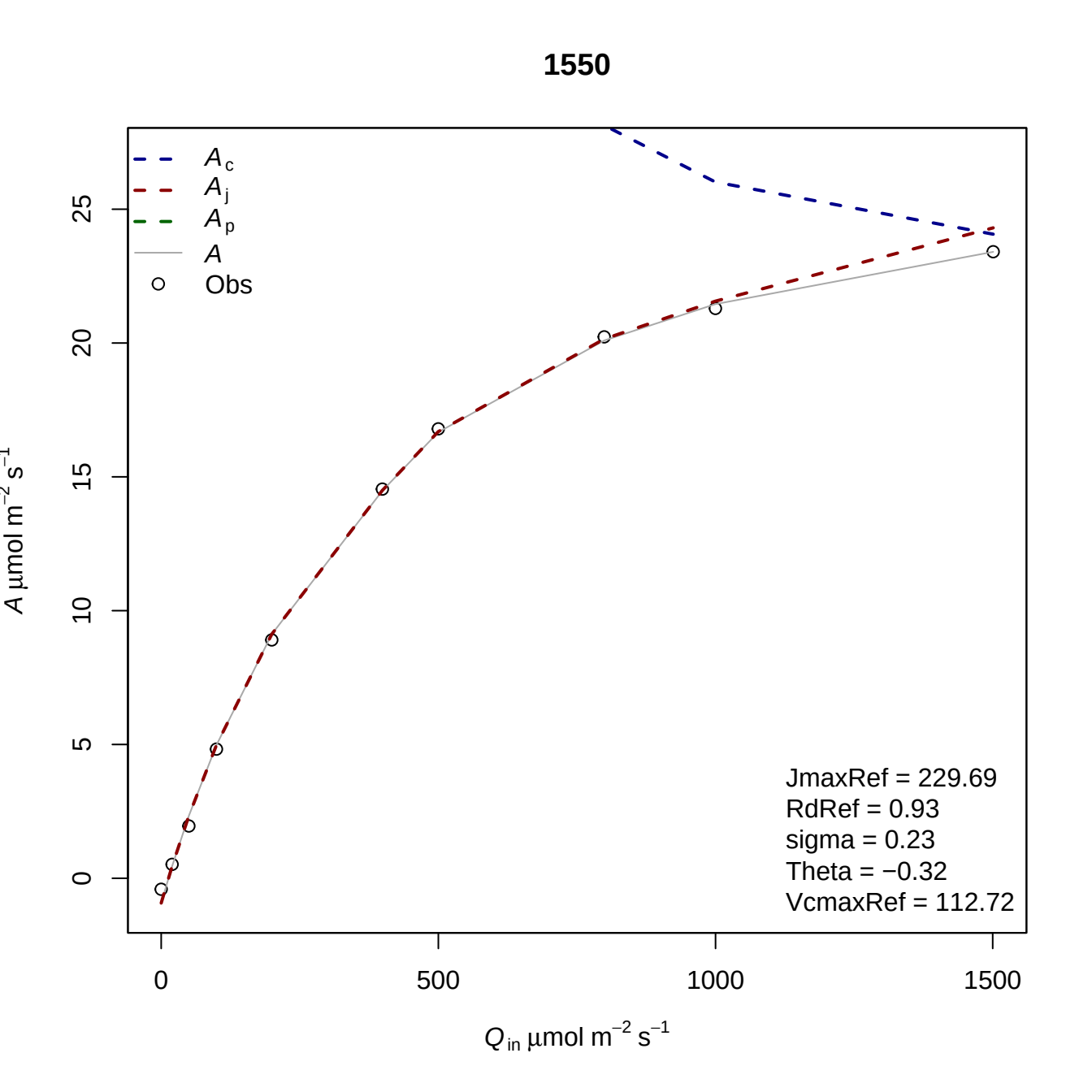

1545
A_c
A_j
A_p
A
Obs
JmaxRef = 81.86
RdRef = 1.1
sigma = 0.11
Theta = 0.44
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

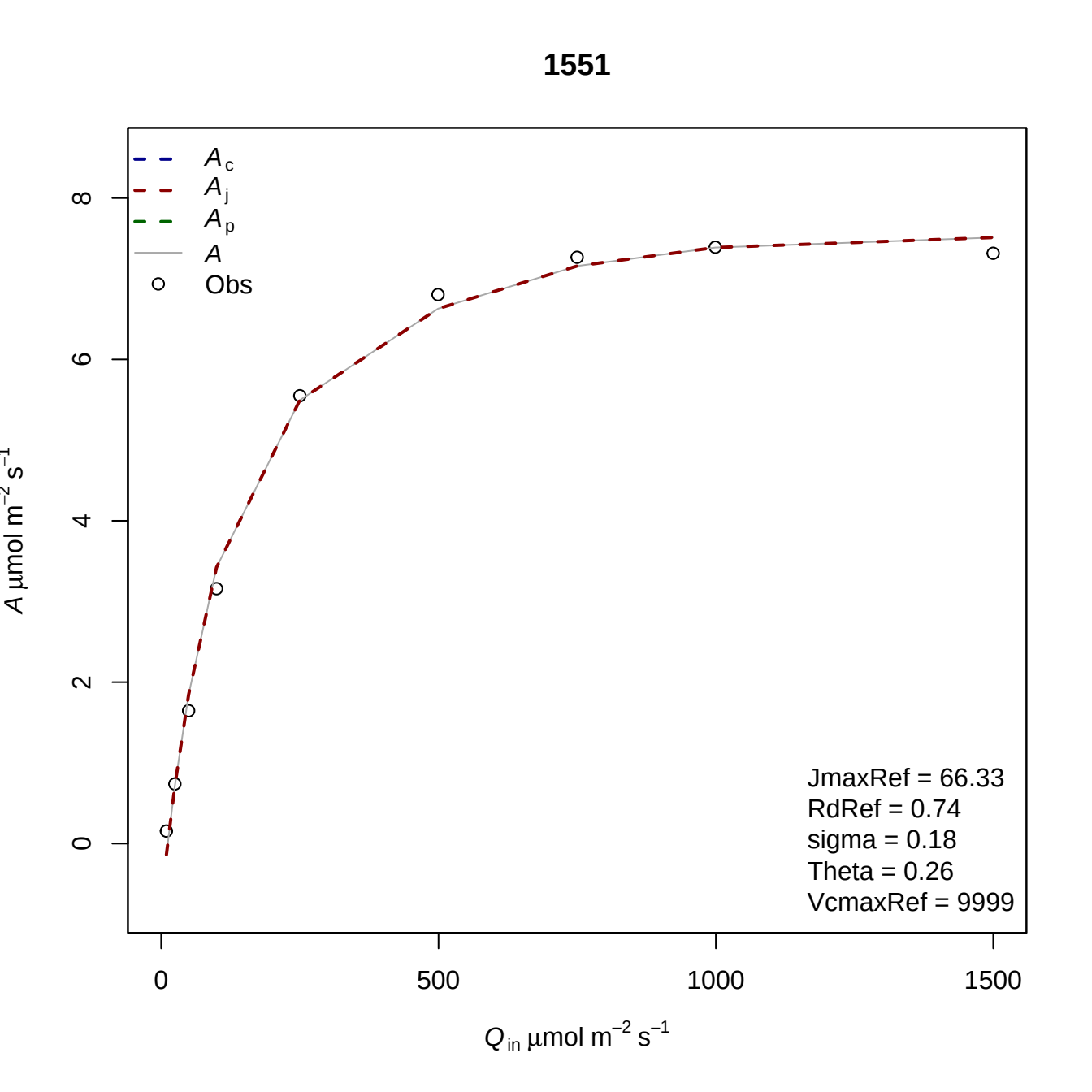

1546
A_c
A_j
A_p
A
Obs
JmaxRef = 106.56
RdRef = 0.84
sigma = 0.16
Theta = 0.32
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

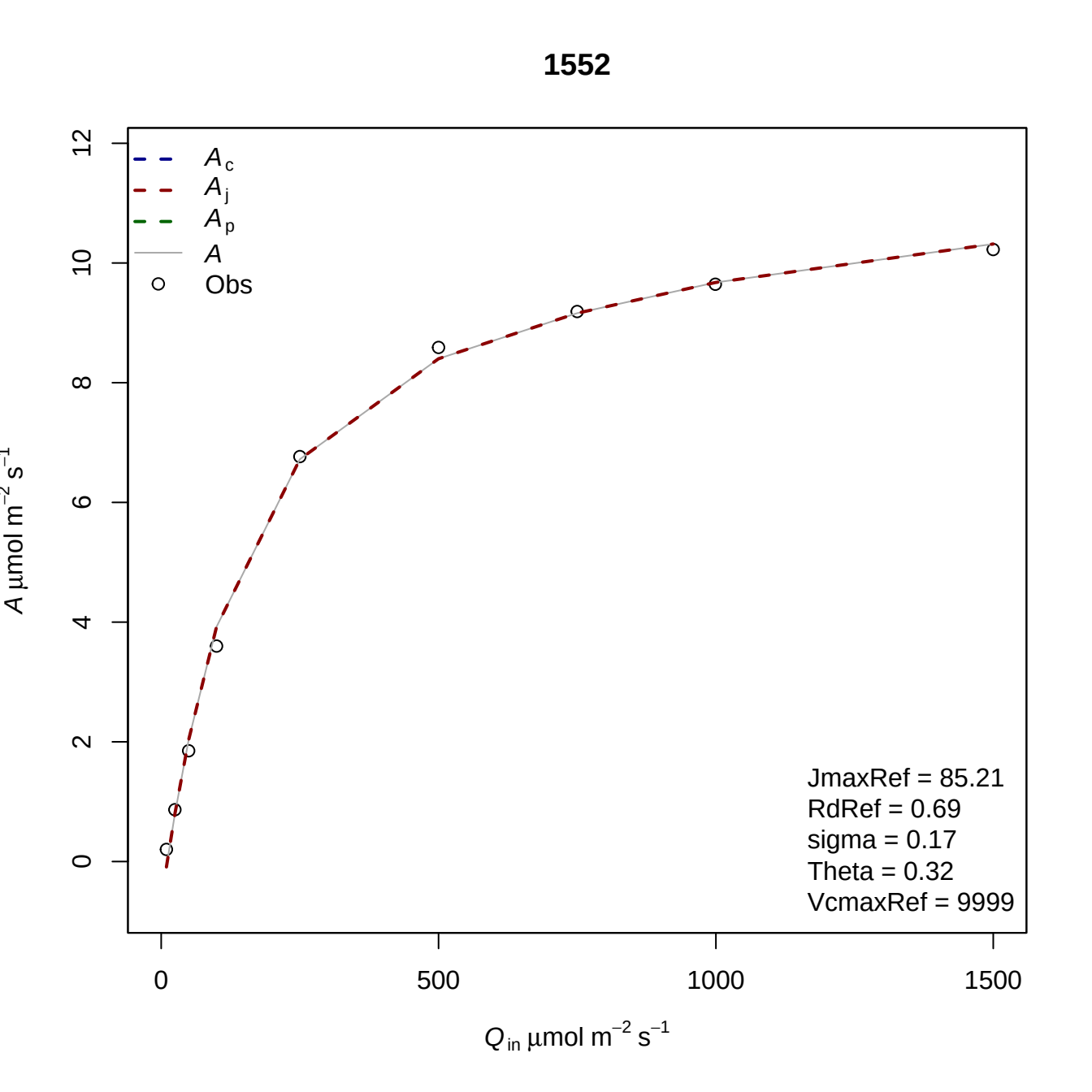

1547
A_c
A_j
A_p
A
Obs
JmaxRef = 156.36
RdRef = 0.95
sigma = 0.11
Theta = 0.25
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

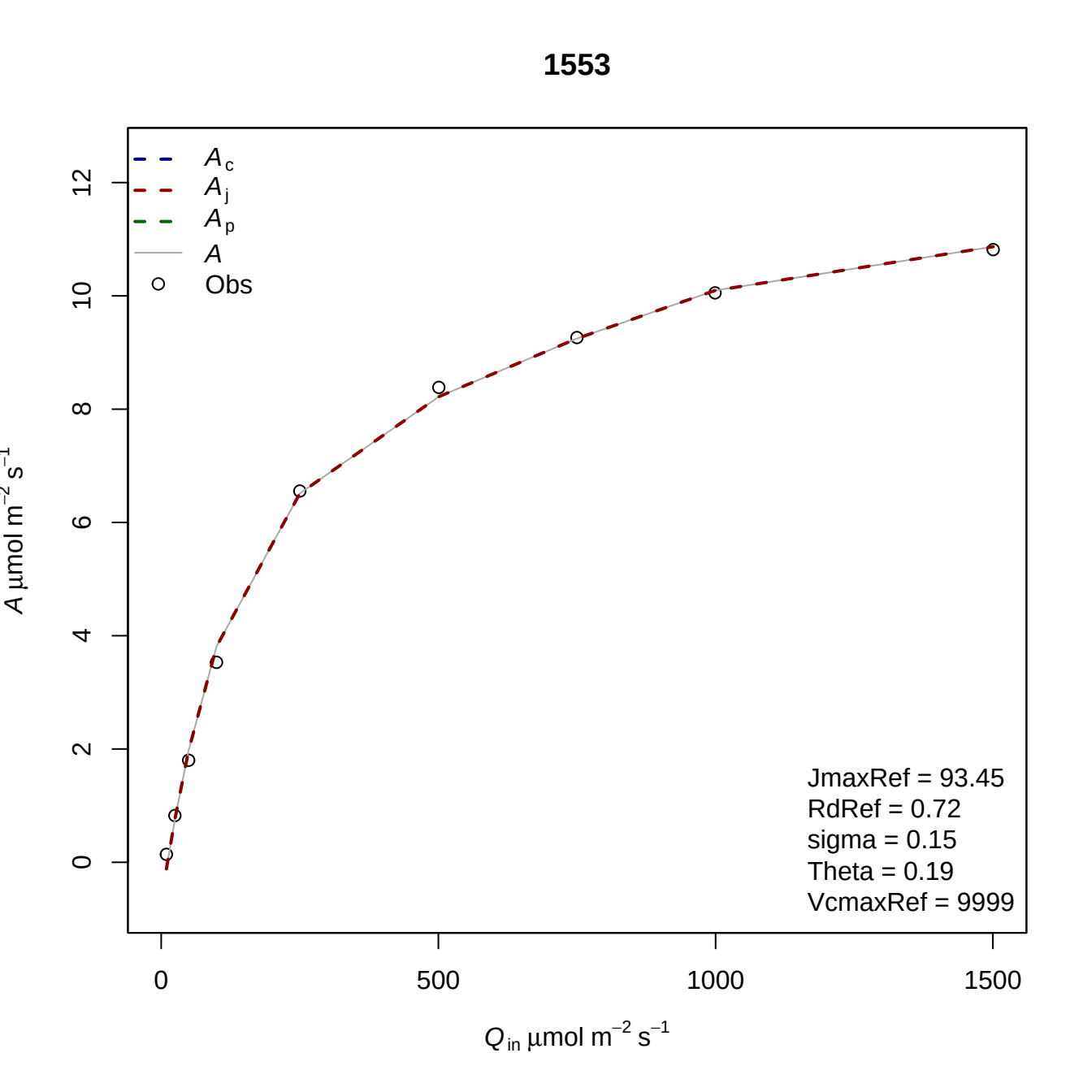

1548
A_c
A_j
A_p
A
Obs
JmaxRef = 193.46
RdRef = 0.91
sigma = 0.18
Theta = 0.17
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

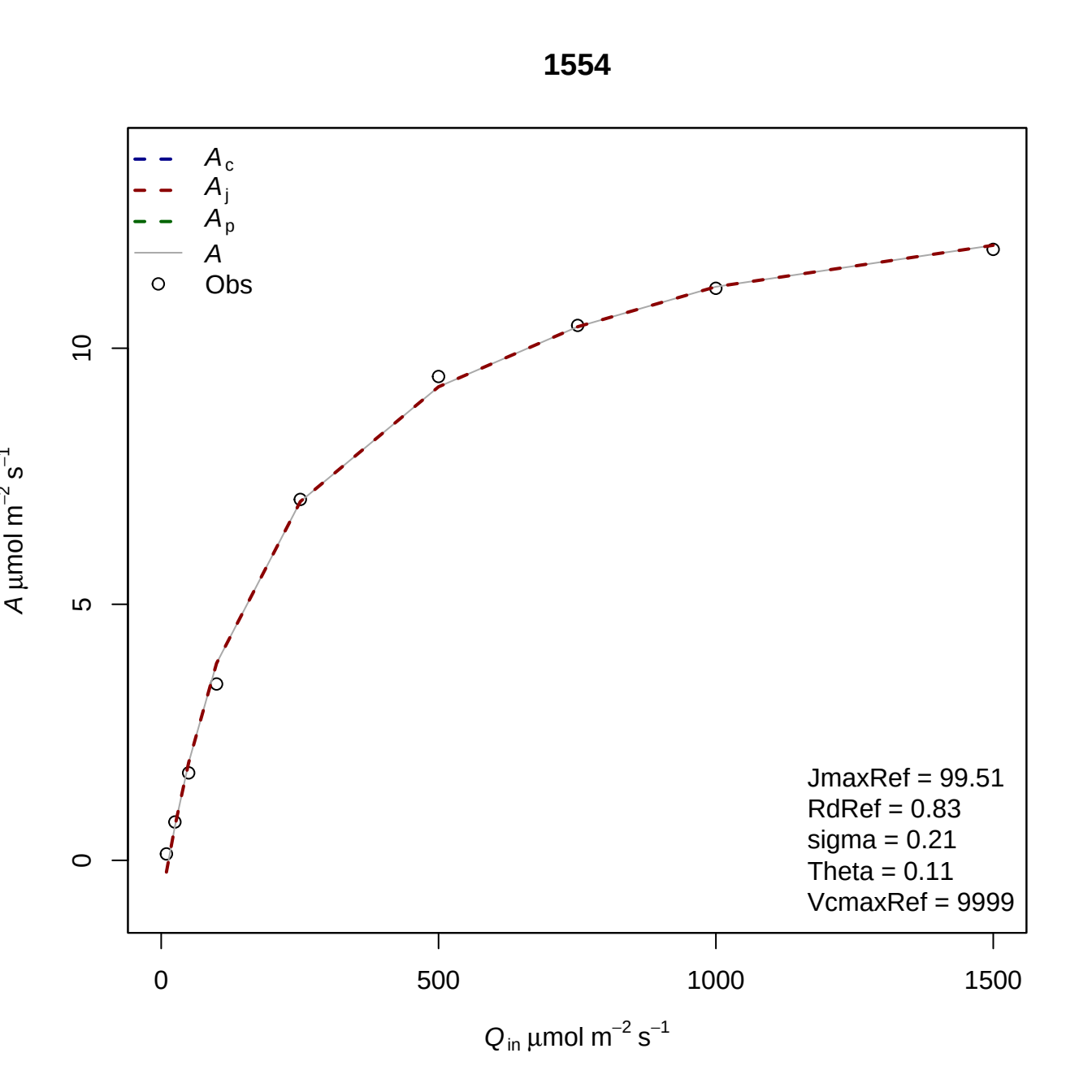

1549
A_c
A_j
A_p
A
Obs
JmaxRef = 137.9
RdRef = 1.19
sigma = 0.13
Theta = 0.38
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

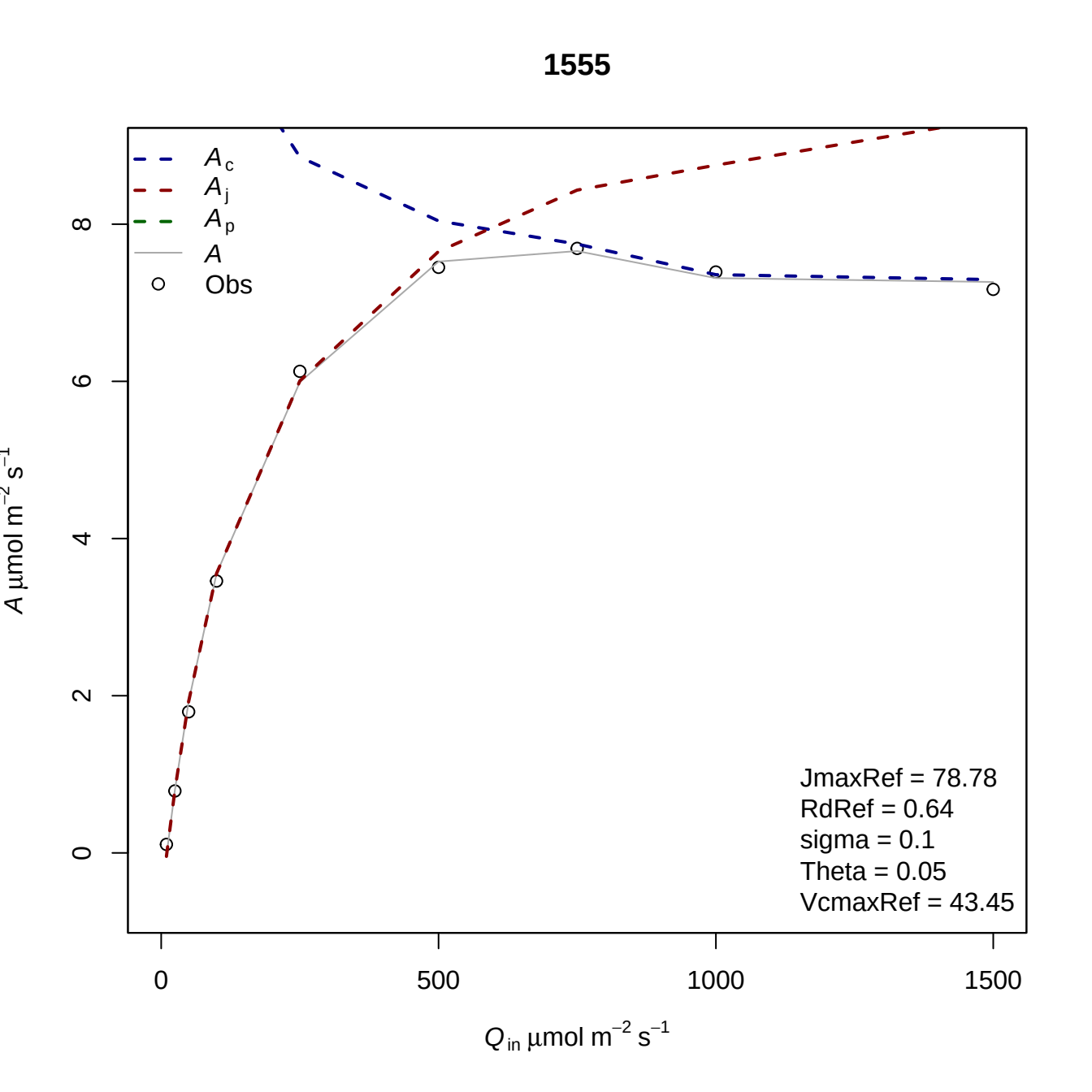

1550
A_c
A_j
A_p
A
Obs
JmaxRef = 229.69
RdRef = 0.93
sigma = 0.23
Theta = −0.32
VcmaxRef = 112.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

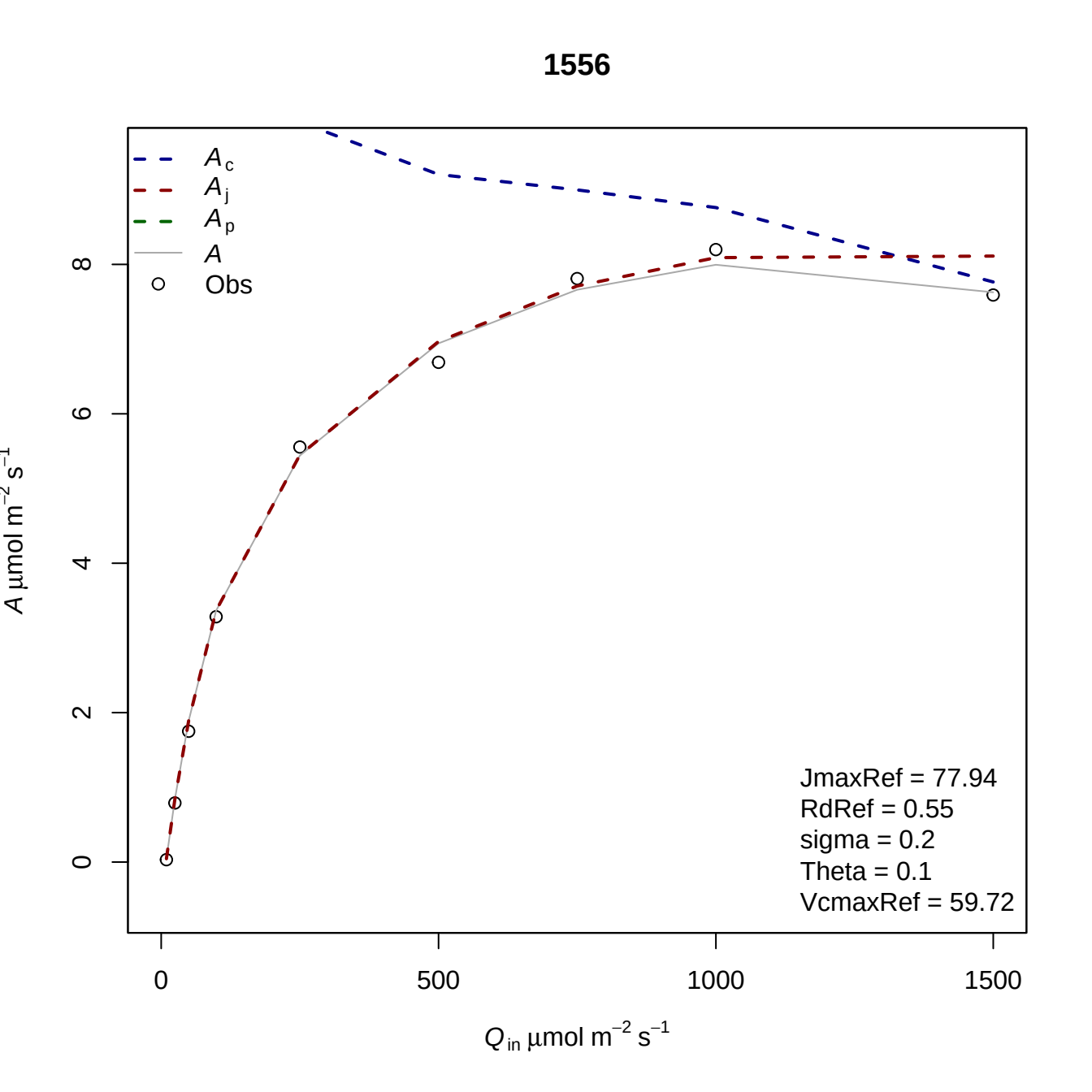

1551
A_c
A_j
A_p
A
Obs
JmaxRef = 66.33
RdRef = 0.74
sigma = 0.18
Theta = 0.26
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

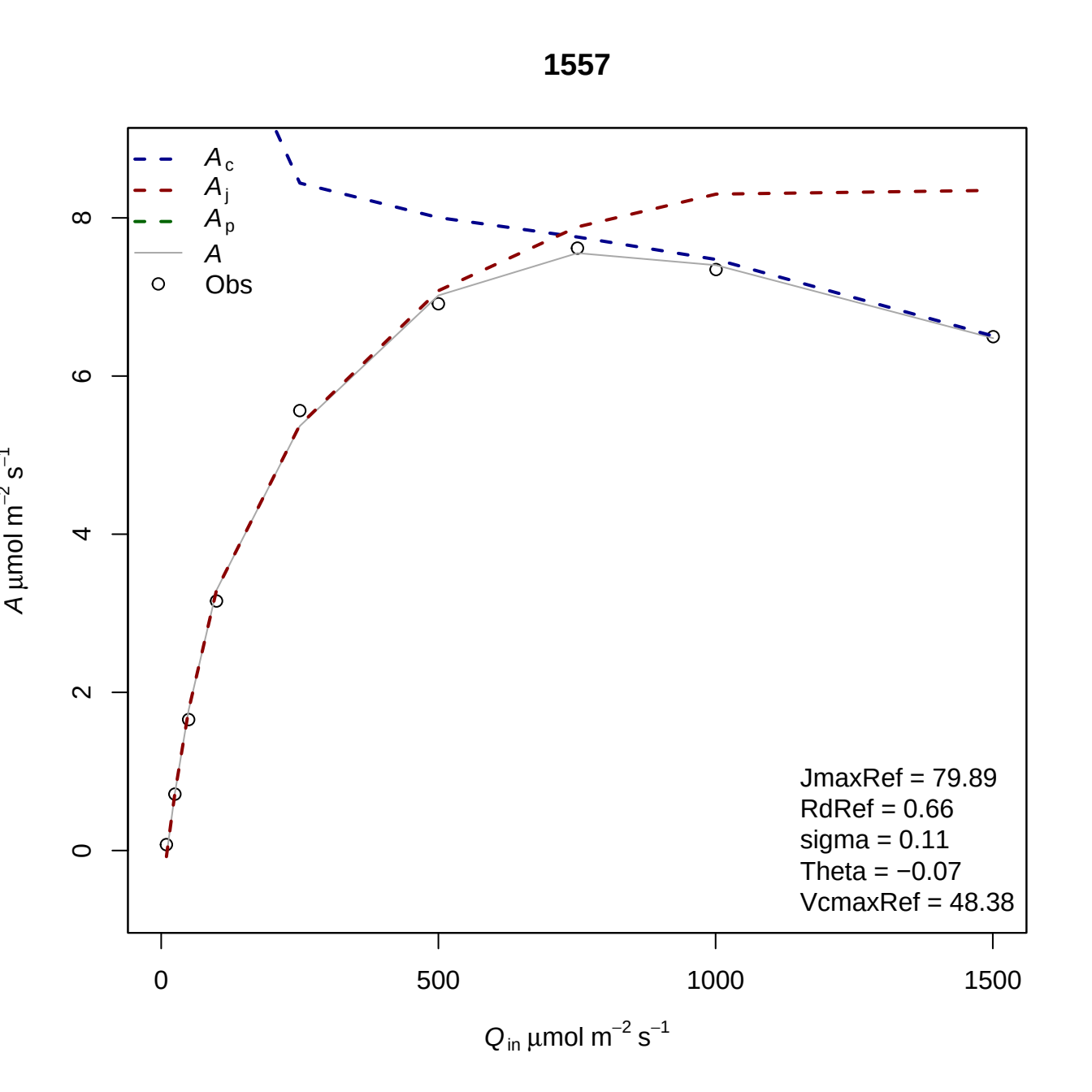

1552
A_c
A_j
A_p
A
Obs
JmaxRef = 85.21
RdRef = 0.69
sigma = 0.17
Theta = 0.32
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

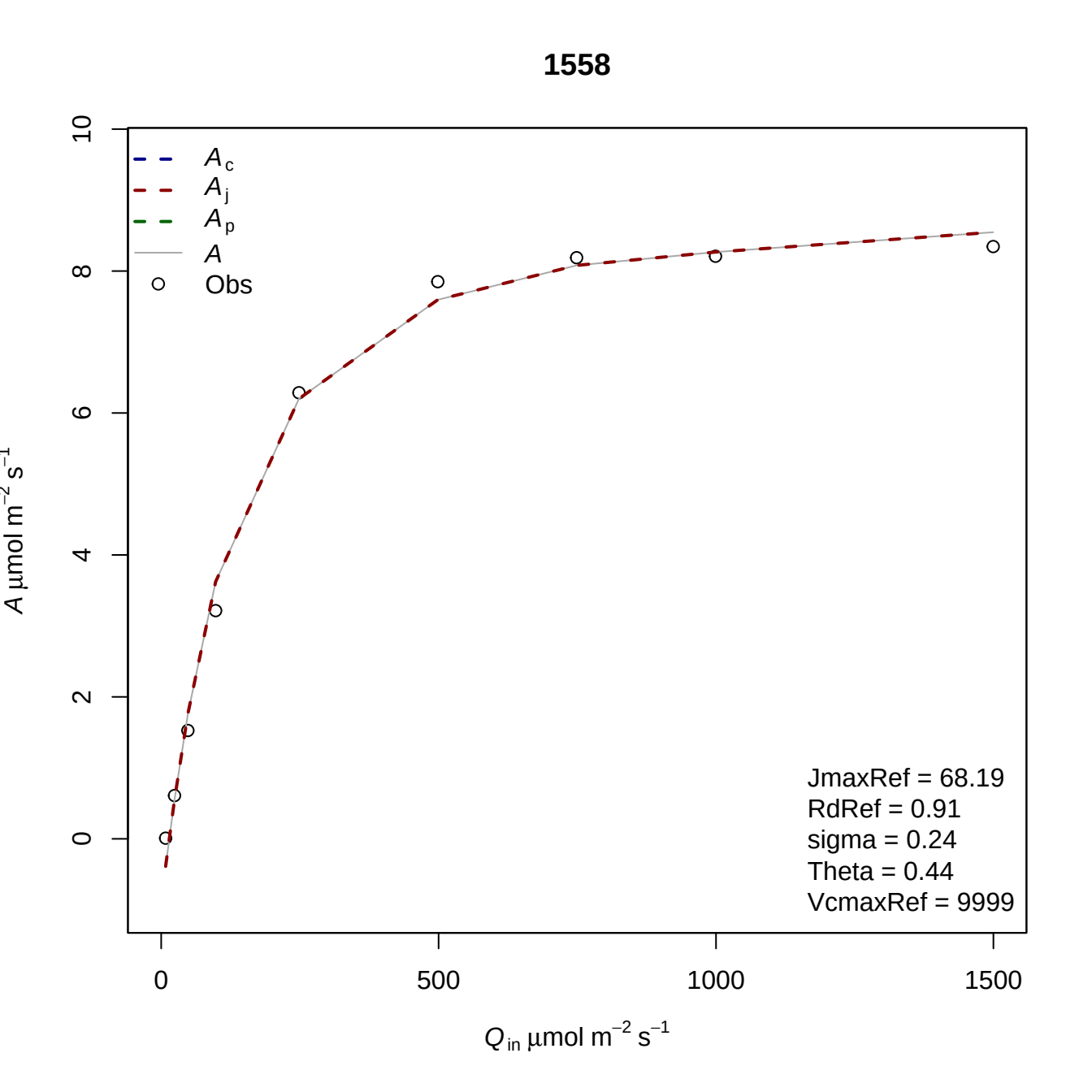

1553
A_c
A_j
A_p
A
Obs
JmaxRef = 93.45
RdRef = 0.72
sigma = 0.15
Theta = 0.19
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

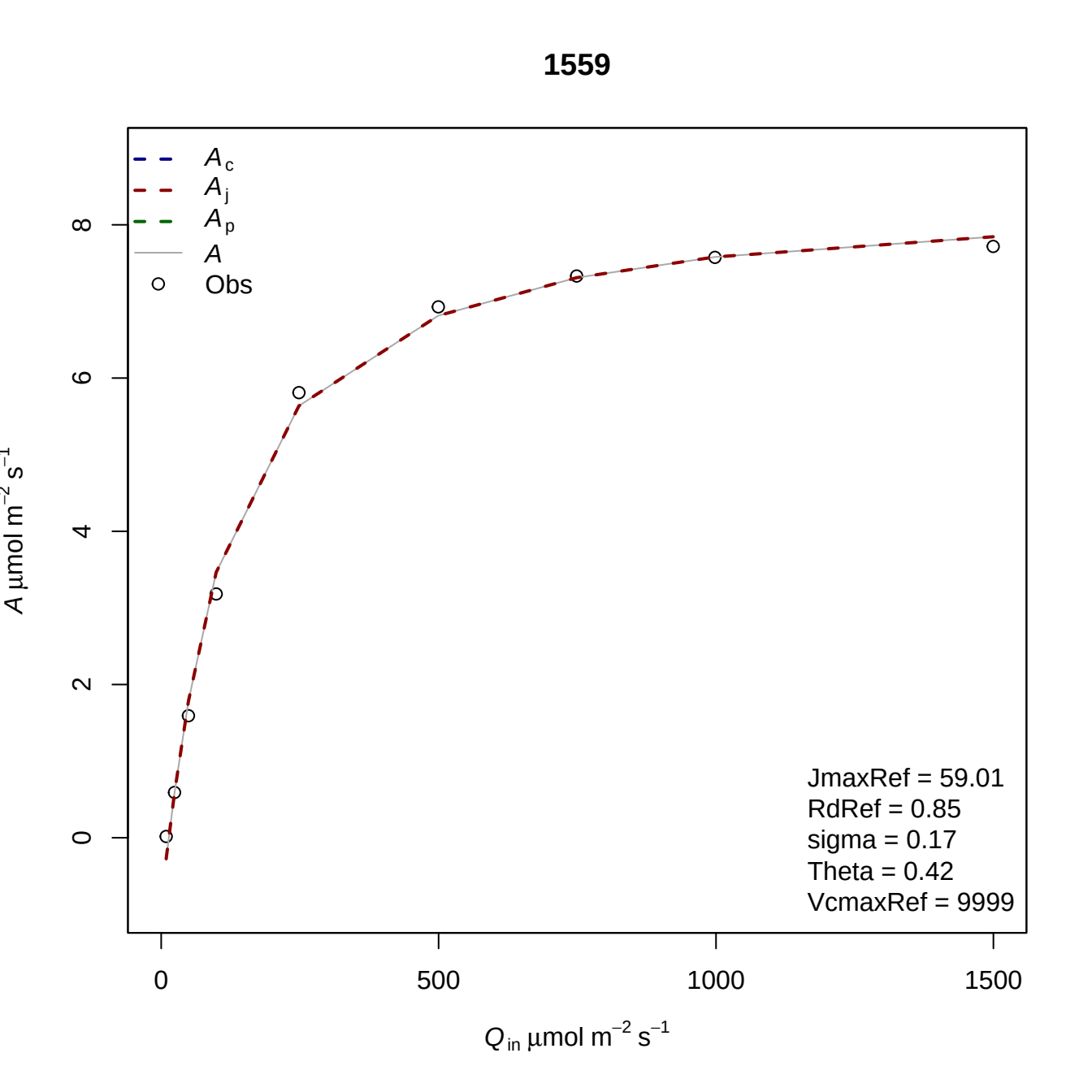

1554
A_c
A_j
A_p
A
Obs
JmaxRef = 99.51
RdRef = 0.83
sigma = 0.21
Theta = 0.11
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

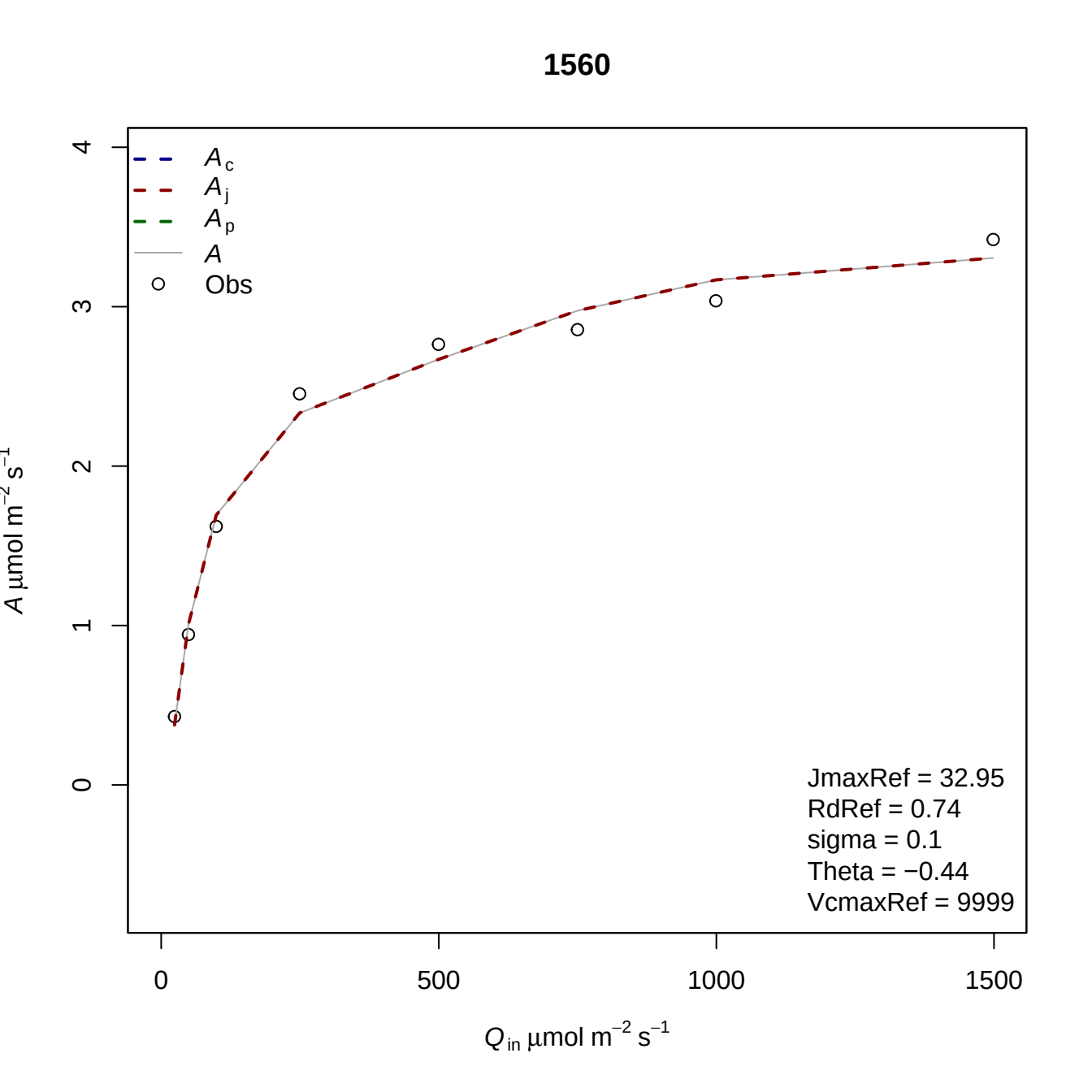

1555
A_c
A_j
A_p
A
Obs
JmaxRef = 78.78
RdRef = 0.64
sigma = 0.1
Theta = 0.05
VcmaxRef = 43.45
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

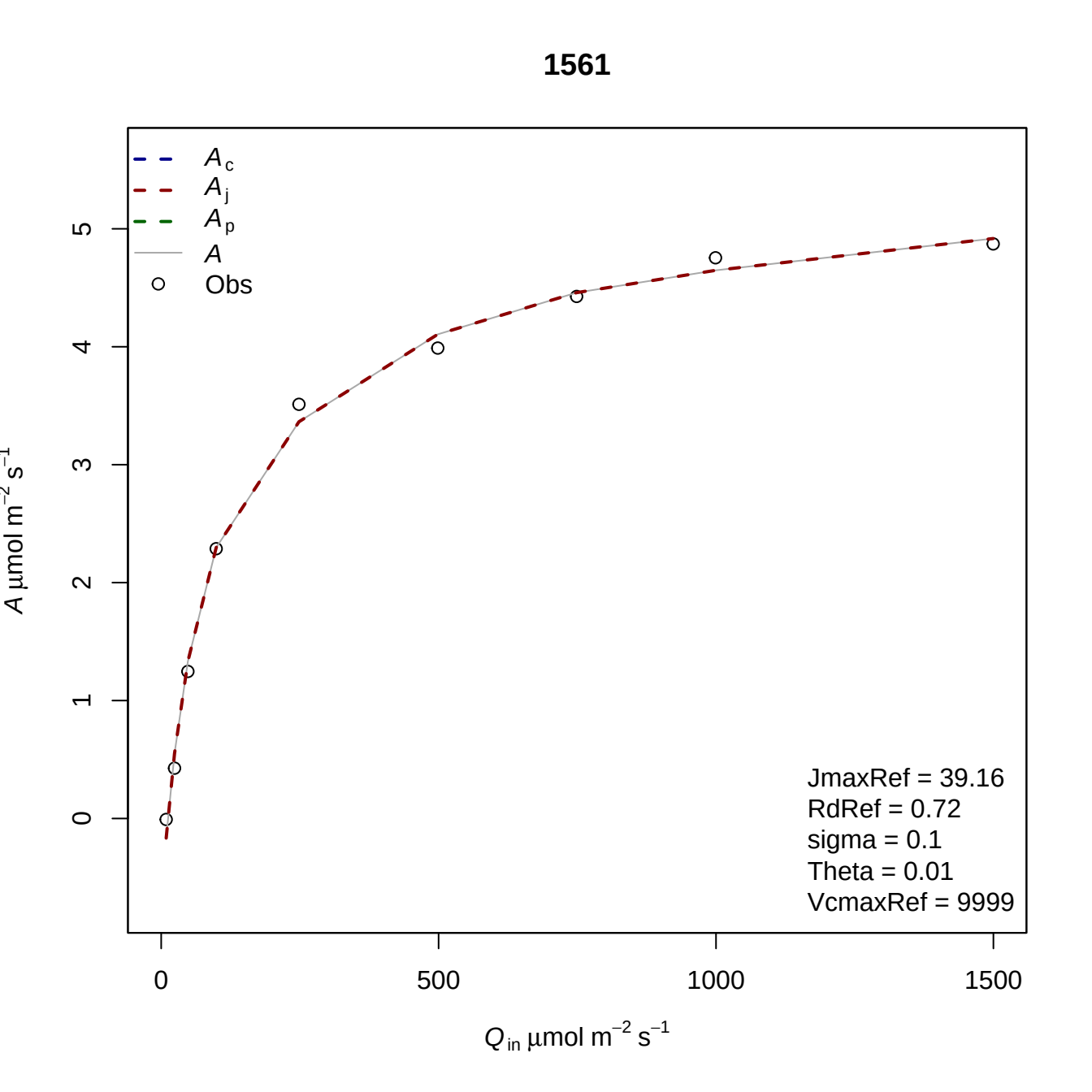

1556
A_c
A_j
A_p
A
Obs
JmaxRef = 77.94
RdRef = 0.55
sigma = 0.2
Theta = 0.1
VcmaxRef = 59.72
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

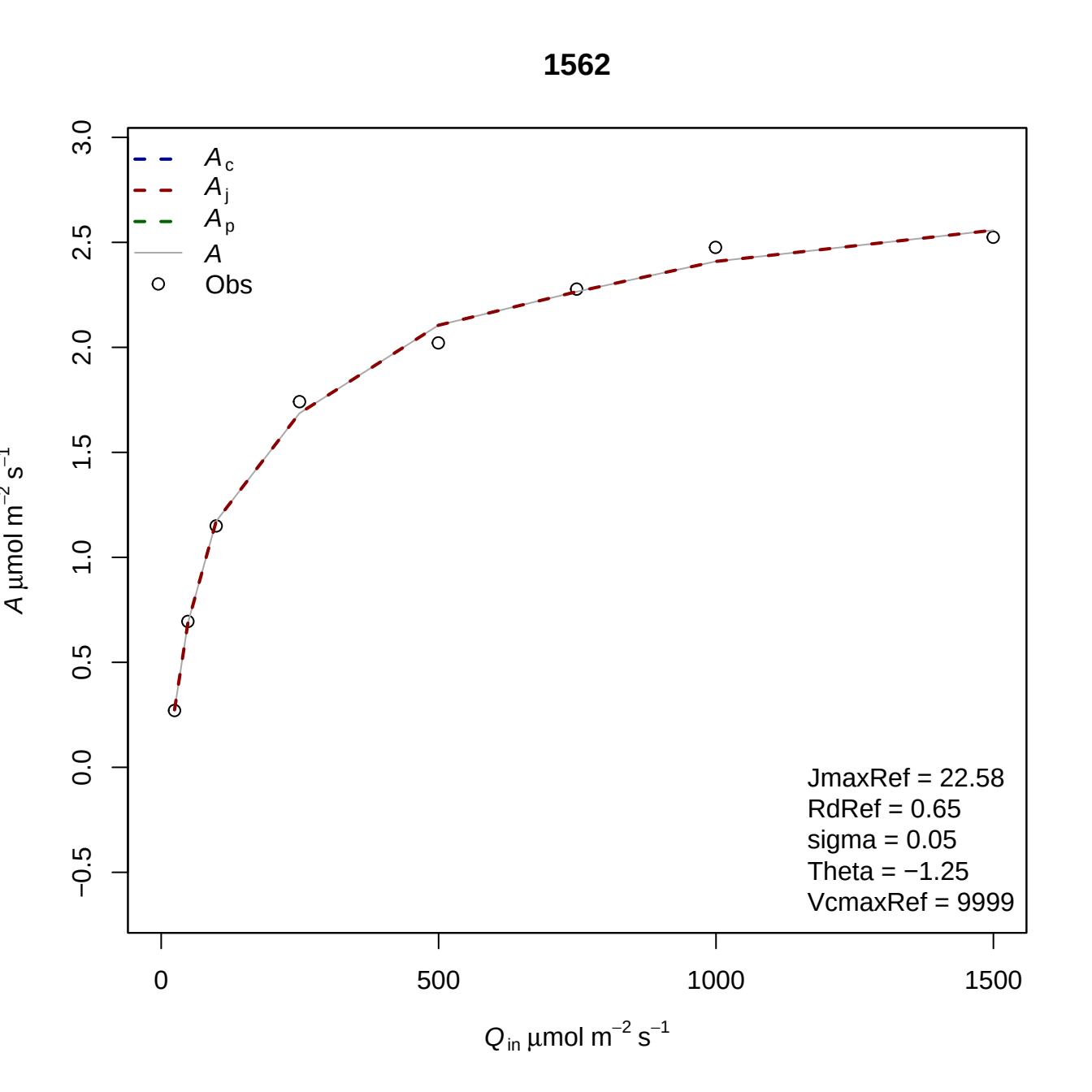

1557
A_c
A_j
A_p
A
Obs
JmaxRef = 79.89
RdRef = 0.66
sigma = 0.11
Theta = −0.07
VcmaxRef = 48.38
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

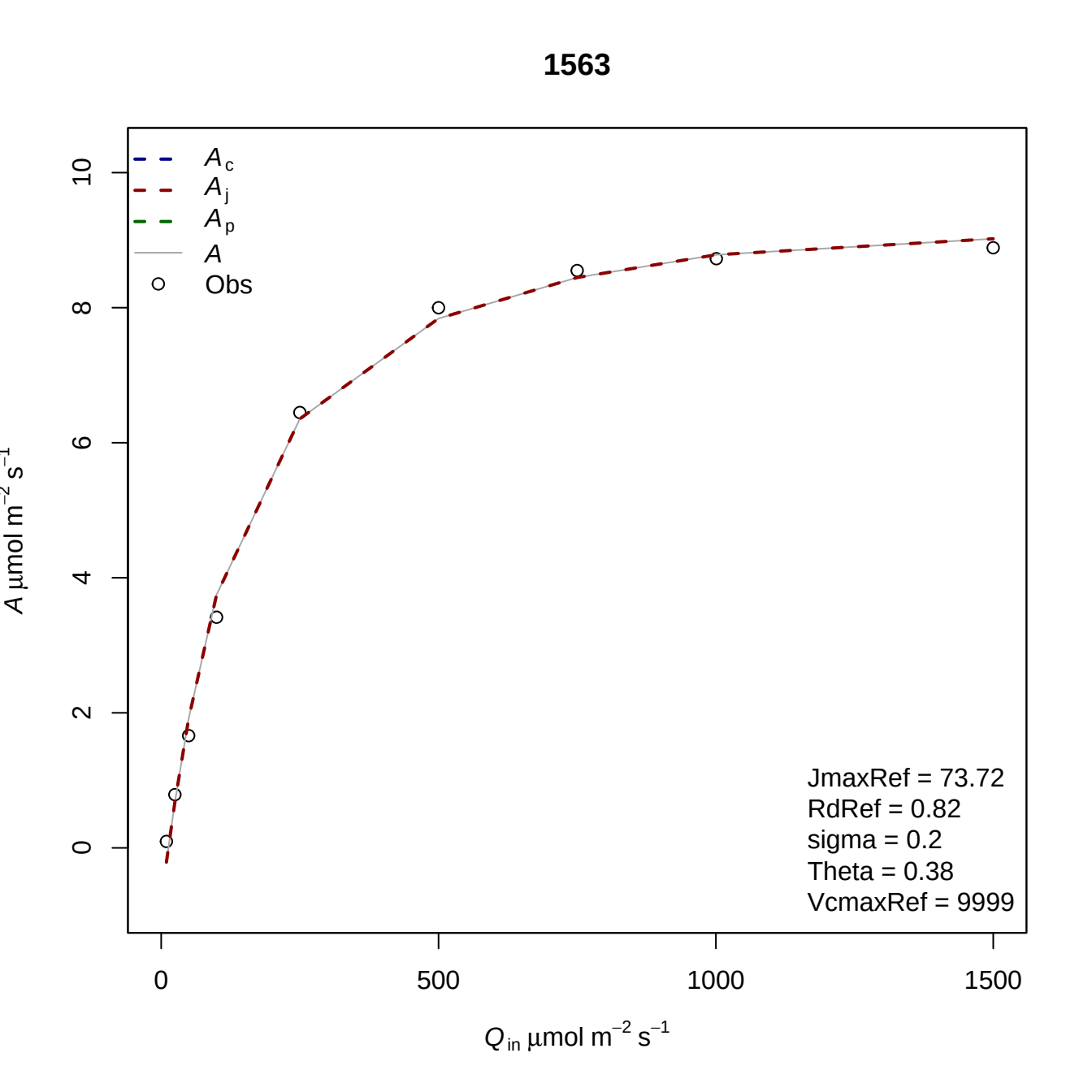

1558
A_c
A_j
A_p
A
Obs
JmaxRef = 68.19
RdRef = 0.91
sigma = 0.24
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

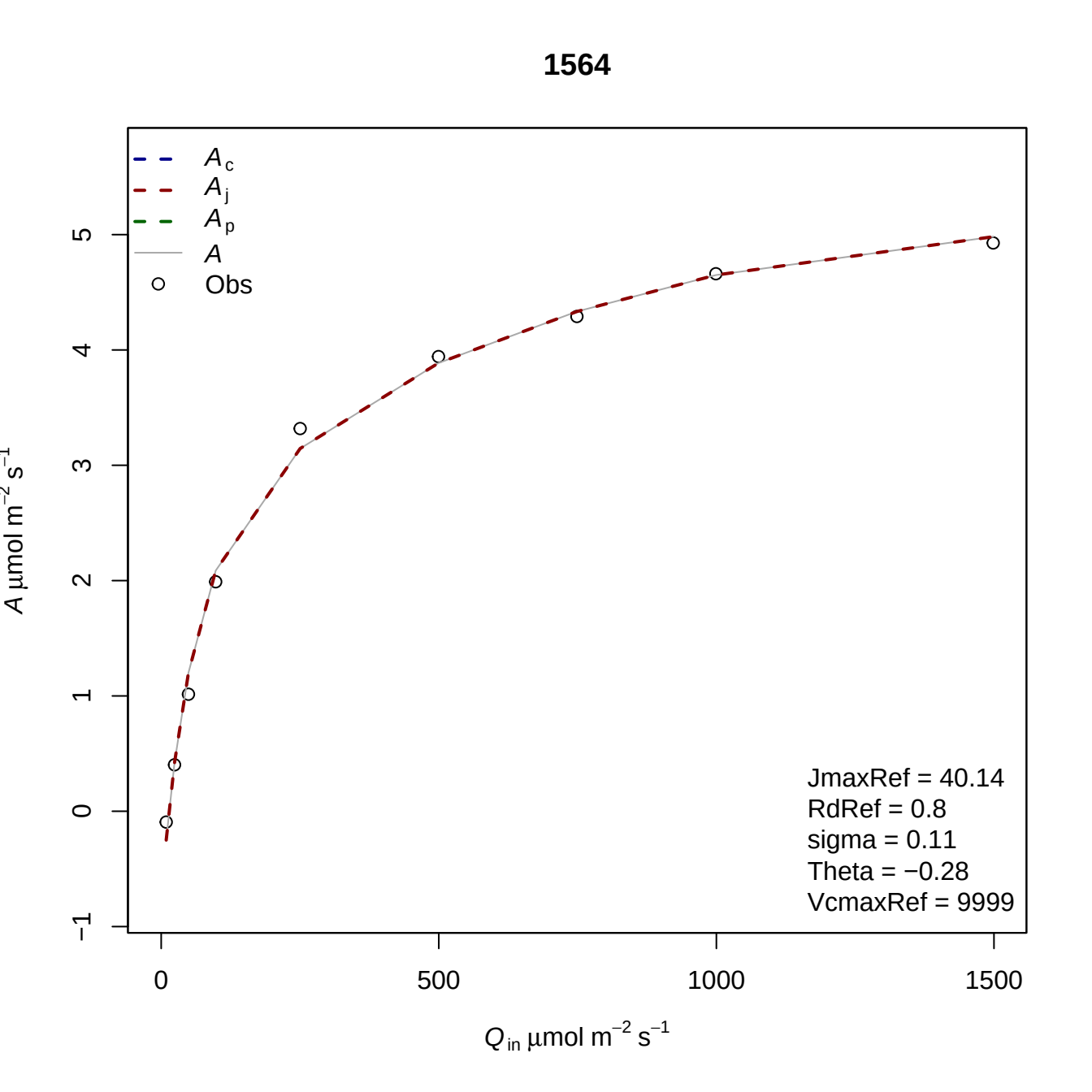

1559
A_c
A_j
A_p
A
Obs
JmaxRef = 59.01
RdRef = 0.85
sigma = 0.17
Theta = 0.42
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

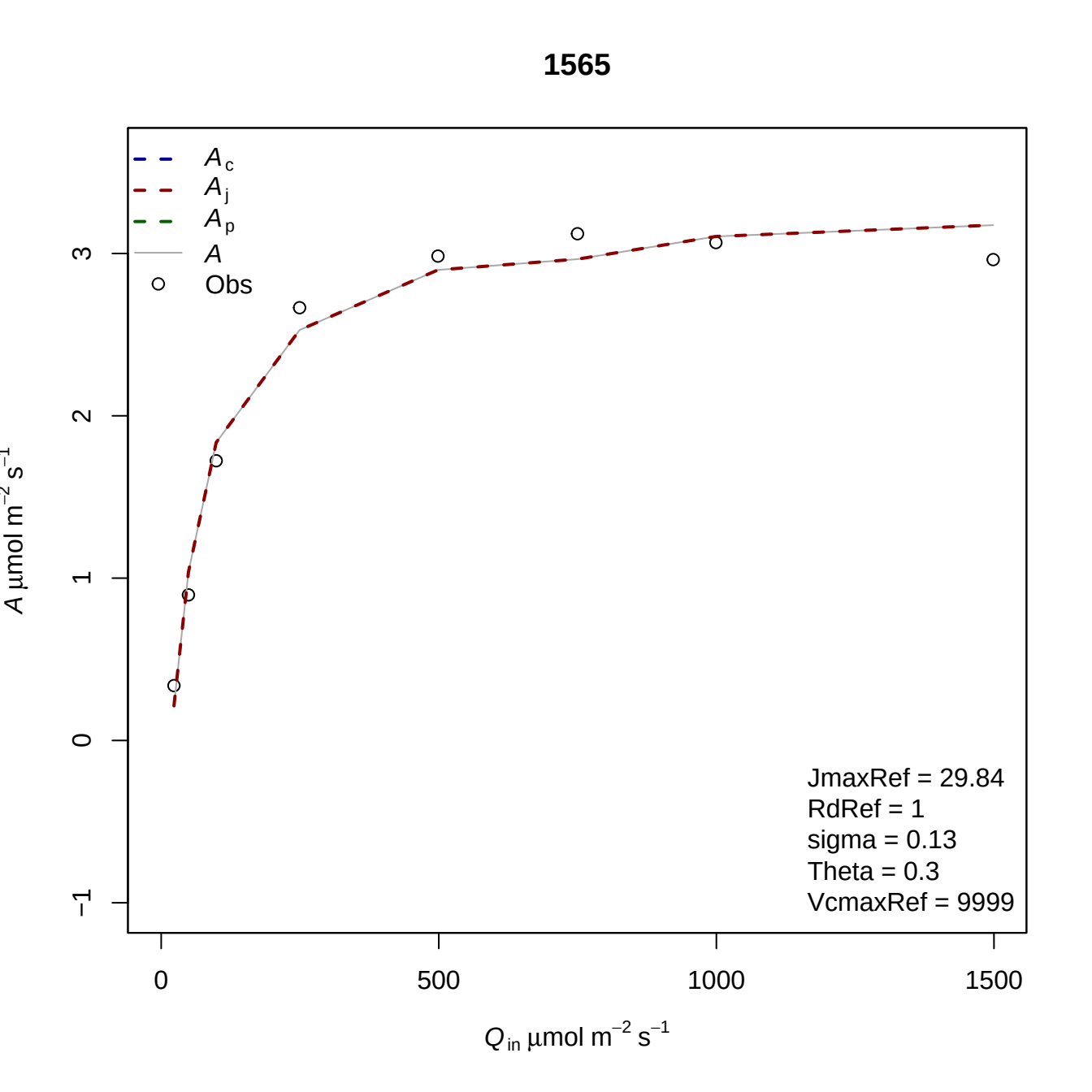

1560
A_c
A_j
A_p
A
Obs
JmaxRef = 32.95
RdRef = 0.74
sigma = 0.1
Theta = −0.44
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

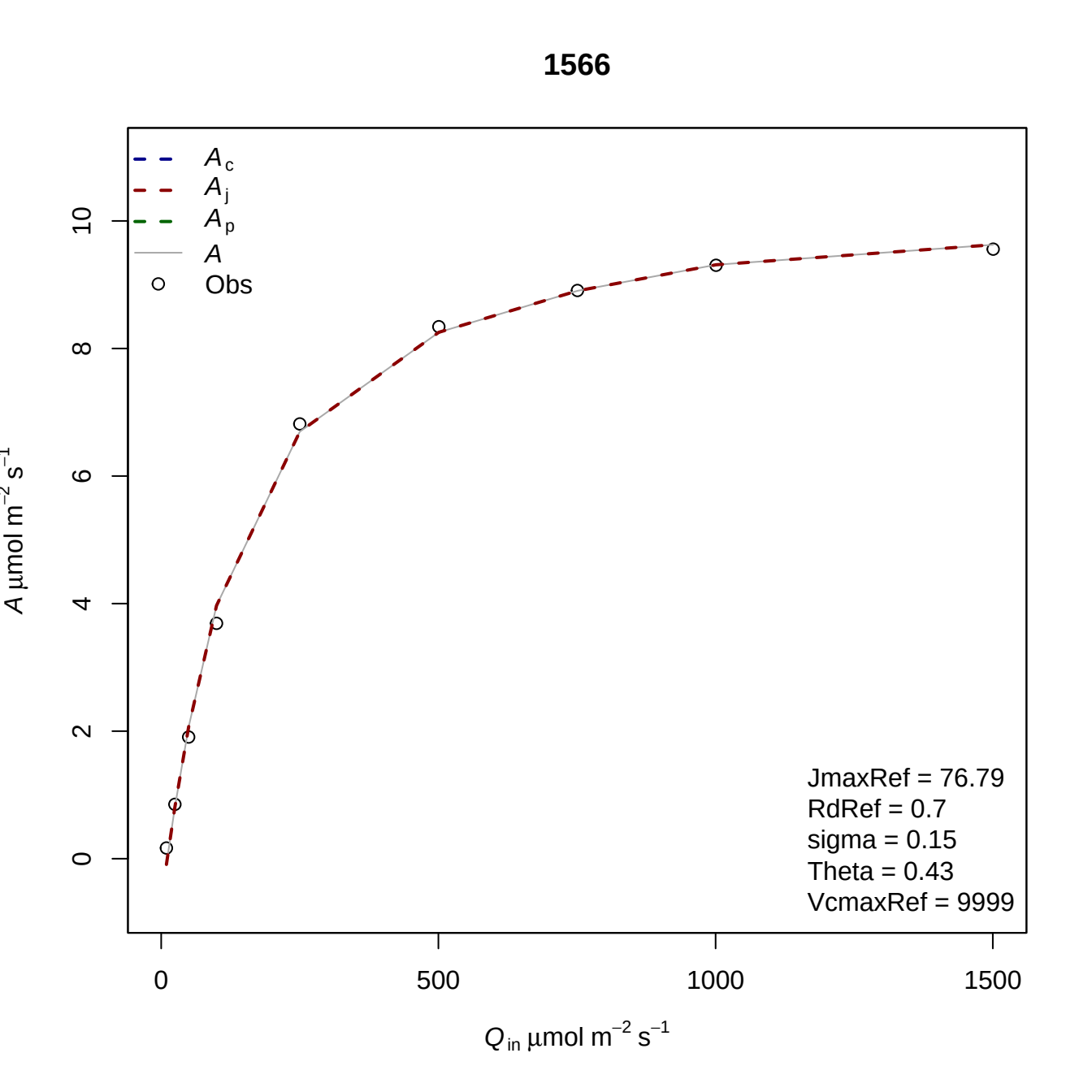

1561
A_c
A_j
A_p
A
Obs
JmaxRef = 39.16
RdRef = 0.72
sigma = 0.1
Theta = 0.01
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

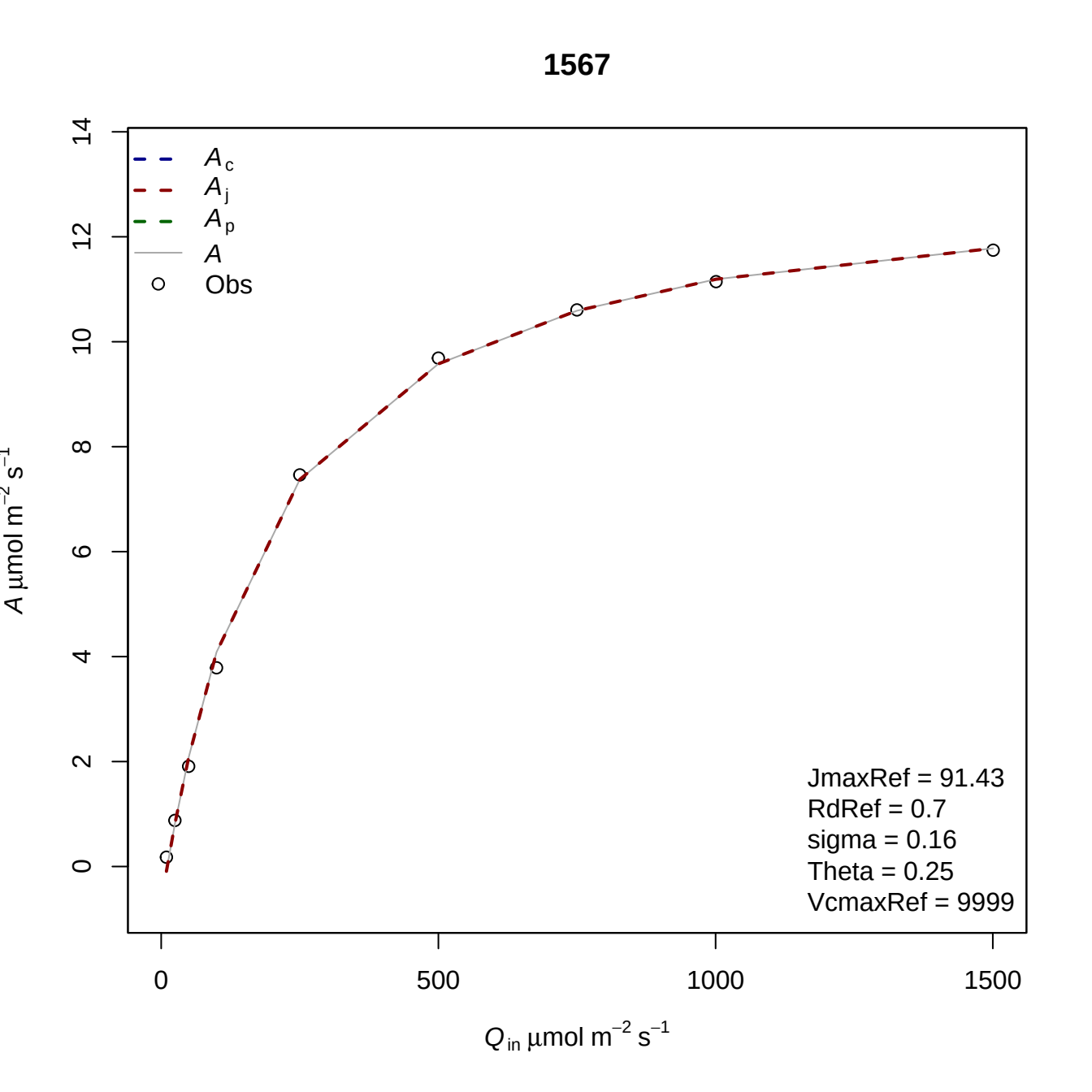

1562
A_c
A_j
A_p
A
Obs
JmaxRef = 22.58
RdRef = 0.65
sigma = 0.05
Theta = −1.25
VcmaxRef = 9999
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

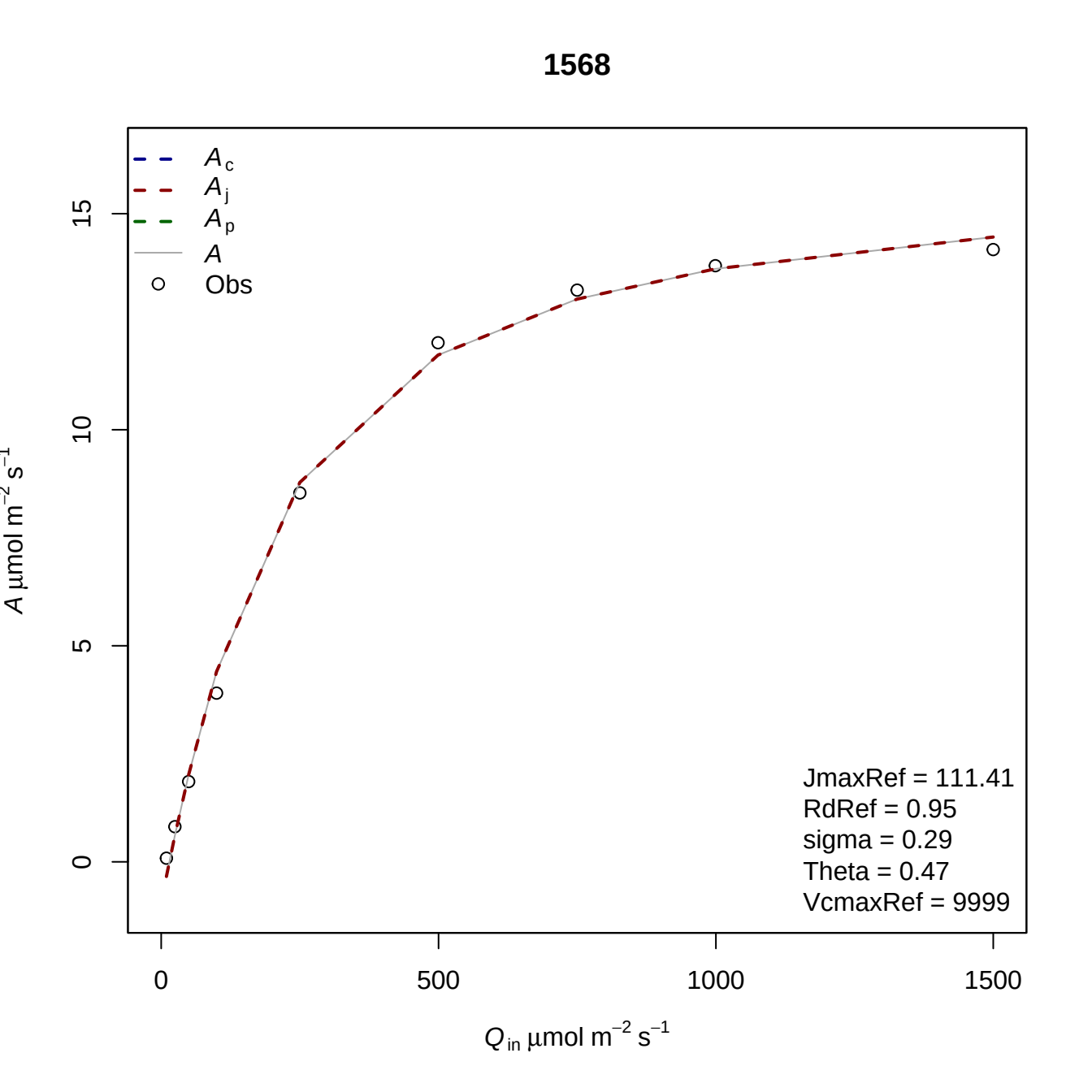

1563
A_c
A_j
A_p
A
Obs
JmaxRef = 73.72
RdRef = 0.82
sigma = 0.2
Theta = 0.38
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

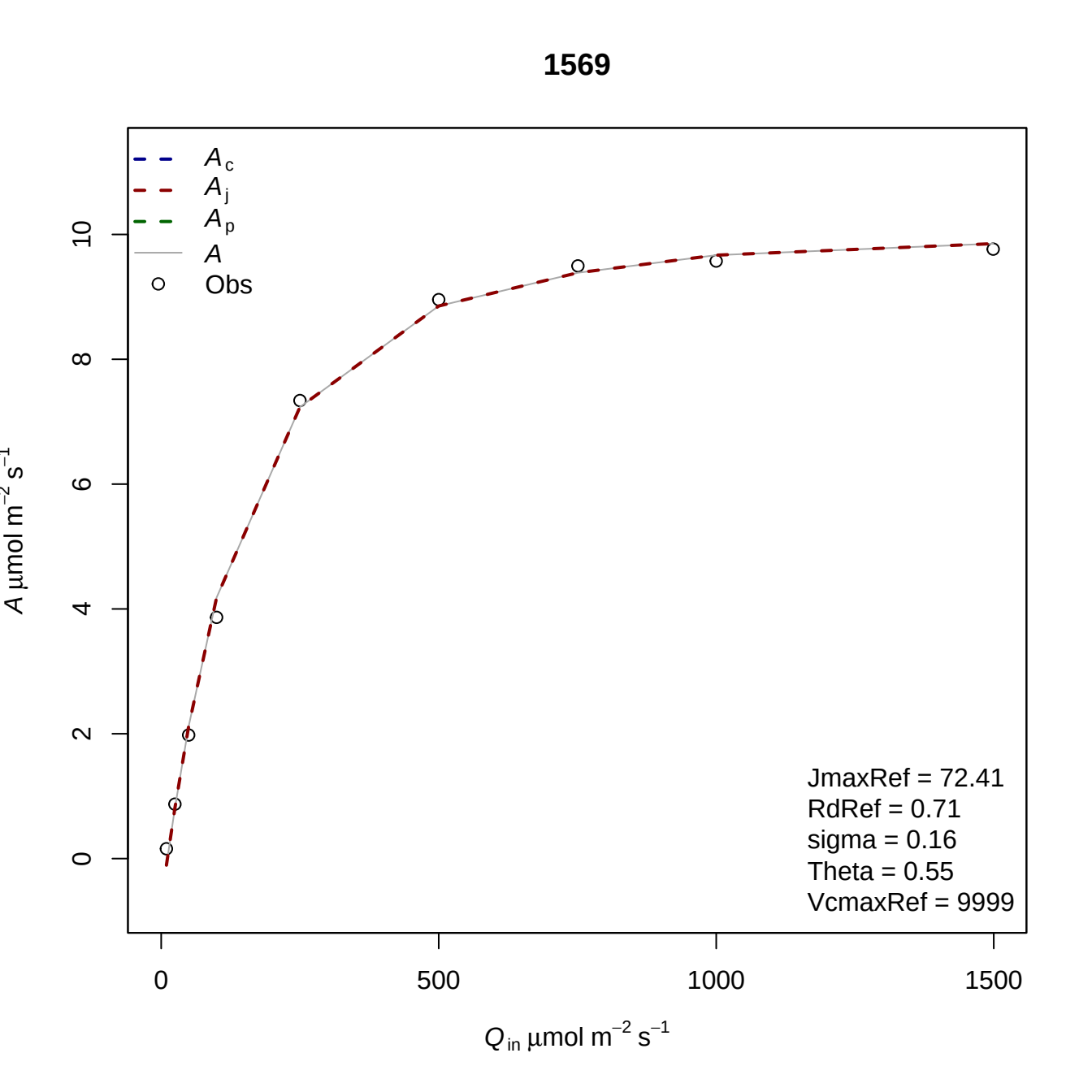

1564
A_c
A_j
A_p
A
Obs
JmaxRef = 40.14
RdRef = 0.8
sigma = 0.11
Theta = −0.28
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

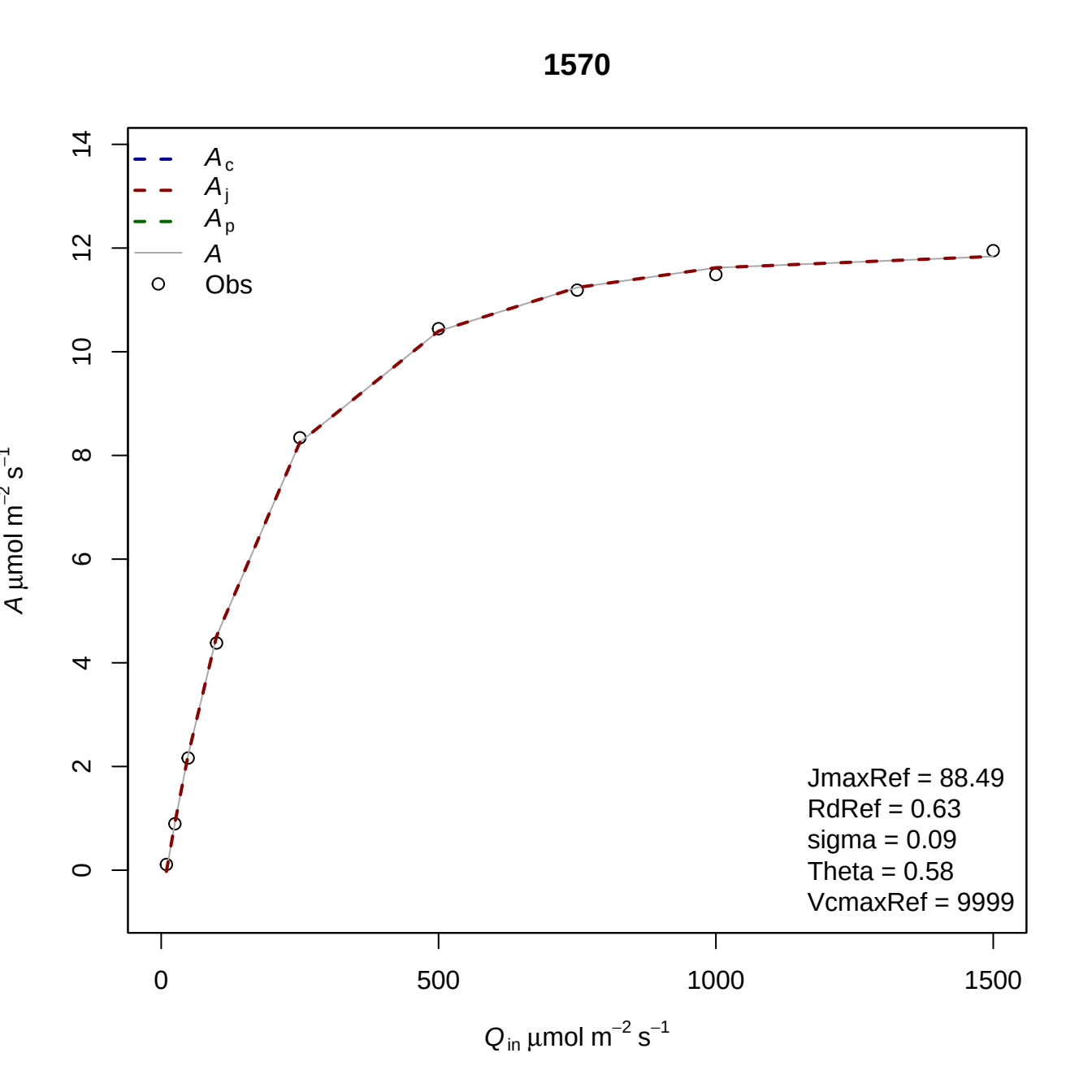

1565
A_c
A_j
A_p
A
Obs
JmaxRef = 29.84
RdRef = 1
sigma = 0.13
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

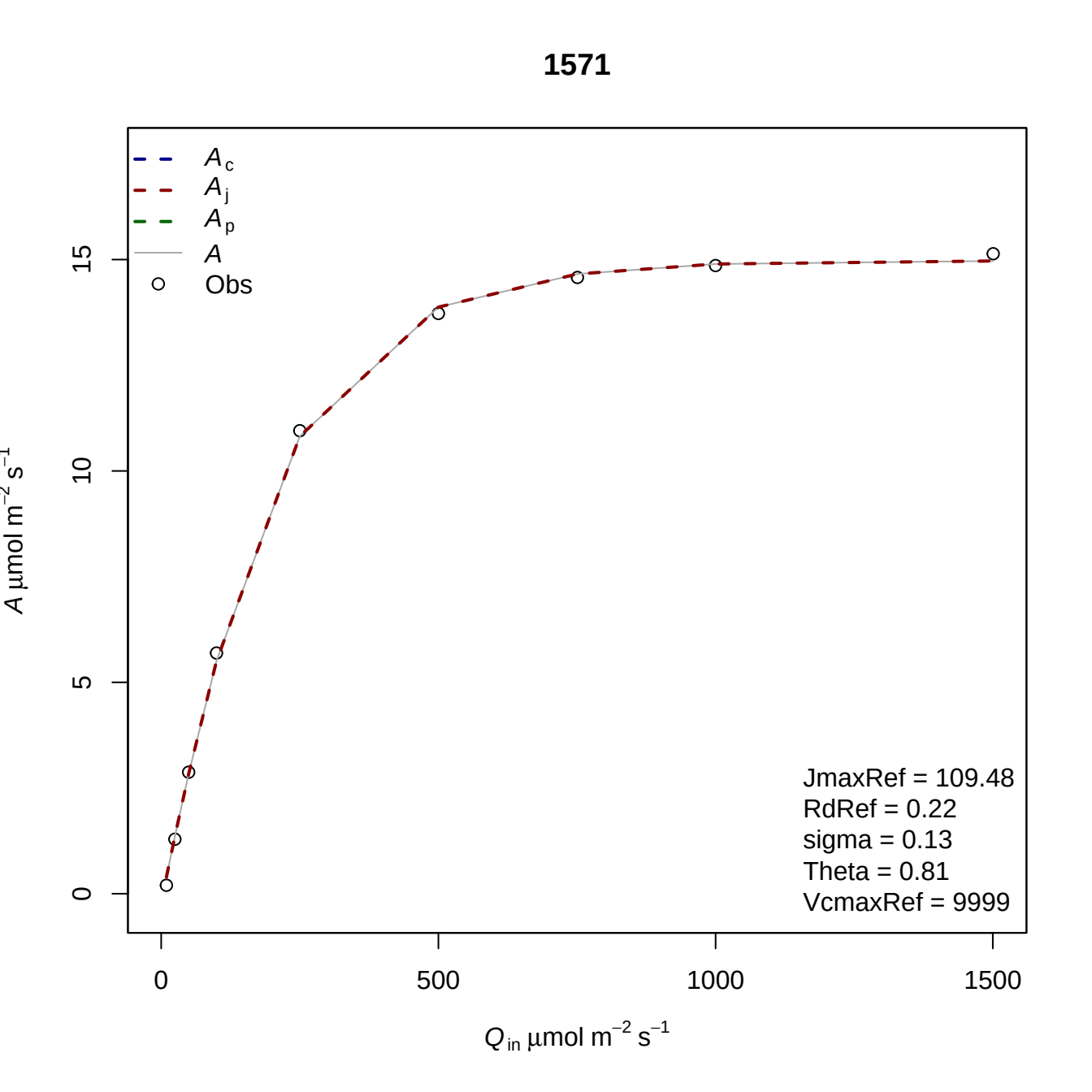

1566
A_c
A_j
A_p
A
Obs
JmaxRef = 76.79
RdRef = 0.7
sigma = 0.15
Theta = 0.43
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

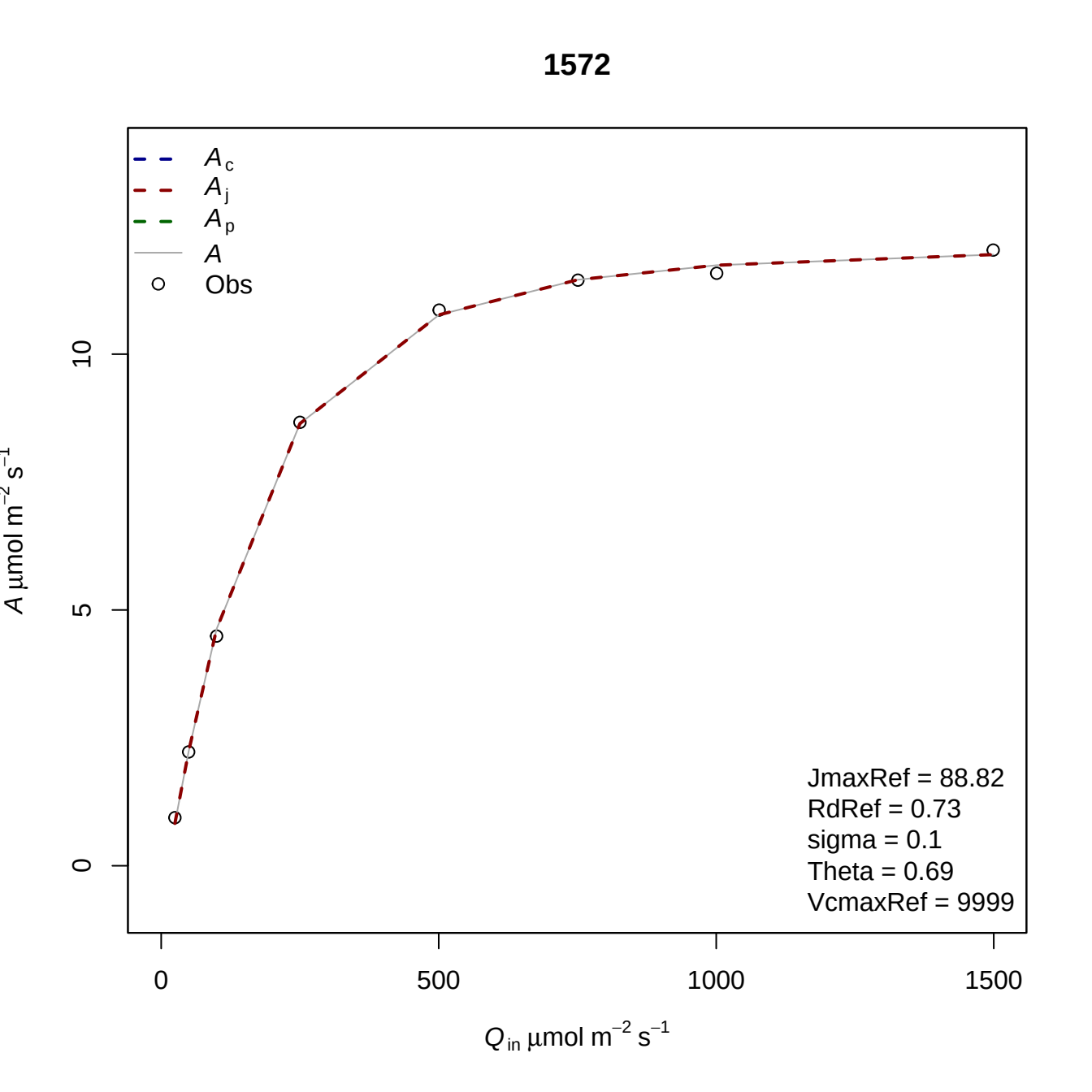

1567
A_c
A_j
A_p
A
Obs
JmaxRef = 91.43
RdRef = 0.7
sigma = 0.16
Theta = 0.25
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

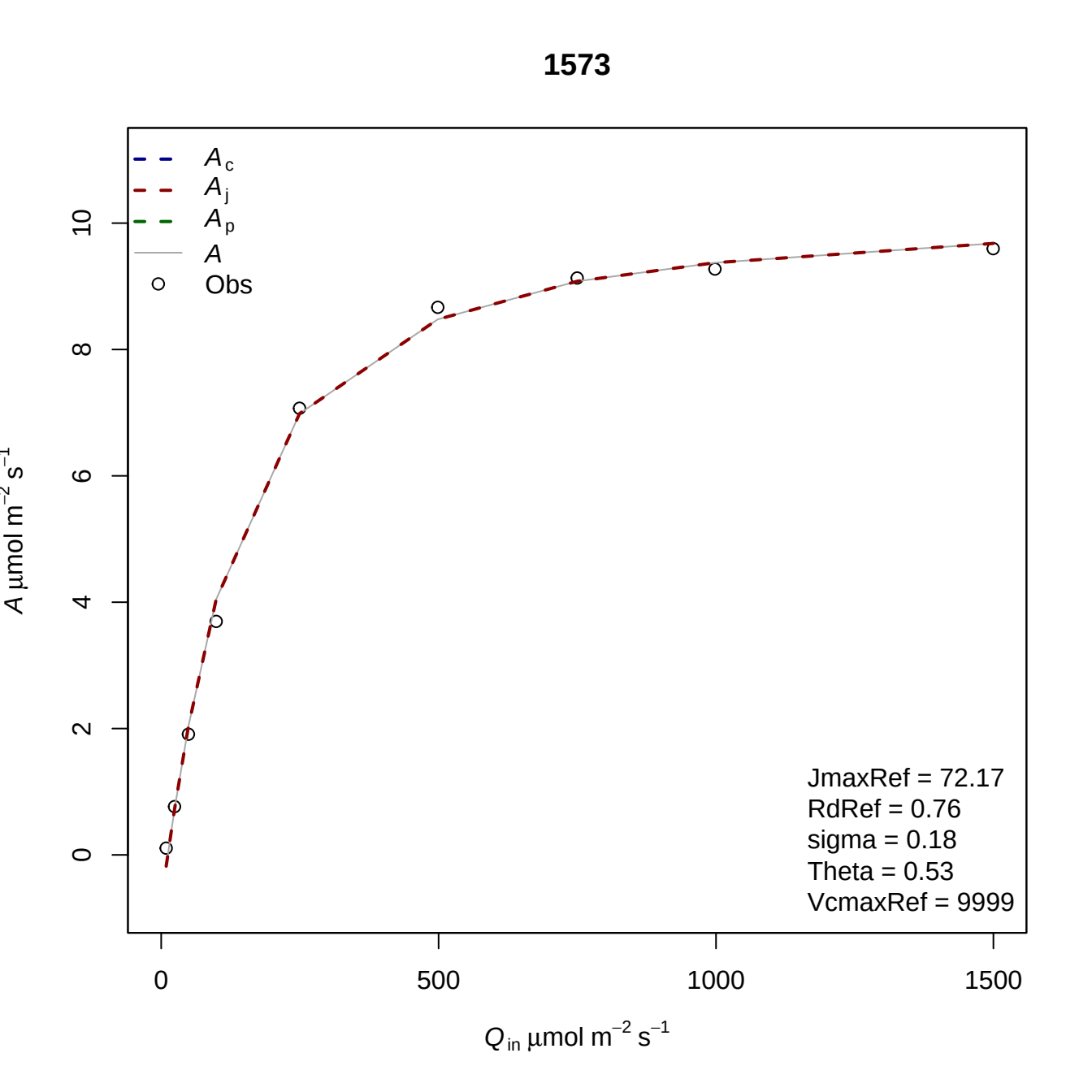

1568
A_c
A_j
A_p
A
Obs
JmaxRef = 111.41
RdRef = 0.95
sigma = 0.29
Theta = 0.47
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

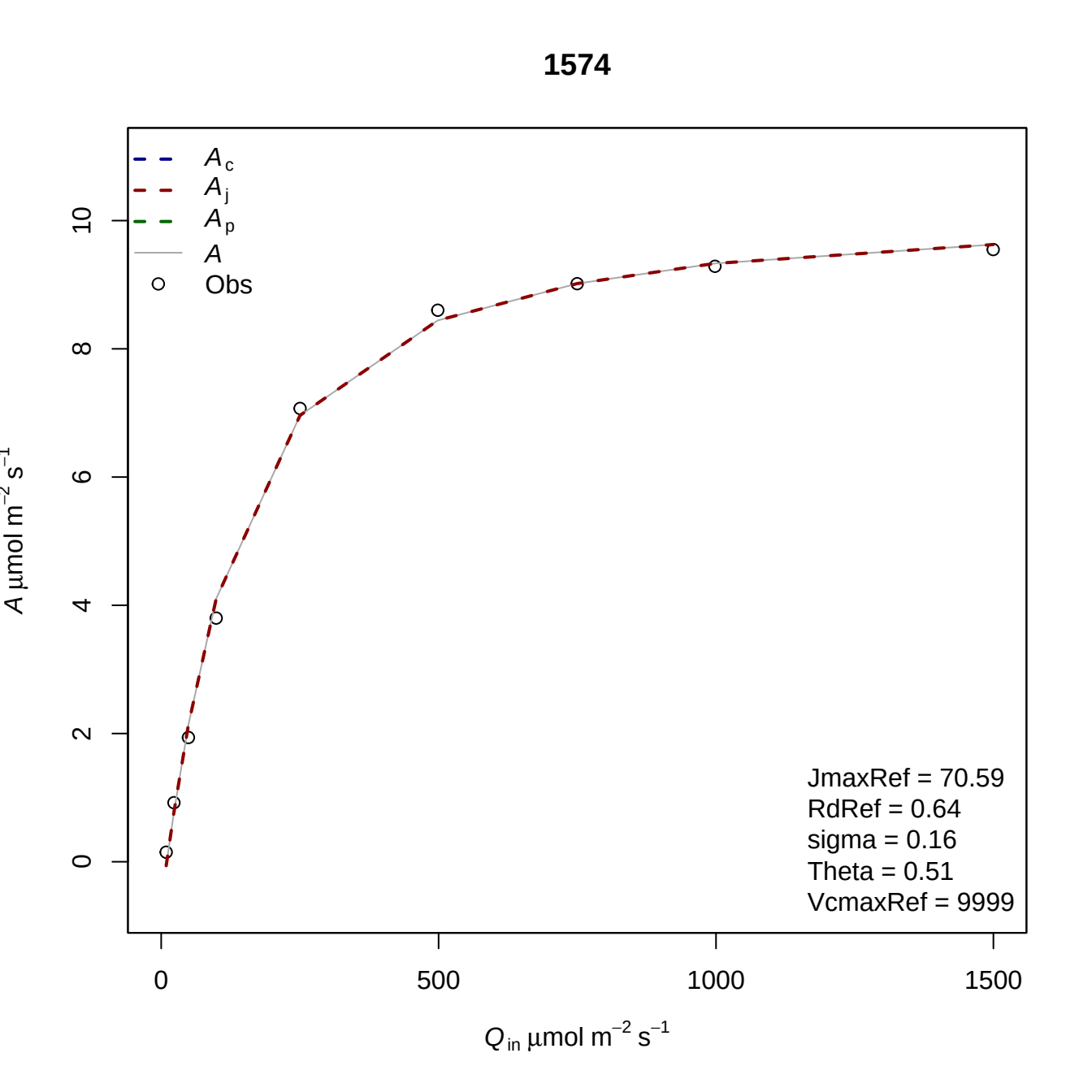

1569
A_c
A_j
A_p
A
Obs
JmaxRef = 72.41
RdRef = 0.71
sigma = 0.16
Theta = 0.55
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

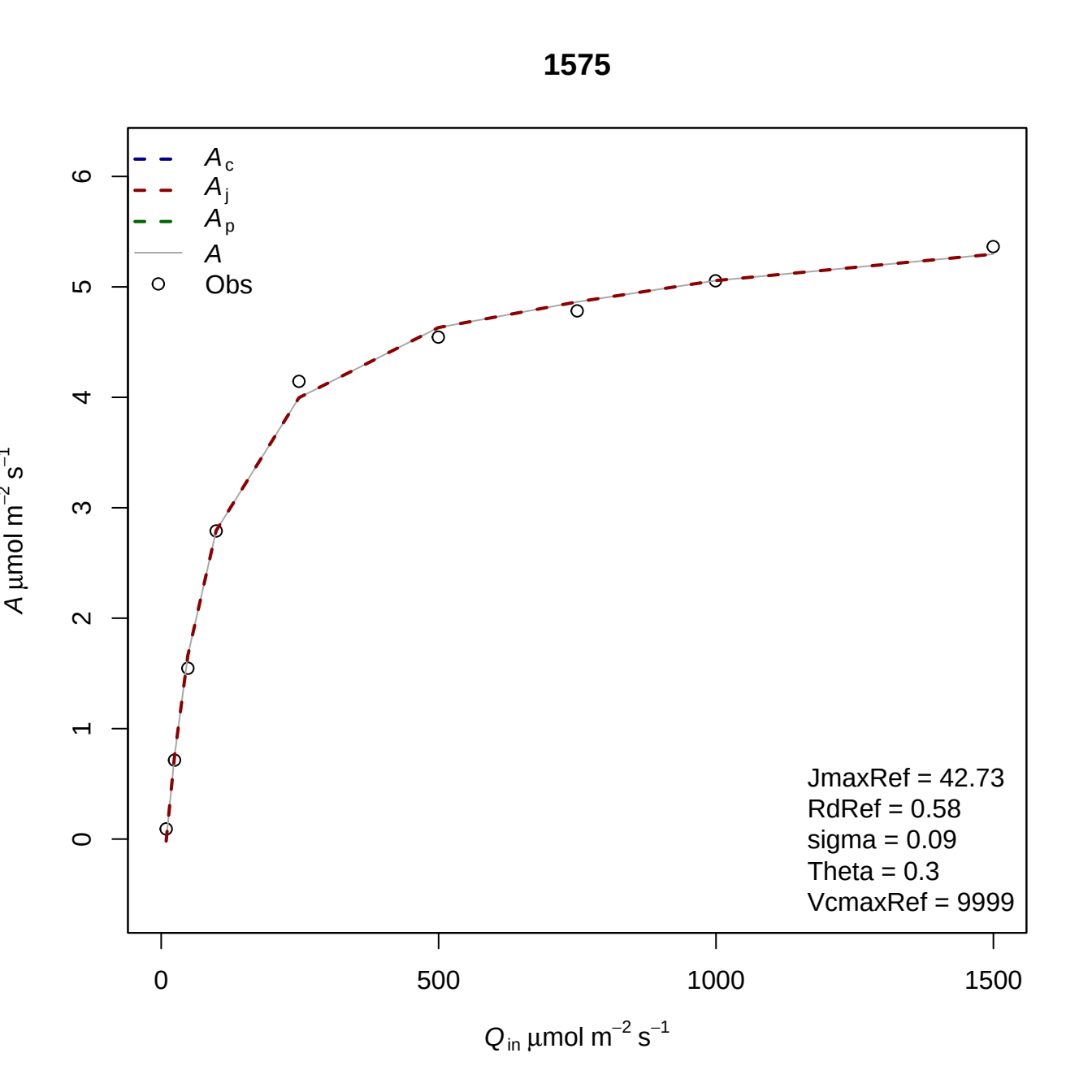

1570
A_c
A_j
A_p
A
Obs
JmaxRef = 88.49
RdRef = 0.63
sigma = 0.09
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

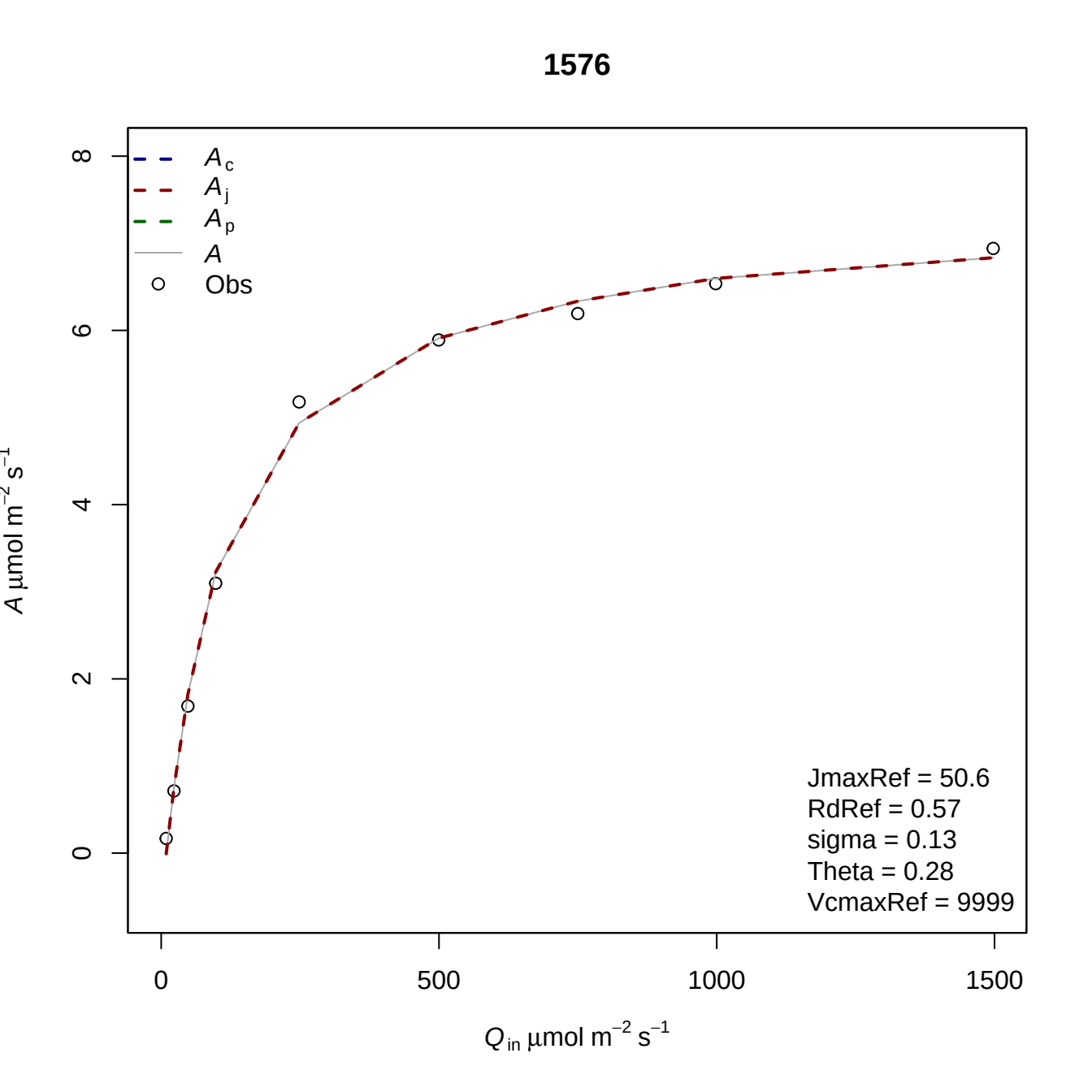

1571
A_c
A_j
A_p
A
Obs
JmaxRef = 109.48
RdRef = 0.22
sigma = 0.13
Theta = 0.81
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

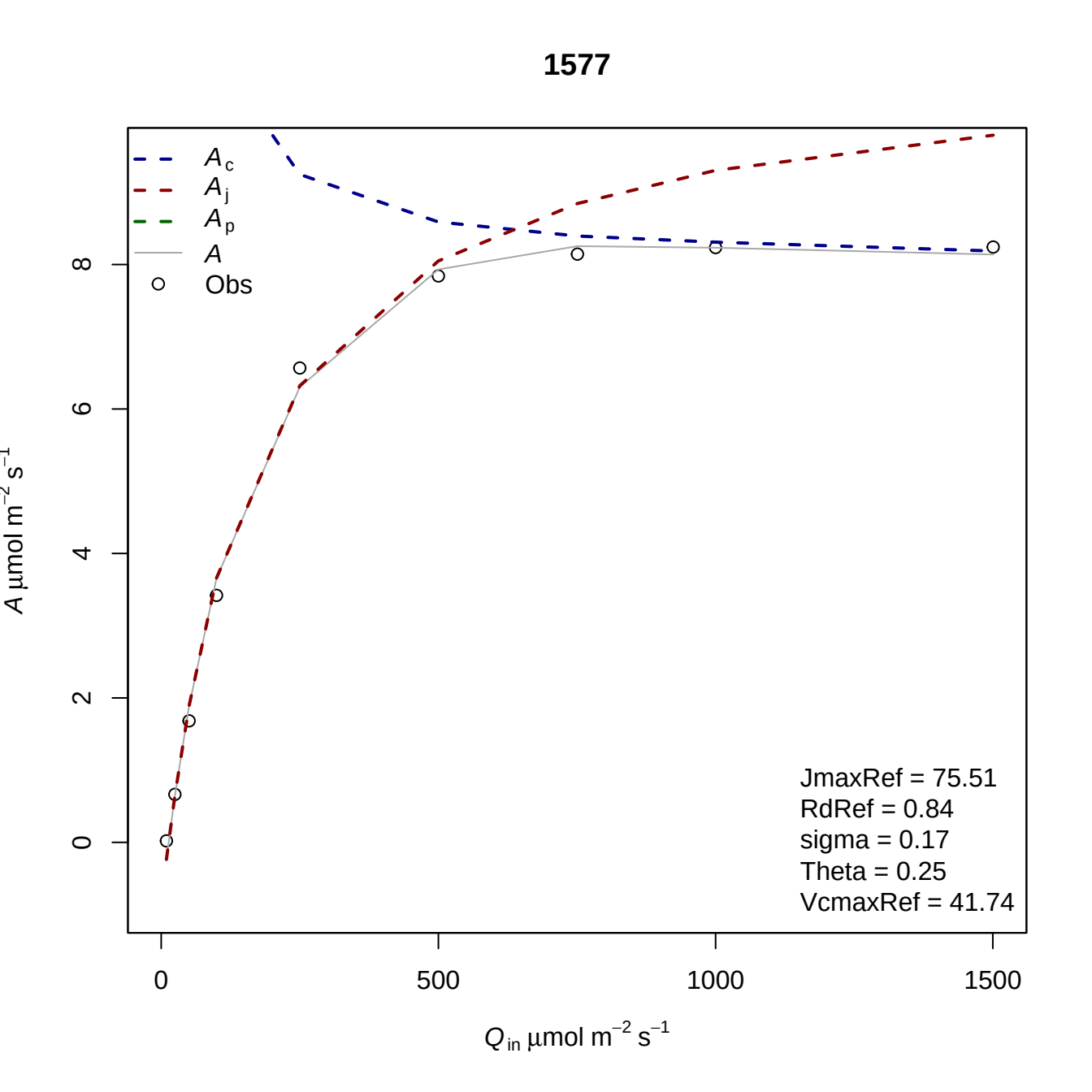

1572
A_c
A_j
A_p
A
Obs
JmaxRef = 88.82
RdRef = 0.73
sigma = 0.1
Theta = 0.69
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

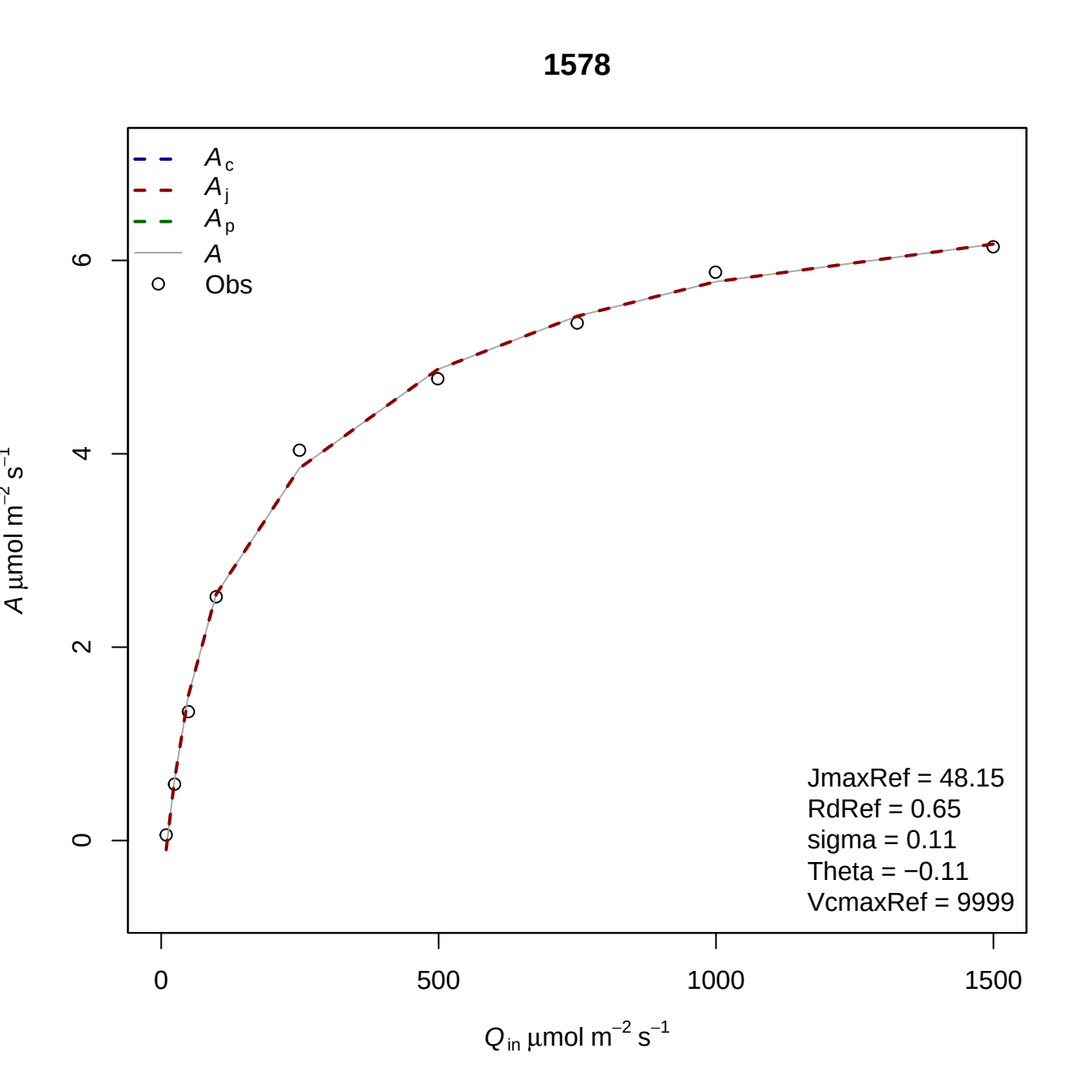

1573
A_c
A_j
A_p
A
Obs
JmaxRef = 72.17
RdRef = 0.76
sigma = 0.18
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

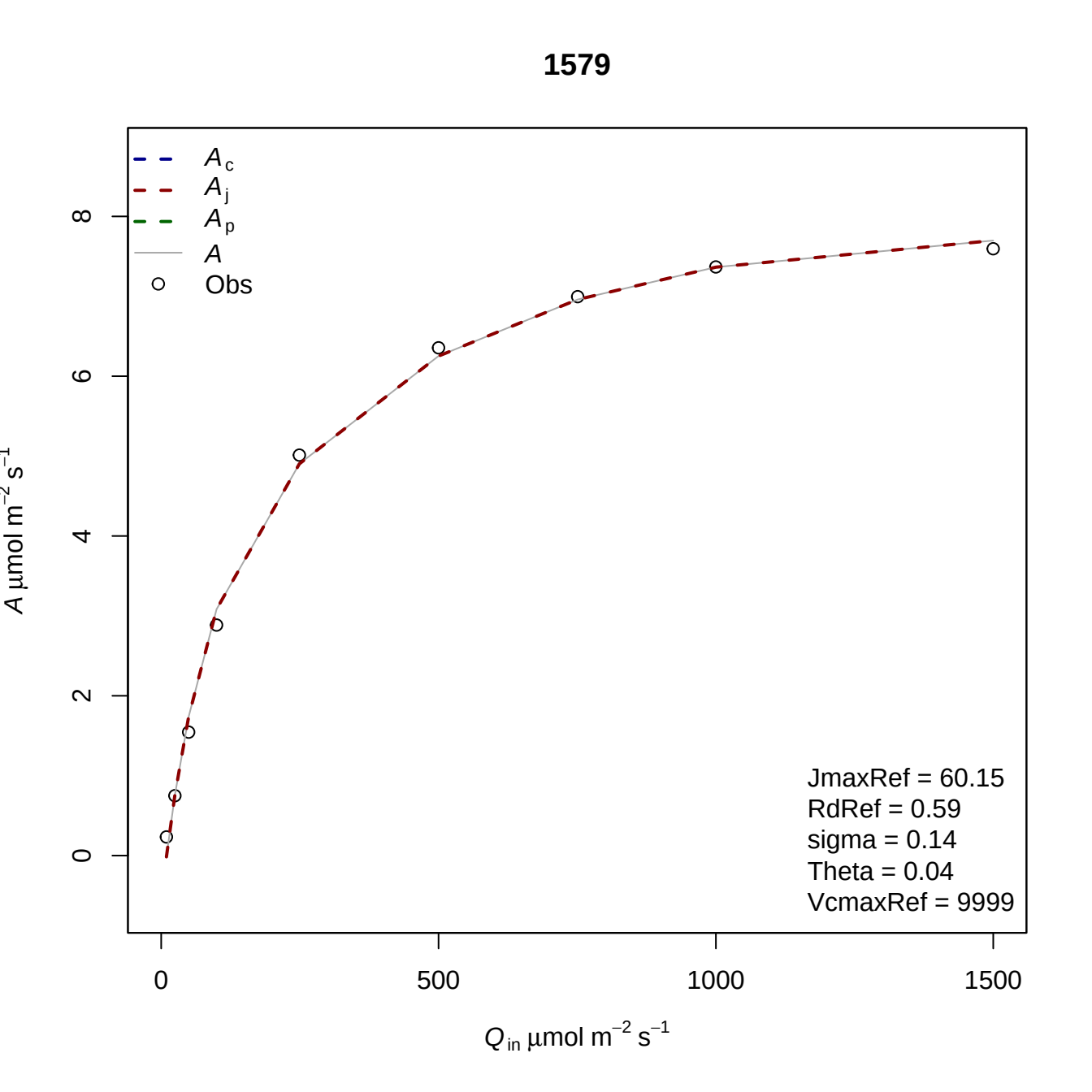

1574
A_c
A_j
A_p
A
Obs
JmaxRef = 70.59
RdRef = 0.64
sigma = 0.16
Theta = 0.51
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

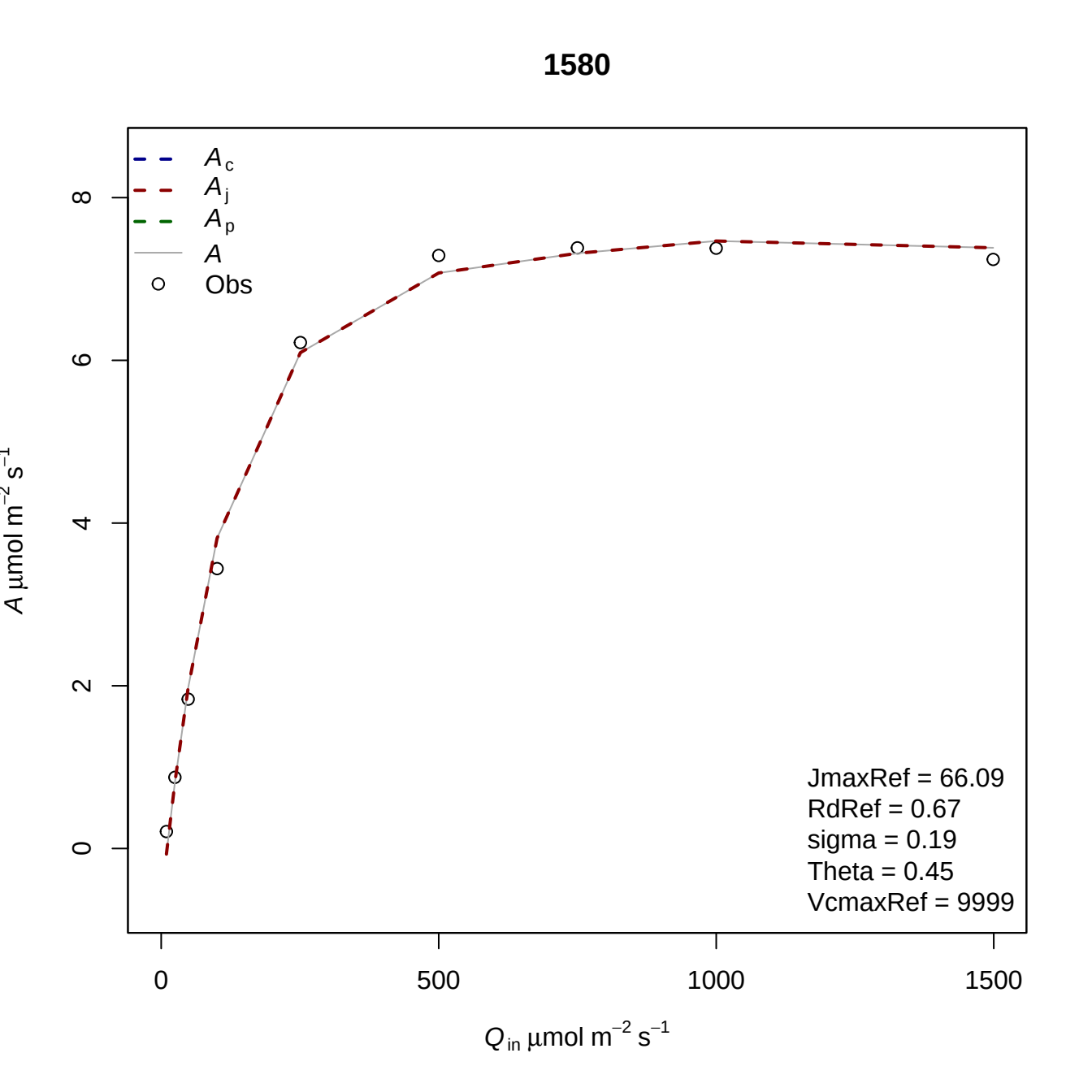

1575
A_c
A_j
A_p
A
Obs
JmaxRef = 42.73
RdRef = 0.58
sigma = 0.09
Theta = 0.3
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

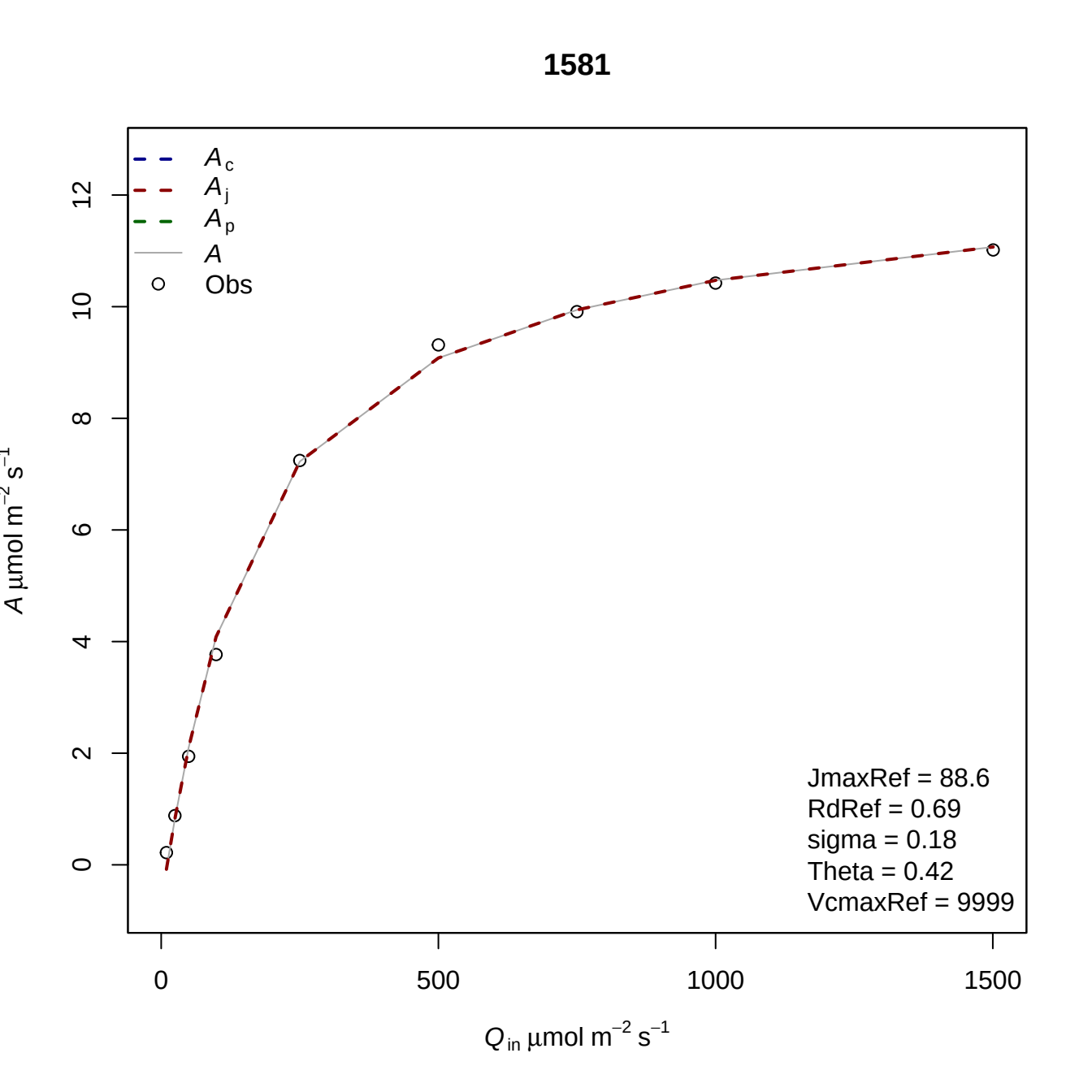

1576
A_c
A_j
A_p
A
Obs
JmaxRef = 50.6
RdRef = 0.57
sigma = 0.13
Theta = 0.28
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

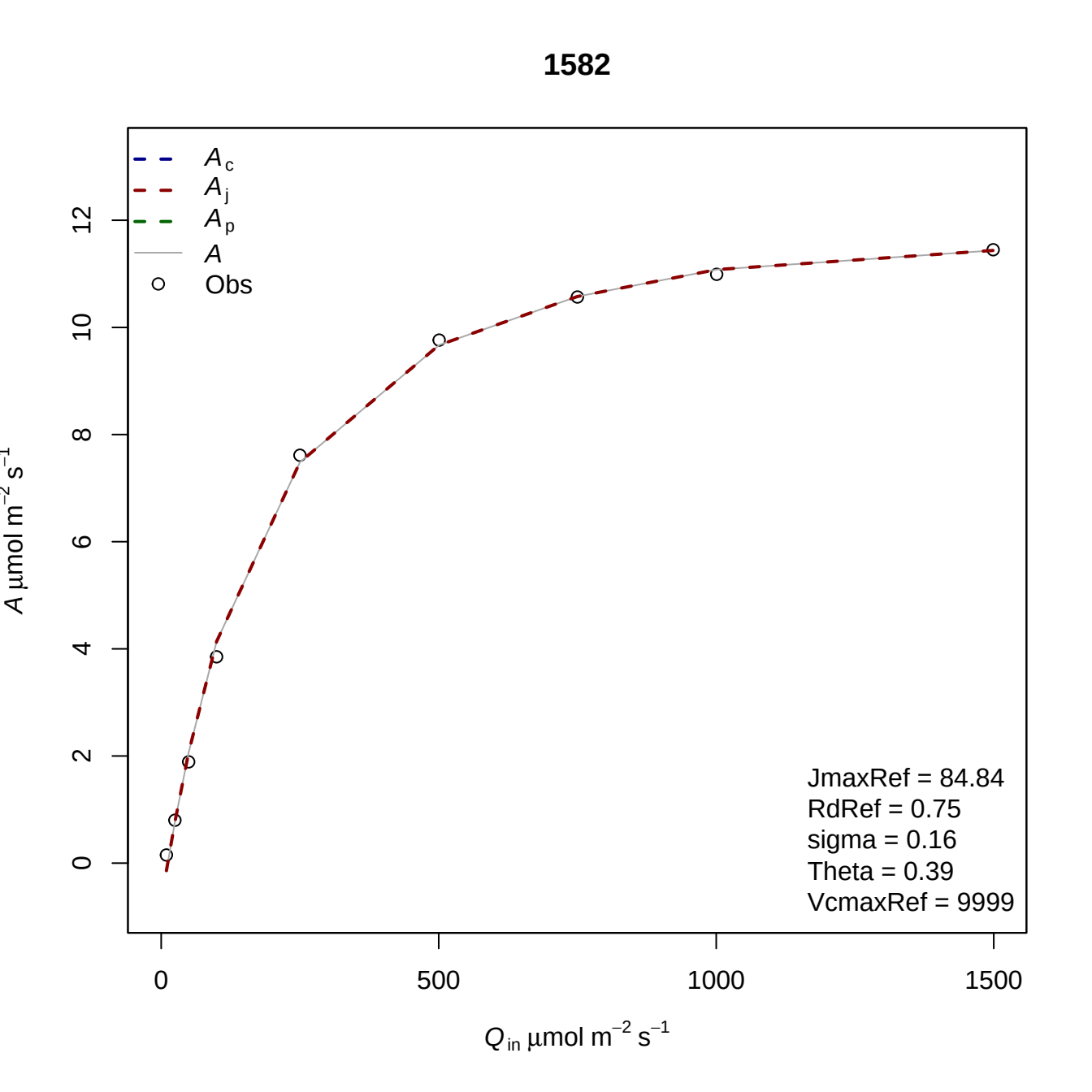

1577
A_c
A_j
A_p
A
Obs
JmaxRef = 75.51
RdRef = 0.84
sigma = 0.17
Theta = 0.25
VcmaxRef = 41.74
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

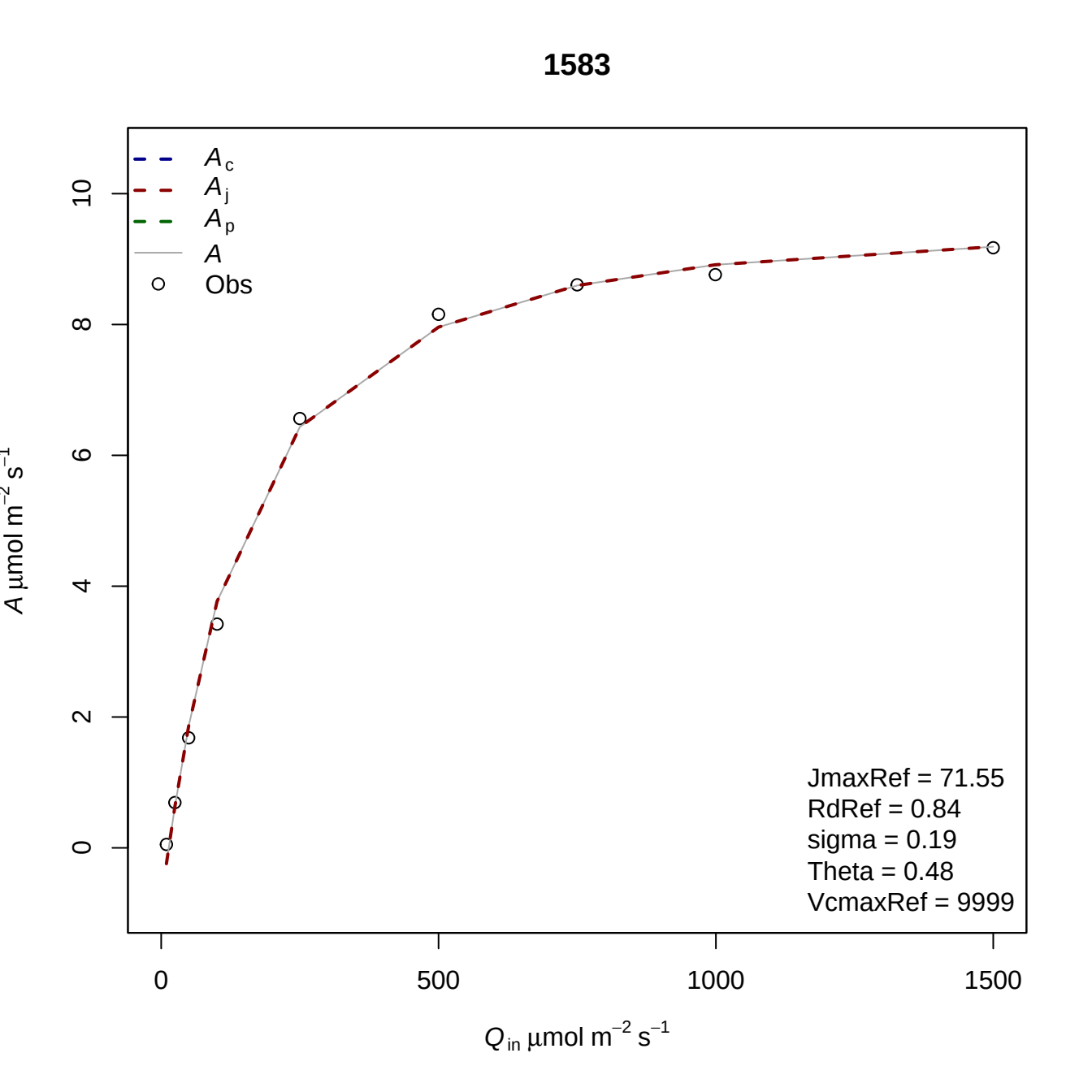

1578
A_c
A_j
A_p
A
Obs
JmaxRef = 48.15
RdRef = 0.65
sigma = 0.11
Theta = −0.11
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

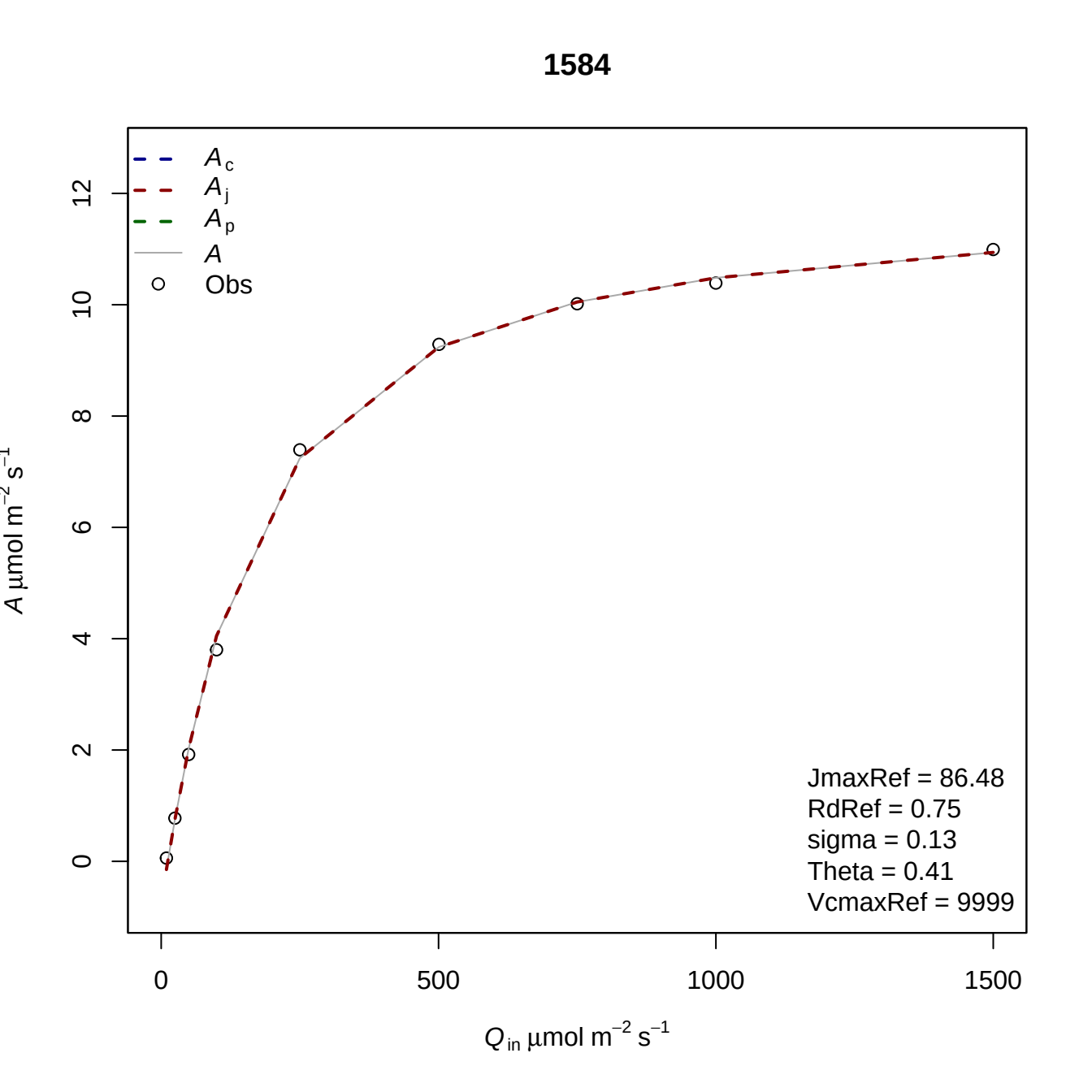

1579
A_c
A_j
A_p
A
Obs
JmaxRef = 60.15
RdRef = 0.59
sigma = 0.14
Theta = 0.04
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

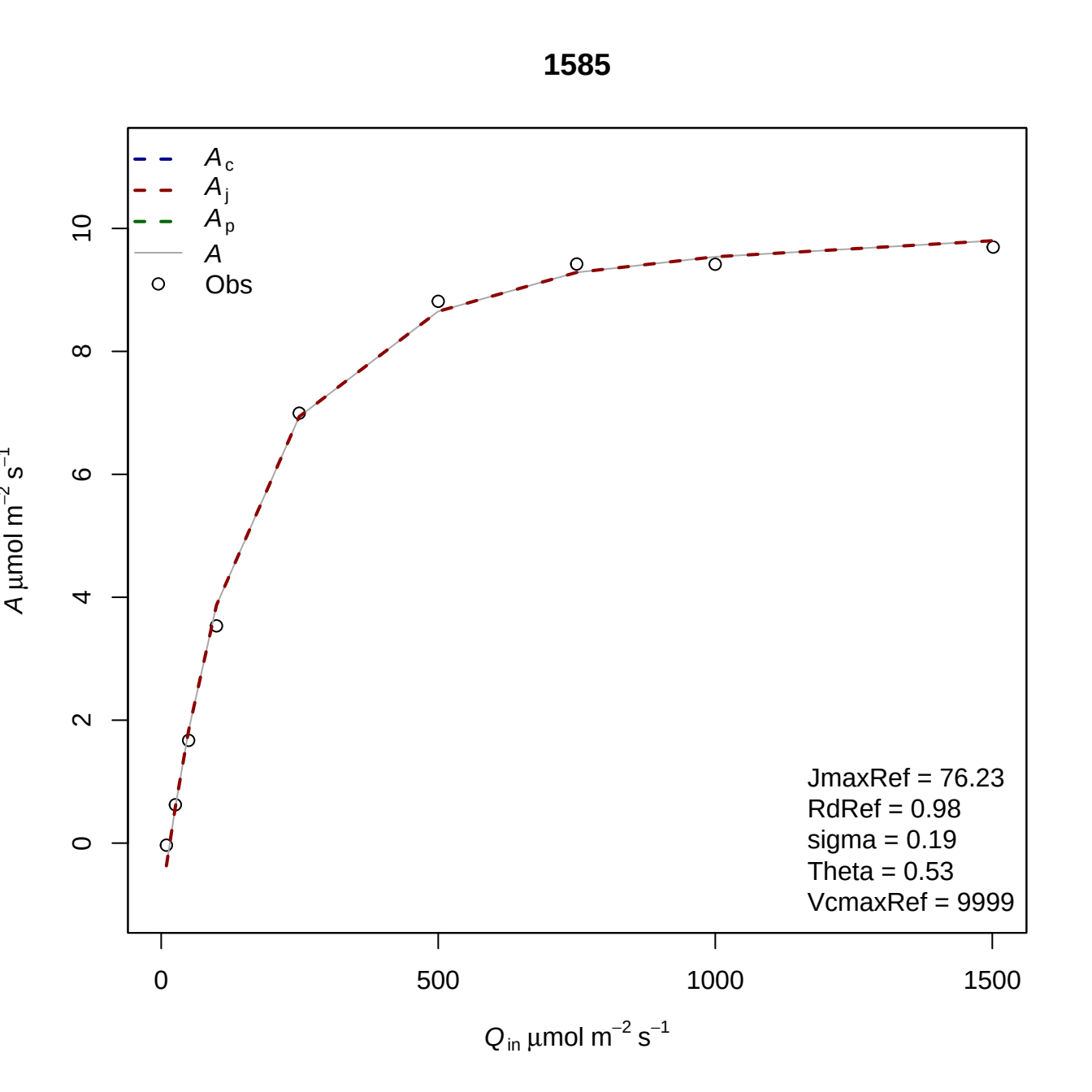

1580
A_c
A_j
A_p
A
Obs
JmaxRef = 66.09
RdRef = 0.67
sigma = 0.19
Theta = 0.45
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

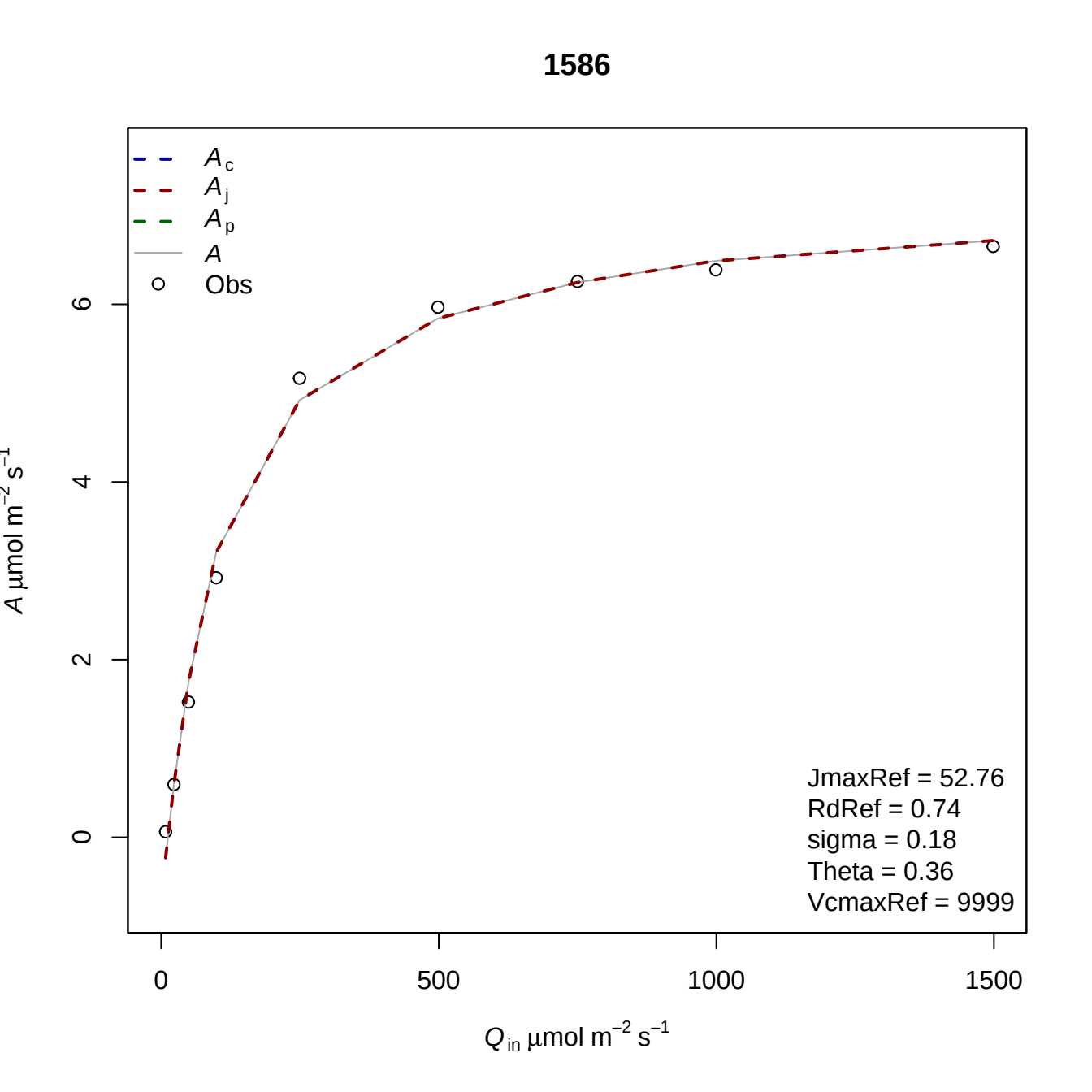

1581
A_c
A_j
A_p
A
Obs
JmaxRef = 88.6
RdRef = 0.69
sigma = 0.18
Theta = 0.42
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

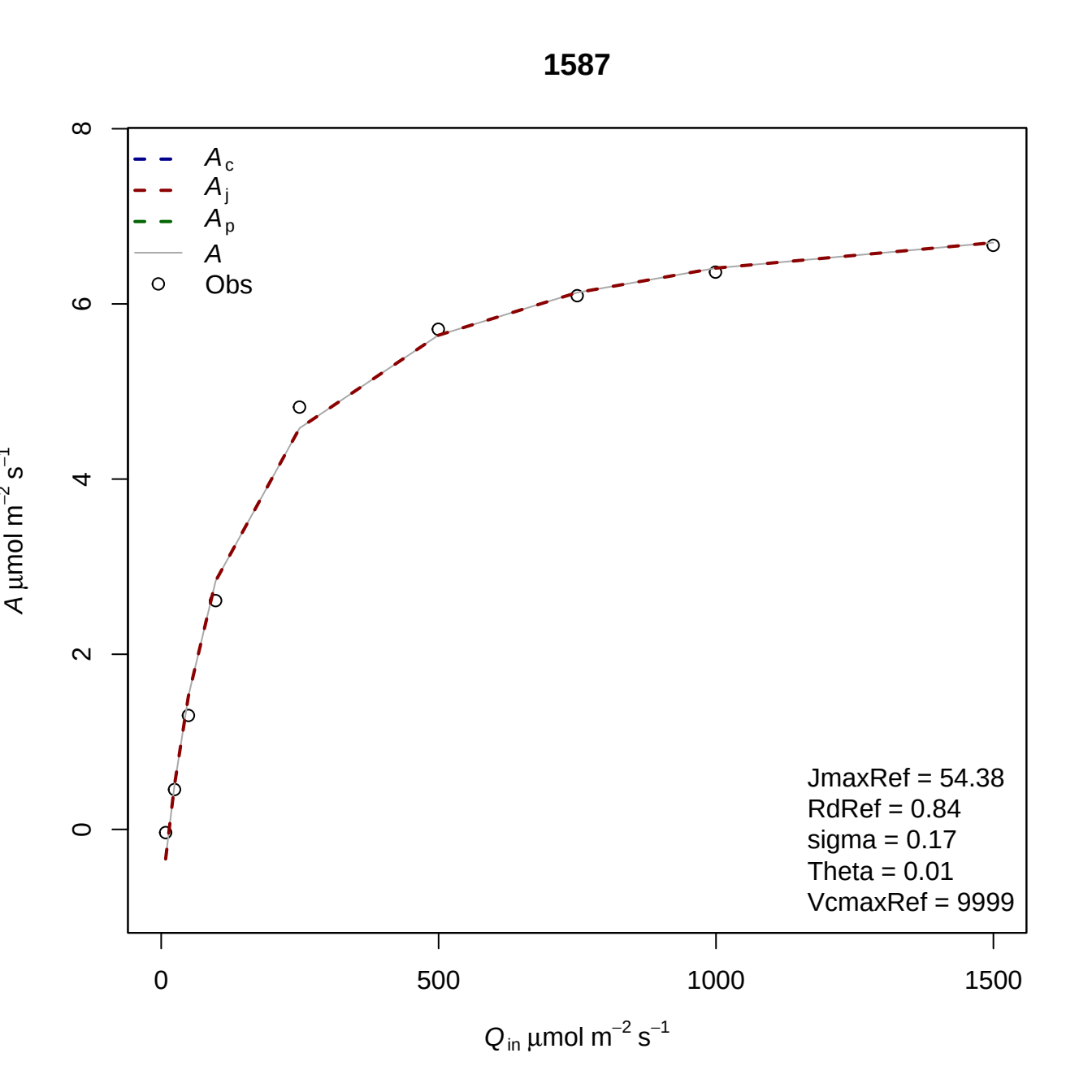

1582
A_c
A_j
A_p
A
Obs
JmaxRef = 84.84
RdRef = 0.75
sigma = 0.16
Theta = 0.39
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

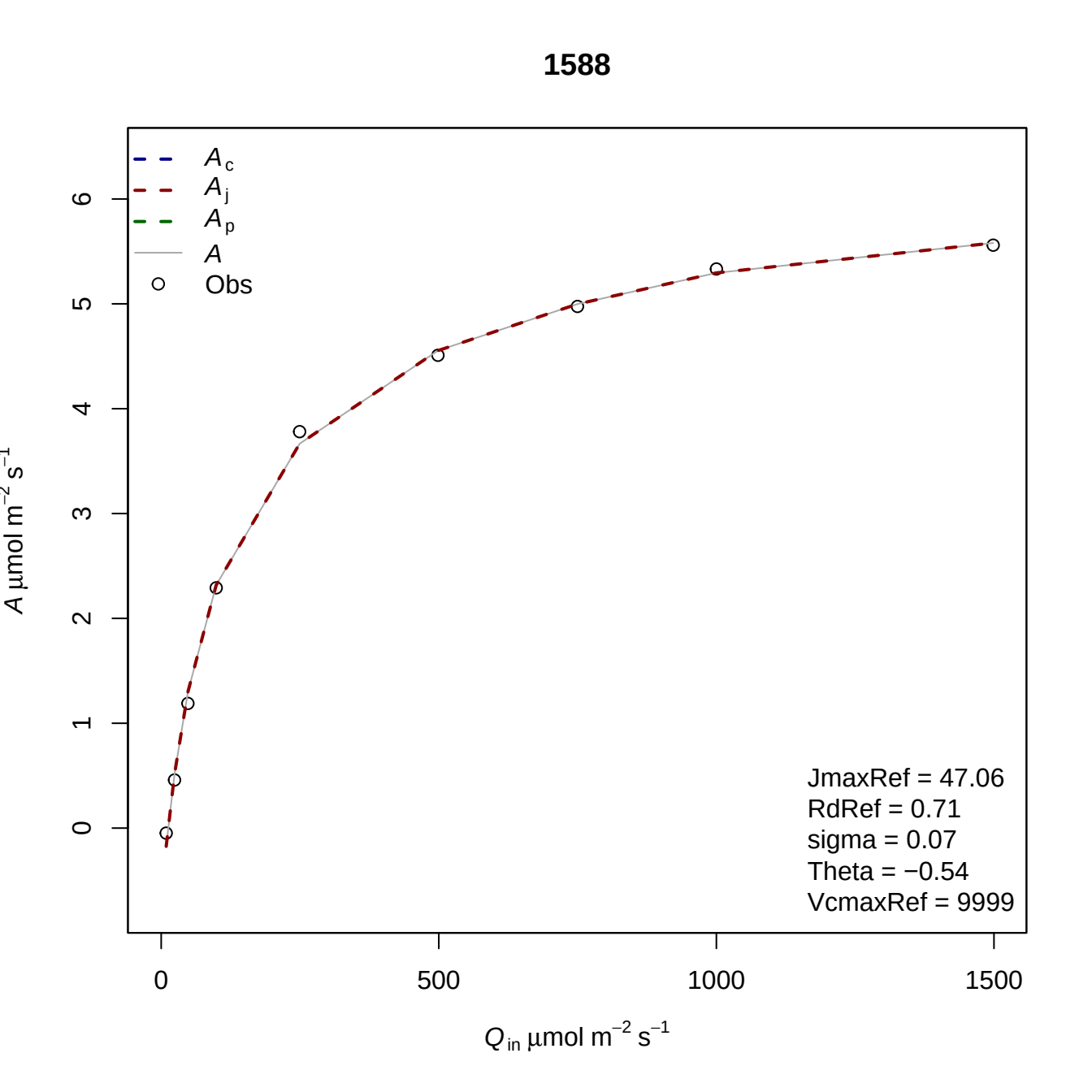

1583
A_c
A_j
A_p
A
Obs
JmaxRef = 71.55
RdRef = 0.84
sigma = 0.19
Theta = 0.48
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

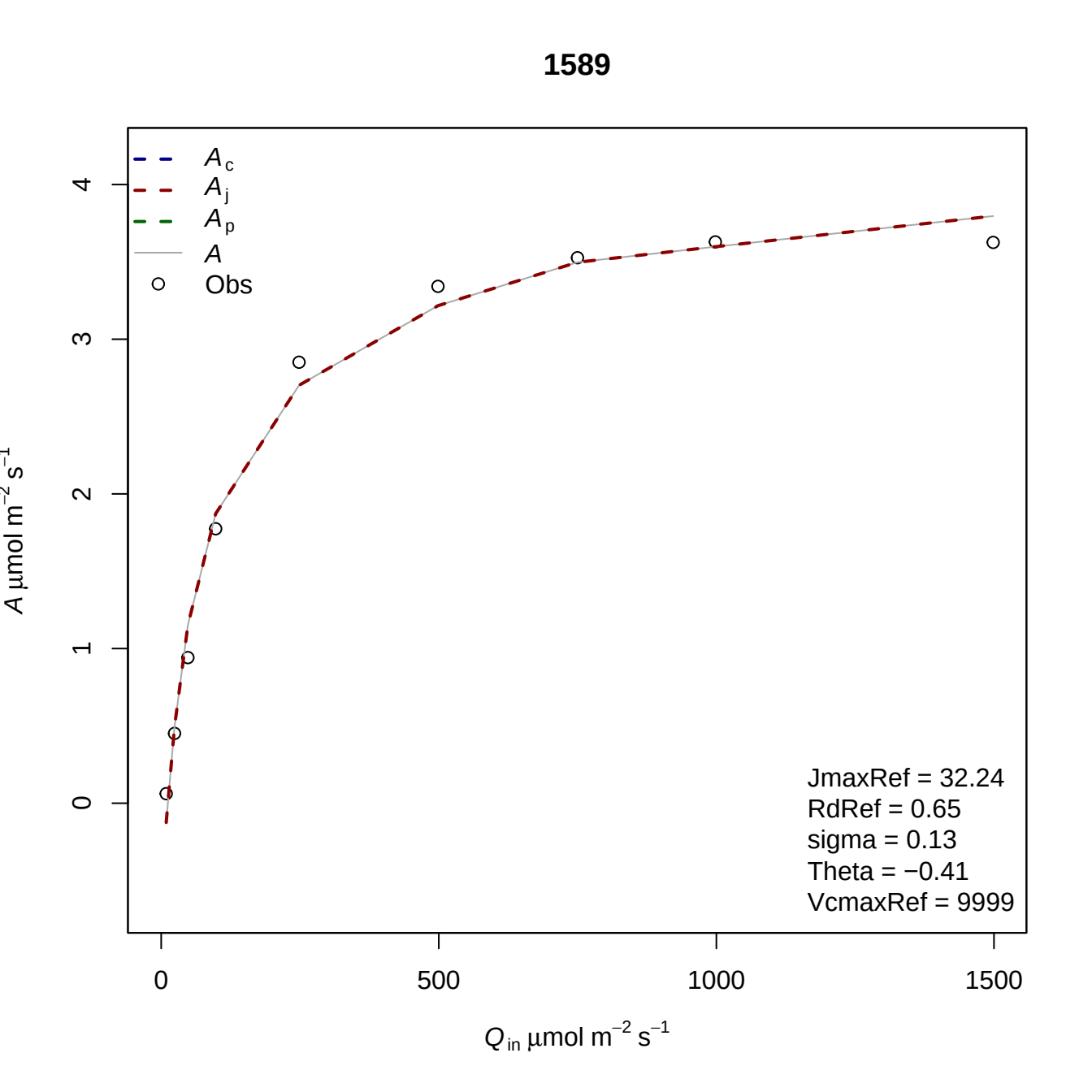

1584
A_c
A_j
A_p
A
Obs
JmaxRef = 86.48
RdRef = 0.75
sigma = 0.13
Theta = 0.41
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

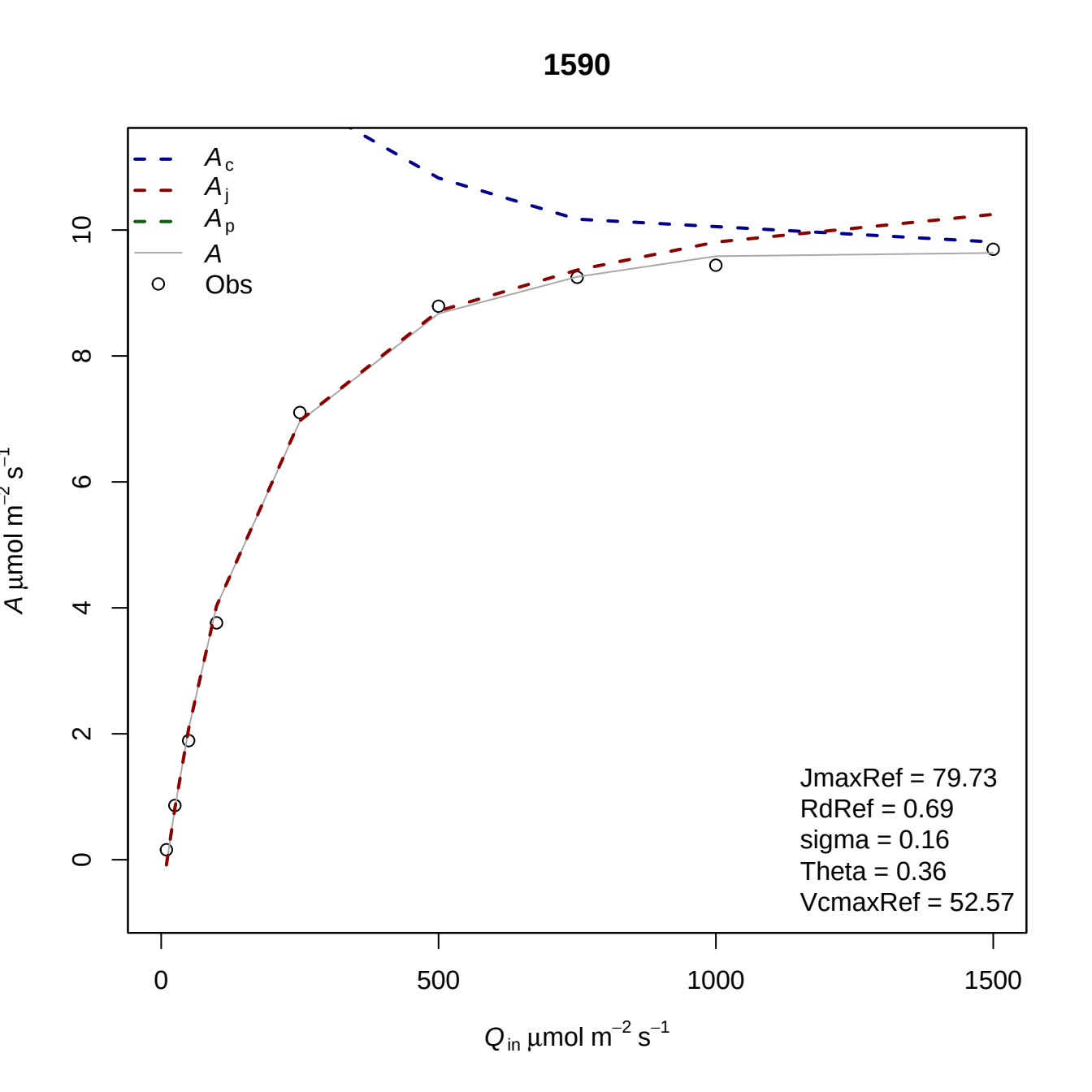

1585
A_c
A_j
A_p
A
Obs
JmaxRef = 76.23
RdRef = 0.98
sigma = 0.19
Theta = 0.53
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

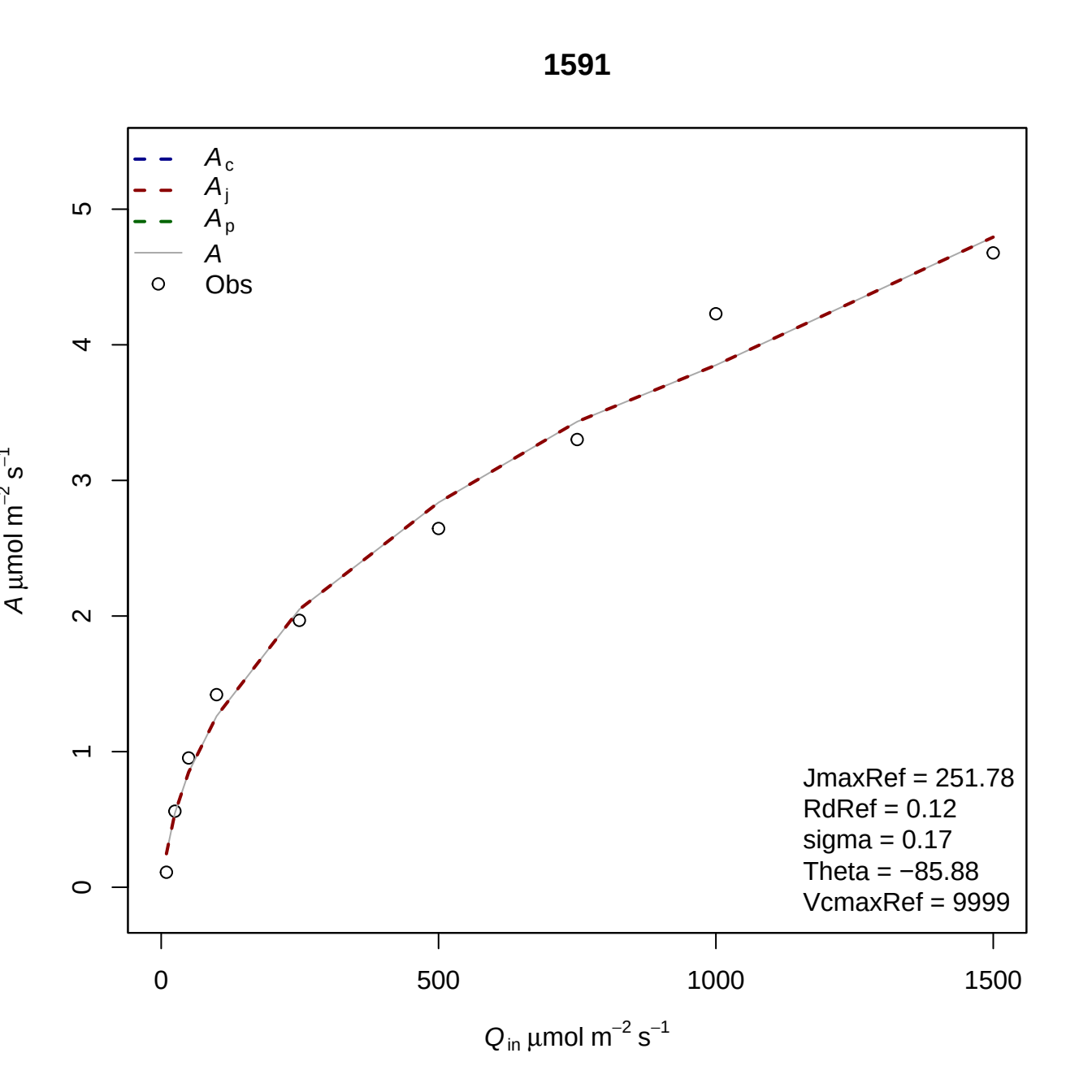

1586
A_c
A_j
A_p
A
Obs
JmaxRef = 52.76
RdRef = 0.74
sigma = 0.18
Theta = 0.36
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

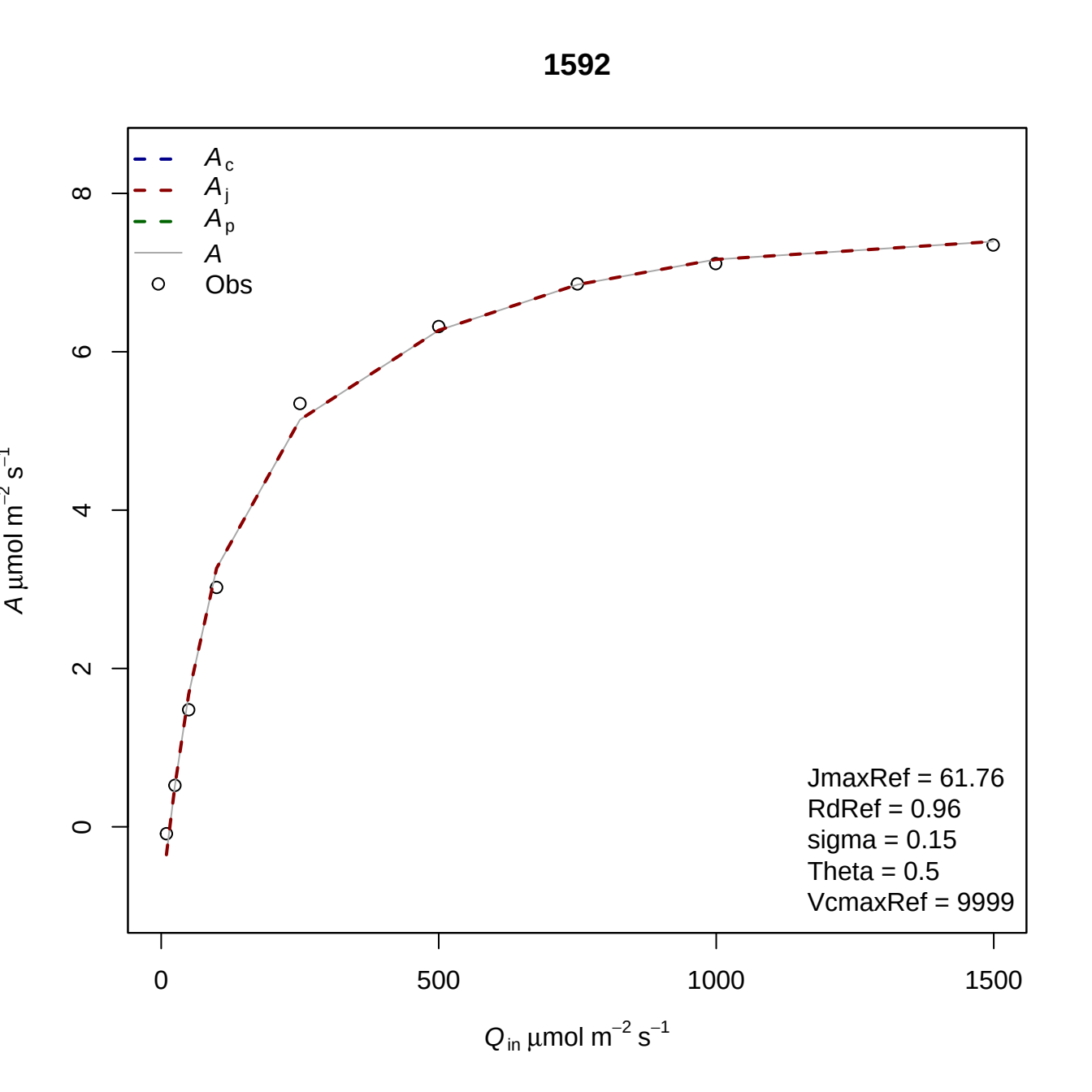

1587
A_c
A_j
A_p
A
Obs
JmaxRef = 54.38
RdRef = 0.84
sigma = 0.17
Theta = 0.01
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

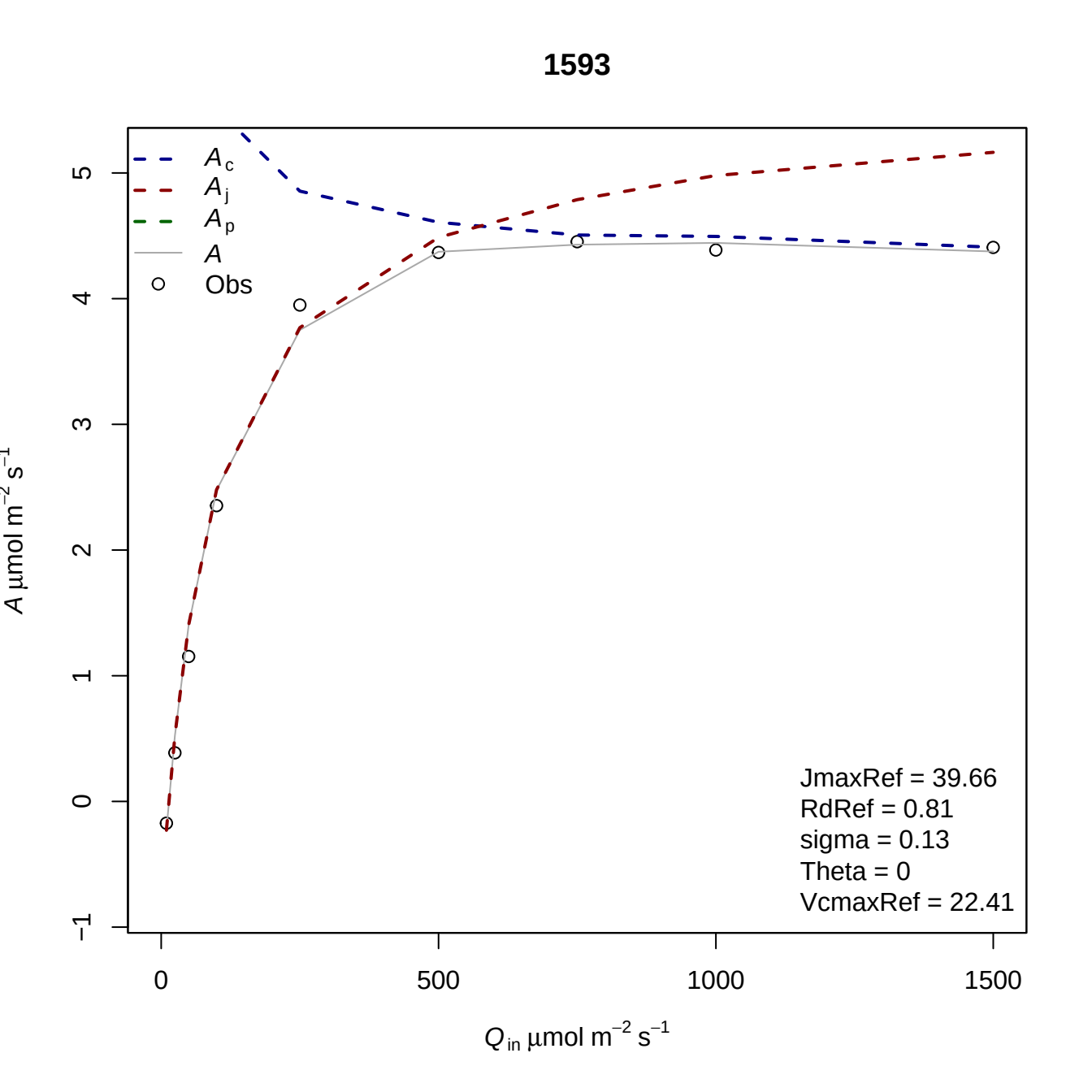

1588
A_c
A_j
A_p
A
Obs
JmaxRef = 47.06
RdRef = 0.71
sigma = 0.07
Theta = −0.54
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

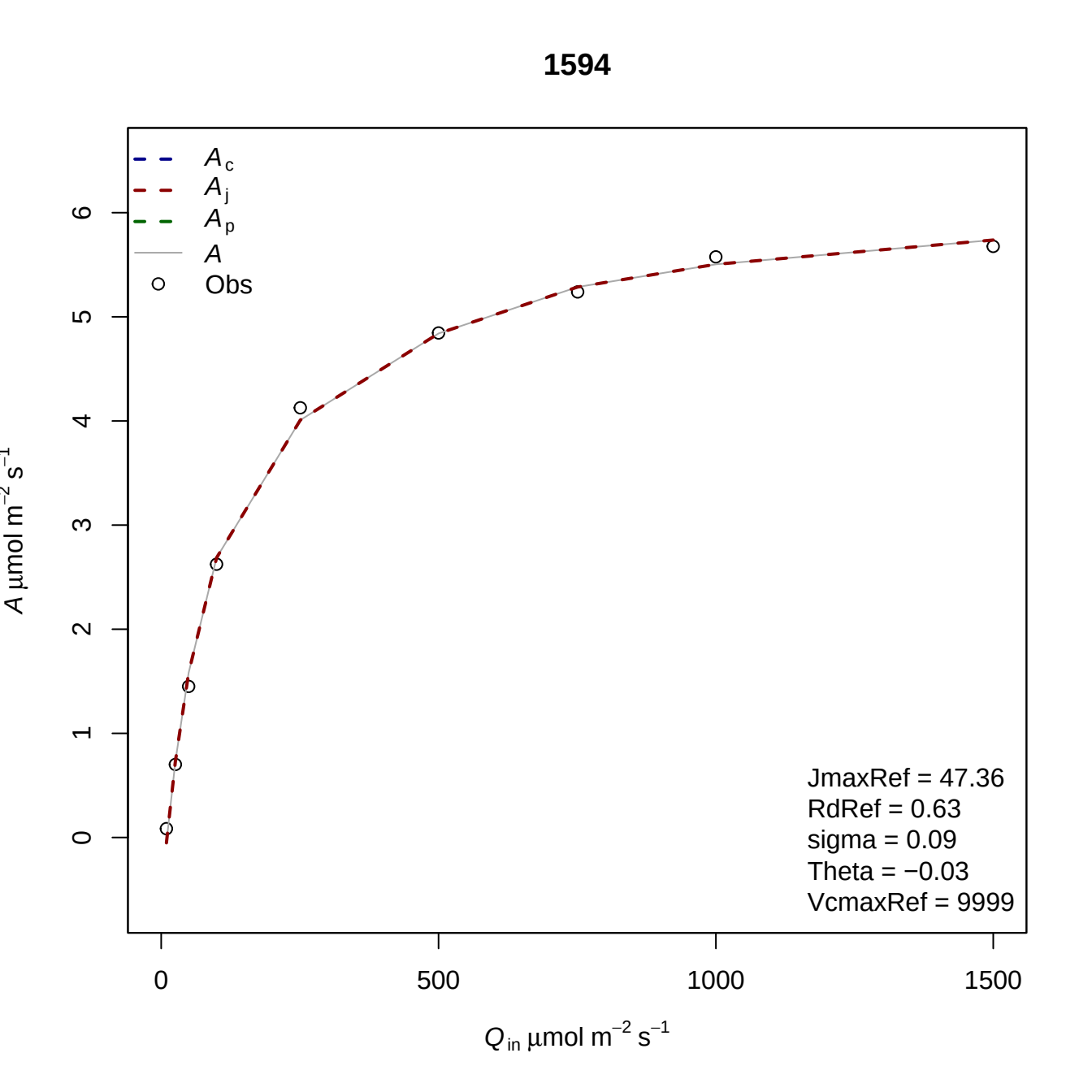

1589
A_c
A_j
A_p
A
Obs
JmaxRef = 32.24
RdRef = 0.65
sigma = 0.13
Theta = −0.41
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

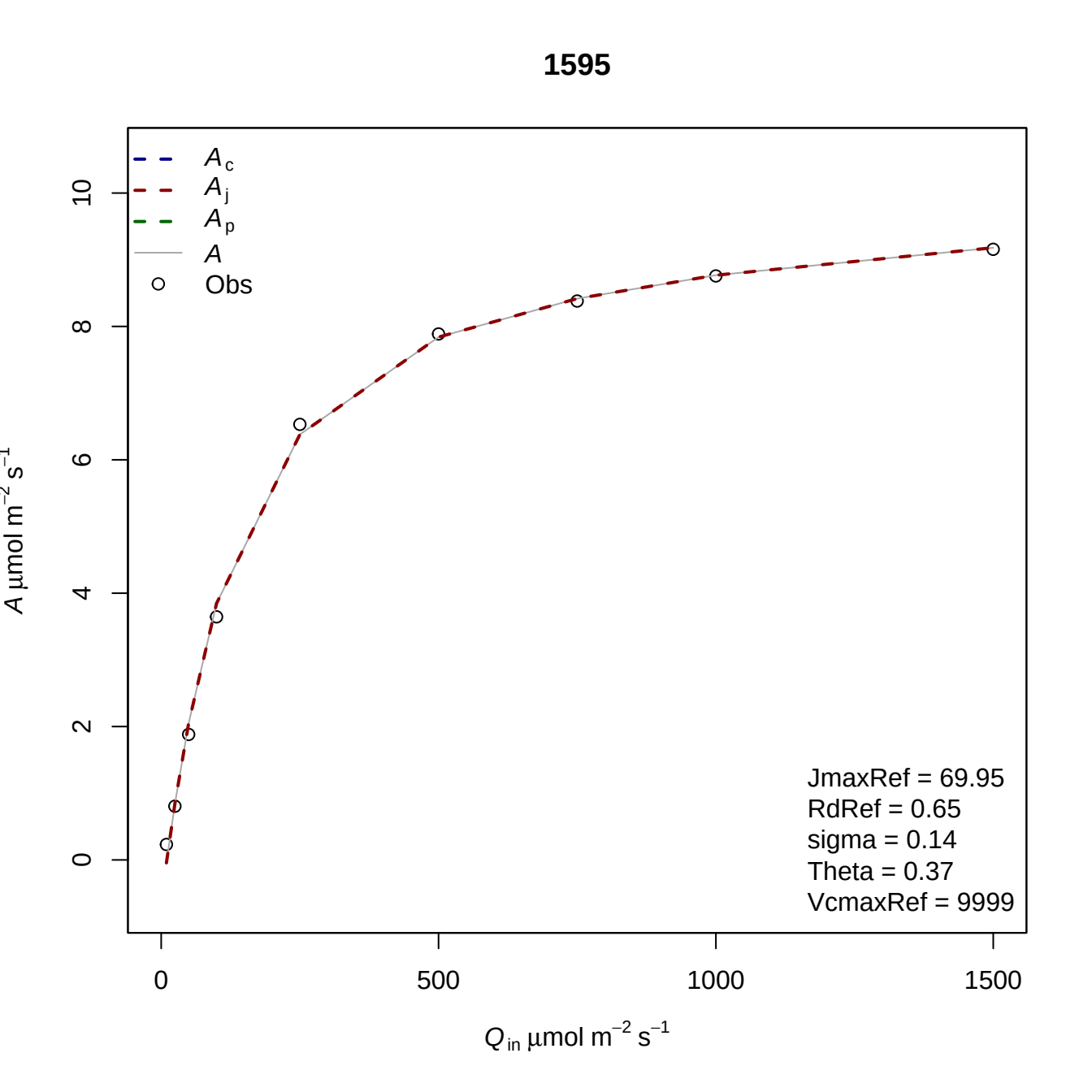

1590
A_c
A_j
A_p
A
Obs
JmaxRef = 79.73
RdRef = 0.69
sigma = 0.16
Theta = 0.36
VcmaxRef = 52.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

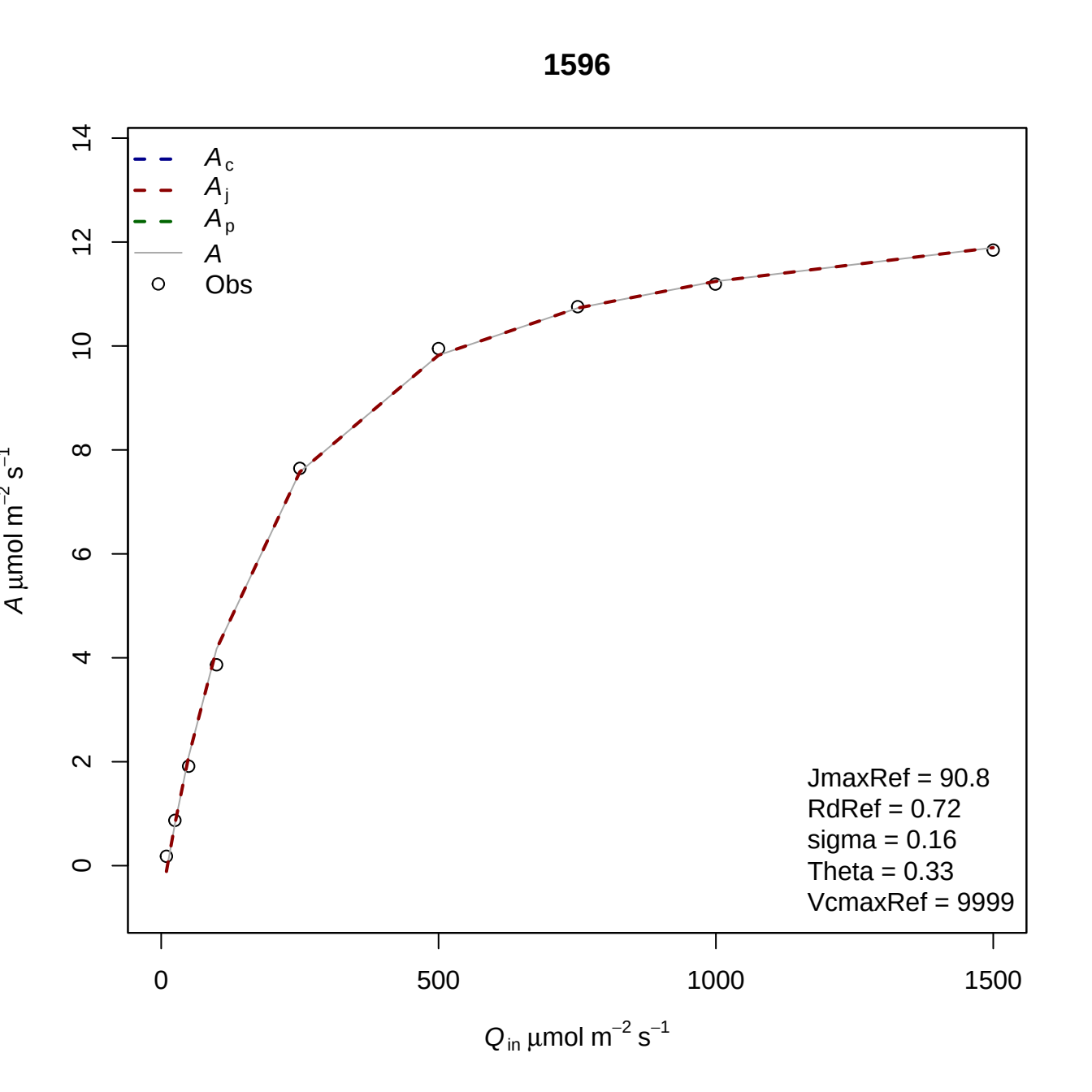

1591
A_c
A_j
A_p
A
Obs
JmaxRef = 251.78
RdRef = 0.12
sigma = 0.17
Theta = −85.88
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

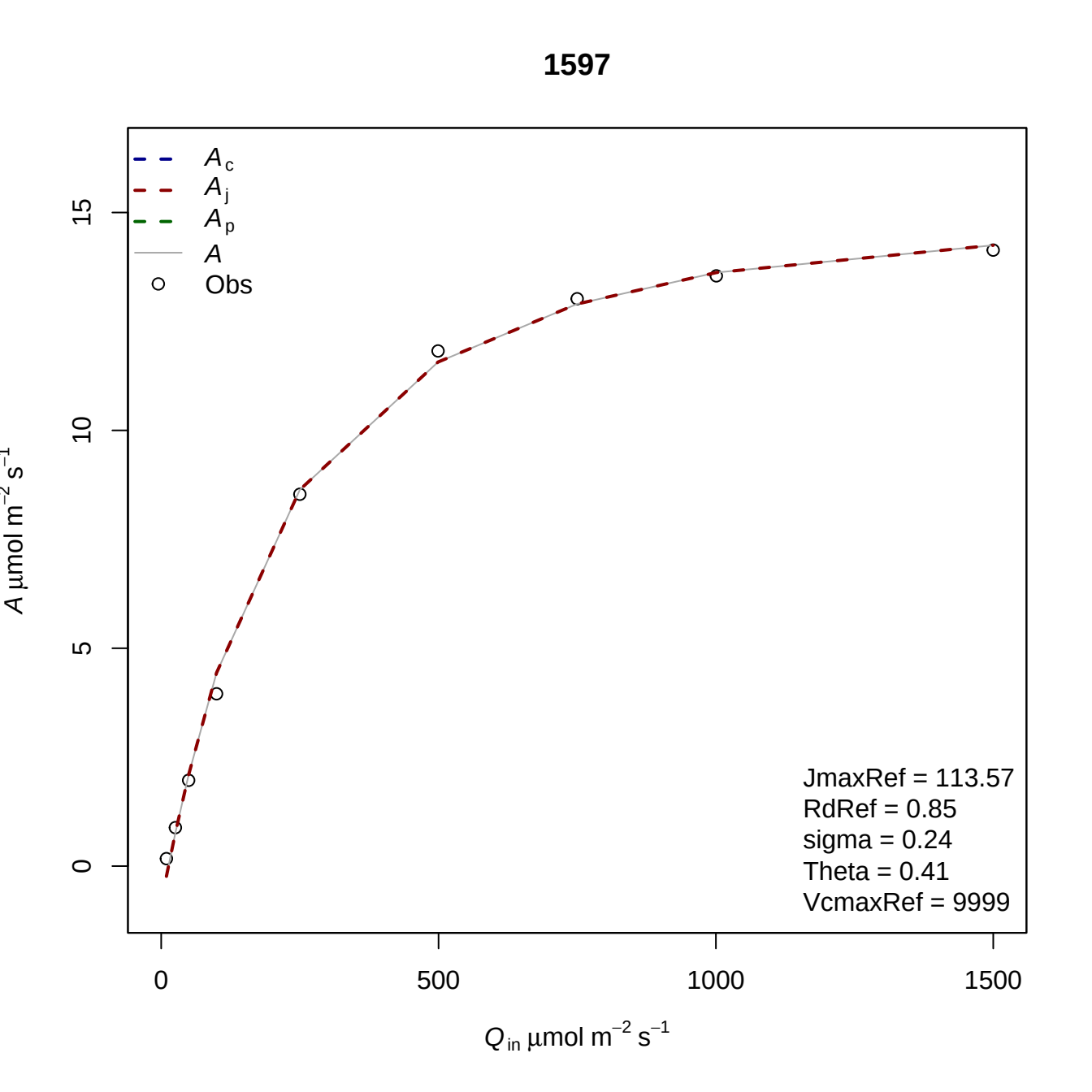

1592
A_c
A_j
A_p
A
Obs
JmaxRef = 61.76
RdRef = 0.96
sigma = 0.15
Theta = 0.5
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

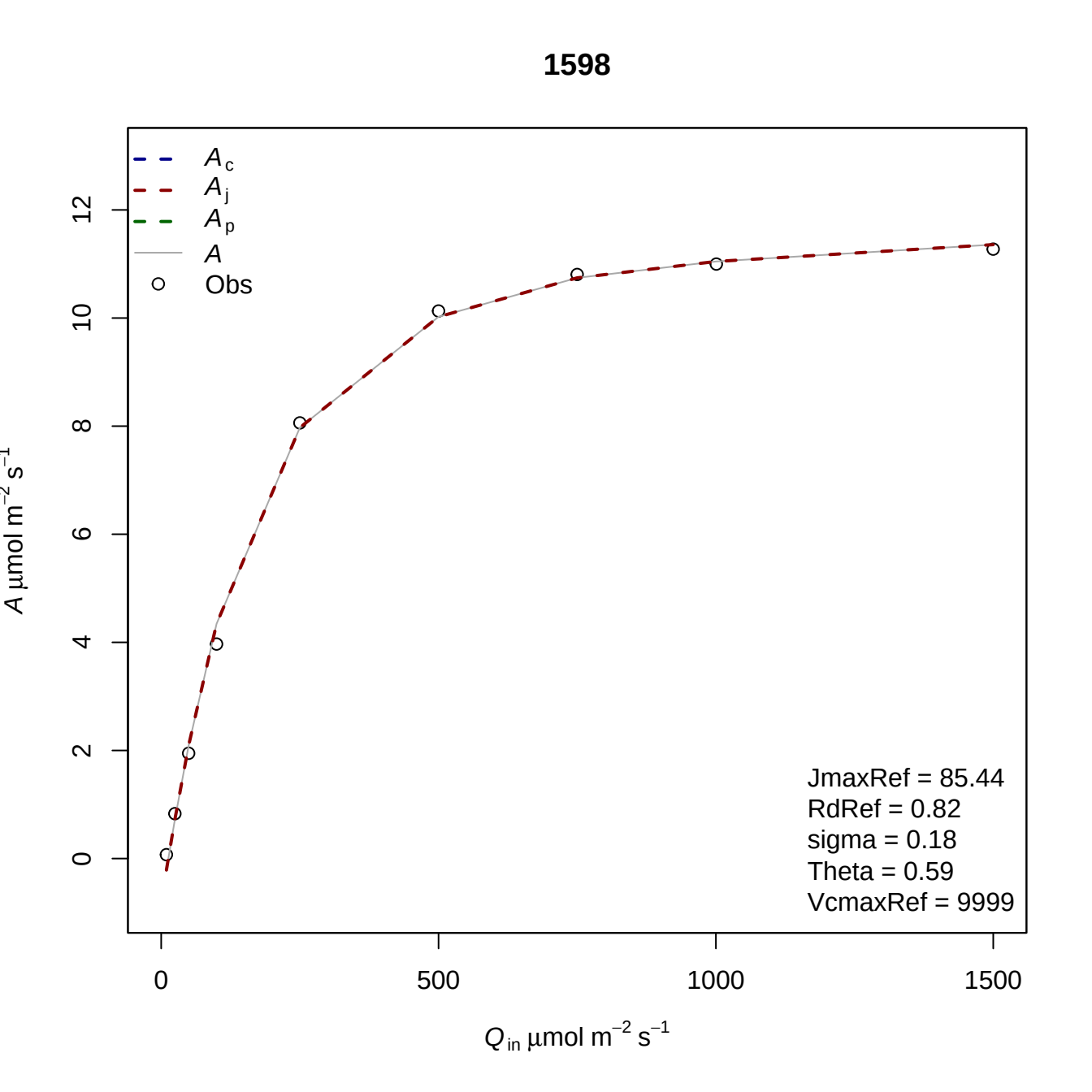

1593
A_c
A_j
A_p
A
Obs
JmaxRef = 39.66
RdRef = 0.81
sigma = 0.13
Theta = 0
VcmaxRef = 22.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

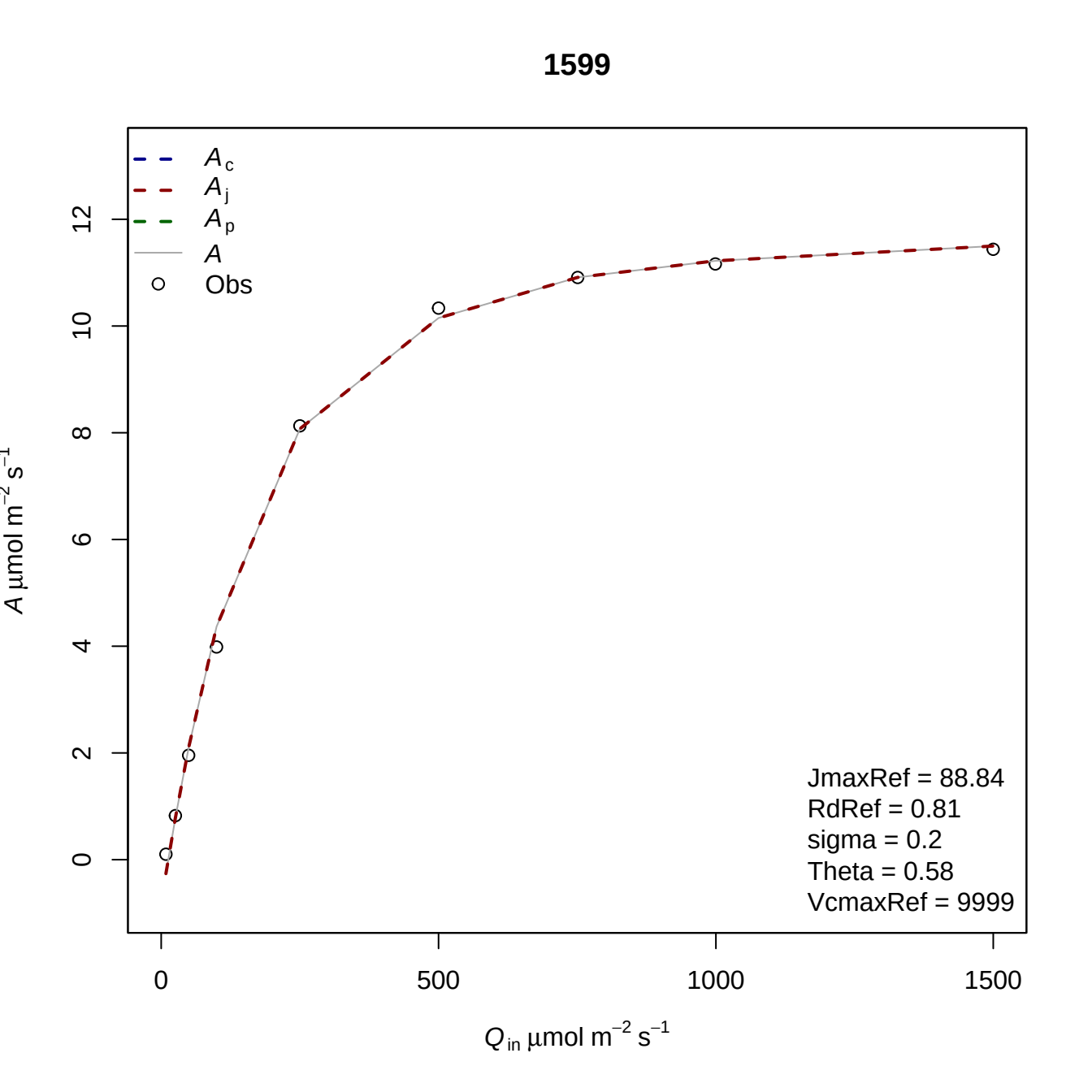

1594
A_c
A_j
A_p
A
Obs
JmaxRef = 47.36
RdRef = 0.63
sigma = 0.09
Theta = −0.03
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

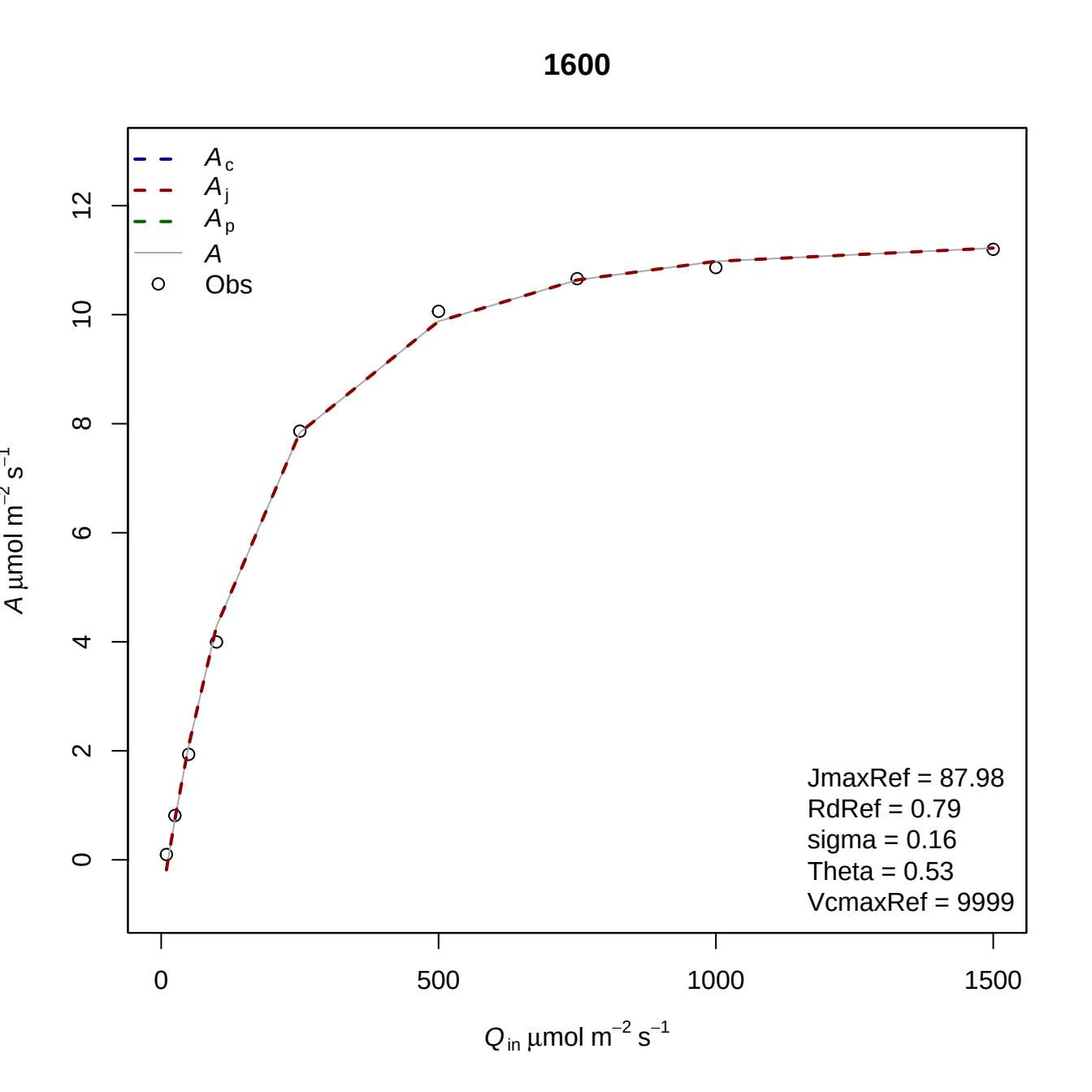

1595
A_c
A_j
A_p
A
Obs
JmaxRef = 69.95
RdRef = 0.65
sigma = 0.14
Theta = 0.37
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

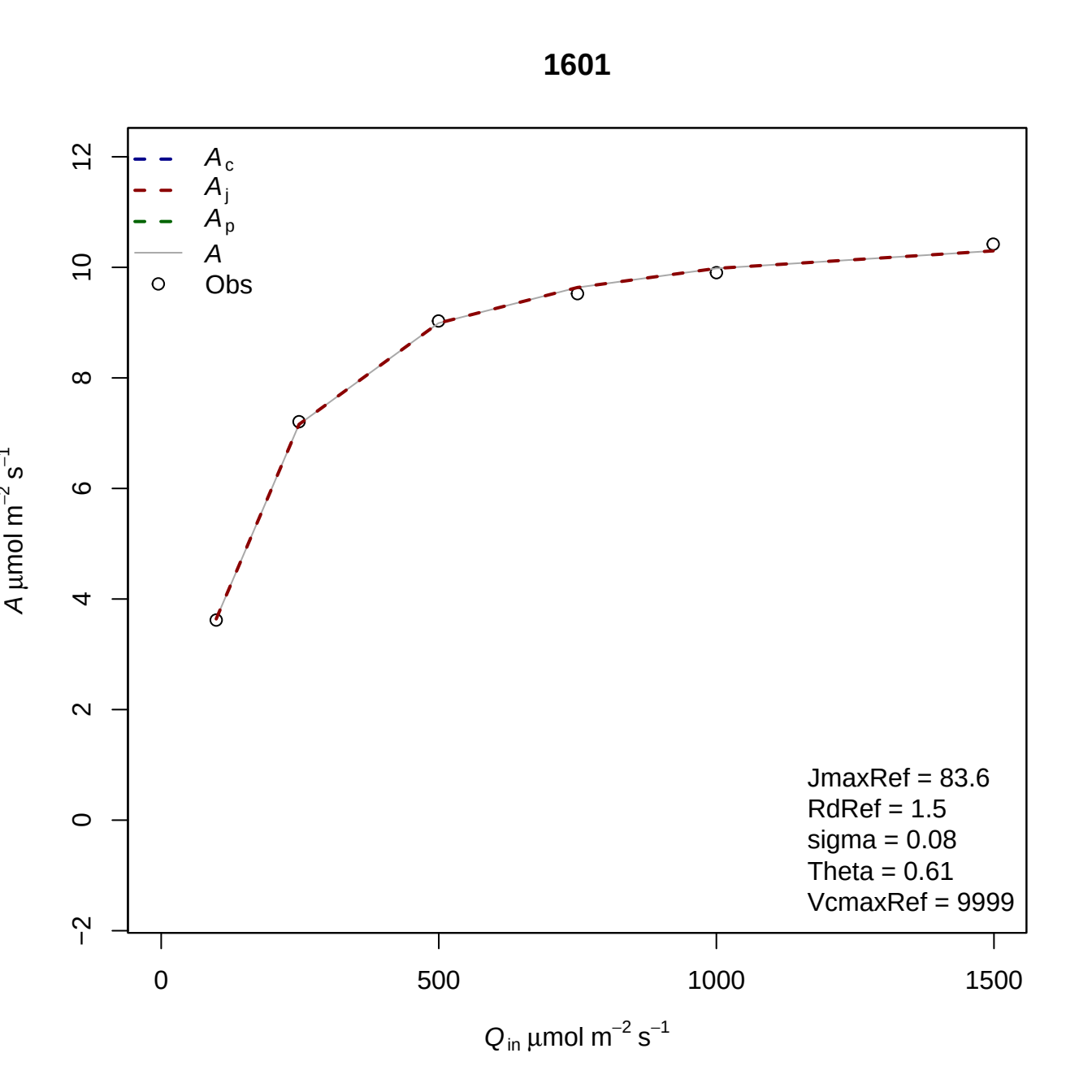

1596
A_c
A_j
A_p
A
Obs
JmaxRef = 90.8
RdRef = 0.72
sigma = 0.16
Theta = 0.33
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

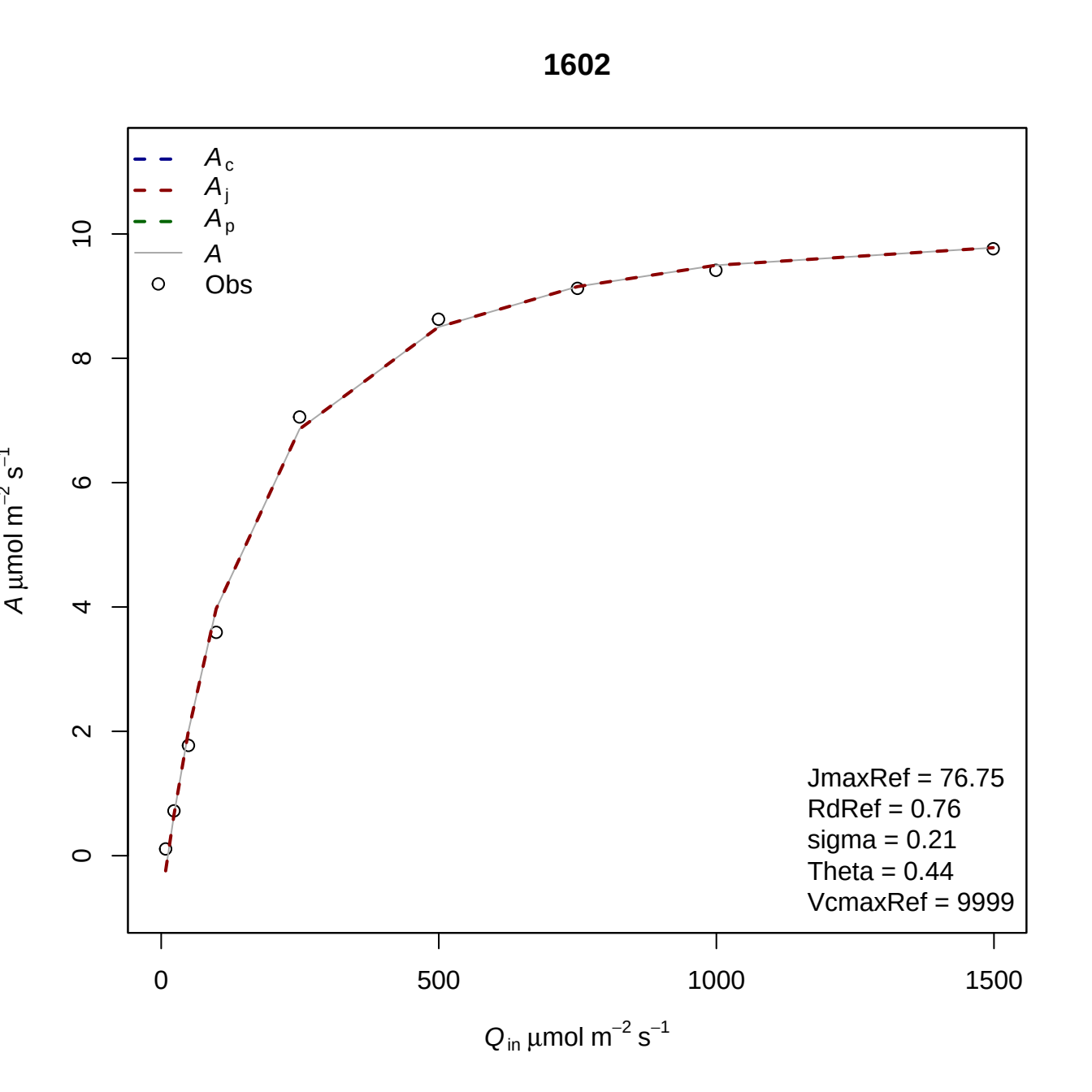

1597
A_c
A_j
A_p
A
Obs
JmaxRef = 113.57
RdRef = 0.85
sigma = 0.24
Theta = 0.41
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

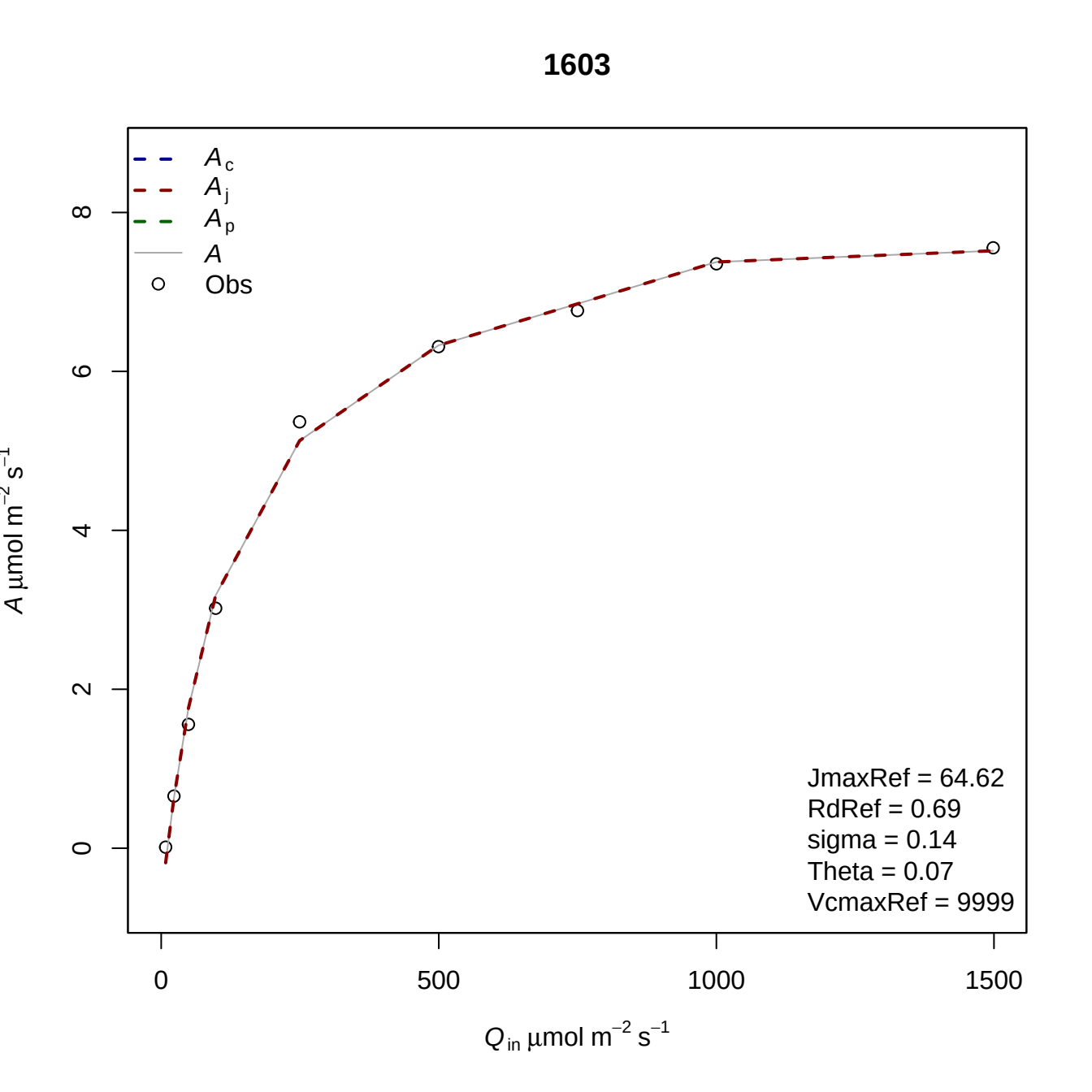

1598
A_c
A_j
A_p
A
Obs
JmaxRef = 85.44
RdRef = 0.82
sigma = 0.18
Theta = 0.59
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

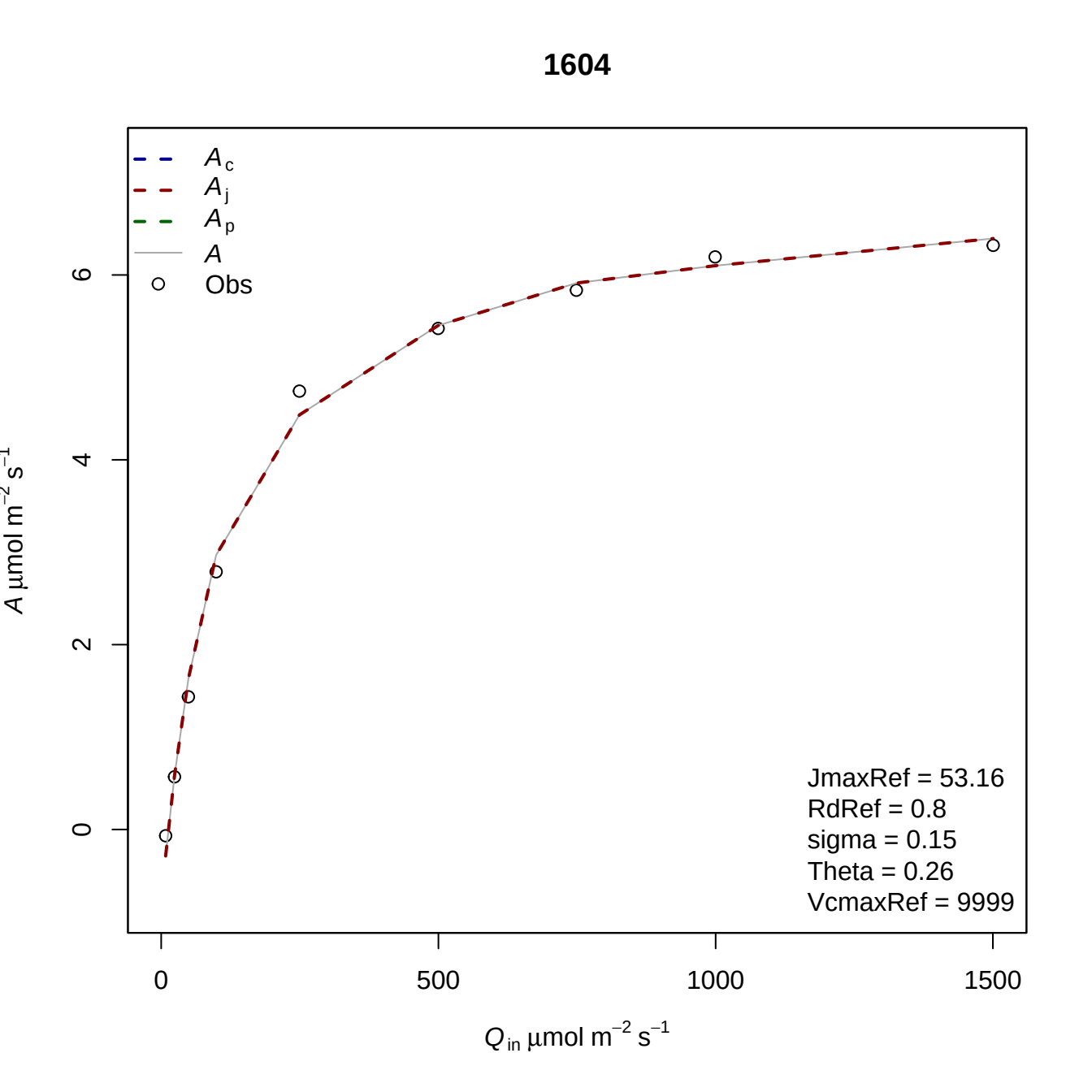

1599
A_c
A_j
A_p
A
Obs
JmaxRef = 88.84
RdRef = 0.81
sigma = 0.2
Theta = 0.58
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

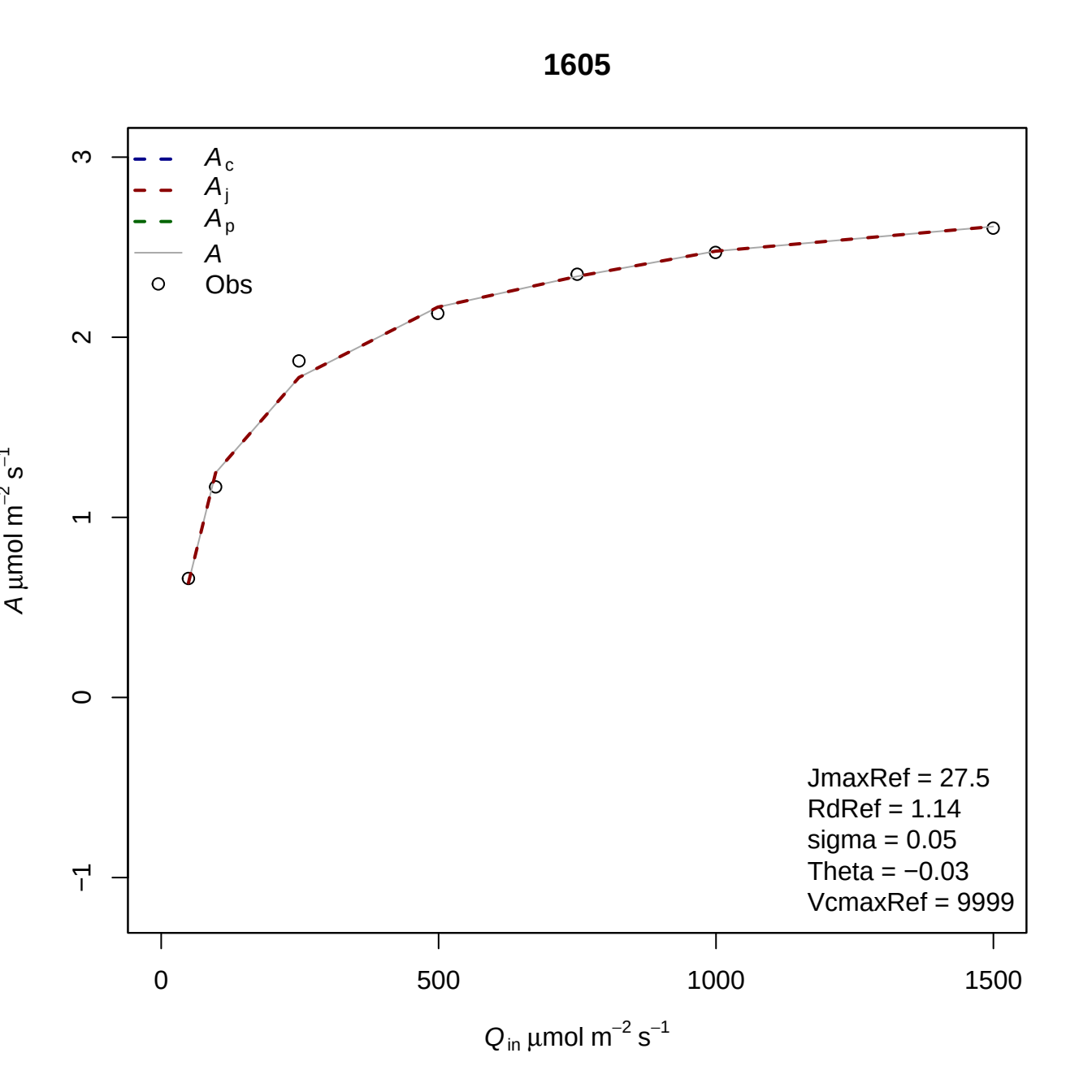

1600
A_c
A_j
A_p
A
Obs
JmaxRef = 87.98
RdRef = 0.79
sigma = 0.16
Theta = 0.53
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

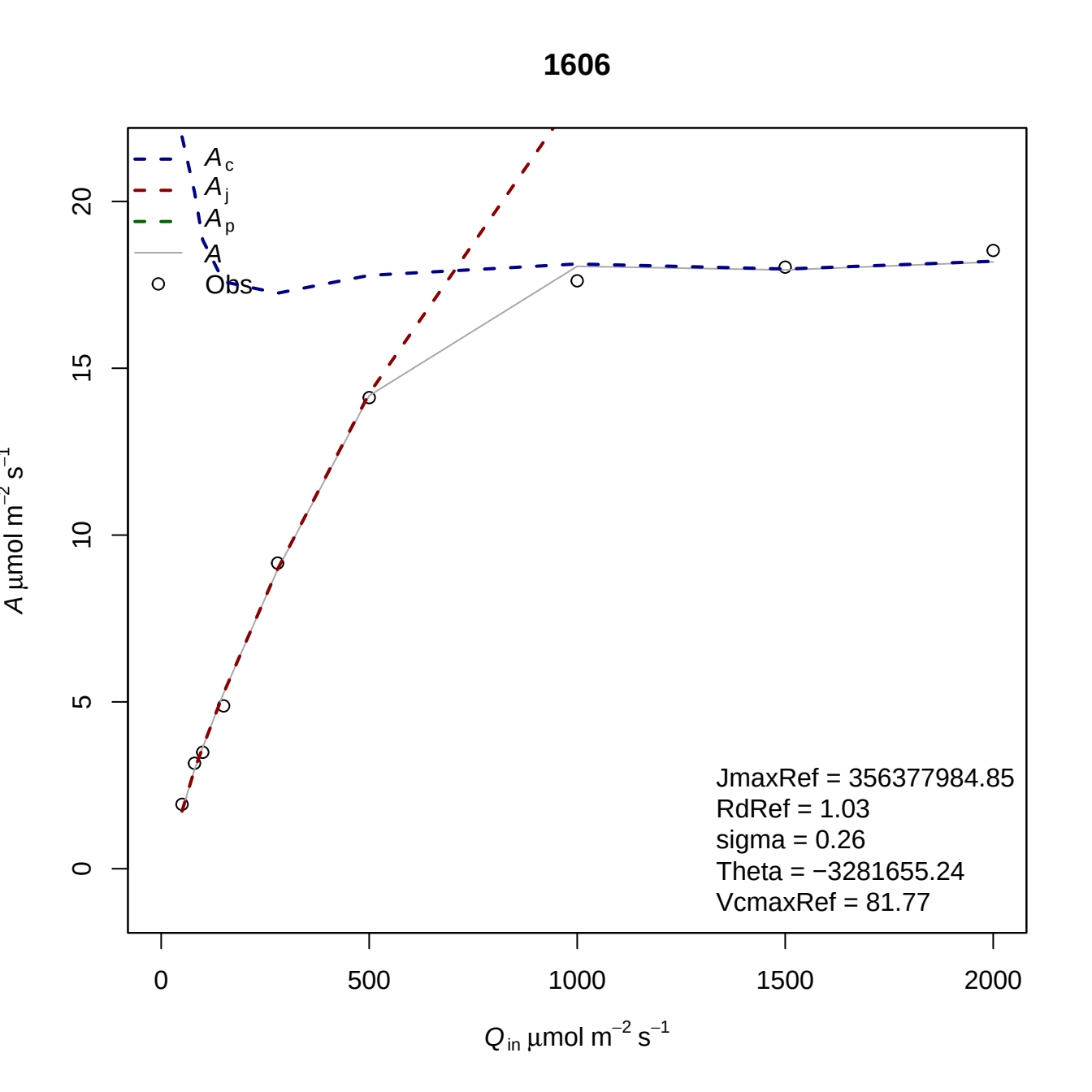

1601
A_c
A_j
A_p
A
Obs
JmaxRef = 83.6
RdRef = 1.5
sigma = 0.08
Theta = 0.61
VcmaxRef = 9999
12
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

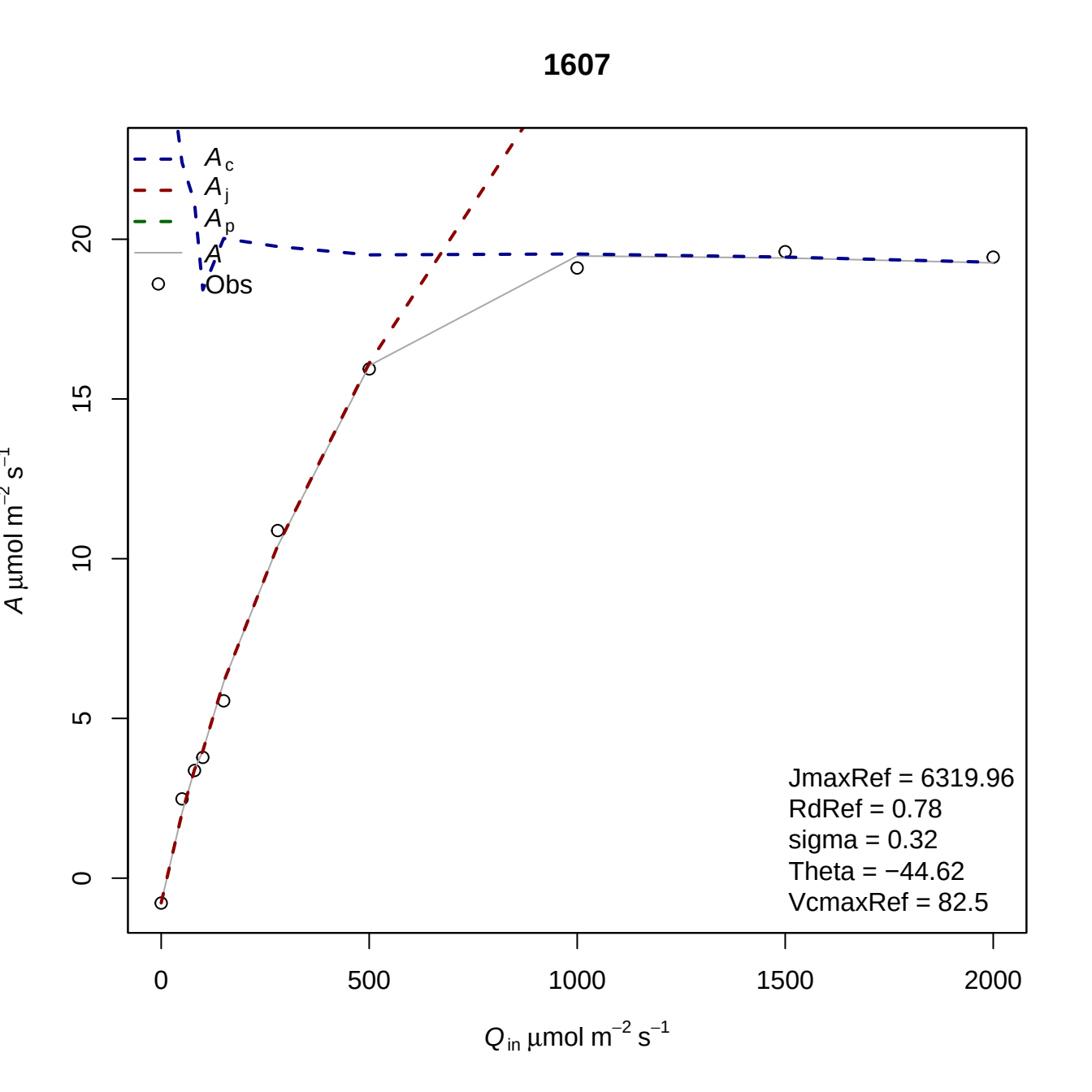

1602
A_c
A_j
A_p
A
Obs
JmaxRef = 76.75
RdRef = 0.76
sigma = 0.21
Theta = 0.44
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

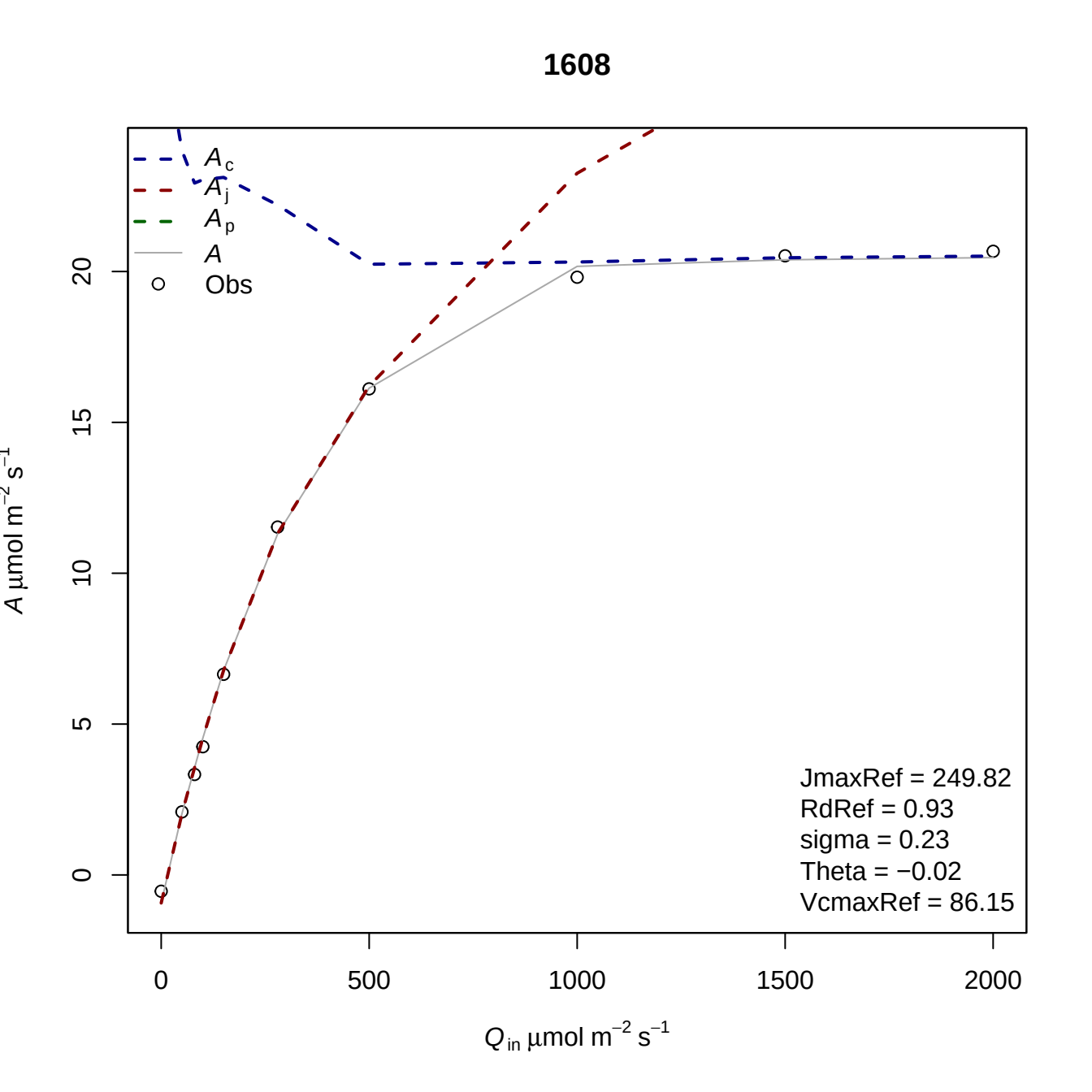

1603
A_c
A_j
A_p
A
Obs
JmaxRef = 64.62
RdRef = 0.69
sigma = 0.14
Theta = 0.07
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

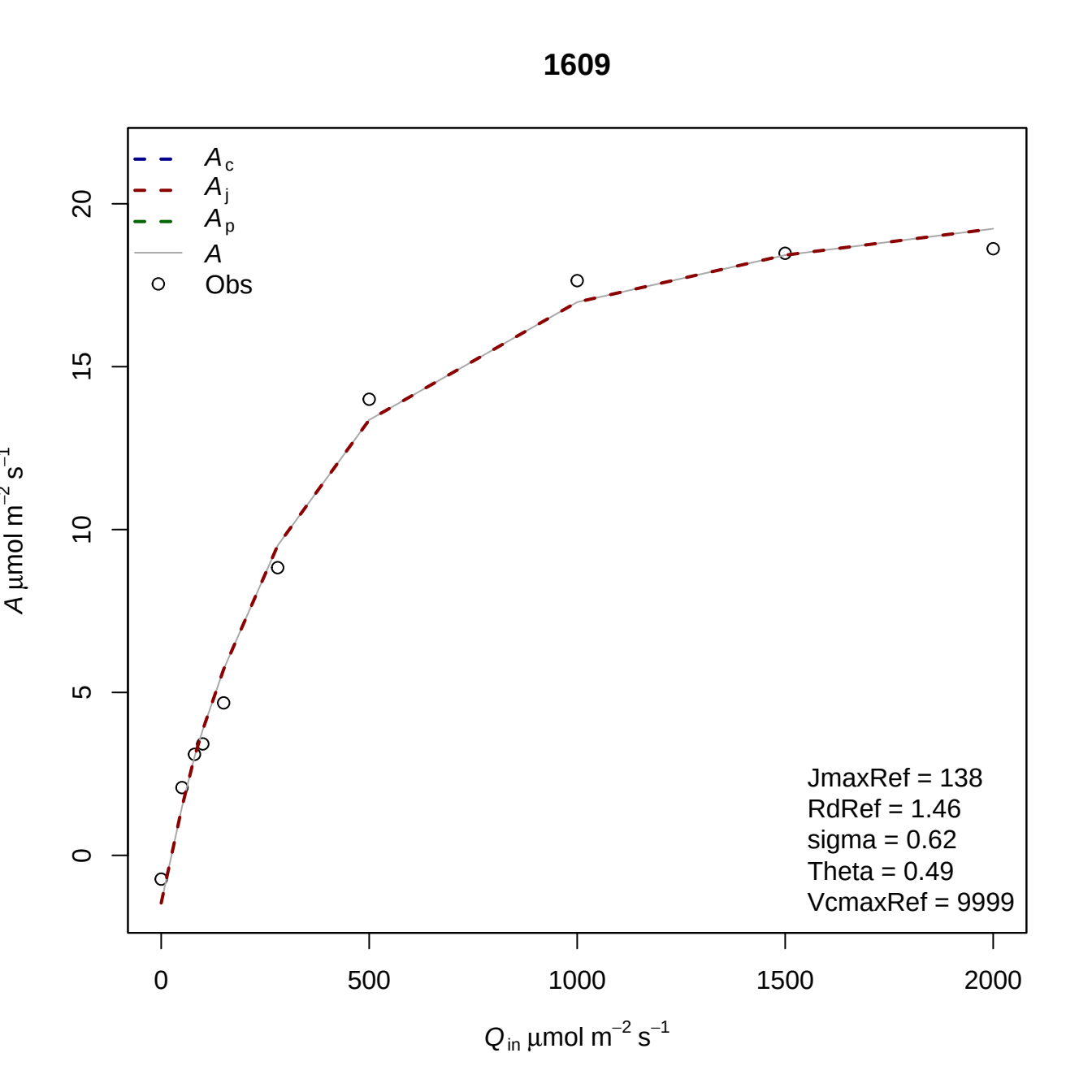

1604
A_c
A_j
A_p
A
Obs
JmaxRef = 53.16
RdRef = 0.8
sigma = 0.15
Theta = 0.26
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

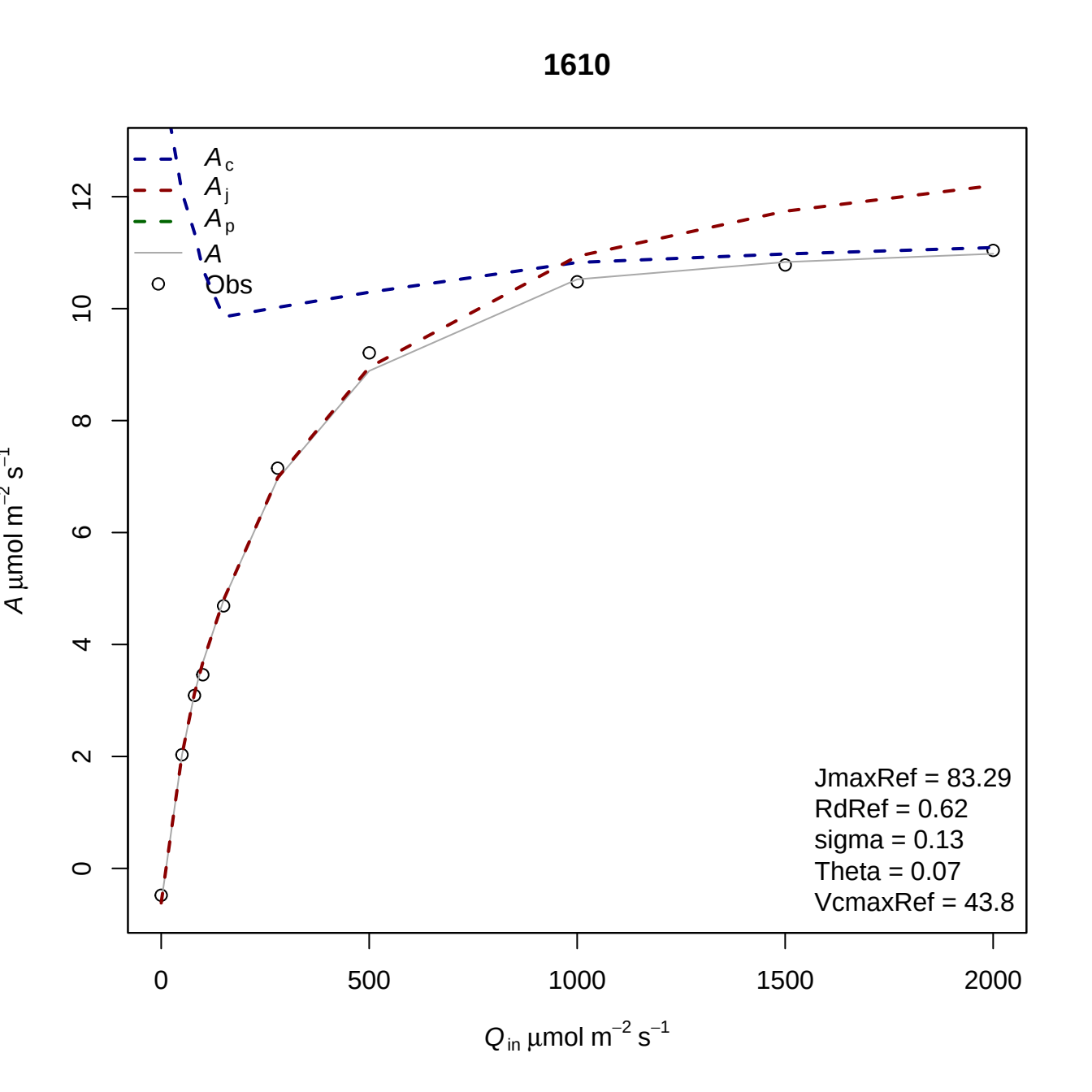

1605
A_c
A_j
A_p
A
Obs
JmaxRef = 27.5
RdRef = 1.14
sigma = 0.05
Theta = −0.03
VcmaxRef = 9999
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

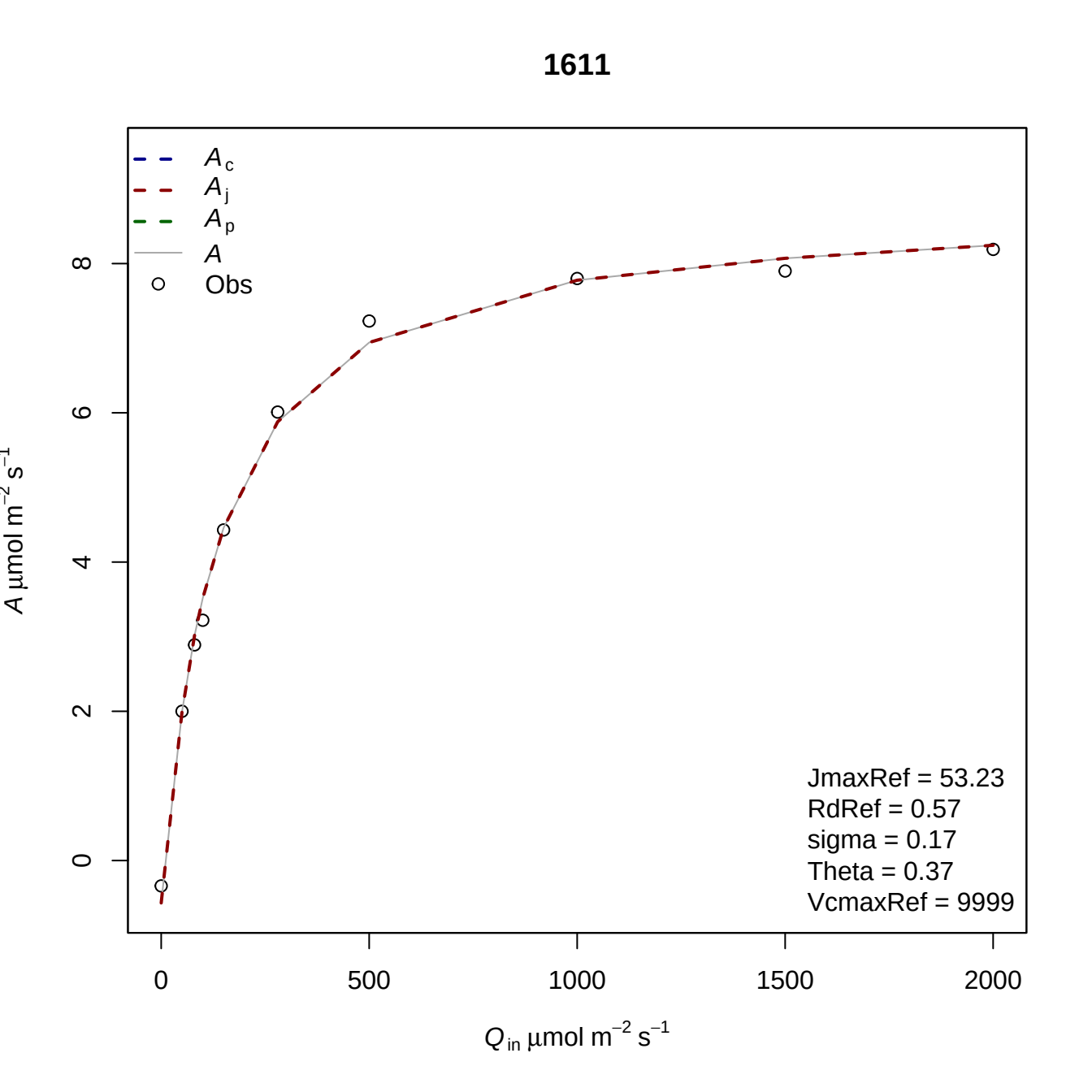

1606
A_c
A_j
A_p
A
Obs
JmaxRef = 356377984.85
RdRef = 1.03
sigma = 0.26
Theta = −3281655.24
VcmaxRef = 81.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

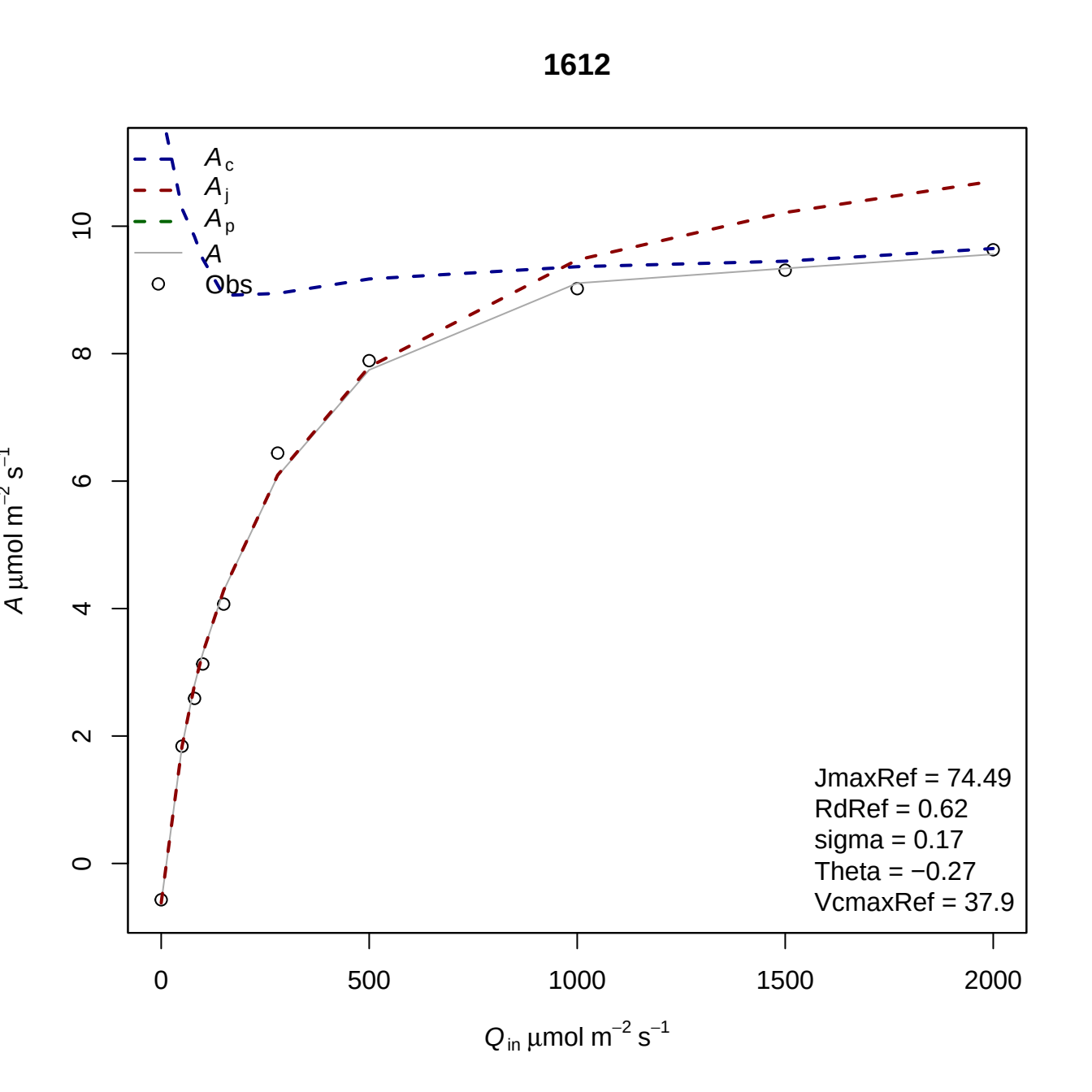

1607
A_c
A_j
A_p
A
Obs
JmaxRef = 6319.96
RdRef = 0.78
sigma = 0.32
Theta = −44.62
VcmaxRef = 82.5
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

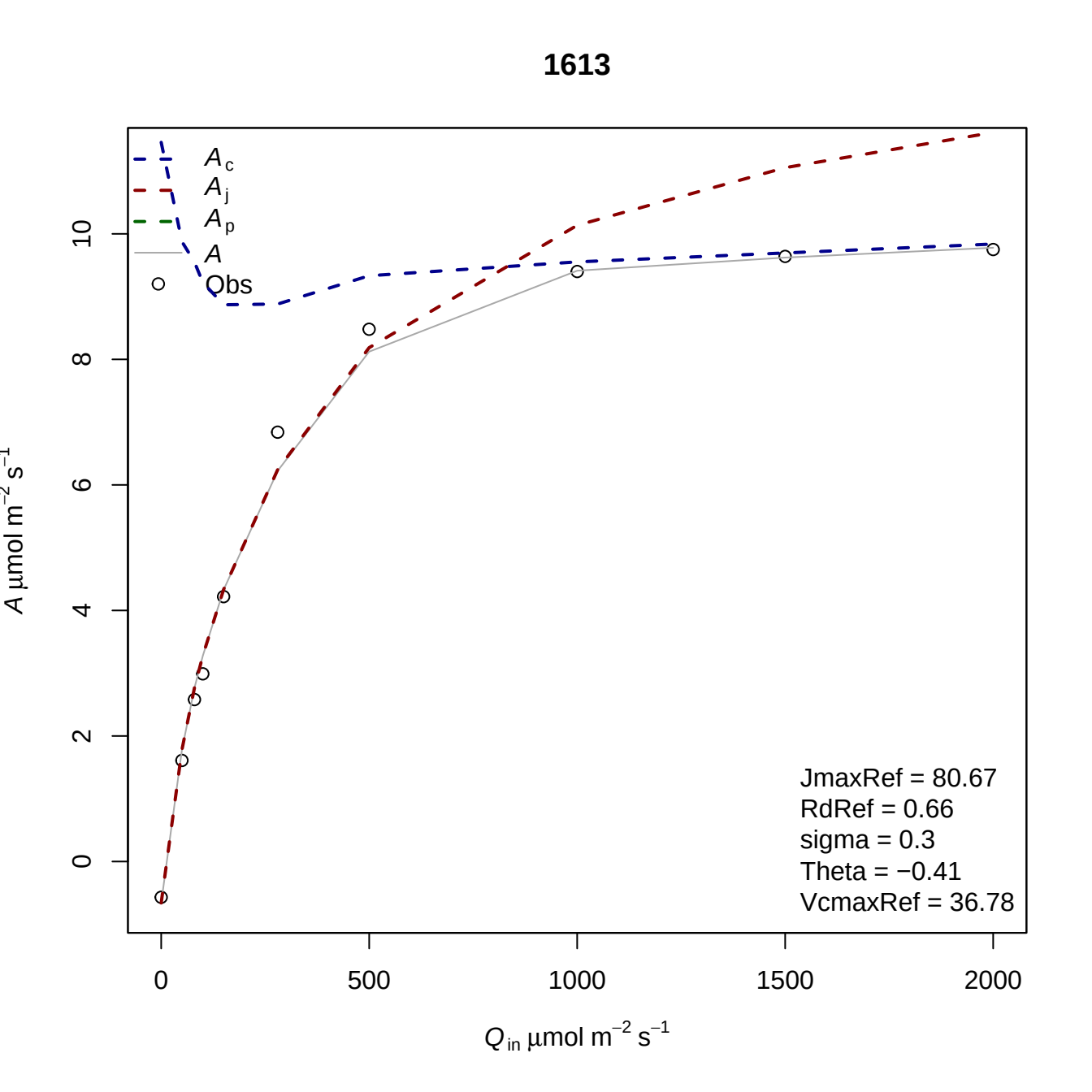

1608
A_c
A_j
A_p
A
Obs
JmaxRef = 249.82
RdRef = 0.93
sigma = 0.23
Theta = −0.02
VcmaxRef = 86.15
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

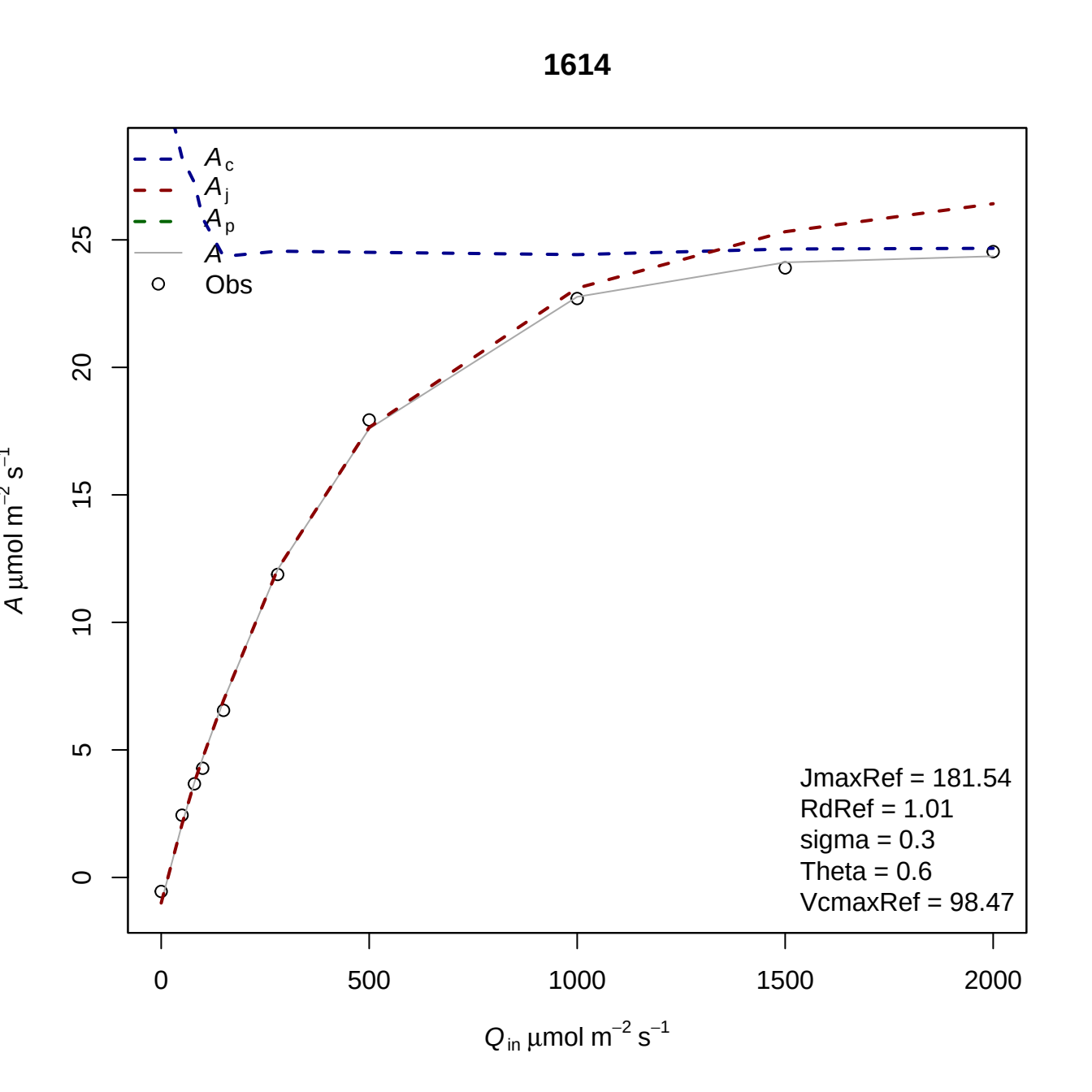

1609
A_c
A_j
A_p
A
Obs
JmaxRef = 138
RdRef = 1.46
sigma = 0.62
Theta = 0.49
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

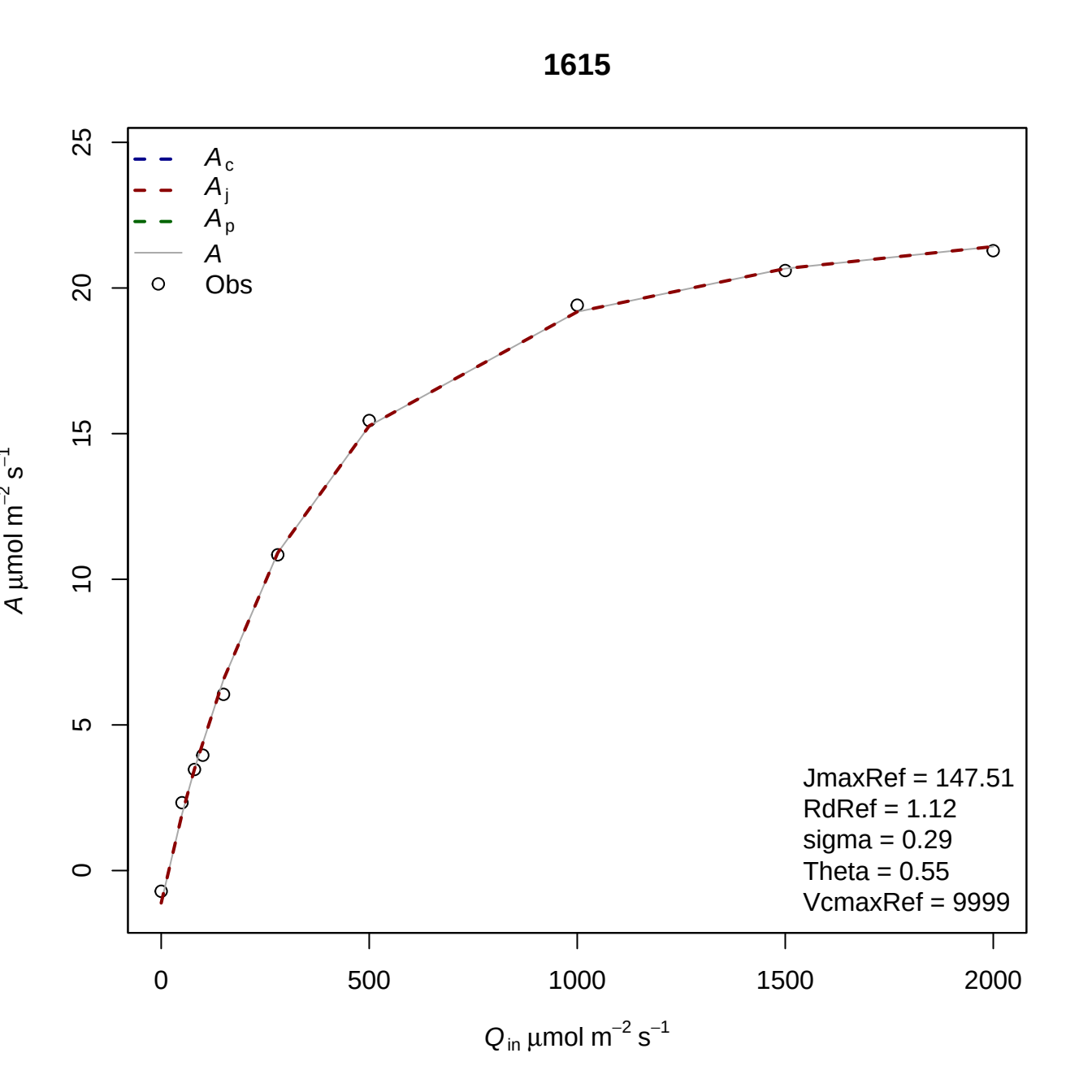

1610
A_c
A_j
A_p
A
Obs
JmaxRef = 83.29
RdRef = 0.62
sigma = 0.13
Theta = 0.07
VcmaxRef = 43.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

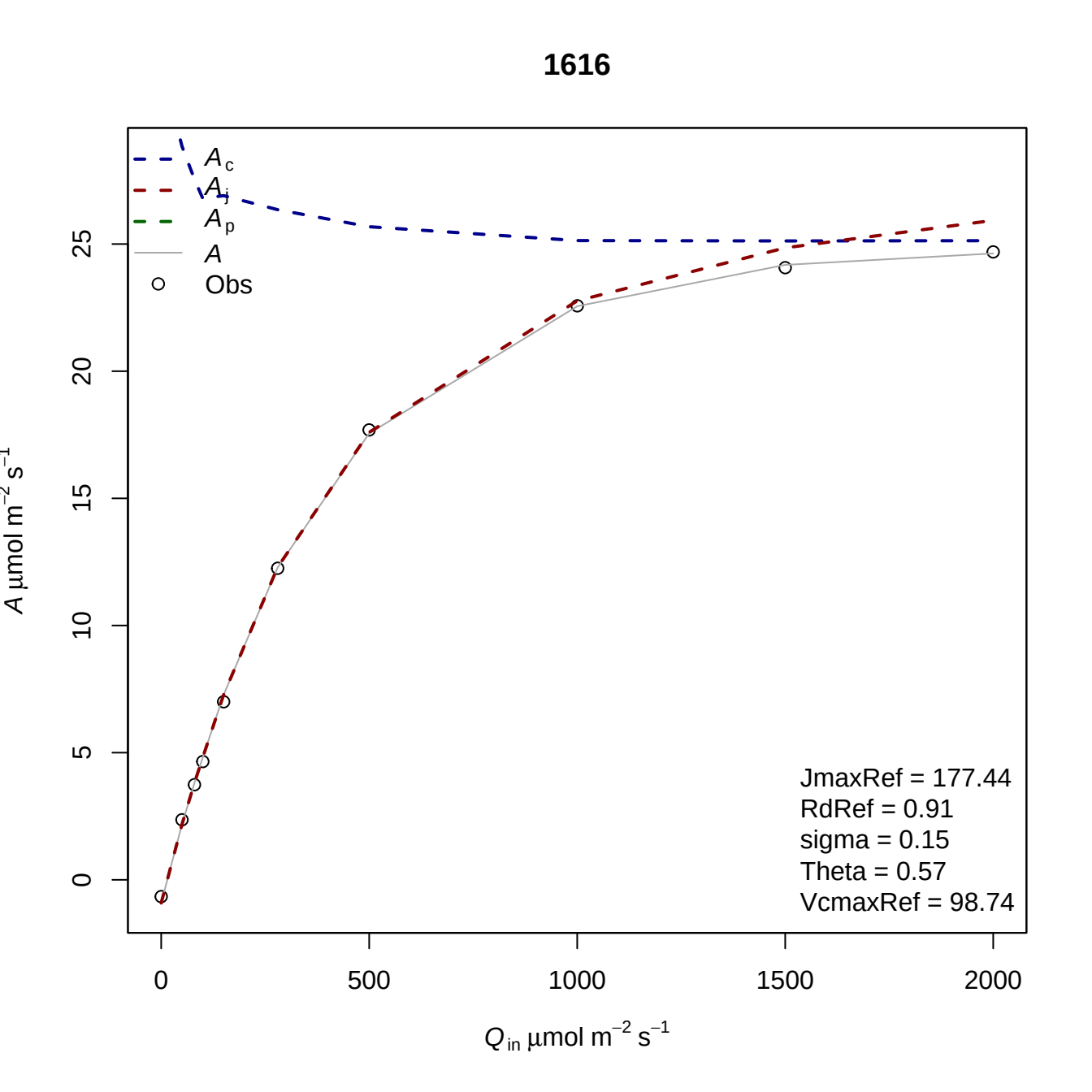

1611
A_c
A_j
A_p
A
Obs
JmaxRef = 53.23
RdRef = 0.57
sigma = 0.17
Theta = 0.37
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

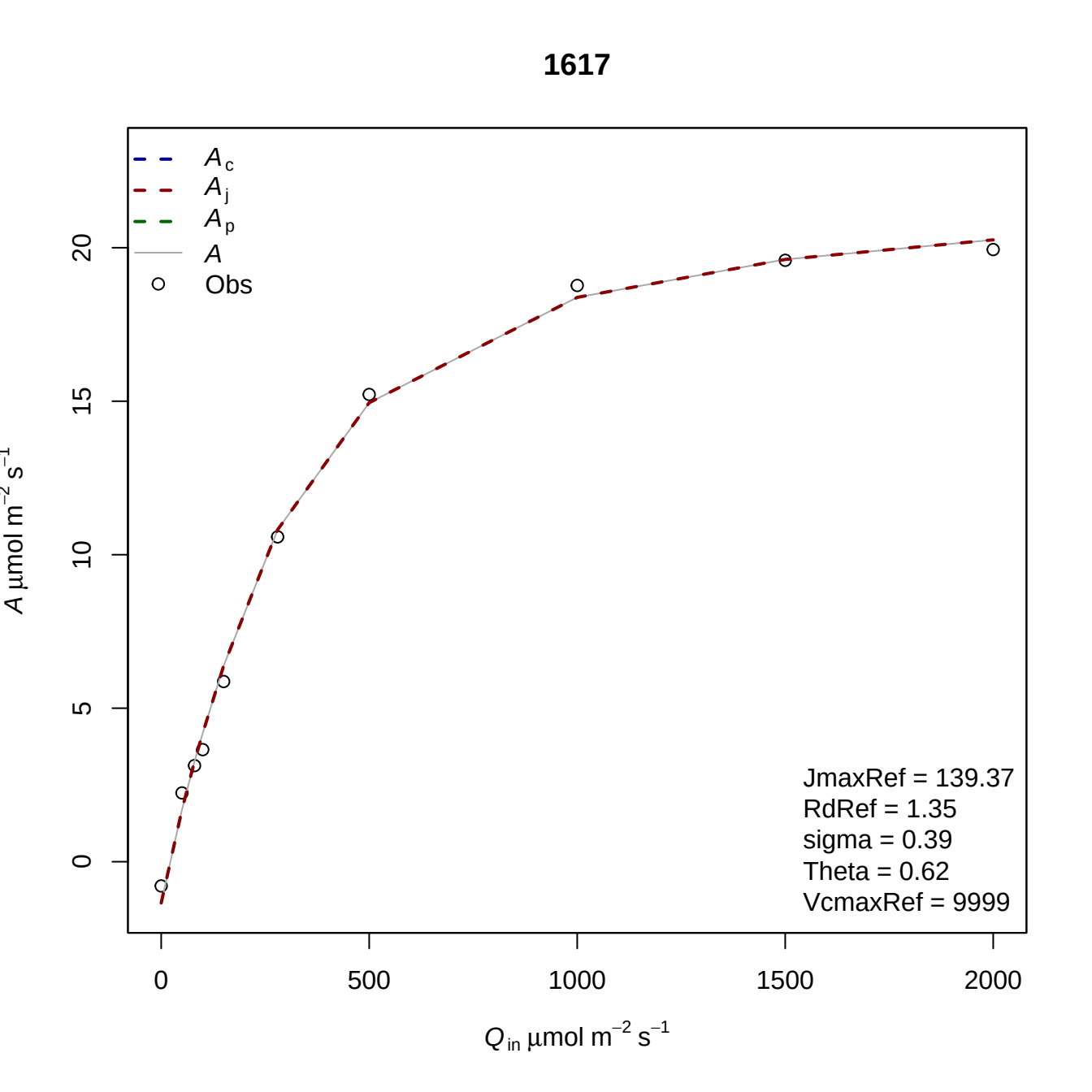

1612
A_c
A_j
A_p
A
Obs
JmaxRef = 74.49
RdRef = 0.62
sigma = 0.17
Theta = −0.27
VcmaxRef = 37.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

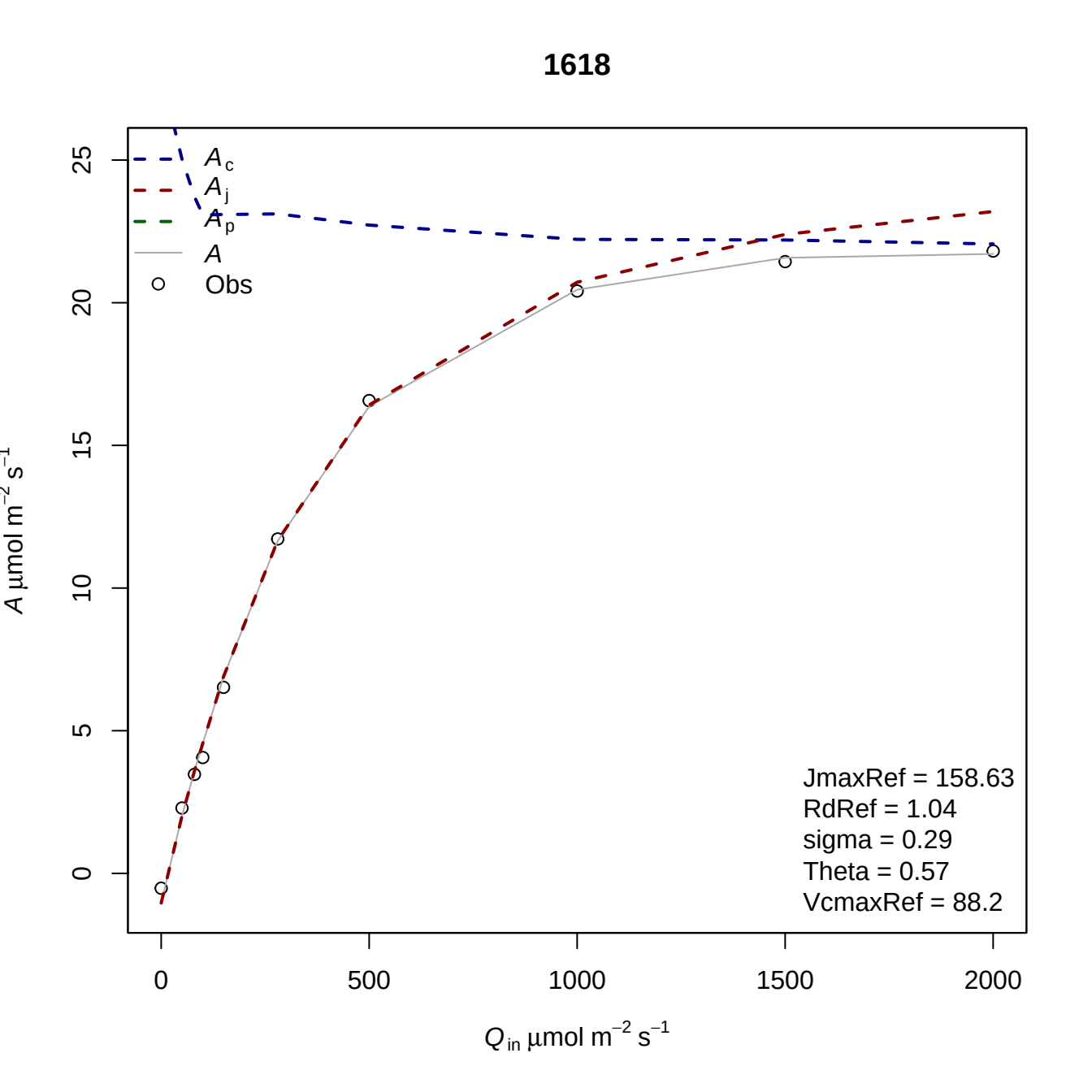

1613
A_c
A_j
A_p
A
Obs
JmaxRef = 80.67
RdRef = 0.66
sigma = 0.3
Theta = −0.41
VcmaxRef = 36.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

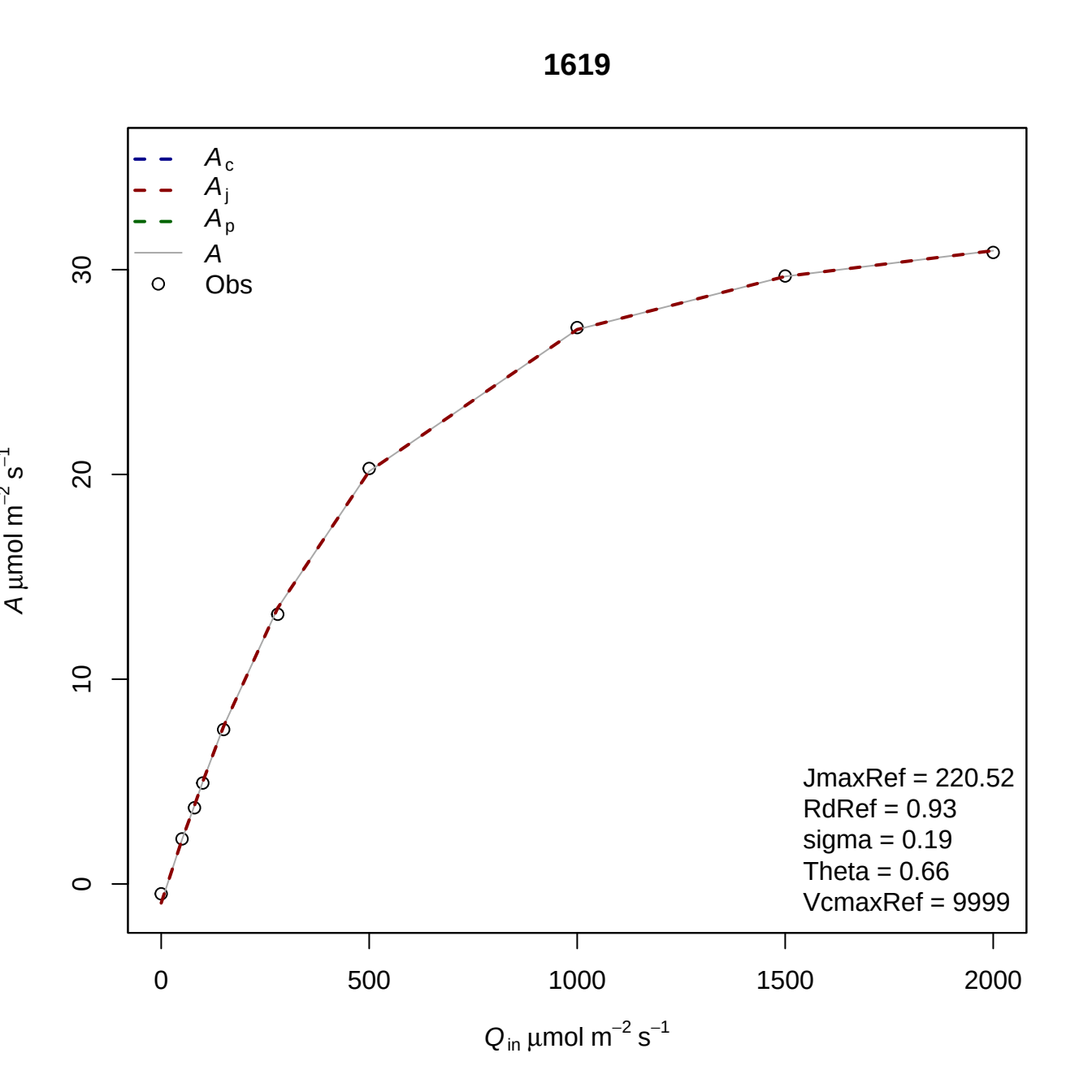

1614
A_c
A_j
A_p
A
Obs
JmaxRef = 181.54
RdRef = 1.01
sigma = 0.3
Theta = 0.6
VcmaxRef = 98.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

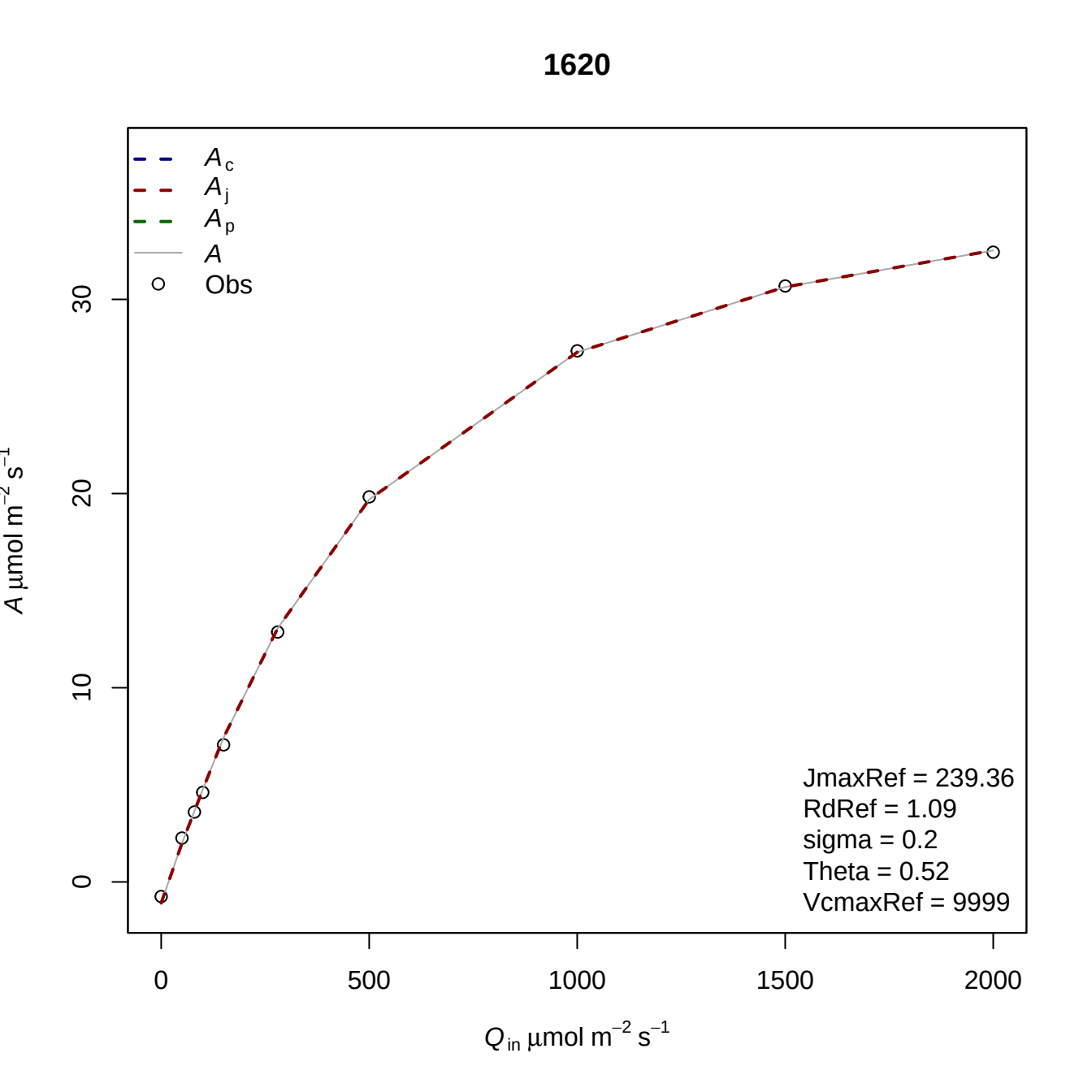

1615
A_c
A_j
A_p
A
Obs
JmaxRef = 147.51
RdRef = 1.12
sigma = 0.29
Theta = 0.55
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

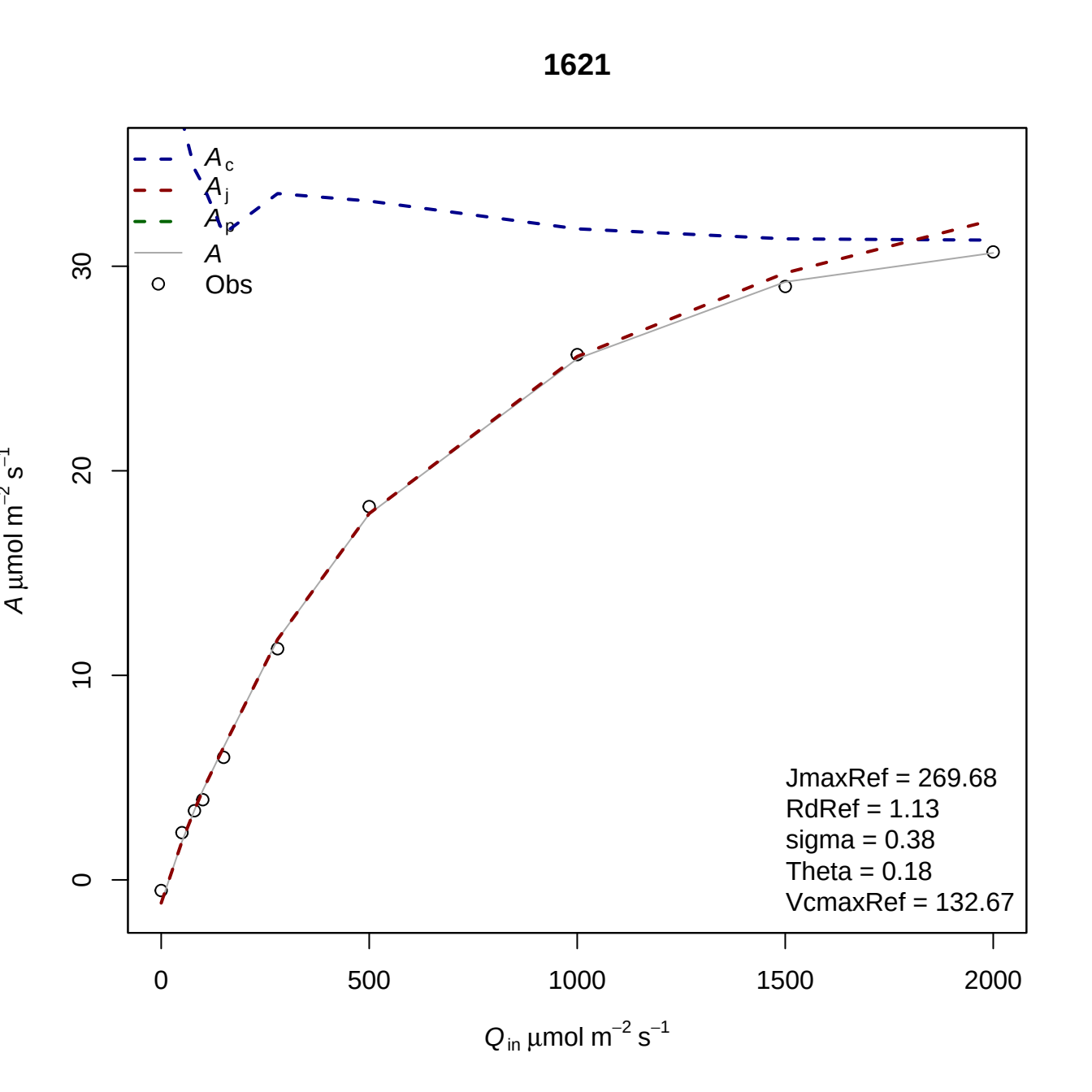

1616
A_c
A_j
A_p
A
Obs
JmaxRef = 177.44
RdRef = 0.91
sigma = 0.15
Theta = 0.57
VcmaxRef = 98.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

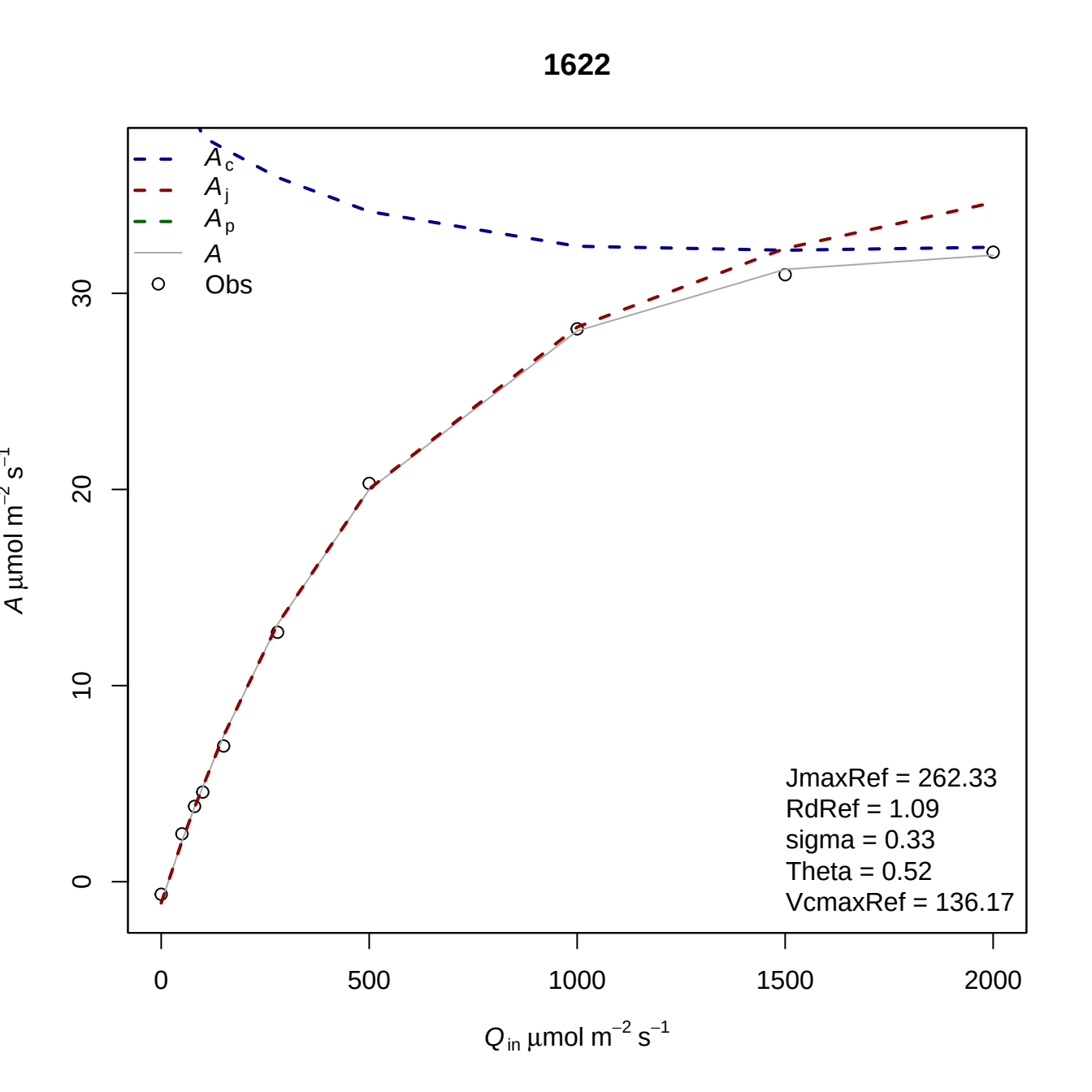

1617
A_c
A_j
A_p
A
Obs
JmaxRef = 139.37
RdRef = 1.35
sigma = 0.39
Theta = 0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

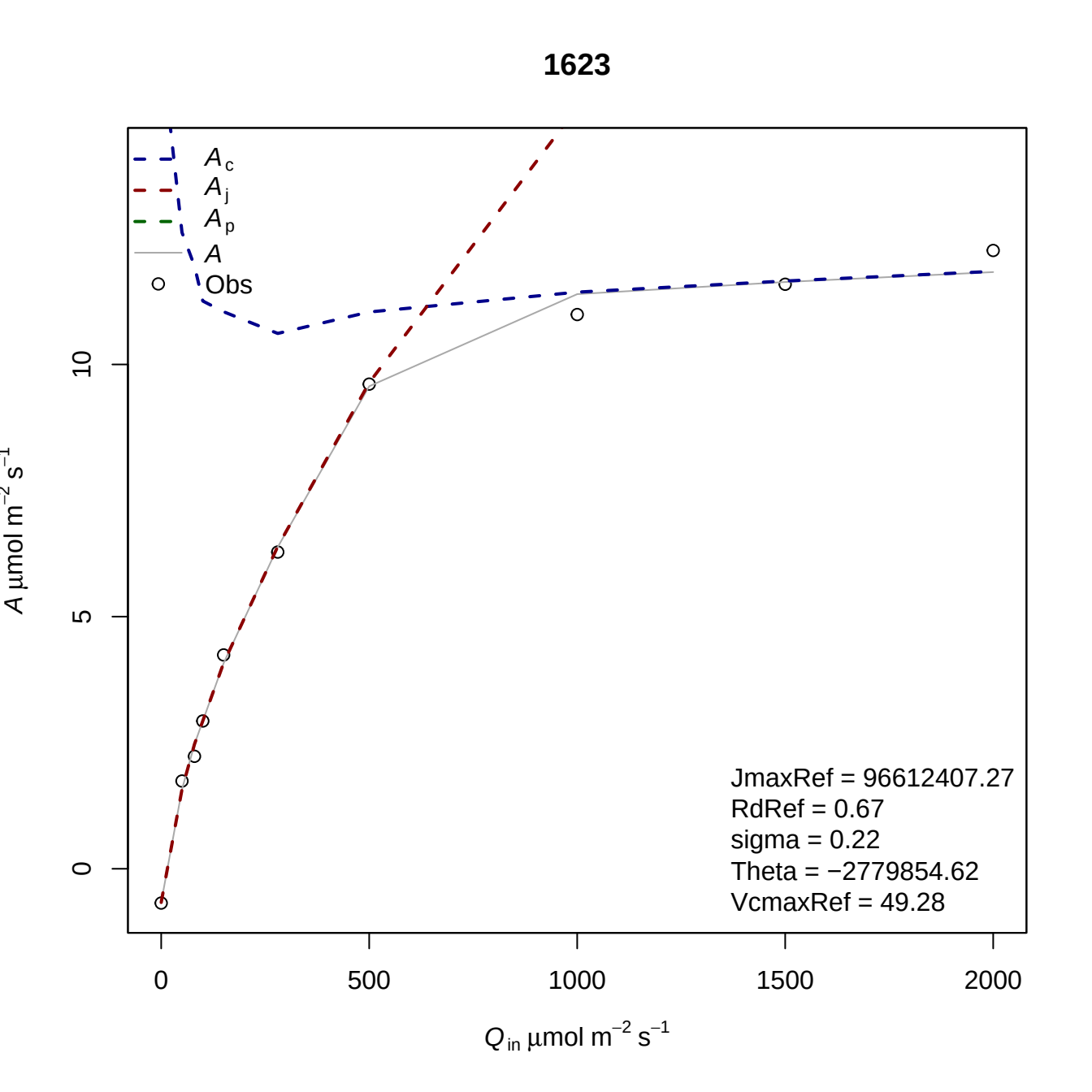

1618
A_c
A_j
A_p
A
Obs
JmaxRef = 158.63
RdRef = 1.04
sigma = 0.29
Theta = 0.57
VcmaxRef = 88.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

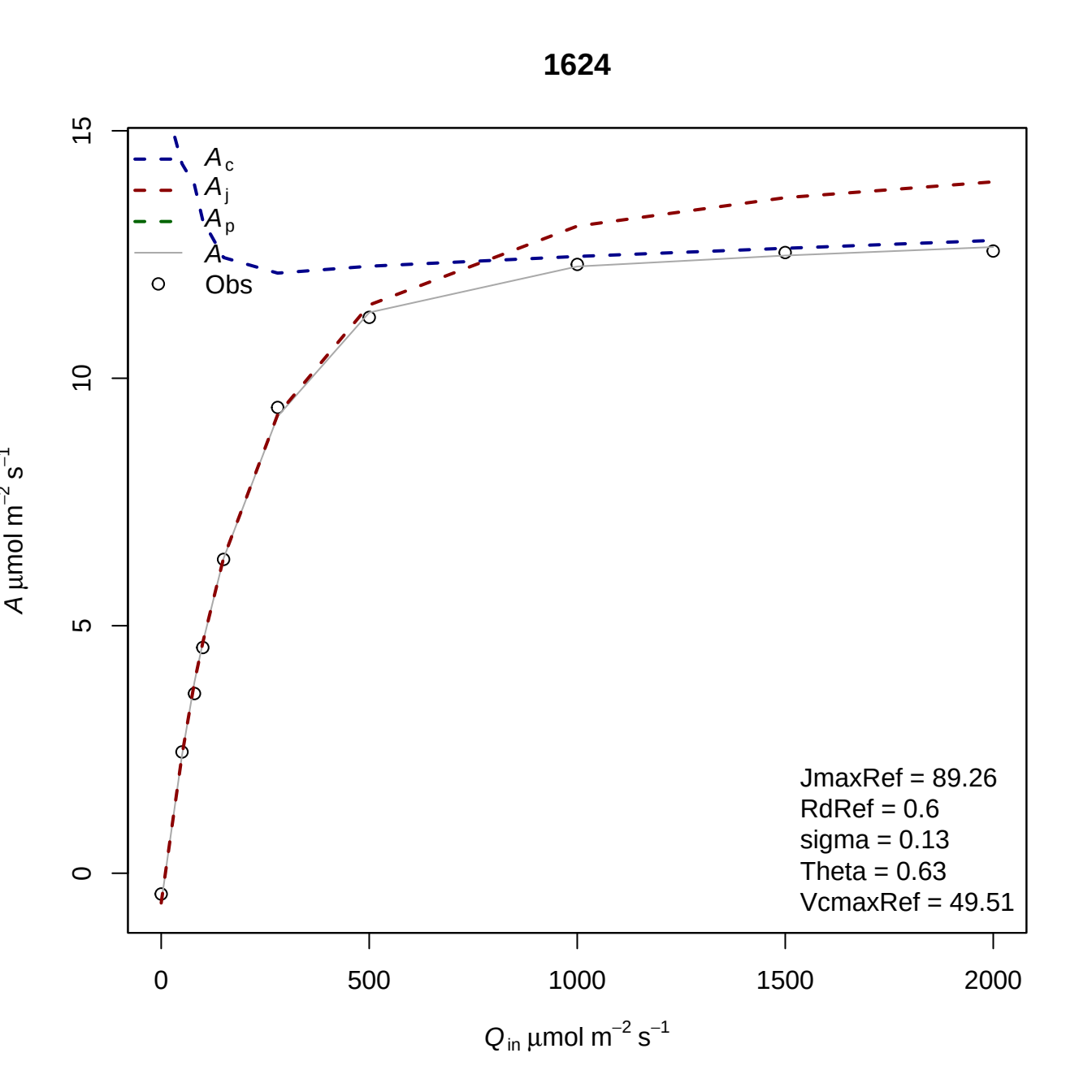

1619
A_c
A_j
A_p
A
Obs
JmaxRef = 220.52
RdRef = 0.93
sigma = 0.19
Theta = 0.66
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

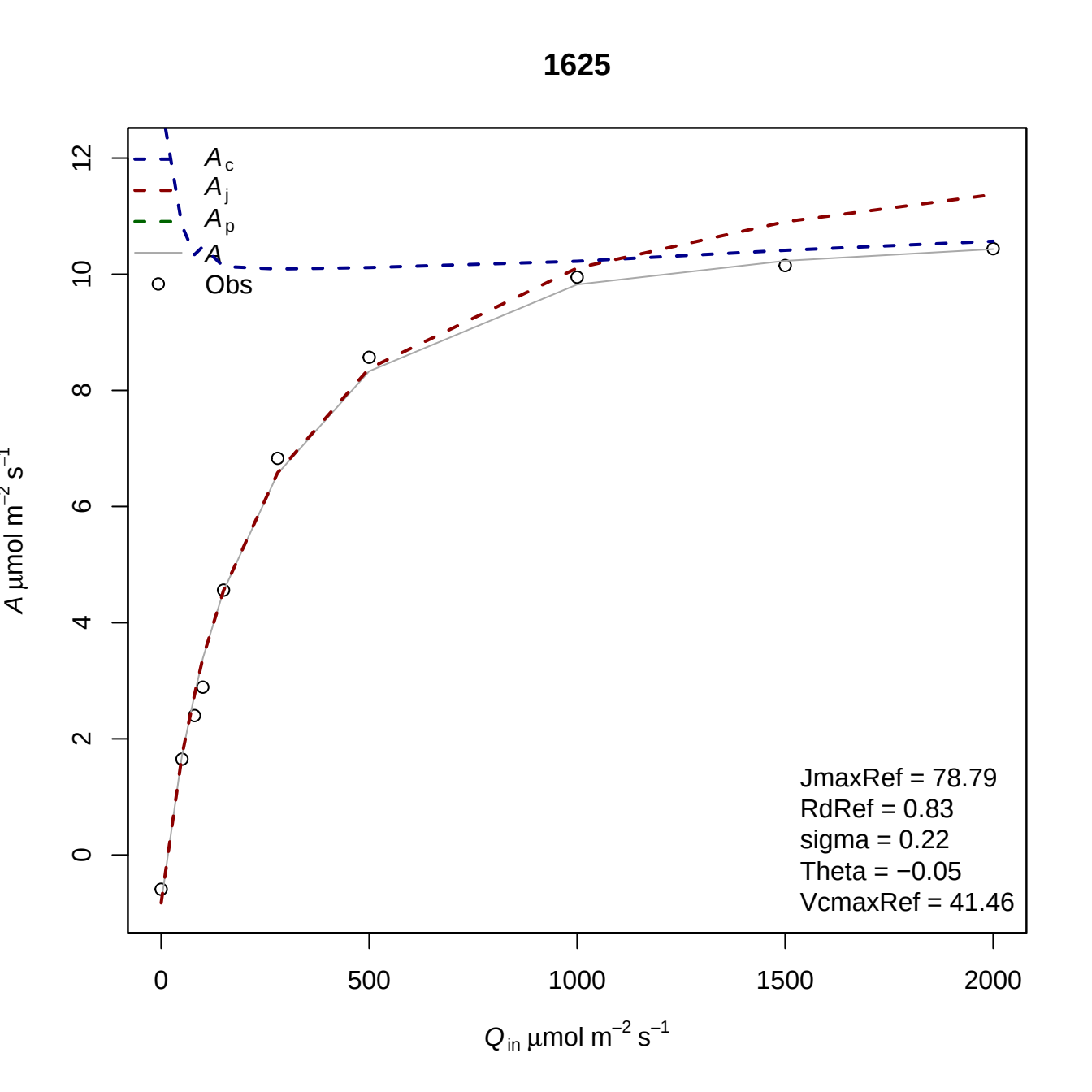

1620
A_c
A_j
A_p
A
Obs
JmaxRef = 239.36
RdRef = 1.09
sigma = 0.2
Theta = 0.52
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

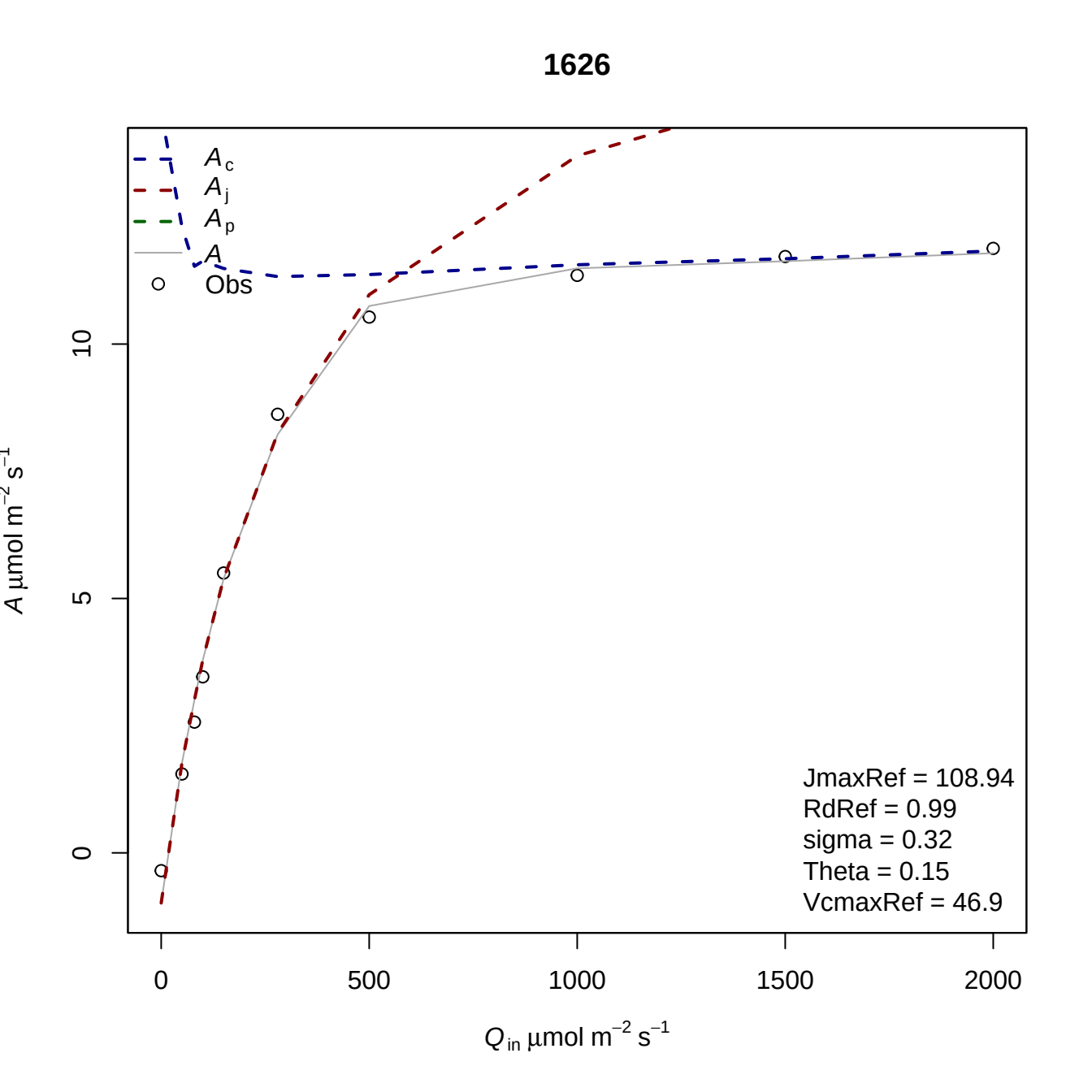

1621
A_c
A_j
A_p
A
Obs
JmaxRef = 269.68
RdRef = 1.13
sigma = 0.38
Theta = 0.18
VcmaxRef = 132.67
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

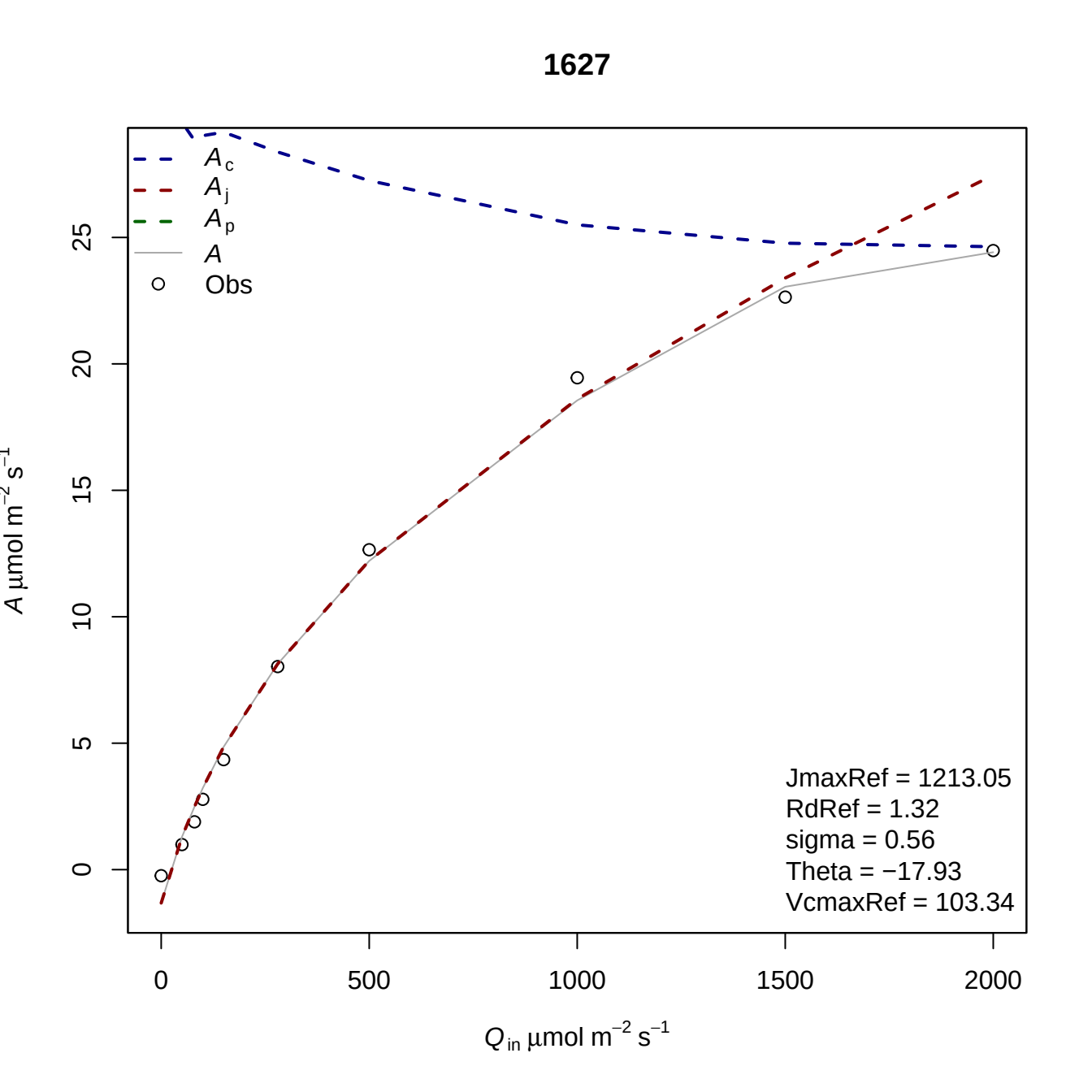

1622
A_c
A_j
A_p
A
Obs
JmaxRef = 262.33
RdRef = 1.09
sigma = 0.33
Theta = 0.52
VcmaxRef = 136.17
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

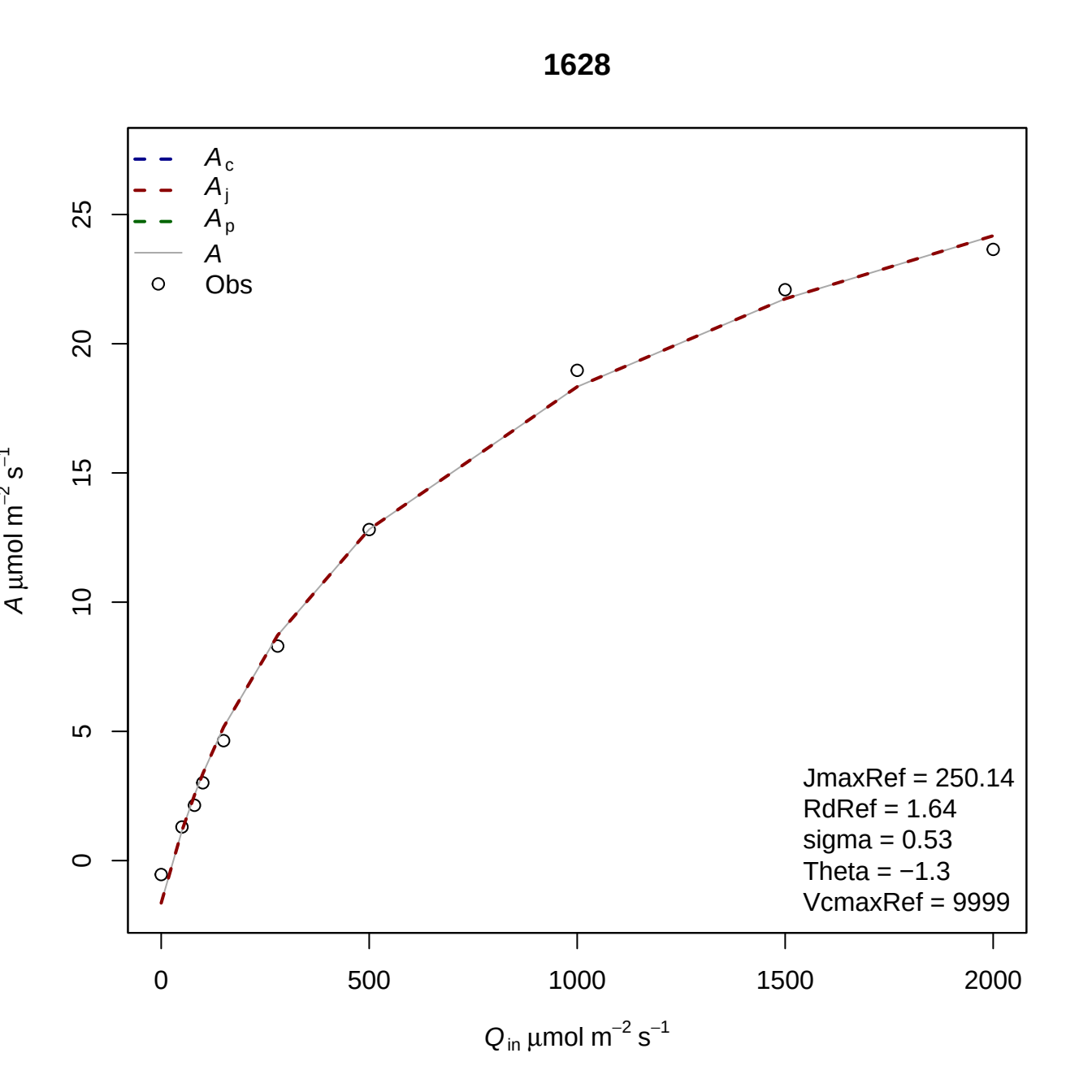

1623
A_c
A_j
A_p
A
Obs
JmaxRef = 96612407.27
RdRef = 0.67
sigma = 0.22
Theta = −2779854.62
VcmaxRef = 49.28
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

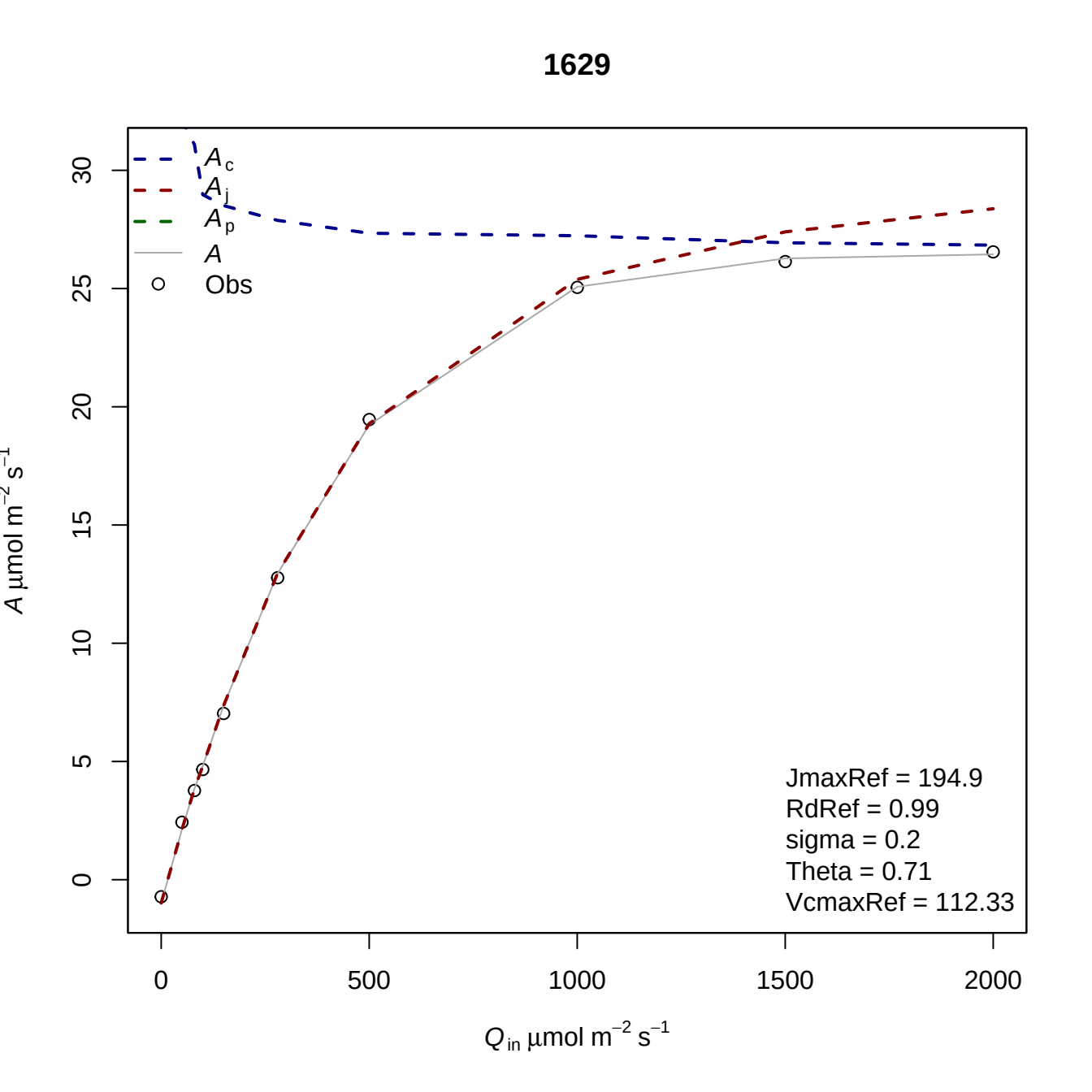

1624
A_c
A_j
A_p
A
Obs
JmaxRef = 89.26
RdRef = 0.6
sigma = 0.13
Theta = 0.63
VcmaxRef = 49.51
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

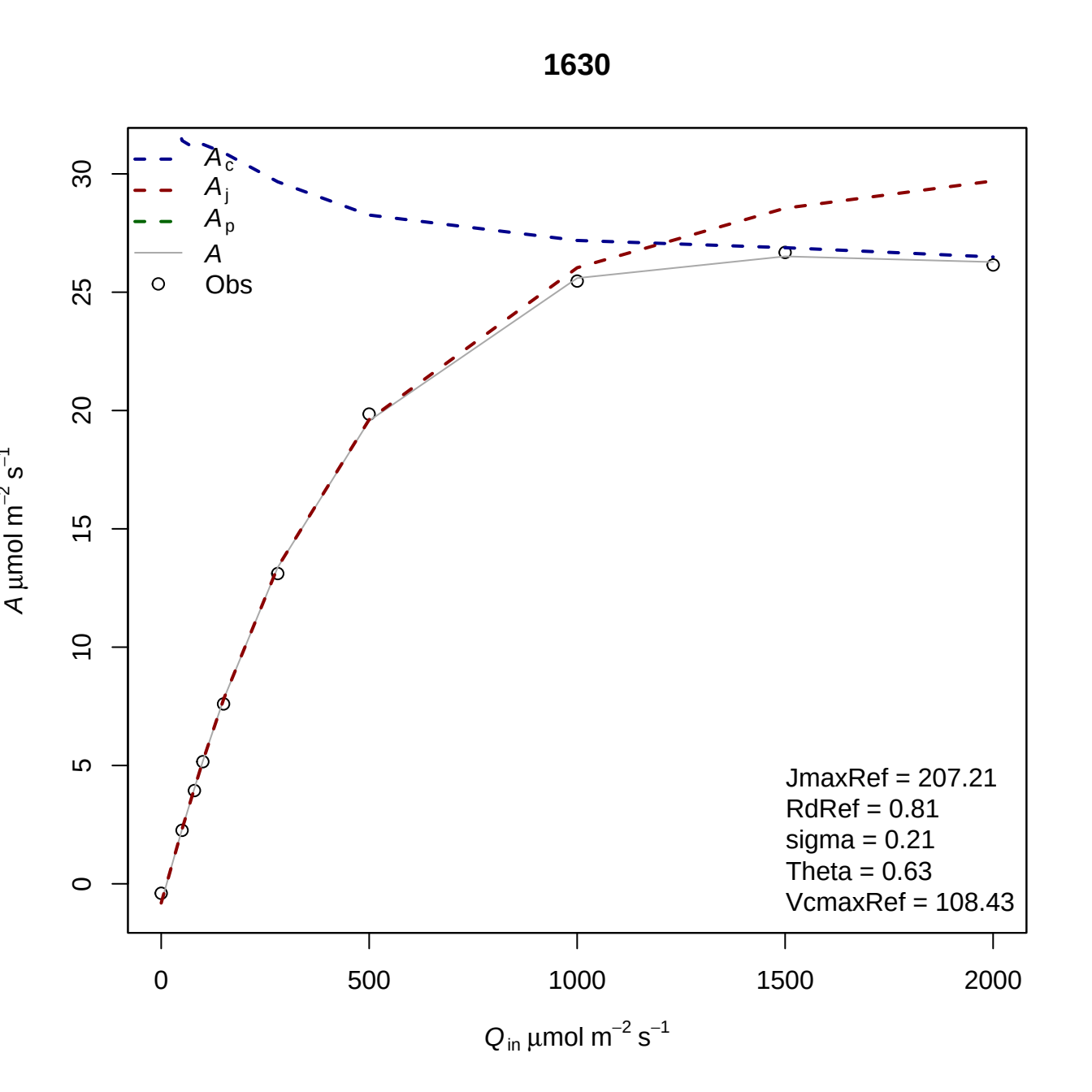

1625
A_c
A_j
A_p
A
Obs
JmaxRef = 78.79
RdRef = 0.83
sigma = 0.22
Theta = −0.05
VcmaxRef = 41.46
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

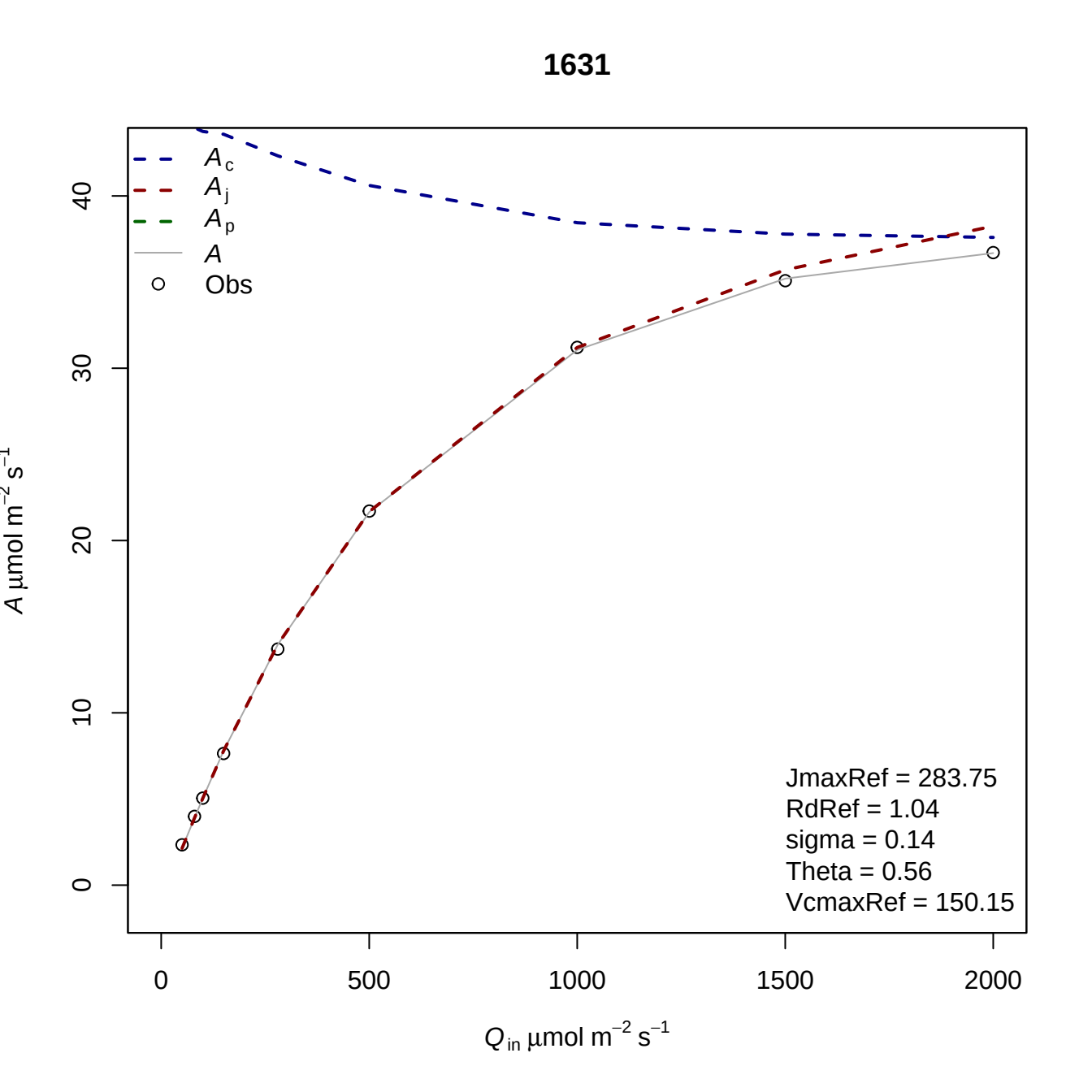

1626
A_c
A_j
A_p
A
Obs
JmaxRef = 108.94
RdRef = 0.99
sigma = 0.32
Theta = 0.15
VcmaxRef = 46.9
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

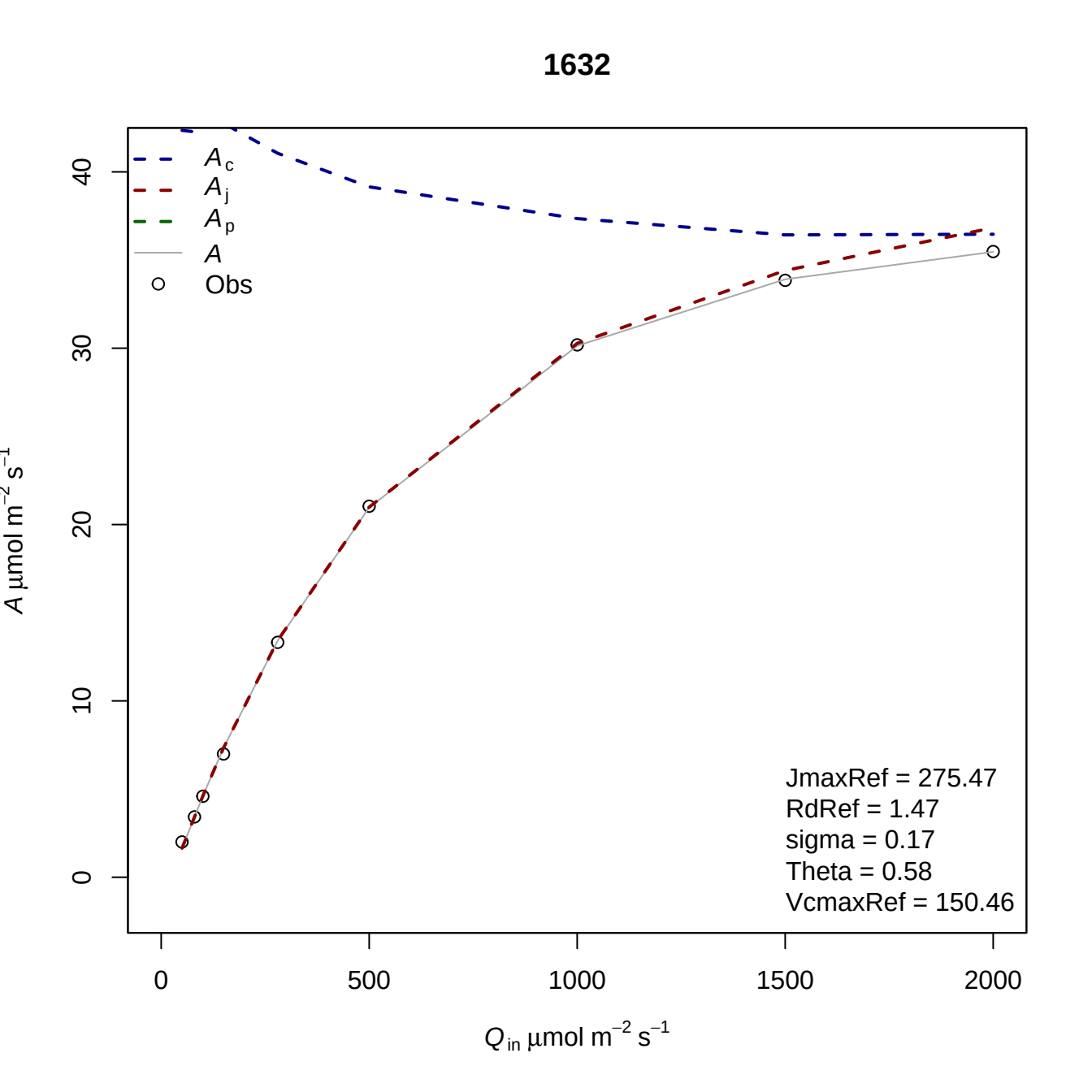

1627
A_c
A_j
A_p
A
Obs
JmaxRef = 1213.05
RdRef = 1.32
sigma = 0.56
Theta = −17.93
VcmaxRef = 103.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

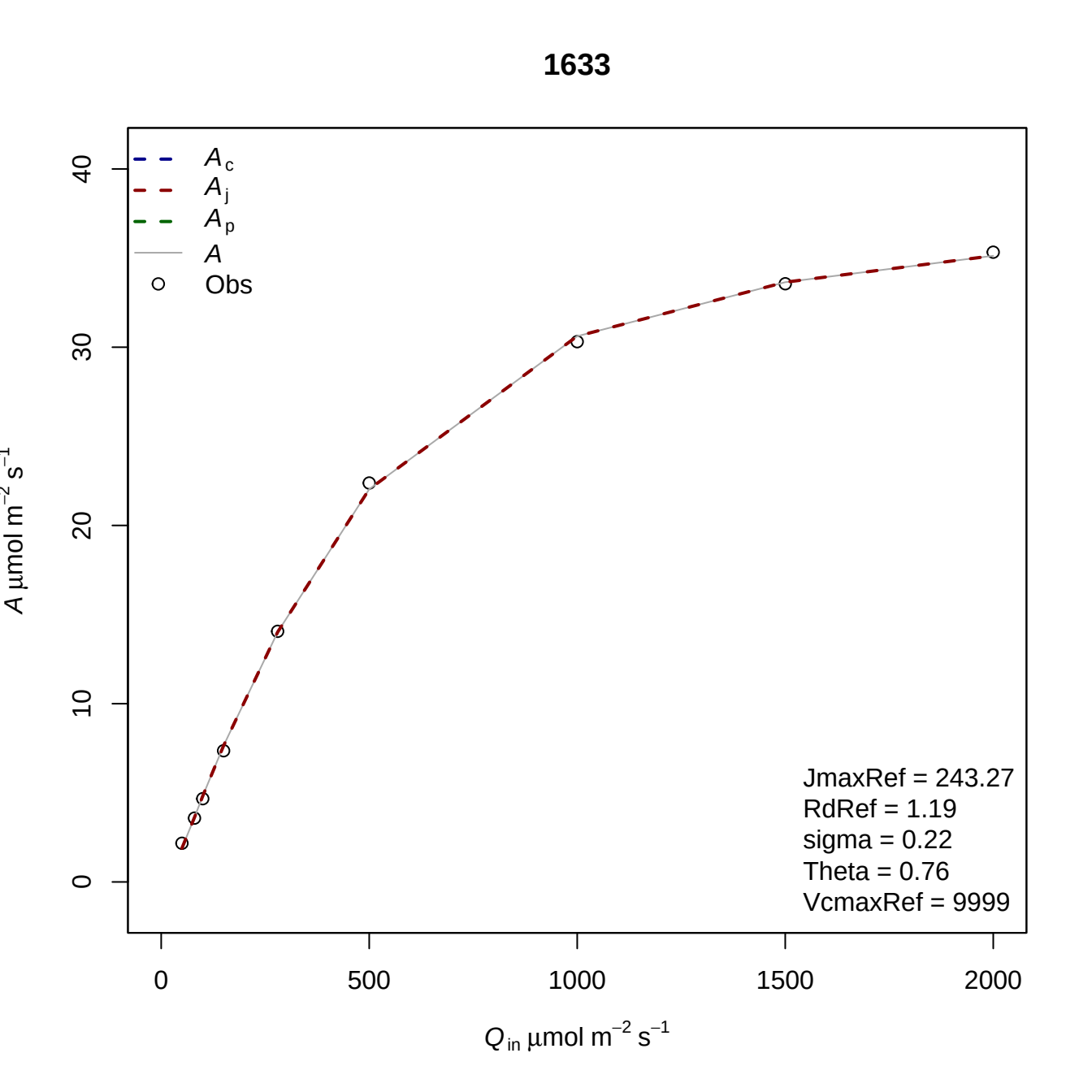

1628
A_c
A_j
A_p
A
Obs
JmaxRef = 250.14
RdRef = 1.64
sigma = 0.53
Theta = −1.3
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

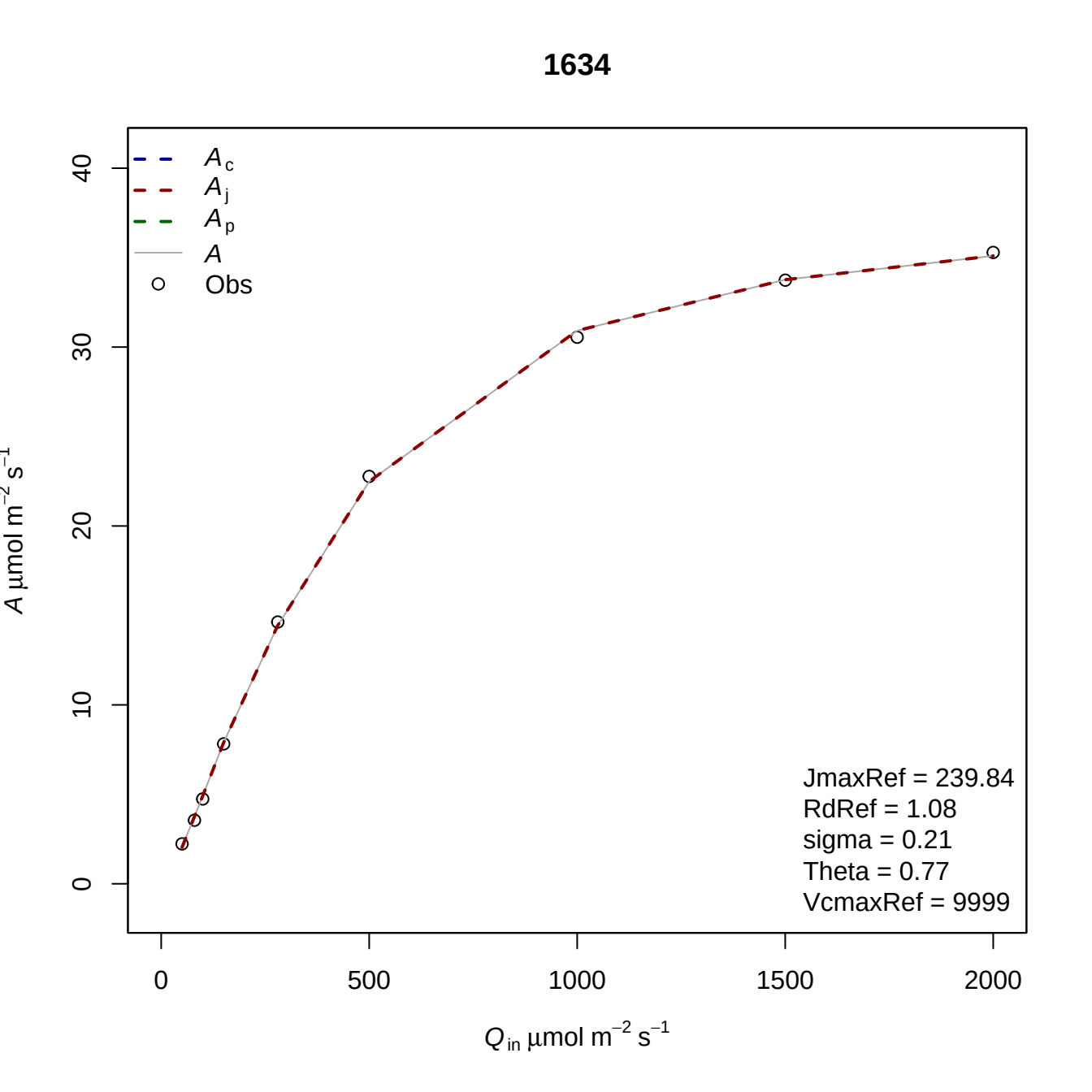

1629
A_c
A_j
A_p
A
Obs
JmaxRef = 194.9
RdRef = 0.99
sigma = 0.2
Theta = 0.71
VcmaxRef = 112.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

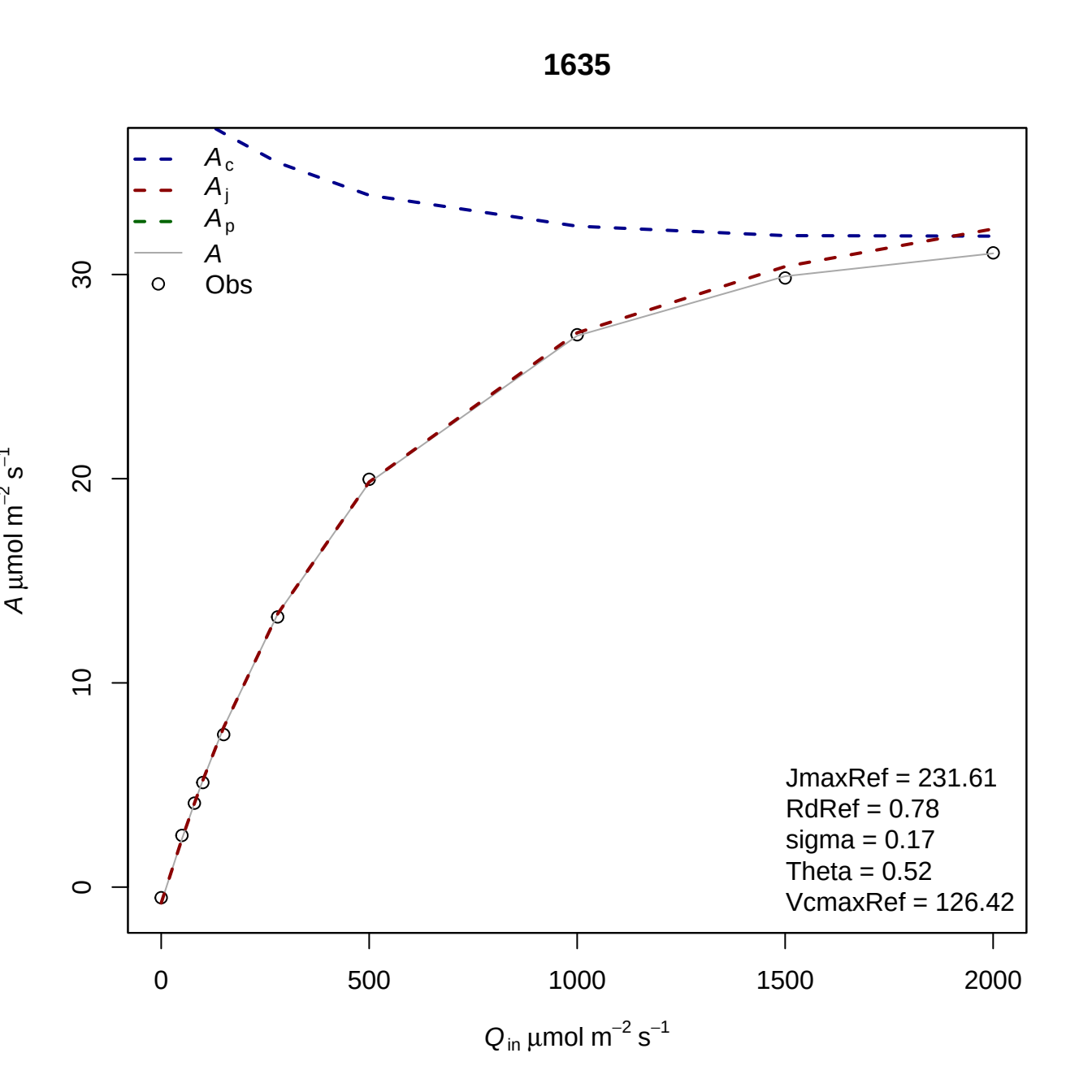

1630
A_c
A_j
A_p
A
Obs
JmaxRef = 207.21
RdRef = 0.81
sigma = 0.21
Theta = 0.63
VcmaxRef = 108.43
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

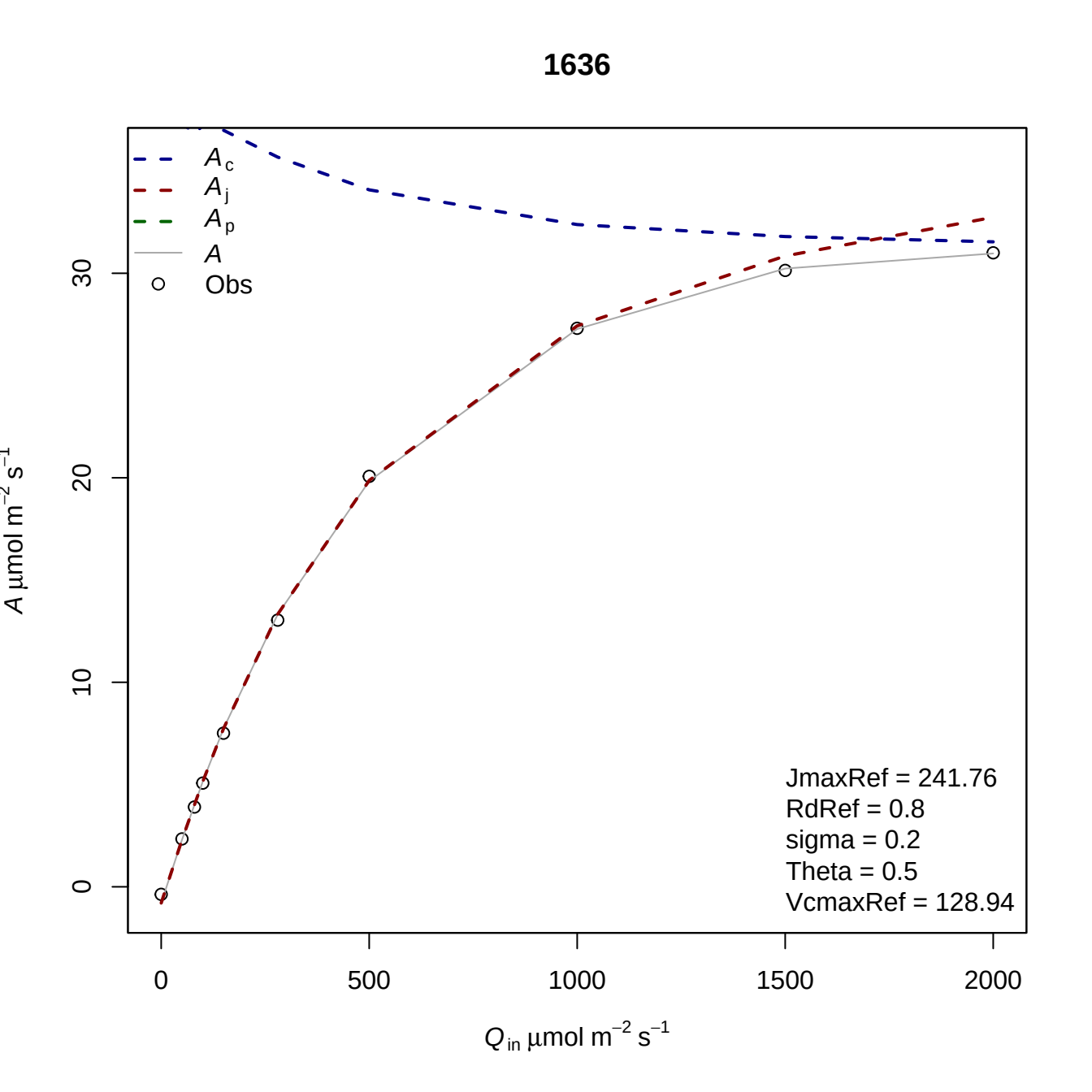

1631
A_c
A_j
A_p
A
Obs
JmaxRef = 283.75
RdRef = 1.04
sigma = 0.14
Theta = 0.56
VcmaxRef = 150.15
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

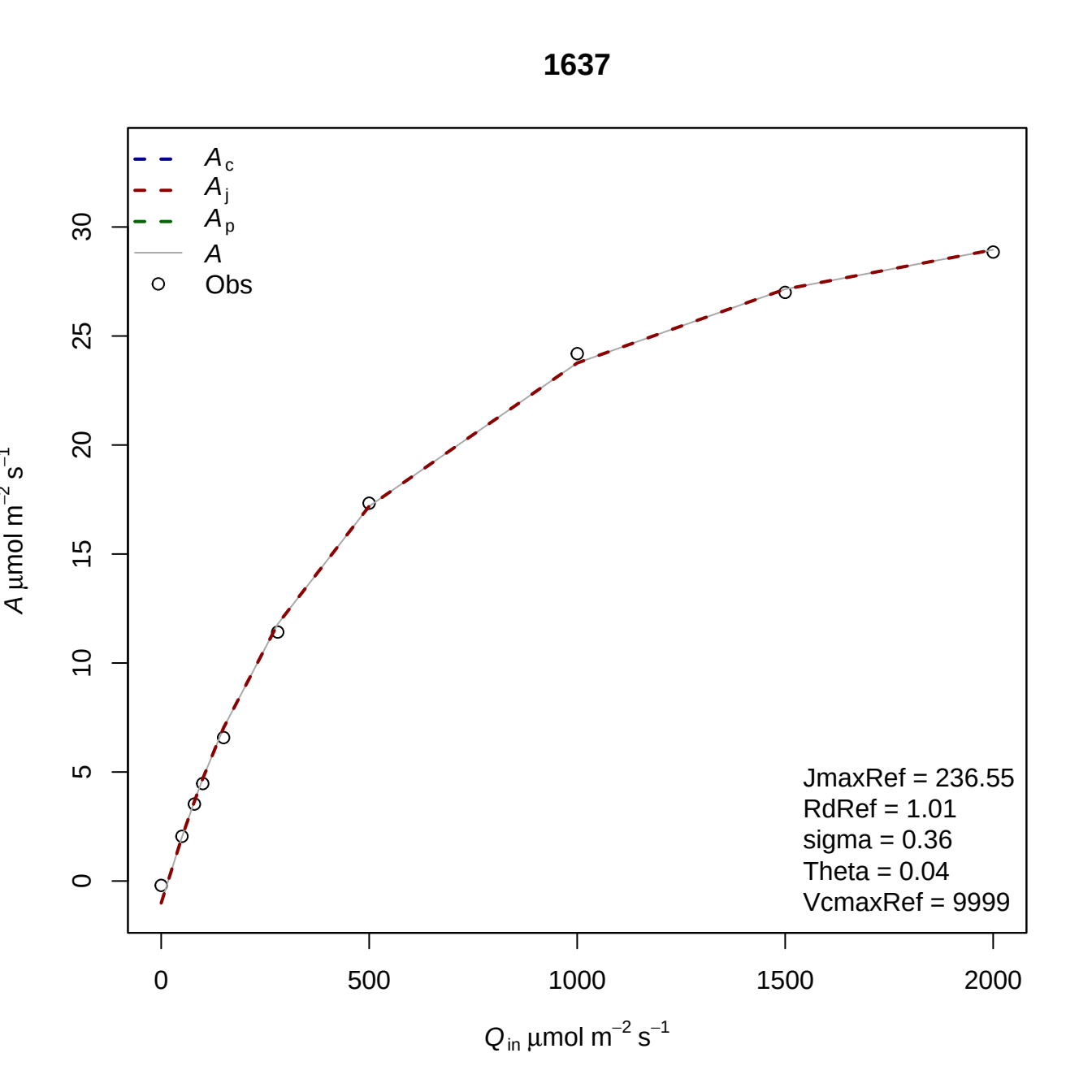

1632
A_c
A_j
A_p
A
Obs
JmaxRef = 275.47
RdRef = 1.47
sigma = 0.17
Theta = 0.58
VcmaxRef = 150.46
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

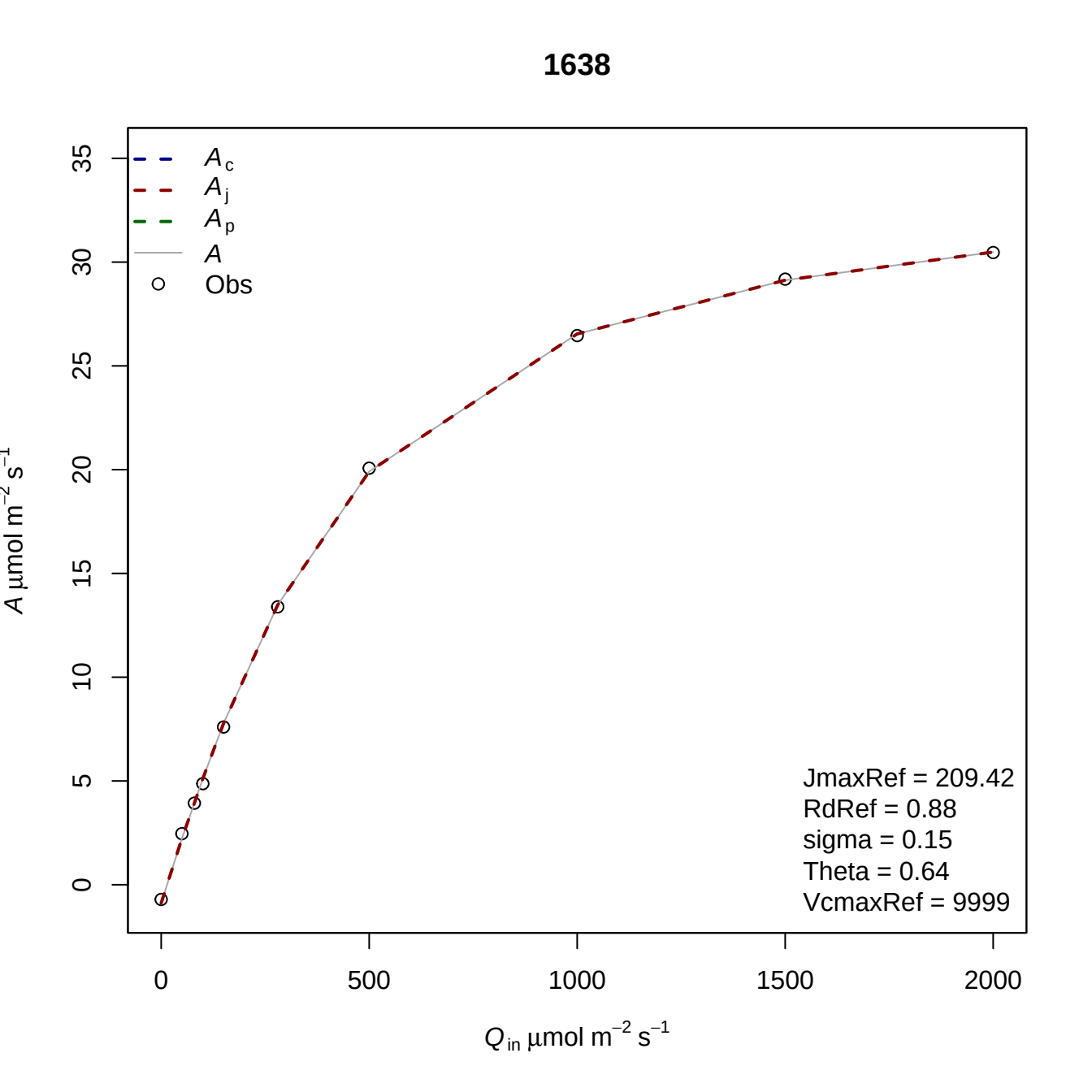

1633
A_c
A_j
A_p
A
Obs
JmaxRef = 243.27
RdRef = 1.19
sigma = 0.22
Theta = 0.76
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

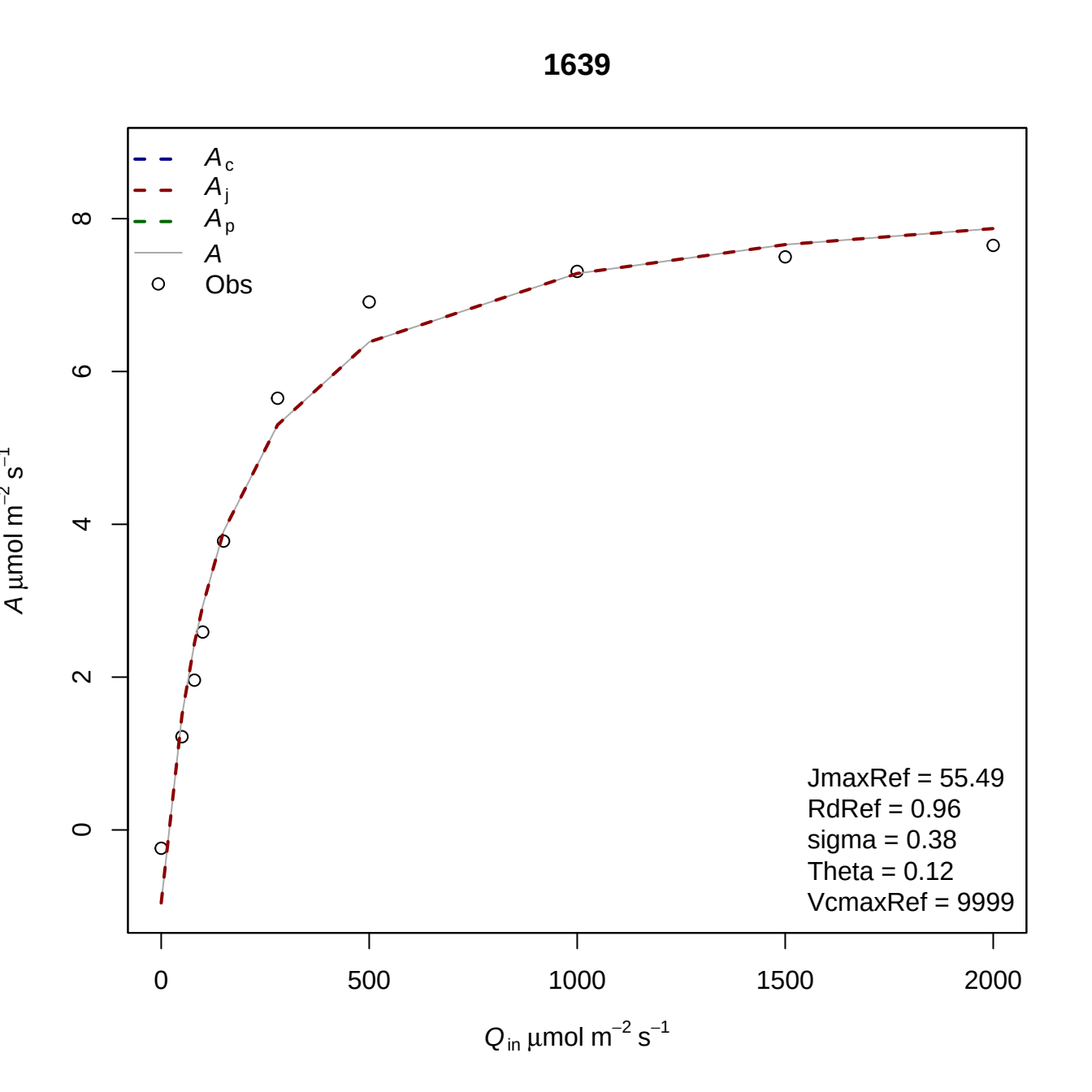

1634
A_c
A_j
A_p
A
Obs
JmaxRef = 239.84
RdRef = 1.08
sigma = 0.21
Theta = 0.77
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

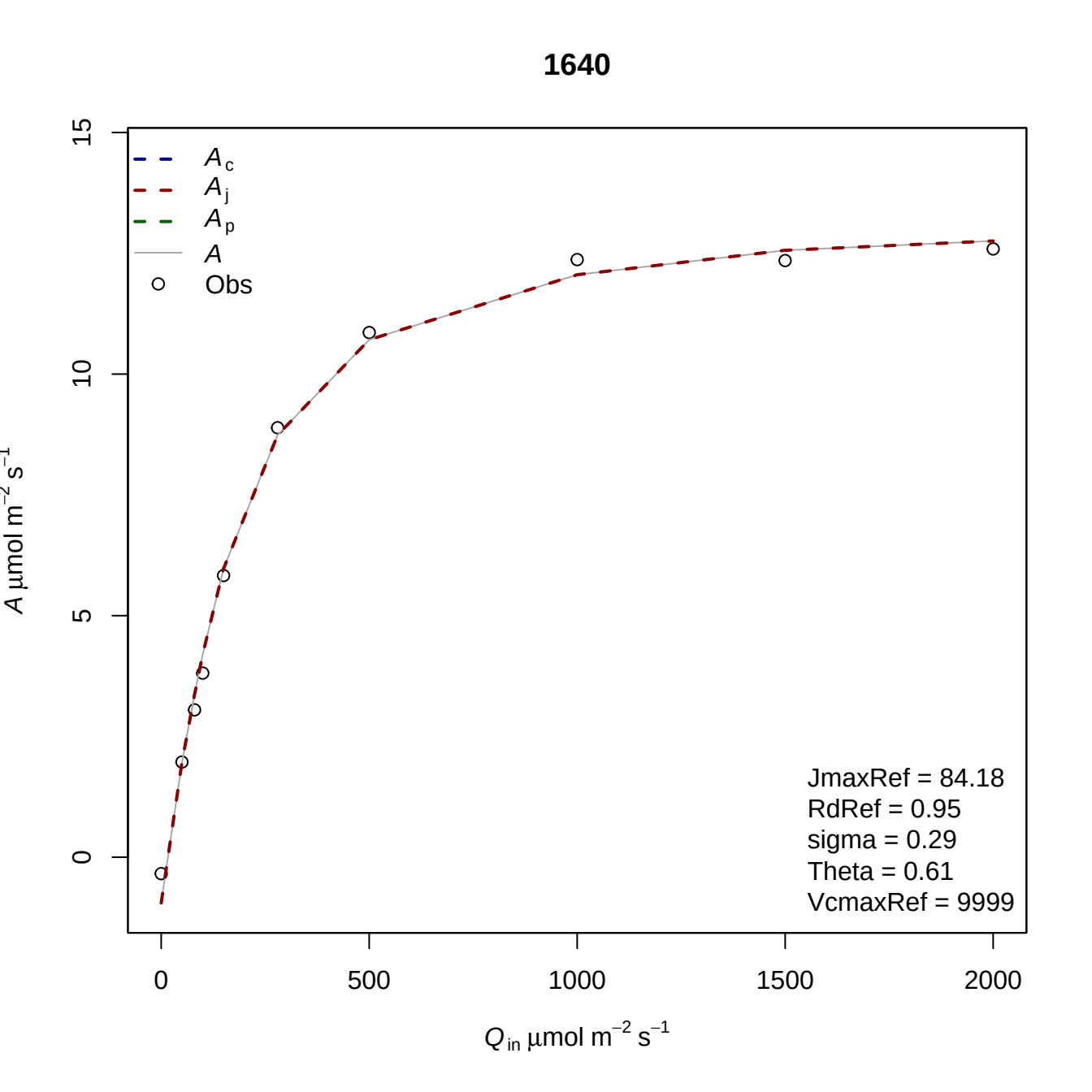

1635
A_c
A_j
A_p
A
Obs
JmaxRef = 231.61
RdRef = 0.78
sigma = 0.17
Theta = 0.52
VcmaxRef = 126.42
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

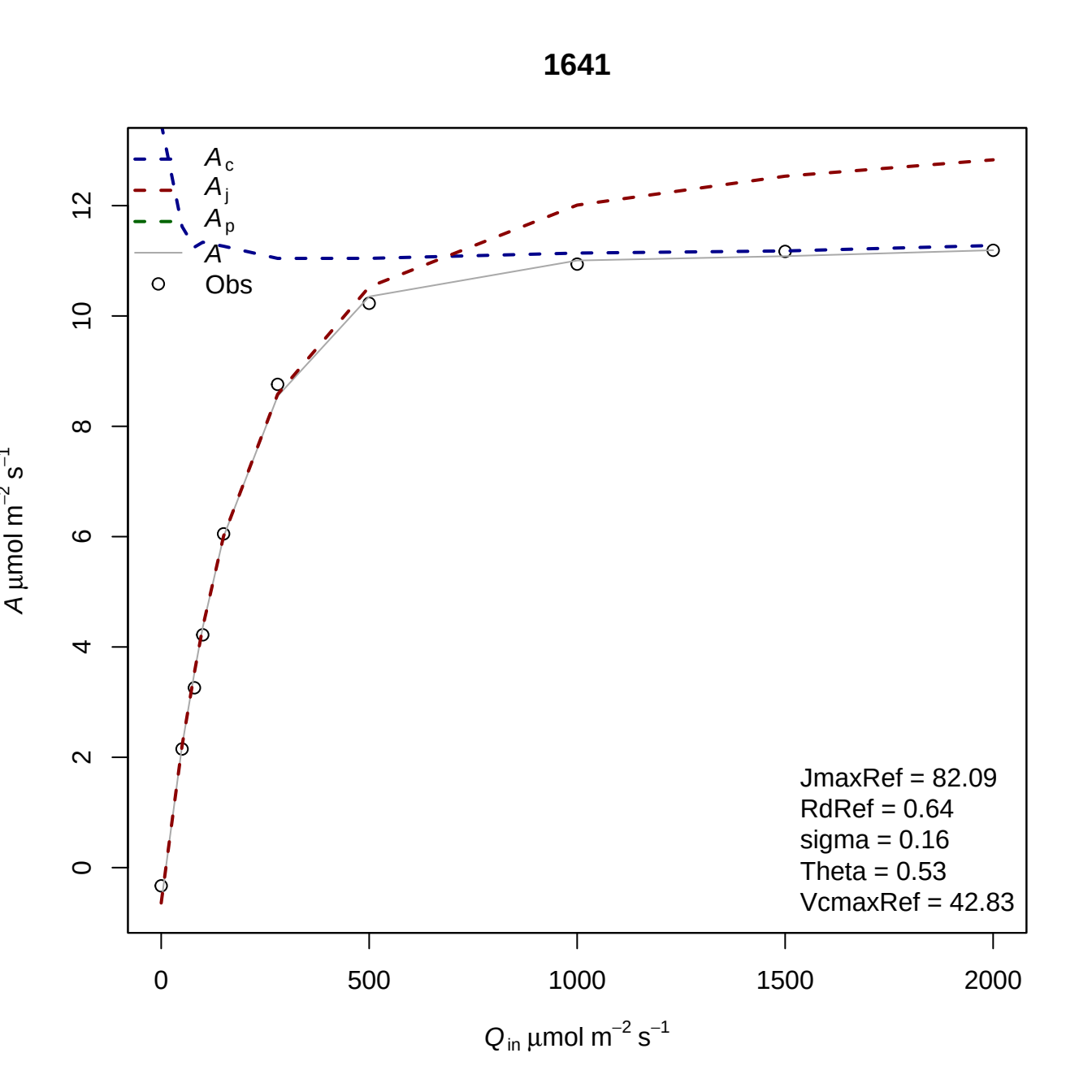

1636
A_c
A_j
A_p
A
Obs
JmaxRef = 241.76
RdRef = 0.8
sigma = 0.2
Theta = 0.5
VcmaxRef = 128.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

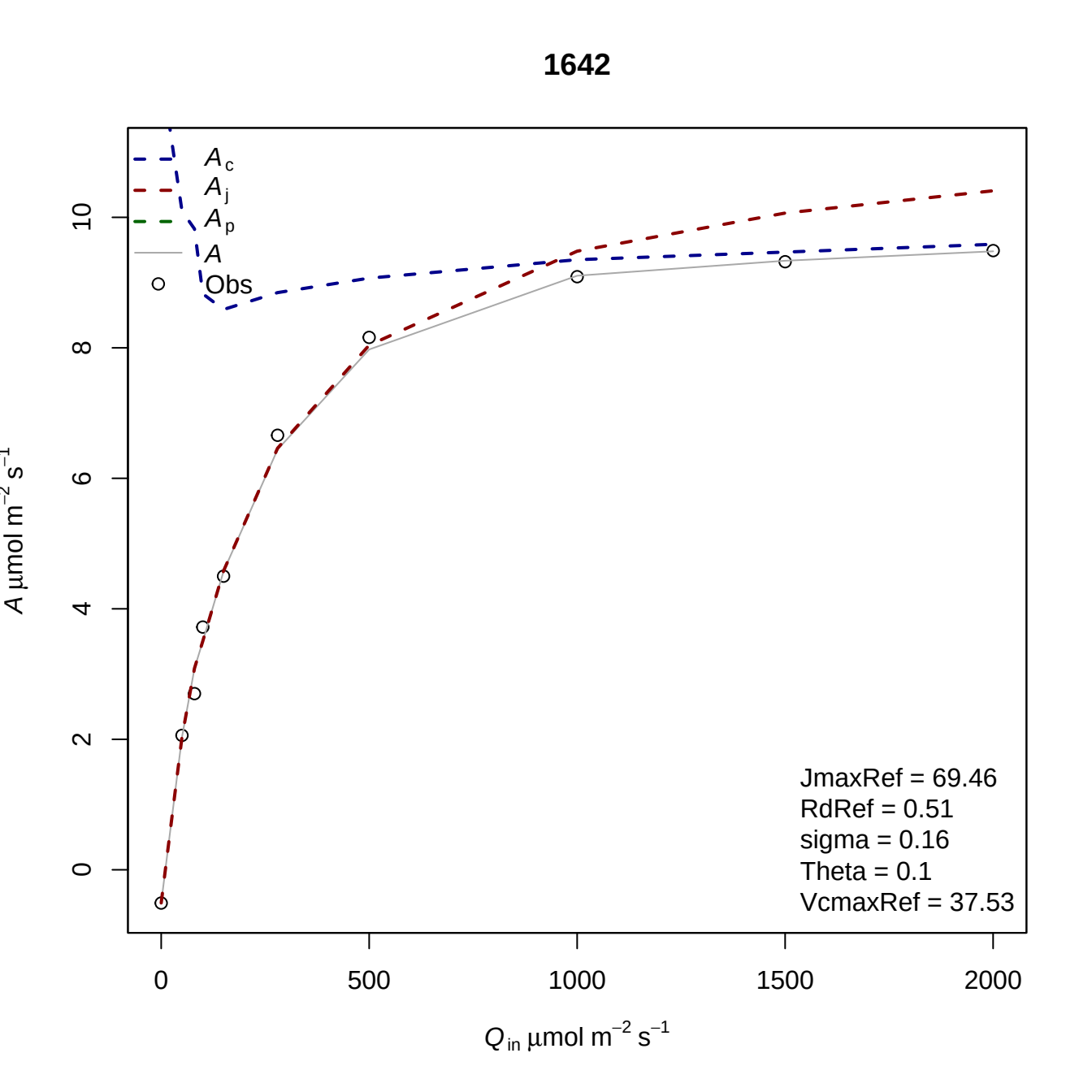

1637
A_c
A_j
A_p
A
Obs
JmaxRef = 236.55
RdRef = 1.01
sigma = 0.36
Theta = 0.04
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

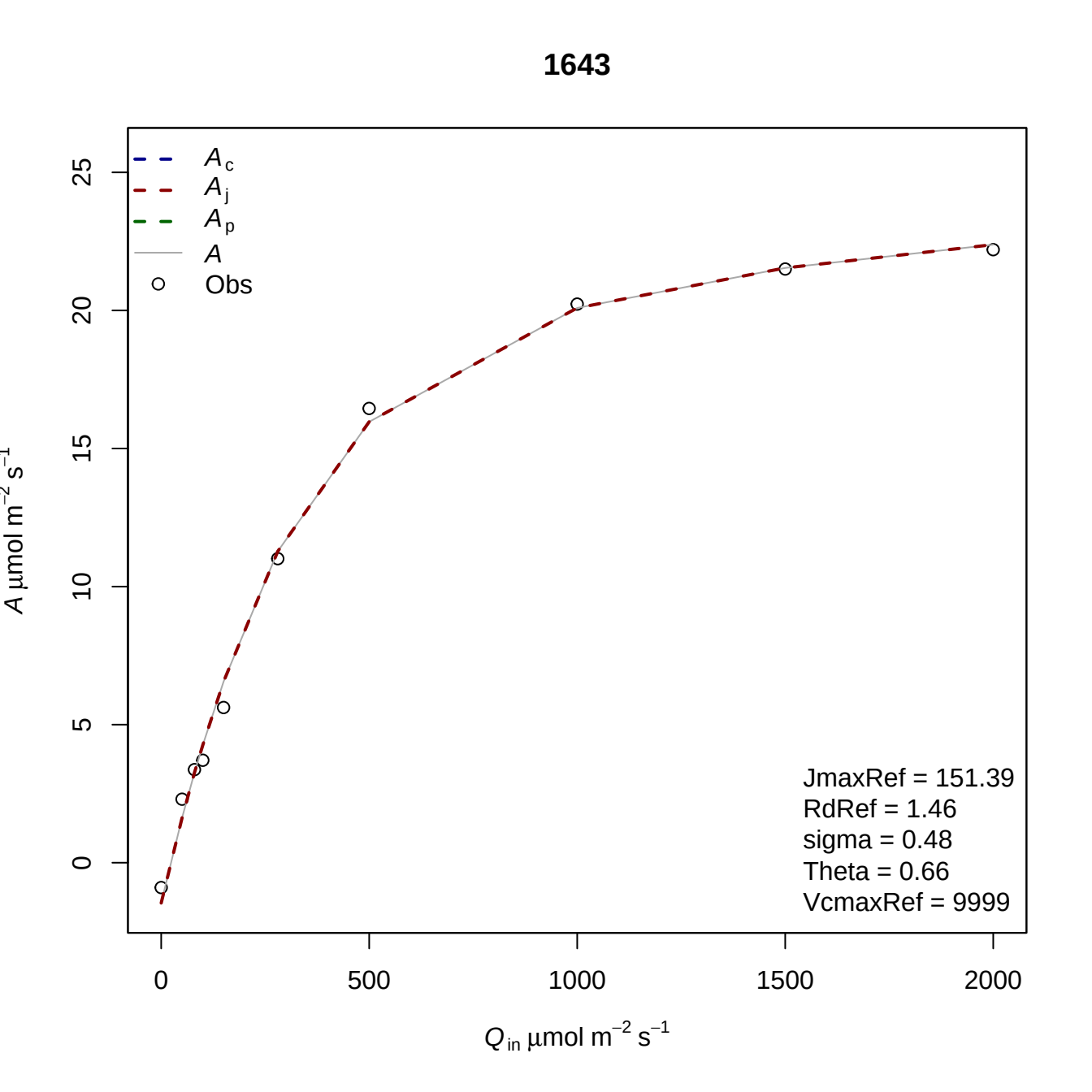

1638
A_c
A_j
A_p
A
Obs
JmaxRef = 209.42
RdRef = 0.88
sigma = 0.15
Theta = 0.64
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

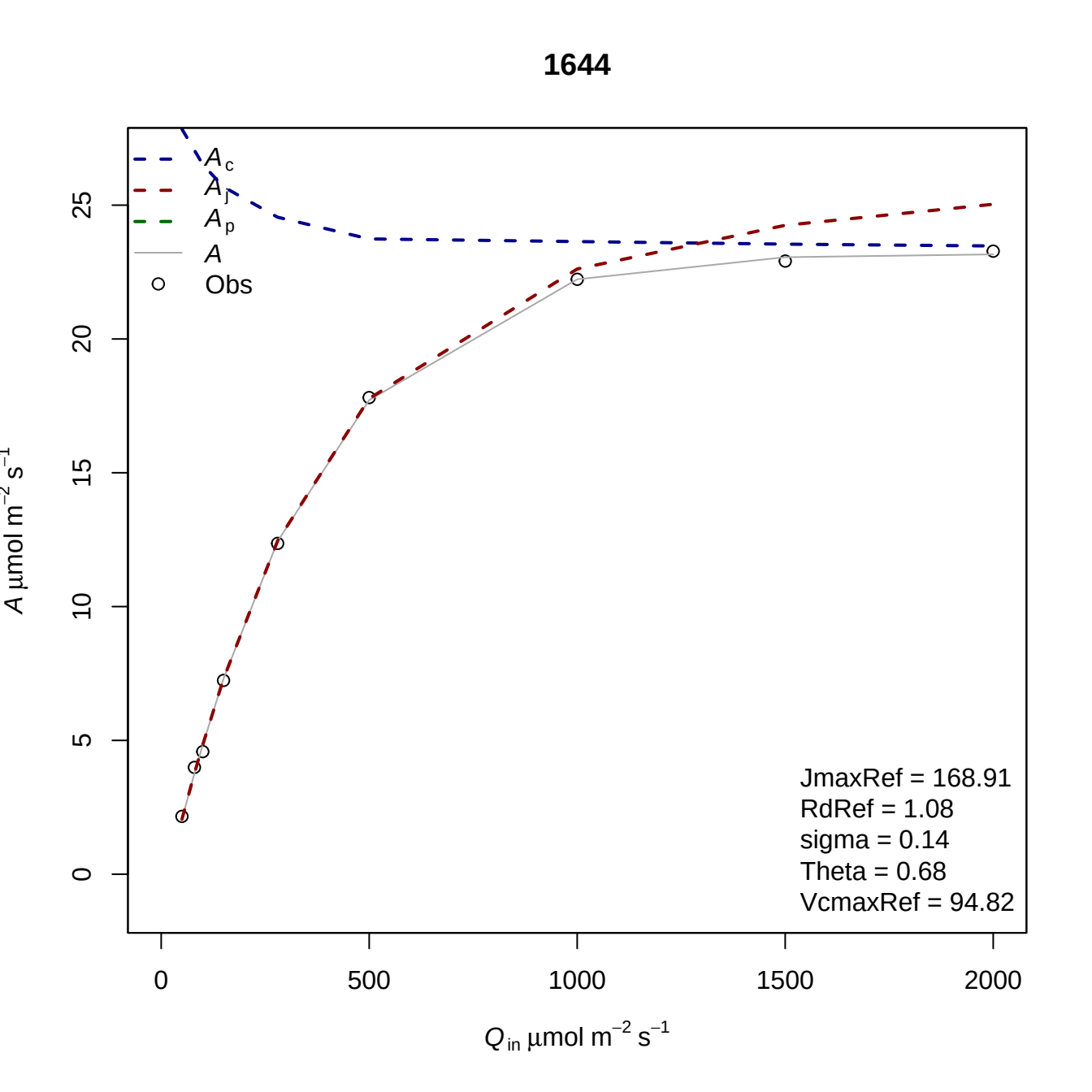

1639
A_c
A_j
A_p
A
Obs
JmaxRef = 55.49
RdRef = 0.96
sigma = 0.38
Theta = 0.12
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

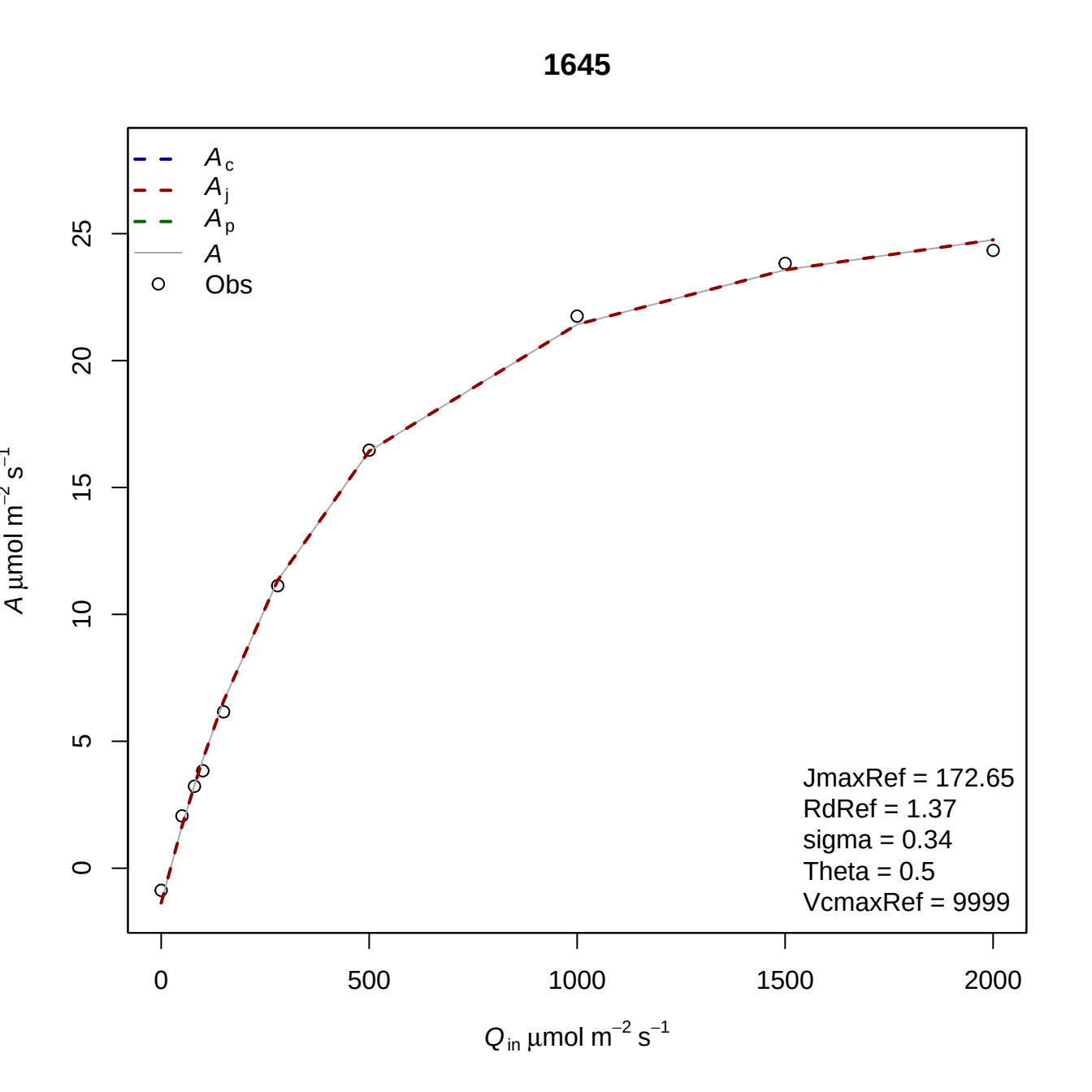

1640
A_c
A_j
A_p
A
Obs
JmaxRef = 84.18
RdRef = 0.95
sigma = 0.29
Theta = 0.61
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

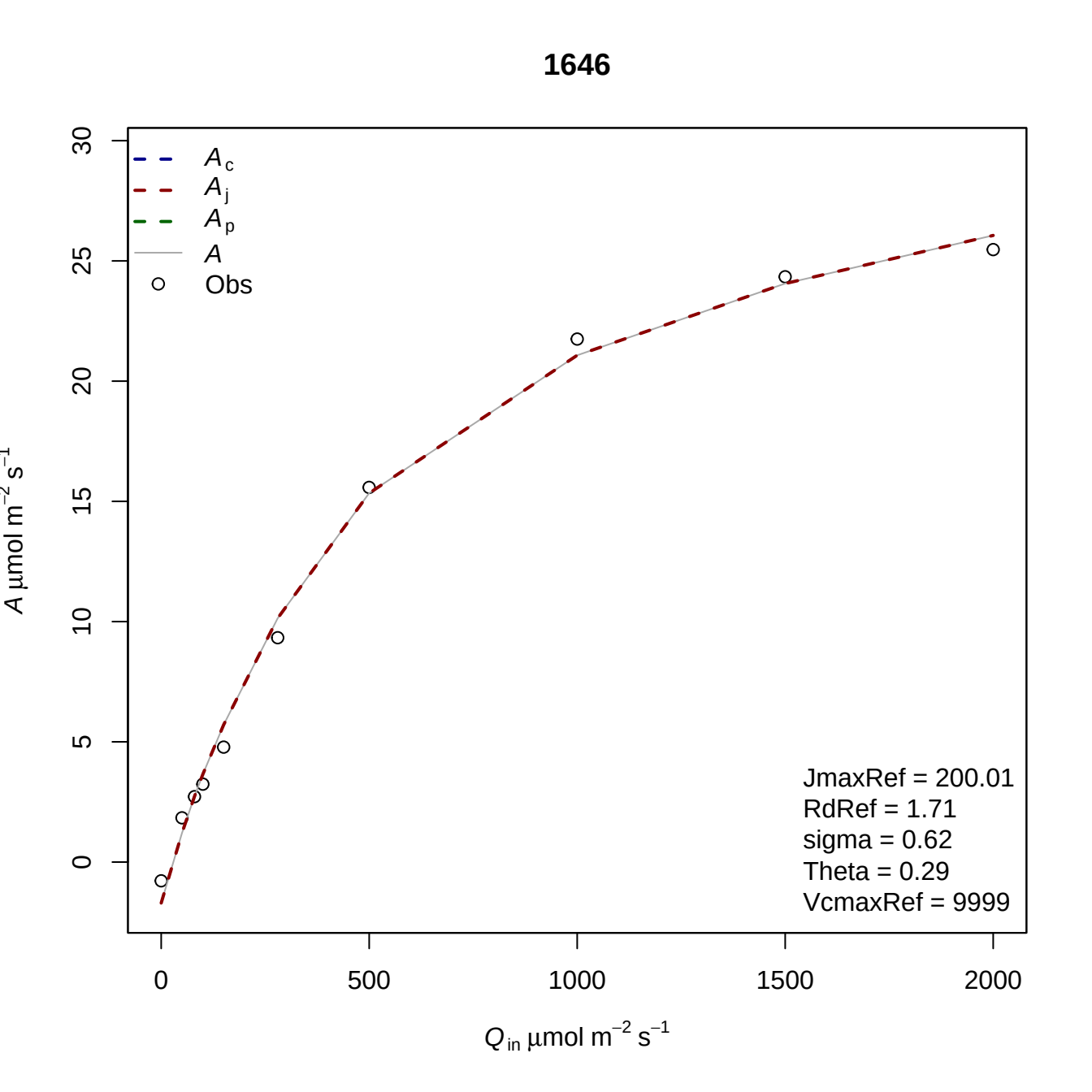

1641
A_c
A_j
A_p
A
Obs
JmaxRef = 82.09
RdRef = 0.64
sigma = 0.16
Theta = 0.53
VcmaxRef = 42.83
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

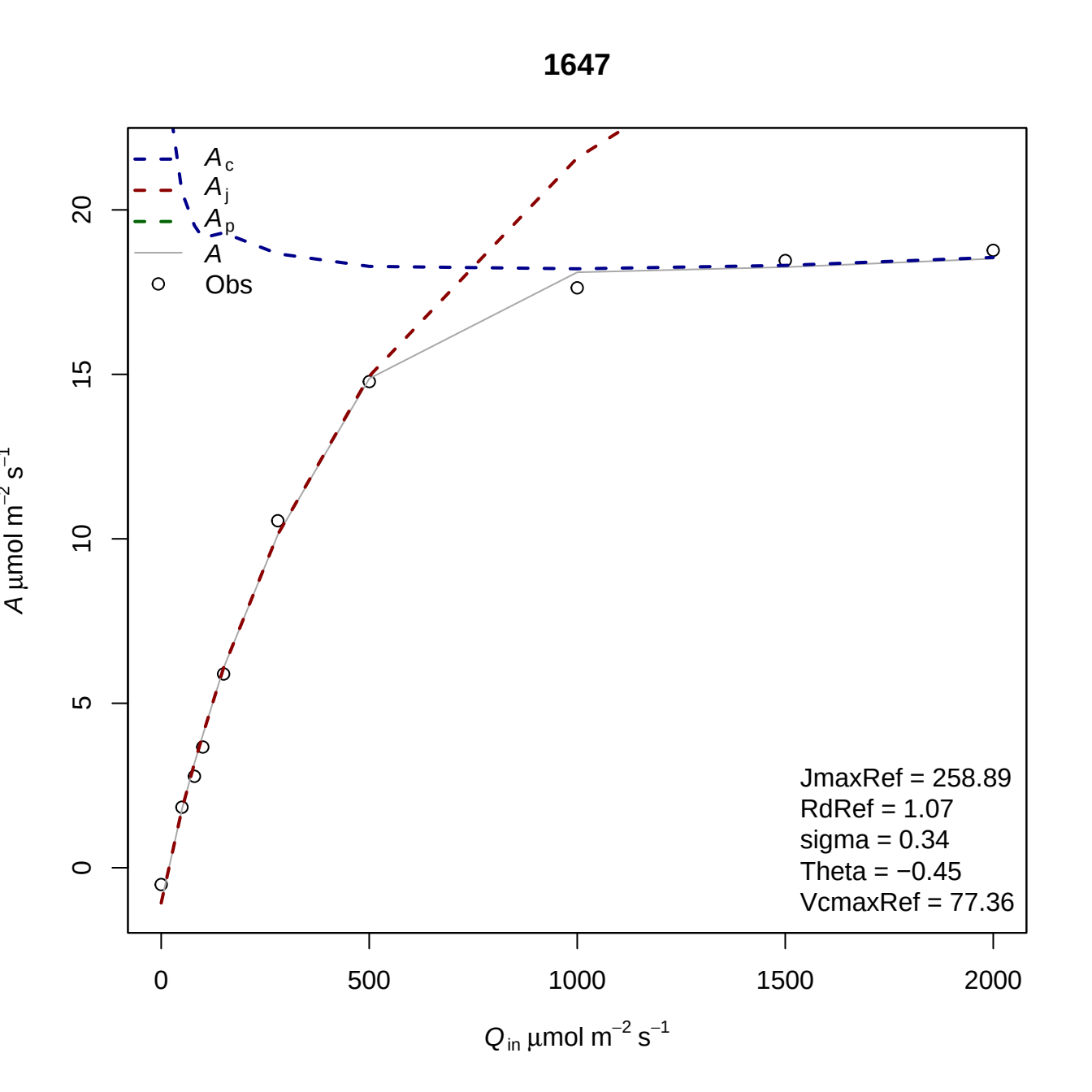

1642
A_c
A_j
A_p
A
Obs
JmaxRef = 69.46
RdRef = 0.51
sigma = 0.16
Theta = 0.1
VcmaxRef = 37.53
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

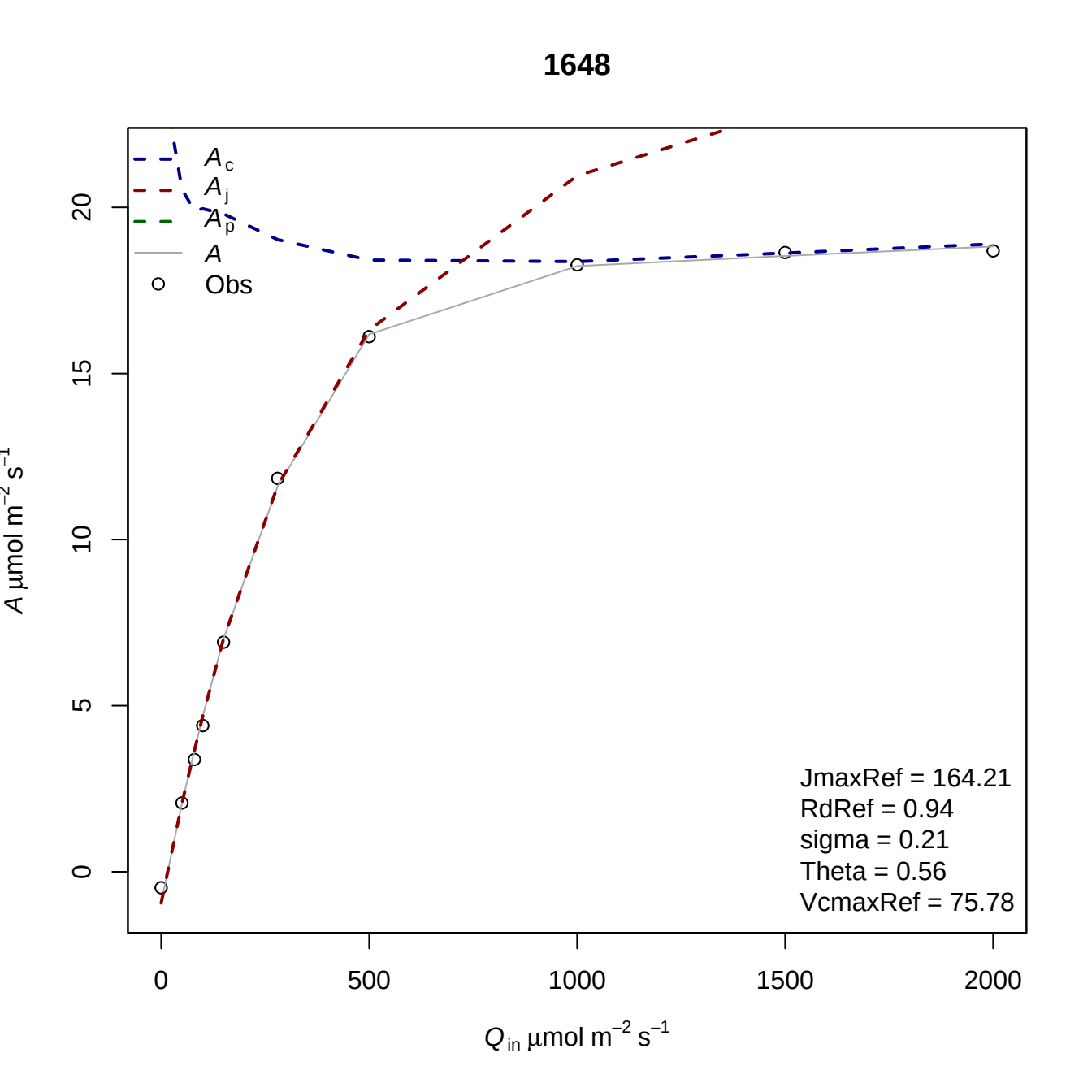

1643
A_c
A_j
A_p
A
Obs
JmaxRef = 151.39
RdRef = 1.46
sigma = 0.48
Theta = 0.66
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

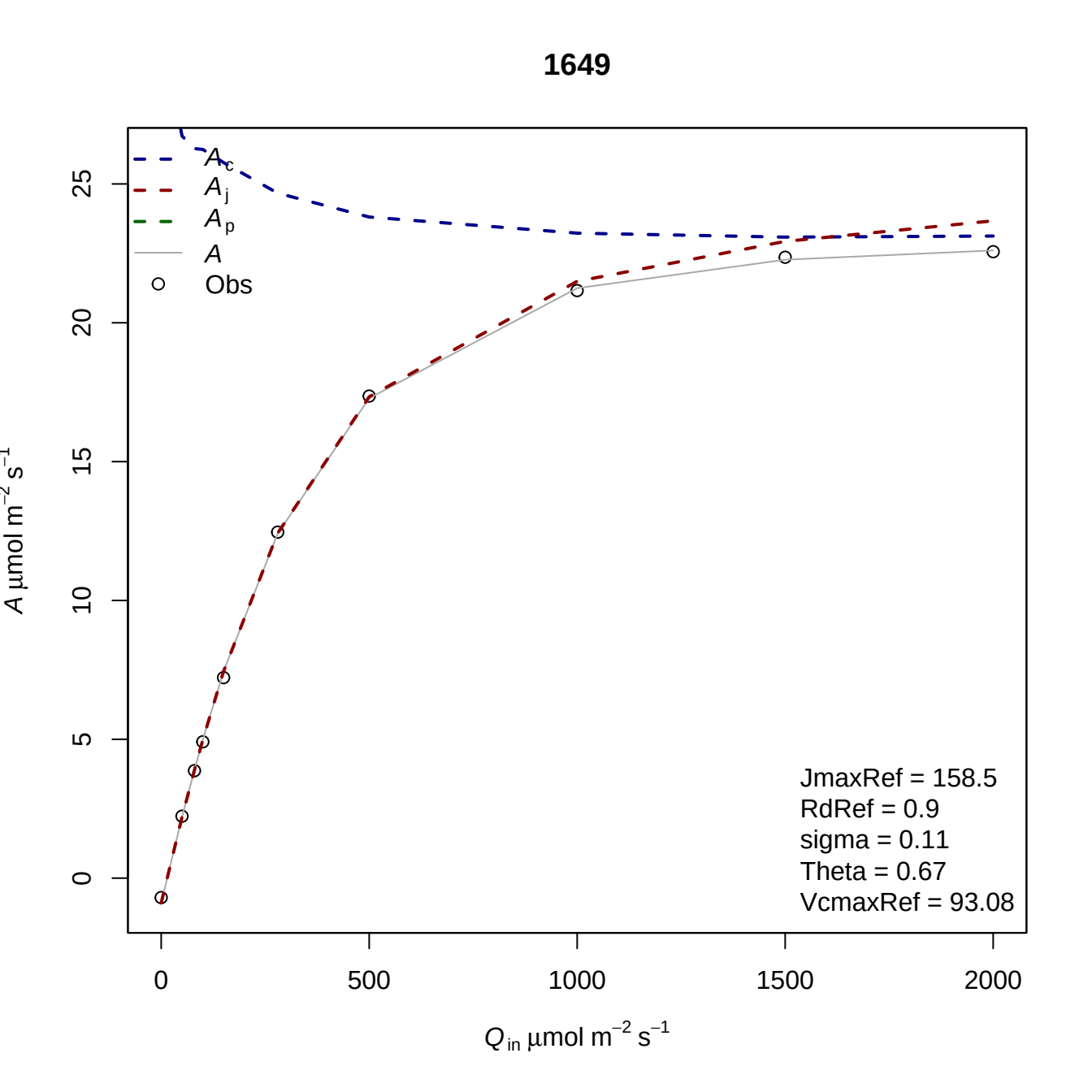

1644
A_c
A_j
A_p
A
Obs
JmaxRef = 168.91
RdRef = 1.08
sigma = 0.14
Theta = 0.68
VcmaxRef = 94.82
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

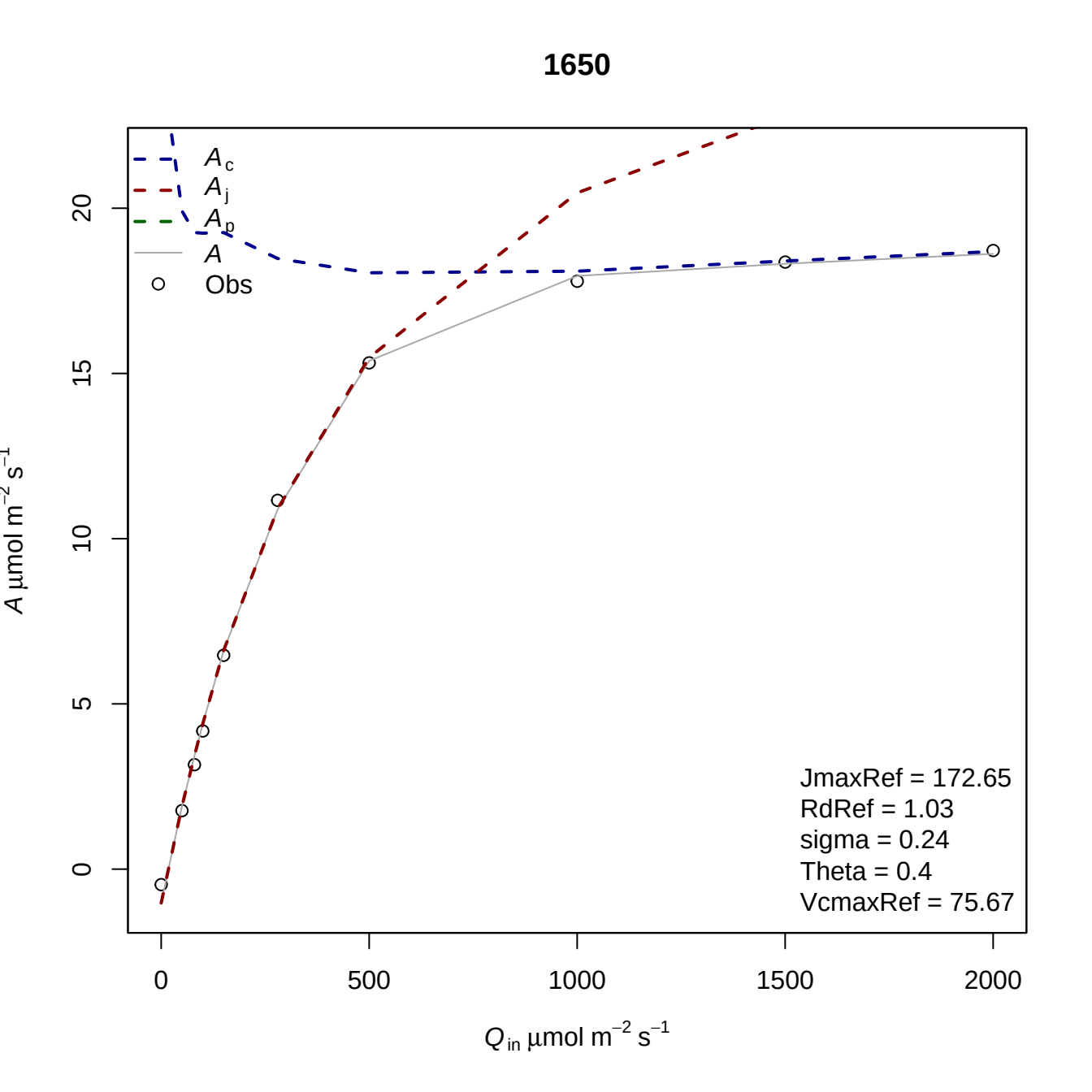

1645
A_c
A_j
A_p
A
Obs
JmaxRef = 172.65
RdRef = 1.37
sigma = 0.34
Theta = 0.5
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

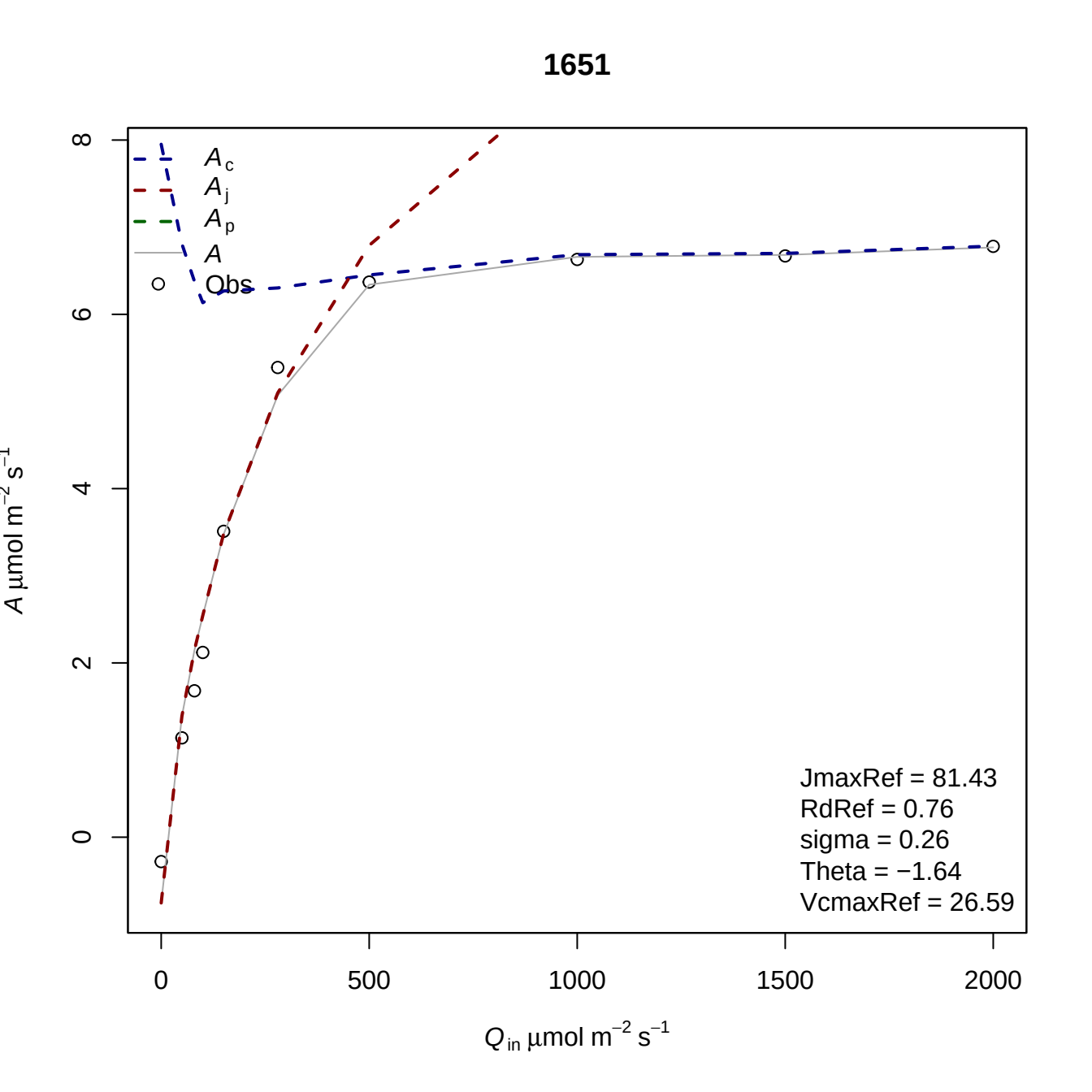

1646
A_c
A_j
A_p
A
Obs
JmaxRef = 200.01
RdRef = 1.71
sigma = 0.62
Theta = 0.29
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

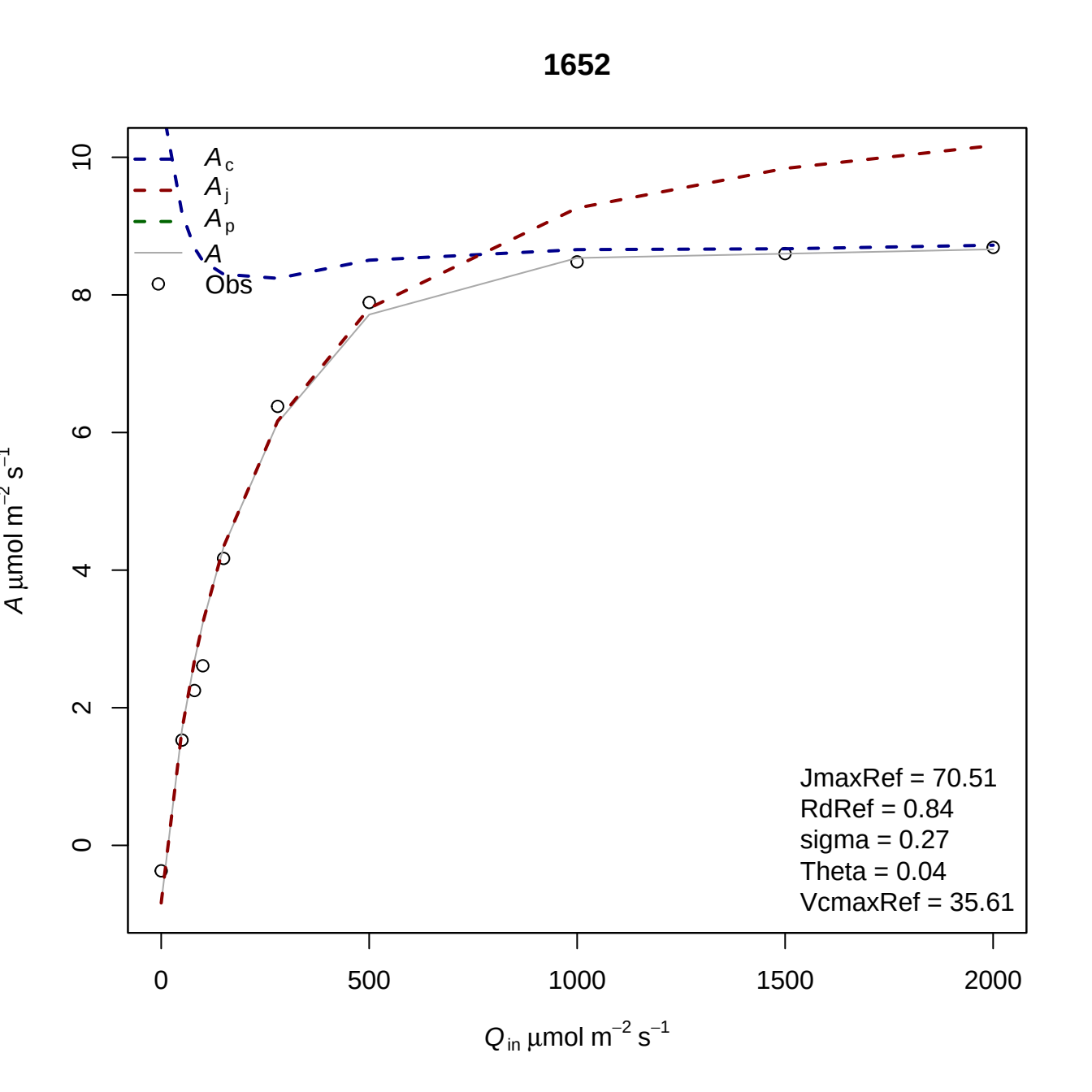

1647
A_c
A_j
A_p
A
Obs
JmaxRef = 258.89
RdRef = 1.07
sigma = 0.34
Theta = −0.45
VcmaxRef = 77.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

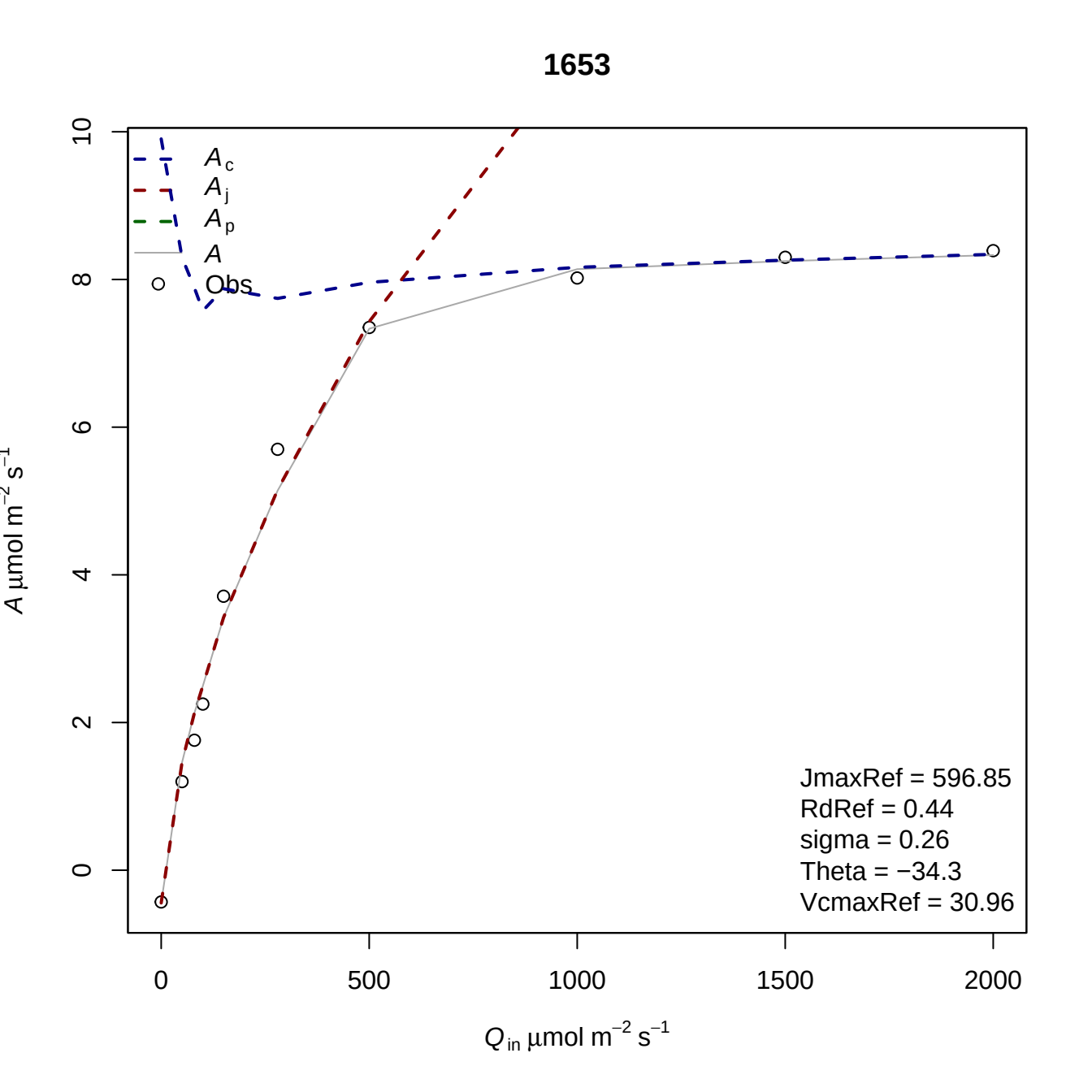

1648
A_c
A_j
A_p
A
Obs
JmaxRef = 164.21
RdRef = 0.94
sigma = 0.21
Theta = 0.56
VcmaxRef = 75.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

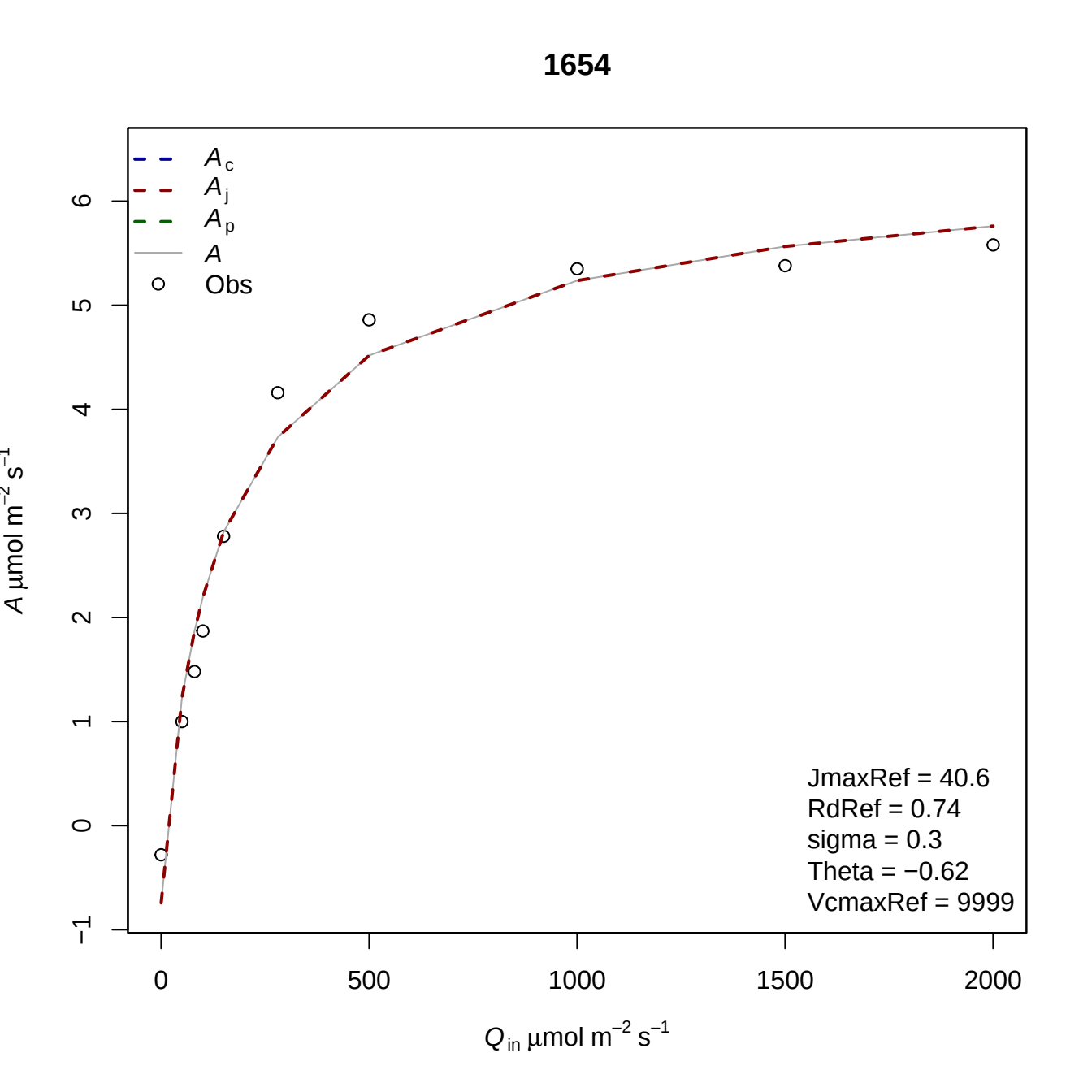

1649
A_c
A_j
A_p
A
Obs
JmaxRef = 158.5
RdRef = 0.9
sigma = 0.11
Theta = 0.67
VcmaxRef = 93.08
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

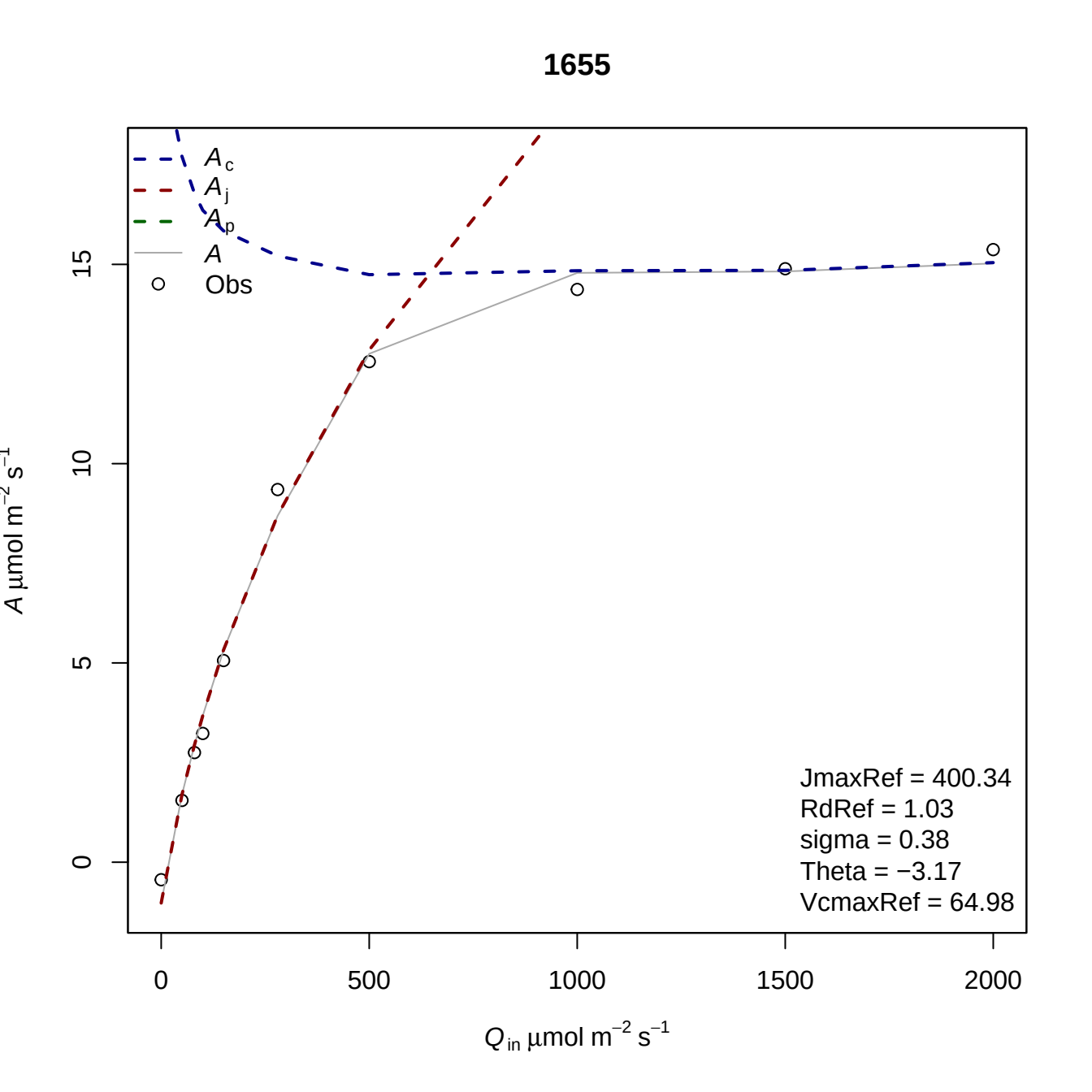

1650
A_c
A_j
A_p
A
Obs
JmaxRef = 172.65
RdRef = 1.03
sigma = 0.24
Theta = 0.4
VcmaxRef = 75.67
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

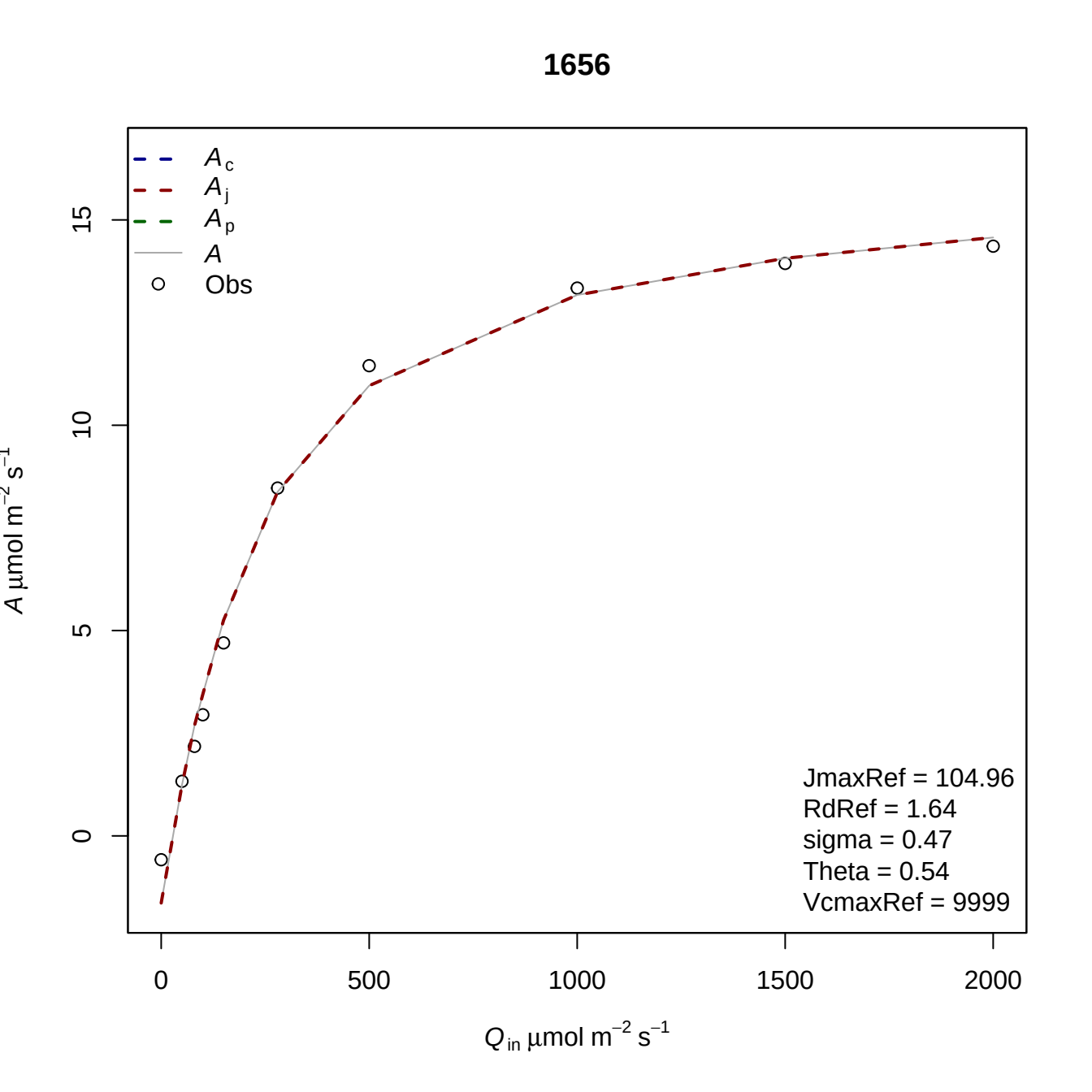

1651
A_c
A_j
A_p
A
Obs
JmaxRef = 81.43
RdRef = 0.76
sigma = 0.26
Theta = −1.64
VcmaxRef = 26.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

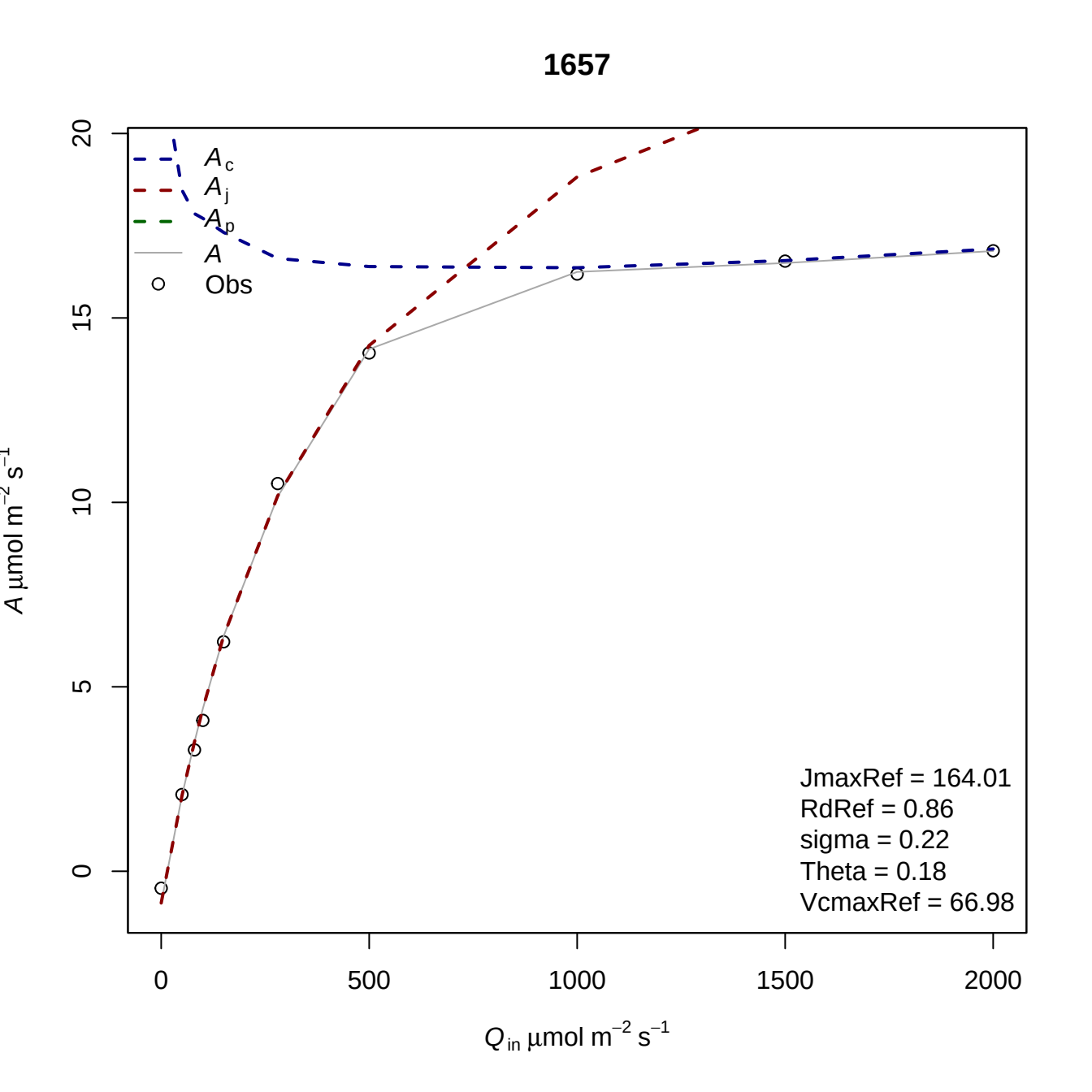

1652
A_c
A_j
A_p
A
Obs
JmaxRef = 70.51
RdRef = 0.84
sigma = 0.27
Theta = 0.04
VcmaxRef = 35.61
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

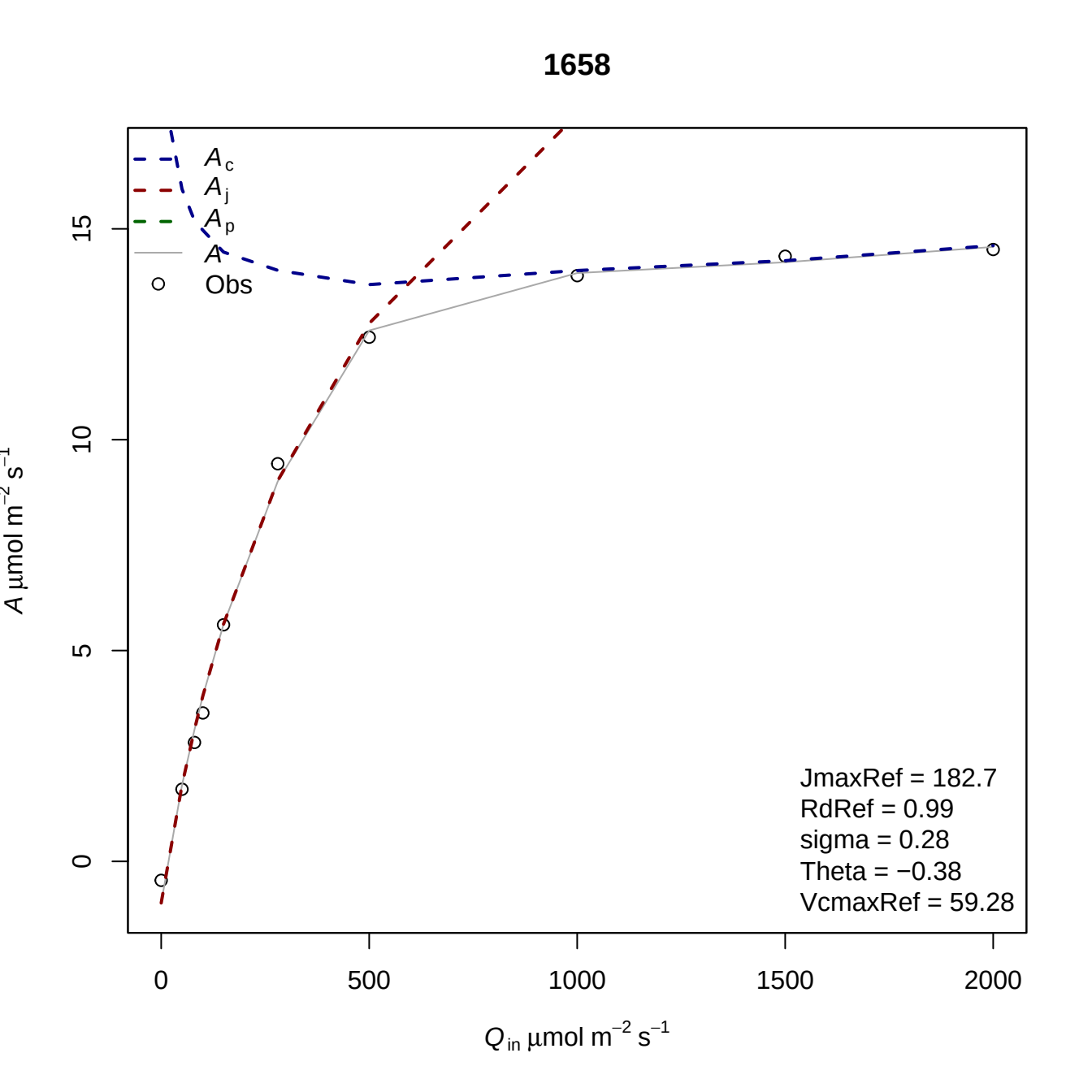

1653
A_c
A_j
A_p
A
Obs
JmaxRef = 596.85
RdRef = 0.44
sigma = 0.26
Theta = −34.3
VcmaxRef = 30.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

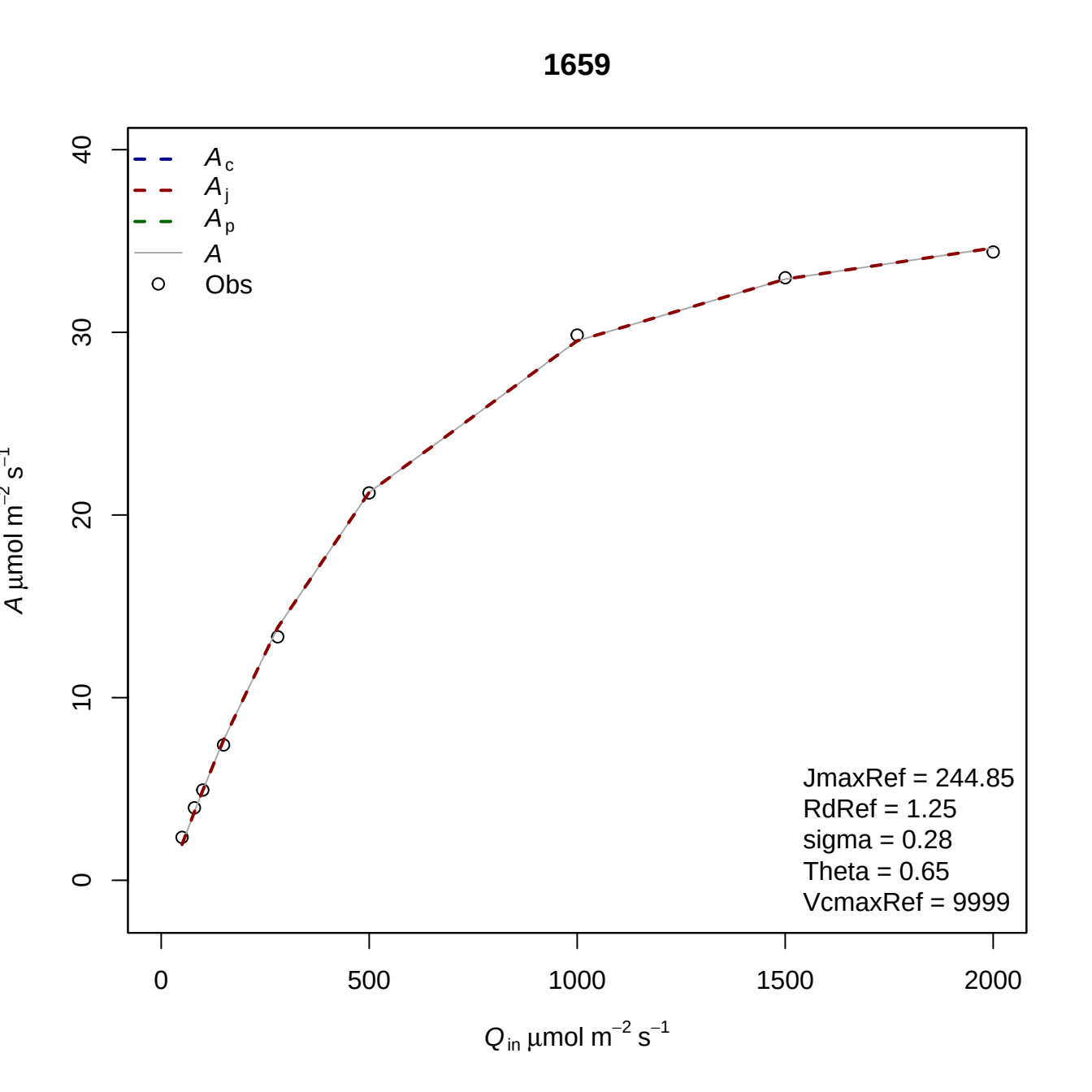

1654
A_c
A_j
A_p
A
Obs
JmaxRef = 40.6
RdRef = 0.74
sigma = 0.3
Theta = −0.62
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

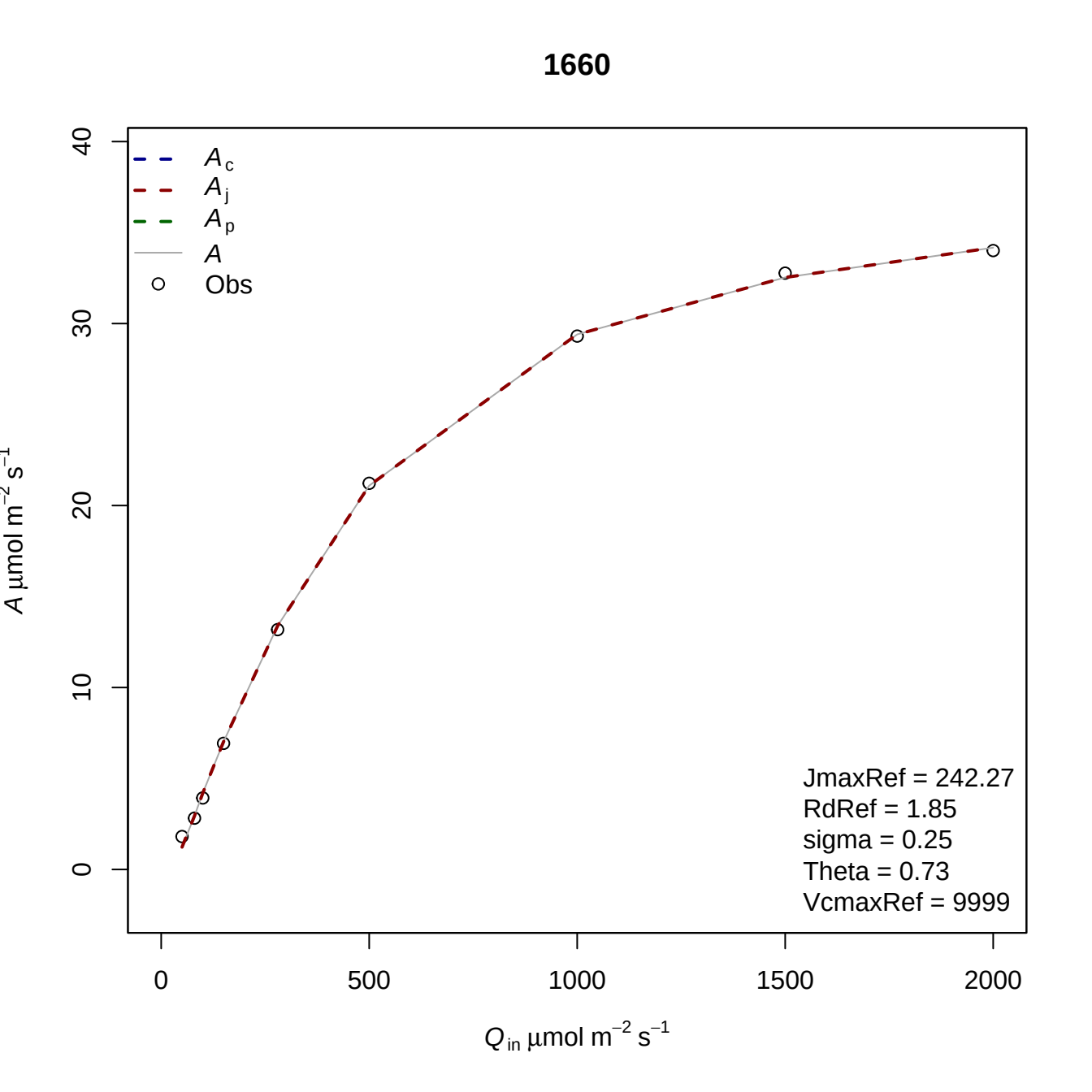

1655
A_c
A_j
A_p
A
Obs
JmaxRef = 400.34
RdRef = 1.03
sigma = 0.38
Theta = −3.17
VcmaxRef = 64.98
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

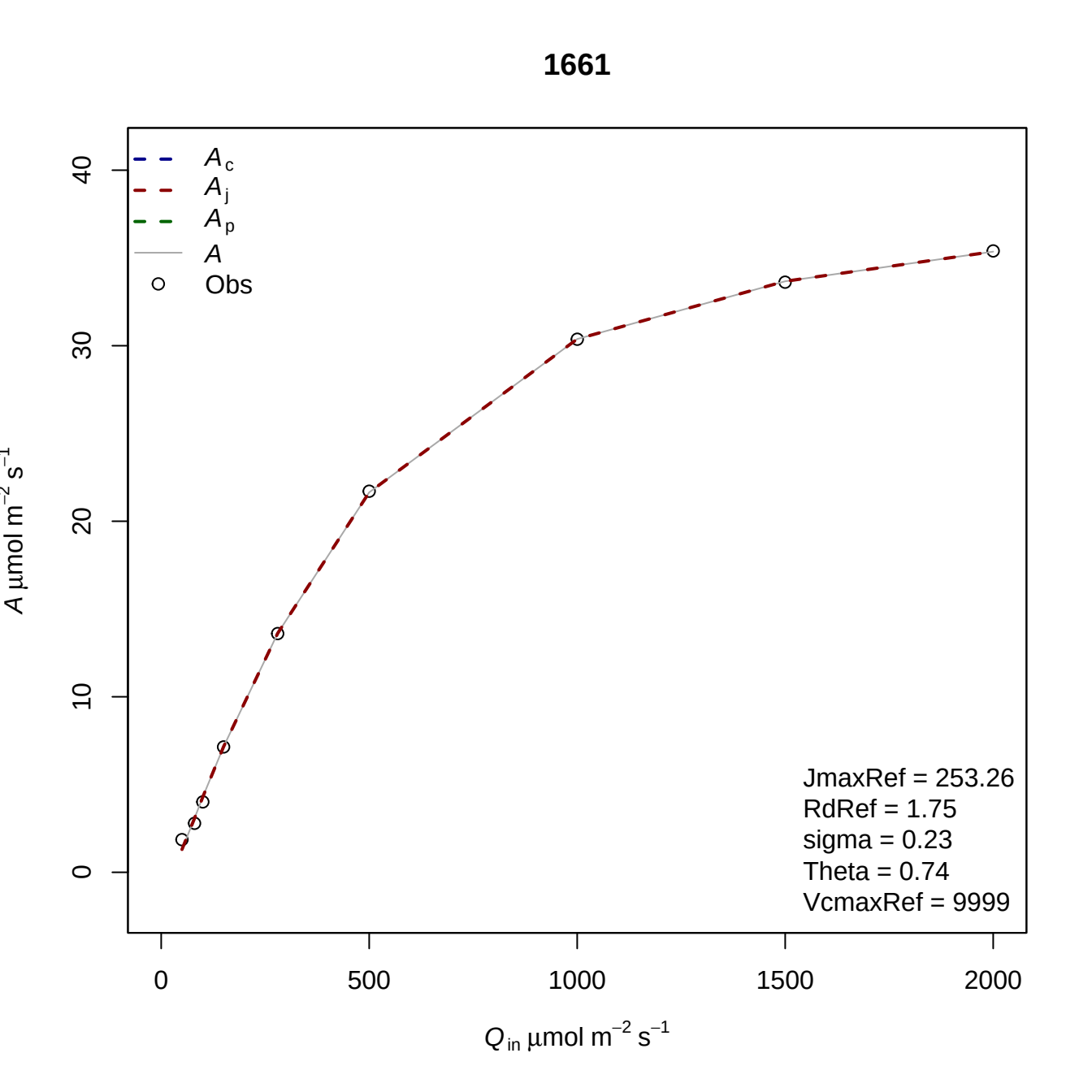

1656
A_c
A_j
A_p
A
Obs
JmaxRef = 104.96
RdRef = 1.64
sigma = 0.47
Theta = 0.54
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

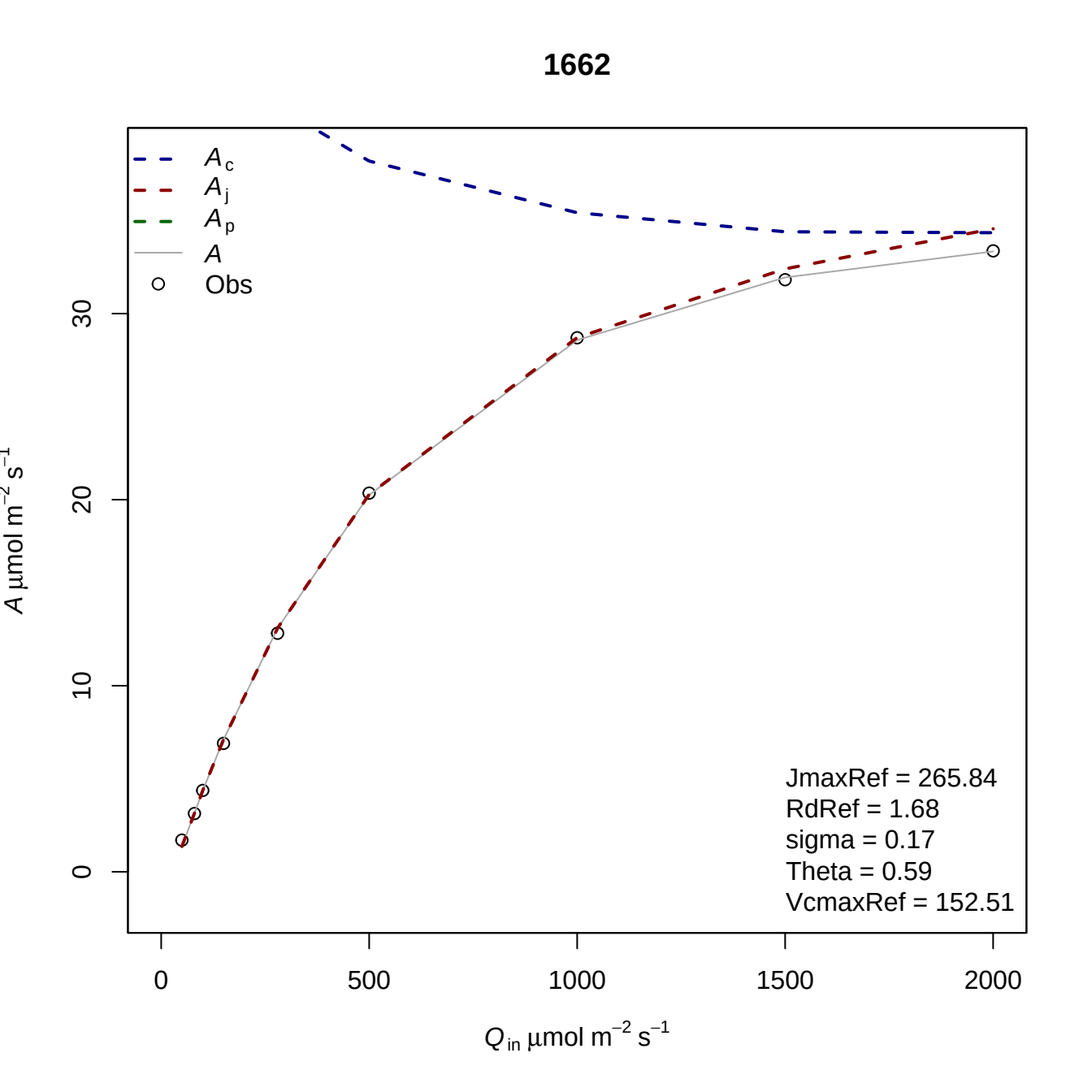

1657
A_c
A_j
A_p
A
Obs
JmaxRef = 164.01
RdRef = 0.86
sigma = 0.22
Theta = 0.18
VcmaxRef = 66.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000


1658
A_c
A_j
A_p
A
Obs
JmaxRef = 182.7
RdRef = 0.99
sigma = 0.28
Theta = −0.38
VcmaxRef = 59.28
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}


1659
A_c
A_j
A_p
A
Obs
JmaxRef = 244.85
RdRef = 1.25
sigma = 0.28
Theta = 0.65
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1660
A_c
A_j
A_p
A
Obs
JmaxRef = 242.27
RdRef = 1.85
sigma = 0.25
Theta = 0.73
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1661
A_c
A_j
A_p
A
Obs
JmaxRef = 253.26
RdRef = 1.75
sigma = 0.23
Theta = 0.74
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1662
A_c
A_j
A_p
A
Obs
JmaxRef = 265.84
RdRef = 1.68
sigma = 0.17
Theta = 0.59
VcmaxRef = 152.51
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}